

CHAPTER 2

AIRFRAME

CHAPTER OVERVIEW

SECTION	TITLE
2-1	Equipment Description and Data
2-2	Troubleshooting
2-3	On Aircraft Inspections
2-4	Maintenance Procedures
2-5	Maintenance Procedures (BENCH)
2-6	Structural Repair

2-1/(2-2 Blank)

SECTION 2-1

EQUIPMENT DESCRIPTION AND DATA

SECTION OVERVIEW

PARAGRAPH	TITLE
2-1-1	Airframe Description and Data

2-1-1 AIRFRAME DESCRIPTION AND DATA.

2-1-1.1 Airframe Overview.

This section contains descriptions of the airframe and associated equipment. The helicopter airframe is divided into six sections (Figure 2-1-1-1):

- (1) Cockpit (Nose Section)
- (2) Cabin
- (3) Transition Section
- (4) Tail Cone
- (5) Tail Rotor Pylon
- (6) Main Rotor Pylon

The primary structure is aluminum alloy. Some titanium and steel are used for firewalls and various fittings. Nonstructural members are primarily made of reinforced plastic. The fuselage is semi-monocoque construction with horizontal anti-plough beams extending through the tub from the cockpit to the transition section. Structural arrangement of each section of the helicopter is shown in Figure 2-1-1-2 through Figure 2-1-1-7.

2-1-1.2 Cockpit.

The cockpit, consisting of an avionics compartment, accommodates the pilot and copilot, and associated systems components. The cockpit is entered through hinged jettisonable cockpit doors on each side of the nose canopy. The avionics compartment is entered through a hinged nose door (Figure 2-1-1-8).

2-1-1.2.1 Nose Door.

A reinforced plastic avionics compartment nose door opens to expose the Interior Systems and components located in the front nose section (Figure 2-1-1-8). The nose door, which is hinged at the top, is opened by pulling out the two latches on the bottom, disengaging latch mechanisms from lockpins. The nose door is held in the open position by a support strut. A watertight seal is installed on the door.

2-1-1.2.2 Windshield and Windows.

The cockpit windshield and windows consist of left and right upper windows, left and right lower windows, pilot and copilot outboard windshields, and a center windshield (Figure 2-1-1-8). The pilot and copilot shatter-resistant windshields are made of electropane panels containing braid lead-in and resistance wiring, used for anti-icing purposes. All other windows are made of stretched Plexiglass. Foam seal tape, installed between all mating surfaces, acts as a sealant to ensure watertightness. The windows of the pilot's and copilot's doors can be jettisoned through mechanisms, which can be actuated by handles on the inside and outside of the doors. Both window jettison handles must be rotated in a circular motion before the windows can be released from the doors.

2-1-1.2.3 Cockpit Doors.

Cockpit doors, one for the pilot and copilot, are hinged at the front edge (Figure 2-1-1-8). Each door contains a fixed as well as a slide-open window made of stretched Plexiglass. A single-action release mechanism in each door allows the door to be jettisoned in an emergency. A watertight seal is installed on each cockpit door.

2-1-1.2.4 Pilot's and Copilot's Seats.

These seats are in the cockpit and are interchangeable with each other (Figure 2-1-1-9). Each seat has a vertical height adjustment and a horizontal seat adjustment. Two tracks on the cockpit floor allow the seat to slide to front or rear. The seat can be locked into position at 1/2-inch spaces. Pilot or copilot restraint is by a shoulder harness, lap belt, and crotch strap. A shoulder harness inertia reel is on the rear center of each seat. The reel allows about 12 to 18-inches of travel. It automatically locks when subjected to a sudden force. Each seat has a rear tilt feature which allows the seat to be disengaged from its tracks and tilted back into the troop compartment. This is performed by pulling the emergency vertical release handle to the right. This will allow the seat to drop to its lowest position. The tilt back release handles must be pushed in toward center. Seats must be in their lowest posi-

tion prior to tilting. Do not operate emergency vertical release mechanisms unless seat is occupied.

2-1-1.2.5 Miscellaneous Furnishings.

The miscellaneous furnishings consist of two portable fire extinguishers, three first aid kits (two located behind the copilot's seat while the other is behind the pilot's seat), three ashtrays, and two map/data cases (Figure 2-1-1-10).

2-1-1.3 Cabin.

The cabin, interconnecting the cockpit and transition section, has two crew chief/crewmember stations and a passenger/cargo compartment. The cabin is entered through rear-sliding passenger/cargo doors on each side of the helicopter. The crew chief stations are rear of the cockpit and to front of the passenger/cargo doors, on each side of the helicopter. There are provisions for six litters behind the crew chief station (Figure 2-1-1-11).

2-1-1.3.1 Passenger/Cargo Doors.

These rear sliding doors, on each side of the helicopter cabin, open to the passenger/cargo compartment (Figure 2-1-1-11). Each door has two Plexiglass windows. A single-action release mechanism incorporated in each door allows both windows to be jettisoned in an emergency.

2-1-1.3.2 Cabin Floor.

There are three removal panels in the midsection of the floor (Figure 2-1-1-12). The panels are made of fiberglass bottom skin, honeycomb core sections of reinforced honeycomb core, and a reinforced plastic (fiberglass) top skin covering. It also has 27 multipurpose fittings for cargo tiedown, passenger/litter installations. The cargo hook is reached through the cargo hook access panel in the floor. Floor panel installation is not required for towing, jacking, tiedown, or mooring. Floor panels are not required providing that no unrestrained equipment is stored in the cabin area during flight or transport of the helicopter. Sidewall interior panels that secure to the floor will require another method of retention or removal during flight or transport operations. During flight or transport of the heli-

copter, personnel or equipment is not allowed in the cabin without the floor panels installed.

2-1-1.3.3 Crew Chief's Sliding Windows.

Two (2) front sliding Plexiglass windows on each side of the helicopter, rear of the cockpit and front of the cabin doors, provide a split hatch covering over the crew chief's station opening. Each window has a seal bonded to the front and rear edge of the frame which interlocks when both windows are closed. A locking mechanism on the rear window is used to secure/unlock both sliding windows (Figure 2-1-1-12).

2-1-1.3.4 Battery Compartment.

The battery compartment houses a 24 VDC, 5.5 amp hour, 20-cell nickel cadmium battery which is identifiable by a blue case or a 24 VDC, 9.5 amp hour, 12 cell sealed lead acid battery which is identifiable by a pink case. The battery is located behind the copilot's seat in the front left corner of the cabin floor.

2-1-1.3.5 Passenger Seats.

The S-70A passenger seats may be set up in any of three different configurations consisting of 12, 14, or 20 seats. Refer to Figure 2-1-1-13, for seat layout. Each seat consists of a tubular frame and nylon cloth and has a seat belt and shoulder harness.

2-1-1.3.6 Soundproofing Panels.

Soundproofing panels, made of an acoustical insulation-type material to lessen noise, are attached to the fuselage structure throughout the helicopter cabin.

2-1-1.4 Transition Section.

The transition section interconnecting the helicopter cabin and tail cone holds the fuel tanks, Fuel Drain Collector System, Environmental Control System (ECS) and the rear avionics compartment. On older models, the transition section is reached from the inside of the passenger/cargo compartment. On newer models, the transition section is reached through the transition section avionics door

mounted on the right side of the transition section (Figure 2-1-1-14).

2-1-1.4.1 Transition Section Avionics Door.

A fiberglass and Kevlar rear transition section avionics door, on the right side of the rear transition section, opens to access the Rear Transition Avionics Systems and components. The door is hinged at the top, and opened by pulling out the two latches on the bottom, disengaging latch mechanisms from lock pins. A support strut holds the door open. A watertight seal is installed on the door and jamb.

2-1-1.4.2 Main Fuel Cell Compartment.

Two fuel cell compartments are located on each side of the transition section. Each compartment, consisting of foam liner panels and an aluminum skinned honeycomb cover panel, houses a fuel cell and Fuel System components. The No. 1 and No. 2 fuel cells consist of a self-sealing rubber compound that protects against leaks caused by punctures. Fuel cell compartments are reached from inside the passenger compartment (Figure 2-1-1-14).

2-1-1.4.3 Environmental Control System.

The Environmental Control System (ECS) consists of a heater/demister and a vapor cycle air conditioner in addition to the standard Heating/Ventilating System. Components are located on each side of the transition section over the fuel cell compartments (Figure 2-1-1-14).

2-1-1.5 Tail Cone.

The tail cone, interconnecting the transition section and tail rotor pylon, supports the tail rotor drive shaft and tail pylon. The tail cone also encloses the tail rotor flight controls and tail landing gear. It is reached from inside the rear passenger compartment. An access cover on each rear end side opens to access the tail pylon attachment bolts, flight controls, and tail landing gear (Figure 2-1-1-15).

The tail cone has two covers on the top exterior. The covers are made of Kevlar material and can be opened to reach the tail drive shafts. Each cover is piano hinged at the right edge. Five quick-release

fasteners on the left edge secure the cover closed (Figure 2-1-1-15).

2-1-1.6 Tail Rotor Pylon.

The tail pylon is a foldable section at the rear end of the helicopter. The tail pylon is supported by and hinged to the tail cone section. The pylon supports the stabilizers, the intermediate and tail gear boxes, connecting drive shaft, the tail rotor assembly, and part of the flight controls. When the stabilator is removed, the tail pylon can be folded along the right side of the tail cone. Removable fairings on the pylon open to the intermediate and tail gear boxes, tail drive shafts, and tail rotor flight controls (Figure 2-1-1-16).

2-1-1.6.1 Tail Rotor Pylon Drive Shaft Cover.

This cover, on the pylon leading edge, opens to the tail drive shafts. The cover consists of nonmetallic honeycomb core sandwiched between reinforced plastic skins, four hinges, and nine quick-release fasteners. The tail pylon drive shaft cover is removed from the pylon by separating the hinge-halves (Figure 2-1-1-16). The drive shaft cover also serves as a VHF/FM antenna.

2-1-1.6.2 Stabilator.

A controllable stabilator is on the tail pylon (Figure 2-1-1-16). The stabilator provides longitudinal stability in forward flight. During hover and low speed forward flight, the stabilator swings down to eliminate nose up attitudes caused by tail rotor downwash hitting the flat stabilator surface. For a complete description of the Stabilator Control System, refer to TM 11-70-23.

2-1-1.6.3 Position Transmitter/Limit Switch Assembly.

The position transmitter/limit switch assembly provides a signal to the stabilator indicators that visually indicates the stabilator position relative to the centerline of the helicopter. The switch assembly also limits stabilator movement to 8° up and 39° down. The position transmitter, which provides the signal to the stabilator indicators, requires 26 VAC excitation. Four limit switches,

two up-limit and two down-limit, limit stabilator movement. The limit switches require 28 VDC operating power (Figure 2-1-1-17).

2-1-1.7 Main Rotor Pylon.

The main rotor pylon attached to the upper cabin and rear fuselage is a protective aerodynamic covering that also gives smooth airflow induction for cooling aircraft major subsystem components. Hinged fiberglass/honeycomb-sandwiched panels and covers open to the internal pylon areas and helicopter component subsystems (Figure 2-1-1-18).

2-1-1.7.1 Engine Cowling and Work Platform.

The engine cowling and work platform is located on each side of the main rotor pylon. When opened, each platform has a flat surface for maintenance use and inspection (Figure 2-1-1-19). Each platform is capable of supporting a Static weight of 400 pounds. When closed in the flight position, the platform confines and directs cooling air into the engine area. The platforms are made of honeycomb aluminum, steel channels, and titanium sheet.

2-1-1.7.2 Main Rotor Pylon Sliding Cover.

The sliding cover is on the front section of the main rotor pylon. Moving the sliding cover to the front exposes the helicopter's Flight Control, Hydraulic, and Heating Systems. The sliding cover consists of a honeycomb aluminum, fiberglass structure, latch assemblies, rollers, and tracks (Figure 2-1-1-20).

2-1-1.8 Wire Strike Protection System **WSPS**.

The Wire Strike Protection System is a simple, Lightweight System that will cut, break, or deflect wires that may strike the helicopter. The system consists of cutters and deflectors located on the fuselage and landing gear. They include the upper cutter on the rear of the sliding fairing, the Pitot cutter/deflector on the front of the sliding fairing, windshield post and wiper deflectors, door hinge deflector, step extension and step deflector, landing gear joint deflector, main landing gear cutter/

deflector, and tail landing gear deflector. The cutters are clamped to landing gear drag beams, and bolted to the airframe and main rotor sliding pylon cover.

2-1-1.9 Vibration Absorbers.

Four vibration absorbers are installed: one in the nose section, one in the cabin overhead just in front of the main transmission, one in the cabin overhead just behind the main transmission and one on the tail pylon (Figure 2-1-1-21). A roll vibration absorber is installed under each drag beam support fairing. Also, a bifilar absorber is installed on top of the main rotor head to dampen rotor vibrations. For a description of the bifilar absorber, refer to PARA 5-1-1.

2-1-1.9.1 Nose Vibration Absorber (Spring-Type).

The nose vibration absorber is behind the glide slope antenna, underneath the avionics compartment lower shelf (Figure 2-1-1-21). The nose vibration absorber is tuned to reduce cockpit centerline vertical 4/rev vibrations. The nose vibration absorber consists of a body (mass), springs bolted to the body, weights (tuning plates), and support fittings. The nose vibration absorber weighs approximately 70 pounds.

2-1-1.9.2 Nose Vibration Absorber (Box-Type).

The nose box frame vibration absorber is behind the glide slope antenna, underneath the avionics compartment lower shelf (Figure 2-1-1-22). The vibration absorber is tuned to reduce cockpit centerline vertical 4/rev vibrations. The vibration absorber consists of an upper and lower housing containing a mass trapped between two sets of springs. The springs are always in compression, even at the limits of the mass travel. The vibration absorber is attached to airframe fittings by four tension bolts and barrel-nuts. The vibration absorber weighs approximately 50 pounds.

2-1-1.9.3 Cabin Vibration Absorber (Spring-Type).

Two vibration absorbers are installed in the cabin (Figure 2-1-1-21). The forward cabin vibration

absorber is under the soundproofing just in front of the main transmission at about station 308. On some helicopters an aft cabin vibration absorber is under the soundproofing just in back of the main transmission at about station 360. The forward cabin vibration absorber is tuned to reduce cabin overhead vertical 4/rev vibrations. The forward cabin vibration absorber consists of a body (mass), springs bolted to the body, weights (tuning plates), and support fittings. The forward cabin vibration absorber weighs approximately 75 pounds.

2-1-1.9.4 Cabin Vibration Absorber (Box-Type).

The cabin box frame vibration absorber is under the soundproofing just in front of the main transmission at approximately station 308 (Figure 2-1-1-22). The vibration absorber is tuned to reduce cabin overhead vertical 4/rev vibrations. The vibration absorber consists of an upper and lower housing containing a mass trapped between two sets of springs. The springs are always in compression, even at the limits of the mass travel. The vibration absorber is attached to airframe fittings by four tension bolts and barrel-nuts. The absorber weighs approximately 94 pounds.

2-1-1.9.5 Stabilator Vibration Absorber.

The stabilator vibration absorber is on the tail pylon (Figure 2-1-1-21). The stabilator vibration

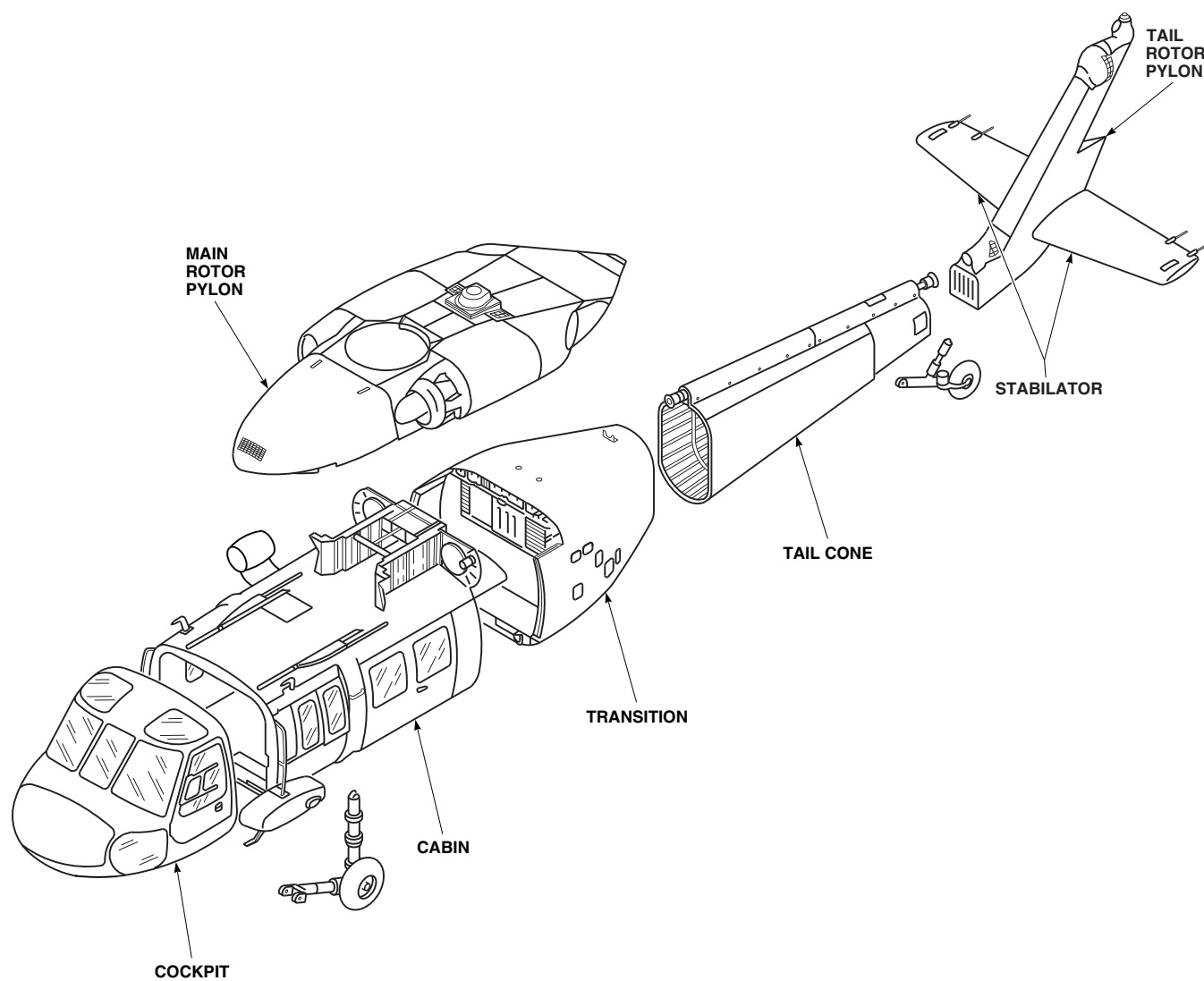
absorber dampens stabilator vibrations induced by main rotor blade downwash. The stabilator vibration absorber consists of an elastomeric bearing attached to the right stabilator attach fitting.

2-1-1.9.6 Roll Vibration Absorber **ROLL .**

A roll vibration absorber is installed under both left and right drag beam support fairings (Figure 2-1-1-21). The roll vibration absorbers are tuned to reduce roll vibrations. The roll vibration absorbers consist of a spring, body (mass) bolted to the spring, weights (tuning plates), and support fitting. Each roll vibration absorber weighs approximately 21 pounds.

2-1-1.10 Fast Rope.

The Fast Rope System is attached to the cabin ceiling at the front end of the cargo/troop door. When the helicopter is hovering, the fast rope is used to exit from the left and right cargo doors. The ropes are jettisoned from the crew chief release handles on the left and right side of the cabin. The fast rope support tubes are retractable when not in use and stowed with quick release pins.

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SA**FIGURE 2-1-1-1. AIRFRAME SECTIONS**

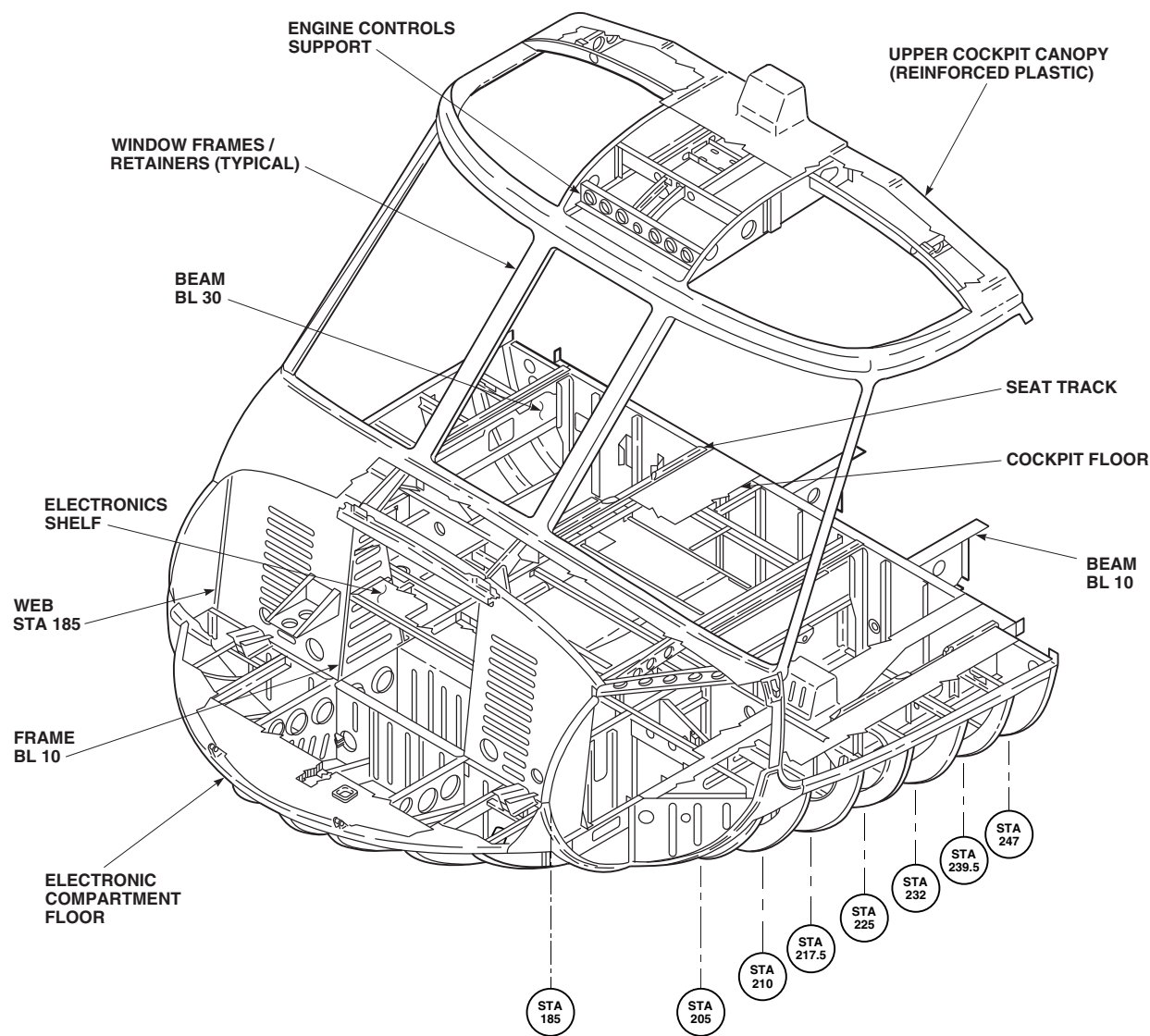


FIGURE 2-1-1-2. COCKPIT STRUCTURE

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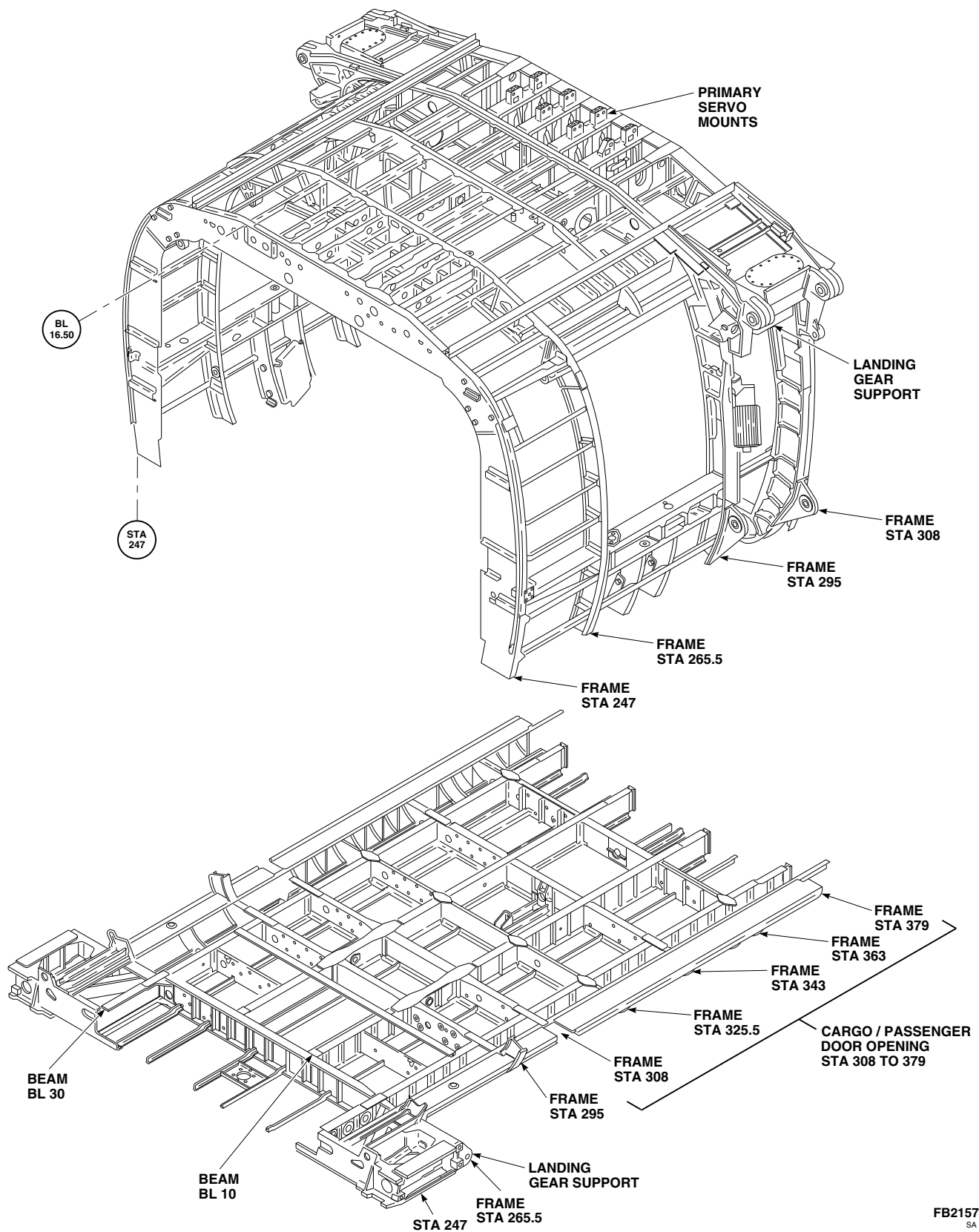
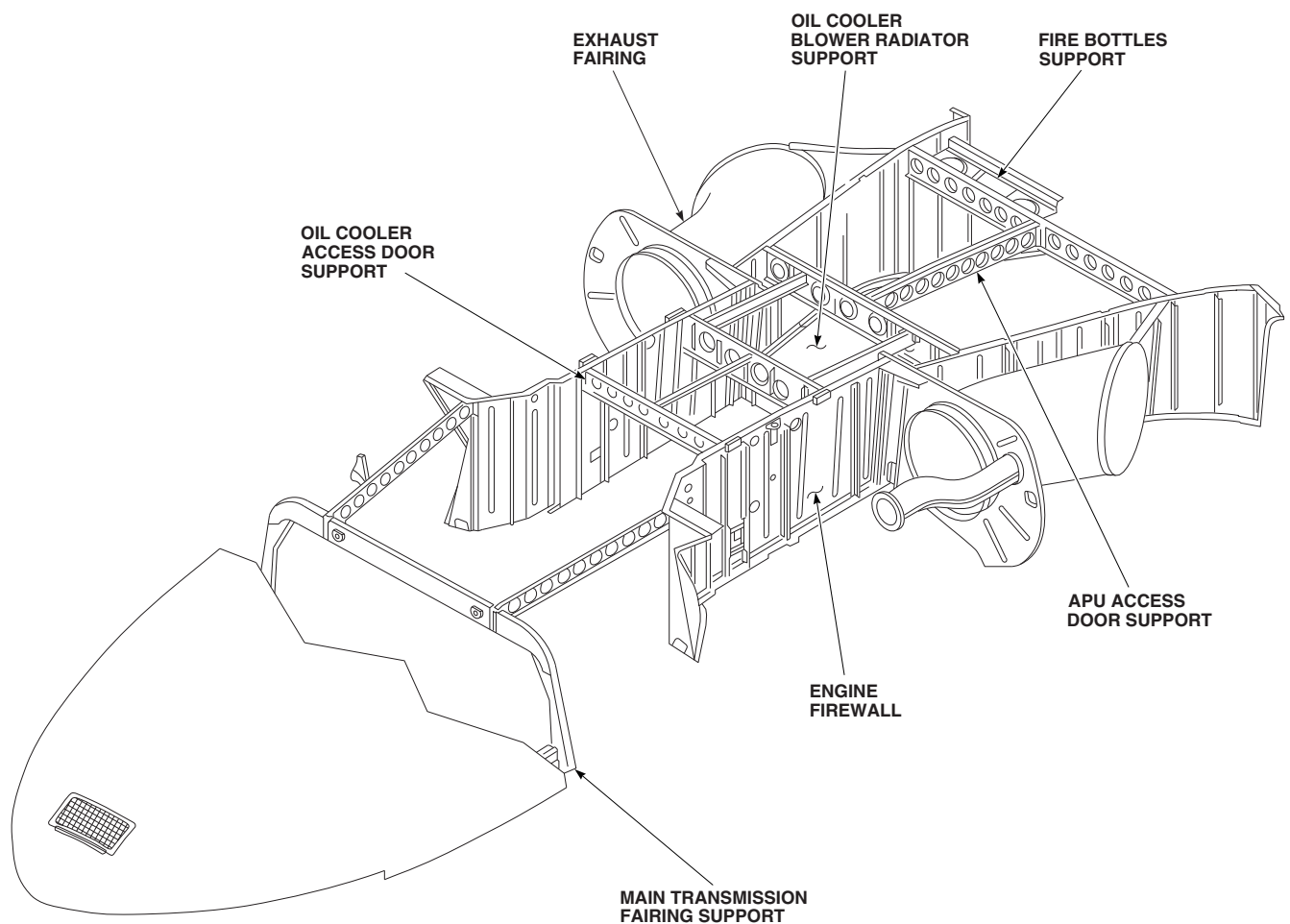
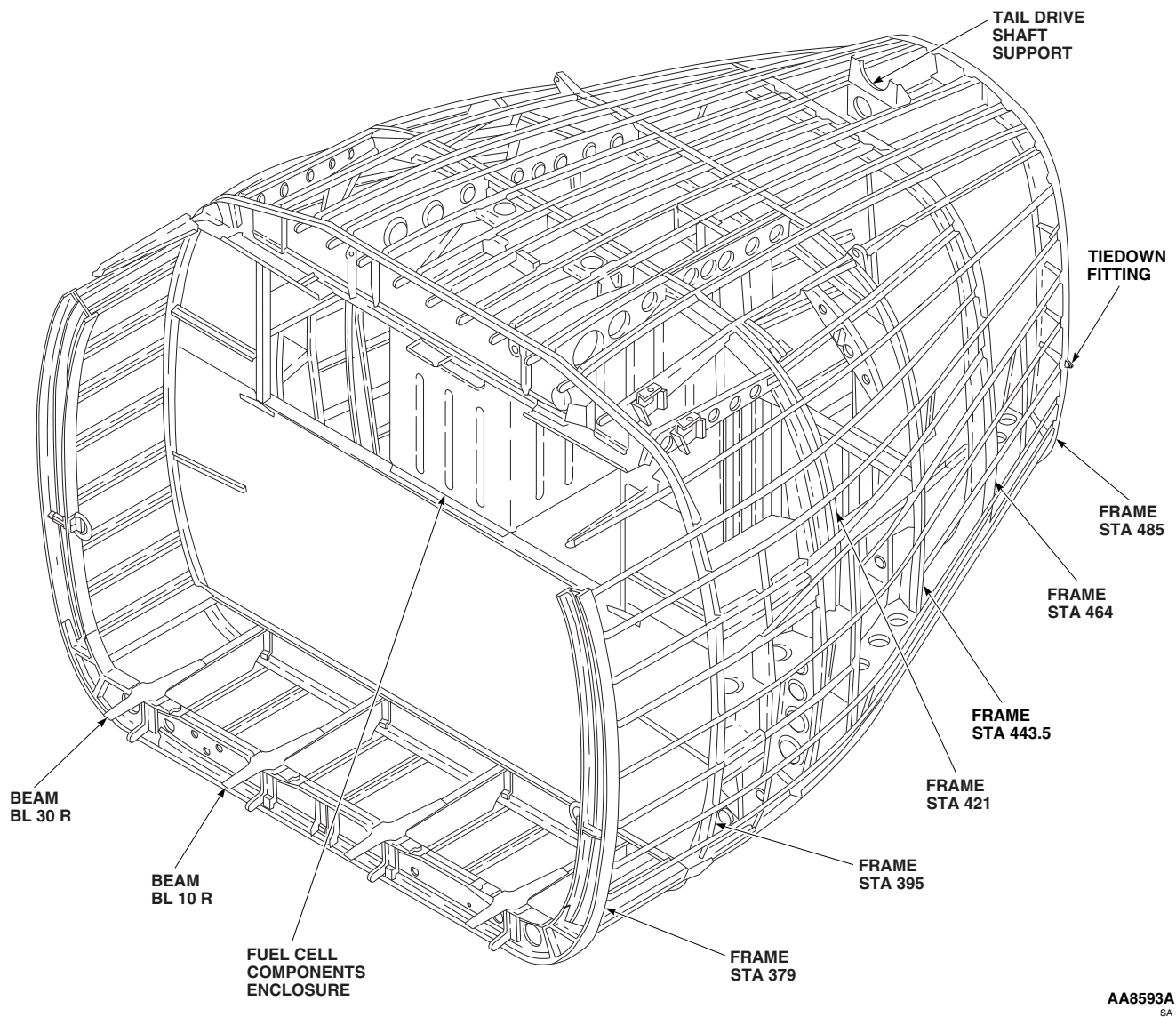
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FIGURE 2-1-1-3. CABIN STRUCTURE



UH60A_45749 (B)
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FIGURE 2-1-1-4. MAIN ROTOR PYLON STRUCTURE

**FIGURE 2-1-1-5. TRANSITION SECTION STRUCTURE**

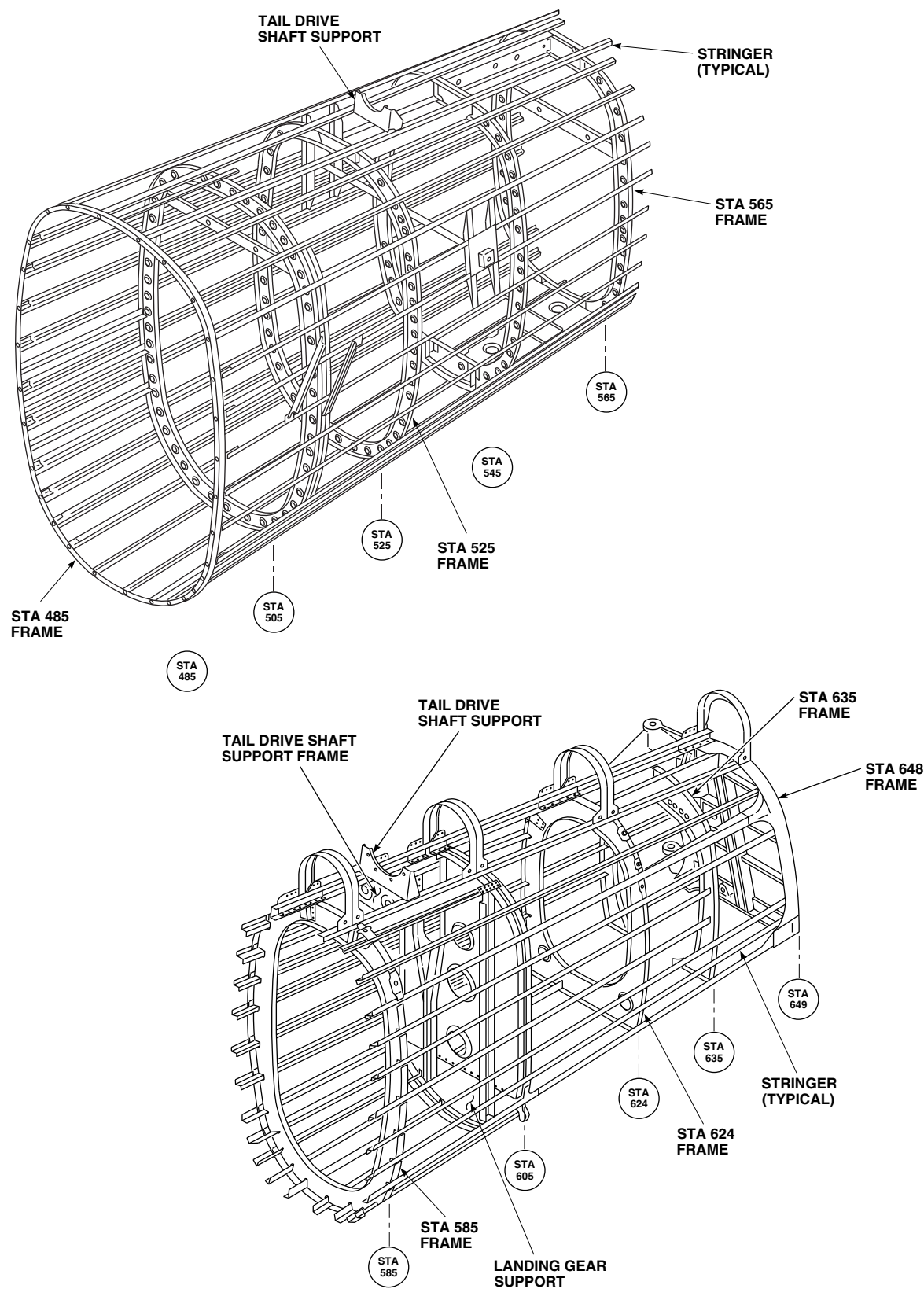


FIGURE 2-1-1-6. TAIL CONE STRUCTURE

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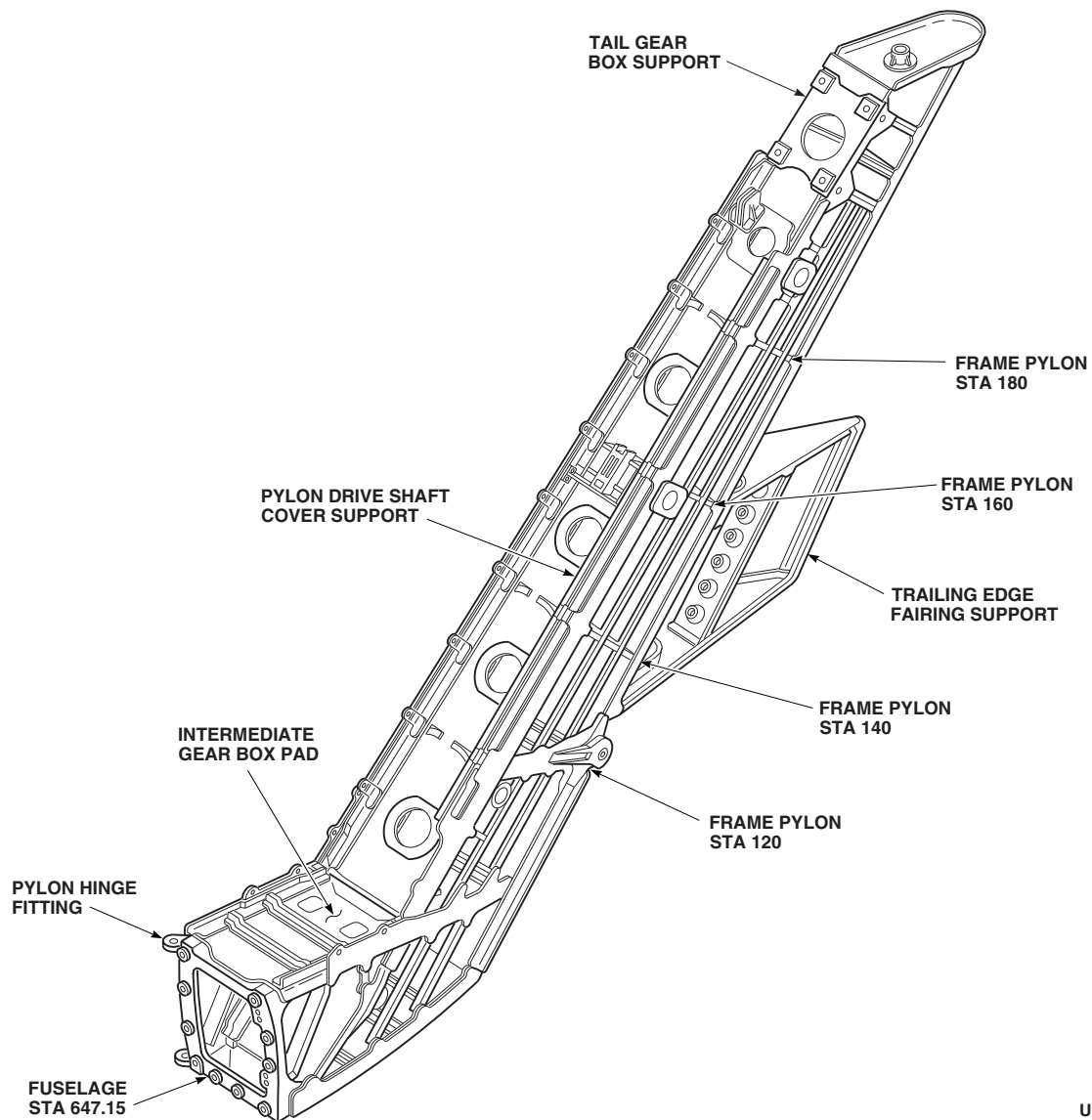
UH60A_45752 (B)
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FIGURE 2-1-1-7. TAIL PYLON STRUCTURE

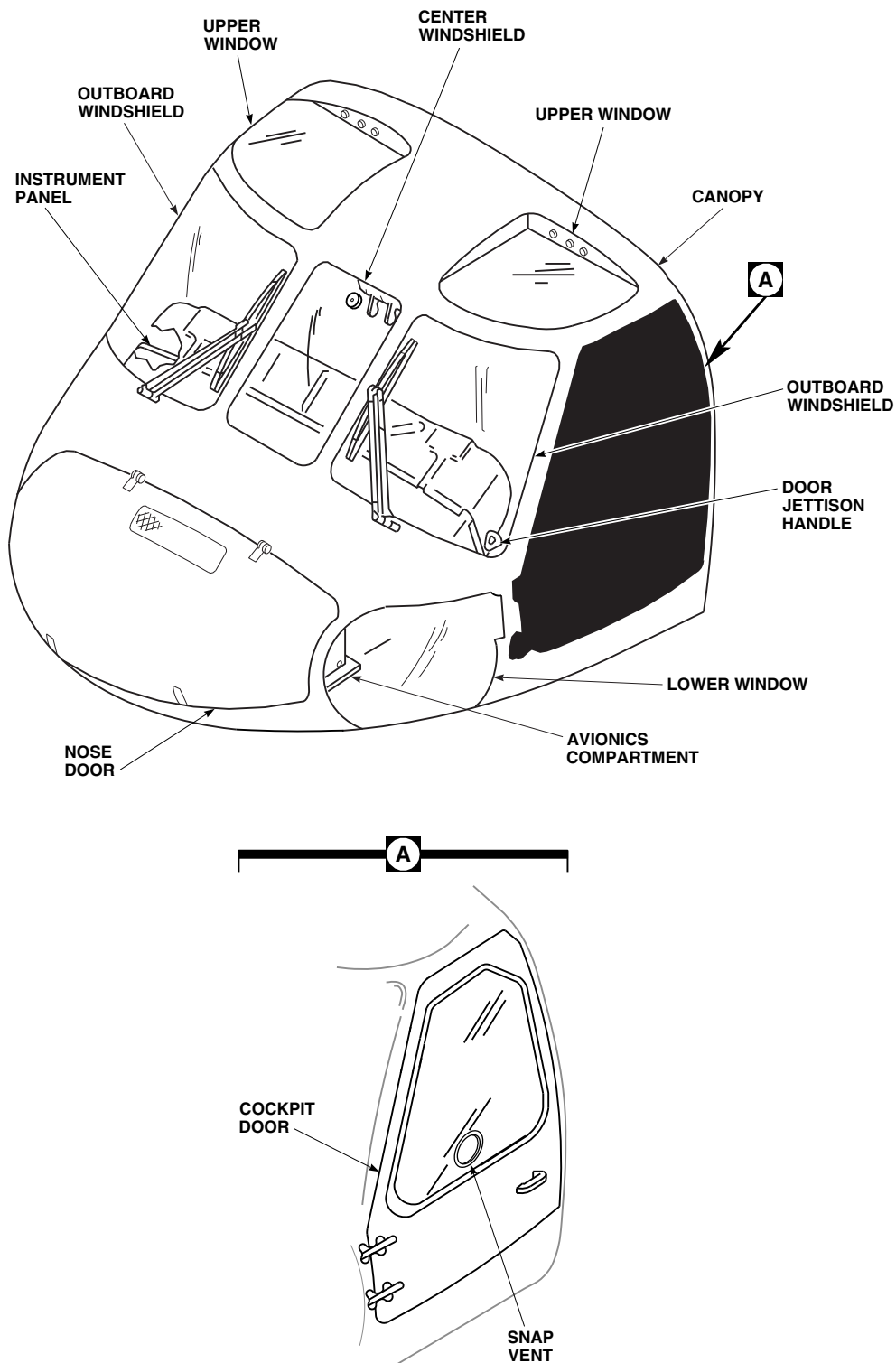
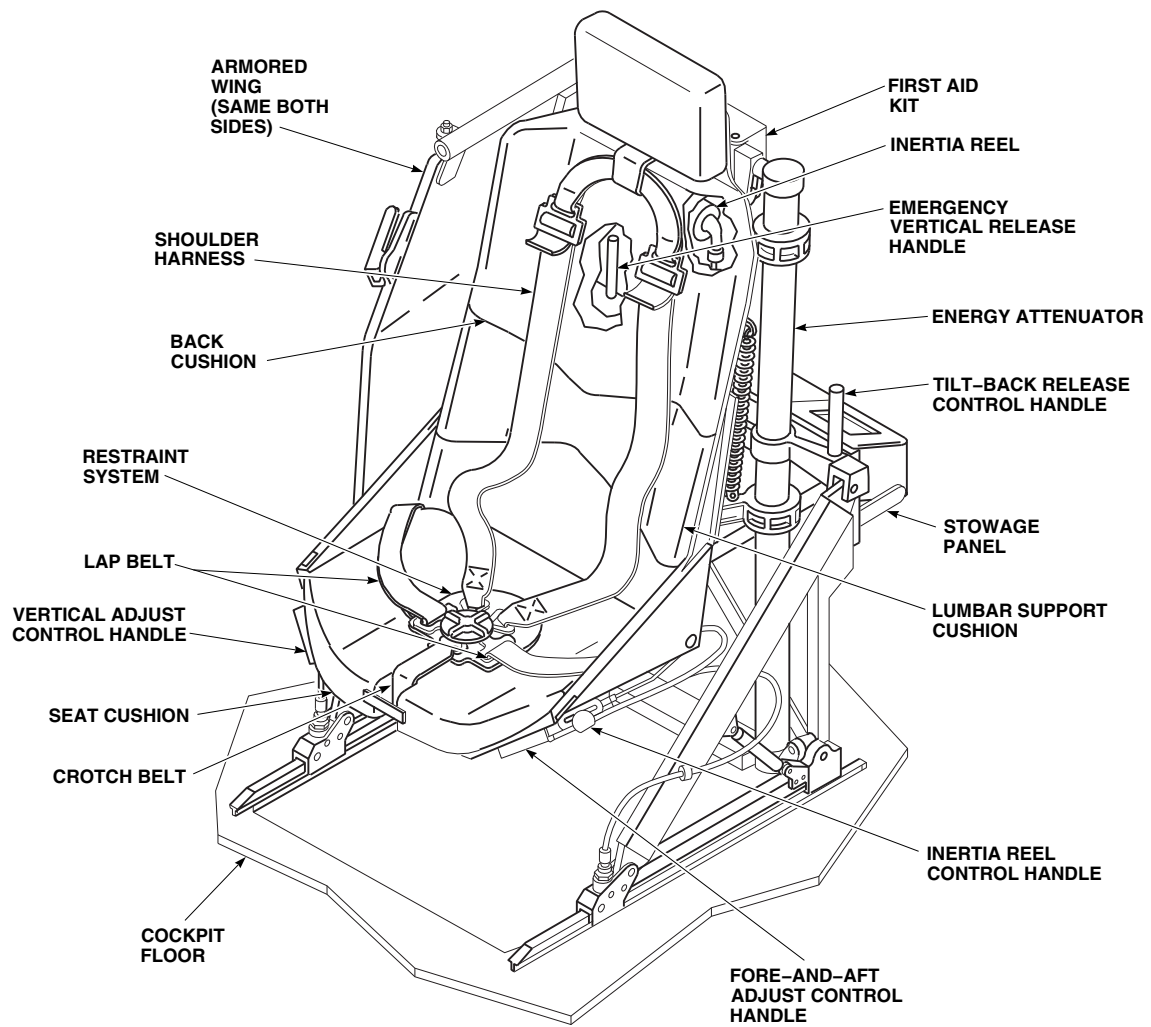


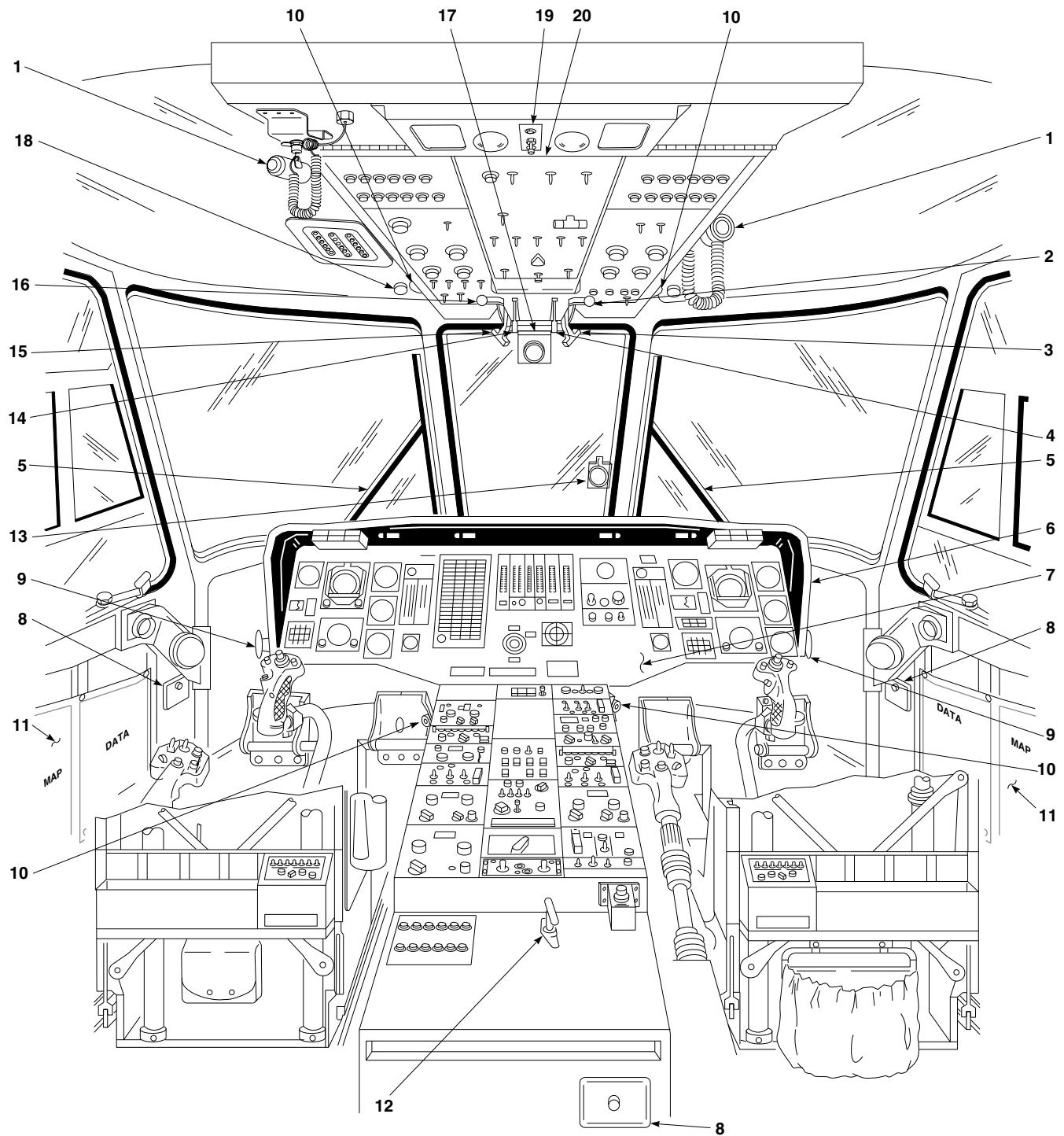
FIGURE 2-1-1-8. COCKPIT (NOSE SECTION)

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FIGURE 2-1-1-9. PILOT'S/COPILOT'S SEATS

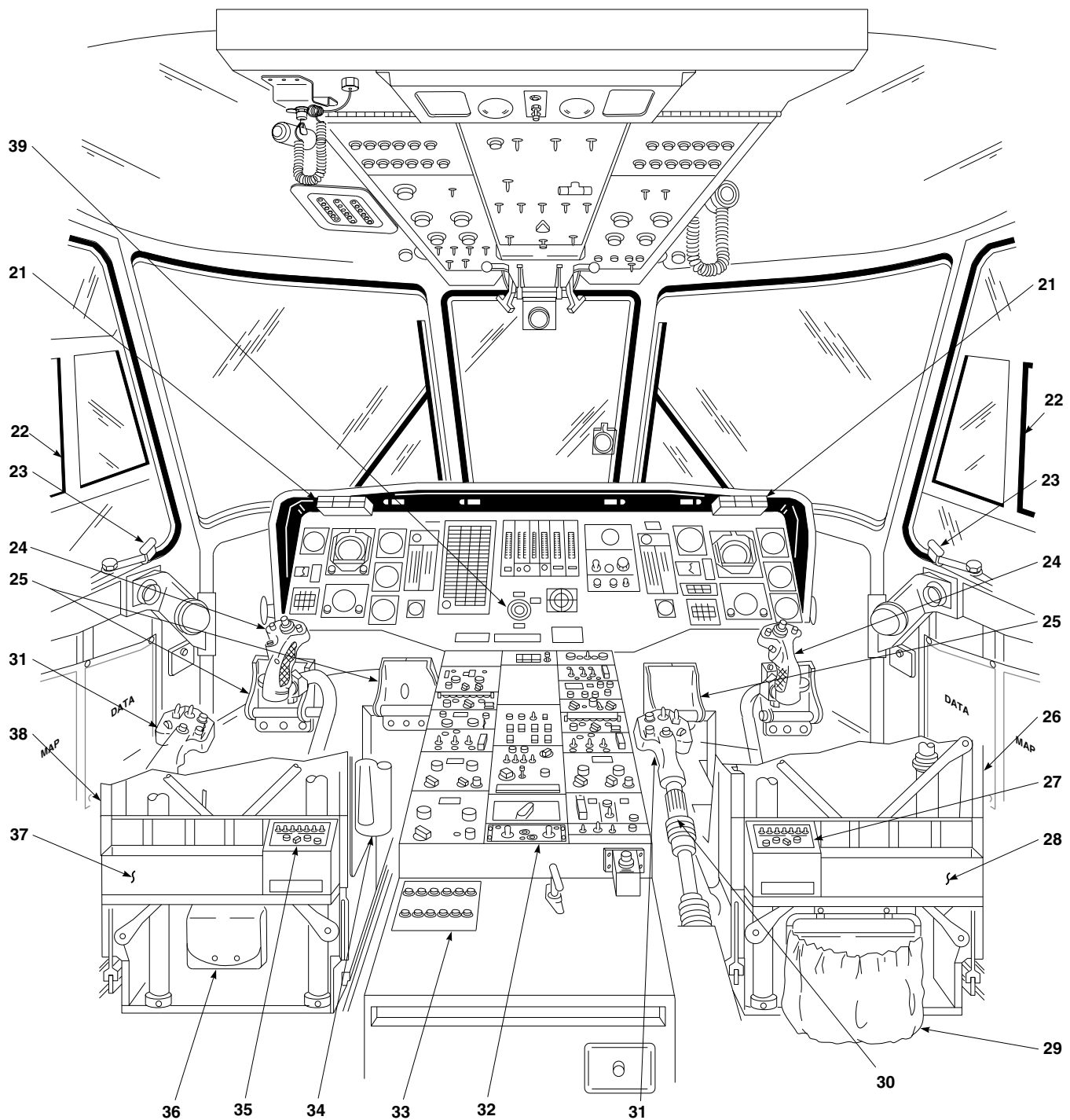


1. UTILITY LIGHT
2. NO. 2 ENGINE FUEL SELECTOR LEVER
3. NO. 2 ENGINE OFF / FIRE T-HANDLE
4. NO. 2 ENGINE POWER CONTROL LEVER
5. WINDSHIELD WIPER
6. INSTRUMENT PANEL GLARE SHIELD
7. INSTRUMENT PANEL
8. ASHTRAY
9. PEDAL ADJUST LEVER
10. VENT
11. MAP / DATA CASE

12. PARKING BRAKE LEVER
13. STANDBY (MAGNETIC) COMPASS
14. NO. 1 ENGINE POWER CONTROL LEVER
15. NO. 1 ENGINE OFF / FIRE T-HANDLE
16. NO. 1 ENGINE FUEL SELECTOR LEVER
17. FREE-AIR THERMOMETER **W/O HCW**
18. FREE-AIR THERMOMETER **HCW**
19. COCKPIT FLOODLIGHT CONTROL
20. UPPER CONSOLE

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FIGURE 2-1-1-10. MISCELLANEOUS COCKPIT FURNISHINGS (TYPICAL) (SHEET 1 OF 2)



- 21. MASTER WARNING PANEL
- 22. SLIDING WINDOW
- 23. COCKPIT DOOR EMERGENCY RELEASE
- 24. CYCLIC STICK
- 25. DIRECTIONAL CONTROL PEDALS
- 26. PILOT'S SEAT
- 27. RIGHT CABIN ICS CONTROL PANEL
- 28. RIGHT CABIN STOWAGE COMPARTMENT
- 29. STOWAGE BAG
- 30. COLLECTIVE STICK FRICTION CONTROL

- 31. COLLECTIVE STICK GRIP
- 32. LOWER CONSOLE
- 33. BATTERY / BATTERY UTILITY BUS CIRCUIT BREAKER PANEL
- 34. FIRE EXTINGUISHER
- 35. LEFT CABIN ICS CONTROL PANEL
- 36. FIRST AID KIT
- 37. LEFT CABIN STOWAGE COMPARTMENT
- 38. COPILOT'S SEAT
- 39. ENGINE IGNITION KEY LOCK

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FIGURE 2-1-1-10. MISCELLANEOUS COCKPIT FURNISHINGS (TYPICAL) (SHEET 2 OF 2)

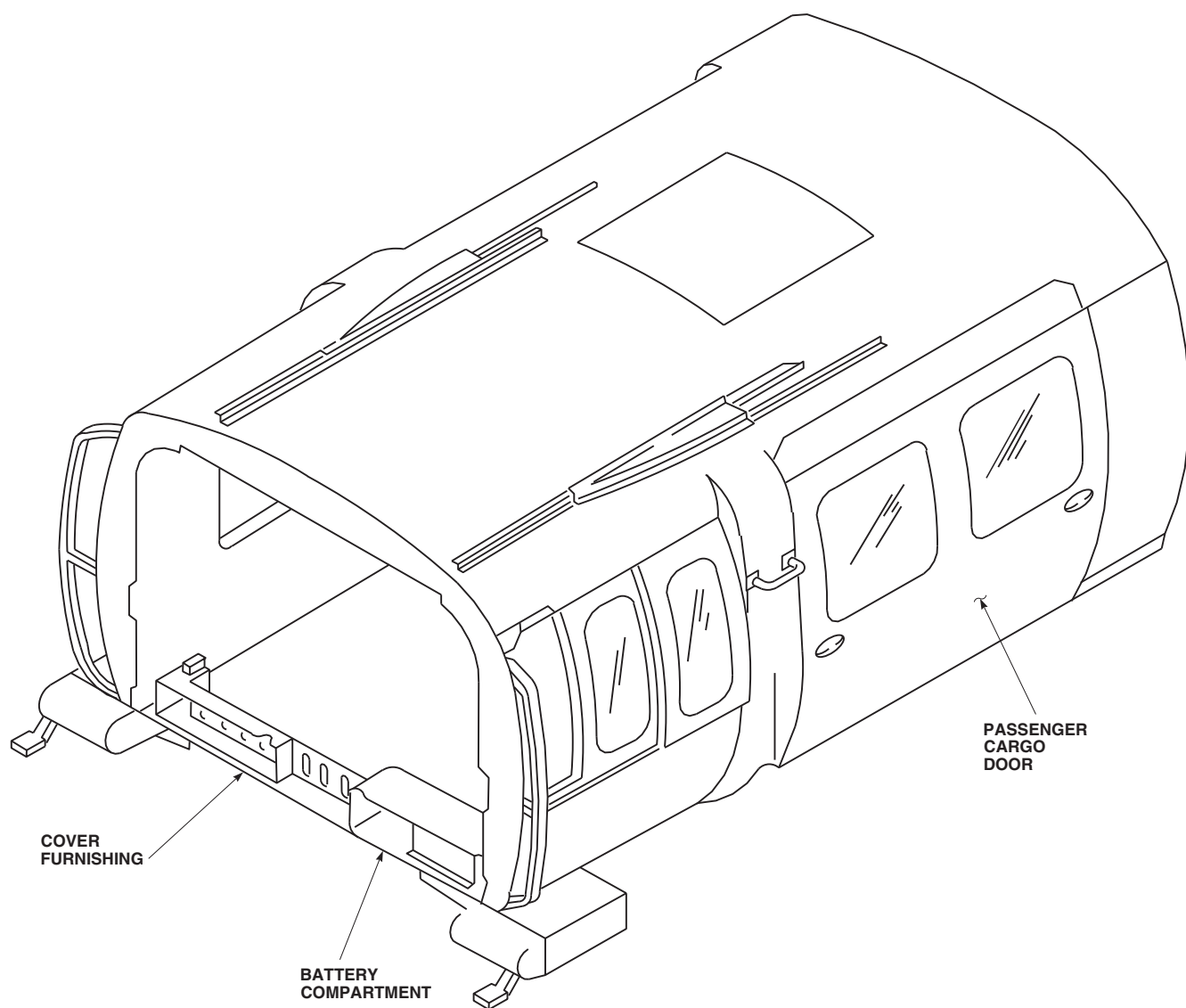


FIGURE 2-1-1-11. CABIN (MIDSECTION)

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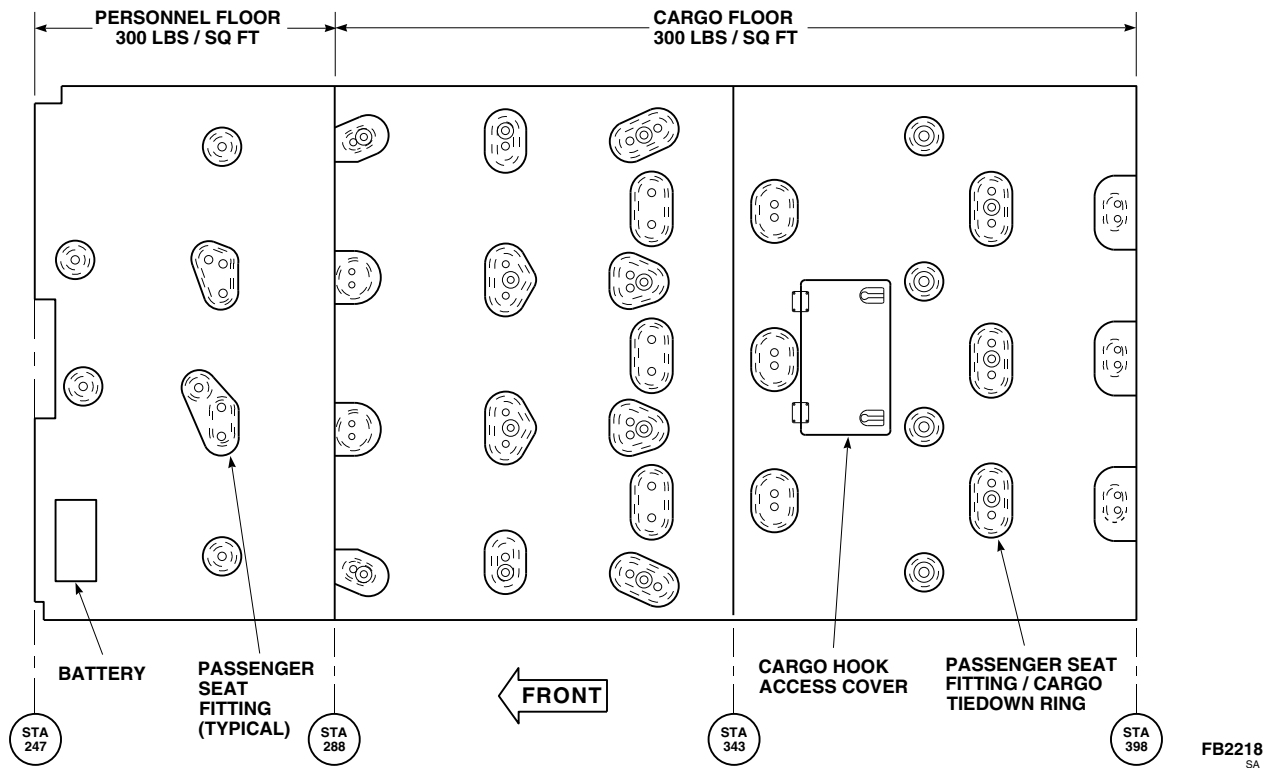


FIGURE 2-1-1-12. CABIN FLOORS

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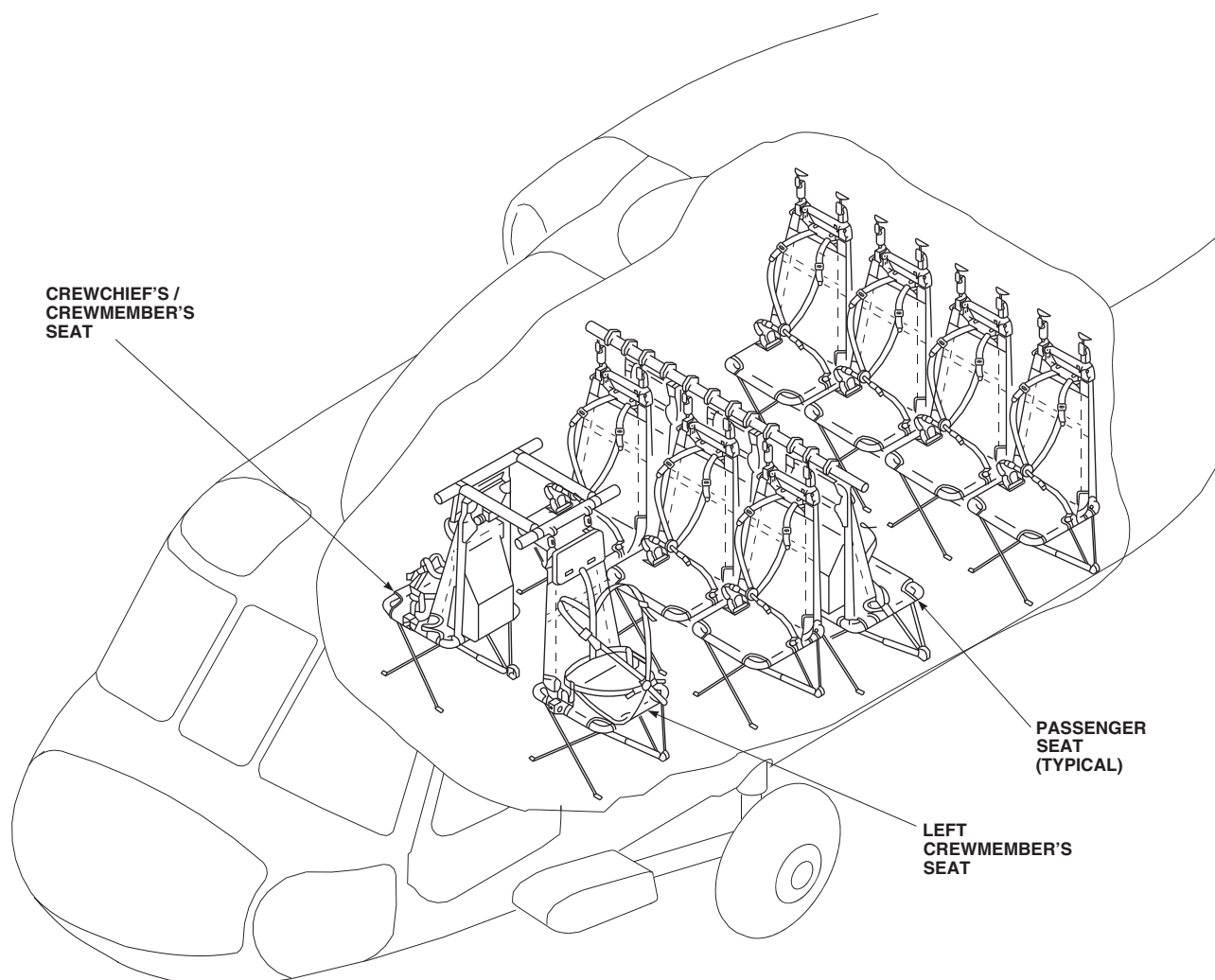
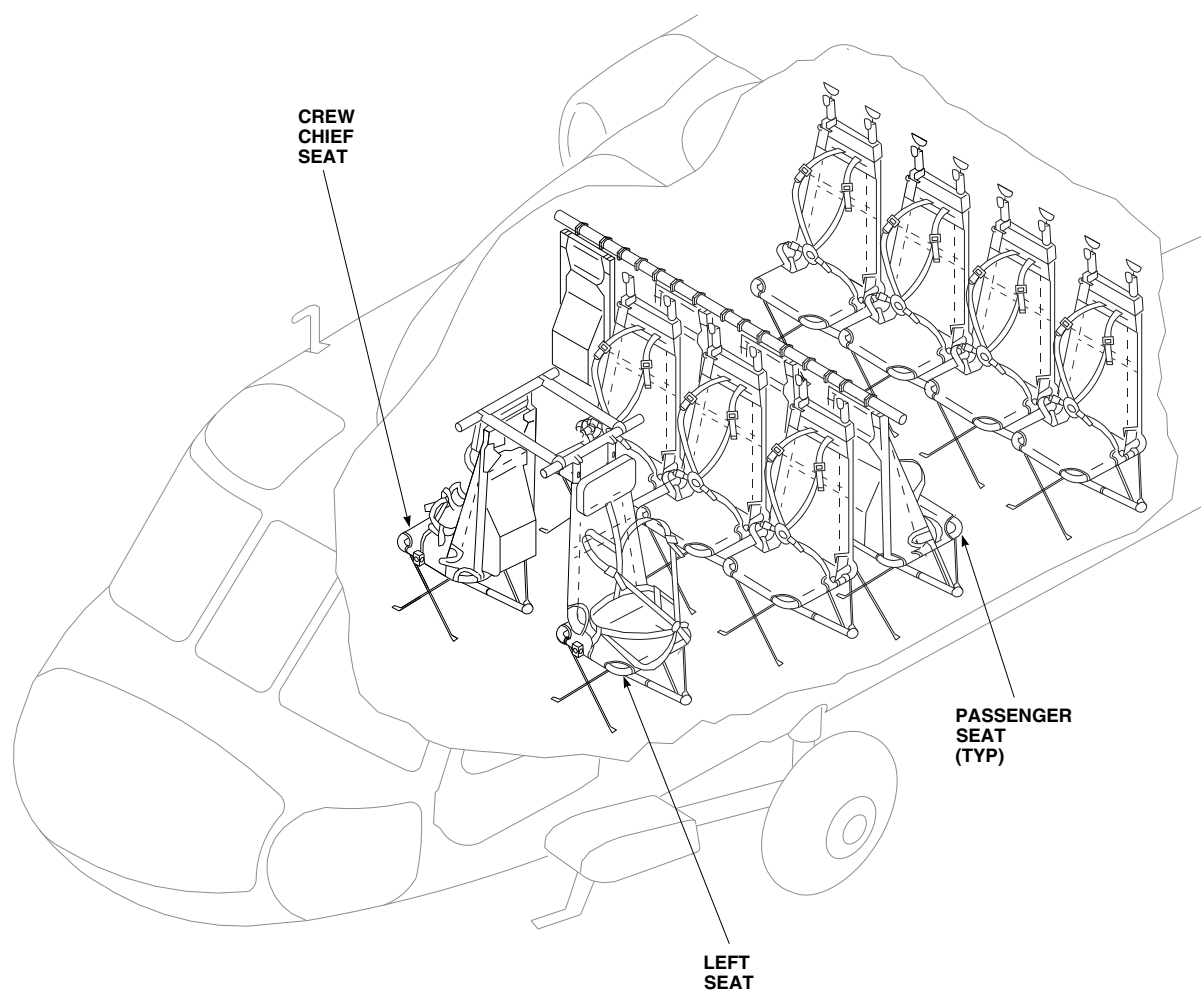


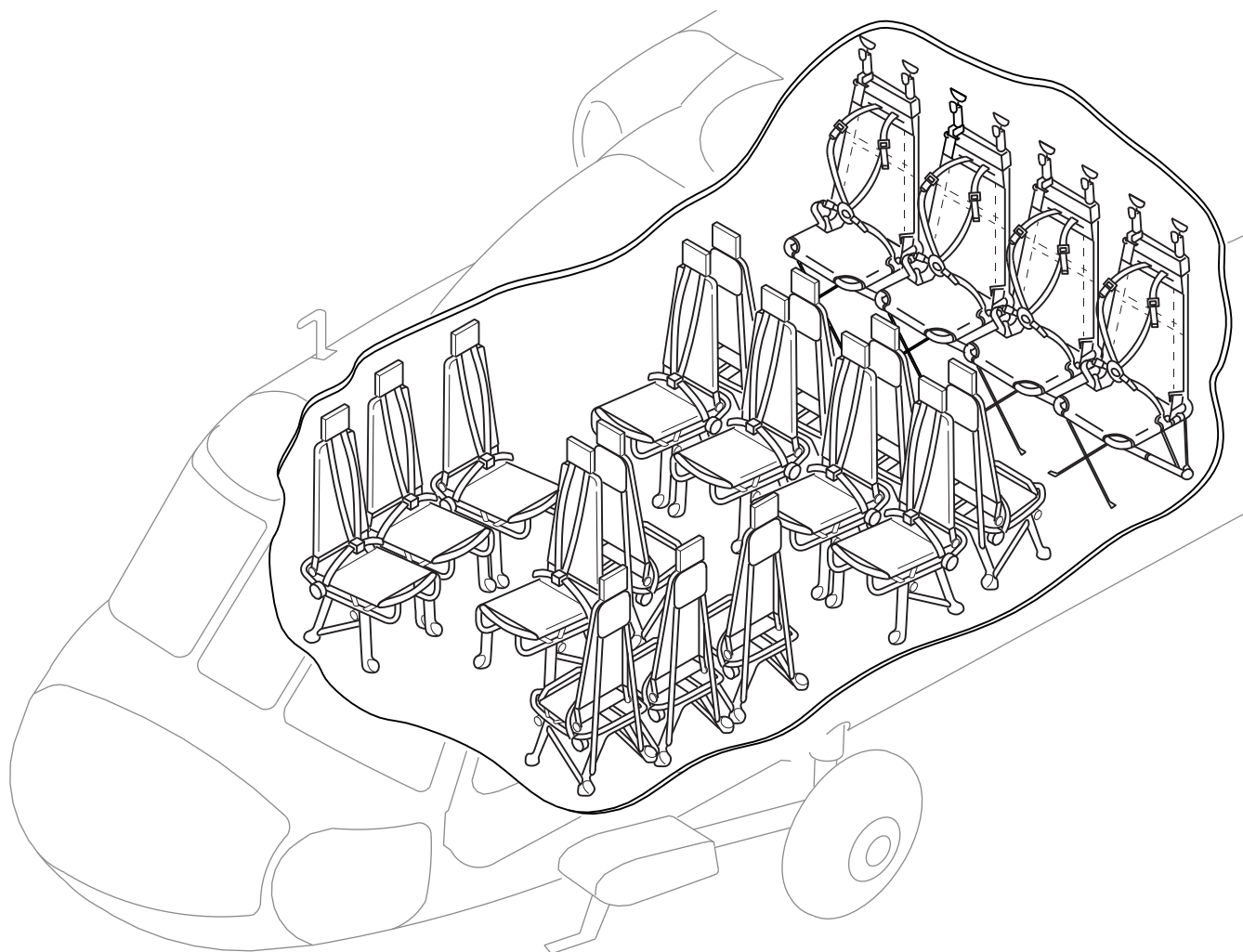
FIGURE 2-1-1-13. PASSENGER SEAT ARRANGEMENT (SHEET 1 OF 3)

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**NOTE**

14 SEAT CONFIGURATION

FB2160_2A
SA**FIGURE 2-1-1-13. PASSENGER SEAT ARRANGEMENT (SHEET 2 OF 3)**



NOTE

20 PASSENGER CONFIGURATION.

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FIGURE 2-1-1-13. PASSENGER SEAT ARRANGEMENT (SHEET 3 OF 3)

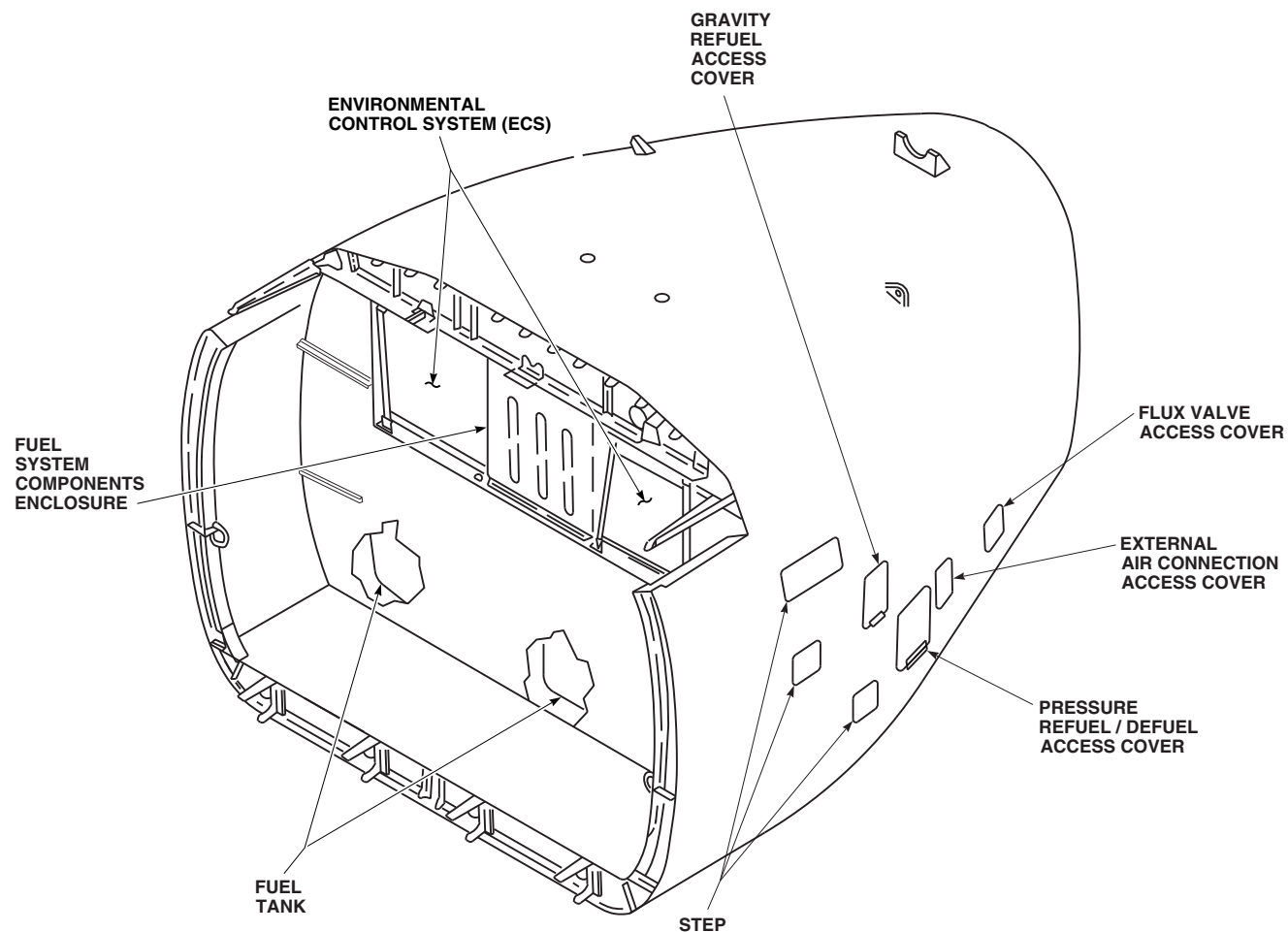
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FIGURE 2-1-1-14. TRANSITION SECTION

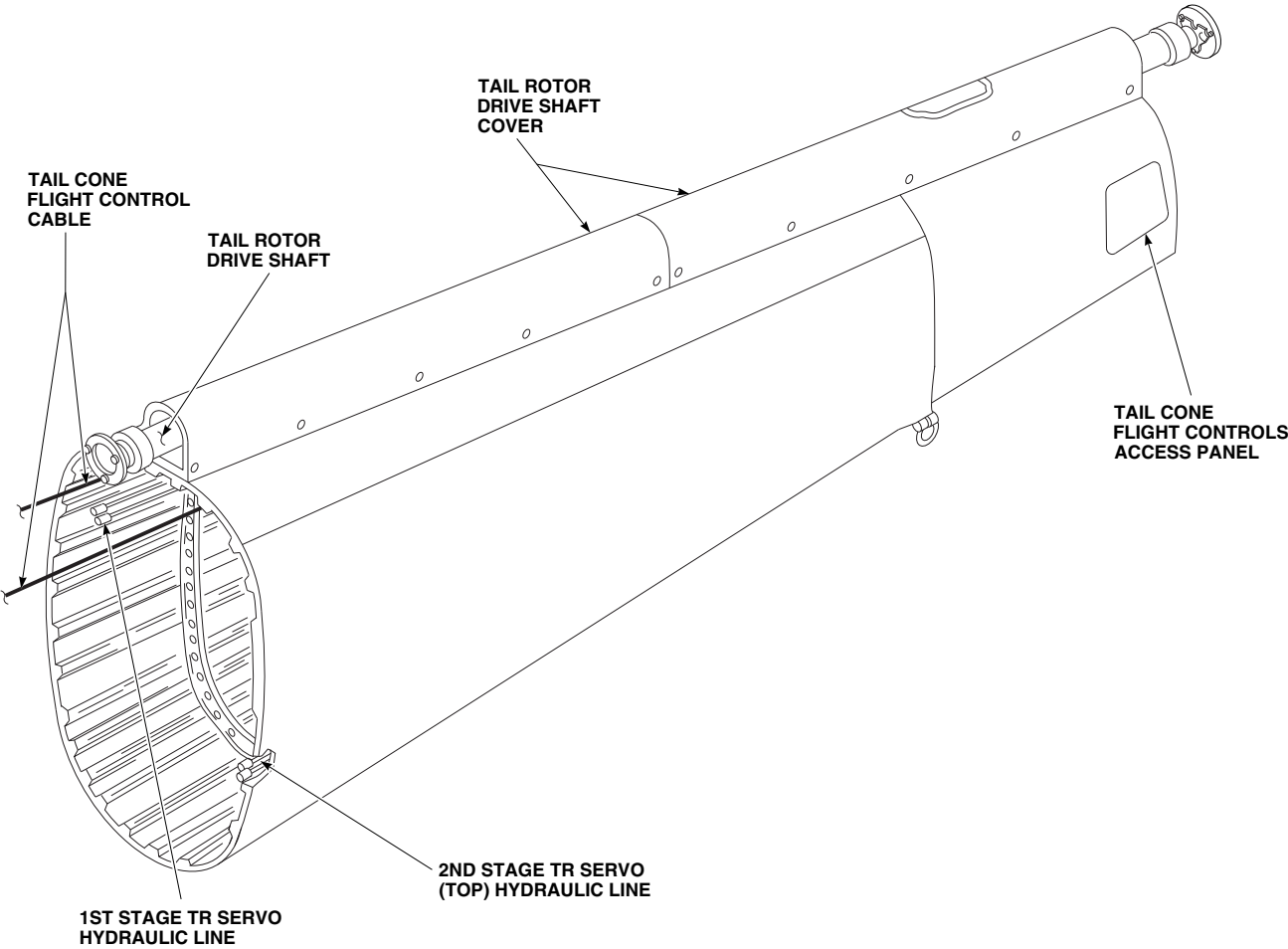


FIGURE 2-1-1-15. TAIL CONE

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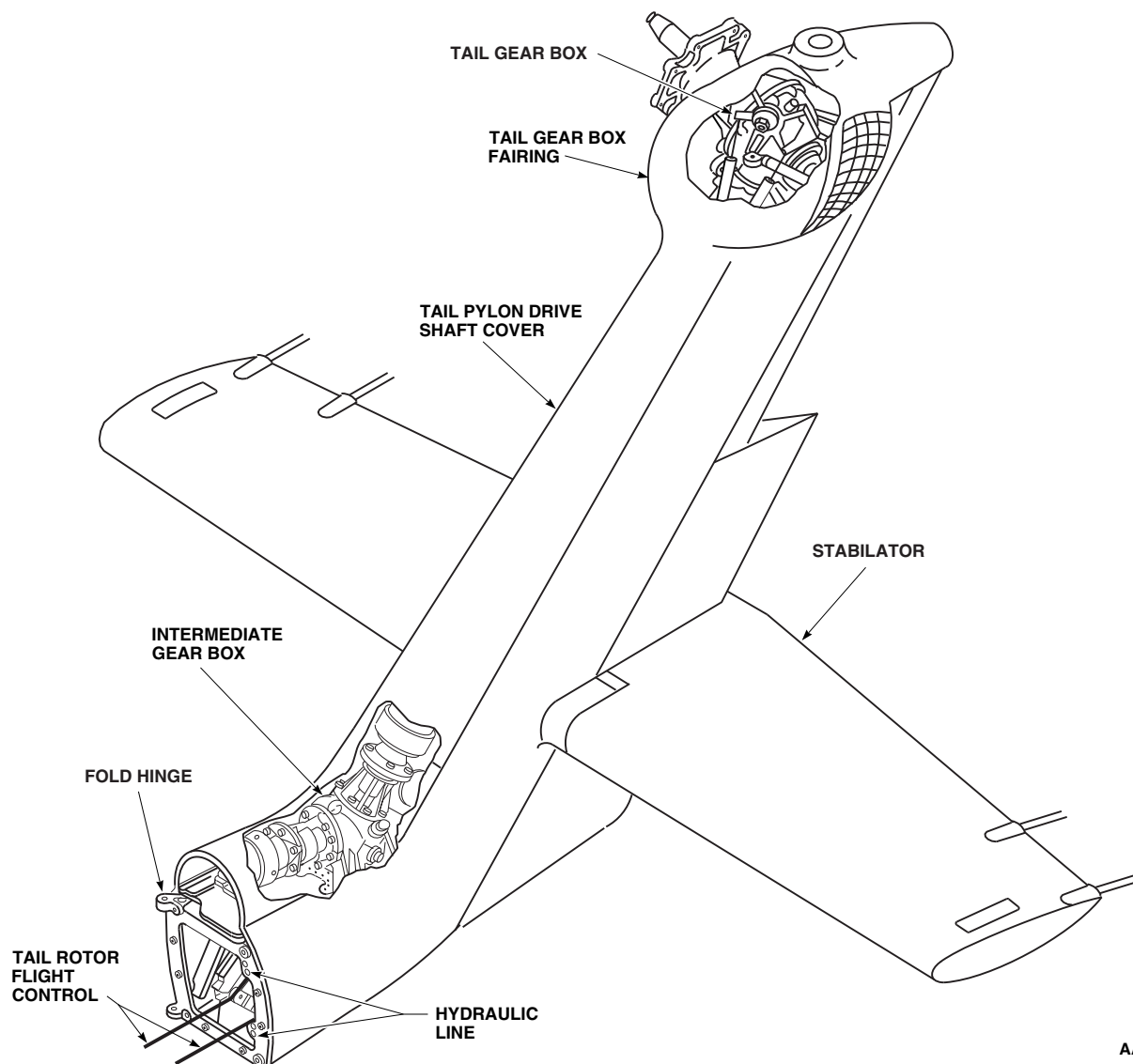
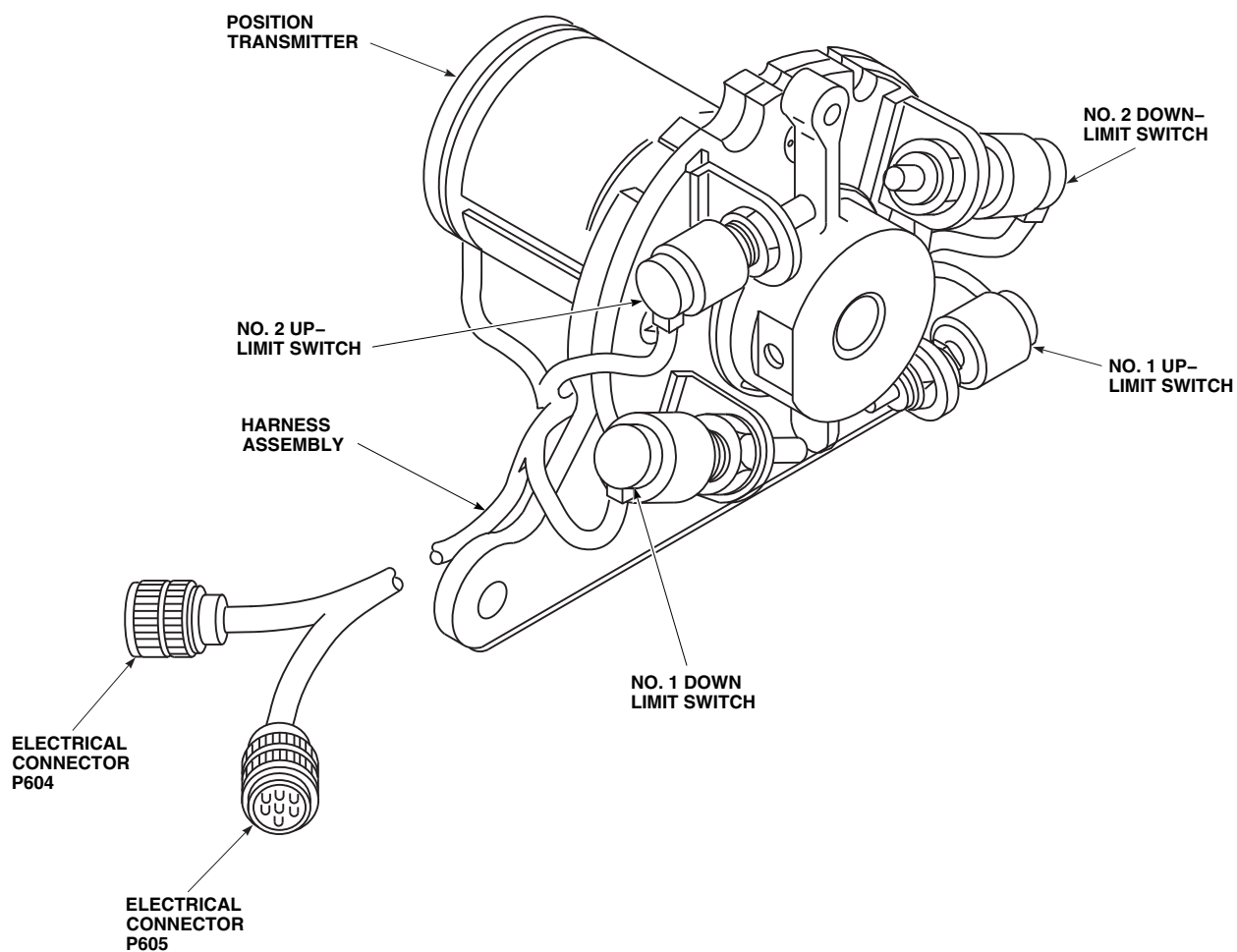
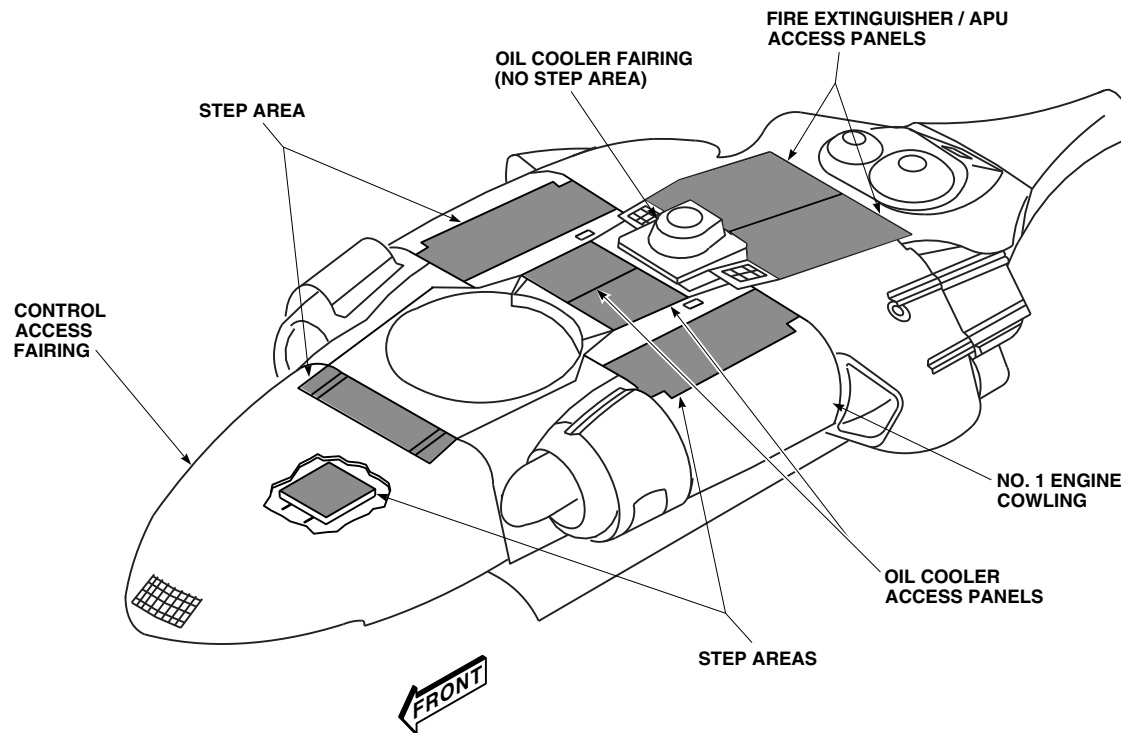
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FIGURE 2-1-1-16. TAIL PYLON



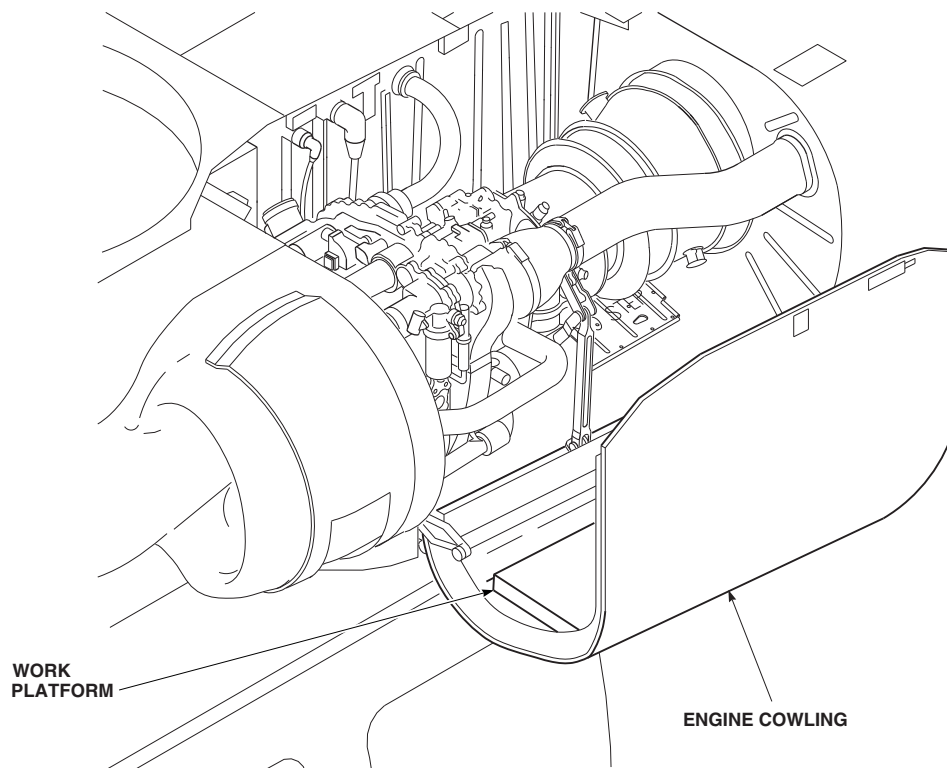
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FIGURE 2-1-1-17. POSITION TRANSMITTER/LIMIT SWITCH ASSEMBLY PARTS LOCATION



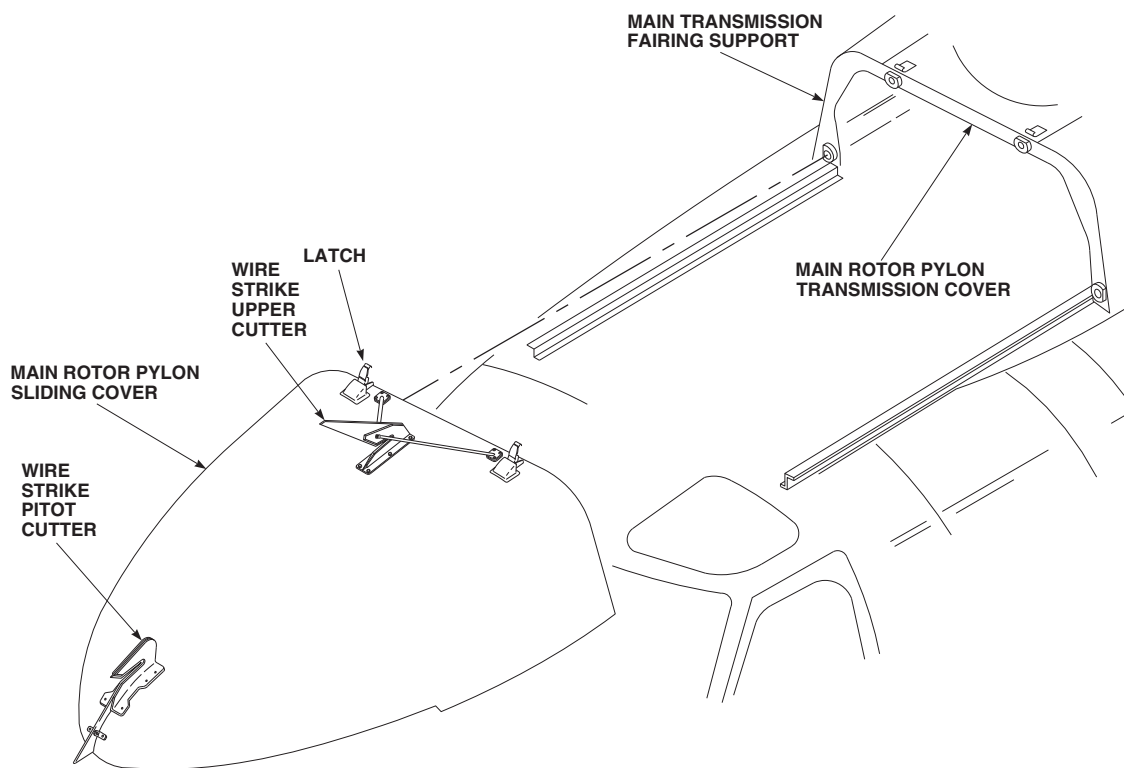
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FIGURE 2-1-1-18. MAIN ROTOR PYLON



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FIGURE 2-1-1-19. ENGINE COWLING AND WORK PLATFORM



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FIGURE 2-1-1-20. MAIN ROTOR PYLON SLIDING COVER

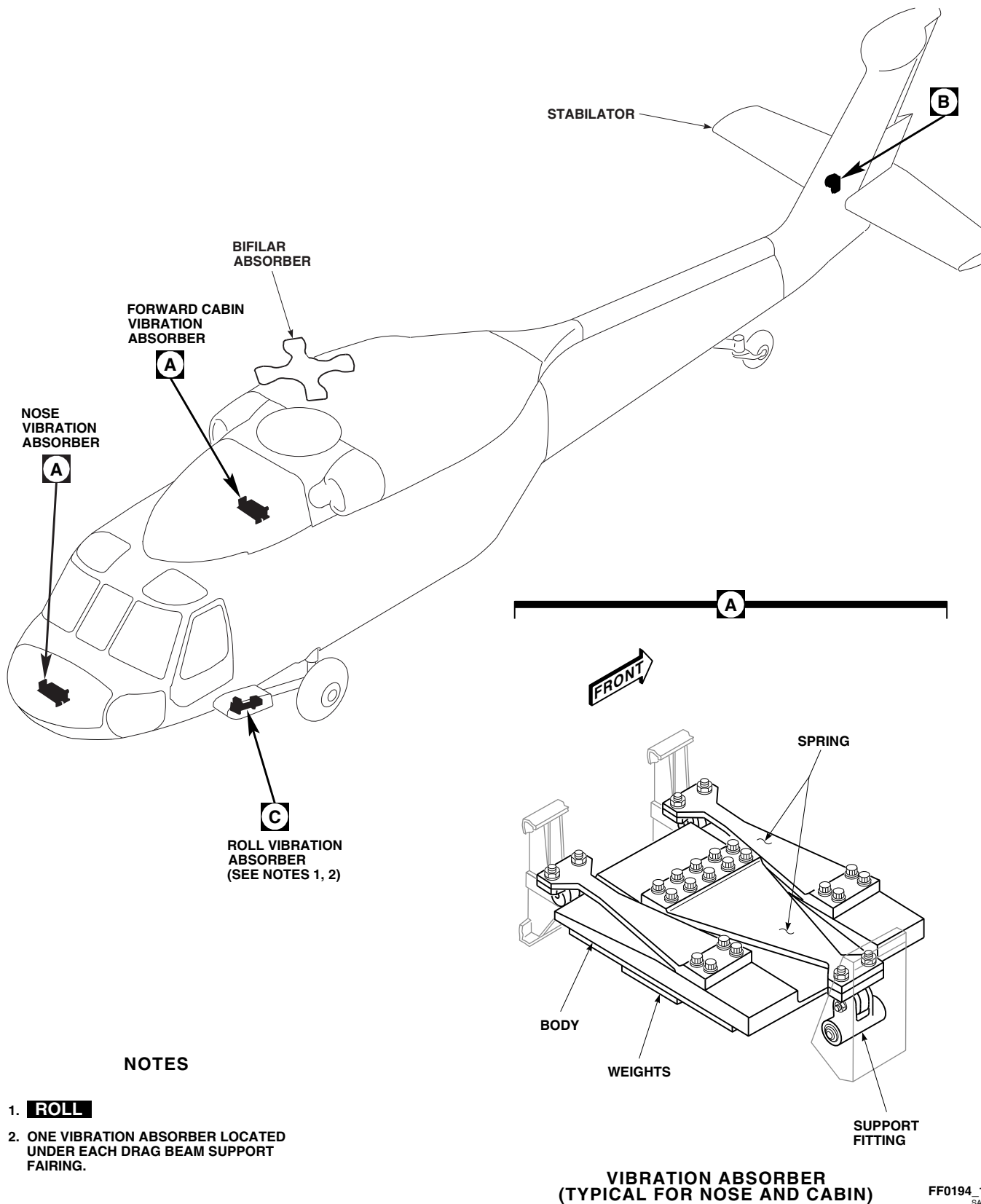


FIGURE 2-1-1-21. VIBRATION ABSORBER (SPRING-TYPE) (SHEET 1 OF 2)

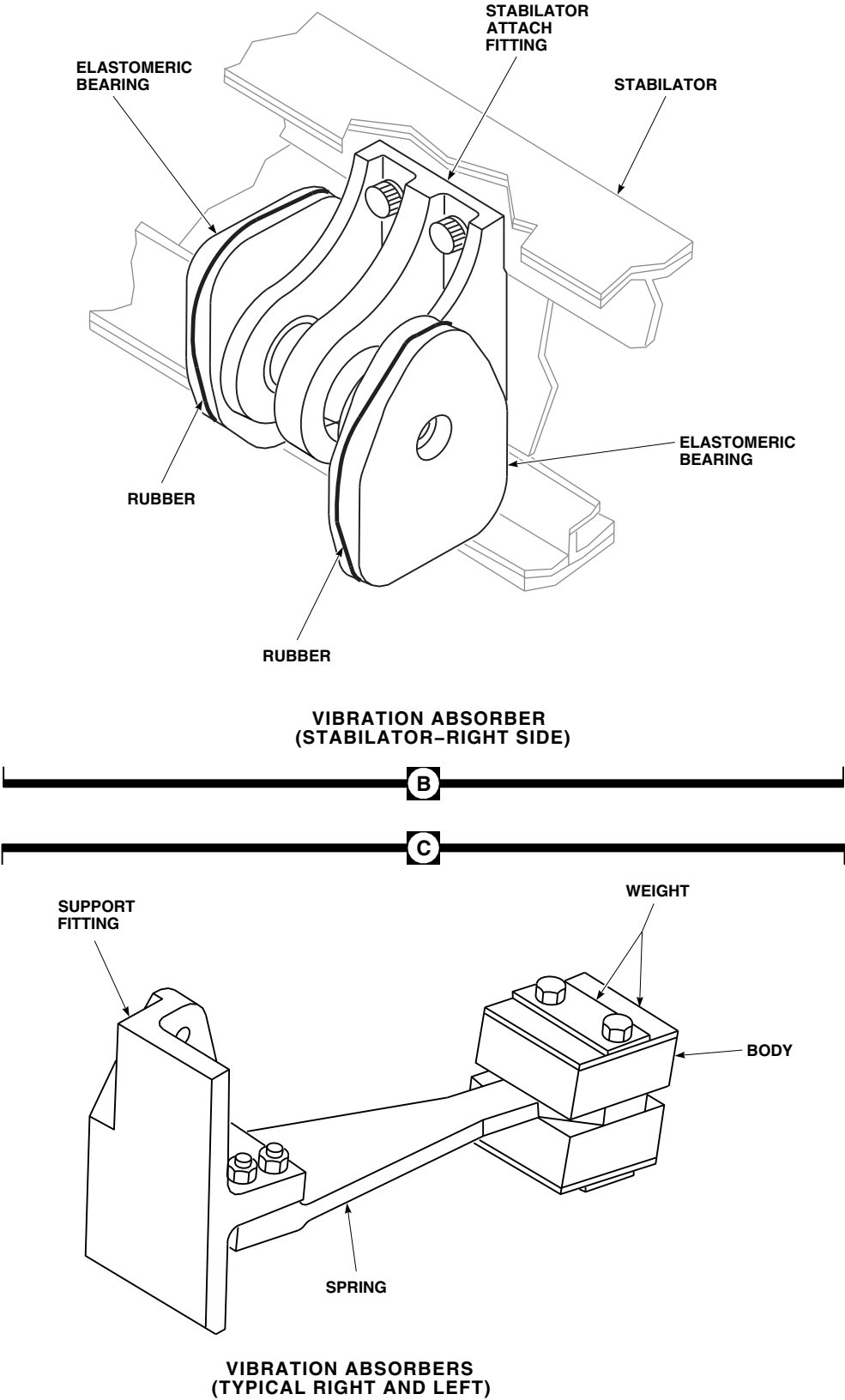


FIGURE 2-1-1-21. VIBRATION ABSORBER (SPRING-TYPE) (SHEET 2 OF 2)

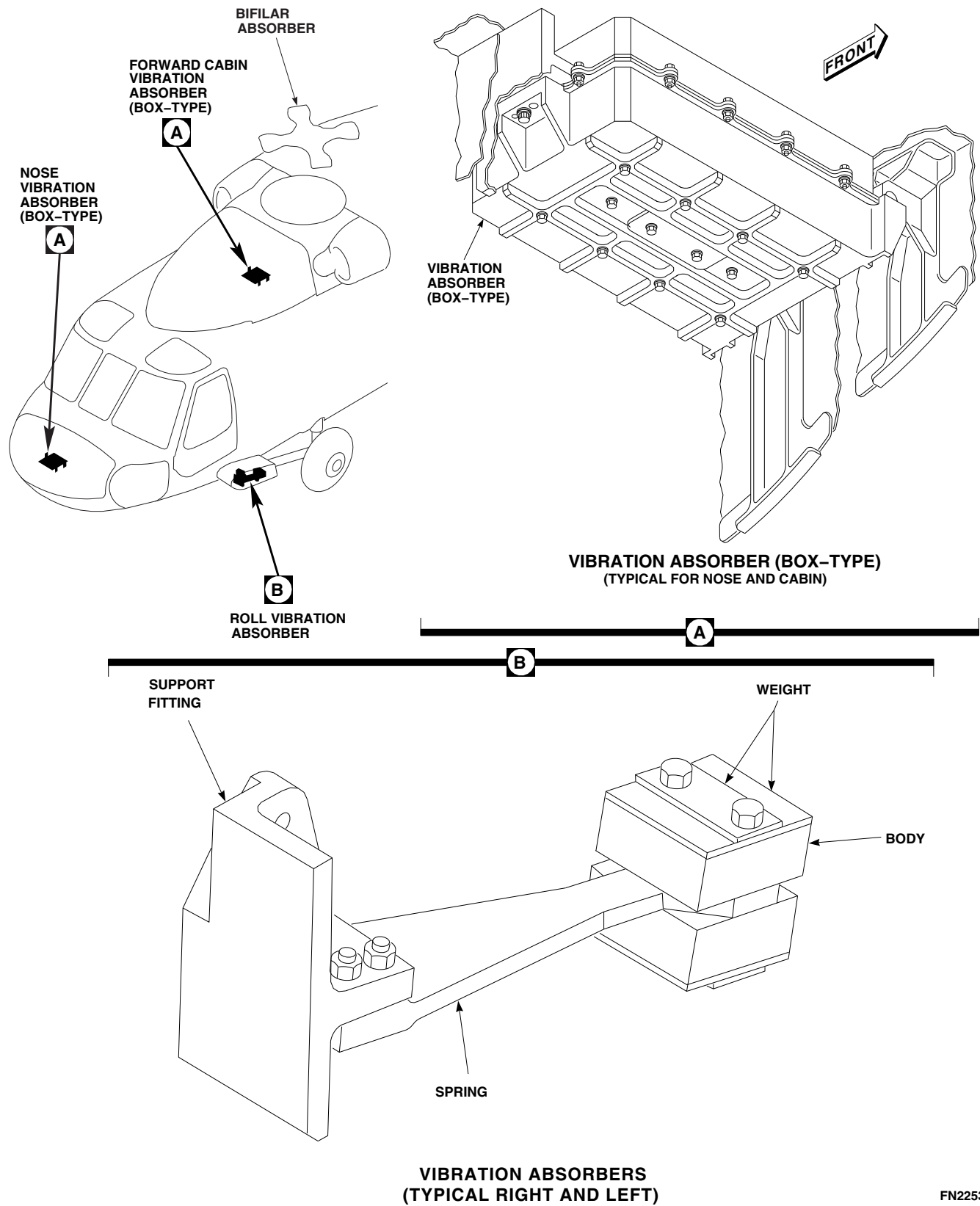
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FIGURE 2-1-1-22. VIBRATION ABSORBER (BOX-TYPE)

SECTION 2-2

TROUBLESHOOTING

SECTION OVERVIEW

PARAGRAPH

TITLE

2-2-1

Position Transmitter/Limit Switch Assembly (BENCH)

2-2-1/(2-2-2 Blank)

2-2-1 STABILATOR POSITION TRANSMITTER/LIMIT SWITCH ASSEMBLY (BENCH).

INITIAL SETUP

Test Equipment

- Angle Position Indicator (API), 8810-F1110
- Stabilator Indicators, ID-2279/A (2 required)

Tools

- Electrical Repairer Toolkit
- Locally-Made Stabilator Position Sensor/Limit Switch Lockpin, Figure H-179, Appendix H
- Locally-Made Stabilator Position Sensor/Limit Switch Test Unit, Figure H-154, Appendix H

- Torque Wrench, S-70 Torque Kit

Materials/Parts

- Alcohol, Isopropyl, Item 72, Appendix D
- Cheesecloth, Item 90, Appendix D
- Machinery Towel, Item 211, Appendix D

References

- Appendix D
- Appendix H

Equipment Conditions

- Bench Power of 27.5 ± 0.5 VDC, 115 VAC, 400 Hz, 1ø

2-2-1.1 Operational/Troubleshooting Procedure

- Remove all dirt, dust, oil, grease, or other foreign matter from stabilator position transmitter, limit switches, wiring, and connectors using cheesecloth dampened with isopropyl alcohol.
- Dry surfaces using machinery towel.
- Inspect for broken or burnt wires, bent or broken connector pins, security of components, and general condition.

RESULT:

- Wires shall not be broken or burnt.
- Connector pins shall not be broken or bent.
- Components shall be secure and in good condition.

- If results are not as specified, repair/replace as required (PARA 2-5-21, PARA 2-5-22, PARA 2-5-23, or PARA 2-5-24).
- Place stabilator position sensor/limit switch test unit (test unit) 28 VDC and 115 VAC ON-OFF switches to OFF.
- Connect transmitter/limit switch assembly electrical connector P604 to test unit J604, and assembly electrical connector P605 to test unit J605.
- Connect stabilator indicators to test unit harness electrical connectors P116 and P471R.
- Connect power supply harness 28 VDC and 115 VAC to bench power source.
- Connect API to corresponding tipjacks as follows:

API	TEST UNIT
S1	S2

API	TEST UNIT
S2	S3
S3	S1
REF HI	REF HI
REF LO	REF LO

- i. Turn API power on by pressing SYN button.
- j. Place test unit 28 VDC and 115 VAC ON-OFF switches to ON.
- k. Press test unit LAMP TEST button and observe that all lamps go on.
- l. While observing stabilator indicators, move pivot arm on limit switch assembly throughout its range of travel.

RESULT: Stabilator indicator pointers shall move smoothly through a range of -8° up to $+40^\circ$ down as pivot assembly is moved.

- If result is not as specified, replace stabilator position transmitter (PARA 2-5-22).
- m. Turn pivot arm to $+10^\circ$ down position and lock arm in place using stabilator position sensor/limit switch lockpin (lockpin) through slot in base and hole in pivot arm.

RESULT: API shall indicate between $351.5^\circ \pm 1.0^\circ$.

- If result is not as specified, adjust transmitter by removing cotter pins, and loosen nuts enough to allow transmitter to turn.
 - (a) Turn transmitter until 10° indication is obtained.
 - (b) If adjustment cannot be obtained, replace stabilator position transmitter (PARA 2-5-22).
 - (c) **TIGHTEN NUTS AND TORQUE TO 20 ± 1 INCH-POUNDS.**

- (d) Install cotter pins (PARA 2-5-22).

CAUTION

Use care when positioning the pivot assembly in the extreme -8° and $+40^\circ$ slot positions. Too much force through the alignment pin may damage the stabilator position transmitter synchro shaft.

- n. Turn pivot arm to -8° up position and lock arm in place using lockpin.

RESULT: #1 UP and #2 UP lights on test unit remain off.

- If result is not as specified, adjust switch until light just goes off.
 - (a) If trouble remains, replace limit switch (PARA 2-5-21).
 - (b) **TIGHTEN NUT AND TORQUE TO 20 ± 1 INCH-POUNDS.**

- o. With pin still installed, turn pivot arm counterclockwise toward up-limit switches.

RESULT: Both #1 UP and #2 UP lights on test unit shall go on.

- If result is not as specified, adjust switch until light just goes on.
 - (a) If trouble remains, replace limit switch (PARA 2-5-21).
 - (b) **TIGHTEN NUTS AND TORQUE TO 20 ± 1 INCH-POUNDS.**

- p. Gently move pivot arm back and forth within notch.

RESULT: The API shall indicate $8^\circ \pm 1^\circ$.

- If result is not as specified, replace base and/or retainer (PARA 2-5-24).
- q. Remove lockpin from pivot arm and base.

- r. Position the pivot arm towards the opposite limit switch to obtain an API indication of $325^\circ \pm 1^\circ$.
- s. Install lockpin in the $+40^\circ$ alignment slot.
- t. Ensure pivot arm stays in contact with the switch plungers and gently move pivot arm back and forth.

RESULT:

- (1) There shall be a deflection of $38^\circ \pm 1^\circ$ on the stabilator indicator.
- (2) Both #1 DN and #2 DN lights on test unit shall remain off.
 - If result (1) is not as specified, ensure that pressure is only applied to the hole end of the pivot arm to hold against limit switch spring force.
 - If result (2) is not as specified, adjust switch until light just goes off.
 - (a) If trouble remains, replace limit switch (PARA 2-5-21).
 - (b) **TIGHTEN NUT AND TORQUE TO 20 ± 1 INCH-POUNDS.**

- u. With pin still installed, turn pivot arm clockwise toward down-limit switches.

RESULT: Both #1 DN and #2 DN lights on test unit shall go on.

- If result is not as specified, adjust switch until light just goes on.
 - (a) If trouble remains, replace limit switch (PARA 2-5-21).
 - (b) **TIGHTEN NUTS AND TORQUE TO 20 ± 1 INCH-POUNDS.**

- v. Gently move pivot arm back and forth within notch.

RESULT: The API shall indicate between 320.5° and 323.5° .

- If result is not as specified, replace base and/or retainer (PARA 2-5-24).

- w. Remove lockpin from pivot arm and base.

- x. Place test unit 28 VDC and 115 VAC ON/OFF switches OFF.

- y. Disconnect all test equipment from stabilator position transmitter/limit switch.

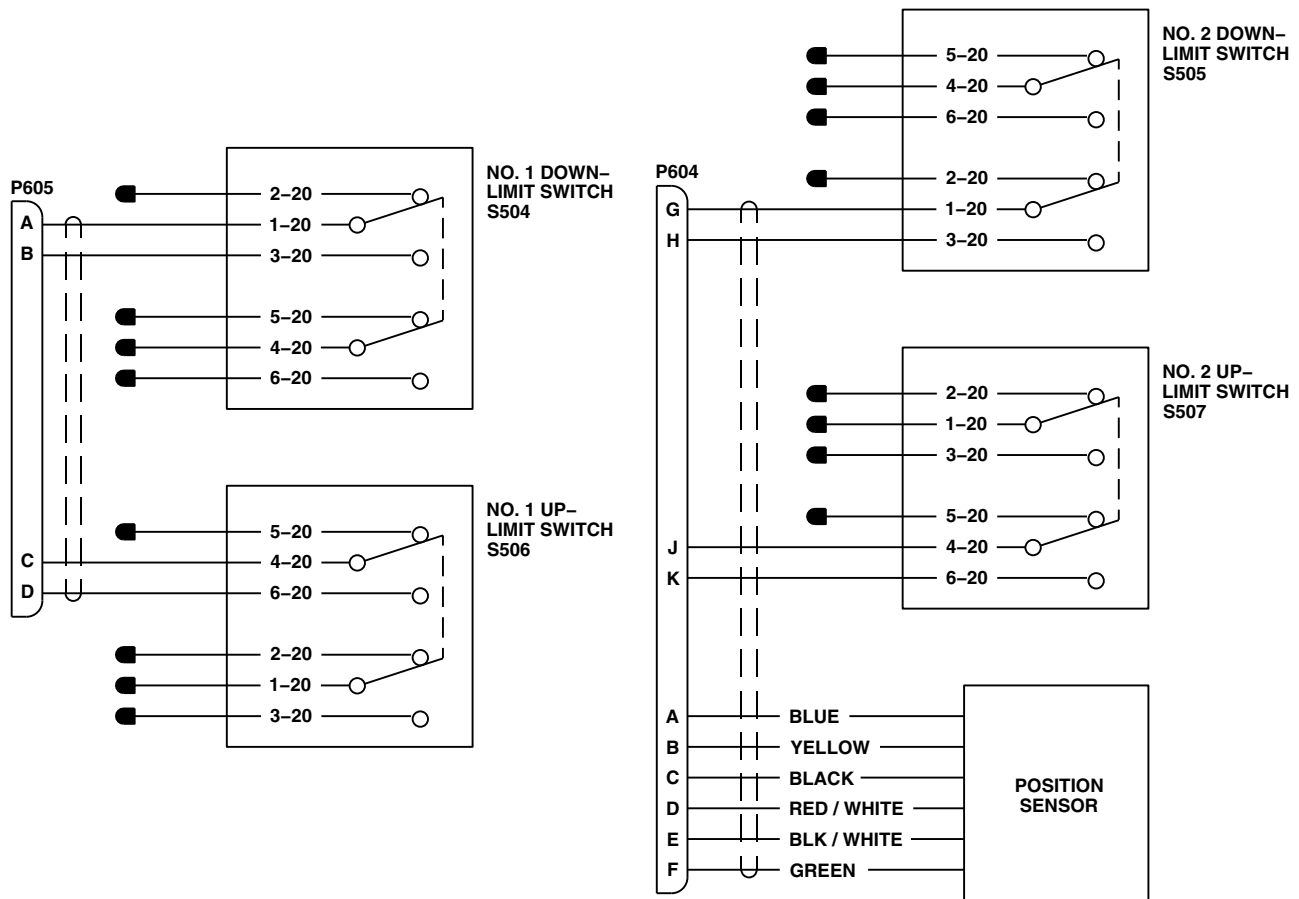
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FIGURE 2-2-1-1. POSITION TRANSMITTER/LIMIT SWITCH ASSEMBLY SCHEMATIC DIAGRAM

SECTION 2-3

ON AIRCRAFT INSPECTIONS

SECTION OVERVIEW

PARAGRAPH	TITLE
2-3-1	Corrosion Inspections
2-3-2	Cockpit Doors Inspection
2-3-3	Window and Windshield Inspections
2-3-4	Pilot's/Copilot's Seat Inspections
2-3-5	Crew Chief's/Passenger's Seat Inspections
2-3-6	Crew Chief's Window Inspections
2-3-7	Cabin Doors Inspections
2-3-8	Cabin Floors Inspection
2-3-9	Wire Strike Protection System Inspections
2-3-10	Main Rotor Pylon Inspections
2-3-11	Engine Cowlings Inspection
2-3-12	Transition Section Inspections
2-3-13	Tail Rotor Drive Shaft Covers Inspection
2-3-14	Tail Rotor Pylon Inspections
2-3-15	Stabilator Inspections

2-3-1 CORROSION INSPECTIONS.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
2-3-1.1	Nose Door Latch Corrosion Inspection	2-3-1-1	2-3-1.2	Cockpit Door Corrosion Inspection	2-3-1-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

References

- PARA 2-4-2
- PARA 2-4-26

2-3-1.1 Nose Door Latch Corrosion Inspection.

NOTE

Corrosion is indicated by red film or pitting.

- Visually inspect latch for corrosion. **NONE ALLOWED.** If corrosion is found, remove corrosion (PARA 2-4-2).
- Raise latch, loosen jamnut, and inspect fit of latch hook and jamnut on threaded stud. Proceed as follows:

- LATCH HOOK OR JAMNUT SHALL NOT BE LOOSE OR WOBBLE ON THREADED STUD.** If loose, replace latch (PARA 2-4-2).

- If fit of latch hook or jamnut is secure, **TIGHTEN JAMNUT 1⁄6 TO 1⁄3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

2-3-1.2 Cockpit Door Corrosion Inspection.

NOTE

Corrosion is indicated by red film or pitting.

- Visually inspect latch for corrosion. **NONE ALLOWED.** If corrosion is found, remove rod end from hinge and remove corrosion (PARA 2-4-26).
- Open and close door to check for proper operation. **ROD END BEARING SHALL NOT BE STICKING OR RATCHETY.** If sticking or ratchety, replace rod end (PARA 2-4-26).

2-3-2 COCKPIT DOORS INSPECTION.

PARAGRAPH BREAKDOWN

		PAGE
2-3-2.1	Cockpit Doors Jettison Inspection	2-3-2-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Foam Padding
- Spring Scale

Materials/Parts

- Lockwire, Item 195, Appendix D

References

- Appendix D
- PARA 2-4-11
- PARA 2-4-14
- PARA 2-4-25
- PARA 2-6-7

2-3-2.1 Cockpit Doors Jettison Inspection.

- Position foam padding on ground to prevent damage to cockpit door during jettison check.
- Have assistants outside helicopter to guide jettisoned cockpit door onto foam padding.
- Make sure cockpit door is closed and latched.
- Connect spring scale to cockpit door jettison handle.
- Pull jettison handle with spring scale against the resistance of the mechanism. **FORCE SHALL BE 27 - 33 POUNDS.** If force is not 27 - 33 pounds, perform the following:
 - Reset jettison handle and adjust hinge screws (PARA 2-4-25).
 - Repeat pull test until force is between 27 - 33 pounds.
- Perform step e. a minimum of four times. After final attempt, perform e. (1) and step e. (2)

- Disconnect spring scale from cockpit door jettison handle.

NOTE

Force required to jettison door from helicopter shall not be so great that an average size person cannot jettison door.

- Pull jettison handle and manually push or kick door from helicopter.
- Inspect cockpit door and jettison mechanism for damage. **NONE ALLOWED.** If damage is found, repair cockpit door (PARA 2-6-7), or replace jettison mechanism components (PARA 2-4-14).
- Install cockpit door (PARA 2-4-11).
- Repeat step d. through step f.
- Position jettison handle in full forward position and lockwire, Item 195, Appendix D, jettison handle to clip on lower front edge of window frame.

2-3-3 WINDOW AND WINDSHIELD INSPECTIONS.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
2-3-3.1	Upper Window Inspection	2-3-3-1	2-3-3.4	Plastic Center Windshield Inspection W/O HCW	2-3-3-3
2-3-3.2	Lower Window Inspection	2-3-3-2			
2-3-3.3	Outboard Windshield Inspection	2-3-3-2	2-3-3.5	Glass Center Windshield Inspection HCW	2-3-3-3

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit

References

- PARA 2-4-37
- PARA 2-4-38

- PARA 2-4-39
- PARA 2-4-40
- PARA 2-4-41

2-3-3.1 Upper Window Inspection.

- Inspect upper window for cracks. **NONE ALLOWED.** If cracked, replace window (PARA 2-4-37).
- Inspect upper window for crazing. **CRAZING SHALL NOT INTERFERE WITH PILOT’S/ COPILOT’S VISIBILITY.** If crazing interferes with pilot’s/copilot’s visibility, replace window (PARA 2-4-37).
- Inspect upper window for scratches. **SCRATCHES OF ANY LENGTH SHALL NOT EXCEED 0.010-INCH DEEP.** If limit is exceeded, replace window (PARA 2-4-37).
- Inspect upper window for chips and nicks. **CHIPS AND/OR NICKS SHALL NOT EXCEED 0.010-INCH DEEP OR 0.093-INCH DIAMETER WITH NO MORE THAN**

FOUR CHIPS OR NICKS WITHIN 6-INCH DIAMETER AREA. If limits are exceeded, replace window (PARA 2-4-37).

- Inspect upper window mounting holes for chips. **CHIPS SHALL NOT EXCEED 0.063-INCH DEEP OR 0.090-INCH LONG EXTENDING FROM EDGE OF MOUNTING HOLES. DAMAGE MUST NOT BE ON MORE THAN SIX HOLES AND NO MORE THAN TWO OUT OF FIVE CONSECUTIVE HOLES.** If limits are exceeded, replace window (PARA 2-4-37).
- Inspect upper window panel edge for chips. **CHIPS SHALL NOT EXCEED ½-INCH LONG OR 0.13-INCH WIDE THROUGH FULL DEPTH OF PANEL ALONG EDGE. NO MORE THAN TWO CHIPS ARE ALLOWED IN ANY 6-INCH LENGTH.** If limits are exceeded, replace window (PARA 2-4-37).

- g. Inspect upper window for overall damage. Proceed as follows: Replace window (PARA 2-4-37) if:

- (1) **DAMAGE PREVENTS SECURE ATTACHMENT.**
- (2) **DAMAGE IS FOUND ON INNER AND OUTER SURFACES OF WINDOW.**
- (3) **DAMAGE INTERFERES WITH PILOT'S/COPILOT'S VISIBILITY.**

- h. Inspect pressure sensitive adhesive tape for cuts and looseness. **SEVERED OR LOOSE TAPE IS NOT ALLOWED. CUTS THAT DO NOT SEVER TAPE ARE REPARABLE.** If cuts do not sever tape, repair tape. If severed or loose, replace tape (PARA 2-4-37).

2-3-3.2 Lower Window Inspection.

- a. Inspect lower window for cracks. **NONE ALLOWED.** If cracked, replace window (PARA 2-4-38).
- b. Inspect lower window for crazing. **CRAZING SHALL NOT INTERFERE WITH PILOT'S/COPILOT'S VISIBILITY.** If crazing interferes with pilot's/copilot's visibility, replace window (PARA 2-4-38).
- c. Inspect lower window for scratches. **SCRATCHES OF ANY LENGTH SHALL NOT EXCEED 0.010-INCH DEEP.** If limit is exceeded, replace window (PARA 2-4-38).
- d. Inspect lower window for chips and nicks. **CHIPS AND/OR NICKS SHALL NOT EXCEED 0.010-INCH DEEP OR 0.093-INCH DIAMETER WITH NO MORE THAN FOUR CHIPS OR NICKS WITHIN 6-INCH DIAMETER AREA.** If limits are exceeded, replace window (PARA 2-4-38).
- e. Inspect lower window mounting holes for chips. **CHIPS SHALL NOT EXCEED 0.063-INCH DEEP OR 0.090-INCH LONG EXTENDING FROM EDGE OF MOUNTING HOLES. DAMAGE MUST NOT BE ON MORE THAN SIX HOLES AND NO MORE**

THAN TWO OUT OF FIVE CONSECUTIVE HOLES. If limits are exceeded, replace window (PARA 2-4-38).

- f. Inspect lower window panel edge for chips. **CHIPS SHALL NOT EXCEED 1/2-INCH LONG OR 0.13-INCH WIDE THROUGH FULL DEPTH OF PANEL ALONG EDGE. NO MORE THAN TWO CHIPS ARE ALLOWED IN ANY 6-INCH LENGTH.** If limits are exceeded, replace window (PARA 2-4-38).

- g. Inspect lower window for overall damage. Replace window (PARA 2-4-38) if:

- (1) **DAMAGE PREVENTS SECURE ATTACHMENT.**
- (2) **DAMAGE IS FOUND ON INNER AND OUTER SURFACES OF WINDOW.**
- (3) **DAMAGE INTERFERES WITH PILOT'S/COPILOT'S VISIBILITY.**

- h. Inspect foil tape for cuts and looseness. **SEVERED OR LOOSE TAPE IS NOT ALLOWED. CUTS THAT DO NOT SEVER TAPE ARE REPARABLE.** If cuts do not sever tape, repair tape. If severed or loose, replace tape (PARA 2-4-38).

2-3-3.3 Outboard Windshield Inspection.

- a. Inspect outboard windshield for cracks. **NONE ALLOWED.** If cracked, replace windshield (PARA 2-4-39).
- b. Inspect outboard windshield for erosion. **EROSION SHALL NOT INTERFERE WITH PILOT'S/COPILOT'S VISIBILITY.** If erosion interferes with pilot's/copilot's visibility, replace windshield (PARA 2-4-39).
- c. Inspect outboard windshield for scratches. **SCRATCHES OF ANY LENGTH SHALL NOT EXCEED 0.005-INCH DEEP.** If limit is exceeded, replace windshield (PARA 2-4-39).
- d. Inspect outboard windshield for chips and nicks on inner and outer layers. **CHIPS AND/OR NICKS SHALL NOT EXCEED 0.010-**

INCH DEEP OR 0.093-INCH DIAMETER WITH NO MORE THAN FOUR CHIPS OR NICKS WITHIN 6-INCH DIAMETER AREA. If limits are exceeded, replace windshield (PARA 2-4-39).

- e. Inspect outboard windshield for delamination of reinforced plastic edging. **DELAMINATION SHALL NOT CAUSE REINFORCED PLASTIC EDGING TO DISBOND FROM GLASS PANEL.** If reinforced plastic edging is disbonded from glass panel, replace windshield (PARA 2-4-39).
- f. Inspect outboard windshield for overall damage. Replace windshield (PARA 2-4-39) if:
 - (1) **DAMAGE PREVENTS SECURE ATTACHMENT.**
 - (2) **DAMAGE IS FOUND ON INNER AND OUTER SURFACES OF WINDOW.**
 - (3) **DAMAGE INTERFERES WITH PILOT'S/COPILOT'S VISIBILITY.**
 - (4) **DEICING IS NOT OPERATIONAL DUE TO CRACKS OR OTHER DAMAGE.**

2-3-3.4 Plastic Center Windshield Inspection **W/O HCW** .

- a. Inspect plastic center windshield for cracks. **NONE ALLOWED.** If cracked, replace windshield (PARA 2-4-41).
- b. Inspect plastic center windshield for crazing. **CRAZING SHALL NOT INTERFERE WITH PILOT'S/COPILOT'S VISIBILITY.** If crazing interferes with pilot's/copilot's visibility, replace windshield (PARA 2-4-41).
- c. Inspect plastic center windshield for scratches. **SCRATCHES OF ANY LENGTH SHALL NOT EXCEED 0.010-INCH DEEP.** If limit is exceeded, replace windshield (PARA 2-4-41).
- d. Inspect plastic center windshield for chips and nicks. **CHIPS AND/OR NICKS SHALL NOT EXCEED 0.010-INCH DEEP OR 0.093-INCH**

DIAMETER WITH NO MORE THAN FOUR CHIPS OR NICKS WITHIN 6-INCH DIAMETER AREA. If limits are exceeded, replace windshield (PARA 2-4-41).

- e. Inspect plastic center windshield mounting holes for chips. **CHIPS SHALL NOT EXCEED 0.063-INCH DEEP OR 0.090-INCH LONG EXTENDING FROM EDGE OF MOUNTING HOLES. DAMAGE MUST NOT BE ON MORE THAN SIX HOLES AND NO MORE THAN TWO OUT OF FIVE CONSECUTIVE HOLES.** If limits are exceeded, replace windshield (PARA 2-4-41).
- f. Inspect plastic center windshield edge for chips. **CHIPS SHALL NOT EXCEED 1/2-INCH LONG OR 0.13-INCH WIDE THROUGH FULL DEPTH OF PANEL ALONG EDGE. NO MORE THAN TWO CHIPS ARE ALLOWED IN ANY 6-INCH LENGTH.** If limits are exceeded, replace window (PARA 2-4-41).
- g. Inspect plastic center windshield for overall damage. Replace windshield (PARA 2-4-41) if:
 - (1) **DAMAGE PREVENTS SECURE ATTACHMENT.**
 - (2) **DAMAGE IS FOUND ON INNER AND OUTER SURFACES OF WINDSHIELD.**
 - (3) **DAMAGE INTERFERES WITH PILOT'S/COPILOT'S VISIBILITY.**

2-3-3.5 Glass Center Windshield Inspection **HCW** .

- a. Inspect center windshield for cracks. **NONE ALLOWED.** If cracked, replace windshield (PARA 2-4-40).
- b. Inspect center windshield for erosion. **EROSION SHALL NOT INTERFERE WITH PILOT'S/COPILOT'S VISIBILITY.** If erosion interferes with pilot's/copilot's visibility, replace windshield (PARA 2-4-40).
- c. Inspect center windshield for scratches. **SCRATCHES OF ANY LENGTH SHALL NOT**

EXCEED 0.005-INCH DEEP. If limit is exceeded, replace windshield (PARA 2-4-40).

- d. Inspect center windshield for chips and nicks on inner and outer layers. **CHIPS AND/OR NICKS SHALL NOT EXCEED 0.010-INCH DEEP OR 0.093-INCH DIAMETER WITH MORE THAN FOUR CHIPS OR NICKS WITHIN 6-INCH DIAMETER AREA.** If limits are exceeded, replace windshield (PARA 2-4-40).

- e. Inspect center windshield for delamination of reinforced plastic edging. **DELAMINATION SHALL NOT CAUSE REINFORCED PLASTIC EDGING TO DISBOND FROM GLASS PANEL.** If reinforced plastic edging is disbonded from glass panel, replace windshield (PARA 2-4-40).

- f. Inspect center windshield for overall damage. Replace windshield (PARA 2-4-40) if:

(1) **DAMAGE PREVENTS SECURE ATTACHMENT.**

(2) **DAMAGE IS FOUND ON INNER AND OUTER SURFACES OF WINDOW.**

(3) **DAMAGE INTERFERES WITH PILOT'S/COPILOT'S VISIBILITY.**

(4) **DEICING IS NOT OPERATIONAL DUE TO CRACKS OR OTHER DAMAGE.**

- g. Inspect pressure sensitive adhesive tape for cuts and looseness. **SEVERED OR LOOSE TAPE IS NOT ALLOWED. CUTS THAT DO NOT SEVER TAPE ARE REPARABLE.** If cuts do not sever tape, repair tape. If severed or loose, replace tape (PARA 2-4-40).

2-3-4 PILOT’S/COPILOT’S SEAT INSPECTIONS.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
2-3-4.1	Pilot’s/Copilot’s Seat - Shoulder Harness Inspection	2-3-4-1	2-3-4.4	Pilot’s/Copilot’s Seat - Trunnion Torque Stripe Inspection	2-3-4-4
2-3-4.2	Pilot’s/Copilot’s Seat - Lap Belt Inspection	2-3-4-2			
2-3-4.3	Pilot’s/Copilot’s Seat - Crotch Belt, Buckle, and Pad Inspection	2-3-4-3			

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- File
- Protection, Eye

- Socket, Modified to fit buckle release handle
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

References

- PARA 2-4-44

2-3-4.1 Pilot’s/Copilot’s Seat - Shoulder Harness Inspection.

WARNING

To prevent injury to personnel, do not release either normal or emergency vertical adjust levers unless someone is sitting in seat. Extension springs are under load at all times. With seat at lowest position, vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, seat will be snapped to highest stop. Anyone leaning over seat, or with hands on guide tubes above linear bearings, will be seriously injured.

- Sit in seat and adjust to full up and full rear position.
- Check shoulder inertia reel mounting hardware for cracks and secure. **NO CRACKS ALLOWED.** If cracks are found or hardware is loose, replace hardware (PARA 2-4-44).
- Check inertia reel surface for smoothness and freedom from sharp edges or burrs that may snag webbing.
- Check operation of control cable and inertia reel.
- Check restraint system webbing as follows:
 - Check for deterioration or discoloration of webbing due to contact with grease, oils,

fuels, or hydraulic fluids. **NONE ALLOWED.** If deterioration or discoloration are found, replace webbing (PARA 2-4-44).

- (2) Check webbing for cuts. **NONE ALLOWED.** If cuts are found, replace webbing (PARA 2-4-44).
- (3) Check webbing for fraying on exterior surface. **FRAYING SHALL NOT EXCEED 20% OF WIDTH OR 2-INCHES IN LENGTH.** If limits are exceeded, replace webbing (PARA 2-4-44).

WARNING

To prevent injury to personnel, do not release either normal or emergency vertical adjust levers unless someone is sitting in seat. Extension springs are under load at all times. With seat at lowest position, vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, seat will be snapped to highest stop. Anyone leaning over seat, or with hands on guide tubes above linear bearings, will be seriously injured.

- f. Sit in seat and adjust full down and full front position.

2-3-4.2 Pilot's/Copilot's Seat - Lap Belt Inspection.

WARNING

To prevent injury to personnel, do not release either normal or emergency vertical adjust levers unless someone is sitting in seat. Extension springs are

under load at all times. With seat at lowest position, vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, seat will be snapped to highest stop. Anyone leaning over seat, or with hands on guide tubes above linear bearings, will be seriously injured.

- a. Sit in seat and adjust to full up and full rear position.
- b. Check lap belt anchor fittings for cracks, snags, and sharp edges. **NONE ALLOWED.** If cracked fittings are found, replace lap belts (PARA 2-4-44). Using file, round off edges and remove snags on serviceable fittings.
- c. Check lap belt inertia reel mounting hardware for cracks and security. **NO CRACKS ALLOWED.** If cracks are found or hardware is loose, replace hardware (PARA 2-4-44).
- d. Check lap belt webbing for cuts. **NONE ALLOWED.** If cuts are found, replace webbing (PARA 2-4-44).

WARNING

To prevent injury to personnel, do not release either normal or emergency vertical adjust levers unless someone is sitting in seat. Extension springs are under load at all times. With seat at lowest position, vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, seat will be snapped to highest stop. Anyone leaning over seat, or with hands on guide tubes above linear bearings, will be seriously injured.

- e. Sit in seat and adjust to full up and full front position.

2-3-4.3 Pilot's/Copilot's Seat - Crotch Belt, Buckle, and Pad Inspection.

WARNING

To prevent injury to personnel, do not release either normal or emergency vertical adjust levers unless someone is sitting in seat. Extension springs are under load at all times. With seat at lowest position, vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, seat will be snapped to highest stop. Anyone leaning over seat, or with hands on guide tubes above linear bearings, will be seriously injured.

- a. Sit in seat and adjust to full up and full rear position.
- b. Check crotch belt anchor fittings for cracks, snags, and sharp edges. **NONE ALLOWED.** If cracked fittings are found, replace crotch belt (PARA 2-4-44). Using file, round off edges and remove snags on serviceable fittings.
- c. Check crotch belt mounting hardware for cracks and security. **NO CRACKS ALLOWED.** If cracks are found or hardware is loose, replace hardware (PARA 2-4-44).

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel

are wearing safety glasses when compressed air is being used.

- d. Check rotary buckle for cleanness and smooth operation. If necessary, blow dirt from buckle mechanisms with dry, compressed air.
- e. Insert shoulder and lap belt harness fittings in rotary buckle. **ENGAGEMENT MUST BE POSITIVE AND SECURE.** If rotary buckle fails engagement test, replace rotary buckle (PARA 2-4-44).
- f. Perform rotary buckle release tension/rotation test as follows:
 - (1) If necessary, modify socket for torque wrench to fit buckle release handle.
 - (2) Insert shoulder and lap belt harness fittings in rotary buckle. Hold buckle stationary without restricting rotary portion of buckle to rotate.
 - (3) Using modified socket and torque wrench, measure buckle release tension (torque) and angle of rotary buckle when harness fittings release. **HARNESS FITTINGS SHALL RELEASE CLEANLY WITH A TORQUE OF NO LESS THAN 15 INCH-POUNDS OR MORE THAN 75 INCH-POUNDS AS HANDLE IS ROTATED 30 - 45° (± 4°) IN EITHER DIRECTION.** Perform step f. (2) and step f. (3) a total of three times.
 - (4) If rotary buckle passes test all three times, test is complete. If rotary buckle fails any one tension or rotation test, replace rotary buckle (PARA 2-4-44), then perform step f. (2) and step f. (3) a total of three times.
- g. Check crotch belt webbing for cuts. **NONE ALLOWED.** If cuts are found, replace webbing (PARA 2-4-44).

WARNING

To prevent injury to personnel, do not release either normal or emergency vertical adjust levers unless someone is sitting in seat. Extension springs are under load at all times. With seat at lowest position, vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, seat will be snapped to high-

est stop. Anyone leaning over seat, or with hands on guide tubes above linear bearings, will be seriously injured.

- h. Sit in seat and adjust full down and full front position.

2-3-4.4 Pilot's/Copilot's Seat - Trunnion Torque Stripe Inspection.

Inspect trunnion torque stripe for slippage. **NONE ALLOWED.** If slippage has occurred, replace threaded insert (PARA 2-4-44).

2-3-5 CREW CHIEF’S/PASSENGER’S SEAT INSPECTIONS.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
2-3-5.1	Passenger’s Seat Restraint System Inspection	2-3-5-1	2-3-5.3	Passenger’s Seat Cable Inspection	2-3-5-3
2-3-5.2	Crew Chief’s Seat Restraint System Inspection	2-3-5-2			

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- File
- Protection, Eye
- Socket, Modified to fit buckle release handle
- Spring Scale

- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

References

- PARA 2-4-45
- PARA 2-4-46
- PARA 2-4-47

2-3-5.1 Passenger’s Seat Restraint System Inspection.

- a. Check webbing for cuts. **NONE ALLOWED.** If cuts are found, replace webbing (PARA 2-4-46).



Injury to personnel will result if precautions are not taken while using compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

- b. Check rotary buckle for cleanness and smooth operation. If necessary, blow dirt from buckle mechanisms with dry, compressed air.
- c. Insert shoulder and lap belt harness fittings in rotary buckle. **ENGAGEMENT MUST BE POSITIVE AND SECURE.** If rotary buckle fails engagement test, replace buckle (PARA 2-4-46).
- d. Perform rotary buckle release tension/rotation test as follows:
 - (1) If necessary, modify socket for torque wrench to fit buckle release handle.
 - (2) Insert shoulder and lap belt harness fittings in rotary buckle. Hold buckle stationary without restricting the rotary portion of the buckle to rotate.

- (3) Using modified socket and torque wrench, measure buckle release tension (torque) and angle of rotary buckle when harness fittings release. **HARNESS FITTINGS SHALL RELEASE CLEANLY WITH A TORQUE OF NOT LESS THAN 15 INCH-POUNDS OR MORE THAN 75 INCH-POUNDS AS HANDLE IS ROTATED 30 - 45° (± 4°) IN EITHER DIRECTION.** Perform step d. (2) and step d. (3) a total of three times.
- (4) If rotary buckle passes test all three times, test is complete. If rotary buckle fails any one tension or rotation test, replace thrust spring, or shim existing thrust spring, to reach proper release tension, then perform step d. (2) and step d. (3) a total of three times.

2-3-5.2 Crew Chief's Seat Restraint System Inspection.

- a. Check restraint seat anchor fittings for cracks, snags, and sharp edges. **NO CRACKS ALLOWED.** If crack are found, replace fittings (PARA 2-4-46). Using file, round off edges and remove snags on serviceable fittings.
- b. Check shoulder and lap belt inertia reel mounting hardware for cracks and security. **NO CRACKS ALLOWED.** If cracks are found or hardware is loose, replace hardware (PARA 2-4-47).
- c. Pull webbing out from shoulder and lap belt inertia reels in sequence and inspect.

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

- d. Check rotary buckle for cleanness and smooth operation. If necessary, blow dirt from buckle mechanisms with dry, compressed air.
- e. Insert shoulder and lap belt harness fittings in rotary buckle. **ENGAGEMENT MUST BE POSITIVE AND SECURE.** If rotary buckle fails engagement test, replace buckle (PARA 2-4-47).
- f. Perform rotary buckle release tension/rotation test as follows:
 - (1) If necessary, modify socket for torque wrench to fit buckle release handle.
 - (2) Insert shoulder and lap belt harness fittings in rotary buckle. Hold buckle stationary without restricting the rotary portion of the buckle to rotate.
 - (3) Using modified socket and torque wrench, measure buckle release tension (torque) and angle of rotary buckle when harness fittings release. **HARNESS FITTINGS SHALL RELEASE CLEANLY WITH A TORQUE OF NOT LESS THAN 15 INCH-POUNDS OR MORE THAN 75 INCH-POUNDS AS HANDLE IS ROTATED 30 - 45° (± 4°) IN EITHER DIRECTION.** Perform step f. (2) and step f. (3) a total of three times.
 - (4) If rotary buckle passes test all three times, test is complete. If rotary buckle fails any one tension or rotation test, replace thrust spring, or shim existing thrust spring, to reach proper release tension, then perform step f. (2) and step f. (3) a total of three times.
- g. Using spring scale, measure shoulder harness and lap belt inertia reel tension against webbing. **AVERAGE TENSION BETWEEN MAXIMUM WEBBING EXTENSION AND RETRACTION TO WITHIN THE LAST 2-INCHES OF TRAVEL IS 2 - 9 POUNDS FOR ALL THREE INERTIA REELS.** If inertia reel(s) fail engagement test, replace inertia reel(s) (PARA 2-4-47).

- h. Give shoulder harness a quick pull. **INERTIA REEL SHALL LOCK.** Release tension. Reel shall smoothly retract or release webbing as required. If inertia reel fails tension check, replace inertia reel (PARA 2-4-47).
- i. Check left and right lap belt inertia reel operation as follows:
 - (1) Pull several inches of webbing from reel.
 - (2) Put reel control in MANUAL-LOCK. Tug on webbing. **REEL SHALL NOT RELEASE ANY MORE THAN ½-INCH OF WEBBING** (compensating for webbing slip and stretch).
 - (3) Release tension. **REEL MUST RETRACT WEBBING BUT LOCK WHEN WEBBING IS PULLED.**
 - (4) Put reel control in AUTO-LOCK. Pull out entire webbing length.
 - (5) Release lap belt. **REEL MUST FULLY RETRACT ENTIRE WEBBING LENGTH.**
 - (6) Give webbing a quick, hard pull. **REEL SHALL LOCK AND HOLD UNTIL TENSION IS RELEASED.**

- (7) Give webbing a quick, hard pull to lock reel. Maintain tension on lock. Put reel control in MANUAL-LOCK, then put reel control in AUTO-LOCK.
- (8) If inertia reel failed any part of this inspection, replace inertia reel (PARA 2-4-47).

2-3-5.3 Passenger's Seat Cable Inspection.

- a. Check strands for breakage. **NO MORE THAN THREE BROKEN STRANDS ALLOWED.** If more than three broken strands are found, replace cable (PARA 2-4-45).
- b. Inspect cables for kinks. **NONE ALLOWED.** If kinks are found, replace cable (PARA 2-4-45).
- c. Inspect sheathing surrounding cable. **CRACKS THROUGH SHEATHING DOWN TO CABLE ARE NOT ALLOWED.** If cracks through sheathing down to cable are found, replace cable (PARA 2-4-45).
- d. Check cable attaching hardware for cracks and security. **NO CRACKS ALLOWED.** If cracks are found or hardware is loose, replace hardware (PARA 2-4-45).

2-3-6 CREW CHIEF’S WINDOW INSPECTIONS.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
2-3-6.1	Crew Chief’s Window Lower Track Inspection	2-3-6-1	2-3-6.2	Crew Chief’s Window Upper Track Inspection	2-3-6-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

References

- PARA 2-4-48
- PARA 2-6-1

- PARA 2-6-5
- PARA 2-6-21
- PARA 2-6-22
- PARA 2-6-23

2-3-6.1 Crew Chief’s Window Lower Track Inspection.

- Remove crew chief’s window (PARA 2-4-48).
- Visually inspect lower track for cracks. **NONE ALLOWED.** If cracked, replace track (PARA 2-6-21).
- Inspect front wear strips for wear and security (Figure 2-3-6-1, Detail A and Detail B). **WEAR SHALL NOT BE MORE THAN 0.010-INCH DEEP.** If wear is more than 0.010-inch deep or wear strip is loose or missing, replace wear strip (PARA 2-6-22).
- Inspect rear wear strips between STA 265.0 and STA 270.0, if installed, for wear (Figure 2-3-6-1, Detail C and Detail D). **WEAR SHALL NOT BE MORE THAN 0.010-INCH DEEP.** If wear is more than 0.010-inch deep, or wear strip is loose or missing, replace wear strip (PARA 2-6-22).
- Inspect lower track for wear. **WEAR SHALL NOT BE MORE THAN 0.025-INCH DEEP,**

WITH NO LIMIT TO LENGTH OR WIDTH. If wear is located between STA 265.0 and STA 270.0, repair track and install locally-made wear strips (PARA 2-6-22). If wear exceeds 0.025-inch at any location, replace track (PARA 2-6-21).

- Inspect areas of track around rivets for corrosion. Remove and treat corrosion (PARA 2-6-1).
- Touch up paint as required (PARA 2-6-5).

2-3-6.2 Crew Chief’s Window Upper Track Inspection.

- Visually inspect upper track for cracks. **NONE ALLOWED.** Replace cracked track (PARA 2-6-23).
- Inspect track lip surfaces for wear by rollers. **MINIMUM MATERIAL THICKNESS ALLOWED IS 0.040-INCH.** Replace track if limit is exceeded (PARA 2-6-23).
- Install crew chief’s window (PARA 2-4-48).

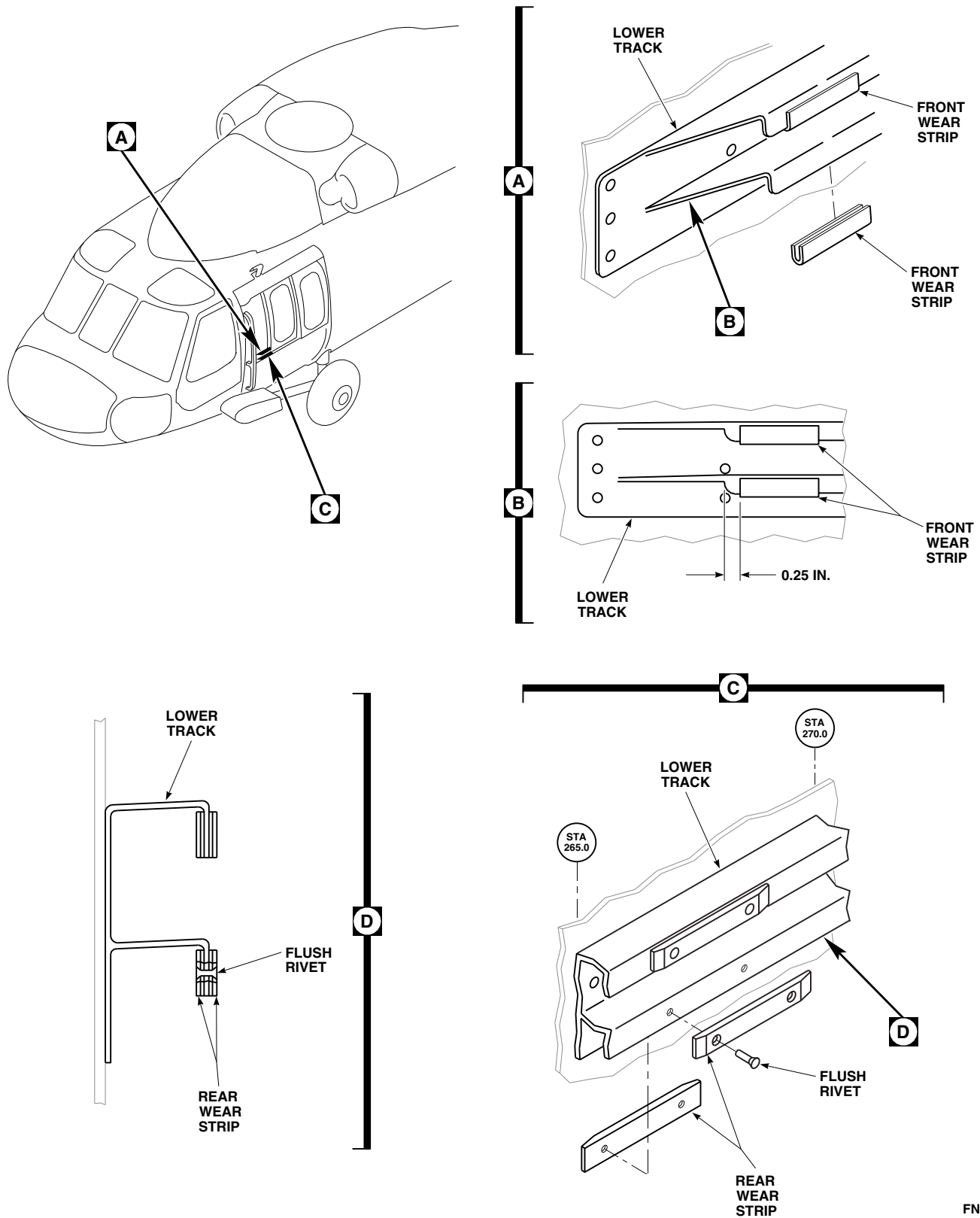


FIGURE 2-3-6-1. INSPECT CREW CHIEF'S WINDOW LOWER TRACK

2-3-7 CABIN DOOR INSPECTIONS.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
2-3-7.1	Cabin Door Lower Track Inspection	2-3-7-1	2-3-7.2	Cabin Door Window Jettison Inspection	2-3-7-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Foam Padding
- Micrometer
- Spring Scale

Materials/Parts

- Masking Tape, 1-inch, Item 212, Appendix D

References

- Appendix D
- PARA 2-4-56
- PARA 2-4-57
- PARA 2-4-67
- PARA 2-4-68
- PARA 2-6-17

2-3-7.1 Cabin Door Lower Track Inspection.

- Inspect cabin door lower track for visual cracks and broken off pieces. **NONE ALLOWED.** If cracks or broken off pieces are found, replace cabin door lower track (PARA 2-4-57).
- Inspect wear strips for wear (Figure 2-3-7-1, Detail B). If wear strip(s) are worn to less than 0.010-inch, replace wear strip(s) (PARA 2-6-17).
- For wear strips that are completely worn through, inspect lower track under wear strip. Using micrometer, measure thickness of lower track at worn areas (Figure 2-3-7-1, Detail B and Detail C). **IF THICKNESS IS LESS THAN 0.080-INCH, REPLACE TRACK** (PARA 2-4-57).
- Inspect screws securing lower track as follows:

- Remove cabin door lower track (PARA 2-4-57).
- Inspect screws for damaged threads and damaged or deformed heads. Inspect washers for deformation or damage (Figure 2-3-7-1, Detail D). **NONE ALLOWED.** If damage or deformation is found, replace screws and washers (PARA 2-4-57).
- Install cabin door lower track (PARA 2-4-57).

2-3-7.2 Cabin Door Window Jettison Inspection.

- Position foam padding on ground to prevent damage to cabin door window during jettison check.

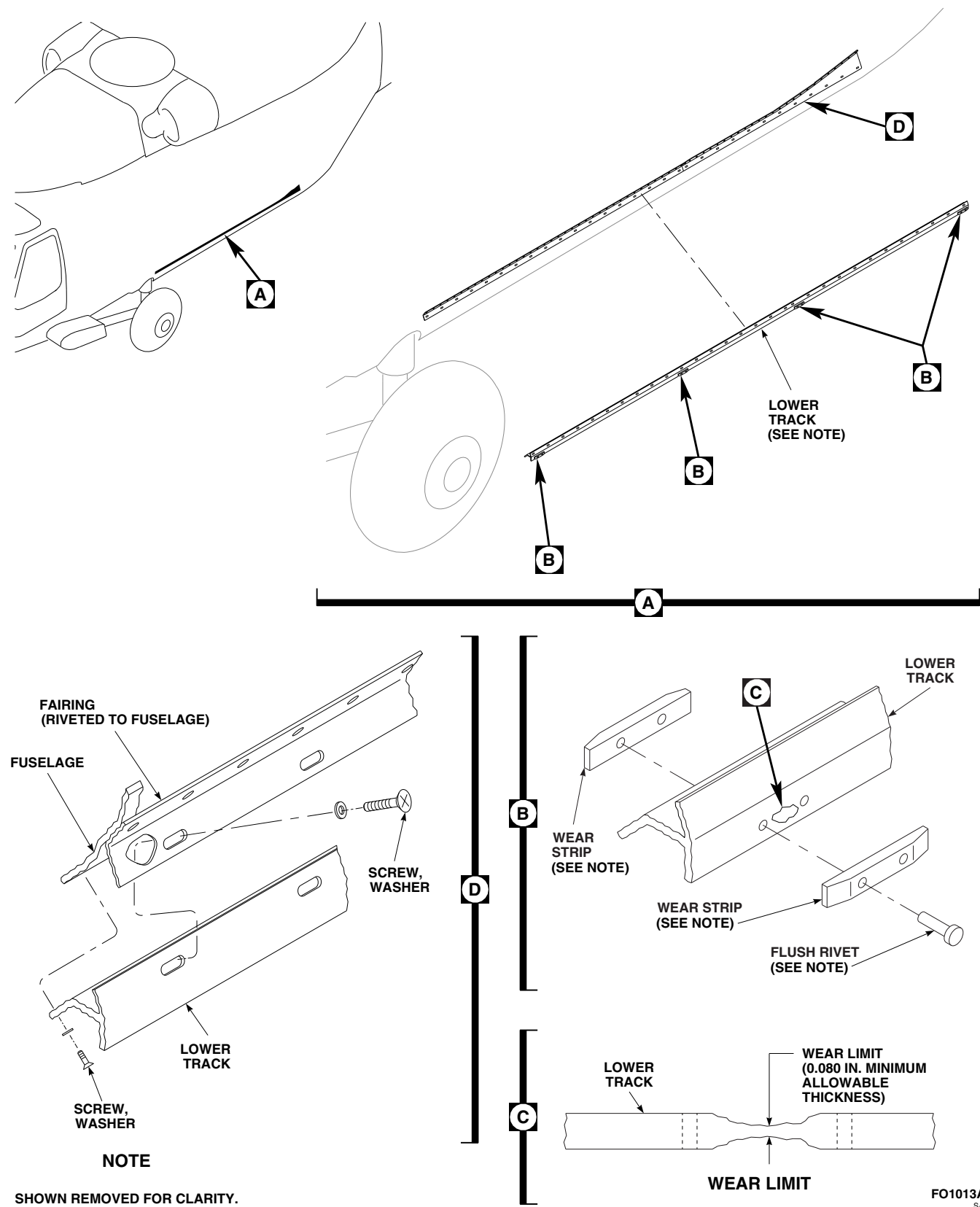


FIGURE 2-3-7-1. INSPECT CABIN DOOR LOWER TRACK

NOTE

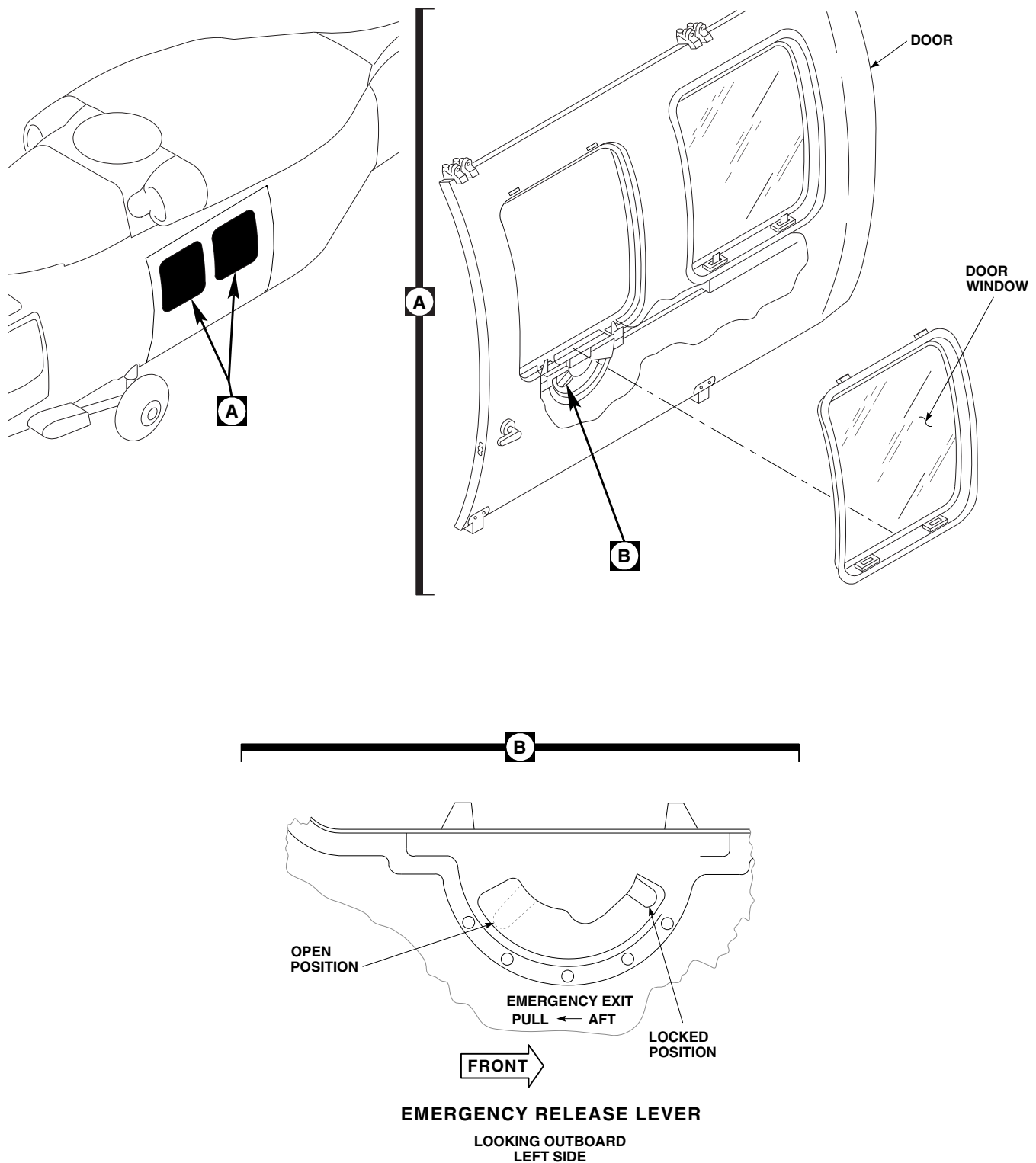
Opening emergency release lever will jettison both windows.

- b. If only one window is to be inspected, secure other window in place with 1-inch masking tape, Item 212, Appendix D.
- c. Remove cabin door window jettison mechanism guard (PARA 2-4-68).
- d. Connect spring scale to cabin door window jettison handle.
- e. Have assistant(s) outside helicopter to guide jettisoned cabin door window onto foam padding.
- f. Pull emergency release lever aft in door to open position with spring scale against resistance of mechanism (Figure 2-3-7-2, Detail B). **FORCE SHALL NOT EXCEED 30 POUNDS.** If force exceeds limits, replace cabin door jettison mechanism (PARA 2-4-67).

NOTE

Force required to jettison window from helicopter shall not be so great that an average size person cannot jettison cabin door window.

- g. Push on bottom of window outward from helicopter (Figure 2-3-7-2, Detail A). **WINDOW SHALL FALL AWAY FROM HELICOPTER.**
- h. Inspect cabin door window and jettison mechanism for damage. **NONE ALLOWED.** If damage is found, replace window (PARA 2-4-56), or jettison mechanism (PARA 2-4-67).
- i. Install cabin door window (PARA 2-4-56).
- j. Install cabin door window jettison mechanism guard (PARA 2-4-68).
- k. If necessary, remove masking tape from other window.



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FIGURE 2-3-7-2. INSPECT CABIN DOOR WINDOW JETTISON

2-3-8 CABIN FLOOR INSPECTIONS.

PARAGRAPH BREAKDOWN

		PAGE
2-3-8.1	Tiedown Rings and Floor Pans Inspections	2-3-8-1

INITIAL SETUP

- PARA 2-6-27

Tools

- Aircraft Mechanic’s Toolkit

References

- PARA 1-6-16
- PARA 2-4-71

2-3-8.1 Tiedown Rings and Floor Pans Inspections.

- Turn off all electrical power (PARA 1-6-16).

NOTE

Corrosion is indicated by pitting.

- Visually inspect cargo tiedown rings and floor pans for corrosion pitting. **PITTING SHALL NOT COVER MORE THAN 25% OF WELD AREA.** If pitting is beyond limit, remove corrosion (PARA 2-6-27).

NOTE

Corrosion is indicated by red film or pitting.

- Visually inspect floor attachment screws, or bolts if installed, for corrosion. **NONE ALLOWED.** If corrosion is evident, perform the following:
 - Replace screws or bolts (PARA 2-4-71) or remove corrosion (PARA 2-6-27).
 - Cut slot in floor shims to drain water away from screws (PARA 2-6-27).
- Visually inspect fasteners, skin under stringers and around bulkheads, and floor attachment fittings for corrosion. **NONE ALLOWED.** If corrosion is present, repair cabin floor (PARA 2-6-27).
- Inspect drain holes for obstructions. **NONE ALLOWED.** If obstructed, remove obstruction from drain hole.

2-3-9 WIRE STRIKE PROTECTION SYSTEM INSPECTIONS.

PARAGRAPH BREAKDOWN

		PAGE
2-3-9.1	Wire Strike Protection System Inspection	2-3-9-1

INITIAL SETUP

- PARA 2-4-85

References

- PARA 2-4-79
 - PARA 2-4-80
 - PARA 2-4-81
 - PARA 2-4-82
 - PARA 2-4-83
 - PARA 2-4-84
- PARA 2-4-86
 - PARA 2-4-87
 - PARA 2-4-88
 - PARA 2-4-89
 - PARA 2-4-90
 - PARA 2-4-91
 - PARA 2-4-92

2-3-9.1 Wire Strike Protection System Inspection.

- a. Inspect indicated components for defects (Figure 2-3-9-1, Sheet 1, Detail A, Detail B, Detail C, Detail D, and Detail E and Sheet 2, Detail F, Detail G, and Detail H).
- b. Visually check the following components for evidence of wire strike (Figure 2-3-9-2):

(1) Upper Cutter. If wire strike is evident, replace wire strike upper cutter (PARA 2-4-79).

(2) Pitot Cutter. If wire strike is evident, replace wire strike pitot cutter (PARA 2-4-80).

(3) Pitot Cutter Guide Channel. If wire strike is evident, replace wire strike pitot cutter guide channel (PARA 2-4-81).

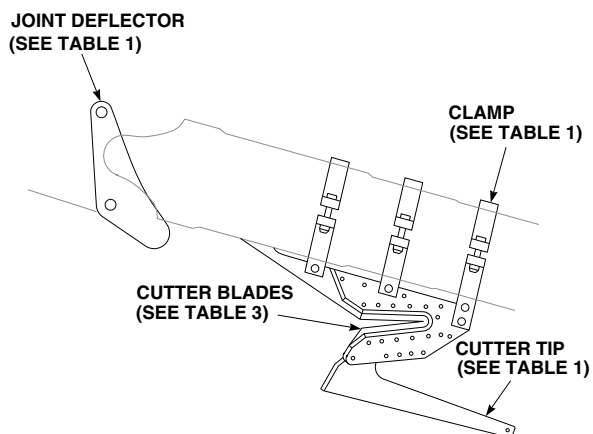
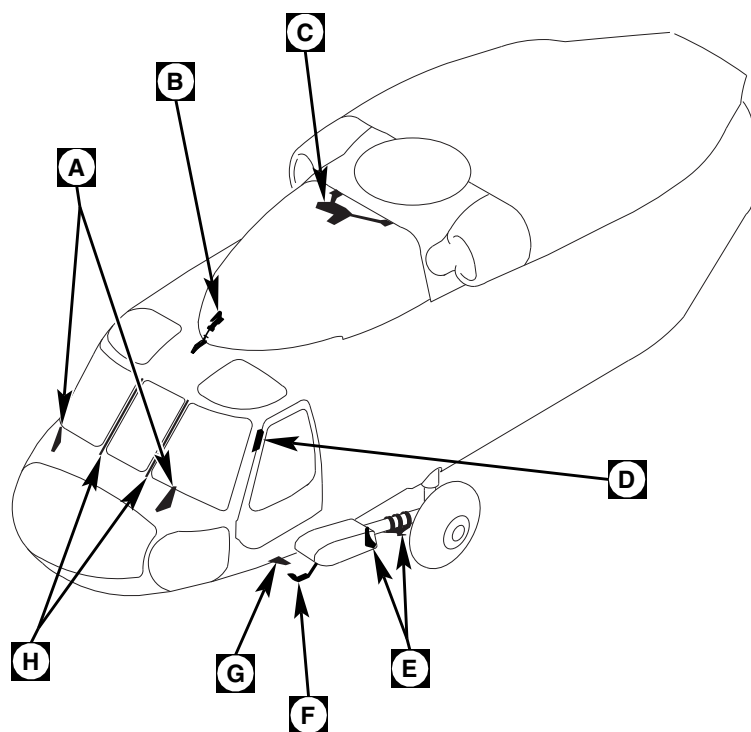
- (4) Windshield Wiper Drive Post Deflector. If wire strike is evident, replace wire strike windshield wiper drive post deflector (PARA 2-4-82).

(5) Main Landing Gear Cutter. If wire strike is evident, replace wire strike main landing gear cutter (PARA 2-4-83).

(6) Main Landing Gear Cutter Clamp. If wire strike is evident, replace wire strike main landing gear cutter clamp (PARA 2-4-84).

(7) Main Landing Gear Joint Deflector. If wire strike is evident, replace wire strike main landing gear joint deflector (PARA 2-4-85).

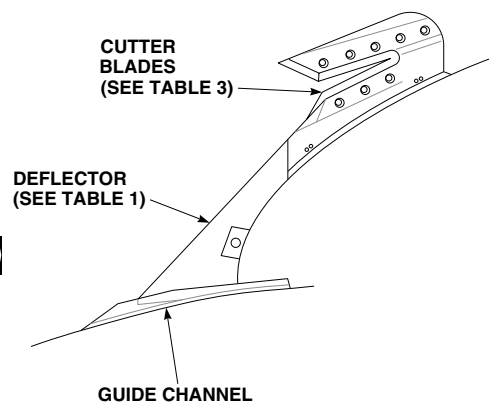
(8) Tail Landing Gear Deflector. If wire strike is evident, replace wire strike tail landing gear deflector (PARA 2-4-86).



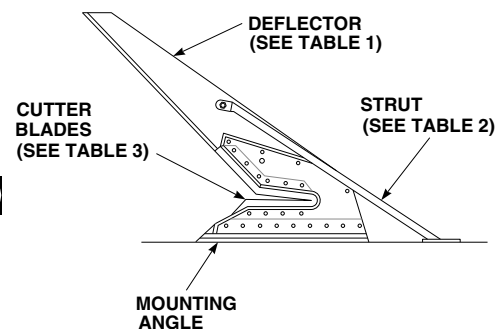
**MAIN LANDING GEAR CUTTER
AND CLAMPS/MAIN LANDING GEAR
JOINT DEFLECTOR**



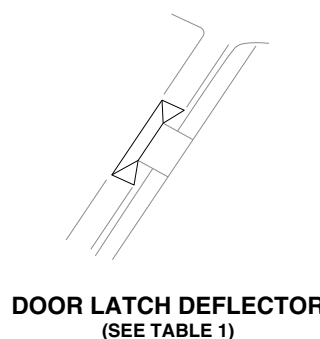
**WINDSHIELD WIPER
POST DEFLECTOR
(SEE TABLE 1)**



**PITOT CUTTER AND
GUIDE CHANNEL**



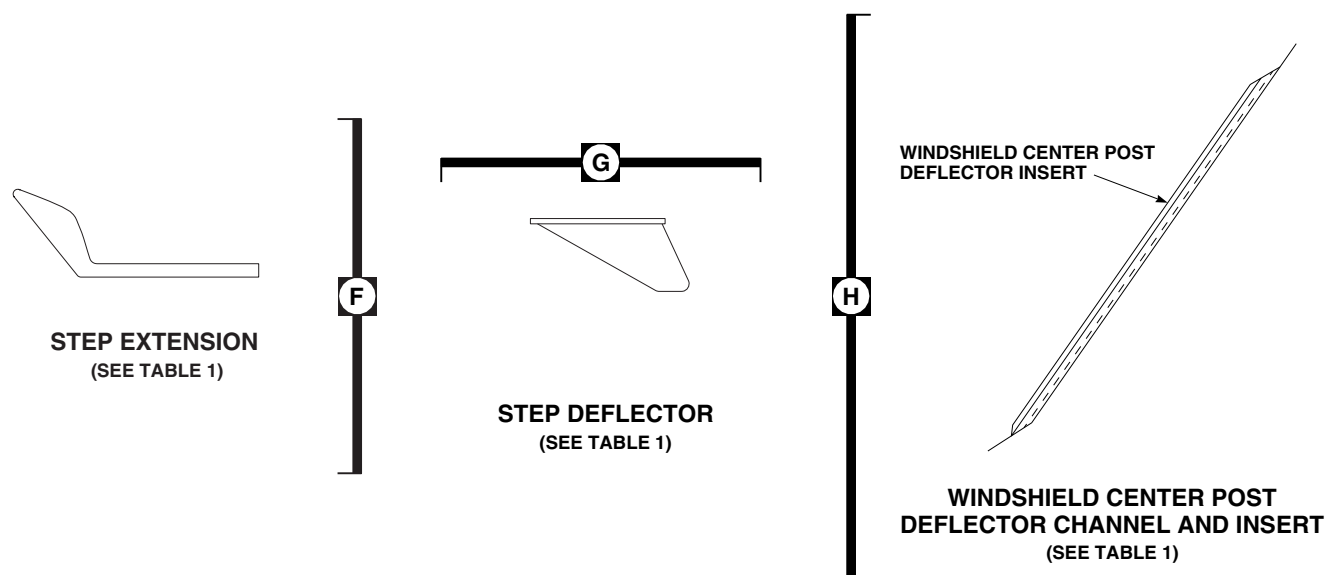
UPPER CUTTER



**DOOR LATCH DEFLECTOR
(SEE TABLE 1)**

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FIGURE 2-3-9-1. INSPECT WIRE STRIKE PROTECTION SYSTEM FOR DEFECTS (SHEET 1 OF 3)



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FIGURE 2-3-9-1. INSPECT WIRE STRIKE PROTECTION SYSTEM FOR DEFECTS (SHEET 2 OF 3)

- | | |
|---|--|
| <p>(9) Tail Landing Gear Deflector Clamp. If wire strike is evident, replace wire strike tail landing gear deflector clamp (PARA 2-4-87).</p> | <p>(12) Door Latch Deflector. If wire strike is evident, replace wire strike door latch deflector (PARA 2-4-90).</p> |
| <p>(10) Step Deflector. If wire strike is evident, replace wire strike step deflector (PARA 2-4-88).</p> | <p>(13) Windshield Center Post Deflector Insert. If wire strike is evident, replace wire strike windshield center post deflector insert (PARA 2-4-91).</p> |
| <p>(11) Step Extension. If wire strike is evident, replace wire strike step extension (PARA 2-4-89).</p> | <p>(14) Windshield Center Post Deflector Channel. If wire strike is evident, replace wire strike windshield center post deflector channel (PARA 2-4-92).</p> |

TABLE 1

DEFECT	ACTION
SCRATCHES, NICKS, GOUGES TO A MAXIMUM OF 0.010 IN.	DRESS OUT AFFECTED AREA USING ABRASIVE CLOTH AND ALUMINUM WOOL. BRUSH ALODINE, PRIME AND PAINT.
PAINT DETERIORATION.	CLEAN AFFECTED AREA USING ALUMINUM WOOL. BRUSH ALODINE, PRIME AND PAINT.
LIGHT CORROSION TO A MAXIMUM OF 0.010 IN.	DRESS OUT AFFECTED AREA USING ABRASIVE CLOTH AND ALUMINUM WOOL. BRUSH ALODINE, PRIME AND PAINT.
MODERATE TO HEAVY CORROSION / WHITE POWDER / MATERIAL BULGING.	REPLACE.
CORRODED / MISSING / INCORRECT / BROKEN HARDWARE.	REPLACE HARDWARE.

TABLE 2

DEFECT	ACTION
SCRATCHES, NICKS, GOUGES TO A MAXIMUM OF 0.005 IN.	DRESS OUT AFFECTED AREA USING ABRASIVE CLOTH AND STEEL WOOL. PRIME AND PAINT.
PAINT DETERIORATION.	CLEAN AFFECTED AREA USING STEEL WOOL. PRIME AND PAINT.
LIGHT CORROSION TO A MAXIMUM OF 0.005 IN.	DRESS OUT AFFECTED AREA USING ABRASIVE CLOTH AND STEEL WOOL. PRIME AND PAINT.
MODERATE TO HEAVY CORROSION / PITTING.	REPLACE.
CORRODED / MISSING / INCORRECT / BROKEN HARDWARE.	REPLACE HARDWARE.

TABLE 3

DEFECT	ACTION
MISSING, PEELING RUBBER.	REMOVE LOOSE RUBBER AND REPLACE WITH SEALANT.
ANY SCRATCHES, NICKS, GOUGES, CHIPPING, ABRASION.	REPLACE.
ANY CORROSION.	REPLACE.
CORRODED / MISSING / INCORRECT / BROKEN HARDWARE.	REPLACE HARDWARE.

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FIGURE 2-3-9-1. INSPECT WIRE STRIKE PROTECTION SYSTEM FOR DEFECTS (SHEET 3 OF 3)

2-3-9-4 Change 4

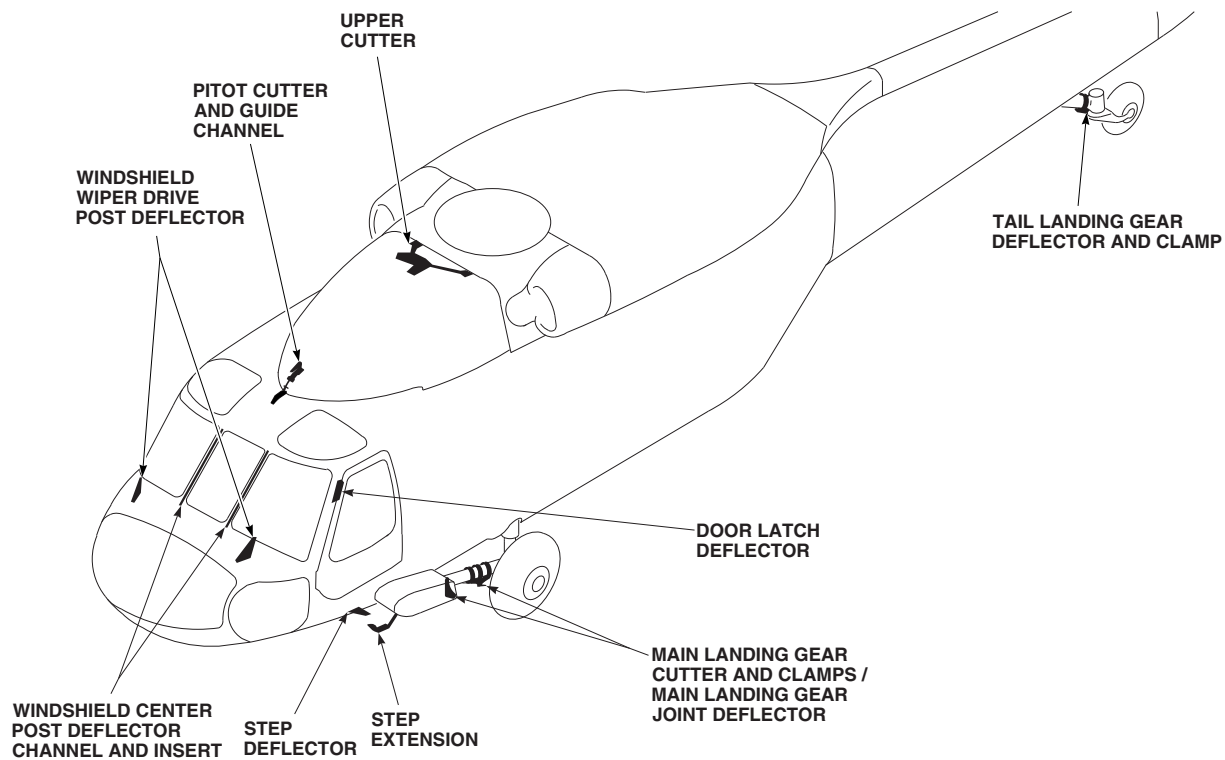


FIGURE 2-3-9-2. INSPECT WIRE STRIKE PROTECTION SYSTEM FOR EVIDENCE OF WIRE STRIKE

2-3-10 MAIN ROTOR PYLON INSPECTIONS.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
2-3-10.1	Main Rotor Pylon Sliding Cover Inspection	2-3-10-1	2-3-10.3	Oil Cooler Compartment Access Doors Inspection	2-3-10-2
2-3-10.2	APU Access Doors Inspection	2-3-10-2			

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Caliper, Vernier
- Caliper Set, Micrometer, Outside

References

- PARA 2-4-101
- PARA 2-4-102

- PARA 2-4-103
- PARA 2-4-104
- PARA 2-4-108
- PARA 2-4-113
- PARA 2-6-1

2-3-10.1 Main Rotor Pylon Sliding Cover Inspection.

NOTE

Corrosion is indicated by red film or pitting.

- Visually inspect main rotor pylon sliding cover latch for corrosion. **NONE ALLOWED.** If corrosion is found, remove corrosion (PARA 2-4-102).
- Raise latch, loosen jamnut, and inspect fit of latch hook and jamnut on threaded stud as follows:
 - LATCH HOOK OR JAMNUT SHALL NOT BE LOOSE OR WOBBLE ON**

THREADED STUD. If latch hook or jamnut is loose or wobbles on threaded stud, replace latch (PARA 2-4-102).

- If fit of latch hook or jamnut is secure, **TIGHTEN JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
- Open main rotor pylon sliding cover.
- Visually inspect roller assemblies for disbonding of nylon rollers from bushings. **NONE ALLOWED.** If disbonded, replace rollers (PARA 2-4-103 or PARA 2-4-104).
- Inspect pin receivers and support structure for cracks. **NONE ALLOWED.** If cracked, repair cracks (PARA 2-6-1).

- f. Inspect pin receiver holes for wear using vernier caliper and outside micrometer caliper from caliper set. **WEAR SHALL NOT EXCEED 0.60-INCH.** If limit is exceeded, repair receiver holes (PARA 2-4-101).
- g. **701029 – SUBQ** Inspect forward locating pin support for wear or damage to nylon edging grommet. 
- h. Close main rotor pylon sliding cover.

2-3-10.2 APU Access Doors Inspection.

- a. With access door closed and catches locked, try to move door horizontally. **MOVEMENT AT HINGE LINE OF 0.060-INCH OR MORE IS NOT ALLOWED.** If limit is exceeded, replace hinges (PARA 2-4-108).
- b. Inspect flush catches for security and damage. **CATCH SHALL NOT OPEN WITH MINOR DOOR MOVEMENT. NO BROKEN CATCH SPRING ALLOWED.** If catch opens with minor door movement or catch spring is broken, replace flush catch (PARA 2-4-108).
- c. Inspect door hinges for cracks and wear. **NO CRACKS ALLOWED. WEAR SHALL NOT EXCEED 10% OF MATERIAL THICKNESS.** If limit is exceeded or door hinge is cracked, replace hinge (PARA 2-4-108).
- d. Inspect hinges, catches, and catch fittings for corrosion. **CORROSION REMOVAL SHALL**

NOT EXCEED 10% OF MATERIAL THICKNESS. If limit is exceeded, replace hinges, catches, and catch fittings (PARA 2-4-108).

2-3-10.3 Oil Cooler Compartment Access Doors Inspection.

- a. With access door closed and catches locked, try to move door horizontally. **MOVEMENT AT HINGE LINE OF 0.060-INCH OR MORE IS NOT ALLOWED.** If limit is exceeded, replace hinges (PARA 2-4-113).
- b. Inspect flush catches for security and damage. **CATCH SHALL NOT OPEN WITH MINOR DOOR MOVEMENT. NO BROKEN CATCH SPRING ALLOWED.** If catch opens with minor door movement or catch spring is broken, replace flush catch (PARA 2-4-113).
- c. Inspect door hinges for cracks and wear. **NO CRACKS ALLOWED. WEAR SHALL NOT EXCEED 10% OF MATERIAL THICKNESS.** If limit is exceeded or door hinge is cracked, replace hinge (PARA 2-4-113).
- d. Inspect hinges, catches, and catch fittings for corrosion. **CORROSION REMOVAL SHALL NOT EXCEED 10% OF MATERIAL THICKNESS.** If limit is exceeded, replace hinges, catches, and catch fittings (PARA 2-4-113).

2-3-11 ENGINE COWLINGS INSPECTION.

PARAGRAPH BREAKDOWN

	PAGE
2-3-11.1 Engine Cowling Latches Inspection	2-3-11-1

INITIAL SETUP

- PARA 2-4-122

Tools

- Aircraft Mechanic’s Toolkit
- Magnifying Glass, 10X

- PARA 2-6-2

- PARA 2-6-29

References

- PARA 2-4-118
- PARA 2-4-120

2-3-11.1 Engine Cowling Latches Inspection.

- Open engine cowling.
- Lift latch handle away from surface of cowling.
- Inspect spring and spacer for proper movement and security (Figure 2-3-11-1, Detail B).
- Inspect flush head rivet forming the center pivot for properly bucked end. **BUCKED END OF RIVET MUST HAVE A MINIMUM DIAMETER OF 0.170-INCH. FLUSH HEADS OF RIVET MUST FILL 75% OF COUNTERSUNK WASHER.** If rivet heads are not bucked properly, replace rivets (PARA 2-4-120).
- Inspect lockpin tubes for kinks or distortion. **NONE ALLOWED.** If kinked or distorted, replace lockpin tubes (PARA 2-4-120).
- Inspect lockpin and lockpin fittings for cracks and galling. **NO CRACKS ALLOWED.**

GALLING SHALL BE NO DEEPER THAN 0.020-INCH. If limit is exceeded or cracks are found, replace lockpin and/or lockpin fittings (PARA 2-4-120).

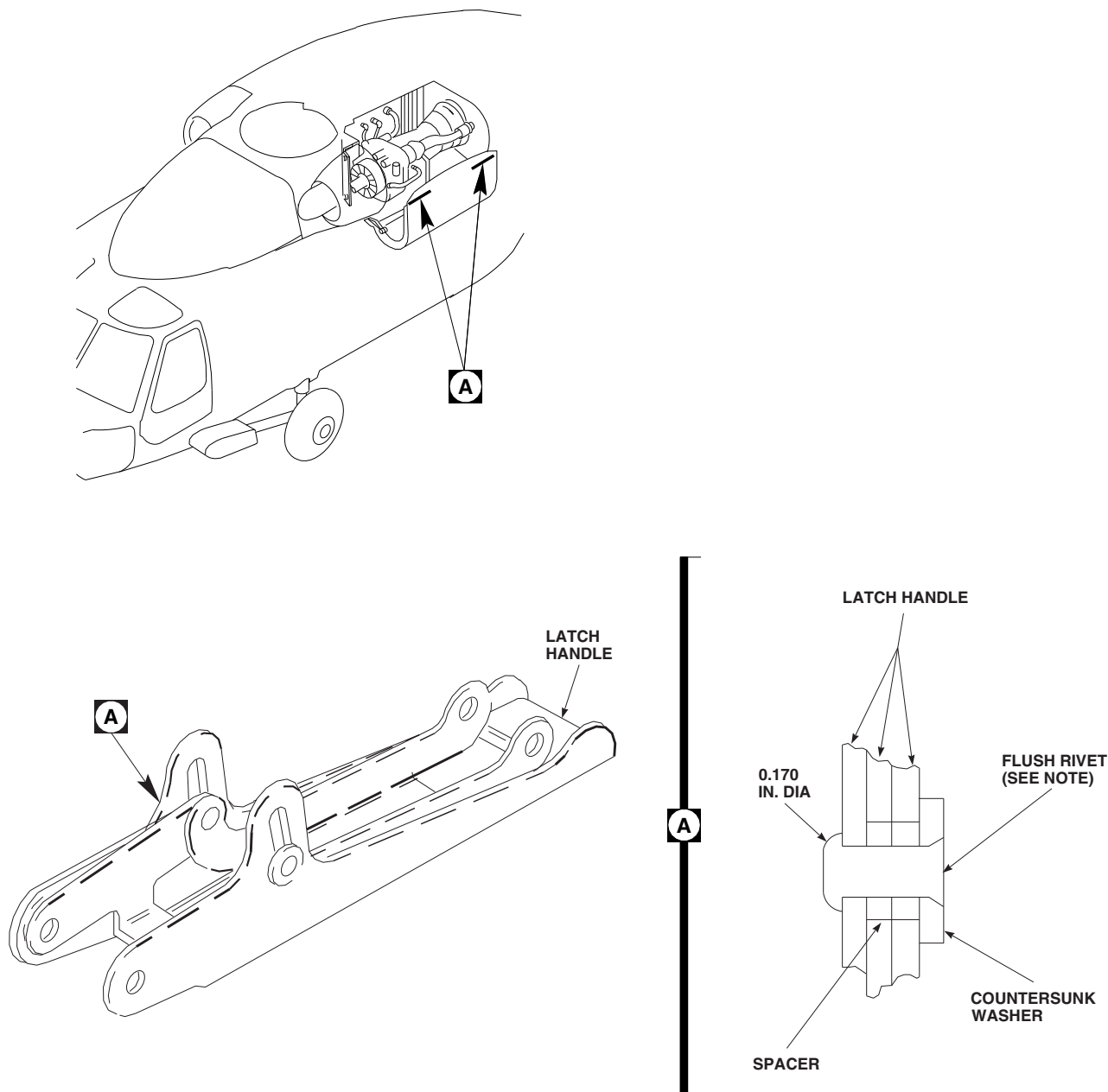
- Inspect latch handle release for ease of movement. **LATCH HANDLE RELEASE MUST MOVE FREELY.** If latch handle release does not move freely, replace latch handle release (PARA 2-4-120).

NOTE

Corrosion is indicated by red film or pitting.

- Visually inspect latch handle assembly for corrosion. **PITS SHALL NOT EXCEED 0.015-INCH DEEP.** If limit is exceeded, remove corrosion (PARA 2-4-120).
- Inspect latch handle and handle support for wear at boltholes and slotted holes. **WEAR SHALL NOT EXCEED 10% OF MATERIAL THICKNESS.** If limit is exceeded, re-

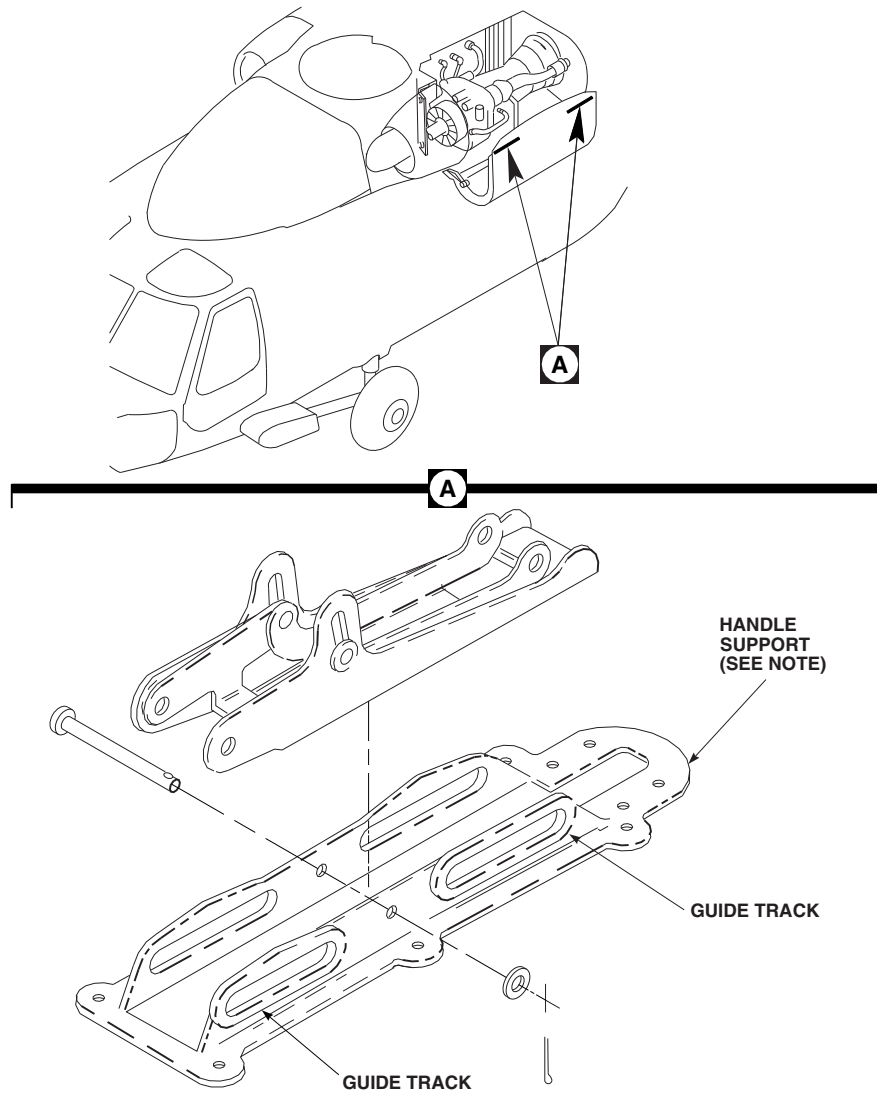
- place latch handle and/or handle support (PARA 2-4-120).
- j. Inspect handle support for cracks across bolt-hole lugs (Figure 2-3-11-2, Detail A). **CRACKS SHALL NOT EXTEND ACROSS BOLTHOLE LUG.** If limit is exceeded, replace handle support (PARA 2-4-120).
- k. Inspect chrome-plated handle guide track. **WEAR UP TO 0.025-INCH IS ALLOWED IF OPERATION IS NOT IMPAIRED.**
- l. Inspect cowling hinges on airframe and adjacent structure for cracks, wear, and security (PARA 2-6-2).
- m. Inspect latch, hinges, and latch fittings for nicks, scratches, and corrosion. **NICKS AND SCRATCHES SHALL NOT EXCEED 10% OF MATERIAL THICKNESS. CORROSION REMOVAL SHALL NOT EXCEED 10% OF MATERIAL THICKNESS.** If limits are exceeded, replace latch, hinges, and latch fittings as required (PARA 2-4-120).
- n. Using 10X magnifying glass, inspect fiberglass top ply for damage at hinge rib location on both ends of engine cowling (Figure 2-3-11-3, Detail A). Proceed as follows:
 - (1) On engine cowling assemblies, 70302-02141-041, -042, -044, -045, -048, and -049, **NO DAMAGE ALLOWED.** If cracks are found through either of the first two fiberglass layers or there is evidence of exposed fiber, replace fittings (PARA 2-4-122), and repair cowling (PARA 2-6-29). If damage extends into third layer, replace cowling (PARA 2-4-118).
 - (2) On engine cowling assemblies, 70070-30308-041 through -046, and 70302-02141-052 and -053, **DAMAGE IS ALLOWED THROUGH FIRST TWO FIBERGLASS LAYERS. NO DAMAGE ALLOWED TO THIRD LAYER.** If damage does not extend into third layer, repair cowling (PARA 2-6-29). If damage extends into third layer, replace cowling (PARA 2-4-118).
- o. Close engine cowling.



NOTE
 FLUSH HEAD OF RIVET MUST FILL 75%
 OF COUNTERSUNK WASHER.

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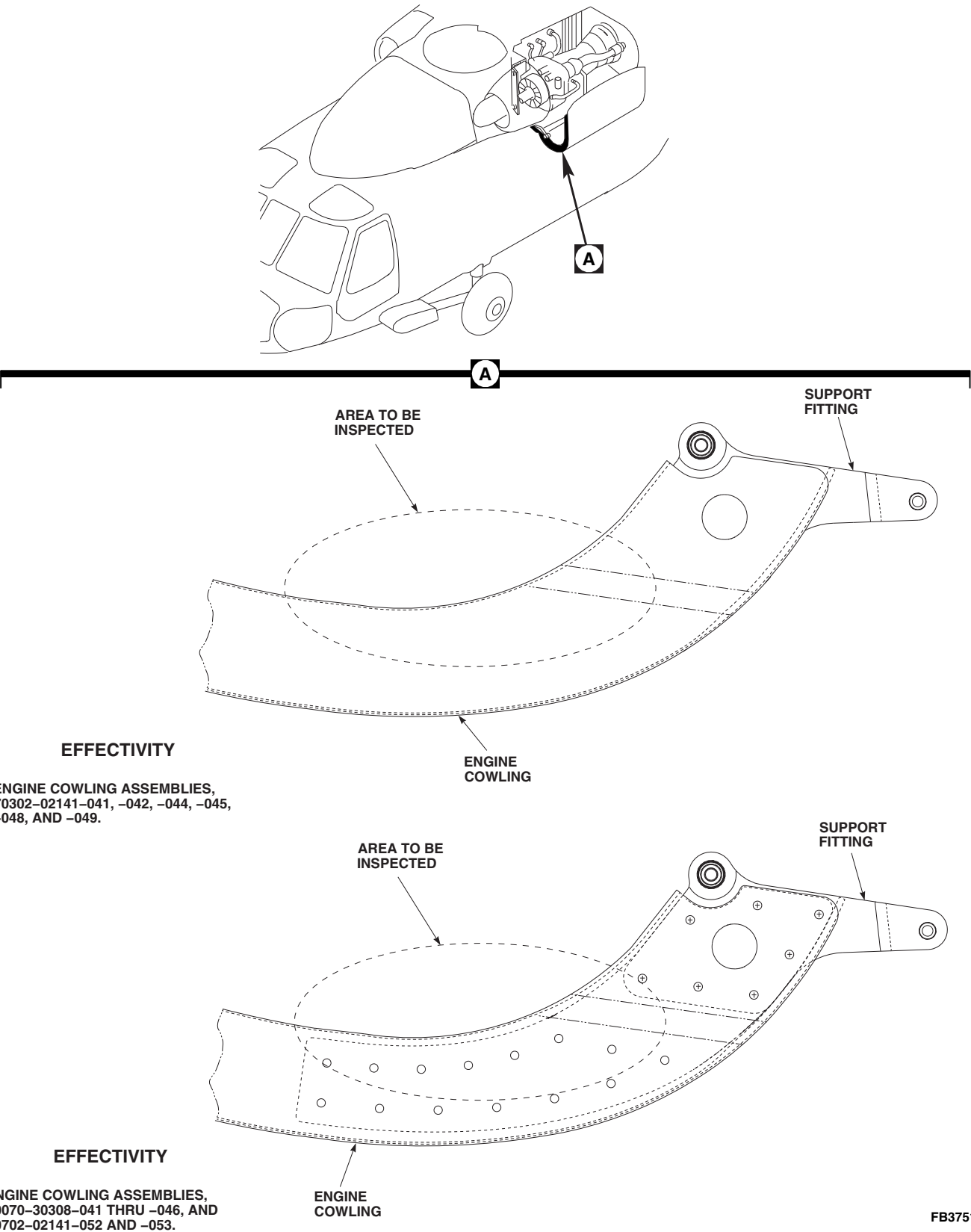
FIGURE 2-3-11-1. INSPECT ENGINE COWLING LATCHES

**NOTE**

LATCH HANDLE SHOWN REMOVED
FOR CLARITY.

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FIGURE 2-3-11-2. INSPECT HANDLE GUIDE TRACKS



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2-3-12 TRANSITION SECTION INSPECTIONS.

PARAGRAPH BREAKDOWN

	PAGE
2-3-12.1 Torque Check Transition Section-to-Tail Cone Bolts	2-3-12-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Feeler Gage
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Washer, MS20002-5

References

- PARA 2-4-45
- PARA 2-4-69
- PARA 2-6-5
- PARA 13-4-21
- PARA 13-4-22
- PARA 13-5-3
- PARA 13-5-6

2-3-12.1 Torque Check Transition Section-to-Tail Cone Bolts.

- Remove rear rear passenger’s seats as required (PARA 2-4-45).
- Remove soundproofing panel from rear bulkhead covering left or right stowage compartment (PARA 2-4-69).
- Remove cargo netting from stowage compartment.
- On helicopters with air conditioning installed, remove the following equipment:
 - Environmental control system plenum (PARA 13-4-21).

- Environmental control system evaporator pallet (PARA 13-5-6), or condenser pallet (PARA 13-5-3).
- Environmental control system supply ducting (PARA 13-4-22).

- Check torque on nuts on transition section-to-tail cone bolts in the sequence shown in Figure 2-3-12-1, Detail A. **TORQUE MUST NOT BE LESS THAN 190 INCH-POUNDS** (Figure 2-3-12-1, Detail B). **IF TORQUE IS 190 - 210 INCH-POUNDS, PROCEED TO STEP F. IF TORQUE IS BELOW 190 INCH-POUNDS, TORQUE NUTS TO 190 - 210 INCH-POUNDS, PERFORM THE FOLLOWING:**
 - REPEAT TORQUE CHECK EVERY 30 FLIGHT HOURS UNTIL TORQUE**

STABILIZES AT 190 INCH-POUNDS MINIMUM.**NOTE**

Nuts may be on front side of stackup for easy access.

- (2) Make sure helicopter logbook is updated to reflect torque check is required.
- f. After torque check, check for proper clamp-up as follows:
 - (1) Apply 80 inch-pounds of torque to nut in a tightening direction without holding bolt.
 - (2) If nut turns, using feeler gage, check for gaps under bolthead and nut.
 - (3) If gap exists, remove nut and add one washer, MS20002-5, under nut.
 - (4) Install nut. **TORQUE NUT TO 190 - 210 INCH-POUNDS.**
- g. Check for proper thread protrusion beyond nut. **BOLT SHALL EXTEND AT LEAST 0.060-INCH BEYOND NUT.**
- h. If torque has stabilized, apply torque stripe (PARA 2-6-5).
- i. On helicopters with air conditioning installed, install the following ECS equipment:
 - (1) Environmental control system plenum (PARA 13-4-21).
 - (2) Environmental control system evaporator pallet (PARA 13-5-6), or condenser pallet (PARA 13-5-3).
 - (3) Environmental control system supply ducting (PARA 13-4-22).
- j. If torque does not stabilize after three consecutive torque checks, perform the following:
 - (1) Remove bolt, countersunk washer, washer(s), and nut.

- (2) Install new bolt, countersunk washer, washer, and nut on forward side of stackup.

- (3) After torque check, check for proper clamp-up as follows:

- (a) Apply 80 inch-pounds of torque to nut in a tightening direction without holding bolt.

- (b) If nut turns, using feeler gage, check for gaps under bolthead and nut.

- (c) If gap exists, remove nut and add one washer, MS20002-5, under nut.

- (d) Install nut. **TORQUE NUT TO 190 - 210 INCH-POUNDS.**

- (4) Check for proper thread protrusion beyond nut. **BOLT SHALL EXTEND AT LEAST 0.060-INCH BEYOND NUT.**

- (5) Repeat torque check every 30 flight hours until stabilized.

- (6) Make sure helicopter logbook is updated to reflect torque check is required.

- k. Make sure area is clean and free of foreign material.

- l. Install cargo netting.

- m. Install soundproofing panel (PARA 2-4-69).

- n. Install rear rear passenger's seats (PARA 2-4-45).

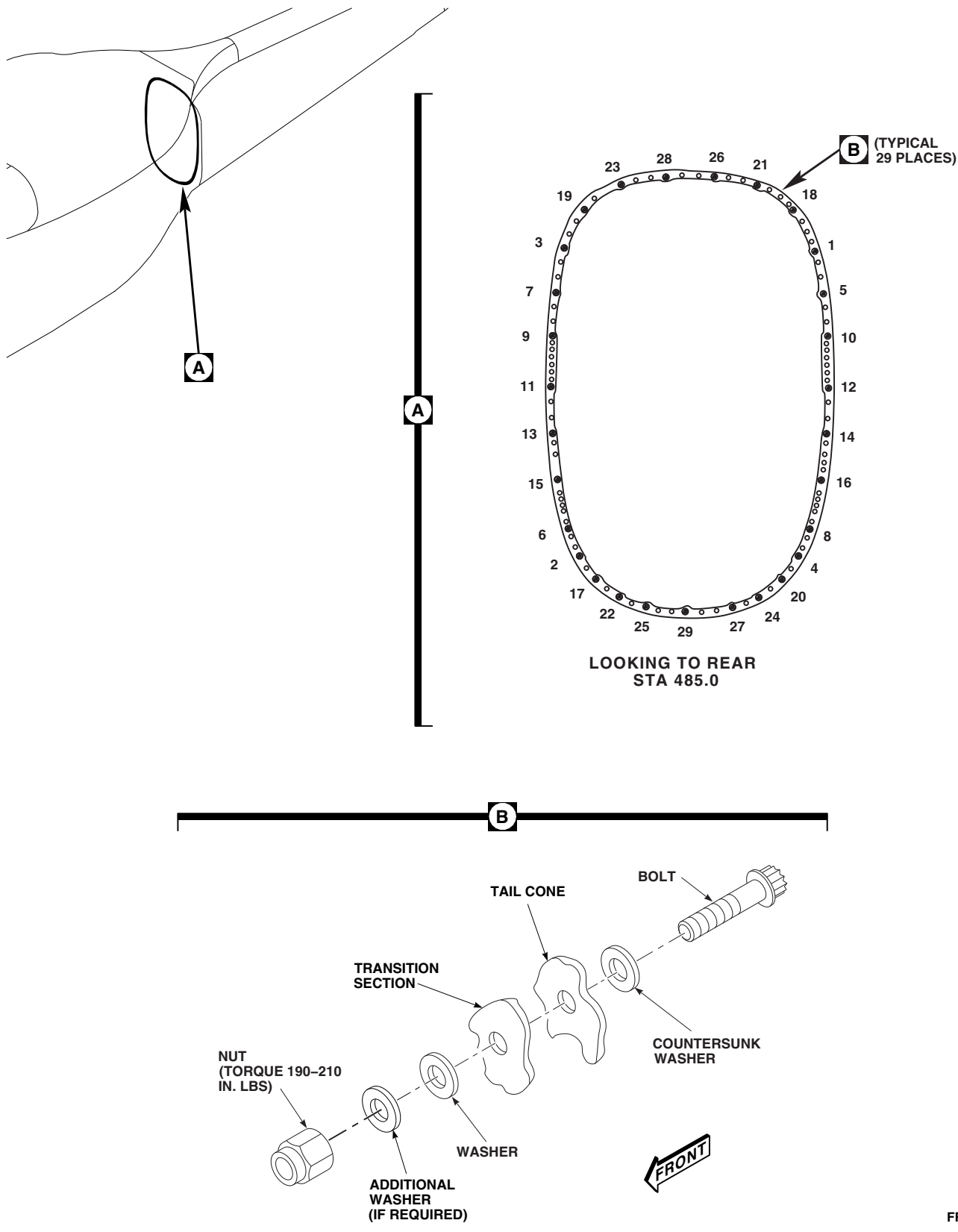


FIGURE 2-3-12-1. TORQUE CHECK TRANSITION SECTION-TO-TAIL CONE BOLTS

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2-3-13 TAIL ROTOR DRIVE SHAFT COVERS INSPECTION.

PARAGRAPH BREAKDOWN

	PAGE
2-3-13.1 Turnlock Fastener and Hinge Inspection	2-3-13-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

References

- PARA 2-4-131

2-3-13.1 Turnlock Fastener and Hinge Inspection.

NOTE

Corrosion is indicated by red film or pitting.

- Visually inspect turnlock fasteners for proper operation and corrosion. **NO INOPERATIVE**

OR CORRODED FASTENERS ALLOWED.

If inoperative or corroded fasteners are found, replace turnlock fasteners.

- Visually inspect pin-type hinges for corrosion. **NONE ALLOWED.** If corrosion is found, remove corrosion (PARA 2-4-131).

2-3-14 TAIL ROTOR PYLON INSPECTIONS.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
2-3-14.1	Tail Gear Box Fitting Inspection	2-3-14-1	2-3-14.2	Tail Rotor Pylon Lower Step Inspection	2-3-14-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Flashlight
- Fluorescent Penetrant Inspection Kit
- Inspection Mirror

References

- ASTM E1417
- PARA 1-6-9
- PARA 1-6-16
- PARA 2-4-141
- PARA 2-6-44

2-3-14.1 Tail Gear Box Fitting Inspection.

- Using fluorescent penetrant inspection kit, inspect tail gear box fitting for surface cracks around lower barrel-nut holes (ASTM E1417). **NONE ALLOWED.** If cracked, repair or replace tail gear box fitting (PARA 2-6-44).
- Inspect skin, especially in area surrounding tail gear box fitting, for cracks. **NONE ALLOWED.** If skin cracks are found, inspect inside surfaces of fitting where secured to front spar cap angles for cracks, using inspection mirror and flashlight through access hole in front spar web. **NONE ALLOWED.** If cracked, one-time flight to depot is allowed.

2-3-14.2 Tail Rotor Pylon Lower Step Inspection.

- Turn off all electrical power (PARA 1-6-16).
- Fold tail rotor pylon 90° (PARA 1-6-9).
- Turn step towards right side of helicopter and lower step from pylon.

- Inspect step for proper operation. If step does not operate properly, perform the following:
 - Inspect spring for security as follows:
 - NO DISCONNECTED SPRING ALLOWED.** If disconnected, connect spring.
 - NO BROKEN SPRING ALLOWED.** If broken, replace spring (PARA 2-4-141).
 - Inspect pin for looseness. **NONE ALLOWED.** If loose, replace tail rotor pylon lower step (PARA 2-4-141).
 - On step assembly, 70207-06052-042 and -043, inspect cotter pin and spacer for presence and security. **NO MISSING COTTER PIN ALLOWED.** If missing, install cotter pin (PARA 2-4-141).
- Using fluorescent penetrant inspection kit, inspect step assembly for cracks (ASTM

E1417). **NONE ALLOWED.** If cracked, replace tail rotor pylon lower step (PARA 2-4-141).

ing is worn to base metal, replace tail rotor pylon lower step (PARA 2-4-141).

- f. Inspect tube for wear at lower support point. **ANODIZED COATING SHALL NOT BE WORN TO BASE METAL.** If anodized coat-

g. Raise and secure step.

h. Unfold tail rotor pylon (PARA 1-6-9).

2-3-15 STABILATOR INSPECTIONS.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
2-3-15.1	Tail Rotor Pylon Stabilator Actuator Attachment Fitting Bearing Inspection	2-3-15-2	2-3-15.9	Right Stabilator-to-Pylon Attachment Fitting Inspection	2-3-15-6
2-3-15.2	Stabilator Spar Inspection	2-3-15-2	2-3-15.10	Right Attachment Bolt Inspection/Repair	2-3-15-8
2-3-15.3	Stabilator Electrostatic Discharger Inspection	2-3-15-2	2-3-15.11	Stabilator Elastomeric Bearing Inspection	2-3-15-8
2-3-15.4	Stabilator Free Play Inspection	2-3-15-3	2-3-15.12	Stabilator Actuator Attachment Fittings Inspection	2-3-15-8
2-3-15.5	Stabilator Actuator Shaft Play Inspection	2-3-15-3	2-3-15.13	Stabilator Actuator Grounding Strap Inspection 701334-SUBQ	2-3-15-9
2-3-15.6	Stabilator Actuator Upper Attachment Inspection	2-3-15-3			
2-3-15.7	Stabilator Actuator Lower Attachment Inspection	2-3-15-5			
2-3-15.8	Left Stabilator-to-Pylon Attachment Fitting Inspection	2-3-15-6			

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Feeler Gage
- Feeler Gage, 0.0015-inch
- Feeler Gage, 0.015-inch
- Flashlight
- Inside Micrometer
- Inspection Mirror

- Micrometer
- Protective Covering
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Corrosion Preventive Compound, Item 114, Appendix D
- Scouring Pad, Item 270, Appendix D

References

- Appendix D
- PARA 2-4-132

- PARA 2-4-133
- PARA 2-4-135
- PARA 2-4-138
- PARA 2-4-142
- PARA 2-4-143
- PARA 2-4-144
- PARA 2-4-146
- PARA 2-4-147
- PARA 2-4-149
- PARA 2-5-20
- PARA 2-6-5
- PARA 2-6-46
- PARA 2-6-47
- PARA 2-6-48
- PARA 2-6-52
- PARA 2-6-53
- PARA 2-6-55

2-3-15.1 Tail Rotor Pylon Stabilator Actuator Attachment Fitting Bearing Inspection.

- a. Check spherical bearing for radial play. **RADIAL PLAY SHALL NOT EXCEED 0.010-INCH.** If limit is exceeded, replace fitting (PARA 2-6-48).

WARNING

- Bonding of liners in center lobe of fitting is a critical characteristic. Proper bonding must be verified.
- Damage to fitting will occur if feeler gage is improperly used. Liner may become disbonded. Use care when using feeler gage to check for liner edge separation.
- b. Visually inspect bonding of liners in center lobe of fitting. Using 0.0015-inch feeler gage, probe for edge separation. **VOIDS MUST NOT EXCEED 5% OF JOINT LINE LENGTH.** If limit is exceeded, replace fitting (PARA 2-6-48).

2-3-15.2 Stabilator Spar Inspection.

- a. Inspect top stabilator surface at front spar, both left and right sides, for skin cracks and/or loose

rivets. If skin cracks are found, repair cracks (PARA 2-6-52 or PARA 2-6-53). If loose rivets are found, replace stabilator (PARA 2-4-142).

- b. **70785-SUBQ** Remove screws and washers securing leading edge inspection plate cover to stabilator. Remove cover. 
- c. Using flashlight and inspection mirror, check top internal spar cap and web for cracks and/or loose rivets. **NONE ALLOWED.** If cracks and/or loose rivets are found, replace stabilator (PARA 2-4-142).
- d. **70785-SUBQ** Install leading edge inspection plate cover on stabilator and secure with screws and washers. 

2-3-15.3 Stabilator Electrostatic Discharger Inspection.

- a. Inspect electrostatic discharger for cracks in rubber coating. If rubber coating is cracked, check for broken or frayed wires in cracked area. **NONE ALLOWED.** Proceed as follows:
 - (1) If broken or frayed wires are found, replace electrostatic discharger (PARA 2-4-143).

- (2) If no broken or frayed wires are found, repair electrostatic discharger rubber coating (PARA 2-4-143).

2-3-15.4 Stabilator Free Play Inspection.

- a. With stabilator at 0°, check stabilator up and down free play at left and right tip caps. **TOTAL VERTICAL MOVEMENT AT EITHER SIDE SHALL BE 5/16-INCH MAXIMUM.**
- b. With stabilator at 0°, check up and down free play at stabilator trailing edge center directly behind stabilator actuator. **TOTAL VERTICAL MOVEMENT SHALL BE 1/4-INCH MAXIMUM** (Figure 2-3-15-1, Sheet 1, Detail A).
- c. If free movement is greater than specified, perform the following:
 - (1) Check endplay and sideplay in stabilator actuator.
 - (2) Check for loose actuator fittings or loose stabilator attachment fittings.
 - (3) Check for loose or worn pivot bolts, bearings, and bushings.
 - (4) Check stabilator elastomeric bearings.

2-3-15.5 Stabilator Actuator Shaft Play Inspection.

- a. Using feeler gage, measure endplay for axial motion of clevis relative to stabilator actuator housing (Figure 2-3-15-1, Sheet 1, Detail B). **ENDPLAY OF EITHER CLEVIS, UNDER REVERSING LOAD OF 20 POUNDS, SHALL NOT EXCEED 0.025-INCH.** If limit is exceeded, replace stabilator actuator (PARA 2-4-146).
- b. Using feeler gage, measure sideplay for radial motion of dual actuator housing relative to fixed clevis ends. **TOTAL SIDEPLAY, UNDER REVERSING LOAD OF 10 POUNDS, SHALL NOT EXCEED 0.025-INCH.** If limit is exceeded, replace stabilator actuator (PARA 2-4-146).

- c. If endplay is within limits, inspect stabilator actuator upper attachment (PARA 2-3-15.6), and stabilator actuator lower attachment (PARA 2-3-15.7).

2-3-15.6 Stabilator Actuator Upper Attachment Inspection.

- a. Open tail rotor pylon leading edge cover/antenna. If cover/antenna interferes with pylon actuator attachment fitting inspection, remove tail rotor pylon leading edge cover/antenna (PARA 2-4-138).
- b. Visually inspect pylon actuator attachment fitting for looseness and wear. Proceed as follows:
 - (1) If pylon actuator attachment fitting is loose, **TORQUE NUTS TO 43 - 47 INCH-POUNDS.**
 - (2) If pylon actuator attachment fitting shows wear, refer to PARA 2-6-48 for repair limits and procedures.
- c. Remove bolt from actuator pylon fitting (PARA 2-4-142).

WARNING

FSCAP

Bolthead shoulder and bolt shank of stabilator actuator attachment bolts are critical surfaces. Dimension limits must not be exceeded.

- d. Using feeler gage, measure clearance between bolt and bushings in tail rotor pylon stabilator actuator attachment fitting. **CLEARANCE SHALL NOT EXCEED 0.005-INCH.** If limit is exceeded, check bolt as follows:
 - (1) If cadmium-plating is worn through to steel, replace bolt (PARA 2-4-142).
 - (2) Using micrometer, measure diameter of bolt. **DIMENSION SHALL NOT BE**

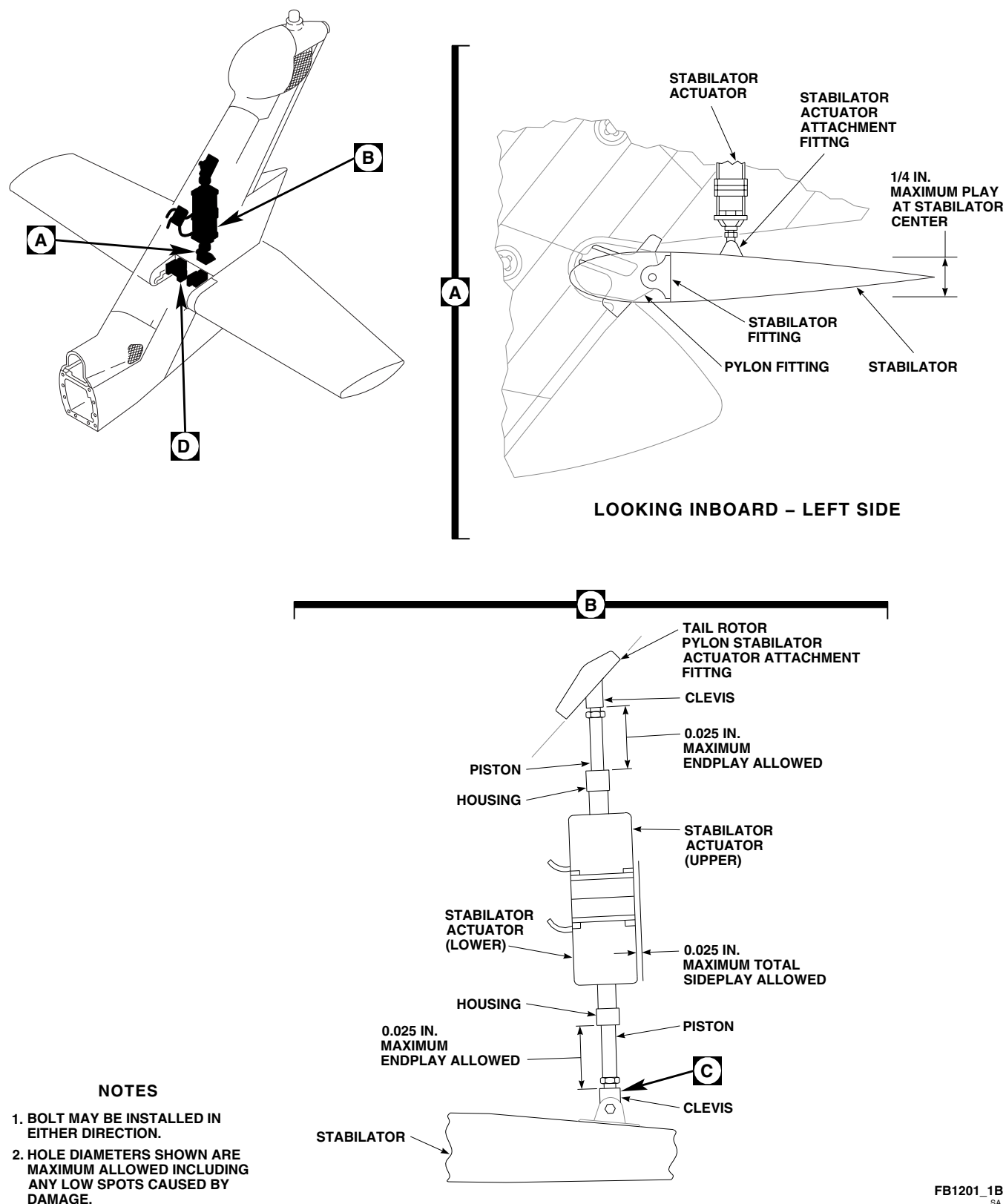
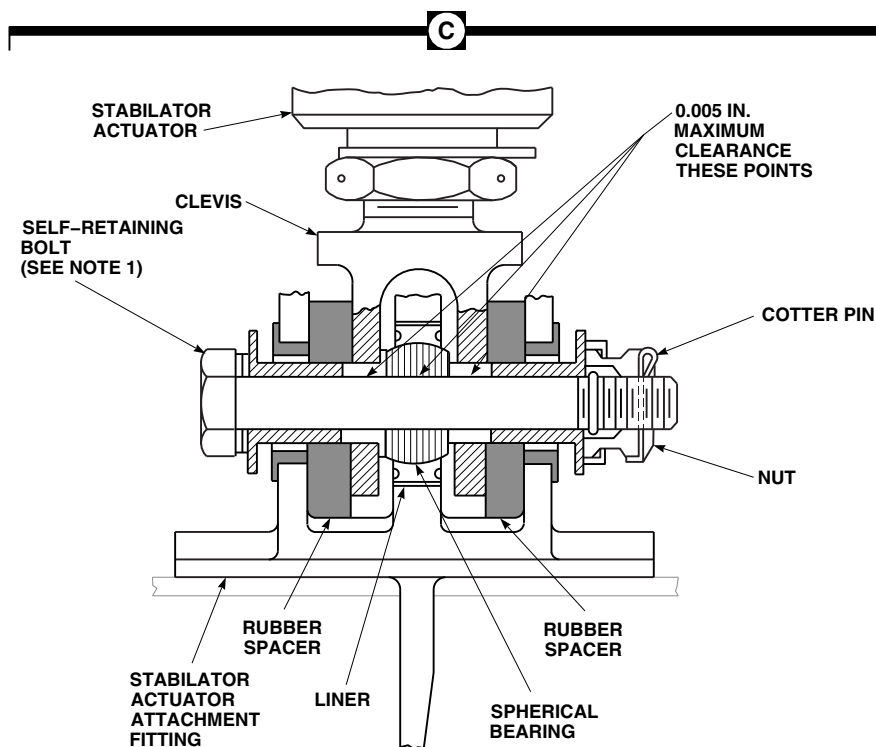


FIGURE 2-3-15-1. INSPECT STABILATOR AND ACTUATOR ATTACHMENT FITTINGS (SHEET 1 OF 4)



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**FIGURE 2-3-15-1. INSPECT STABILATOR AND ACTUATOR ATTACHMENT FITTINGS
(SHEET 2 OF 4)**

LESS THAN 0.370-INCH. If bolt diameter is not within limit, replace bolt (PARA 2-4-142).

- (3) If bolt diameter is within limit, replace tail rotor pylon stabilator actuator attachment fitting (PARA 2-6-48).
- e. Using feeler gage, measure clearance between bolt and upper clevis bushing bore. **CLEARANCE SHALL NOT EXCEED 0.005-INCH.** If limit is exceeded and bolt diameter is within limit, replace upper clevis (PARA 2-4-147).
- f. Remove bolt from stabilator lower attachment fitting (PARA 2-4-142) and check bolt.
- g. If removed, install tail rotor pylon leading edge cover/antenna (PARA 2-4-138).
- h. Close tail rotor pylon leading edge cover/antenna.

2-3-15.7 Stabilator Actuator Lower Attachment Inspection.

- a. If cambered fairing interferes with lower actuator attachment fitting inspection, remove tail rotor pylon cambered fairing (PARA 2-4-135).
- b. If outside diameter and shoulder of bolt show signs of nicks, dents, scratches, and tool or scuff marks, replace bolt (PARA 2-4-142).
- c. Using micrometer, measure diameter of bolt. **DIMENSION SHALL NOT BE LESS THAN 0.370-INCH.** If bolt diameter is not within limit, replace bolt (PARA 2-4-142).
- d. Check spherical bearing in stabilator actuator attachment fitting for radial play (Figure 2-3-15-1, Sheet 2, Detail C). **RADIAL PLAY SHALL NOT EXCEED 0.012-INCH.** If limit is exceeded, replace spherical bearing (PARA 2-6-55).

WARNING**FSCAP**

- Bonding of liners in center lobe of fitting is a critical characteristic. Proper bonding must be verified.
 - Damage to fitting will occur if feeler gage is improperly used. Liner may become disbonded. Use care when using feeler gage to check for liner edge separation.
- e. Visually inspect bonding of liners in center lobe of stabilator actuator attachment fitting. Using 0.0015-inch feeler gage, probe for edge separation. **VOIDS MUST NOT EXCEED 5% OF JOINT LINE LENGTH.** If limit is exceeded, replace stabilator (PARA 2-4-142).
 - f. Using feeler gage, measure clearance between bolt and lower clevis bushing bore. **CLEARANCE SHALL NOT EXCEED 0.005-INCH.** If limit is exceeded and bolt diameter is within limit, replace lower clevis (PARA 2-4-147).
 - g. Using feeler gage, measure clearance between bolt and bearing/bushing bore in stabilator actuator attachment fitting. **CLEARANCE SHALL NOT EXCEED 0.005-INCH.** If limit is exceeded and bolt diameter is within limit, replace stabilator (PARA 2-4-142).
 - h. If removed, install tail rotor pylon cambered fairing (PARA 2-4-135).

2-3-15.8 Left Stabilator-to-Pylon Attachment Fitting Inspection.

- a. Remove tail rotor pylon lower fairing (PARA 2-4-133).
- b. Check left pylon attachment fitting for looseness (Figure 2-3-15-1, Sheet 3, Detail D). **NONE ALLOWED.** If loose, replace tail rotor pylon (PARA 2-4-132).
- c. Check left stabilator attachment fitting for looseness. **NONE ALLOWED.** If loose, tighten bolts (PARA 2-4-144).

- d. Remove stabilator (PARA 2-4-142), and check left bolt as follows:

WARNING**FSCAP**

Bolthead shoulder and bolt shank of stabilator attachment bolts are critical surfaces. Dimension limits must not be exceeded.

- (1) Check cadmium-plating on bolt for wear. **CADMIUM-PLATING SHALL NOT BE WORN THROUGH TO STEEL.** If cadmium-plating is worn through to steel, replace bolt (PARA 2-4-142).
 - (2) Using micrometer, measure diameter of bolt. **DIMENSION SHALL NOT BE LESS THAN 0.6215-INCH.** If bolt diameter is not within limit, replace bolt (PARA 2-4-142).
- e. Check spherical bearing in left stabilator attachment fitting for radial play. **RADIAL PLAY SHALL NOT EXCEED 0.012-INCH.** If limit is exceeded, replace bearing (PARA 2-6-47).
 - f. Using feeler gage, measure clearance between bolt and bore in stabilator spherical bearing. **CLEARANCE SHALL NOT EXCEED 0.005-INCH.** If limit is exceeded and bolt diameter is within limits, replace fitting (PARA 2-4-144).
 - g. Using feeler gage, measure clearance between bolt and bushing bore in pylon attachment fitting. **CLEARANCE SHALL NOT EXCEED 0.005-INCH.** If limit is exceeded and bolt diameter is within limits, replace pylon fitting bushings (PARA 2-6-46).
 - h. Install tail rotor pylon lower fairing (PARA 2-4-133).

2-3-15.9 Right Stabilator-to-Pylon Attachment Fitting Inspection.

- a. Remove tail rotor pylon lower fairing (PARA 2-4-133).

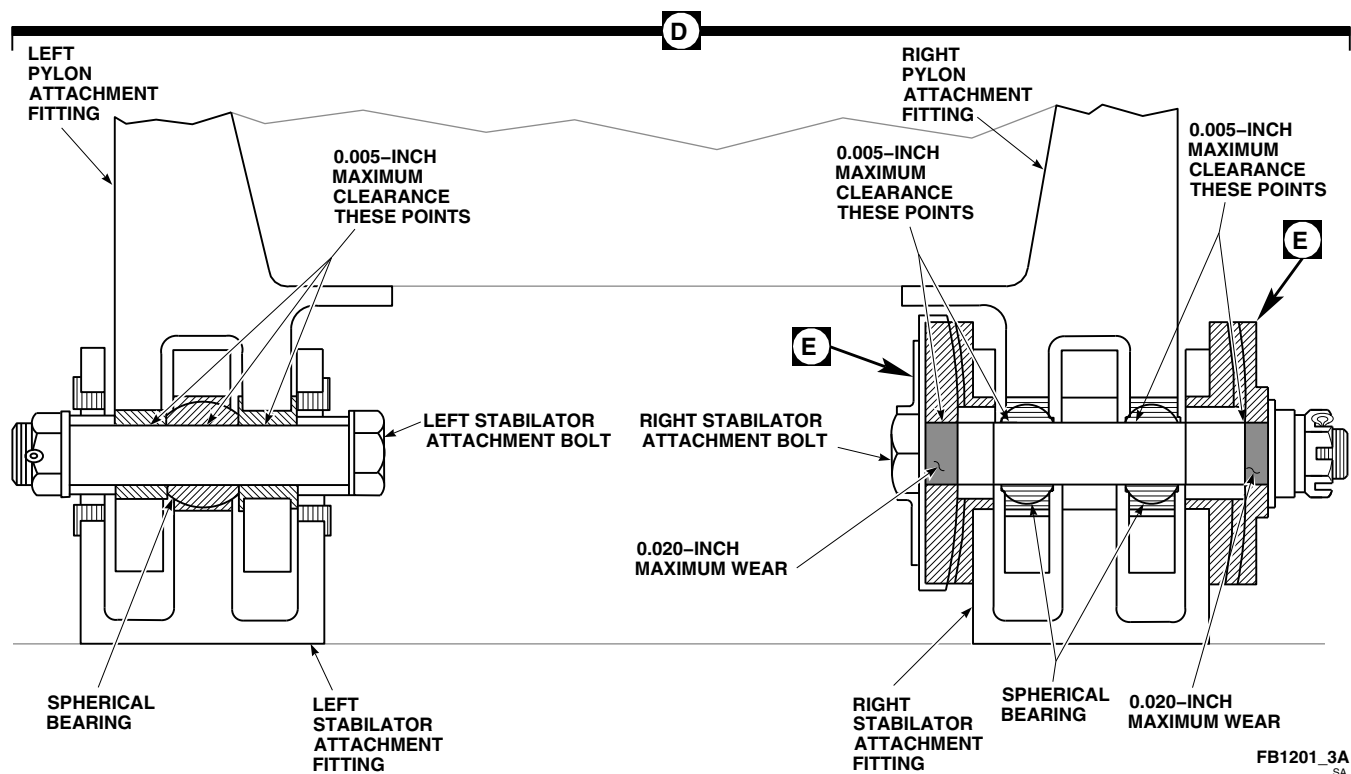


FIGURE 2-3-15-1. INSPECT STABILATOR AND ACTUATOR ATTACHMENT FITTINGS (SHEET 3 OF 4)

- b. Check right pylon attachment fitting for looseness (Figure 2-3-15-1, Sheet 3, Detail D). **NONE ALLOWED.** If loose, replace tail rotor pylon (PARA 2-4-132).
- c. Check right stabilator attachment fitting for looseness. **NONE ALLOWED.** If loose, tighten bolts (PARA 2-4-142).
- d. Remove stabilator (PARA 2-4-142).
- e. Check spherical bearing in right stabilator attachment fitting for radial play. **RADIAL PLAY SHALL NOT EXCEED 0.012-INCH IN EITHER BEARING.** If limit is exceeded, replace bearing (PARA 2-6-47).
- f. Using feeler gage, measure clearance between bolt and bearing bore in right pylon attachment fitting. **CLEARANCE SHALL NOT EXCEED 0.005-INCH.** If limit is exceeded and bolt diameter is within limit, replace bearing (PARA 2-6-46).
- g. Using feeler gage, measure clearance between bolt and bore in right stabilator attachment fitting elastomeric bearing. **CLEARANCE SHALL NOT EXCEED 0.005-INCH.** If limit is exceeded and bolt diameter is within limits, replace fitting (PARA 2-4-144).
- h. Install stabilator (PARA 2-4-142).

- i. Install tail rotor pylon lower fairing (PARA 2-4-133).

2-3-15.10 Right Attachment Bolt Inspection/Repair.

WARNING

FSCAP

Bolthead shoulder and unthreaded shank of right attachment bolt are critical surfaces that must not be damaged. Use extreme care when removing and installing. Use protective covering when removed.

- a. Using micrometer, measure right attachment bolt diameter where bolt contacts inboard and outboard elastomeric bearings (Figure 2-3-15-1, Sheet 3, Detail D). **WEAR IN THIS AREA SHALL NOT EXCEED 0.020-INCH.** If wear exceeds limit, replace bolt.
- b. Using micrometer, measure attachment bolt diameter throughout shank, except in areas shaded in Figure 2-3-15-1, Sheet 3, Detail D. **BOLT DIAMETER SHALL NOT BE LESS THAN 0.7465-INCH.** If bolt diameter is not within limit, replace bolt (PARA 2-4-142).
- c. Visually inspect bolt for scuffs and scratches. **NICKS, DENTS, SCRATCHES AND SCUFF MARKS, NOT MORE THAN 0.003-INCH WIDE, ARE ALLOWED.** Damage blending is permitted within limits of step a.

NOTE

- Right stabilator attachment bolt, 70209-07057-101, may become discolored (reddish-brown) while in service. Surface discoloration to bolt is not cause for replacement.
- Corrosion is indicated by red film or pitting.
- d. Visually inspect bolthead and threaded portion of shank for corrosion. **PITS SHALL NOT EXCEED 0.010-INCH IN DIAMETER.**

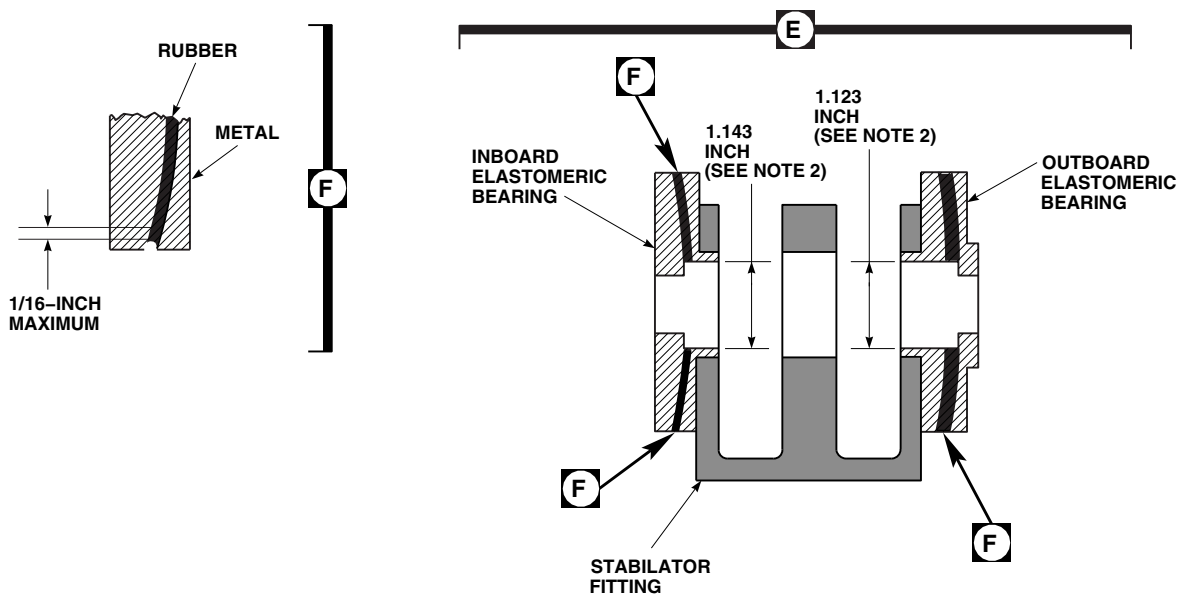
- e. If necessary, remove corrosion from noncritical areas of bolt manually, using scouring pad, Item 270, Appendix D.
- f. Coat bolt with corrosion preventive compound, Item 114, Appendix D.

2-3-15.11 Stabilator Elastomeric Bearing Inspection.

- a. Using inside micrometer, measure large inside diameter of inboard elastomeric bearing (Figure 2-3-15-1, Sheet 4, Detail E). **DIAMETER SHALL NOT EXCEED 1.143-INCHES, INCLUDING ANY LOW SPOTS CAUSED BY DAMAGE.** If limit is exceeded, replace stabilator elastomeric bearing (PARA 2-5-20).
- b. Using inside micrometer, measure large inside diameter of outboard elastomeric bearing. **DIAMETER SHALL NOT EXCEED 1.123-INCHES, INCLUDING ANY LOW SPOTS CAUSED BY DAMAGE.** If limit is exceeded, replace stabilator elastomeric bearing (PARA 2-5-20).
- c. Check elastomeric bearings for cracking or separation. Using 0.015-inch feeler gage, measure gaps between rubber and metal (Figure 2-3-15-1, Sheet 4, Detail F). **CRACKS OR SEPARATION BETWEEN METAL AND RUBBER SHALL NOT EXCEED 1/16-INCH DEEP.** If limit is exceeded, replace stabilator elastomeric bearing (PARA 2-5-20).
- d. Visually inspect stabilator attachment fitting for disbonding between mating surface of bearing and fitting. **UP TO 50% DISBONDING OF MATING SURFACE IS ALLOWED.** If disbonding is 50% or greater, repair bond (PARA 2-5-20).

2-3-15.12 Stabilator Actuator Attachment Fittings Inspection.

- a. Check stabilator actuator attachment fittings on tail rotor pylon and stabilator for nicks, dents, scratches, and tool or scuff marks in lug area. **NONE ALLOWED.**
- b. Check fittings for cracks in lugs, base, and attachment holes as follows:



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**FIGURE 2-3-15-1. INSPECT STABILATOR AND ACTUATOR ATTACHMENT FITTINGS
(SHEET 4 OF 4)**

- (1) IF TAIL ROTOR PYLON STABILATOR ACTUATOR ATTACHMENT FITTING IS CRACKED, REPLACE FITTING (PARA 2-6-48).
 - (2) IF STABILATOR ACTUATOR ATTACHMENT FITTING IS CRACKED, REPLACE STABILATOR (PARA 2-4-142).
 - (3) IF LEFT OR RIGHT PYLON ATTACHMENT FITTING IS CRACKED, REPLACE TAIL ROTOR PYLON (PARA 2-4-132).
 - (4) IF LEFT OR RIGHT STABILATOR ATTACHMENT FITTING IS CRACKED, REPLACE FITTING (PARA 2-4-144).
- c. Check fittings for nicks, scratches, gouges, and tool marks. **REPAIR DAMAGE LESS THAN 0.005-INCH DEEP, 0.040-INCH WIDE, AND 0.500-INCH LONG BY BLENDING OUT TO 0.008-INCH MAXIMUM DEPTH, WITH MINIMUM BLEND RADIUS OF $\frac{3}{8}$ -INCH.**
- d. **DAMAGE GREATER THAN 0.005-INCH DEEP, 0.040-INCH WIDE, AND 0.500-INCH LONG, REQUIRES ENGINEERING EVALUATION FOR REPAIR OR REPLACEMENT.**

2-3-15.13 Stabilator Actuator Grounding Strap Inspection **701334-SUBQ** .

- a. Check grounding strap for cracks or tears in shrink tubing. **NONE ALLOWED.** If damage is found, replace stabilator actuator grounding strap (PARA 2-4-149).
- b. Check for disbonding of shrink tubing at strap end fittings. **NONE ALLOWED.** If disbonding is found, replace stabilator actuator grounding strap (PARA 2-4-149).
- c. Inspect chemical conversion-coated mating surfaces of grounding strap, grounding bracket, and rear spar. Touch up chemical conversion coating as required (PARA 2-6-5).

SECTION 2-4

MAINTENANCE PROCEDURES

SECTION OVERVIEW

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2-4-2	Nose Door Latches
2-4-3	Nose Door Strut
2-4-4	Nose Door Strut Catch Assembly
2-4-5	Nose Door Hinge Fittings
2-4-6	Nose Door Lock
2-4-7	Nose Door Cover and Screen
2-4-8	Nose Door Seals
2-4-9	Nose Door Stops
2-4-10	Nose Door Drain Hoses
2-4-11	Cockpit Doors
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PARAGRAPH	TITLE
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2-4-105	Main Rotor Pylon Work Platform
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2-4-107	Main Rotor Pylon Transmission Cover
2-4-108	APU Access Doors
2-4-109	APU Screens
2-4-110	APU Access Door Beam
2-4-111	APU Exhaust Fairing
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2-4-113	Oil Cooler Compartment Access Doors
2-4-114	IR Jammer Fairing
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2-4-116	Main Rotor Pylon Rear Fairing
2-4-117	Yaw Pushrod Boot and Shroud
2-4-118	Engine Cowlings
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PARAGRAPH	TITLE
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2-4-143	Stabilator Electrostatic Dischargers
2-4-144	Stabilator Attachment Fittings
2-4-145	Stabilator Tip Caps

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2-4-147	Stabilator Actuator Repair
2-4-148	Stabilator Actuator Adjustment
2-4-149	Stabilator Actuator Grounding Strap 701334-SUBQ
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2-4-151	Position Transmitter Limit Switch
2-4-152	Crew Chief's/Passenger's Seat - 14 Seat Configuration
2-4-153	Crew Chief's/Passenger's Seat - 20 Seat Configuration
2-4-154	Crew Chief's/Passenger's Seat Pallets - 20 Seat Configuration
2-4-155	Auxiliary Searchlight Support
2-4-156	Fast Rope System

2-4-1 NOSE DOOR.

PARAGRAPH BREAKDOWN

	PAGE
2-4-1.1 Replace Nose Door	2-4-1-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Temporary Nose Door Support

Materials/Parts

- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-2
- PARA 2-4-6
- PARA 2-4-42
- PARA 2-6-5
- TM 11-70-23

2-4-1.1 Replace Nose Door.

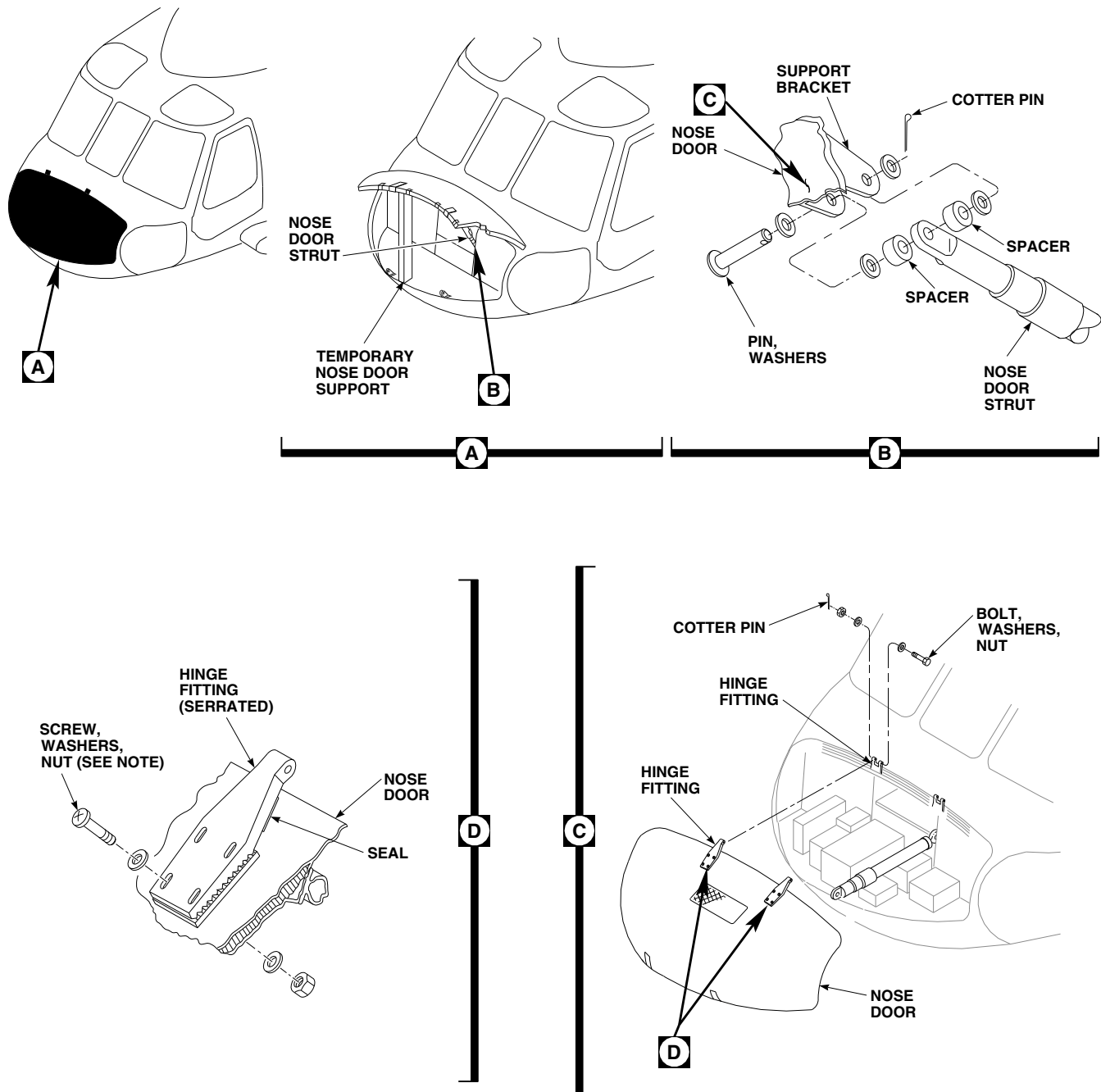
2-4-1.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Unlatch and open nose door. Lock nose door strut in open position.
- If installed, remove weather radar system nose receiver, antennas, and associated wiring from nose door (TM 11-70-23).
- Install temporary nose door support to support door in open position (Figure 2-4-1-1, Detail A).
- Remove pin, washers, and spacers securing strut to support bracket (Figure 2-4-1-1, Detail B).

- Remove temporary nose door support and lower door to closed position (Figure 2-4-1-1, Detail A).
- Remove bolts, washers, and nuts securing hinge fittings together. Remove door (Figure 2-4-1-1, Detail C).
- Remove door lock assembly from door (PARA 2-4-6). Retain for installation.

2-4-1.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Position nose door on helicopter (Figure 2-4-1-1, Detail C).
- Line up hinge fittings and install bolts, washers, nuts, and cotter pins.

**NOTE**

HARDWARE SHOWN EXPLODED FOR CLARITY.

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FIGURE 2-4-1-1. REPLACE NOSE DOOR

- d. Install temporary nose door support to support door in open position (Figure 2-4-1-1, Detail A).
- e. Position end of nose door strut in support bracket. Install washer and spacer on each side of strut end. Install pin, washers, and cotter pin securing strut to support bracket (Figure 2-4-1-1, Detail B).
- f. Remove temporary nose door support and close door (Figure 2-4-1-1, Detail A). Inspect door for proper centering within door opening. If adjustment is required, perform the following (Figure 2-4-1-1, Detail D):
 - (1) Remove sealing compound from perimeter of hinge fitting using machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D.
 - (2) Loosen screws and nuts securing upper serrated hinge fitting to nose door.
 - (3) Remove sealing compound from exposed portion of screws using machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D.
 - (4) Reposition door as necessary.
 - (5) Apply sealing compound, Item 292, Appendix D, to hinge fitting screwholes and screws. **TIGHTEN SCREWS AND NUTS.**
 - (6) Wipe off excess sealing compound using clean machinery towel, Item 211, Appendix D.

- (7) Apply fillet of sealing compound, Item 292, Appendix D, to perimeter of hinge fitting and seal.
- (8) Wipe off excess sealing compound using clean machinery towel, Item 211, Appendix D.
- (9) Touch up paint as required (PARA 2-6-5).

NOTE

To keep locksets uniform when replacing nose door, install lock removed from old door.

- g. Install door lock assembly (PARA 2-4-6).
- h. Check door for adjustment as follows:
 - (1) Press down on door.
 - (2) Pull outward on latch. If latch hooks can be pulled out from bracket, adjust latch (PARA 2-4-2).
- i. If removed, install weather radar system nose receiver, antennas, and associated wiring on nose door (TM 11-70-23).
- j. Make sure area is clean and free of foreign material before closing up.
- k. Unlock nose door strut. Close nose door and secure latches.
- l. Perform nose door leak test (PARA 2-4-42).

2-4-2 NOSE DOOR LATCHES.

PARAGRAPH BREAKDOWN

		PAGE
2-4-2.1	Replace Nose Door Latches	2-4-2-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Feeler Gage

Materials/Parts

- Adhesive, Item 32, Appendix D
- Corrosion Compound, Item 108, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D

- Paint, Polyurethane, Item 222A, Appendix D
- Plastic Sheet, Item 235, Appendix D
- Scouring Pad, Item 270, Appendix D
- Water

References

- Appendix D
- PARA 2-4-9
- PARA 2-6-5

2-4-2.1 Replace Nose Door Latches.

NOTE

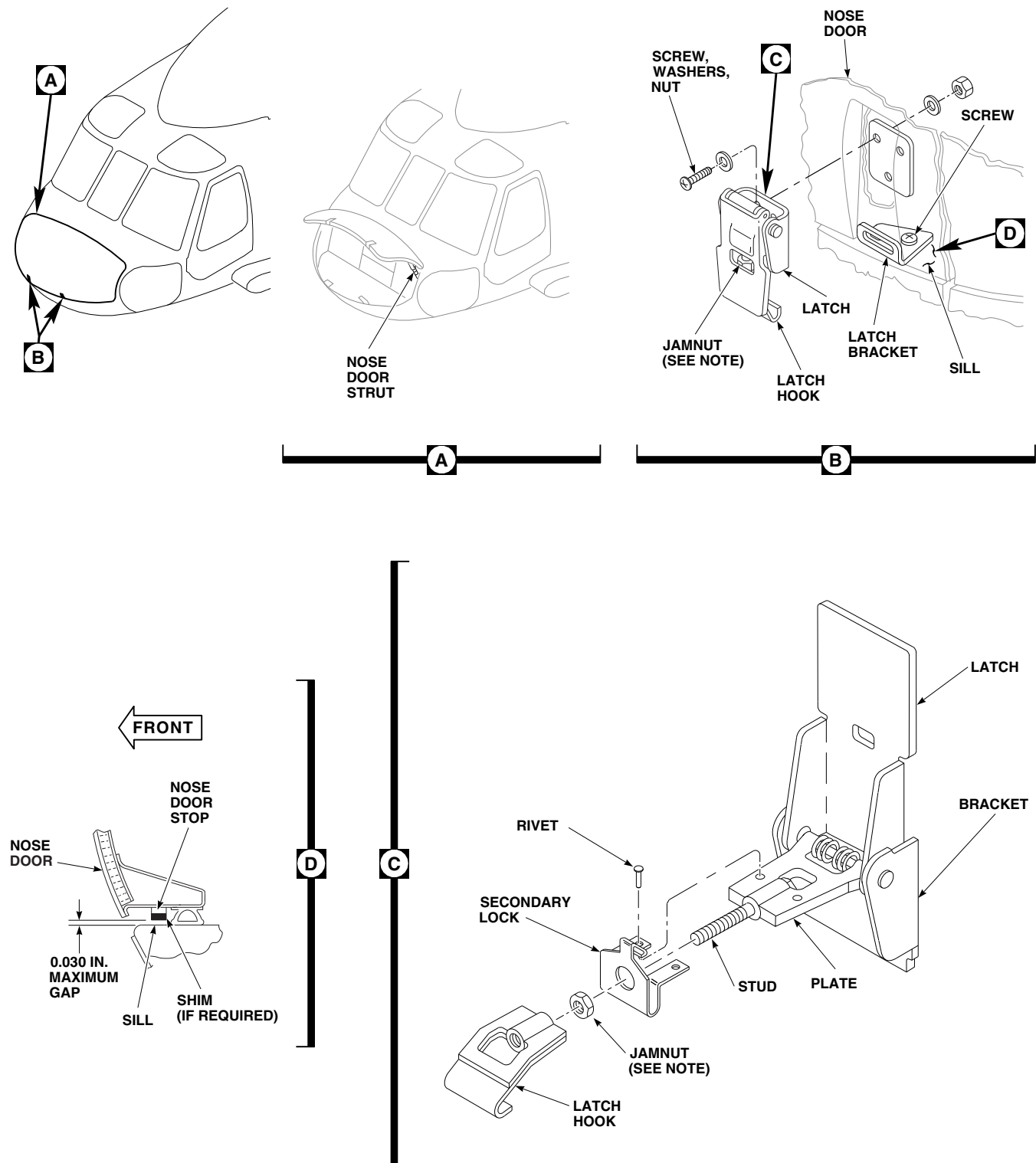
Replacement of both nose door latches is the same.

2-4-2.1.1 Remove.

- Unlatch and open nose door. Lock nose door strut in open position (Figure 2-4-2-1, Detail A).
- Remove screws, washers, and nuts securing latch to nose door (Figure 2-4-2-1, Detail B). Remove latch.

2-4-2.1.2 Inspect/Repair.

- Inspect latch for corrosion. **NONE ALLOWED.** If corrosion is found, remove as follows:
 - Loosen jamnut and remove latch hook from stud. Remove jamnut from stud (Figure 2-4-2-1, Detail C).
 - Remove corrosion from latch, using scouring pad, Item 270, Appendix D. **PITS ON NONTHEADED AREAS OF LATCH SHALL NOT EXCEED 0.005-INCH AFTER REWORK.**



NOTE

TIGHTEN JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.

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FIGURE 2-4-2-1. REPLACE NOSE DOOR LATCHES

- (3) Mix solution of one part corrosion compound, Item 108, Appendix D, to one part water.
- (4) Apply compound solution to corroded areas of latch. Rinse thoroughly with clean water.
- (5) Using epoxy primer coating kit, Item 135, Appendix D, touch up paint with two coats of epoxy primer and top coat of polyurethane paint, Item 222A, Appendix D (PARA 2-6-5).
- (6) Install jamnut on stud. Install latch hook on stud.
- b. Inspect secondary lock for damage and function. **NO DAMAGE ALLOWED.** If necessary, replace secondary lock as follows:
 - (1) Loosen jamnut and remove latch hook from stud. Remove jamnut from stud.
 - (2) Remove rivets securing secondary lock to plate. Remove lock.
 - (3) Position replacement secondary lock on plate and secure with rivets.
 - (4) Install jamnut on stud. Install latch hook on stud.
- b. Open latch and loosen jamnut on latch hook (Figure 2-4-2-1, Detail B).
- c. Turn latch hook to adjust latch and secure both latches. Nose door will draw down on seal.
- d. Check contact of stops (Figure 2-4-2-1, Detail D). **AT LEAST ONE STOP ON EITHER SIDE OF LATCHES MUST CONTACT SILL.** If necessary, trim stop as follows:
 - (1) Determine which stop on either side of latch is closest to making contact with sill.
 - (2) Unlatch and open nose door. Lock nose door strut in open position (Figure 2-4-2-1, Detail A).
 - (3) Trim stop as necessary.
 - (4) Close nose door and secure latches. Recheck contact of stops. Repeat step d. (2) through d. (4) until contact is made.
- e. Using feeler gage, check gaps between other two stops and sill. **MAXIMUM GAP ALLOWED IS 0.030-INCH** (Figure 2-4-2-1, Detail D).

2-4-2.1.3 Install.

- a. Position latch on nose door and secure with screws, washers, and nuts (Figure 2-4-2-1, Detail B).
- b. Loosen two screws securing latch bracket and move bracket to center latch hook in bracket slot. **TIGHTEN BRACKET SCREWS.**
- c. Unlock nose door strut. Close nose door and secure latches (Figure 2-4-2-1, Detail A).
- f. Open latch.
- g. **TIGHTEN JAMNUT 1/4 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT** (Figure 2-4-2-1, Detail B).
- h. Secure latch. Repeat step e. If gap(s) still exceeds limit, perform the following:
 - (1) Note measurement exceeding gap limit.
 - (2) Make shim from plastic sheet, Item 235, Appendix D, to same thickness as noted measurement.
 - (3) Bond shim to stop with adhesive, Item 32, Appendix D (Figure 2-4-2-1, Detail D).

2-4-2.1.4 Adjust.

- a. Verify that all four nose door stops are installed (Figure 2-4-2-1, Detail D). If stop is missing, install nose door stop (PARA 2-4-9).

2-4-3 NOSE DOOR STRUT.

PARAGRAPH BREAKDOWN

	PAGE
2-4-3.1 Replace Nose Door Strut	2-4-3-1

INITIAL SETUP

- Torque Wrench, 30 - 150 in. lbs

Tools

- Aircraft Mechanic’s Toolkit
- Feeler Gage
- Temporary Nose Door Support

2-4-3.1 Replace Nose Door Strut.

2-4-3.1.3 Install.

2-4-3.1.1 Remove.

- Unlatch and open nose door. Lock nose door strut in open position.
- Install temporary nose door support to support door in open position (Figure 2-4-3-1).
- Remove pin, washers, and spacers securing front end of strut to support bracket on door (Figure 2-4-3-1, Detail A).
- Remove bolt, washers, and nut securing rear end of strut to support fitting on compartment bulkhead.
- Remove strut and washer from compartment.

2-4-3.1.2 Inspect.

Using feeler gage, measure play between rod end bearing and race. **IF PLAY IS 0.012-INCH OR MORE, REPLACE ROD END.**

- Position rear end of strut on left side of support fitting. Place washer between strut and fitting on compartment bulkhead. Secure strut to fitting with bolt, washers, and nut. **TORQUE NUT TO 42 - 47 INCH-POUNDS** (Figure 2-4-3-1, Detail A).
- Install front end of strut in clevis of support bracket on nose door. Install washer and spacer on each side of strut end. Install pin and washers. Install cotter pin.
- Remove temporary nose door support (Figure 2-4-3-1).
- Make sure area is clean and free of foreign material.
- Unlock nose door strut. Close nose door and secure latches.

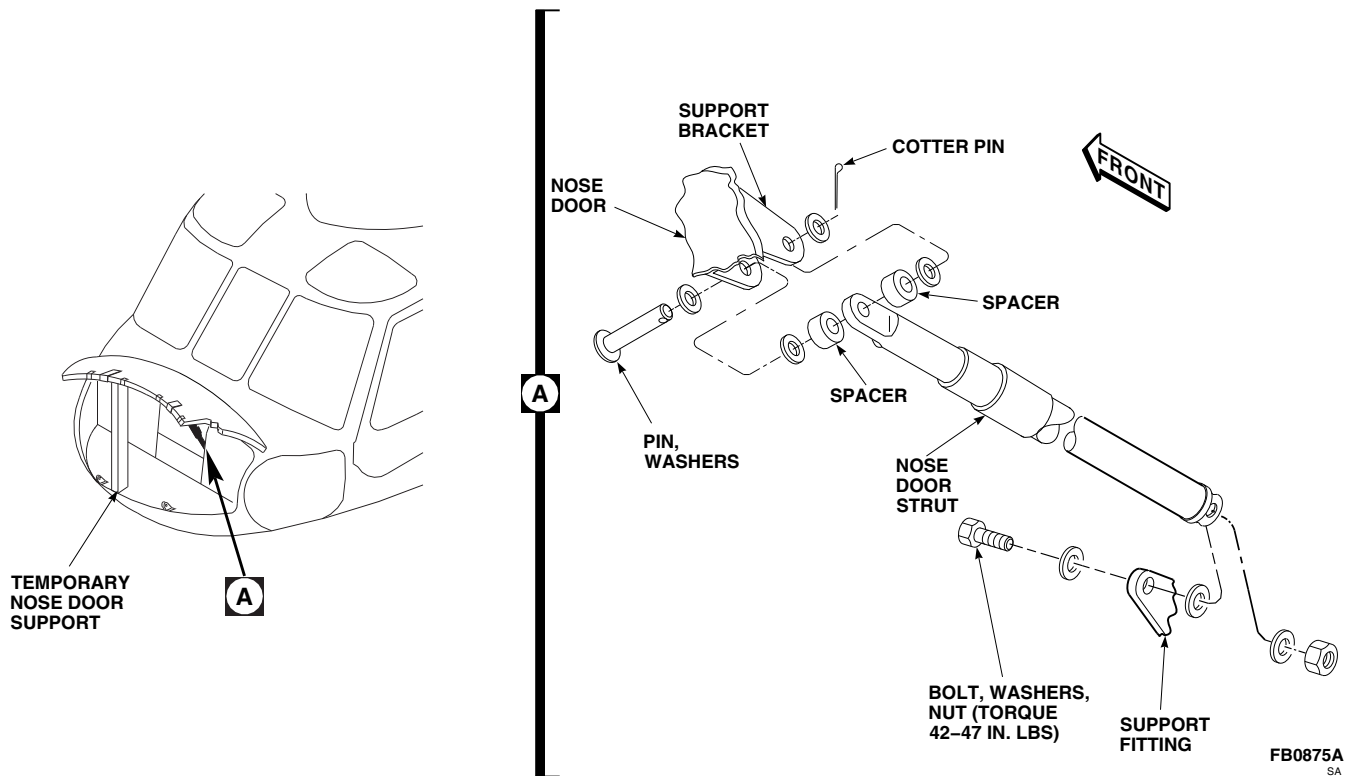


FIGURE 2-4-3-1. REPLACE NOSE DOOR STRUT

2-4-4 NOSE DOOR STRUT CATCH ASSEMBLY.

PARAGRAPH BREAKDOWN

		PAGE
2-4-4.1	Replace Nose Door Strut Catch Assembly	2-4-4-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Temporary Nose Door Support

References

- PARA 1-6-16

2-4-4.1 Replace Nose Door Strut Catch Assembly.

2-4-4.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Unlatch and open nose door. Lock nose door strut in open position (Figure 2-4-4-1).
- Install temporary nose door support to support door in open position.
- Remove screw, washers, and nut securing catch assembly to clevis on strut (Figure 2-4-4-1, Detail A).

2-4-4.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Secure catch assembly to clevis with screw, washers, and nut (Figure 2-4-4-1, Detail A).
- Check that catch assembly works properly.
- Remove temporary nose door support (Figure 2-4-4-1).
- Make sure area is clean and free of foreign material.
- Unlock nose door strut. Close nose door and fasten latches (Figure 2-4-4-1).

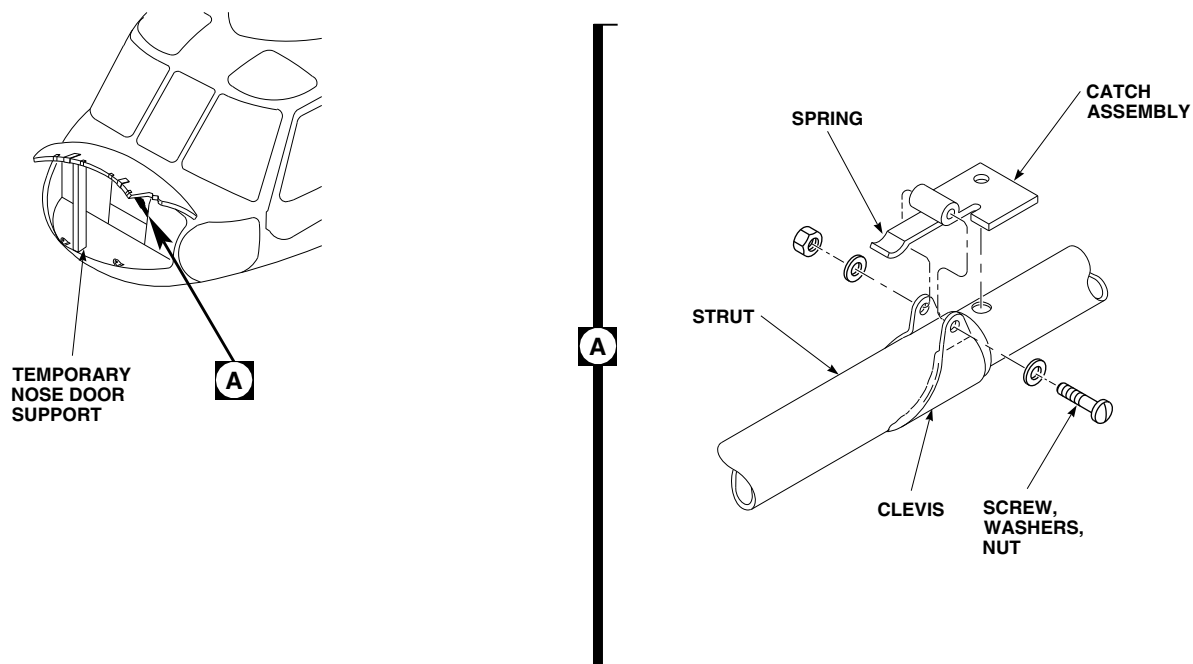
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FIGURE 2-4-4-1. REPLACE NOSE DOOR STRUT CATCH ASSEMBLY

2-4-5 NOSE DOOR HINGE FITTINGS.

PARAGRAPH BREAKDOWN

	PAGE
2-4-5.1 Replace Nose Door Hinge Fittings	2-4-5-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Magnifying Glass, 10X
- Nonmetallic Scraper, Locally-Made
- Temporary Nose Door Support

Materials/Parts

- Abrasive Cloth, Item 5, Appendix D
- Adhesive, Item 37, Appendix D

- Cheesecloth, Item 90, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 2-6-5

2-4-5.1 Replace Nose Door Hinge Fittings.

NOTE

Replacement of both nose door hinge fittings is the same.

2-4-5.1.1 Remove.

- Unlatch and open nose door. Lock nose door strut in open position (Figure 2-4-2-1, Detail A).
- Install temporary nose door support to support door in open position.
- Remove bolt, washers, and nut securing hinge fittings (Figure 2-4-5-1, Detail B).
- Remove screws, washers, and nuts securing hinge fitting to door. Remove hinge fitting from door (Figure 2-4-5-1, Detail C).

- Remove serrated plate and shims.
- If installed, remove seal(s) using nonmetallic scraper.

2-4-5.1.2 Inspect.

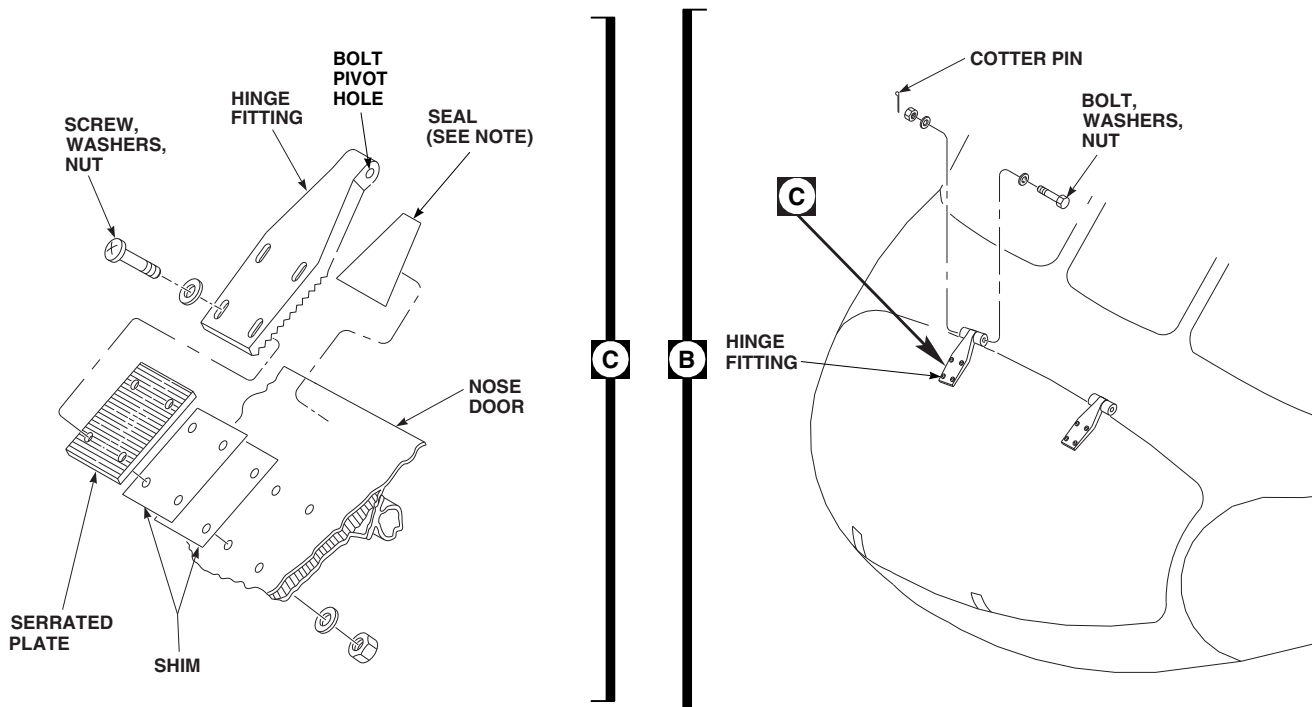
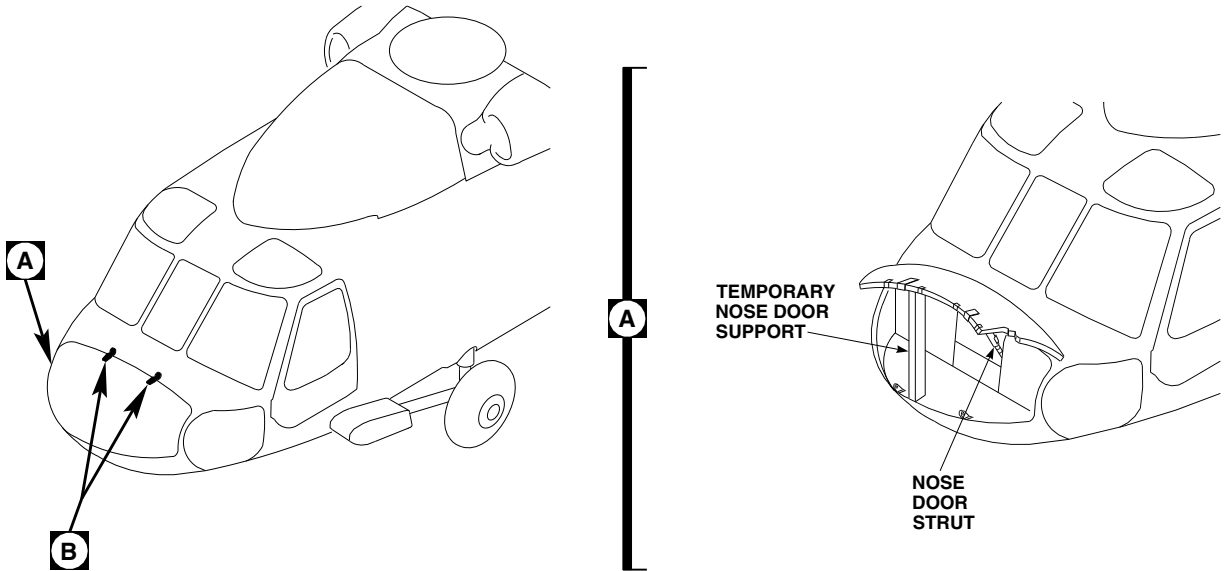
NOTE

Pay particular attention to bolt pivot hole areas.

Using 10X magnifying glass, inspect hinge fitting for cracks, paying particular attention to bolt pivot hole area (Figure 2-4-5-1, Detail C). **NONE ALLOWED.** If cracks are found, replace hinge fitting.

2-4-5.1.3 Install.

- Secure hinge fittings together using bolt, washers, and nut. Install cotter pin (Figure 2-4-5-1, Detail B).



NOTE

INSTALL A MAXIMUM OF TWO SEALS
AS REQUIRED TO FILL GAP BETWEEN
HINGE FITTING AND NOSE DOOR.

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FIGURE 2-4-5-1. REPLACE NOSE DOOR HINGE FITTING

- b. Position door to align with holes in hinge fitting and temporarily install serrated plates and hinge fitting. Add or remove shims to line up door with contour of nose. No more than two shims may be used (Figure 2-4-5-1, Detail C).
- c. If necessary, install seal(s) as follows:
 - (1) Wipe seal(s) and door clean using cheesecloth, Item 90, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D.
- e. Apply sealing compound, Item 292, Appendix D, to hinge fitting screwholes and screws.
- f. Install hinge fitting, serrated plate, and shim(s) on door with screws, washers, and nuts.
- g. Remove temporary nose door support and close door (Figure 2-4-5-1, Detail A). Inspect door for proper centering within door opening. If adjustment is required, perform the following (Figure 2-4-5-1, Detail C):
 - (1) Loosen screws and nuts securing upper serrated hinge fitting.
 - (2) Reposition door as necessary.
 - (3) **TIGHTEN SCREWS AND NUTS.**
- h. Wipe off extra sealing compound with clean machinery towel, Item 211, Appendix D.
- i. Apply a fillet of sealing compound, Item 292, Appendix D, to perimeter of hinge fitting and seal.
- j. Touch up paint as required (PARA 2-6-5).
- k. Make sure area is clean and free of foreign material.
- l. Unlock nose door strut. Close nose door and fasten latches (Figure 2-4-2-1, Detail A).
- d. Wipe hinge parts clean using machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D.

NOTE

Do not sand through paint or primer.

- (2) Lightly scuff mating surface on door with abrasive cloth, Item 5, Appendix D. Wipe door clean using cheesecloth, Item 90, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D.
- (3) Apply a thin, uniform coat of adhesive, Item 37, Appendix D, to seal(s) and to mating surface.

2-4-6 NOSE DOOR LOCK.

PARAGRAPH BREAKDOWN

	PAGE
2-4-6.1 Replace Nose Door Lock	2-4-6-1

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit

References

- PARA 1-6-16

2-4-6.1 Replace Nose Door Lock.

NOTE

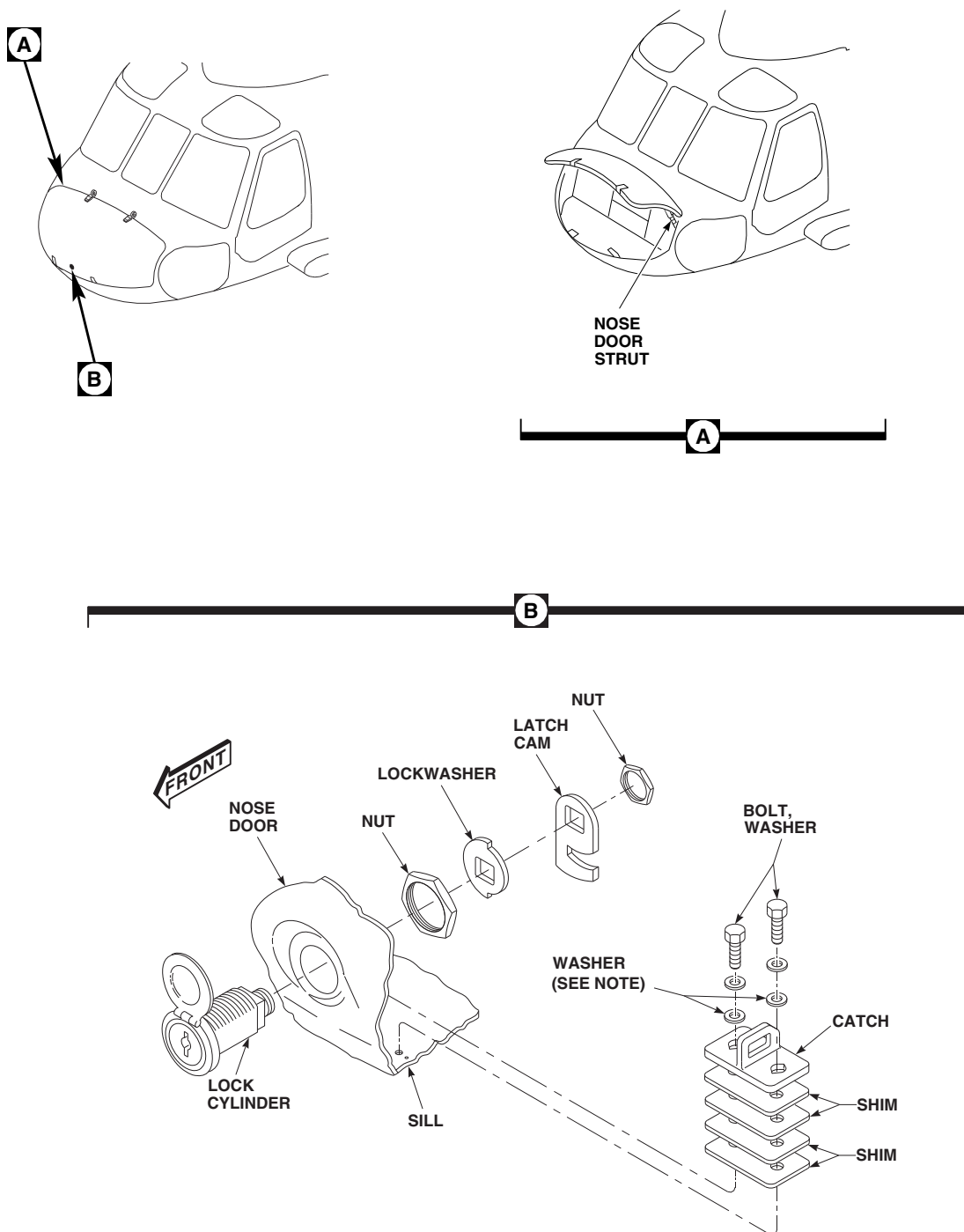
Lock cylinders are available only in sets. When replacing nose door lock cylinder, replace lock cylinders on all helicopter doors.

2-4-6.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Unlatch and open nose door. Lock nose door strut in open position (Figure 2-4-6-1, Detail A).
- Remove nut from back of lock cylinder. Remove latch cam and lockwasher from lock cylinder (Figure 2-4-6-1, Detail B).
- Remove nut securing lock cylinder to nose door. Push lock cylinder out of door.
- Inspect catch for damage. **NONE ALLOWED.** If damaged, remove catch as follows:
 - Remove bolts and washers securing catch to nose door.
 - Remove shims.

2-4-6.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Insert lock cylinder through front of nose door and secure with nut (Figure 2-4-6-1, Detail B).
- Line up square end of lock cylinder with sill and install lockwasher and latch cam on back of lock cylinder and secure with nut.
- If removed, install catch as follows:
 - Install catch and shims on sill and secure with bolts and washers. Catch must line up with latch cam when lock cylinder is locked.
 - If required, loosen bolts and adjust position of catch.
- Check to make sure latch cam engages catch without force when locking nose door. If required, adjust as follows:
 - Remove bolts and washers securing catch and shims to sill.
 - Add or remove shims as required.

**NOTE**

ADD ONE WASHER UNDER BOLTHEAD
FOR EACH SHIM REMOVED TO TAKE UP
BOLT SHANK LENGTH.

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FIGURE 2-4-6-1. REPLACE NOSE DOOR LOCK

NOTE

Add one washer under bolthead for each shim removed to take up bolt shank length.

- (3) Install catch and shims to sill and secure with bolts and washers.

- f. Make sure area is clean and free of foreign material.
- g. Unlock nose door strut. Close nose door and secure latches (Figure 2-4-2-1, Detail A).

2-4-7 NOSE DOOR COVER AND SCREEN.

PARAGRAPH BREAKDOWN

	PAGE
2-4-7.1 Replace Nose Door Cover and Screen	2-4-7-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

References

- PARA 1-6-16

2-4-7.1 Replace Nose Door Cover and Screen.

2-4-7.1.1. Remove.

- a. Turn off all electrical power (PARA 1-6-16).

NOTE

Remove cover when air temperature above 32°F (0°C) makes it desirable to allow air flow into nose area.

- b. Remove screws and washers securing cover, if installed, and screen to nose door (Figure 2-4-7-1, Detail A). Remove cover and screen.

- c. Store cover if it is not going to be reinstalled.

2-4-7.1.2. Install.

- a. Turn off all electrical power (PARA 1-6-16).

NOTE

Install cover when air temperature below 32°F (0°C) makes it desirable to block air flow into nose area.

- b. Install screen and cover, if required, on nose door and secure with screws and washers (Figure 2-4-7-1, Detail A).

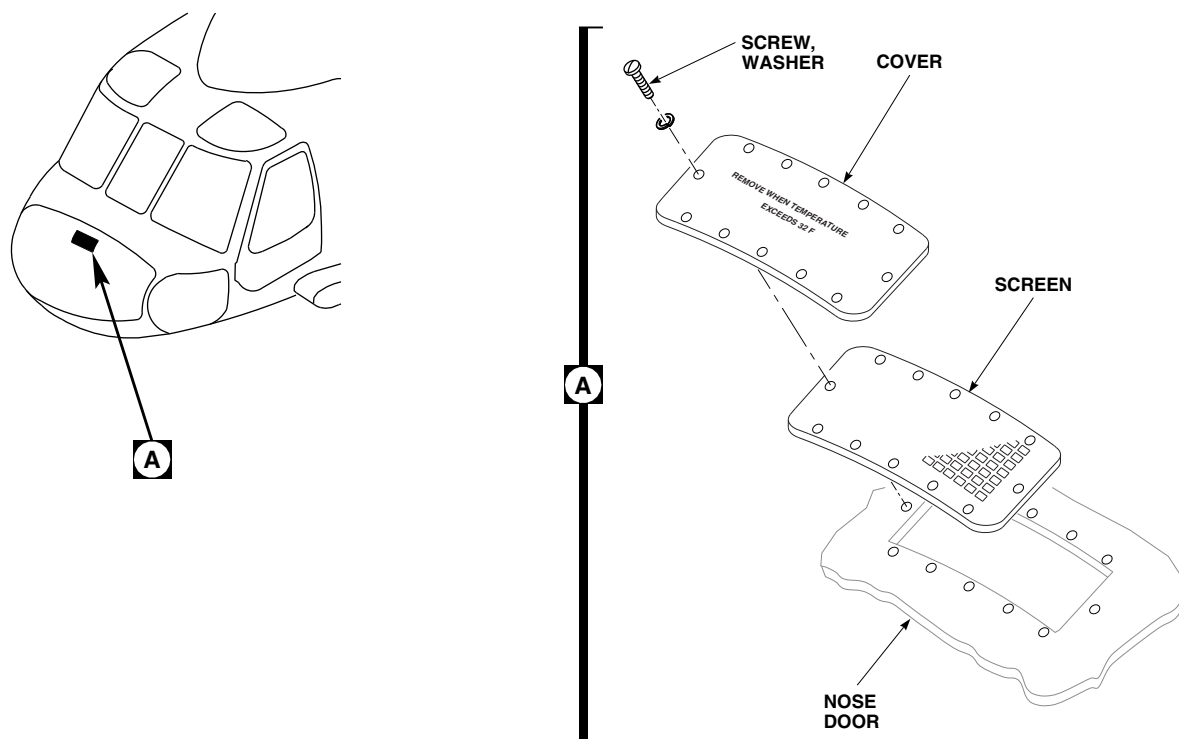
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FIGURE 2-4-7-1. REPLACE NOSE DOOR COVER AND SCREEN

2-4-8 NOSE DOOR SEALS.

PARAGRAPH BREAKDOWN

	PAGE
2-4-8.1 Replace Nose Door Seals	2-4-8-1

INITIAL SETUP Tools - Aircraft Mechanic’s Toolkit - Nonmetallic Scraper, Locally-Made Materials/Parts - Abrasive Paper, Item 15, Appendix D - Adhesive, Item 30, Appendix D - Adhesive, Item 55, Appendix D	- Machinery Towel, Item 211, Appendix D - Methyl Ethyl Ketone, Item 217, Appendix D - Plastic Strip, Item 237, Appendix D - Stirring Stick, Beverage, Item 315C, Appendix D References - Appendix D - PARA 1-6-16 - PARA 2-4-1
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2-4-8.1 Replace Nose Door Seals.

2-4-8.1.1 Remove.

- a. Remove nose door (PARA 2-4-1).
- b. Remove nose door seal from nose door using nonmetallic scraper (Figure 2-4-8-1, Detail A).
- c. Remove remaining adhesive, using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, and nonmetallic scraper.

2-4-8.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Position nose door seals on nose door with spliced breaks where shown (Figure 2-4-8-1, Detail A).



- Damage to equipment will result if tube section of seal is cut into. Cut only the flange.
- c. To keep seals from bunching up or stretching around corners, make 30°- 40° cut-outs on inner radius flange and slits on outer radius flange (Figure 2-4-8-1, Detail B).
 - d. Slit flange of seals in bolt recess area to prevent disbonding (Figure 2-4-8-1, Detail C).
 - e. Roughen all adhesive contact surfaces of seals with abrasive paper, Item 15, Appendix D.
 - f. Wipe all adhesive contact areas with clean machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.

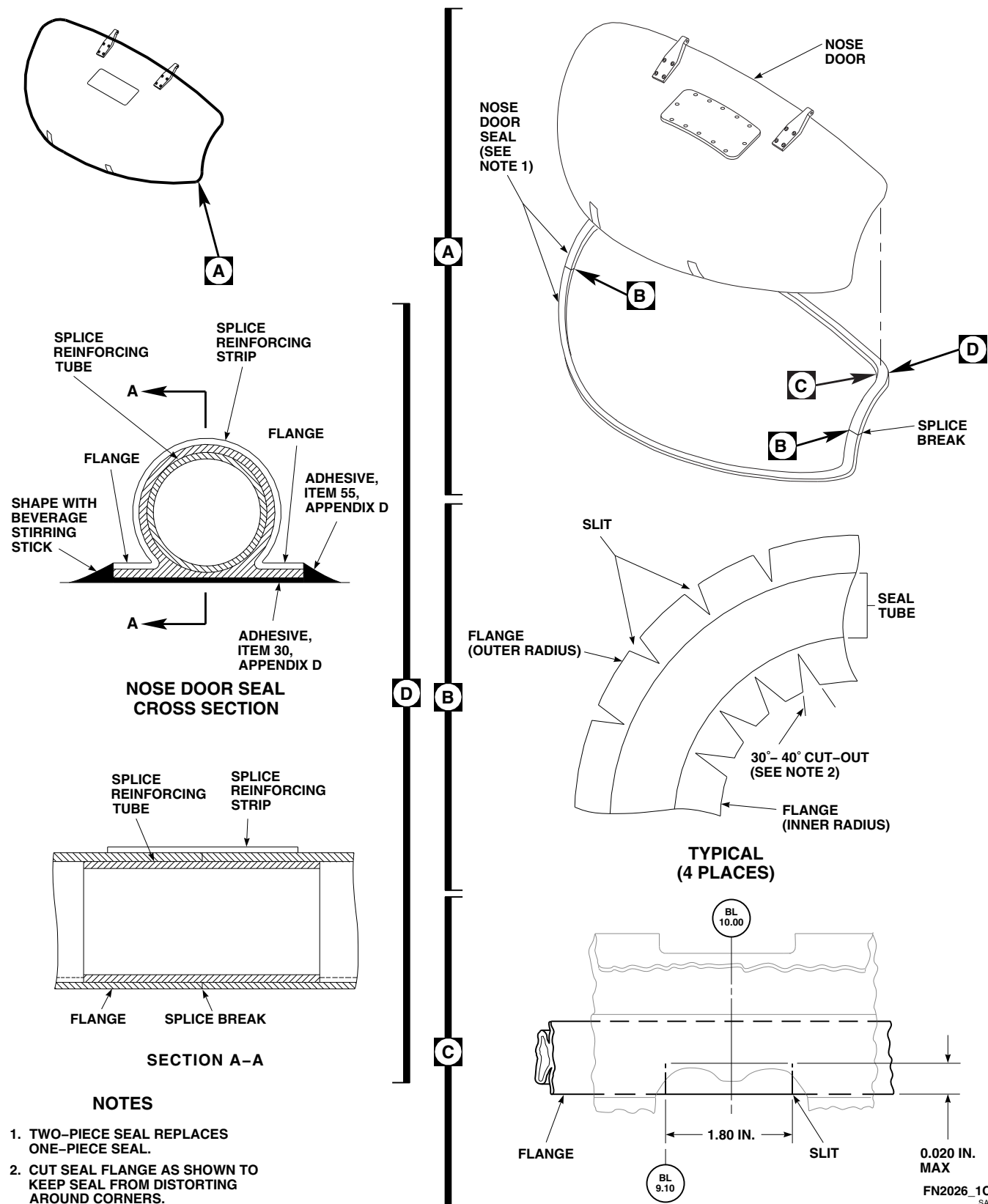


FIGURE 2-4-8-1. REPLACE NOSE DOOR SEALS

- g. Wipe inside seal ends and splice reinforcing tube with clean machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.
- h. Apply a light coat of adhesive, Item 55, Appendix D, to inside of seal ends and outside surface of splice reinforcing tube.
- i. Immediately install splice reinforcing tube in seal ends and close up gap (Figure 2-4-8-1, Detail D).
- j. Line up seal joints on a flat surface covered with plastic strip, Item 237, Appendix D, and hold for 10 - 15 minutes while adhesive sets up.
- k. Roughen splice reinforcing strip and surface of splice joint with abrasive paper, Item 15, Appendix D.
- l. Wipe roughened areas with clean machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.
- m. Apply adhesive, Item 55, Appendix D, to inside of splice reinforcing strip and splice joint. Install reinforcing strip over splice joint. Work out any gaps or blisters.
- n. Mix adhesive, Item 30, Appendix D, following manufacturer's instructions.
- o. Apply an even coat of adhesive, Item 30, Appendix D, to door and seal joining surfaces. Allow adhesive to air dry for 30 minutes or until tacky.
- p. Install seal on door (Figure 2-4-8-1, Detail A). Work out gaps, trapped air, and strained areas all the way around seal installation. Remove extra adhesive with clean machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.
- q. Mix and apply a bead of adhesive, Item 55, Appendix D, around the inner and outer flanges of seal (Figure 2-4-8-1, Detail D).
- r. Using beverage stirring stick, Item 315C, Appendix D, shape bead of adhesive as shown.
- s. Allow adhesive to cure at least 24 hours.
- t. Install nose door (PARA 2-4-1).

2-4-9 NOSE DOOR STOPS.

PARAGRAPH BREAKDOWN

	PAGE
2-4-9.1 Replace Nose Door Stops	2-4-9-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Feeler Gage
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Adhesive, Item 32, Appendix D
- Adhesive, Item 66, Appendix D

- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D

References

- Appendix D
- PARA 2-4-2
- PARA 2-6-5

2-4-9.1 Replace Nose Door Stops.

NOTE

Replacement of all nose door stops is the same.

2-4-9.1.1 Remove.

- Unlatch and open nose door. Lock nose door strut in open position (Figure 2-4-9-1, Detail A).
- Using nonmetallic scraper and machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D, remove loose nose door stop and old adhesive (Figure 2-4-9-1, Detail B).

2-4-9.1.2 Install.

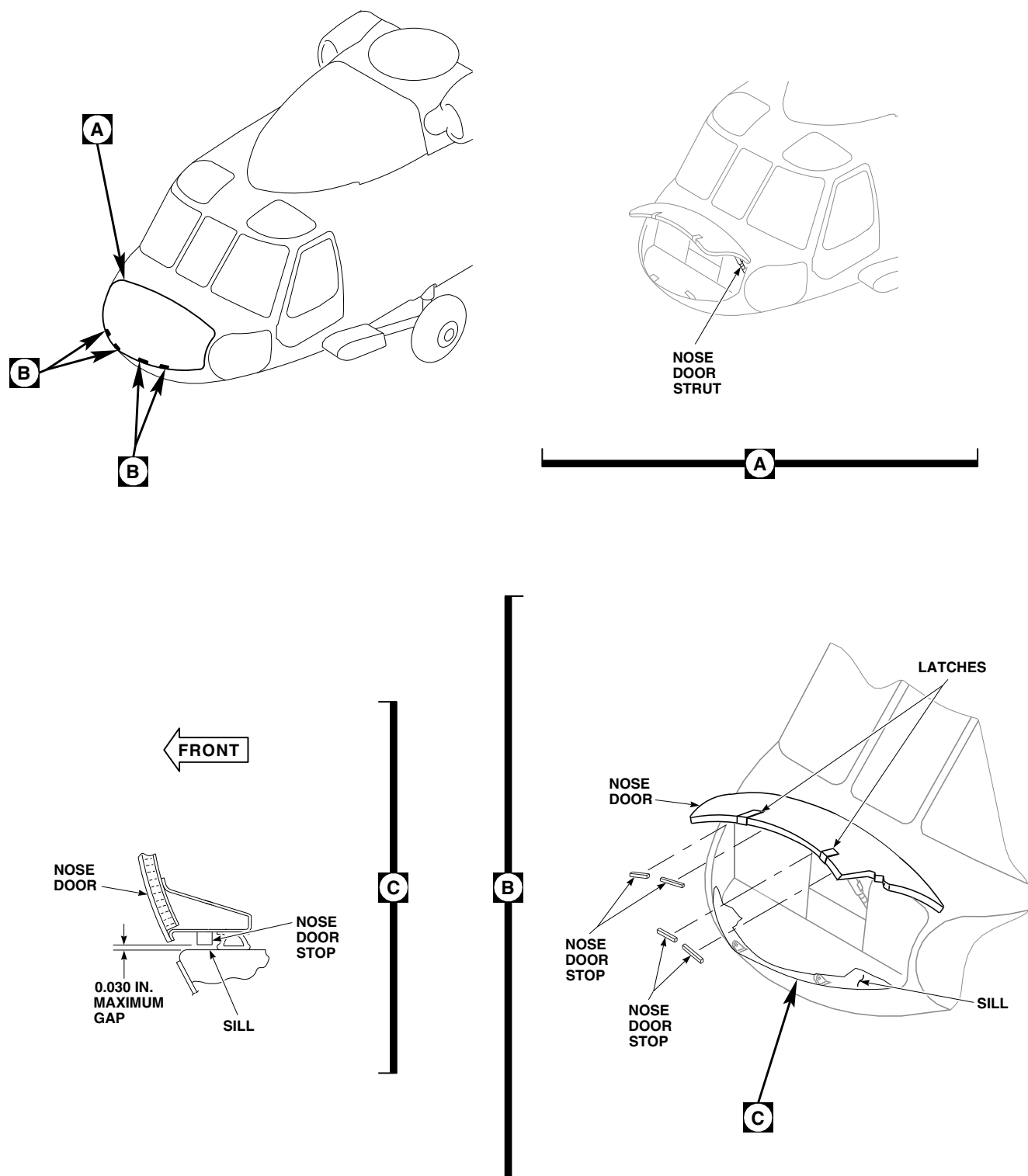
- Apply light coat of adhesive, Item 66, Appendix D, to bonding surface of nose door. Al-

low adhesive to dry at 65° - 80°F (18° - 27°C) for at least 1 hour.

NOTE

Nose door stops, 70217-01010-102 or -109, may be used as an alternate to nose door stop, 70217-01010-112.

- Using adhesive, Item 32, Appendix D, mix 100 parts by weight of part A to 22 parts by weight of part B. Apply light coat of adhesive to door and stop.
- Immediately install stop on door (Figure 2-4-9-1, Detail B and Detail C). Allow adhesive to dry at room temperature for at least 24 hours or for 2 hours at 110° - 130°F (43° - 54°C).
- Unlock nose door strut. Close nose door and fasten latches. Verify stop installation as follows:
 - Check contact of stops (Figure 2-4-9-1, Detail C). **AT LEAST ONE STOP ON**



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FIGURE 2-4-9-1. REPLACE NOSE DOOR STOPS

EITHER SIDE OF LATCHES MUST CONTACT SILL.

- (a) If proper contact is achieved, proceed to step d. (2).
- (b) If proper contact is not achieved, adjust nose door latches (PARA 2-4-2).

- (2) Using feeler gage, measure gaps between other two stops and sill. **MAXIMUM GAP ALLOWED IS 0.030-INCH.**

- (a) If proper gap is achieved, proceed to step e.

- (b) If proper gap is not achieved, adjust nose door latches (PARA 2-4-2).

- e. Unlock nose door and open door. Lock nose door strut in open position (Figure 2-4-9-1, Detail A).

- f. Touch up paint as required (PARA 2-6-5).

- g. Unlock nose door strut. Close nose door and fasten latches.

2-4-10 NOSE DOOR DRAIN HOSES.

PARAGRAPH BREAKDOWN

	PAGE
2-4-10.1 Replace Nose Door Drain Hoses	2-4-10-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

References

- PARA 1-6-16

2-4-10.1 Replace Nose Door Drain Hoses.

NOTE

Replacement of both nose door drain hoses is the same.

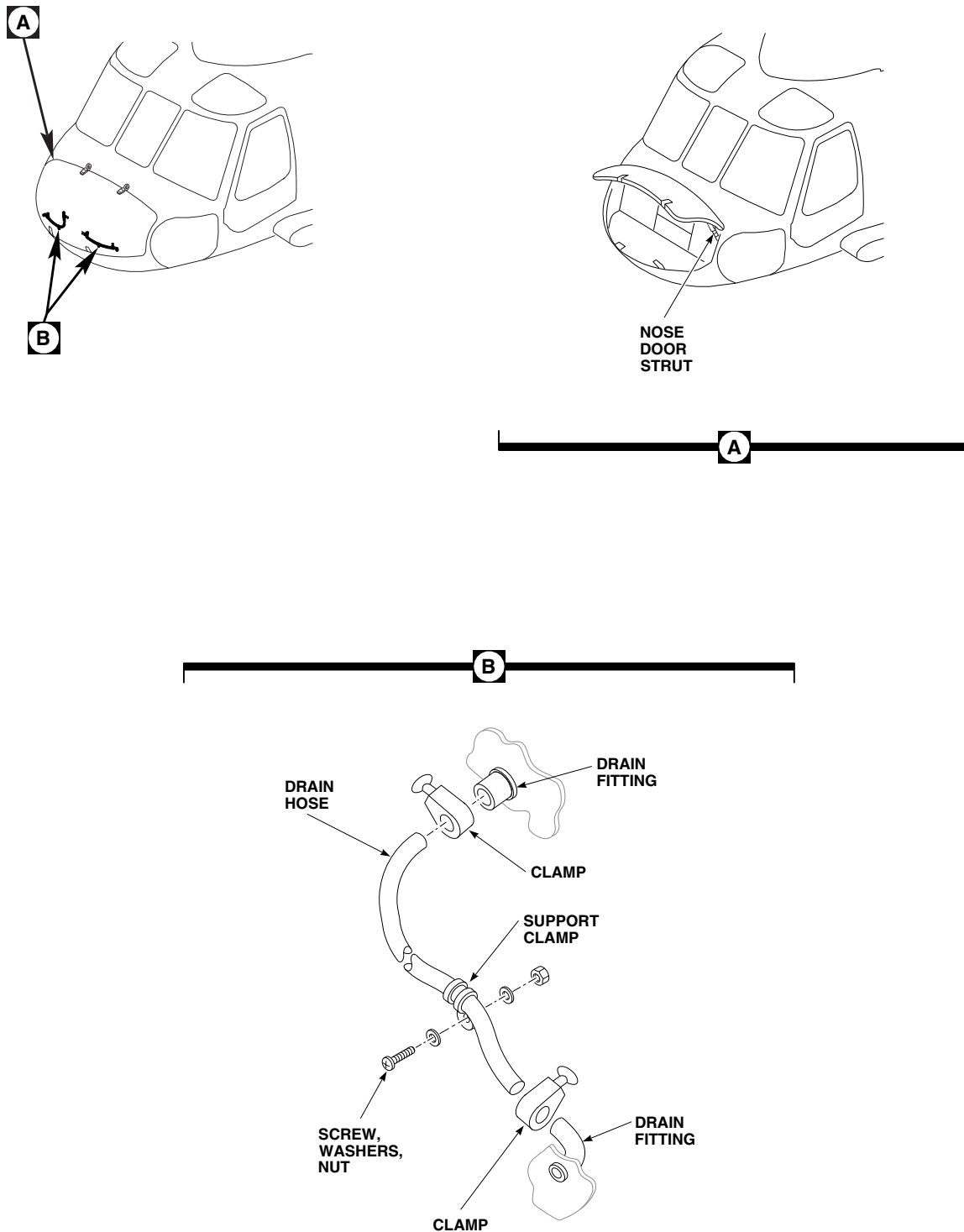
2-4-10.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Unlatch and open nose door. Lock nose door strut in open position (Figure 2-4-10-1, Detail A).
- Loosen clamps securing drain hose to upper and lower drain fittings. Remove screw, wash-

ers, and nut on support clamp. Remove hose (Figure 2-4-10-1, Detail B).

2-4-10.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Loosely assemble support clamp and hose to door using screw, washers, and nut. Fit hose to upper and lower drain fittings and cut to correct length (Figure 2-4-10-1, Detail B).
- Install clamps on hoses, and tighten screw on support clamp.
- Unlock nose door strut. Close nose door and fasten latches (Figure 2-4-10-1, Detail A).



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FIGURE 2-4-10-1. REPLACE NOSE DOOR DRAIN HOSES

2-4-11 COCKPIT DOORS.

PARAGRAPH BREAKDOWN

	PAGE
2-4-11.1 Replace Cockpit Doors	2-4-11-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Feeler Gage

Materials/Parts

- Gasket, 70217-01000-142, -143, and -144
- Grease, Item 151A, Appendix D
- Machinery Towel, Item 211, Appendix D
- Sealing Compound, Item 280, Appendix D

- Tetrachloroethylene, Item 324, Appendix D

References

- Appendix D
- PARA 2-3-2
- PARA 2-4-12
- PARA 2-4-27
- PARA 2-6-5
- PARA 13-4-10

2-4-11.1 Replace Cockpit Doors.

NOTE

Replacement of left and right cockpit doors is the same.

2-4-11.1.1 Remove.

- Open cockpit door.
- While supporting cockpit door, pull jettison handle to disengage door hinges and jettison pin. Remove door from helicopter (Figure 2-4-11-1, Detail A and B).
- Check gaskets under upper and lower door hinges for damage (Figure 2-4-11-1, Detail C). **NONE ALLOWED.** If damaged, remove from hinge.

- If cockpit door is being replaced, remove cockpit door lock (PARA 2-4-27). Retain for installation.

2-4-11.1.2 Install.

NOTE

To keep locksets uniform when replacing cockpit door, install lock removed from old door.

- If removed, install cockpit door lock on replacement door (PARA 2-4-27).

NOTE

Additional gasket under hinge assemblies may be used for a tighter fit.

- If removed, place replacement gaskets, 70217-01000-142, 70217-01000-143, or 70217-

01000-144 (in combination, if required) under hinge (Figure 2-4-11-1, Detail C).

- c. Have assistant pull jettison handle to secure arm assemblies in open position (Figure 2-4-11-1, Detail B).

NOTE

It may be necessary to push on jettison lever to engage arms with hinge screws.

- d. Place cockpit door in position and move jettison handle forward, allowing arms to engage with screws of hinges.
- e. If door edge touches frame, trim edge until edge is clear of frame when door is fully closed.
- f. If cockpit door edge was trimmed, touch up paint on trimmed edge (PARA 2-6-5).
- g. After arms are installed in hinge screws, using feeler gage, measure for 0.010-inch gap between screwheads and arm (Figure 2-4-11-1, Detail D).
- h. If necessary to adjust gap, perform the following:
 - (1) Remove screws.
 - (2) Clean screws and tapped hole using machinery towel, Item 211, Appendix D, dampened with tetrachloroethylene, Item 324, Appendix D, and dry.
 - (3) Coat screw threads with sealing compound, Item 280, Appendix D, and install in hinge.
 - (4) Apply thin coat of grease, Item 151A, Appendix D, between screwheads and arm.

(5) Adjust screws to obtain 0.010-inch gap between screwheads and arm.

- i. Remove heating and ventilation system cockpit door duct (PARA 13-4-10).
- j. Remove screws and washers securing cover to inside of cockpit door, and remove cover (Figure 2-4-11-1, Detail A).
- k. Install jettison pin through check lever (Figure 2-4-11-1, Detail B).
- l. With aid of assistant, perform functional check of door jettison mechanism.
- m. Install cockpit door, inserting jettison pin through check lever and into fitting in door.
- n. Open and close door a few times to check for proper operation. Door shall operate smoothly without binding. If door is binding, adjust door (PARA 2-4-12).
- o. Close cockpit door and check that door latch has engaged door frame. Rotate door handle to latch door and check that door is securely closed and locked. If door does not securely close and lock, perform the following:
 - (1) Adjust cockpit door (PARA 2-4-12).
 - (2) Spray water on door and check for leaks.
 - (3) Repeat step o. (1) and step o. (2) until door closes and locks properly, and no leaks are found.
- p. Perform cockpit door jettison check (PARA 2-3-2).
- q. Install cover on inside of door and secure with screws and washers (Figure 2-4-11-1, Detail A).
- r. Install heating and ventilation system cockpit door duct (PARA 13-4-10).

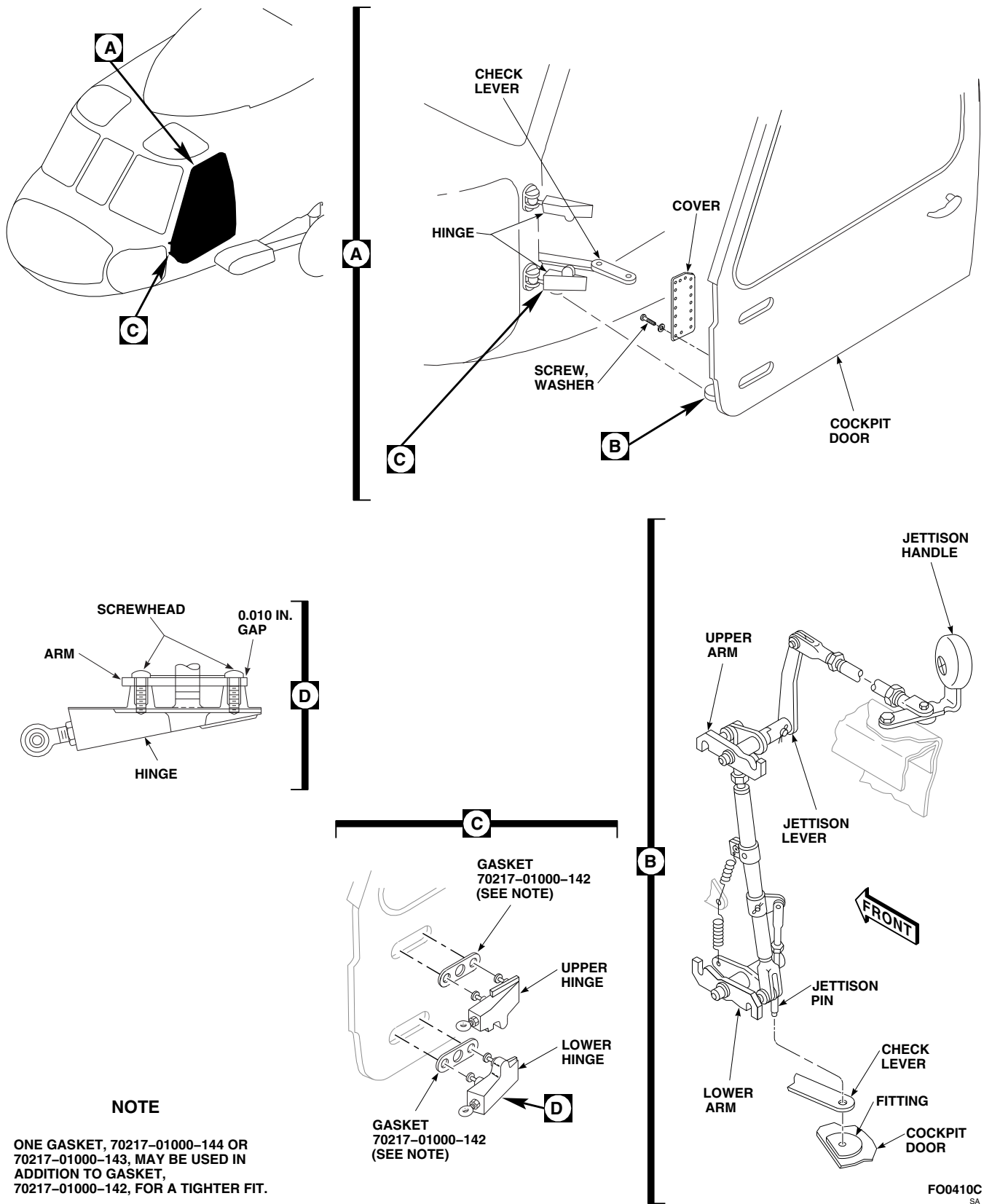


FIGURE 2-4-11-1. REPLACE COCKPIT DOORS

2-4-12 COCKPIT DOOR ADJUSTMENT.

PARAGRAPH BREAKDOWN

	PAGE
2-4-12.1 Adjust Hinges, Latches, and Striker Plates	2-4-12-0

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Shim, 70217-01005-107

References

- PARA 2-3-2
- PARA 2-4-11
- PARA 2-4-20
- PARA 2-4-25

2-4-12.1 Adjust Hinges, Latches, and Striker Plates.

2-4-12.1.1. Adjust Hinges.

NOTE

- Adjustment of right and left cockpit doors is the same.
 - Adjusting both hinge rod ends equally in same direction in or out will move door front and rear.
 - Adjusting hinge rod ends equally in opposite direction to each other will move rear end of door up or down to line up latches with striker plates.
 - To move hinge end of door in to compress seal against frame requires peeling shims under door hinge fittings (PARA 2-4-25).
- Remove cockpit door (PARA 2-4-11).
 - Remove headed pin and washer from hinge to be adjusted (Figure 2-4-12-1, Detail A).

- Loosen jamnut and adjust rod end to lengthen or shorten rod end as required.
- Remove hinge fittings (PARA 2-4-25).

NOTE

- A maximum of 2 new shims may be used under each hinge fitting.
- Add or remove shims to adjust door.
 - Install hinge fittings (PARA 2-4-25).
 - Reinstall headed pin. Do not install cotter pin yet.
 - Open and close door to check adjustment. Continue to adjust door until latchbolt engages and releases properly at striker plate.
 - Once adjusted, tighten jamnut on rod end. **TIGHTEN 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.** Install washer and cotter pin.
 - Install cockpit door (PARA 2-4-11).

2-4-12.1.2. Adjust Latches.

- a. Open cockpit door.

NOTE

If required for proper latch bolt engagement with striker plate, a laminated shim may be installed on the latch housing (PARA 2-4-20).

- b. Operate inside/outside handle and check latchbolt action. Latchbolt must be flush with latch with door handle in full open position. **LATCHBOLT TRAVEL MUST BE AT LEAST 0.600-INCH BETWEEN OPEN AND LOCK POSITIONS.** Handle must turn freely into LOCK (over center) position without preloading linkage (Figure 2-4-12-1, Detail B).
- c. Adjust long vertical rod and short rod to keep both latchbolts same measurement from latch.
- d. Adjust lower horizontal rod to position latchbolt flush with latch when handle is in full open position. Vertical and short rods may have to be adjusted at same time.
- e. Adjust upper horizontal rod with handle in locked position.
- f. Remove headed pin from rod to be adjusted at adjustable end. Loosen jamnut and adjust rod end to lengthen or shorten rod. Install headed pin.
- g. Operate handle and check latchbolt action.

- h. Continue adjustment until latchbolt and lock action meet requirements.

- i. **TIGHTEN JAMNUTS 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.** Install washers and cotter pins.

- j. Shut cockpit door and check latchbolt engagement in striker plate.

- k. Spray water on door and check for leaks.

- l. Perform cockpit door jettison check (PARA 2-3-2).

2-4-12.1.3. Adjust Striker Plates.

- a. Open door.
- b. Close door and make sure striker plate is positioned properly. Make sure door latches and locks without too much force, yet compresses seal enough to prevent leakage.
- c. Loosen screws, move striker plate in or out under serrated plate as necessary. **TIGHTEN SCREWS** (Figure 2-4-12-1, Detail B and Detail C).
- d. Check latch engagement in striker plate (Figure 2-4-12-1, Detail C).
- e. Latchbolt should extend into striker plate 0.250 - 0.500-inch. If necessary add shim, 70217-01005-107, under striker plate.
- f. Perform cockpit door jettison check (PARA 2-3-2).

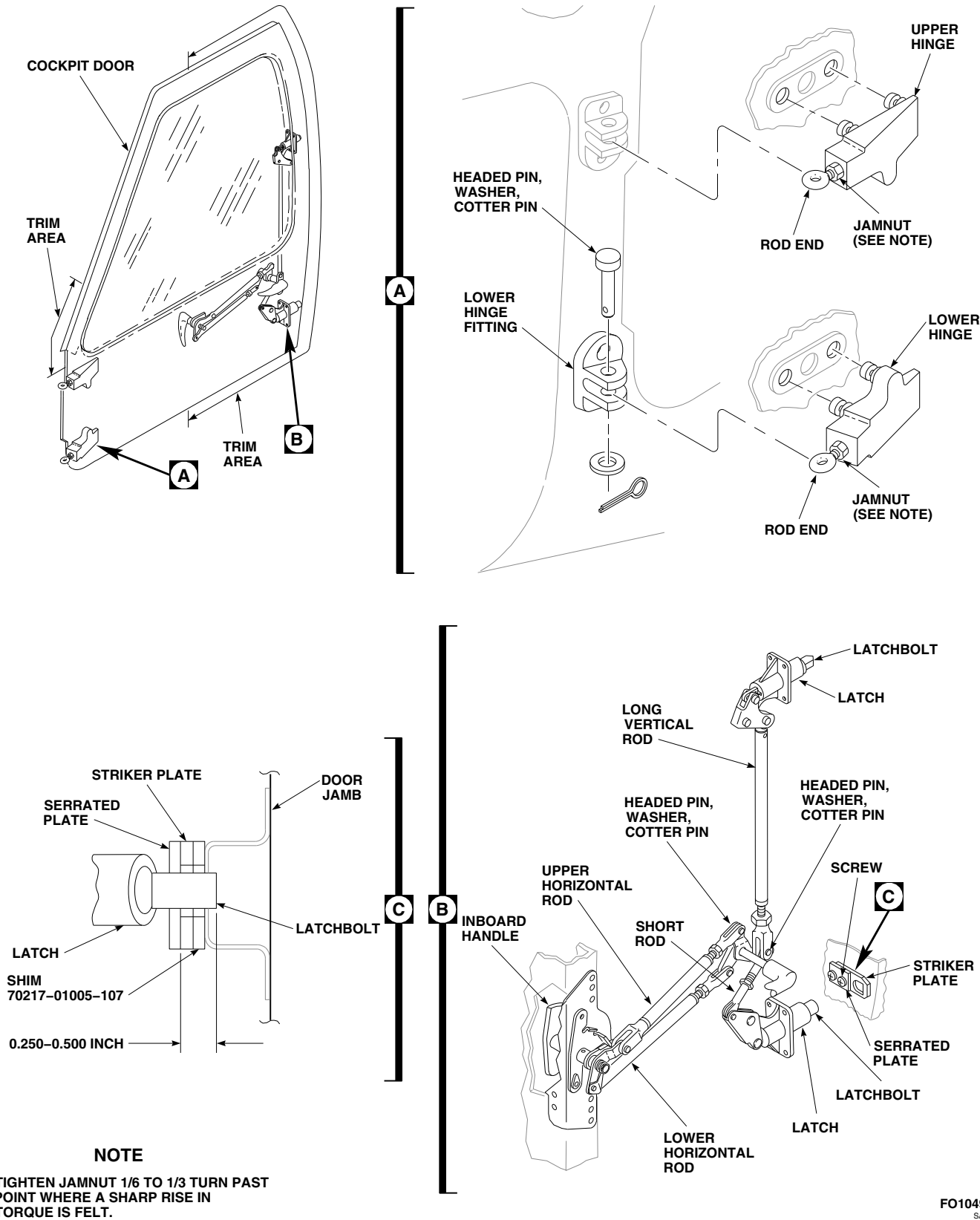


FIGURE 2-4-12-1. ADJUST HINGES, LATCHES, AND STRIKER PLATES

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2-4-13 COCKPIT DOOR CHECK COMPONENTS.

PARAGRAPH BREAKDOWN

	PAGE
2-4-13.1 Replace Cockpit Door Check Components	2-4-13-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Drill Set
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D

- Primer, Item 257, Appendix D
- Sealing Compound, Item 292, Appendix D
- Trichloroethane 1,1,1, Item 341, Appendix D

References

- Appendix D
- PARA 2-3-2
- PARA 2-4-12
- PARA 13-4-10

2-4-13.1 Replace Cockpit Door Check Components.

NOTE

Replacement of right and left cockpit door check components is the same.

2-4-13.1.1. Remove.

- Open cockpit door.
- Remove bolt, washers, and nut attaching lever assembly to door check fitting (Figure 2-4-13-1, Detail A).
- If replacing lever assembly, remove bolt, washers, and nut attaching lever assembly to lever.
- If replacing lever, perform the following:

- Remove bolt, washers, and nut attaching lever assembly to lever.
- Pull jettison pin to release lever assembly from door.
- Remove cotter pins and washers attaching spring to fitting. Pull spring out of fitting.
- If worn, press out bushings in fitting (Figure 2-4-13-1, Detail B).
- If fitting needs replacement, remove as follows:
 - Drill out rivets fastening flange of fitting to frame.
 - Remove swage lockpins fastening fitting body to frame.

- (3) Remove fitting from frame.

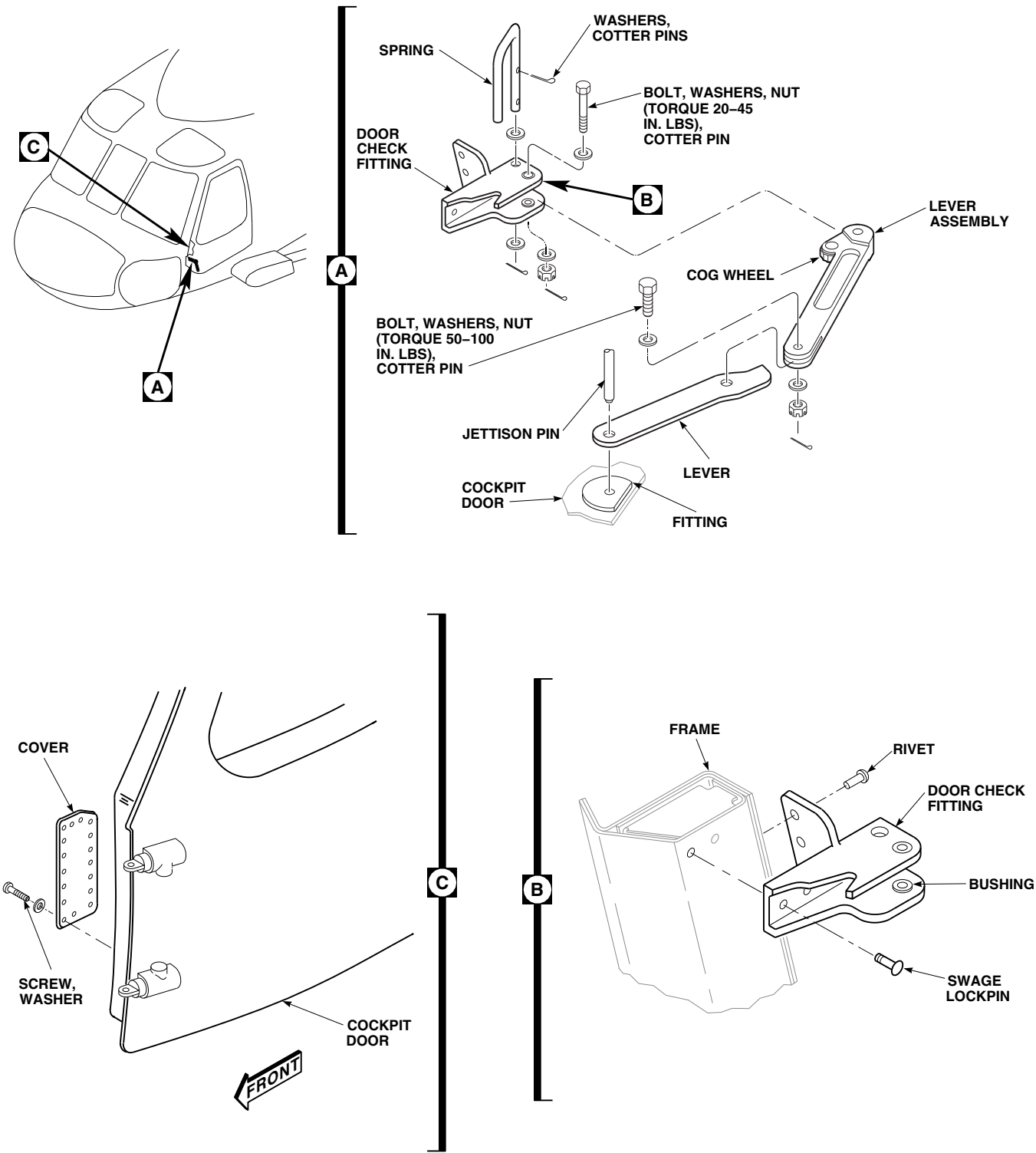
2-4-13.1.2. Install.

- a. If fitting has been removed from frame, perform the following (Figure 2-4-13-1, Detail B):
 - (1) Thoroughly clean mating surfaces of fitting and frame using cheesecloth, Item 90, Appendix D, wet with trichloroethane 1,1,1, Item 341, Appendix D.
 - (2) Wipe dry with clean cheesecloth.
 - (3) Apply sealing compound, Item 292, Appendix D, or using epoxy primer coating kit, Item 135, Appendix D, apply epoxy primer to all mating surfaces and fasteners.
 - (4) Install fitting while coating is still wet.
 - (5) Install 2 swage lockpins.
 - (6) Install 3 rivets.
 - (7) Fair coating squeeze out into fillet where possible.
- b. If removed, coat outside diameter of replacement bushings, with primer, Item 257, Appendix D, and press into fitting while primer is wet.
- c. Install a washer on leg of spring that has holes in it. Insert that leg of spring into forward hole of fitting. Install remaining washer and secure with cotter pin (Figure 2-4-13-1, Detail A).
- d. If lever assembly was replaced, install bolt, washers, and nut attaching lever assembly to lever. **TORQUE NUT TO 50 - 100 INCH-POUNDS.** Install cotter pin.
- e. Attach lever assembly to fitting with bolt, washers, and nut. **TORQUE NUT TO 20 - 45 INCH-POUNDS.** Install cotter pin.
- f. If lever was replaced, perform the following:
 - (1) Remove heating and ventilation system cockpit door duct (PARA 13-4-10).
 - (2) Remove screws and washers securing cover to inside of door, and remove cover (Figure 2-4-13-1, Detail C).
 - (3) Connect lever to door by inserting jettison pin through lever and into fitting in door (Figure 2-4-13-1, Detail A).
 - (4) Install bolt, washers, and nut attaching lever assembly to lever. **TORQUE NUT TO 50 - 100 INCH-POUNDS.**
 - (5) Install cover on inside of door and secure with screws and washers (Figure 2-4-13-1, Detail C).
 - (6) Install heating and ventilation system cockpit door duct (PARA 13-4-10).

NOTE

Make sure cog wheel on lever assembly will engage spring when door is open.

- g. Open and close cockpit door a few times to check for proper operation. Adjust door if required (PARA 2-4-12).
- h. Close door and check that door latch has engaged the door frame. Rotate door handle to latch door and check that door is securely closed and locked. If door does not securely close and lock, adjust door (PARA 2-4-12).
- i. Spray water on door and check for leaks.
- j. Perform cockpit door jettison check (PARA 2-3-2).



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FIGURE 2-4-13-1. REPLACE COCKPIT DOOR CHECK COMPONENTS

2-4-14 COCKPIT DOOR JETTISON COMPONENTS.

PARAGRAPH BREAKDOWN

	PAGE
2-4-14.1 Replace Cockpit Door Jet- tison Components	2-4-14-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Gasket, 70217-01000-142, -143, and -144

References

- PARA 2-4-11
- PARA 2-4-12
- PARA 13-4-10

2-4-14.1 Replace Cockpit Door Jettison Components.

NOTE

Replacement of right and left cockpit door jettison components is the same.

2-4-14.1.1. Remove.

- Remove heating and ventilation system cockpit door duct (PARA 13-4-10).
- Remove screws and washers securing cover to inside of door, and remove cover (Figure 2-4-14-1, Detail A).
- With assistant holding door, pull jettison handle. If door does not jettison, reach through door opening and disengage arms and jettison pin by pulling up on tube assembly (Figure 2-4-14-1, Detail B).
- Remove cockpit door from helicopter.

- Check condition of gaskets under door hinge. If damaged, remove from hinge.
- Remove headed pin and washer attaching control rod to jettison handle.
- Remove headed pin and washer attaching control rod to jettison lever. Remove control rod.
- Disconnect springs from tube assembly and lower arm.
- Remove headed pin and washer securing upper end of tube assembly to upper arm.
- Remove headed pin and washer securing lower end of tube assembly to lower arm. Remove tube assembly.
- Remove headed pin and washer attaching jettison pin to tube assembly.
- Remove headed pin attaching jettison lever to arm pin A. Remove jettison lever.

- m. Remove retaining ring securing arm pin A in door. Remove arm pin and upper arm.
- n. Remove headed pin attaching sleeve to arm pin B. Remove sleeve.
- o. Remove retaining ring securing arm pin B in door. Remove arm pin and lower arm.
- p. Remove bolt, spacer, washers, and nut and remove jettison handle.

2-4-14.1.2. Install.

- a. Install jettison handle with bolt, spacer, washers, and nut. **TORQUE NUT TO 95 - 105 INCH-POUNDS.** Install cotter pin (Figure 2-4-14-1, Detail B).
- b. Install arm pin B in door and secure with retaining ring.
- c. Install lower arm and sleeve on arm pin B and fasten sleeve on arm pin with headed pin. Install cotter pin.
- d. Install arm pin A in door and secure with retaining ring.
- e. Install upper arm and jettison lever on arm pin A and fasten lever on arm pin A with headed pin. Install cotter pin.
- f. Install jettison pin in fitting on bottom of door and attach jettison pin clevis to tube assembly with headed pin and washer. Install cotter pin.
- g. Connect lower end of tube assembly to lower arm with headed pin and washer. Install cotter pin.
- h. Connect upper end of tube assembly to upper arm with headed pin and washer. Install cotter pin.
- i. Connect springs to tube assembly and to lower arm.
- j. Attach control rod to jettison lever with headed pin and washer. Install cotter pin.
- k. Attach control rod to jettison handle with headed pin and washer. Install cotter pin.

- l. Adjust jettison components, as follows:

- (1) Loosen jamnuts on control rod and turn rod to increase or decrease length until jettison handle is in full forward position. **TIGHTEN JAMNUTS.**

NOTE

An additional gasket under hinge assemblies may be used for a tighter fit.

- (2) If removed, place replacement gaskets, 70217-01000-142, 70217-01000-143, or 70217-01000-144 (in combination, if required), under hinge.
- (3) Have assistant pull jettison handle to hold arm assemblies in open position.

NOTE

It may be necessary to push on jettison lever to engage arms with hinge screws.

- (4) Place door in position on helicopter and place hinges in door, move jettison handle forward allowing arms to engage with screws of hinges.
- (5) Check for 0.010-inch gap between screw-heads and arm. If necessary, adjust screw (PARA 2-4-11).
- (6) If necessary, remove headed pin and washer connecting tube assembly to upper arm, loosen jamnut, and adjust clevis to put both arms in the engaged position. Tighten jamnut and connect tube assembly to upper arm with headed pin and washer. Install cotter pin.
- m. Test jettison mechanism for operation and full engagement of arms and of jettison pin in door check lever. Adjust if necessary (PARA 2-4-12).
- n. Install cockpit door (PARA 2-4-11).
- o. Install heating and ventilation system cockpit door duct (PARA 13-4-10).

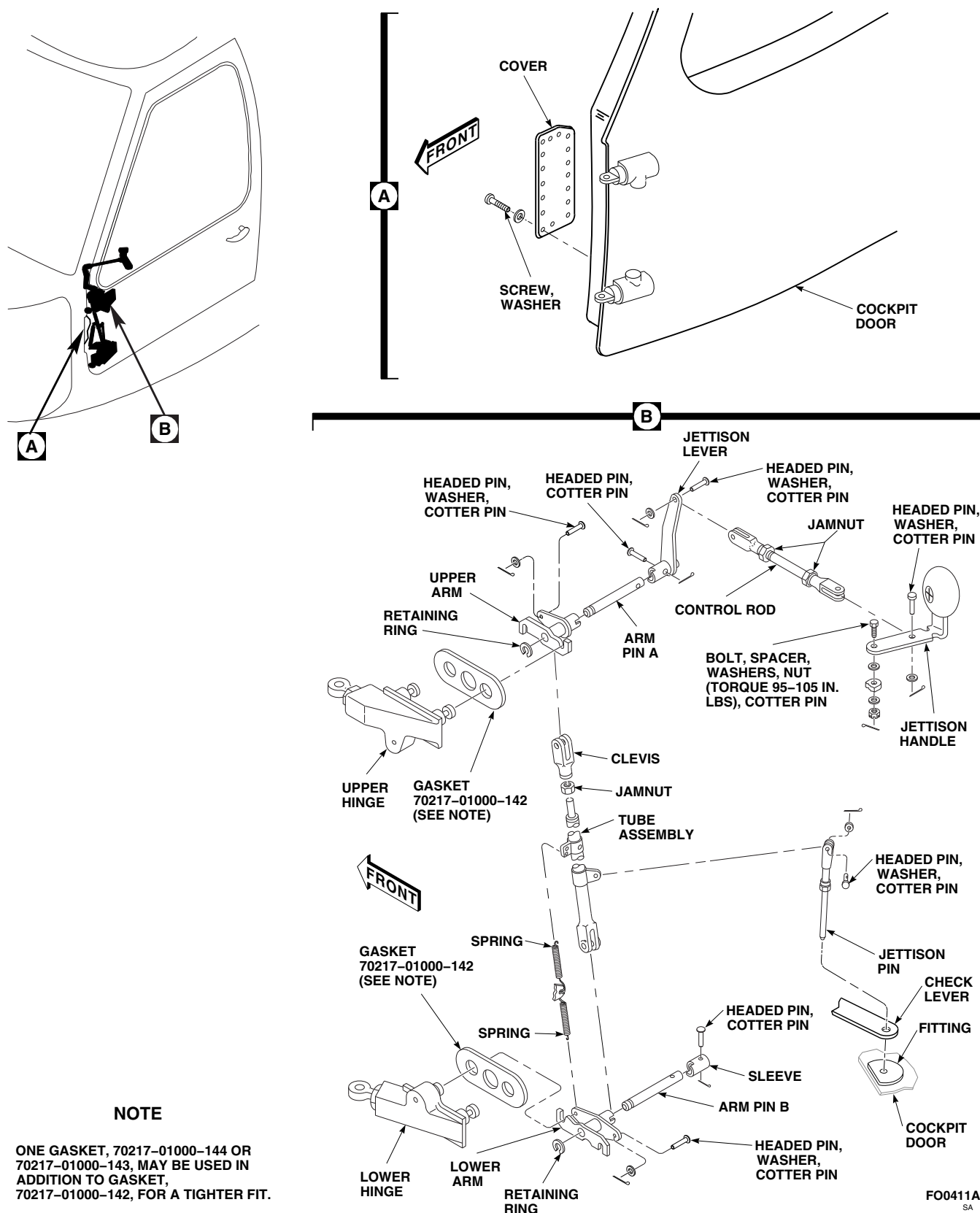


FIGURE 2-4-14-1. REPLACE COCKPIT DOOR JETTISON COMPONENTS

Change 1

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2-4-15 COCKPIT DOOR RETAINER AND CLIP.

PARAGRAPH BREAKDOWN

	PAGE
2-4-15.1 Replace Cockpit Door Retainer and Clip	2-4-15-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Clecos, 3⁄16-inch diameter
- Drill
- Drill Bit, No. 11 (0.19-inch diameter)
- Nonmetallic Scraper, Locally-Made
- Pin Punch, 5⁄32-inch diameter

Materials/Parts

- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 2-3-2
- PARA 2-4-12
- PARA 2-6-5

2-4-15.1 Replace Cockpit Door Retainer and Clip.

NOTE

Replacement of left and right cockpit door retainer and clip is the same.

2-4-15.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Open cockpit door.
- Remove screws, washers, and nuts securing retainer plate and shims to cockpit door. Remove retainer plate and shims (Figure 2-4-15-1, Detail A).

- Check condition of clip riveted to canopy structure for damage. **NONE ALLOWED.** If damaged, replace clip as follows:

- Using drill with No. 11 drill bit and 5⁄32-inch pin punch, carefully drill out and remove four rivets securing clip to canopy structure.
- Using nonmetallic scraper and machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D, remove sealing compound between clip and structure.
- Remove clip and clean up rivet debris.

- (4) Using drill with No. 11 drill bit, open pilot holes in clip. Deburr holes.
- (5) Position clip on canopy structure and secure in place using three $\frac{3}{16}$ -inch diameter clecos.
- (6) Check open/close function of cockpit door for clearance of retainer plate in clip slot.
- (7) Install blind rivet in remaining hole.
- (8) Remove one cleco from diagonal corner hole and install rivet.
- (9) Remove remaining clecos and install rivets.
- (10) Using sealing compound, Item 292, Appendix D, seal all gaps between clip and structure.

2-4-15.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Position retainer plate and shims on cockpit door. Install short screw, washer, and nut in front hole, with washer and nut against retainer plate. Install remaining screws, washers, and nuts with screwheads against retainer plate (Figure 2-4-15-1, Detail A).

- c. Adjust retainer plate as follows:
 - (1) Close cockpit door.
 - (2) Check clearance between outer edge of retainer plate and slot in clip.
 - (3) Peel shims as necessary to make clearance 0.03 - 0.09-inch (Figure 2-4-15-1, Detail B).
 - (4) Mount retainer plate loosely with screws, washers, and nuts (Figure 2-4-15-1, Detail A).
 - (5) Close cockpit door.
 - (6) Adjust retainer plate until tip is 0.25 - 0.31-inch beyond clip (Figure 2-4-15-1, Detail B).
 - (7) **TIGHTEN SCREWS.**
- d. Open and close cockpit door a few times to check operation. If required, adjust door (PARA 2-4-12).
- e. Spray water on cockpit door and check for leaks.
- f. Perform cockpit door jettison inspection (PARA 2-3-2).
- g. Touch up paint in repaired area (PARA 2-6-5).

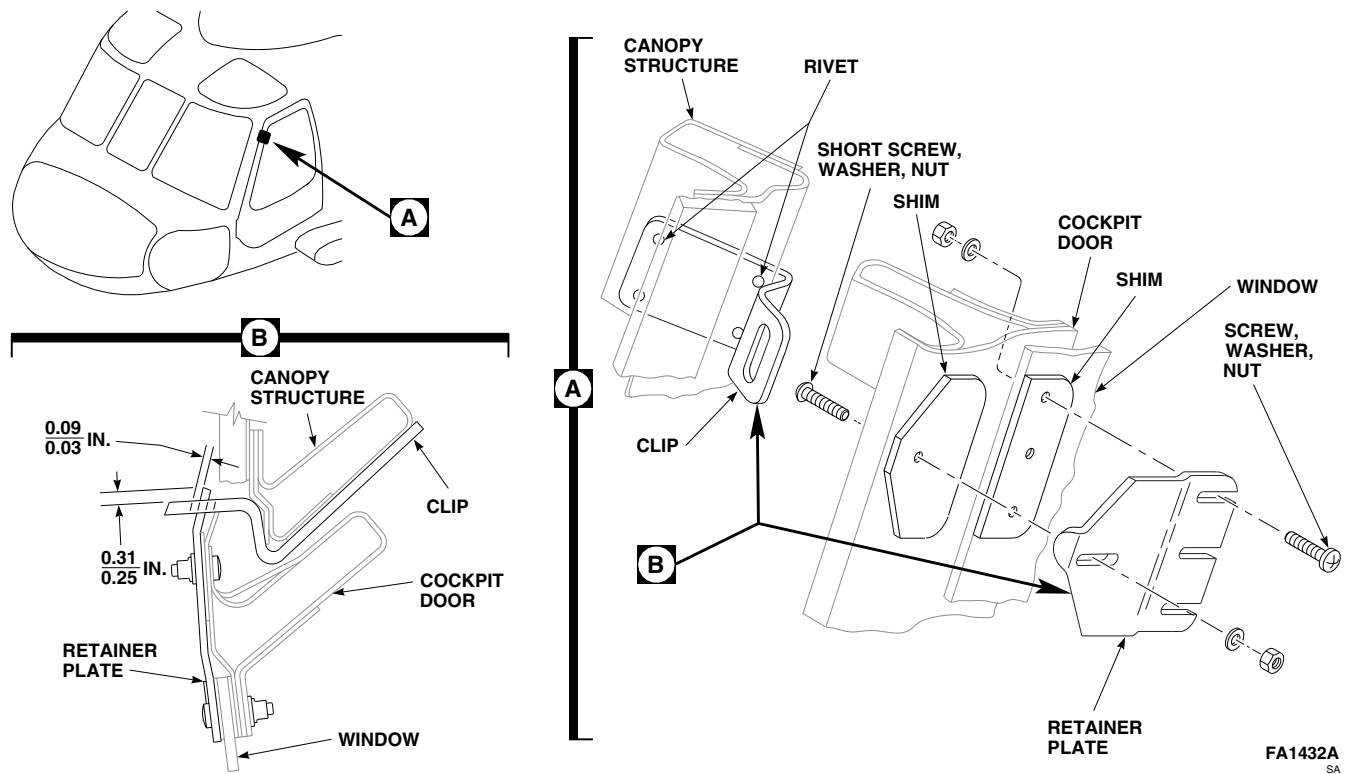


FIGURE 2-4-15-1. REPLACE COCKPIT DOOR RETAINER AND CLIP

2-4-16 COCKPIT DOOR STATIONARY WINDOW.

PARAGRAPH BREAKDOWN

	PAGE
2-4-16.1 Replace Cockpit Door Stationary Window	2-4-16-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Drill
- Drill Bit, 7/32-inch (0.2188-inch diameter)
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Lockwire, Item 195, Appendix D
- Paint Remover, Item 231, Appendix D

- Pressure Sensitive Adhesive Tape, Item 244C, Appendix D

- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-15
- PARA 2-4-17
- PARA 2-6-5

2-4-16.1 Replace Cockpit Door Stationary Window.

NOTE

Replacement of left and right cockpit door stationary window is the same.

2-4-16.1.1 Remove.

- Remove cockpit door vent window (PARA 2-4-17).
- Remove lockwire (breakaway wire) securing jettison handle to clip on lower front edge of cockpit door window frame.

- Remove screws, washers, and nuts securing window and clip to cockpit door frame (Figure 2-4-16-1, Detail A).
- Loosen screws, washers, and nuts securing upper cockpit door retainer to door. Do not remove from door (PARA 2-4-15).
- Remove window and clip from cockpit door.
- Remove pressure sensitive adhesive tape from cockpit door and clean flanges with paint remover, Item 231, Appendix D. Wipe dry.

2-4-16.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).

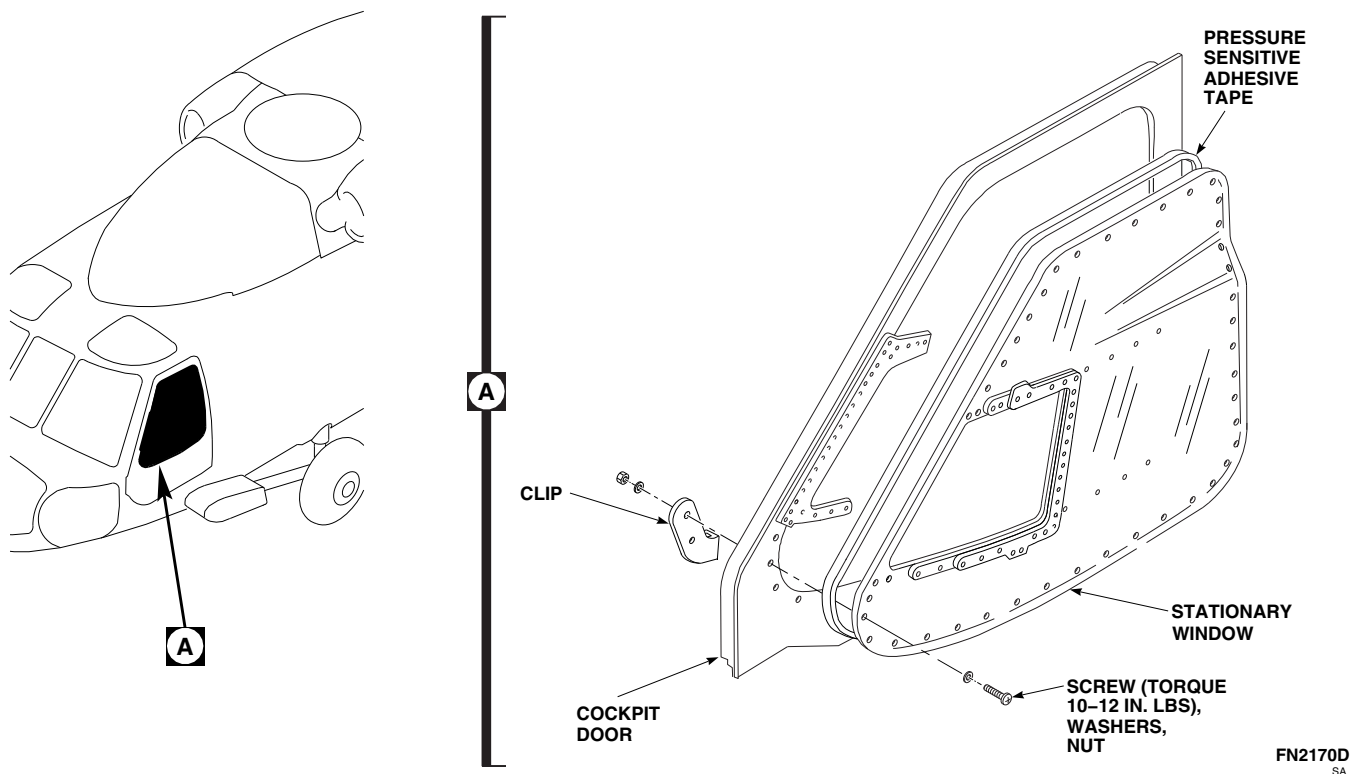


FIGURE 2-4-16-1. REPLACE COCKPIT DOOR STATIONARY WINDOW

b. Trial fit window in frame. Trim window, if necessary, to seat in window frame (Figure 2-4-16-1, Detail A).

c. If holes in window and door frame do not line up, perform the following:

- (1) Mark holes in frame that do not line up.
- (2) Fill holes and touch up paint area (PARA 2-6-5).
- (3) Using replacement window as template, mark new holes.
- (4) Using drill with $\frac{7}{32}$ -inch diameter drill bit, drill new holes through frame.

d. Apply pressure sensitive adhesive tape, Item 244C, Appendix D, on flanges of replacement window, with sticky side against flange. Trim tape as necessary.

e. Coat pressure sensitive adhesive tape with a thin coating (about 0.003-inch) of sealing compound, Item 292, Appendix D.

CAUTION

To prevent window from cracking, do not overtighten securing screws.

- f. Position window on cockpit door. Place clip on lower front edge of door window frame. Install screws, washers, and nuts. Tighten screws until window touches tape. **TORQUE SCREWS TO 10 - 12 INCH-POUNDS.**
- g. Tighten screws, washers, and nuts securing upper cockpit door retainer to door (PARA 2-4-15).
- h. Fill any gaps between window and frame with sealing compound, Item 292, Appendix D. Smooth out sealing compound until surface is flush with window and frame.
- i. Install cockpit door vent window (PARA 2-4-17).
- j. With assistant inside, spray window with water and check for leaks, especially around window

edge. Seal any leaks with sealing compound, Item 292, Appendix D.

- k. Using lockwire, Item 195, Appendix D, lockwire jettison handle to clip on lower front edge of door window frame.

2-4-17 COCKPIT DOOR VENT WINDOW.

PARAGRAPH BREAKDOWN

	PAGE
2-4-17.1 Replace Cockpit Door Vent Window	2-4-17-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Drill
- Drill Bit, No. 21
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Abrasive Cloth, Item 3, Appendix D
- Adhesive, Item 37, Appendix D

- Adhesive, Item 57, Appendix D
- Cheesecloth, Item 90, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 275, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-4-18
- PARA 1-6-16
- PARA 2-6-2

2-4-17.1 Replace Cockpit Door Vent Window.

NOTE

Replacement of left and right cockpit door vent windows is the same.

2-4-17.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Open cockpit door. Move vent window to open position.

- Remove screws, washers, and nuts securing upper vent window retainer and radius block to window (Figure 2-4-17-1, Sheet 1, Detail B).
- Remove screws, washers, and nuts securing lower vent window retainer and radius block to window (Figure 2-4-17-1, Sheet 1, Detail C).
- Remove screws, sealing washers, washers, and nuts securing vent window lower track to window (Figure 2-4-17-1, Sheet 2, Detail D). Remove track.
- Lift vent window and pull inboard slightly to remove window from upper track.

- g. Remove screws, sealing washers, washers, and nuts securing upper track to window. Remove track.

2-4-17.1.2 Inspect.

- a. Check condition of sheet metal doubler (Figure 2-4-17-1, Sheet 2, Detail E). If damaged, repair (PARA 2-6-2) or replace doubler (PARA 2-4-17.1.3).
- b. Check condition of vent window seals. If damaged or dried out, replace seals (PARA 2-4-17.1.4).

2-4-17.1.3 Replace Doubler.

- a. Remove screws, washers, and nuts securing large doubler to window (Figure 2-4-17-1, Sheet 2, Detail D).
- b. Using drill with No. 21 drill bit, carefully drill out rivets securing doubler to door and window.

NOTE

Do not discard doubler. Replacement doubler does not have rivet holes pre-drilled. Doubler will be used as a template for replacement doubler.

- c. Remove doubler from window and clean up rivet debris.
- d. Remove small doublers from large doubler using nonmetallic scraper.
- e. If required, replace cockpit door vent window seals (PARA 2-4-17.1.4).
- f. Position replacement doubler on window and secure with screws, washers, and nuts.
- g. Mark 11 rivet holes from window into rear edge of doubler.
- h. Remove screws, washers, nuts, and doubler from window.
- i. Using old doubler as a template, mark 18 rivet holes in front edge of replacement doubler.

- j. Using drill with No. 21 drill bit, drill marked rivet holes on front and rear of large doubler.

- k. Trial fit doubler to window as follows:

- (1) Install doubler with screws, washers, and nuts.
- (2) If rivet holes do not line up, perform the following:

- (a) Remove screws, washers, nuts, and doubler from window.
- (b) Screw holes may be opened up to 0.250-inch, as needed, to allow for proper alignment of holes.
- (c) Repeat step k. (1) and k. (2) until holes line up.

- (3) With doubler still secured to window, check for voids between doubler and window. If voids are found, perform the following:

- (a) Mark voids on doubler.
- (b) Remove screws, washers, nuts, and doubler from window.
- (c) Apply sealing compound, Item 275, Appendix D, to large doubler, as required, to fill voids and provide flush fit between doubler and window.
- (d) Install doubler with screws, washers, and nuts.
- (e) Repeat step k. (3) until no voids exist.

- l. Apply sealing compound, Item 292, Appendix D, to area of window that mates with large doubler.

- m. Position small doublers on large doubler and secure to window with screws, washers, and nuts.

- n. Secure doubler to door and window with rivets.

2-4-17.1.4 Replace Seals.

- a. Remove doubler (PARA 2-4-17.1.3).
- b. Using nonmetallic scraper, remove damaged seals from tracks or doubler.

CAUTION

Plastic windows are damaged by contact with adhesives or adhesive solvents. Do not spill adhesive compounds or solvents on windows.

- c. Remove old adhesive using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.
- d. Trim new seals to correct size.
- e. Roughen replacement seals, tracks, and doubler contact surfaces using abrasive cloth, Item 3, Appendix D.

NOTE

Oil from fingers will weaken adhesive bond. Avoid touching adhesive contact surfaces after cleanup or after application of adhesive compound.

- f. Remove debris from roughened parts with clean cheesecloth, Item 90, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D. Wipe parts clean and dry.
- g. Apply thin, even coat of adhesive, Item 37, Appendix D, to seal and track or doubler bonding surfaces. Allow adhesive to air-dry for 5 - 10 minutes or until tacky.

NOTE

Do not stretch seal when installing around corners of doubler.

- h. Align and secure seals to bonding surfaces. Press seals firmly in place. Work out wrinkles or trapped air.

- i. Allow seal to cure 24 hours at 68°F (20°C).
- j. Clean window as required (PARA 1-4-18).
- k. Install doubler (PARA 2-4-17.1.3).

2-4-17.1.5 Install.

- a. Turn off all electrical power (PARA 1-6-16).

CAUTION

To prevent cracking window, do not overtighten attaching nuts.

- b. Spot-bond plastic shims to window with adhesive, Item 57, Appendix D (Figure 2-4-17-1, Sheet 2, Detail D).
- c. Apply sealing compound, Item 292, Appendix D, under sealing washers on cockpit door window screws.
- d. Position upper track in place. Install upper retainer, radius block, screws with sealing washers under screwheads, washers, and nuts. **TORQUE NUTS TO 7 - 9 INCH-POUNDS.**
- e. Place vent window into upper track in rear (open) position.
- f. Position lower track in place. Install lower retainer and radius block, screws with sealing washers under screwheads, washer, and nuts. **TORQUE NUTS TO 7 - 9 INCH-POUNDS.**
- g. Slide vent window open and closed to check operation.
- h. If window binds in track, trim top and bottom of window as required, or in lieu of trimming vent window, drill out 0.210-inch diameter screwholes in upper track and retainer, as required, to a maximum of 0.250-inch diameter, so track can be adjusted to eliminate vent window binding.

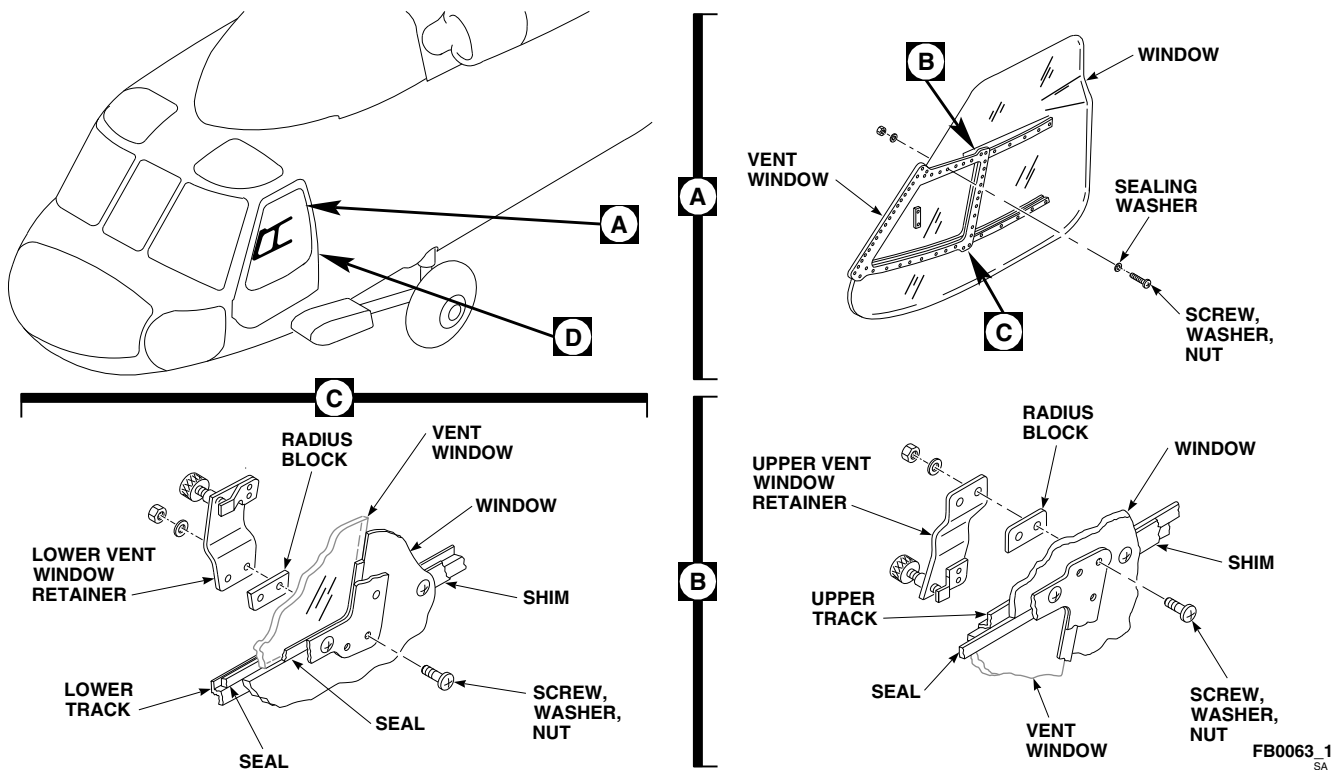


FIGURE 2-4-17-1. REPLACE COCKPIT DOOR VENT WINDOW (SHEET 1 OF 2)

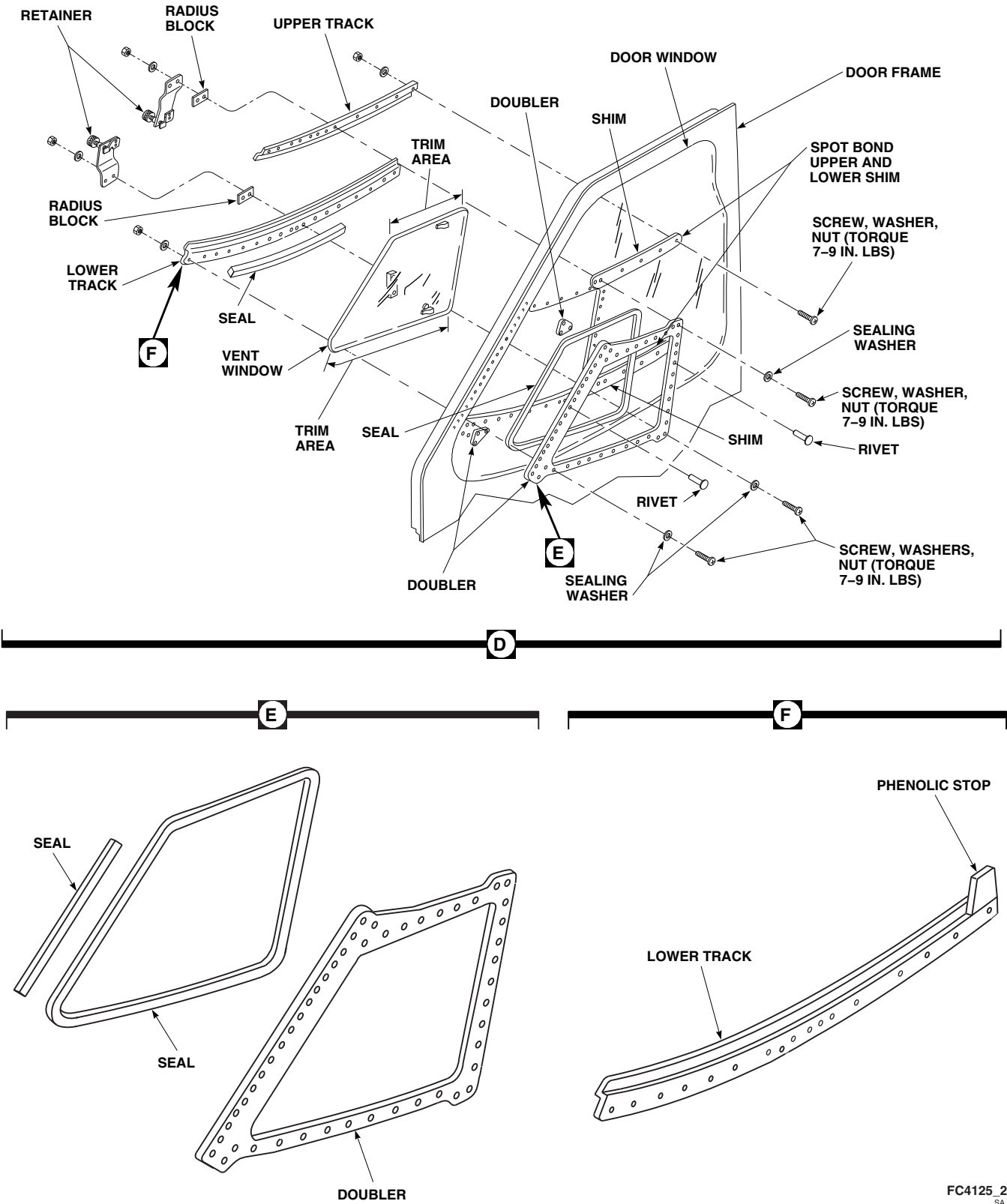


FIGURE 2-4-17-1. REPLACE COCKPIT DOOR VENT WINDOW (SHEET 2 OF 2)

2-4-18 COCKPIT DOOR JETTISONABLE WINDOW.

PARAGRAPH BREAKDOWN

		PAGE
2-4-18.1	Replace Cockpit Door Jet-tisonable Window	2-4-18-1
2-4-18.2	Replace Cockpit Door Jet-tisonable Window As-sembly Frame	2-4-18-5

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Abrasive Cloth, Item 6, Appendix D
- Adhesive, Item 47, Appendix D
- Adhesive, Item 66E, Appendix D
- Brush, Acid, Item 84, Appendix D
- Cheesecloth, Item 90, Appendix D
- Fastener Tape, Pile, Item 137B, Appendix D

- Flannel Cloth, Item 140, Appendix D
- Lockwire, Item 195, Appendix D
- Masking Tape, 1-inch, Item 212, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Paint Remover, Item 231, Appendix D
- Pressure Sensitive Adhesive Tape, Item 244C, Appendix D
- Sealing Compound, Item 292, Appendix D
- Stirring Stick, Beverage, Item 315C, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-15
- PARA 2-6-5

2-4-18.1 Replace Cockpit Door Jettisonable Window.

NOTE

Replacement of left and right cockpit door jettisonable windows is the same.

2-4-18.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Open cockpit door. Have assistant hold door during removal of window.

CAUTION

To prevent damage to jettisonable window, extreme care must be taken during removal.

- c. Pull emergency strap inboard to disengage jettisonable window from seal (Figure 2-4-18-1, Detail B).
- d. Push jettisonable window outboard, through window opening, and remove from helicopter.
- e. Close cockpit door.

2-4-18.1.2 Inspect/Repair.

- a. Visually check condition of adhesive application on interface of jettisonable window and seal as follows:
 - (1) **NO ADHESIVE ALLOWED BETWEEN JETTISONABLE WINDOW AND SEAL.** If adhesive is present between jettisonable window and seal, remove jettisonable window and replace seal.
 - (2) **BEAD OF ADHESIVE SHALL NOT EXCEED $\frac{3}{16}$ -INCH.** If adhesive exceeds limit, replace seal.
- b. Check condition of jettisonable window seal. If damaged or dried out, replace seal as follows:
 - (1) Starting at twelve o'clock position, pull seal from door frame (Figure 2-4-18-1, Detail A).

WARNING

Improper installation of window seal may result in window being blown off helicopter and subsequent damage to helicopter and/or injury to personnel.

NOTE

- Long lip of seal must rest against window when window is installed.

- Make sure split in replacement seal is at twelve o'clock position on door frame.
- (2) While making sure long lip of seal will rest on window, position split in seal at twelve o'clock, push replacement seal over door frame, continue applying seal around entire frame (Figure 2-4-18-1, Detail A and Detail C).
- c. Check condition of emergency strap. If damaged, replace strap as follows:
 - (1) Remove screws, washers, doublers, and nuts securing strap to jettisonable window (Figure 2-4-18-1, Detail D).
 - (2) Remove adhesive from window.
 - (3) Apply adhesive, Item 66E, Appendix D, to screws and mating surfaces of doublers.
 - (4) Position replacement strap on window, and secure with screws, washers, doublers, and nuts. Wet install screws.
 - d. Check condition of pile fastener tape on cockpit door frame for damage and separation. **NO DAMAGE ALLOWED. SEPARATIONS MUST BE REPAIRED.** Repair or replace as follows:
 - (1) If pile fastener tape is not damaged but separated from door frame, repair as follows (Figure 2-4-18-1, Detail B):
 - (a) Using cheesecloth, Item 90, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D, clean exposed door frame surface.
 - (b) Using acid brush, Item 84, Appendix D, apply adhesive, Item 47, Appendix D, to door frame.
 - (c) Using light hand pressure, install loose pile fastener tape against door frame.
 - (2) If pile fastener tape is damaged, replace as follows:

- (a) Using nonmetallic scraper, remove pile fastener tape from door frame.
- (b) Using cheesecloth, Item 90, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D, remove adhesive residue from door frame.
- (c) Using abrasive cloth, Item 6, Appendix D, lightly abrade surface of door frame where pile fastener tape is located.
- (d) Using cheesecloth, Item 90, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D, wipe surface.
- (e) Using acid brush, Item 84, Appendix D, apply adhesive, Item 47, Appendix D, to door frame surface where pile fastener tape will be located.
- (f) Using light hand pressure, install pile fastener tape, Item 137B, Appendix D, on door frame.

2-4-18.1.3 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Open cockpit door. Have assistant hold door during installation of window.

WARNING

Improper installation of window seal may result in window being blown off helicopter and subsequent damage to helicopter and/or injury to personnel.

CAUTION

- To prevent damage to jettisonable window, extreme care must be taken during installation.

- Make sure window is installed into seal on inboard side of door frame.

NOTE

- Make sure long lip of seal rests on window (Figure 2-4-18-1, Detail C).
 - Make sure split in seal is at twelve o'clock position.
 - Do not use lubricant when installing jettisonable window. Powdery residue present on seal is adequate for installation.
- c. Starting in area of door latch, insert window from inboard side of door into seal. Flex and manipulate window as required to properly engage seal. If necessary, use beverage stirring stick, Item 315C, Appendix D, to lift seal over window (Figure 2-4-18-1, Detail A and Detail C).
 - d. Check seal for any area that may have lifted from jettisonable window during installation. If seal has lifted, push seal towards center of jettisonable window until seal has fully engaged window.
 - e. Make sure area of interface of jettisonable window and seal is clean.
 - f. Apply $\frac{3}{16}$ -inch bead of adhesive, Item 66E, Appendix D, around entire perimeter of jettisonable window and seal.

- g. Allow adhesive to dry for 3 - 4 hours.

NOTE

If window frame is to be painted, apply adhesive after frame has been painted.

- h. If necessary, paint window frame (PARA 2-6-5).
- i. Make sure area of interface of window frame and seal is clean.

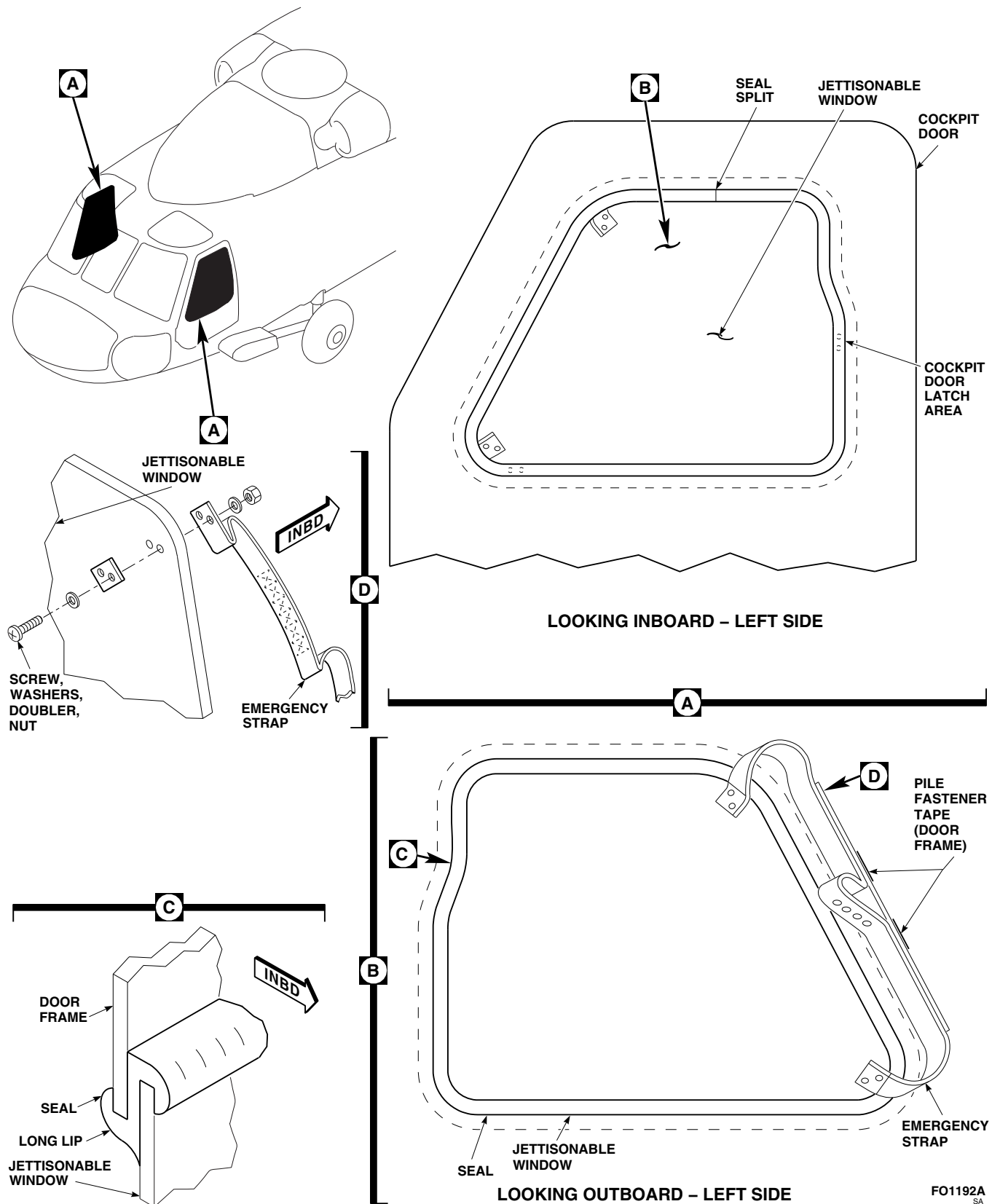


FIGURE 2-4-18-1. REPLACE COCKPIT DOOR JETTISONABLE WINDOW



Improper adhesive application to interface of jettisonable window and seal may cause jettisonable window to malfunction during emergency egression. Make sure bead of adhesive does not exceed $\frac{3}{16}$ -inch and that adhesive is not located between window and seal.

- j. Apply $\frac{3}{16}$ -inch bead of adhesive, Item 66E, Appendix D, around entire perimeter of window frame and seal.
- k. Allow 3 - 4 hours for adhesive to dry.
- l. Close cockpit door.

2-4-18.2 Replace Cockpit Door Jettisonable Window Assembly Frame.

NOTE

Replacement for left and right cockpit door jettisonable window assembly frames is the same.

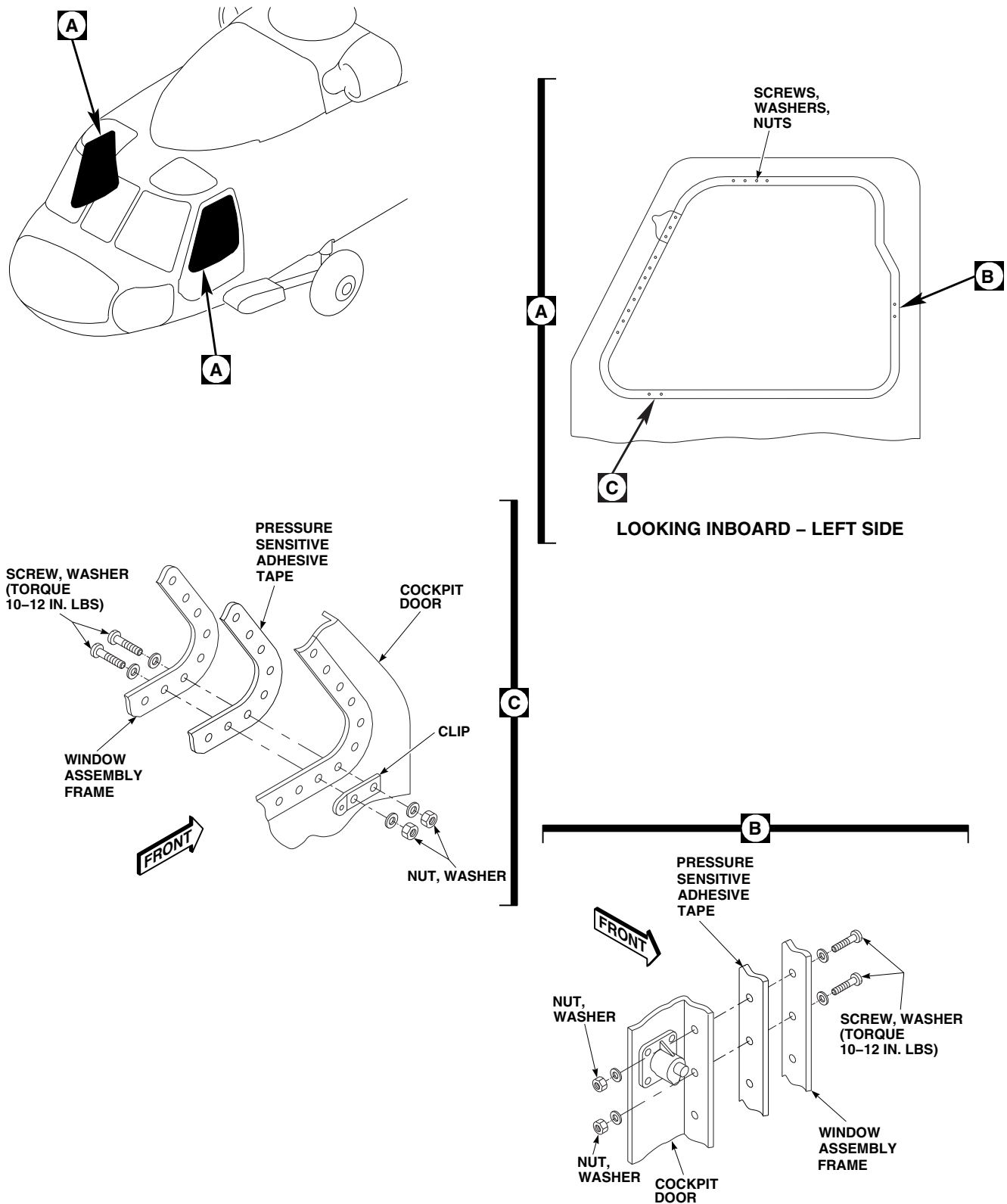
2-4-18.2.1 Remove.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Open cockpit door.
- c. If installed, remove jettisonable window (PARA 2-4-18.1.1).
- d. Remove cockpit door retainer (PARA 2-4-15).
- e. Remove screws, washers, and nuts securing window frame assembly and clip to door (Figure 2-4-18-2, Detail A, Detail B, and Detail C).
- f. Remove pressure sensitive adhesive tape from cockpit door and clean flanges using paint remover, Item 231, Appendix D. Wipe dry.

- g. Clean window assembly frame using clean flannel cloth, Item 140, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D. Wipe dry with clean flannel cloth.

2-4-18.2.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Using 1-inch masking tape, Item 212, Appendix D, tape off door flange and jettisonable window assembly.
- c. Apply pressure sensitive adhesive tape, Item 244C, Appendix D, on flanges of replacement window, with sticky side against flange. Trim tape as necessary.
- d. Coat pressure sensitive adhesive tape with thin coating (about 0.003-inch) of sealing compound, Item 292, Appendix D.
- e. Position window assembly frame on door and secure with clip, screws, washers, and nuts while sealing compound is workable. **TORQUE SCREWS TO 10 - 12 INCH-POUNDS** (Figure 2-4-18-2, Detail A, Detail B, and Detail C).
- f. Install cockpit door retainer (PARA 2-4-15).
- g. Fillet seal area around upper door retainer with sealing compound, Item 292, Appendix D, to a smooth surface.
- h. Touch up paint in repaired area (PARA 2-6-5).
- i. Remove masking tape.
- j. Install cockpit door jettisonable window (PARA 2-4-18.1.3).
- k. Spray water on cockpit door and check for leaks.
- l. Using lockwire, Item 195, Appendix D, lockwire jettison handle to clip on lower front edge of door frame.



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FIGURE 2-4-18-2. REPLACE COCKPIT DOOR JETTISONABLE WINDOW ASSEMBLY FRAME

2-4-19 COCKPIT DOOR FRONT CRANK MECHANISM.

PARAGRAPH BREAKDOWN

		PAGE
2-4-19.1	Replace Cockpit Door Front Crank Mechanism	2-4-19-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Drill Set

References

- PARA 1-6-16
- PARA 2-3-2
- PARA 2-4-12

2-4-19.1 Replace Cockpit Door Front Crank Mechanism.

NOTE

Replacement of right and left cockpit door front crank mechanism is the same.

2-4-19.1.1. Remove.

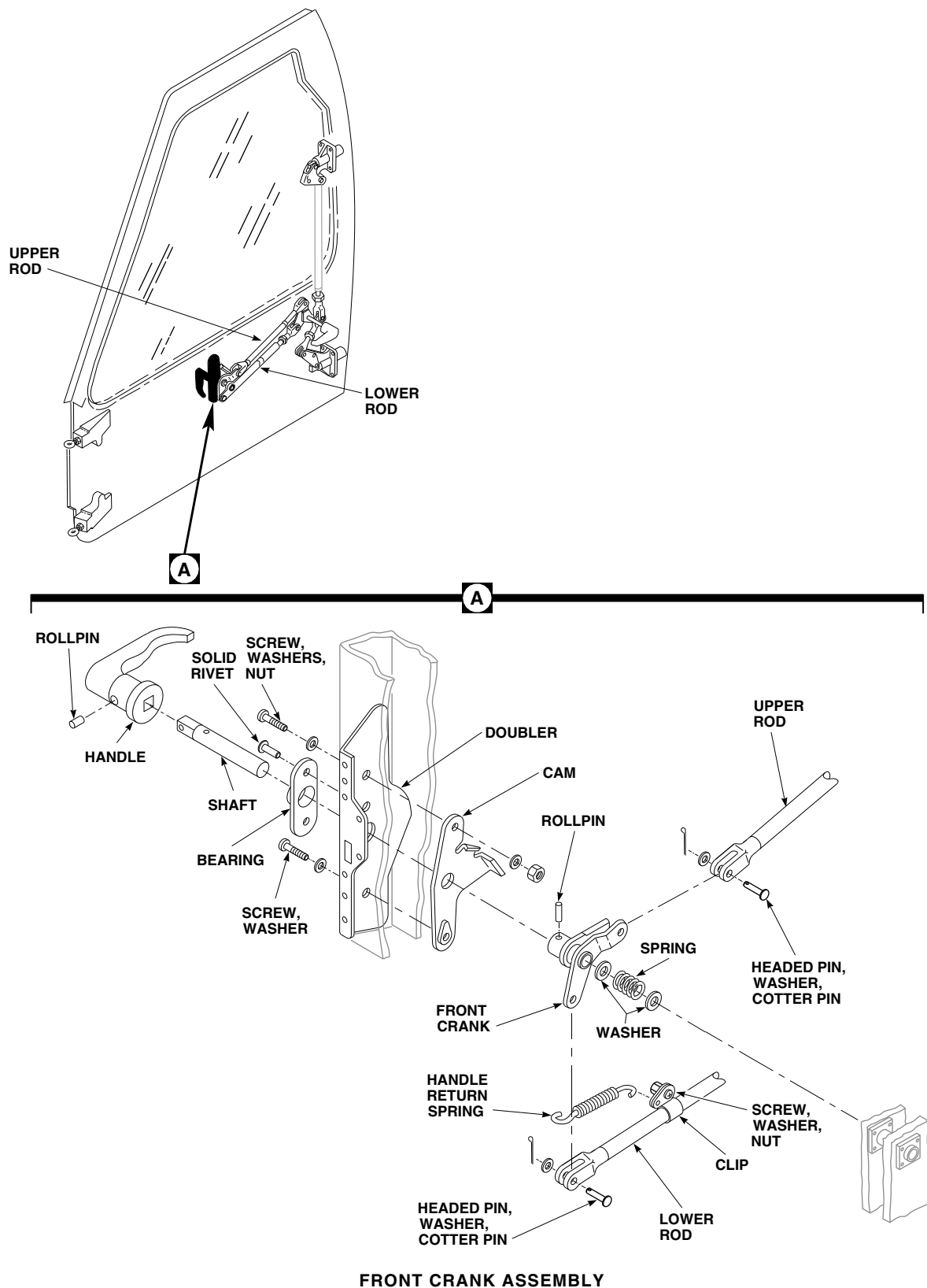
- Turn off all electrical power (PARA 1-6-16).
- Open cockpit door.
- Drive out rollpin attaching inside door handle shaft to front crank (Figure 2-4-19-1, Detail A).
- Unhook handle return spring from cam and clip on lower rod.
- Pull handle and shaft out of door. Remove spring and washers from between door and front crank.
- Push crank outboard to get at clevis attaching points.
- Remove headed pins attaching upper and lower rods to front crank.

- Remove front crank from door.
- Remove screws, washers, and nut attaching cam to doubler. Remove cam.
- Drive out rollpin and remove handle from shaft.
- Drill out rivets and remove doubler from door.
- Drill out rivets and remove bearing from doubler.

2-4-19.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Rivet bearing to doubler (Figure 2-4-19-1, Detail A).
- Rivet doubler to door.
- Position cam in door. Attach cam to door and doubler with screw and washer, and screw, washer, and nut.
- Place front crank in door. Connect upper and lower rods to front crank with headed pins, washers, and cotter pins.

- f. Install front crank boss in doubler bearing. Place spring and washers in door behind crank.
- g. Attach handle to shaft with rollpin.
- h. Install handle and shaft through bearing, front crank, washers, and spring.
- i. Attach shaft to front crank with rollpin.
- j. Hook handle return spring to cam and clip on lower rod.
- k. Open and close door a few times to check for correct operation. Adjust door as required (PARA 2-4-12).
- l. Spray water on door and check for leaks.
- m. Perform cockpit door jettison check (PARA 2-3-2).



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FIGURE 2-4-19-1. REPLACE COCKPIT DOOR FRONT CRANK MECHANISM

2-4-19-3/(2-4-19-4 Blank)

2-4-20 COCKPIT DOOR LATCHES.

PARAGRAPH BREAKDOWN

	PAGE
2-4-20.1 Replace Cockpit Door Latches	2-4-20-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit

Materials/Parts

- Sealing Compound, Item 276, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 2-3-2
- PARA 2-4-12

2-4-20.1 Replace Cockpit Door Latches.

NOTE

NOTE

Replacement of both cockpit door latches is the same for both cockpit doors, except as noted.

Screw installed in upper outboard location of upper latch bolt housing.

- f. Remove bolts, washers, and screw from upper or lower latchbolt housing. Remove latchbolt housing and shim, if installed, from door.

2-4-20.1.1. Remove.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Open cockpit door.
- c. Remove headed pins, washers, and cotter pins. Remove vertical rod from upper bellcrank and remove bellcrank from latchbolt housing and latchbolt (Figure 2-4-20-1, Detail A).
- d. Remove headed pins, washers, and cotter pins. Remove link terminal from lower bellcrank and bellcrank from latchbolt housing and latchbolt (Figure 2-4-20-1, Detail C).
- e. Push latchbolt into housing, turn bolt to line up slotted end with slot in housing, and remove latchbolt and spring from housing (Figure 2-4-20-1, Detail B).

2-4-20.1.2. Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Apply sealing compound, Item 276, Appendix D, to threads of bolts and screw.

NOTE

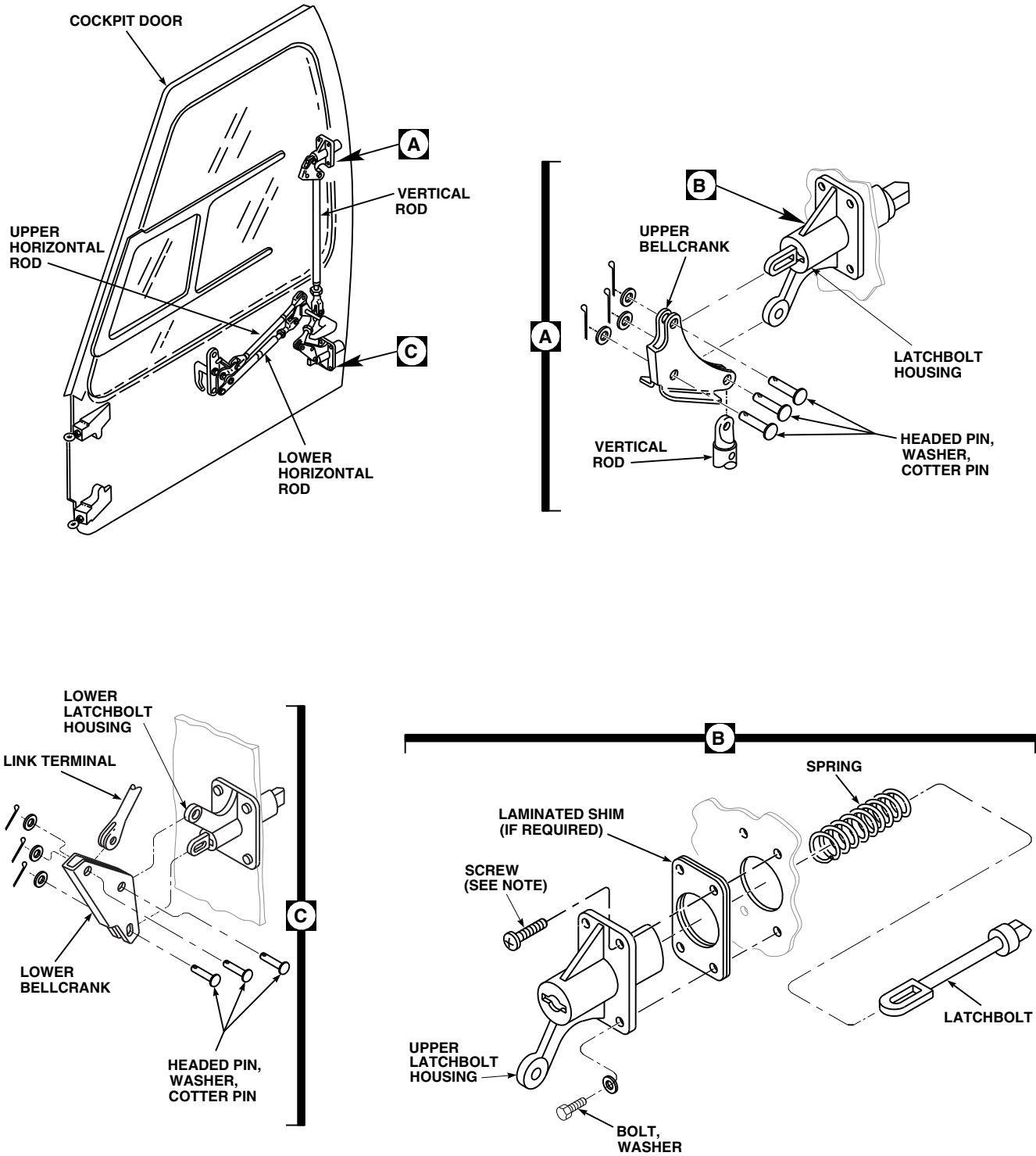
If required for proper latchbolt engagement with striker plate, laminated shim may be used.

- c. Install latchbolt housing and shim, if required, in door frame (Figure 2-4-20-1, Detail B). Position upper latchbolt housing with bellcrank arm down. Position lower latchbolt housing with bellcrank arm up. Secure as follows:

NOTE

Screw is to be installed in upper outboard location.

- (1) On upper latchbolt housing, secure with screw, bolts, and washers.
 - (2) On lower latchbolt housing, secure with bolts and washers.
- d. Install spring and latchbolt in latch housing. Push latchbolt slotted end through slot in housing and turn 90°. Face beveled side of latchbolt inboard.
 - e. Install lower bellcrank on lower latchbolt housing arm and on latchbolt slotted pin hole. Install headed pins, washers, and cotter pins (Figure 2-4-20-1, Detail C).
 - f. Install link terminal in lower bellcrank with headed pin, washer, and cotter pin.
 - g. Install upper bellcrank on upper latch housing arm and on latchbolt, with slotted pin hole on latchbolt. Install headed pins, washers, and cotter pins (Figure 2-4-20-1, Detail A).
 - h. Install vertical rod in upper bellcrank with headed pin, washer, and cotter pin.
 - i. Close door and check that latchbolts engage striker plates correctly. Adjust if necessary (PARA 2-4-12).
 - j. Spray water on door and check for leaks.
 - k. Perform cockpit door jettison check (PARA 2-3-2).



NOTE

SCREW IS USED ON UPPER LATCHBOLT HOUSING ONLY.

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FIGURE 2-4-20-1. REPLACE COCKPIT DOOR LATCHES

2-4-21 COCKPIT DOOR REAR CRANK MECHANISM.

PARAGRAPH BREAKDOWN

		PAGE
2-4-21.1	Replace Cockpit Door Rear Crank Mechanism	2-4-21-1

INITIAL SETUP

- PARA 2-4-12

Tools

- Aircraft Mechanic’s Toolkit

References

- PARA 1-6-16
- PARA 2-3-2

2-4-21.1 Replace Cockpit Door Rear Crank Mechanism.

NOTE

Replacement of right and left cockpit door rear crank mechanism is the same.

2-4-21.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Open cockpit door.
- Remove screws, washers, lockwashers, and nuts attaching support to inside of door (Figure 2-4-21-1, Detail A). Remove support from door.
- Remove screws, washers, and nuts attaching door handle to door. Remove handle.
- Remove headed pin and washer connecting vertical rod, link, and bellcrank.
- Remove headed pins and washers attaching upper and lower rods to rear crank and bellcrank.

- Remove retaining ring and washer from shaft of rear crank.
- Remove rear crank and bellcrank from door. Slide inboard until clear of door structure.
- Remove laminated shim washer or plain washer from rear crank and slide bellcrank off rear crank.

2-4-21.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Slide rear crank into bellcrank and install laminated shim washer or plain washer on shaft of rear crank (Figure 2-4-21-1, Detail A).
- Install rear crank in door and install washer and retaining ring.
- Connect upper rod to bellcrank and lower rod to rear crank with headed pins, washers, and cotter pins.
- Connect vertical rod and link to bellcrank with headed pin, washer, and cotter pin.

- f. Install door handle on shaft of rear crank and fasten to door with screws, washers, and nuts.
- g. Place support on rear crank and inside of door. Fasten with screws, washers, lockwashers, and nuts.
- h. Open and close door a few times to check for proper operation. Adjust latches if required (PARA 2-4-12).
- i. Spray water on door and check for leaks.
- j. Perform cockpit door jettison check (PARA 2-3-2).

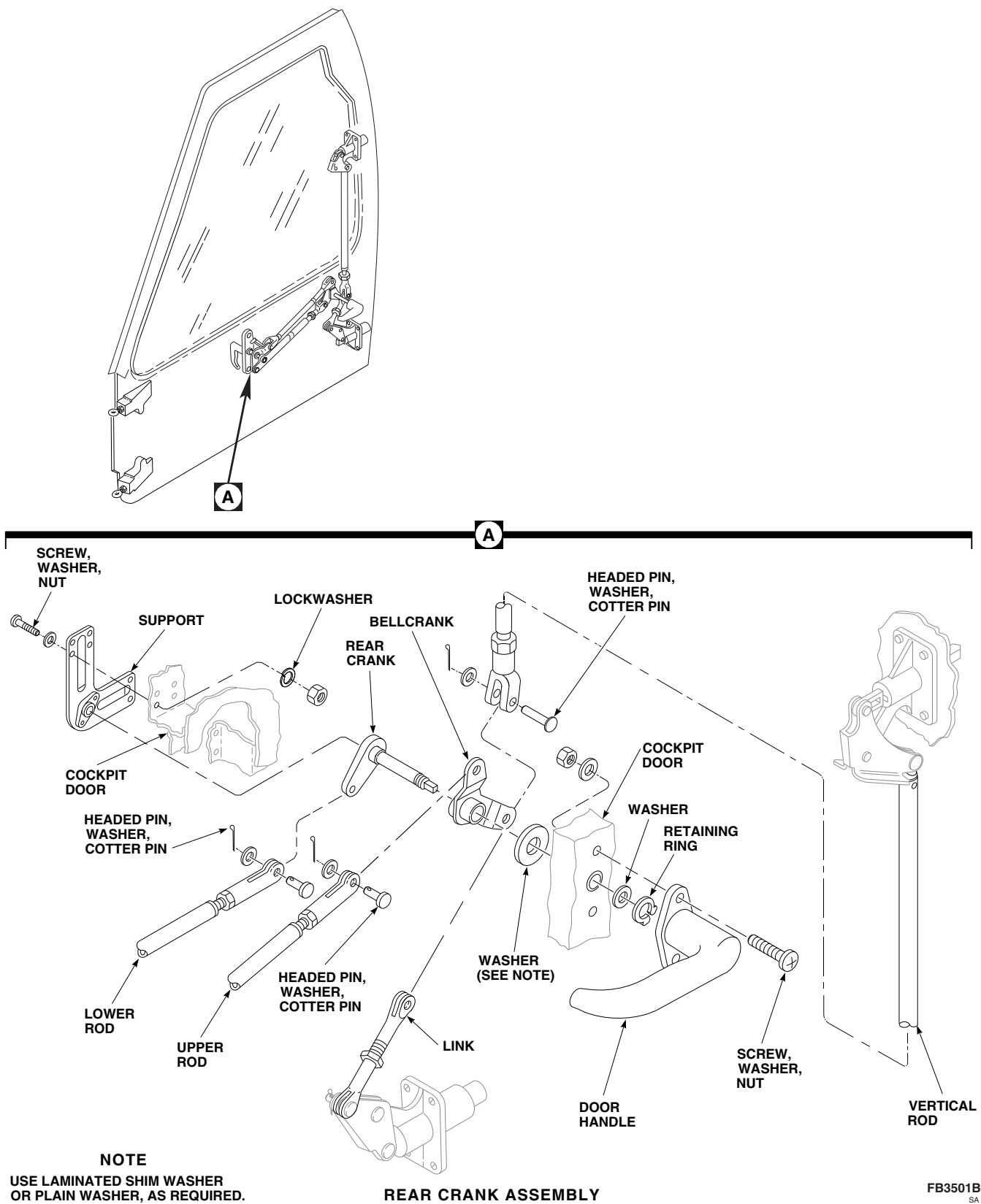


FIGURE 2-4-21-1. REPLACE COCKPIT DOOR REAR CRANK MECHANISM

2-4-22 COCKPIT DOOR STRIKER PLATES.

PARAGRAPH BREAKDOWN

	PAGE
2-4-22.1 Replace Cockpit Door Striker Plates	2-4-22-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Acetone, Item 25, Appendix D
- Corrosion Preventive Compound, Item 115, Appendix D

- Machinery Towel, Item 211, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 2-3-2
- PARA 2-4-12

2-4-22.1 Replace Cockpit Door Striker Plates.

NOTE

Each cockpit door has two striker plates. Replacement of both striker plates on left and right cockpit doors is the same.

clean off any oil or grease from serrated plate, striker plate, and shim.

NOTE

Do not allow sealing compound to contact serrated side of plate.

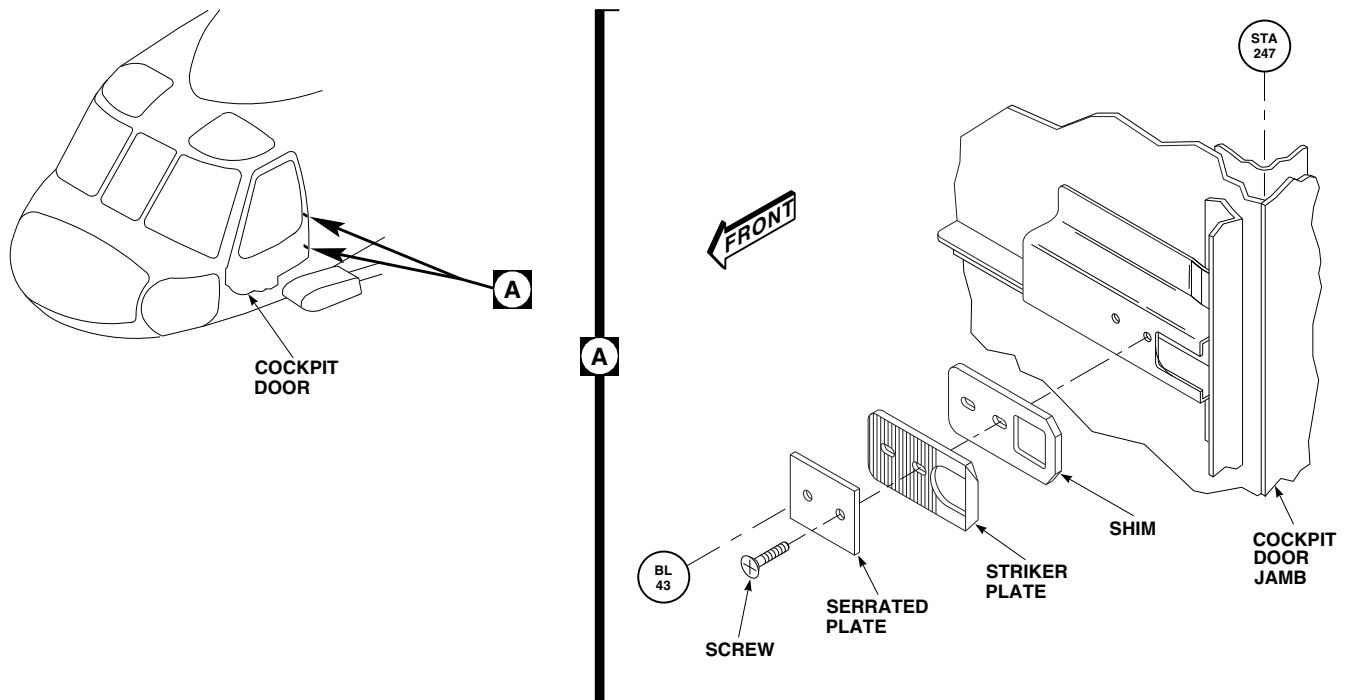
2-4-22.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Open cockpit door.
- Remove screws and remove serrated plate, striker plate, and shim from cockpit door jamb (Figure 2-4-22-1, Detail A).

2-4-22.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Using machinery towel, Item 211, Appendix D, dampened with acetone, Item 25, Appendix D,

- Apply sealing compound, Item 292, Appendix D, to smooth side of striker plate.
- Press striker plate and shim together with fingers. Using clean machinery towel, Item 211, Appendix D, remove any squeezeout.
- Apply corrosion preventive compound, Item 115, Appendix D, to serrated sides of striker plate and serrated plate.
- Install shim, striker plate, and serrated plate on cockpit door jamb with screws (Figure 2-4-22-1, Detail A).

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SA**FIGURE 2-4-22-1. REPLACE COCKPIT DOOR STRIKER PLATES**

- g. Open and close cockpit door a few times to check for proper operation. If required, adjust cockpit door (PARA 2-4-12).
- h. Spray water on cockpit door and check for leaks.
- i. Perform cockpit door jettison inspection (PARA 2-3-2).

2-4-23 COCKPIT DOOR SEAL.

PARAGRAPH BREAKDOWN

		PAGE
2-4-23.1	Replace Cockpit Door Seal	2-4-23-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Abrasive Cloth, Item 3, Appendix D
- Adhesive, Item 37, Appendix D
- Adhesive, Item 44, Appendix D

- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Scouring Pad, Item 270, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-11

2-4-23.1 Replace Cockpit Door Seal.

NOTE

Replacement of left and right cockpit door seals is the same.

2-4-23.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Remove cockpit door (PARA 2-4-11).
- Using nonmetallic scraper, remove damaged seal from cockpit door (Figure 2-4-23-1, Detail A).

CAUTION

Damage to window will result if methyl ethyl ketone contacts it. Do not allow methyl ethyl ketone to contact window.

- Remove adhesive from cockpit door seal flange using machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D. Using nonmetallic scraper, remove remaining adhesive.
- Clean any adhesive residue from flange using machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217,

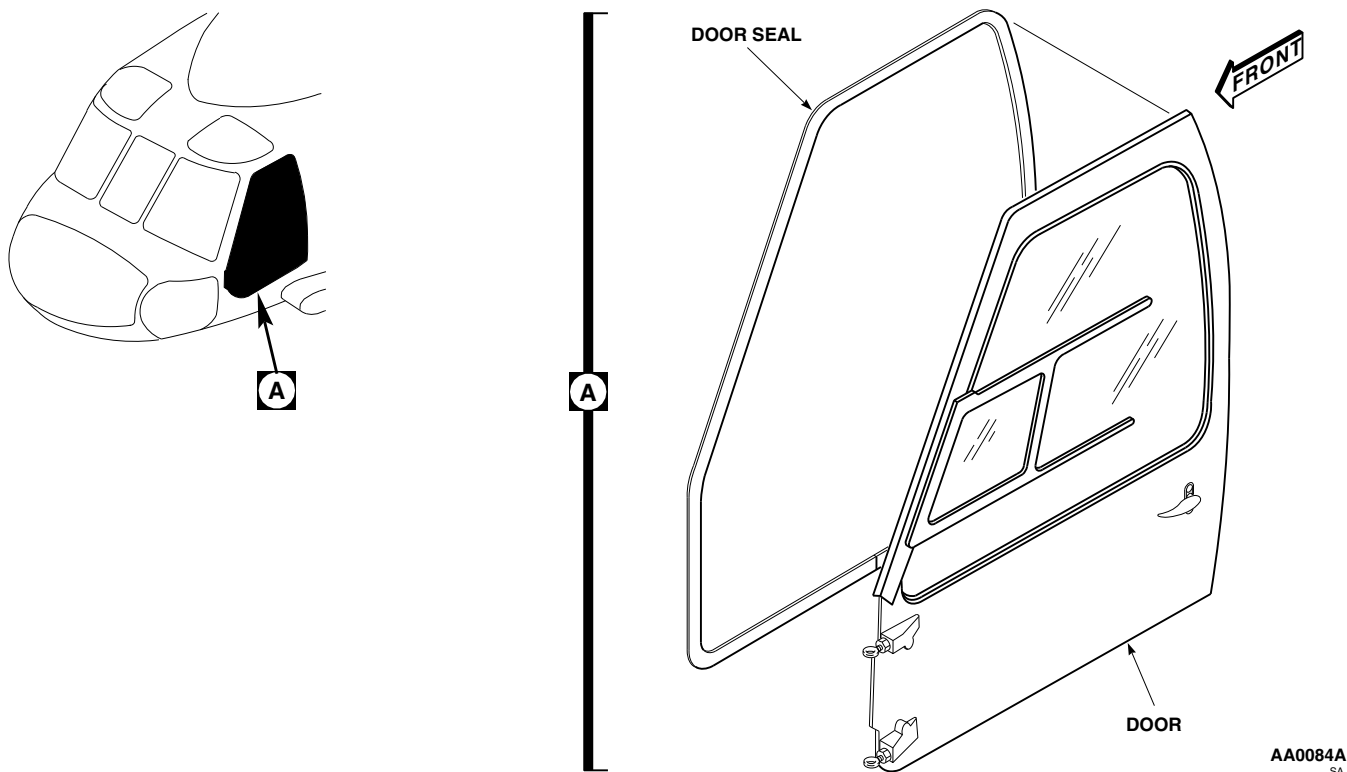


FIGURE 2-4-23-1. REPLACE COCKPIT DOOR SEAL

Appendix D. Wipe dry using clean machinery towel.

2-4-23.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Roughen mating surface of replacement seal lightly, using abrasive cloth, Item 3, Appendix D, or scouring pad, Item 270, Appendix D.
- c. Wipe seal clean using machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D. Wipe dry with clean machinery towel.
- d. Apply thin, even coat of adhesive, Item 37 or Item 44, Appendix D, to cockpit door seal flange and mating surface of replacement seal.

NOTE

Do not stretch seal while installing, especially around curved parts of door.

- e. Without stretching seal, starting at bottom center of cockpit door, install seal to door flange area (Figure 2-4-23-1, Detail A).
- f. Trim seal to fit and join at bottom of cockpit door.
- g. Apply fillet of sealing compound, Item 292, Appendix D, to seal joining area.
- h. Install cockpit door (PARA 2-4-11).

2-4-24 COCKPIT DOOR CONDUCTIVE SEALS.

PARAGRAPH BREAKDOWN

		PAGE
2-4-24.1	Replace Cockpit Door Conductive Seals	2-4-24-0

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Drill Set

References

- PARA 1-6-16
- PARA 2-3-2
- PARA 2-4-12

2-4-24.1 Replace Cockpit Door Conductive Seals.

NOTE

Replacement of right and left cockpit door conductive seals is the same.

2-4-24.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- To remove conductive seal from edge of door, remove screws, plates, and angle securing seal to door (Figure 2-4-24-1, Detail B). Remove conductive seal.
- To remove conductive seal from door jamb, perform the following:
 - Drill out and remove rivets (Figure 2-4-24-1, Detail A).
 - Remove plates, conductive seals, and angles from door jamb.

2-4-24.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- To install conductive seal to edge of door, perform the following (Figure 2-4-24-1, Detail B):
 - Install conductive seal, plates, and angle with screws.
 - Adjust seal by loosening screws, sliding seal.
- To install conductive seal on door jamb, install angle, conductive seals, and plates on door jamb with rivets (Figure 2-4-24-1, Detail A).
- Open and close door a few times to check for proper operation. Adjust door if required (PARA 2-4-12).
- Spray water on door and check for leaks.
- Perform cockpit door jettison check (PARA 2-3-2).

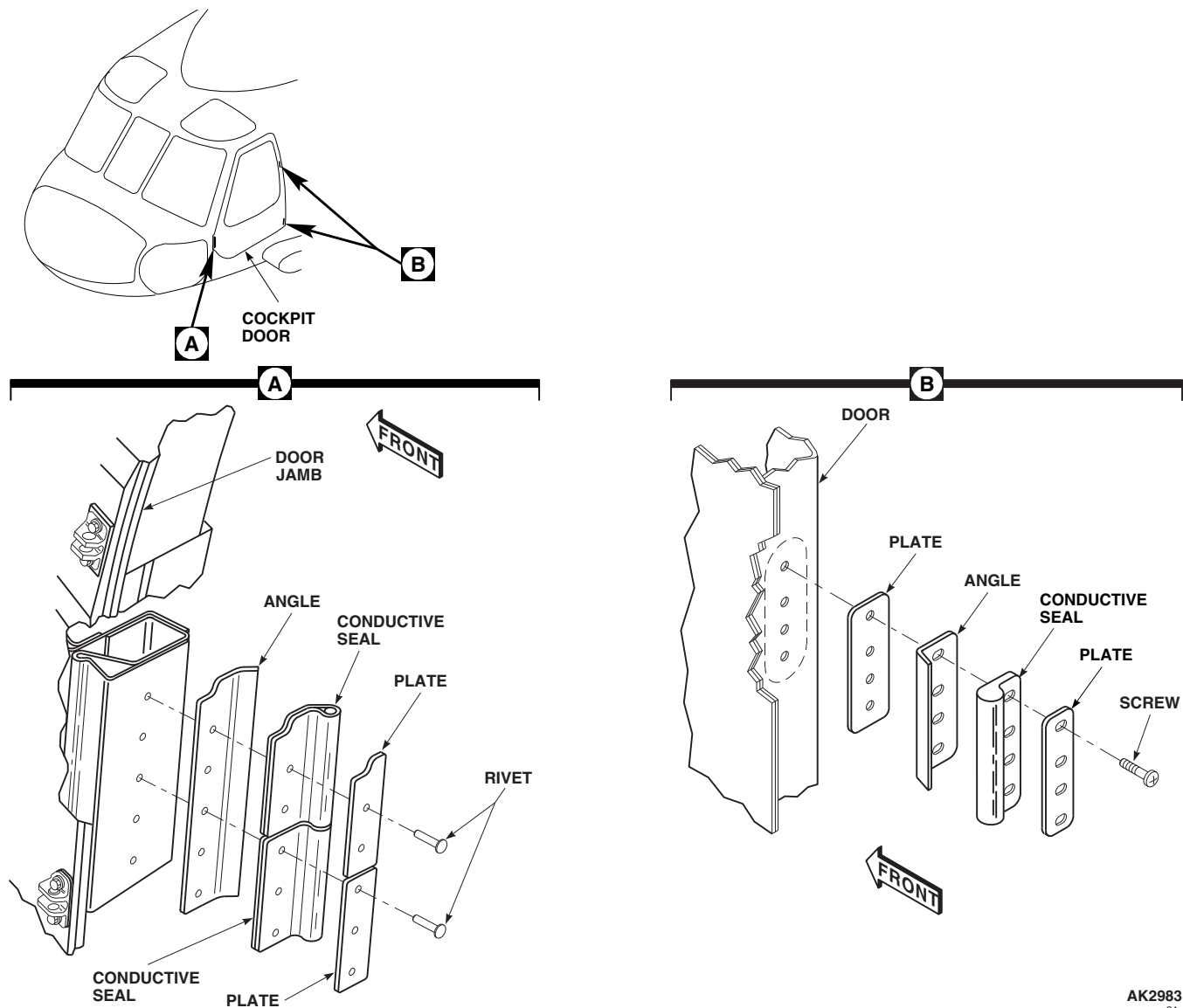


FIGURE 2-4-24-1. REPLACE COCKPIT DOOR CONDUCTIVE SEALS

2-4-25 COCKPIT DOOR HINGE FITTINGS.

PARAGRAPH BREAKDOWN

		PAGE
2-4-25.1	Replace Cockpit Door Hinge Fittings	2-4-25-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Arbor Press
- Drift
- Nonmetallic Scraper, Locally-Made
- Reamer Set
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Alcohol, Isopropyl, Item 72, Appendix D
- Dry Ice, Item 125, Appendix D

- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Primer, Item 257, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 2-3-2
- PARA 2-4-12

2-4-25.1 Replace Cockpit Door Hinge Fittings.

NOTE

Replacement of upper and lower hinge fittings on both left and right cockpit doors is the same, except as noted.

2-4-25.1.1. Remove.

NOTE

Remove one hinge fitting at a time so as not to disturb door adjustment.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Using assistant inside cockpit, make sure door is closed and latched.
- c. Remove headed pin, washer, and cotter pin attaching hinge to hinge fitting (Figure 2-4-25-1, Sheet 1, Detail A).
- d. Using nonmetallic scraper, remove sealing compound from hinge fitting.
- e. **70583-701247** Remove bolts, washers, and nuts attaching hinge fitting and laminated shim(s) to cockpit structure. Keep laminated shim(s) for reinstallation (Figure 2-4-25-1, Sheet 1, Detail B or Sheet 2, Detail B).
- f. **701248-SUBQ** Remove bolts, washers, wedge-shaped washers (lower hinge fitting

only), and nuts attaching hinge fitting, serrated plate, and laminated shim(s) to cockpit structure. Keep laminated shim(s) for reinstallation (Figure 2-4-25-1, Sheet 1, Detail B or Sheet 2, Detail B).◀

- g. Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean sealing compound from cockpit structure, shim(s), serrated plate, hinge fitting, wedge-shaped washer, and bolts.

2-4-25.1.2. Repair.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Using arbor press and drift, press bushings from hinge fitting (Figure 2-4-25-1, Sheet 1, Detail B or Sheet 2, Detail B).
- c. Place new bushings in a container of dry ice, Item 125, Appendix D, for about 15 minutes.
- d. Remove bushing from dry ice. Apply a thin coat of primer, Item 257, Appendix D, to bushing outside diameter.
- e. Using arbor press, and suitable drift, press bushings into hinge fitting.
- f. Using reamer set, line-ream bushings 0.3746 to 0.3751-inch diameter.

2-4-25.1.3. Install.

- a. Turn off all electrical power (PARA 1-6-16).

NOTE

When applying sealing compound to shank of bolt, make sure no sealing compound gets on the threads.

- b. Apply a light coat (approximately 0.005-inch) of sealing compound, Item 292, Appendix D, on both faces of shim(s) and boltshank.

- c. Clean surface with cloth moistened with isopropyl alcohol, Item 72, Appendix D. Allow to dry completely.

- d. **70583-701247** ▶ Position laminated shim(s) on hinge fitting and place hinge fitting on cockpit structure. Line up holes in fitting and structure. Install bolt in hinge fitting (Figure 2-4-25-1, Sheet 1, Detail B).◀

- e. **701248-SUBQ** ▶ Position laminated shim(s) on serrated plate, place serrated plate on hinge fitting, line up holes, and place on cockpit structure. Install bolts in hinge fitting. For lower hinge fitting, install wedge-shaped washers under boltheads (Figure 2-4-25-1, Sheet 2, Detail B).◀

- f. Install washers and nuts on bolts. **TORQUE NUT TO 95 - 105 INCH-POUNDS.**

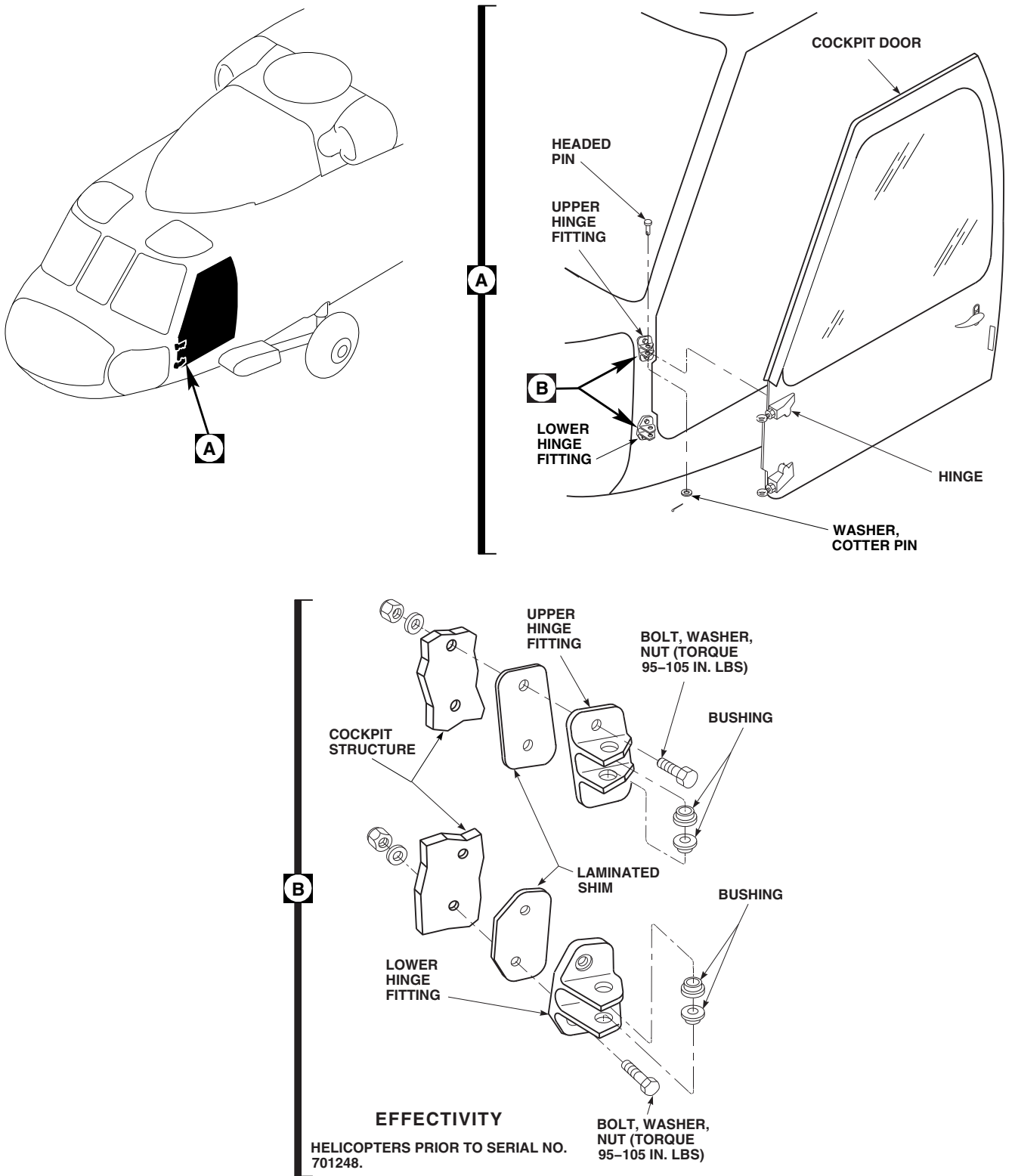
- g. Apply a continuous bead of sealing compound, Item 292, Appendix D, around hinge fitting.

- h. Connect hinge to hinge fitting with headed pin, washer, and cotter pin (Figure 2-4-25-1, Sheet 1, Detail A).

- i. Open and close door a few times to check for proper operation. Adjust door if required (PARA 2-4-12).

- j. Spray water on door and check for leaks.

- k. Perform cockpit door jettison check (PARA 2-3-2).



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FIGURE 2-4-25-1. REPLACE COCKPIT DOOR HINGE FITTINGS (SHEET 1 OF 2)

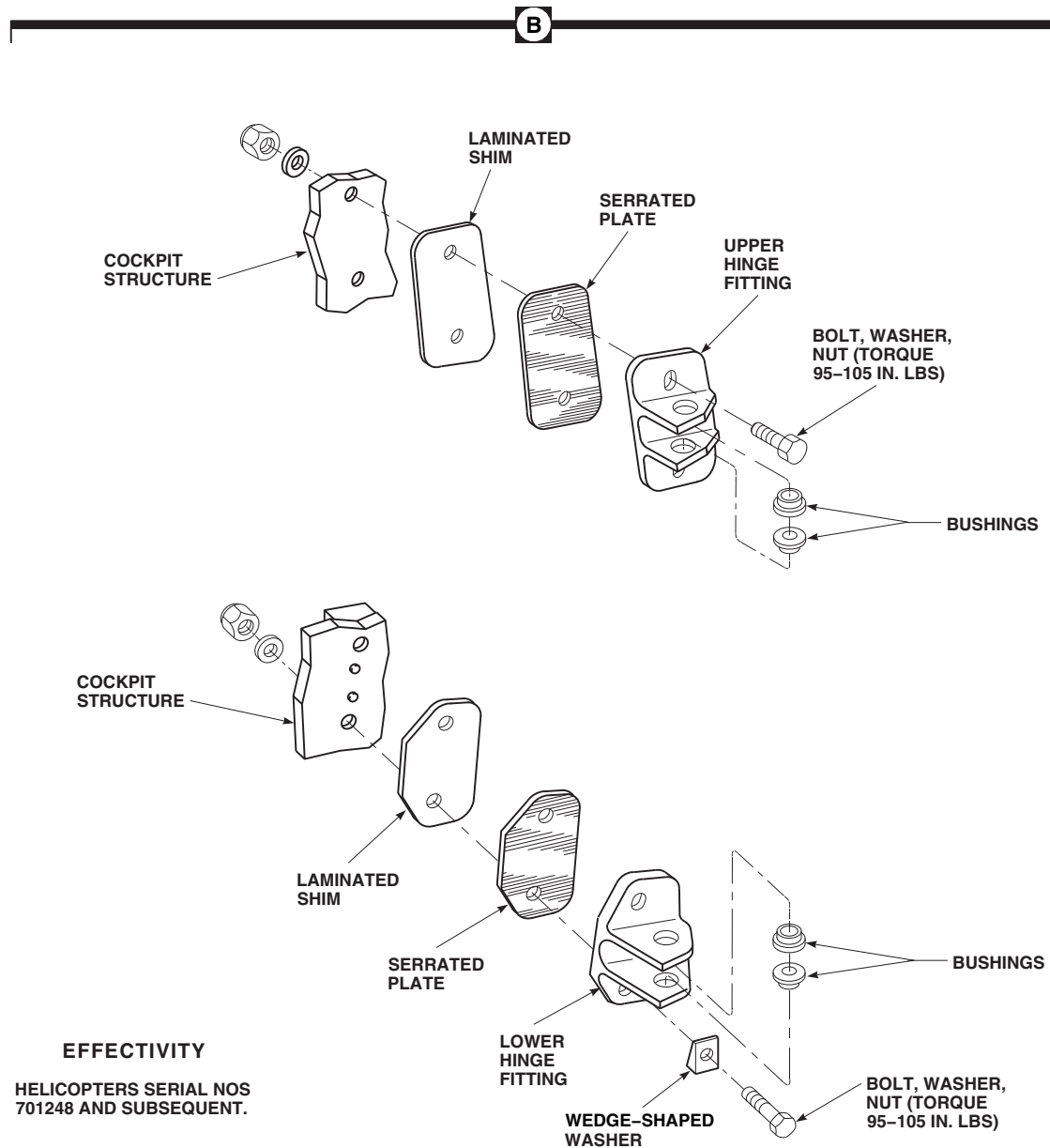
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FIGURE 2-4-25-1. REPLACE COCKPIT DOOR HINGE FITTINGS (SHEET 2 OF 2)

2-4-25-4 Change 3

2-4-26 COCKPIT DOOR HINGE ROD ENDS.

PARAGRAPH BREAKDOWN

		PAGE
2-4-26.1	Replace Cockpit Door Hinge Rod Ends	2-4-26-0

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Corrosion Compound, Item 108, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Masking Tape, 1-inch, Item 212, Appendix D

- Paint, Polyurethane, Item 222A, Appendix D
- Scouring Pad, Item 270, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-11
- PARA 2-6-5

2-4-26.1 Replace Cockpit Door Hinge Rod Ends.

NOTE

Replacement of both cockpit door hinge rod ends is the same for both cockpit doors.

2-4-26.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Remove cockpit door (PARA 2-4-11).
- Remove cotter pin, headed pin, and washer from hinge fitting and remove hinge (Figure 2-4-26-1, Detail A).
- Measure distance between hole center of rod end and face of hinge (Figure 2-4-26-1, Detail B). Record dimension.
- Loosen jamnut on rod end.

- Remove rod end from hinge. Remove jamnut from rod end (Figure 2-4-26-1, Detail C).

2-4-26.1.2. Repair.

- Remove corrosion manually, using scouring pad, Item 270, Appendix D.
- Mix solution of 1 part corrosion compound, Item 108, Appendix D, to 1 part water.



Corrosion removal and treatment compound solution can damage rod end bearing. Do not allow solution to contact bearing.

- Mask rod end bearing to prevent contact with corrosion removal and treatment compound solution, using 1-inch masking tape, Item 212, Appendix D.

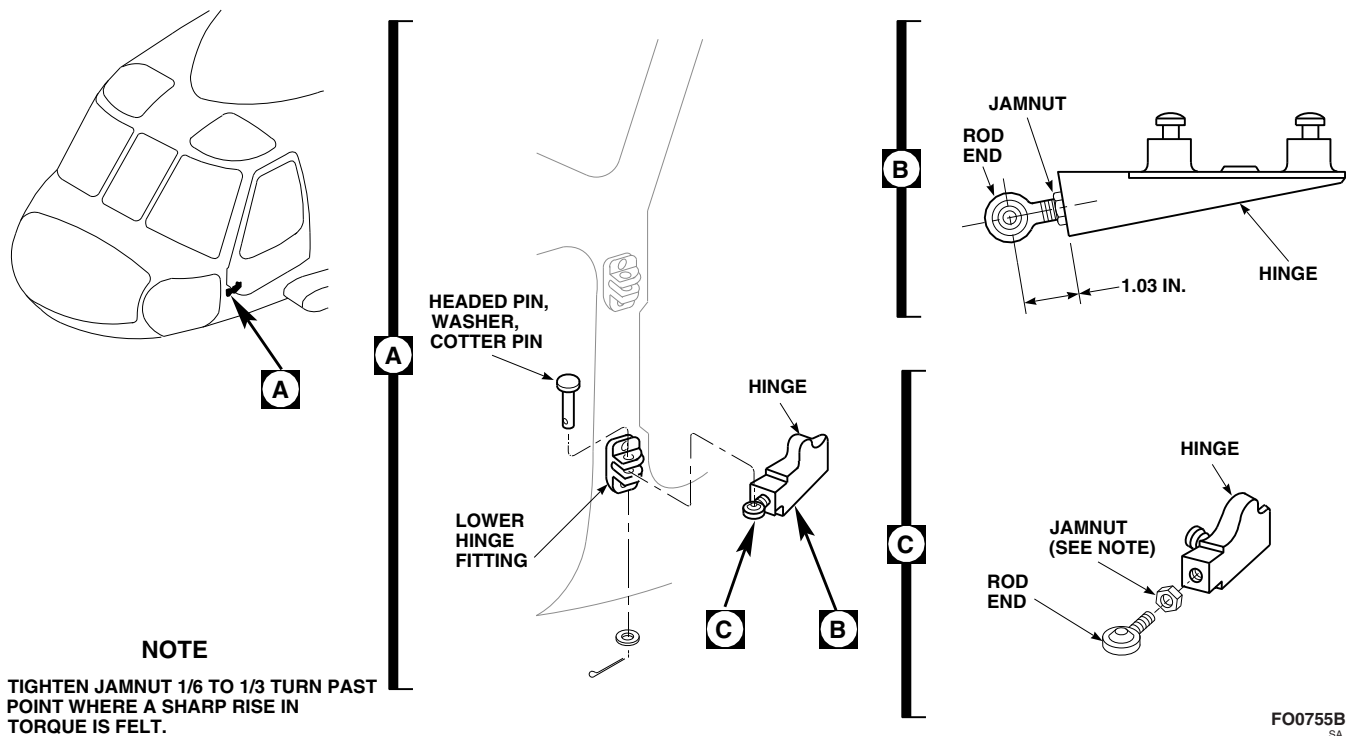


FIGURE 2-4-26-1. REPLACE COCKPIT DOOR HINGE ROD ENDS

- d. Apply corrosion removal and treatment compound solution to corroded areas of rod end.
- e. Rinse thoroughly with clean water.
- f. Remove masking tape from bearing.
- g. Using epoxy primer coating kit, Item 135, Appendix D, touch up paint with two coats of epoxy primer and top coat of polyurethane paint, Item 222A, Appendix D (PARA 2-6-5).

2-4-26.1.3. Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Install rod end in hinge as follows:
 - (1) Install jamnut on rod end (Figure 2-4-26-1, Detail C).

- (2) Install rod end in hinge.

- (3) Adjust rod end length to dimension recorded. **IF DIMENSION WAS NOT RECORDED, SET ROD END LENGTH TO 1.03-INCH FROM CENTER OF ROD END TO FACE OF HINGE** (Figure 2-4-26-1, Detail B).

- (4) **TIGHTEN JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT** (Figure 2-4-26-1, Detail C).

- c. Install hinge on hinge fitting. Fasten with headed pin and cotter pin (Figure 2-4-26-1, Detail A).

- d. Install cockpit door (PARA 2-4-11).

2-4-27 COCKPIT DOOR LOCK.

PARAGRAPH BREAKDOWN

		PAGE
2-4-27.1	Replace Cockpit Door Lock	2-4-27-0

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Rubber Mallet
- Screwdriver, Thin-Bladed

References

- PARA 1-6-16

2-4-27.1 Replace Cockpit Door Lock.

NOTE

- Lock cylinders are available only in sets. When replacing cockpit door lock cylinder, replace lock cylinders on all aircraft doors.
- Replacement of right and left cockpit door lock is the same.

2-4-27.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Open cockpit door.
- Remove screws, washers, lockwashers, and nuts securing support to inside of door. Remove support (Figure 2-4-27-1, Detail A).
- Remove screws, washers, and nuts attaching handle to door. Remove handle.
- Remove snapring and washers fastening handle to escutcheon. Remove escutcheon from handle (Figure 2-4-27-1, Detail B).

- Gently tap lock case with rubber mallet to remove it from handle.
- Remove 2 pins securing shell in lock case. Remove shell from lock case.
- Using thin-bladed screwdriver, depress retaining ring into lock cylinder groove and remove lock cylinder from shell.

2-4-27.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Compress retaining ring, insert lock cylinder into shell, and snap into place (Figure 2-4-27-1, Detail B).
- Insert shell into lock case and install 2 pins. Pins must be flush with surface of lock case.
- Install lock case in handle.
- Install washer on handle and handle in escutcheon.
- Install small washers and snapring on handle.
- Install handle on door with screws, washers, and nuts (Figure 2-4-27-1, Detail A).

- h. Make sure area is clean and free of foreign material before closing up.
- i. Install support on inside of door and secure with screws, washers, lockwashers, and nuts.
- j. Close and open door a few times to check operation of handle.
- k. Close cockpit door.

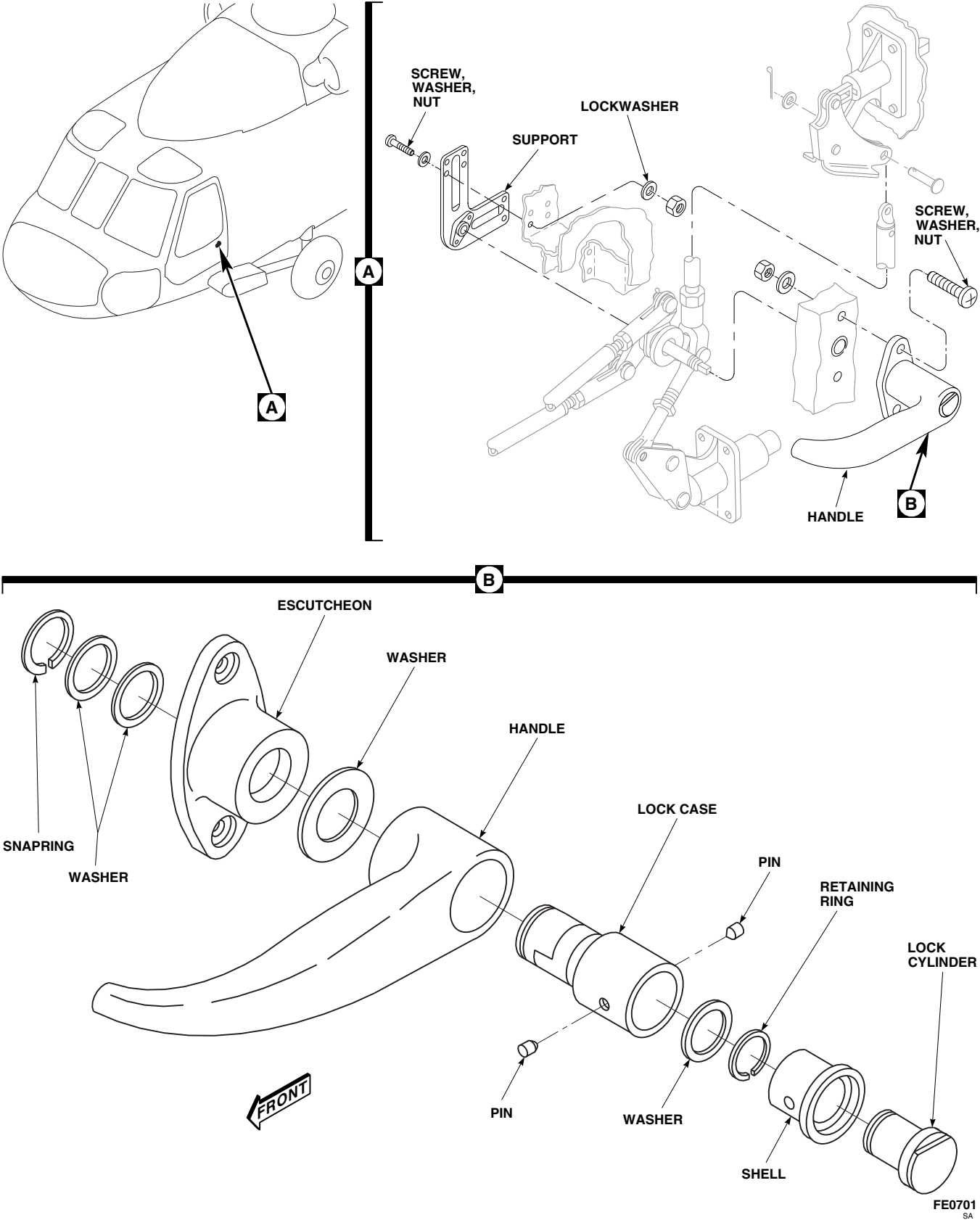


FIGURE 2-4-27-1. REPLACE COCKPIT DOOR LOCK

2-4-27-3/(2-4-27-4 Blank)

2-4-28 GLARE SHIELD.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
2-4-28.1	Replace Glare Shield	2-4-28-1	2-4-28.2	Repair Glare Shield	2-4-28-3

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Abrasive Cloth, Item 2, Appendix D
- Adhesive, Item 37, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D

- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D
- Tiedown Straps, Item 334, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-6-3
- PARA 8-4-24

2-4-28.1 Replace Glare Shield.

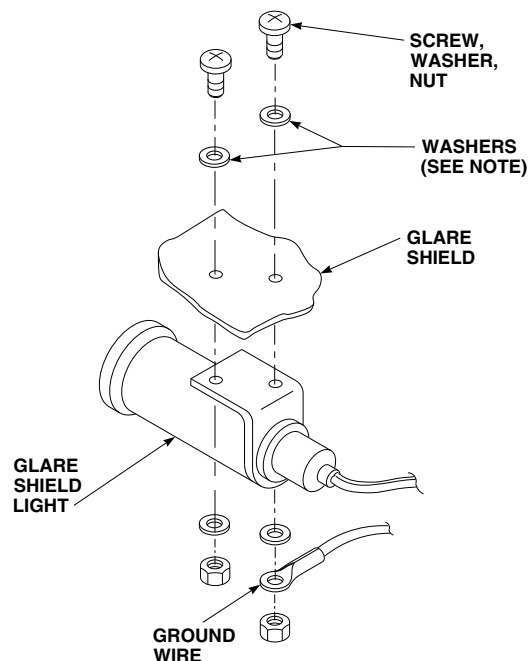
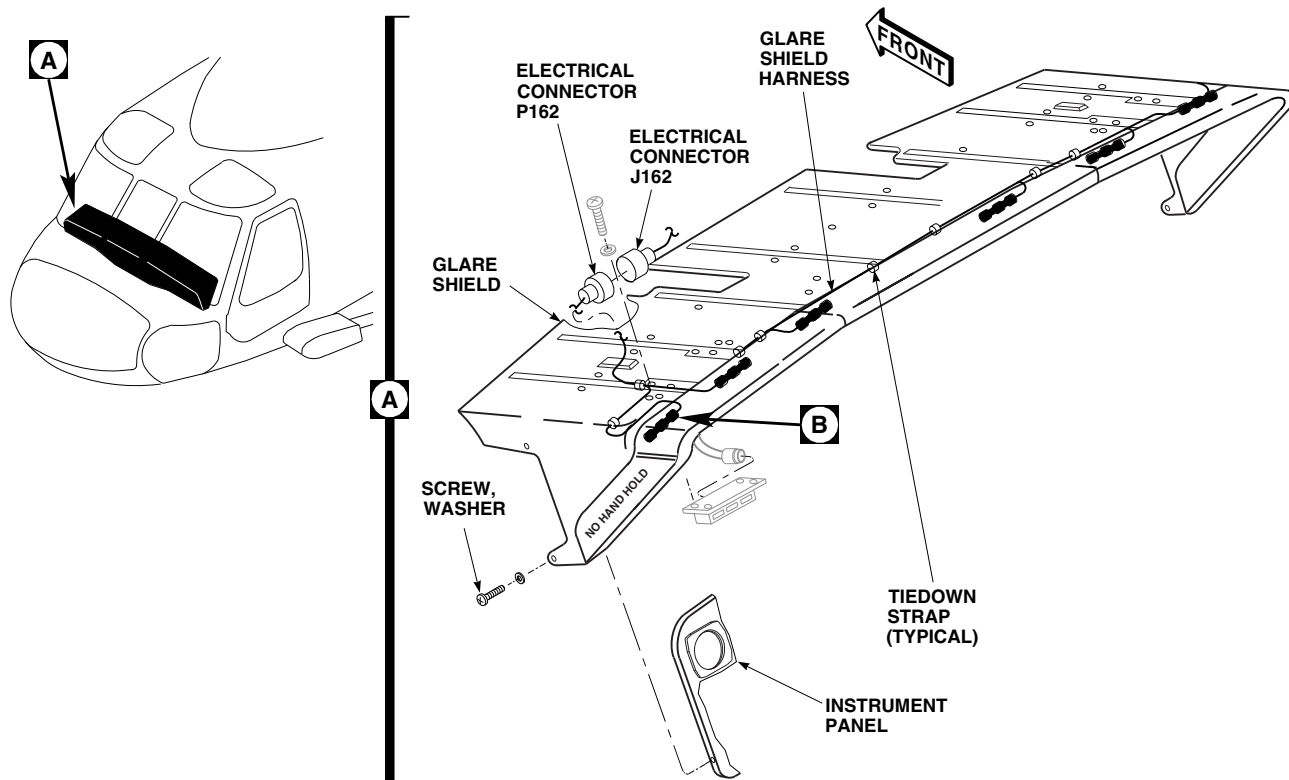
2-4-28.1.1 Remove.

- Remove pilot’s and copilot’s master warning panel (PARA 8-4-24).
- Remove screws and washers securing glare shield to instrument panel (Figure 2-4-28-1, Detail A).
- Carefully slide glare shield from instrument panel enough to access electrical connectors, P162 and J162.
- Disconnect electrical connector, P162 from J162, and remove glare shield from helicopter.
- Install protective caps and plugs on electrical connectors.

- If glare shield is being replaced, remove glare shield lights and harness as follows:
 - Remove screws, washers, and nuts securing glare shield lights and ground wires to glare shield (Figure 2-4-28-1, Detail B).
 - Cut tiedown straps securing glare shield harness to glare shield. Remove harness (Figure 2-4-28-1, Detail A).

2-4-28.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- If installing new glare shield, install glare shield lights and harness as follows:
 - Install glare shield harness to glare shield using tiedown straps, Item 334, Appendix D (Figure 2-4-28-1, Detail A).

**NOTE**

WASHERS ARE INSTALLED UNDER SCREWHEAD ON OUTBOARD GLARE SHIELD LIGHT ASSEMBLIES ONLY.

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FIGURE 2-4-28-1. REPLACE GLARE SHIELD

NOTE

Washers are installed under screwheads on outboard glare shield light assemblies only.

- (2) Install glare shield lights and ground wires on glare shield with screws, washers, and nuts. Make sure ground wire is installed under nut (Figure 2-4-28-1, Detail B).
- c. Remove protective caps and plugs from electrical connectors.
- d. Inspect and treat electrical connectors (PARA 1-7-20).

NOTE

Prior to positioning glare shield, make sure electrical connectors for master warning panels are hanging over top of instrument panel.

- e. If necessary, position electrical connectors for master warning panels so they hang over top of instrument panel.
- f. Position glare shield on instrument panel so that electrical connectors, P162 and J162, can be accessed (Figure 2-4-28-1, Detail B).
- g. Connect electrical connector, P162 to J162.
- h. Carefully slide glare shield forward and position on instrument panel and supports.
- i. Install screws and washers securing glare shield to instrument panel.
- j. Install pilot's and copilot's master warning panel (PARA 8-4-24).

2-4-28.2 Repair Glare Shield.

2-4-28.2.1 Repair Structural Damage.

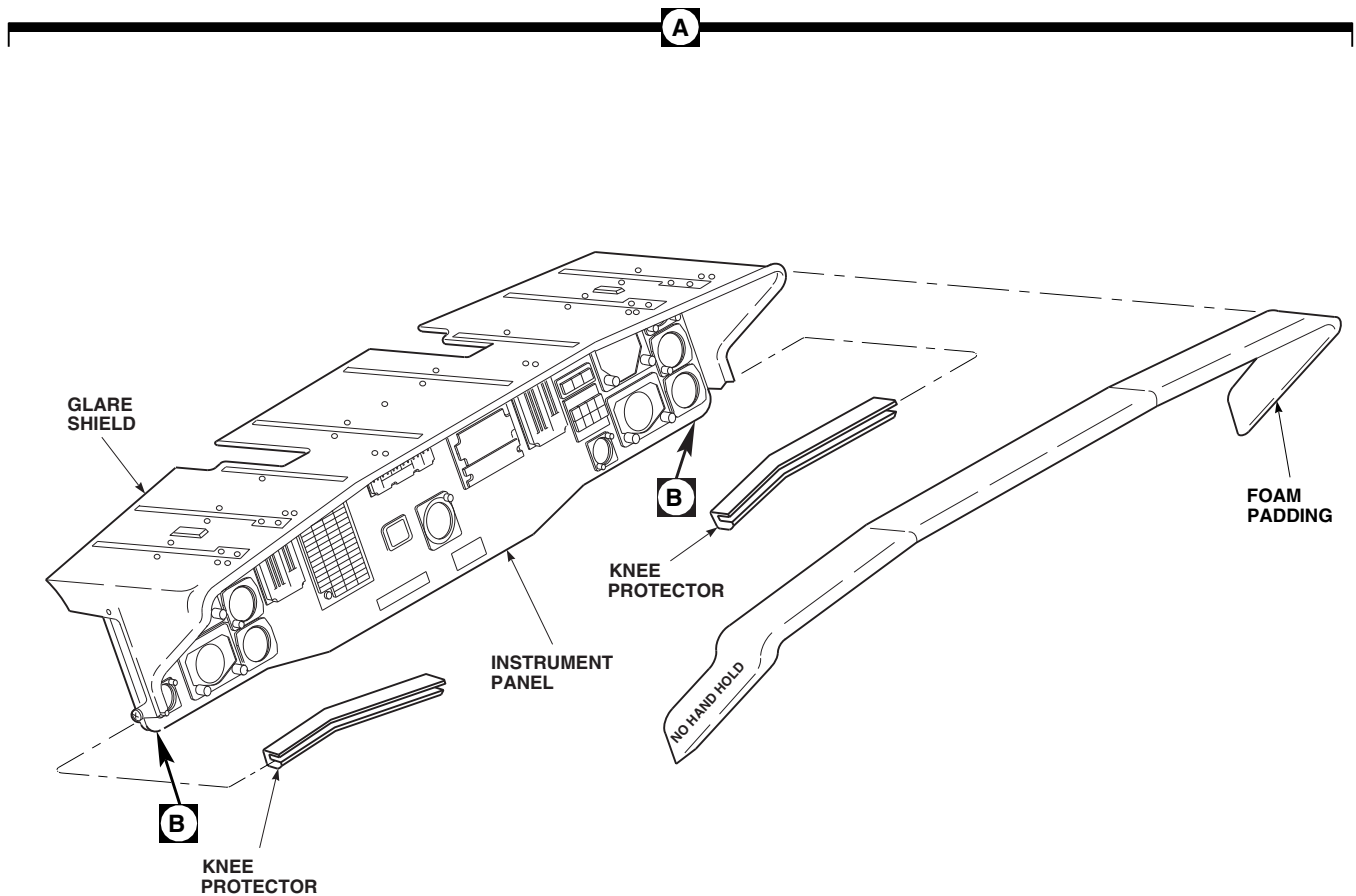
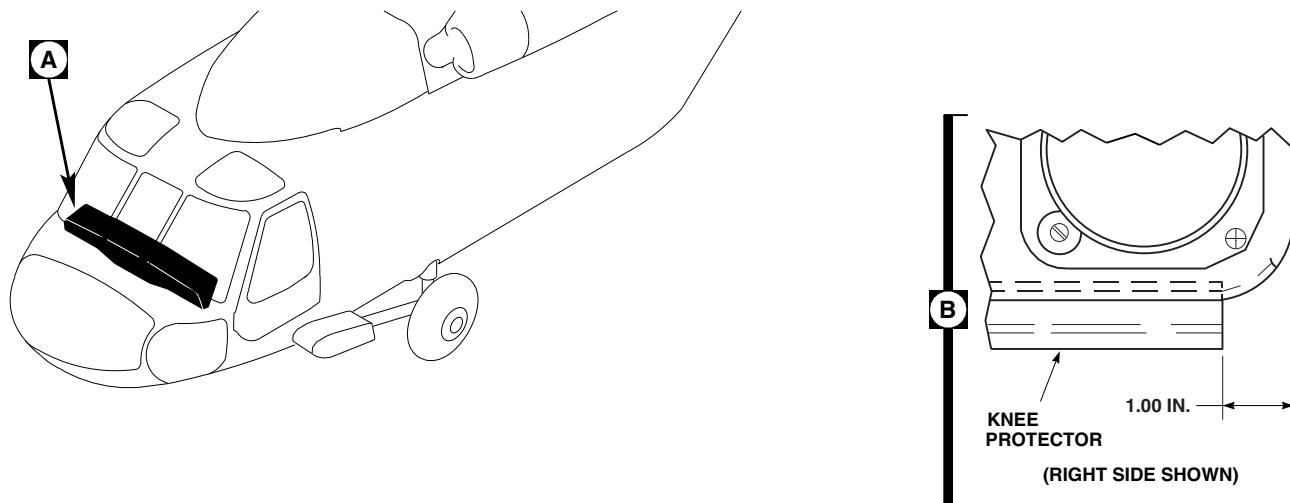
Repair structure of glare shield (PARA 2-6-3).

2-4-28.2.2 Replace Foam Padding and Knee Protector.

NOTE

Replacement of foam padding and knee protectors is the same, except as noted.

- a. Using nonmetallic scraper, remove loose or damaged foam padding/knee protector (Figure 2-4-28-2, Detail A).
- b. Clean remaining adhesive from glare shield using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.
- c. Wipe edge of glare shield dry with clean machinery towel, Item 211, Appendix D.
- d. If required, foam padding/knee protector may be spliced in transition areas, joints to be bonded with adhesive, Item 37, Appendix D.
- e. Lightly scuff inside surfaces of foam padding/knee protector using abrasive cloth, Item 2, Appendix D.
- f. Apply even coat of adhesive, Item 37, Appendix D, to inside surfaces of foam padding/knee protector and mating surfaces of glare shield.
- g. Allow adhesive to air-dry for 5 - 10 minutes until tacky.
- h. If installing foam padding/knee protector, position on front edge of glare shield. Make sure surfaces of foam padding/knee protector and glare shield are in direct contact.
- i. If installing knee protector, position 1-inch from edge of instrument panel (Figure 2-4-28-2, Detail B). Make sure surfaces of knee protector and instrument panel are in direct contact.
- j. Apply bead of sealing compound, Item 292, Appendix D, to where top and side edges of foam padding/knee protector contact glare shield.



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FIGURE 2-4-28-2. REPAIR GLARE SHIELD

2-4-29 INSTRUMENT PANEL.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
2-4-29.1	Replace Instrument Panel	2-4-29-1	2-4-29.2	Repair Instrument Panel	2-4-29-3

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Digital Low-Resistance Ohmmeter
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Corrosion Resistant Coating, Item 118, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-28
- PARA 2-6-2

- PARA 2-6-5
- PARA 8-4-4
- PARA 8-4-5
- PARA 8-4-6
- PARA 8-4-10
- PARA 8-4-16
- PARA 8-4-19
- PARA 8-4-28
- PARA 9-4-96
- PARA 12-4-27
- PARA 12-4-29
- PARA 12-4-46
- QQ-A-250
- TM 11-70-23

2-4-29.1 Replace Instrument Panel.

2-4-29.1.1 Remove.

- Remove glare shield (PARA 2-4-28).
- Remove the following from instrument panel:
 - Airspeed Indicators (PARA 8-4-5).
 - Barometric Altimeters (PARA 8-4-6).

- Vertical Speed Indicators (PARA 8-4-4).
- Pilot Display Units (PARA 8-4-16).
- Clocks (PARA 8-4-28).
- Caution/Advisory Panel (PARA 8-4-19).
- Central Display Unit (PARA 8-4-10).
- Blade Deice Control Panel (PARA 12-4-29).

- (9) Blade Deice Test Panel (PARA 12-4-46).
- (10) Icing Rate Meter (PARA 12-4-27).
- (11) Engine Ignition Switch (PARA 9-4-96).
- c. Remove the following from instrument panel (TM 11-70-23):
 - (1) Stabilator Indicator.
 - (2) HSI/VSI Mode Selectors.
 - (3) Vertical Situation Indicators.
 - (4) Horizontal Situation Indicators.
 - (5) Radar Altimeter Receiver/Transmitter Indicator.
 - (6) Radar Signals Indicator.
 - (7) Height Indicator (Radar Altimeter).
 - (8) Battery Alert Horn.
 - (9) Battery Test Switch.
 - (10) Emergency Power Arm Switch.
- d. Remove bolts and washers securing instrument panel to end supports and top supports (Figure 2-4-29-1, Sheet 1, Detail A).
- e. If installed, remove bolts, washers, and bonding strap securing instrument panel to lower console (Figure 2-4-29-1, Sheet 2, Detail B).
- f. Remove instrument panel.

2-4-29.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Install instrument panel as follows:
 - (1) If installed, inspect chemical conversion coated surfaces where bonding strap mate with instrument panel (PARA 2-6-5).
 - (2) Position instrument panel and install bonding strap at RBL 1.5 and LBL 4.0, if

removed, to lower console with bolts, lockwashers, and washers (Figure 2-4-29-1, Sheet 1, Detail A and Sheet 2, Detail B). Do not torque bolts now.

- (3) Install remaining bolts and washers securing instrument panel to end supports, top supports, and lower console (Figure 2-4-29-1, Sheet 1, Detail A). **TORQUE ALL BOLTS TO 45 - 55 INCH-POUNDS.**
- (4) Perform EME resistance check as follows:

NOTE

Resistance measurements must be taken at least three times.

- (a) Using digital low-resistance ohmmeter, perform EME resistance check between instrument panel and lower console, right/left end supports, and top support. **RESISTANCE SHALL BE 5.0 MILLIOHMS OR LESS.**
- (b) If resistance is greater than 5.0 milliohms, remove instrument panel and replace chemical conversion coating on instrument panel and lower console (PARA 2-6-5).

- c. Install the following in instrument panel:

- (1) Airspeed Indicators (PARA 8-4-5).
- (2) Barometric Altimeters (PARA 8-4-6).
- (3) Vertical Speed Indicators (PARA 8-4-4).
- (4) Pilot Display Units (PARA 8-4-16).
- (5) Clocks (PARA 8-4-28).
- (6) Caution/Advisory Panel (PARA 8-4-19).
- (7) Central Display Unit (PARA 8-4-10).
- (8) Blade Deice Control Panel (PARA 12-4-29).

2-4-29-2 Change 5

- (9) Blade Deice Test Panel (PARA 12-4-46).
- (10) Icing Rate Meter (PARA 12-4-27).
- (11) Engine Ignition Switch (PARA 9-4-96).
- d. Install the following in instrument panel (TM 11-70-23):
 - (1) Stabilator Indicator.
 - (2) HSI/VSI Mode Selectors.
 - (3) Vertical Situation Indicators.
 - (4) Horizontal Situation Indicators.
 - (5) Radar Altimeter Receiver/Transmitter Indicator.
 - (6) Radar Signals Indicator.
 - (7) Height Indicator (Radar Altimeter).
 - (8) Battery Alert Horn.
- (9) Battery Test Switch.
- (10) Emergency Power Arm Switch.
- e. Install glare shield (PARA 2-4-28).
- f. Using digital low-resistance ohmmeter, perform EME resistance check between bottom edge of instrument panel and lower console. **RESISTANCE SHALL BE 5.0 MILLIOHMS OR LESS.**
- g. Perform operational checks (TM 11-70-23).

2-4-29.2 Repair Instrument Panel.

- a. Repair structure of instrument panel (PARA 2-6-2).
- b. Inspect areas with chemical conversion coating (Figure 2-4-29-2).
- c. Apply corrosion resistant coating, Item 118, Appendix D, as required (PARA 2-6-5).

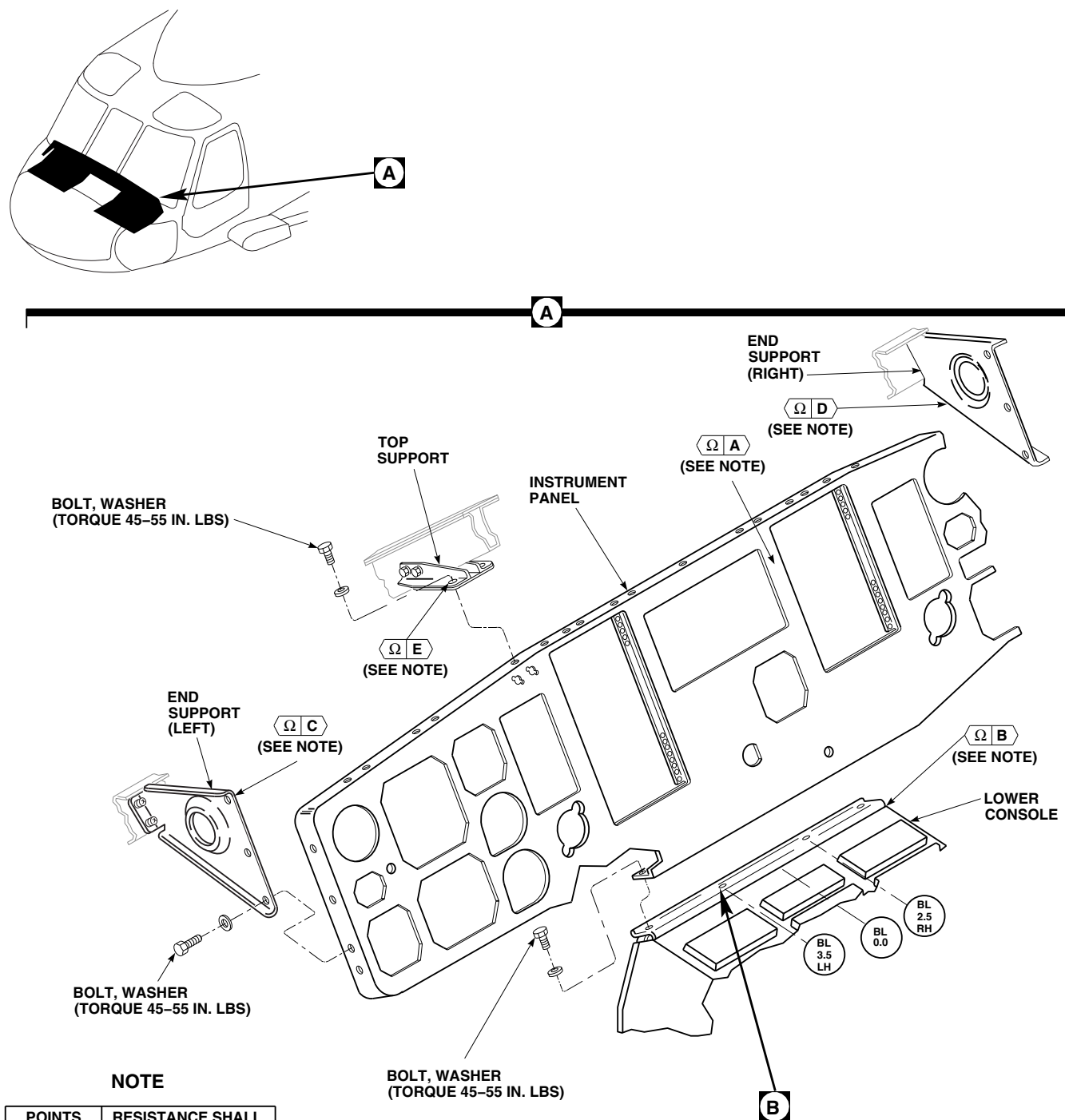


FIGURE 2-4-29-1. REPLACE INSTRUMENT PANEL (SHEET 1 OF 2)

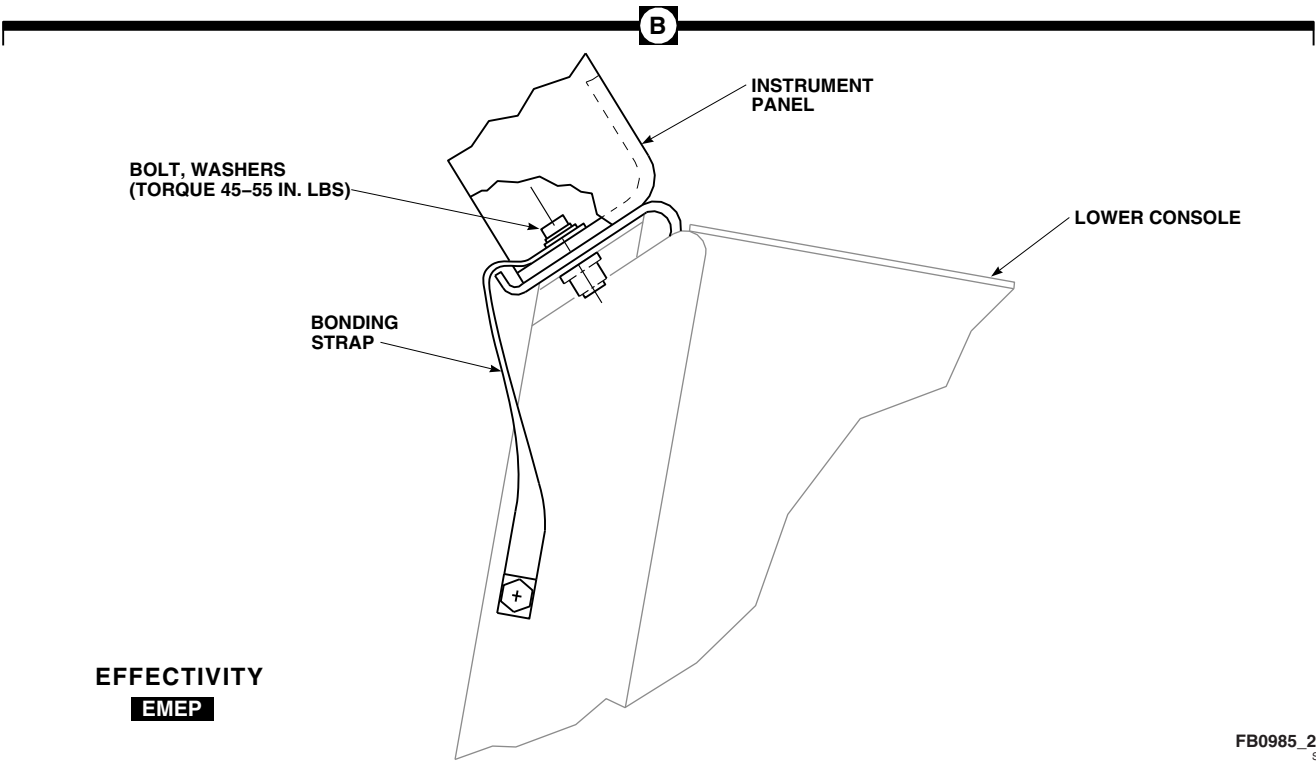


FIGURE 2-4-29-1. REPLACE INSTRUMENT PANEL (SHEET 2 OF 2)

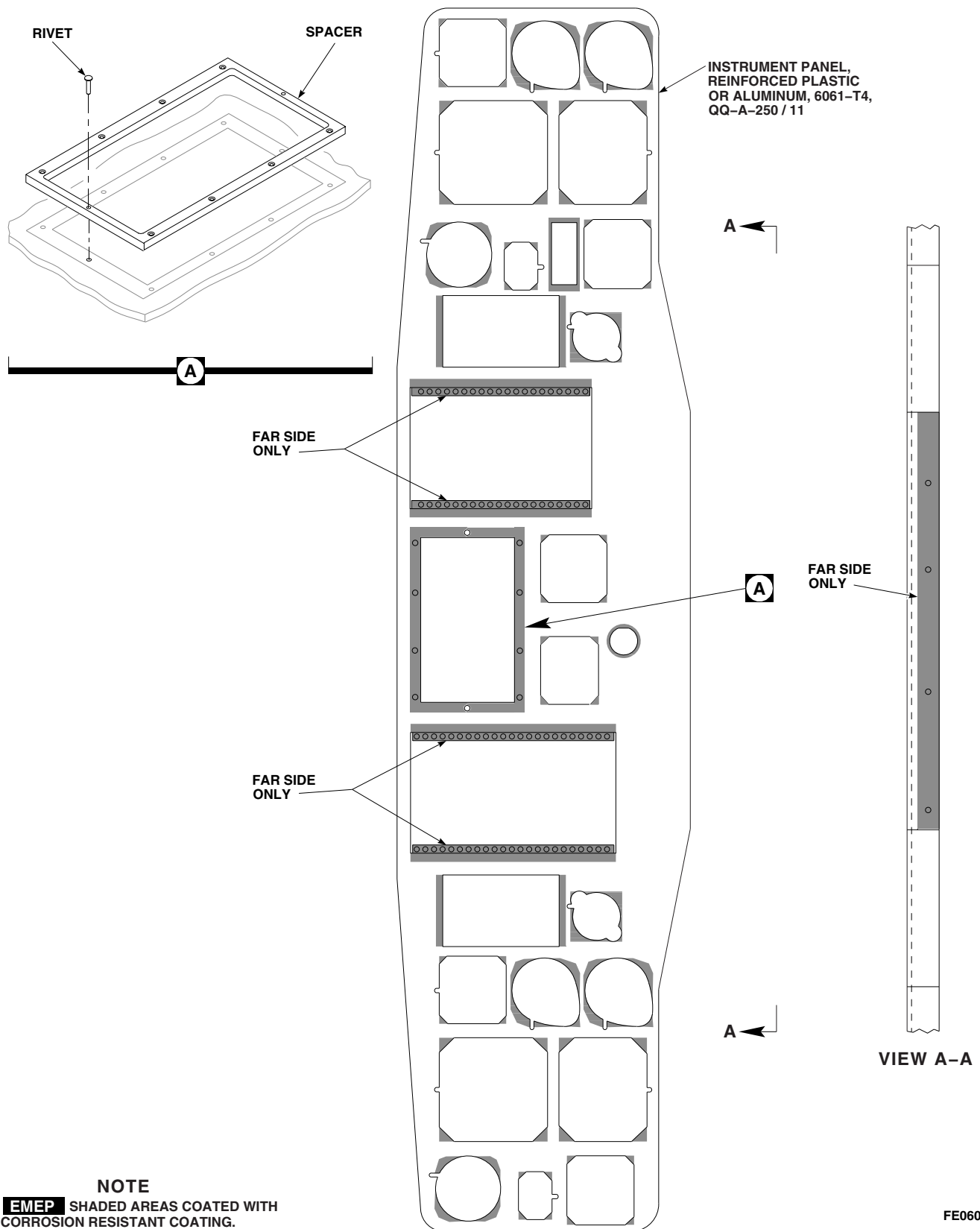


FIGURE 2-4-29-2. REPAIR INSTRUMENT PANEL

2-4-30 NOSE SECTION AVIONIC COMPARTMENT COVER.

PARAGRAPH BREAKDOWN

		PAGE
2-4-30.1	Replace Nose Section Avionic Compartment Cover	2-4-30-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Rivet Gun

Materials/Parts

- Fastener Tape, Hook, Item 137A, Appendix D
- Fastener Tape, Pile, Item 137B, Appendix D

- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D

References

- Appendix D
- PARA 1-6-16

2-4-30.1 Replace Nose Section Avionic Compartment Cover.

2-4-30.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Open nose door.
- Pull avionic compartment cover from bottom, tip down from top, and remove cover from hook fastener tape on structure (Figure 2-4-30-1, Detail A).

2-4-30.1.2 Repair.

- Remove rivets securing hinge to avionic compartment cover (Figure 2-4-30-1, Detail A). Remove door.
- Remove rivets securing hinge to door. Remove hinge.

- Pull damaged pile fastener tape from cover and/or door. Using machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D, wipe surface clean of remaining adhesive.
- Pull damaged hook fastener tape from structure and/or cover. Using machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D, wipe surface clean of remaining adhesive.
- Install hook fastener tape, Item 137A, Appendix D, where removed on structure and/or cover.
- Install pile fastener tape, Item 137B, Appendix D, where removed on cover and/or door.
- Position hinge on door and using rivet gun, secure with rivets.

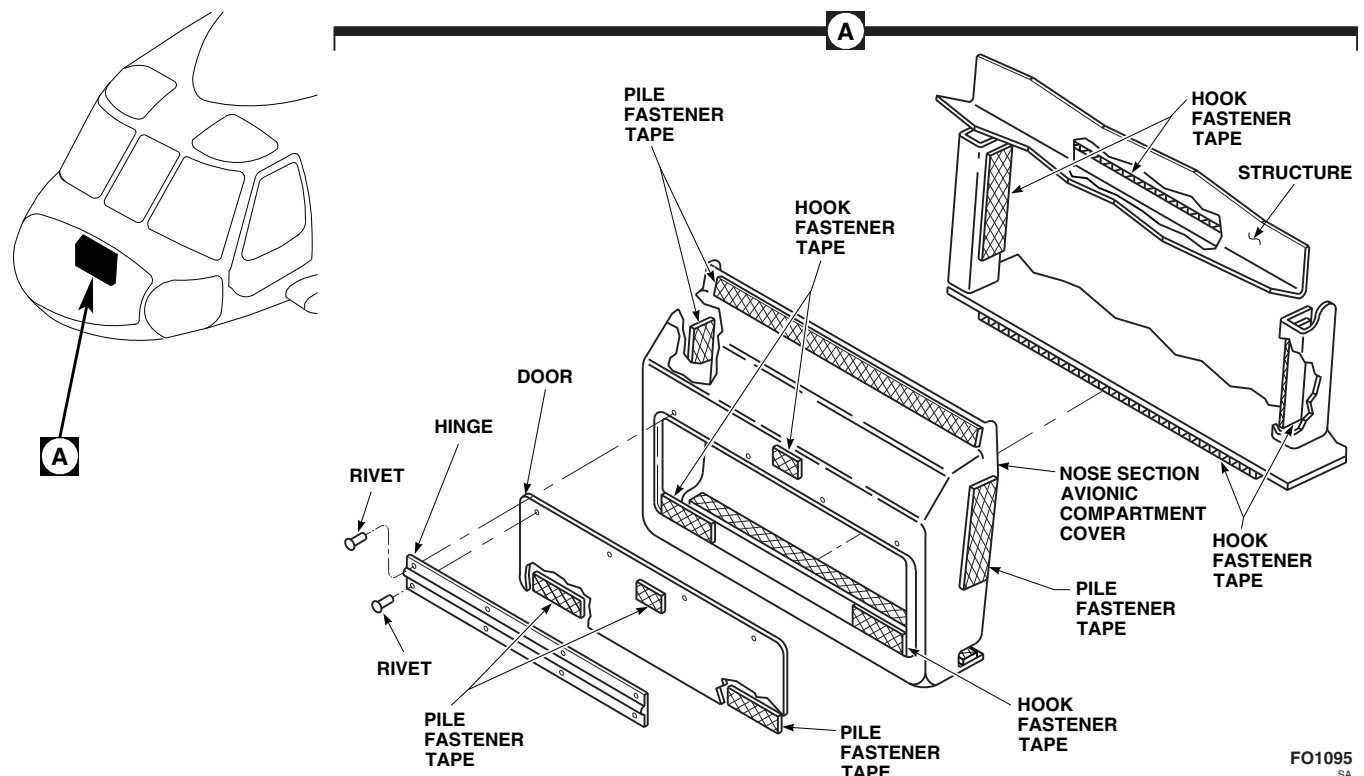
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FIGURE 2-4-30-1. REPLACE NOSE SECTION AVIONIC COMPARTMENT COVER

- h. Position door on cover and using rivet gun, secure hinge to cover with rivets.

flange and push in bottom and sides. Engage pile fastener tape with hook fastener tape (Figure 2-4-30-1, Detail A).

2-4-30.1.3 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Install cover on structure in avionics compartment. Place top of cover behind upper structure

- c. Close nose door and secure latches.

2-4-31 NOSE SECTION THERMAL BARRIERS.

PARAGRAPH BREAKDOWN

		PAGE
2-4-31.1	Replace Nose Section Thermal Barriers	2-4-31-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit

Materials/Parts

- Adhesive, Item 49, Appendix D
- Adhesive, Item 61, Appendix D
- Adhesive Primer, Item 67, Appendix D

- Fastener Tape, Hook, Item 137A, Appendix D
- Fastener Tape, Pile, Item 137B, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 2-6-3

2-4-31.1 Replace Nose Section Thermal Barriers.

NOTE

Replacement of right and left nose section thermal barriers is the same.

2-4-31.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Remove screws and washers, and pull thermal barrier from supports and hook fastener tape (Figure 2-4-31-1, Detail A and Detail B).

2-4-31.1.2. Repair.

- Repair thermal barrier structure (PARA 2-6-3).
- Pull damaged pile fastener tape from barrier and hook fastener tape from supports (Figure 2-4-31-1, Detail A and Detail B).

- Pull damaged grommet from barrier.
- Using machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D, clean remaining adhesive from barrier and supports.
- Coat mating surface of grommet and top of barrier with adhesive, Item 49, Appendix D, and bond grommet to barrier. Trim grommet to edge of barrier.
- Apply a light coat of adhesive primer, Item 67, Appendix D, to thermal barrier and supports. Allow primer to dry for 10 minutes or until tacky.
- Apply a light coat of adhesive, Item 61, Appendix D, to thermal barrier, supports, pile fastener tape, Item 137B, Appendix D, and hook fastener tape, Item 137A, Appendix D. Allow adhesive to become tacky.

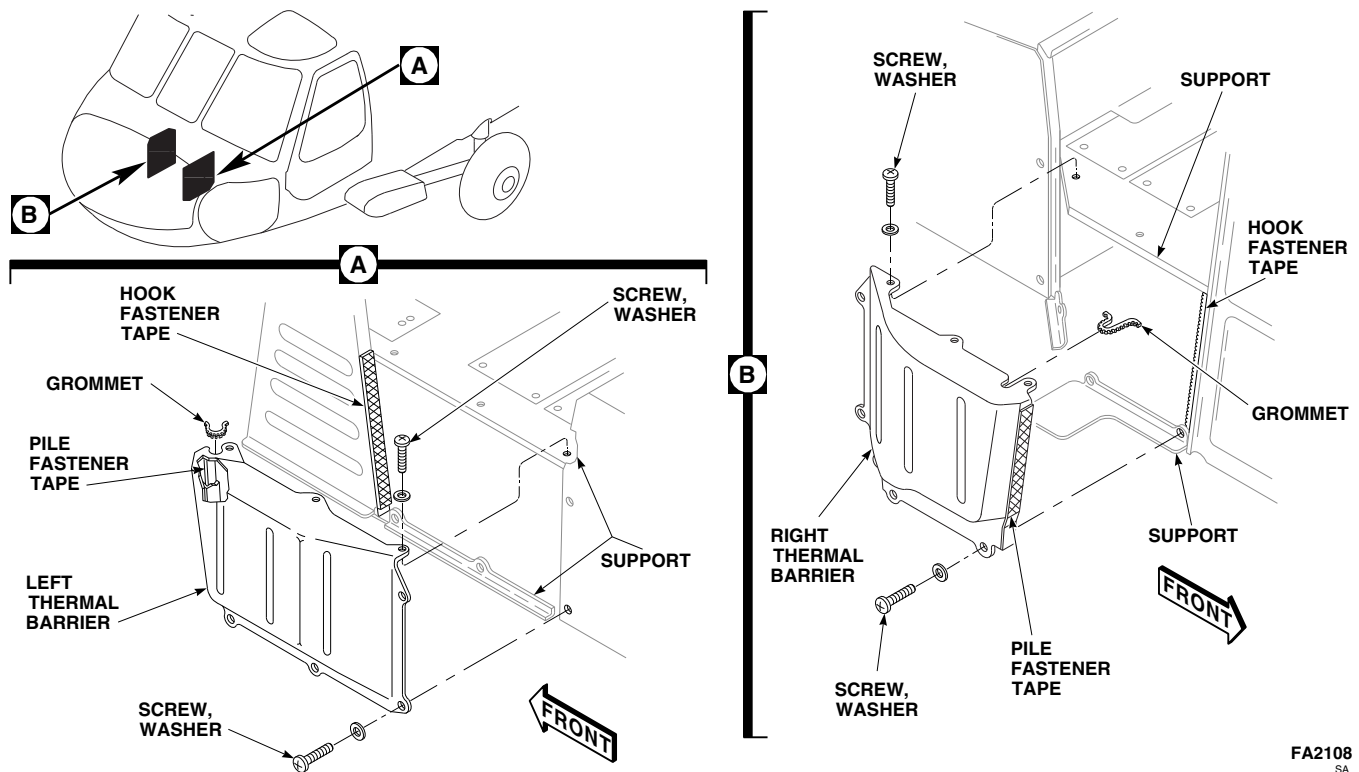
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FIGURE 2-4-31-1. REPLACE NOSE SECTION THERMAL BARRIERS

- h. Bond pile fastener tape, Item 137B, Appendix D, to barrier and hook fastener tape, Item 137A, Appendix D, to supports.

2-4-31.1.3. Install.

- a. Turn off all electrical power (PARA 1-6-16).

- b. Place thermal barrier against hook fastener tape and supports, guide wire bundle through grommet, and secure barrier with screws and washers (Figure 2-4-31-1, Detail A and Detail B).

2-4-32 NOSE VIBRATION ABSORBER (SPRING-TYPE).

PARAGRAPH BREAKDOWN

		PAGE			PAGE
2-4-32.1	Replace Nose Vibration Absorber (Spring-Type)	2-4-32-2	2-4-32.3	Adjust Nose Vibration Absorber Tuning Weights	2-4-32-10
2-4-32.2	Inspect/Repair Nose Vibration Absorber (Spring-Type)	2-4-32-5			

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Dial Indicator
- Feeler Gage
- Fluorescent Penetrant Inspection Kit
- Protection, Eye, Skin and Breathing
- Torque Wrench, 0 - 16 in. oz
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Abrasive Cloth, Item 8, Appendix D
- Abrasive Cloth, Item 13, Appendix D
- Bolt, NAS1305-24DH

- Bolt, NAS1305-27D
- Bolt, NAS1305-68D
- Grease, Item 151A, Appendix D
- Machinery Towel, Item 211, Appendix D
- Sealing Compound, Item 280, Appendix D
- Sealing Compound, Item 282, Appendix D
- Tags
- Tetrachloroethylene, Item 324, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 2-5-6
- PARA 2-6-5
- PARA 2-6-8
- TM 1-70-VIB
- TM 11-70-23

2-4-32.1 Replace Nose Vibration Absorber (Spring-Type).

2-4-32.1.1 Remove.

WARNING

Nose vibration absorber lower fitting bushing, 70219-02150-102, and nose vibration absorber airframe fitting bushing, 70219-02150-104, are made of beryllium copper metal, which can be a health hazard. Avoid breathing dust while reworking damaged areas. Use a dust respirator if necessary. Grinding is prohibited. Keep beryllium copper out of open wounds. Use skin and eye protection if necessary. Clean up visible dust piles and wash hands and arms after maintenance.

- Remove glide slope antenna (TM 11-70-23).
- Remove nuts and washers from bolts securing front and rear left fittings of nose vibration absorber to left airframe fittings (Figure 2-4-32-1, Detail A). Do not remove bolts now.
- Remove nut and washer from bolt securing right lower fitting of absorber to right airframe fitting. Do not remove bolt now.

WARNING

Injury to personnel or damage to equipment will result if absorber is not supported when removing attachment hardware. Spring-type vibration absorber weighs approximately 75 pounds. Do not remove attachment hardware unless absorber is supported.

- With assistant supporting absorber, remove bolts and washers securing left fittings of absorber to left airframe fittings.
- Lower left side of absorber onto structure.

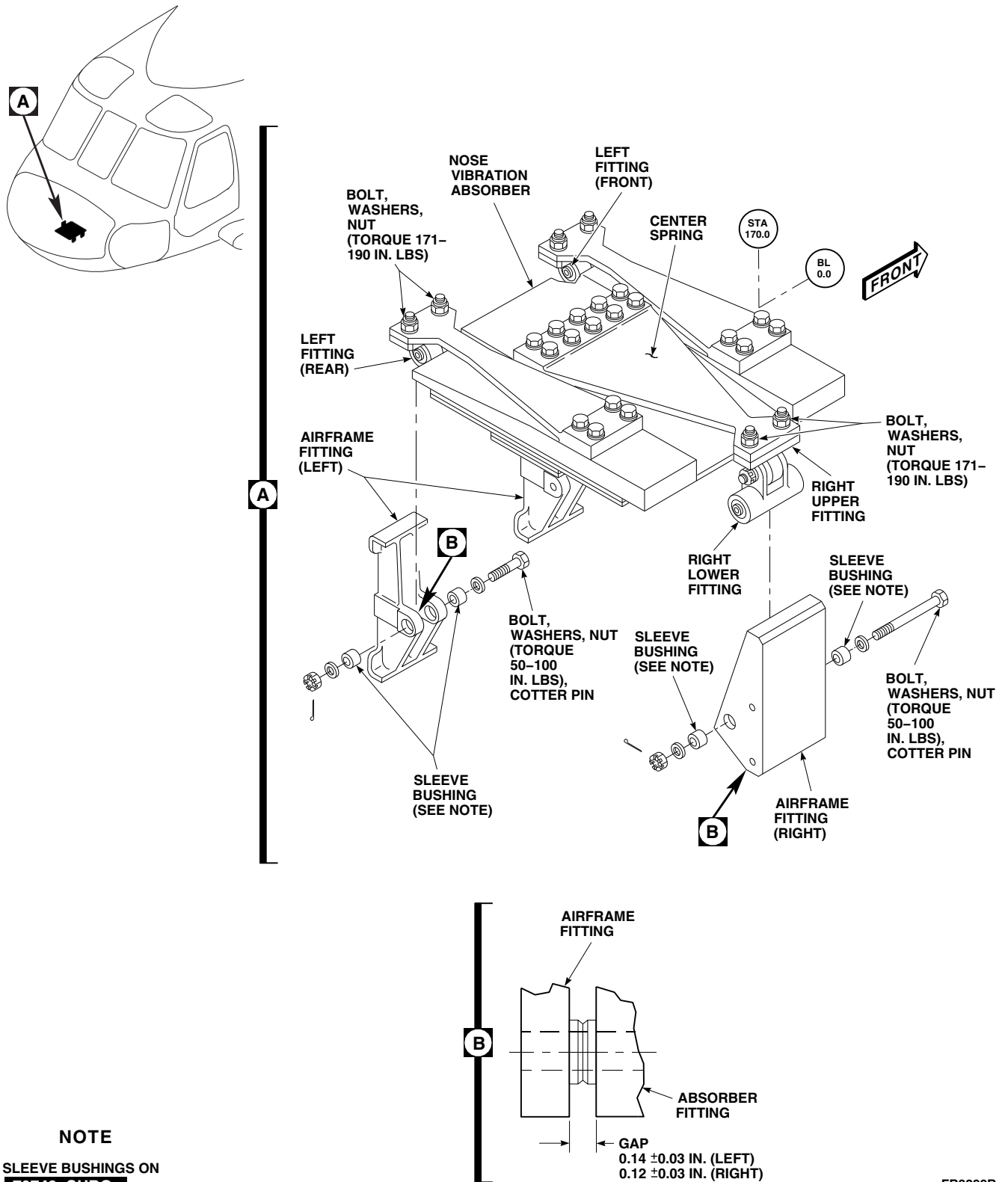
- With assistant supporting absorber, remove bolt securing right lower fitting to right airframe fitting.
- While still supporting absorber, lower right side.
- Raise left side and move to left until right side clears structure. Continue to lower right side until absorber is removed from structure.
- Inspect/repair nose vibration absorber (PARA 2-4-32.2.2).

2-4-32.1.2 Install.

WARNING

Nose vibration absorber lower fitting bushing, 70219-02150-102, and nose vibration absorber airframe fitting bushing, 70219-02150-104, are made of beryllium copper metal, which can be a health hazard. Avoid breathing dust while reworking damaged areas. Use a dust respirator if necessary. Grinding is prohibited. Keep beryllium copper out of open wounds. Use skin and eye protection if necessary. Clean up visible dust piles and wash hands and arms after maintenance.

- Lubricate sleeve bushings with grease, Item 151A, Appendix D.
- With aid of assistant, position absorber in nose structure and raise into place, keeping left side of absorber higher than right side.
- Move absorber to left as far as possible to give clearance on right side.
- Raise right side so right lower fitting engages right airframe fitting. Install sleeve bushings in right airframe fitting. Install bolt and washer with bolt head facing front (Figure 2-4-32-1, Detail A).
- Engage left fittings of absorber with left airframe fittings. Install sleeve bushings in left



airframe fittings. Install bolts and washers with boltheads facing front.

- f. Install washers and nuts on bolts in left and right airframe fittings.
- g. Torque nuts on left and right airframe fittings as follows:
 - (1) **TORQUE NUTS TO 80 - 95 INCH-POUNDS.** Loosen torque back to 0 inch-pounds.
 - (2) **TORQUE NUTS TO 50 - 100 INCH-POUNDS.** Install cotter pins.
- h. Check center spring fitting alignment with left rear and right airframe fittings as follows:
 - (1) Rotate left front fitting bolt two full turns.
 - (2) Rotate left rear and right attachment bolts two full turns.
 - (3) Using feeler gage, measure gaps between absorber fitting faces and airframe fitting faces (Figure 2-4-32-1, Detail B). **FOR RIGHT AIRFRAME FITTING, GAPS MUST BE 0.09 - 0.15-INCH. FOR LEFT REAR AIRFRAME FITTING, GAPS MUST BE 0.11 - 0.17-INCH.** If gap(s) is beyond limit, adjust clearance as required (step j. and/or step k.).
- i. Check for absorber binding as follows:
 - (1) Grasp right side of absorber body. Rock absorber back-and-forth. **MAKE SURE SLEEVE BUSHINGS SLIDE FROM SIDE-TO-SIDE IN AIRFRAME FITTINGS.** If right sleeve bushings do not slide in airframe fitting, check torque on nut and/or slip-fit of sleeve bushings.
 - (2) Check breakout torque on left rear and right side attachment bolts using torque wrench. **BREAKOUT TORQUE MUST NOT EXCEED 10 INCH-POUNDS THROUGH FULL 360° TURN.** If breakout torque exceeds limit, check torque on nut and slip-fit of sleeve bushings.

- (3) Check breakout torque on left front side attachment bolt using torque wrench. **BREAKOUT TORQUE MUST EXCEED 50 INCH-POUNDS.** If breakout torque is 50 inch-pounds or less, check buildup and torque on nut.

NOTE

If front left fitting of absorber locates against flanged bushing in front left airframe fitting, no adjustment is required.

- j. Adjust right upper fitting of absorber in right airframe fitting as follows (Figure 2-4-32-1, Detail A):
 - (1) Make sure absorber is installed properly.
 - (2) Loosen bolts securing right upper fitting to center spring.
 - (3) **ADJUST RIGHT LOWER FITTING TO GAP OF 0.09 - 0.15-INCH EACH SIDE** (Figure 2-4-32-1, Detail B).
 - (4) **TORQUE NUTS TO 171 - 190 INCH-POUNDS** (Figure 2-4-32-1, Detail A).
 - (5) Rotate right airframe fitting attachment bolt two full turns.
 - (6) Recheck gaps between right lower fitting faces and right airframe fitting faces. If limits are not met, repeat step j. (2) through step j. (6) until gap is obtained.
- k. Adjust left fitting of absorber in left rear airframe fitting as follows:
 - (1) Make sure vibration absorber is installed properly.
 - (2) Loosen bolts securing fitting to side spring.
 - (3) **ADJUST LEFT FITTING TO GAP OF 0.11 - 0.17-INCH EACH SIDE** (Figure 2-4-32-1, Detail B).
 - (4) **TORQUE NUTS TO 171 - 190 INCH-POUNDS** (Figure 2-4-32-1, Detail A).

- (5) Rotate rear left airframe fitting attachment bolt two full turns.
- (6) Recheck gaps between left fitting faces and left rear airframe fitting faces. If limits are not met, repeat step k. (2) through step k. (6) until gap is obtained.
- l. Install glide slope antenna (TM 11-70-23).
- m. Perform vibration check of absorber (TM 1-70-VIB).

2-4-32.2 Inspect/Repair Nose Vibration Absorber (Spring-Type).

2-4-32.2.1 Disassemble.

- a. If not previously done, remove nose vibration absorber (PARA 2-4-32.1.1).

WARNING

Nose vibration absorber lower fitting bushing, 70219-02150-102, and nose vibration absorber airframe fitting bushing, 70219-02150-104, are made of beryllium copper metal, which can be a health hazard. Avoid breathing dust while reworking damaged areas. Use a dust respirator if necessary. Grinding is prohibited. Keep beryllium copper out of open wounds. Use skin and eye protection if necessary. Clean up visible dust piles and wash hands and arms after maintenance.

- b. Remove right upper fitting and right lower fitting as follows:
 - (1) Remove bolt, washers, and nut securing right upper fitting to right lower fitting (Figure 2-4-32-2, Sheet 1). Keep hardware for installation.
 - (2) Remove bolts, washers, and nuts securing right upper fitting and liner to center spring. Keep hardware for installation.

- c. Remove bolts, washers, and nuts securing left fittings and liners to side springs. Keep hardware for installation.
- d. Remove bolts and washers securing tuning weights to center weight and body weight. Tag and record each tuning weight for installation.
- e. Remove bolts and washers securing center weight to body weight.
- f. Remove bolts, countersunk washers, washers, and nuts securing center spring and liner to body weight.
- g. Remove bolts, countersunk washers, washers, and nuts securing side springs and liners to body weight.

2-4-32.2.2 Inspect/Repair.

WARNING

Nose vibration absorber lower fitting bushing, 70219-02150-102, and nose vibration absorber airframe fitting bushing, 70219-02150-104, are made of beryllium copper metal, which can be a health hazard. Avoid breathing dust while reworking damaged areas. Use a dust respirator if necessary. Grinding is prohibited. Keep beryllium copper out of open wounds. Use skin and eye protection if necessary. Clean up visible dust piles and wash hands and arms after maintenance.

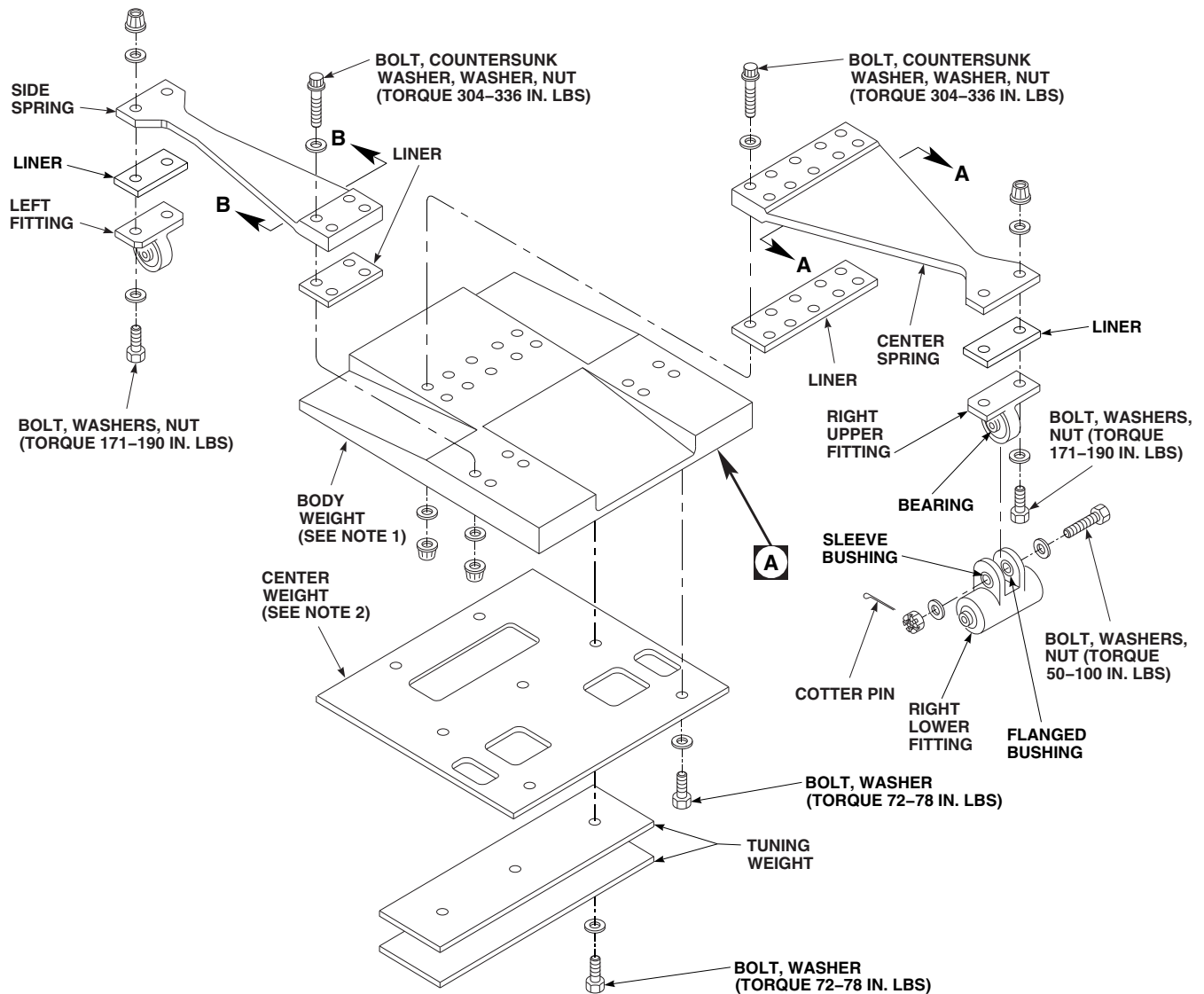
- a. Check bearings in absorber fittings for wear as follows:
 - (1) Set up dial indicator with stylus tracking bearing inner race radial movement.
 - (2) **MAXIMUM RADIAL PLAY ALLOWED IS 0.005-INCH.** If radial play exceeds limit, replace bearing (PARA 2-5-6).
- b. Check bearing in absorber fittings for breakout torque as follows:

- (1) Install new bolt, NAS1305-27D, in bearing with enough washers on bolt to seat nut. Install and tighten nut so washers do not turn.
 - (2) Rotate bearing inner race four complete turns.
 - (3) Using torque wrench, turn bearing. **BREAKOUT TORQUE MUST NOT EXCEED 10 INCH-OUNCES THROUGH FULL 360° TURN.**
- c. Check all absorber attachment bushings for wear as follows:
- (1) Install all absorber attachment bushings in respective airframe fitting bushing bores.
 - (2) Install a new bolt, NAS1305-27D, through bushings in left front airframe fitting.
 - (3) Set up dial indicator with indicator stylus touching bolt.
 - (4) Take measurements as close to bushings as possible.
 - (5) **MAXIMUM RADIAL PLAY ALLOWED IS 0.002-INCH.** If radial play exceeds limit, replace bushing (PARA 2-5-6). Repeat step c. (1) through step c. (5).
 - (6) If replacing bushing does not correct problem, replace fitting (PARA 2-6-8).
 - (7) Install a new bolt, NAS1305-27D, through bushings in left rear airframe fitting.
 - (8) Repeat step c. (3) through step c. (6).
 - (9) Install a new bolt, NAS1305-68D, through bushings in right airframe fitting.
 - (10) Repeat step c. (3) through step c. (6).
- d. Check sleeve bushings in airframe fittings as follows:
- (1) Remove sleeve bushing from airframe fittings (Figure 2-4-32-1, Detail A).
 - (2) Coat outer surface of sleeve bushing with grease, Item 151A, Appendix D.
 - (3) Install sleeve bushing in airframe fitting.
 - (4) Grasp sleeve bushing ends between thumb and forefinger and slide bushing side-to-side in sleeve bushing hole. If sleeve bushing does not move freely, perform the following as required:
 - (a) Replace sleeve bushing. Repeat step d. (1) through step d. (4).
 - (b) If replacing sleeve bushing does not correct problem, inspect sleeve bushing holes in airframe fitting for burrs. If burrs are found, remove burrs using abrasive cloth, Item 13, Appendix D. Repeat step d. (1) through step d. (4).
 - (c) If replacing sleeve bushings and deburring sleeve bushing hole does not correct the problem, replace fitting (PARA 2-6-8).
 - (5) Remove sleeve bushing from airframe fitting.
- e. Check springs for corrosion, gouges, nicks, and cracks as follows:

NOTE

Damage rework will remove shotpeen coverage and is considered acceptable in order to extend component service life.

- (1) **DAMAGE TO SIDE SPRINGS, 70219-02106-110, AND CENTER SPRING, 70219-02106-111, MUST NOT EXCEED 0.315-INCH MINIMUM THICKNESS** (Figure 2-4-32-2, Sheet 1, Section A-A or B-B). If necessary, blend damage using abrasive cloth, Item 8, Appendix D. If damage exceeds limit, replace spring.
- (2) **DAMAGE TO SIDE SPRINGS, 70219-02106-114, AND CENTER SPRING, 70219-02106-115, MUST NOT EXCEED**



NOTES

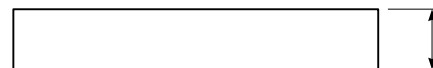
1. BODY WEIGHT, 70219-02107-103, ON VIBRATION ABSORBERS, 70219-02107-141, -042, AND -044; BODY WEIGHT, 70219-02107-109, ON VIBRATION ABSORBERS, 70219-02107-043 AND -046.
2. CENTER WEIGHT ON VIBRATION ABSORBERS 70219-02107-141, -042 (2 USED), -044 (2 USED), -045 AND -046.

SECTION A-A



70219-02106-111 = 0.315 IN. MINIMUM THICKNESS
70219-02106-115 = 0.300 IN. MINIMUM THICKNESS

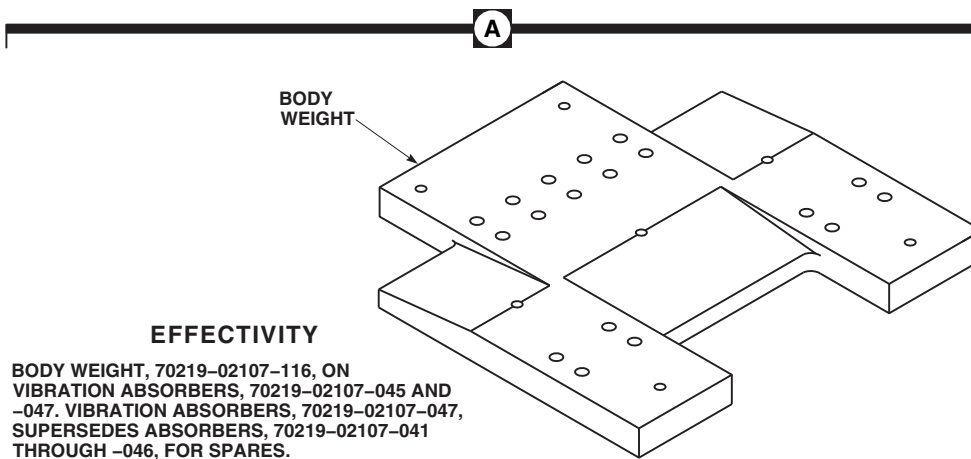
SECTION B-B



70219-02106-110 = 0.315 IN. MINIMUM THICKNESS
70219-02106-114 = 0.300 IN. MINIMUM THICKNESS

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**FIGURE 2-4-32-2. INSPECT/REPAIR NOSE VIBRATION ABSORBER (SPRING-TYPE)
(SHEET 1 OF 2)**

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**FIGURE 2-4-32-2. INSPECT/REPAIR NOSE VIBRATION ABSORBER (SPRING-TYPE)
(SHEET 2 OF 2)**

0.300-INCH MINIMUM THICKNESS.

If necessary, blend damage using abrasive cloth, Item 8, Appendix D. If damage exceeds limit, replace spring.

- (3) **DAMAGE TO ANY CROSS-SECTIONAL AREA MUST NOT EXCEED 25% OF PERIMETER.**
- (4) **NO DAMAGE ALLOWED WITHIN A RADIUS EQUAL TO ONE DIAMETER AT ANY ATTACHMENT HOLE.**
- (5) **IF DAMAGE IS FOUND IN OR AROUND ATTACHMENT HOLES, CONTACT ENGINEERING AUTHORITY FOR DAMAGE ANALYSIS AND RECOMMENDED REPAIR OR REPLACEMENT.**
- (6) Touch up finish (PARA 2-6-5).

- f. Check attachment bolts and hardware for corrosion and wear. **NO CORROSION ALLOWED. MAXIMUM WEAR ALLOWED IS 0.002-INCH.** If corroded, or wear exceeds limit, replace hardware.
- g. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect airframe fittings around attachment holes for cracks and corrosion (ASTM E1417). **NONE ALLOWED.** If cracked or corroded, replace or repair fittings (PARA 2-6-8).

- h. Check surrounding structure for cracks, dents, nicks, or other obvious damage. **NONE ALLOWED.** If damaged, repair (PARA 2-6-8).

2-4-32.2.3 Assemble.

WARNING

Nose vibration absorber lower fitting bushing, 70219-02150-102, and nose vibration absorber airframe fitting bushing, 70219-02150-104, are made of beryllium copper metal, which can be a health hazard. Avoid breathing dust while reworking damaged areas. Use a dust respirator if necessary. Grinding is prohibited. Keep beryllium copper out of open wounds. Use skin and eye protection if necessary. Clean up visible dust piles and wash hands and arms after maintenance.

- a. Clean threads of all bolts and nuts with tetrachloroethylene, Item 324, Appendix D. Allow to air-dry.
- b. Install center weight as follows (Figure 2-4-32-2, Sheet 1):
 - (1) Position center weight on body weight with boltholes lined up.

- (2) Apply sealing compound, Item 282, Appendix D, to threads of bolts.
 - (3) Allow to dry for 5 - 15 minutes.
 - (4) Apply sealing compound, Item 280, Appendix D, to threads of bolts.
 - (5) Install bolts and washers securing center weight to body weight. **TORQUE BOLTS TO 72 - 78 INCH-POUNDS.**
- c. Install side springs as follows:
- (1) Position liners and side springs on body weight with boltholes lined up.
 - (2) Install countersunk washers on bolts with countersunk side facing boltheads.
 - (3) Apply sealing compound, Item 282, Appendix D, to threads of bolts and nuts.
 - (4) Allow sealing compound to dry for 5 - 15 minutes.
 - (5) Apply sealing compound, Item 280, Appendix D, to threads of bolts and nuts.
 - (6) Install bolts (with countersunk washers installed and boltheads seated on side springs), washers, and nuts securing side springs and liners to body weight. **TORQUE NUTS TO 304 - 336 INCH-POUNDS.**
- d. Install center spring as follows:
- (1) Position liner and center spring on body weight with boltholes lined up.
 - (2) Install countersunk washers on bolts with countersunk side facing boltheads.
 - (3) Apply sealing compound, Item 282, Appendix D, to threads of bolts and nuts.
 - (4) Allow sealing compound to dry for 5 - 15 minutes.
 - (5) Apply sealing compound, Item 280, Appendix D, to threads of bolts and nuts.

- (6) Install bolts (with countersunk washers installed and boltheads seated on side springs), washers, and nuts securing center spring and liner to body weight. **TORQUE NUTS TO 304 - 336 INCH-POUNDS.**

- e. Install tuning weights as follows:

NOTE

If installing new absorber, use same amount of tuning weights as on old absorber.

- (1) Remove tags from tuning weights and position weights on center weight and body weight with boltholes lined up.
- (2) Apply sealing compound, Item 282, Appendix D, to threads of bolts. Allow sealing compound to dry for 5 - 15 minutes.
- (3) Allow sealing compound to dry for 5 - 15 minutes.
- (4) Apply sealing compound, Item 280, Appendix D, to threads of bolts.
- (5) Install bolts and washers securing tuning weights to center weight. **TORQUE BOLTS TO 72 - 78 INCH-POUNDS.**

- f. Install left fittings as follows:

- (1) Apply sealing compound, Item 282, Appendix D, to threads of bolts and nuts.
- (2) Allow sealing compound to dry for 5 - 15 minutes.
- (3) Apply sealing compound, Item 280, Appendix D, to threads of bolts and nuts.
- (4) Position left fittings and liners on side springs. Use any required combination of thick and thin washers.
- (5) Install bolts, washers, and nuts securing left fittings and liners to side springs. **TORQUE NUTS TO 171 - 190 INCH-POUNDS.**

g. Install right upper fitting and right lower fitting as follows:

- (1) Install right upper fitting and liner on center spring.
- (2) Install bolt, washers, and nuts securing right upper fitting and liner to center spring. **TORQUE NUTS TO 171 - 190 INCH-POUNDS.**

NOTE

Lower fitting front end is on the side of flanged bushing.

- (3) Install right lower fitting on right upper fitting and secure with washer and new bolt, NAS1305-24DH. Make sure washer under bolthead is against flanged bushing. Use any required combination of thick and thin washers.
- (4) Install washer and nut. **TORQUE NUT TO 50 - 100 INCH-POUNDS.**
- (5) Make sure bearing in right upper fitting is sandwiched between flange and sleeve bushings, and does not move from side-to-side. Install cotter pin.

h. Install nose vibration absorber (PARA 2-4-32.1.2).

2-4-32.3 Adjust Nose Vibration Absorber Tuning Weights.

- a. Determine weight change (TM 1-70-VIB).
- b. Remove glide slope antenna (TM 11-70-23).

c. Remove bolts and washers securing tuning weights to center weight and body weight (Figure 2-4-32-2, Sheet 1).

d. Tune vibration absorber (TM 1-70-VIB).

e. Check for proper bolt length as follows:

- (1) Stack up tuning weights and insert bolts through tuning weight holes.
- (2) Count number of threads protruding beyond tuning weights. If less than five threads are visible, use longer bolts. If unthreaded portion of bolt is visible, use shorter bolts.
- (3) Remove bolts.

f. Clean threads of tuning weight bolts with machinery towel, Item 211, Appendix D, wet with tetrachloroethylene, Item 324, Appendix D. Allow to air-dry.

g. Apply sealing compound, Item 282, Appendix D, to threads of bolts.

h. Allow to dry for 5 - 15 minutes.

i. Apply sealing compound, Item 280, Appendix D, to threads of bolts.

j. Position tuning weights on center weight and align boltholes. Install bolts and washers. **TORQUE BOLTS TO 72 - 78 INCH-POUNDS.**

k. Install glide slope antenna (TM 11-70-23).

2-4-33 NOSE VIBRATION ABSORBER (BOX-TYPE).

PARAGRAPH BREAKDOWN

		PAGE
2-4-33.1	Replace Nose Vibration Absorber (Box-Type)	2-4-33-2
2-4-33.2	Inspect/Repair Nose Vibration Absorber (Box-Type)	2-4-33-3

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Fluorescent Penetrant Inspection Kit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Vise, Soft-Jawed

Materials/Parts

- Lockwire, Item 197, Appendix D
- Machinery Towel, Item 211, Appendix D

- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 301, Appendix D
- Tags

References

- Appendix D
- ASTM E1417
- PARA 2-6-8
- TM 1-70-VIB
- TM 11-70-23

2-4-33.1 Replace Nose Vibration Absorber (Box-Type).

2-4-33.1.1 Remove.

- a. Remove glide slope antenna (TM 11-70-23).

WARNING

Injury to personnel or damage to equipment will result if absorber is not supported when removing attachment hardware. Box-type vibration absorber weighs approximately 50 pounds. Do not remove attachment hardware unless absorber is supported.

- b. With assistant supporting absorber, remove bolts and washers securing absorber to left and right airframe fittings (Figure 2-4-33-1, Detail A).
- c. With aid of assistant, lower absorber until clear of structure.
- d. Remove barrel-nuts and retainers from airframe fittings.
- e. Inspect/repair nose vibration absorber (PARA 2-4-33.2).

2-4-33.1.2 Install.

CAUTION

Damage to recess in airframe fittings will result if barrel-nuts are incorrectly installed. To prevent damage, make sure barrel-nuts are installed with barrel (round side) facing bolthead.

- a. Install barrel-nuts and retainers into airframe fittings (Figure 2-4-33-1, Detail A).
- b. Apply sealing compound, Item 301, Appendix D, to mating surfaces of airframe fittings and absorber.

WARNING

Injury to personnel or damage to equipment will result if absorber is not supported when installing attachment hardware. Box-type vibration absorber weighs approximately 50 pounds.

- c. With aid of assistant, lift and position absorber in airframe fittings.
- d. Make sure boltholes line up, and serrated plates on airframe fittings and absorber are engaged.
- e. Apply sealing compound, Item 301, Appendix D, to bolts.
- f. Install bolts and washers. **TORQUE BOLTS TO 385 - 425 INCH-POUNDS.** Using lockwire, Item 197, Appendix D, lockwire bolts.
- g. Install glide slope antenna (TM 11-70-23).
- h. Check cockpit main rotor 4/rev vibration utilizing data measurement procedure in TM 1-70-VIB, WP 008 00, as follows:

NOTE

TM 1-70-VIB primarily covers main rotor 1/rev tuning, but also measures and checks cockpit main rotor 4/rev vibration.

- (1) Refer to the Data Sheet, Figure 3 or 4, then PARA 8 - Main Rotor 1/Rev Flight Tuning, Recalling Data, to determine measured 4/rev levels and limits.
- (2) If the procedures in TM 1-70-VIB can not be done (for example due to the utilization of different test equipment), cockpit main rotor 4/rev vibration can be checked by acquiring the following measurements:
 - (a) Cockpit 4/rev A+B at 140 knots/100% Nr.
 - (b) The limit for cockpit 4/rev A+B at 140 knots is 0.40 ips, and the limit

for pilot heel vertical at both 120 knots and 140 knots is 0.60 ips.

- (c) If cockpit main rotor 4/rev vibration is within limits, no absorber tuning is required. If cockpit main rotor 4/rev vibration is not within limits, contact a Sikorsky Field Service Representative, or Sikorsky Aircraft Corporation for assistance.

2-4-33.2 Inspect/Repair Nose Vibration Absorber (Box-Type).

2-4-33.2.1 Disassemble.

WARNING

Injury to personnel and damage to equipment will occur if housing attachment bolts are loosened or removed. Box-type vibration absorber springs are always in compression. Make sure that housing attachment bolts are not loosened or removed.

NOTE

Disassembly of absorber is limited to removal of tuning weights and serrated plate.

- a. If removing serrated plate, perform the following:
 - (1) If not previously done, remove nose vibration absorber (PARA 2-4-33.1.1).
 - (2) Remove screws securing serrated plate to absorber (Figure 2-4-33-1, Detail A). Remove serrated plate.
- b. Remove bolts and washers securing short tuning weights to long tuning weights and absorber. Tag and record each tuning weight for installation.
- c. Remove bolts and washers securing long tuning weights to absorber. Tag and record each tuning weight for installation.

- d. Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean sealing compound from all hardware.

2-4-33.2.2 Inspect/Repair.

- a. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect absorber upper and lower housings for cracks and corrosion (ASTM E1417). **NO CRACKS ALLOWED.** If cracks are found, replace absorber.
- b. Check serrated plate for damage to serrations. **NONE ALLOWED.** If damage is found, replace serrated plate.
- c. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect airframe fittings for cracks and corrosion (ASTM E1417). **NONE ALLOWED.** If cracked or corroded, replace or repair fittings (PARA 2-6-8).
- d. Check barrel-nuts and retainers for damage and corrosion. **NONE ALLOWED.** If damaged or corroded, replace barrel-nuts and retainers.
- e. Test barrel-nuts as follows:

NOTE

Make sure that only square barrel-nuts are used.

- (1) Place barrel-nut in soft-jawed vise.
- (2) Thread bolt into barrel-nut until two threads are showing beyond nut.
- (3) Using torque wrench, back out bolt. **BREAKAWAY TORQUE MUST NOT BE LESS THAN 10 INCH-POUNDS.** If breakaway torque is less than 10 inch-pounds, replace barrel-nut.
- (4) Repeat step e. (1) through e. (3) for other barrel-nut.
- f. Check surrounding structure for cracks, dents, nicks or other obvious damage. If damage is found, repair as required (PARA 2-6-8).

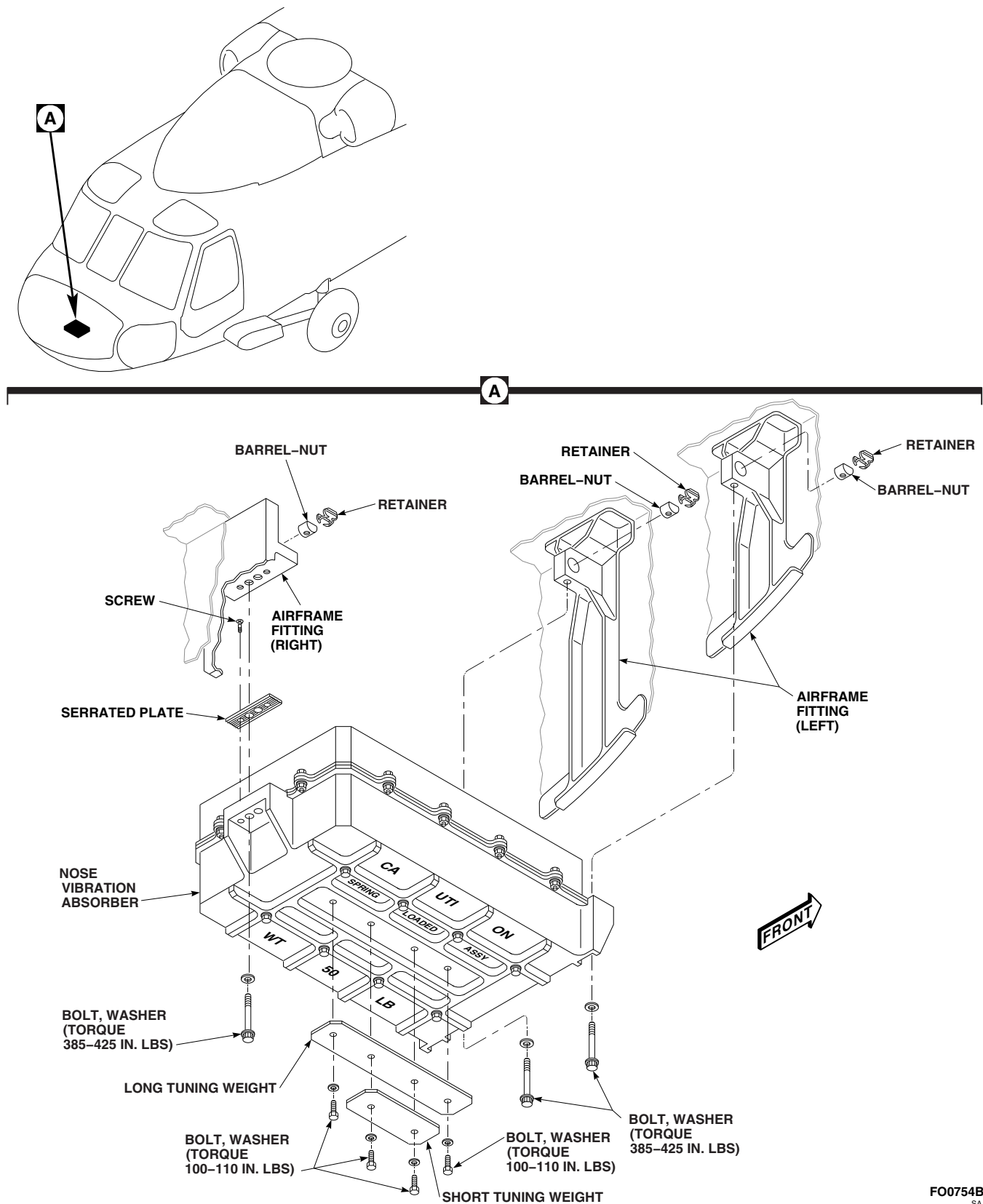
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FIGURE 2-4-33-1. REPLACE/REPAIR NOSE VIBRATION ABSORBER (BOX-TYPE)

2-4-33-4 Change 11

2-4-33.2.3 Assemble.

- a. Apply sealing compound, Item 301, Appendix D, to threads of screws.
- b. Install serrated plate on absorber and secure with screws (Figure 2-4-33-1, Detail A).
- c. Remove tags from short and long tuning weights. Position weights on absorber with boltholes lined up.
- d. Apply sealing compound, Item 301, Appendix D, to threads of bolts.
- e. Install bolts and washers securing tuning weights to absorber. **TORQUE BOLTS TO 100 - 110 INCH-POUNDS.**
- f. If removed, install nose vibration absorber (PARA 2-4-33.1.2).

2-4-34 CABIN VIBRATION ABSORBER (SPRING-TYPE).

PARAGRAPH BREAKDOWN

		PAGE			PAGE
2-4-34.1	Replace Cabin Vibration Absorber (Spring-Type)	2-4-34-2	2-4-34.3	Adjust Cabin Vibration Absorber Tuning Weights	2-4-34-8
2-4-34.2	Inspect/Repair Cabin Vibration Absorber (Spring-Type)	2-4-34-4			

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Dial Indicator
- Feeler Gage
- Fluorescent Penetrant Inspection Kit
- Protection, Eye, Skin, and Breathing
- Torque Wrench, 0 - 16 in. oz
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Wooden Tuning Wedge, Locally-Made, Figure H-159, Appendix H

Materials/Parts

- Abrasive Cloth, Item 8, Appendix D
- Abrasive Cloth, Item 13, Appendix D
- Bolt, NAS1306-38D

- Bolt, NAS1306-39D
- Grease, Item 151A, Appendix D
- Machinery Towel, Item 211, Appendix D
- Sealing Compound, Item 280, Appendix D
- Sealing Compound, Item 282, Appendix D
- Tags
- Tetrachloroethylene, Item 324, Appendix D

References

- Appendix D
- Appendix H
- ASTM E1417
- PARA 2-4-69
- PARA 2-5-7
- PARA 2-5-8
- PARA 2-6-5
- PARA 2-6-11
- TM 1-70-VIB

2-4-34.1 Replace Cabin Vibration Absorber (Spring-Type).

2-4-34.1.1 Remove.

WARNING

Cabin vibration absorber airframe fitting bushing, 70219-02150-101, and cabin vibration absorber spring bushing, 70219-02150-103, are made of beryllium copper metal, which can be a health hazard. Avoid breathing dust while reworking damaged areas. Use dust respirator if necessary. Grinding is prohibited. Keep beryllium copper out of open wounds. Use skin and eye protection if necessary. Clean up visible dust piles and wash hands and arms after maintenance.

- Remove soundproofing panel(s) from cabin ceiling, forward of drip pan, as required (PARA 2-4-69).
- Remove nuts and thick washers from bolts securing front and rear springs of absorber to left airframe fittings (Figure 2-4-34-1, Detail A). Do not remove bolts now.
- Remove nut and thick washer from bolt securing right lower fitting of absorber to right airframe fitting. Do not remove bolt now.
- Remove bolts, washers, and nuts securing stop to right airframe fitting.

WARNING

Injury to personnel or damage to equipment will result if absorber is not supported when removing attachment hardware. Spring-type vibration absorber weighs approximately 75 pounds. Do not remove attachment hardware unless absorber is supported.

- With assistant supporting absorber, remove bolts and thin washers securing front and rear springs of absorber to left airframe fittings.

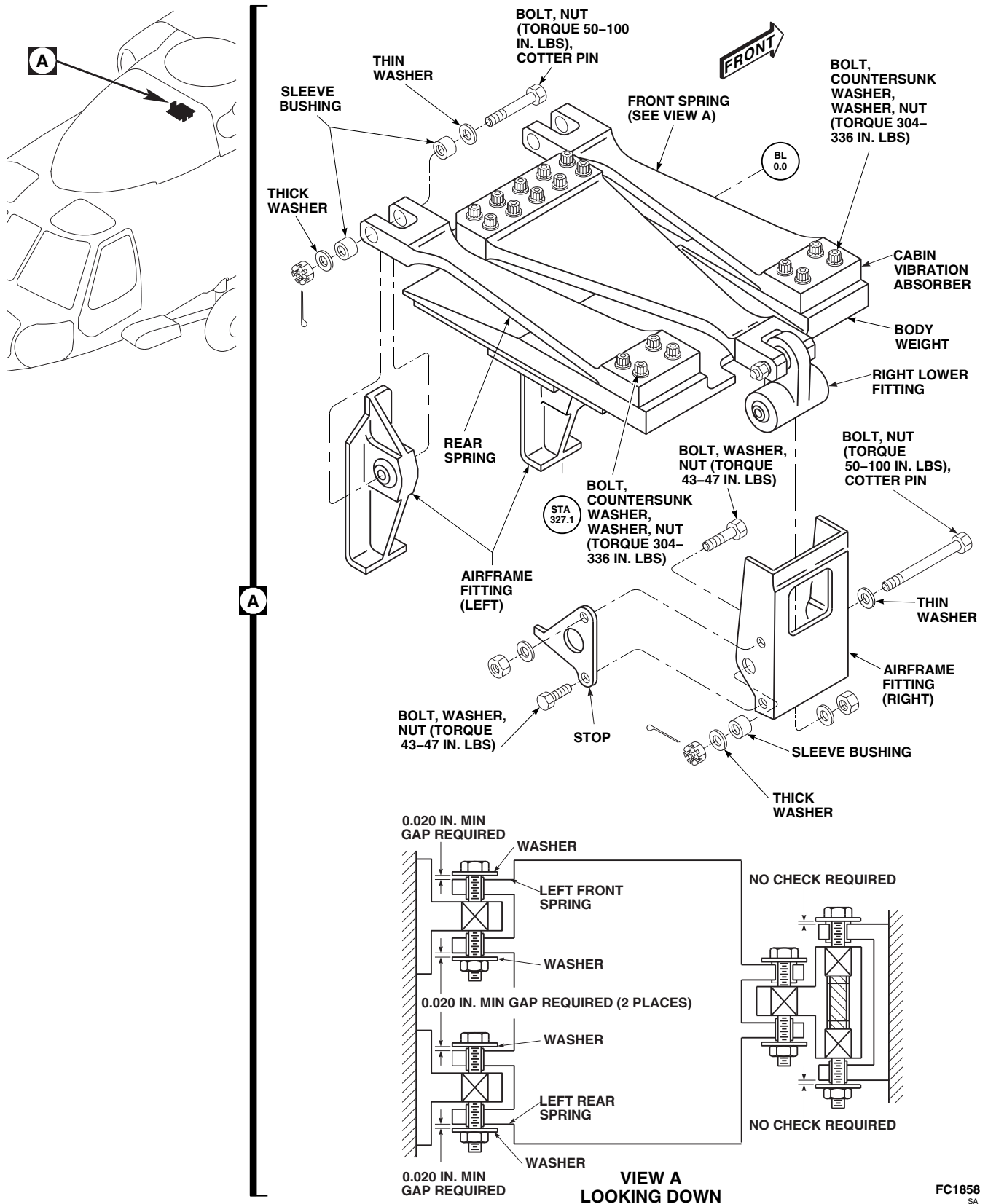
- Lower left side of absorber onto structure.
- With assistant supporting absorber, remove bolt and thin washer securing right lower fitting to right airframe fitting.
- While still supporting absorber, lower right side.
- Raise left side and move to left until right side clears structure. Continue to lower right side until absorber is clear of structure.
- Inspect/repair cabin vibration absorber (PARA 2-4-34.2.2).

2-4-34.1.2 Install.

WARNING

Cabin vibration absorber airframe fitting bushing, 70219-02150-101, and cabin vibration absorber spring bushing, 70219-02150-103, are made of beryllium copper metal, which can be a health hazard. Avoid breathing dust while reworking damaged areas. Use dust respirator if necessary. Grinding is prohibited. Keep beryllium copper out of open wounds. Use skin and eye protection if necessary. Clean up visible dust piles and wash hands and arms after maintenance.

- Lubricate sleeve bushings with grease, Item 151A, Appendix D.
- With aid of assistant, position absorber in structure and raise into place, keeping left side of absorber higher than right side.
- Move absorber left as far as possible to give clearance on right side.
- Raise right side so right lower fitting engages right airframe fitting. Install sleeve bushings in right airframe fitting. Install bolt, washers, and nut, with bolthead facing front (Figure 2-4-34-1, Detail A).
- Engage front and rear springs of absorber with left airframe fittings. Install sleeve bushings in



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front and rear springs. Install bolts, washers, and nuts, with boltheads facing front.

- f. Torque nuts on left and right airframe fittings as follows:

- (1) **TORQUE NUTS TO 72 - 78 INCH-POUNDS.** Loosen torque back to 0 inch-pounds.
- (2) **TORQUE NUTS TO 50 - 100 INCH-POUNDS.** Install cotter pins.

- g. Secure stop on right airframe fitting with bolts, washers, and nuts. **TORQUE NUTS TO 43 - 47 INCH-POUNDS.**

- h. Using feeler gage, measure for 0.020-inch gap on front and rear spring, between spring yoke and washers (Figure 2-4-34-1, View A). If gap on either side of spring yoke, between spring yoke and washers, is not 0.020-inch, align front and rear springs as follows:

NOTE

- Center spring (right airframe fitting) is resting on a flanged bushing and cannot be out of alignment.
- Alignment procedure is the same for front and rear springs.

- (1) Make sure absorber is installed properly.
- (2) Loosen four nuts securing spring to body weight. Loosen nuts enough to allow bolts and nuts to turn together with moderate wrench force on nut.

- (3) Push or pull on spring to obtain 0.020-inch gap between both sides of spring yoke and washers.

- (4) When spring alignment is correct, tighten two spring bolts.

- (5) Recheck gaps between both sides of spring yoke and washers. If limits are not met, repeat step h. (2) through step h. (4) until gap is obtained.

- (6) Once clearance is obtained, **TORQUE SPRING NUTS TO 304 - 336 INCH-POUNDS** (Figure 2-4-34-1, Detail A).

- i. Check absorber attachment bolts for breakout torque as follows:

- (1) Check breakout torque on left side attachment bolts using torque wrench. **BREAKOUT TORQUE MUST NOT EXCEED 10 INCH-POUNDS THROUGH FULL 360° TURN.** If breakout torque exceeds 10 inch-pounds, check torque on nuts and slip-fit of sleeve bushings.

- (2) Check breakout torque on right side attachment bolt using torque wrench. **BREAKOUT TORQUE MUST NOT EXCEED 50 INCH-POUNDS.** If breakout torque is more than 50 inch-pounds, check buildup and torque on nut.

- j. Perform vibration check of absorber (TM 1-70-VIB).

- k. Install soundproofing panel(s) (PARA 2-4-69).

2-4-34.2 Inspect/Repair Cabin Vibration Absorber (Spring-Type).

2-4-34.2.1 Disassemble.

- a. If not previously done, remove cabin vibration absorber (PARA 2-4-34.1.1).

WARNING

Cabin vibration absorber airframe fitting bushing, 70219-02150-101, and cabin vibration absorber spring bushing, 70219-02150-103, are made of beryllium copper metal, which can be a health hazard. Avoid breathing dust while reworking damaged areas. Use dust respirator if necessary. Grinding is prohibited. Keep beryllium copper out of open wounds. Use skin and eye protection if necessary. Clean up visible dust piles and wash hands and arms after maintenance.

- b. Remove bolt, countersunk washer, washer, nut and remove right lower fitting from center spring (Figure 2-4-34-2).
- c. Remove bolts and washers securing tuning weights to center weight and body weight. Tag and record each tuning weight for installation.
- d. Remove bolts and washers securing center weight to body weight.
- e. Remove bolts, countersunk washers, washers, and nuts securing center spring and liner to body weight.
- f. Remove bolts, countersunk washers, washers, and nuts securing side springs and liners to body weight.

2-4-34.2.2 Inspect/Repair.

WARNING

Cabin vibration absorber airframe fitting bushing, 70219-02150-101, and cabin vibration absorber spring bushing, 70219-02150-103, are made of beryllium copper metal, which can be a health hazard. Avoid breathing dust while reworking damaged areas. Use dust respirator if necessary. Grinding is

prohibited. Keep beryllium copper out of open wounds. Use skin and eye protection if necessary. Clean up visible dust piles and wash hands and arms after maintenance.

- a. Check bearings in right lower fitting and left airframe fittings for wear as follows:
 - (1) Set up dial indicator with stylus tracking bearing inner race radial movement.
 - (2) **MAXIMUM RADIAL PLAY ALLOWED IS 0.005-INCH.** If radial play exceeds limit, replace bearing (PARA 2-5-8).
- b. Check bearings in right lower fitting and left airframe fittings for breakout torque as follows:
 - (1) Install new bolt, NAS1306-39D, in left airframe fitting bearing with enough washers on bolt to seat nut. Install and tighten nut so washers do not turn.
 - (2) Rotate bearing inner race four complete turns.
 - (3) Turn bearing inner race with torque wrench. **BREAKOUT TORQUE MUST NOT EXCEED 10 INCH-OUNCES THROUGH A FULL 360° TURN.**
- c. Check sleeve bushings in side springs and right airframe fitting as follows:
 - (1) Remove sleeve bushings from side springs and right airframe fitting (Figure 2-4-34-1, Detail A).
 - (2) Coat outer surface of sleeve bushing with grease, Item 151A, Appendix D.
 - (3) Install sleeve bushing in side springs and right airframe fitting.
 - (4) Grasp sleeve bushing ends between thumb and forefinger and slide bushing side-to-side in sleeve bushing hole. If sleeve bushing does not move freely, perform the following as required:

- (a) Replace sleeve bushing. Repeat step c. (2) through step c. (4).
 - (b) If replacing sleeve bushing does not correct problem, inspect sleeve bushing holes in side springs and right airframe fitting for burrs. If burrs are found, remove burrs using abrasive cloth, Item 13, Appendix D. Repeat step c. (2) through step c. (4).
 - (c) If replacing sleeve bushings and deburring sleeve bushing hole does not correct the problem, replace side spring(s) and/or right airframe fitting (PARA 2-6-11).
- (5) Remove sleeve bushings.
- d. Check springs for corrosion, gouges, nicks, and cracks as follows:

NOTE

Damage rework will remove shotpeen coverage and is considered acceptable in order to extend component service life.

- (1) **DAMAGE TO SIDE SPRINGS, 70219-02106-110, AND CENTER SPRING, 70219-02106-111, MUST NOT EXCEED 0.315-INCH MINIMUM THICKNESS** (Figure 2-4-34-2, Section A-A). If necessary, blend damage using abrasive cloth, Item 8, Appendix D. If damage exceeds limits, replace spring.
- (2) **DAMAGE TO SIDE SPRINGS, 70219-02106-114, AND CENTER SPRING, 70219-02106-115, MUST NOT EXCEED 0.300-INCH MINIMUM THICKNESS** (Figure 2-4-34-2, Section B-B). If necessary, blend damage using abrasive cloth, Item 8, Appendix D. If damage exceeds limits, replace spring.
- (3) **DAMAGE TO ANY CROSS-SECTIONAL AREA MUST NOT EXCEED 25% OF PERIMETER.**
- (4) **NO DAMAGE ALLOWED WITHIN A RADIUS EQUAL TO ONE DIAMETER AT ANY ATTACHMENT HOLE.**

- (5) **IF DAMAGE IS FOUND IN OR AROUND ATTACHMENT HOLES, CONTACT ENGINEERING AUTHORITY FOR DAMAGE ANALYSIS AND RECOMMENDED REPAIR OR REPLACEMENT.**

- (6) Touch up finish (PARA 2-6-5).

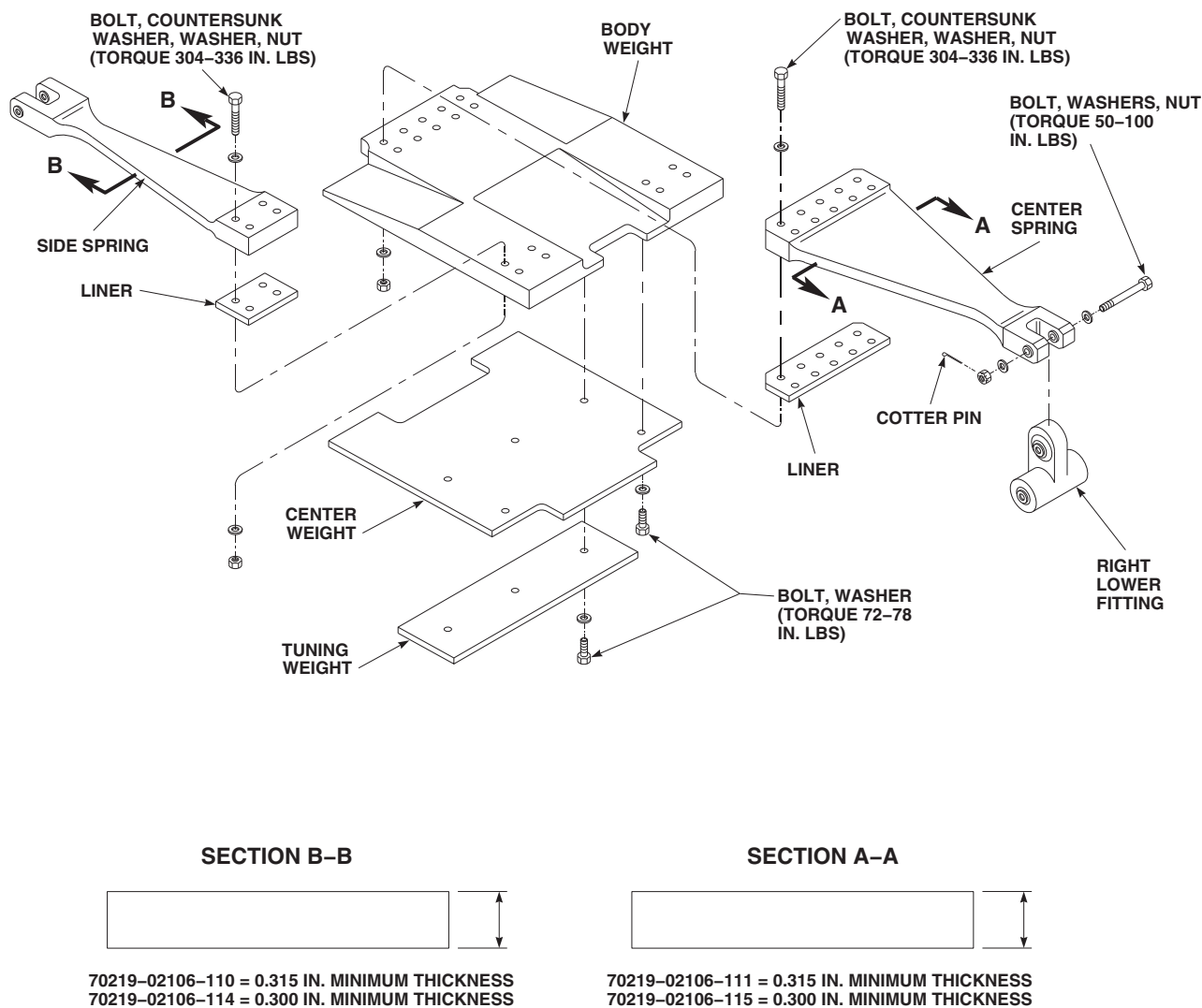
- e. Check attachment bolts and hardware for corrosion and wear. **NO CORROSION ALLOWED. MAXIMUM WEAR ALLOWED IS 0.002-INCH.** If corroded, or wear exceeds limit, replace hardware.
- f. Fluorescent penetrant inspect airframe fittings around bearings and bushings for cracks and corrosion using fluorescent penetrant inspection kit (ASTM E1417). **NONE ALLOWED.** If cracked or corroded, replace or repair fittings (PARA 2-5-7 or PARA 2-5-8).
- g. Check surrounding structure for cracks, dents, nicks, or other obvious damage. **NONE ALLOWED.** If damaged, repair (PARA 2-6-11).

2-4-34.2.3 Assemble.

WARNING

Cabin vibration absorber airframe fitting bushing, 70219-02150-101, and cabin vibration absorber spring bushing, 70219-02150-103, are made of beryllium copper metal, which can be a health hazard. Avoid breathing dust while reworking damaged areas. Use dust respirator if necessary. Grinding is prohibited. Keep beryllium copper out of open wounds. Use skin and eye protection if necessary. Clean up visible dust piles and wash hands and arms after maintenance.

- a. Clean threads of all bolts and nuts with tetrachloroethylene, Item 324, Appendix D. Allow to air-dry.
- b. Install center weight as follows (Figure 2-4-34-2):



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FIGURE 2-4-34-2. INSPECT/REPAIR CABIN VIBRATION ABSORBER (SPRING-TYPE)

- (1) Position center weight on body weight with boltholes lined up.
- (2) Apply sealing compound, Item 282, Appendix D, to threads of bolts.
- (3) Allow sealing compound to dry for 5 - 15 minutes.
- (4) Apply sealing compound, Item 280, Appendix D, to threads of bolts.
- (5) Install bolts and washers securing center weight to body weight. **TORQUE BOLTS TO 72 - 78 INCH-POUNDS.**
- c. Install side springs as follows:
 - (1) Position liners and side springs on body weight with boltholes lined up.
 - (2) Install countersunk washers on bolts with countersunk side facing boltheads.
 - (3) Apply sealing compound, Item 282, Appendix D, to threads of bolts and nuts.
 - (4) Allow sealing compound to dry for 5 - 15 minutes.
 - (5) Apply sealing compound, Item 280, Appendix D, to threads of bolts and nuts.

- (6) Install bolts (with countersunk washers installed and boltheads seated on side springs), washers, and nuts securing side springs and liners to body weight. **TORQUE NUTS TO 304 - 336 INCH-POUNDS.**

d. Install center spring as follows:

- (1) Position liner and center spring on body weight with boltholes lined up.
- (2) Install countersunk washers on bolts with countersunk side facing boltheads.
- (3) Apply sealing compound, Item 282, Appendix D, to threads of bolts and nuts.
- (4) Allow sealing compound to dry for 5 - 15 minutes.
- (5) Apply sealing compound, Item 280, Appendix D, to threads of bolts and nuts.
- (6) Install bolts (with countersunk washers installed and boltheads seated on side springs), washers, and nuts securing center spring and liner to body weight. **TORQUE NUTS TO 304 - 336 INCH-POUNDS.**

e. Install tuning weights as follows:

NOTE

If installing new absorber, use same amount of tuning weights as on old absorber.

- (1) Remove tags from tuning weights and position weights on center weight with boltholes lined up.
- (2) Apply sealing compound, Item 282, Appendix D, to threads of bolts.
- (3) Allow sealing compound to dry for 5 - 15 minutes.
- (4) Apply sealing compound, Item 280, Appendix D, to threads of bolts.

- (5) Install bolts and washers securing tuning weights to center weight and body weight. **TORQUE BOLTS TO 72 - 78 INCH-POUNDS.**

f. Install right lower fitting as follows:

- (1) Install new bolt, NAS1306-38D, and washer. Make sure bolthead washer is against flanged bushing.

NOTE

Front of fitting is end with inspection hole.

- (2) Install washer and nut. **TORQUE NUT TO 50 - 100 INCH-POUNDS.**
- (3) Make sure fitting bearing is sandwiched between flanged bushing and sleeve bushing, and does not move side-to-side in yoke. Install cotter pin.

g. Install cabin vibration absorber (PARA 2-4-34.1.2).

2-4-34.3 Adjust Cabin Vibration Absorber Tuning Weights.

NOTE

- Center weight weighs approximately 5 pounds. Thick tuning weight weighs approximately 1 pound. Thin tuning weight weighs approximately ½ pound.
 - For in-flight adjustments, wooden tuning wedge must be made before in-flight adjustments are made.
- a. Remove soundproofing panel(s) from cabin ceiling, forward of drip pan, as required (PARA 2-4-69).
 - b. If absorber has less than 3 pounds of tuning weights and a 5-pound center weight, remove center weight and install 5 pounds of tuning weights.
 - c. If vibration absorber tuning is to be performed in-flight, perform the following:

- (1) Locally-make wooden tuning wedge (Figure H-159, Appendix H).
 - (2) Insert wooden wedge into absorber to reduce absorber vibrations while making in-flight tuning weight changes.
- d. Tune vibration absorber (TM 1-70-VIB).
 - e. Check for proper bolt length as follows:

NOTE

- Bolts used during in-flight adjustments with less than five threads showing must be changed after landing.
 - If unthreaded portion of bolt is visible, and shorter bolts are unavailable, return to original weight and make adjustment on ground.
- (1) Stack up tuning weights and insert bolts through tuning weight holes.
 - (2) Count the number of threads protruding beyond tuning weights. If less than five threads are visible, use longer bolts. If unthreaded portion of bolt is visible, use shorter bolts.

- (3) Remove bolts.

- f. Clean threads of tuning weight bolts using machinery towel, Item 211, Appendix D, wet with tetrachloroethylene, Item 324, Appendix D. Allow to air-dry.

NOTE

For in-flight adjustments, torquing and sealing compound application to bolts should be done after landing. Weights must not move on absorber body after tightening bolts.

- g. Apply sealing compound, Item 282, Appendix D, to threads of bolts.
- h. Allow sealing compound to dry for 5 - 15 minutes.
- i. Apply sealing compound, Item 280, Appendix D, to threads of bolts.
- j. Position tuning weights on center weight. Install bolts and washers. **TORQUE BOLTS TO 72 - 78 INCH-POUNDS.**
- k. Install soundproofing panel(s) (PARA 2-4-69).

2-4-35 CABIN VIBRATION ABSORBER (BOX-TYPE).

PARAGRAPH BREAKDOWN

		PAGE
2-4-35.1	Replace Cabin Vibration Absorber (Box-Type)	2-4-35-1
2-4-35.2	Inspect/Repair Cabin Vibration Absorber (Box-Type)	2-4-35-3

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Fluorescent Penetrant Inspection Kit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Vibration Absorber Support Assembly, Locally-Made, Figure H-220, Appendix H
- Vise, Soft-Jawed

Materials/Parts

- Lockwire, Item 197, Appendix D
- Machinery Towel, Item 211, Appendix D

- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 301, Appendix D
- Tags

References

- Appendix D
- Appendix H
- ASTM E1417
- PARA 1-6-16
- PARA 2-4-69
- PARA 2-6-11
- TM 1-70-VIB

2-4-35.1 Replace Cabin Vibration Absorber (Box-Type).

2-4-35.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).

- Remove soundproofing panel(s) from cabin ceiling, forward of drip pan, as required (PARA 2-4-69).
- Locally make vibration absorber support assembly (Figure H-220, Appendix H).

WARNING

Injury to personnel or damage to equipment will result if absorber is not supported when removing attachment hardware. Box-type vibration absorber weighs approximately 90 pounds. Do not remove attachment hardware unless absorber is supported with vibration absorber support assembly and two assistants.

- d. Have assistants support absorber using vibration absorber support assembly.
- e. Remove bolts and washers securing absorber to left and right airframe fittings (Figure 2-4-35-1, Detail A).
- f. With two assistants supporting absorber, remove vibration absorber support assembly and remove absorber from helicopter.
- g. Remove barrel-nuts and retainers from airframe fittings.
- h. Inspect/repair cabin vibration absorber (PARA 2-4-35.2.2).

2-4-35.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).

CAUTION

Damage to recess in airframe fittings will result if barrel-nuts are incorrectly installed. To prevent damage, make sure barrel-nuts are installed with barrel (round side) facing bolthead.

- b. Install barrel-nuts and retainers into airframe fittings (Figure 2-4-35-1, Detail A).
- c. Apply sealing compound, Item 301, Appendix D, to mating surfaces of fittings and absorber.

WARNING

Injury to personnel or damage to equipment will result if absorber is not supported when installing attachment hardware. Box-type vibration absorber weighs approximately 90 pounds. Use at least two assistants to lift absorber and support with vibration absorber support assembly.

- d. With two assistants supporting absorber, lift and position absorber in airframe fittings, and support using vibration absorber support assembly.
- e. Make sure boltholes line up, and serrated plates on airframe fitting and absorber are engaged.
- f. Apply sealing compound, Item 301, Appendix D, to bolts.
- g. Install bolts and washers. **TORQUE BOLTS TO 385 - 425 INCH-POUNDS.** Using lockwire, Item 197, Appendix D, lockwire bolts.
- h. Install soundproofing panel(s) (PARA 2-4-69).
- i. Check cockpit main rotor 4/rev vibration utilizing data measurement procedure in TM 1-70-VIB, WP 008 00, as follows:

NOTE

TM 1-70-VIB primarily covers main rotor 1/rev tuning, but also measures and checks cockpit main rotor 4/rev vibration.

- (1) Refer to the Data Sheet, Figure 3 or 4, then paragraph 8 - Main Rotor 1/Rev Flight Tuning, Recalling Data, to determine measured 4/rev levels and limits.
- (2) If the procedures in TM 1-70-VIB can not be performed (for example due to the utilization of different test equipment), cockpit main rotor 4/rev vibration can be

checked by acquiring the following measurements:

- (a) Cockpit 4/rev A+B at 140 knots/100% Nr.
- (b) The limit for cockpit 4/rev A+B at 140 knots is 0.40 ips, and the limit for pilot heel vertical at both 120 knots and 140 knots is 0.60 ips.
- (c) If cockpit main rotor 4/rev vibration is within limits, no absorber tuning is required. If cockpit main rotor 4/rev vibration is not within limits, contact a Sikorsky Field Service Representative, or Sikorsky Aircraft Corporation for assistance.

2-4-35.2 Inspect/Repair Cabin Vibration Absorber (Box-Type).

2-4-35.2.1 Disassemble.

WARNING

Injury to personnel and damage to equipment will occur if housing attachment bolts are loosened or removed. Box-type vibration absorber springs are always in compression. Make sure that housing attachment bolts are not loosened or removed.

NOTE

Disassembly of absorber is limited to removal of tuning weights and serrated plates.

- a. If removing serrated plates, perform the following:
 - (1) If not previously done, remove cabin vibration absorber (PARA 2-4-35.1.1).
 - (2) Remove screws securing serrated plate to absorber (Figure 2-4-35-1, Detail A). Remove serrated plate.

- b. Remove bolts and washers securing short tuning weights to absorber. Tag and record each tuning weight for reinstallation.
- c. Remove bolts and washers securing long tuning weights to absorber. Tag and record each tuning weight for reinstallation.
- d. Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean sealing compound from all hardware.

2-4-35.2.2 Inspect/Repair.

- a. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect absorber upper and lower housings for cracks and corrosion (ASTM E1417). **NONE ALLOWED.** If cracked, replace absorber.
- b. Check serrated plate for damage to serrations. **NONE ALLOWED.** If damaged, replace serrated plate.
- c. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect airframe fittings for cracks and corrosion (ASTM E1417). **NONE ALLOWED.** If cracked or corroded, replace or repair fittings (PARA 2-6-11).
- d. Check barrel-nuts and retainers for damage and corrosion. **NONE ALLOWED.** If damaged or corroded, replace barrel-nuts and retainers.
- e. Test barrel-nuts as follows:

NOTE

Make sure that only square barrel-nuts are used.

- (1) Place barrel-nut in soft-jawed vise.
- (2) Thread bolt into barrel-nut until two threads are showing beyond nut.
- (3) Using torque wrench, back out bolt. **BREAKAWAY TORQUE MUST NOT BE LESS THAN 10 INCH-POUNDS.** If breakaway torque is less than 10 inch-pounds, replace barrel-nut.

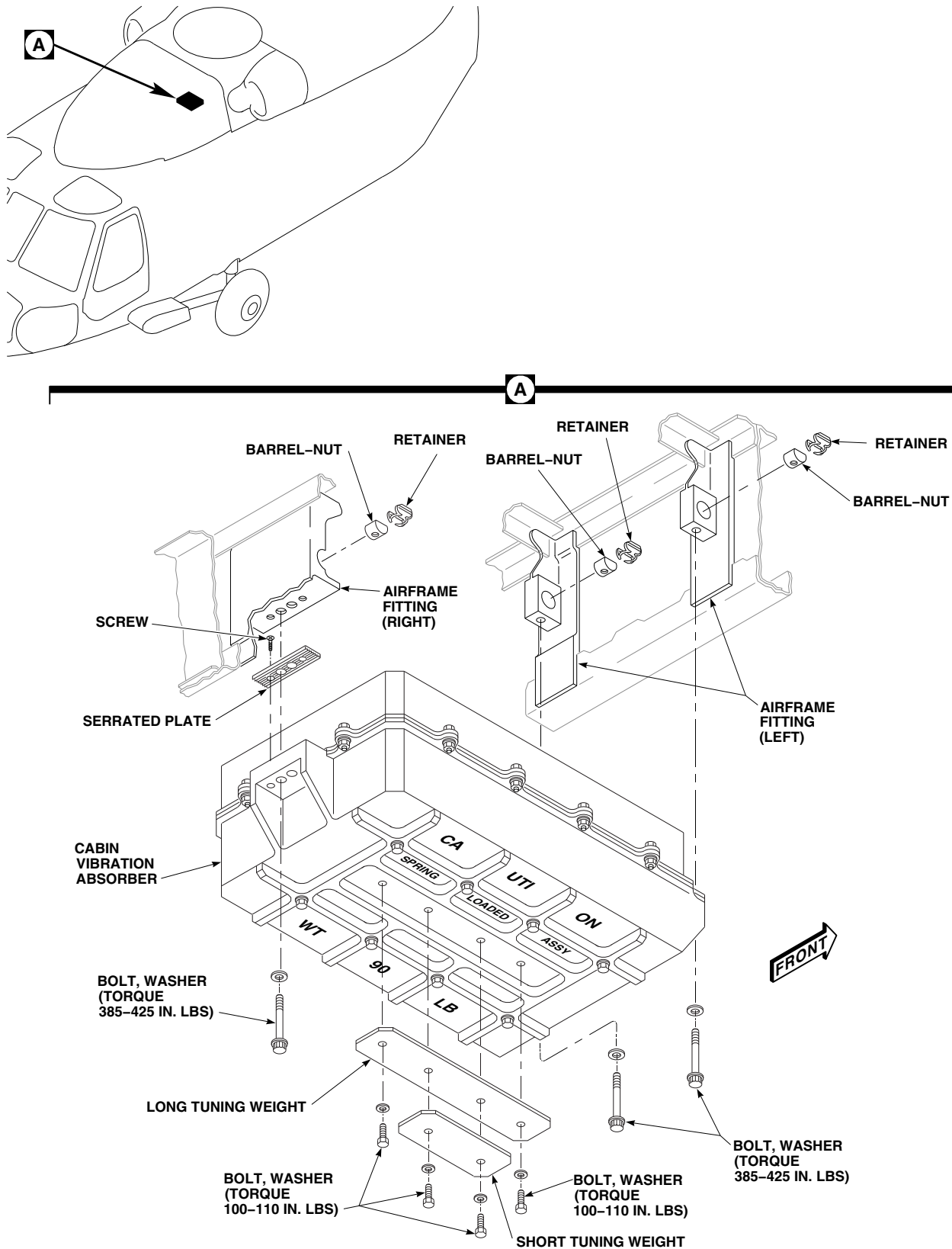
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FIGURE 2-4-35-1. REPLACE/REPAIR CABIN VIBRATION ABSORBER (BOX-TYPE)

- (4) Repeat step e. (1) through e. (3) for other barrel-nut.
- f. Check surrounding structure for cracks, dents, nicks or other obvious damage. If damage is found, repair as required (PARA 2-6-11).
- c. Remove tags from short and long tuning weights. Position weights on absorber with boltholes lined up.
- d. Apply sealing compound, Item 301, Appendix D, to threads of bolts.

2-4-35.2.3 Assemble.

- a. Apply sealing compound, Item 301, Appendix D, to threads of screws.
- b. Install serrated plate on absorber and secure with screws (Figure 2-4-35-1, Detail A).
- e. Install bolts and washers securing tuning weights to absorber. **TORQUE BOLTS TO 100 - 110 INCH-POUNDS.**
- f. If removed, install cabin vibration absorber (PARA 2-4-35.1.2).

2-4-36 ROLL VIBRATION ABSORBER **ROLL** .

PARAGRAPH BREAKDOWN

	PAGE
2-4-36.1 Replace Roll Vibration Absorber	2-4-36-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Fluorescent Penetrant Inspection Kit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 700 - 1600 in. lbs
- Vise, Soft-Jawed

Materials/Parts

- Corrosion Preventive Compound, Item 115, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 2-4-89
- PARA 2-4-94
- PARA 2-6-2
- PARA 2-6-12

2-4-36.1 Replace Roll Vibration Absorber.

NOTE

Replacement of left and right roll vibration absorbers is the same.

2-4-36.1.1 Remove.

- Remove drag beam support fairing (PARA 2-4-94).

NOTE

Upper bolt, washer, and barrel-nut were removed when drag beam support fairing was removed.

- Remove lower bolt and washer from roll vibration absorber. Remove absorber (Figure 2-4-36-1, Detail A).

- Remove barrel-nut and retainer from support.

2-4-36.1.2 Inspect.

- Using fluorescent penetrant inspection kit (ASTM E1417), fluorescent penetrant inspect roll vibration absorber for corrosion, gouges, nicks, and cracks. **NONE ALLOWED.** If cracked, replace roll vibration absorber.
- Using fluorescent penetrant inspection kit (ASTM E1417), fluorescent penetrant inspect bolts, washers, and barrel-nuts for corrosion. **NONE ALLOWED.** If corrosion is found, replace hardware.
- Test upper and lower bolts and barrel-nuts as follows:

NOTE

Make sure only square barrel-nuts are used.

- (1) Place barrel-nut in soft-jawed vise.
- (2) Thread bolt into barrel-nut until two threads are showing beyond nut.
- (3) Using torque wrench, back off bolt. **BREAKAWAY TORQUE SHALL NOT BE LESS THAN 18 INCH-POUNDS.** If breakaway torque is less than 18 inch-pounds, perform the following:
 - (a) If breakaway torque is less than 18 inch-pounds, replace bolt. Repeat step c. (1) through step c. (3).
 - (b) If breakaway torque is still less than 18 inch-pounds, replace barrel-nut. Repeat step c. (1) through step c. (3).

- d. Using fluorescent penetrant inspection kit (ASTM E1417), fluorescent penetrant inspect surrounding structure for cracks, dents, nicks, or other obvious damage. **NONE ALLOWED.** If necessary, repair damage (PARA 2-6-2 and PARA 2-6-12).

2-4-36.1.3 Install.

CAUTION

Damage to recess in support will result if barrel-nut is incorrectly installed. To

prevent damage, make sure barrel-nut is installed with barrel (round side) facing bolthead.

- a. Install barrel-nuts and retainers in support with barrel sides facing boltheads (Figure 2-4-36-1, Detail A).
- b. Secure roll vibration absorber with bolt and washer in top and bottom holes. **TORQUE BOTTOM BOLT TO 845 - 935 INCH-POUNDS.** Remove bolt and washer from top hole.
- c. Install drag beam support fairing (PARA 2-4-94). Do not install cockpit step at this time.
- d. Insert cockpit step through hole in drag beam support fairing and position step between support and vibration absorber. Install bolt and washer. **TORQUE BOLT TO 845 - 935 INCH-POUNDS** (Figure 2-4-36-1, Detail B).
- e. Install screws and washers securing cockpit step to drag beam support fairing.
- f. Apply corrosion preventive compound, Item 115, Appendix D, to fairing hardware.
- g. Secure cover on drag beam support fairing with screws and washers.
- h. Install wire strike step extension (PARA 2-4-89).

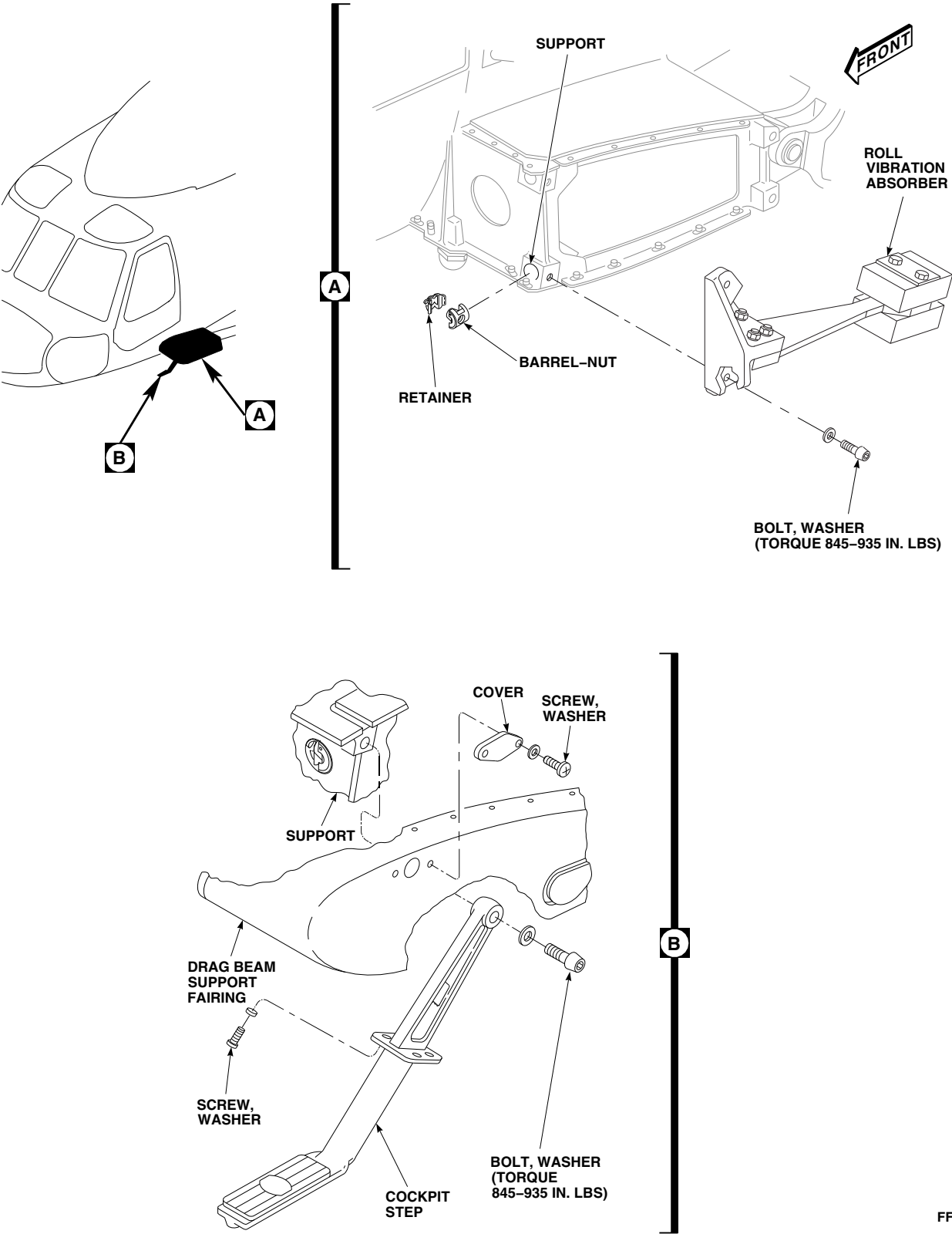


FIGURE 2-4-36-1. REPLACE ROLL VIBRATION ABSORBER

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2-4-37 UPPER WINDOWS.

PARAGRAPH BREAKDOWN

	PAGE
2-4-37.1 Replace Upper Windows	2-4-37-1

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Paint Remover, Item 230, Appendix D

- Pressure Sensitive Adhesive Tape, Item 248, Appendix D
- Pressure Sensitive Adhesive Tape, Item 250A, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 2-4-42
- PARA 8-4-30

2-4-37.1 Replace Upper Windows.

NOTE

Replacement of left and right upper windows is the same.



To prevent damage to plastic windows, do not allow paint remover to contact plastic.

2-4-37.1.1 Remove.

- 70949-SUBQ** Remove free-air thermometer (PARA 8-4-30).
- Remove screws and packing retainers and carefully remove upper window and rain gutter from flanges (Figure 2-4-37-1, Detail A).
- Remove pressure sensitive adhesive tape from flanges.
- Clean flanges with paint remover, Item 230, Appendix D, and nonmetallic scraper.
- Clean paint remover residue with machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D. Wipe dry with clean machinery towel, Item 211, Appendix D.

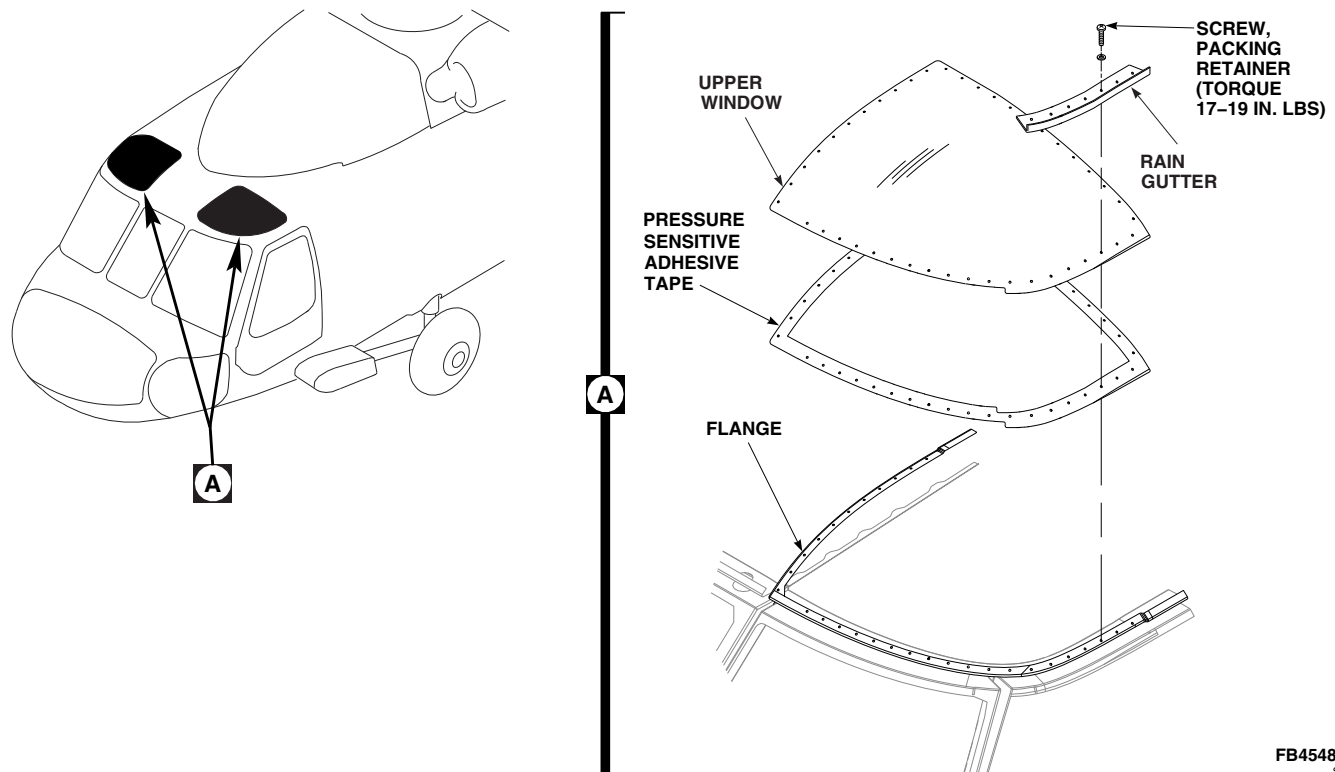
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FIGURE 2-4-37-1 REPLACE UPPER WINDOWS

2-4-37.1.2 Inspect.

Inspect packing retainers for missing or unserviceable seals. **NONE ALLOWED.** If packing retainers have missing or unserviceable seals, replace packing retainers.

2-4-37.1.3 Install.**NOTE**

For better fit, cut and join tape at sharp corners.

- Apply pressure sensitive adhesive tape, Item 248, Appendix D, to flanges of frame with sticky surface against flange (Figure 2-4-37-1, Detail A).
- Apply additional layer of pressure sensitive adhesive tape, Item 250A, Appendix D, in areas of joggles to ensure proper window installation.

NOTE

- Window installation screws should be gradually tightened in alternating pattern

as required to provide uniform squeeze-out of sealing compound about 0.100-inch wide.

- Torque should be at least 17 inch-pounds, but no more than 19 inch-pounds, due to wide range of breakaway torque possible for anchor nuts.
- Torquing of all screws to same value is not recommended, as this can result in waviness of window edge.
- Coat screws with sealing compound, Item 292, Appendix D, place window and rain gutter against flanges, and wet install screws and packing retainers. **TORQUE SCREWS TO 17 - 19 INCH-POUNDS.**
- Using nonmetallic scraper, clean area along window edges to be sealed.
- Fill all gaps around installed window with sealing compound, Item 292, Appendix D, to make sure surface is smooth and flush.

- f. **70949-SUBQ** Install free-air thermometer (PARA 8-4-30). 
- g. Perform upper window leak test (PARA 2-4-42).

2-4-38 LOWER WINDOWS.

PARAGRAPH BREAKDOWN

	PAGE
2-4-38.1 Replace Lower Windows	2-4-38-1

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Cleaning Compound, Item 92, Appendix D
- Foil Tape, Locally-Made, Figure H-170, Appendix H
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D

- Paint Remover, Item 230, Appendix D
- Pressure Sensitive Adhesive Tape, Item 248, Appendix D
- Pressure Sensitive Adhesive Tape, Item 250A, Appendix D
- Sealing Compound, Item 292, Appendix D
- Sealing Compound, Item 300, Appendix D

References

- Appendix D
- Appendix H
- PARA 2-4-42

2-4-38.1 Replace Lower Windows.

NOTE

Replacement of left and right lower windows is the same.

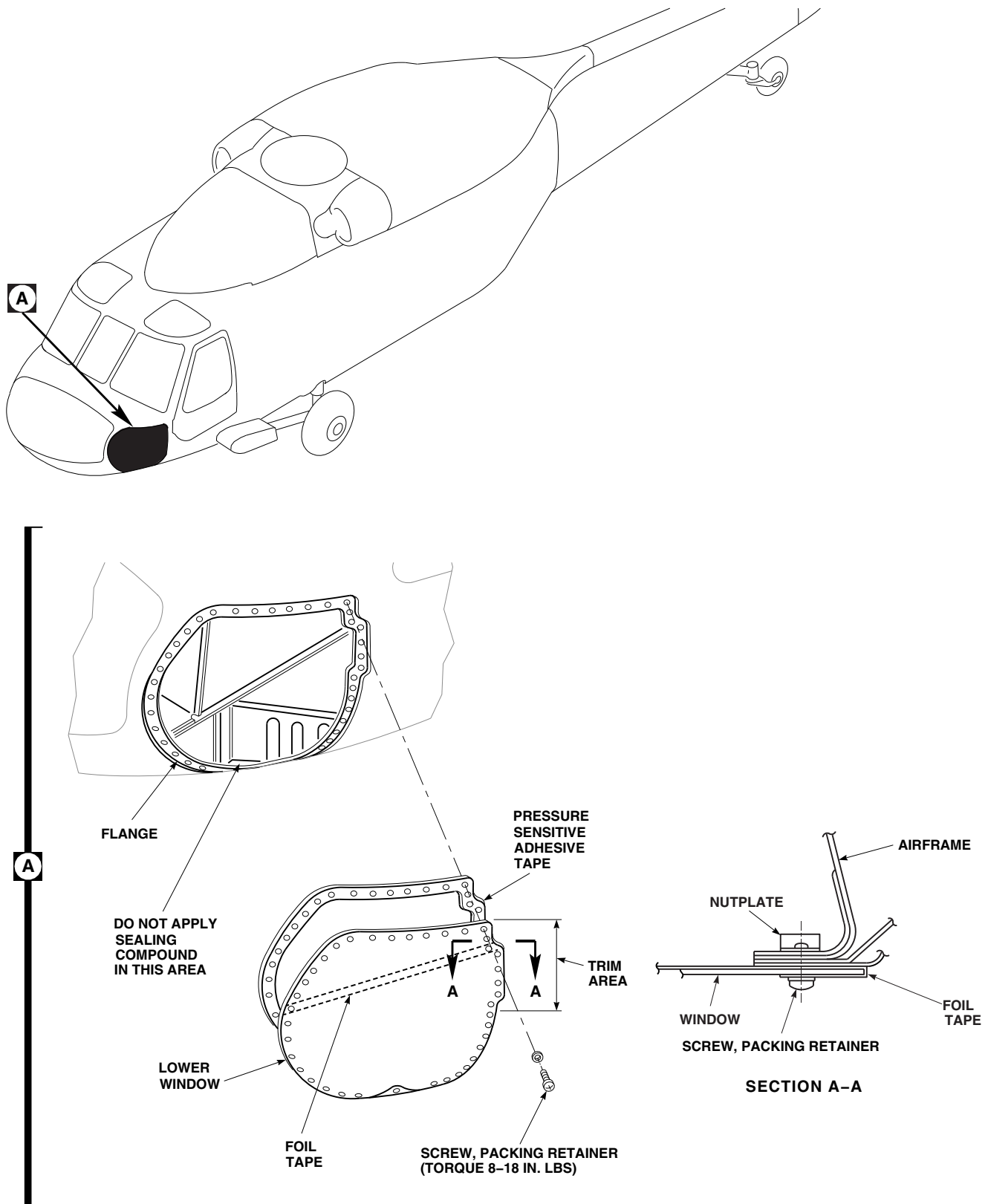
2-4-38.1.1 Remove.

- Remove screws and packing retainers and carefully remove lower window from flanges (Figure 2-4-38-1, Detail A).
- Remove pressure sensitive adhesive tape from flanges.

CAUTION

To prevent damage to plastic windows, do not allow paint remover to contact plastic.

- Clean flanges with paint remover, Item 230, Appendix D, and nonmetallic scraper.
- Clean paint remover residue with machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D. Wipe dry with clean machinery towel, Item 211, Appendix D.



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FIGURE 2-4-38-1. REPLACE LOWER WINDOWS

2-4-38-2

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- e. Inspect foil tape for damage. Repair (PARA 2-4-38.1.2) or replace (PARA 2-4-38.1.3) foil tape as necessary.

2-4-38.1.2 Repair Foil Tape.

Repair cuts in foil tape by painting over cut area with sealing compound, Item 300, Appendix D.

2-4-38.1.3 Replace Foil Tape.



To prevent damage to window, take extreme care when using tools to remove tape.

- a. Remove foil tape using nonmetallic scraper.
- b. Clean foil tape location with machinery towel, Item 211, Appendix D, moistened with cleaning compound, Item 92, Appendix D. Wipe dry with clean machinery towel, Item 211, Appendix D.
- c. Locally make foil tape (Figure H-170, Appendix H).
- d. Peel backing from new foil tape. Apply foil tape to inside of window, overlapping outside about 1.5-inches centered on mounting holes. Press firmly to make sure there is a good bond.

2-4-38.1.4 Install.

- a. Trial fit replacement window. Trim window edges to fit opening, if needed (Figure 2-4-38-1, Detail A).

NOTE

For a better fit, cut and join tape at sharp corners.

- b. Apply pressure sensitive adhesive tape, Item 248, Appendix D, to flanges of frame with

sticky surface against flange. Trim tape as necessary.

- c. Apply additional layer of pressure sensitive adhesive tape, Item 250A, Appendix D, in areas of joggles to ensure proper window installation.
- d. Place window against flanges (Figure 2-4-38-1, Detail A). Wrap foil tape around window edge so washer can make good contact with tape (Figure 2-4-38-1, Section A-A).
- e. Inspect packing retainers for missing or unserviceable seals. **NONE ALLOWED.** If washers have missing or unserviceable seals, replace packing retainers.

NOTE

Window installation screws should be gradually tightened in alternating pattern as required to provide for uniform squeezeout of sealing compound about 0.100-inch wide. Torque should be at least 8 inch-pounds, but no more than 18 inch-pounds, because of wide range of breakaway torque possible for anchor nuts. Torquing of all screws to same value is not recommended, as this can result in waviness of window edge.

- f. Coat screws with sealing compound, Item 292, Appendix D, and wet install screws and packing retainers (Figure 2-4-38-1, Detail A). **TORQUE SCREWS TO 8 - 18 INCH-POUNDS.**
- g. Using nonmetallic scraper, clean area along window edges to be sealed.
- h. Fill all gaps around installed window with sealing compound, Item 292, Appendix D, to make sure surface is smooth and flush.
- i. Check for continuity between foil tape and canopy structure.
- j. Check window for water leaks (PARA 2-4-42).

2-4-39 OUTBOARD WINDSHIELDS.

PARAGRAPH BREAKDOWN

		PAGE
2-4-39.1	Replace Outboard Windshields	2-4-39-1

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Drill Set
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Paint Remover, Item 230, Appendix D
- Sealing Compound, Item 275, Appendix D
- Sealing Compound, Item 292, Appendix D

- Tags
- Windshield Shim Kit, 70206-01003-043

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-28
- PARA 2-4-42
- PARA 2-4-90
- PARA 2-4-92
- PARA 2-6-3
- PARA 12-2-4
- PARA 12-4-19

2-4-39.1 Replace Outboard Windshields.

NOTE

NOTE

Replacement of left and right outboard windshields is the same, except as noted.

It is not necessary to remove glare shield on helicopters with nutplates installed.

2-4-39.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).

- If necessary, remove glare shield (PARA 2-4-28).
- Disconnect upper wiring as follows (Figure 2-4-39-1, Sheet 2, Detail C):

- (1) Pull back insulation boots to expose wiring connections on terminal blocks.
 - (2) Remove screws and lockwashers securing wiring to terminal blocks.
 - (3) Remove and tag wires.
- d. Hold wiper arm away from outboard windshield. Insert cotter pin in holdoff pin hole in wiper arm to lock wiper blade away from windshield surface (Figure 2-4-39-1, Sheet 1, Detail A). If necessary, remove wiper arm (PARA 12-4-19).

NOTE

Screws used to secure windshield are not all the same length or same alloy steel. Tag or identify screws as they are removed.

- e. Remove screws and packing retainers from windshield (Figure 2-4-39-1, Sheet 1, Detail B).
- f. Remove rain gutter and wire strike deflectors as follows:
 - (1) Remove windshield center post deflector channel (PARA 2-4-92).
 - (2) If installed, remove cockpit door latch deflector (PARA 2-4-90).
 - (3) Remove screws and packing retainers securing rain gutter to windshield (Figure 2-4-39-1, Detail A).
- g. Carefully lift windshield from frame.
- h. Disconnect lower wiring as follows (Figure 2-4-39-1, Sheet 2, Detail C):
 - (1) Pull back insulation boots to expose wiring connections on terminal blocks.
 - (2) Remove screws and lockwashers securing wiring to terminal blocks.
 - (3) Remove and tag wires.

- i. Check insulation boots on wiring for cracks or tears. **NONE ALLOWED.** If cracked or torn, replace insulation boots.

NOTE

Sealing compound is in the form of tape.

- j. Remove shim tape and sealing compound from frame flanges (Figure 2-4-39-1, Sheet 1, Detail B).

CAUTION

To prevent damage to windshield, do not allow paint remover to contact glass.

- k. Clean flanges with paint remover, Item 230, Appendix D, and nonmetallic scraper.
- l. Clean paint remover residue with machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D. Wipe dry with clean machinery towel.

2-4-39.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Trial-fit replacement window in canopy.
- c. If holes in window and canopy do not line up, perform the following:
 - (1) Mark holes in canopy that do not line up.
 - (2) Fill holes (PARA 2-6-3).
 - (3) Using replacement window as template, mark and drill new holes.
- d. Place new windshield on cleaned window opening. Line up windshield with mounting holes (Figure 2-4-39-1, Sheet 1, Detail B).

NOTE

Do not install screws in holes 18 and 41 at this time.

- e. Loosely install screws in holes 1, 7, 17, and 21 (Figure 2-4-39-1, Sheet 2, Detail D).

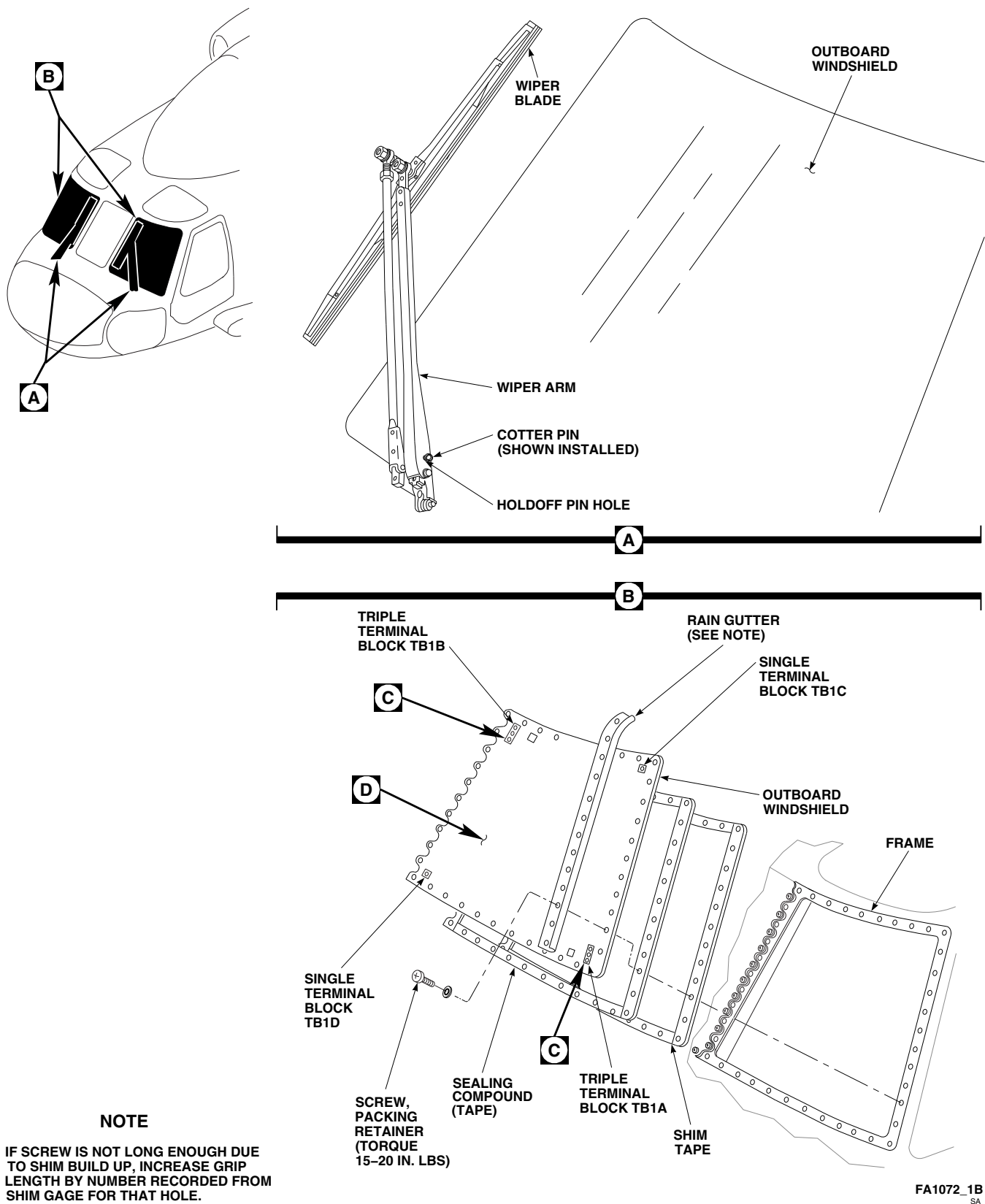
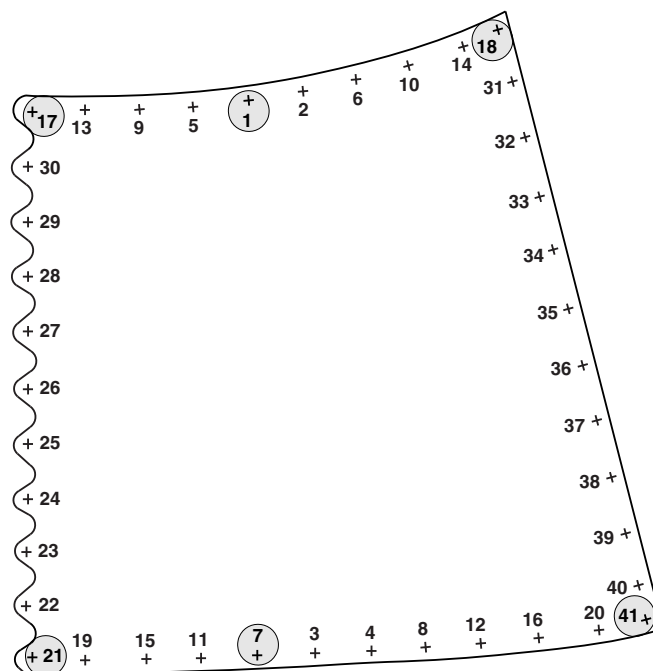
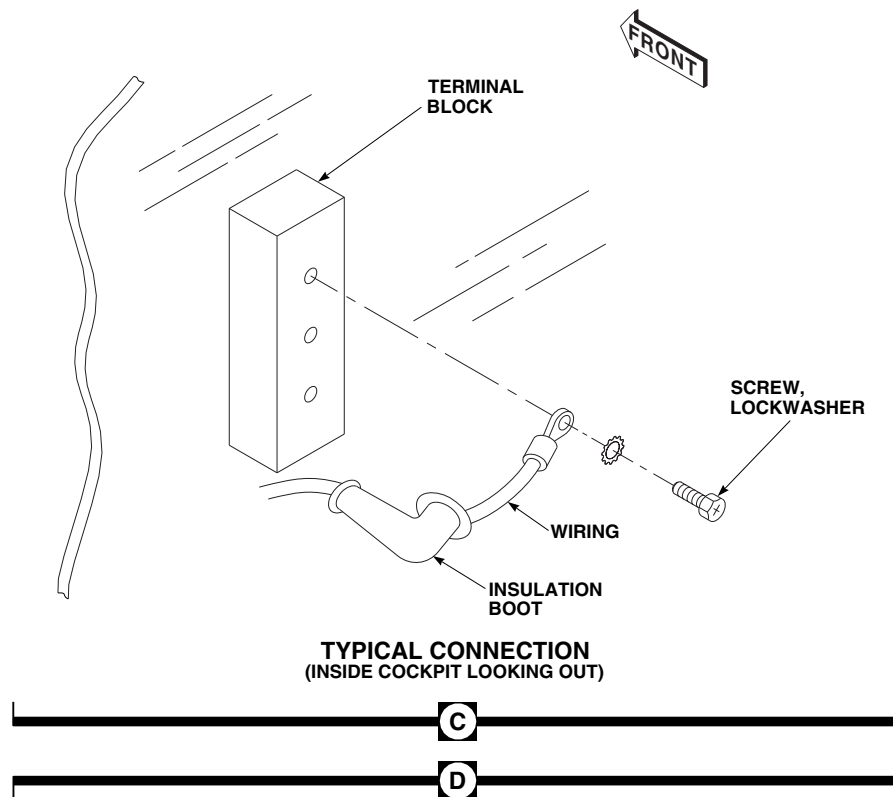


FIGURE 2-4-39-1. REPLACE OUTBOARD WINDSHIELDS (SHEET 1 OF 2)



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FIGURE 2-4-39-1. REPLACE OUTBOARD WINDSHIELDS (SHEET 2 OF 2)

- f. Push windshield down on inboard edge at holes 22 through 30 until windshield rests flush on canopy.

NOTE

If gage reading is on a line, use next higher number as gap reading.

- g. Using shim gage provided in windshield shim kit, measure gap at each windshield mounting hole along outboard and bottom edges of windshield. Remove screws at holes 21 and 7, measure gap, then reinstall screws.
- h. Record number from gage on windshield shim record sheet provided in windshield shim kit.
- i. Release windshield and move screws from holes 17 and 21 to 18 and 41.
- j. Push windshield down on outboard edge at holes 31 and 40 until windshield rests on canopy.

NOTE

If gage reading is on a line, use next higher number as gap reading.

- k. Measure gap at each windshield mounting hole along top edge and center post section of windshield using shim gage provided in windshield shim kit. Remove screws at holes 1 and 18, measure gap, then reinstall screws.
- l. Record number from gage on windshield shim record sheet provided in windshield shim kit.
- m. Release windshield, remove screws, and remove windshield.

NOTE

- If center windshield is installed, trim shim tape to fit openings at center post.
- 1/8-inch shim tape may be substituted for 1/16-inch shim tape by using one layer of 1/8-inch for each two layers of 1/16-inch shim tape.

- Use 3-inch hole spacing shim tape for top edge of windshield.
- Use 3.125-inch hole spacing shim tape for other edges of windshield.
- Stretch, compress, or cut shim tape as required to line up holes.

- n. Install shim tape as follows (Figure 2-4-39-1, Detail A):

- (1) Using appropriate shim tape, install one layer of shim tape around canopy opening.

NOTE

Number recorded from gage readings equals number of layers of shim tape including first layer.

- (2) Using appropriate shim tape, build up layers equal to number on shim record sheet recorded during step h. and step l.

- o. Apply layer of sealing compound, Item 275, Appendix D, over shim tape.
- p. Inspect packing retainers for missing or unserviceable seals. **NONE ALLOWED.** If packing retainers have missing or unserviceable seals, replace packing retainers.
- q. Coat threads of screws with sealing compound, Item 292, Appendix D. Position windshield on canopy and wet install screws and packing retainers loosely to position windshield (Figure 2-4-39-1, Sheet 1, Detail B). Do not tighten screws at this time.

NOTE

- Do not install rain gutter or wire strike deflectors at this time.
 - If any screw is not long enough due to shim tape build up, increase grip length by number recorded from shim gage for that hole.
- r. Using sequence numbers shown in Figure 2-4-39-1, Detail B, **TIGHTEN ALL SCREWS IN**

5 INCH-POUND INCREMENTS UNTIL EVEN CONTACT BETWEEN WINDSHIELD AND PACKING RETAINERS IS ACHIEVED.

- s. After previous windshield torque has stabilized, using same sequence, **RETORQUE ALL SCREWS TO 15 - 20 INCH-POUNDS** (Figure 2-4-39-1, Detail A).
- t. Fill all gaps and edges around windshield with additional sealing compound, Item 292, Appendix D, making sure surface is smooth and flush.
- u. Allow sealing compound to dry.
- v. Carefully remove screws and packing retainers used to install rain gutter and wire strike deflectors.
- w. Install heater wire, H4302A14C (pilot's windshield), or H4307A14C (copilot's windshield), on lower center terminal board, TB1D-6, with screw, lockwasher, and washer.
- x. Install rain gutter and wire strike deflectors as follows:
 - (1) Brush mating surfaces of rain gutter with sealing compound, Item 292, Appendix D.
 - (2) Inspect packing retainers for missing or unserviceable seals. **NONE ALLOWED.** If packing retainers have missing or unserviceable seals, replace packing retainers.
- (4) If removed, install cockpit door latch deflector (PARA 2-4-90).
- (5) Install windshield center post deflector channel (PARA 2-4-92).
- y. Fill all gaps and edges around windshield with additional sealing compound, Item 292, Appendix D, making sure surface is smooth and flush.
- z. Remove cotter pin from holdoff pin hole in wiper arm and ease wiper blade back to outboard windshield (Figure 2-4-39-1, Sheet 1, Detail A), or if removed, install wiper arm (PARA 12-4-19).
- aa. Perform outboard windshield leak test (PARA 2-4-42).
- ab. Connect wiring to pilot's windshield terminal blocks with screws and lockwashers. Slide insulation boots over wiring connections on terminal blocks. Connect as follows (Figure 2-4-39-1, Sheet 2, Detail C):
 - (1) H4301B18B to TB1A-1
 - (2) H4305B20 to TB1A-4
 - (3) H4304B20 to TB1A-3
 - (4) H4300A14A to TB1C-2
 - (5) H4300A14B to TB1B-5
 - (6) H4305A20 to TB1B-7
 - (7) H4305B20 to TB1B-7
 - (8) H4304A20 to TB1B-8
- ac. Connect wiring to copilot's windshield terminal blocks with screws and lockwashers. Slide insulation boots over wiring connections on terminal blocks. Connect as follows (Figure 2-4-39-1, Sheet 2, Detail C):
 - (1) H4308B18B to TB1A-1
 - (2) H4310B20 to TB1A-4

NOTE

Do not install screws and packing retainers in cockpit door latch deflector location on rain gutter (Figure 2-4-39-1, Sheet 1, Detail B).

- (3) Coat threads of screws with sealing compound, Item 292, Appendix D, and install rain gutter on outboard side of window with screws and packing retainers. **TORQUE ALL SCREWS TO 15 - 20 INCH-POUNDS.**

(3) H4311B20 to TB1A-3

(4) H4309A14C to TB1C-2

(5) H4308A14B to TB1B-5

(6) H4310A20 to TB1B-7

(7) H4310B20 to TB1B-7

(8) H4311A20 to TB1B-8

ad. Perform operational check of windshield anti-ice system (PARA 12-2-4).

ae. Remove tags from wires.

af. If removed, install glare shield (PARA 2-4-28).

2-4-40 CENTER WINDSHIELD (GLASS).

PARAGRAPH BREAKDOWN

	PAGE
2-4-40.1 Replace Center Windshield (Glass)	2-4-40-1

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Paint Remover, Item 230, Appendix D
- Pressure Sensitive Adhesive Tape, Item 248, Appendix D

- Sealing Compound, Item 275, Appendix D
- Sealing Compound, Item 292, Appendix D
- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-42
- PARA 2-4-91
- PARA 2-4-92
- PARA 12-2-4

2-4-40.1 Replace Center Windshield (Glass).

2-4-40.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect upper wiring as follows (Figure 2-4-40-1, Detail C):
 - Pull back insulation boots to expose wiring connections on terminal blocks.
 - Remove screws and lockwashers securing wiring to terminal blocks.
 - Tag and remove wires.

- Remove windshield center post deflector channels (PARA 2-4-92).

NOTE

Screws used to secure windshield are not all the same length or same alloy steel. Tag and identify screws as they are removed.

- Remove remaining screws and packing retainers from windshield (Figure 2-4-40-1, Detail A).
- Carefully lift windshield from frame.

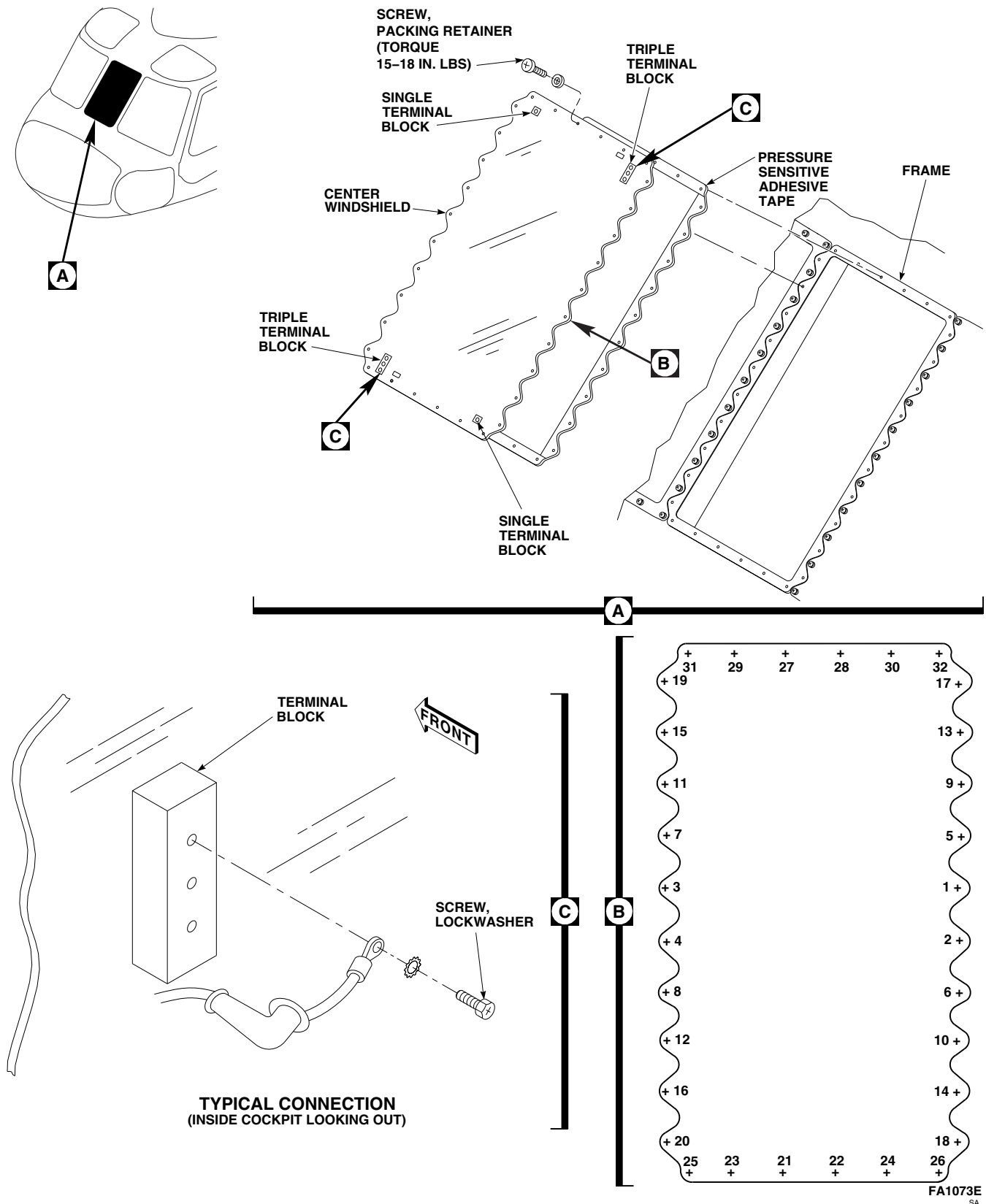


FIGURE 2-4-40-1. REPLACE CENTER WINDSHIELD (GLASS)

- f. Disconnect lower wiring as follows (Figure 2-4-40-1, Detail C):
 - (1) Pull back insulation boots to expose wiring connections on terminal blocks.
 - (2) Remove screws and lockwashers securing wiring to terminal blocks.
 - (3) Tag and remove wires.
- g. Check insulation boots for cracks or tears. **NONE ALLOWED.** If cracked or torn, replace insulation boots.
- h. Remove pressure sensitive adhesive tape from frame flanges.



To prevent damage to windshield, do not allow paint remover to contact glass.

- i. Using nonmetallic scraper and paint remover, Item 230, Appendix D, clean flanges.
- j. Clean paint remover residue with machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D. Wipe dry with clean machinery towel.

2-4-40.1.2 Inspect.

Inspect packing retainers for missing or unserviceable seals. **NONE ALLOWED.** If packing retainers have missing or unserviceable seals, replace packing retainers.

2-4-40.1.3 Install.

- a. Turn off all electrical power (PARA 1-6-16).

NOTE

For better fit, cut and join pressure sensitive tape at sharp corners.

- b. Apply pressure sensitive adhesive tape, Item 248, Appendix D, to frame flanges with sticky surface against flange (Figure 2-4-40-1, Detail A).

- c. Position windshield on frame.
- d. Connect lower wiring as follows (Figure 2-4-40-1, Detail C):
 - (1) Install wires to terminal blocks with screws and lockwashers.
 - (2) Slide insulation boots over wiring connections on terminal blocks.

NOTE

When installing windshield center post deflector channels, do not install windshield center post deflector inserts.

- e. Install windshield center post deflector channels (PARA 2-4-92). Do not install insert or tighten screws at this time.

NOTE

Install windshield securing screws in sequence shown in Figure 2-4-40-1, Detail B.

- f. Coat threads of screws with sealing compound, Item 292, Appendix D, and secure top and bottom of windshield with screws and packing retainers (Figure 2-4-40-1, Detail A). Do not tighten screws at this time.
- g. Using sequence numbers shown in Figure 2-4-40-1, Detail B, **TIGHTEN ALL SCREWS JUST ENOUGH FOR EVEN CONTACT BETWEEN WINDSHIELD AND PRESSURE SENSITIVE ADHESIVE TAPE.**
- h. Using same sequence, **TORQUE ALL SCREWS TO 15 - 18 INCH-POUNDS.**
- i. Fill all gaps and edges around windshield with additional sealing compound, Item 275, Appendix D, making sure surface is smooth and flush.
- j. Allow sealing compound to dry.
- k. Perform center windshield leak test (PARA 2-4-42).

- l. Install windshield center post deflector inserts (PARA 2-4-91).
- m. Connect upper wiring as follows (Figure 2-4-40-1, Detail C):
 - (1) Install wires to terminal blocks with screws and lockwashers.
 - (2) Slide insulation boots over wiring connections on terminal blocks.
- n. Remove tags from wires.
- o. Perform operational check of windshield anti-ice system (PARA 12-2-4).

2-4-41 CENTER WINDSHIELD (PLASTIC).

PARAGRAPH BREAKDOWN

	PAGE
2-4-41.1 Replace Center Windshield (Plastic)	2-4-41-1

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Drill Set
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D

- Paint Remover, Item 230, Appendix D
- Tags

References

- Appendix D
- PARA 2-4-28
- PARA 2-4-40
- PARA 2-4-92
- PARA 8-4-30

2-4-41.1 Replace Center Windshield (Plastic).

NOTE

Plastic center windshield must be replaced by heated glass center windshield.

2-4-41.1.1 Remove.

- Remove free-air thermometer (FAT) from plastic center windshield (PARA 8-4-30). Retain FAT and hardware for installation on glass center windshield.

NOTE

It is not necessary to remove glare shield on helicopters with nutplates installed.

- If necessary, remove glare shield (PARA 2-4-28).
- Remove windshield center post deflector channels (PARA 2-4-92).

NOTE

Screws used to secure windshield are not all the same length or same alloy steel. Tag or identify screws as they are removed.

- Remove remaining screws and packing retainers from center windshield (Figure 2-4-41-1, Detail A).
- Carefully lift center windshield from frame.
- Remove pressure sensitive adhesive tape from frame flanges.

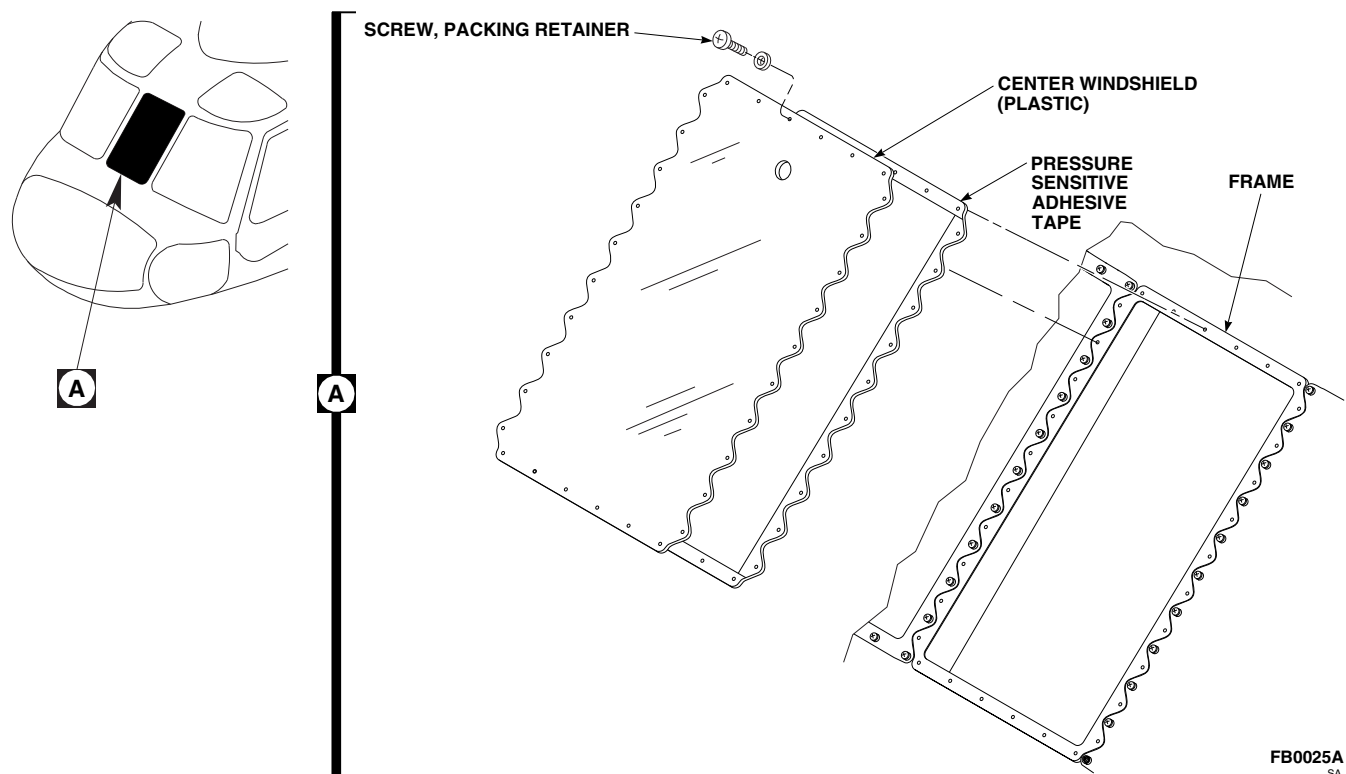


FIGURE 2-4-41-1. REMOVE CENTER WINDSHIELD (PLASTIC)

CAUTION

To prevent damage to windshield, do not allow paint remover to contact plastic.

- g. Clean flanges with paint remover, Item 230, Appendix D, and nonmetallic scraper.
- h. Clean paint remover residue with machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D. Wipe dry with clean machinery towel.

2-4-41.1.2 Install.

NOTE

First time installation of heated glass center windshield requires installation of

one free-air thermometer in each upper window.

- a. Install glass center windshield (PARA 2-4-40).
- b. Using drill, drill 0.455 to 0.460-inch diameter hole in each upper window, centered 3-inches from inboard side and front window edge.
- c. Install bonding jumpers (PARA 8-4-30).
- d. Install free-air thermometers (PARA 8-4-30).
- e. If removed, install glare shield (PARA 2-4-28).

2-4-42 WINDOW, WINDSHIELD, AND DOOR LEAK TEST.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
2-4-42.1	Nose Door Leak Test	2-4-42-1	2-4-42.3	Crew Chief's Window Leak Test	2-4-42-3
2-4-42.2	Center/Outboard Windshield and Upper/Lower Window Leak Test	2-4-42-2	2-4-42.4	Cabin Door and Window Leak Test	2-4-42-3

INITIAL SETUP

Tools

- Airframe Repairer's Toolkit
- Garden Hose
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Barrier Material, Item 80, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-6-16

- PARA 2-4-7
- PARA 2-4-8
- PARA 2-4-12
- PARA 2-4-37
- PARA 2-4-38
- PARA 2-4-39
- PARA 2-4-40
- PARA 2-4-41
- PARA 2-4-48
- PARA 2-4-51
- PARA 2-4-58
- PARA 2-6-8

2-4-42.1 Nose Door Leak Test.

- Turn off all electrical power (PARA 1-6-16).
- Unlatch and open nose door. Lock nose door strut in open position.
- Mask components in nose section avionics compartment using barrier material, Item 80, Appendix D.

- Unlock nose door strut. Close nose door and fasten.
- If not installed, install nose door cover (PARA 2-4-7), or mask screen on nose door using barrier material, Item 80, Appendix D.
- Spray water from garden hose. Concentrate spray on sealed edges.

- g. Unlatch nose door, and open nose door. Lock strut in open position.
- h. Check for leakage. If no leakage is found, leak test is complete, proceed to step i. If leakage is found, perform the following:
 - (1) Loosen jamnuts and turn each latch hook an equal number of turns to allow door to draw down tighter on seal. **TIGHTEN JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
 - (2) Unlock nose door strut. Close nose door and fasten.
 - (3) Spray door again, concentrating spray on sealed edges.
 - (4) Unlatch nose door, and open nose door. Lock strut in open position.
 - (5) Check for leakage. If no leakage is found, leak test is complete, proceed to step i. If leakage continues and nose door stops are not bottomed on sill, repeat step h. (1) through h. (5). If leakage continues and nose door stops are bottomed on sill, perform the following:
 - (a) Inspect nose door seal for damage. If damaged, replace seal (PARA 2-4-8).
 - (b) Inspect sill for damage. If damaged, repair sill (PARA 2-6-8).
 - (c) Repeat leak test.
- i. Remove barrier material from nose section avionics compartment.
- j. Unlock nose door strut. Close nose door and fasten.
- k. If used, remove barrier material from screen on nose door, or remove cover (PARA 2-4-7).

2-4-42.2 Center/Outboard Windshield and Upper/Lower Window Leak Test.

NOTE

Leak test for both outboard windshields, upper windows, and lower windows is the same, except as noted.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Have assistant in cockpit to note possible leakage.
- c. Close cockpit door.
- d. Spray water from garden hose. Concentrate spray on sealed edges. If no leakage is found, leak test is complete. If leakage is found, perform the following:
 - (1) Check torque of screws securing window/windshield in leakage area as follows:
 - (a) For upper and lower windows, **TORQUE SCREWS TO 8 - 18 INCH-POUNDS.**
 - (b) For center windshield, **TORQUE SCREWS TO 15 - 18 INCH-POUNDS.**
 - (c) For outboard windshields, **TORQUE SCREWS TO 15 - 20 INCH-POUNDS.**
 - (2) Spray area where leakage was found. If leakage continues, dry area and apply sealing compound, Item 292, Appendix D, to leaky area making sure area is smooth and flush. Allow to dry.
 - (3) Spray area where leakage was found again. If leakage still continues, perform the following:
 - (a) Remove applicable window (PARA 2-4-37 or PARA 2-4-38) or windshield (PARA 2-4-39 or PARA 2-4-41).

NOTE

If a plastic center windshield was removed, it must be replaced by heated glass center windshield.

- (b) Inspect frame for damage. If no damage on frame is found, replace pressure sensitive adhesive tape and install applicable window (PARA 2-4-37 or PARA 2-4-38) or windshield (PARA 2-4-39 or PARA 2-4-40). If damage on frame is found, repair frame (PARA 2-6-8).

- (c) Repeat leak test.

2-4-42.3 Crew Chief's Window Leak Test.

NOTE

Leak test for both crew chief's windows is the same.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Have assistant in cabin to note possible leakage.
- c. Close cabin door.
- d. Spray water from garden hose. Concentrate spray on sealed edges. If no leakage is found,

leak test is complete. If leakage is found, perform the following:

- (1) Replace window (PARA 2-4-48) or window seals (PARA 2-4-51) as required.
- (2) Repeat leak test.

2-4-42.4 Cabin Door and Window Leak Test.

NOTE

Leak test for both cabin doors and windows is the same.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Have assistant in cabin to note possible leakage.
- c. Close cabin door.
- d. Spray water from garden hose. Concentrate spray on sealed edges around door and/or windows. If no leakage is found, leak test is complete. If leakage is found, perform the following:
 - (1) Replace door seals and/or window seals (PARA 2-4-58) as required.
 - (2) Repeat leak test.

2-4-43 COCKPIT FLOORS.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
2-4-43.1	Replace/Repair Cockpit Floor Covers	2-4-43-1	2-4-43.3	Replace/Repair Access Panel	2-4-43-2
2-4-43.2	Repair Cockpit Floors	2-4-43-2			

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Nonmetallic Scraper, Locally-Made
- Putty Knife

Materials/Parts

- Adhesive, Item 61, Appendix D
- Fastener Tape, Hook, Item 137A, Appendix D

- Fastener Tape, Pile, Item 137B, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Thread, Item 326, Appendix D

References

- Appendix D
- PARA 2-6-1
- PARA 2-6-12

2-4-43.1 Replace/Repair Cockpit Floor Covers.

NOTE

Replace/repair procedures are the same for both pilot’s and copilot’s floor covers.

2-4-43.1.1. Remove.

- Using release lever, recline pilot’s/copilot’s seat.
- Start at sides of floor cover and gently pull up on cover until pile fastener tape is free of hook fastener tape (Figure 2-4-43-1, Detail A).
- Remove cover.

2-4-43.1.2. Repair.

- Remove damaged hook fastener tape from cockpit floor and front cover using nonmetallic scraper (Figure 2-4-43-1, Detail A).
- Using machinery towel, Item 211, Appendix D, dampened in methyl ethyl ketone, Item 217, Appendix D, and nonmetallic scraper, remove remaining adhesive from cockpit floor and front cover.
- Within 15 minutes after cleaning, apply coat of adhesive, Item 61, Appendix D, to hook fastener tape, Item 137A, Appendix D, cockpit floor, and front cover with putty knife.
- After adhesive becomes tacky, press hook fastener tape, Item 137A, Appendix D, to cockpit floor and front cover. Allow adhesive

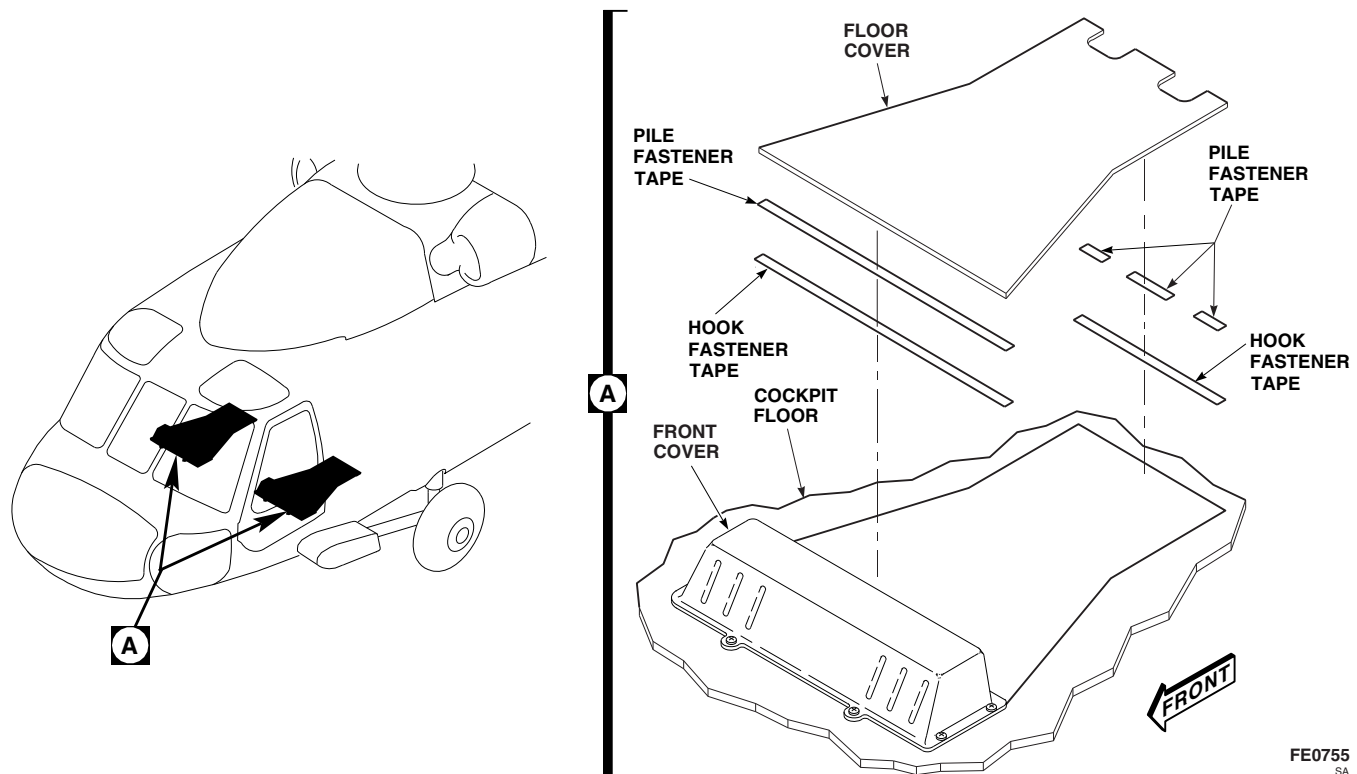


FIGURE 2-4-43-1. REPLACE/REPAIR COCKPIT FLOOR COVERS

to cure for 16 hours before installing floor cover.

- e. Repair or install pile fastener tape, Item 137B, Appendix D, on floor cover by sewing with thread, Item 326, Appendix D.

2-4-43.1.3. Install.

- a. Position pile fastener tape of floor cover over hook fastener tape on front cover and gently press down on cover. Make sure pile fastener tape makes good contact with hook fastener tape (Figure 2-4-43-1, Detail A).
- b. Use same procedure in step a. for securing rear of floor cover.
- c. Make sure area is clean and free of foreign material.
- d. Return pilot's/copilot's seat to upright position.

2-4-43.2 Repair Cockpit Floors.

Refer to PARA 2-6-12 for repair instructions.

2-4-43.3 Replace/Repair Access Panel.

NOTE

Replace/repair procedures are the same for both pilot's and copilot's access panels.

2-4-43.3.1. Remove.

- a. Using release lever, recline pilot's/copilot's seat.
- b. Remove screws and washers securing access panel to cockpit floor (Figure 2-4-43-2, Detail A).
- c. Remove access panel.

2-4-43.3.2. Repair.

Inspect panels for corrosion. If required, treat for corrosion (PARA 2-6-1).

2-4-43.3.3. Install.

- a. Make sure area is clean and free of foreign material.

- b. Position access panel on cockpit floor.
- c. Secure access panel to cockpit floor using screws and washers (Figure 2-4-43-2, Detail A).
- d. Return pilot's/copilot's seat to upright position.

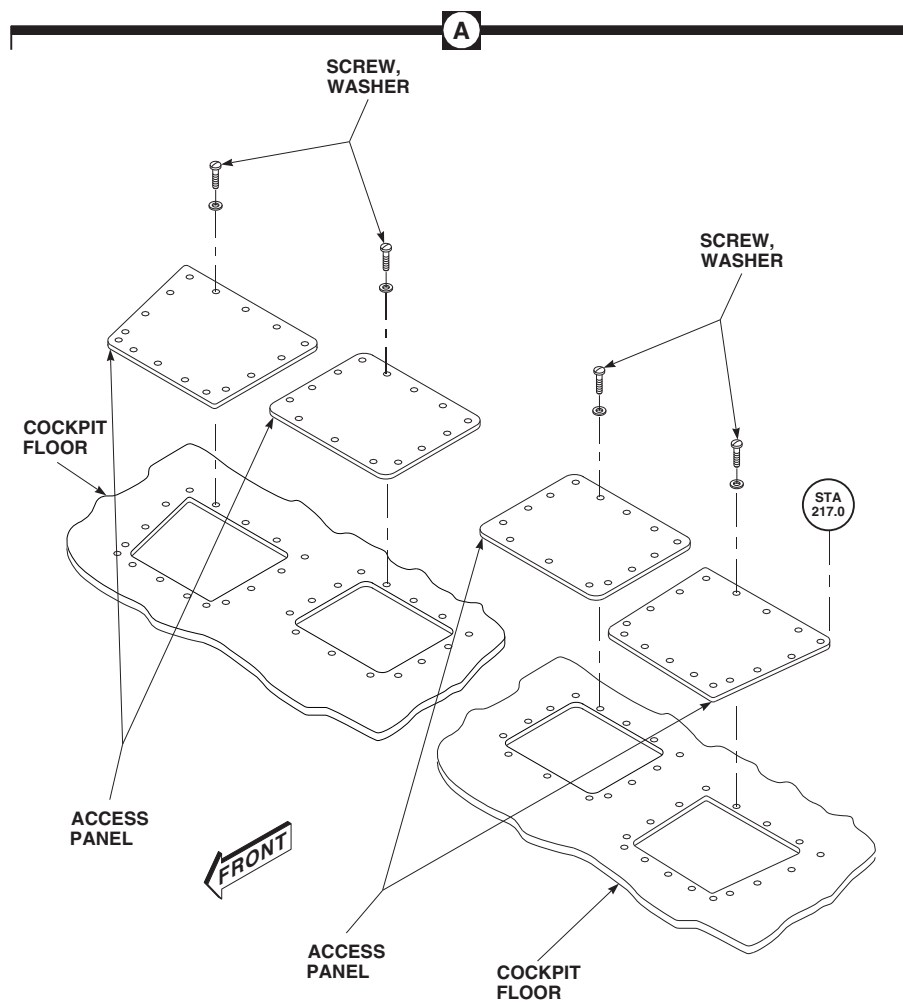
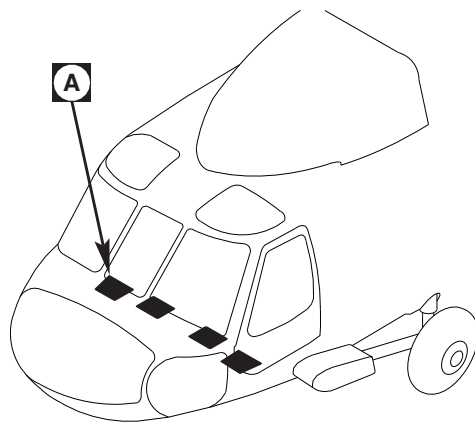


FIGURE 2-4-43-2. REPLACE/REPAIR ACCESS PANELS

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2-4-43-4

2-4-44 PILOT'S/COPILOT'S SEATS, RA 30525-1, 100900-1, AND 100900-5.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
2-4-44.1	Replace Inertia Reel	2-4-44-3	2-4-44.16	Replace Up-and-Down Guide Tube	2-4-44-31
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2-4-44.5	Replace Crotch Belt, Buckle, and Pad	2-4-44-7	2-4-44.20	Replace Seat Frame Tilt Handle Jacket	2-4-44-40
2-4-44.6	Replace Seat Cushions	2-4-44-9	2-4-44.21	Replace Seat Frame Diagonal Strap	2-4-44-41
2-4-44.7	Replace Front-to-Rear Adjust Cables and Adjust Handle Baseplate	2-4-44-13	2-4-44.22	Replace Seat Frame Diagonal Strut and Strut Support	2-4-44-43
2-4-44.8	Replace Up-and-Down Extension Spring and Retainer	2-4-44-16	2-4-44.23	Replace Seat Frame Frangible Strut	2-4-44-45
2-4-44.9	Replace Up-and-Down Attenuator	2-4-44-17	2-4-44.24	Replace Seat Frame Track Rollers and Bearings	2-4-44-47
2-4-44.10	Replace Up-and-Down Yoke and Cable	2-4-44-19	2-4-44.25	Replace Pilot's Stowage Box, Panel, and Bag	2-4-44-49
2-4-44.11	Replace Up-and-Down Release Handle and Baseplate	2-4-44-23	2-4-44.26	Replace Copilot's Stowage Box, Panel, and Bag	2-4-44-51
2-4-44.12	Replace Up-and-Down Vertical Adjustment Track	2-4-44-25	2-4-44.27	Replace Armored Wing	2-4-44-53
2-4-44.13	Replace Seat Bucket	2-4-44-27	2-4-44.28	Replace Pilot's/Copilot's Seat	2-4-44-57
2-4-44.14	Replace Shoulder Harness Rubbing Bar	2-4-44-27	2-4-44.29	Replace Trunnion Threaded Insert	2-4-44-63
2-4-44.15	Replace Seat Frame Upper Tube and Supports	2-4-44-30			

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Allen Wrench
- Allen Wrench, $\frac{5}{64}$ -inch
- Allen Wrench, $\frac{7}{32}$ -inch
- Clamp (or Suitable Bracing)
- Die and Tap Set
- Drill
- Drill Bit, No. 21 (0.1590-inch diameter)
- Feeler Gage
- Feeler Gage, 0.010-inch
- File
- Heat Gun
- Insert Installation Tool
- Mallet
- Nonmetallic Scraper, Locally-Made
- Pallet
- Pin Punch (Alternate for Tang Break Off Tool)
- Putty Knife
- Stabilizer Strut, Locally-Made, Figure H-182, Appendix H
- Tang Break Off Tool (Alternate for Pin Punch)
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

- Wood Block (or equivalent)
- Wood Block, Padded

Materials/Parts

- Abrasive Paper, Item 20, Appendix D
- Acetone, Item 25, Appendix D
- Adhesive, Item 37, Appendix D
- Adhesive, Item 38, Appendix D
- Adhesive, Item 40, Appendix D
- Adhesive, Item 42, Appendix D
- Adhesive, Item 46, Appendix D
- Adhesive, Item 58, Appendix D
- Brush, Item 83, Appendix D
- Cleaning Compound, Item 96, Appendix D
- Dishwashing Compound, Item 122, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Etching Solution, Item 136, Appendix D
- Fastener Tape, Pile, Item 137B, Appendix D
- Lockwire, Item 197, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Primer, Item 257, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 271A, Appendix D
- Sealing Compound, Item 278, Appendix D
- Sealing Compound, Item 284, Appendix D

- Sealing Compound, Item 285, Appendix D
- Tiedown Strap, Item 329, Appendix D

References

- Appendix D
- Appendix H
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-45
- PARA 2-6-2
- PARA 2-6-3
- PARA 2-6-5
- PARA 17-4-1
- TM 11-70-23

2-4-44.1 Replace Inertia Reel.

2-4-44.1.1 Remove.

WARNING

To prevent injury to personnel, do not release either normal or emergency vertical adjust levers unless someone is sitting in seat. Extension springs are under load at all times. With seat at lowest position, vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, seat will be snapped to highest stop. Anyone leaning over seat, or with hands on guide tubes above linear bearings, will be seriously injured.

- Sit in seat and adjust to full up and full front position.
- Remove seat back cushions to access inertia reel attachment screws (PARA 2-4-44.6).
- Set inertia reel control to AUTO-LOCK (rear) position (Figure 2-4-44-1, Detail A).
- Remove screws, washers, and nuts, and pull inertia reel from seat back (Figure 2-4-44-1, Detail B).

CAUTION

To prevent damage to inertia reel due to uncontrolled release of spring tension, do not allow Allen wrench to become disengaged from reel shaft until spring has been unwound.

- Hold inertia reel and pull shoulder harness fully out. Place $\frac{7}{32}$ -inch Allen wrench in hole on end of reel housing, and hold reel shaft from turning.
- Disconnect control cable knurled nut from inertia reel.
- Set inertia reel control to MANUAL-LOCK (front) position (Figure 2-4-44-1, Detail A).
- Remove control handle cable from inertia reel (Figure 2-4-44-1, Detail B).
- Push belt retaining insert and shoulder harness down and out of reel shaft and lower slot (Figure 2-4-44-1, Detail C).
- Remove belt retaining insert and pull belt from reel shaft.
- Install belt retaining insert 90° from normal installed position. Slowly back off Allen wrench until power spring holds insert.

2-4-44.1.2 Install.

CAUTION

Replacement reel is spring-wound. Spring is pre-wound by manufacturer. Belt retaining insert is installed 90° from normal installation for shipping purposes only.

- a. Install 7/32-inch Allen wrench in hole on end of replacement inertia reel housing. Rotate Allen wrench counterclockwise just enough to remove belt retaining insert (Figure 2-4-44-1, Detail B).

NOTE

Make sure lap belt seam of shoulder harness is up before installing.

- b. While securing inertia reel, slip end of shoulder harness through slot in shaft and through bottom opening in housing.
- c. Install belt retaining insert on shoulder harness end and pull up into main reel shaft. Maintain tension on harness and remove Allen wrench from shaft. Allow harness to slowly wind on shaft until fully retracted (Figure 2-4-44-1, Detail B and Detail C).
- d. Install inertia reel on seat back and secure with screws, washers, and nuts. **TORQUE NUTS TO 20 - 25 INCH-POUNDS** (Figure 2-4-44-1, Detail B).
- e. Connect control handle cable and control cable knurled nut to inertia reel.
- f. Install seat back cushion (PARA 2-4-44.6).

2-4-44.2 Replace Inertia Reel Control.**2-4-44.2.1 Remove.**

- a. Disconnect control handle cable from inertia reel control by loosening control cable knurled nut (Figure 2-4-44-2, Detail A).

- b. Remove screws and washers and remove inertia reel control and washers from seat bracket.

2-4-44.2.2 Install.

- a. Position replacement inertia reel control on seat bracket. Place two washers, as spacers, between inertia reel control housing and bracket at each mounting hole. Secure control to bracket with screws and washers (Figure 2-4-44-2, Detail A).
- b. Secure control handle cable to inertia reel control with knurled nut.
- c. Check operation of control handle cable and inertia reel.

2-4-44.3 Replace Shoulder Harness.

WARNING

To prevent injury to personnel, do not release either normal or emergency vertical adjust levers unless someone is sitting in seat. Extension springs are under load at all times. With seat at lowest position, vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, seat will be snapped to highest stop. Anyone leaning over seat, or with hands on guide tubes above linear bearings, will be seriously injured.

2-4-44.3.1 Remove.

- a. Release shoulder harness from rotary buckle (Figure 2-4-44-3).
- b. Set inertia reel control to AUTO-LOCK (rear) position (Figure 2-4-44-3, Detail C).

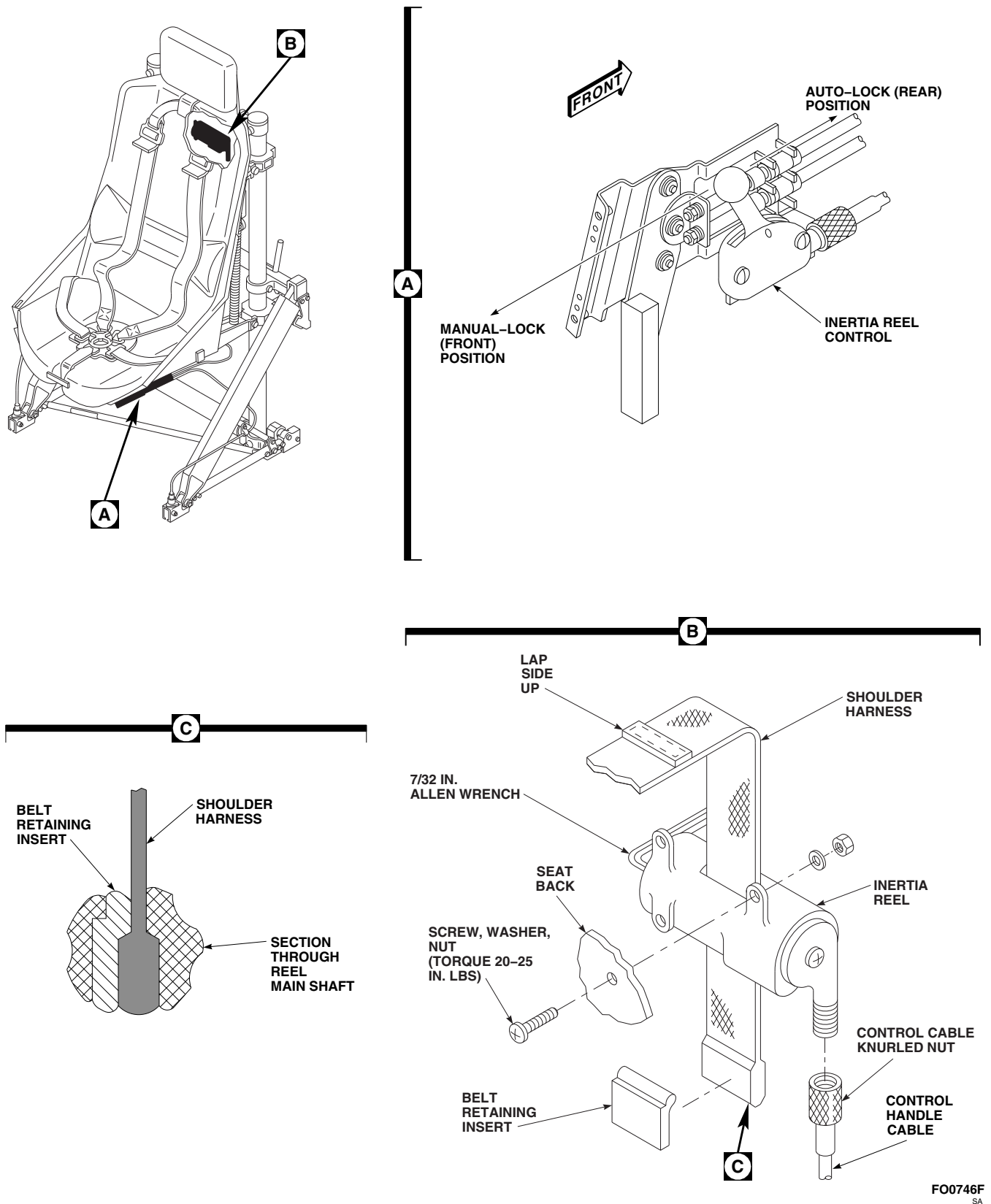
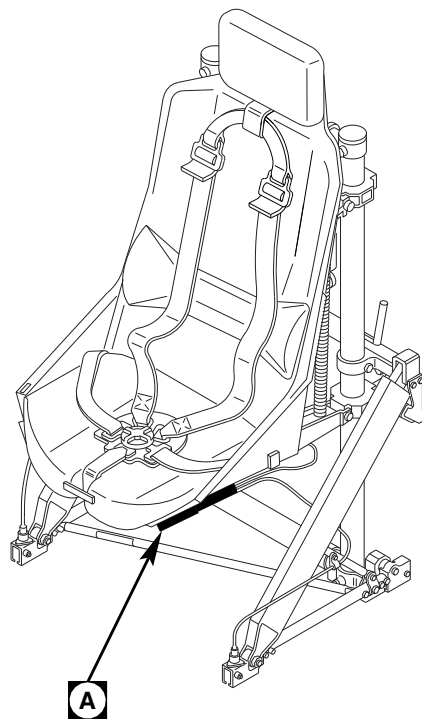


FIGURE 2-4-44-1. REPLACE INERTIA REEL



A

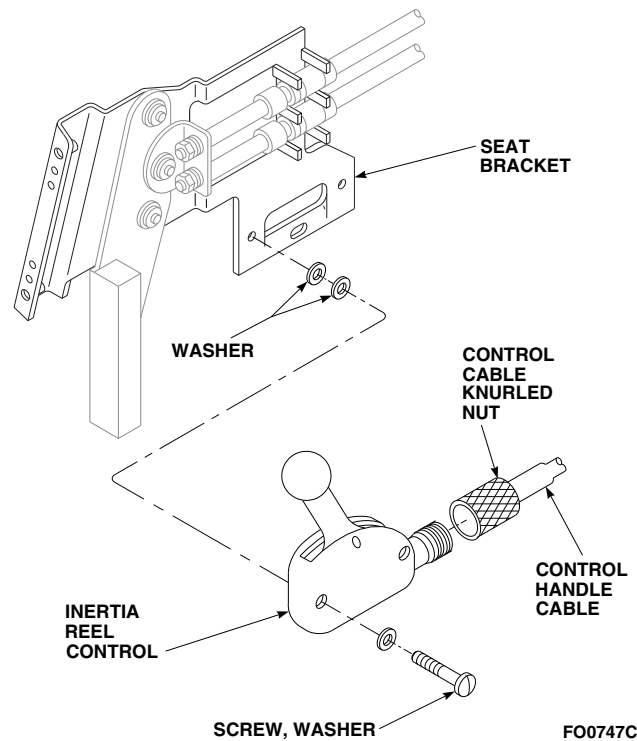
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FIGURE 2-4-44-2. REPLACE INERTIA REEL CONTROL

CAUTION

To prevent damage to inertia reel due to uncontrolled release of spring tension, do not allow Allen wrench to become disengaged from reel shaft until harness is replaced.

- c. Hold inertia reel and pull shoulder harness fully out. Place $\frac{7}{32}$ -inch Allen wrench in hole on end of reel housing, and hold reel shaft from turning (Figure 2-4-44-3, Detail A).
- d. Push belt retaining insert and shoulder harness down and out of reel shaft and lower slot (Figure 2-4-44-3, Detail B).
- e. Remove belt retaining insert and pull belt from reel shaft and through belt slot in seat back (Figure 2-4-44-3, Detail A).

2-4-44.3.2 Install.

- a. Insert replacement shoulder harness through belt slot in seat back with overlapping side of belt up (Figure 2-4-44-3, Detail A).

NOTE

On ARA seat, D-3801, belt is to be routed through head rest guide.

- b. Insert belt end through upper slot in reel housing and through slot in shaft until belt end sticks out through lower slot.
- c. Install belt retaining insert on belt end. Pull belt up to seat insert and belt in reel main shaft. Maintain tension on belt (Figure 2-4-44-3, Detail B).
- d. Remove $\frac{7}{32}$ -inch Allen wrench from inertia reel housing. Release belt tension and let belt rewind on main shaft.

- e. Check that shoulder harness end fittings lock when engaged in rotary buckle (Figure 2-4-44-3).
- f. Check operation of inertia reel.

2-4-44.4 Replace Lap Belts.

WARNING

To prevent injury to personnel, do not release either normal or emergency vertical adjust levers unless someone is sitting in seat. Extension springs are under load at all times. With seat at lowest position, vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, seat will be snapped to highest stop. Anyone leaning over seat, or with hands on guide tubes above linear bearings, will be seriously injured.

2-4-44.4.1 Remove.

- a. Release lap belts from rotary buckle (Figure 2-4-44-4).
- b. Lift seat cushion to access attachment hardware. Remove screws, washers, and nuts and remove lap belts from bucket (Figure 2-4-44-4, Detail A).

2-4-44.4.2 Install.

- a. Install replacement lap belt fittings on inside of bucket and secure with screws, washers, and nuts. Install screws, from outside of seat, and a washer on each side of fittings (Figure 2-4-44-4).
- b. Tighten nut, then back off one turn to allow free rotation of belt fitting.
- c. Lower seat cushion.
- d. Engage lap belt in rotary buckle.

2-4-44.4.3 Inspect.

NOTE

Improperly installed lap belt adjuster retaining ring(s) may cause lap belt adjustments to be difficult to complete. Make sure lap belt adjuster retaining rings are properly installed.

- a. Fold back edges of lap belt on lap belt adjuster and inspect retaining ring(s) for proper installation (Figure 2-4-44-5). Evaluate as follows:
 - (1) If retaining ring(s) is not present, replace lap belt.
 - (2) If retaining ring(s) is elevated above round axle or belt, push down on retaining ring(s) until audible click is heard. If retaining ring(s) position cannot be corrected, replace lap belt.

2-4-44.5 Replace Crotch Belt, Buckle, and Pad.

WARNING

To prevent injury to personnel, do not release either normal or emergency vertical adjust levers unless someone is sitting in seat. Extension springs are under load at all times. With seat at lowest position, vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, seat will be snapped to highest stop. Anyone leaning over seat, or with hands on guide tubes above linear bearings, will be seriously injured.

2-4-44.5.1 Remove.

- a. Release shoulder harness and lap belts from rotary buckle (Figure 2-4-44-6).
- b. Remove pin-rivets, washers, and nuts securing crotch belt fitting to front of seat. Slide belt up through loop on seat cushion and remove

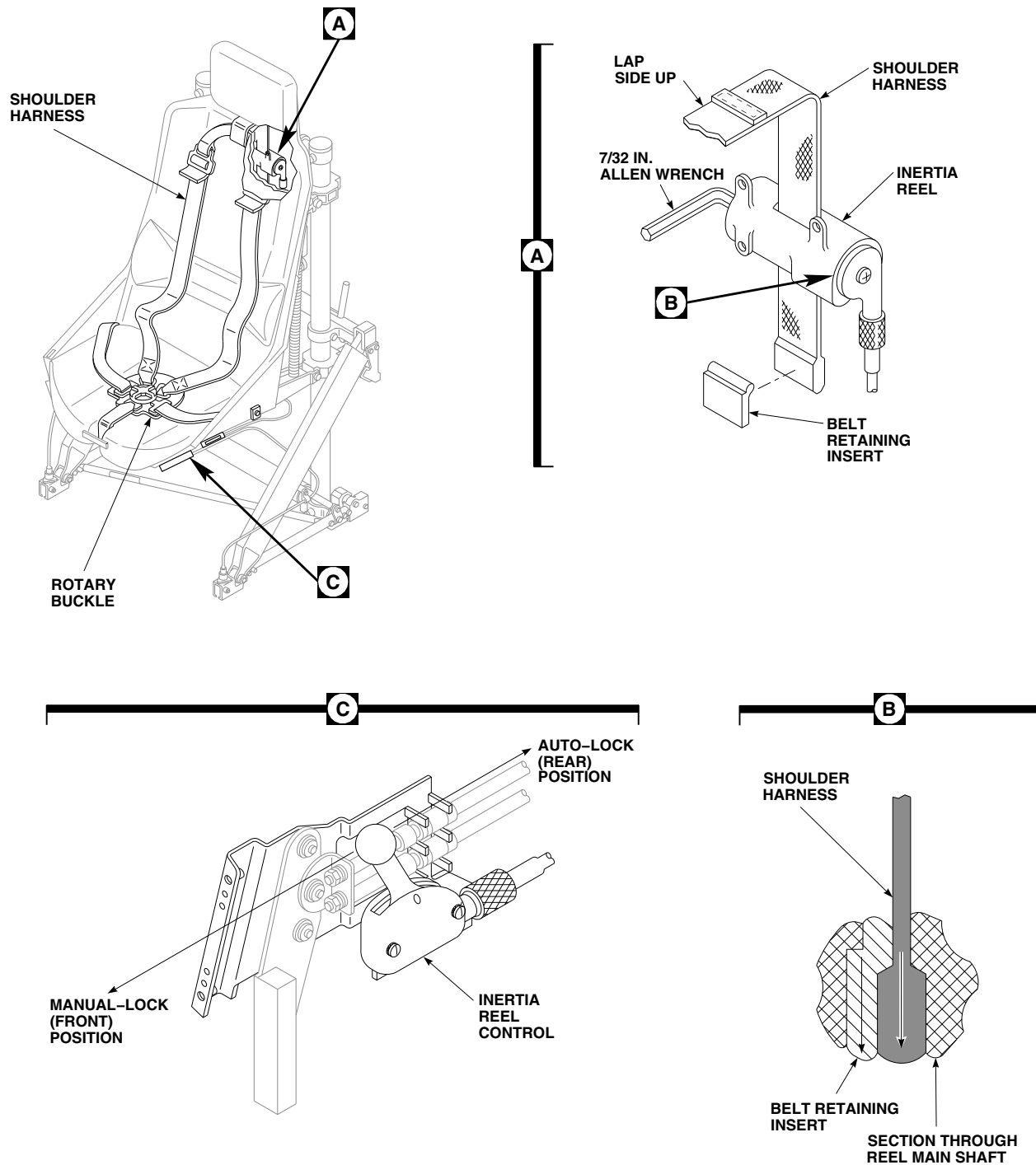
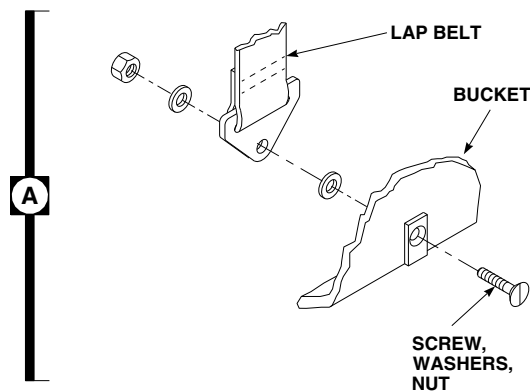
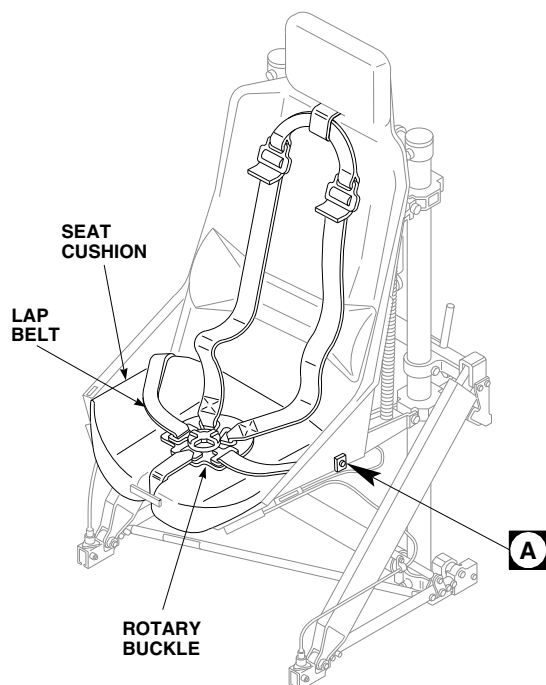


FIGURE 2-4-44-3. REPLACE SHOULDER HARNESS

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FIGURE 2-4-44-4. REPLACE LAP BELTS

crotch belt and buckle (Figure 2-4-44-6, Detail A).

- c. Open buckle pad and remove screws and pad plate from rotary buckle (Figure 2-4-44-6, Detail B).

2-4-44.5.2 Install.

WARNING

Pilot's and copilot's seat restraint buckles must have same type of release mechanism. Different types of release mechanisms could hinder an emergency egression. If installing replacement buckle, make sure both pilot's and copilot's seat restraint buckles have same type of release mechanism.

- a. Place pad on replacement rotary buckle. Install pad plate inside pad and secure to rotary buckle with screws (Figure 2-4-44-6, Detail B).

- b. Slide crotch belt down through loop on seat cushion and secure belt fitting to front of seat with pin-rivets, washers, and nuts. **TORQUE NUTS TO 20 - 25 INCH-POUNDS** (Figure 2-4-44-6, Detail A).

- c. Engage shoulder harness and lap belts in rotary buckle and check rotary buckle for correct latching and unlatching operation (Figure 2-4-44-6).

2-4-44.6 Replace Seat Cushions.

NOTE

Replacement of seat, back, and headrest cushions is the same.

2-4-44.6.1 Remove.

Starting at corner of cushion, pull cushion from seat bucket.

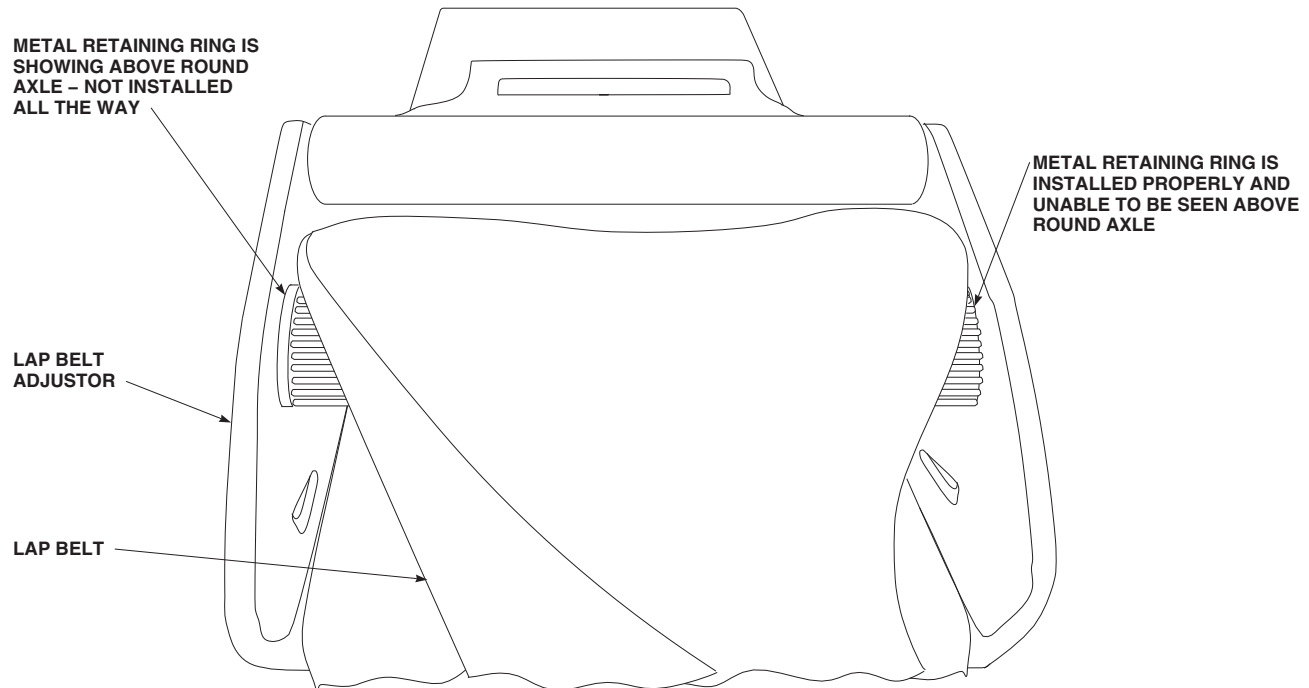
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FIGURE 2-4-44-5. INSPECT LAP RETAINING RING

2-4-44.6.2 Clean/Repair.

- a. Clean and/or repair seat cushions as follows:
 - (1) Remove cover from cushions.
 - (2) Using brush, Item 83, Appendix D, brush fabric to remove excess dirt.
 - (3) Mix 3 ounces of dishwashing compound, Item 122, Appendix D, to each gallon of water.

NOTE

Water temperature should not exceed 100°F (37°C).

- (4) Immerse one cover at a time in soap solution for 20 - 25 minutes. Hand wash for 5 minutes.
- (5) Rinse cover in clean water until all soap is removed.

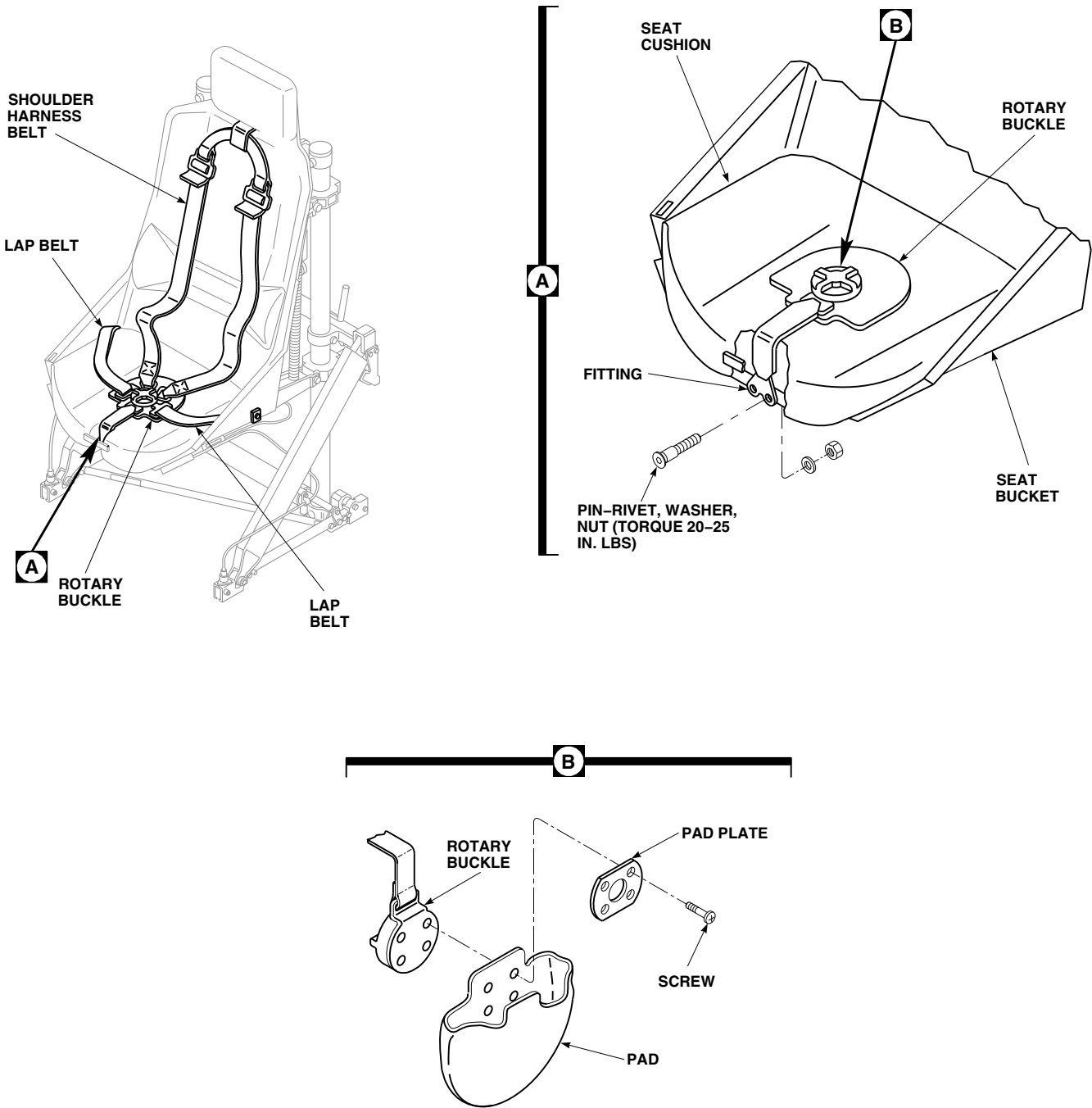
- (6) Spread cover on clean surface out of direct sunlight to dry.

- (7) Install cover on cushions.

- b. Inspect pile fastener tape on seat cushion for damage or looseness. Replace or glue pile fastener tape, Item 137B, Appendix D, as required.
- c. Inspect pile fastener tape on seat bucket for damage, misalignment, or foreign matter. If necessary, install 2-inch wide pile fastener tape, Item 137B, Appendix D, on seat bucket as required using adhesive, Item 58, Appendix D (Figure 2-4-44-7, Detail A and Detail B).

2-4-44.6.3 Install.

- a. Install cushions in seat bucket by lining up and pressing down on cushion in area of fastener tape.



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FIGURE 2-4-44-6. REPLACE CROTCH BELT, BUCKLE, AND PAD

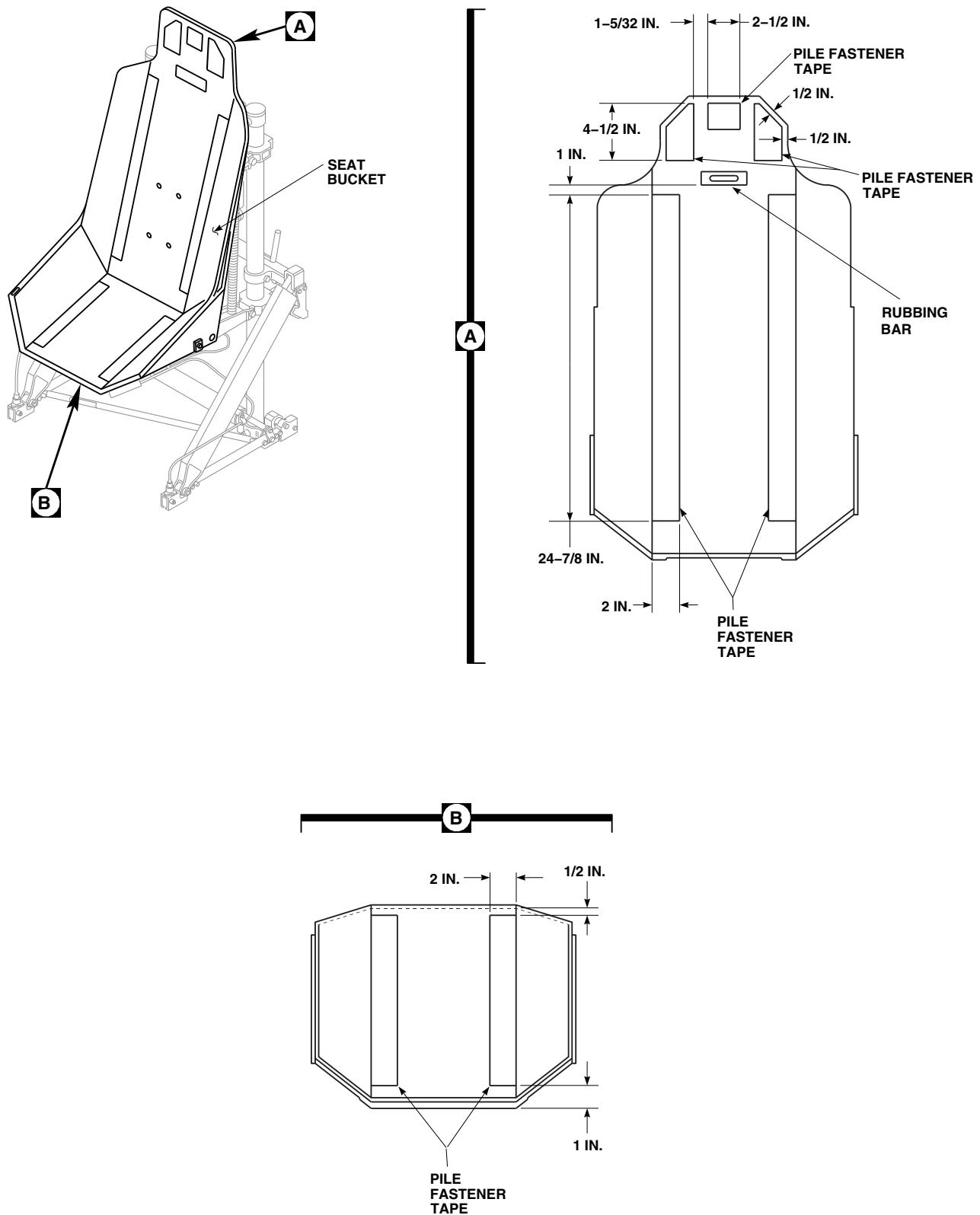
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FIGURE 2-4-44-7. REPAIR SEAT CUSHIONS

2-4-44.7 Replace Front-to-Rear Adjust Cables and Adjust Handle Baseplate.

2-4-44.7.1 Remove.

WARNING

To prevent injury to personnel, do not release either normal or emergency vertical adjust levers unless someone is sitting in seat. Extension springs are under load at all times. With seat at lowest position, vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, seat will be snapped to highest stop. Anyone leaning over seat, or with hands on guide tubes above linear bearings, will be seriously injured.

- a. Sit in seat and adjust to full up position.
- b. Remove inertia reel control (PARA 2-4-44.2).
- c. Remove bolts securing lockpin housing to front fitting (Figure 2-4-44-8, Sheet 1, Detail A).
- d. Remove screw, washers, and nut securing cable and clamp to diagonal strut. Retain clamp, screw, washers, and nut until ready to replace cable and clamp.
- e. On seat, remove nuts and washers securing release cables to bracket on release handle (Figure 2-4-44-8, Sheet 2, Detail C).
- f. Remove screw, shaft clamp, and release cables from adjust handle baseplate. Remove cable and lockpin housing from helicopter.
- g. Remove seat cushion (PARA 2-4-44.6).
- h. Remove screws, washers, and nuts securing baseplate to seat bucket. Remove baseplate.

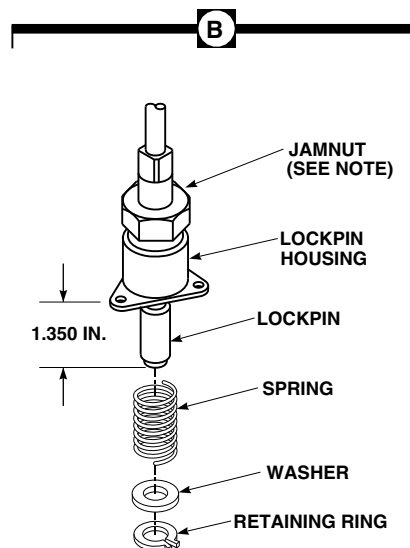
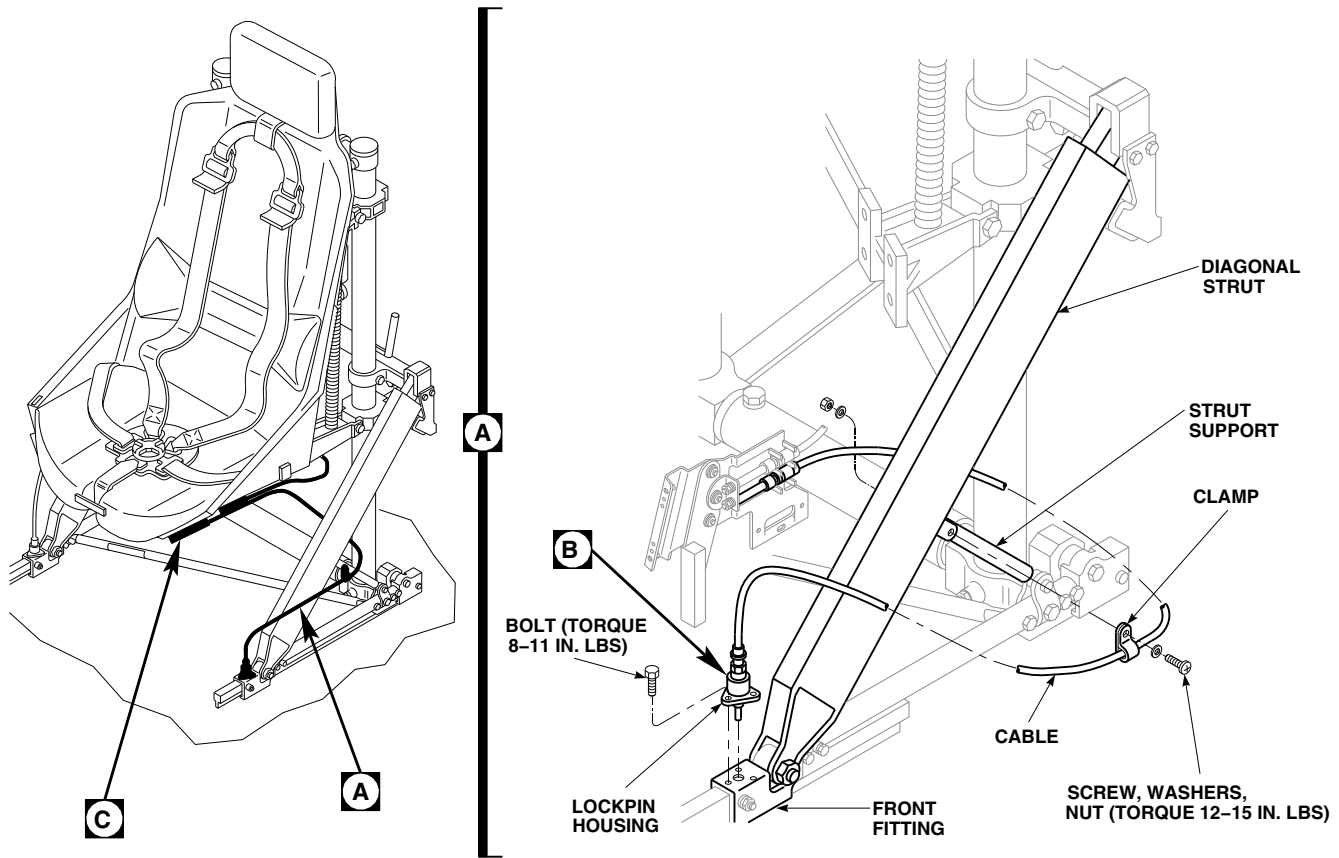
2-4-44.7.2 Repair Adjust Cable.

- a. Loosen jamnut on lockpin housing and turn housing up on cable until all tension is released on spring (Figure 2-4-44-8, Sheet 1, Detail B).

- b. Remove retaining ring from lockpin and remove washer and spring.
- c. Clean housing and lockpin using machinery towel, Item 211, Appendix D, moistened with dry-cleaning solvent, Item 124, Appendix D. Wipe dry with clean machinery towel, Item 211, Appendix D.
- d. Install replacement spring on lockpin. Install washer and retaining ring on lockpin.
- e. Turn housing down on lockpin until dimension from bottom face of housing to tip of lockpin is 1.350-inches. **TIGHTEN JAMNUT $\frac{1}{8}$ TO $\frac{1}{3}$ PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
- f. Check operation of cable and pin.
- g. Using feeler gage, measure clearance between lockpin and seat rail, while securing handle full forward. **CLEARANCE MUST BE BETWEEN 0.010 - 0.050-INCH.** If clearance is not within limits, adjust as required by shimming between lockpin housing and front fitting.

2-4-44.7.3 Repair Adjust Handle Baseplate.

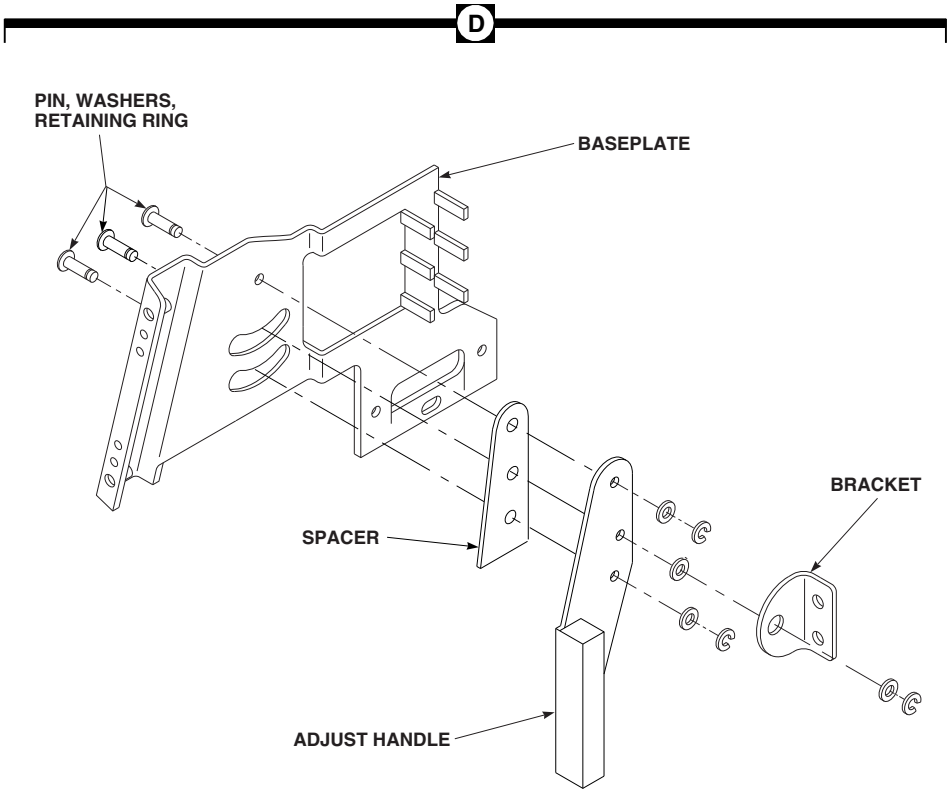
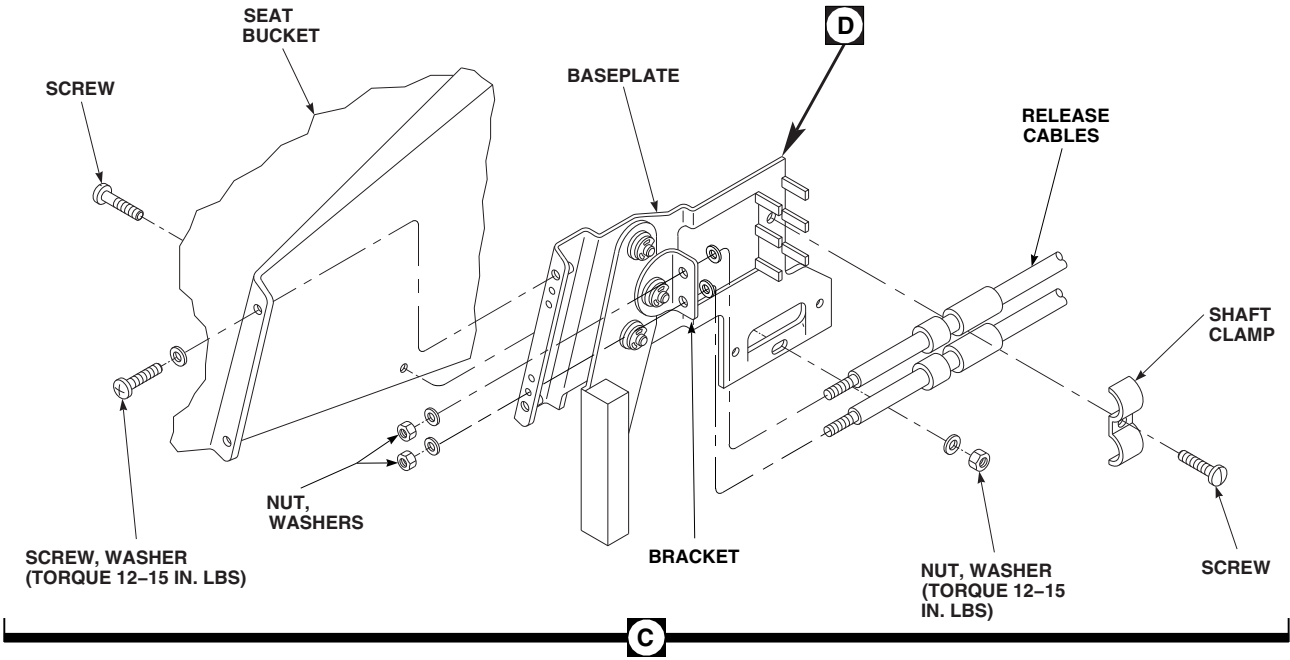
- a. Inspect parts for cracks, bends, corrosion, or hole elongation. **NONE ALLOWED.** If damaged, replace parts as necessary.
- b. Remove retaining ring, washers, and pin, and remove bracket from adjust handle (Figure 2-4-44-8, Sheet 2, Detail D).
- c. Remove retaining rings, washers, pins, and spacer, and remove adjust handle from baseplate.
- d. Secure adjust handle and spacer on baseplate with pins, washers, and retaining ring, with pinhead against baseplate, spacer between adjust handle and baseplate, and washer under retaining ring.
- e. Secure bracket on adjust handle with pin, washers, and retaining ring, with pinhead against adjust handle. Install washer under pin head, one between adjust handle and bracket and one under retaining ring.

**NOTE**

TIGHTEN JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE A SHARP RISE IN TORQUE IS FELT.

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FIGURE 2-4-44-8. REPLACE FRONT-TO-REAR ADJUST CABLES AND ADJUST HANDLE BASEPLATE (SHEET 1 OF 2)



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FIGURE 2-4-44-8. REPLACE FRONT-TO-REAR ADJUST CABLES AND ADJUST HANDLE BASEPLATE (SHEET 2 OF 2)

2-4-44.7.4 Install.

- a. Place adjust handle and baseplate on side of seat and secure to front of seat with screws and washers. Secure rear of baseplate to seat with screw, washer, and nut, with nut and washer against baseplate. **TORQUE FRONT SCREWS AND NUT TO 12 - 15 INCH-POUNDS** (Figure 2-4-44-8, Sheet 2, Detail C).
- b. Install seat cushion (PARA 2-4-44.6).
- c. Install lockpin housing in front track fitting and secure with bolts. **TORQUE BOLTS TO 8 - 11 INCH-POUNDS** (Figure 2-4-44-8, Sheet 1, Detail A).
- d. Route cable around outside of diagonal strut and install end of cable in adjust handle bracket and cable shaft in baseplate clips. Clamp cable to base with shaft clamp and screw (Figure 2-4-44-8, Sheet 2, Detail C).
- e. Install nuts and washers on cable ends, with a washer on each side of bracket.
- f. Remove screw, washers, and nut from diagonal strut and strut support (Figure 2-4-44-8, Sheet 1, Detail A). Secure cable to diagonal strut with clamp, screw, washers, and nut. **TORQUE NUT TO 12 - 15 INCH-POUNDS.**
- g. Using feeler gage, measure clearance between lockpin and seat rail, while securing handle full forward. **CLEARANCE MUST BE BETWEEN 0.010 - 0.050-INCH.** If clearance is not within limits, adjust as required by shimming between lockpin housing and front fitting.
- h. Check operation of cable and stop.
- i. Install inertia reel control on adjust baseplate (PARA 2-4-44.2).
- j. Make sure area is clean and free of foreign material.

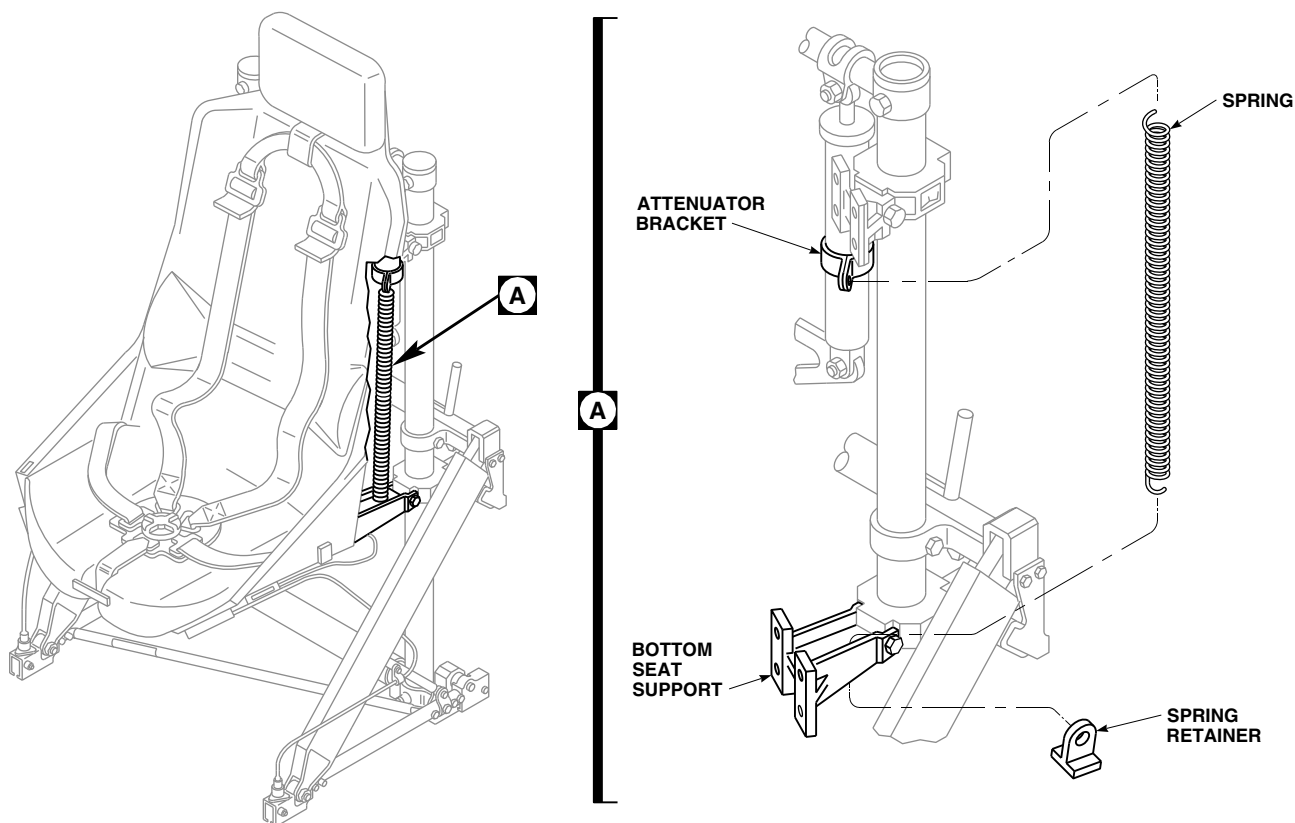
2-4-44.8 Replace Up-and-Down Extension Spring and Retainer.**2-4-44.8.1 Remove.****WARNING**

To prevent injury to personnel, do not release either normal or emergency vertical adjust levers unless someone is sitting in seat. Extension springs are under load at all times. With seat at lowest position, vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, seat will be snapped to highest stop. Anyone leaning over seat, or with hands on guide tubes above linear bearings, will be seriously injured.

- a. Sit in seat and adjust to full up position.
- b. While securing spring retainer up, pull down on spring and unhook spring from spring retainer. Remove retainer from bottom seat support (Figure 2-4-44-9, Detail A).
- c. Unhook spring from attenuator bracket and remove spring.

2-4-44.8.2 Install.

- a. Hook replacement spring on attenuator bracket (Figure 2-4-44-9, Detail A).
- b. Push retainer up through hole in bottom seat bracket, with loop on spring retainer leaning to rear.
- c. Pull down on spring and hook it to spring retainer.
- d. Make sure area is clean and free of foreign material.



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FIGURE 2-4-44-9. REPLACE UP-AND-DOWN EXTENSION SPRING AND RETAINER

2-4-44.9 Replace Up-and-Down Attenuator.

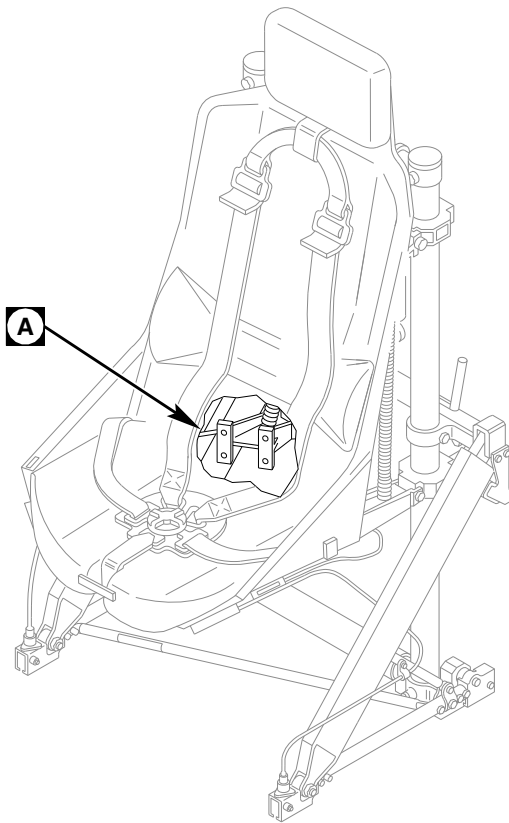
2-4-44.9.1 Remove.

WARNING

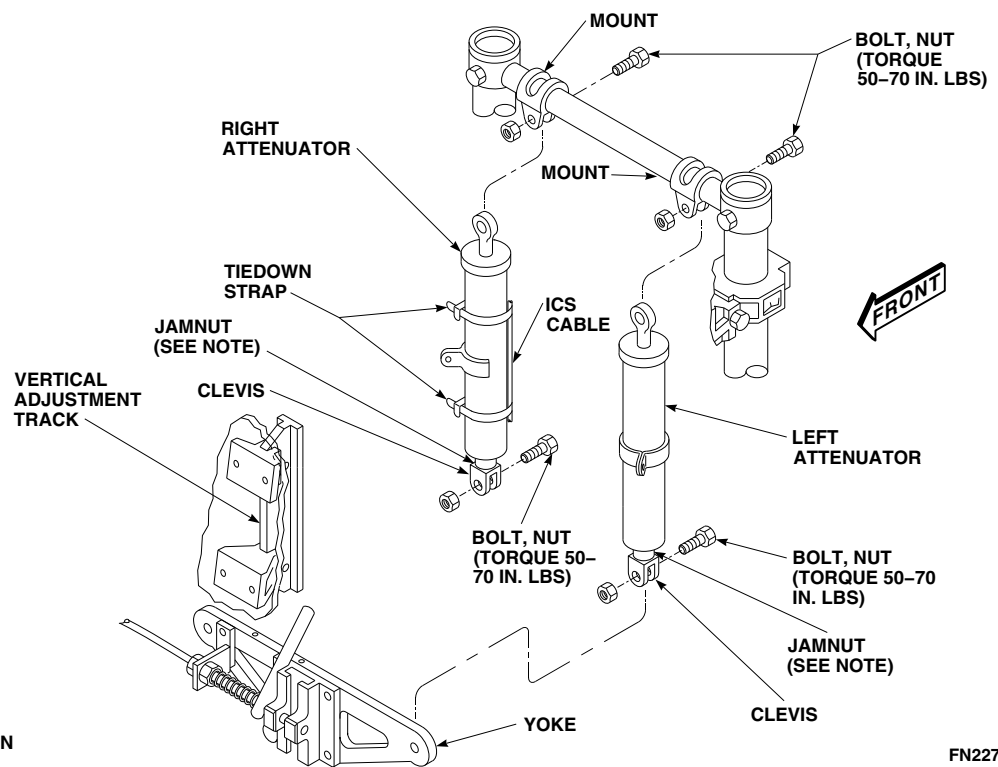
To prevent injury to personnel, do not release either normal or emergency vertical adjust levers unless someone is sitting in seat. Extension springs are under load at all times. With seat at lowest position, vertical preload on seat

could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, seat will be snapped to highest stop. Anyone leaning over seat, or with hands on guide tubes above linear bearings, will be seriously injured.

- Sit in seat and adjust to full up position.
- Remove first aid kit (PARA 17-4-1).
- Remove tiedown straps and ICS cable from right attenuator (Figure 2-4-44-10, Detail A).



A



NOTE

TIGHTEN JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.

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FIGURE 2-4-44-10. REPLACE UP-AND-DOWN ATTENUATOR

- d. Provide support for seat bucket by placing wood block, or equivalent, across seat rails in front of diagonal strut supports.
- e. Unhook extension springs from attenuators (PARA 2-4-44.8).
- f. Support seat bucket and remove bolts and nuts securing attenuators to yoke. Lower seat and rest on wood block.
- g. Remove bolts and nuts securing attenuator to upper mounts and remove attenuators.

2-4-44.9.2 Install.

- a. Install replacement attenuators in mounts on upper support and secure with bolts and nuts. **TORQUE NUTS TO 50 - 70 INCH-POUNDS** (Figure 2-4-44-10, Detail A).
- b. Loosen jamnuts on attenuator clevis. Face spring brackets to front. Lift seat and line up clevis holes with yoke holes. Install bolts and nuts. **TORQUE NUTS TO 50 - 70 INCH-POUNDS. HOLD CLEVIS AND TIGHTEN JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
- c. Hook extension springs to brackets on attenuators (PARA 2-4-44.8).
- d. Remove support from seat rails.
- e. Place ICS cable next to right attenuator and secure to attenuator with tiedown straps, Item 329, Appendix D.
- f. Install first aid kit (PARA 17-4-1).
- g. Check operation of up-and-down adjustment.
- h. Make sure area is clean and free of foreign material.

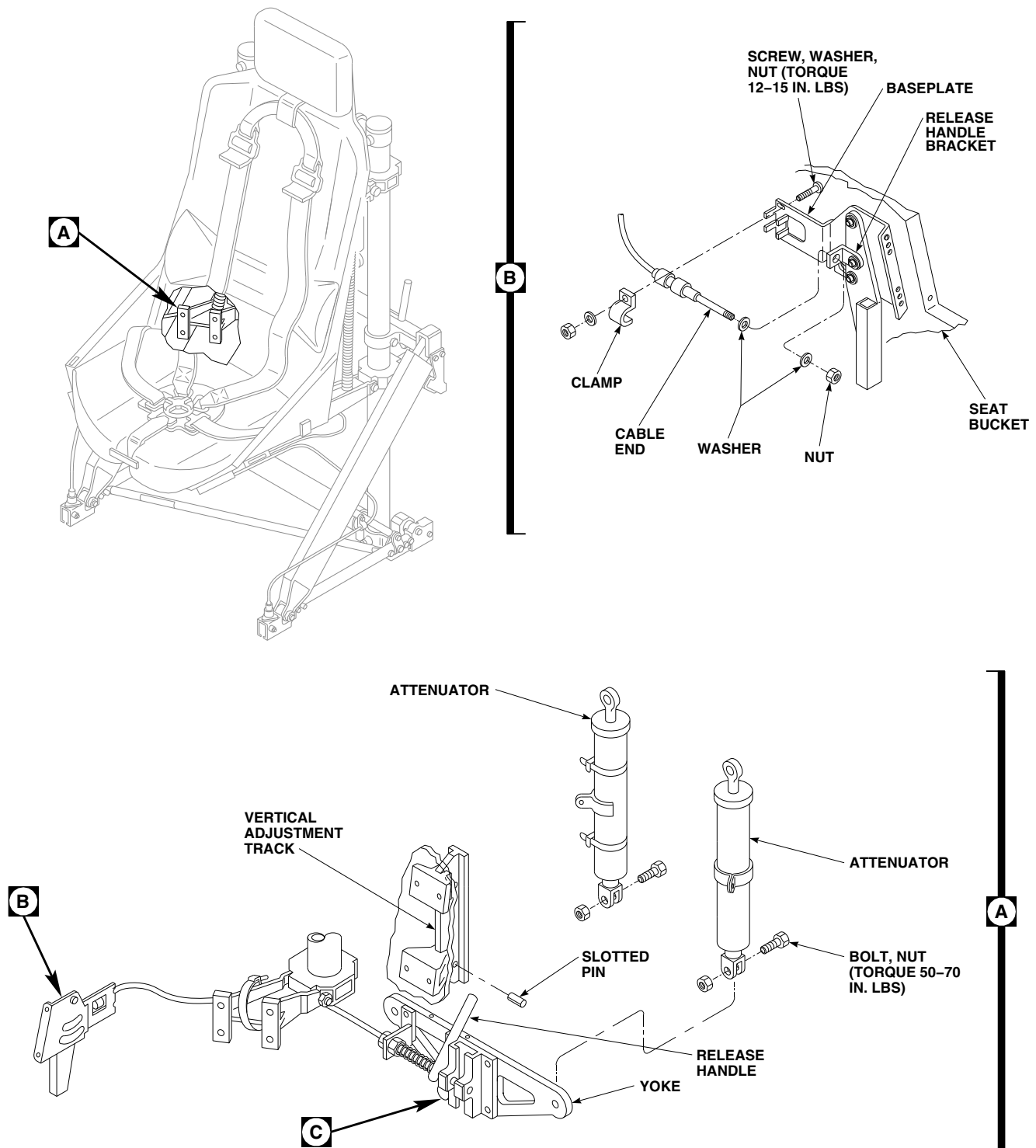
2-4-44.10 Replace Up-and-Down Yoke and Cable.

2-4-44.10.1 Remove.

WARNING

To prevent injury to personnel, do not release either normal or emergency vertical adjust levers unless someone is sitting in seat. Extension springs are under load at all times. With seat at lowest position, vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, seat will be snapped to highest stop. Anyone leaning over seat, or with hands on guide tubes above linear bearings, will be seriously injured.

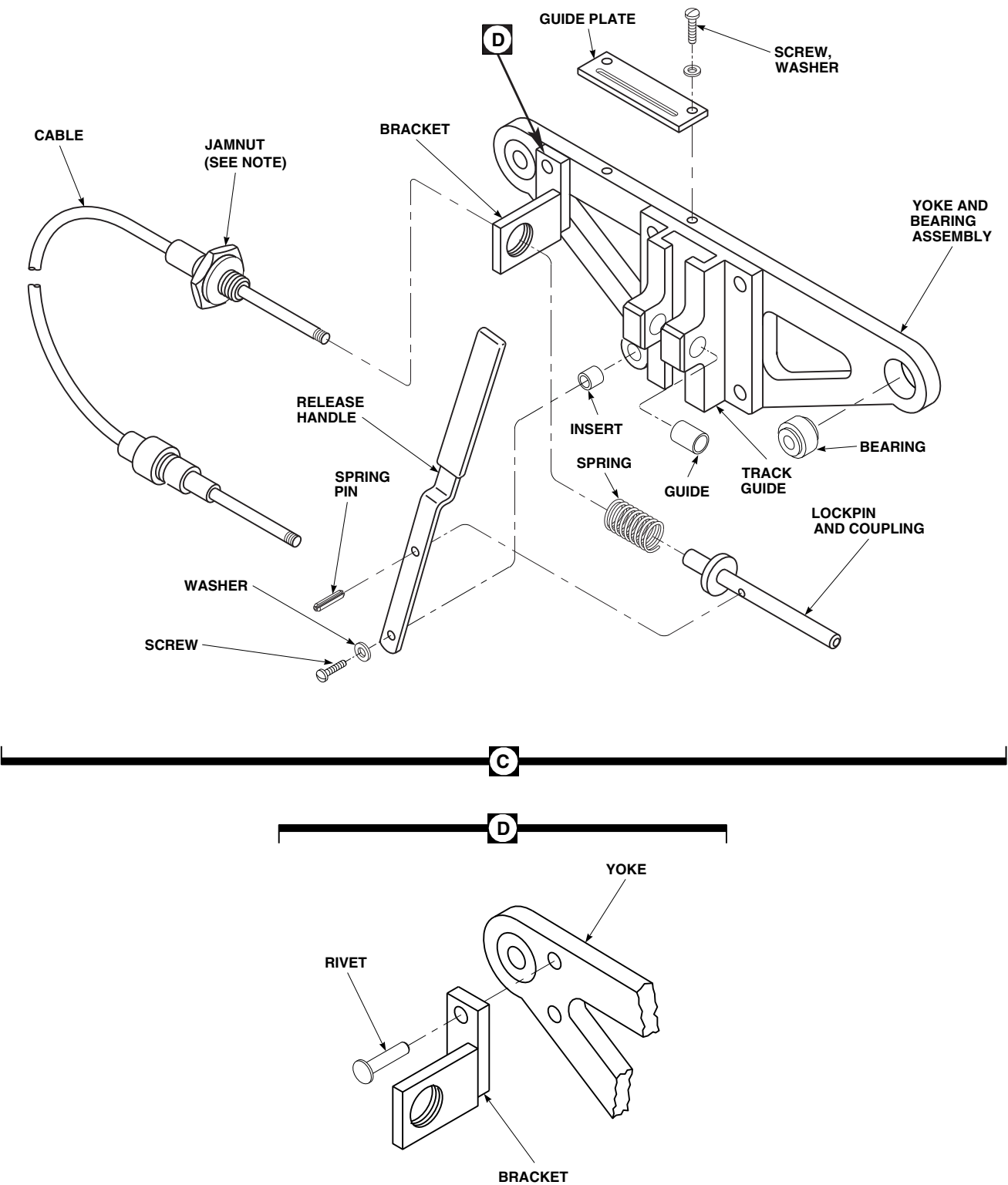
- a. Sit in seat and adjust to full up position.
- b. Provide support for seat bucket by placing wood block, or equivalent, across seat rails in front of diagonal strut supports.
- c. Unhook extension springs from attenuators (PARA 2-4-44.8).
- d. Support seat bucket and remove nuts and bolts securing attenuators to yoke. Lower seat and rest it on wood block (Figure 2-4-44-11, Sheet 1, Detail A).
- e. Remove slotted pin from vertical adjustment track. Pull release handle and pull yoke down and off track.
- f. Lift seat cushion on right side. Hold screw in seat bucket and remove nut and washer securing clamp and baseplate to seat. Remove clamp (Figure 2-4-44-11, Sheet 1, Detail B).
- g. Remove nut, washer, and cable end from release handle bracket. Remove washer from

**NOTE**

TIGHTEN JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.

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FIGURE 2-4-44-11. REPLACE UP-AND-DOWN YOKE AND CABLE (SHEET 1 OF 2)



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FIGURE 2-4-44-11. REPLACE UP-AND-DOWN YOKE AND CABLE (SHEET 2 OF 2)

cable end. Slide cable through tiedown strap on seat support. Remove yoke and cable.

2-4-44.10.2 Disassemble.

- a. Remove screws and washers securing guide plate to yoke and bearing assembly (Figure 2-4-44-11, Sheet 2, Detail C).
- b. Remove spring pin securing release handle to lockpin and coupling.
- c. Remove screw and washer securing release handle to insert on yoke and bearing assembly, and remove handle. Remove insert from yoke and bearing assembly. Remove guide plate from handle.
- d. Holding cable, thread lockpin and coupling from cable. Remove spring. Remove lockpin and coupling from guide. Remove guide from yoke and bearing assembly.
- e. Loosen jamnut and unthread cable from bracket on yoke.

2-4-44.10.3 Inspect.

- a. Check yoke and track guide for deformation, deep pitting corrosion damage to release handle pivot, and elongation of mounting holes. **NONE ALLOWED.** If damaged, replace yoke and bearing assembly.
- b. Check cable bracket for cracks, corrosion, damaged threads, and loose rivets. **NONE ALLOWED.** If damaged, replace bracket.
- c. Check lockpin bushings in track guide for hole elongation or grooved wear that would prevent free operation. **NONE ALLOWED.** If elongation or wear is found, replace bushing.
- d. Check lockpin and coupling for condition of threads, bends, and grooved wear that would prevent free operation and elongation of spring pin hole. **NONE ALLOWED.** If wear is found, replace lockpin and coupling.
- e. Check spring for deformation, corrosion, or compressed set that would prevent complete return of lockpin. **NONE ALLOWED.** If

deformation or corrosion is found, replace spring.

- f. Check cable for kinks, corrosion, condition of threads, and free operation. **NONE ALLOWED.** If damaged, replace cable.
- g. Check handle for deformation, deep pitting corrosion, elongation of mounting holes, and cracked or torn grip. **NONE ALLOWED.** If damaged, replace as necessary.
- h. Check guide plate for deformation, deep pitting corrosion, and elongation of mounting holes. **NONE ALLOWED.** If damaged, replace guide plate.
- i. Check bearing for damage. **NONE ALLOWED.** If damaged, replace bearing.

2-4-44.10.4 Repair.

- a. If required, drill out rivets and remove bracket. Place replacement bracket on yoke and install solid rivets (Figure 2-4-44-11, Sheet 2, Detail D).
- b. Remove minor surface corrosion, treat and touch up paint (PARA 2-6-5).
- c. If lockpin, coupling, and/or cable are not to be replaced, remove retaining compound from threads with No. 10-32 tap and die.
- d. If required, replace bearing in yoke assembly. **I**

2-4-44.10.5 Assemble.

- a. Install jamnut on cable housing. Thread cable housing into bracket on yoke. **TIGHTEN JAMNUT 1/6 TO 1/3 PAST POINT WHERE SHARP RISE IN TORQUE IS FELT** (Figure 2-4-44-11, Sheet 2, Detail C).
- b. Install guide in yoke and bearing assembly. Install lockpin and coupling in guide.
- c. Coat threads of cable with sealing compound, Item 285, Appendix D. Slip spring over cable, and thread lockpin and coupling on cable end until it bottoms.

- d. Place guide plate on release handle so mounting holes are under grip.
- e. Install release handle on yoke. With release handle grip over yoke, secure to yoke with washer and screw. Line up spring pin holes in lockpin and handle, and install spring pin.
- f. Secure guide plate to yoke with screws and washers.
- g. Check that lockpin slides in and out of track guide fitting freely, and spring returns lockpin to fully locked position when handle is released.

2-4-44.10.6 Install.

- a. Place replacement yoke behind seat bucket. Feed cable up through tiedown strap on seat support.
- b. Secure cable end in release handle bracket with washers and nut. Make sure cable end is seated on clips on baseplate. Secure clamp on cable with screw, washer, and nut. **TORQUE NUT TO 12 - 15 INCH-POUNDS.** Push seat cushion down in place (Figure 2-4-44-11, Sheet 1, Detail B).
- c. Pull release handle and install yoke on vertical adjustment track. Release handle and allow locking pin to engage lowest locking hole.
- d. Install slotted pin in vertical adjustment track.
- e. Lift seat and secure attenuator fittings on yoke with bolts and nuts (Figure 2-4-44-11, Sheet 1, Detail A). **TORQUE NUTS TO 50 - 70 INCH-POUNDS.**
- f. Hook extension springs on attenuator brackets (PARA 2-4-44.8).
- g. Remove support from seat rails.
- h. Check operation of yoke and cable. Lock in each stop hole.
- i. Make sure area is clean and free of foreign material.

2-4-44.11 Replace Up-and-Down Release Handle and Baseplate.

2-4-44.11.1 Remove.

WARNING

To prevent injury to personnel, do not release either normal or emergency vertical adjust levers unless someone is sitting in seat. Extension springs are under load at all times. With seat at lowest position, vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, seat will be snapped to highest stop. Anyone leaning over seat, or with hands on guide tubes above linear bearings, will be seriously injured.

- a. Sit in seat and adjust to full up position.
- b. Remove seat cushion (PARA 2-4-44.6).
- c. Remove screw, washer, and nut securing cable clamp and baseplate to seat bucket. Remove clamp (Figure 2-4-44-12, Detail A).
- d. Remove nut and washer from end release handle bracket. Pull cable from baseplate and bracket. Remove washer from cable end.
- e. Remove screws, washers, and nuts securing baseplate to front of seat bucket, and remove baseplate.
- f. Remove retaining ring, washers, and pin securing bracket to release handle (Figure 2-4-44-12, Detail B).
- g. Remove retaining rings, washers, spacer, and pins securing release handle to baseplate.

2-4-44.11.2 Inspect.

Inspect parts for cracks, bends, corrosion, or hole elongation. **NONE ALLOWED.** If damaged, replace parts as necessary.

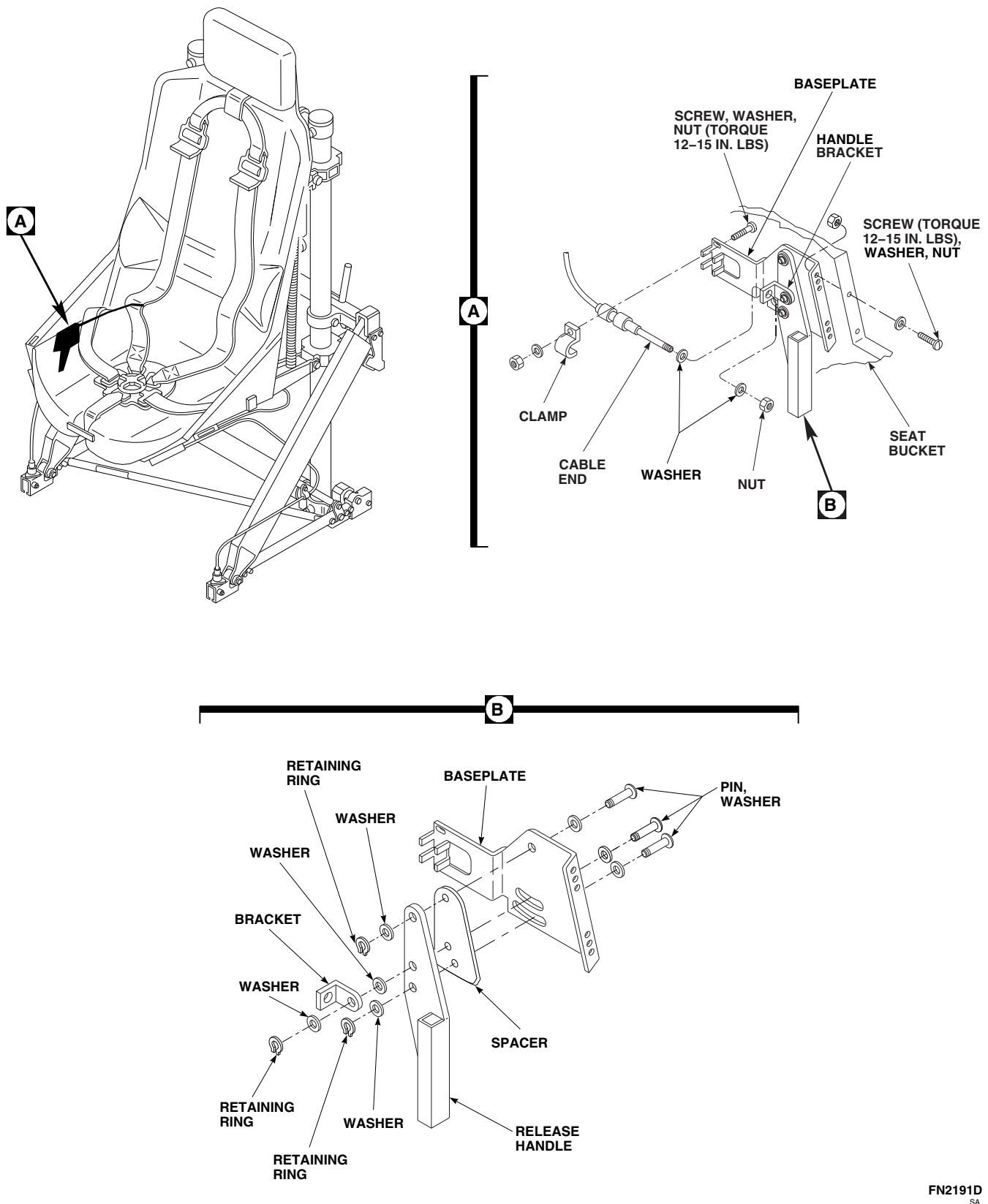


FIGURE 2-4-44-12. REPLACE UP-AND-DOWN RELEASE HANDLE AND BASEPLATE

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2-4-44.11.3 Install.

- a. Secure release handle and spacer to baseplate with pins, washers, and retaining rings (Figure 2-4-44-12, Detail B).
- b. Secure bracket to release handle, spacer, and baseplate with pin, washers, and retaining ring.
- c. Secure baseplate to front of seat bucket with screws, washers, and nuts (Figure 2-4-44-12, Detail A). **TORQUE SCREWS TO 12 - 15 INCH-POUNDS.**
- d. Install washer on cable end. Install cable end in handle bracket and cable shaft in baseplate with nut and washer.
- e. Secure clamp on cable, baseplate, and seat bucket with screw, washer, and nut. **TORQUE NUT TO 12 - 15 INCH-POUNDS.**
- f. Install seat cushion (PARA 2-4-44.6).
- g. Check operation of up-and-down adjustment.
- h. Make sure area is clean and free of foreign material.

2-4-44.12 Replace Up-and-Down Vertical Adjustment Track.

2-4-44.12.1 Remove.

WARNING

To prevent injury to personnel, do not release either normal or emergency vertical adjust levers unless someone is sitting in seat. Extension springs are under load at all times. With seat at lowest position, vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, seat will be snapped to highest stop. Anyone leaning over seat, or with hands on guide tubes above linear bearings, will be seriously injured.

- a. Sit in seat and adjust to full up position.

- b. Provide support for seat bucket by placing wood block, or equivalent, across seat rails in front of diagonal strut supports.
- c. Unhook extension spring from attenuators (PARA 2-4-44.8).
- d. Support seat and remove bolts and nuts securing attenuators to yoke. Remove attenuator fittings from yoke (PARA 2-4-44.9). Lower seat and rest it on wood block.
- e. Remove slotted pin from vertical adjustment track (Figure 2-4-44-13, Detail A). Pull release lever and pull yoke down and off track.
- f. Remove seat back cushion (PARA 2-4-44.6).
- g. Remove screws, washers, and nuts securing vertical adjustment track to seat. Remove track.

2-4-44.12.2 Install.

- a. Remove slotted pin from bottom of replacement vertical adjustment track (Figure 2-4-44-13, Detail A).
- b. Secure vertical adjustment track on back of seat with screws, washers, and nuts. Install screws from front of seat back. **TORQUE NUTS TO 50 - 70 INCH-POUNDS.**
- c. Install seat back cushion (PARA 2-4-44.6).
- d. Pull release handle and install yoke on vertical adjustment track. Release handle and allow locking pin to engage lowest locking hole.
- e. Install slotted pin in vertical adjustment track.
- f. Lift seat and secure attenuator fittings on yoke with bolts and nuts (PARA 2-4-44.9). **TORQUE NUTS TO 50 - 70 INCH-POUNDS.**
- g. Hook extension springs on attenuator brackets (PARA 2-4-44.8).
- h. Remove support from seat rails.
- i. Check up and down operation of seat. Lock in each stop hole.

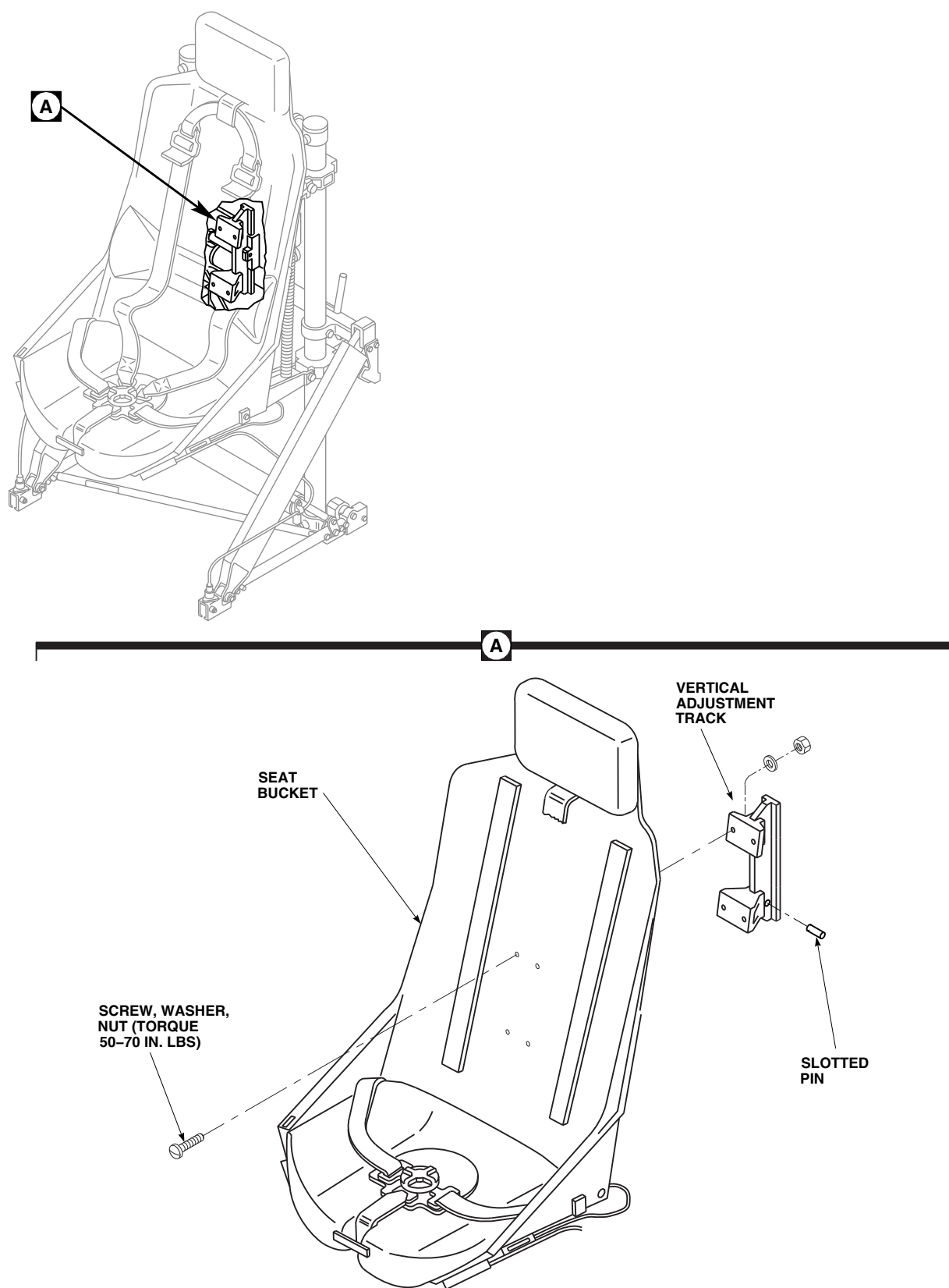
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FIGURE 2-4-44-13. REPLACE UP-AND-DOWN VERTICAL ADJUSTMENT TRACK

- j. Make sure area is clean and free of foreign material.

2-4-44.13 Replace Seat Bucket.

2-4-44.13.1 Remove.

WARNING

To prevent injury to personnel, do not release either normal or emergency vertical adjust levers unless someone is sitting in seat. Extension springs are under load at all times. With seat at lowest position, vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, seat will be snapped to highest stop. Anyone leaning over seat, or with hands on guide tubes above linear bearings, will be seriously injured.

- a. Sit in seat and adjust to full up position.
- b. Remove seat cushions (PARA 2-4-44.6).
- c. Remove shoulder harness (PARA 2-4-44.3), lap belts (PARA 2-4-44.4), and rotary buckle (PARA 2-4-44.5).
- d. Remove inertia reel control (PARA 2-4-44.2).
- e. Remove inertia reel (PARA 2-4-44.1).
- f. Remove front-to-rear release handle and baseplate (PARA 2-4-44.7).
- g. Remove vertical adjustment release handle and baseplate (PARA 2-4-44.11).
- h. Remove screws, washers, and nuts securing seat bucket to vertical adjustment track (Figure 2-4-44-14).
- i. Support seat bucket and remove bolts and washers securing seat to support brackets on frame. Remove seat bucket.

2-4-44.13.2 Install.

- a. Install replacement seat bucket in frame and secure to support brackets with bolts and wash-

ers (Figure 2-4-44-14). **TORQUE BOLTS TO 100 - 140 INCH-POUNDS.**

- b. Secure vertical adjustment track to bucket with screws, washers, and nuts. **TORQUE NUTS TO 50 - 70 INCH-POUNDS.**
- c. Install vertical adjustment release handle and baseplate (PARA 2-4-44.11).
- d. Install front-to-rear release handle and baseplate (PARA 2-4-44.7).
- e. Install inertia reel (PARA 2-4-44.1).
- f. Install inertia reel control (PARA 2-4-44.2).
- g. Install shoulder harness (PARA 2-4-44.3), lap belts (PARA 2-4-44.4), and rotary buckle (PARA 2-4-44.5).
- h. Install seat cushions (PARA 2-4-44.6).
- i. Check operation of seat and inertia reel.
- j. Make sure area is clean and free of foreign material.

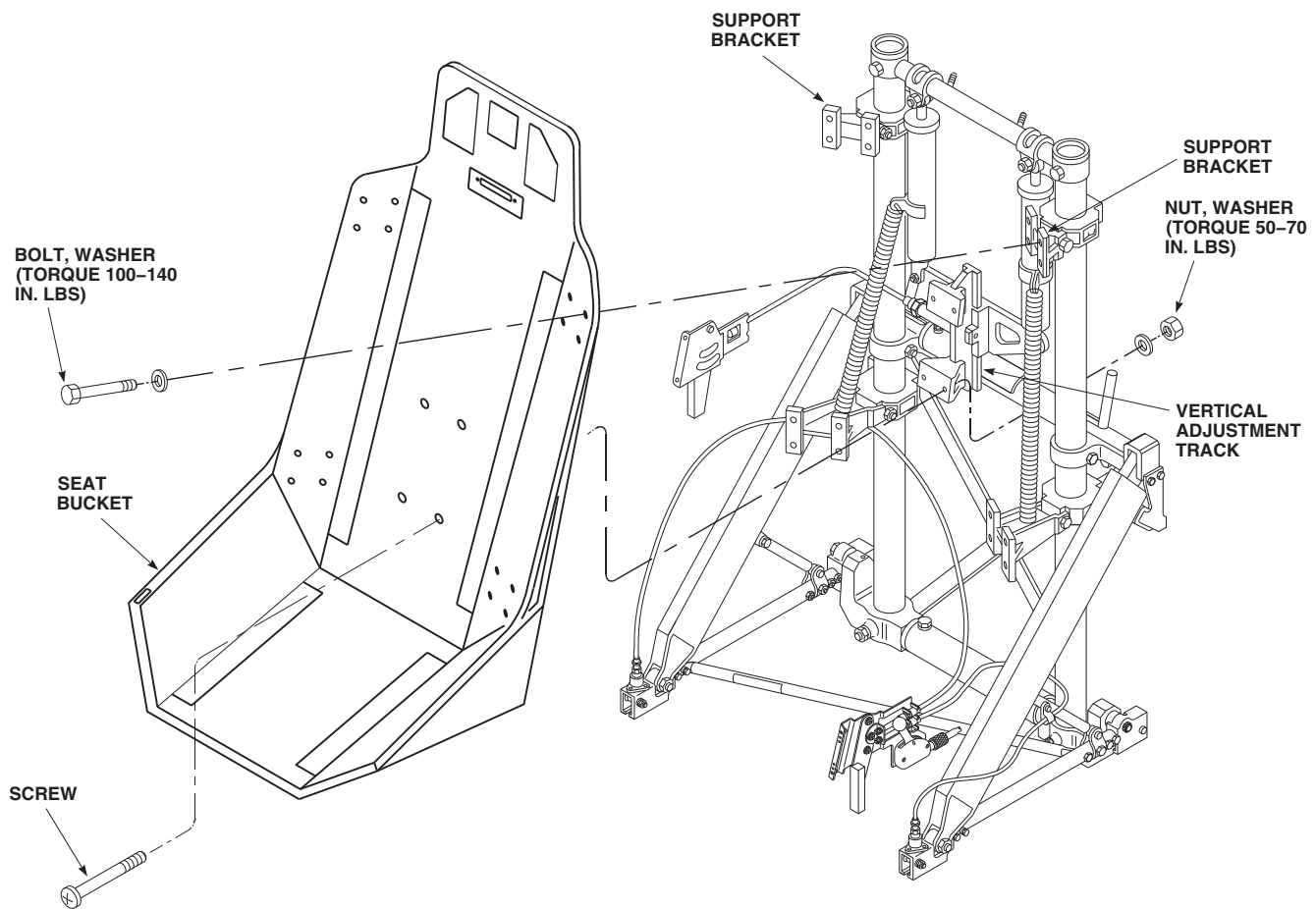
2-4-44.14 Replace Shoulder Harness Rubbing Bar.

2-4-44.14.1 Remove.

WARNING

To prevent injury to personnel, do not release either normal or emergency vertical adjust levers unless someone is sitting in seat. Extension springs are under load at all times. With seat at lowest position, vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, seat will be snapped to highest stop. Anyone leaning over seat, or with hands on guide tubes above linear bearings, will be seriously injured.

- a. Sit in seat and adjust to full up position.

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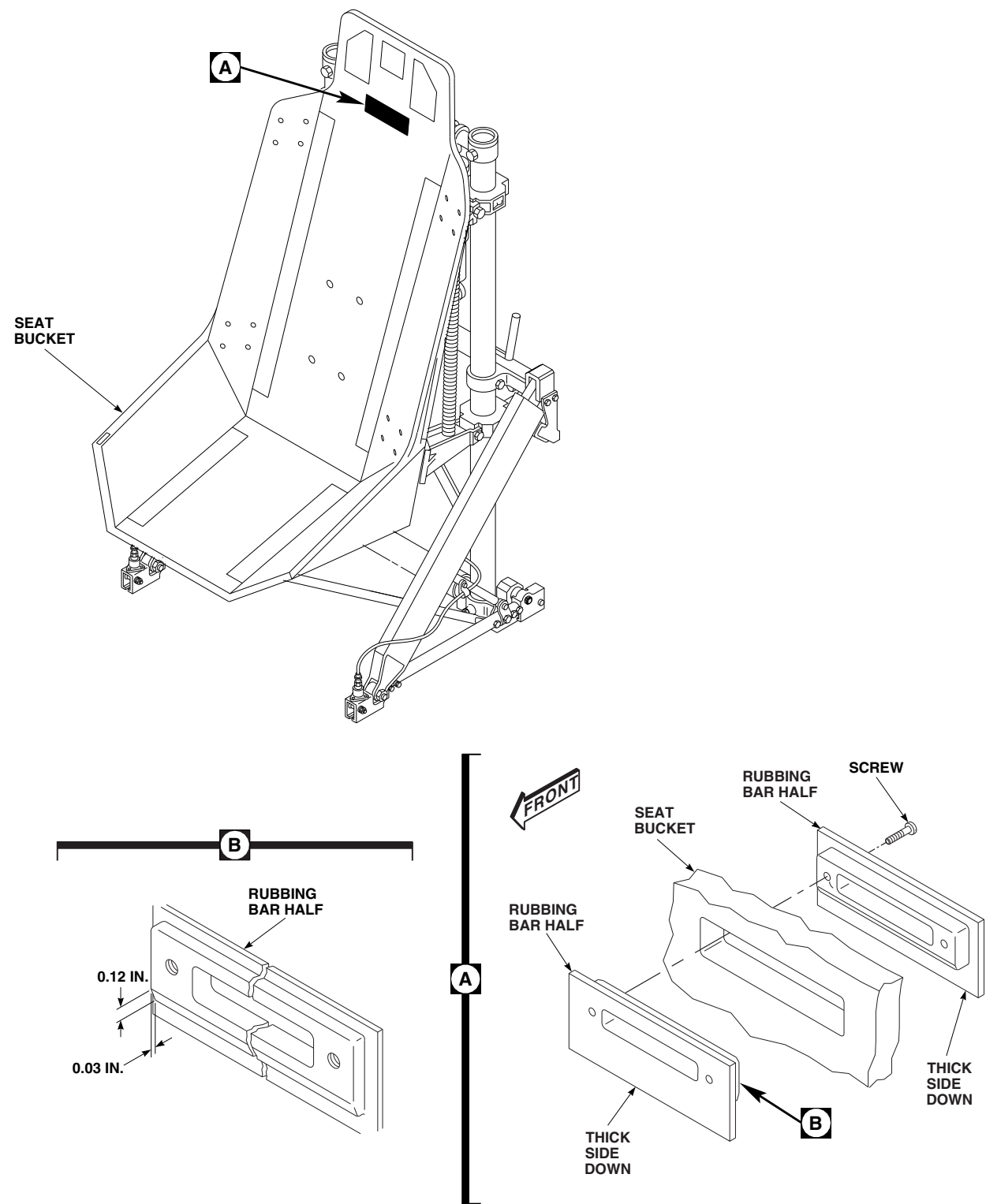
- b. Remove headrest and seat back cushions (PARA 2-4-44.6).
- c. Remove shoulder harness (PARA 2-4-44.3).
- d. Remove screws and rubbing bar halves from seat bucket (Figure 2-4-44-15, Detail A).

2-4-44.14.2 Install.**NOTE**

Flanged piece of rubbing bar is inserted from rear of armored seat bucket with thick side down. Flat piece of rubbing

bar is positioned on front of armored seat bucket with thick side down.

- a. Check fit of flat piece of rubbing bar with seat bucket. Using file, file flat piece of rubbing bar as necessary to eliminate any interference with seat bucket (Figure 2-4-44-15, Detail B).
- b. Apply sealing compound, Item 278, Appendix D, to threads of screws. Install screws from rear of seat bucket through flanged piece into flat piece of rubbing bar (Figure 2-4-44-15, Detail A).
- c. Install shoulder harness (PARA 2-4-44.3).



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FIGURE 2-4-44-15. REPLACE SHOULDER HARNESS RUBBING BAR

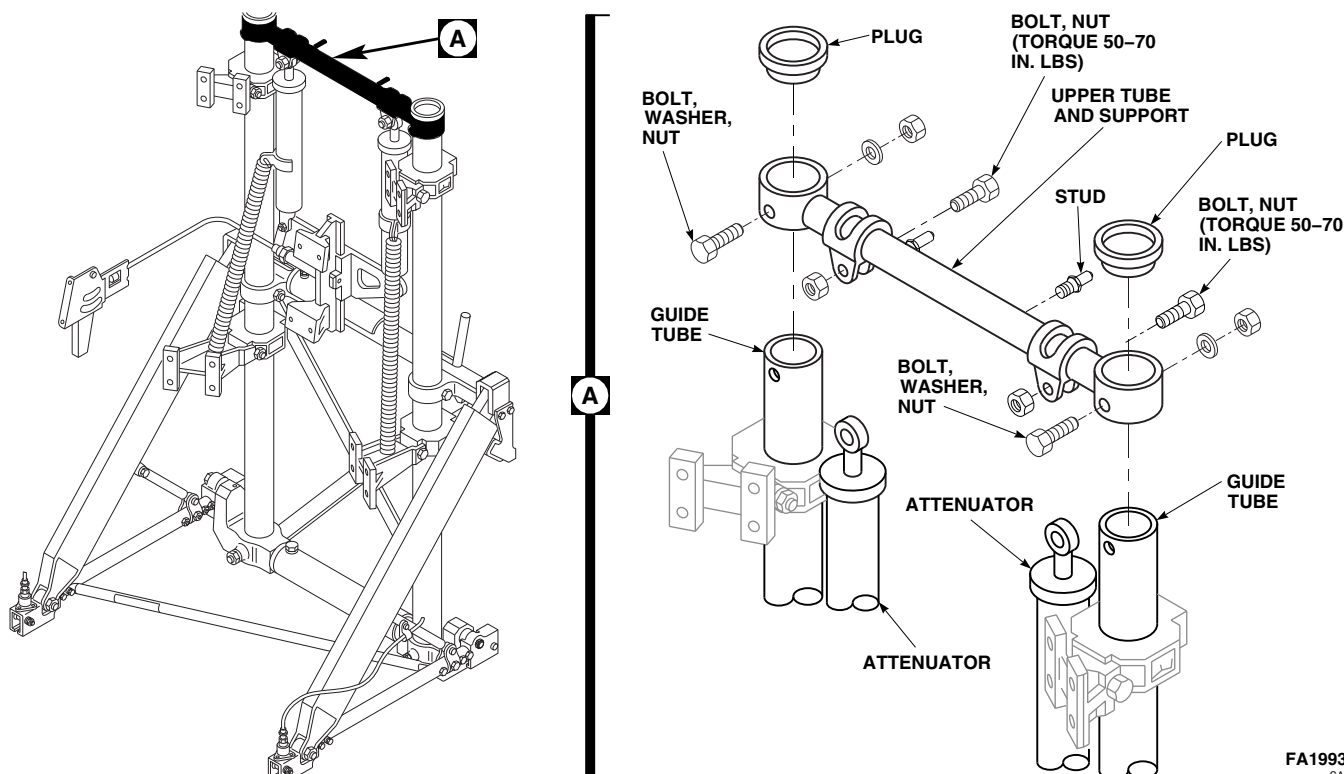


FIGURE 2-4-44-16. REPLACE SEAT FRAME UPPER TUBE AND SUPPORTS

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- d. Install seat back and headrest cushions (PARA 2-4-44.6).
- e. Make sure area is clean and free of foreign material.

2-4-44.15 Replace Seat Frame Upper Tube and Supports.

2-4-44.15.1 Remove.

WARNING

To prevent injury to personnel, do not release either normal or emergency vertical adjust levers unless someone is sitting in seat. Extension springs are under load at all times. With seat at lowest position, vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, seat will be snapped to highest stop. Anyone leaning over seat, or

with hands on guide tubes above linear bearings, will be seriously injured.

- a. Sit in seat and adjust to full up position.
- b. Unhook extension springs from attenuator brackets (PARA 2-4-44.8).
- c. Unclamp ICS cable from upper tube. Cut tiedown straps securing ICS cable to right attenuator and move ICS cable to one side (Figure 2-4-44-10, Detail A).
- d. Provide support for seat bucket by placing wood block, or equivalent, across seat rails in front of diagonal strut supports.
- e. Support seat and remove attenuator upper bolts, and nuts (Figure 2-4-44-16, Detail A). Lower seat to rest on support.
- f. Pry plugs from top of guide tubes.
- g. Remove bolts, washers, and nuts securing supports to top of guide tubes.

- h. Remove supports from guide tubes by tapping up one side, then other, with mallet.

2-4-44.15.2 Install.

- a. Install replacement upper tube and supports on guide tubes, with studs on tube facing rear, by tapping down one side, then other, with mallet until attachment holes line up (Figure 2-4-44-16, Detail A).
- b. Install bolts, washers, and nuts with boltheads toward front. **TIGHTEN NUTS UNTIL THERE IS NO PLAY, THEN BACK OFF UNTIL THERE IS 0.004 - 0.010-INCH END-PLAY.** Install plugs in top of guide tubes.
- c. Lift seat and secure attenuator rod ends in upper tube straps with bolts and nuts. **TORQUE NUTS TO 50 - 70 INCH-POUNDS.**
- d. Hook extension springs on attenuator brackets (PARA 2-4-44.8).
- e. Clamp ICS cable to upper tube. Secure ICS cable to right attenuator with two tiedown straps, Item 329, Appendix D, one at each end of attenuator (Figure 2-4-44-10, Detail A).
- f. Remove support from seat rails.
- g. Make sure area is clean and free of foreign material.

2-4-44.16 Replace Up-and-Down Guide Tube.

2-4-44.16.1 Remove.

WARNING

To prevent injury to personnel, do not release either normal or emergency vertical adjust levers unless someone is sitting in seat. Extension springs are under load at all times. With seat at lowest position, vertical preload on seat

could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, seat will be snapped to highest stop. Anyone leaning over seat, or with hands on guide tubes above linear bearings, will be seriously injured.

- a. If seat can be operated up, sit in seat and adjust to full up position. If not, be careful not to touch release handle on side of seat or emergency handle at rear of seat before removing extension springs.
- b. Unhook extension springs from attenuator brackets if required (PARA 2-4-44.8).
- c. Remove upper tube and support brackets (PARA 2-4-44.15).
- d. Remove midframe support bolt and nut, and trunnion bolt and nut (Figure 2-4-44-17).
- e. Twist, tap, and pull guide tube up from trunnion and through midframe support.

2-4-44.16.2 Install.

- a. Install tube through upper and lower brackets, trunnion, and midframe supports. Line up bolt-holes.
- b. Install trunnion bolt through diagonal strap and trunnion. Install nut. **TORQUE NUT TO 200 - 250 INCH-POUNDS** (Figure 2-4-44-17).
- c. Secure midframe support bolt and nut. **TORQUE NUT TO 160 - 190 INCH-POUNDS.**
- d. Install upper tube and support brackets (PARA 2-4-44.15).
- e. Hook extension springs on attenuator brackets (PARA 2-4-44.8).
- f. Check operation of up-and-down seat adjust.
- g. Make sure area is clean and free of foreign material.

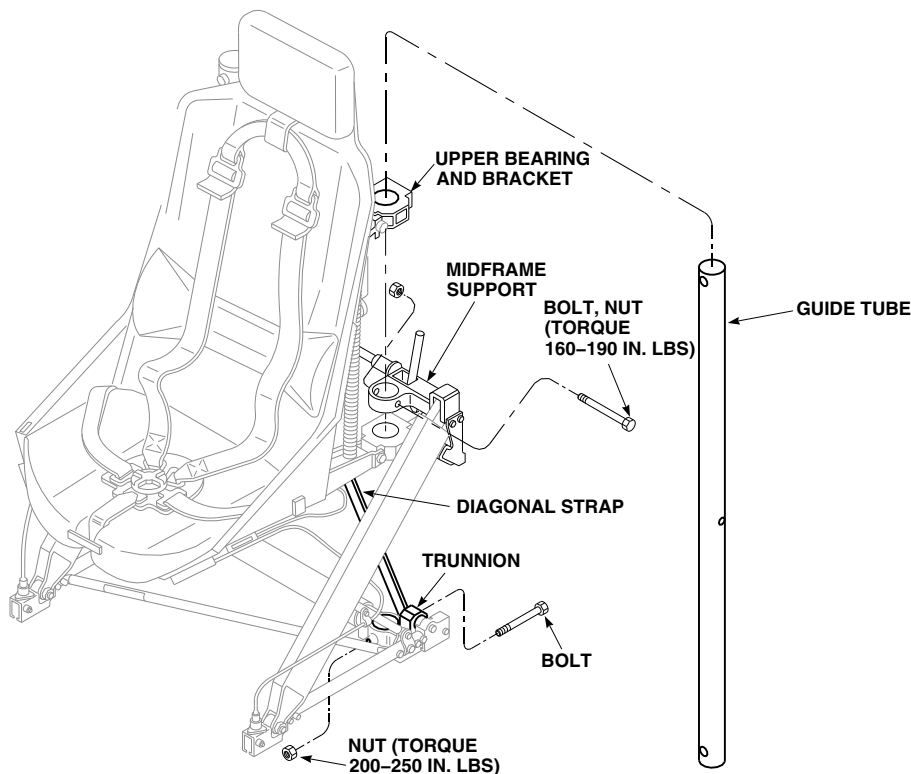
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FIGURE 2-4-44-17. REPLACE UP-AND-DOWN GUIDE TUBE

2-4-44.17 Replace Midframe Support and Tube.**2-4-44.17.1 Remove.****WARNING**

To prevent injury to personnel, do not release either normal or emergency vertical adjust levers unless someone is sitting in seat. Extension springs are under load at all times. With seat at lowest position, vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, seat will be snapped to highest stop. Anyone leaning over seat, or with hands on guide tubes above linear bearings, will be seriously injured.

- a. Sit in seat and adjust to full up position.
- b. Remove stowage panel from seat (PARA 2-4-44.25 or PARA 2-4-44.26).
- c. Remove up-and-down guide tube from side on which support is to be replaced (PARA 2-4-44.16).
- d. Remove bolt and washer securing stowage panel support bracket to midframe support and remove stowage panel support bracket (Figure 2-4-44-18, Sheet 1, Detail A).
- e. Remove bolt, washer, shim washers (if installed), and nut securing upper end of strap to midframe support (PARA 2-4-44.21).
- f. Remove screw, washer, and nut securing cable clamp and strut support to diagonal strut. Remove cable clamp and strut support.
- g. Remove bolt, washer, and nut securing midframe support to tube. Disengage tilt release pin from diagonal strut and turn midframe support to clear diagonal strut. Lift diagonal strut from support. Remove support from tube.
- h. Remove bolt, washer, nut, and tube from other midframe support. Slide tube from support.

2-4-44.17.2 Disassemble.

- a. Unscrew tilt release handle from lockpin and remove guide (Figure 2-4-44-18, Sheet 2, Detail B).
- b. Remove setscrew, spring, and ball from support housing, and slide lockpin from housing.

2-4-44.17.3 Inspect.

- a. Inspect housing for cracks, deformation, corrosion, surface blemishes, and condition of paint. Check all threaded holes for damaged threads. **NO CRACKS, DEFORMATION, DAMAGED THREADS, OR DEEP CORROSION PITTING ALLOWED.** If cracks, deformation, or deep corrosion pitting is found, reject housing.
- b. Inspect bushing and guide inserts for grooved wear that would prevent free operation of lockpin. **NONE ALLOWED.** If wear is found, replace as necessary.
- c. Inspect setscrew, spring, and ball for wear, corrosion, deformed or compressed spring, and damaged threads. **NONE ALLOWED.** If damaged, replace hardware.

NOTE

Normal wear along length of lockpin is acceptable if grooves or ridges are not seen.

- d. Inspect lockpin and guide for bends, wear, grooves that would prevent lockpin from sliding freely in bushing or guides, corrosion, or damaged threads. Replace for any damage, except damaged threads. Threads can be repaired by retapping. Replace lockpin with stripped threads.
- e. Inspect handle for bends, corrosion, and damaged threads. Inspect jacket for cracking, peeling, or security. Replace as necessary. Damaged threads can be repaired. Replace handle with stripped threads.

2-4-44.17.4 Repair.

- a. Treat minor corrosion and touch up nicks, scratches, and paint (PARA 2-6-5).

- b. Use 5/16-18 UNC tap in lockpin to clean out old adhesive and repair any damaged threads.
- c. Use 5/16-18 UNC die to clean old adhesive and repair damaged threads on lockpin handle.
- d. Use No. 10-32 tap to retap threaded holes in housing.
- e. Press out guide inserts and bushing from housing (Figure 2-4-44-18, Sheet 2, Detail B).
- f. Coat replacement bushing with primer, Item 257, Appendix D, and press it into housing while primer is wet. End of bushing to be 0.055 - 0.060-inch from window end of bore (Figure 2-4-44-18, Sheet 2, Section A-A).
- g. Using drill with No. 21 drill bit, position in setscrew hole and drill through one side of bushing. Deburr hole in bushing. Flush out chips with dry-cleaning solvent, Item 124, Appendix D, and blow dry.
- h. Coat outside diameters and face of flange on replacement guide inserts with primer, Item 257, Appendix D, and press into housing while primer is wet (Figure 2-4-44-18, Sheet 2, Detail B).

2-4-44.17.5 Assemble.

- a. Slide replacement lockpin in housing and into bushing and guide inserts. Install guide on lockpin (Figure 2-4-44-18, Sheet 2, Detail B).
- b. Coat threads of handle with sealing compound, Item 285, Appendix D. Thread handle into lockpin until it bottoms.
- c. Check that lockpin moves freely in and out.



To prevent damage to spring, do not tighten setscrew beyond first feel of solid contact.

- d. Install ball, spring, and setscrew in housing. **TIGHTEN SETSCREW UNTIL SPRING**

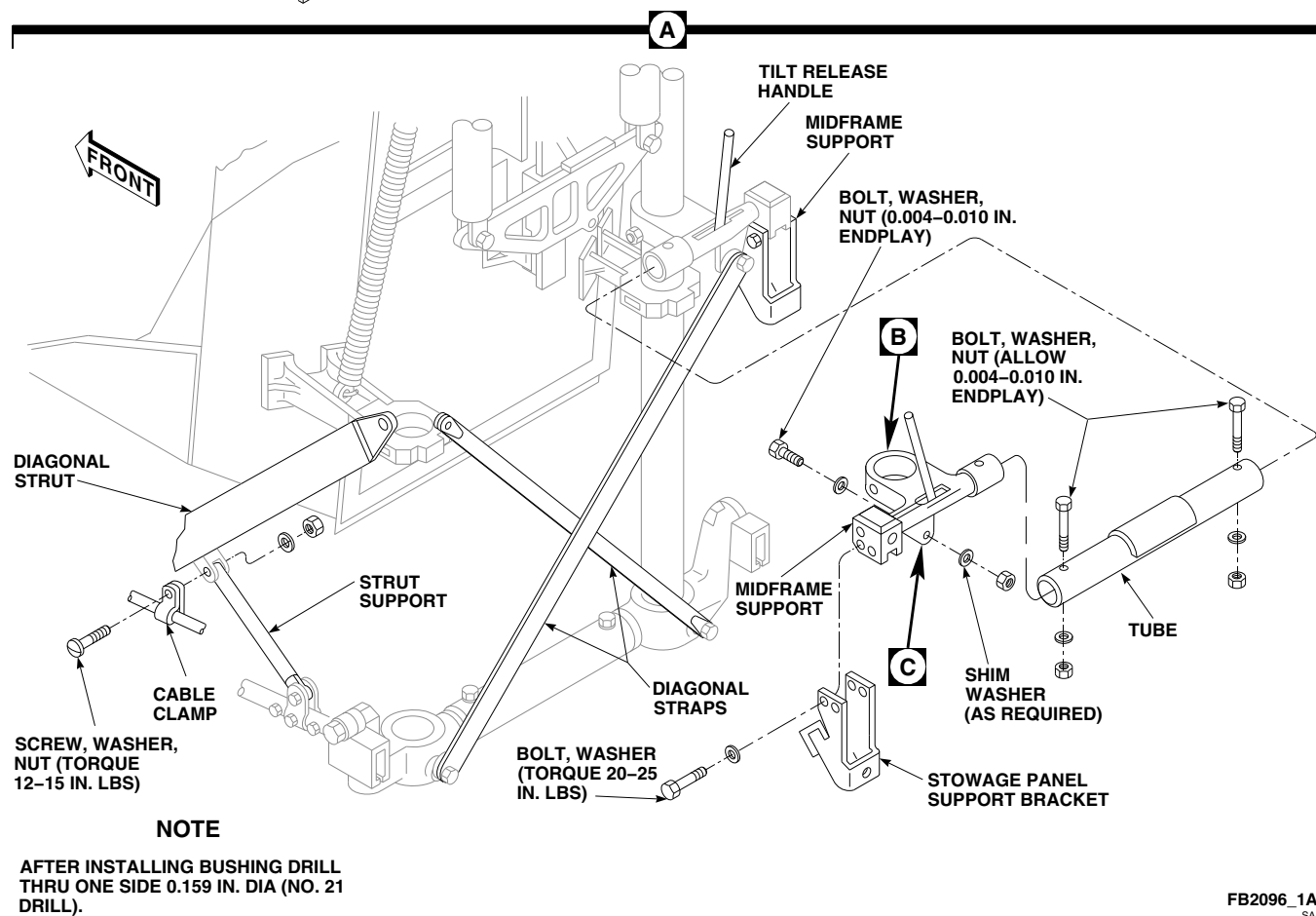
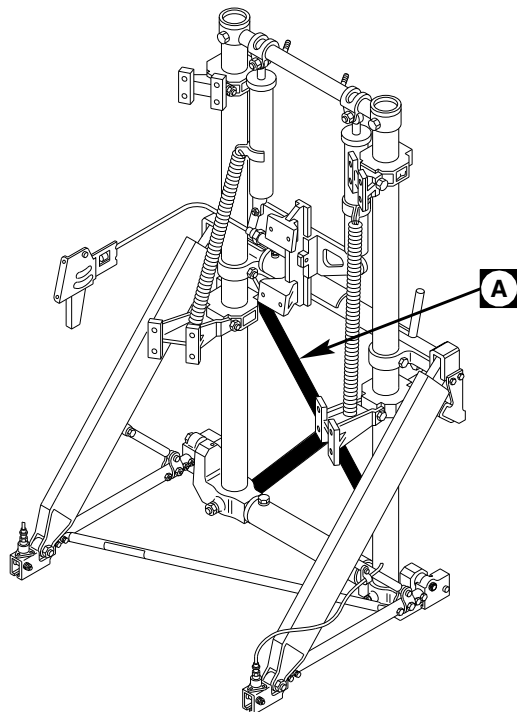


FIGURE 2-4-44-18. REPLACE MIDFRAME SUPPORT AND TUBE (SHEET 1 OF 2)

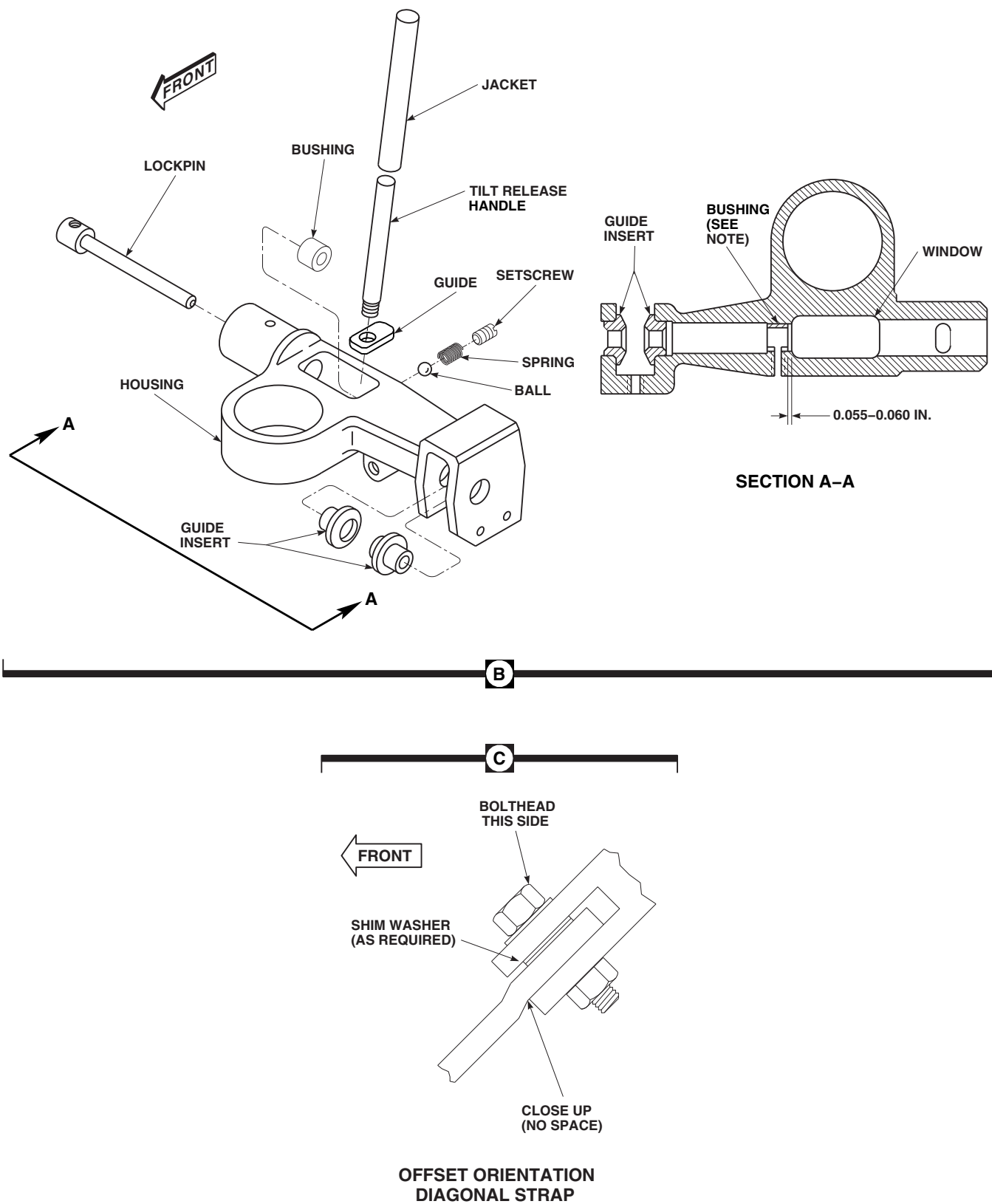


FIGURE 2-4-44-18. REPLACE MIDFRAME SUPPORT AND TUBE (SHEET 2 OF 2)

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**BOTTOMS (FULLY COMPRESSED),
THEN BACK OFF 1 ½ TO 2 TURNS.**

- e. Make sure lockpin moves freely, except for extra force needed to unseat ball when pulling lockpin from insert guides.

2-4-44.17.6 Install.

- a. Slide replacement tube on midframe support on seat. Install bolt, washer, and nut (Figure 2-4-44-18, Sheet 1, Detail A).
- b. Slide replacement support in tube, turn support and install diagonal strut in support. Engage tilt release lockpin in strut.
- c. Install bolts, washers, and nuts in tube and support. **TIGHTEN BOTH NUTS UNTIL THERE IS NO ENDPLAY, THEN BACK OFF UNTIL THERE IS 0.004 - 0.010-INCH ENDPLAY.**
- d. Install strut support in diagonal strut clevis, secure cable on clevis with clamp, screw, washer, and nut. **TORQUE NUT TO 12 - 15 INCH-POUNDS.**
- e. Secure upper end of replacement strap in clevis of midframe support with bolt, washer, shim washers (as required), and nut to eliminate play between strap and clevis (Figure 2-4-44-18, Sheet 2, Detail C). **TIGHTEN NUT UNTIL THERE IS NO ENDPLAY, THEN BACK OFF UNTIL THERE IS 0.004 - 0.010-INCH ENDPLAY.**
- f. Secure stowage panel support bracket to midframe support with bolts and washers (Figure 2-4-44-18, Sheet 1, Detail A). **TORQUE BOLTS TO 20 - 25 INCH-POUNDS.**
- g. Install up-and-down guide tube (PARA 2-4-44.16).
- h. Install stowage panel on midframe supports (PARA 2-4-44.25 or PARA 2-4-44.26).
- i. Check operation of tilt release and up-and-down adjust system.
- j. Make sure area is clean and free of foreign material.

2-4-44.18 Replace Trunnion and Tube.

2-4-44.18.1 Remove.

WARNING

To prevent injury to personnel, do not release either normal or emergency vertical adjust levers unless someone is sitting in seat. Extension springs are under load at all times. With seat at lowest position, vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, seat will be snapped to highest stop. Anyone leaning over seat, or with hands on guide tubes above linear bearings, will be seriously injured.

- a. Sit in seat and adjust to full up position.
- b. Remove upper tube and supports (PARA 2-4-44.15).
- c. Remove both midframe support bolts, washers, and nuts (Figure 2-4-44-19, Detail A).
- d. Remove bolts and nuts from diagonal straps, trunnions, and guide tubes.
- e. Twist and lift both guide tubes up far enough to clear trunnions.
- f. Remove bolts securing trunnions to track fittings. Remove trunnion spacer and shims. Record number of shims removed from each side. Remove tube and trunnions.
- g. Remove bolts, washers, and nuts, and remove trunnions from tube.

2-4-44.18.2 Install.

- a. Install replacement tube on replacement trunnions and secure with bolts, washers, and nuts. Install one washer under boltheads, and two washers under each nut. **TIGHTEN NUT UNTIL THERE IS NO ENDPLAY, THEN BACK OFF NUT UNTIL THERE IS 0.004 - 0.010-INCH ENDPLAY** (Figure 2-4-44-19, Detail A).

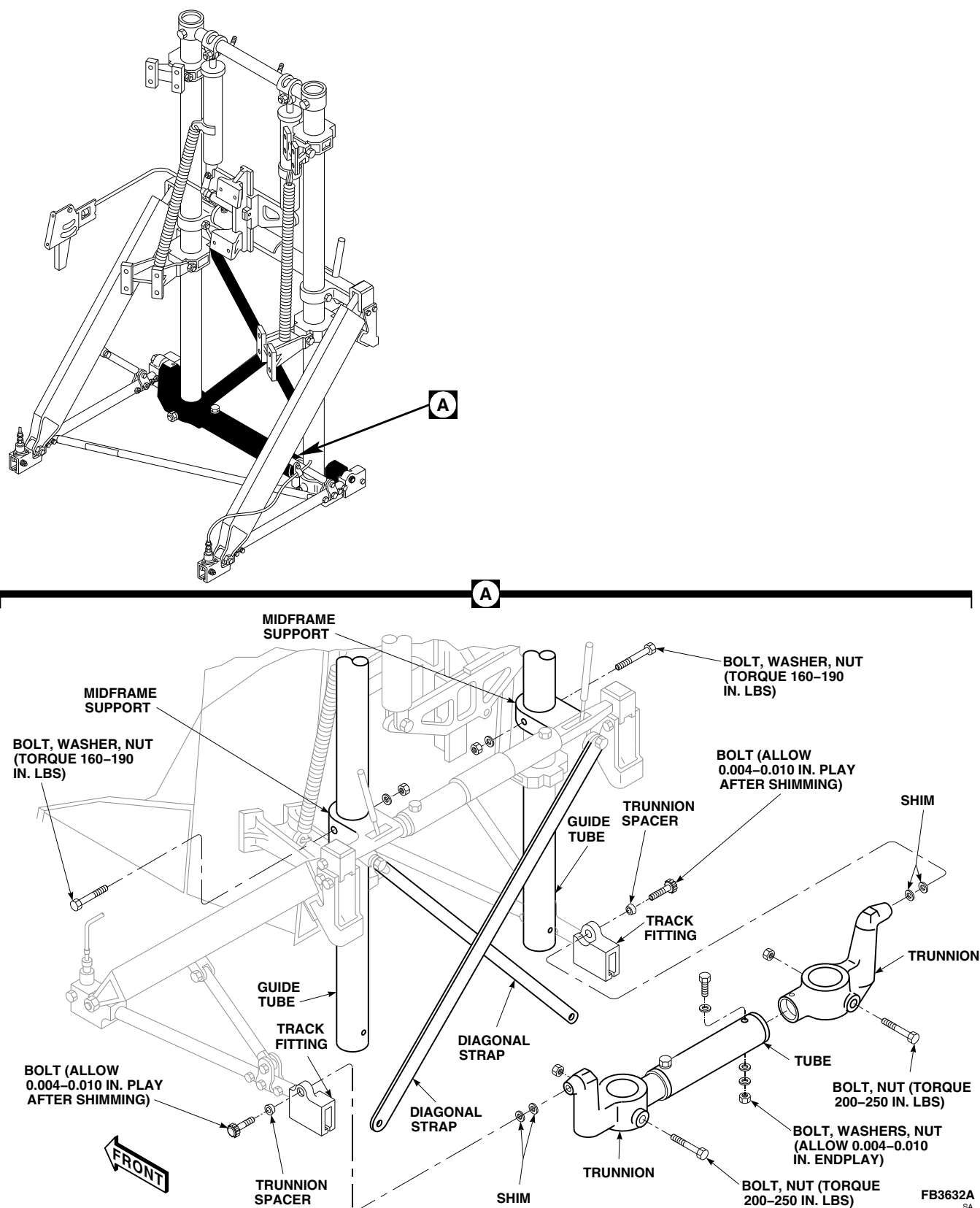


FIGURE 2-4-44-19. REPLACE TRUNNION AND TUBE

- b. Install replacement tube and trunnions between seat track fittings at spherical bearings and perform the following:

NOTE

If it is unknown how many shims are installed on each side, split amount of shims equally on both sides.

- (1) Install shims on each side as removed, between bearings and trunnions.
- (2) Using clean machinery towel, Item 211, Appendix D, dampened with acetone, Item 25, Appendix D, clean trunnion bolts.

NOTE

It should take approximately 100 - 150 inch-pounds of torque to eliminate end-play.

- (3) Install trunnion spacers and bolts. **TIGHTEN BOLTS UNTIL THERE IS NO ENDPLAY, THEN BACK OFF $\frac{1}{8}$ TO $\frac{1}{4}$ TURN UNTIL THERE IS 0.004 - 0.010-INCH ENDPLAY.**

- c. Tap guide tubes down into trunnions and line up attaching boltholes in trunnions, midframe supports, and guide tubes.
- d. Secure diagonal straps, trunnions, and guide tubes with bolts and nuts. **TORQUE NUTS TO 200 - 250 INCH-POUNDS.**
- e. Install bolts, washers, and nuts in midframe supports. **TORQUE NUTS TO 160 - 190 INCH-POUNDS.**
- f. Install upper tube and supports (PARA 2-4-44.15).
- g. Move seat to rear and check rear seat track fittings for binding. Perform the following:
 - (1) If binding is found, adjust shims between trunnions and spherical bearing in track fittings as follows:

NOTE

Shim one side at a time and try to have same number of shims on each side.

- (a) Remove trunnion bolt and spacer. Add or remove shims as necessary to free track fitting from binding on track.

NOTE

Do not remove shims.

- (b) Keep adjusting until seat rolls freely for entire length of track. Remove trunnion bolt and spacer.
- (c) Apply sealing compound, Item 271A, Appendix D, to threads of trunnion bolt.

NOTE

It should take approximately 100 - 150 inch-pounds of torque to eliminate end-play.

- (d) Install trunnion bolt and spacer. **TIGHTEN BOLT UNTIL THERE IS NO ENDPLAY, THEN BACK OFF $\frac{1}{8}$ TO $\frac{1}{4}$ TURN UNTIL THERE IS 0.004 - 0.010-INCH ENDPLAY.**

NOTE

Make sure torque stripe is applied horizontally.

- (e) Apply torque stripe (PARA 2-6-5).
- (2) If no binding is found, perform the following:

NOTE

Do not remove shims.

- (a) Remove trunnion bolt and spacer.
- (b) Apply sealing compound, Item 271A, Appendix D, to threads of trunnion bolt.

NOTE

It should take approximately 100 - 150 inch-pounds of torque to eliminate end-play.

- (c) Install trunnion bolt and spacer.
TIGHTEN BOLT UNTIL THERE IS NO ENDPLAY, THEN BACK OFF $\frac{1}{8}$ TO $\frac{1}{4}$ TURN UNTIL THERE IS 0.004 - 0.010-INCH ENDPLAY.

NOTE

Make sure torque stripe is applied horizontally.

- (d) Apply torque stripe (PARA 2-6-5).
- h. Make sure area is clean and free of foreign material.

2-4-44.19 Replace Bearings and Brackets.

2-4-44.19.1 Remove.

- a. To remove upper bearing and bracket, perform the following:

WARNING

To prevent injury to personnel, do not release either normal or emergency vertical adjust levers unless someone is sitting in seat. Extension springs are under load at all times. With seat at lowest position, vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, seat will be snapped to highest stop. Anyone leaning over seat, or with hands on guide tubes above linear bearings, will be seriously injured.

- (1) Sit in seat and adjust to full up position.
- (2) Remove upper tube and supports (PARA 2-4-44.15).

- (3) Remove seat back cushion (PARA 2-4-44.6).
- (4) Remove bolts and washers securing seat bucket to upper bracket. Slide bearing and bracket from guide tube (Figure 2-4-44-20).
- (5) Remove bolt, washers, nut, and bearing housing from upper bracket (Figure 2-4-44-20, Detail A).
- b. To remove lower bearing and bracket, perform the following:
 - (1) Remove upper tube and supports (PARA 2-4-44.15).
 - (2) Remove guide tube (PARA 2-4-44.16).
 - (3) Remove seat back cushion (PARA 2-4-44.6).
 - (4) Remove bolts and washers securing seat bucket to support bracket. Slide bearing and bracket down from guide tube (Figure 2-4-44-20).

NOTE

On right support, remove tiedown strap securing up-and-down adjust cable.

- (5) Remove bolt, washers, nut, and bearing housing from lower bracket (Figure 2-4-44-20, Detail B).

2-4-44.19.2 Install.

- a. To install upper bearing and bracket, perform the following:
 - (1) Install replacement bearing housing in replacement upper bracket, and secure with bolt, washers, and nut (Figure 2-4-44-20, Detail A). **TIGHTEN NUT UNTIL THERE IS NO ENDPLAY, THEN BACK OFF NUT UNTIL THERE IS 0.005 - 0.010-INCH END-PLAY.**
 - (2) Slide bearing and bracket over guide tube. Install bolts and washers through seat

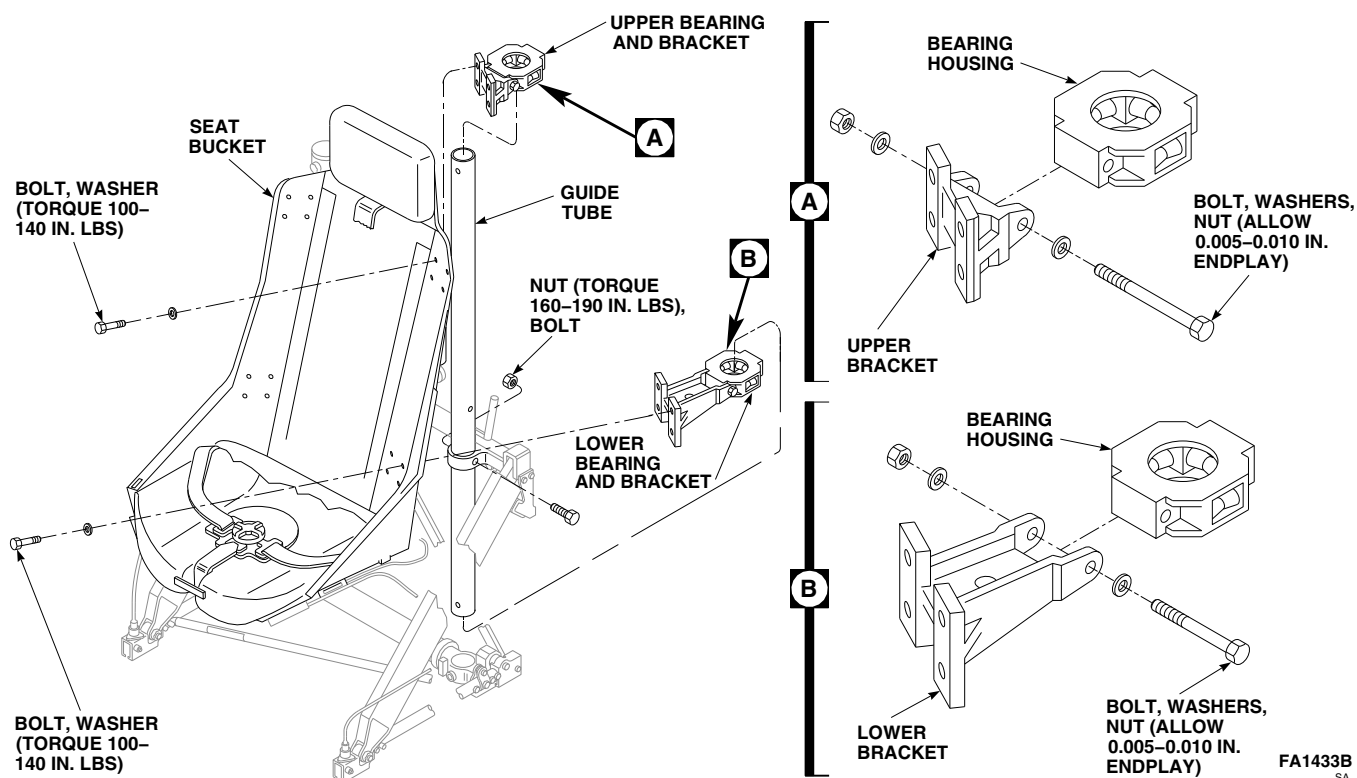


FIGURE 2-4-44-20. REPLACE BEARINGS AND BRACKETS

bucket and secure support to seat bucket. **TORQUE BOLTS TO 100 - 140 INCH-POUNDS** (Figure 2-4-44-20).

- (3) Install seat back cushion (PARA 2-4-44.6).
- (4) Install upper tube and supports (PARA 2-4-44.15).
- (5) Make sure area is clean and free of foreign material.

b. To install lower bearing and support bracket, perform the following:

- (1) Secure bearing housing in lower bracket with bolt, washers, and nut (Figure 2-4-44-20, Detail B). **TIGHTEN NUT UNTIL THERE IS NO ENDPLAY, THEN BACK OFF NUT UNTIL THERE IS 0.005 - 0.010-INCH ENDPLAY.**
- (2) Slide bearing and support bracket up on guide tube. Secure support on seat bucket

with bolts and washers through seat bucket. **TORQUE BOLTS TO 100 - 140 INCH-POUNDS** (Figure 2-4-44-20).

- (3) Install guide tube (PARA 2-4-44.16).
- (4) Install seat back cushion (PARA 2-4-44.6).
- (5) Install upper tube and support (PARA 2-4-44.15).
- (6) Make sure area is clean and free of foreign material.

2-4-44.20 Replace Seat Frame Tilt Handle Jacket.

2-4-44.20.1 Remove.

- a. Cut handle jacket and peel from tilt release handle (Figure 2-4-44-21, Detail A).
- b. Using machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D, and nonmetallic scraper, remove old adhesive from handle.

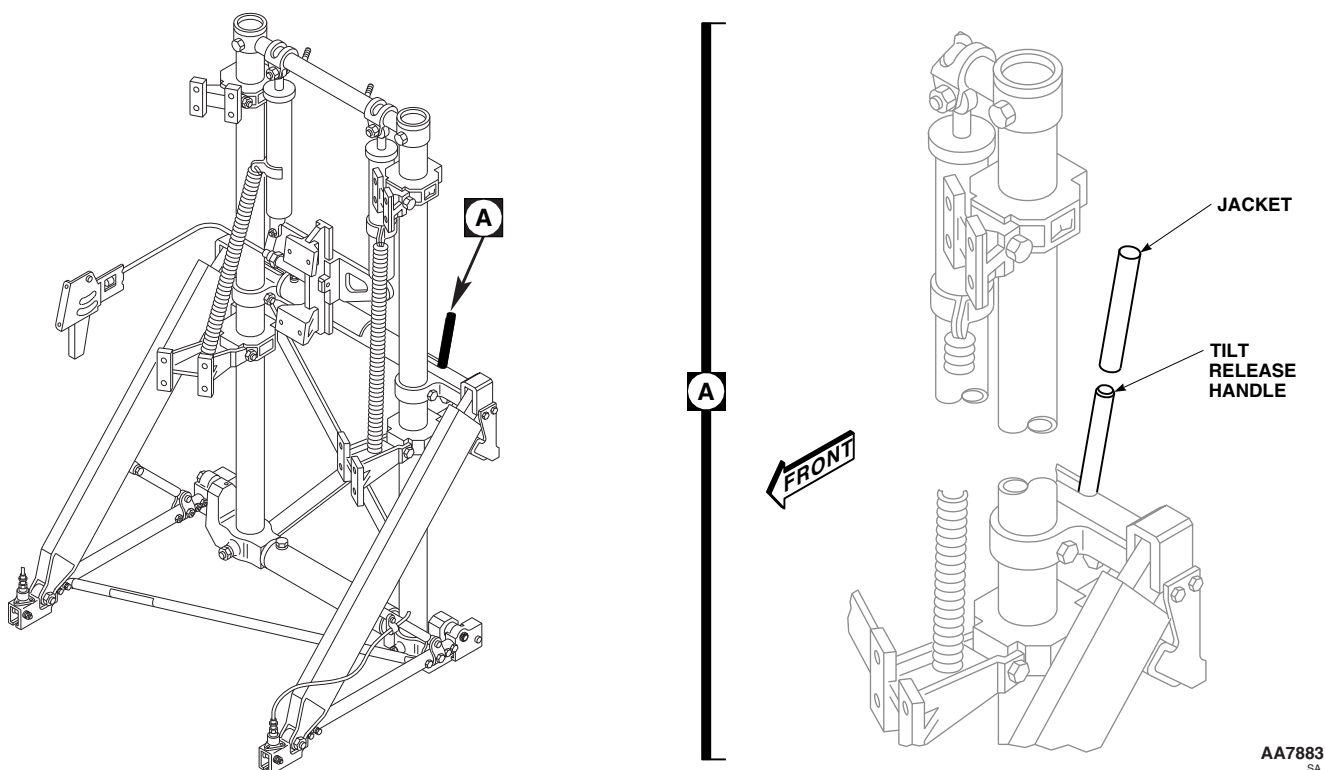


FIGURE 2-4-44-21. REPLACE SEAT FRAME TILT HANDLE JACKET

2-4-44.20.2 Install.

- Apply thin coat of adhesive, Item 46, Appendix D, to entire surface of handle. Allow to air-dry for 10 minutes.
- Slide replacement jacket over handle and press down firmly (Figure 2-4-44-21, Detail A).
- Using machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D, remove any extra adhesive and clean handle.

2-4-44.21 Replace Seat Frame Diagonal Strap.

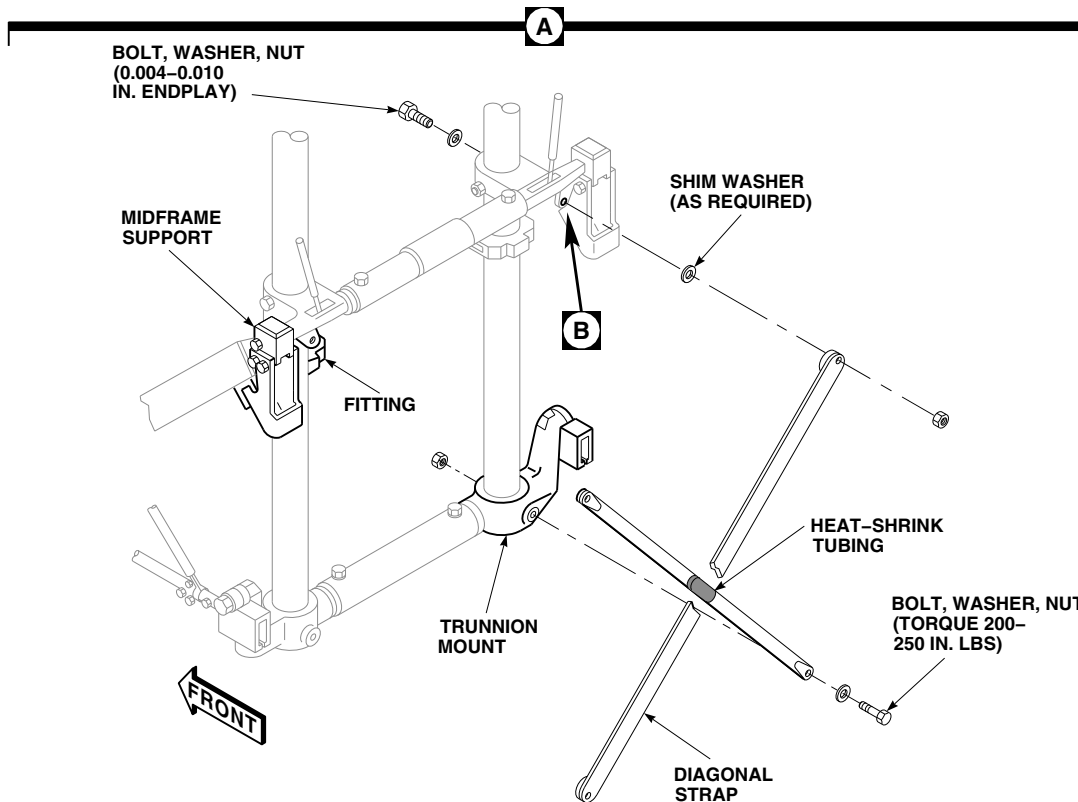
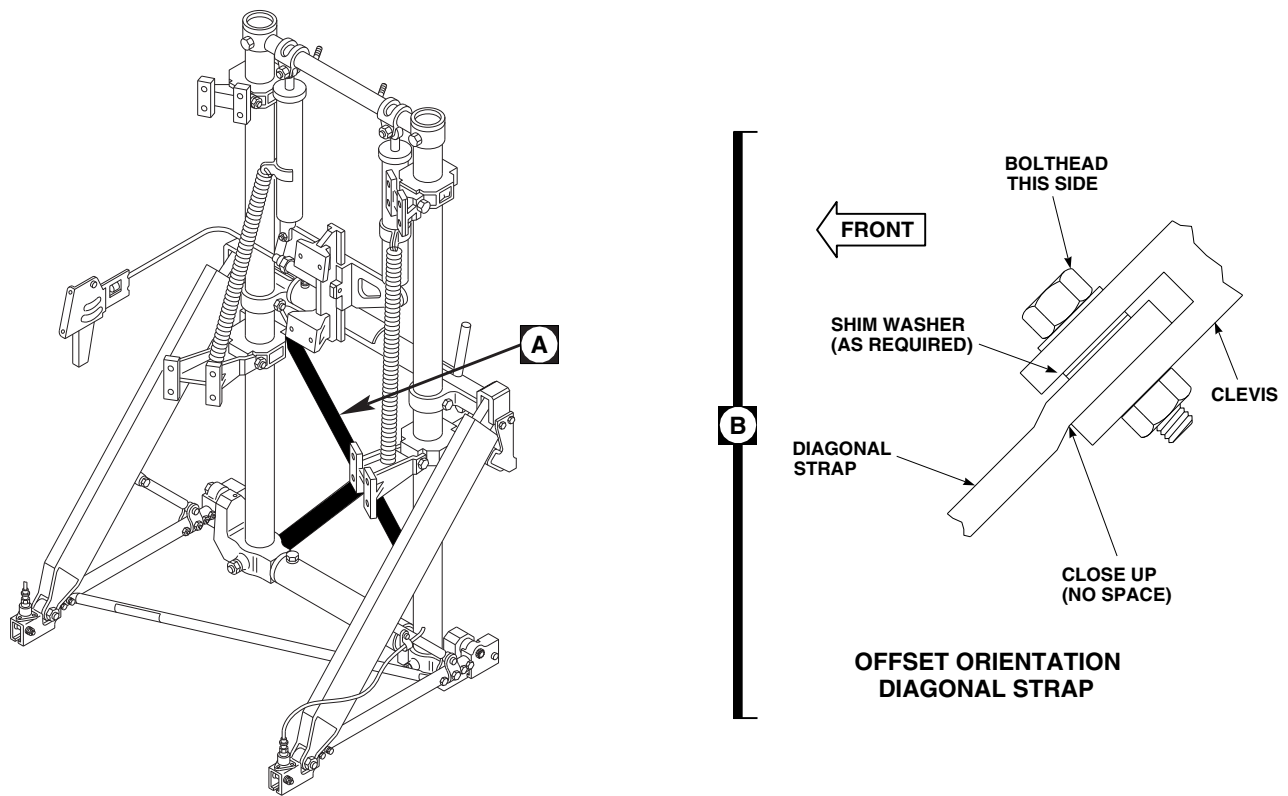
2-4-44.21.1 Remove.

WARNING

To prevent injury to personnel, do not release either normal or emergency

vertical adjust levers unless someone is sitting in seat. Extension springs are under load at all times. With seat at lowest position, vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, seat will be snapped to highest stop. Anyone leaning over seat, or with hands on guide tubes above linear bearings, will be seriously injured.

- Sit in seat and adjust to 3 - 4 stops below full up position.
- Remove stowage box and panel (PARA 2-4-44.25 or PARA 2-4-44.26)
- Remove clamps securing ICS cable to diagonal strap.
- Remove bolt, washer, shim washers (if installed), and nut securing upper end of diagonal strap to midframe support (Figure 2-4-44-22, Detail A).



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FIGURE 2-4-44-22. REPLACE SEAT FRAME DIAGONAL STRAP

NOTE

Remove diagonal straps one at a time from lower mounts to prevent guide tubes from slipping down into trunnions.

- e. Remove bolt, washer, and nut securing lower end of diagonal strap to trunnion mount. Remove strap. If new strap is not going to be installed now, install bolt to keep guide tube and trunnion mount in place.

2-4-44.21.2 Repair.

If necessary, replace wear strip on diagonal strap using suitable heat-shrink tubing.

2-4-44.21.3 Install.

WARNING

Injury to personnel and damage to equipment will result if diagonal straps are installed improperly. Diagonal straps offset portion must face towards seat bucket. Make sure diagonal straps are installed correctly.

NOTE

Replacement diagonal straps, 104460-2, will be manufactured as a one-piece design. When used as replacement for previous straps, 100018-001, with welded doubler attachment ends, they will require replacement as pairs (no intermixing of strap configurations). Straps, 104460-2, have an "offset" on upper attachment and must be installed on seat attachment point with protruding portion of "offset" positioned facing towards seat bucket.

- a. Secure bottom end of replacement strap on trunnion mount with bolt, washer, and nut. **TORQUE NUT TO 200 - 250 INCH-POUNDS.** Make sure rubber spacer faces other diagonal strap (Figure 2-4-44-22, Detail A).
- b. Secure upper end of replacement strap in clevis of midframe support with bolt, washer, shim

washers (as required), and nut to eliminate play between strap and clevis (Figure 2-4-44-22, Detail A and Detail B). **TIGHTEN NUT UNTIL THERE IS NO ENDPLAY, THEN BACK OFF 1/8 TO 1/4 TURN UNTIL THERE IS 0.004 - 0.010-INCH ENDPLAY** (Figure 2-4-44-22, Detail A).

- c. Using clamp, secure ICS cable to diagonal strap.
- d. Install stowage box and ICS control panel (PARA 2-4-44.25 or PARA 2-4-44.26).
- e. Make sure area is clean and free of foreign material.

2-4-44.22 Replace Seat Frame Diagonal Strut and Strut Support.

2-4-44.22.1 Remove.

- a. Remove strut support as follows:

WARNING

To prevent injury to personnel, do not release either normal or emergency vertical adjust levers unless someone is sitting in seat. Extension springs are under load at all times. With seat at lowest position, vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, seat will be snapped to highest stop. Anyone leaning over seat, or with hands on guide tubes above linear bearings, will be seriously injured.

- (1) Sit in seat and adjust to full up position.
- (2) Remove screw, washer, and nut securing release cable, clamp, and strut support to diagonal strut (Figure 2-4-44-23, Detail A).
- (3) Remove screw and nut securing strut support to track rail strap, and remove strut.
- b. Remove diagonal strut as follows:

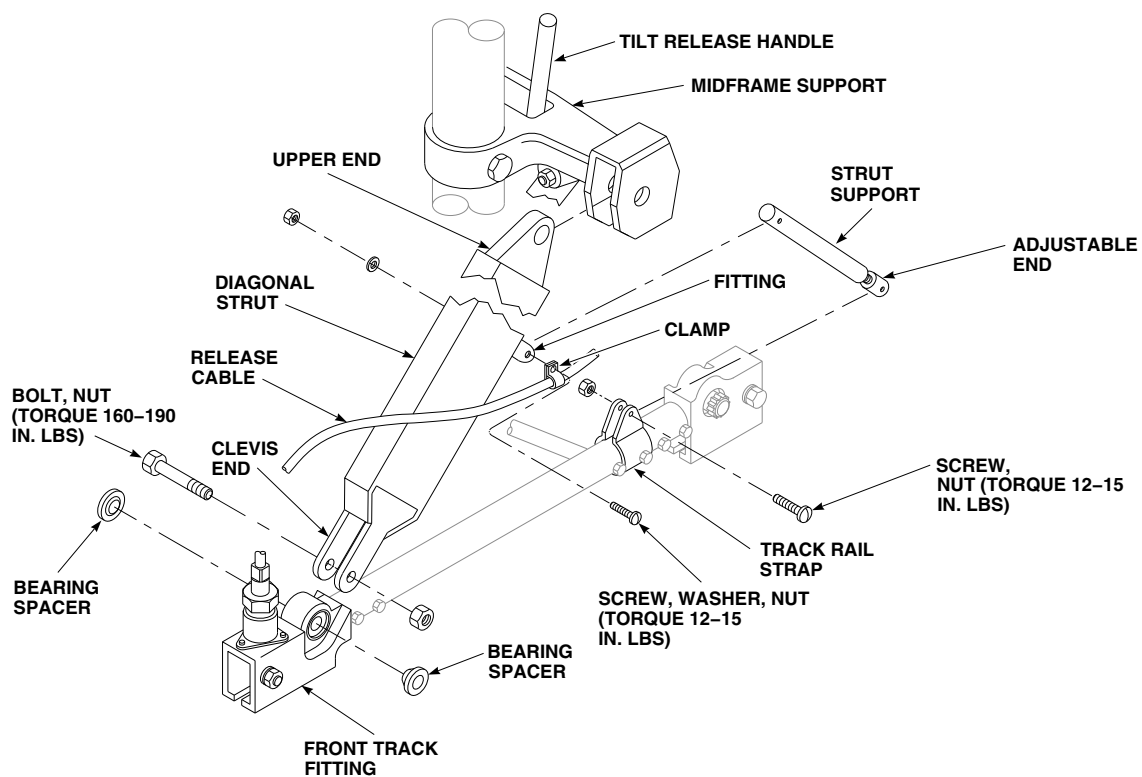
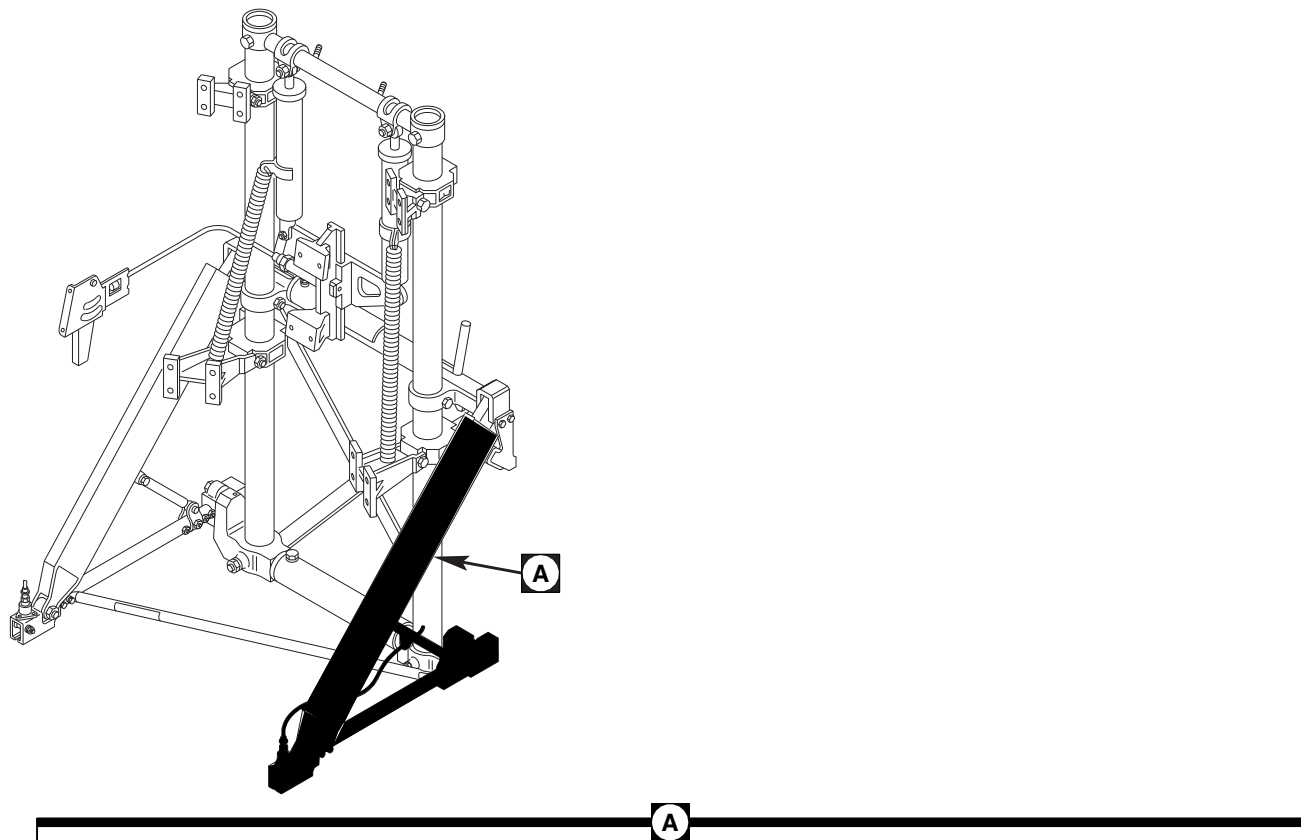
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FIGURE 2-4-44-23. REPLACE SEAT FRAME DIAGONAL STRUT AND STRUT SUPPORT

- (1) Remove screw, washer, and nut securing release cable, clamp, and strut support to diagonal strut.
- (2) Remove bolt and nut securing lower end of diagonal strut to front track fitting. Using Allen wrench in bolt shank, hold bolt while removing nut.
- (3) Pull tilt release handle and disengage upper end of diagonal strut from midframe support. Remove strut from track fitting. Remove bearing spacers.
- (4) Make sure area is clean and free of foreign material.

2-4-44.22.2 Install.

- a. Install diagonal strut as follows:
 - (1) Install bearing spacers in front track fitting. Slide clevis end of diagonal strut over spacers. Make sure support fitting faces down (Figure 2-4-44-23, Detail A).
 - (2) Pull tilt release handle and install upper end of diagonal strut in midframe support. Engage tilt release pin.
 - (3) Secure clevis end of diagonal strut with bolt and nut. Hold bolt shank with Allen wrench. **TORQUE NUT TO 160 - 190 INCH-POUNDS.**
 - (4) Install strut support in diagonal strut fitting. Secure clamp and release cable on outboard side of fitting with screw, washer, and nut. **TORQUE NUT TO 12 - 15 INCH-POUNDS.**
- b. Install strut support as follows:
 - (1) Install replacement support strut between track rail strap and diagonal strut fitting with adjustable end to track rail strap. Install clamp and release cable on outboard side of diagonal strut fitting. Secure clamp and support to diagonal strut fitting with screw, washer, and nut. **TORQUE NUT TO 12 - 15 INCH-POUNDS.**

- (2) If necessary, adjust end of support until attaching holes line up.
- (3) Secure strut support to track rail strap with screw and nut. **TORQUE NUT TO 12 - 15 INCH-POUNDS.**
- (4) Make sure area is clean and free of foreign material.

2-4-44.23 Replace Seat Frame Frangible Strut.

2-4-44.23.1 Remove.

WARNING

To prevent injury to personnel, do not release either normal or emergency vertical adjust levers unless someone is sitting in seat. Extension springs are under load at all times. With seat at lowest position, vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, seat will be snapped to highest stop. Anyone leaning over seat, or with hands on guide tubes above linear bearings, will be seriously injured.

- a. Sit in seat and adjust to full up position.
- b. Remove bolts, washers, and nuts from both ends of frangible strut, and remove strut (Figure 2-4-44-24, Detail A).

2-4-44.23.2 Install.

NOTE

All bolts are installed with heads toward left side of seat frame.

- a. Secure replacement frangible strut between track rails, with jogged end fitting on left track rail, with bolts, washers, and nuts (Figure 2-4-44-24, Detail A). **TORQUE RIGHT SIDE NUTS TO 20 - 25 INCH-POUNDS. TORQUE LEFT SIDE NUTS TO 30 - 35 INCH-POUNDS.**

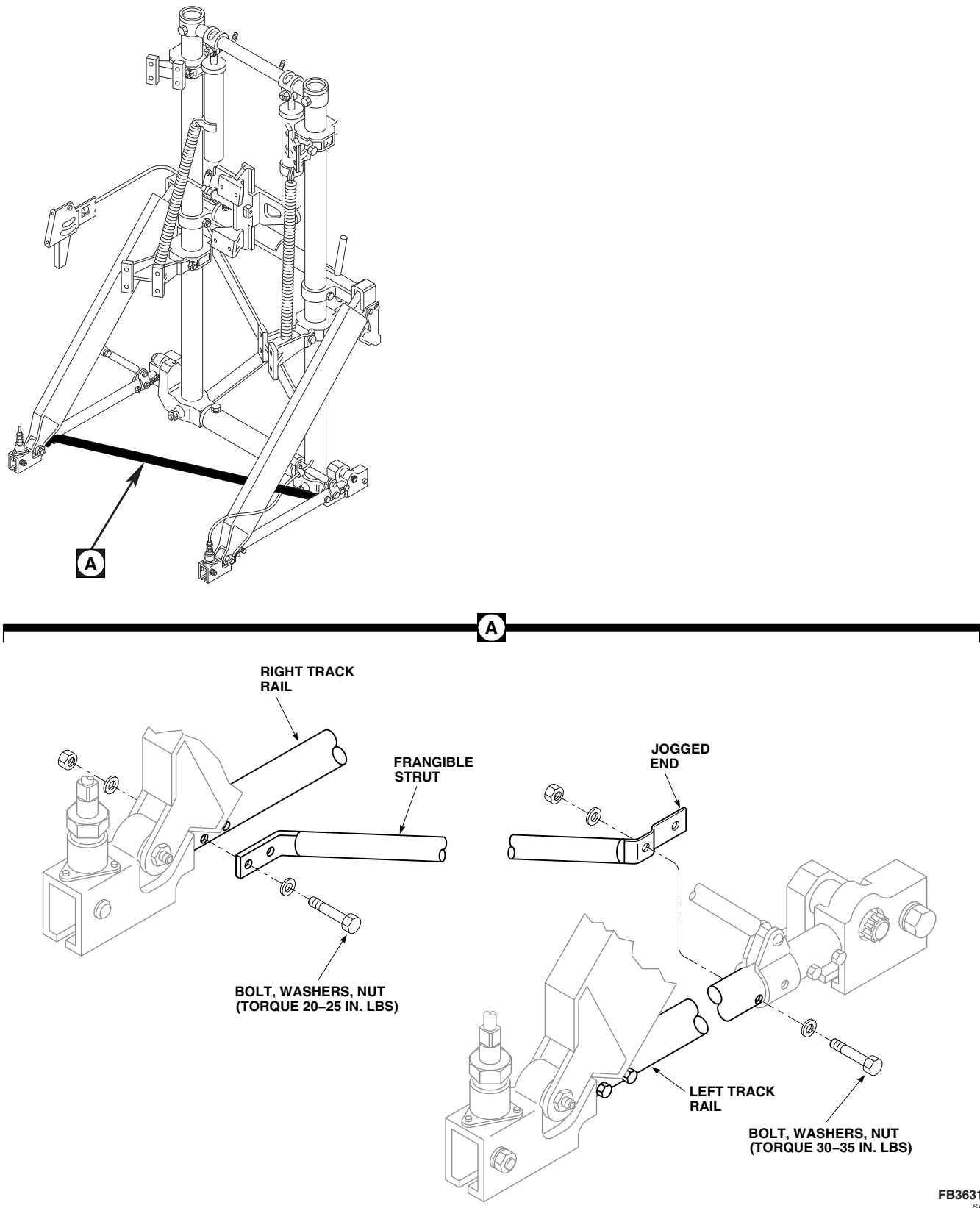


FIGURE 2-4-44-24. REPLACE SEAT FRAME FRANGIBLE STRUT

- b. Make sure area is clean and free of foreign material.

2-4-44.24 Replace Seat Frame Track Rollers and Bearings.

2-4-44.24.1 Remove.

WARNING

To prevent injury to personnel, do not release either normal or emergency vertical adjust levers unless someone is sitting in seat. Extension springs are under load at all times. With seat at lowest position, vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, seat will be snapped to highest stop. Anyone leaning over seat, or with hands on guide tubes above linear bearings, will be seriously injured.

- a. Sit in seat and adjust to full up position.
- b. Remove stowage box and panel (PARA 2-4-44.25 or PARA 2-4-44.26).
- c. Remove seat (PARA 2-4-44.28).

CAUTION

To prevent damage to tilt handles or seat frame, make sure to place support at midframe supports.

- d. Provide wood block with padding to place under midframe support. Carefully lay seat on its back on support blocks.
- e. Remove bolts and pin housing from front track fitting (Figure 2-4-44-25, Detail A).
- f. Remove bolt and nut securing strut support to track strap. Remove bolts, washers, and nuts securing frangible strut to track.
- g. Remove bolt and nut securing diagonal strut to front track fitting. Remove spacers.

NOTE

Record number of shims removed.

- h. Remove bolt securing rear track fitting to trunnion. Remove spacer and shims. Remove track.

2-4-44.24.2 Disassemble.

- a. Install Allen wrench in front bolt shank, hold bolt and remove nut and washer. Tap bolt from front track fitting. Remove roller with eccentric bushing from inside fitting. Push sliding bushing from nut side of fitting (Figure 2-4-44-25, Detail B).
- b. Repeat step a. for rear track fitting.
- c. Remove retainers, seals, and eccentric bushings from rollers.
- d. Remove retaining rings and press spherical bearings from fittings.

2-4-44.24.3 Inspect.

- a. Clean all parts using machinery towel, Item 211, Appendix D, moistened with dry-cleaning solvent, Item 124, Appendix D. Wipe dry with clean machinery towel, Item 211, Appendix D.
- b. Inspect track tube and track fittings for distortion, cracks, deep pitting corrosion, deep gouges, and hole elongations. **NONE ALLOWED.** If damaged, replace track assembly.
- c. Inspect rollers, eccentric bushings, sliding bushings, and bolts for distortion, cracks, deep pitting corrosion, deep gouges, and hole elongations. **NONE ALLOWED.** If damaged, replace parts as necessary.
- d. Inspect front and rear bolts for damaged threads, worn bushing, and bent or broken eccentric bushing drive pin. **NONE ALLOWED.** If damaged, replace bolts.
- e. Inspect threaded holes in front track fitting for thread damage.
- f. Inspect spherical bearings and needle bearings in rollers.

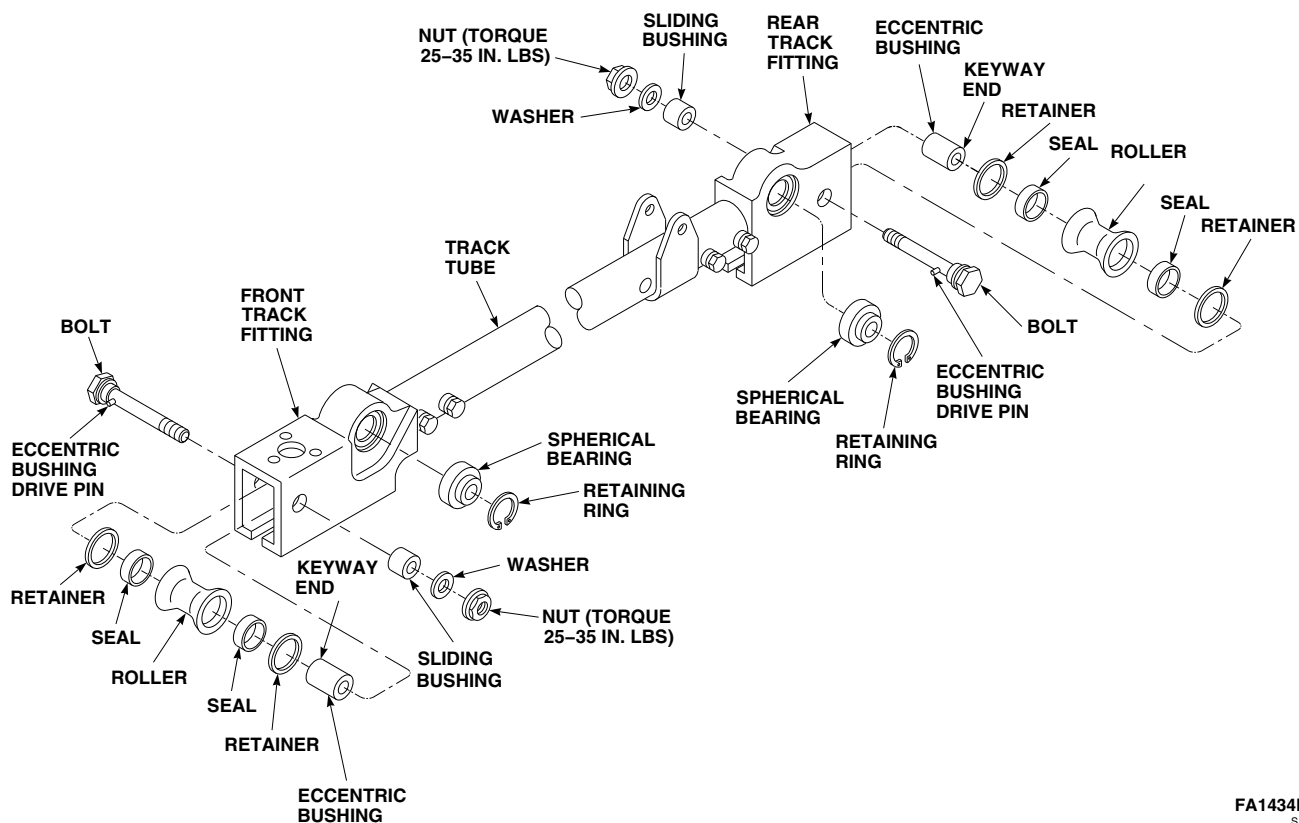
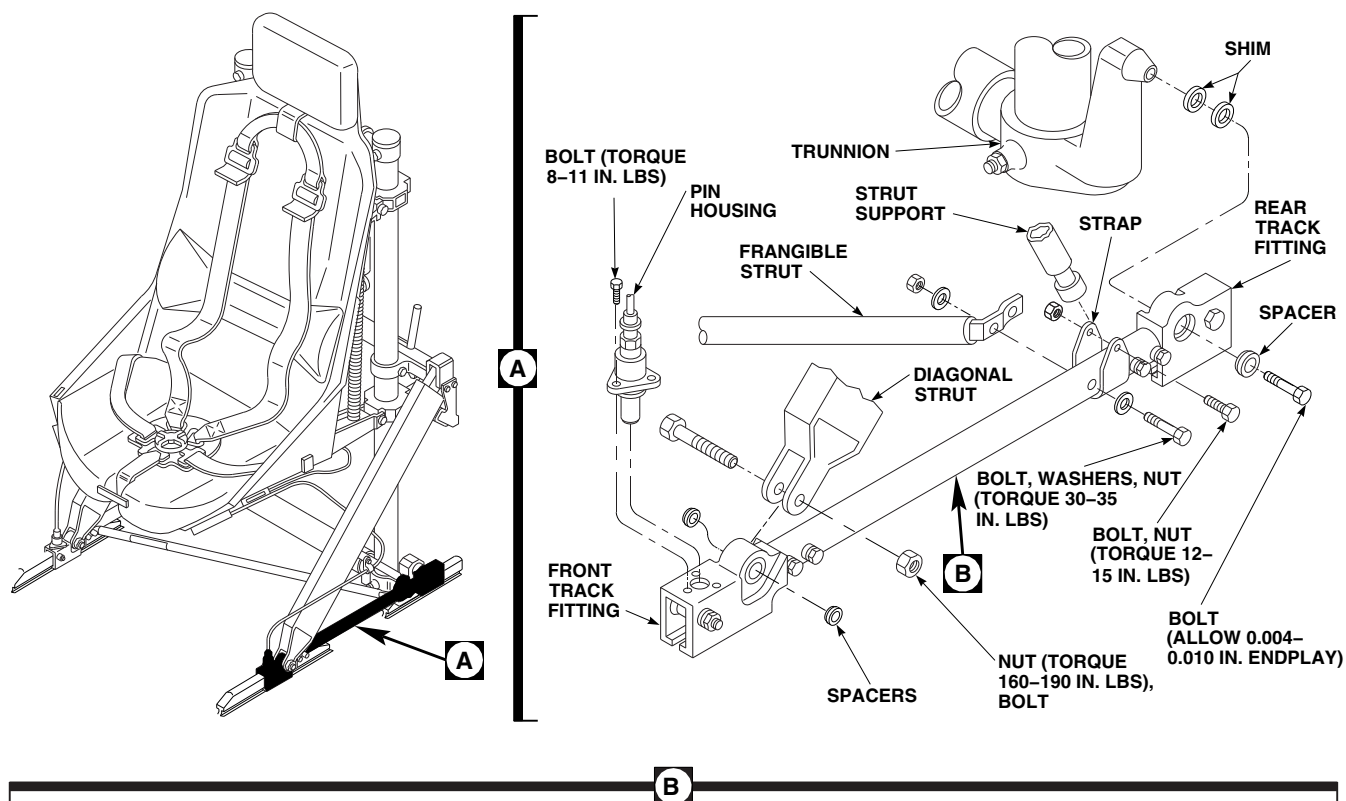
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FIGURE 2-4-44-25. REPLACE SEAT FRAME TRACK ROLLERS AND BEARINGS

2-4-44.24.4 Repair.

- a. Treat minor corrosion and touch up paint nicks, and scratches (PARA 2-6-5).
- b. Using No. 6-32 UNC-2B tap, retap three screw-
holes in front track, if threads are damaged. If
threads are stripped, replace track.

2-4-44.24.5 Assemble.

- a. Press replacement spherical bearings in track
fittings and install retaining rings (Figure 2-4-
44-25, Detail B).
- b. Install seals and retainers in roller. Install ec-
centric bushing in roller.
- c. Install roller with eccentric bushing in front
track fitting with keyway end of eccentric
bushing facing side opposite to spherical bear-
ing retaining ring. Face high lobe of eccentric
bushing toward front. Install bolt through fit-
ting and roller, and engage drive pin with
keyway in eccentric bushing.
- d. Install sliding bushing in front track fitting.
Install washer and nut on bolt. **TORQUE NUT
TO 25 - 35 INCH-POUNDS.**

NOTE

Rollers will be adjusted when seat is in-
stalled in helicopter.

- e. Install roller with eccentric bushing in rear
track fitting with keyway end of eccentric
bushing facing same side as spherical bearing
retaining ring. Face high lobe of eccentric
bushing toward rear. Install bolt through fitting
and roller, and engage drive pin with keyway
in eccentric bushing.
- f. Install sliding bushing in rear track fitting.
Install washer and nut on bolt. **TORQUE NUT
TO 25 - 35 INCH-POUNDS.**

2-4-44.24.6 Install.

- a. Clean track fittings using machinery towel,
Item 211, Appendix D, moistened with dry-
cleaning solvent, Item 124, Appendix D. Wipe

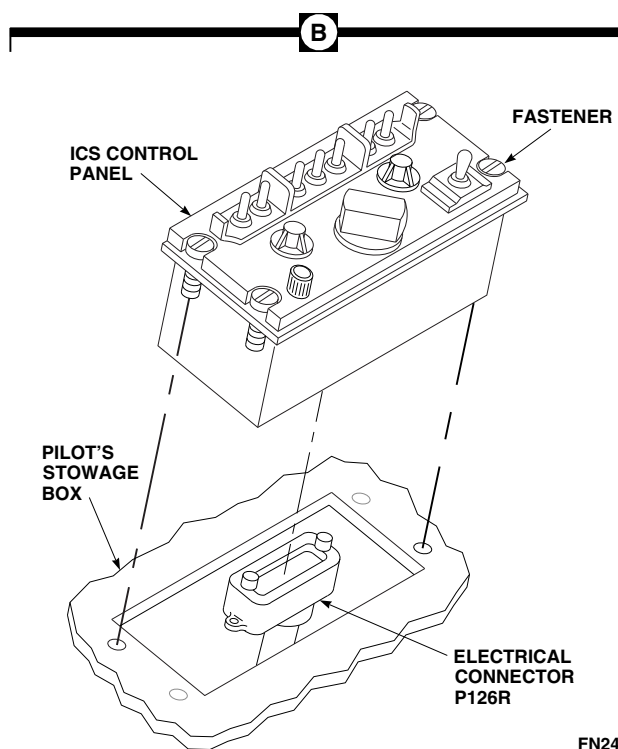
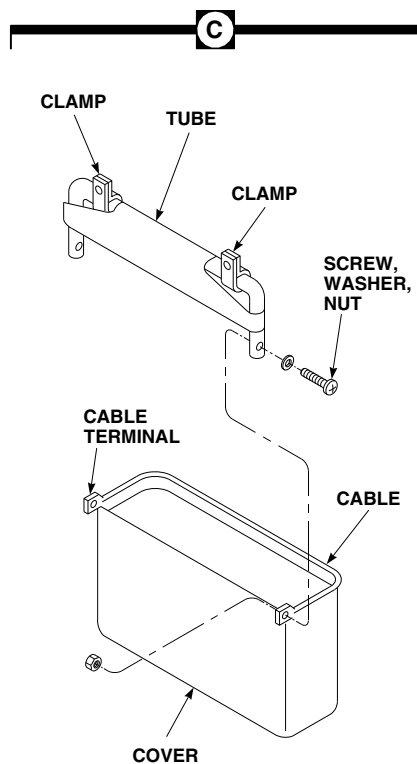
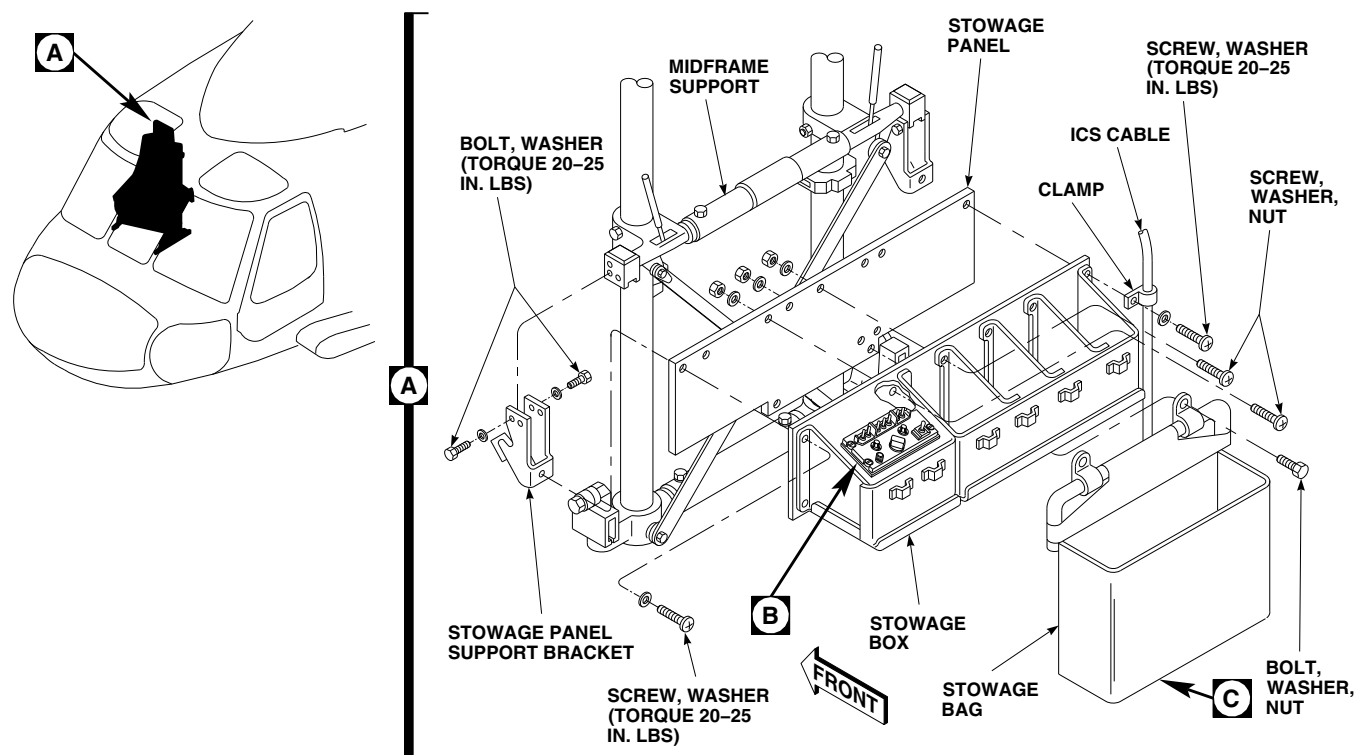
dry using clean machinery towel Item 211, Ap-
pendix D.

- b. Install rear track fitting bearing against trun-
nion. Place same number of shim washers
between trunnion and bearing as removed.
Install spacer against bearing and secure fitting
to trunnion with bolt. **TIGHTEN BOLT
UNTIL THERE IS NO ENDPLAY, THEN
BACK OFF UNTIL THERE IS 0.004 TO
0.010-INCH TOTAL PLAY BETWEEN
TRUNNION AND BOLT HEAD** (Figure 2-4-
44-25, Detail A).
- c. Install spacers on each side of spherical bearing
in front track fitting with smaller diameter
toward bearing. Secure fitting in diagonal strut
with bolt and nut. **TORQUE NUT TO 160 -
190 INCH-POUNDS.**
- d. Secure diagonal strut support in track bracket
with bolt and nut. **TORQUE NUT 12 - 15
INCH-POUNDS.**
- e. Secure end of frangible strut on track tube with
bolts, washers, and nuts. **TORQUE NUTS TO
30 - 35 INCH-POUNDS.**
- f. Secure pin housing in front track fitting with
bolts. **TORQUE BOLTS TO 8 - 11 INCH-
POUNDS.**
- g. Adjust seat to full upright position.
- h. Install stowage panel and box (PARA 2-4-
44.25 or PARA 2-4-44.26).
- i. Install seat in helicopter (PARA 2-4-44.28).
- j. Make sure area is clean and free of foreign
material.

2-4-44.25 Replace Pilot's Stowage Box, Panel, and Bag.

2-4-44.25.1 Remove.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Remove bolts, washers, and nuts securing
stowage bag to stowage panel (Figure 2-4-44-
26, Detail A).



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FIGURE 2-4-44-26. REPLACE PILOT'S STOWAGE BOX, PANEL, AND BAG

- c. Loosen fasteners and lift ICS control panel enough to access electrical connector. Disconnect electrical connector, P126R, from control panel and remove control panel. Pull connector through bottom of stowage box (Figure 2-4-44-26, Detail B).
- d. Install protective caps and plugs on electrical connectors.
- e. On right side of stowage panel, remove screw and washer securing stowage panel and ICS cable and clamps to stowage panel support bracket (Figure 2-4-44-26, Detail A).
- f. On left side of stowage panel, remove screw and washer securing stowage box and panel to stowage panel support bracket. Remove panel.
- g. Remove screws, washers, and nuts and remove stowage box from panel.
- h. Remove bolts and washers and remove stowage panel support brackets from midframe supports.

2-4-44.25.2 Repair.

- a. Turn off all electrical power (PARA 1-6-16).
- b. If necessary, repair stowage box and stowage panel (PARA 2-6-3).
- c. Remove screws, washers, and nuts securing tube and cover to cable terminals. Remove cable from cover. If clamps are damaged, remove from tube (Figure 2-4-44-26, Detail C).
- d. Install replacement cable in cover. Connect tube and cover to cable terminals with screws, washers, and nuts. Install replacement clamps on tube.
- e. Repair minor rips or tears in covers by stitching.

2-4-44.25.3 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Install stowage panel support brackets on midframe supports with bolts and washers.

TORQUE BOLTS TO 20 - 25 INCH-POUNDS (Figure 2-4-44-26, Detail A).

- c. Install stowage box on panel with screws, washers, and nuts.
- d. Install stowage panel on stowage panel support brackets with screws and washers. On right side, install ICS cable and clamps. **TORQUE SCREWS TO 20 - 25 INCH-POUNDS.**
- e. Remove protective cap and plug from electrical connectors.
- f. Inspect and treat electrical connector (PARA 1-7-20).
- g. Pull electrical connector up through stowage box and connect electrical connector, P126R, to ICS control panel. Install control panel in stowage box and secure with fasteners (Figure 2-4-44-26, Detail B).
- h. Install stowage bag on replacement stowage panel with bolts, washers, and nuts.
- i. Perform operational check of ICS (TM 11-70-23).

2-4-44.26 Replace Copilot's Stowage Box, Panel, and Bag.

2-4-44.26.1 Remove.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Remove first aid kit (PARA 17-4-1).
- c. Remove fire extinguisher from support and remove screws, washers, and nuts securing support to stowage panel (Figure 2-4-44-26, Detail A).
- d. **701029-SUBQ** Pull back rubber boots. Disconnect electrical wires from utility light terminal block. 
- e. Loosen fasteners and lift ICS control panel enough to gain access to electrical connector. Disconnect electrical connector, P127R, from control panel and remove control panel. Pull connector through bottom of stowage box (Figure 2-4-44-27, Detail B).

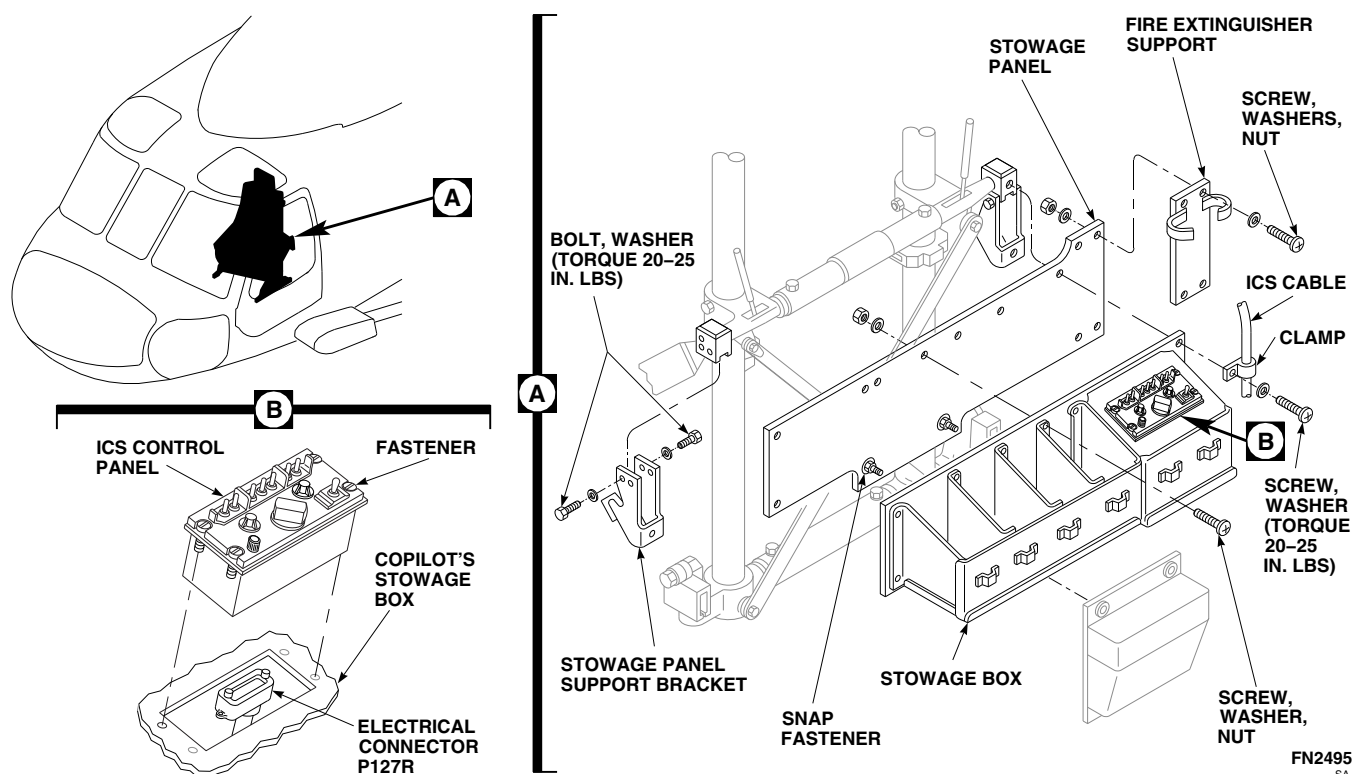


FIGURE 2-4-44-27. REPLACE COPILOT'S STOWAGE BOX, PANEL, AND BAG

- f. Install protective caps and plugs on electrical connectors.
- g. On right side of stowage panel, remove screws and washers securing stowage panel and ICS cable and clamps to midframe support (Figure 2-4-44-27, Detail A).
- h. On left side of panel, remove screws and washers securing stowage box and panel to midframe support. Remove box and panel.
- i. Remove screws, washers, and nuts and remove stowage box assembly from panel.
- j. Remove bolts and washers and remove stowage panel support brackets from midframe supports.

2-4-44.26.2 Repair.

- a. Turn off all electrical power (PARA 1-6-16).
- b. If necessary, repair stowage box and stowage panel (PARA 2-6-3). Replace snap fasteners in stowage panel.

2-4-44.26.3 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Install stowage panel support brackets on midframe supports with bolts and washers. **TORQUE BOLTS TO 20 - 25 INCH-POUNDS** (Figure 2-4-44-27, Detail A).
- c. Install stowage box assembly on stowage panel with screws, washers, and nuts.
- d. Install replacement stowage panel on midframe supports with screws and washers. On right side, install ICS cable and clamps. **TORQUE SCREWS TO 20 - 25 INCH-POUNDS.**
- e. **701029-SUBQ** Connect electrical wires to utility light terminal block. Cover terminal ends with rubber boots.
- f. Remove protective caps and plugs from electrical connectors.
- g. Inspect and treat electrical connectors (PARA 1-7-20).

- h. Pull electrical connector, P127R, up through stowage box and connect electrical connector to ICS control panel. Install control panel in stowage box and secure using fasteners (Figure 2-4-44-27, Detail B).
- i. Install fire extinguisher support on stowage panel and secure with screws, washers, and nuts. Replace fire extinguisher in support (Figure 2-4-44-27, Detail A).
- j. Install first aid kit (PARA 17-4-1).
- k. Perform operational check of ICS (TM 11-70-23).

2-4-44.27 Replace Armored Wing.

2-4-44.27.1 Remove.

- a. Remove pin and washer from upper cable end. Loosen jamnut, remove upper cable end from lever support (Figure 2-4-44-28, Sheet 1, Detail B).
- b. Remove jamnut from lower end of cable. Pull cable out of bracket. Remove cable from helicopter (Figure 2-4-44-28, Sheet 1, Detail C).
- c. Remove spring and washer from bracket.
- d. Remove spring pins from upper and lower slide tubes (Figure 2-4-44-28, Sheet 1, Detail A).
- e. Slide movable wing to front and remove from helicopter.
- f. Remove screws, washers, and nuts securing stationary wing to brackets. Remove stationary wing (Figure 2-4-44-28, Sheet 2, Detail D and Detail G).

2-4-44.27.2 Inspect.

- a. Check release cable lever and support for damage and elongated holes. Replace as necessary.
- b. Check tubes for dents. **DENTED TUBES SHALL NOT BIND IN BEARINGS.** If dented tubes bind in bearings, replace tubes.

- c. Check Teflon bearings for wear (Figure 2-4-44-28, Sheet 2, Detail F). **WORN TUBES SHALL NOT BIND IN BEARINGS.** If worn to point where tube binds in bearing, replace bearings (PARA 2-4-44.27.3, step h.).
- d. Check cable for binding. **CABLE MUST SLIDE FREELY IN SLEEVE.** If cable does not slide freely in sleeve, replace cable.
- e. Check brackets for damage. Repair or replace as necessary (PARA 2-6-2).

2-4-44.27.3 Repair.

- a. Remove bolt, washer, nut, and lever from movable wing (Figure 2-4-44-28, Sheet 1, Detail B).
- b. Remove screw, washers, nut, and rivets securing lever support to movable wing.
- c. Remove nut, washer, and upper tube from movable wing (Figure 2-4-44-28, Sheet 2, Detail D).
- d. Remove nuts, washers, and arm from movable wing (Figure 2-4-44-28, Sheet 2, Detail E).
- e. Remove nuts, washers, and lower tube from movable wing (Figure 2-4-44-28, Sheet 2, Detail D).
- f. Remove nuts, washers, and rivets, and remove bracket and bearing housings from stationary wing.
- g. Remove rivets and bearing housings from brackets (Figure 2-4-44-28, Sheet 2, Detail F).
- h. Replace Teflon bearings in bearing housings as follows:
 - (1) Cut damaged bearing from housing.
 - (2) Using cleaning compound, Item 96, Appendix D, remove old adhesive.
 - (3) Using machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D, clean housing. Wipe dry using clean machinery towel, Item 211, Appendix D.

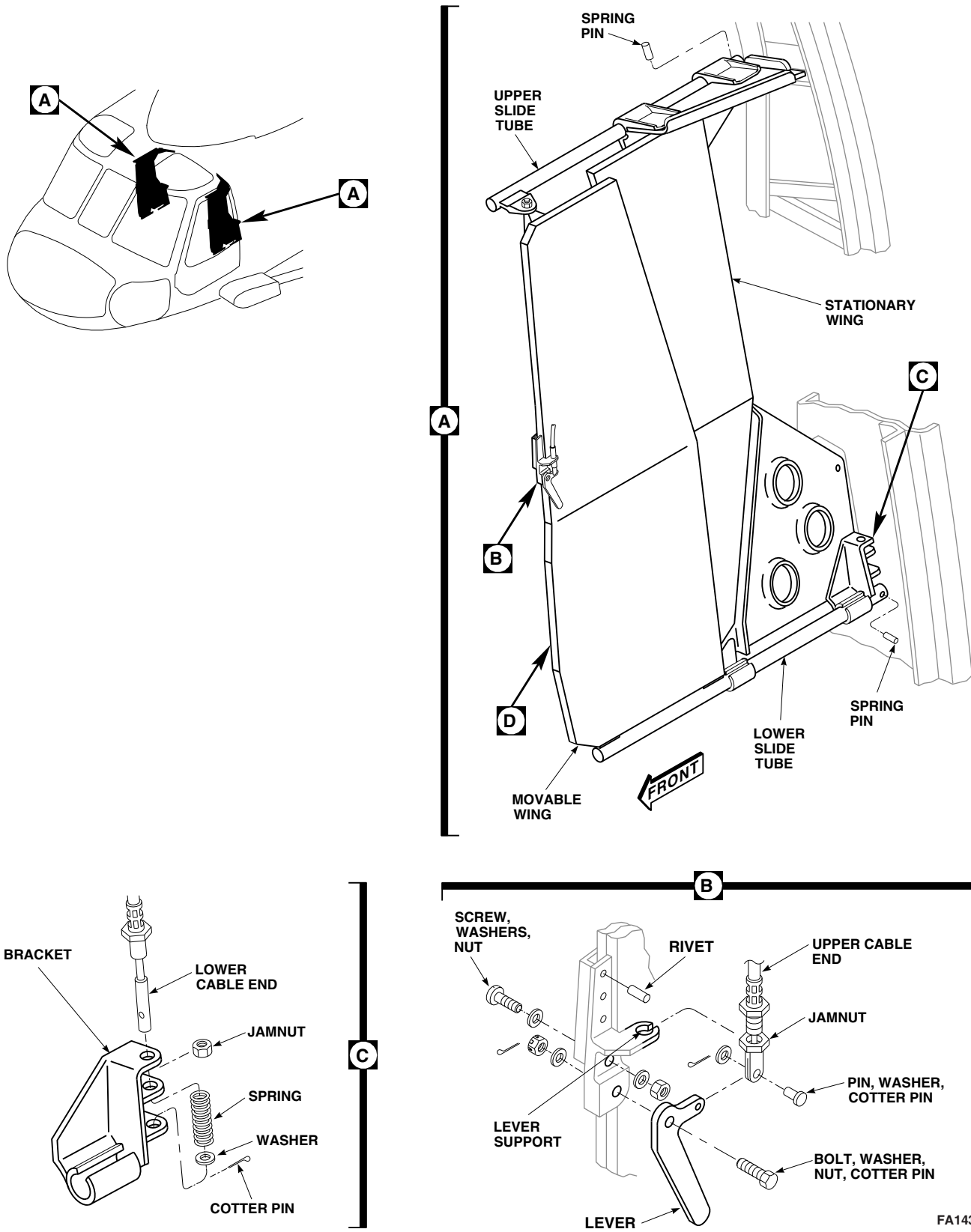


FIGURE 2-4-44-28. REPLACE ARMORED WING (SHEET 1 OF 2)

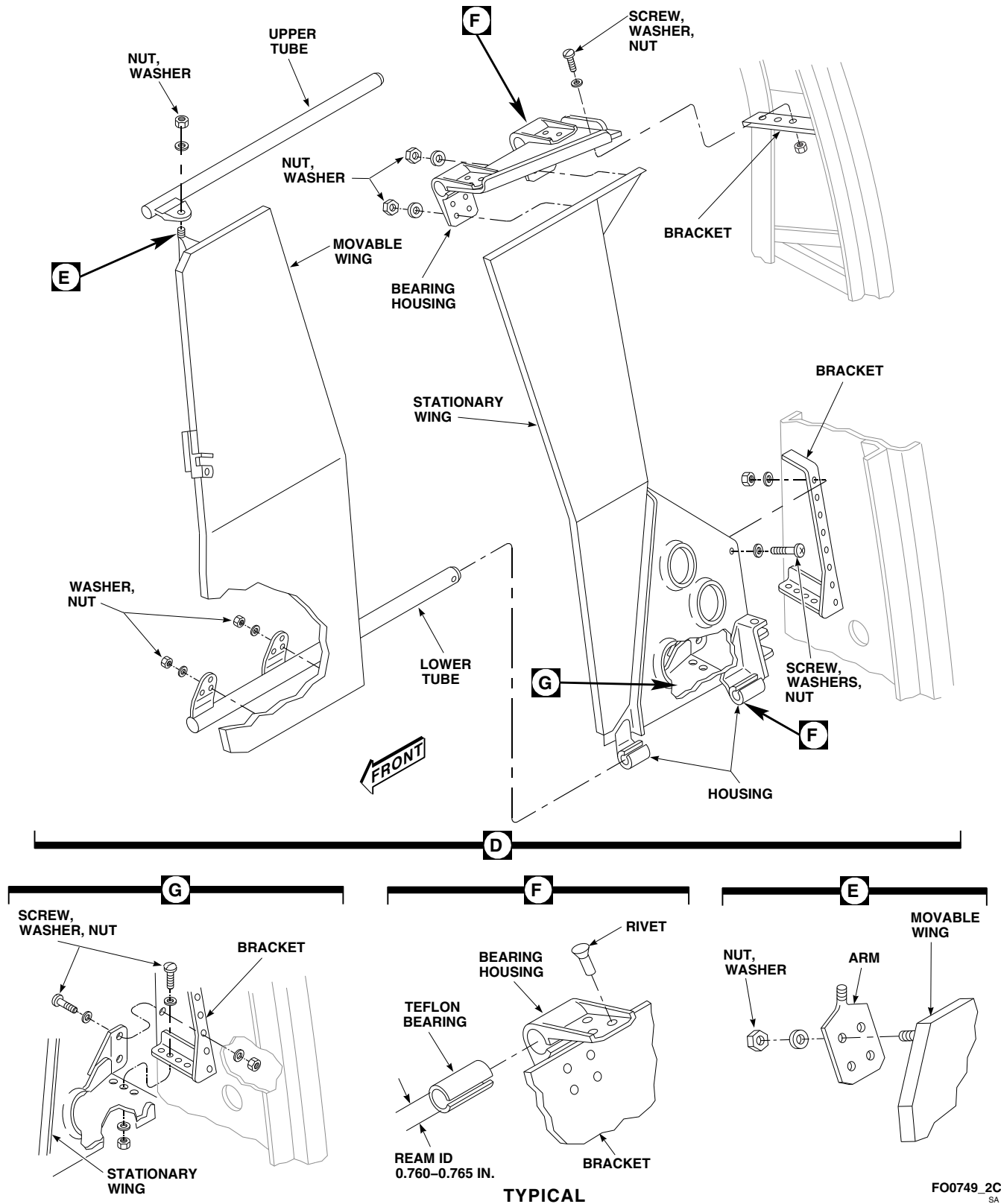


FIGURE 2-4-44-28. REPLACE ARMORED WING (SHEET 2 OF 2)

- (4) Before bonding, etch outside of bearing with etching solution, Item 136, Appendix D, following manufacturer's instructions.
- (5) Apply coat of adhesive, Item 38, Appendix D, to bearing housing.
- (6) Press Teflon bearing into housing and let cure for 12 hours at 70° - 85°F (21° - 29°C).
- (7) If adhesive, Item 38, Appendix D, is not available, apply coat of adhesive, Item 40, Appendix D, to bearing housing.
- (8) Press Teflon bearing into housing and using heat gun, apply heat 235° - 265°F (113° - 129°C) to bearing and housing and allow adhesive to cure for 60 minutes.
- (9) After completing bonding, ream Teflon bearing inside diameter to 0.760 - 0.765-inch.
- (10) Cut slot in Teflon bearing to conform to slotted bearing housings.
 - i. Install lower bearing housings on brackets with solid rivets.
 - j. Install upper bearing housings and bracket on stationary wing with solid rivets and nuts and washers.
 - k. Install lower tube on movable wing with nuts and washers (Figure 2-4-44-28, Sheet 2, Detail D).
 - l. Install arm with nuts and washers on movable wing (Figure 2-4-44-28, Sheet 2, Detail E).
 - m. Install upper tube on movable wing with nut and washer (Figure 2-4-44-28, Sheet 2, Detail D).
 - n. Install lever support on movable wing with screw, washer, nut, and solid rivets (Figure 2-4-44-28, Sheet 1, Detail B).
 - o. Install lever on movable wing with bolt, washers, nut, and cotter pin.

2-4-44.27.4 Repair Wing Panels.

NOTE

It is not necessary to remove stationary panel for repair. Movable panels should be removed to facilitate workspace for stationary panel.

- a. Remove movable panel (PARA 2-4-44.27.1).
- b. Using putty knife, peel back canvas cover from panel.
- c. For center bond separations, perform the following:
 - (1) Using abrasive paper, Item 20, Appendix D, and machinery towel Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean area.
 - (2) Gently spread open disbonded area and inject with adhesive, Item 38, Appendix D.
 - (3) Clamp area closed with clamps or suitable bracing. Allow 24 hours for adhesive to cure.
- d. For edge separations, inject with adhesive, Item 42, Appendix D. Allow 24 hours for adhesive to cure.
- e. Using adhesive, Item 37, Appendix D, secure canvas cover. Install panel (PARA 2-4-44.27.5).
- f. Touch up paint as necessary (PARA 2-6-5).

2-4-44.27.5 Install.

- a. Install stationary wing to fuselage with screws, washers, and nuts (Figure 2-4-44-28, Sheet 2, Detail D and Detail G).
- b. Slide upper and lower tubes into Teflon bearings on stationary wing and install movable wing (Figure 2-4-44-28, Sheet 1, Detail A).
- c. Press spring pins in upper and lower tubes.

- d. Install upper end of cable in lever with pin, washer, and cotter pin (Figure 2-4-44-28, Sheet 1, Detail B).
- e. Slide cable end in lever support. **TIGHTEN JAMNUT.** Using lockwire, Item 197, Appendix D, lockwire jamnuts.
- f. Install spring with washer on bottom in bracket on stationary wing (Figure 2-4-44-28, Sheet 1, Detail C).
- g. Slide lower end of cable through bracket, jamnut, spring, and washer. Compress spring and install cotter pin in end of cable. **TIGHTEN JAMNUT.**
- h. Check armored wing for smooth operation.

2-4-44.28 Replace Pilot's/Copilot's Seat.

2-4-44.28.1 Remove.

- a. Turn off all electrical power (PARA 1-6-16).

WARNING

To prevent injury to personnel, do not release either normal or emergency vertical adjust levers unless someone is sitting in seat. Extension springs are under load at all times. With seat at lowest position, vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, seat will be snapped to highest stop. Anyone leaning over seat, or with hands on guide tubes above linear bearings, will be seriously injured.

- b. Sit in seat. Under front right of seat, release VERT adjust lever and adjust seat to full up position. Under front left of seat, pull LONG adjust lever and slide seat fully rear.
- c. Remove passenger's seats next to cabin door on side through which seat will be removed (PARA 2-4-45).
- d. Remove stowage box and panel from seat (PARA 2-4-44.25 or PARA 2-4-44.26).

- e. Remove first aid kit (PARA 17-4-1).
- f. On copilot's seat, remove fire extinguisher and mounting bracket. On both seats, remove stowage box and panel.
- g. Remove screws, washers, and nuts, and remove clamps and spacer securing ICS cable to seat top support tube. Remove tiedown straps securing ICS cable to right attenuator (Figure 2-4-44-29, Sheet 1, Detail A, Detail B, Detail C, and Detail D).

CAUTION

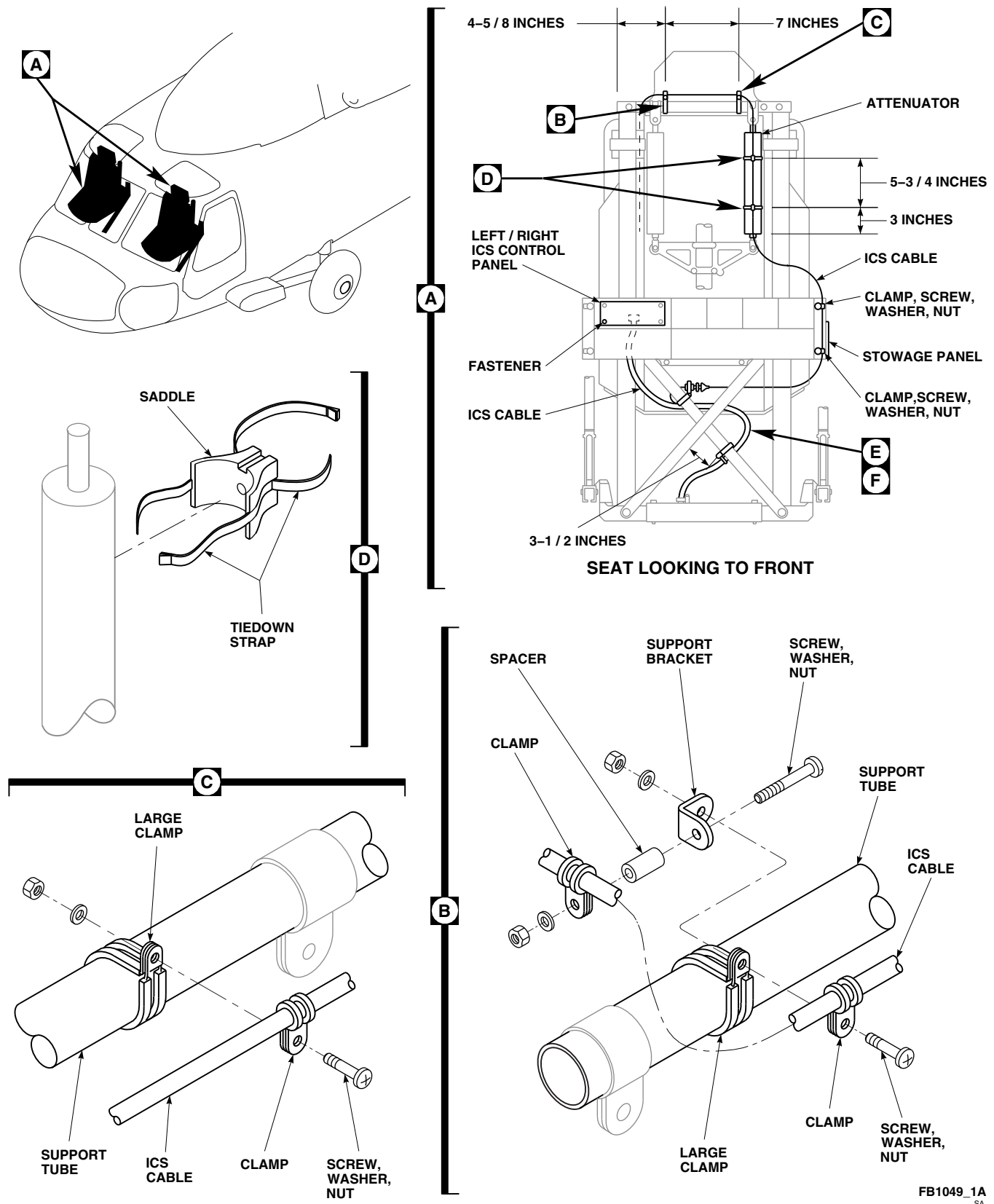
To prevent damage to ICS cable, make sure to move it out of way when removing seat.

- h. Remove screws, washers, and nuts, and remove clamps and ICS cable from cross brace (Figure 2-4-44-29, Sheet 2, Detail E and Detail F).
- i. Move ICS cable out of way so it will not be damaged when removing and installing seat. Keep clamps, screws, washers, and nuts with ICS cable.
- j. Remove screws securing raised floor panel at front of seat tracks, and remove panel.
- k. Remove bolts and stops from front of front seat tracks (Figure 2-4-44-30).
- l. If necessary, locally make stabilizer strut (Figure H-182, Appendix H).

CAUTION

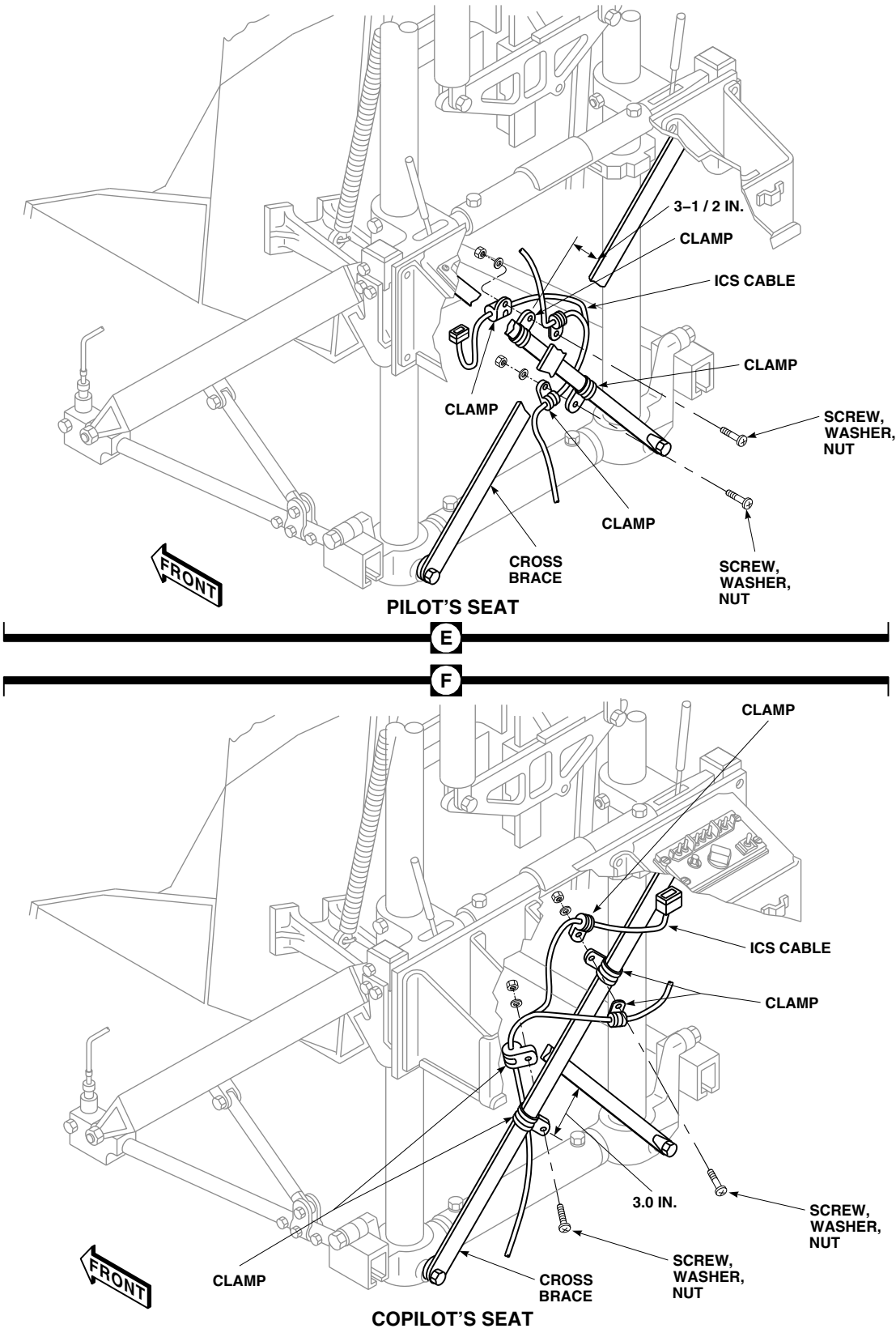
To prevent damage to seat, do not remove seat from tracks unless red stabilizer strut is installed.

- m. Secure red stabilizer strut on front seat rail fittings with screws (Figure 2-4-44-30, Detail A).



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FIGURE 2-4-44-29. REPLACE PILOT'S/COPILOT'S SEAT (SHEET 1 OF 2)



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FIGURE 2-4-44-29. REPLACE PILOT'S/COPILOT'S SEAT (SHEET 2 OF 2)

CAUTION

- Seat components can be bent or otherwise damaged by careless handling. Do not lift seat by frangible strut or track fittings.
 - Avoid damage to diagonal strut when tilting seat backward. Do not bump, push on, or apply twisting loads to strut.
 - Abrasive damage can be caused by carelessness in moving seat. Carry seat to new location. Do not drag it across ground or floor on track fittings, lower cross frame fittings, seat sides, or seat back.
- n. Under front left side of seat bucket, pull LONG adjust lever and roll seat forward until track fittings clear tracks. Lift seat by guide tubes, middle or lower cross members in back of seat, and forward edge of bucket. If necessary for clearance, tilt seat back into cabin by using TILT RELEASE levers. With aid of assistant, lift seat through cabin and out cabin door.
- o. Place seat on pallet for transporting.

2-4-44.28.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).

WARNING

To prevent injury to personnel, do not release either normal or emergency vertical adjust levers unless someone is sitting in seat. Extension springs are under load at all times. With seat at lowest position, vertical preload on seat could be as high as 150 pounds. If no one is in seat and vertical adjust lever(s) is released, seat will be snapped to highest stop. Anyone leaning over seat, or with hands on guide tubes above linear bearings, will be seriously injured.

- b. Sit in replacement seat. Under front right of seat, release VERT adjust lever and adjust seat to full up position.

CAUTION

Track fittings can be damaged through improper engagement with seat tracks if width between fittings is incorrect. Do not install seat unless red stabilizer strut is in place.

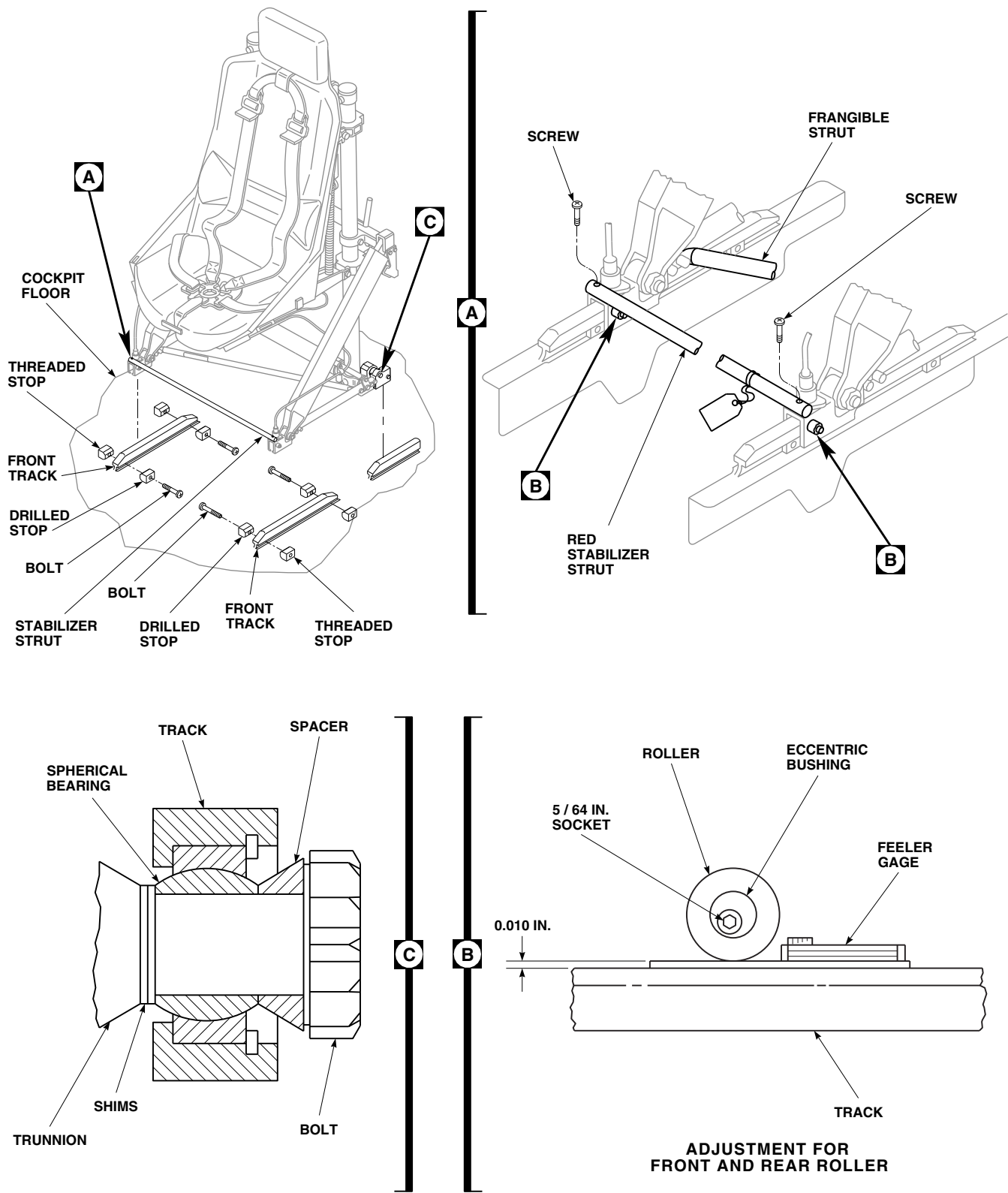
- c. If necessary, secure red stabilizer strut on front seat track fittings with screws (Figure 2-4-44-30, Detail A).

WARNING

Pilot's and copilot's seat restraint buckles must have same type of release mechanism. Different types of release mechanisms could hinder an emergency egression. If installing replacement seat, make sure both pilot's and copilot's seat restraint buckles have same type of release mechanism.

CAUTION

- Seat components can be bent or otherwise damaged by careless handling. Do not lift seat by frangible strut or track fittings.
 - Avoid damage to diagonal strut when tilting seat backward. Do not bump, push on, or apply twisting loads to strut.
 - Abrasive damage can be caused by carelessness in moving seat. Carry seat to new location. Do not drag it across ground or floor on track fittings, lower cross frame fittings, seat sides, or seat back.
- d. Lift replacement seat by guide tubes, middle or lower cross members in back of seat, and front



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FIGURE 2-4-44-30. ADJUST PILOT'S/COPILOT'S SEAT

edge of bucket. Install seat through cabin door, place in cockpit with track fittings in front of and in line with tracks on cockpit floor.

- e. Pull LONG adjust lever and roll seat onto tracks against rear stops.
- f. If seat rear track fittings do not line up with floor tracks, or if they bind when seat is moved to rear, adjust shims between trunnions and spherical bearings in track fitting as follows (Figure 2-4-44-30, Detail C):

NOTE

Adjust track fittings one side at a time and try to have same number of shims on each side. Keep adjusting until seat rolls freely for entire length of track.

- (1) Remove bolt and spacer and add or remove shims as necessary to free track fitting from binding on track.
- (2) Install spacer and bolt. **TIGHTEN BOLT UNTIL THERE IS NO ENDPLAY, THEN BACK OFF UNTIL THERE IS 0.004 TO 0.010-INCH PLAY BETWEEN PARTS.**

NOTE

If replacing seat stop, maximum of 0.062-inch may be removed from stop length, if required.

701323-SUBQ Rear stops must be properly oriented. On seats, 100900-1 and 100900-5, face red end of stop to front. On seats, RA 30525-1, face red end of stop to rear. ■

- g. Secure seat stops (chamfered side against track) on front track with bolts (Figure 2-4-44-30).
- h. Secure raised floor panel in front of seat with screws.
- i. Pull LONG adjust lever and move seat forward against stops.

- j. Route ICS cables and clamp to cross braces (Figure 2-4-44-29, Sheet 1, Detail A, and Sheet 2, Detail E and Detail F). Allow at least 1 1/2-inch slack in ICS cable between first clamp on cross brace and where cable comes out of floor at STA 247.0.

WARNING

Injury to personnel and damage to equipment will occur if improper tiedown straps are used to secure ICS cable to attenuator. Attenuator movement will be restricted if incorrect tiedown straps are used. Use only tiedown straps, Item 329, Appendix D, to secure ICS cable to attenuator.

- k. Route ICS cable up along right attenuator, continue across seat top support tube, to left side of seat back cushion. Secure ICS cable to attenuator with two tiedown straps, Item 329, Appendix D (Figure 2-4-44-29, Sheet 1, Detail A).
- l. Secure two saddle assemblies to right attenuator using tiedown straps, Item 329, Appendix D (Figure 2-4-44-29, Sheet 1, Detail D).
- m. Secure ICS cable to saddle assemblies using tiedown straps, Item 329, Appendix D.
- n. Secure ICS cable to right side of seat top support with clamps, screw, washer, and nut (Figure 2-4-44-29, Sheet 1, Detail C).

NOTE

ICS cable should be 16-inches in length from final attachment clamp.

- o. Secure ICS cable to left side of seat top support tube with clamps, support bracket, spacer, screw, washer, and nut. Position so ICS cable goes over left side of seat back cushion without touching headrest (Figure 2-4-44-29, Sheet 1, Detail B).
- p. Install stowage box and panel (PARA 2-4-44.25 or PARA 2-4-44.26). At same time,

secure ICS cable to right side stowage panel with clamps, bolts, washers, and nuts. Install washers under bolthead and nuts.

- q. Install first aid kit (PARA 17-4-1).
- r. Inspect for excessive grease and dirt buildup on tracks and rollers. Clean as follows:
 - (1) Remove front stops from seat tracks. Move seat to front of tracks (Figure 2-4-44-30).
 - (2) Tilt seat to rear to expose bottom of seat rail guides and horizontal lockpins. Clean using machinery towel, Item 211, Appendix D, moistened with dry-cleaning solvent, Item 124, Appendix D. Wipe dry with clean machinery towel, Item 211, Appendix D.
 - (3) Install seat on track.
 - (4) Install front stops.
- s. Adjust track rollers as follows:
 - (1) Loosen nut on track roller bolt.
 - (2) Insert 0.010-inch feeler gage between roller and track rail (Figure 2-4-44-30, Detail B).

CAUTION

To prevent damage to roller, fitting, or bolt, do not try to turn adjustment bolt 360°. Roller will jam against fitting and damage cross pin in bolt.

- (3) Using $\frac{5}{64}$ -inch Allen wrench in nut end of bolt, turn eccentric bushing until track fitting touches underside of track rail.
- (4) **WHILE MAINTAINING ADJUSTMENT WITH ALLEN WRENCH,**

TORQUE NUT TO 25 - 35 INCH-POUNDS.

- (5) Remove feeler gage and Allen wrench.
- t. Pull LONG adjust lever and roll seat stop to stop. Make sure seat rolls freely on track.
- u. Install passenger's seats (PARA 2-4-45).
- v. Perform operational check of ICS (TM 11-70-23).
- w. Make sure area is clean and free of foreign material.

2-4-44.29 Replace Trunnion Threaded Insert.

NOTE

- Replacement of both trunnion threaded inserts is the same.
- Do not remove trunnions from tube.
- a. Remove trunnion and tube assembly (PARA 2-4-44.18).
- b. Using insert extraction tool, remove threaded insert (Figure 2-4-44-31).
- c. Using epoxy primer coating kit, Item 135, Appendix D, sparingly apply primer to threaded hole in trunnion.
- d. Using insert installation tool, install insert while primer is wet. Insert is to be installed 0.0375 - 0.0750-inch below surface of trunnion.
- e. Using tang break off tool or suitable size pin punch, remove insert tang.
- f. Install trunnion and tube assembly (PARA 2-4-44.18).

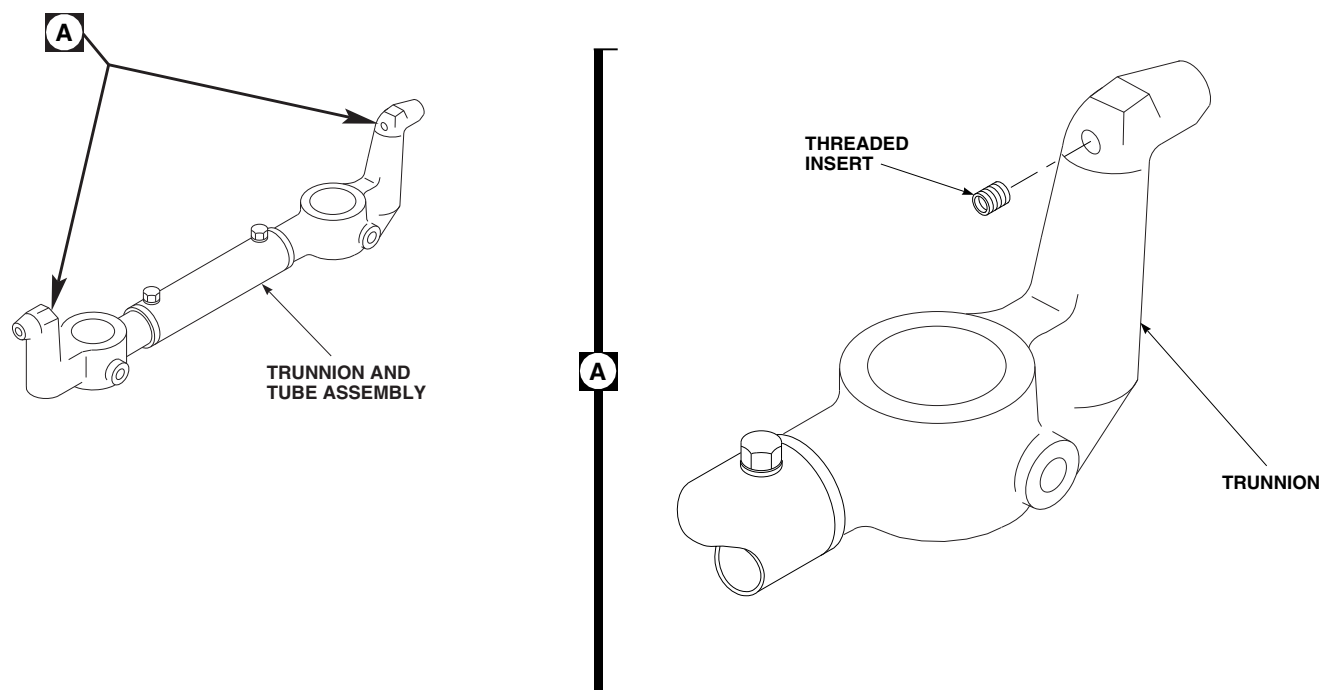
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FIGURE 2-4-44-31. REPLACE TRUNNION THREADED INSERT

2-4-45 CREW CHIEF’S/PASSENGER’S SEATS.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
2-4-45.1	Replace Seat Head Rest	2-4-45-2	2-4-45.8	Replace Disconnect Fitting	2-4-45-10
2-4-45.2	Replace Lower Energy Attenuator	2-4-45-2	2-4-45.9	Replace Seat Upper Supports	2-4-45-12
2-4-45.3	Replace Upper Energy Attenuator	2-4-45-2	2-4-45.10	Replace Seat Upper Support Fittings	2-4-45-14
2-4-45.4	Replace Cable	2-4-45-6	2-4-45.11	Repair Seat	2-4-45-17
2-4-45.5	Replace Anti-Sag Device	2-4-45-6	2-4-45.12	Replace Seat	2-4-45-17
2-4-45.6	Replace Seat Cover	2-4-45-8			
2-4-45.7	Replace Latch	2-4-45-10			

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Blind Rivet Gun
- Feeler Gage
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Adhesive, Item 38, Appendix D
- Cheesecloth, Item 90, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Tags

References

- Appendix D
- MIL-S-18729
- MIL-S-5000
- PARA 1-6-16
- PARA 2-4-46
- PARA 2-4-47
- PARA 2-4-69
- PARA 2-6-1
- PARA 2-6-2
- PARA 2-6-4
- PARA 2-6-5
- QQ-A-200
- QQ-A-250

- WW-T-700

2-4-45.1 Replace Seat Head Rest.

NOTE

Replacement of all seat head rests is the same.

2-4-45.1.1 Remove.

- Remove screws and washers securing head rest to brackets and seat frame (Figure 2-4-45-1, Detail A).
- Remove head rest from seat frame.

2-4-45.1.2 Install.

- Position replacement head rest on seat frame (Figure 2-4-45-2, Detail A).
- Secure head rest to brackets and seat frame with screws and washers (Figure 2-4-45-1, Detail A).

2-4-45.2 Replace Lower Energy Attenuator.

NOTE

Replacement of all lower energy attenuators is the same.

2-4-45.2.1 Remove.

- Disengage base of disconnect fitting from stud in cabin floor (Figure 2-4-45-2, Detail A).
- Remove bolt, washers, and nut securing attenuator to seat frame (Figure 2-4-45-2, Detail C).
- Remove washer and headed pin securing lever, attenuator, spring, shim washers, and washer(s) to disconnect fitting. Keep shim washers for installation (Figure 2-4-45-2, Detail A).

2-4-45.2.2 Install.

- ADJUST ATTENUATOR LENGTH TO 21.43 - 21.49-INCHES MEASURED**

BETWEEN END FITTING HOLE CENTERS (Figure 2-4-45-2, Detail D).

- Center rod end of replacement attenuator in lever. Install $\frac{3}{8}$ -inch shim washer on each side of rod end (Figure 2-4-45-2, Detail A and Detail B).

NOTE

Up to three washers can be used to obtain proper seating of attenuator.

- Install $\frac{1}{2}$ -inch shim washer between lever and disconnect fitting and partially insert headed pin. Install spring and $\frac{1}{2}$ -inch plain washer. Fully insert headed pin.
- CHECK FOR 0.005 - 0.010-INCH ENDPLAY BETWEEN LEVER AND ATTENUATOR ROD END. ALSO 0.005 - 0.010-INCH ENDPLAY BETWEEN LEVER AND DISCONNECT FITTING.** Adjust washers as required to obtain endplay.
- Hook spring onto lever while under tension.
- After adjusting shims, install washer and cotter pin (Figure 2-4-45-2, Detail A).
- Secure attenuator to seat frame with bolt, washers, and nut. **TIGHTEN NUT SLIGHTLY. ATTENUATOR MUST BE FREE TO PIVOT** (Figure 2-4-45-2, Detail C).
- Engage disconnect fitting to stud on cabin floor (Figure 2-4-45-2, Detail A).

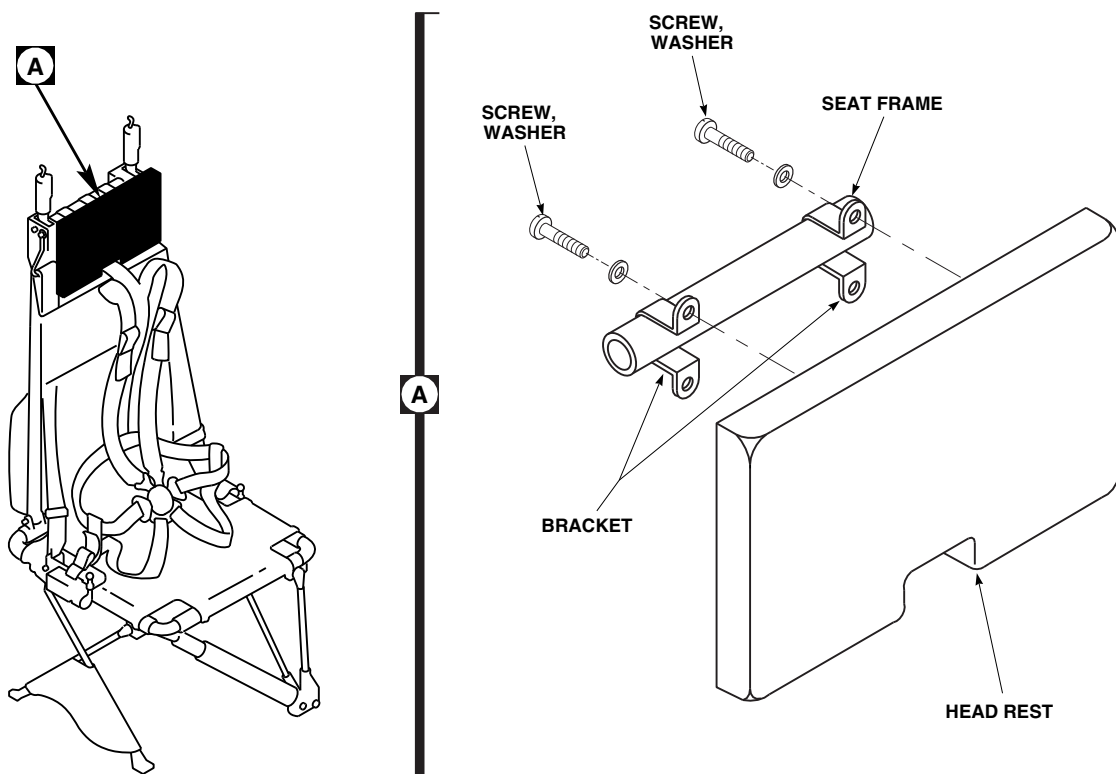
2-4-45.3 Replace Upper Energy Attenuator.

NOTE

Replacement of all upper energy attenuators is the same, except as noted.

2-4-45.3.1 Remove.

- Remove latch (PARA 2-4-45.7.1).



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FIGURE 2-4-45-1. REPLACE SEAT HEAD REST

- b. On rear-facing passenger's seats, remove turnbuckle locking clips and release tension on cable.
- c. Remove bolts, washers, and nuts from support tube and ring (Figure 2-4-45-3, Detail A). On rear-facing passenger's seats, retain upper attach link on roller retaining bolt.
- d. Withdraw hanger wire with rollers and washers.

2-4-45.3.2 Inspect.

Inspect upper energy attenuator rollers and hanger wires for corrosion. If found, remove minor corrosion.

2-4-45.3.3 Install.

WARNING

Injury to personnel and damage to equipment will result if retention bolt

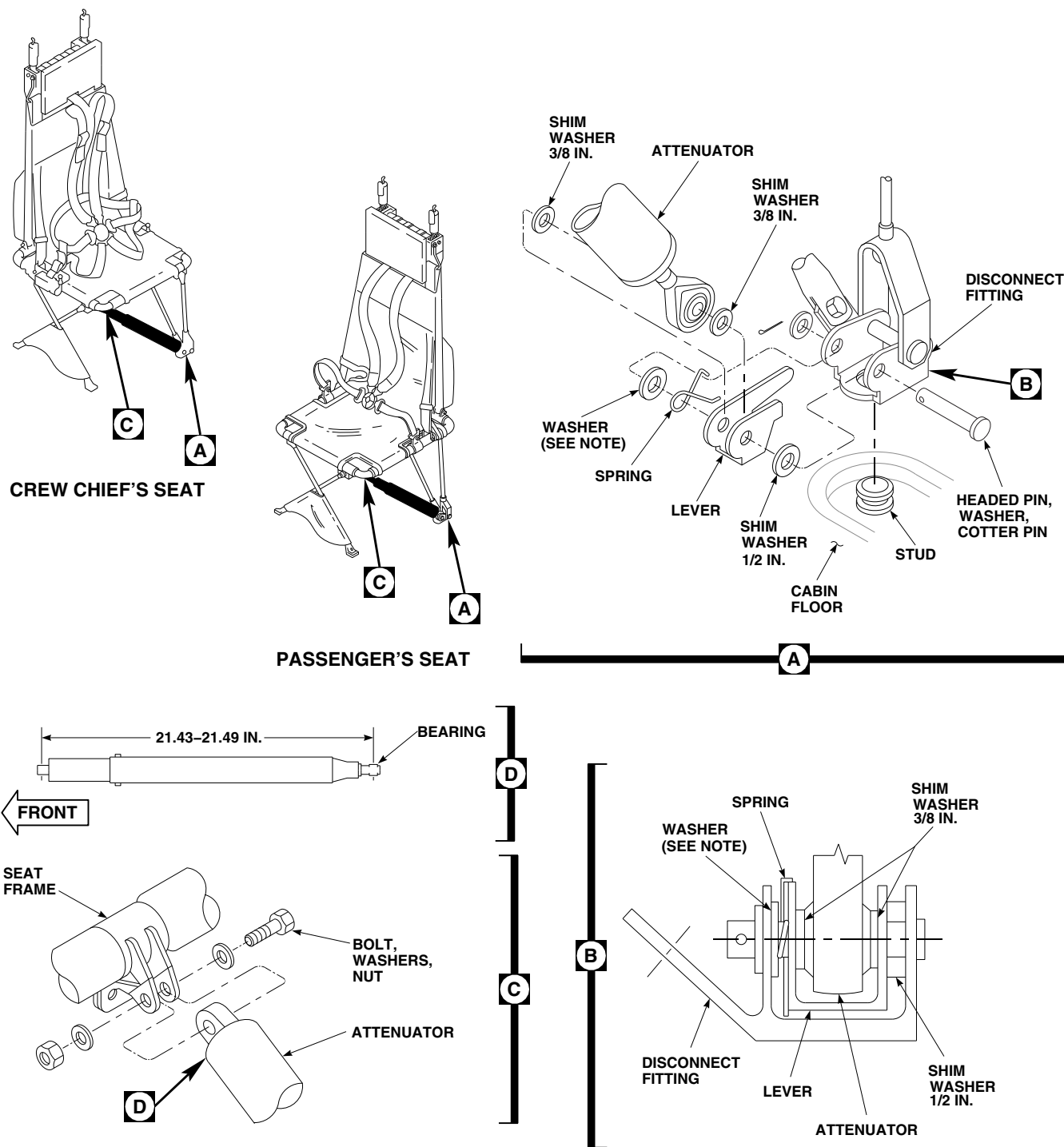
nuts are excessively tightened. Crushing of support tube and roller binding will result. Make sure shimming and tightening procedures are followed.

- a. Determine roller assembly endplay as follows:

NOTE

Roller endplay shall be calculated without hanger wires installed.

- (1) Place two washers along with roller(s) into support tube. Temporarily install retaining bolt through washers and each roller. Slide rollers and washers to one side of support tube.
- (2) Using feeler gage, measure distance between support tube and washer/roller stackup.
- (3) Using dimension obtained from step a. (2), add washers until clearance does not exceed 0.030-inch.



NOTE

UP TO 3 WASHERS CAN BE USED TO OBTAIN PROPER SEATING OF ATTENUATOR.

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FIGURE 2-4-45-2. REPLACE LOWER ENERGY ATTENUATOR

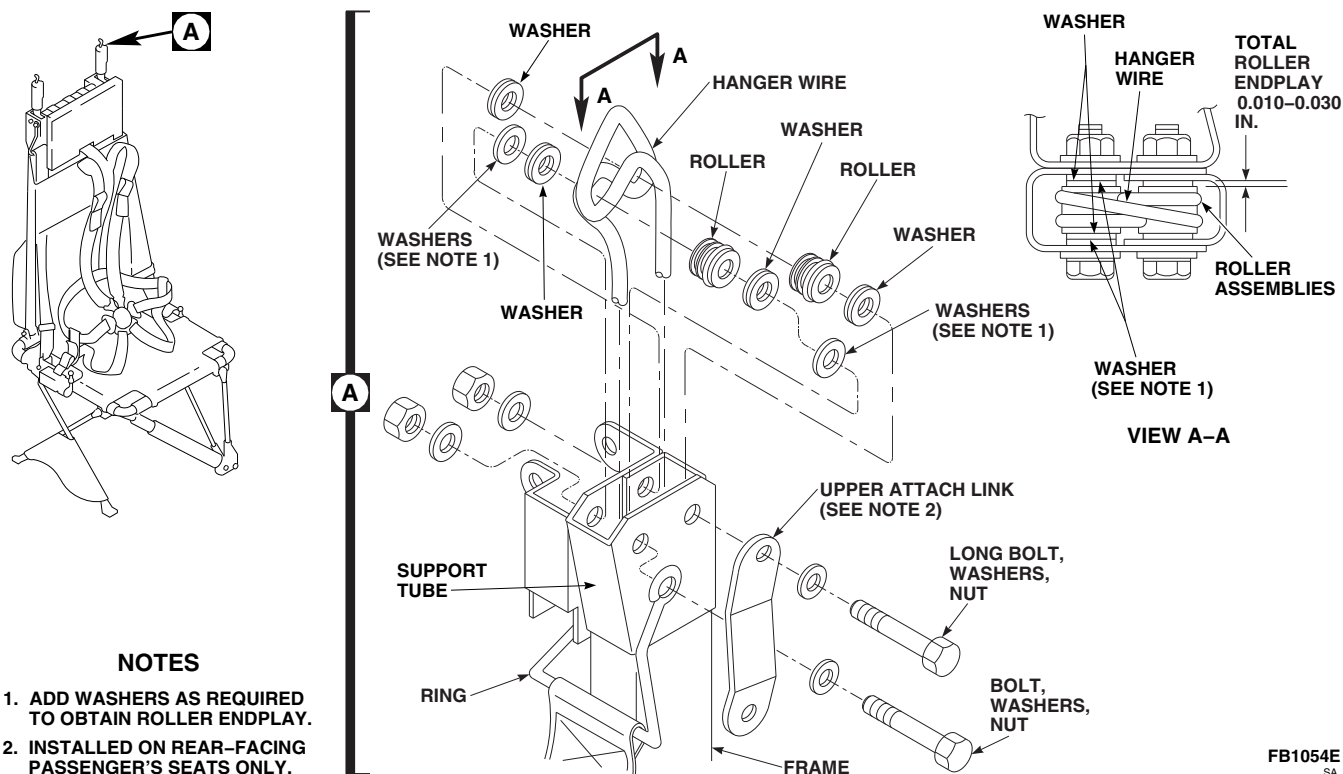


FIGURE 2-4-45-3. REPLACE UPPER ENERGY ATTENUATOR

- (4) Place washers equally and adjacent to each respective roller into support tube and install retaining bolt.
 - (5) Make sure rollers are free to move and clearance is between 0.010-inch and 0.030-inch (Figure 2-4-45-3, View A-A).
 - (6) Remove roller assemblies from support tube.
- b. Install hanger wire and roller assembly into support tube as follows:
- (1) Place roller through each wire loop.
- (3) Install roller assemblies into frame as follows (Figure 2-4-45-3, Detail A):
 - (a) Install front roller assembly, ring, bolt, washers, and nut. **DO NOT TIGHTEN NUT AT THIS TIME.**
 - (b) Install rear roller assembly, bolt, washers, and nut. **DO NOT TIGHTEN NUT AT THIS TIME.**
 - (c) On rear-facing passenger's seats, check that upper attach link is installed on rear roller retaining bolt.

NOTE

Washers are located adjacent to roller.

- (2) Assemble each roller assembly with previously determined hardware stackup.

NOTE

Washers adjacent to rollers shall be able to rotate with minimum pressure.

- (4) Make sure washers rotate with minimum pressure.

CAUTION

Damage to support tube and binding of roller assemblies will occur if roller retaining bolt nuts are over tightened. Do not over tighten roller retaining bolt nuts.

- (5) **TIGHTEN NUTS SO WASHERS ROTATE AND ROLLER ASSEMBLIES HAVE 0.010 - 0.030-INCH ENDPLAY** (Figure 2-4-45-3, View A-A).

- (6) Install latch (PARA 2-4-45.7.2).

- (7) On rear-facing passenger's seats, adjust turnbuckle tension to remove slack on cable assembly and install locking clips.

2-4-45.4 Replace Cable.

NOTE

Replacement of all cables is the same, except as noted.

2-4-45.4.1 Remove.

- a. Remove bolt, washers, and nut securing diagonal cable to seat support (Figure 2-4-45-4, Detail A).
- b. On diagonal cables with end clips, perform the following (Figure 2-4-45-4, Detail C):
 - (1) Disconnect end clip from stud in cabin floor.
 - (2) Remove bolt, washers, and nut securing cable and spring to end clip. Keep clip and spring for installation.
- c. On cables that secure to disconnect fittings, perform the following (Figure 2-4-45-4, Detail B):
 - (1) Disengage disconnect fitting securing stud in cabin floor.
 - (2) Remove headed pin and washer securing vertical cable to disconnect fitting.

- (3) Remove bolt, washers, and nut securing diagonal cables to fitting.

2-4-45.4.2 Install.

- a. On cables that secure to disconnect fitting, perform the following (Figure 2-4-45-4, Detail B):
 - (1) Place replacement vertical cable on disconnect fitting and install headed pin, washer, and cotter pin.
 - (2) Connect diagonal cable to disconnect fitting with bolt, washers, and nut. **TIGHTEN NUT SLIGHTLY; CABLE MUST BE FREE TO PIVOT.**
 - (3) Engage disconnect fitting with stud in cabin floor.
- b. On diagonal cables with end clips, perform the following (Figure 2-4-45-4, Detail C):
 - (1) Secure end clip and spring to replacement cable with bolt, washers, and nut. **TIGHTEN NUT SLIGHTLY; CABLE MUST BE FREE TO PIVOT.**
 - (2) Engage end clip with stud in cabin floor.
- c. Install cable to seat support with bolt, washers, and nut. **TIGHTEN NUT SLIGHTLY; CABLE MUST BE FREE TO PIVOT** (Figure 2-4-45-4, Detail A).

2-4-45.5 Replace Anti-Sag Device.

NOTE

- Anti-sag device is installed on rear-facing passenger's seats only.
- Replacement of all anti-sag devices is the same.

2-4-45.5.1 Remove.

- a. Remove turnbuckle locking clips and release tension on cable (Figure 2-4-45-5, Detail B).
- b. Release locking lever on latches.

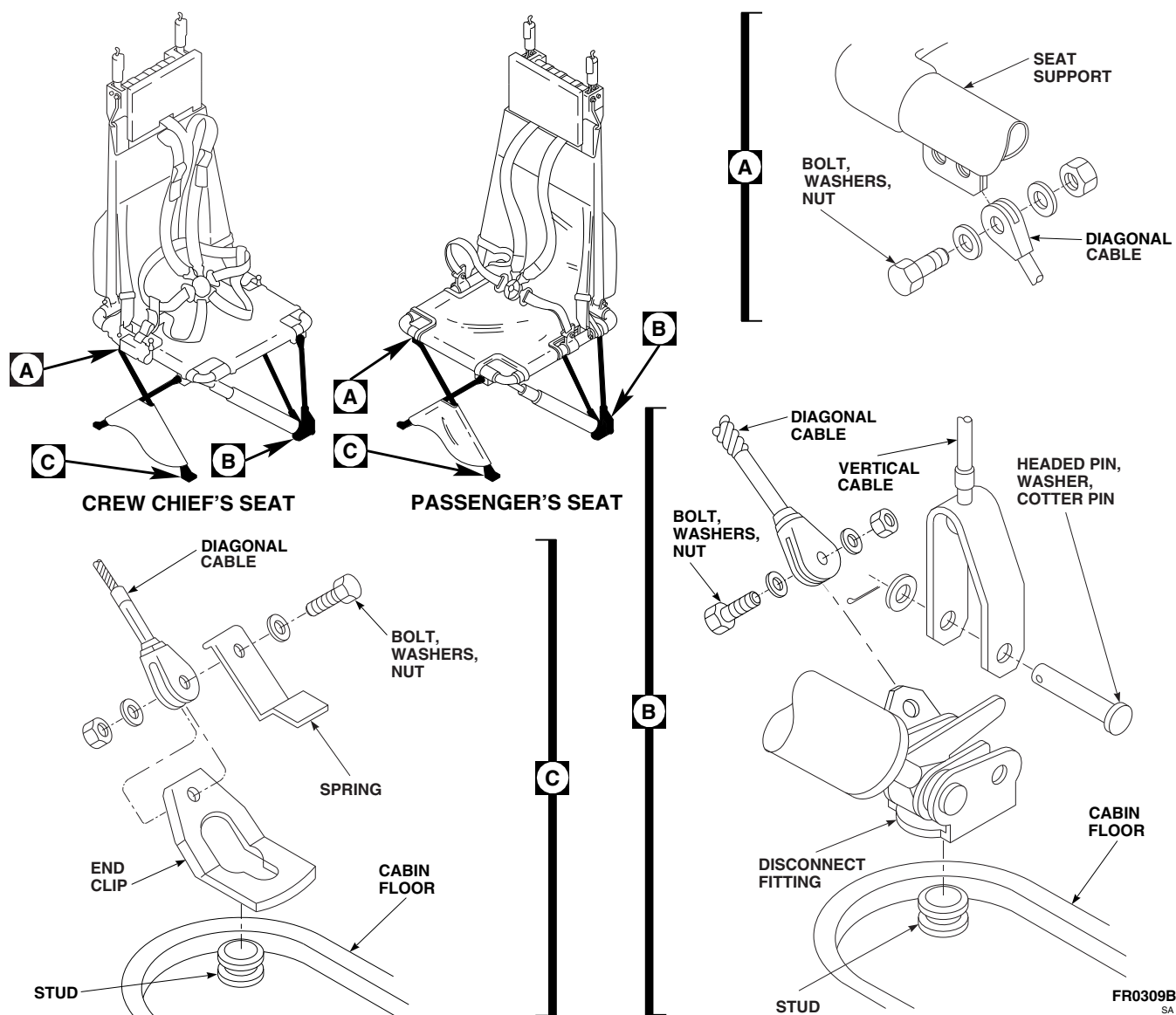


FIGURE 2-4-45-4. REPLACE CABLE

WARNING

Injury to personnel and damage to equipment will result if rollers are released when removing roller retaining bolt. Make sure roller assembly remains in place when removing bolt.

- c. Remove roller retaining bolt, nut, and washers (Figure 2-4-45-5, Detail A).

- d. Remove upper attach link from roller retaining bolt. Temporarily install roller retaining bolt in seat frame.
- e. Remove bolt, washers, and nut from web retainer, lower attach link, and seat frame.
- f. If required, disconnect cable from upper and lower attach links by removing pins and washers.

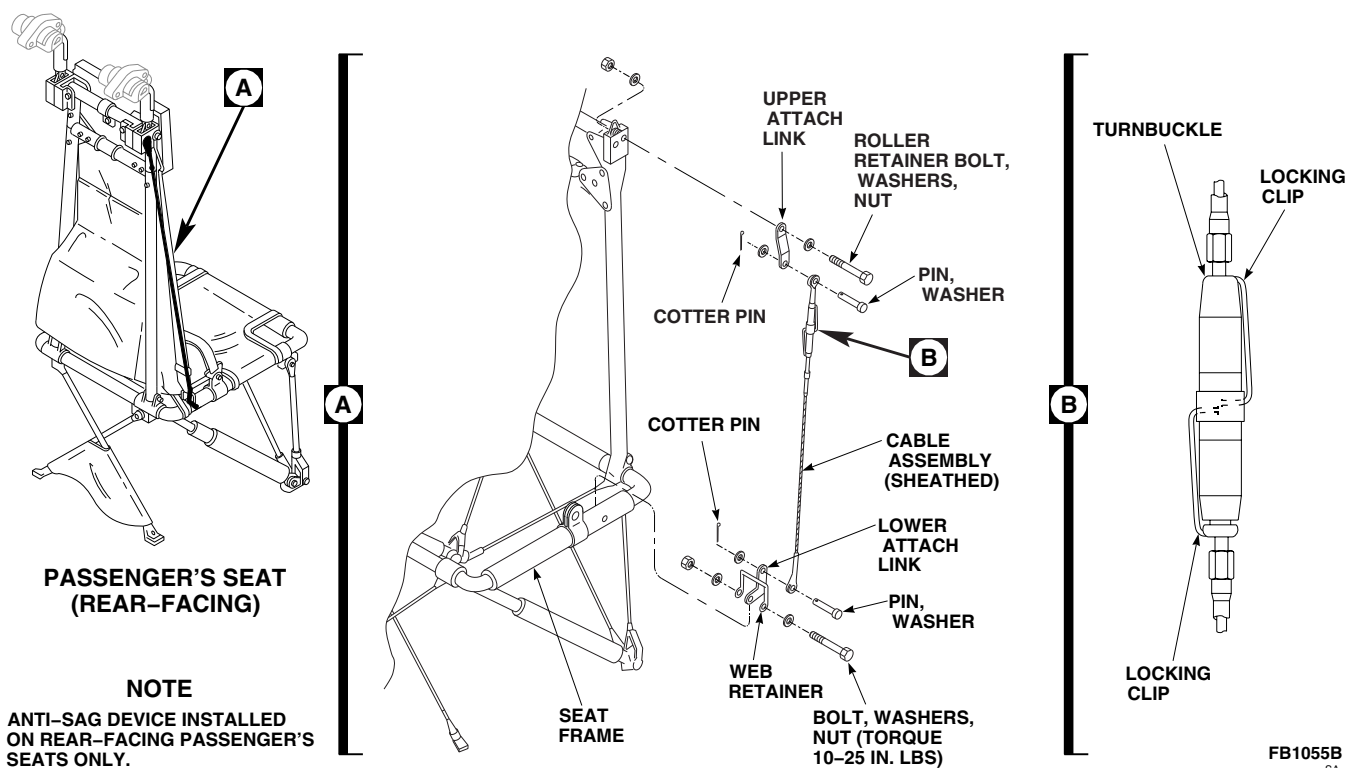


FIGURE 2-4-45-5. REPLACE ANTI-SAG DEVICE

2-4-45.5.2 Install.**WARNING**

Injury to personnel and damage to equipment will result if rollers are released when removing roller retaining bolt. Make sure roller assembly remains in place when removing bolt.

- Remove temporarily installed roller retaining bolt. Install upper attach link on roller retainer bolt and install bolt, washers, and nut (Figure 2-4-45-5, Detail A).
- If cable was disconnected from upper attach link, connect cable to upper attach link using pin, washer, and cotter pin.
- Install lower attach link and web retainer on seat support frame with bolt, washers, and nut. **TORQUE NUT TO 10 - 25 INCH-POUNDS.**
- If cable was disconnected from lower attach link, connect cable to lower attach link using pin, washer, and cotter pin.

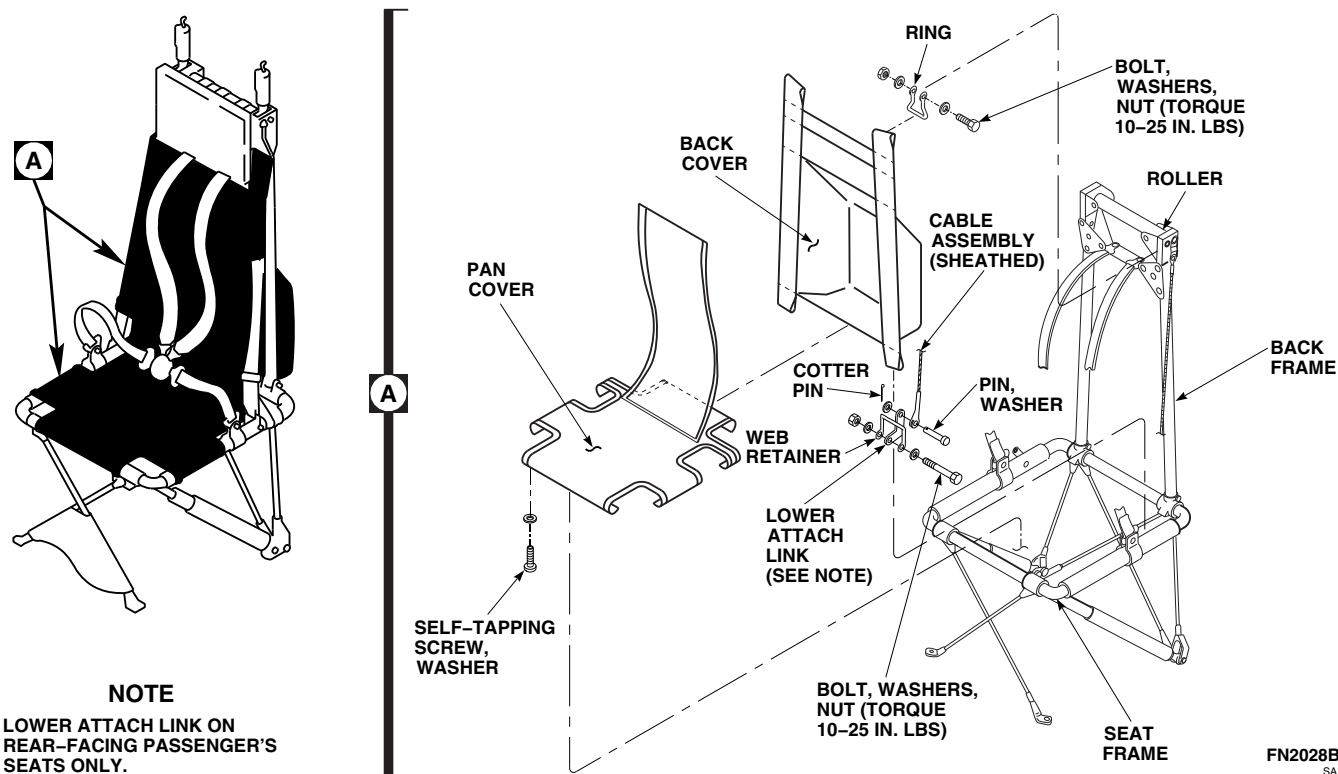
- Lock locking lever on latch.
- Tighten turnbuckle until slack is removed from cable.
- Insert turnbuckle locking clips (Figure 2-4-45-5, Detail B).

2-4-45.6 Replace Seat Cover.**NOTE**

Replacement of all seat covers is the same.

2-4-45.6.1 Remove.

- Remove passenger's seat cover as follows:
 - Remove self-tapping screws and washers securing pan cover to seat frame. Remove cover (Figure 2-4-45-6, Detail A).
 - Remove bolts, washers, and nuts securing web retainers to seat frame. On rear-facing passenger's seats, remove pin and washer securing lower attach link, and remove lower attach link.



WARNING

Injury to personnel and damage to equipment will result if rollers are released when removing ring attachment bolt. Make sure roller assembly remains in place when removing bolt.

- (3) Remove bolts, washers, and nuts securing rings to back frame.
 - (4) Remove back cover from seat frame.
 - (5) Remove all web retainers and rings from back cover.
- b. Remove crew chief's seat cover as follows:
- (1) Remove self-tapping screws and washers securing pan cover to seat frame. Remove cover.
 - (2) Remove bolts, washers, and nuts securing rings to back frame.

- (3) Remove headed pin and washers securing web retainer and belt reel yoke to seat frame.
- (4) Remove back cover from seat frame.
- (5) Remove all web retainers and rings from cover.

2-4-45.6.2 Install.

NOTE

All fasteners in seat frames covered by seat cover fabric must be smooth and free of burrs prior to installation of seat cover.

- a. Install passenger's seat cover as follows:

- (1) Slip retainers and rings through ends of straps of replacement back cover (Figure 2-4-45-6, Detail A).
- (2) Place rings on back frame and install bolts, washers, and nuts. **CLEARANCE**

SHALL BE 0.010 - 0.030-INCH BETWEEN ROLLERS AND FRAME.

- (3) Place web retainers and lower part of back cover on seat frame. On rear-facing passenger's seats, position lower attach link between seat frame and retainer. Install bolt, washers, and nut. **TORQUE NUT TO 10 - 25 INCH-POUNDS.**

NOTE

If holes for self-tapping screws become oversized, replace self-tapping screws with machine screws and rivnuts.

- (4) Place pan cover around tubes of seat frame. Pull cover to slightly tighten material. Install self-tapping screws and washers.
- b. Install crew chief's seat cover as follows:
- (1) Slip retainers and rings through ends of straps of replacement back cover.
 - (2) Place rings on upper back frame and install bolts, washers, and nuts. **CLEARANCE SHALL BE 0.010 - 0.030-INCH BETWEEN ROLLERS AND FRAME.**
 - (3) Place web retainers, belt reel yoke, and lower part of back cover on seat frame. Install headed pin, washers, and cotter pin.

NOTE

If holes for self-tapping screws become oversized, replace with machine screws and rivnuts.

- (4) Position pan cover around tubes of seat frame. Pull cover to slightly tighten material. Install self-tapping screws and washers.

2-4-45.7 Replace Latch.

NOTE

Replacement of all latches is the same.

2-4-45.7.1 Remove.

WARNING

Tension in anti-sag device cable can cause personal injury when releasing locking lever. Relieve cable tension before releasing locking lever.

- a. Release locking levers on latches to relieve seat tension (Figure 2-4-45-7, Detail A).
- b. Unhook latch from support bracket in cabin overhead.
- c. Remove bolt, washers, and nut, and remove latch and bushing from seat hanger wire.

2-4-45.7.2 Install.

- a. Place bushing (flat edge down) in hanger wire. Position latch in place and install bolt, washers, and nut. **TIGHTEN NUT SLIGHTLY; LATCH MUST BE FREE TO PIVOT** (Figure 2-4-45-7, Detail A).
- b. Hook latch onto support bracket in cabin overhead.
- c. Engage locking lever on latch to add tension to seat.

2-4-45.8 Replace Disconnect Fitting.

NOTE

Replacement of all disconnect fittings is the same.

2-4-45.8.1 Remove.

- a. Release locking lever on latch.
- b. Disengage disconnect fitting from stud in cabin floor (Figure 2-4-45-8, Detail A).
- c. Disconnect lower energy attenuator from disconnect fitting (PARA 2-4-45.2.1).
- d. Remove bolt, washers, and nut securing diagonal cable to disconnect fitting.

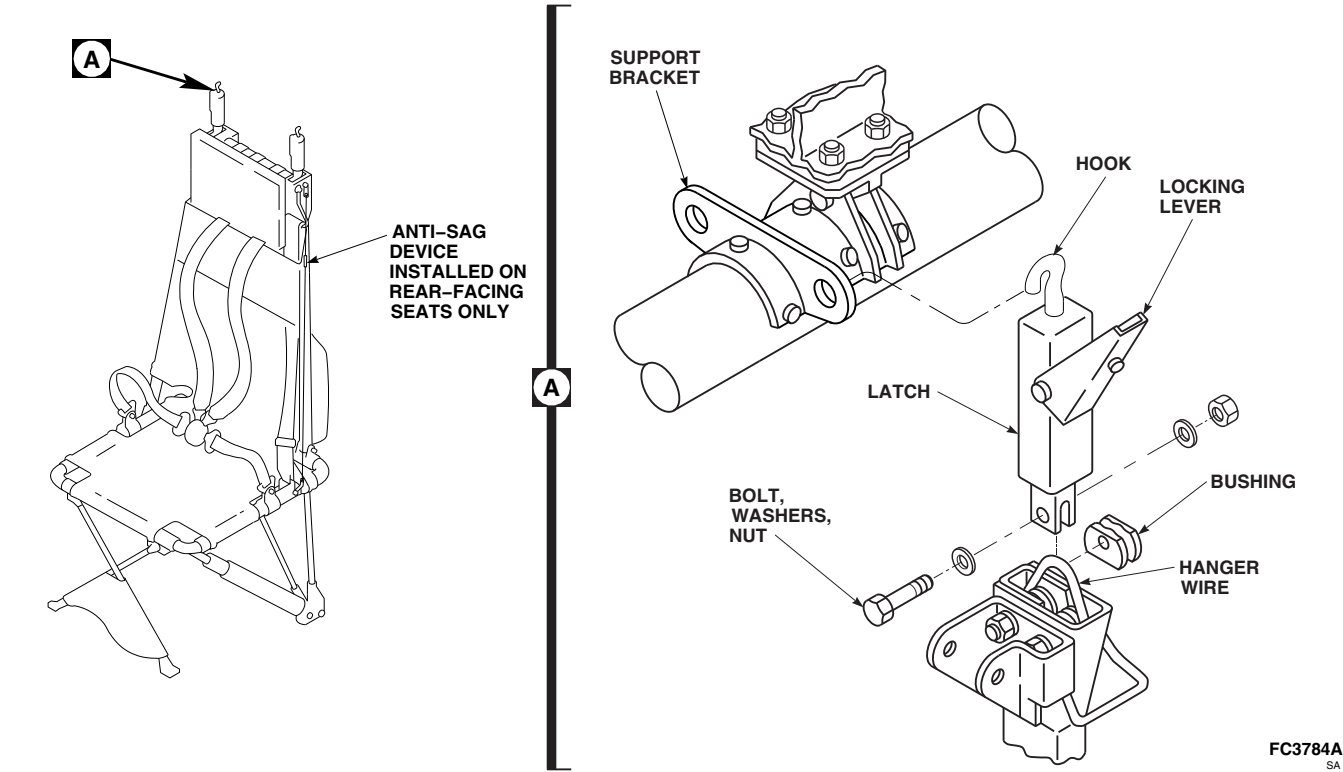


FIGURE 2-4-45-7. REPLACE LATCH

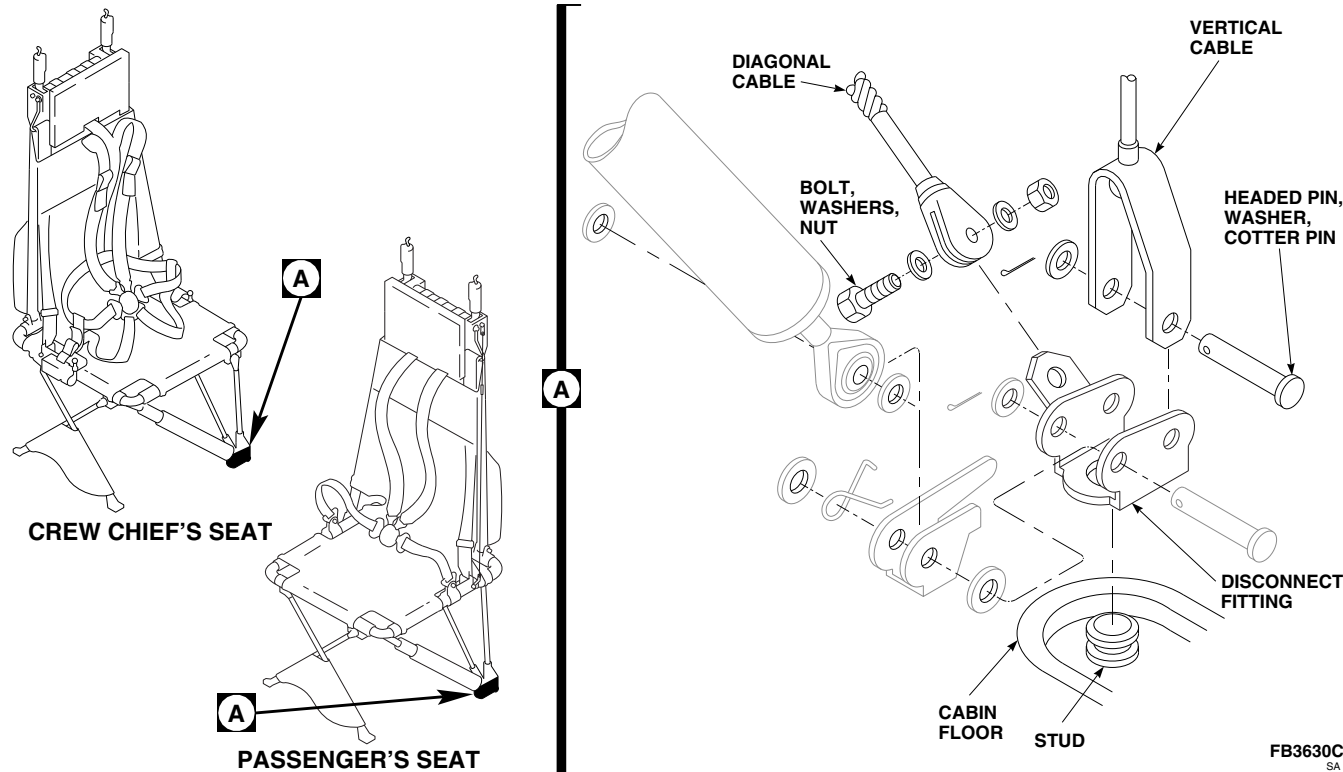


FIGURE 2-4-45-8. REPLACE DISCONNECT FITTING

- e. Remove headed pin and washer securing vertical cable to disconnect fitting.

2-4-45.8.2 Install.

- a. Place vertical cable on replacement disconnect fitting. Install headed pin, washer, and cotter pin (Figure 2-4-45-8, Detail A).
- b. Connect diagonal cable to disconnect fitting with bolt, washers, and nut. **TIGHTEN NUT SLIGHTLY; CABLE MUST BE FREE TO PIVOT.**
- c. Connect lower energy attenuator to disconnect fitting (PARA 2-4-45.2.2).
- d. Engage disconnect fitting with stud in cabin floor.
- e. Engage locking lever on latch to add tension to seat.

2-4-45.9 Replace Seat Upper Supports.

2-4-45.9.1 Remove.

CAUTION

Damage to upper support and support fittings will result if support is allowed to hang from end support fitting. Upper support will bend and damage support fitting. Upper support must be completely removed.

- a. To remove crew chief's seats upper support, perform the following:
 - (1) Remove crew chief's seats (PARA 2-4-45.14.1).
 - (2) Disconnect ICS cables at disconnects (Figure 2-4-45-9).
 - (3) Remove covers from support tube ends.
 - (4) Remove bolt, washers, and nut from upper support and remove spacers from fittings.

- (5) If replacing upper support, remove bolts, clamps, and ICS cables from support tubes.

- b. To remove passenger's seat upper support, perform the following:

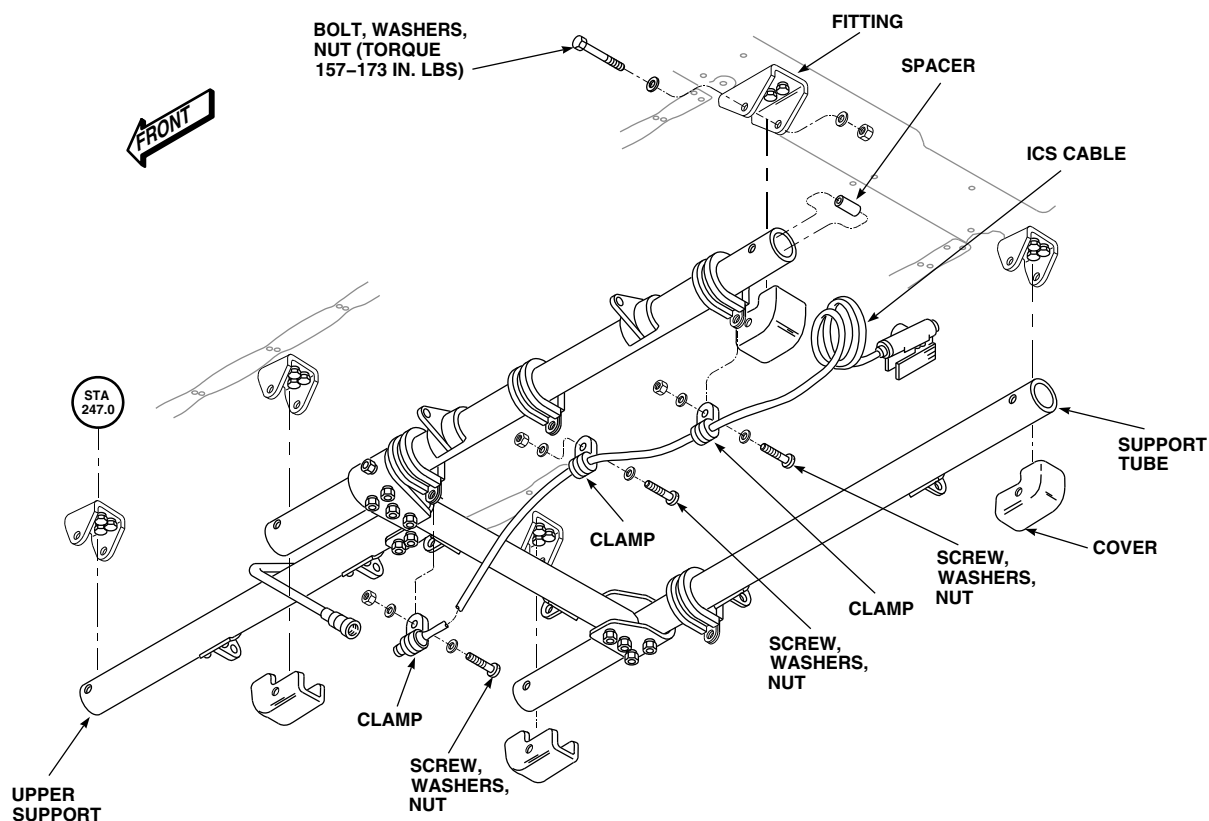
- (1) Remove center passenger's seats (PARA 2-4-45.14.1).
- (2) Remove bolt, washers, and nut securing upper support to fittings.

2-4-45.9.2 Install.

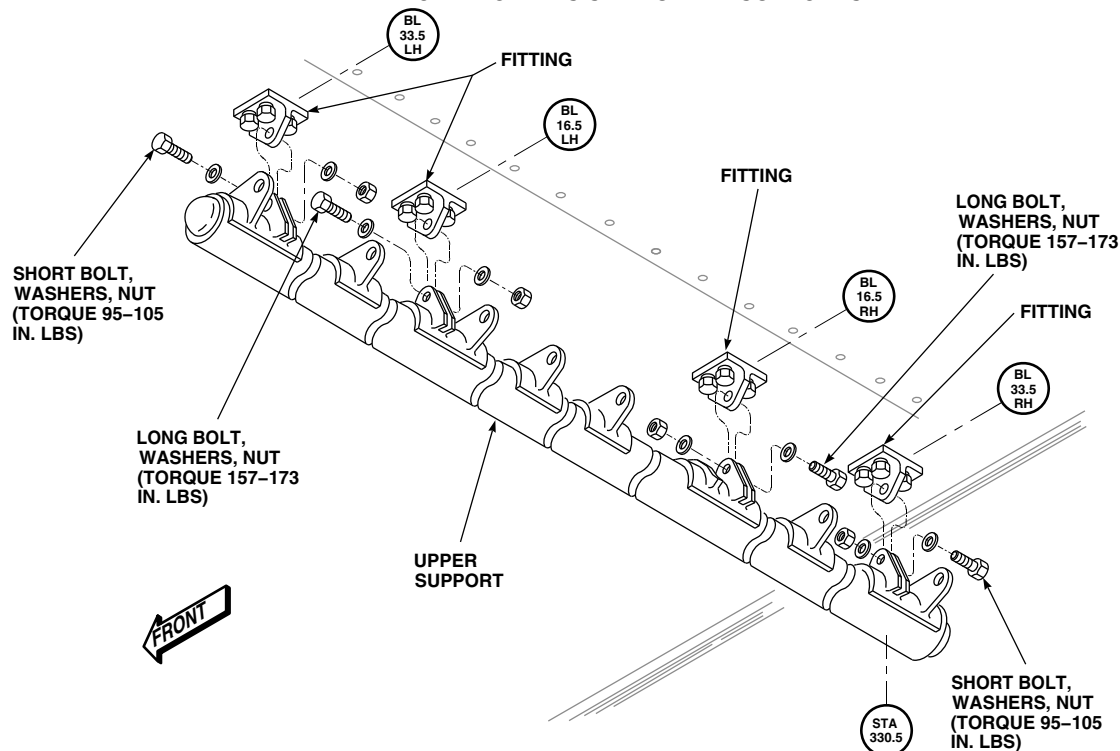
- a. To install crew chief's seat upper support, perform the following:

- (1) If ICS cables on upper support were removed, install cables using clamps, screws, washers, and nuts (Figure 2-4-45-9).
- (2) Position upper support in fittings, and install spacers in ends of tubes.
- (3) Line up holes in fittings, spacers, and tubes. Install bolts, washers, and nuts. **TORQUE NUTS TO 157 - 173 INCH-POUNDS.**
- (4) Clean mating surfaces of cover and tube using cheesecloth, Item 90, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D.
- (5) Apply thin coat of adhesive, Item 38, Appendix D, to mating surfaces inside tube and outside cover.
- (6) Install cover in tube and make 0.050-inch fillet of adhesive squeezeout.
- (7) Allow adhesive to cure for 12 hours at 70° - 85°F (21° - 29°C).
- (8) Connect ICS cables at disconnects.
- (9) Install seats (PARA 2-4-45.14.3).

- b. To install passenger's seat upper support, perform the following:



CREW CHIEF'S SEAT UPPER SUPPORTS



PASSENGER'S SEAT UPPER SUPPORTS

FIGURE 2-4-45-9. REPLACE SEAT UPPER SUPPORTS

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- (1) Position upper support in fittings and install short bolts, washers, and nuts in fittings near ends of support (BL 33.50). **TORQUE NUTS TO 95 - 105 INCH-POUNDS.**
- (2) Install long bolts, washers, and nuts in fittings near center of support (BL 16.50). **TORQUE NUTS TO 157 - 173 INCH-POUNDS.**
- (3) Install center passenger's seats (PARA 2-4-45.14.3).

2-4-45.10 Replace Seat Upper Support Fittings.

2-4-45.10.1 Remove.

- a. Remove crew chief's/passenger's seat upper supports (PARA 2-4-45.9.1).
- b. Remove soundproofing panels as required (PARA 2-4-69).
- c. Remove bolts, washers, and radius block securing support fitting at STA 247.0, BL 6.5 (Figure 2-4-45-10, Sheet 1, Detail A).
- d. Remove bolts, washers, and radius blocks securing support fittings at STA 265.5 (left and right).

NOTE

Fitting at STA 295.0, RBL 10.0, receives three bolts, AN3-7A, and one bolt, NAS1303-8. Tag and identify bolts as they are removed for proper installation.

- e. Remove bolts, washers, radius blocks, and nuts securing support fittings at STA 295.0, BL 10.0 (left and right).
- f. Remove bolts, washers, and nuts securing support fittings at STA 330.5, BL 16.5 (left and right) (Figure 2-4-45-10, Sheet 2, Detail B).
- g. Remove bolts and washers securing support fitting at STA 330.5, LBL 33.5.

- h. Remove blind rivets securing support fitting at STA 330.5, RBL 33.5.
- i. Remove bolts, washers, nuts, angle bracket, and retaining ring securing seat support fittings at STA 398.0, BL 35.0 (left and right) (Figure 2-4-45-10, Sheet 2, Detail C).
- j. Remove bolts, washers, strap fastener loop, and retaining ring securing seat support fittings at STA 398.0, BL 25.0, BL 15.0, and BL 5.0 (Figure 2-4-45-10, Sheet 2, Detail D).

2-4-45.10.2 Install.

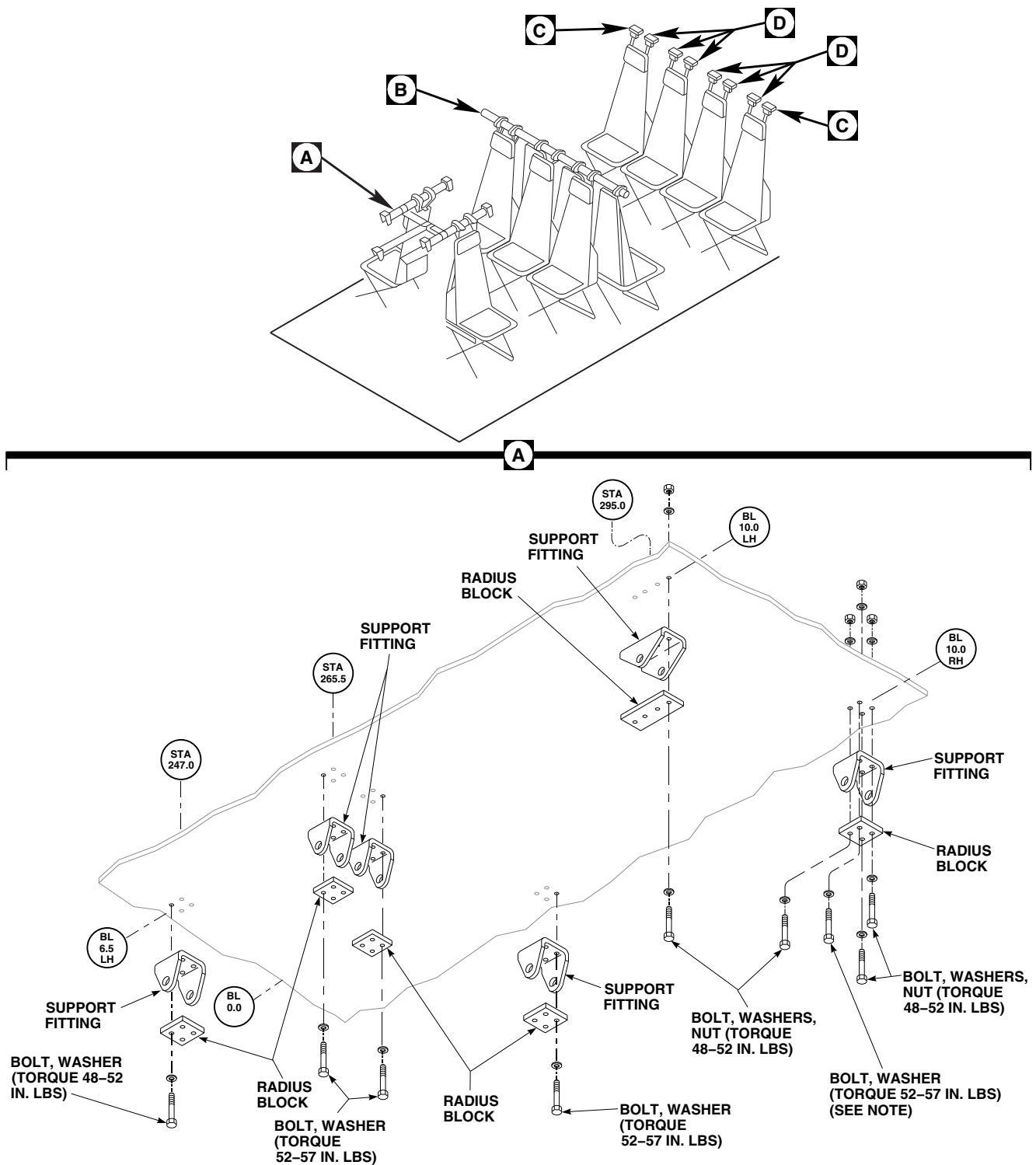
WARNING

- Injury to personnel and damage to equipment will result if improper length bolts are used to install support fittings. Bolts can bottom out on nutplates or nuts before proper torque is reached. Make sure proper length bolts are used.
- Minimum of 1.5 threads must show through nutplates or nuts. Up to three washers may be used under boltheads to obtain proper thread engagement.

CAUTION

Damage to mounting hardware, fittings and radius blocks will occur if radius blocks are not properly installed. Make sure radius blocks are installed so contours on radius block match radius of inside corners of support fittings.

- a. Check self-locking feature of nutplates or nuts.
- b. Install support fitting at STA 247.0, BL 6.5, and STA 330.5, LBL 33.5 with bolts, washers, and radius block. **TORQUE BOLTS TO 48 - 52 INCH-POUNDS** (Figure 2-4-45-10, Sheet 1, Detail A and Sheet 2, Detail B).
- c. Install support fitting at STA 330.5, RBL 33.5 with blind rivets (Figure 2-4-45-10, Sheet 2, Detail B).



NOTE

USE UP TO THREE WASHERS OF ANY THICKNESS UNDER BOLTHEAD TO PROVIDE CORRECT THREAD PROTRUSION.

CREW CHIEF'S SEAT UPPER SUPPORT FITTINGS

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FIGURE 2-4-45-10. REPLACE SEAT UPPER SUPPORT FITTINGS (SHEET 1 OF 2)

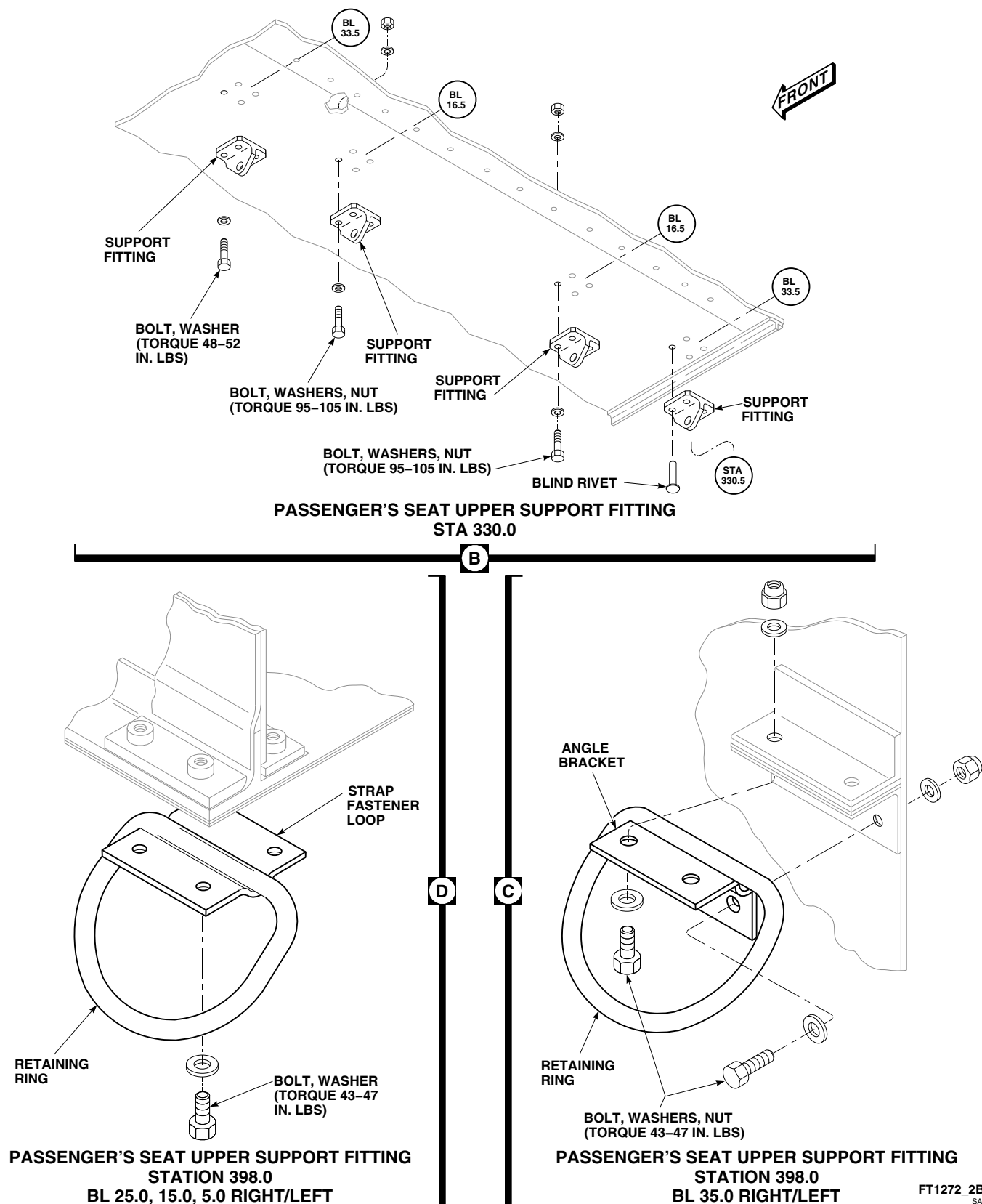


FIGURE 2-4-45-10. REPLACE SEAT UPPER SUPPORT FITTINGS (SHEET 2 OF 2)

- d. Install support fittings at STA 330.5, BL 16.5 (left and right) with bolts, washers, and nuts. **TORQUE NUTS TO 95 - 105 INCH-POUNDS.**
- e. Install support fittings at STA 265.5 (left and right) with bolts, washers, and radius blocks (Figure 2-4-45-10, Sheet 1, Detail A). **TORQUE BOLTS TO 52 - 57 INCH-POUNDS.**

NOTE

- Fitting at STA 295.0, RBL 10.0 receives three bolts, AN3-7A, and one bolt, NAS1303-8. Remove tags when installing bolts.
- Up to three washers of any thickness can be used under bolthead of bolt, NAS1303-8, to obtain proper thread extrusion.
- f. Install support fittings at STA 295.0, BL 10.0 (left and right) with bolts, washers, nuts, and radius blocks. **TORQUE BOLTS, AN3-7A, TO 48 - 52 INCH-POUNDS. ON FITTING AT STA 295.0, RBL 10.0, TORQUE BOLT, NAS1303-8, TO 52 - 57 INCH-POUNDS.**
- g. Install support fittings at STA 398.0, BL 35.0 (left and right) with bolts, washers, nuts, angle brackets, and retaining rings (Figure 2-4-45-10, Sheet 2, Detail C). **TORQUE NUTS TO 43 - 47 INCH-POUNDS.**
- h. Install seat support fittings at STA 398.0, BL 25.0, BL 15.0, and BL 5.0 with bolts, washers, strap fastener loops, and retaining rings (Figure 2-4-45-10, Sheet 2, Detail D). **TORQUE BOLTS TO 43 - 47 INCH-POUNDS.**
- i. Install soundproofing panels (PARA 2-4-69).
- j. Install crew chief's/passenger's seat upper supports (PARA 2-4-45.9.2).

2-4-45.11 Repair Seat.

NOTE

- Repair for all seats is the same.

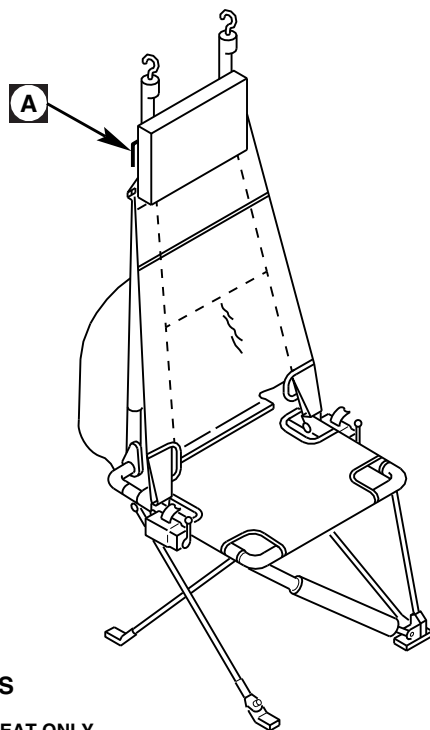
- Crew chief's/passenger's seats are tubular aluminum alloy framework covered with fabric and held rigid by diagonal cable assemblies. Quick-disconnect floor fittings, upper support tubes, and support fittings are formed or extruded aluminum alloy. Restraint fittings and upper latch attachment fittings are 4130 steel welded assemblies (Figure 2-4-45-11, Sheet 1, Sheet 2, and Sheet 3).

- a. Evaluate damage to aluminum parts (PARA 2-6-1).
- b. Use PARA 2-6-2 and PARA 2-6-4 to repair or replace aluminum parts.
- c. Evaluate damage or wear on steel welded assemblies using PARA 2-6-1 limits. Blend out minor damage and restore cadmium-plating. Replace welded assemblies damaged or worn beyond negligible damage limits.
- d. Use PARA 2-4-45.11 for damage limits or repair on straps, webbing, and covers.
- e. To repair crew chief's and passenger's seat pan covers, damaged by rivets on seat frame, sew gray webbing, 2-inches wide, to inner surface of pan cover (Figure 2-4-45-11, Sheet 2, Detail B). Stitch webbing.
- f. To reduce wear on seat cover, replace protruding head blind rivet with flush head rivet, in two places (Figure 2-4-45-11, Sheet 2, Detail A).
- g. Use standard procedures for tubing repair.
- h. Touch up paint as required (PARA 2-6-5).

2-4-45.12 Replace Seat.

NOTE

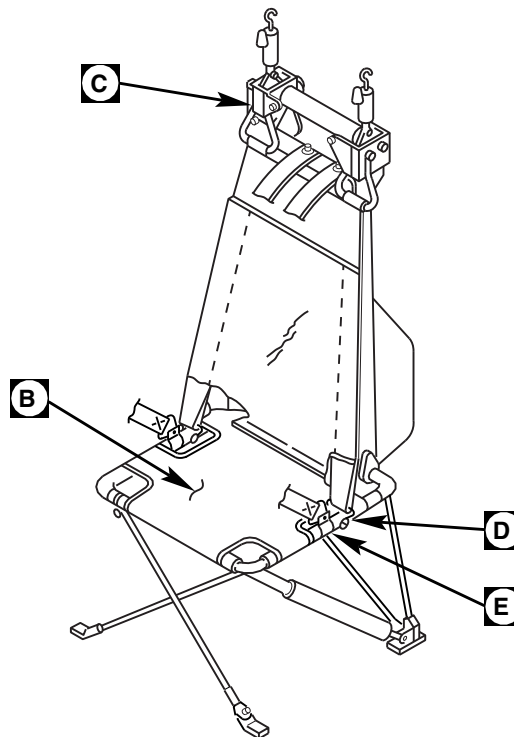
- Do not place tool boxes, tools, or oily rags on seat covers while performing maintenance.
- Replacement of all seats is the same, except as noted.



NOTES

1. CREW CHIEF'S SEAT ONLY.
2. PASSENGER'S SEAT ONLY.

CREW CHIEF'S SEAT



PASSENGER'S SEAT

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FIGURE 2-4-45-11. REPAIR SEAT (SHEET 1 OF 3)

2-4-45.12.1 Remove.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Release locking lever on latches (Figure 2-4-45-12, Sheet 1, Detail B).
- c. Disengage disconnect fitting and cable clips from studs on cabin floor (Figure 2-4-45-12, Sheet 3, Detail D and Detail E).

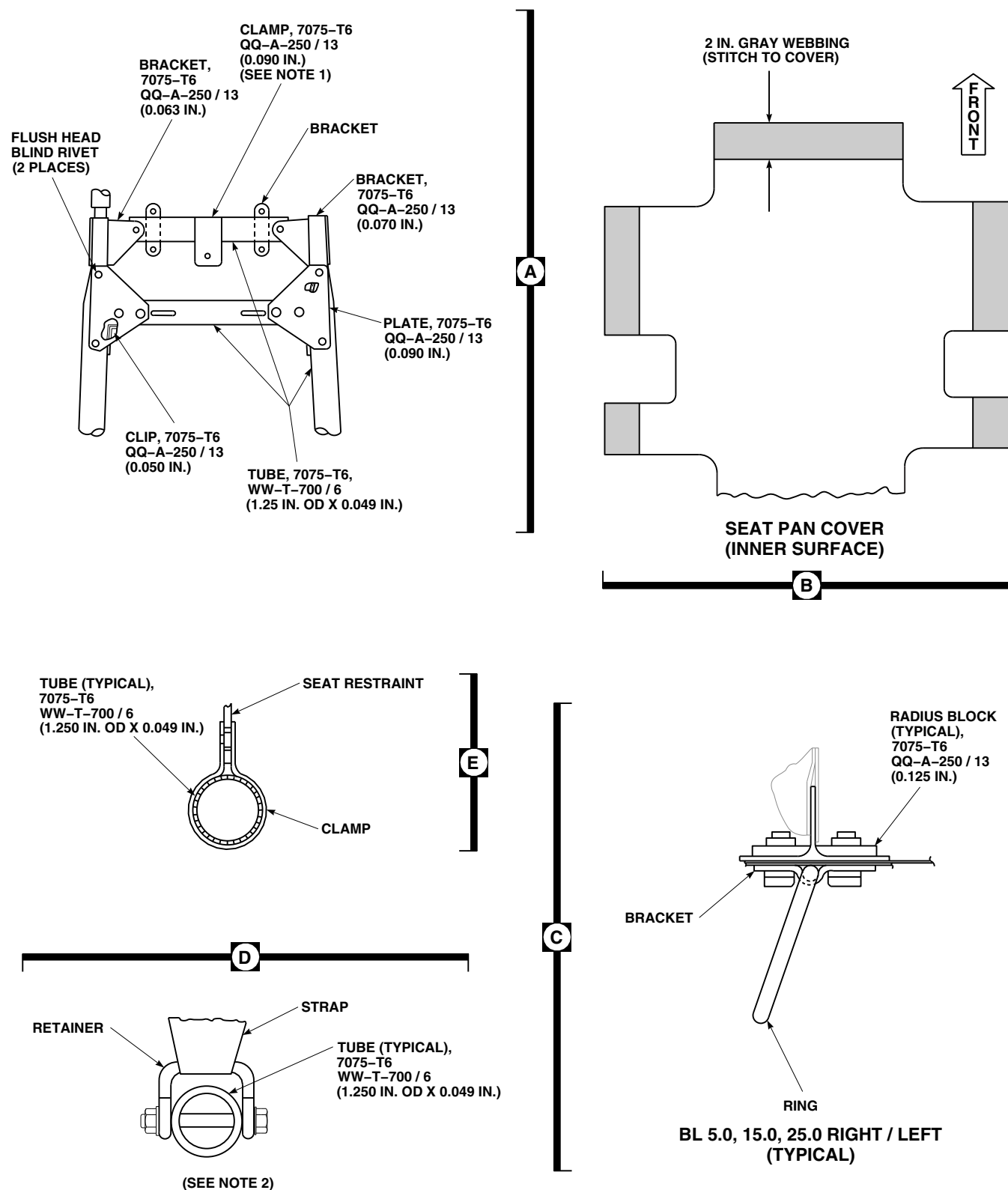
CAUTION

Damage to hanger wire will occur if passenger's seat is not properly handled and stored. Mishandling will cause wire to bend, twist and not be centered on roller depression. Make sure to use care when handling or storing passenger's seat.

- d. Unhook latch from support fittings or support brackets in cabin overhead. Remove seat (Figure 2-4-45-12, Sheet 1, Detail B).

2-4-45.12.2 Inspect.

- a. Check upper latches and latch locking mechanism for cracks and broken springs. **NONE ALLOWED.** If damaged, replace latch (PARA 2-4-45.7).
- b. Check hanger wire for cracks, wear, flatness (wire diameter), deformation, or other damage. **NO CRACKS ALLOWED. WIRE DAMAGE SHALL BE LIMITED TO MINIMUM DIAMETER OF 0.090-INCH.** If damaged, replace hanger wire (PARA 2-4-45.3).
- c. Check seat back, pan frames, and frame fittings for cracks, loose, sheared, and missing fasteners. **NONE ALLOWED.** If damaged, replace frame.
- d. On rear-facing passenger's seats, check anti-sag device for broken or loose strands, cable sheathing for tears or damage, and attaching hardware for cracks and security. **NONE ALLOWED.** If damaged, replace necessary parts (PARA 2-4-45.5).



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FIGURE 2-4-45-11. REPAIR SEAT (SHEET 2 OF 3)

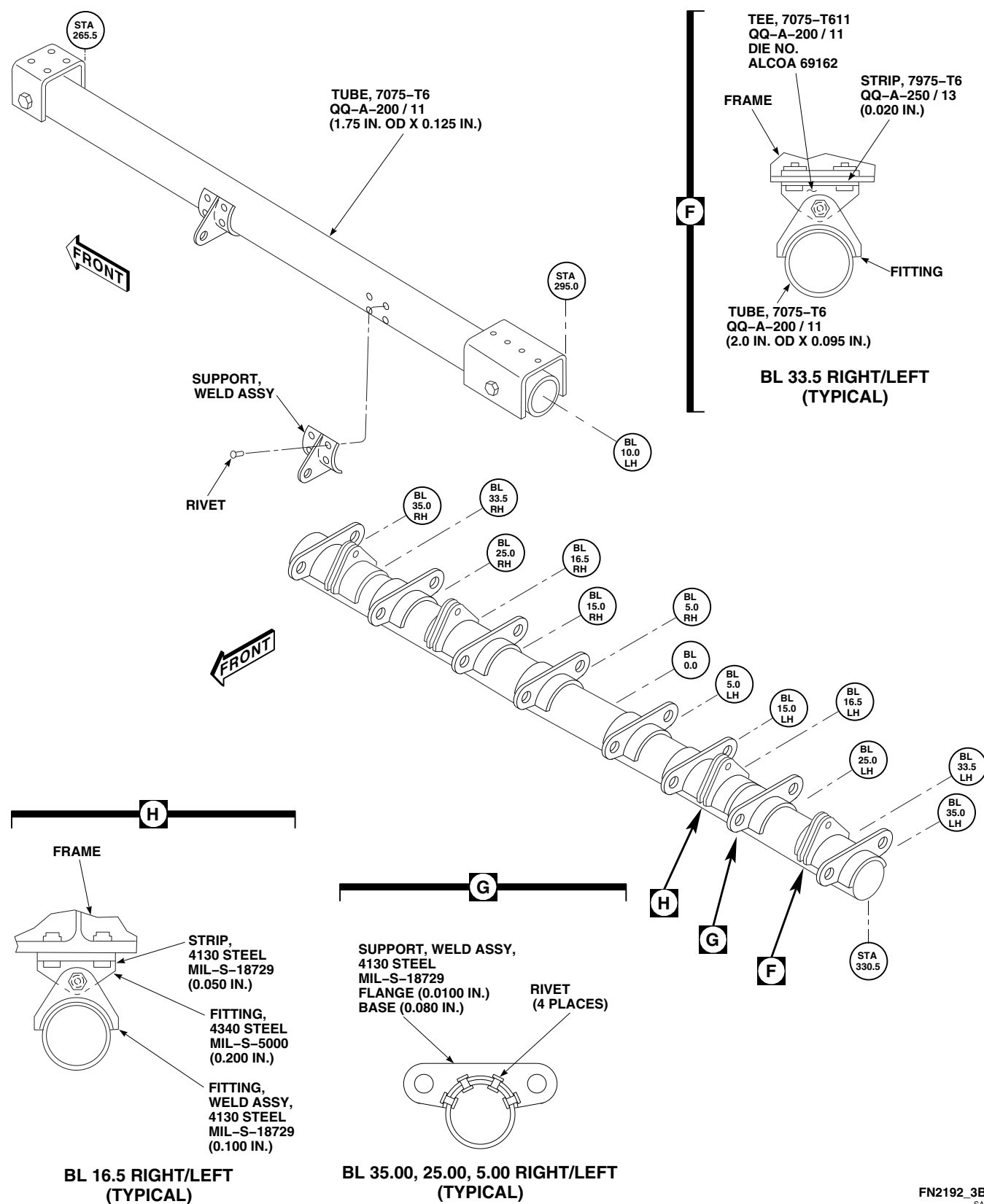


FIGURE 2-4-45-11. REPAIR SEAT (SHEET 3 OF 3)

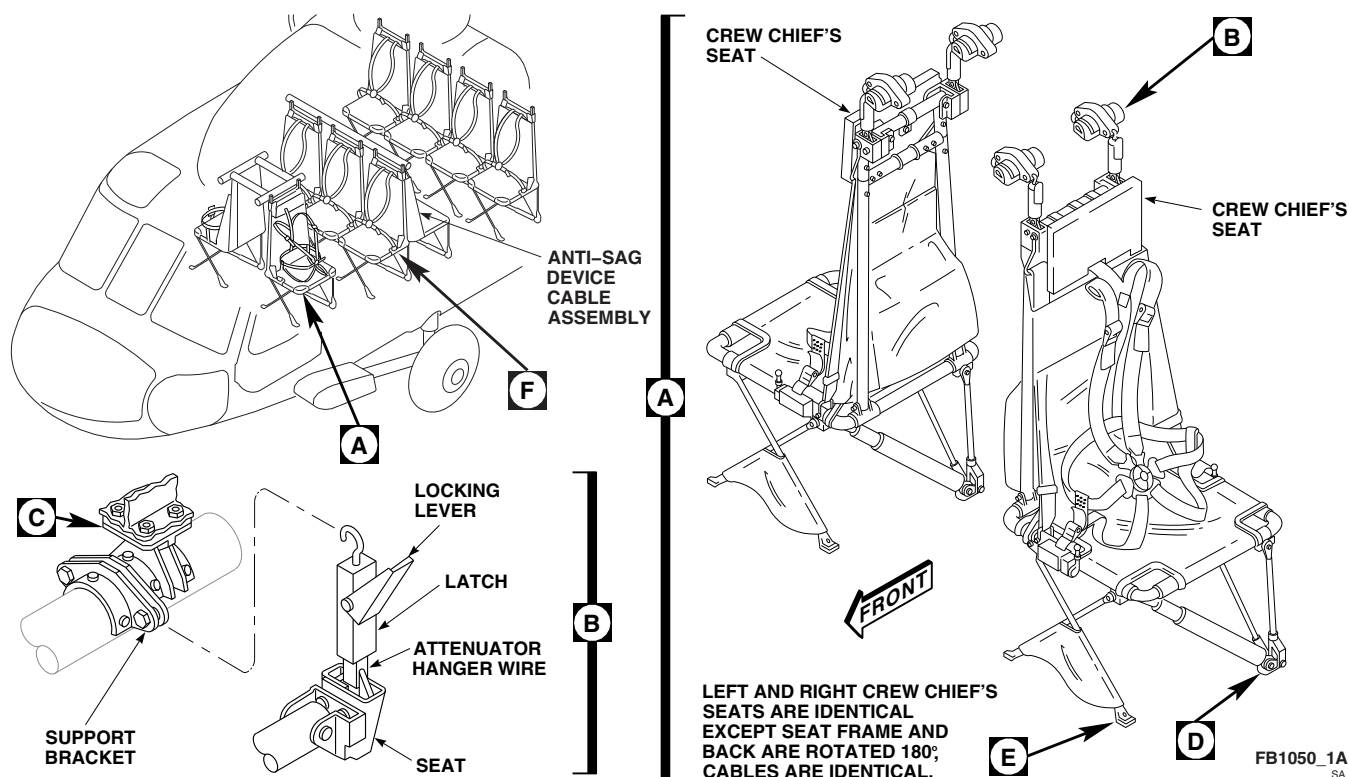


FIGURE 2-4-45-12. REPLACE SEAT (SHEET 1 OF 3)

- e. Check seat covers for tears, holes, and loose or missing stitching. Check pan cover for loose or missing fabric retaining screws. **NONE ALLOWED.** If retaining screws are loose or missing, tighten or replace screws. If damaged, repair or replace seat covers (PARA 2-4-45.6).
- f. Check area between upper seat cover ring and seat frame for cracks and wear. If damaged, replace ring (PARA 2-4-45.6).
- g. Check lower energy attenuator tube for misalignment, dents, cracks, and tube extension. **NO DAMAGE ALLOWED.** If damaged, replace lower energy attenuator (PARA 2-4-45.2).
- h. Check lower energy attenuator tube end fittings for loose, sheared, or missing fasteners. **NONE ALLOWED.** Check bearing end for free movement. If loose, sheared, or missing fasteners are found, replace necessary fasteners (PARA 2-4-45.2). If bearing is damaged, replace bearing.
- i. Check disconnect hardware for cracks, wear, and missing or broken parts. **NONE ALLOWED.** If damaged, replace necessary parts.
- j. Check cables for broken strands. If three or more strands are broken, replace cable (PARA 2-4-45.4).
- k. Check cables for kinks. If cable cannot be straightened by hand, replace cable (PARA 2-4-45.4).
- l. On rear-facing passenger's seats, check to make sure anti-sag device is installed. If missing, install anti-sag device (PARA 2-4-45.5).
- m. On rear-facing passenger's seats, check attaching hardware on anti-sag device for cracks and security. **NONE ALLOWED.** If damaged, replace hardware (PARA 2-4-45.5).
- n. Inspect passenger's seat restraint system (PARA 2-4-46) or crew chief's seat restraint system (PARA 2-4-47).

2-4-45.12.3 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Hook latches onto support fittings or support brackets in cabin overhead. Do not engage

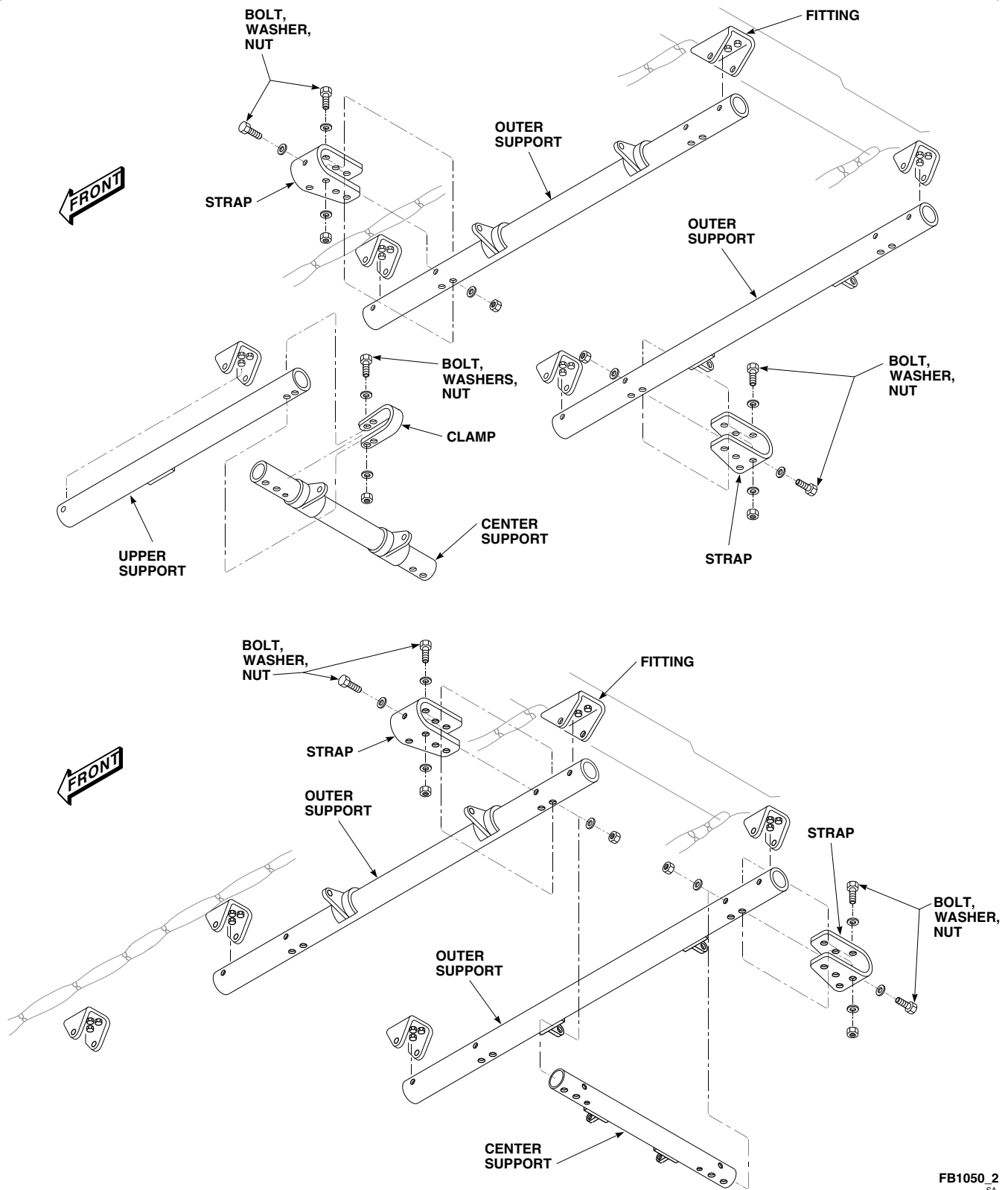
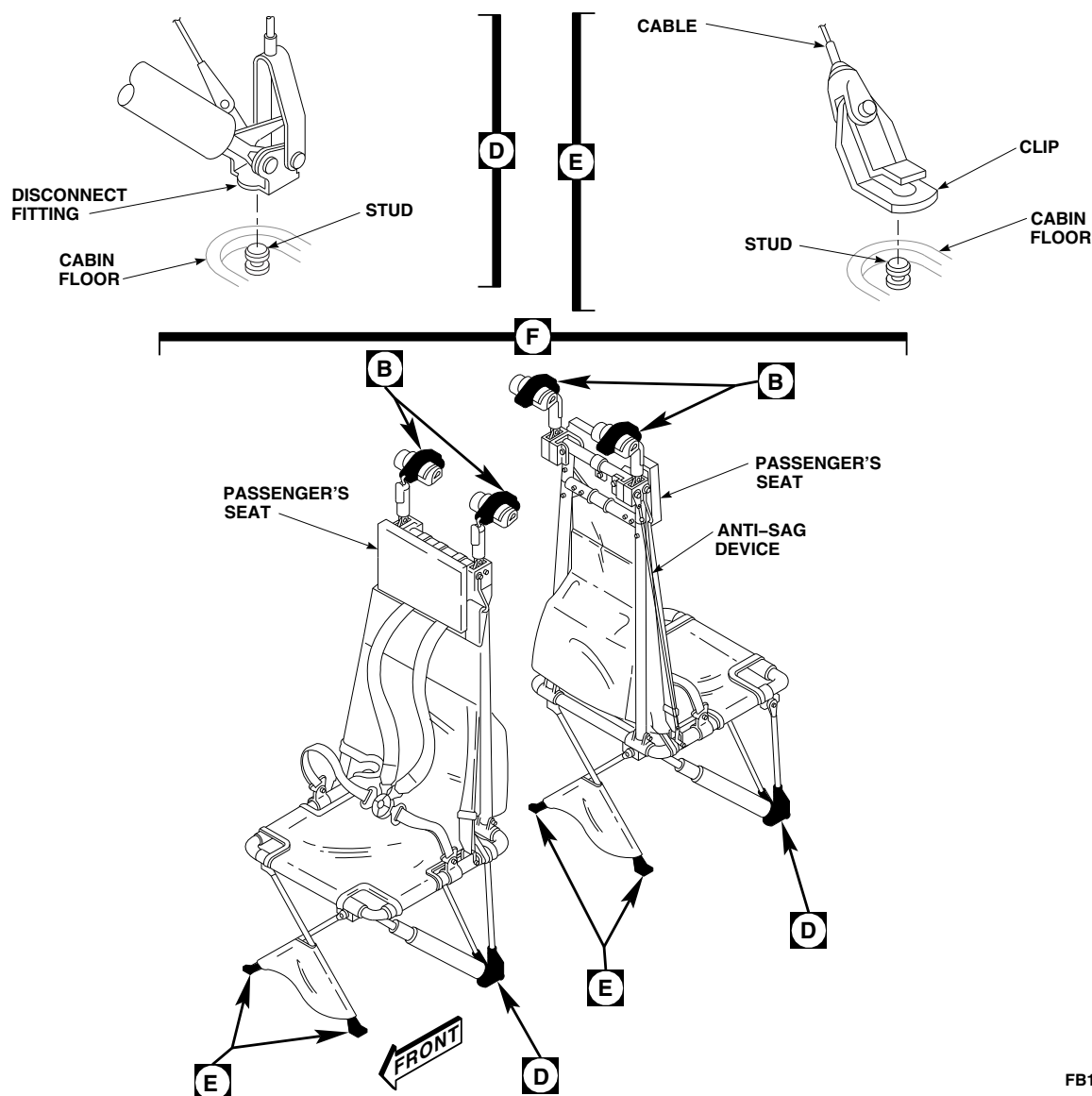
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FIGURE 2-4-45-12. REPLACE SEAT (SHEET 2 OF 3)



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FIGURE 2-4-45-12. REPLACE SEAT (SHEET 3 OF 3)

locking levers (Figure 2-4-45-12, Sheet 1, Detail B).

WARNING

To reduce personal injury in a hard landing, make sure attenuators are installed with small ends pointing up and toward nose of helicopter.

- c. Connect disconnect fitting and cable clips to studs on cabin floor (Figure 2-4-45-12, Sheet 3, Detail D and Detail E).
- d. Screw in upper seat latch hooks to take up cable slack.
- e. Lock locking lever on latch (Figure 2-4-45-12, Sheet 1, Detail B).

2-4-46 PASSENGER’S SEAT RESTRAINT SYSTEM.

PARAGRAPH BREAKDOWN

	PAGE
2-4-46.1 Replace Passenger’s Seat Restraint System	2-4-46-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

References

- PARA 1-6-16

2-4-46.1 Replace Passenger’s Seat Restraint System.

NOTE

Replacement procedure for all passenger’s seat restraints is the same.

2-4-46.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Remove bolt, washers, and nut securing lap belt fittings to clamp on seat frame on each side of seat (Figure 2-4-46-1, Detail B).
- Remove bolts, washers, nuts, spacers, and clamps securing shoulder strap fittings to rear of seat frame (Figure 2-4-46-1, Detail A).
- Remove restraint system from seat.

2-4-46.1.2. Install.

- Turn off all electrical power (PARA 1-6-16).
- Place restraint system on seat.
- Attach shoulder straps as follows (Figure 2-4-46-1, Detail A):

NOTE

Position clamps with counterbore ends facing rear. Spacers are installed in front locations.

- Install shoulder strap fittings in clamps and secure with bolts, washers, spacers, and nuts.
- Position clamps parallel to floor with bolts orientated perpendicular to floor.
- TORQUE NUTS TO 90 - 100 INCH-POUNDS.**

- Install lap belt fittings to clamps on sides of seat frame with bolts, washers, and nuts. **BELT MUST BE FREE TO PIVOT** (Figure 2-4-46-1, Detail B).

2-4-46.1.3. Test.

- Sit in passenger seat. Attach lap belt and shoulder harness to buckle.
- Adjust lap belt and shoulder harness to maximum tightness. Lean hard against restraints. **ADJUSTMENTS MUST HOLD.** Harnesses which loosen must be replaced or repaired.

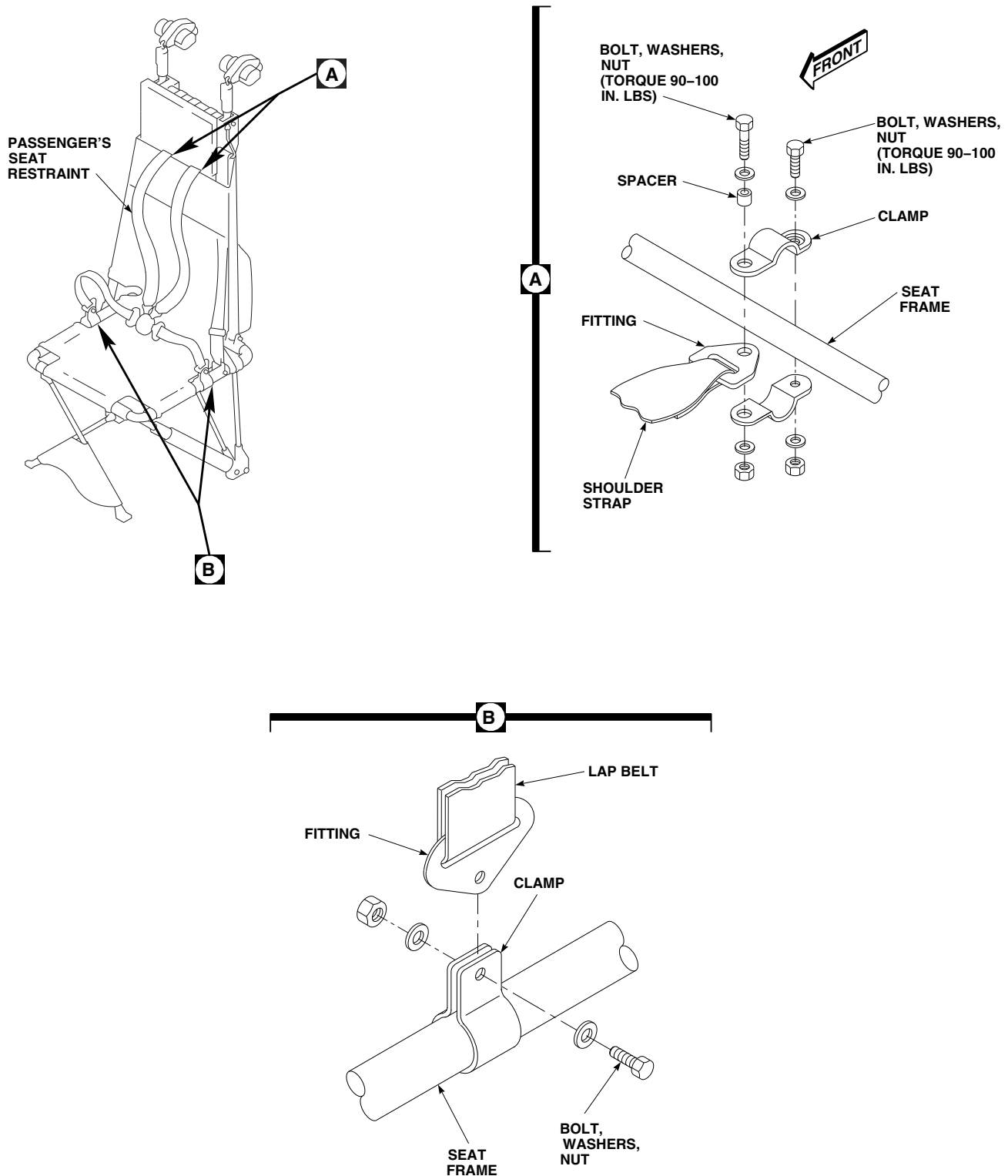
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FIGURE 2-4-46-1. REPLACE PASSENGER'S SEAT RESTRAINT SYSTEM

- c. Lean into restraints while operating buckle release. **RESTRAINT RELEASE SHALL REQUIRE RELEASE PRESSURE NO**

GREATER THAN CAN BE APPLIED WITH ONE FINGER. Replace buckles failing release test.

2-4-47 CREW CHIEF’S SEAT RESTRAINT SYSTEM.

PARAGRAPH BREAKDOWN

	PAGE
2-4-47.1 Replace Crew Chief’s Seat Restraint System	2-4-47-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Hex Wrench, 7/32-inch

References

- PARA 2-4-45

2-4-47.1 Replace Crew Chief’s Seat Restraint System.

2-4-47.1.1 Remove.

- Remove head rest (PARA 2-4-45).
- Remove bolts, washers, nuts, and bushing securing lap belt reels to seat frame (Figure 2-4-47-1, Detail A).
- Remove screw, washers, and nut securing shoulder strap reel to rear of seat frame (Figure 2-4-47-1, Detail C).
- Remove restraint from seat.

WARNING

Accidental roller release can cause personal injury and reel damage. Use care to prevent accidental release. Replace a sprung reel.

NOTE

Shoulder harness is sewn onto inertia reel roller. Remove and replace shoulder harness and reel as a unit.

- To remove and install lap belt on inertia reel roller, perform the following:
 - Remove reel cover.
 - Fully extend belt.
 - Insert a 7/32-inch hex wrench in reel. Hold roller in position (Figure 2-4-47-1, Detail B).
 - Push belt and insert out of roller.
 - Withdraw belt. Do not release roller.
 - Secure roller.
- To replace belt on roller, perform the following:
 - Carefully wind spring until tight. Back off ½ to 1 ½ turns.

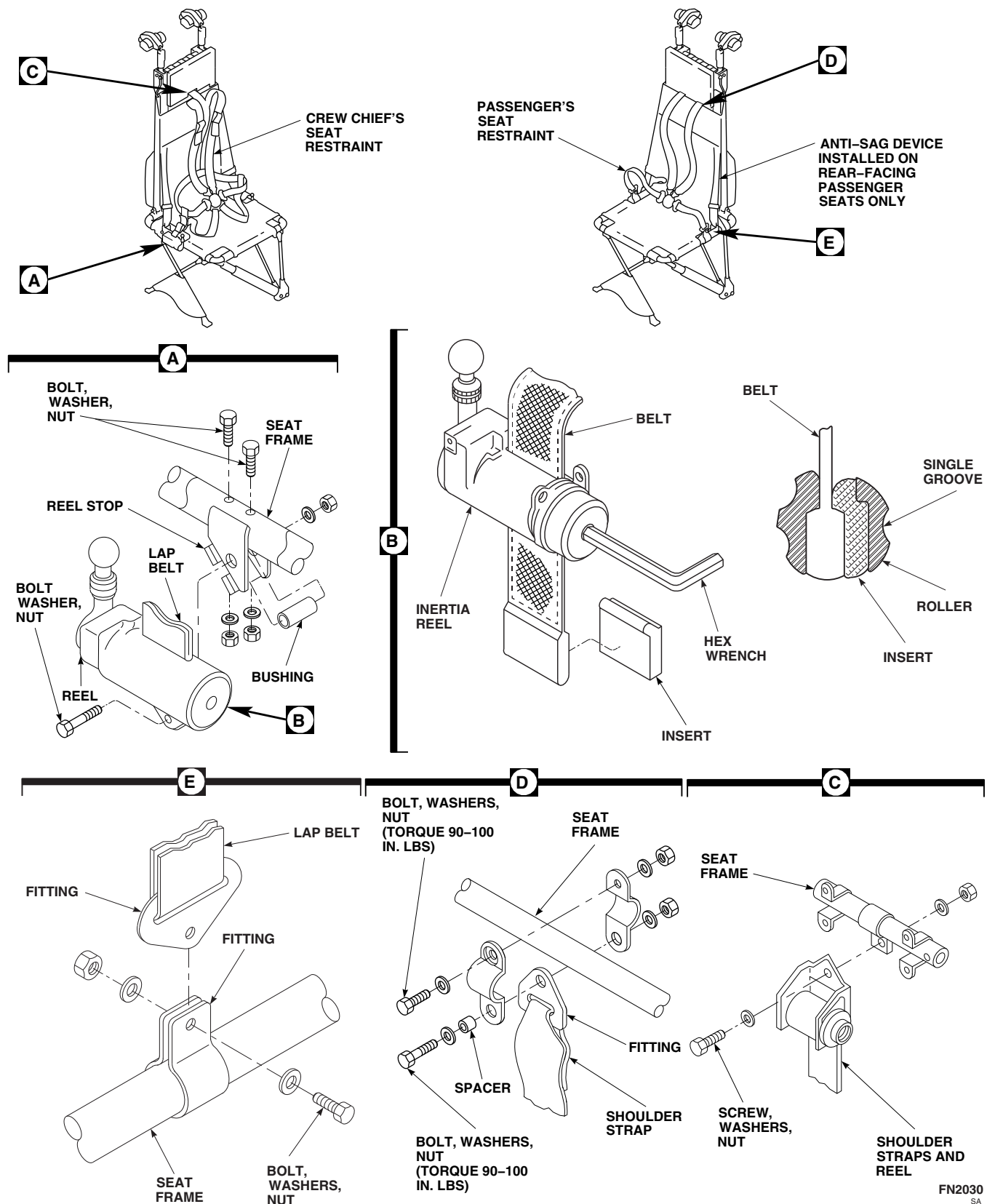


FIGURE 2-4-47-1. REPLACE CREW CHIEF'S SEAT RESTRAINT SYSTEM

- (2) Install belt and insert with insert closest to single roller groove.

2-4-47.1.2 Install.

- a. Place restraint system on seat.
- b. Secure shoulder strap and reel to rear of seat frame using screws, washers, and nut (Figure 2-4-47-1, Detail C).
- c. Install lap belt reels to seat frame using bolts, washers, bushing, and nuts. **REELS MUST BE FREE TO PIVOT** (Figure 2-4-47-1, Detail A).
- d. Install head rest (PARA 2-4-45).

2-4-47.1.3 Test.

- a. Sit in crew chief's seat. Attach lap belt and shoulder harness to buckle.

- b. Adjust lap belt and shoulder harness to maximum tightness.
- c. Put lap belt inertia reels in MANUAL-LOCK position.
- d. Lean hard into restraint system. **HARNESS ADJUSTMENTS MUST HOLD.** Harnesses which loosen must be replaced or repaired. Some webbing will come out of inertia reels as webbing tightens on rollers.
- e. Lean into restraints while operating buckle release. **RESTRAINT RELEASE SHALL REQUIRE RELEASE PRESSURE NO GREATER THAN CAN BE APPLIED WITH ONE FINGER.** Replace buckles failing release test.

2-4-48 CREW CHIEF’S WINDOW.

PARAGRAPH BREAKDOWN

	PAGE
2-4-48.1 Replace Crew Chief’s Window	2-4-48-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- String

References

- PARA 1-6-16
- PARA 2-4-42

- PARA 2-4-49
- PARA 2-6-24
- TM 11-70-23

2-4-48.1 Replace Crew Chief’s Window.

NOTE

Replacement of left and right crew chief’s windows is the same.

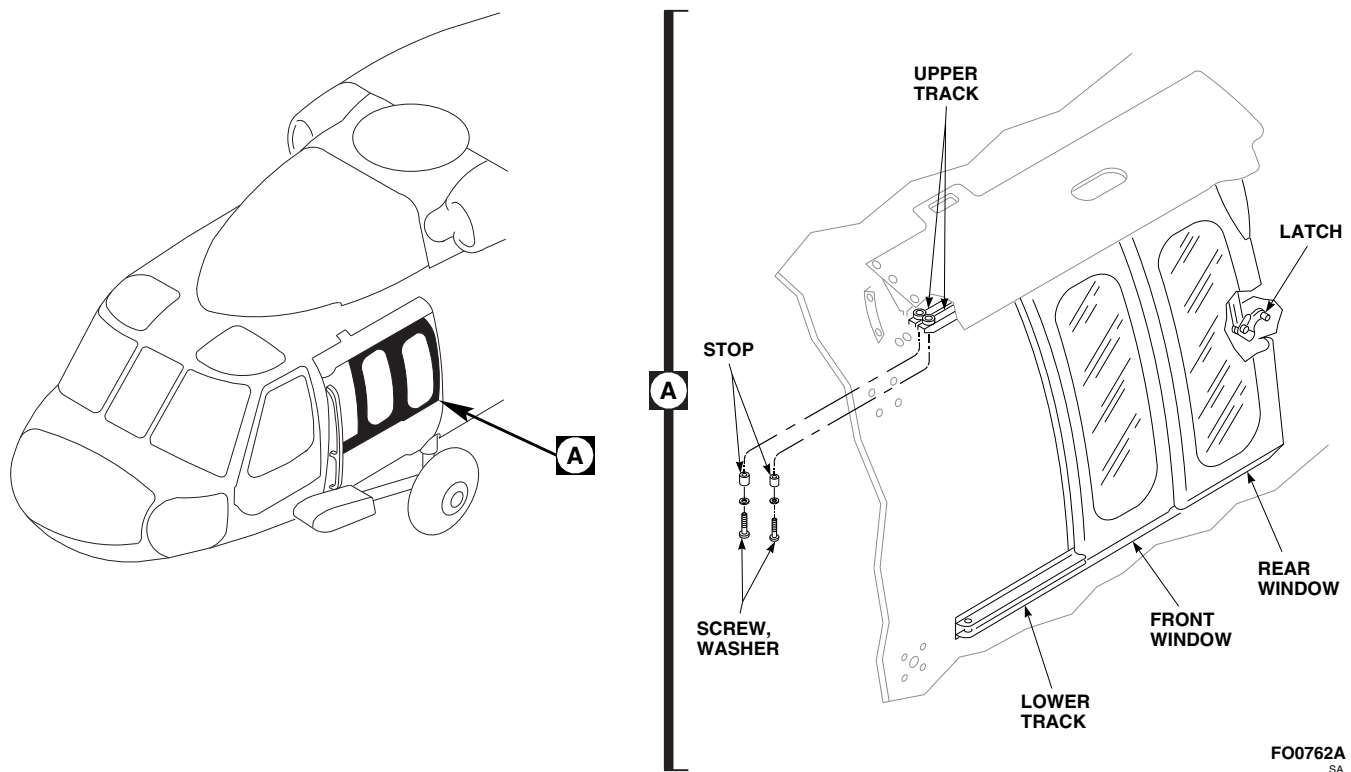
2-4-48.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Remove VHF/FM homing antenna (TM 11-70-23).
- Tie string to antenna lead and push lead back into helicopter.
- Remove fairing cap from upper track fairing (PARA 2-6-24).
- Remove screws, washers, and stops from upper tracks (Figure 2-4-48-1, Detail A).

- Unlatch rear window. Slide both windows forward, out of tracks, and remove from helicopter.

2-4-48.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Line up rear window with tracks and start window on track about 6-inches. Start front window on tracks and slide both windows to closed position (Figure 2-4-48-1, Detail A).
- Check clearance between window assemblies and fuselage to avoid chafing of outer skin. **MINIMUM CLEARANCE ALLOWED IS 0.040-INCH.**
- If rear window frame binds against forward window, trim rear frame.
- Latch rear window. Check that rear seal is tight against frame. **NO GAPS ALLOWED.** If gap is found, proceed as follows:

FO0762A
SA**FIGURE 2-4-48-1. REPLACE CREW CHIEF'S WINDOW**

- (1) On helicopters with windows, 70000-02400-041/-042, adjust pin block (PARA 2-4-49).
 - (2) On helicopters with windows, 70216-02402-047 through -050, adjust pivot plate (PARA 2-4-49).
- f. Install screws, washers, and stops in upper tracks.
 - g. Perform crew chief's window leak test (PARA 2-4-42).
 - h. Install fairing cap on upper track fairing (PARA 2-6-24).
 - i. Retrieve antenna lead from inside helicopter by pulling on string. Remove string from antenna lead.
 - j. Make sure area is clean and free of foreign material.
 - k. Install VHF/FM homing antenna (TM 11-70-23).
 - l. Perform functional check of VHF/FM homing system (TM 11-70-23).

2-4-49 CREW CHIEF’S WINDOW LATCH ASSEMBLY.

PARAGRAPH BREAKDOWN

	PAGE
2-4-49.1 Replace Crew Chief’s Window Latch Assembly	2-4-49-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Adhesion Promoter, Item 25A, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D

- Machinery Towel, Item 211, Appendix D
- Pressure Sensitive Adhesive Tape, Item 243A, Appendix D
- Primer, Item 258, Appendix D
- Washer, NAS1149C0332R

References

- Appendix D
- PARA 2-6-25

2-4-49.1 Replace Crew Chief’s Window Latch Assembly.

NOTE

Replacement of left and right crew chief’s window latch assembly is the same.

2-4-49.1.1 Remove.

- On helicopters with windows, 70000-02400-041 and -042, perform the following (Figure 2-4-49-1, Sheet 1, Detail A):
 - Open crew chief’s rear window.
 - Disconnect latch spring from latch.
 - Remove screws securing weldment to rear window.

- Remove weldment, latch and handle, and washers from rear window and fitting.
 - Remove bolts, washers, and nuts securing pin block and plate to window frame. Remove pin block and plate.
- On helicopters with windows, 70216-02402-041 through -044, perform the following (Figure 2-4-49-1, Sheet 1, Detail A):
 - Open crew chief’s rear window.
 - Disconnect latch spring from latch.
 - Remove screws securing weldment to rear window.
 - Remove weldment, latch and handle, and washers from rear window and fitting.

- (5) Remove cotter pin, headed pin, and washers from window frame.

remove rivets securing catch to crew chief's window and remove catch.

- c. On helicopters with windows, 70216-02402-047 and -048, perform the following (Figure 2-4-49-1, Sheet 1, Detail A):

- (1) Open crew chief's rear window.
- (2) Disconnect latch spring from latch.
- (3) Remove bolts and washers securing pivot plate and spacer plate to rear window.
- (4) Remove bolt and washer securing pivot plate, spacer plate, and latch and handle to rear window and fitting.
- (5) Remove pivot plate, spacer plate, latch and handle, and washers from rear window and fitting.
- (6) Remove cotter pin, headed pin, and washers from window frame.

- d. On helicopters with windows, 70216-02402-049 and -050, perform the following (Figure 2-4-49-1, Sheet 2, Detail A):

- (1) Open crew chief's rear window.
- (2) Disconnect latch spring from latch.
- (3) Remove bolts and washers securing pivot plate and spacer plate to rear window.
- (4) Remove screw securing pivot plate, spacer plate, latch and handle, and retainer to rear window and fitting.
- (5) Remove pivot plate, spacer plate, latch and handle, and washers from rear window and fitting.
- (6) Remove screw securing retainer to latch and handle. Remove retainer and separate latch from handle.
- (7) Remove cotter pin, headed pin, and washers from window frame.
- (8) Inspect catch for damage and security. **NONE ALLOWED.** If damaged or loose,

2-4-49.1.2 Inspect.

Check latch parts, mounting hardware, and window frame for cracks, misalignment, binding, interference, and loose or missing parts. **NONE ALLOWED.** If loose, missing, crooked, bent, or broken parts are found, repair or replace as necessary (PARA 2-6-25).

2-4-49.1.3 Repair.

- a. If required, remove pressure sensitive adhesive tape covering from latch handle using nonmetallic scraper and dry-cleaning solvent, Item 124, Appendix D. Wipe surface with clean machinery towel, Item 211, Appendix D.
- b. Spray on one coat of adhesion promoter, Item 25A, Appendix D, and install new pressure sensitive adhesive tape, Item 243A, Appendix D. Trim tape to shape of handle.

2-4-49.1.4 Install.

- a. On helicopters with windows, 70000-02400-041 and -042, perform the following (Figure 2-4-49-1, Sheet 1, Detail A):
 - (1) Position latch and handle, and washers between fitting and rear window.

NOTE

Make sure shaft of weldment is installed through latch and handle, washers, and fitting.

- (2) Position weldment on rear window and secure with screws.
- (3) Connect latch spring to latch.
- (4) Install pin block and plate on window frame and secure with bolts, washers, and nuts. **TORQUE NUTS TO 43 - 47 INCH-POUNDS.**
- (5) Open and close window a few times to check for correct latch operation. Check

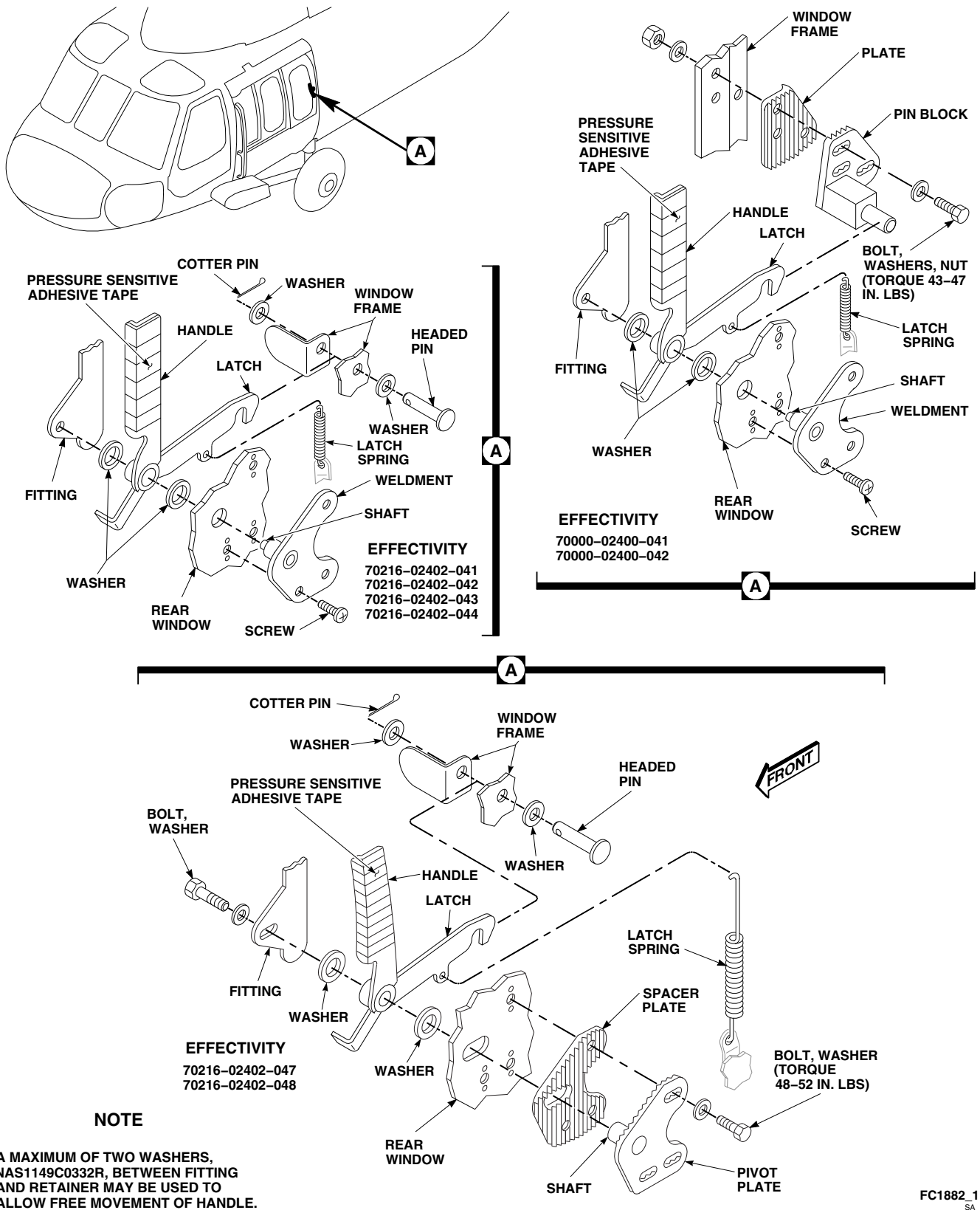


FIGURE 2-4-49-1. REPLACE CREW CHIEF'S WINDOW LATCH ASSEMBLY (SHEET 1 OF 2)

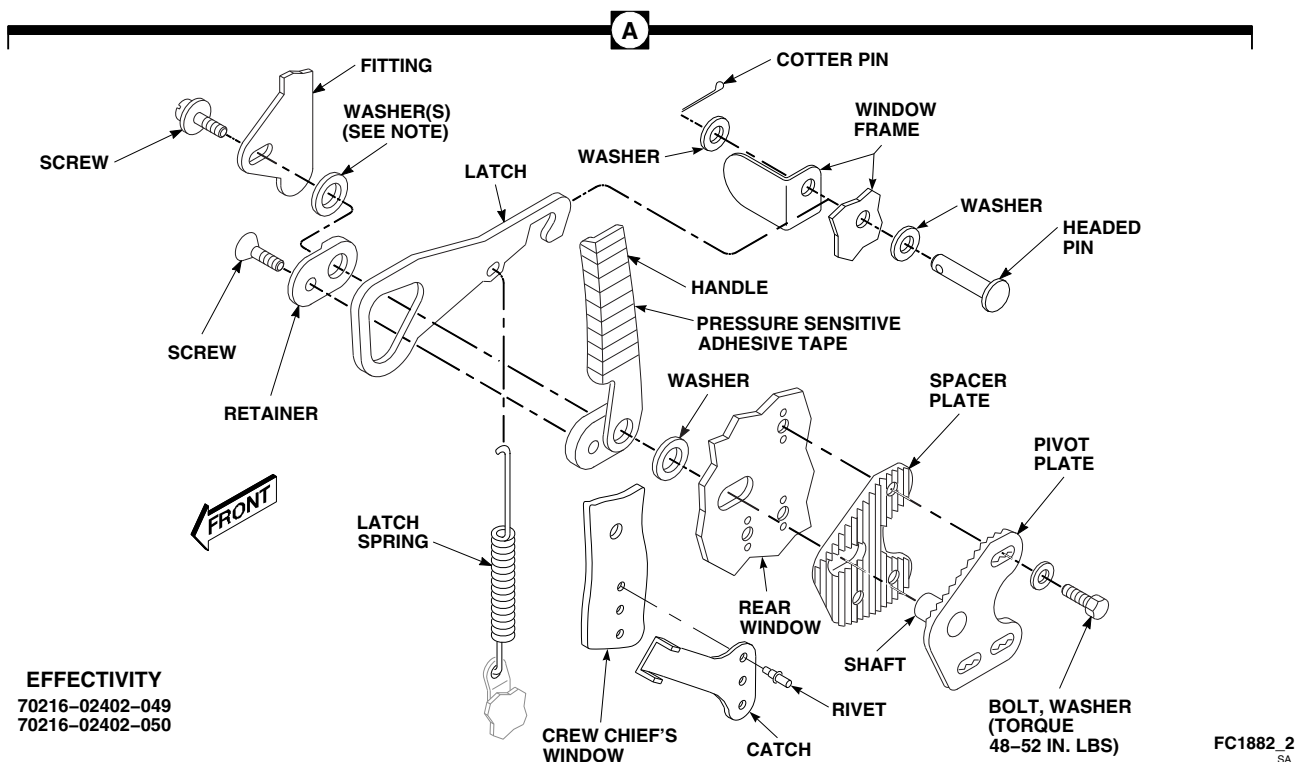


FIGURE 2-4-49-1. REPLACE CREW CHIEF'S WINDOW LATCH ASSEMBLY (SHEET 2 OF 2)

that rear seal is tight against window frame. **NO GAPS ALLOWED.** If gap is found, adjust latch assembly (PARA 2-4-49.1.5).

- b. On helicopters with windows, 70216-02402-041 through -044, perform the following (Figure 2-4-49-1, Sheet 1, Detail A):

- (1) Position latch and handle, and washers between fitting and rear window.

NOTE

Make sure shaft of weldment is installed through latch and handle, washers, and fitting.

- (2) Position weldment on rear window and secure with screws.
- (3) Connect latch spring to latch.
- (4) Install headed pin and washers in window frame and secure with cotter pin.

- (5) Open and close window a few times to check for correct latch operation. Check that rear seal is tight against window frame. **NO GAPS ALLOWED.** If gap is found, adjust latch assembly (PARA 2-4-49.1.5).

- c. On helicopters with windows, 70216-02402-047 and -048, perform the following (Figure 2-4-49-1, Sheet 1, Detail A):

- (1) Position latch and handle, and washers between fitting and rear window.

NOTE

Make sure shaft of pivot plate is installed through latch and handle, washers, and fitting.

- (2) Position pivot plate and spacer plate on rear window and secure with bolts and washers. **TORQUE BOLTS TO 48 - 52 INCH-POUNDS.**

- (3) Secure latch and handle to fitting, rear window, and pivot plate with bolt and washer.
- (4) Connect latch spring to latch.
- (5) Install headed pin and washers in window frame and secure with cotter pin.
- (6) Open and close window a few times to check for correct latch operation. Check that rear seal is tight against window frame. **NO GAPS ALLOWED.** If gap is found, adjust latch assembly (PARA 2-4-49.1.5).

d. On helicopters with windows, 70216-02402-049 and -050, perform the following (Figure 2-4-49-1, Sheet 2, Detail A):

- (1) If catch was removed, position replacement catch on crew chief's window and secure with rivets.
- (2) Assemble latch and handle and secure with retainer and screw.

NOTE

A maximum of two washers, NAS1149C0332R, between fitting and retainer may be used to allow free movement of handle.

- (3) Position latch and handle, and washer(s) between fitting and rear window.

NOTE

Make sure shaft of pivot plate is installed through latch and handle, washers, retainer, and fitting.

- (4) Position pivot plate and spacer plate on rear window and secure with bolts and washers. **TORQUE BOLTS TO 48 - 52 INCH-POUNDS.**
- (5) Apply primer, Item 258, Appendix D, to screw used to secure handle and latch to fitting, rear window, and pivot plate.

- (6) Secure latch and handle to fitting, rear window, and pivot plate with screw.
- (7) Connect latch spring to latch.
- (8) Install headed pin and washers in window frame and secure with cotter pin.
- (9) Open and close window a few times to check for correct latch operation. Check that rear seal is tight against window frame. **NO GAPS ALLOWED.** If gap is found, adjust latch assembly (PARA 2-4-49.1.5).

2-4-49.1.5 Adjust.

a. On helicopters with windows, 70000-02400-041/-042, adjust pin block as follows (Figure 2-4-49-1, Sheet 1, Detail A):

- (1) Unlatch and open window.
- (2) Loosen nuts and bolts securing pin block, and move pin block one serration toward window.
- (3) Close and latch window. Check if rear seal is compressed against window frame with no obvious gaps.
- (4) Repeat step a. (1) through step a. (3) until rear seal is compressed against window frame with no obvious gaps.
- (5) Tighten nuts and bolts.
- (6) **TORQUE NUTS TO 43 - 47 INCH-POUNDS.**

b. On helicopters with windows, 70216-02402-047 through -050, adjust pivot plate as follows (Figure 2-4-49-1, Sheet 1, Detail A and Sheet 2, Detail A):

- (1) Unlatch and open window.
- (2) Loosen bolts securing pivot plate, and move pivot plate front or rear as needed to compress rear seal against window frame.
- (3) Close and latch window. Check if rear seal is compressed against window frame with no obvious gaps.

- (4) Repeat step a. (1) through step a. (3) until rear seal is compressed against window frame with no obvious gaps.
- (5) Tighten bolts.

- (6) **TORQUE BOLTS TO 48 - 52 INCH-
POUNDS.**

2-4-50 CREW CHIEF’S WINDOW SLIDE.

PARAGRAPH BREAKDOWN

		PAGE
2-4-50.1	Replace Crew Chief’s Window Slide	2-4-50-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

2-4-50.1 Replace Crew Chief’s Window Slide.

NOTE

Replacement of right and left crew chief’s window slide is the same.

2-4-50.1.1. Remove.

- Remove screws and washers attaching slide assembly to window (Figure 2-4-50-1, Detail A).

- Remove slide assembly.

2-4-50.1.2. Install.

- Place replacement slide assembly on window (Figure 2-4-50-1, Detail A).
- Install screws and washers.

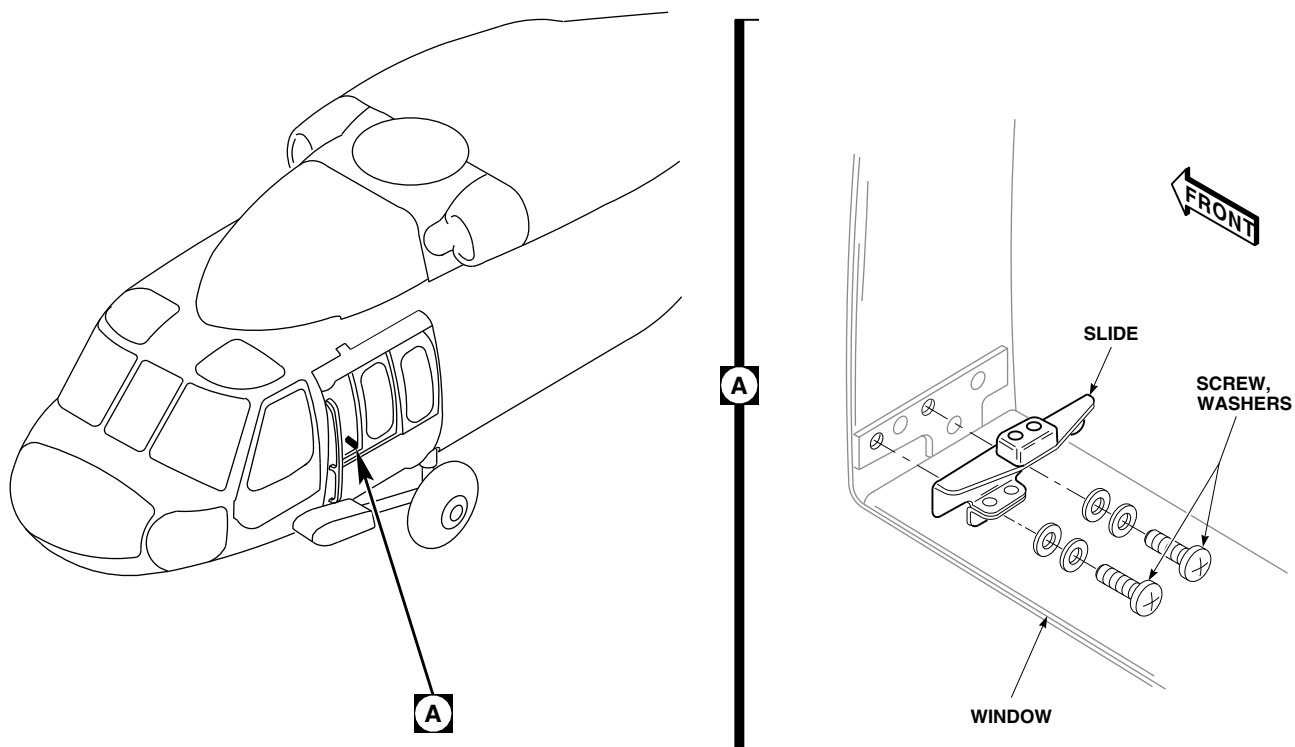
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FIGURE 2-4-50-1. REPLACE CREW CHIEF'S WINDOW SLIDE

2-4-50-2

2-4-51 CREW CHIEF’S WINDOW SEALS.

PARAGRAPH BREAKDOWN

	PAGE
2-4-51.1 Replace Crew Chief’s Window Seals	2-4-51-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Drill Set
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Adhesive, Item 37, Appendix D
- Adhesive, Item 57, Appendix D
- Crew Chief’s Window Brush Seal, Locally-Made, Figure H-183, Appendix H

- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Petrolatum, Item 234, Appendix D

References

- Appendix D
- Appendix H
- PARA 2-4-48
- PARA 2-6-5

2-4-51.1 Replace Crew Chief’s Window Seals.

NOTE

Replacement of right and left crew chief’s window seals is the same.

2-4-51.1.1 Remove.

- Remove crew chief’s window(s) (PARA 2-4-48).

NOTE

A rubber seal bonded to a vertical angle bears against rear surface of front window forward seal with window

closed. Remove and install this seal in same way as seals bonded to windows.

- Remove damaged rear seal(s) from window using nonmetallic scraper.
- Remove front seal and retainer by drilling out rivets (Figure 2-4-51-1, Detail A).
-  Drill out rivets and remove clip (Figure 2-4-51-1, Detail B).
- Remove damaged seal plug from lower window track using nonmetallic scraper.
- Remove damaged brush seals from lower track and rear window by pushing out of retainer

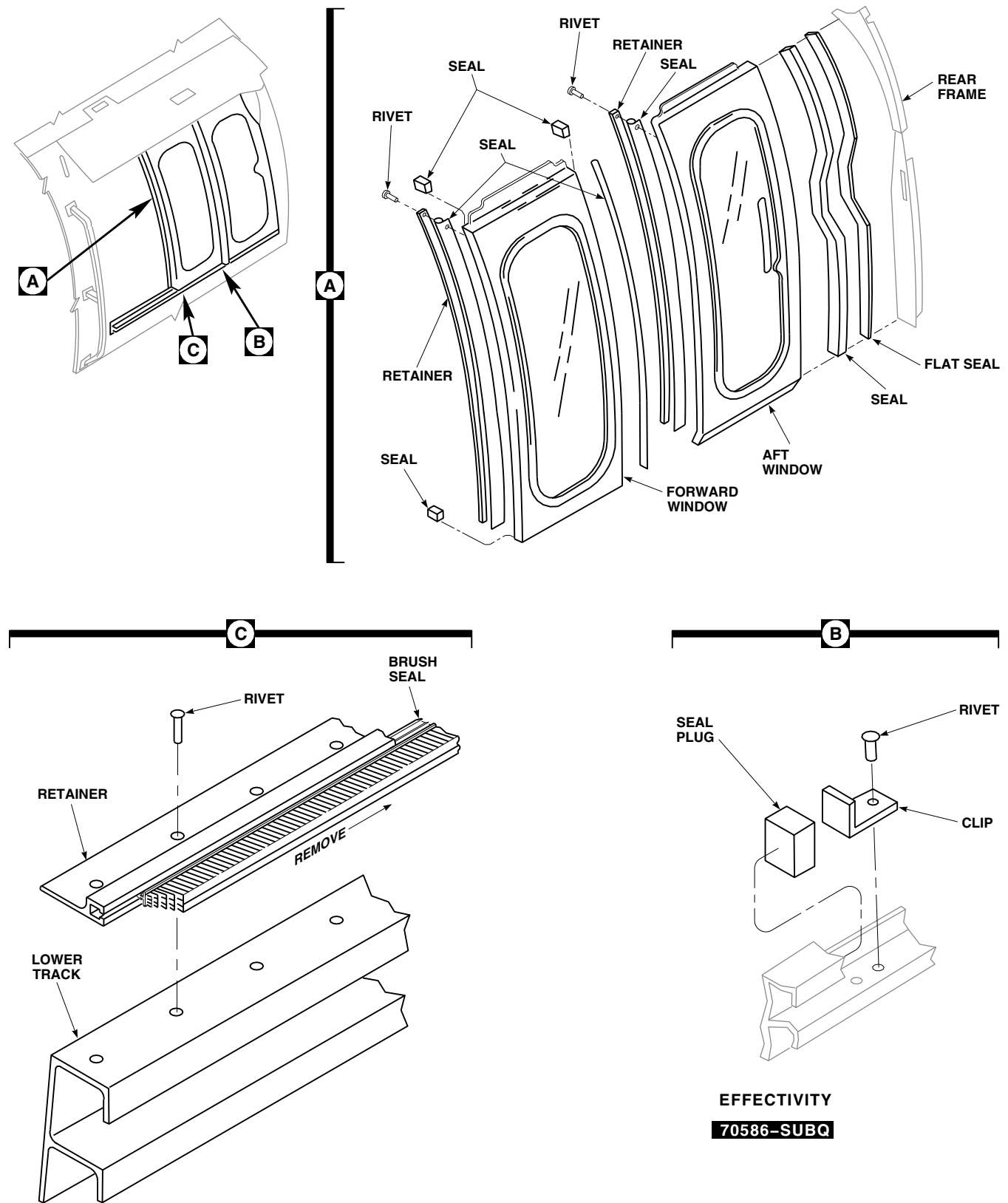


FIGURE 2-4-51-1. REPLACE CREW CHIEF'S WINDOW SEALS

grooves, using nonmetallic scraper (Figure 2-4-51-1, Detail C).

2-4-51.1.2 Install.



Damage to plastic windows will result if methyl ethyl ketone contacts them. Do not allow methyl ethyl ketone to contact plastic windows.

- a. Clean adhesive from window seal bonding surfaces with machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D, and nonmetallic scraper. Wipe dry with clean machinery towel.
- b. Place replacement seal(s) on window, rear frame, and track. Bond seals in position with adhesive, Item 37, Appendix D. Fill any gaps between seal and window with adhesive.
- c. Bond vertical seal to angle, attached to rear of front door with adhesive, Item 57, Appendix D.

- d. Bond vertical seals to angles attached to fuselage at STA 265.0 with adhesive, Item 57, Appendix D.
- e. Install front seal and retainer with rivets (Figure 2-4-51-1, Detail A).
- f. **70587-SUBQ** Install clip with rivets (Figure 2-4-51-1, Detail B).
- g. Locally make crew chief's window brush seal (Figure H-183, Appendix H).
- h. Slide replacement seal into retainer groove, using petrolatum, Item 234, Appendix D, for lubrication if required (Figure 2-4-51-1, Detail C).
- i. Install crew chief's window(s) (PARA 2-4-48).

NOTE

All seals are to be free of primer and paint overspray.

- j. Touch up paint, as required (PARA 2-6-5).

2-4-52 CREW CHIEF’S WINDOW UPPER TRACK SEALS.

PARAGRAPH BREAKDOWN

		PAGE
2-4-52.1	Replace Crew Chief’s Window Upper Track Seals	2-4-52-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Adhesive, Item 37, Appendix D
- Crew Chief’s Window Brush Seal, Locally-Made, Figure H-183, Appendix H

- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Petrolatum, Item 234, Appendix D
- Washer, 70500-02056-102

References

- Appendix D
- Appendix H
- PARA 2-6-5

2-4-52.1 Replace Crew Chief’s Window Upper Track Seals.

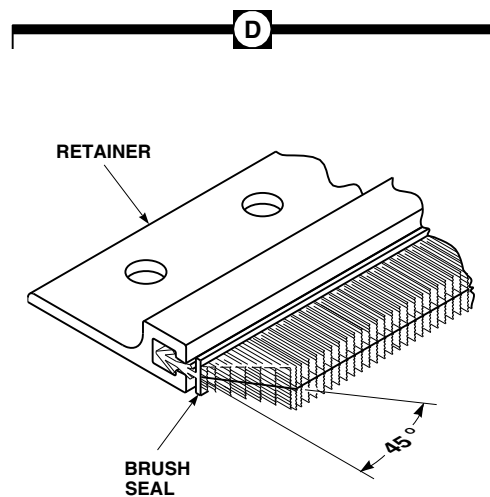
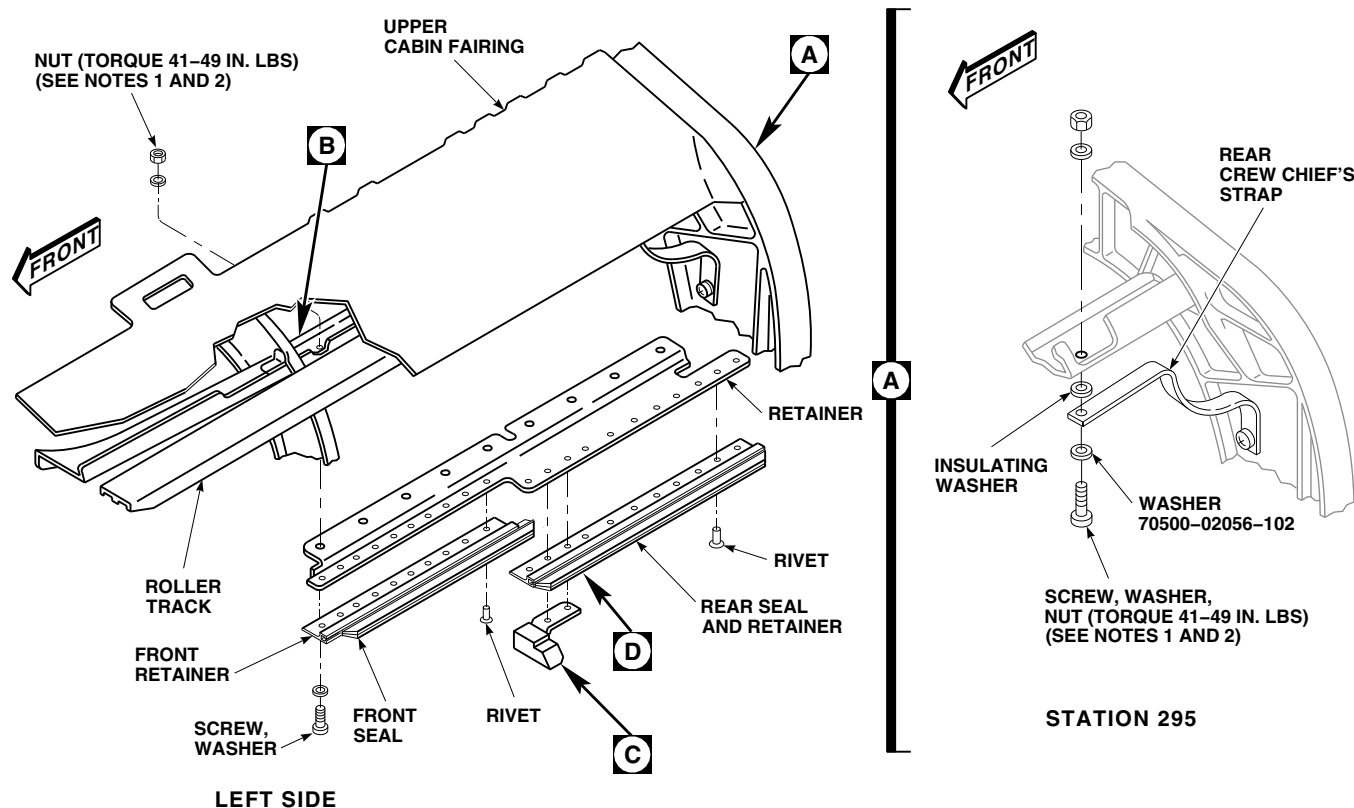
NOTE

Replacement of right and left crew chief’s window upper track seals is the same.

2-4-52.1.1. Remove.

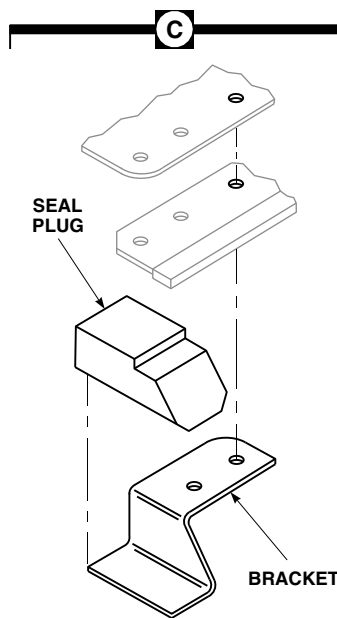
- Slide both windows fully open.
- 70583-70665** Remove screws, washers, and nuts attaching seal assembly, front crew chief’s strap and rear crew chief’s strap to airframe structure. Remove seal assembly (Figure 2-4-52-1, Detail A and Detail B).

- 70666-SUBQ** Remove screws and washers attaching seal assembly, front crew chief’s strap and rear crew chief’s strap to airframe structure. Remove seal assembly (Figure 2-4-52-1, Detail A and Detail B).
- Remove damaged brush seals from retainer by pushing out of retainer grooves, using nonmetallic scraper (Figure 2-4-52-1, Detail D).
- 70923-SUBQ** Replace seal plug as follows:
 - Remove seal plug from bracket using nonmetallic scraper (Figure 2-4-52-1, Detail C).



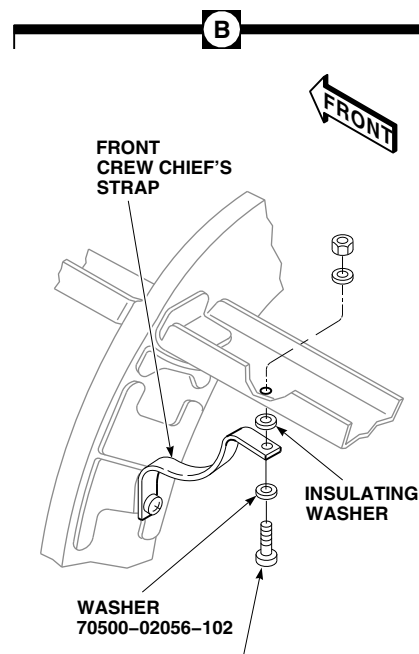
NOTES

1. NUTS, WASHERS INSTALLED ON **70583-70665**
2. NUTPLATES INSTALLED ON **70666-SUBQ**



EFFECTIVITY

70923-SUBQ



SCREW, WASHER, NUT (TORQUE 41-49 IN. LBS) (SEE NOTES 1 AND 2)

STATION 265.5

FB1064
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FIGURE 2-4-52-1. REPLACE CREW CHIEF'S WINDOW UPPER TRACK SEALS

CAUTION

Damage to plastic windows will result if methyl ethyl ketone contacts them. Do not allow methyl ethyl ketone to contact plastic windows.

- (2) Remove adhesive residue with machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D, and nonmetallic scraper. Wipe dry with clean machinery towel.

2-4-52.1.2. Install.

- a. Apply adhesive, Item 37, Appendix D, to mating surfaces of seal plug and bracket. Let dry 10 minutes.
- b. Press seal firmly into position on bracket (Figure 2-4-52-1, Detail C).
- c. Trim seal plug to 0.10 inch minimum interference with rear end of front door.
- d. Locally make crew chief's window brush seal (Figure H-183, Appendix H).
- e. Slide replacement seal into retainer groove using petrolatum, Item 234, Appendix D, if required (Figure 2-4-52-1, Detail D).

NOTE

- Long screws are installed through crew chief's straps.
- Large washer, 70500-02056-102, is installed with round edge against straps.
- Insulating washer is installed between strap and structure.

- f. **70583-70665** Loosely attach seal assembly to structure with screws, washers, and nuts. Make sure rear crew chief's strap is attached to second screw from rear and front crew chief's strap is attached to second screw from front (Figure 2-4-52-1, Detail A and Detail B).◀
- g. **70666-SUBQ** Loosely attach seal assembly to structure with screws and washers. Make sure rear crew chief's strap is attached to second screw from rear and front strap is attached to second screw from front.◀
- h. Slide windows closed, and adjust seals to make contact with windows. **TIGHTEN SCREWS FROM INSIDE CABIN.**
- i. **70880-SUBQ** Remove rubber seal plug between upper track and fuselage, near forward end of track.◀

CAUTION

Damage to plastic windows will result if methyl ethyl ketone contacts them. Do not allow methyl ethyl ketone to contact plastic windows.

- j. Remove old seal with machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D, and nonmetallic scraper. Wipe dry with clean machinery towel.
- k. Coat contacting surfaces of new seal, track, and fuselage with adhesive, Item 37, Appendix D. Let dry for about 10 minutes and press seal firmly into place.
- l. Touch up paint, as required (PARA 2-6-5).

2-4-53 CREW CHIEF’S WINDOW UPPER AND LOWER ROLLERS.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
2-4-53.1	Replace Roller Assembly	2-4-53-0	2-4-53.3	Replace Upper Rollers	2-4-53-0
2-4-53.2	Replace Lower Rollers	2-4-53-0			

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Spring Scale

References

- PARA 2-4-48

2-4-53.1 Replace Roller Assembly.

NOTE

Replacement of all roller assemblies is the same.

2-4-53.1.1. Remove.

- If necessary, trim angle on upper seal to provide additional clearance for roller bracket.
- Remove crew chief’s window(s) (PARA 2-4-48).
- On helicopters with windows, 70216-02401-047, 70216-02401-048, 70216-02401-051, and 70216-02401-052; 70216-02402-043, 70216-02402-044, and 70216-02402-047 through 70216-02402-050, remove screws, washers, and nuts attaching roller to window(s). Remove roller (Figure 2-4-53-1, Detail A).
- On helicopters with window, 70216-02402-041 through 70216-02402-044, remove screws attaching roller to window. Remove roller.

2-4-53.1.2. Install.

- On helicopters with windows, 70216-02401-047, 70216-02401-048, 70216-02401-051, and 70216-02401-052; 70216-02402-043, 70216-02402-044, and 70216-02402-047 through 70216-02402-050, place replacement roller on window. Fasten roller to window with screws, washers, and nuts (Figure 2-4-53-1, Detail A).
- On helicopters with window, 70216-02402-041, through 70216-02402-044, place replacement roller on window. Fasten roller to window with screws.
- Install crew chief’s window(s) (PARA 2-4-48).
- Slide window to front and rear, and check for correct operation.
- Using spring scale, close window. **MAXIMUM FORCE TO CLOSE WINDOW NOT TO EXCEED 15 POUNDS.**
- On helicopters with windows, 70216-02401-043 through 70216-02401-048, 70216-02401-051, and 70216-02401-052, adjust window(s)

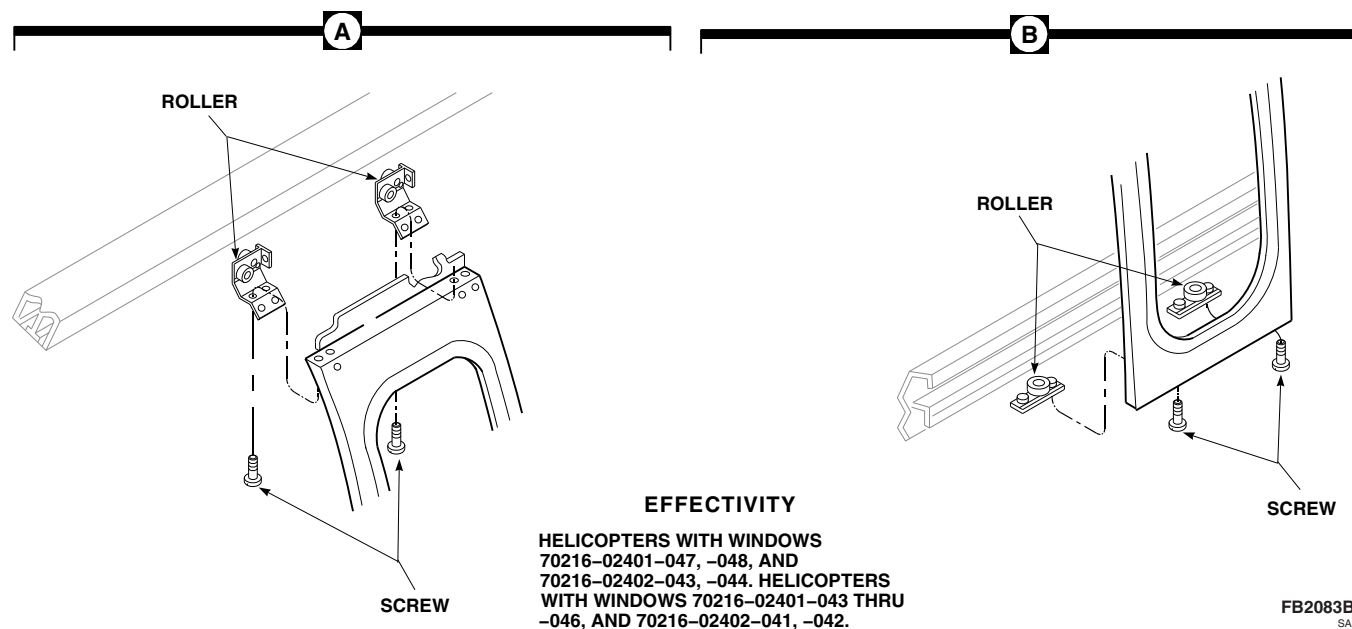
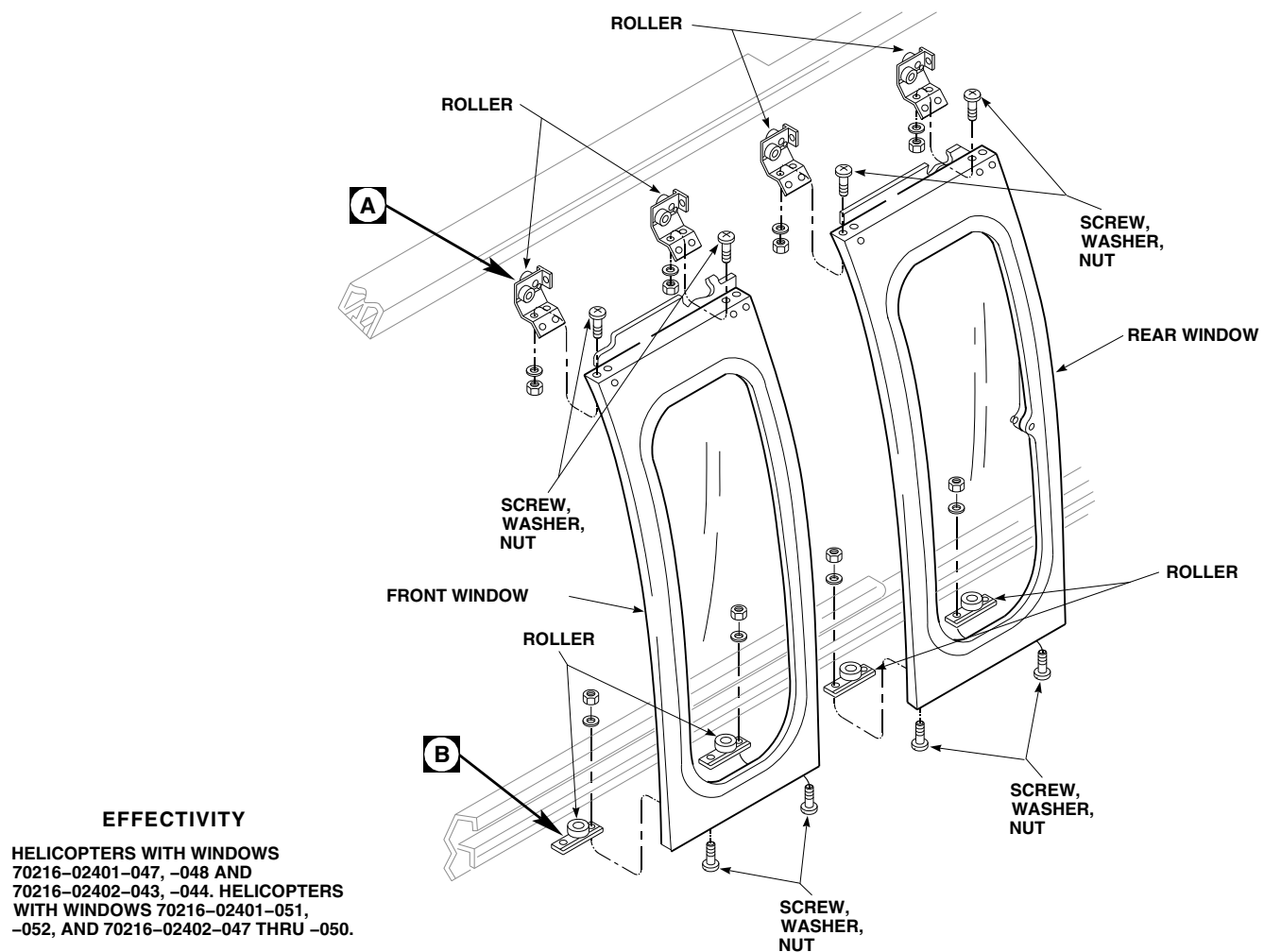
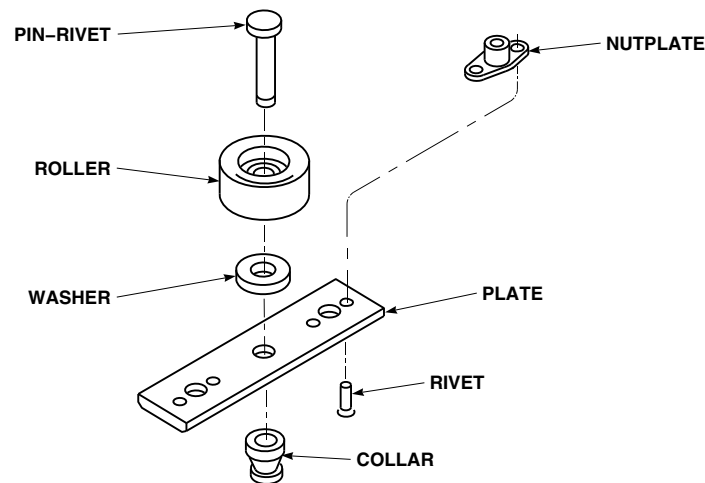


FIGURE 2-4-53-1. REPLACE CREW CHIEF'S WINDOW UPPER/LOWER ROLLERS

2-4-53-2



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**FIGURE 2-4-53-2. REPAIR CREW CHIEF'S WINDOW UPPER/LOWER ROLLERS
(SHEET 1 OF 2)**

as necessary. To do this, loosen roller screws and nuts, move window in or out. **TIGHTEN SCREWS AND NUTS.**

2-4-53.2 Replace Lower Rollers.

- a. On helicopters with windows, 70216-02401-043 through 70216-02401-048, 70216-02401-051, 70216-02401-052; 70216-02402-049 and 70216-02402-050, remove lower roller assembly from window (PARA 2-4-53.1.1).
- b. Remove pin-rivet and collar from lower roller assembly. Remove roller and washer (Figure 2-4-53-2, Sheet 1).
- c. On helicopters with windows, 70216-02402-049 and 70216-02402-050, remove rivet retaining nutplate to plate, if required.
- d. Install roller and washer to plate with pin-rivet and collar.
- e. On helicopters with windows, 70216-02402-049 and 70216-02402-050, install nutplates with rivets.
- f. Install roller assemblies on windows (PARA 2-4-53.1.2).

2-4-53.3 Replace Upper Rollers.

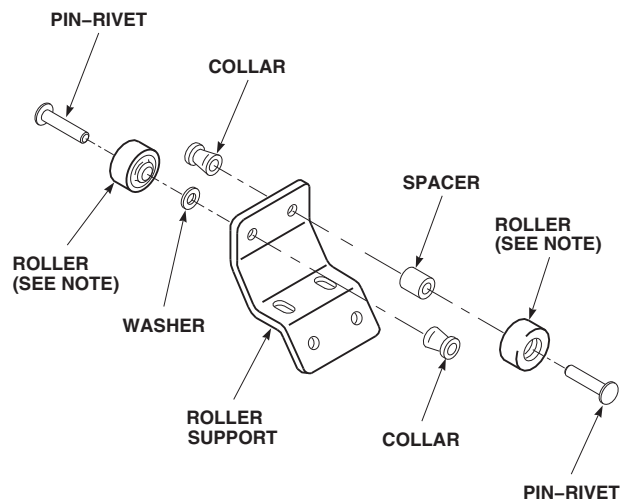
- a. Remove upper roller assembly from window (PARA 2-4-53.1).

- b. On roller assemblies, 70216-02407-045 through 70216-02407-050, and 70216-02407-054, remove pin-rivets and collars securing rollers washer, and spacer to collar support (Figure 2-4-53-2, Sheet 2).
- c. On roller assemblies, 70216-02407-052 and 70216-02407-053, remove pin-rivets and collars securing rollers, washers and tee to roller support (Figure 2-4-53-2, Sheet 2).
- d. Remove rivets securing seal and plate to tee if required.

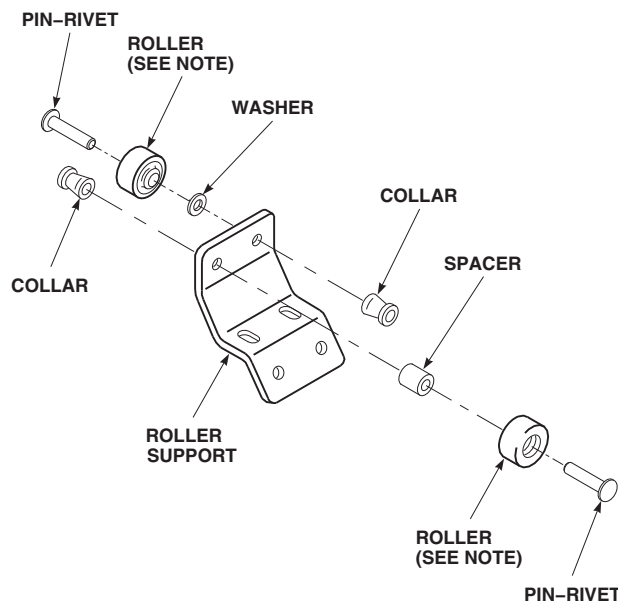
NOTE

On roller assemblies, 70216-02407-045 through 70216-02407-050, and 70216-02407-054, an additional washer, may be used to improve fit of roller.

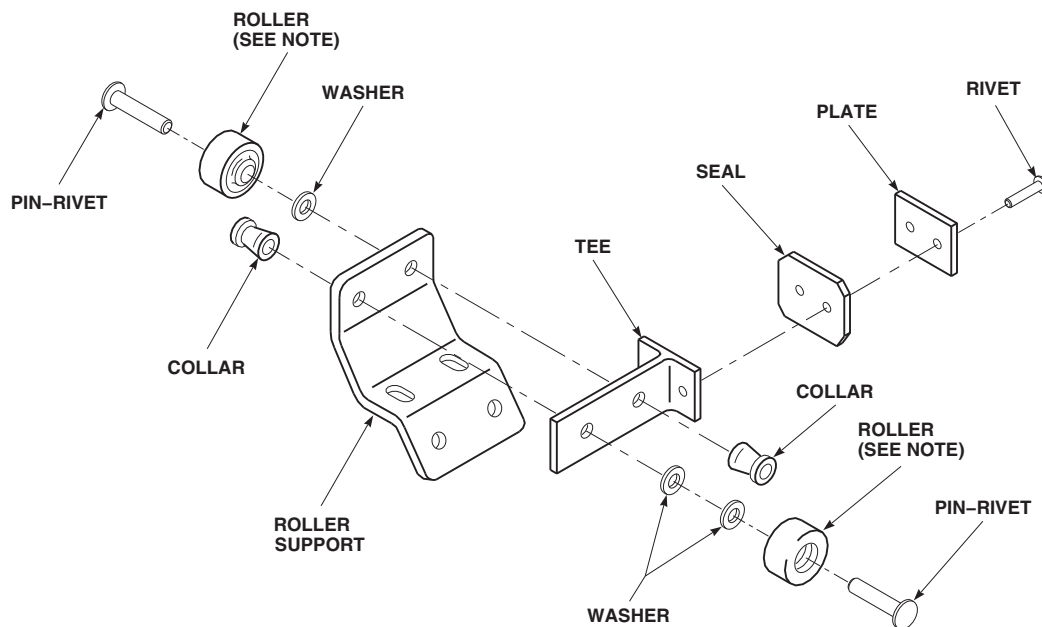
- e. On roller assemblies, 70216-02407-045 through 70216-02407-050, and 70216-02407-054, install rollers, spacer, washer, additional washer if required, to roller support with pin-rivet and collar (Figure 2-4-53-2, Sheet 2). **MAXIMUM DISTANCE ACROSS FACE OF ROLLERS SHALL NOT BE MORE THAN 1.08-INCHES.**
- f. On roller assemblies, 70216-02407-052 and 70216-02407-053, install seal and plate to tee with rivets, if required (Figure 2-4-53-2, Sheet 2).

**EFFECTIVITY**

ROLLER ASSEMBLY 70216-02407-045,
70216-02407-047 THROUGH
70216-02407-049.

**EFFECTIVITY**

ROLLER ASSEMBLY 70216-02407-046,
70216-02407-050, 70216-02407-054.

**EFFECTIVITY**

ROLLER ASSEMBLY 70216-02407-052,
70216-02407-053.

NOTE:
AN ADDITIONAL WASHER MAY BE
USED UNDER BOTH
ROLLERS TO IMPROVE FIT.

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**FIGURE 2-4-53-2. REPAIR CREW CHIEF'S WINDOW UPPER/LOWER ROLLERS
(SHEET 2 OF 2)**

NOTE

On roller assemblies, 70216-02407-052 and 70216-02407-053, an additional washer may be used to improve fit of roller.

- g. On roller assemblies, 70216-02407-052 and 70216-02407-053, install rollers, washers, and

tee to roller support with pin-rivet and collar.
MAXIMUM DISTANCE ACROSS FACE OF ROLLERS SHALL NOT BE MORE THAN 1.09-INCHES.

- h. Install roller assemblies on windows (PARA 2-4-53.1).

2-4-54 CREW CHIEF’S WINDOW SECURITY DEVICE.

PARAGRAPH BREAKDOWN

		PAGE
2-4-54.1	Replace Crew Chief’s Window Security Device	2-4-54-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

References

- PARA 2-4-96

2-4-54.1 Replace Crew Chief’s Window Security Device.

- Remove nuts, washers, and pin from window frame.

NOTE

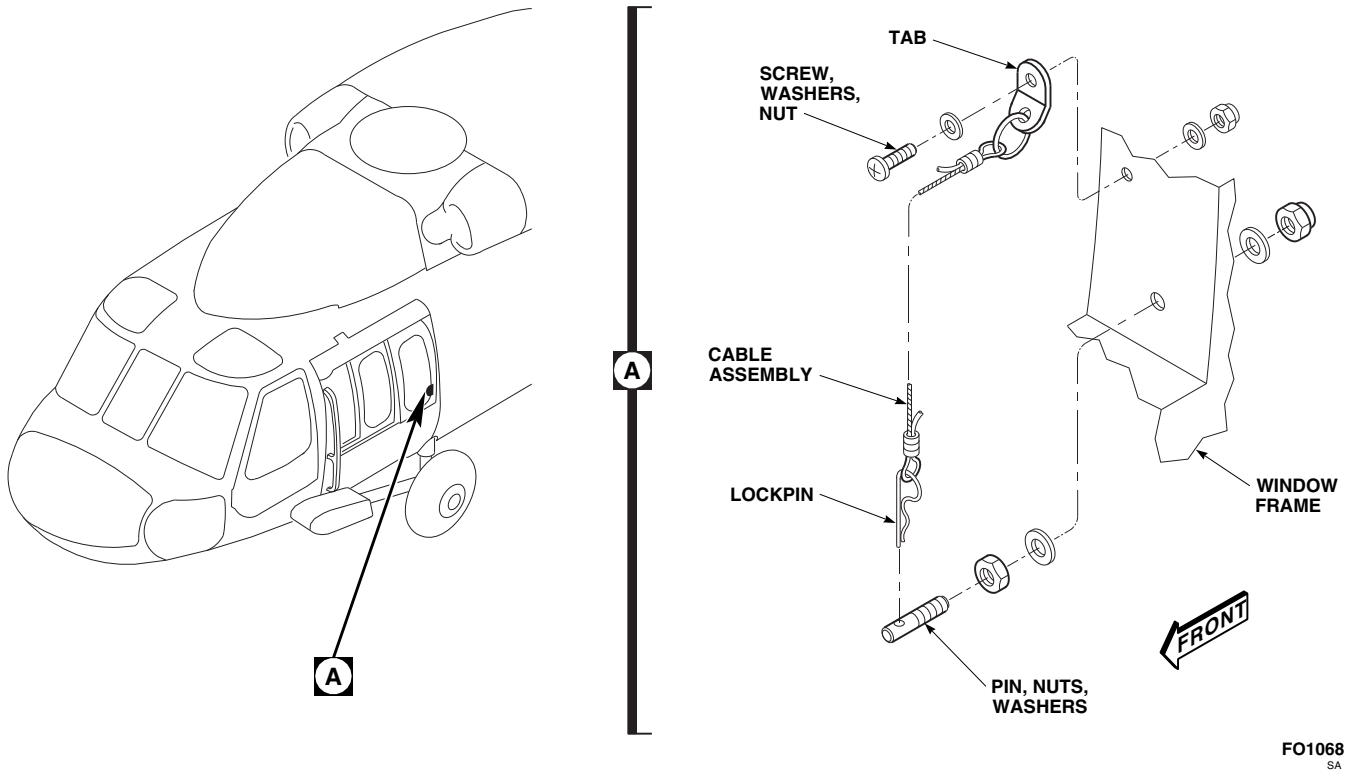
70785-SUBQ Replacement of left and right security devices is the same.

2-4-54.1.2 Install.

2-4-54.1.1 Remove.

- Remove upper main landing gear fairing (PARA 2-4-96).
- Open crew chief’s rear window.
- Remove screw, washers, and nut securing tab of cable assembly to window frame (Figure 2-4-54-1, Detail A).
- Remove lockpin from pin and remove cable assembly.

- Install nut and washer on replacement pin and install pin into window frame (Figure 2-4-54-1, Detail A).
- Secure pin to window frame with remaining nut and washer.
- Install tab of cable assembly to window frame and secure with screw, washers, and nut.
- Install lockpin in pin.
- Install upper main landing gear fairing (PARA 2-4-96).
- Close crew chief’s rear window.



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FIGURE 2-4-54-1. REPLACE CREW CHIEF'S WINDOW SECURITY DEVICE

2-4-55 CABIN DOORS.

PARAGRAPH BREAKDOWN

	PAGE
2-4-55.1 Replace Cabin Doors	2-4-55-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Molybdenum Disulfide, Item 219, Appendix D
- Petrolatum, Item 234, Appendix D

References

- Appendix D
- PARA 1-7-15
- PARA 2-3-7
- PARA 2-4-42
- PARA 2-4-64
- PARA 2-4-66

2-4-55.1 Replace Cabin Doors.

NOTE

Replacement of right and left cabin door is the same.

- With aid of assistant, slide door rearward, out of upper and lower tracks, and remove it from helicopter (Figure 2-4-55-1, Detail A).
- If door is being replaced, remove door lock (PARA 2-4-66).

2-4-55.1.1. Remove.

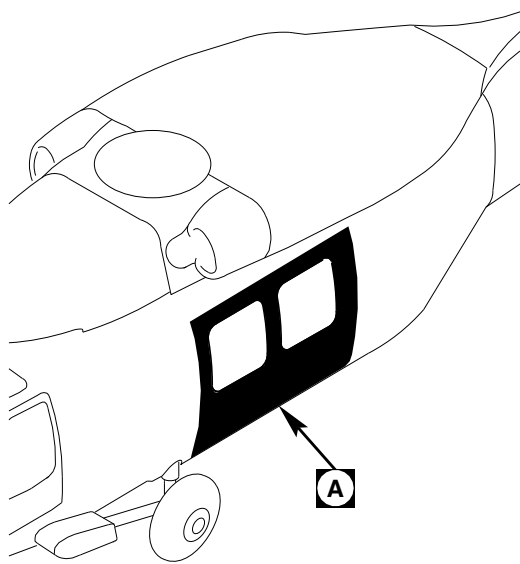
- Remove aft stop (1 thick stop or 2 thin stops) from upper track by removing bolt(s), washers, and nuts (Figure 2-4-55-1, Detail B).
- Remove stop from lower track by removing screw (Figure 2-4-55-1, Detail D).
- Open door. As door reaches full open, manually keep open stop from engaging by holding door stop open.

2-4-55.1.2. Inspect.

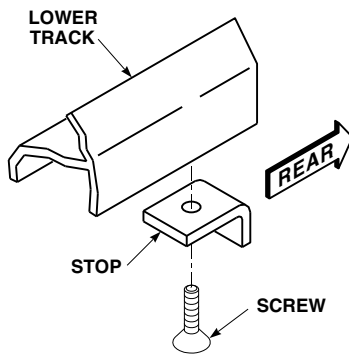
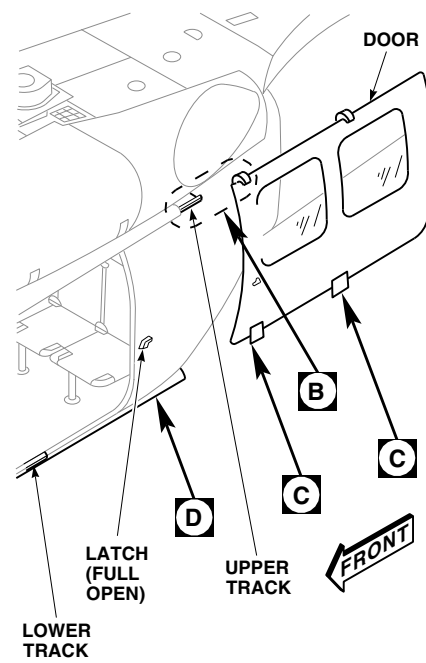
- Inspect upper and lower roller support assemblies for wear. If wear is found, refer to PARA 2-4-64.
- Inspect upper door track (PARA 1-7-15) and lower door track (PARA 2-3-7) for wear.

2-4-55.1.3. Install.

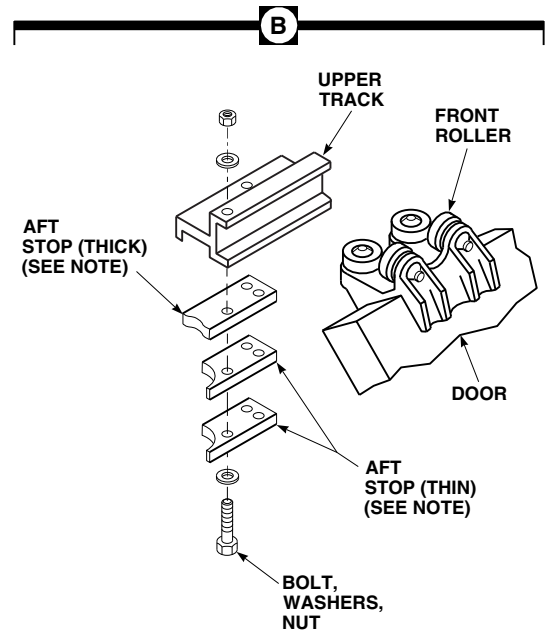
- If new door is being installed, install door lock removed from old door (PARA 2-4-66).



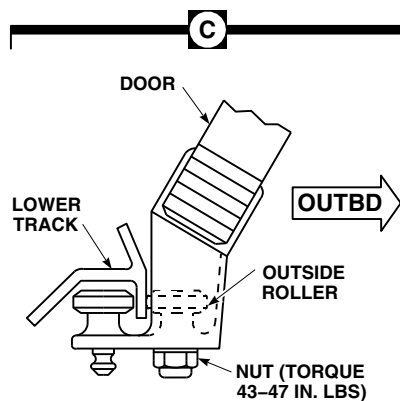
A



D



B



C

NOTE

THICK STOP REPLACES THIN STOPS. DO NOT INTERCHANGE.

FIGURE 2-4-55-1. REPLACE CABIN DOORS

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CAUTION

Do not close door with handle in locked position, as damage to latching mechanism may result.

NOTE

Up to 35 pounds of pressure may be needed to roll door forward on tracks.

- b. Lightly apply molybdenum disulfide, Item 219, Appendix D, to upper and lower tracks to ease installation. Apply molybdenum disulfide only to surface of tracks where rollers on door make contact.
- c. Apply a thin coat of petrolatum, Item 234, Appendix D, to upper and lower track seals.
- d. With aid of assistant, line up front rollers on door with upper and lower tracks. Guide front seal flap into inboard bottom angle of upper track. Guide rear seal flap under inboard bottom edge of upper track. Open door latch and carefully slide door forward until all rollers are in tracks, and close door (Figure 2-4-55-1, Detail A).

NOTE

Thick stop replaces thin stops. Do not interchange stops.

- e. Install aft stop in upper track and secure with bolt(s), washers, and nuts (Figure 2-4-55-1, Detail B).

- f. Install stop on lower track, with stop leg toward rear, and secure with screw (Figure 2-4-55-1, Detail D).

WARNING

To prevent injury to personnel and damage to equipment, make sure that lower rollers on door are engaged in lower track on helicopter through entire range of travel. If not fully engaged, cabin door may derail. Use care when opening and closing door to avoid injury.

- g. Open and close door a few times to check operation. Door shall operate smoothly without binding.
- h. Close cabin door.
- i. Check that door latch has engaged the door frame. Rotate door handle to latch door and check that door is securely closed.
- j. Perform cabin door window jettison check (PARA 2-3-7).
- k. If necessary, adjust outside roller in lower roller assemblies by loosening nut and moving roller against lower track to prevent door rattle. Move roller away from lower track to prevent binding. **TORQUE NUT TO 43 - 47 INCH-POUNDS** (Figure 2-4-55-1, Detail C).
- l. Check door for water leaks (PARA 2-4-42).

2-4-56 CABIN DOOR WINDOWS.

PARAGRAPH BREAKDOWN

		PAGE
2-4-56.1	Replace Cabin Door Windows	2-4-56-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit

Materials/Parts

- Lockwire, Item 195, Appendix D
- Masking Tape, 1-inch, Item 212, Appendix D

References

- Appendix D
- PARA 2-3-7
- PARA 2-4-42
- PARA 2-4-68
- PARA 2-6-5

2-4-56.1 Replace Cabin Door Windows.

NOTE

Replacement of both cabin door windows is the same for left and right cabin doors.

2-4-56.1.1 Remove.

- Close cabin door.



Opening emergency release lever will jettison both windows.

- If only one cabin door window is to be removed, secure other window in place with 1-inch masking tape, Item 212, Appendix D.
- Remove cabin door window jettison mechanism guard (PARA 2-4-68).

- With an assistant supporting cabin door window(s) from outside helicopter, from inside helicopter, turn emergency release lever in cabin door to open position (Figure 2-4-56-1, Detail B).
- Push out on bottom of cabin door window and remove from helicopter (Figure 2-4-56-1, Detail A).

2-4-56.1.2 Install.

- From outside helicopter, have assistant lift cabin door window(s) into position (Figure 2-4-56-1, Detail A).
- Line up locators at top of cabin door window with slots in window frame.
- From outside helicopter, have assistant push in at bottom of cabin door window.
- Turn emergency release lever to locked position (Figure 2-4-56-1, Detail B).

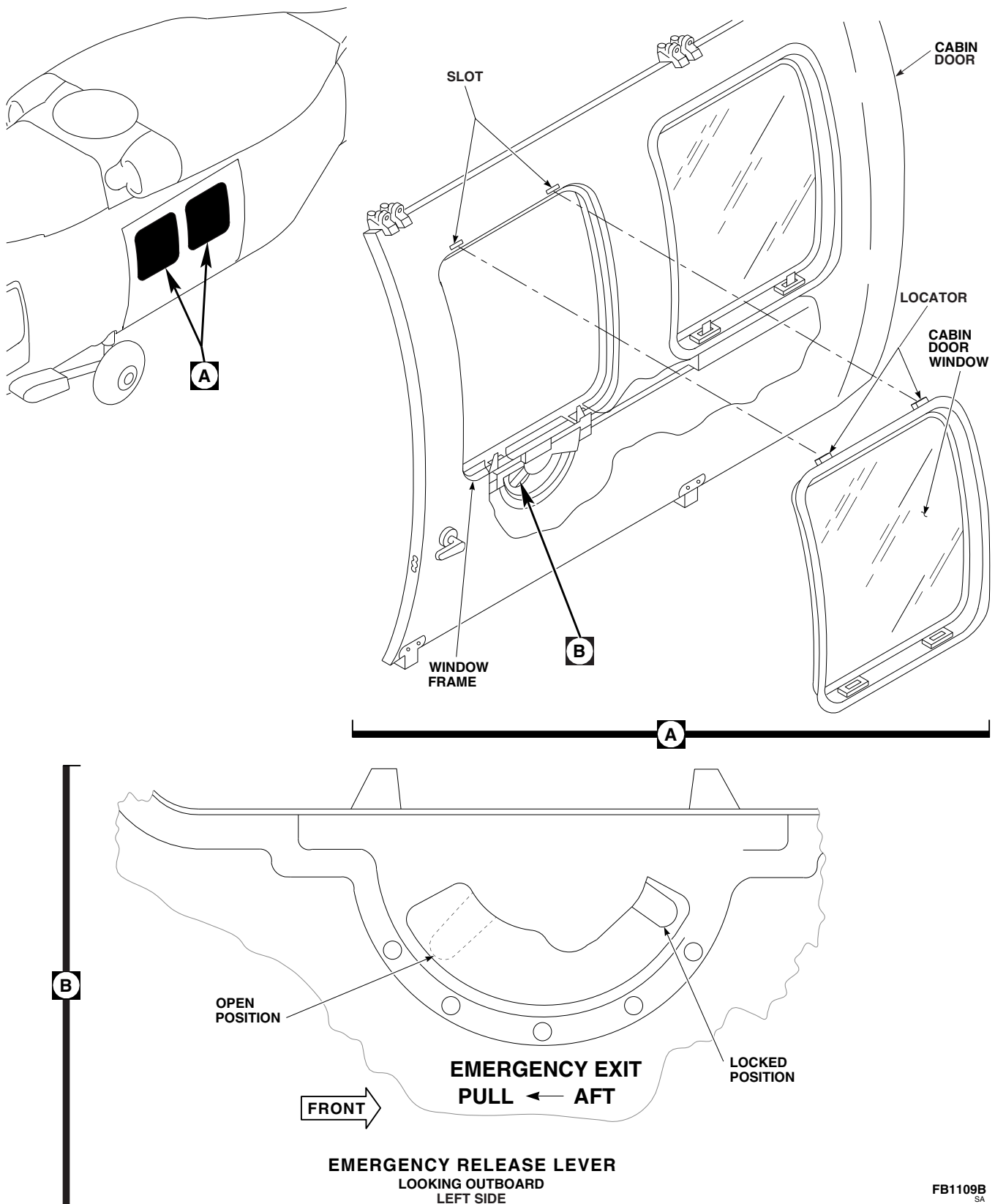


FIGURE 2-4-56-1. REPLACE CABIN DOOR WINDOWS

- e. Perform cabin door window jettison inspection (PARA 2-3-7).
- f. Using lockwire, Item 195, Appendix D, lockwire jettison handle.
- g. Install cabin door window jettison mechanism guard (PARA 2-4-68).
- h. If necessary, identify cabin door window as emergency exit (PARA 2-6-5).
- i. If necessary, remove masking tape from other cabin door window.
- j. Perform cabin door window leak test (PARA 2-4-42).

2-4-57 CABIN DOOR LOWER TRACK.

PARAGRAPH BREAKDOWN

		PAGE
2-4-57.1	Replace Cabin Door Lower Track	2-4-57-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit

Materials/Parts

- Corrosion Preventive Compound, Item 115, Appendix D

References

- Appendix D
- PARA 2-4-55
- PARA 2-4-95
- PARA 2-6-5

2-4-57.1 Replace Cabin Door Lower Track.

NOTE

Replacement of left and right cabin door lower track is the same.

2-4-57.1.1 Remove.

- Remove cabin door (PARA 2-4-55).
- Remove lower main landing gear fairing (PARA 2-4-95).
- Remove screws and washers securing cabin door lower track to fairing (Figure 2-4-57-1, Detail B).
- Remove screws and washers securing lower track to fuselage. Remove spacers.
- Remove lower track.

2-4-57.1.2 Install.

- If installing new cabin door lower track, trial-fit new lower track on fairing with screws and

washers (Figure 2-4-57-1, Detail B). Inspect fairing for mounting holes that do not line up with nutplates in lower track. If necessary, repair mismatched mounting hole(s) in fairing as follows:

- Mark mounting hole(s) in fairing that do not line up with nutplate(s) in lower track.
- Remove screws and washers securing lower track to fairing and remove lower track.
- Slot marked holes in fairing to dimensions shown in Figure 2-4-57-1, Detail C. Slot may be made in any direction provided minimum edge distance is maintained.
- Repeat step a. until proper fit is achieved.
- Touch up paint in slotted area(s) (PARA 2-6-5).

- Apply corrosion preventive compound, Item 115, Appendix D, to fairing attachment hardware.

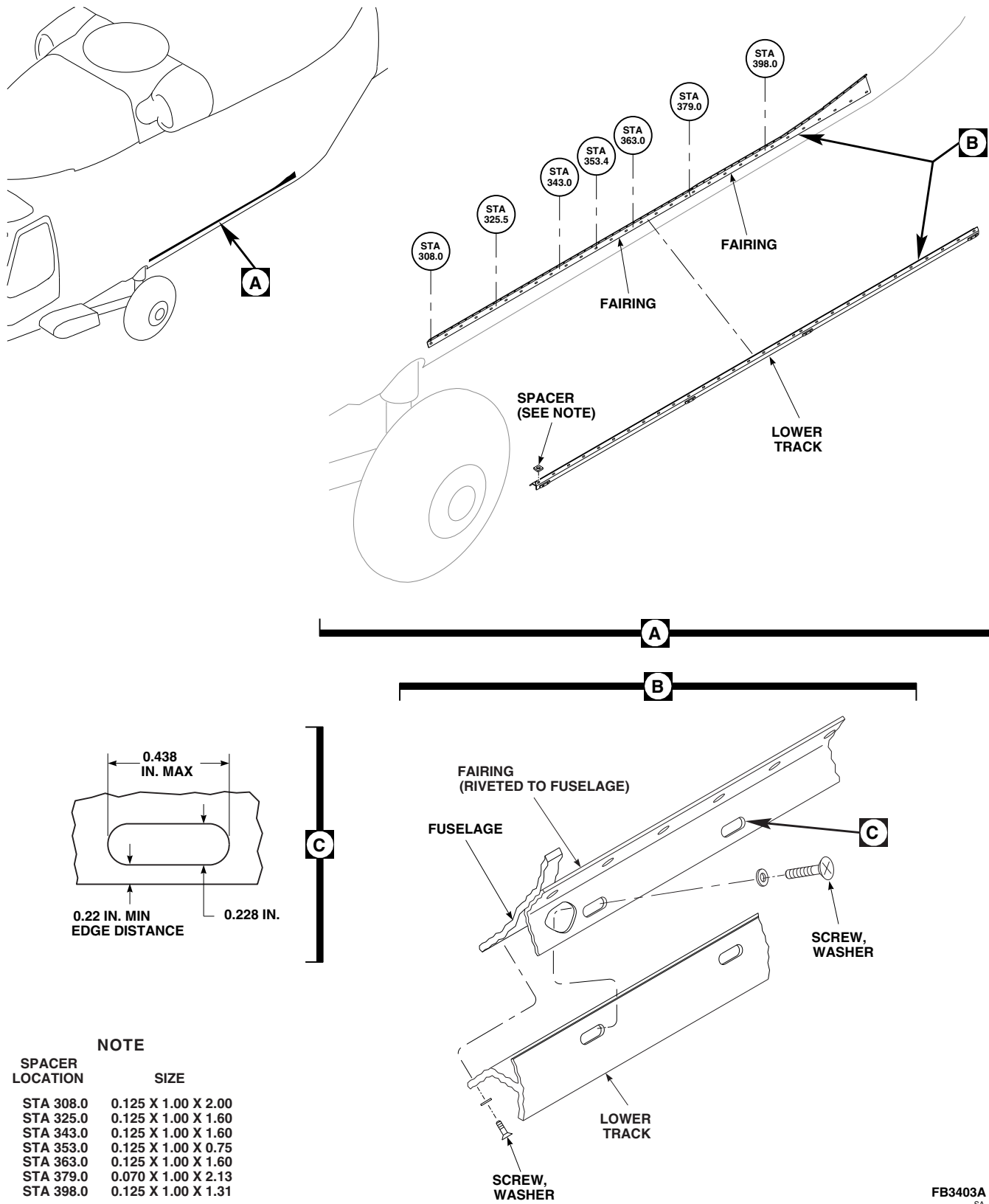


FIGURE 2-4-57-1. REPLACE CABIN DOOR LOWER TRACK

- c. Position cabin lower track on fuselage.

NOTE

Track spacers vary in size. Correct spacer location is shown in Figure 2-4-57-1, Detail A.

- d. Place spacers between lower track and fairing and secure lower track with screws and washers (Figure 2-4-57-1, Detail A and Detail B).

- e. Install lower main landing gear fairing (PARA 2-4-95).

- f. Install cabin door (PARA 2-4-55).

2-4-58 CABIN DOOR SEALS.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
2-4-58.1	Replace Window Seals	2-4-58-2	2-4-58.5	Replace Lower Track Seal Block	2-4-58-7
2-4-58.2	Replace Door Front Seals	2-4-58-3	2-4-58.6	Replace Upper Door Seal	2-4-58-7
2-4-58.3	Replace Upper Door Frame Seal Block	2-4-58-6	2-4-58.7	Replace Door Fairing Seal	2-4-58-8
2-4-58.4	Replace Door Bottom Seal or Seal Blocks/Sheath Seal	2-4-58-6	2-4-58.8	Replace Fuselage Rear Door Seal	2-4-58-8

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Blind Riveter, Hand
- Clamps
- Clecos
- Clecos, 1⁄8-inch diameter
- Drill
- Drill Bit, 5⁄32-inch diameter
- Nonmetallic Scraper, Locally-Made
- Pin Punch
- Sand Bags
- Saw Horses, Padded

Materials/Parts

- Abrasive Cloth, Item 9, Appendix D
- Acetone, Item 25, Appendix D

- Adhesive, Item 30, Appendix D
- Adhesive, Item 37, Appendix D
- Adhesive, Item 57, Appendix D
- Cabin Door Seal, Locally-Made, Figure H-76, Appendix H
- Cheesecloth, Item 90, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- Appendix H
- PARA 2-4-42
- PARA 2-4-55
- PARA 2-4-56
- PARA 2-6-1
- PARA 2-6-5

2-4-58.1 Replace Window Seals.**NOTE**

Replacement procedure for all cabin door window seals is the same for both cabin doors.

2-4-58.1.1 Remove.

- a. Remove window(s) from cabin door (PARA 2-4-56).
- b. Using nonmetallic scraper, remove damaged bulb seal from window opening flange in door (Figure 2-4-58-1).
- c. Using cheesecloth, Item 90, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean remaining adhesive from window opening flange.

NOTE

70583-70725 Have a seal on the window flange at top of window only. **70726-SUBQ** Have additional seals for front and rear window flanges.

- d. Using nonmetallic scraper, remove damaged seal(s) and rubber strip from window flange.
- e. Using cheesecloth, Item 90, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean remaining adhesive from window flange.

2-4-58.1.2 Install.**NOTE**

Bulb of seal is to face outboard.

- a. Place replacement bulb seal on window opening flange in door. Trim length to fit opening (Figure 2-4-58-1).
- b. Mark location of window jettison levers and cut seal to fit. Remove seal from window opening flange.
- c. Apply a thin even coat of adhesive, Item 37, Appendix D, to window opening flange and mating surfaces of bulb seal.

- d. Starting at bottom of window flange, install bulb seal on flange. Do not stretch seal around curved parts of frame.
- e. **70583-70725** Place replacement seal around top of window flange and cut seal to size of old seal.
- f. Lightly roughen mating surface of seal using abrasive cloth, Item 9, Appendix D. Clean roughened surface of seal using clean cheesecloth, Item 90, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.
- g. Apply a light coat of adhesive, Item 30, Appendix D, on window flange and roughened surface of seal. Allow adhesive to dry until tacky.
- h. Install seal on window flange. Do not stretch seal around corners of window. Do not let seal lose contact with window flange.
- i. Allow seal installation to cure at room temperature for 24 hours.
- j. **70726-SUBQ** Lightly roughen mating surface of replacement seal(s) using abrasive cloth, Item 9, Appendix D. Clean roughened surface of seal using clean cheesecloth, Item 90, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.
- k. Apply a light coat of adhesive, Item 30, Appendix D, to roughened surface of seal(s) and window flange. Allow to air dry until tacky.
- l. Install seal(s) and rubber strip on window flange. Do not stretch top seal around corners of window flange. Do not let seal(s) lose contact with window flange.
- m. Allow seal installation to cure at room temperature for 24 hours.
- n. Touch up paint as required (PARA 2-6-5).
- o. Install window(s) in cabin door (PARA 2-4-56).
- p. Perform cabin door leak test (PARA 2-4-42).

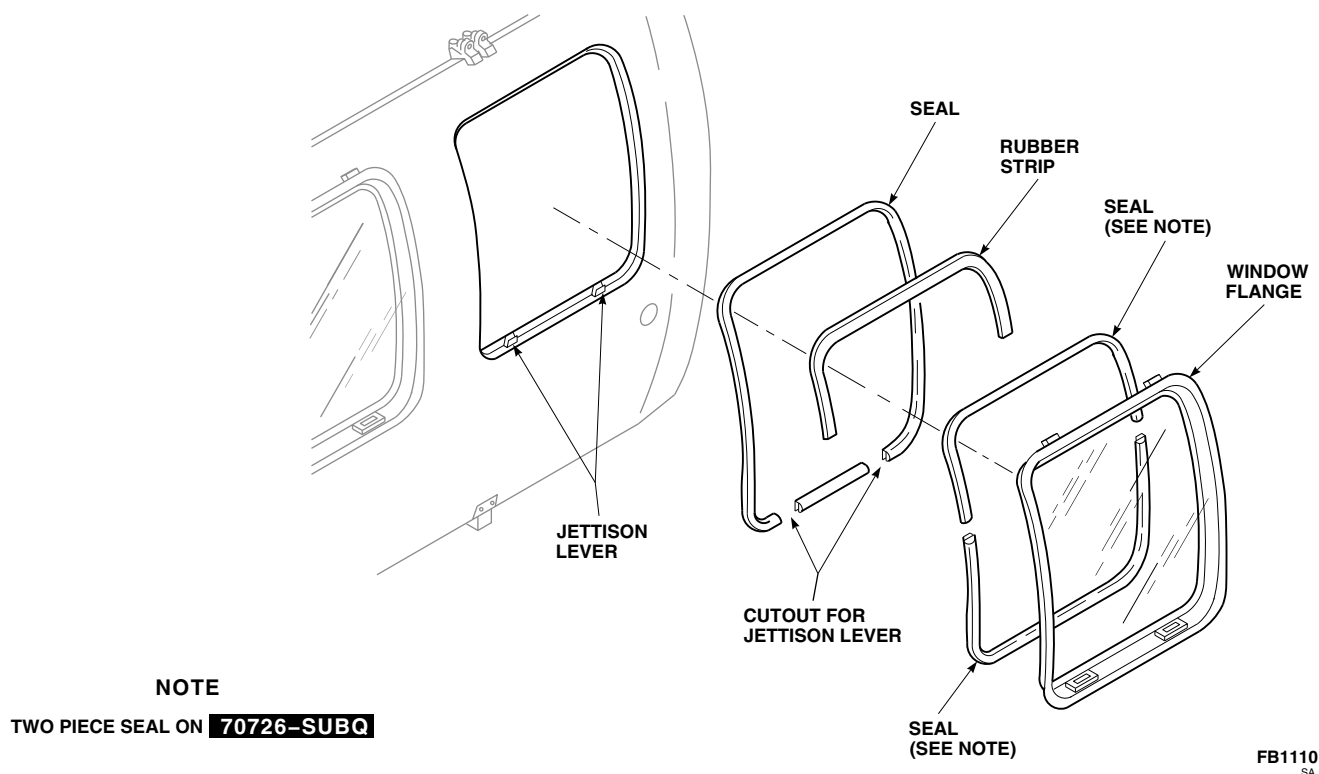


FIGURE 2-4-58-1. REPLACE WINDOW SEALS

2-4-58.2 Replace Door Front Seals.

2-4-58.2.1 Remove.

- Cut away old bulb seal(s) to seal retainers (Figure 2-4-58-2, Sheet 2, Detail D).
- Using drill, remove seal retainer(s) by drilling out rivets.
- Pull damaged seal(s) from door frame.
- Remove seal(s) and adhesive debris using nonmetallic scraper and cheesecloth, Item 90, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D.

2-4-58.2.2 Install.

- Place replacement bulb seal(s) on door frame and trim to fit (Figure 2-4-58-2, Sheet 2, Detail D). Trim latch seal to fit, if necessary (Figure 2-4-58-2, Sheet 1, Detail A).
- On helicopters with doors, 70217-02701-043 through 70217-02701-046, drill holes in seal and seal retainer as follows:

- Using clecos, attach seal and seal retainer to door. Using drill, drill rivet holes and countersink as required.

- Remove seal and seal retainer. Deburr as required.

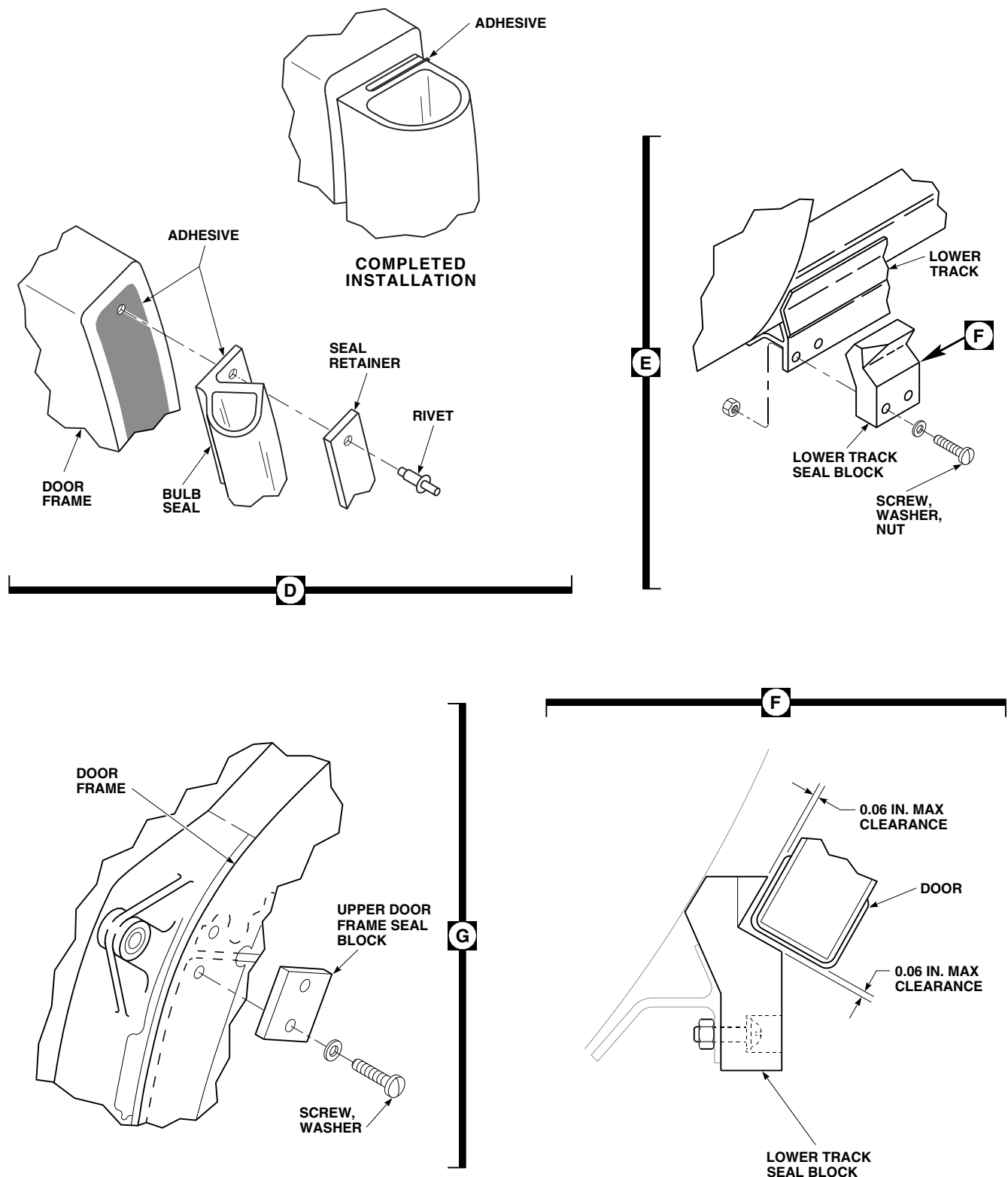
- Wipe door, seal, and seal retainer clean using cheesecloth, Item 90, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.

- Apply thin even coat of adhesive, Item 30 or Item 57, Appendix D, on door frame and mating surfaces of seals.

- On helicopters with doors, 70217-02701-043 through 70217-02701-046, hold seal retainer and seal assemblies together with clecos.

- Line up latch seal with latch opening and press seal in place.

- Line up and press bulb seal on door frame starting at latch seal and moving to end of door. Do not stretch seal while installing it. Wipe excess sealing compound off installation.



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FIGURE 2-4-58-2. REPLACE FRONT AND LOWER SEALS (SHEET 2 OF 2)

- g. On helicopters with doors, 70217-02701-043 through 70217-02701-046, install standard or oversized rivets.
- h. Apply even coat of adhesive, Item 57, Appendix D, over seal retainer and seal joint face.
- i. Close and press seal over seal retainer.
- j. Using cloth dampened with acetone, Item 25, Appendix D, remove excess adhesive.
- k. Touch up paint as required (PARA 2-6-5).
- l. Close cabin door and latch.

2-4-58.3 Replace Upper Door Frame Seal Block.

2-4-58.3.1 Remove.

- a. At fuselage upper door frame, remove screws and washers and remove upper door frame seal block from door frame (Figure 2-4-58-2, Sheet 2, Detail G).
- b. Using nonmetallic scraper, scrape remaining adhesive and parts of seal block from frame.
- c. Using cheesecloth, Item 90, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D, clean remaining adhesive from door frame. Wipe dry.

2-4-58.3.2 Install.

- a. Apply thin even coat of adhesive, Item 37, Appendix D, on mating surface of door frames and largest flat area of seal blocks.
- b. Face angled side of seal block outboard.
- c. Press seal block on door frame and secure with screws and washers (Figure 2-4-58-2, Sheet 2, Detail G).
- d. Using cheesecloth, Item 90, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D, clean excess adhesive from door frame.
- e. Touch up paint as required (PARA 2-6-5).

2-4-58.4 Replace Door Bottom Seal or Seal Blocks/Sheath Seal.

2-4-58.4.1 Remove.

- a. On doors, 70217-02701-045 and 70217-02701-046, remove bottom seal as follows:
 - (1) Remove door (PARA 2-4-55).
 - (2) Remove screws, washers, seal retainer, and seal from inside edge of door (Figure 2-4-58-2, Sheet 1, Detail C).
- b. On doors, 70217-02701-041 through 70217-02701-044, remove bottom seal blocks as follows:
 - (1) Pull off damaged seal block(s) (Figure 2-4-58-2, Sheet 1, Detail B).
 - (2) Scrape off remaining parts of seal block, using nonmetallic scraper.
 - (3) Using cloth wet with methyl ethyl ketone, Item 217, Appendix D, clean remaining adhesive from cabin door and roller fittings.
- c. On doors, 70217-02701-041 through 70217-02701-044, remove sheath seal as follows:
 - (1) Remove fasteners securing sheath seal section(s). Remove old seal (Figure 2-4-58-2, Sheet 1, Detail B).
 - (2) Remove old sealant, as required using cloth wet with methyl ethyl ketone, Item 217, Appendix D, and nonmetallic scraper.

2-4-58.4.2 Install.

- a. On doors, 70217-02701-045 and 70217-02701-046, install bottom seal as follows:
 - (1) Locally-make cabin door seal (Figure H-76, Appendix H).
 - (2) Place replacement seal on inside edge of door (Figure 2-4-58-2, Sheet 1, Detail C).

- (3) Place seal retainer on seal with flange toward bottom of door and install screws and washers.
 - (4) Apply a bead of sealing compound, Item 292, Appendix D, along outside edge of door where it touches seal.
 - (5) Install door (PARA 2-4-55).
- b. On doors, 70217-02701-041 through 70217-02701-044, install bottom seal blocks as follows:
- (1) Apply a thin even coat of adhesive, Item 37, Appendix D, on mating area of seal blocks and door and roller fittings.
 - (2) Press seal block on door and roller fitting with chamfered edge toward inside corner of sheath seal.
- c. On doors, 70217-02701-041 through 70217-02701-044, install sheath seal as follows:
- (1) Line up holes in new sheath seal with existing holes in door. Use hole holders to hold in position. Using drill, drill holes as required. Remove sheath seal.
 - (2) Clean door and seal joining surfaces using clean cloth dampened with methyl ethyl ketone, Item 217, Appendix D.
 - (3) Apply a thin even coat of sealing compound, Item 292, Appendix D, to joining surfaces of door and sheath seal.
 - (4) Use clecos to hold seal in place on door. Install rivets using blind riveter.
- d. Touch up paint as required (PARA 2-6-5).

2-4-58.5 Replace Lower Track Seal Block.

2-4-58.5.1 Remove.

- a. Remove screws, washers, nuts, and seal block from lower track (Figure 2-4-58-2, Sheet 2, Detail E).
- b. Using nonmetallic scraper, scrape remaining sealing compound from fuselage and lower track.

- c. Using cheesecloth, Item 90, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D, clean remaining sealing compound from lower track. Wipe dry.

2-4-58.5.2 Install.

- a. Place replacement seal block on lower track. Face angled side of seal block outboard and secure with screws, washers, and nuts (Figure 2-4-58-2, Sheet 2, Detail E).
- b. Close door over seal, measure, and if necessary, trim seal to allow no more than 0.06-inch clearance between door and seal (Figure 2-4-58-2, Sheet 2, Detail F).
- c. Apply bead of sealing compound, Item 292, Appendix D, along seal edges contacting track and fuselage.
- d. Touch up paint as required (PARA 2-6-5).

2-4-58.6 Replace Upper Door Seal.

2-4-58.6.1 Remove.

- a. Remove cabin door (PARA 2-4-55). Place door on padded saw horses or sandbags.
- b. Using drill, drill out rivets securing seal and seal retainer to door (Figure 2-4-58-3, Detail C). Remove screws and washers, if installed (Figure 2-4-58-3, Detail B).
- c. Inspect seal retainer for damage. If damaged beyond limits, replace seal retainer (PARA 2-6-1).
- d. Using nonmetallic scraper, remove old seal.
- e. Using cheesecloth, Item 90, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean remaining sealing compound from door.

2-4-58.6.2 Install.

- a. Clamp old seal retainer as a template over replacement seal retainer and seal. Using drill with $\frac{5}{32}$ -inch diameter drill bit, drill rivet holes as required. Deburr seal retainer (Figure 2-4-58-3, Detail C).

- b. Using cheesecloth, Item 90, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D, clean parts and door bonding area.
- c. Apply an even coat of adhesive, Item 37, Appendix D, to door.

NOTE

If rivet holes are oversized and cannot hold seal, use rivnut, MS27130S25K.

- d. Place seal retainer on seal and install rivets, or screws and washers (Figure 2-4-58-3, Detail B).
- e. Apply bead of sealing compound, Item 292, Appendix D, around all edges of installation (Figure 2-4-58-3, Detail A).
- f. Using acetone, Item 25, Appendix D, remove excess sealing compound.
- g. Install cabin door (PARA 2-4-55).
- h. Touch up paint as required (PARA 2-6-5).

2-4-58.7 Replace Door Fairing Seal.**2-4-58.7.1 Remove.**

- a. Remove cabin door (PARA 2-4-55). Place on padded saw horses or sand bags.
- b. Using drill, drill out rivets securing seal and seal retainer to door fairing (Figure 2-4-58-3, Detail D).
- c. Using small pin punch, drive rivets out of holes.
- d. Strip seal retainer and seal from fairing.
- e. Remove old adhesive using cheesecloth, Item 90, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, and nonmetallic scraper.

2-4-58.7.2 Install.

- a. Line up new seal and seal retainer over existing holes in door fairing. Use clecos to hold in position (Figure 2-4-58-3, Detail D).

- b. Using drill, drill holes as required.
- c. Remove seal and seal retainer.
- d. Roughen seal and fairing joint surfaces with abrasive cloth, Item 9, Appendix D, and remove all grit using cheesecloth, Item 90, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D.
- e. Apply coat of adhesive, Item 37, Appendix D, to door and seal joining surfaces. Let adhesive dry for at least 10 minutes.
- f. Fasten seal and seal retainer together and install on door fairing with 1/8-inch diameter cleco in every third hole.
- g. Rivet seal and seal retainer to door fairing.
- h. Remove 1/8-inch diameter clecos and install remaining rivets.
- i. Install cabin door (PARA 2-4-55).
- j. Touch up paint as required (PARA 2-6-5).

2-4-58.8 Replace Fuselage Rear Door Seal.**2-4-58.8.1 Remove.**

- a. Remove cabin door (PARA 2-4-55).
- b. Using drill, carefully drill out and remove blind rivets securing seal retainer and seal to fuselage skin (Figure 2-4-58-4, Detail A).
- c. Clean up rivet debris.
- d. Inspect seal retainer for damage. If damaged beyond limits, replace seal retainer (PARA 2-6-1).

2-4-58.8.2 Install.

- a. Position new seal and seal retainer and temporarily hold in place using 1/8-inch diameter clecos in every third hole (Figure 2-4-58-4, Detail A).
- b. Using hand blind riveter, install blind rivets.

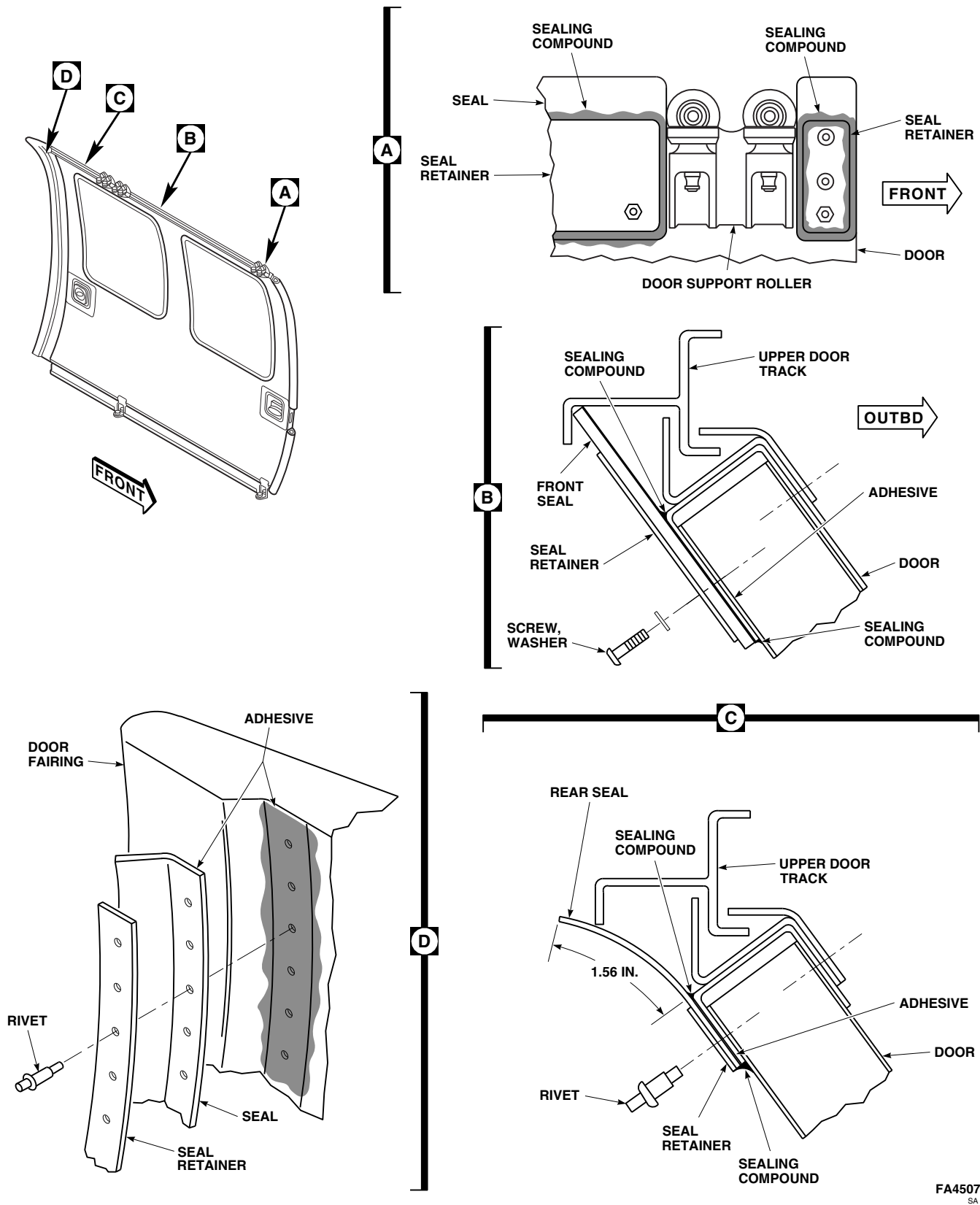
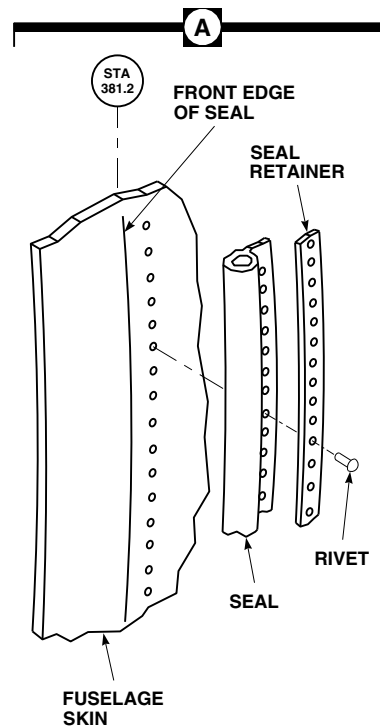
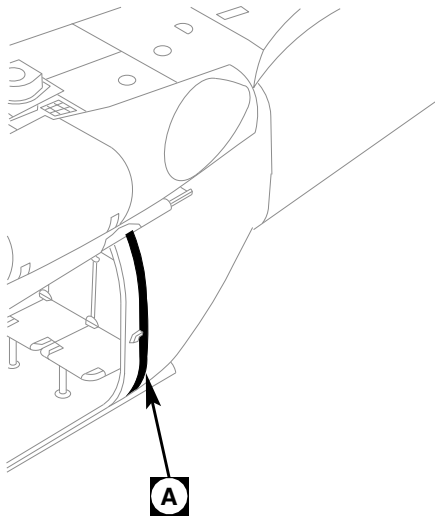


FIGURE 2-4-58-3. REPLACE REAR AND UPPER SEALS

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FA4508
SA**FIGURE 2-4-58-4. REPLACE FUSELAGE REAR DOOR SEAL**

- c. Remove 1/8-inch diameter clecos and install remaining blind rivets.
- d. Touch up paint as required (PARA 2-6-5).
- e. Install cabin door (PARA 2-4-55).

2-4-59 CABIN DOOR FAIRING SEAL.

PARAGRAPH BREAKDOWN

	PAGE
2-4-59.1 Replace Cabin Door Fairing Seal	2-4-59-0

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Clecos, 1⁄8-inch diameter
- Drill Set

Materials/Parts

- Abrasive Cloth, Item 9, Appendix D
- Cheesecloth, Item 90, Appendix D

- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 2-4-55
- PARA 2-6-5

2-4-59.1 Replace Cabin Door Fairing Seal.

NOTE

Replacement of right and left upper cabin door track fairing seals is the same.

2-4-59.1.1. Remove.

- Remove cabin door (PARA 2-4-55).
- Drill out and remove rivets securing upper door track fairing seal and seal retainer to fairing (Figure 2-4-59-1, Detail A).
- Remove retainer and seal from fairing.
- Remove all old sealing compound from fairing using cheesecloth, Item 90, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D.

2-4-59.1.2. Install.

- Locate and line up pilot holes in retainer with existing rivet holes in fairing (Figure 2-4-59-1, Detail A).
- Hold retainer and seal in position using 1⁄8-inch diameter clecos.
- Open rivet holes to correct rivet size using fairing rivet holes as a pattern.
- Remove seal and retainer from fairing.
- Roughen new seal and fairing bonding surfaces with abrasive cloth, Item 9, Appendix D.
- Wipe clean with cheesecloth, Item 90, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D. Wipe dry with clean cheesecloth.

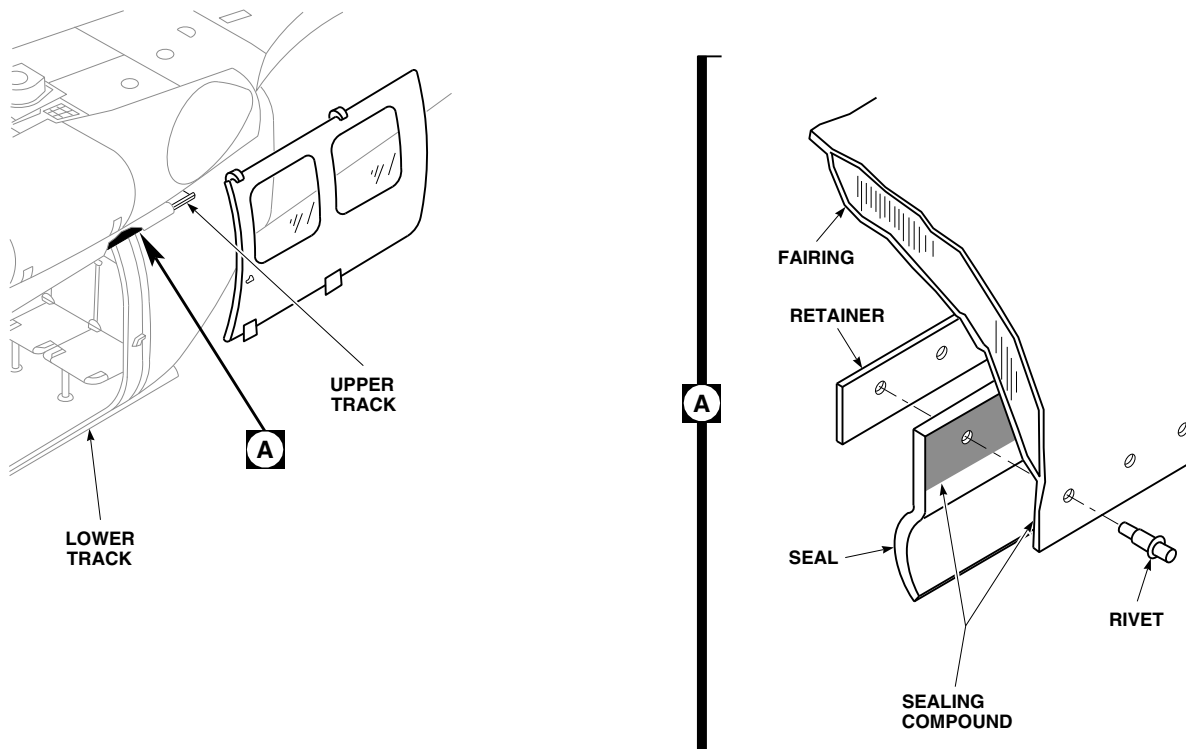
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FIGURE 2-4-59-1. REPLACE CABIN DOOR FAIRING SEAL

- g. Fasten seal and retainer together with $\frac{1}{8}$ -inch diameter clecos.
- h. Apply coat of sealing compound, Item 292, Appendix D, on seal and fairing bonding surfaces.
- i. Rivet seal to fairing while sealing compound is wet.
- j. Install cabin door (PARA 2-4-55).
- k. Touch up paint as required (PARA 2-6-5).

2-4-60 CABIN DOOR HANDLES.

PARAGRAPH BREAKDOWN

		PAGE
2-4-60.1	Replace Cabin Door Handles	2-4-60-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit

Materials/Parts

- Sealing Compound, Item 276, Appendix D

References

- Appendix D
- PARA 2-4-66

2-4-60.1 Replace Cabin Door Handles.

NOTE

Replacement of inside and outside cabin door handles is the same for both cabin doors.

2-4-60.1.1. Remove.

- Remove two screws attaching outside handle to door and remove handle (Figure 2-4-60-1, Detail A).

NOTE

Do not pull spindle out of latch when removing inside handle, as internal parts will come loose.

- Remove two screws attaching inside handle to door, and remove handle and spacer (Figure 2-4-60-1, Detail B).
- If outside handle is being replaced, remove door lock (PARA 2-4-66).

2-4-60.1.2. Install.

- Coat threads of screws with sealing compound, Item 276, Appendix D.

NOTE

Make sure inside handle is in the horizontal position.

- Place inside handle and spacer on door. Line up handle with spindle. Secure with screws (Figure 2-4-60-1, Detail B).
- For replacement outside handle, install door lock (PARA 2-4-66).
- Coat threads of screws with sealing compound, Item 276, Appendix D.
- Place outside handle on door. Secure with screws (Figure 2-4-60-1, Detail A).
- Open and close door using both inside and outside handles to check operation.

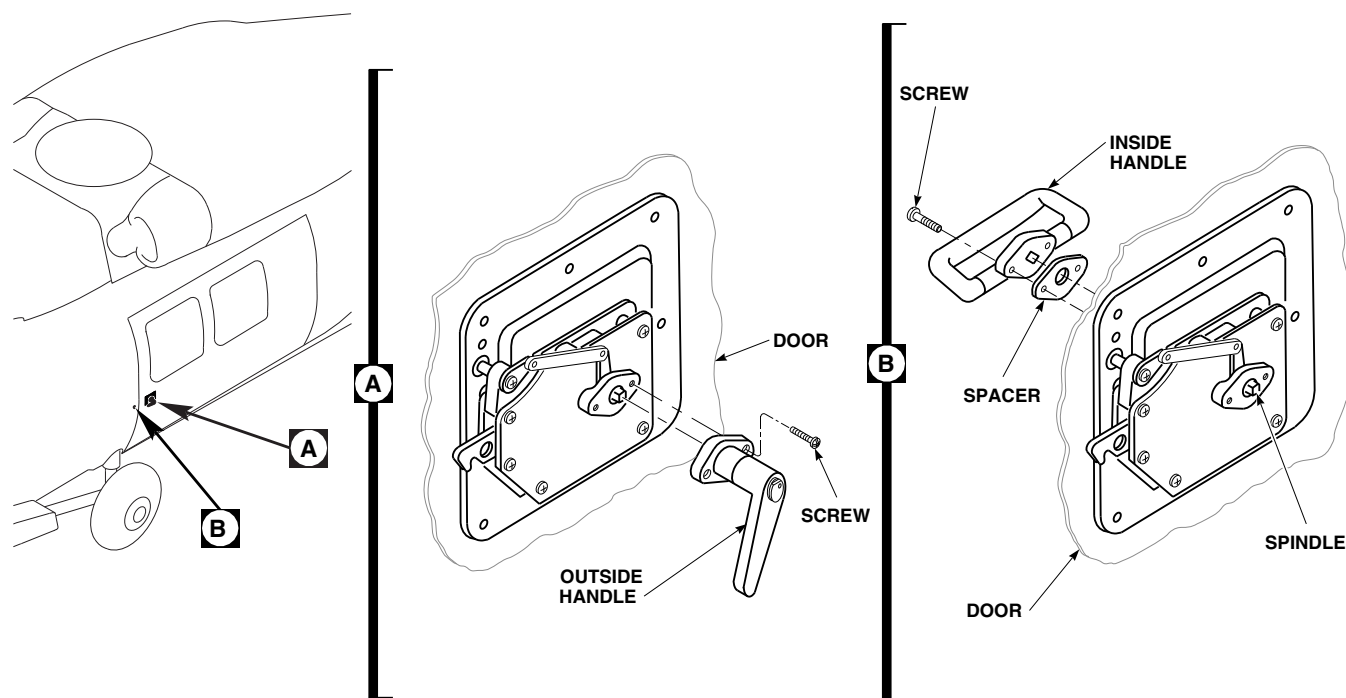
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FIGURE 2-4-60-1. REPLACE CABIN DOOR HANDLES

2-4-61 CABIN DOOR LATCH ASSEMBLY.

PARAGRAPH BREAKDOWN

	PAGE
2-4-61.1 Replace Cabin Door Latch Assembly	2-4-61-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Arbor Press
- Bushing Installation Kit
- C-Clamps
- Drill
- Drill Bit, No. 30 (0.1285-inch diameter)
- Nonmetallic Scraper, Locally-Made
- Weights

Materials/Parts

- Adhesive, Item 32, Appendix D
- Cheesecloth, Item 90, Appendix D
- Lubricant, Solid Film, Item 201, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 276, Appendix D
- Sealing Compound, Item 280, Appendix D
- Sealing Compound, Item 292, Appendix D
- Trichloroethylene, Technical, Item 343, Appendix D

References

- Appendix D
- PARA 2-4-60

2-4-61.1 Replace Cabin Door Latch Assembly.

NOTE

Replacement of left and right cabin door latch assemblies is the same.

2-4-61.1.1 Remove.

- Open cabin door enough to access both sides of latch (about halfway open).
- Block door to keep it from sliding open or closed while removing latch.

NOTE

Do not pull spindle out of latch, as internal parts will come loose.

- Remove inside and outside handles, leaving spindle in latch (PARA 2-4-60).
- Remove screws and washers securing pan and latch assembly to inside surface of cabin door (Figure 2-4-61-1, Sheet 1, Detail A).
- Remove screw and washer securing catch shaft to crank assembly. Remove catch shaft and Te-

flon washers (Figure 2-4-61-1, Sheet 2, Detail C).

- f. Remove screws and washers securing latch assembly to pan.
- g. If replacing sleeve bearing, perform the following:

- (1) Using nonmetallic scraper, remove old sealing compound from edges of sleeve bearing.
- (2) Using arbor press and tools from bushing installation kit, press sleeve bearing from pan.
- (3) Clean remaining sealing compound from pan using cheesecloth, Item 90, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D.
- (4) Using arbor press and tools from bushing installation kit, press new sleeve bearing in pan.
- (5) Apply bead of sealing compound, Item 292, Appendix D, around edge of flange of bearing and pan, and around joint of pan and sleeve of bearing.

- h. If replacing laminated shim (Figure 2-4-61-1, Sheet 1, Detail B), perform the following:

- (1) Soften adhesive securing shim to door, using cheesecloth, Item 90, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D, and remove shim with nonmetallic scraper.
- (2) Clean off all adhesive residue, using methyl ethyl ketone, Item 217, Appendix D, and cheesecloth, Item 90, Appendix D. Wipe dry with clean cheesecloth.
- (3) Apply adhesive, Item 32, Appendix D, to mating surfaces of shim and door.
- (4) Install shim on door, aligning all fastener holes and hold in place with C-clamps or weights.

- (5) Cure adhesive for 24 hours at 70° - 85°F (22° - 29°C) for 1 hour at 140° - 160°F (60° - 71°C).

- (6) Peel laminated shim, as required, to eliminate interference of latch hook with side of receiver and of catch with door stop.

2-4-61.1.2 Disassemble.

- a. Remove screws securing outside handle support to sideplate (Figure 2-4-61-1, Sheet 2, Detail D). Remove outside handle support.
- b. Slide arm and crank assembly off spindle.
- c. Remove bolt, washers, and nut.
- d. Position latch on bench with screwheads facing up.
- e. Remove screws and washers securing sideplate and remove sideplate from sideplate assembly.
- f. Remove pin from latch hook and clevis end of lower spring (Figure 2-4-61-1, Sheet 3, Detail E).
- g. Remove lower spring.
- h. Remove pin from clevis and cam.
- i. Remove clevis, upper spring, and washer from latch hook.
- j. Remove latch hook, washers, and bushing from sideplate assembly.
- k. Remove spindle from sideplate assembly (Figure 2-4-61-1, Sheet 3, Detail F).
- l. Remove bushings and cam from sideplate assembly.
- m. Remove spacers from standoffs.
- n. Remove screws securing support assembly to sideplate assembly and remove support assembly (Figure 2-4-61-1, Sheet 3, Detail E).
- o. Remove screws securing spacer to sideplate assembly. Remove spacer from sideplate assembly (Figure 2-4-61-1, Sheet 3, Detail F).

- p. Inspect sideplate and spacer for wear. **NONE ALLOWED.** If sideplate or spacer bushings are worn, replace as required.
- q. Inspect all parts for signs of binding, excessive wear, loose bushings, and cracks. **NONE ALLOWED.** If required, replace parts failing inspection.

2-4-61.1.3 Repair.

- a. If required, replace latch guard as follows (Figure 2-4-61-2, Detail A and Detail B):
 - (1) Remove rivets from latch guard.
 - (2) Remove latch guard.
 - (3) Position replacement latch guard vertically on cabin door so upper end of guard is 2-inches above latch hook, and mark rivet holes.
 - (4) Using drill with No. 30 drill bit, drill pre-marked holes in latch guard.
 - (5) Position latch guard (with bend facing outboard) on cabin door, and install rivets.

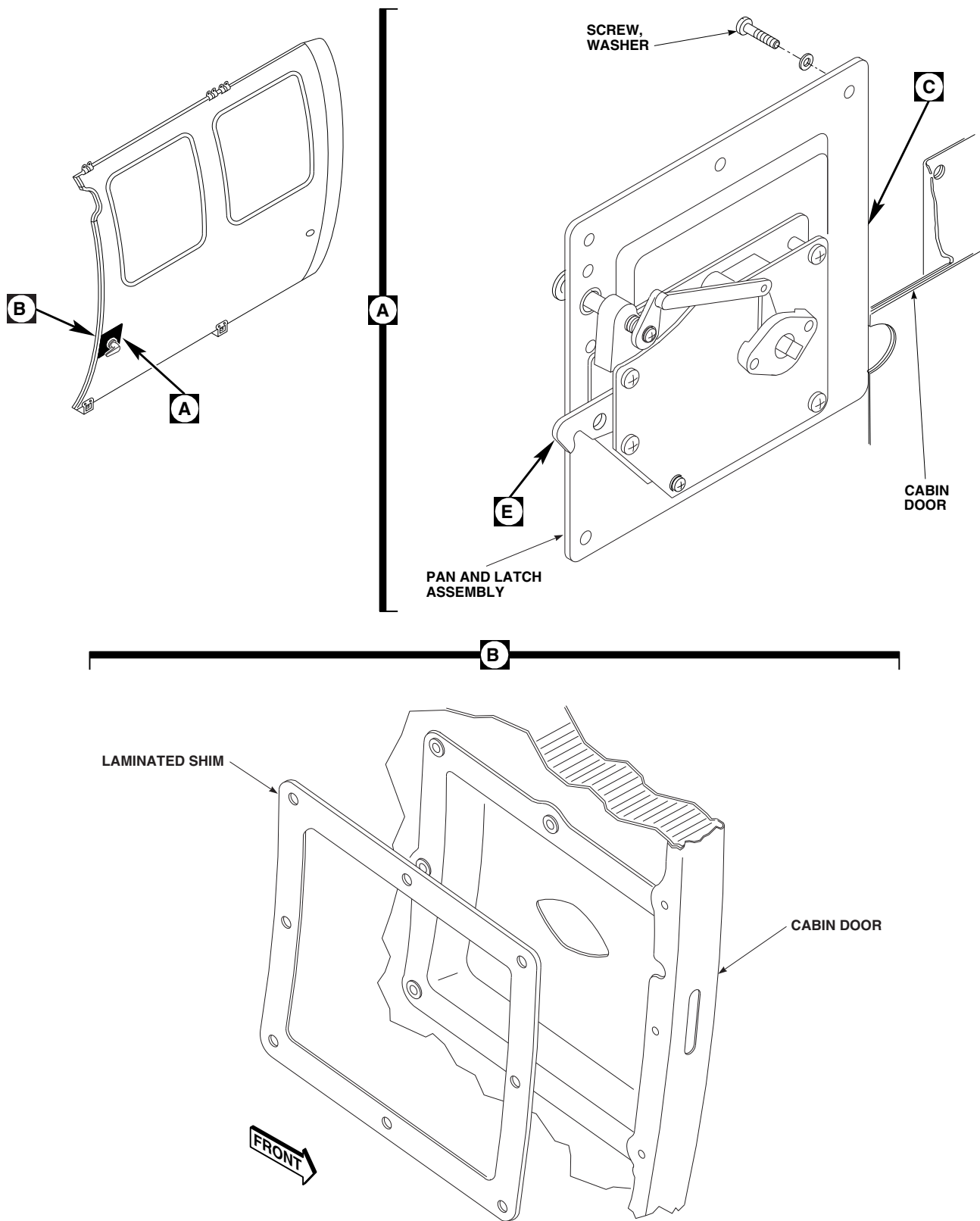
2-4-61.1.4 Assemble.

- a. Lubricate all parts with solid film lubricant, Item 201, Appendix D, before assembly.
- b. Clean outside diameter of bushings and inner bore of sideplates using cheesecloth, Item 90, Appendix D, and technical trichloroethylene, Item 343, Appendix D.
- c. Apply $\frac{1}{16}$ -inch bead of sealing compound, Item 280, Appendix D, to leading edge of bushing outside diameter and edge of bore in sideplate and sideplate assembly.
- d. Carefully press bushings in sideplate and sideplate assembly (Figure 2-4-61-1, Sheet 2, Detail D and Sheet 3, Detail F).
- e. Wipe excess sealing compound from inner bore of bushings using cheesecloth, Item 90, Appendix D, and technical trichloroethylene, Item 343, Appendix D.

- f. Press bushing into spacer (Figure 2-4-61-1, Sheet 3, Detail F).
- g. Secure spacer on sideplate assembly with screws.
- h. Install spindle through bushing in spacer.
- i. Install spacers on standoffs.
- j. Install cam on spindle and sideplate assembly.
- k. Install clevis on cam with pin (Figure 2-4-61-1, Sheet 3, Detail E).
 - l. Position support assembly on sideplate assembly. Coat threads of screws with sealing compound, Item 280, Appendix D, and install screws.
- m. Install upper spring and washer on clevis.
- n. Connect clevis end of lower spring assembly to latch hook with pin.
- o. Position washers and latch hook on sideplate assembly, engaging end of clevis in latch hook tab hole.
- p. Install bushing in center hole of latch hook.
- q. Place end of lower spring assembly in hole in sideplate assembly.
- r. Position sideplate on spacers, engaging lower spring assembly in holes (Figure 2-4-61-1, Sheet 2, Detail D).
- s. Coat threads of screws with sealing compound, Item 280, Appendix D, and install screws and washers.
- t. Install bolt and washer through sideplates and latch hook. Install washer and nut on bolt.
- u. Install arm and crank assembly on spindle.
- v. Secure outside handle support on sideplate with screws.
- w. Coat threads of small screws with sealing compound, Item 280, Appendix D, and install screws in sideplate and handle support.

2-4-61.1.5 Install.

- a. Secure latch assembly to pan with screws and washers (Figure 2-4-61-1, Sheet 2, Detail C).
- b. Insert catch shaft through sleeve bearing in pan, catch support, and Teflon washers. Coat threads of screw with sealing compound, Item 276, Appendix D. Install square hole of crank assembly onto catch shaft with washer and screw.
- c. Coat threads of screws with sealing compound, Item 276, Appendix D, and secure pan and latch assembly to inside surface of door with screws and washers.
- d. Install inside and outside handles (PARA 2-4-60).
- e. Apply fillet of sealing compound, Item 292, Appendix D, at least 1/8-inch wide, in angle of outside handle base and door.
- f. Close cabin door.
- g. Check that door latch has engaged door frame. Rotate door handle to latch door and check that door is securely closed.



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FIGURE 2-4-61-1. REPLACE CABIN DOOR LATCH ASSEMBLY (SHEET 1 OF 3)

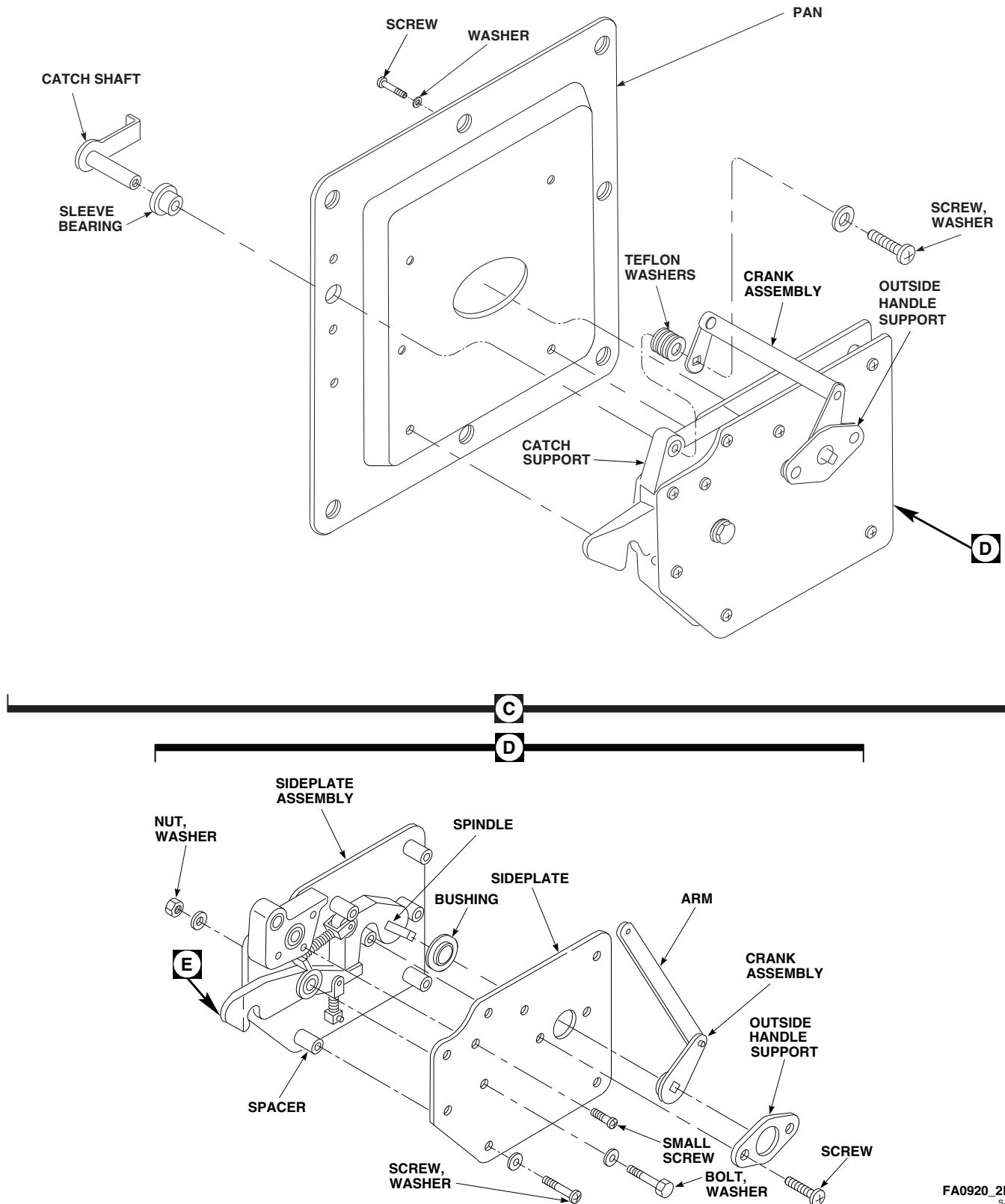
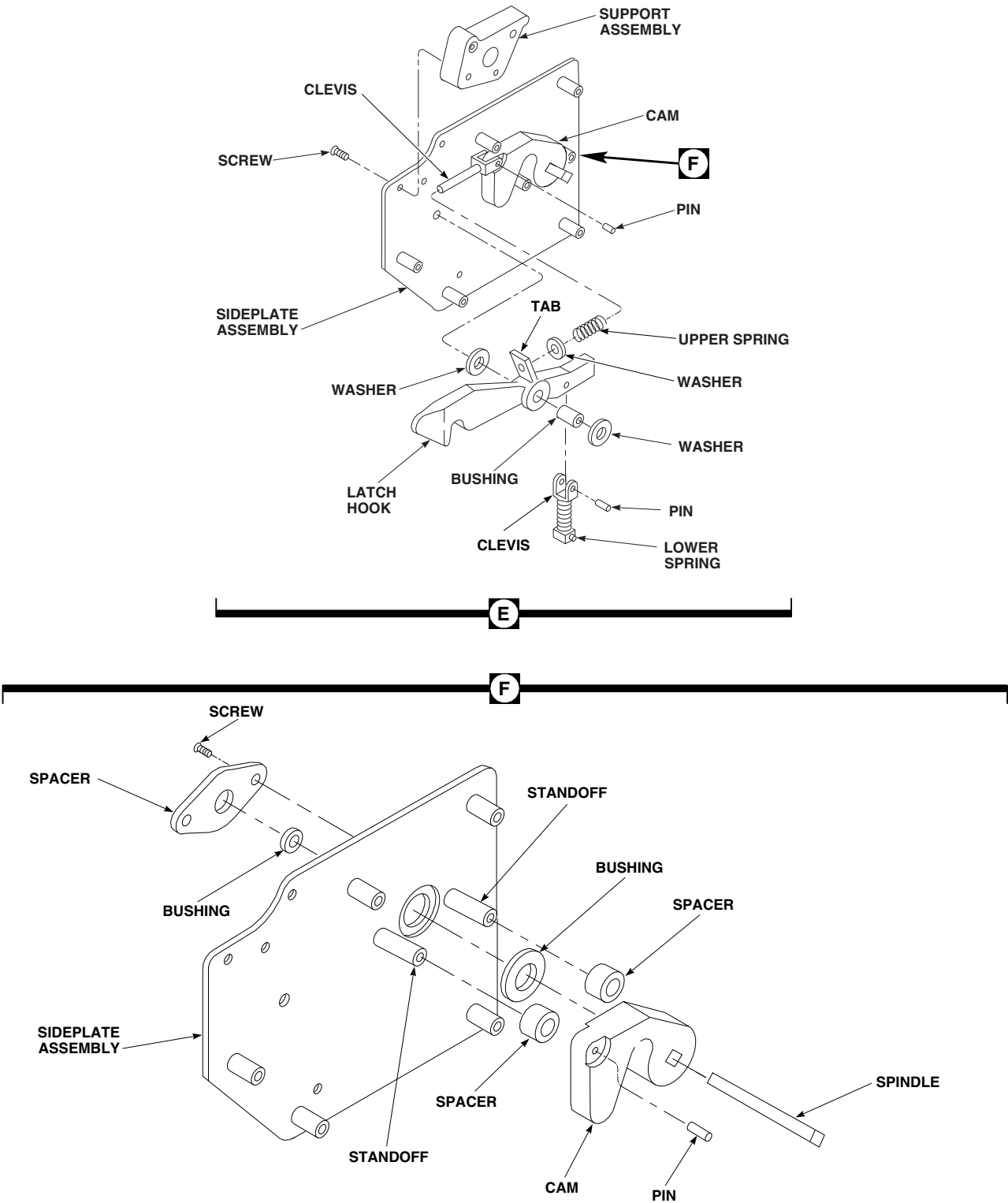


FIGURE 2-4-61-1. REPLACE CABIN DOOR LATCH ASSEMBLY (SHEET 2 OF 3)



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FIGURE 2-4-61-1. REPLACE CABIN DOOR LATCH ASSEMBLY (SHEET 3 OF 3)

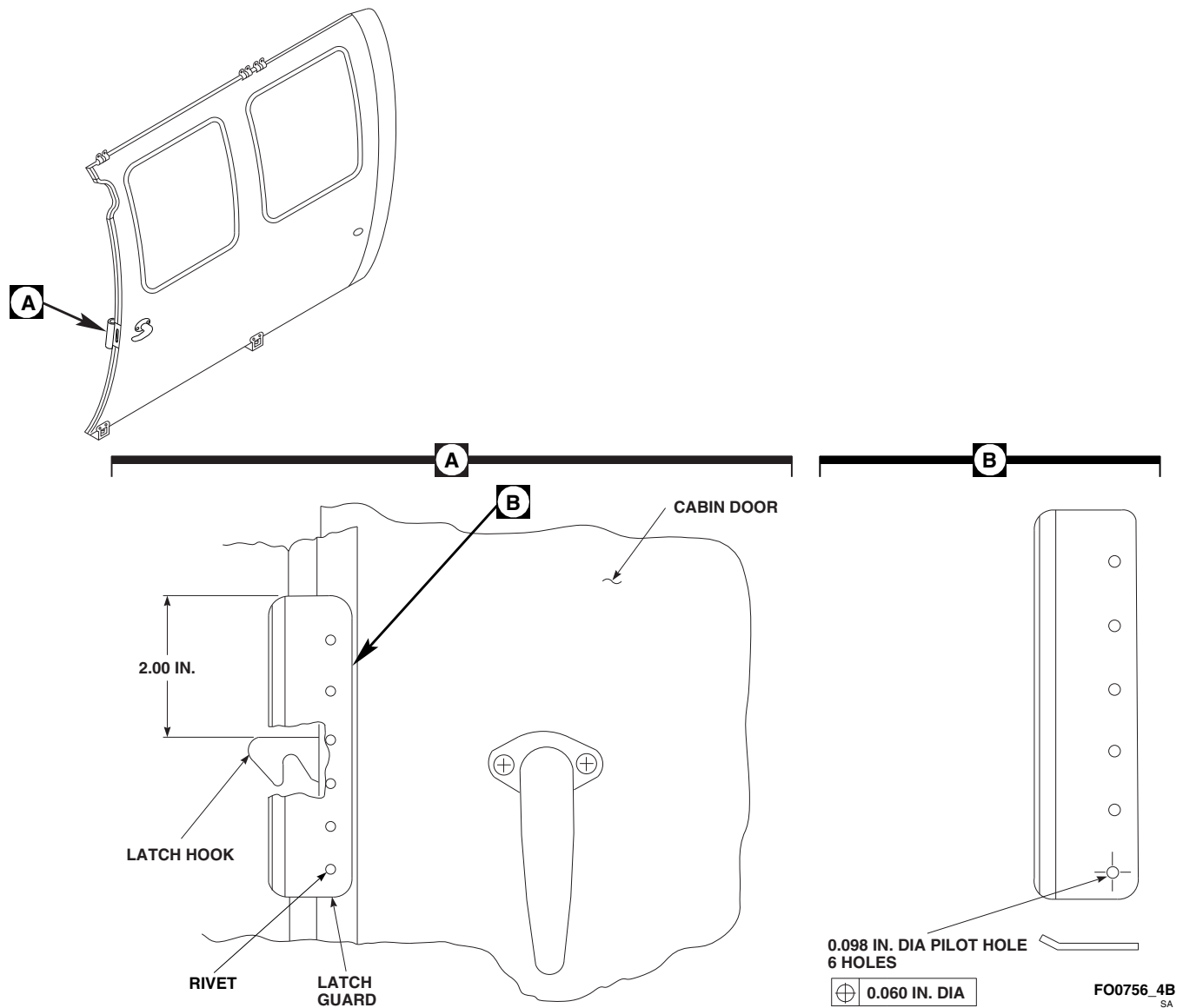


FIGURE 2-4-61-2. REPLACE LATCH GUARD

2-4-62 CABIN DOOR RECEIVER.

PARAGRAPH BREAKDOWN

		PAGE
2-4-62.1	Replace Cabin Door Receiver	2-4-62-1

INITIAL SETUP

- Airframe Repairer’s Toolkit

Tools

- Aircraft Mechanic’s Toolkit

2-4-62.1 Replace Cabin Door Receiver.

NOTE

Replacement of left and right cabin door receivers is the same.

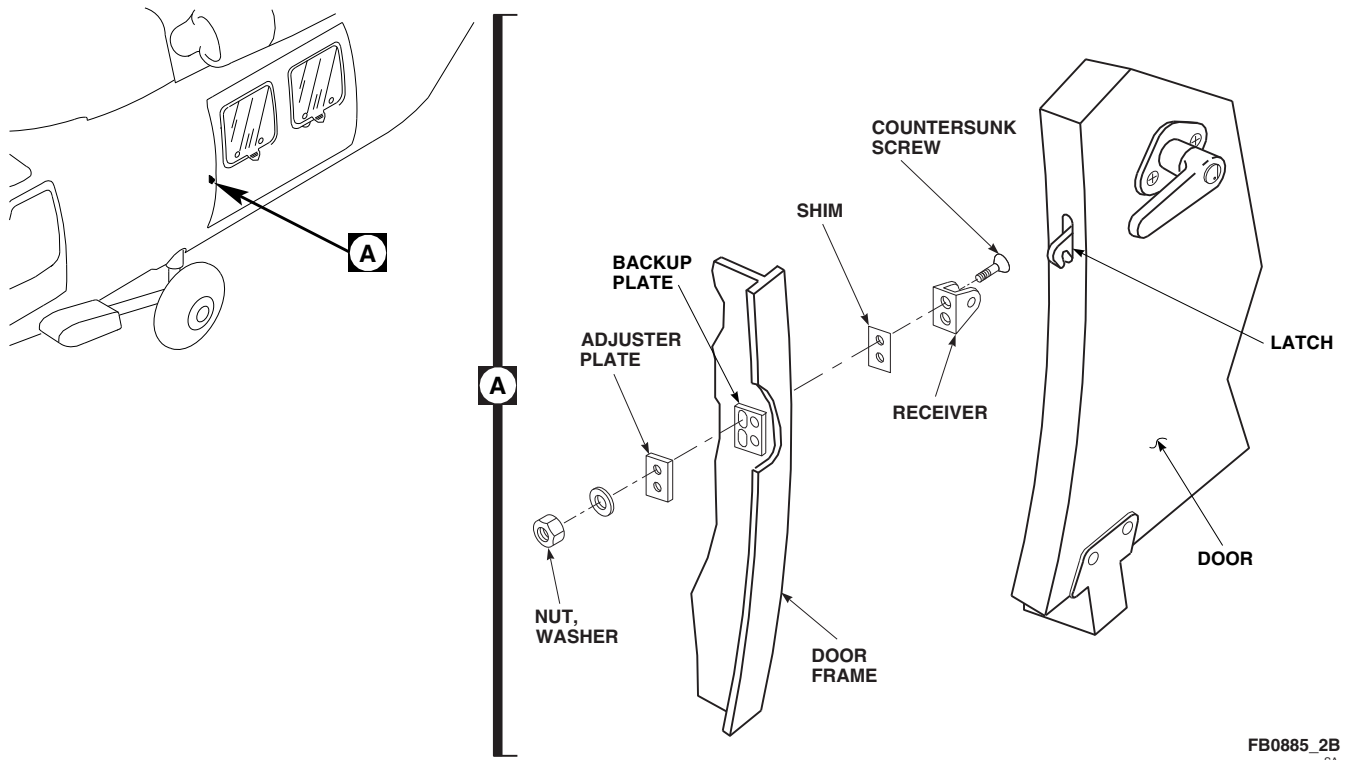
2-4-62.1.1. Remove.

- Open cabin door.
- Remove countersunk screws, washers, and nuts securing adjuster plate, shim(s), and receiver to door frame (Figure 2-4-62-1, Detail A). Remove adjuster plate, shim(s), and receiver.

2-4-62.1.2. Install.

- Install shim and receiver on door frame and install countersunk screws (Figure 2-4-62-1, Detail A).
- On other side of door frame, install adjuster plate on screws, with serrated side against serrated backup plate.
- Install nuts and washers.
- Close cabin door and check position of latch in receiver and compression of seal.

- If necessary, adjust receiver as follows:
 - Loosen nuts to allow movement of receiver.
 - Lower receiver to its lowest limit.
 - With latch horizontal, raise receiver until pin touches inside of latch hook without raising latch.
 - TIGHTEN NUTS WHILE CHECKING THAT LATCH DOES NOT MOVE.**
 - If latch moves, drop adjuster plate one or two serrations.
 - TIGHTEN NUTS.**
- With cabin door fully closed, door seal should not be compressed more than ¼-inch. If seal compression is not met, perform the following:
 - Remove receiver (PARA 2-4-62.1.1).
 - Add shim under receiver.
 - Install receiver (PARA 2-4-62.1.2).
- Close cabin door.



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FIGURE 2-4-62-1. REPLACE CABIN DOOR RECEIVER

2-4-63 CABIN DOOR STOP MECHANISM.

PARAGRAPH BREAKDOWN

	PAGE
2-4-63.1 Replace Cabin Door Stop Mechanism	2-4-63-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit

References

- PARA 2-4-55

2-4-63.1 Replace Cabin Door Stop Mechanism.

NOTE

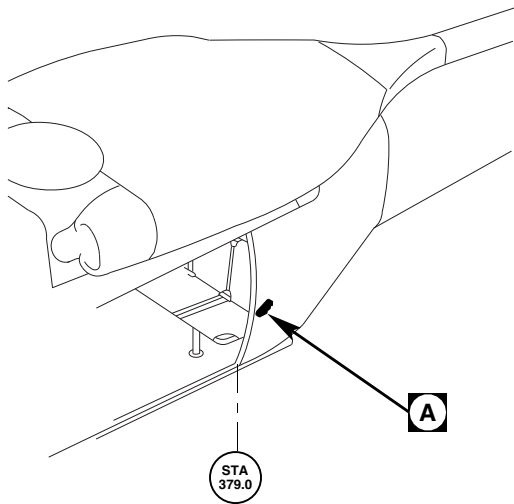
Replacement of right and left cabin door stop mechanisms is the same.

2-4-63.1.1. Remove.

- Remove cabin door (PARA 2-4-55).
- Remove screws and washers attaching stop to frame at STA 379.0 (Figure 2-4-63-1, Detail A).
- Remove stop, spacers, spring, and Teflon washers.

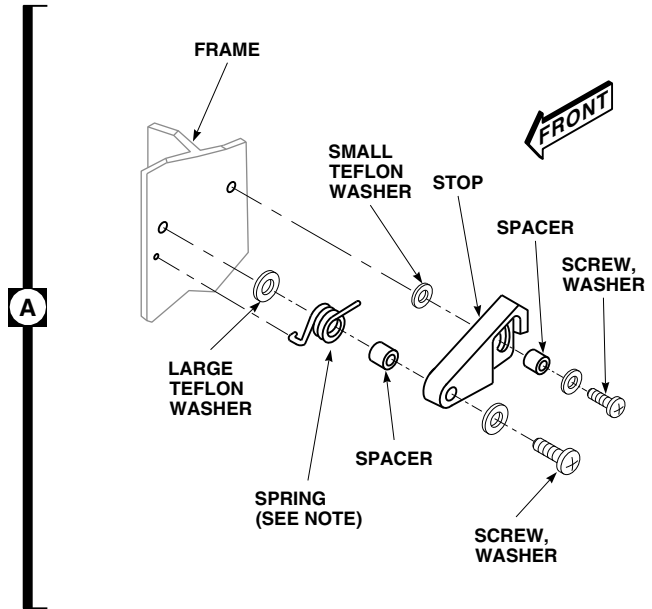
2-4-63.1.2. Install.

- Place new Teflon washers between stop and frame at STA 379.0 (Figure 2-4-63-1, Detail A).
- Place stop in position and insert spacers and spring. Install bent leg of spring into hole in frame.
- Install screws and washers.
- Install cabin door (PARA 2-4-55).
- Open door fully and check stop operation.



NOTE

BENT LEG OF SPRING INSTALLED IN FRAME.



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FIGURE 2-4-63-1. REPLACE CABIN DOOR STOP MECHANISM

2-4-64 CABIN DOOR UPPER/LOWER ROLLER SUPPORTS AND ROLLERS.

PARAGRAPH BREAKDOWN

		PAGE
2-4-64.1	Replace Cabin Door Upper/Lower Roller Supports and Rollers	2-4-64-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Depth Micrometer
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Abrasive Cloth, Item 9, Appendix D
- Cheesecloth, Item 90, Appendix D

- Dry-Cleaning Solvent, Item 124, Appendix D
- Tags
- Washer, NAS1149D0316K
- Washer, NAS1149D0363K

References

- Appendix D
- PARA 2-4-55

2-4-64.1 Replace Cabin Door Upper/Lower Roller Supports and Rollers.

NOTE

Replacement of left and right upper/lower roller supports is the same.

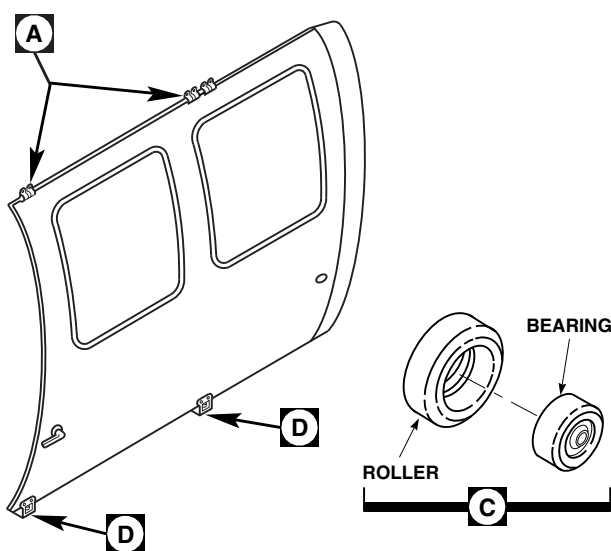
2-4-64.1.1. Remove.

- Remove cabin door (PARA 2-4-55).
- Remove screws, washers, and nuts attaching roller support fittings to door (Figure 2-4-64-1, Detail A).

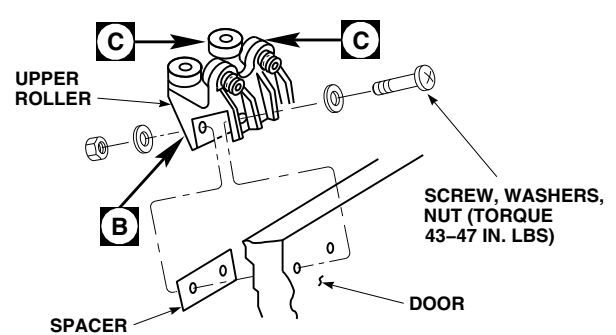
- Remove roller support fitting from door. Tag each roller support for location upon removal. Keep spacer for installation of replacement roller support.

2-4-64.1.2. Disassemble.

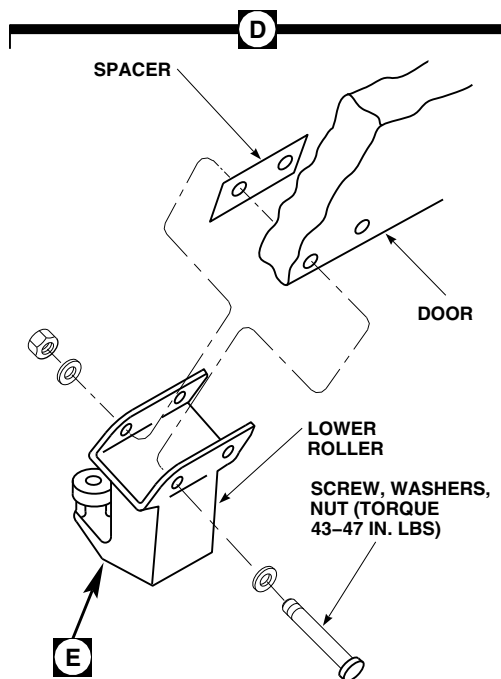
- Remove pins, washers, and collars and remove rollers from upper roller support fitting (Figure 2-4-64-1, Detail B).
- Remove pins, washers, collar, and nut and remove rollers from lower roller support fitting (Figure 2-4-64-1, Detail E).



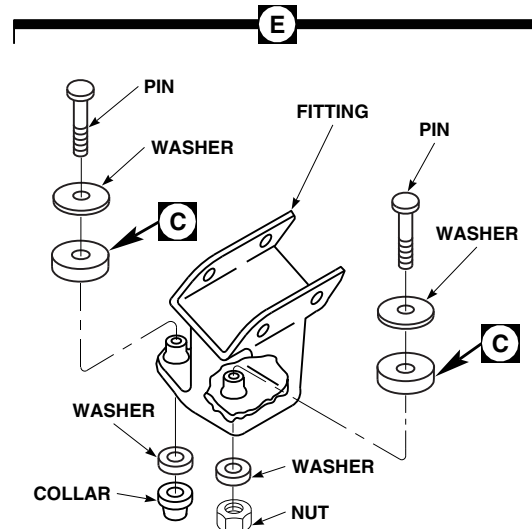
A



B



E

**EFFECTIVITY**

THIRD ROLLER FOUND ON
HELICOPTERS WITH DOORS
70217-02701-047 THROUGH
70217-02701-050.

NOTES

1. ADDITIONAL WASHERS ON REPAIRED ROLLER SUPPORT ASSEMBLY, 70217-02717-041, AND ROLLER SUPPORT ASSEMBLIES 70217-02717-043 AND 70217-02717-044.
2. WASHERS USED ONLY ON ROLLER SUPPORT 70217-02717-044.

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FIGURE 2-4-64-1. REPLACE CABIN DOOR UPPER/LOWER ROLLER SUPPORTS AND ROLLERS

2-4-64-2

2-4-64.1.3. Inspect.



Bearings will be damaged by loss of lubricant if soaked in solvent. Keep solvent away from bearings when cleaning parts.

- a. Clean all parts with clean cloth wet with dry-cleaning solvent, Item 124, Appendix D, or equivalent, to avoid saturation of bearings. Inspect bearings.

- b. On helicopters with doors, 70217-02701-041 through 70217-02701-046, inspect roller support fittings, 70217-02717-041 and 70217-02717-042, as follows:

- (1) Inspect upper front roller support fitting for groove on angled surface using depth micrometer. **MAXIMUM DAMAGE IS 0.030 INCH.**
- (2) Inspect upper rear roller support fitting for groove on angled surface using depth micrometer. **NO WEAR ALLOWED.**
- (3) Inspect lower front roller support fitting for groove between rollers using depth micrometer. **MAXIMUM DAMAGE IS 0.060 INCH.**
- (4) Inspect lower rear roller support fitting for groove between rollers using depth micrometer. **MAXIMUM DAMAGE IS 0.025 INCH.**

- c. On helicopter with doors, 70217-02701-047 and 70217-02701-048, inspect roller support fittings, 70217-02717-041 and 70217-02717-042, as follows:

- (1) Inspect upper front roller support fitting for groove on angled surface using depth micrometer. **MAXIMUM WEAR IS 0.030 INCH.**

- (2) Inspect middle and rear upper roller support fitting for groove on angled surface using depth micrometer. **MAXIMUM WEAR IS 0.015 INCH.**

- (3) Inspect lower front roller support fitting for groove between rollers using depth micrometer. **MAXIMUM WEAR IS 0.060 INCH.**

- (4) Inspect lower rear roller support fitting for groove between rollers using depth micrometer. **MAXIMUM WEAR IS 0.060 INCH.**

- (5) If wear limits are exceeded, replace roller support fitting.

- (6) Repair damage by blending to remove groove using abrasive cloth, Item 9, Appendix D.

- d. Inspect support fittings for cracks, distortion, and broken or missing pieces. Replace damaged fittings with fittings having same part number.

- e. Check roller bearings for smooth operation. A slight rolling resistance caused by lubricant and seal drag is normal. Replace dry, loose, rough, seized, or sticking bearings and bearings having loose, open, missing, bent, or dented seals.

- f. Check bearing radial wear. **REPLACE BEARINGS HAVING RADIAL WEAR OF OVER 0.0025 INCH.**

- g. Check rollers for cracks, flatspots, and missing pieces. Replace damaged rollers.

2-4-64.1.4. Assemble.

- a. On nonreworked upper roller support, 70217-02717-041, install rollers with large chamfered edge toward mounting boss. Install pins, washers, and collars, with a washer under pin head and collar (Figure 2-4-64-1, Detail B).

WARNING

Injury to personnel and damage to equipment will result if all upper roller assemblies do not have the same part number or additional washers under the horizontal rollers. Door could separate from helicopter. All upper roller assemblies must have the same part number or be reworked to add additional washers under the horizontal rollers.

- b. On repaired roller support, 70217-02717-041, and roller supports, 70217-02717-043 and 70217-02717-044, install horizontal rollers, with large chamfered edge toward mounting boss, with washers, NAS1149D0363K and NAS1149D0316K, between mounting boss roller. Install pins, washers, and collars with washer under pin head and collar (Figure 2-4-64-1, Detail B).
- c. Install vertical rollers on upper support with large chamfered edge toward mounting boss. Install pins, washers, and collars with washer under pin head and collar.
- d. Install rollers on lower roller support fitting, with large chamfered edge toward mounting boss. Install pin, washers, and collar in inboard roller with a washer under pin head and collar. Install pin, washers, and nut in outboard roller

with washer under pin head and nuts (Figure 2-4-64-1, Detail E).

2-4-64.1.5. Install.**WARNING**

Injury to personnel and damage to equipment will result if all upper roller assemblies do not have the same part number or additional washers under the horizontal rollers. Door could separate from helicopter. All upper roller assemblies must have the same part number or be reworked to add additional washers under the horizontal rollers.

- a. Place roller support on door in same location as removed (Figure 2-4-64-1, Detail A and Detail D).
- b. Slide spacer in between support and inside of door.
- c. Install screws, washers, and nuts attaching roller support to door. **TORQUE NUTS TO 43 - 47 INCH-POUNDS.**
- d. Wipe door tracks clean using cheesecloth, Item 90, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D. Wipe dry with clean cheesecloth.
- e. Install cabin door (PARA 2-4-55).

2-4-65 CABIN DOOR UPPER FAIRINGS.

PARAGRAPH BREAKDOWN

	PAGE
2-4-65.1 Replace Cabin Door Upper Fairings	2-4-65-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Drill Set
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Abrasive Cloth, Item 9, Appendix D
- Acetone, Item 25, Appendix D
- Adhesive, Item 37, Appendix D

- Cheesecloth, Item 90, Appendix D
- Corrosion Preventive Compound, Item 115, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 2-4-55
- PARA 2-4-59
- PARA 2-4-118
- PARA 2-6-3
- PARA 2-6-5

2-4-65.1 Replace Cabin Door Upper Fairings.

NOTE

Replacement of left and right cabin door upper fairings is the same.

2-4-65.1.1. Remove.

- Remove screws, washers, and upper door track fairing between STA 420.0 and STA 428.0 (Figure 2-4-65-1, Detail A).
- Remove cabin door (PARA 2-4-55).

- Remove engine cowling (PARA 2-4-118).
- Remove bolt, screws, and washers securing inlet shroud fairing. Remove fairing (Figure 2-4-65-1, Detail B).
- Remove screws and washers from engine fairing and upper fuselage fairing between STA 308.0 and STA 398.0. Remove fairings with rubber seal attached.
- If necessary, remove rubber seal from fairings (PARA 2-4-59).

- g. On engine fairing, if necessary, remove block seal from clip with nonmetallic scraper (Figure 2-4-65-1, Detail C).
- h. Clean off old adhesive using cheesecloth, Item 90, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.
- i. Check fairings for damage and repair as necessary (PARA 2-6-3).

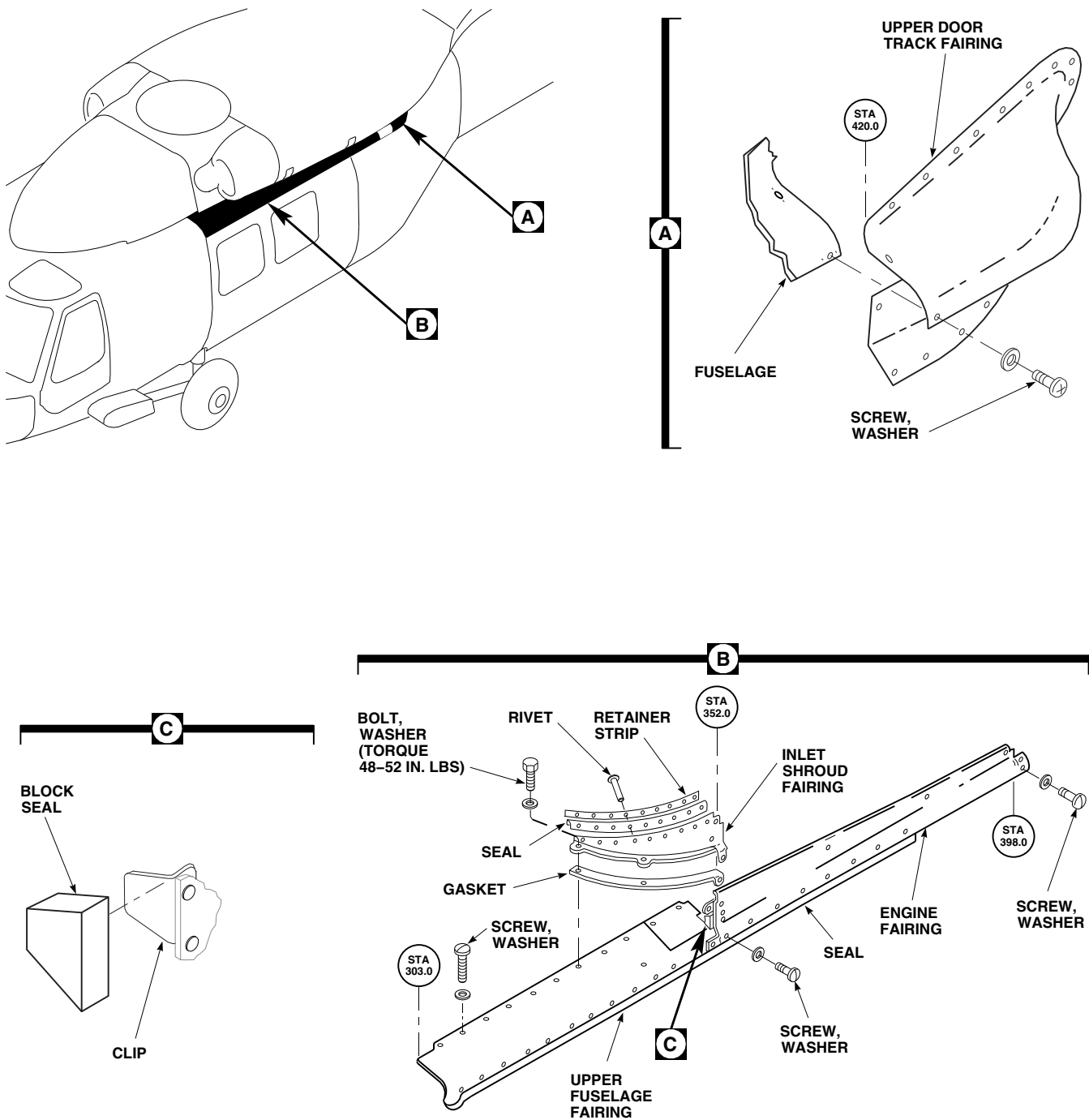
2-4-65.1.2. Repair.

- a. On inlet shroud fairing, drill out rivets that secure retainer strip. Remove strip and seal (Figure 2-4-65-1, Detail B).
- b. Remove gasket from fairing flange using nonmetallic scraper.
- c. Clean old adhesive from fairing using cheesecloth, Item 90, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.
- d. Drill rivet holes through new seal, using retainer strip as a template.
- e. Make sure fairing surface that contacts seal is wiped clean. Fasten seal and retainer strip in place with rivets.
- f. Apply a thin, even coat of adhesive, Item 37, Appendix D, to joining surfaces of fairing flange and gasket. Let adhesive dry for 10 minutes.
- g. Bond gasket to fairing flange.

2-4-65.1.3. Install.

- a. Apply corrosion preventive compound, Item 115, Appendix D, to fairing attachment hardware.
- b. Install engine fairing with screws and washers (Figure 2-4-65-1, Detail B).
- c. If necessary, install block seal as follows:
 - (1) Apply adhesive, Item 37, Appendix D, to mating surfaces of block seal and clip. Let dry for 10 minutes.

- (2) Position seal and bond to clip (Figure 2-4-65-1, Detail C).
- (3) Seal front side of seal with sealing compound, Item 292, Appendix D.
- d. Sand upper fuselage fairing and fuselage joining surfaces with abrasive cloth, Item 9, Appendix D.
- e. Wipe joining surfaces using cheesecloth, Item 90, Appendix D, wet with acetone, Item 25, Appendix D. Before solvent dries, wipe surfaces with clean cheesecloth, turning cloth with each wipe.
- f. Apply sealing compound, Item 292, Appendix D, to fairing and fuselage joining surfaces.
- g. Apply corrosion preventive compound, Item 115, Appendix D, to fairing attachment hardware.
- h. Install upper fuselage fairing and fasten with screws and washers (Figure 2-4-65-1, Detail B).
- i. Apply sealing compound, Item 292, Appendix D, around top of fairing, all edges, fasteners, and bottom where fairing meets fuselage frame.
- j. Apply corrosion preventive compound, Item 115, Appendix D, to fairing attachment hardware.
- k. Install inlet shroud fairing with screws, washers, and bolt. **TORQUE BOLT TO 48 - 52 INCH-POUNDS.**
- l. If removed, install rubber seal (PARA 2-4-59).
- m. Install engine cowling (PARA 2-4-118).
- n. Check that engine cowling does not bind against fairing. If binding occurs, perform the following:
 - (1) Remove engine cowling (PARA 2-4-118).
 - (2) Slit fairing lengthwise and remove piece of fairing to shorten distance between top



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FIGURE 2-4-65-1. REPLACE CABIN DOOR UPPER FAIRINGS

- of fairing and mounting screws (PARA 2-6-3).
- (3) Repaint fairing (PARA 2-6-5).
- (4) Install engine cowling (PARA 2-4-118).
- o. Install cabin door (PARA 2-4-55).
- p. Apply corrosion preventive compound, Item 115, Appendix D, to fairing attachment hardware.
- q. Install upper door track fairing between STA 420.0 and STA 428.0 with screws and washers (Figure 2-4-65-1, Detail A).

2-4-66 CABIN DOOR LOCK.

PARAGRAPH BREAKDOWN

	PAGE
2-4-66.1 Replace Cabin Door Lock	2-4-66-0

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Rubber Mallet

- Screwdriver, Thin-Bladed

Materials/Parts

- Sealing Compound, Item 281, Appendix D

References

- Appendix D

2-4-66.1 Replace Cabin Door Lock.

NOTE

- Lock cylinders are available only in sets. When replacing cabin door lock cylinder, replace lock cylinders on all doors.
- Replacement of left and right cabin door locks is the same.

2-4-66.1.1. Remove.

- Unlock and open cabin door.
- Remove screws attaching outside handle to door. Remove handle (Figure 2-4-66-1).
- Remove snapping and small washer fastening handle to escutcheon. Remove escutcheon and large washer from handle (Figure 2-4-66-1, Detail A).
- Gently tap lock case with rubber mallet to remove it from handle.
- Remove two pins securing shell in lock case. Remove shell from lock case. Make sure wave washer remains inside lockcase.

- Using thin-bladed screwdriver, depress retaining ring into cylinder groove and remove lock cylinder from shell.

2-4-66.1.2. Install.

- Compress retaining ring, insert cylinder into shell, and snap into place (Figure 2-4-66-1, Detail A).
- Make sure wave washer remains inside lock-case and insert shell into lock case and install two pins. Pins must be flush with surface of lock case.
- Install lock case in handle.
- Install large washer on handle and handle in escutcheon.
- Install small washer and snapping on handle.
- Place handle on door. Coat threads of screws with sealing compound, Item 281, Appendix D, and install (Figure 2-4-66-1).
- Open and close door a few times to check operation.

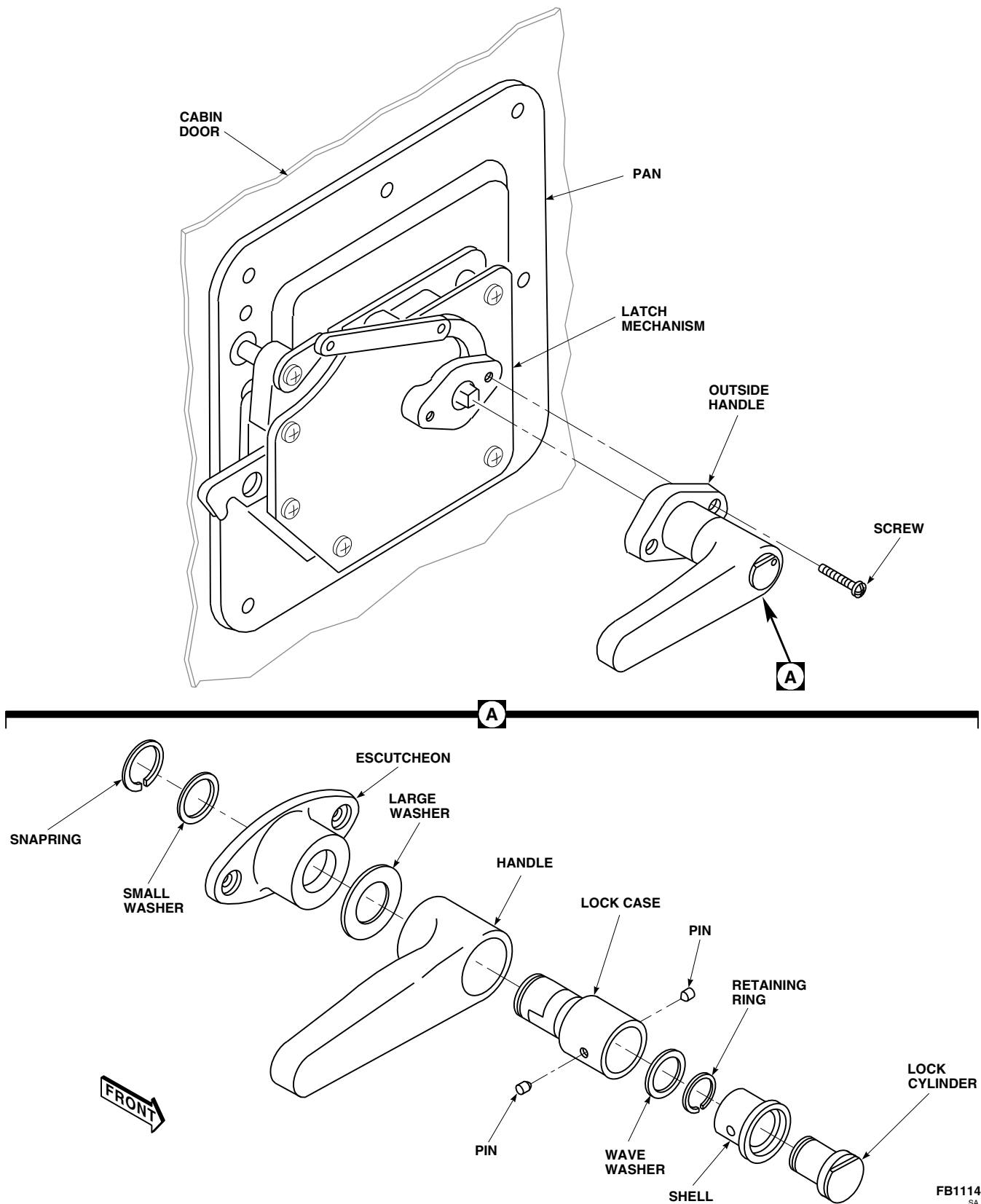


FIGURE 2-4-66-1. REPLACE CABIN DOOR LOCK

2-4-66-2

- h. Close and lock cabin door.

2-4-67 CABIN DOOR WINDOW JETTISON MECHANISM.

PARAGRAPH BREAKDOWN

		PAGE
2-4-67.1	Replace Cabin Door Window Jettison Mechanism	2-4-67-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Nonmetallic Scraper, Locally-Made
- Rivet Gun

Materials/Parts

- Adhesive, Item 32, Appendix D
- Cheesecloth, Item 90, Appendix D

- Methyl Ethyl Ketone, Item 217, Appendix D

References

- Appendix D
- PARA 2-4-56

2-4-67.1 Replace Cabin Door Window Jettison Mechanism.

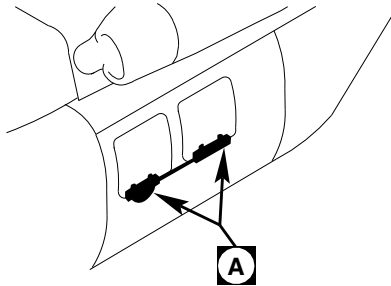
NOTE

Replacement of left and right cabin door window jettison mechanisms is the same.

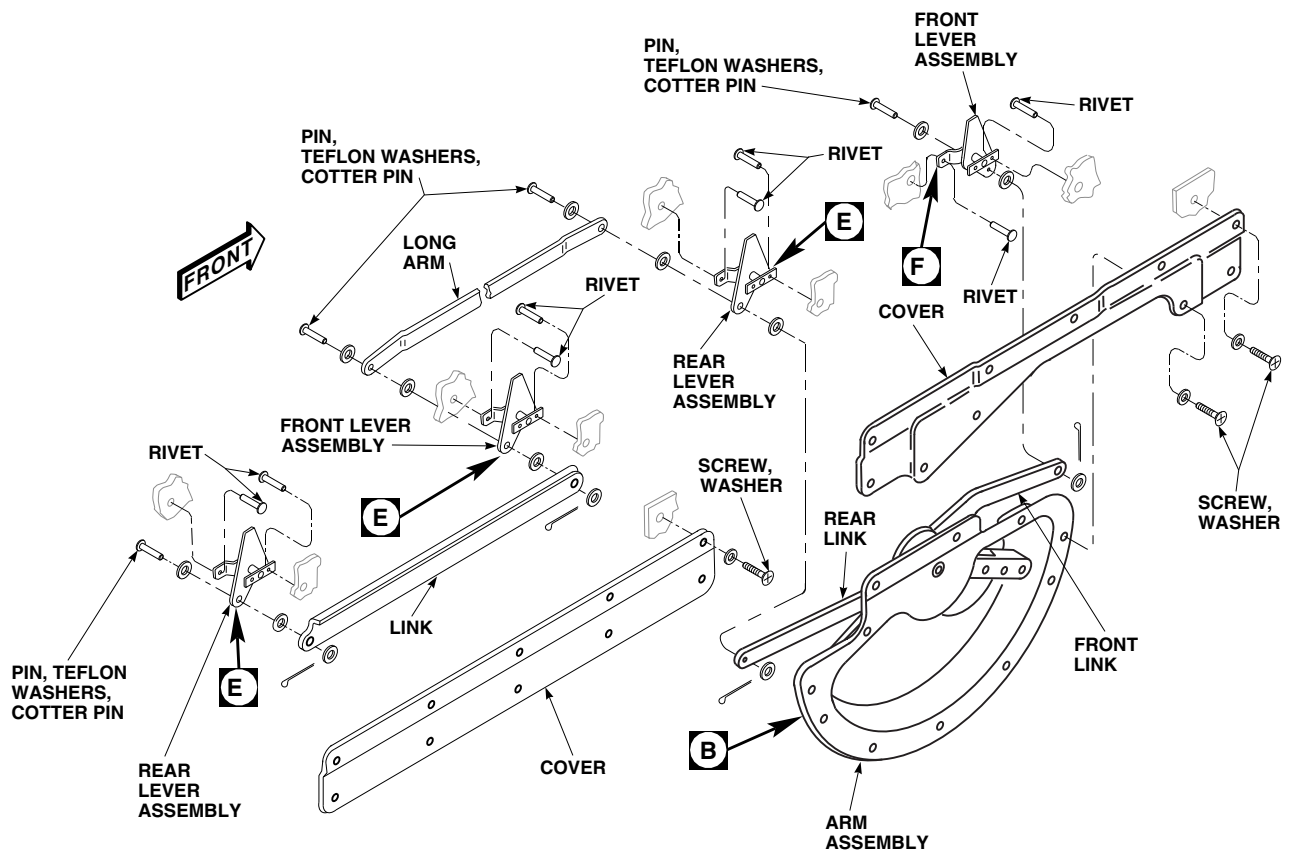
2-4-67.1.1 Disassemble Mechanism.

- If installed, remove windows (PARA 2-4-56).
- Remove screws and washers and remove front and rear access covers from inside of cabin door (Figure 2-4-67-1, Sheet 1, Detail A).
- Disconnect long arm and rear link from rear lever assembly in front window by removing pin and Teflon washers.

- Disconnect long arm and link from front lever assembly in rear window by removing pin and Teflon washers and remove long arm.
- Disconnect link from rear lever assembly in rear window by removing pin and Teflon washers.
- Disconnect front link from front lever assembly in front window by removing pin and Teflon washers.
- Remove arm assembly from cabin door by removing screws and washers.
- Remove rivets securing front and rear lever assemblies to cabin door. Remove assemblies.



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**FIGURE 2-4-67-1. REPLACE CABIN DOOR WINDOW JETTISON MECHANISM
(SHEET 1 OF 2)**

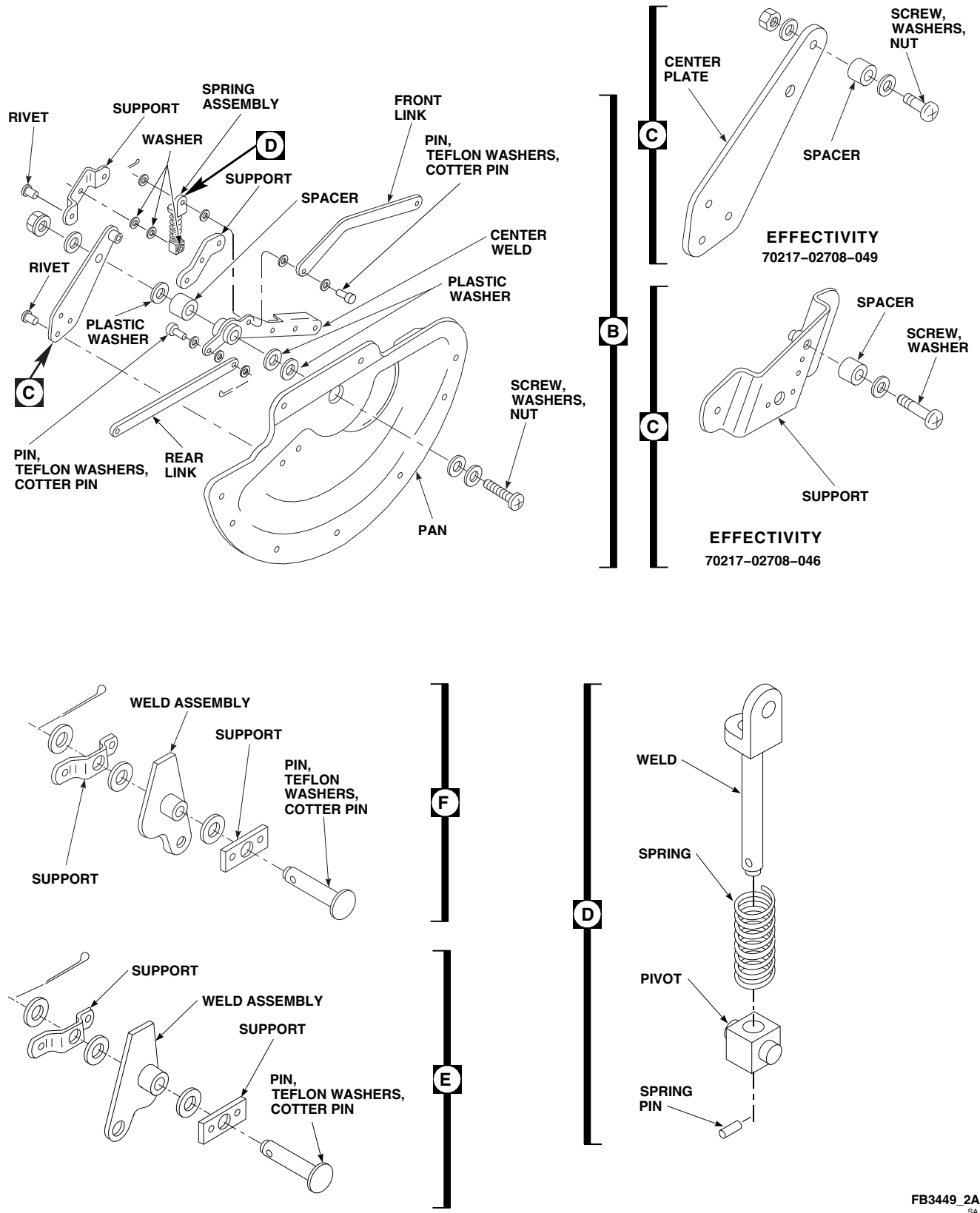


FIGURE 2-4-67-1. REPLACE CABIN DOOR WINDOW JETTISON MECHANISM (SHEET 2 OF 2)

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NOTE

- Disassembly of all lever assemblies is the same.
 - When removing pin from lever, Teflon washers will fall from pin. Retain Teflon washers for reassembly.
- i. Disassemble front and rear lever assemblies by removing cotter pin and sliding pin from supports, washers, and weld assembly (Figure 2-4-67-1, Sheet 2, Detail E and Detail F).

2-4-67.1.2 Disassemble Arm Assembly.

- a. Remove rivets securing supports and spring assembly to center weld (Figure 2-4-67-1, Sheet 2, Detail B).
- b. Remove rivets securing center plate to pan.

NOTE

When removing screw from center weld, plastic and Teflon washers will fall from screw. Do not lose washers.

- c. Remove screw, washers, plastic washers, spacer, and nut securing center plate and center weld to pan.
- d. Remove center weld from arm, 70217-02708-046, by removing screw from center weld and support.
- e. Remove screw and spacer from support.
- f. Remove pin and Teflon washers securing spring assembly and front link to center weld.
- g. Remove screw, washers, and nut securing spacer to center plate (Figure 2-4-67-1, Sheet 2, Detail C).
- h. Remove spring pin from weld and remove spring and pivot (Figure 2-4-67-1, Sheet 2, Detail D).

2-4-67.1.3 Assemble Arm Assembly.

- a. Install spring and pivot on weld and install spring pin in bottom of weld (Figure 2-4-67-1, Sheet 2, Detail D).

- b. Install spacer on center plate with screw, washers, and nut (Figure 2-4-67-1, Sheet 2, Detail C).
- c. Install rear link on center weld as follows (Figure 2-4-67-1, Sheet 2, Detail B):
 - (1) Install Teflon washer under head of pin.
 - (2) Install pin (head of pin outboard) in center weld.
 - (3) Install Teflon washer, rear link, and Teflon washer on pin. Install cotter pin.
- d. Install front link and spring assembly on center weld as follows:
 - (1) Install Teflon washer under head of pin and install pin in link.
 - (2) Install Teflon washer on pin, and install pin with link (head of pin inboard) in center weld.
 - (3) Install Teflon washer, spring assembly, and Teflon washer on pin. Install cotter pin.
- e. Install center weld on pan as follows:
 - (1) If washer is disbonded from pan, bond washer as follows:
 - (a) Clean old adhesive from pan, using cheesecloth, Item 90, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D, and nonmetallic scraper.
 - (b) Bond washer to pan using adhesive, Item 32, Appendix D.
 - (2) Install washer under screwhead and install screw in pan.
 - (3) Install plastic washers on screw.
 - (4) Install center weld, spacer, and plastic washer on screw.
 - (5) Install center plate (with screwhead facing inboard) on center weld.

(6) Install washer and nut on screw.

- f. Line up rivet holes in center plate and pan and install rivets.

2-4-67.1.4 Assemble Mechanism.

a. Assemble front and rear lever assemblies as follows (Figure 2-4-67-1, Sheet 2, Detail E and Detail F):

(1) Slide pin through support and install Teflon washer on pin.

(2) Install weld assembly on pin.

(3) Install Teflon washer, support, and Teflon washer on pin. Install cotter pin.

b. Install lever assemblies in cabin door as follows (Figure 2-4-67-1, Sheet 1, Detail A):

(1) Position lever assembly in cabin door with flat support facing inboard.

NOTE

Install round head rivets from outside of door and flush head rivets from inside of door.

(2) Secure lever assembly to cabin door with rivets.

c. Install arm assembly on door using screws and washers.

d. Connect front link to front lever assembly in front window with pin (head facing outboard), Teflon washers, and cotter pin.

e. Connect link to rear lever assembly in rear window as follows:

(1) Install Teflon washer under head of pin and install pin (head facing outboard) through lever.

(2) Install Teflon washer, link, Teflon washer, and cotter pin.

f. Connect long arm and link to front lever assembly in rear window as follows:

(1) Install Teflon washer under head of pin and install pin (head facing outboard) through long arm.

(2) Install Teflon washer on pin and install pin through lever.

(3) Install Teflon washer, link, Teflon washer, and cotter pin.

g. Connect long arm and rear link to rear lever assembly in front window as follows:

(1) Install Teflon washer under head of pin.

(2) Install pin (head facing outboard) through arm, Teflon washer, and lever assembly.

(3) Install Teflon washer, rear link, and Teflon washer. Install cotter pin.

h. Secure front and rear access covers with screws and washers.

i. If removed, install windows (PARA 2-4-56).

2-4-68 CABIN DOOR WINDOW JETTISON MECHANISM GUARD.

PARAGRAPH BREAKDOWN

	PAGE
2-4-68.1 Replace Cabin Door Window Jettison Mechanism Guard	2-4-68-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Adhesive, Item 61, Appendix D
- Cheesecloth, Item 90, Appendix D

- Fastener Tape, Hook, Item 137A, Appendix D
- Fastener Tape, Pile, Item 137B, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D

References

- Appendix D

2-4-68.1 Replace Cabin Door Window Jet-tison Mechanism Guard.

NOTE

Replacement of left and right cabin door window jettison handle guards is the same.

2-4-68.1.1. Remove.

Pull handle guard from door (Figure 2-4-68-1, Detail A).

2-4-68.1.2. Repair.

- Using nonmetallic scraper, peel hook fastener tape from guard and pile fastener tape from jet-tison arm and access cover (Figure 2-4-68-1, Detail A).

- Remove all adhesive with cheesecloth, Item 90, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.
- Measure and cut hook fastener tape, Item 137A, Appendix D, and pile fastener tape, Item 137B, Appendix D, to length (Figure 2-4-68-1). Trim tapes to match upper and front edges of handle guard.
- Brush a thin even coat of adhesive, Item 61, Appendix D, on jettison arm/access cover and pile fastener tape bonding surfaces. Allow adhesive to air-dry for at least 10 minutes.
- Carefully press pile fastener tape into position on jettison arm and access cover (Figure 2-4-68-1, Detail A). Allow 16 hours for adhesive to dry before handling.
- Brush a thin even coat of adhesive, Item 61, Appendix D, on handle guard and hook

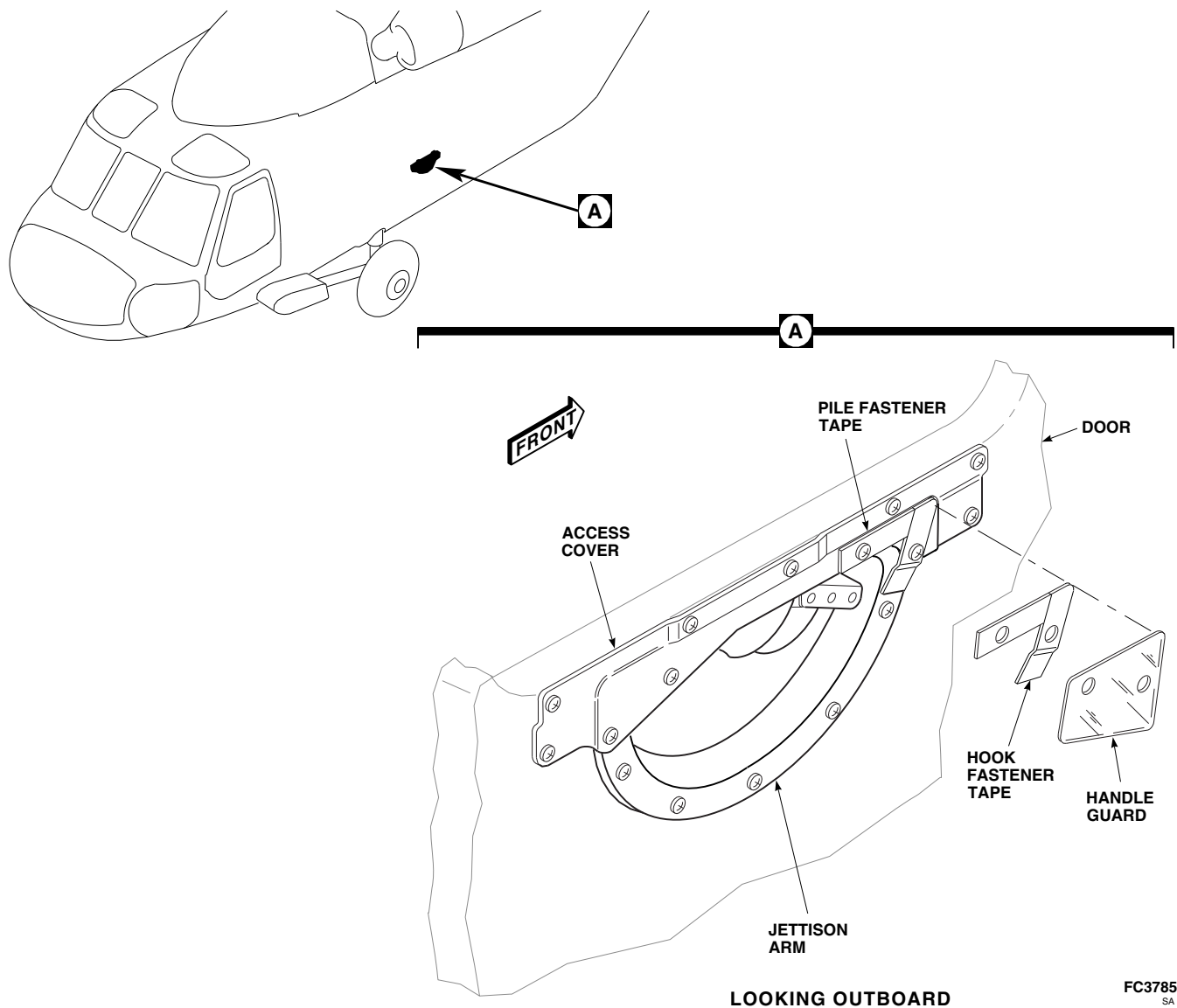


FIGURE 2-4-68-1. REPLACE CABIN DOOR WINDOW JETTISON MECHANISM GUARD

fastener tape bonding surfaces. Allow adhesive to dry for at least 10 minutes.

- g. Carefully press hook fastener tape into position on guard. Allow 16 hours for adhesive to dry before handling.
- h. Cut 0.5-inch diameter holes in both hook fastener tape and pile fastener tape to match holes in handle guard.

2-4-68.1.3. Install.

Install handle guard by pressing in place, engaging hook fastener tape and pile fastener tape (Figure 2-4-68-1, Detail A).

2-4-69 SOUNDPROOFING PANELS.

PARAGRAPH BREAKDOWN

		PAGE
2-4-69.1	Replace Soundproofing Panels	2-4-69-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

References

- PARA 2-4-45

2-4-69.1 Replace Soundproofing Panels.

2-4-69.1.2 Install.

NOTE

Replacement of all soundproofing panels is the same, except as noted.

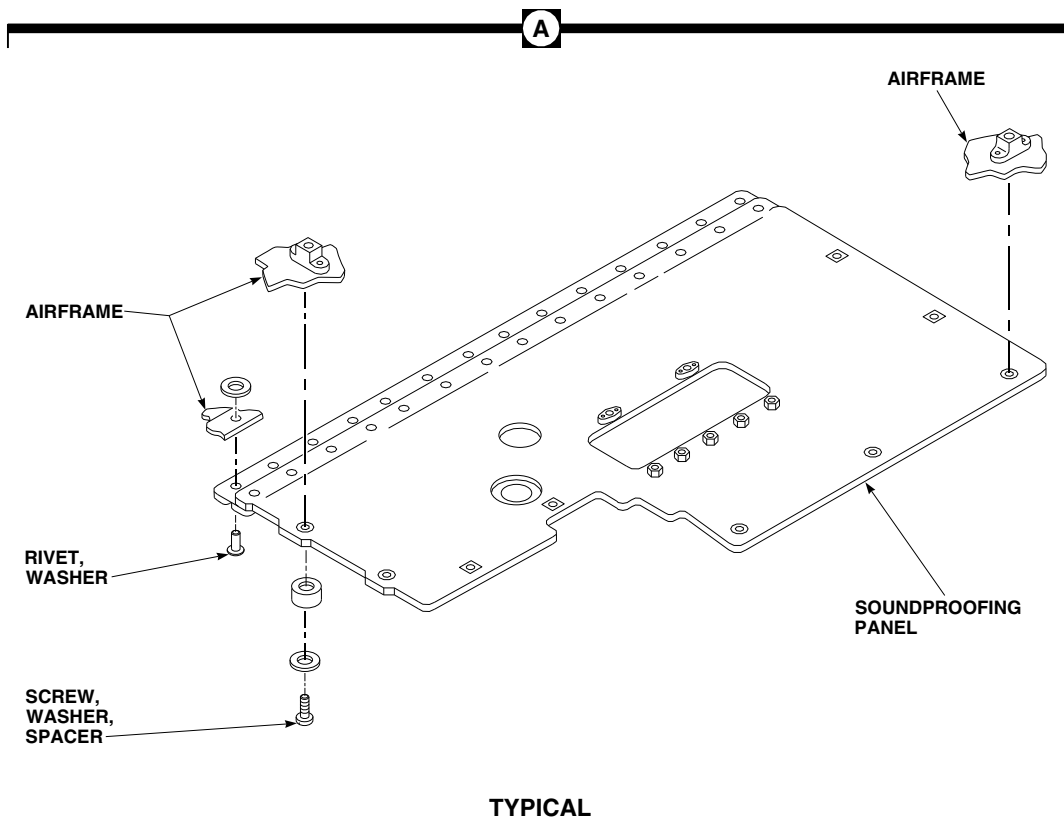
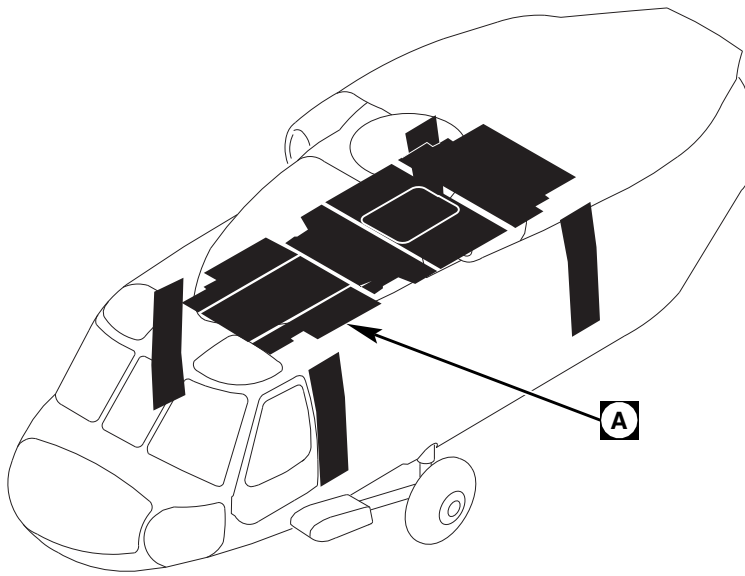
2-4-69.1.1 Remove.

- If required, remove any furnishings/equipment.
- If required, remove crew chief’s seat upper support fittings (PARA 2-4-45).
- Remove attaching hardware and loosen ¼-turn fasteners, if necessary, securing soundproofing panel to cabin, ceiling, rear bulkhead, or sidewalls (Figure 2-4-69-1, Detail A).

CAUTION

To avoid possible interference between flight control rods and soundproofing panels, make sure panel over vertical flight control rods overlaps cover (STA 247.0, WL 206.75 - WL 215.0).

- Install soundproofing panels in cabin, ceiling, rear bulkhead, or sidewalls with attaching hardware and ¼-turn fasteners, if necessary (Figure 2-4-69-1, Detail A).
- If required, install crew chief’s seat upper support fittings (PARA 2-4-45).
- If required, install furnishings/equipment.



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FIGURE 2-4-69-1. REPLACE SOUNDPROOFING PANELS

2-4-70 MAIN TRANSMISSION DRIP PAN.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
2-4-70.1	Replace Drain Tubes	2-4-70-1	2-4-70.4	Replace Drip Pan Cover Seals	2-4-70-6
2-4-70.2	Replace Drip Pan and Support	2-4-70-4	2-4-70.5	Replace Support Seal	2-4-70-8
2-4-70.3	Replace Drip Pan Seals	2-4-70-6			

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Drill
- Drill Bit, No. 40 (0.098-inch diameter)
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Abrasive Cloth, Item 11, Appendix D
- Adhesive, Item 30, Appendix D
- Adhesive, Item 37, Appendix D
- Adhesive, Item 56, Appendix D

- Alcohol, Ethyl, Item 71, Appendix D
- Brush, Acid, Item 84, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 2-4-45
- PARA 2-4-69
- PARA 2-4-95
- PARA 2-4-96
- PARA 2-4-98
- PARA 2-6-3

2-4-70.1 Replace Drain Tubes.

2-4-70.1.1 Remove.

- Remove soundproofing panels at STA 308.0, left and right side, and at STA 383.0 left side (PARA 2-4-69).
- Remove cabin ceiling drain tubes as follows:

- Remove screws, washers, and nuts securing clamps and tube to soundproofing panel (Figure 2-4-70-1, Sheet 1, Detail A).
- Loosen hose clamps at both ends of tube and disconnect tube from drain fitting (on drip pan), and bulkhead fitting or tube

coupling (Figure 2-4-70-1, Sheet 1, Detail A, Detail B, and Detail C).

- (3) Remove tube. Remove hose clamps and clamps from tube.

c. Remove STA 308.0 drain tubes as follows:

- (1) Remove upper ESSS fitting cap fairing (PARA 2-4-98), upper main landing gear fairing (PARA 2-4-96), and lower main landing gear cap fairing (PARA 2-4-95).
- (2) Working inside strut well, remove screws, washers, spacers, and nuts securing clamps and tube to bulkhead (Figure 2-4-70-1, Sheet 1, Detail D and Sheet 2, Detail E).
- (3) Loosen hose clamp and disconnect tube from bulkhead fitting (Figure 2-4-70-1, Sheet 1, Detail C and Detail D).
- (4) Remove tube. Remove hose clamps and clamps from tube.

d. Remove STA 383.0 drain tube as follows:

- (1) Remove screws, washers, and nuts securing clamps and tube to airframe (Figure 2-4-70-1, Sheet 2, Detail G).
- (2) Loosen hose clamps at both ends of tube and disconnect tube from tube coupling and airframe fitting (Figure 2-4-70-1, Sheet 1, Detail B and Sheet 2, Detail G).
- (3) Remove tube. Remove hose clamps and clamps from tube.

2-4-70.1.2 Install.

- a. Slip support clamps and hose clamps on replacement tube as required.

b. Install cabin ceiling drain tubes as follows:

- (1) Connect drain tube to drain fitting (on drip pan), and bulkhead fitting or tube coupling and secure with hose clamps (Figure 2-4-70-1, Sheet 1, Detail A, Detail B, and Detail C). **TIGHTEN HOSE CLAMPS.**

- (2) Secure clamps and tube to soundproofing panel using screws, washers, and nuts (Figure 2-4-70-1, Sheet 1, Detail A). Install washers to keep drain tube sloping from drip pan fitting to bulkhead fitting or tube coupling.

c. Install STA 308.0 drain tubes as follows:

- (1) Connect drain tube to bulkhead fitting and secure with hose clamp (Figure 2-4-70-1, Sheet 1, Detail C). **TIGHTEN HOSE CLAMP.**
- (2) Working inside strut well, secure clamps and tube to airframe using screws, washers, spacers, and nuts (Figure 2-4-70-1, Sheet 1, Detail D and Sheet 2, Detail E).
- (3) Leave 2 1/2-inches of tubing extended below grommet in lower main landing gear fairing. Cut away extra length on a 45° opening facing rear (Figure 2-4-70-1, Sheet 2, Detail F).
- (4) Apply utility drain identification tape (Figure 2-4-70-1, Sheet 1, Detail D).
- (5) Install upper ESSS fitting cap fairing (PARA 2-4-98), upper main landing gear fairing (PARA 2-4-96), and lower main landing gear cap fairing (PARA 2-4-95).

d. Install STA 383.0 drain tubes as follows:

- (1) Connect drain tube to tube coupling and airframe fitting and secure with hose clamps (Figure 2-4-70-1, Sheet 1, Detail B and Sheet 2, Detail G). **TIGHTEN HOSE CLAMPS.**
- (2) Secure clamps and tube to airframe using screws, washers, and nuts (Figure 2-4-70-1, Sheet 2, Detail G).

- e. Make sure area is clean and free of foreign material before closing up.

- f. Install soundproofing panels (PARA 2-4-69).

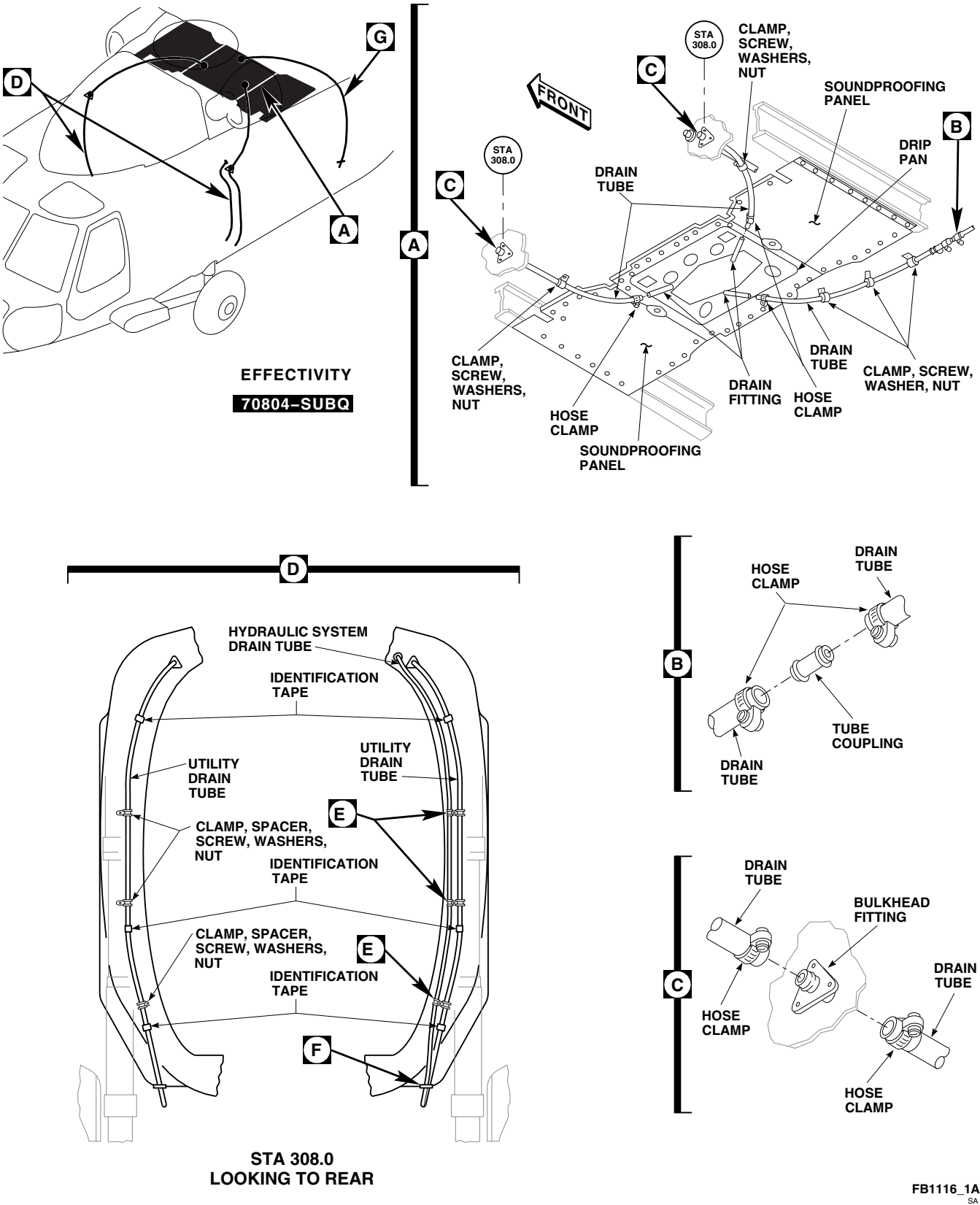


FIGURE 2-4-70-1. REPLACE DRAIN TUBES (SHEET 1 OF 2)

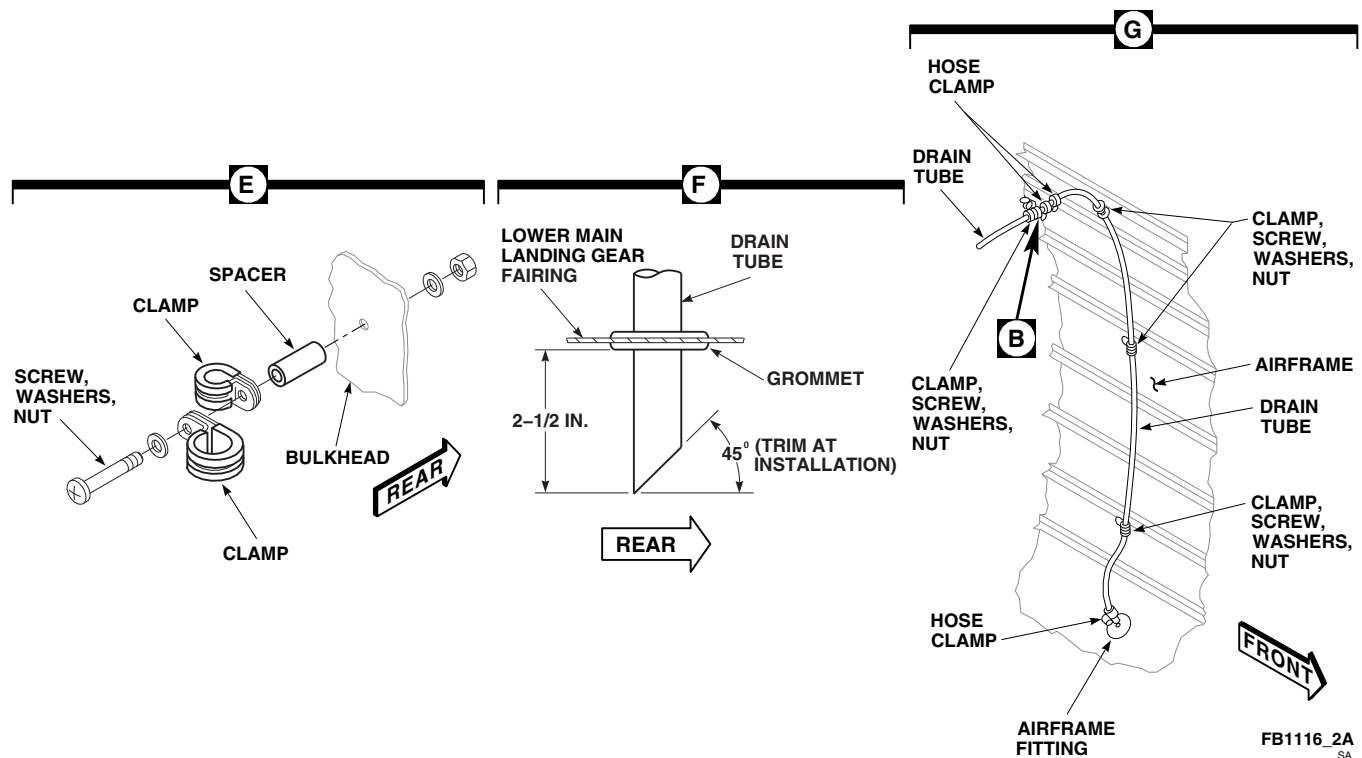


FIGURE 2-4-70-1. REPLACE DRAIN TUBES (SHEET 2 OF 2)

2-4-70.2 Replace Drip Pan and Support.**2-4-70.2.1 Remove.****NOTE**

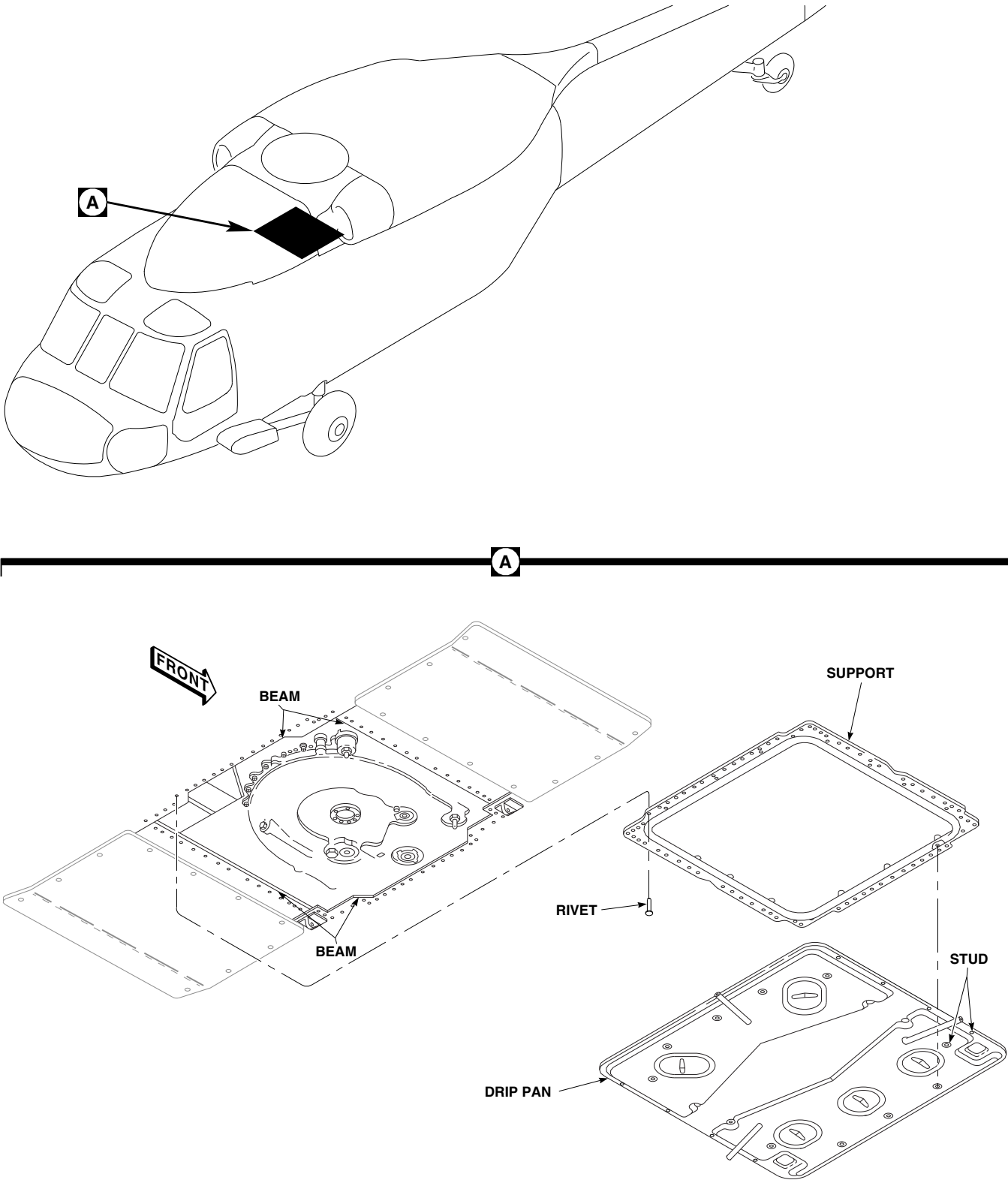
As an alternate to removing passenger's seat upper supports, proceed to step b.

- a. Remove passenger's seat upper supports (PARA 2-4-45).
- b. As an alternate method of removing drip pan without removing upper seat supports, perform the following:
 - (1) Cut a slot at left/right corners, 1.6-inches in from outboard edge, 2.2-inches wide by 5.2-inches long.
 - (2) Repair edges of slot (PARA 2-6-3).
- c. Loosen hose clamps and disconnect cabin ceiling drain tubes from drain fittings on drip pan (Figure 2-4-70-1, Sheet 1, Detail A).

- d. Support drip pan. Unlock studs fastening drip pan to support (Figure 2-4-70-2, Detail A). Remove drip pan.
- e. Remove rivets and remove support from beams.
- f. Remove sealing compound from beams using nonmetallic scraper.

2-4-70.2.2 Install.

- a. Seal all overlapping surfaces between beam flanges and support with sealing compound Item 292, Appendix D.
- b. Install support on beams and secure with rivets (Figure 2-4-70-2, Detail A).
- c. Line up studs of drip pan with holes in support. Secure drip pan to support.
- d. Connect cabin ceiling drain tubes to drain fittings on drip pan. **TIGHTEN HOSE CLAMPS** (Figure 2-4-70-1, Sheet 1, Detail A).



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FIGURE 2-4-70-2. REPLACE DRIP PAN AND SUPPORT

- e. If removed, install passenger's seat upper supports (PARA 2-4-45).

2-4-70.3 Replace Drip Pan Seals.

NOTE

Replacement of all drip pan seals is the same.

2-4-70.3.1 Remove.

- a. Remove drip pan from support (PARA 2-4-70.2).
- b. Remove seal from drip pan using nonmetallic scraper (Figure 2-4-70-3, Detail A).
- c. Clean off any remaining adhesive using machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D. Wipe dry with clean machinery towel, Item 211, Appendix D.

2-4-70.3.2 Install.

- a. Clean replacement seal using machinery towel, Item 211, Appendix D, moistened with ethyl alcohol, Item 71, Appendix D. Wipe dry with clean machinery towel.
- b. Thoroughly scuff up bonding surface of seal and drip pan with abrasive cloth, Item 11, Appendix D. Wipe off all dust with clean machinery towel, Item 211, Appendix D.
- c. Perform final cleaning of surfaces to be bonded with clean machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D.
- d. Within 15 minutes after cleaning surfaces to be bonded, brush one coat of adhesive, Item 37, Appendix D, on drip pan and one coat on replacement seal.
- e. Allow adhesive to air-dry for 10 - 15 minutes.
- f. Install replacement seal on drip pan. Join together firmly while making sure entire mating surfaces are in direct contact. Do not move or disturb bonded surfaces after mating (Figure 2-4-70-3, Detail A).

- g. Wipe off extra adhesive with machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D.

- h. Allow adhesive to cure for at least 12 hours.

- i. Install drip pan (PARA 2-4-70.2).

2-4-70.4 Replace Drip Pan Cover Seals.

NOTE

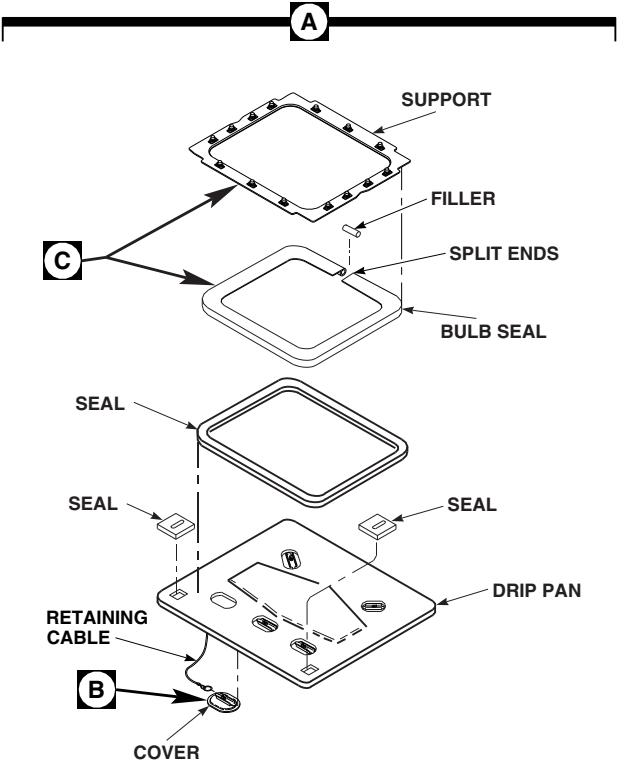
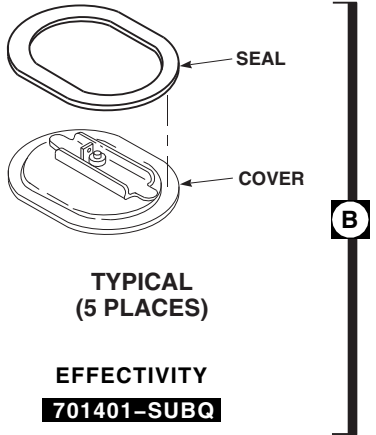
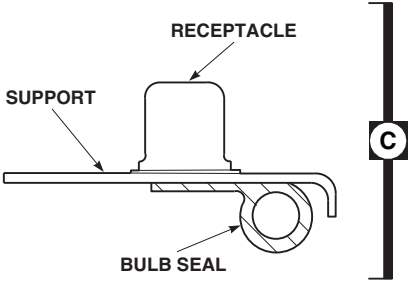
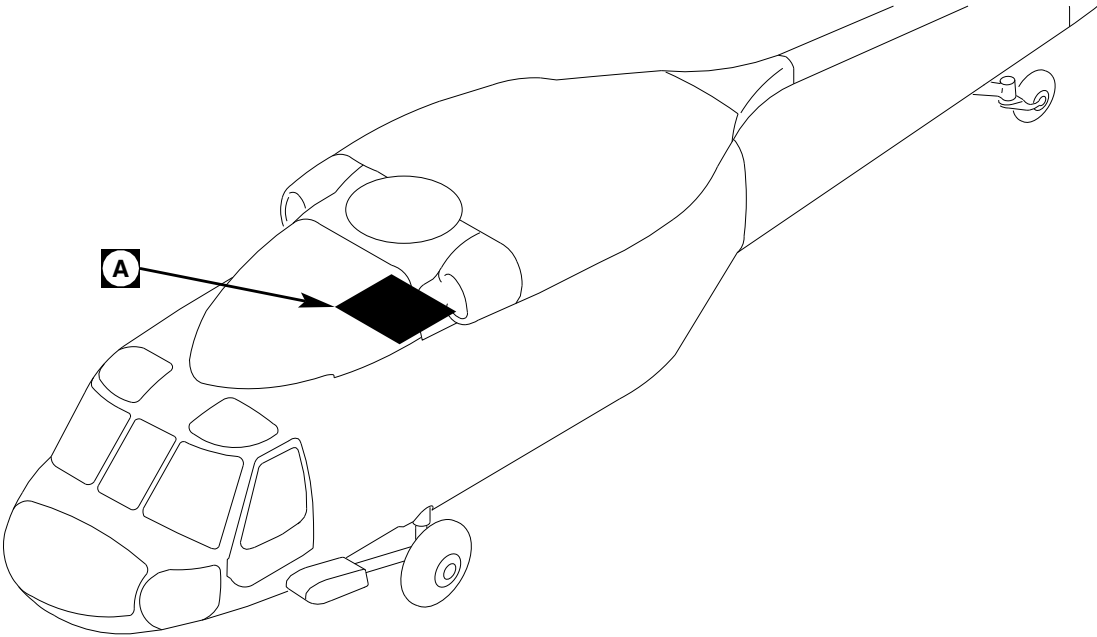
Replacement of all drip pan cover seals is the same.

2-4-70.4.1 Remove.

- a. Remove cover from drip pan and disconnect retaining cable from cover.
- b. Remove old seal from cover using nonmetallic scraper (Figure 2-4-70-3, Detail B).
- c. Clean off any remaining adhesive using machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D. Wipe dry with clean towel.

2-4-70.4.2 Install.

- a. Scuff up bonding surface of seal with abrasive cloth, Item 11, Appendix D.
- b. Wipe off all dust with clean machinery towel, Item 211, Appendix D.
- c. Clean seal bonding surface with machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D. Let air dry.
- d. Apply even coat of adhesive, Item 30 or Item 56, Appendix D, to bonding surfaces of both seal and cover, using clean acid brush, Item 84, Appendix D.
- e. Install seal on cover and apply moderate pressure to make sure of firm contact at all points (Figure 2-4-70-3, Detail B).
- f. Wipe off excess adhesive with machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D.



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FIGURE 2-4-70-3. REPLACE DRIP PAN SEALS

- g. Allow adhesive to cure for 24 hours at room temperature. For adhesive, Item 30, Appendix D, alternate curing time is 1 hour at 140° - 160°F (60° - 71°C).
- h. Connect retaining cable to cover and install cover in drip pan.

2-4-70.5 Replace Support Seal.

2-4-70.5.1 Remove.

- a. Remove drip pan and support (PARA 2-4-70.2.1).
- b. Remove old bulb seal from support using nonmetallic scraper (Figure 2-4-70-3, Detail C).
- c. Clean off any remaining adhesive using machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D. Wipe dry with clean machinery towel.

2-4-70.5.2 Install.

- a. Clean replacement bulb seal using machinery towel, Item 211, Appendix D, moistened with ethyl alcohol, Item 71, Appendix D. Wipe dry with clean machinery towel.
- b. Thoroughly scuff up surface of bulb seal and support to be bonded with abrasive cloth, Item 11, Appendix D. Wipe off all dust with clean machinery towel, Item 211, Appendix D.
- c. Perform final cleaning of surfaces to be bonded with clean machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D.

- d. Before coating bulb seal or support with adhesive, perform the following:

- (1) Position bulb seal on support (Figure 2-4-70-3, Detail C).
- (2) Place ends of bulb seal at rear end of support half way between center of support and right side. Install filler, an equal distance in each end of bulb seal.
- (3) Slit flat part of bulb seal as necessary to allow seal to bend at corners.
- (4) Using drill with No. 40 (0.098-inch diameter) drill bit, punch or drill four 0.098-inch diameter holes through only outboard surface of bulb seal. Hole should be at about center of each side of support to allow seal to compress.

- e. Coat surfaces of bulb seal and support to be bonded with adhesive, Item 30, Appendix D.
- f. Allow adhesive to air-dry for about 1/2 to 1 hour, or until sticky.
- g. Install bulb seal on support. Join ends together firmly, while making sure entire mating surfaces are in direct contact. Do not move or disturb bonded surfaces after mating.
- h. Wipe off extra adhesive with machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D.
- i. Allow adhesive to cure for at least 24 hours. For greatest strength, cure about 7 days. Cure time can be accelerated by applying heat at 110° - 130°F (43° - 54°C) for 16 hours.
- j. Install drip pan and support (PARA 2-4-70.2.2).

2-4-71 CABIN FLOOR.

PARAGRAPH BREAKDOWN

	PAGE
2-4-71.1 Replace Cabin Floor	2-4-71-2

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Drill Set
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, Range to Support Seating Torque of Cabin Floor Hardware

Materials/Parts

- Adhesive, Item 32, Appendix D
- Adhesive, Item 66, Appendix D
- Alcohol, Isopropyl, Item 72, Appendix D
- Bolt, AN3-5A
- Cleaning Compound, Solvent, Item 99C, Appendix D
- Cloth, Cleaning, Item 102, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Paint, Polyurethane, Item 228, Appendix D
- Paint Remover, Item 231, Appendix D
- Pressure Sensitive Adhesive Tape, Item 250B, Appendix D

- Pressure Sensitive Adhesive Tape, Item 250C, Appendix D
- Pressure Sensitive Adhesive Tape, Item 250D, Appendix D
- Primer, Item 257, Appendix D
- Sealing Compound, Item 275, Appendix D
- Sealing Compound, Item 301A, Appendix D
- Sealing Compound, Item 302D, Appendix D
- Shim, 70214-02555-109
- Tags
- Washer, NAS1149F0332P

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-45
- PARA 2-6-3
- PARA 2-6-5
- PARA 2-6-27
- PARA 9-4-34
- PARA 9-4-106
- PARA 9-4-107

2-4-71.1 Replace Cabin Floor.**NOTE**

Replacement of all floor panels is the same, except as noted.

2-4-71.1.1 Remove.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Remove passenger's seats from applicable floor panel (PARA 2-4-45).
- c. If removing front floor panel (STA 247.0 - 288.0), perform the following:
 - (1) Remove battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
 - (2) Remove sealed lead-acid battery junction box (PARA 9-4-107).

NOTE

- On front floor panel (STA 247.0 - 288.0), a combination of bolt, AN3-5A, and washer, NAS1149F0332P, may be used as an alternate to screw only in locations shown in Figure 2-4-71-1, Sheet 2, Detail C.
 - On front floor panel, not all screws securing floor pans to support structure use washers. Screws and washers, where used, are of different sizes. Hardware must be tagged indicating location.
 - On middle and rear floor panels, screws and washers securing floor pans to support structure are the same within each panel. If removing multiple panels, hardware must be tagged indicating location.
- d. Remove screws, or bolts and washers (if installed), securing floor panel to support structure (Figure 2-4-71-1, Sheet 1, Detail A or Sheet 2, Detail C). Remove screws and washers securing floor pans to support structure (Figure 2-4-71-1, Sheet 2, Detail B). Remove floor panel from helicopter.

- e. Remove sealing compound from edges of floor panel, areas of floor panel that contact support structure, and adjacent floor panel edge using paint remover, Item 231, Appendix D, and nonmetallic scraper.
- f. Remove sealing compound and tape from attachment flanges of support structure using nonmetallic scraper. Remove any remaining residue with paint remover, Item 231, Appendix D.

2-4-71.1.2 Inspect.

- a. Visually inspect fasteners, skin under stringers and around bulkheads, and floor attachment fittings for corrosion.
- b. Inspect tub section for presence of corrosion protection as follows:
 - (1) If corrosion protection is present, clean tub section with cleaning cloth, Item 102, Appendix D, moistened with isopropyl alcohol, Item 72, Appendix D.
 - (2) If corrosion protection is not present, perform the following:
 - (a) Clean interior tub section with cleaning cloth, Item 102, Appendix D, moistened with isopropyl alcohol, Item 72, Appendix D. Allow to dry completely.
 - (b) Apply one coat of clear polyurethane paint, Item 228, Appendix D, to the entire area. Allow to dry completely.

2-4-71.1.3 Install.

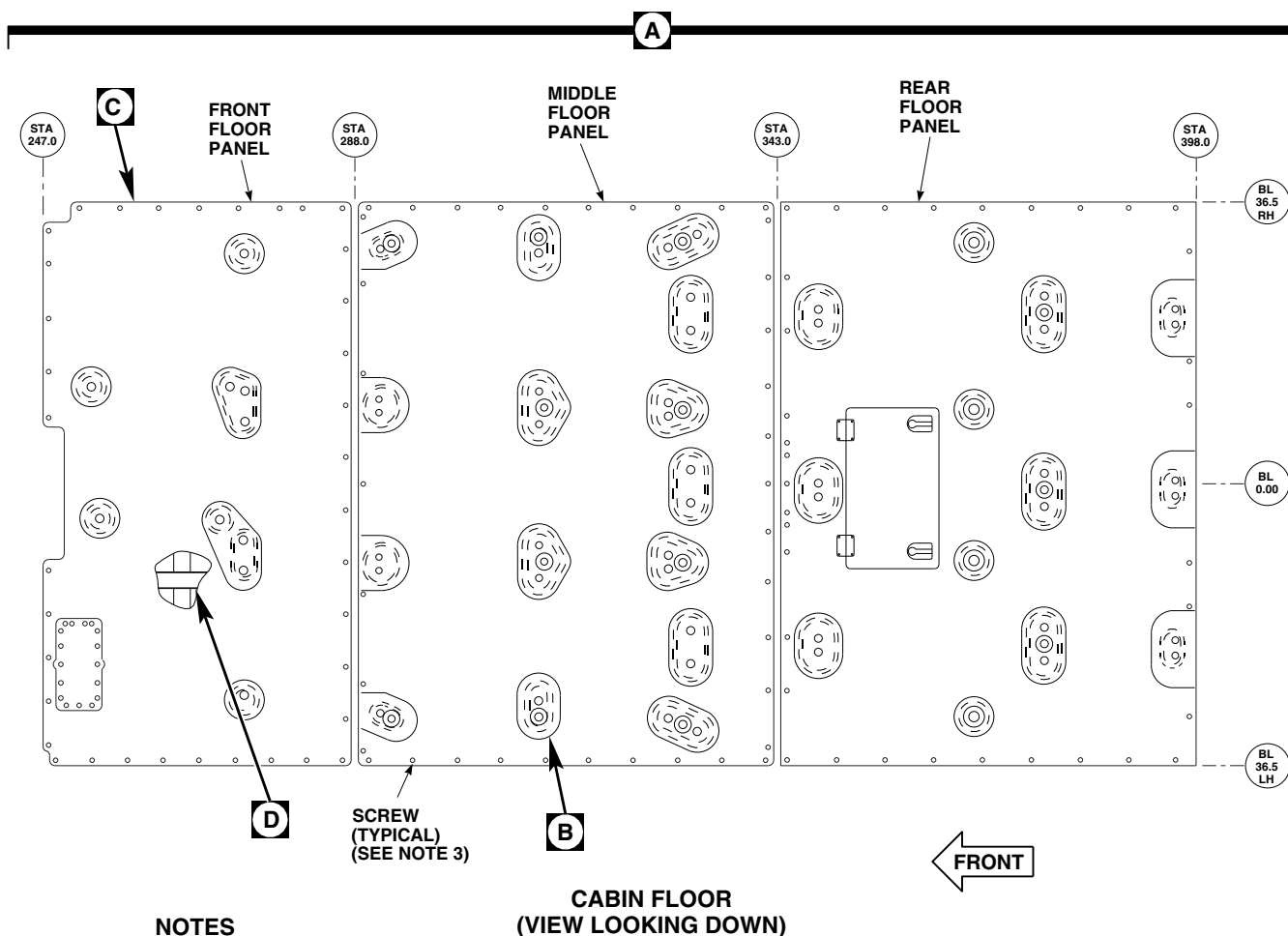
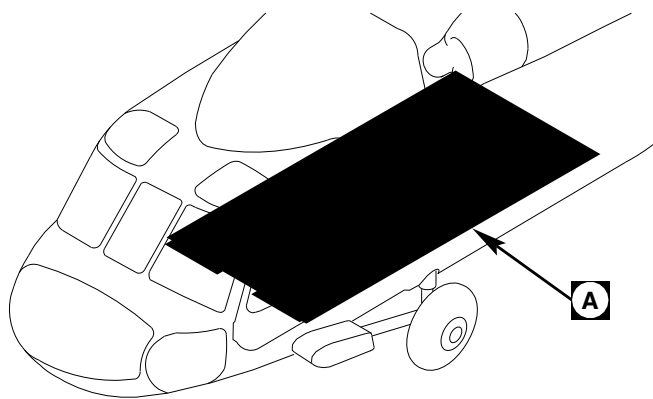
- a. Turn off all electrical power (PARA 1-6-16).
- b. Inspect pans for missing or disbonded shims. If shims are missing or have become disbonded from pan, perform the following as required:

NOTE

Two shims, 70214-02555-109, replace one shim, 70214-02555-107.

- (1) If shim is missing, perform the following:

2-4-71-2 Change 5

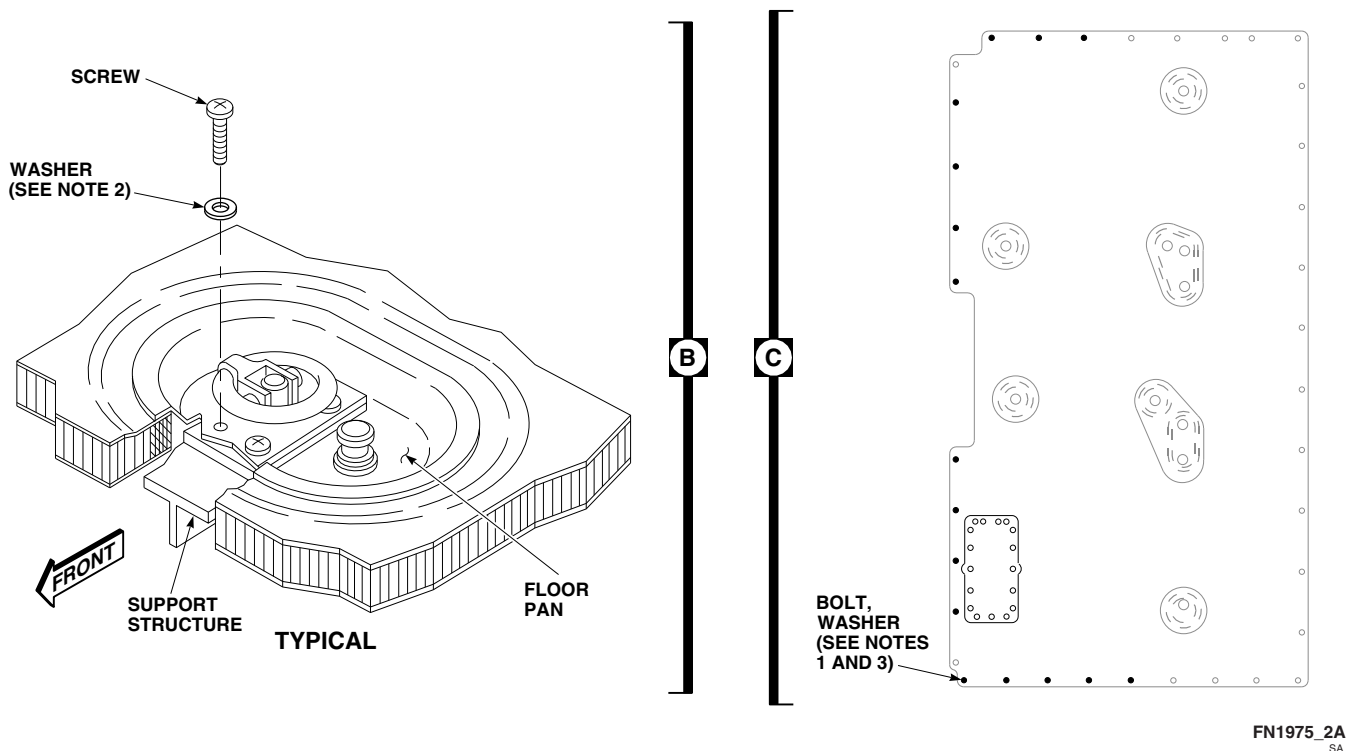


NOTES

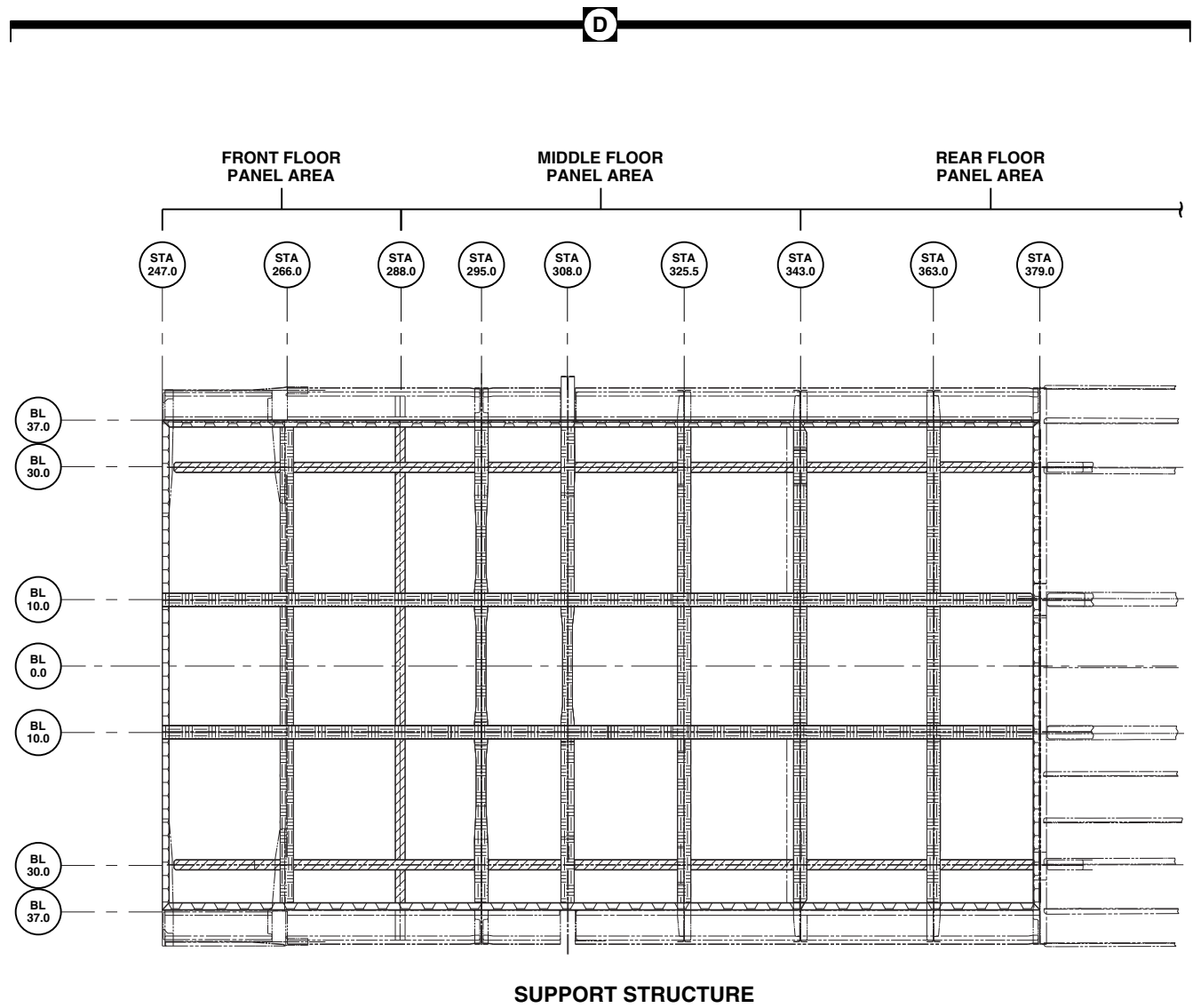
1. THE COMBINATION OF BOLT, AN3-5A, AND WASHER, NAS1149F0332P, CAN BE USED AS AN ALTERNATE TO SCREW ONLY IN LOCATIONS SHOWN.
2. ON FRONT FLOOR PANEL, NOT ALL SCREWS SECURING FLOOR PANS TO SUPPORT STRUCTURE USE WASHERS.
3. TORQUE SCREWS, OR BOLTS WHERE SPECIFIED, NO MORE THAN 10 IN. LBS OVER SEATING TORQUE.

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FIGURE 2-4-71-1. REPLACE CABIN FLOOR (SHEET 1 OF 3)

FN1975_2A
SA**FIGURE 2-4-71-1. REPLACE CABIN FLOOR (SHEET 2 OF 3)**

- (a) Peel replacement shim so that there is a maximum gap of 0.010-inch between bottom of floor panel and top of shim. When using solid shim, machine to obtain gap.
- (b) Bond new shim to bottom of pan using bonding procedure in step b. (2).
- (2) If shim is disbonded, perform the following:
 - (a) Wipe bottom of pan and solid side of shim using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.
 - (b) Check for corrosion on pan and shim. If required, remove corrosion and repair area (PARA 2-6-27).
 - (c) Remove any remaining adhesive from pan and shim using nonmetallic scraper. Clean area using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.
 - (d) Prime solid side of shim using adhesive, Item 66, Appendix D. Allow primer to dry at room temperature for at least 1 hour.
 - (e) Apply a light coat of adhesive, Item 32, Appendix D, to solid side of shim.
 - (f) Place shim on bottom of pan, with holes in shim and pan lined up.
 - (g) Allow adhesive, Item 32, Appendix D, to cure at room temperature for at least 1 hour.
- c. Apply two coats of primer, Item 257, Appendix D, to attachment flanges of support structure. Allow primer to dry.
- d. Apply pressure sensitive adhesive tape to support structure as follows:



LEGEND

-  TAPE, ITEM 250B, APPENDIX D
-  TAPE, ITEM 250C, APPENDIX D
-  TAPE, ITEM 250D, APPENDIX D

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FIGURE 2-4-71-1. REPLACE CABIN FLOOR (SHEET 3 OF 3)

- (1) Using solvent cleaning compound, Item 99C, Appendix D, and cleaning cloth, Item 102, Appendix D, clean attachment flanges of support structure. Wipe dry with clean, cleaning cloth.
- (2) Apply pressure sensitive adhesive tape, Item 250B, Item 250C, and Item 250D, Appendix D, to attachment flanges of support structure, using as many long, continuous lengths as possible. Refer to Figure 2-4-71-1, Sheet 3, Detail D for location of tape. Do not remove backing from top side of tape.

NOTE

Leakage may occur if gaps exist between pieces of tape.

- (3) Where multiple pieces of tape must be used, overlap tape in corners 1/4-inch. **NO GAPS ALLOWED BETWEEN PIECES OF TAPE.**

- (4) Trim tape so that all edges are flush with floor panels (± 0.060 -in.).

- e. If installing new floor panel, trial-fit new panel on support structure with screws, or bolts and washers (if removed) (Figure 2-4-71-1, Sheet 1, Detail A or Sheet 2, Detail B and Detail C). Inspect panel for mounting holes that do not line up with nutplates in support structure. If necessary, repair mismatched mounting hole(s) as follows:

- (1) Mark mounting hole(s) in panel that do not line up with nutplate(s) in support structure. Mark panel with correct location of mounting hole(s).
- (2) Remove screws, or bolts and washers (if installed), securing panel to support structure and remove panel.
- (3) Fill marked hole(s) (PARA 2-6-3).
- (4) Using drill, drill and countersink new mounting hole(s) in panel.
- (5) Touch up paint on panel as required (PARA 2-6-5).

- f. Make sure area is clean and free of foreign material before closing up.
- g. Apply sealing compound, Item 275, Appendix D, to attaching flanges on support structure over pressure sensitive adhesive tape.
- h. Apply sealing compound, Item 302D, Appendix D, to edges of floor panel (joint seams).
- i. Position floor panel on support structure and line up mounting holes (Figure 2-4-71-1, Sheet 1, Detail A).
- j. Secure floor panel as follows:

NOTE

- On front floor panel (STA 247.0 - 288.0), a combination of bolt, AN3-5A, and washer, NAS1149F0332P, may be used as an alternate to screw only in locations shown in Figure 2-4-71-1, Sheet 2, Detail C.

- On front floor panel, not all screws securing floor pans to support structure use washers. Screws and washers, where used, are of different sizes. Remove tags as hardware is installed.

- On middle and rear floor panels, screws and washers securing floor pans to support structure are the same within each panel. Remove tags as hardware is installed.

- (1) Install screws, or bolts and washers (if removed), securing floor panel to support structure wet with sealing compound, Item 301A, Appendix D (Figure 2-4-71-1, Sheet 1, Detail A or Sheet 2, Detail C).

- (2) Install screws and washers securing floor pans to support structure wet with sealing compound, Item 301A, Appendix D (Figure 2-4-71-1, Sheet 2, Detail B).

- (3) Tighten hardware until seated.

- (4) **TORQUE SCREWS, OR BOLTS WHERE SPECIFIED, NO MORE**

**THAN 10 INCH-POUNDS OVER
SEATING TORQUE.**

- k. Remove any sealing compound squeeze out from edges of floor panel (joint seams).
- l. If front floor panel (STA 247.0 - 288.0) was installed, perform the following:
 - (1) Install battery (PARA 9-4-34 (NiCad) or PARA 9-4-106 (SLAB)).
 - (2) Install sealed lead-acid battery junction box (PARA 9-4-107).
- m. Install passenger's seats (PARA 2-4-45).

2-4-72 CARGO HOOK DOOR FLOOR SEAL.

PARAGRAPH BREAKDOWN

		PAGE
2-4-72.1	Replace Cargo Hook Door Strap Seal	2-4-72-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Abrasive Cloth, Item 9, Appendix D
- Adhesive, Item 37, Appendix D

- Cargo Hook Access Seal, Locally-Made, Figure H-60, Appendix H
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D

References

- Appendix D
- Appendix H
- PARA 2-4-75

2-4-72.1 Replace Cargo Hook Door Strap Seal.

2-4-72.1.1. Remove.

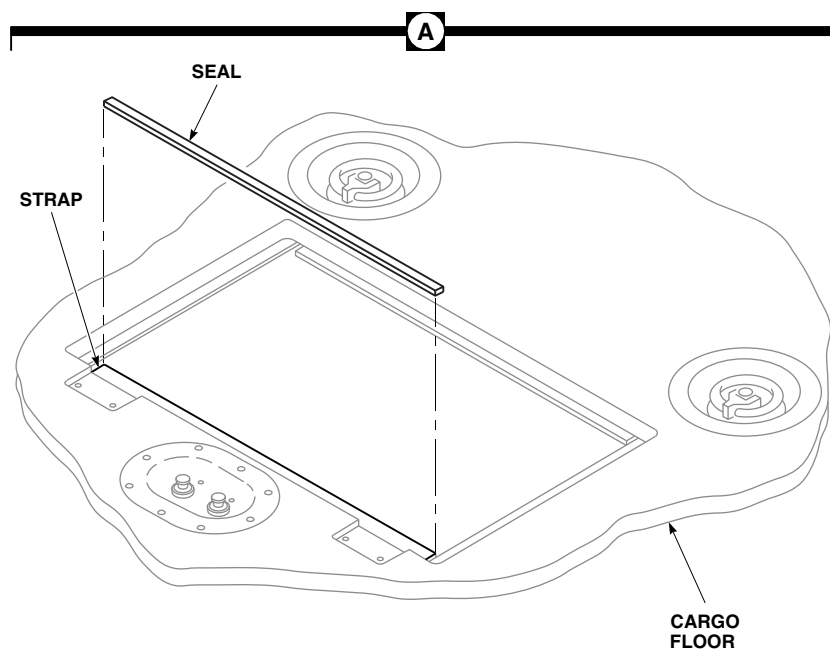
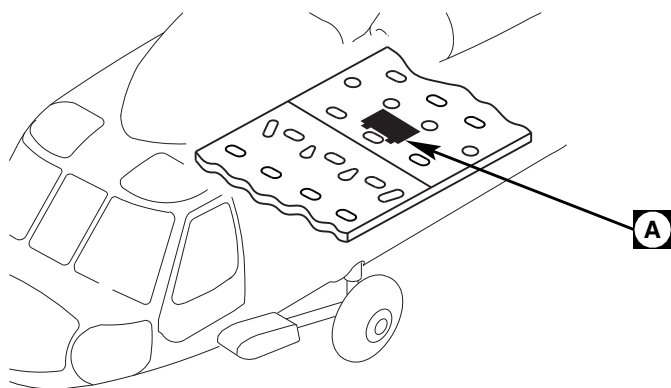
- Remove cargo hook door (PARA 2-4-75).
- Remove seal on strap as follows:
 - Remove damaged seal from strap using nonmetallic scraper (Figure 2-4-72-1, Detail A).
 - Clean any remaining adhesive from strap using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.

2-4-72.1.2. Install.

- Install seal on strap as follows:

- Locally make new cargo hook access seal (Figure H-60, Appendix H).
- Roughen side of new seal using abrasive cloth, Item 9, Appendix D. Clean roughened surface of seal using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.
- Apply a light coat of adhesive, Item 37, Appendix D, to roughened side of seal and edge of strap. Allow adhesive to dry for about 10 minutes or until it is tacky.
- Install seal on strap, making sure all mat-ing surfaces are in contact. Allow adhesive to dry at room temperature for 24 hours (Figure 2-4-72-1, Detail A).

- Install cargo hook door (PARA 2-4-75).

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SA**FIGURE 2-4-72-1. CARGO HOOK DOOR STRAP SEAL**

2-4-73 CABIN FLOOR SEAT STUDS.

PARAGRAPH BREAKDOWN

	PAGE
2-4-73.1 Replace Cabin Floor Seat Studs	2-4-73-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Drill Set
- Nonmetallic Scraper, Locally-Made

- Rivet Gun

Materials/Parts

- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 2-4-71

2-4-73.1 Replace Cabin Floor Seat Studs.

NOTE

Replacement of all indicated cabin floor seat studs is the same, except as noted.

2-4-73.1.1 Remove.

- Remove applicable floor panel from support structure (PARA 2-4-71).
- Using nonmetallic scraper, remove sealing compound from one keeper rivet.
- Using drill, drill out rivet (Figure 2-4-73-1, Detail B).
- Remove retaining ring securing seat stud to floor pan.
- Turn keeper to one side so seat stud can be removed. Remove seat stud.

NOTE

Washers are used with seat studs located at STA 395.5.

- If seat stud from STA 395.5 was removed, remove washer from seat stud.

2-4-73.1.2 Install.

NOTE

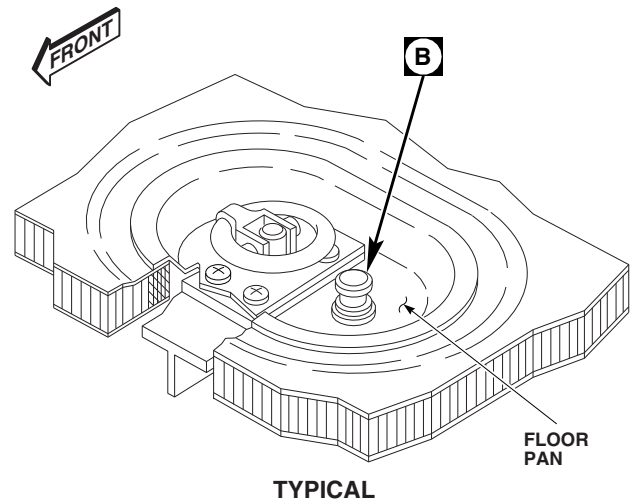
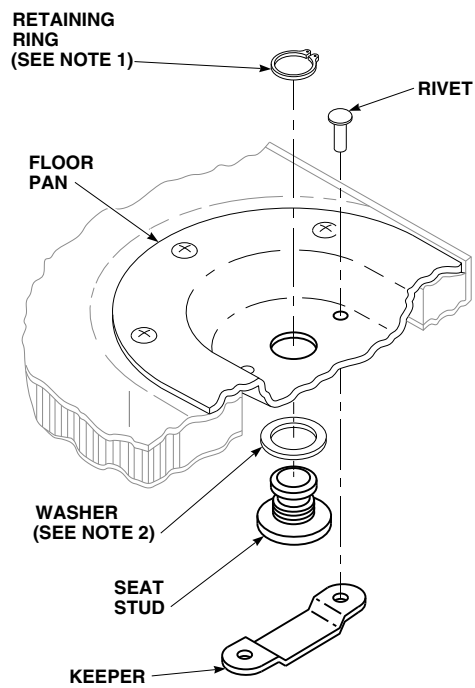
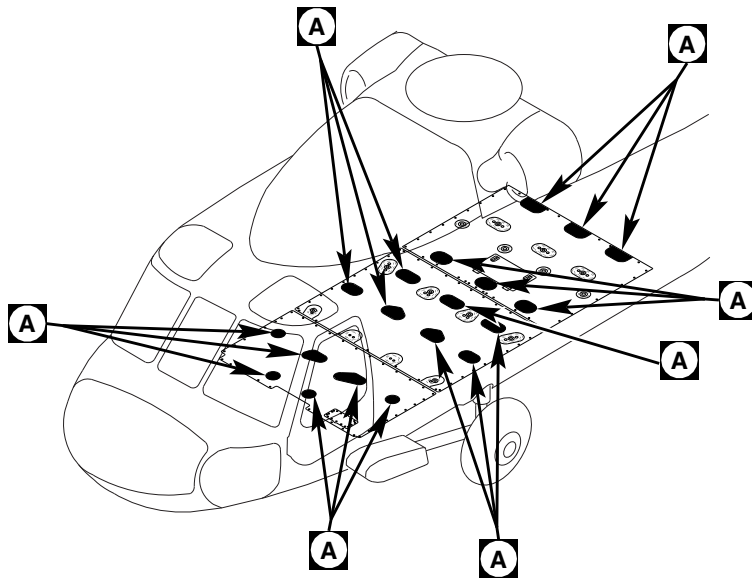
Washers are used with seat studs located at STA 395.5

- If installing seat stud at STA 395.5, install washer on seat stud (Figure 2-4-73-1, Detail B).
- Install seat stud, with washer if necessary, in floor pan and position keeper over rivet hole.

NOTE

Make sure retaining ring is installed with flat side up.

- Install retaining ring securing seat stud to floor pan.
- Using rivet gun, install rivet securing keeper to floor pan.



NOTES

1. INSTALL RETAINING RING FLAT SIDE UP.
2. WASHERS ARE USED WITH STUDS LOCATED AT STA 395.5.

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FIGURE 2-4-73-1. REPLACE CABIN FLOOR SEAT STUDS

- e. Apply sealing compound, Item 292, Appendix D, to rivet.
- f. Install floor panel (PARA 2-4-71).

2-4-74 CABIN FLOOR LEVELING PLATE.

PARAGRAPH BREAKDOWN

	PAGE
2-4-74.1 Replace Cabin Floor Leveling Plate	2-4-74-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Bubble Protractor
- Nonmetallic Scraper, Locally-Made
- Plumb Bob
- Spirit Level

Materials/Parts

- Adhesive, Item 32, Appendix D
- Cheesecloth, Item 90, Appendix D

- Dry Ice, Item 125, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D

References

- Appendix D
- PARA 1-6-3
- PARA 2-4-69
- PARA 2-6-5

2-4-74.1 Replace Cabin Floor Leveling Plate.

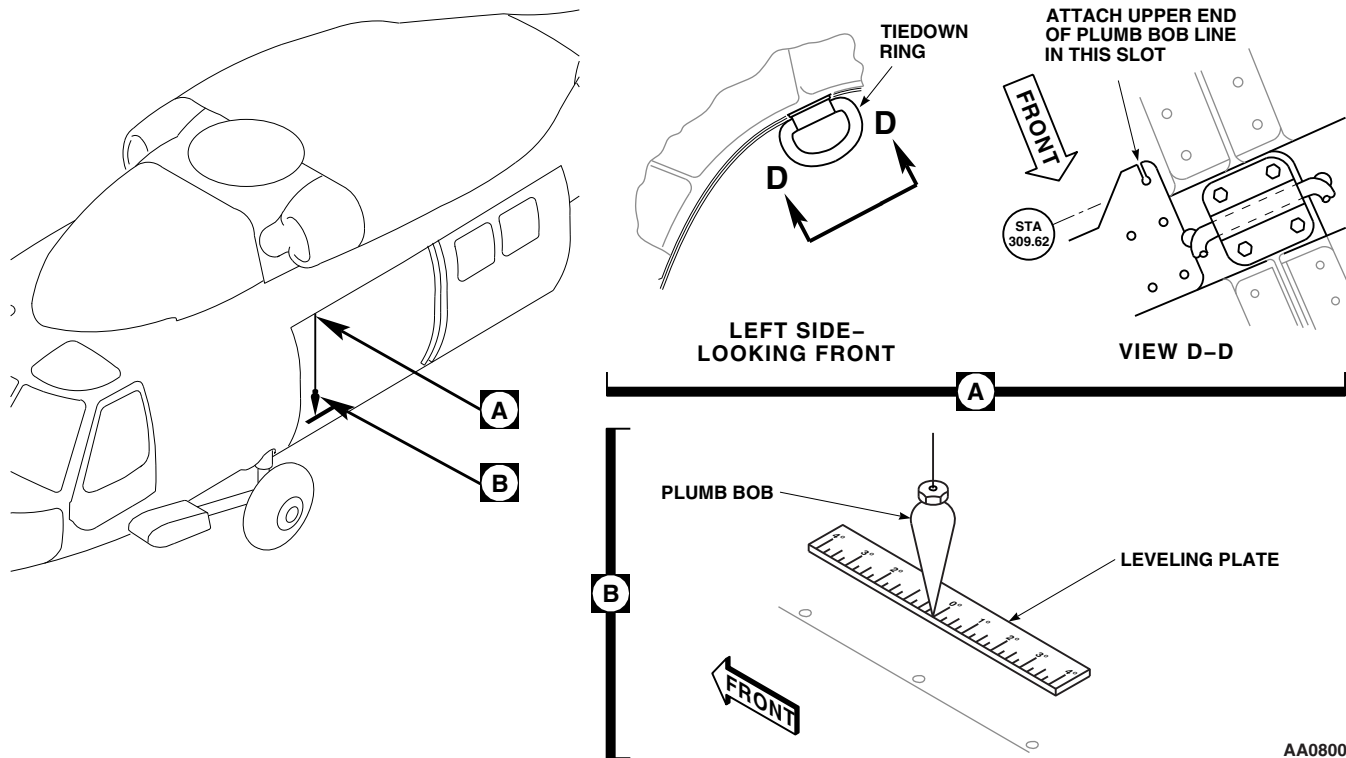
2-4-74.1.1. Remove.

Remove old leveling plate using nonmetallic scraper and dry ice, Item 125, Appendix D.

2-4-74.1.2. Install.

- Open engine cowling work platform.
- Remove soundproofing panel below main transmission and directly above floor panel leveling location (PARA 2-4-69).
- Jack helicopter (PARA 1-6-3).

- Hang plumb bob from slot in fitting in cabin overhead beam at STA 309.62 and BL 36.0 (Figure 2-4-74-1, Detail A).
- Place spirit level or bubble protractor on transmission support beam near transmission.
- While observing spirit level or protractor, adjust jacks to bring helicopter to level attitude both lengthwise and laterally.
- Clean leveling plate and floor panel plate using cheesecloth, Item 90, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D.
- Coat underside of leveling plate with adhesive, Item 32, Appendix D.

AA0800
SA**FIGURE 2-4-74-1. REPLACE CABIN FLOOR LEVELING PLATE**

- i. Position plate on floor with zero mark at edge exactly under point of plumb bob, and with long scale line parallel to centerline of helicopter (Figure 2-4-74-1, Detail B).
- j. Remove plumb bob and install soundproofing panel (PARA 2-4-69).
- k. Remove level or protractor from transmission support beam and close engine cowling work platform.
- l. Lower helicopter and remove jacks (PARA 1-6-3).
- m. Touch up paint around leveling plate (PARA 2-6-5).

2-4-75 CARGO HOOK DOOR.

PARAGRAPH BREAKDOWN

	PAGE
2-4-75.1 Replace Cargo Hook Door	2-4-75-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Machinery Towel, Item 211, Appendix D

- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 301A, Appendix D

References

- Appendix D
- PARA 2-4-45
- PARA 2-4-77
- PARA 2-6-3

2-4-75.1 Replace Cargo Hook Door.

2-4-75.1.1 Remove.

- Remove passenger’s seats as required (PARA 2-4-45).
- Open cargo hook door and support in open position.

NOTE

Do not remove screws, washers, and nuts securing hinges to rear floor panel.

- Remove screws, washers, and nuts securing hinges to cargo hook door (Figure 2-4-75-1, Detail A). Remove door.
- Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean sealing compound from door, hinge, and hardware.

2-4-75.1.2 Inspect/Repair.

Inspect cargo hook door for skin or structural damage. **NONE ALLOWED.** If damage is found, repair door (PARA 2-6-3).

2-4-75.1.3 Install.

- If new cargo hook door is being installed, install seals as follows:
 - Clean edges of replacement door using cheesecloth, Item 90, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D.
 - Install new seals to fit front, rear, and sides of door (PARA 2-4-77).
- Position door on hinges. Install screws, washers, and nuts, wet with sealing compound, Item 301A, Appendix D, to secure door (Figure 2-4-75-1, Detail A).

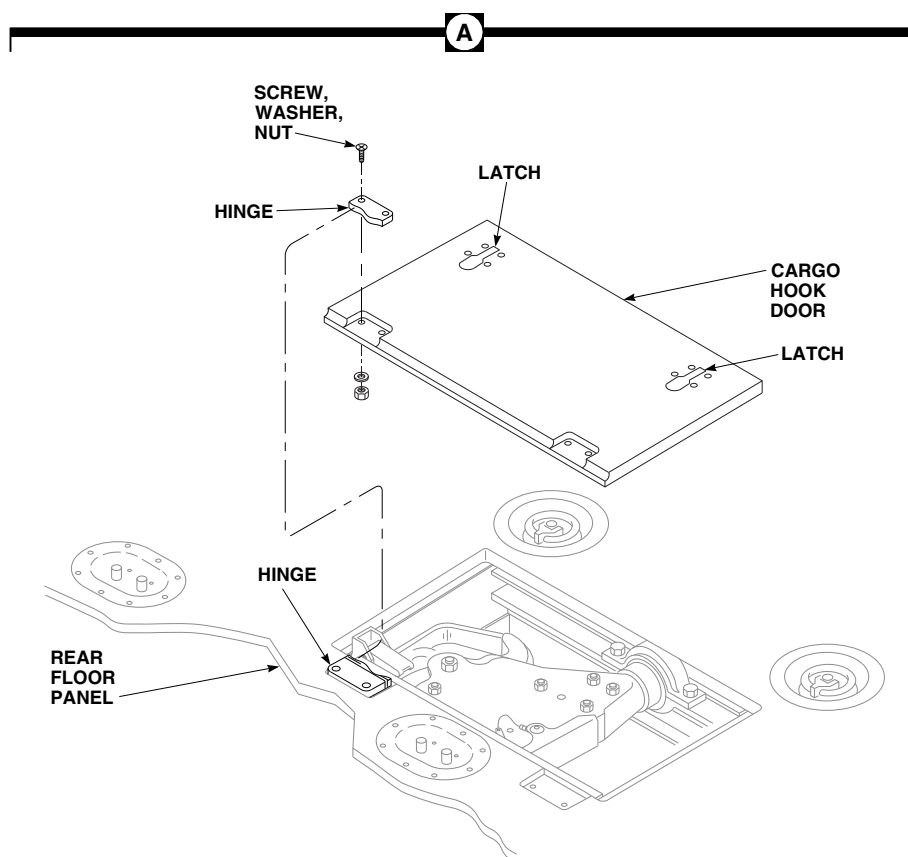
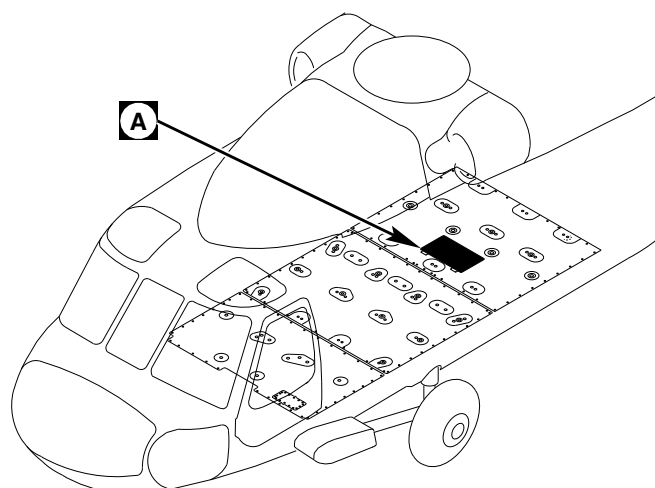


FIGURE 2-4-75-1. REPLACE CARGO HOOK DOOR

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- c. Close and latch cargo hook door.
- d. Install passenger's seats (PARA 2-4-45).

2-4-76 CARGO HOOK DOOR LATCHES.

PARAGRAPH BREAKDOWN

	PAGE
2-4-76.1 Replace Cargo Hook Door Latches	2-4-76-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
 - Airframe Repairer’s Toolkit
 - Clecos, 3/32-inch diameter
- Drill
 - Drill Bit, No. 40 (0.098-inch diameter)
 - Pin Punch, 3/32-inch drive
 - Rivet Gun

2-4-76.1 Replace Cargo Hook Door Latches.

NOTE

Replacement of both cargo hook door latches is the same.

2-4-76.1.1 Remove.

- a. Carefully drill out and remove four rivets securing damaged latch to cargo hook door using drill with No. 40 drill bit and 3/32-inch drive pin punch (Figure 2-4-76-1, Detail A).
 - b. Remove latch and make sure area is clean and free of foreign material.

2-4-76.1.2 Install.

- a. If required, open pilot holes using drill with No. 40 drill bit. Deburr holes.
 - b. Position latch in cargo door and hold in place using three 3/32-inch diameter clecos.
 - c. Check operation of door latch.
 - d. Install one rivet in empty hole (Figure 2-4-76-1, Detail A).
 - e. Remove one cleco from diagonal corner hole and install rivet.
 - f. Remove remaining two clecos and install rivets.

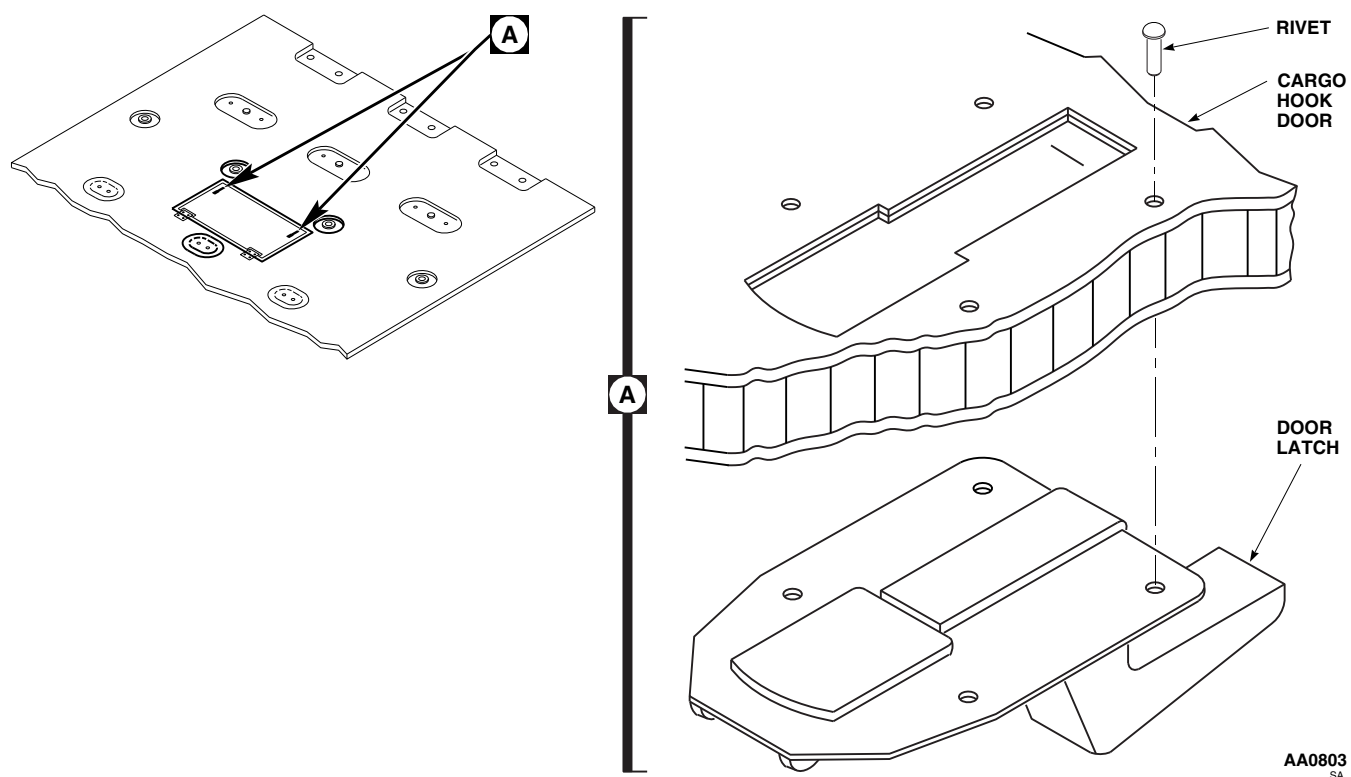


FIGURE 2-4-76-1. REPLACE CARGO HOOK DOOR LATCHES

2-4-77 CARGO HOOK DOOR SEALS.

PARAGRAPH BREAKDOWN

	PAGE
2-4-77.1 Replace Cargo Hook Door Seals	2-4-77-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Abrasive Cloth, Item 8, Appendix D
- Adhesive, Item 37, Appendix D

- Cheesecloth, Item 90, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D

References

- Appendix D
- PARA 2-4-75

2-4-77.1 Replace Cargo Hook Door Seals.

NOTE

Replacement of all cargo hook door seals is the same.

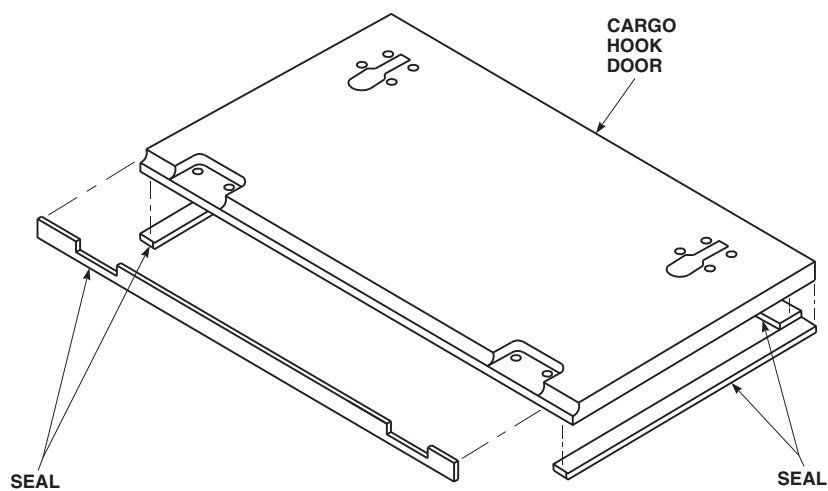
2-4-77.1.1. Remove.

- Remove cargo hook door (PARA 2-4-75).
- Remove damaged seal using nonmetallic scraper, being careful not to damage other seals (Figure 2-4-77-1).
- Clean any remaining adhesive from cargo hook door using cheesecloth, Item 90, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D.

2-4-77.1.2. Install.

- Measure and cut new seal.

- Roughen side of new seal using abrasive cloth, Item 8, Appendix D.
- Clean roughened surface of seal using cheesecloth, Item 90, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D.
- Apply a light coat of adhesive, Item 37, Appendix D, to roughened side of seal and edge of door. Allow adhesive to dry for about 10 minutes or until tacky.
- Install seal on edge of door, making sure all mating surfaces are in contact (Figure 2-4-77-1).
- Allow adhesive to dry at room temperature for 24 hours.
- Install cargo hook door (PARA 2-4-75).

AA0804
SA**FIGURE 2-4-77-1 REPLACE CARGO HOOK DOOR SEALS**

2-4-78 CARGO HOOK DOOR HINGES.

PARAGRAPH BREAKDOWN

	PAGE
2-4-78.1 Replace Cargo Hook Door Hinges	2-4-78-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit

Materials/Parts

- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D

- Sealing Compound, Item 301A, Appendix D

References

- Appendix D
- PARA 2-4-45

2-4-78.1 Replace Cargo Hook Door Hinges.

NOTE

Replacement of both cargo hook door hinges is the same.

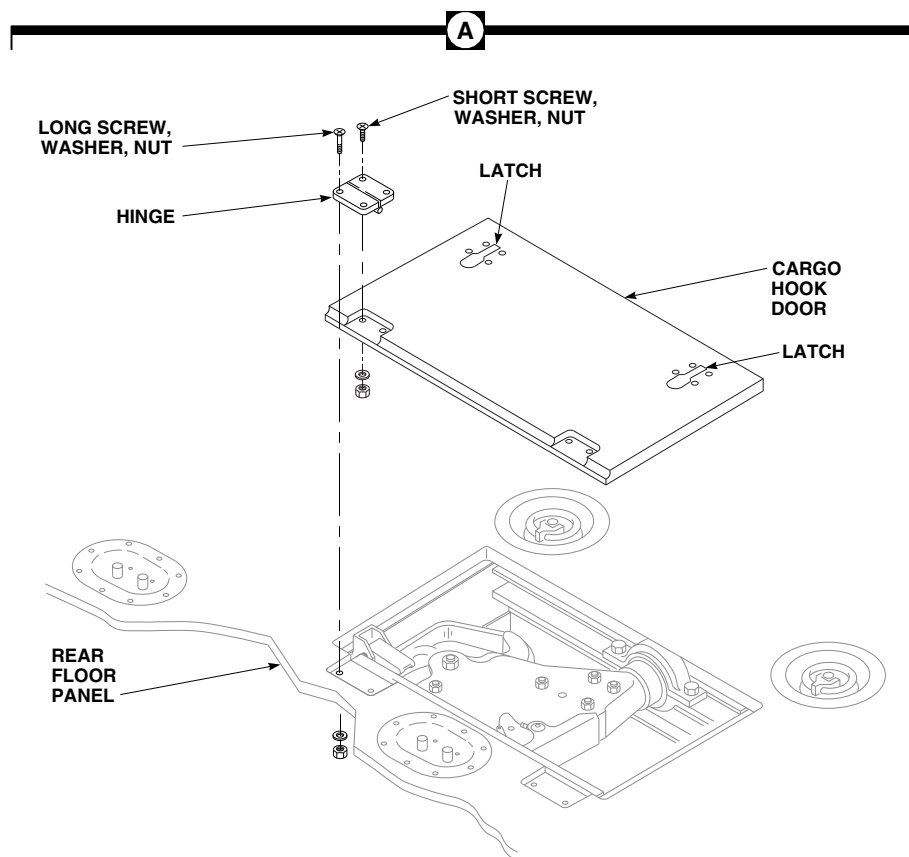
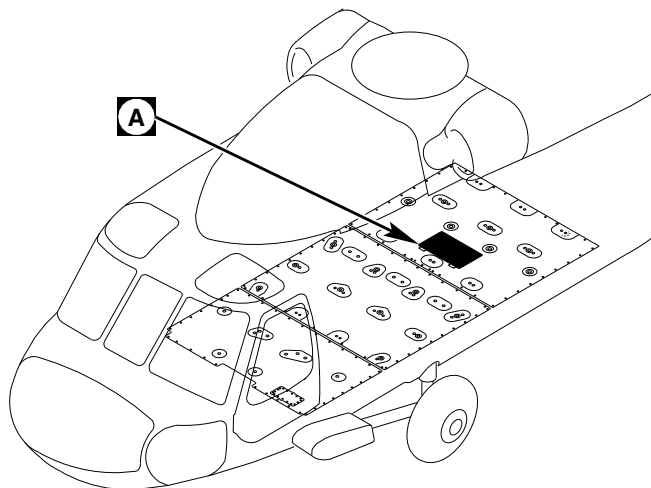
2-4-78.1.1 Remove.

- Remove passenger’s seats as required (PARA 2-4-45).
- Open cargo hook door and support in open position.
- Remove screws, washers, and nuts securing hinge to rear floor panel and cargo hook door. Remove hinge (Figure 2-4-78-1, Detail A).

- Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, clean sealing compound from door, hinge, and hardware.

2-4-78.1.2 Install.

- Position hinge on cabin floor and secure with long screws, washers, and nuts wet with sealing compound, Item 301A, Appendix D (Figure 2-4-78-1, Detail A).
- Position other end of hinge on cargo hook door and secure with short screws, washers, and nuts wet with sealing compound, Item 301A, Appendix D.
- Close and latch cargo hook door.
- Install passenger’s seats (PARA 2-4-45).



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FIGURE 2-4-78-1. REPLACE CARGO HOOK DOOR HINGES

2-4-79 WIRE STRIKE UPPER CUTTER.

PARAGRAPH BREAKDOWN

	PAGE
2-4-79.1 Replace Wire Strike Upper Cutter	2-4-79-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 2-3-9

2-4-79.1 Replace Wire Strike Upper Cutter.

2-4-79.1.1 Remove.

- Open main rotor pylon sliding cover.
- Using nonmetallic scraper, remove sealing compound from bolts and nuts.
- Remove bolt, washer, and nut securing upper cutter struts to upper cutter (Figure 2-4-79-1, Detail A).
- Remove screws, washers, shims, and nuts securing upper cutter struts to main rotor pylon sliding cover (Figure 2-4-79-1, Detail B).
- Remove bolts and washers securing upper cutter to main rotor pylon sliding cover (Figure 2-4-79-1, Detail A).
- Using nonmetallic scraper, remove sealing compound from main rotor pylon sliding cover, struts, and cutter.

2-4-79.1.2 Inspect.

Inspect wire strike upper cutter and struts (PARA 2-3-9).

2-4-79.1.3 Install.

- Secure upper cutter onto main rotor pylon sliding cover using bolts and washers. Do not torque bolts now (Figure 2-4-79-1, Detail A).
- Apply sealing compound, Item 292, Appendix D, around upper cutter strut mounting holes, and to shank of bolt.
- Secure upper cutter struts onto main rotor pylon sliding cover using screws, washers, shims, and nuts. Do not torque nuts now (Figure 2-4-79-1, Detail B).
- Secure upper cutter struts to upper cutter using bolt, washer, and nut. **TORQUE NUT TO 43 - 47 INCH-POUNDS.** Apply sealing compound, Item 292, Appendix D, around bolthead and nut (Figure 2-4-79-1, Detail A).
- Tighten nuts on upper cutter strut screws.
- TORQUE UPPER CUTTER BOLTS TO 48 - 52 INCH-POUNDS.**

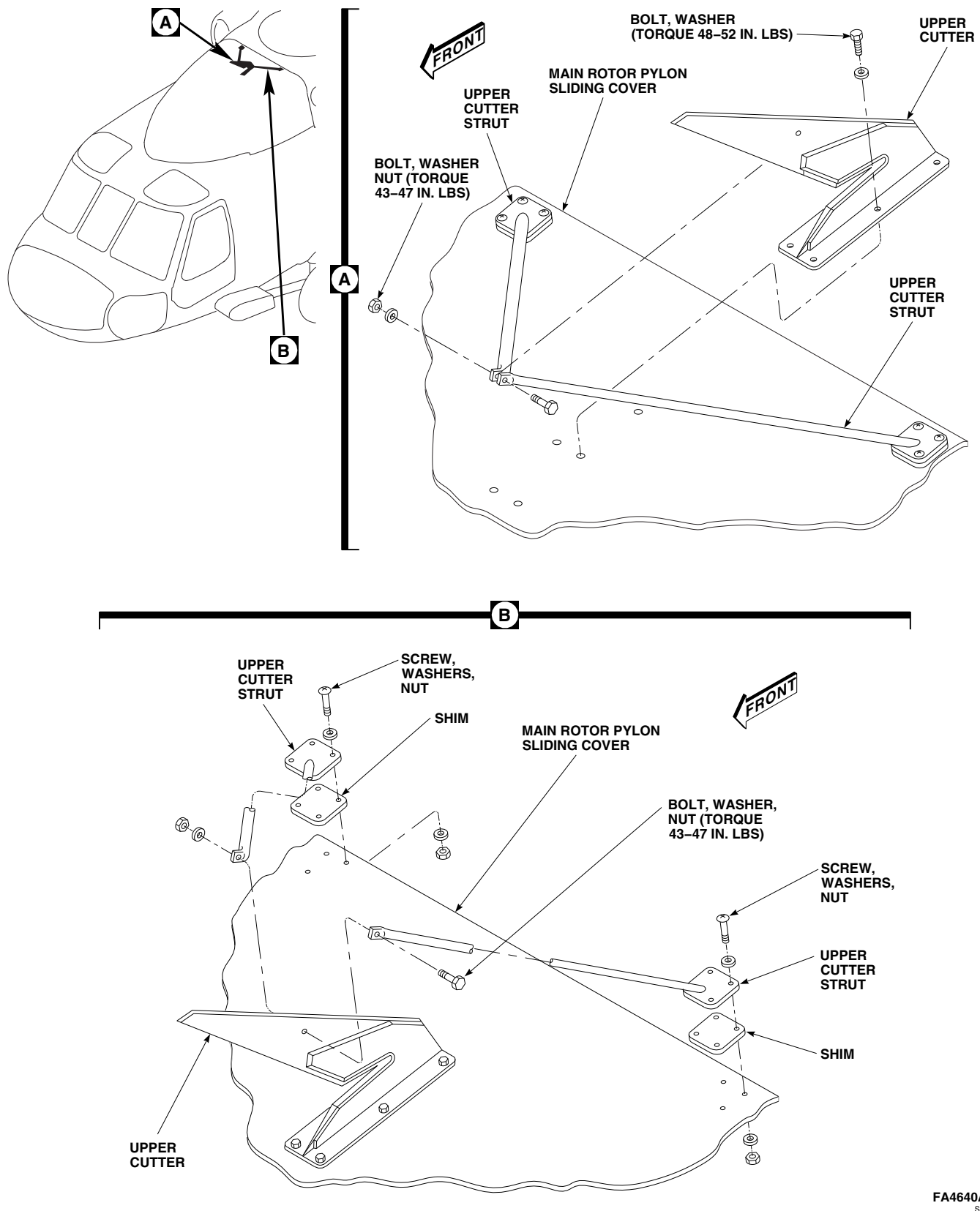
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FIGURE 2-4-79-1. REPLACE WIRE STRIKE UPPER CUTTER

- g. Using sealing compound, Item 292, Appendix D, seal cutter struts to main rotor pylon sliding cover.
- h. Close and latch main rotor pylon sliding cover.

2-4-80 WIRE STRIKE PITOT CUTTER.

PARAGRAPH BREAKDOWN

	PAGE
2-4-80.1 Replace Wire Strike Pitot Cutter	2-4-80-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 2-3-9
- PARA 2-4-101

2-4-80.1 Replace Wire Strike Pitot Cutter.

2-4-80.1.1 Remove.

- Remove main rotor pylon sliding cover (PARA 2-4-101).
- Remove sealing compound from screws, washers, lockwashers, and nuts with nonmetallic scraper.
- Remove screws, lockwashers, washers, and nuts securing pitot cutter to main rotor pylon sliding cover (Figure 2-4-80-1, Detail A). Remove pitot cutter.
- Remove sealing compound from main rotor pylon sliding cover with nonmetallic scraper.

2-4-80.1.2 Inspect.

Inspect wire strike pitot cutter (PARA 2-3-9).

2-4-80.1.3 Install.

- Install screws and washers in two front holes. (Figure 2-4-80-1, Detail A):
- Install screws and washers in four middle holes. **MAXIMUM PROTRUSION OF SCREWS BELOW NUTPLATES IS 0.09-INCH.** Adjust, as required, by adding washers under screwheads.
- Install screws, washers, lockwashers, and nuts in two rear holes with screwhead down.
- Seal screwheads, washers, lockwashers, and nuts with sealing compound, Item 292, Appendix D.
- Seal pitot cutter to main rotor pylon sliding cover with sealing compound, Item 292, Appendix D.
- Install main rotor pylon sliding cover (PARA 2-4-101).

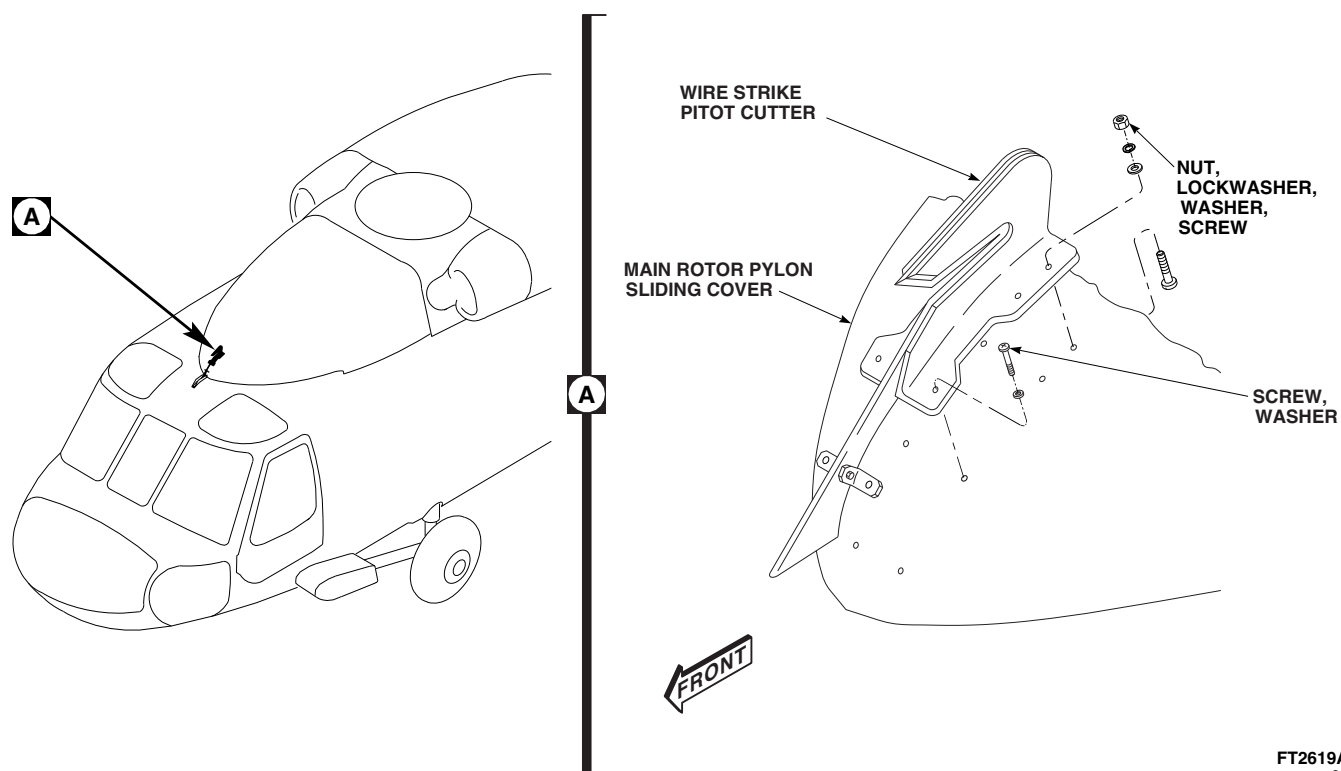
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FIGURE 2-4-80-1. REPLACE WIRE STRIKE PITOT CUTTER

2-4-81 WIRE STRIKE PITOT CUTTER GUIDE CHANNEL.

PARAGRAPH BREAKDOWN

	PAGE
2-4-81.1 Replace Wire Strike Pitot Cutter Guide Channel	2-4-81-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 2-3-9
- PARA 2-4-80

2-4-81.1 Replace Wire Strike Pitot Cutter Guide Channel.

2-4-81.1.1 Remove.

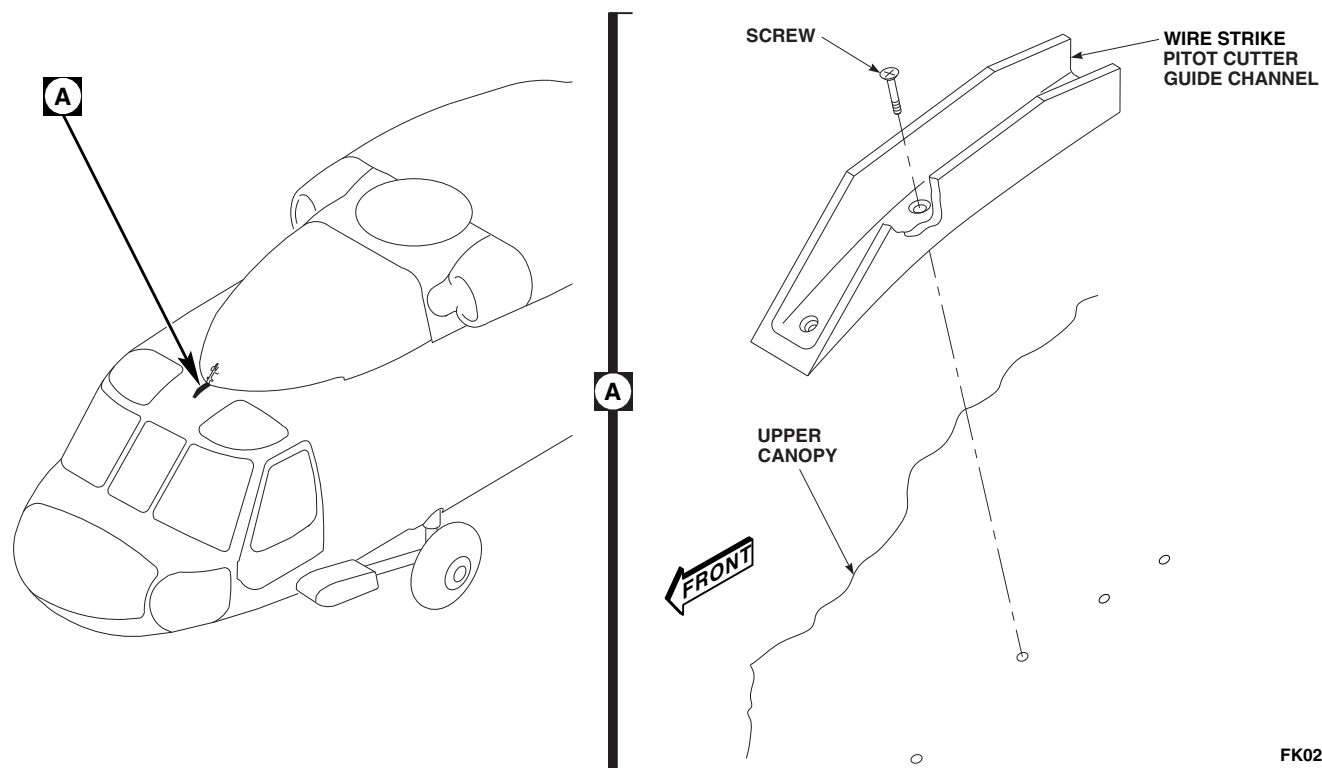
- Remove wire strike pitot cutter (PARA 2-4-80).
- Remove sealing compound from screws with nonmetallic scraper.
- Remove screws and remove pitot cutter guide channel from upper canopy (Figure 2-4-81-1, Detail A).
- Remove sealing compound from upper canopy with nonmetallic scraper.

2-4-81.1.2 Inspect.

Inspect wire strike pitot cutter guide channel (PARA 2-3-9).

2-4-81.1.3 Install.

- Install pitot cutter guide channel onto upper canopy with screws (Figure 2-4-81-1, Detail A).
- Seal screwheads with sealing compound, Item 292, Appendix D.
- Seal guide channel to canopy with sealing compound, Item 292, Appendix D.
- Install wire strike pitot cutter (PARA 2-4-80).

FK0262
SA**FIGURE 2-4-81-1. REPLACE WIRE STRIKE PITOT CUTTER GUIDE CHANNEL**

2-4-82 WIRE STRIKE WINDSHIELD WIPER DRIVE POST DEFLECTORS.

PARAGRAPH BREAKDOWN

		PAGE
2-4-82.1	Replace Wire Strike Windshield Wiper Drive Post Deflectors	2-4-82-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 2-3-9

2-4-82.1 Replace Wire Strike Windshield Wiper Drive Post Deflectors.

NOTE

Replacement of pilot’s and copilot’s windshield wiper drive post deflectors is the same.

2-4-82.1.1 Remove.

- Remove sealing compound from screws and washers with nonmetallic scraper.
- Remove screws and washers and remove windshield wiper drive post deflector from nose structure (Figure 2-4-82-1, Detail A).

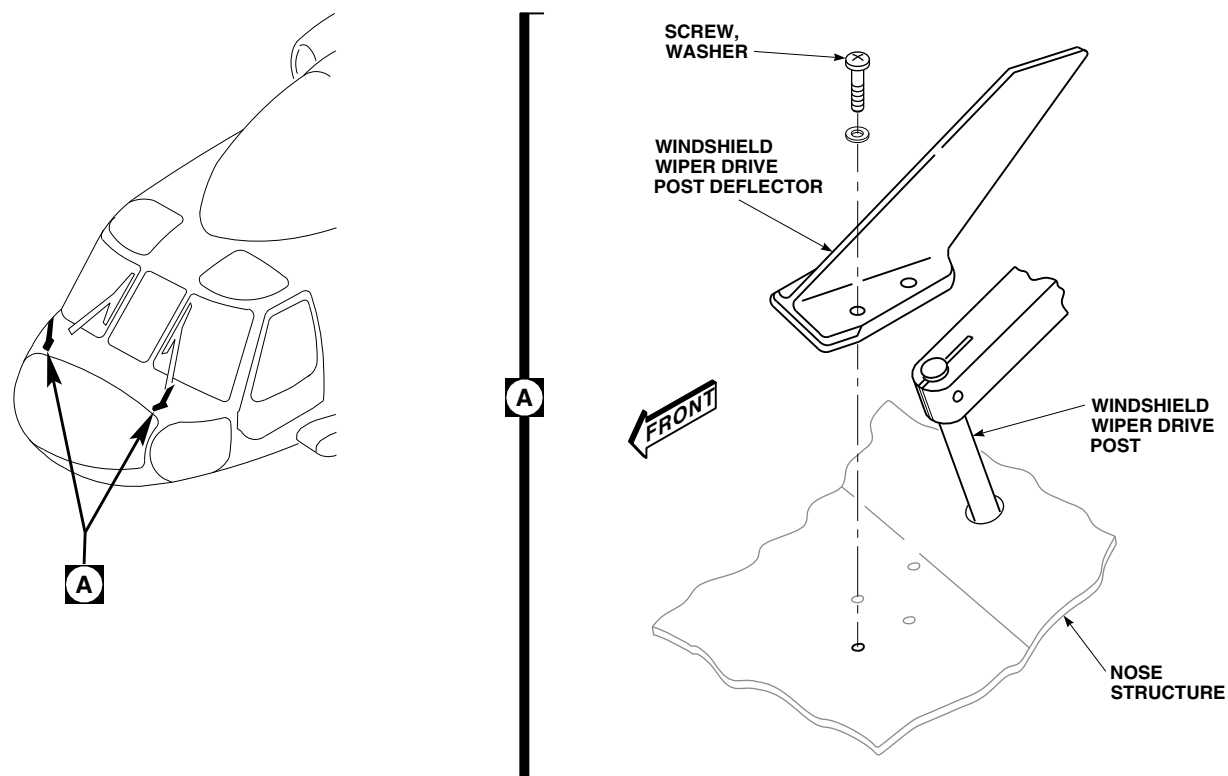
- Remove sealing compound from nose structure and deflector with nonmetallic scraper.

2-4-82.1.2 Inspect.

Inspect wire strike windshield wiper drive post deflector (PARA 2-3-9).

2-4-82.1.3 Install.

- Install windshield wiper drive post deflector onto nose structure with screws and washers (Figure 2-4-82-1, Detail A).
- Seal screwheads and washers with sealing compound, Item 292, Appendix D.
- Seal deflector to nose structure with sealing compound, Item 292, Appendix D.

AA0809
SA**FIGURE 2-4-82-1. REPLACE WIRE STRIKE WINDSHIELD WIPER DRIVE POST DEFLECTORS**

2-4-83 WIRE STRIKE MAIN LANDING GEAR CUTTERS.

PARAGRAPH BREAKDOWN

	PAGE
2-4-83.1 Replace Wire Strike Main Landing Gear Cutters	2-4-83-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

References

- PARA 2-3-9

2-4-83.1 Replace Wire Strike Main Landing Gear Cutters.

NOTE

Replacement of left and right main landing gear cutters is the same.

2-4-83.1.1 Remove.

- Remove bolts, washers, and nuts securing breakaway tip to main landing gear cutter and remove tip (Figure 2-4-83-1, Detail A).
- Remove bolts, washers, and nuts securing main landing gear cutter to clamps and remove cutter.

2-4-83.1.2 Inspect.

Inspect wire strike main landing gear cutter (PARA 2-3-9).

2-4-83.1.3 Install.

- Position main landing gear cutter in clamps and align holes in cutter with holes in clamps (Figure 2-4-83-1, Detail A).
- Secure cutter to clamps with bolts, washers, and nuts. **TORQUE BOLTS TO 95 - 105 INCH-POUNDS.**
- Install breakaway tip on main landing gear cutter with bolts, washers, and nuts. **TORQUE BOLTS TO 85 - 95 INCH-POUNDS.**

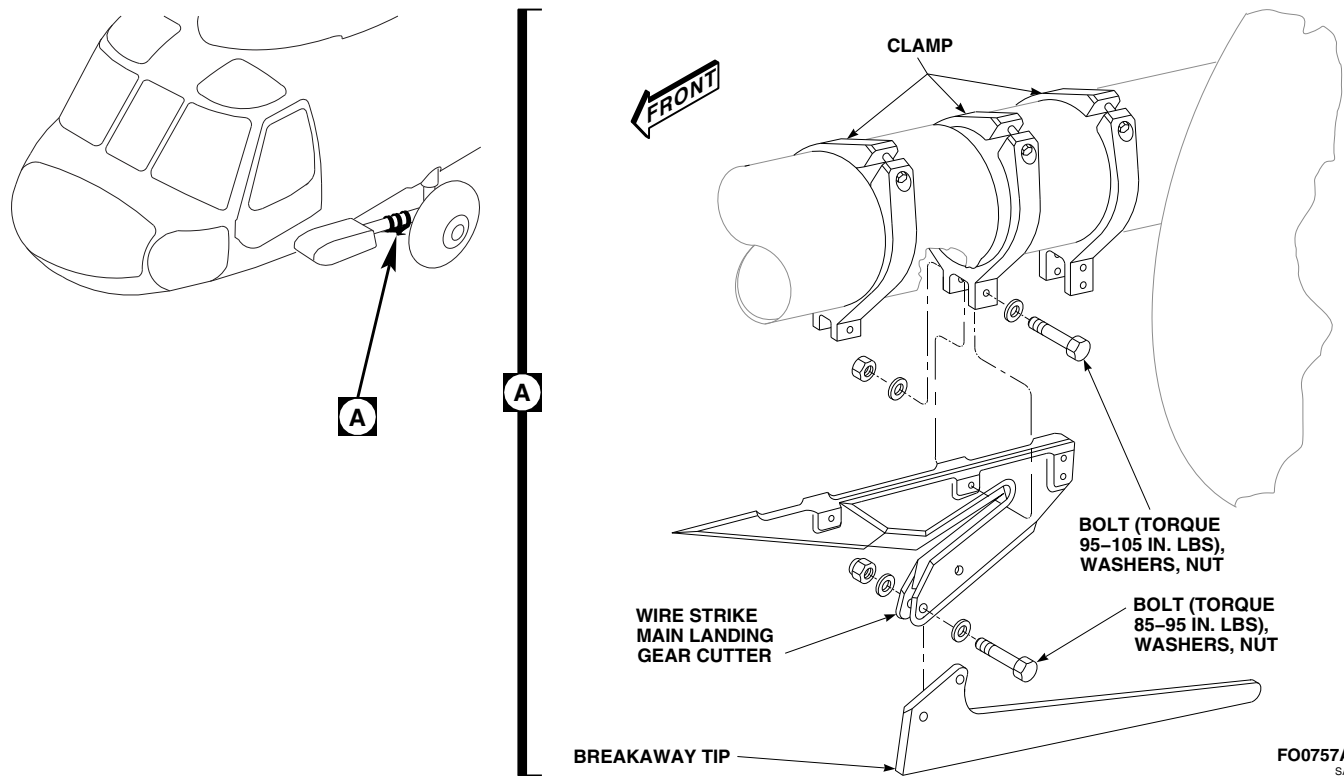


FIGURE 2-4-83-1. REPLACE WIRE STRIKE MAIN LANDING GEAR CUTTERS

2-4-84 WIRE STRIKE MAIN LANDING GEAR CUTTER CLAMPS.

PARAGRAPH BREAKDOWN

	PAGE
2-4-84.1 Replace Wire Strike Main Landing Gear Cutter Clamps	2-4-84-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 2-3-9
- PARA 2-4-83

2-4-84.1 Replace Wire Strike Main Landing Gear Cutter Clamps.

NOTE

Replacement of all main landing gear cutter clamps on both left and right main landing gear is the same.

2-4-84.1.1 Remove.

WARNING

FSCAP

Shotpeening of exterior surfaces of main landing gear drag beam is a critical characteristic. Use extreme care to avoid damage to these surfaces during removal

of wire strike main landing gear cutter clamps.

- If removing front, middle, or all three clamps, perform the following:
 - Remove wire strike main landing gear cutter (PARA 2-4-83).
 - Remove bolts, washers, and nuts and remove clamp(s) from drag beam (Figure 2-4-84-1, Detail A).
- If removing rear clamp only, remove bolts, washers, and nuts that secure wire strike main landing gear cutter to clamp (Figure 2-4-84-1, Detail B).

2-4-84.1.2 Inspect.

Inspect wire strike main landing gear cutter clamp (PARA 2-3-9).

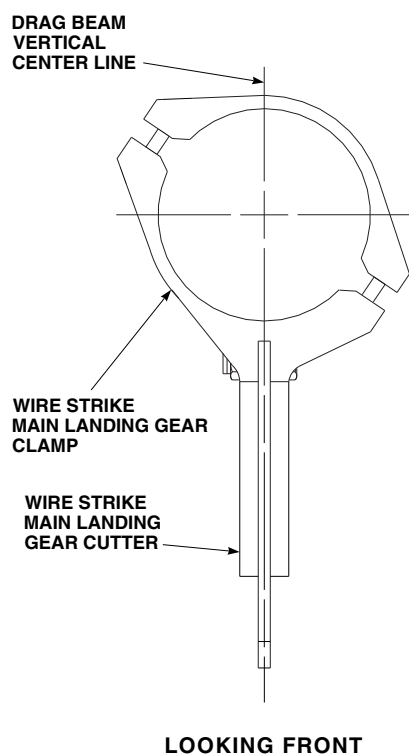
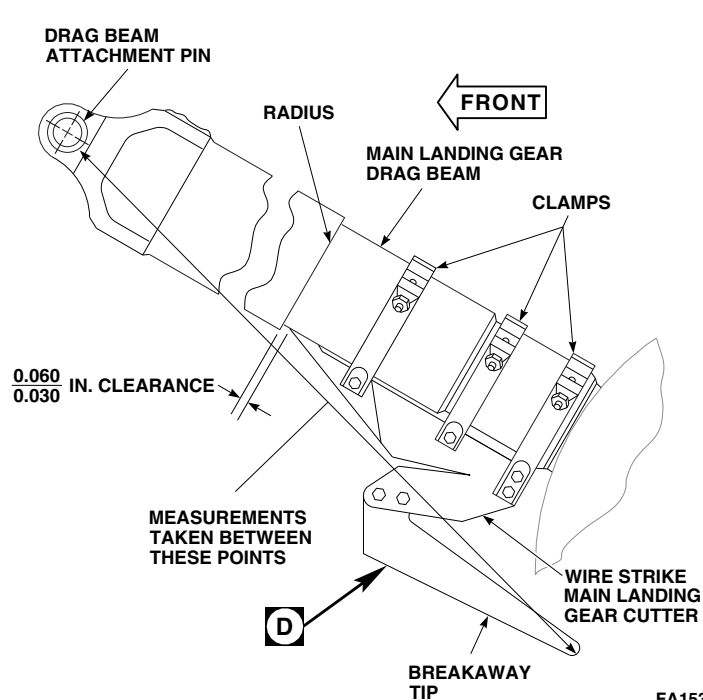
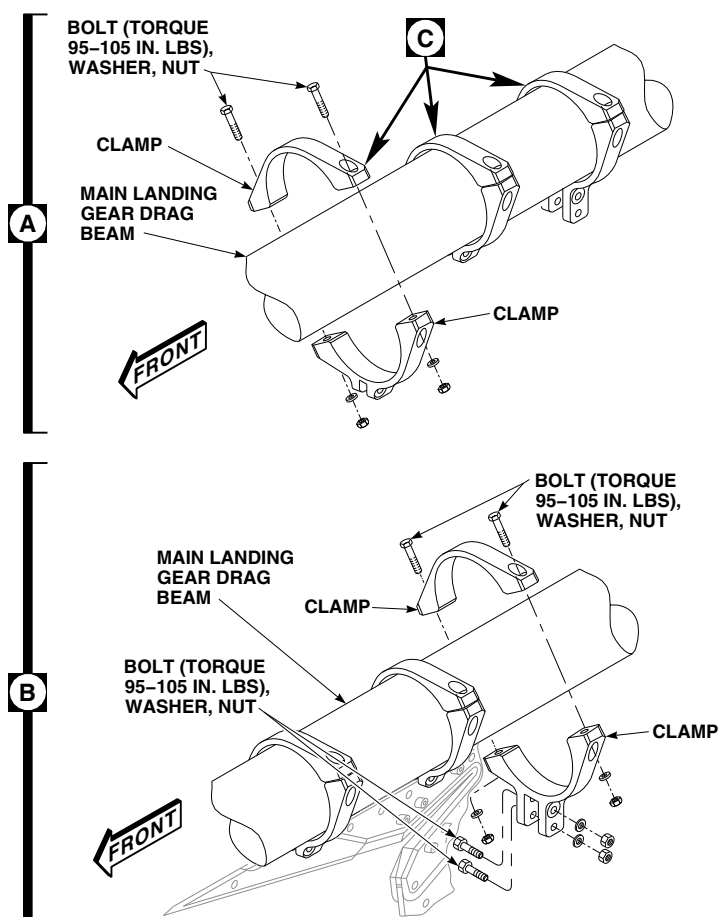
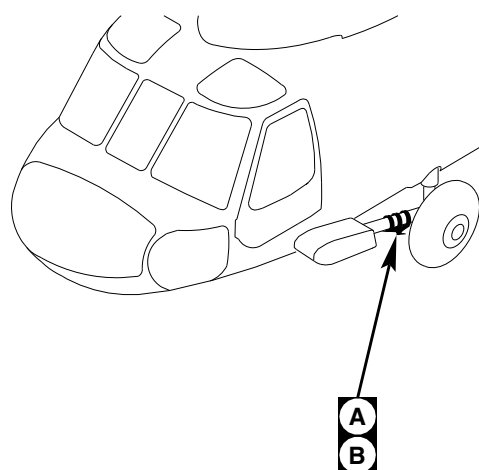


FIGURE 2-4-84-1. REPLACE WIRE STRIKE MAIN LANDING GEAR CUTTER CLAMPS

2-4-84.1.3 Install.

WARNING

FSCAP

Shotpeening of exterior surfaces of main landing gear drag beam is a critical characteristic. Use extreme care to avoid damage to these surfaces during installation of wire strike main landing gear cutter clamps.

- a. Apply sealing compound, Item 292, Appendix D, to mating surfaces of wire strike main landing gear cutter clamp and drag beam.
- b. Position clamp(s) onto drag beam, and install bolts, washers and nuts. **TIGHTEN NUTS HANDTIGHT** (Figure 2-4-84-1, Detail A or Detail B).
- c. If installing forward, middle, or all three clamps, install wire strike main landing gear cutter (PARA 2-4-83).

- d. If installing rear clamp only, install bolts, washers, and nuts securing cutter to clamp (Figure 2-4-84-1, Detail B). **TORQUE BOLTS TO 95 - 105 INCH-POUNDS.**
- e. If all three clamps were removed, adjust cutter by as follows:
 - (1) Slide cutter and clamps as required to position cutter tip to within 0.030 to 0.060-inch of radius on drag beam (Figure 2-4-84-1, Detail C).
 - (2) Measure distance from edge of drag beam attachment pin to bend radius of break-away tip. Take measurements from both inboard and outboard sides of drag beam. **MEASUREMENTS MUST BE WITHIN 1/8-INCH OF EACH OTHER.**
 - (3) Align cutter tip to vertical centerline ($\pm 2^\circ$) of drag beam (Figure 2-4-84-1, Detail D).
- f. **TORQUE BOLTS SECURING CLAMP TO DRAG BEAM TO 95 - 105 INCH-POUNDS** (Figure 2-4-84-1, Detail A or Detail B).

2-4-85 WIRE STRIKE MAIN LANDING GEAR JOINT DEFLECTORS.

PARAGRAPH BREAKDOWN

	PAGE
2-4-85.1 Replace Wire Strike Main Landing Gear Joint Deflectors	2-4-85-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 700 - 1600 in. lbs

- Vise, Soft-Jawed

References

- PARA 2-3-9
- PARA 2-4-94

2-4-85.1 Replace Wire Strike Main Landing Gear Joint Deflectors.

2-4-85.1.3 Install.

NOTE

Replacement of left and right main landing gear joint deflectors is the same.

2-4-85.1.1 Remove.

- Remove drag beam support fairing (PARA 2-4-94).
- Remove bolt, washer, and main landing gear joint deflector from fuselage (Figure 2-4-85-1, Detail A).
- Remove barrel-nuts and retainers from fitting.

2-4-85.1.2 Inspect.

Inspect wire strike main landing gear joint deflector (PARA 2-3-9).

NOTE

Make sure that only square barrel-nuts are used.

- Test attachment bolts and barrel-nuts as follows:
 - Place barrel-nut in soft-jawed vise.
 - Thread bolt into barrel-nut until two threads are showing beyond nut.
 - Back off bolt using torque wrench. **BREAKAWAY TORQUE SHALL NOT BE LESS THAN 18 INCH-POUNDS.**
 - If breakaway torque is less than 18 inch-pounds, replace bolt. Repeat steps a. (1) through a. (3).
 - If breakaway torque is less than 18 inch-pounds, replace barrel-nut. Repeat steps a. (1) through a. (3).

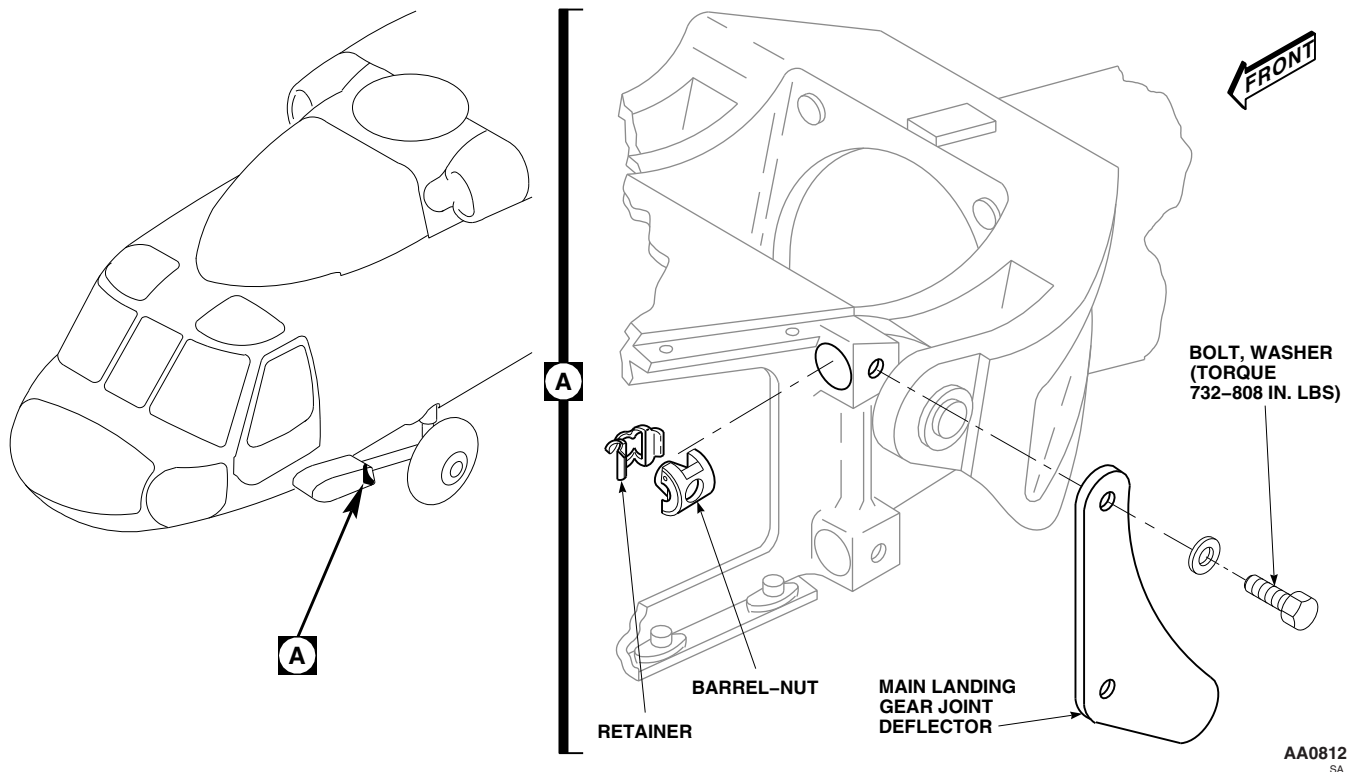


FIGURE 2-4-85-1. REPLACE WIRE STRIKE MAIN LANDING GEAR JOINT DEFLECTORS

CAUTION

Damage to fitting will result if barrel-nut is incorrectly installed. To prevent damage make sure barrel-nut is installed with barrel (round side) towards bolt-head.

- b. Install barrel-nuts and retainers in fitting with barrel sides towards boltheads (Figure 2-4-85-1, Detail A).
- c. Install main landing gear joint deflector onto fuselage with bolt and washer. **TORQUE BOLTS TO 732 - 808 INCH-POUNDS.**
- d. Install drag beam support fairing (PARA 2-4-94).

2-4-86 WIRE STRIKE TAIL LANDING GEAR DEFLECTOR.

PARAGRAPH BREAKDOWN

	PAGE
2-4-86.1 Replace Wire Strike Tail Landing Gear Deflector	2-4-86-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

References

- PARA 2-3-9
- PARA 2-4-87

2-4-86.1 Replace Wire Strike Tail Landing Gear Deflector.

2-4-86.1.1 Remove.

- Remove bolts, washers, and nuts securing tail landing gear deflector to clamp (Figure 2-4-86-1, Detail A). Remove deflector.

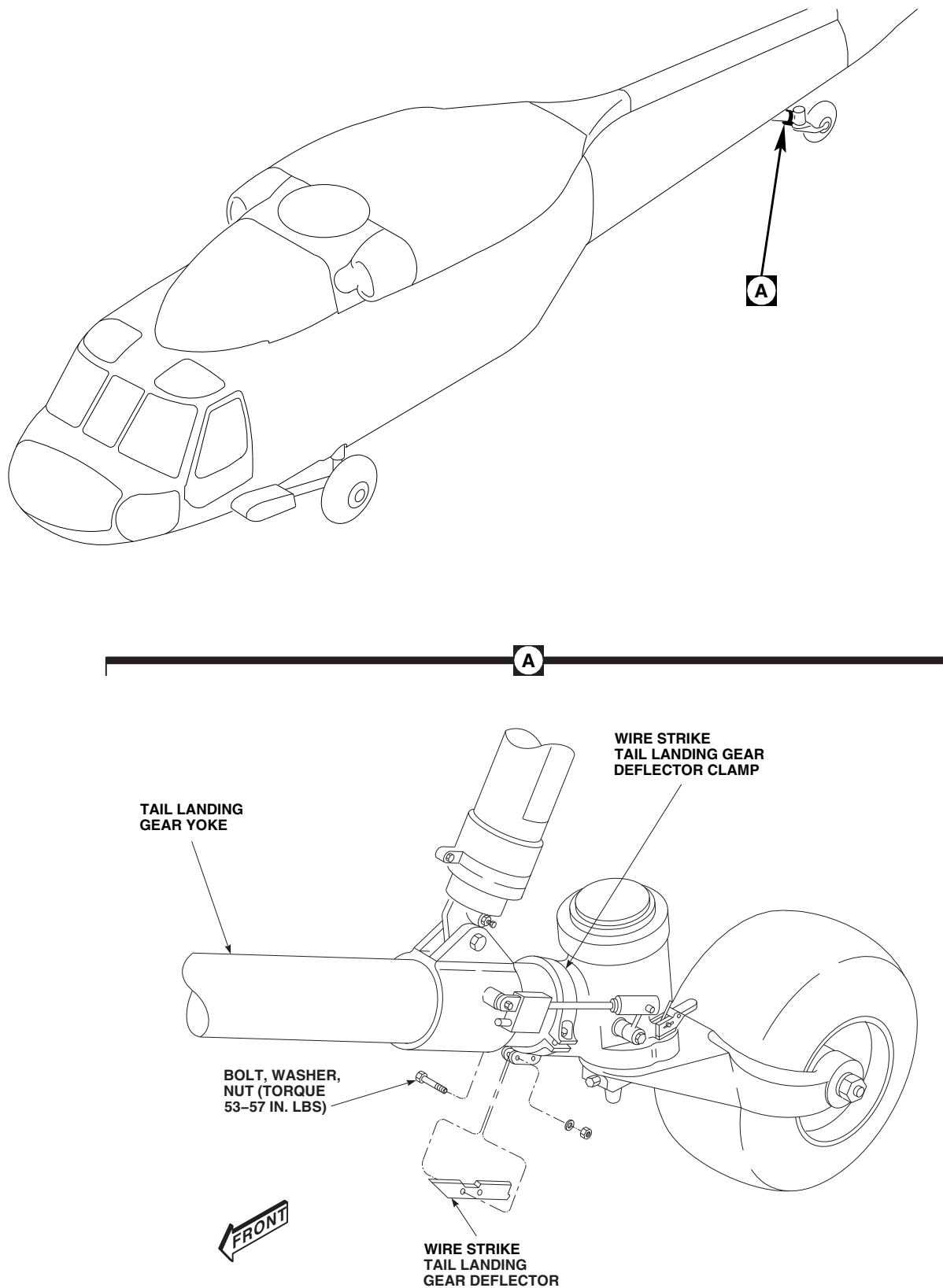
2-4-86.1.2 Inspect.

Inspect wire strike tail landing gear deflector (PARA 2-3-9).

2-4-86.1.3 Install.

- Position wire strike tail landing gear deflector in clamp and align holes in deflector with holes in clamp (Figure 2-4-86-1, Detail A).

- Install bolts, washers, and nuts. Maintain minimum possible clearance between deflector and yoke.
- Position tail landing gear deflector so that there is 0.070-inch clearance between deflector and tail landing gear fork. If 0.070-inch clearance cannot be obtained, reposition tail landing gear deflector clamp (PARA 2-4-87).
- TORQUE BOLTS TO 53 - 57 INCH-POUNDS.**

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SA**FIGURE 2-4-86-1. REPLACE WIRE STRIKE TAIL LANDING GEAR DEFLECTOR**

2-4-87 WIRE STRIKE TAIL LANDING GEAR DEFLECTOR CLAMP.

PARAGRAPH BREAKDOWN

		PAGE
2-4-87.1	Replace Wire Strike Tail Landing Gear Deflector Clamp	2-4-87-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Feeler Gage
- Torque Wrench, 30 - 150 in. lbs

References

- PARA 2-3-9
- PARA 2-4-86

2-4-87.1 Replace Wire Strike Tail Landing Gear Deflector Clamp.

2-4-87.1.1 Remove.

- Remove wire strike tail landing gear deflector (PARA 2-4-86).
- Remove bolts, washers, and nuts securing wire strike tail landing gear deflector clamp to tail landing gear yoke (Figure 2-4-87-1, Detail A). Remove clamp.

2-4-87.1.2 Inspect.

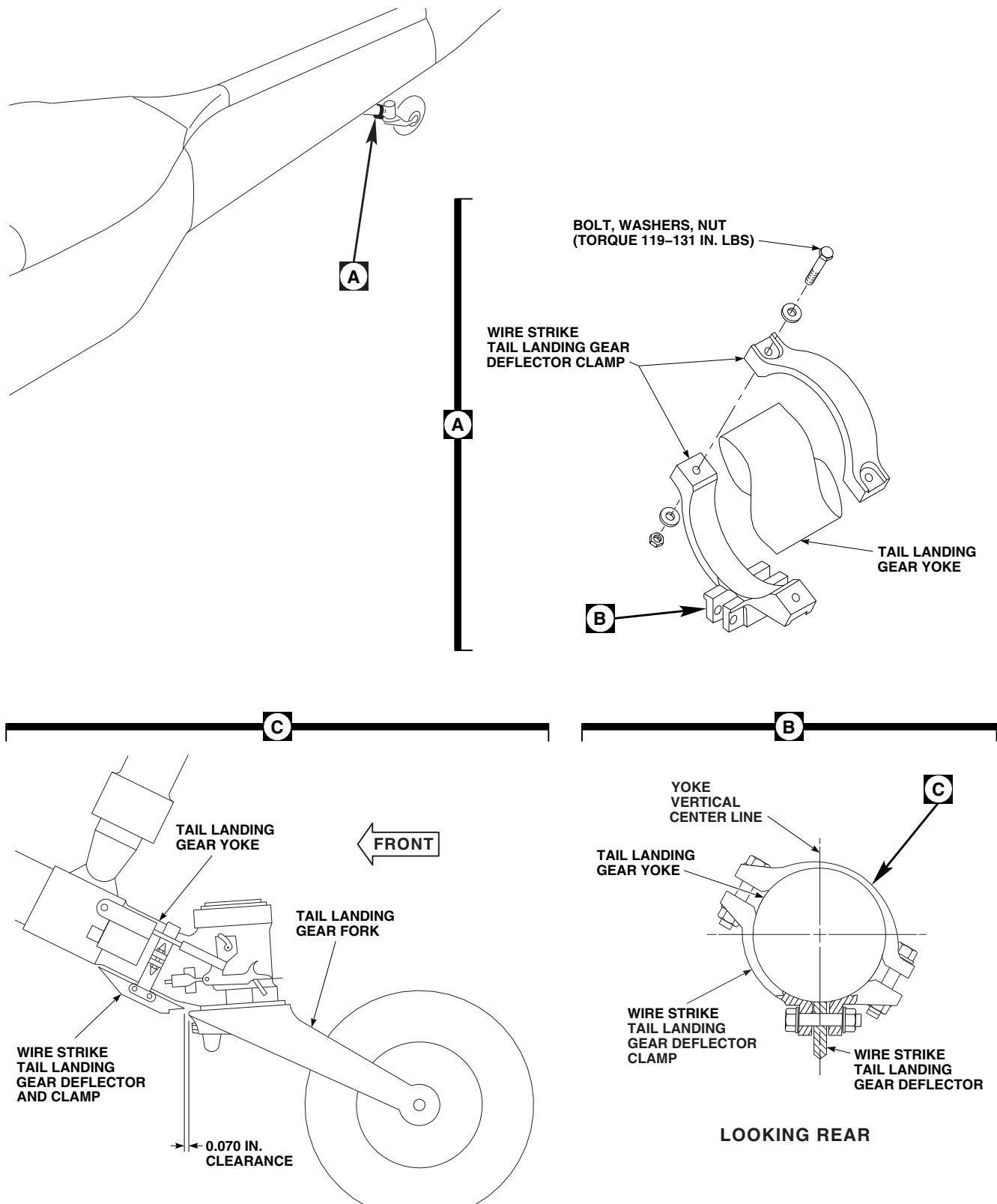
Inspect wire strike tail landing gear deflector clamp (PARA 2-3-9).

2-4-87.1.3 Install.

- Position wire strike tail landing gear deflector clamp on tail landing gear yoke, and secure us-

ing bolts, washers, and nuts. **TIGHTEN NUTS HANDTIGHT** (Figure 2-4-87-1, Detail A).

- Install wire strike tail landing gear deflector (PARA 2-4-86).
- Align deflector with vertical centerline ($\pm 2^\circ$) of yoke (Figure 2-4-87-1, Detail B).
- Using feeler gage, measure for 0.070-inch clearance between deflector and tail landing gear fork (Figure 2-4-87-1, Detail C). If necessary, move deflector clamp along yoke to obtain 0.070-inch clearance between deflector and tail landing gear fork.
- TORQUE TAIL LANDING GEAR DEFLECTOR CLAMP BOLTS TO 119 - 131 INCH-POUNDS** (Figure 2-4-87-1, Detail A).



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FIGURE 2-4-87-1. REPLACE WIRE STRIKE TAIL LANDING GEAR DEFLECTOR CLAMP

2-4-88 WIRE STRIKE STEP DEFLECTORS.

PARAGRAPH BREAKDOWN

	PAGE
2-4-88.1 Replace Wire Strike Step Deflectors	2-4-88-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 2-3-9

2-4-88.1 Replace Wire Strike Step Deflectors.

NOTE

Replacement of left and to right step deflectors is the same.

2-4-88.1.1 Remove.

- Remove sealing compound from screwheads and washers with nonmetallic scraper.
- Remove screws and washers and remove step deflector from fuselage (Figure 2-4-88-1, Detail A).

- Remove sealing compound from fuselage with nonmetallic scraper.

2-4-88.1.2 Inspect.

Inspect wire strike step deflector (PARA 2-3-9).

2-4-88.1.3 Install.

- Install step deflector onto fuselage with screws and washers (Figure 2-4-88-1, Detail A).
- Seal screwheads and washers with sealing compound, Item 292, Appendix D.
- Seal step deflector to fuselage with sealing compound, Item 292, Appendix D.

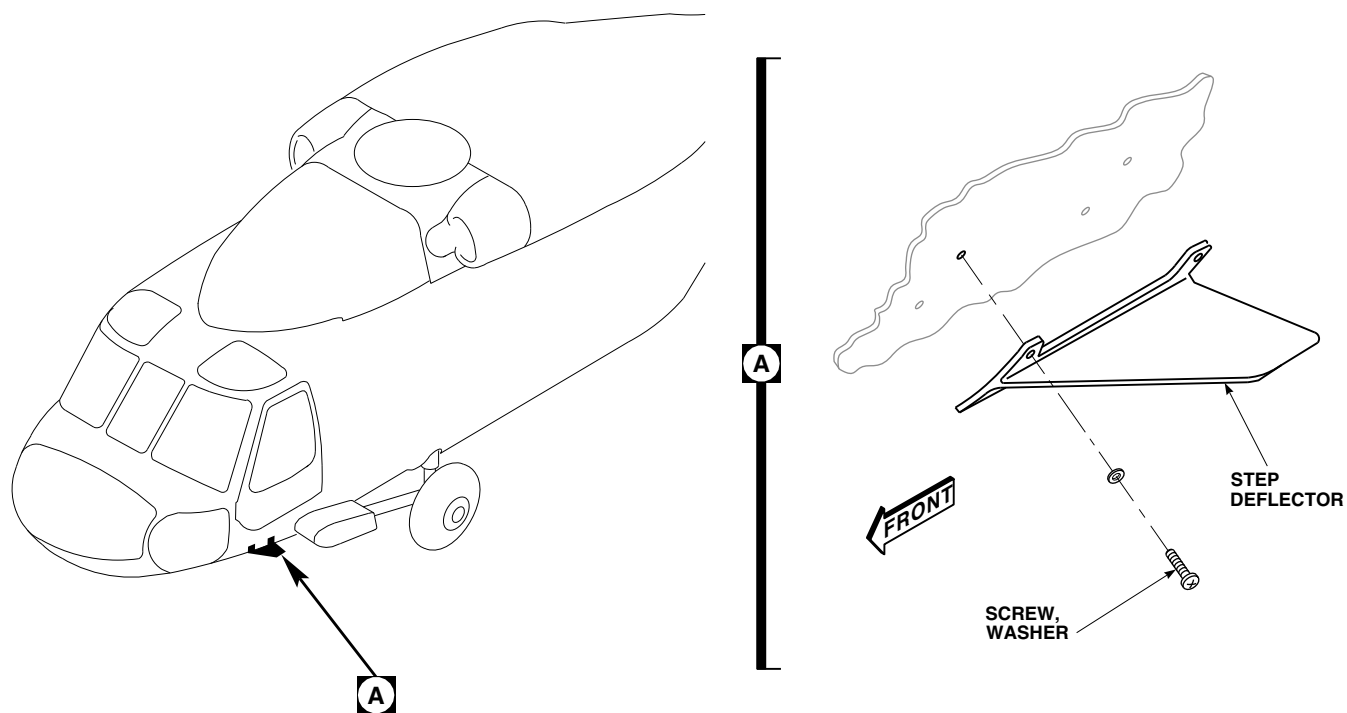
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FIGURE 2-4-88-1. REPLACE WIRE STRIKE STEP DEFLECTORS

2-4-89 WIRE STRIKE STEP EXTENSIONS.

PARAGRAPH BREAKDOWN

		PAGE
2-4-89.1	Replace Wire Strike Step Extensions	2-4-89-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

References

- PARA 2-3-9

2-4-89.1 Replace Wire Strike Step Extensions.

NOTE

Replacement of left and to right step extensions is the same.

2-4-89.1.1 Remove.

- Remove screws, lockwashers, and washers securing plate to step extension (Figure 2-4-89-1, Detail A).

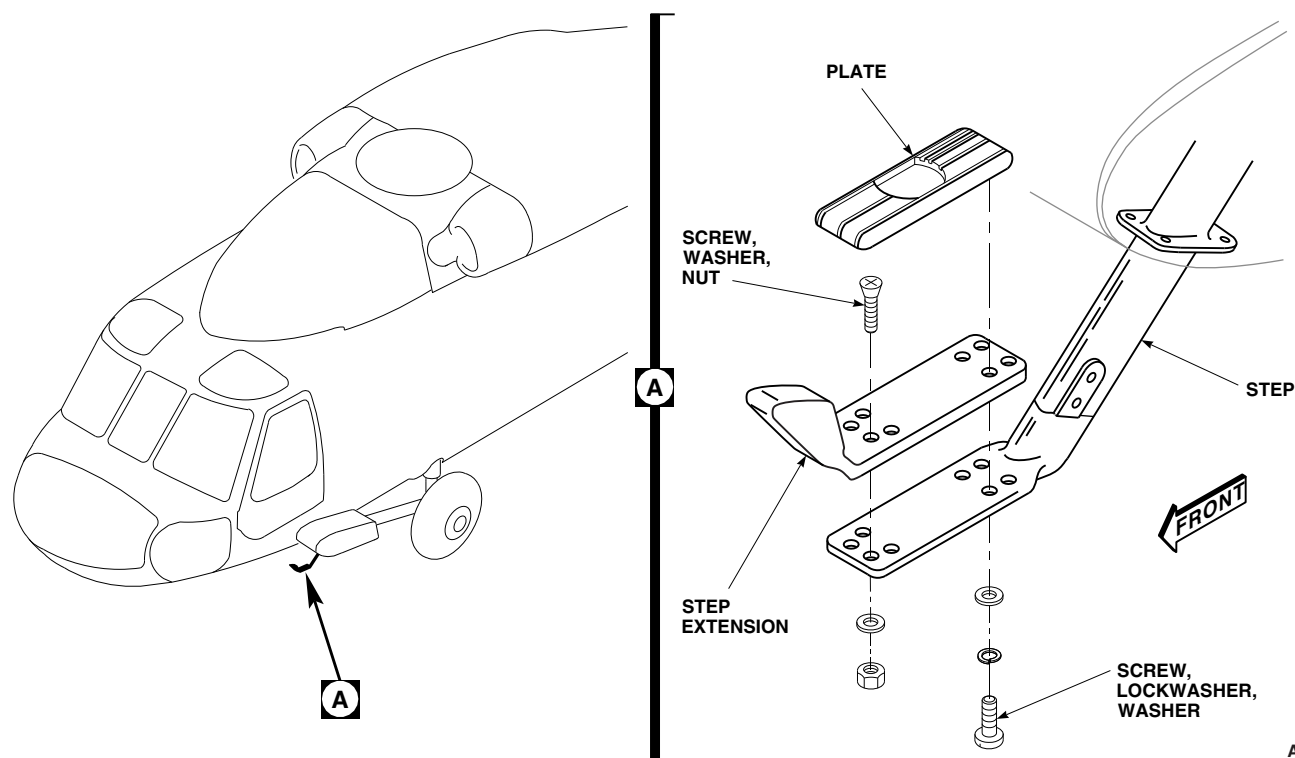
- Remove screws, washers, and nuts securing step extension to step. Remove step extension.

2-4-89.1.2 Inspect.

Inspect wire strike step extension (PARA 2-3-9).

2-4-89.1.3 Install.

- Install step extension onto step with screws, washers, and nuts (Figure 2-4-89-1, Detail A).
- Install plate onto step extension with screws, lockwashers, and washers.

AA0816
SA**FIGURE 2-4-89-1. REPLACE WIRE STRIKE STEP EXTENSIONS**

2-4-90 WIRE STRIKE COCKPIT DOOR LATCH DEFLECTORS.

PARAGRAPH BREAKDOWN

		PAGE
2-4-90.1	Replace Wire Strike Cockpit Door Latch Deflectors	2-4-90-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Petrolatum, Item 234, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 2-3-9

2-4-90.1 Replace Wire Strike Cockpit Door Latch Deflectors.

NOTE

Replacement of left and right wire strike cockpit door latch deflectors is the same.

2-4-90.1.1 Remove.

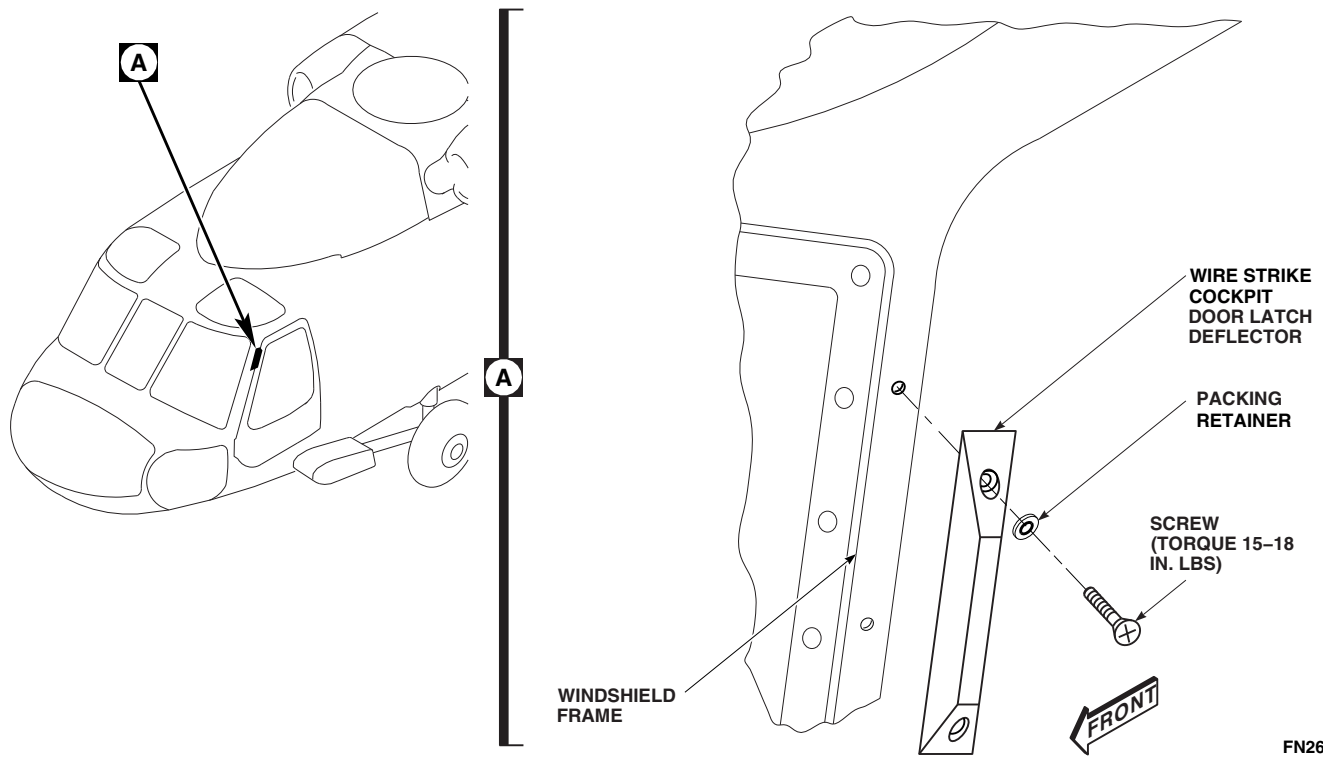
- Using nonmetallic scraper, remove sealing compound from screwheads.
- Remove screws and packing retainers securing wire strike cockpit door latch deflector to windshield frame (Figure 2-4-90-1, Detail A).
- Using nonmetallic scraper, remove sealing compound from windshield frame.

2-4-90.1.2 Inspect.

- Inspect wire strike cockpit door latch deflector (PARA 2-3-9).
- Inspect packing retainers for missing or unserviceable seals. **NONE ALLOWED.** If packing retainers have missing or unserviceable seals, replace packing retainers.

2-4-90.1.3 Install.

- Coat fastening screws with petrolatum, Item 234, Appendix D, to prevent damaging or folding over rubber seals of packing retainers.
- Install wire strike cockpit door latch deflector onto windshield frame with screws and packing retainers. **TORQUE SCREWS 15 - 18 INCH-POUNDS** (Figure 2-4-90-1, Detail A).

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SA**FIGURE 2-4-90-1. REPLACE WIRE STRIKE COCKPIT DOOR LATCH DEFLECTORS**

- c. Seal screwheads and packing retainers with sealing compound, Item 292, Appendix D.
- d. Seal wire strike cockpit door latch deflector to windshield frame with sealing compound, Item 292, Appendix D.

2-4-91 WIRE STRIKE WINDSHIELD CENTER POST DEFLECTOR INSERTS.

PARAGRAPH BREAKDOWN

		PAGE
2-4-91.1	Replace Wire Strike Windshield Center Post Deflector Inserts	2-4-91-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

References

- PARA 2-3-9

2-4-91.1 Replace Wire Strike Windshield Center Post Deflector Inserts.

NOTE

Replacement of pilot’s and copilot’s windshield center post deflector inserts is the same.

2-4-91.1.1 Remove.

Remove windshield center post deflector insert from channel by removing screws, washers, and nuts (Figure 2-4-91-1, Detail A).

2-4-91.1.2 Inspect.

Inspect wire strike windshield center post deflector inserts (PARA 2-3-9).

2-4-91.1.3 Install.

Position windshield center post deflector insert into channel and secure with screws, washers, and nuts (Figure 2-4-91-1, Detail A).

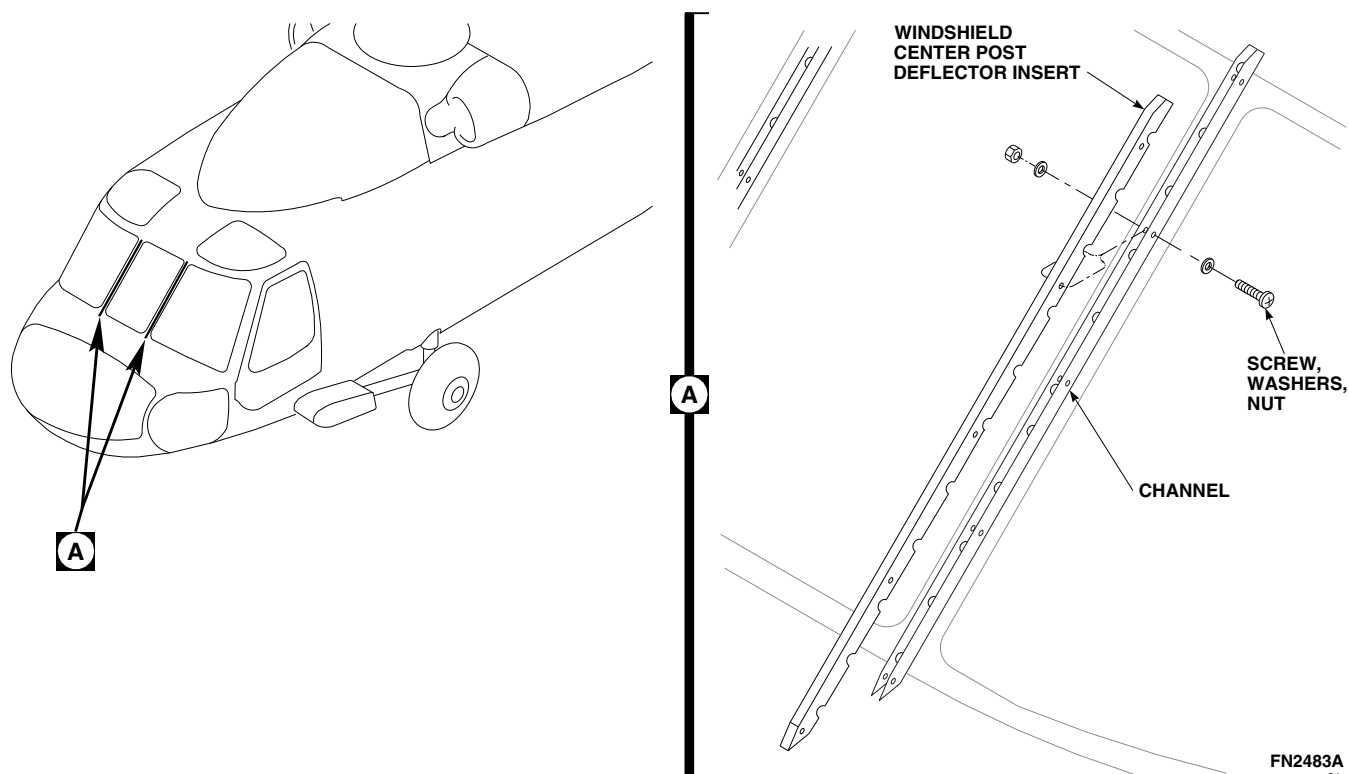


FIGURE 2-4-91-1. REPLACE WIRE STRIKE WINDSHIELD CENTER POST DEFLECTOR INSERTS

2-4-92 WIRE STRIKE WINDSHIELD CENTER POST DEFLECTOR CHANNELS.

PARAGRAPH BREAKDOWN

		PAGE
2-4-92.1	Replace Wire Strike Windshield Center Post Deflector Channels	2-4-92-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Drill
- Drill Bit, No. 11 (0.191-inch diameter)
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Sealing Compound, Item 292, Appendix D
- Tags

References

- Appendix D
- PARA 2-3-9
- PARA 2-4-91

2-4-92.1 Replace Wire Strike Windshield Center Post Deflector Channels.

NOTE

Replacement of pilot’s and copilot’s wire strike windshield center post deflector channels is the same.

2-4-92.1.1 Remove.

- Remove wire strike windshield center post deflector insert (PARA 2-4-91).

NOTE

Screws used to secure channel are not all the same length or same alloy steel. Tag or identify screws as they are removed.

- Remove screws and packing retainers securing windshield center post deflector channel to windshield center post. Remove channel (Figure 2-4-92-1, Detail A).
- Using nonmetallic scraper, remove sealing compound from channel and windshield center post.

2-4-92.1.2 Inspect.

- Inspect wire strike windshield center post deflector channel (PARA 2-3-9).
- Inspect packing retainers for missing or unserviceable seals. **NONE ALLOWED.** If packing retainers have missing or unserviceable seals, replace packing retainers.

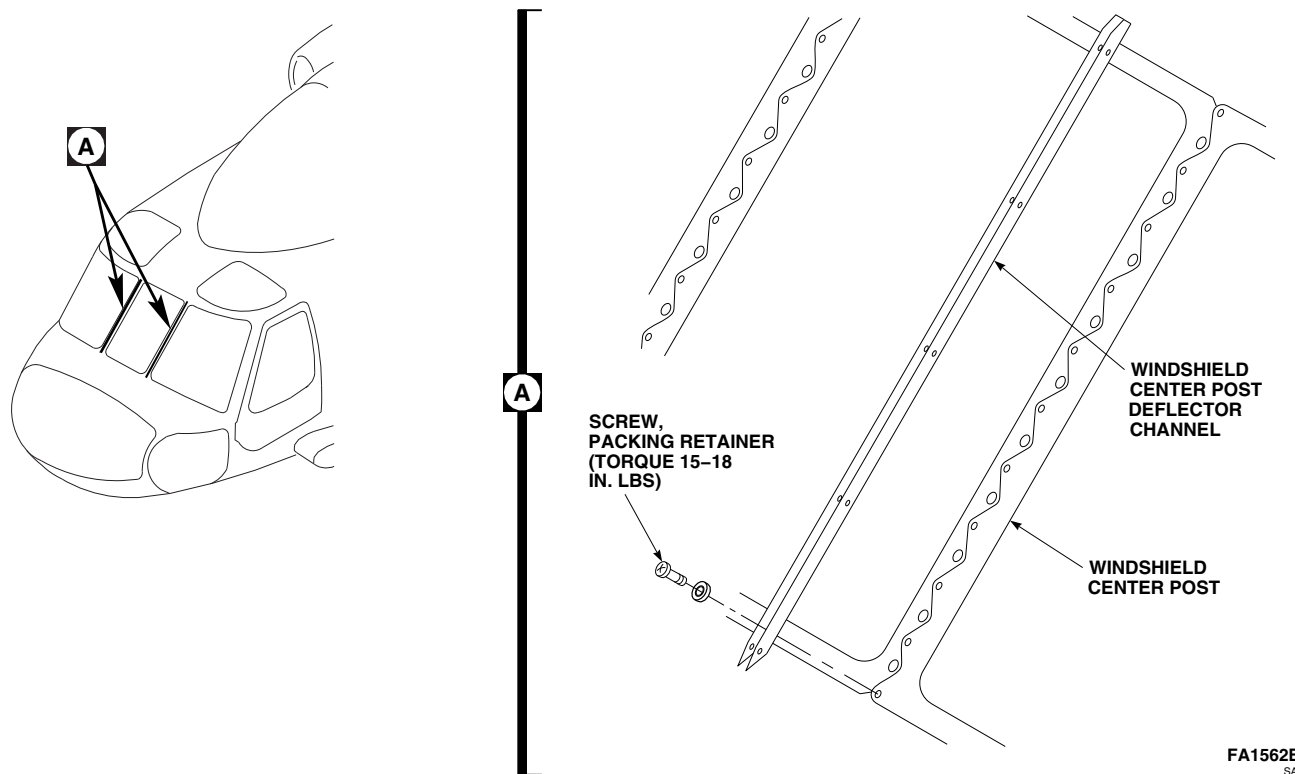
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FIGURE 2-4-92-1. REPLACE WIRE STRIKE WINDSHIELD CENTER POST DEFLECTOR CHANNELS

2-4-92.1.3 Install.

NOTE

Windshield center post deflector channel may need to have screwholes enlarged.

- If necessary, enlarge screwholes in channel into transverse slots, using drill with No. 11 drill bit. **LENGTH OF SLOT SHALL NOT EXCEED INSIDE WIDTH OF CHANNEL.**
- Apply 1/8-inch wide bead of sealing compound, Item 292, Appendix D, around edges of chan-

nel, and position channel over windshield center post (Figure 2-4-92-1, Detail A).

- Coat threads of screws with sealing compound, Item 292, Appendix D, and secure channel with screws and packing retainers. **TORQUE SCREWS TO 15 - 18 INCH-POUNDS.**
- Using nonmetallic scraper, remove excess sealing compound from edges of deflector and screwheads.
- Install wire strike windshield center post deflector insert (PARA 2-4-91).

2-4-93 FUSELAGE JACKPADS.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
2-4-93.1	Replace Drag Beam Support Jackpad	2-4-93-1	2-4-93.2	Replace Tail Cone Jackpad	2-4-93-2

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Drill Set
- Drive Wrench
- Lockring Removal Tool
- Nonmetallic Scraper, Locally-Made
- Swage Tool
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Antiseize Compound, Item 74, Appendix D
- Cheesecloth, Item 90, Appendix D
- Corrosion Preventive Compound, Item 115, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D

2-4-93.1 Replace Drag Beam Support Jackpad.

NOTE

Replacement of left and right drag beam support jackpad is the same.

2-4-93.1.1 Remove.

- a. Remove jackpad from drag beam support (Figure 2-4-93-1, Detail A).
- b. Wipe stud and threads clean using cheesecloth, Item 90, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D.

2-4-93.1.2 Inspect/Repair.

- a. Check jackpad stud for misalignment and thread damage. **NO BENT STUD OR DAMAGED THREADS ALLOWED.** If stud is bent, replace jackpad. If threads are damaged, repair threads.
- b. Check pad threaded insert for damage and security. **NO DAMAGE OR LOOSENESS ALLOWED.** If damaged or loose, replace inserts.
- c. Replace threaded insert as follows:
 - (1) Using lockring removal tool, drill out insert lockring to 0.082-inch deep.

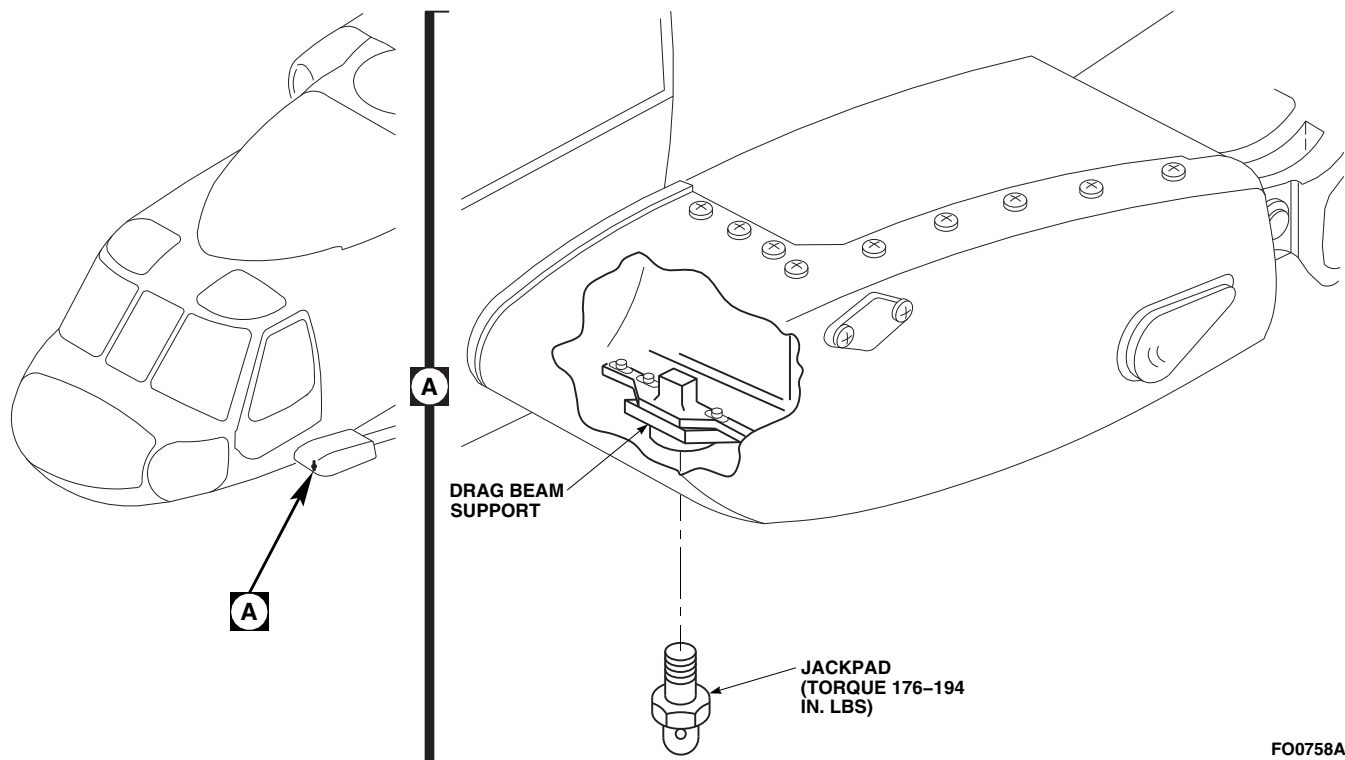
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FIGURE 2-4-93-1. REPLACE DRAG BEAM SUPPORT JACKPAD

- (2) Remove remainder of insert with drive wrench tool.
- (3) Clean insert hole and threads of any foreign material and chips.
- (4) Install new insert in hole with drive wrench, until top of insert is 0.010 - 0.020-inch below surface.
- (5) Swage locking onto insert with swage tool, until shoulder of tool touches surface of fitting. Lockring should be 0.005 - 0.010-inch below surface.

2-4-93.1.3 Install.

- a. Coat jackpad stud threads with antiseize compound, Item 74, Appendix D.
- b. Install jackpad in drag beam support (Figure 2-4-93-1, Detail A). **TORQUE JACKPAD TO 176 - 194 INCH-POUNDS.**

2-4-93.2 Replace Tail Cone Jackpad.

2-4-93.2.1 Remove.

- a. Remove screws and washers from front fairing, of tail landing gear fairing, to helicopter (Figure 2-4-93-2, Detail A). Remove fairing.
- b. Remove pins and collars securing jackpad to structure. Discard used hardware.
- c. Using nonmetallic scraper, separate bond between jackpad and structure. Remove jackpad.
- d. Remove sealing compound debris using cheesecloth, Item 90, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, and nonmetallic scraper.

2-4-93.2.2 Install.

- a. Using clean cheesecloth, Item 90, Appendix D, wet with methyl ethyl ketone, Item 217, Ap-

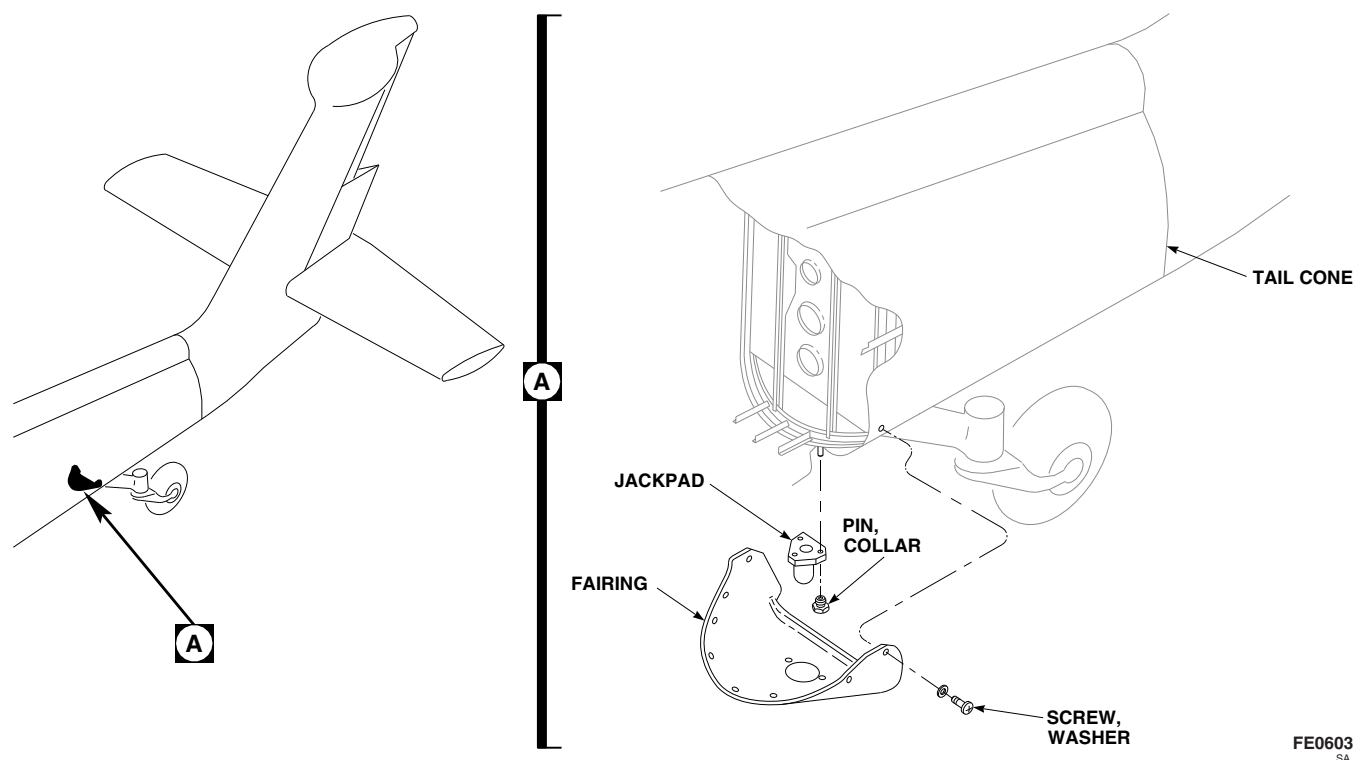


FIGURE 2-4-93-2. REPLACE TAIL CONE JACKPAD

- pendix D, wipe jackpad, fuselage mating surfaces, fastener holes, and fasteners, until clean.
- b. Coat new fastener pins, jackpad, and fuselage joint surfaces with sealing compound, Item 292, Appendix D (Figure 2-4-93-2, Detail A).
- c. Install jackpad with pins and collars while sealing compound is wet.
- d. Apply corrosion preventive compound, Item 115, Appendix D, to fairing attachment hardware.
- e. Position front fairing, covering jack point forward of yoke, on helicopter and secure with screws and washers.

2-4-94 DRAG BEAM SUPPORT FAIRINGS.

PARAGRAPH BREAKDOWN

	PAGE
2-4-94.1 Replace Drag Beam Support Fairings	2-4-94-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Abrasive Cloth, Item 9, Appendix D
- Adhesive, Item 37, Appendix D
- Cheesecloth, Item 90, Appendix D
- Corrosion Preventive Compound, Item 115, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D

- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-99
- PARA 2-6-3
- PARA 9-2-14
- PARA 9-4-78
- PARA 9-4-79

2-4-94.1 Replace Drag Beam Support Fairings.

NOTE

Replacement of left and right drag beam support fairing is the same, except as noted.

2-4-94.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Remove cockpit step (PARA 2-4-99).
- Remove screws and washers, and separate drag beam support fairing from fuselage enough to

access electrical connectors for position lights (Figure 2-4-94-1, Detail A).

- Disconnect electrical connector, P254 (left) or P253 (right), from incandescent position light connector, J254 (left) or J253 (right).
- Disconnect electrical connector, P340 (left) or P341 (right), from infrared position light connector, J340 (left) or J341 (right).
- Install protective caps and plugs on electrical connectors.
- Remove fairing.

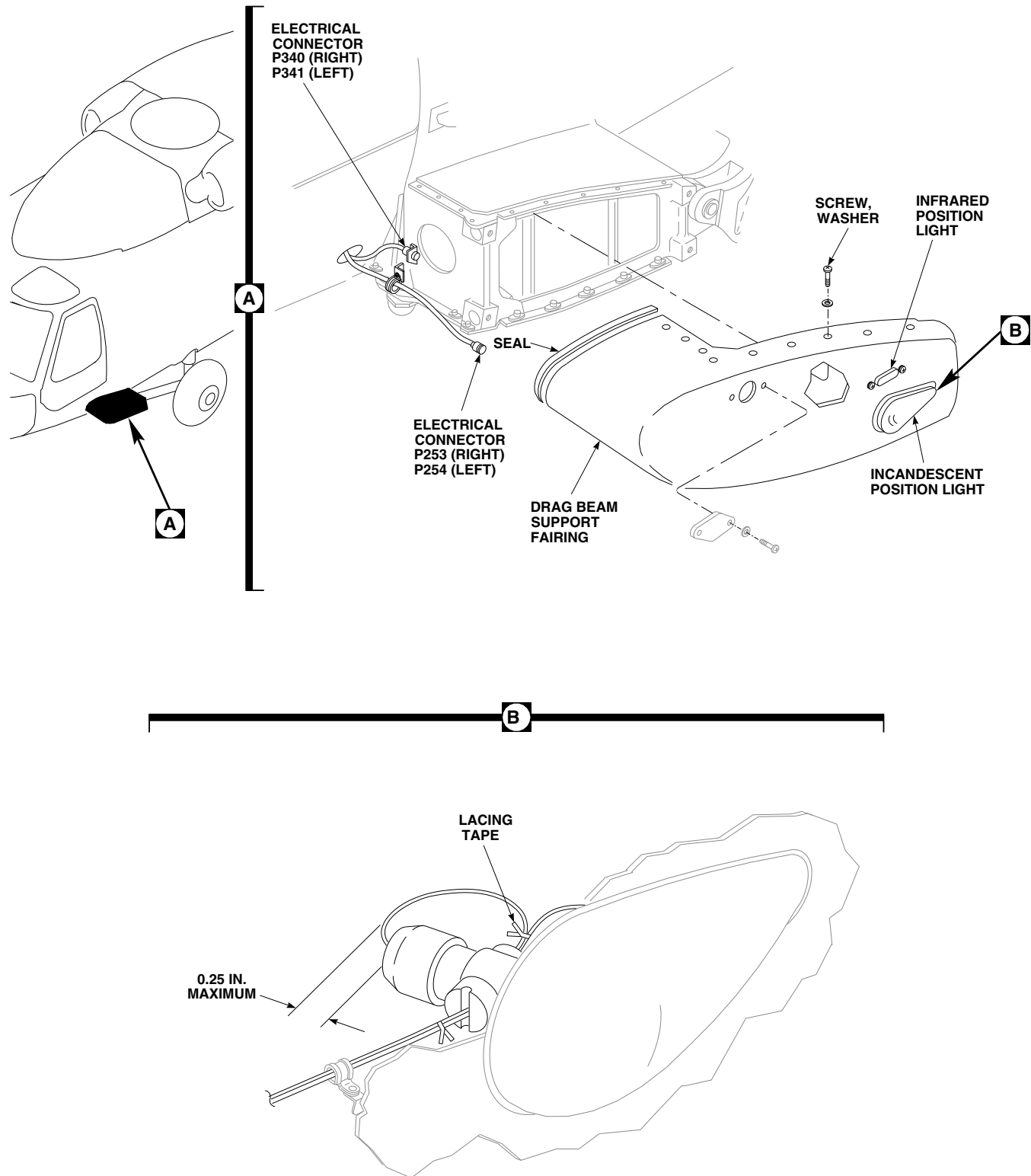


FIGURE 2-4-94-1. REPLACE DRAG BEAM SUPPORT FAIRINGS

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- h. If new fairing is to be installed, remove drag beam support fairing incandescent position light (PARA 9-4-78) and infrared position light (PARA 9-4-79).

2-4-94.1.2 Inspect/Repair.

- a. Inspect fairing for damage. If necessary, repair fairing (PARA 2-6-3).
- b. Inspect seal for damage. **NONE ALLOWED.** If damaged, replace seal as follows:
 - (1) Remove damaged seal, seal debris, and adhesive using nonmetallic scraper.
 - (2) Roughen seal and fairing bonding surfaces using abrasive cloth, Item 9, Appendix D.
 - (3) Wipe bonding surfaces clean with cheesecloth, Item 90, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D. Wipe dry.
 - (4) Apply thin even coat of adhesive, Item 37, Appendix D, to seal and fairing bonding surfaces. Allow adhesive to become tacky.
 - (5) Bond seal to fairing. Work out wrinkles or blisters. Check seal installation for complete contact.
 - (6) Remove extra adhesive.

2-4-94.1.3 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. If installing new fairing, install drag beam support fairing incandescent position light (PARA 9-4-78) and infrared position light (PARA 9-4-79), then proceed to step e.

- c. If same fairing is being used, perform the following:
 - (1) Remove protective caps and plugs from electrical connectors.
 - (2) Position fairing on helicopter enough to connect electrical connectors.
 - (3) Inspect and treat electrical connectors (PARA 1-7-20).
 - (4) Connect electrical connector, P254 (left) or P253 (right), to incandescent position light connector, J254 (left) or J253 (right).
 - (5) Connect electrical connector, P340 (left) or P341 (right), to infrared position light connector, J340 (left) or J341 (right).
 - (6) Check position light wire harness for clearance (Figure 2-4-94-1, Detail B).
- d. Make sure area is clean and free of foreign material before closing up.
- e. Position fairing on helicopter, lining up screw holes.
- f. Apply corrosion preventive compound, Item 115, Appendix D, to fairing attachment hardware.
- g. Secure fairing to fuselage with screws and washers (Figure 2-4-94-1, Detail A).
- h. Seal joining area between fairing and fuselage with sealing compound, Item 292, Appendix D.
- i. Install cockpit step (PARA 2-4-99).
- j. Perform operational check of position lights (PARA 9-2-14).

2-4-95 LOWER MAIN LANDING GEAR FAIRINGS **ESSS** .

PARAGRAPH BREAKDOWN

	PAGE
2-4-95.1 Replace Lower Main Landing Gear Fairings	2-4-95-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Drill Set

Materials/Parts

- Corrosion Preventive Compound, Item 115, Appendix D

References

- Appendix D
- PARA 2-6-3

2-4-95.1 Replace Lower Main Landing Gear Fairings.

NOTE

Replacement of left and right side lower main landing gear fairings is the same.

2-4-95.1.1. Remove.

- Unlock studs, and remove lower cap fairings from ESSS fittings (Figure 2-4-95-1, Detail A).
- Remove screws, washers, and cover from lower fairing.
- Remove screws, washers, and lower fairing from fuselage.

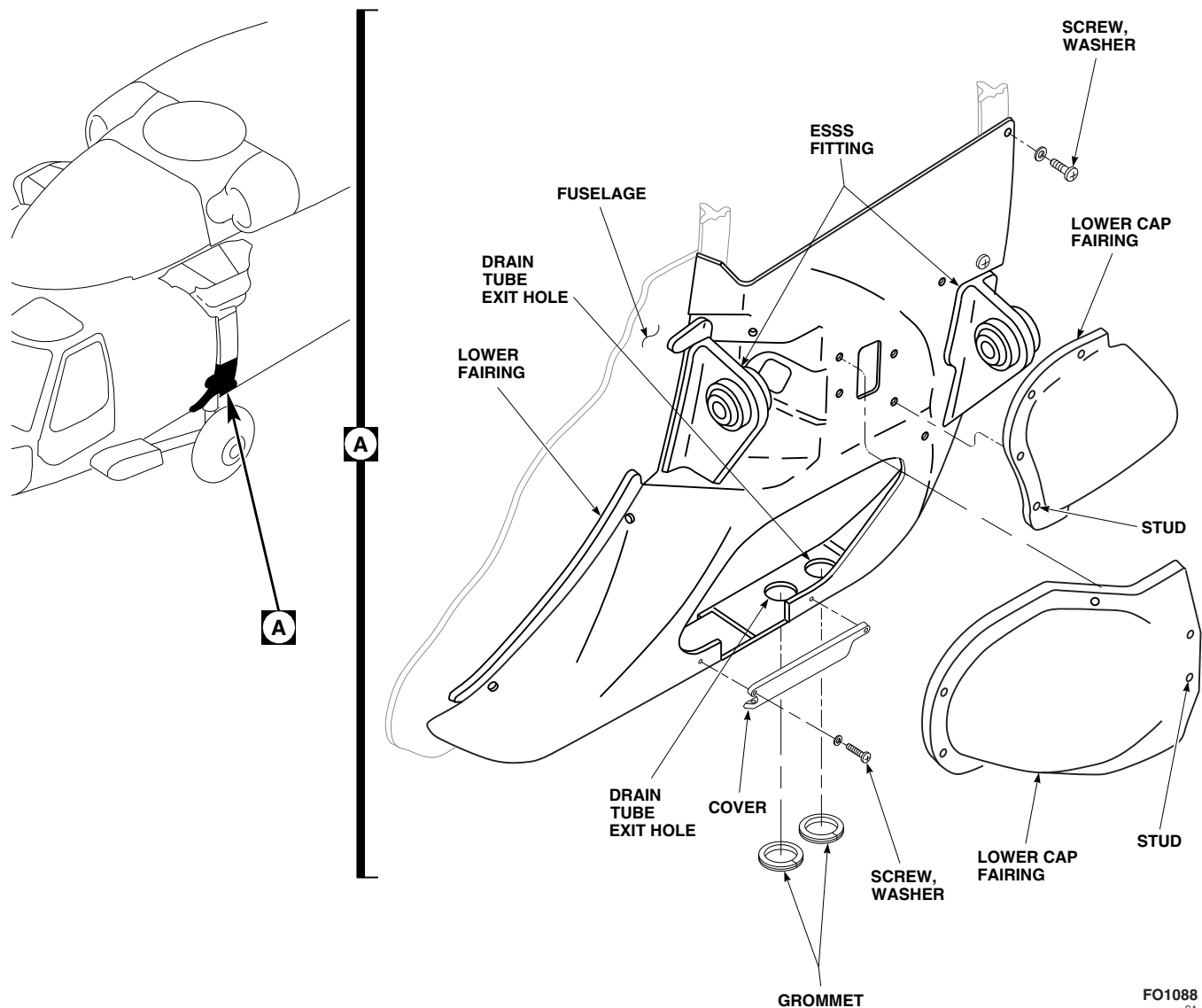
2-4-95.1.2. Inspect.

- Check drain tube exit holes for missing grommets. Replace grommets as necessary (Figure 2-4-95-1, Detail A).

- For left fairing, cut grommet to correct length and install. **GAP BETWEEN CUT ENDS MUST NOT BE OVER 0.025 INCH.**
- For right fairing, split round grommet and insert in hole.

2-4-95.1.3. Install.

- If installing new fairings, perform the following:
 - Trial-fit replacement fairing on fuselage.
 - Mark holes that do not line up with nut-plates.
 - Fill marked holes (PARA 2-6-3).
 - Locate and drill new holes in fairing, using fuselage as template.
- Make sure area is clean and free of foreign material before closing up.
- Apply corrosion preventive compound, Item 115, Appendix D, to fairing attachment hardware.

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SA**FIGURE 2-4-95-1. REPLACE LOWER MAIN LANDING GEAR FAIRINGS**

- d. Fasten fairing to fuselage with screws and washers (Figure 2-4-95-1, Detail A).
- e. Insert drain tubes through fairing exit holes.
- f. Apply corrosion preventive compound, Item 115, Appendix D, to fairing attachment hardware.
- g. Install cover on fairing and fasten in place with screws and washers.
- h. Install lower cap fairings over ESSES fittings with studs.

2-4-96 UPPER MAIN LANDING GEAR FAIRINGS **ESSS** .

PARAGRAPH BREAKDOWN

		PAGE
2-4-96.1	Replace Upper Main Landing Gear Fairings	2-4-96-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Corrosion Preventive Compound, Item 115, Appendix D

References

- Appendix D

2-4-96.1 Replace Upper Main Landing Gear Fairings.

NOTE

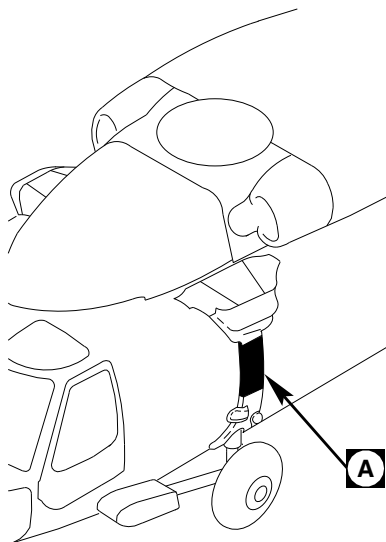
Replacement of left and right upper main landing gear fairings is the same.

2-4-96.1.1. Remove.

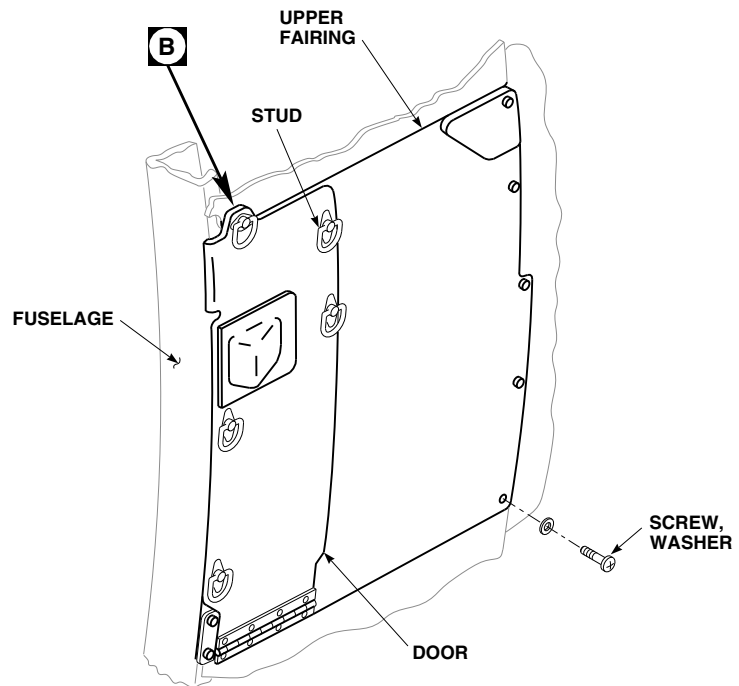
- Unfasten studs securing door to fairing and fuselage, and remove screws and washers securing fairing to fuselage (Figure 2-4-96-1, Detail A or Detail B).
- Remove upper fairing.

2-4-96.1.2. Install.

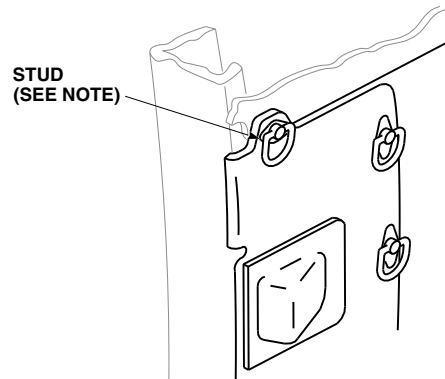
- Make sure area is clean and free of foreign material before closing up.
- Apply corrosion preventive compound, Item 115, Appendix D, to fairing attachment hardware.
- Place fairing into position on fuselage (Figure 2-4-96-1, Detail A or Detail B).
- Attach fairing to fuselage with screws and washers.
- Fasten studs securing door to fairing and fuselage.



A



B



NOTE

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FIGURE 2-4-96-1. REPLACE UPPER MAIN LANDING GEAR FAIRINGS

2-4-97 TAIL LANDING GEAR FAIRING.

PARAGRAPH BREAKDOWN

		PAGE
2-4-97.1	Replace Tail Landing Gear Fairing	2-4-97-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Corrosion Preventive Compound, Item 115, Appendix D

References

- Appendix D
- PARA 2-6-2
- PARA 2-6-3

2-4-97.1 Replace Tail Landing Gear Fairing.

NOTE

Tail landing gear fairing consists of five pieces.

2-4-97.1.1 Remove.

- Remove screws and washers securing front fairing to helicopter (Figure 2-4-97-1, Detail A). Remove fairing.
- Remove screws and washers securing front plate to fairing. Remove plate from fairing.
- Remove screws and washers securing rear plate to fairing. Remove plate from fairing.
- Remove screws and washers securing left and right fairings to helicopter. Remove fairings from helicopter.

2-4-97.1.2 Inspect.

Inspect fairings and plates for damage. **NONE ALLOWED.** If damaged, repair (PARA 2-6-2 and PARA 2-6-3).

2-4-97.1.3 Install.

- Make sure area is clean and free of foreign material before closing up.
- Apply corrosion preventive compound, Item 115, Appendix D, to fairing attachment hardware.
- Position left and right fairings on helicopter fuselage and secure with screws and washers (Figure 2-4-97-1, Detail A).
- Apply corrosion preventive compound, Item 115, Appendix D, to fairing attachment hardware.
- Position rear plate on fairing and secure with screws and washers.
- Apply corrosion preventive compound, Item 115, Appendix D, to fairing attachment hardware.
- Position front plate on fairing and secure with screws and washers.

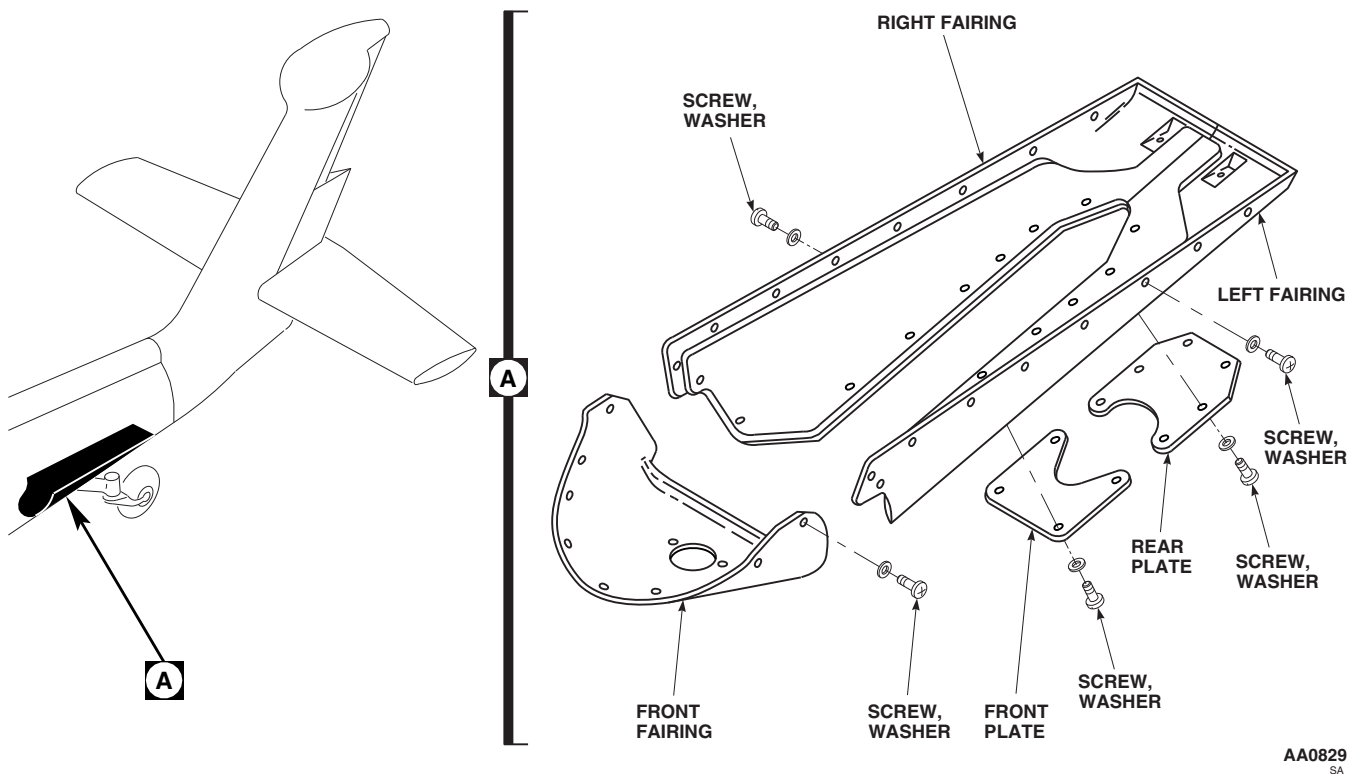
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FIGURE 2-4-97-1. REPLACE TAIL LANDING GEAR FAIRING

- h. Apply corrosion preventive compound, Item 115, Appendix D, to fairing attachment hardware.
- i. Position front fairing, covering jack point forward of yoke, on helicopter and secure with screws and washers.

2-4-98 UPPER ESSS FITTING FAIRING AND PLATFORM **ESSS** .

PARAGRAPH BREAKDOWN

		PAGE
2-4-98.1	Replace Upper ESSS Fitting Fairing and Platform	2-4-98-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Corrosion Preventive Compound, Item 115, Appendix D

References

- Appendix D
- PARA 2-6-2
- PARA 2-6-3

2-4-98.1 Replace Upper ESSS Fitting Fairing and Platform.

NOTE

Replacement of left and right upper ESSS fitting fairings and platforms is the same.

2-4-98.1.1 Remove.

- Unlock studs and remove upper cap fairing from upper ESSS fittings (Figure 2-4-98-1, Detail A).
- Unfasten studs and remove screws and washers securing front fairing panel. Remove fairing panel.
- Unfasten all studs and remove screws securing center platform and hinged rear fairing. Remove fairing.

2-4-98.1.2 Inspect.

Inspect fairings, platform, and panel for damage. **NONE ALLOWED.** If damaged, repair (PARA 2-6-2 and PARA 2-6-3).

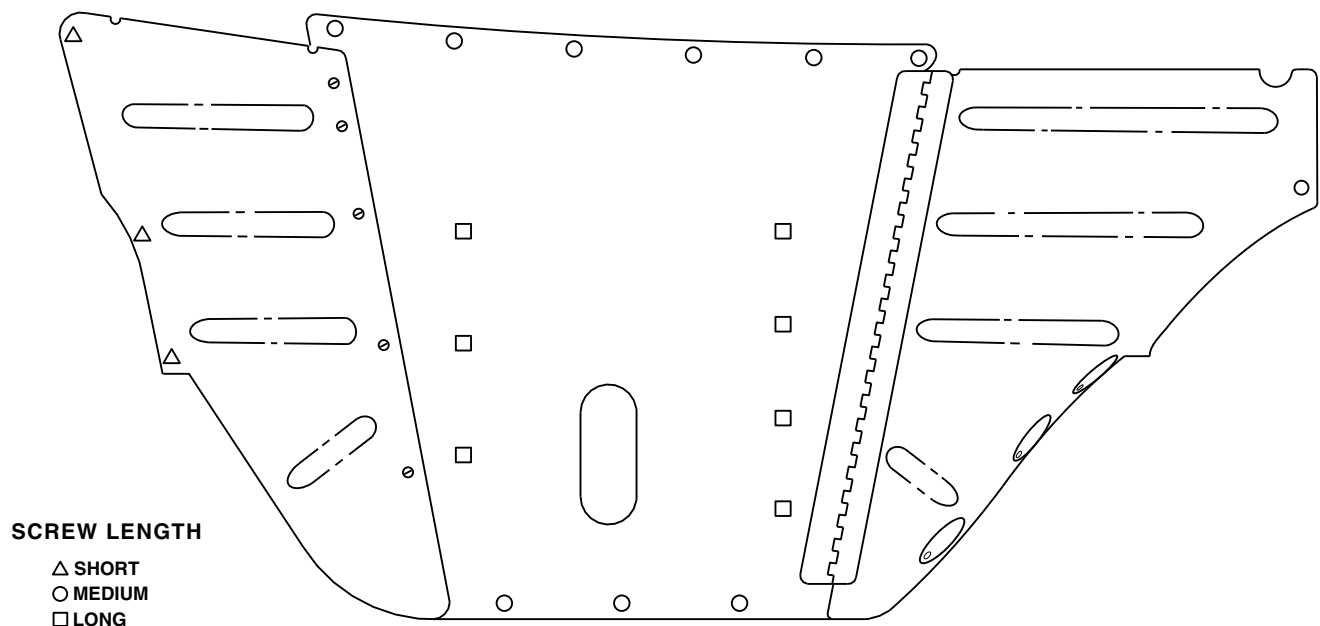
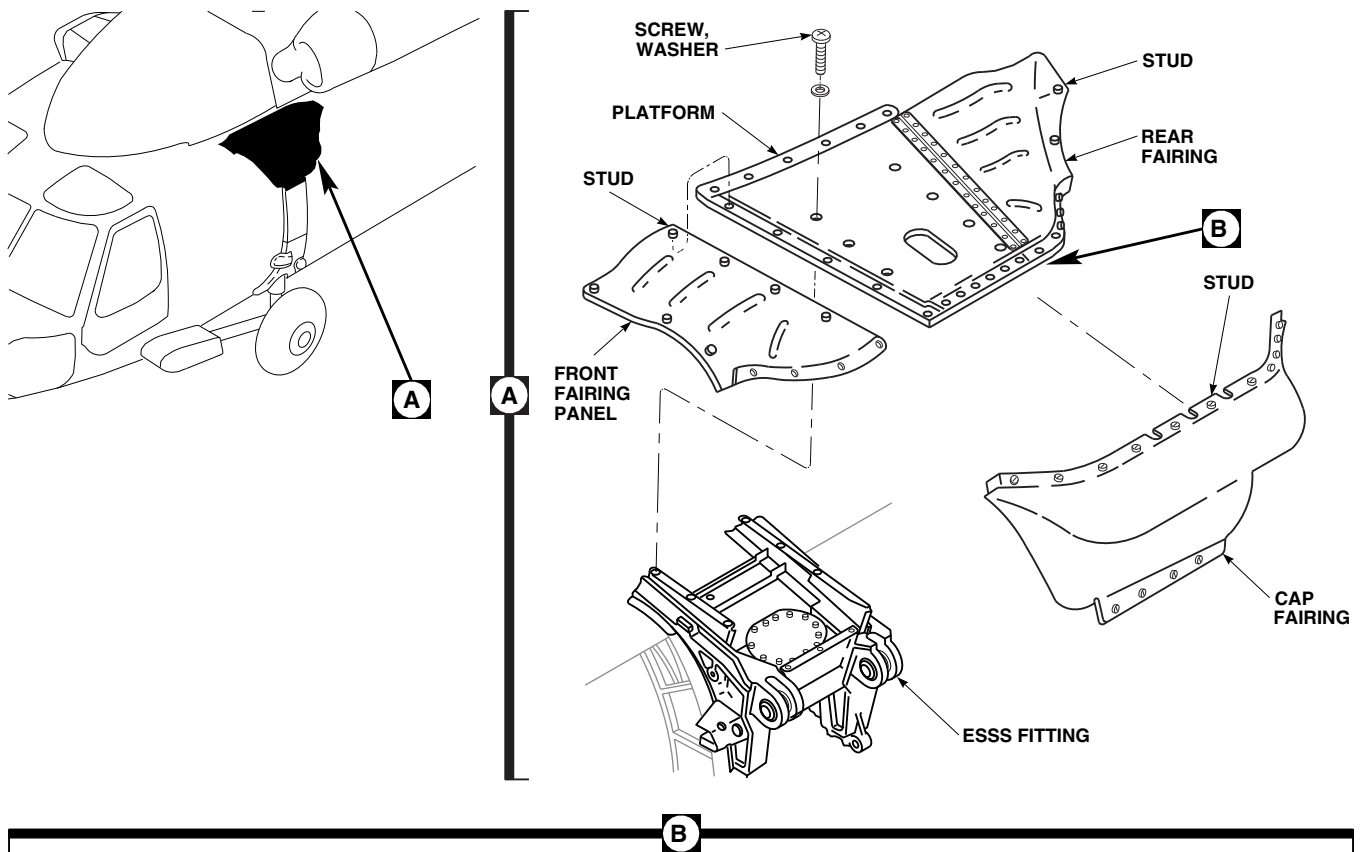
2-4-98.1.3 Install.

- Line up center and rear fairing panels over fastener holes.
- Apply corrosion preventive compound, Item 115, Appendix D, to fairing attachment hardware.

NOTE

Refer to Figure 2-4-98-1, Detail B, for fastener size and location.

- Install screws and washers in center fairing platform followed by rear hinged panel studs (Figure 2-4-98-1, Detail A).
- Line up front fairing panel over fastener holes. Install screws and secure studs.
- Install upper cap fairing on upper ESSS fittings and secure with studs.



UPPER ESSS FAIRING
TOP VIEW

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FIGURE 2-4-98-1. REPLACE UPPER ESSS FITTING FAIRING AND PLATFORM

2-4-99 COCKPIT STEPS.

PARAGRAPH BREAKDOWN

	PAGE
2-4-99.1 Replace Cockpit Steps	2-4-99-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Arbor Press
- Drift
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 700 - 1600 in. lbs

- Vise, Soft-Jawed

Materials/Parts

- Corrosion Preventive Compound, Item 115, Appendix D

References

- Appendix D
- PARA 2-4-89
- PARA 2-4-94

2-4-99.1 Replace Cockpit Steps.

NOTE

Replacement of left and right cockpit steps is the same.

2-4-99.1.1 Remove.

- Remove wire strike step extension (PARA 2-4-89).
- Remove screws and washers securing cover to fairing (Figure 2-4-99-1, Detail A).
- Remove screws and washers securing step to drag beam support fairing.
- Remove bolt and washer securing step to support, using access from removed cover. Remove step.
- Remove drag beam support fairing (PARA 2-4-94).

- Remove barrel-nut and retainer from support.

2-4-99.1.2 Inspect/Repair.

- 701015-SUBQ** Using arbor press and suitable drift, remove bushing from cockpit step (Figure 2-4-99-1, Detail B).
- 701015-SUBQ** Using arbor press and suitable drift, press bushing in cockpit step.
- Test bolt and barrel-nut as follows:

NOTE

Make sure that only square barrel-nuts are used.

- Place barrel-nut in soft-jawed vise.
- Thread bolt into barrel-nut until two threads are showing beyond nut.
- Using torque wrench, back off bolt. **BREAKAWAY TORQUE SHALL NOT**

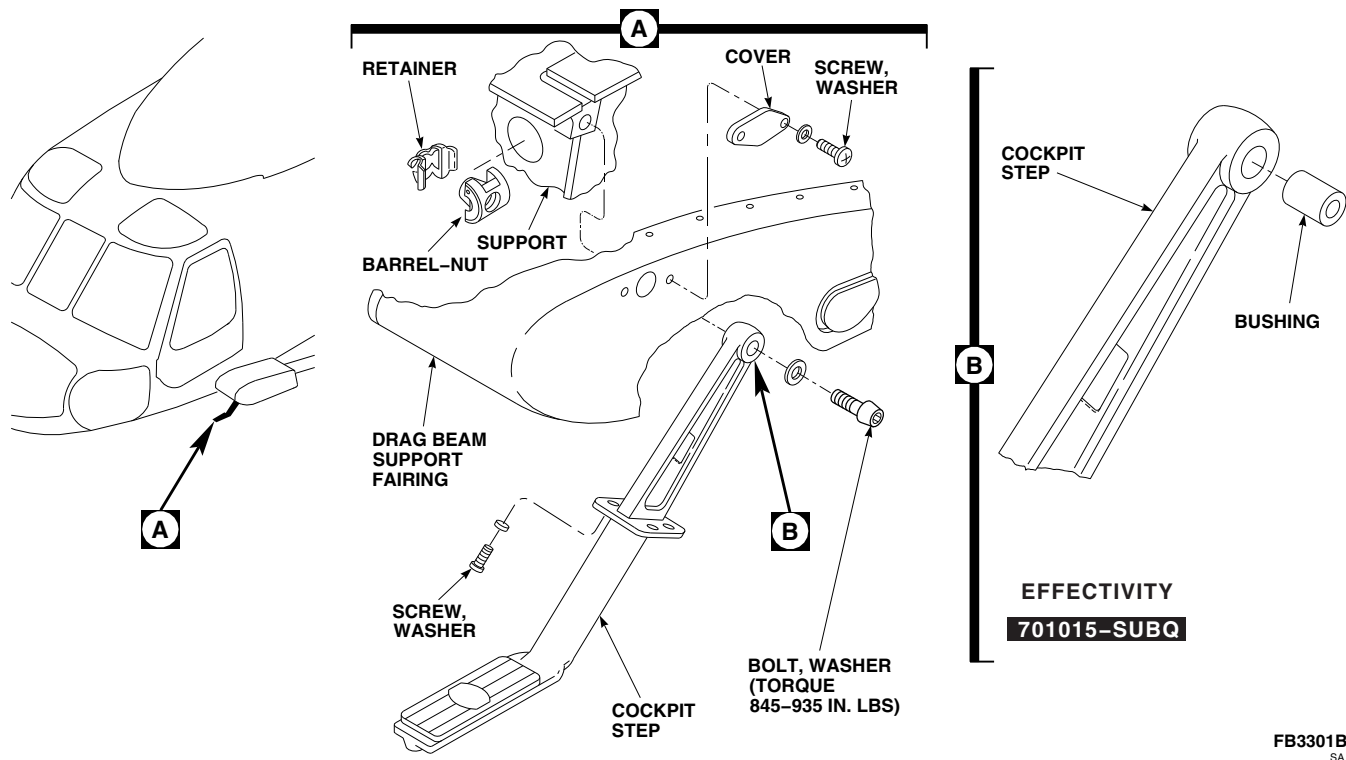


FIGURE 2-4-99-1. REPLACE COCKPIT STEPS

BE LESS THAN 18 INCH-POUNDS. If breakaway torque is less than 18 inch-pounds, perform the following:

- (a) Replace bolt. Repeat step c. (1) through step c. (3).
- (b) If breakaway torque is still less than 18 inch-pounds, replace barrel-nut. Repeat step c. (1) through step c. (3).

2-4-99.1.3 Install.

CAUTION

Damage to recess in support will result if barrel-nut is incorrectly installed. To prevent damage, make sure barrel-nut is installed with barrel (round side) facing bolthead.

- a. Install barrel-nut and retainer in support with barrel side toward bolthead (Figure 2-4-99-1, Detail A).
- b. Install drag beam support fairing (PARA 2-4-94).
- c. Insert cockpit step through hole in drag beam support fairing and position step on support. Install bolt and washer. **TORQUE BOLT TO 845 - 935 INCH-POUNDS.**
- d. Install screws and washers securing cockpit step to drag beam support fairing.
- e. Apply corrosion preventive compound, Item 115, Appendix D, to fairing hardware.
- f. Install cover on drag beam support fairing with screws and washers.
- g. Install wire strike step extension (PARA 2-4-89).

2-4-100 CABIN STEPS.

PARAGRAPH BREAKDOWN

	PAGE
2-4-100.1 Replace Cabin Steps	2-4-100-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Drill Set
- Feeler Gage
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Adhesive, Item 37, Appendix D
- Cheesecloth, Item 90, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D

References

- Appendix D

2-4-100.1 Replace Cabin Steps.

NOTE

Replacement procedures are the same for left and right side.

2-4-100.1.1. Remove.

- Open step fairing.
- Remove bolts, washers, nuts, laminated shims, filler, and angle from bulkhead (Figure 2-4-100-1, Detail A).
- If required, remove filler, laminated shims, and old adhesive using nonmetallic scraper and cheesecloth, Item 90, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D.
- Remove shaft and washers from support.
- Remove step and shaft from helicopter.
- If necessary, remove hinge pin, washers, and step from shaft.

2-4-100.1.2. Install.

- If step was removed, install on shaft with hinge pin, washers, and cotter pins (Figure 2-4-100-1, Detail A).
- Place shaft and washer in support.
- Install washers on top of shaft.
- If required, install filler and laminated shim on bulkhead using adhesive, Item 37, Appendix D. Drill fastener holes.
- Install shaft with washers in angle, and fasten to bulkhead with bolts, washers, and nuts.
- Check to make sure angle is aligned with support. Peel laminated shim as required to get alignment.
- Measure clearance between washers on shaft and angle using feeler gage. **GAP MUST BE 0.020 - 0.060 INCH.**

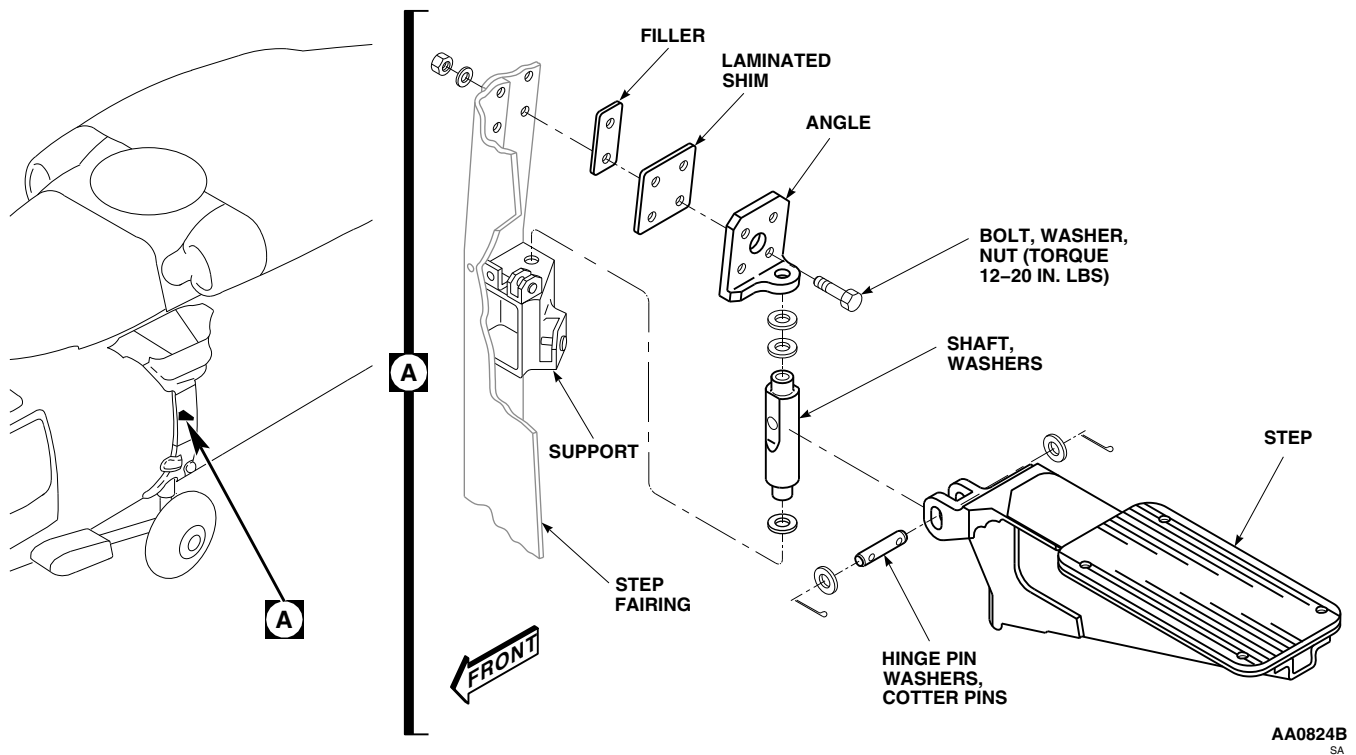


FIGURE 2-4-100-1. REPLACE CABIN STEPS

- h. Add or subtract washers as necessary to get proper gap. **TORQUE NUTS TO 12 - 20 INCH-POUNDS.**
- i. Fold step.
- j. Close and fasten fairing.

2-4-101 MAIN ROTOR PYLON SLIDING COVER.

PARAGRAPH BREAKDOWN

	PAGE
2-4-101.1 Replace Main Rotor Pylon Sliding Cover	2-4-101-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Drill Set
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Adhesive, Item 32, Appendix D
- Adhesive, Item 66, Appendix D
- Cheesecloth, Item 90, Appendix D

- Grommet, Nonmetallic, Item 152, Appendix D
- Lockwire, Item 193, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D

References

- Appendix D
- PARA 2-3-10
- PARA 2-4-79
- PARA 2-4-80
- PARA 2-4-102
- PARA 2-6-3

2-4-101.1 Replace Main Rotor Pylon Sliding Cover.

WARNING

To prevent injury to personnel, keep hands away from sharp edges of main rotor pylon sliding cover near track and mating edge, and ventilation blower inlet hole, when opening and closing cover. Use care to avoid injury.

2-4-101.1.1 Remove.

- Unlatch and slide main rotor pylon sliding cover forward. Slide cover against stops (Figure 2-4-101-1, Detail A and Detail B).
- With aid of an assistant, lift up on quick-release pins in stops and slide cover forward (Figure 2-4-101-1, Detail B). Remove cover from track.
- Place cover on a padded surface.
- If cover is to be replaced, perform the following:

- (1) Remove wire strike pitot cutter (PARA 2-4-80).
- (2) Remove wire strike upper cutter (PARA 2-4-79).

2-4-101.1.2 Inspect/Repair.

- a. Perform main rotor pylon sliding cover inspection (PARA 2-3-10).
- b. If required, replace pin support grommet as follows (Figure 2-4-101-2, Detail A):
 - (1) Remove damaged grommet using nonmetallic scraper.
 - (2) Remove adhesive residue from pin support and clean bonding surfaces thoroughly, using cheesecloth, Item 90, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D.
 - (3) Apply thin coat of adhesive, Item 66, Appendix D, to bonding surfaces of pin support.
 - (4) Allow primer to cure for at least 1 hour at room temperature.
 - (5) Cut piece of nonmetallic grommet, Item 152, Appendix D, to fit edge of pin support.
 - (6) Mix adhesive, Item 32, Appendix D, and apply to bonding surfaces of pin support and grommet.
 - (7) Immediately install grommet on edge of pin support and press in place.

NOTE

Cotter pin may be used as an alternate for spring roll pin.

- (1) If spring roll pin is installed, check for engagement and security of spring roll pin

in quick-release pin. If engagement and/or security is loose, perform the following:

- (a) Remove spring roll pin and install new pin.
- (b) Insert lockwire, Item 193, Appendix D, through the spring roll pin and wrap around the retention pin (over washer on the spring side), back again around the retention pin, and twist off.
- (2) If cotter pin is installed, check for engagement and security of cotter pin in quick-release pin. If engagement and/or security is loose, perform the following:
 - (a) Remove cotter pin and install new pin, MS24665-153.
 - (b) Fold over rear end of cotter pin against retention pin.

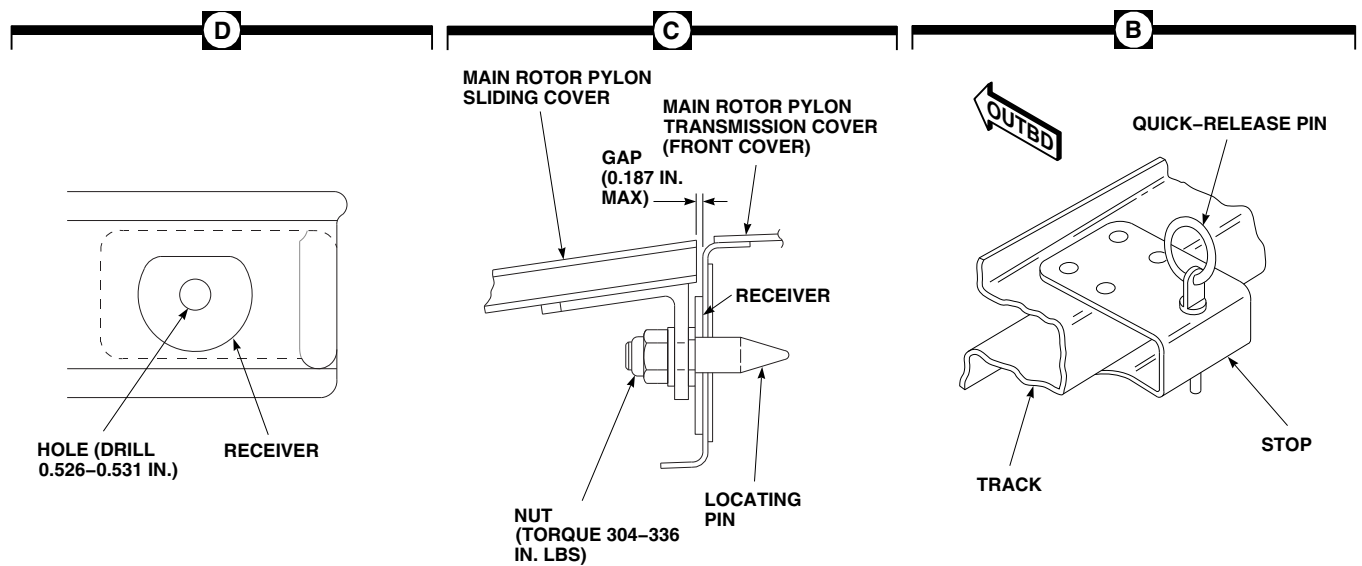
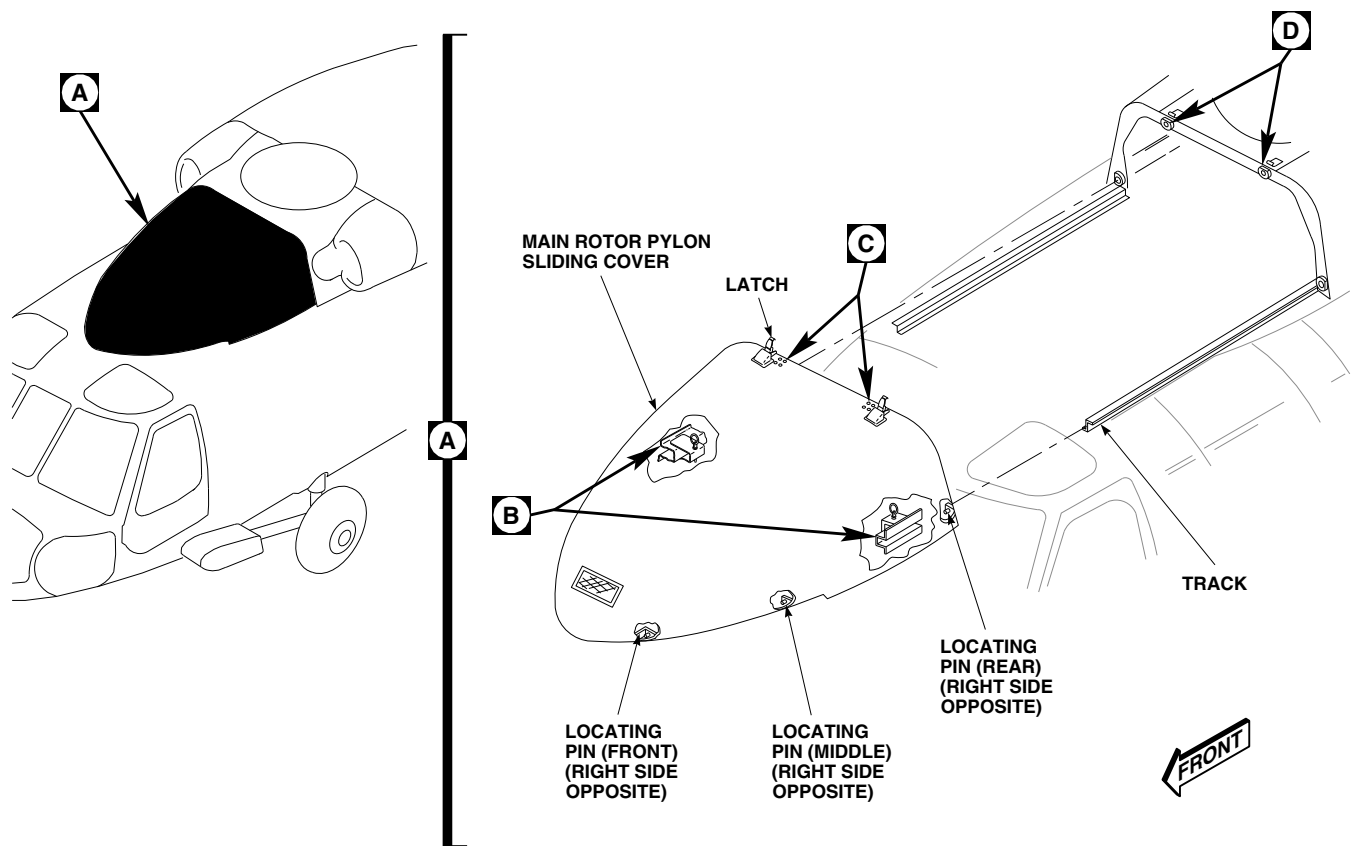
2-4-101.1.3 Install.

- a. If installing new main rotor pylon sliding cover, perform the following:
 - (1) Install wire strike pitot cutter (PARA 2-4-80).
 - (2) Install wire strike upper cutter (PARA 2-4-79).
- b. Loosen nuts securing six locating pins on main rotor pylon sliding cover. **TIGHTEN FIN-GERTIGHT** (Figure 2-4-101-1, Detail C).

WARNING

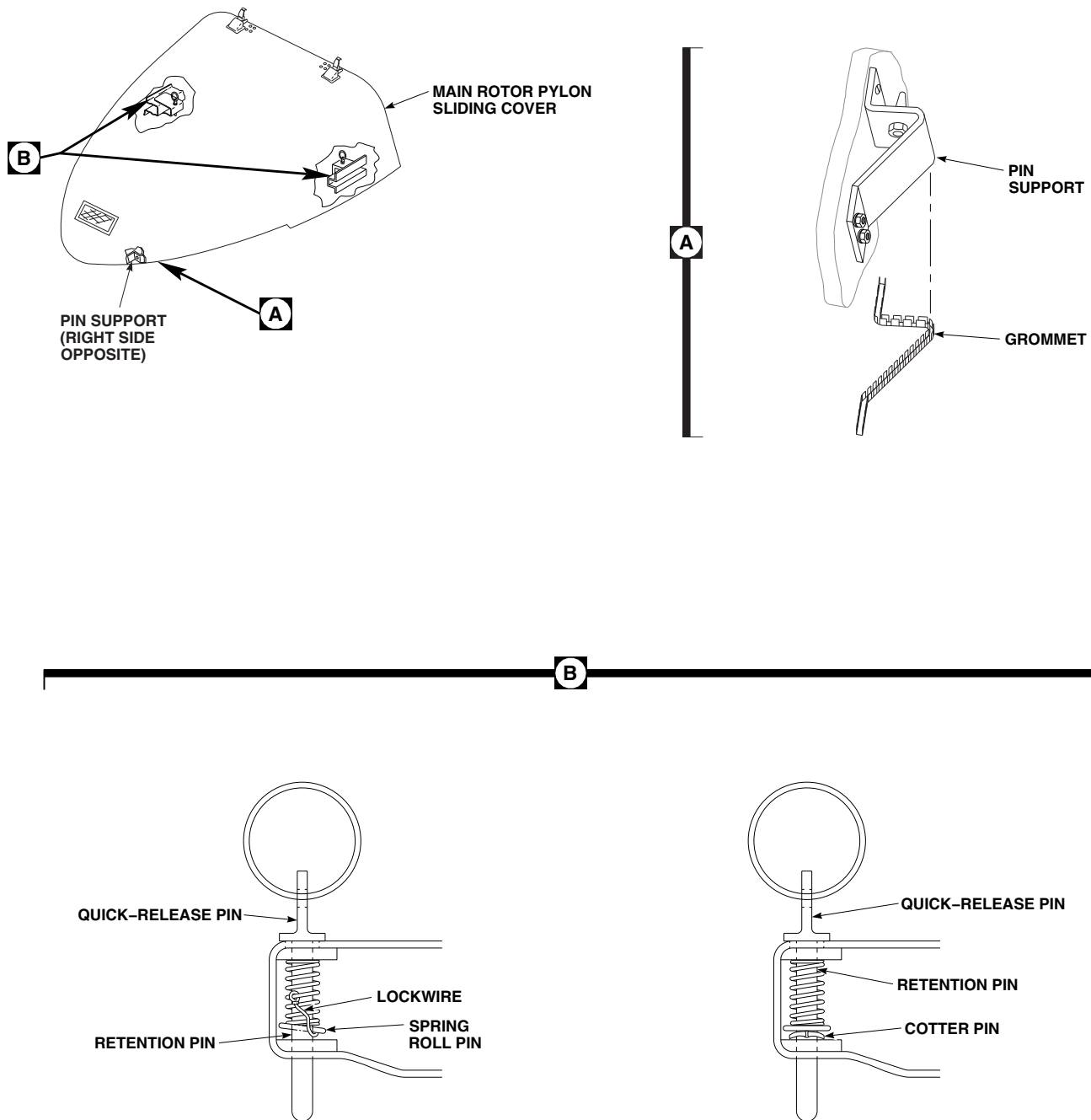
Injury to personnel and damage to equipment will result if rollers are damaged. Care must be taken to make sure rollers properly engage tracks during installation of main rotor pylon sliding cover. Damaged rollers will result in separation of sliding cover from helicopter.

- c. With aid of an assistant, position cover on tracks. Lift up on quick-release pins and slide



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FIGURE 2-4-101-1. REPLACE MAIN ROTOR PYLON SLIDING COVER



EFFECTIVITY

QUICK-RELEASE PIN WITH SPRING ROLL PIN INSTALLED.

EFFECTIVITY

QUICK-RELEASE PIN WITH COTTER PIN INSTALLED.

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FIGURE 2-4-101-2. INSPECT/REPAIR MAIN ROTOR PYLON SLIDING COVER

cover rearward (Figure 2-4-101-1, Detail A and Detail B). Release pins.

d. Slide cover open and closed a few times to check for proper operation.

e. If receiver holes are binding, drill receiver holes to 0.526 - 0.531-inch (Figure 2-4-101-1, Detail D).

f. After checking for proper line up and operation, **TORQUE NUTS TO 304 - 336 INCH-POUNDS** (Figure 2-4-101-1, Detail C).

g. With latches fully engaged, measure gap between shoulder of locating pin and surface of receiver to the following limits:

(1) Upper Rear: **NO GAP ALLOWED.**

(2) Lower Rear: **MAXIMUM GAP ALLOWED IS 0.120-INCH.**

(3) Middle and Front: **MAXIMUM GAP ALLOWED IS 0.250-INCH.**

h. With latches fully engaged, measure gap between main rotor pylon sliding cover and front cover of main rotor pylon transmission cover. **MAXIMUM GAP ALLOWED IS 0.187-INCH.** If gap exceeds limit, build up rear of main rotor pylon sliding cover (PARA 2-6-3).

i. Make sure area is clean and free of foreign material before closing up.

j. Close and fasten latches. If necessary, adjust latches (PARA 2-4-102).

2-4-102 MAIN ROTOR PYLON SLIDING COVER LATCHES.

PARAGRAPH BREAKDOWN

	PAGE
2-4-102.1 Replace Main Rotor Pylon Sliding Cover Latches	2-4-102-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit

Materials/Parts

- Corrosion Compound, Item 108, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D

- Paint, Polyurethane, Item 222, Appendix D

- Scouring Pad, Item 270, Appendix D

References

- Appendix D
- PARA 2-6-5

2-4-102.1 Replace Main Rotor Pylon Sliding Cover Latches.

NOTE

Replacement of both main rotor sliding cover latches is the same.

2-4-102.1.1 Remove.

- Unlatch and slide main rotor pylon sliding cover forward a little.
- Remove screws, washers, and nuts securing latch to sliding cover (Figure 2-4-102-1, Detail A). Remove latch.

2-4-102.1.2 Inspect/Repair.

- Inspect latch for corrosion. If corrosion is found, remove as follows:
 - Loosen jamnut and remove latch hook from stud. Remove jamnut from stud (Figure 2-4-102-1, Detail B).

- Remove corrosion manually from latch, using scouring pad, Item 270, Appendix D. **PITS ON NONTHREADED AREAS OF LATCH SHALL NOT EXCEED 0.005-INCH AFTER REWORK.**
- Mix solution of 1 part corrosion compound, Item 108, Appendix D, to 1 part water.
- Apply compound solution to corroded areas of latch. Rinse thoroughly with clean water.
- Using epoxy primer coating kit, Item 135, Appendix D, touch up paint with two coats of epoxy primer and top coat of polyurethane paint, Item 222, Appendix D (PARA 2-6-5).
- Install jamnut on stud. Install latch hook on stud.
- Install and adjust latch on sliding cover.

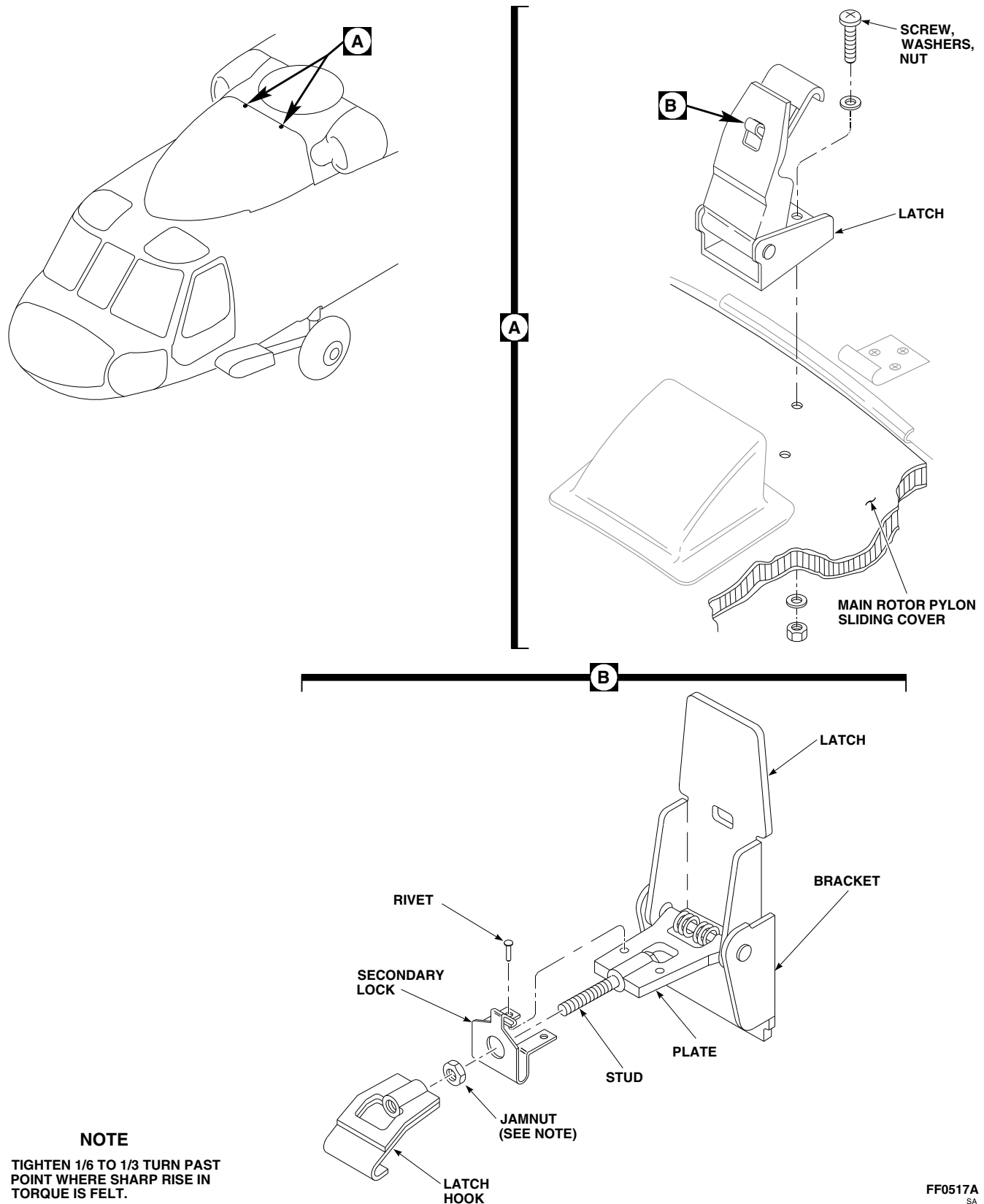


FIGURE 2-4-102-1. REPLACE MAIN ROTOR PYLON SLIDING COVER LATCHES

- b. Inspect secondary lock for damage and function. If necessary, replace secondary lock as follows:

- (1) Loosen jamnut and remove latch hook from stud. Remove jamnut from stud.
- (2) Remove rivets securing secondary lock to plate. Remove lock.
- (3) Position replacement secondary lock on plate and secure with rivets.
- (4) Install jamnut on stud. Install latch hook on threaded stud.
- (5) Install and adjust latch on main rotor pylon sliding cover.

2-4-102.1.3 Install.

- a. Position latch on main rotor pylon sliding cover and secure with screws, washers, and nuts (Figure 2-4-102-1, Detail A).

- b. Make sure area is clean and free of foreign material before closing up.

- c. Slide cover rearward and fasten latches.

- d. If latch does not lock firmly, adjust as follows:

- (1) Loosen jamnut on stud (Figure 2-4-102-1, Detail B).
- (2) Screw latch hook on or off stud until latch locks firmly.
- (3) **TIGHTEN JAMNUT $\frac{1}{6}$ TO $\frac{1}{3}$ TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

2-4-103 MAIN ROTOR PYLON SLIDING COVER VERTICAL ROLLERS.

PARAGRAPH BREAKDOWN

	PAGE
2-4-103.1 Replace Main Rotor Pylon Sliding Cover Vertical Rollers	2-4-103-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

References

- PARA 2-4-101

2-4-103.1 Replace Main Rotor Pylon Sliding Cover Vertical Rollers.

NOTE

Replacement of all main rotor pylon sliding cover vertical rollers is the same.

2-4-103.1.1 Remove.

- Remove main rotor pylon sliding cover (PARA 2-4-101).
- Place cover on padded surface.

- Remove bolt, washers, and bushing securing vertical roller to web (Figure 2-4-103-1, Detail A). Remove roller.

2-4-103.1.2 Install.

- Insert bushing in vertical roller (Figure 2-4-103-1, Detail A).
- Install large washer and roller (with bushing installed) on web and secure with bolt and small washer. **TORQUE BOLT TO 105 - 115 INCH-POUNDS.**
- Install main rotor pylon sliding cover (PARA 2-4-101).

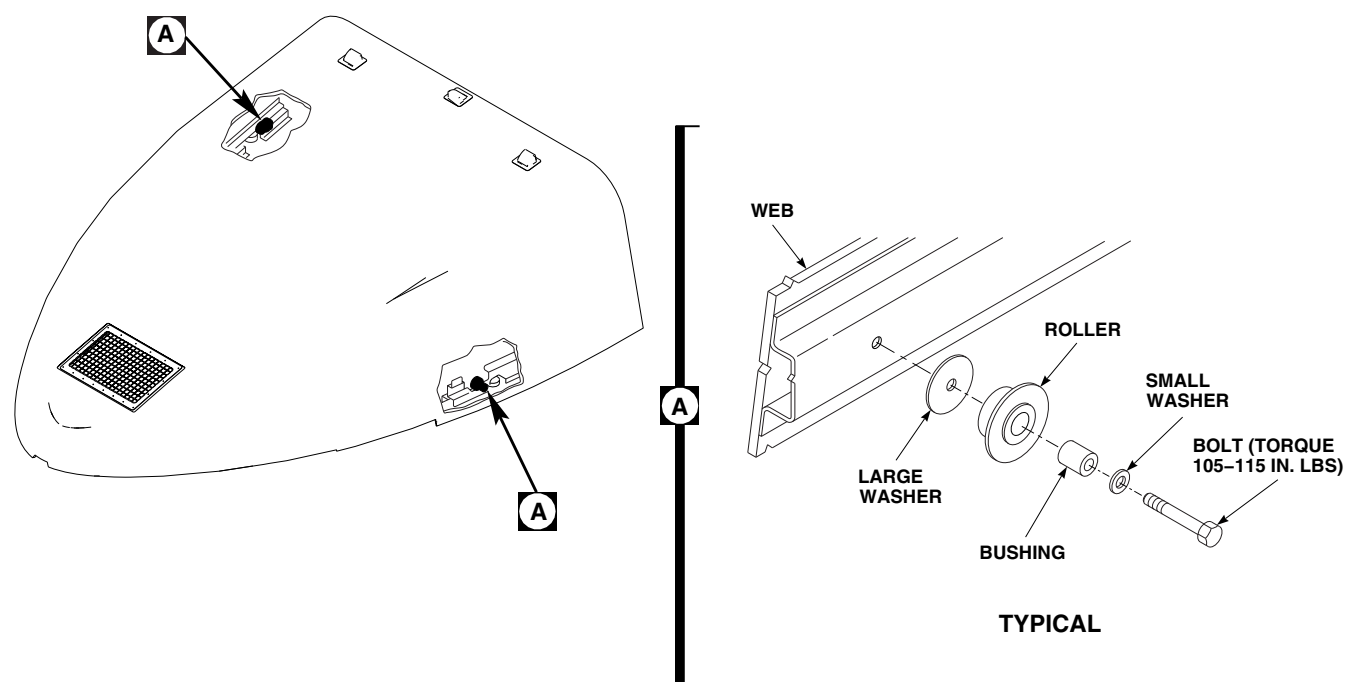
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FIGURE 2-4-103-1. REPLACE MAIN ROTOR PYLON SLIDING COVER VERTICAL ROLLERS

2-4-104 MAIN ROTOR PYLON SLIDING COVER HORIZONTAL ROLLERS.

PARAGRAPH BREAKDOWN

	PAGE
2-4-104.1 Replace Main Rotor Pylon Sliding Cover Horizontal Rollers	2-4-104-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

References

- PARA 2-4-101

2-4-104.1 Replace Main Rotor Pylon Sliding Cover Horizontal Rollers.

NOTE

Replacement of all main rotor pylon sliding cover horizontal rollers is the same.

2-4-104.1.1 Remove.

- Remove main rotor pylon sliding cover (PARA 2-4-101).
- Place cover on padded surface.

- Remove bolt, washers, spacer, bushing, and nut securing roller to web (Figure 2-4-104-1, Detail A). Remove roller.

2-4-104.1.2 Install.

- Insert bushing in roller (Figure 2-4-104-1, Detail A).
- Install washer, spacer, and roller (with bushing installed) on web and secure with bolt, washers, and nut. **TORQUE NUT TO 105 - 115 INCH-POUNDS.**
- Install main rotor pylon sliding cover (PARA 2-4-101).

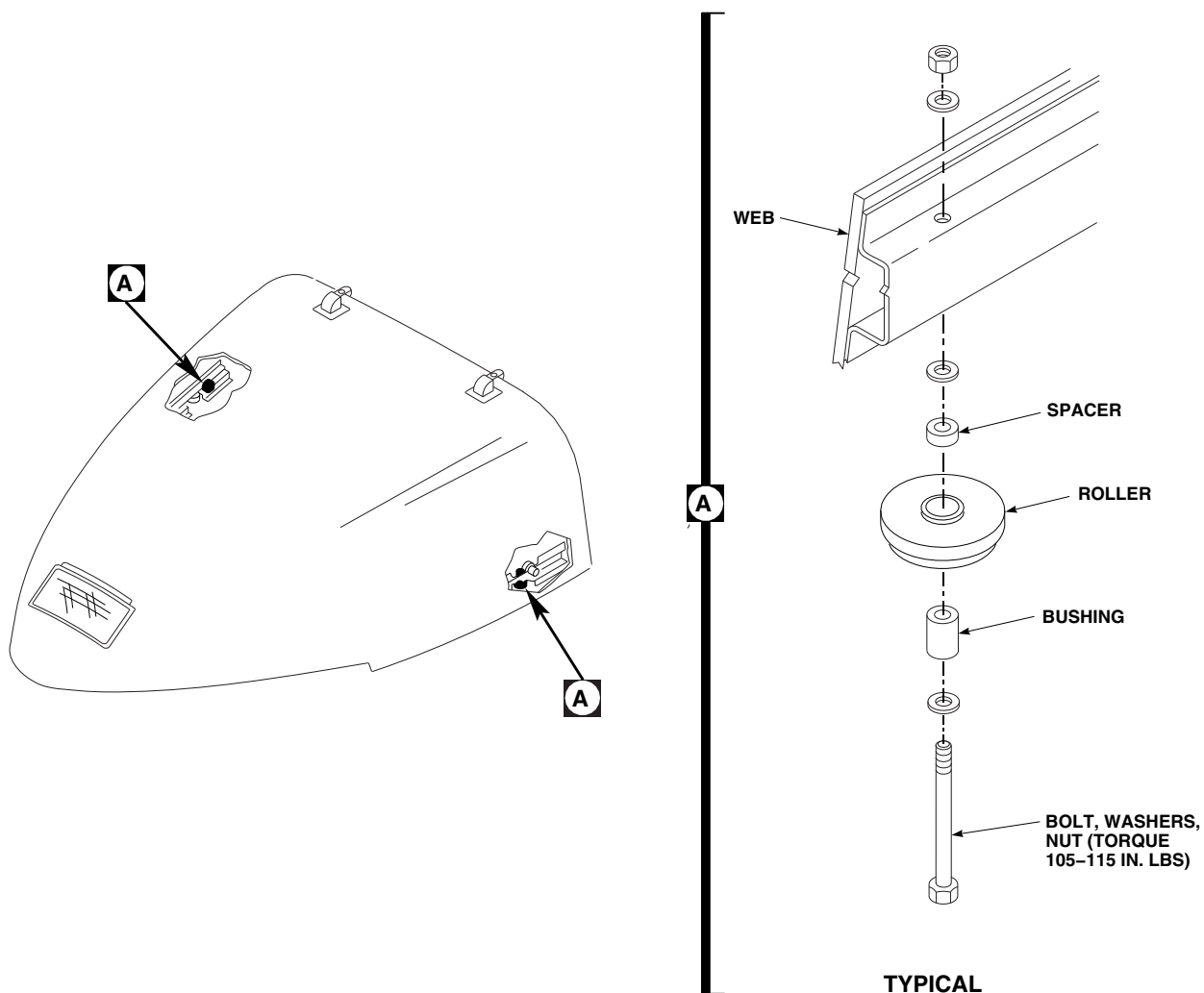
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FIGURE 2-4-104-1. REPLACE MAIN ROTOR PYLON SLIDING COVER HORIZONTAL ROLLERS

2-4-105 MAIN ROTOR PYLON WORK PLATFORM.

PARAGRAPH BREAKDOWN

	PAGE
2-4-105.1 Replace Main Rotor Pylon Work Platform	2-4-105-1

INITIAL SETUP

- Torque Wrench, 30 - 150 in. lbs

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit

2-4-105.1 Replace Main Rotor Pylon Work Platform.

2-4-105.1.1. Remove.

- Open main rotor pylon sliding cover.
- Remove quick-release pin from aft stop, place stop in remove position and install pin (Figure 2-4-105-1, Sheet 2, Detail C).
- Push handle down and disengage arms from lower tracks (Figure 2-4-105-1, Sheet 1, Detail B).
- Slide work platform forward and remove from lower track (Figure 2-4-105-1, Sheet 1, Detail A).
- Remove bolts and washers attaching left and right tee tracks to platform (Figure 2-4-105-1, Sheet 1, Detail B). Remove tracks.
- Remove bolt, washers, and nut attaching aft stop to bracket on platform (Figure 2-4-105-1, Sheet 2, Detail C).
- Remove rivets securing bracket to platform. Remove bracket.

2-4-105.1.2. Disassemble.

- Remove rivets securing arm and handle to lower tracks. Remove arm and handle (Figure 2-4-105-1, Sheet 1, Detail B).
- Remove springs from arms.

2-4-105.1.3. Assemble.

- Install springs on arms (Figure 2-4-105-1, Sheet 1, Detail B).
- Secure arm and handle on lower tracks with rivets.

2-4-105.1.4. Install.

- Install aft stop bracket on work platform with rivets (Figure 2-4-105-1, Sheet 2, Detail C).
- Install aft stop on bracket with bolt, washers, and nut. Install cotter pin. Place stop and quick-release pin in remove position and install quick-release pin.
- Install bolts and washers attaching left and right tee tracks to platform. **TORQUE BOLTS TO 47 - 53 INCH-POUNDS** (Figure 2-4-105-1, Sheet 1, Detail B).

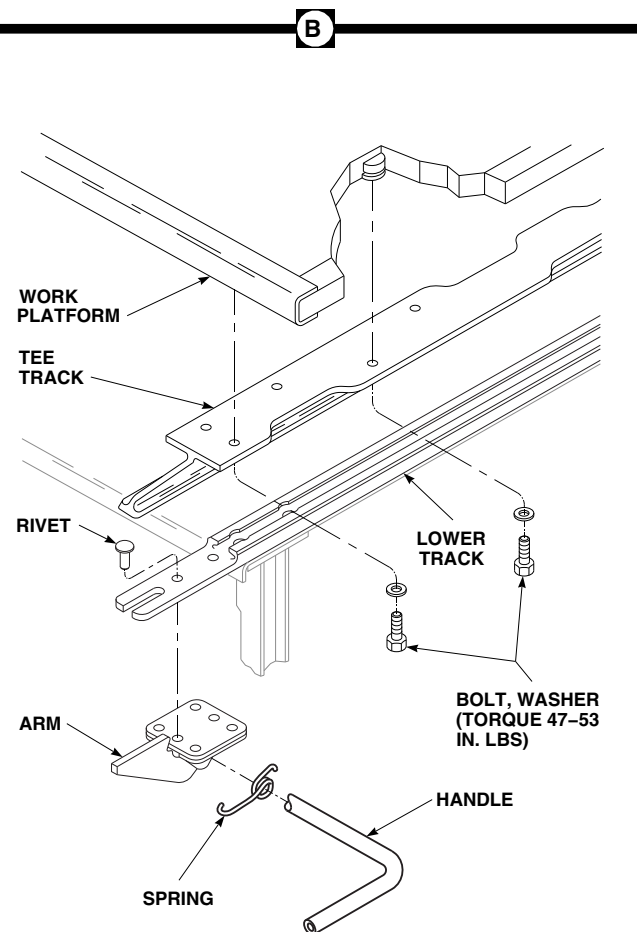
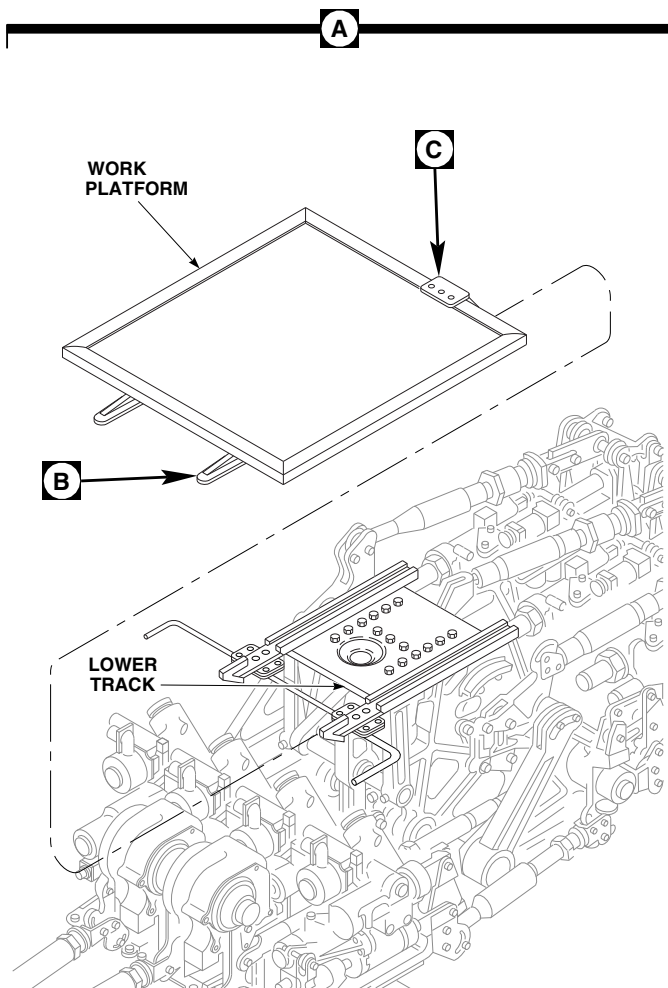
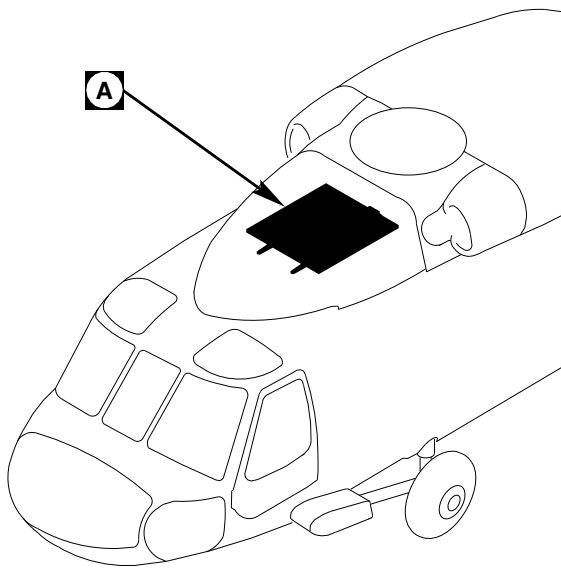
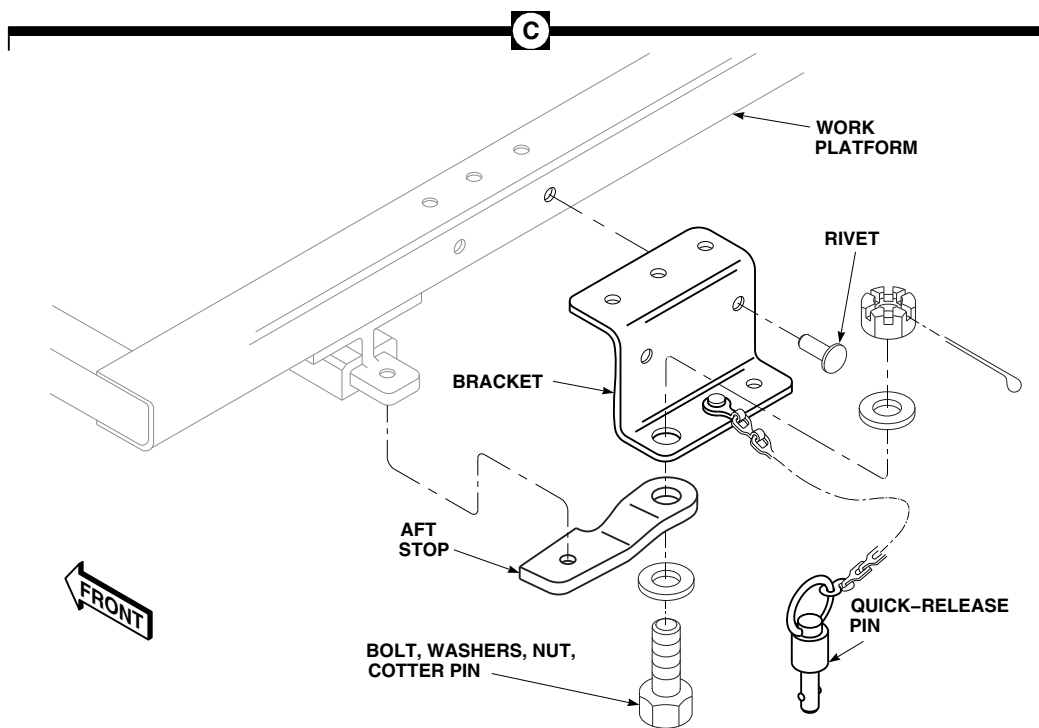
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FIGURE 2-4-105-1. REPLACE MAIN ROTOR PYLON WORK PLATFORM (SHEET 1 OF 2)

2-4-105-2

Change 2



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FIGURE 2-4-105-1. REPLACE MAIN ROTOR PYLON WORK PLATFORM (SHEET 2 OF 2)

- d. Place work platform on front end of lower track (Figure 2-4-105-1, Sheet 1, Detail A).
- e. Push handle down and slide platform back.
- f. Release handle to allow arms to engage in slots in lower tracks.
- g. Remove quick-release pin from remove position and install in stop position.
- h. Make sure area is clean and free of foreign material before closing sliding cover.
- i. Close main rotor pylon sliding cover.

2-4-106 MAIN ROTOR PYLON AIR INLET FAIRINGS.

PARAGRAPH BREAKDOWN

	PAGE
2-4-106.1 Replace Main Rotor Pylon Air Inlet Fairings	2-4-106-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Corrosion Preventive Compound, Item 115, Appendix D

- Lockwire, Item 194, Appendix D

References

- Appendix D
- PARA 2-4-101
- PARA 4-4-4
- PARA 10-4-38

2-4-106.1 Replace Main Rotor Pylon Air Inlet Fairings.

NOTE

Replacement of left and right main rotor pylon air inlet fairing is the same.

2-4-106.1.1. Remove.

- Remove main rotor pylon sliding cover (PARA 2-4-101).
- Remove engine air inlet (PARA 4-4-4).
- ESSS** Remove fixed fuel lines and disconnects (PARA 10-4-38).
- Remove bolt, screws, and washers securing inlet shroud fairing. Remove fairing (Figure 2-4-106-1, Detail A).
- Remove bolt, washers, and nut securing clevis of support tube to fairing beam fitting.

- Remove screws and washers securing edge of main rotor pylon air inlet fairing to upper fuselage fairing and main rotor pylon transmission cover.

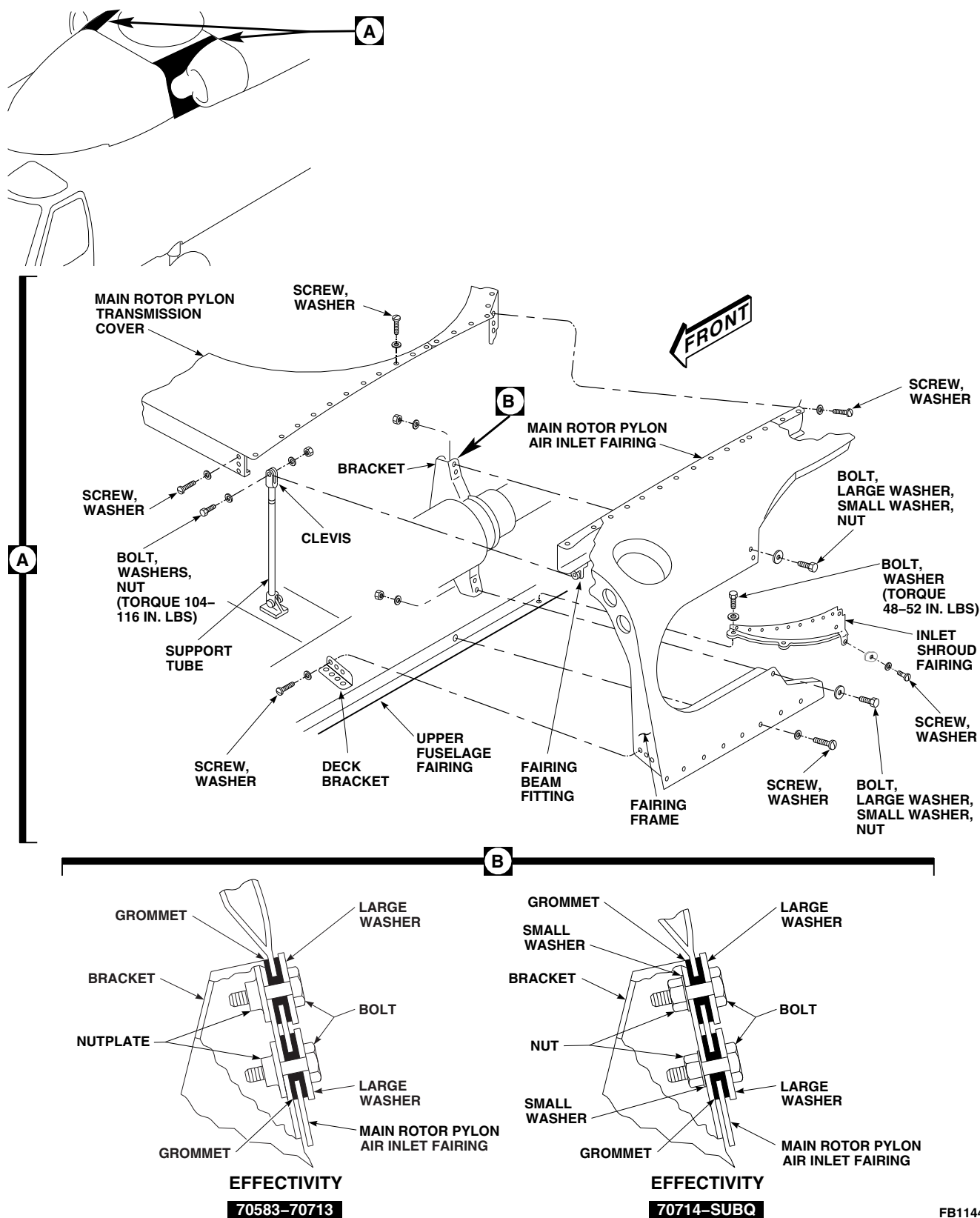
- 70583-70713** Remove bolts and washers securing air inlet fairing to brackets (Figure 2-4-106-1, Detail B).

- 70714-SUBQ** Remove bolts, large and small washers, and nuts securing air inlet fairing to brackets (Figure 2-4-106-1, Detail B).

- Remove screws and washers securing fairing frame and beam ends to deck bracket. Remove air inlet fairing (Figure 2-4-106-1, Detail A).

2-4-106.1.2. Install.

- Apply corrosion preventive compound, Item 115, Appendix D, to fairing hardware as hardware is installed.



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FIGURE 2-4-106-1. REPLACE MAIN ROTOR PYLON AIR INLET FAIRINGS

2-4-106-2

Change 3

- b. Place fairing frame against deck bracket and install screws and washers (Figure 2-4-106-1, Detail A).
- c. **70583-70713** Secure main rotor pylon air inlet fairing to brackets with bolts and washers (Figure 2-4-106-1, Detail B). Tighten bolts only until grommet starts to compress.■
- d. **70714-SUBQ** Secure main rotor pylon air inlet fairing to brackets with bolts, large and small washers, and nuts (Figure 2-4-106-1, Detail B). Tighten bolts only until grommet starts to compress.■
- e. Lockwire, Item 194, Appendix D, bolts.
- f. Secure edges of air inlet fairing to main rotor pylon transmission cover and upper fuselage fairing with screws and washers (Figure 2-4-106-1, Detail A).
- g. Position clevis of support tube to fairing beam fitting. If boltholes in clevis do not line up with boltholes in fitting, adjust clevis to align with fitting.
- h. Secure clevis to fairing beam fitting with bolt washers, and nut. **TORQUE NUT TO 104 - 116 INCH-POUNDS.**
- i. Install inlet shroud fairing with screws, washers, and bolt. **TORQUE BOLT TO 48 - 52 INCH-POUNDS.**
- j. Install engine air inlet (PARA 4-4-4).
- k. **ESSS** Install fixed fuel lines and disconnects (PARA 10-4-38).■
- l. Install main rotor pylon sliding cover (PARA 2-4-101).

2-4-107 MAIN ROTOR PYLON TRANSMISSION COVER.

PARAGRAPH BREAKDOWN

	PAGE
2-4-107.1 Replace Main Rotor Pylon Transmission Cover	2-4-107-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

References

- PARA 2-4-102

2-4-107.1 Replace Main Rotor Pylon Trans-
mission Cover.

NOTE

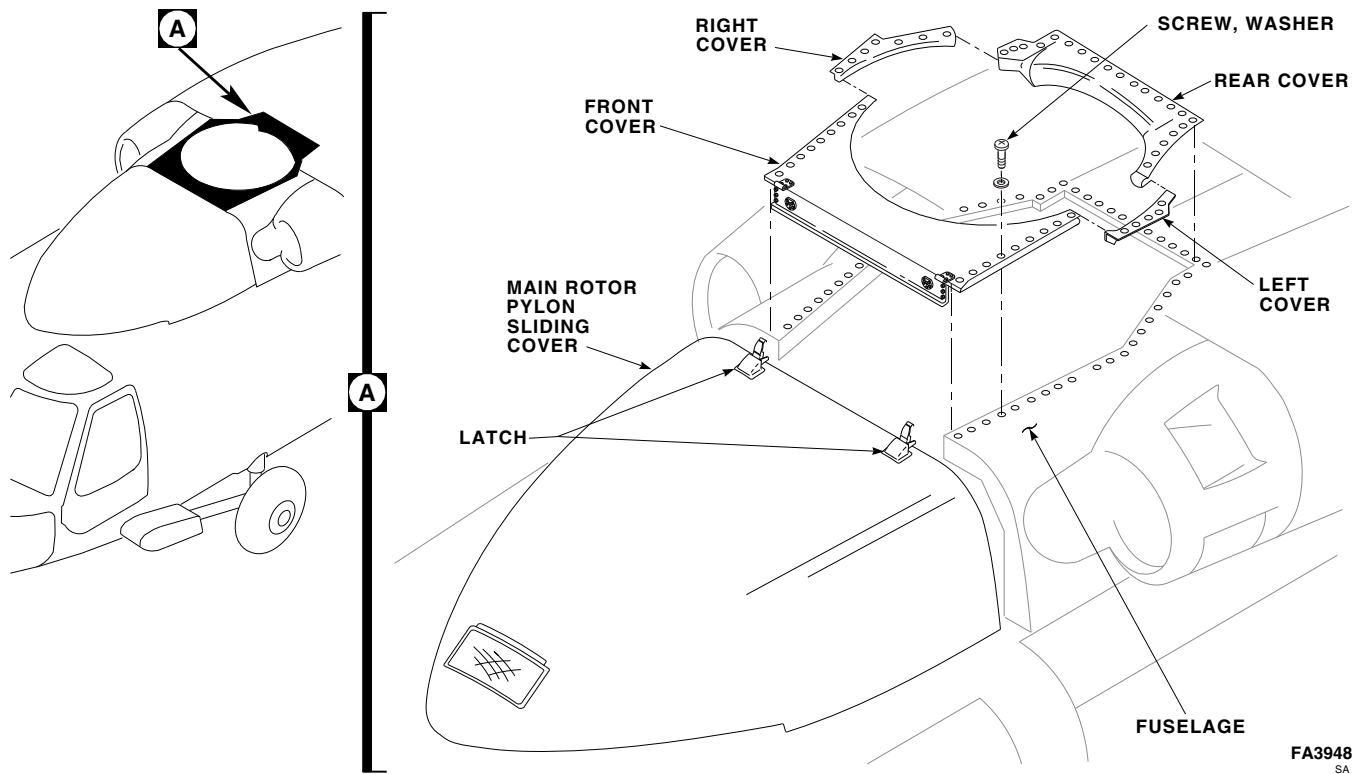
Main rotor pylon transmission cover consists of four covers. Replacement of all covers is the same, except as noted.

2-4-107.1.1. Remove.

- If replacing front cover, open main rotor pylon sliding cover enough to fully disengage latches from front cover (Figure 2-4-107-1, Detail A).
- Remove screws and washers securing cover to fuselage. Remove cover.

2-4-107.1.2. Install.

- Make sure area is clean and free of foreign material.
- Position replacement cover on fuselage and install attaching screws and washers (Figure 2-4-107-1, Detail A).
- If front cover was replaced, perform the following:
 - Check latches on main rotor pylon sliding cover for proper operation. If necessary, adjust latches (PARA 2-4-102).
 - Close main rotor pylon sliding cover.

FA3948
SA**FIGURE 2-4-107-1. REPLACE MAIN ROTOR PYLON TRANSMISSION COVER**

2-4-108 APU ACCESS DOORS.

PARAGRAPH BREAKDOWN

	PAGE
2-4-108.1 Replace APU Access Doors	2-4-108-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Medium Cut File
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Adhesive, Item 37, Appendix D
- Chafing Strip, Locally-Made, Figure H-57, Appendix H
- Cheesecloth, Item 90, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D

- Pylon Door Chafing Strip, Locally-Made, Figure H-28, Appendix H
- Rubber Sheet, 0.062-inch thick, Item 268, Appendix D
- Scouring Pad, Item 270, Appendix D

References

- Appendix D
- Appendix H
- PARA 2-4-110
- PARA 2-6-3
- PARA 2-6-5
- PARA 2-6-29
- PARA 2-6-38

2-4-108.1 Replace APU Access Doors.

NOTE

Replacement of right and left APU access door is the same, except as noted.

2-4-108.1.1. Remove.

- On left APU compartment access door, release catches (Figure 2-4-108-1, Detail A).
- Squeeze hinge-release pins together at each hinge (Figure 2-4-108-1, Detail B). Remove access door.

2-4-108.1.2. Inspect/Repair.

- Visually inspect hinges, catches, and catch fittings for cracks, wear, corrosion, and security. Proceed as follows:
 - NO CRACKS ALLOWED.** If crack is detected, replace cracked item (PARA 2-4-110, PARA 2-6-29, or PARA 2-6-38), or APU access door, as required.
 - If corrosion is detected, remove corrosion and blend out wear, using scouring pad, Item 270, Appendix D. **MAXIMUM**

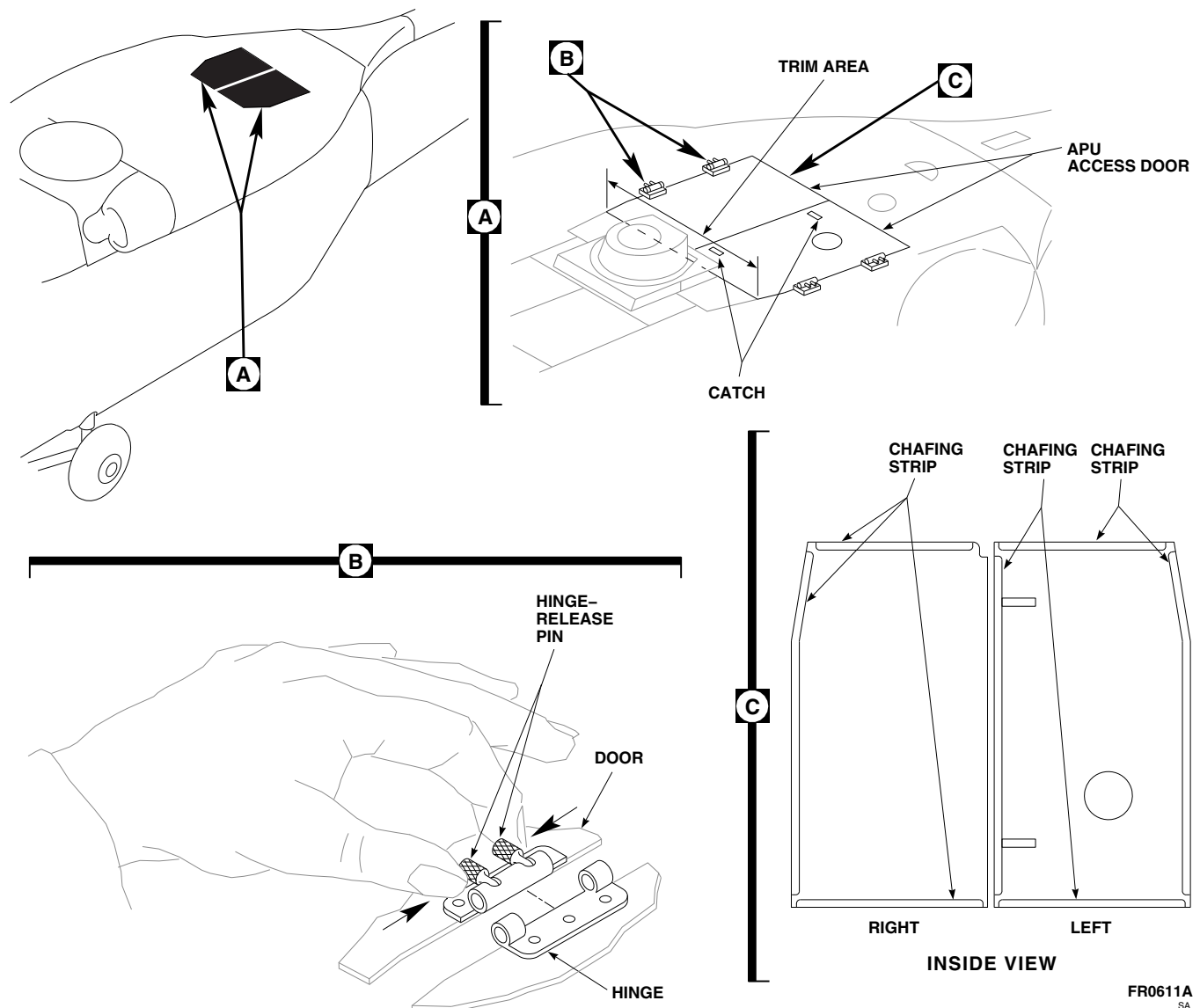


FIGURE 2-4-108-1. REPLACE APU ACCESS DOORS

DEPTH OF CLEANUP ALLOWED IS 10 % OF ORIGINAL MATERIAL THICKNESS. If limit is exceeded, replace part.

b. If chafing strip(s) require replacement, perform the following:

- (1) Remove chafing strip and old adhesive, using nonmetallic scraper and cheesecloth, Item 90, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D (Figure 2-4-108-1, Detail C).

- (2) Make replacement strip from 0.062-inch thick rubber sheet, Item 268, Appendix D, to specifications in Figures H-28 and H-57, Appendix H.

- (3) Lightly abrade bonding surface of strip, using scouring pad, Item 270, Appendix D.

- (4) Thoroughly clean bonding surfaces of chafing strip and door, using cheesecloth, Item 90, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D.

- (5) Apply even coat of adhesive, Item 37, Appendix D, to both bonding surfaces.
 - (6) Let adhesive dry for about 10 minutes until tacky.
 - (7) Install chafing strip on door and press firmly in place.
- c. Touch up paint as required (PARA 2-6-5).
- (1) Using medium cut file, trim contact area on door to a maximum of 1/4-inch.
 - (2) If trimmed area does not extend into honeycomb core, touch up paint as required (PARA 2-6-5). If trimmed area extends into honeycomb core, repair core (PARA 2-6-3).
- d. Check door for fit. If door edge binds against frames, trim area (PARA 2-6-3).
- e. Make sure area is clean and free of foreign material.

2-4-108.1.3. Install.

- a. Place door over compartment. Squeeze hinge-release pins together at each hinge, put hinge halves together, and release pins (Figure 2-4-108-1, Detail A and Detail B).
- b. If right door was installed, close door.
- c. Check IR jammer wire bundle for chafing by door. If wire bundle is chafed by door, perform the following:
 - f. Close left APU compartment access door and fasten catches (Figure 2-4-108-1, Detail A).

NOTE

Installed access doors may be warped 0.75-inch above surface of surrounding pylon. Hand pressure up to 16 pounds may be used to hold door closed for fastening.

2-4-109 APU SCREENS.

PARAGRAPH BREAKDOWN

	PAGE
2-4-109.1 Replace APU Screens	2-4-109-1

INITIAL SETUP

- Airframe Repairer’s Toolkit

Tools

- Aircraft Mechanic’s Toolkit

2-4-109.1 Replace APU Screens.

- b. Remove screen.

NOTE

Replacement of both APU screens is the same.

2-4-109.1.2. Install.

2-4-109.1.1. Remove.

- a. Remove screws and washers attaching APU screen to fuselage (Figure 2-4-109-1, Detail A).

- a. Make sure area is clean and free of foreign material before closing up.
- b. Place APU screen in position (Figure 2-4-109-1, Detail A).
- c. Install screws and washers attaching screen to fuselage.

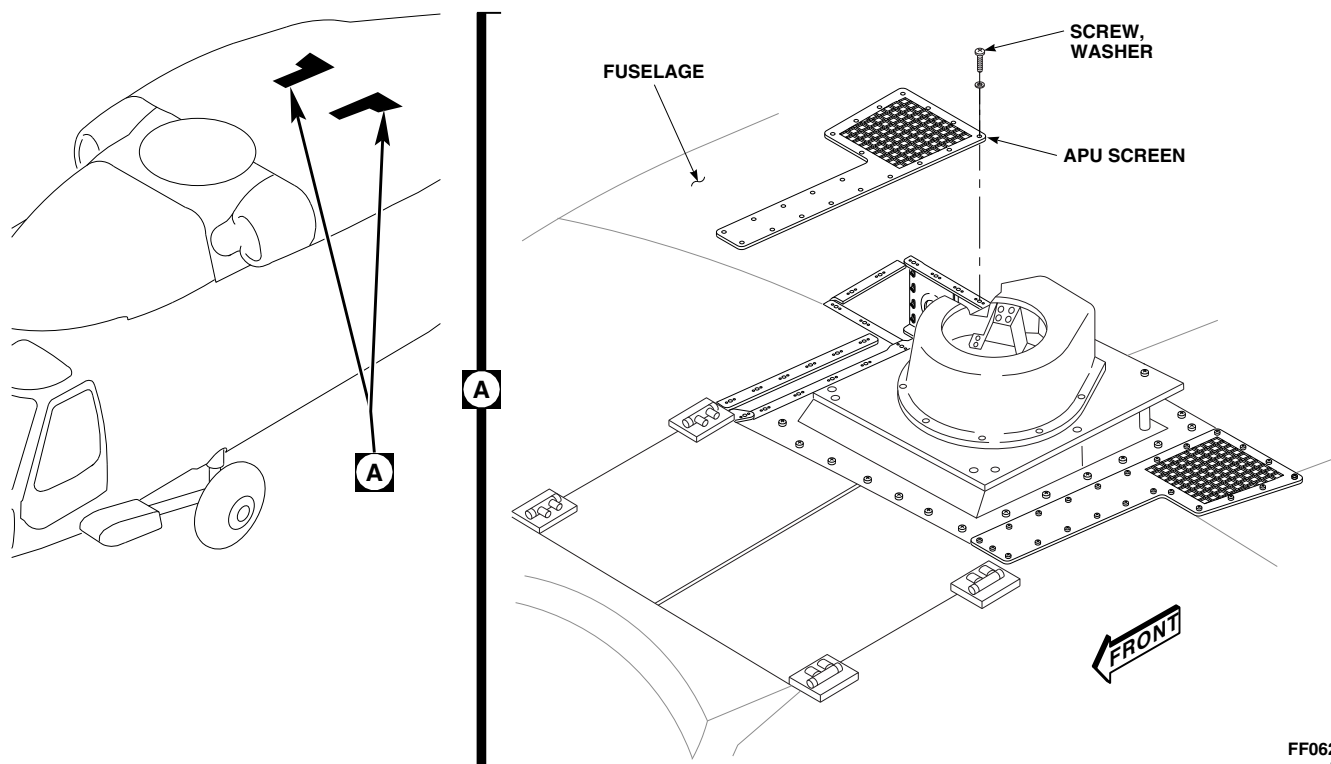


FIGURE 2-4-109-1. REPLACE APU SCREENS

2-4-110 APU ACCESS DOOR BEAM.

PARAGRAPH BREAKDOWN

	PAGE
2-4-110.1 Replace APU Access Door Beam	2-4-110-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit

Materials/Parts

- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D

- Sealing Compound, Item 283, Appendix D
- Sealing Primer, Item 306, Appendix D

References

- Appendix D
- PARA 2-4-108

2-4-110.1 Replace APU Access Door Beam.

2-4-110.1.1. Remove.

- Remove APU access doors (PARA 2-4-108).
- Remove screws and washers securing APU access door beam to support structure and remove beam (Figure 2-4-110-1, Detail A).

2-4-110.1.2. Install.

- Clean beam attaching screws and support structure nutplate threads, using machinery

towel, Item 211, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D.

- Apply sealing primer, Item 306, Appendix D, to beam attaching screws and nutplates. Allow primer to dry for five minutes.
- Apply sealing compound, Item 283, Appendix D, to beam attaching screws and nutplates.
- Secure beam to support structure with attaching screws and washers (Figure 2-4-110-1, Detail A).
- Install APU access doors (PARA 2-4-108).

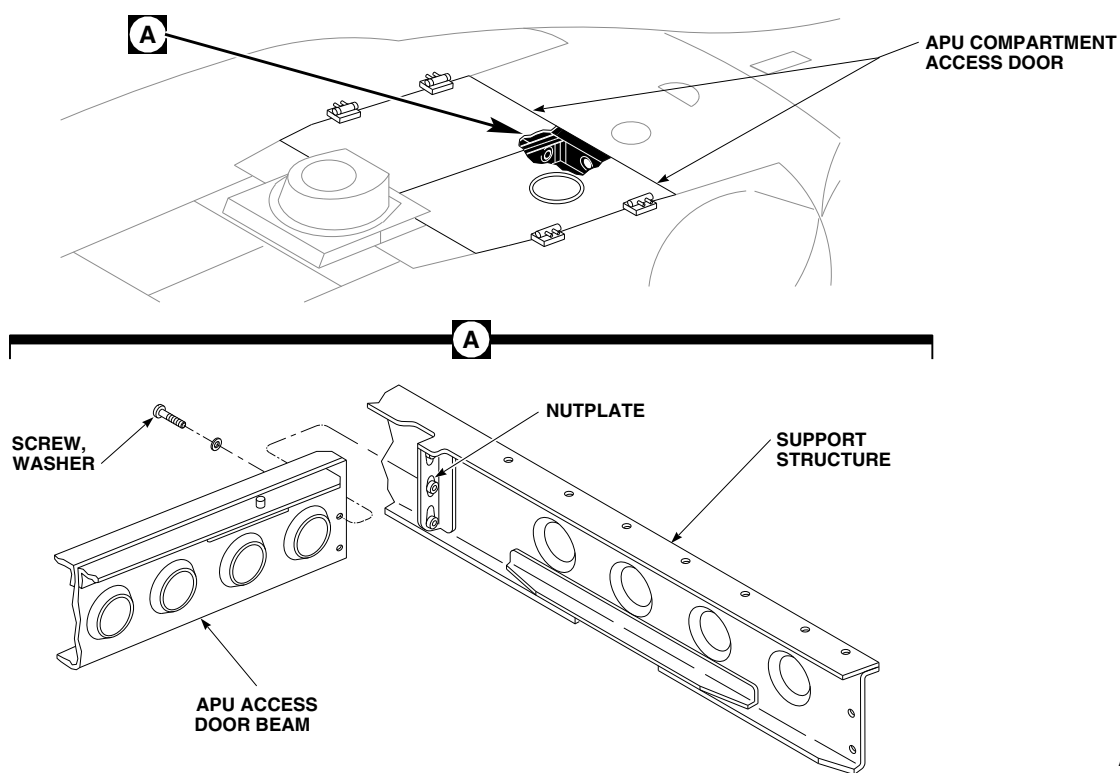


FIGURE 2-4-110-1. REPLACE APU ACCESS DOOR BEAM

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2-4-111 APU EXHAUST FAIRING.

PARAGRAPH BREAKDOWN

	PAGE
2-4-111.1 Replace APU Exhaust Fairing	2-4-111-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Drill Set

Materials/Parts

- Corrosion Preventive Compound, Item 115, Appendix D

References

- Appendix D
- PARA 2-6-3
- PARA 2-6-5

2-4-111.1 Replace APU Exhaust Fairing.

2-4-111.1.1. Remove.

- Remove screws and washers securing APU exhaust fairing to engine exhaust module, APU panel, and main rotor pylon rear fairing (Figure 2-4-111-1, Detail A).
- Remove APU exhaust fairing.

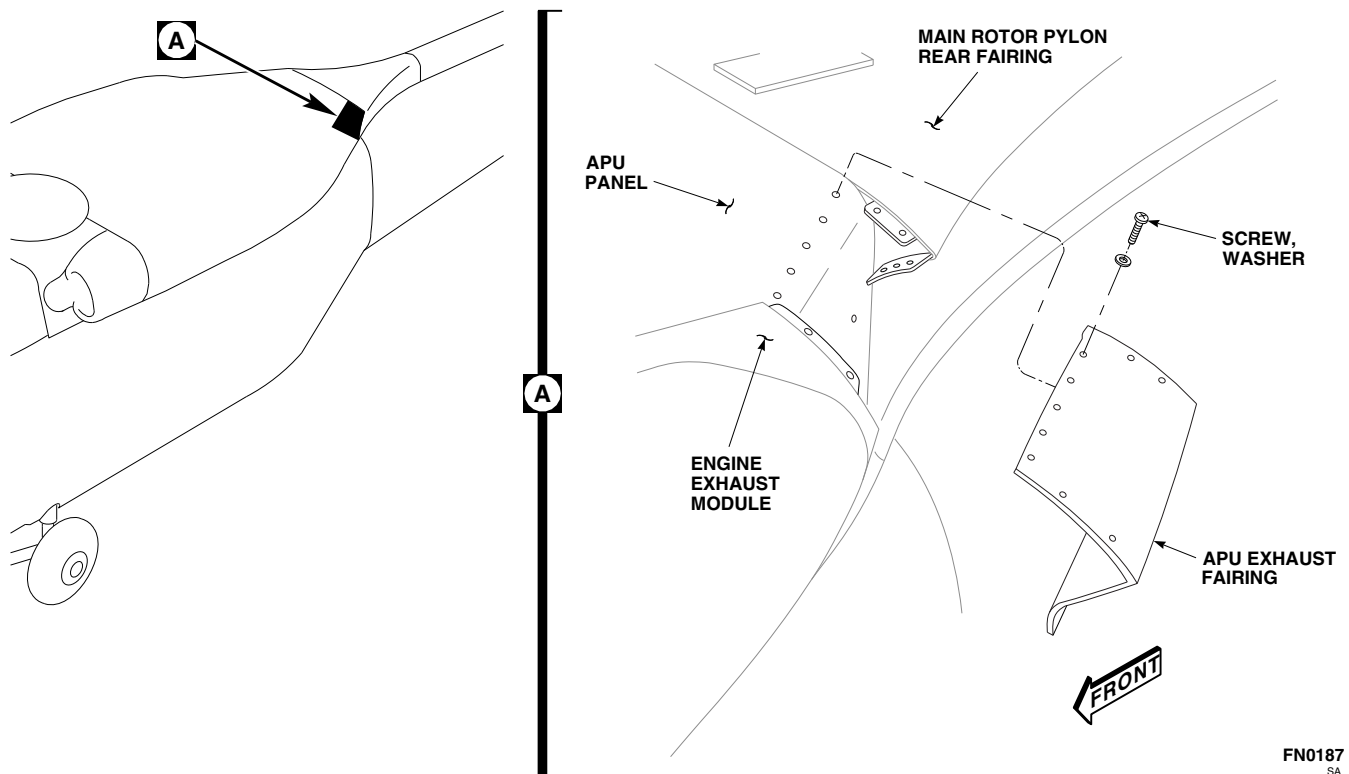
2-4-111.1.2. Install.

- If installing new fairing, trial-fit fairing with screws and washers (Figure 2-4-111-1, Detail A). Inspect fairing for mounting holes that do not line up with mounting holes in exhaust module, APU panel, and main rotor pylon rear fairing. If necessary, repair mismatched mounting hole(s) as follows:
 - Mark mounting hole(s) in fairing that do not line up with mounting hole(s) in module, panel, and rear fairing. Mark fair-

ing with correct location of mounting hole(s).

- Remove screws and washers securing fairing and remove fairing.
- Fill in hole(s) in fairing as marked (PARA 2-6-3).
- Using drill, drill new mounting hole(s) in fairing as marked.
- Touch up paint on fairing as required (PARA 2-6-5).

- Make sure area is clean and free of foreign material before closing up.
- Position fairing on exhaust module, APU panel, and main rotor pylon rear fairing. Line up mounting holes in fairing with module, panel, and rear fairing (Figure 2-4-111-1, Detail A).

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SA**FIGURE 2-4-111-1. REPLACE APU EXHAUST FAIRING**

- d. Apply corrosion preventive compound, Item 115, Appendix D, to fairing attachment hardware.
- e. Install screws and washers securing fairing to exhaust module, APU panel, and main rotor pylon rear fairing.

2-4-112 APU PANEL.

PARAGRAPH BREAKDOWN

	PAGE
2-4-112.1 Replace APU Panel	2-4-112-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Drill Set

Materials/Parts

- Corrosion Preventive Compound, Item 115, Appendix D
- Protective Caps
- Tags

References

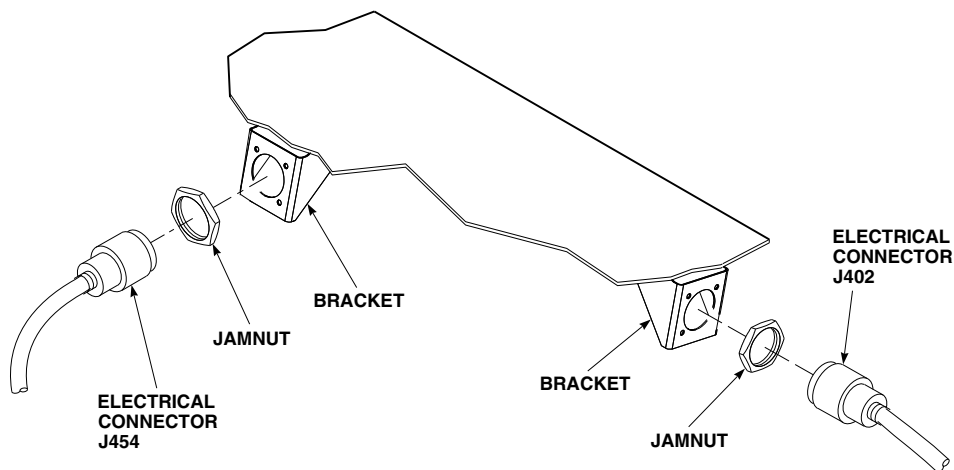
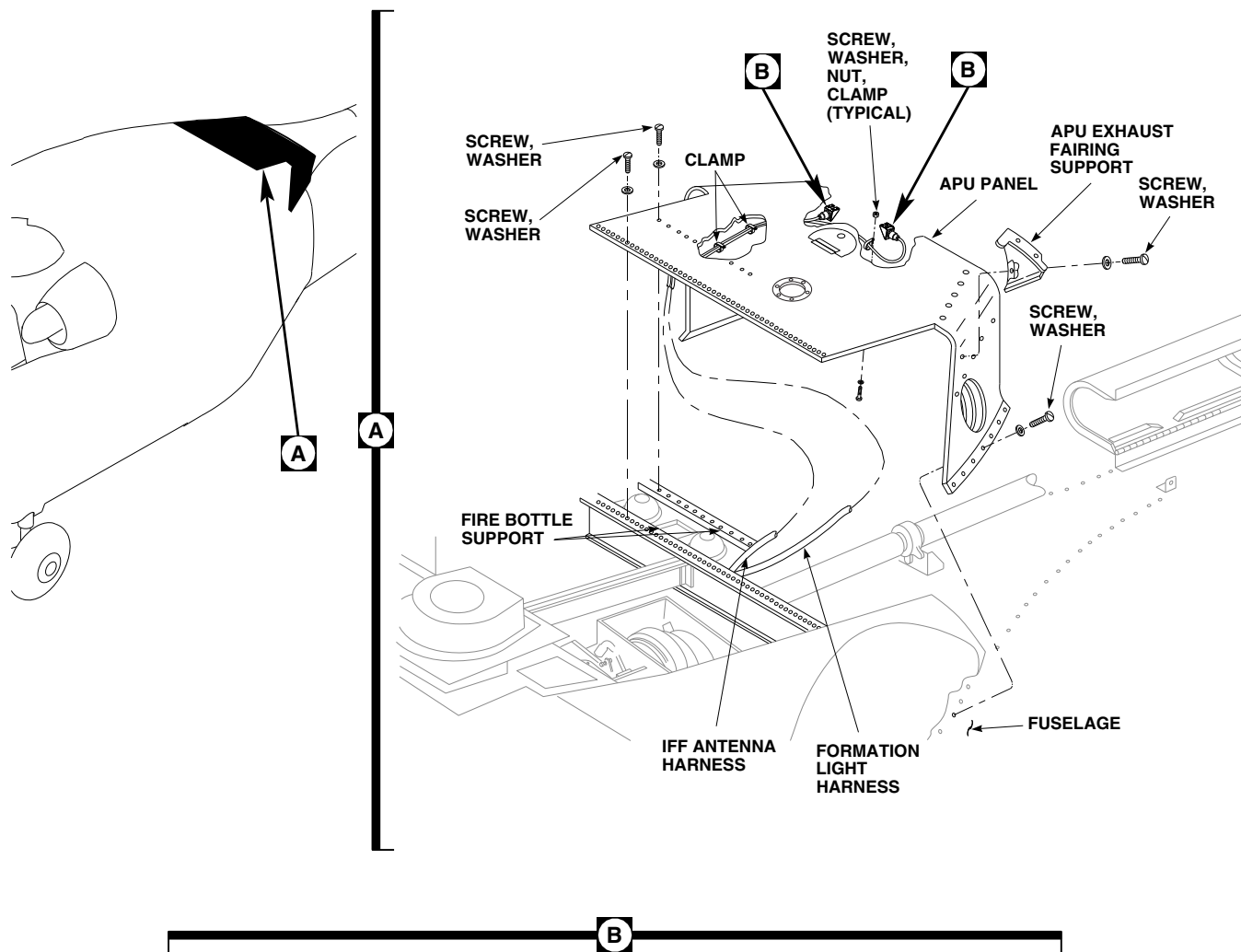
- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-108
- PARA 2-4-111
- PARA 2-4-116
- PARA 2-6-3
- PARA 15-4-2
- TM 11-70-23

2-4-112.1 Replace APU Panel.

2-4-112.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Remove APU exhaust fairing (PARA 2-4-111).
- Remove screws and washers securing APU exhaust fairing support to APU panel (Figure 2-4-112-1, Detail A). Remove fairing support.
- Remove APU access doors (PARA 2-4-108).
- Remove top IFF antenna (TM 11-70-23).
- Remove main rotor pylon rear fairing (PARA 2-4-116).

- Remove jamnuts securing electrical connectors, J402 and J454, to brackets (Figure 2-4-112-1, Detail B). Remove electrical connectors from brackets. Tag electrical connectors.
- Install protective caps on electrical connectors.
- Remove screws, washers, nuts, and clamps securing formation light and IFF antenna harnesses to underside of APU panel (Figure 2-4-112-1, Detail A).
- Remove screws and washers securing APU panel to fire bottle support.
- Remove APU exhaust duct (PARA 15-4-2).
- Remove remaining screws and washers securing APU panel to fuselage.



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FIGURE 2-4-112-1. REPLACE APU PANEL

- m. Remove APU panel.

2-4-112.1.2. Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. If installing new APU panel, perform the following:
 - (1) Install APU access doors (PARA 2-4-108).
 - (2) Trial-fit new panel on fuselage and fire bottle support with screws and washers. Inspect panel for proper fit. If necessary, trim panel as follows
 - (a) Mark areas on panel that bind against firewall or APU access doors.
 - (b) Remove screws and washers securing panel to fuselage and fire bottle support and remove panel.
 - (c) Trim marked areas (PARA 2-6-3).
 - (d) Repeat step b. (2) (a) through b. (2) (c) until proper fit is achieved.
 - (e) Remove screws and washers securing panel to fuselage and fire bottle support and remove panel.
 - (3) Install main rotor pylon rear fairing with two screws (PARA 2-4-116).
 - (4) Trial-fit panel on fuselage and fire bottle support with screws and washers.
 - (5) If required, trim panel as follows:
 - (a) Mark areas on panel that bind against main rotor pylon rear fairing.
 - (b) Remove screws and washers securing panel to fuselage and fire bottle support and remove panel.
 - (c) Trim marked areas (PARA 2-6-3).
 - (d) Repeat step b. (4) through step b. (5) (c) until proper fit is achieved.
 - (6) With panel installed, mark all holes that do not line up with attachment holes in fuselage or fire bottle support.
 - (7) Remove screws and washers securing panel to fuselage and fire bottle support and remove panel.
 - (8) Remove main rotor pylon rear fairing (PARA 2-4-116).
 - (9) Fill in marked holes on panel (PARA 2-6-3).
 - (10) Install panel on fuselage and fire bottle support with screws and washers. Mark and pilot-drill fastener holes where required.
 - (11) Remove screws and washers securing panel to fuselage and fire bottle support and remove panel.
 - (12) Enlarge pilot holes for fastening screws.
 - c. Apply corrosion preventive compound, Item 115, Appendix D, to APU panel attachment hardware.
 - d. Make sure area is clean and free of foreign material before closing up.
 - e. Install APU panel on fuselage and fire bottle support with screws and washers (Figure 2-4-112-1, Detail A).
 - f. Install APU exhaust duct (PARA 15-4-2).
 - g. Install formation light and IFF antenna harnesses to underside of APU panel with screws, washers, nuts, and clamps.
 - h. Remove protective caps from electrical connectors
 - i. Inspect and treat electrical connectors (PARA 1-7-20).
 - j. Install electrical connectors, J402 and J454, in brackets and secure with jamnuts (Figure 2-4-112-1, Detail B). Remove tags.

- k. Install main rotor pylon rear fairing (PARA 2-4-116).
- l. Install top IFF antenna (TM 11-70-23).
- m. If new APU panel was not installed, install APU access doors (PARA 2-4-108).
- n. Apply corrosion preventive compound, Item 115, Appendix D, to APU panel attachment hardware.
- o. Install APU exhaust fairing support to APU panel with screws and washers (Figure 2-4-112-1, Detail A).
- p. Install APU exhaust fairing (PARA 2-4-111).

2-4-113 OIL COOLER COMPARTMENT ACCESS DOORS.

PARAGRAPH BREAKDOWN

	PAGE
2-4-113.1 Replace Oil Cooler Compartment Access Doors	2-4-113-0

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Adhesive, Item 37, Appendix D
- Chafing Strip, Locally-Made, Figure H-57, Appendix H
- Cheesecloth, Item 90, Appendix D

- Methyl Ethyl Ketone, Item 217, Appendix D
- Pylon Door Chafing Strip, Locally-Made, Figure H-28, Appendix H
- Scouring Pad, Item 270, Appendix D

References

- Appendix D
- Appendix H
- PARA 2-3-10
- PARA 2-6-3
- PARA 2-6-5

2-4-113.1 Replace Oil Cooler Compartment Access Doors.

NOTE

Replacement of right and left oil cooler compartment access door is the same.

2-4-113.1.1. Remove.

- Unfasten oil cooler compartment access door.
- Squeeze hinge-release pins together at each hinge (Figure 2-4-113-1, Detail A). Remove access door from helicopter.
- Inspect oil cooler compartment access door (PARA 2-3-10).

2-4-113.1.2. Inspect/Repair.

- Visually inspect hinges, catches, and catch fittings for cracks, wear, corrosion, and security. **NO CRACKS ALLOWED.** If crack is detected, replace cracked item or door assembly, as required.
- Remove corrosion and blend out wear, using scouring pad, Item 270, Appendix D. **MAXIMUM DEPTH OF CLEANUP ALLOWED IS 10% OF ORIGINAL MATERIAL THICKNESS.** Replace part if limit is exceeded.
- If chafing strip(s) require replacement, perform the following:

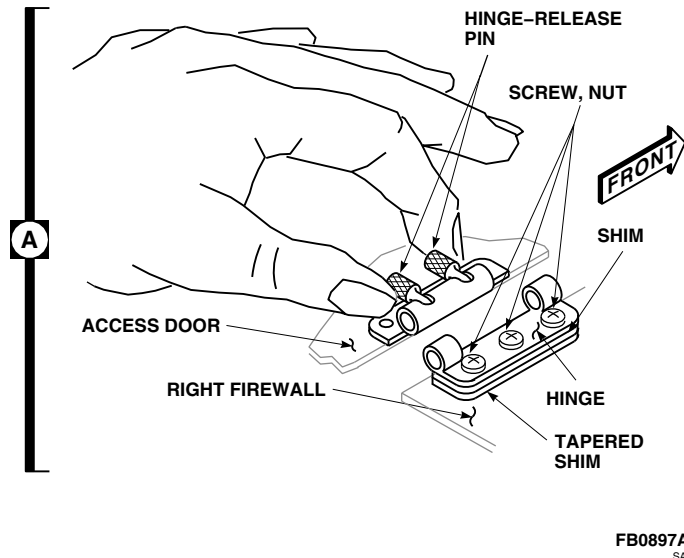
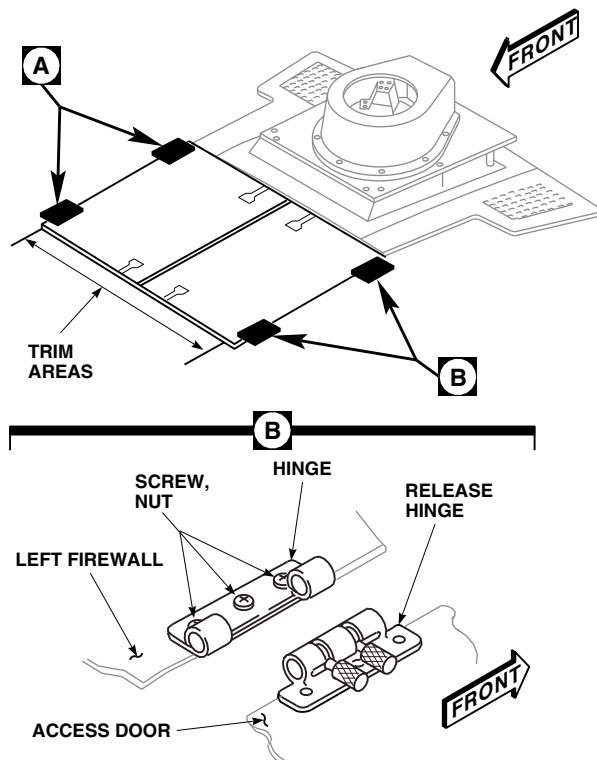
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FIGURE 2-4-113-1. REPLACE OIL COOLER COMPARTMENT ACCESS DOORS

- (1) Remove chafing strip and old adhesive, using nonmetallic scraper and cheesecloth, Item 90, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D.
 - (2) Locally make strip (Figure H-28 and H-57, Appendix H).
 - (3) Lightly abrade bonding surface of strip, using scouring pad, Item 270, Appendix D.
 - (4) Thoroughly clean bonding surfaces of chafing strip and door, using cheesecloth, Item 90, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D.
 - (5) Apply even coat of adhesive, Item 37, Appendix D, to both bonding surfaces.
 - (6) Let adhesive dry for about 10 minutes until tacky.
 - (7) Install chafing strip on door and press firmly in place.
- d. Touch up paint as required (PARA 2-6-5).

2-4-113.1.3. Install.

- a. Place door over compartment. Squeeze hinge-release pins together at each hinge, put hinge halves together, and release pins (Figure 2-4-113-1, Detail A).
- b. Fit doors between firewalls. If necessary, loosen firewall hinge half attaching screws and nuts. Adjust to doors, then tighten screws and nuts (Figure 2-4-113-1, Detail B).
- c. Check door for fit. If door edge binds against frame, trim area shown in Figure 2-4-113-1. Trim edge (PARA 2-6-3).
- d. Make sure area is clean and free of foreign material.

2-4-113-2

NOTE

Installed access doors may be warped 0.30-inch above surface of surrounding pylon. Hand pressure up to 16 pounds

may be used to hold door closed for fastening.

- e. Close and fasten oil cooler compartment access door.

2-4-114 IR JAMMER FAIRING.

PARAGRAPH BREAKDOWN

	PAGE
2-4-114.1 Replace IR Jammer Fairing	2-4-114-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit

Materials/Parts

- Corrosion Preventive Compound, Item 115, Appendix D

References

- Appendix D
- PARA 1-6-16

2-4-114.1 Replace IR Jammer Fairing.

2-4-114.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Remove screws and washers securing IR jammer fairing to IR jammer panel (Figure 2-4-114-1, Detail A).

NOTE

Make sure IR jammer transmitter wiring is clear of fairing when removing.

- Carefully remove IR jammer fairing.

2-4-114.1.2 Install.

- Make sure area is clean and free of foreign material.
- Position IR jammer fairing over IR jammer transmitter mounting brackets. Make sure transmitter wiring is clear of fairing and is through slot in fairing (Figure 2-4-114-1, Detail A).
- Apply corrosion preventive compound, Item 115, Appendix D, to fairing attachment hardware.
- Install screws and washers securing fairing to IR jammer panel.

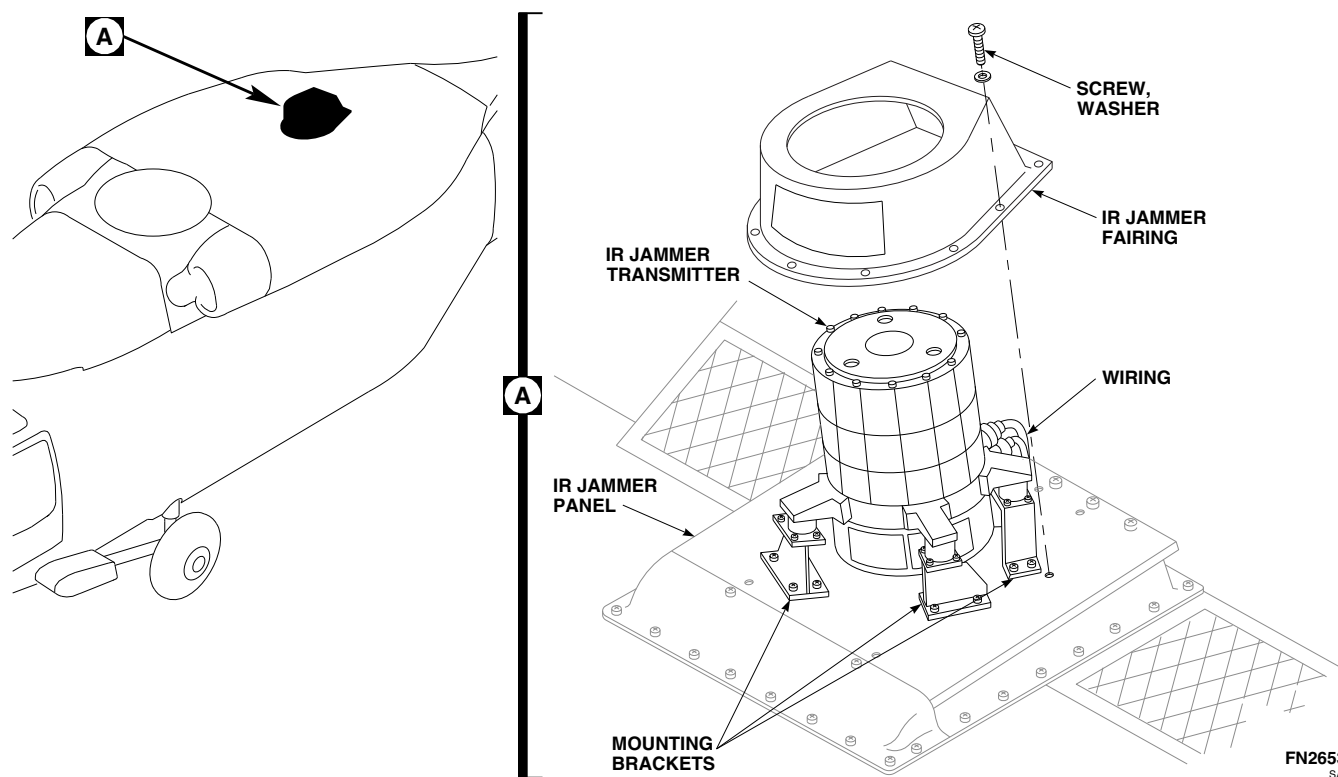


FIGURE 2-4-114-1. REPLACE IR JAMMER FAIRING

2-4-115 IR JAMMER PANEL.

PARAGRAPH BREAKDOWN

	PAGE
2-4-115.1 Replace IR Jammer Panel	2-4-115-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit

References

- PARA 2-4-114
- TM 11-70-23

2-4-115.1 Replace IR Jammer Panel.

2-4-115.1.1 Remove.

- If installed, remove IR jammer (TM 11-70-23).
- If IR jammer is not installed, remove IR jammer fairing (PARA 2-4-114).
- Remove screws, washers, and spacers attaching IR jammer panel to airframe. Remove panel from oil cooler support (Figure 2-4-115-1, Detail A).
- Remove screws, washers, and nuts attaching mount support brackets to panel.

2-4-115.1.2 Install.

- Install mount support brackets to IR jammer panel with screws, washers, and nuts (Figure 2-4-115-1, Detail A).

- Position panel on airframe (Figure 2-4-115-1, Detail A).
- Position spacers, with a washer between spacer and panel, and a washer between spacer and oil cooler support, under aft end of panel. Install screws and washers securing panel and spacers to airframe.
- Install remaining screws and washers attaching panel to support.
- If removed, install IR jammer (TM 11-70-23).
- If IR jammer is not to be installed, install IR jammer fairing (PARA 2-4-114).

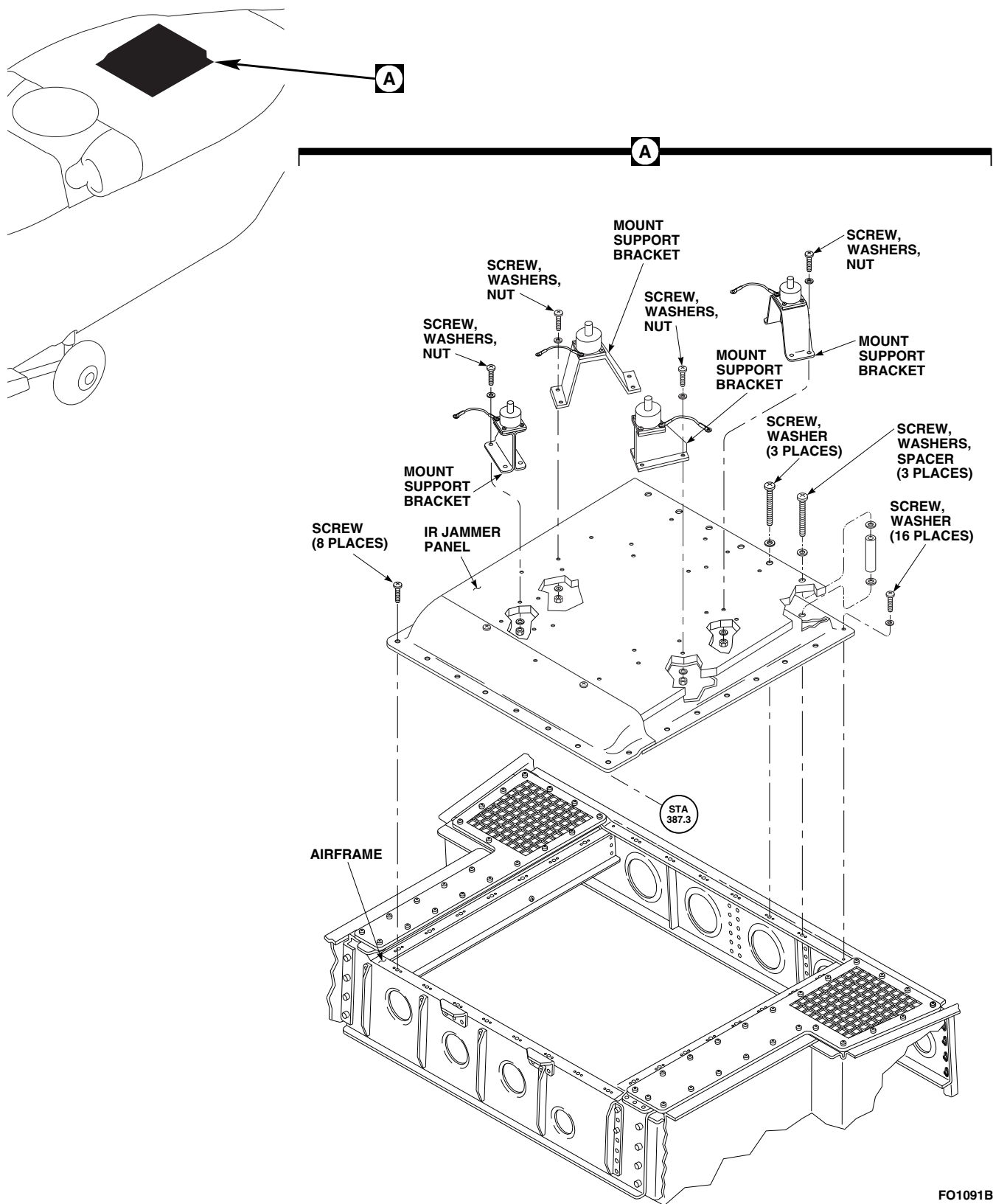
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FIGURE 2-4-115-1. REPLACE IR JAMMER PANEL

2-4-116 MAIN ROTOR PYLON REAR FAIRING.

PARAGRAPH BREAKDOWN

	PAGE
2-4-116.1 Replace Main Rotor Pylon Rear Fairing	2-4-116-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Drill Set

Materials/Parts

- Corrosion Preventive Compound, Item 115, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-6-3
- PARA 2-6-5
- PARA 9-2-12
- PARA 9-4-70
- PARA 9-4-71

2-4-116.1 Replace Main Rotor Pylon Rear Fairing.

2-4-116.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Open front tail rotor drive shaft cover.
- Open access panel on APU panel.
- Disconnect electrical connector, P402 and P454 from J402 and J454, on APU panel (Figure 2-4-116-1, Sheet 1, Detail B).
- Install protective caps and plugs on electrical connectors.

NOTE

- When removing screws and washers securing rear fairing to APU panel, remove only forward row of screws and washers. Do not remove rear row of screws and washers securing straps to top and vertical sides, inside the rear fairing.
- Remove screws and washers securing rear fairing to fuselage and APU panel. Remove fairing (Figure 2-4-116-1, Sheet 1, Detail A).
 - If new fairing is to be installed, remove main rotor pylon electro-luminescent formation light (PARA 9-4-70) and infrared formation light (PARA 9-4-71).

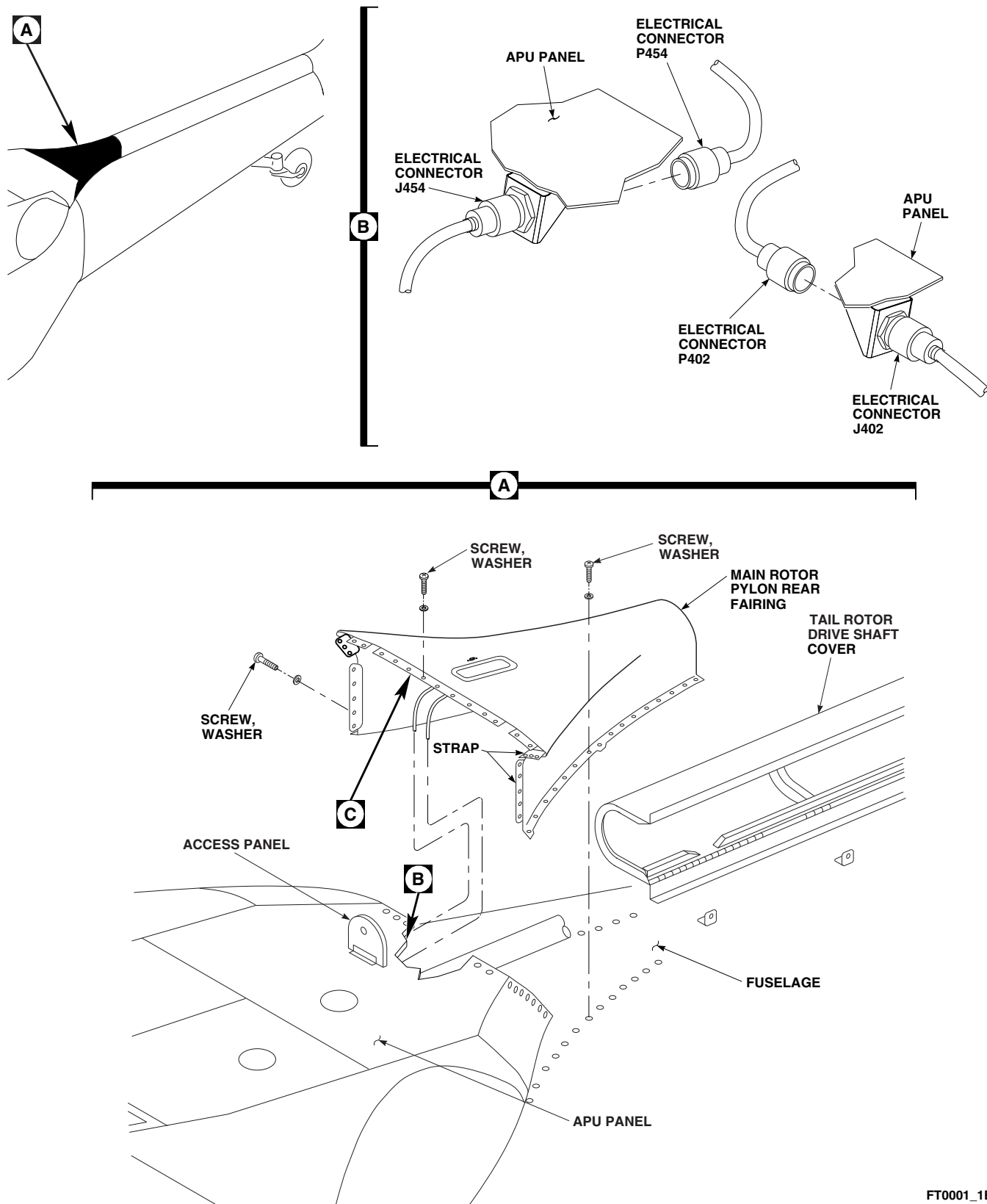
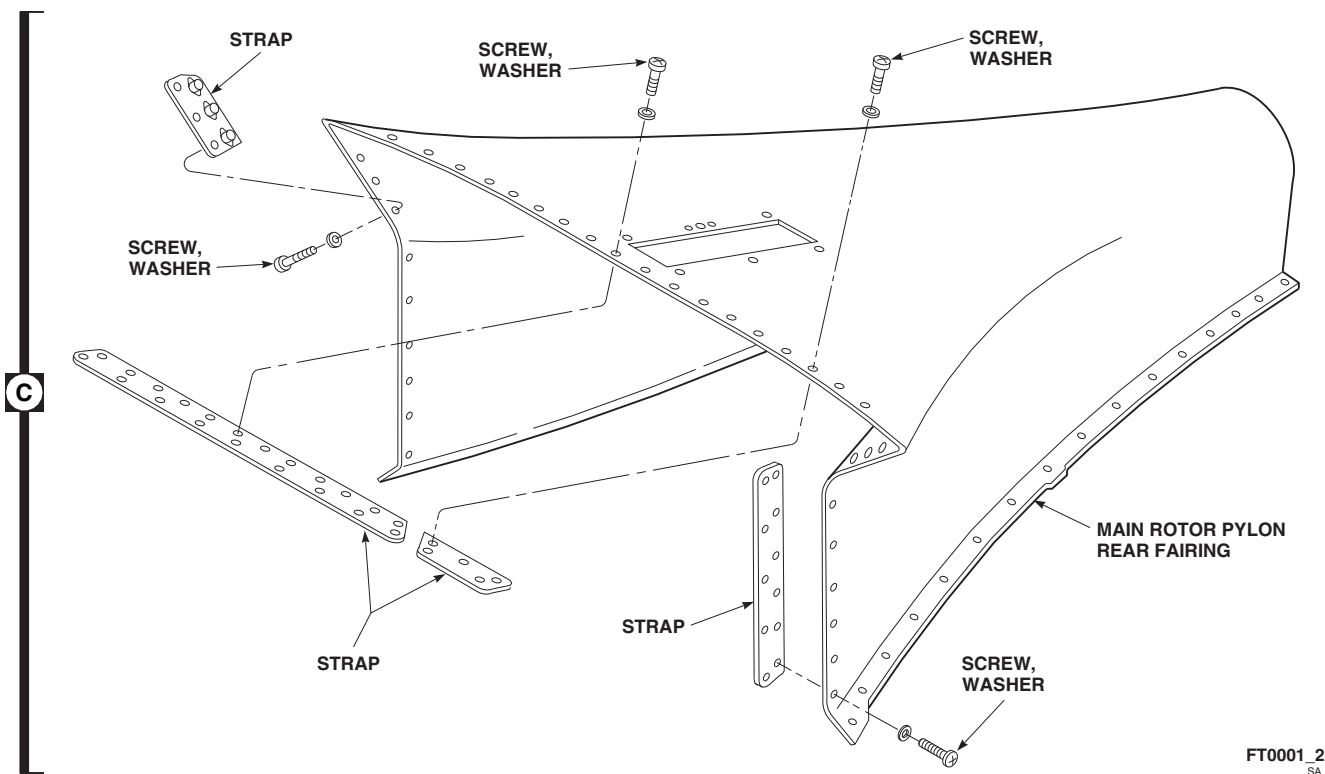
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FIGURE 2-4-116-1. REPLACE MAIN ROTOR PYLON REAR FAIRING (SHEET 1 OF 2)



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FIGURE 2-4-116-1. REPLACE MAIN ROTOR PYLON REAR FAIRING (SHEET 2 OF 2)

2-4-116.1.2. Install.

- a. If installing new fairing, trial-fit new fairing on fuselage with screws and washers (Figure 2-4-116-1, Sheet 1, Detail A). Inspect fairing for mounting holes that do not line up with nut-plates in fuselage. If necessary, repair mismatched mounting hole(s) as follows:

- (1) Mark mounting hole(s) in fairing that do not line up with mounting hole(s) in fuselage. Mark fairing with correct location of mounting hole(s).
- (2) Remove screws and washers securing fairing to fuselage and remove fairing.
- (3) Remove screws and washers securing straps to fairing and remove straps (Figure 2-4-116-1, Sheet 2, Detail C).
- (4) Fill in hole(s) in fairing as marked (PARA 2-6-3).
- (5) Using drill, drill new mounting hole(s) in fairing as marked.

- (6) Install all straps (removed from fairing) in their corresponding locations on APU panel with screws and washers.

NOTE

Secure fairing to fuselage only with screws and washers. Do not install screws and washers securing fairing to straps installed on APU panel.

- (7) Install fairing on fuselage and secure with screws and washers. Fairing should now install to fuselage with no mismatched holes (Figure 2-4-116-1, Sheet 1, Detail A).
- (8) With straps installed to APU panel and fairing installed on fuselage, inspect holes in straps that do not line up with mounting holes in fairing. If necessary, perform the following:
 - (a) Mark mounting hole(s) in fairing that do not line up with hole(s) in

- straps. Mark fairing with correct location of mounting hole(s).
- (b) Remove screws and washers securing fairing to fuselage and remove fairing (Figure 2-4-116-1, Sheet 1, Detail A).
 - (c) Fill in hole(s) in fairing as marked (PARA 2-6-3).
 - (d) Using drill, drill new mounting hole(s) in fairing as marked.
- (9) Touch up paint on fairing as required (PARA 2-6-5).
- (10) Remove screws and washers securing straps to APU panel. Remove straps.
- (11) Install straps in their corresponding locations on fairing with screws and washers (Figure 2-4-116-1, Sheet 2, Detail C).
- (12) Install main rotor pylon electroluminescent formation light (PARA 9-4-70) and main rotor pylon infrared formation light (PARA 9-4-71).
- b. Make sure area is clean and free of foreign material before closing up.
 - c. Position fairing on fuselage. Line up mounting holes in fairing with fuselage (Figure 2-4-116-1, Sheet 1, Detail A).
 - d. Apply corrosion preventive compound, Item 115, Appendix D, to fairing attachment hardware.
 - e. Install screws and washers securing fairing to fuselage and APU panel.
 - f. Remove protective caps and plugs from electrical connectors.
 - g. Inspect and treat electrical connectors (PARA 1-7-20).
 - h. Connect electrical connector, P402 and P454 to J402 and J454, on APU panel (Figure 2-4-116-1, Sheet 1, Detail B).
 - i. Close and fasten access panel on APU panel.
 - j. Close front tail rotor drive shaft cover.
 - k. Perform operational check of formation light (PARA 9-2-12).

2-4-117 YAW PUSHROD BOOT AND SHROUD.

PARAGRAPH BREAKDOWN

	PAGE
2-4-117.1 Replace Yaw Pushrod Boot and Shroud	2-4-117-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Abrasive Cloth, Item 1A, Appendix D
- Adhesive, Item 32, Appendix D

- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 2-4-105
- PARA 7-4-24
- PARA 11-4-83

2-4-117.1 Replace Yaw Pushrod Boot and Shroud.

2-4-117.1.1 Remove.

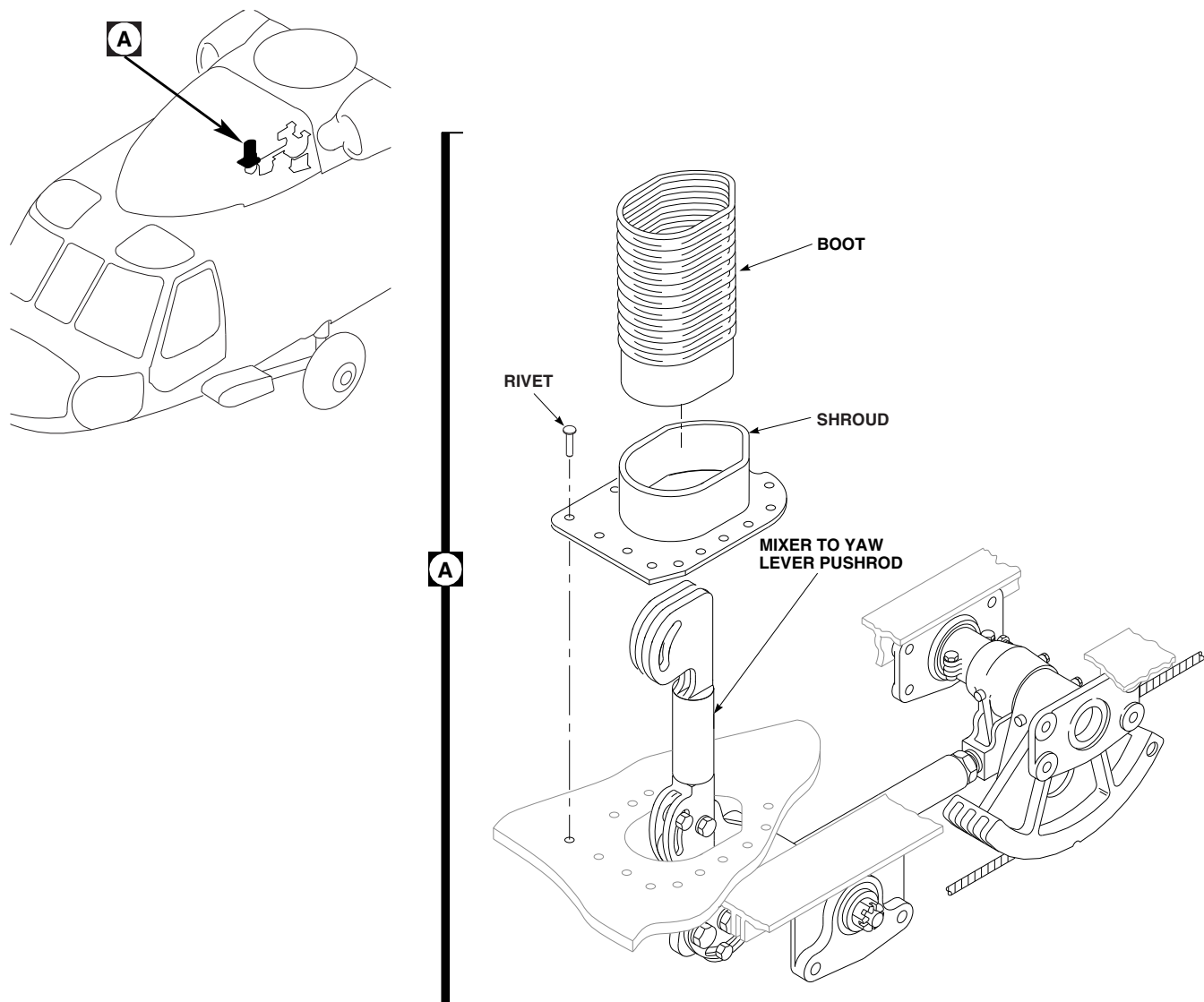
- Remove No. 1 transfer module manifold (PARA 7-4-24).
- Remove main rotor pylon work platform (PARA 2-4-105).
- Disconnect yaw pushrod from mixer lever (PARA 11-4-83).
- Remove boot from shroud using nonmetallic scraper (Figure 2-4-117-1, Detail A). Lift boot over mixer to yaw lever pushrod and remove boot.
- If replacing boot only, remove remaining adhesive from shroud using machinery towel, Item 211, Appendix D, wet with methyl ethyl

ketone, Item 217, Appendix D, and nonmetallic scraper.

- Remove rivets securing shroud to deck. Remove shroud.
- Remove remaining sealing compound from helicopter fuselage using machinery towel, Item 211, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D, and nonmetallic scraper.

2-4-117.1.2 Install.

- Lightly abrade inner bonding surface of boot using abrasive cloth, Item 1A, Appendix D.
- Wipe abraded area using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.

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SA**FIGURE 2-4-117-1. REPLACE YAW PUSHROD BOOT AND SHROUD**

- c. Wipe bonding surface of shroud with machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.
- d. Mix 100 parts by weight of part A to 22 parts by weight of part B of adhesive, Item 32, Appendix D.
- e. Apply a light coat of adhesive, Item 32, Appendix D, to bonding surface of boot and shroud.
- f. Install boot over mixer to yaw lever pushrod and position in shroud, making sure of good contact between bonding surfaces (Figure 2-4-117-1, Detail A).
- g. Allow adhesive, Item 32, Appendix D, to cure at room temperature for a minimum of 12 hours.
- h. Apply a light coat of sealing compound, Item 292, Appendix D, to bottom of shroud.
- i. Position boot and shroud on deck.
- j. Secure shroud to deck with rivets wet with sealing compound, Item 292, Appendix D.

- | | |
|---|---|
| <ul style="list-style-type: none"> k. Connect yaw pushrod to mixer lever (PARA 11-4-83). l. Make sure area is clean and free of foreign material. | <ul style="list-style-type: none"> m. Install main rotor pylon work platform (PARA 2-4-105). n. Install No. 1 transfer module manifold (PARA 7-4-24). |
|---|---|

2-4-118 ENGINE COWLINGS.

PARAGRAPH BREAKDOWN

	PAGE
2-4-118.1 Replace Engine Cowlings	2-4-118-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Magnifying Glass, 10X
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Abrasive Cloth, Item 13, Appendix D
- Scouring Pad, Item 270, Appendix D

References

- Appendix D
- PARA 2-4-122
- PARA 2-6-3
- PARA 2-6-29

2-4-118.1 Replace Engine Cowlings.

NOTE

Replacement of left and right engine cowlings is the same.

2-4-118.1.1 Remove.

CAUTION

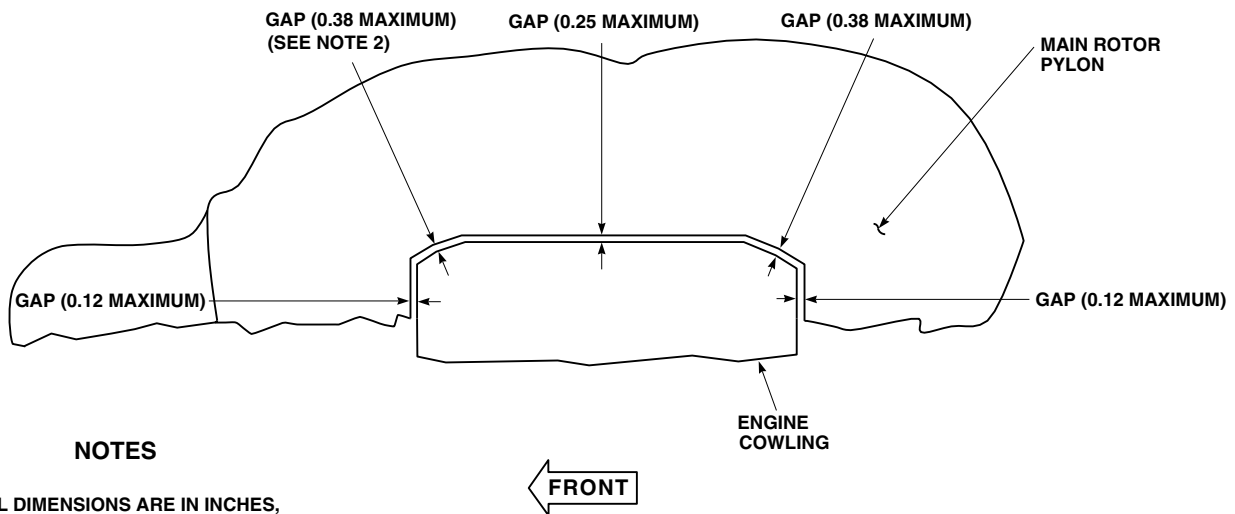
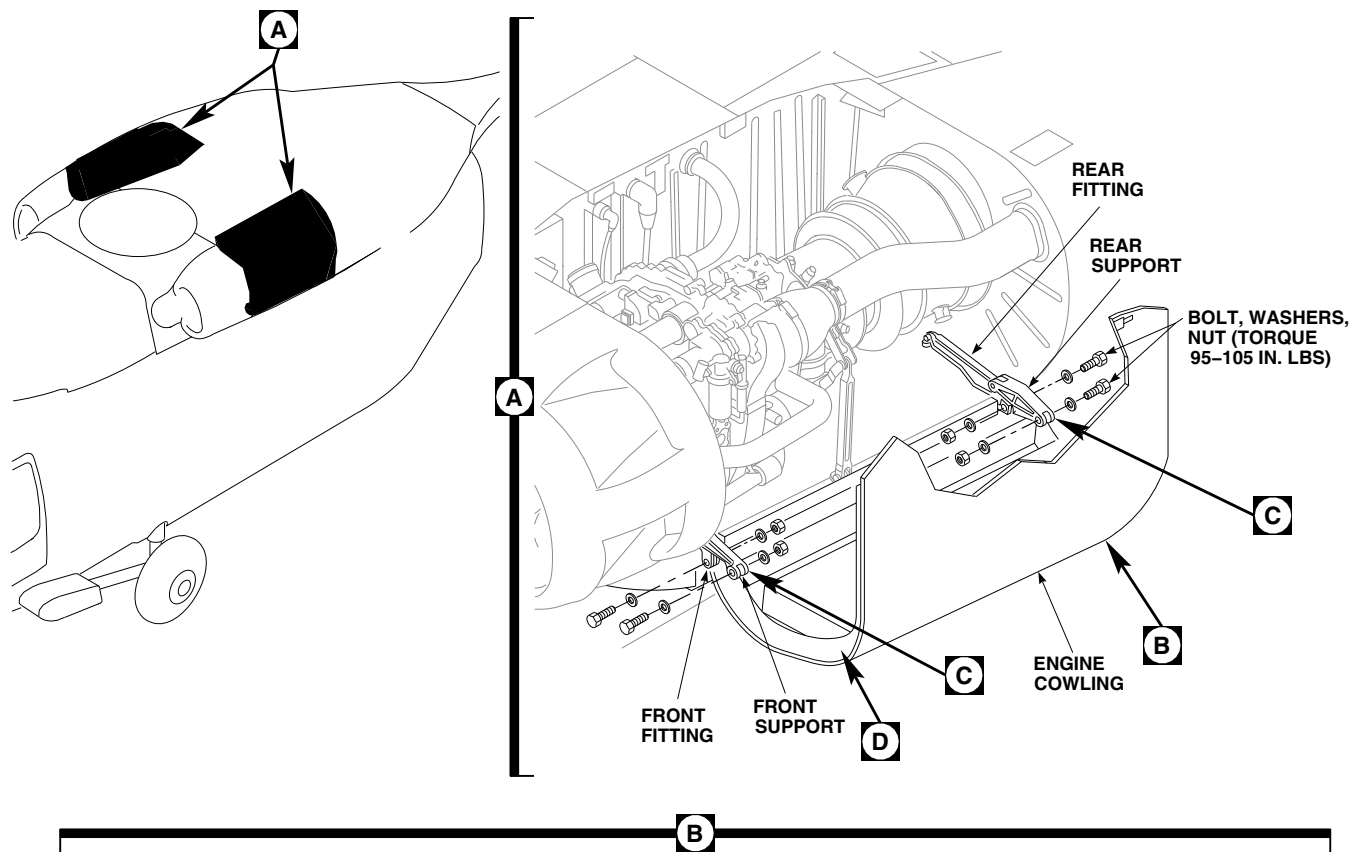
To prevent damage to helicopter, do not let engine cowling drop to open position.

- Open engine cowling (Figure 2-4-118-1, Sheet 1, Detail A).

CAUTION

To prevent cowling from pivoting down causing possible damage to helicopter, support cowling when removing hardware from front and rear supports.

- With aid of assistant, remove bolts, washers, and nuts securing front and rear supports to cowling.
- With aid of assistant, remove bolts, washers, and nuts securing cowling to front and rear fittings. Remove cowling from helicopter.



NOTES

1. ALL DIMENSIONS ARE IN INCHES, EXCEPT AS NOTED.
2. 1.00 MAXIMUM GAP RIGHT SIDE COWLING
3. ADD CHAMFER BY FILING OR MACHINING.

LOOKING DOWN-LEFT SIDE

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FIGURE 2-4-118-1. REPLACE ENGINE COWLINGS (SHEET 1 OF 3)

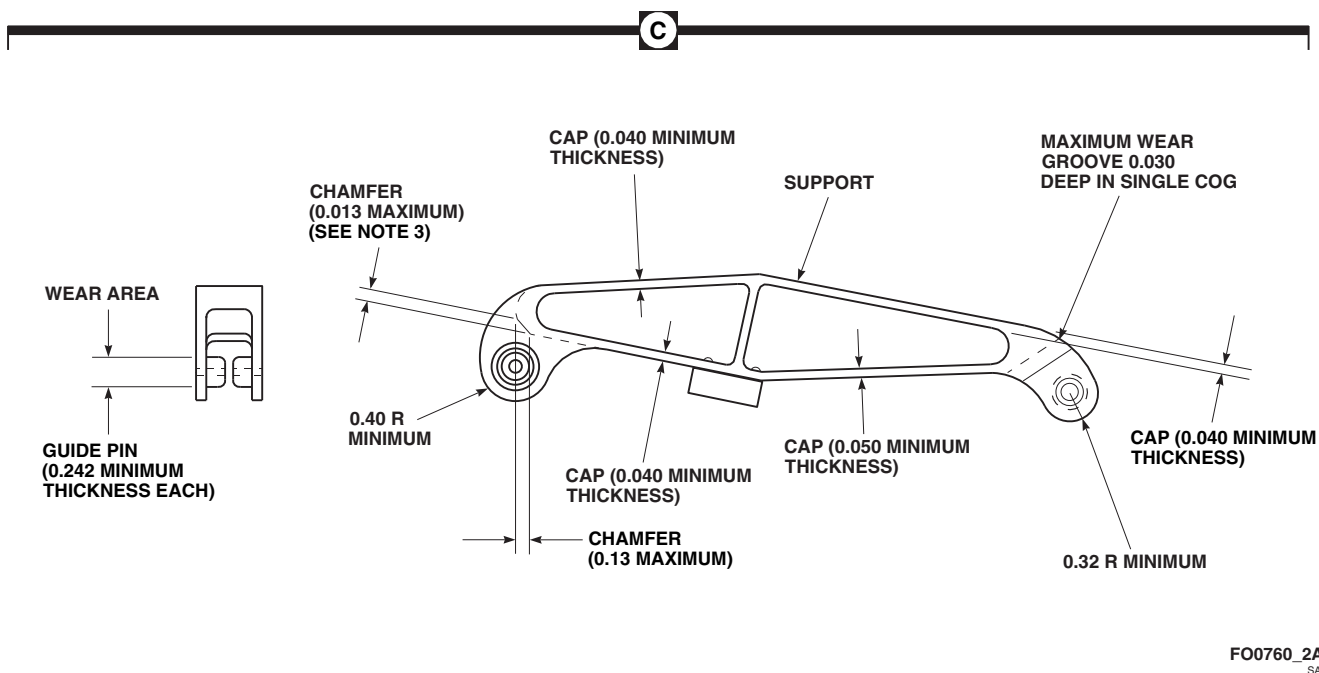


FIGURE 2-4-118-1. REPLACE ENGINE COWLINGS (SHEET 2 OF 3)

2-4-118.1.2 Inspect/Repair.

NOTE

- If possible, replace corroded bolts. Cadmium-plating will be removed during corrosion removal.
 - Corrosion is indicated by pitting.
- a. Visually inspect hinge bolts for corrosion. **NONE ALLOWED.** If corrosion is found, remove using scouring pad, Item 270, Appendix D. **NO PITS ALLOWED AFTER CLEANUP.**
 - b. Inspect front and rear supports for wear and damage. **NO DAMAGE BEYOND LIMITS ALLOWED AS SHOWN IN FIGURE 2-4-118-1, SHEET 2, DETAIL C.** If worn or damaged beyond limits, replace front or rear support.
 - c. Inspect front and rear support guide pin for wear. **GUIDE PIN DIAMETER SHALL NOT BE LESS THAN 0.242-INCH.** If diameter is below limit, replace front or rear support.
 - d. Inspect front and rear supports for wear and damage. **NONE ALLOWED.** If worn or damaged, blend front or rear supports using abrasive cloth, Item 13, Appendix D. **DO NOT BLEND TO DEPTH BEYOND DAMAGE.**
 - e. Using 10X magnifying glass, inspect fiberglass top ply for damage at hinge rib location on both ends of engine cowling (Figure 2-4-118-1, Sheet 3, Detail D). Proceed as follows:
 - (1) On engine cowling assemblies, 70302-02141-041, -042, -044, -045, -048, and -049, **NO DAMAGE ALLOWED.** If cracks are found through either of the first two fiberglass layers or there is evidence of exposed fiber, replace fittings (PARA 2-4-122), and repair cowling (PARA 2-6-29). If damage extends into third layer, replace cowling.
 - (2) On engine cowling assemblies, 70070-30308-041 through -046, and 70302-02141-052 and -053, **DAMAGE IS ALLOWED THROUGH FIRST TWO FIBERGLASS LAYERS. NO DAMAGE**

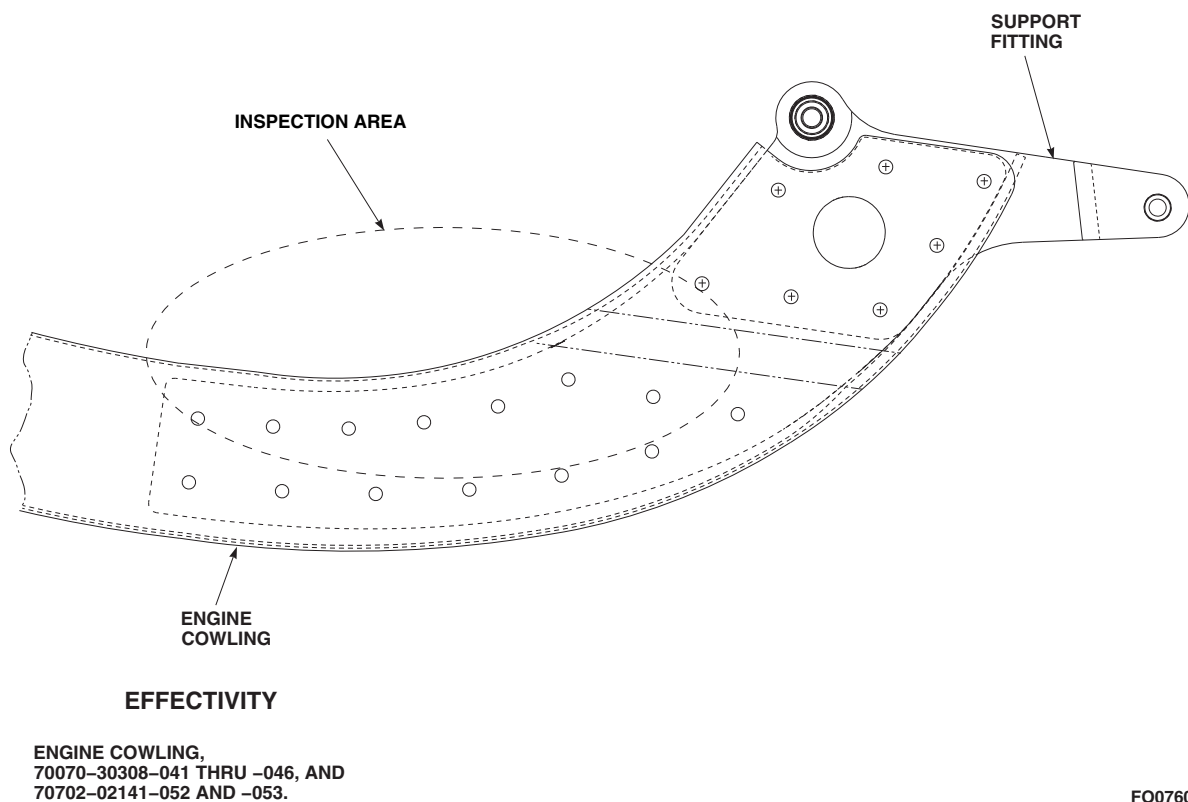
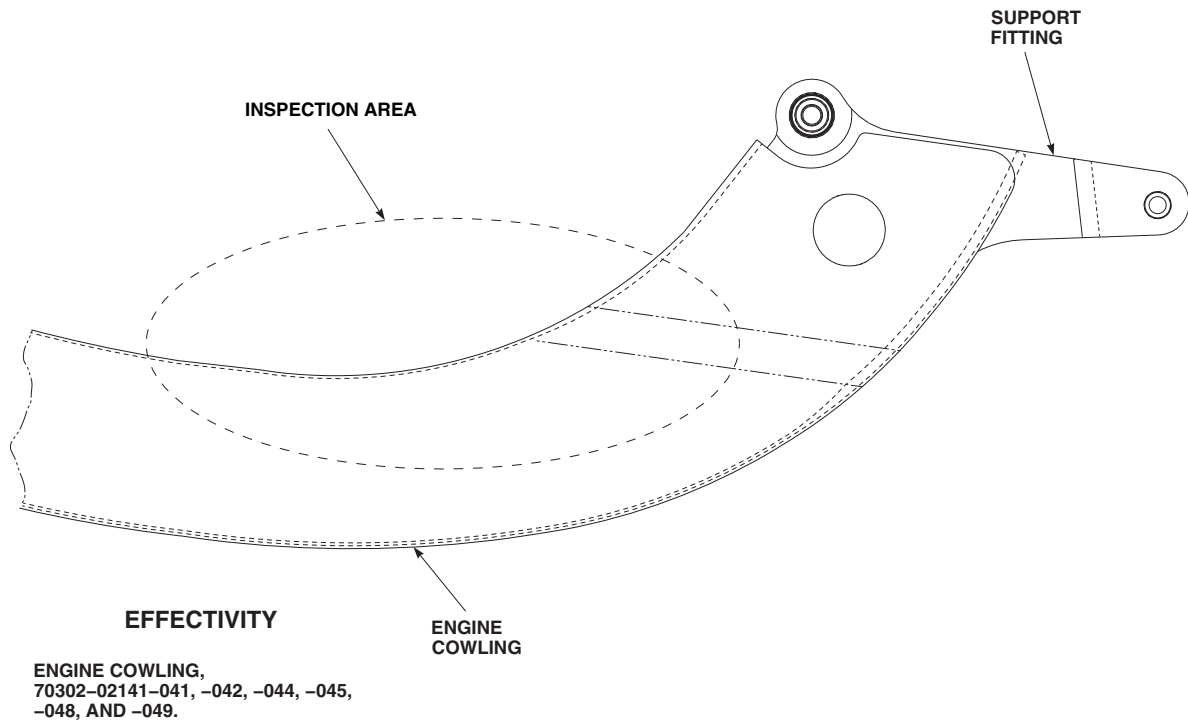
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FIGURE 2-4-118-1. REPLACE ENGINE COWLINGS (SHEET 3 OF 3)

ALLOWED TO THIRD LAYER. If damage does not extend into third layer, repair cowling (PARA 2-6-29). If damage extends into third layer, replace cowling.

2-4-118.1.3 Install.

- a. With aid of assistant, position cowling on front and rear fittings and secure with bolts, washers, and nuts. **TORQUE NUTS TO 95 - 105 INCH-POUNDS** (Figure 2-4-118-1, Sheet 1, Detail A).
- b. Secure front and rear support to cowling with bolts, washers, and nuts. **TORQUE NUTS TO 95 - 105 INCH-POUNDS.**

NOTE

Make sure waterwash line does not contact front fitting of work platform.

- c. Route waterwash lines over anti-ice duct on No. 1 engine and over sensor tube on No. 2 engine. Adjust clamps accordingly to make sure of proper clearance.

NOTE

Gaps are same for left and right engine cowling, except as noted.

- d. Check fit of cowling using gap tolerances provided in Figure 2-4-118-1, Sheet 1, Detail B. If gap is too small, trim edges of cowling (PARA 2-6-3).
- e. Make sure area is clean and free of foreign material before closing up.
- f. Close and latch engine cowling.

2-4-119 ENGINE COWLING BLANKET INSULATION.

PARAGRAPH BREAKDOWN

	PAGE
2-4-119.1 Replace Engine Cowling Blanket Insulation	2-4-119-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

2-4-119.1 Replace Engine Cowling Blanket Insulation.

NOTE

Replacement of right and left engine cowling blanket insulation is the same.

2-4-119.1.1. Remove.

- Open engine cowling.
- Remove screws, washers, and remove blanket insulation from cowling (Figure 2-4-119-1, Detail A).

2-4-119.1.2. Install.

- Place blanket insulation on cowling and install washers and screws (Figure 2-4-119-1, Detail A).
- Make sure area is clean and free of foreign material before closing up.
- Close and latch engine cowling.

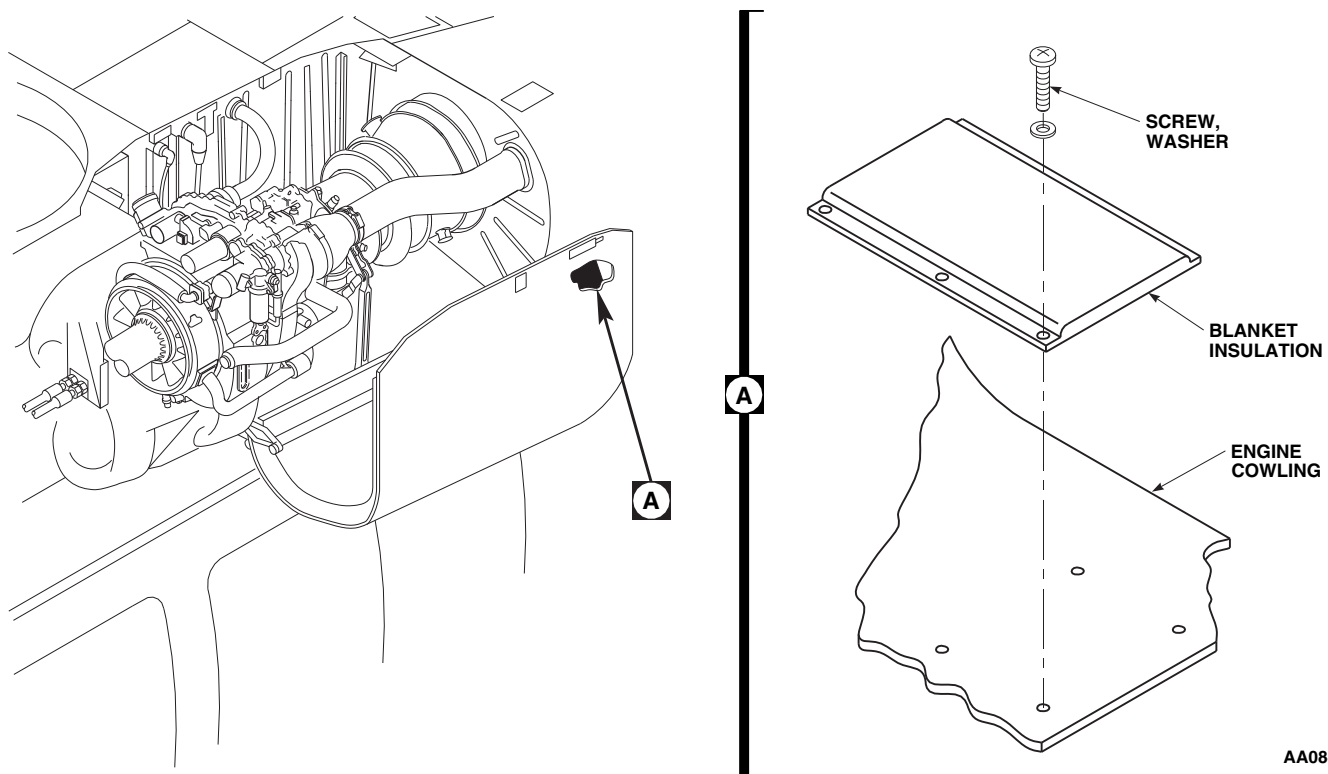


FIGURE 2-4-119-1. REPLACE ENGINE COWLING BLANKET INSULATION

2-4-120 ENGINE COWLING LATCH.

PARAGRAPH BREAKDOWN

	PAGE
2-4-120.1 Replace Engine Cowling Latch	2-4-120-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Corrosion Compound, Item 108, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D

- Paint, Polyurethane, Item 222A, Appendix D
- Scouring Pad, Item 270, Appendix D

References

- Appendix D
- PARA 2-3-11
- PARA 2-6-5

2-4-120.1 Replace Engine Cowling Latch.

NOTE

Replacement of right and left engine cowling latch is the same.

2-4-120.1.1 Remove.

- Open engine cowling.
- Remove clevis bolts, washers, bushings, and nuts attaching lockpin tubes to latch handle assembly. Remove lockpin tubes (Figure 2-4-120-1).
- Remove screws, washers, and nuts attaching latch handle release to engine cowling. Remove latch handle release.
- Remove screws, washers, and nuts attaching latch handle assembly to engine cowling. Remove latch handle assembly.

- Remove screws, washers, and nuts attaching lockpin fittings to cowling. Remove fittings.
- Inspect engine cowling latch (PARA 2-3-11).

2-4-120.1.2 Disassemble.

- Disassemble latch handle assembly as follows:
 - Remove clevis pin, washer, and cotter pin attaching latch handle to handle support (Figure 2-4-120-1, Detail A).
 - Remove latch handle.
- Disassemble lockpin tubes as follows:
 - Remove clevis bolts, washers, nuts, and cotter pins attaching lockpins to lockpin tubes (Figure 2-4-120-1).
 - Remove lockpins.

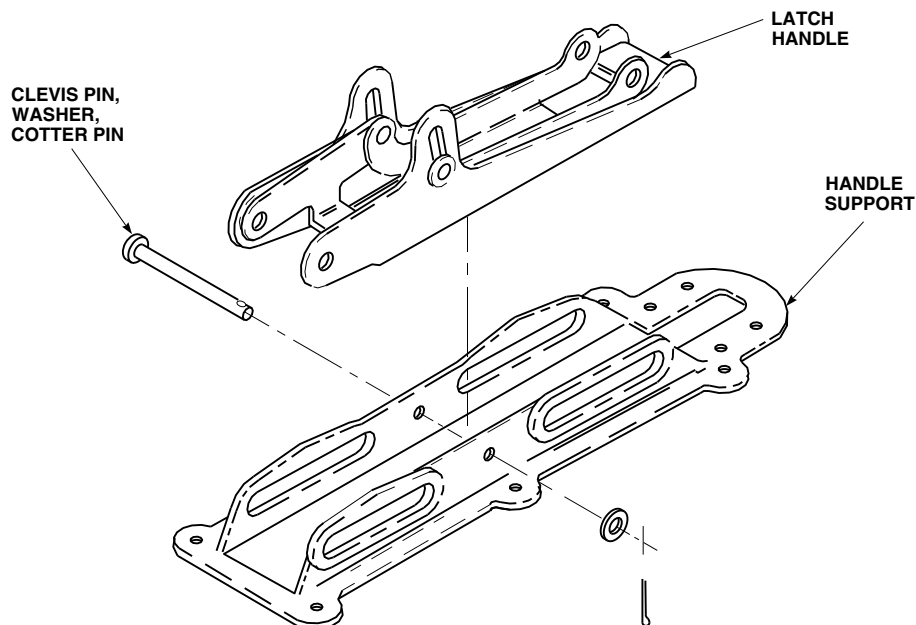
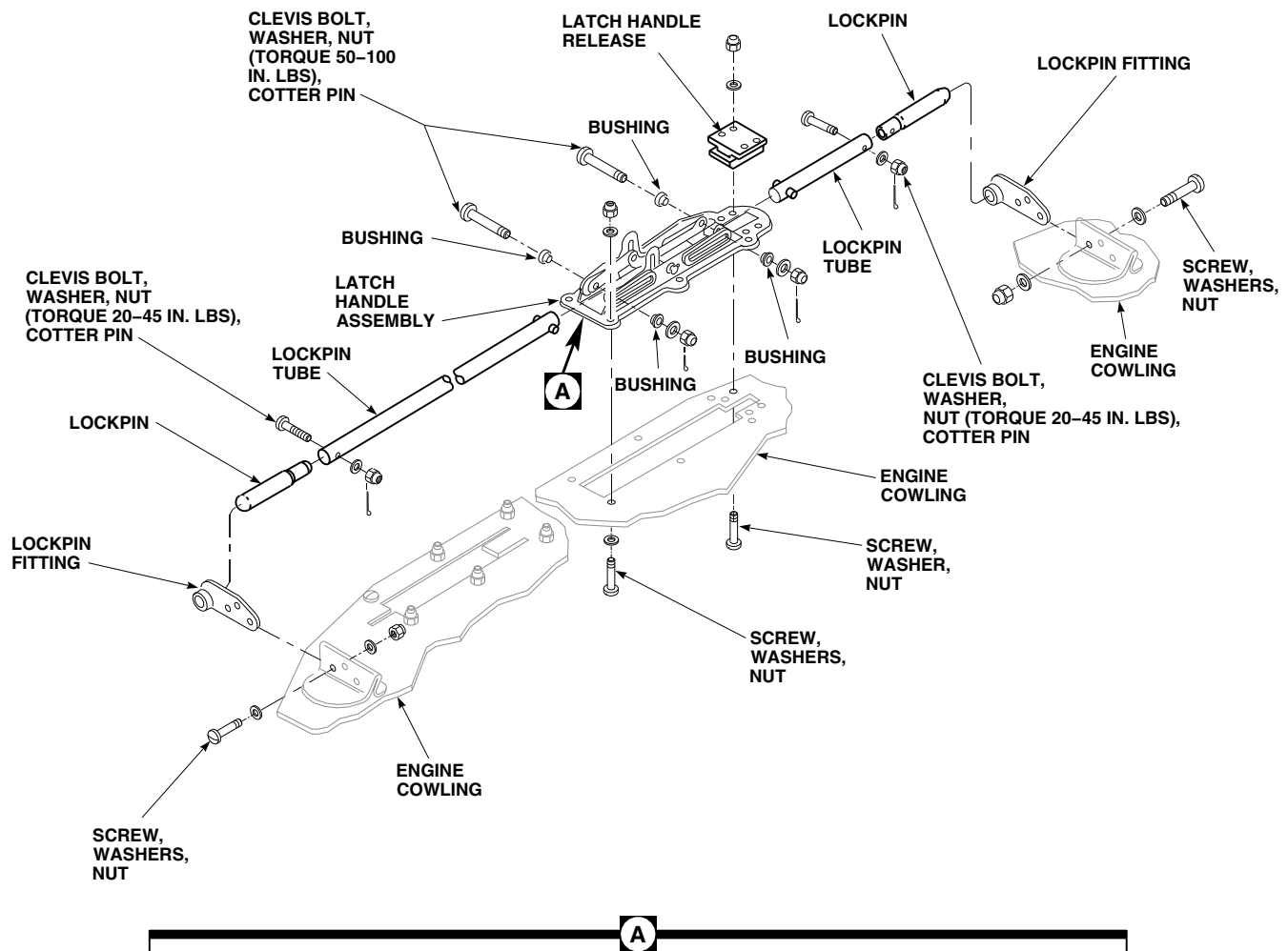


FIGURE 2-4-120-1. REPLACE ENGINE COWLING LATCH

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2-4-120.1.3 Repair.

- a. Remove corrosion manually, using scouring pad, Item 270, Appendix D.
- b. Mix solution of 1 part corrosion compound, Item 108, Appendix D, to 1 part water.
- c. Apply corrosion compound solution to corroded areas of latch handle. Rinse thoroughly with clean water.
- d. Touch up paint nicks, scratches, and corroded areas, using epoxy primer coating kit, Item 135, Appendix D, and polyurethane paint, Item 222A, Appendix D (PARA 2-6-5).

2-4-120.1.4 Assemble.

- a. Assemble latch handle as follows:
 - (1) Position latch handle in handle support (Figure 2-4-120-1, Detail A).
 - (2) Install clevis pin, washer, and cotter pin.
- b. Assemble lockpin tubes as follows:
 - (1) Install lockpins in lockpin tubes using clevis bolts, washers, and nuts (Figure 2-4-120-1). **TORQUE NUTS TO 20 - 45 INCH-POUNDS.**

- (2) Install cotter pins.

2-4-120.1.5 Install.

- a. Install latch handle assembly on engine cowling with screws, washers, and nuts (Figure 2-4-120-1).
- b. Install latch handle release on engine cowling using screws, washers, and nuts.
- c. Install lockpin fittings on engine cowling using screws, washers, and nuts.
- d. Insert lockpin tubes in fittings and other ends in latch handle assembly.
- e. Install clevis bolts, bushings, and nuts. **TORQUE NUTS TO 50 - 100 INCH-POUNDS.**
- f. Install cotter pins.
- g. Check action of latch handle and lockpins for smooth operation.
- h. Make sure area is clean and free of foreign material.
- i. Close and latch engine cowling several times to make sure of proper operation.

2-4-121 ENGINE COWLING FITTINGS.

PARAGRAPH BREAKDOWN

	PAGE
2-4-121.1 Replace Engine Cowling Fittings	2-4-121-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Abrasive Cloth, Item 12, Appendix D
- Acetone, Item 25, Appendix D
- Adhesive, Item 60, Appendix D

- Cheesecloth, Item 90, Appendix D
- Wear Strip, Locally-Made, Figure H-201, Appendix H

References

- Appendix D
- Appendix H
- PARA 2-4-118
- PARA 2-6-5

2-4-121.1 Replace Engine Cowling Fittings.

NOTE

Replacement of front and rear engine cowling fitting is the same for both engine cowlings, except as noted.

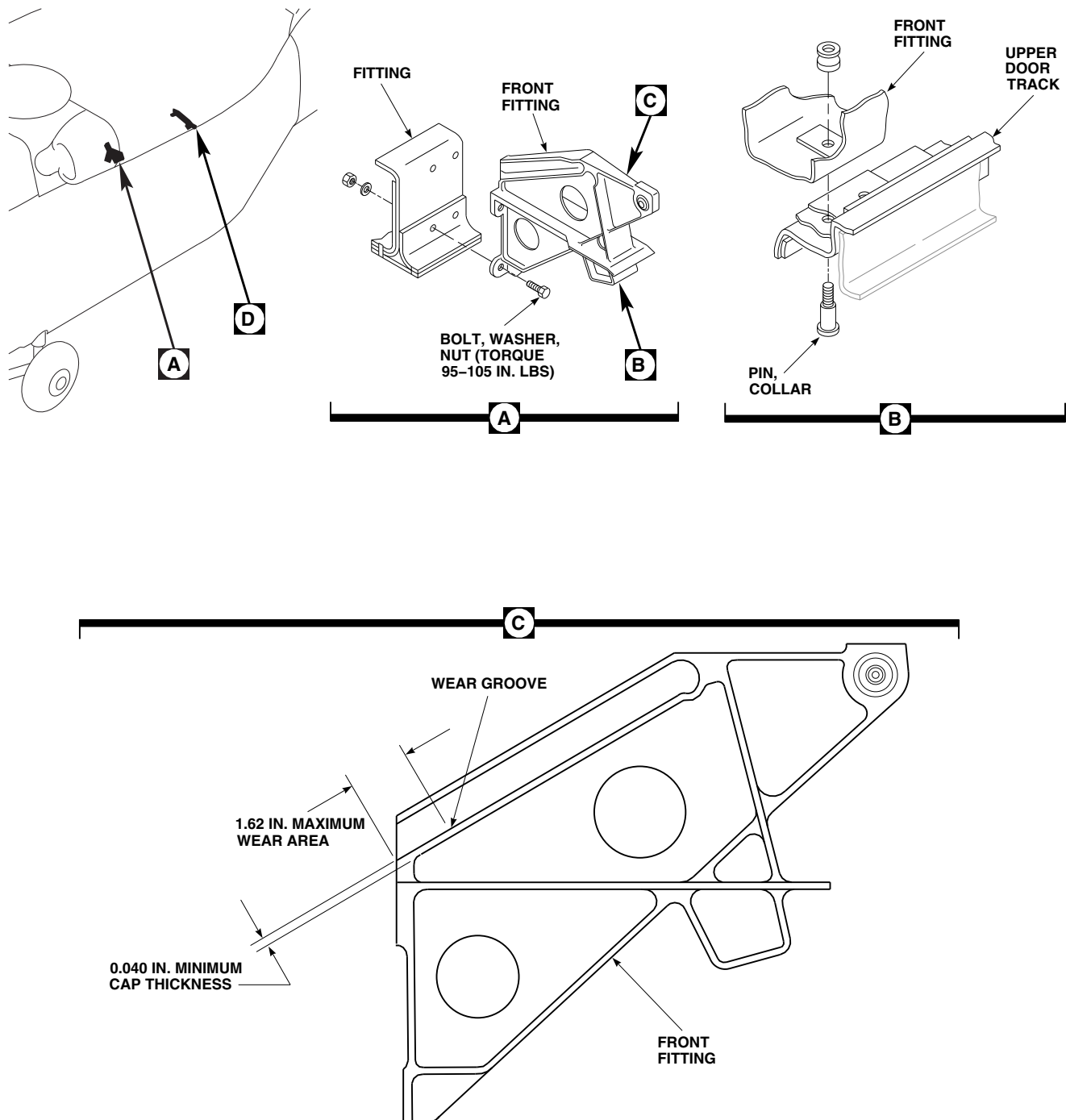
2-4-121.1.1 Remove.

- Remove engine cowling (PARA 2-4-118).
- Remove pin and collar securing front fitting to upper door track (Figure 2-4-121-1, Sheet 1, Detail B).
- Remove bolt, washer, and nut securing rear fitting to upper door track (Figure 2-4-121-1, Sheet 2, Detail E).

- Remove bolts, washers, and nuts securing front or rear fitting to fittings on engine deck (Figure 2-4-121-1, Sheet 1, Detail A or Sheet 2, Detail D).
- Remove fitting.

2-4-121.1.2 Inspect.

- Inspect front fitting for wear. **CAP THICKNESS SHALL NOT BE LESS THAN 0.040-INCH IN WEAR AREA** (Figure 2-4-121-1, Sheet 1, Detail C). If not within limit, replace or repair fitting as required.
- Inspect rear fitting for wear. **CAP THICKNESS SHALL NOT BE LESS THAN 0.055-INCH IN ZONE A AND 0.065-INCH IN ZONE B** (Figure 2-4-121-1, Sheet 2, Detail F).



NOTE

FILL WITH ADHESIVE, ITEM 60,
APPENDIX D.

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FIGURE 2-4-121-1. REPLACE ENGINE COWLING FITTINGS (SHEET 1 OF 2)

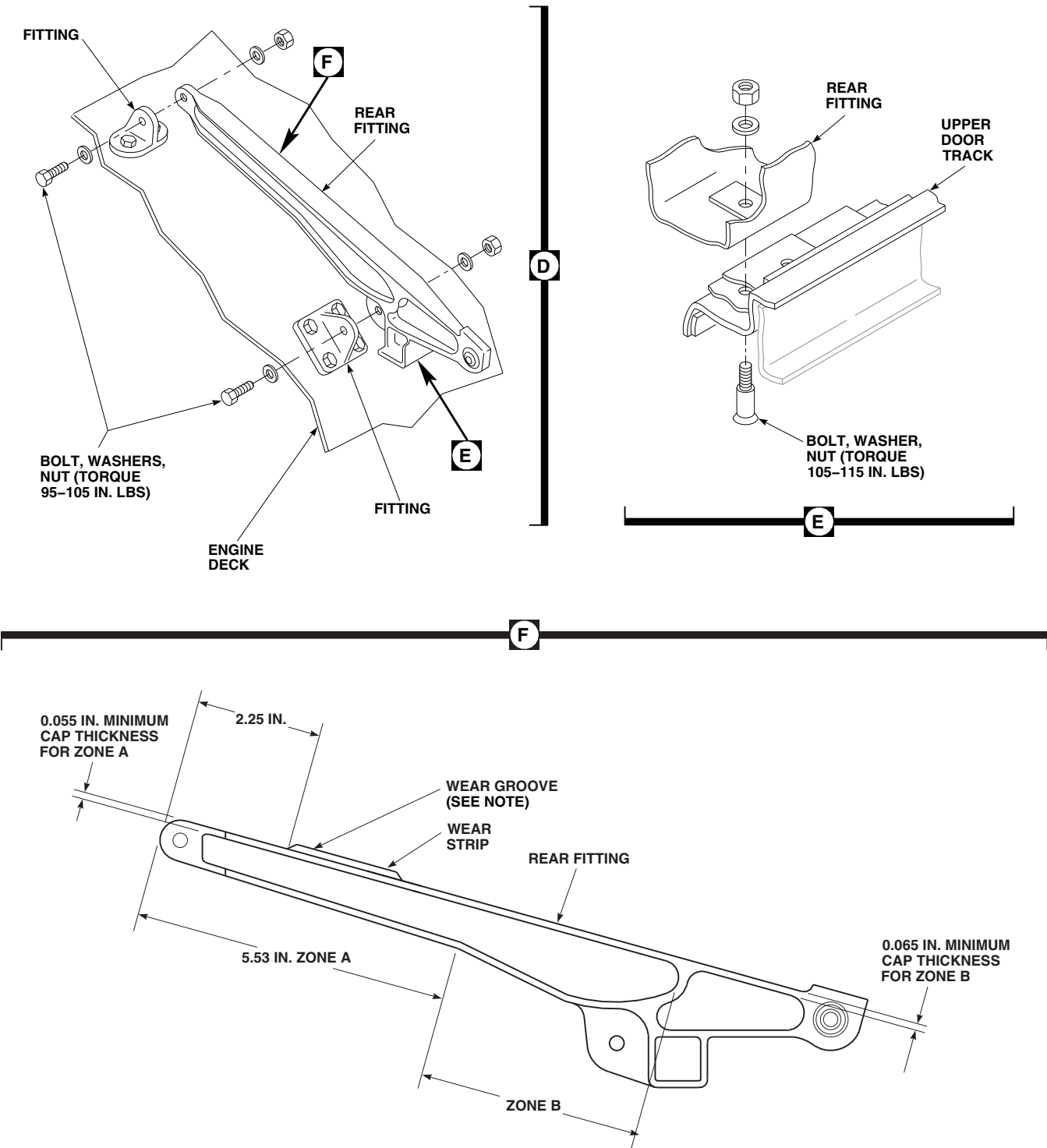


FIGURE 2-4-121-1. REPLACE ENGINE COWLING FITTINGS (SHEET 2 OF 2)

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If not within limits, replace or repair fitting as required.

2-4-121.1.3 Repair.

- a. Using abrasive cloth, Item 12, Appendix D, blend wear to fitting. **DO NOT BLEND TO DEPTH BEYOND DAMAGE.**
- b. Bond wear strip to rear fitting as follows:
 - (1) Locally make wear strip (Figure H-201, Appendix H).
 - (2) Clean bonding surface of rear fitting and wear strip using cheesecloth, Item 90, Appendix D, moistened with acetone, Item 25, Appendix D.
 - (3) Apply adhesive, Item 60, Appendix D, to each surface to be bonded.
 - (4) Allow accelerator to air-dry at room temperature for a minimum of 3 minutes.
 - (5) Apply bead of adhesive, Item 60, Appendix D, to rear fitting.
 - (6) Position wear strip on rear fitting (Figure 2-4-121-1, Sheet 2, Detail F).

(7) Press wear strip firmly in place.

(8) Allow to cure at room temperature for 24 hours.

(9) Touch up paint as required (PARA 2-6-5).

2-4-121.1.4 Install.

- a. Place replacement fitting on engine deck.
- b. Install bolt, washer, and nut securing rear fitting to upper door track. **TORQUE NUT TO 105 - 115 INCH-POUNDS** (Figure 2-4-121-1, Sheet 2, Detail E).
- c. Install pin and collar securing front fitting to upper door track (Figure 2-4-121-1, Sheet 1, Detail B).
- d. Install bolts, washers, and nuts securing front or rear fitting to fitting(s) on engine deck. **TORQUE NUTS TO 95 - 105 INCH-POUNDS** (Figure 2-4-121-1, Sheet 1, Detail A or Sheet 2, Detail D).
- e. Install engine cowling (PARA 2-4-118).

2-4-122 ENGINE COWLING SUPPORT FITTINGS.

PARAGRAPH BREAKDOWN

	PAGE
2-4-122.1 Replace Engine Cowling Support Fittings	2-4-122-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- C-Clamps
- Drill
- Drill Bit, No. 10
- Nonmetallic Drift
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 0 - 50 in. lbs
- Weights

Materials/Parts

- Acetone, Item 25, Appendix D
- Adhesive, Item 34, Appendix D
- Cheesecloth, Item 90, Appendix D
- Ink Marker
- Machinery Towel, Item 211, Appendix D
- Masking Tape, 1-inch, Item 212, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 302A, Appendix D

References

- Appendix D
- PARA 2-6-3
- PARA 2-6-5

2-4-122.1 Replace Engine Cowling Support Fittings.

NOTE

Replacement of front and rear fittings is the same.

2-4-122.1.1 Remove.

NOTE

Rivets are installed on engine cowling assemblies, 70070-30308-041 through 70070-30308-046, and 70302-02141-052 and -053.

- If present, carefully drill out blind bolts/pin-rivets securing support fitting to engine cowl-ing using drill and suitable size drill bit (Figure 2-4-122-1, Sheet 2, Section A-A).

NOTE

- On engine cowling assemblies, 70070-30308-041 through 70070-30308-046, and 70302-02141-052 and -053, force may be required in order to remove sup-port fitting from engine cowling.
- Remove screws, washers, and nuts securing support fitting and laminated shims to engine

cowling. Remove fitting and shims (Figure 2-4-122-1, Sheet 1, Detail A).

- c. If fitting is to be reused, and shims are bonded to fitting, perform the following:
 - (1) Soften adhesive securing shim to fitting, using cheesecloth, Item 90, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D.
 - (2) Remove shim from fitting using nonmetallic scraper.
 - (3) Clean remaining adhesive residue using cheesecloth, Item 90, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D. Wipe dry using clean, dry cheesecloth.

NOTE

On engine cowling assemblies, 70070-30308-041 through 70070-30308-046, and 70302-02141-052 and -053, sealing compound is used to fill cavity that surrounds support fitting.

- d. If present, remove all sealing compound from engine cowling cavity using nonmetallic scraper.

2-4-122.1.2 Install.

NOTE

- On engine cowling assemblies, 70302-02141-041, -042, -044, -045, -048, and -049, support fitting, 70302-02157-041, must be replaced with support fitting, 70070-30308-047. Support fitting, 70302-02157-042, must be replaced with support fitting, 70070-30308-048.
- On engine cowling assemblies, 70302-02141-041, -042, -044, -045, -048, and -049, a nonstructural foam insert located in cowling cavity may prevent fitting from being fully installed. Repeatedly push in and pull out fitting, to displace foam insert, until fitting can be properly

positioned. In order to displace foam insert, aggressive force may be required when pushing in fitting.

- a. Prepare support fitting for installation as follows:
 - (1) Position fitting in engine cowling and mark location of screwholes using ink marker (Figure 2-4-122-1, Sheet 1, Detail A).
 - (2) Remove fitting and drill seven screwholes in marked locations using drill and No. 10 drill bit.
- b. Prepare shims for installation as follows:
 - (1) Using fitting as a template, position shim on fitting and mark location of screwholes using ink marker.
 - (2) Using drill and No. 10 drill bit, drill seven screwholes in marked locations.
 - (3) Repeat for other shim.
- c. On engine cowling assemblies, 70302-02141-041, -042, -044, -045, -048, and -049, perform the following:
 - (1) Create template by locally reproducing Figure 2-4-122-2 at 200%.
 - (2) Using template, mark location of blind bolt/pin-rivet holes on engine cowling.

NOTE

- There are two different configuration engine cowlings (Figure 2-4-122-1, Sheet 2, Section A-A). On configuration 2 cowlings, use care when drilling pin-rivet holes in order not to drill into adjacent side.
- Make sure to keep drill perpendicular with side of engine cowling. Failure to do so may result in the inability to install and secure blind bolts/pin-rivets.
- Make sure holes are drilled inline. Failure to do so may result in the in-

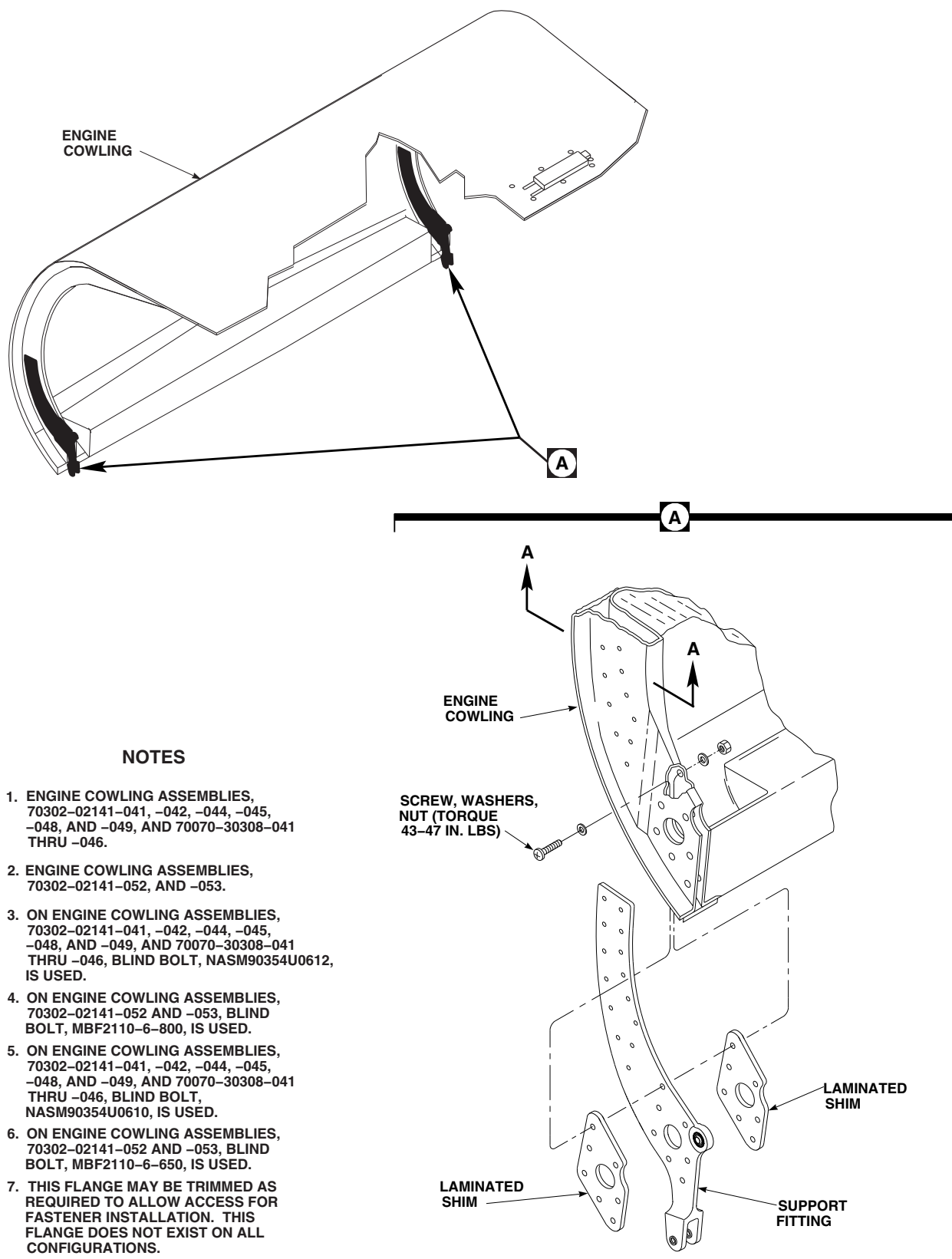
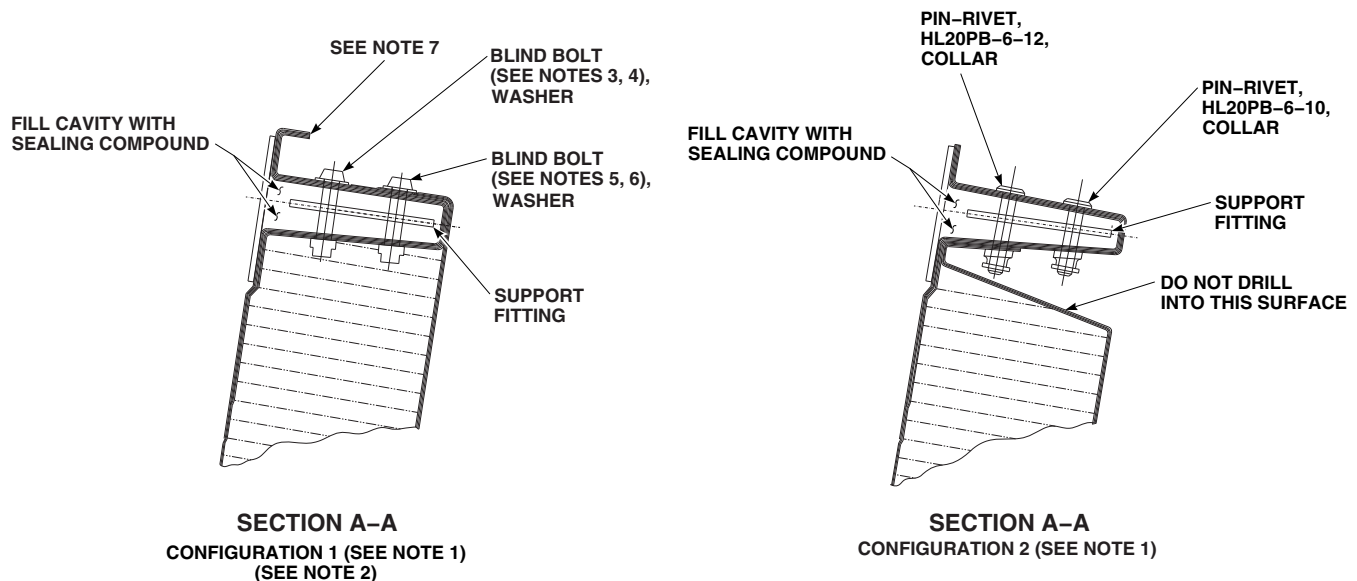


FIGURE 2-4-122-1. REPLACE ENGINE COWLING SUPPORT FITTINGS (SHEET 1 OF 2)

FB3749_2
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ability to install and secure blind bolts/
pin-rivets.

- (3) Drill holes in marked locations using drill and suitable size drill bit as follows:

- (a) On configuration 1 cowlings, holes are to be 0.200 - 0.202-inch.
- (b) On configuration 2 cowlings, holes are to be 0.190 - 0.193-inch.

- d. Prepare support fitting for installation as follows:

- (1) Position fitting in engine cowling and secure with screws, washers, and nuts (Figure 2-4-122-1, Sheet 1, Detail A).

NOTE

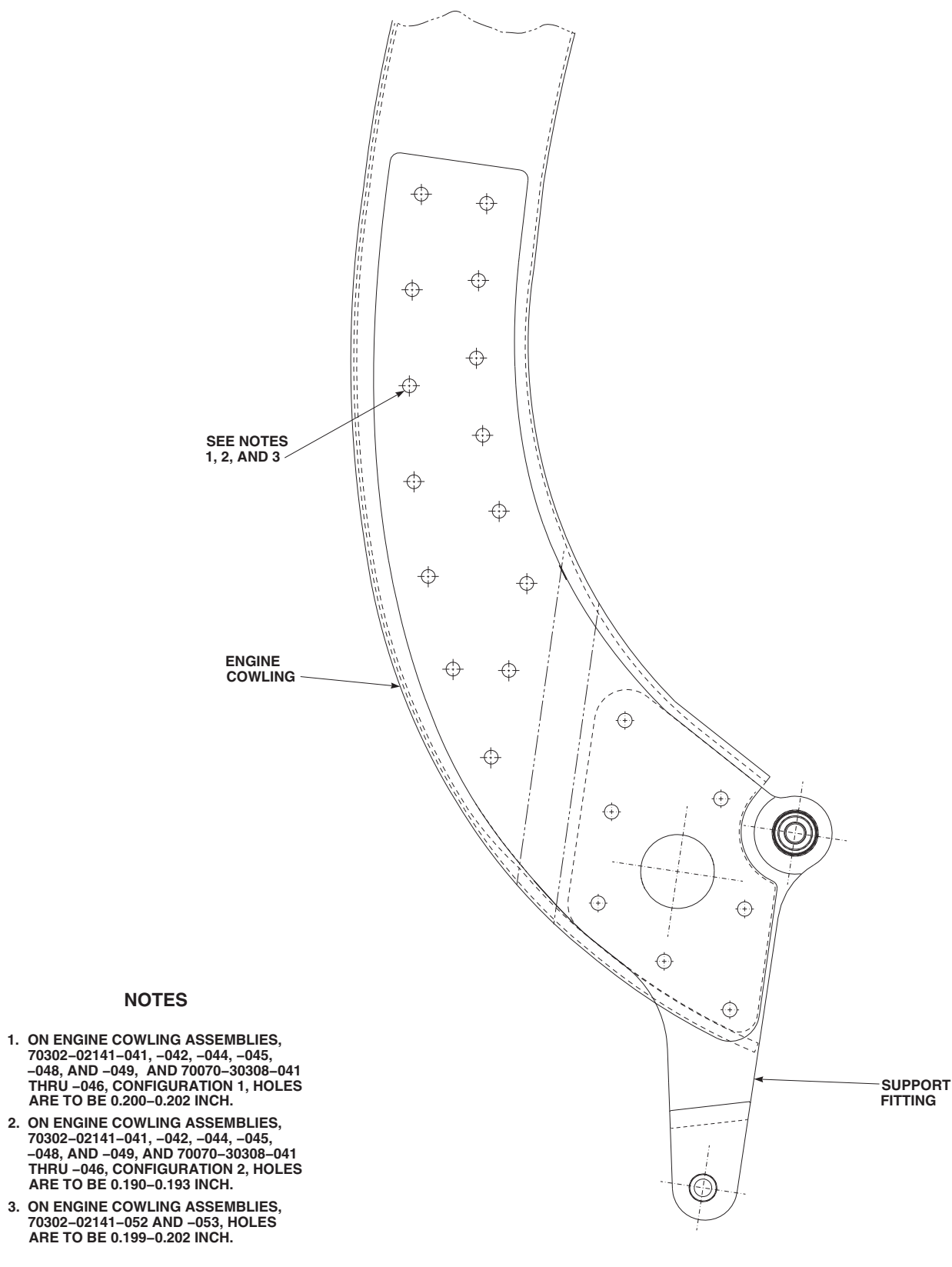
Make sure to accurately mark location of blind bolt/pin rivet holes. Failure to do so may result in the inability to install and secure blind bolts/pin-rivets.

- (2) Mark location of blind bolt/pin-rivet holes using ink marker.
- (3) Remove screws, washers, and nuts securing fitting to cowling and remove fitting.

NOTE

On engine cowling assemblies, 70302-02141-041, -042, -044, -045, -048, and -049, and 70070-30308-041 through -046, two different hole sizes are used depending on engine cowling configuration. Refer to Figure 2-4-122-1, Sheet 2, Section A-A, to determine configuration.

- (4) Drill holes in marked locations using drill and suitable size drill bit as follows:
 - (a) For configuration 1 cowlings, holes are to be 0.200 - 0.202-inch.
 - (b) For configuration 2 cowlings, holes are to be 0.190 - 0.193-inch.



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**FIGURE 2-4-122-2. ENGINE COWLING SUPPORT FITTING TEMPLATE
(LOCALLY REPRODUCE AT 200%)**

- (c) On engine cowling assemblies, 70302-02141-052 and -053, holes are to be 0.199 - 0.202-inch.
- e. Install shims on fitting as follows (Figure 2-4-122-1, Sheet 1, Detail A):
 - (1) Apply adhesive, Item 34, Appendix D, to mating surfaces of shims and fittings.
 - (2) Install shims on fitting, making sure to align all screwholes. Hold shims in place with C-clamps or weights.
 - (3) Allow adhesive to cure for 24 hours.

NOTE

Peel shims, as required, to eliminate gaps between fitting and cowling.

- f. Install fitting and shims in cowling and secure with screws, washers, and nuts. **TORQUE NUTS 43 - 47 INCH-POUNDS.**
- g. Mask all but the end two blind bolt/pin-rivet holes with 1-inch masking tape, Item 212, Appendix D.
- h. Inject sealing compound, Item 302A, Appendix D, into cavity that surrounds support fitting (Figure 2-4-122-1, Sheet 2, Section A-A). Inject sealing compound until compound squeezes out the two unmasked holes.
- i. Allow sealing compound to set for a minimum of 20 - 30 minutes.
- j. Remove masking tape. Using suitable sized nonmetallic drift, clean sealing compound from holes.
- k. Allow sealing compound to fully cure for 72 hours at 65° - 80°F (18.3° - 26.6°C), or for 2 hours at 65° - 80°F (18.3° - 26.6°C), then 3 hours at 145° - 155°F (62.8° - 68.3°C).

NOTE

- Refer to step d. (4) for correct drill bit size.

- Use care when cleaning out rivet holes. Failure to do so may result in elongation of rivet holes.
- l. If required, clean sealing compound from rivet holes using drill and suitable size drill bit.
- m. Using nonmetallic scraper, remove any sealing compound squeezeout. Remove residue using machinery towel, Item 211, Appendix D, dampened with acetone, Item 25, Appendix D. Wipe dry with clean machinery towel, Item 211, Appendix D.

NOTE

In order to install blind bolts, flange may require trimming (Figure 2-4-122-1, Sheet 2, Section A-A).

- n. If required, trim flange as necessary (PARA 2-6-3).

NOTE

- On engine cowling assemblies, 70302-02141-041, -042, -044, -045, -048, and -049, and 70070-30308-041 through -046, hardware used to secure fitting is dependent upon engine cowling configuration. Configuration 1 uses blind bolts and washers to secure support fitting. Configuration 2 uses pin-rivets and collars to secure support fitting. Refer to Figure 2-4-122-1, Sheet 2, Section A-A to determine configuration.
- On engine cowling assemblies, 70302-02141-052 and -053, support fitting is secured using blind bolts and washers.
- o. Secure support fitting to engine cowling using blind bolts and washers or pin-rivets and collars (Figure 2-4-122-1, Sheet 2, Section A-A).
- p. Touch up paint as required (PARA 2-6-5).
- q. If engine cowling was modified per step c., reidentify cowling as follows:
 - (1) 70302-02141-041 becomes 70070-30308-041

- | | |
|---|---|
| (2) 70302-02141-042 becomes 70070-30308-042 | (5) 70302-02141-048 becomes 70070-30308-045 |
| (3) 70302-02141-044 becomes 70070-30308-043 | (6) 70302-02141-049 becomes 70070-30308-046 |
| (4) 70302-02141-045 becomes 70070-30308-044 | |

2-4-123 ENGINE COMPARTMENT HEAT SHIELDS.

PARAGRAPH BREAKDOWN

	PAGE
2-4-123.1 Replace Engine Compartment Heat Shields	2-4-123-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Protective Caps and Plugs

2-4-123.1 Replace Engine Compartment Heat Shields.

NOTE

Replacement of right and left engine compartment heat shields is the same.

2-4-123.1.1 Remove.

- Open engine cowling.
- Disconnect engine manifold drain hose from elbow on top side of engine compartment rear deck (Figure 2-4-123-1, Detail A).
- Disconnect engine combustion drain hose from elbow on top side of engine compartment rear deck.
- Install protective caps and plugs on elbows and hoses.
- Remove screws and washers securing engine compartment heat shield to deck (Figure 2-4-123-1, Detail B).
- Slightly bend rear end of heat shield up enough to clear elbows in deck. Remove heat shield, spacers, washers, and packings from deck.

2-4-123.1.2 Install.

- Slightly bend rear end of engine compartment heat shield up enough to clear elbows in engine compartment deck and position heat shield over elbows on deck (Figure 2-4-123-1, Detail B).
- Position spacers, washers, and packing beneath heat shield at each attachment hole and secure with screws and washers.
- Remove protective caps and plugs from elbows and hoses.
- Connect engine combustion drain hose to elbow on top side of engine compartment rear deck. **TORQUE B-NUT TO 135 - 150 INCH-POUNDS** (Figure 2-4-123-1, Detail A).
- Connect engine manifold drain hose to elbow on top side of engine compartment rear deck. **TORQUE B-NUT TO 144 - 232 INCH-POUNDS.**
- Make sure area is clean and free of foreign material before closing up.
- Close and latch engine cowling.

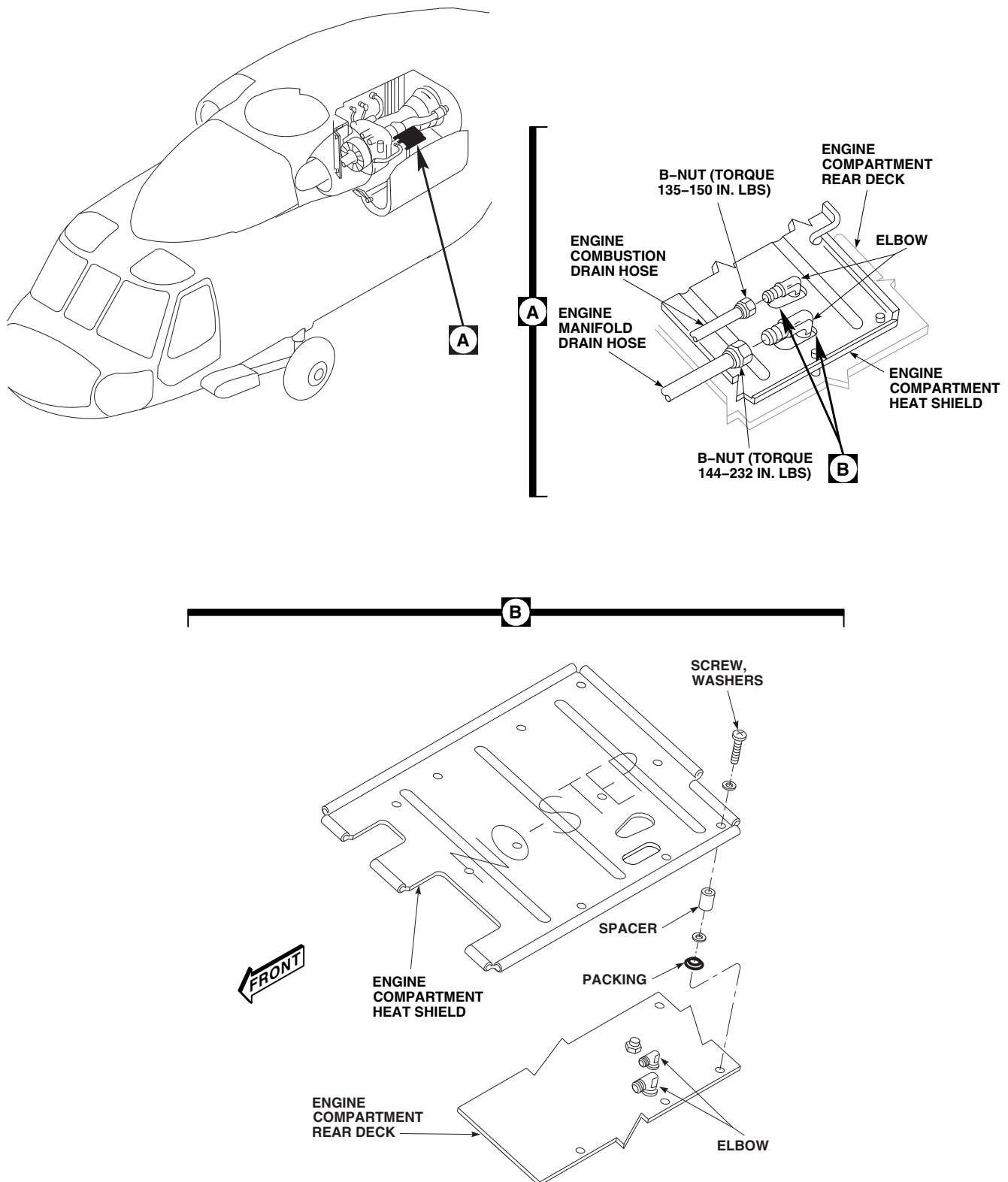


FIGURE 2-4-123-1. REPLACE ENGINE COMPARTMENT HEAT SHIELDS

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2-4-123-2

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2-4-124 TRANSITION SECTION AVIONICS DOOR.

PARAGRAPH BREAKDOWN

	PAGE
2-4-124.1 Replace Transition Section Avionics Door	2-4-124-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Digital Low-Resistance Ohmmeter

References

- PARA 1-6-16
- PARA 2-4-130

2-4-124.1 Replace Transition Section Avionics Door.

2-4-124.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Unlatch and open transition section avionics door (Figure 2-4-124-1, Detail A).
- Remove screws and washers securing retainers to inside of airframe and remove retainers (Figure 2-4-124-1, Detail B).
- Support door in open position.
- Disengage outboard end of strut from lock fitting on avionics door.
- Lower door to closed position.
- Remove clevis pins securing hinge halves together and remove avionics door.

2-4-124.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).

- Place avionics door in open position on helicopter, engaging hinge halves.
- Line up hinges with airframe. Install clevis pins (Figure 2-4-124-1, Detail B).
- Install retainers on inside of airframe and secure with screws and washers.
- Support door in open position.
- Attach outboard end of strut to lock fitting on avionics door.
- Make sure area is clean and free of foreign material before closing up.
- Close and latch avionics door.
- Perform EME resistance check of avionics door as follows (Figure 2-4-124-1, Detail A):

NOTE

Resistance measurements must be taken at least three times.

- Using digital low-resistance ohmmeter, measure resistance from structure

between latches on inside of avionics door and adjacent door jamb. **RESISTANCE SHALL BE 22.5 MILLIOHMS OR LESS.**

- (2) If resistance is more than 22.5 milliohms, replace avionics door.

- j. Check if door is centered within door opening. If centering is required, adjust door by loosening screws and nuts that hold door hinge fittings and reposition door as necessary. **RETIGHTEN SCREWS.**

- k. If required, adjust door latch (PARA 2-4-130). ■

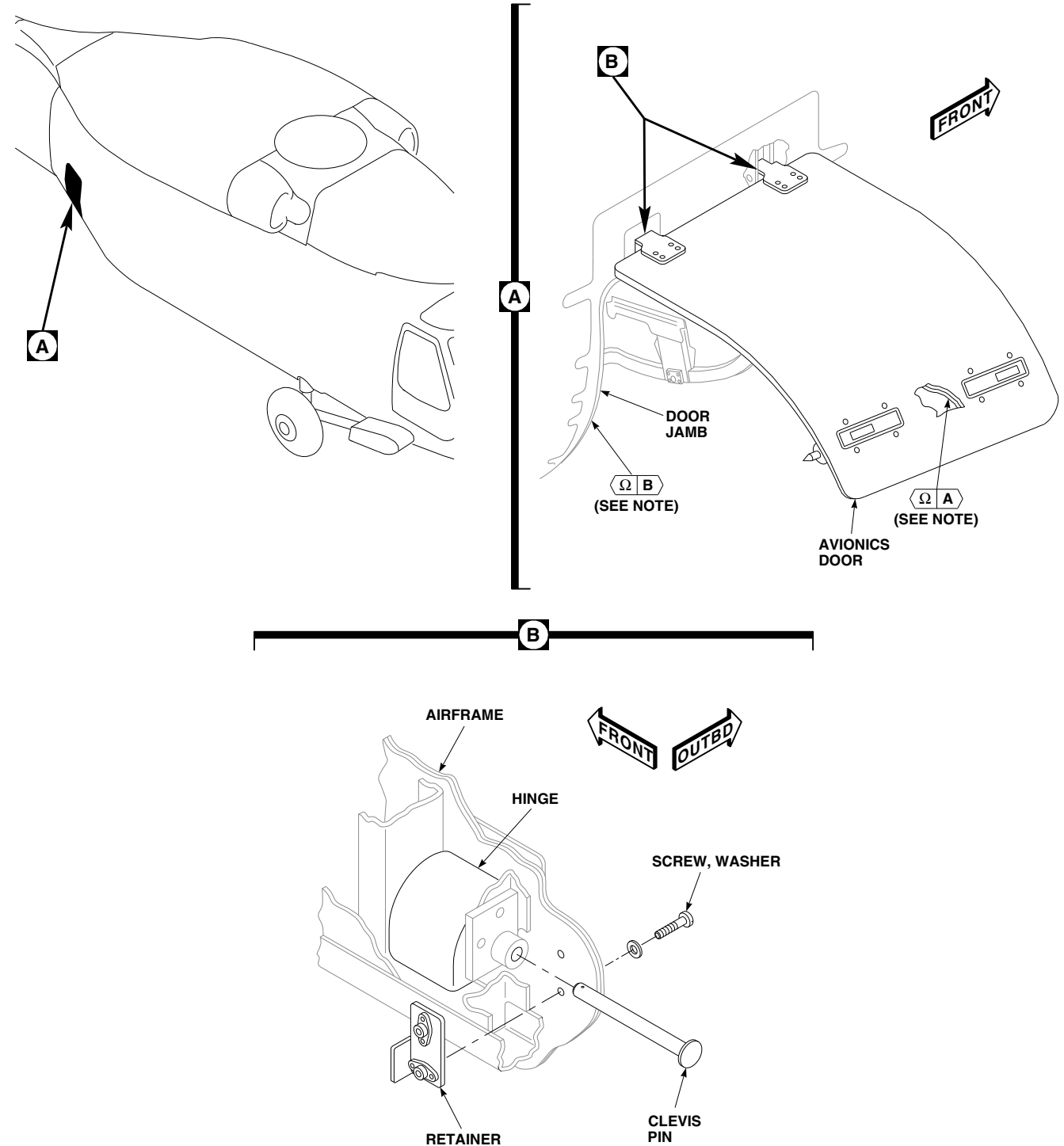


FIGURE 2-4-124-1. REPLACE TRANSITION SECTION AVIONICS DOOR

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2-4-125 TRANSITION SECTION AVIONICS DOOR STRUT.

PARAGRAPH BREAKDOWN

	PAGE
2-4-125.1 Replace Transition Section Avionics Door Strut	2-4-125-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit

References

- PARA 1-6-16

2-4-125.1 Replace Transition Section Avionics Door Strut.

2-4-125.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Unlatch and open transition section avionics door (Figure 2-4-125-1, Detail A).
- Support door in open position.
- Disengage outboard end of strut from lock fitting on avionics door.
- Remove clevis pin and washer from inboard end of strut from eyebolt on fitting. Remove strut.

2-4-125.1.2. Install.

- Install clevis end of strut on eyebolt on support with pin. Install washer and cotter pin (Figure 2-4-125-1, Detail A).
- Support door in open position.
- Attach outboard end of strut to lock fitting on avionics door.
- Make sure area is clean and free of foreign material before closing up.
- Close door and fasten latches.

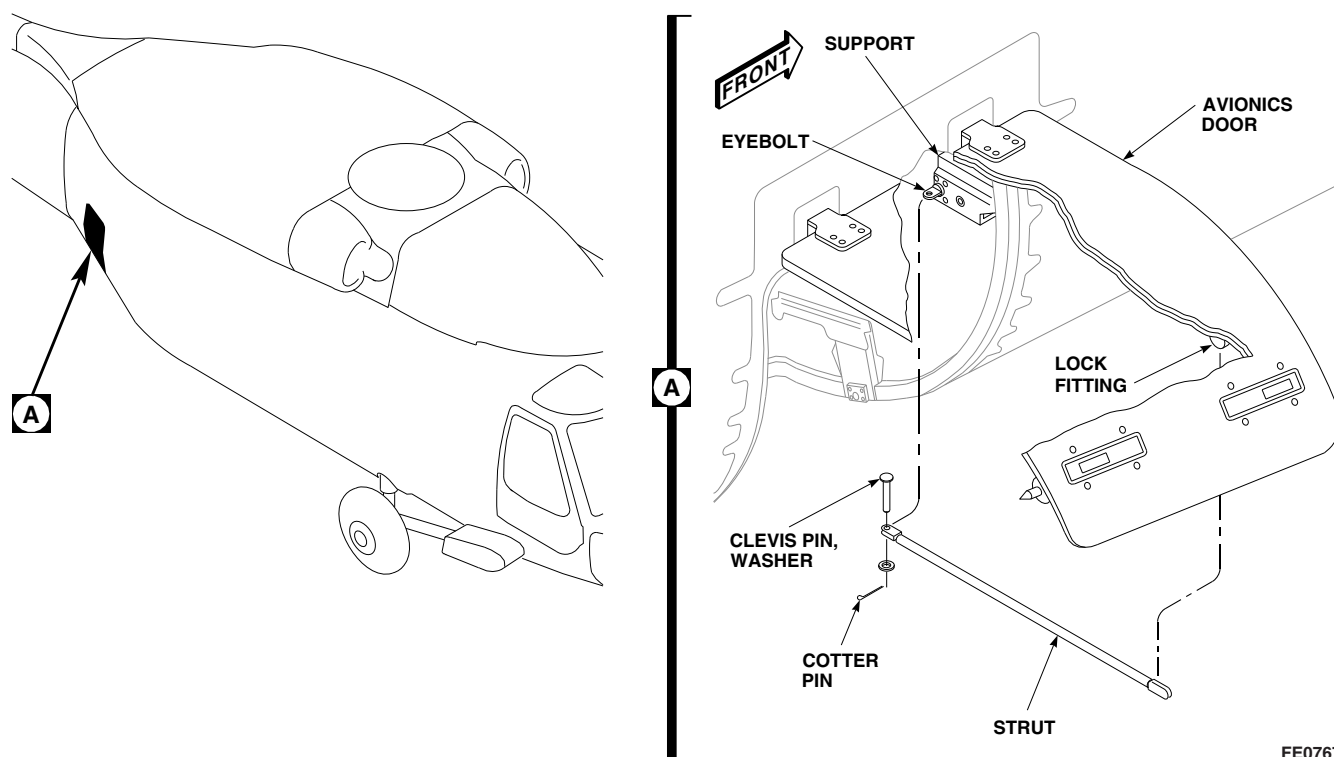


FIGURE 2-4-125-1. REPLACE TRANSITION SECTION AVIONICS DOOR STRUT

2-4-126 TRANSITION SECTION AVIONICS DOOR EYEBOLT AND SUPPORT.

PARAGRAPH BREAKDOWN

	PAGE
2-4-126.1 Replace Transition Section Avionics Door Eyebolt and Support	2-4-126-0

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Drill Set

References

- PARA 2-4-125

2-4-126.1 Replace Transition Section Avionics Door Eyebolt and Support.

Eyebolt should turn freely after cotter pin is installed.

2-4-126.1.1. Remove.

- Remove avionics door strut (PARA 2-4-125).
- Drill out rivets securing support to bulkhead (Figure 2-4-126-1, Detail B). Remove support with eyebolt.
- Remove nut, washers, and eyebolt from support.

- Install eyebolt, washers, and nut on support. **TIGHTEN NUT FINGERTIGHT.** If cotter pin hole in eyebolt does not line up with slots in nut, loosen nut to next slot to install cotter pin. Install cotter pin (Figure 2-4-126-1, Detail B).
- Install support on bulkhead and rivet in place.
- Install avionics door strut (PARA 2-4-125).

2-4-126.1.2. Install.



Equipment can be damaged if eyebolt nut is too tight. Do not overtighten nut.

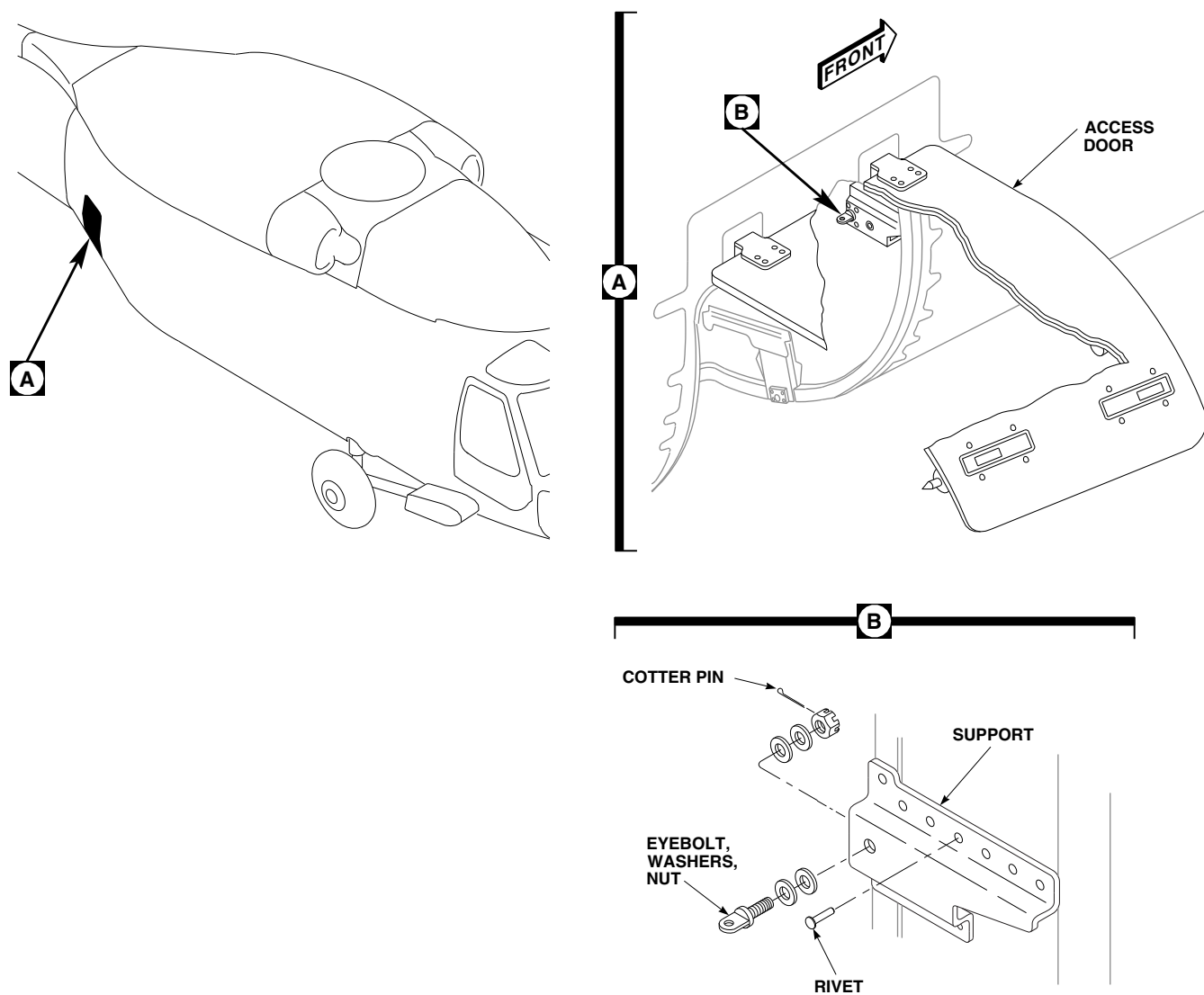
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FIGURE 2-4-126-1. REPLACE TRANSITION SECTION AVIONICS DOOR EYEBOLT AND SUPPORT

2-4-127 TRANSITION SECTION AVIONICS DOOR STRUT LOCK FITTING.

PARAGRAPH BREAKDOWN

	PAGE
2-4-127.1 Replace Transition Section Avionics Door Strut Lock Fitting	2-4-127-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Lockwire, Item 196, Appendix D

References

- Appendix D

2-4-127.1 Replace Transition Section Avionics Door Strut Lock Fitting.

2-4-127.1.1. Remove.

- Unlatch and open transition section avionics door (Figure 2-4-127-1, Detail A).
- Support door in open position while replacing strut lock fitting.
- Disengage outboard end of strut from lock fitting on avionics door.
- Remove bolts, washers, and lock fitting from avionics door (Figure 2-4-127-1, Detail B).

2-4-127.1.2. Install.

- Install strut lock fitting on avionics door with bolts and washers (Figure 2-4-127-1, Detail B).
- TORQUE BOLTS TO 105 - 115 INCH-POUNDS.** Lockwire, Item 196, Appendix D, boltheads together.
- Attach outboard end of strut to lock fitting on avionics door.
- Make sure area is clean and free of foreign material.
- Close door and fasten latches.

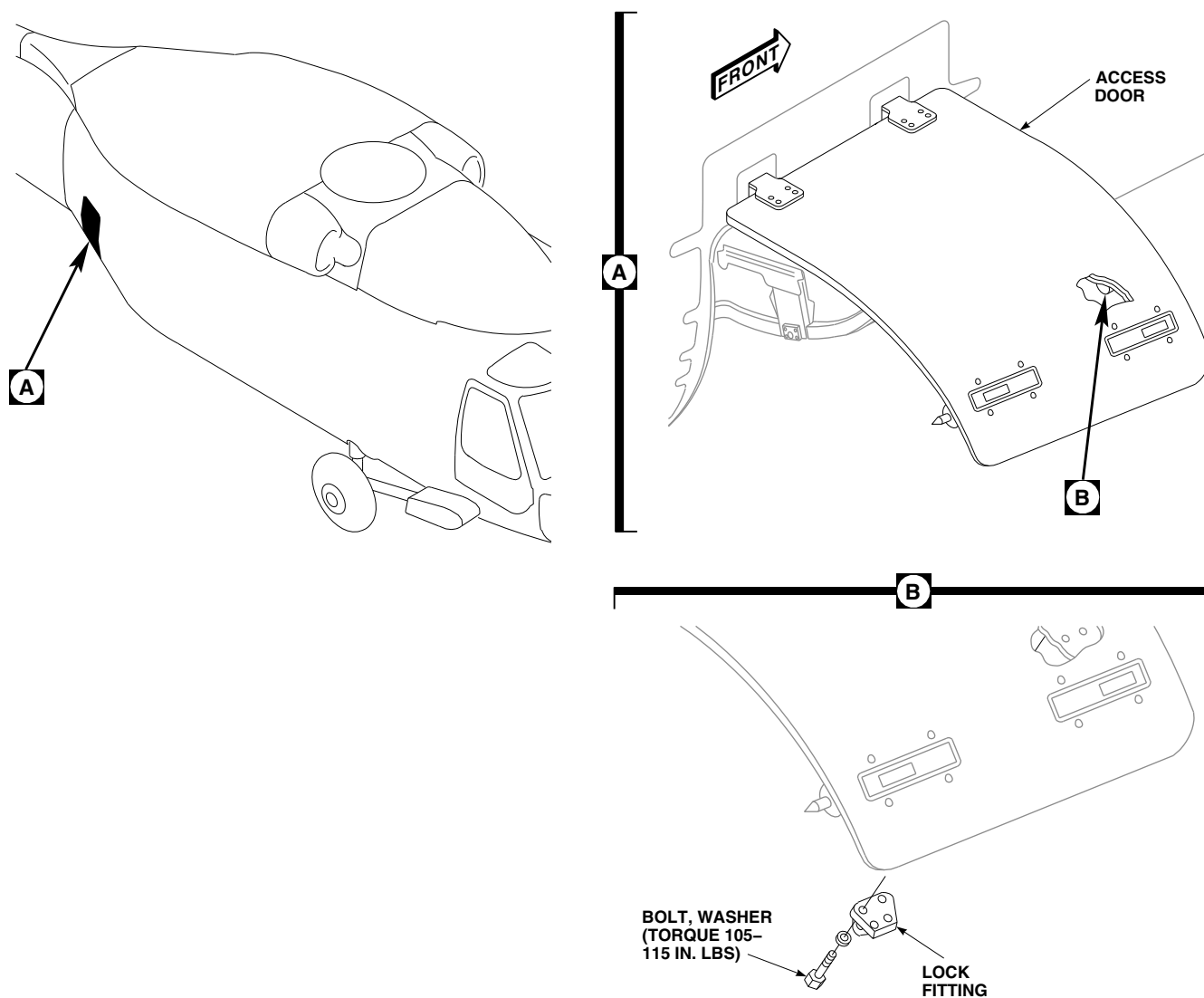
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FIGURE 2-4-127-1. REPLACE TRANSITION SECTION AVIONICS DOOR STRUT LOCK FITTING

2-4-128 TRANSITION SECTION AVIONICS DOOR HINGES.

PARAGRAPH BREAKDOWN

	PAGE
2-4-128.1 Replace Transition Section Avionics Door Hinges	2-4-128-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Adhesive, Item 38, Appendix D
- Machinery Towel, Item 211, Appendix D

- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-130

2-4-128.1 Replace Transition Section Avionics Door Hinges.

NOTE

Replacement of both transition section avionics door hinges is the same.

2-4-128.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Unlatch and open transition section avionics door (Figure 2-4-128-1, Detail A).
- Remove screws and washers securing retainer to inside of airframe and remove retainer (Figure 2-4-128-1, Detail B).
- Support door in open position when replacing hinge.

- Remove clevis pin securing hinge halves together.
- Using nonmetallic scraper, remove sealing compound from hardware if necessary.
- Remove screws, washers, sealing washers, and nuts securing hinge and laminated shim, if installed, to avionics door and remove hinge and shim (Figure 2-4-128-1, Detail C).

2-4-128.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Position door on aft transition section of helicopter.
- Support door in open position.
- Line up hinge with airframe. Install clevis pin (Figure 2-4-128-1, Detail B).

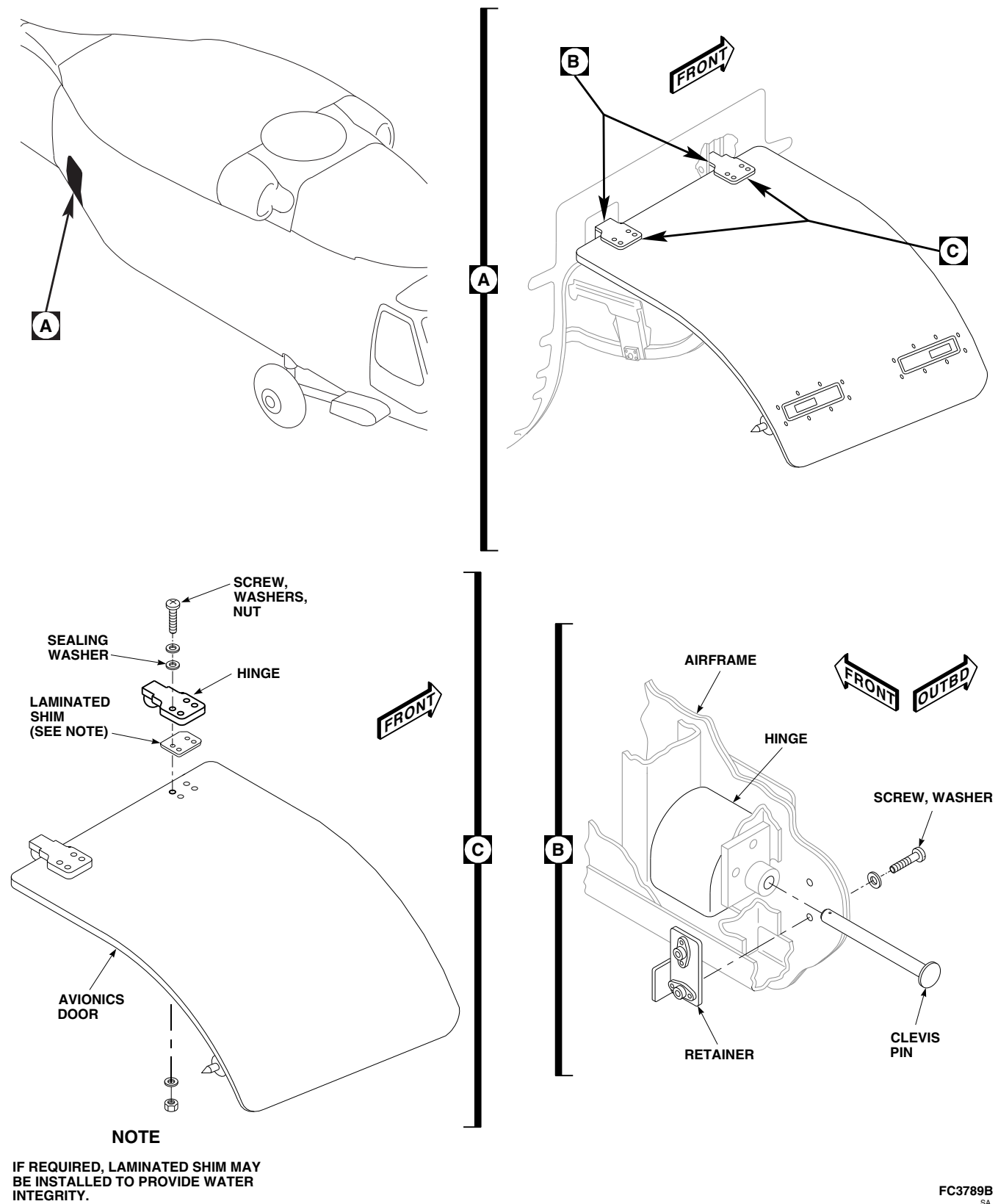


FIGURE 2-4-128-1. REPLACE TRANSITION SECTION AVIONICS DOOR HINGES

- e. Install retainer on inside of airframe and secure with screws and washers.
- f. Wipe hinge and hardware clean with machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.
- g. Inspect sealing washers for missing or unserviceable seals. **NONE ALLOWED.** If washers have missing or unserviceable seals, replace sealing washers.
- i. Coat threads of screws with sealing compound, Item 292, Appendix D.
- j. Immediately install hinge and laminated shim, if required, on avionics door with screws, washers, sealing washers, and nuts (Figure 2-4-128-1, Detail C).
- k. Make sure area is clean and free of foreign material before closing up.

NOTE

If required, laminated shim may be installed to provide water integrity.

- h. Apply adhesive, Item 38, Appendix D, to raised pad area on avionics door. Adhesive is applied between hinge and door or laminated shim as required.
- l. Close and latch transition section avionics door and check if door is centered within door opening. If centering is required, adjust door by loosening screws and nuts that hold door hinge fittings and reposition door as necessary. **RETIGHTEN SCREWS.**
- m. If required, adjust door latch (PARA 2-4-130).

2-4-129 TRANSITION SECTION AVIONICS DOOR SEALS.

PARAGRAPH BREAKDOWN

	PAGE
2-4-129.1 Replace Transition Section Avionics Door Seals	2-4-129-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Abrasive Cloth, Item 9, Appendix D
- Adhesive, Item 37, Appendix D
- Adhesive, Item 55, Appendix D

- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Plastic Strip, Item 237, Appendix D
- Sealing Compound, Item 292, Appendix D
- Stirring Stick, Beverage, Item 315C, Appendix D

References

- Appendix D
- PARA 2-4-124

2-4-129.1 Replace Transition Section Avionics Door Seals.

2-4-129.1.1 Remove.

- Remove transition section avionics door (PARA 2-4-124).
- Using nonmetallic scraper, remove old seals from door and jamb (Figure 2-4-129-1, Detail A).
- Using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, and nonmetallic scraper, remove remaining adhesive.

2-4-129.1.2 Install.

- Fit new seal to door with spliced joint at bottom center. Locate and remove 30° - 45° sectional cuts to keep seal from bunching or stretching around corners (Figure 2-4-129-1, Detail A).
- Split door seal to prevent disbonding in bolt recess area.
- Fit new seal to jamb with spliced joints at top and bottom center.
- Using abrasive cloth, Item 9, Appendix D, roughen all adhesive contact surfaces of new seals.

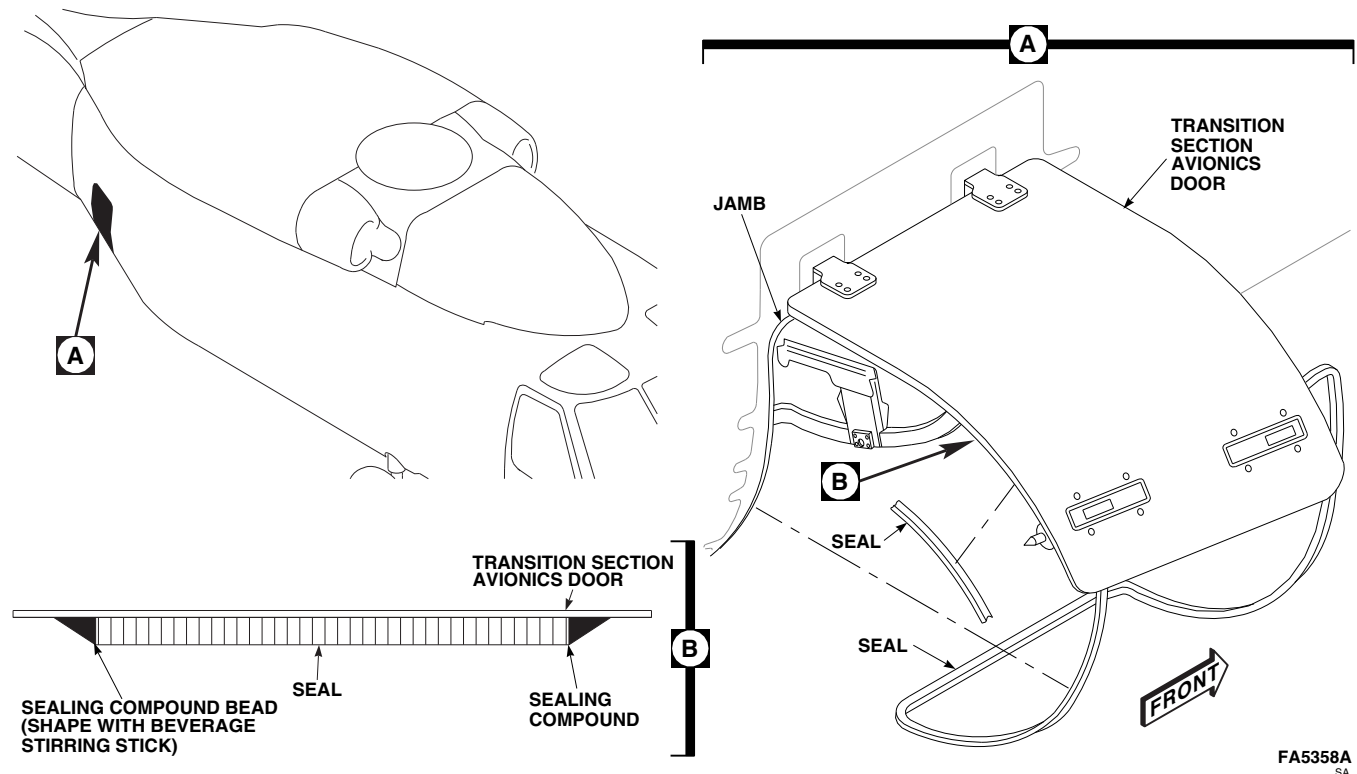


FIGURE 2-4-129-1. REPLACE TRANSITION SECTION AVIONICS DOOR SEALS

- e. Wipe all adhesive contact areas with machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.
- f. Line up jamb seal joints on flat surface covered with plastic strip, Item 237, Appendix D.
- g. Wipe areas with machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.
- h. Apply adhesive, Item 55, Appendix D, to seal splice.
- i. Apply even coat of adhesive, Item 37, Appendix D, to door, jamb and seal joining surfaces. Allow adhesive to air-dry for 30 minutes or until tacky.
- j. Secure seal to door and to jamb. Work out gaps, trapped air, and strained areas all the way around seal installation.
- k. Remove extra adhesive with machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.
- l. Mix and apply bead of sealing compound, Item 292, Appendix D, around inner and outer edges of door seal installation.
- m. Using beverage stirring stick, Item 315C, Appendix D, shape bead of sealing compound as necessary (Figure 2-4-129-1, Detail B).
- n. Allow sealing compound to cure at least 24 hours.
- o. Install transition section avionics door (PARA 2-4-124).

2-4-130 TRANSITION SECTION AVIONICS DOOR LATCHES.

PARAGRAPH BREAKDOWN

	PAGE
2-4-130.1 Replace Transition Section Avionics Door Latches	2-4-130-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Grease Pencil
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Grease, Item 150, Appendix D
- Machinery Towel, Item 211, Appendix D

- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 283, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-6-16

2-4-130.1 Replace Transition Section Avionics Door Latches.

NOTE

Replacement of both transition section avionics door latches is the same.

2-4-130.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Unlatch and open transition section avionics door.
- Remove screws, washers, and sealing washers securing cover to avionics door. Remove cover (Figure 2-4-130-1, Sheet 1, Detail B).

- If necessary, remove sealing compound from hardware using nonmetallic scraper.
- Remove screws, washers, sealing washers, and nuts securing latch and support angles to avionics door. Remove latch and support angles as an assembly.

2-4-130.1.2 Repair.

- Remove screws, washers, radius blocks, and nuts securing latch. Separate support angles and remove latch (Figure 2-4-130-1, Sheet 1, Detail C).
- Remove laminated shims from support angles.
- Lubricate moving parts in latch with grease, Item 150, Appendix D.

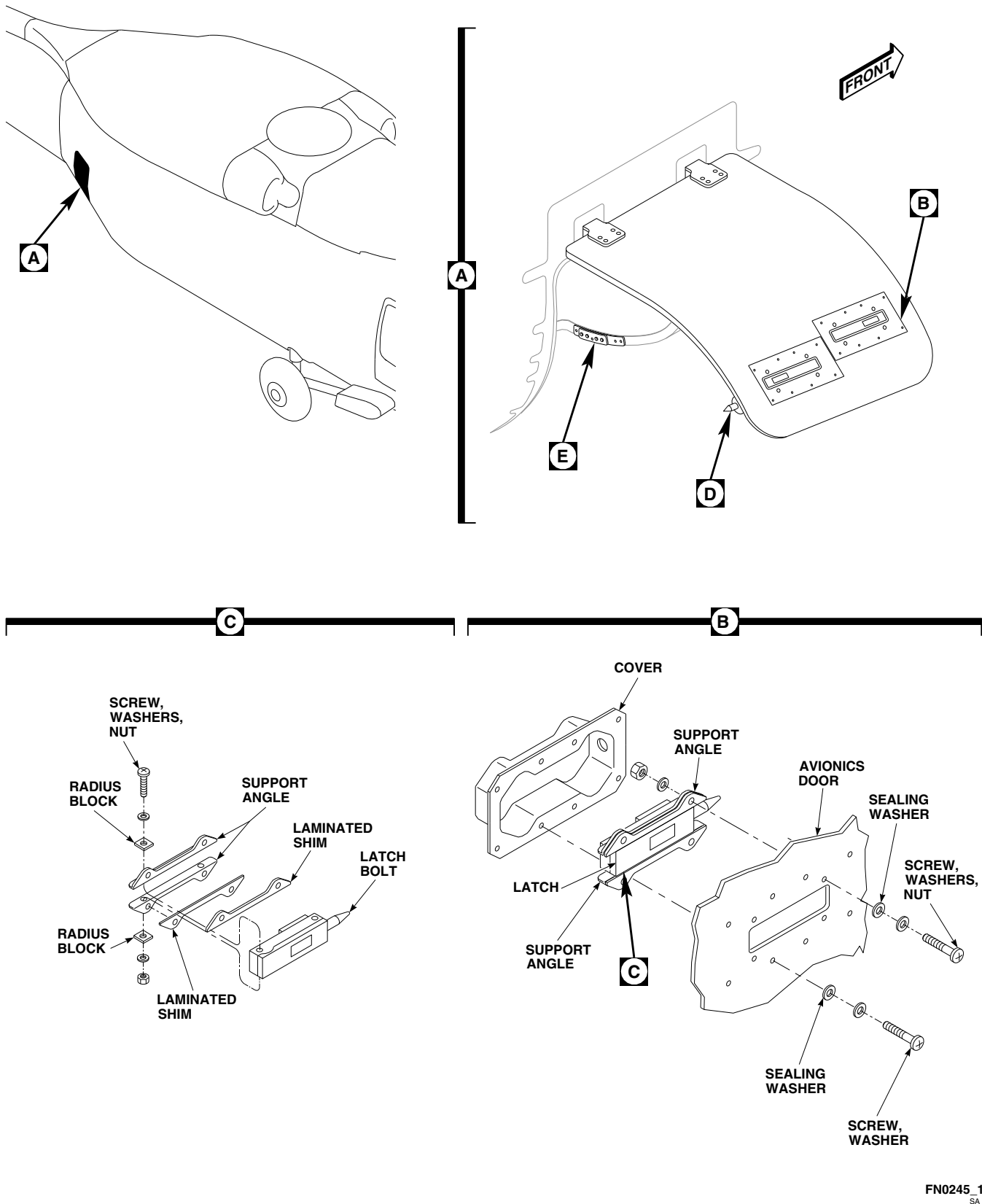


FIGURE 2-4-130-1. REPLACE TRANSITION SECTION AVIONICS DOOR LATCHES
(SHEET 1 OF 2)

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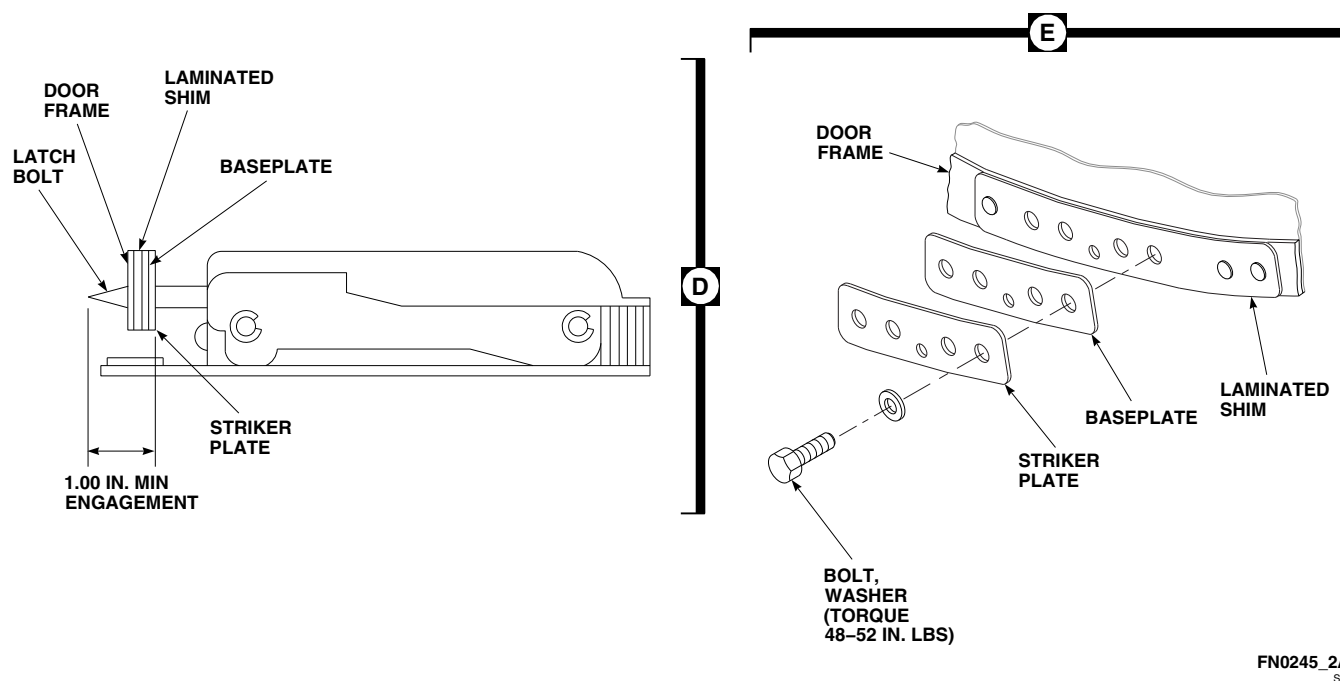


FIGURE 2-4-130-1. REPLACE TRANSITION SECTION AVIONICS DOOR LATCHES (SHEET 2 OF 2)

- d. Install laminated shims on support angles. Assemble latch and support angles and secure with screws, washers, radius blocks, and nuts.

2-4-130.1.3 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Wipe hardware clean using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.
- c. Inspect sealing washers for missing or unserviceable seals. **NONE ALLOWED.** If washers have missing or unserviceable seals, replace sealing washers.
- d. Coat threads of screws with sealing compound, Item 292, Appendix D.
- e. Position latch and support angles, as an assembly, on avionics door and secure with screws, washers, sealing washers, and nuts (Figure 2-4-130-1, Sheet 1, Detail B).

- f. Position cover over latch and secure to avionics door with screws, washers, and sealing washers.
- g. Apply fillet of sealing compound, Item 283, Appendix D, to edge of latch and avionics door.
- h. Check, and if necessary, adjust door (PARA 2-4-130.1.4).

2-4-130.1.4 Check/Adjust.

- a. Coat latch bolts with grease pencil to provide mark.
- b. Close avionics door and fully engage latches. Reopen door. Measure mark on latch bolts. **LATCH BOLTS MUST ENGAGE STRIKER PLATES ON DOOR FRAME A MINIMUM OF 1.00-INCH** (Figure 2-4-130-1, Sheet 2, Detail D). If limit is met, close and latch transition section avionics door. If limit is not met, adjust latch(es) as follows:

- (1) Determine difference between actual engagement of latch bolt and required minimum engagement.
- (2) Remove bolts and washers securing striker plate and baseplate to door frame. Remove plates and if installed, remove laminated shim from under baseplate (Figure 2-4-130-1, Sheet 2, Detail E).

NOTE

New laminated shim is 1/8-inch thick and consists of 0.002-inch laminations.

- (3) Peel new laminated shim so latch bolt will engage 1.00-inch or more beyond striker

plate surface (Figure 2-4-130-1, Sheet 2, Detail D).

NOTE

Serrated faces of striker and baseplate must face each other. Baseplate has largest latch bolthole.

- (4) Position laminated shim, baseplate, and striker plate on door frame and secure with bolts and washers.
- (5) **TORQUE BOLTS TO 48 - 52 INCH-POUNDS.**
- (6) Repeat step a. and step b. until proper latch bolt engagement is obtained.

2-4-131 TAIL ROTOR DRIVE SHAFT COVERS.

PARAGRAPH BREAKDOWN

	PAGE
2-4-131.1 Replace Tail Rotor Drive Shaft Covers	2-4-131-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Drill Set
- Rivet Gun

Materials/Parts

- Corrosion Compound, Item 108, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Lubricating Oil, Item 203, Appendix D

- Paint, Polyurethane, Item 222A, Appendix D
- Scouring Pad, Item 270, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 2-6-3
- PARA 9-4-72
- PARA 9-4-73

2-4-131.1 Replace Tail Rotor Drive Shaft Covers.

NOTE

Replacement of both tail rotor drive shaft covers is the same, except as noted.

2-4-131.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Unfasten studs and open tail rotor drive shaft cover (Figure 2-4-131-1, Sheet 1, Detail A).
- Remove nut, washer, and screw securing stop (Figure 2-4-131-1, Sheet 2, Detail B). Remove stop.

- If removing rear cover, perform the following:
 - Remove electro-luminescent formation light (PARA 9-4-72).
 - Remove infrared formation light (PARA 9-4-73).
- Remove hinge pin and tail rotor drive shaft cover.

2-4-131.1.2 Repair.

- Manually remove corrosion from hinges, using scouring pad, Item 270, Appendix D. **PITS ON HINGES SHALL NOT EXCEED 0.005-INCH DEEP AFTER CLEANUP.**
- Mix solution of 1 part corrosion compound, Item 108, Appendix D, and 1 part water.

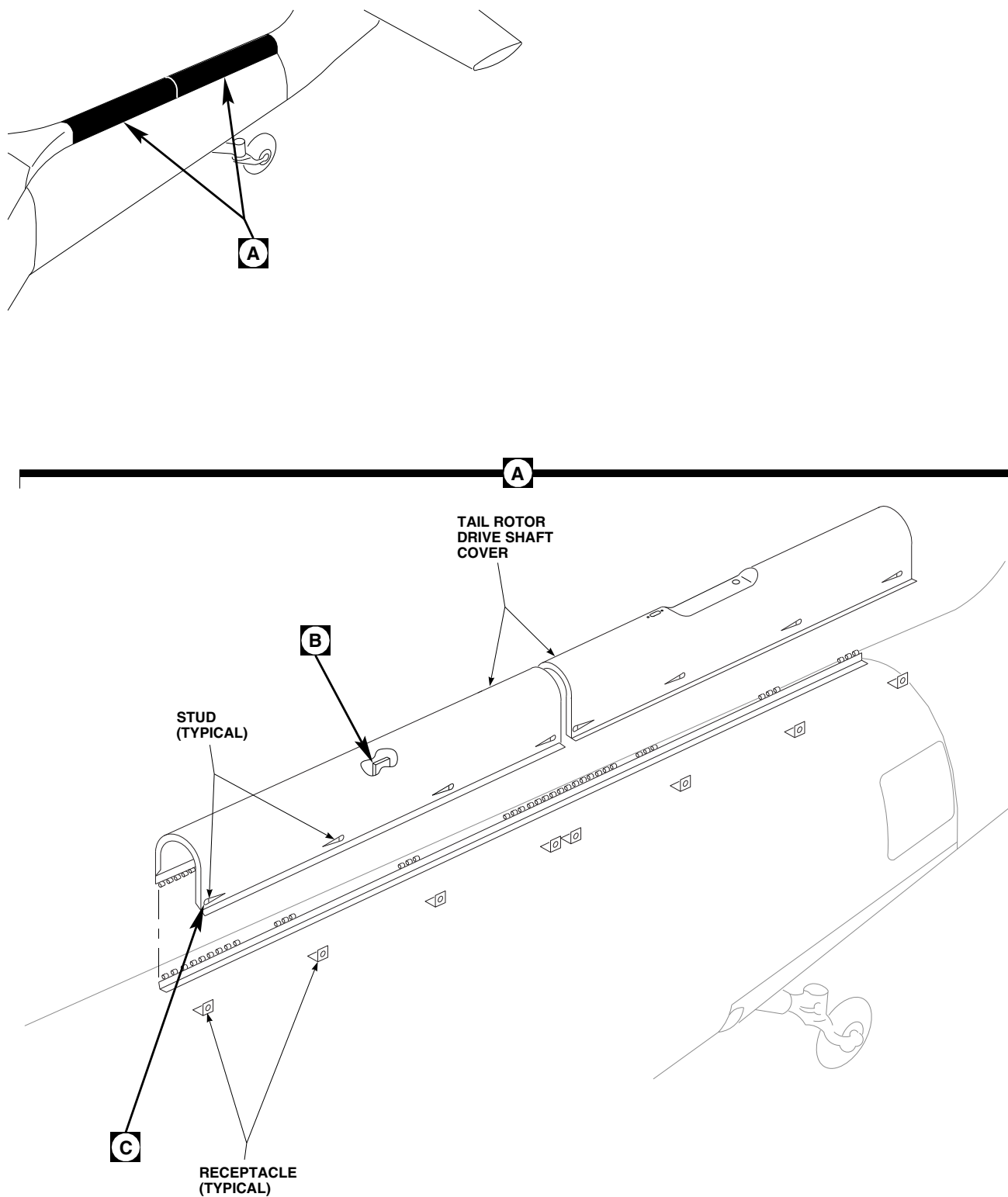
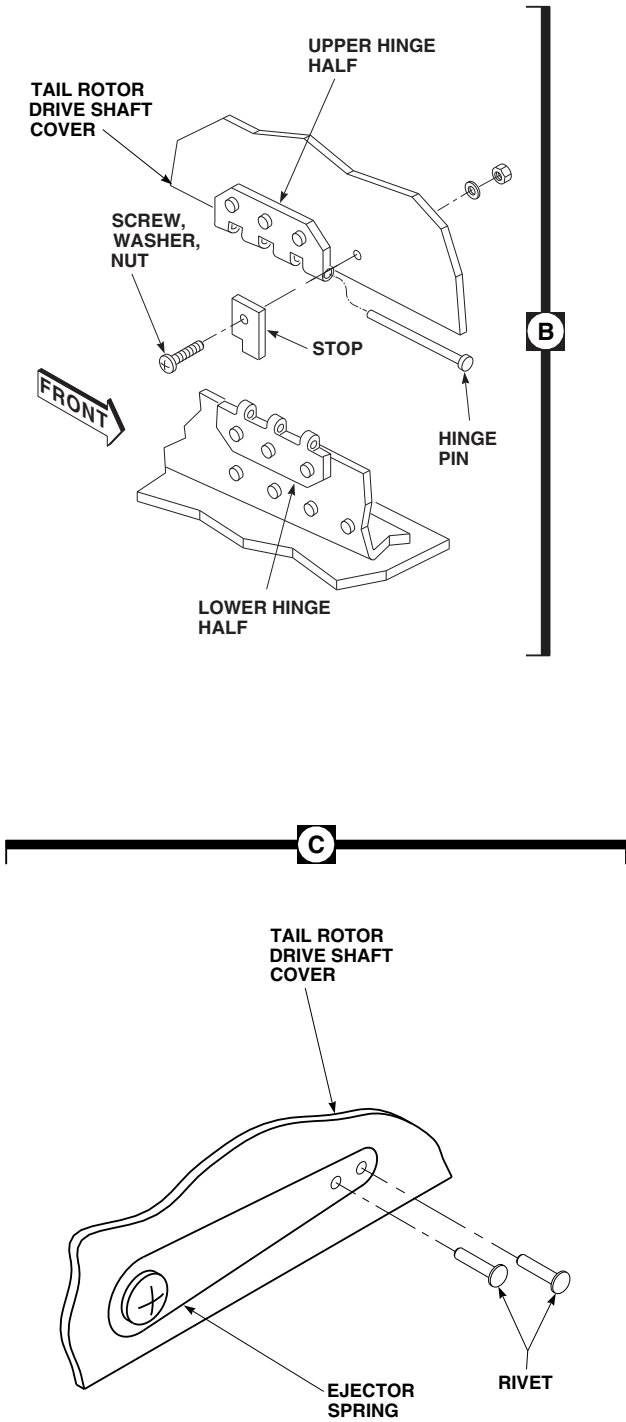
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FIGURE 2-4-131-1. REPLACE TAIL ROTOR DRIVE SHAFT COVERS (SHEET 1 OF 2)



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FIGURE 2-4-131-1. REPLACE TAIL ROTOR DRIVE SHAFT COVERS (SHEET 2 OF 2)

- c. Apply corrosion compound solution to corroded areas of hinges.
- d. Rinse thoroughly with clean water.

CAUTION

Paint on bearing surfaces can cause improper operation of hinges and fasteners. Keep paint off of bearing surfaces.

- e. Touch up paint using epoxy primer coating kit, Item 135, Appendix D, and polyurethane paint, Item 222A, Appendix D.
- f. Coat hinge pins with lubricating oil, Item 203, Appendix D.

2-4-131.1.3 Install.

- a. Line up upper hinge half on tail rotor drive shaft cover with lower hinge half and install hinge pin with swaged end on pin stop side (Figure 2-4-131-1, Sheet 2, Detail B).
- b. Close tail rotor drive shaft cover and check that studs line up with receptacles (Figure 2-4-131-1, Sheet 1, Detail A). If they do not, perform the following:
 - (1) Unfasten studs and open tail rotor drive shaft cover.
 - (2) Remove hinge pin and tail rotor drive shaft cover (Figure 2-4-131-1, Sheet 2, Detail B).
 - (3) Using drill, drill out rivets securing ejector spring to tail rotor drive shaft cover (Figure 2-4-131-1, Sheet 2, Detail C).
 - (4) Fill in rivet holes (PARA 2-6-3).

- (5) Line up hinge on tail rotor drive shaft cover with lower hinge half and install hinge pin with swaged end on pin stop side (Figure 2-4-131-1, Sheet 2, Detail B).
- (6) Close tail rotor drive shaft cover and secure studs (Figure 2-4-131-1, Sheet 1, Detail A).
- (7) Mark new holes using ejector spring as template.
- (8) Remove tail rotor drive shaft cover per step b. (1) and step b. (2).
- (9) Drill new holes.
- (10) Using rivet gun, install rivets securing ejector spring to tail rotor drive shaft cover (Figure 2-4-131-1, Sheet 2, Detail C).
- (11) Install tail rotor drive shaft cover per step b. (5) and step b. (6).
- c. If installing rear cover, perform the following:
 - (1) Install electro-luminescent formation light (PARA 9-4-72).
 - (2) Install infrared formation light (PARA 9-4-73).
- d. Secure stop using screw, washer, and nut (Figure 2-4-131-1, Sheet 2, Detail B).
- e. Make sure area is clean and free of foreign material.
- f. Check open/close function of tail rotor drive shaft cover.
- g. Close tail rotor drive shaft cover and secure studs.

2-4-132 TAIL ROTOR PYLON.

PARAGRAPH BREAKDOWN

	PAGE
2-4-132.1 Replace Tail Rotor Pylon	2-4-132-2

INITIAL SETUP	• ASTM E1417	
Tools	• PARA 1-6-16	
• Aircraft Mechanic’s Toolkit	• PARA 1-7-11	
• Airframe Repairer’s Toolkit	• PARA 1-7-20	
• Cargo Straps	• PARA 2-4-133	
• Feeler Gage	• PARA 2-4-134	
• Fluorescent Penetrant Inspection Kit	• PARA 2-4-135	
• Hoist (Capacity to Support Tail Rotor Pylon)	• PARA 2-4-136	
• Sawhorses (or equivalent)	• PARA 2-4-137	
• Torque Wrench, 0 - 30 in. lbs	• PARA 2-4-138	
• Torque Wrench, 700 - 1600 in. lbs	• PARA 2-4-139	
• Vise, Soft-Padded	• PARA 2-4-140	
Materials/Parts	• PARA 2-4-141	
• Cleaning Cloth, Item 102A, Appendix D	• PARA 2-4-142	
• Protective Caps and Plugs	• PARA 2-6-5	
• Sealing Compound, Item 295A, Appendix D	• PARA 2-6-42	
• Sealing Compound, Item 301A, Appendix D	• PARA 6-4-31	
References	• PARA 6-4-41	
• Appendix D	• PARA 6-4-52	
• Appendix G	• PARA 11-4-106	
• Appropriate MTF		

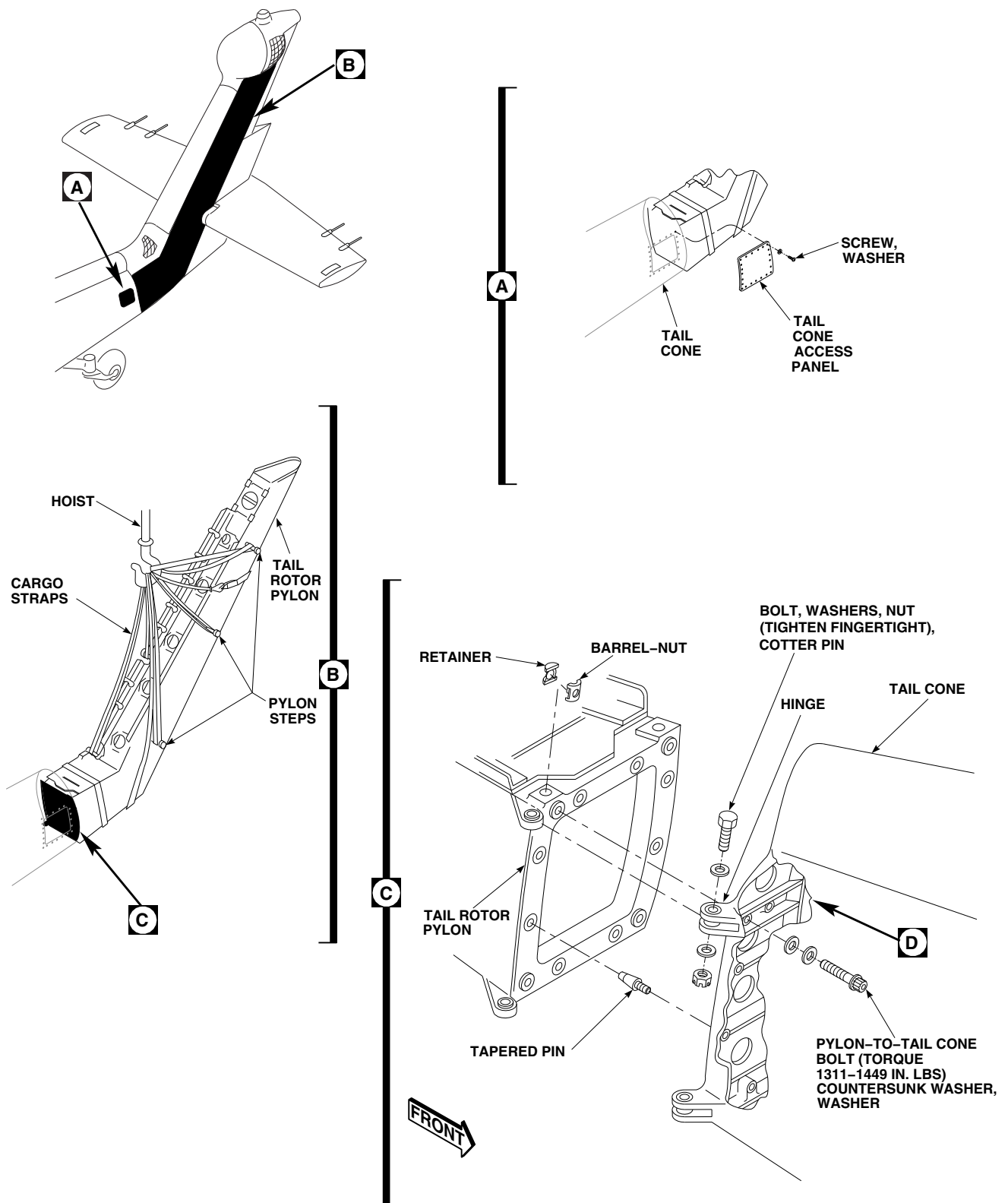
2-4-132.1 Replace Tail Rotor Pylon.**2-4-132.1.1 Remove.**

- a. Turn off all electrical power (PARA 1-6-16).
- b. Remove all pylon fairings and covers/antennas from pylon as follows:
 - (1) Tail rotor pylon lower fairing (PARA 2-4-133)
 - (2) Tail rotor pylon stabilator-to-pylon fairing (PARA 2-4-134)
 - (3) Tail rotor pylon cambered fairing (PARA 2-4-135)
 - (4) Tail rotor pylon trailing edge fairing/antenna (PARA 2-4-136)
 - (5) Tail gear box fairing (PARA 2-4-137)
 - (6) Tail rotor pylon leading edge cover/antenna (PARA 2-4-138)
 - (7) Intermediate gear box fairing (PARA 2-4-139)
- c. Remove screws and washers securing left and right tail cone access panels to tail cone (Figure 2-4-132-1, Sheet 1, Detail A).
- d. Remove stabilator (PARA 2-4-142).
- e. Remove tail rotor drive shaft section IV (PARA 6-4-41).
- f. Remove tail gear box (PARA 6-4-52).
- g. Remove intermediate gear box (PARA 6-4-31).
- h. Remove tail rotor pylon flight control cables (PARA 11-4-106).
- i. Disconnect electrical connectors, coaxial connectors, and bonding jumpers at pylon-to-tail cone disconnect.
- j. Install protective caps and plugs on electrical connectors.

- k. Extend upper pylon steps and secure cargo straps to pylon (Figure 2-4-132-1, Sheet 1, Detail B).
- l. Position hoist over pylon and secure hoist to cargo straps.

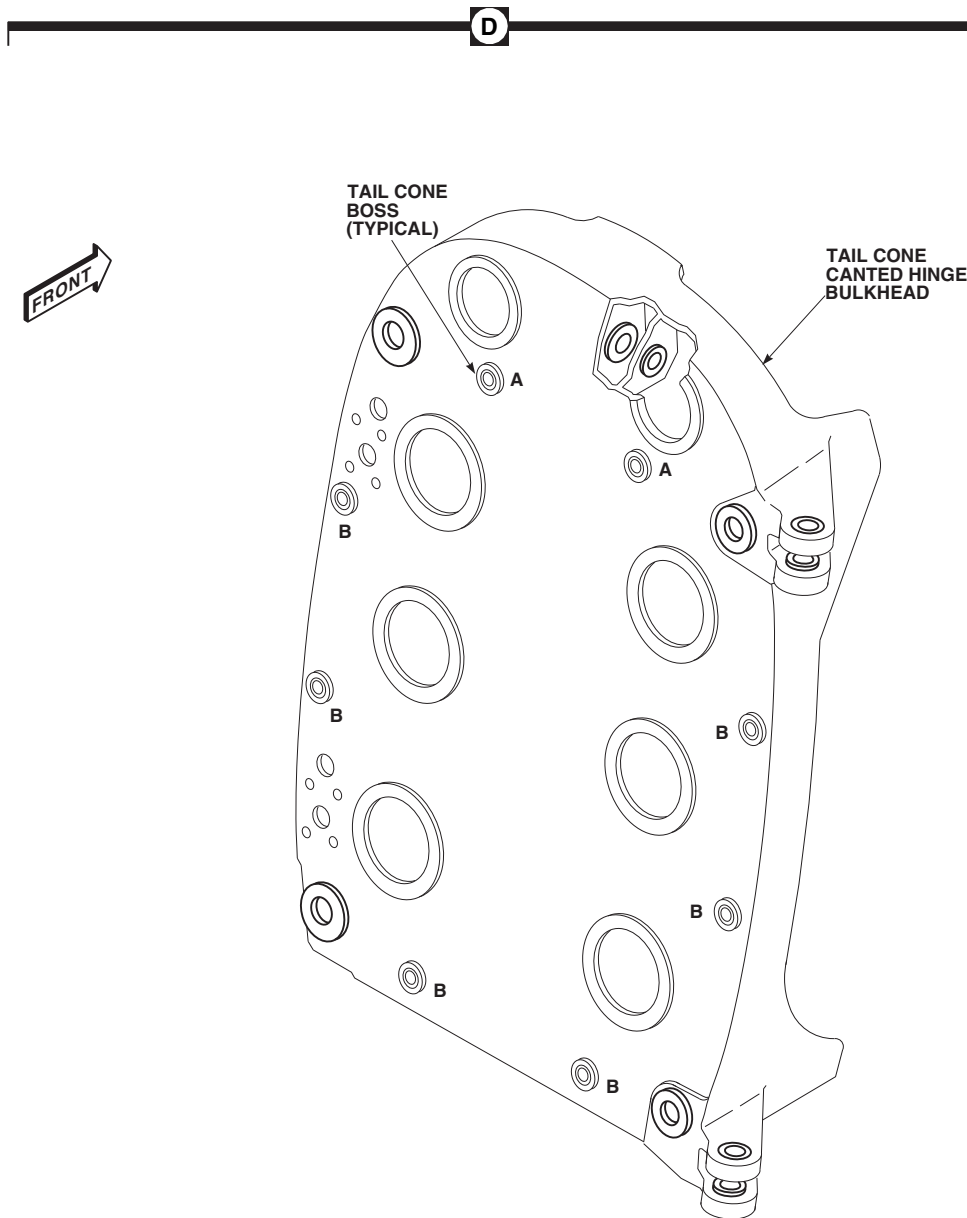
WARNING

- Injury to personnel and damage to equipment will result if caution is not taken when hoisting tail rotor pylon. Keep area clear when hoisting tail rotor pylon.
 - Hoist or hoist cable may break, causing injury to personnel or damage to equipment, if an improper capacity hoist is used. Pylon weighs 176 pounds. Select hoist capable of supporting tail rotor pylon structure.
 - Hoist cable may break, causing injury to personnel or damage to equipment, if hoist is used to free a sticking component being removed from helicopter. Do not use too much force when removing a component.
 - Hoist cable may be damaged or broken, causing injury to personnel or damage to equipment, if hook is allowed to contact pulley when removing component from helicopter. Do not allow hook assembly to contact pulley.
- m. Raise up on hoist to provide tension on straps, taking weight of pylon off pylon-to-tail cone bolts.
 - n. Remove bolts, washers, and nuts in pylon hinges (Figure 2-4-132-1, Sheet 1, Detail C).
 - o. Working within tail cone, back off tapered pins from pylon.
 - p. With one assistant operating hoist and another to guide pylon, remove pylon-to-tail cone bolts, countersunk washers, and washers.



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FIGURE 2-4-132-1. REPLACE TAIL ROTOR PYLON (SHEET 1 OF 2)

FA5432_2
SA**FIGURE 2-4-132-1. REPLACE TAIL ROTOR PYLON (SHEET 2 OF 2)**

NOTE

Adjust strap tension as required to balance pylon.

- q. Carefully separate pylon from tail cone, making sure all electrical and hydraulic connections have been disconnected.
- r. Slowly hoist pylon away from tail cone. Position pylon on padded sawhorses or equivalent.
- s. Disconnect straps from hoist and remove straps from pylon.
- t. Remove tapered pins from tail cone.
- u. Remove barrel-nuts and retainers from pylon.
- v. Remove sealing compounds from barrel-nuts, retainers, and tapered pins.
- w. If pylon is being replaced, remove tail rotor pylon steps (PARA 2-4-140) and tail rotor pylon lower step (PARA 2-4-141).

2-4-132.1.2 Inspect.

- a. Using fluorescent penetrant inspection kit (ASTM E1417), fluorescent penetrant inspect pylon attachment bolts and washers, tapered pins, barrel-nuts, and pylon hinge bolts, washers, and nuts for cracks. **NONE ALLOWED.** If cracked, replace part(s) as necessary.
- b. Check all four pylon-to-tail cone bolts and barrel-nuts for minimum breakaway torque as follows:

WARNING

FSCAP

Verification of barrel-nut and retainer installation and breakaway torque are critical characteristics. Verification of correct installation and torque is required.

NOTE

Make sure only square barrel-nuts are used.

- (1) Place barrel-nut in soft-padded vise.
- (2) Install bolt into barrel-nut until two threads protrude past nut.
- (3) Check breakaway torque by backing off bolt. **IF TORQUE NEEDED TO REMOVE BOLT IS LESS THAN 24 INCH-POUNDS, REPLACE BARREL-NUT AND RECHECK TORQUE. IF TORQUE IS STILL LESS THAN 24 INCH-POUNDS, REPLACE BOLT.**

WARNING

FSCAP

Radius pad installation is a critical characteristic. Make sure installation inspection is performed.

- c. Inspect canted bulkhead for damage and radius pad installation (PARA 2-6-42).

2-4-132.1.3 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. If replacement pylon is being installed, install tail rotor pylon steps (PARA 2-4-140) and tail rotor pylon lower step (PARA 2-4-141).
- c. Apply sealing compound, Item 301A, Appendix D, to ID of tail rotor pylon bores, barrel-nuts, retainers, and tapered pins.
- d. Remove excess sealing compound with cleaning cloth, Item 102A, Appendix D.
- e. Apply sealing compound, Item 295A, Appendix D, to accessible face of cavity.

WARNING**FSCAP**

Installation of barrel-nuts and retainers is a critical characteristic. Verification of correct installation and torque is required.

CAUTION

Serious damage to barrel-nut recess will result when bolt is torqued if barrel-nut is incorrectly installed. To prevent damage make sure barrel-nut is installed with barrel (round side) towards bolt-head.

NOTE

Make sure only barrel-nuts, RMLH2577, are used. These barrel-nuts are identified with letters "EN" or "SPS" stamped on an end (additional marking may also include "LH2577").

- f. Install barrel-nuts and retainers wet with sealing compound, Item 301A, Appendix D, in pylon. Barrel side of nut must face bolthead (Figure 2-4-132-1, Sheet 1, Detail C).
- g. Install tapered pins wet with sealing compound, Item 301A, Appendix D, in tail cone. Make sure pins are backed off to prevent interference when closing pylon.
- h. Secure cargo straps to pylon (Figure 2-4-132-1, Sheet 1, Detail B).
- i. Position hoist over pylon and secure hoist to straps.

WARNING

- Injury to personnel and damage to equipment will result if caution is not

taken when hoisting tail rotor pylon. Keep area clear when hoisting tail rotor pylon.

- Hoist or hoist cable may break, causing injury to personnel or damage to equipment, if an improper capacity hoist is used. Pylon weighs 176 pounds. Select hoist capable of supporting tail rotor pylon structure.
- Hoist cable may be damaged or broken, causing injury to personnel or damage to equipment, if hook is allowed to contact pulley when installing component on helicopter. Do not allow hook assembly to contact pulley.
- j. With one assistant operating hoist and another guiding pylon, carefully position pylon against tail cone.
- k. Coat mating surface of washers and pylon-to-tail cone bolts with sealing compound, Item 301A, Appendix D.
- l. Install countersunk washers (countersunk side facing bolthead) and washers on pylon-to-tail cone bolts (Figure 2-4-132-1, Sheet 1, Detail C).

WARNING**FSCAP**

Installation of pylon-to-tail cone bolts and washers and bolt torque are critical characteristics. Verification of correct installation and torque is required.

- m. Install four pylon-to-tail cone bolts, countersunk washers, and washers securing pylon to tail cone. Run bolts down until pylon evenly contacts tail cone. **DO NOT TORQUE MORE THAN 480 INCH-POUNDS AT THIS TIME.**
- n. Turn all tapered pins into mating bushings in pylon.

WARNING

FSCAP

Installation of pylon-to-tail cone bolts and washers and bolt torque are critical characteristics. Verification of correct installation and torque is required.

NOTE

Be careful not to damage tail rotor servo lines when torquing bolts.

- o. **TORQUE PYLON-TO-TAIL CONE BOLTS TO 1311 - 1449 INCH-POUNDS.**
- p. Apply torque stripe to bolts (PARA 2-6-5).
- q. Perform tail cone/tail rotor pylon gap check as follows (Figure 2-4-132-1, Sheet 2, Detail D):

NOTE

No gaps may exceed stated maximum allowable gap.

- (1) Working within tail cone, back off tapered pins from pylon.
- (2) Using feeler gage, measure gap between tail pylon and tail cone bosses marked "A". **MAXIMUM ALLOWABLE GAP TO BE 0.012-INCH.**
- (3) Using feeler gage, measure gap between tail pylon and tail cone bosses marked "B". **MAXIMUM ALLOWABLE GAP TO BE 0.030-INCH. NO MORE THAN TWO NON-ADJACENT LOCATIONS CAN BE GREATER THAN 0.015-INCH.**
- (4) If gap exceeds limits, contact Sikorsky ILS for recommended disposition.
- (5) Turn all tapered pins into mating bushings in pylon (Figure 2-4-132-1, Sheet 1, Detail C).

- r. Install bolts, washers, and nuts in hinge of tail rotor pylon. **TIGHTEN NUTS FINGER-TIGHT.** Install cotter pins.
- s. Disconnect straps from hoist and remove straps from pylon, and push in upper pylon steps (Figure 2-4-132-1, Sheet 1, Detail B).
- t. Paint and stencil pylon as required (PARA 2-6-5).
- u. Install tail rotor pylon flight control cables (PARA 11-4-106).
- v. Install intermediate gear box (PARA 6-4-31).
- w. Install tail gear box (PARA 6-4-52).
- x. Install tail rotor drive shaft section IV (PARA 6-4-41).
- y. Install stabilator (PARA 2-4-142).
- z. Remove protective caps and plugs from electrical connectors.
- aa. Inspect and treat electrical connectors (PARA 1-7-20).
- ab. Connect all electrical connectors, coaxial connectors, and bonding jumpers.
- ac. Make sure area is clean and free of foreign material before closing up.
- ad. Install left and right tail cone access covers on tail cone and secure with screws and washers (Figure 2-4-132-1, Sheet 1, Detail A).
- ae. Install all pylon fairings and covers/antennas on pylon as follows:
 - (1) Tail rotor pylon lower fairing (PARA 2-4-133)
 - (2) Tail rotor pylon stabilator-to-pylon fairing (PARA 2-4-134).
 - (3) Tail rotor pylon cambered fairing (PARA 2-4-135)
 - (4) Tail rotor pylon trailing edge fairing/antenna (PARA 2-4-136)

- (5) Tail gear box fairing (PARA 2-4-137)
 - (6) Tail rotor pylon leading edge cover/antenna (PARA 2-4-138)
 - (7) Intermediate gear box fairing (PARA 2-4-139)
- af. Check helicopter logbook to make sure pylon to tail cone torque check is shown as due 30 flight hours after installation (PARA 1-7-11).
- ag. Perform weight and balance check (Appendix G).
- ah. Perform maintenance test flight (Appropriate MTF).

2-4-133 TAIL ROTOR PYLON LOWER FAIRING.

PARAGRAPH BREAKDOWN

	PAGE
2-4-133.1 Replace Tail Rotor Pylon Lower Fairing	2-4-133-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Drill Set

Materials/Parts

- Corrosion Preventive Compound, Item 115, Appendix D
- Protective Caps and Plugs

- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-6-3
- PARA 2-6-5
- TM 11-70-23

2-4-133.1 Replace Tail Rotor Pylon Lower Fairing.

2-4-133.1.1 Remove.

- Turn on electrical power (PARA 1-6-16).
- On STABILATOR CONTROL/AUTO FLT CONT panel, use MAN SLEW switch to position stabilator as required to access lower stabilator actuator attachment hardware.
- Turn off all electrical power (PARA 1-6-16).
- Loosen studs and open cover on bottom of fairing (Figure 2-4-133-1, Detail A).
- If new fairing is to be installed, remove APR-39 aft radar receiver and aft spiral antennas, if equipped (TM 11-70-23). If same fairing is to

be installed, and helicopter is equipped with aft radar receiver, perform the following:

- Disconnect bonding jumper from aft radar receiver by removing screw, lockwasher, and washer.
- Tag and disconnect coaxial connectors from receptacles on aft radar receiver as follows:
 - P649R from J3
 - P650R from J4
 - P648R from J5
- Install protective caps and plugs on coaxial connectors.

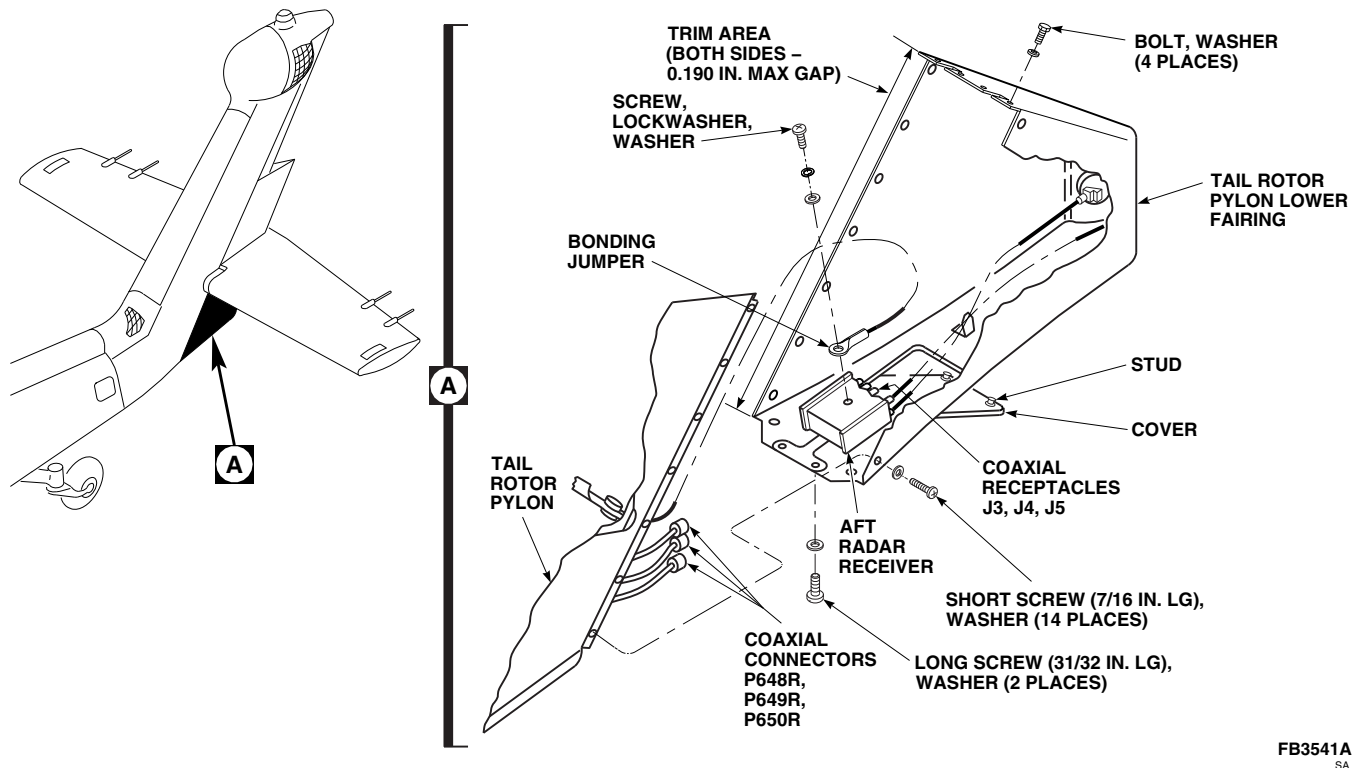
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FIGURE 2-4-133-1. REPLACE TAIL ROTOR PYLON LOWER FAIRING

NOTE

If fairing has been previously repaired with new mounting holes drilled, fairing is not readily interchangeable. If possible, retain with original tail rotor pylon.

- f. Remove screws, bolts, and washers securing fairing to tail rotor pylon. Remove fairing.

2-4-133.1.2 Install.

- a. Make sure stabilator is in neutral position or 0°.
- b. Turn off all electrical power (PARA 1-6-16).
- c. If installing new fairing, perform the following:
 - (1) Trial-fit new fairing on tail rotor pylon with screws, bolts, and washers. Inspect fairing for proper fit. If necessary, trim fairing as follows (Figure 2-4-133-1, Detail A):
 - (a) Mark areas on fairing, in the allowed trimming areas, that require trimming.

- (b) Remove screws, bolts, and washers securing fairing to pylon and remove fairing.
- (c) Trim marked areas (PARA 2-6-3). **DO NOT ALLOW MORE THAN 0.190-INCH GAP.**
- (d) Repeat step c. (1) until proper fit is achieved.

- (2) Trial-fit fairing on tail rotor pylon with screws, bolts, and washers. Inspect fairing for mounting holes that do not line up with nutplates in pylon. If necessary, repair mismatched mounting hole(s) as follows:
 - (a) Mark mounting hole(s) in fairing that do not line up with nutplate(s) in pylon. Mark fairing with correct location of mounting hole(s).
 - (b) Remove screws, bolts, and washers securing fairing to pylon and remove fairing.

- (c) Fill marked hole(s) (PARA 2-6-3).
 - (d) Using drill, drill new mounting hole(s) in fairing.
 - (e) Touch up paint fairing as required (PARA 2-6-5).
- d. Make sure area is clean and free of foreign material before closing up.
- e. Position fairing on pylon. Line up mounting holes in fairing with pylon.
- f. Apply corrosion preventive compound, Item 115, Appendix D, to fairing attachment hardware as it is installed.
- g. Install long screws and washers in bottom front holes in fairing (Figure 2-4-133-1, Detail A).
- h. Install short screws and washers in side holes in fairing.
- i. Install bolts and washers in rear top holes in fairing.
- j. If new fairing was installed, install APR-39 aft radar receiver and aft spiral antennas, if equipped (TM 11-70-23). If same fairing was installed, and helicopter is equipped with aft radar receiver, perform the following:
 - (1) Connect bonding jumper to aft radar receiver with screw, lockwasher, and washer.
 - (2) Remove protective caps and plugs on coaxial connectors.
 - (3) Inspect and treat coaxial connectors (PARA 1-7-20).
 - (4) Connect coaxial connectors to receptacles on aft radar receiver as follows:
 - (a) P649R to J3
 - (b) P650R to J4
 - (c) P648R to J5
 - (d) Remove tags.
- k. Close cover in bottom of fairing and secure studs.
- l. If aft radar receiver is installed, check operation of radar warning system, APR-39 (TM 11-70-23).

2-4-134 TAIL ROTOR PYLON STABILATOR-TO-PYLON FAIRING.

PARAGRAPH BREAKDOWN

	PAGE
2-4-134.1 Replace Tail Rotor Pylon Stabilator-to-Pylon Fairing	2-4-134-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit

Materials/Parts

- Corrosion Preventive Compound, Item 115, Appendix D

References

- Appendix D

2-4-134.1 Replace Tail Rotor Pylon Stabilator-to-Pylon Fairing.

NOTE

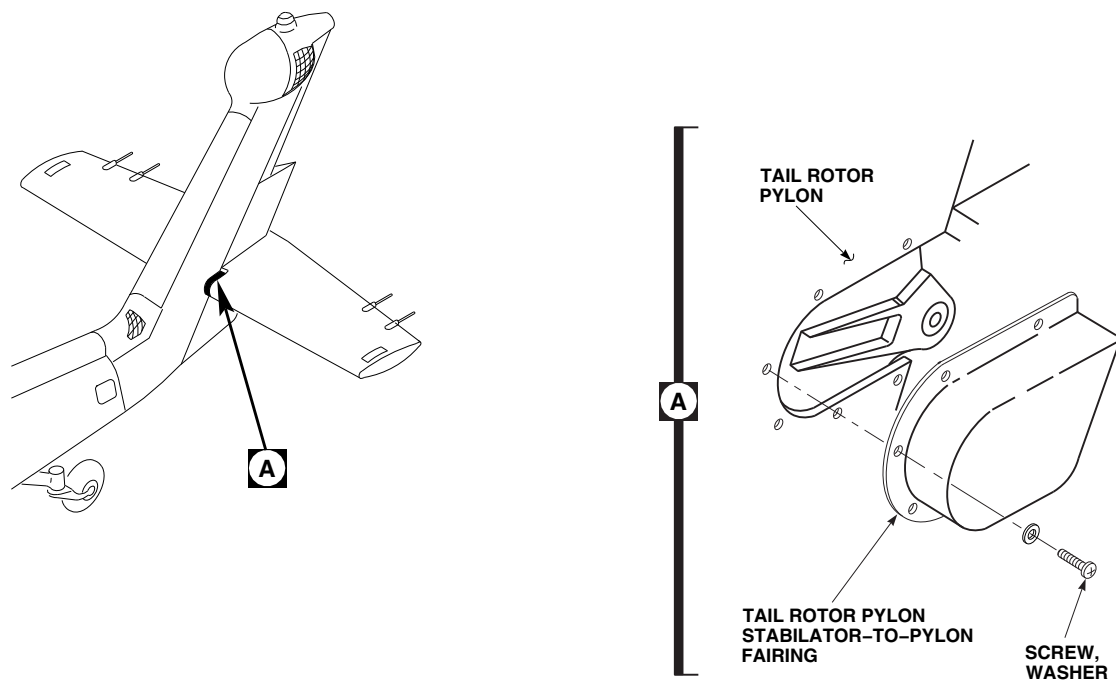
Replacement of left or right tail rotor pylon stabilator-to-pylon fairings is the same.

2-4-134.1.1. Remove.

- a. Remove screws and washers securing stabilator-to-pylon fairing to tail rotor pylon (Figure 2-4-134-1, Detail A).
- b. Remove fairing from pylon.

2-4-134.1.2. Install.

- a. Make sure area is clean and free of foreign material before closing up.
- b. Apply corrosion preventive compound, Item 115, Appendix D, to fairing hardware.
- c. Position stabilator-to-pylon fairing in position on tail rotor pylon (Figure 2-4-134-1, Detail A).
- d. Install screws and washers securing fairing to pylon.

FA3643
SA**FIGURE 2-4-134-1. REPLACE TAIL ROTOR PYLON STABILATOR-TO-PYLON FAIRING**

2-4-135 TAIL ROTOR PYLON CAMBERED FAIRING.

PARAGRAPH BREAKDOWN

	PAGE
2-4-135.1 Replace Tail Rotor Pylon Cambered Fairing	2-4-135-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Drill Set
- Maintenance Work Platform
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Corrosion Preventive Compound, Item 115, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 2-6-3
- PARA 2-6-5

2-4-135.1 Replace Tail Rotor Pylon Cambered Fairing .

2-4-135.1.1 Remove.

- Remove bolts and washers securing bottom of fairing to tail rotor pylon (Figure 2-4-135-1, Detail A).
- Position suitable maintenance work platform next to fairing or extend pylon steps.
- Remove bolts and washers securing top of fairing to pylon.

NOTE

If fairing has been previously repaired with new mounting holes drilled, fairing is not readily interchangeable. If possible, retain with original tail rotor pylon.

- Remove screws and washers securing fairing to pylon from both sides of fairing. Remove fairing.

2-4-135.1.2 Install.

- If installing new fairing, perform the following:
 - Trial-fit new fairing on tail rotor pylon with bolts and washers. Inspect fairing for proper fit. If necessary, trim fairing as follows (Figure 2-4-135-1, Detail A):
 - Mark areas on fairing, in allowed trimming areas, that require trimming.
 - Remove bolts and washers securing fairing to pylon and remove fairing.
 - Trim marked areas (PARA 2-6-3). **DO NOT ALLOW MORE THAN 0.190-INCH GAP.**

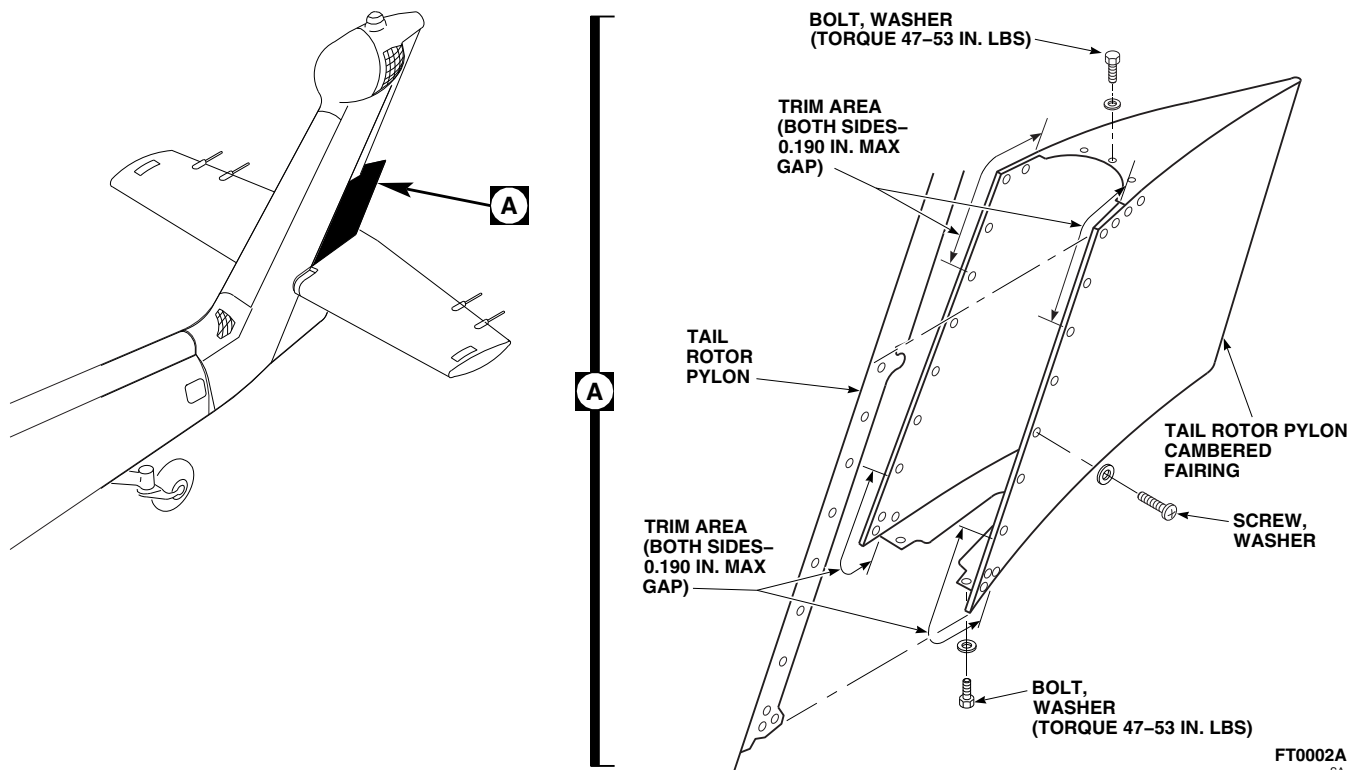
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FIGURE 2-4-135-1. REPLACE TAIL ROTOR PYLON CAMBERED FAIRING

- (d) Repeat step a. (1) until proper fit is achieved.
- (2) Trial-fit fairing on tail rotor pylon with bolts and washers. Inspect fairing for mounting holes that do not line up with nutplates in pylon. If necessary, repair mismatched mounting hole(s) as follows:
 - (a) Mark mounting hole(s) in fairing that do not line up with nutplate(s) in pylon. Mark fairing with correct location of mounting hole(s).
 - (b) Remove bolts and washers securing fairing to pylon and remove fairing.
 - (c) Fill marked hole(s) (PARA 2-6-3).
 - (d) Using drill, drill new mounting hole(s) in fairing.
 - (e) Touch up fairing paint as required (PARA 2-6-5).
 - (f) Apply corrosion preventive compound, Item 115, Appendix D, to fairing attachment hardware while installing.
 - (g) Make sure area is clean and free of foreign material before closing up.
 - (h) Position fairing on pylon. Line up mounting holes in fairing with pylon.
 - (i) Install bolts and washers securing top of fairing to pylon. **TORQUE BOLTS TO 47 - 53 INCH-POUNDS.**
 - (j) Install bolts and washers securing bottom of fairing to pylon. **TORQUE BOLTS TO 47 - 53 INCH-POUNDS.**
 - (k) Install screws securing fairing to pylon on both sides of fairing.
 - (l) Apply sealing compound, Item 292, Appendix D, to all joining of fairing, pylon, and securing hardware.

2-4-136 TAIL ROTOR PYLON TRAILING EDGE FAIRING/ANTENNA.

PARAGRAPH BREAKDOWN

	PAGE
2-4-136.1 Replace Tail Rotor Pylon Trailing Edge Fairing/ Antenna	2-4-136-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Drill Set
- Maintenance Work Platform

Materials/Parts

- Corrosion Preventive Compound, Item 115, Appendix D
- Protective Caps and Plugs

- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-6-1
- PARA 2-6-3
- PARA 2-6-5
- TM 11-70-23

2-4-136.1 Replace Tail Rotor Pylon Trailing Edge Fairing/Antenna.

2-4-136.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Position suitable maintenance platform next to tail rotor pylon trailing edge fairing/antenna.
- Remove screws, lockwashers, and washers securing fairing/antenna to tail rotor pylon (Figure 2-4-136-1, Detail A).
- Carefully remove antenna from pylon enough to gain access to coaxial connector.

- Disconnect coaxial connector from receptacle on fairing/antenna.
- Install protective caps and plugs on coaxial connector and receptacle.

NOTE

If fairing has been previously repaired with new mounting holes drilled, fairing is not readily interchangeable. If possible, retain with original tail rotor pylon.

Remove fairing/antenna from tail rotor pylon.

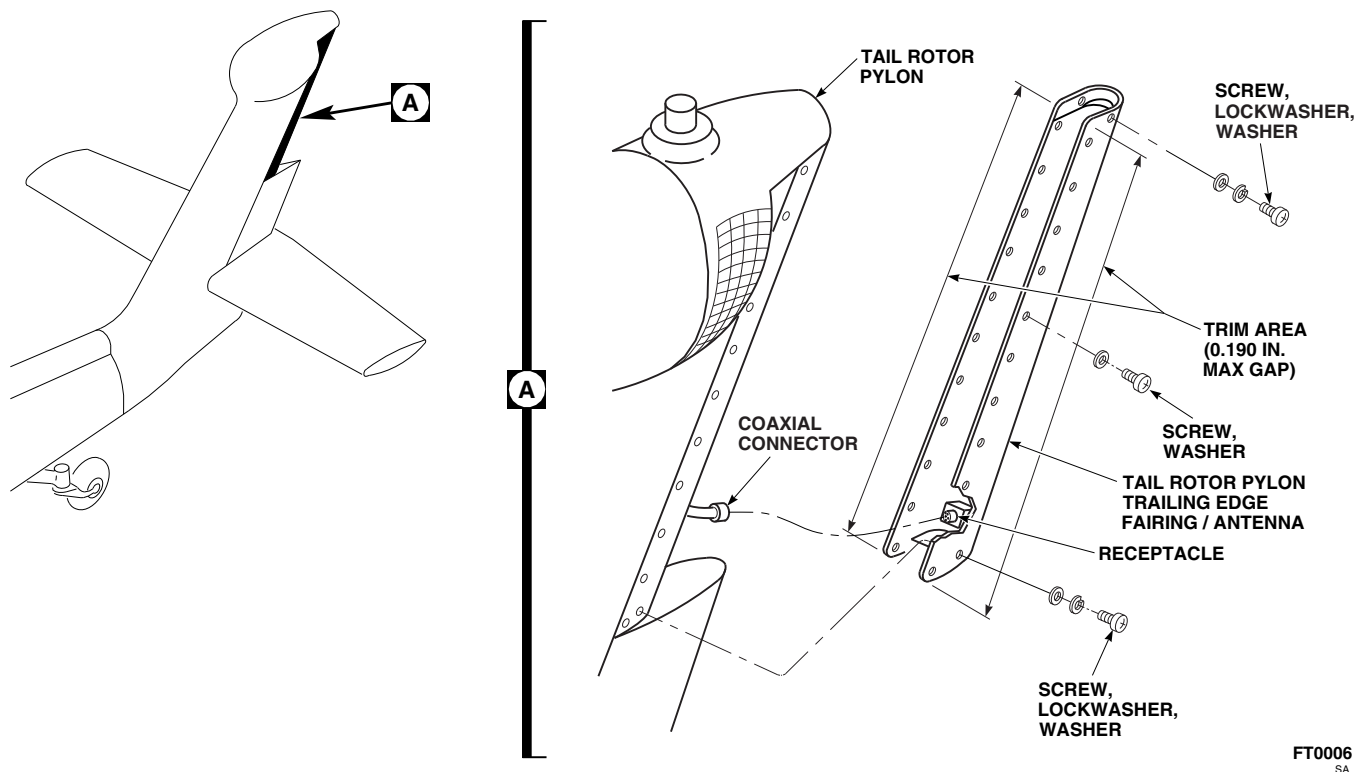
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FIGURE 2-4-136-1. REPLACE TAIL ROTOR PYLON TRAILING EDGE FAIRING/ANTENNA

2-4-136.1.2 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. If installing new tail rotor pylon trailing edge fairing/antenna, perform the following:
 - (1) Trial-fit new fairing/antenna on tail rotor pylon with screws and washers. Inspect fairing/antenna for proper fit. If necessary, trim fairing/antenna as follows (Figure 2-4-137-1, Detail A):
 - (a) Mark areas on fairing/antenna, in the allowed trimming areas, that require trimming.
 - (b) Remove screws and washers securing fairing/antenna to pylon and remove fairing/antenna.
 - (c) Trim marked areas (PARA 2-6-3). **DO NOT ALLOW MORE THAN 0.190-INCH GAP.**
 - (d) Repeat step b. (1) until proper fit is achieved.
 - (2) Trial-fit fairing/antenna on tail rotor pylon with screws and washers. Inspect fairing/antenna for mounting holes that do not line up with nutplates in pylon. If necessary, repair mismatched mounting hole(s) as follows:
 - (a) Mark mounting hole(s) in fairing/antenna that do not line up with nutplate(s) in pylon. Mark fairing/antenna with correct location of mounting hole(s).
 - (b) Remove screws and washers securing fairing/antenna to pylon and remove fairing/antenna.
 - (c) Fill marked hole(s) (PARA 2-6-3).
 - (d) Using drill, drill new mounting hole(s) in fairing/antenna.
 - (e) Touch up paint fairing/antenna as required (PARA 2-6-5).
- c. Make sure area is clean and free of foreign material before closing up.

- d. Remove protective caps and plugs from coaxial connector and receptacle.
- e. Inspect and treat coaxial connector and receptacle (PARA 1-7-20).
- f. Connect coaxial connector to receptacle on fairing/antenna.
- g. Position fairing/antenna on pylon. Line up mounting holes in fairing with pylon.
- h. Apply corrosion preventive compound, Item 115, Appendix D, to fairing attachment hardware.
- i. Install screws, lockwashers, and washers securing fairing/antenna to pylon.
- j. Apply sealing compound, Item 292, Appendix D, where edges of fairing meet pylon skin and to fastening screwheads, to form watertight seal (PARA 2-6-1).
- k. Perform operational check of portable VHF/FM antenna in Intercommunication Set (TM 11-70-23).

NOTE

Make sure three screws at top of fairing and two screws at bottom have lockwashers installed under head of screw.

2-4-137 TAIL GEAR BOX FAIRING.

PARAGRAPH BREAKDOWN

	PAGE
2-4-137.1 Replace Tail Gear Box Fairing	2-4-137-2

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Drill Set
- Maintenance Platform
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Abrasive Cloth, Item 4, Appendix D
- Adhesion Promoter, Item 25A, Appendix D
- Adhesive, Item 37, Appendix D
- Cloth, Cleaning, Item 102, Appendix D
- Corrosion Preventive Compound, Item 115, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D

- Pressure Sensitive Tape, Item 251A, Appendix D
- Protective Caps and Plugs
- Scouring Pad, Item 270, Appendix D
- Sealing Compound, Item 292, Appendix D
- Tags

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-6-3
- PARA 2-6-5
- PARA 9-2-14
- PARA 9-2-16
- PARA 9-4-83
- PARA 9-4-84
- PARA 9-4-86

2-4-137.1 Replace Tail Gear Box Fairing.**NOTE**

- Tail gear box fairing is a two piece fairing consisting of a front and rear fairing.
- Front fairing must be removed prior to removing rear fairing.

2-4-137.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Position suitable maintenance platform next to tail gear box fairing.
- Unfasten spring-type fasteners securing front fairing to tail rotor pylon (Figure 2-4-137-1, Detail A). Remove fairing.
- Remove rear fairing as follows:
 - Disconnect electrical connector, P602 from J602, on anticollision light (Figure 2-4-137-1, Detail B).
 - Install protective caps and plugs on electrical connectors.
 - Disconnect electrical connector, P616 from J616, on tail rotor pylon (Figure 2-4-137-1, Detail C).
 - Slide rubber covers back on tail incandescent position light to gain access to wire connections.
 - Tag wires. Remove screws and washers securing wires to tail incandescent position light.
 - Remove screws and washers attaching fairing to tail rotor pylon (Figure 2-4-137-1, Detail A). Remove fairing.
 - If fairing is being replaced, remove anticollision light (PARA 9-4-86), tail incandescent position light (PARA 9-4-83), and tail infrared position light (PARA 9-4-84) from fairing.

2-4-137.1.2 Inspect/Repair.

- Inspect tail gear box fairing for missing or damaged seals. If seal(s) are missing, or damaged so that they do not prevent chafing between gear box and fairing, replace seal(s) as follows:
 - Remove damaged seal using nonmetallic scraper.
 - Clean any remaining adhesive on edge of fairing using methyl ethyl ketone, Item 217, Appendix D.
 - Roughen inside edges of new seal using abrasive cloth, Item 4, Appendix D. Clean roughened edges of seal using methyl ethyl ketone, Item 217, Appendix D.
 - Apply a light coat of adhesive, Item 37, Appendix D, to roughened edges of seal and edge of tail gear box. Allow adhesive to dry for about 10 minutes or until it is tacky.
 - Install seal on edge of fairing, making sure all mating surfaces are in contact.
 - Allow adhesive to dry for 24 hours.
- 70785-SUBQ** Inspect rear fairing handhold flap and strap for damage (Figure 2-4-137-2). If damage is found, replace flap and strap as follows:
 - Drill out rivets securing flap and strap to fairing.
 - Remove flap and strap from fairing.
 - Position replacement flap and strap on fairing and install with washers and rivets.
- Inspect gear box fairing for missing or worn chafing tape. If tape is missing or worn so that chafing is not prevented, replace tape as follows:
 - Remove worn tape using nonmetallic scraper.
 - Clean dirt and grease from fairing using cleaning cloth, Item 102, Appendix D,

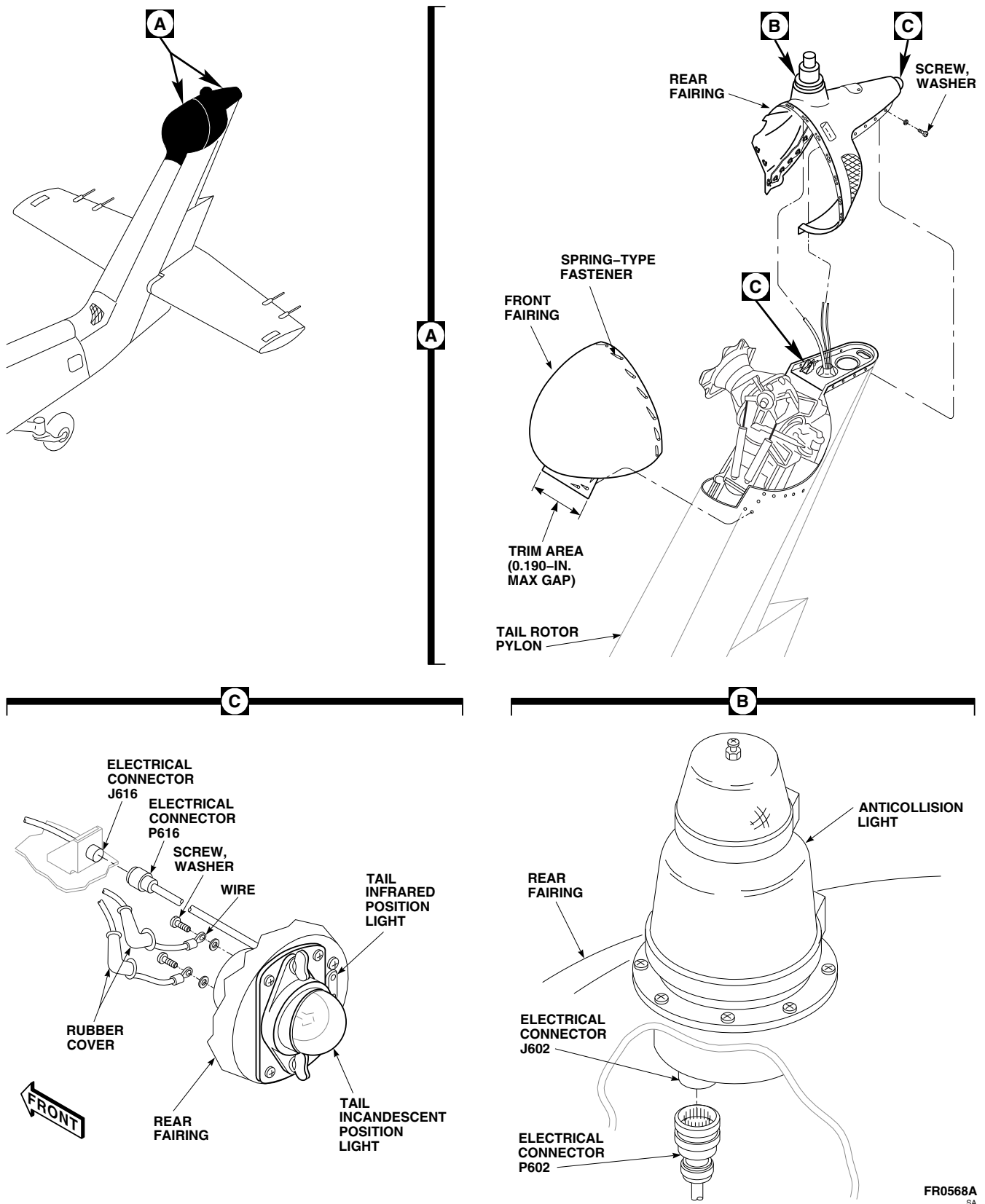
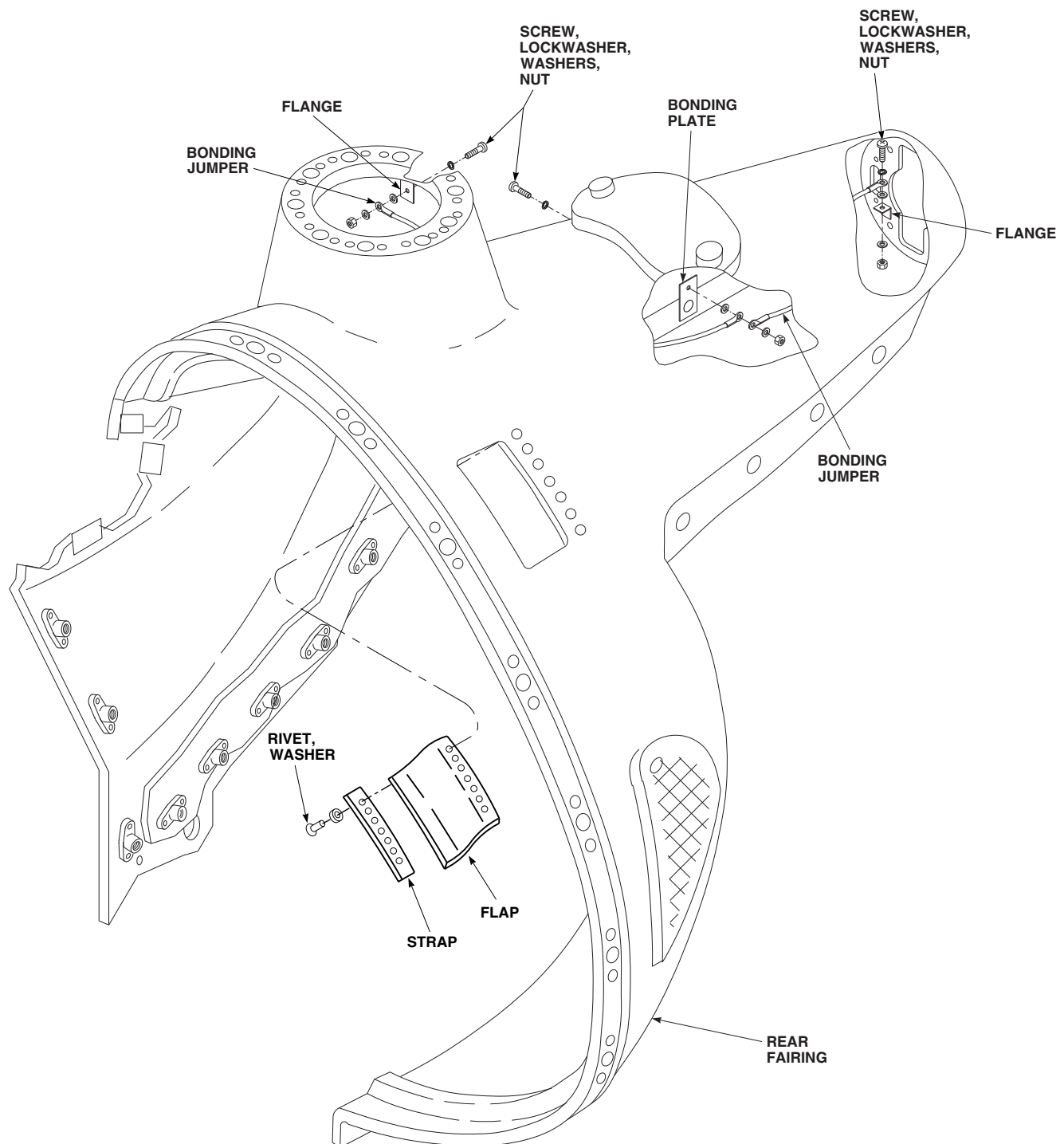


FIGURE 2-4-137-1. REPLACE TAIL GEAR BOX FAIRING

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dampened with methyl ethyl ketone, Item 217, Appendix D.

- (3) Lightly roughen surface with scouring pad, Item 270, Appendix D.
 - (4) Clean roughened surface of fairing using cleaning cloth, Item 102, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.
 - (5) Measure and cut new pressure sensitive tape, Item 251A, Appendix D.
 - (6) Spray a light coat of adhesion promoter, Item 25A, Appendix D, on area where tape is to be installed.
 - (7) Install pressure sensitive tape, Item 251A, Appendix D, on fairing applying pressure to make sure of good bond.
- d. On rear fairing, inspect bonding jumpers for damage. **NONE ALLOWED.** If damaged, replace bonding jumper(s) as follows:
- (1) Remove screws, lockwashers, washers, and nuts securing bonding jumper to flange and bonding plate. Remove bonding jumper.
 - (2) Using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D, clean mating surfaces of bonding jumper, flange, and bonding plate. Using clean, dry machinery towel, dry surface.
 - (3) Install bonding jumper to flange and bonding plate, and secure with screws, lockwashers, washers, and nuts.
 - (4) Seal bonded connections with sealing compound, Item 292, Appendix D.

2-4-137.1.3 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. If installing new fairing, perform the following:
 - (1) Trial-fit new fairing on tail rotor pylon with screws and washers. Inspect fairing

for proper fit. If necessary, trim fairing as follows (Figure 2-4-137-1, Detail A):

- (a) Mark areas on fairing, in the allowed trimming areas, that require trimming.
 - (b) Remove screws and washers securing fairing to pylon and remove fairing.
 - (c) Trim marked areas (PARA 2-6-3). **DO NOT ALLOW MORE THAN 0.190-INCH GAP.**
 - (d) Repeat step b. (1) until proper fit is achieved.
- (2) Trial-fit fairing on tail rotor pylon with screws and washers. Inspect fairing for mounting holes that do not line up with nutplates in pylon. If necessary, repair mismatched mounting hole(s) as follows:
- (a) Mark mounting hole(s) in fairing that do not line up with nutplate(s) in pylon. Mark fairing with correct location of mounting hole(s).
 - (b) Remove screws and washers securing fairing to pylon and remove fairing.
 - (c) Fill marked hole(s) (PARA 2-6-3).
 - (d) Using drill, drill new mounting hole(s) in fairing.
 - (e) If spring-type fastener(s) does not line up, remove and move receptacle(s) (PARA 2-6-3).
 - (f) Touch up paint fairing as required (PARA 2-6-5).
- c. Make sure area is clean and free of foreign material.
- d. Install rear fairing as follows:
- (1) If not installed, install anticollision light (PARA 9-4-86), tail incandescent position

- light (PARA 9-4-83), and tail infrared position light (PARA 9-4-84) on fairing.
- (2) Position fairing on pylon (Figure 2-4-137-1, Detail A).
 - (3) Remove protective caps and plugs from electrical connectors.
 - (4) Inspect and treat electrical connectors (PARA 1-7-20).
 - (5) Connect electrical connector, P616 to J616, on tail rotor pylon (Figure 2-4-137-1, Detail C).
 - (6) Connect electrical connector, P602 to J602, on anticollision light (Figure 2-4-137-1, Detail B).
 - (7) Apply corrosion preventive compound, Item 115, Appendix D, to fairing attachment hardware.
 - (8) Install screws and washers securing fairing to tail rotor pylon (Figure 2-4-137-1, Detail A).
- e. Install front fairing and fasten spring-type fasteners.
 - f. Perform operational check of position and anticollision lights (PARA 9-2-14 and PARA 9-2-16).

2-4-138 TAIL ROTOR PYLON LEADING EDGE COVER/ANTENNA.

PARAGRAPH BREAKDOWN

	PAGE
2-4-138.1 Replace Tail Rotor Pylon Leading Edge Cover/Antenna	2-4-138-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Maintenance Work Platform
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Abrasive Cloth, Item 6, Appendix D
- Adhesive, Item 32, Appendix D
- Cloth, Cleaning, Item 102, Appendix D
- Epoxy Coating Kit, Item 126A, Appendix D

- Lockwire, Item 197, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Pressure Sensitive Adhesive Tape, Item 247, Appendix D
- Protective Caps and Plugs
- Trichloroethane 1,1,1, Item 341, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 1-7-20
- TM 11-70-23

2-4-138.1 Replace Tail Rotor Pylon Leading Edge Cover/Antenna.

2-4-138.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).

NOTE

To prevent damage to cover/antenna, make sure tail rotor pylon steps are not extended when cover is being hinged open.

- If necessary, retract pylon steps.

- Working from maintenance work platform, unfasten spring-type fasteners securing cover/antenna to pylon and open cover/antenna (Figure 2-4-138-1, Detail A).
- Disconnect coaxial connector from coaxial receptacle.
- Install protective cap and plug on coaxial connector.
- Lift up on cover/antenna to disengage hinge pins from hinge fittings and remove cover/antenna from tail rotor pylon.

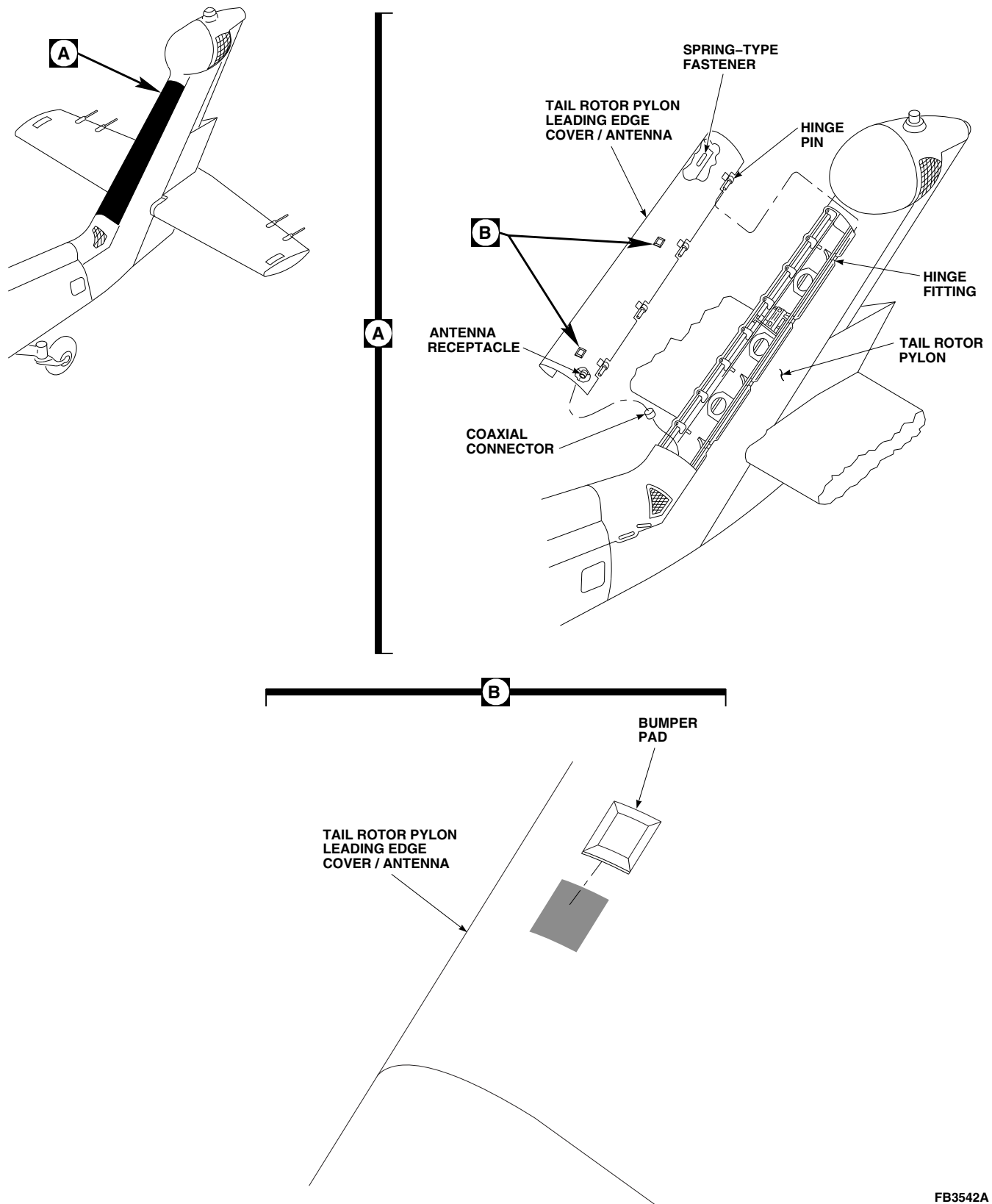
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FIGURE 2-4-138-1. REPLACE TAIL ROTOR PYLON LEADING EDGE COVER/ANTENNA

2-4-138.1.2 Repair.

NOTE

- Replacement of upper and lower bumper pads is the same.
- a. Mask area around bumper pad with pressure sensitive adhesive tape, Item 247, Appendix D (Figure 2-4-138-1, Detail B).
- b. Using nonmetallic scraper and cleaning cloth, Item 102, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D, carefully remove damaged bumper pad.
- c. Apply light coat of adhesive, Item 32, Appendix D, to large flat surface of bumper pad.
- d. Install bumper pad on leading edge cover/antenna.

NOTE

Spot painting may be done with cover/antenna installed on helicopter. If antenna requires total repainting, remove cover/antenna.

- e. If cover/antenna requires total repainting, remove tail rotor pylon leading edge cover/antenna (PARA 2-4-138.1.1).
- f. Using cleaning cloth, Item 102, Appendix D, soaked with trichloroethane 1,1,1, Item 341, Appendix D, clean outer surface.

CAUTION

Damage to antenna elements will occur if antenna surface is sanded excessively. Use extreme care when sanding antenna surface.

- g. Lightly hand sand cover/antenna with abrasive cloth, Item 6, Appendix D.
- h. Using cleaning cloth, Item 102, Appendix D, soaked with trichloroethane 1,1,1, Item 341, Appendix D, reclean surface.

NOTE

Allow at least ½ hour between coats.

- i. Using epoxy coating kit, apply two coats of epoxy coating, Item 126A, Appendix D.
- j. If removed, install tail rotor pylon leading edge cover/antenna (PARA 2-4-138.1.3).

2-4-138.1.3 Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Working from maintenance work platform, insert cover/antenna hinge pins in mating hinge fittings on tail rotor pylon (Figure 2-4-138-1, Detail A).
- c. Remove protective cap and plug from coaxial connector.
- d. Inspect and treat coaxial connectors (PARA 1-7-20).
- e. Connect coaxial connector, P1085R, to antenna receptacle.
- f. Lockwire coaxial connector to flange as follows:
 - (1) Cut piece of lockwire, Item 197, Appendix D, approximately 4-inches.
 - (2) Wrap lockwire around coupling ring of coaxial connector and twist strands (10 turns per inch) clockwise until hole in flange is reached.
 - (3) Insert one strand of lockwire through hole in flange and position the other end over top of flange.
 - (4) Connect strands by twisting together clockwise (10 turns per inch) for ½-inch.
 - (5) Remove any excess lockwire.
- g. Make sure area is clean and free of foreign material before closing up.
- h. Close cover/antenna and secure cover/antenna to tail rotor pylon with spring-type fasteners.

- i. Perform operational check of No. 1 VHF/FM antenna (TM 11-70-23).

2-4-139 INTERMEDIATE GEAR BOX FAIRING.

PARAGRAPH BREAKDOWN

	PAGE
2-4-139.1 Replace Intermediate Gear Box Fairing	2-4-139-1
INITIAL SETUP	• Methyl Ethyl Ketone, Item 217, Appendix D
Tools	• Pressure Sensitive Tape, Item 251A, Appendix D
• Aircraft Mechanic’s Toolkit • Airframe Repairer’s Toolkit • File • Nonmetallic Scraper, Locally-Made	References • Appendix D • PARA 2-6-5
Materials/Parts	
• Adhesion Promoter, Item 25A, Appendix D • Cloth, Cleaning, Item 102, Appendix D	
2-4-139.1 Replace Intermediate Gear Box Fairing.	(1) Remove worn tape using nonmetallic scraper.
2-4-139.1.1 Remove.	(2) Clean dirt and grease from fairing using cleaning cloth, Item 102, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.
a. Unfasten spring-type fasteners securing intermediate gear box fairing to tail rotor pylon (Figure 2-4-139-1, Detail A).	(3) Measure and cut new pressure sensitive tape, Item 251A, Appendix D.
b. Remove fairing from pylon.	(4) Spray a light coat of adhesion promoter, Item 25A, Appendix D, on area where tape is to be installed.
2-4-139.1.2 Inspect/Repair.	(5) Install pressure sensitive tape, Item 251A, Appendix D, on fairing.
a. Inspect intermediate gear box fairing for evidence of chafing with tail rotor pylon. If chafing exists, replace worn or missing tape as follows (Figure 2-4-139-1, Section B-B):	

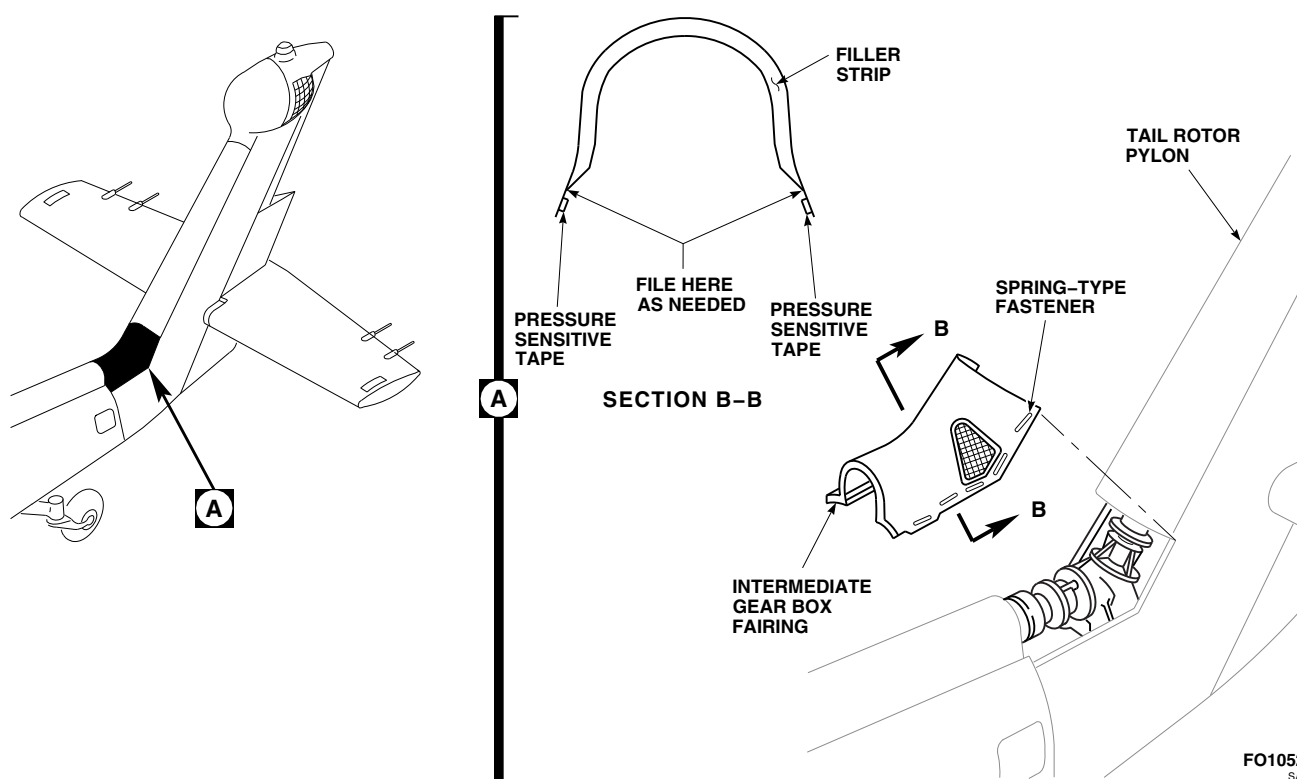


FIGURE 2-4-139-1. REPLACE INTERMEDIATE GEAR BOX FAIRING

2-4-139.1.3 Install.

NOTE

On some replacement fairings the tapered ends of filler strips may contact skin of tail rotor pylon, making installation difficult. Fairings with this condition must be repaired.

- a. If installing new fairing, trial-fit new fairing on tail rotor pylon and inspect tapered ends of filler strips for contact with pylon (Figure 2-4-139-1, Detail A). If tapered pins make contact with pylon, repair as follows:

- (1) Remove fairing from pylon.

- (2) Using file, file tapered ends as needed to let fairing seat properly (Figure 2-4-139-1, Section B-B).

- (3) Trial-fit fairing on pylon and inspect tapered ends for contact with pylon (Figure 2-4-139-1, Detail A).

- (4) Repeat step a. (1) through a. (3) until proper fit is achieved.

- (5) Touch up paint fairing as required (PARA 2-6-5).

- b. Make sure area is clean and free of foreign material before closing up.

- c. Install fairing on pylon and fasten spring-type fasteners (Figure 2-4-139-1, Detail A).

2-4-140 TAIL ROTOR PYLON STEPS.

PARAGRAPH BREAKDOWN

	PAGE
2-4-140.1 Replace Tail Rotor Pylon Steps	2-4-140-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Spring Tester

Materials/Parts

- Primer, Item 257, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-141

2-4-140.1 Replace Tail Rotor Pylon Steps.

NOTE

Replacement of all pylon steps is the same, except the lower step. To replace lower step, refer to PARA 2-4-141.

2-4-140.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Remove screws and washers securing step to tail rotor pylon (Figure 2-4-140-1, Detail A).
- Remove step from tail rotor pylon.

2-4-140.1.2. Disassemble.

WARNING

Tread is installed with approximately 16 pounds of spring pressure. Personal

injury may occur when removing pin from tread and housing. Make sure step is fully extended before removing pin.

- Extend step fully and remove pin from tread and housing (Figure 2-4-140-1, Detail B).
- Remove tread from housing.
- Remove rivets securing ring to tread. Remove ring from tread.
- Remove rivets securing cap to tread. Remove cap from tread.
- Remove spring from housing.
- Check spring tension as follows:
 - Using spring tester, apply 16 - 17 pounds of pressure to compress spring.
 - Measure compressed length of spring. **LENGTH OF SPRING MUST BE 1.79-INCHES.**

- (3) If length of spring is less than 1.79-inches, replace spring.

2-4-140.1.3. Assemble.

- a. Install spring in housing (Figure 2-4-140-1, Detail B).
- b. Install cap in tread and line up holes.
- c. Install rivets and grind heads flush.
- d. Install ring in tread and line up holes.
- e. Install rivets and grind heads flush.
- f. Install tread in housing. Line up hole in tread with slot in housing.

- g. Press pin into tread. Stake pin 2 places on each end.
- h. Check for smooth operation by alternately extending and pushing in tread.

2-4-140.1.4. Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Prime tail rotor pylon-to-step contact surfaces with primer, Item 257, Appendix D.
- c. Install step in tail rotor pylon and fasten with screws and washers (Figure 2-4-140-1, Detail A).
- d. Extend and push in tread a few times to check operation.

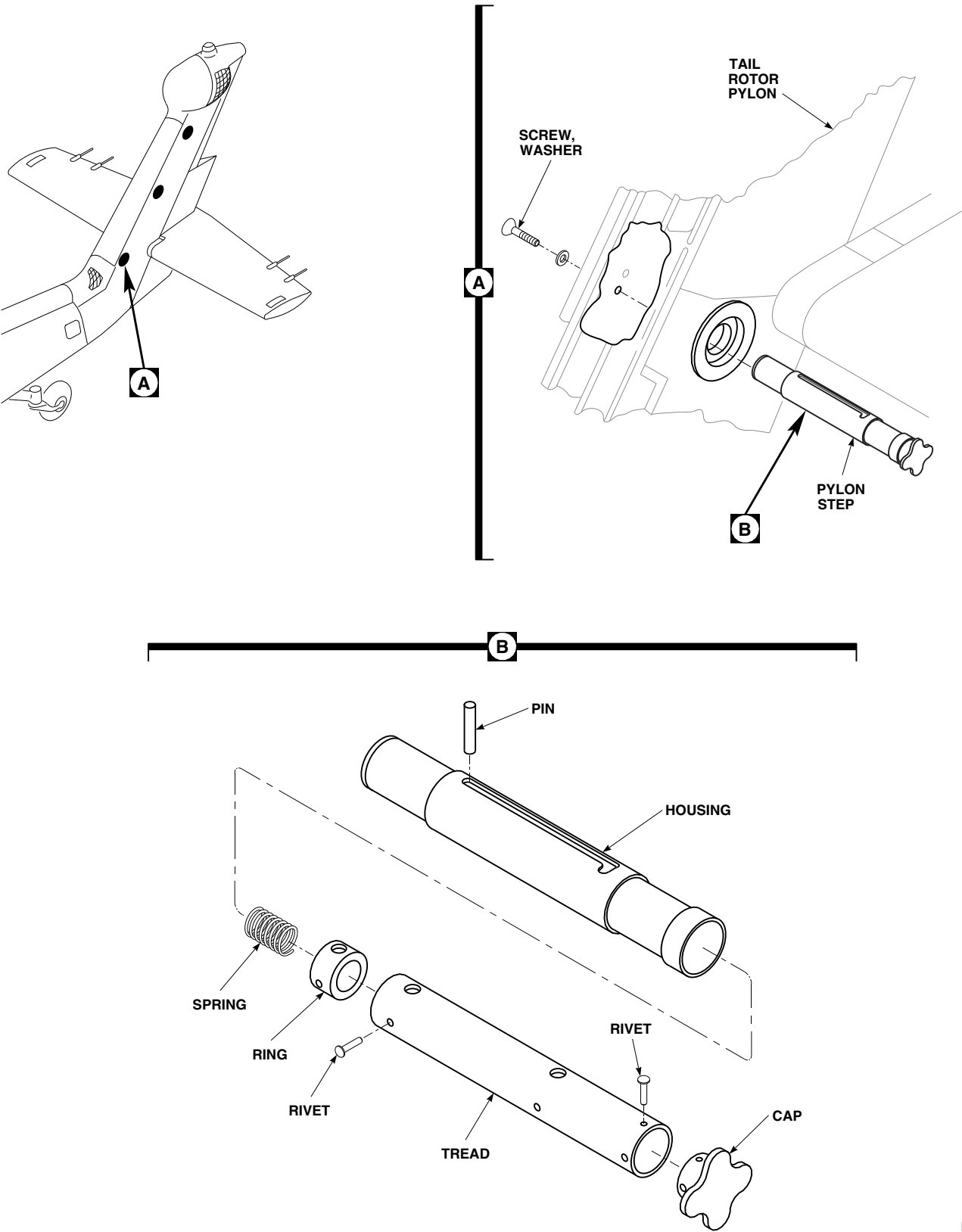


FIGURE 2-4-140-1. REPLACE TAIL ROTOR PYLON STEPS

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2-4-141 TAIL ROTOR PYLON LOWER STEP.

PARAGRAPH BREAKDOWN

	PAGE
2-4-141.1 Replace Tail Rotor Pylon Lower Step	2-4-141-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit

References

- PARA 1-6-9
- PARA 1-6-16

2-4-141.1 Replace Tail Rotor Pylon Lower Step.

- (2) Install eyebolt in tube using washer, spacer, and nut.

2-4-141.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Fold pylon 90° (PARA 1-6-9).
- Push up and turn step towards right side of helicopter and lower step from pylon.
- Remove bolts, washers, and nuts attaching step to tube. Remove step (Figure 2-4-141-1, Detail A).
- Disconnect spring from eyebolt.
- Pull tube up and out of pylon.
- If tube will be reinstalled, check condition of eyebolt in tube. Replace eyebolt if damaged or worn as follows:
 - Remove nut, spacer, and washer securing eyebolt to tube. Remove eyebolt.

2-4-141.1.2. Install.

- If eyebolt is not installed, install it in tube using washer, spacer, and nut (Figure 2-4-141-1, Detail A).
- From inside of pylon, insert tube through pylon into position. Make sure pin of tube is in pylon fitting (detent position).
- Slide tube down to its lowered position.
- Connect spring to eyebolt.
- Slide step on tube and install bolts, washers, and nuts.
- Check operation of step assembly a few times.
- Push step up and turn to its stowed position.
- Unfold pylon (PARA 1-6-9).

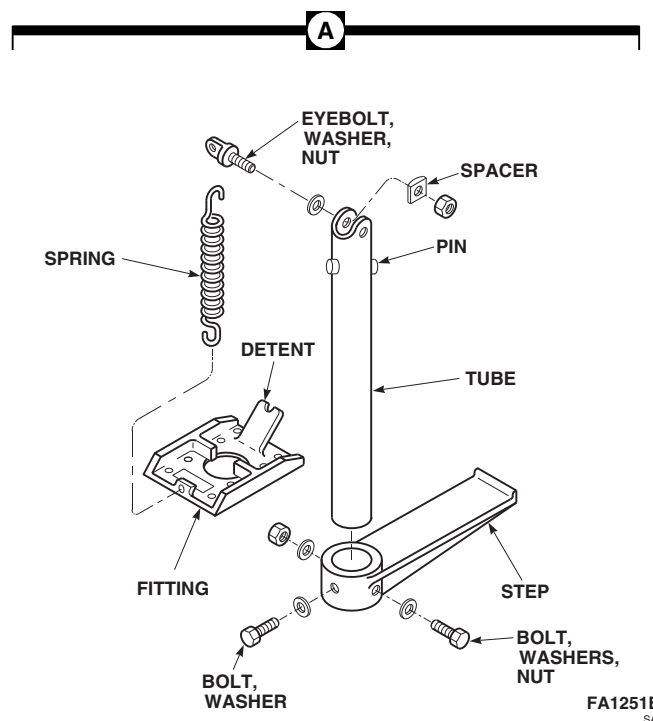
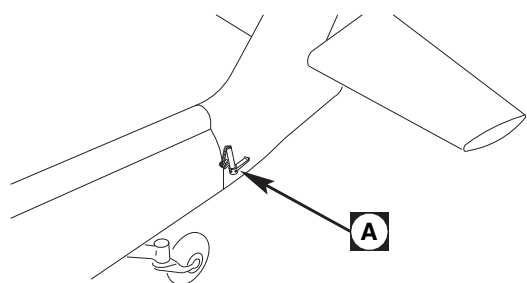


FIGURE 2-4-141-1. REPLACE TAIL ROTOR PYLON LOWER STEP

2-4-142 STABILATOR.

PARAGRAPH BREAKDOWN

	PAGE
2-4-142.1 Replace Stabilator	2-4-142-2

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Cradle Assembly, 70700-20435-041
- Flashlight
- Inspection Mirror
- Protective Covering
- Stabilator Bolt Removal Tool, Locally-Made, Figure H-169, Appendix H
- Stabilator Drift Pin, Locally-Made, Figure H-251, Appendix H
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs
- Wrench

Materials/Parts

- Antiseize Compound, Item 76, Appendix D

- Corrosion Preventive Compound, Item 114, Appendix D
- Corrosion Preventive Compound, Item 115, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Petrolatum, Item 234, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- Appendix H
- Appropriate MTF
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-133
- PARA 2-6-5
- PARA 2-6-53
- TM 11-70-23

2-4-142.1 Replace Stabilator.**2-4-142.1.1 Remove.**

- a. Remove accessible screws and washers securing left and right stabilator-to-pylon fairings (Figure 2-4-142-1, Sheet 1, Detail A).
- b. Connect external electrical power (PARA 1-6-16).
- c. On upper console, place EXT PWR switch ON.
- d. With assistant watching stabilator, place MAN SLEW to -8° (trailing edge up).
- e. Turn off all electrical power (PARA 1-6-16).
- f. Pull out the following circuit breakers:
 - (1) STAB CONTR (pilot's circuit breaker panel - NO. 2 AC PRI BUS)
 - (2) STAB CONTR (pilot's circuit breaker panel - AC ESNTL BUS)
 - (3) STAB PWR (pilot's circuit breaker panel - NO. 2 DC PRI BUS)
 - (4) STAB PWR (upper console - DC ESNTL BUS)
- g. Remove remaining screws and washers securing left and right stabilator-to-pylon fairings, and remove fairings.
- h. Remove tail rotor pylon lower fairing (PARA 2-4-133).
- i. Disconnect electrical connectors, P702 (right) and P703 (left), from receptacles on tail rotor pylon.
- j. Install protective caps and plugs on electrical connectors.
- k. Remove cotter pin, nut, and washer from self-retaining bolt securing position transmitter/limit switch assembly actuator rod to stabilator fitting. Do not remove bolt, washer(s), and bushing now (Figure 2-4-142-1, Sheet 3, Detail D).
- l. Remove cotter pins, nuts, and washers from stabilator attachment bolts. Do not remove bolts, bushings, bolt retainer, and countersunk washer now (Figure 2-4-142-1, Sheet 1, Detail A).
- m. Turn on electrical power (PARA 1-6-16).
- n. Push in the following circuit breakers:
 - (1) STAB PWR (pilot's circuit breaker panel - NO. 2 DC PRI BUS)
 - (2) STAB PWR (upper console - DC ESNTL BUS)

NOTE

STABILATOR AUTO CONTROL
RESET will not light.

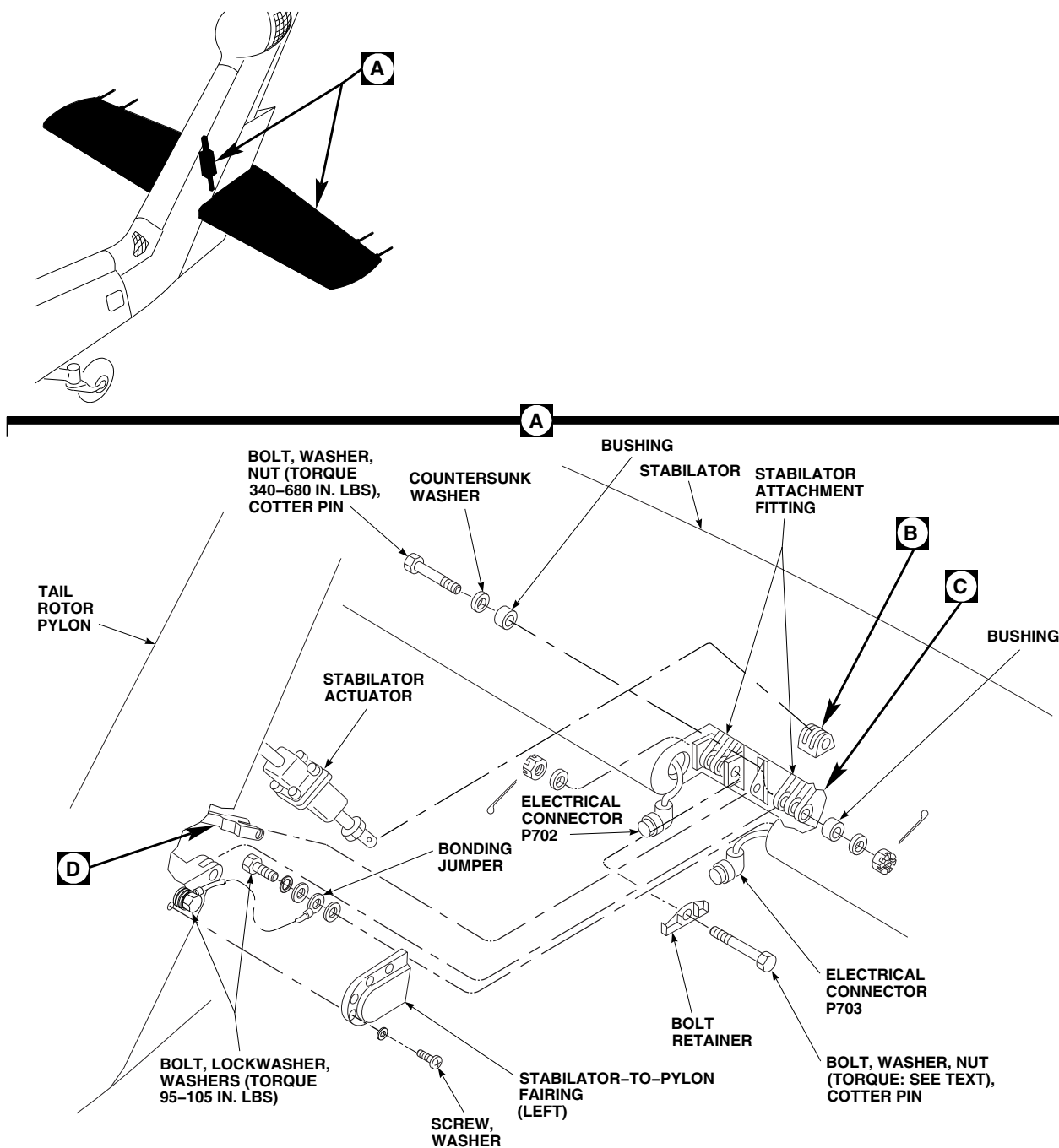
- o. With assistant watching stabilator, place MAN SLEW to 0° (level position) to access stabilator actuator self-retaining bolt.
- p. Pull out the following circuit breakers:
 - (1) STAB PWR (pilot's circuit breaker panel - NO. 2 DC PRI BUS)
 - (2) STAB PWR (upper console - DC ESNTL BUS)
- q. Turn off all electrical power (PARA 1-6-16).

CAUTION

Forcible removal of self-retaining bolt will damage equipment. Use stabilator bolt removal tool.

- r. Locally make stabilator bolt removal tool, Figure H-169, Appendix H.
- s. Using stabilator bolt removal tool, press release pin in center of bolthead while removing position transmitter/limit switch assembly actuator rod attachment bolt.
- t. Remove self-retaining bolt, washer(s), and bushing securing position transmitter/limit

2-4-142-2 Change 8



NOTES

1. APPLY A LIGHT COAT OF PETROLATUM, ITEM 234, APPENDIX D, TO FACE OF RUBBER SPACERS FOR EASE OF INSTALLATION.
2. INSTALL WASHERS UNDER BOLTHEAD (3 MAXIMUM), TO MAKE SURE NUT BOTTOMS AGAINST BUSHING AND NOT ON BOLT.
3. BOLTHEAD MAY BE INSTALLED IN EITHER DIRECTION.
4. USE 1 OR 2 WASHERS AS REQUIRED.

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FIGURE 2-4-142-1. REPLACE STABILATOR (SHEET 1 OF 3)

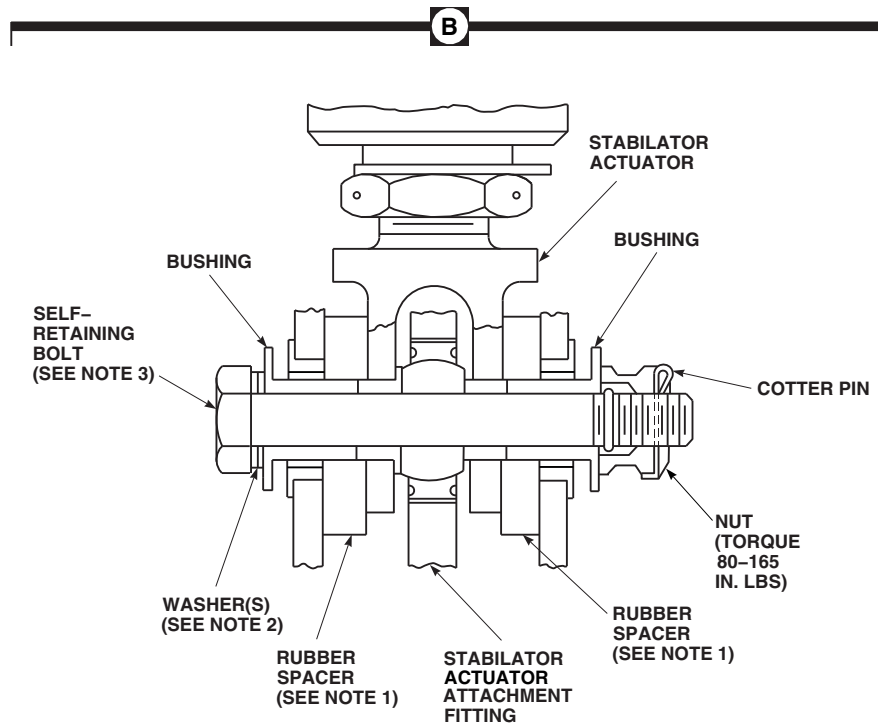
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FIGURE 2-4-142-1. REPLACE STABILATOR (SHEET 2 OF 3)

switch assembly actuator rod to stabilator fitting. Disengage rod from stabilator fitting (Figure 2-4-142-1, Sheet 3, Detail D).

WARNING**FSCAP**

Outside diameter and bolthead shoulder of actuator self-retaining bolts and stabilator actuator attachment fitting lug areas are critical surfaces which must not be damaged during and after removal of actuator. Use extreme caution when removing actuator and provide protective covering after removal.

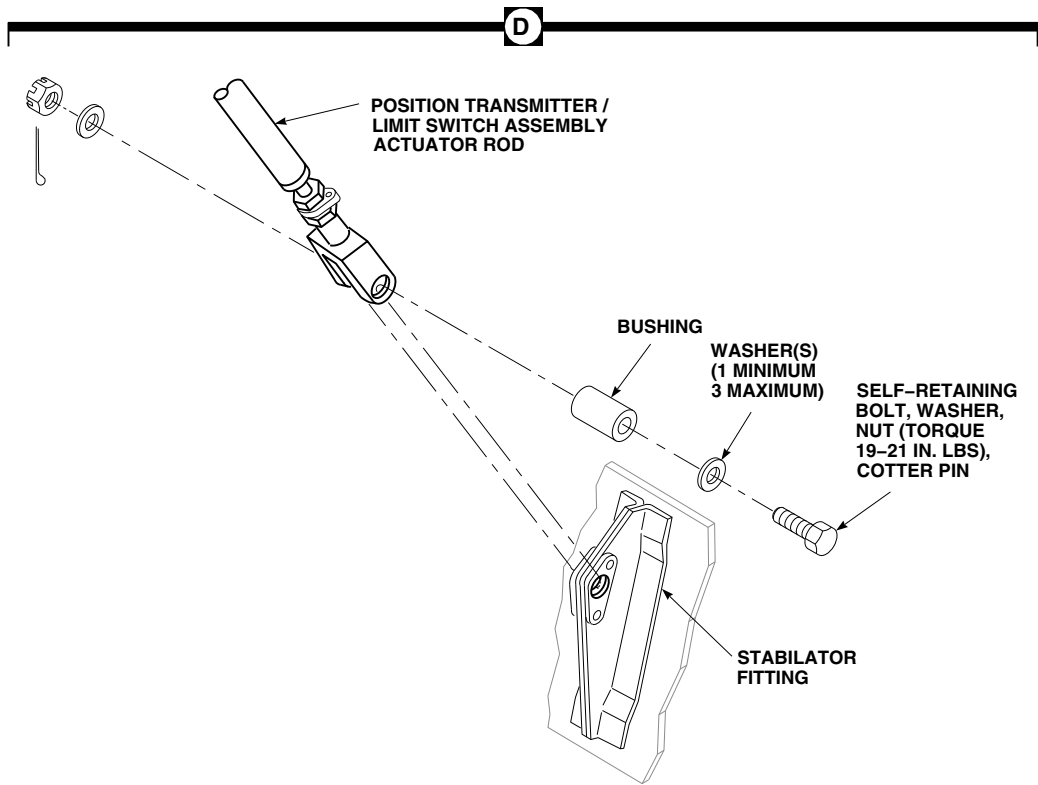
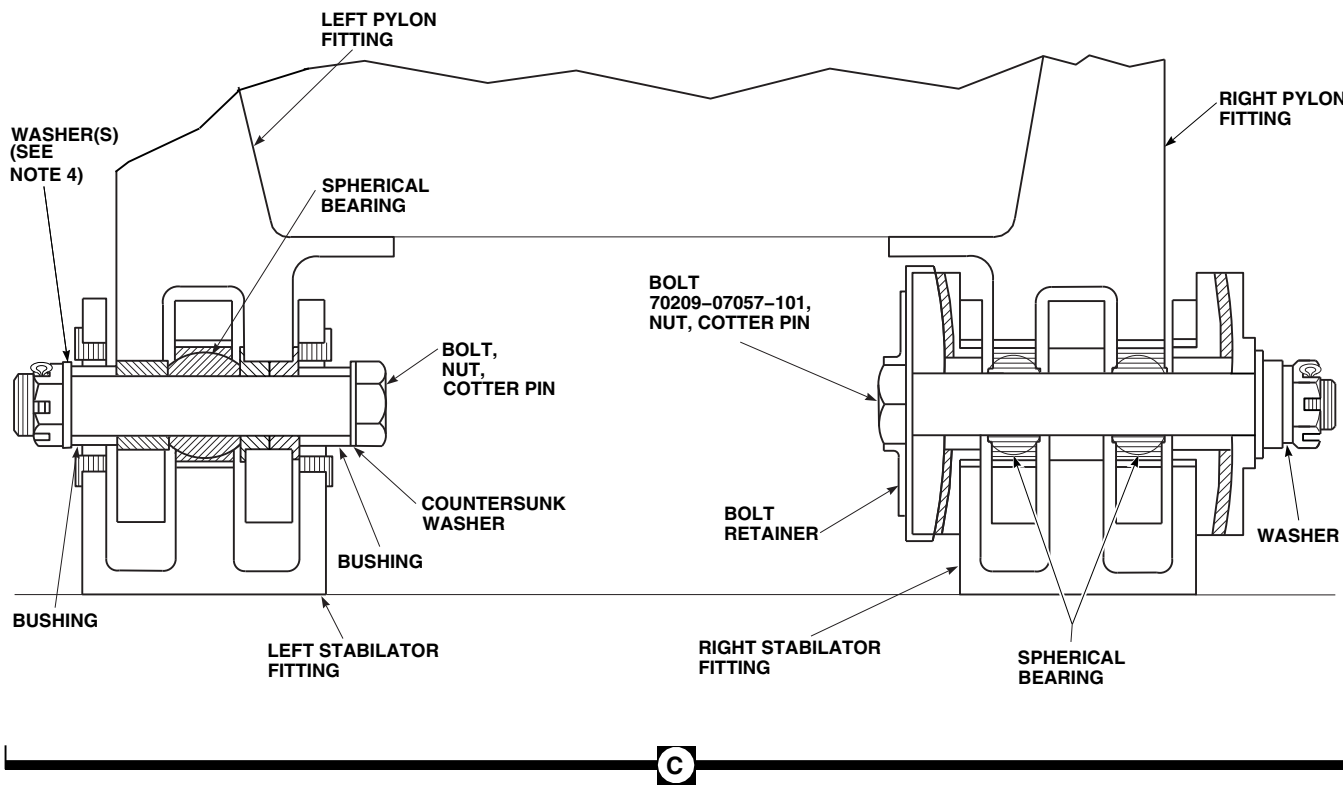
- u. With two assistants supporting stabilator, remove self-retaining bolt, washer(s), bushings, rubber spacers, and nut securing stabilator actuator to stabilator actuator attachment fitting (Figure 2-4-142-1, Sheet 2, Detail B).

- v. Remove bolts, lockwashers, and washers and disconnect bonding jumpers from stabilator (Figure 2-4-142-1, Sheet 1, Detail A).
- w. Remove left bolt, countersunk washer, and bushing securing stabilator to pylon.

NOTE

Stabilator bolt removal tool may be used on right and left stabilator attachment bolts.

- x. If corroded or frozen bolt cannot be removed by hand, use stabilator bolt removal tool and remove as follows:
 - (1) Select hexnut of proper length from tool and run it onto stabilator bolt.
 - (2) Place steel plate of tool against inside of stabilator rib, opposite bolt.
 - (3) Hold bolthead with wrench and back off tool hexnut with second wrench, with nut bearing against steel plate of tool.



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FIGURE 2-4-142-1. REPLACE STABILATOR (SHEET 3 OF 3)

- (4) When bolt has broken free, remove hex-nut and steel plate, and remove bolt.

NOTE

Right stabilator attachment bolt, 70209-07057-101, may become discolored (reddish-brown) while in service. Surface discoloration to bolt is not cause for replacement.

- y. Gently rocking stabilator to aid in removal, remove right bolt and bolt retainer securing stabilator to pylon. If corroded or frozen bolt cannot be removed by hand, perform step x.
- z. Remove stabilator and install in cradle assembly, 70700-20435-041.

2-4-142.1.2 Inspect.

- a. Inspect bonding jumper and hardware for corrosion. Replace bonding jumper and attaching hardware if necessary. If bonding jumper is to be replaced, remove bolt, lockwasher, and washers securing bonding jumper to pylon (Figure 2-4-142-1, Sheet 1, Detail A).
- b. Inspect top stabilator surface at front spar, both left and right sides, for skin cracks and/or loose rivets. If necessary, repair cracks (PARA 2-6-53) or replace stabilator.
- c. **70785-SUBQ** Remove screws, washers, and leading edge inspection plate cover.■
- d. Using flashlight and inspection mirror, inspect top internal spar cap and web for cracks and loose rivets. **NONE ALLOWED**. If cracks or loose rivets are found, replace stabilator.
- e. Inspect actuator upper and lower rubber spacers for damage or deterioration. **NONE ALLOWED**. If damaged or deteriorated, replace rubber spacers.

2-4-142.1.3 Install.

WARNING

FSCAP

Stabilator fittings on pylon, stabilator attachment bolts, stabilator actuator attachment fitting and self-retaining bolt have critical areas that must not be damaged during installation. Installation of stabilator attachment bolts and actuator self-retaining bolt have critical characteristics. Verification of proper installation and torque limits must be accomplished.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Pull inner sleeve bushing out a little from left pylon fitting to allow better clearance for positioning of attaching parts.
- c. Clean areas where bonding jumpers are attached using clean cloth and dry-cleaning solvent, Item 124, Appendix D.
- d. Have assistants place stabilator on helicopter, being careful not to hit position transmitter/limit switch assembly actuator rod.
- e. Coat attachment fittings with antiseize compound, Item 76, Appendix D.

NOTE

Use locally-made drift pin to help line up parts before installing attaching parts.

- f. Locally make stabilator drift pin (Figure H-251, Appendix H).
- g. Push drift pin through inboard side of right attachment fittings.
- h. Coat shank of right bolt with antiseize compound, Item 76, Appendix D.

CAUTION

Do not use force while inserting bolts through fittings, as damage may result from possible out-of-line condition.

- i. Install bolt retainer. Push bolt through retainer and inboard bearing, pushing drift pin out with bolt. Have assistants rock stabilator to aid in installation. Do not install nut now (Figure 2-4-142-1, Sheet 1, Detail A and Sheet 3, Detail C).
- j. Coat boltshank with antiseize compound, Item 76, Appendix D.
- k. Push left bolt, countersunk washer, and bushing through left stabilator fittings. Do not force bolt. Have assistants gently rock stabilator to aid installation. Do not install nut now (Figure 2-4-142-1, Sheet 3, Detail C).
- l. Connect bonding jumpers to stabilator (and pylon, if bonding jumper(s) was replaced) with bolts, lockwashers, and washers. **TORQUE BOLTS TO 95 - 105 INCH-POUNDS** (Figure 2-4-142-1, Sheet 1, Detail A).
- m. Apply light coat of petrolatum, Item 234, Appendix D, to face of rubber spacers for ease of installation.

WARNING

FSCAP

Outside diameter and bolthead shoulder of actuator self-retaining bolts and stabilator actuator attachment fitting lug areas on stabilator are critical surfaces which must not be damaged during installation of actuator. After removal of protective covering, use extreme caution when installing actuator.

- n. Place stabilator actuator in stabilator actuator attachment fitting (Figure 2-4-142-1, Sheet 1, Detail A and Sheet 2, Detail B).

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

CAUTION

Do not use force while inserting bolts through stabilator actuator attachment fitting, as damage may result from possible out-of-line condition.

NOTE

- If any attaching hardware for stabilator actuator is lost, replace all attaching hardware. Attaching hardware cannot be intermixed.
 - Install up to three washers under bolt-head to maintain proper grip length of bolt.
- o. Install self-retaining bolt, washer(s) (under bolthead), bushings, rubber spacers, and nut securing actuator to stabilator actuator attachment fitting. **TORQUE NUT TO 80 - 165 INCH-POUNDS**. Install cotter pin.
 - p. **PROOF TORQUE BOLTHEAD TO 45 - 47 INCH-POUNDS**. Apply torque stripe (PARA 2-6-5).
 - q. Coat position transmitter/limit switch assembly actuator bolt and bushing with corrosion preventive compound, Item 114, Appendix D. Do not apply compound to bolt threads.

WARNING**FSCAP**

Proper hardware installation, torque, proof torque, and safetying of self-retaining bolt are critical characteristics and must be verified.

CAUTION

To prevent damage to equipment, do not attach position transmitter/limit switch assembly actuator rod to stabilator unless stabilator actuator is in place.

NOTE

- Install up to three washers under bolt-head to maintain proper grip length of bolt.
 - Before installing nut, make sure no portion of self-retaining element of bolt (ring or balls) is inside bolthead.
- r. Position transmitter/limit switch assembly actuator rod on stabilator fitting (Figure 2-4-142-1, Sheet 3, Detail D). Install bushing, self-retaining bolt, washers, and nut. **TORQUE TO 19 - 21 INCH-POUNDS.** Install cotter pin.
- s. **PROOF TORQUE BOLTHEAD TO 10 - 12 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).
- t. Turn on all electrical power (PARA 1-6-16).
- u. Push in the following circuit breakers:
- (1) STAB PWR (pilot's circuit breaker panel - NO. 2 DC PRI BUS)
 - (2) STAB PWR (upper console - DC ESNTL BUS)
- v. With assistant watching stabilator, place MAN SLEW to -8° (trailing edge up).

- w. Turn off all electrical power (PARA 1-6-16).
- x. Pull out the following circuit breakers:
- (1) STAB PWR (pilot's circuit breaker panel - NO. 2 DC PRI BUS)
 - (2) STAB PWR (upper console - DC ESNTL BUS)
- y. Install washer and nut on right stabilator attachment bolt (Figure 2-4-142-1, Sheet 3, Detail C).

WARNING**FSCAP**

Right stabilator attachment bolt installation has critical characteristics. Verification of nut tightening procedure and cotter pin installation are critical characteristics.

CAUTION

To prevent damage to fitting, make sure retainer is properly aligned with elastomeric bearing and bolthead before torquing nut. Hold bolthead securely while torquing nut.

- z. To torque right nut, perform the following:
- (1) **TIGHTEN NUT TO 25 - 50 INCH-POUNDS.**
 - (2) **IF SLOT IN NUT IS LINED UP WITH COTTER PIN HOLE IN BOLT.** Install cotter pin. Apply torque stripe (PARA 2-6-5).
 - (3) **IF SLOT IN NUT IS NOT LINED UP WITH COTTER PIN HOLE IN BOLT, KEEP TIGHTENING NUT UNTIL HOLE AND SLOT ARE LINED UP. DO NOT EXCEED 50 INCH-POUNDS.** Install cotter pin. Apply torque stripe (PARA 2-6-5).

WARNING

FSCAP

Left stabilator attachment bolt installation has critical characteristics. Verification of nut torque and cotter pin installation are critical characteristics.

- aa. Install bushings, washers, and nut on left stabilator attachment bolt. **TORQUE TO 340 - 680 INCH-POUNDS.** Install cotter pin (Figure 2-4-142-1, Sheet 1, Detail A and Sheet 3, Detail C). Apply torque stripe (PARA 2-6-5).
- ab. Remove protective caps and plugs from electrical connectors.
- ac. Inspect and treat electrical connectors (PARA 1-7-20).
- ad. Connect electrical connectors to receptacles on pylon (Figure 2-4-142-1, Sheet 1, Detail A).
- ae. Install tail rotor pylon lower fairing (PARA 2-4-133).
- af. Apply corrosion preventive compound, Item 115, Appendix D, to fairing attachment hardware.
- ag. Install left and right stabilator-to-ylon fairings with accessible screws and washers.
- ah. Turn on all electrical power (PARA 1-6-16).
- ai. Push in the following circuit breakers:
 - (1) STAB PWR (pilot's circuit breaker panel - NO. 2 DC PRI BUS)
 - (2) STAB PWR (upper console - DC ESNTL BUS)
- aj. Adjust position transmittter/limit switch assembly actuator rod (TM 11-70-23).
- ak. With assistant watching stabilator, check operation by manually slewing through full travel. Leave stabilator fully down.
- al. Push in the following circuit breakers:
 - (1) STAB CONTR (pilot's circuit breaker panel - NO. 2 AC PRI BUS)
 - (2) STAB CONTR (pilot's circuit breaker panel - AC ESNTL BUS)
- am. Remove external power from helicopter (PARA 1-6-16).
- an. Apply corrosion preventive compound, Item 115, Appendix D, to fairing attachment hardware.
- ao. Install remaining screws and washers in stabilator-to-ylon fairings.

NOTE

If stabilator has been removed from and reinstalled on same serial numbered helicopter without repair or adjustment of stabilator, no maintenance test flight is required.

- ap. Perform maintenance test flight (Appropriate MTF).

2-4-143 STABILATOR ELECTROSTATIC DISCHARGERS.

PARAGRAPH BREAKDOWN

	PAGE
2-4-143.1 Replace Stabilator Electrostatic Dischargers	2-4-143-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Heat Gun

Materials/Parts

- Insulation Sleeving, Heat-Shrinkable, Item 162, Appendix D

References

- Appendix D

2-4-143.1 Replace Stabilator Electrostatic Dischargers.

NOTE

Replacement of all electrostatic dischargers is the same.

2-4-143.1.1. Remove.

Remove screws, washers, and nut securing electrostatic discharger to stabilator. Remove electrostatic discharger (Figure 2-4-143-1, Detail A).

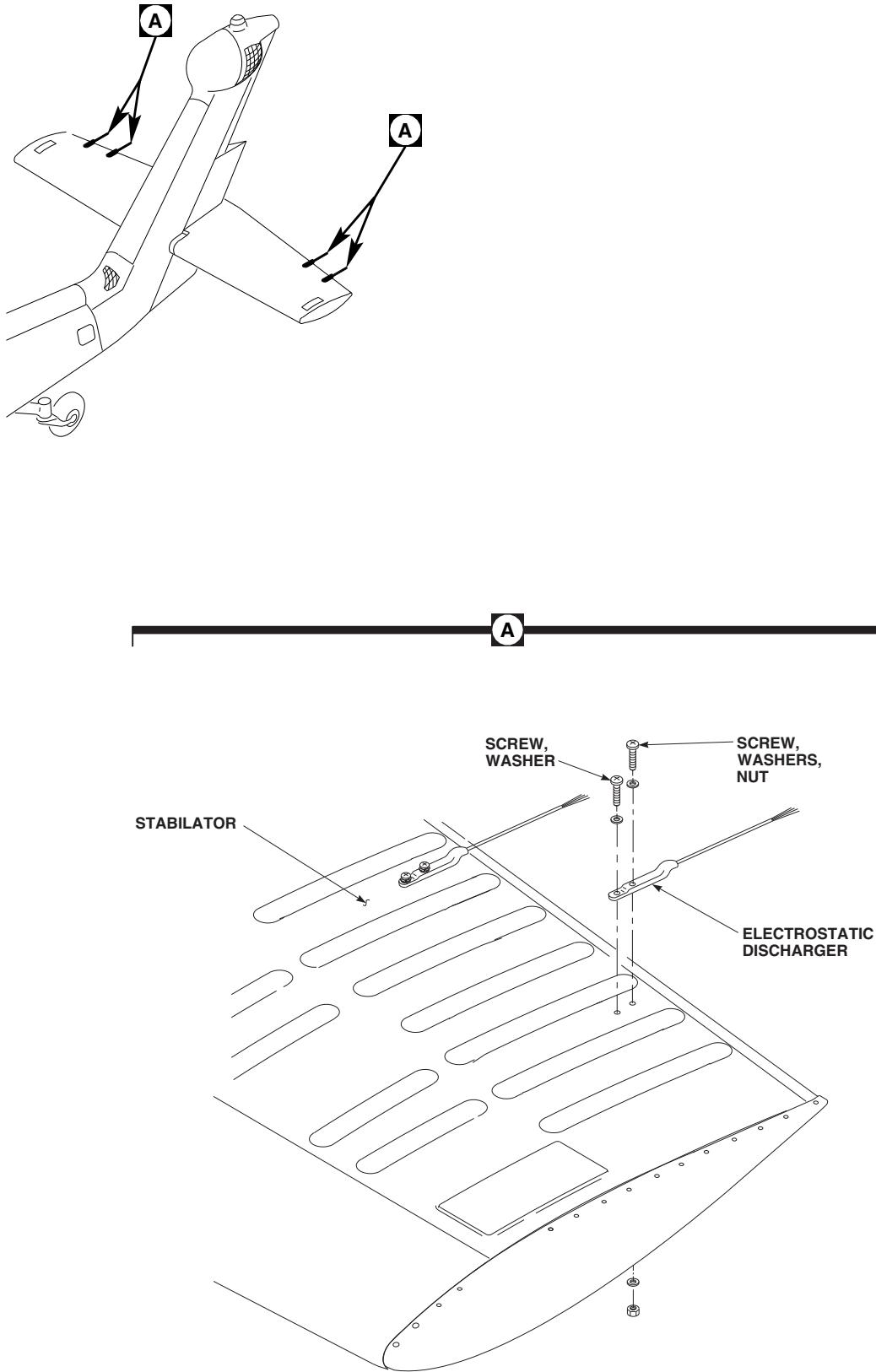
2-4-143.1.2. Repair.

- Repair cracks in electrostatic discharger rubber coating as follows:

- Cut a piece of heat-shrinkable insulation sleeving, Item 162, Appendix D, 1 ½-inches long.
- Slide insulation sleeving over electrostatic discharger. Make sure insulation sleeving extends ¾-inch beyond both sides of crack.
- Shrink insulation sleeving around electrostatic discharger using heat gun.

2-4-143.1.3. Install.

- Position electrostatic discharger on stabilator and secure with screw and washer in forward hole (Figure 2-4-143-1, Detail A).
- Install screw, washers, and nut in rear hole.



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FIGURE 2-4-143-1. REPLACE STABILATOR ELECTROSTATIC DISCHARGERS

2-4-143-2

Change 1

2-4-144 STABILATOR ATTACHMENT FITTINGS.

PARAGRAPH BREAKDOWN

	PAGE
2-4-144.1 Replace Stabilator Attachment Fittings	2-4-144-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Drill
- Drill Bit, 1⁄8-inch diameter
- Drill Bit, 1⁄4-inch diameter
- Feeler Gage
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Adhesive, Item 37, Appendix D
- Cheesecloth, Item 90, Appendix D

- Corrosion Compound, Item 108, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Paint, Polyurethane, Item 222A, Appendix D
- Scouring Pad, Item 270, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-7-11
- PARA 1-7-16
- PARA 2-4-142
- PARA 2-6-5

2-4-144.1 Replace Stabilator Attachment Fittings.

NOTE

Replacement of both stabilator attachment fittings is the same, except as noted.

2-4-144.1.1 Remove.

- Remove stabilator (PARA 2-4-142).

- Remove screws and washers securing access covers to underside of stabilator (Figure 2-4-144-1, Detail A). Remove access covers.
- Remove bolts, washers, square washers, countersunk washers, and nuts. Remove fitting (Figure 2-4-144-1, Detail A and Detail B).
- Using nonmetallic scraper, remove laminated shim from stabilator. Discard shim.
- Using cheesecloth, Item 90, Appendix D, soaked in methyl ethyl ketone, Item 217, Ap-

pendix D, remove any remaining adhesive from stabilator.

2-4-144.1.2 Inspect.

NOTE

Corrosion is indicated by red film or pitting.

Visually inspect right side stabilator attachment fitting for corrosion. **PITS SHALL NOT EXCEED 0.050-INCH IN DIAMETER.** If corrosion is found, repair (PARA 2-4-144.1.3).

2-4-144.1.3 Repair.

- a. Using scouring pad, Item 270, Appendix D, manually remove corrosion.
- b. Mix solution of one part corrosion compound, Item 108, Appendix D, to one part water.
- c. Apply compound solution to corroded areas of fitting. Rinse thoroughly with clean water.
- d. Using epoxy primer coating kit, Item 135, Appendix D, and polyurethane paint, Item 222A, Appendix D, touch up paint (PARA 2-6-5).

2-4-144.1.4 Install.

- a. Prepare laminated shim as follows:

(1) Drill locator pin hole as follows:

- (a) Position stabilator attachment fitting and laminated shim on stabilator with four corner bolts, square washers, washers, and nuts. Position square washers under boltheads, with radius sides facing left to right and toward fitting (Figure 2-4-144-1, Detail A and Detail B).
- (b) Tighten bolts only enough to secure shim to stabilator and leave a dimple impression of locator pin.
- (c) Remove bolts, square washers, washers, and nuts and remove fitting and laminated shim.

- (d) Using drill and 1/8-inch diameter drill bit, drill locator pin hole through impression on laminated shim.

(2) Drill middle mounting holes as follows:

- (a) Position laminated shim on fitting and secure to stabilator with four corner bolts, square washers, washers, and nuts. Position square washers under boltheads, with radius sides facing left to right and toward fitting.
- (b) Tighten bolts only enough to secure shim to stabilator.
- (c) Using remaining fitting boltholes as a template, mark laminated shim for drilling.
- (d) Remove bolts, square washers, washers, and nuts and remove fitting and laminated shim.
- (e) Using drill with 1/4-inch diameter drill bit, drill boltholes through laminated shim where marked.

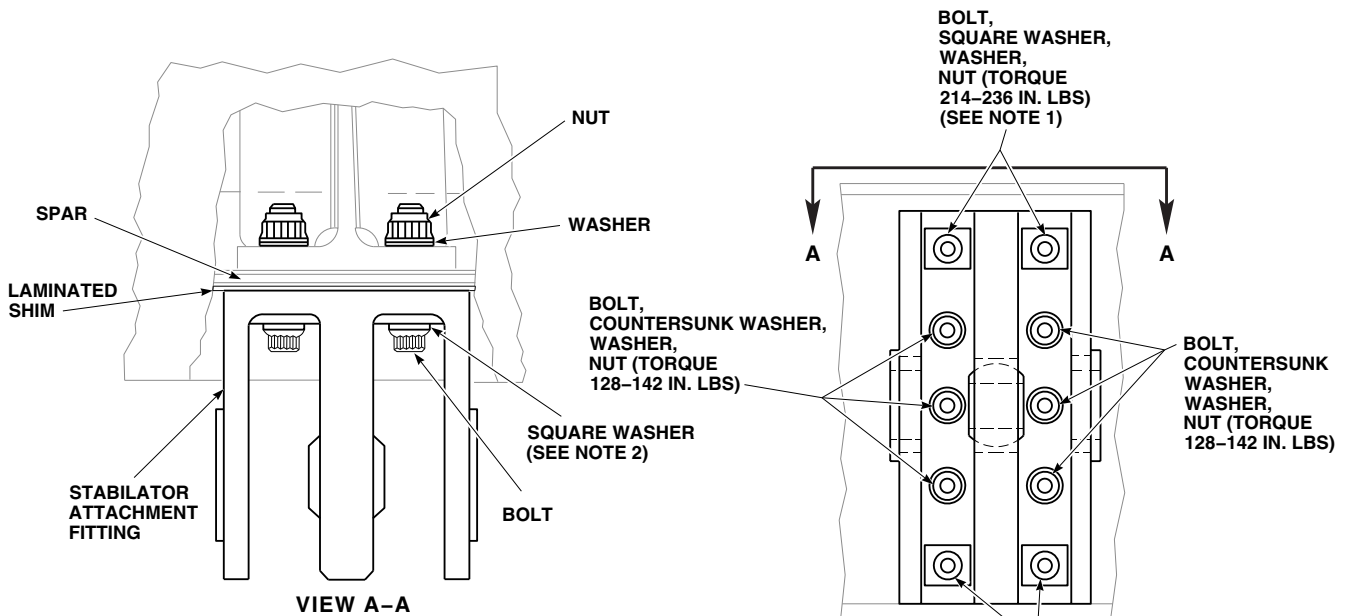
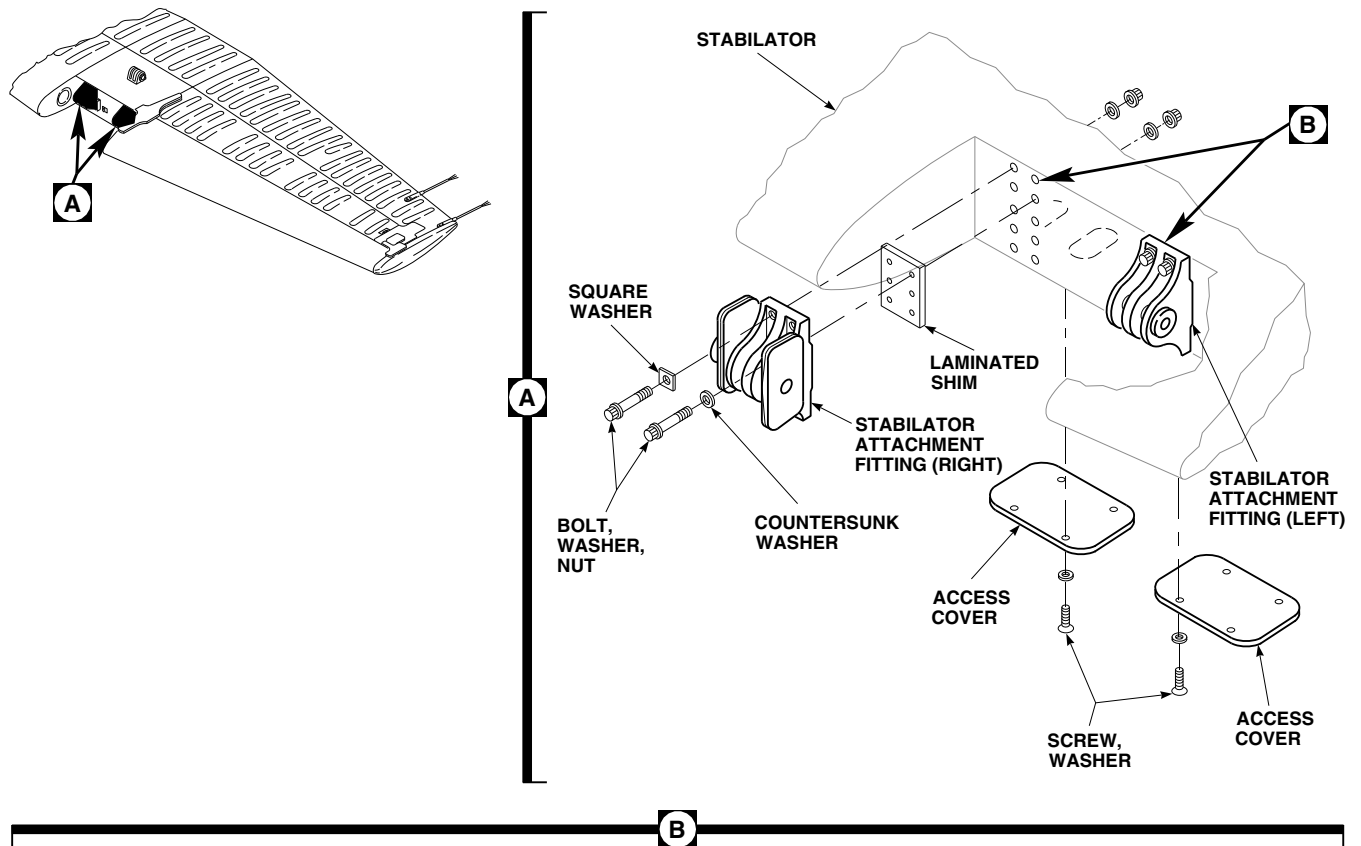
- b. Apply light coat of adhesive, Item 37, Appendix D, to solid side of laminated shim.

- c. Position laminated shim on stabilator with holes in shim lining up with six inner holes in stabilator.

CAUTION

Damage to equipment could result if locator pin in right stabilator attachment fitting is missing. Make sure pin has not fallen out and that pin mates properly with locator hole in stabilator.

- d. Position fitting on stabilator.
- e. Install four corner bolts, square washers, washers, and nuts. Position square washers under boltheads, with radius sides facing left to right and toward fitting.



NOTES

1. EITHER SQUARE WASHER, 70201-07056-119 OR 70201-07056-120, MAY BE USED.
2. MATE RADIUS OF SQUARE WASHERS (RADIUS BLOCKS) WITH RADIUS OF STABILATOR ATTACHMENT FITTING.

FIGURE 2-4-144-1. REPLACE STABILATOR ATTACHMENT FITTINGS

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- f. Install middle bolts, countersunk washers, washers, and nuts. Install countersunk washers under bolthead, with countersunk side toward bolthead.
- g. Tighten top and bottom corner bolts in a diagonal pattern (top left corner, bottom right corner, top right corner, bottom left corner) and center bolts in a diagonal pattern (top left center, bottom right center, top right center, bottom left center, then two middle bolts) so fitting fits snug against laminated shim and stabilator.
- h. Using feeler gage, check gap between fitting and spar in stabilator (Figure 2-4-144-1, Detail B). **NONE ALLOWED.** If no gap is found, proceed to step i. If gap is found, perform the following:
 - (1) Remove bolts, square washers, countersunk washers, washers, and nuts and remove fitting.
 - (2) Peel laminated shim as required to eliminate gap between fitting and stabilator.
 - (3) Repeat step d. through step h. until gap between fitting and spar is eliminated.
- i. **TORQUE TOP AND BOTTOM CORNER BOLTS IN A DIAGONAL PATTERN (TOP**

LEFT CORNER, BOTTOM RIGHT CORNER, TOP RIGHT CORNER, BOTTOM LEFT CORNER) 214 - 236 INCH-POUNDS.

- j. Check helicopter log book to make sure top and bottom corner bolts torque check is shown due 30 flight hours after installation (PARA 1-7-11).
- k. **TORQUE CENTER BOLTS IN A DIAGONAL PATTERN (TOP LEFT CENTER, BOTTOM RIGHT CENTER, TOP RIGHT CENTER, BOTTOM LEFT CENTER; THEN TWO MIDDLE BOLTS) 128 - 142 INCH-POUNDS.**
 - l. Check helicopter log book to make sure center bolts torque check is shown due 250 flight hours after installation (PARA 1-7-16).
- m. Using feeler gage, check between fitting and spar in stabilator to make sure gap has been eliminated.
- n. Apply fillet of sealing compound, Item 292, Appendix D, to seal joining area of fitting.
- o. Install access covers on stabilator and secure with screws and washers (Figure 2-4-144-1, Detail A).
- p. Install stabilator (PARA 2-4-142).

2-4-145 STABILATOR TIP CAPS.

PARAGRAPH BREAKDOWN

	PAGE
2-4-145.1 Replace Stabilator Tip Caps	2-4-145-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Clecocos, 1⁄8-inch diameter
- Drill
- Drill Bit, 1⁄8-inch diameter
- Nonmetallic Scraper, Locally-Made
- Pin Punch, 3⁄32-inch diameter drive

Materials/Parts

- Cloth, Cleaning, Item 102, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 2-6-1
- PARA 2-6-5

2-4-145.1 Replace Stabilator Tip Caps.

NOTE

Replacement of both stabilator tip caps is the same.

2-4-145.1.1 Remove.

- Carefully drill out and remove rivets and washers securing tip cap to stabilator using drill with 1⁄8-inch diameter drill bit and 3⁄32-inch diameter drive pin punch (Figure 2-4-145-1, Detail A).
- Remove tip cap using nonmetallic scraper.
- Using nonmetallic scraper, remove old sealing compound.

- Clean up rivet debris and wipe area clean with cleaning cloth, Item 102, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D.

2-4-145.1.2 Install.

- Make sure area is clean and free of foreign material.
- Apply sealing compound, Item 292, Appendix D, to mating surface of stabilator and to inside edge of tip cap.
- Position tip cap on stabilator and temporarily hold in place using 1⁄8-inch diameter clecocos (Figure 2-4-145-1, Detail A).
- Install rivets as follows:

NOTE

All rivets must be installed wet with sealing compound, Item 292, Appendix D.

- (1) 8 blind rivets, NAS1738B4-2, and washers, SS5072-T4H, in holes 1 through 4 from leading edge, upper and lower surfaces (Figure 2-4-145-1, Detail A).
- (2) 20 blind rivets, NAS1738B4-1, and washers, SS5072-T4H, in holes 5 through 14 from leading edge, upper and lower surfaces.
- (3) 1 blind rivet, MS20470AD3, and 2 washers, SS5072-3H, in hole 15 at trailing edge.

- e. Install one rivet of required length between each cleco, top and bottom.
- f. Install second rivet of required length between each cleco, top and bottom.
- g. Remove clecos and install remaining rivets.
- h. Inspect rivets for acceptable condition.
- i. Apply sealing compound, Item 292, Appendix D, around edges of tip cap (PARA 2-6-1). Thickness of sealing compound after curing shall be 0.005-inch maximum.
- j. Touch up paint finish as required (PARA 2-6-5).

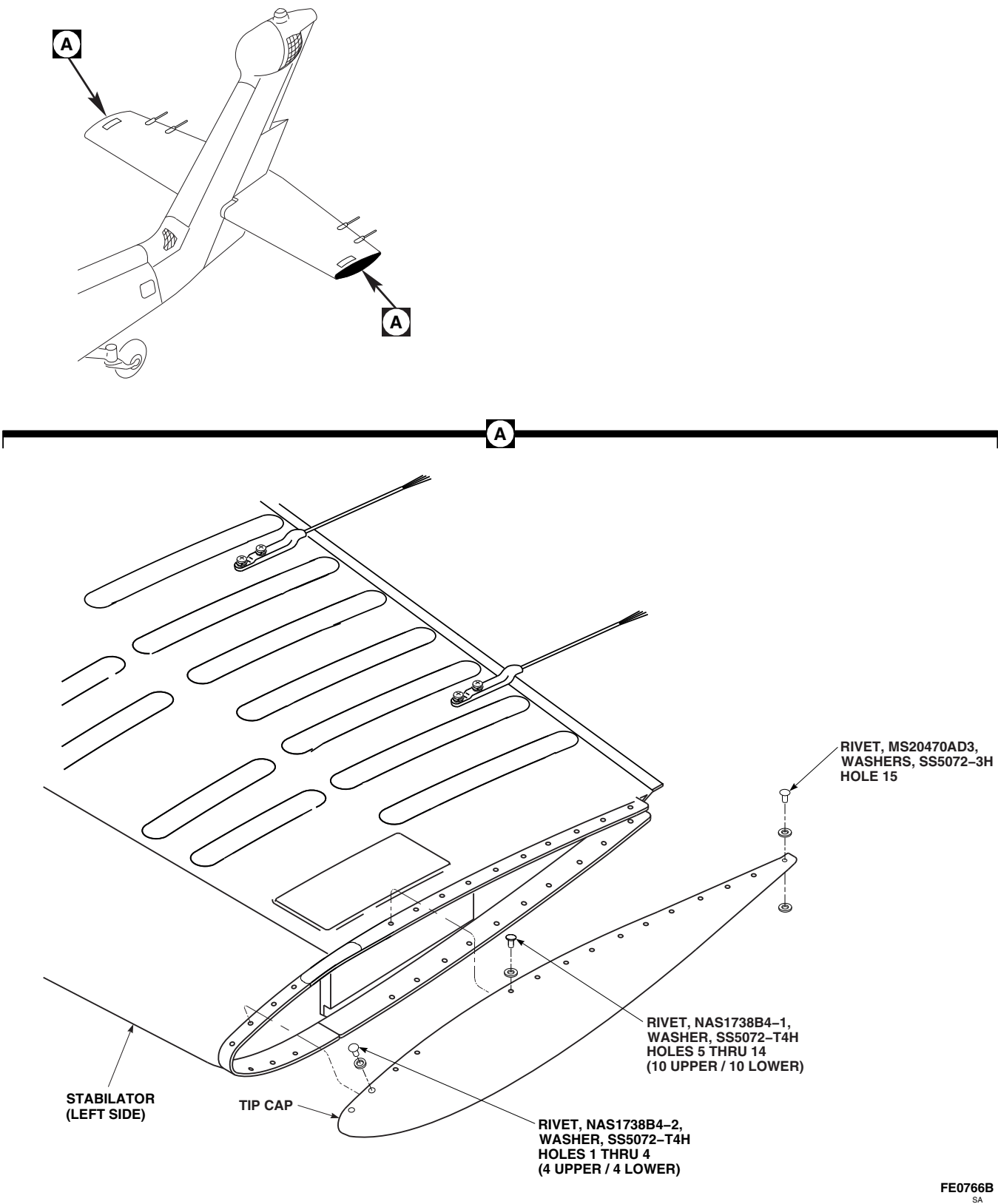


FIGURE 2-4-145-1. REPLACE STABILATOR TIP CAPS

2-4-146 STABILATOR ACTUATOR.

PARAGRAPH BREAKDOWN

	PAGE
2-4-146.1 Replace Stabilator Actuator	2-4-146-2

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Digital Low-Resistance Ohmmeter
- Electrical Repairer’s Toolkit
- Fluorescent Penetrant Inspection Kit
- Nonmetallic Scraper, Locally-Made
- Padded Support
- Protective Sleeve
- Torque Wrench, 30 - 150 in. lbs
- Torque Wrench, 150 - 750 in. lbs

Materials/Parts

- Abrasive Cloth, Item 6, Appendix D
- Abrasive Cloth, Item 8, Appendix D
- Abrasive Cloth, Item 10, Appendix D
- Abrasive Mat, Item 14, Appendix D
- Corrosion Compound, Item 108, Appendix D

- Lubricant, Solid Film, Item 201, Appendix D
- Machinery Towel, Item 211, Appendix D
- Petrolatum, Item 234, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-3-15
- PARA 2-4-135
- PARA 2-4-138
- PARA 2-4-148
- PARA 2-6-5
- PARA 2-6-48
- PARA 2-6-55
- TM 11-70-23

2-4-146.1 Replace Stabilator Actuator.**NOTE****2-4-146.1.1 Remove.**

- a. Pull out the following circuit breakers:
 - (1) STAB CONTR (pilot's circuit breaker panel - NO. 2 AC PRI BUS)
 - (2) STAB CONTR (pilot's circuit breaker panel - AC ESNTL BUS)
- b. Make sure the following circuit breakers are pushed in:
 - (1) STAB IND (pilot's circuit breaker panel - AC ESNTL BUS)
 - (2) STAB PWR (upper console - DC ESNTL BUS)
- c. Turn on all electrical power (PARA 1-6-16).
- d. Establish ICS communication between cockpit and maintenance personnel at stabilator. Shut off beeping tone in headsets by pressing, then releasing, pilot's MASTER CAUTION PRESS TO RESET capsule.
- e. On STABILATOR CONTROL/AUTO FLT CONT panel, use MAN SLEW switch to position stabilator as required to access lower stabilator actuator attachment hardware.
- f. Turn off all electrical power (PARA 1-6-16).
- g. Open tail rotor pylon leading edge cover/antenna. If cover/antenna interferes with removal of stabilator actuator, remove cover/antenna (PARA 2-4-138).
- h. **W/O EMEP** Remove screws securing connector bracket to tail rotor pylon and lift bracket from under tail rotor drive shaft section IV (Figure 2-4-146-1, Sheet 1, Detail A, and Detail B).
- i. **701503** **701523 - SUBQ** Remove screws and separate connector bracket halves (Figure 2-4-146-1, Sheet 2, Detail D).

EMEP Pin filter adapters are considered part of harness assembly. Do not remove pin filter adapters from electrical connectors, J606 and J607 (Figure 2-4-146-1, Sheet 1, Detail B).

- j. Disconnect electrical connectors, P606 from J606, and P607 from J607 (Figure 2-4-146-1, Sheet 1, Detail A).
- k. Install protective caps and plugs on electrical connectors.
 - l. If tail rotor pylon cambered fairing interferes with removal of stabilator actuator, remove tail rotor pylon cambered fairing (PARA 2-4-135).
- m. Using nonmetallic scraper, remove sealing compound from bolts, washers, and nuts securing grounding strap to grounding bracket and aft spar.

NOTE

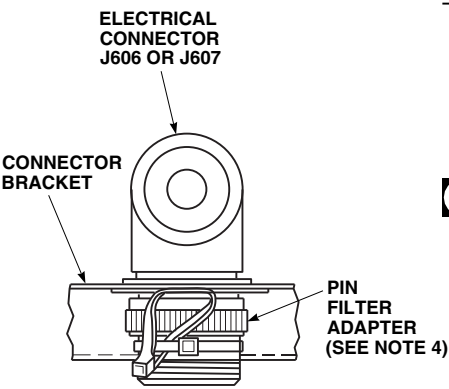
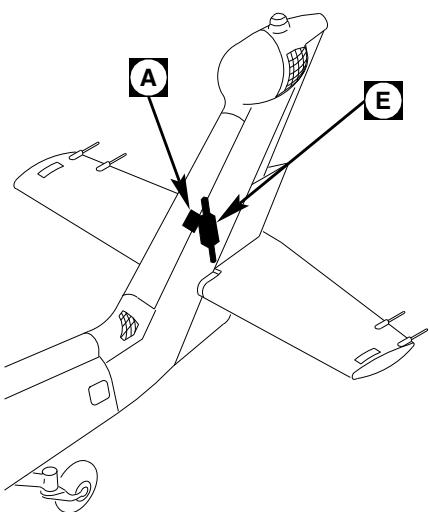
Bolt and washer securing grounding strap will remain with grounding bracket once nut, lockwasher, washer, and grounding strap are removed.

- n. **701334-SUBQ** Remove nut, lockwasher, washer, and grounding strap from bolt and washer on grounding bracket (Figure 2-4-146-1, Sheet 2, Detail C).

CAUTION

To prevent damage to limit switches connected to stabilator, position sensor actuator rod, stabilator, or lower fairing; make sure support is high enough to prevent stabilator from going further than 35° trailing edge down. To prevent damage to stabilator trailing edge, make sure support is well padded.

- o. Place padded support below stabilator trailing edge.



NOTES

- 1. APPLY A LIGHT COAT OF PETROLATUM TO FACE OF RUBBER SPACERS FOR EASE OF INSTALLATION.
- 2. INSTALL WASHERS UNDER BOLTHEAD (3 MAXIMUM), TO MAKE SURE NUT BOTTOMS AGAINST BUSHING AND NOT ON BOLT.
- 3. BOLT MAY BE INSTALLED FROM EITHER DIRECTION.
- 4. **EMEP**
- 5.

POINTS FROM TO		RESISTANCE SHALL BE
A	B	7.5 MILLIOHMS OR LESS

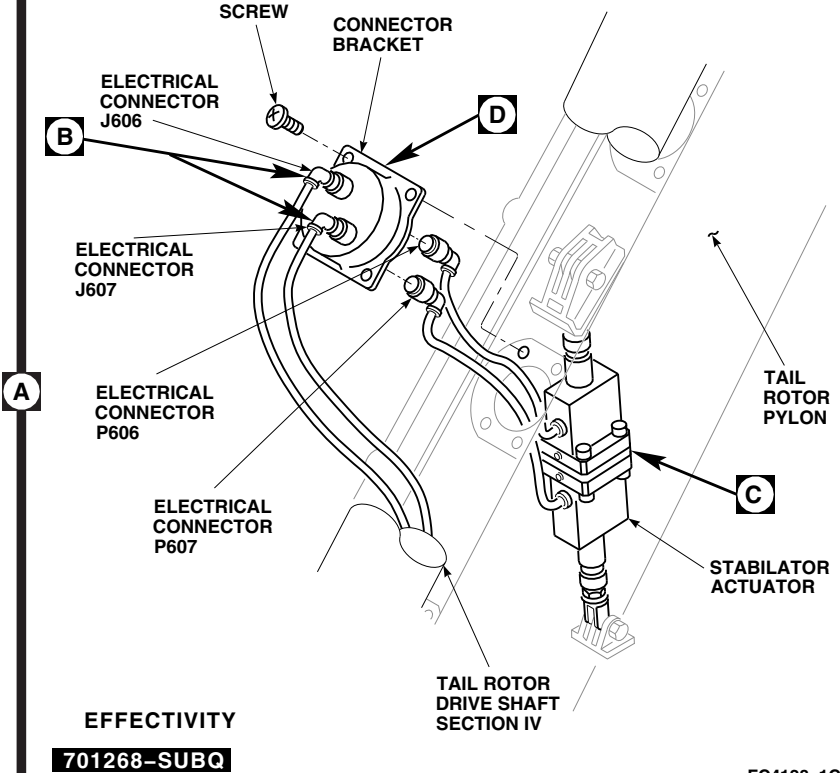
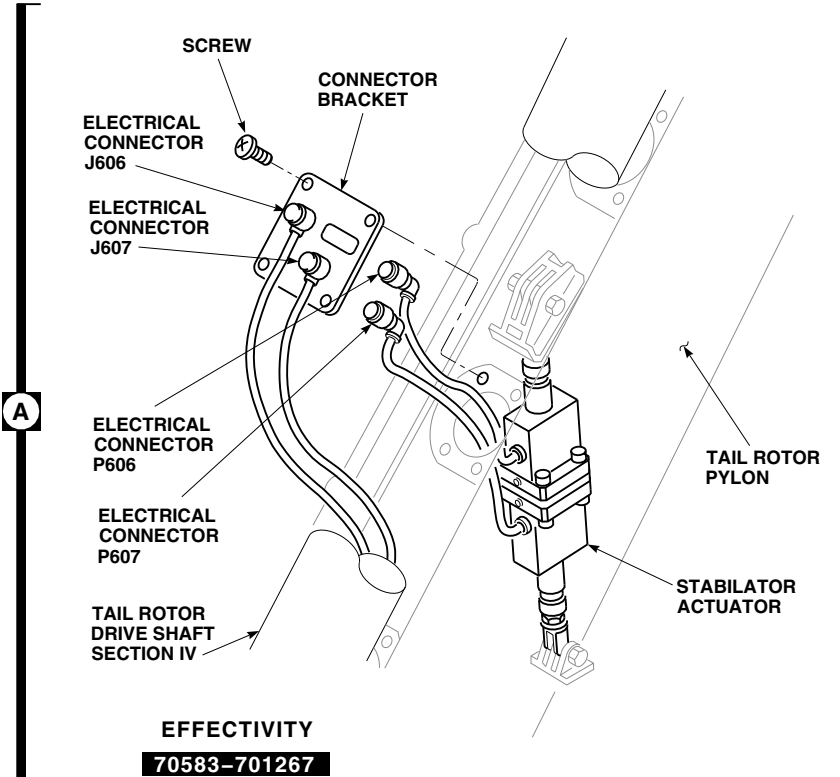
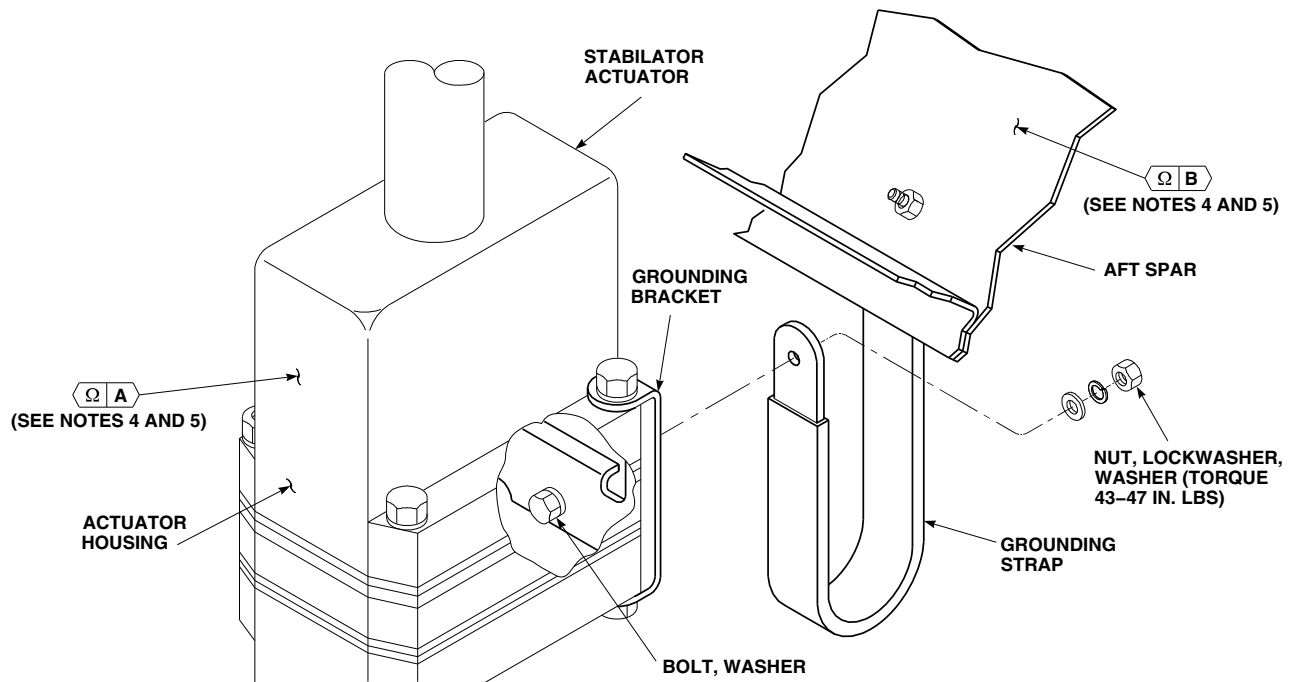


FIGURE 2-4-146-1. REPLACE STABILATOR ACTUATOR (SHEET 1 OF 3)

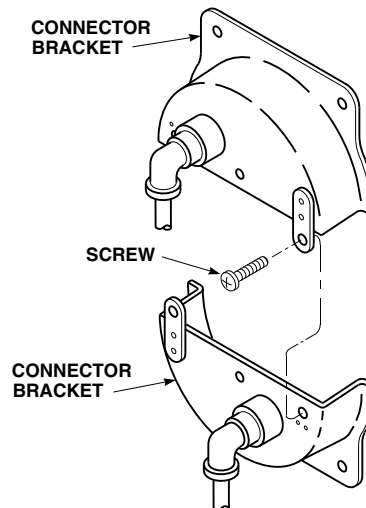


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EFFECTIVITY

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FIGURE 2-4-146-1. REPLACE STABILATOR ACTUATOR (SHEET 2 OF 3)

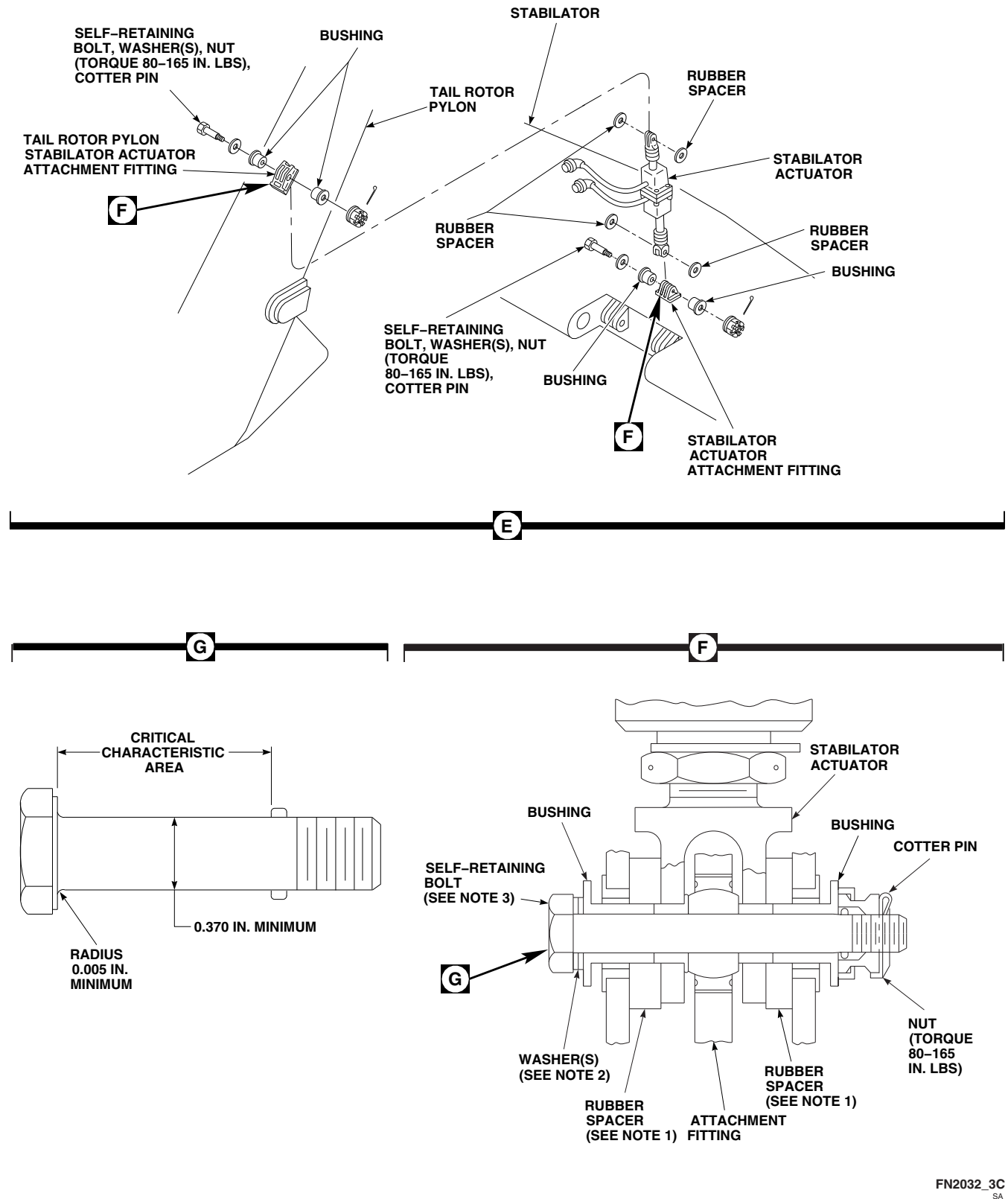


FIGURE 2-4-146-1. REPLACE STABILATOR ACTUATOR (SHEET 3 OF 3)

WARNING**FSCAP**

Outside diameter and bolthead shoulder of self-retaining bolts and stabilator actuator attachment fitting lug areas are critical surfaces which must not be damaged during and after removal of actuator. Use extreme caution when removing actuator and provide protective covering after removal.

- p. Remove self-retaining bolt, washer(s), bushings, rubber spacers, and nut securing stabilator actuator to stabilator actuator attachment fitting. Rest stabilator on padded support (Figure 2-4-146-1, Sheet 3, Detail E and Detail F).
- q. Install protective sleeve on self-retaining bolt.

WARNING

Injury to personnel and damage to equipment will result if moving parts of stabilator actuator come in contact with personnel or helicopter. Make sure actuator shafts do not come in contact with personnel or any part of helicopter when positioning shafts.

- r. Position stabilator actuator to ease removal as follows:
 - (1) Remove protective caps and plugs from electrical connectors.
 - (2) Temporarily connect electrical connectors, P606 to J606, and P607 to J607.
 - (3) Turn on all electrical power (PARA 1-6-16).
 - (4) On STABILATOR CONTROL/AUTO FLT CONT panel, use MAN SLEW switch to position stabilator as required to access lower stabilator actuator attachment hardware.

- (5) Turn off all electrical power (PARA 1-6-16).
- (6) Disconnect electrical connectors, P606 from J606, and P607 from J607.
- (7) Install protective caps and plugs on electrical connectors.

WARNING**FSCAP**

Outside diameter and bolthead shoulder of self-retaining bolts and pylon attachment fitting lug areas are critical surfaces which must not be damaged during and after removal of stabilator actuator. Use extreme caution when removing actuator and provide protective covering after removal.

- s. Remove self-retaining bolt, washer(s), rubber spacers, bushings, and nut securing stabilator actuator to tail rotor pylon stabilator actuator attachment fitting. Remove actuator.
- t. Install protective sleeve on self-retaining bolt.

2-4-146.1.2 Inspect.

- a. Check stabilator actuator cables and electrical connectors, P606 and P607, for damage. **NONE ALLOWED.**

WARNING

Proper bonding of liner in center lug of pylon attachment fitting is a critical characteristic and must be verified.

- b. Inspect tail rotor pylon stabilator actuator attachment fitting for liner edge separation (PARA 2-6-48 and PARA 2-3-15).
- c. Inspect stabilator actuator attachment fitting lug areas (PARA 2-6-55 and PARA 2-3-15).

- d. Check stabilator actuator housing for damage and cracks. **NONE ALLOWED.** Minor scratches are acceptable.
- e. Inspect attachment hardware for damage and scratches. **NO SCRATCHES, NICKS, BURRS, FRETTING, OR SCORING ALLOWED ON BOLTS OR BUSHINGS.** If damaged or scratched, replace parts as necessary.
- f. Visually inspect outside diameter and shoulder of bolts for nicks, dents, and scratches. **SCRATCHES NOT THROUGH CADMIUM-PLATING ARE ALLOWED.** If damaged, replace bolts.

NOTE

Repair self-retaining bolt(s) immediately upon discovery of damage and replace bolt(s) at earliest possible opportunity.

- g. If damage to self-retaining bolt(s) has penetrated into parent metal and new bolt(s) is not available for replacement, perform temporary repair as follows:

WARNING

FSCAP

Minimum shank diameter and head-to-shank radius, as given in Figure 2-4-146-1, Sheet 3, Detail G, must be retained.

- (1) Using abrasive cloth, Item 6 or Item 8, Appendix D, or abrasive mat, Item 14, Appendix D, blend damaged area. **DEPTH OF BLENDED DAMAGED AREA SHALL NOT EXCEED 0.003-INCH.** Blend reworked area to smooth transition into adjacent areas.
- (2) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect bolt for cracks (ASTM E1417). **NONE ALLOWED.** If cracked, replace bolt.

- (3) Apply corrosion compound, Item 108, Appendix D. Rinse thoroughly with water and allow to dry.
- (4) If process is available, brush cadmium-plate repaired area of bolt. As an alternate, apply solid film lubricant, Item 201, Appendix D.
- h. Inspect stabilator actuator upper and lower rubber spacers for damage or deterioration. **NONE ALLOWED.** If damaged or deteriorated, replace rubber spacers.
- i. Inspect stabilator actuator shaft for surface corrosion. If surface corrosion is found, remove surface corrosion from actuator shafts with abrasive mat, Item 14, Appendix D. Wipe shafts clean with machinery towel, Item 211, Appendix D.

NOTE

Stabilator actuator shafts must be fully extended for complete inspection of shafts.

- j. If shafts are not fully extended, extend shafts as follows:
 - (1) Remove protective caps and plugs from electrical connectors.

WARNING

Injury to personnel and damage to equipment will result if moving parts of stabilator actuator come in contact with personnel or helicopter. Make sure actuator shafts do not come in contact with personnel or any part of helicopter when extending shafts.

- (2) Temporarily connect electrical connectors, P606 to J606, and P607 to J607.
- (3) Turn on all electrical power (PARA 1-6-16).

- (4) On STABILATOR CONTROL/AUTO FLT CONT panel, place MAN SLEW switch to DN position to extend shafts.
- (5) Turn off all electrical power (PARA 1-6-16).
- (6) Disconnect electrical connectors, P606 from J606, and P607 from J607.
- (7) Install protective caps and plugs on electrical connectors.

- k. Check stabilator actuator output shafts for scratches, nicks, and burrs. **SCRATCHES OF ANY LENGTH SHALL NOT EXCEED 0.010-INCH DEEP. NICKS AND BURRS SHALL NOT EXCEED 0.250-INCH LONG, 0.050-INCH WIDE, AND 0.010-INCH-DEEP.** If necessary, using abrasive cloth, Item 10, Appendix D, blend out all rough and raised edges.
- l. **701334-SUBQ** Inspect stabilator actuator grounding strap (PARA 2-3-15).■

2-4-146.1.3 Install.

NOTE

- Stabilator actuator adjustment is required when one or both actuators have been replaced.
 - Stabilator actuator is made up of two halves. If actuator, 70400-06641-117, is not available, actuators, 70400-06641-115 and 70400-06641-117, may be used in any combination.
- a. If required, adjust both stabilator actuators (PARA 2-4-148).
 - b. On pilot's circuit breaker panel, make sure both STAB CONTR circuit breakers are still pulled out.
 - c. Turn on electrical power (PARA 1-6-16).
 - d. Establish ICS communication between cockpit and maintenance personnel at stabilator. Shut off beeping tone in headsets by pressing, then releasing, pilot's MASTER CAUTION PRESS TO RESET capsule.

WARNING

Injury to personnel and damage to equipment will result if moving parts of stabilator actuator come in contact with personnel or helicopter. Make sure actuator shafts do not come in contact with personnel or any part of helicopter when extending shafts.

CAUTION

Damage to limit switch will occur if stabilator is allowed to go further than 35° trailing edge down. To prevent damage to limit switch, do not allow stabilator to exceed 35° trailing edge down.

- e. Have assistant support trailing edge of stabilator and help line up boltholes in stabilator actuator lower clevis and stabilator actuator attachment fitting. On STABILATOR CONTROL/AUTO FLT CONT panel, use MAN SLEW switch to position actuator to line up boltholes.

WARNING

FSCAP

Outside diameter and bolthead shoulder of self-retaining bolts and pylon attachment fitting lug areas are critical surfaces which must not be damaged during installation of stabilator actuator. After removal of protective covering, use extreme caution when installing actuator.

- f. Position upper clevis (long clevis) of stabilator actuator in tail rotor pylon stabilator actuator attachment fitting.
- g. Apply light coat of petrolatum, Item 234, Appendix D, to face of rubber spacers for ease of installation.

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

CAUTION

Do not use force while inserting bolts through tail rotor pylon stabilator actuator attachment fitting, as damage may result from possible out-of-line condition.

NOTE

- If any attaching hardware for stabilator actuator is lost, replace all attaching hardware. Attaching hardware cannot be intermixed.
- Install up to three washers under bolt-head to maintain proper grip length of bolt.
- h. Install upper self-retaining bolt, washer(s) (under bolthead), bushings, rubber spacers, and nut securing stabilator actuator to tail rotor pylon stabilator actuator attachment fitting. **TORQUE NUT TO 80 - 165 INCH-POUNDS.** Install cotter pin.
- i. **PROOF TORQUE BOLTHEAD TO 45 - 47 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).

WARNING

FSCAP

The outside diameter and bolthead shoulder of the self-retaining bolts and stabilator actuator attachment fitting lug

areas on stabilator are critical surfaces which must not be damaged during installation of actuator. After removal of protective covering, use extreme caution when installing actuator.

- j. Position lower clevis (short clevis) of stabilator actuator in stabilator actuator attachment fitting.
- k. Apply light coat of petrolatum, Item 234, Appendix D, to face of rubber spacers for ease of installation.

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

CAUTION

Do not use force while inserting bolts through stabilator actuator attachment fitting, as damage may result from possible out-of-line condition.

NOTE

- If any attaching hardware for stabilator actuator is lost, replace all attaching hardware. Attaching hardware cannot be intermixed.
- Install up to three washers under bolt-head to maintain proper grip length of bolt.
- l. Install lower self-retaining bolt, washer(s) (under bolthead), bushings, rubber spacers, and nut securing stabilator actuator to stabilator actuator attachment fitting. **TORQUE NUT TO 80 - 165 INCH-POUNDS.** Install cotter pin.

- m. **PROOF TORQUE BOLT HEAD TO 45 - 47 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).
- n. Remove padded support from under stabilator.
- o. **701334-SUBQ** Position grounding strap on bolt and washer in bracket and secure with washer, lockwasher, and nut. **TORQUE NUT TO 43 - 47 INCH-POUNDS** (Figure 2-4-146-1, Sheet 2, Detail C).
- p. **EMEP** Perform EME resistance check between stabilator actuator housing and aft spar as follows:

NOTE

Resistance check is required only if stabilator actuator is made up of two actuators, 70400-06641-117.

- (1) Using digital low-resistance ohmmeter, measure resistance between stabilator actuator housing and aft spar. **RESISTANCE SHALL BE 7.5 MILLIOHMS OR LESS.**
- (2) If resistance is more than 7.5 milliohms, clean surface between pin filter adapters and stabilator actuator electrical connectors.
- q. Cover nuts, bolts, and washers securing grounding strap to aft spar and bracket with sealing compound, Item 292, Appendix D.
- r. Remove protective caps and plugs from electrical connectors.
- s. Inspect and treat electrical connectors (PARA 1-7-20).
- t. Connect electrical connectors, P606 to J606, and P607 to J607 (Figure 2-4-146-1, Sheet 1, Detail A).
- u. **701503** **701523 - SUBQ** Fasten connector bracket halves to each other with screws (Figure 2-4-146-1, Sheet 2, Detail D).
- v. Fasten connector bracket to tail rotor pylon with screws (Figure 2-4-146-1, Sheet 1, Detail A).

- w. Make sure area is clean and free of foreign material.
- x. If removed, install tail rotor pylon cambered fairing (PARA 2-4-135).
- y. Close tail rotor pylon leading edge cover/antenna. If removed, install cover/antenna (PARA 2-4-138).
- z. Adjust stabilator position transmitter (TM 11-70-23). Leave test set hooked up to do next step.
- aa. Adjust stabilator amplifier bias (TM 11-70-23).
- ab. Check stabilator auto mode as follows:

NOTE

- Make sure all personnel are clear of stabilator.
- Stabilator auto mode check does not require hook up of test set.
- (1) On STABILATOR CONTROL/AUTO FLT CONT panel, press and release AUTO CONTROL RESET switch. Verify the following:
 - (a) STAB POS indicators should move to 34° to 42° DN.
 - (b) STABILATOR and MASTER CAUTION capsules should be off.
 - (c) Beeping tone in headsets should be off.
 - (d) Visually verify stabilator position.
- (2) Note position of stabilator as indicated on both STAB POS indicators. Press and hold TEST switch for about 3 seconds. Verify the following:
 - (a) STAB POS indicators should move 5° to 12° up from noted position.
 - (b) AUTO CONTROL RESET switch ON legend should go off.

- (c) STABILATOR and MASTER CAUTION capsules should be on.
- (d) Beeping tone in headsets should be audible.
- (3) Press MASTER CAUTION PRESS TO RESET capsule. Verify the following:
 - (a) Beeping tone should go off.
 - (b) Both MASTER CAUTION capsules should go off.
- (4) Press and release AUTO CONTROL RESET switch. Verify the following:
 - (a) AUTO CONTROL RESET switch ON legend should go on.
 - (b) STABILATOR capsule should be off.
 - (c) STAB POS indicators should move to 34° to 42° DN.
 - (d) Visually verify stabilator position.
- (5) Place MAN SLEW switch to UP. Hold until stabilator stops moving. Verify the following:
 - (a) STAB POS indicators should move to 6° to 10° up within about 7 seconds.
 - (b) STABILATOR and MASTER CAUTION capsules should go on.
- (c) AUTO CONTROL RESET switch ON legend should go off.
- (d) Beeping tone should be audible.
- (e) Visually verify stabilator position.
- (6) Place MAN SLEW switch to DN. Hold until STAB POS indicators read 0°. Visually verify stabilator position.
- (7) Press and release AUTO CONTROL RESET switch. Verify the following:
 - (a) AUTO CONTROL RESET switch ON legend should go on.
 - (b) STABILATOR and MASTER CAUTION capsules should go off.
 - (c) STAB POS indicators should move to 34° to 42° DN.
 - (d) Beeping tone in headsets should go off.
 - (e) Visually verify stabilator position.
- (8) If any part of stabilator checks fail, refer to TM 11-70-23.
 - ac. Turn off all electrical power (PARA 1-6-16).
 - ad. Make sure area is clean and free of foreign material.

2-4-147 STABILATOR ACTUATOR REPAIR.

PARAGRAPH BREAKDOWN

	PAGE
2-4-147.1 Repair Stabilator Actuator	2-4-147-2
INITIAL SETUP	
Tools	
- Aircraft Mechanic’s Toolkit - Borescope - Digital Low-Resistance Ohmmeter - Electrical Repairer’s Toolkit - Heat Gun - Magnetic Particle Inspection Kit - Nonmetallic Scraper, Locally-Made - Torque Wrench, 30 - 150 in. lbs - Torque Wrench, 150 - 750 in. lbs	- Corrosion Preventive Compound, Item 113, Appendix D - Corrosion Removing Compound, Item 116, Appendix D - Dry-Cleaning Solvent, Item 124, Appendix D - Epoxy Primer Coating Kit, Item 135, Appendix D - Lockwire, Item 197, Appendix D - Machinery Towel, Item 211, Appendix D - Masking Tape, 1-inch, Item 212, Appendix D - Sealing Compound, Item 292, Appendix D
Materials/Parts	
Abrasive Cloth, Item 10, Appendix D	References - Appendix D - ASTM E1444 - PARA 2-4-148 - PARA 2-6-5

2-4-147.1 Repair Stabilator Actuator.**NOTE**

Stabilator actuator is made up of two halves. If actuator, 70400-06641-117, is not available, actuators, 70400-06641-115 and 70400-06641-117, may be used in any combination.

2-4-147.1.1 Disassemble.**NOTE**

Disassemble stabilator actuator only to the extent necessary to replace damaged parts.

- a. Back off jamnuts from each clevis (Figure 2-4-147-1, Sheet 1, Detail A and Sheet 2, Detail A).

NOTE

Note number of exposed threads and turns required to unscrew each clevis to aid in installation.

- b. Note number of exposed threads on each clevis. Remove upper and lower clevises from actuator, noting number of turns required to unscrew each clevis. Remove jamnuts and lockwashers.
- c. Remove bolts, washers, and nuts from actuators. **701334-SUBQ** Remove grounding bracket and washers.
- d. Separate upper and lower actuators. Remove boot.

2-4-147.1.2 Inspect.**NOTE**

Inspection of actuators equipped with 70400-26638 series clevis assemblies is not required.

- a. Inspect $\frac{3}{8}$ -inch inside diameter of threaded shank (bore) of clevises for application of epoxy primer. **PRIMER MUST BE**

EVIDENT. If primer is not evident, proceed as follows:

- (1) Using nonmetallic scraper, remove corrosion from bore.
- (2) Wash with dry-cleaning solvent, Item 124, Appendix D. Rinse thoroughly with water and dry.
- (3) Using borescope, inspect bore for corrosion pits. If pitting is found, replace clevis.
- (4) Apply corrosion removing compound, Item 116, Appendix D, per manufacturer's instructions.
- (5) Using magnetic particle inspection kit, magnetic particle inspect clevis (ASTM E1444).
- (6) Mask threads with 1-inch masking tape, Item 212, Appendix D.
- (7) Fill bore using epoxy primer coating kit, Item 135, Appendix D. Empty excess primer and allow to dry.
- (8) Using epoxy primer coating kit, Item 135, Appendix D, apply second coat of epoxy primer. Empty excess primer and allow to dry.
- (9) Remove masking tape from threads.

2-4-147.1.3 Assemble.

- a. Coat threads of upper and lower clevis with corrosion preventive compound, Item 113, Appendix D.
- b. Install upper clevis and jamnut on upper actuator as follows (Figure 2-4-147-1, Sheet 1, Detail A and Sheet 2, Detail A):
 - (1) Inspect lockwasher for broken tang. **NO BROKEN TANG ALLOWED.** If tang is broken, replace lockwasher.
 - (2) Thread upper clevis into actuator same number of turns recorded during disas-

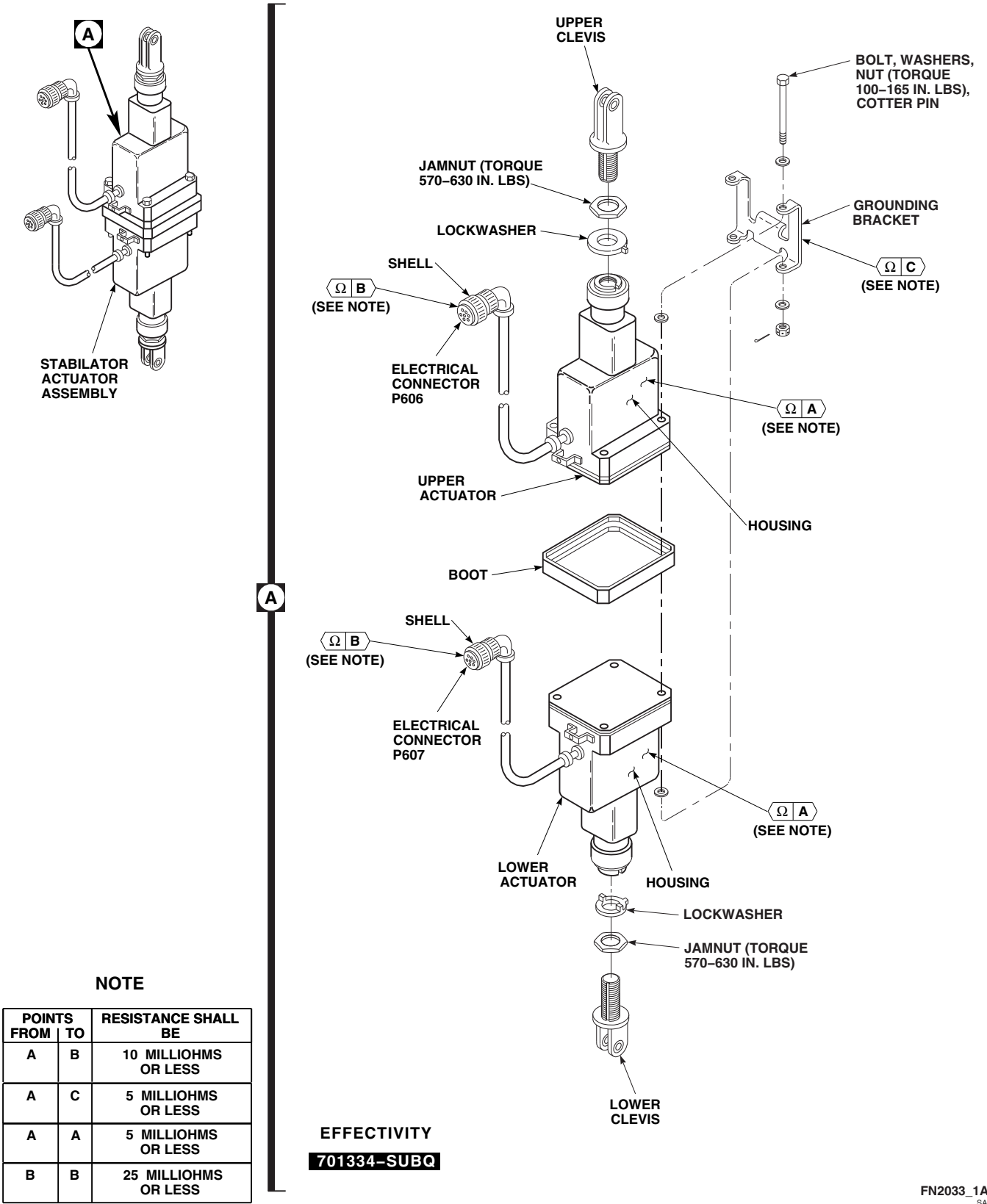
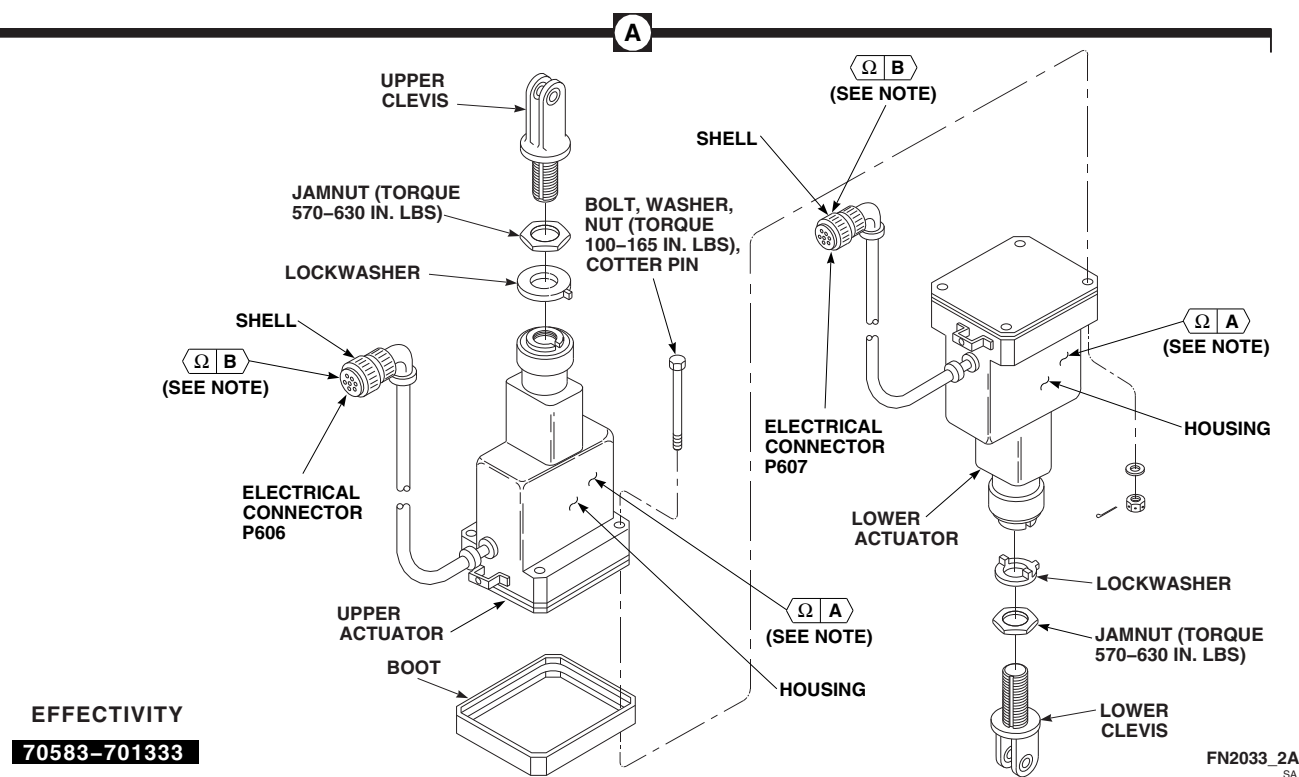


FIGURE 2-4-147-1. REPAIR STABILATOR ACTUATOR ASSEMBLY (SHEET 1 OF 2)



EFFECTIVITY

70583-701333

FIGURE 2-4-147-1. REPAIR STABILATOR ACTUATOR ASSEMBLY (SHEET 2 OF 2)

sembly. **MAXIMUM NUMBER OF EXPOSED THREADS ON CLEVIS SHALL NOT EXCEED FOUR.**

- (3) Seat lockwasher in key slot in actuator.

WARNING

FSCAP

Verification of proper hardware installation, torque, and safety installation are critical characteristics.

- (4) **TORQUE JAMNUT TO 570 - 630 INCH-POUNDS.** Using lockwire, Item 197, Appendix D, lockwire jamnut.
- (5) Apply torque stripe (PARA 2-6-5).

- c. Install lower clevis on lower actuator as follows:

- (1) Inspect lockwasher for broken tang. **NO BROKEN TANG ALLOWED.** If tang is broken, replace lockwasher.

- (2) Thread lower clevis into actuator same number of turns recorded during disassembly. **MAXIMUM NUMBER OF EXPOSED THREADS ON CLEVIS SHALL NOT EXCEED FOUR.**

- (3) Seat lockwasher in key slot in actuator.

WARNING

FSCAP

Verification of proper hardware installation, torque and safety installation are critical characteristics.

- (4) **TORQUE JAMNUT TO 570 - 630 INCH-POUNDS.** Using lockwire, Item 197, Appendix D, lockwire jamnut.

- (5) Apply torque stripe (PARA 2-6-5).

NOTE

The following step is required on actuator, 70400-06641-117, only.

- d. **701334-SUBQ** Inspect chemical conversion-coated actuator mating surfaces, bolthole spot-faces, and grounding bracket mating surfaces for compliance with electrical bonding requirements (PARA 2-6-5).■
- e. **701334-SUBQ** Clean actuator mating surfaces using machinery towel, Item 211, Appendix D, dampened with dry-cleaning solvent, Item 124, Appendix D.■
- f. Assemble actuators as follows:

NOTE

- Resistance check is required on actuator, 70400-06641-117, only.
 - Assemble actuators with electrical cables on same side.
 - Boltheads must be on upper actuator.
- (1) Perform EME resistance check between shell of electrical connector and actuator housing as follows:

NOTE

Resistance measurements must be taken at least three times.

- (a) Using digital low-resistance ohm-meter, measure resistance between shell of electrical connector and actuator housing. **RESISTANCE SHALL BE 10.0 MILLIOHMS OR LESS.**
- (b) If resistance is more than 10.0 milliohms, replace actuator.
- (2) Position actuator boot on upper actuator as required to leave mating surface exposed.

- (3) **701334-SUBQ** Position grounding bracket and washers across both actuators.■
- (4) Secure actuators together with bolts, washers, and nuts. **TORQUE NUTS TO 100 - 165 INCH-POUNDS.** Install cotter pin.

NOTE

701334-SUBQ Resistance check is required only if stabilator actuator is made up of two actuators, 70400-06641-117.■

- (5) **701334-SUBQ** Perform EME resistance check between grounding bracket and each actuator housing as follows:■

NOTE

Resistance measurements must be taken at least three times.

- (a) Using digital low-resistance ohm-meter, measure resistance between grounding bracket and each actuator housing. **RESISTANCE SHALL BE 5.0 MILLIOHMS OR LESS.**
- (b) If resistance is more than 5.0 milliohms, disassemble actuators and replace chemical conversion-coated surfaces (PARA 2-6-5).

NOTE

Resistance check is required only if stabilator actuator is made up of two actuators, 70400-06641-117.

- (6) Perform EME resistance check between both actuator housings as follows:

NOTE

Resistance measurements must be taken at least three times.

- (a) Using digital low-resistance ohm-meter, measure resistance between

both actuator housings. **RESISTANCE SHALL BE 5.0 MILLIOHMS OR LESS.**

- (b) If resistance is more than 5.0 milliohms, disassemble actuators and replace chemical conversion-coated surfaces (PARA 2-6-5).

NOTE

Resistance check is required only if stabilator actuator is made up of two actuators, 70400-06641-117.

- (7) Perform EME resistance check between shells of both electrical connectors as follows:

NOTE

Resistance measurements must be taken at least three times.

- (a) Using digital low-resistance ohmmeter, measure resistance between shells of both electrical connectors. **RESISTANCE SHALL BE 25.0 MILLIOHMS OR LESS.**
- (b) If resistance is more than 25.0 milliohms, disassemble actuators and repeat step f. (1) through step f. (5).
- (c) If resistance is still more than 25.0 milliohms, replace actuator assembly.

- (8) Apply sealing compound, Item 292, Appendix D, over mounting hardware and to area where actuators meet. Allow to cure for 24 hours at room temperature, or 2 hours using heat gun at 150° - 180°F (66° - 82°C).

- (9) Position actuator boot on area where actuators meet after sealing compound cures.

- g. Check both connectors for identification tapes. If necessary, install new identification tapes. Upper actuator connector is marked "P606 to J606 UPR STAB ACT." Lower actuator connector is marked "P607 to J607 LWR STAB ACT."

CAUTION

On actuator, 70400-06641-117, damage to wiring may result if connector is rotated more than 90°. On all other actuators, connector must not rotate more than 180°.

- h. Check actuator housings for minor surface scratches. Polish out any superficial scratches using abrasive cloth, Item 10, Appendix D.
- i. Touch up paint as required (PARA 2-6-5).
- j. Adjust stabilator actuators (PARA 2-4-148).

2-4-148 STABILATOR ACTUATOR ADJUSTMENT.

PARAGRAPH BREAKDOWN

	PAGE
2-4-148.1 Adjust Stabilator Actuator	2-4-148-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Digital Protractor
- Electrical Repairer’s Toolkit
- Protractor Fixture, Locally-Made
- Torque Wrench, 150 - 750 in. lbs
- Universal Propeller Protractor

Materials/Parts

- Lockwire, Item 197, Appendix D
- Protective Caps and Plugs

References

- Appendix D
- MIL-STD-1686
- PARA 1-6-16
- PARA 2-4-69
- PARA 2-4-146
- PARA 2-4-147
- PARA 2-6-5

2-4-148.1 Adjust Stabilator Actuator.

jamnuts only with actuator removed from helicopter.

WARNING

Injury to personnel and damage to equipment will result due to loss of jam-nut torque. Do not torque actuator upper or lower clevises while installed on helicopter. Adjust clevises and torque

CAUTION

Actuator, 70400-06641-117, contains parts and assemblies sensitive to damage by electrostatic discharge (ESD). Use ESD precautionary procedures when

touching, removing or inserting (MIL-STD-1686).

NOTE

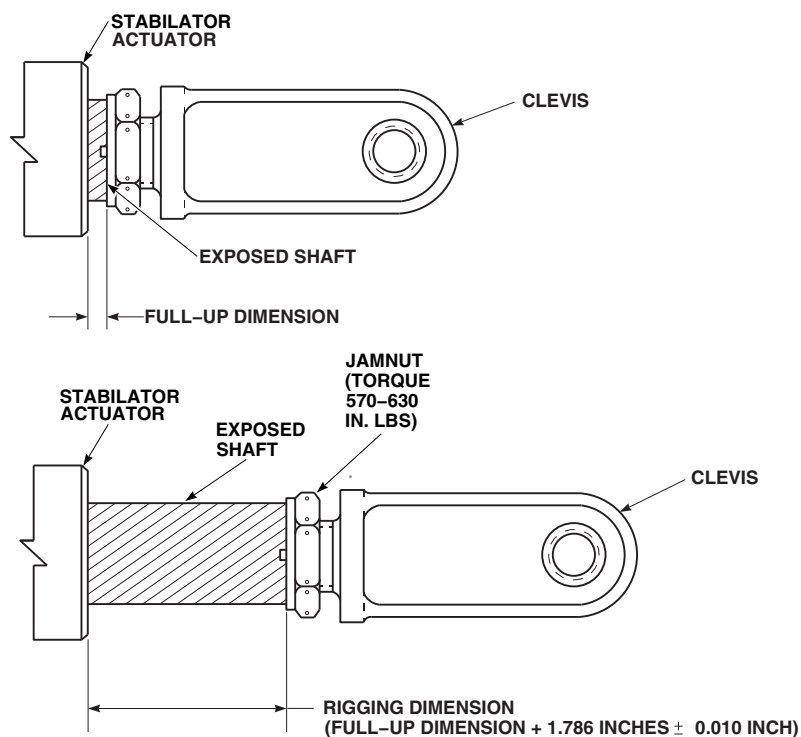
Adjust actuators when upper and lower clevises are not threaded equally into actuators, or when one or both actuators are being replaced.

- a. Assemble actuators (PARA 2-4-147).
- b. Turn on electrical power (PARA 1-6-16).
- c. Place stabilator actuator close to tail rotor pylon and connect electrical connector, P606 to J606, on pylon.
- d. On STABILATOR CONTROL/AUTO FLT CONT panel, place MAN SLEW switch to UP position until actuator stops. Measure and record length of exposed end of shaft. Add 1.786-inches to this dimension to get rigging dimension for upper actuator (Figure 2-4-148-1).
- e. On STABILATOR CONTROL/AUTO FLT CONT panel, place MAN SLEW switch to DOWN position until exposed length of shaft is same as rigging dimension $\pm$ 0.010-inch.
- f. Disconnect electrical connector, P606 from J606.
- g. Install protective caps and plugs on electrical connectors.
- h. Remove protective caps and plugs and connect electrical connector, P607 to J607, on pylon.
- i. On STABILATOR CONTROL/AUTO FLT CONT panel, place MAN SLEW switch to UP position until actuator stops. Measure and record exposed end of shaft. Add 1.786-inches to this dimension to get rigging dimension for lower actuator.
- j. On STABILATOR CONTROL/AUTO FLT CONT panel, place MAN SLEW switch to DOWN position until exposed length of shaft is same as rigging dimension $\pm$ 0.010-inch.
- k. Disconnect electrical connector, P607 from J607, on pylon.
- l. Install protective caps and plugs on electrical connectors.
- m. Loosen jamnuts and thread upper and lower clevises into their shafts.
- n. Back both clevises out of their shafts four turns. **TIGHTEN JAMNUTS HANDTIGHT.**
- o. Install stabilator actuator (PARA 2-4-146). Do not torque attachment nuts now.
- p. Place protractor fixture on top of stabilator between BL 9.0 (right or left), with lip of fixture plate over forward edge of stabilator.
- q. Pull back fixture hook and secure to trailing edge of stabilator so lip of fixture fits snugly against forward edge of stabilator.
- r. Remove soundproofing panels from cabin ceiling between STA 360.0 and STA 398.0 (PARA 2-4-69).

NOTE

Make sure to orient protractor face in the same direction as will be used at the stabilator bar.

- s. Place universal propeller protractor or digital protractor against bottom side of right transmission beam and set protractor scale at 0°.
- t. Remove universal propeller protractor or digital protractor from beam and place on protractor fixture or bar through fold fittings.
- u. **ANGLE OF STABILATOR SHOULD BE 7.5° - 8.5° (TRAILING EDGE UP).** If stabilator is at proper angle, proceed to step v. If stabilator is not at proper angle, perform the following:
 - (1) Remove stabilator actuator (PARA 2-4-146).
 - (2) Thread both upper and lower clevises in or out at same time to get proper reading.



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FIGURE 2-4-148-1. ADJUST STABILATOR ACTUATOR CLEVIS

MAXIMUM NUMBER OF EXPOSED THREADS ON EITHER CLEVIS SHALL NOT EXCEED FOUR.

v. If not already done, position stabilator at 7.5° - 8.5° and perform the following:

(1) Remove stabilator actuator (PARA 2-4-146).

(2) Perform final torque on upper and lower clevises as follows:

(a) Make sure lockwasher is seated in key slot in both upper and lower actuators.

(b) **TORQUE JAMNUTS TO 570 - 630 INCH-POUNDS.**

(c) Lockwire, Item 197, Appendix D, jamnuts.

(d) Apply torque stripes (PARA 2-6-5).

w. Remove protractor fixture from stabilator.

x. Turn off all electrical power (PARA 1-6-16).

y. Install stabilator actuator (PARA 2-4-146).

z. Install soundproofing panels from cabin ceiling between STA 360.0 and STA 398.0 (PARA 2-4-69).

WARNING

FSCAP

Verification of proper hardware installation, torque, and safety installation are critical characteristics.

2-4-149 STABILATOR ACTUATOR GROUNDING STRAP 701334-SUBQ .

PARAGRAPH BREAKDOWN

	PAGE
2-4-149.1 Replace Stabilator Actuator Grounding Strap	2-4-149-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Digital Low-Resistance Ohmmeter
- Electrical Repairer’s Toolkit
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 2-3-15
- PARA 2-4-135
- PARA 2-6-5

2-4-149.1 Replace Stabilator Actuator Grounding Strap.

2-4-149.1.1 Remove.

- Turn off all electrical power (PARA 1-6-16).
- Remove tail rotor pylon cambered fairing (PARA 2-4-135).
- Using nonmetallic scraper, remove sealing compound from bolts, washers, and nuts securing grounding strap to grounding bracket and aft spar.

NOTE

Bolt and washer securing grounding strap will remain with grounding bracket once nut, lockwasher, washer, and grounding strap are removed.

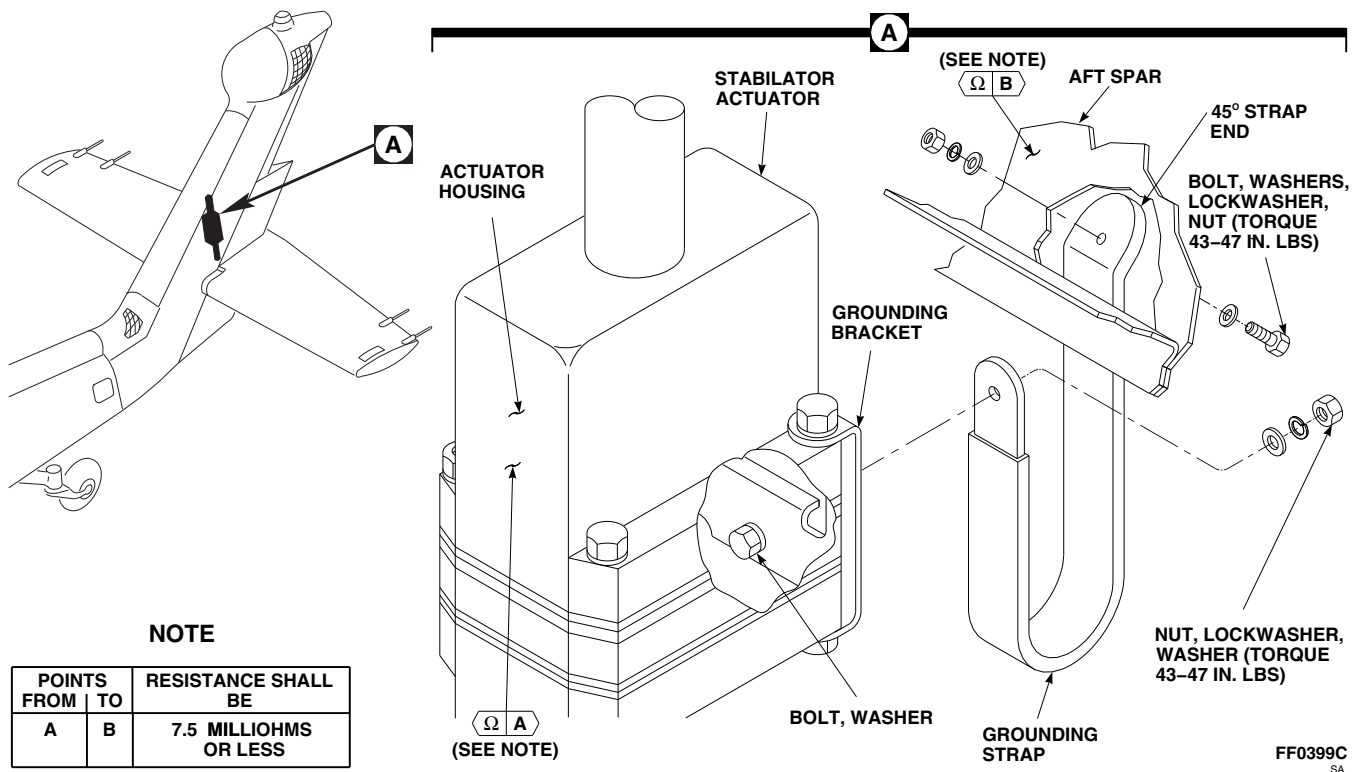
- Remove nut, lockwasher, washer, and grounding strap from bolt and washer on grounding bracket (Figure 2-4-149-1, Detail A).
- Remove bolt, washers, lockwasher, and nut securing grounding strap to aft spar.
- Remove grounding strap.

2-4-149.1.2 Inspect.

Inspect stabilator actuator grounding strap (PARA 2-3-15).

2-4-149.1.3 Install.

- Position 45° strap end fitting on aft spar and secure with bolt, washers, lockwasher, and nut. **TORQUE NUT TO 43 - 47 INCH-POUNDS** (Figure 2-4-149-1, Detail A).



- b. Position grounding strap on bolt and washer in bracket and secure with washer, lockwasher, and nut. **TORQUE NUT TO 43 - 47 INCH-POUNDS.**
- c. Perform EME resistance check between actuator housing and aft spar as follows:
 - (1) Using digital low-resistance ohmmeter, measure resistance between actuator housing and aft spar. **RESISTANCE SHALL BE 7.5 MILLIOHMS OR LESS.**
 - (2) If resistance is more than 7.5 milliohms, inspect chemical conversion-coated surfaces (PARA 2-6-5).
- d. Cover nuts, bolts, and washers securing grounding strap to aft spar and bracket with sealing compound, Item 292, Appendix D.
- e. Install tail rotor pylon cambered fairing (PARA 2-4-135).

2-4-150 DELETED.

I

2-4-151 POSITION TRANSMITTER LIMIT SWITCH.

PARAGRAPH BREAKDOWN

	PAGE
2-4-151.1 Replace Position Transmitter Limit Switch	2-4-151-2

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Electrical Repairer’s Toolkit
- File
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Adhesive, Item 32, Appendix D
- Adhesive, Item 37, Appendix D
- Adhesive, Item 66, Appendix D
- Cheesecloth, Item 90, Appendix D
- Corrosion Preventive Compound, Item 114, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D

- Lockwire, Item 196, Appendix D
- Lockwire, Item 197, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Protective Caps and Plugs
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- Appropriate MTF
- PARA 1-6-16
- PARA 1-7-20
- PARA 2-2-1
- PARA 2-4-133
- PARA 2-4-134
- PARA 2-6-1
- PARA 2-6-5
- PARA 6-4-41
- TM 11-70-23

2-4-151.1 Replace Position Transmitter Limit Switch.

2-4-151.1.1 Remove.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Remove left and right tail rotor pylon stabilator-to-ptylon fairings (PARA 2-4-134).
- c. Remove tail rotor pylon lower fairing (PARA 2-4-133).
- d. Remove self-retaining bolt, bushing, washer(s), and nut securing actuator rod to stabilator fitting (Figure 2-4-151-1, Detail A).
- e. Mark jamnut location on actuator rod and record number of free threads above jamnut (Figure 2-4-151-1, Detail A and Detail C).
- f. Loosen jamnut. Remove rod end, locking device, and jamnut from actuator rod (Figure 2-4-151-1, Detail A).
- g. Remove tail rotor drive shaft section IV (PARA 6-4-41).

NOTE

If bolts are used to secure cover, replace with screws at installation.

- h. Remove screws or bolts (if installed), washers, and spacers securing cover to housing. Remove cover.
- i. Disconnect electrical connectors, P604 and P605, from J604 and J605 (Figure 2-4-151-1, Detail B).
- j. Install protective caps and plugs on electrical connectors.
- k. Remove screw, washer, and nut securing clamp and electrical harness to clip on tail rotor pylon.
- l. Remove bolt, washer, shim, and nut securing sensor in housing track. Remove sensor from housing.

- m. Remove self-retaining bolt, washers, nut, spacer, and bushing securing actuator rod to sensor. Remove actuator rod.
- n. Remove screws and washers securing housing to pylon. Remove housing from pylon.

2-4-151.1.2 Inspect.

- a. Inspect housing for dents, cracks, corrosion, and broken welds. **NONE ALLOWED.** If dents, cracks, corrosion, or broken welds are found, replace housing.
- b. Inspect gasket on cover for looseness or damage. **NONE ALLOWED.** If necessary, rebond gasket to cover with sealing compound, Item 292, Appendix D.
- c. Visually inspect actuator rod for cracks, distortion, and thread damage. **NONE ALLOWED.** If cracks, distortion, or thread damage are found, replace actuator rod.
- d. Inspect bushings for corrosion and wear. **NONE ALLOWED.** If corrosion or wear is found, replace bushing(s).

2-4-151.1.3 Install.

- a. Install housing as follows:
 - (1) Using nonmetallic scraper or file, remove any paint or other solid material from mating surfaces of housing lugs and pylon spar to avoid abrasive residue.
 - (2) Using cheesecloth, Item 90, Appendix D, moistened with dry-cleaning solvent, Item 124, Appendix D, thoroughly clean mating surfaces.
 - (3) Using clean cheesecloth, Item 90, Appendix D, wipe dry.
 - (4) Apply corrosion resistant coating, Item 118, Appendix D, to all mating surfaces.
 - (5) Install housing on pylon spar and secure with screws and washers (Figure 2-4-151-1, Detail B).

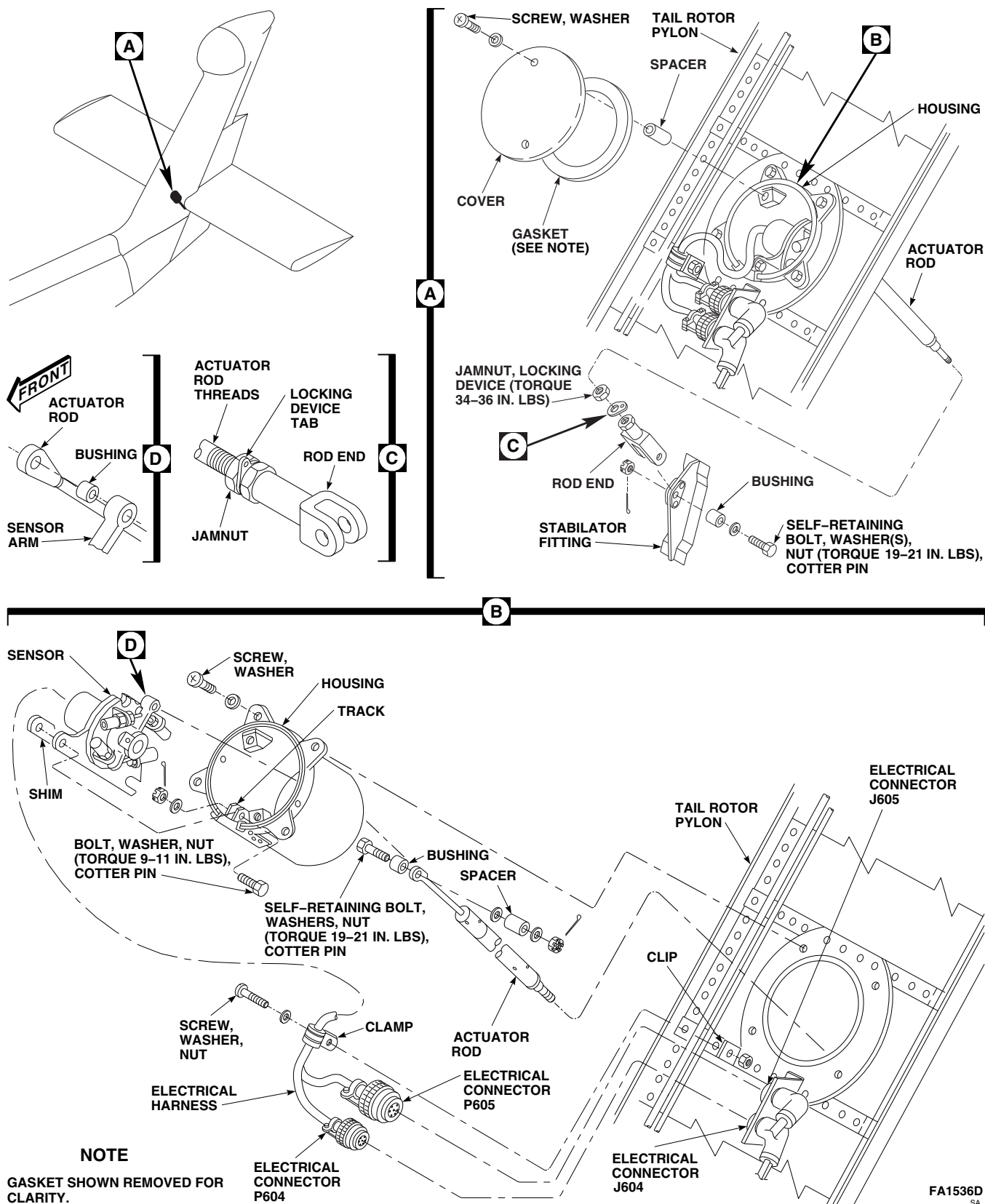


FIGURE 2-4-151-1. REPLACE POSITION TRANSMITTER LIMIT SWITCH

- (6) Seal all edges of mating surfaces with sealing compound, Item 292, Appendix D (PARA 2-6-1).

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

- b. Check that lower end of actuator rod has mark for jamnut on it. If mark is not present, perform the following:
- (1) Install jamnut, locking device, and rod end on rod (Figure 2-4-151-1, Detail A and Detail C).
 - (2) Thread rod end on rod so distance between rod end holes is 15 23/64-inches.
 - (3) Note number of free threads, and mark location of jamnut.
 - (4) Remove lower rod end, jamnut, and locking device from rod end.
- c. Adjust position sensor and limit switches (PARA 2-2-1).

NOTE

Actuator rod end is installed inboard of sensor arm.

- d. Install actuator rod on sensor as follows (Figure 2-4-151-1, Detail B and Detail D):
- (1) Coat self-retaining bolt and bushing with corrosion preventive compound, Item 114, Appendix D. Do not apply compound to bolt threads.

NOTE

Install bushing between actuator rod and sensor arm.

- (2) Position actuator rod on sensor arm and secure with self-retaining bolt, bushing, spacer, and washers as follows:

NOTE

- Do not place more than three washers under nut. Do not place any washers under bolthead.
 - Self-retaining element of bolt must ride in spacer grooves without being restrained or depressed. With selected washer(s) in place and nut not installed, perceptible axial play must be present.
- (a) Position washers, in any required combination, between sensor arm and spacer, between spacer and nut, or some in both locations.

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

- (3) Install nut on bolt. **TORQUE NUT TO 19 - 21 INCH-POUNDS.** Install cotter pin.
- e. Slide sensor base and shim into housing track (Figure 2-4-151-1, Detail B). Check for side-play. **SIDEPLAY MUST BE LESS THAN 0.003-INCH.** If necessary, peel shim to obtain clearance.
- f. Bond shim to sensor as follows:
- (1) Using machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D, clean bond surfaces.

NOTE

Either adhesive, Item 32, Appendix D, or adhesive, Item 37, Appendix D, may be used.

- (2) If adhesive, Item 37, Appendix D, is used, perform the following:

- (a) Apply even coat of adhesive, Item 37, Appendix D, to both bond surfaces.
- (b) Allow adhesive to dry for 10 minutes until tacky.
- (c) Install shim on sensor.
- (d) Allow adhesive to cure for at least 1 hour.

- (3) If adhesive, Item 32, Appendix D, is used, perform the following:

- (a) Apply thin, even coat of adhesive, Item 66, Appendix D, to both bond surfaces.
- (b) Allow primer to dry for 1 hour.
- (c) Apply even coat of adhesive, Item 32, Appendix D, to both bond surfaces.
- (d) Immediately install shim on sensor.

- (e) Allow adhesive to cure for 2 hours at 110° - 130°F (43° - 54°C).

- g. Install bolt, washer, shim, and nut securing sensor to housing track. **TORQUE NUT TO 9 - 11 INCH-POUNDS.** Install cotter pin. Using lockwire, Item 197, Appendix D, lockwire bolt-head to sensor base.

- h. Pass length of lockwire through holes in tail rotor pylon that actuator rod will pass through. Secure wire to rod end. Have assistant stand behind pylon to help guide rod end through hole in pylon rear spar.

- i. Install clamp and wiring harness to clip on pylon and secure with screw, washer, and nut.

- j. Remove protective caps and plugs from electrical connectors.

- k. Inspect and treat electrical connectors (PARA 1-7-20).

- l. Connect electrical connectors, P604 to J604, and P605 to J605.

NOTE

If bolts were used to secure cover, replace with screws.

- m. Install cover on housing and secure with screws, washers, and spacers (Figure 2-4-151-1, Detail A).
- n. Remove wire used to guide rod end through hole in pylon rear spar from pylon rear spar and rod end.
- o. Install jamnut on lower end of actuator rod to mark, leaving number of free threads above nut as recorded during removal (Figure 2-4-151-1, Detail A and Detail C).

WARNING

FSCAP

Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

- p. Install rod end as follows:

- (1) Install locking device against jamnut.

- (2) Thread rod end onto actuator rod, engaging locking device lugs in rod end slots so final tab position is snug against jamnut.

- (3) **TORQUE JAMNUT TO 34 - 36 INCH-POUNDS.**

- (4) Verify thread engagement by inserting lockwire, Item 196, Appendix D, into rod

end inspection hole nearest jamnut (some rod ends have two inspection holes). If lockwire passes through, review rod end adjustment.

- (5) Using lockwire, Item 196, Appendix D, lockwire jamnut to locking device tab.
- q. Position rod end on stabilator fitting (Figure 2-4-151-1, Detail A).
- r. Apply corrosion preventive compound, Item 114, Appendix D, to actuator rod upper attachment self-retaining bolt and bushing. Do not apply compound to bolt threads.
- s. Install self-retaining bolt, washer(s) (under bolthead), and bushing as follows:
 - (1) Install self-retaining bolt with washers under bolthead (three or less) to make sure of correct grip length and cotter pin position.
 - (2) Make sure self-retaining bolt has noticeable axial motion and self-retaining element is not restrained or depressed. Change washer stackup for correct installation.
- t. Install nut on bolt. **TORQUE NUT TO 19 - 21 INCH-POUNDS.** Install cotter pin.
- u. **PROOF TORQUE BOLTHEAD TO 10 - 12 INCH-POUNDS.** Apply torque stripe (PARA 2-6-5).
- v. Adjust position transmitter/limit switch assembly actuator rod (TM 11-70-23).
- w. Make sure area is clean and free of foreign material.
- x. Install tail rotor drive shaft section IV (PARA 6-4-41).
- y. Install tail rotor pylon lower fairing (PARA 2-4-133).
- z. Install left and right tail rotor pylon stabilator-to-pylon fairings (PARA 2-4-134).
- aa. Check helicopter logbook to make sure tail rotor drive shaft section IV installation torque check is shown as due 9 to 11 flight hours after installation.
- ab. Perform maintenance test flight (Appropriate MTF).

2-4-152 CREW CHIEF’S/PASSENGER’S SEAT - 14 SEAT CONFIGURATION.

PARAGRAPH BREAKDOWN

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2-4-152.1 Replace Crew Chief’s/ Passenger’s Seat - 14 Seat Configuration	2-4-152-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit

References

- PARA 1-6-16
- PARA 2-4-46
- PARA 2-4-47

2-4-152.1 Replace Crew Chief’s/Passenger’s Seat - 14 Seat Configuration.

NOTE

- Do not place tool boxes, tools, or oily rags on seat covers while performing maintenance.
- Replace procedure for all 14 seats is the same.

and stored. Mishandling will cause wire to bend, twist and not be centered on roller depression. Make sure to use care when handling or storing passenger’s seat.

- Unhook seat latches from upper support brackets. Remove seat (Figure 2-4-152-1, Sheet 1, Detail B).

2-4-152.1.1. Remove.

- Turn off all electrical power (PARA 1-6-16).
- Release seat latch locking levers (Figure 2-4-152-1, Sheet 1, Detail B).
- Disengage disconnect fittings and cable clips from floor attachment studs (Figure 2-4-152-1, Sheet 2, Detail C and Detail D).



Damage to attenuator wire will occur if passenger’s seat is not properly handled

2-4-152.1.2. Inspect.

- Check upper latches and latch locking mechanism for cracks and broken springs. **NO CRACKS ALLOWED.** Replace cracked latches.
- Check attenuator wire for wear, cracks, flatness (wire diameter), deformation, or other damage. **NO CRACKS ALLOWED. WIRE DAMAGE SHALL BE LIMITED TO A MINIMUM DIAMETER OF 0.090 INCH.** Replace damaged attenuator wire.
- Check seat back, pan frames, and frame fittings for cracks, loose, sheared, and missing fasten-

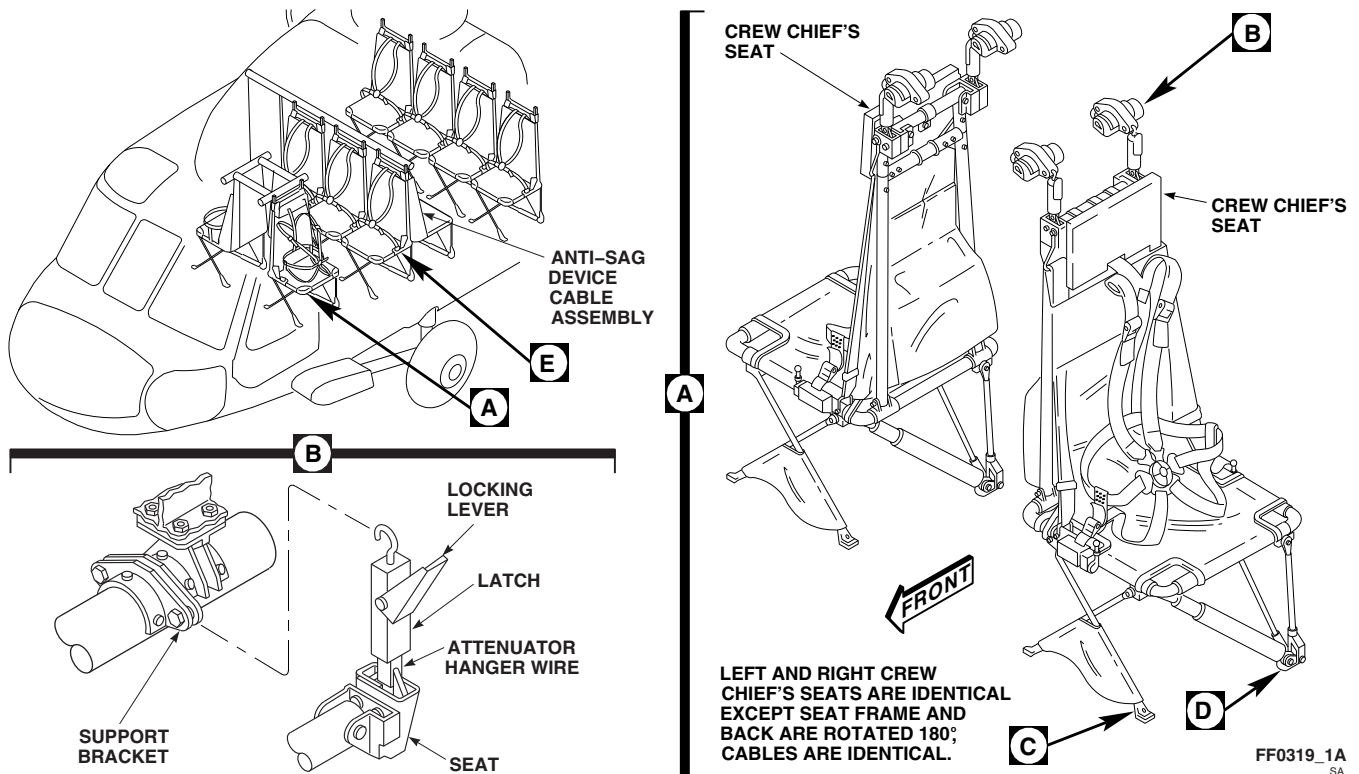
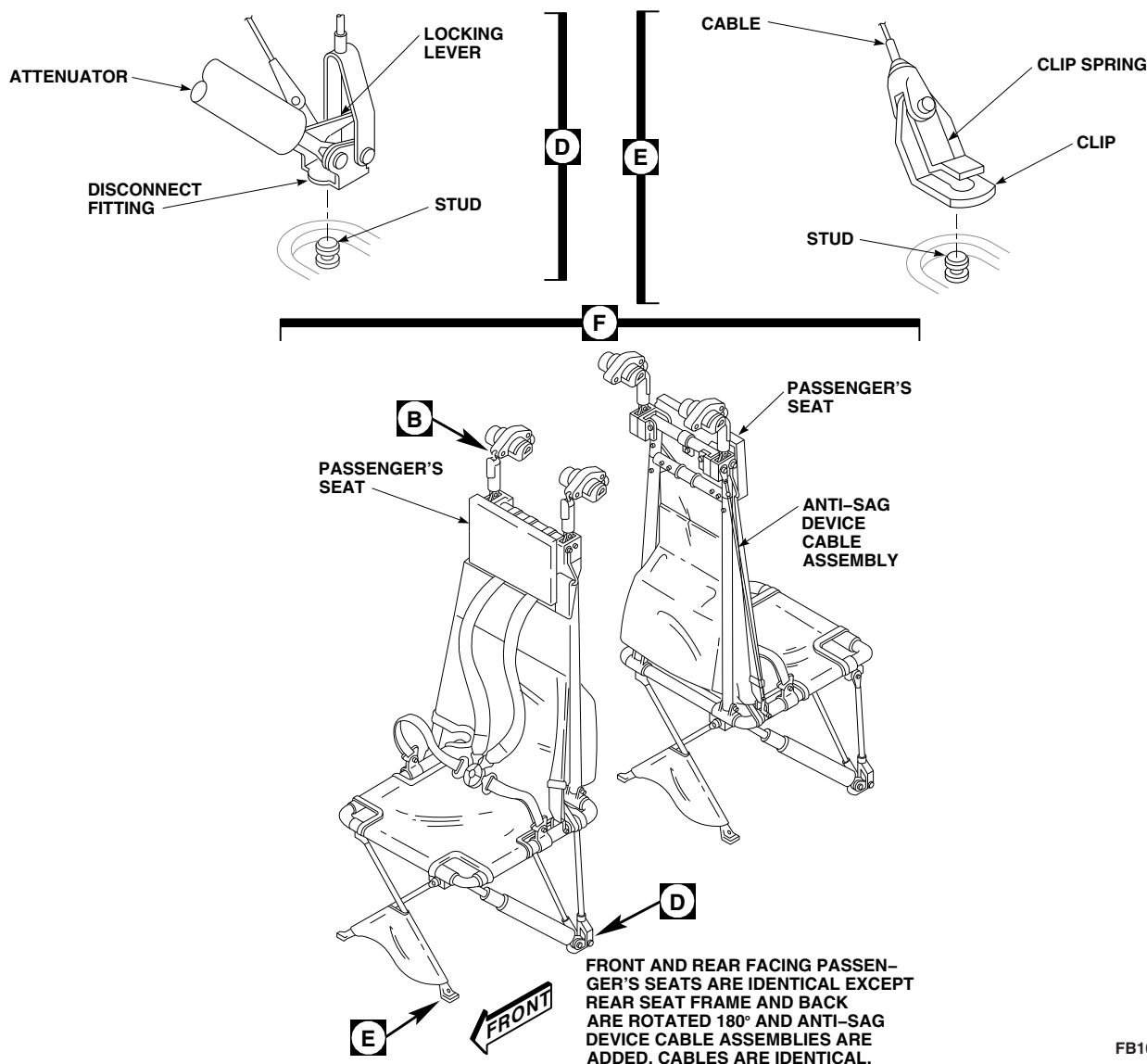


FIGURE 2-4-152-1. REPLACE CREW CHIEF'S/PASSENGER'S SEAT - 14 SEAT CONFIGURATION (SHEET 1 OF 2)



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FIGURE 2-4-152-1. REPLACE CREW CHIEF'S/PASSENGER'S SEAT - 14 SEAT CONFIGURATION (SHEET 2 OF 2)

- ers. **NO CRACKS ALLOWED.** Repair or replace damaged frames.
- d. Check anti-sag device cable assembly on rear-facing passenger's seats for broken or loose strands, cable sheathing for tears or damage, and attaching hardware for cracks and security. Replace damaged and worn parts.
- e. Check fabric for tears, holes, and loose or missing stitching. Check seat pan for loose or missing fabric retaining screws. Replace missing screws. Repair or replace damaged seat coverings.
- f. Check area between upper seat cover ring and seat frame for cracks and wear. Replace damaged ring.
- g. Check attenuator tubing for misalignment, dents, cracks, and torque stripes for tube extension. **NO BENDS, CRACKS, OR TUBE EXTENSION ALLOWED.** Replace bent, cracked, or extended attenuators.
- h. Check attenuator tube end fittings for loose, sheared, or missing fasteners. Check bearing end for free movement. Repair loose, sheared,

or missing fasteners. Replace damaged bearings.

- i. Check disconnect hardware for cracks, wear, and missing or broken parts. **NO CRACKS ALLOWED.** Replace broken disconnect parts.
- j. Check cables for broken strands. If three or more strands are broken, replace cable.
- k. Check cables for kinks. If cables cannot be straightened by hand, replace damaged cables.

NOTE

Check seat configuration to make sure that anti-sag device cable assembly is installed on rear facing passenger's seats.

- l. Check cable attaching hardware for cracks and security. **NO CRACKS ALLOWED.** Replace damaged hardware.
- m. Inspect passenger's seat restraint system (PARA 2-4-46) or crew chief's seat restraint system (PARA 2-4-47).

2-4-152.1.3. Install.

- a. Turn off all electrical power (PARA 1-6-16).
- b. Hook seat latches on overhead support brackets. Do not lock lever (Figure 2-4-152-1, Sheet 1, Detail B).

WARNING

To reduce personal injury in a hard landing, make sure attenuators are installed with small ends pointing up and toward nose of aircraft.

- c. Install attenuator and cable clip fittings on floor attachment studs (Figure 2-4-152-1, Sheet 2, Detail C and Detail D).
- d. Screw in upper seat latch hooks to take up cable slack (Figure 2-4-152-1, Sheet 1, Detail B).
- e. Add tension to seat by locking upper latch lever.

2-4-153 CREW CHIEF’S/PASSENGER’S SEAT - 20 SEAT CONFIGURATION.

PARAGRAPH BREAKDOWN

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2-4-153.1 Replace Crew Chief’s/ Passenger’s Seat - 20 Seat Configuration	2-4-153-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit

References

- PARA 2-4-45

2-4-153.1 Replace Crew Chief’s/Passenger’s Seat - 20 Seat Configuration.

NOTE

Replacement of all 20 seats is the same, except as noted.

2-4-153.1.1 Remove.

- To remove crew chief’s/passenger’s seat, 70500-02800-101 and -102, perform the following (Figure 2-4-153-1, Sheet 2, Detail B, Detail C, and Detail D):
 - Disconnect all four seat tiedown fittings from seat tracks by pulling up sleeve on each leg. Rotate sleeve 90° to disconnect each leg.
 - Slide seat to next hole in seat track. Remove seat.
- To remove rear, front-facing passenger’s seat, 70500-02151-045, at STA 379.0, refer to PARA 2-4-45.

2-4-153.1.2 Inspect.

- Check seat, back, and frame fittings for cracks, loose, sheared, and missing fasteners. **NONE ALLOWED.**
- Check fabric for tears, holes, loose or missing stitching. **NONE ALLOWED.** If damaged, repair or replace seat covers. For rear, front-facing seat, at STA 379.0, refer to PARA 2-4-45 for replacing seat cover.
- Inspect rear, front-facing seats at STA 379.0 (PARA 2-4-45).

2-4-153.1.3 Install.

NOTE

Make sure left front passenger’s seat is installed first, to prevent seat from being installed in wrong position. Note distance between legs is narrower than all other free standing passenger’s seats (Figure 2-4-153-1, Sheet 2, Detail C).

- To install crew chief’s/passenger’s seat, 70500-02800-101 and -102, perform the following

(Figure 2-4-153-1, Sheet 2, Detail B, Detail C, and Detail D):

- (1) Install seat in seat tracks on seat pallets as indicated by arrows in Figure 2-4-153-1, Sheet 1, Detail A.

NOTE

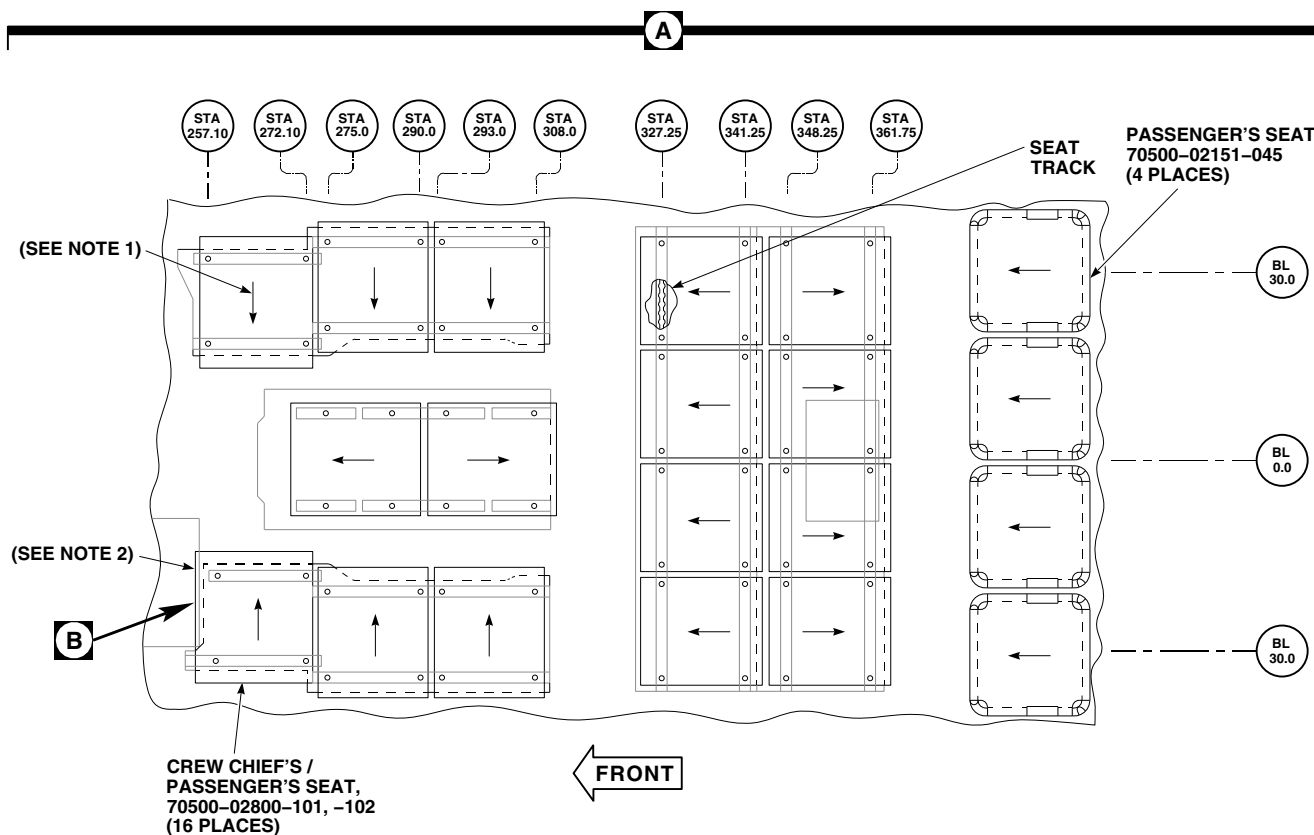
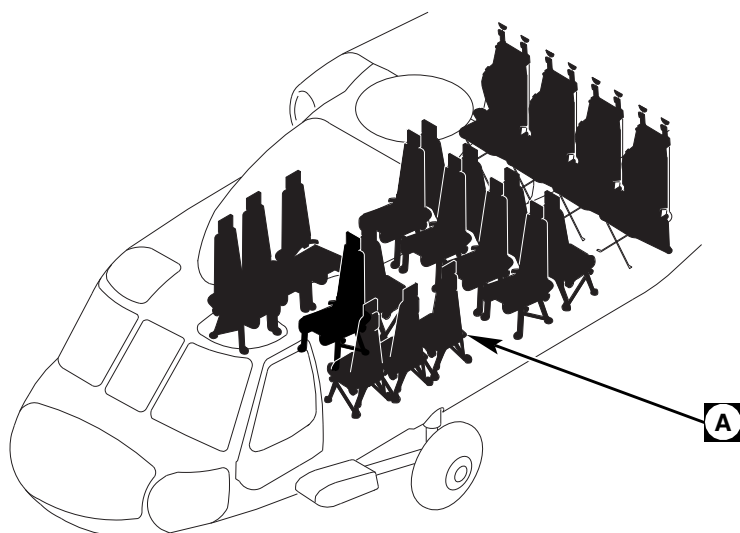
Seat tiedown fittings will disconnect if moved to next full size hole in seat track.

- (2) Slide seats in seat track about 1-inch to connect seat tiedown fitting in track (Figure 2-4-153-1, Sheet 2, Detail D).

- (3) Rotate sleeve on tiedown fitting of each leg until slots allow sleeve to drop into track.

- (4) Check each leg on each seat for 1/8-inch white band around tiedown fitting. This verifies that each leg is locked into seat track.

- b. To install rear, front-facing passenger's seat, 70500-02151-045, at STA 379.0, refer to PARA 2-4-45.



NOTES

1. ARROW INDICATES DIRECTION SEAT IS FACING.
2. SEAT WITH NARROW LEGS INSTALLED IN THIS POSITION (DETAIL "B").
3. 0.12 IN. WHITE BAND WILL SHOW WHEN SEAT IS LOCKED INTO SEAT TRACK TIEDOWN FITTING.

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FIGURE 2-4-153-1. REPLACE CREW CHIEF'S/PASSENGER'S SEAT - 20 SEAT CONFIGURATION (SHEET 1 OF 2)

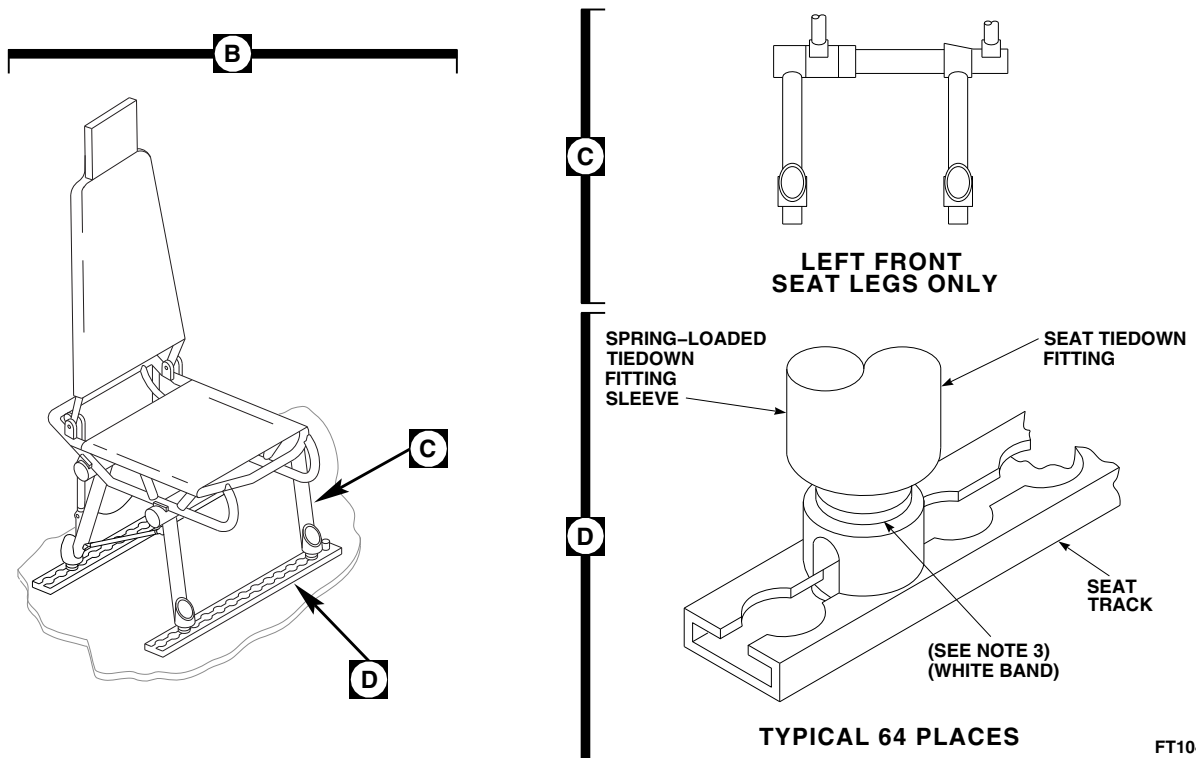


FIGURE 2-4-153-1. REPLACE CREW CHIEF'S/PASSENGER'S SEAT - 20 SEAT CONFIGURATION (SHEET 2 OF 2)

2-4-154 CREW CHIEF’S/PASSENGER’S SEAT PALLETS - 20 SEAT CONFIGURATION.

PARAGRAPH BREAKDOWN

	PAGE
2-4-154.1 Replace Crew Chief’s/ Passenger’s Seat Pallets - 20 Seat Configuration	2-4-154-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit

- Torque Wrench, Range to Support Seating Torque of Seat Pallet Hardware

References

- PARA 2-4-153

2-4-154.1 Replace Crew Chief’s/Passenger’s Seat Pallets - 20 Seat Configuration.

2-4-154.1.1 Remove.

NOTE

Replacement of all seat pallets is the same.

- Remove crew chief’s/passenger’s seats - 20 seat configuration (PARA 2-4-153).
- Remove screws securing seat pallets to floor panels. Remove seat pallets from floor panels (Figure 2-4-154-1, Detail A).

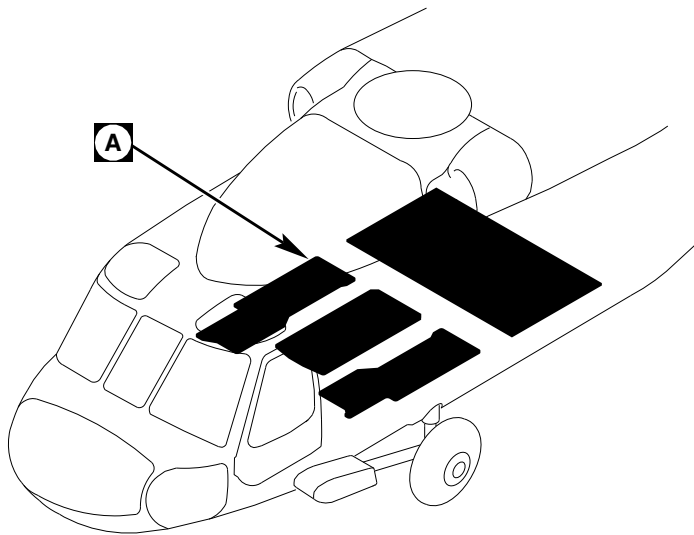
2-4-154.1.2 Inspect.

- Inspect floor panels for any damage.

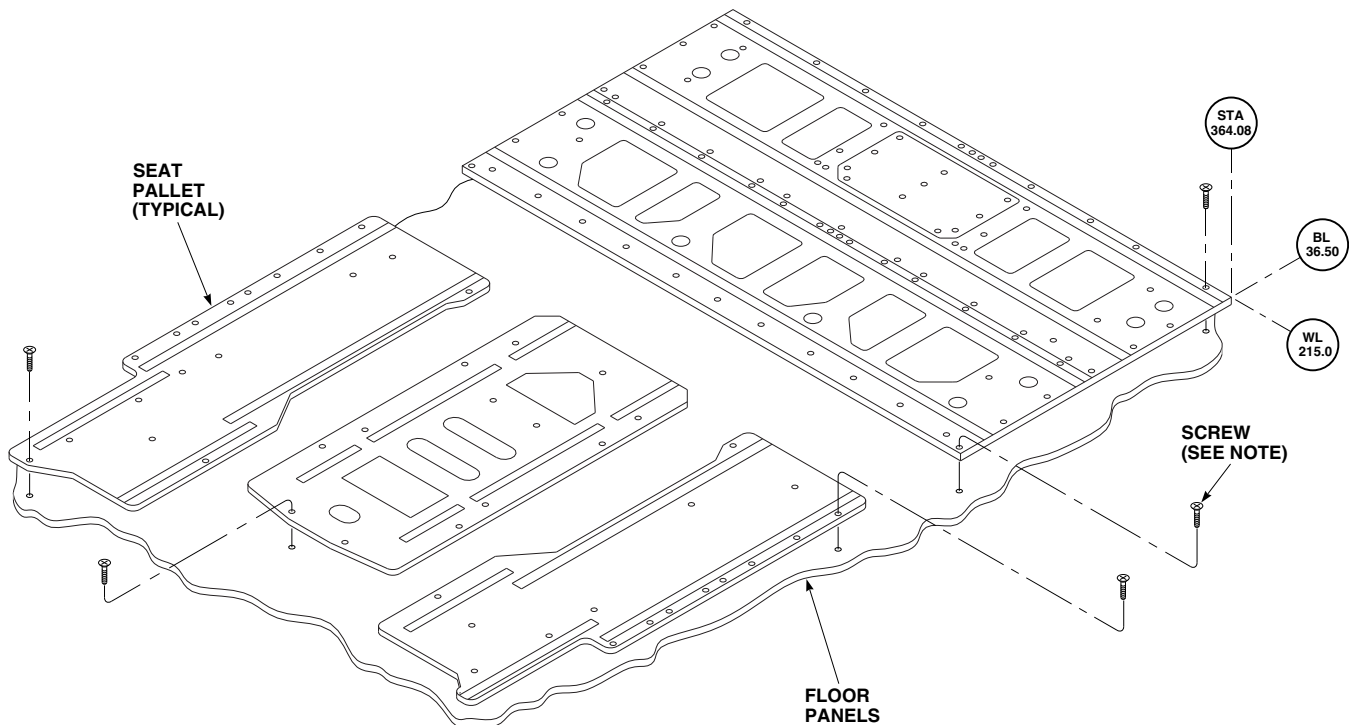
- Inspect floor panels for foreign material. **FLOOR PANELS MUST BE CLEAN AND FREE OF FOREIGN MATERIAL.** If necessary, clean floor panel.

2-4-154.1.3 Install.

- Position seat pallets on floor panels (Figure 2-4-154-1, Detail A).
- Line up screw holes with floor panels and install screws. **TORQUE SCREWS NO MORE THAN 10 INCH-POUNDS GREATER THAN TORQUE NEEDED TO SEAT SCREWS.**
- Install crew chief’s/passenger’s seats - 20 seat configuration (PARA 2-4-153).



A

**NOTE**

TORQUE SCREWS NO MORE THAN 10
IN. LBS GREATER THAN TORQUE
NEEDED TO SEAT SCREWS.

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**FIGURE 2-4-154-1. REPLACE CREW CHIEF'S/PASSENGER'S SEAT PALLETS - 20 SEAT
CONFIGURATION**

2-4-155 AUXILIARY SEARCHLIGHT SUPPORT.

PARAGRAPH BREAKDOWN

	PAGE
2-4-155.1 Replace Auxiliary Searchlight Support	2-4-155-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-6-16
- PARA 1-7-20
- PARA 2-4-94
- PARA 9-4-103

2-4-155.1 Replace Auxiliary Searchlight Support.

2-4-155.1.1 Remove.

- Remove auxiliary searchlight assembly (PARA 9-4-103).
- Remove drag beam support fairing (PARA 2-4-94).
- Remove screws, washers, and nuts securing clamps and electrical harnesses to support (Figure 2-4-155-1, Sheet 1, Detail B and Detail C).
- Disconnect electrical connectors, P340A and P254A, from infrared and incandescent position lights (Figure 2-4-155-1, Sheet 2, Detail E).
- Install protective caps and plugs on electrical connectors.
- Remove screw, washers, and nut securing clamps and electrical harnesses to strut (Figure 2-4-155-1, Sheet 1, Detail D).

- Remove bolts, washers, and cover assembly from searchlight support fitting (Figure 2-4-155-1, Sheet 1, Detail A).
- Remove bolt, washers, and nut securing outboard end of strut to support fitting.
- Remove bolt, washers, and nut securing inboard end of strut to inboard support fitting and remove strut.
- Remove bolts, washers, and nuts securing support to support fitting.
- Remove bolts, washers, and nuts securing support to support fittings and remove support.

2-4-155.1.2 Install.

- Turn off all electrical power (PARA 1-6-16).
- Install support in support fittings with bolts, washers, and nuts (Figure 2-4-155-1, Detail A). **TORQUE NUTS TO 50 - 100 INCH-POUNDS.** Install cotter pins.
- Install searchlight support fitting on support with bolts, washers, and nuts. **TORQUE NUTS**

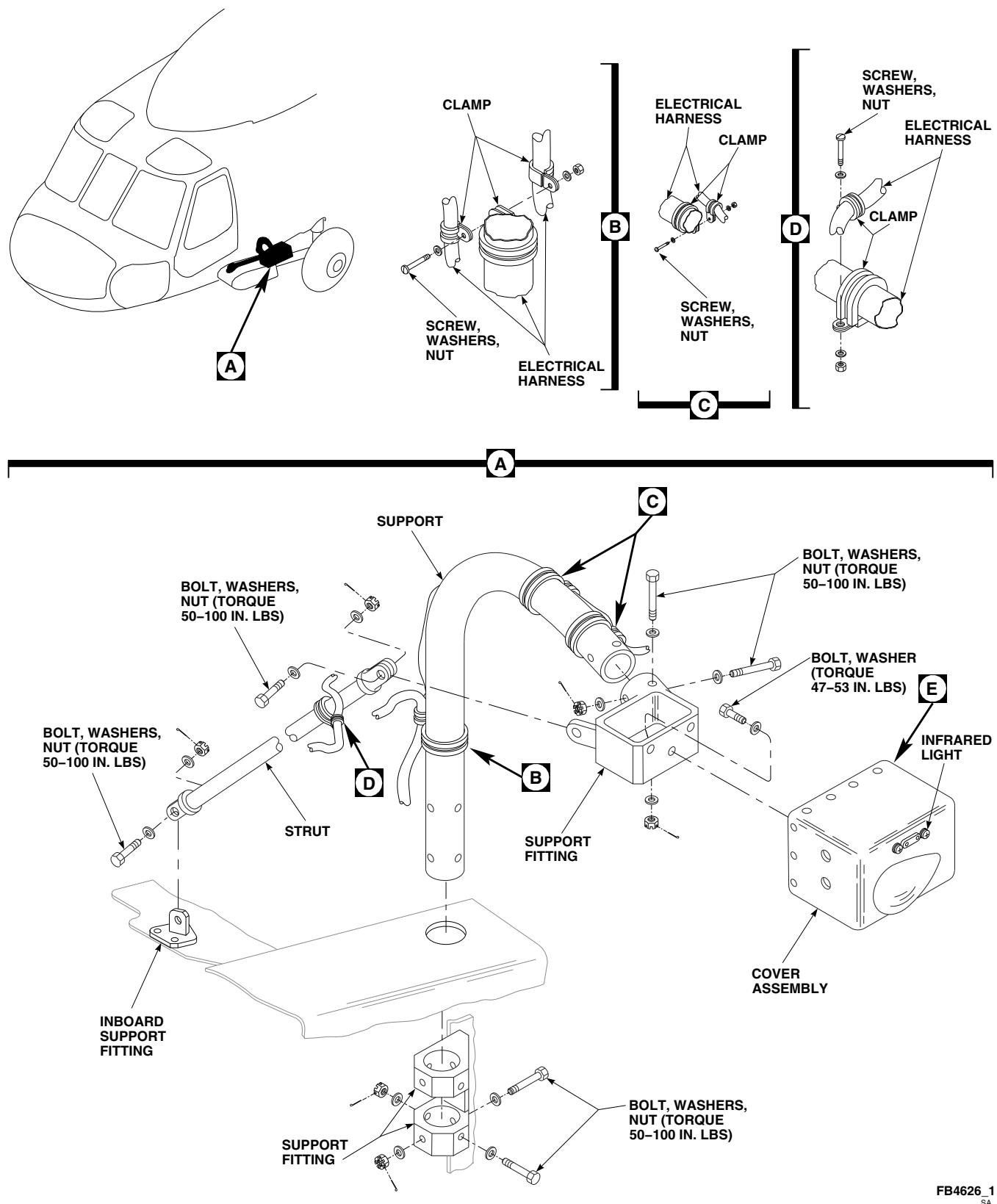
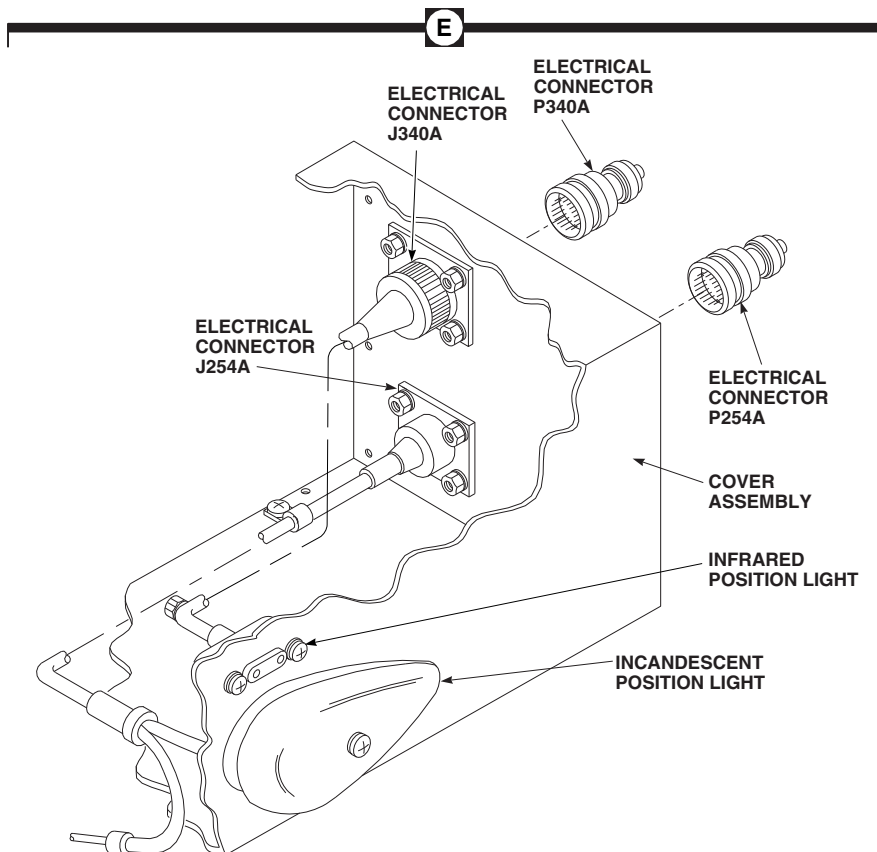


FIGURE 2-4-155-1. REPLACE AUXILIARY SEARCHLIGHT SUPPORT (SHEET 1 OF 2)



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FIGURE 2-4-155-1. REPLACE AUXILIARY SEARCHLIGHT SUPPORT (SHEET 2 OF 2)

- TO 50 - 100 INCH-POUNDS. Install cotter pins.
- d. Insert inboard end of strut to inboard support fitting with bolt, washers, and nut. Do not torque nut at this time.
- e. Insert outboard end of strut to support fitting with bolt, washers, and nut. **TORQUE NUTS AT BOTH ENDS OF STRUT TO 50 - 100 INCH-POUNDS.** Install cotter pins.
- f. Install cover assembly to support fitting with bolts and washers. **TORQUE BOLTS TO 43 - 47 INCH-POUNDS.**
- g. Install electrical harnesses on strut using clamp, screw, washers, and nut (Figure 2-4-155-1, Sheet 1, Detail D).
- h. Remove protective caps and plugs from electrical connectors.
- i. Inspect and treat electrical connectors (PARA 1-7-20).
- j. Connect electrical connectors, P340A and P254A, to infrared and incandescent position lights (Figure 2-4-155-1, Sheet 2 Detail E).

- k. Install electrical harnesses on support using clamps, screws, washers, and nuts (Figure 2-4-155-1, Sheet 1, Detail B and Detail C).
- l. Install drag beam support fairing (PARA 2-4-94).
- m. Install auxiliary searchlight assembly (PARA 9-4-103).

2-4-156 FAST ROPE SYSTEM.

PARAGRAPH BREAKDOWN

	PAGE		PAGE
2-4-156.1 Replace Fries Bar	2-4-156-1	2-4-156.4 Replace Cable	2-4-156-6
2-4-156.2 Replace Fries Bar Support	2-4-156-3	2-4-156.5 Replace Crew Chief's Release	2-4-156-6
2-4-156.3 Replace Fries Bar Cap	2-4-156-3		

INITIAL SETUP

Tools

- Aircraft Mechanic's Toolkit
- Torque Wrench, 0 - 30 in. lbs
- Torque Wrench, 30 - 150 in. lbs

- Torque Wrench, 150 - 750 in. lbs
- Torque Wrench, 700 - 1,600 in. lbs

2-4-156.1 Replace Fries Bar.

2-4-156.1.2 Install.

2-4-156.1.1 Remove.

- Remove cable (PARA 2-4-156.4).

WARNING

Injury to personnel or damage to equipment will result if fries bar is not supported during removal. Use at least three assistants to support fries bar during removal. Do not remove bolts without at least three assistants supporting fries bar.

- With at least three assistants supporting fries bar, remove bolts, thin washers, and thick washers (Figure 2-4-156-1, Detail A).
- Remove fries bar and spacers from cabin ceiling.

WARNING

Injury to personnel or damage to equipment will result if fries bar is not supported during installation. Use at least three assistants to support fries bar during installation. Do not install fries bar without at least three assistants supporting fries bar.

- With the aid of at least three assistants, position spacers and fries bar against cabin ceiling (Figure 2-4-156-1, Detail A).

NOTE

Bolts and barrel-nuts must be replaced after 15 installations or removals.

- With at least three assistants supporting fries bar in position, install bolts, thin washers, and

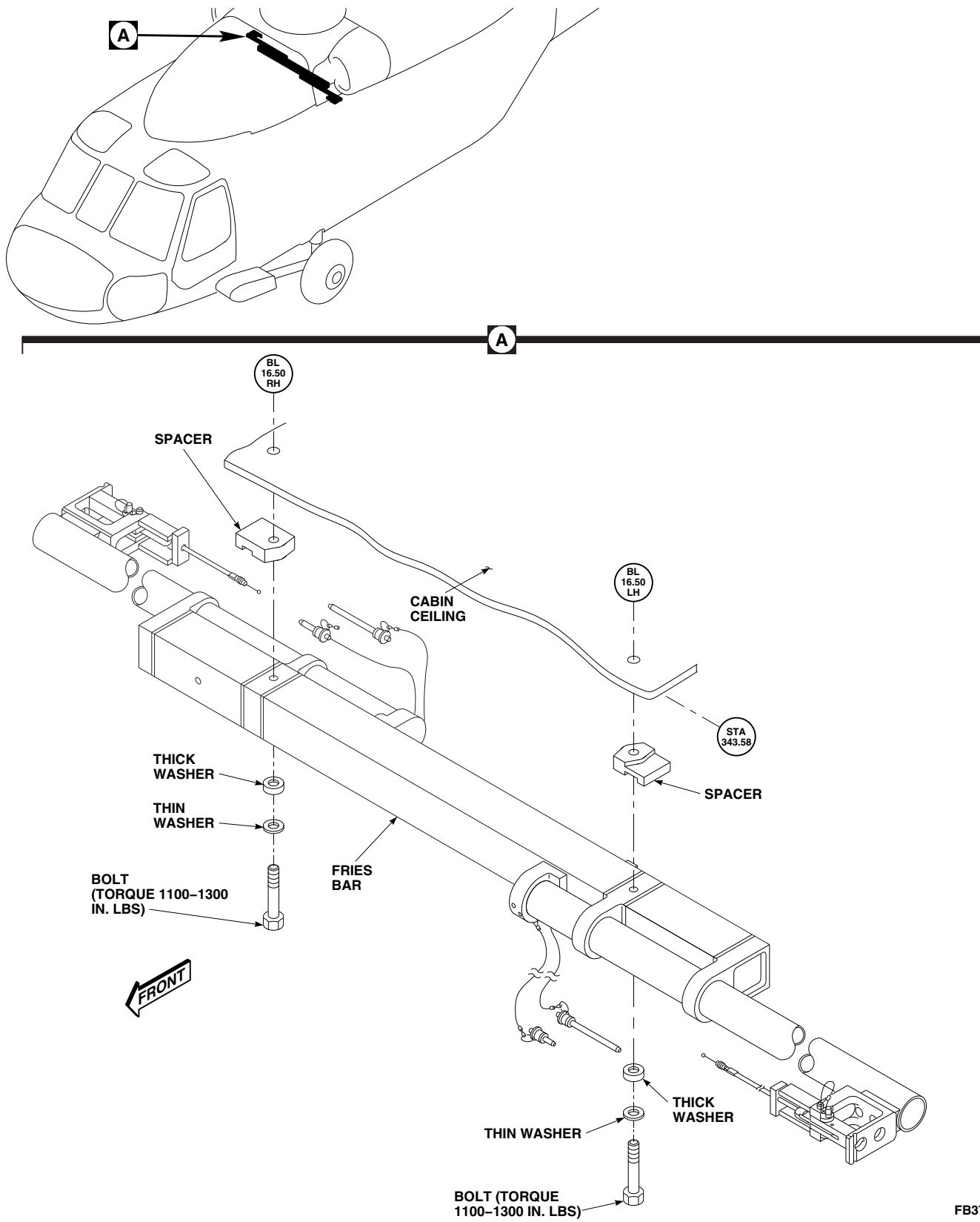


FIGURE 2-4-156-1. REPLACE FRIES BAR

thick washers through fries bar and into barrel-nuts.

- c. **ALTERNATELY TORQUE BOLTS TO 1,100 - 1,300 INCH-POUNDS.**

- d. Install cable (PARA 2-4-156.4).

2-4-156.2 Replace Fries Bar Support.

NOTE

Replacement of right and left fries bar supports is the same.

2-4-156.2.1 Remove.

- a. Remove cable (PARA 2-4-156.4).
- b. Remove mount as follows:
 - (1) Remove bolt, washers, and nut securing release bar and quick-release pin lanyard to mount (Figure 2-4-156-2, Sheet 1, Detail B).
 - (2) Remove quick-release pin from mount.
 - (3) Remove release bar from mount.
 - (4) Remove bolt, wedge, spacer, washer, and self-locking nut from mount (Figure 2-4-156-2, Sheet 2, Detail C).
- c. Slide support out of fries bar (Figure 2-4-156-2, Sheet 1, Detail D).
- d. Remove fries bar cap (PARA 2-4-156.3).

2-4-156.2.2 Disassemble.

- a. Disassemble release bar as follows:
 - (1) Remove screw, washers, and self-locking nut securing cable anchor to release arm (Figure 2-4-156-2, Sheet 2, Detail E).
 - (2) Remove cable anchor from release arm.
- b. Remove spring pin and remove release arm from body.

- c. Remove retainer screw, compression spring, and shoulder pin from body.

2-4-156.2.3 Assemble.

- a. Install shoulder pin, compression spring, and retainer screw in body. **TIGHTEN RETAINER SCREW UNTIL FLUSH WITH BODY** (Figure 2-4-156-2, Sheet 2, Detail E).
- b. Install release arm in body with spring pin. **PRESS SPRING PIN INTO BODY UNTIL FLUSH WITH SURFACE.**
- c. Install cable anchor on release arm and secure with screw, washers, and self-locking nut.

2-4-156.2.4 Install.

- a. Install fries bar cap (PARA 2-4-156.3).
- b. Slide support through fries bar (Figure 2-4-156-2, Sheet 1, Detail D).
- c. Install mount as follows:
 - (1) Position mount on support and install bolt, wedge, spacer, washer, and self-locking nut (Figure 2-4-156-2, Sheet 1, Detail C). **TORQUE SELF-LOCKING NUT TO 99 - 121 INCH-POUNDS.**
 - (2) Install release bar in mount and install quick-release pin (Figure 2-4-156-2, Sheet 2, Detail B).
 - (3) Secure release bar and lanyard to mount with bolt, washers, and nut. **TORQUE NUT TO 247 - 302 INCH-POUNDS.**
- d. Install cable (PARA 2-4-156.4).

2-4-156.3 Replace Fries Bar Cap.

NOTE

Replacement of right and left fries bar caps is the same.

2-4-156.3.1 Remove.

- a. Remove cable (PARA 2-4-156.4).

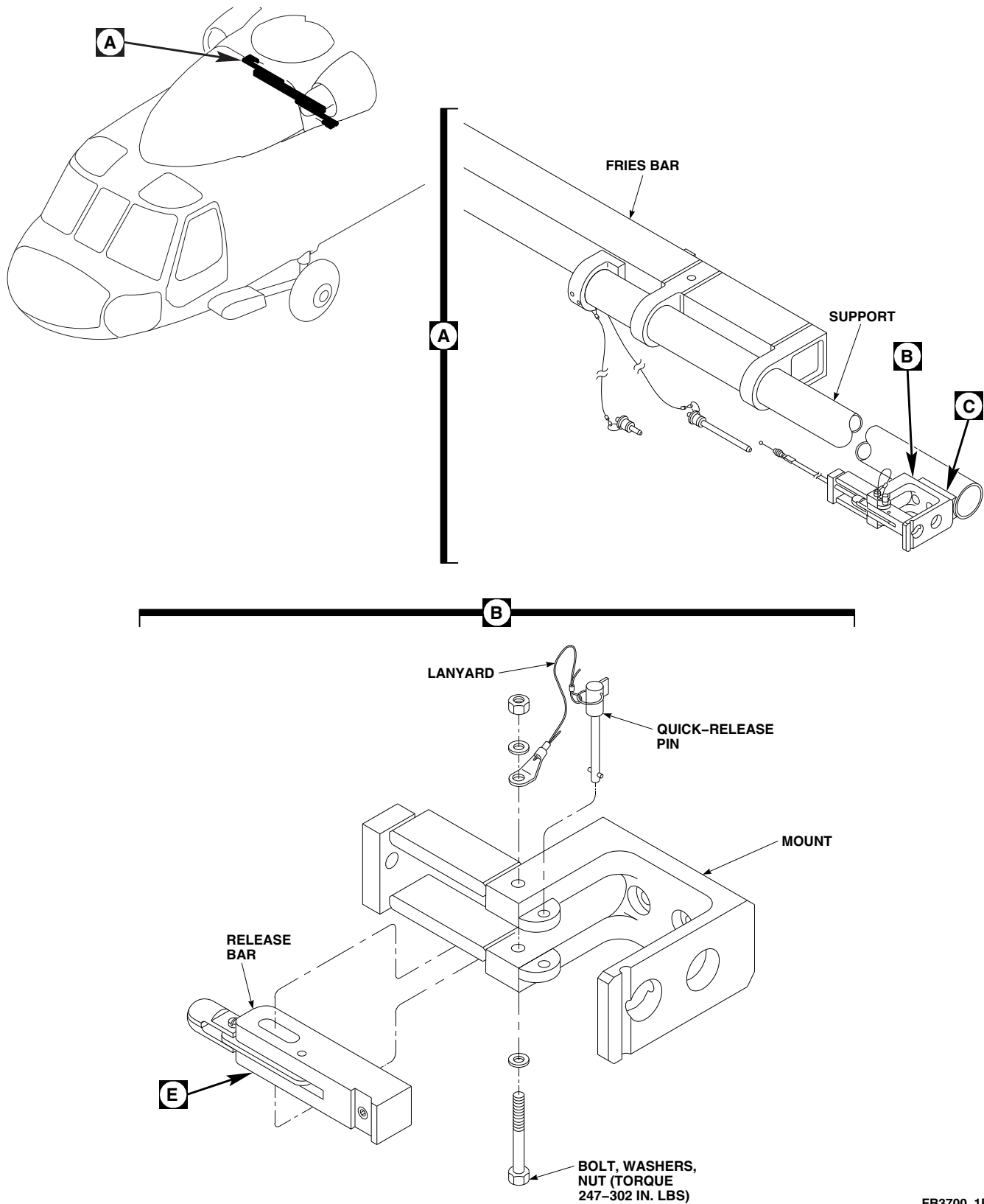
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FIGURE 2-4-156-2. REPLACE FRIES BAR SUPPORT (SHEET 1 OF 2)

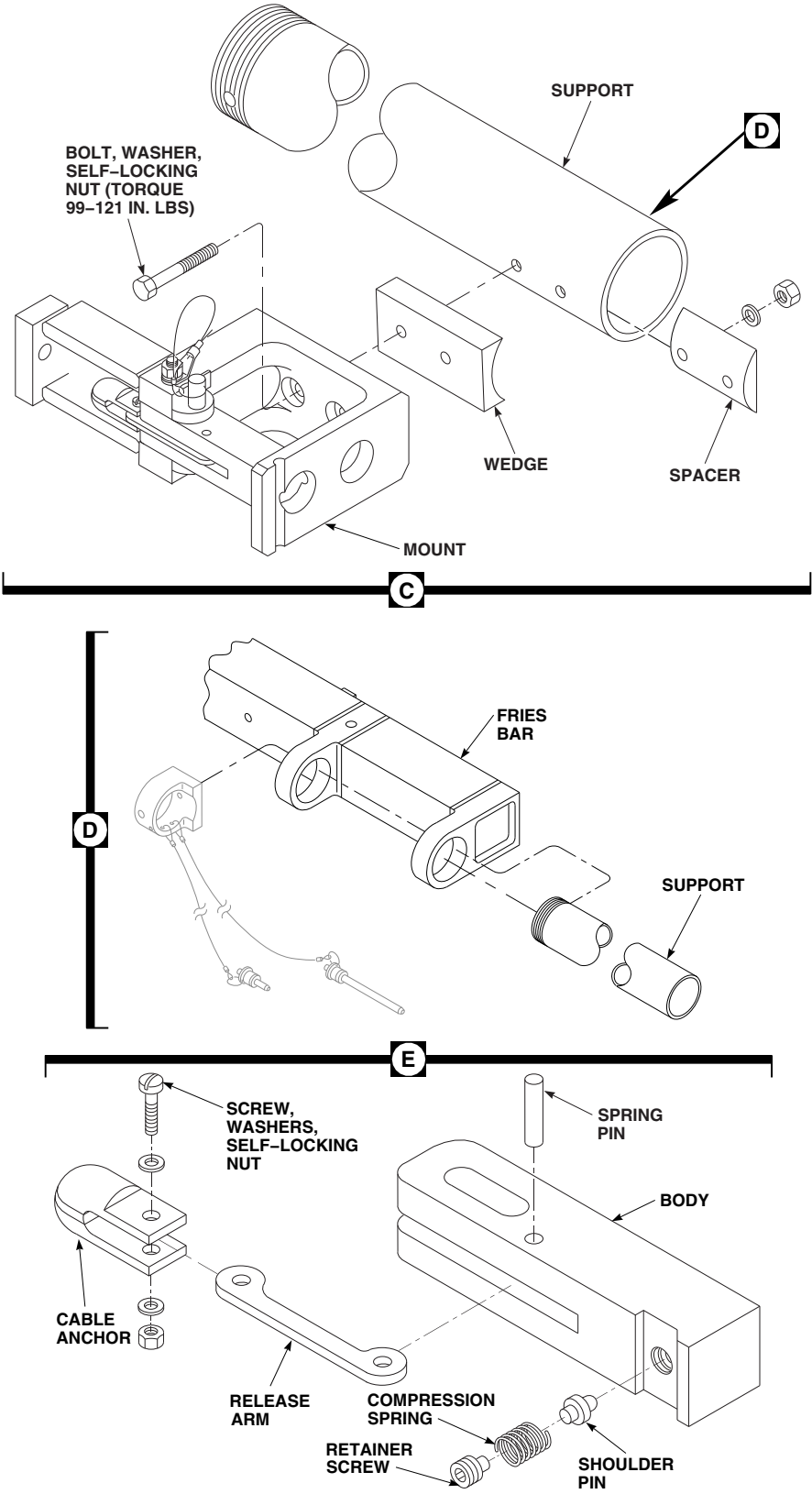


FIGURE 2-4-156-2. REPLACE FRIES BAR SUPPORT (SHEET 2 OF 2)

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- b. Remove quick-release pins, if installed (Figure 2-4-156-3, Detail A).
- c. Remove support (PARA 2-4-156.2).
- d. Remove rivet securing cap to support.
- e. Remove fries bar cap from support.

2-4-156.3.2 Install.

- a. Install fries bar cap onto support until rivet hole in cap and support line up (Figure 2-4-156-3, Detail A).
- b. Install rivet.
- c. Install support (PARA 2-4-156.2).
- d. Install quick-release pins, if required.
- e. Install cable (PARA 2-4-156.4).

2-4-156.4 Replace Cable.

NOTE

Replacement of right and left cables are the same.

2-4-156.4.1 Remove.

- a. Remove all clamps and hardware securing cable (Figure 2-4-156-4, Detail A).
- b. Remove cable from cable anchor on release bar (Figure 2-4-156-4, Detail B).
- c. Remove hardware securing cable to crew chief's release.
- d. Remove cable from cable anchor and slide up through hole in crew chief's release (Figure 2-4-156-4, Detail A).

2-4-156.4.2 Install.

- a. Install cable down through hole in crew chief's release (Figure 2-4-156-4, Detail A).
- b. Install lower nut on cable. **TIGHTEN NUT SO AT LEAST TWO THREADS ARE PROTRUDING.**

- c. Route cable and loosely install existing clamps and hardware.
- d. Install cable in cable anchor on release bar (Figure 2-4-156-4, Detail B).
- e. Tighten all existing clamps and hardware (Figure 2-4-156-4, Detail A).

2-4-156.4.3 Adjust.

- a. Check quick-release mechanism for proper operation.
- b. Adjust cable by changing position of upper and lower nuts to alter cable length (Figure 2-4-156-4, Detail A).

2-4-156.5 Replace Crew Chief's Release.

NOTE

Replacement of right and left crew chief's releases are the same.

2-4-156.5.1 Remove.

- a. Remove lower nut securing cable to crew chief's release (Figure 2-4-156-4, Detail A).
- b. Remove cable from cable anchor, slide cable up through hole in crew chief's release, temporarily reinstall lower nut on cable.
- c. Remove pins and collars securing crew chief's release to airframe (Figure 2-4-156-5, Sheet 1, Detail A).

2-4-156.5.2 Disassemble.

- a. Remove rivets securing zeos to channel (Figure 2-4-156-5, Sheet 1, Detail B).
- b. Remove bolts and washers securing release handle mount to channel.
- c. Remove screw, washers, and nut securing cable anchor to lever (Figure 2-4-156-5, Sheet 2, Detail C).
- d. Remove screw and grip from lever.

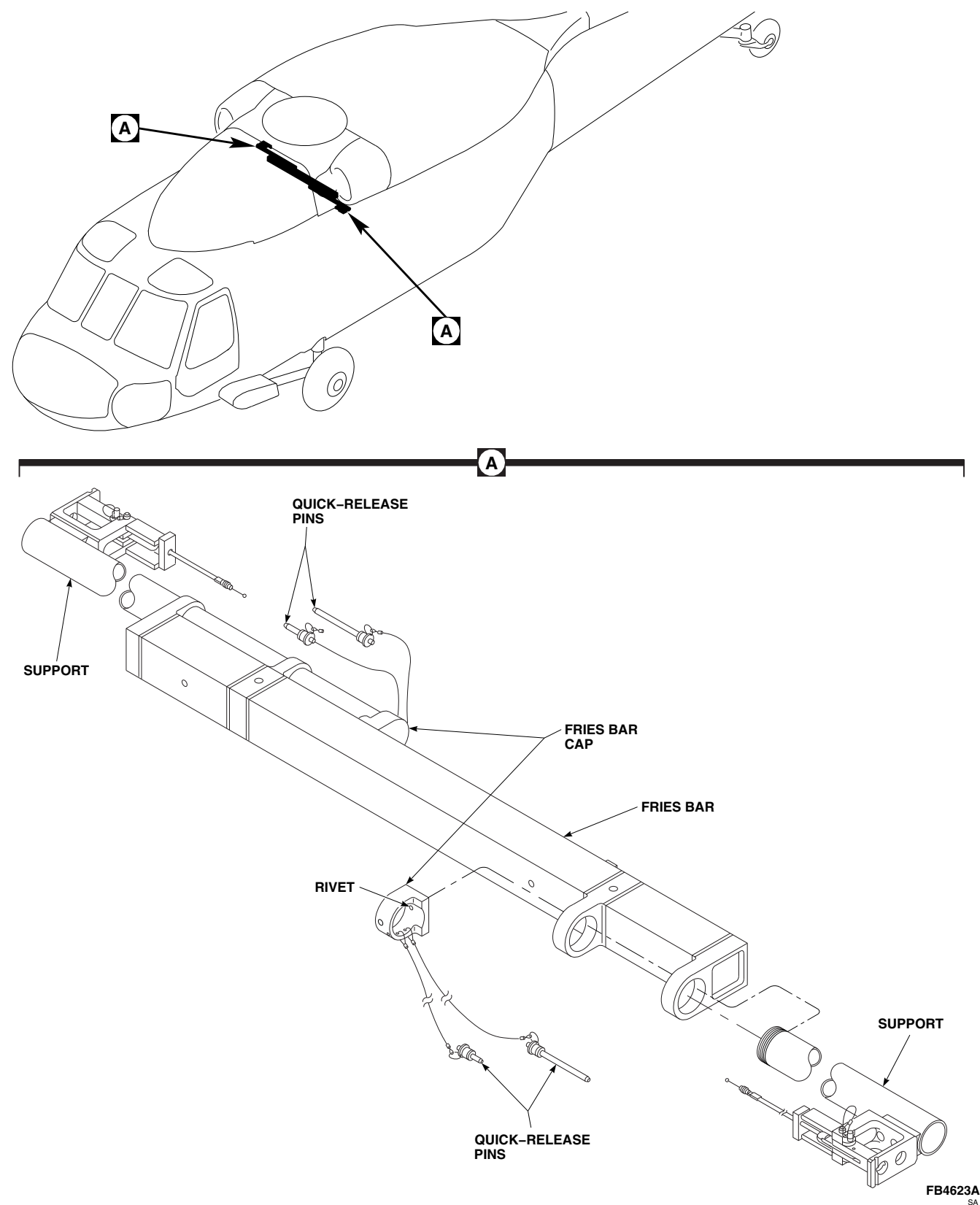


FIGURE 2-4-156-3. REPLACE FRIES BAR CAP

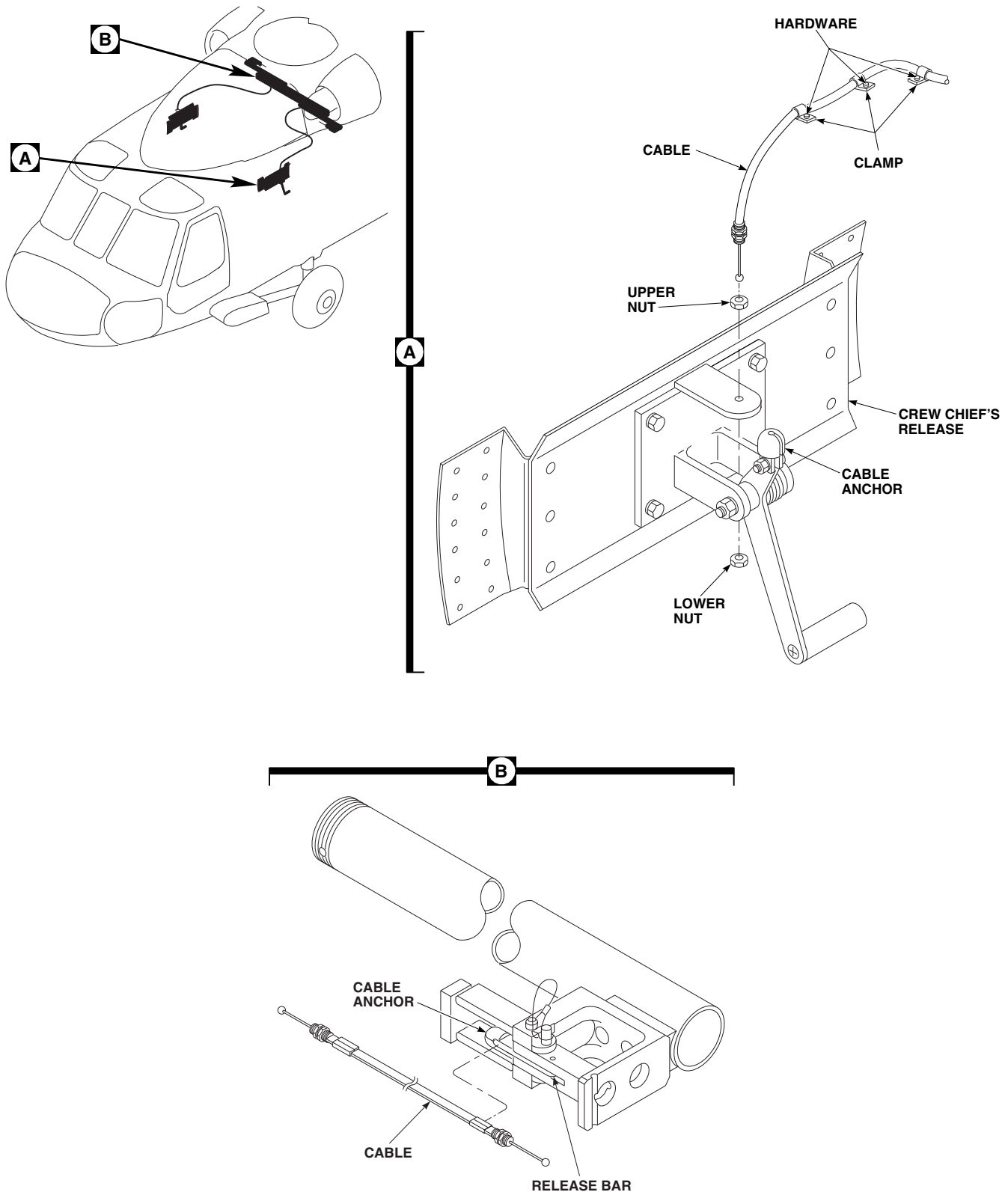


FIGURE 2-4-156-4. REPLACE CABLE

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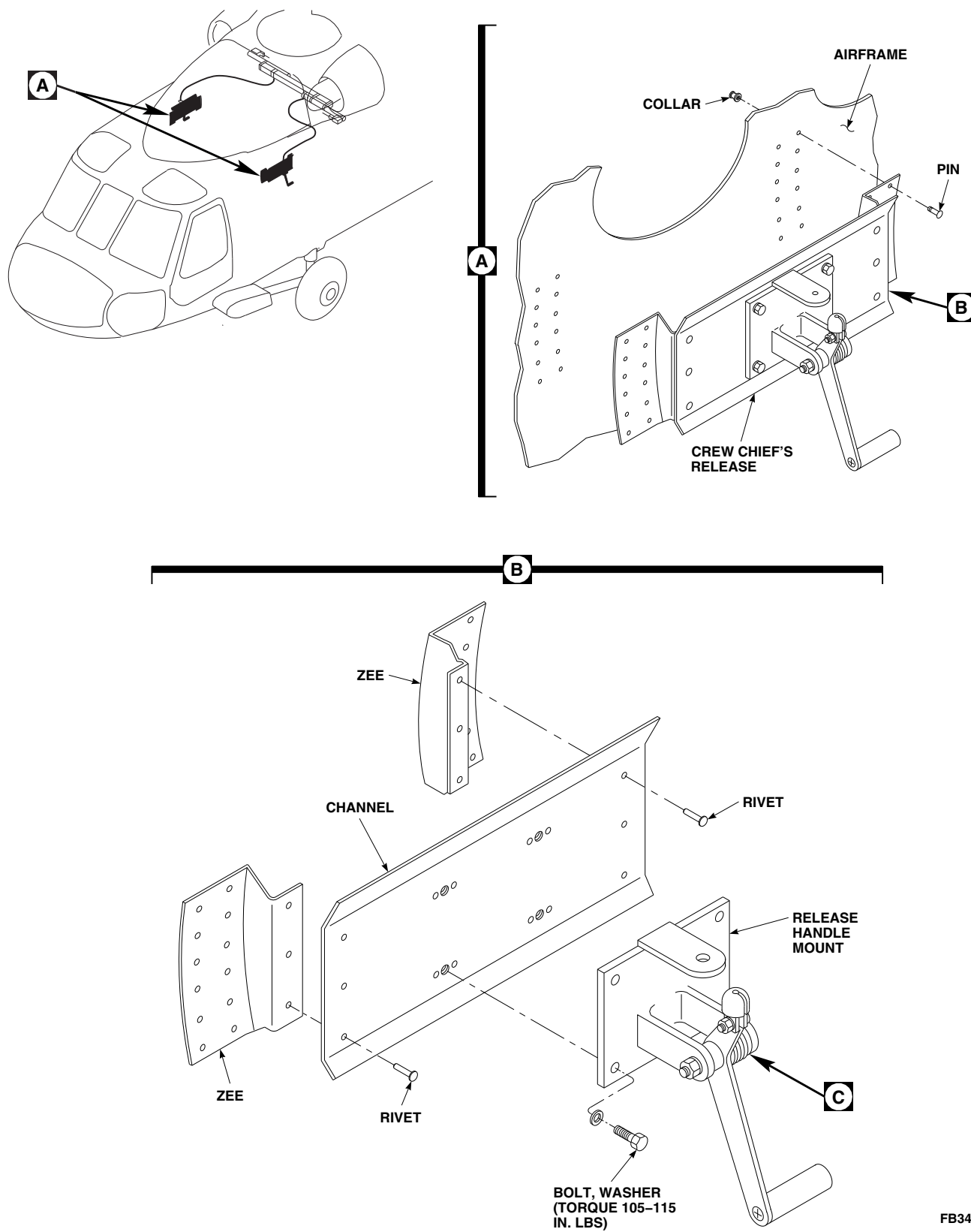


FIGURE 2-4-156-5. REPLACE CREW CHIEF'S RELEASE (SHEET 1 OF 2)

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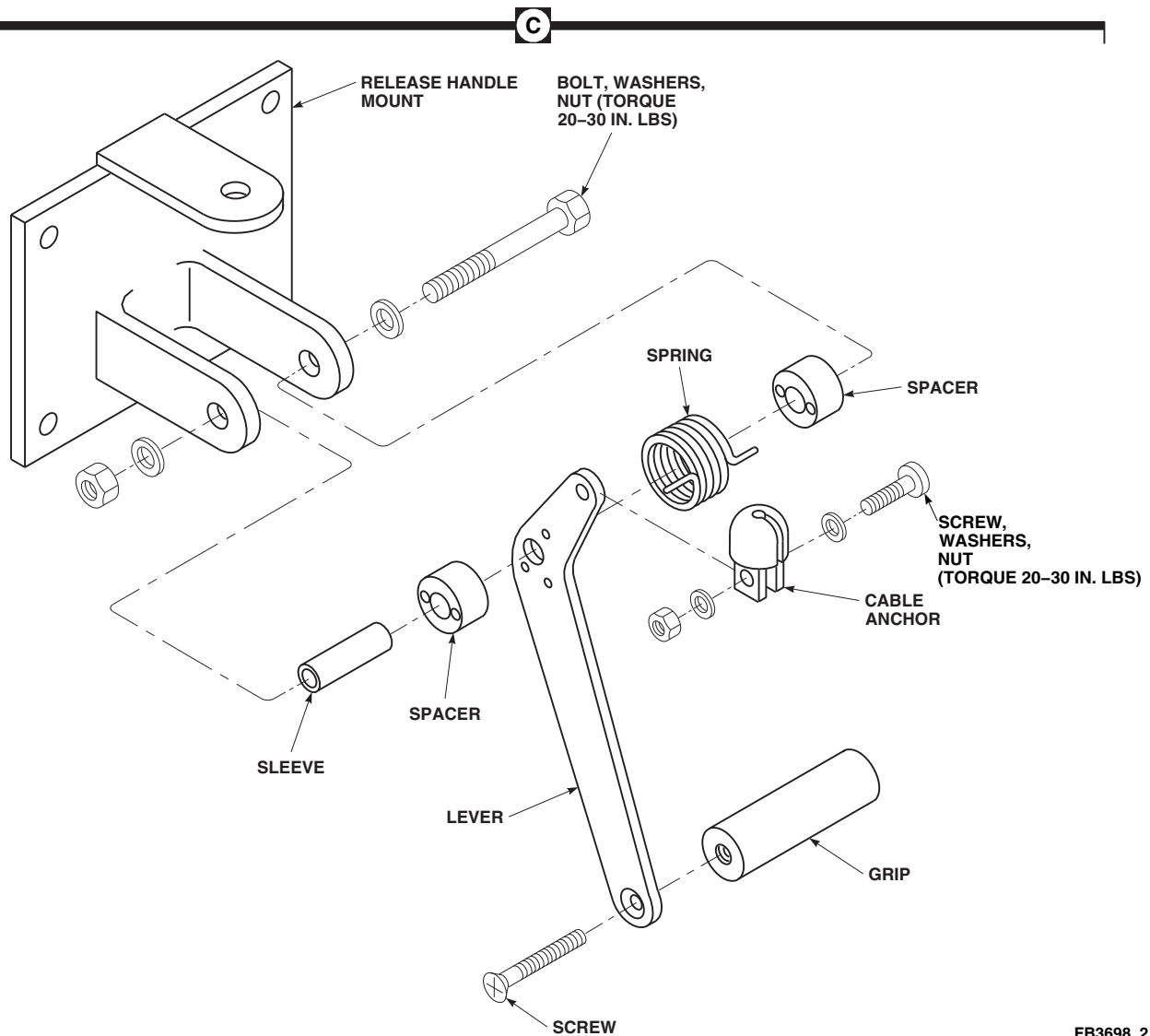
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FIGURE 2-4-156-5. REPLACE CREW CHIEF'S RELEASE (SHEET 2 OF 2)

- e. Remove bolt, washers, spacers, spring, sleeve, and nut securing lever to release handle mount.

2-4-156.5.3 Assemble.

- Install lever on release handle mount with bolt, washers, spacers, spring, sleeve, and nut (Figure 2-4-156-5, Sheet 2, Detail C). **TORQUE NUT TO 20 - 30 INCH-POUNDS.**
- Install grip on lever with screw. **HANDTIGHTEN SCREW.**
- Install cable anchor on lever with screw, washers, and nut. **TORQUE NUT TO 20 - 30 INCH-POUNDS.**
- Install release handle mount to channel with bolts and washers (Figure 2-4-156-5, Sheet 1, Detail B). **TORQUE BOLT TO 105 - 115 INCH-POUNDS.**
- Install zeos on channel using rivets.

2-4-156.5.4 Install.

- a. Install crew chief's release on airframe with pins and collars (Figure 2-4-156-5, Sheet 1, Detail A).
- b. Remove lower nut from cable, and slide cable down through hole in release handle mount.

- c. Install lower nut on cable (Figure 2-4-156-4, Detail A). **TIGHTEN NUT SO AT LEAST TWO THREADS ARE PROTRUDING.**
- d. Adjust cable, if required (PARA 2-4-156.4).

SECTION 2-5

MAINTENANCE PROCEDURES (BENCH)

SECTION OVERVIEW

PARAGRAPH	TITLE
2-5-1	Cockpit Door Hinge Support Fitting (BENCH)
2-5-2	Nose Vibration Absorber Lower Fitting Bushings (BENCH)
2-5-3	Nose Vibration Absorber Fitting Bearings (BENCH)
2-5-4	Cabin Vibration Absorber Fitting Bearings (BENCH)
2-5-5	Cabin Vibration Absorber Spring Bushings (BENCH)
2-5-6	Vibration Absorber Airframe Nose Fitting Bushings (BENCH)
2-5-7	Vibration Absorber Airframe Cabin Fitting Bushings (BENCH)
2-5-8	Vibration Absorber Airframe Cabin Fitting Bearings (BENCH)
2-5-9	Main Landing Gear Fuselage Fitting Spherical Bearing (BENCH)
2-5-10	Cabin Door Window (BENCH)
2-5-11	Cabin Door Window Frame (BENCH)
2-5-12	Soundproofing Panel (BENCH)
2-5-13	Engine Cowling Support Fitting Bearing (BENCH)
2-5-14	Engine Cowling Support Fitting Bushings (BENCH)
2-5-15	Engine Cowling Fitting Bearing (BENCH)
2-5-16	Oil Cooler Support (BENCH)
2-5-17	Oil Cooler Forward Support (BENCH)
2-5-18	Oil Cooler Rear Support (BENCH)
2-5-19	Tail Rotor Drive Shaft Support (BENCH)
2-5-20	Stabilator Elastomeric Bearing (BENCH)

	PARAGRAPH	TITLE
I	2-5-21	Stabilator Position Transmitter/Limit Switches (BENCH)
	2-5-22	Stabilator Position Transmitter (BENCH)
I	2-5-23	Stabilator Position Transmitter/Limit Switch Electrical Connector (BENCH)
I	2-5-24	Stabilator Position Transmitter/Limit Switch Assembly (BENCH)

2-5-1 COCKPIT DOOR HINGE SUPPORT FITTING (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
2-5-1.1	Repair Cockpit Door Hinge Support Fitting (BENCH)	2-5-1-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Container
- Drift
- Hammer, Slide
- Pliers
- Puller Set

Materials/Parts

- Applicator, Item 78, Appendix D
- Cheesecloth, Item 90, Appendix D

- Dry Ice, Item 125, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Primer, Item 257, Appendix D
- Sealing Compound, Item 274, Appendix D
- Sealing Primer, Item 305, Appendix D
- Trichloroethylene, Technical, Item 343, Appendix D

References

- AMS 5463
- Appendix D
- PARA 2-4-11
- PARA 2-4-14

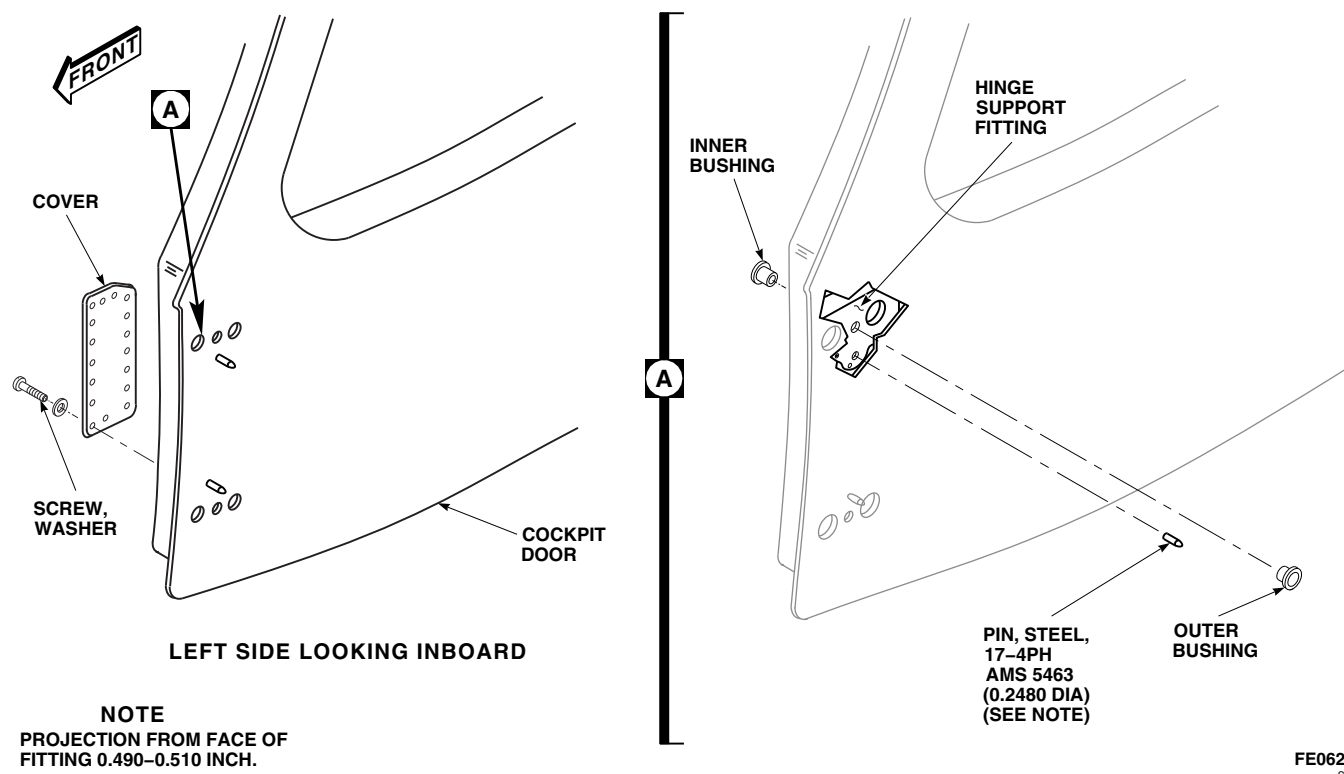
2-5-1.1 Repair Cockpit Door Hinge Support Fitting (BENCH).

2-5-1.1.1. Remove.

- Remove hinge assembly and gasket (PARA 2-4-11).
- Remove screws and washers securing cover to inside of door, and remove cover (Figure 2-5-1-1).
- Remove components of door jettison mechanism necessary to access hinge support fitting (PARA 2-4-14).

2-5-1.1.2. Repair.

- To replace bushings, perform the following:
 - Remove one bushing using puller from puller set and slide hammer (Figure 2-5-1-1, Detail A).
 - Using slide hammer and drift, remove second bushing.
 - Remove primer residue from outer surfaces of fitting and inside of bore with cheesecloth, Item 90, Appendix D, wet



- with methyl ethyl ketone, Item 217, Appendix D.
 - (4) Put two new bushings in container of dry ice, Item 125, Appendix D, for 30 minutes.
 - (5) Remove one bushing from dry ice, and apply thin coat of primer, Item 257, Appendix D, to all outer surfaces that will contact hinge support fitting.
 - (6) While primer is still wet, install bushing in inner side of fitting, using puller from puller set and slide hammer.
 - (7) Remove second bushing from dry ice and apply thin coat of primer, Item 257, Appendix D, to all outer surfaces that will contact hinge support fitting.
 - (8) While primer is still wet, install second bushing in outer side of fitting, using appropriate drift and slide hammer.
- b. To replace pin, perform the following:
- (1) Grip pin with pliers, twist to break bond, and remove from fitting.
 - (2) Remove adhesive residue from inside of pin socket with applicator, Item 78, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D.
 - (3) Thoroughly clean inside of pin socket and surface of new pin with technical trichloroethylene, Item 343, Appendix D. Use applicator, Item 78, Appendix D, for socket and cheesecloth, Item 90, Appendix D, for pin surface.
 - (4) Apply a thin coat of sealing primer, Item 305, Appendix D, on inside of socket and on mating pin surface to 0.50-inch from tip. Let dry for 5 minutes.
 - (5) Coat inside of socket, and mating pin surface to 0.50-inch from tip, with sealing compound, Item 274, Appendix D.
 - (6) Insert pin into socket until tip is 0.490 - 0.510-inch from fitting surface.

2-5-1.1.3. Install.

- a. Install components of door jettison mechanism (PARA 2-4-14).
- b. Install cover on inside of door with screws and washers (Figure 2-5-1-1).
- c. Install door hinge assembly and gasket (PARA 2-4-11).

2-5-2 NOSE VIBRATION ABSORBER LOWER FITTING BUSHINGS (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
2-5-2.1	Replace Nose Vibration Absorber Lower Fitting Bushings (BENCH)	2-5-2-1

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Arbor Press
- Bushing Installation Kit
- Powertrain Repairer’s Toolkit
- Protection, Skin and Breathing

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Machinery Towel, Item 211, Appendix D
- Sealing Compound, Item 273, Appendix D

References

- Appendix D
- PARA 2-6-5

2-5-2.1 Replace Nose Vibration Absorber Lower Fitting Bushings (BENCH).

2-5-2.1.1. Remove.



- Bushings, 70219-02150-101 through 70219-02150-104, are made of beryllium copper metal, which can be a health hazard. Avoid breathing dust while reworking damaged areas. Use a dust respirator if necessary. Grinding is prohibited. Keep beryllium copper out of open wounds. Use skin protection if necessary. Clean up visible dust piles

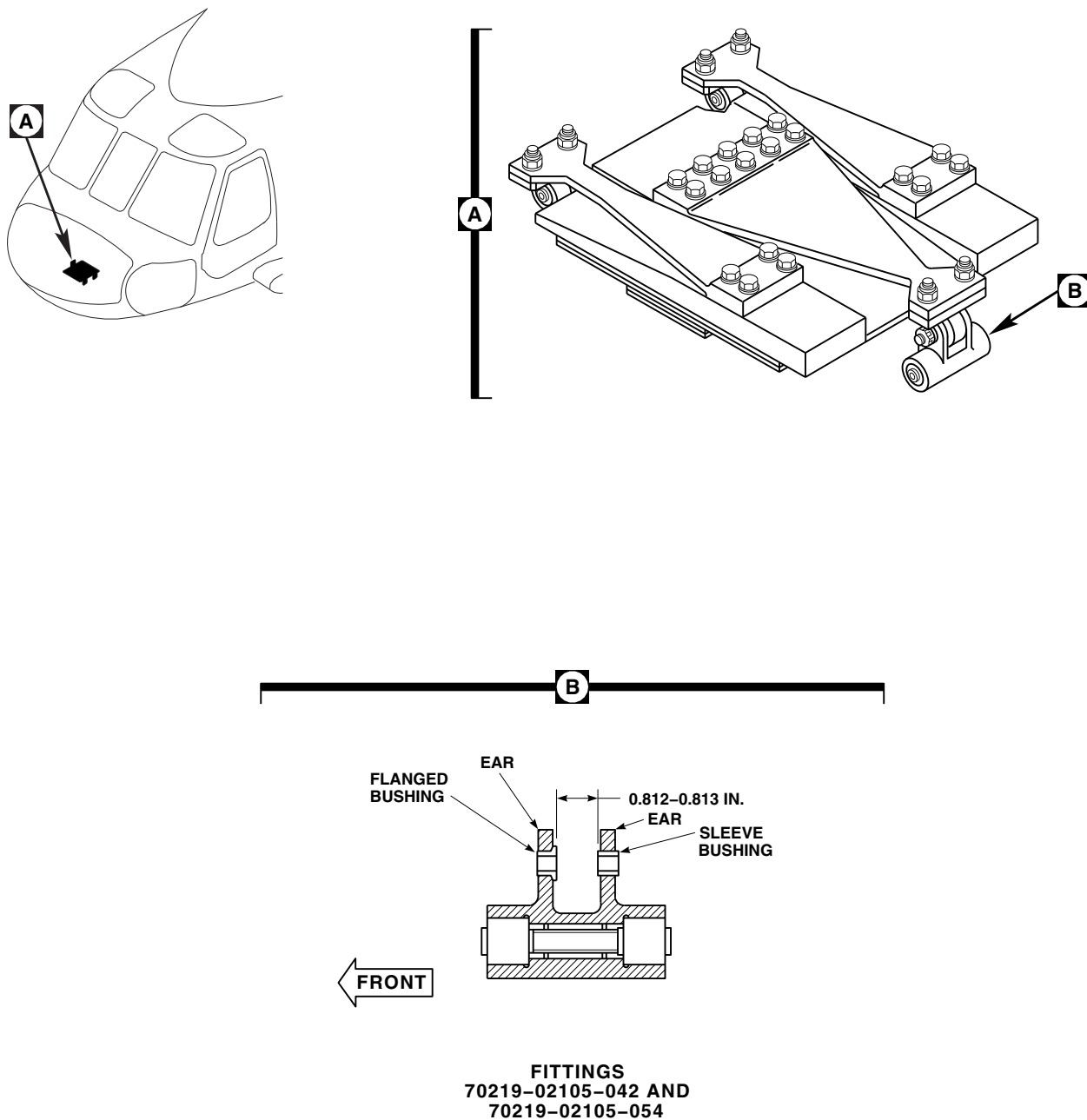
and wash hands and arms after maintenance.

- No minor repair or corrosion removal allowed on beryllium copper components.

On fittings, 70219-02105-042 and 70219-02105-054, remove bushings using bushing installation kit (Figure 2-5-2-1, Detail B).

2-5-2.1.2. Install.

- a. Wipe bushings and fitting holes clean, using machinery towel, Item 211, Appendix D, wet with dry-cleaning solvent, Item 124, Appendix D. Avoid getting solvent into bearing. Wipe dry.



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FIGURE 2-5-2-1. REPLACE NOSE VIBRATION ABSORBER LOWER FITTING BUSHINGS (BENCH)

2-5-2-2 Change 2

- b. Coat bushings and fitting holes with sealing compound, Item 273, Appendix D.

NOTE

Make sure flanged bushing is installed in location noted during removal.

- c. Using bushing installation kit and arbor press, install flanged bushing in fitting lug with flange inside (Figure 2-5-2-1, Detail B).

- d. Using bushing installation kit and arbor press, press sleeve bushing in fitting lug until it is 0.812 - 0.813-inch from flanged bushing.

- e. Touch up paint as required (PARA 2-6-5).

2-5-3 NOSE VIBRATION ABSORBER FITTING BEARINGS (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
2-5-3.1	Replace Nose Vibration Absorber Fitting Bearings (BENCH)	2-5-3-2

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Arbor Press
- Bushing Installation Kit
- Fluorescent Penetrant Inspection Kit
- Powertrain Repairer’s Toolkit
- Reamer Set
- Staking Tools, Locally-Made
- Torque Wrench, 0 - 16 in. oz
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Abrasive Cloth, Item 6, Appendix D
- Applicator, Item 78, Appendix D

- Cheesecloth, Item 90, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Grease, Item 151A, Appendix D
- Machinery Towel, Item 211, Appendix D
- Masking Tape, 1-inch, Item 212, Appendix D
- Sealing Compound, Item 302C, Appendix D
- Sealing Primer, Item 305, Appendix D
- Trichloroethane 1,1,1, Item 341, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 2-4-32
- PARA 2-6-5

2-5-3.1 Replace Nose Vibration Absorber Fitting Bearings (BENCH).

GREATER THAN 5 INCH-OUNCES THROUGH FULL 360° TURN.

2-5-3.1.1 Remove Fitting Assembly Bearing.

- a. Using bushing installation kit and arbor press, remove bearing from fittings, 70219-02105-041, 70219-02105-053, and 70219-02127-042 (Figure 2-5-3-1, Detail A).
- b. On fittings, 70219-02105-041 and 70219-02105-053, perform the following:
 - (1) Ream bearing bore 1.2510 - 1.2535-inch diameter (Figure 2-5-3-1, Detail B).
 - (2) Blend out staking protrusions.
 - (3) Using fluorescent penetrant inspection kit, inspect reworked areas for cracks (ASTM E1417). **NONE ALLOWED.** If cracks are found, replace fitting (PARA 2-4-32).
- c. Clean fitting bore, using applicator, Item 78, Appendix D, moistened with trichloroethane 1,1,1, Item 341, Appendix D.
- d. Remove retaining compound residue, using abrasive cloth, Item 6, Appendix D.
- e. Remove abrasive residue, using cheesecloth, Item 90, Appendix D, moistened with trichloroethane 1,1,1, Item 341, Appendix D.

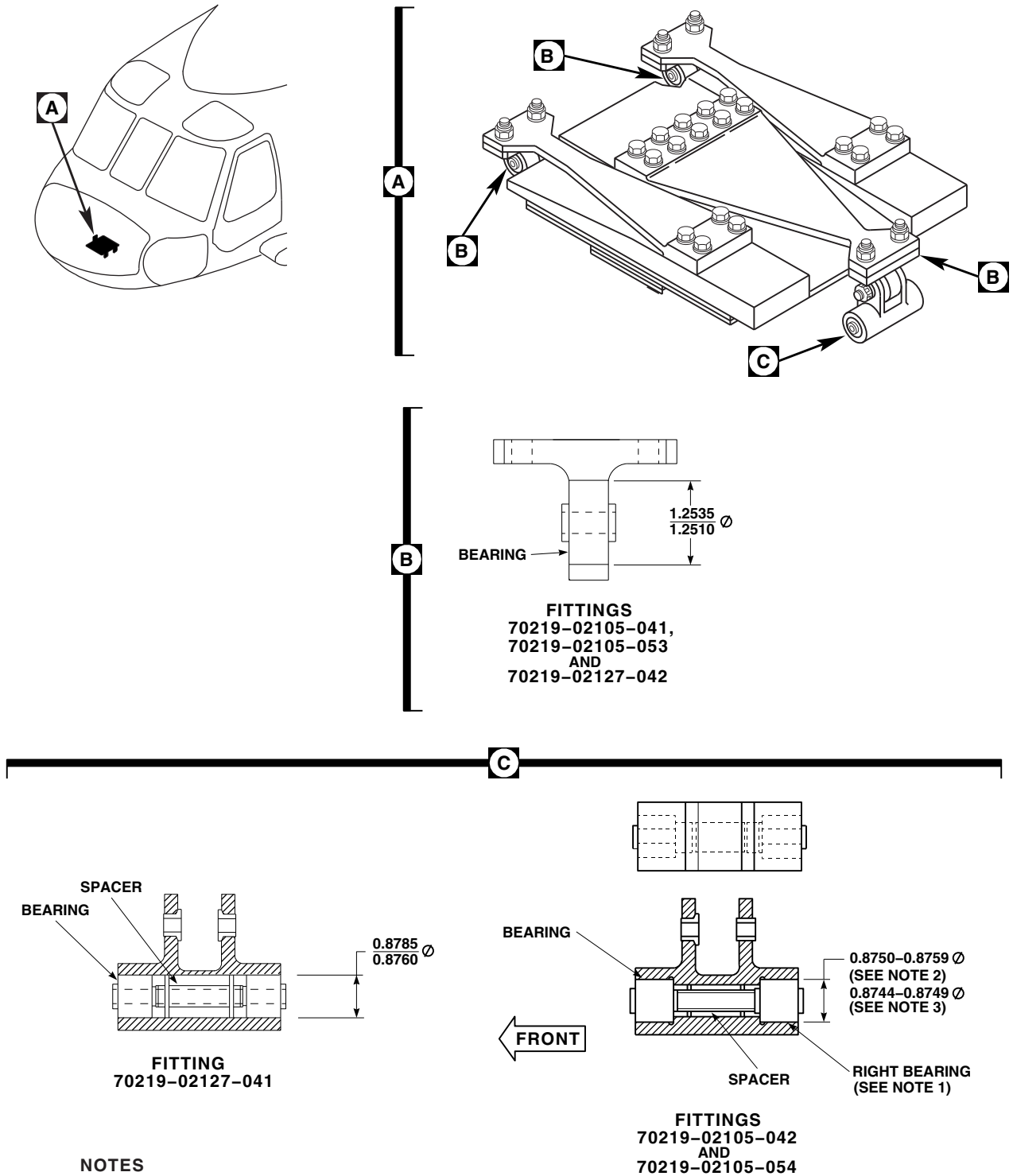
- (4) Remove nut, washers, and bolt from bearing.
- b. Prepare bonding surfaces of bearing and bearing bore as follows:
 - (1) Mask bearing interior against contamination using 1-inch masking tape, Item 212, Appendix D.
 - (2) Sand bonding surface of bearing and bearing bore lightly, using abrasive cloth, Item 6, Appendix D.
 - (3) Clean off abrasive residue using cheesecloth, Item 90, Appendix D, moistened with trichloroethane 1,1,1, Item 341, Appendix D.
 - (4) Remove masking tape from bearing.
- c. Clean bonding surfaces of fitting and bearing, using cheesecloth, Item 90, Appendix D, moistened with trichloroethane 1,1,1, Item 341, Appendix D.
- d. Apply light coat of sealing primer, Item 305, Appendix D, to bearing outside diameter and inside diameter of fitting bore, using applicator, Item 78, Appendix D.

2-5-3.1.2 Install Fitting Assembly Bearing.

- a. Prior to installing upper bearing, perform breakout torque check, as follows:
 - (1) Install bolt through bearing. Install enough washers on bolt to seat nut. **INSTALL AND TIGHTEN NUT SO THAT WASHERS DO NOT ROTATE.**
 - (2) Rotate bearing inner race 4 complete turns.
 - (3) Using torque wrench rotate bearing. **BREAKOUT TORQUE MUST NOT BE**



- Protect primed surfaces from contamination during drying.
- Do not touch primed surfaces with bare hands.
- e. Let sealing primer dry for 15 minutes minimum. If any break in primer coat occurs, repeat steps c. through e.



NOTES

1. STAKE RIGHT BEARING ONLY.
2. DIMENSION ON FITTING 70219-02105-054.
3. DIMENSION ON FITTING 70219-02105-042.

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FIGURE 2-5-3-1. REPLACE NOSE VIBRATION ABSORBER FITTING BEARINGS (BENCH)

CAUTION

If compound is removed from container, do not return unused portion to container. Do not dip applicator in container. Contamination of retaining compound will make it unusable.

- f. Apply even coat of sealing compound, Item 302C, Appendix D, to outside diameter of bearing and bonding surface of fitting only, using applicator, Item 78, Appendix D.

CAUTION

To avoid weak bond, bearing must be installed in fitting within 4 minutes of retaining compound application. If this time is exceeded, step c. through step f. must be repeated.

- g. Immediately install bearing in fitting with slight twisting motion to distribute retaining compound. Bearing must be centered in fitting bore within 0.020-inch. Observe squeeze out at bond line and wipe off excess immediately (Figure 2-5-3-1, Detail B).
- h. Allow fitting assembly to cure for 24 hours minimum.
- i. On reworked fittings, 70219-02105-041 and 70219-02105-053, perform the following:
 - (1) Apply corrosion resistant coating, Item 118, Appendix D, to any exposed reworked surfaces.
 - (2) Touch up paint as required (PARA 2-6-5).
- j. Perform breakout torque check, as follows:
 - (1) Install bolt through bearing. Install enough washers on bolt to seat nut. **INSTALL AND TIGHTEN NUT SO THAT WASHERS DO NOT ROTATE.**
 - (2) Rotate bearing inner race 4 complete turns.

- (3) Using torque wrench, rotate bearing. **BREAKOUT TORQUE MUST NOT BE GREATER THAN 10 INCH-OUNCES THROUGH FULL 360° TURN WHEN MISALIGNING INNER AND OUTER RACES BY 5°.**

- (4) Remove nut, washers, and bolt from bearing.

2-5-3.1.3 Remove Fitting Assembly Bearings and Spacer.

- a. Using bushing installation kit and arbor press, remove bearings and spacer from fittings, 70219-02105-042, 70219-02105-054, and 70219-02127-041 (Figure 2-5-3-1, Detail C).
- b. On fittings, 70219-02105-042 and 70219-02105-054, clean fitting bores and spacer using machinery towel, Item 211, Appendix D, wet with dry-cleaning solvent, Item 124, Appendix D. Wipe dry.
- c. On fitting, 70219-02127-041, clean fitting bores and spacer, as follows:
 - (1) Clean fitting bores, using applicator, Item 78, Appendix D, moistened with trichloroethane 1,1,1, Item 341, Appendix D.
 - (2) Remove retaining compound residue from fitting bores, using abrasive cloth, Item 6, Appendix D.
 - (3) Remove abrasive residue from fitting bores using cheesecloth, Item 90, Appendix D, moistened with trichloroethane 1,1,1, Item 341, Appendix D.
 - (4) Clean spacer using machinery towel, Item 211, Appendix D, wet with dry-cleaning solvent, Item 124, Appendix D. Wipe dry.

2-5-3.1.4 Install Fitting Assembly Bearings and Spacer.

- a. Prior to installing bearings, perform breakout torque check, as follows:
 - (1) Install bolt in bearings with enough washers on bolt to seat nut. **INSTALL AND**

TIGHTEN NUT SO THAT WASHERS DO NOT ROTATE.

- (2) Rotate bearing inner race 4 complete turns.
 - (3) Using torque wrench, rotate bearing inner race. **BREAKOUT TORQUE MUST NOT BE GREATER THAN 4 INCH-OUNCES THROUGH FULL 360° TURN.**
 - (4) Remove bolt, washers, and nut.
- b. On fitting assembly, 70219-02105-042, install bearings and spacer as follows:
- (1) Apply grease, Item 151A, Appendix D, to outside diameter of bearings.

NOTE

Use correct sizing tools and arbor press so inner and outer races of bearing contact at the same time.

- (2) Using bushing installation kit and arbor press, press right bearing into fitting until bearing seats against shoulder of housing (Figure 2-5-3-1, Detail C).
- (3) Install spacer.



Bearing preload will result in premature bearing wear. Do not attempt to bottom left bearing outer race against housing shoulder.

- (4) Using bushing installation kit and arbor press, press left bearing into fitting until bearing inner race just touches spacer.
- (5) Perform breakout torque check, as follows:
 - (a) Install bolt through bearings with enough washers to seat nut. Install

nut on bolt. **TORQUE NUT TO 100 INCH-POUNDS.**

- (b) Rotate bearing inner races 4 complete turns.
 - (c) Using torque wrench, rotate bearing. **BREAKOUT TORQUE THROUGH FULL 360° TURN MUST NOT EXCEED 10 INCH-OUNCES.**
 - (d) Remove bolt, washers, and nut.
- (6) Using staking tools, pressure stake right bearing only.
- c. On fitting, 70219-02105-054, install bearings and spacer as follows (Figure 2-5-3-1, Detail C):
- (1) Measure outside diameter of bearing. **OUTSIDE DIAMETER MUST BE 0.8750-INCH +0.0000/-0.0004-INCH.** Replace bearing if limits are exceeded.
 - (2) Measure diameter of fitting bearing bores. **DIAMETER MUST BE 0.8750 - 0.8759-INCH.**
 - (3) If fitting bearing bore is greater than 0.8759-inch, replace fitting.
 - (4) If fitting bearing bore is less than 0.8750-inch line-ream bearing bores to 0.8750 - 0.8759-inch.
 - (5) Inspect bearing and fitting bearing bore for burrs. Remove burrs if necessary.
 - (6) Apply grease, Item 151A, Appendix D, to outside diameter of bearings.

NOTE

During bearing installation, if a press fit condition exists, do not attempt to install the bearing. The bearing must be a slip fit.

- (7) Install right bearing into fitting.

- (8) Install spacer.

NOTE

- During bearing installation, if a press fit condition exists, do not attempt to install the bearing. The bearing must be a slip fit.
- Right side of fitting can be identified by stake marks from previously installed bearing and dimension indicated in Figure 2-5-3-1, Detail C.

- (9) Install left bearing into fitting.

- (10) Perform breakout torque check, as follows:

- (a) Install bolt through bearings with enough washers to seat nut. Install nut on bolt. **TORQUE NUT TO 100 INCH-POUNDS.**

- (b) Rotate bearing inner races 4 complete turns.

- (c) Using torque wrench, rotate bearing. **BREAKOUT TORQUE THROUGH FULL 360° TURN MUST NOT EXCEED 10 INCH-OUNCES.**

- (d) Remove bolt, washers, and nut.

- (11) Using locally-made staking tools, pressure stake right bearing only.

- d On fitting, 70219-02127-041, install bearings and spacer as follows:

- (1) Prepare bonding surfaces of bearing and bearing bore as follows:

- (a) Mask bearing interior against contamination using 1-inch masking tape, Item 212, Appendix D.
- (b) Sand bonding surface of bearing and bearing bore lightly, using abrasive cloth, Item 6, Appendix D.

- (c) Clean off abrasive residue using cheesecloth, Item 90, Appendix D, moistened with trichloroethane 1,1,1, Item 341, Appendix D.

- (d) Remove masking tape from bearing.

- (2) Clean bonding surfaces of fitting and bearing using cheesecloth, Item 90, Appendix D, moistened with trichloroethane 1,1,1, Item 341, Appendix D.

- (3) Apply light coat of sealing primer, Item 305, Appendix D, to bearing outside diameter and inside diameter of fitting bores, using applicator, Item 78, Appendix D.

CAUTION

- Protect primed surfaces from contamination during drying.
- Do not touch primed surfaces with bare hands.
- (4) Let primer dry for 15 minutes minimum. If any break in primer coat occurs, repeat step c. (2) through step c. (4).
- (5) Install spacer in fitting bore. Make sure spacer is located approximately center of fitting bore.

CAUTION

If compound is removed from container, do not return unused portion to container. Do not dip applicator in container. Contamination of retaining compound will make it unusable.

- (6) Apply even coat of sealing compound, Item 302C, Appendix D, to outside diameter of bearings and inside diameter of fitting bores, using applicator, Item 78, Appendix D.



To avoid weak bond, bearing must be installed in fitting within 4 minutes of retaining compound application. If this time is exceeded, step c. (2) through step c. (6) must be repeated.

- (7) Immediately install bearings in fitting bores with slight twisting motion to distribute retaining compound.

NOTE

Adhesive sets in five minutes.

- (8) Prior to adhesive set, perform the following:
 - (a) Install bolt through bearings with enough washers to seat nut. Install

nut on bolt. **TORQUE NUT TO 100 INCH-POUNDS.**

- (b) Adjust position by centering bearings and spacer within fitting bore.
- (9) Allow fitting assembly to cure for 24 hours minimum.
- (10) Perform breakout torque check, as follows:
 - (a) Rotate bearing inner races 4 complete turns.
 - (b) Using torque wrench, rotate bearing. **BREAKOUT TORQUE THROUGH FULL 360° TURN MUST NOT EXCEED 10 INCH-OUNCES.**
 - (c) Remove bolt, washers, and nut.

2-5-4 CABIN VIBRATION ABSORBER FITTING BEARINGS (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
2-5-4.1	Replace Cabin Vibration Absorber Fitting Bearings (BENCH)	2-5-4-2

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Arbor Press
- Bushing Installation Kit
- Fluorescent Penetrant Inspection Kit
- Powertrain Repairer’s Toolkit
- Reamer Set
- Staking Tools, Locally-Made
- Torque Wrench, 0 - 16 in. oz
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Abrasive Cloth, Item 6, Appendix D
- Applicator, Item 78, Appendix D

- Cheesecloth, Item 90, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Grease, Item 151A, Appendix D
- Machinery Towel, Item 211, Appendix D
- Masking Tape, 1-inch, Item 212, Appendix D
- Sealing Compound, Item 302C, Appendix D
- Sealing Primer, Item 305, Appendix D
- Trichloroethane 1,1,1, Item 341, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 2-4-34
- PARA 2-6-5

2-5-4.1 Replace Cabin Vibration Absorber Fitting Bearings (BENCH).

2-5-4.1.1 Remove Fitting Assembly Bearing.

- a. Using bushing installation kit and arbor press, remove upper bearing from fittings, 70219-02116-042, 70219-02116-049, and 70219-02116-055 (Figure 2-5-4-1, Detail B).
- b. On fittings, 70219-02116-042 and 70219-02116-049, perform the following:
 - (1) Ream bearing bore 1.4385 - 1.4410-inch diameter.
 - (2) Blend out staking protrusions.
 - (3) Using fluorescent penetrant inspection kit, inspect reworked areas for cracks (ASTM E1417). **NONE ALLOWED.** If cracks are found, replace fitting (PARA 2-4-34).
- c. Clean fitting bore, using applicator, Item 78, Appendix D, moistened with trichloroethane 1,1,1, Item 341, Appendix D.
- d. Remove retaining compound residue, using abrasive cloth, Item 6, Appendix D.
- e. Remove abrasive residue, using cheesecloth, Item 90, Appendix D, moistened with trichloroethane 1,1,1, Item 341, Appendix D.

2-5-4.1.2 Install Fitting Assembly Bearing.

- a. Prior to installing upper bearing, perform breakout torque check, as follows:
 - (1) Install bolt through bearing. Install enough washers on bolt to seat nut. **INSTALL AND TIGHTEN NUT SO THAT WASHERS DO NOT ROTATE** (Figure 2-5-4-1, Detail B).
 - (2) Rotate bearing inner race 4 complete turns.
 - (3) Using torque wrench, rotate bearing inner race. **BREAKOUT TORQUE MUST**

NOT BE GREATER THAN 5 INCH-OUNCES THROUGH FULL 360° TURN.

- (4) Remove nut, washers, and bolt from bearing.
- b. Prepare bonding surfaces of bearing and bearing bore as follows:
 - (1) Mask bearing interior against contamination using 1-inch masking tape, Item 212, Appendix D.
 - (2) Lightly sand bonding surface of bearing in circumferential direction, using abrasive cloth, Item 6, Appendix D.
 - (3) Clean off abrasive residue using cheesecloth, Item 90, Appendix D, moistened with trichloroethane 1,1,1, Item 341, Appendix D.
 - (4) Remove masking tape from bearing.
- c. Clean bonding surfaces of fitting and bearing using cheesecloth, Item 90, Appendix D, moistened with trichloroethane 1,1,1, Item 341, Appendix D.
- d. Apply light coat of sealing primer, Item 305, Appendix D, to bearing outside diameter and inside diameter of fitting bore, using applicator, Item 78, Appendix D.

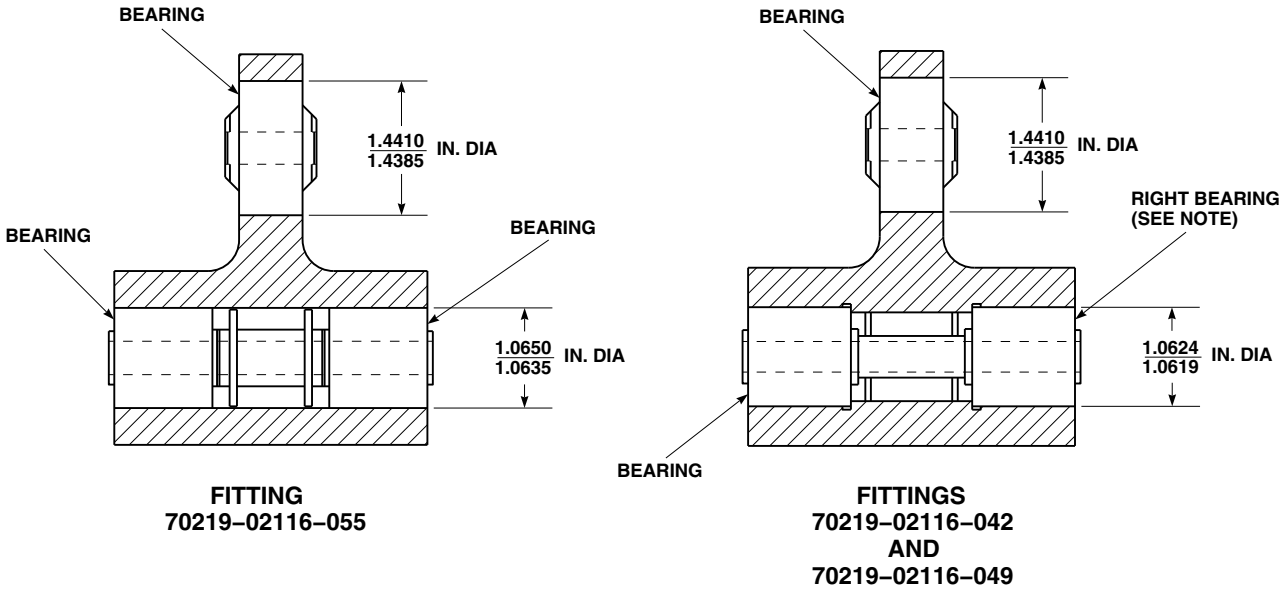
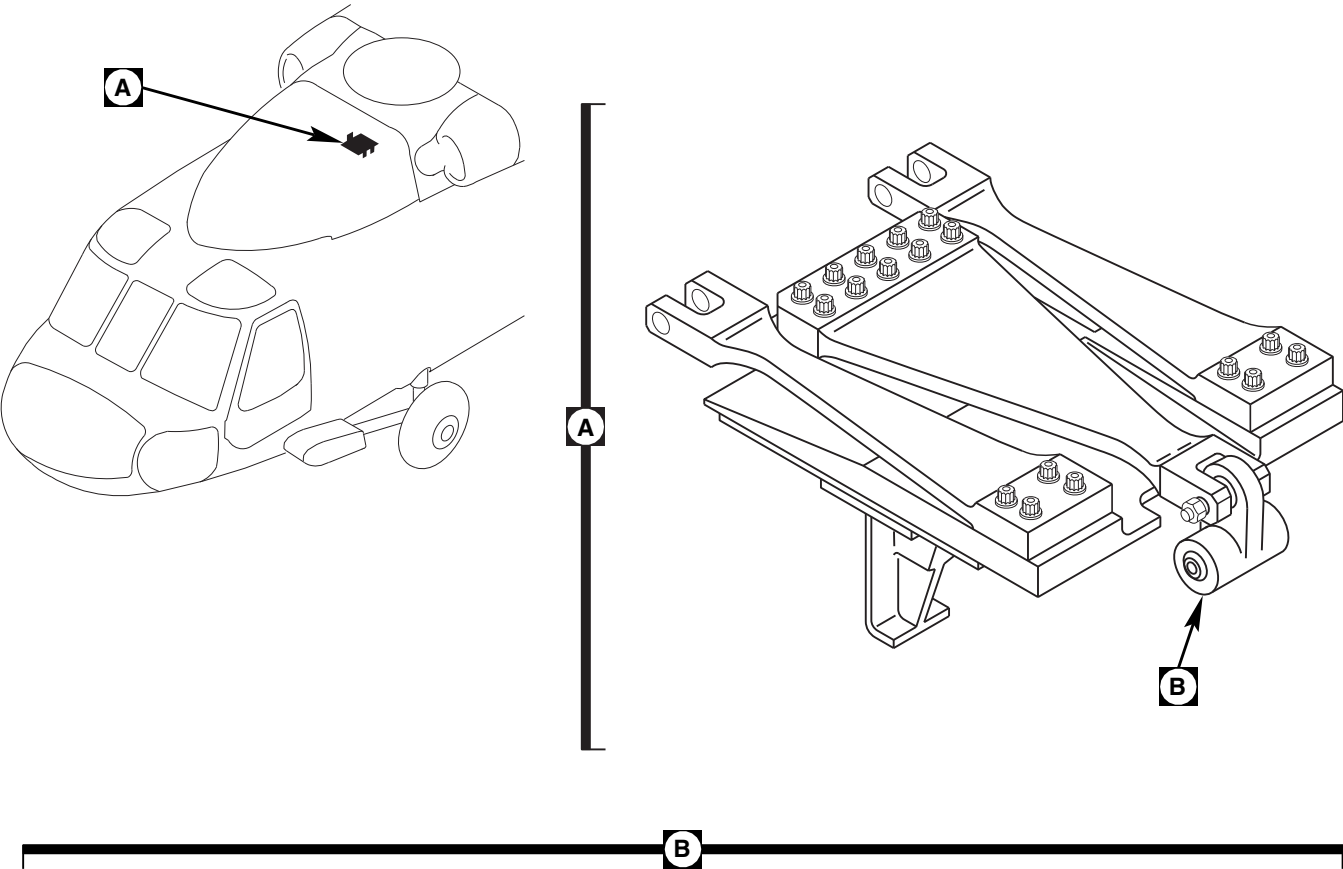
CAUTION

Make sure primed surfaces are protected from contamination during drying.

NOTE

Do not touch primed surfaces with bare hands.

- e. Allow sealing primer to dry for 15 minutes minimum. If any break in primer coat occurs, repeat step c. through step e.



NOTE
 ON FITTING 70219-02116-042 STAKE
 RIGHT BEARING ONLY.

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FIGURE 2-5-4-1 REPLACE CABIN VIBRATION ABSORBER FITTING BEARINGS (BENCH)

CAUTION

If compound is removed from container, do not return unused portion to container. Do not dip applicator in container. Contamination of retaining compound will make it unusable.

- f. Apply even coat of sealing compound, Item 302C, Appendix D, to outside diameter of bearing and inside diameter of fitting bore, using applicator, Item 78, Appendix D.

CAUTION

To avoid weak bond, bearing must be installed in fitting within 4 minutes of retaining compound application. If this time is exceeded, step c. through step f. must be repeated.

- g. Immediately install bearing in fitting with slight twisting motion to distribute retaining compound. Bearing must be centered in fitting bore within 0.020-inch. Observe squeeze out at bond line and wipe off excess immediately.
- h. Allow fitting assembly to cure for 24 hours minimum.
- i. On reworked fittings, 70219-02116-042 and 70219-02116-049, perform the following:
 - (1) Apply corrosion resistant coating, Item 118, Appendix D, to any exposed reworked surfaces.
 - (2) Touch up paint as required (PARA 2-6-5).

2-5-4.1.3 Remove Fitting Assembly Bearings and Spacer.

- a. Using bushing installation kit and arbor press, remove lower bearings and spacer from fittings, 70219-02116-042, 70219-02116-049, and 70219-02116-055 (Figure 2-5-4-1, Detail B).

- b. On fittings, 70219-02116-042 and 70219-02116-049, clean fitting bores and spacer using machinery towel, Item 211, Appendix D, wet with dry-cleaning solvent, Item 124, Appendix D. Wipe dry.
- c. On fitting, 70219-02116-055, clean fitting bores and spacer, as follows:
 - (1) Clean fitting bores, using applicator, Item 78, Appendix D, moistened with trichloroethane 1,1,1, Item 341, Appendix D.
 - (2) Remove retaining compound residue from fitting bores, using abrasive cloth, Item 6, Appendix D.
 - (3) Remove abrasive residue from fitting bores using cheesecloth, Item 90, Appendix D, moistened with trichloroethane 1,1,1, Item 341, Appendix D.
 - (4) Clean spacer using machinery towel, Item 211, Appendix D, wet with dry-cleaning solvent, Item 124, Appendix D. Wipe dry.

2-5-4.1.4 Install Fitting Assembly Bearings and Spacer.

- a. Prior to installing bearings, perform breakout torque check, as follows:
 - (1) Install bolt in bearings with enough washers on bolt to seat nut. **INSTALL AND TIGHTEN NUT SO THAT WASHERS DO NOT ROTATE.**
 - (2) Rotate bearing inner race 4 complete turns.
 - (3) Using torque wrench, rotate bearing inner race. **BREAKOUT TORQUE MUST NOT BE GREATER THAN 4 INCH-OUNCES THROUGH FULL 360° TURN.**
 - (4) Remove bolt, washers, and nut.

NOTE

On modified fitting, 70219-02116-049, staking is not required.

- b. On fittings, 70219-02116-042 and 70219-02116-049, install bearings and spacer, as follows (Figure 2-5-4-1, Detail B):
 - (1) Apply grease, Item 151A, Appendix D, to outside diameter of bearings.

NOTE

- Use correct sizing tools and arbor press so inner and outer races of bearing contact at the same time.
 - Right side of fitting can be identified by dimension indicated in Figure 2-5-4-1, Detail B.
- (2) Using bushing installation kit and arbor press, press right bearing into fittings, 70219-02116-042 and 70219-02116-049, until bearing seats against shoulder of housing.
 - (3) Install spacer.



Bearing preload will result in premature bearing wear. Do not attempt to bottom left bearing outer race against housing shoulder.

- (4) On fitting assembly, 70219-02116-042, use bushing installation kit and arbor press, press left bearing into fitting until bearing inner race just touches spacer. Use inspection hole to verify.

NOTE

During bearing installation, if a press fit condition exists, do not attempt to install the bearing. The bearing must be a slip fit.

- (5) On fitting, 70219-02116-049, install left bearing into fitting.
- (6) Perform breakout torque check, as follows:
 - (a) Install bolt through bearings with enough washers to seat nut. Install nut on bolt. **TORQUE NUT TO 100 INCH-POUNDS.**
 - (b) Rotate bearing inner races 4 complete turns.
 - (c) Using torque wrench, rotate bearing. **BREAKOUT TORQUE THROUGH FULL 360° TURN MUST NOT EXCEED 10 INCH-OUNCES.**
 - (d) Remove bolt, washers, and nut.

- (7) On fitting, 70219-02116-042, using locally made staking tools, pressure stake right bearing only.
- c. On fitting, 70219-02116-055, install bearings and spacer, as follows:
 - (1) Prepare bonding surfaces of bearing and bearing bore as follows:
 - (a) Mask bearing interior against contamination using 1-inch masking tape, Item 212, Appendix D.
 - (b) Lightly sand bonding surface of bearing in circumferential direction, using abrasive cloth, Item 6, Appendix D.
 - (c) Clean off abrasive residue using cheesecloth, Item 90, Appendix D, moistened with trichloroethane 1,1,1, Item 341, Appendix D.
 - (d) Remove masking tape from bearing.
 - (2) Clean bonding surfaces of fitting and bearing using cheesecloth, Item 90, Appendix D, moistened with trichloroethane 1,1,1, Item 341, Appendix D.

- (3) Apply light coat of sealing primer, Item 305, Appendix D, to bearing outside diameter and inside diameter of fitting bores, using applicator, Item 78, Appendix D.

CAUTION

- Protect primed surfaces from contamination during drying.
 - Do not touch primed surfaces with bare hands.
- (4) Let primer dry for 15 minutes minimum. If any break in primer coat occurs, repeat step c. (2) through step c. (4).
- (5) Install spacer in fitting bore. Make sure spacer is located approximately center of fitting bore.

CAUTION

If compound is removed from container, do not return unused portion to container. Do not dip applicator in container. Contamination of retaining compound will make it unusable.

- (6) Apply even coat of sealing compound, Item 302C, Appendix D, to outside diameter of bearings and inside diameter of fitting bores, using applicator, Item 78, Appendix D.

CAUTION

To avoid weak bond, bearing must be installed in fitting within 4 minutes of retaining compound application. If this time is exceeded, step c. (2) through step c. (6) must be repeated.

- (7) Immediately install bearings in fitting bores with slight twisting motion to distribute retaining compound.

NOTE

Adhesive sets in five minutes.

- (8) Prior to adhesive set, perform the following:
- (a) Install bolt through bearings with enough washers to seat nut. Install nut on bolt. **TORQUE NUT TO 100 INCH-POUNDS.**
 - (b) Adjust position by centering bearings and spacer within fitting bore.
- (9) Allow fitting assembly to cure for 24 hours minimum.
- (10) Perform breakout torque check, as follows:
- (a) Rotate bearing inner races 4 complete turns.
 - (b) Using torque wrench, rotate bearing. **BREAKOUT TORQUE THROUGH FULL 360° TURN MUST NOT EXCEED 10 INCH-OUNCES.**
 - (c) Remove bolt, washers, and nut.

2-5-5 CABIN VIBRATION ABSORBER SPRING BUSHINGS (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
2-5-5.1	Replace Cabin Vibration Absorber Spring Bushings (BENCH)	2-5-5-2

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Bushing Installation Kit
- Powertrain Repairer’s Toolkit
- Protection, Skin and Breathing

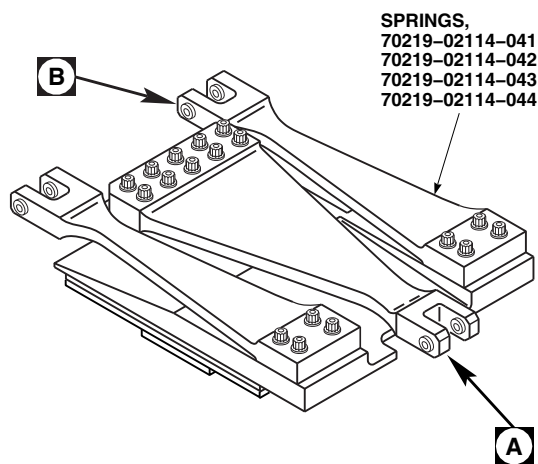
Materials/Parts

- Epoxy Primer Coating Kit, Item 135, Appendix D
- Machinery Towel, Item 211, Appendix D

- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 273, Appendix D

References

- Appendix D
- PARA 2-4-32



NOTES

1. ALL DIMENSIONS ± 0.010 INCH EXCEPT AS NOTED.
2. DIMENSIONS DO NOT APPLY TO 70219-02114-043 AND 70219-02114-044 (WITH SLIP FIT SLEEVE BUSHINGS).

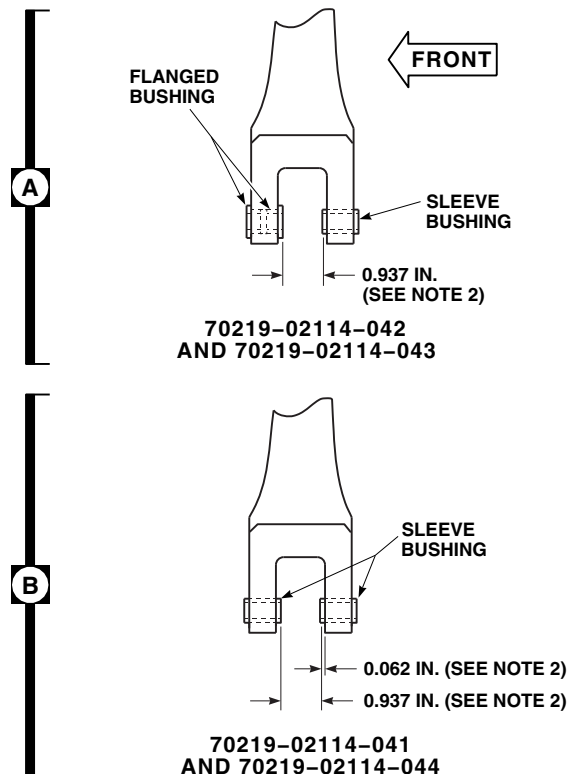
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FIGURE 2-5-5-1. REPLACE CABIN VIBRATION ABSORBER SPRING BUSHINGS (BENCH)

2-5-5.1 Replace Cabin Vibration Absorber Spring Bushings (BENCH).

2-5-5.1.1. Remove.

WARNING

- Bushing, 70219-02150-103, is made of beryllium copper metal, which can be a health hazard. Avoid breathing dust while reworking damaged areas. Use a dust respirator if necessary. Grinding is prohibited. Keep beryllium copper out of open wounds. Use skin protection if necessary. Clean up visible dust piles and wash hands and arms after maintenance.
- No minor repair or corrosion removal allowed on beryllium copper components.

Remove bushings using extractor from bushing installation kit (Figure 2-5-5-1, Detail A and Detail B).

2-5-5.1.2. Install.

- a. Wipe replacement bushings and spring bushing holes clean using machinery towel, Item 211, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D. Wipe dry.

NOTE

Use sealing compound, Item 273, Appendix D, on all fitting assemblies with shoulder bushings. Coat bushings and spring bushing holes.

- b. If required, using epoxy primer coating kit, Item 135, Appendix D, wet bushing and spring bushing holes.

NOTE

On springs, 70219-02114-043 and 70219-02114-044, loose sleeve bushings are installed at time of vibration absorber installation (PARA 2-4-32).

- c. On spring, 70219-02114-041, press in front sleeve bushing to protrude 0.062-inch inward from yoke ear (Figure 2-5-5-1, Detail B).

- d. On springs, 70219-02114-042 and 70219-02114-043, press flanged bushings into front yoke ear (Figure 2-5-5-1, Detail A).
- e. Press sleeve bushing into opposite ear to dimensions shown in (Figure 2-5-5-1).
- f. Leave squeezed out primer in place.

2-5-6 VIBRATION ABSORBER AIRFRAME NOSE FITTING BUSHINGS (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
2-5-6.1	Replace Vibration Absorber Airframe Nose Fitting Bushings (BENCH)	2-5-6-1

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Arbor Press
- Bushing Installation Kit
- Powertrain Repairer’s Toolkit
- Protection, Skin, and Breathing

Materials/Parts

- Dry-Cleaning Solvent, Item 124, Appendix D
- Grease, Item 145, Appendix D
- Machinery Towel, Item 211, Appendix D
- Sealing Compound, Item 273, Appendix D

References

- Appendix D

2-5-6.1 Replace Vibration Absorber Airframe Nose Fitting Bushings (BENCH).

2-5-6.1.1 Remove.

WARNING

- Bushings, 70219-02150-102 and 70219-02150-104, are made of beryllium copper metal, which can be a health hazard. Avoid breathing dust while reworking damaged areas. Use a dust respirator if necessary. Grinding is prohibited. Keep beryllium copper out of open wounds. Use skin protection if necessary. Clean

up visible dust piles and wash hands and arms after maintenance.

- No minor repair or corrosion removal allowed on beryllium copper components.

Using extractor from bushing installation kit, remove bushings from fitting (Figure 2-5-6-1, Sheet 1 and Sheet 2). Discard bushings.

2-5-6.1.2 Install.

- Wipe new bushings and fitting holes clean using machinery towel, Item 211, Appendix D, wet with dry-cleaning solvent, Item 124, Appendix D. Wipe dry.

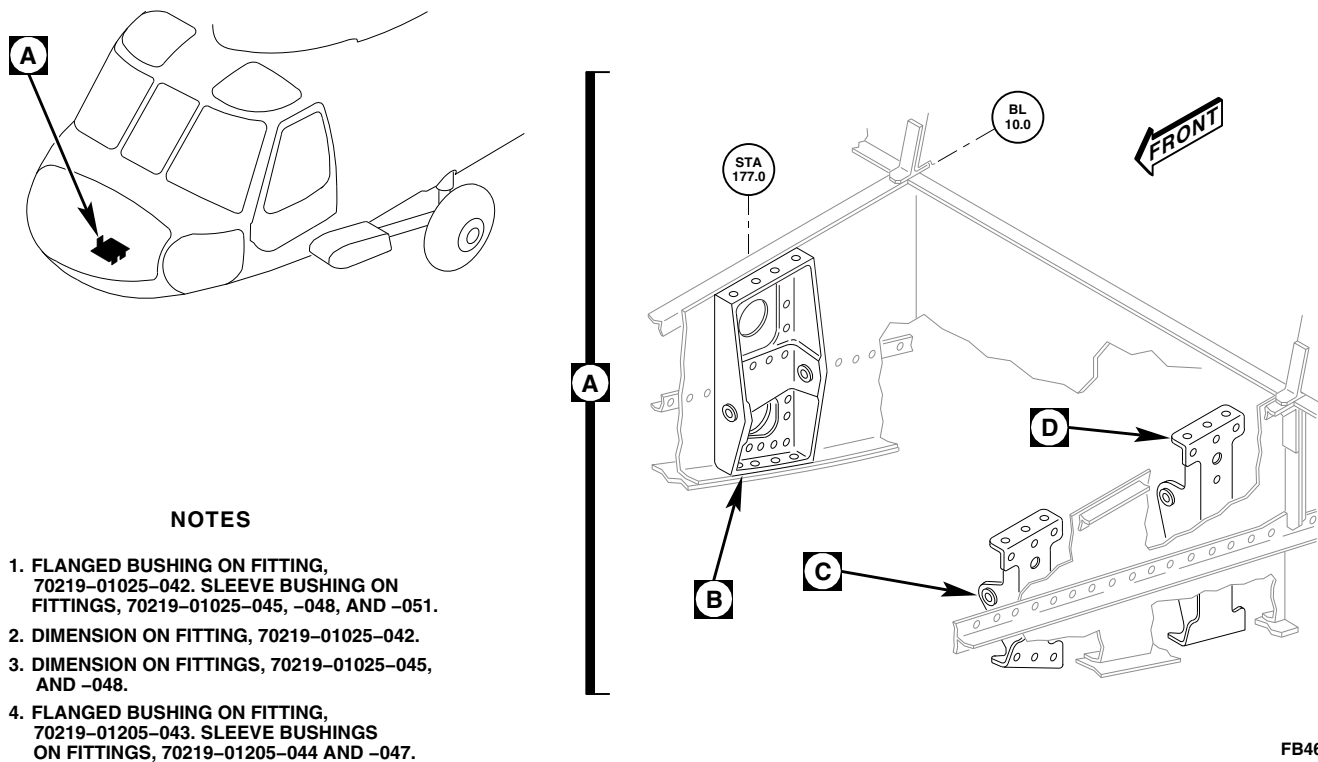


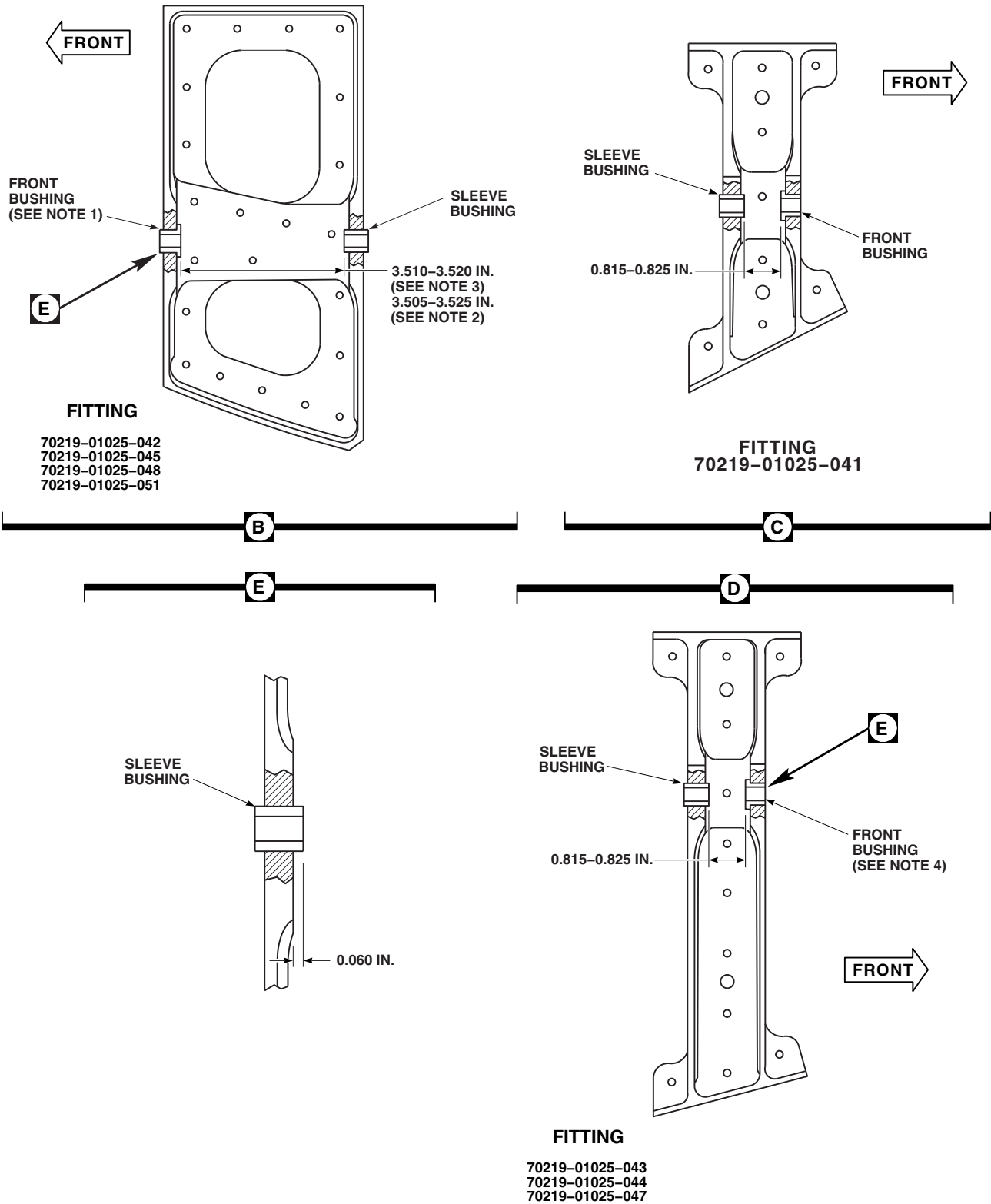
FIGURE 2-5-6-1. REPLACE VIBRATION ABSORBER AIRFRAME NOSE FITTING BUSHINGS (BENCH) (SHEET 1 OF 2)

NOTE

70785 - SUBQ Do not use sealing compound on bushings, 70319-02150-102 and 70219-02150-104. ■

- 70785 - SUBQ** Install all slip-fit sleeve bushings, 70219-02150-102 and 70219-02150-104, with outer surface coated with grease, Item 145, Appendix D. ■
- Apply sealing compound, Item 273, Appendix D, on all flanged bushings and fitting holes.

- Using bushing installation kit and arbor press, press flanged bushing into front ear with flange inside (Figure 2-5-6-1, Sheet 2, Detail B, Detail C, and Detail D).
- Using bushing installation kit and arbor press, on fittings with two sleeve bushings, press front bushing in fitting to proper dimension (Figure 2-5-6-1, Sheet 2, Detail B and Detail E).
- Using bushing installation kit and arbor press, press plain bushing in opposite fitting ear using sizing block.



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FIGURE 2-5-6-1. REPLACE VIBRATION ABSORBER AIRFRAME NOSE FITTING BUSHINGS (BENCH) (SHEET 2 OF 2)

2-5-7 VIBRATION ABSORBER AIRFRAME CABIN FITTING BUSHINGS (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
2-5-7.1	Replace Vibration Absorber Airframe Cabin Fitting Bushings (BENCH)	2-5-7-1

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Arbor Press
- Bushing Installation Kit
- Powertrain Repairer’s Toolkit
- Protection, Skin and Breathing
- Reamer Set

Materials/Parts

- Corrosion Preventive Compound, Item 112, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Grease, Item 150, Appendix D
- Machinery Towel, Item 211, Appendix D

References

- Appendix D

2-5-7.1 Replace Vibration Absorber Airframe Cabin Fitting Bushings (BENCH).

2-5-7.1.1. Remove.

WARNING

- Bushing, 70219-02150-101, is made of beryllium copper metal, which can be a health hazard. Avoid breathing dust while reworking damaged areas. Use a dust respirator if necessary. Grinding is prohibited. Keep beryllium copper out of open wounds. Use skin protection if necessary. Clean up visible dust piles

and wash hands and arms after maintenance.

- No minor repair or corrosion removal allowed on beryllium copper components.

Remove bushings using extractor from bushing installation kit (Figure 2-5-7-1, Sheet 1 and Sheet 2). Discard old bushings.

2-5-7.1.2. Inspect.

- Measure diameter of fitting bushing bores. **DIAMETER MUST BE 0.6241 TO 0.6251-INCH.** Replace any fitting that exceeds 0.6251-inch.

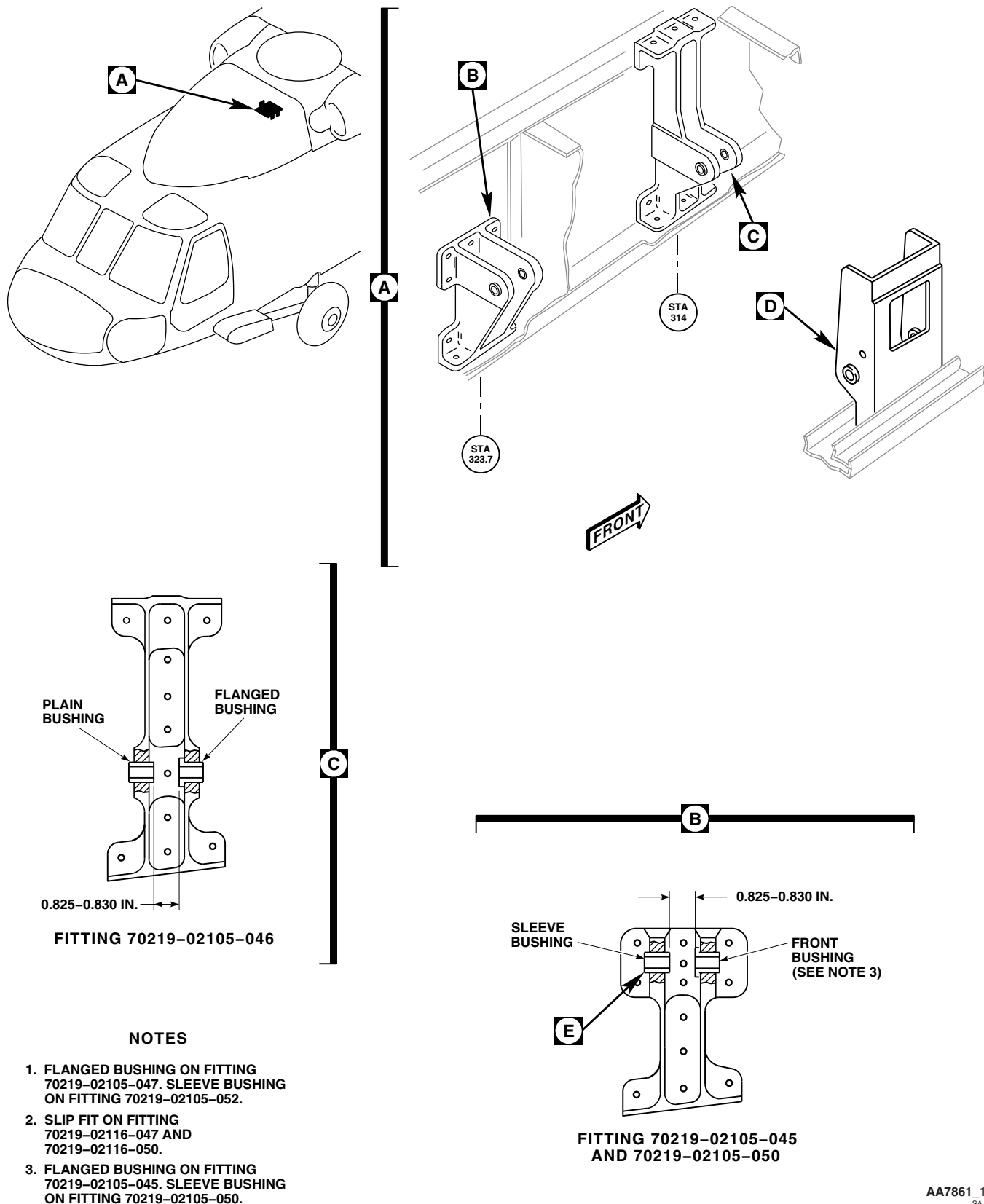
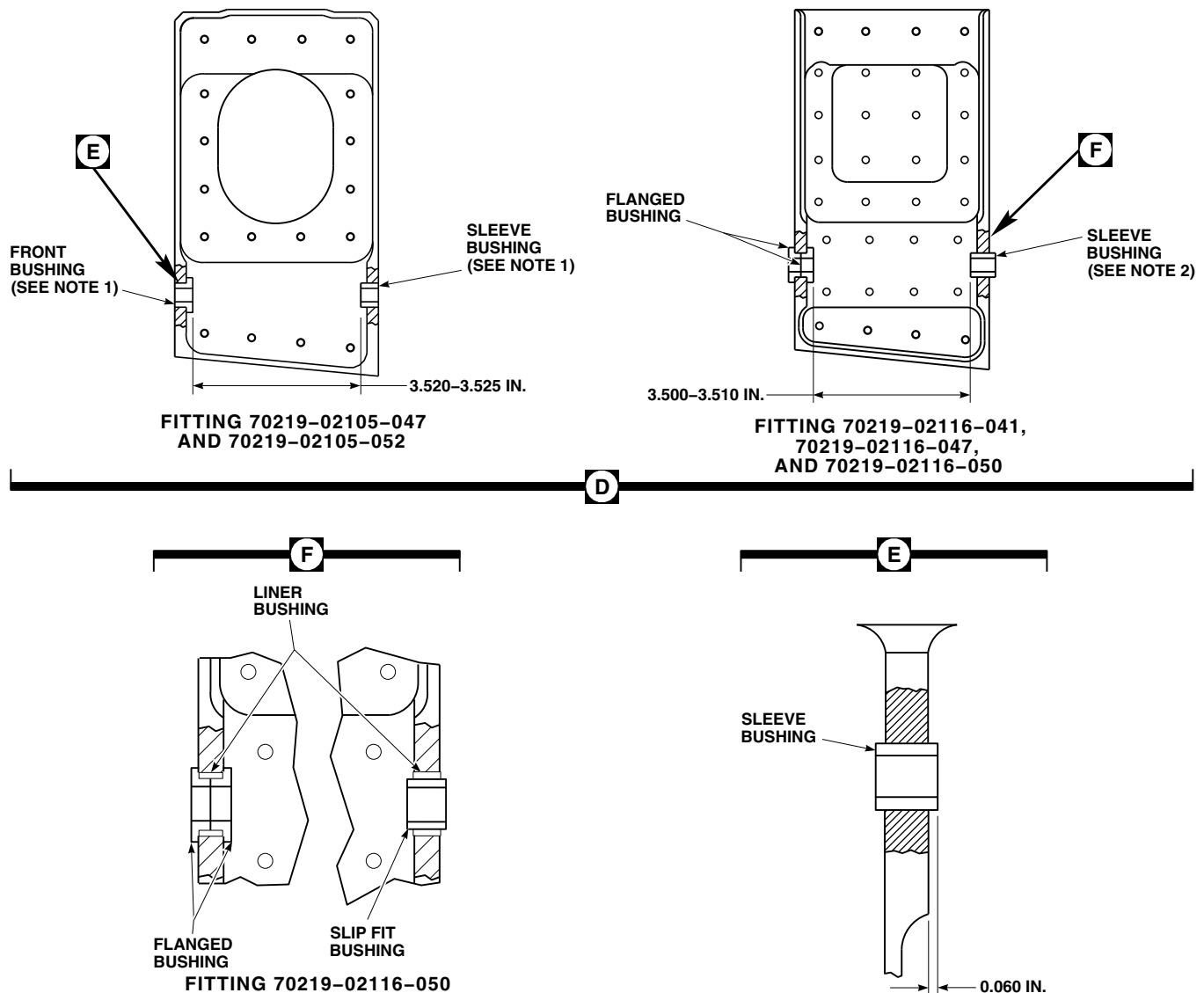
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FIGURE 2-5-7-1. REPLACE VIBRATION ABSORBER AIRFRAME CABIN FITTING BUSHINGS (BENCH) (SHEET 1 OF 2)

- b. If fitting is less than 0.6241-inch, line-ream to 0.6241 to 0.6251-inch.

2-5-7.1.3. Install.

- a. Wipe new bushings and fitting holes clean using machinery towel, Item 211, Appendix D, wet with dry-cleaning solvent, Item 124, Appendix D. Wipe dry.
- b. Wipe bushing outer surfaces and fitting holes with corrosion preventive compound, Item 112, Appendix D.
- c. **70785 – SUBQ** Install all slip-fit sleeve bushings, 70219-02150-101, with outer surface coated with grease, Item 150, Appendix D. 
- d. On fittings, 70219-02105-045, 70219-02105-046, and 70219-02105-047, press flanged bushings into fittings with flange inside, using bushing installation kit and arbor press (Figure 2-5-7-1, Sheet 1 and Sheet 2).
- e. On fittings with two sleeve bushings, and using bushing installation kit and arbor press, press front bushing in fitting to dimension shown in (Figure 2-5-7-1, Sheet 1).
- f. Using bushing installation kit and arbor press, press plain bushing into opposite fitting ear to dimension shown in (Figure 2-5-7-1, Sheet 2).



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FIGURE 2-5-7-1. REPLACE VIBRATION ABSORBER AIRFRAME CABIN FITTING BUSHINGS (BENCH) (SHEET 2 OF 2)

2-5-8 VIBRATION ABSORBER AIRFRAME CABIN FITTING BEARINGS (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
2-5-8.1	Replace Vibration Absorber Airframe Cabin Fitting Bearings (BENCH)	2-5-8-2

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Anvil
- Arbor Press
- Drill Press
- Drill Set
- Fluorescent Penetrant Inspection Kit
- Fly Cutter
- Powertrain Repairer’s Toolkit
- Reamer Set
- Staking Tools, Locally-Made
- Torque Wrench, 0 - 16 in. oz

Materials/Parts

- Abrasive Cloth, Item 6, Appendix D
- Applicator, Item 78, Appendix D
- Cheesecloth, Item 90, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D
- Masking Tape, 1-inch, Item 212, Appendix D
- Sealing Compound, Item 302C, Appendix D
- Sealing Primer, Item 305, Appendix D
- Trichloroethane 1,1,1, Item 341, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 2-6-5

2-5-8.1 Replace Vibration Absorber Airframe Cabin Fitting Bearings (BENCH).

2-5-8.1.1 Remove.

- a. Drill out rivets and remove fitting from structure (Figure 2-5-8-1, Detail A).

NOTE

On modified fittings, 70219-02116-045 and 70219-02116-046, removal of staking is not required. Proceed to step c.

- b. On fittings, 70219-02116-045 and 70219-02116-046, remove staking using fly cutter and drill press.
- c. Press bearing out of fitting, using arbor press and properly sized arbor and anvil.

NOTE

On modified fittings, 70219-02116-045 and 70219-02116-046, blending of staking protrusions is not required. Proceed to step d. (3) after reaming (Figure 2-5-8-1, Detail B and Detail C).

- d. On fittings, 70219-02116-045 and 70219-02116-046, perform the following:
 - (1) Ream bearing bore 1.4385 - 1.4410-inch diameter.
 - (2) Blend out staking protrusions.
 - (3) Using fluorescent penetrant inspection kit, inspect reworked areas for cracks (ASTM E1417). **NONE ALLOWED.**
- e. Clean fitting bore, using applicator, Item 78, Appendix D, moistened with trichloroethane 1,1,1, Item 341, Appendix D.
- f. Remove retaining compound residue, using abrasive cloth, Item 6, Appendix D.
- g. Remove abrasive residue, using cheesecloth, Item 90, Appendix D, moistened with trichloroethane 1,1,1, Item 341, Appendix D.

2-5-8.1.2 Install.

- a. Prior to installing upper bearing, perform breakout torque check, as follows:

- (1) Install bolt through bearing. Install enough washers on bolt to seat nut. **INSTALL AND TIGHTEN NUT SO WASHERS DO NOT ROTATE.**

- (2) Rotate bearing inner race 4 complete turns.

- (3) Using torque wrench rotate bearing inner race. **BREAKOUT TORQUE MUST NOT BE GREATER THAN 5 INCH-OUNCES THROUGH FULL 360° TURN.**

- (4) Remove nut, washers, and bolt from bearing.

- b. Prepare bonding surfaces of bearing and bearing bore as follows:

- (1) Mask bearing interior against contamination using 1-inch masking tape, Item 212, Appendix D.

- (2) Lightly sand bonding surface of bearing in circumferential direction, using abrasive cloth, Item 6, Appendix D.

- (3) Clean off abrasive residue using cheesecloth, Item 90, Appendix D, moistened with trichloroethane 1,1,1, Item 341, Appendix D.

- (4) Remove masking tape from bearing.

- c. Clean bonding surfaces of fitting and bearing using cheesecloth, Item 90, Appendix D, moistened with trichloroethane 1,1,1, Item 341, Appendix D.

- d. Apply light coat of sealing primer, Item 305, Appendix D, to bearing outside diameter and inside diameter of fitting bore, using applicator, Item 78, Appendix D.

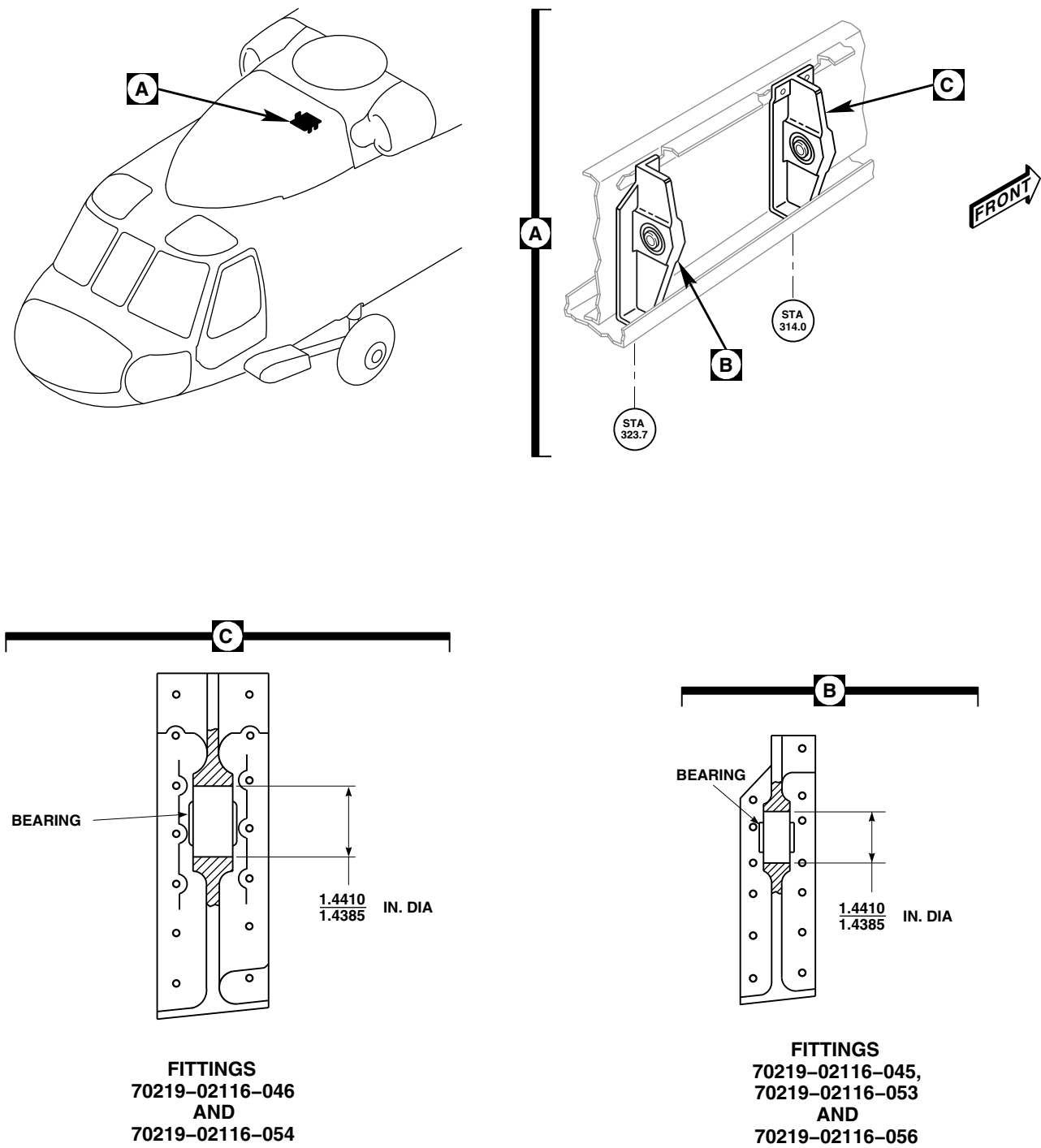


FIGURE 2-5-8-1. REPLACE VIBRATION ABSORBER AIRFRAME CABIN FITTING BEARINGS (BENCH)

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CAUTION

- Protect primed surfaces from contamination during drying.
 - Do not touch primed surfaces with bare hands.
- e. Let primer dry for 15 minutes minimum. If any break in primer coat occurs, repeat step c. through step e.

CAUTION

If retaining compound is removed from container, do not return unused portion to container. Do not dip applicator in container. Contamination of retaining compound will make it unusable.

- f. Apply even coat of sealing compound, Item 302C, Appendix D, to outside diameter of bearing and inside diameter of fitting bore, using applicator, Item 78, Appendix D.

CAUTION

To avoid weak bond, bearing must be installed in fitting within 4 minutes of retaining compound application. If time is exceeded, step c. through step f. must be repeated.

- g. Immediately install bearing in fitting with slight twisting motion to distribute retaining compound. Bearing must be centered in fitting bore within 0.020-inch. Observe squeeze out at bond line and wipe off excess immediately (Figure 2-5-8-1, Detail B and Detail C).
- h. Allow fitting assembly to cure for 24 hours minimum.
- i. On reworked fitting assemblies, 70219-02116-045 and 70219-02116-046, perform the following:
- (1) Apply corrosion resistant coating, Item 118, Appendix D, to any exposed reworked surfaces.
 - (2) Touch up paint as required (PARA 2-6-5).

2-5-9 MAIN LANDING GEAR FUSELAGE FITTING SPHERICAL BEARING (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
2-5-9.1	Replace Main Landing Gear Fuselage Fitting Spherical Bearing (BENCH)	2-5-9-2

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Anvil
- Arbor Press
- Clamp
- Drill Press
- Driver
- Fluorescent Penetrant Inspection Kit
- Fly Cutter
- Nonmetallic Scraper, Locally-Made
- Powertrain Repairer’s Toolkit
- Spinning Tool

Materials/Parts

- Abrasive Cloth, Item 10, Appendix D
- Cheesecloth, Item 90, Appendix D
- Corrosion Preventive Compound, Item 113, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Petrolatum, Item 234, Appendix D
- Primer, Item 257, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 2-6-5

2-5-9.1 Replace Main Landing Gear Fuselage Fitting Spherical Bearing (BENCH).

2-5-9.1.1 Remove.

CAUTION

Damage to fitting chamfer will occur if staked lip of bearing is improperly removed. Be careful not to damage fitting chamfer when machining off staked lip of bearing.

- a. Clamp fitting assembly to baseplate of drill press. Using fly cutter, machine off part of bearing outer race swaged over chamfer on fitting.
- b. Press out bearing using arbor press, anvil, and driver combination.

2-5-9.1.2 Repair.

- a. Clean primer residue from bore and chamfer using cheesecloth, Item 90, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D, and nonmetallic scraper.
- b. Check bearing bore lug for galling or scoring. **NO MORE THAN 0.010-INCH IF 25% OF CONTACT AREA IS AFFECTED.**
- c. Blend out galling or scoring in bearing bore lug using abrasive cloth, Item 10, Appendix D.
- d. Using fluorescent penetrant inspection kit, perform fluorescent penetrant inspection of bearing bore lug (ASTM E1417).

2-5-9.1.3 Install.

- a. Make sure fitting bore, chamfers, and adjacent surfaces are clean and dry.
- b. Apply a light coat of primer, Item 257, Appendix D, to inside of fitting bore and outer race of bearing.

- c. Press bearing into fitting using arbor press, anvil, and driver combination.
- d. Clean primer, if any, from bearing V-grooves using cheesecloth, Item 90, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D, and wipe dry.
- e. Fix fitting with bearing on drill press base with proper spinning tool and anvil.

CAUTION

Do not allow petrolatum on ball bearing, or on exposed edge of the fabric liner. If petrolatum contacts these surfaces, remove petrolatum immediately.

- f. Lubricate V-grooves of bearing and spinning tool with petrolatum, Item 234, Appendix D.

CAUTION

Do not allow spinning tool to make contact with ball side of groove as binding may result.

- g. Swage V-groove lip over chamfer in fitting bore.
- h. Turn fitting over and repeat swaging process on other side.
- i. Clean petrolatum from V-groove using cheesecloth, Item 90, Appendix D, wet with dry-cleaning solvent, Item 124, Appendix D.
- j. Touch up bearing surface with corrosion preventive compound, Item 113, Appendix D.
- k. Touch up paint as required (PARA 2-6-5).

2-5-10 CABIN DOOR WINDOW (BENCH).

PARAGRAPH BREAKDOWN

	PAGE
2-5-10.1 Replace Cabin Door Window (BENCH)	2-5-10-1

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Drill Set
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Abrasive Cloth, Item 9, Appendix D
- Adhesive, Item 30, Appendix D

- Cheesecloth, Item 90, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Sealing Compound, Item 274, Appendix D

References

- Appendix D
- PARA 2-6-5

2-5-10.1 Replace Cabin Door Window (BENCH).

NOTE

Replacement of all cabin door windows is the same.

not allow methyl ethyl ketone to contact plexiglass window.

2-5-10.1.1 Remove.

- If installed, using nonmetallic scraper, remove seal from window (Figure 2-5-10-1, Detail A).
- Drill out countersunk rivets, remove strap and using nonmetallic scraper, carefully work plexiglass window loose from frame.

CAUTION

Damage to plexiglass window will result if methyl ethyl ketone contacts it. Do

- Using nonmetallic scraper and cheesecloth, Item 90, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D, remove sealing compound and adhesive from frame. Wipe dry with clean cheesecloth.

2-5-10.1.2 Install.

- Position strap over replacement window (Figure 2-5-10-1, Detail A).
- Using strap as template, drill enough holes in plexiglass window to position plexiglass window and strap on frame.
- Apply sealing compound, Item 274, Appendix D, to area on frame where window will be installed.

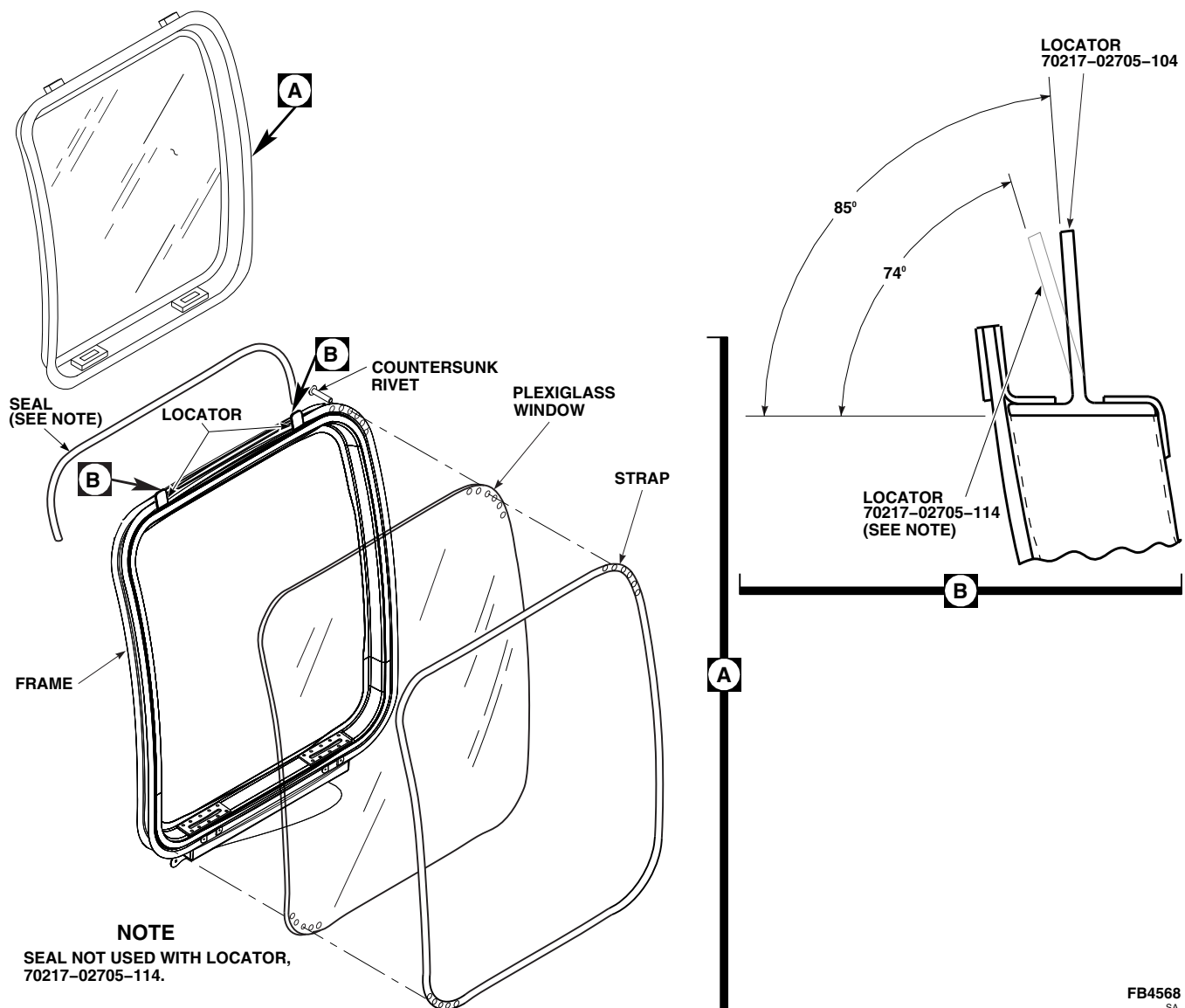


FIGURE 2-5-10-1. REPLACE CABIN DOOR WINDOW (BENCH)

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- d. Install strap and plexiglass window on frame and secure with countersunk rivets.

NOTE

Seal is used on window with locator, 70217-02705-104 (bend angle 85°), installed (Figure 2-5-10-1, Detail B). Do not install seal on window with locator, 70217-02705-114 (bend angle 74°), installed.

- e. If locator, 70217-02705-104, is installed on window, perform the following:

- (1) Using abrasive cloth, Item 9, Appendix D, roughen surface of new seal.
- (2) Bond new seal in place with adhesive, Item 30, Appendix D.

- f. Touch up paint, as required (PARA 2-6-5).

2-5-11 CABIN DOOR WINDOW FRAME (BENCH).

PARAGRAPH BREAKDOWN

	PAGE
2-5-11.1 Repair Cabin Door Window Frame (BENCH)	2-5-11-1

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Drill Set

References

- PARA 2-6-2
- PARA 2-6-5
- PARA 2-6-26

2-5-11.1 Repair Cabin Door Window Frame (BENCH).

NOTE

Repair of all cabin door window frames is the same, except as noted.

2-5-11.1.1 General Repair.

NOTE

Strap bonded to inside of frame opposite window may be cut in four places to obtain required fit on assembly.

Evaluate damage and repair (PARA 2-6-26 and PARA 2-6-2).

2-5-11.1.2 Frame Repair.

- Carefully drill out rivets and remove locators from frame (Figure 2-5-11-1, Detail A).
- Carefully drill out rivets and remove cams and doublers from frame.

NOTE

Upper and lower straps are used only on frame, 70217-02705-043.

- On frame, 70217-02705-043, carefully drill out rivets and remove upper and lower straps from frame.

NOTE

Use blind rivets in locations noted in Figure 2-5-11-1, Detail B.

- Install locators in frame and secure with rivets (Figure 2-5-11-1, Detail A).
- Install doublers and cams on bottom of frame and secure with rivets.

NOTE

Upper and lower straps are installed only on frame, 70217-02705-043.

- On frame, 70217-02705-043, install upper and lower straps on corners of frame and secure with rivets.
- Touch up paint (PARA 2-6-5).

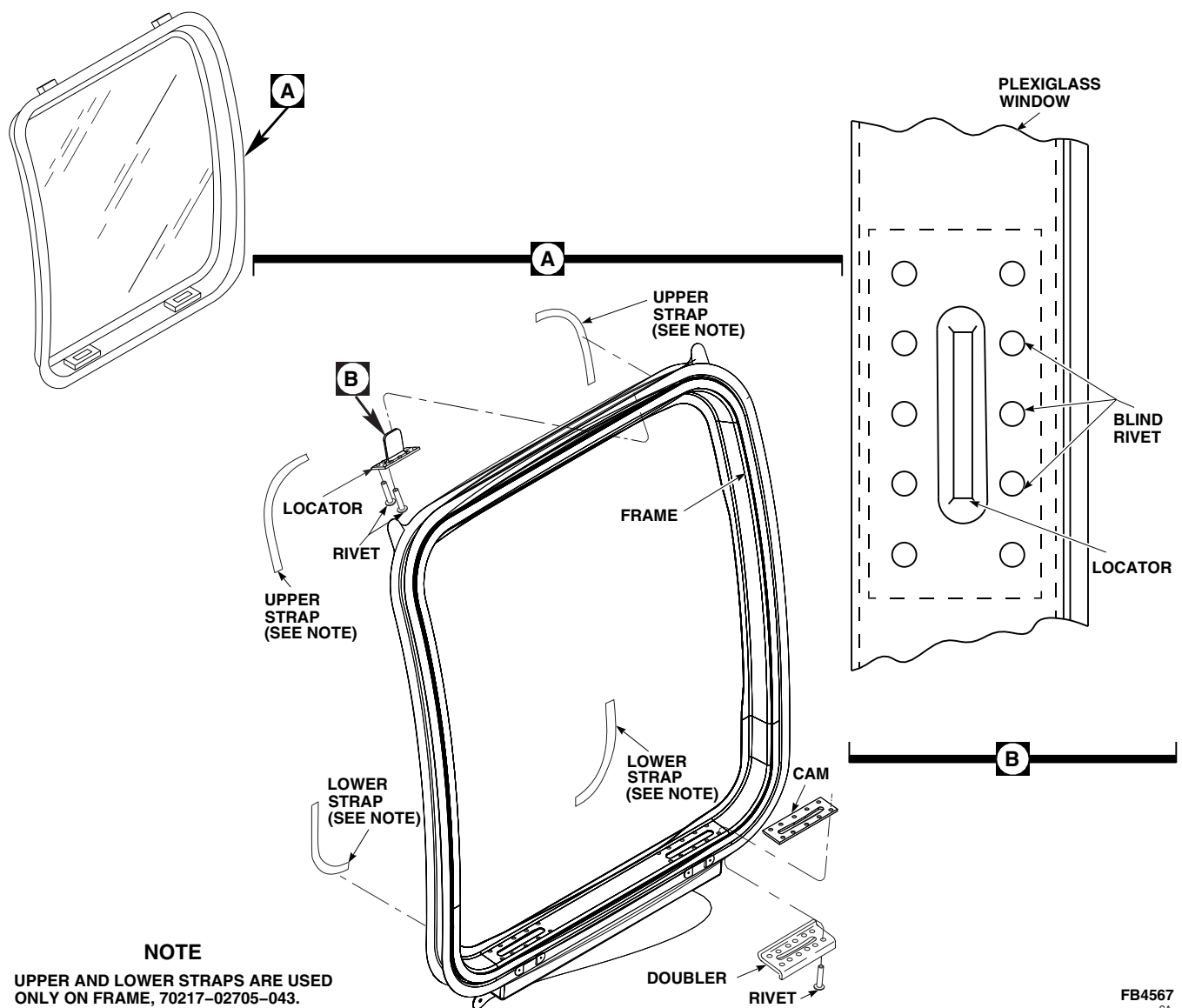


FIGURE 2-5-11-1 REPAIR CABIN DOOR WINDOW FRAME (BENCH)

2-5-12 SOUNDPROOFING PANEL (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
2-5-12.1	Repair Soundproofing Panel (BENCH)	2-5-12-2

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Steam Iron

Materials/Parts

- Adhesive, Item 37, Appendix D
- Adhesive, Item 58, Appendix D
- Adhesive Primer, Item 67, Appendix D
- Cheesecloth, Item 90, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Rubber Sheet, Item 267, Appendix D
- Rubber Sheet, Item 268, Appendix D

References

- AMS 3195
- AMS 4260

- Appendix D
- MIL-F-9084
- MIL-P-15035
- MIL-P-25690
- MIL-R-6855
- MIL-R-9300
- MIL-S-7998
- MIL-STD-670
- MIL-W-4088
- PARA 2-4-69
- PARA 2-6-2
- PARA 2-6-3
- QQ-A-200
- QQ-A-225
- QQ-A-250
- WW-T-700

2-5-12.1 Repair Soundproofing Panel (BENCH).

NOTE

- Repair of all soundproofing panels is the same, except as noted.
 - Soundproofing panels are bonded sandwich construction of Tedlar-coated Nomex facing cloth, foam, aluminum stiffening panel, backing foam, and Tedlar-coated Nomex backing cloth. Layers are stitched and bonded to form soundproofing blanket. In some areas, Kevlar/epoxy laminates are used as panel material in place of aluminum (Figure 2-5-12-1, Sheet 1 through Sheet 8).
- a. Remove soundproofing panel (PARA 2-4-69).
 - b. Tears and rips in outer cloth are repaired by bonding Tedlar/Nomex facing or backing cloth over torn area. Repair as follows:

(1) Primary Method.

NOTE

Use facing cloth for interior patching and backing cloth for patching backside of panel.

- (a) Cut cloth patch 1-inch larger, on all sides, than tear dimensions.
- (b) Lay film adhesive ($\frac{1}{8}$ -inch larger than patch size) on repair area and center over tear.
- (c) Position cloth patch and center over film adhesive.
- (d) Using steam iron heated to 310°F (154°C), iron on cloth patch with steam.

(2) Alternate Method.

NOTE

Use facing cloth for interior patching and backing cloth for patching backside of panel.

- (a) Cut cloth patch 1-inch larger on all sides than tear dimensions.

NOTE

Apply adhesive carefully to prevent discoloring of panel outside repair area.

- (b) Apply adhesive, Item 37, or Item 58, Appendix D, to cloth patch.
- (c) Position cloth patch over tear.
- (d) Apply pressure to create sufficient bond.

NOTE

Fasteners pulling through stiffening panels are repaired per PARA 2-6-2 for aluminum, and PARA 2-6-3 for Kevlar/epoxy laminates.

- c. Peel back soundproofing blanket to expose damaged area and apply patch 2 x 2-inches over hole. Rebond foam and cloth using noted adhesives in step b., and install fasteners.
- d. Tedlar coating on facing or backing cloth is oil-proof and may be cleaned using cheesecloth, Item 90, Appendix D, moistened with dry-cleaning solvent, Item 124, Appendix D.
- e. Rebond loosened seals and loosened hook or pile fastener tape using adhesive, Item 37, Appendix D.
- f. Repair honeycomb core damage (PARA 2-6-3).
- g. Install soundproofing panel (PARA 2-4-69).

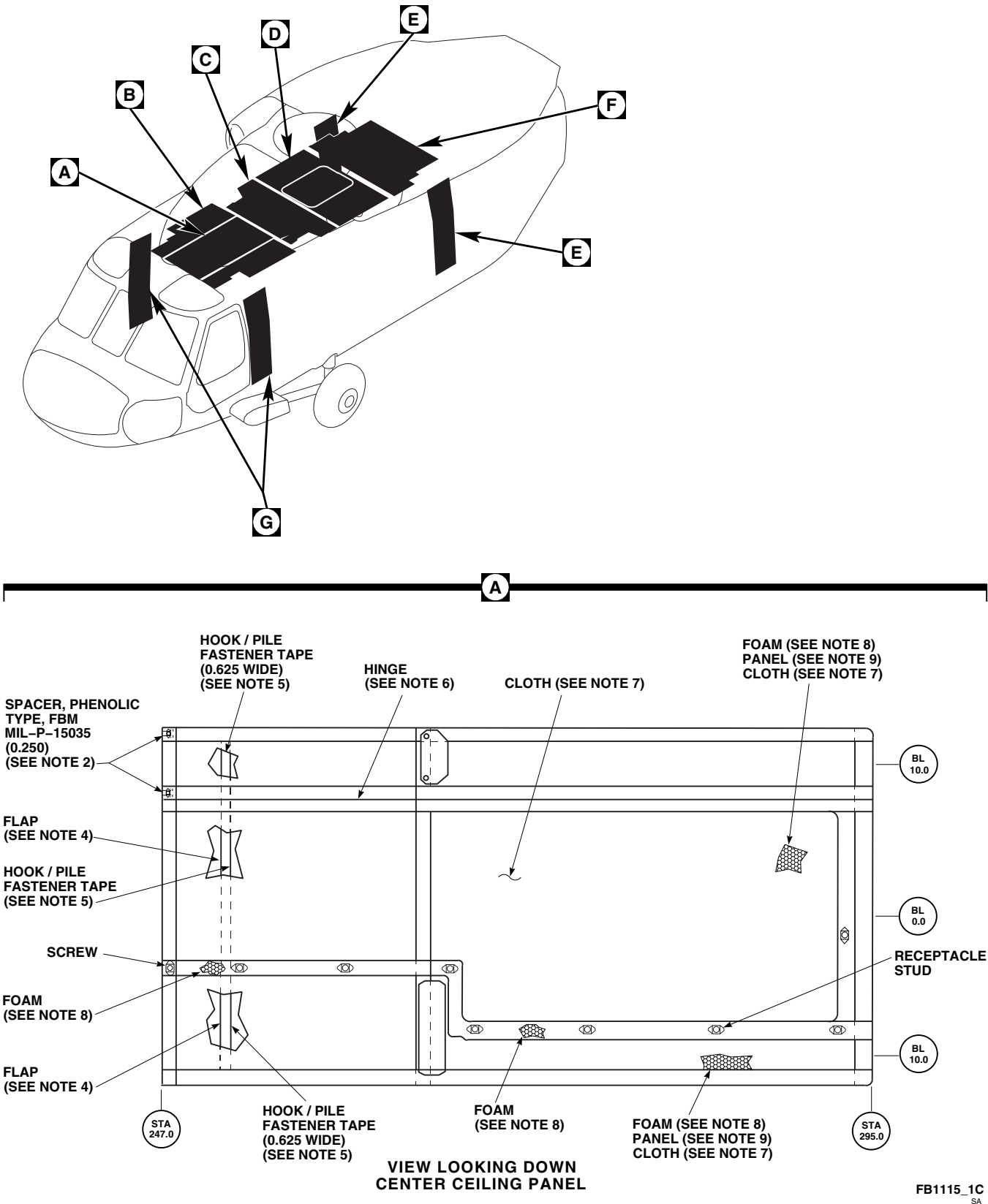


FIGURE 2-5-12-1. REPAIR SOUNDPROOFING PANELS (BENCH) (SHEET 1 OF 8)

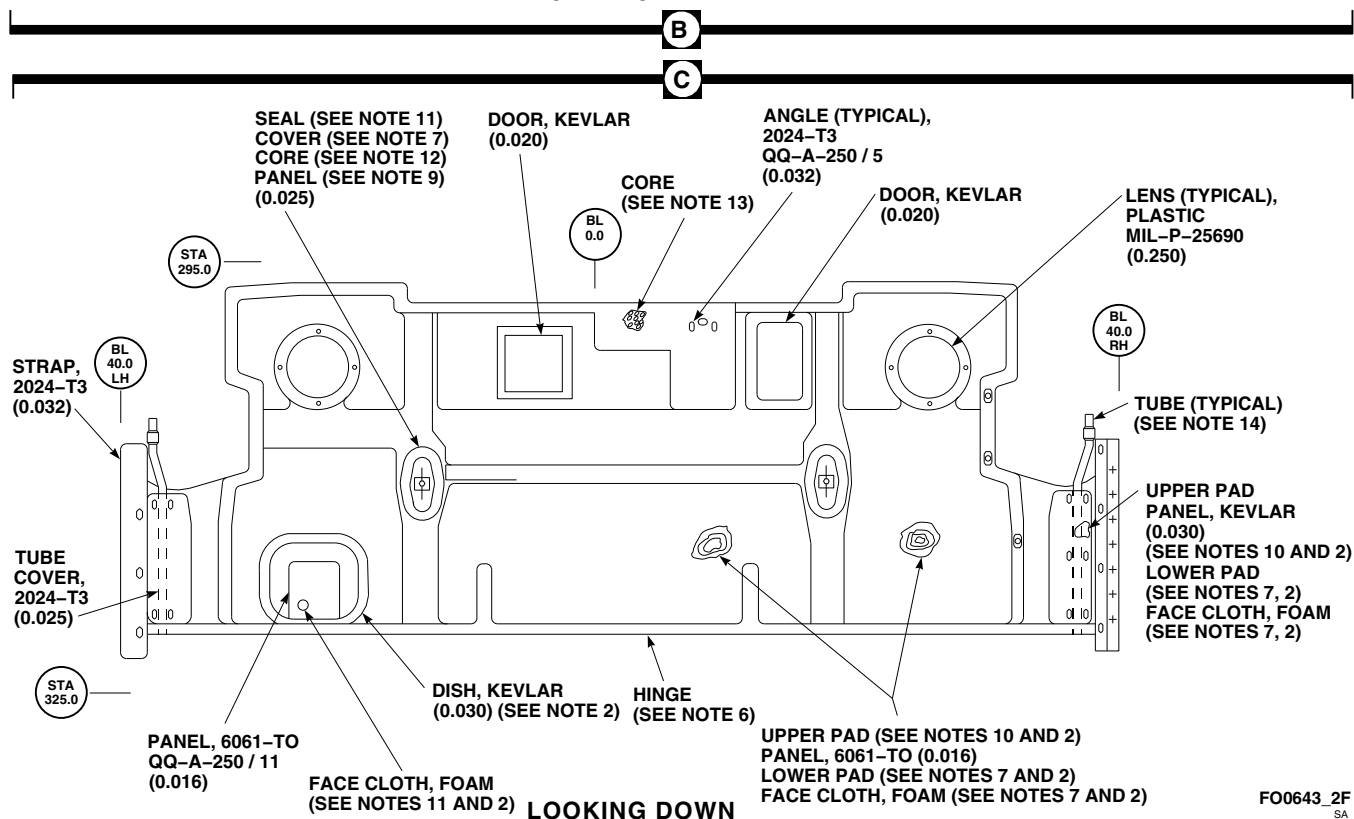
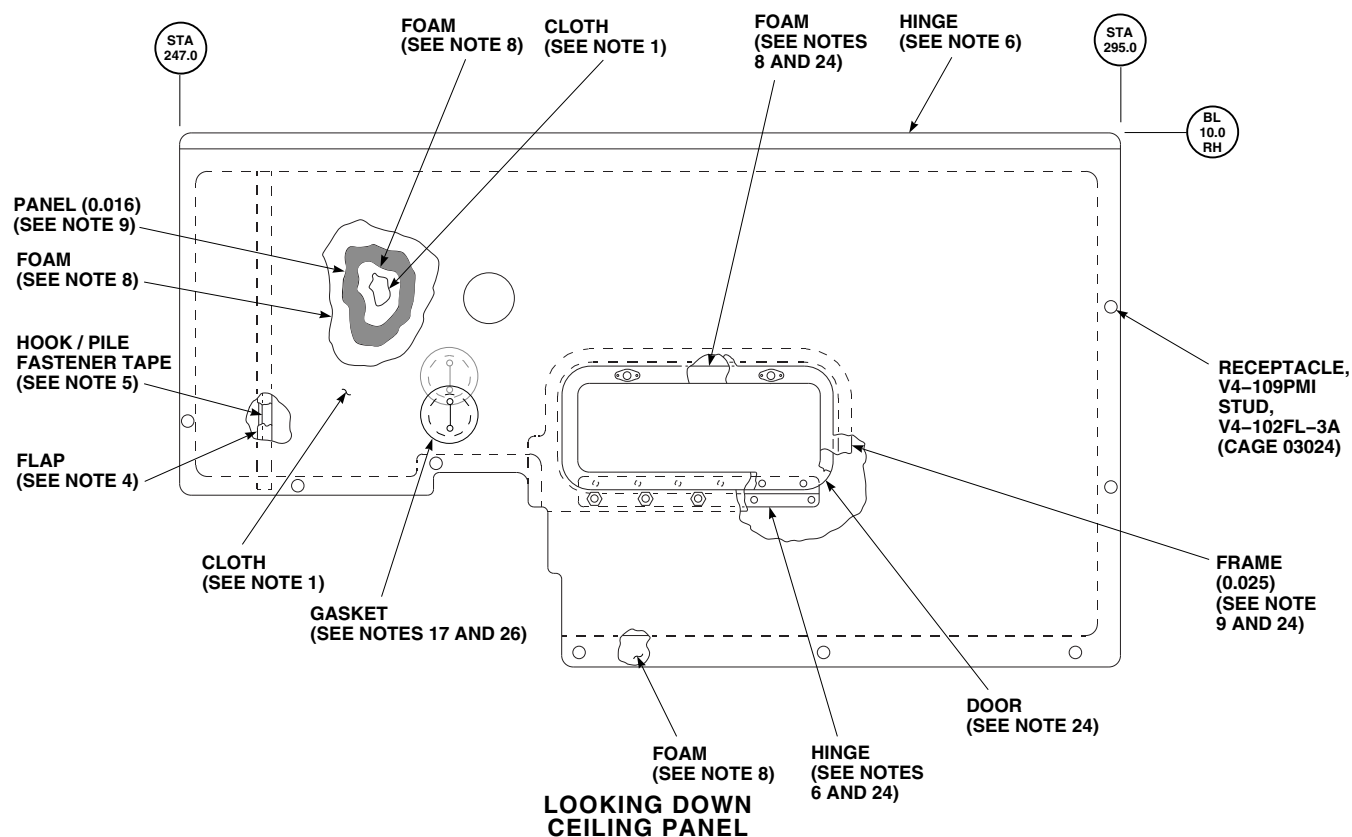
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FIGURE 2-5-12-1. REPAIR SOUNDPROOFING PANELS (BENCH) (SHEET 2 OF 8)

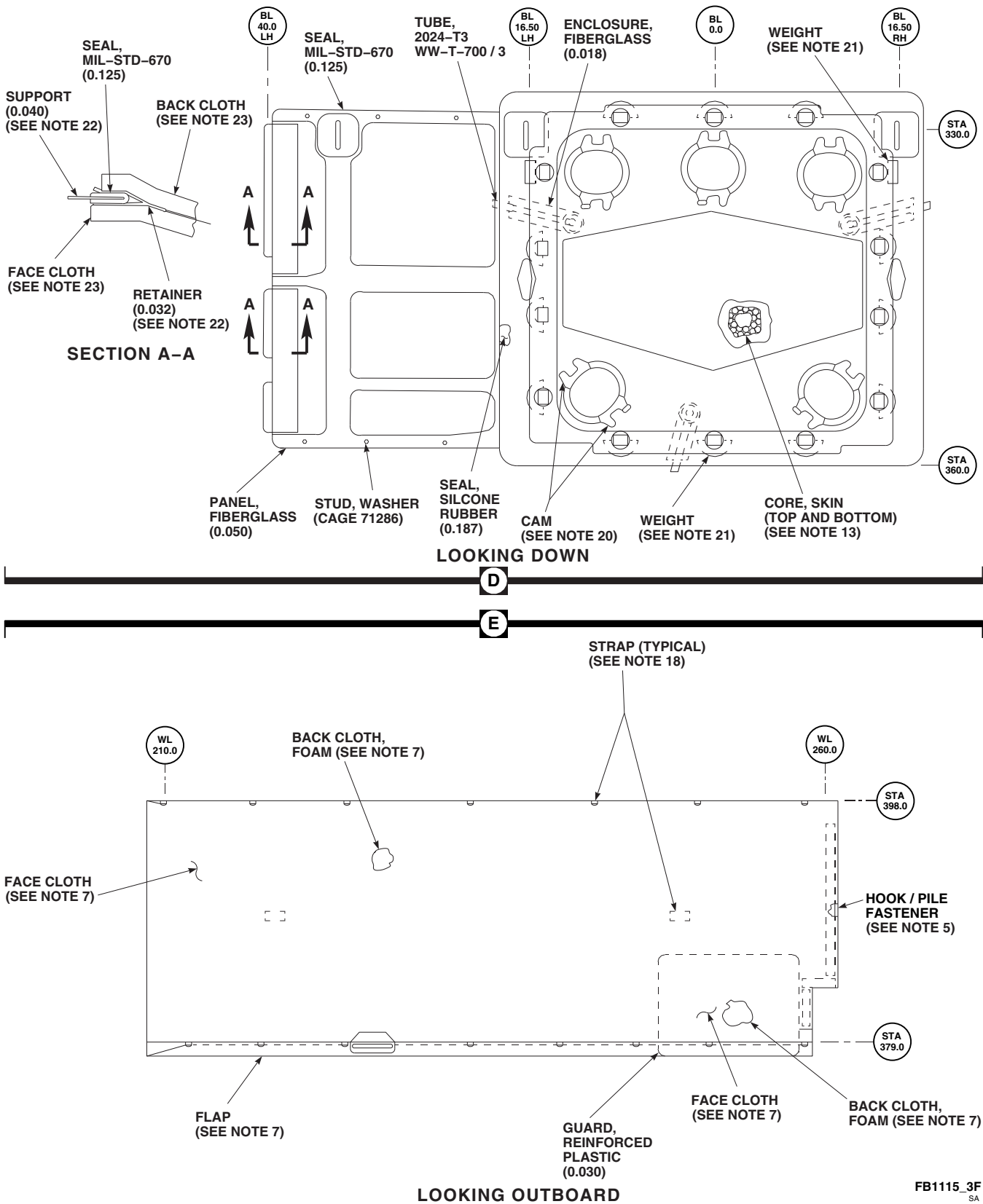


FIGURE 2-5-12-1. REPAIR SOUNDPROOFING PANELS (BENCH) (SHEET 3 OF 8)

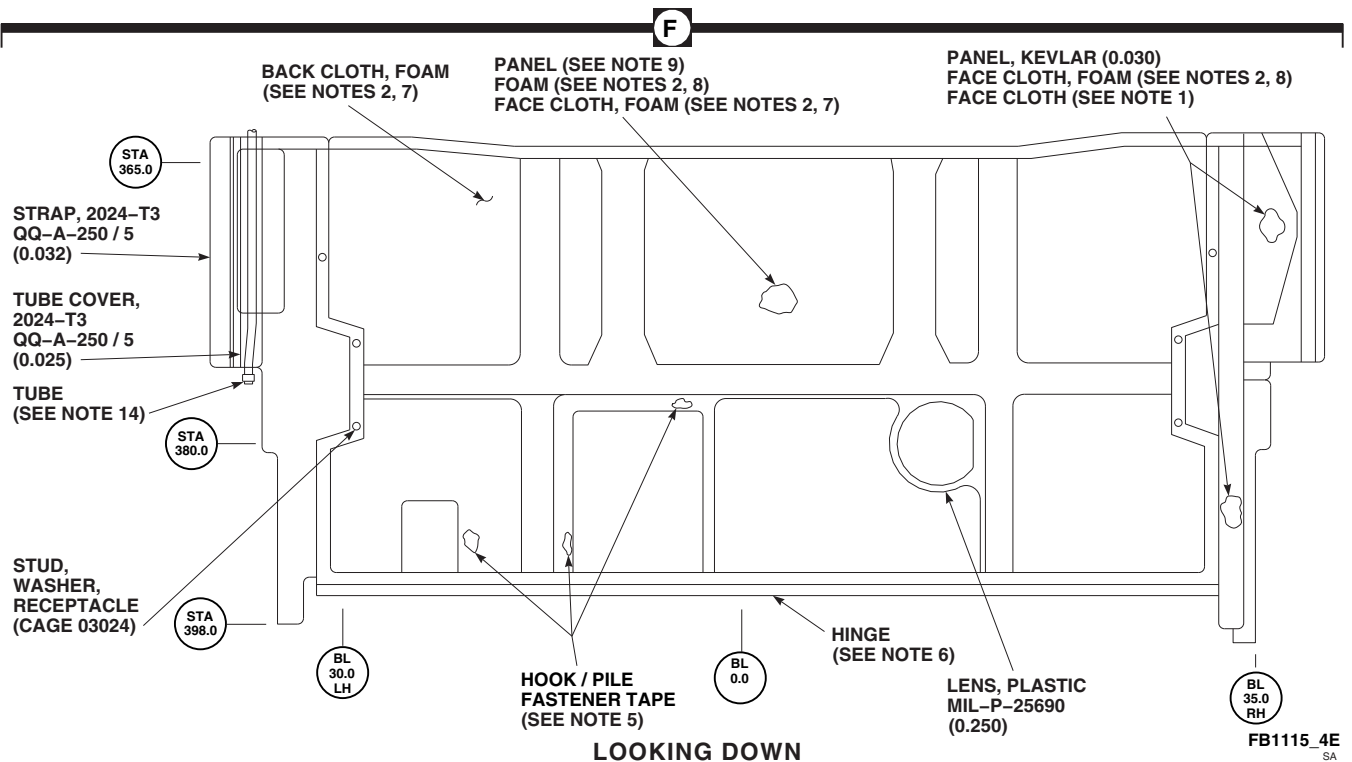
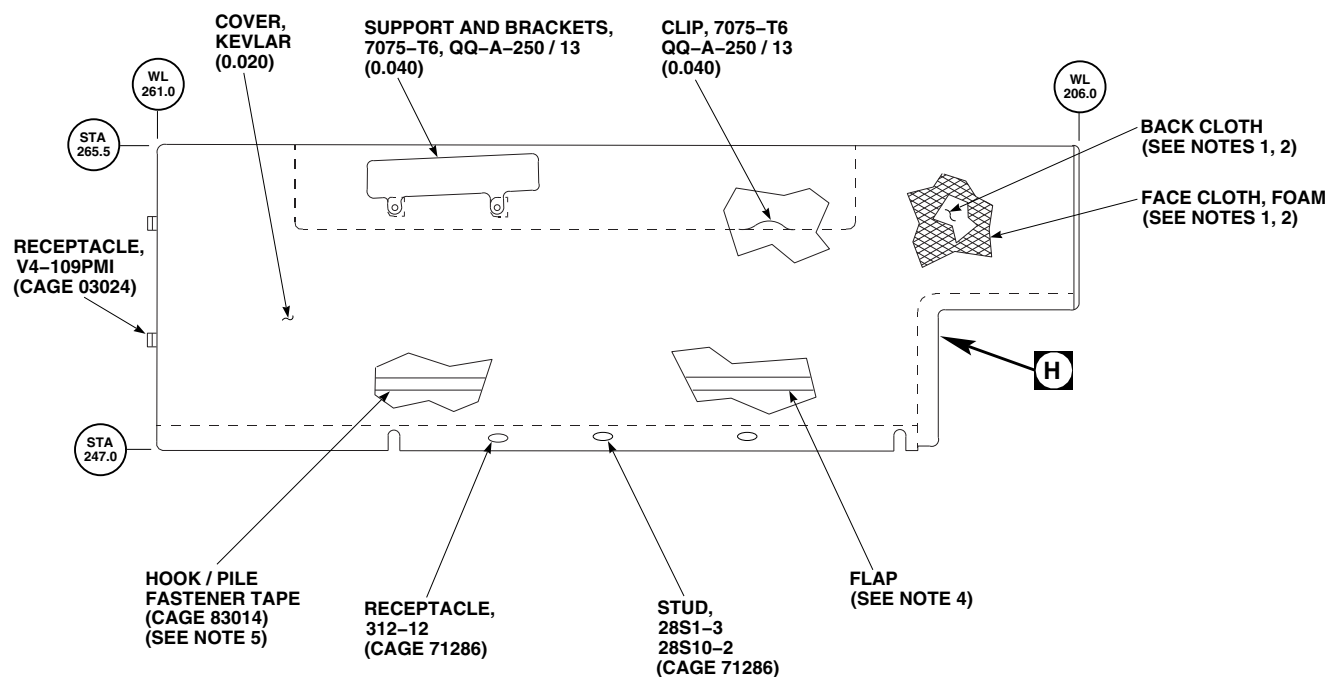
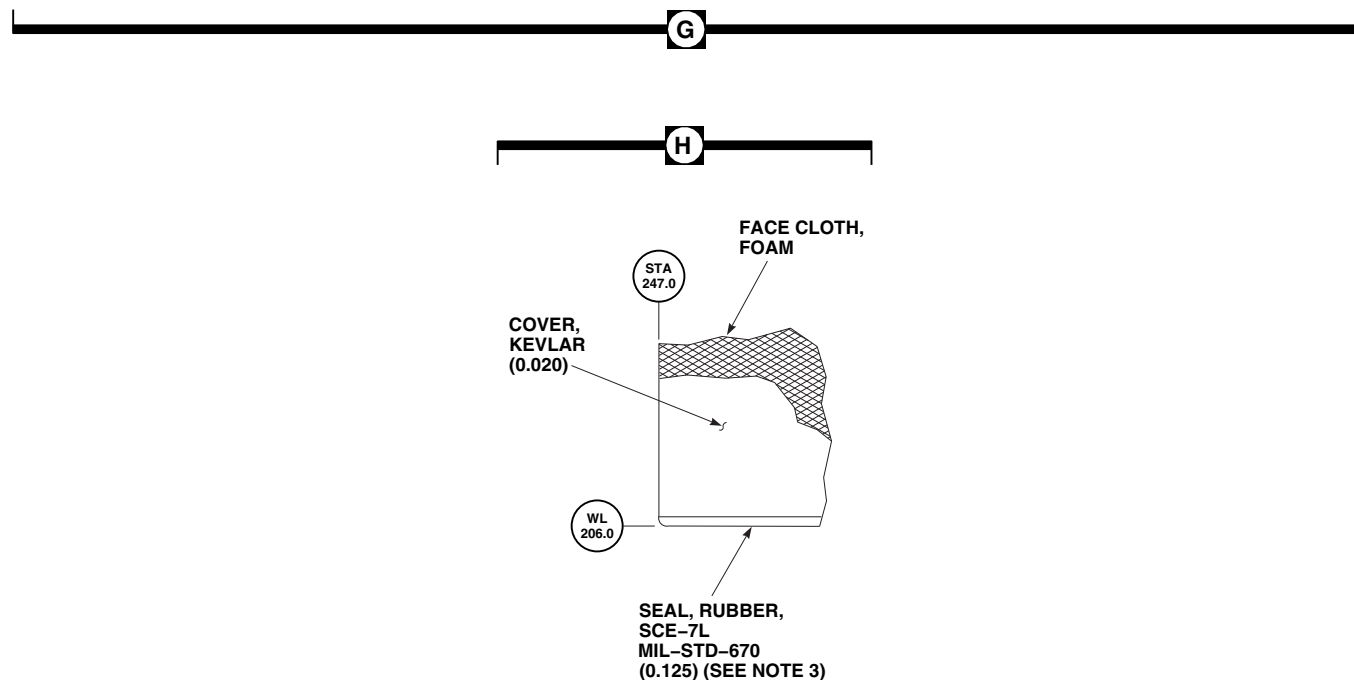


FIGURE 2-5-12-1. REPAIR SOUNDPROOFING PANELS (BENCH) (SHEET 4 OF 8)



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LOOKING OUTBOARD



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FIGURE 2-5-12-1. REPAIR SOUNDPROOFING PANELS (BENCH) (SHEET 5 OF 8)

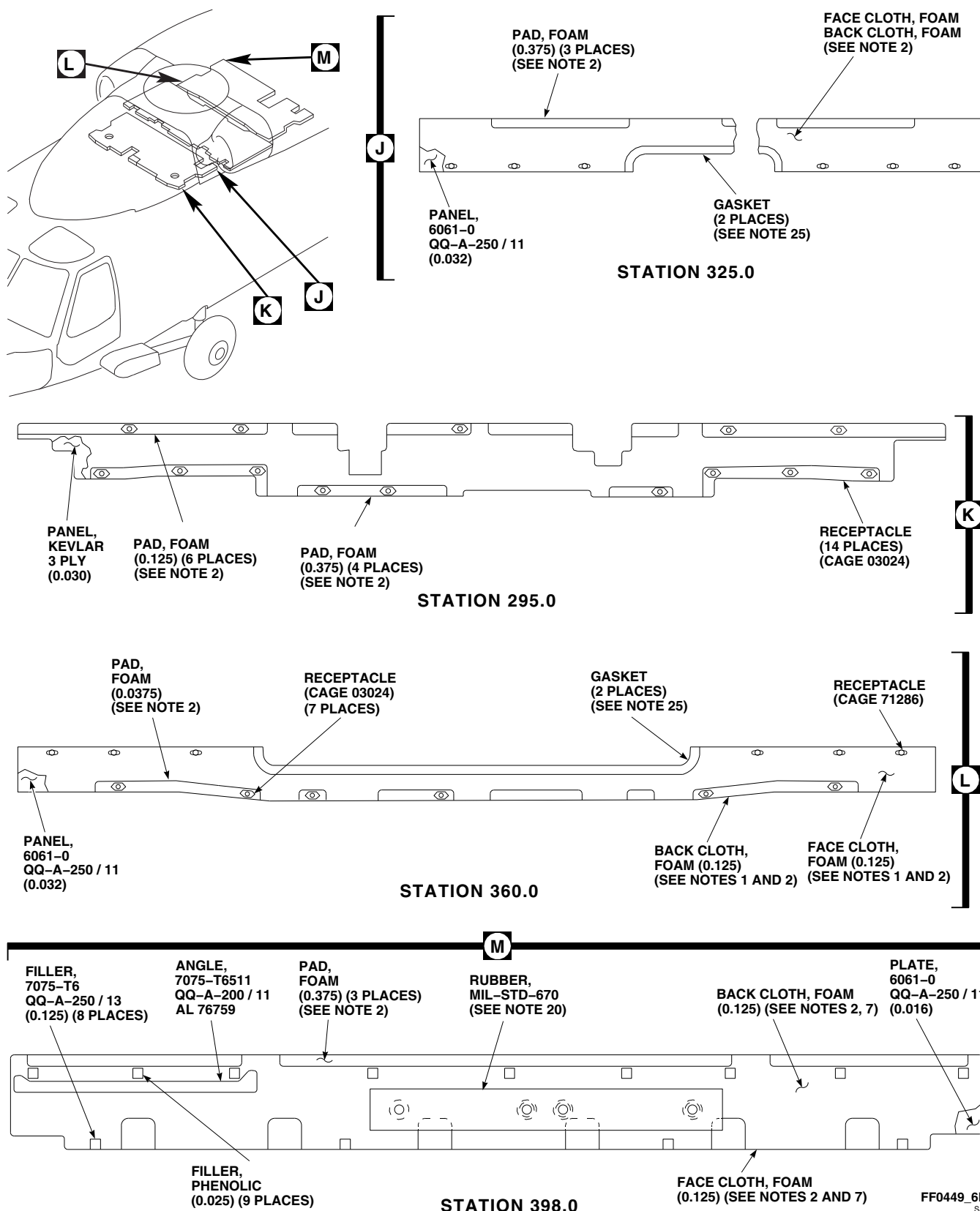
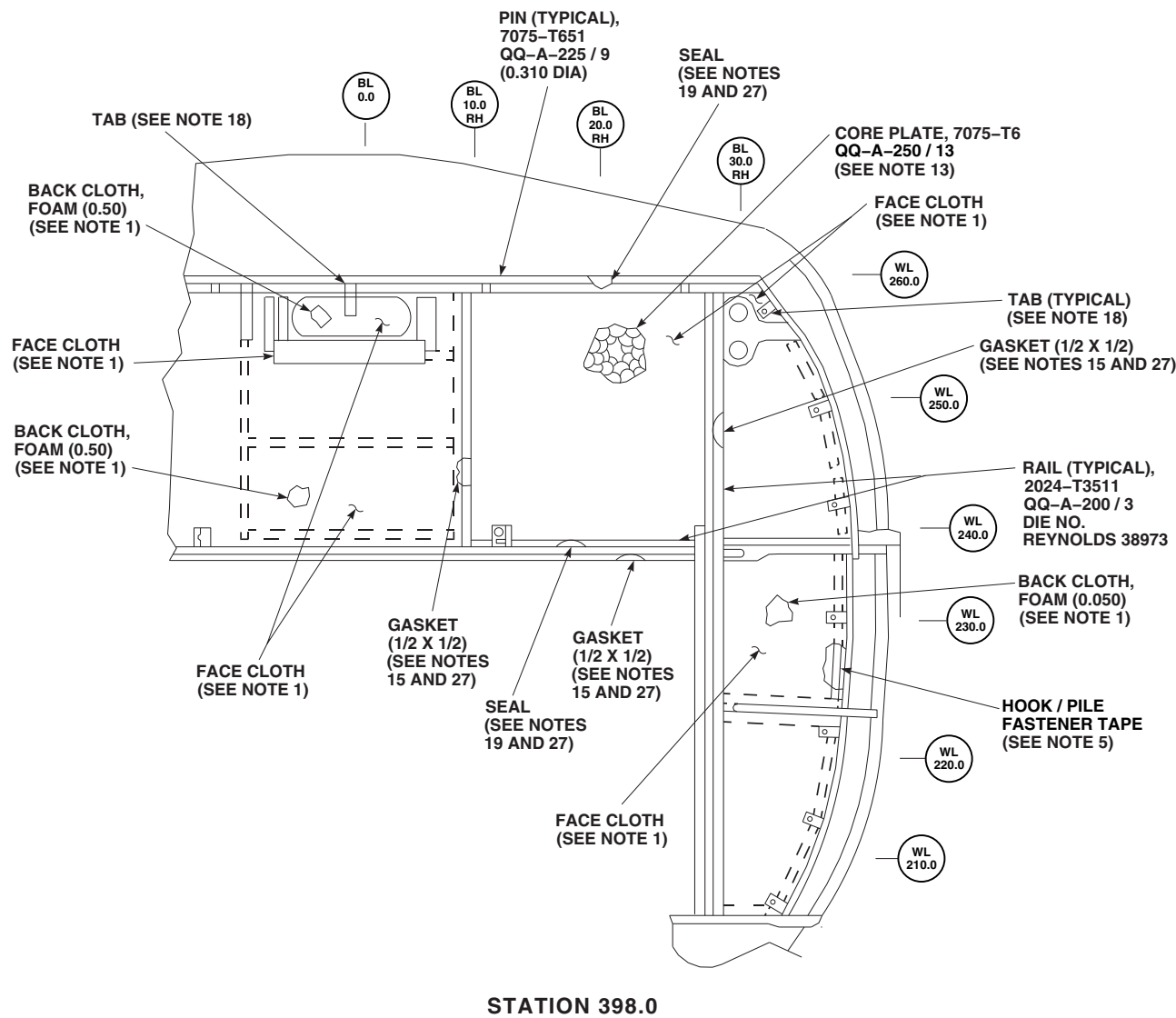


FIGURE 2-5-12-1. REPAIR SOUNDPROOFING PANELS (BENCH) (SHEET 6 OF 8)



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FIGURE 2-5-12-1. REPAIR SOUNDPROOFING PANELS (BENCH) (SHEET 7 OF 8)

NOTES

1. COLOR SAGE GREEN.
2. BOND FOAM TO PANELS AND COATED NOMEX CLOTH TO FOAM WITH ADHESIVE, ITEM 37, APPENDIX D.
3. BOND SEAL TO COVER WITH ADHESIVE, ITEM 37, APPENDIX D.
4. TUFF-TARP, VINYL-COATED NYLON, NO. 620 LIGHT GULL GRAY.
5. BOND PILE FASTENER TAPE TO STRUCTURE AND HOOK FASTENER TAPE TO PANELS WITH ADHESIVE, ITEM 67, APPENDIX D.
6. NYLON HINGE, GRAY (0.100 x 1.5 WIDE).
7. POLYESTER POLYURETHANE-COATED NOMEX, TYPE 1, CLASS 1, GREEN BOTH SIDES.
8. RUBBER SHEET, ITEM 267, APPENDIX D.
9. ALUMINUM ALLOY, 6061-T6, QQ-A-250 / 11 (0.016).
10. FIBERGLASS BATTING, RESIN, MIL-R-9300 AND GLASS FABRIC, MIL-F-9084.
11. PYRELL FOAM (0.12), CAGE 78112.
12. BALSA CORE, PLANK, MIL-S-7998.
13. FIBERGLASS / HONEYCOMB / FLYSCREEN CORE, HRH-10 NOMEX, NO. 1507 (0.50).
14. POLYETHYLENE TUBE, TYPE 3 (0.62 O. D. x 0.06 WALL).
15. NITRILE RUBBER, MIL-R-6855.
16. HONEYCOMB / FLYSCREEN CORE, HRH-10 NOMEX, NO. 1505 (0.50).
17. RUBBER SHEET, ITEM 268, APPENDIX D.
18. NYLON WEBBING, GRAY, MIL-W-4088, TYPE 1A, BUTTON-MS27980-IN, SOCKET-MS27980-6N.
19. SILICONE SPONGE RUBBER, AMS 3195 (3/8 x 1/2).
20. CASTING, AMS 4260, 356-T6.
21. STEEL 4130 COND A OR N.
22. 7075-T6, QQ-A-250 / 13.
23. MAKE FROM SS9232-025.
24. RIGHT SIDE ONLY.
25. MAKE FROM MIL-STD-670 SCEIL.
26. RIGHT SIDE GASKET LOCATION SHOWN IN PHANTOM.
27. BOND TO PANEL USING ADHESIVE PRIMER, ITEM 67, APPENDIX D.
28. ALL DIMENSIONS ARE IN INCHES, UNLESS NOTED.

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2-5-13 ENGINE COWLING SUPPORT FITTING BEARING (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
2-5-13.1	Replace Engine Cowling Support Fitting Bearing (BENCH)	2-5-13-2

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Anvil
- Arbor Press
- Clamp
- Drill Press
- Driver
- Fly Cutter
- Nonmetallic Scraper, Locally-Made

- Spinning Tool

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Corrosion Preventive Compound, Item 113, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Petrolatum, Item 234, Appendix D
- Primer, Item 257, Appendix D

References

- Appendix D

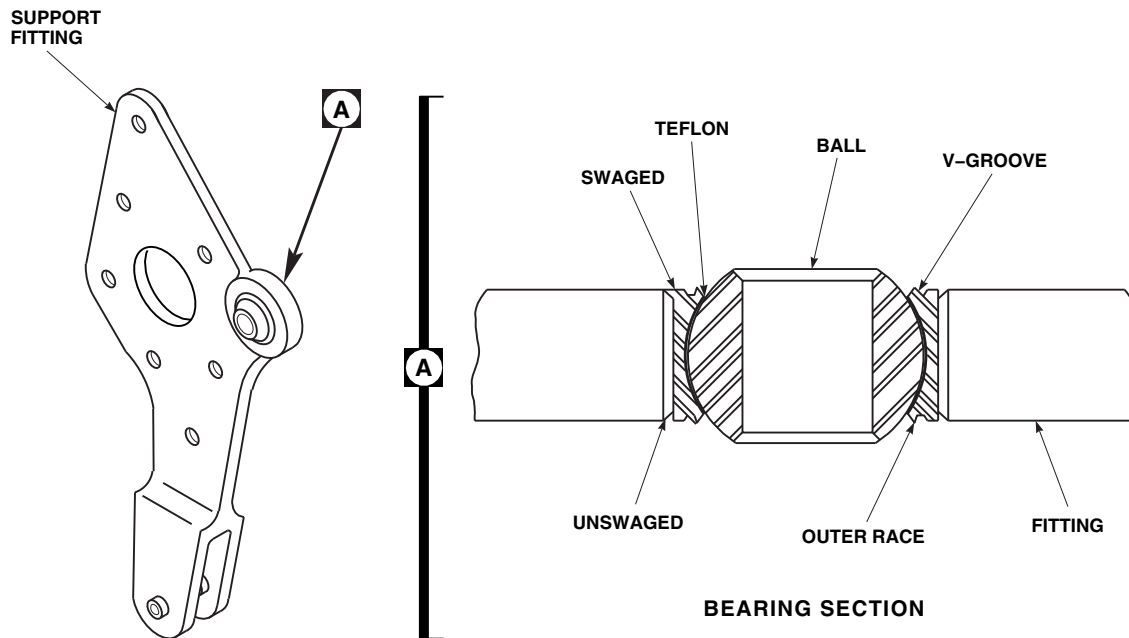
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FIGURE 2-5-13-1. REPLACE ENGINE COWLING SUPPORT FITTING BEARING (BENCH)

2-5-13.1 Replace Engine Cowling Support Fitting Bearing (BENCH).**NOTE**

Bearing replacement is the same for both front and rear support fittings.

2-5-13.1.1. Remove.**CAUTION**

Damage to fitting chamfer will occur if staked lip is improperly removed. Be careful not to damage fitting chamfer when machining off staked lip of bearing.

- Clamp fitting to drill press baseplate.
- Using fly cutter, machine off the part of bearing outer race swaged over chamfer on fitting (Figure 2-5-13-1, Detail A).

- Press out bearing using arbor press and suitable driver and anvil combination.
- Clean primer residue from bore and chamfer, using cheesecloth, Item 90, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D, and nonmetallic scraper.

2-5-13.1.2. Inspect.

- Measure diameter of fitting bearing bore. **DIAMETER MUST BE 0.6248 TO 0.6253-INCH.** Replace fitting if measurement exceeds 0.6253-inch.
- If diameter of fitting bearing bore is less than 0.6248-inch. Ream fitting bore to 0.6248 to 0.6253-inch.

2-5-13.1.3. Install.

- Make sure that fitting bore, chamfers, and adjacent surfaces are clean and dry.

2-5-13-2

- b. Apply light coat of primer, Item 257, Appendix D, to inside of fitting bore and outside surface of bearing outer race.
- c. Press bearing into fitting using arbor press and suitable driver and anvil combination (Figure 2-5-13-1, Detail A).
- d. Clean primer, if any, from bearing V-grooves using cheesecloth, Item 90, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D. Wipe dry.
- e. Fix fitting on drill press baseplate, with proper spinning tool and anvil.

CAUTION

To prevent damage to Teflon, keep petrolatum from bearing ball and Teflon liner. Remove petrolatum immediately if it gets on these surfaces.

- f. Lubricate V-groove of bearing and spinning tool with petrolatum, Item 234, Appendix D.

CAUTION

Deformation of outer race can cause binding of bearing. Do not let spinning tool contact ball side of V-groove.

- g. Swage V-groove lip over chamfer in fitting bore.
- h. Turn fitting over and repeat swaging process on other side.
- i. Clean petrolatum, Item 234, Appendix D, from V-groove with dry-cleaning solvent, Item 124, Appendix D, on cheesecloth, Item 90, Appendix D.
- j. Touch up bearing surfaces with corrosion preventive compound, Item 113, Appendix D.
- k. Touch up fitting surfaces with primer, Item 257, Appendix D.

2-5-14 ENGINE COWLING SUPPORT FITTING BUSHINGS (BENCH).

PARAGRAPH BREAKDOWN

	PAGE
2-5-14.1 Replace Engine Cowling Support Fitting Bushings (BENCH)	2-5-14-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Arbor Press
- Bushing Installation Kit
- Spacer Blocks

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Primer, Item 257, Appendix D

References

- Appendix D

2-5-14.1 Replace Engine Cowling Support Fitting Bushings (BENCH).

NOTE

Bushing replacement is the same for both front and rear support fittings.

2-5-14.1.1. Remove.

- Press bushings out of fitting, using arbor press and bushing installation kit (Figure 2-5-14-1, Detail A).
- Remove primer residue from surfaces of fitting clevis and inside of bore using cheesecloth, Item 90, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D.

2-5-14.1.2. Inspect.

- Measure support fitting bushing bore diameters. **DIAMETER OF BORES MUST BE 0.3742 TO 0.3751-INCH.** Replace fitting if any diameter exceeds 0.3751-inch.

- If any bushing bore diameter is less than 0.3742-inch, line-ream bushing bores to 0.3742 to 0.3751-inch.

2-5-14.1.3. Install.

- Make sure that insides of fitting clevis bores are clean and dry.
- Apply thin coat of primer, Item 257, Appendix D, to bushing outer surface (on flanged bushing, coat under surface of flange).
- Apply thin coat of primer, Item 257, Appendix D, to inside of clevis bore.
- On rear fitting, using arbor press and bushing installation kit, install flanged bushing first and use appropriate spacer blocks to achieve gap dimension specified in Figure 2-5-14-1, Detail A when installing plain bushing.
- On front fitting, using arbor press and bushing installation kit, install front plain bushing to

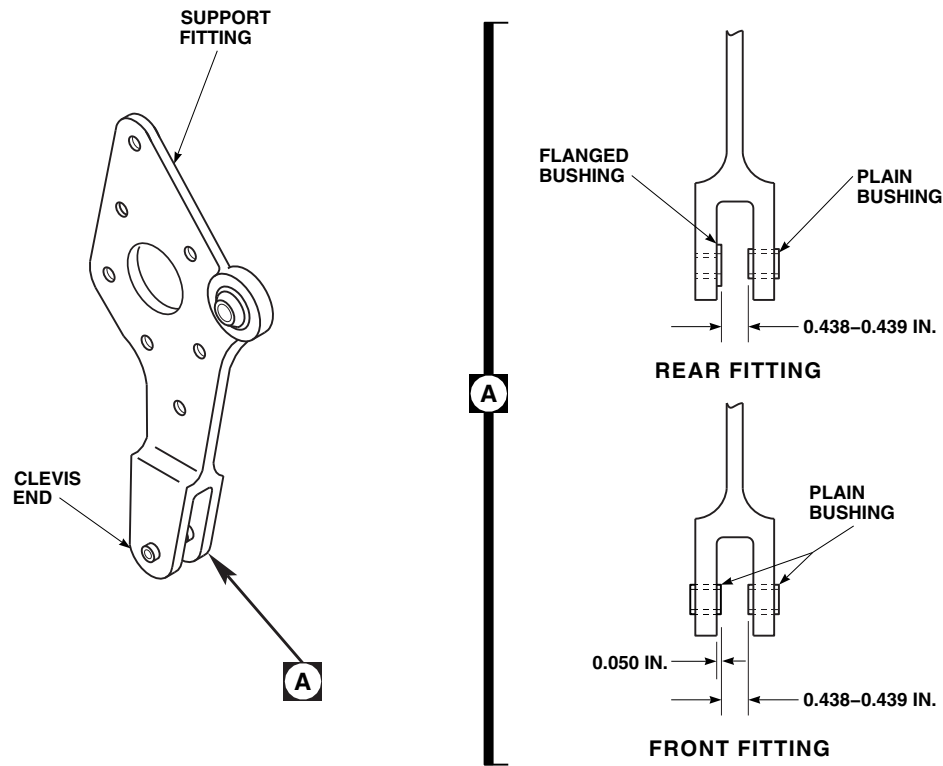
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FIGURE 2-5-14-1. REPLACE ENGINE COWLING SUPPORT FITTING BUSHINGS (BENCH)

protrude 0.050-inch into clevis slot and use spacer blocks achieve gap dimension specified

in Figure 2-5-14-1, Detail A when installing rear plain bushing.

2-5-15 ENGINE COWLING FITTING BEARING (BENCH).

PARAGRAPH BREAKDOWN

	PAGE
2-5-15.1 Replace Engine Cowling Fitting Bearing (BENCH)	2-5-15-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Anvil
- Arbor Press
- Clamp
- Drill Press
- Driver
- Fly Cutter
- Nonmetallic Scraper, Locally-Made
- Spinning Tool

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Corrosion Preventive Compound, Item 113, Appendix D
- Dry-Cleaning Solvent, Item 124, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Petrolatum, Item 234, Appendix D
- Primer, Item 257, Appendix D

References

- Appendix D

2-5-15.1 Replace Engine Cowling Fitting Bearing (BENCH).

NOTE

Bearing replacement is the same for both front and rear cowling fittings.

2-5-15.1.1. Remove.

CAUTION

Damage to fitting chamfer will occur if staked lip is improperly removed. Be careful not to damage chamfer of fitting

when machining off staked lip of bearing.

- Clamp fitting to baseplate of drill press.
- Using fly cutter, machine off lip of bearing outer race that is swaged over chamfer on fitting (Figure 2-5-15-1, Detail A).
- Press out bearing using arbor press, suitable driver, and anvil combination.
- Clean primer residue from fitting bore and chamfer, using cheesecloth, Item 90, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D, and nonmetallic scraper.

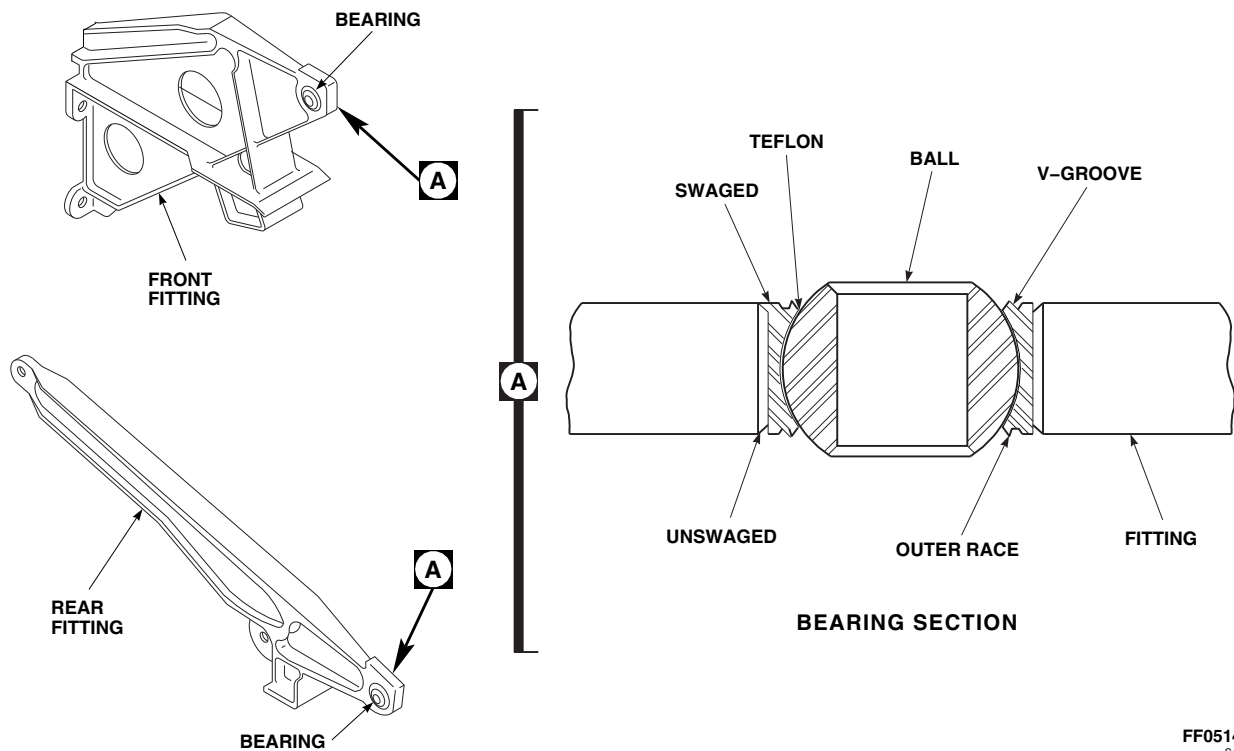
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FIGURE 2-5-15-1. REPLACE ENGINE COWLING FITTING BEARING (BENCH)

2-5-15.1.2. Install.

- Make sure that fitting bore, chamfer, and adjacent surfaces are clean and dry.
- Apply light coat of primer, Item 257, Appendix D, to inside of fitting bore and to outside surfaces of bearing outer race.
- Press bearing into fitting using arbor press, suitable driver, and anvil combination (Figure 2-5-15-1, Detail A).
- Clean primer, if any, from bearing V-grooves using cheesecloth, Item 90, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D. Wipe dry.
- Fix fitting on drill press baseplate, with proper spinning tool and anvil.

CAUTION

To prevent damage to Teflon, keep petrolatum from bearing ball and Teflon

liner. Remove petrolatum immediately if it gets on these surfaces.

- Lubricate V-groove of bearing and spinning tool with petrolatum, Item 234, Appendix D.

CAUTION

Deformation of outer race can cause binding of bearing. Do not let spinning tool contact ball side of V-groove.

- Swage V-groove lip over chamfer in fitting bore.
- Turn fitting over and repeat swaging process on other side.
- Clean petrolatum, Item 234, Appendix D, from V-groove with dry-cleaning solvent, Item 124, Appendix D, on cheesecloth, Item 90, Appendix D.
- Touch up bearing surfaces with corrosion preventive compound, Item 113, Appendix D.

2-5-15-2

- k. Touch up fitting surfaces with primer, Item 257, Appendix D.

2-5-16 OIL COOLER SUPPORT (BENCH)

PARAGRAPH BREAKDOWN

	PAGE
2-5-16.1 Repair Oil Cooler Support (BENCH)	2-5-16-1

INITIAL SETUP Tools - Airframe Repairer’s Toolkit - Powertrain Repairer’s Toolkit - Rivet Gun Materials/Parts - Sealing Compound, Item 292, Appendix D	References - Appendix D - PARA 2-6-1 - PARA 2-6-2 - PARA 2-6-4 - PARA 2-6-5 - QQ-A-200 - QQ-A-250
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2-5-16.1 Repair Oil Cooler Support (BENCH).



NOTE - Oil cooler blower forward and rear supports include formed and extruded members made from 7075-T6, QQ-A-250/13 Alclad sheet stock and 7075-T6511, QQ-A-200/11 aluminum alloy extrusion (Figure 2-5-16-1, Detail A). - Support members are primary structure. a. Evaluate damage to support members (PARA 2-6-1).	All repairs to oil cooler blower supports must maintain correct alignment with tail rotor drive shaft. b. Refer to PARA 2-6-2 and PARA 2-6-4 for typical support member repair. c. Inspect forward support for loose or damaged nutplate(s). NONE ALLOWED. If loose or damaged nutplate(s) are found, replace nutplate(s) using rivets. d. For corrosion protection and watertightness, seal all support parts which secure to structure
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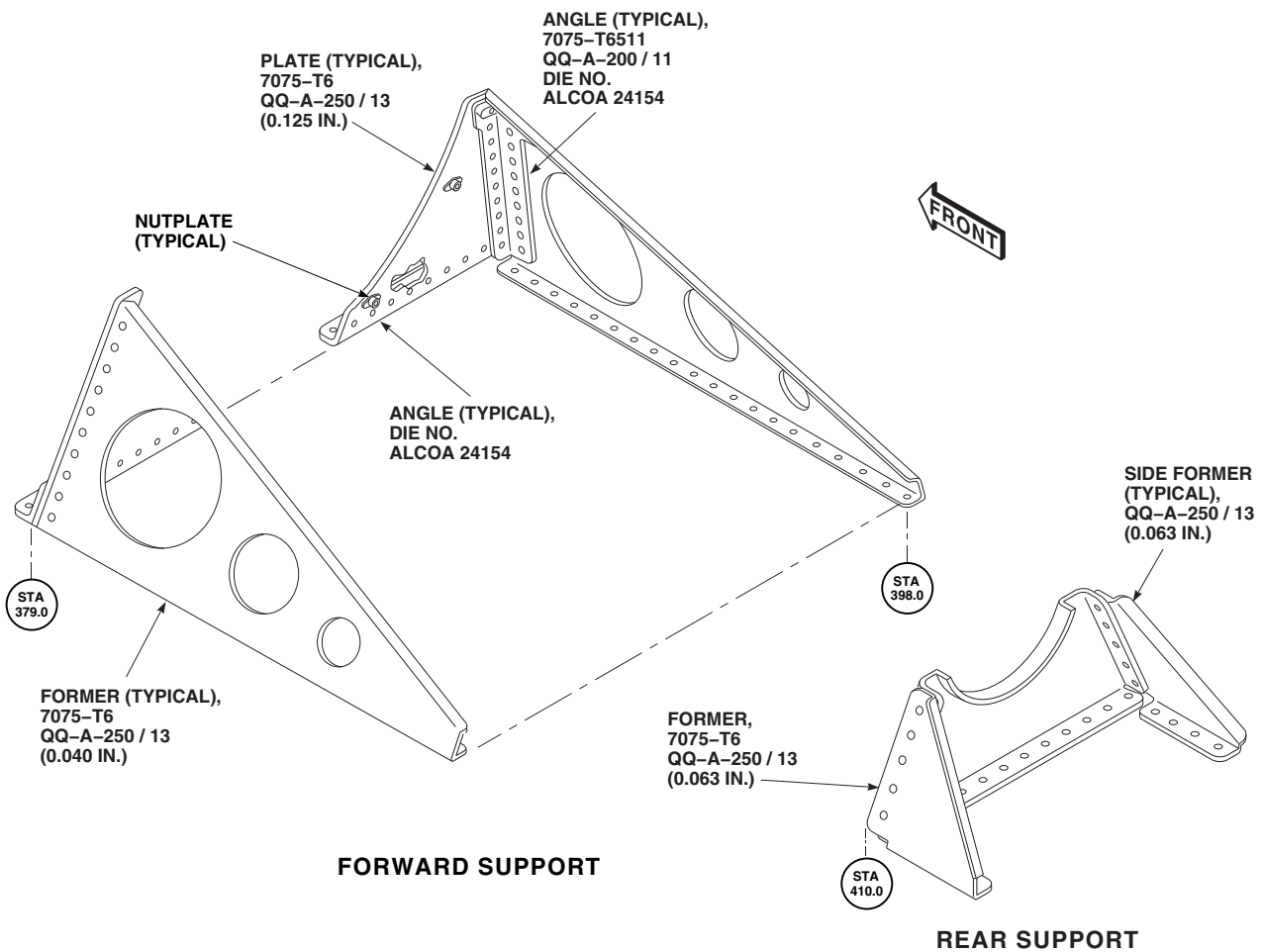
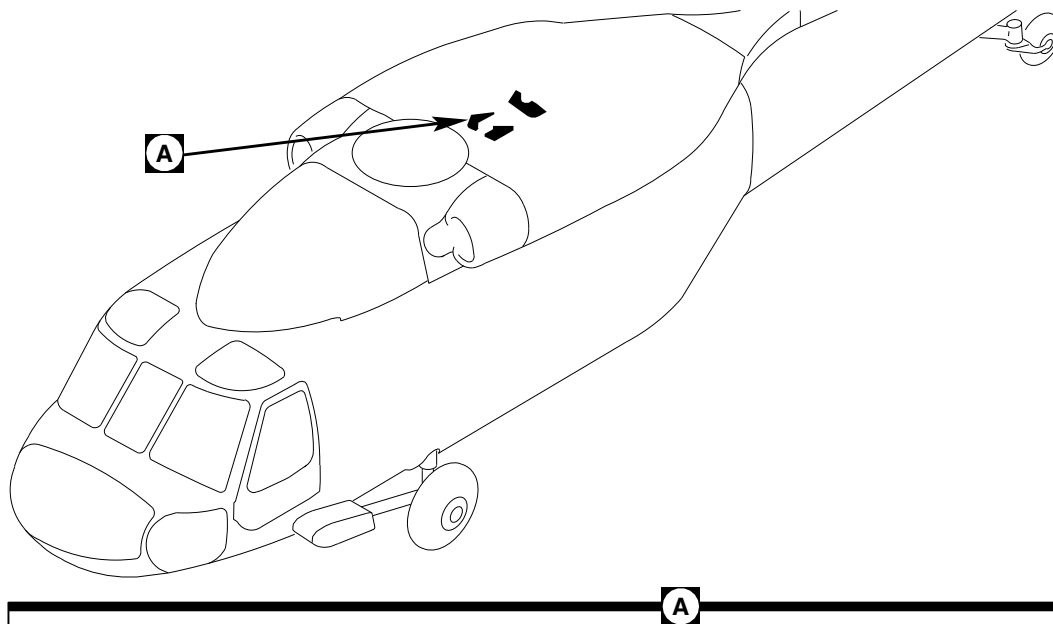
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FIGURE 2-5-16-1. REPAIR OIL COOLER SUPPORT (BENCH)

with sealing compound, Item 292, Appendix D
(PARA 2-6-1).

e. Touch up paint as required (PARA 2-6-5).

2-5-17 OIL COOLER FORWARD SUPPORT (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
2-5-17.1	Replace Oil Cooler Forward Support (BENCH)	2-5-17-2

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Blind Rivet Gun Kit
- Clamp
- Clecos
- Drill Bit, 17/64 -inch diameter (0.2656-inch diameter)
- Drill Set
- Nonmetallic Mallet
- Powertrain Repairer’s Toolkit
- Tail Rotor Drive Shaft Alignment Fixture Set, 70700-20480-041
- Target Wire

Materials/Parts

- Collar, NAS1080-5
- Collar, NAS1080-6
- Lockbolt, HL2086

- Lockbolt, NAS1465-5
- Lockbolt, NAS1465-6
- Rivet, MS20470AD5
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-3-20
- PARA 1-3-21
- PARA 2-4-45
- PARA 2-4-69
- PARA 2-4-113
- PARA 2-6-1
- PARA 2-6-5
- PARA 6-4-39
- PARA 6-4-40
- PARA 6-4-50
- TM 1-70-VIB
- TM 11-70-23

2-5-17.1 Replace Oil Cooler Forward Support (BENCH).

2-5-17.1.1 Remove.

CAUTION

- After removal of left or right oil cooler forward support, remaining forward support will be used to reinstall removed forward support.
- If left and right forward support assemblies must be removed, contact depot maintenance.

NOTE

The following instructions apply to both left and right forward support assemblies.

- a. Defuel main fuel tanks (PARA 1-3-20 or PARA 1-3-21).
- b. Remove rear passenger's seats (PARA 2-4-45).
- c. Remove soundproofing panels from left and right stowage compartments and center structure (PARA 2-4-69).
- d. Remove No. 1 and No. 2 stabilator amplifiers (TM 11-70-23).
- e. Remove oil cooler compartment access doors (PARA 2-4-113).
- f. Remove oil cooler (PARA 6-4-50), tail rotor drive shaft section II (PARA 6-4-39), and tail rotor drive shaft section III (PARA 6-4-40).
- g. Cut lockbolt collars and remove lockbolts from lower support angles (Figure 2-5-17-1, Detail A).
- h. Carefully drill out and remove remaining rivets in lower support angles and rivets securing former to structure.
- i. Remove forward support assembly.

- j. Make sure area is clean and free of foreign material.

2-5-17.1.2 Repair.

Remove and replace damaged parts as required (Figure 2-5-17-2, Sheet 1 and Sheet 2). If plates are replaced, refer to drill oil cooler installation holes (PARA 2-5-17.1.4).

2-5-17.1.3 Install.

CAUTION

Forward oil cooler support will be misaligned if target wire is not centered. Make sure target wire is centered in front adapter plate.

- a. Install target wire in front adapter plate.
- b. Install front adapter plate of tail rotor drive shaft alignment fixture set to main transmission tail takeoff flange (Figure 2-5-17-3).
- c. Install rear adapter plate of tail rotor drive shaft alignment fixture set to intermediate gear box input flange.

CAUTION

Target wire will break if overtightened. Use care when turning rear adapter plate hand knob to tighten target wire.

NOTE

To prevent inadvertent tracking of target wire within forward support alignment fixture, make sure rear adapter plate hand knob does not bottom out while adjusting target wire tension.

- d. Install target wire in rear adapter plate. Turn hand knob on rear adapter plate to tighten target wire.
- e. Seal for dissimilar metals insulation during installation, using sealing compound, Item 292, Appendix D (PARA 2-6-1).

2-5-17-2 Change 9

- f. Position oil cooler forward support alignment fixture on oil cooler forward support assembly and secure with installation bolts.
- g. Carefully position assembled alignment fixture and oil cooler forward support on structure at STA 379.0.
- h. Carefully place target wire in hole of front and rear target (Figure 2-5-17-3, Detail A).
- i. While making sure target wire remains centered in hole of front and rear targets, temporarily secure oil cooler support to structure using clecos.
- j. Insert required lockbolt pins in lower support angles between clecos and install collars from inside helicopter structure (Figure 2-5-17-1, Detail A and Figure 2-5-17-2, Sheet 1).
- k. Install required rivets between clecos attaching former and lower support angle to structure.
- l. Remove clecos and install remaining rivets and lockbolts.
- m. Turn hand knob to relieve tension on target wire. Remove target wire (Figure 2-5-17-3).
- n. Remove hardware securing front adapter plate to main transmission tail takeoff flange. Remove adapter plate.
- o. Remove hardware securing rear adapter plate to intermediate gear box input flange. Remove adapter plate.
- p. Remove bolts securing oil cooler forward support alignment fixture to oil cooler forward support assembly. Remove alignment fixture.
- q. Remove tools, clean up repair area, and check parts for secure attachment.
- r. Touch up paint as required (PARA 2-6-5).
- s. Apply fillet of sealing compound, Item 292, Appendix D, around periphery of forward oil cooler support (PARA 2-6-1).
- t. Install oil cooler (PARA 6-4-50), tail rotor drive shaft section II (PARA 6-4-39), and tail rotor drive shaft section III (PARA 6-4-40).

- u. Install oil cooler compartment access doors (PARA 2-4-113).
- v. Install No. 1 and No. 2 stabilator amplifiers (TM 11-70-23).
- w. Perform oil cooler vibration check (TM 1-70-VIB).
- x. Install soundproofing panels (PARA 2-4-69).
- y. Install rear passenger's seats (PARA 2-4-45).

2-5-17.1.4 Drill Oil Cooler Installation Holes.

- a. Remove oil cooler compartment access doors (PARA 2-4-113).
- b. Remove oil cooler (PARA 6-4-50), tail rotor drive shaft section II (PARA 6-4-39), and tail rotor drive shaft section III (PARA 6-4-40).

CAUTION

- Oil cooler will be misaligned if installation holes in oil cooler forward support are drilled improperly. Make sure oil cooler forward support alignment fixture is used to locate installation holes.
 - Forward oil cooler support will be misaligned if target wire is not centered. Make sure target wire is centered in front adapter plate.
- c. Install target wire in front adapter plate.
 - d. Install front adapter plate to main transmission tail takeoff flange (Figure 2-5-17-3).
 - e. Install rear adapter plate to intermediate gear box input flange.

CAUTION

Target wire will break if overtightened. Use care when turning hand knob to tighten target wire.

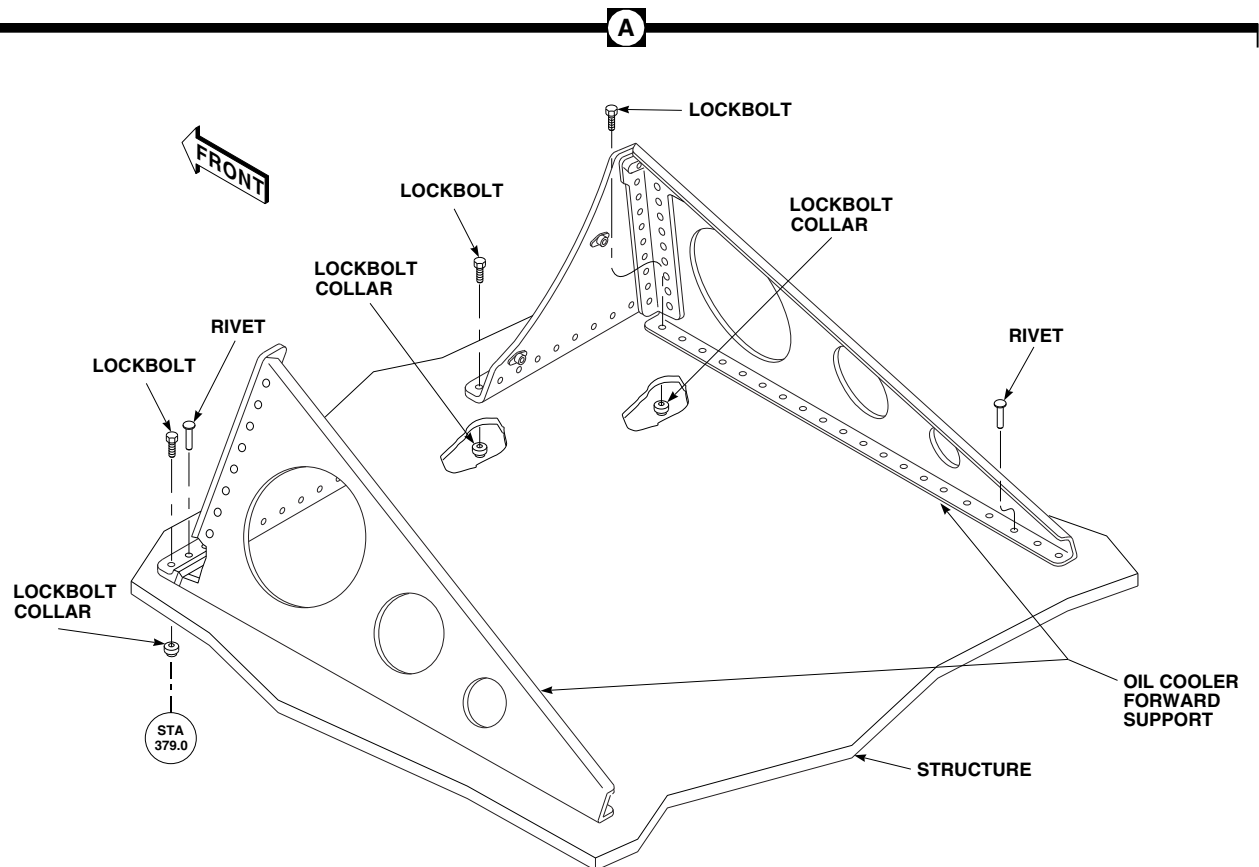
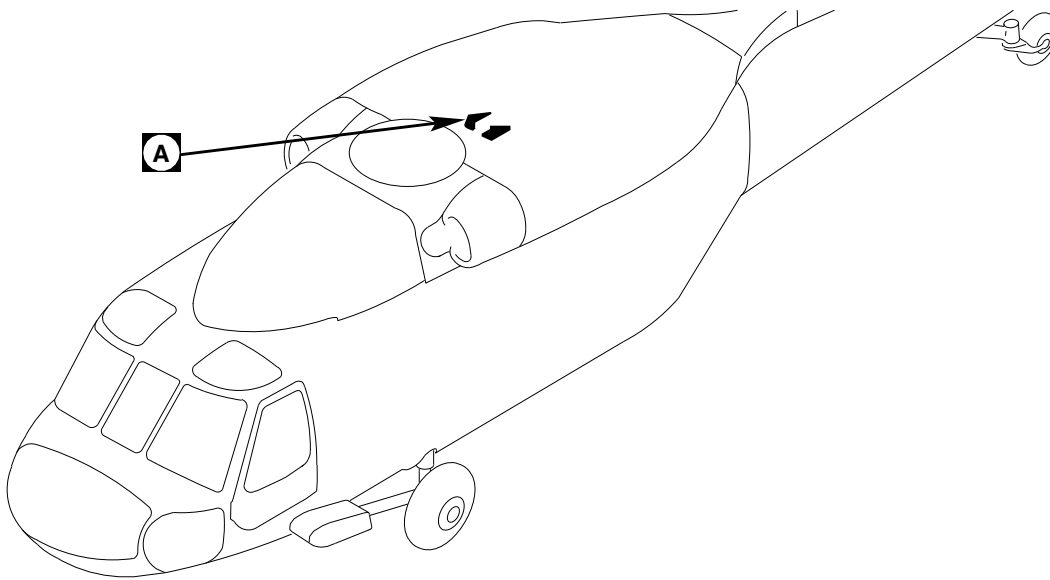
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FIGURE 2-5-17-1. REPLACE OIL COOLER FORWARD SUPPORT (BENCH)

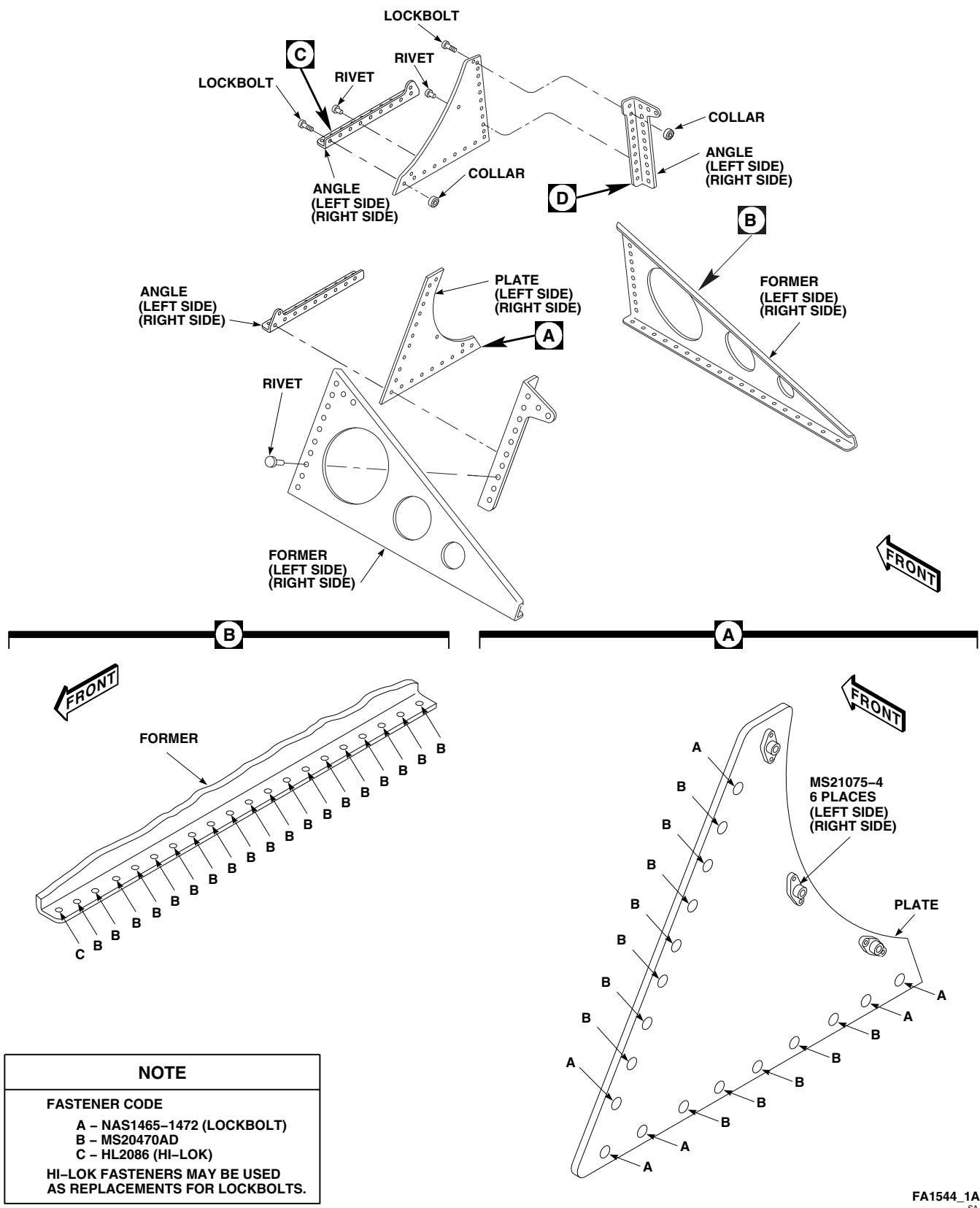


FIGURE 2-5-17-2. REPAIR OIL COOLER FORWARD SUPPORT (BENCH) (SHEET 1 OF 2)

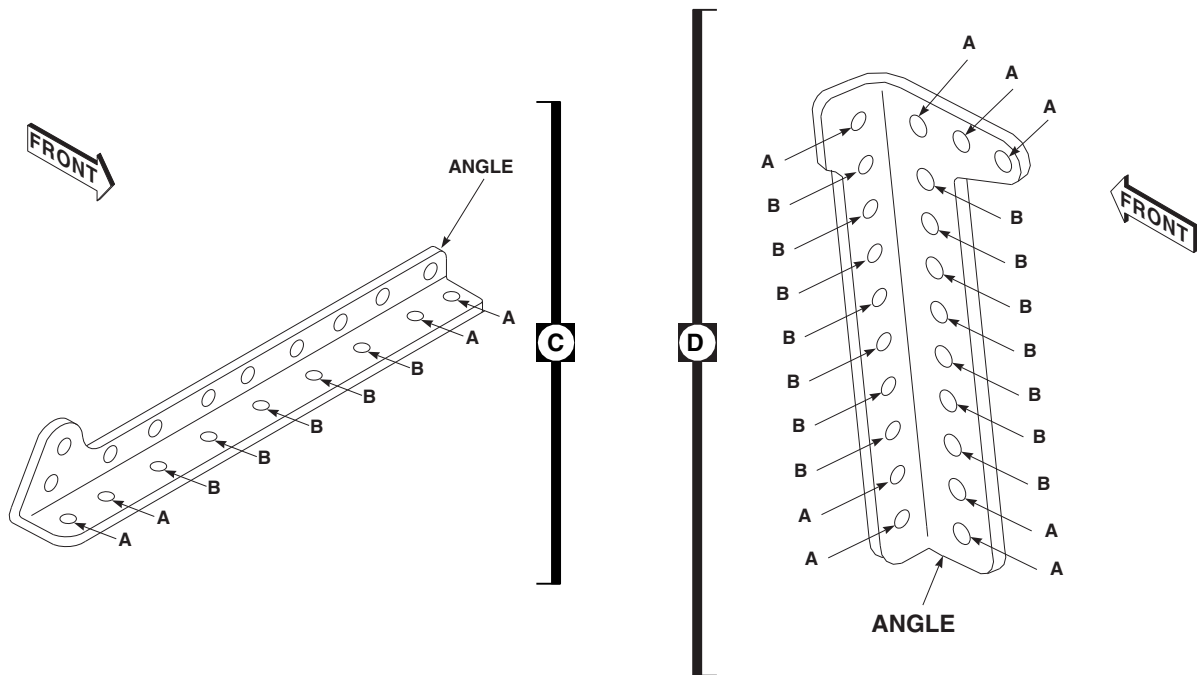
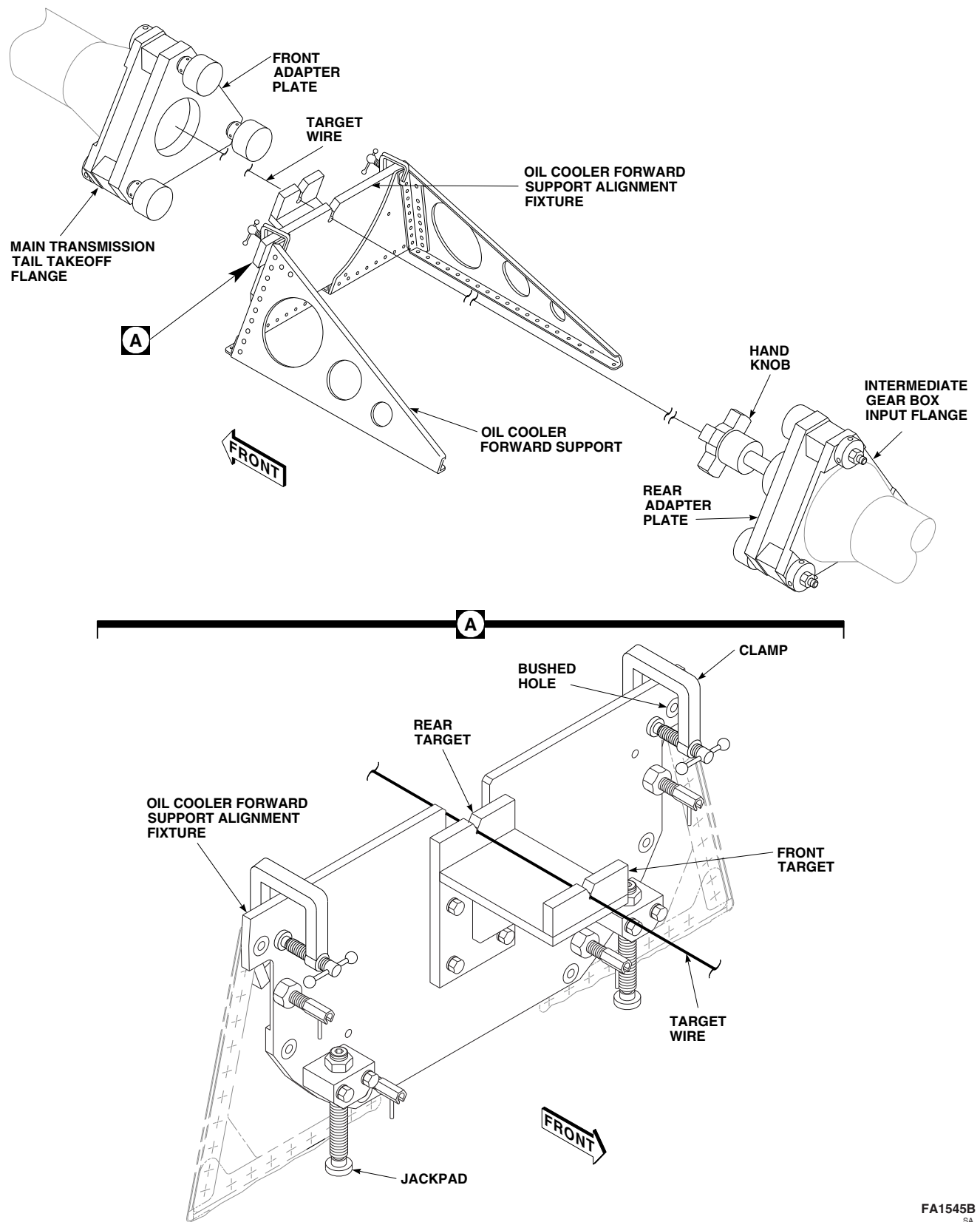
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FIGURE 2-5-17-2. REPAIR OIL COOLER FORWARD SUPPORT (BENCH) (SHEET 2 OF 2)

NOTE

To prevent inadvertent tracking of target wire within forward support alignment fixture, make sure rear adapter plate hand knob does not bottom out while adjusting target wire tension.

- f. Install target wire in rear adapter plate. Turn hand knob on rear adapter plate to tighten target wire.
- g. Carefully place oil cooler forward support alignment fixture against plates of oil cooler forward support and lightly clamp alignment fixture to plates.
- h. If only one plate was replaced, perform the following:
 - (1) Using a combination of jackpad adjustments, for vertical positioning, and gentle taps with nonmetallic mallet, for horizontal positioning, align existing installation holes in plate with bushed holes in alignment fixture.
- i. If both plates were replaced, perform the following:
 - (2) While making sure target wire is centered in front and rear targets, secure alignment fixture to plate using installation bolts.
 - (3) Drill three holes in plate through bushed holes in alignment fixture using drill with 17/64-inch diameter drill bit. **HOLE DIAMETER SHALL BE +0.004-INCH/-0.001-INCH FOR A 17/64-INCH DRILLED HOLE.**
 - (4) Remove bolts securing alignment fixture to plate. Remove alignment fixture.
 - (5) Using rivets, secure nutplates to plate.



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FIGURE 2-5-17-3. ALIGN OIL COOLER FORWARD SUPPORT (BENCH)

- (2) Drill three holes in each plate through bushed holes in alignment fixture using drill with 17/64-inch diameter drill bit. **HOLE DIAMETER SHALL BE +0.004-INCH/-0.001-INCH FOR A 17/64-INCH DRILLED HOLE.**
- (3) Remove clamps securing alignment fixture to oil cooler forward support assembly. Remove alignment fixture.
- (4) Using rivets, secure nutplates to plate.
- j. Turn hand knob to relieve tension on target wire. Remove target wire.
- k. Remove hardware securing front adapter plate to main transmission tail takeoff flange. Remove adapter plate.
- l. Remove hardware securing rear adapter plate to intermediate gear box input flange. Remove adapter plate.
- m. Remove tools, clean up repair area, and check parts for secure attachment.
- n. Touch up paint as required (PARA 2-6-5).
- o. Install oil cooler (PARA 6-4-50), tail rotor drive shaft section II (PARA 6-4-39), and tail rotor drive shaft section III (PARA 6-4-40).
- p. Install oil cooler compartment access doors (PARA 2-4-113).
- q. Perform oil cooler vibration check (TM 1-70-VIB).
- r. Install soundproofing panels (PARA 2-4-69).
- s. Install rear passenger's seats (PARA 2-4-45).

2-5-18 OIL COOLER REAR SUPPORT (BENCH).

PARAGRAPH BREAKDOWN

	PAGE
2-5-18.1 Replace Oil Cooler Rear Support (BENCH)	2-5-18-1

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Blind Rivet Gun
- Clecos, No. 40
- Drill
- Drill Bit, No. 24 (0.152-inch diameter)
- Drill Bit, 17/64-inch diameter (0.2656-inch diameter)
- Pin Punch, 1/8-inch diameter drive
- Powertrain Repairer’s Toolkit
- Tail Drive Shaft Support Alignment Kit, 70350-05000-012, T-046

Materials/Parts

- Epoxy Coating Kit, Item 126A, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-3-20
- PARA 1-3-21
- PARA 2-4-45
- PARA 2-4-69
- PARA 2-6-1
- PARA 2-6-5
- PARA 6-4-38
- PARA 6-4-39
- PARA 6-4-50

2-5-18.1 Replace Oil Cooler Rear Support (BENCH).

2-5-18.1.1 Remove.

- Defuel main fuel tanks (PARA 1-3-20 or PARA 1-3-21).
- Remove rear passenger’s seats (PARA 2-4-45).
- Remove soundproofing panel from left and right stowage compartments and center structure (PARA 2-4-69).

- Remove oil cooler fan (PARA 6-4-50).
- Remove tail rotor drive shaft section I (PARA 6-4-38).
- Remove tail rotor drive shaft section II (PARA 6-4-39).
- Unlock studs securing left and right panels on top of fuel cells. Remove panels.
- Remove screws and front and top panels from center structure.

- i. From top side, using drill with No. 24 drill bit and 1/8-inch diameter drive pin punch, carefully drill out and remove 22 rivets securing rear former and side formers to skin (Figure 2-5-18-1, Detail A). Remove support assembly.
- j. Clean up rivet debris.

2-5-18.1.2 Repair.

Inspect components for damage. If damaged, replace components as required.

2-5-18.1.3 Install.

CAUTION

Do not drill four upper holes in support used to attach support housing until support is installed. To avoid damage to equipment through misalignment, these holes must be drilled using tail drive shaft support alignment fixture.

- a. Seal for dissimilar metals insulation during installation, using epoxy coating kit, Item 126A, Appendix D (PARA 2-6-1).
- b. Position rear support on skin at STA 410.54 and secure in place with No. 40 clecos (Figure

2-5-18-1, Detail A). Perform shaft support alignment and shimming for STA 410.54.

- c. Install rivets in between clecos using blind rivet gun. Remove clecos and install remaining rivets.
- d. Set up tail drive shaft support alignment kit and using drill with 17/64-inch diameter drill bit, drill four holes in support.
- e. Remove tools and make sure area is clean and free of foreign material.
- f. Remove and store tail drive shaft support alignment kit.
- g. Touch up paint as required (PARA 2-6-5).
- h. Using sealing compound, Item 292, Appendix D, seal for watertightness.
- i. Install oil cooler fan (PARA 6-4-50).
- j. Install tail rotor drive shaft section I (PARA 6-4-38).
- k. Install tail rotor drive shaft section II (PARA 6-4-39).
- l. Install center structure and stowage compartment soundproofing panels (PARA 2-4-69).
- m. Install rear passenger's seats (PARA 2-4-45).

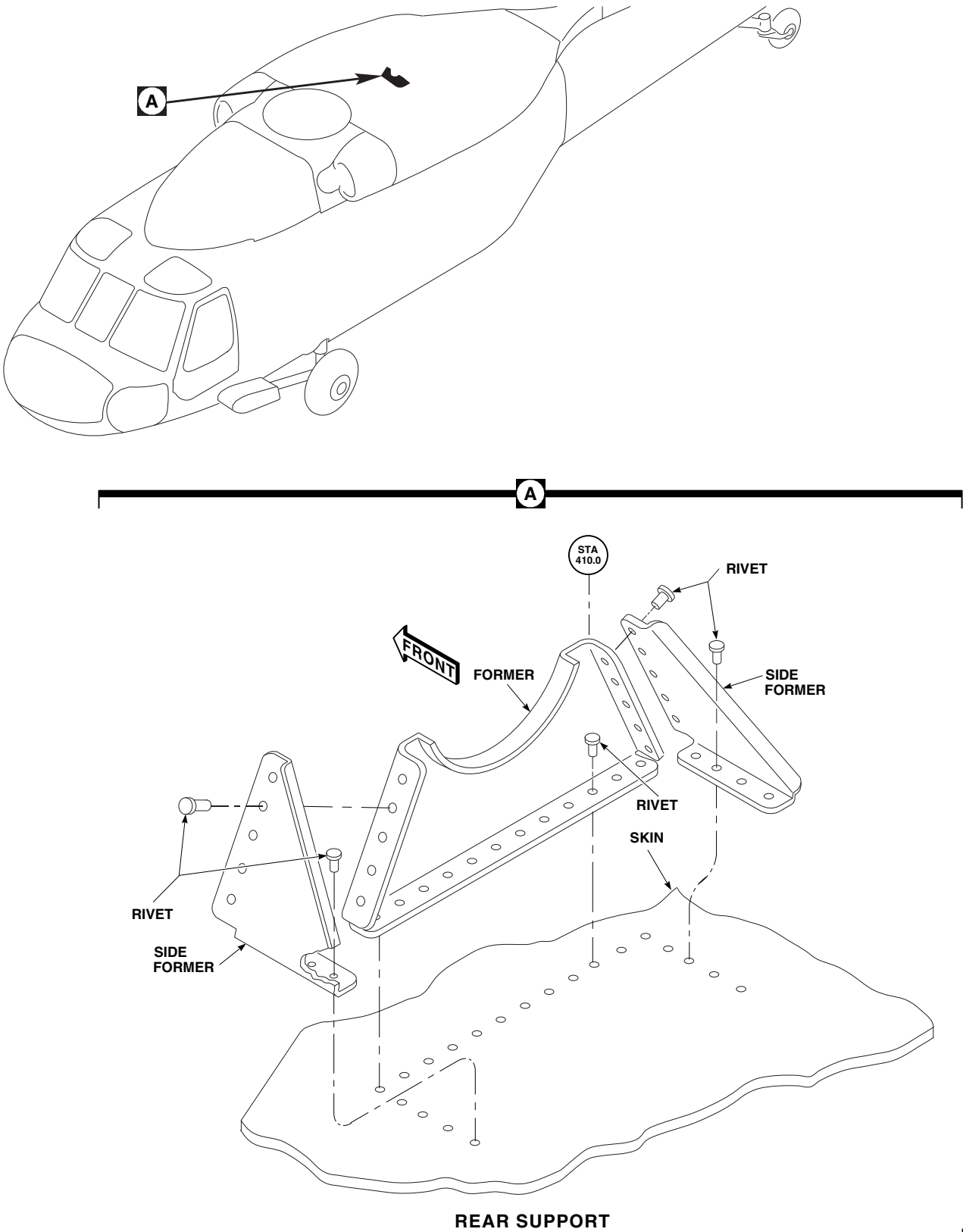


FIGURE 2-5-18-1. REPLACE OIL COOLER REAR SUPPORT (BENCH)

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2-5-19 TAIL ROTOR DRIVE SHAFT SUPPORT (BENCH).

PARAGRAPH BREAKDOWN

		PAGE			PAGE
2-5-19.1	Replace Tail Rotor Drive Shaft Support (One-Piece Configuration) (BENCH)	2-5-19-2	2-5-19.2	Replace Tail Rotor Drive Shaft Support (Two-Piece Configuration) (BENCH)	2-5-19-8

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Clamps
- Clecos, No. 40
- Drill
- Drill Bit, Letter F (0.2570-inch diameter)
- Drill Bit, No. 21 (0.159-inch diameter)
- Drill Bit, 17/64-inch diameter (0.2656-inch diameter)
- Rivet Gun
- Tail Rotor Drive Shaft Alignment Fixture Set, 70700-20480-041
- Target Wire

Materials/Parts

- Shim Kit, 70070-20232-011

References

- PARA 1-3-20
- PARA 1-3-21
- PARA 2-4-45
- PARA 2-4-69
- PARA 2-6-1
- PARA 2-6-2
- PARA 2-6-5
- PARA 6-4-38
- PARA 6-4-39
- PARA 6-4-40
- PARA 6-4-50
- TM 1-70-VIB

2-5-19.1 Replace Tail Rotor Drive Shaft Support (One-Piece Configuration) (BENCH).

NOTE

Replacement of tail rotor drive shaft supports at STA 410.54, STA 470.98, STA 531.43, and STA 591.88, is the same.

2-5-19.1.1 Remove.

NOTE

One and two-piece configuration installations of tail rotor drive shaft supports can be intermixed on a helicopter. Components of a configuration can not be installed with components of another configuration.

- a. Remove rear passenger's seats (PARA 2-4-45).
- b. Remove left or right rear soundproofing panel (PARA 2-4-69).
- c. Remove cargo netting from storage compartment.
- d. Open tail rotor drive shaft cover.
- e. Remove oil cooler (PARA 6-4-50).
- f. Remove tail rotor drive shaft section I (PARA 6-4-38).
- g. Remove tail rotor drive shaft section II (PARA 6-4-39).
- h. Remove tail rotor drive shaft section III (PARA 6-4-40).
- i. Have assistant inside transition section or tail cone to remove lockbolts, buck rivets, and clean up debris.
- j. Cut collars and remove lockbolt pins and rivets securing support angles to support and upper plating (Figure 2-5-19-1, Sheet 1).
- k. Remove rivets securing tail rotor drive shaft support to upper plating.

1. Clean up debris.

2-5-19.1.2 Inspect.

- a. Inspect support angles, tail rotor drive shaft support, bracket, and upper plating for damage (PARA 2-6-1). If damaged, repair components as required (PARA 2-6-2).
- b. Inspect holes in viscous damper bearing mount for elongation. **HOLE DIAMETER SHALL BE +0.004/-0.001-INCH FOR A 17/64-INCH DRILLED HOLE.** If one hole is beyond limit, replace mount.

2-5-19.1.3 Align Shaft Supports.

- a. Defuel main fuel tanks (PARA 1-3-20 or PARA 1-3-21).
- b. If tail drive shaft support has been removed at STA 531.43 or STA 591.88, measure helicopter skin thickness in area of removed support as follows:
 - (1) If helicopter thickness is 0.025-inch, install, 70070-20266-011 between STA 525.0 and STA 545.0.
 - (2) Install 70070-20266-012 between STA 585.0 and STA 605.0.
- c. Seal for dissimilar metals insulation during installation of support angles, tail rotor drive shaft support, shims, and fasteners (PARA 2-6-1).

CAUTION

Tail rotor drive shaft supports will be misaligned if installed improperly. Make sure tail rotor drive shaft support alignment tool is used to locate tail rotor drive shaft supports.

NOTE

Shimming procedures are different at each station location. If base of support angle extends beyond shim, add filler shims and radius blocks as required.

- d. Position support angles on supports for STA 410.54, STA 470.98, STA 531.43, and STA 591.88. Hold each angle in place with two No. 40 clecos.
- e. Position supports on upper plating at STA 410.54, STA 470.98, STA 531.43, and STA 591.88. Hold in place with one No. 40 cleco in center hole of support angle and two No. 40 clecos in support.



Front oil cooler support will be misaligned if target wire is not centered. Make sure target wire is centered in front adapter plate.

- f. Install target wire of tail rotor drive shaft alignment fixture set in front adapter plate.
- g. Install front adapter plate to main transmission tail takeoff flange (Figure 2-5-19-1, Sheet 1).
- h. Install rear adapter plate to intermediate gear box input flange.



Target wire will break if overtightened. Use care when turning hand knob to tighten target wire.

NOTE

To prevent inadvertent tracking of target wire within forward support alignment fixture, make sure rear adapter plate hand knob does not bottom out while adjusting target wire tension.

- i. Install target wire in rear adapter plate. Turn hand knob on rear adapter plate to tighten target wire.
- j. Loosely clamp target plate to front side of tail rotor drive shaft support.

- k. Position target plate as follows:

- (1) Install plate so shoulders of plate line up with top of tail rotor drive shaft support.
- (2) Line up drill bushings in target plate with tail rotor drive shaft support holes if previously drilled. Clamp in place.
- (3) Position target wire in hole of target plate.
- (4) Check wire in hole of target plate for proper position. **WIRE MUST BE IN CENTER OF REAR TARGET AND NOT DEFLECT OFF EDGE OF FRONT TARGET** (Figure 2-5-19-1, Sheet 2, Detail A).
- (5) If wire is in proper position at all station positions, proceed to PARA 2-5-19.1.7.

2-5-19.1.4 Shim STA 410.54 Support.

- a. If wire is not in proper position at STA 410.54, perform the following:
 - (1) Unclamp and remove target plate.
 - (2) Remove support angles and support.
 - (3) Position replacement support.
 - (4) Install target plate so shoulders of plate line up with top of support. Clamp in place (Figure 2-5-19-1, Sheet 1).
 - (5) Move support as required and position target wire in target plate hole.

NOTE

- Use doubler, 70070-20232-101, if shimming up to 0.093-inch is required.
- Use doubler, 70070-20232-101, and add one thickness of shims, 70070-20232-102 and 70070-20232-103, if shimming up to 0.125-inch is required.
- Use doubler, 70070-20232-101, and add one thickness of shims, 70070-20232-

104 and 70070-20232-105, if shimming up to 0.156-inch is required.

- (6) Position doubler and shims under support as required to position target wire in target plate hole.
- (7) Mark holes on doubler, shims, and support through holes in upper plating as required. Remove doubler, shims, support, and target plate.
- (8) Drill holes in doubler, shims, and support using drill with No. 21 drill bit. Deburr holes.
- (9) Position doubler, shims, and support on upper plating. Hold in place with clecos.
- (10) Install target plate so shoulders of plate line up with top of support. Clamp in place.
- (11) Make sure target wire is positioned correctly in target plate hole.
- (12) Position shims and support angles on upper plating and support. Mark holes on shims, support angles, and support. Remove support angles and shims.
- (13) Drill holes in shims, support angles, and support using drill with No. 21 drill bit. Deburr holes.
- (14) Position shims and support angles on upper plating and support. Hold in place with clecos.
- (15) Proceed to PARA 2-5-19.1.7.

2-5-19.1.5 Shim STA 470.98 Support.

- a. If wire is not in proper position at STA 470.98, perform the following:
 - (1) Unclamp and remove target plate.
 - (2) Remove support angles and support.
 - (3) Position replacement support.

- (4) Install target plate so shoulders of plate line up with top of support. Clamp in place (Figure 2-5-19-1, Sheet 1).
- (5) Move support as required and position target wire in target plate hole.

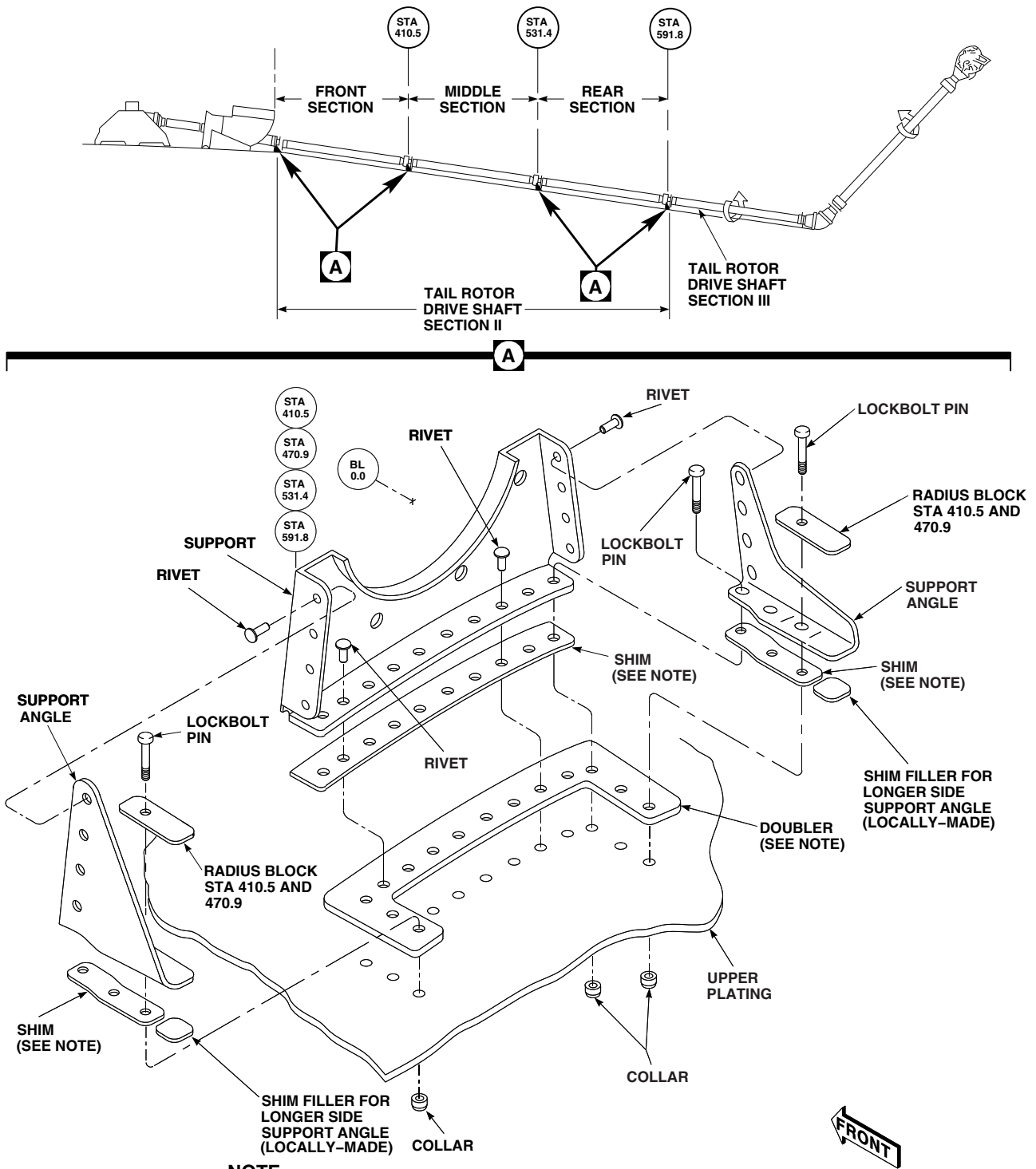
NOTE

Use one thickness of shims, 70070-20232-106 and 70070-20232-107, if shimming up to 0.063-inch is required.

- (6) Position shim under support as required to position target wire in target plate hole.
- (7) Mark holes on shim and support through holes in upper plating as required. Remove shim, support, and target plate.
- (8) Drill holes in shim and support using drill with No. 21 drill bit. Deburr holes.
- (9) Position shim and support on upper plating. Hold in place with clecos.
- (10) Install target plate so shoulders of plate line up with top of support. Clamp in place.
- (11) Make sure target wire is positioned correctly in target plate hole.
- (12) Position shims and support angles on upper plating and support. Mark holes on shims and support angles and support. Remove support angles and shims.
- (13) Drill holes in shims, support angle, and support using drill with No. 21 drill bit. Deburr holes.
- (14) Position shims and support angle on upper plating and support. Hold in place with clecos.
- (15) Proceed to PARA 2-5-19.1.7.

2-5-19.1.6 Shim STA 531.43 or STA 591.88 Support.

- a. If wire is not in proper position at STA 531.43 or STA 591.88, perform the following:



NOTE

DOUBLER	70070-20232-101	USED AT STA 410.5 - 0.090 SHIM STOCK
SHIM	70070-20232-102	USED AT STA 410.5 - 0.032 SHIM STOCK
SHIM	70070-20232-103	USED AT STA 410.5 - 0.032 SHIM STOCK
SHIM	70070-20232-104	USED AT STA 410.5 - 0.063 SHIM STOCK
SHIM	70070-20232-105	USED AT STA 410.5 - 0.063 SHIM STOCK
SHIM	70070-20232-106	USED AT STA 470.9 - 0.063 SHIM STOCK
SHIM	70070-20232-107	USED AT STA 470.9, 531.4, AND 591.3 - 0.063 SHIM STOCK
SHIM	70070-20232-108	USED AT STA 591.8 - 0.063 SHIM STOCK

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FIGURE 2-5-19-1. TAIL ROTOR DRIVE SHAFT SUPPORT (ONE-PIECE CONFIGURATION) (BENCH) (SHEET 1 OF 2)

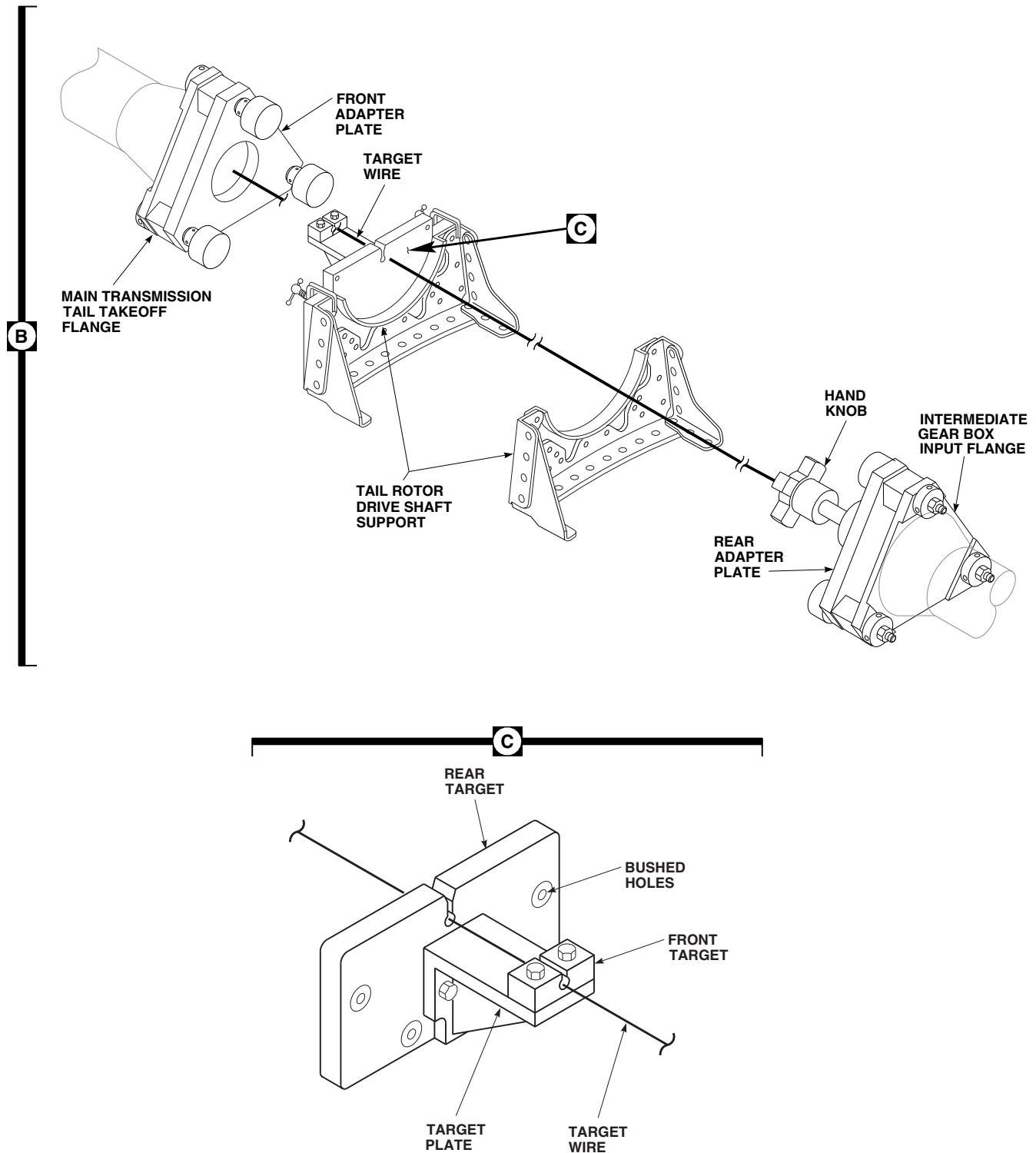
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FIGURE 2-5-19-1. TAIL ROTOR DRIVE SHAFT SUPPORT (ONE-PIECE CONFIGURATION) (BENCH) (SHEET 2 OF 2)

- (1) Unclamp and remove target plate.
- (2) Remove support angles and support.
- (3) Position replacement support.
- (4) Install target plate so shoulders of plate line up with top of support. Clamp in place (Figure 2-5-19-1, Sheet 1).
- (5) Move support as required to position target wire in target plate hole.

NOTE

Use one thickness of shims, 70070-20232-107 and 70070-20232-108, if shimming up to 0.063-inch is required.

- (6) Position shim under support as required to position target wire in target plate hole.
- (7) Mark holes on shim and support through holes in upper plating as required. Remove shim, support, and target plate.
- (8) Drill holes in shim and support using drill with No. 21 drill bit. Deburr holes.
- (9) Position shim and support on upper plating. Hold in place with clecos.
- (10) Install target plate so shoulders of plate line up with top of support. Clamp in place.
- (11) Make sure target wire is positioned correctly in target plate hole.
- (12) Position shims and support angles on upper plating and support. Mark holes on shims and support angles and support. Remove support angles and shims.
- (13) Drill holes in shims, support angle, and support using drill with No. 21 drill bit. Deburr holes.
- (14) Position shims and support angle on upper plating and support. Hold in place with clecos.

- (15) Proceed to PARA 2-5-19.1.7.

2-5-19.1.7 Install Fasteners.

- a. Install support angles (with shims, if required) to upper plating with lockbolt pins and collars. Remove clecos and install remaining lockbolt pins and collars (Figure 2-5-19-1, Sheet 1).
- b. Install support (with shims and doubler, if required) to upper plating with rivets. Remove clecos and install remaining rivets.
- c. Turn hand knob to relieve tension on target wire. Remove target wire.

2-5-19.1.8 Drill Viscous Damper Bearing Holes.



- Tail rotor drive shaft will be misaligned if holes in viscous damper bearing are drilled improperly. Make sure tail rotor drive shaft support alignment fixture is used to locate holes in viscous damper bearing.
- Target wire will break if overtightened. Use care when turning hand knob to tighten target wire.

NOTE

- To prevent inadvertent tracking of target wire within front support alignment fixture, make sure rear adapter plate hand knob does not bottom out while adjusting target wire tension.
 - Drilling procedures are the same for tail rotor drive shaft supports at STA 410.54, STA 470.98, STA 531.43, and STA 591.88.
- a. Install target wire in front and rear adapter plates. Turn hand knob on rear adapter plate to tighten target wire.
 - b. Loosely clamp target plate to front side of tail rotor drive shaft support.

- c. Position target plate so shoulders of plate line up with top of support and target wire passes through center of target plate hole. **TIGHTEN CLAMPS** (Figure 2-5-19-1, Sheet 1).
- d. Drill four holes in tail rotor drive shaft support through drill bushings in target plate using drill with 17/64-inch diameter drill bit (refer to PARA 2-5-19.1.2 for hole tolerance).
- e. Turn hand knob to relieve tension on target wire. Remove target wire.
- f. Remove hardware securing front adapter plate to main transmission tail takeoff flange. Remove adapter plate.
- g. Remove hardware securing rear adapter plate to intermediate gear box input flange. Remove adapter plate.
- h. Loosen clamps attaching target plate to tail rotor drive shaft support. Remove target plate. Deburr holes.
- i. Make sure area is clean and free of foreign material.
- j. Touch up protective finish (PARA 2-6-5).
- k. Install oil cooler (PARA 6-4-50).
- l. Install tail rotor drive shaft section I (PARA 6-4-38).
- m. Install tail rotor drive shaft section II (PARA 6-4-39).
- n. Install tail rotor drive shaft section III (PARA 6-4-40).
- o. Close tail rotor drive shaft covers.
- p. Perform oil cooler vibration check (TM 1-70-VIB).
- q. Install cargo netting in stowage compartment.
- r. Install rear soundproofing panel (PARA 2-4-69).
- s. Install rear passenger's seats (PARA 2-4-45).

2-5-19.2 Replace Tail Rotor Drive Shaft Support (Two-Piece Configuration) (BENCH).

NOTE

Replacement of tail rotor drive shaft supports at STA 410.54, STA 470.98, STA 531.43, and STA 591.88, is the same.

2-5-19.2.1 Remove.

NOTE

One and two-piece configuration installations of tail rotor drive shaft supports can be intermixed on a helicopter. Components of a configuration can not be installed with components of another configuration. Only two-piece configuration installation of tail rotor drive shaft support shall be used to replace an existing configuration.

- a. If oil cooler, 70361-03005-103 through -106, is installed, remove the following:
 - (1) Oil cooler (PARA 6-4-50)
 - (2) Tail rotor drive shaft section II (PARA 6-4-39)
 - (3) Tail rotor drive shaft section III (PARA 6-4-40)
- b. If oil cooler, 70361-03005-107, is installed, remove the following:
 - (1) Tail rotor drive shaft section I (PARA 6-4-38)
 - (2) Tail rotor drive shaft section II (PARA 6-4-39)
 - (3) Tail rotor drive shaft section III (PARA 6-4-40)
- c. Remove rivets securing tail rotor drive shaft support to bracket (Figure 2-5-19-2, Sheet 1, Detail A).
- d. Clean up debris.

2-5-19.2.2 Inspect.

- a. Inspect support angles, tail rotor drive shaft support, bracket, and upper plating for damage (PARA 2-6-1). If damaged, repair components as required (PARA 2-6-2).
- b. Inspect holes in viscous damper bearing mount for elongation. **MAXIMUM HOLE DIAMETER SHALL NOT EXCEED 0.2696-INCH.** If one hole is beyond limit, replace mount.

2-5-19.2.3 Align Shaft Supports.

- a. Defuel main fuel tanks (PARA 1-3-20 or PARA 1-3-21).
- b. If tail rotor drive shaft support was removed at STA 531.43 or STA 591.88, measure helicopter skin thickness in area of removed support as follows:
 - (1) If helicopter skin thickness is 0.025-inch, install, 70212-05005-011, between STA 525.0 and STA 545.0.
 - (2) Install, 70212-05005-012, between STA 585.0 and STA 605.0.
- c. Install tail rotor drive shaft alignment fixture as follows:

CAUTION

Front oil cooler support will be misaligned if target wire is not centered. Make sure target wire is centered in front adapter plate.

- (1) Install target wire of tail rotor drive shaft alignment fixture set in front adapter plate.
- (2) Seal for dissimilar metals insulation during installation of support angles, tail ro-

tor drive shaft support, bracket, and fasteners (PARA 2-6-1).

CAUTION

Tail rotor drive shaft supports will be misaligned if installed improperly. Make sure tail rotor drive shaft support alignment tool is used to locate tail rotor drive shaft supports.

- (3) Install front adapter plate to main transmission tail takeoff flange (Figure 2-5-19-2, Sheet 2, Detail B).
- (4) Install rear adapter plate to intermediate gear box input flange.

CAUTION

Target wire will break if overtightened. Use care when turning hand knob to tighten target wire.

NOTE

To prevent inadvertent tracking of target wire within front support alignment fixture, make sure rear adapter plate hand knob does not bottom out while adjusting target wire tension.

- (5) Install target wire in rear adapter plate. Turn hand knob on rear adapter plate to tighten target wire.
- (6) Loosely clamp target plate to front side of tail rotor drive shaft support.
- d. Position target plate as follows:
 - (1) Install plate so shoulders of plate line up with top of tail rotor drive shaft support.

CAUTION

All rivet pilot holes in tail rotor drive shaft support shall line up with rivet holes in bracket.

- (2) Line up drill bushings in target plate with tail rotor drive shaft support holes if previously drilled. Clamp in place.
- (3) Position target wire in hole of target plate.
- (4) Check wire in hole of target plate for proper position. **WIRE MUST BE IN CENTER OF REAR TARGET AND NOT DEFLECT OFF EDGE OF FRONT TARGET** (Figure 2-5-19-2, Sheet 2, Detail C).
- (5) If wire is in proper position at all station positions, proceed to PARA 2-5-19.2.4.

2-5-19.2.4 Install Fasteners.

- a. Using rivet gun, secure tail rotor drive shaft support to bracket with rivets.
- b. Turn hand knob to relieve tension on target wire. Remove target wire.

2-5-19.2.5 Drill Viscous Damper Bearing Holes.

CAUTION

- Tail rotor drive shaft will be misaligned if holes in viscous damper bearing are drilled improperly. Make sure tail rotor drive shaft support alignment fixture is used to locate holes in viscous damper bearing.
- Target wire will break if overtightened. Use care when turning hand knob to tighten target wire.

NOTE

- To prevent inadvertent tracking of target wire within front support alignment

fixture, make sure rear adapter plate hand knob does not bottom out while adjusting target wire tension.

- Drilling procedures are the same for tail rotor drive shaft supports at STA 410.54, STA 470.98, STA 531.43, and STA 591.88.
- a. Install target wire in front and rear adapter plates. Turn hand knob on rear adapter plate to tighten target wire.
- b. Loosely clamp target plate to front side of tail rotor drive shaft support.
- c. Position target plate so shoulders of plate line up with top of tail rotor drive shaft support and target wire passes through center of target plate hole. **TIGHTEN CLAMPS** (Figure 2-5-19-2, Sheet 2, Detail B).
- d. Drill four holes in tail rotor drive shaft support through drill bushings in target plate using drill with Letter F drill bit (refer to PARA 2-5-19.2.2 for hole tolerance).
- e. Turn hand knob to relieve tension on target wire. Remove target wire.
- f. Remove hardware securing front adapter plate to main transmission tail takeoff flange. Remove adapter plate.
- g. Remove hardware securing rear adapter plate to intermediate gear box input flange. Remove adapter plate.
- h. Loosen clamps securing target plate to tail rotor drive shaft support. Remove target plate. Deburr holes.
- i. Make sure area is clean and free of foreign material.
- j. Touch up protective finish (PARA 2-6-5).
- k. If oil cooler, 70361-03005-103 through -106, was removed, install the following:
 - (1) Oil cooler (PARA 6-4-50)

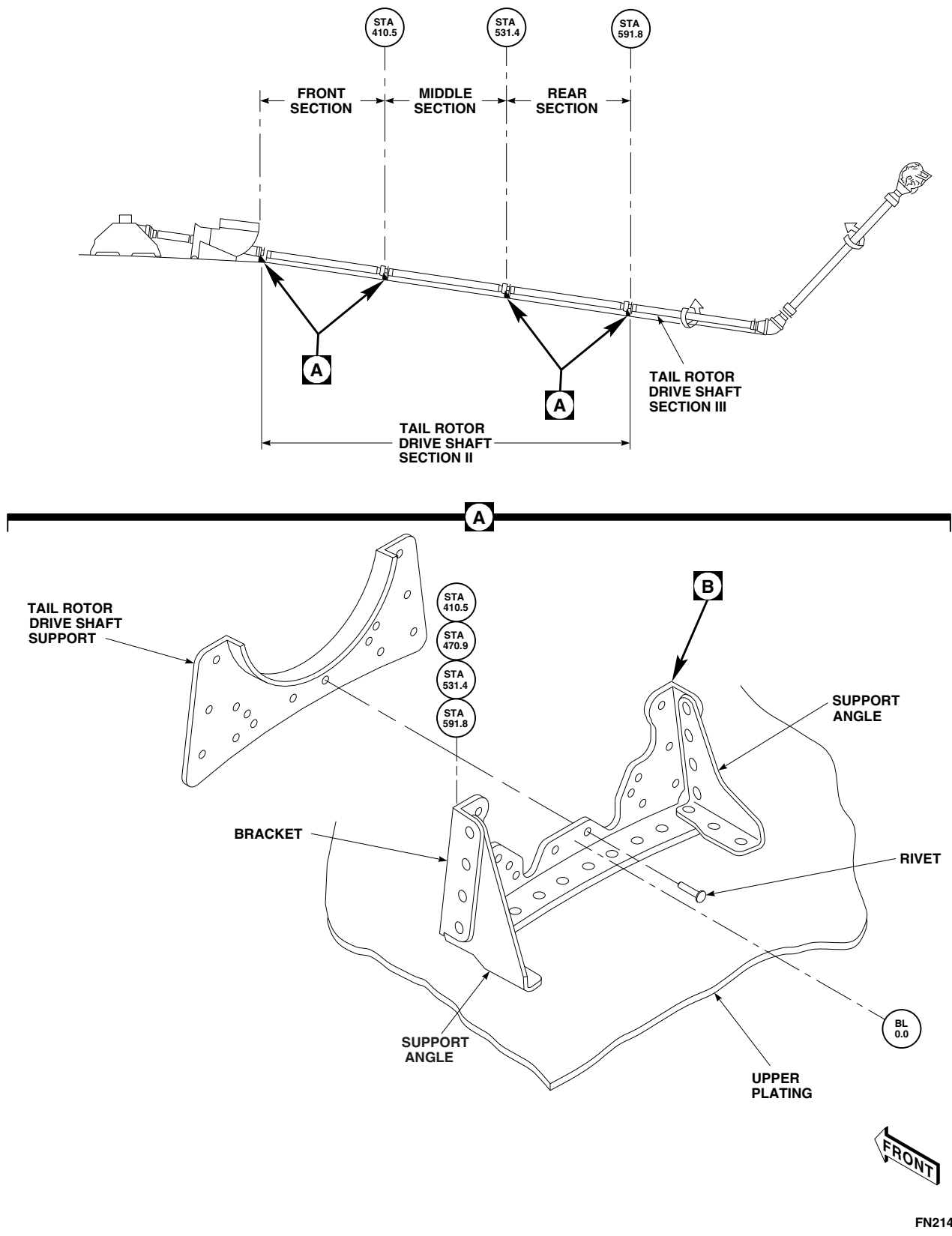
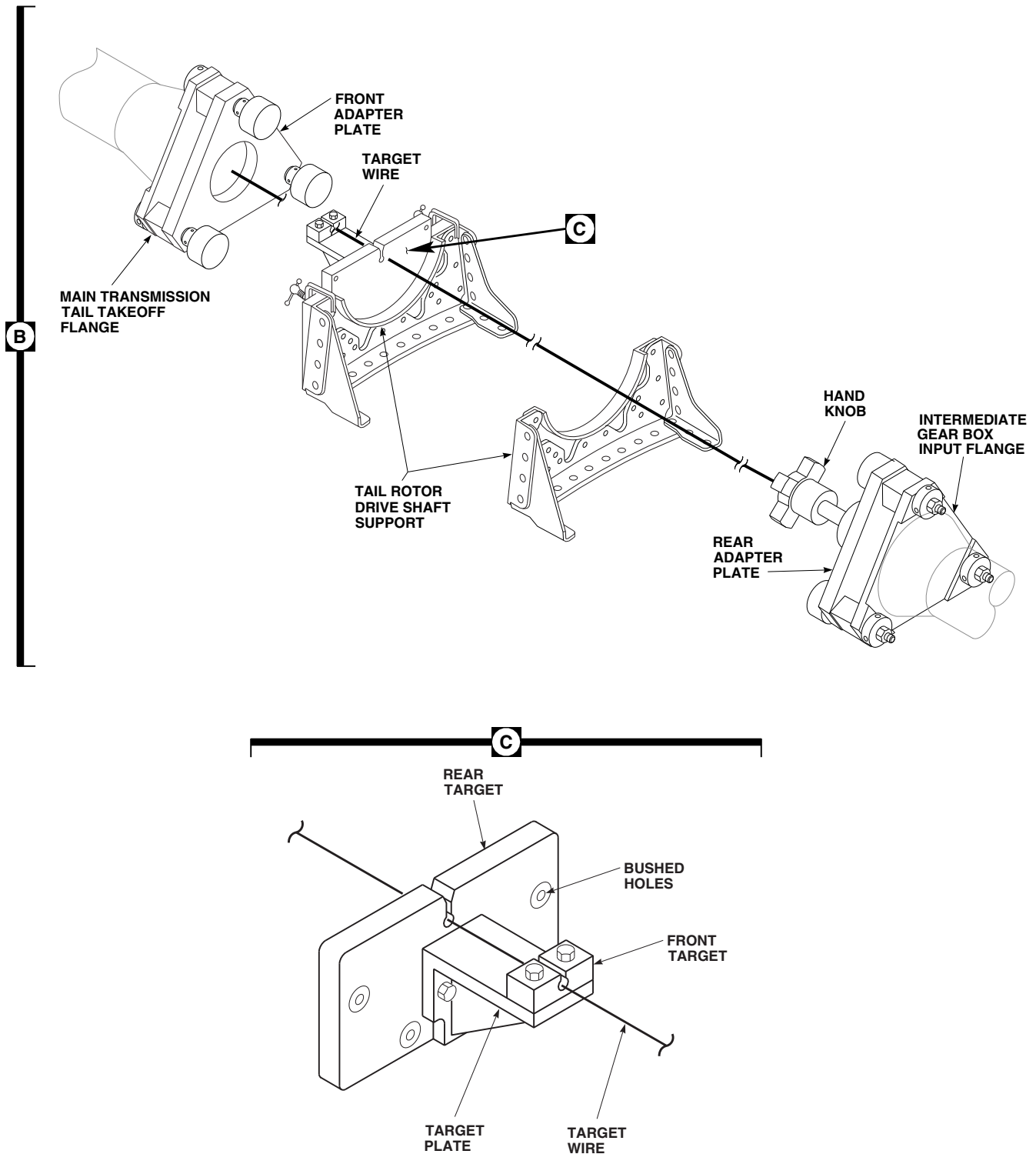


FIGURE 2-5-19-2. TAIL ROTOR DRIVE SHAFT SUPPORT (TWO-PIECE CONFIGURATION) (BENCH) (SHEET 1 OF 2)



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FIGURE 2-5-19-2. TAIL ROTOR DRIVE SHAFT SUPPORT (TWO-PIECE CONFIGURATION) (BENCH) (SHEET 2 OF 2)

(2) Tail rotor drive shaft section II (PARA 6-4-39)

(3) Tail rotor drive shaft section III (PARA 6-4-40)

1. If oil cooler, 70361-03005-107, is installed, install the following:

(1) Tail rotor drive shaft section I (PARA 6-4-38)

(2) Tail rotor drive shaft section II (PARA 6-4-39)

(3) Tail rotor drive shaft section III (PARA 6-4-40)

m. Perform oil cooler vibration check (TM 1-70-VIB).

2-5-20 STABILATOR ELASTOMERIC BEARING (BENCH).

PARAGRAPH BREAKDOWN

	PAGE
2-5-20.1 Replace Stabilator Elastomeric Bearing (BENCH)	2-5-20-1

INITIAL SETUP

Tools

- Arbor Press
- C-Clamp
- Nonmetallic Drift, Locally-Made
- Powertrain Repairer’s Toolkit
- Stabilator Fitting Support, Locally-Made, Figure H-171, Appendix H

Materials/Parts

- Abrasive Cloth, Item 6, Appendix D
- Abrasive Cloth, Item 8, Appendix D

- Adhesive, Item 32, Appendix D
- Adhesive, Item 66, Appendix D
- Cheesecloth, Item 90, Appendix D
- Mat, Reinforcing, Fibrous, Item 216, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Plastic Strip, Item 237A, Appendix D
- Pressure Sensitive Adhesive Tape, Item 247, Appendix D

References

- Appendix D
- Appendix H
- PARA 2-6-5

2-5-20.1 Replace Stabilator Elastomeric Bearing (BENCH).

NOTE

Replacement of both stabilator elastomeric bearings is the same.

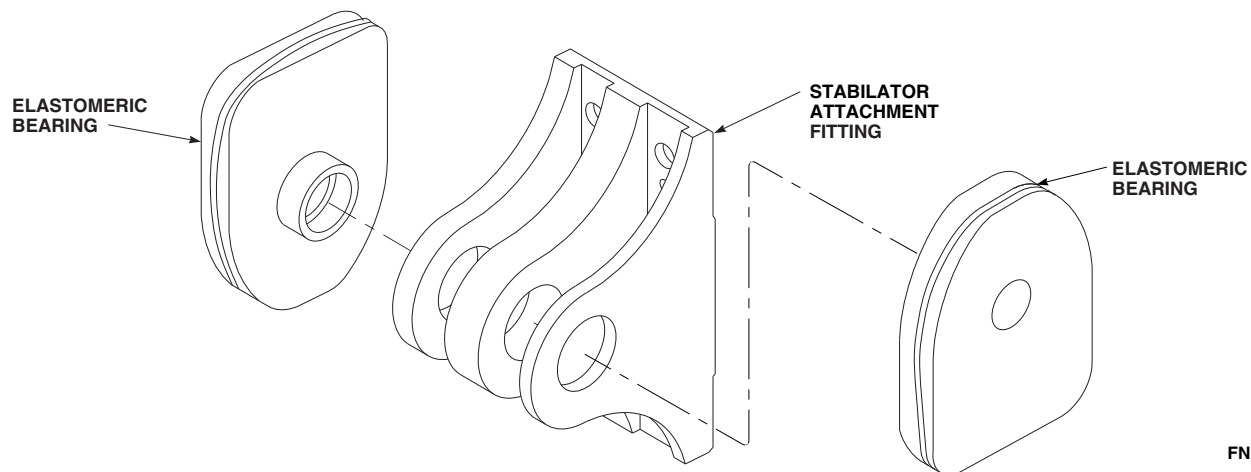
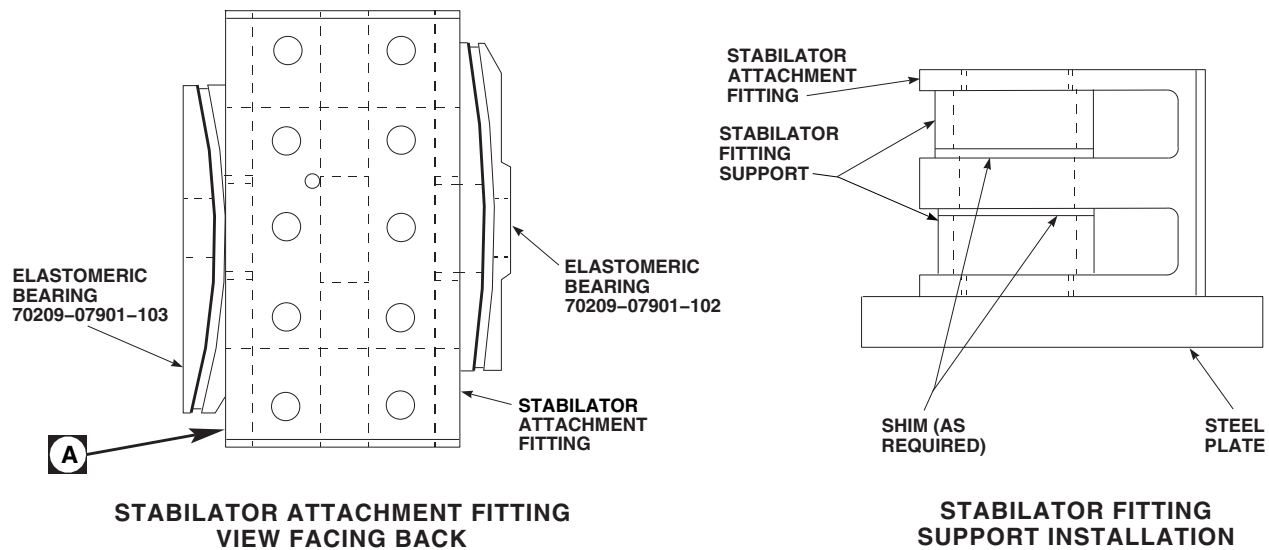
2-5-20.1.1 Remove.

- Place stabilator attachment fitting on workbench so elastomeric bearing is on edge of bench.
- Using nonmetallic drift, alternately tap and twist ends of elastomeric bearing.

- Remove elastomeric bearing from stabilator attachment fitting (Figure 2-5-20-1, Detail A).

2-5-20.1.2 Install.

- If rebonding existing elastomeric bearing to stabilator attachment fitting, perform the following:
 - Using pressure sensitive adhesive tape, Item 247, Appendix D, cover exposed rubber on ends of bearing.
 - Using abrasive cloth, Item 6, Appendix D, carefully remove remaining adhesive

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- from mating surface of fitting and bearing.
- (3) Using cheesecloth, Item 90, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D, clean mating surface of fitting and bearing. Remove masking tape from ends of bearing. Proceed to step c.
- b. If replacing elastomeric bearing, perform the following:
 - (1) Using abrasive cloth, Item 8, Appendix D, carefully remove adhesive from mating surface of stabilator attachment fitting.
 - (2) Using cheesecloth, Item 90, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D, clean mating surface of fitting.
 - c. Apply light coat of adhesive, Item 66, Appendix D, to mating surface of stabilator attachment fitting and elastomeric bearing.
 - d. Allow adhesive to cure at room temperature for at least 1 hour.
 - e. Locally make two stabilator fitting supports (Figure H-171, Appendix H).

CAUTION

Damage to lugs of stabilator attachment fitting will result if stabilator fitting supports are not tightly shimmed. Lugs of fitting will bend. Make sure supports are tightly shimmed before pressing in elastomeric bearing.

- f. Install stabilator fitting supports and shims using plastic strip, Item 237A, Appendix D, between center and outer lugs of stabilator attachment fitting (Figure 2-5-20-1, Detail B).
- g. Mix adhesive, Item 32, Appendix D, per manufacturer's instructions (100 parts by weight of part A to 22 parts by weight of part B) and apply light coat of adhesive to mating surface of stabilator attachment fitting and elastomeric bearing.
- h. Install fibrous reinforcing mat, Item 216, Appendix D, to mating area of stabilator attachment fitting. Cut out mat to diameter of fitting hole and trim to contour of fitting lug.

CAUTION

Damage to liners will occur if elastomeric bearing is not properly seated in stabilator attachment fitting. Liners will become unseated. Use care when pressing bearing in fitting.

NOTE

To properly position elastomeric bearings on stabilator attachment fitting, place fitting on bench with lugs facing down and alignment pin towards top left side. Bearing, 70209-07901-103, will be bonded to side of fitting nearest alignment pin with chamfered end of bearing down and facing rear. Refer to Figure 2-5-20-1, to correctly install bearings.

- i. Position elastomeric bearing on stabilator attachment fitting. Using arbor press, carefully press bearing into fitting. Make sure liners do not become unseated (Figure 2-5-20-1, Detail A).
- j. Remove stabilator attachment fitting from arbor press and clean extra adhesive from around fitting and elastomeric bearing, using cheesecloth, Item 90, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D.
- k. Maintain pressure on stabilator attachment fitting and elastomeric bearing by installing C-clamps on ends of bearing and lugs of fitting.
- l. Allow adhesive to dry at least 24 hours at room temperature or 2 hours at 110° - 130°F (43° - 54°C).
- m. When adhesive is dry, remove C-clamps, stabilator fitting supports, and shims.
- n. Touch up unprotected areas (PARA 2-6-5).

2-5-21 STABILATOR POSITION TRANSMITTER/LIMIT SWITCHES (BENCH).

PARAGRAPH BREAKDOWN

	PAGE
2-5-21.1 Replace Stabilator Position Transmitter/Limit Switches (BENCH)	2-5-21-1

INITIAL SETUP

Tools

- Electrical Connector Maintenance/Repair Kit
- Electrical Repairer’s Toolkit
- Heat Gun
- Insert/Extract Tool

Materials/Parts

- Adhesive, Item 49, Appendix D

- Insulation Sleeving, Electrical, Item 160C, Appendix D
- Insulation Sleeving, Heat-Shrinkable, Item 177, Appendix D
- Lockwire, Item 196, Appendix D
- Tags
- Tiedown Straps, Item 329, Appendix D

References

- Appendix D
- PARA 2-2-1

2-5-21.1 Replace Stabilator Position Transmitter/Limit Switches (BENCH).

NOTE

Replacement of No. 1 and No. 2 up and down stabilator position transmitter/limit switches is the same, except as noted.

2-5-21.1.1 Remove.

- If removing No. 1 down stabilator position transmitter/limit switch, remove screw, washers, nut, and small clamp secured to large clamp on No. 1 down limit switch (Figure 2-5-21-1, Sheet 1, Detail A). If No. 1 down limit

switch is being replaced, remove large and small clamps.

- Remove nut and lockwasher securing limit switch to stabilator position transmitter/limit switch assembly.
- Remove limit switch from stabilator position transmitter/limit switch assembly and remove keyway washer.
- Remove tiedown straps from harness (Figure 2-5-21-1, Sheet 1, Detail A and Sheet 2, Detail B).
- Remove identification strap from harness going to limit switch.

- f. Loosen backshell and slide down harness away from electrical connector (Figure 2-5-21-1, Sheet 2, Detail B and Detail C).
- g. Tag wires. Using insert/extract tool, remove applicable wires and pins from electrical connector (Table 2-5-21-1) (Figure 2-5-21-1, Sheet 2, Detail C).
- h. Remove limit switch and harness from stabilator position transmitter/limit switch assembly (Figure 2-5-21-1, Sheet 1, Detail A).

2-5-21.1.2 Install.

- a. Cut limit switch wires to length shown in Table 2-5-21-1 for applicable limit switch.
- b. Cut four limit switch wires not being used to length (Figure 2-5-21-1, Sheet 3, Detail D).
- c. Install electrical insulation sleeving, Item 160C, Appendix D, on four shorter limit switch wires. Using heat gun, heat-shrink end caps to switch wires.
- d. Slide $\frac{3}{8}$ -inch diameter x 5-inch long heat-shrinkable insulation sleeving, Item 177, Appendix D, over wires and bond insulation sleeving to limit switch with adhesive, Item 49, Appendix D.
- e. Slide $\frac{1}{4}$ -inch-diameter insulation sleeving, Item 177, Appendix D, over wires and into $\frac{3}{8}$ -inch diameter sleeving so they overlap by 2 inches. Bond overlap with adhesive, Item 49, Appendix D.
- f. Install new electrical connector pins on two long wires from limit switch being replaced (Figure 2-5-21-1, Sheet 2, Detail C).
- g. Install limit switch and keyway washer in stabilator position transmitter/limit switch assembly and secure with lockwasher and nut

(Figure 2-5-21-1, Sheet 1, Detail A). Do not lockwire until switch activation point is determined when performing operational check in PARA 2-2-1.

- h. If installing No. 1 down limit switch, install large clamp over limit switch.
- i. Route all limit switch wires in bundle through small clamp secured to large clamp and secure with screw, washers, and nut.
- j. Slide limit switch wires through backshell of electrical connector (Figure 2-5-21-1, Sheet 2, Detail C).
- k. Using insert/extract tool, install pins into applicable electrical connector contacts (Table 2-5-21-1). Remove tags.
- l. Slide backshell on electrical connector. **TIGHTEN BACKSHELL.**
- m. Perform operational check of stabilator position transmitter/limit switch assembly (PARA 2-2-1).
- n. Using heat gun, heat-shrink insulation sleeving to limit switch (Figure 2-5-21-1, Sheet 3, Detail D).
- o. Secure harness with tiedown straps, Item 329, Appendix D. Maintain at least $\frac{1}{4}$ -inch spacing between harness and nuts/cotter pins on stabilator position transmitter (Figure 2-5-21-1, Sheet 1, Detail A and Sheet 2, Detail B).
- p. Install identification strap on harness.
- q. Perform operational check of stabilator position transmitter/limit switch assembly (PARA 2-2-1).
- r. Using lockwire, Item 196, Appendix D, lockwire stabilator position transmitter/limit switch.

Table 2-5-21-1. Wiring Requirements

Limit Switch	Wire Code	Electrical Connector No. - Pin No.	Wire Lengths (Inches)
NO. 1 DOWN	1 - 20	P605 - A	27
	3 - 20	P605 - B	27
NO. 1 UP	4 - 20	P605 - C	23
	6 - 20	P605 - D	23
NO. 2 DOWN	1 - 20	P604 - G	21
	3 - 20	P604 - H	21
NO. 2 UP	4 - 20	P604 - J	25
	6 - 20	P604 - K	25

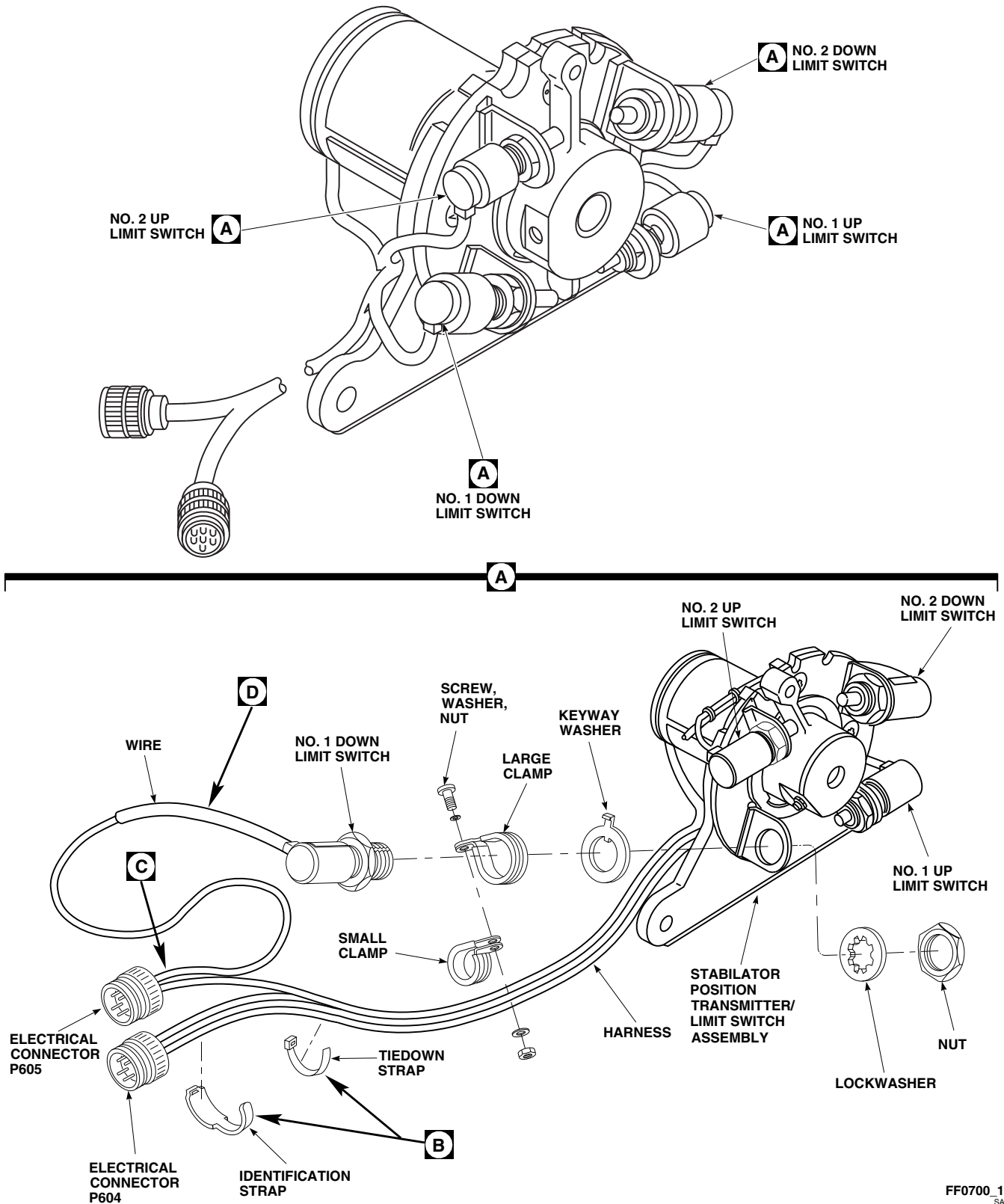
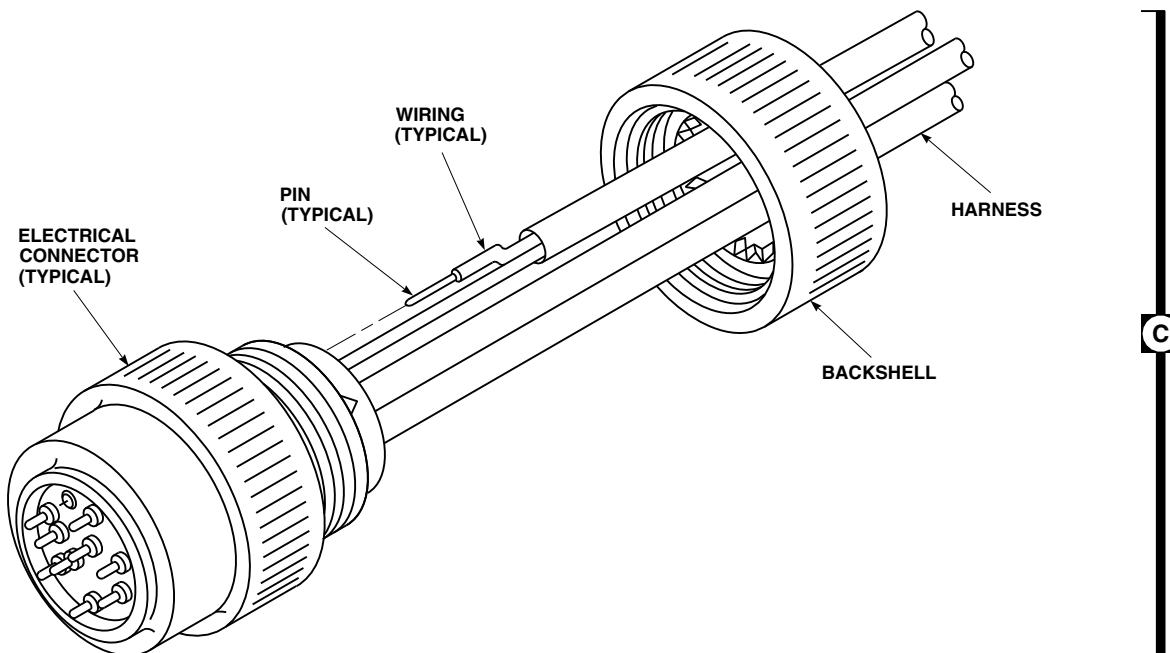
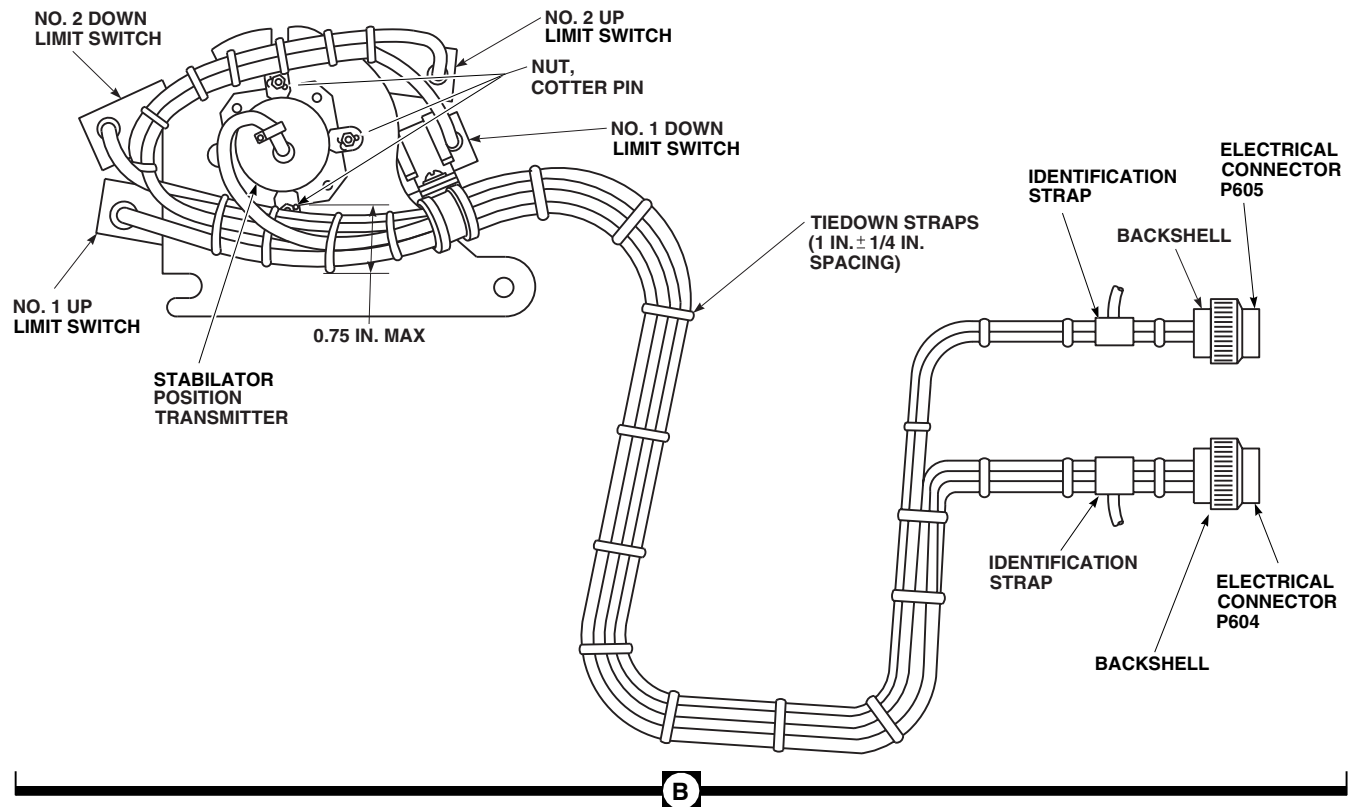
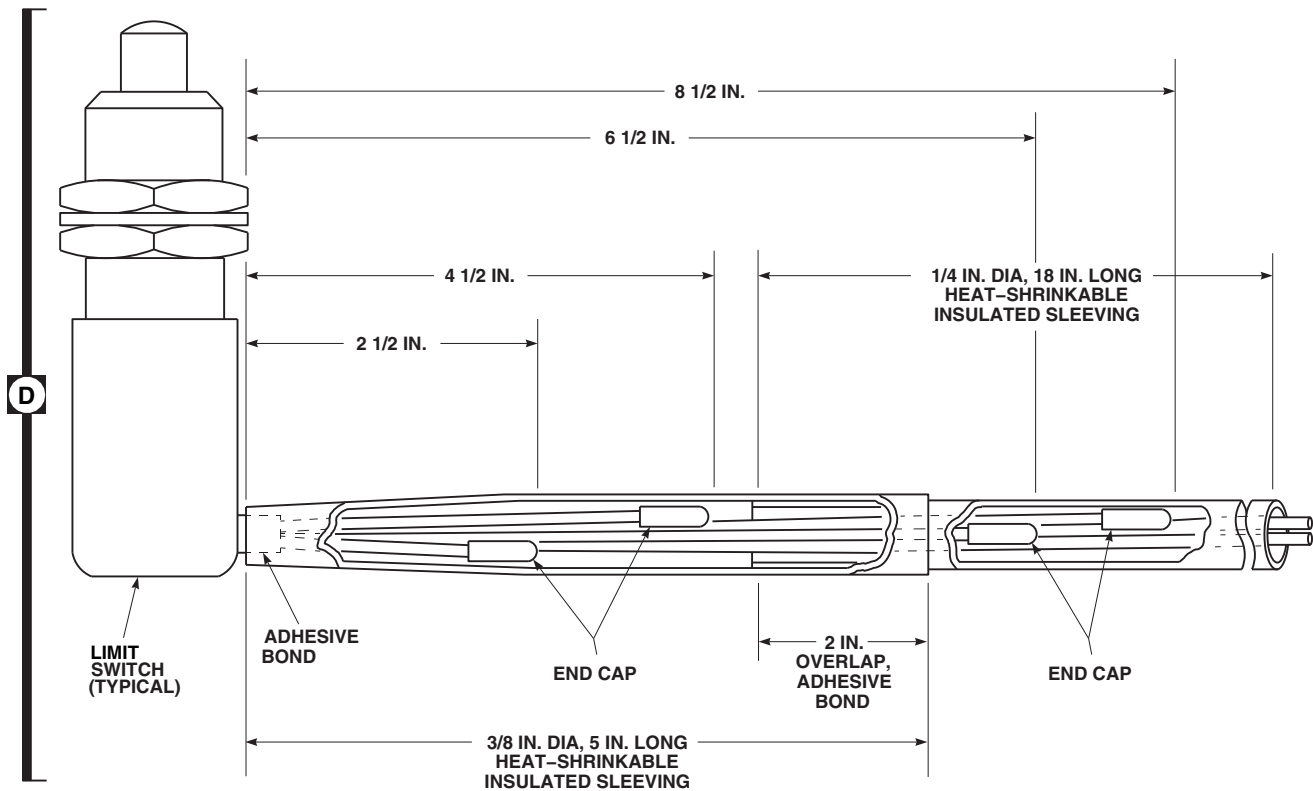


FIGURE 2-5-21-1. REPLACE STABILATOR POSITION TRANSMITTER/LIMIT SWITCHES (SHEET 1 OF 3) (BENCH)



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FIGURE 2-5-21-1. REPLACE STABILATOR POSITION TRANSMITTER/LIMIT SWITCHES (SHEET 2 OF 3) (BENCH)

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**FIGURE 2-5-21-1. REPLACE STABILATOR POSITION TRANSMITTER/LIMIT SWITCHES
(SHEET 3 OF 3) (BENCH)**

2-5-22 STABILATOR POSITION TRANSMITTER (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
2-5-22.1	Replace Stabilator Position Transmitter (BENCH)	2-5-22-1

INITIAL SETUP

Tools

- Electrical Connector Maintenance/Repair Kit
- Electrical Repairer’s Toolkit
- Heat Gun
- Insert/Extract Tool
- Torque Wrench, 0 - 30 in. lbs

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Insulation Sleeving, Heat-Shrinkable, Item 177, Appendix D

- Lockwire, Item 196, Appendix D
- Sealing Compound, Item 280, Appendix D
- Sealing Compound, Item 282, Appendix D
- Tags
- Tetrachloroethylene, Item 324, Appendix D
- Tiedown Straps, Item 329, Appendix D

References

- Appendix D
- PARA 2-2-1

2-5-22.1 Replace Stabilator Position Transmitter (BENCH).

2-5-22.1.1 Remove.

- Loosen jamnut and setscrew on pivot assembly (Figure 2-5-22-1, Sheet 1, Detail A).
- Remove screws, washers, synchro cleats, and nuts, and shim washers (if installed) from stabilator position transmitter (Figure 2-5-22-1, Sheet 1, Detail A and Sheet 2, Detail C).
- Pull stabilator position transmitter away from stabilator position transmitter/limit switch assembly.

- Remove screw, washer, nut, and small clamp from large clamp on No. 1 down limit switch.
- On electrical connector, P604, loosen backshell and slide down harness away from electrical connector (Figure 2-5-22-1, Sheet 1, Detail B and Sheet 2, Detail D).
- Tag wires. Using insert/extract tool, remove wires and pins from contact A through contact F on electrical connector (Figure 2-5-22-1, Sheet 1, Detail B).
- Remove identification strap and tiedown straps from harness (Figure 2-5-22-1, Sheet 1, Detail A and Sheet 2, Detail D).

- h. Remove stabilator position transmitter from stabilator position transmitter/limit switch assembly.

2-5-22.1.2 Inspect.

Visually inspect mechanical parts (positioner, retainer, and baseplate) for bends or distortions.

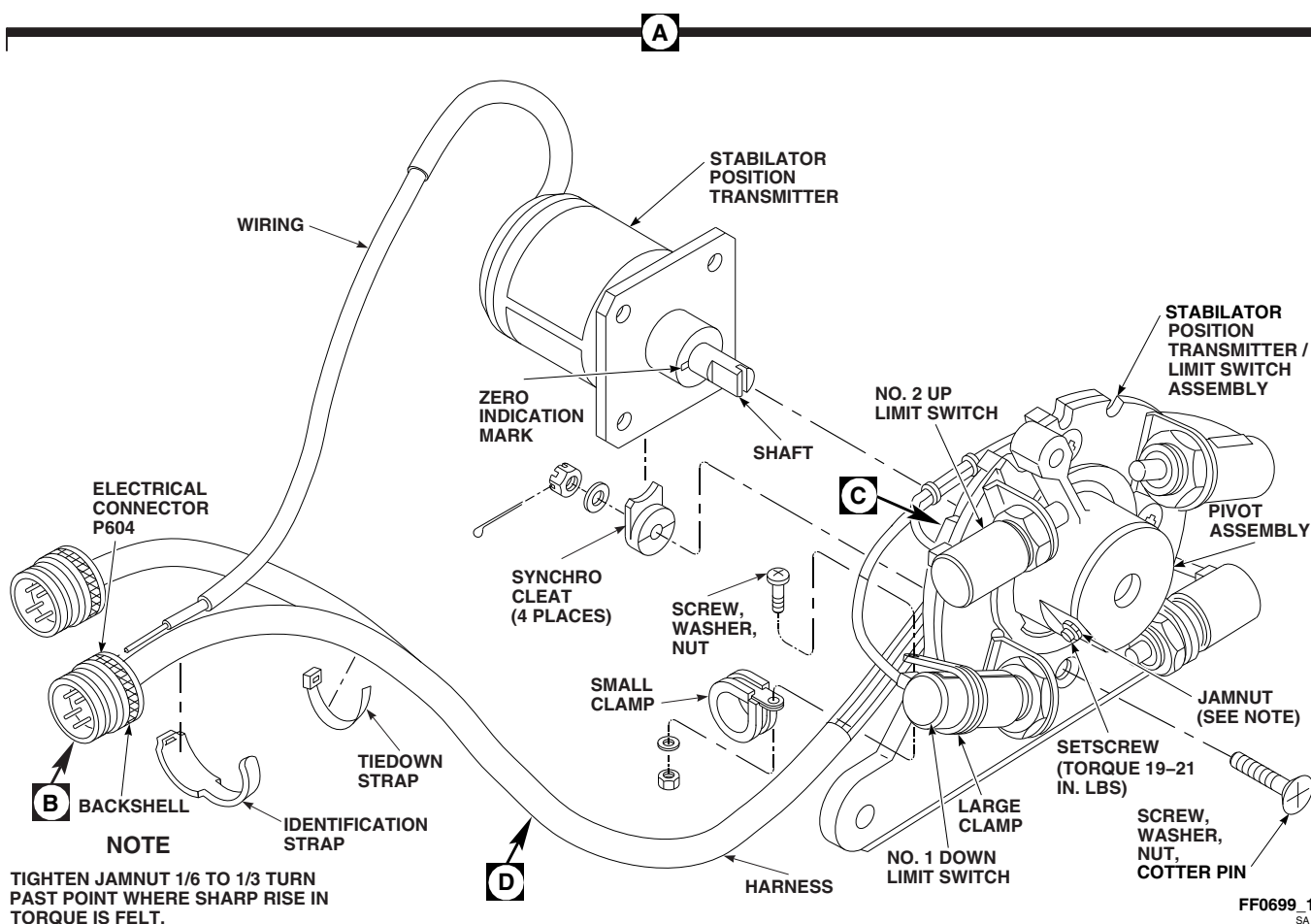
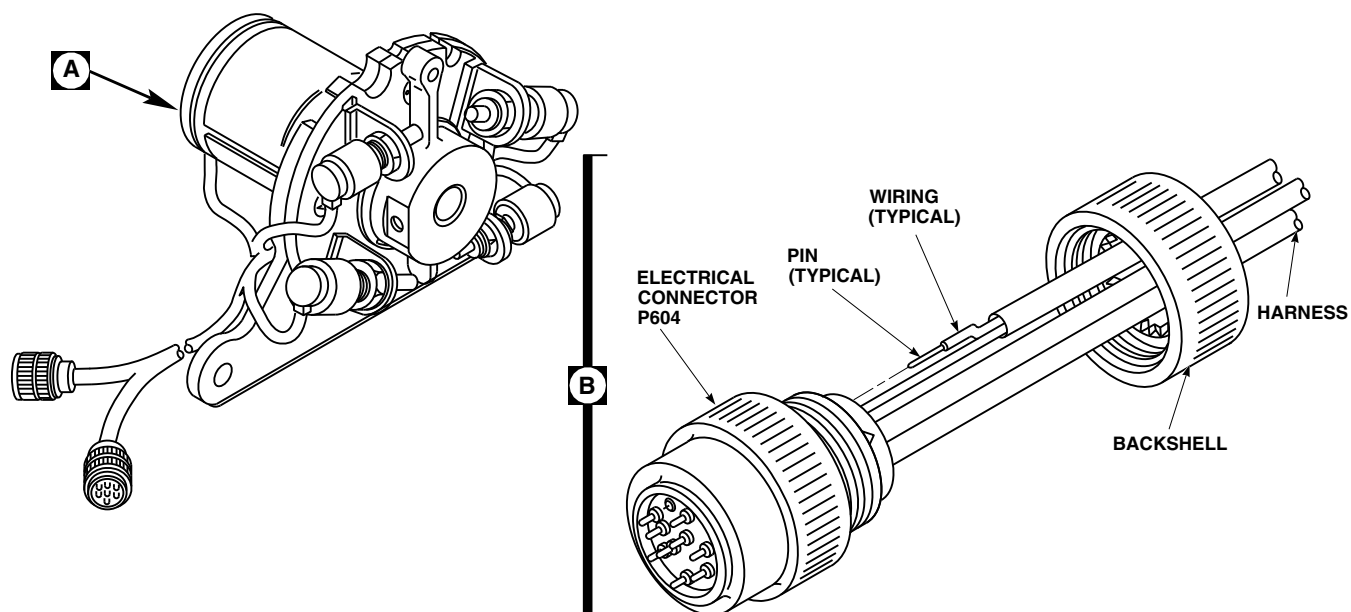
BENDS OR DISTORTIONS MUST NOT INTERFERE WITH FIT OR FUNCTION. If bends or distortion interfere with fit or function, replace part(s) as necessary.

2-5-22.1.3 Install.

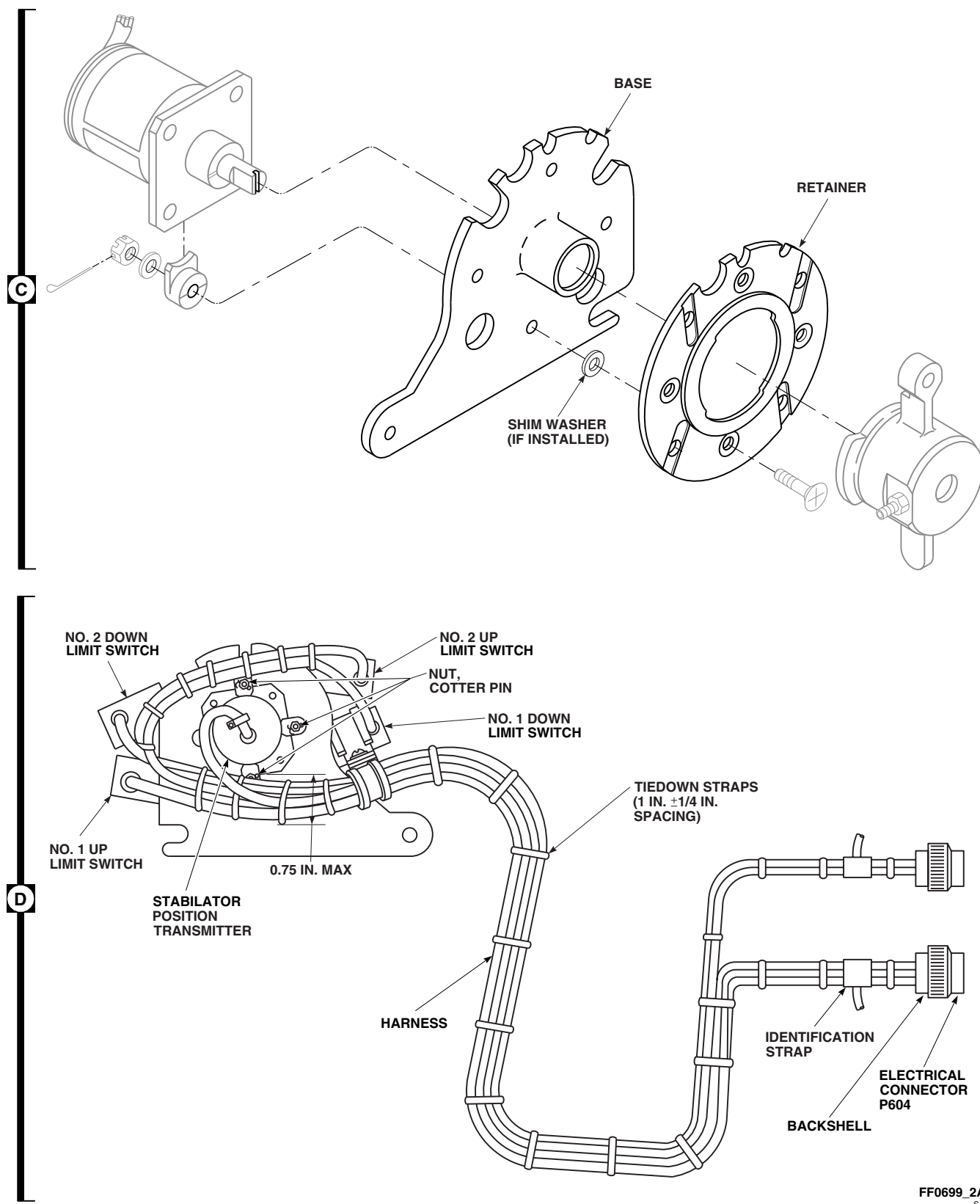
- a. Insert shaft of stabilator position transmitter in stabilator transmitter/limit switch assembly (Figure 2-5-22-1, Sheet 1, Detail A). Orient zero indication mark on shaft collar with screw, located between No. 1 down limit switch and No. 2 up limit switch.
- b. Secure position transmitter to stabilator position transmitter/limit switch assembly with screws, washers, synchro cleats, and nuts. Install cotter pins.
- c. Tighten nuts and check for binding in pivot assembly. **NONE ALLOWED.** If binding occurs, install shim washers on screws between retainer and base (Figure 2-5-22-1, Sheet 2, Detail C). Use fewest shims permitting free play of pivot assembly. **SHIM THICKNESS SHALL NOT EXCEED 0.030-INCH.**
- d. Tighten setscrew on pivot assembly (Figure 2-5-22-1, Sheet 1, Detail A). **TORQUE SETSCREW TO 19 - 21 INCH-POUNDS.**
- e. Remove jamnut. Using cheesecloth, Item 90, Appendix D, wet with tetrachloroethylene, Item 324, Appendix D, clean inside threads of jamnut and threads of setscrew. Wipe dry with clean cheesecloth, Item 90, Appendix D.
- f. Apply thin coat of sealing compound, Item 282, Appendix D, to threads of jamnut and setscrew. Allow to dry.
- g. Apply thin coat of sealing compound, Item 280, Appendix D, to threads of jamnut and setscrew.
- h. Install jamnut on setscrew. **TIGHTEN JAM-NUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
- i. Slide 1/4-inch-diameter, 3-inch long section of heat-shrink insulation sleeving, Item 177, Appendix D, over end of wiring. Leave 1 1/2-inches of wires exposed for installation of electrical connector pins on wire ends.
- j. Install pins on stabilator position transmitter wires (Figure 2-5-22-1, Sheet 1, Detail B).
- k. Slide 3-inch long section of heat-shrink insulation sleeving, Item 177, Appendix D, over exposed wires, leaving connector pins exposed. Using heat gun, heat shrink insulation sleeving over harness.
- l. Thread wires through backshell of electrical connector. Using insert/extract tool, install pins in electrical connector, P604, as follows, then remove tags:

WIRE COLOR	P604 CONTACT
Yellow	A
Blue	B
Black	C
Red/White	D
Black/White	E
Green	F
- m. Slide backshell on electrical connector, P604. **TIGHTEN BACKSHELL.**
- n. Adjust stabilator position transmitter (PARA 2-2-1).
- o. Using lockwire, Item 196, Appendix D, lockwire nut and setscrew to pivot assembly and install cotter pins (Figure 2-5-22-1, Sheet 1, Detail A).
- p. Secure harness with tiedown straps, Item 329, Appendix D. Maintain at least 1/4-inch spacing between harness and nuts/cotter pins on stabilator position transmitter (Figure 2-5-22-1, Detail A and Sheet 2, Detail D).

- q. Install identification strap to harness (Figure 2-5-22-1, Detail A and Sheet 2, Detail D).
- r. Install small clamp on harness and secure to large clamp on No. 1 down limit switch with screw, washer, and nut.
- s. Perform operational check of stabilator position transmitter/limit switch assembly (PARA 2-2-1).



**FIGURE 2-5-22-1. REPLACE STABILATOR POSITION TRANSMITTER (BENCH)
(SHEET 1 OF 2)**



**FIGURE 2-5-22-1. REPLACE STABILATOR POSITION TRANSMITTER (BENCH)
(SHEET 2 OF 2)**

2-5-23 STABILATOR POSITION TRANSMITTER/LIMIT SWITCH ELECTRICAL CONNECTOR (BENCH).

PARAGRAPH BREAKDOWN

		PAGE
2-5-23.1	Replace Stabilator Position Transmitter/Limit Switch Electrical Connector (BENCH)	2-5-23-1

INITIAL SETUP

Tools

- Electrical Connector Maintenance/Repair Kit
- Electrical Repairer’s Toolkit
- Insert/Extract Tool

Materials/Parts

- Tags

References

- PARA 2-2-1

2-5-23.1 Replace Stabilator Position Transmitter/Limit Switch Electrical Connector (BENCH).

NOTE

Replacement of stabilator position transmitter/limit switch electrical connectors, P604 and P605, is the same.

2-5-23.1.1 Remove.

- Loosen backshell and slide down harness away from electrical connector (Figure 2-5-23-1, Detail A).
- Tag wires. Using insert/extract tool, remove wires and pins from electrical connector.

- Remove electrical connector.

2-5-23.1.2 Install.

- Using insert/extract tool, install pins in electrical connector (Figure 2-5-23-1, Detail A). Remove tags.
- Slide backshell on electrical connector. **TIGHTEN BACKSHELL.**
- Perform operational check of stabilator position sensor/limit switch assembly (PARA 2-2-1).

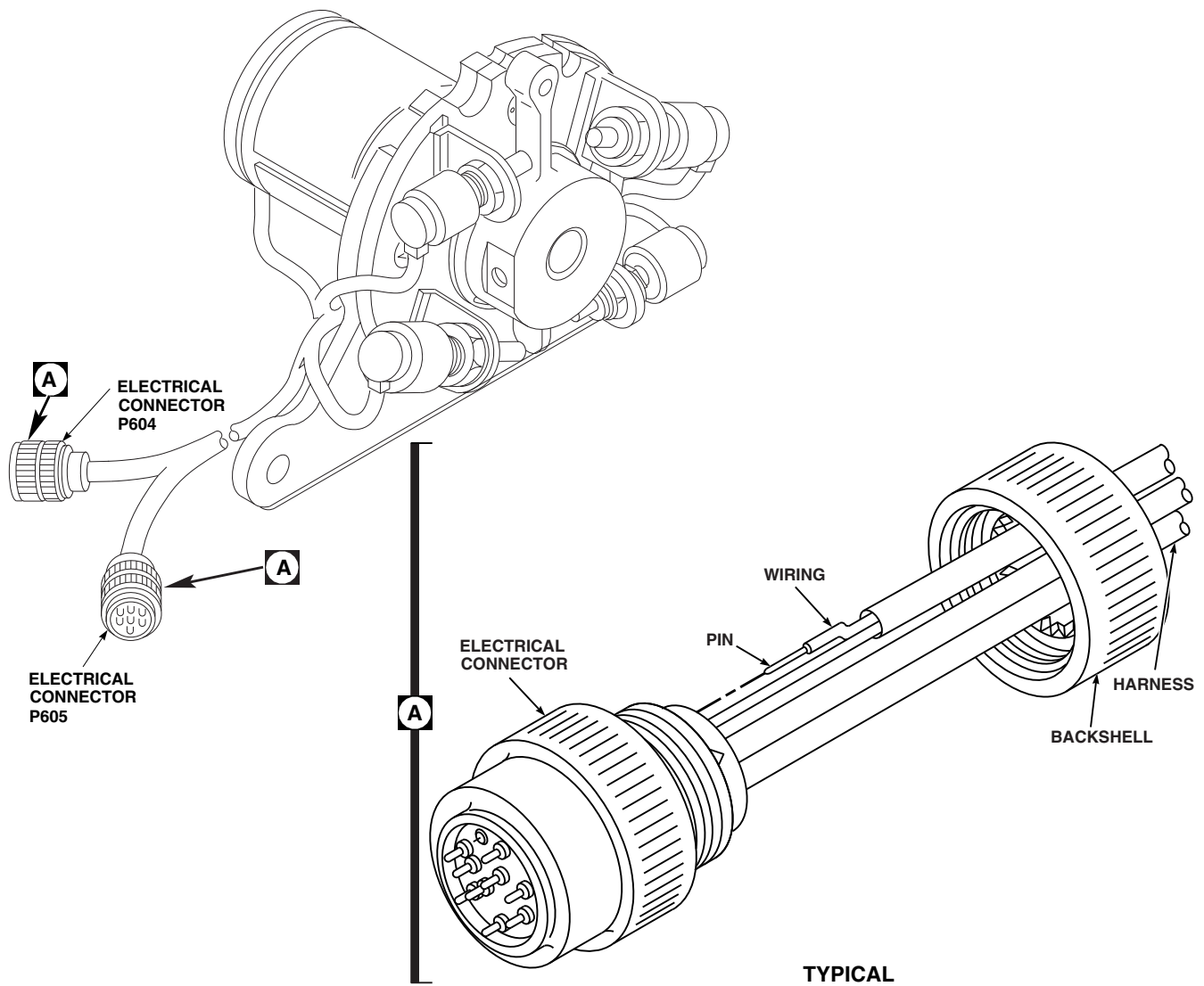
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FIGURE 2-5-23-1. REPLACE STABILATOR POSITION TRANSMITTER/LIMIT SWITCH ELECTRICAL CONNECTOR (BENCH)

2-5-24 STABILATOR POSITION TRANSMITTER/LIMIT SWITCH ASSEMBLY (BENCH).

PARAGRAPH BREAKDOWN

	PAGE
2-5-24.1 Replace Mechanical Parts (BENCH)	2-5-24-1

INITIAL SETUP

Tools

- Electrical Connector Maintenance/Repair Kit
- Electrical Repairer’s Toolkit

References

- PARA 2-2-1

2-5-24.1 Replace Mechanical Parts (BENCH).

NOTE

Disassemble stabilator position sensor/limit switch assembly only as far as necessary to remove malfunctioning or damaged part.

- Visually inspect the following mechanical parts. If damaged or corroded beyond fit or function, replace part (Figure 2-5-24-1, Detail A).

- Pivot assembly
- Retainer

- Positioners
- Base
- Synchro cleats
- Clamps
- Hardware
- Shim

- If pivot assembly is to be removed, any two diagonally opposite limit switches and their positioners must be removed.
- Perform operational check of stabilator position transmitter/limit switch assembly (PARA 2-2-1).

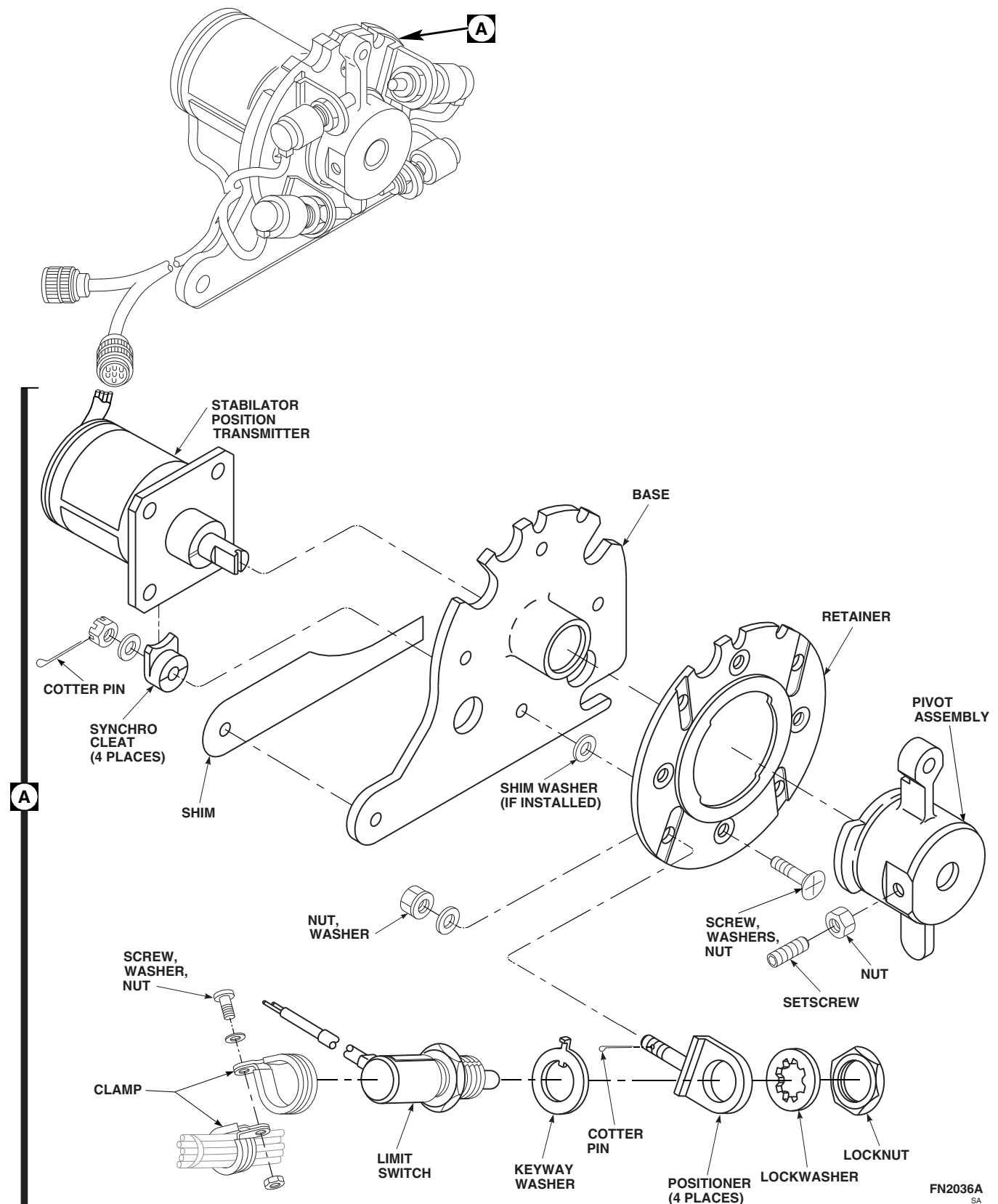


FIGURE 2-5-24-1. REPLACE MECHANICAL PARTS (BENCH)

2-5-24-2

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SECTION 2-6

STRUCTURAL REPAIR

SECTION OVERVIEW

PARAGRAPH	TITLE
2-6-1	Structural Repair General Information
2-6-2	Metal Structure Repair
2-6-3	Composite Structure Repair
2-6-4	Extrusion Charts
2-6-5	Helicopter Painted Surfaces - Touch Up Repair
2-6-6	Nose Door Repair
2-6-7	Cockpit Door Repair
2-6-8	Nose Section Framing Repair
2-6-9	Nose Section Plating Repair
2-6-10	Cabin Section Forward Framing Repair
2-6-11	Cabin Section Upper Framing Repair
2-6-12	Cabin Section Lower Framing Repair
2-6-13	Cabin Section Plating Repair
2-6-14	Cabin Section Beaded Inner Fuselage Plating - STA 295.0 through STA 308.0 Repair
2-6-15	Cabin Door Upper Track Repair
2-6-16	Cabin Door Lower Track Fairing Repair
2-6-17	Cabin Door Lower Track Wear Strips Repair
2-6-18	External Power Receptacle Latch, Hinge, and Cover
2-6-19	Rescue Hoist Airframe Fitting Repair
2-6-20	External Stores Support System Fuselage Fitting Bearing

PARAGRAPH

TITLE

2-6-21	Crew Chief's Window Lower Track
2-6-22	Crew Chief's Window Lower Track Repair
2-6-23	Crew Chief's Window Upper Track
2-6-24	Window Upper Track Fairing
2-6-25	Crew Chief's Window Repair
2-6-26	Cabin Door Repair
2-6-27	Cabin Floor Repair
2-6-28	Main Rotor Pylon Framing Repair
2-6-29	Main Rotor Pylon Fairing Repair
2-6-30	Transition Section Framing Repair
2-6-31	Transition Section Plating Repair
2-6-32	Transition Section Step Door Hinge - WL 210.0 and WL 230.0 Repair
2-6-33	Transition Section Step Door Hinge - WL 250.0
2-6-34	Transition Section Fuel Sump Drain Door
2-6-35	Transition Section Gravity Refuel Door
2-6-36	Transition Section Pressure Refuel Door
2-6-37	Transition Section Pneumatic Ground Start Door
2-6-38	Transition Section APU Access Door Screws Repair
2-6-39	Grounding Receptacle
2-6-40	Tail Cone Framing Repair
2-6-41	Tail Cone Plating Repair
2-6-42	Tail Cone Canted Hinge Bulkhead Repair
2-6-43	Tail Rotor Drive Shaft Covers Repair
2-6-44	Tail Rotor Pylon Framing Repair
2-6-45	Tail Rotor Pylon Plating Repair

PARAGRAPH	TITLE	
2-6-46	Tail Rotor Pylon Left Stabilator Attachment Fitting	I
2-6-47	Tail Rotor Pylon Right Stabilator Attachment Fitting	I
2-6-48	Tail Rotor Pylon Stabilator Actuator Attachment Fitting	
2-6-49	Tail Rotor Pylon Stabilator Actuator Attachment Fitting Bushings and Bearing	
2-6-50	Tail Rotor Pylon Cable Pulley Supports Repair	
2-6-51	Tail Rotor Pylon Canted Hinge Frame Repair	
2-6-52	Stabilator Framing Repair	
2-6-53	Stabilator Plating Repair	
2-6-54	Stabilator Tip Cap Repair	
2-6-55	Stabilator Actuator Attachment Fitting Repair	
2-6-56	HIRSS Fairing Repair	

2-6-1 **STRUCTURAL REPAIR GENERAL INFORMATION.**

PARAGRAPH BREAKDOWN

		PAGE			PAGE
2-6-1.1	General Information	2-6-1-2	2-6-1.6	Fire Damage Evaluation	2-6-1-4
2-6-1.2	Damage Evaluation	2-6-1-2	2-6-1.7	Negligible Damage Limitations	2-6-1-5
2-6-1.3	Classification of Damage and Types of Repair	2-6-1-2	2-6-1.8	Corrosion Protection and Water Integrity Sealing	2-6-1-10
2-6-1.4	Cleaning Out Damaged Area	2-6-1-3	2-6-1.9	Inspection of Camloc Fasteners	2-6-1-15
2-6-1.5	Inspection of Damage	2-6-1-4			

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Brush, Metal-Bound, Flat
- Drill
- Drill Bit, 1⁄8-inch diameter
- Fluorescent Penetrant Inspection Kit
- Paint Roller
- Protective Covers
- Sealant Gun
- Soft Mallet
- Vacuum Cleaner

Materials/Parts

- Acetone, Item 25, Appendix D
- Alkaline Cleaner
- Barrier Material, Item 80, Appendix D

- Cheesecloth, Item 90, Appendix D
- Corrosion Preventive Compound, Item 111, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D
- Distilled Water, Item 123, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Litmus Paper, Item 190, Appendix D
- Sealing Compound, Item 292, Appendix D
- Tape

References

- Appendix D
- ASTM E1417
- MIL-DTL-53039
- PARA 1-6-5
- PARA 1-6-16
- PARA 2-6-2

- PARA 2-6-3
- PARA 2-6-4
- PARA 2-6-5

2-6-1.1 General Information.

- a. Section 2-6 contains general instructions and information necessary for maintenance personnel to repair the primary and secondary structure.
- b. Use standard general helicopter maintenance repair methods for fastener installation and repair materials.
- c. Refer to PARA 2-6-1.8 for specific corrosion protection and watertight sealing information.
- d. Refer to PARA 2-6-5 for specific paint requirements.
- e. Stop-drilling cracks is a temporary repair. It requires a follow-up repair with another reinforcing repair at the next 500-hour inspection.

2-6-1.2 Damage Evaluation.

- a. Damage to helicopter is caused by impact, wear, or malfunctioning parts. Impact damage will vary with force and location of impact and type of object causing the damage. Wear of parts is due to normal usage and/or misalignment of movable surfaces. Parts can be destroyed by fire, or may malfunction as a result of fatigue, corrosion, overstress, or previously undetected damage. Damage not caused by impact must be evaluated and corrected to prevent it from recurring. This may require reinforcement of the structure or additional corrosion prevention measures. Increased inspections and periodic maintenance may also be required. Each part in the damaged area

must be inspected and the amount of damage determined and classified.

- b. Structural repairs described in this section are intended for use in conjunction with general helicopter maintenance repair procedures.

2-6-1.3 Classification of Damage and Types of Repair.

2-6-1.3.1 General.

Airframe damage consists of bending, twisting, breaking, or reduction of material in the cross-section of a structural member. Damage may be a complete break, hole, wrinkle, twist, nick, scratch, crack, or material eaten away by corrosion. In case of fire, it may be pitted, blistered, burned, or an eaten away area of the skin or skeleton. After examining damage, classify it as either negligible damage, repairable damage, or damage requiring replacement. This will aid in determining the type and size of the repair needed.

2-6-1.3.2 Negligible Damage.

NOTE

Permissible damage limits for structural aluminum alloy members are explained and illustrated in PARA 2-6-1.7. Permissible damage limits for reinforced plastic are described in PARA 2-6-3.

Negligible damage is damage which may be permitted to remain as is or which may be corrected by a simple repair procedure such as stop-drilling cracks, blending out nicks or scratches, or touching up protective finish on parts.

2-6-1.3.3 Repairable Damage.

NOTE

Recommended typical repairs for aluminum alloy structural members are shown in PARA 2-6-2 and PARA 2-6-4. Recommended typical repairs for reinforced plastic components are shown in PARA 2-6-3. These repair examples may be used as shown, or used in any combination necessary to complete a repair.

- a. Damage greater than negligible limits that requires the addition of a patch, splice, or reinforcement to restore full load-carrying capacity of the structure is classified as repairable damage.

NOTE

When patching, use same gage, type, temper material, and same fasteners as original part for patch material.

- b. Cracks and scratches deeper than 10% of material thickness are stop-drilled using drill with 1/8-inch diameter drill bit and patched.
- c. Nicks or gouges deeper than 10% of material thickness are drilled out and patched.
- d. Punctures, deep dents, and corroded areas of skin and webs are trimmed out and patched.
- e. Smooth dents are tapped out using soft mallet and the area reinforced with stiffening angles.
- f. Extensive damage to a long structural member is repaired by cutting away damaged section and splicing in a piece identical in shape and material. Cut replacement member as follows:
 - (1) Make splices long enough to include no less than five fasteners on each side of cut.
 - (2) Allow 0.030 - 0.060-inch clearance on each end of inserted member.

2-6-1.3.4 Damage Requiring Replacement.

Damage that requires replacement includes damage which cannot be repaired by available equipment, or extensive damage which requires replacement of an entire component. Small doublers, splices, clips, brackets, and short stiffening members, damaged beyond negligible limits, are usually replaced because repair of these members is impracticable. Some highly-stressed structural members cannot be repaired without reducing the established margin of safety for the structure. Such members, when damaged beyond negligible limits, must be replaced. Sometimes, where damaged members have a shape which is difficult to repair, it may be better, in view of man-hours and cost, to replace the damaged member rather than attempt a repair. Damage to a removed structural component should be evaluated for possible repair at a higher level of maintenance. Replacement members shall be the same type material and temper as the original member and are installed using the same type fasteners. Information for replacement or recommended substitution of extrusion is found in PARA 2-6-4, Extrusion Charts.

2-6-1.4 Cleaning Out Damaged Area.

CAUTION

If a part or section of any supporting structure is removed for replacement or repair, make sure nearby structure is properly supported to prevent further distortion and damage.

NOTE

After a thorough investigation of damage, a course of action should be taken to restore damaged area to its original condition. Clean out damaged area first. Take care to prevent further damage during this operation. The following procedure for cleaning out a damaged area applies to all types of damage, except main and tail rotor blade damage.

- a. Restore all dented skin and supporting structure to its original shape and condition wherever possible.
- b. Cut away all jagged edges and severely distorted material, removing enough material to eliminate any radiating cracks.
- c. Leave as much radius as possible in corners of cutouts. Smooth out all edges.
- d. Round out holes in flat or contoured sheets to form a smooth outline.
- e. If nicks or abrasions are greater than negligible limitations, they should be treated as cracks and prepared for patching.
- f. Do not leave isolated cracks unattended even though area is considered non-stressed. Using drill with 1/8-inch diameter drill bit, stop-drill crack.

2-6-1.5 Inspection of Damage.

2-6-1.5.1 Preliminary External Inspection.

- a. Clean up structure and make thorough visual inspection of damaged members and nearby structure for local and secondary damage.
- b. Check for secondary damage resulting from shock loads being transmitted to other structures.
- c. Check nearby structure for dents, gouges, punctures, cracks, loose joints, and distortion.
- d. Check all riveted or bolted joints for evidence of pulled fasteners or distorted holes. If extent of damage is in doubt, remove enough fasteners to remove parts, and inspect framing for condition and alignment.
- e. Check all wrinkles in skin and webs to decide whether they are stress wrinkles. List all items to be removed or repaired. Note all suspect areas of conditions which will call for closer detailed inspection when actual repair procedures are started.

2-6-1.5.2 Detailed Inspection.

Inspect each part in damaged area and classify all damage to determine whether parts can be repaired and reused, or due to damage limitations or extent of repair, if they should be replaced. For major damage to highly-stressed areas of airframe or components, and following "sudden stoppage" of rotor blades, submit a detailed report to local engineering authority, or to contractor, for engineering analysis and specific repair recommendations. "Sudden stoppage" is defined as shock load to drive train or rotor system, with or without power applied. This shock load may result from blade strike, FOD, or compressor stall.

2-6-1.6 Fire Damage Evaluation.

NOTE

First step in evaluation of structural damage caused by fire is a thorough cleaning of the damaged area. Evaluation of damage, which may be caused either by the high temperature or by subsequent corrosion, involves both visual and optical inspection, plus the use of test equipment and test procedures. Hardness tests will determine the approximate ultimate strength of a material, but it does not eliminate the possibility of hidden damage. When in doubt as to the serviceability of a part or component, replace it.

2-6-1.6.1 Cleanup of Damaged Areas.

CAUTION

Further damage to helicopter may result if fire damage is not removed quickly. Clean up as soon as possible after a fire, to reduce damage due to corrosion.

- a. Shut off power source (PARA 1-6-16).
- b. Remove batteries, fuel, loose equipment, and materials or interior accessories that are in the way or that may be damaged during cleanup and repair.

- c. Remove access panels, drill out rivets and open seams if necessary, to prepare areas for complete flushing and draining.
- d. Use preventive measures to protect areas not affected, using tape, barrier material, Item 80, Appendix D, and protective covers (PARA 1-6-5).
- e. Remove all traces of foam fire extinguishing residue by flushing with clean, fresh water. Flushing of applicable area or part must be complete, internally and externally.
- f. Use approved alkaline cleaner and remove all oil, soot, extinguishing products and fire residue from damaged area. Flush thoroughly with clean water and drain all areas completely. Wipe internal surfaces with clean cloths until dry. Use forced warm air to hasten drying.

2-6-1.6.2 Visual Inspection.

- a. Inspect area for visible fire damage, such as distortion, burning, or pitting.
- b. Inspect parts near fire for blistered or discolored paint. These parts shall be inspected by nondestructive inspection methods before being returned to service.

2-6-1.6.3 Nondestructive Inspection.

Hardness tests, using portable equipment, should be made on all materials suspected of exposure to high temperatures. Hardness tests are not entirely conclusive; however, they give a good indication of the reliability of the part. Check suspect areas for cracks, especially in the radius of formed parts or extruded members, using fluorescent penetrant inspection kit (ASTM E1417). If reliability of any part is doubtful, replace part.

2-6-1.6.4 Corrosion Damage Inspection.

NOTE

Extinguishing agents used to fight fire may be corrosive when in contact with high temperatures and water. Materials

consumed by fire may also release corrosive gases or by-products which are highly corrosive when combined with water. Check all areas that have been reached by smoke or fumes for evidence of corrosion.

- a. Clean all affected areas with an approved cleaner.
- b. Using distilled water, Item 123, Appendix D, and litmus paper, Item 190, Appendix D, inspect affected area. Pay particular attention to skin laps and joining surfaces for evidence of corrosive acid residue.
- c. Using an alkaline wash, neutralize areas that turn blue litmus paper red.

2-6-1.7 Negligible Damage Limitations.

2-6-1.7.1 Damage Measurement.

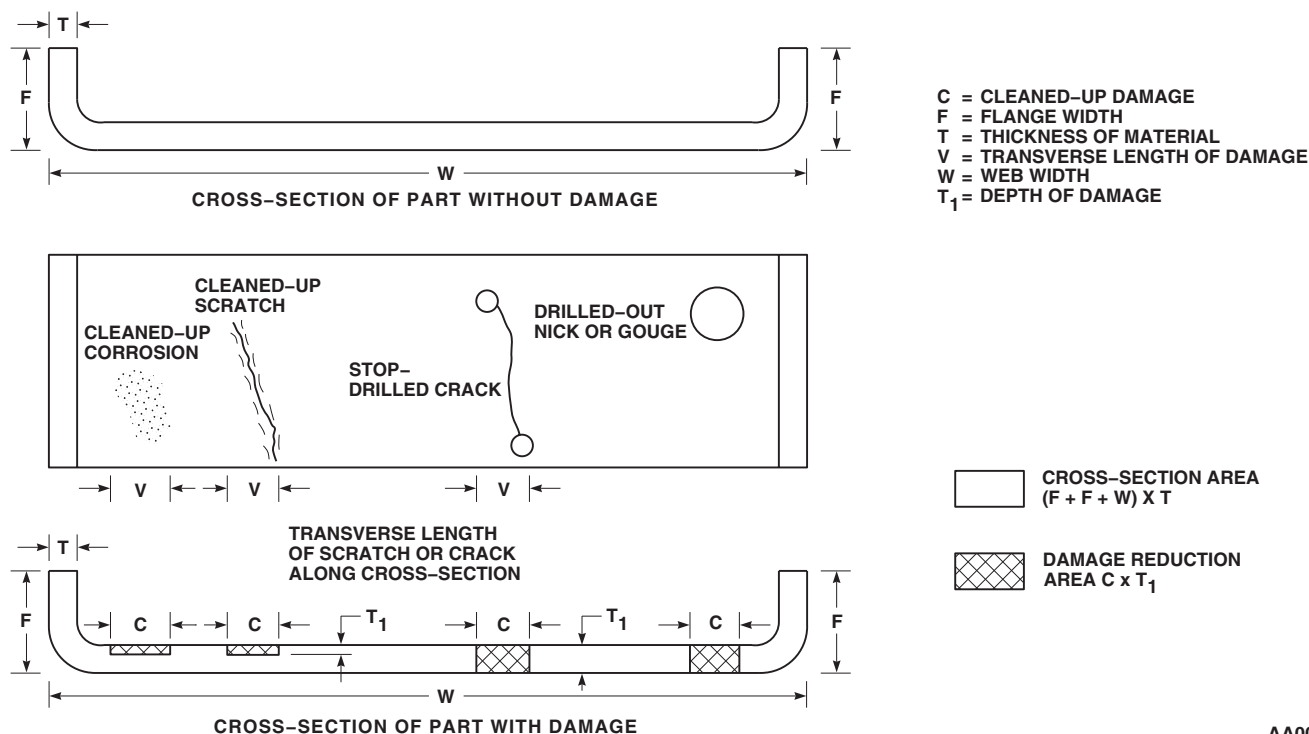
- a. Damage is measured by comparing normal cross-section of a part with depth and area affected because of holes, cracks, scratches, nicks, and corrosion. Amount of material loss is called damage reduction. Any damage reduction will affect strength of part.
- b. Cross-section area of a part is found by adding the width of flanges and web, and multiplying this total by the material thickness (Figure 2-6-1-1). Use equation:

$$(F + F + W) \times T = \text{area of cross section.}$$

- c. Damage reduction area is found by measuring the dimension of damage (C) along the cross section and multiplying this dimension by total depth of the cleaned-up damage. Use equation:

$$C \times T_1 = \text{damage reduction area.}$$

- d. On high-stressed structural members, if the damage reduction area is not more than 5% of cross-section area, damage is classified as negligible. On low-stressed structural members, if damage reduction area is not more than 10% of the cross-section area, damage is classified as negligible. Exceptions to this rule are found in linear dimensions.

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e. Negligible damage limit examples:

- (1) Dimension of cross section: 1.0-inch flanges, 3.0-inch web, and 0.040-inch material thickness. Cross-section area would be 0.200 ($1.0 + 1.0 + 3.0 = 5.0$, then $5.0 \times 0.040 = 0.200$). Negligible damage limit for high-stress frames (5% of cross-section area) would be 0.010 ($0.200 \times 0.05 = 0.010$). Negligible damage limit for low-stress frames (10% of cross-section area) would be 0.020 ($0.200 \times 0.10 = 0.020$).
- (2) A scratch or corrosion, whose maximum depth after cleanup is 0.008-inch and transverse length (V) is 1.5-inches. Damage reduction area would be 0.012 ($0.008 \times 1.5 = 0.012$). Damage would be within limits for low-stress frames, but greater than allowable limits for high-stress frames, which would need patching.
- (3) A stop-drilled crack with transverse length (V) of 1.0-inch would have a damage reduction area of 0.040 ($1.0 \times 0.040 =$

0.040). This would be greater than permitted negligible damage limits for both high and low-stress frames, and each would need patching.

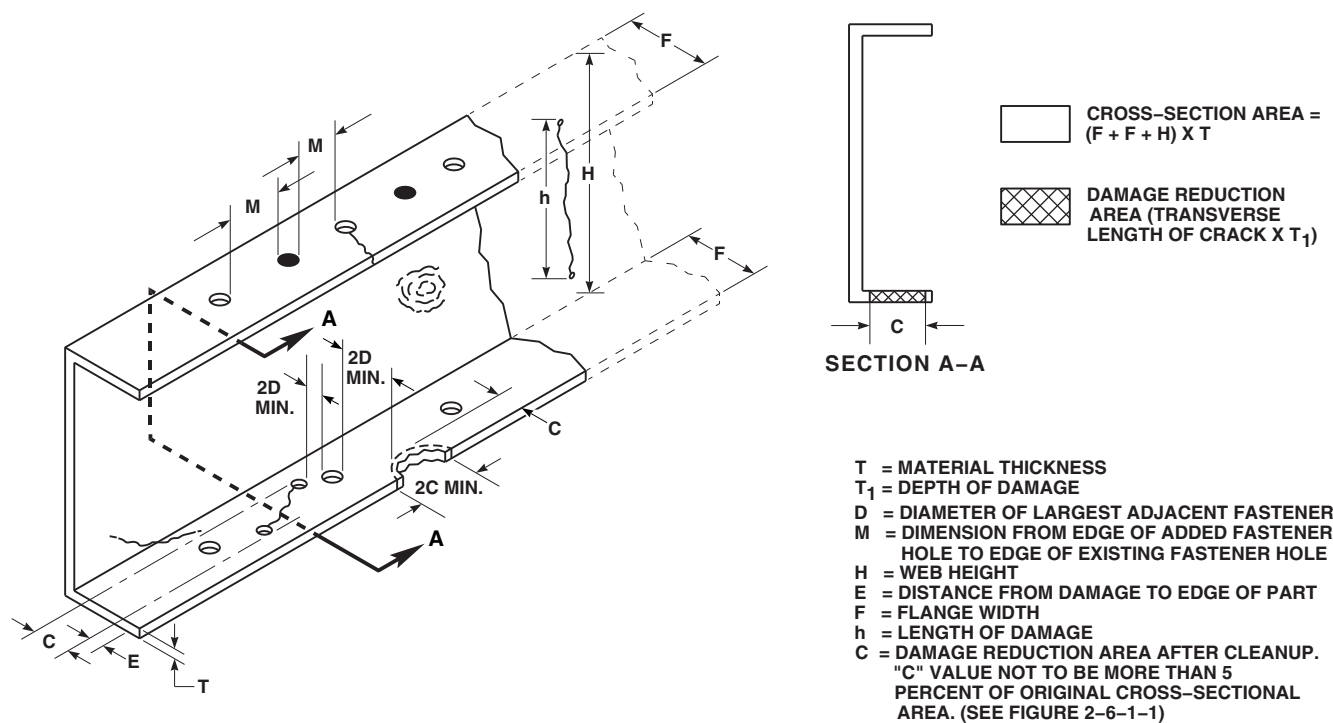
- (4) A nick or gouge drilled out using 1/4-inch diameter drill bit and polished would have a damage reduction area of 0.010 ($0.250 \times 0.040 = 0.010$). This would be within permitted negligible damage limits for both high and low-stress frames. Patching would be needed if a larger drill bit is used.

2-6-1.7.2 Negligible Damage Limits for Structural High-Stress Frames.

- a. Nicks, Scratches, or Corrosion. Do not remove more material than necessary. Determine cross-sectional area of part, using equation:

$$(F + F + H) \times T = \text{area of cross section}$$

If damage reduction area clean up (Figure 2-6-1-2, Section A-A) is no more than 5% of cross section area, damage is negligible. Treat nicks



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FIGURE 2-6-1-2. STRUCTURAL FRAME DAMAGE

and scratches as cracks if reduction area is more than 5% of cross section area.

- Nicks and Cracks. No nick or crack is allowed in radius. Stop-drill cracks, or nicks and scratches over damage limit, using drill with $\frac{1}{8}$ -inch diameter drill bit. Cleaned-up damage to be not over damage reduction limits. Consider transverse length of damage when figuring damage reduction (Figure 2-6-1-1). If distance "E" is less than "D", consider damage goes to the edge of part. Nicks and gouges in flange edge, not over negligible limits, are blended out to not more than twice the depth of damage (Figure 2-6-1-2).
- Dents. Smooth dents, free from sharp projections or abrasions, not over $\frac{1}{8}$ -inch in depth and 1-inch in diameter, are considered negligible. Treat all dents over $\frac{1}{8}$ -inch in depth as holes, or those with nicks or gouges. Trim out damage, sand to a smooth rounded hole, and patch (PARA 2-6-2).
- Cracks in Flange Edge. When crack ends at a fastener hole, another fastener of the same type

may be added on each side of damage hole, providing "M" dimension is not less than "2D". If "M" dimension is less than "2D" repair flange (PARA 2-6-2).

2-6-1.7.3 Negligible Damage Limits for Extrusions.

- Cracks. Cracks are not allowed in extrusion radius (Figure 2-6-1-3). Stop-drill cracks using drill with $\frac{1}{8}$ -inch diameter drill bit. Minimum distance allowable from adjacent damage or fastener hole is 2D. Consider transverse length of damage and cleanup when calculating damage reduction (Figure 2-6-1-1). If distance "E" is less than "D", consider damage goes to edge of part.
- Nicks and Gouges. Refer to negligible damage limits for structural (high-stress) frames. If nick or gouge is over negligible limits, use drill bit large enough to remove entire damage. Blend out nicks and gouges in flange edge, not greater than negligible limits, to not more than twice depth of damage.

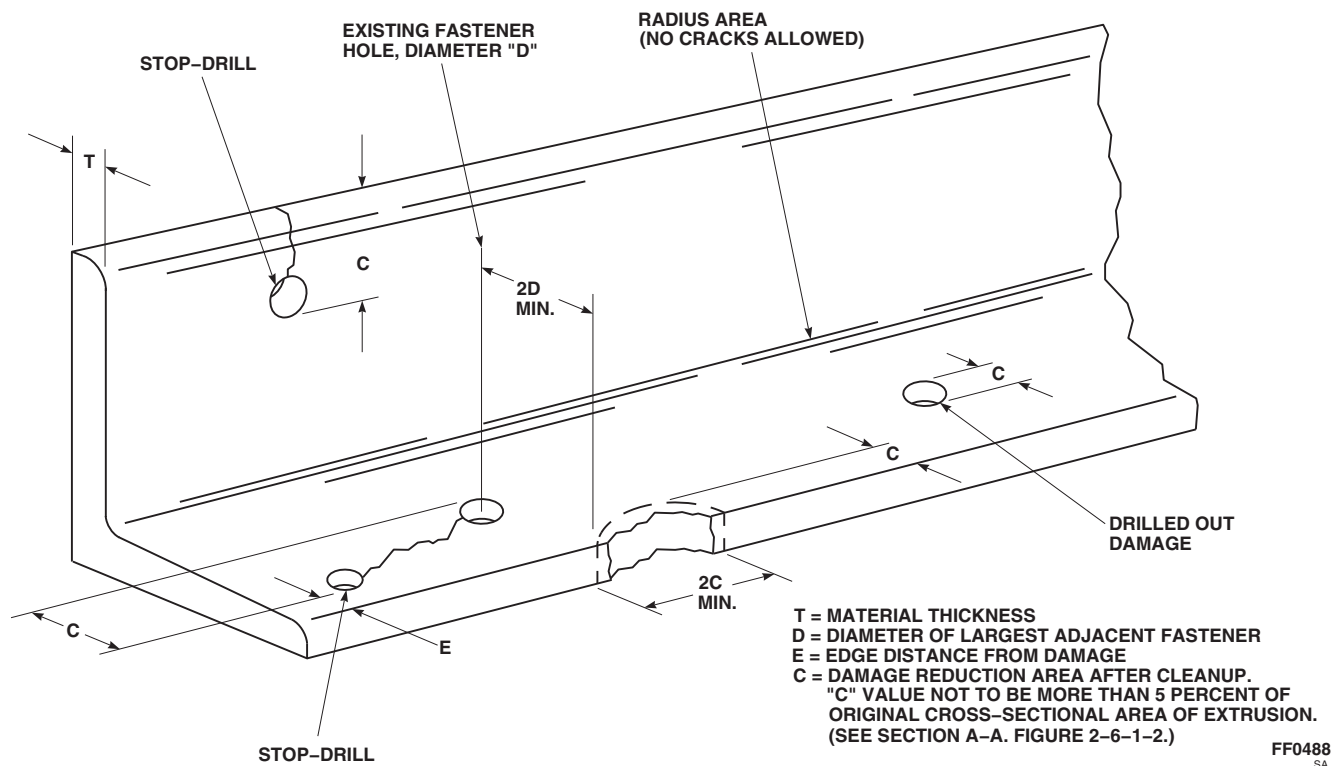


FIGURE 2-6-1-3. EXTRUSION DAMAGE

- c. Radius Damage. Nicks, scratches, and/or corrosion in extrusion radius not over 25% of "T" after cleanup. Blend damage to not more than twice damage depth.

Width (b) - $2 \frac{1}{2}$ times material thickness T.

Length (h) - 40% of H. Treat nicks and scratches as cracks if over limits T_1 , b, or h.

2-6-1.7.4 Negligible Damage Limits for Primary Skin.

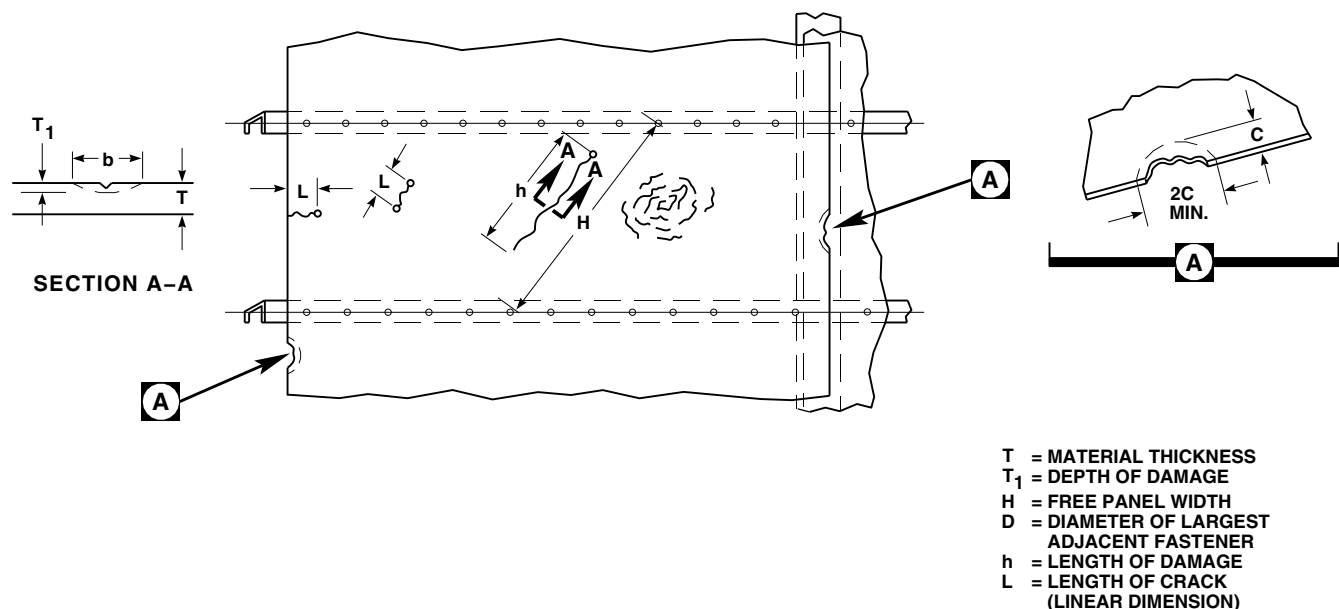
- Cracks Not Ending at Fastener Holes. Stop-drill using drill with $\frac{1}{8}$ -inch diameter drill bit. Length of crack "L" not to be over $\frac{1}{2}$ -inch (Figure 2-6-1-4).
- Edge Damage. Cleaned-up area not to be over $\frac{1}{2}$ -inch deep. Blend out damage to smooth rounded contour, not more than twice depth (Figure 2-6-1-4, Detail A).
- Nicks, Scratches, and Corrosion. Do not remove more material than necessary. Cleaned-up area (Figure 2-6-1-4, Section A-A) is not to be over the following limits:

Depth (T_1) - 15% of T.

- Dents. Smooth dents, free from sharp projections or abrasions, not over $\frac{1}{4}$ -inch deep. Treat as holes all dents over $\frac{1}{4}$ -inch deep, or those with nicks or gouges. Trim out damage, sand to a smooth rounded hole, and patch (PARA 2-6-2).

2-6-1.7.5 Negligible Damage Limits for Secondary Skin.

- Cracks Not Ending at Fastener Holes. Stop-drill using drill with $\frac{1}{8}$ -inch diameter drill bit. Length of crack "L" not to be over 1-inch.
- Edge Damage. Cleaned-up area not to be over 1-inch deep. Blend out damage to smooth rounded contour to no more than twice depth (Figure 2-6-1-4, Detail A).
- Nicks, Scratches, and Corrosion. Same as primary skin, except depth can be 35% of "T".



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FIGURE 2-6-1-4. SKIN DAMAGE

- d. Dents. Same as primary skin.
- e. Punctures and Cracks. Trim damage to a smooth rounded contour. Longest dimension not to be over 1-inch. Patch cleaned-up damage over 1-inch (PARA 2-6-2).

2-6-1.7.6 Negligible Damage Limits for Non-structural Low-Stress Frames.

- a. Nicks, Scratches, or Corrosion. Do not remove more material than necessary. Determine cross-sectional area of part, using equation:

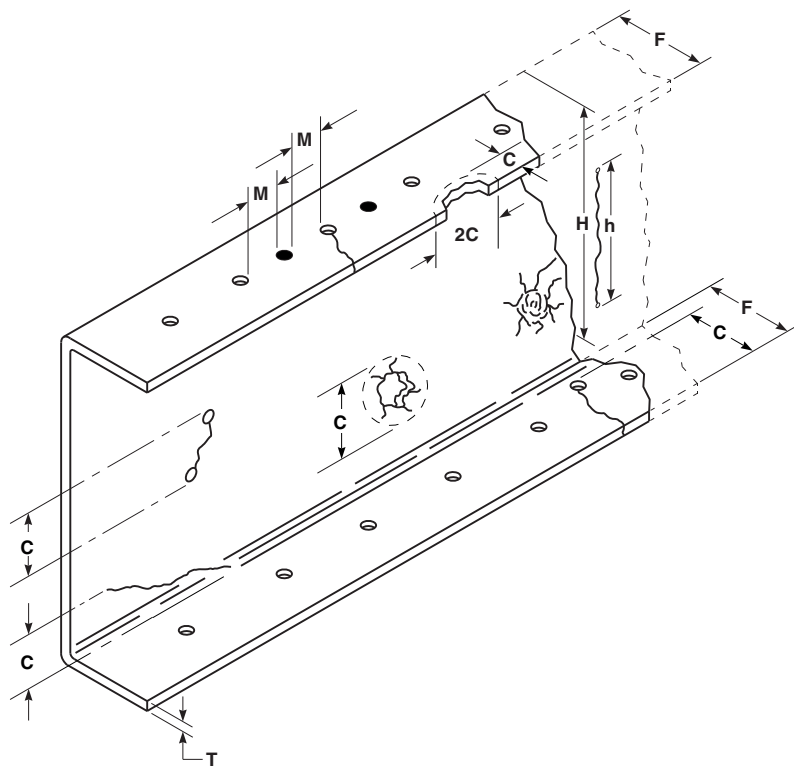
$$(F + F + W) \times T = \text{area of cross section}$$

If damage reduction area (cleaned-up area, Figure 2-6-1-2, Section A-A) is no more than 10% of cross-section area, damage is classified as negligible. Treat nicks and scratches as cracks if reduction area is more than 10% of cross section area.

- b. Cracks. No cracks allowed in radius area. Stop-drill cracks, or nicks and scratches over damage limit, using drill with 1/8-inch diameter drill

bit. Cleaned-up damage not to be more than "C" damage reduction (Figure 2-6-1-5). Consider transverse length of damage when figuring damage reduction (Figure 2-6-1-1). Nicks and gouges in flange edge not over 1/4-inch deep are blended out to not more than twice damage depth (Figure 2-6-1-2).

- c. Punctures. Cleaned-up area (including radiating cracks) to be not more than "C" damage reduction. Patch holes over "C" damage limits (PARA 2-6-2).
- d. Dents. Smooth dents, free from sharp projections or abrasions, not over 1/4-inch deep and 1-inch in diameter, are considered negligible. Treat as holes all dents over 1/4-inch deep or those with nicks or gouges. Trim out damage, sand to a smooth rounded contour, and patch (PARA 2-6-2).
- e. Cracks in Flange Edge. When a crack ends at fastener hole, same type fastener may be added on each side of damaged hole, providing "M" dimension is not less than "2D". If "M" dimension



T = MATERIAL THICKNESS
 D = DIAMETER OF LARGEST ADJACENT FASTENER
 M = DIMENSION FROM EDGE OF ADDED FASTENER HOLE TO EDGE OF EXISTING FASTENER HOLE
 H = WEB HEIGHT
 F = FLANGE WIDTH
 h = LENGTH OF DAMAGE
 C = DAMAGE REDUCTION AREA AFTER CLEANUP. "C" VALUE NOT TO BE MORE THAN 10 PERCENT OF ORIGINAL CROSS-SECTIONAL AREA. (SEE FIGURE 2-6-1-1)

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FIGURE 2-6-1-5. NONSTRUCTURAL FRAME DAMAGE

sion is less than "2D", repair flange (PARA 2-6-2).

2-6-1.8 Corrosion Protection and Water Integrity Sealing.

2-6-1.8.1 Description.

- a. Without proper protection, metals used in any helicopter structure are subject to various types of corrosion damage. During manufacture and assembly, primers, sealants, and coatings are applied to the entire structure to shield against moisture, salts, chemicals, harmful vapors, and the electrolytic action caused by dissimilar metals in direct contact with each other. Sealants also prevent rain and splashed water from leaking through the skin, plating, and window joints of the helicopter. MIL-DTL-53039 paints used for interior and exterior color schemes of late series helicopters resist chemical, gas, and radiation absorption.

NOTE

On earlier series helicopters, some areas will be found that were not originally

sealed with sealing compound, Item 292, Appendix D. However, for making repairs, use the following sealing requirements for all helicopters, regardless of serial number.

- b. When making repairs, the corrosion protection and water integrity of the entire structure must be carefully preserved. If old parts are being repaired or replaced, sealants and coatings which were broken or removed in cleanup must be reapplied. Likewise, when new joints, fasteners, and overlaps are added as would be done in making a skin patch, the specified corrosion protection and sealant must be carefully and thoroughly applied.
- c. Clean surfaces and correct use of materials are essential for sealants to adhere properly. Special instructions for all helicopter series are given in the following paragraphs. For paint touch up instructions, refer to PARA 2-6-5.

2-6-1.8.2 Cleaning.



Metal chips, filings, and grit may damage metal surfaces when wiping. Make sure to remove them before wiping surface.

- a. Remove metal chips, filings, and grit from area using vacuum cleaner.
- b. Wipe all surfaces clean using cheesecloth, Item 90, Appendix D, moistened with acetone, Item 25, Appendix D. When cleaning large areas, change or refold cheesecloth frequently.
- c. Clean one small section at a time and do not allow acetone, Item 25, Appendix D, to air dry.
- d. Dry surfaces with clean cheesecloth, Item 90, Appendix D, wiping in one direction only.

2-6-1.8.3 Surface Preparation and Priming.

Unless otherwise noted and before sealing compound is applied, treat all bare metal surfaces with corrosion resistant coating, Item 118, Appendix D, and prime (PARA 2-6-5).

2-6-1.8.4 Application of Sealing Compound.

- a. Use sealing compound, Item 292, Appendix D, unless otherwise specified.
- b. When using sealing compound, Item 292, Appendix D, select the grade that will allow repair parts to be assembled and fastened while compound is still wet.
- c. Refer to Figure 2-6-1-6, Sheet 1, for examples of different methods of sealing compound application.
- d. When hole tolerances are too close for use of sealing compound, Item 292, Appendix D, install bushings, bolts, inserts, etc., with a wet coat of epoxy primer, using epoxy primer coating kit, Item 135, Appendix D.

- e. Refer to Figure 2-6-1-6, Sheet 2, for structural examples of sealing compound application. Apply sealing compound according to example which matches or is most like the repair being made.
- f. Before installation, apply coat of sealing compound, Item 292, Appendix D, to all rivets, bolts, and other fasteners which secure skins, fittings, plating, and accessories to the airframe exterior. Install fasteners while sealing compound is still wet.
- g. When installing skins, patches, fittings and accessories on any exterior surface of the airframe, seal all overlapping and mating surfaces. Brush or roll on coat of sealing compound, Item 292, Appendix D, to make a finished seal 0.002 - 0.005-inch maximum thickness.
- h. On entire exterior of airframe, seal all edges of skins, panels, and plating to mating structure with sealing compound, Item 292, Appendix D. Spread compound with brush or roller to make a finished seal 0.002 - 0.005-inch thick. Inboard locations of items that attach to structure do not need brush sealing. At all hole locations, along edges, and in inboard areas, install rivets and fasteners, and coat of sealing compound (Figure 2-6-1-7).
- i. Apply sealing compound, Item 292, Appendix D, between gussets, doublers, and mating skin surfaces. Install fasteners for these items with coat of sealing compound.
- j. After completing a repair, but before painting top coat, fill all gaps, joints, voids, and pockets which could trap and hold water with a fillet of sealing compound, Item 292, Appendix D.
- k. After priming, seal all repaired surfaces on interior of structure from STA 162.0 - STA 485.0 and below waterlines as shown (Figure 2-6-1-8) with two spray coats of corrosion preventive compound, Item 111, Appendix D. Do not apply corrosion preventive compound to top sides of cockpit and cabin floors.
- l. Refer to PARA 2-6-5, to find paint type and color for helicopter.

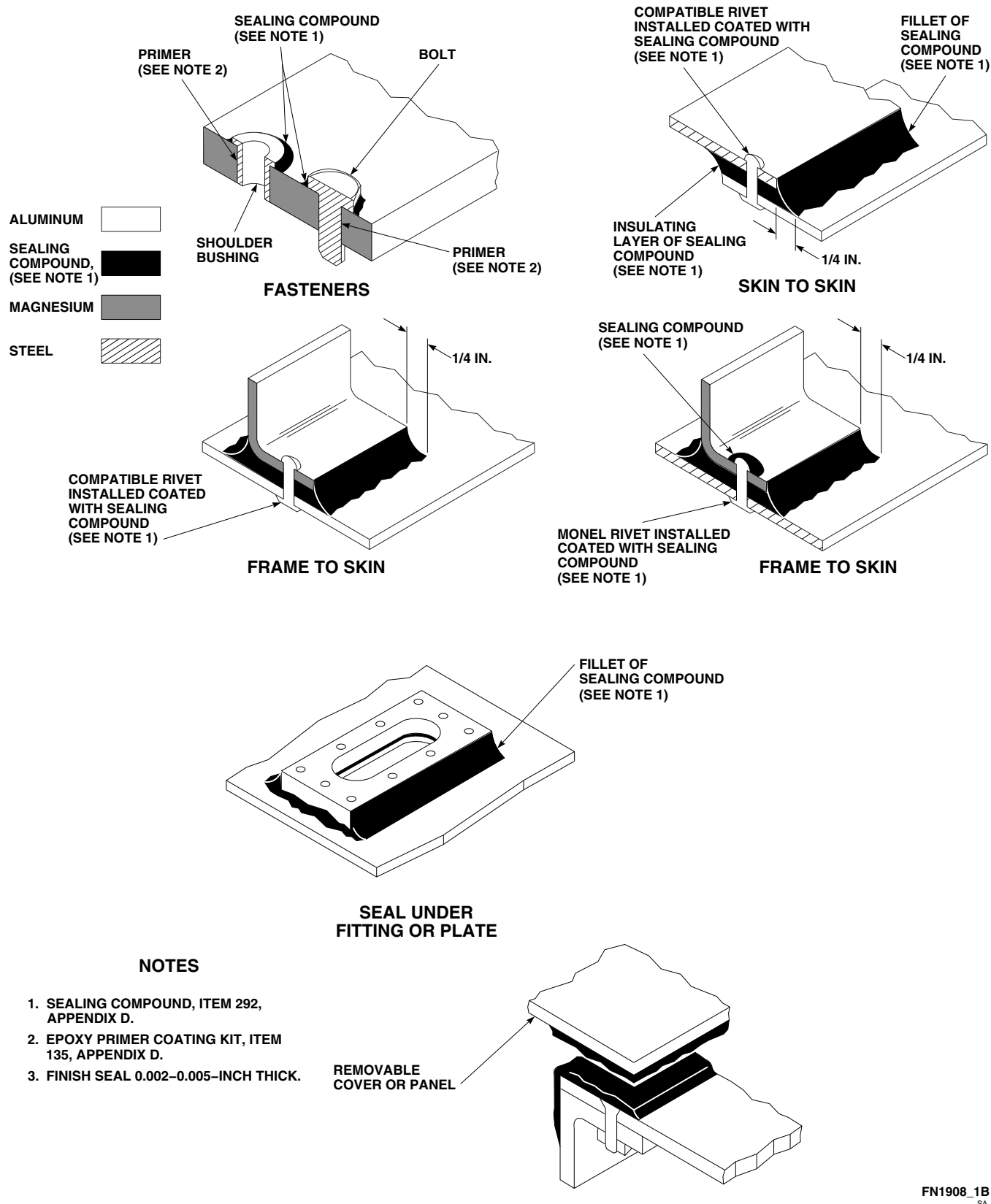
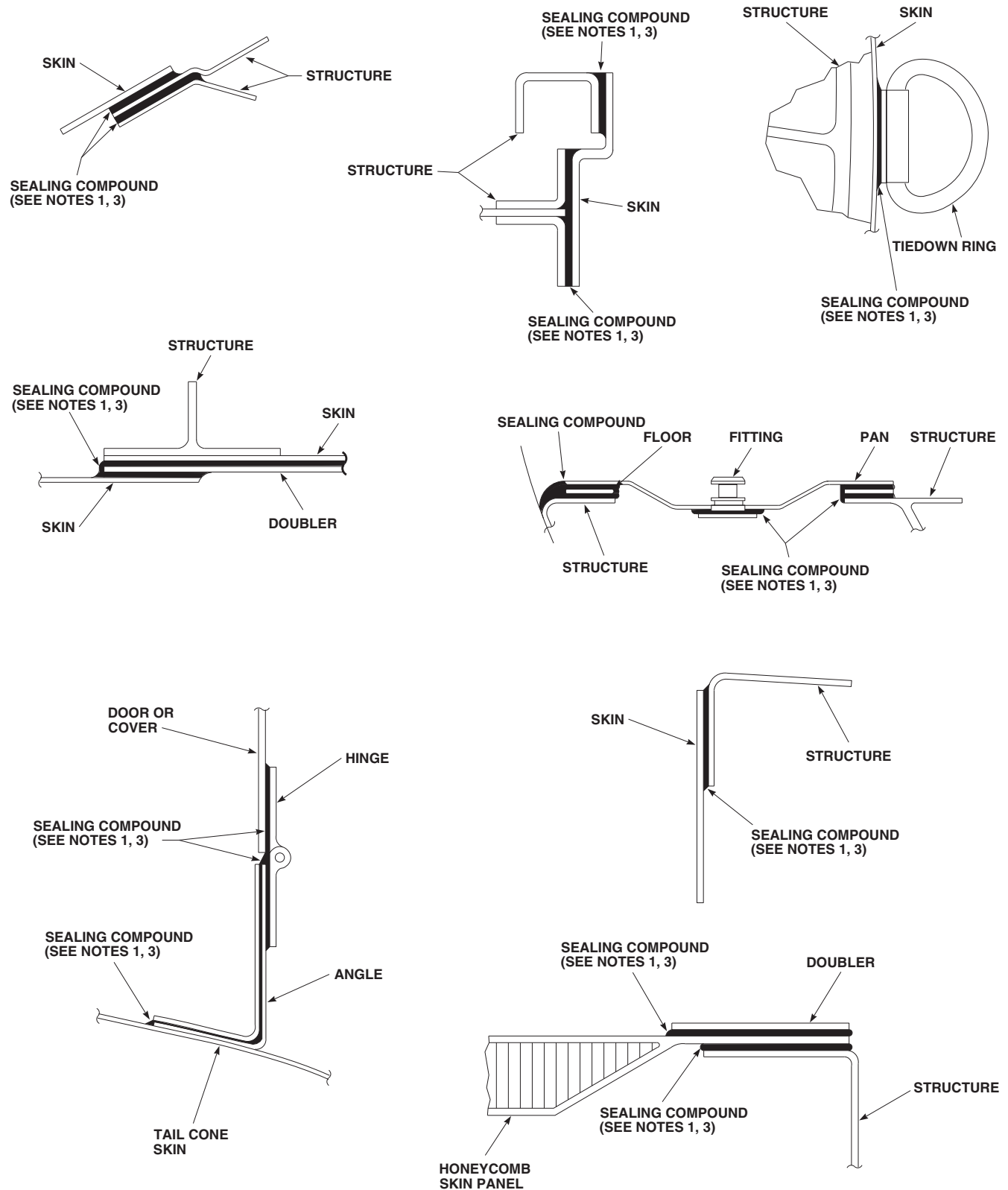
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FIGURE 2-6-1-6. SEALING COMPOUND APPLICATION (SHEET 1 OF 2)



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FIGURE 2-6-1-6. SEALING COMPOUND APPLICATION (SHEET 2 OF 2)

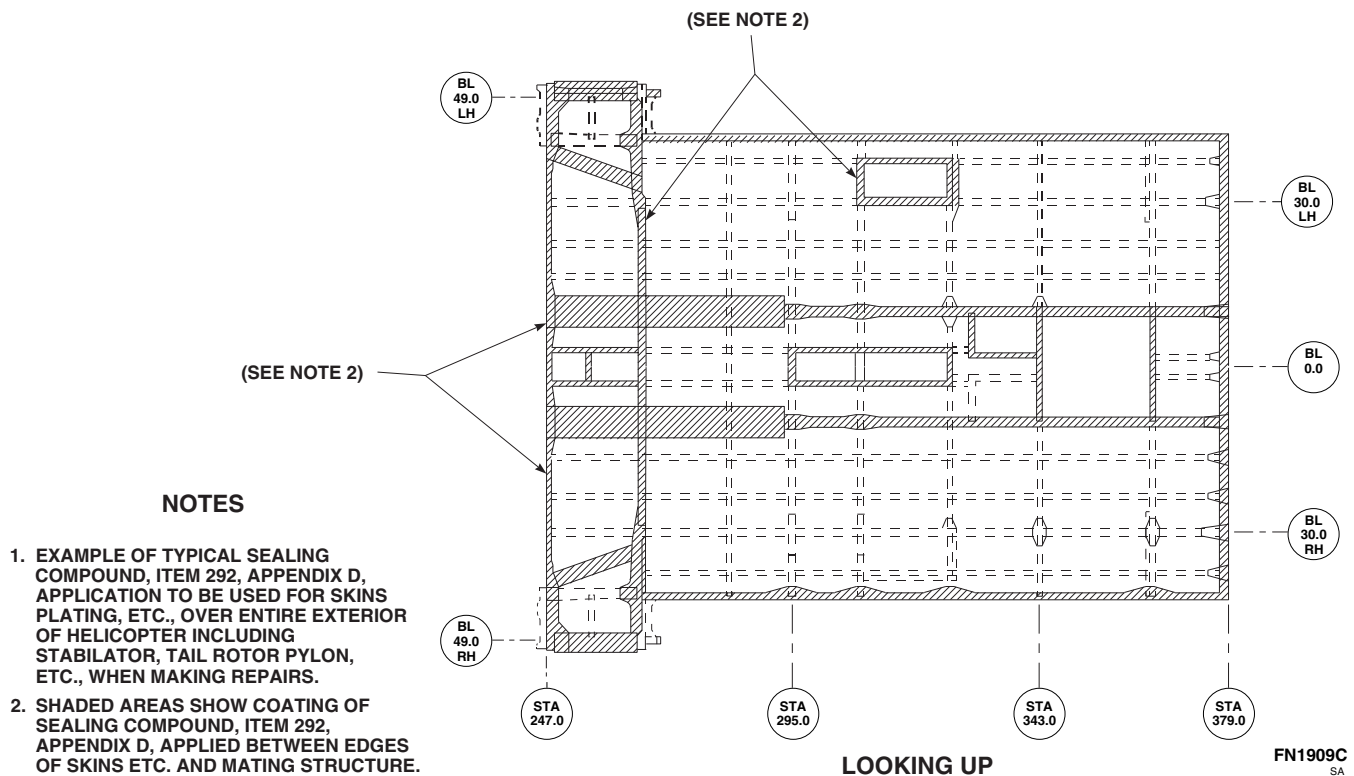


FIGURE 2-6-1-7. EXTERIOR PLATING SEALING

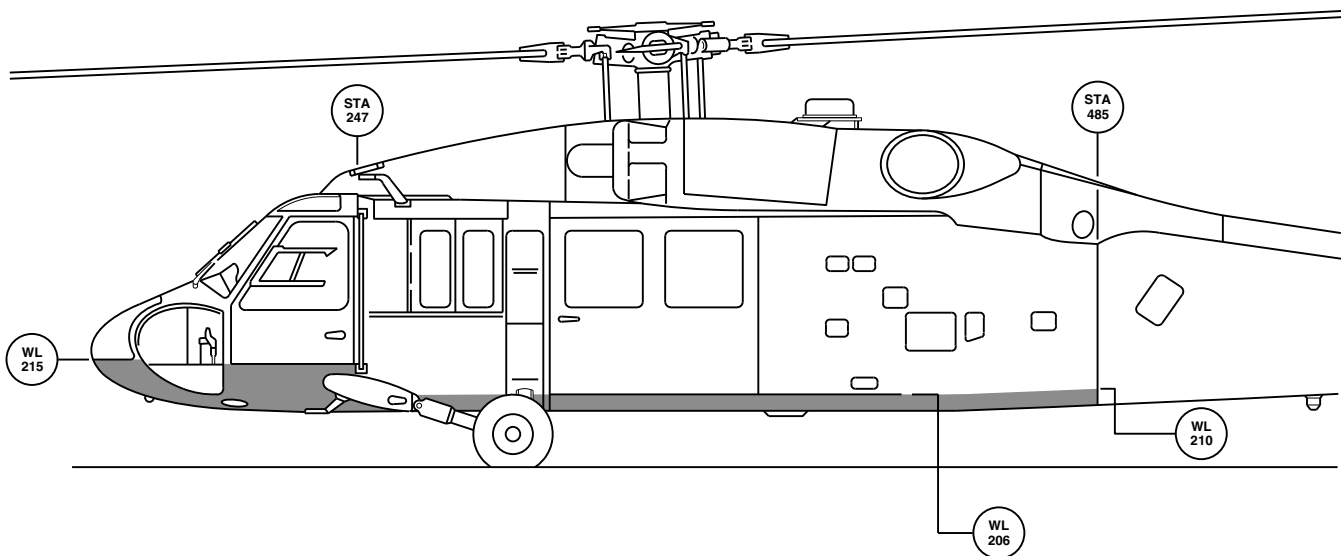


FIGURE 2-6-1-8. CORROSION PREVENTION SPRAY

2-6-1.9 Inspection of Camloc Fasteners.

- a. Inspect for broken Camloc fasteners. **NO MORE THAN 20% OF CAMLOC FASTENERS ON ANY FAIRING ARE ALLOWED TO BE BROKEN.**
- b. Each broken fastener must have serviceable fasteners on both sides. **NO TWO ADJACENT FASTENERS CAN BE BROKEN.**
- c. When fasteners are grouped in a small area, the following damage limits apply:
 - (1) Quantity of 1 - 3 fasteners. **NO FASTENERS SHALL BE BROKEN.**
 - (2) Quantity of 4 - 9 fasteners. **ONE FASTENER CAN BE BROKEN.**
 - (3) Quantity of 10 or more fasteners. **20% OF FASTENERS CAN BE BROKEN.**

2-6-2 METAL STRUCTURE REPAIR.**PARAGRAPH BREAKDOWN**

		PAGE			PAGE
2-6-2.1	Primary Structure Skin Patch - Non-Flush Patch	2-6-2-2	2-6-2.14	Repair Damaged Long-eron	2-6-2-17
2-6-2.2	Primary Structure Skin Patch - Non-Flush Formed Patch	2-6-2-2	2-6-2.15	Repair Damaged Frame	2-6-2-17
			2-6-2.16	Repair Damaged Lightning Holes	2-6-2-17
2-6-2.3	Secondary Structure Skin Patch	2-6-2-2	2-6-2.17	Repair Damaged Hat Section	2-6-2-17
2-6-2.4	Riveted Flush Patch Repair	2-6-2-2	2-6-2.18	Repair Damaged Channel Stiffener	2-6-2-17
2-6-2.5	Temporary Aluminum Tape Patch Repair	2-6-2-2	2-6-2.19	Repair Damaged Tee and Web	2-6-2-19
2-6-2.6	Repair Damaged Beaded Section	2-6-2-5	2-6-2.20	Repair Damaged Beam Caps	2-6-2-19
2-6-2.7	Repair Skin Close to a Frame	2-6-2-5	2-6-2.21	Repair Damaged Skin and Frame	2-6-2-19
2-6-2.8	Repair Formed and Extruded Angle	2-6-2-7	2-6-2.22	Repair Damaged Skin and Stringer	2-6-2-21
2-6-2.9	Repair Damaged Tee	2-6-2-11	2-6-2.23	Repair Damaged Skin, Frame, and Stringer	2-6-2-21
2-6-2.10	Repair Damaged Former	2-6-2-13	2-6-2.24	Repair Damaged Bulkhead Web and Skin	2-6-2-21
2-6-2.11	Repair Damaged Rib	2-6-2-13	2-6-2.25	Miscellaneous Repairs	2-6-2-21
2-6-2.12	Repair Damaged Channel	2-6-2-13			
2-6-2.13	Repair Damaged Zee	2-6-2-13			

INITIAL SETUP

Tools

- Airframe Repairer's Toolkit
- Drill
- Drill Bit, 1/8 (0.125)-Inch Diameter

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Hi-Shear Rivet, HS52P6
- Masking Tape, 1-inch, Item 212, Appendix D

- Methyl Ethyl Ketone, Item 217, Appendix D
- Pressure Sensitive Tape, Item 251B, Appendix D
- Rivet, MS20426AD
- Rivet, MS20470AD
- Rivet, MS20600AD
- Rivet, MS20615M
- Rivet, NAS1097AD

References

- Appendix D

2-6-2.1 Primary Structure Skin Patch - Non-Flush Patch.

- Refer to Figure 2-6-2-1.
- On aluminum alloy use rivets, MS20470AD.
- On stainless steel or titanium use rivets, MS20615M.

2-6-2.2 Primary Structure Skin Patch - Non-Flush Formed Patch.

- Refer to Figure 2-6-2-2.
- On aluminum alloy use rivets, MS20470AD.
- On stainless steel or titanium use rivets, MS20615M.

2-6-2.3 Secondary Structure Skin Patch.

- Refer to Figure 2-6-2-3.
- On aluminum alloy use rivets, MS20470AD.
- On stainless steel or titanium use rivets, MS20615M.

2-6-2.4 Riveted Flush Patch Repair.

- Refer to Figure 2-6-2-4.

- Skin filler - same gage and material as original skin.
- Skin patch - one gage thicker of same material as original skin.
- For flush installations, dimple gages under 0.050 and machine countersink gages 0.050 and greater. Double dimple or dimple thin skin into countersunk heavy skin. Rivet head style will be flush type.

2-6-2.5 Temporary Aluminum Tape Patch Repair.

- Refer to Figure 2-6-2-5.
- Damage 1-inch or less in diameter after trimming may be repaired by patching with pressure sensitive tape, Item 251B, Appendix D.
- Bend edge of hole slightly inward to provide smooth, flat surface for patch.
- Mask off rounded area on skin 1 1/8-inches from and around edge of trimmed hole, using 1-inch masking tape, Item 212, Appendix D.
- Remove paint and clean bare metal using cheesecloth, Item 90, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D.

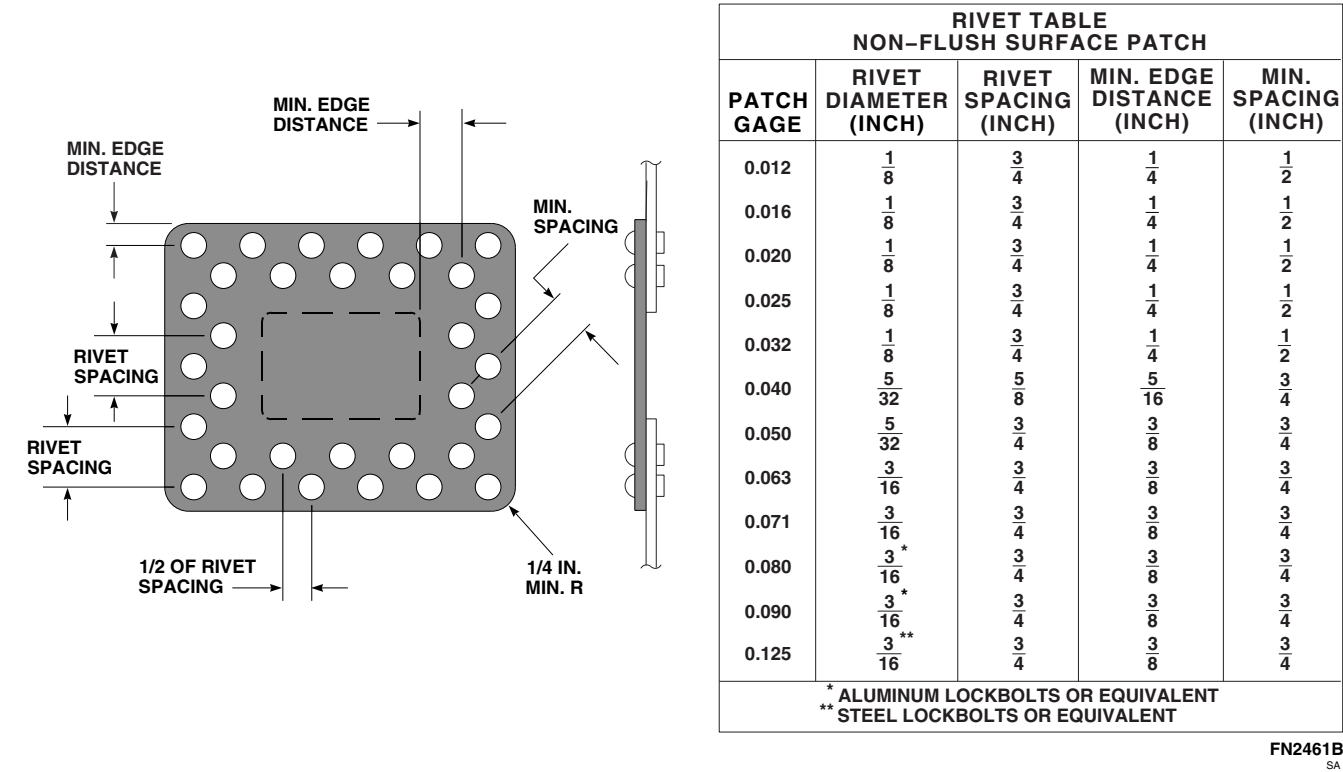


FIGURE 2-6-2-1. PRIMARY STRUCTURE SKIN PATCH - NON-FLUSH PATCH

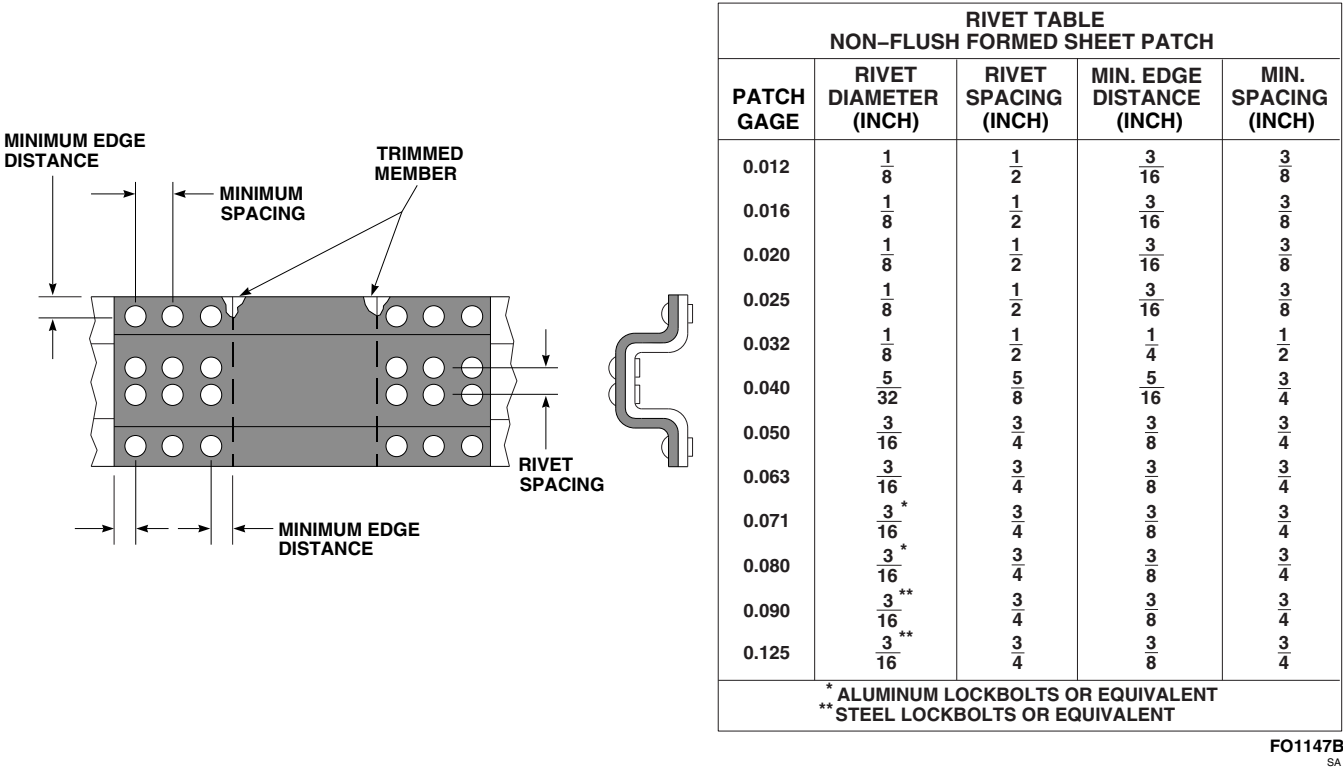
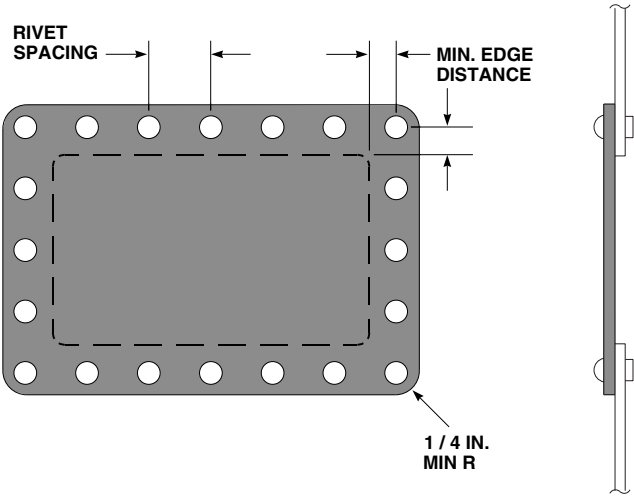


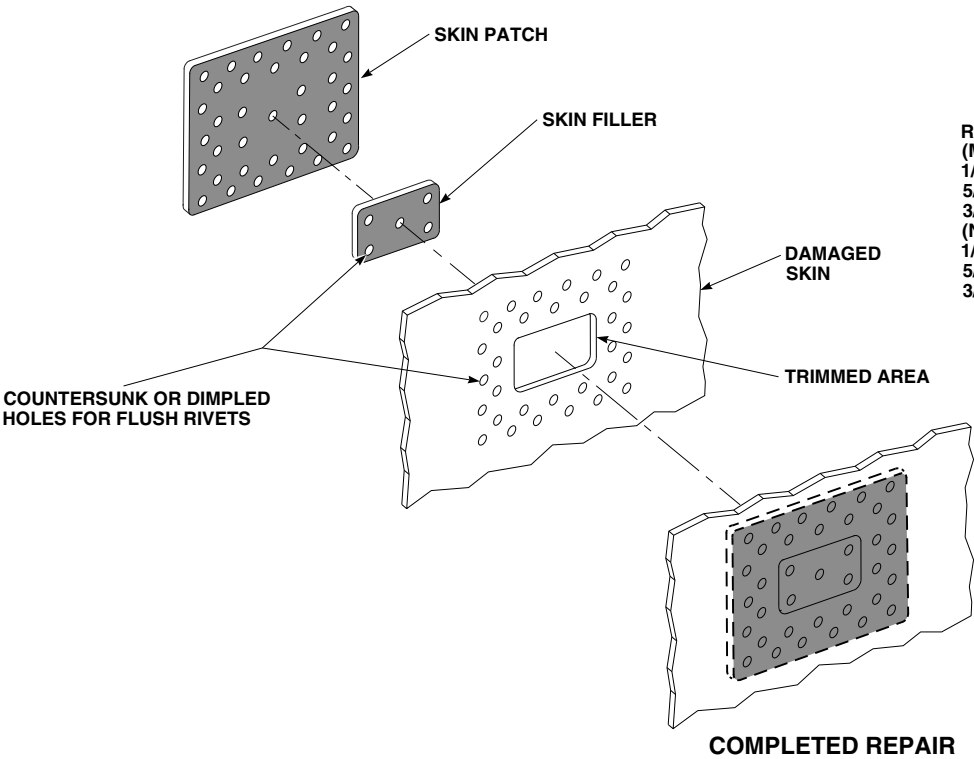
FIGURE 2-6-2-2. PRIMARY STRUCTURE SKIN PATCH - NON-FLUSH FORMED PATCH



RIVET TABLE			
PATCH GAGE	RIVET DIAMETER (INCH)	RIVET SPACING (INCH)	MIN. EDGE DISTANCE (INCH)
0.012	$\frac{1}{8}$	$\frac{3}{4}$	$\frac{3}{16}$
0.016	$\frac{1}{8}$	$\frac{3}{4}$	$\frac{3}{16}$
0.020	$\frac{1}{8}$	$\frac{3}{4}$	$\frac{3}{16}$
0.025	$\frac{1}{8}$	$\frac{3}{4}$	$\frac{3}{16}$
0.032	$\frac{1}{8}$	$\frac{7}{8}$	$\frac{3}{16}$
0.040	$\frac{5}{32}$	$\frac{7}{8}$	$\frac{5}{16}$
0.050	$\frac{5}{32}$	$\frac{7}{8}$	$\frac{5}{16}$
0.063	$\frac{3}{16}$	1	$\frac{3}{8}$
0.071	$\frac{3}{16}$	1	$\frac{3}{8}$
0.080	$\frac{3}{16}$	1	$\frac{3}{8}$
0.090	$\frac{3}{16}$	1	$\frac{3}{8}$
0.125	$\frac{3}{16}$	1	$\frac{3}{8}$

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FIGURE 2-6-2-3. SECONDARY STRUCTURE SKIN PATCH

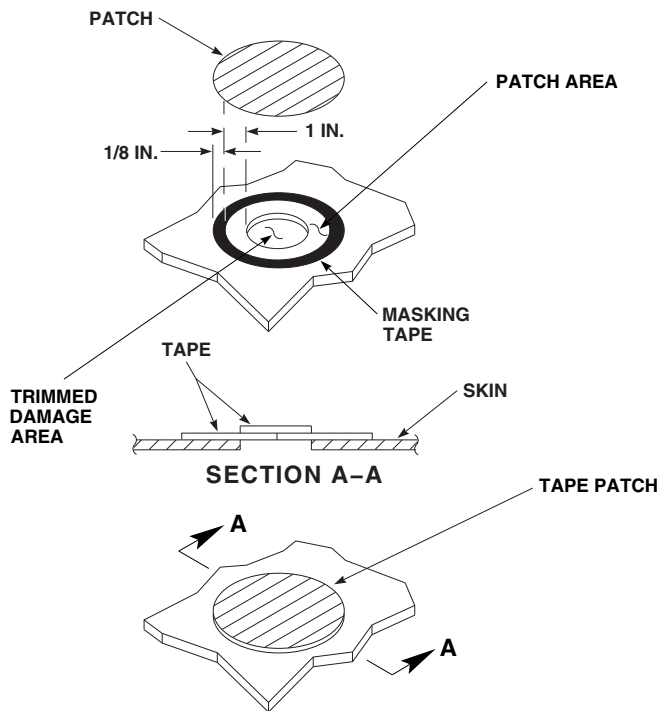


FOR COUNTERSINKING

RIVET SIZE (INCH) (MS20426AD)	MATERIAL THICKNESS (MINIMUM) (INCH)
1/8	0.050
5/32	0.063
3/16	0.080
(NAS1097AD)	
1/8	0.040
5/32	0.050
3/16	0.063

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FIGURE 2-6-2-4. RIVETED FLUSH PATCH REPAIR



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FIGURE 2-6-2-5. TEMPORARY ALUMINUM TAPE PATCH REPAIR

CAUTION

- Pressure sensitive tape adhesive will become contaminated if not properly handled. Do not place adhesive side of pressure sensitive tape patch against template to prevent adhesive contamination.
 - Pressure sensitive tape will not adhere to skin if surface of skin is too cold. Temperature of skin should be at least 50°F (10°C) before aluminum tape patch is applied. Warm skin area to proper temperature if temperature of skin is below minimum.
- f. Make paper patch template same size as cleaned patch area inside masking tape. Trim template 1/8-inch short of masking tape edges and cut pressure sensitive tape, Item 251B, Appendix D, to shape and size of template.
 - g. To maintain patch overlap requirements on wide repairs, use two pieces of tape side-by-

side with edges touching over center of hole. Add another strip of tape in same direction to cover tape joint (Figure 2-6-2-5, Section A-A).

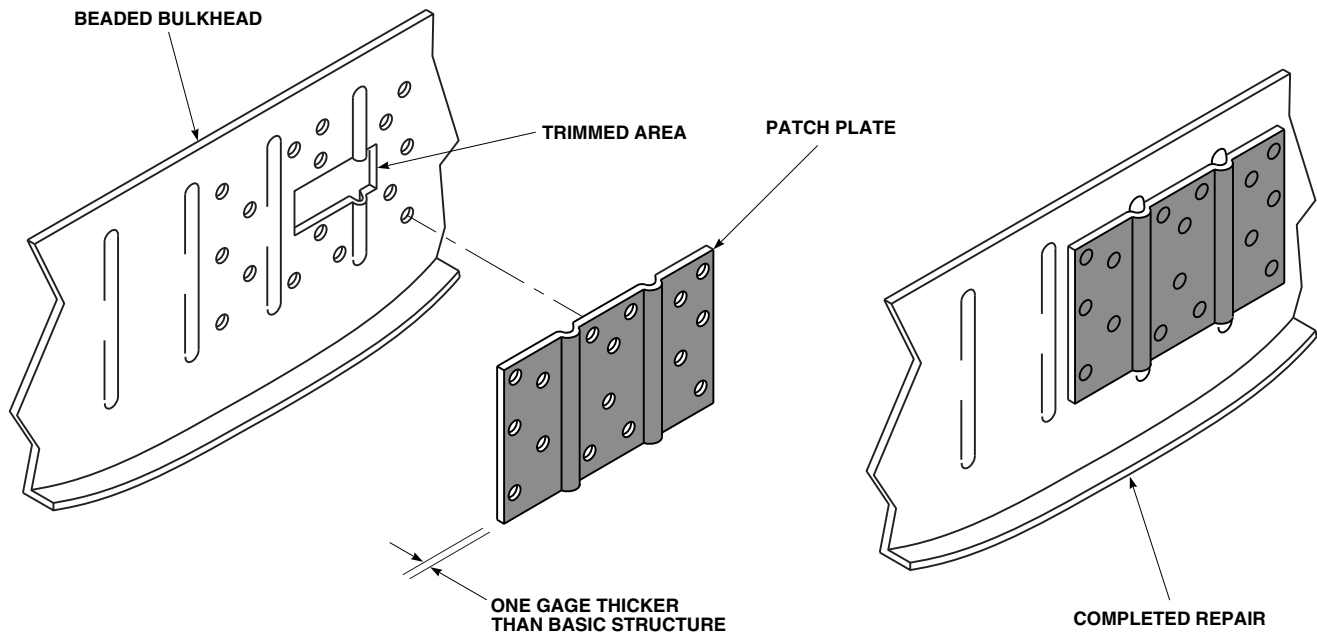
- h. Carefully place pressure sensitive tape patch in place over damaged area on skin, 1/8-inch away from edges of masking tape. Rub patch with clean pad of cheesecloth, Item 90, Appendix D. Do not rub or depress tape patch over area of trimmed hole.

2-6-2.6 Repair Damaged Beaded Section.

- a. Refer to Figure 2-6-2-6.
- b. Patch plate bead radii should nest over or inside bulkhead beads.
- c. Rivet heads should not contact bead radii.

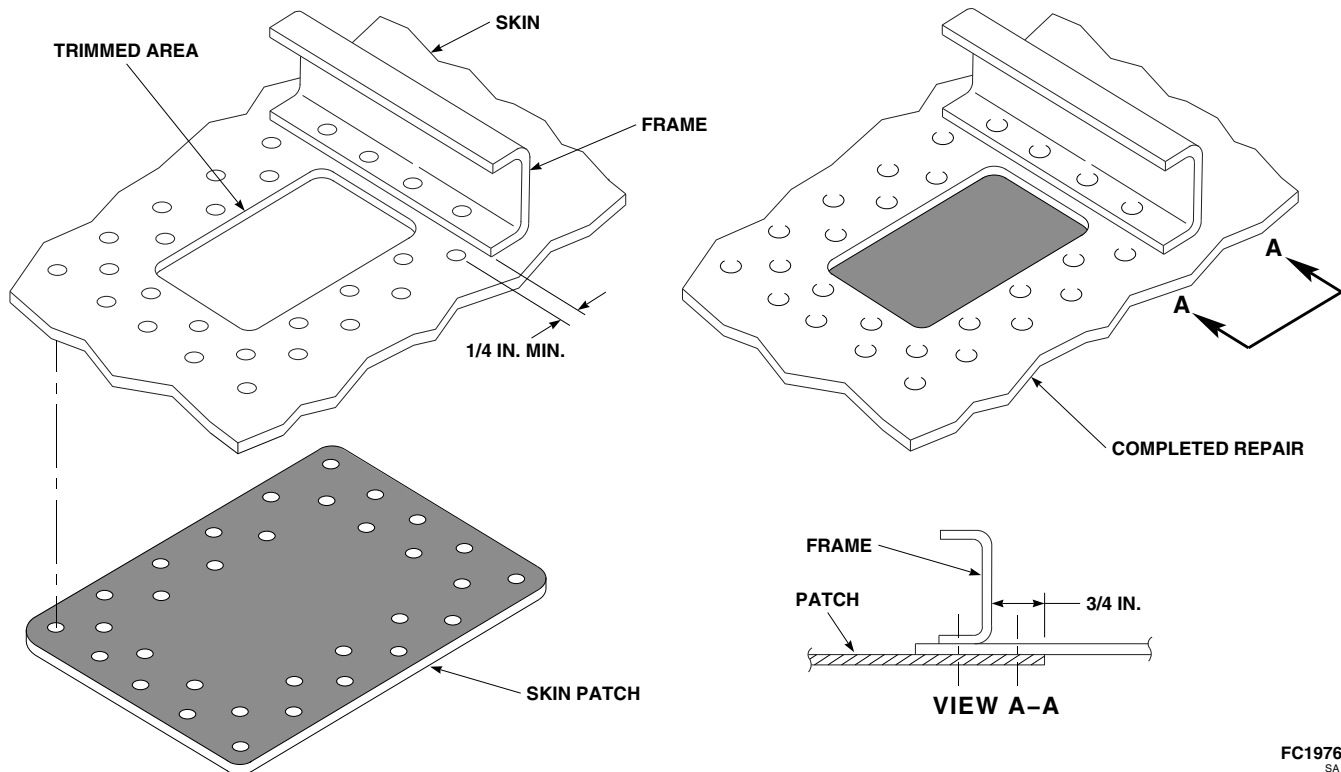
2-6-2.7 Repair Skin Close to a Frame.

- a. Refer to Figure 2-6-2-7.
- b. Skin patch - one gage thicker of same material as original skin.



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FIGURE 2-6-2-6. REPAIR DAMAGED BEADED SECTION



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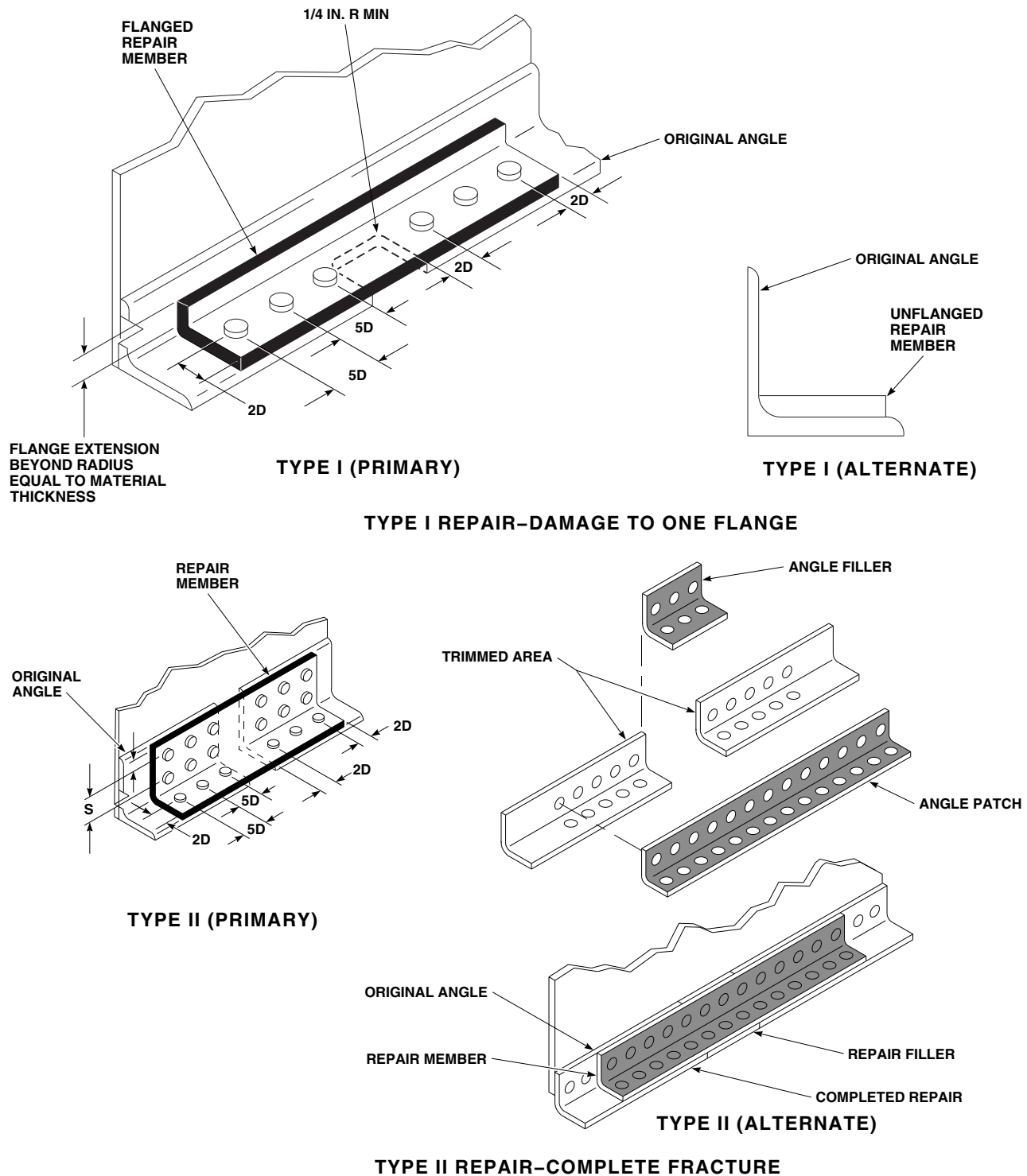
FIGURE 2-6-2-7. REPAIR SKIN CLOSE TO FRAME

- c. Skin patch must extend $\frac{3}{4}$ -inch beyond frame (Figure 2-6-2-7, View A-A).

2-6-2.8 Repair Formed and Extruded Angle.

- a. Refer to Figure 2-6-2-8 and Table 2-6-2-1.
- b. Type I Repair, Damage to One Flange - **DAMAGE MUST NOT BE MORE THAN 60% OF FLANGE WIDTH**. If damage exceeds limit, use Type II Repair.
- c. Repair member to be next heavier gage of same material as original angle (clad sheet permissible for extrusion).
- d. Type I Repair (Primary) - Use flanged repair member. Flange extension beyond radius must be equal to material thickness of repair member.
- e. Type I Repair (Alternate) - Use unflanged repair member.

- f. Lay out rivet pattern around existing original attachments. Replace original attachments with same diameter and type attachments. Use aluminum alloy fillers as necessary.
- g. Repair member radii should nest or be a practicable minimum. Rivet heads should not contact bend radius.
- h. Blind rivets, MS20600AD (or equivalent), may be substituted for same diameter solid rivets, provided the number of blind rivets required is increased at a ratio of 3 to 2.
- i. Rivet requirements for 7075-T6 clad material are tabulated as follows:
 - (1) If original material is 2024-T4 clad, multiply tabulated rivet requirements by 0.8.
 - (2) If original material is extruded, multiply tabulated rivet requirements by 1.15.



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FIGURE 2-6-2-8. REPAIR FORMED AND EXTRUDED ANGLE

Table 2-6-2-1. Rivet Requirements for Formed and Extruded Angle Repair

Angle Gage (in.) Rivets Size Lengthwise Spacing (4D) (in.) Minimum Crosswise Spacing (in.) Minimum Edge Distance (2D) (in.)	Number of Rivets Required When Extent of Damage Across Angle is:				
	0.50-inch	0.75-inch	1-inch	1.25-inches	1.50-inches
Angle Gage - 0.025 Rivet Size - MS20470AD4 Lengthwise Spacing - 0.63 Minimum Crosswise Spacing - 0.55 Minimum Edge Distance - 0.25	3	3	4	5	6
Angle Gage - 0.032 Rivet Size - MS20470AD4 Lengthwise Spacing - 0.63 Minimum Crosswise Spacing - 0.55 Minimum Edge Distance - 0.25	3	4	5	6	8
Angle Gage - 0.040 Rivet Size - MS20470AD5 Lengthwise Spacing - 0.78 Minimum Crosswise Spacing - 0.67 Minimum Edge Distance - 0.31	3	3	4	5	6
Angle Gage - 0.050 Rivet Size - MS20470AD5 Lengthwise Spacing - 0.78 Minimum Crosswise Spacing - 0.67 Minimum Edge Distance - 0.31	3	4	5	6	8
Angle Gage - 0.063 Rivet Size - MS20470AD5 Lengthwise Spacing - 0.78 Minimum Crosswise Spacing - 0.67 Minimum Edge Distance - 0.31	3	5	6	8	9

Table 2-6-2-1. Rivet Requirements for Formed and Extruded Angle Repair (Cont)

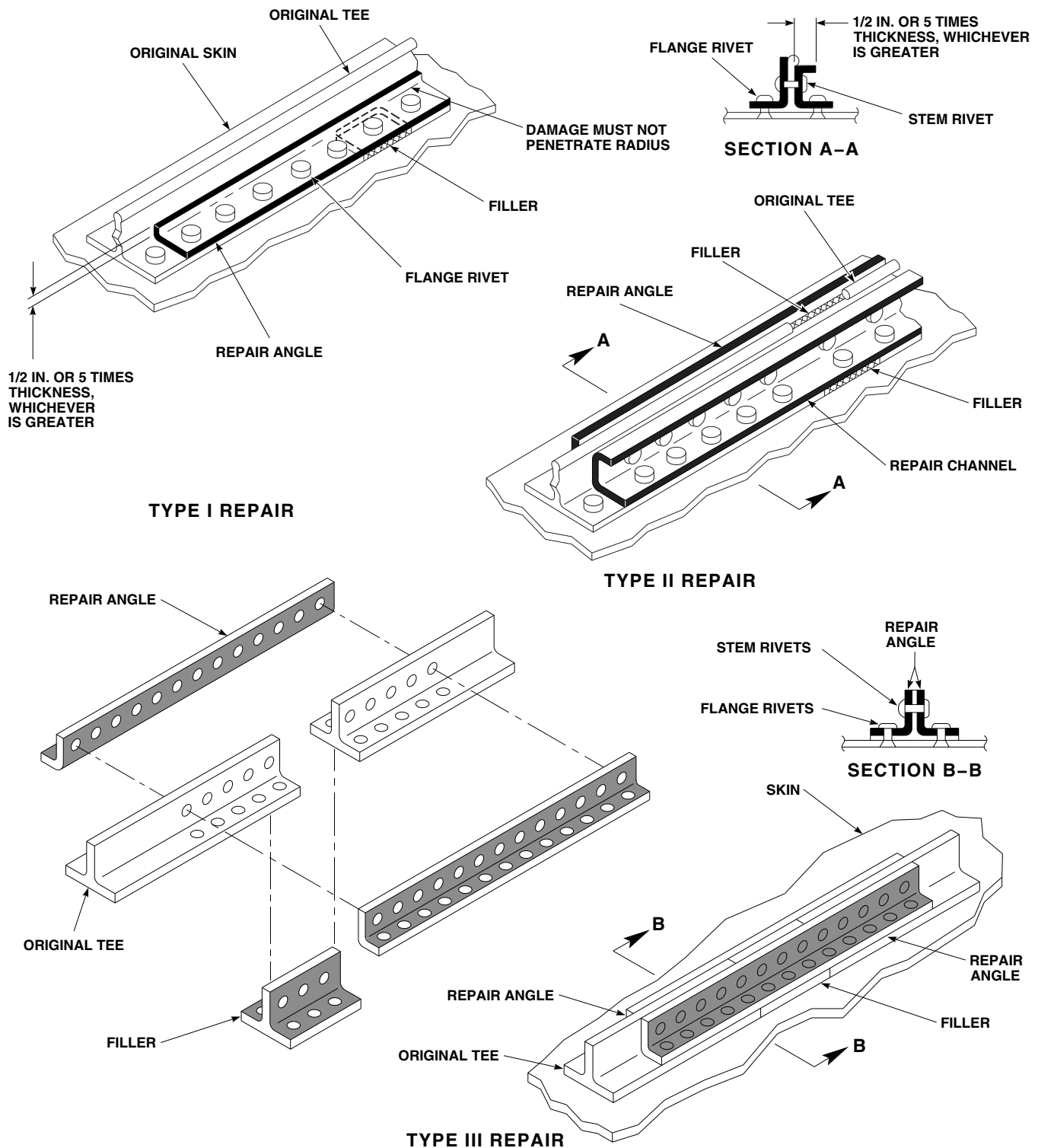
Angle Gage (in.) Rivets Size Lengthwise Spacing (4D) (in.) Minimum Crosswise Spacing (in.) Minimum Edge Distance (2D) (in.)	Number of Rivets Required When Extent of Damage Across Angle is:				
Angle Gage - 0.063 Rivet Size - MS20470AD6 Lengthwise Spacing - 0.94 Minimum Crosswise Spacing - 0.81 Minimum Edge Distance - 0.38	3	4	5	6	7

2-6-2.9 Repair Damaged Tee.

- a. Refer to Figure 2-6-2-9.
- b. Repairs are applicable to two types of tee extrusions shown in Figure 2-6-2-9, Sections A-A and B-B.
- c. Make repair angles and fillers from same material as original section. Gage for repair angles to be one gage heavier than flanges of original section, but not less than one-half thickness of stem of original section.
- d. Lay out rivet pattern around any original attachments. Replace original attachments using fasteners shown in Table 2-6-2-2.
- e. If extent of damage exceeds 0.500-inch, insert fillers and continue rivet distribution through damaged area.
- f. Repair member radii should nest or be practicable minimum. Rivet heads should not contact bend radius.
- g. Stem Rivets, MS20470AD:
 - (1) For 7075-T6, use 5 rivets on each side of damaged area.
 - (2) For 2024-T4, use 4 rivets on each side of damaged area.
- h. Flange Rivets, MS20426AD, Hi-Loks, or Lockbolts:
 - (1) For 7075-T6, use 5 rivets on each side of damaged area in each row.
 - (2) For 2024-T4, use 4 rivets on each side of damaged area in each row.

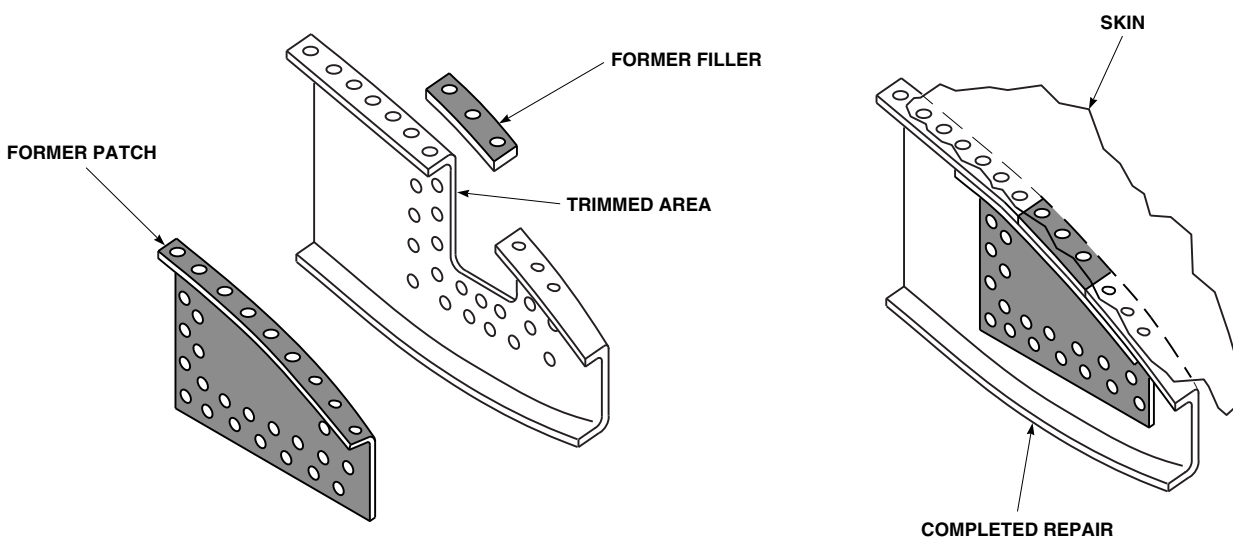
Table 2-6-2-2. Attachment Requirements for Damaged Tee Repair

Original Material Thickness	Rivet Diameter (D)	Hi-Lok or Lockbolt Diameter (D)	2D Edge Distance
0.032	4	-	0.25
0.040	5	-	0.31
0.050	5	-	0.31
0.063	6	-	0.38
0.071	6	-	0.38
0.080	-	6	0.38
0.090	-	6	0.38
0.100	-	6	0.38
0.125	-	6	0.38
0.160	-	6	0.38
0.190	-	6	0.38



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FIGURE 2-6-2-9. REPAIR DAMAGED TEE



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FIGURE 2-6-2-10. REPAIR DAMAGED FORMER

2-6-2.10 Repair Damaged Former.

- Refer to Figure 2-6-2-10.
- Former filler same material and thickness as former.
- Former patch same material as former, one gage thicker.
- Former patch bend radius should nest or be practicable minimum. Rivet heads should not contact bend radius.
- Minimum of 3 rivets required on each side of damage.

- Rivet heads should not contact bend radii.

- Example One Repair - Minimum of 3 rivets in each row are required on each side of damage.

2-6-2.12 Repair Damaged Channel.

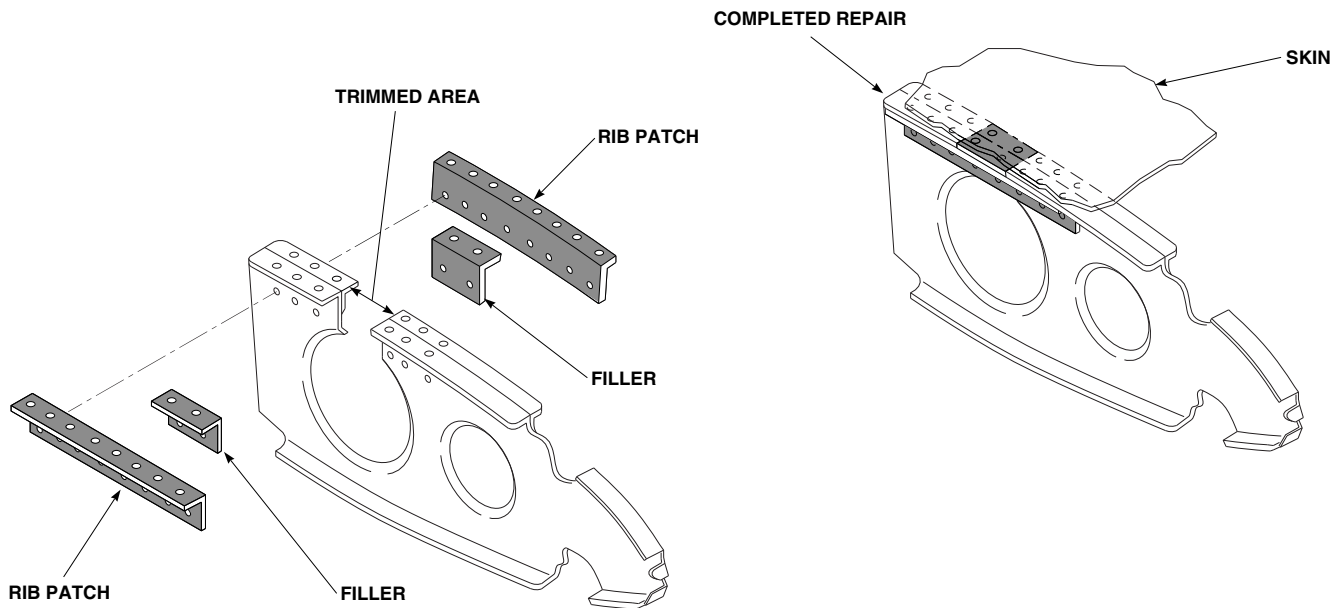
- Refer to Figure 2-6-2-12.
- Channel patch same material as channel, one gage thicker.
- Channel patch bend radii should nest or be practicable minimum.
- Rivet heads should not contact bend radii.

2-6-2.11 Repair Damaged Rib.

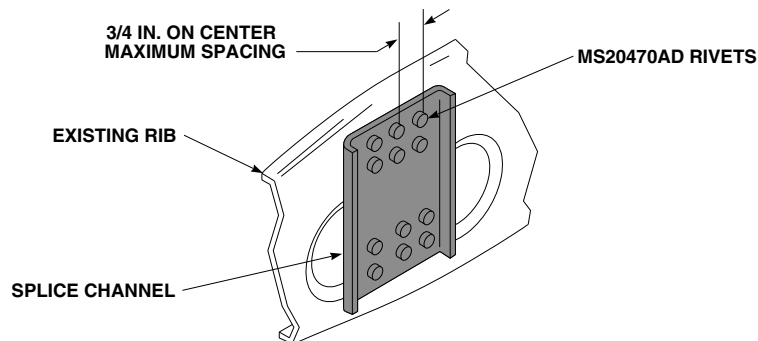
- Refer to Figure 2-6-2-11.
- Rib Patch - Pick up existing rivet locations through bulkhead flanges. Rivets should be added between existing rivets when minimum of 4D rivet spacing can be maintained.

2-6-2.13 Repair Damaged Zee.

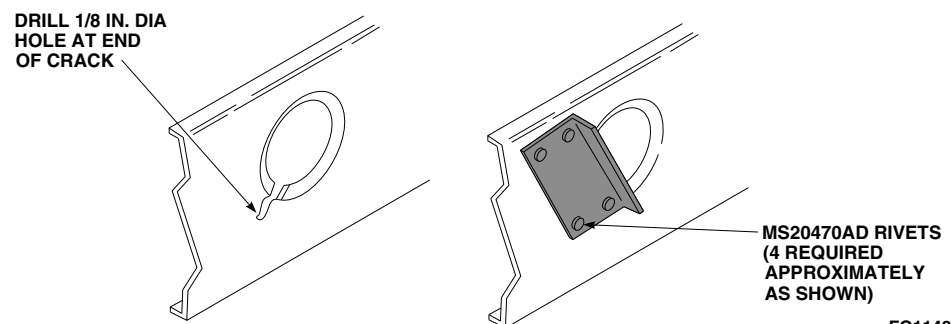
- Refer to Figure 2-6-2-13, Sheets 1 and 2, and Table 2-6-2-3.
- Repair material to be same as existing material.
- Rivet Requirements:



EXAMPLE ONE



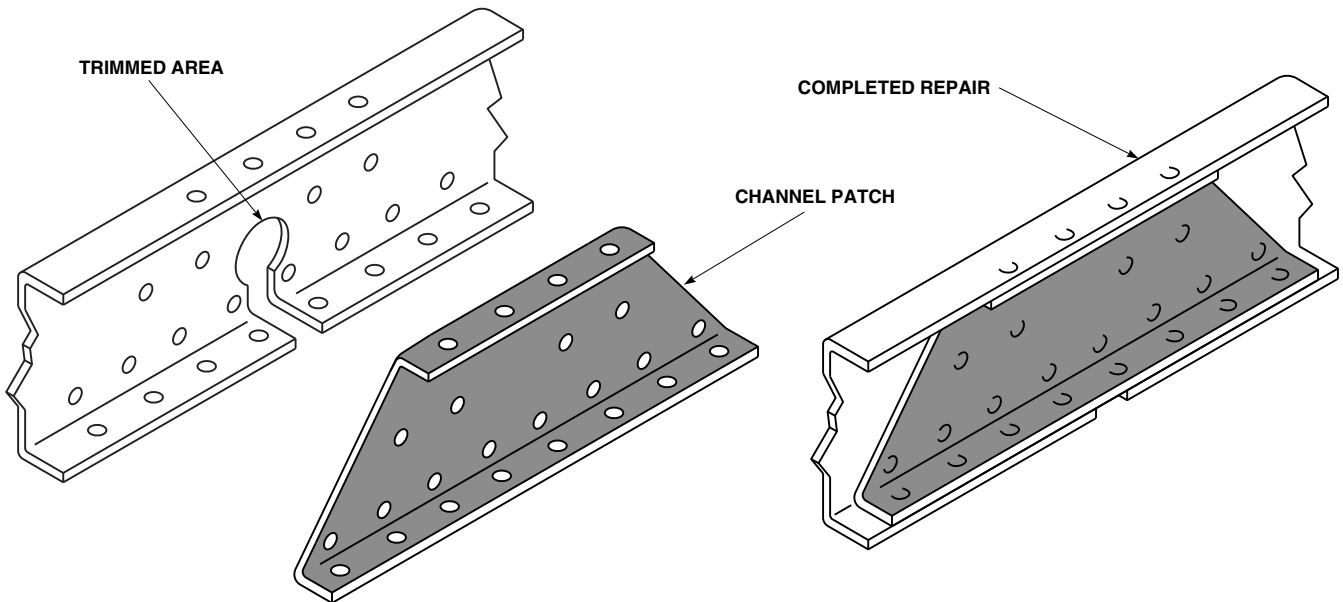
EXAMPLE TWO



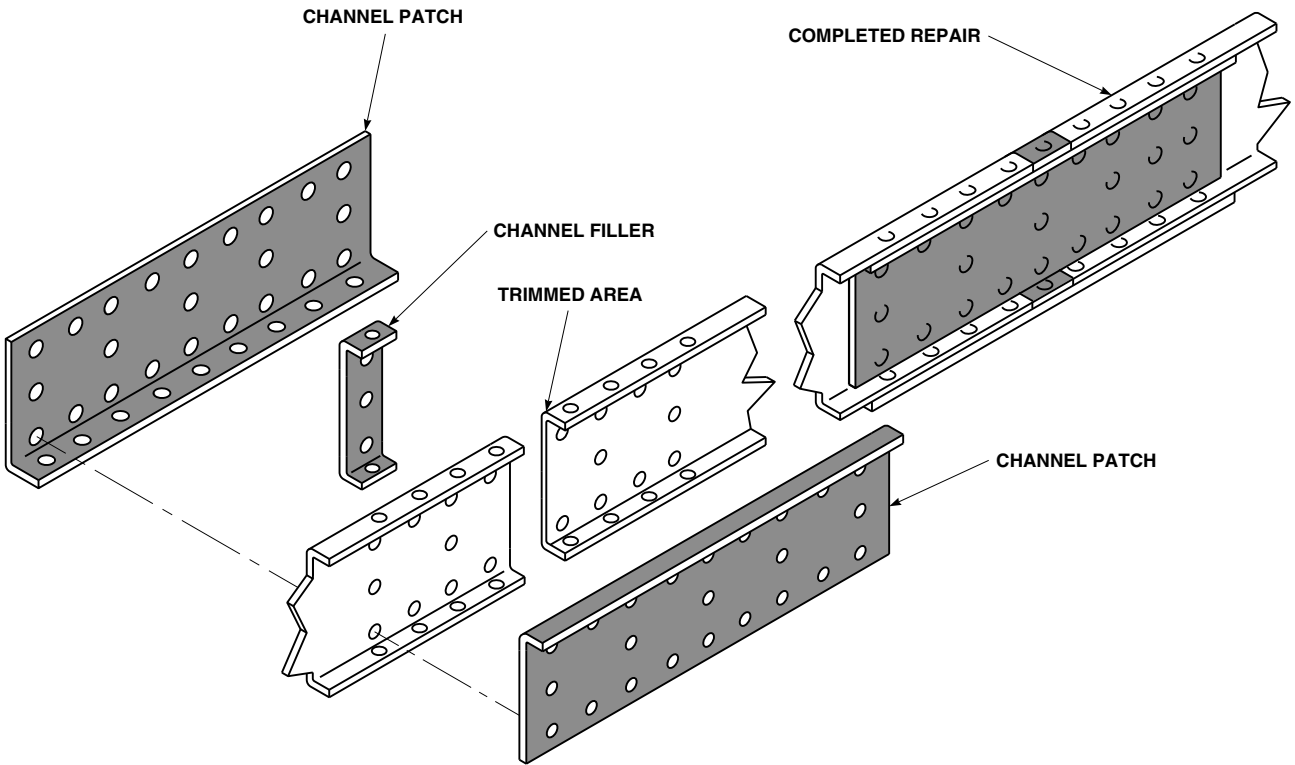
EXAMPLE THREE

FIGURE 2-6-2-11. REPAIR DAMAGED RIB

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PATCHING CHANNEL CRACKS



CHANNEL INSERTION

FIGURE 2-6-2-12. REPAIR DAMAGED CHANNEL

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- (1) Use same type of rivets in skin flange as originally installed. If there is no skin attached to member, use rivets, MS20470AD.
 - (2) Use existing skin flange rivet spacing if flange is drilled. If member was not originally drilled, use 5 times rivet diameter for center-to-center spacing.
 - (3) Use original hole and rivet size as required. If hole is elongated, use larger diameter.
- d. Filler member to be same gage and material as original zee.
 - e. Use same number of rows of attachments per flange as in original zee.
 - f. Flange angle with web may vary.
 - g. When L2 is less than 10 times diameter of required rivets, form single zee instead of two angles and leave off web strap. Use only one row of rivets through web.
 - h. Repair angle and zee patch bend radii should nest or be practicable minimum.
 - i. Rivet heads should not contact bend radii.

Table 2-6-2-3. Rivet Requirements for Damaged Zee Repair

Zee Gage (in.) Repair Member Gage (in.) Rivet Size	Number of Rivets Required After Calculating L1 + L2 + L3 (See Note):				
	2- Inches	2.50- Inches	3- Inches	3.5- Inches	4- Inches
Zee Gage - 0.025 Repair Member Gage - 0.032 Rivet Size - MS20470AD4	10	13	15	18	20
Zee Gage - 0.032 Repair Member Gage - 0.040 Rivet Size - MS20470AD4	20	15	18	22	25
Zee Gage - 0.040 Repair Member Gage - 0.050 Rivet Size - MS20470AD5	-	13	15	18	20
Zee Gage - 0.050 Repair Member Gage - 0.063 Rivet Size - MS20470AD5	-	15	18	21	24
Zee Gage - 0.063 Repair Member Gage - 0.071 Rivet Size - MS20470AD6	-	19	23	27	30

Table 2-6-2-3. Rivet Requirements for Damaged Zee Repair (Cont)				
Zee Gage (in.) Repair Member Gage (in.) Rivet Size	Number of Rivets Required After Calculating L1 + L2 + L3 (See Note):			
NOTE Number of rivets shown is for each side of damage. Divide equally among all rows. Add to number in table as necessary to make it divisible by number of rows used. For in-between values of L1 + L2 + L3 use next higher column.				

2-6-2.14 Repair Damaged Longeron.

- a. Refer to Figure 2-6-2-14.
- b. Repair member bend radii should nest or be practicable minimum.
- c. Rivet heads should not contact bend radii.
- d. Filler - Same gage and material as longeron.
- e. Top splice zee, top splice, and bottom splice - Same material as longeron, one gage thicker.

2-6-2.15 Repair Damaged Frame.

- a. Refer to Figure 2-6-2-15.
- b. Repair member bend radii should nest or be practicable minimum.
- c. Rivet heads should not contact bend radii.
- d. Filler - Same gage and material as frame.
- e. Patch angle and patch- Same material as frame, one gage thicker.
- f. Use MS20470AD-type rivet.

2-6-2.16 Repair Damaged Lightning Holes.

- a. Refer to Figure 2-6-2-16.
- b. Repair member bend radii should nest or be practicable minimum.

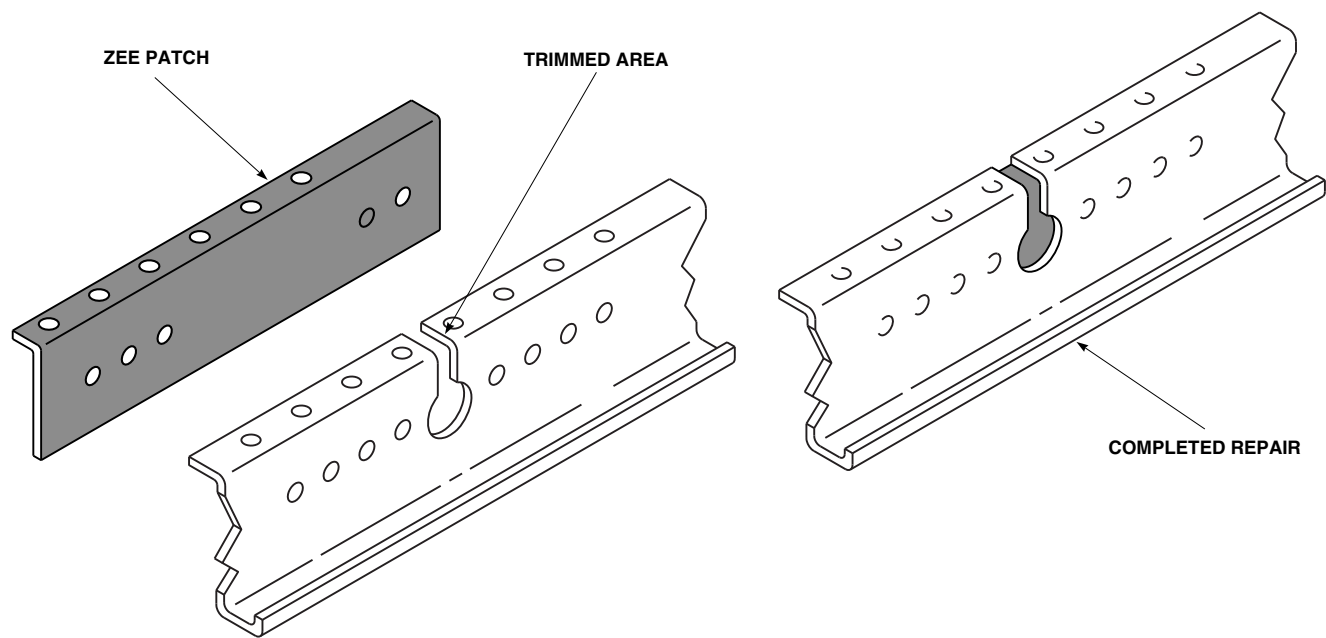
- c. Rivet heads should not contact bend radii.
- d. Filler - Same gage and material as frame.
- e. Frame patch - Same material as frame, one gage thicker.

2-6-2.17 Repair Damaged Hat Section.

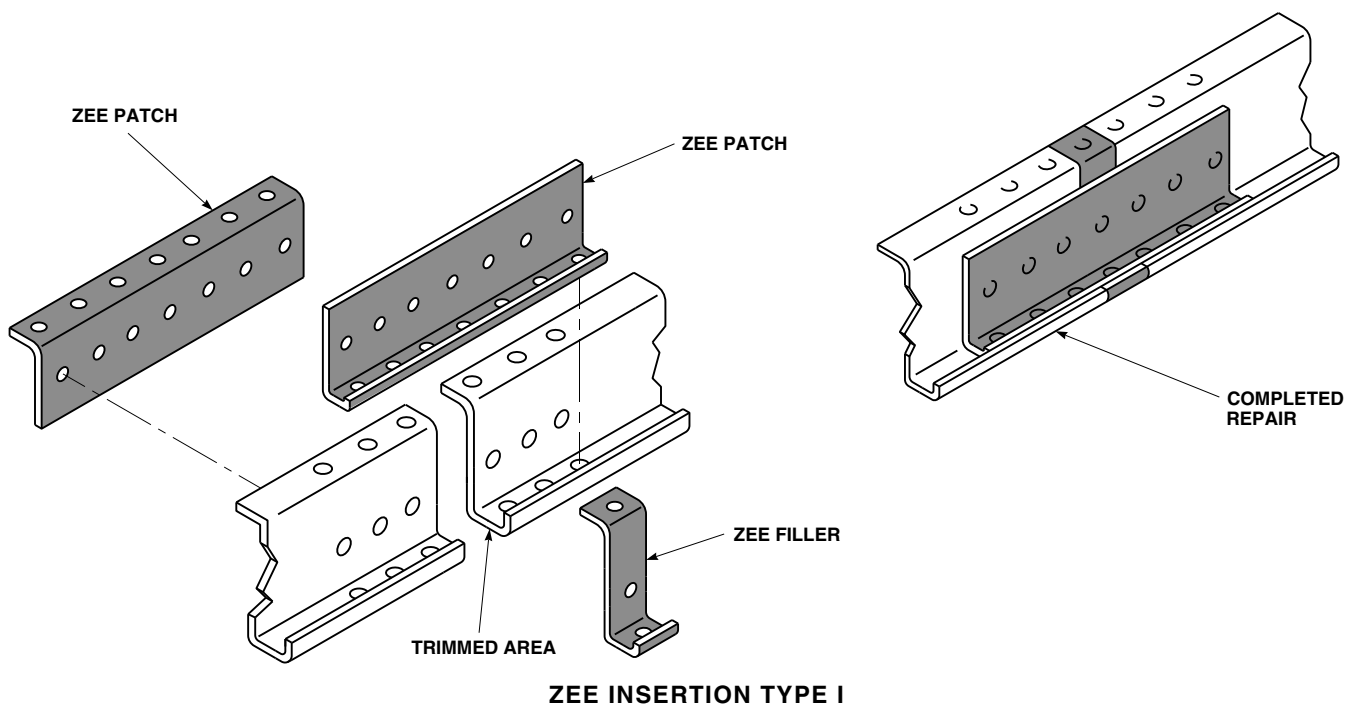
- a. Refer to Figure 2-6-2-17.
- b. Hat section patch bend radii should nest or be practicable minimum.
- c. Rivet heads should not contact bend radii.
- d. Skin filler - Same gage and material as skin.
- e. Hat section filler - Same gage and material as hat section.
- f. Skin patch - Same material as skin, one gage thicker.
- g. Hat section patch- Same material as hat section, one gage thicker.

2-6-2.18 Repair Damaged Channel Stiffener.

- a. Refer to Figure 2-6-2-18.
- b. Repair member bend radii should nest or be practicable minimum.
- c. Rivet heads should not contact bend radii.
- d. Channel filler - Same gage and material as channel.



PATCHING ZEE CRACKS



ZEE INSERTION TYPE I

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FIGURE 2-6-2-13. REPAIR DAMAGED ZEE (SHEET 1 OF 2)

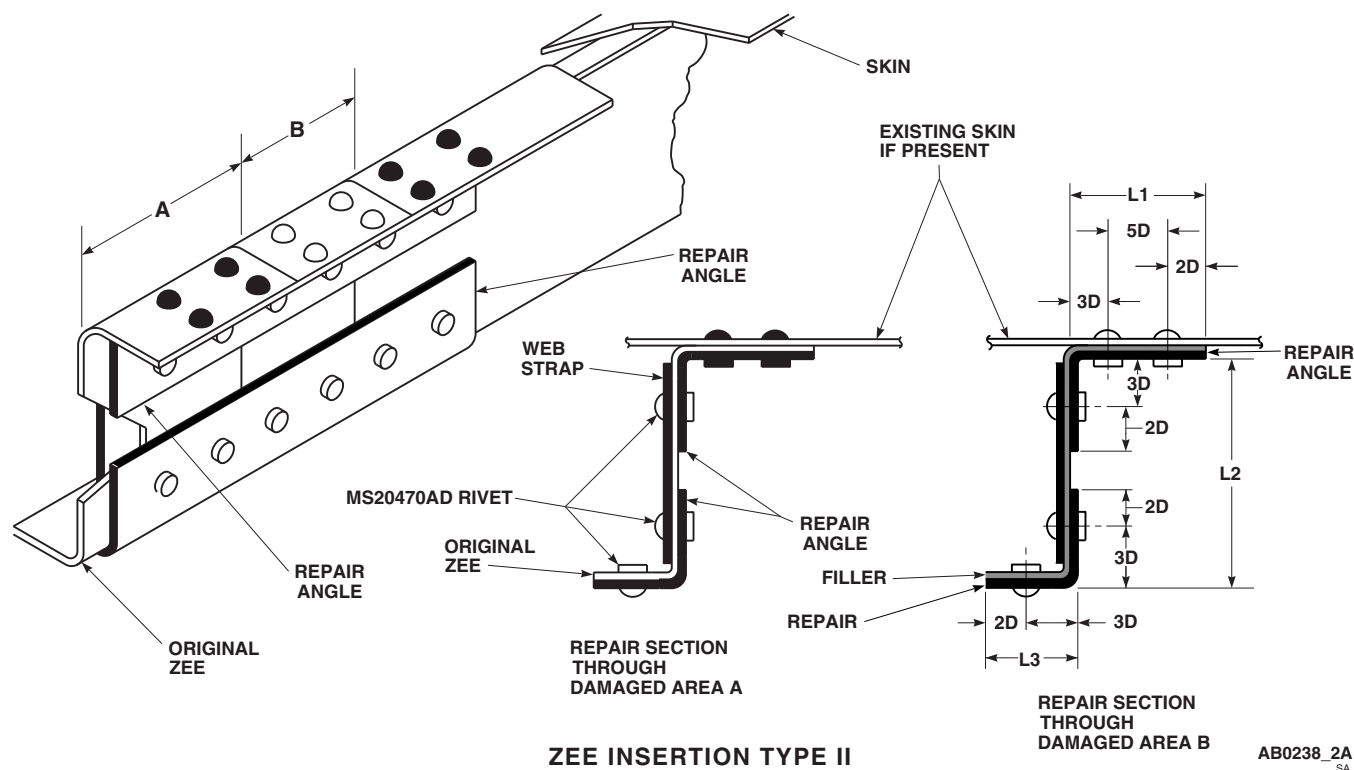


FIGURE 2-6-2-13. REPAIR DAMAGED ZEE (SHEET 2 OF 2)

- e. Skin filler - Same gage and material as skin.
- f. Skin patch - Same material as skin, one gage thicker.

2-6-2.19 Repair Damaged Tee and Web.

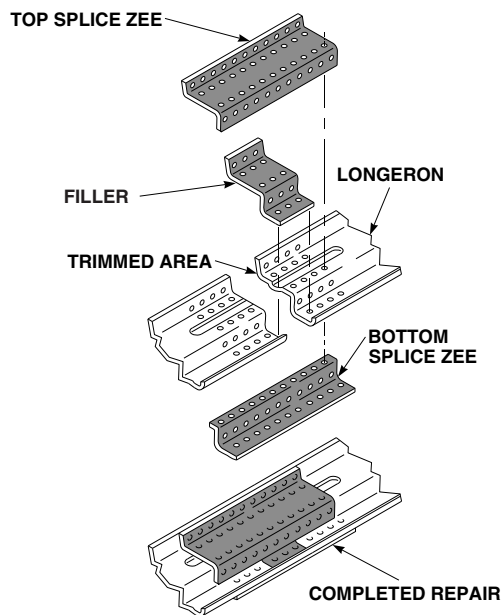
- a. Refer to Figure 2-6-2-19.
- b. Repair member bend radii should nest or be practicable minimum.
- c. Rivet heads should not contact bend radii.
- d. Web filler - Same gage and material as web.
- e. Tee filler - Same gage and material as tee.
- f. Patch angle - 0.125 gage, same material as tee.
- g. Skin filler - Same gage and material as skin.
- h. Skin Patch - Same material as skin, one gage thicker.
- i. Use Hi-Shear rivets, HS52P6.

2-6-2.20 Repair Damaged Beam Caps.

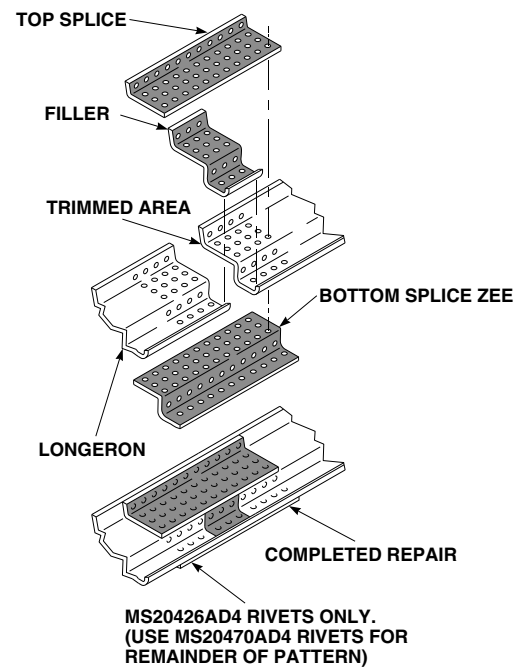
- a. Refer to Figure 2-6-2-20.
- b. Strap patch edge should have radius to nest within radius of angle.
- c. Rivet heads should not contact bend radii.
- d. Skin filler - Same gage and material as skin.
- e. Angle filler - Same gage and material as angle.
- f. Strap patch - Same material as angle, one gage thicker.
- g. Skin patch - Same material as skin, one gage thicker.

2-6-2.21 Repair Damaged Skin and Frame.

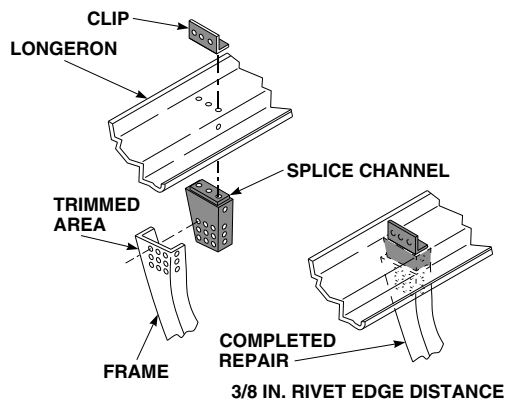
- a. Refer to Figure 2-6-2-21.
- b. Repair member bend radii should nest or be practicable minimum.
- c. Rivet heads should not contact bend radii.



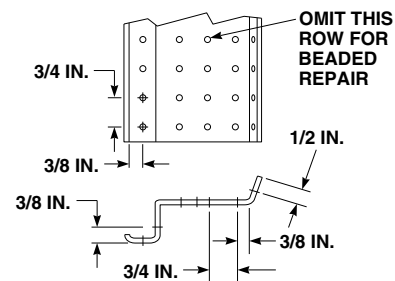
BEADED REPAIR



UNBEADED REPAIR



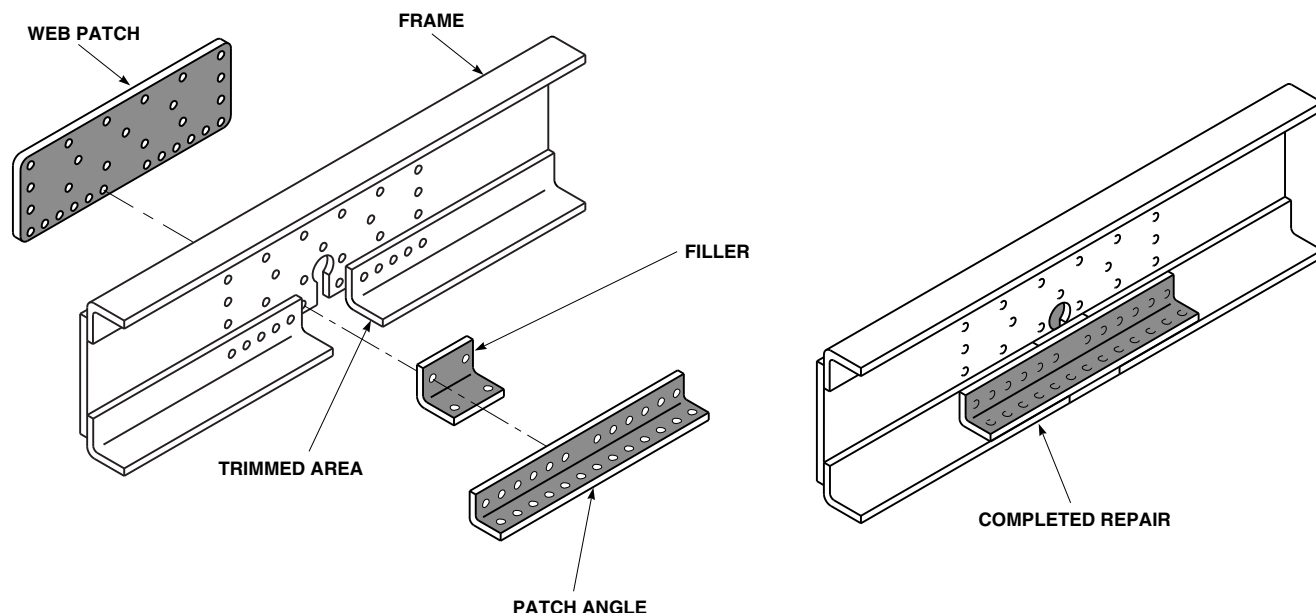
FRAME AND LONGERON REPAIR



RIVET PATTERN (TYPICAL)

FIGURE 2-6-2-14. REPAIR DAMAGED LONGERON

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FIGURE 2-6-2-15. REPAIR DAMAGED FRAME

- d. Fillers - Same gage and material as original member.
- e. Patches - Same material as original member, one gage thicker.

2-6-2.22 Repair Damaged Skin and Stringer.

- a. Refer to Figure 2-6-2-22.
- b. Stringer patch bend radii should nest or be practicable minimum.
- c. Rivet heads should not contact bend radii.
- d. Fillers - Same gage and material as original member.
- e. Patches - Same material as original member, one gage thicker.

2-6-2.23 Repair Damaged Skin, Frame, and Stringer.

- a. Refer to Figure 2-6-2-23.
- b. Repair member bend radii should nest or be practicable minimum.

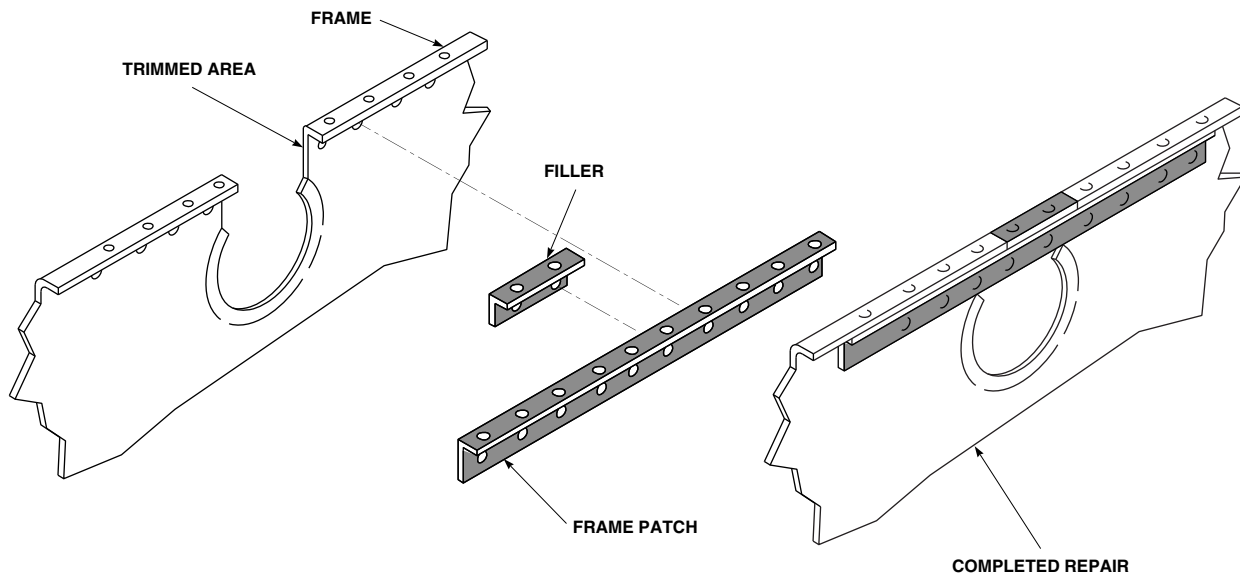
- c. Rivet heads should not contact bend radii.
- d. Fillers - Same gage and material as original member.
- e. Splices and patches - Same material as original member, one gage thicker.

2-6-2.24 Repair Damaged Bulkhead Web and Skin.

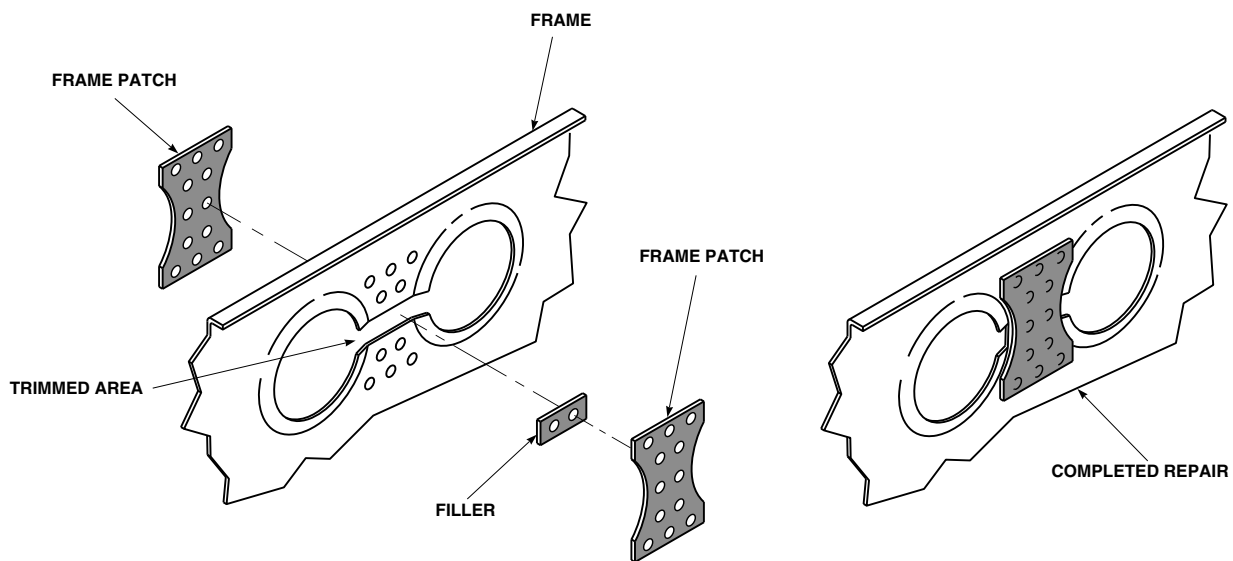
- a. Refer to Figure 2-6-2-24.
- b. Repair member bend radii should nest or be practicable minimum.
- c. Rivet heads should not contact bend radii.
- d. Fillers - Same gage and material as original member.
- e. Patches - Same material as original member, one gage thicker.

2-6-2.25 Miscellaneous Repairs.

- a. Refer to Figure 2-6-2-25, Sheet 1 and Sheet 2.



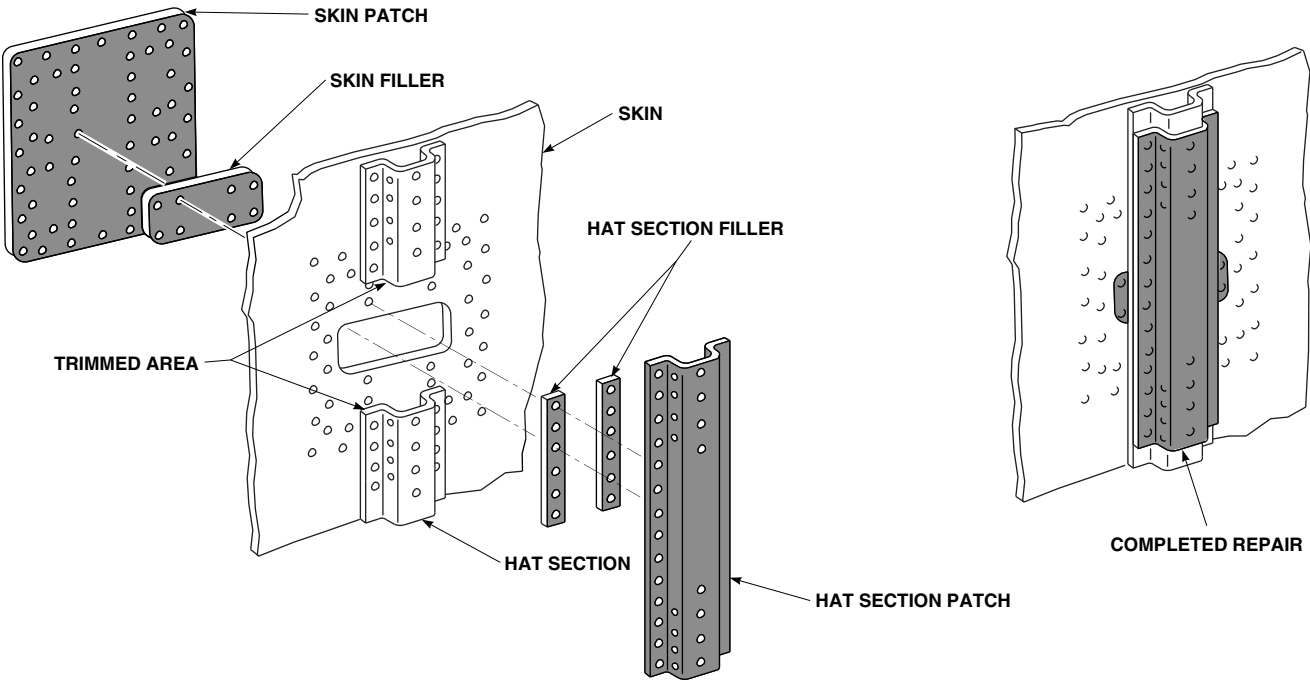
REPAIR OF LIGHTENING HOLE AND FLANGE



REPAIR BETWEEN LIGHTENING HOLES

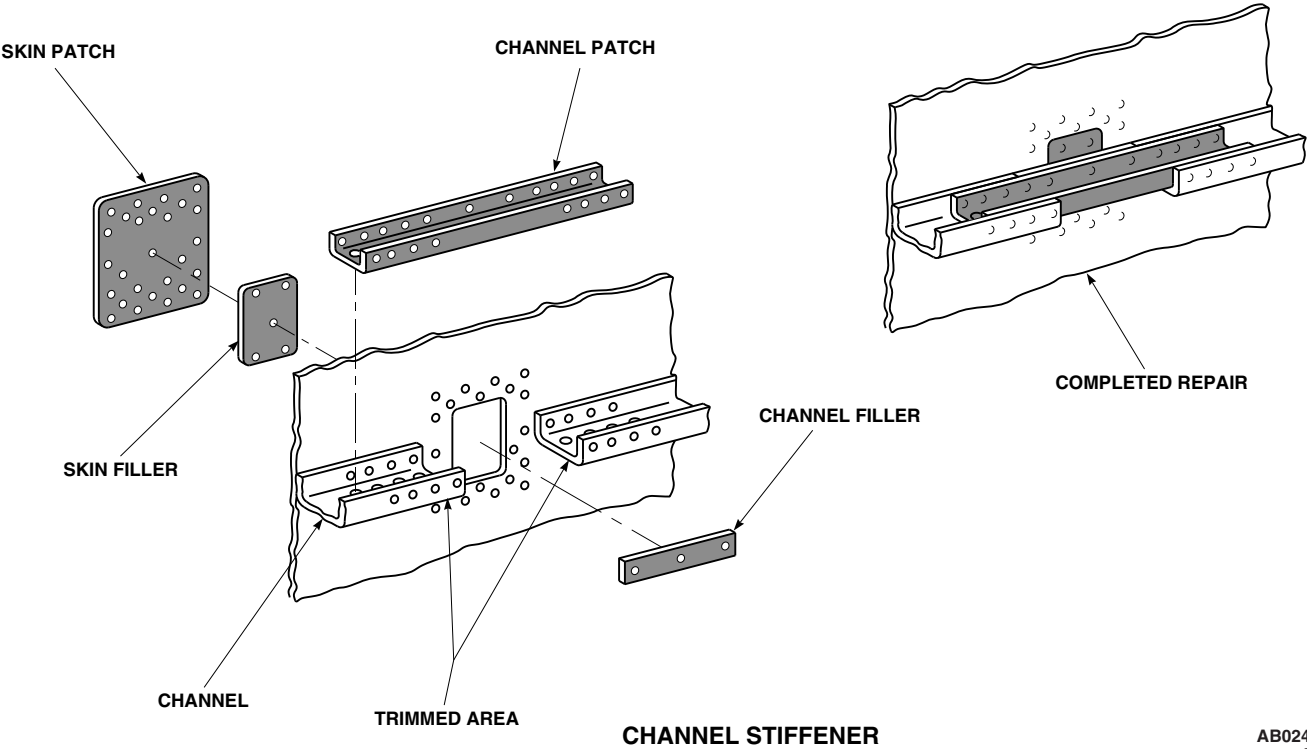
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FIGURE 2-6-2-16. REPAIR DAMAGED LIGHTENING HOLES



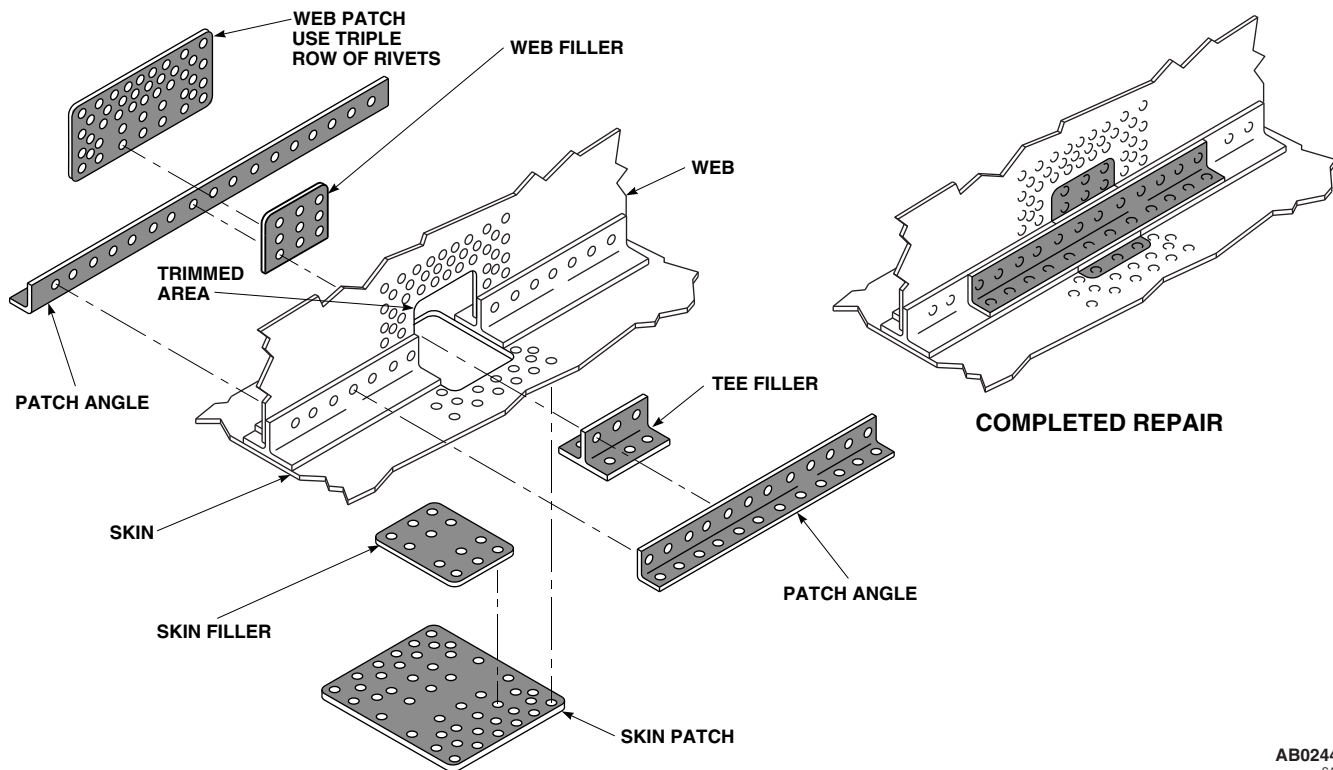
AB0242
SA

FIGURE 2-6-2-17. REPAIR DAMAGED HAT SECTION



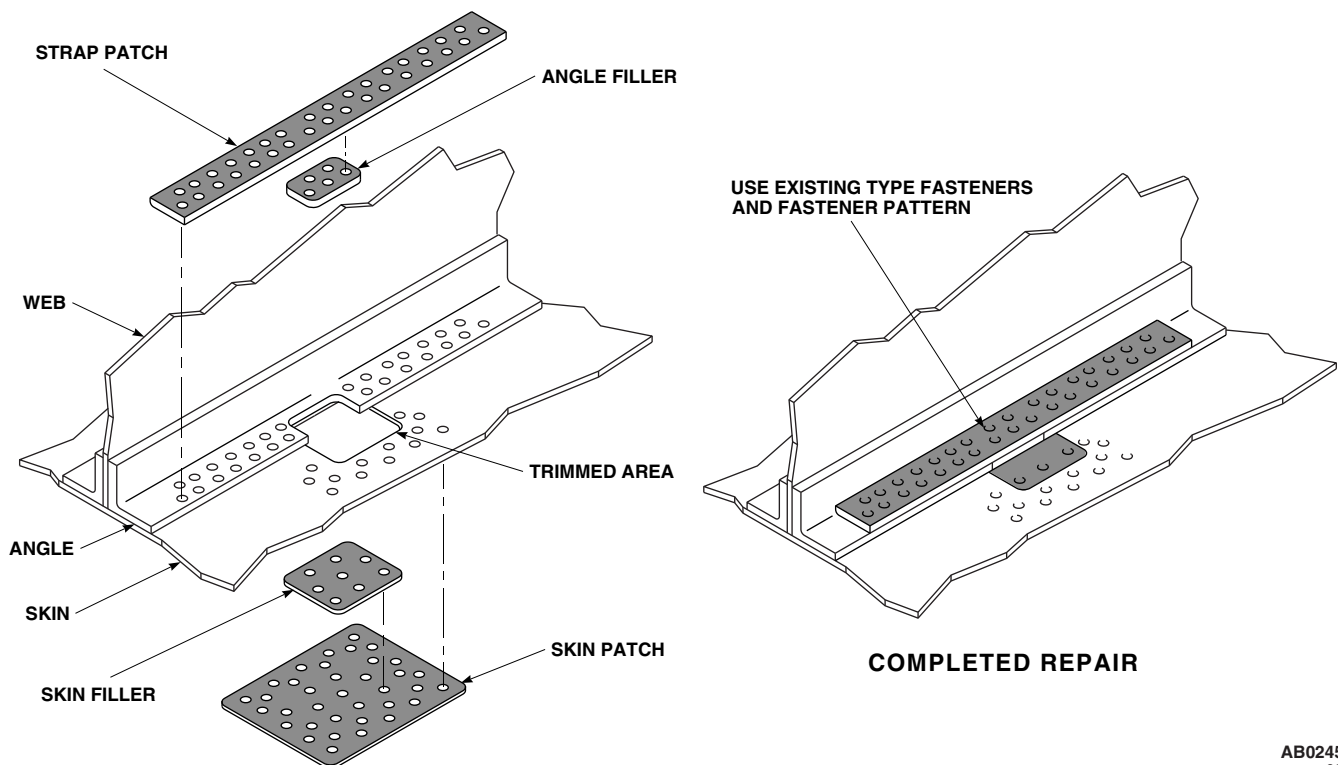
AB0243
SA

FIGURE 2-6-2-18. REPAIR DAMAGED CHANNEL STIFFENER



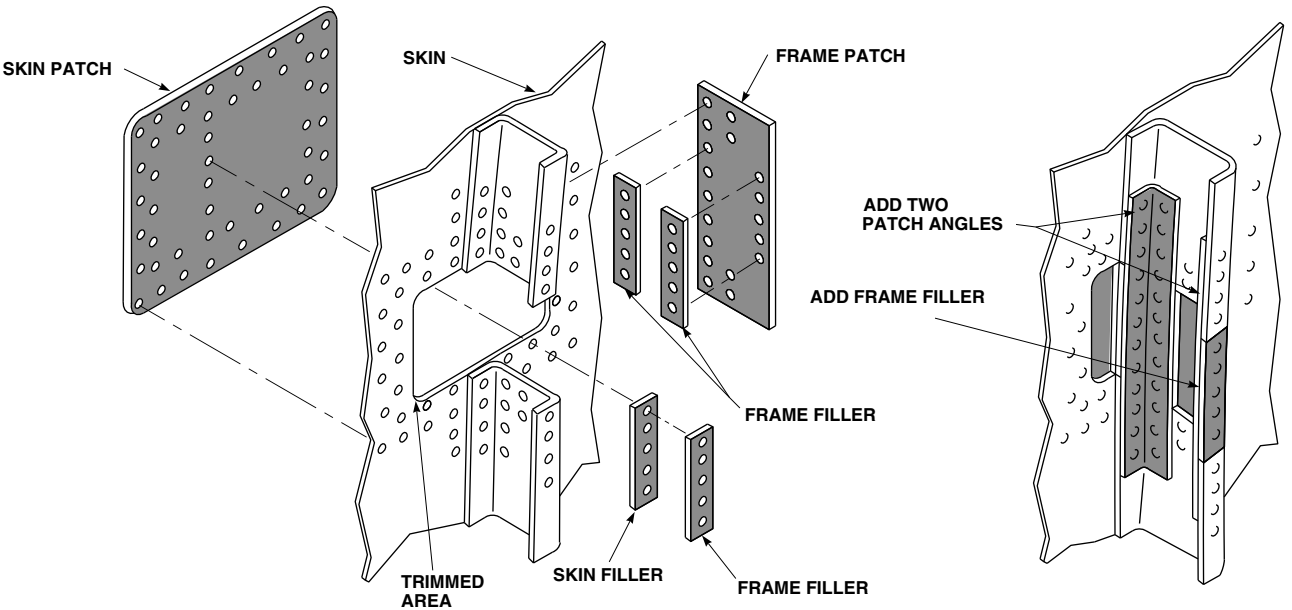
AB0244
SA

FIGURE 2-6-2-19. REPAIR DAMAGED TEE AND WEB



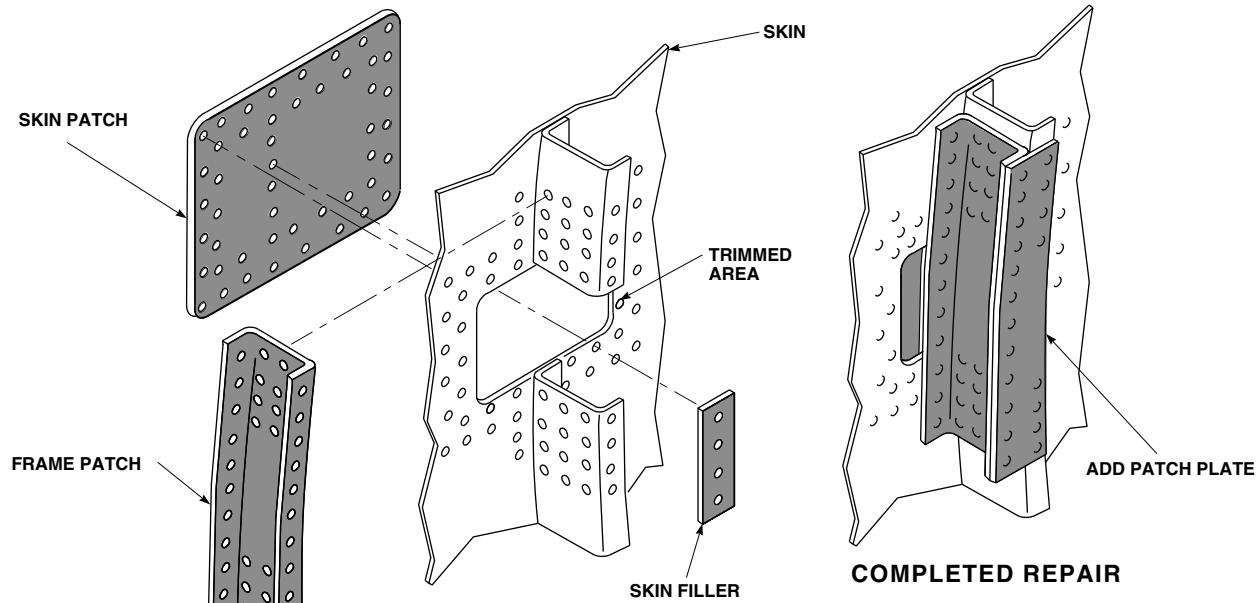
AB0245
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FIGURE 2-6-2-20. REPAIR DAMAGED BEAM CAPS



FRAME AND SKIN REPAIR

COMPLETED REPAIR

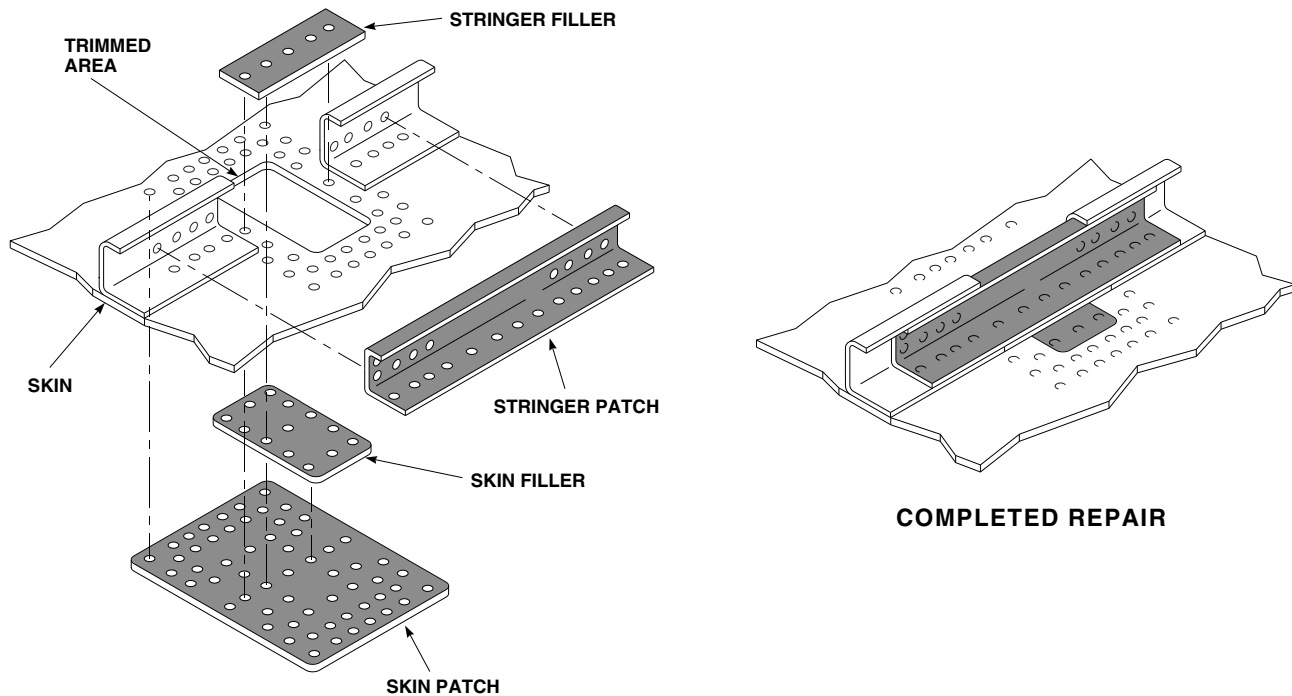


ALTERNATE FRAME AND SKIN REPAIR

COMPLETED REPAIR

FIGURE 2-6-2-21. REPAIR DAMAGED SKIN AND FRAME

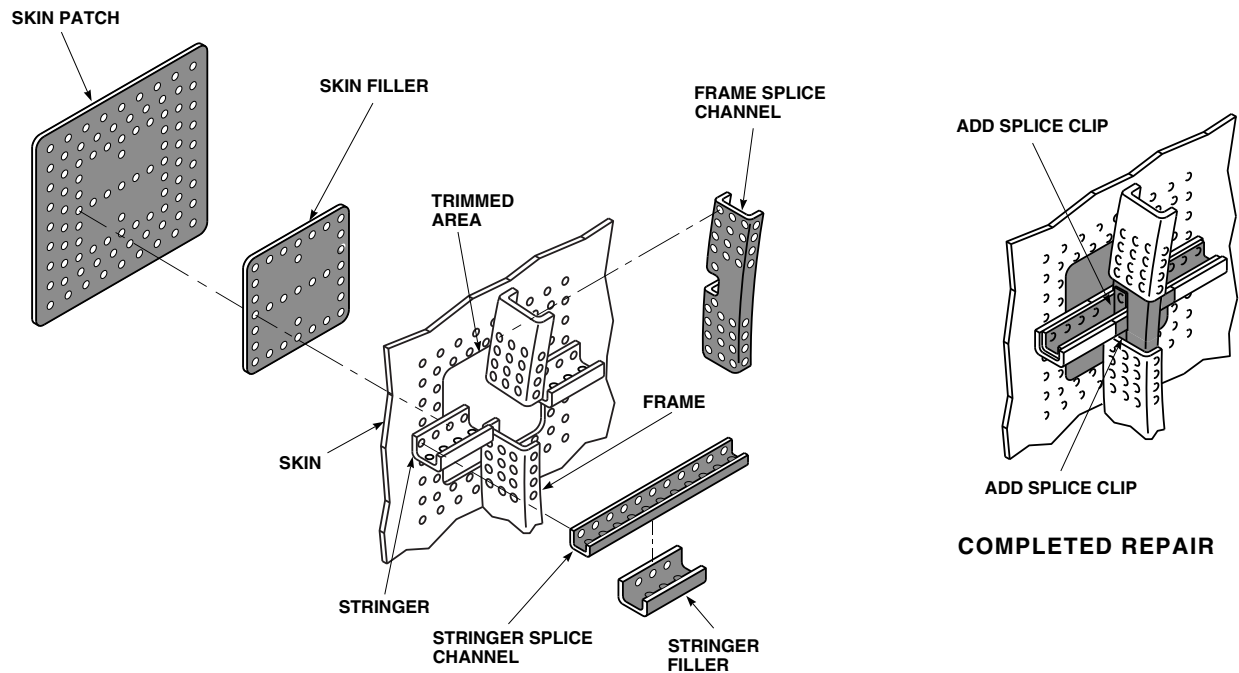
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SA**FIGURE 2-6-2-22. REPAIR DAMAGED SKIN AND STRINGER****b. Corner Crack:**

- (1) **LENGTH OF CRACK SHALL NOT EXCEED 25 TIMES THICKNESS OF DAMAGED MEMBER.**
- (2) Stop-drill hole using drill with $\frac{1}{8}$ (0.125)-inch diameter drill bit.
- (3) Repair member - Same gage and material as original member.

c. Crack at Stringer Cutout (Table 2-6-2-4):

- (1) **LENGTH OF CRACK SHALL NOT EXCEED 20 TIMES THICKNESS OF DAMAGED MEMBER.**
- (2) Stop-drill hole using drill with $\frac{1}{8}$ (0.125)-inch diameter drill bit.
- (3) Repair member - Same gage and material as original member.



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FIGURE 2-6-2-23. REPAIR DAMAGED SKIN, FRAME, AND STRINGER

Table 2-6-2-4. Rivet Requirements for Crack at Stringer Cutout

Original Material Gage (in.)	Rivet Type and Size
0.020	MS20470AD3
0.025	MS20470AD4
0.032, 0.040	MS20470AD5
0.050, 0.063, 0.071	MS20470AD6
0.080, 0.090	MS20470AD8

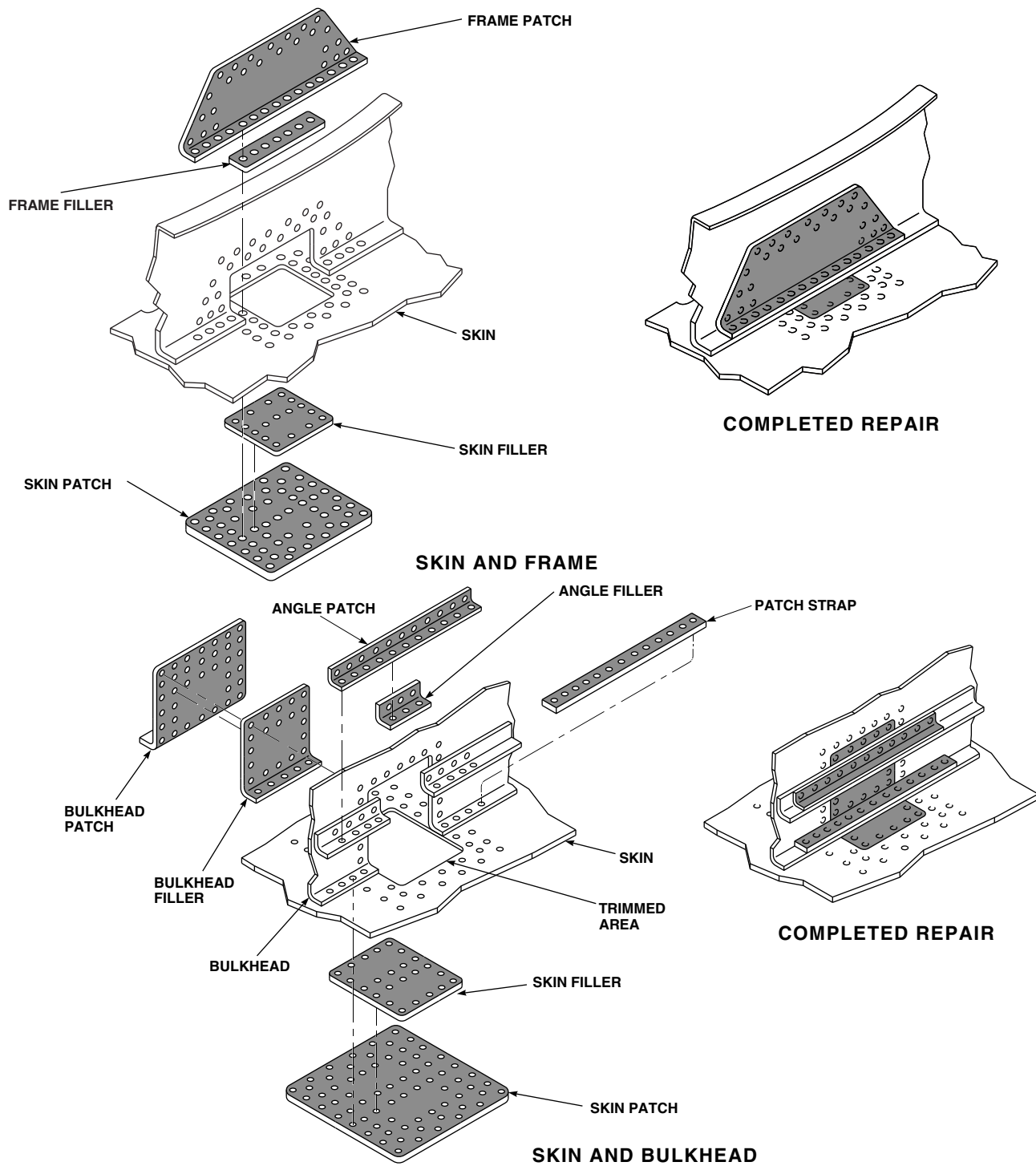
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FIGURE 2-6-2-24. REPAIR DAMAGED BULKHEAD WEB AND SKIN

d. Crack in Bend Radius:

- (1) **LENGTH OF CRACK SHALL NOT EXCEED 25 TIMES THICKNESS OF DAMAGED MEMBER.**

- (2) Stop-drill hole using drill with $\frac{1}{8}$ (0.125)-inch diameter drill bit.

- (3) Repair member - Same gage and material as original member.

e. Rivet Having Short Edge Distance:

NOTE

This repair is not to be used for skin splices.

- (1) **REPAIR IS LIMITED TO NOT MORE THAN 5 CONSECUTIVE ATTACHMENTS OR MORE THAN 50% OF THE TOTAL RIVET PATTERN.**

- (2) Install 2 rivets, one on each side of faulty rivet. Use same type, but one diameter smaller than faulty rivet.

- (3) Repair member - Same gage and material as original member.

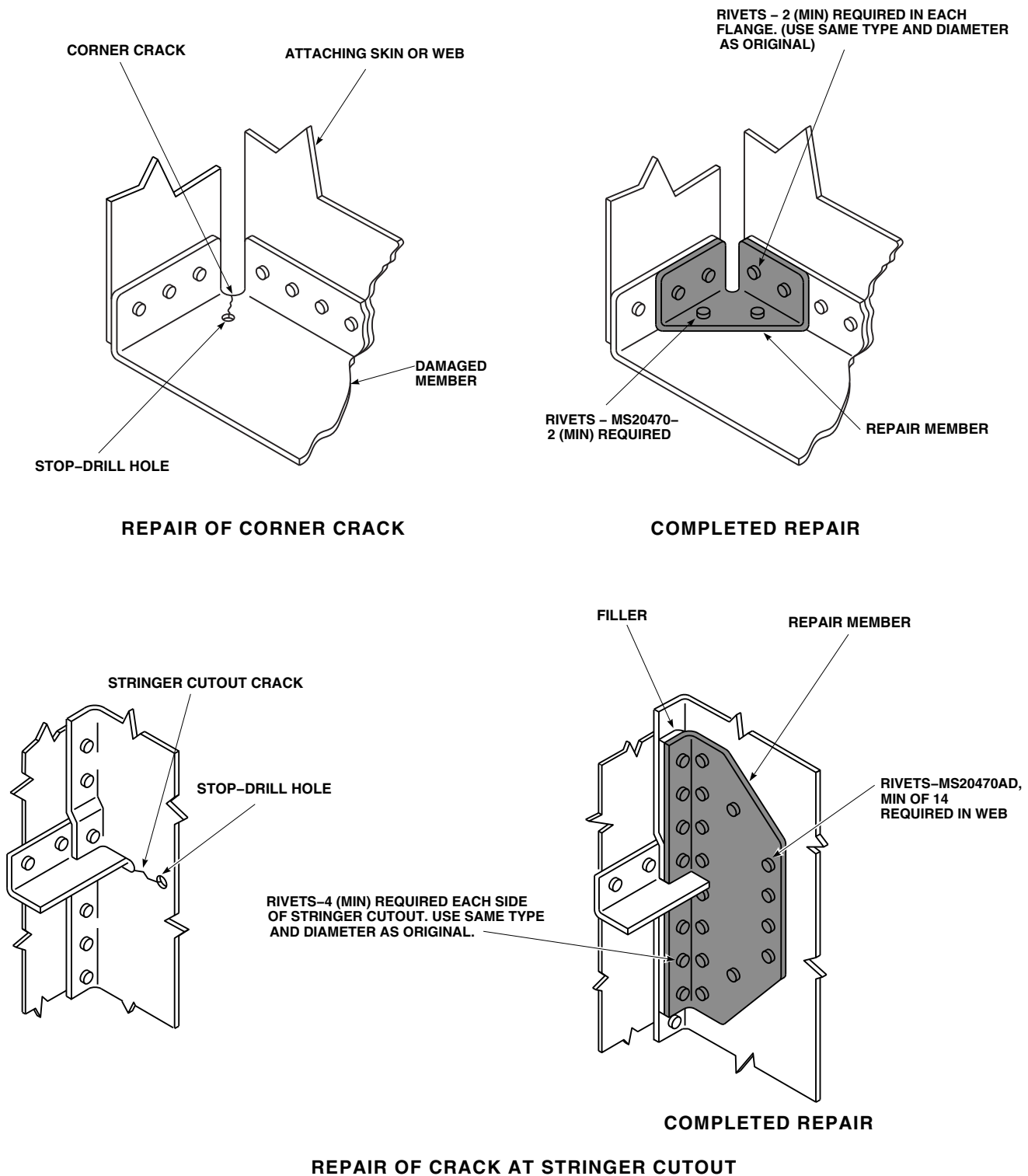
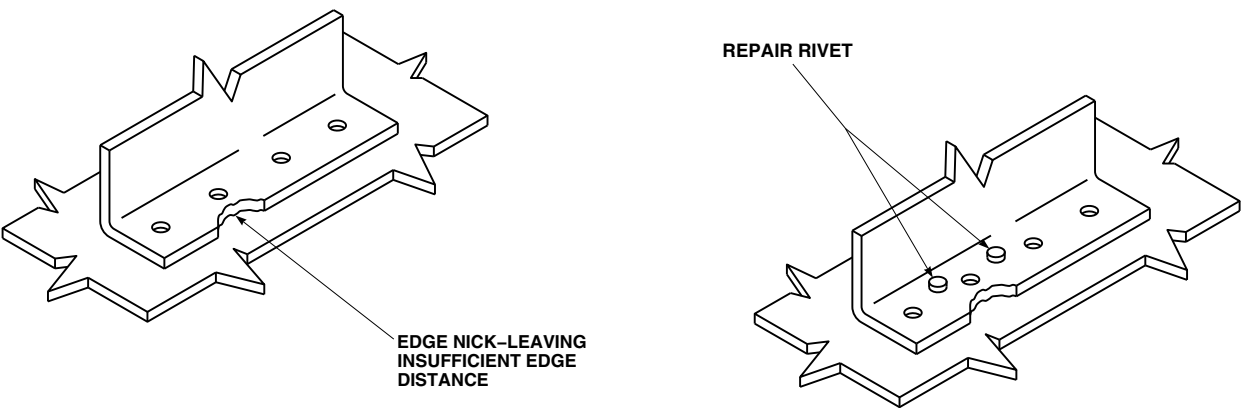
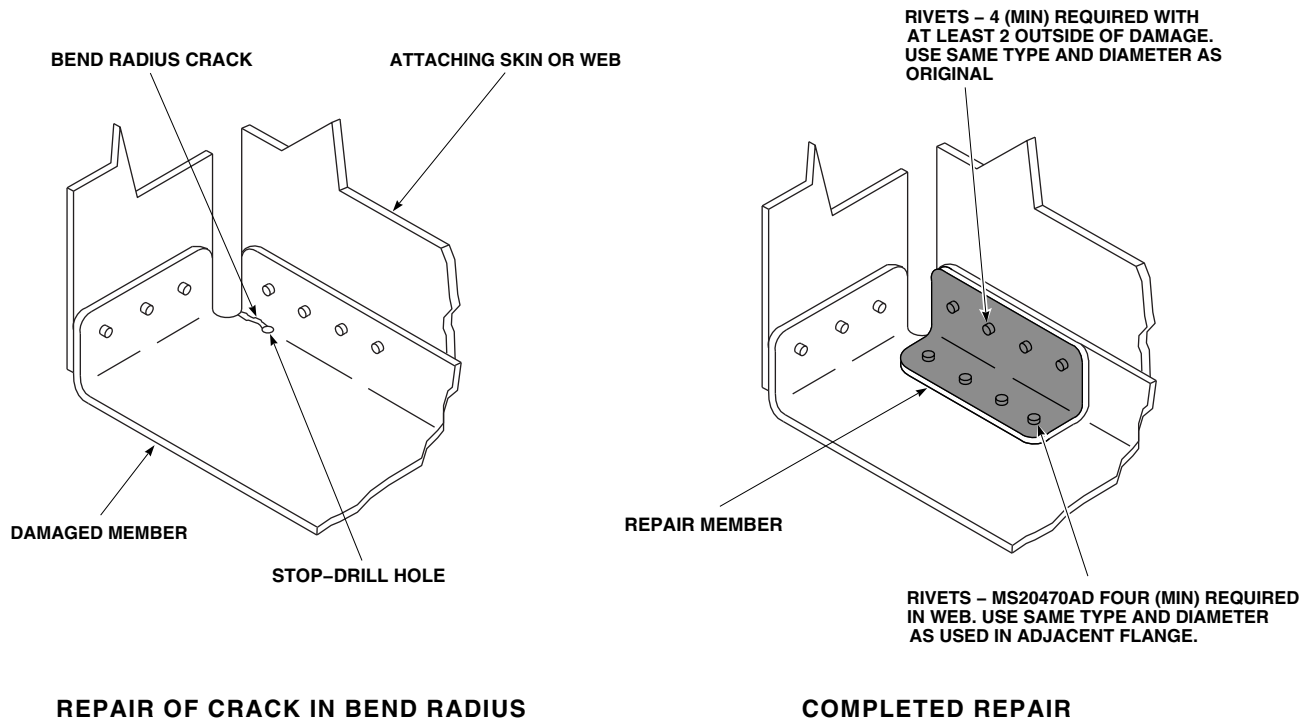


FIGURE 2-6-2-25. MISCELLANEOUS REPAIRS (SHEET 1 OF 2)

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REPAIR OF RIVET HAVING SHORT EDGE DISTANCE

FIGURE 2-6-2-25. MISCELLANEOUS REPAIRS (SHEET 2 OF 2)

AB0250_2A
SA

2-6-3 COMPOSITE STRUCTURE REPAIR.**PARAGRAPH BREAKDOWN**

		PAGE			PAGE
2-6-3.1	General	2-6-3-3	2-6-3.14	Repair Honeycomb Core - Composite Skin Panel, Impact Damage Not Penetrating to Honeycomb Core	2-6-3-20
2-6-3.2	Types of Damage	2-6-3-4			
2-6-3.3	Types of Resin for Repair	2-6-3-4			
2-6-3.4	Type I Repairable Damage	2-6-3-8	2-6-3.15	Repair Honeycomb Core - Composite Skin Panel, Damage to One Skin Penetrating Into Honeycomb Core	2-6-3-21
2-6-3.5	Type II Repairable Damage	2-6-3-8			
2-6-3.6	Repair Holes and Attachment Points 0.098-Inch or Less, 12 Plies or Less	2-6-3-10	2-6-3.16	Repair Honeycomb Core - Composite Skin Panel, Damage Penetrating Both Skins and Honeycomb Core	2-6-3-23
2-6-3.7	Repair Holes and Attachment Points Over 0.098-Inch, Under $\frac{3}{4}$ -Inch, Thickness 6 Plies or Less	2-6-3-11	2-6-3.17	Repair Honeycomb Core - Composite Skin Panel, Skin-to-Core Disbond	2-6-3-24
2-6-3.8	Repair Holes and Attachment Points Over 0.098-Inch, Under $\frac{3}{4}$ -Inch, Thickness 7 - 12 Plies	2-6-3-12	2-6-3.18	Repair Honeycomb Core - Composite Skin Panel, Repair Core Edge	2-6-3-27
2-6-3.9	Repair Broken Corner or Tab	2-6-3-14	2-6-3.19	Repair Honeycomb Core - Metal Skin Panel, Repair Inner Skin-to-Core Disbond	2-6-3-29
2-6-3.10	Repair Damaged Foam Core Rib, Damage Less Than 2-Inches Long	2-6-3-14	2-6-3.20	Repair Honeycomb Core - Metal Skin Panel, Repair Closeout Disbond	2-6-3-29
2-6-3.11	Repair Damaged Foam Core, Crushed Rib Repair	2-6-3-17	2-6-3.21	Repair Honeycomb Core - Metal Skin Panel, Repair Inner Skin Puncture, Under $\frac{1}{2}$ -Inch	2-6-3-31
2-6-3.12	Repair/Replace Mounting Hole in Honeycomb Panel	2-6-3-18			
2-6-3.13	Repair Honeycomb Core - Composite Skin Panel, Damage to Skin Not Penetrating to Honeycomb Core	2-6-3-19			

	PAGE		PAGE
2-6-3.22	Repair Honeycomb Core - Metal Skin Panel, Repair Inner Skin Puncture, 1/2 - 1 1/2-Inches	2-6-3-32	2-6-3.23
			Repair Honeycomb Core - Metal Skin Panel, Repair Outer Skin Puncture
			2-6-3-33

INITIAL SETUP

Tools

- Airframe Repairer's Toolkit
- Allen Wrench
- C-Clamp
- Clamp
- Clamping Plates (Alternate for Vacuum Bags)
- Coin, Soft Copper (or equivalent metal), 1 1/2 - 2-inches diameter $\times$ 1/8 - 3/16-inch thick, Procured or Locally-Made
- Drill
- Drill Bit, No. 40 (0.098-inch diameter)
- Drill Bit, 3/8 (0.375-inch diameter)
- Heat Gun
- Heat Lamps
- Hole Saw
- Hypodermic Syringe, Locally-Made, Figure H-197, Appendix H
- Metal, Plastic, or Wood Strip (used for fabricating mold for composite repair)
- Nonmetallic Scraper, Locally-Made
- Pressure Plates
- Protection, Eye, Skin, and Breathing
- Protective Cotton Gloves

- Protective Plastic Gloves
- Router
- Scribe
- Spring Clamps
- Squeegee
- Thin Plastic Tool (Alternate for Beverage Stirring Stick)
- Vacuum Bags
- Vacuum Cleaner
- Weights

Materials/Parts

- Abrasive Cloth, Item 2, Appendix D
- Abrasive Cloth, Item 6, Appendix D
- Abrasive Paper, Item 18, Appendix D
- Acetone, Item 25, Appendix D
- Adhesive, Item 32, Appendix D
- Adhesive, Item 34, Appendix D
- Adhesive, Item 35, Appendix D
- Adhesive, Item 66A, Appendix D
- Adhesive, Item 66B, Appendix D
- Adhesive, Item 66D, Appendix D
- Adhesive Hardener, Item 66G, Appendix D

- Brush, Acid, Item 84, Appendix D
 - Cheesecloth, Item 90, Appendix D
 - Cloth, Satin, Item 105, Appendix D
 - Corrosion Resistant Coating, Item 118, Appendix D
 - Curing Agent, Item 120, Appendix D
 - Diethylenetriamine, Item 121A, Appendix D
 - Doubler, Anodized 2024-T3 Aluminum Sheet, 0.016-inch thick, Locally-Made
 - Filler, Metal Surface Dent, Item 139A, Appendix D
 - Insulating Compound, Electrical, Item 182A, Appendix D
 - Masking Tape, 1-inch, Item 212, Appendix D
 - Methyl Ethyl Ketone, Item 217, Appendix D
 - Plastic, Polyvinyl Sheet, Item 237B, Appendix D
- Sealing Compound, Item 272, Appendix D
 - Sealing Compound, Item 302A, Appendix D
 - Sealing Compound, Item 302B, Appendix D
 - Tack Rag
 - Stirring Stick, Beverage, Item 315C, Appendix D
- References**

 - Appendix D
 - Appendix H
 - MIL-F-9084
 - MIL-R-9300
 - PARA 2-6-5
 - PARA 5-3-11

2-6-3.1 General.

2-6-3.1.1 Equipment and Supplies.

NOTE

- Equipment and material used in repair of reinforced plastic should duplicate as closely as possible those used in the manufacture of the original part.
 - The following repair procedures are not to be used for rotor blade repair.
 - For Kevlar repair, follow instructions in PARA 2-6-3.1.2 in addition to these procedures.
- Recommended satin fabric used for reinforced plastic repairs shall conform to MIL-F-9084, Class 2, and resin systems shall conform to MIL-R-9300.
 - Use backup forms to repair large rounded areas.

- Use vacuum bags or clamping plates to provide pressure during cure.
- Use heat lamps or heat guns whenever possible to cure repair areas.



To avoid poor bond and paint adhesion, do not use waxes or similar products as release materials. They cannot be removed completely from repair surface. Use only film release materials.

- Use release materials such as polyvinyl plastic sheet, Item 237B, Appendix D.
- Assemble equipment for storing resins, measuring, weighing, cutting fabric, cutting cured laminates, sanding, and finishing.

WARNING

Reinforced plastic dust is a health hazard that requires protection of eyes, respiratory system, and exposed skin. Wear protective clothing, eye protection, and dust mask when working with reinforced plastic.

CAUTION

Reinforced plastic dust is highly abrasive and can damage equipment and contaminate parts with which it comes in contact. Cover or remove equipment when sanding reinforced plastic.

- g. Have an exhaust system for removing reinforced plastic dust resulting from cutting and sanding operations.
- h. Have solvents (free of contaminants) such as acetone, Item 25, Appendix D, or methyl ethyl ketone, Item 217, Appendix D, available for cleaning equipment and parts.
- i. Control storage temperature and shelf life of resins for resin reliability. Resins should be stored at 75°F (24°C), unless otherwise indicated on containers.

2-6-3.1.2 Kevlar.**CAUTION**

To prevent damage to Kevlar laminate, do not allow paint removers to contact exposed fibers. Do not allow solvents to remain in contact with Kevlar laminate.

NOTE

Kevlar fibers fray when sanded, cut, or drilled.

- a. To prevent surface fraying and provide a smooth finish, mask surface as close to repair

dimensions as possible, using 1-inch masking tape, Item 212, Appendix D.

- b. Trim edges using abrasive cloth, Item 2, Appendix D. Use abrasive cloth, Item 6, Appendix D, to remove resin gloss.
- c. Do not remove masking tape until after final sanding.

2-6-3.2 Types of Damage.

- a. There are two major types of repairable damage:
 - (1) Type I repair is for laminates with surface damage such as scratches or pits that do not penetrate first ply (PARA 2-6-3.4).
 - (2) Type II repair is for laminates with damage extending into laminate fabric, or through part (PARA 2-6-3.5).
- b. Damage to holes and attachment points may be repaired (PARA 2-6-3.6).
- c. Broken corners or tabs may be repaired (PARA 2-6-3.9).
- d. Damage to foam ribs may be repaired (PARA 2-6-3.10).

2-6-3.3 Types of Resin for Repair.**WARNING**

- Resin bases, curing agents, times, and temperatures listed in Table 2-6-3-1, are for repair of these laminates: glass fabric/polyester resin, glass fabric/epoxy resin, and Kevlar/epoxy resin. Type I resin is for low temperature curing and Type II resin is for high temperature curing.
- When using or mixing resins and curing agents, fumes from these materials cause both a fire and health hazard.
- Have good ventilation and breathing protection. Wear protective clothing and

eye shield. Post NO SMOKING signs in area and keep materials away from spark or flame. Wash contacted skin with soap and water. If materials contact eyes, flush out eyes with clean water and seek immediate medical attention.

- Use properly grounded and explosion-proof electrical equipment in working area.

- Wear protective cotton and plastic gloves when handling materials.
- Wash hands before eating or smoking.

Resins and curing agents shown in Table 2-6-3-1 are to be used during repairs.

Table 2-6-3-1. Resins and Curing Agents

Mix Ratio, Resin Base-to-Curing Agent (by Weight)					
Resin Base and Curing Agent Type	Resin Base	Curing Agent	Pot Life at 70°F (21°C) Minutes	Notes	Cure Cycle Minimum Time
Resin Base and Curing Agent: Adhesive, Item 35, Appendix D (Type II)	100 ± 2	57 - 59	25 - 30	2, 5	24 hrs at 70°F (21°C) or 1 hr at 240° - 260°F (116° - 126°C)
Resin Base: Sealing Compound, Item 302A, Appendix D (Type II) Curing Agent: Electrical Insulating Compound, Item 182A, Appendix D	100 ± 2	13 - 15	20	3, 5	48 hrs at 70°F (21°C)
Resin Base: Sealing Compound, Item 302A, Appendix D (Type II) Curing Agent: Sealing Compound, Item 302B, Appendix D	100 ± 2	13 - 15	30	3, 5	336 hrs at 70°F (21°C) or 16 hrs at 70°F (21°C) plus 1 hr at 195° - 205°F (91° - 96°C)

Table 2-6-3-1. Resins and Curing Agents (Cont)

Mix Ratio, Resin Base-to-Curing Agent (by Weight)					
Resin Base: Sealing Compound (Epocast 1652 A), Item 302A, Appendix D (Type II) Curing Agent: Sealing Compound (Epocast 1652 B), Item 302A, Appendix D	100 ± 2	11 - 13	40	3, 5	2 hrs at 70°F (21°C) plus 1 - 3 hrs at 145° - 155°F (63° - 68°C)
Resin Base: Sealing Compound, Item 272, Appendix D (Type I) Curing Agent:	100 ± 5	48 - 52	60 - 120	2, 3, 4	Room temp plus 2 hrs at 140° - 160°F (60° - 71°C) or 24 hrs at 70° - 80°F (20° - 26°C)
Resin Base: Adhesive, Item 66A, Appendix D (Type I) Curing Agent: Electrical Insulating Compound, Item 182A, Appendix D	100 ± 2	14 - 16	15 - 20	2, 4	16 hrs at 70°F (21°C) or 2 hrs at 140° - 160°F (60° - 71°C)
Resin Base: Adhesive, Item 66A, Appendix D (Type I) Curing Agent: Adhesive Hardener, Item 66G, Ap- pendix D	100 ± 2	10 - 12	40 - 70	2, 4	24 hrs at 70°F (21°C) or 2 hrs at 140° - 160°F (60° - 71°C)

Table 2-6-3-1. Resins and Curing Agents (Cont)

Mix Ratio, Resin Base-to-Curing Agent (by Weight)					
Resin Base: Adhesive, Item 66B, Appendix D (Type I) Curing Agent: Curing Agent, Item 120, Appendix D	100 ± 2	23 - 30	15 - 40	1, 4, 6	24 hrs at 70°F (21°C) or 1 hr at 140° - 160°F (60° - 71°C)
Resin Base: Adhesive, Item 66D, Appendix D (Type I) Curing Agent: Diethylenetriamine, Item 121A, Ap- pendix D	100 ± 2	8 - 12	15 - 30	1, 3, 4, 6	24 hrs at 70°F (21°C) or 1 hr at 140° - 160°F (60° - 71°C)
Resin Base: Adhesive, Item 66D, Appendix D (Type I) Curing Agent: Curing Agent, Item 120, Appendix D	100 ± 2	23 - 30	15 - 40	1, 3, 4, 6	24 hrs at 70°F (21°C) or 1 hr at 140° - 160°F (60° - 71°C)

NOTES

1. Exterior use only.
2. High temperature cure follows 1 hour at room temperature.
3. Syntactic foam.
4. Type I (General Purpose). Kevlar/epoxy: 160°F (71°C) continuous, 250°F (121°C) short-term; fiberglass/epoxy: 160° (71°C) continuous, 300°F (149°C) short-term.

Table 2-6-3-1. Resins and Curing Agents (Cont)

5. Type II (High Temperature). Kevlar/epoxy: 425°F (218°C) maximum service; fiberglass/epoxy: 450°F (232°C) maximum service.
6. Do not use in fuel contact area.

2-6-3.4 Type I Repairable Damage.**NOTE**

For Kevlar repair, follow instructions in PARA 2-6-3.1.2 in addition to these procedures.

- a. Surface damage consisting of scratches or pits in resin coat, not affecting laminate fabric ply (Figure 2-6-3-1).
- b. Using cheesecloth, Item 90, Appendix D, moistened with acetone, Item 25, Appendix D, or methyl ethyl ketone, Item 217, Appendix D, remove paint finish in damaged area.
- c. Using abrasive cloth, Item 6, Appendix D, sand damaged area, plus ½-inch beyond on all sides.
- d. Using cheesecloth, Item 90, Appendix D, moistened with acetone, Item 25, Appendix D, or methyl ethyl ketone, Item 217, Appendix D, wipe sanded area.

NOTE

Observe all warnings and cautions in PARA 2-6-3.1 and PARA 2-6-3.3.

- e. Mix resin/curing agent (Table 2-6-3-1), adding chopped satin cloth, Item 105, Appendix D, to make putty-like mixture.
- f. Apply thin, smooth coat of mixed resin to damaged area, using beverage stirring stick, Item 315C, Appendix D, and squeegee to uniform thickness.
- g. Cure repair (Table 2-6-3-1).

- h. Lightly sand surface smooth using abrasive cloth, Item 6, Appendix D.

- i. Touch up paint as required (PARA 2-6-5).

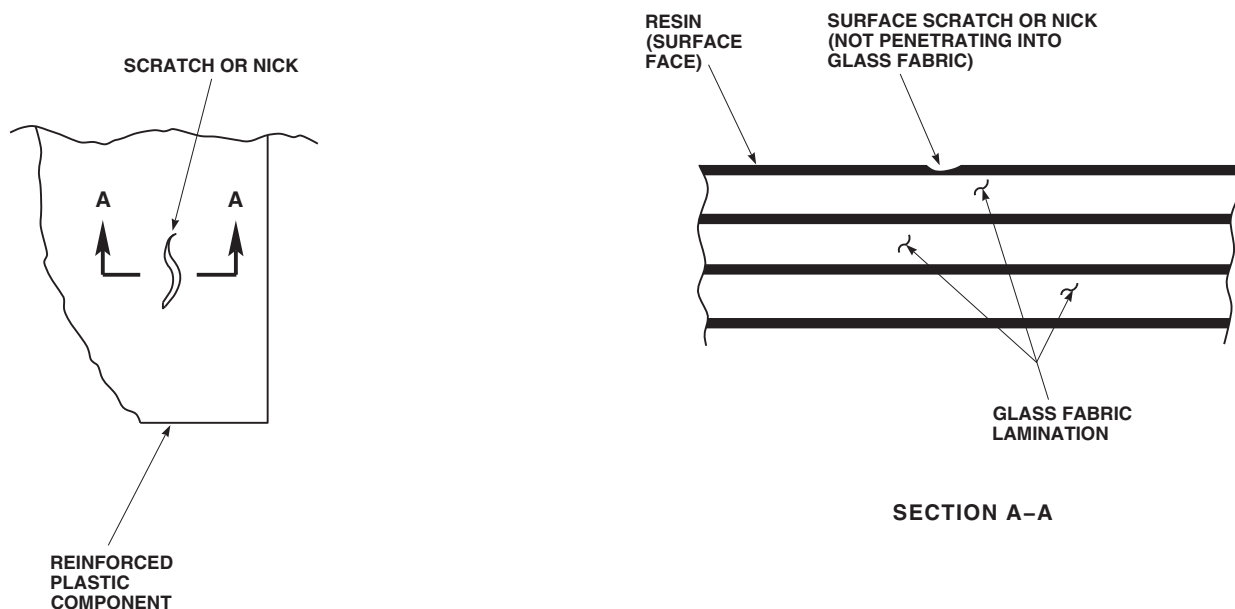
2-6-3.5 Type II Repairable Damage.**NOTE**

For Kevlar repair, follow instructions in PARA 2-6-3.1.2 in addition to these procedures.

- a. Remove paint from surface in repair area.

NOTE

- Observe all warnings and cautions (PARA 2-6-3.1 and PARA 2-6-3.3).
 - Do not mark on work with grease or graphite pencil, or ink pen.
- b. Determine total size of repair, allowing at least ½-inch step (or blend) in all directions for each ply penetrated (Figure 2-6-3-2). Final patch must overlap at least 1-inch in all directions.
 - c. Using abrasive cloth, Item 2, Appendix D, sand away plies as indicated.
 - d. Lightly sand top surface to limits of final patch.
 - e. Thoroughly clean entire repair area using cheesecloth, Item 90, Appendix D, moistened with acetone, Item 25, Appendix D, or methyl ethyl ketone, Item 217, Appendix D.
 - f. Protect surfaces from contamination while allowing to air-dry.



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FIGURE 2-6-3-1. TYPE I REPAIRABLE DAMAGE

NOTE

Cut each repair ply so weave direction matches its laminate ply.

- g. Cut plies of satin cloth, Item 105, Appendix D, to fit stepped plies sanded away.
- h. Lay repair plies on flat surface covered with polyvinyl plastic sheet, Item 237B, Appendix D.
- i. Mix resin/curing agent (Table 2-6-3-1).
- j. Using acid brush, Item 84, Appendix D, apply coat of mixed resin to repair area.
- k. Saturate repair plies of satin cloth, Item 105, Appendix D, with mixed resin using acid brush, Item 84, Appendix D, or beverage stirring stick, Item 315C, Appendix D. Resin content of fabric, when thoroughly impregnated, must be about 50% by weight.
- l. Place repair plies in damaged area, matching weave direction to laminate plies. Each repair

ply overlaps preceding ply by $\frac{1}{2}$ -inch in all directions. Final ply overlaps 1-inch and adds 1-ply thickness to laminate (2-ply thickness when repairing both sides).

- m. Cover repair with polyvinyl plastic sheet, Item 237B, Appendix D. Using squeegee, work out all trapped air and excess resin.
- n. Apply pressure to repair plies by vacuum bag, clamping plates, or system of weights and pressure plates.



Improper control of heat source can damage laminate or weaken bond. Pay strict attention to temperature and curing time.

- o. Cure repair (Table 2-6-3-1).
- p. Remove pressure and heating equipment and plastic strip.

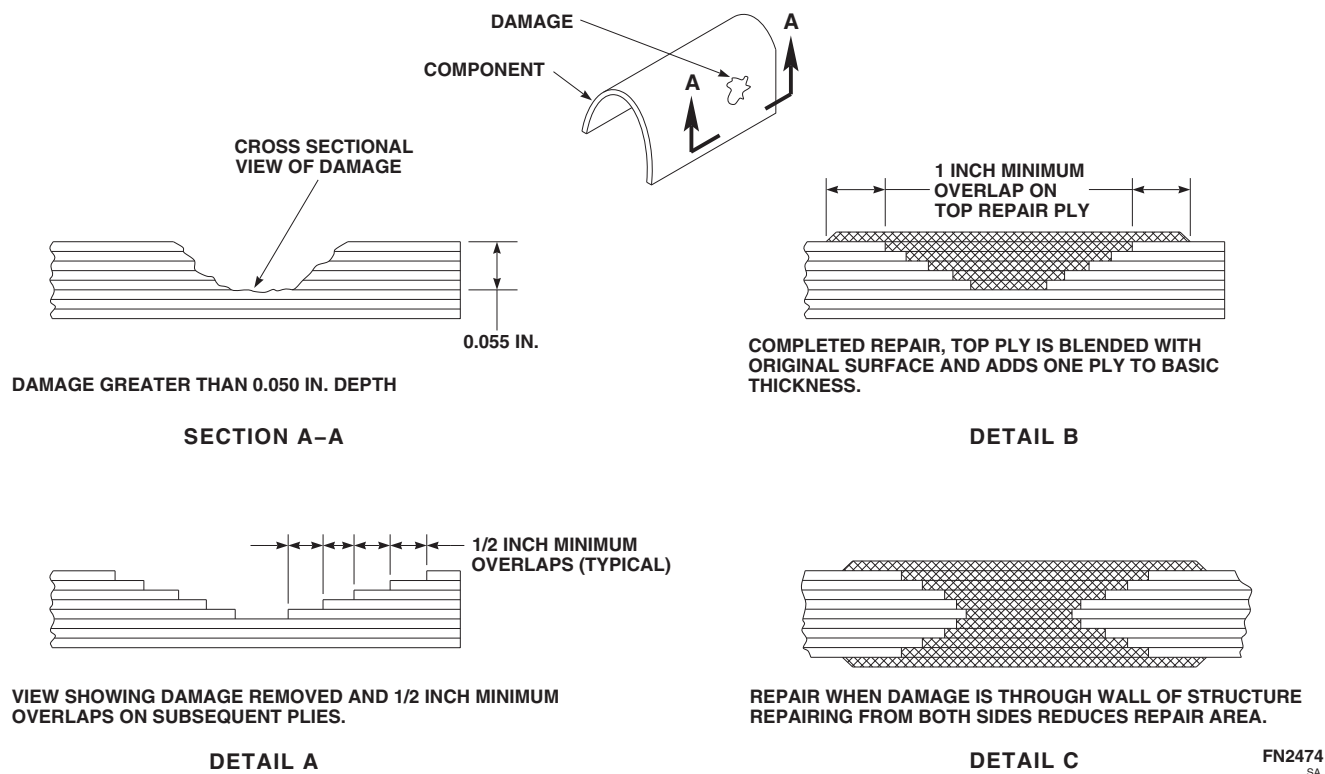


FIGURE 2-6-3-2. TYPE II REPAIRABLE DAMAGE

- q. Using abrasive cloth, Item 6, Appendix D, blend repair to surface of part.
- r. Touch up paint as required (PARA 2-6-5).

2-6-3.6 Repair Holes and Attachment Points 0.098-Inch or Less, 12 Plies or Less.

NOTE

For Kevlar repair, follow instructions in PARA 2-6-3.1.2 in addition to these procedures.

- a. For holes not more than 0.098-inch in diameter, with centers at least 1/8-inch from edge of part, in laminates of 12 plies or less, repair as follows:
 - (1) Clean repair area thoroughly using cheesecloth, Item 90, Appendix D,

moistened with methyl ethyl ketone, Item 217, Appendix D, or acetone, Item 25, Appendix D. Wipe dry with clean cheesecloth.

- (2) Mix adhesive, Item 32, Appendix D (Table 2-6-3-2).
- (3) Using beverage stirring stick, Item 315C, Appendix D, fill holes with mixed adhesive.
- (4) Cover repair area with polyvinyl plastic sheet, Item 237B, Appendix D.
- (5) Cure adhesive (Table 2-6-3-2).
- (6) Sand smooth to adjacent skin using abrasive cloth, Item 6, Appendix D.
- b. Touch up paint as required (PARA 2-6-5).

Table 2-6-3-2. Fillers and Accelerators

Mix Ratio, Filler Base-to-Accelerator (by Weight)				
Filler Base and Accelerator	Filler Base	Accelerator	Pot Life at 70°F (21°C) (Minutes)	Cure Cycle Minimum Time
Filler Base and Accelerator: Adhesive, Item 32, Appendix D	100 ± 5	21 - 23	30	24 hrs at 70° - 85°F (21° - 29°C) for handling of bonded parts, or 5 days at 70° - 85°F (21° - 29°C) for com- plete cure. Com- plete cure can also be achieved through an ac- celerated cure of 1 hour at 140° - 160°F (60° - 71°C)
Filler Base and Accelerator: Adhesive, Item 34, Appendix D	100 ± 5	31 - 35	60	1 hr at 190° - 210°F (88° - 99°C)
Filler Base and Accelerator: Metal Surface Dent Filler, Item 139A, Appendix D	100 ± 5	2	10	2 hrs at 70°F (21°C)

**2-6-3.7 Repair Holes and Attachment Points
Over 0.098-Inch, Under ¾-Inch, Thickness 6
Plies or Less.**

NOTE

For Kevlar repair, follow instructions in
PARA 2-6-3.1.2 in addition to these
procedures.

- a. For holes larger than 0.098-inch and less than
¾-inch in diameter, if part thickness is 6 plies
or less, repair as follows (Figure 2-6-3-3):

- (1) On both surfaces of part, mark an outline
for repair allowing ½-inch minimum
overlap for edge of hole in all directions
(allow 1-inch for ½ - ¾-inch diameter
hole).
- (2) Lightly sand repair areas using abrasive
cloth, Item 2, Appendix D.
- (3) Clean repair area thoroughly using
cheesecloth, Item 90, Appendix D,
moistened with methyl ethyl ketone, Item

217, Appendix D, or acetone, Item 25, Appendix D. Wipe dry with clean cheesecloth.

- (4) Cut ply of satin cloth, Item 105, Appendix D, for each side of part, following outline on part surface. Match weave direction of repair ply and surface ply.
- (5) Mix resin/curing agent (Table 2-6-3-1) and saturate both repair plies using beverage stirring stick, Item 315C, Appendix D.
- (6) Apply coat of mixed resin to one repair surface of part. Apply repair ply and cover with polyvinyl plastic sheet, Item 237B, Appendix D.
- (7) Mix resin/curing agent (Table 2-6-3-1) adding chopped satin cloth, Item 105, Appendix D, to obtain putty-like consistency.
- (8) Using beverage stirring stick, Item 315C, Appendix D, fill holes with resin/cloth mixture.
- (9) Repeat step a. (5) and step a. (6) on other side of part.
- (10) Using squeegee, carefully work out all air and excess resin.
- (11) Apply pressure to repair plies by vacuum bag, clamping plates, or system of weights and pressure plates.
- (12) Cure repair (Table 2-6-3-1).
- (13) Remove vacuum bags, clamping plates, or system of weights and pressure plates, remove plastic strip.
- (14) Sand to remove excess resin and blend repair into adjoining surface using abrasive cloth, Item 6, Appendix D.

b. Touch up paint as required (PARA 2-6-5).

2-6-3.8 Repair Holes and Attachment Points Over 0.098-Inch, Under 3/4-Inch, Thickness 7 - 12 Plies.

NOTE

For Kevlar repair, follow instructions in PARA 2-6-3.1.2 in addition to these procedures.

- a. For holes larger than 0.098-inch and less than 3/4-inch in diameter, if part is 7 - 12-ply thick, repair as follows (Figure 2-6-3-3):
 - (1) On both surfaces of part, mark an outline for repair allowing 1 1/2-inches minimum overlap from edge of hole in all directions (allow 2-inches for 1/2 - 3/4-inch diameter hole).
 - (2) Lightly sand repair areas using abrasive cloth, Item 2, Appendix D.
 - (3) Mark outline for removal of 1 ply (from both surfaces of part) a minimum of 1/2-inch from edge of hole in all directions (allow 3/4-inch for 1/2 - 3/4-inch diameter hole).
 - (4) Remove 1 ply from each surface of part within marked outline, using abrasive cloth, Item 2, Appendix D.
 - (5) Clean repair area thoroughly using cheesecloth, Item 90, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D, or acetone, Item 25, Appendix D. Wipe dry with clean cheesecloth.

NOTE

Cut each repair ply so weave direction matches its laminate ply.

- (6) Cut plies of satin cloth, Item 105, Appendix D, to match outlines on part surfaces and to fit step cuts in surface layers.
- (7) Mix resin/curing agent (Table 2-6-3-1) and saturate all repair plies, using beverage stirring stick, Item 315C, Appendix D.

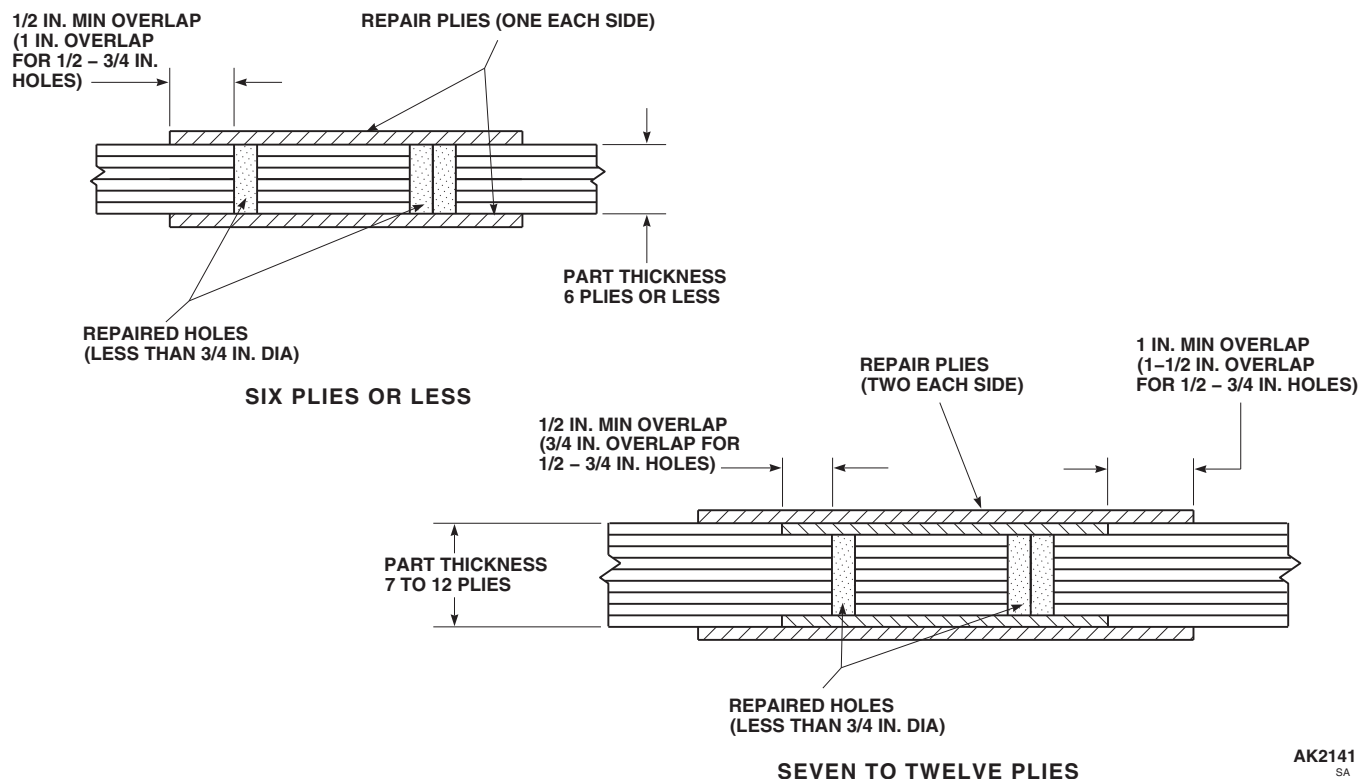


FIGURE 2-6-3-3. REPAIR HOLE

- age stirring stick, Item 315C, Appendix D.
- (8) Apply coat of mixed resin to repair area of one surface. Apply step back ply into surface ply and outer ply on top of it. Match weave direction to laminate plies.
 - (9) Cover outer repair ply with polyvinyl plastic sheet, Item 237B, Appendix D.
 - (10) Mix resin/curing agent (Table 2-6-3-1) adding chopped satin cloth, Item 105, Appendix D, to make putty-like mixture.
 - (11) Fill holes with mixed resin using beverage stirring stick, Item 315C, Appendix D.
 - (12) Repeat step a. (8) and step a. (9) on other side of part.
 - (13) Using squeegee, carefully work out all air and excess resin.
 - (14) Apply pressure to repair by vacuum bag, clamping plates, or system of weights and pressure plates.
 - (15) Cure repair (Table 2-6-3-1).
 - (16) Remove vacuum bags, clamping plates, or system of weights and pressure plates, remove plastic strip.
 - (17) Sand to remove excess resin and blend repair into adjoining surface using abrasive cloth, Item 6, Appendix D.

CAUTION

To prevent damage to repaired part when using heat lamp or heat gun, pay strict attention to temperature range. Adjust distance of heat source to part, as necessary.

- b. Touch up paint as required (PARA 2-6-5).

2-6-3.9 Repair Broken Corner or Tab.

NOTE

- For Kevlar repair, follow instructions in PARA 2-6-3.1.2 in addition to these procedures.
 - Damage extending $\frac{3}{4}$ -inch or less from edge of part may be repaired. More extensive damage requires part replacement.
 - Repair increases edge thickness by adding one ply to each side of part. If this will cause interference, do not repair part, replace it.
- a. Prepare part for repair by stepping back plies as follows (Figure 2-6-3-4):
- (1) For edge less than 4-ply thick, step back from one side only.
 - (2) For edge 4-ply thick or more, step back from both sides.
 - (3) Allow $\frac{1}{2}$ -inch overlap for each repair ply, plus 1-inch overlap for outer repair ply on each side.
 - (4) Mark outline of ply areas for removal.
 - (5) Using abrasive cloth, Item 2, Appendix D, remove laminate plies, as required.
 - (6) Clean repair area thoroughly using cheesecloth, Item 90, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D, or acetone, Item 25, Appendix D. Allow to air-dry.
- b. Cut repair plies of satin cloth, Item 105, Appendix D, to fit removed areas of laminate, extending them beyond edge of part to allow trimming to size after curing. Match weave direction of repair ply to laminate ply.
- c. Mix resin/curing agent (Table 2-6-3-1).

- d. Saturate repair plies with mixed resin, using beverage stirring stick, Item 315C, Appendix D, or acid brush, Item 84, Appendix D.
- e. Apply coat of mixed resin to repair areas of part with acid brush, Item 84, Appendix D.
- f. Put all repair plies in place, including outer plies on each side.
- g. If necessary, use metal strip, covered with polyvinyl plastic sheet, Item 237B, Appendix D, held to one side of part by spring clamps, to keep repair buildup in alignment.
- h. Cover repair with polyvinyl plastic sheet, Item 237B, Appendix D. Work out trapped air and excess resin using squeegee.
- i. Apply pressure by vacuum bag, clamping plates, or system of weights and pressure plates.
- j. Cure repair (Table 2-6-3-1).
- k. Remove vacuum bags, clamping plates, or system of weights and pressure plates, remove plastic strip.
- l. Trim edge of repair to match part.
- m. Using abrasive cloth, Item 6, Appendix D, blend outer plies into adjoining surfaces.
- n. Touch up paint as required (PARA 2-6-5).

2-6-3.10 Repair Damaged Foam Core Rib, Damage Less Than 2-Inches Long.

NOTE

For Kevlar repair, follow instructions in PARA 2-6-3.1.2 in addition to these procedures.

- a. Stiffening ribs are formed by laminated plies of glass cloth or Kevlar over foam cores. Ribs with covering plies cut, cracked, delaminated, punctured, or torn, with damage to foam core, may be repaired. Damaged area must be less than 2-inches long; only 2 repairs, not less than 1-foot apart, may be made in the same rib.

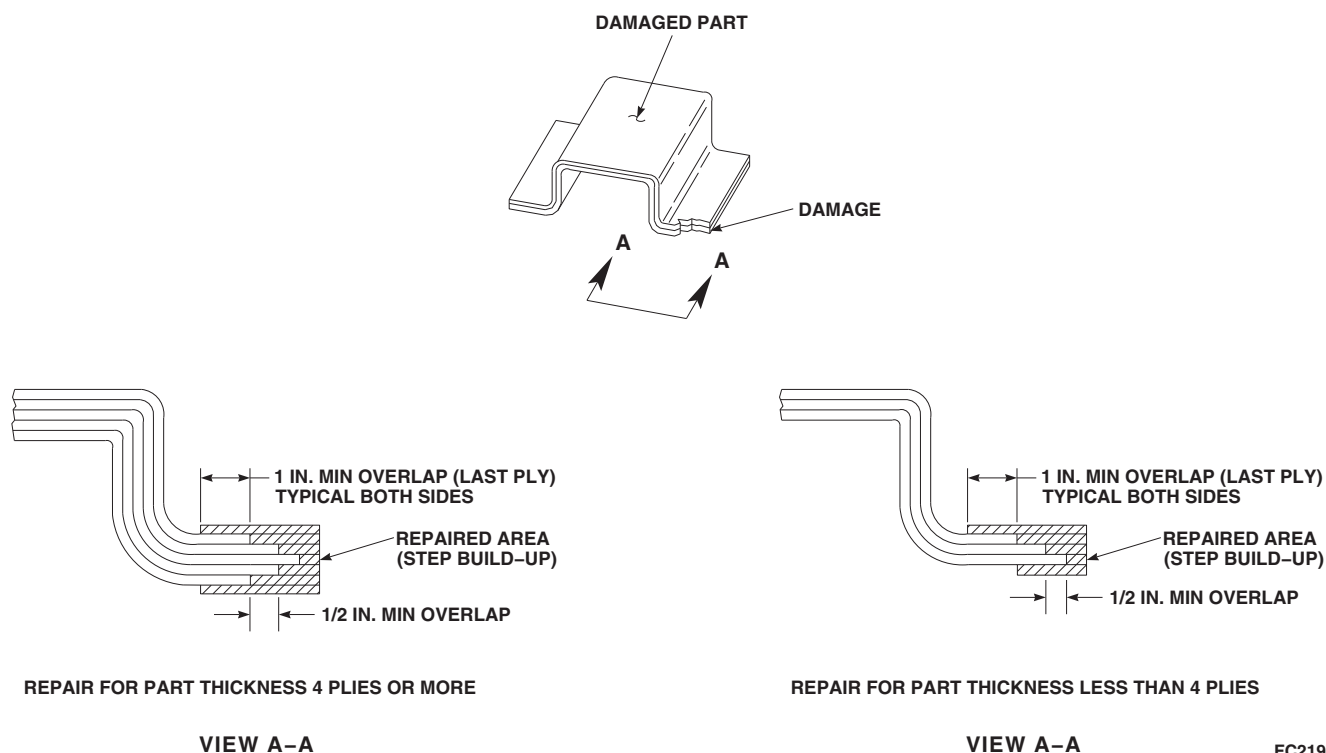
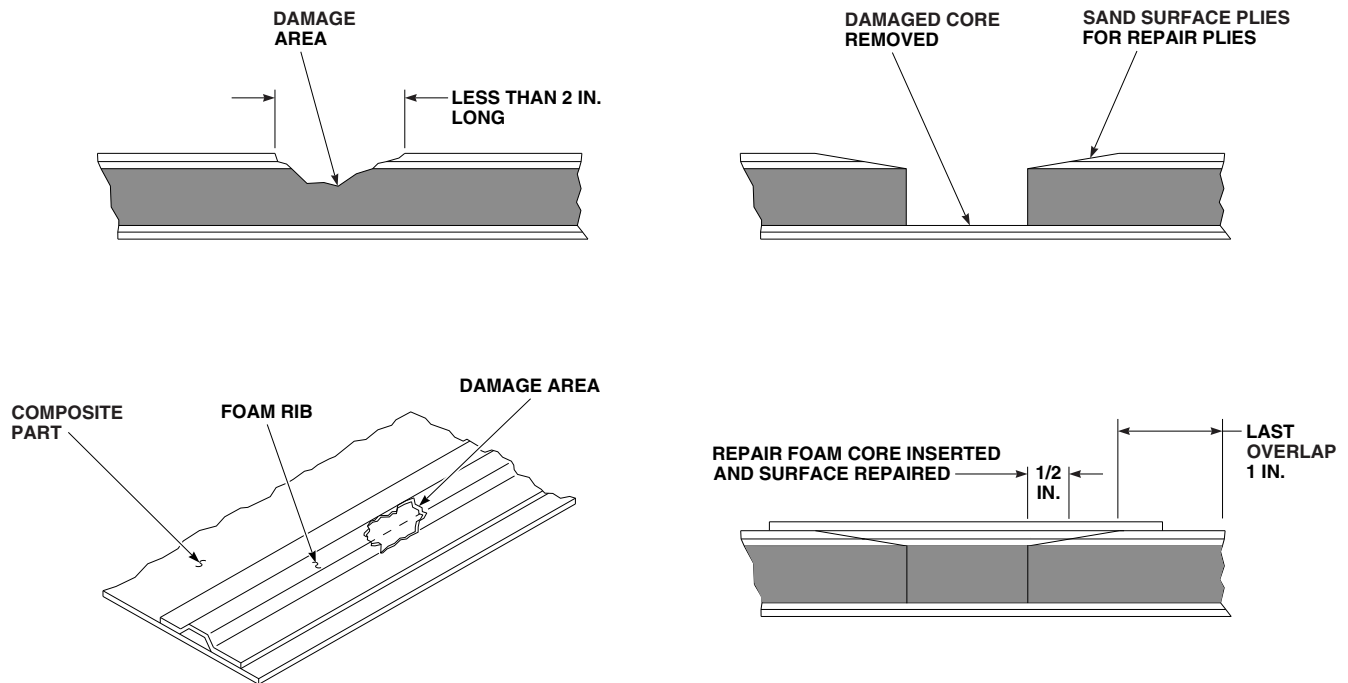


FIGURE 2-6-3-4. REPAIR CORNER OR TAB

CAUTION

Foam core material can be damaged by contact with methyl ethyl ketone, Item 217, Appendix D, or acetone, Item 25, Appendix D. Do not allow methyl ethyl ketone or acetone to contact foam core.

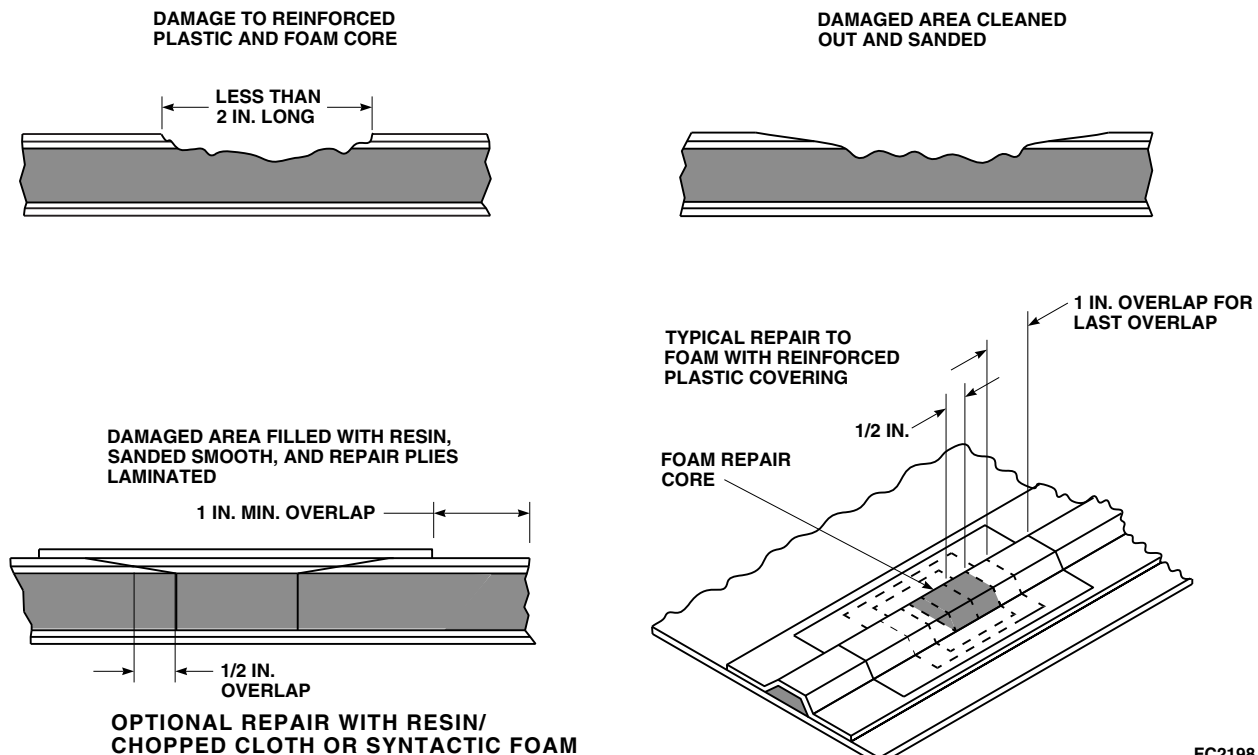
- b. Clean damaged area and adjacent surfaces using cheesecloth, Item 90, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D, or acetone, Item 25, Appendix D.
- c. Mark points where rib can be cut to remove damaged material (Figure 2-6-3-5, Sheet 1).
- d. Mark limits of required laminate blending, allowing for patch overlaps of 1/2-inch per ply.
- e. Cut away damage and blend laminate using abrasive cloth, Item 2, Appendix D.
- f. Cut repair core from foam strip that matches rib core.
- g. Clean repair area, except for foam surfaces, using cheesecloth, Item 90, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D, or acetone, Item 25, Appendix D. Allow to air-dry.
- h. Mix resin/curing agent (Table 2-6-3-1).
- i. Apply coat of mixed resin to repair core and cutaway section of rib with acid brush, Item 84, Appendix D.
- j. Insert repair core into rib cutout.
- k. For optional method, add chopped satin cloth, Item 105, Appendix D, to mixed resin, or use syntactic foam, to repair core (Table 2-6-3-1) (Figure 2-6-3-5, Sheet 2).
- l. Cure resin bond of inserted core, or built-up repair of resin/chopped cloth or syntactic foam (Table 2-6-3-1).
- m. Sand repair to match undamaged core, using abrasive cloth, Item 6, Appendix D.



REPAIR WITH FOAM CORE SECTION INSERTED

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FIGURE 2-6-3-5. REPAIR DAMAGED RIB (SHEET 1 OF 2)



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FIGURE 2-6-3-5. REPAIR DAMAGED RIB (SHEET 2 OF 2)

- n. Cut repair plies of satin cloth, Item 105, Appendix D, to size, allowing 1/2-inch minimum overlap on corresponding ply of rib.
- o. Cut extra top ply, allowing 1-inch minimum overlap on top patch ply.
- p. Saturate all patch plies with mixed resin, using acid brush, Item 84, Appendix D, or beverage stirring stick, Item 315C, Appendix D.
- q. Apply coat of mixed resin to repair area of rib.
- r. Place all repair plies in position and cover with polyvinyl plastic sheet, Item 237B, Appendix D.
- s. Using squeegee, work out trapped air and excess resin.
- t. Apply pressure to patches by vacuum bag, clamping plates, or system of weights and pressure plates.
- u. Cure repair (Table 2-6-3-1).
- v. When cure is complete, remove vacuum bag, clamping plates, or weights; remove plastic strip.
- w. Remove excess resin and blend repair into adjacent surface using abrasive cloth, Item 6, Appendix D.
- x. Touch up paint as required (PARA 2-6-5).

2-6-3.11 Repair Damaged Foam Core, Crushed Rib Repair.

NOTE

For Kevlar repair, follow instructions in PARA 2-6-3.1.2 in addition to these procedures.

- a. Crushed ribs may be repaired when one side is undistorted and other side and/or top, has been pushed in less than 1/4-inch. No part of rib may project beyond normal dimension (Figure 2-6-3-6).
- b. Determine outside limits of repair area:

- (1) Total number of repair plies must be the same as number of plies in rib.
- (2) First ply must overlap beyond crushed area by at least 1/2-inch; each succeeding ply overlaps by the same amount.
- (3) Final ply must overlap preceding ply by at least 1-inch.
- (4) Outline limits of final ply.
- c. Scuff entire repair surface lightly using abrasive cloth, Item 2, Appendix D.
- d. Clean repair area thoroughly using cheesecloth, Item 90, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D, or acetone, Item 25, Appendix D. Allow to air-dry.
- e. To make mold for repair, secure wood or metal strips to rib sides with 1-inch masking tape, Item 212, Appendix D, or clamps. Cover inside of mold with polyvinyl plastic sheet, Item 237B, Appendix D.
- f. Mix syntactic foam (Table 2-6-3-1).
- g. Fill cavity between deformed rib and mold with syntactic foam by pouring or with beverage stirring stick, Item 315C, Appendix D. Release trapped air by working beverage stirring stick up and down. Completely fill cavity.
- h. Cure repair (Table 2-6-3-1).
- i. Remove mold and shape repair to undamaged part of rib using abrasive cloth, Item 2, Appendix D.
- j. Clean repair area using cheesecloth, Item 90, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D, or acetone, Item 25, Appendix D. Allow to air-dry.
- k. Cut repair plies from satin cloth, Item 105, Appendix D, in sizes to meet requirement for 1/2-inch overlaps (1-inch for final patch).
- l. Mix resin/curing agent (Table 2-6-3-1).
- m. Saturate all repair plies with mixed resin using acid brush, Item 84, Appendix D, or beverage stirring stick, Item 315C, Appendix D.

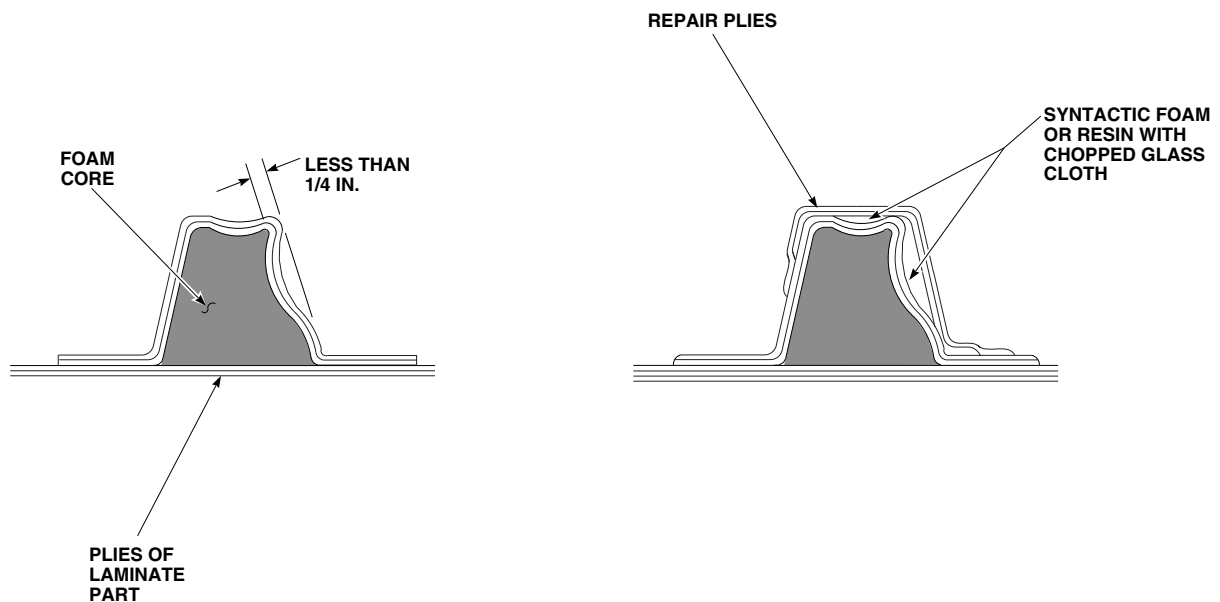
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FIGURE 2-6-3-6. REPAIR CRUSHED RIB

- n. Coat repair area of rib with mixed resin.
- o. Match all repair plies in position and cover with polyvinyl plastic sheet, Item 237B, Appendix D.
- p. Using squeegee, work out trapped air and excess resin.
- q. Apply pressure to repair by vacuum bag, clamping plates, or system of weights and pressure plates.
- r. Cure repair (Table 2-6-3-1).
- s. Remove vacuum bag, clamping plates, or system of weights and pressure plates, and plastic strip.
- t. Use abrasive cloth, Item 6, Appendix D, to remove excess resin and blend repair into adjoining surfaces.
- u. Touch up paint as required (PARA 2-6-5).

2-6-3.12 Repair/Replace Mounting Hole In Honeycomb Panel.

NOTE

For Kevlar repair, follow instructions in PARA 2-6-3.1.2 in addition to these procedures.

- a. Mark hole location on top skin of honeycomb panel. Extend lines or reference points out at least 4-inches from center.
- b. Mark outline of repair plies at least 1-inch in all directions beyond drilled hole and mask around repair area with 1-inch masking tape, Item 212, Appendix D.
- c. Drill hole through panel skin using drill with $\frac{3}{8}$ -inch diameter drill bit.
- d. Using ground-down Allen wrench, or suitable size L-shaped tool attached to drill motor, rout out honeycomb core material through top skin hole. Make cavity 1 $\frac{1}{2}$ times larger than drilled hole (Figure 2-6-3-7).

- e. Fill cavity with syntactic foam, as follows:
 - (1) Cover skin opening on one side of panel with polyvinyl plastic sheet, Item 237B, Appendix D, backed by wood or metal retaining strip held in place by clamps or 1-inch masking tape, Item 212, Appendix D.
 - (2) Mix syntactic foam (Table 2-6-3-1).
 - (3) Fill cavity of panel with syntactic foam by pouring or with beverage stirring stick, Item 315C, Appendix D.
 - (4) Release trapped air by working beverage stirring stick up and down. Completely fill cavity, slightly beyond outer surface of skin to allow for shrinkage.
 - (5) Cure syntactic foam (Table 2-6-3-1).
 - (6) Remove retaining strip and plastic strip.
- f. Sand exposed syntactic foam flush with skin surface, using abrasive cloth, Item 2, Appendix D.
- g. Cut satin cloth, Item 105, Appendix D, for repair plies to extend at least 1-inch in all directions beyond edge of drilled hole. Use same number of plies as in skin. Match ply weave direction to skin.
- h. Apply and cure repair plies as follows:
 - (3) Spread repair plies on flat surface covered with polyvinyl plastic sheet, Item 237B, Appendix D.
 - (4) Saturate repair plies with mixed resin using acid brush, Item 84, Appendix D, or beverage stirring stick, Item 315C, Appendix D.
 - (5) Apply coat of mixed resin to repair area on both sides of panel, using acid brush, Item 84, Appendix D.
 - (6) Position repair plies in place, cover with polyvinyl plastic sheet, Item 237B, Appendix D, and using squeegee, force out trapped air and excess resin.
 - (7) Apply pressure to repair plies by vacuum bag, clamping plates, or system of weights and pressure plates.
 - (8) Cure repair (Table 2-6-3-1).
- i. After curing, remark hole location lines on patch to pick up center of plug and drill appropriate size hole for intended fastener through repaired area.
- j. Sand repair to blend with surrounding skin using abrasive cloth, Item 6, Appendix D.
- k. Remove masking tape.
- l. Touch up paint as required (PARA 2-6-5).

CAUTION

Delamination of honeycomb panel will result if too much solvent is allowed to seep into panel. Use solvent sparingly to prevent delamination.

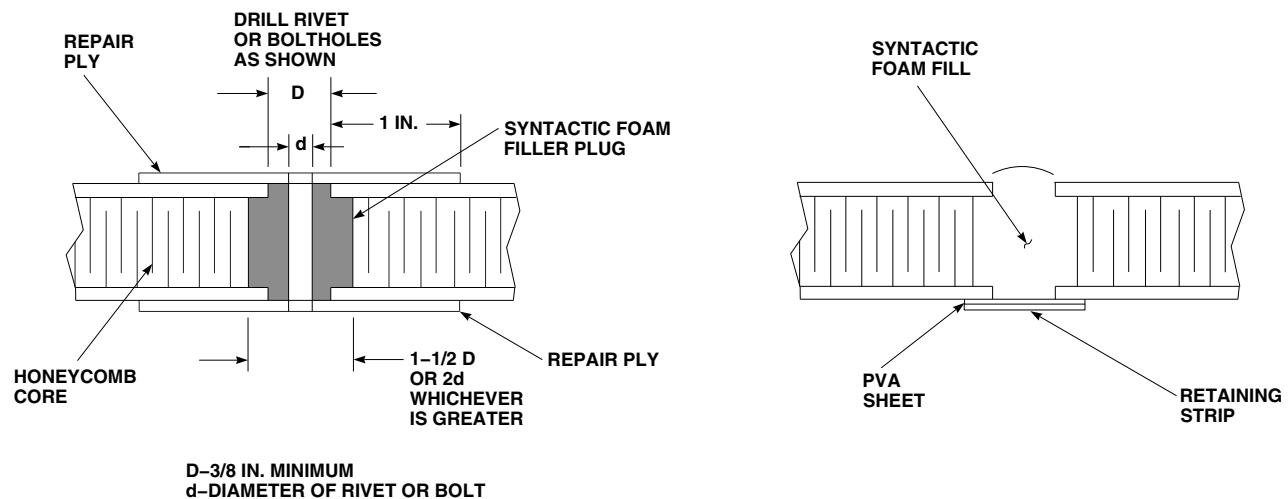
- (1) Clean repair area using cheesecloth, Item 90, Appendix D, moistened with acetone, Item 25, Appendix D, or methyl ethyl ketone, Item 217, Appendix D.
- (2) Mix resin/curing agent (Table 2-6-3-1).

2-6-3.13 Repair Honeycomb Core - Composite Skin Panel, Damage to Skin Not Penetrating to Honeycomb Core.

NOTE

For Kevlar repair, follow instructions in PARA 2-6-3.1.2 in addition to these procedures.

- a. Repair minor damage to skin covering (PARA 2-6-3.4).
- b. Repair damage to skin covering affecting one or more plies (PARA 2-6-3.5).



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FIGURE 2-6-3-7. REPAIR HONEYCOMB PANEL MOUNTING HOLE

2-6-3.14 Repair Honeycomb Core - Composite Skin Panel, Impact Damage Not Penetrating to Honeycomb Core.

NOTE

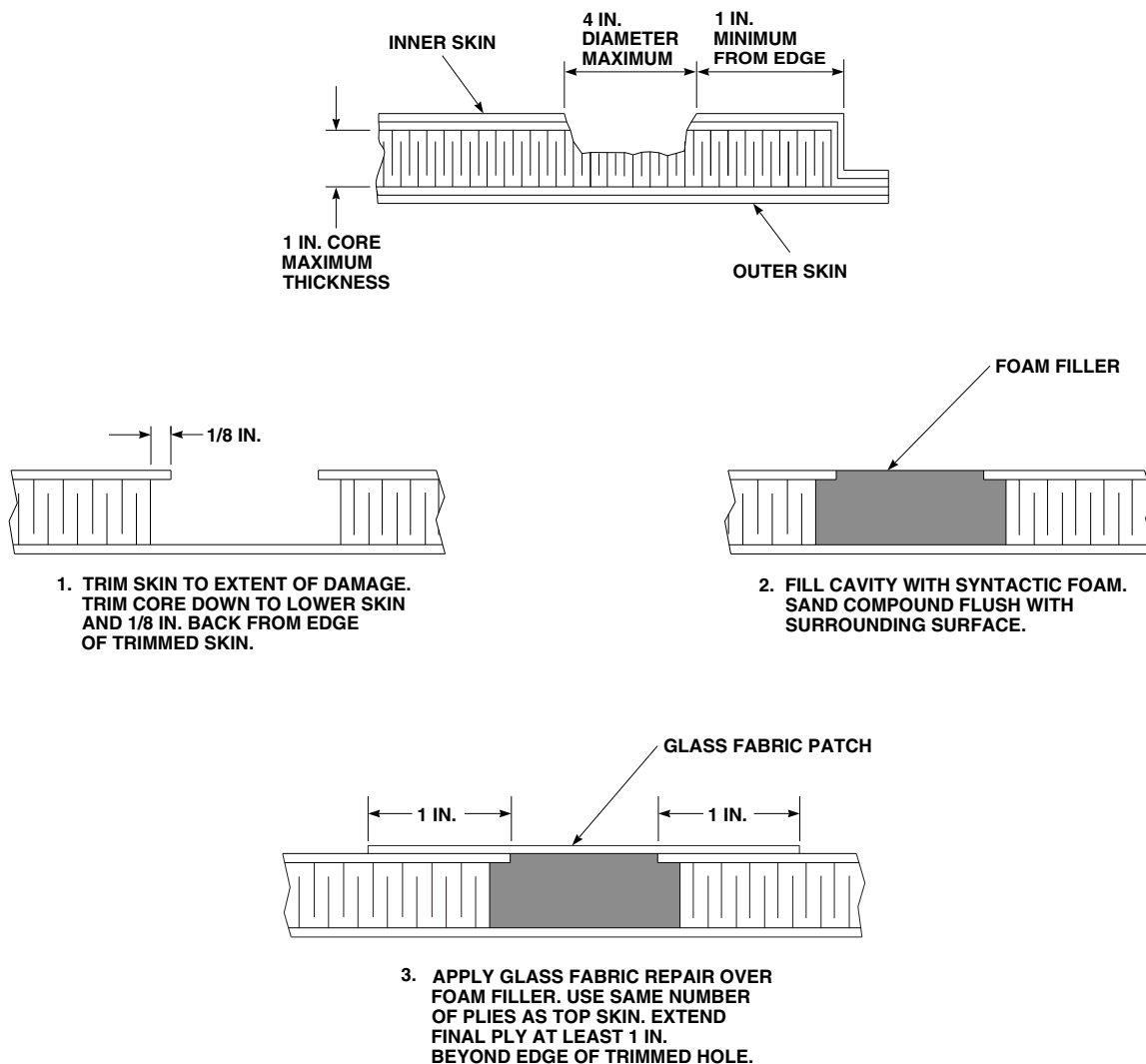
For Kevlar repair, follow instructions in PARA 2-6-3.1.2 in addition to these procedures.

- 2-6-3.14 Repair Honeycomb Core - Composite Skin Panel, Impact Damage Not Penetrating to Honeycomb Core.**

NOTE

For Kevlar repair, follow instructions in PARA 2-6-3.1.2 in addition to these procedures.

 - a. Damage less than 1/8-inch deep, not puncturing skin, with skin not disbonded from core, may be repaired.
 - b. Using coin-tapping method, inspect damaged area to verify absence or disbonding (PARA 5-3-11).
 - c. Mark outline of damaged area with at least 1-inch added in all directions.
 - d. Mask surface beyond marked area with 1-inch masking tape, Item 212, Appendix D.
 - e. Lightly sand repair area using abrasive cloth, Item 6, Appendix D.
 - f. Thoroughly clean repair area using cheesecloth, Item 90, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D, or acetone, Item 25, Appendix D. Allow to air-dry.
 - g. Mix metal surface dent filler, Item 139A, Appendix D (Table 2-6-3-2).
 - h. Apply metal surface dent filler, Item 139A, Appendix D, with beverage stirring stick, Item 315C, Appendix D.
 - i. Using squeegee, smooth filler and remove excess.
 - j. Allow filler to dry for at least 2 hours.
 - k. Sand repair to blend into adjoining surface using abrasive cloth, Item 6, Appendix D.



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FIGURE 2-6-3-8. REPAIR DAMAGE TO ONE SKIN AND CORE

1. Touch up paint as required (PARA 2-6-5).

2-6-3.15 Repair Honeycomb Core - Composite Skin Panel, Damage to One Skin Penetrating Into Honeycomb Core.

NOTE

- Indentations deeper than 1/8-inch can be repaired by this procedure.

- For Kevlar repair, follow instructions in PARA 2-6-3.1.2 in addition to these procedures.

- a. Punctures from one surface, less than 4-inches in length or diameter, located at least 1-inch from edge of honeycomb core, may be repaired (core not more than 1-inch thick) (Figure 2-6-3-8).

CAUTION

Damage to inner skin will result if cutting tools are used improperly. Use care when using cutting tools.

- b. Open up puncture (or dent) with drill or router, removing ragged edges and broken fibers.
- c. Clean out crushed core, undercutting skin by at least 1/8-inch. Remove debris with vacuum cleaner.
- d. Allowing at least 1-inch in all directions beyond trimmed edge of damage, mark perimeter of repair patch.
- e. Mask surface just outside marked area with 1-inch masking tape, Item 212, Appendix D.

CAUTION

Delamination of honeycomb panel will result if too much solvent is allowed to seep into panel. Use solvent sparingly to prevent delamination.

- f. Remove paint finish inside masked area and clean surface using cheesecloth, Item 90, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D, or acetone, Item 25, Appendix D.
- g. Mix syntactic foam (Table 2-6-3-1).
- h. Fill cavity of panel with syntactic foam by pouring or with beverage stirring stick, Item 315C, Appendix D. Release trapped air by working beverage stirring stick up and down. Completely fill cavity slightly beyond skin surface to allow for shrinkage.
- i. Cure syntactic foam (Table 2-6-3-1).
- j. Sand cured foam flush with skin, using abrasive cloth, Item 2, Appendix D.

- k. Clean repair area using cheesecloth, Item 90, Appendix D, moistened with acetone, Item 25, Appendix D, or methyl ethyl ketone, Item 217, Appendix D.
- l. Cut repair plies from satin cloth, Item 105, Appendix D, to shape of marked area, number of plies to equal number of plies in skin.
- m. Mix resin/curing agent (Table 2-6-3-1).
- n. Spread repair plies on flat surface covered with polyvinyl plastic sheet, Item 237B, Appendix D.
- o. Saturate repair plies with mixed resin using acid brush, Item 84, Appendix D, or beverage stirring stick, Item 315C, Appendix D.
- p. Apply coat of mixed resin to unmasked skin and foam areas of part, using acid brush, Item 84, Appendix D.
- q. Put repair plies in place over cavity, overlapping undamaged skin at least 1-inch in all directions.
- r. Cover repair with polyvinyl plastic sheet, Item 237B, Appendix D.
- s. Squeegee trapped air and excess resin out of repair plies.
- t. Apply pressure to repair by vacuum bag, clamping plates, or system of weights and pressure plates.
- u. Cure repair (Table 2-6-3-1).
- v. Remove vacuum bag, clamping plates, or system of weights and pressure plates and plastic strip.
- w. Sand to remove excess resin and blend repair into adjoining surface, using abrasive cloth, Item 6, Appendix D.
- x. Touch up paint as required (PARA 2-6-5).

2-6-3.16 Repair Honeycomb Core - Composite Skin Panel, Damage Penetrating Both Skins and Honeycomb Core.

NOTE

For Kevlar repair, follow instructions in PARA 2-6-3.1.2 in addition to these procedures.

- a. Penetrations completely through panel (including misdrilled holes), less than 1-inch in length or diameter and at least 2 1/2-inches from honeycomb core edge, may be repaired.

CAUTION

Damage to inner skin will result if cutting tools are used improperly. Use care when using cutting tools.

- b. Open up puncture, with drill, router, or hole saw to remove ragged edges and broken fibers, to a maximum of 1-inch diameter.
- c. Counterbore hole in exterior skin to 1 1/2 times original hole diameter. Do not exceed 1 1/2-inches (Figure 2-6-3-9).
- d. Clean out crushed core, undercutting at least 1/8-inch below exterior skin. Remove debris with vacuum cleaner.
- e. Mark outline of repair at least 1-inch beyond hole edge, on both exterior and interior skins.
- f. Mask around outlines with 1-inch masking tape, Item 212, Appendix D.

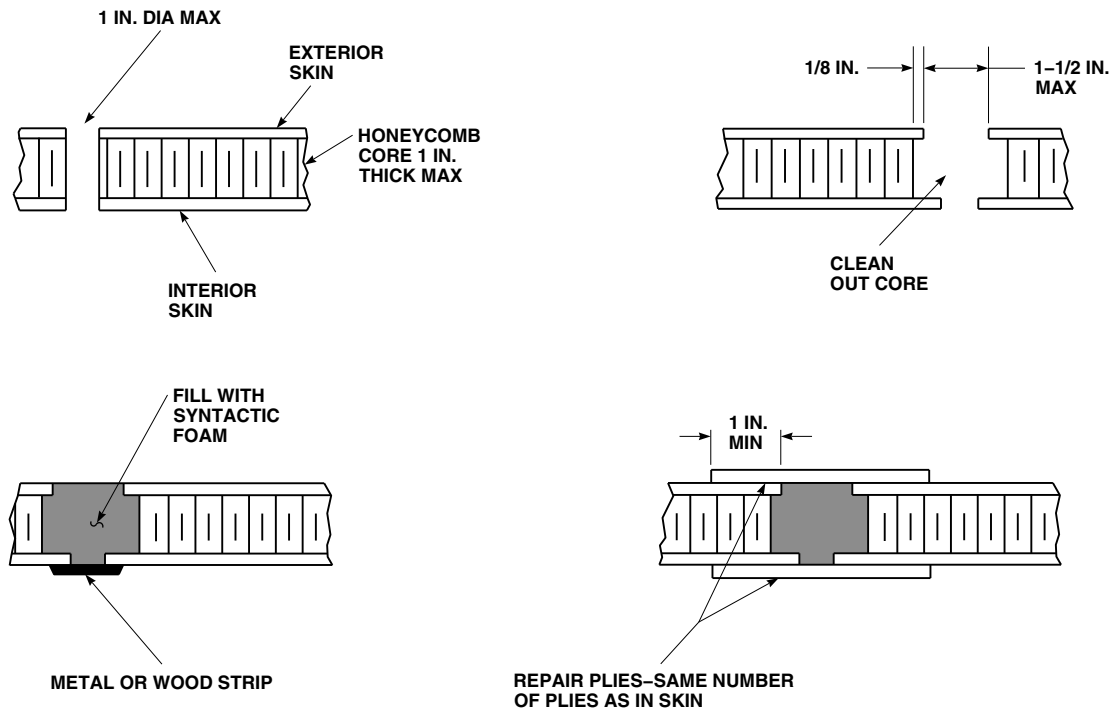
CAUTION

Delamination of honeycomb panel will result if too much solvent is allowed to seep into panel. Use solvent sparingly to prevent delamination.

- g. Remove paint finish inside masked areas and clean skin using cheesecloth, Item 90, Ap-

pendix D, moistened with acetone, Item 25, Appendix D, or methyl ethyl ketone, Item 217, Appendix D.

- h. Cover opening in interior skin with polyvinyl plastic sheet, Item 237B, Appendix D, backed up by wood or metal strip held in place by clamps or 1-inch masking tape, Item 212, Appendix D.
- i. Mix syntactic foam with curing agent (Table 2-6-3-1).
- j. Fill cavity with syntactic foam by pouring or with beverage stirring stick, Item 315C, Appendix D.
- k. Release trapped air by working beverage stirring stick up and down. Completely fill cavity, slightly beyond skin surface to allow for shrinkage.
- l. Cure syntactic foam (Table 2-6-3-1).
- m. Sand foam repair even with exterior skin using abrasive cloth, Item 6, Appendix D.
- n. Remove wood or metal strip and plastic strip, from interior skin surface.
- o. Clean foam repair and both skins inside masked areas using cheesecloth, Item 90, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D, or acetone, Item 25, Appendix D.
- p. Prepare resin mix (Table 2-6-3-1).
- q. Cut repair plies of satin cloth, Item 105, Appendix D, to fit areas on both skins within masked area. Number of plies in each repair must equal number of plies in skin.
- r. Place repair plies on flat surface protected by polyvinyl plastic sheet, Item 237B, Appendix D.
- s. Saturate repair plies with resin mix, using acid brush, Item 84, Appendix D, or beverage stirring stick, Item 315C, Appendix D.
- t. Apply coat of resin mix to repair area of both skins, using acid brush, Item 84, Appendix D.

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- u. Put repair plies in place, cover with polyvinyl plastic sheet, Item 237B, Appendix D, and force out trapped air and excess resin using squeegee.
- v. Apply pressure to repair plies by vacuum bag, clamping plates, or system of weights and pressure plates.
- w. Cure repair (Table 2-6-3-1).
- x. Remove pressure and heat equipment, plastic sheet, and 1-inch masking tape.
- y. Blend repair into surrounding skin surfaces using abrasive cloth, Item 6, Appendix D.
- z. Touch up paint as required (PARA 2-6-5).

2-6-3.17 Repair Honeycomb Core - Composite Skin Panel, Skin-to-Core Disbond.

NOTE

For Kevlar repair, follow instructions in PARA 2-6-3.1.2 in addition to these procedures.

- a. Using coin-tapping method, determine extent of disbond (PARA 5-3-11).
- b. Disbonded areas not more than 2-inches in diameter are repairable by method A or method B.

2-6-3.17.1 Repair Method A.

CAUTION

Honeycomb core can be easily damaged while cutting through skin. Use extreme care to limit damage.

- a. Cut through skin along marked outline of disbonded area, using router bit. Remove and discard cutout skin.
- b. Allowing 1/2-inch blend per ply, mark outside limit of repair.
- c. Blend skin plies using abrasive cloth, Item 2, Appendix D (Figure 2-6-3-10, Sheet 1).

- d. Mark outline of overlap ply extending 1-inch beyond blended area.
- e. Sand overlap area lightly using abrasive cloth, Item 6, Appendix D.
- f. Remove loose particles from cavity with vacuum cleaner and wipe up sanding dust with clean tack rag.
- g. Mask surface just outside marked area with 1-inch masking tape, Item 212, Appendix D.

CAUTION

Delamination of honeycomb panel will result if too much solvent is allowed to seep into panel. Use solvent sparingly to prevent delamination.

- h. Clean skin inside masked area using cheesecloth, Item 90, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D, or acetone, Item 25, Appendix D.
- i. Mix sealing compound, Item 272 or Item 302A, Appendix D (Table 2-6-3-1).
- j. Fill exposed core with sealing compound, Item 272 or Item 302A, Appendix D, using beverage stirring stick, Item 315C, Appendix D. Force sealing compound into honeycomb to eliminate air pockets, filling core slightly higher than original surface to allow for shrinkage.

CAUTION

Damage to part will result from expansion of wet sealing compound when heated. Allow sealing compound to harden to touch before heat-curing.

- k. Cure sealing compound, Item 272 or Item 302A, Appendix D (Table 2-6-3-1).

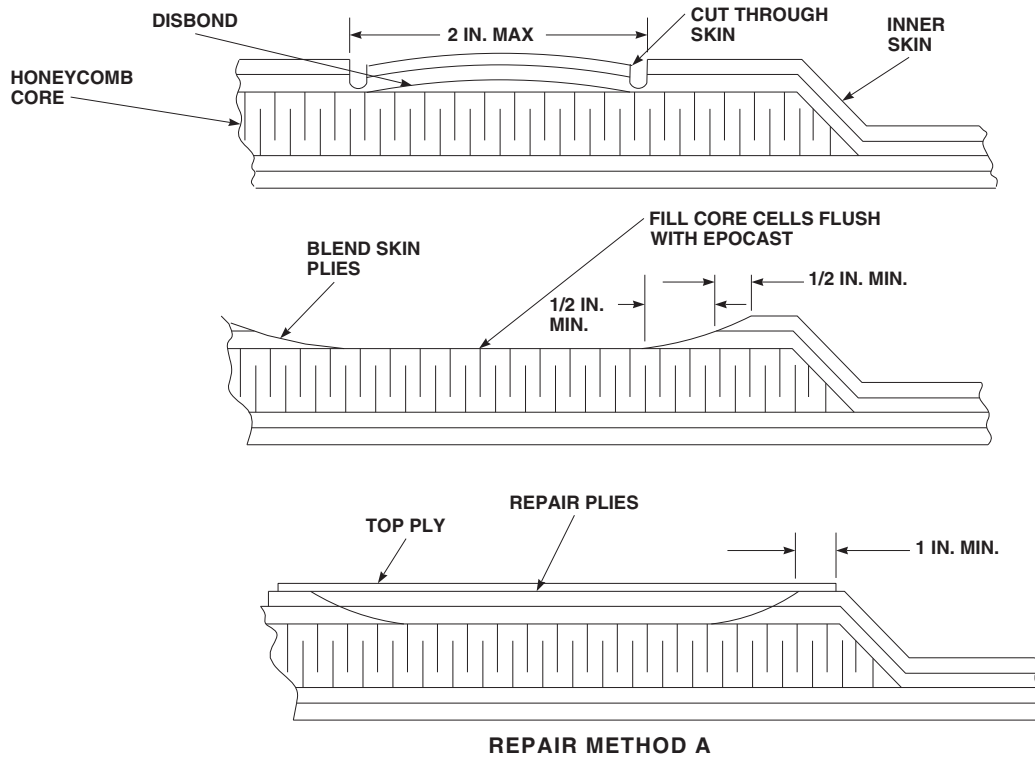
NOTE

Cut weave of each repair ply to match its laminate ply.

- l. Cut the same number of repair plies as in original skin from satin cloth, Item 105, Appendix D.
- m. Cut top ply from satin cloth, Item 105, Appendix D, to overlap repair by 1-inch, with fabric direction the same as top skin ply.
- n. Mix resin/curing agent (Table 2-6-3-1).
- o. Spread repair plies on flat surface covered with polyvinyl plastic sheet, Item 237B, Appendix D.
- p. Saturate repair plies with mixed resin using acid brush, Item 84, Appendix D, or beverage stirring stick, Item 315C, Appendix D.
- q. Apply coat of mixed resin to unmasked skin and foam areas of part, using acid brush, Item 84, Appendix D.
- r. Put repair plies in place with top ply, overlapping undamaged skin at least 1-inch in all directions.
- s. Cover repair with polyvinyl plastic sheet, Item 237B, Appendix D.
- t. Using squeegee, work out trapped air and excess resin out of repair patch.
- u. Apply pressure to repair by vacuum bag, clamping plates, or system of weights and pressure plates.
- v. Cure repair (Table 2-6-3-1).
- w. Remove vacuum bag, clamping plates, or system of weights and pressure plates, and sheet.
- x. Sand to remove excess resin and blend repair into adjoining surface, using abrasive cloth, Item 6, Appendix D.
- y. Touch up paint as required (PARA 2-6-5).

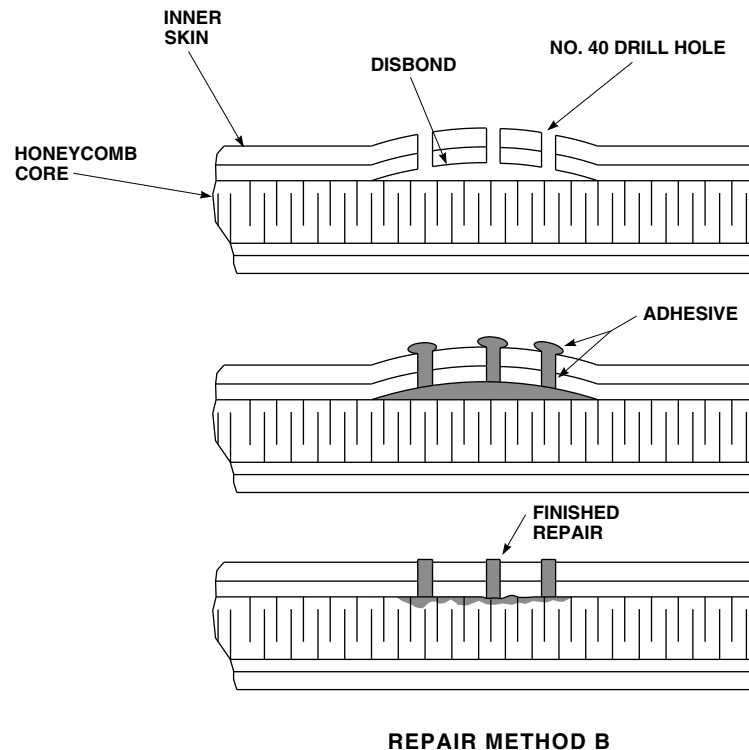
2-6-3.17.2 Repair Method B.

- a. Drill two or more holes, through disbanded skin only, using drill with No. 40 drill bit, at least 1/2-inch apart (Figure 2-6-3-10, Sheet 2).



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FIGURE 2-6-3-10. REPAIR SKIN-TO-CORE DISBOND (SHEET 1 OF 2)



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FIGURE 2-6-3-10. REPAIR SKIN-TO-CORE DISBOND (SHEET 2 OF 2)

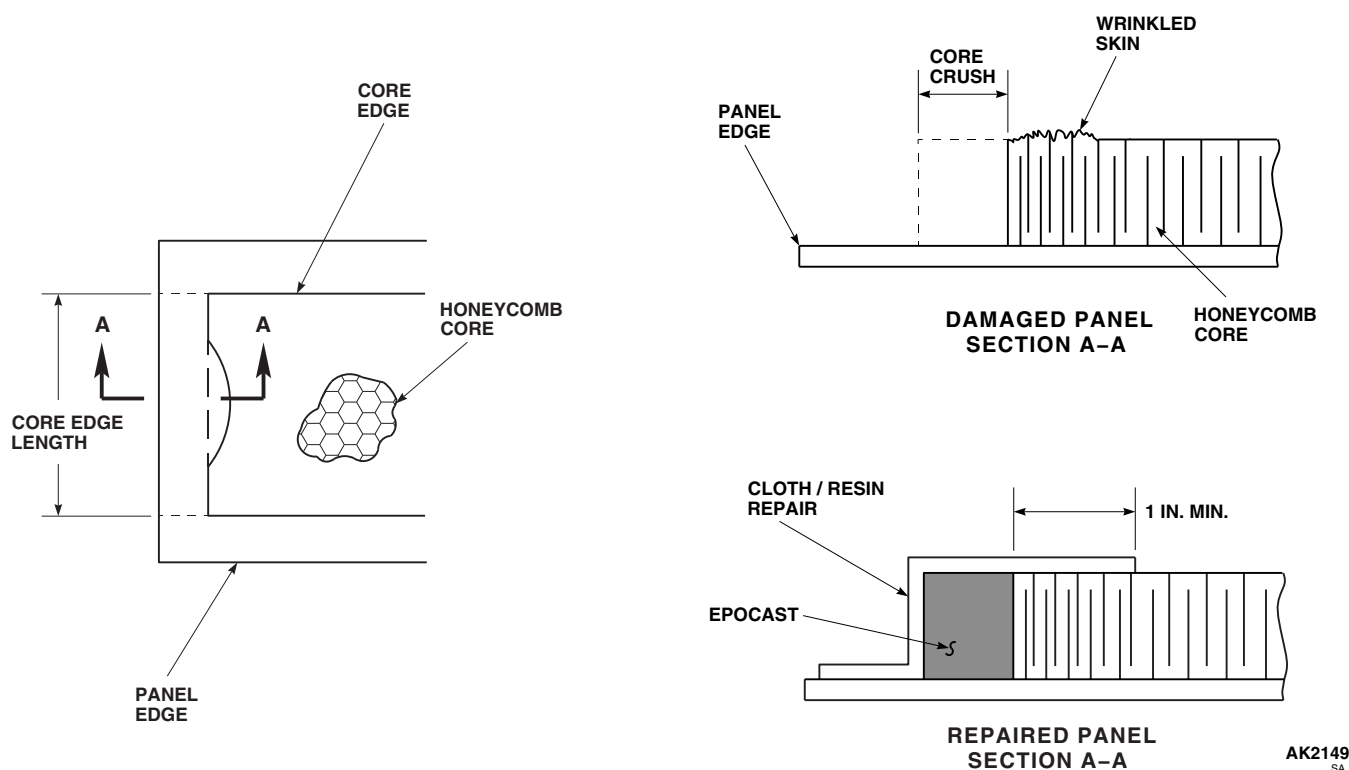


FIGURE 2-6-3-11. REPAIR CRUSHED CORE EDGE

- b. Remove loose particles in and around drill holes with vacuum cleaner.
- c. Select adhesive, Item 34, Item 35, or Item 66D, Appendix D. Mix adhesive, Item 34 or Item 35, Appendix D, per Table 2-6-3-2, or adhesive, Item 66D, Appendix D, per Table 2-6-3-1.
- d. Locally make hypodermic syringe (Figure H-197, Appendix H).
- e. Using hypodermic syringe, inject adhesive into each drilled hole until it overflows.
- f. Remove excess adhesive from skin surface and place polyvinyl plastic sheet, Item 237B, Appendix D, over repair.
- g. Apply pressure to disbonded area by vacuum bag, clamping plates, or system of weights and pressure plates.
- h. Cure repair (Table 2-6-3-1 or Table 2-6-3-2).
- i. Remove vacuum bag, clamping plates, or system of weights and pressure plates. Remove plastic strip.
- j. Sand to remove excess resin and blend repair into adjoining surface, using abrasive cloth, Item 6, Appendix D.
- k. Touch up paint as required (PARA 2-6-5).

2-6-3.18 Repair Honeycomb Core - Composite Skin Panel, Repair Core Edge.

NOTE

- Core edges crushed inward more than 1/2-inch but less than 1 1/2-inches, not exceeding 50% of total core edge length, are repairable by this method.
 - For Kevlar repair, follow instructions in PARA 2-6-3.1.2 in addition to these procedures.
- a. Mark outline of repair area, indicating original contour of core, using scribe (Figure 2-6-3-11).
 - b. Adding at least 1-inch in all directions for feathering repair, mask outside of area with 1-inch masking tape, Item 212, Appendix D.

- c. Sand smooth any wrinkles of inner skin caused by crushing of core, using abrasive cloth, Item 2 and Item 6, Appendix D.
- d. Using abrasive cloth, Item 6, Appendix D, lightly sand entire repair area.
- e. Clean repair area thoroughly using cheesecloth, Item 90, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D, or acetone, Item 25, Appendix D. Allow to air-dry.
- f. To make mold for repair, secure wood or metal strip across crushed area with 1-inch masking tape, Item 212, Appendix D. Cover inside of strip with polyvinyl plastic sheet, Item 237B, Appendix D.
- g. Mix sealing compound, Item 272 or Item 302A, Appendix D (Table 2-6-3-1).
- h. Apply sealing compound with beverage stirring stick, Item 315C, Appendix D, to fill any exposed core and build core up to original contour.

CAUTION

Damage to part will result from expansion of wet sealing compound when heated. Allow sealing compound to harden to touch before heat-curing.

- i. Cure sealing compound, Item 272 or Item 302A, Appendix D (Table 2-6-3-1).
- j. Remove mold strip and plastic strip.
- k. Sand sealing compound flush with adjacent core edges using abrasive cloth, Item 2 and Item 6, Appendix D.
- l. Clean repair and overlap area using cheesecloth, Item 90, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D, or

acetone, Item 25, Appendix D. Allow to air-dry.

NOTE

Match weave direction of each repair ply to its laminate ply.

- m. Cut repair plies from satin cloth, Item 105, Appendix D, to extend at least 1-inch beyond actual repair in all directions. Use same number of plies as in original skin.
- n. Mix resin/curing agent (Table 2-6-3-1).
- o. Spread repair plies on flat surface covered with polyvinyl plastic sheet, Item 237B, Appendix D.
- p. Saturate repair plies with mixed resin using acid brush, Item 84, Appendix D, or beverage stirring stick, Item 315C, Appendix D.
- q. Coat repair area and 1-inch overlap with mixed resin, using acid brush, Item 84, Appendix D, or beverage stirring stick, Item 315C, Appendix D.
- r. Place all repair plies in position and cover with polyvinyl plastic sheet, Item 237B, Appendix D.
- s. Using squeegee, work out trapped air and excess resin.
- t. Apply pressure to repair by vacuum bag, clamping plates, or system of weights and pressure plates.
- u. Cure repair (Table 2-6-3-1).
- v. Remove vacuum bag, clamping plates, or system of weights and pressure plates, and plastic strip.
- w. Use abrasive cloth, Item 6, Appendix D, to remove excess resin and blend repair into adjoining surfaces.

- x. Touch up paint as required (PARA 2-6-5).

2-6-3.19 Repair Honeycomb Core - Metal Skin Panel, Repair Inner Skin-to-Core Disbond.

NOTE

Disbond between inner skin and honeycomb core, not more than 2 square-inches in area, is repairable. Multiple disbonds on one panel, at least 4-inches between areas, are repairable, but total area must not exceed 10% of total panel surface.

- a. Using coin-tapping method, locate and outline disbond areas (PARA 5-3-11).

CAUTION

Helicopter fuel contacting repair can adversely affect adhesive. Do not make this repair on bonded panels exposed to helicopter fuel.

- b. Drill two or more holes through inner skin in disbanded area, using drill with No. 40 drill bit. Space holes at least 1/2-inch apart (Figure 2-6-3-12).
- c. Mix adhesive, Item 32, Appendix D (Table 2-6-3-2).
- d. Locally make hypodermic syringe (Figure H-197, Appendix H).
- e. Using hypodermic syringe, inject adhesive into each drilled hole until it overflows.
- f. Apply 1-inch masking tape, Item 212, Appendix D, over all drilled holes.
- g. Turn panel upside down to allow adhesive flow throughout disbanded area for 10 - 15 minutes.
- h. Apply pressure to disbanded area by vacuum bag, clamping plates, or system of weights and pressure plates.

- i. Cure repair (Table 2-6-3-2).
- j. Remove vacuum bag, clamping plates, or weights and pressure plates.
- k. Remove masking tape.
- l. Remove excess adhesive using nonmetallic scraper and cheesecloth, Item 90, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D. Wipe dry with clean cloth.
- m. If skin surface has been abraded during repair, apply corrosion resistant coating, Item 118, Appendix D, to skin surface and dry thoroughly.
- n. Touch up paint as required (PARA 2-6-5).

2-6-3.20 Repair Honeycomb Core - Metal Skin Panel, Repair Closeout Disbond.

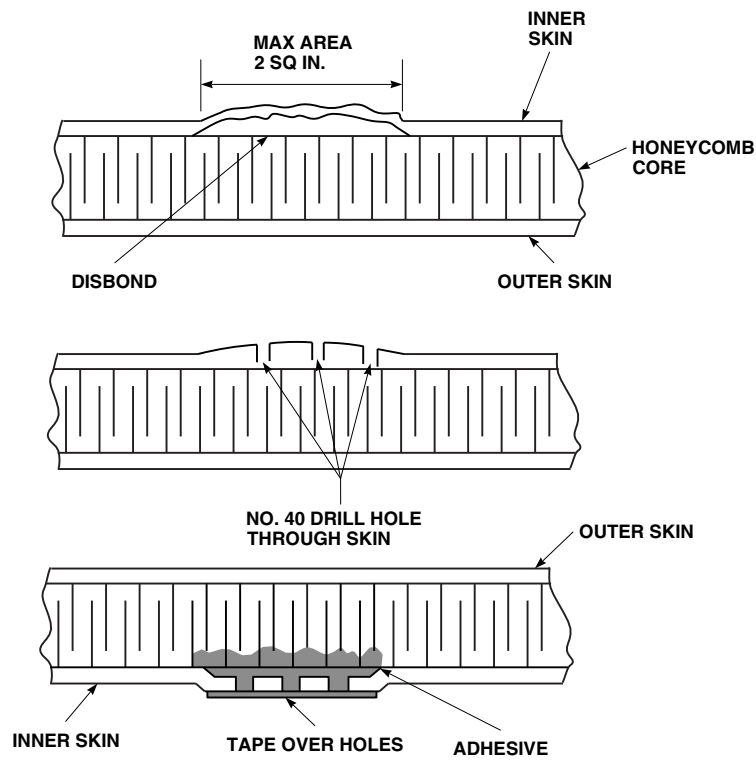
CAUTION

- Helicopter fuel contacting repair can adversely affect adhesive. Do not make this repair on bonded panels exposed to helicopter fuel.
- Use care in slotting closeout to avoid damaging skin or cutting into bonded area.

NOTE

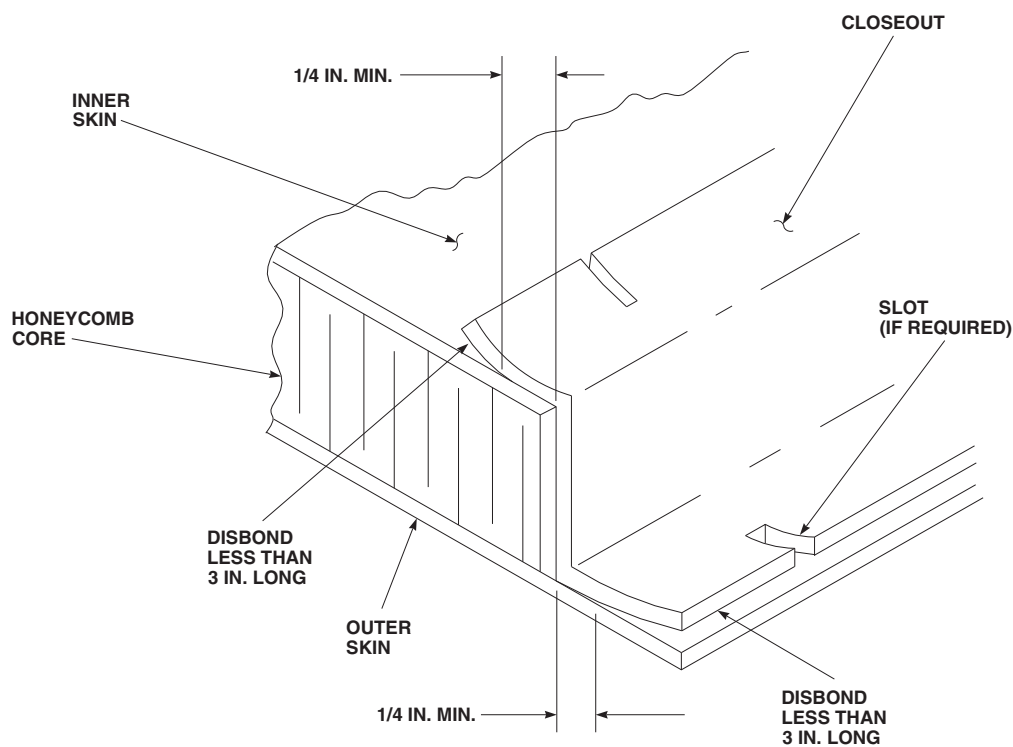
Disbonds with at least 1/4-inch contact (measured from vertical core edge toward outer edge of closeout) and less than 3-inches long are repairable.

- a. If disbanded area of closeout bulges away from panel, cut slot near center of disbond to permit flattening closeout (Figure 2-6-3-13).
- b. Mix adhesive, Item 32, Appendix D (Table 2-6-3-2).
- c. Force adhesive into crack of disbond, using beverage stirring stick, Item 315C, Appendix D (sharpened, if required) or thin plastic tool, until disbond is filled.



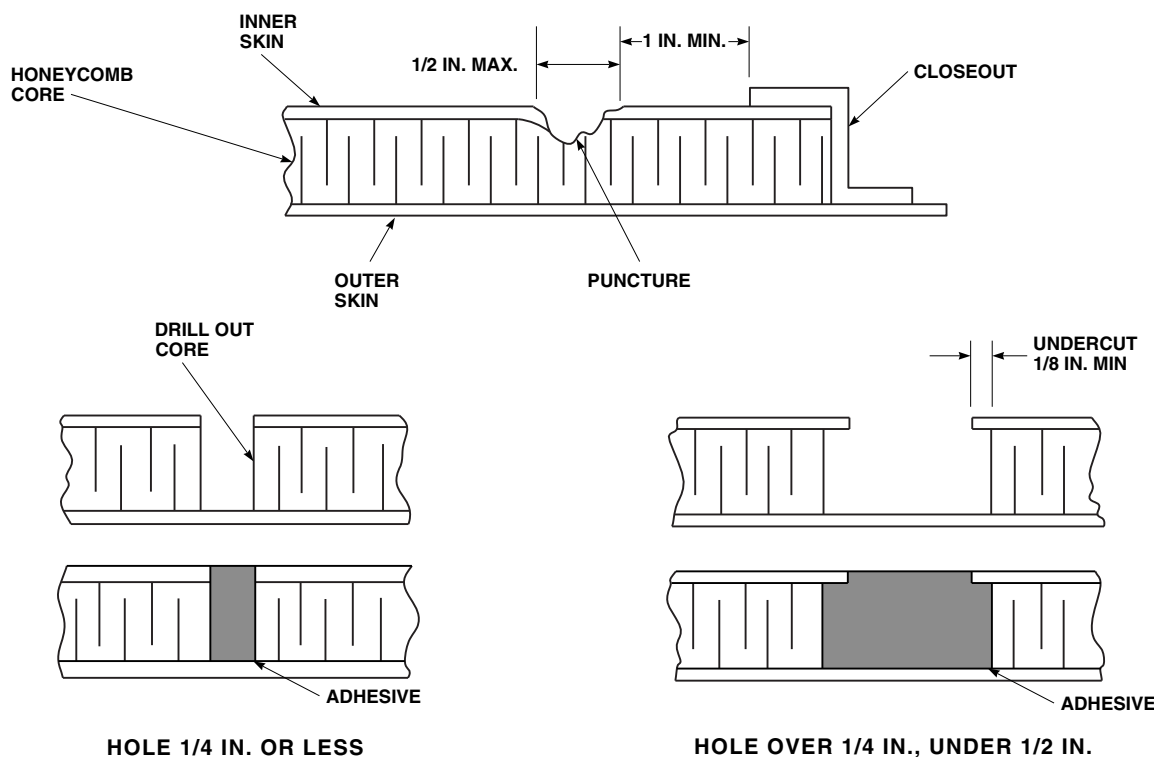
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FIGURE 2-6-3-12. REPAIR INNER SKIN-TO-CORE DISBOND



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FIGURE 2-6-3-13. REPAIR CLOSEOUT DISBOND



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FIGURE 2-6-3-14. REPAIR INNER SKIN PUNCTURE (SHEET 1 OF 2)

- d. Press closeout tightly against panel, using C-clamps to maintain pressure.
- e. Remove excess adhesive using cheesecloth, Item 90, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D.
- f. Cure repair (Table 2-6-3-2).
- g. Remove clamps.
- h. Using coin-tapping method, verify bond (PARA 5-3-11).
- i. Touch up paint as required (PARA 2-6-5).

2-6-3.21 Repair Honeycomb Core - Metal Skin Panel, Repair Inner Skin Puncture, Under 1/2-Inch.

CAUTION

- Helicopter fuel contacting repair can adversely affect adhesive. Do not make

this repair on bonded panels exposed to helicopter fuel.

- If outer skin is punctured during work, do not complete repair. Use extreme care to avoid this.

NOTE

Punctures to inner skin less than 1/2-inch in length or diameter, located at least 1-inch from nearest panel closeout, are repairable by this method.

- a. Clean up puncture with drill or hole saw to remove ragged edges and eliminate cracks. Deburr hole edge (Figure 2-6-3-14, Sheet 1).
- b. If finished hole is more than 1/4-inch in diameter, clean out crushed core with router and undercut skin at least 1/8-inch.
- c. Remove all loose particles from cavity with vacuum cleaner.
- d. Apply corrosion resistant coating, Item 118, Appendix D, to cut skin and dry thoroughly.

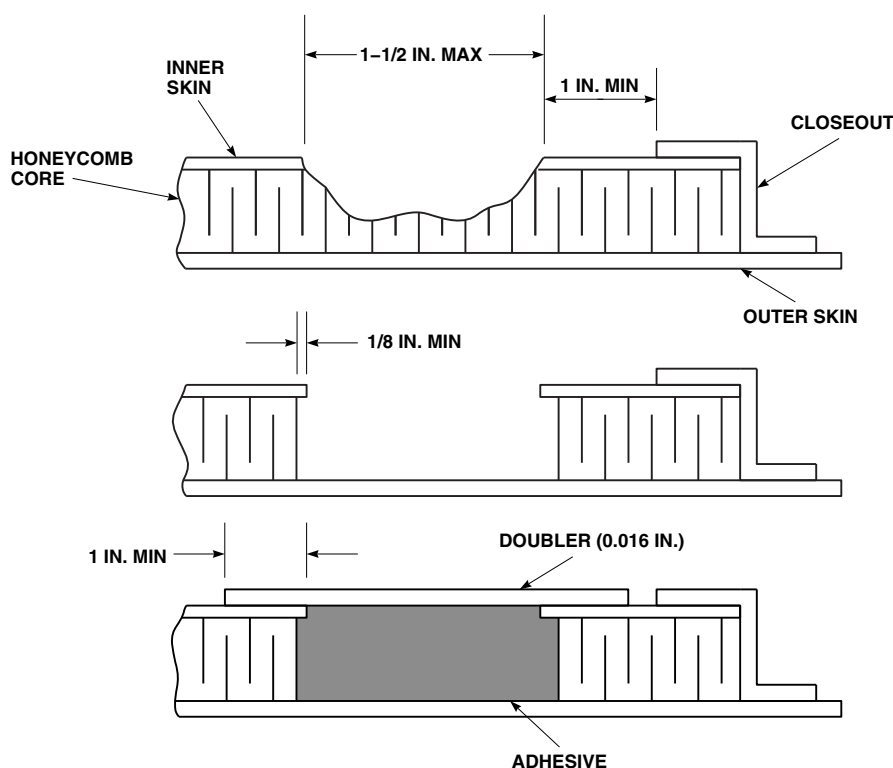
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FIGURE 2-6-3-14. REPAIR INNER SKIN PUNCTURE (SHEET 2 OF 2)

- e. Mix adhesive, Item 32, Appendix D (Table 2-6-3-2).
- f. Fill cavity completely with adhesive, slightly above skin surface to allow for shrinkage.
- g. Cure repair (Table 2-6-3-2).
- h. Lightly sand repair flush with skin surface, using abrasive paper, Item 18, Appendix D.
- i. If surrounding skin is scratched or abraded, apply corrosion resistant coating, Item 118, Appendix D, and dry thoroughly.
- j. Touch up paint as required (PARA 2-6-5).

2-6-3.22 Repair Honeycomb Core - Metal Skin Panel, Repair Inner Skin Puncture, 1/2 - 1 1/2-Inches.

CAUTION

- Helicopter fuel contacting repair can adversely affect adhesive. Do not make

this repair on bonded panels exposed to helicopter fuel.

- If outer skin is punctured during work, do not complete repair. Use extreme care to avoid this condition.

NOTE

Punctures to inner skin from 1/2 - 1 1/2-inches in length or diameter, located at least 1-inch from nearest panel closeout, are repairable by this method.

- a. Clean up puncture with drill or hole saw to remove ragged edges and eliminate cracks. De-burr hole edge (Figure 2-6-3-14, Sheet 2).
- b. Using router, remove damaged core and undercut inner skin by at least 1/8-inch.
- c. Remove all loose particles from cavity with vacuum cleaner.
- d. Apply corrosion resistant coating, Item 118, Appendix D, to cut skin and dry thoroughly.

- e. Make doubler as follows:
 - (1) Cut from anodized 2024-T3 aluminum sheet, 0.016-inch-thick, extending at least 1-inch beyond skin opening in all directions.
 - (2) If doubler is rectangular, round corners to 0.120-inch radius, minimum.
 - (3) Deburr all edges.
 - (4) Apply corrosion resistant coating, Item 118, Appendix D, to cut edges and dry thoroughly.
- f. Thoroughly clean skin area to be covered by doubler using cheesecloth, Item 90, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D. Air dry.
- g. Mix adhesive, Item 32, Appendix D (Table 2-6-3-2).
- h. Fill cavity with adhesive until level with skin surface.
- i. Spread adhesive over skin surface to be covered by doubler, using beverage stirring stick, Item 315C, Appendix D, or acid brush, Item 84, Appendix D.
- j. Press doubler firmly in place.
- k. Remove adhesive squeeze out, using nonmetallic scraper, and cover with polyvinyl plastic sheet, Item 237B, Appendix D.
- l. Apply pressure to doubler by vacuum bag, clamping plates, or weight and pressure plate.
- m. Cure repair (Table 2-6-3-2).

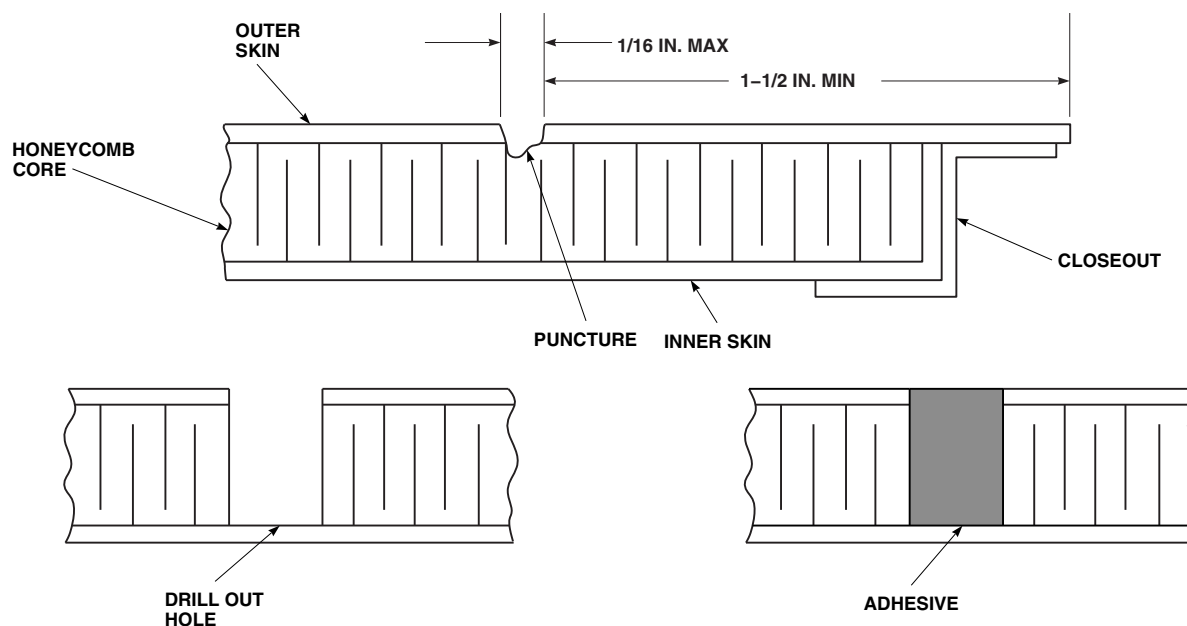
- n. Remove pressure equipment and plastic strip.
- o. Touch up paint as required (PARA 2-6-5).

2-6-3.23 Repair Honeycomb Core - Metal Skin Panel, Repair Outer Skin Puncture.

NOTE

Punctures to outer skin $\frac{1}{16}$ -inch or less in length or diameter, located at least 1 $\frac{1}{2}$ -inches from skin edge, are repairable by this method.

- a. Using drill, enlarge puncture hole enough to remove ragged edges and eliminate any cracks (Figure 2-6-3-15). Deburr hole edge.
- b. Remove all loose particles from cavity with vacuum cleaner.
- c. Apply corrosion resistant coating, Item 118, Appendix D, to cut skin and dry thoroughly.
- d. Mix adhesive, Item 32, Appendix D (Table 2-6-3-2).
- e. Fill cavity completely with adhesive, flush with outer skin surface.
- f. Cure repair (Table 2-6-3-2).
- g. Lightly sand repair flush with outer skin surface, using abrasive paper, Item 18, Appendix D.
- h. If surrounding skin surface is scratched or abraded, apply corrosion resistant coating, Item 118, Appendix D, and dry thoroughly.
- i. Touch up paint as required (PARA 2-6-5).

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2-6-4 EXTRUSION CHARTS.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
2-6-4.1	Typical Rivet Spacing - Sheet Metal Substitute	2-6-4-1	2-6-4.7	Closed Angle Tee	2-6-4-2
2-6-4.2	Equal Leg Angle	2-6-4-2	2-6-4.8	Zee Section	2-6-4-2
2-6-4.3	Unequal Leg Angle	2-6-4-2	2-6-4.9	Channel	2-6-4-2
2-6-4.4	Tee Section	2-6-4-2	2-6-4.10	Closed Leg Angle	2-6-4-2
2-6-4.5	Unequal Thickness Angle	2-6-4-2	2-6-4.11	Open Leg Angle	2-6-4-2
2-6-4.6	Tapered Leg Angle	2-6-4-2	2-6-4.12	Miscellaneous Shapes	2-6-4-2

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Drill Bit, No. 21 (0.1590-inch diameter)
- Drill Bit, No. 30 (0.1285-inch diameter)
- Drill Set

Materials/Parts

- Hi-Loks
- Lockbolts, NAS1414 through NAS6932

- Rivet, MS20470AD4
- Rivet, MS20470AD5

References

- QQ-A-200

2-6-4.1 Typical Rivet Spacing - Sheet Metal Substitute.

- a. Use required rivet spacing and edge distance, when making sheet stock substitution for extrusion (Figure 2-6-4-1).
- b. Use No. 30 drill bit for rivets, MS20470AD4, and No. 21 drill bit for rivets, MS20470AD5.
- c. Use flush rivets, as required, to reduce interference with adjacent structural members.

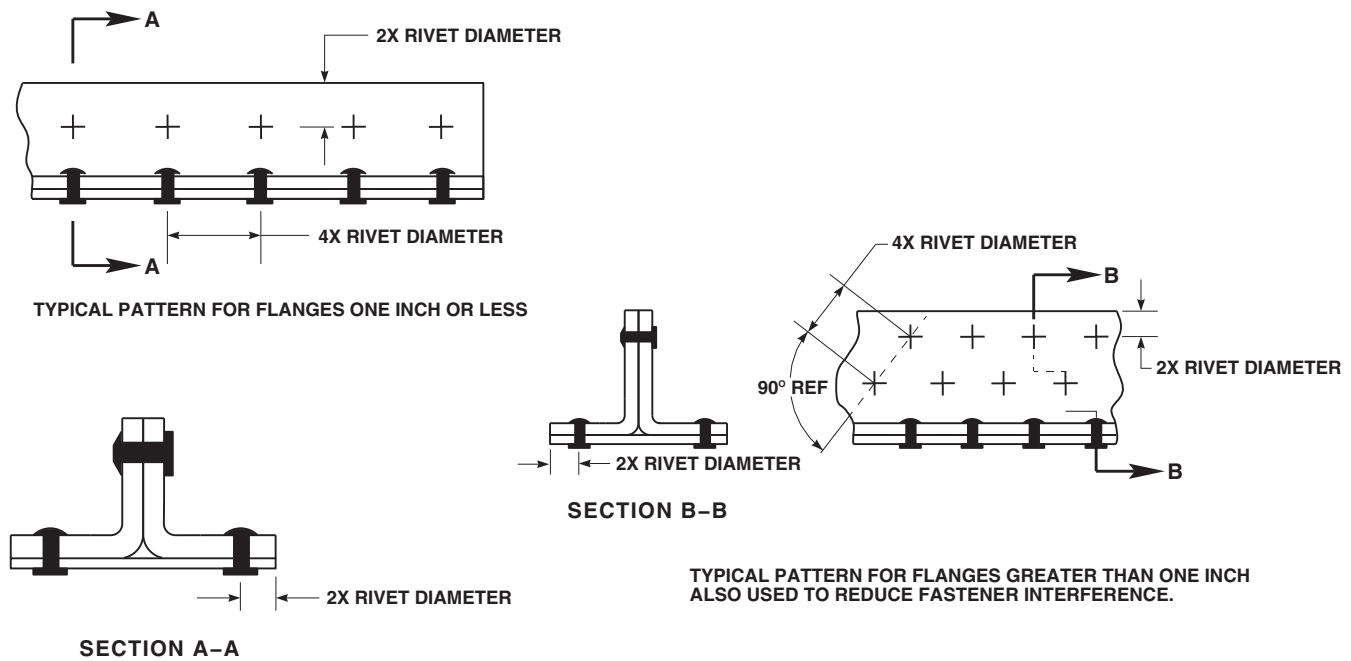
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FIGURE 2-6-4-1. TYPICAL RIVET SPACING AND EDGE DISTANCE

2-6-4.2 Equal Leg Angle.

Use Figure 2-6-4-2, Sheet 1 and Sheet 2, for equal leg angle.

2-6-4.3 Unequal Leg Angle.

Use Figure 2-6-4-3, Sheet 1 through Sheet 3, for unequal leg angle.

2-6-4.4 Tee Section.

Use Figure 2-6-4-4, Sheet 1 through Sheet 3, for tee section.

2-6-4.5 Unequal Thickness Angle.

Use Figure 2-6-4-5 for unequal thickness angle.

2-6-4.6 Tapered Leg Angle.

Use Figure 2-6-4-6 for tapered leg angle.

2-6-4.7 Closed Angle Tee.

Use Figure 2-6-4-7 for closed angle tee.

2-6-4.8 Zee Section.

Use Figure 2-6-4-8 for zee section.

2-6-4.9 Channel.

Use Figure 2-6-4-9 for channel.

2-6-4.10 Closed Leg Angle.

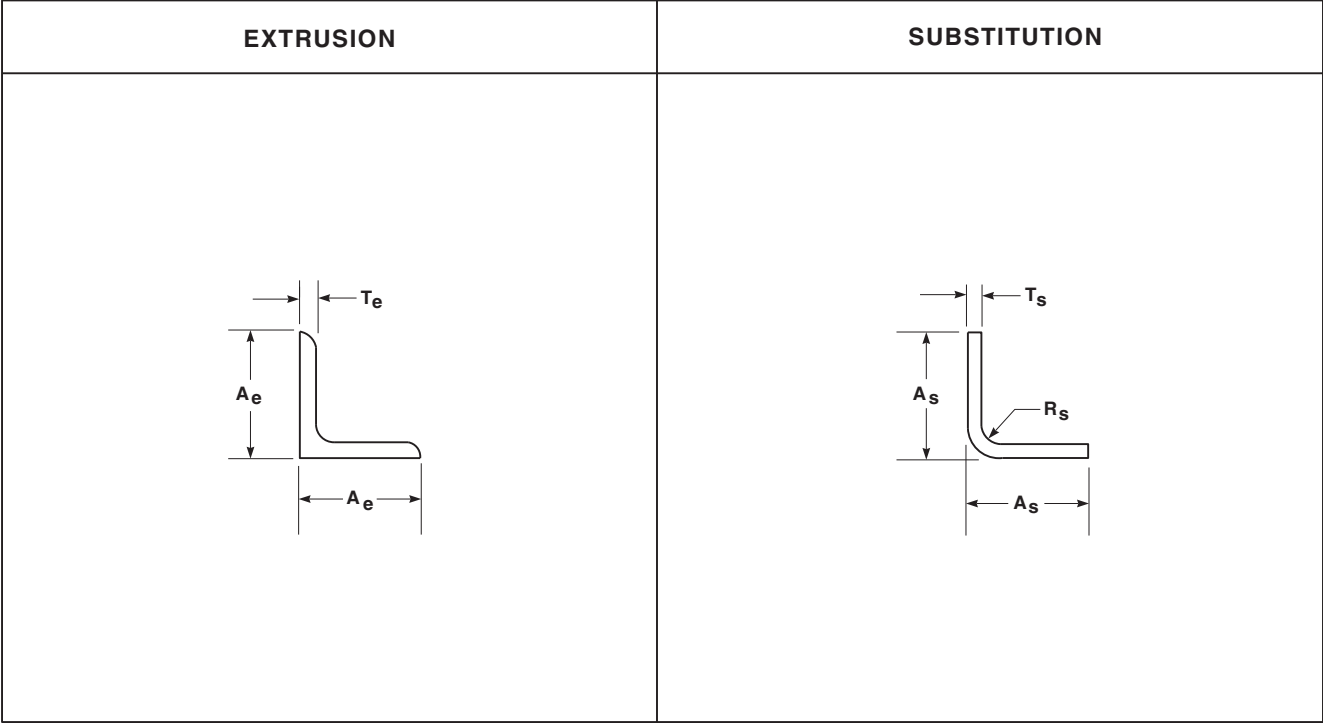
Use Figure 2-6-4-10 for closed leg angle.

2-6-4.11 Open Leg Angle.

Use Figure 2-6-4-11 for open leg angle.

2-6-4.12 Miscellaneous Shapes.

Use Figure 2-6-4-12, Sheet 1 through Sheet 5, for miscellaneous shapes.



EXTRUSION				SUBSTITUTION		
DIE NUMBER	MATERIAL	A _e	T _e	A _s	T _s	R _s
AND10133-0601	7075-T6511	0.75	0.063	0.750	0.071	7/32
AND10133-0701	7075-T6511	0.875	0.063	0.875	0.071	7/32
AND10133-0702	7075-T6	0.875	0.094	0.875	0.125	3/8
AND10133-1001	7075-T6	1.000	0.063	1.000	0.071	7/32
AND10133-1002	7075-T6511	1.000	0.094	1.000	0.125	3/8
AND10133-1003*	7075-T6	1.000	0.125	1.000	0.125	0.188**
AND10133-1201	7075-T6	1.250	0.063	1.250	0.071	7/32
AND10133-1202	7075-T6	1.250	0.094	1.250	0.125	3/8
AND10133-1401	7075-T6511	1.500	0.063	1.500	0.071	7/32
AND10133-1404	7075-T6	1.500	0.188	MACHINE FROM STOCK		
ALCOA 77U*	7075-T6	1.750	0.188	1.750	0.188	0.188**
ALCOA 78F	7075-T6	1.000	0.094	1.000	0.125	3/8

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FIGURE 2-6-4-2. EQUAL LEG ANGLE (SHEET 1 OF 2)

(CONT)

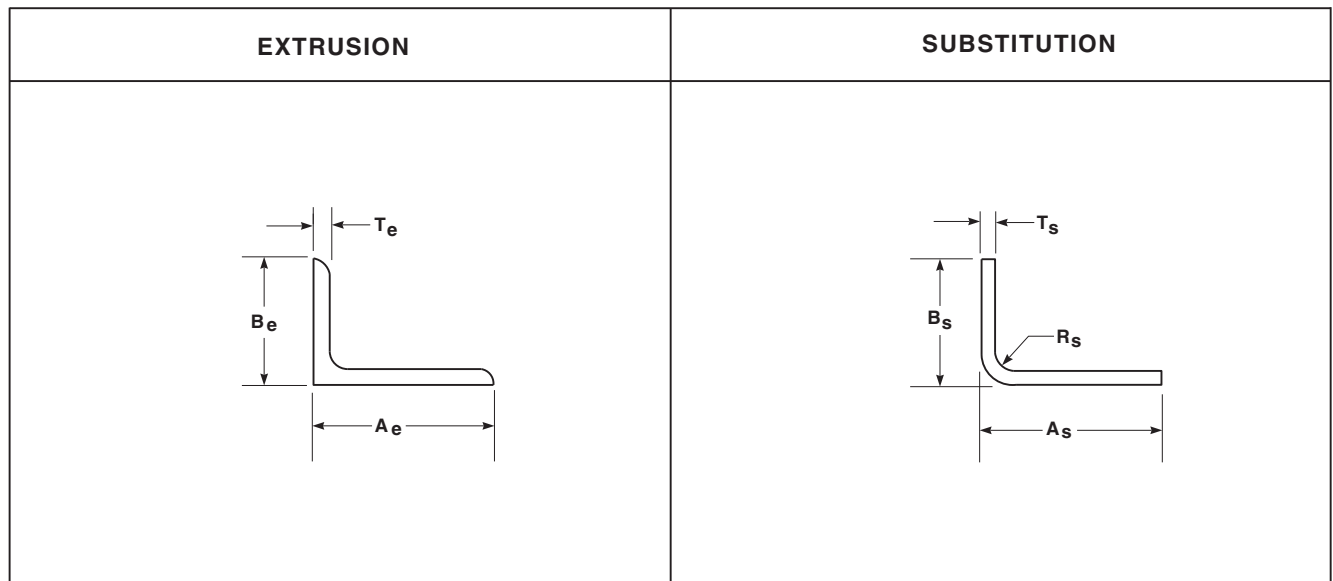
EXTRUSION				SUBSTITUTION		
DIE NUMBER	MATERIAL	A _e	T _e	A _s	T _s	R _s
ALCOA 78J	7075-T6511	1.000	0.062	1.000	0.071	7/32
ALCOA 78L	7075-T6511	1.750	0.094	1.750	0.125	3/8
ALCOA 79CC	7075-T6	1.000	0.094	1.000	0.125	3/8
ALCOA 5401	7075-T6511	0.750	0.062	0.750	0.071	7/32
ALCOA 9802	7075-T6511	0.750	0.050	0.750	0.063	3/16
ALCOA 10854	7075-T6511	0.625	0.040	0.625	0.050	5/32
ALCOA 22477	7075-T6511	0.750	0.040	0.750	0.050	5/32
ALCOA 23299	7075-T6511	0.875	0.063	0.875	0.071	7/32
ALCOA 26290*	7075-T6511	2.000	0.156	2.000	0.156	0.156**
ALCOA 30845	7075-T6	1.000	0.076	1.000	0.090	9/32
ALCOA 32059	7075-T6	0.875	0.081	0.875	0.090	9/32
ALCOA 44076	7075-T6511	0.750	0.050	0.750	0.063	3/16
ALCOA 59409	7075-T6	1.000	0.050	1.000	0.063	3/16
ALCOA 67580	7075-T6511	0.750	0.050	0.750	0.063	3/16
ALCOA 71369	7075-T6	1.000	0.072	1.000	0.090	9/32
ALCOA 82300	7075-T6511	1.000	0.062	1.000	0.071	7/32
HARVEY 22264	7075-T6511	1.000	0.080	1.000	0.090	9/32
HARVEY 50276	7075-T6	1.000	0.040	1.000	0.050	5/32
REYNOLDS 19237	7075-T6511	0.875	0.050	0.875	0.063	3/16
REYNOLDS 22647	7075-T6511	0.875	0.040	0.875	0.050	5/32
REYNOLDS 23888	7075-T6511	1.000	0.090	1.000	0.100	5/16
TIERNAY 60-4645	7075-T6511	1.000	0.100	1.000	0.125	3/8

* MACHINE SUBSTITUTE FROM BAR STOCK OF SAME MATERIAL.

** MACHINED SUBSTITUTE RADIUS.

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FIGURE 2-6-4-2. EQUAL LEG ANGLE (SHEET 2 OF 2)



EXTRUSION					SUBSTITUTION			
DIE NUMBER	MATERIAL	A _e	B _e	T _e	A _s	B _s	T _s	R _s
AND10134-0501	7075-T6511	0.625	0.500	0.063	0.625	0.500	0.071	7/32
AND10134-0702	7075-T6511	0.875	0.750	0.063	0.875	0.750	0.071	7/32
AND10134-1001	7075-T6	1.000	0.625	0.063	1.000	0.625	0.071	7/32
AND10134-1002*	7075-T6	1.000	0.625	0.125	1.000	0.625	0.063	0.125**
AND10134-1003	7075-T6511	1.000	0.750	0.063	1.000	0.750	0.071	7/32
AND10134-1004*	7075-T6	1.000	0.750	0.125	1.000	0.750	0.125	0.125**
AND10134-1005	7075-T6	1.000	0.875	0.063	1.000	0.875	0.071	7/32
AND10134-1006	7075-T6511	1.000	0.875	0.094	1.000	0.875	0.125	3/8
AND10134-1204	7075-T6	1.250	1.000	0.063	1.250	1.000	0.071	7/32
AND10134-1205	7075-T6511	1.250	1.000	0.094	1.250	1.000	0.125	3/8
AND10134-1206*	7075-T6	1.250	1.000	0.125	1.250	1.000	0.125	0.125**
AND10134-1401	7075-T6	1.500	0.750	0.094	1.500	0.750	0.125	3/8
AND10134-1407	7075-T6511	1.500	1.250	0.125	NO SUBSTITUTION			
AND10134-1604	7075-T6511	1.750	1.500	0.156	NO SUBSTITUTION			
AND10134-2402	7075-T6511	2.500	1.250	0.188	NO SUBSTITUTION			

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FIGURE 2-6-4-3. UNEQUAL LEG ANGLE (SHEET 1 OF 3)

EXTRUSION					SUBSTITUTION			
DIE NUMBER	MATERIAL	A _e	B _e	T _e	A _s	B _s	T _s	R _s
ALCOA 1288	7075-T6	1.000	0.750	0.070	1.000	0.750	0.080	1/4
ALCOA 6494	7075-T6511	1.750	1.125	0.064	1.750	1.125	0.080	1/4
ALCOA 9413	7075-T6	0.750	2.000	0.125	0.750	2.000	0.125	3/8
ALCOA 10188	7075-T6	2.500	1.250	0.062	1.500	1.250	0.071	7/32
ALCOA 11472	7075-T6	1.500	1.000	0.062	1.500	1.000	0.071	7/32
ALCOA 12888	7075-T6511	0.750	0.625	0.051	0.750	0.625	0.063	3/16
ALCOA 13633	7075-T6511	4.000	2.500	0.125	MACHINE FROM STOCK			
ALCOA 14263	7075-T6511	1.750	1.000	0.094	1.750	1.000	0.125	3/8
ALCOA 22385	7075-T6	1.000	0.750	0.062	1.000	0.750	0.071	7/32
ALCOA 22509	7075-T6511	1.000	0.750	0.040	1.000	0.750	0.050	5/32
ALCOA 22526	7075-T6	1.000	0.750	0.050	1.000	0.750	0.063	3/32
ALCOA 22580	7075-T6	1.250	1.125	0.062	1.250	1.125	0.071	7/32
ALCOA 22878	7075-T6511	0.875	0.750	0.062	0.875	0.750	0.071	7/32
ALCOA 22885	7075-T6511	1.000	0.750	0.062	1.000	0.750	0.071	7/32
ALCOA 22990	7075-T6	1.500	0.750	0.062	1.500	0.750	0.071	7/32
ALCOA 24262	7075-T6	0.875	0.625	0.051	0.875	0.625	0.063	3/32
ALCOA 24920	7075-T6	1.250	1.000	0.051	1.250	1.000	0.063	3/32
ALCOA 27076	7075-T6511	1.250	0.750	0.062	1.250	0.750	0.071	7/32
ALCOA 44562	7075-T6	0.750	0.500	0.050	0.750	0.500	0.063	3/16
ALCOA 46675	7075-T6511	1.250	1.000	0.080	1.250	1.000	0.090	9/32
ALCOA 52106	7075-T6	0.750	0.680	0.040	0.750	0.680	0.050	5/32
ALCOA 52147	7075-T6511	1.250	1.000	0.109	1.250	1.000	0.125	3/8
ALCOA 52421		1.000	1.250	0.120	NO SUBSTITUTION			
ALCOA 52735	7075-T6	1.250	0.750	0.051	1.250	0.750	0.063	3/16
ALCOA 57238	7075-T6511	3.000	1.250	0.125	3.000	1.250	0.125	3/8
ALCOA 59128	7075-T6511	1.120	1.000	0.094	1.120	1.000	0.125	3/8
ALCOA 59194	7075-T6	1.000	0.800	0.050	1.000	0.800	0.063	3/16
ALCOA 59195	7075-T6511	1.750	1.420	0.064	1.750	1.420	0.071	7/32
ALCOA 59342	7075-T6511	1.000	0.830	0.060	1.000	0.830	0.071	7/32
ALCOA 60242	7075-T6	1.700	1.000	0.070	1.700	1.000	0.080	1/4
ALCOA 61992	7075-T6	1.800	3.000	0.062	1.800	3.000	0.071	7/32

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FIGURE 2-6-4-3. UNEQUAL LEG ANGLE (SHEET 2 OF 3)

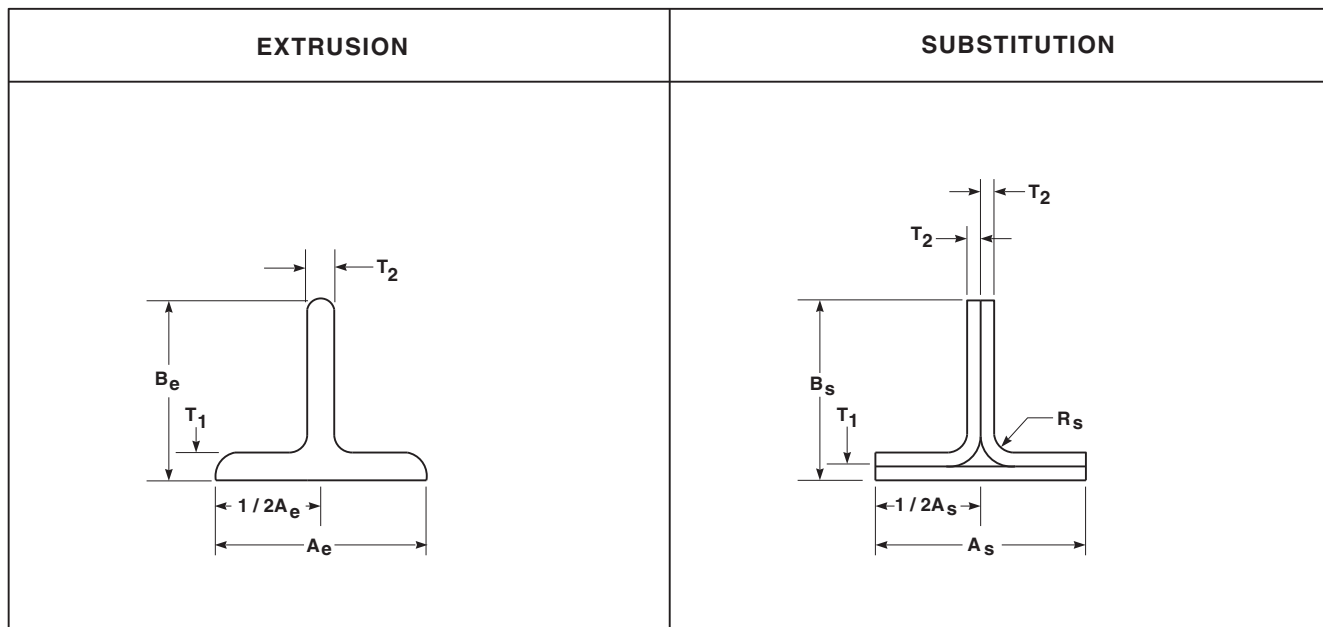
EXTRUSION					SUBSTITUTION			
DIE NUMBER	MATERIAL	A _e	B _e	T _e	A _s	B _s	T _s	R _s
ALCOA 66084	7075-T6511	1.000	0.830	0.050	1.000	0.830	0.063	3/16
ALCOA 78849	7075-T6	0.870	0.750	0.040	0.870	0.750	0.050	5/32
ALCOA 80415	7075-T6	1.812	0.938	0.062	1.812	0.938	0.071	7/32
ALCOA 91655	7075-T6	2.500	1.000	0.078	2.500	1.000	0.090	9/32
ALCOA 91655	2024-T3511	2.500	1.000	0.078	2.500	1.000	0.090	9/32
ALCOA 141532	7075-T6	2.500	1.062	0.125	NO SUBSTITUTION			
HARVEY 14518	7075-T6511	1.000	0.750	0.078	1.000	0.750	0.090	9/32
HARVEY 15078	7075-T6	0.875	0.750	0.051	0.875	0.750	0.063	3/16
HARVEY 27407	7075-T6511	0.800	0.750	0.062	0.800	0.750	0.071	7/32
PIONEER 4191	7075-T6	1.000	1.180	0.050	1.000	1.180	0.063	3/16
REYNOLDS 9204	7075-T6	1.125	1.000	0.062	1.125	1.000	0.071	7/32
REYNOLDS 16381	7075-T6511	0.875	0.750	0.051	0.875	0.750	0.063	3/16
REYNOLDS 18275	7075-T6511	2.625	0.875	0.125	NO SUBSTITUTION			
REYNOLDS 19767	7075-T6511	2.500	1.250	0.094	2.500	1.250	0.125	3/8
S6105-2012-2	7075-T6	1.562	0.750	0.062	1.562	0.750	0.071	7/32
TIERNAY 60-4645	7075-T6511	1.000	1.000	0.100	1.000	1.000	0.125	3/8

* MACHINE SUBSTITUTE FROM BAR STOCK OF SAME MATERIAL.

** MACHINED SUBSTITUTE RADIUS.

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FIGURE 2-6-4-3. UNEQUAL LEG ANGLE (SHEET 3 OF 3)



EXTRUSION						SUBSTITUTION				
DIE NUMBER	MATERIAL	A_e	B_e	T_1	T_2	A_s	B_s	T_1	T_2	R_s
AND10136-1302	7075-T6	1.375	1.000	0.063	0.063	1.375	1.000			
AND10136-1303	7075-T6	1.375	1.000	0.078	0.078	1.375	1.000			
AND10136-1304	7075-T6	1.375	1.000	0.094	0.094	1.375	1.000			
AND10136-1401	7075-T6	1.500	0.750	0.063	0.063	1.500	0.750			
AND10136-1405	7075-T6511	1.500	1.250	0.078	0.078	1.500	1.250			
AND10136-1407	7075-T6511	1.500	1.250	0.125	0.125		MACHINE FROM STOCK			
AND10136-1504	7075-T6	1.625	1.125	0.125	0.125	1.625	1.125	0.125	0.125	3/8
AND10136-1603	7075-T6511	1.750	1.125	0.094	0.094	1.750	1.125	0.090	0.090	9/32
AND10136-2001	7075-T6511	2.000	1.000	0.078	0.078	2.000	1.000	0.080	0.080	1/4
AND10136-2002	7075-T6	2.000	1.000	0.094	0.094	2.000	1.000	0.090	0.090	9/32
AND10136-2003	7075-T6	2.000	1.000	0.125	0.125	2.000	1.000	0.125	0.125	3/8
AND10136-2004	7075-T6511	2.000	1.250	0.078	0.078	2.000	1.250	0.080	0.080	1/4
AND10136-2005	7075-T6511	2.000	1.250	0.094	0.094	2.000	1.250	0.090	0.090	9/32
AND10136-2006	7075-T6	2.000	1.250	0.125	0.125	2.000	1.250	0.125	0.125	3/8
AND10136-2008	7075-T6	2.000	1.750	0.125	0.125	2.000	1.750	0.125	0.125	3/8
AND10136-2402	7075-T6	2.500	1.250	0.125	0.125	2.500	1.250	0.125	0.125	3/8

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FIGURE 2-6-4-4. TEE SECTION (SHEET 1 OF 3)

EXTRUSION						SUBSTITUTION				
DIE NUMBER	MATERIAL	A _e	B _e	T ₁	T ₂	A _s	B _s	T ₁	T ₂	R _s
AND10136-2404	7075-T6	2.500	1.625	0.094	0.094	2.500	1.625	0.090	0.090	9/32
AND10136-2407	7075-T6511	2.000	1.756	0.094	0.094	MACHINE FROM STOCK				
AND10136-2408	7075-T6511	2.500	2.000	0.156	0.156	MACHINE FROM STOCK				
AND10136-2409	7075-T6	2.500	2.000	0.188	0.188	MACHINE FROM STOCK				
ALCOA 3094	7075-T6	1.250	1.550	0.050	0.050	1.250	1.550	0.050	0.050	5/32
ALCOA 11485	7075-T6511	2.000	3.000	0.093	0.093	2.000	3.000	0.090	0.090	9/32
ALCOA 13630	7075-T6511	1.937	3.250	0.125	0.125	1.937	3.250	0.125	0.125	3/8
ALCOA 16136	7075-T6	1.500	0.750	0.062	0.062	1.500	0.750	0.063	0.063	5/32
ALCOA 22675	7075-T6	2.375	2.625	0.080	0.080	2.375	2.625	0.080	0.080	1/4
ALCOA 22899	7075-T6511	1.875	1.000	0.050	0.050	1.875	1.000	0.050	0.050	5/32
ALCOA 22982	7075-T6	1.500	0.812	0.050	0.050	1.500	0.812	0.050	0.050	5/32
ALCOA 27427	7075-T6511	1.750	3.500	0.156	0.125	NO SUBSTITUTION				
ALCOA 28957	7075-T6	1.500	0.750	0.050	0.050	1.500	0.750	0.050	0.050	1/8
ALCOA 29119	7075-T6	2.000	2.000	0.094	0.094	2.000	2.000	0.090	0.090	9/32
ALCOA 31510	7075-T6	2.000	2.562	0.125	0.125	2.000	2.562	0.125	0.125	3/8
ALCOA 33481	7075-T6	2.250	1.750	0.062	0.062	2.250	1.750	0.063	0.063	5/32
ALCOA 39689	7075-T6511	1.750	2.500	0.091	0.091	1.750	2.500	0.090	0.090	9/32
ALCOA 43025*	7075-T6511	2.600	3.850	0.380	0.190	2.600	3.850	0.380	0.190	0.120**
ALCOA 43874	7075-T6511	2.500	3.500	0.250	0.125	NO SUBSTITUTION				
ALCOA 44103*	7075-T6511	2.625	4.562	0.500	0.125	2.625	4.562	0.500	0.125	0.250**
ALCOA 44282	7075-T6	2.140	3.700	0.100	0.100	2.140	3.700	0.100	0.100	5/16
ALCOA 44532	7075-T6511	4.000	2.000	0.250	0.250	MACHINE FROM STOCK				
ALCOA 45180	7075-T6	3.000	1.750	0.094	0.094	3.000	1.750	0.090	0.090	9/32
ALCOA 45277	2024-0	2.680	1.780	0.064	0.064	2.680	1.780	0.063	0.063	3/16
ALCOA 51777	7075-T6	1.562	1.000	0.062	0.062	1.562	1.000	0.063	0.063	3/16
ALCOA 52414	7075-T6	2.000	1.200	0.062	0.062	2.000	1.200	0.063	0.063	3/16
ALCOA 52986	7075-T6	3.700	2.850	0.125	0.125	3.700	2.850	0.125	0.125	3/8
ALCOA 55780	7075-T6	1.760	2.000	0.062	0.062	1.760	2.000	0.063	0.063	3/16
ALCOA 55919	7075-T6511	3.240	1.460	0.078	0.078	3.240	1.460	0.080	0.080	1/4
ALCOA 55956	7075-T6511	2.800	2.200	0.078	0.078	2.800	2.200	0.080	0.080	1/4
ALCOA 59051	7075-T6511	1.500	1.125	0.050	0.050	1.500	1.125	0.050	0.050	5/32
ALCOA 59223	7075-T6	1.800	3.250	0.080	0.080	1.800	3.250	0.080	0.080	1/4

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FIGURE 2-6-4-4. TEE SECTION (SHEET 2 OF 3)

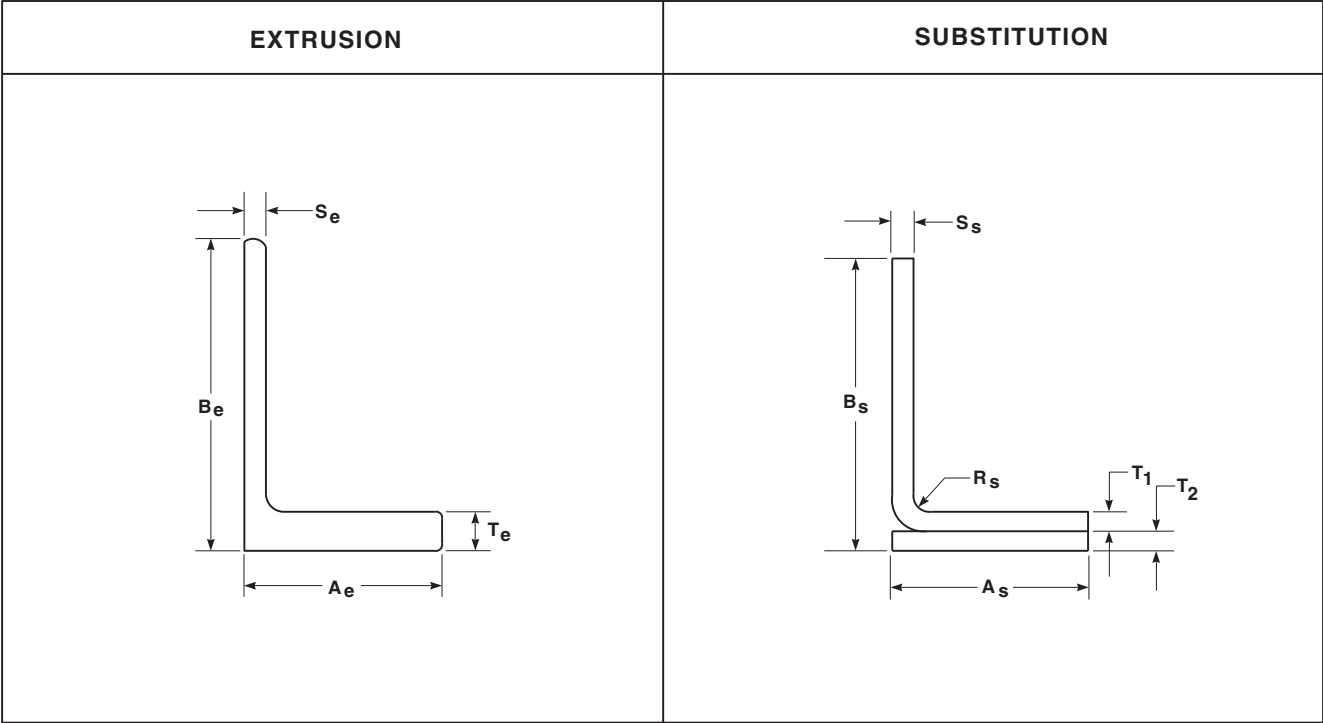
EXTRUSION						SUBSTITUTION				
DIE NUMBER	MATERIAL	A _e	B _e	T ₁	T ₂	A _s	B _s	T ₁	T ₂	R _s
ALCOA 59862	7075-T6	1.500	1.500	0.062	0.062	1.500	1.500	0.063	0.063	3/16
ALCOA 60038	7075-T6	1.750	1.750	0.125	0.125	1.750	1.750	0.125	0.125	3/8
ALCOA 61633	7075-T6	1.260	2.250	0.060	0.060	1.260	2.250	0.063	0.063	3/16
ALCOA 66336	7075-T6511	2.120	3.000	0.125	0.125	2.120	3.000	0.125	0.125	3/8
ALCOA 69162	7075-T6511	2.600	2.000	0.100	0.100	2.600	2.000	0.100	0.100	5/16
ALCOA 71132	7075-T6	2.600	2.030	0.080	0.080	2.600	2.030	0.080	0.080	1/4
ALCOA 72221	7075-T6511	1.780	2.875	0.078	0.078	1.780	2.875	0.080	0.080	1/4
ALCOA 76759	2024-T3511	1.500	0.800	0.040	0.040	1.500	0.800	0.040	0.040	1/8
ALCOA 80760	7075-T6511	2.250	1.000	0.050	0.050	2.250	1.000	0.050	0.050	5/32
ALCOA 85740	7075-T6511	3.000	3.250	0.090	0.090	3.000	3.250	0.090	0.090	9/32
ALCOA 156895	7075-T6	2.000	0.750	0.040	0.040	2.000	0.750	0.040	0.040	1/8
HARVEY 14712	7075-T6511	1.750	1.750	0.050	0.050	1.750	1.750	0.050	0.050	5/32
HARVEY 14678	7075-T6511	2.000	3.000	0.125	0.125	2.000	3.000	0.125	0.125	3/8
HARVEY 18648	7075-T6511	2.000	2.600	0.188	0.100	MACHINE FROM STOCK				
HARVEY 21423	7075-T6511	1.780	3.125	0.094	0.094	1.780	3.125	0.090	0.090	9/32
HARVEY 21701	7075-T6	2.000	1.100	0.050	0.050	2.000	1.100	0.050	0.050	5/32
HARVEY 22398	7075-T6511	4.000	1.500	0.062	0.062	4.000	1.500	0.063	0.063	3/16
PIONEER 6568	7075-T6511	1.500	1.000	0.062	0.062	1.500	1.000	0.063	0.063	3/16
REYNOLDS 11416	7075-T6511	2.000	1.000	0.062	0.062	2.000	1.000	0.063	0.063	3/16
REYNOLDS 16917	7075-0	9.943	1.250	0.063	0.063	9.943	1.250	0.063	0.063	3/16
REYNOLDS 11731	7075-T6511	2.000	3.000	0.125	0.100	2.000	3.000	0.125	0.100	3/8
REYNOLDS 19274	7075-T6	1.500	2.500	0.062	0.062	1.500	2.500	0.063	0.063	3/16
REYNOLDS 19446	7075-T6	1.700	2.250	0.062	0.062	1.700	2.250	0.063	0.063	3/16
REYNOLDS 23203	7075-T6	2.000	1.000	0.062	0.062	2.000	1.000	0.063	0.063	3/16
REYNOLDS 23243	7075-T6511	1.500	2.500	0.125	0.102	1.500	2.500	0.125	0.125	9/32
REYNOLDS 39689	7075-T6511	1.750	2.500	0.091	0.091	1.750	2.500	0.090	0.090	9/32
TIERNAY 60-2617	7075-T6511	1.500	1.000	0.050	0.050	1.500	1.000	0.050	0.050	5/32
TIERNAY 60-4394	7075-T6511	2.450	1.450	0.050	0.050	2.450	1.450	0.050	0.050	5/32
TIERNAY 60-5243	7075-T6511	1.500	2.125	0.125	0.125	1.500	2.125	0.125	0.125	3/8
TIERNAY 60-6081	7075-T6511	1.500	3.375	0.125	0.062	1.500	3.375	0.125	0.125	3/8

* MACHINE SUBSTITUTE FROM BAR STOCK OF SAME MATERIAL.

** MACHINED SUBSTITUTE RADIUS.

AK5004.3
SA

FIGURE 2-6-4-4. TEE SECTION (SHEET 3 OF 3)

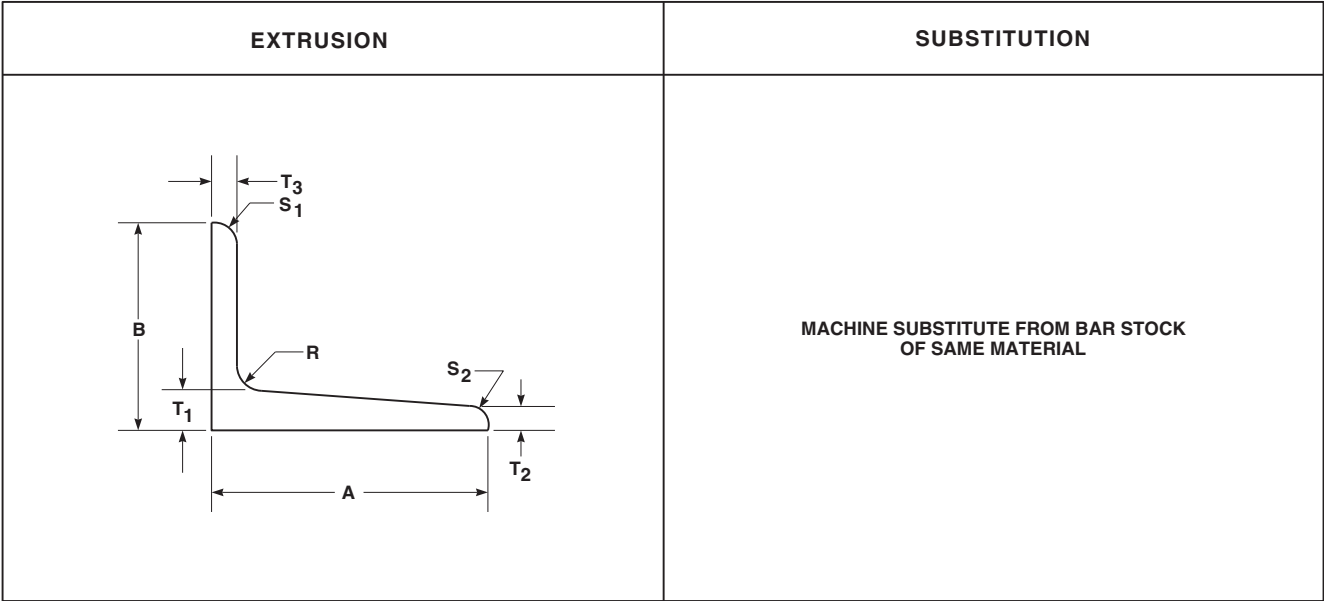


EXTRUSION						SUBSTITUTION				
DIE NUMBER	MATERIAL	A _e	B _e	S _e	T _e	A _s	B _s	T ₁	T ₂	R _s
ALCOA 14800	7075-T6511	1.625	3.250	0.094	0.188	1.000	MACHINE FROM STOCK			3/8
ALCOA 27159	7075-T6511	1.000	2.500	0.094	0.156		2.500	0.125	0.063	
ALCOA 30737	7075-76511	2.312	4.094	0.125	0.188		MACHINE FROM STOCK			
ALCOA 44350	7075-T6	1.625	3.250	0.094	0.188		MACHINE FROM STOCK			
ALCOA 46202	7075-T6	2.000	1.500	0.375	0.312	A MACHINE FROM STOCK				
ALCOA 52301	7075-T6511	1.000	2.500	0.094	0.125	1.000	2.500	0.125	0.050	3/8
ALCOA 76145	7075-T6	1.250	1.400	0.080	0.150	1.250	1.400	0.100	0.100	5/16
REYNOLDS 7825	7075-T6511	1.250	1.125	0.188	0.125	A MACHINE FROM STOCK				
REYNOLDS 8334*	7075-T6511	1.500	1.750	0.125	0.188	1.500	1.750	0.188	0.125	0.125**
REYNOLDS 8334*	7075-T2024-T4	1.500	1.750	0.125	0.188	1.500	1.750	0.188	0.125	0.125**
REYNOLDS 17676*	7075-T6	2.000	1.500	0.375	0.312	MACHINE FROM STOCK				

* MACHINE SUBSTITUTE FROM BAR STOCK OF SAME MATERIAL.

** MACHINED SUBSTITUTE RADIUS.

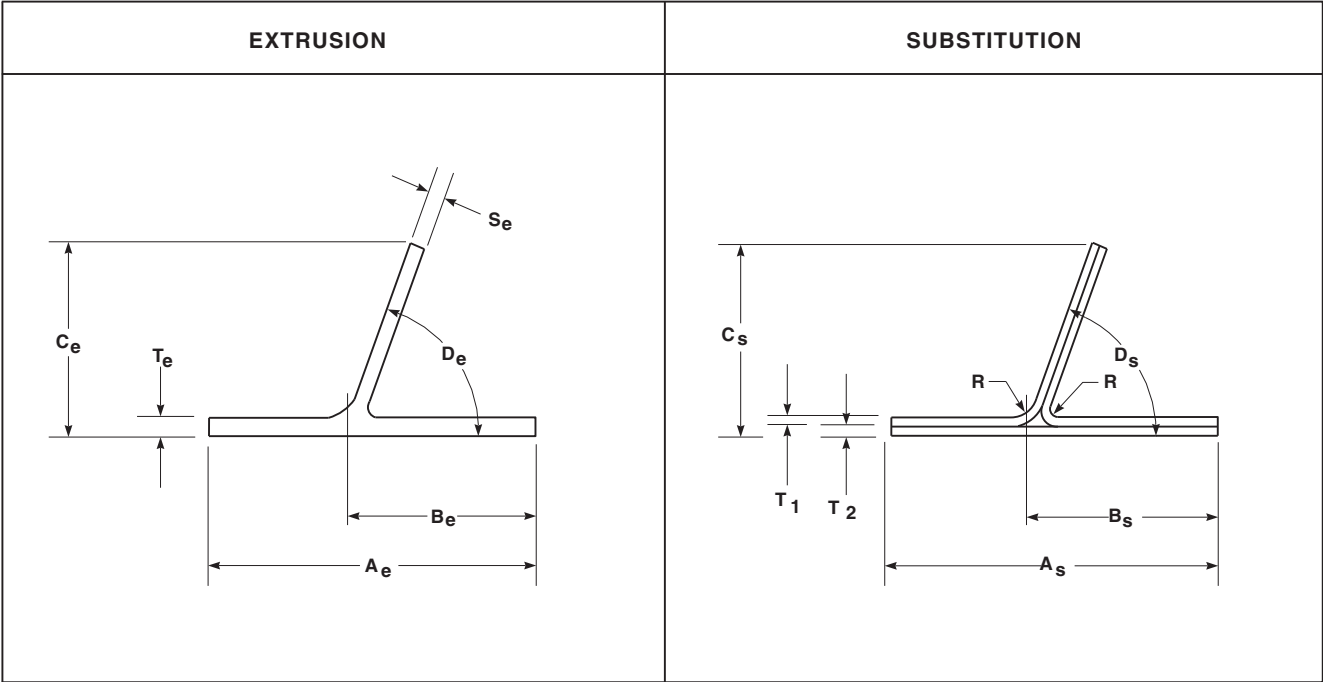
FIGURE 2-6-4-5. UNEQUAL THICKNESS ANGLE



EXTRUSION						SUBSTITUTION			
DIE NUMBER	MATERIAL	A	B	T ₁	T ₂	T ₃	R	S ₁	S ₂
ALCOA 14030	7075-T6	2.000	1.500	0.078	0.047	0.100	0.125	0.100	0.047
ALCOA 14030	2024-T4	2.000	1.500	0.078	0.047	0.100	0.125	0.100	0.047
ALCOA 22384	7075-T6	3.125	1.125	0.188	0.078	0.188	0.188		
ALCOA 24698	7075-T6	3.125	1.062	0.288	0.094	0.312	0.250	0.031	0.031
ALCOA 31928	7075-T6	6.500	1.625	0.375	0.094	0.500	0.500	0.062	0.062
ALCOA 60169	7075-T6511	2.000	1.125	0.125	0.062	0.125	0.125	0.031	0.031
ALCOA 76114	7075-T6	2.800	1.250	0.180	0.080	0.300	0.180	0.031	0.031

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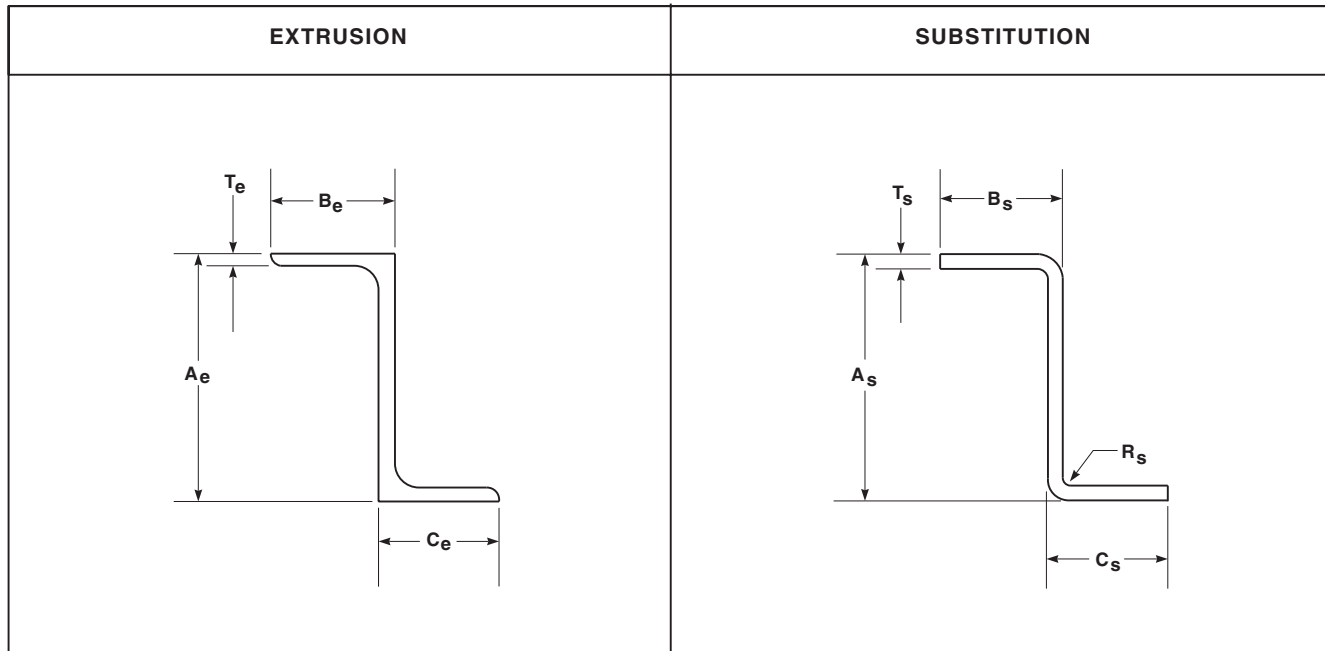
FIGURE 2-6-4-6. TAPERED LEG ANGLE



EXTRUSION								SUBSTITUTION						
DIE NUMBER	MATERIAL	A _e	B _e	C _e	D _e	T _e	S _e	A _s	B _s	C _s	D _s	T ₁	T ₂	R _s
ALCOA 44271	7075-T6511	2.000	1.000	1.121	85°	0.125	0.125	2.000	1.000	1.121	85°			
ALCOA 76941	7075-T6511	2.290	1.274	2.498	88°33'	0.064	0.064	2.290	1.274	2.498	88°33'	0.071	0.063	7/32
ALCOA 93560	2024-T4	2.250	1.125	1.997	94°30'	0.062	0.062	2.250	1.125	1.997	94°30'	0.071	0.071	7/32
PIONEER 11032	7075-T6511	1.500	0.750	1.994	95°	0.094	0.094	1.500	0.750	1.994	95°	0.125	0.125	3/8
SIKORSKY 38010-70026	7075-T6511	2.000	1.025	0.920	48°42'	0.050	0.050	NO SUBSTITUTION						

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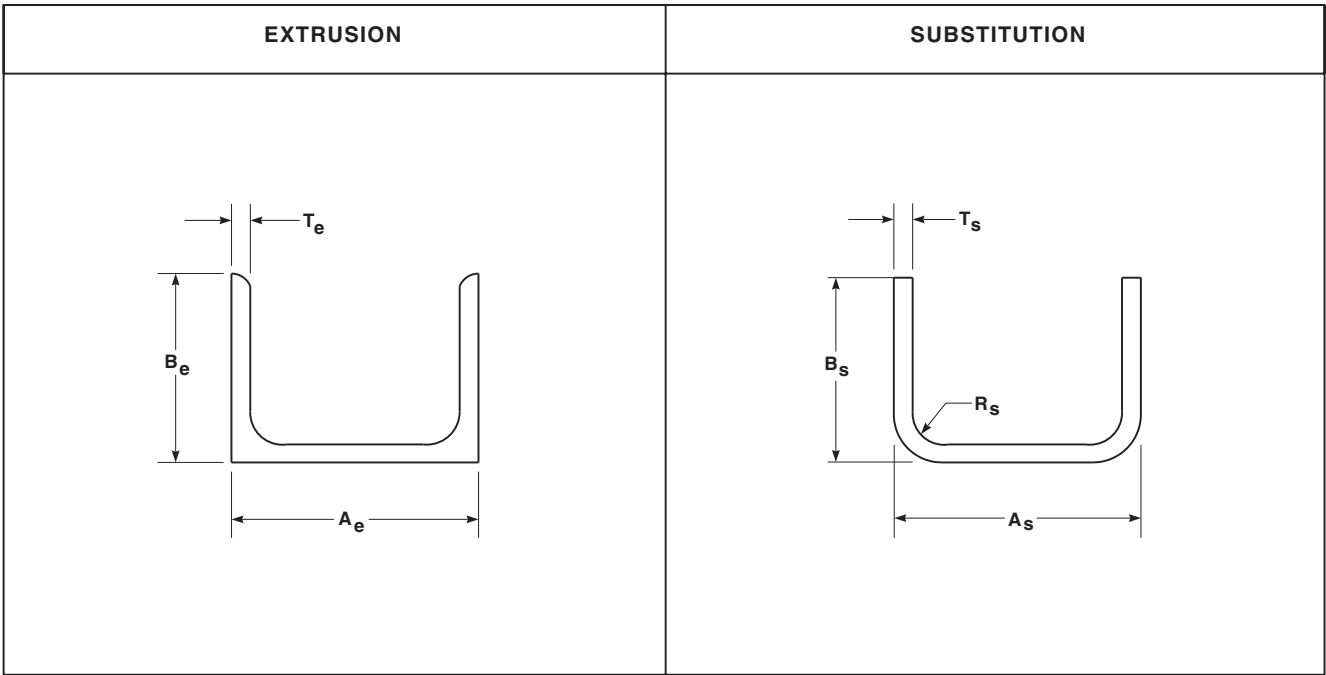
FIGURE 2-6-4-7. CLOSED ANGLE TEE



EXTRUSION						SUBSTITUTION				
DIE NUMBER	MATERIAL	A _e	B _e	C _e	T _e	A _s	B _s	C _s	T _s	R _s
AND 10138-1004	7075-T6511	1.000	0.750	0.750	0.063	1.000	0.750	0.750	0.071	7/32
AND 10138-1402	7075-T6511	1.500	1.000	1.000	0.078	1.500	1.000	1.000	0.090	9/32
AND 10139-1205	7075-T6511	1.250	0.750	1.250	0.078	1.250	0.750	1.250	0.090	9/32
ALCOA 3172	7075-T6511	0.563	0.875	1.500	0.125	MACHINE FROM STOCK				
REYNOLDS 18226	7075-T6511	0.354	0.880	0.500	0.064	0.354	0.880	0.500	0.071	7/32
REYNOLDS 38973	2024-T3511	0.922	0.625	0.625	0.040	0.922	0.625	0.625	0.050	5/32

AA0059
SA

FIGURE 2-6-4-8. ZEE SECTION



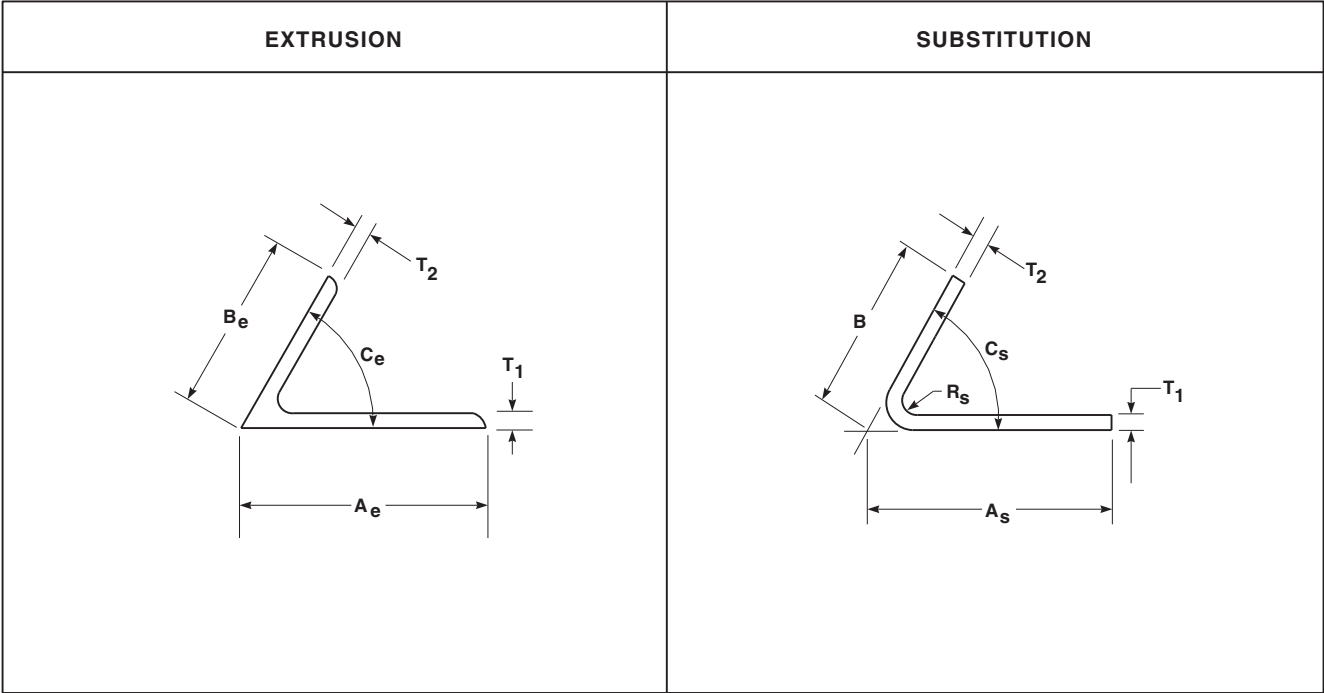
EXTRUSION					SUBSTITUTION			
DIE NUMBER	MATERIAL	A _e	B _e	T _e	A _s	B _s	T _s	R _s
AND10137-1012*	7075-T6	1.000	1.250	0.125	1.000	1.250	0.125	0.125**
SIKORSKY 38010-70025	7075-T6511	0.587	0.937	0.050	0.587	0.937	0.063	3/16

* MACHINE SUBSTITUTE FROM BAR STOCK OF SAME MATERIAL.

** MACHINED SUBSTITUTE RADIUS.

AK5005
SA

FIGURE 2-6-4-9. CHANNEL



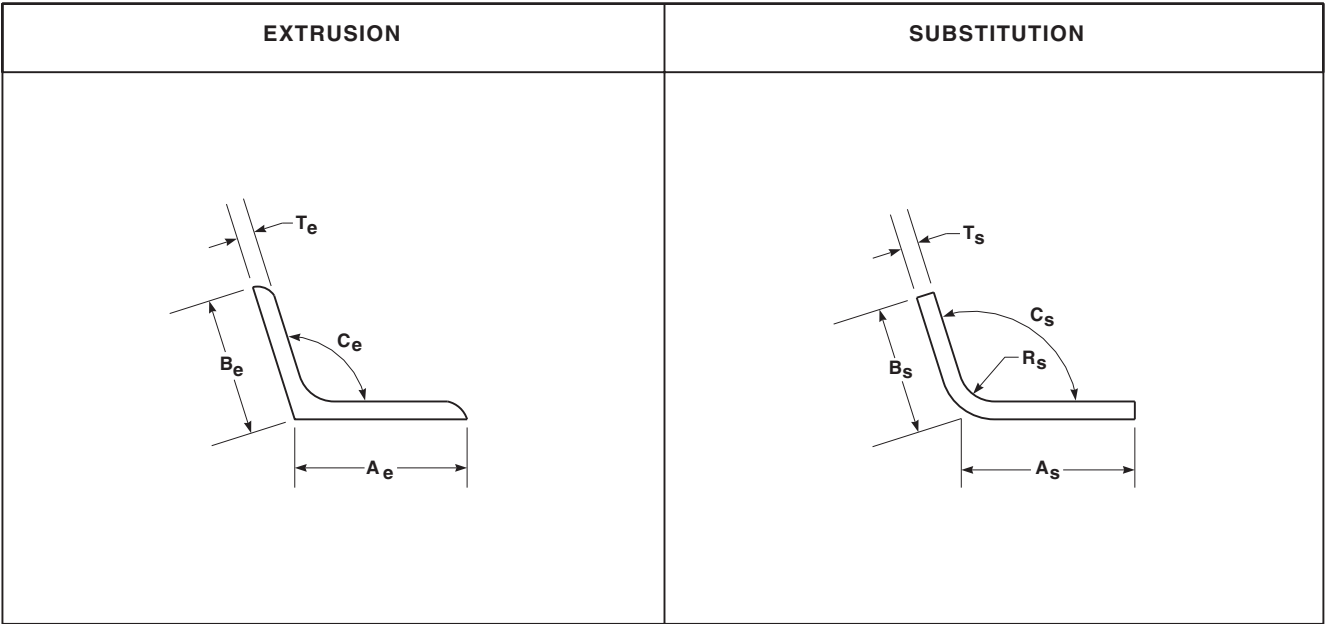
EXTRUSION							SUBSTITUTION					
DIE NUMBER	MATERIAL	A _e	B _e	C _e	T ₁	T ₂	A _s	B _s	C _s	T ₁	T ₂	R _s
38010-70029-001*	7075-T6511	1.300	1.300	80°30'	0.125	1.125	1.30	1.300	88°30'	0.125	120.125	0.156**
HARVEY 16161*	7075-T6511	1.880	1.000	88°30'	0.090	0.188	1.880	1.000	88°30'	0.090	0.188	0.156**
REYNOLDS 19234	7075-T6511	1.000	1.000	84°	0.076	0.076	1.000	1.000	84°	0.090	0.090	9/32

* MACHINE SUBSTITUTE FROM BAR STOCK OF SAME MATERIAL.

** MACHINE SUBSTITUTE RADIUS.

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SA

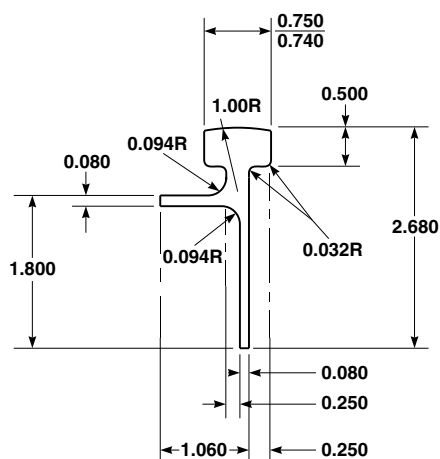
FIGURE 2-6-4-10. CLOSED LEG ANGLE



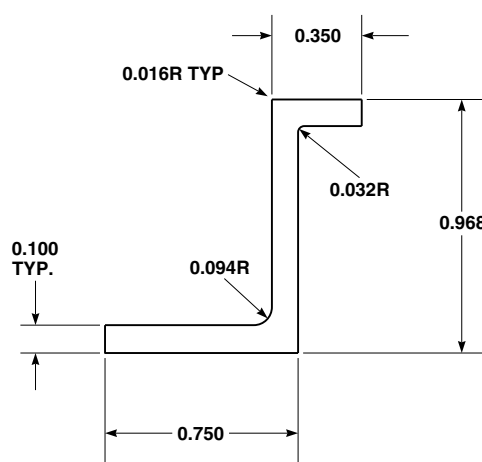
EXTRUSION						SUBSTITUTION				
DIE NUMBER	MATERIAL	A _e	B _e	C _e	T _e	A _s	B _s	C _s	T _s	R
REYNOLDS 11465	7075-T6	0.950	0.840	95°30'	0.050	0.950	0.840	95°30'	0.063	3/16
REYNOLDS 19272	7075-T6511	1.750	1.250	115°	0.125	2.000	MACHINE FROM STOCK			3/8
SIKORSKY 38010-70014	7075-T6511	2.500	1.750	109°	0.125		A MACHINE FROM STOCK			
SIKORSKY 38010-70020	7075-T6511	2.000	1.250	100° 45'	0.094		1.250	100° 45'	0.125	
SIKORSKY 38010-70021	7075-T6511	1.094	1.688	132°	0.153		A MACHINE FROM STOCK			

AA0062
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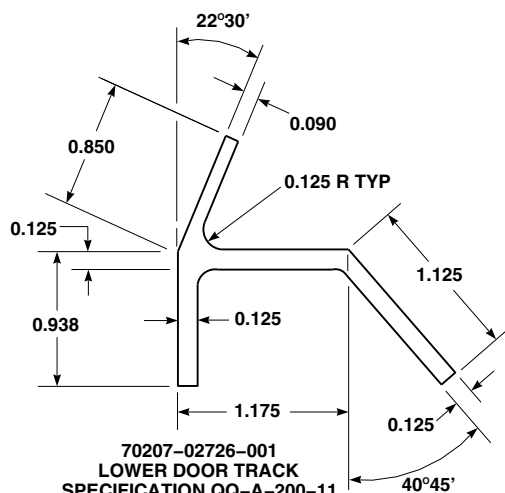
FIGURE 2-6-4-11. OPEN LEG ANGLE



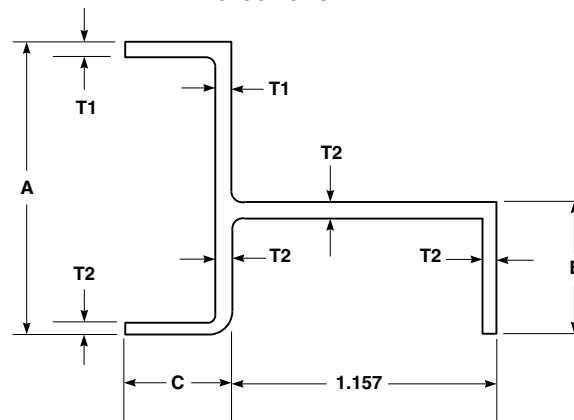
70202-01010-001
SEAT TRACK EXTRUSION
SPECIFICATION QQ-A-200-11
MATERIAL 7075-T6511
MACHINE SUBSTITUTE FROM
BAR STOCK OF SAME MATERIAL



70209-04204-001
FWD TRACK - MAIN ROTOR PYLON
SPECIFICATION QQ-A-200-3
MATERIAL 2024-T3511
MACHINE SUBSTITUTE FROM
BAR STOCK OF SAME MATERIAL



70207-02726-001
LOWER DOOR TRACK
SPECIFICATION QQ-A-200-11
MATERIAL 7075-T6511
MACHINE SUBSTITUTE FROM
BAR STOCK OF SAME MATERIAL



70207-02725-001
70207-02725-002
UPPER DOOR TRACK
SPECIFICATION QQ-A-200-11
MATERIAL 7075-T6511
MACHINE SUBSTITUTE FROM
BAR STOCK OF SAME MATERIAL

AA0063_1
SA

FIGURE 2-6-4-12. MISCELLANEOUS SHAPES (SHEET 1 OF 5)

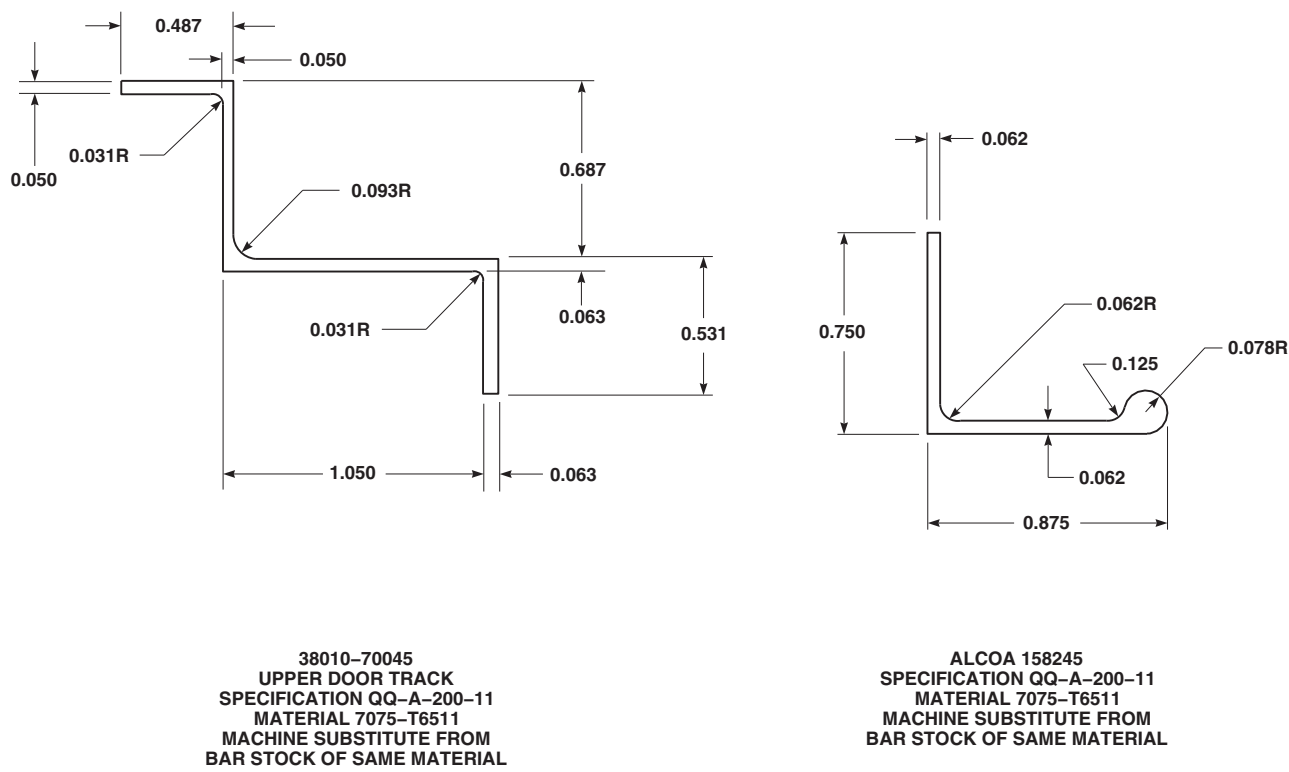
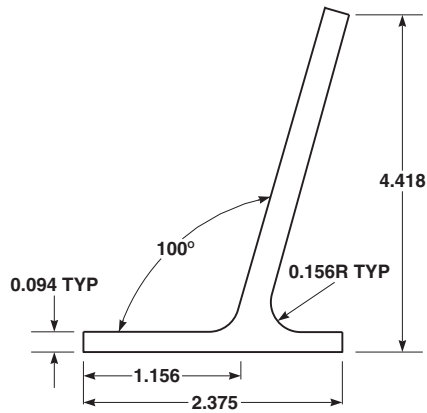
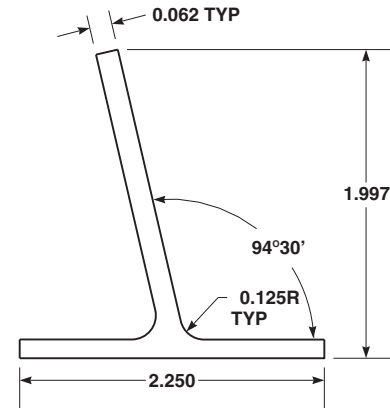


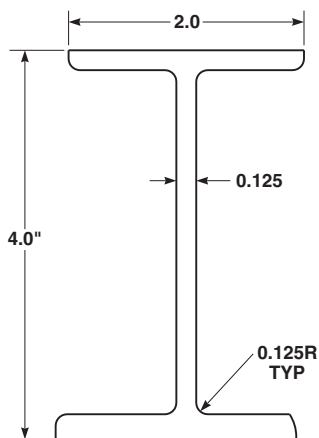
FIGURE 2-6-4-12. MISCELLANEOUS SHAPES (SHEET 2 OF 5)



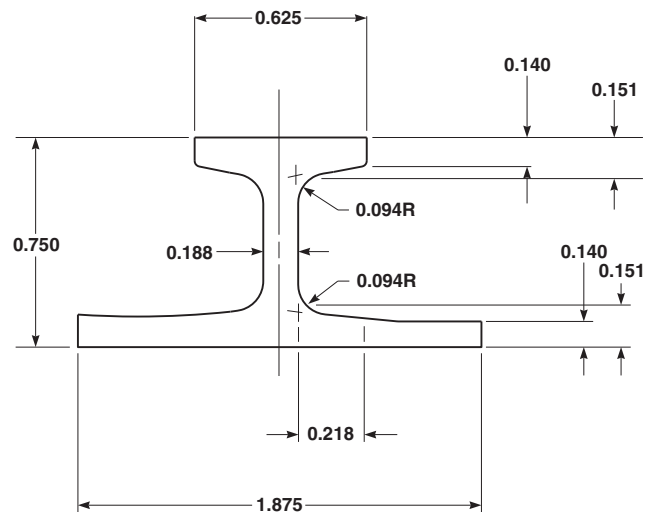
ALCOA 41129
SPECIFICATION QQ-A-200-11
MATERIAL 7075-T6
MACHINE SUBSTITUTE FROM
BAR STOCK OF SAME MATERIAL



ALCOA 93560
SPECIFICATION QQ-A-200-3
MATERIAL 2024-T4
MACHINE SUBSTITUTE FROM
BAR STOCK OF SAME MATERIAL



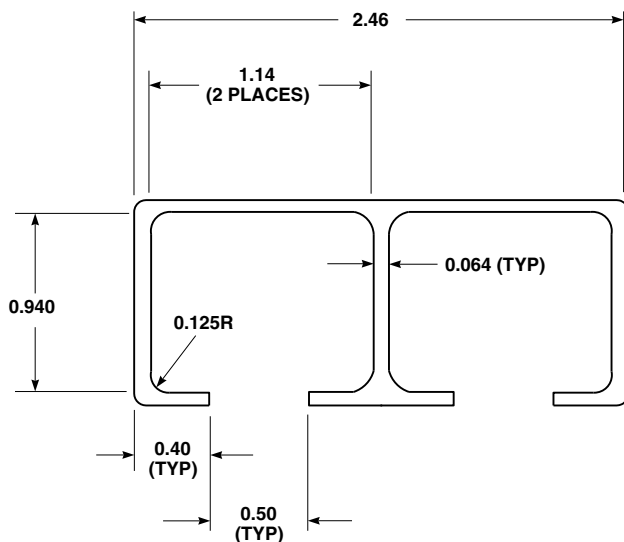
AND10140-4003
SPECIFICATION QQ-A-200-11
MATERIAL 7075-T6
MACHINE SUBSTITUTE FROM
BAR STOCK OF SAME MATERIAL



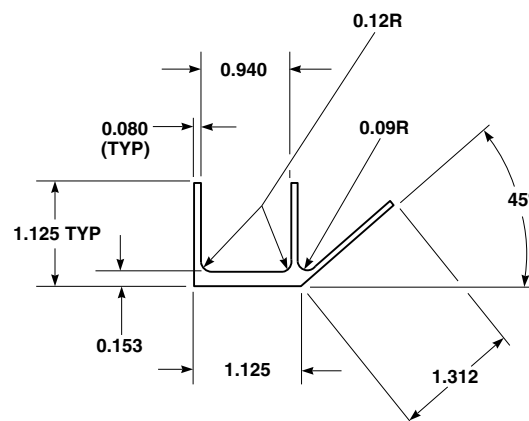
REYNOLDS 34557
SPECIFICATION QQ-A-200-11
MATERIAL 7075-T6
MACHINE SUBSTITUTE FROM
BAR STOCK OF SAME MATERIAL

AA0063_3
SA

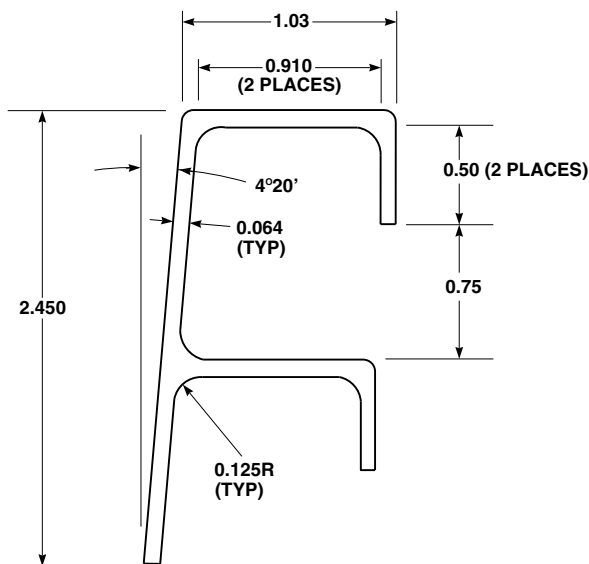
FIGURE 2-6-4-12. MISCELLANEOUS SHAPES (SHEET 3 OF 5)



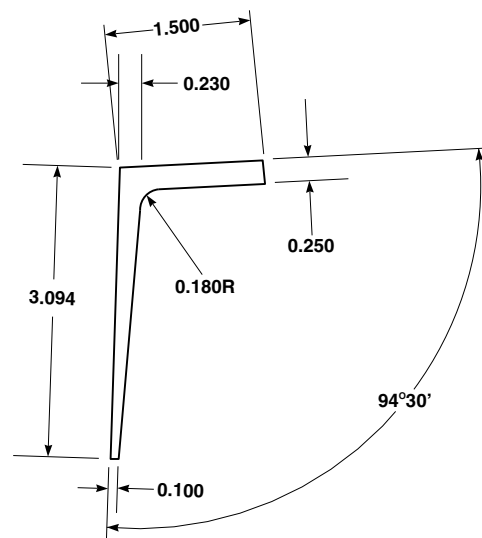
30810-70016
SPECIFICATION QQ-A-200-11
MATERIAL 7075-T6511
MACHINE SUBSTITUTE FROM
BAR STOCK OF SAME MATERIAL



30810-70022
SPECIFICATION QQ-A-200-11
MATERIAL 7075-T6511
MACHINE SUBSTITUTE FROM
BAR STOCK OF SAME MATERIAL



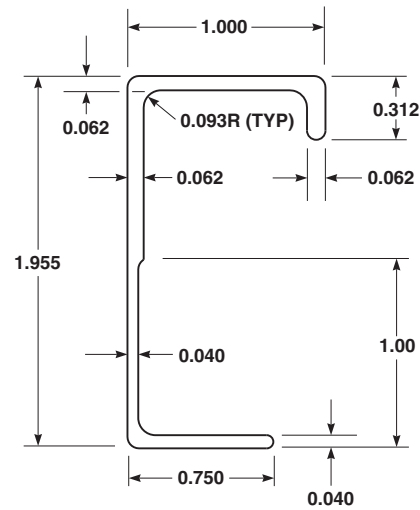
38010-70017
SPECIFICATION QQ-A-200-11
MATERIAL 7075-T6511
MACHINE SUBSTITUTE FROM
BAR STOCK OF SAME MATERIAL



38010-70019
SPECIFICATION QQ-A-200-11
MATERIAL 7075-T6511
MACHINE SUBSTITUTE FROM
BAR STOCK OF SAME MATERIAL

AA0063_4
SA

FIGURE 2-6-4-12. MISCELLANEOUS SHAPES (SHEET 4 OF 5)



38010-70001
 SPECIFICATION QQ-A-200-11
 MATERIAL 7075-T6511
 MACHINE SUBSTITUTE FROM
 BAR STOCK OF SAME MATERIAL

EXTRUSION					SUBSTITUTION	
DIE NUMBER	MATERIAL	A	B	C	T ₁	T ₂
70207-02725-001*	7075-T6511	1.218	0.531	0.437	0.050	0.063
70207-02725-002*	7075-T6511	1.294	0.607	0.513	0.16	0.139

* MACHINE SUBSTITUTE FROM BAR STOCK OF SAME MATERIAL.

AA0063_5
 SA

FIGURE 2-6-4-12. MISCELLANEOUS SHAPES (SHEET 5 OF 5)

2-6-5 HELICOPTER PAINTED SURFACES - TOUCH UP REPAIR.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
2-6-5.1	Touch Up Helicopter Painted Surfaces	2-6-5-2	2-6-5.3	Corrosion Resistant Coating, MIL-C-5541, Class 3	2-6-5-7
2-6-5.2	Touch Up Helicopter Conductive Paint	2-6-5-7	2-6-5.4	Torque Stripe Application	2-6-5-7

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Bristle Brush
- Nonmetallic Scraper, Locally-Made
- Protection, Eye

Materials/Parts

- Abrasive Cloth, Item 4, Appendix D
- Abrasive Cloth, Item 7, Appendix D
- Abrasive Pad
- Cheesecloth, Item 90, Appendix D
- Cleaning Cloth, Item 102A, Appendix D
- Cleaning Compound, Item 93, Appendix D
- Corrosion Preventive Compound, Item 113, Appendix D
- Corrosion Resistant Coating, Item 117, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D

- Epoxy Coating Kit, Item 126A, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Non-Skid Coating, Dark Gull Grey
- Sealing Compound, Item 300, Appendix D
- Thinner, Acrylic, Item 324B, Appendix D

References

- AMS-C-83231
- Appendix D
- MIL-C-5541
- MIL-C-85054
- MIL-DTL-53039
- MIL-L-81352
- MIL-PRF-22750
- MIL-PRF-85285
- PARA 2-6-1
- PARA 5-4-31
- PARA 5-4-38

2-6-5.1 Touch Up Helicopter Painted Surfaces.

NOTE

- This information does not apply to touchup repair of main and tail rotor blades. Finishes and materials for rotor blades are covered in PARA 5-4-31 and PARA 5-4-38.
 - Touch up protective finishes (metal treatment and paint) whenever finish has been damaged or removed by repair, corrosion, or wear.
- a. To touch up protective finishes (metal treatment and paint), prepare surfaces as follows:
- (1) Protect areas not to be refinished with suitable covers to prevent damage.

CAUTION

Do not allow paint, corrosion removers, solvents, or corrosion resistant coating to contact transparent plastic panels, bonded joints, reinforced plastic surfaces, or rubber seals.

- (2) Treat bare aluminum, cut or trimmed edges, and drilled holes with corrosion resistant coating, Item 118, Appendix D, before applying primer.
 - (3) Seal and prime sheet metal parts before assembly.
- b. To touch up composite plastic structures, prepare surfaces as follows:

CAUTION

Methyl ethyl ketone will damage bonded joints and bond lines of reinforced plastic components. Do not allow methyl ethyl ketone to run into bonded joints and bond lines or bond

may be weakened or separated. Use methyl ethyl ketone sparingly.

- (1) Clean surface using clean cheesecloth, Item 90, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D.
- (2) Dry surface with another clean cheesecloth, Item 90, Appendix D, by wiping in single strokes, then turning and refolding cheesecloth to fresh side for next stroke.
- (3) Sand surface with abrasive cloth, Item 4 and Item 7, Appendix D.
- (4) Make solution of cleaning compound, Item 93, Appendix D, mixing two parts water to one part cleaner. Use solution at room temperature.
- (5) Using bristle brush or abrasive pad, scrub surface thoroughly with solution.
- (6) Thoroughly rinse solution from surface with clear cold tap water.

WARNING

Injury to personnel will result if precautions are not taken while using compressed air. Compressed air will propel debris from hoses and tubes. Point hoses and tubes in safe direction before blowing compressed air. Do not point hose towards personnel when blowing compressed air through hose. Make sure all personnel are wearing safety glasses when compressed air is being used.

- (7) Dry with filtered compressed air or clean cheesecloth, Item 90, Appendix D.
- (8) Perform water-break test on cleaned surface.
- (9) Let surface dry for minimum of 1 hour.
- (10) Prime per Table 2-6-5-1, as follows:

- (a) Mix primer per manufacturer's instructions.
- (b) Make sure surface is dry.
- (c) Wipe surface to be primed with methyl ethyl ketone, Item 217, Appendix D.
- (d) Allow surface to dry a minimum of 15 minutes.
- (e) Apply epoxy primer using epoxy primer coating kit, Item 135, Appendix D, to surface.
- (f) Allow primer to dry a minimum of 1 hour.

Table 2-6-5-1. Paint Requirements

Area or Details	Helicopter Effectiv- ity	Paint Color, Type, and Specification	Number of Coats
Interior and Exterior - Overall and All Details	All Serial Nos	Epoxy Primer, Item 135, Appendix D	One
Interior Airframe of Tub (STA 162.0 to STA 485.0)	All Serial Nos	Corrosion Preventive Compound, MIL-C- 85054 or Clear Polyurethane, MIL- PRF-85285	One
Exterior Top Coat- Overall	Serial Nos 70583 and Subsequent	Aircraft Green Aliphatic Polyurethane, No. 34031, MIL-DTL- 53039	Two
Cabin Interior Top Coat	Serial Nos Prior To 70785	Dark Gull Gray, Acrylic Lacquer, No. 36231, A-A-3164	Two
	Serial Nos 70785 and Subsequent	Gray, Aliphatic Polyurethane, MIL- DTL-53039	Two
Cockpit Interior Top Coat	Serial Nos Prior To 70785	Instrument Black, Acrylic Lacquer, No. 5157A, MIL-DTL- 53039	Two
	Serial Nos 70785 and Subsequent	Black , Aliphatic Polyurethane, No. 37038, MIL-DTL- 53039	Two

Table 2-6-5-1. Paint Requirements (Cont)

Area or Details	Helicopter Effectivity	Paint Color, Type, and Specification	Number of Coats
Cabin Emergency Handles and Letter Markings	Serial Nos Prior To 70785	Orange Yellow Nitrocellous Lacquer, No. 13538, A-A-3164	One
	Serial Nos 70785 and Subsequent	Orange Yellow Aliphatic Polyurethane, No. 33538, MIL-DTL-53039	One
Oil Level Gage Handle	All Serial Nos	Yellow Epoxy Paint, No. 13655, MIL-PRF-22750	Two
Cockpit Emergency Handles and Letter Markings, Emergency Exit Markings	Serial Nos Prior To 70785	Orange Yellow Nitrocellous Lacquer, No. 33538, A-A-3164	One
	Serial Nos 70785 and Subsequent	Orange Yellow Aliphatic Polyurethane, No. 33538, MIL-DTL-53039	One
Cockpit Glare Shield Stencil	All Serial Nos	White Epoxy Paint, No. 37875, MIL-PRF-22750	One
Stencils and Step Markings	Serial Nos Prior To 70785	Lusterless Black, Acrylic Lacquer, No. 37038, MIL-DTL-53039	Two
	Serial Nos 70785 and Subsequent	Lusterless Black Aliphatic Polyurethane, No. 37038, MIL-DTL-53039	Two

Table 2-6-5-1. Paint Requirements (Cont)

Area or Details	Helicopter Effectiv- ity	Paint Color, Type, and Specification	Number of Coats
Stabilator Tips Danger Markings	Serial Nos Prior To 70785	Camouflage Red Acrylic Lacquer, No. 31136, MIL-L- 81352,	Two
	Serial Nos 70785 and Subsequent	Camouflage Red Aliphatic Polyurethane, No. 31136, MIL-DTL- 53039	Two
Engine Inlet Duct and Exhaust Area (NOTE: Black paint is not required on exhaust area with HIRSS module in- stalled)	Serial Nos Prior To 70785	Lusterless Black, Epoxy Polyamide, No. 37038, MIL- PRF-22750	Two
	Serial Nos 70785 and Subsequent	Lusterless Black Aliphatic Polyurethane, No. 37038, MIL-DTL- 53039	Two
Oil Cooler Support Installation	All Serial Nos	Lusterless Black Aliphatic Polyurethane, No. 37038, MIL-DTL- 53039	Two
Battery Area	All Serial Nos	Protective Neoprene Coating, No. N700- 9, AMS-C-83231	Two
Walkway Areas: Cabin Floor Strips, Steps, and Crew Chief's Window Sill	All Serial Nos	Dark Gull Gray Non-Skid Coating, No. 36231	Two
Exterior Skin Under Pylon Sliding Cover	Serial Nos Prior To 70785	Yellow Acrylic Lacquer, No. 33531, MIL-L-81352,	Two

Table 2-6-5-1. Paint Requirements (Cont)

Area or Details	Helicopter Effectivity	Paint Color, Type, and Specification	Number of Coats
	Serial Nos 70785 and Subsequent	Tan Aliphatic Polyurethane, No. 33446, MIL-DTL-53039	Two
Internal Mechanical Parts, Control System Linkages and Supports, Bell Cranks, Levers, Servos, Etc.	All Serial Nos	Yellow, MIL-DTL-53039 or Lusterless Gray, MIL-DTL-53039, No. 36231 (SEE NOTE)	Two
Transmission and Gear Housings	All Serial Nos	Silver Gloss Epoxy Polyamide Coating, MIL-PRF-22750, or Gull Gray Aliphatic Polyurethane, No. 36231, MIL-DTL-53039	Two
Main Transmission Oil Level Gage Handle	All Serial Nos	Yellow Epoxy Polyamide, No. 13655, MIL-PRF-22750	Two

NOTE

Late model helicopter parts and replacement parts are painted gray, MIL-DTL-53039. These parts and older parts painted yellow, MIL-DTL-53039, are interchangeable and need not be repainted in the field. Yellow painted parts may be touched up with gray, MIL-DTL-53039 if first primed using epoxy primer coating kit, Item 135, Appendix D.

CAUTION

Interruption of proper airflow can affect engine operation. Do not block any slots or holes with paint in the engine inlet duct or exhaust areas.

NOTE

- Black anodized surfaces in engine inlet duct areas are not to be painted.

- Paint must be applied within 8 hours of primer application. If paint is not applied within this time, component must be reprimed per step b. (10).
 - To prevent a sheen on part surface, reduce component B of paint by 10%.
- c. Paint per Table 2-6-5-1, as follows:
- (1) Mix paint per manufacturer's instructions.
 - (2) Apply paint to surface.

- d. After primer touch up, treat repairs on interior surfaces of structure below floor levels with corrosion preventive compound, Item 113, Appendix D (PARA 2-6-1).

2-6-5.2 Touch Up Helicopter Conductive Paint.

- a. Using abrasive cloth, Item 7, Appendix D, lightly sand repaired area.
- b. Wipe area using cheesecloth, Item 90, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D.
- c. Using epoxy primer coating kit, Item 135, Appendix D, apply one coat of epoxy primer. Allow primer to dry at least 1 hour.
- d. Apply three coats of sealing compound, Item 300, Appendix D, to make 0.003 - 0.004 inch thick coat.
- e. Install any grounding attachments.
- f. Touch up paint area (PARA 2-6-5.1).
- g. Remove grounding attachments.

2-6-5.3 Corrosion Resistant Coating, MIL-C-5541, Class 3.

2-6-5.3.1 Apply.

- a. Remove paint and corrosion resistant coating from area slightly larger than surface to be treated.
- b. Clean surface to be treated.
- c. Mix corrosion resistant coating, Item 117, Appendix D, per manufacturer's instructions.
- d. Apply corrosion resistant coating solution per manufacturer's instructions.

2-6-5.3.2 Inspect.

Inspect coated surface for breaks or scratches. **NONE ALLOWED.** If breaks or scratches are found, repeat steps in PARA 2-6-5.3.1.

2-6-5.4 Torque Stripe Application.

NOTE

- Application of torque stripe is required as specified by the maintenance manual, regardless of requirements for lockwire, cotter pins, and other safety devices.
- Cracking or chipping of torque stripe caused by vibration, use, or accidental contact is not reason for reapplication of torque value and torque stripe.
- Breakage of torque stripe is unacceptable. Broken torque stripe shall be removed and the item retorqued whether or not there are signs of fastener rotation.
- a. Remove old torque stripe using nonmetallic scraper and cheesecloth, Item 90, Appendix D, moistened with acrylic thinner, Item 324B, Appendix D.
- b. After final required torque value has been obtained, clean areas to be torque striped, using cheesecloth, Item 90, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D.

NOTE

- Torque stripe shall be applied on head end. If head end is inaccessible, then nut end shall be torque striped.
- If part is subsequently sealed or painted, torque stripe shall be reapplied per step f.
- For critical characteristic torques, two stripes intersecting in a V-shape shall be applied. The V-shape may have rounded vertex in order to avoid release pins, safety wire, cotter pins, or other safety devices (Figure 2-6-5-2). There shall be an angle of no less than 45° between straight portions of stripe.
- For non-critical characteristic torques, single torque stripe is sufficient, broken

into two halves if there are lockwire, release pins, cotter pins, or other safety devices within assembly (Figure 2-6-5-1).

- For critical and non-critical characteristic torques, torque stripe shall not cover and shall be kept clear of release pins on self-retaining bolts. Torque stripe, used to indicate torque has been applied, shall not cover and shall be kept clear of any safety wire, cotter pins, or other safety devices.
 - For pivoting bolt/joint, torque stripe shall extend only to end of washer.
- c. Using epoxy coating kit, Item 126A, Appendix D, apply torque stripe width about half of fastener diameter (not more than $\frac{3}{16}$ -inch), extending at least $\frac{1}{2}$ -inch on base part, or to edge of part, whichever is smaller (Figure 2-6-5-1).
 - d. Touch up paint as required (PARA 2-6-5.1).
 - e. Seal parts as required (PARA 2-6-1).
 - f. If torque stripe was covered due to subsequent touchup, sealing, or caulking, reapply torque stripe as follows:
 - (1) Using clean, lint-free cleaning cloth, Item 102A, Appendix D, clean areas to be torque striped.
 - (2) Repeat step c.
 - g. If torque stripe is broken, perform the following:

NOTE

- Breakage of torque stripe (due to fastener rotation) is unacceptable. Broken torque stripe shall be removed and the item retorqued whether or not there are signs of fastener rotation.
- Do not use torque wrench to loosen castellated nuts or fasteners.
 - (1) For castellated nut with cotter pin, perform the following:
 - (a) Remove and discard cotter pin.
 - (b) Back off nut enough so that torquing procedure for castellated nuts can be accomplished.
 - (c) For self-locking nuts, check prevailing and breakaway torque before re-torquing. Self-locking fastener will have to be backed off enough to break contact with mating parts.
 - (2) For all other fasteners, perform the following:
 - (a) Back off fastener and torque to specified value.
 - (b) For self-locking fasteners, check prevailing and breakaway torque before retorquing. Self-locking fastener will have to be backed off enough to break contact with mating parts.
 - (3) Repeat step b. through step f.

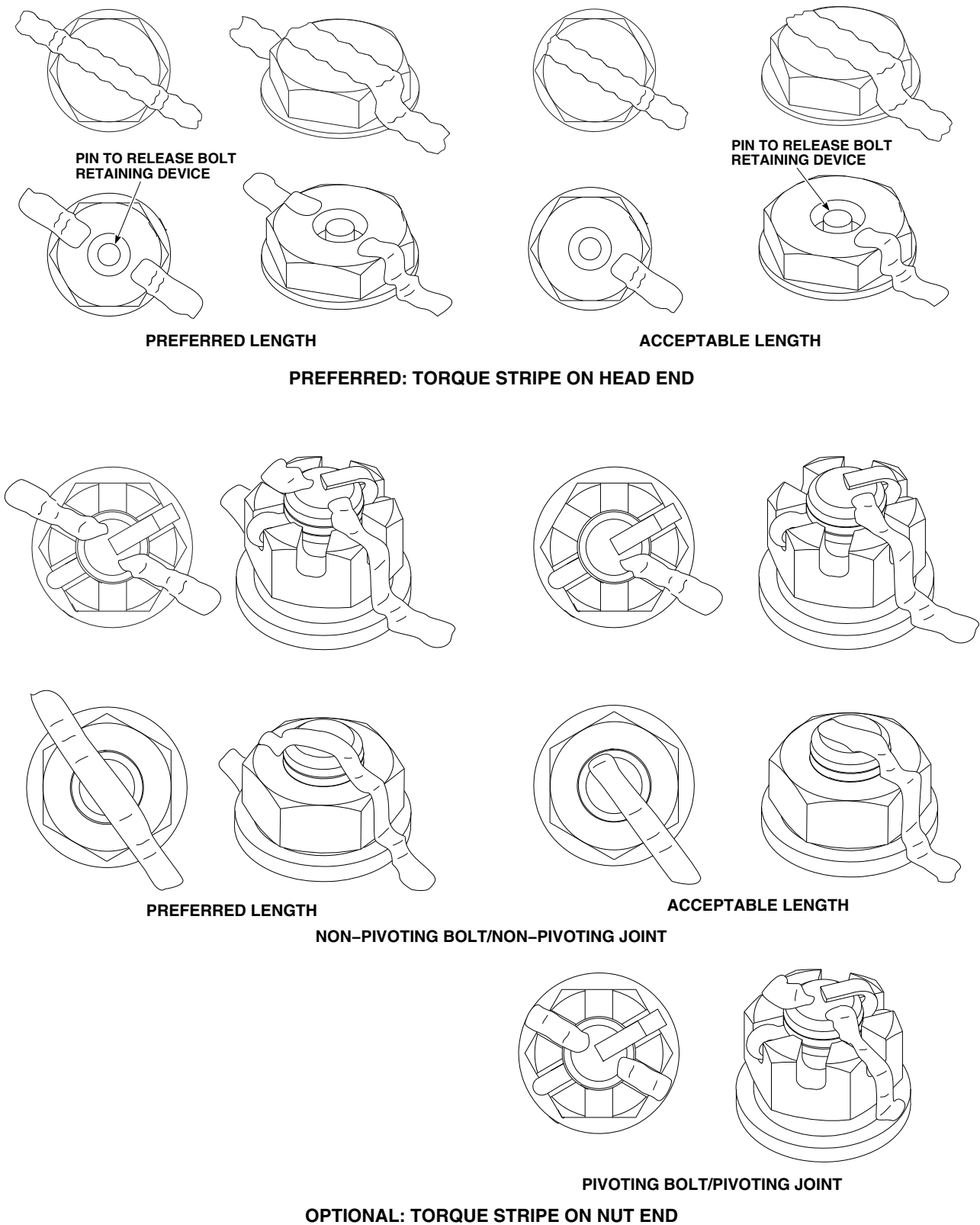


FIGURE 2-6-5-1. TORQUE STRIPE APPLICATION FOR NON-CRITICAL CHARACTERISTIC TORQUES

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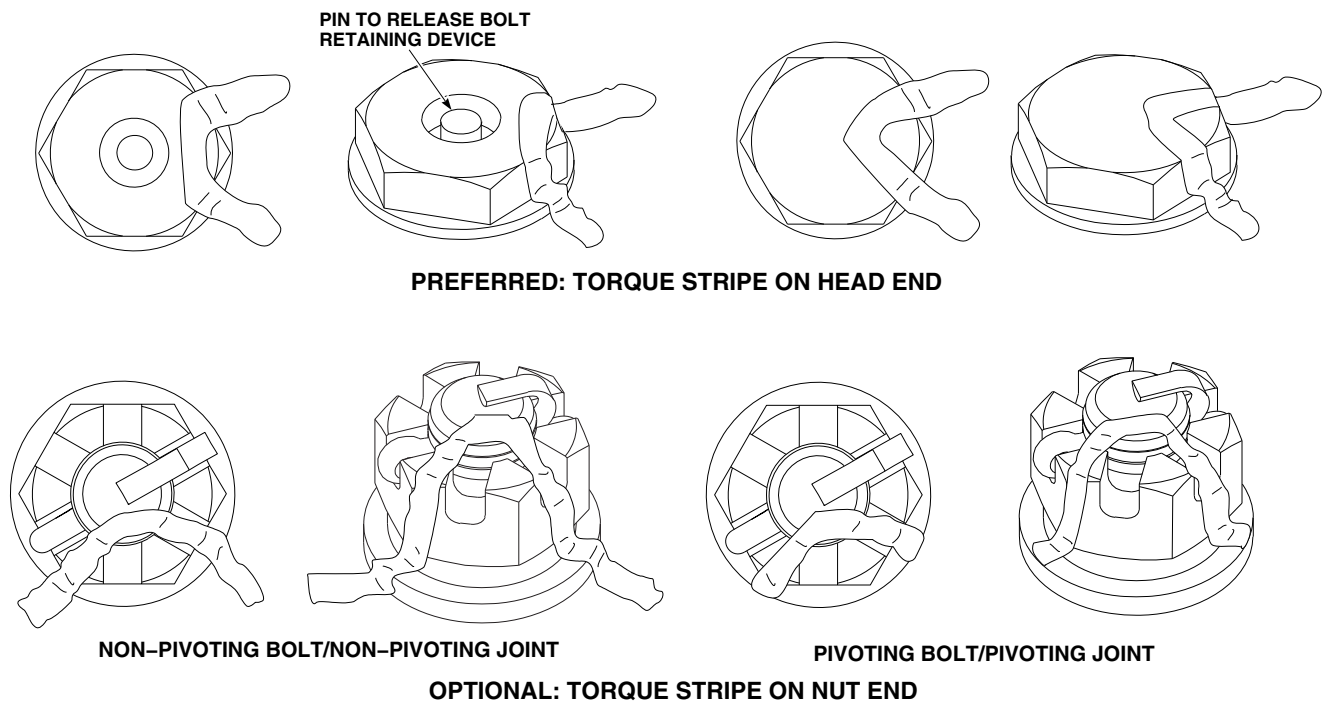
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FIGURE 2-6-5-2. TORQUE STRIPE APPLICATION FOR CRITICAL CHARACTERISTIC TORQUES

2-6-6 NOSE DOOR REPAIR.

PARAGRAPH BREAKDOWN

	PAGE
2-6-6.1 Repair Nose Door	2-6-6-1
INITIAL SETUP	MIL-I-24768
Tools	MIL-R-6855
Airframe Repairer’s Toolkit	PARA 2-6-1
Materials/Parts	PARA 2-6-2
Abrasive Cloth, Item 1A, Appendix D	PARA 2-6-3
Sealing Compound, Item 292, Appendix D	PARA 2-6-4
References	PARA 2-6-5
A-A-59588	QQ-A-200
Appendix D	QQ-A-225
MIL-I-631	QQ-A-250

2-6-6.1 Repair Nose Door.

- a.

Nose door is bonded sandwich-constructed assembly made from aluminum honeycomb core and Kevlar (reinforced plastic) skins. Outboard edges of nose door are graphite reinforced with Kevlar covering (Figure 2-6-6-1, Sheet 2, Detail B). Cockpit cooling ducts are part of bonded assembly. Metal components include hinges, angle brackets, and latches (Figure 2-6-6-1, Sheet 1, View A-A, and Sheet 2, Detail A).
- b.

Refer to PARA 2-6-2 and PARA 2-6-3 for evaluating damage and repairing nose door.
- c.

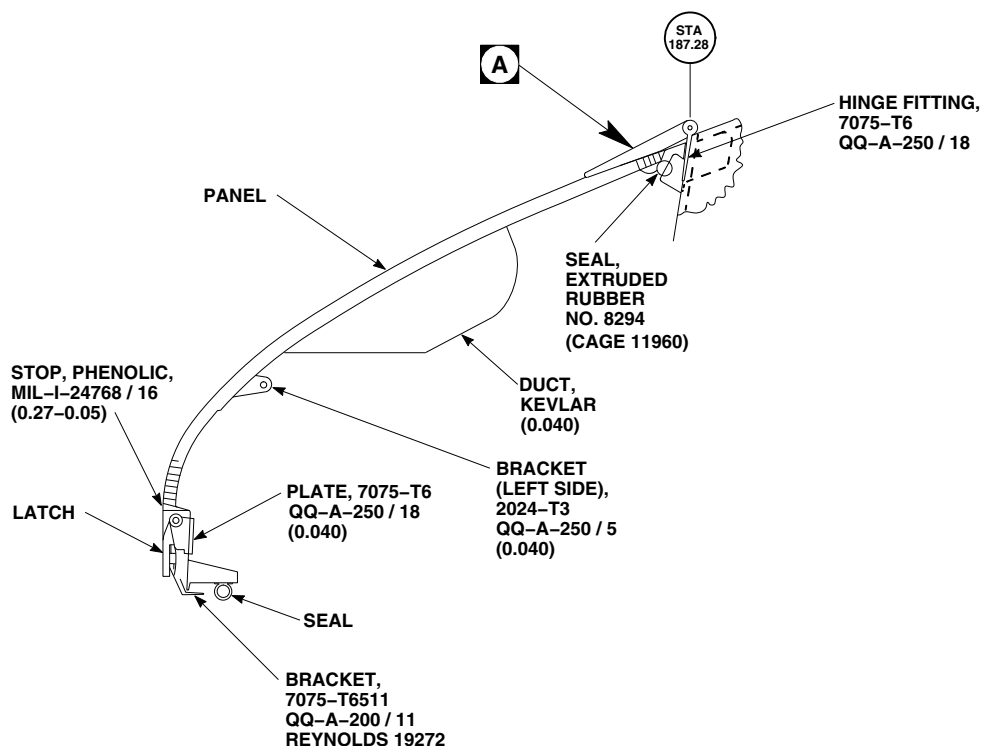
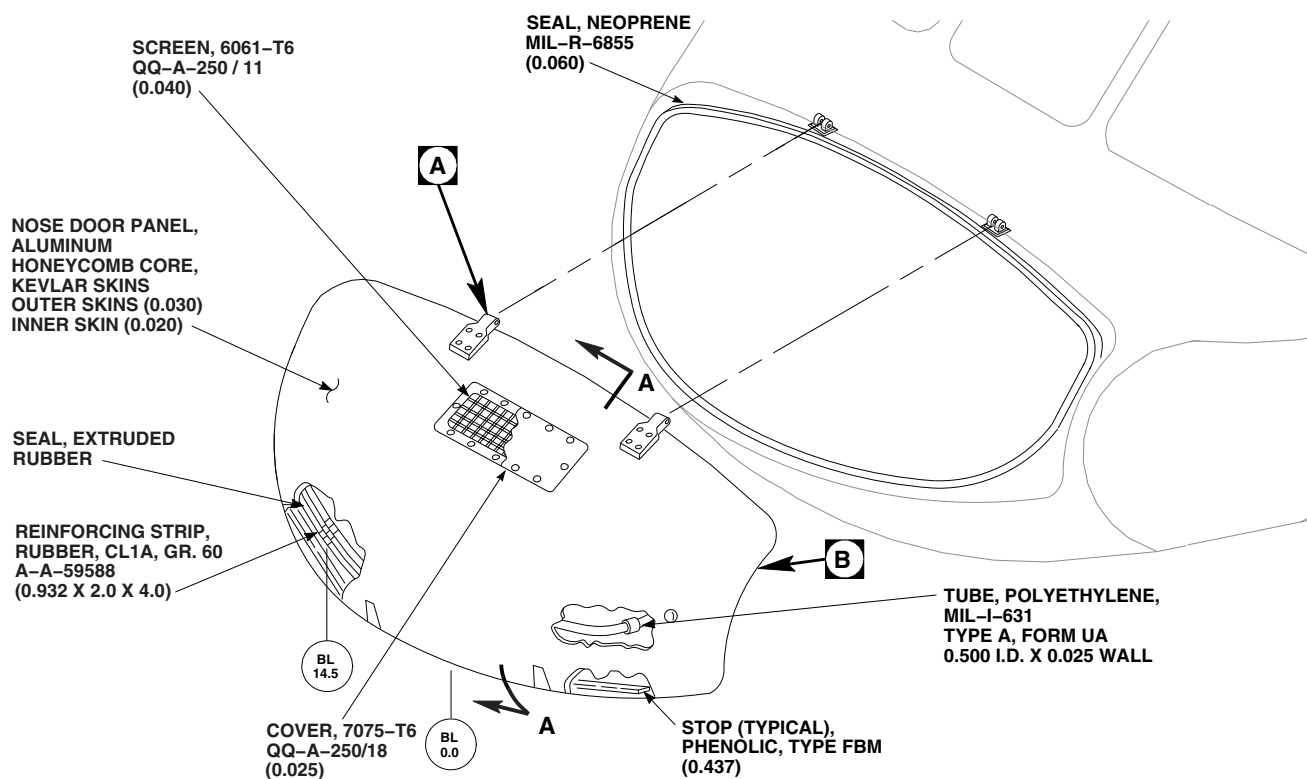
Repair of graphite is beyond the scope of this manual.
- d.

Evaluate damage to metal components (PARA 2-6-1). Negligible damage to hinge fittings is limited to nicks, scratches, and corrosion not over 5% of material thickness at point of damage within 1-inch of hinge line, and 10% of material thickness at point of damage beyond 1-inch of hinge line. Blend and polish damage with abrasive cloth, Item 1A, Appendix D, to same finish as surrounding area.
- e.

Replace aluminum alloy extrusions (PARA 2-6-4).
- f.

Seal repaired area for watertightness (PARA 2-6-1).
- g.

Touch up paint in repaired area (PARA 2-6-5).



VIEW A-A

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FIGURE 2-6-6-1. REPAIR NOSE DOOR (SHEET 1 OF 3)

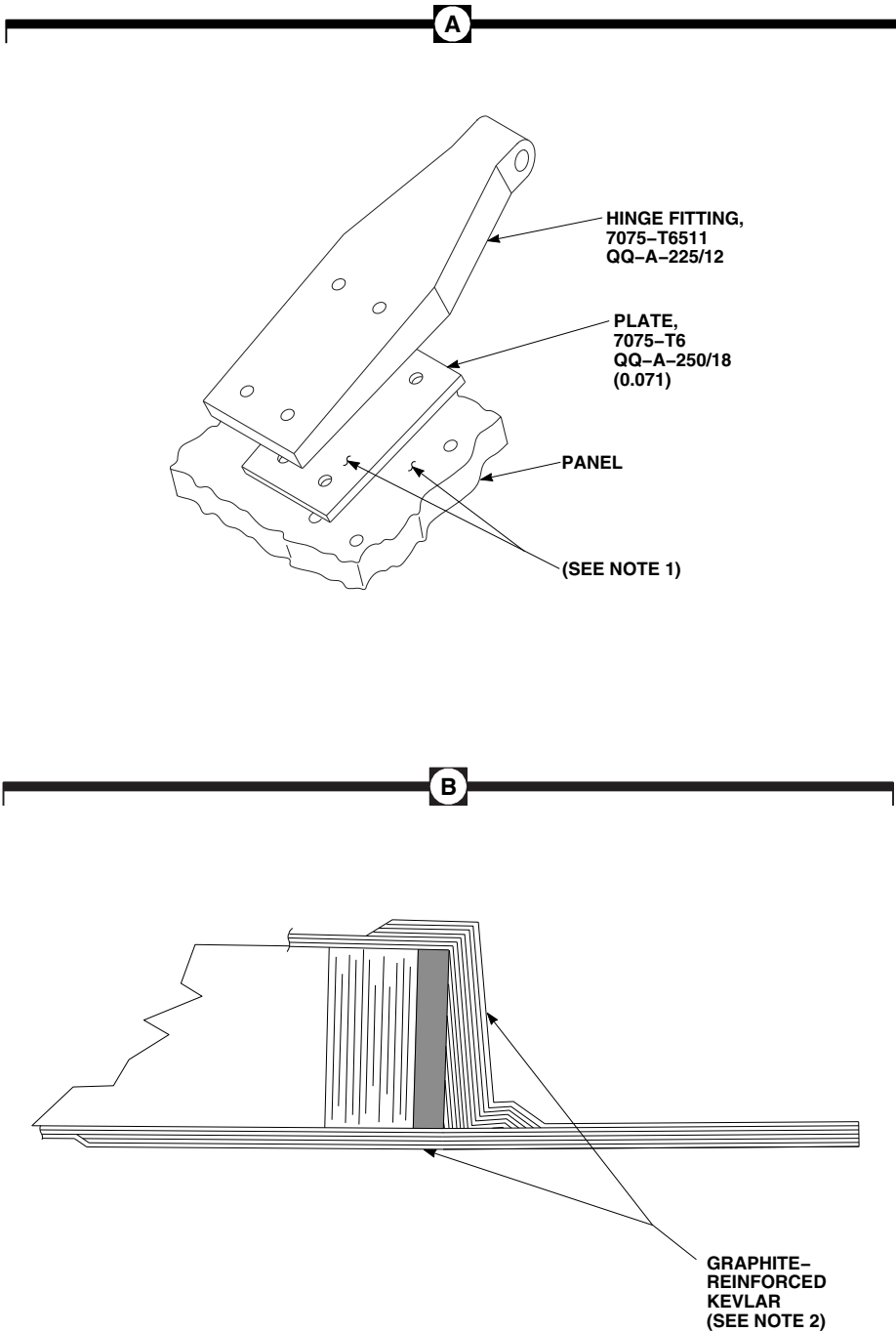


FIGURE 2-6-6-1. REPAIR NOSE DOOR (SHEET 2 OF 3)

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NOTES

1. APPLY SEALING COMPOUND, ITEM 292, APPENDIX D, PER PARA 2-6-1.
2. INNER PLIES ARE GRAPHITE CONSTRUCTION. OUTER COVERING IS ONE PLY KEVLAR.
REPAIR KEVLAR COVERING PER PARA 2-6-3.
3. ALL DIMENSIONS ARE IN INCHES, UNLESS NOTED.

FA1311_3A
SA**FIGURE 2-6-6-1. REPAIR NOSE DOOR (SHEET 3 OF 3)**

2-6-7 COCKPIT DOOR REPAIR.

PARAGRAPH BREAKDOWN

	PAGE
2-6-7.1 Repair Cockpit Door	2-6-7-1
INITIAL SETUP	MIL-I-24768/16
Tools	MIL-P-25690
Airframe Repairer’s Toolkit	MIL-PRF-25690
Materials/Parts	MIL-S-5059
Adhesive, Item 32, Appendix D	MIL-S-6758
Adhesive, Item 37, Appendix D	MIL-STD-670
Adhesive, Item 57, Appendix D	PARA 2-6-1
Sealing Compound, Item 292, Appendix D	PARA 2-6-2
Sealing Compound, Item 300, Appendix D	PARA 2-6-3
References	PARA 2-6-4
AMS 4260	PARA 2-6-5
AMS 5463	QQ-A-200
Appendix D	QQ-A-250
	QQ-A-367

2-6-7.1 Repair Cockpit Door.

- a. Bonded door assembly is made of Kevlar (reinforced plastic) framing and skins, with aluminum angles, plates, and channels. Hinge and latch mechanism attachment points are stiffened with Kevlar channels and honeycomb (Figure 2-6-7-1, Sheet 1, Detail A and Sheet 2, View A-A). Door window is made from acrylic sheet plastic. Door seal is extruded rubber bonded to door with adhesive, Item 37, Appendix D. Hook and pile fastener tape sewn to data case cover keeps it closed. Door structure is coated with sealing compound, Item 300, Appendix D. Door is electrically bonded to airframe with grounding seals (Figure 2-6-7-1, Sheet 2).
- b. Evaluate damage to aluminum parts (PARA 2-6-1).

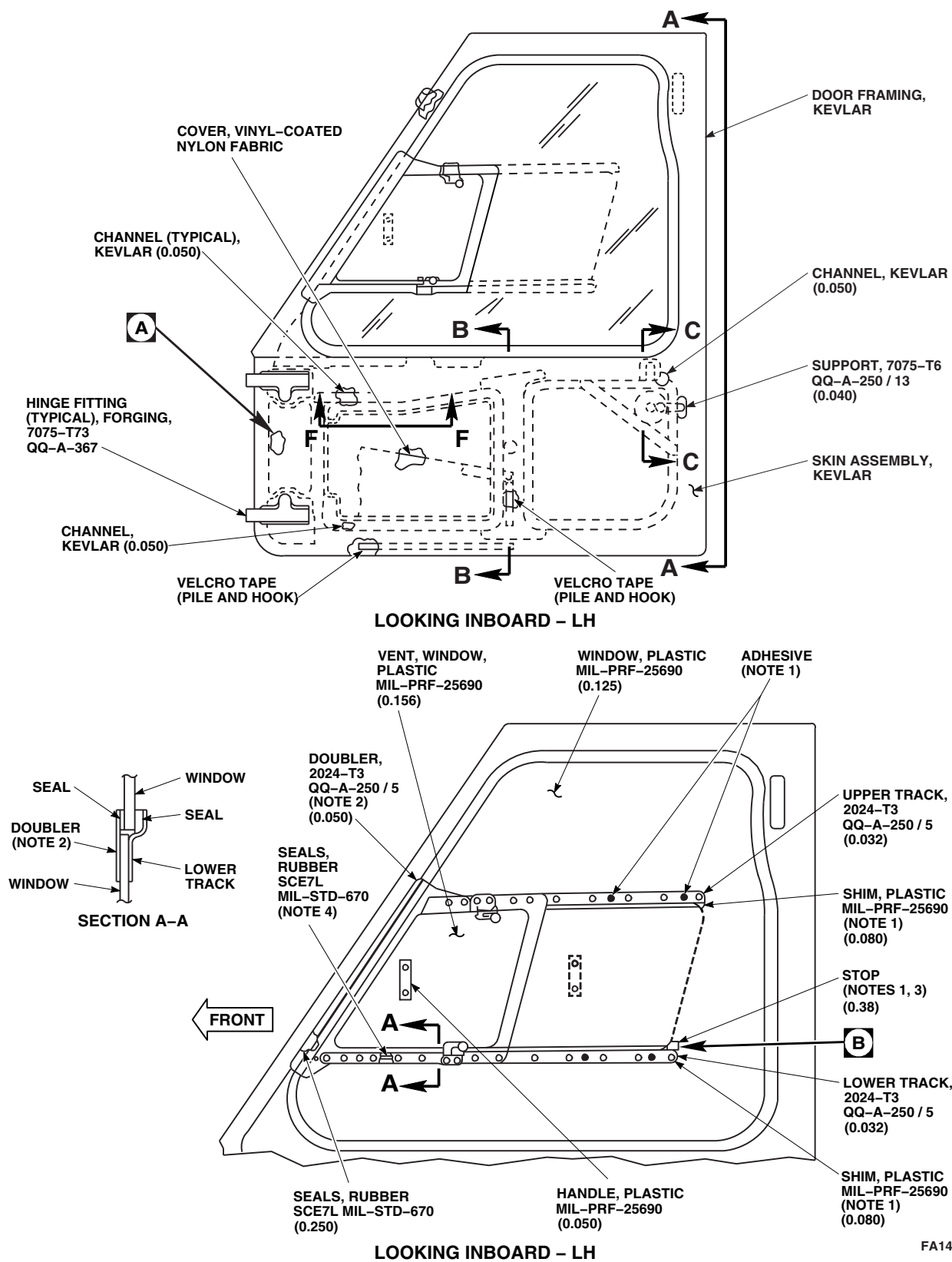


FIGURE 2-6-7-1. REPAIR COCKPIT DOOR (SHEET 1 OF 4)

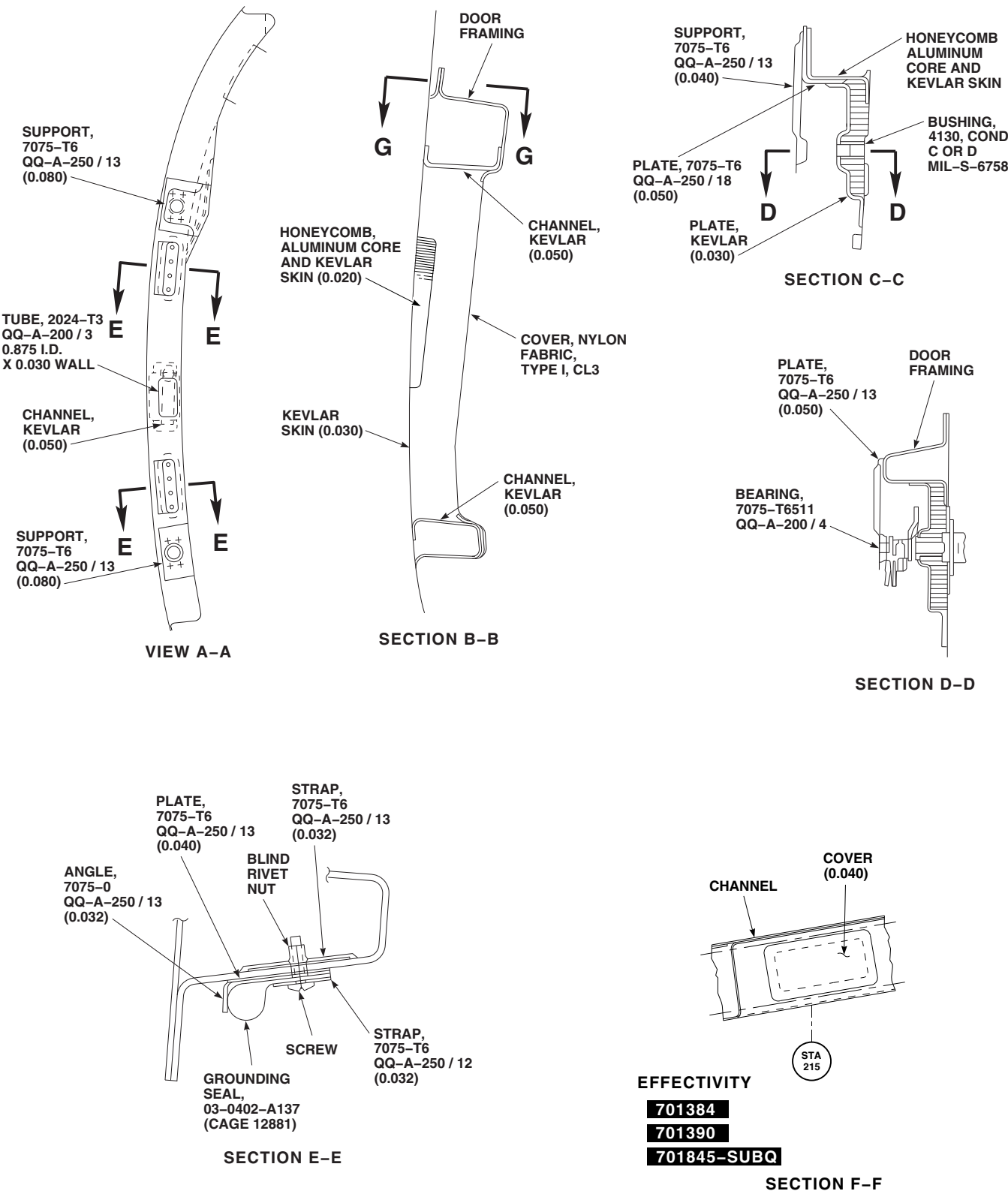


FIGURE 2-6-7-1. REPAIR COCKPIT DOOR (SHEET 2 OF 4)

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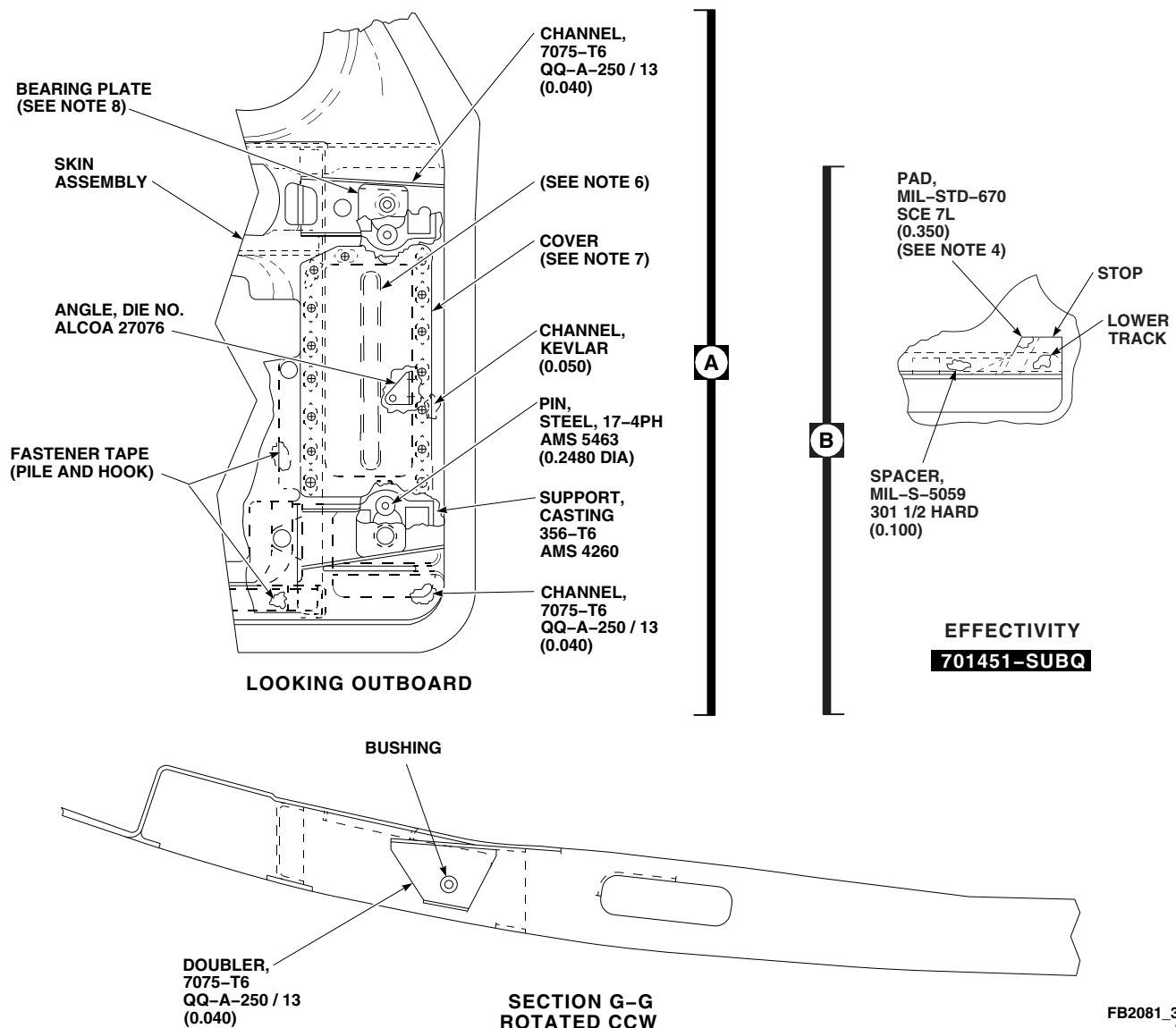


FIGURE 2-6-7-1. REPAIR COCKPIT DOOR (SHEET 3 OF 4)

- c. Use PARA 2-6-2 and PARA 2-6-4 for typical repairs to aluminum parts.
- d. Evaluate damage and make repairs to Kevlar and composite plastic components (PARA 2-6-3).
- e. Clean, polish, wax, and maintain acrylic plastic windows.
- f. Touch up paint in repaired area (PARA 2-6-5).

NOTES

1. BONDED TO WINDOW WITH ADHESIVE, ITEM 57, APPENDIX D IN 1/4 INCH DIAMETER SPOTS WHERE SHOWN.
2. SEAL MATING SURFACES WITH SEALING COMPOUND, ITEM 292, APPENDIX D.
3. **S70-I** PLASTIC MIL-P-25690. **S70-II** PHENOLIC MIL-I-24768 / 16, TYPE FMB.
4. **S70-I** (0.062). **S70-II** (0.125).
5. BOND WITH ADHESIVE, ITEM 32, APPENDIX D.
6. SKIN MAY OR MAY NOT BE BEADED.
7. EITHER KEVLAR (0.060) OR 7075-T6, QQ-A-250 / 13, (0.125).
8. **S70-I** 7075-T6 QQ-A-250 / 13 (0.125). **S70-II** 4130 COND A OR N (0.062).
9. BOND TO STOP AND TRACK WITH ADHESIVE, ITEM 37, APPENDIX D.
10. ALL DIMENSIONS ARE IN INCHES, UNLESS NOTED.

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FIGURE 2-6-7-1. REPAIR COCKPIT DOOR (SHEET 4 OF 4)

2-6-8 NOSE SECTION FRAMING REPAIR.

PARAGRAPH BREAKDOWN

	PAGE
2-6-8.1 Repair Nose Section Framing	2-6-8-1
INITIAL SETUP	• PARA 2-6-1
Tools	• PARA 2-6-2
• Airframe Repairer’s Toolkit	• PARA 2-6-3
Materials/Parts	• PARA 2-6-4
• Sealing Compound, Item 292, Appendix D	• PARA 2-6-5
References	• PARA 2-6-12
• Appendix D	• QQ-A-200
• AMS 4036	• QQ-A-225
• AMS-A-21180	• QQ-A-230
• AMS-DTL-22499	• QQ-A-250
• MIL-C-7438	• QQ-A-367
• MIL-S-18729	• WW-T-700

2-6-8.1 Repair Nose Section Framing.

- a. All nose section framing, except reinforced plastic members, is primary structure. Beams at BL 10.0, right and left side, are reinforced with Boron straps bonded to cap angles. Damage to angles bonded to Boron straps is limited to negligible damage repairs using PARA 2-6-1. Repair of Boron is beyond the scope of this section. Fittings are considered critical primary structure. Negligible damage to fittings is limited to nicks, scratches, and corrosion no deeper than 5% of material thickness at point

of damage. Blend damage smoothly into surrounding area using long, even strokes. Polish out all surface imperfections and restore protective finish. Damage to fittings greater than negligible limits must be reviewed by engineering authority for determining repair or replacement. All repairs to nose section framing must maintain design strength and alignment (Figure 2-6-8-1, Sheet 1 through Sheet 21).

- b. Evaluate damage in noncritical areas (PARA 2-6-1).

- c. Refer to PARA 2-6-2 and PARA 2-6-4 for typical framing repairs.
- d. Refer to PARA 2-6-3 for evaluating damage and repairing reinforced plastic.
- e. For corrosion protection and watertight sealing, seal all repairs between framing and plating (PARA 2-6-1).
- f. Touch up paint as required (PARA 2-6-5).

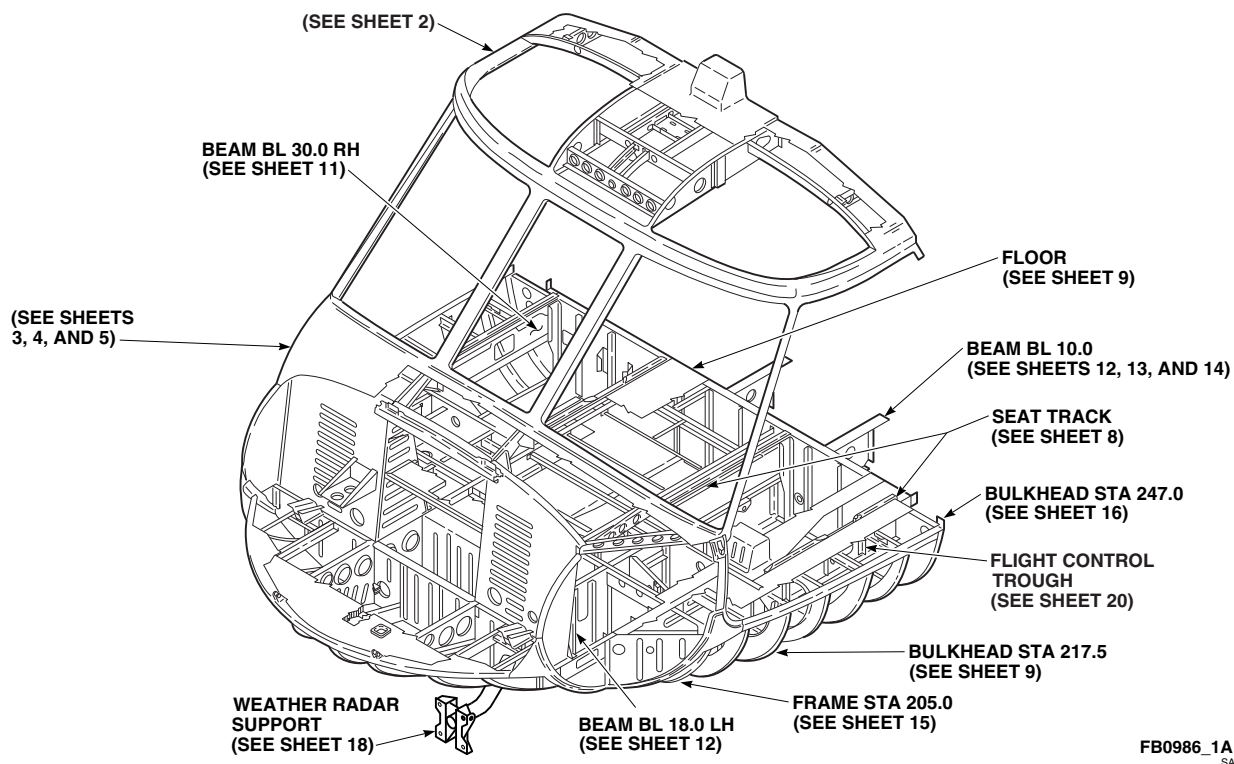


FIGURE 2-6-8-1. REPAIR NOSE SECTION FRAMING (SHEET 1 OF 21)

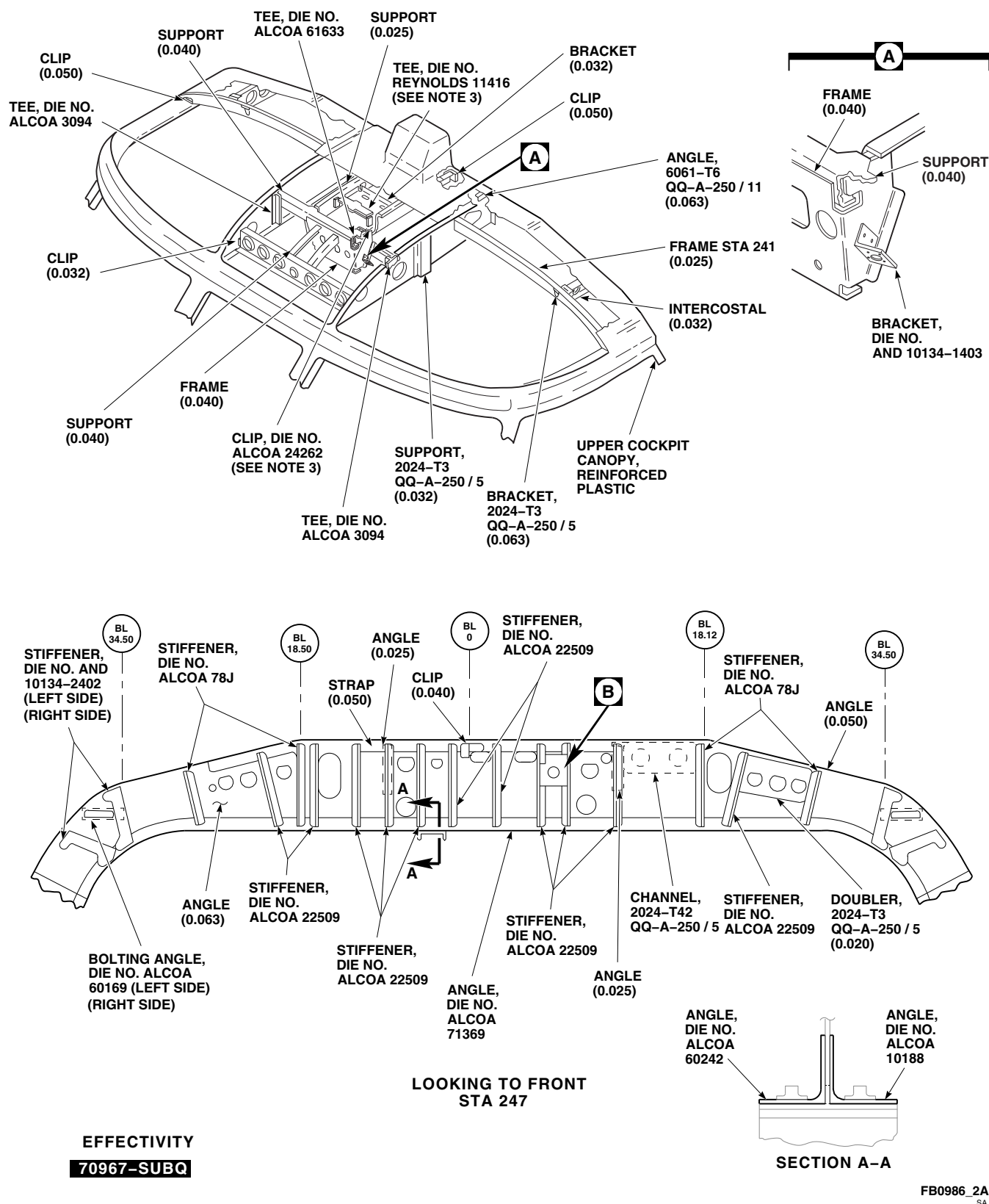


FIGURE 2-6-8-1. REPAIR NOSE SECTION FRAMING (SHEET 2 OF 21)

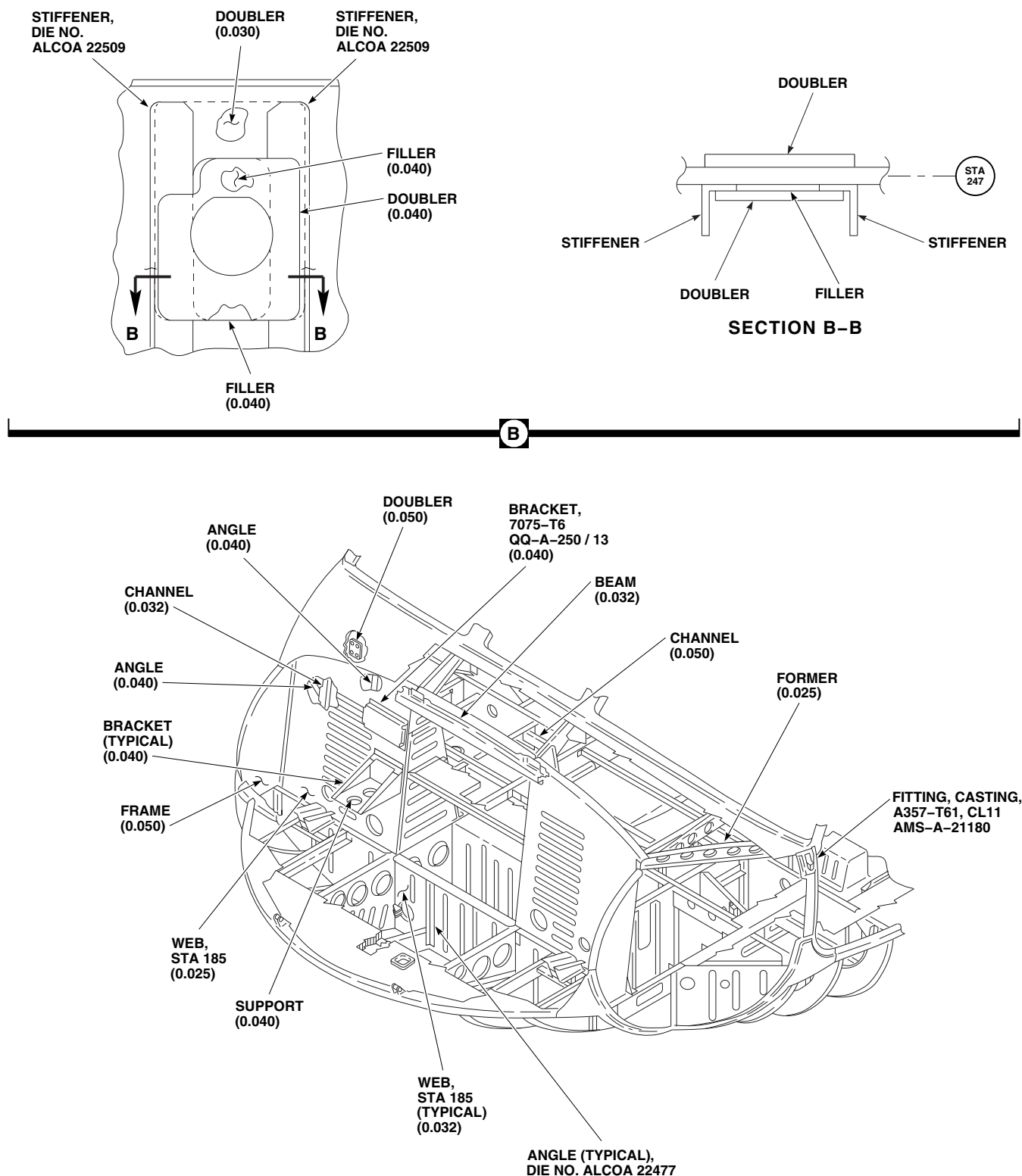


FIGURE 2-6-8-1. REPAIR NOSE SECTION FRAMING (SHEET 3 OF 21)

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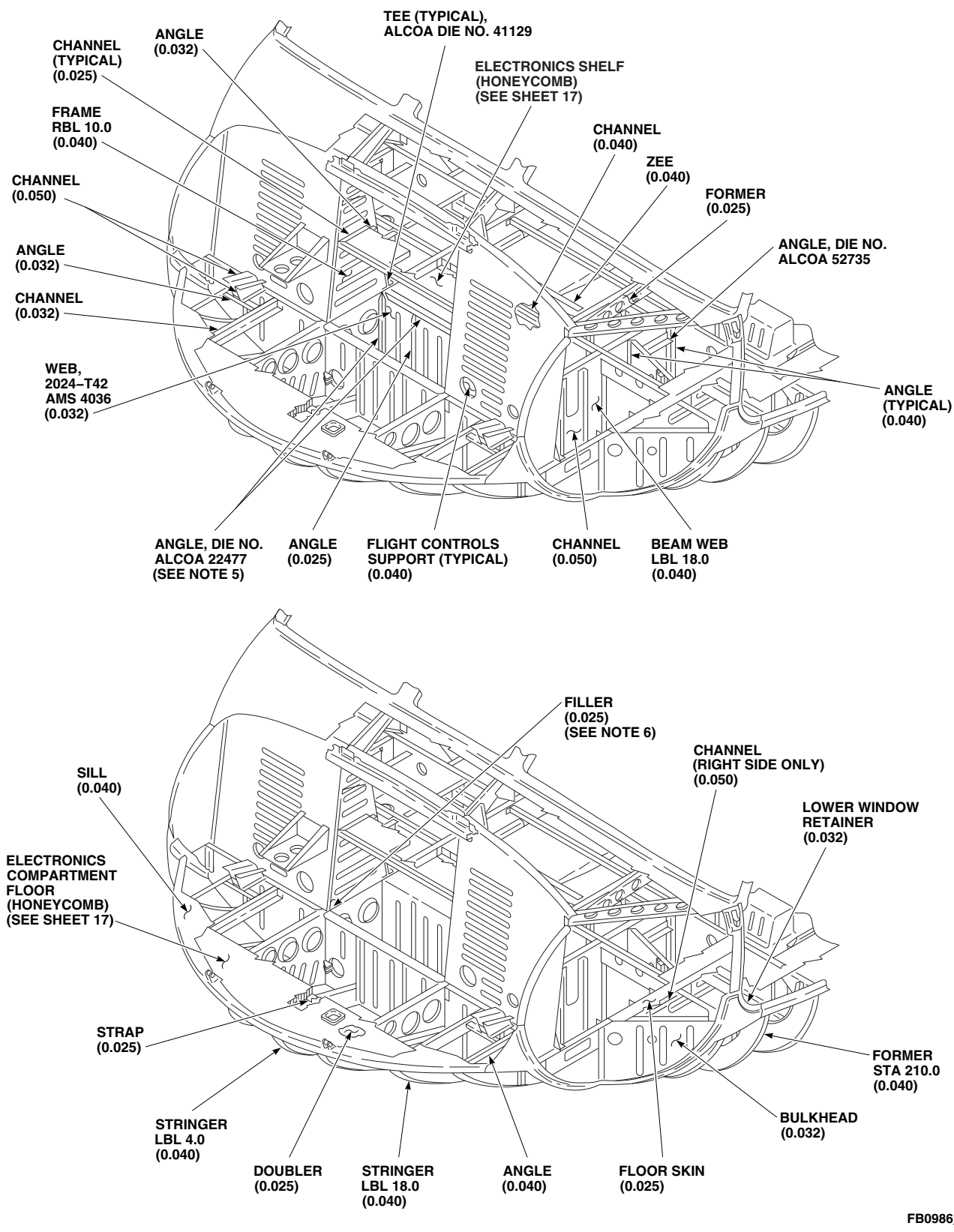
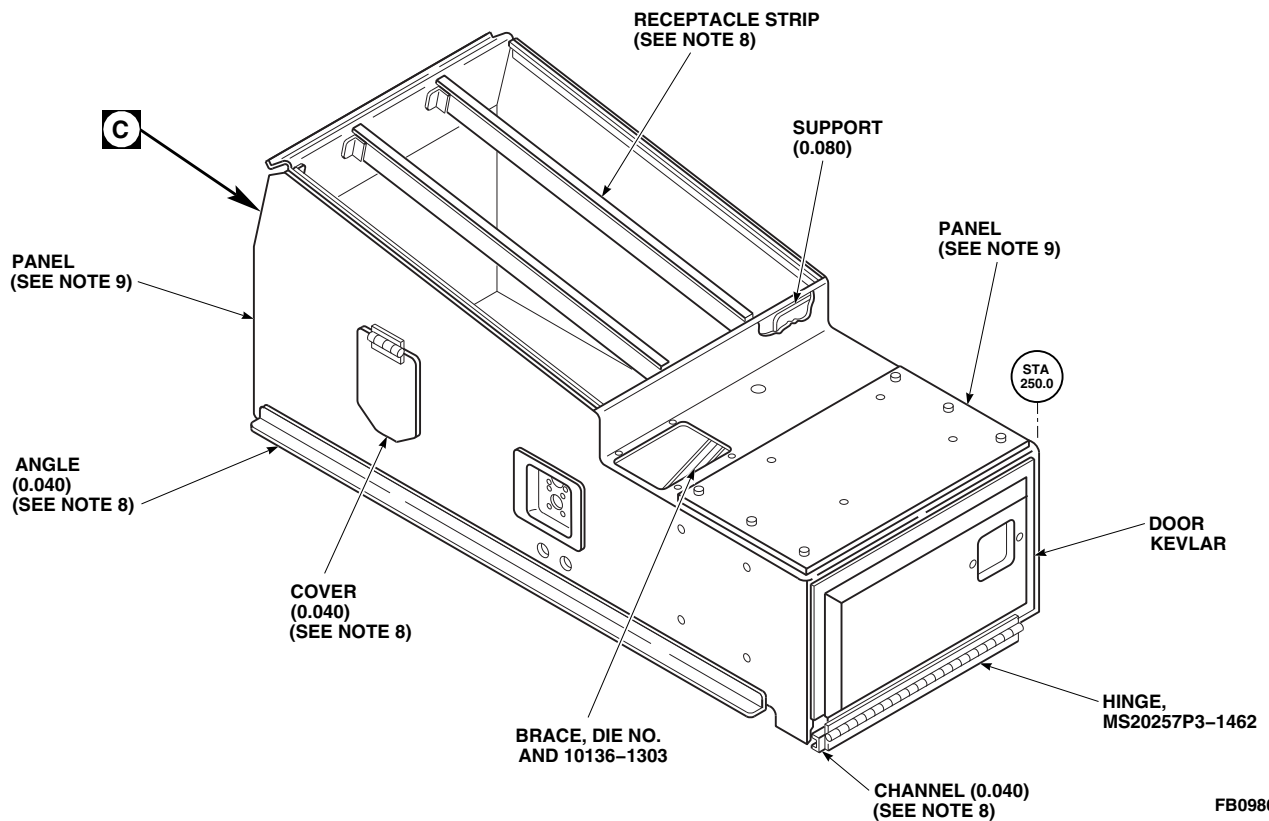
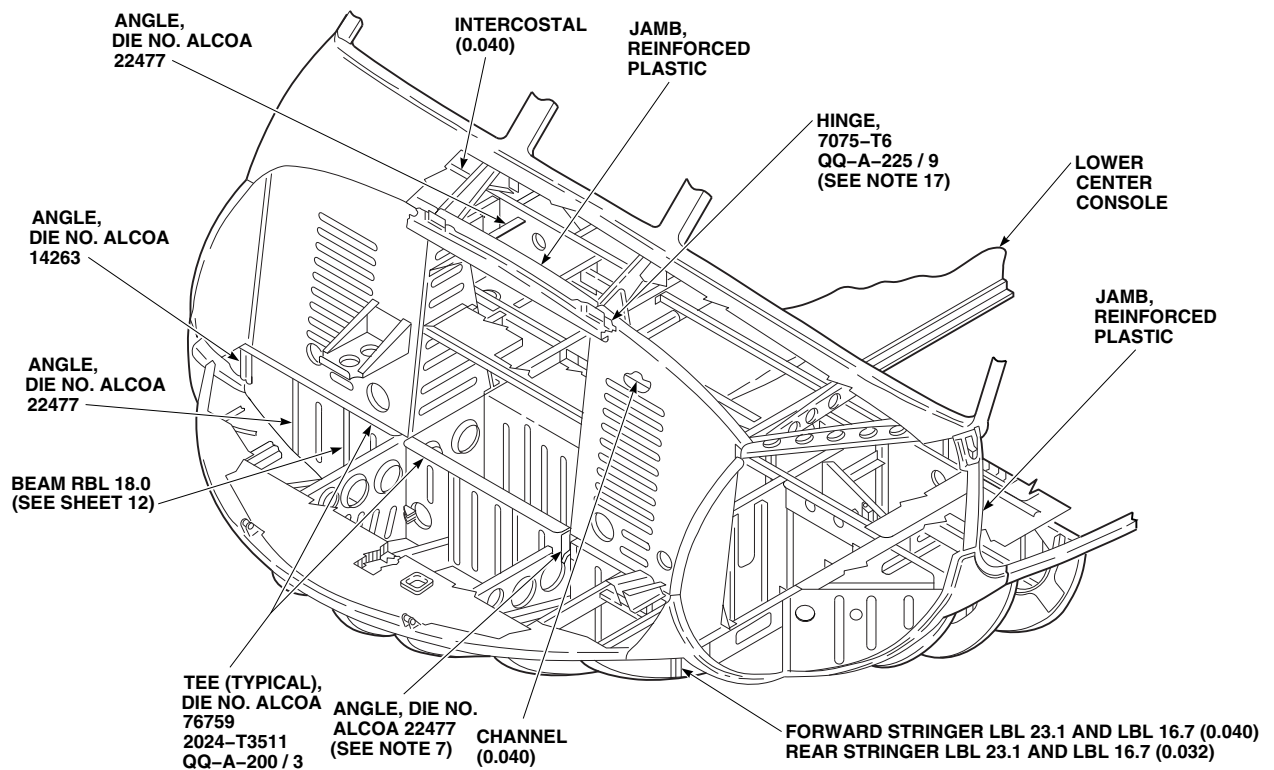


FIGURE 2-6-8-1. REPAIR NOSE SECTION FRAMING (SHEET 4 OF 21)



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FIGURE 2-6-8-1. REPAIR NOSE SECTION FRAMING (SHEET 5 OF 21)

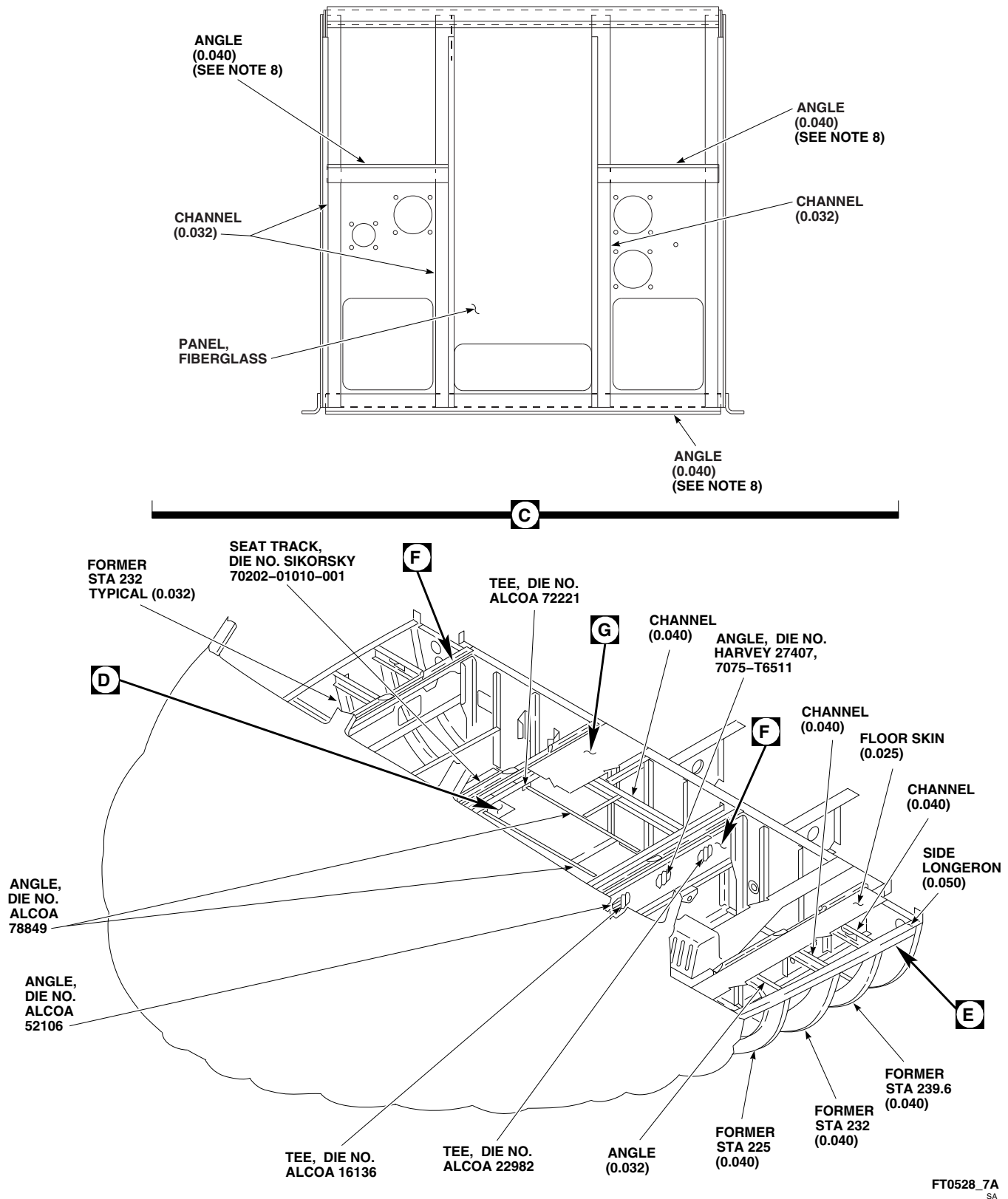
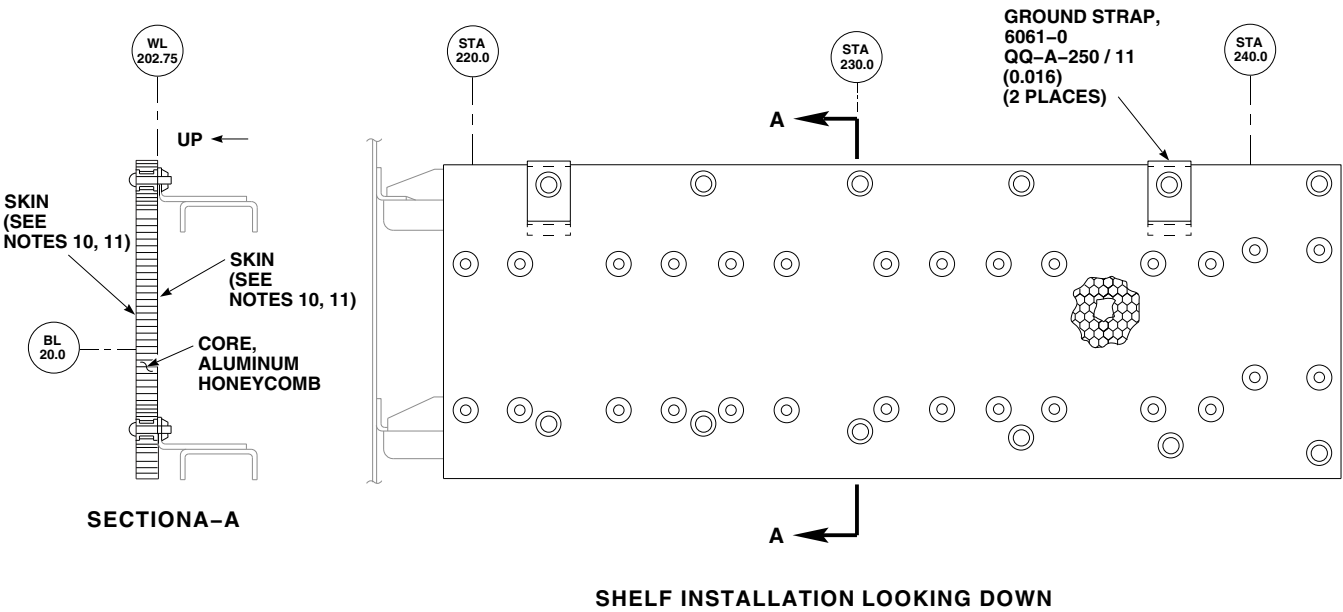


FIGURE 2-6-8-1. REPAIR NOSE SECTION FRAMING (SHEET 6 OF 21)



EFFECTIVITY
70789-SUBQ

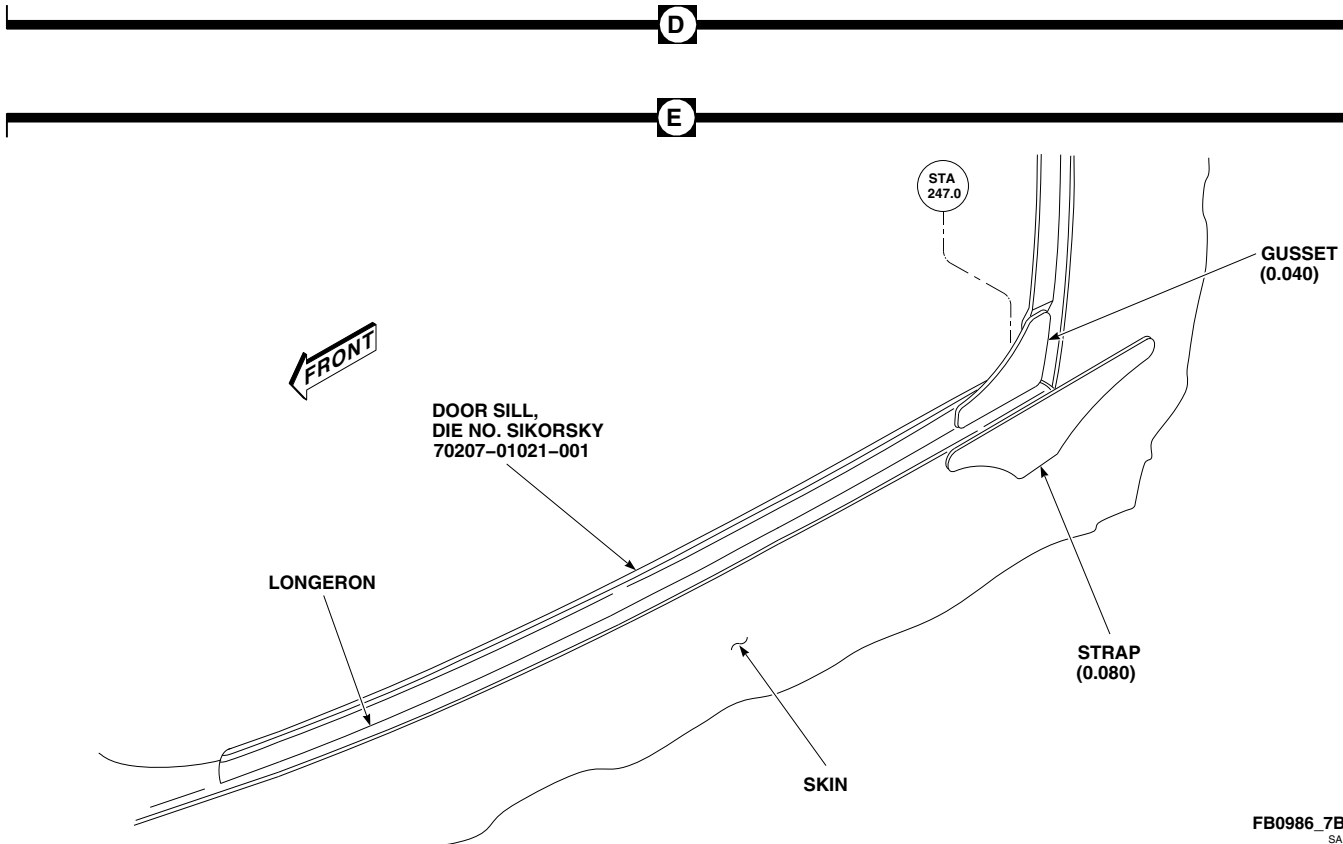


FIGURE 2-6-8-1. REPAIR NOSE SECTION FRAMING (SHEET 7 OF 21)

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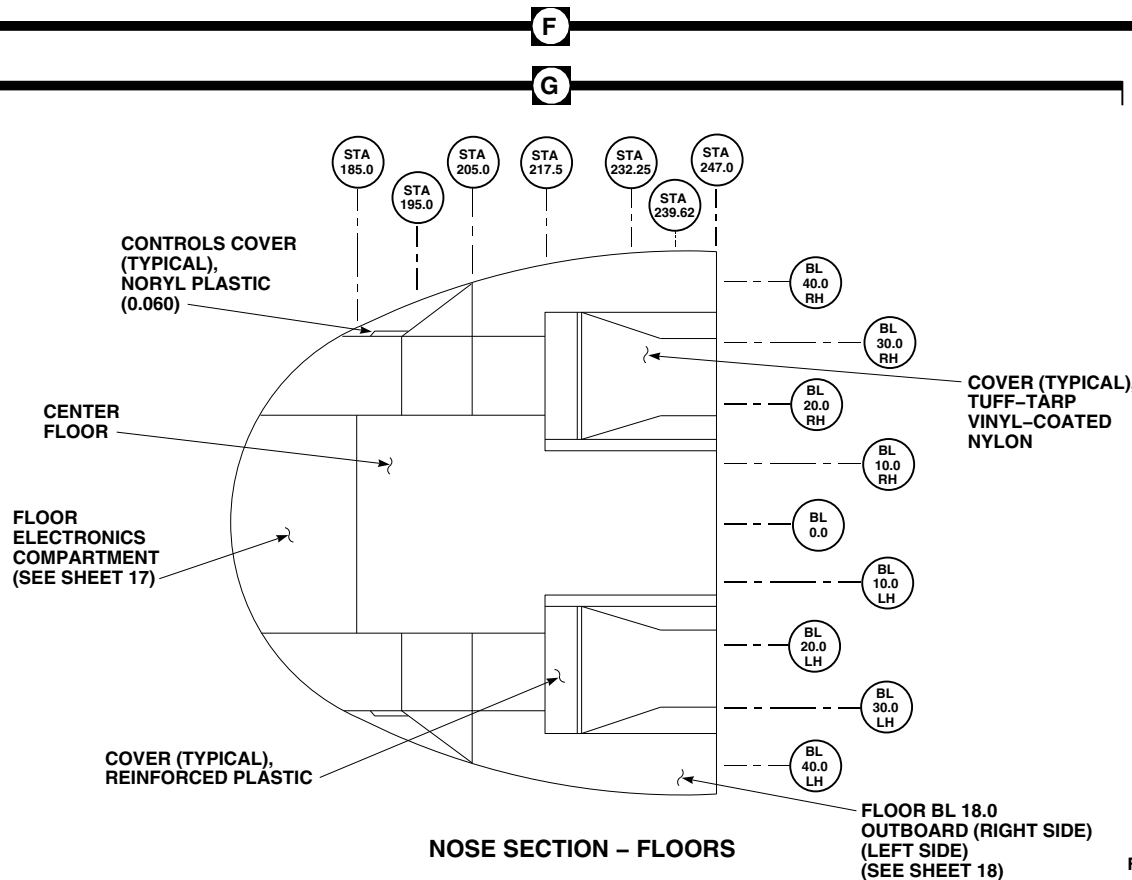
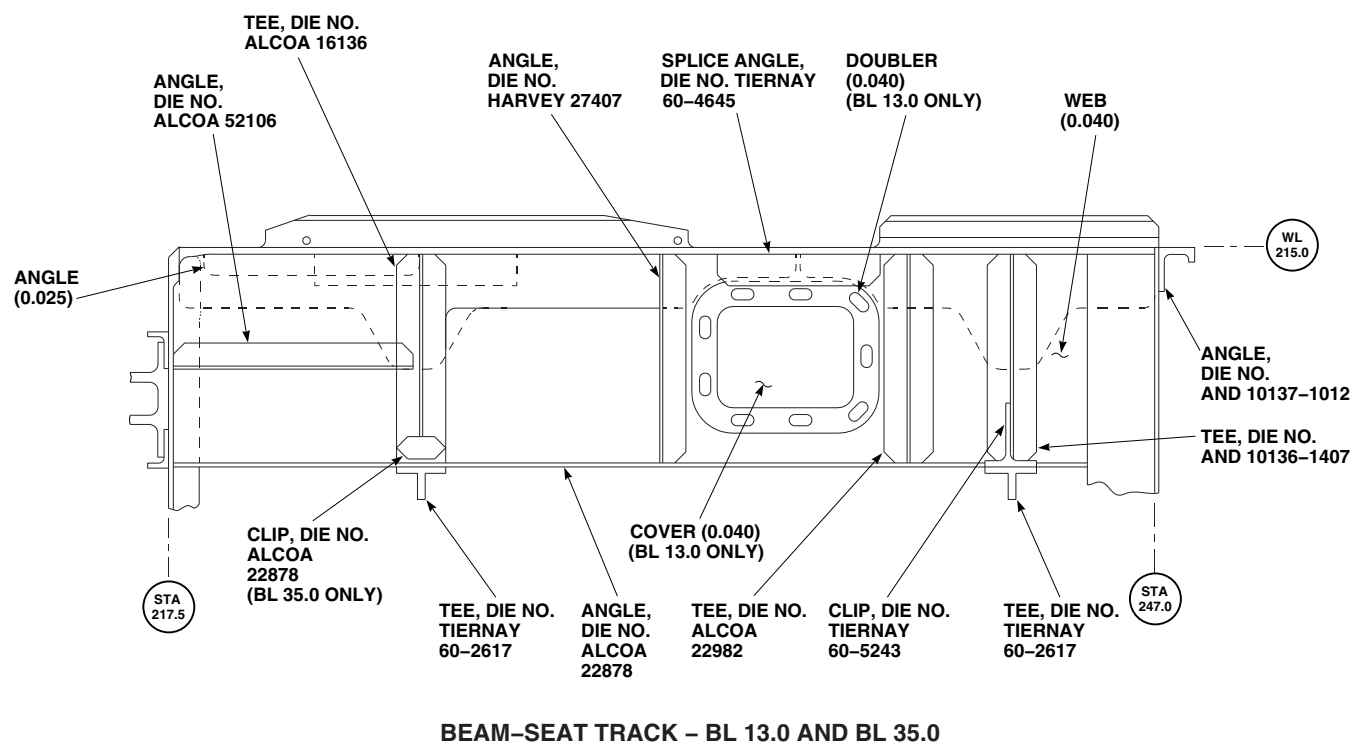
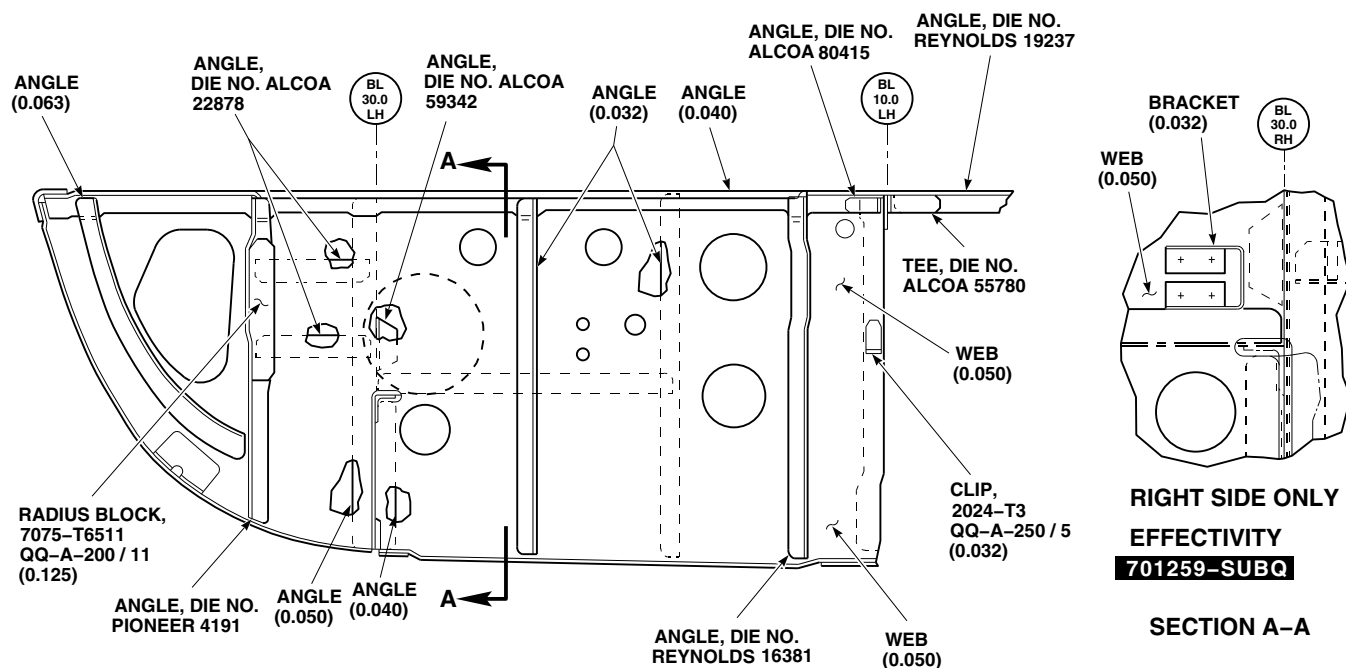


FIGURE 2-6-8-1. REPAIR NOSE SECTION FRAMING (SHEET 8 OF 21)

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**BULKHEAD STA 217.5
LEFT SIDE SHOWN, RIGHT SIDE OPPOSITE**

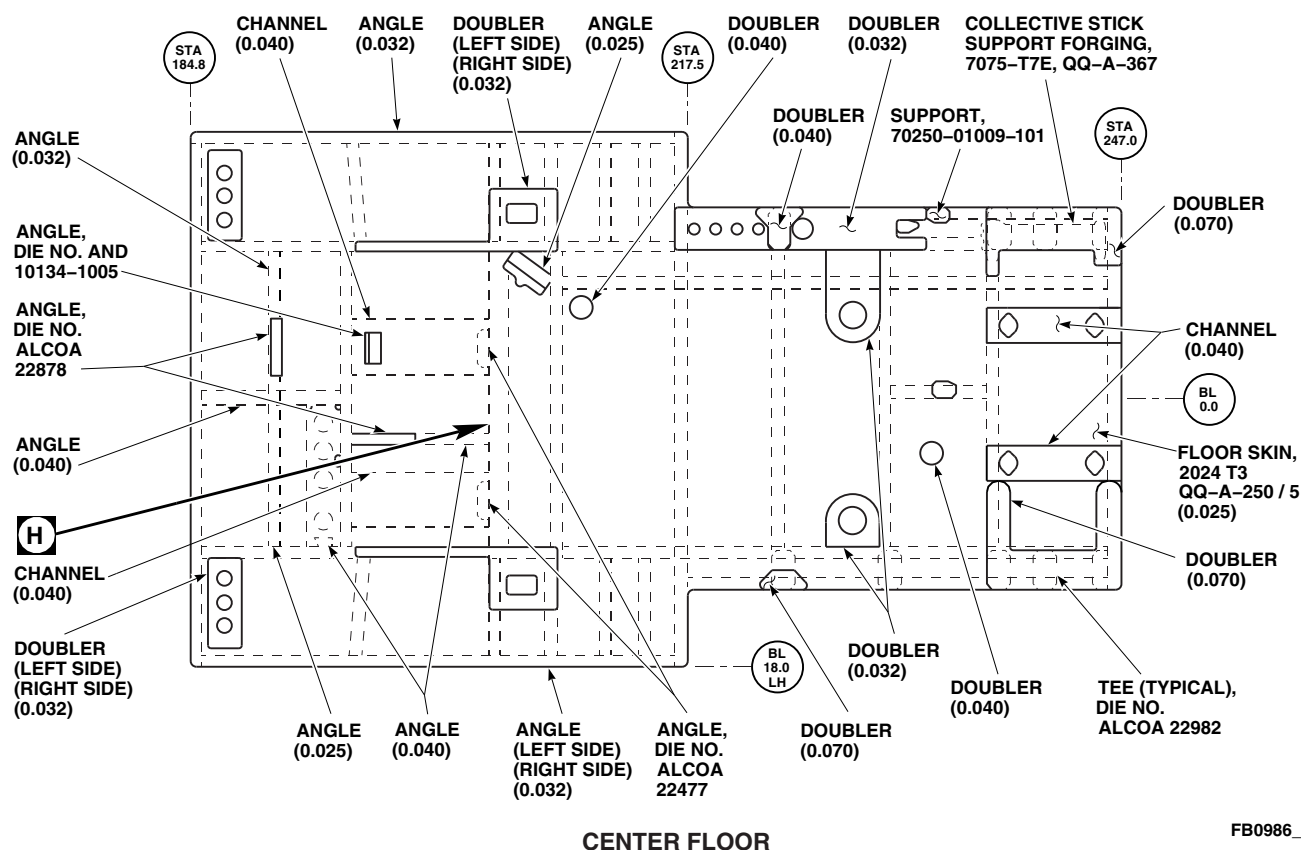
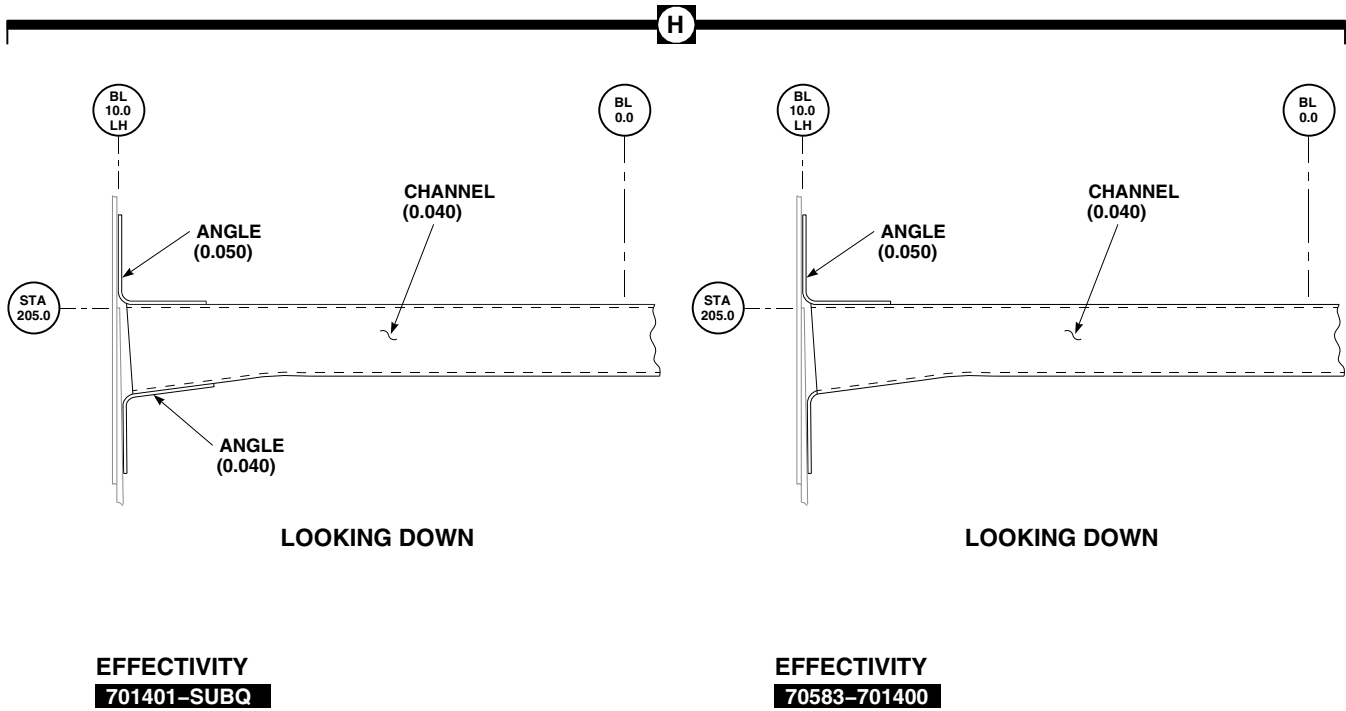


FIGURE 2-6-8-1. REPAIR NOSE SECTION FRAMING (SHEET 9 OF 21)



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FIGURE 2-6-8-1. REPAIR NOSE SECTION FRAMING (SHEET 10 OF 21)

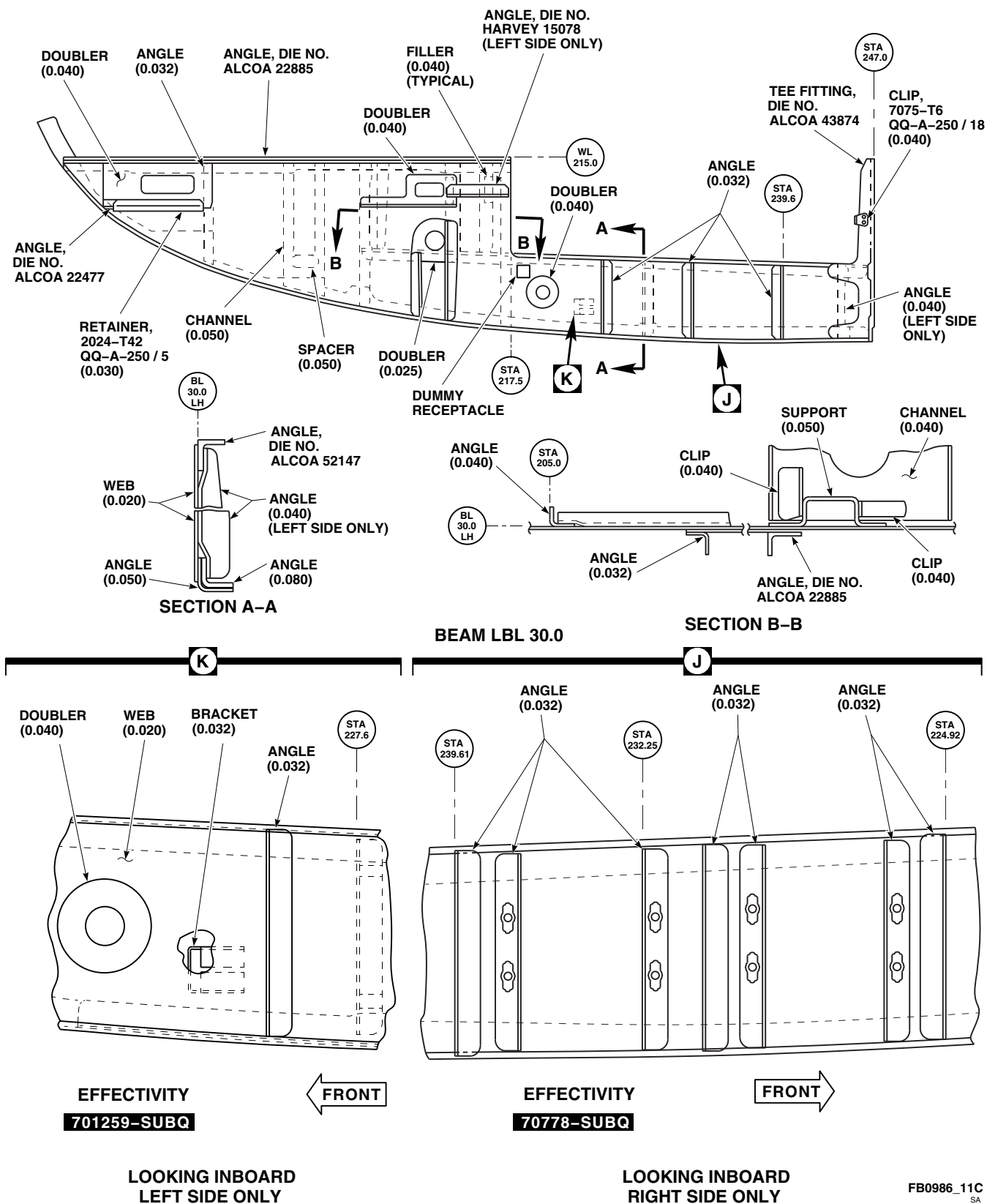


FIGURE 2-6-8-1. REPAIR NOSE SECTION FRAMING (SHEET 11 OF 21)

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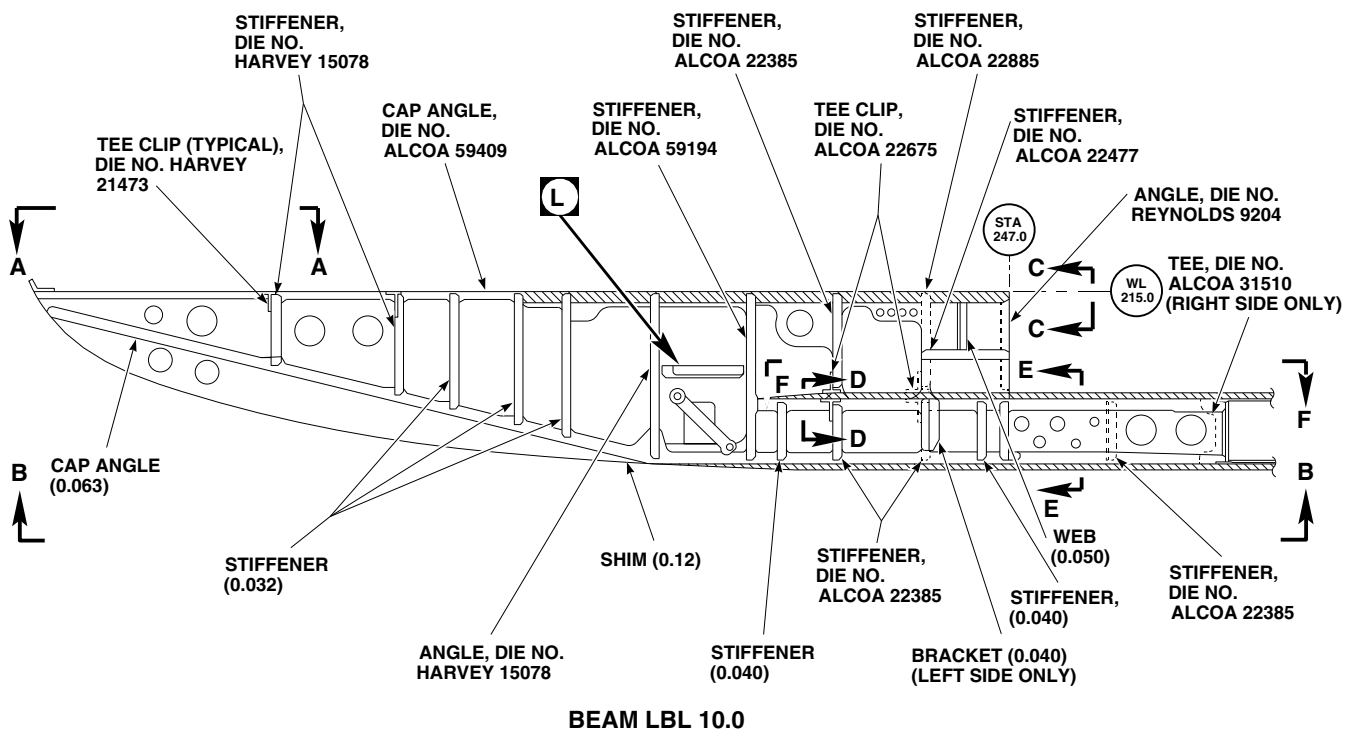
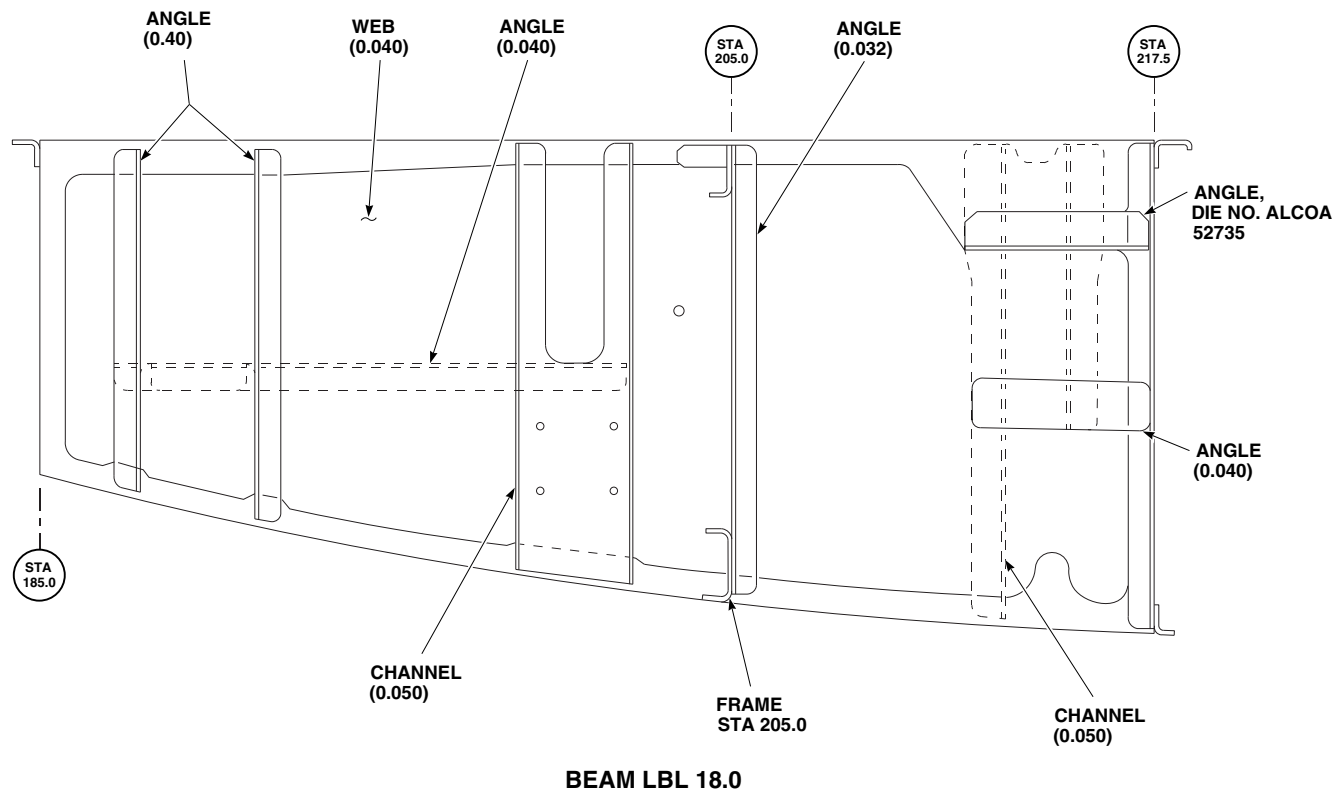


FIGURE 2-6-8-1. REPAIR NOSE SECTION FRAMING (SHEET 12 OF 21)

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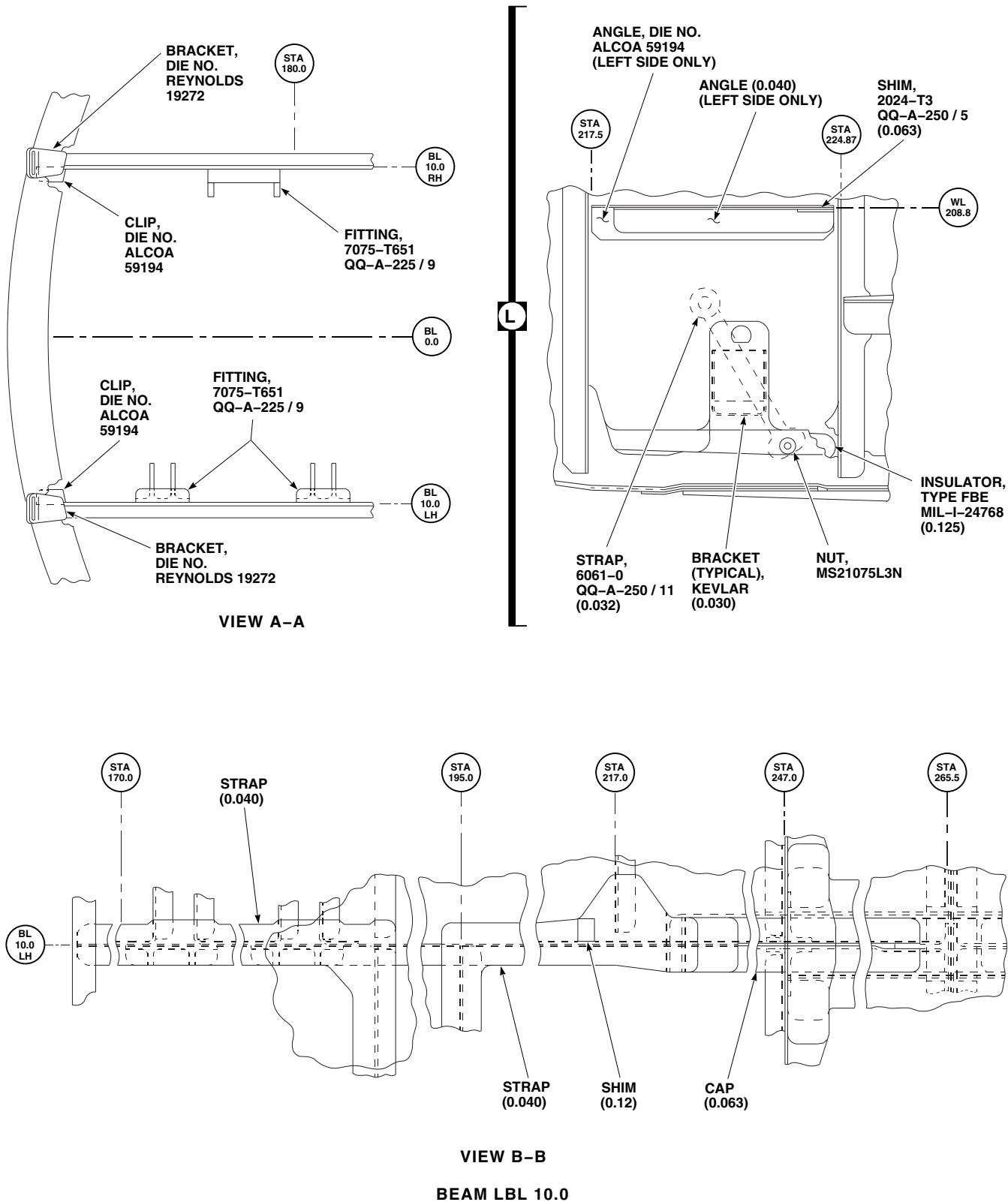
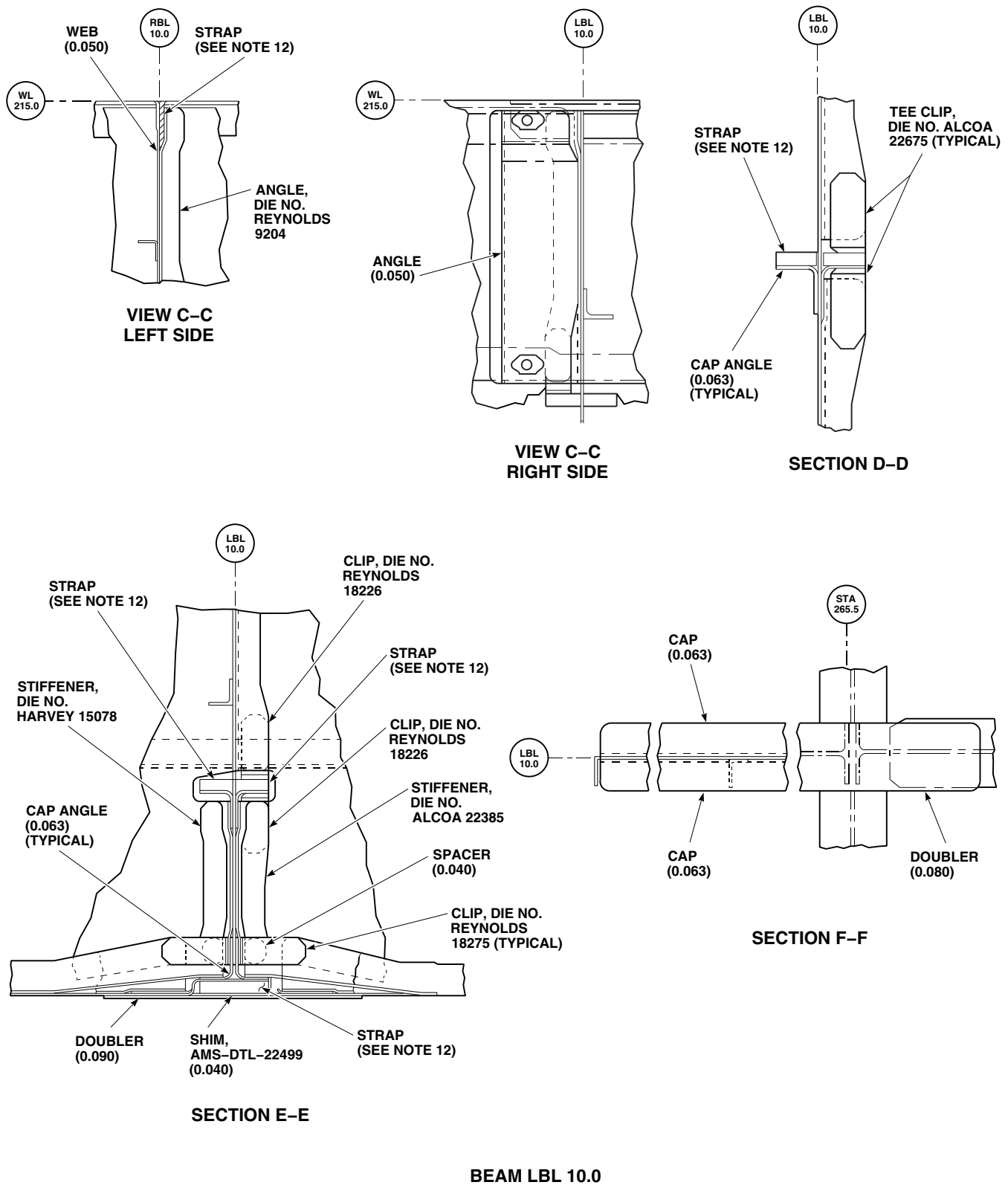


FIGURE 2-6-8-1. REPAIR NOSE SECTION FRAMING (SHEET 13 OF 21)

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FIGURE 2-6-8-1. REPAIR NOSE SECTION FRAMING (SHEET 14 OF 21)

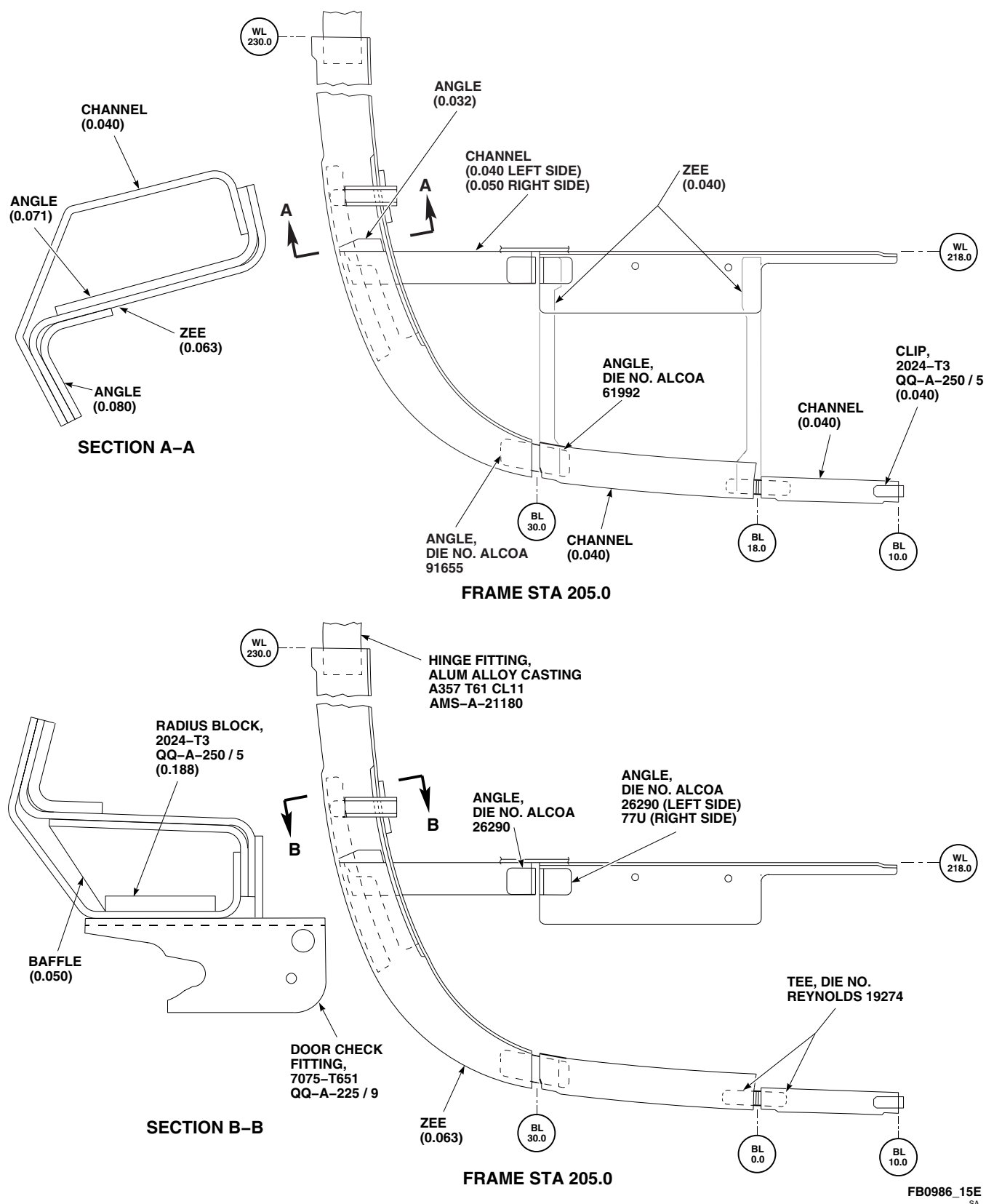
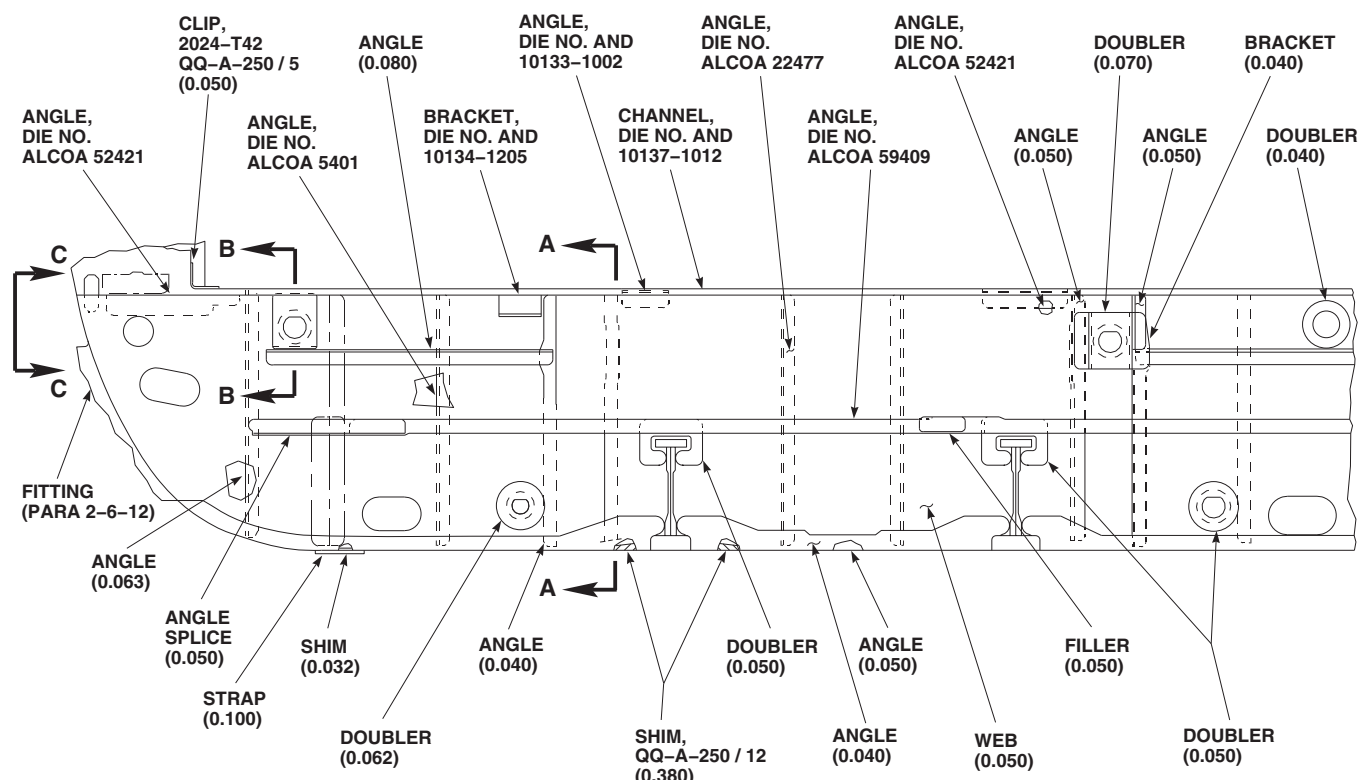
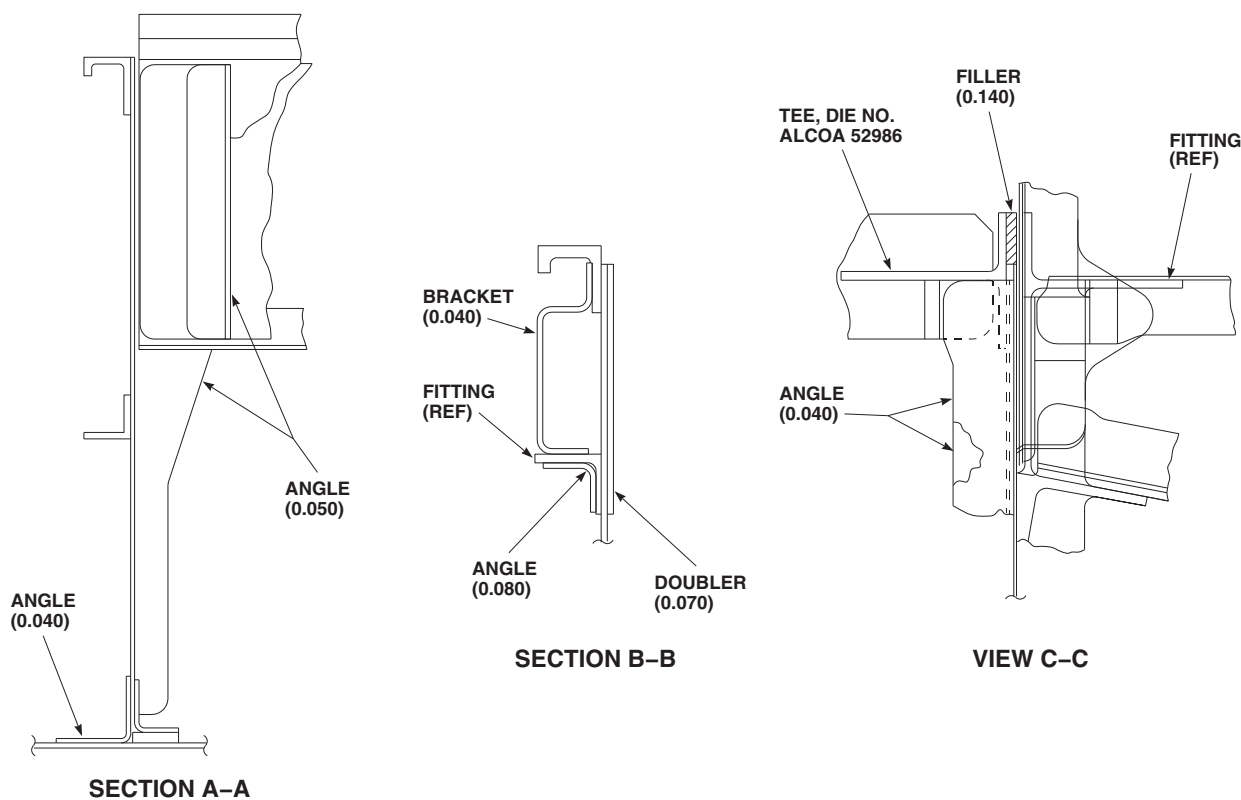


FIGURE 2-6-8-1. REPAIR NOSE SECTION FRAMING (SHEET 15 OF 21)



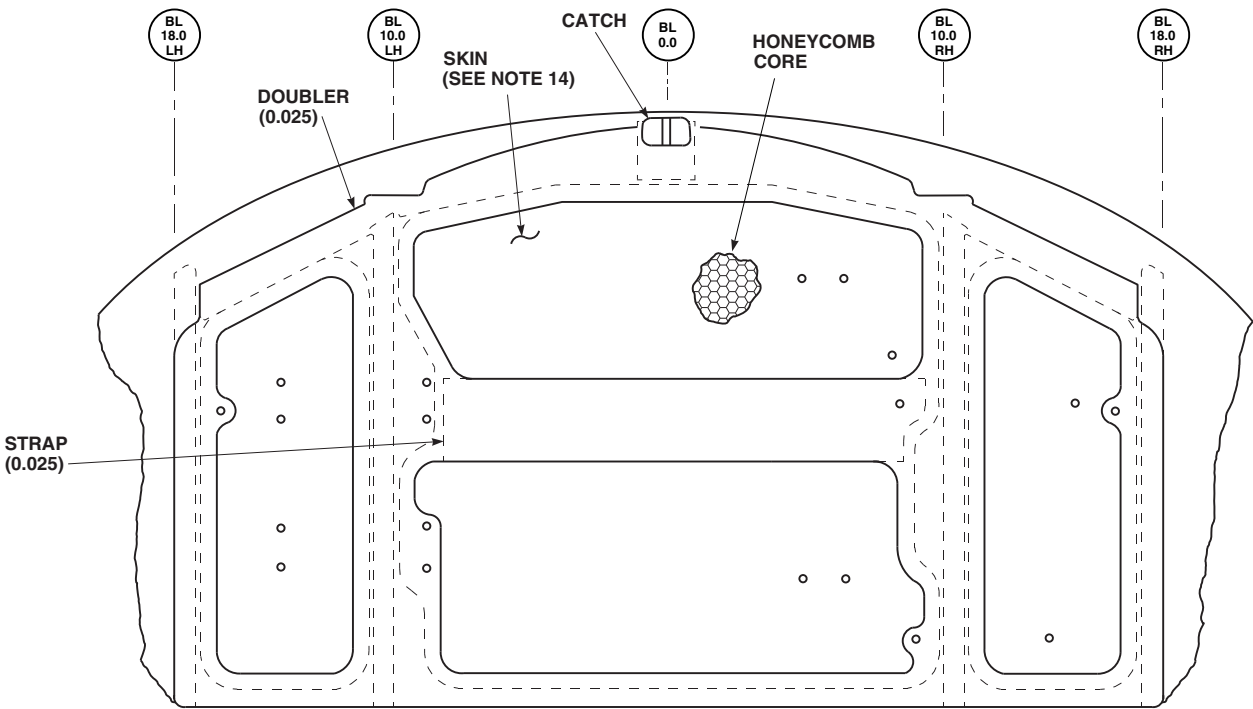
BULKHEAD STATION 247.0



BULKHEAD STATION 247.0

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FIGURE 2-6-8-1. REPAIR NOSE SECTION FRAMING (SHEET 16 OF 21)



LOOKING DOWN
ELECTRONICS COMPARTMENT FLOOR

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FIGURE 2-6-8-1. REPAIR NOSE SECTION FRAMING (SHEET 17 OF 21)

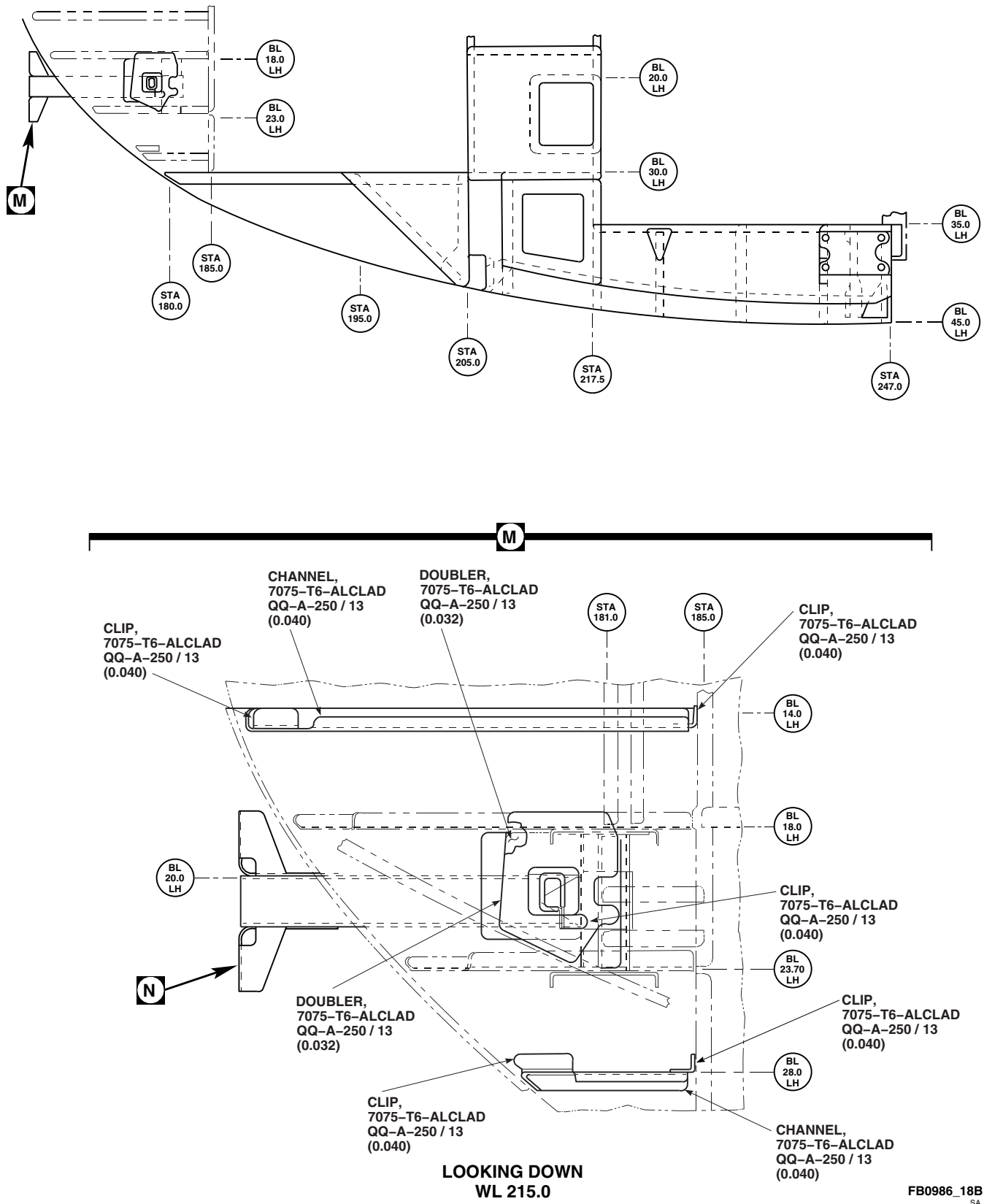


FIGURE 2-6-8-1. REPAIR NOSE SECTION FRAMING (SHEET 18 OF 21)

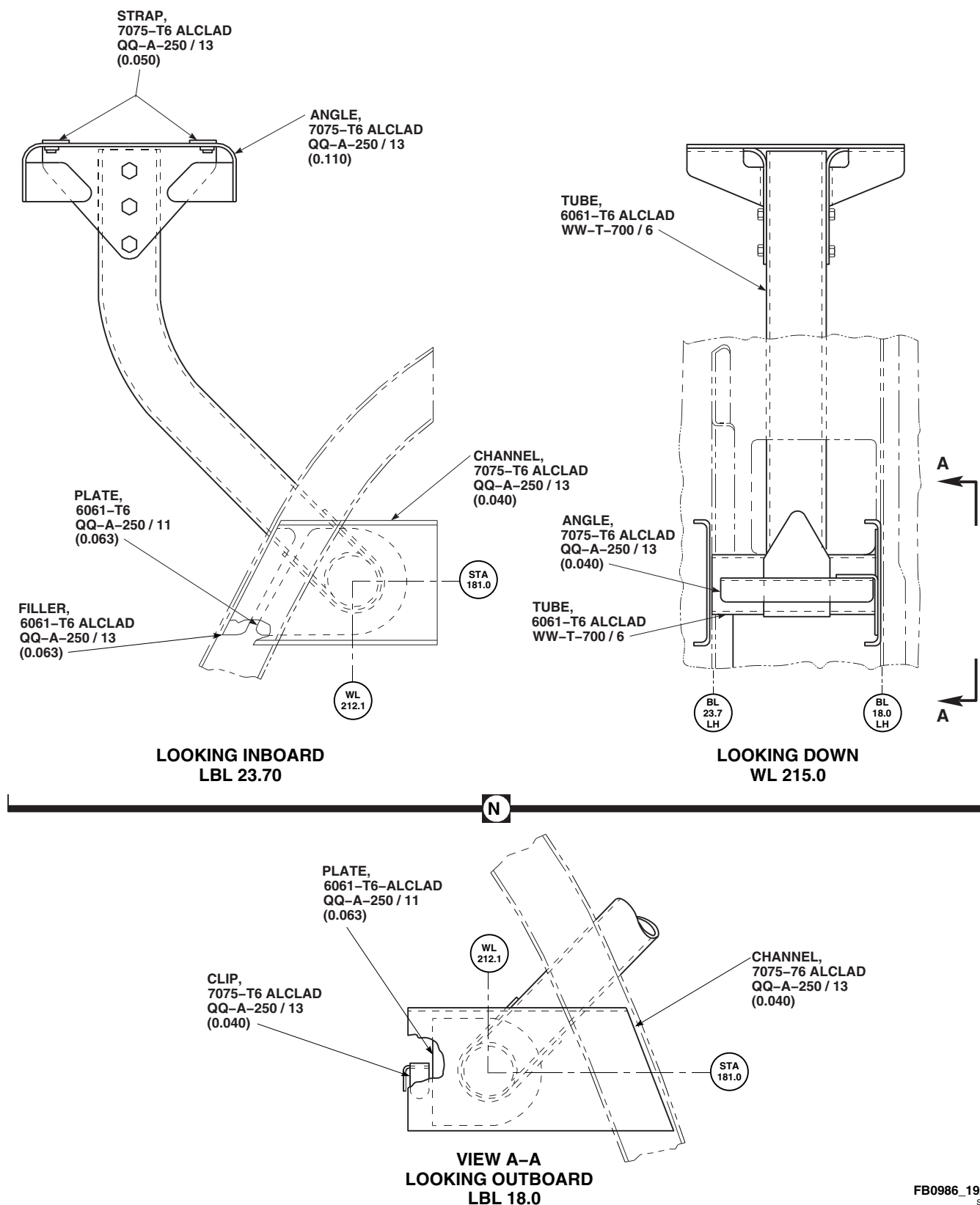


FIGURE 2-6-8-1. REPAIR NOSE SECTION FRAMING (SHEET 19 OF 21)

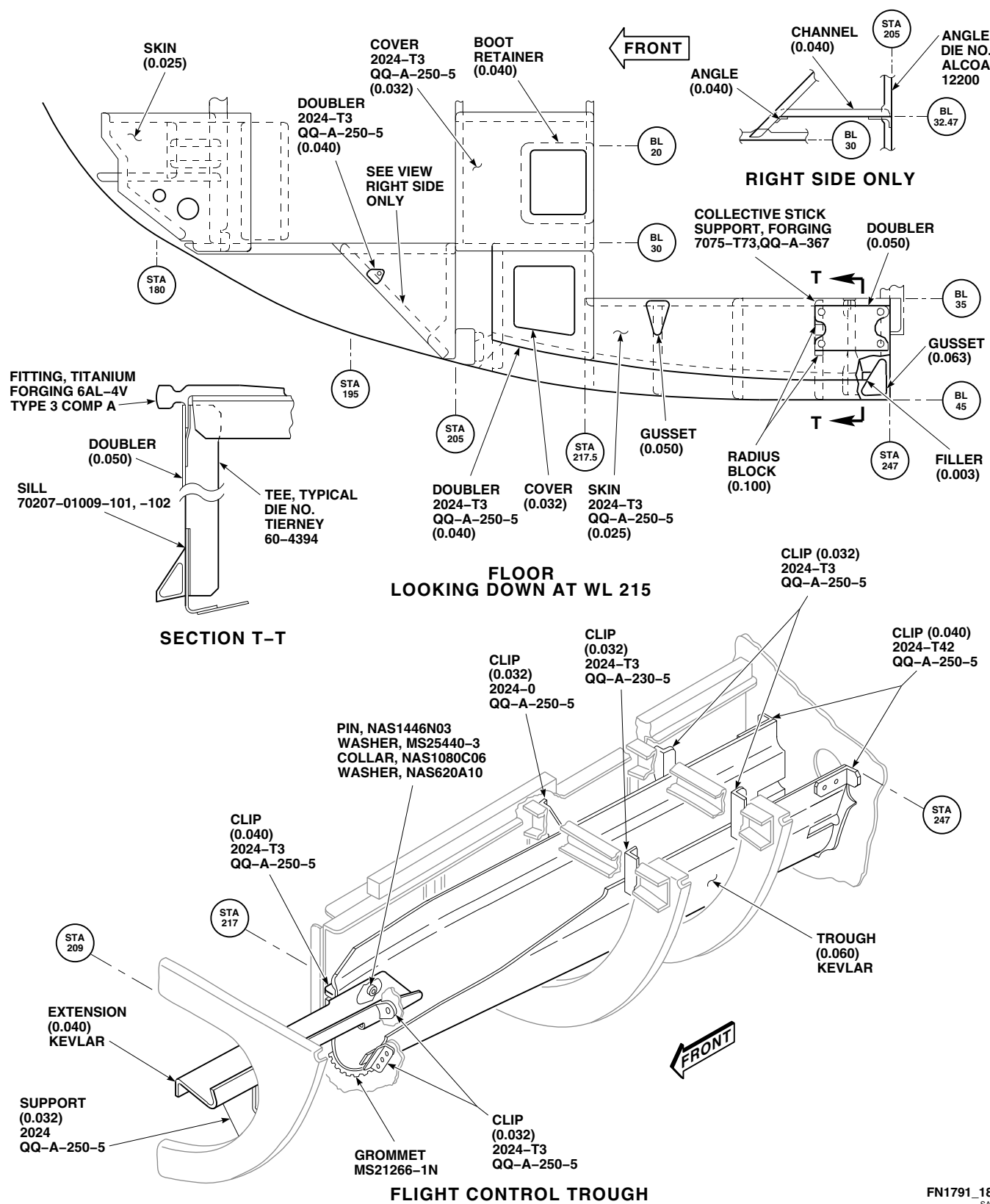


FIGURE 2-6-8-1. REPAIR NOSE SECTION FRAMING (SHEET 20 OF 21)

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NOTES

1. UNLESS OTHERWISE NOTED ON SHEETS OF THIS FIGURE, ALL SHEET METAL PARTS SHOWN ARE ALCLAD 7075-T6, QQ-A-250-13, AND ALL EXTRUSIONS ARE ALUMINUM ALLOY, 7075-T6511, QQ-A-200-11.
2. LEFT AND RIGHT SIDES ARE THE SAME EXCEPT WHEN NOTED.
3. **701015-SUBQ**
4. **701220-SUBQ**
5. **701334-SUBQ**
6. **701323-701428** (0.063).◀
7. **701361-SUBQ**
8. ALCLAD 2024-T3, QQ-A-250-5.
9. SANDWICH CONSTRUCTION: ALUMINUM HONEYCOMB MIL-C-7438 AND ALUMINUM SKINS.
10. **70778-701062** TOP SKIN IS 7075-T6, QQ-A-250-13 (0.020);
BOTTOM SKIN IS 7075-T6, QQ-A-250-13 (0.016).◀
11. **701188-SUBQ** TOP AND BOTTOM SKINS ARE 2024-T3, QQ-A-250-5 (0.020).◀
12. **701073-SUBQ**
13. STEEL, 4130 COND N, MIL-S-18729 (0.090).
14. STEEL (0.016) AND ALUMINUM (0.190).
15. BOTTOM SKIN: 6061-T6, QQ-A-250-11 (0.020).
16. **701828-SUBQ**
17. APPLY SEALING COMPOUND, ITEM 292, APPENDIX D, AROUND ENTIRE PERIMETER OF HINGE, WHERE ACCESSIBLE.

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FIGURE 2-6-8-1. REPAIR NOSE SECTION FRAMING (SHEET 21 OF 21)

2-6-9 NOSE SECTION PLATING REPAIR.

PARAGRAPH BREAKDOWN

	PAGE
2-6-9.1 Repair Nose Section Plating	2-6-9-1

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit

Materials/Parts

- Insulating Compound, Electrical, Item 182A, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 2-6-1
- PARA 2-6-2
- PARA 2-6-3
- PARA 2-6-5
- QQ-A-250

2-6-9.1 Repair Nose Section Plating.

- All plating with the exception of reinforced plastic covers is primary structure (Figure 2-6-9-1, Sheet 1 through Sheet 5). Evaluate damage (PARA 2-6-1).
- Refer to PARA 2-6-2 for typical skin repairs.

- Refer to PARA 2-6-3 for typical reinforced plastic repairs.
- For corrosion protection and watertightness, seal all plating repairs (PARA 2-6-1).
- Touch up paint (PARA 2-6-5).

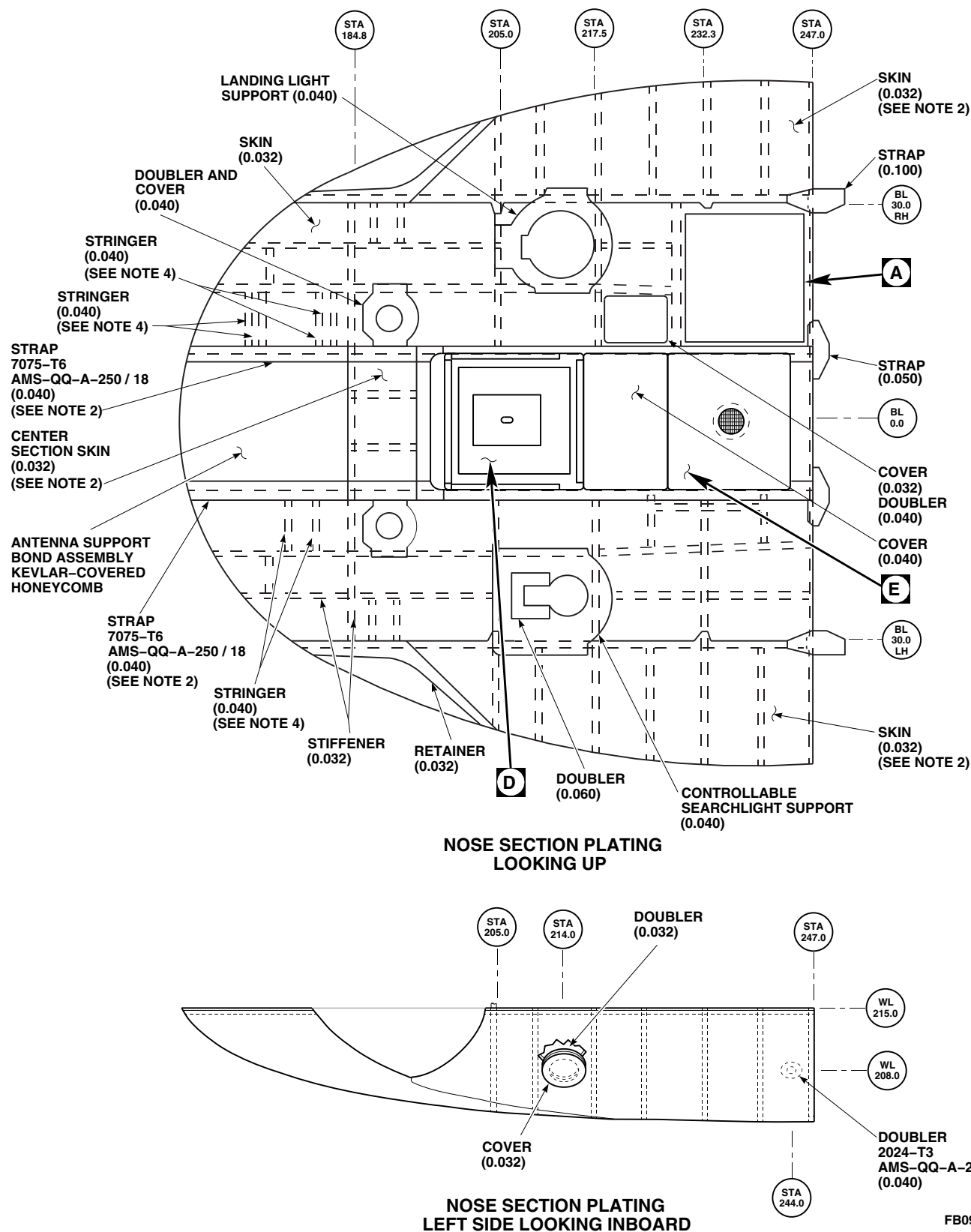
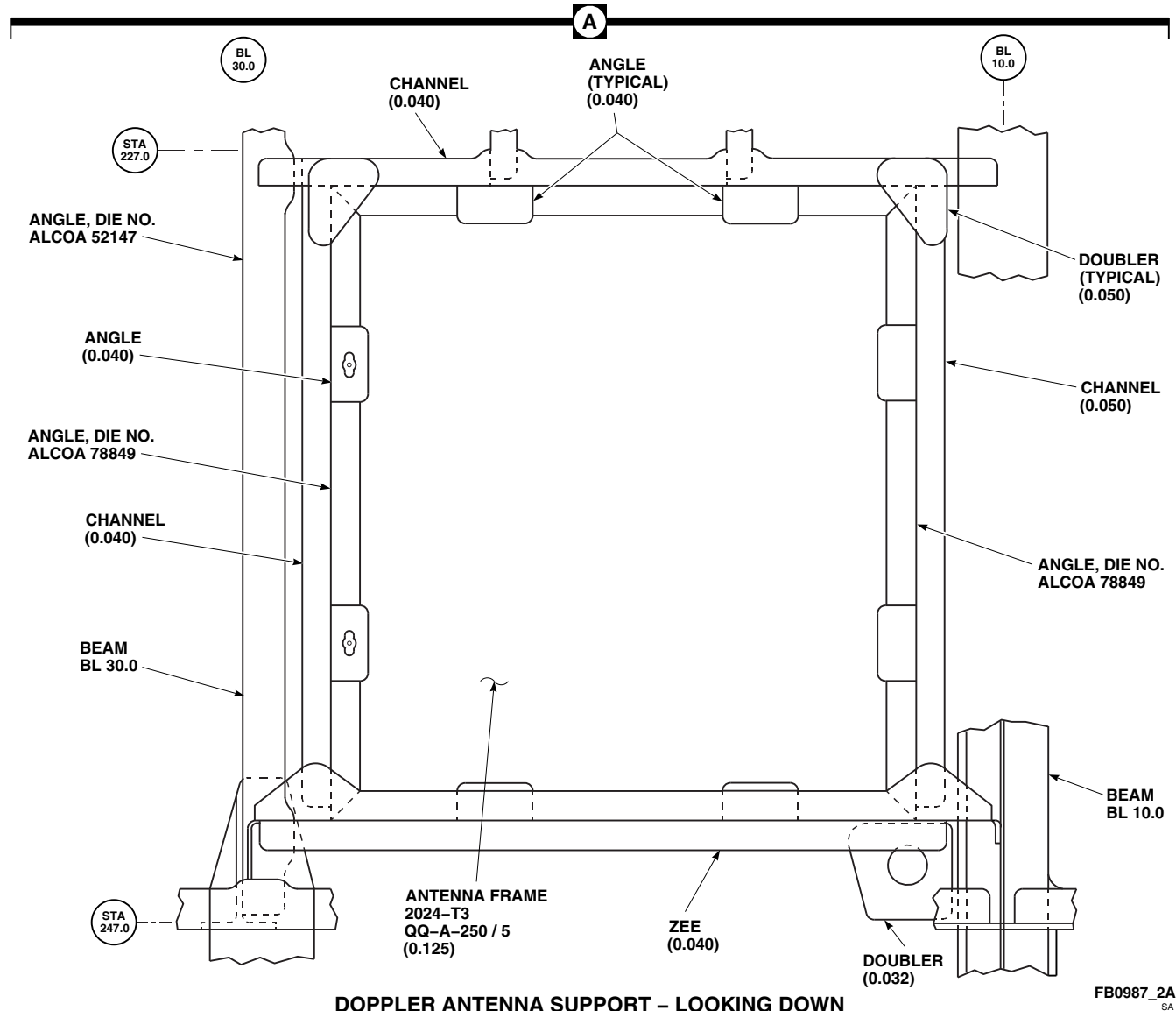
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FIGURE 2-6-9-1. REPAIR NOSE SECTION PLATING (SHEET 1 OF 5)



DOPPLER ANTENNA SUPPORT - LOOKING DOWN
FIGURE 2-6-9-1. REPAIR NOSE SECTION PLATING (SHEET 2 OF 5)

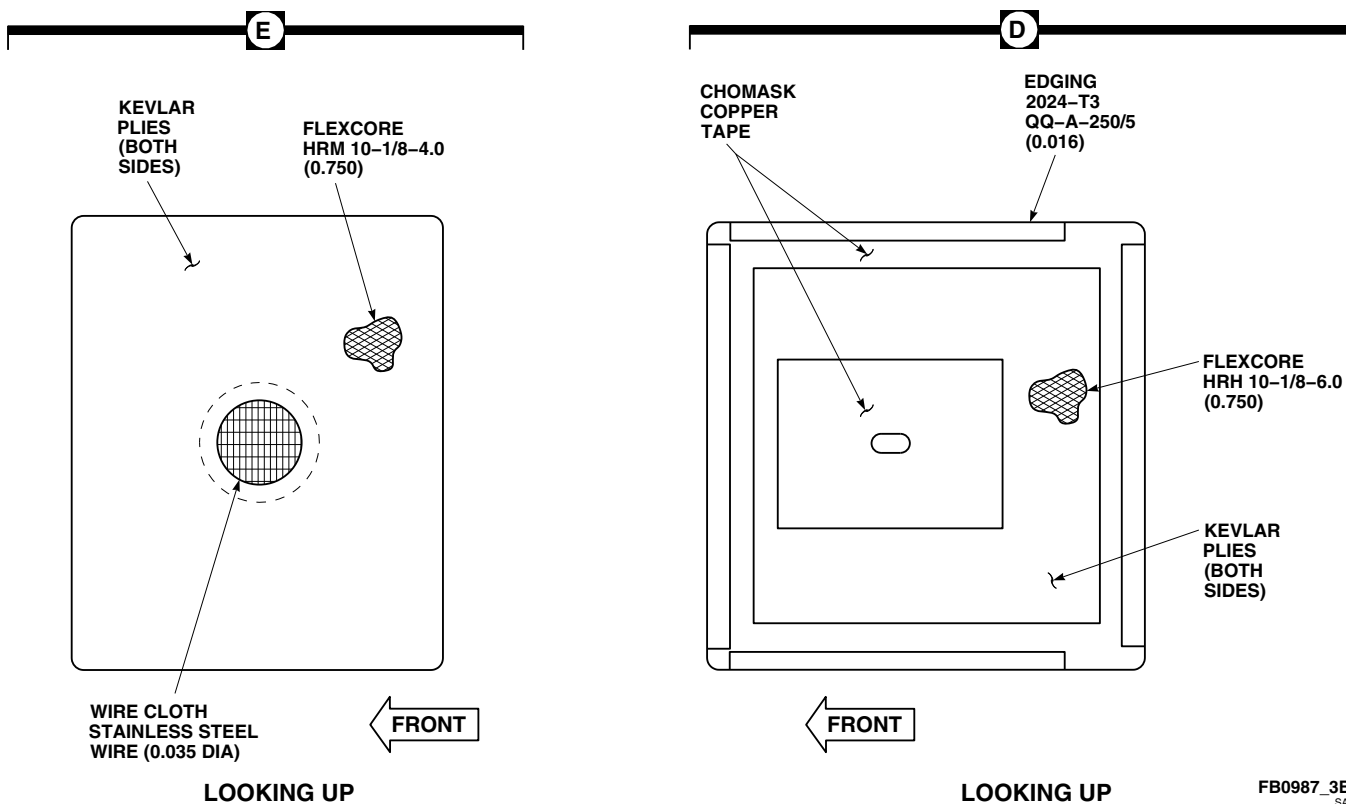
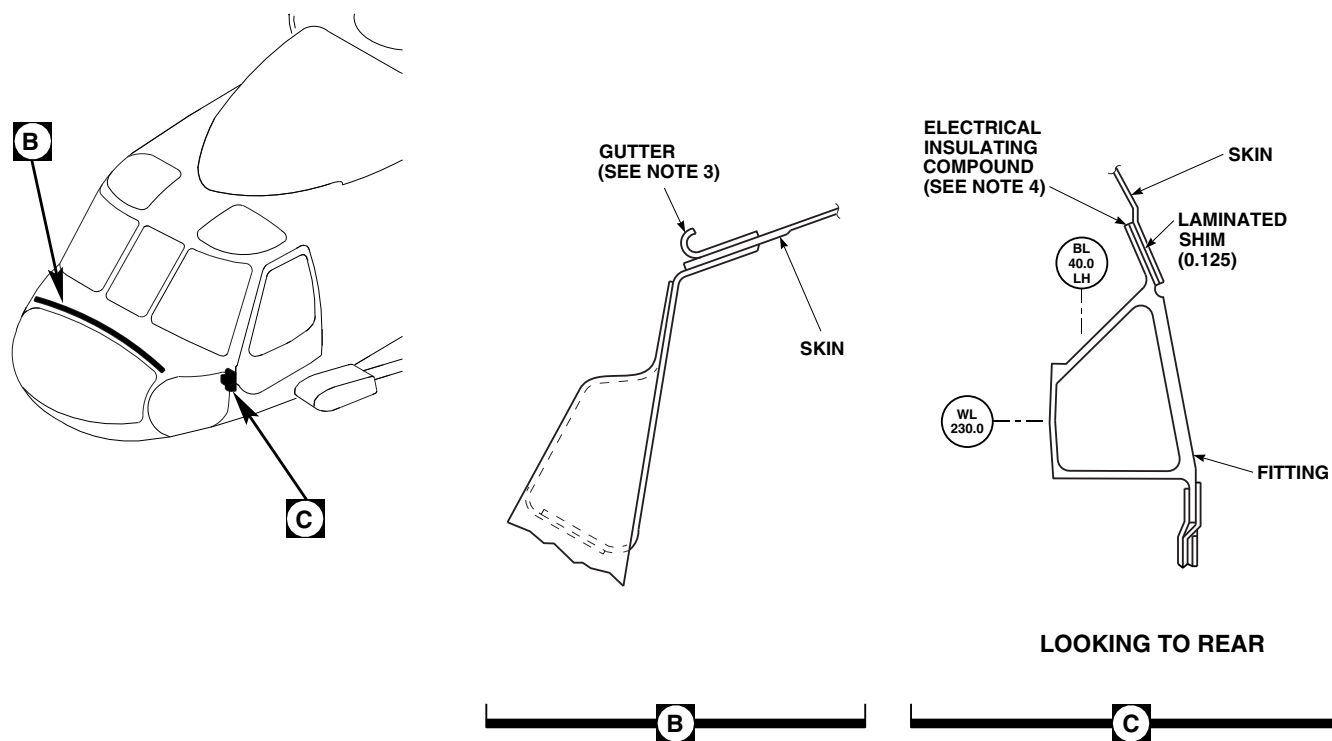


FIGURE 2-6-9-1. REPAIR NOSE SECTION PLATING (SHEET 3 OF 5)

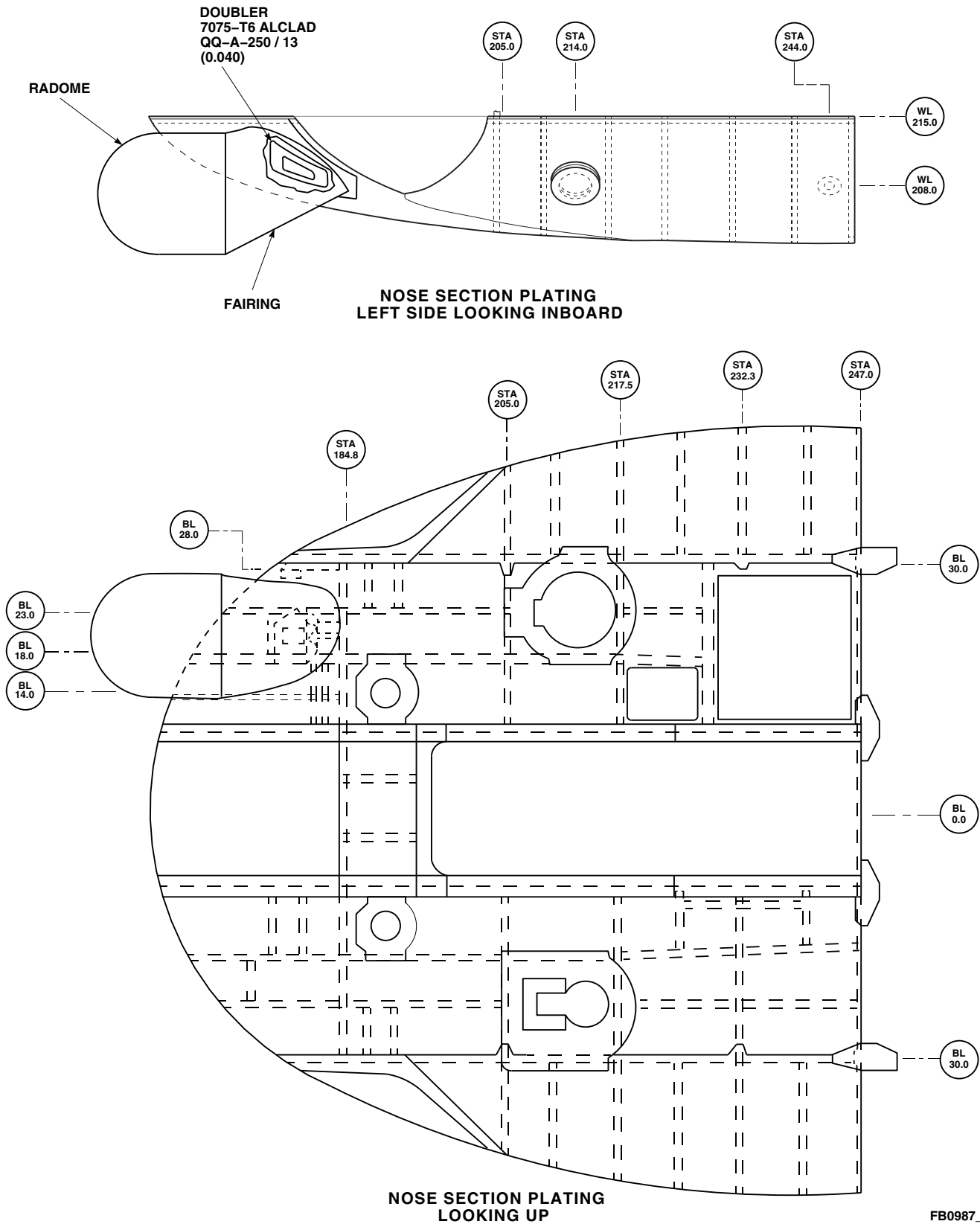


FIGURE 2-6-9-1. REPAIR NOSE SECTION PLATING (SHEET 4 OF 5)

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NOTES

1. SEAL WITH SEALING COMPOUND, ITEM 292, APPENDIX D. REFER TO PARA 2-6-1.
2. **70808-SUBQ**
3. **701134-SUBQ**
4. SHIM WITH ELECTRICAL INSULATING COMPOUND, ITEM 182A, APPENDIX D, AS REQUIRED.
5. ALL DIMENSIONS ARE IN INCHES, UNLESS NOTED.

FB0987_5C
SA**FIGURE 2-6-9-1. REPAIR NOSE SECTION PLATING (SHEET 5 OF 5)**

2-6-10 CABIN SECTION FORWARD FRAMING REPAIR.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
2-6-10.1	Repair Cabin Section Forward Framing	2-6-10-2	2-6-10.2	Inspect/Repair Primary Servo Beam Rail Fittings	2-6-10-2

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Coin, Soft Copper (or equivalent metal), 3/4-inch diameter x 1/8 - 3/16-inch thick, Procured or Locally-Made
- Feeler Gage, 0.0015-inch
- Fluorescent Penetrant Inspection Kit
- Grease Pencil
- Tapping Tool, Copper or Brass Rod (Rockwell "B" hardness of 70 or less), sized to fit into 0.750-inch diameter hole, Procured or Locally-Made

Materials/Parts

- Abrasive Cloth, Item 6, Appendix D
- Acetone, Item 25, Appendix D
- Adhesive, Item 32, Appendix D
- Cheesecloth, Item 90, Appendix D
- Scouring Pad, Item 270, Appendix D

- Sealing Compound, Item 292, Appendix D

References

- AMS 5643
- AMS 6359
- Appendix D
- ASTM E1417
- MIL-S-22499
- MIL-STD-670
- PARA 2-6-1
- PARA 2-6-2
- PARA 2-6-4
- PARA 2-6-5
- PARA 11-4-87
- QQ-A-200
- QQ-A-225
- QQ-A-250
- QQ-A-367

2-6-10.1 Repair Cabin Section Forward Framing.

- a. Forward cabin framing includes formed and extruded members made from 7075-T6, QQ-A-250/13, Alclad sheet stock, and 7075-T6511, QQ-A-200/11, aluminum alloy extrusion. Landing gear attachment fittings, bellcrank support fittings, and frame fittings at STA 295.0 and STA 308.0 are made from forged aluminum alloy and heat-treated to 7075-T73, QQ-A-367 condition. Tracks for sliding cover (main rotor pylon) and crew chief's window are made from 7075-T6511 extruded sections. Fittings under crew chief's window are made from 4340 and 4130 steel (Figure 2-6-10-1, Sheet 1 through Sheet 16).
- b. All forward cabin framing is primary structure. Fittings, however, are considered critical primary structure. Negligible damage to fittings is limited to nicks, scratches, and corrosion, no deeper than 5% of material thickness at point of damage. Blend damage smoothly to surrounding area, using long, even strokes. Polish out all surface imperfections and restore protective finish. Damage greater than negligible limits in critical areas must be reviewed by engineering authority for damage evaluation, and either repair or replacement. All repairs to forward cabin structure must maintain design strength and alignment.
- c. Evaluate damage in noncritical areas (PARA 2-6-1).
- d. Refer to PARA 2-6-2 and PARA 2-6-4 for typical framing repairs.
- e. For corrosion protection and watertightness, seal all framing repairs (PARA 2-6-1).
- f. Touch up paint as required (PARA 2-6-5).

2-6-10.2 Inspect/Repair Primary Servo Beam Rail Fittings.

- a. If required, remove primary servo (PARA 11-4-87).
- b. Inspect primary servo beam rail fitting(s) for wear strip and barrel-nut liner disbonding as

follows (Figure 2-6-10-2, Sheet 1, Detail A and Detail B and Sheet 2, Detail C and Detail D):

NOTE

- Wear strip disbonding inspection includes wear strips located on top of primary servo beam rail fitting and barrel-nut hole liners.
- Inspection of wear strips for disbonding is not applicable for helicopters with servo beam fittings, 70219-02134-046 (-105), -048 (-106), -050 (-107), and -052 (-108) installed. Inspection of barrel-nut liners for disbonding is required.
 - (1) Clean exposed fitting surfaces using cheesecloth, Item 90, Appendix D, dampened with acetone, Item 25, Appendix D.

NOTE

Coin is to be smooth disc of soft copper (or equivalent metal), 3/4-inch diameter and 1/8 - 3/16-inch thick. Round and smooth edges of coin.

- (2) Procure or locally make coin.

NOTE

Tapping tool is to be made from copper or brass rod having Rockwell "B" hardness of 70 or less with blunt mass at tip. Tip is to be sized to fit into 0.750-inch diameter hole and still allow for adequate tapping motion.

- (3) Procure or locally make tapping tool.

NOTE

- Inspection for voids and disbonds should only be performed by personnel experienced with coin-tapping inspection techniques.
- Tapping tool is used to inspect barrel-nut liners.

- (4) Inspect wear strips and barrel-nut liners for voids and disbonds as follows:
 - (a) Hold coin or tapping tool loosely between thumb and forefinger. Tap lightly along bond line or areas suspected of having voids or disbonds, and listen for variations in tapping sound. A "sharp-solid" sound indicates good bond and a "dull-dead" sound indicates bond separation.
 - (b) Tap across suspected area in horizontal, vertical, and diagonal lines. Using grease pencil, mark points on wear strip where sound varies. Continue tapping and marking until outline of bond separation is completed.
 - (c) If voids have been detected, determine extent of void using 0.0015-inch feeler gage. **BONDS WITH EDGE VOIDS DEEPER THAN 0.040-INCH EXTENDING OVER MORE THAN 5 % OF EDGE OF BOND PERIPHERY SHALL BE REJECTED.**
 - (d) **VOIDS SHALL NOT EXCEED 10 % OF BOND AREA. NO INDIVIDUAL VOID SHALL BE MORE THAN 1-INCH IN LENGTH.**
 - (e) If limits are exceeded, replace primary servo beam rail fitting at approved Depot maintenance facility.
- c. Inspect primary servo beam rail fitting(s) for damage as follows:

NOTE

Wear strip damage inspection is not applicable for helicopters with servo beam fittings, 70219-02134-046 (-105), -048 (-106), -050 (-107), and -052 (-108) installed.

- (1) Visually inspect servo beam wear strips for nicks and scratches as follows:

- (a) **NICKS ARE LIMITED TO 0.020-INCH DEEP AND 0.250-INCH DIAMETER AFTER BLENDING.**
- (b) **SCRATCHES ARE LIMITED TO 0.010-INCH DEEP, 0.120-INCH WIDE, AND 0.500-INCH LONG AFTER BLENDING.**

NOTE

Servo footprint is area within 0.500-inch radius of center of each servo beam fitting vertical hole.

- (c) **TOTAL SURFACE DAMAGE AREA OF NICKS AND SCRATCHES AFTER BLENDING SHALL NOT EXCEED 0.060 SQUARE-INCH ON WEAR STRIP UNDER SERVO FOOTPRINT. LIMIT DOES NOT APPLY OUTSIDE SERVO FOOTPRINT AREA.**
 - (d) Remove damage using scouring pad, Item 270, Appendix D.
 - (e) If limits are exceeded, replace primary servo beam rail fitting at approved Depot maintenance facility.
- (2) Visually inspect servo beam barrel-nut liners for nicks, scratches, and gouges as follows:
 - (a) **NICKS ARE LIMITED TO 0.012-INCH DEEP AFTER BLENDING, PROVIDED LINER INSIDE DIAMETER DOES NOT EXCEED 0.380-INCH.**
 - (b) **SCRATCHES AND GOUGES ARE LIMITED TO 0.003-INCH DEEP, 0.250-INCH WIDE, AND 0.900-INCH LONG AFTER BLENDING.**
 - (c) Remove damage using scouring pad, Item 270, Appendix D.
 - (d) If limits are exceeded, replace primary servo beam rail fitting at approved Depot maintenance facility.

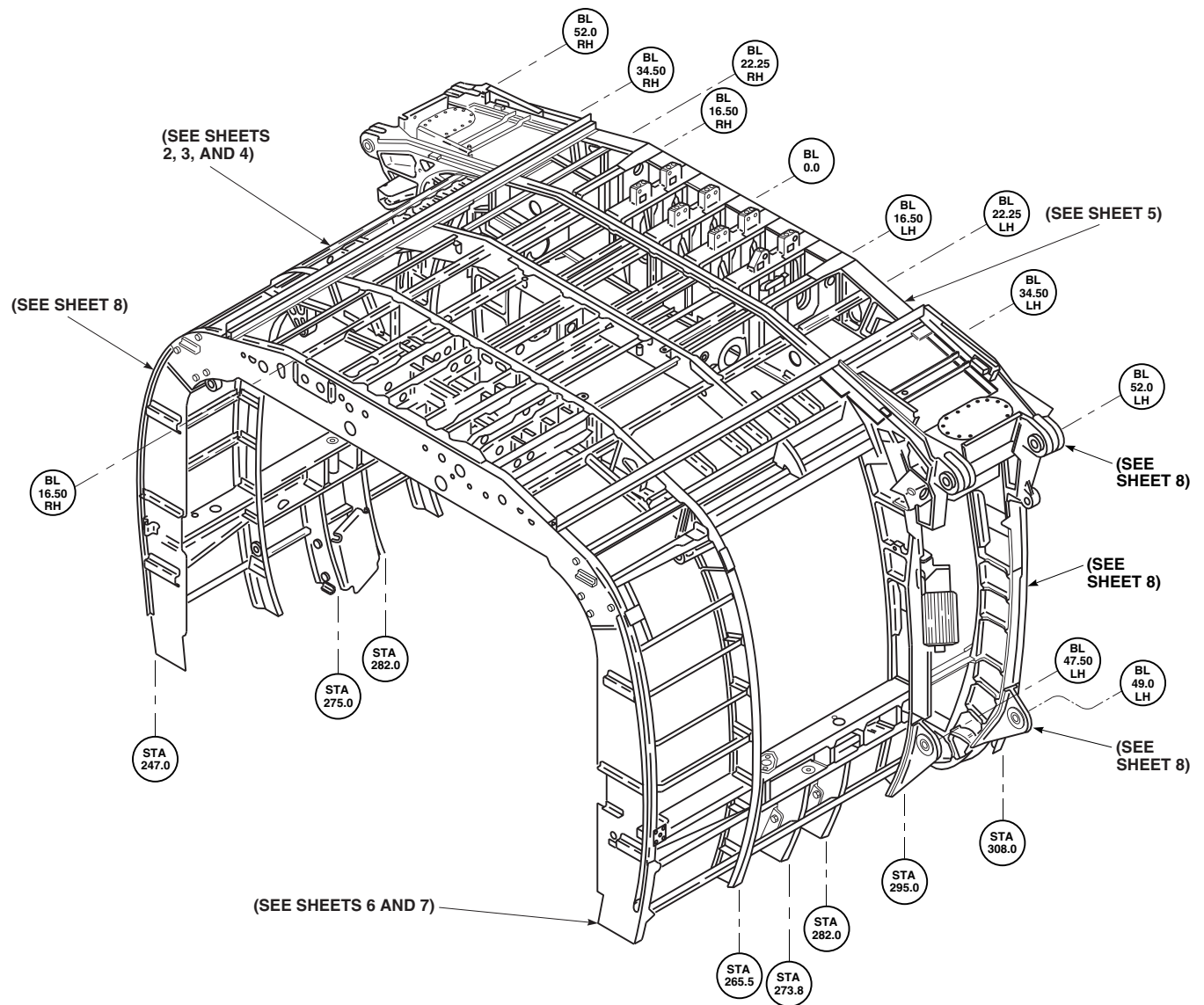
- (3) Visually inspect servo beam rail fittings for nicks, scratches, cracks, and corrosion as follows:

- (a) **NICKS, SCRATCHES, AND CORROSION FOUND ON CRITICAL FACES OF PAD ARE LIMITED TO 0.001-INCH DEEP, 0.250-INCH WIDE, AND 1.000-INCH LONG AFTER BLENDING.**
- (b) **NICKS, SCRATCHES, AND CORROSION FOUND ON REMAINING AREAS OF PAD ARE LIMITED TO 0.003-INCH DEEP, 1.000-INCH WIDE, AND 1.000-INCH LONG AFTER BLENDING.**
- (c) **ANY DAMAGE FOUND ON ROOT RADIUS IS LIMITED TO 0.001-INCH DEEP, 0.120-INCH WIDE, AND 1.000-INCH LONG AFTER BLENDING.**
- (d) Remove damage using abrasive cloth, Item 6, Appendix D.

WARNING**FSCAP**

Fluorescent penetrant inspection acceptance is a critical characteristic.

- (e) Using fluorescent penetrant inspection kit, inspect entire portion of servo beam fitting that extends above upper cabin skin for cracks (ASTM E1417). **NONE ALLOWED.**
 - (f) If limits are exceeded, replace primary servo beam rail fitting at approved Depot maintenance facility.
 - (g) Touch up paint as required (PARA 2-6-5).
- d. If required, install primary servo (PARA 11-4-87).



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FIGURE 2-6-10-1. REPAIR CABIN SECTION FORWARD FRAMING (SHEET 1 OF 16)

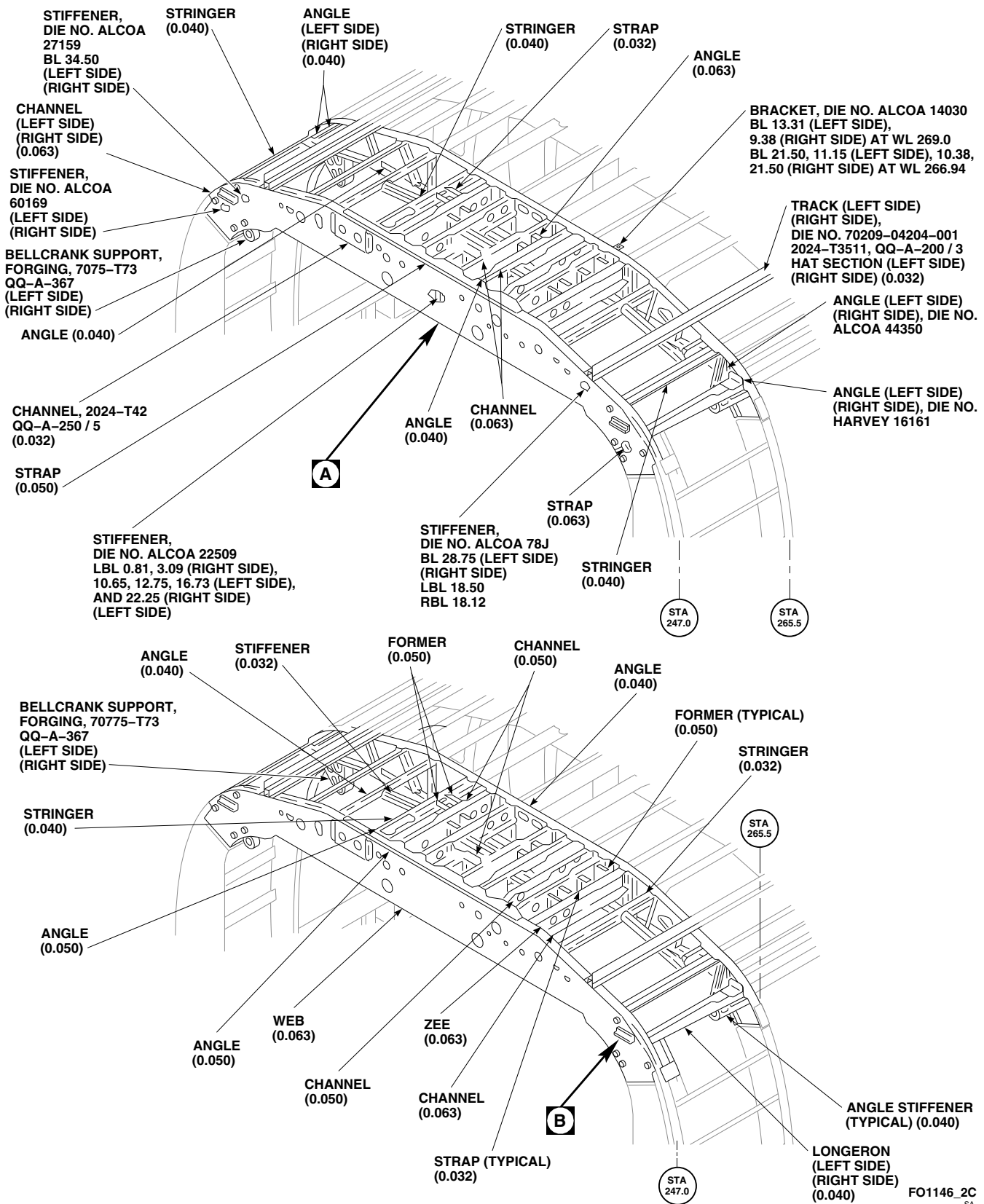


FIGURE 2-6-10-1. REPAIR CABIN SECTION FORWARD FRAMING (SHEET 2 OF 16)

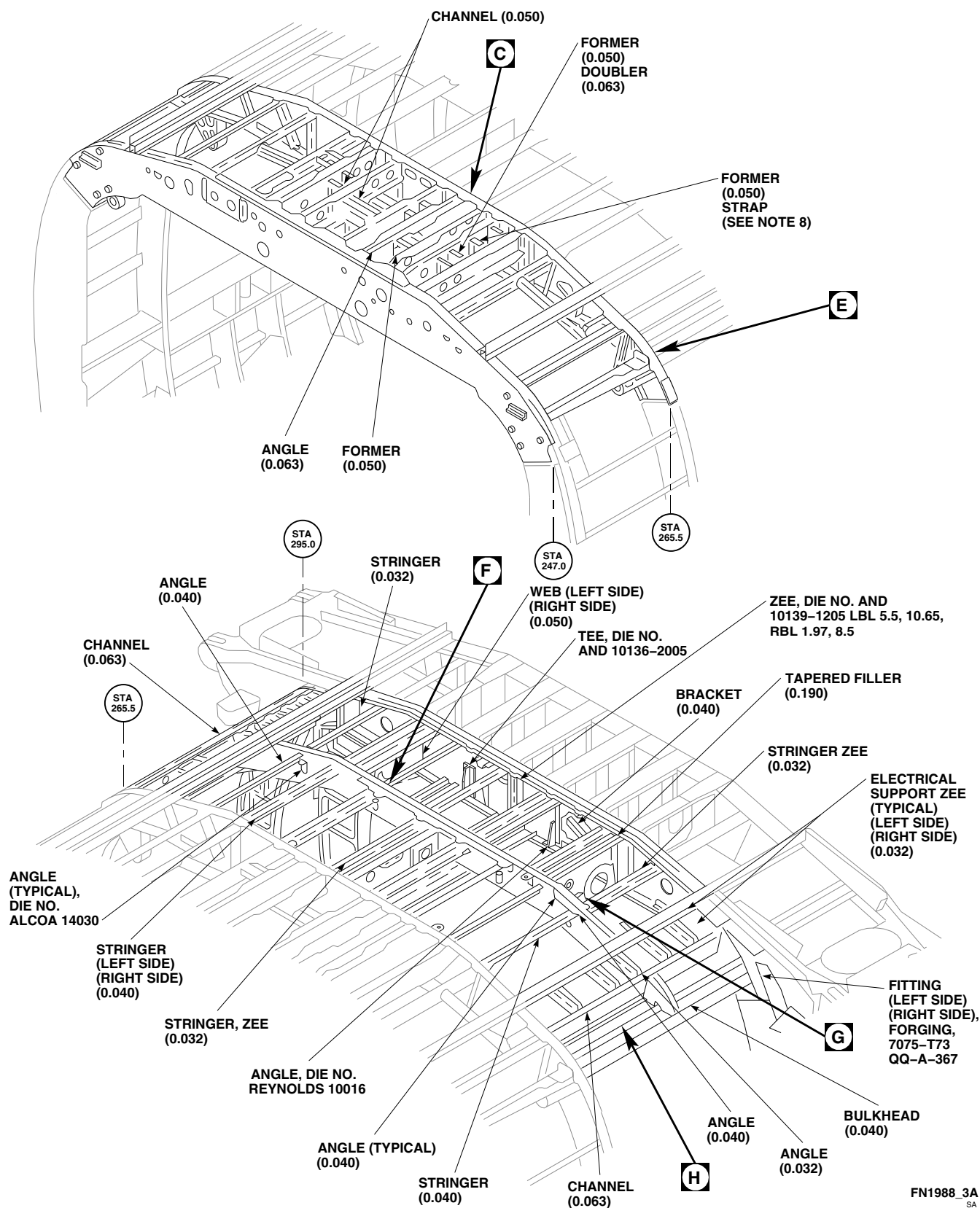


FIGURE 2-6-10-1. REPAIR CABIN SECTION FORWARD FRAMING (SHEET 3 OF 16)

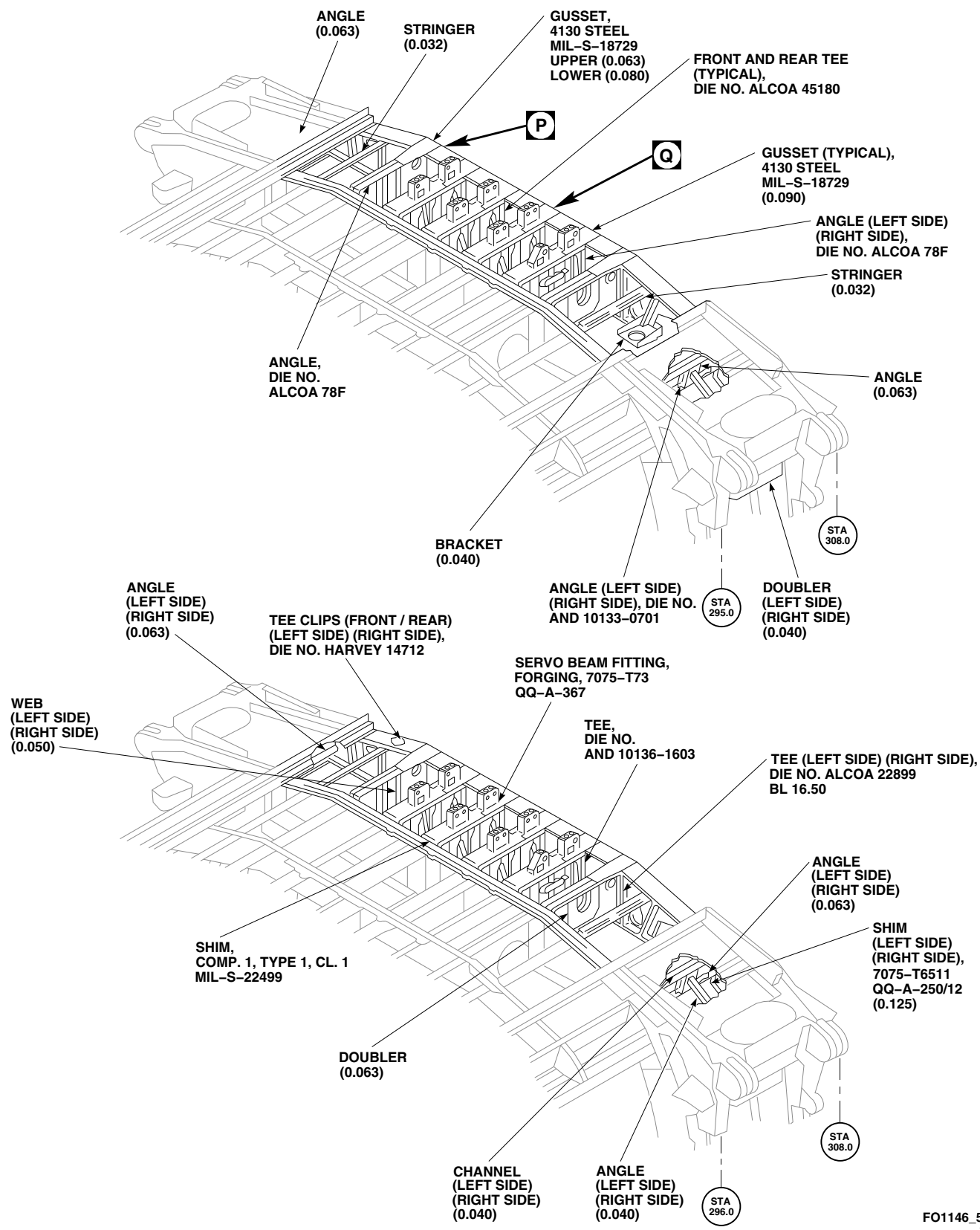


FIGURE 2-6-10-1. REPAIR CABIN SECTION FORWARD FRAMING (SHEET 5 OF 16)

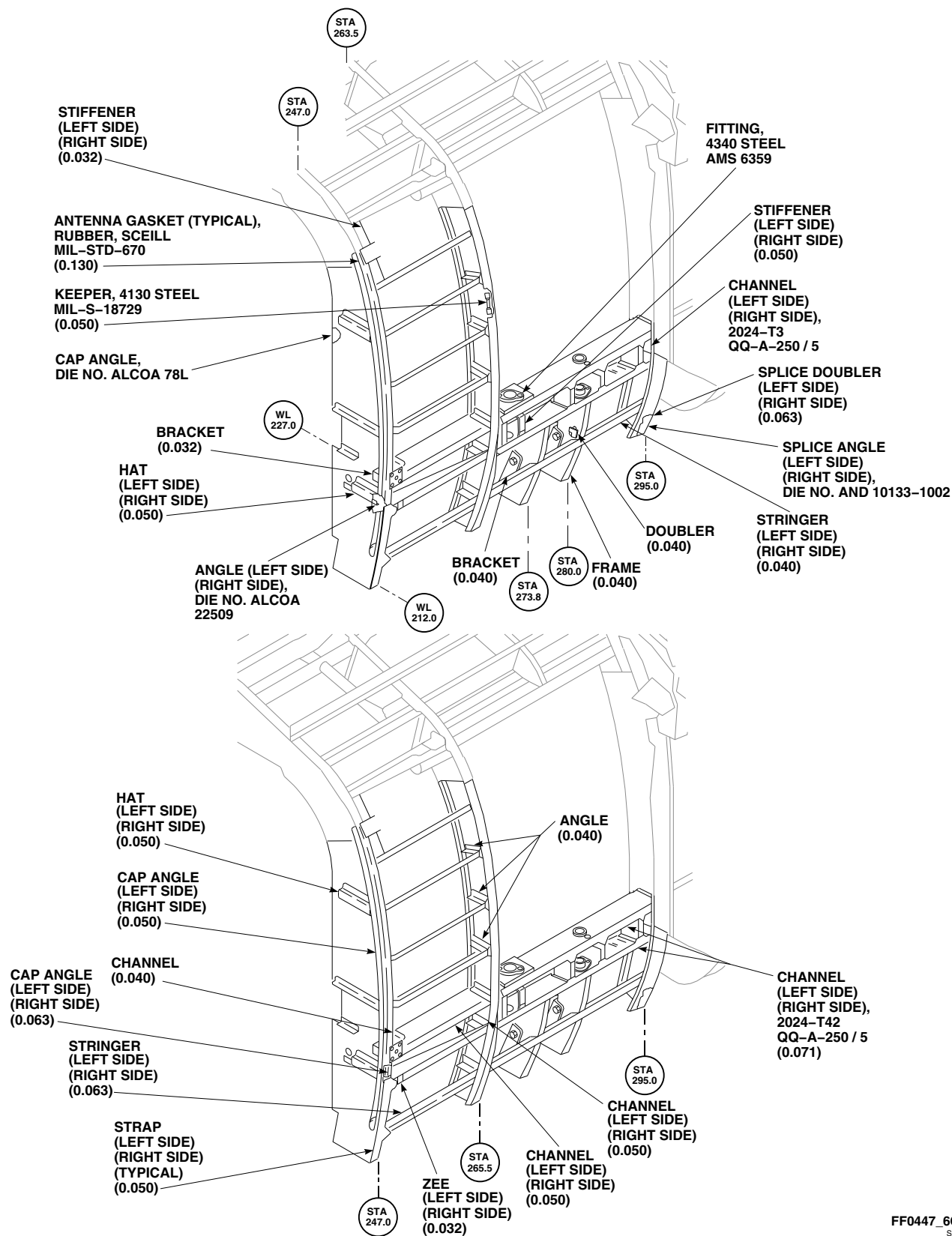


FIGURE 2-6-10-1. REPAIR CABIN SECTION FORWARD FRAMING (SHEET 6 OF 16)

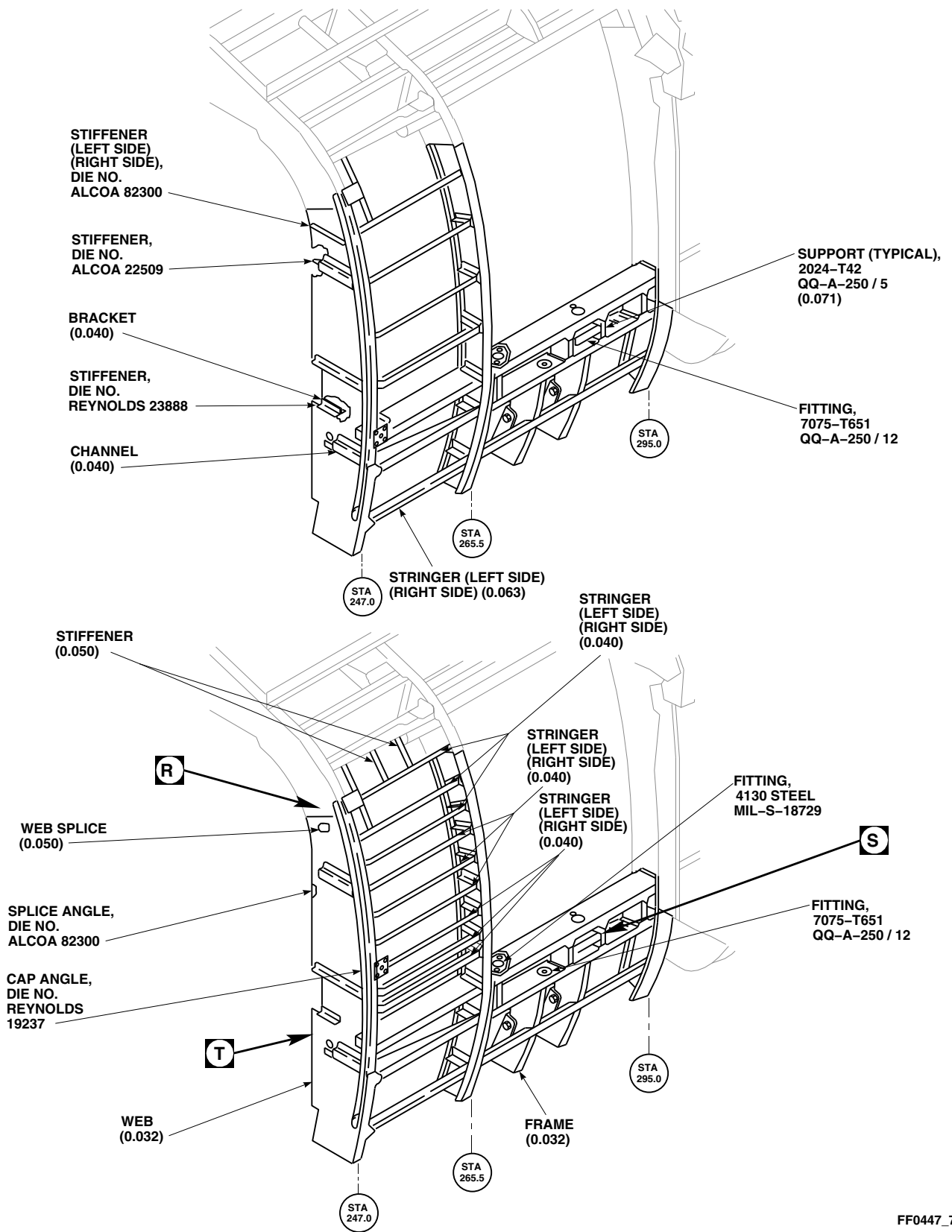


FIGURE 2-6-10-1. REPAIR CABIN SECTION FORWARD FRAMING (SHEET 7 OF 16)

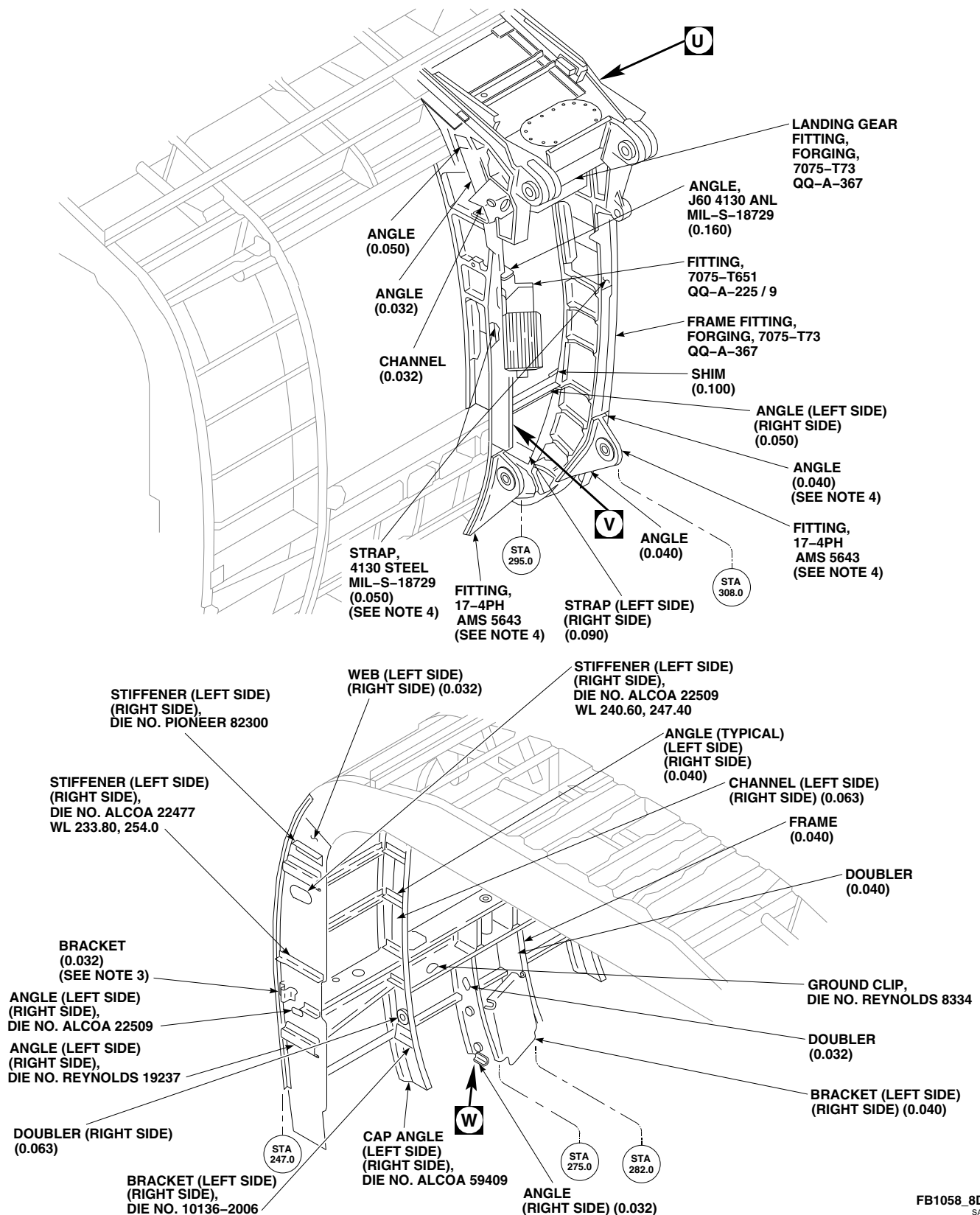


FIGURE 2-6-10-1. REPAIR CABIN SECTION FORWARD FRAMING (SHEET 8 OF 16)

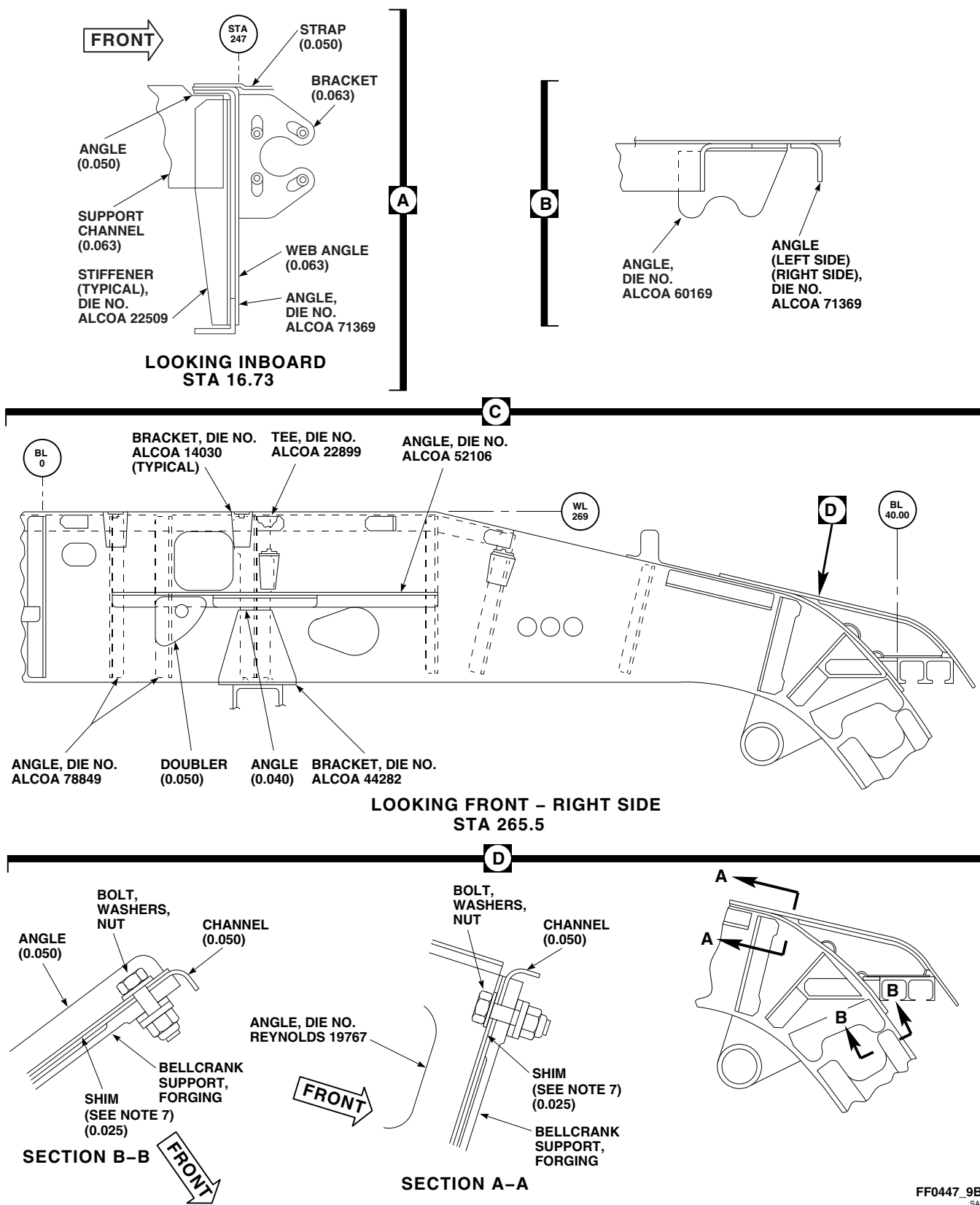


FIGURE 2-6-10-1. REPAIR CABIN SECTION FORWARD FRAMING (SHEET 9 OF 16)

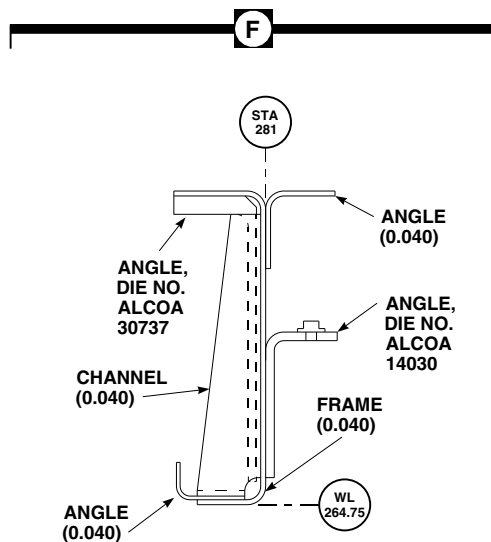
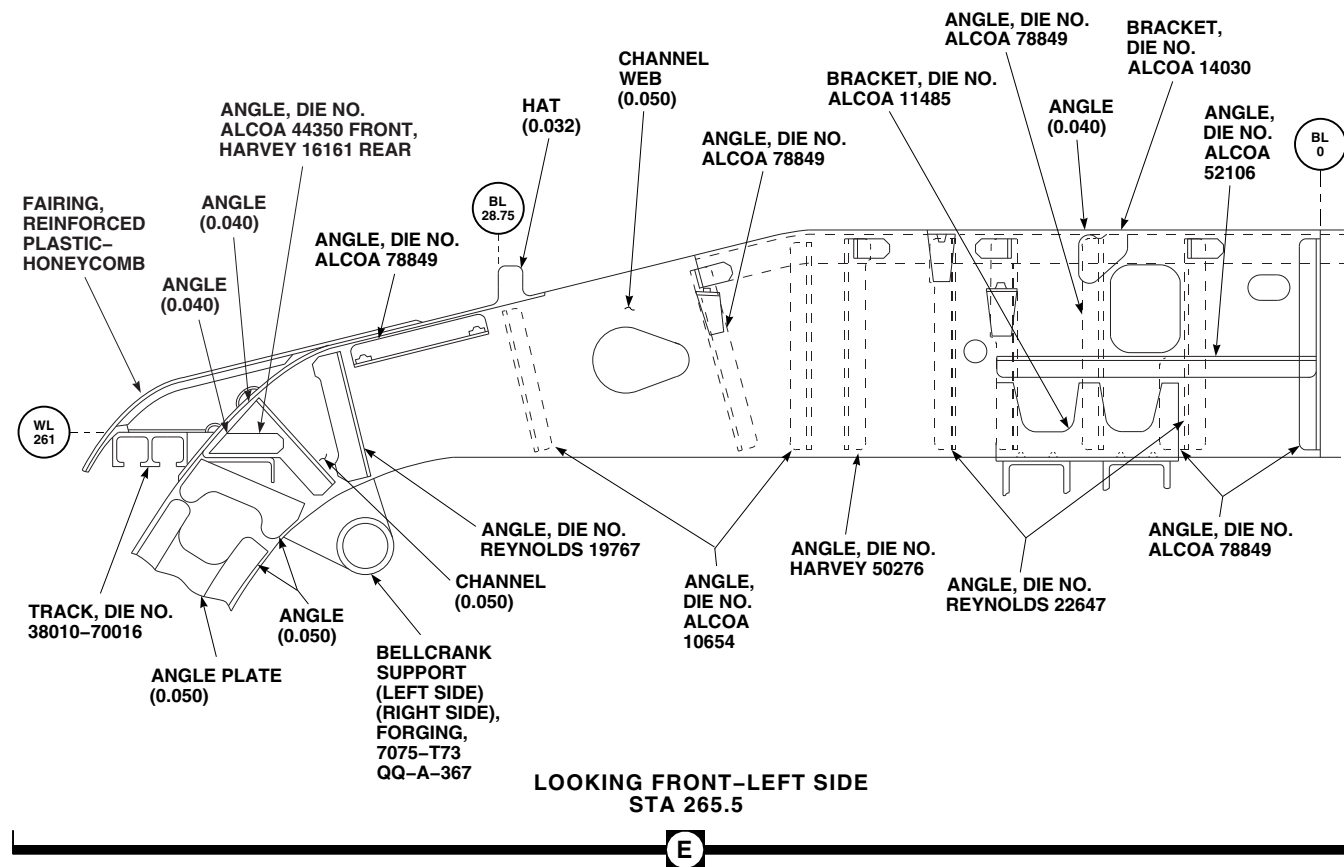
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FIGURE 2-6-10-1. REPAIR CABIN SECTION FORWARD FRAMING (SHEET 10 OF 16)

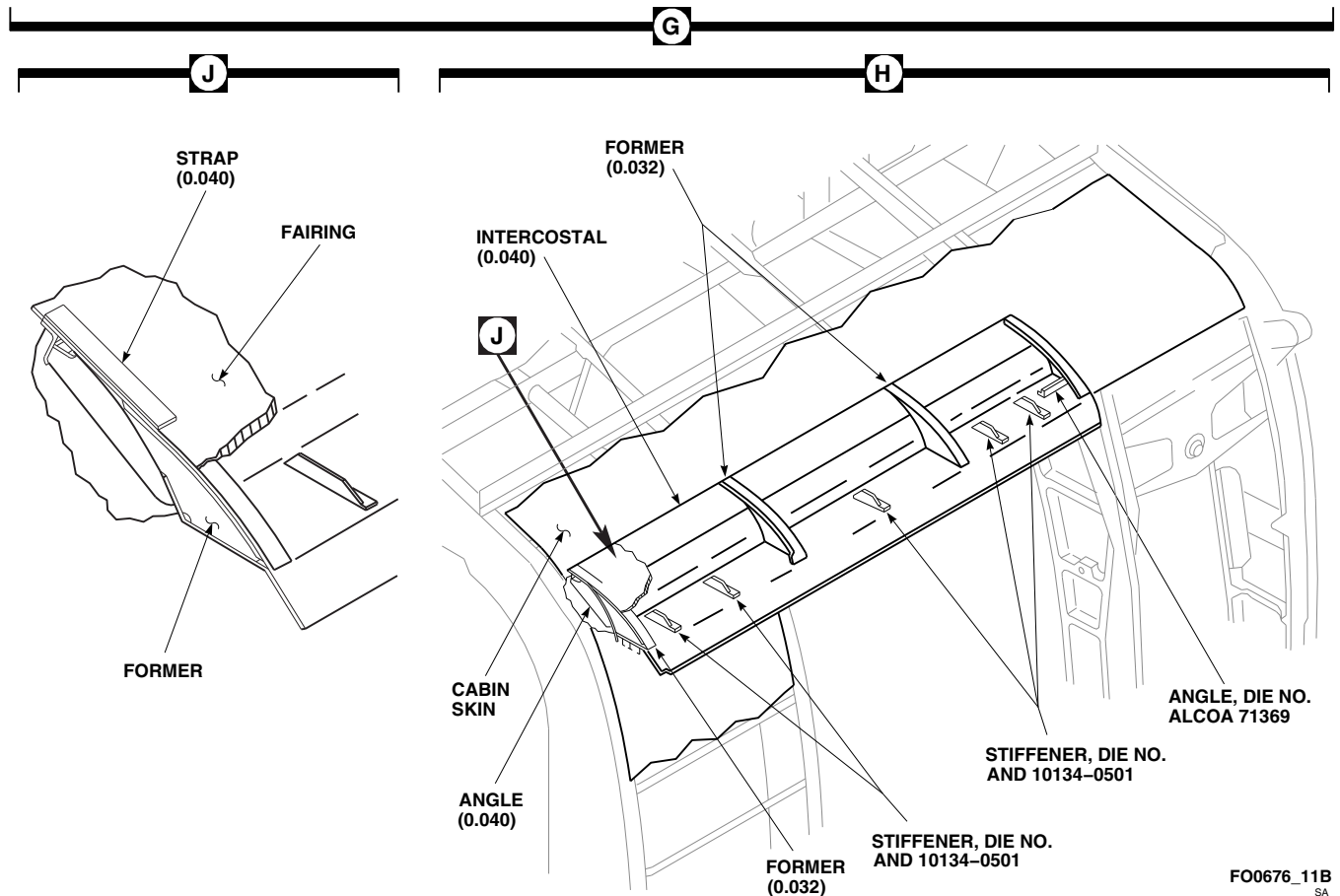
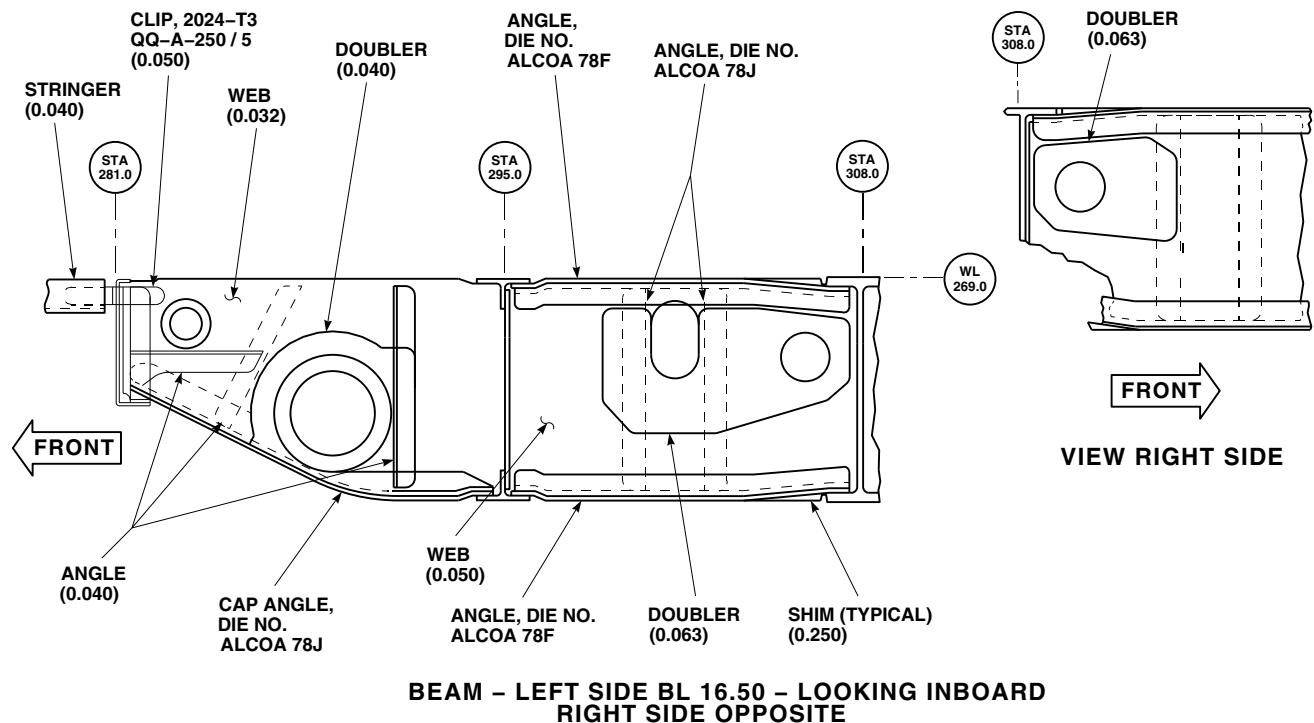


FIGURE 2-6-10-1. REPAIR CABIN SECTION FORWARD FRAMING (SHEET 11 OF 16)

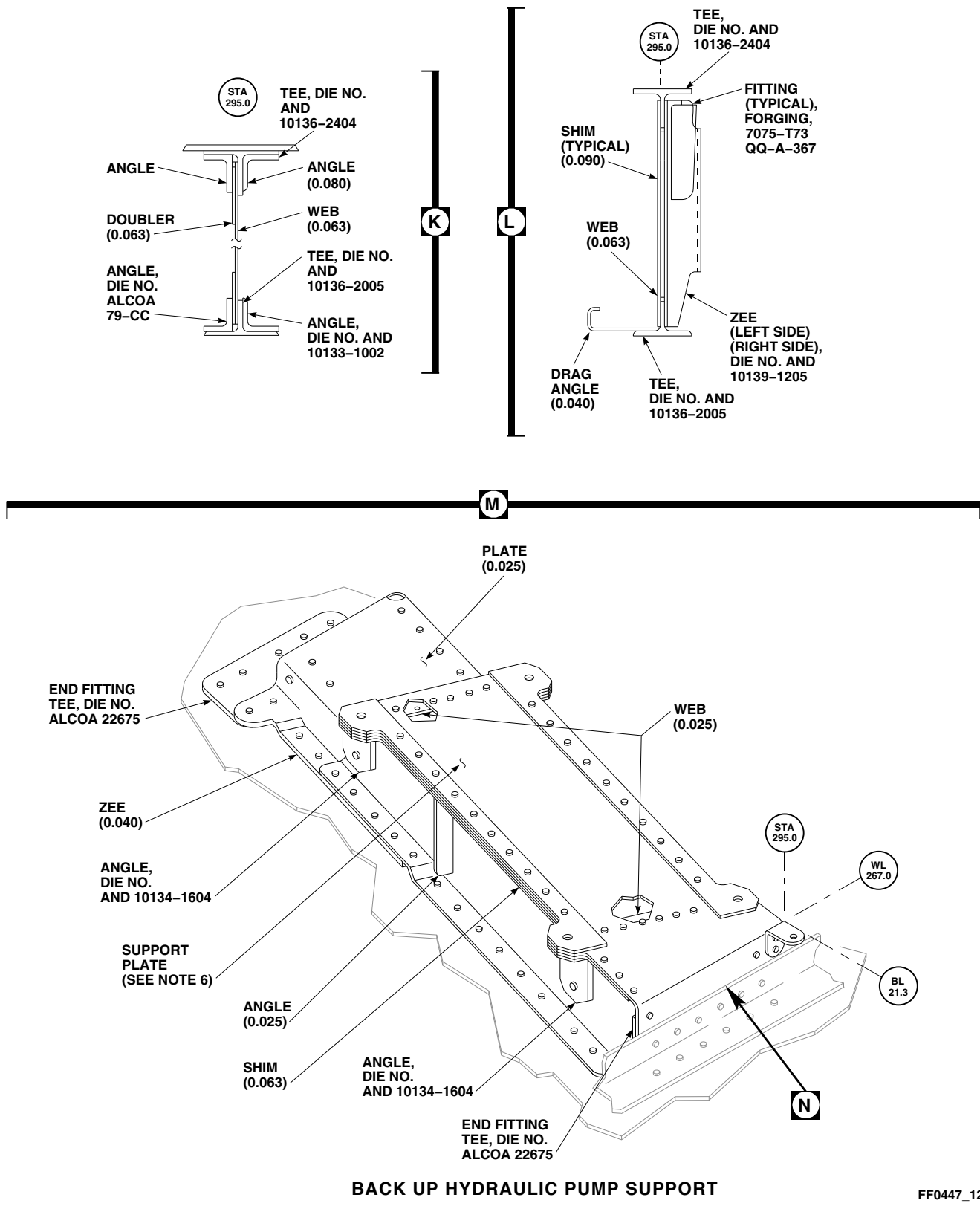


FIGURE 2-6-10-1. REPAIR CABIN SECTION FORWARD FRAMING (SHEET 12 OF 16)

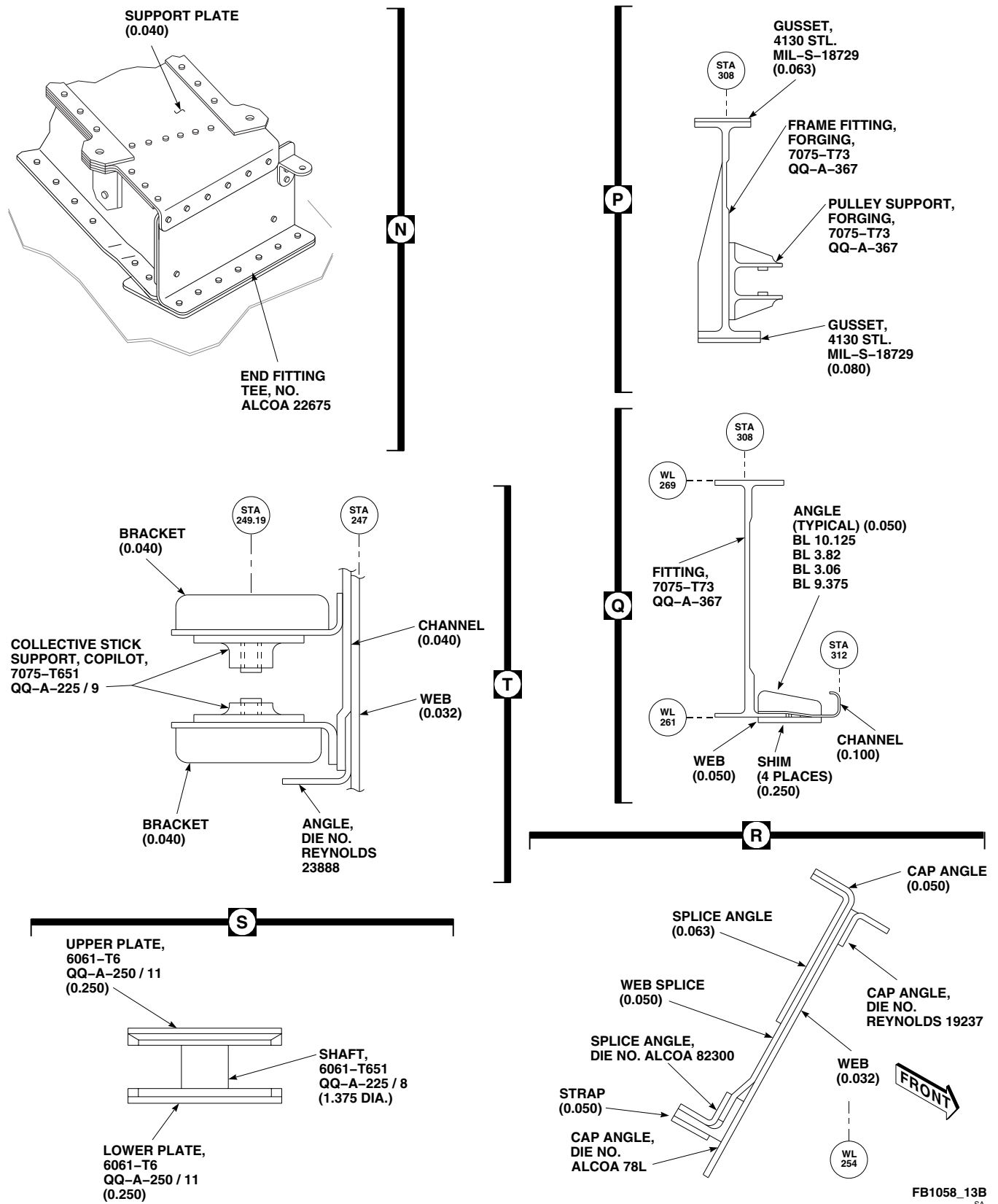
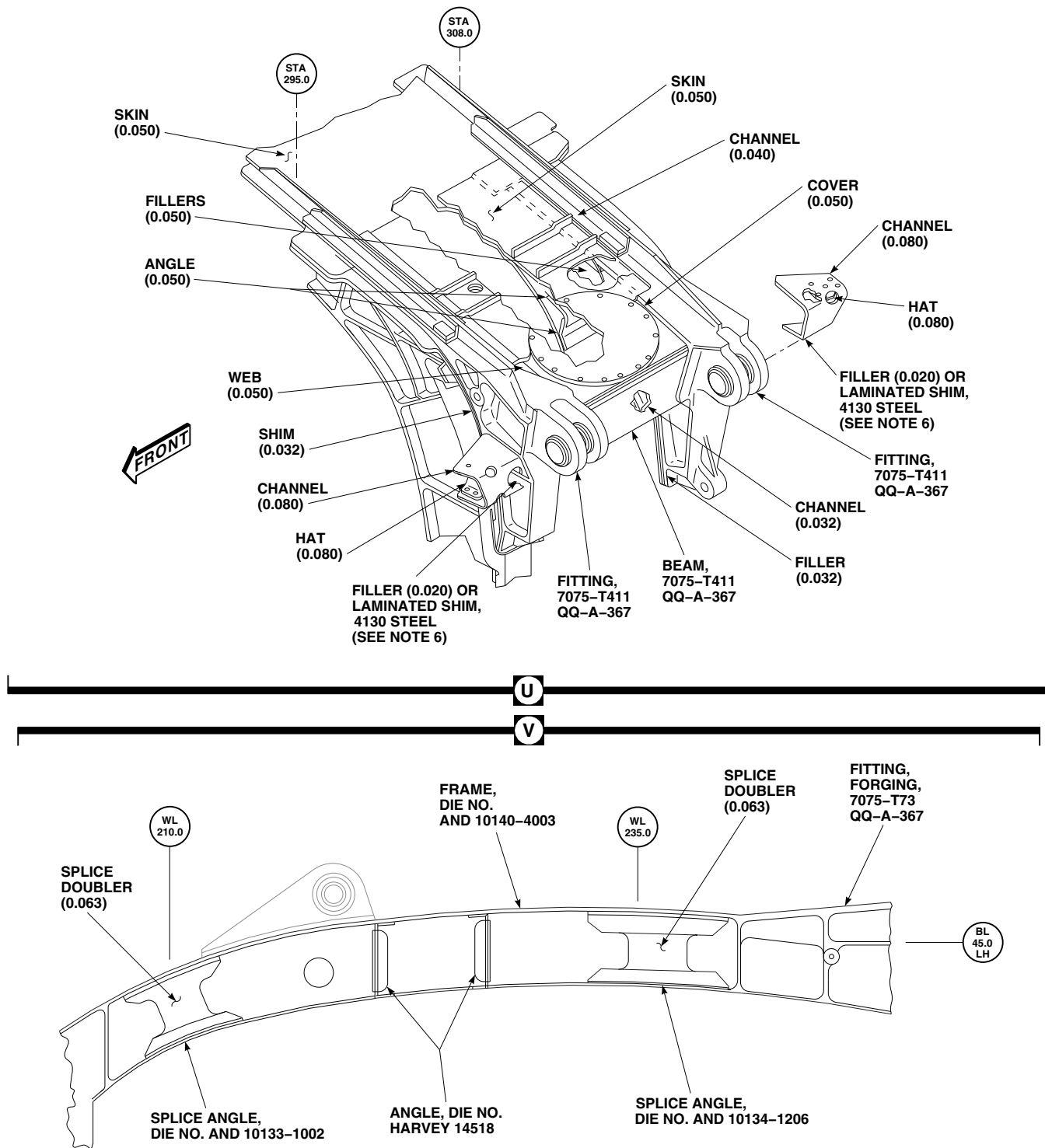


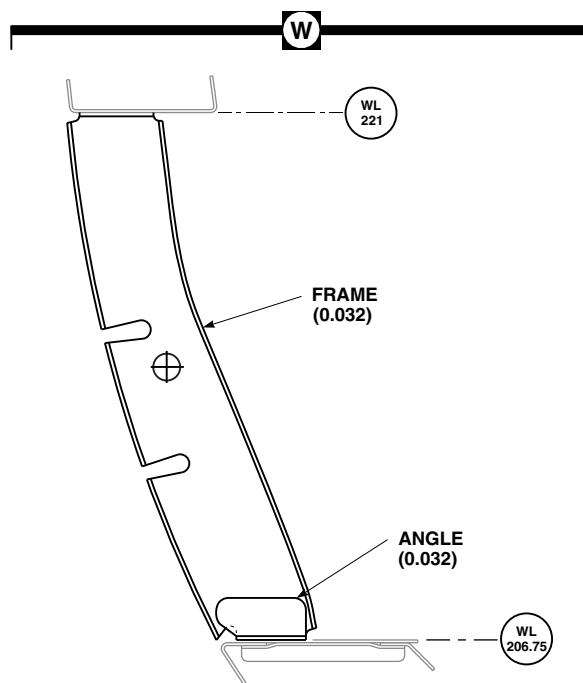
FIGURE 2-6-10-1. REPAIR CABIN SECTION FORWARD FRAMING (SHEET 13 OF 16)



STATION 295.0, LEFT SIDE LOOKING FRONT

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FIGURE 2-6-10-1. REPAIR CABIN SECTION FORWARD FRAMING (SHEET 14 OF 16)



LOOKING REAR – RIGHT SIDE – STA 275

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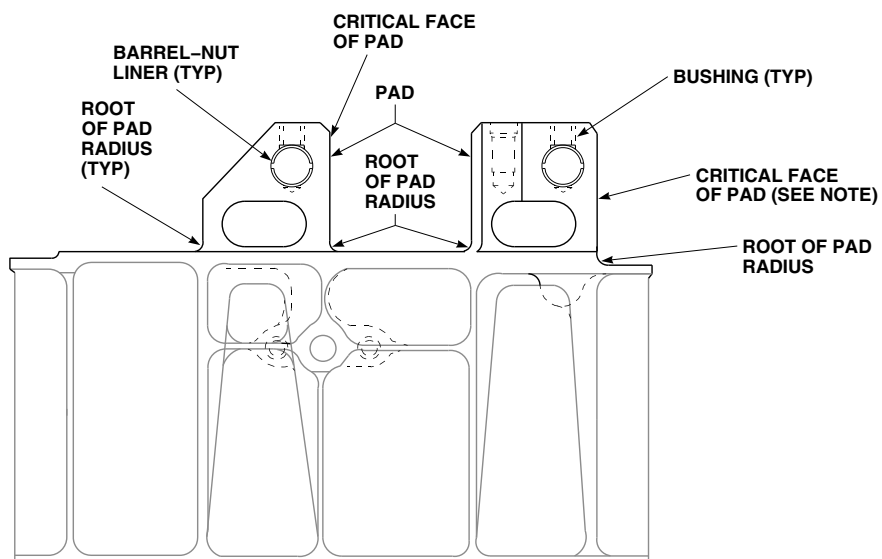
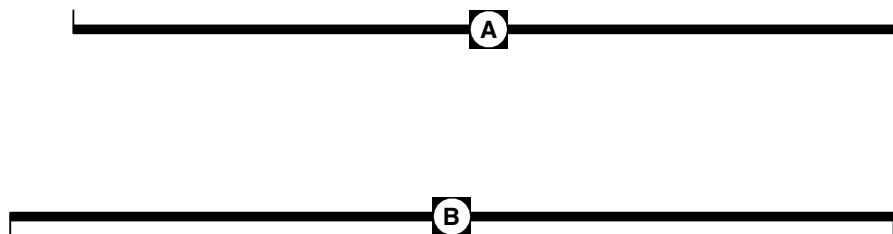
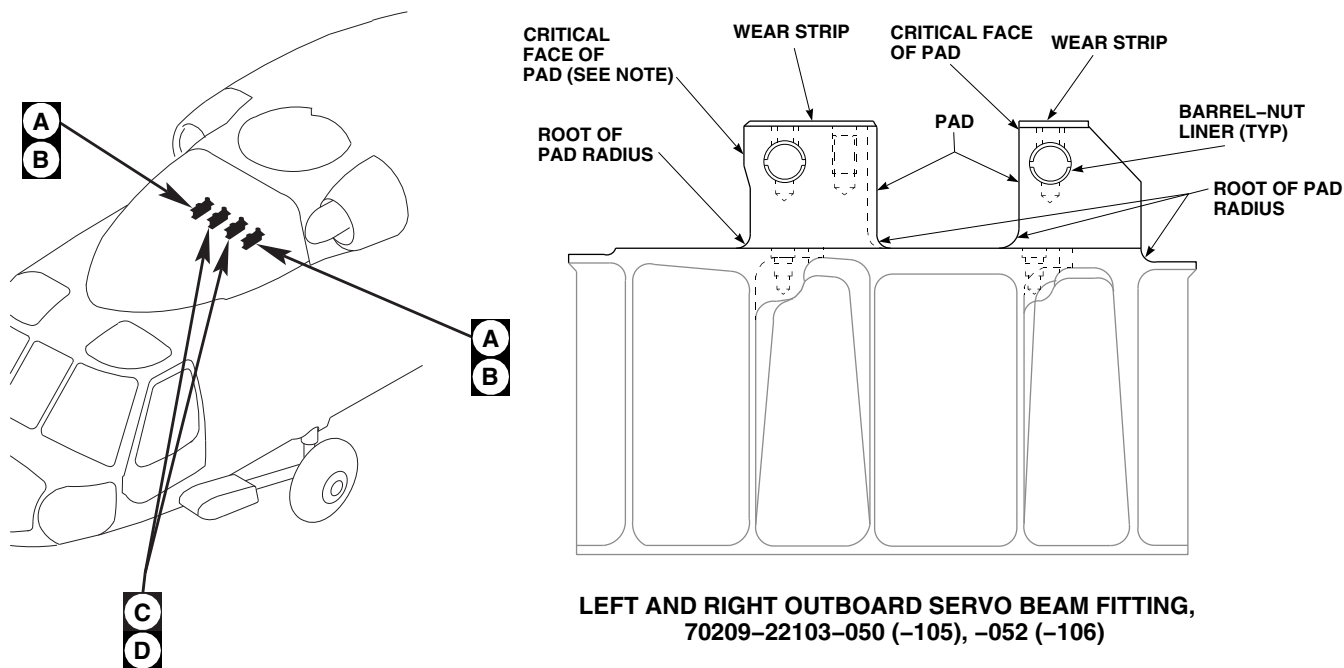
FIGURE 2-6-10-1. REPAIR CABIN SECTION FORWARD FRAMING (SHEET 15 OF 16)

NOTES

1. UNLESS OTHERWISE NOTED ON SHEETS OF THIS FIGURE, ALL SHEET METAL PARTS ARE ALCLAD 7075-T6, QQ-A-250 / 13, AND ALL EXTRUSIONS ARE ALUMINUM ALLOY 7075-T6511, QQ-A-200 / 11.
2. LEFT AND RIGHT SIDES ARE THE SAME, EXCEPT WHEN NOTED.
3. REFER TO PARA 2-6-5 FOR CONDUCTIVE PAINT AND GROUNDING REQUIREMENTS.
4. SEAL BETWEEN SURFACES OF STRAPS, FITTINGS, AND OTHER PARTS WITH SEALING COMPOUND, ITEM 292, APPENDIX D. USE ENOUGH SEALING COMPOUND TO FILL GAPS BETWEEN SURFACES.
5. APPLY ENOUGH SEALING COMPOUND, ITEM 292, APPENDIX D TO SQUEEZE OUT INTO GAP BETWEEN FITTING AND STRUCTURE (PARA 2-6-1).
6. USE 0.020 SOLID STOCK OR LAMINATED SHIM, MIL-S-22499, COMP 1, TYPE 1, CLASS 2, AS REQUIRED TO FILL GAP.
7. LEFT SIDE OPPOSITE.
8. BOND TO FRAME WITH ADHESIVE, ITEM 32, APPENDIX D.
9. ALL DIMENSIONS ARE IN INCHES, UNLESS NOTED.

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FIGURE 2-6-10-1. REPAIR CABIN SECTION FORWARD FRAMING (SHEET 16 OF 16)



NOTE
ALL PAD SURFACES ABOVE
BOTTOM OF BARREL NUT HOLES ARE
CRITICAL FACES (BOTH PADS).

LEFT AND RIGHT OUTBOARD SERVO BEAM FITTING,
70219-02134-046 (-105), -048 (-106)

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**FIGURE 2-6-10-2. INSPECT/REPAIR PRIMARY SERVO BEAM RAIL FITTINGS
(SHEET 1 OF 2)**

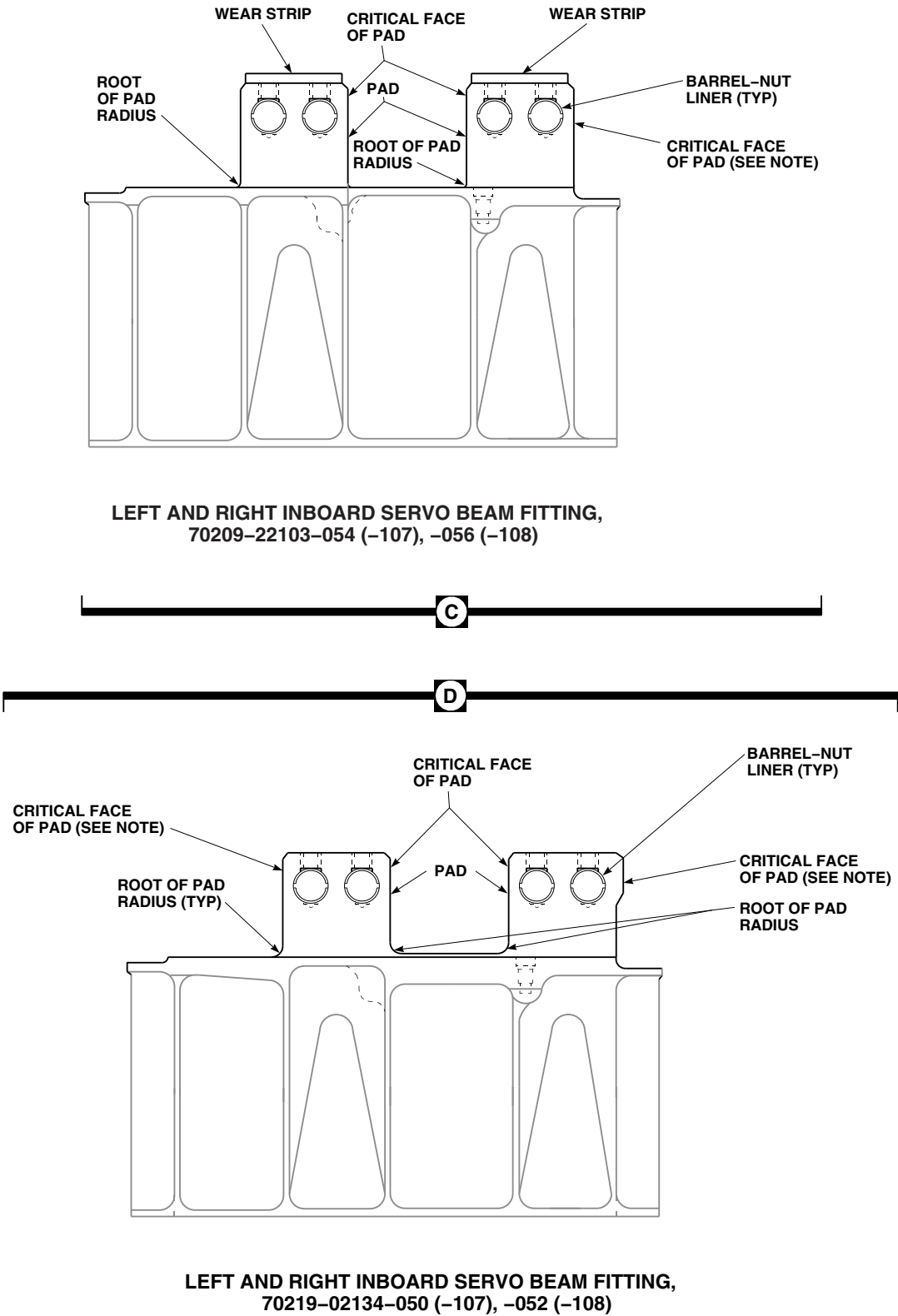


FIGURE 2-6-10-2. INSPECT/REPAIR PRIMARY SERVO BEAM RAIL FITTINGS
 (SHEET 2 OF 2)

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2-6-11 CABIN SECTION UPPER FRAMING REPAIR.

PARAGRAPH BREAKDOWN

	PAGE
2-6-11.1 Repair Cabin Section Upper Framing	2-6-11-1
INITIAL SETUP	MIL-S-18729
Tools	MIL-S-22499
Airframe Repairer’s Toolkit	PARA 2-6-1
Materials/Parts	PARA 2-6-2
Acetone, Item 25, Appendix D	PARA 2-6-4
Sealing Compound, Item 275, Appendix D	PARA 2-6-5
Sealing Compound, Item 292, Appendix D	QQ-A-200
References	QQ-A-225
AMS 4149	QQ-A-250
Appendix D	

2-6-11.1 Repair Cabin Section Upper Framing.

- Upper cabin framing (STA 308.0 to STA 398.0) includes formed and extruded members made from 7075-T6, QQ-A-250/13, Alclad sheet stock and 7075-T6511, QQ-A-200/11, aluminum alloy extrusion. Gussets at STA 308.0 are made from 4130 steel, MIL-S-18729. Main transmission support structure and the STA 379.0 frame fitting are forged aluminum alloy, 7175-T736, AMS 4149 (Figure 2-6-11-1, Sheet 1 through Sheet 11).
- All upper cabin framing is primary structure, however, the main gear box support structure

and STA 379.0 frame fitting are considered critical primary structure. Negligible damage to the forged fittings is limited to nicks, scratches, and corrosion, no deeper than 5% of material thickness, at point of damage. Blend damage smoothly to surrounding area, using long, even strokes. Polish out all surface imperfections and restore protective finish. Damage greater than negligible limits in critical areas must be reviewed by engineering authority for damage evaluation and either repair or replacement. All repairs to the upper cabin framing must maintain the design strength and alignment.

- Evaluate damage in noncritical areas (PARA 2-6-1).

- d. Use PARA 2-6-2 and PARA 2-6-4 for typical framing repairs.
- e. For corrosion protection and watertightness, seal all repairs between framing and plating (PARA 2-6-1).
- f. Touch up paint as required (PARA 2-6-5).

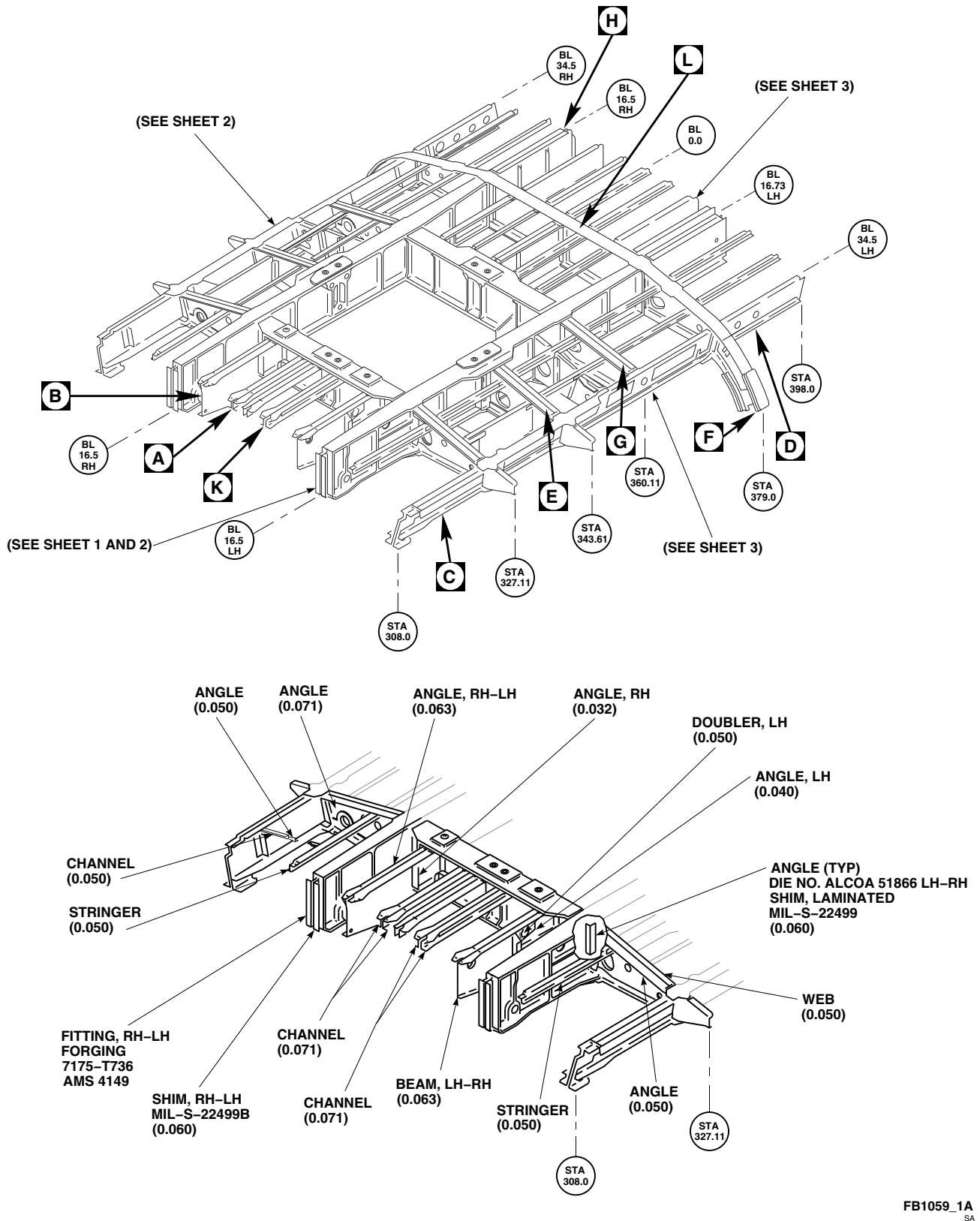


FIGURE 2-6-11-1. REPAIR CABIN SECTION UPPER FRAMING (SHEET 1 OF 11)

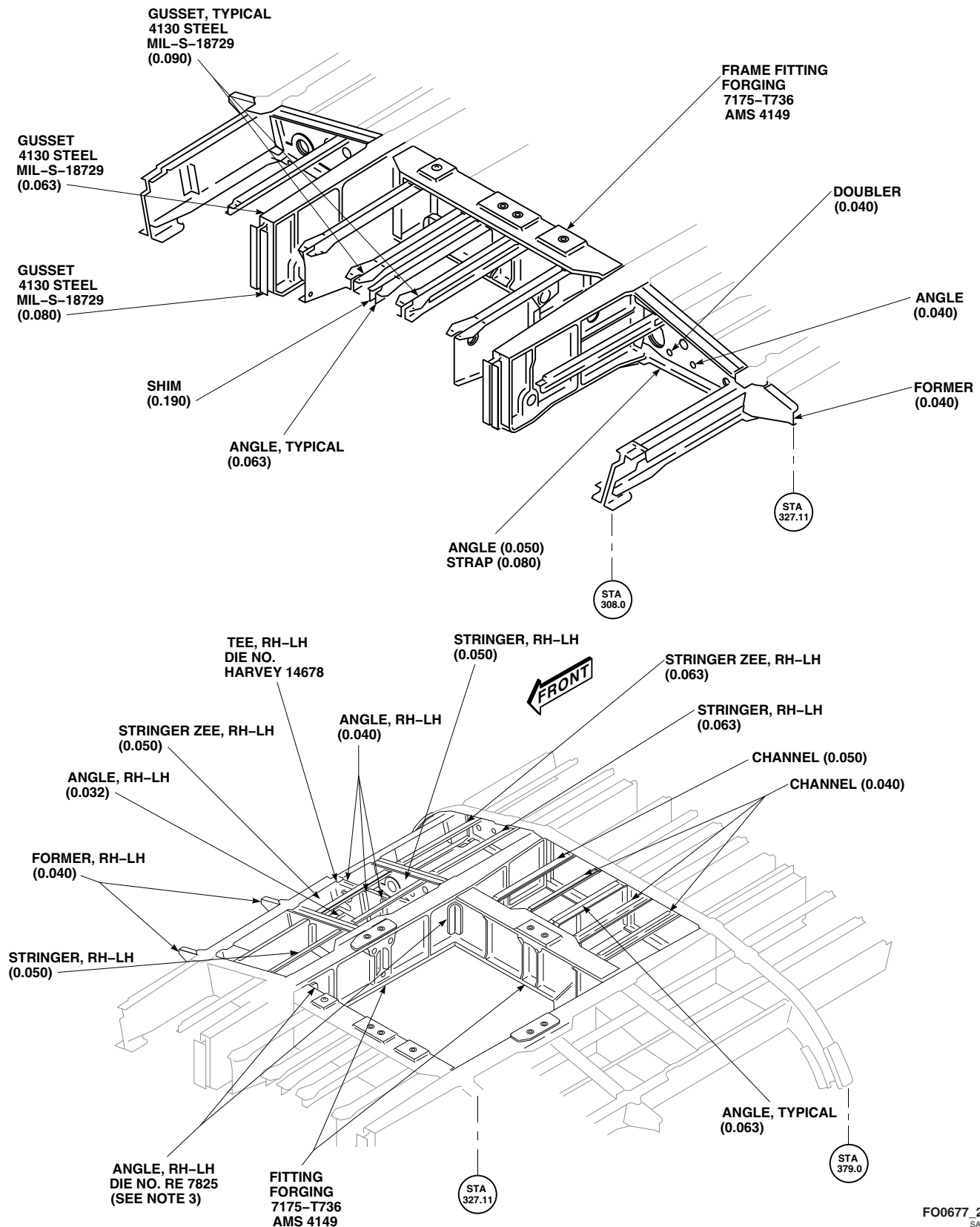
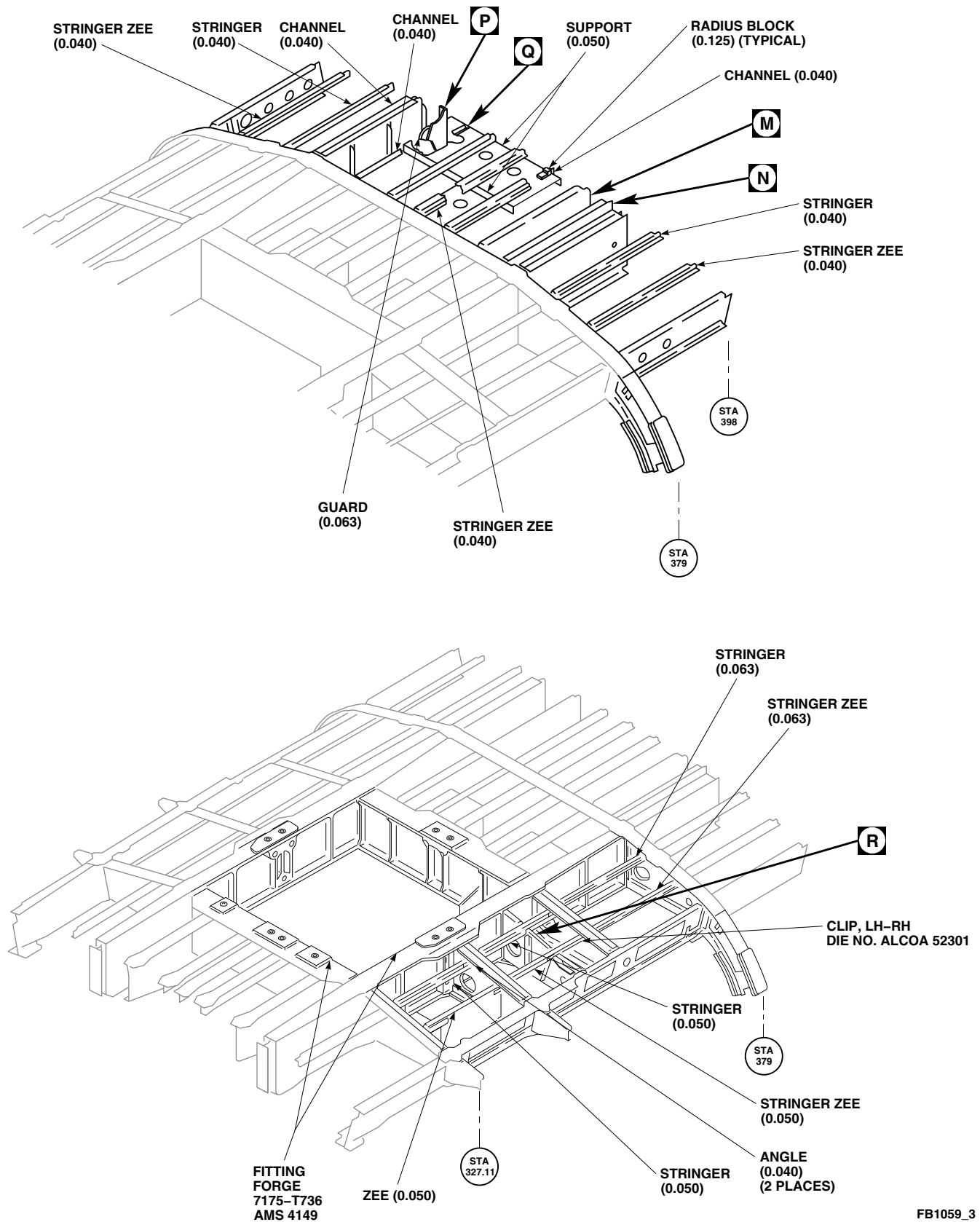
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FIGURE 2-6-11-1. REPAIR CABIN SECTION UPPER FRAMING (SHEET 2 OF 11)

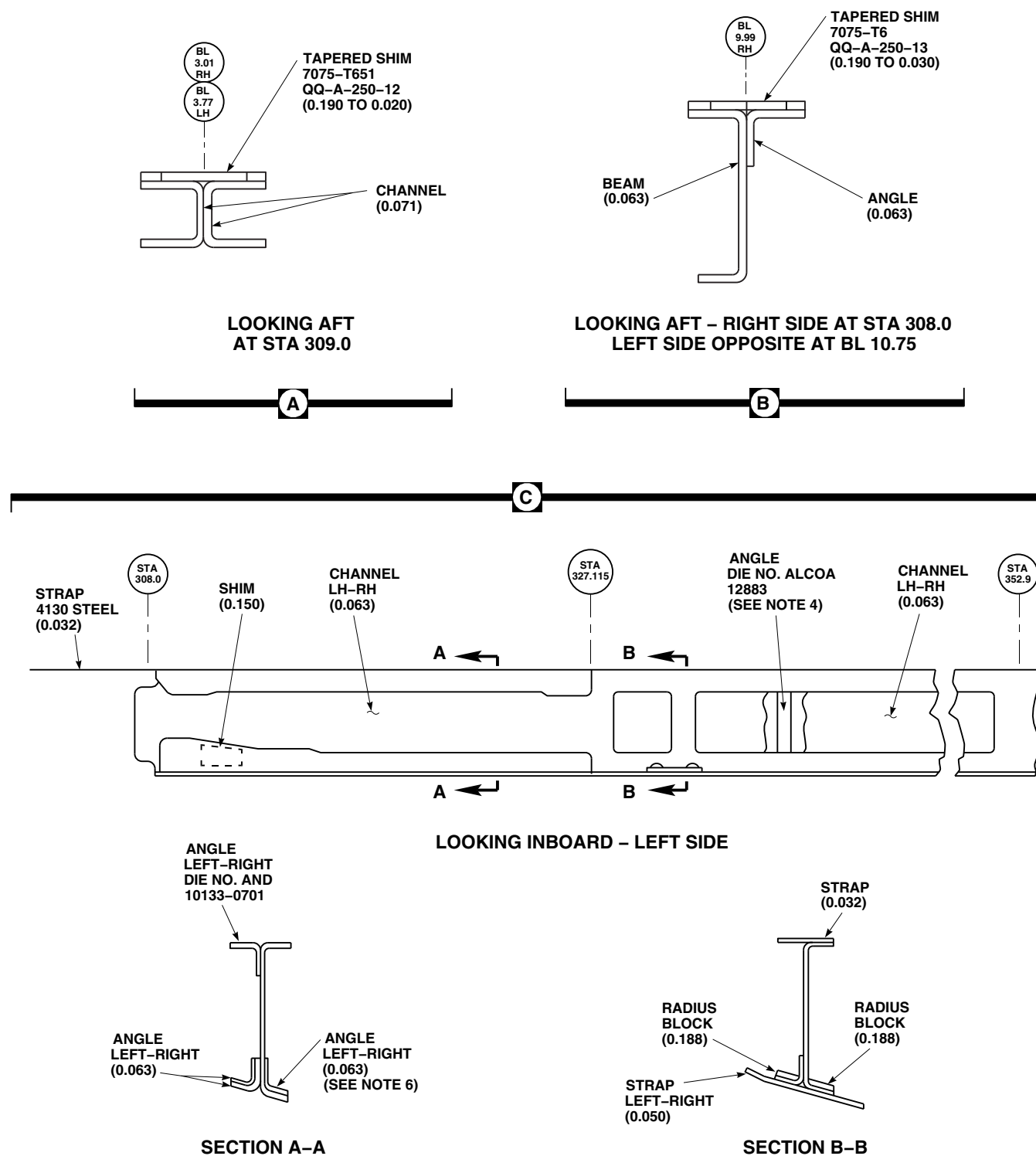


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FIGURE 2-6-11-1. REPAIR CABIN SECTION UPPER FRAMING (SHEET 3 OF 11)

Change 3

2-6-11-5



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FIGURE 2-6-11-1. REPAIR CABIN SECTION UPPER FRAMING (SHEET 4 OF 11)

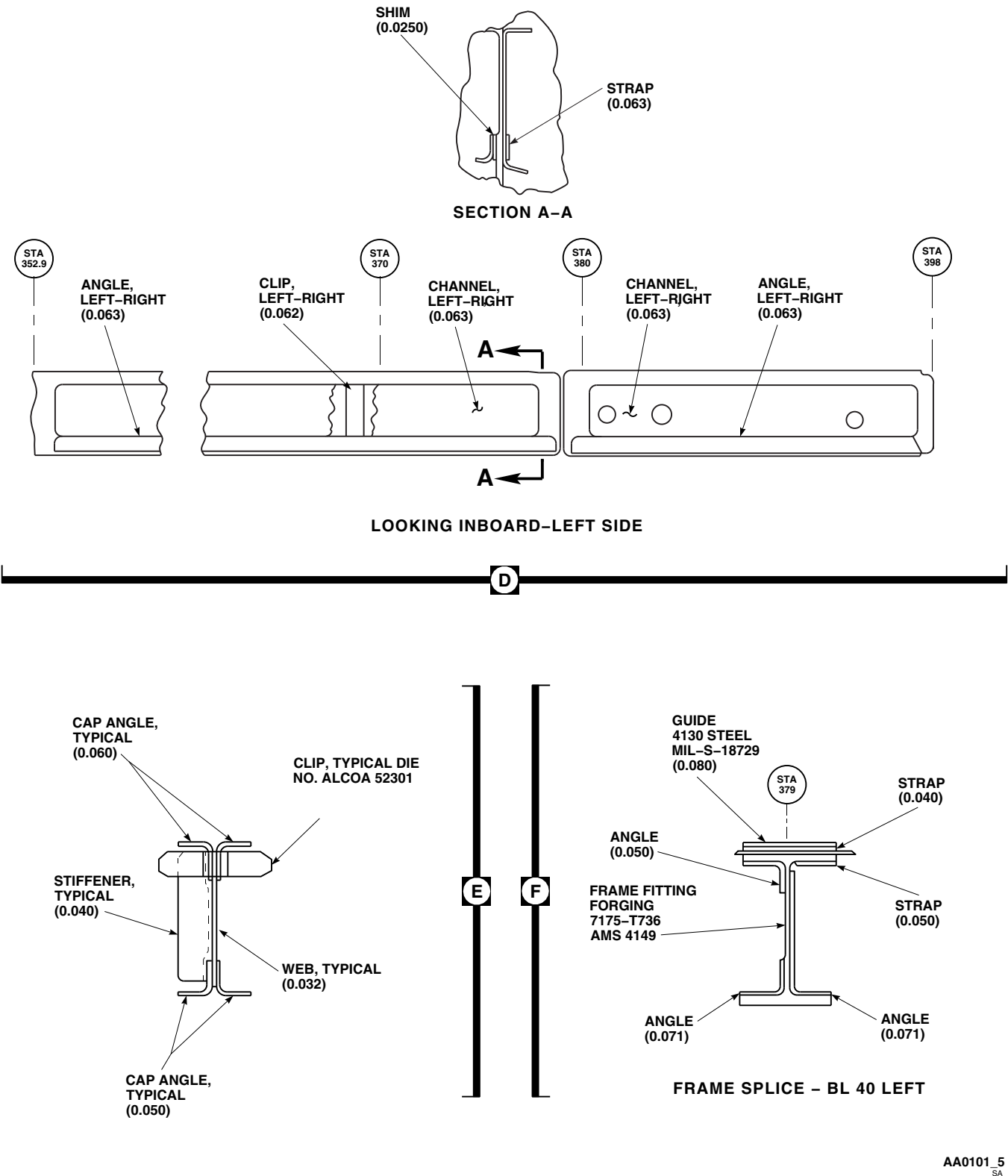


FIGURE 2-6-11-1. REPAIR CABIN SECTION UPPER FRAMING (SHEET 5 OF 11)

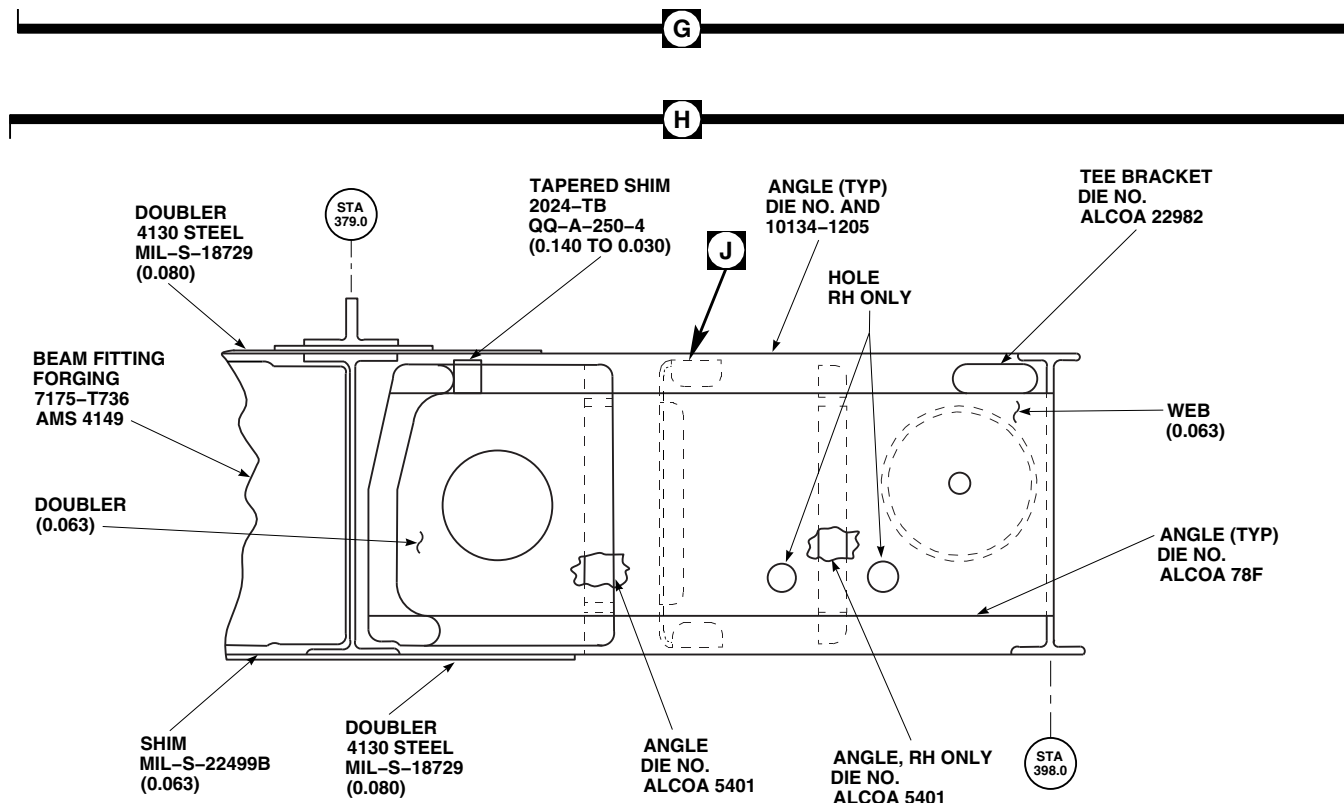
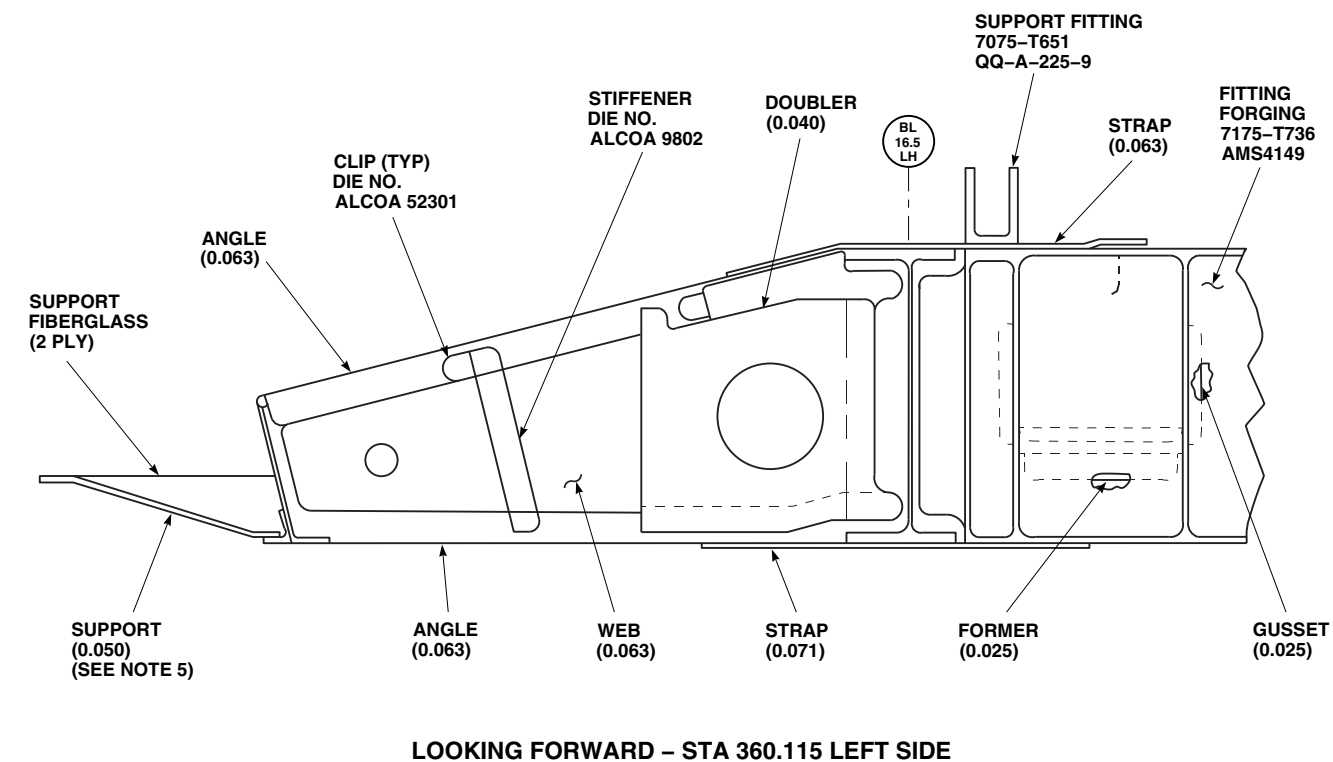
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FIGURE 2-6-11-1. REPAIR CABIN SECTION UPPER FRAMING (SHEET 6 OF 11)

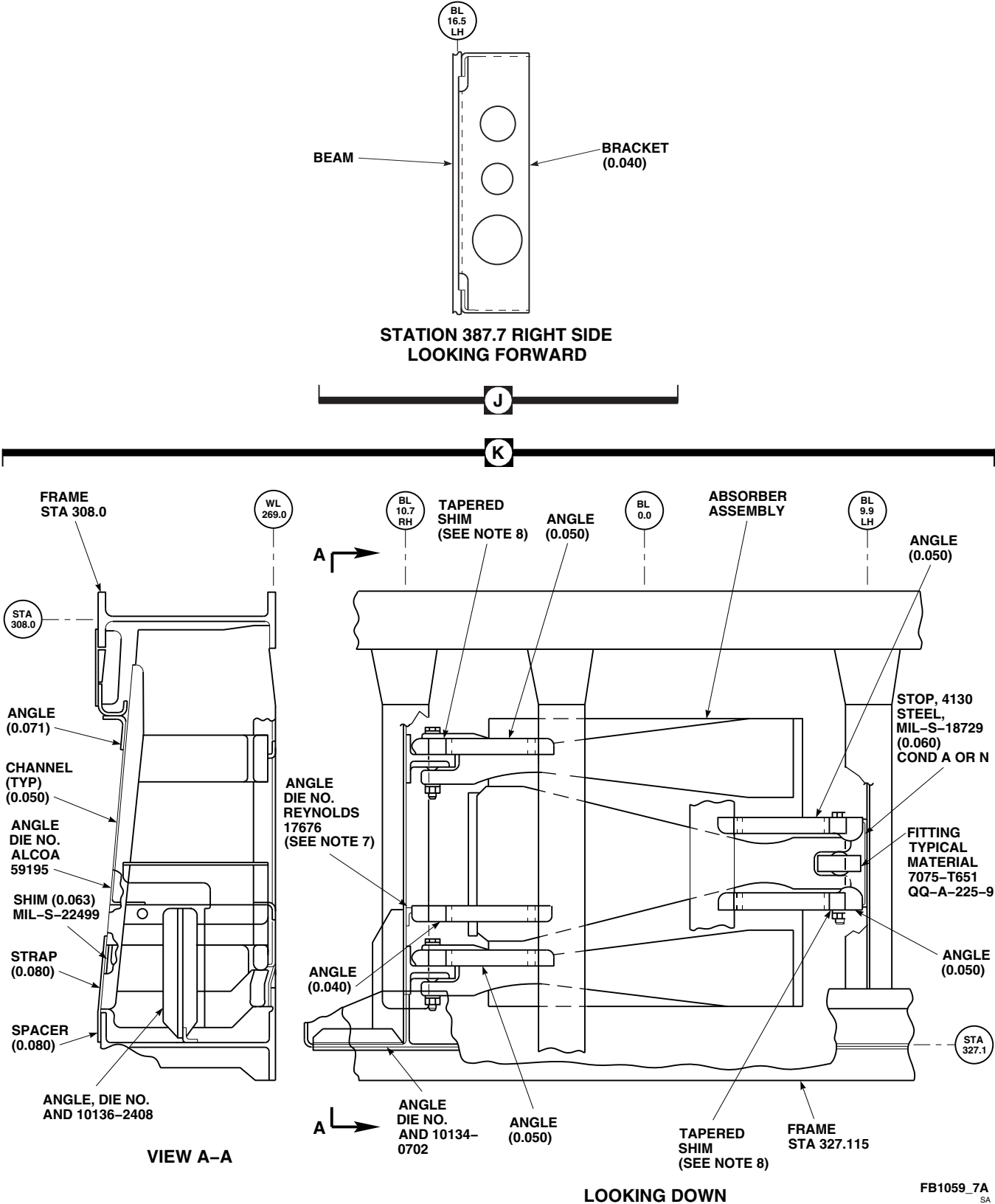


FIGURE 2-6-11-1. REPAIR CABIN SECTION UPPER FRAMING (SHEET 7 OF 11)

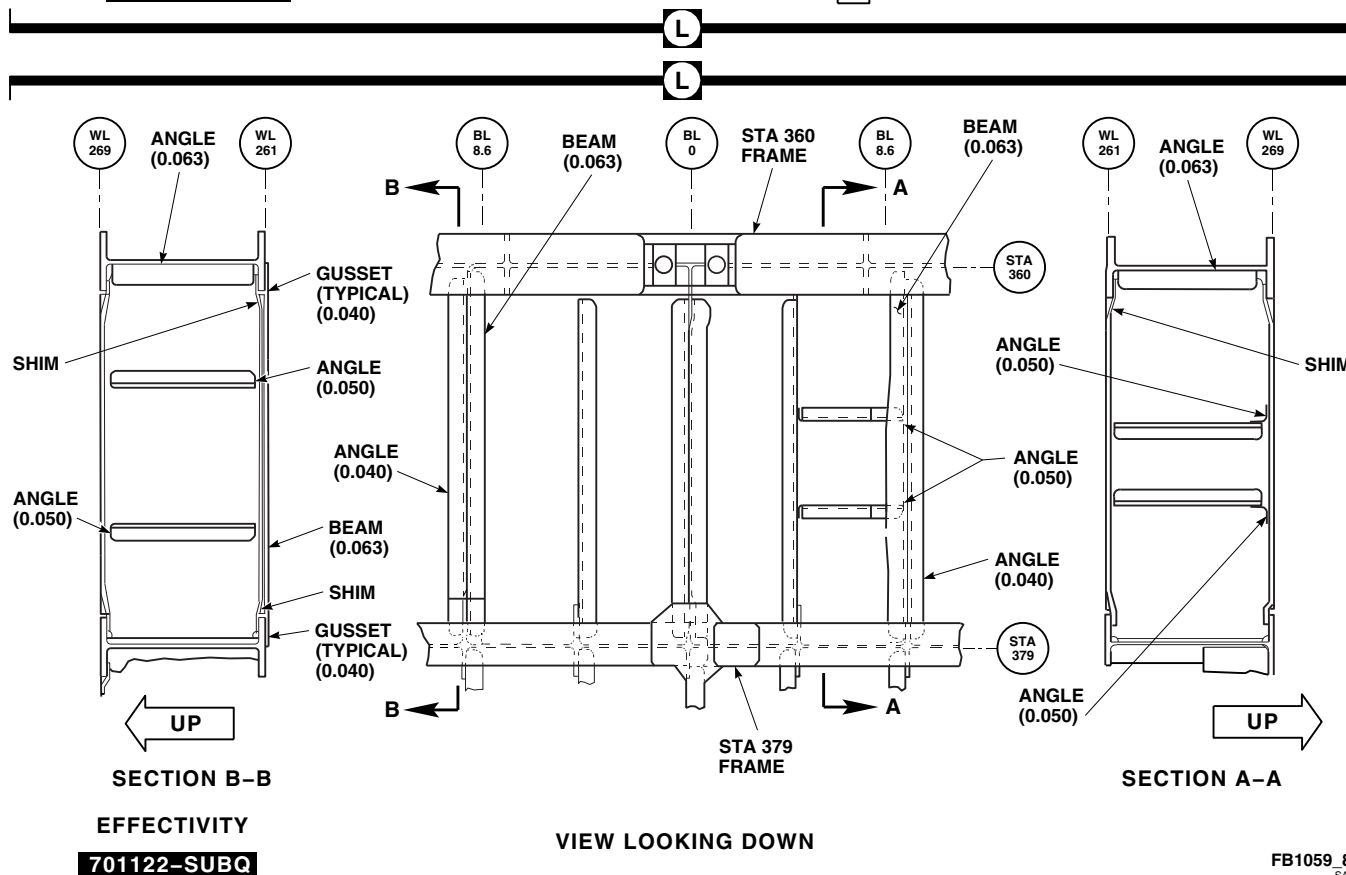
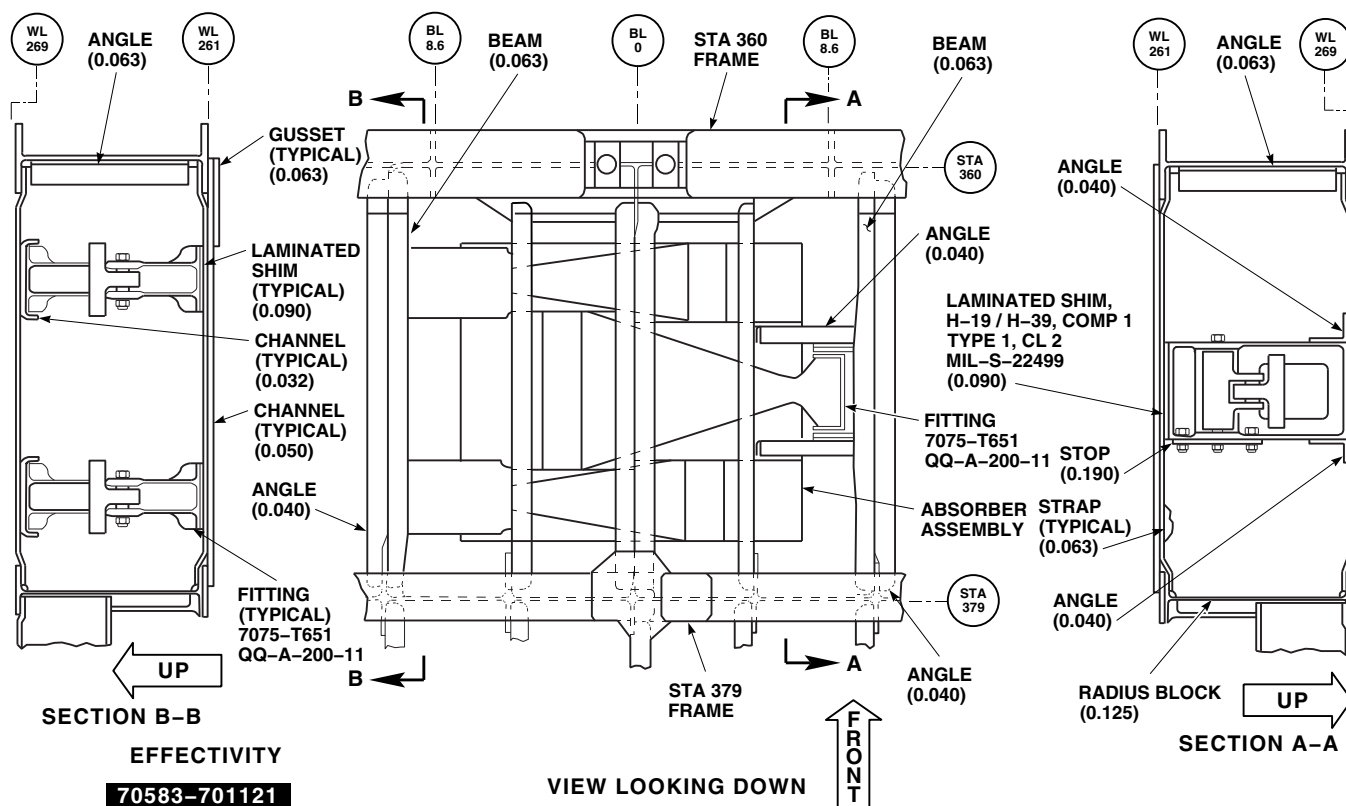
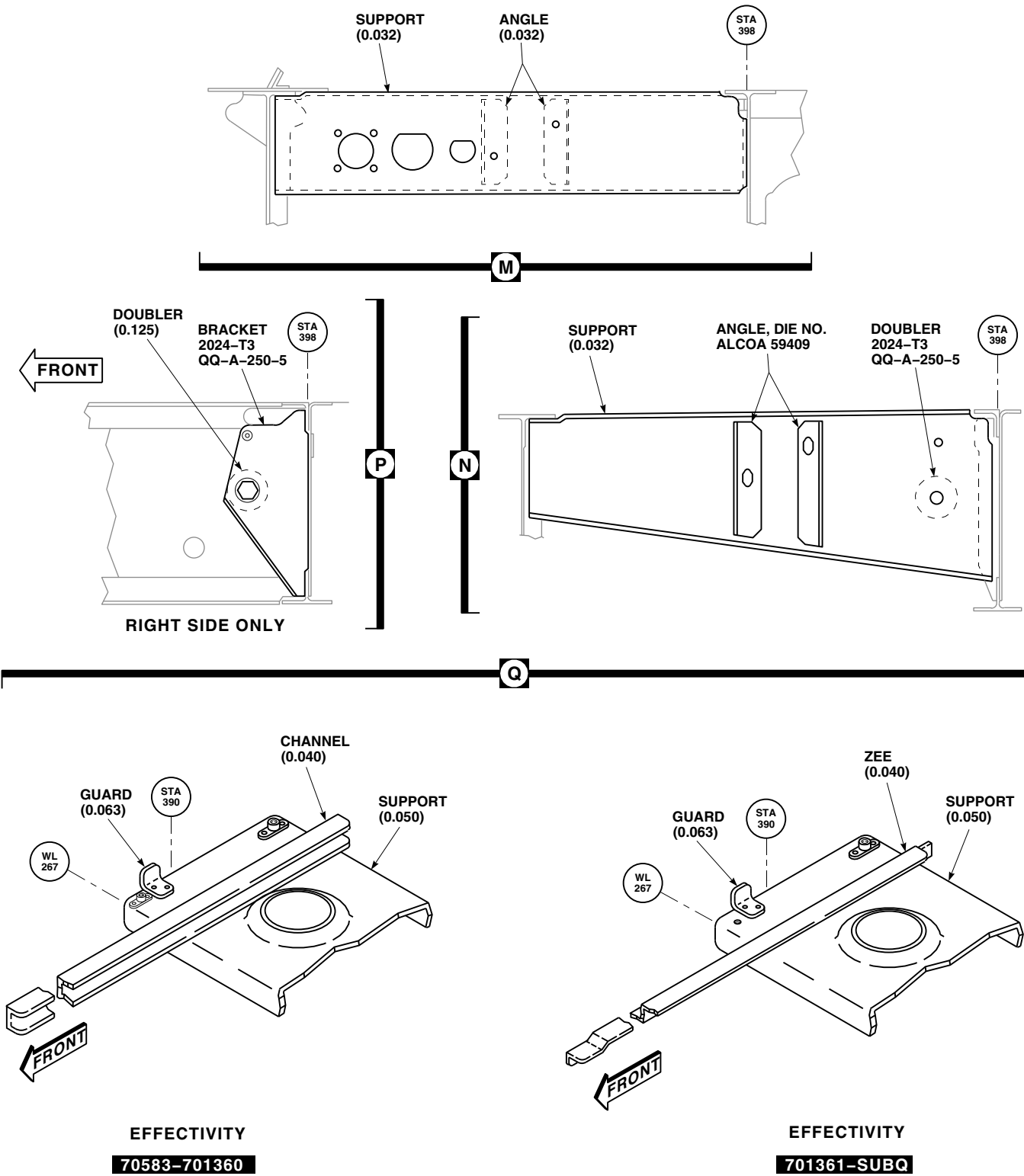


FIGURE 2-6-11-1. REPAIR CABIN SECTION UPPER FRAMING (SHEET 8 OF 11)



RIGHT SIDE ONLY

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SA

FIGURE 2-6-11-1. REPAIR CABIN SECTION UPPER FRAMING (SHEET 9 OF 11)

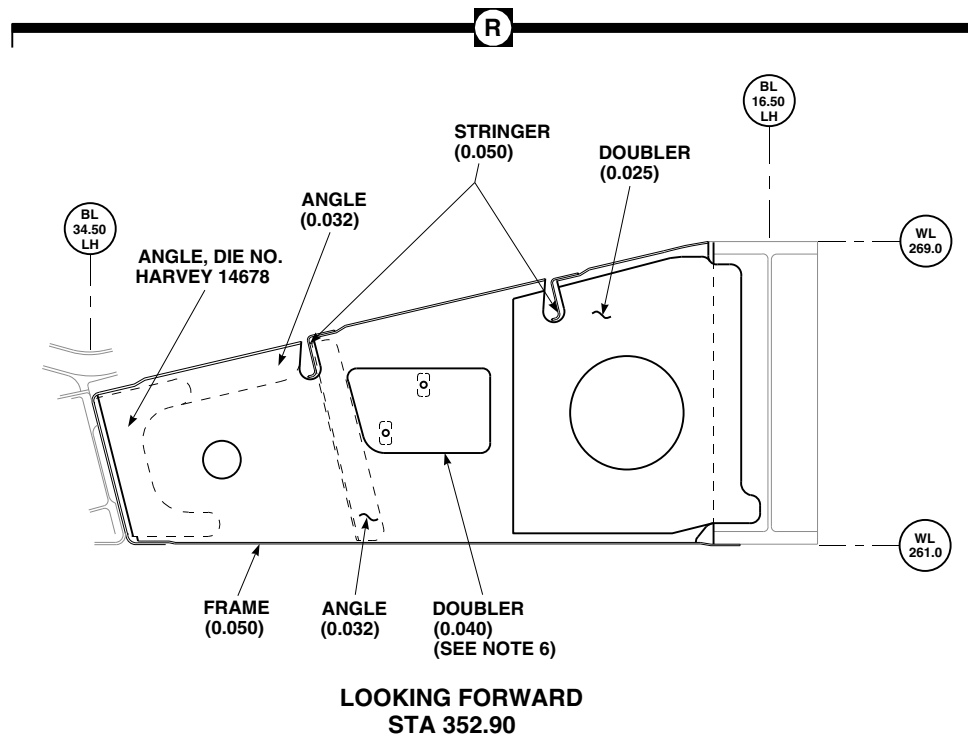
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SA

FIGURE 2-6-11-1. REPAIR CABIN SECTION UPPER FRAMING (SHEET 10 OF 11)

NOTES

1. UNLESS OTHERWISE NOTED ON SHEETS OF THIS FIGURE, ALL SHEET METAL PARTS SHOWN ARE ALCLAD 7075-T6, QQ-A-250/13, AND ALL EXTRUSIONS ARE ALUMINUM ALLOY 7075-T6511, QQ-A-200/11.
2. LEFT AND RIGHT SIDES ARE THE SAME, EXCEPT WHEN NOTED.
3. APPLY SEALING COMPOUND, ITEM 292, APPENDIX D, TO ALL MATING SURFACES. AFTER INSTALLING FASTENERS, RECLEAN AREA, PLUG GAPS WITH SEALING COMPOUND, ITEM 275, APPENDIX D, AND APPLY SEALING COMPOUND, ITEM 292, APPENDIX D, OVER PLUGS AND AROUND EDGES OF MATING PARTS.
4. **W/O ESSS**
5. **70583-70704**
6. **70705-SUBQ**
7. **S70-II**
8. LIQUID SHIM BETWEEN ANGLE AND MATING SURFACE, UP TO 0.060 INCH THICKNESS, USING ACETONE, ITEM 25, APPENDIX D.

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SA

FIGURE 2-6-11-1. REPAIR UPPER CABIN FRAMING (SHEET 11 OF 11)

2-6-12 CABIN SECTION LOWER FRAMING REPAIR.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
2-6-12.1	Repair Cabin Section Lower Framing	2-6-12-2	2-6-12.2	Inspect/Repair Drag Beam Support	2-6-12-2

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Fluorescent Penetrant Inspection Kit

Materials/Parts

- Abrasive Cloth, Item 6, Appendix D
- Adhesive, Item 37, Appendix D
- Adhesive, Item 38, Appendix D
- Brush, Acid, Item 84, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D
- Polyurethane Coating (Clear), Item 240A, Appendix D
- Pressure Sensitive Adhesive Tape, Item 247, Appendix D
- Sealing Compound, Item 273, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- AMS 4149
- Appendix D
- ASTM A231
- ASTM E1417
- MIL-A-21180
- MIL-R-6855
- MIL-S-5059
- MIL-S-6758
- MIL-T-9046
- PARA 2-6-1
- PARA 2-6-2
- PARA 2-6-4
- PARA 2-6-5
- QQ-A-200
- QQ-A-250
- QQ-A-367

2-6-12.1 Repair Cabin Section Lower Framing.

- a. Lower cabin framing (STA 247.0 to STA 379.0) includes formed and extruded members made from 7075-T6, QQ-A-250/13, Alclad sheet stock and 7075-T6511, QQ-A-200/11, aluminum alloy extrusion. Landing gear support fittings and frame fittings between STA 247.0 and STA 295.0 are forged aluminum alloy, 7075-T73, QQ-A-367. Support fittings for cargo hook are forged aluminum alloy, 7175-T736, AMS 4149 (Figure 2-6-12-1, Sheet 1 through Sheet 10).
- b. All lower cabin framing is primary structure; however, support and frame fittings, and fittings in cargo hook area, are considered critical primary structure. Negligible damage to forged fittings is limited to nicks, scratches, and corrosion, no deeper than 5% of material thickness at point of damage. Blend damage smoothly to surrounding area, using long, even strokes. Polish out all surface imperfections and restore protective finish. Damage greater than negligible limits in critical areas must be reviewed by engineering authority for damage evaluation and either repair or replacement. All repairs to lower cabin framing must maintain design strength and alignment.
- c. Inspect lower cabin framing for corrosion. Remove surface corrosion using abrasive cloth, Item 6, Appendix D. **CORROSION REMOVAL SHALL NOT EXCEED 5% OF MATERIAL THICKNESS.**
- d. Inspect for corroded fasteners. Remove corroded fasteners. Remove corrosion beneath fasteners using abrasive cloth, Item 6, Appendix D. **CORROSION REMOVAL SHALL NOT EXCEED 5% OF MATERIAL THICKNESS.** Replace fasteners.
- e. Evaluate damage in noncritical areas (PARA 2-6-1).

- f. Refer to PARA 2-6-2 and PARA 2-6-4 for typical framing repairs.
- g. For corrosion protection and watertightness, seal all repairs between framing and plating (PARA 2-6-1).
- h. Touch up paint as required (PARA 2-6-5).

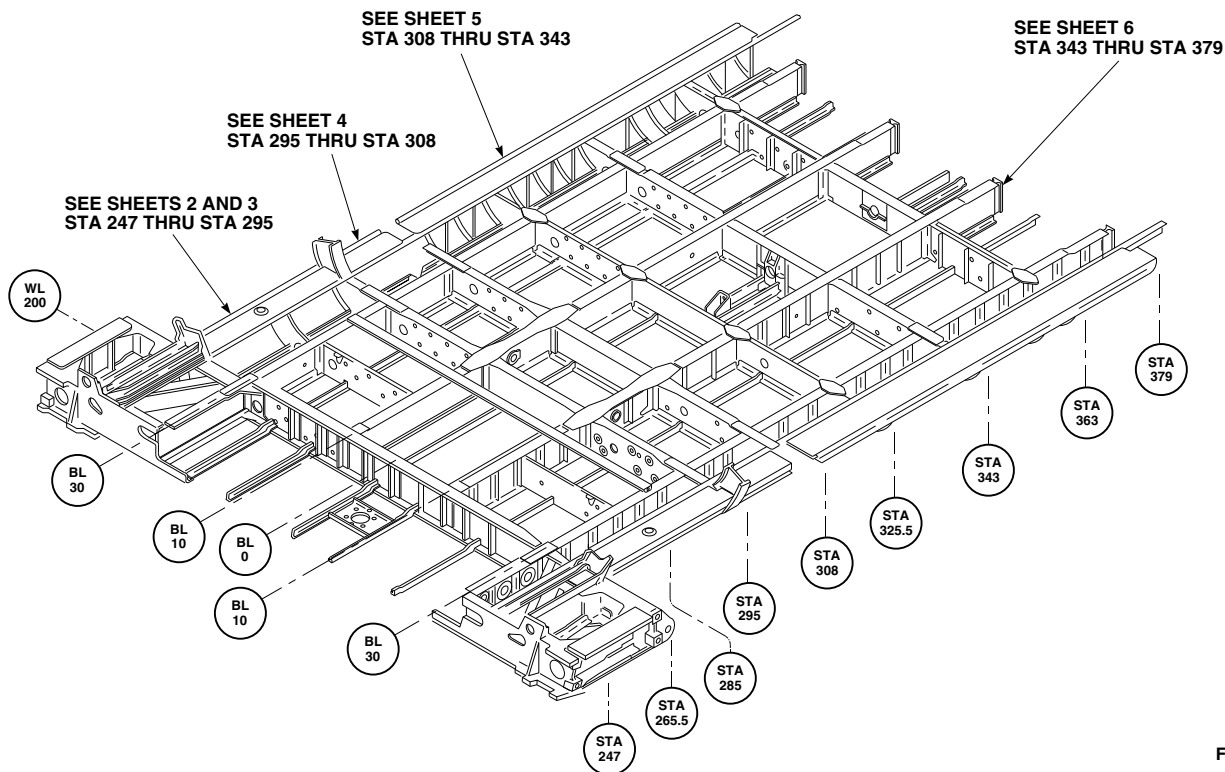
2-6-12.2 Inspect/Repair Drag Beam Support.

- a. Inspect lower, rear area of left and right drag beam supports for damage caused by interference with drag beam or shock strut at STA 265.0 (Figure 2-6-12-2, Detail A). If no damage is found, inspection is complete. If damage is found, proceed to step b.
- b. Mask fitting bearing bores with pressure sensitive adhesive tape, Item 247, Appendix D.

NOTE

Rework does not affect structural integrity of component.

- c. Remove damaged material as necessary from fuselage fitting by blending or machining off bottom rib between drag beam attachment lugs (shaded area). Break all edges 0.005 - 0.015-inch. Blend smoothly to surrounding area using long, even strokes.
- d. Using fluorescent penetrant inspection kit, inspect reworked area for cracks (ASTM E1417).
- e. Apply corrosion resistant coating, Item 118, Appendix D, to reworked area with acid brush, Item 84, Appendix D.
- f. Touch up paint as required (PARA 2-6-5).
- g. Remove pressure sensitive adhesive tape.



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FIGURE 2-6-12-1. REPAIR CABIN SECTION LOWER FRAMING (SHEET 1 OF 10)

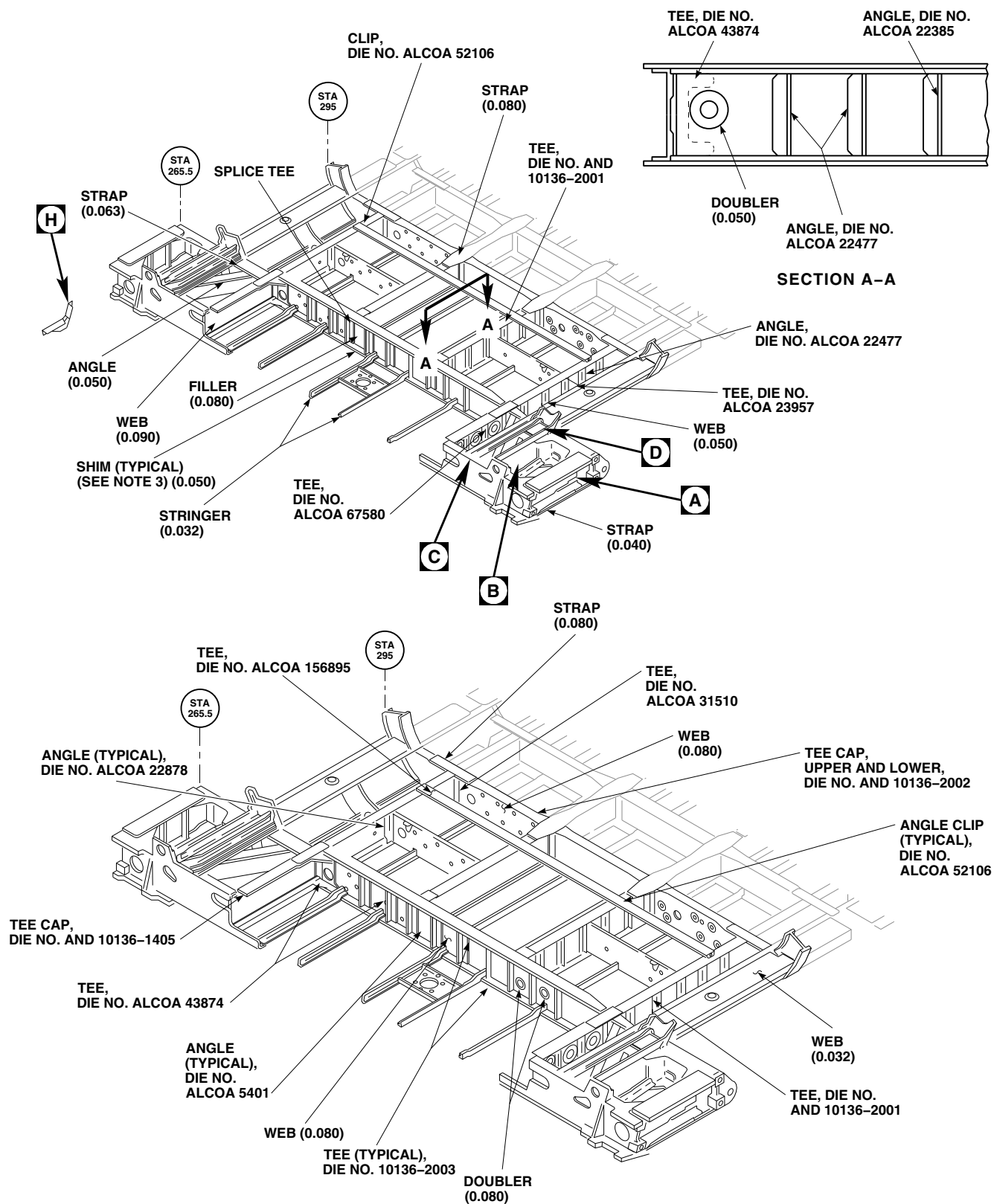
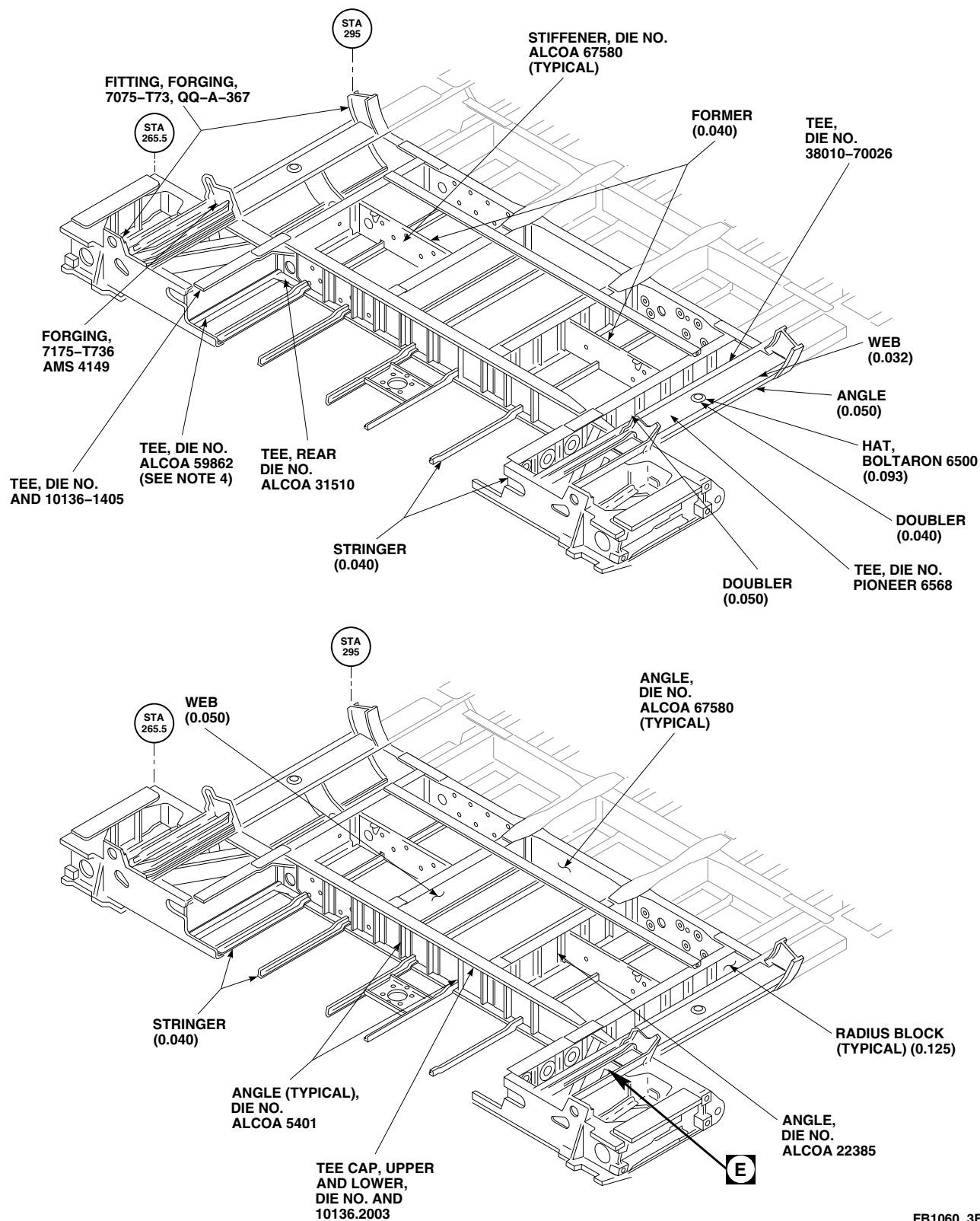
FB1060_2B
SA

FIGURE 2-6-12-1. REPAIR CABIN SECTION LOWER FRAMING (SHEET 2 OF 10)



FB1060_3B
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FIGURE 2-6-12-1. REPAIR CABIN SECTION LOWER FRAMING (SHEET 3 OF 10)

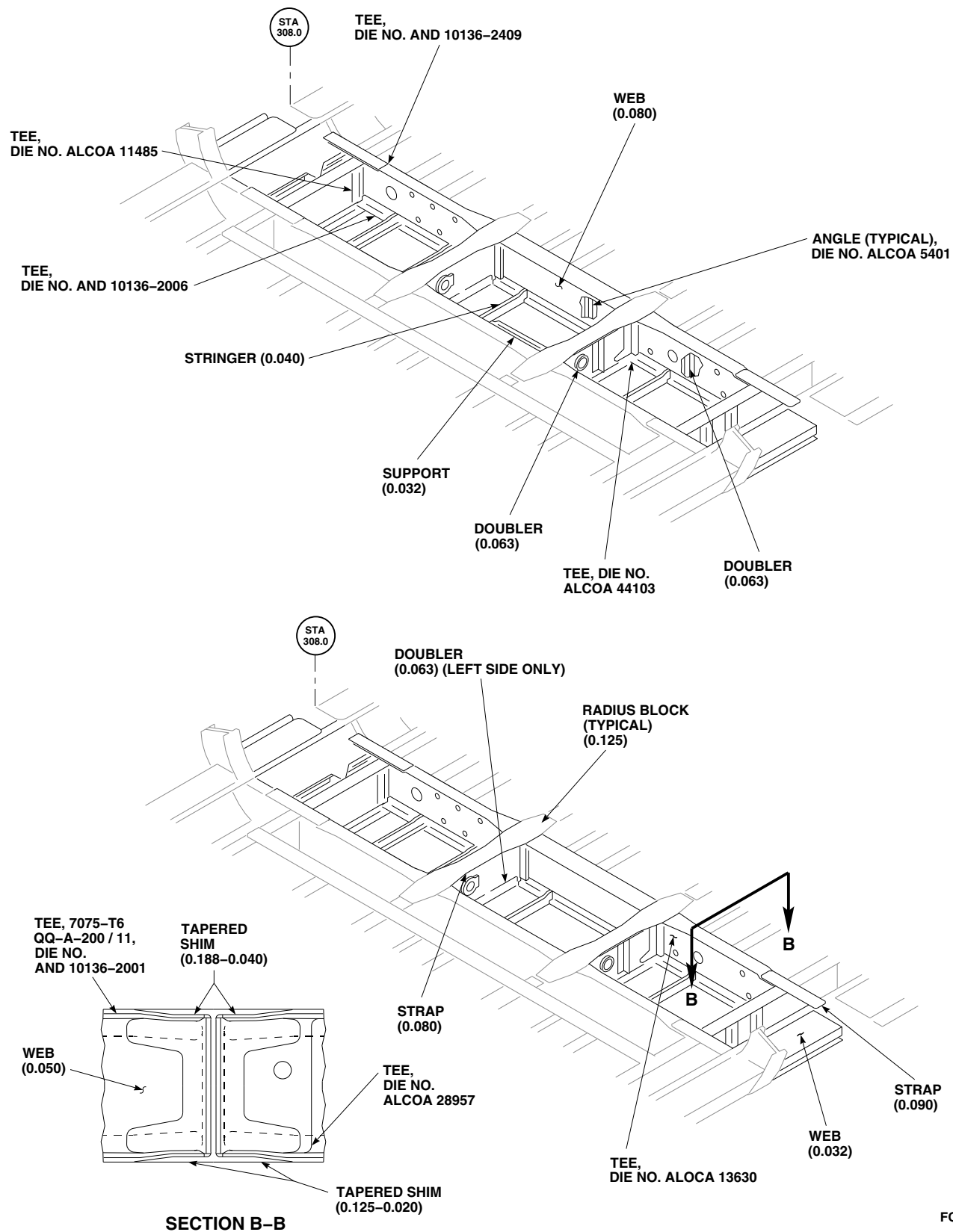
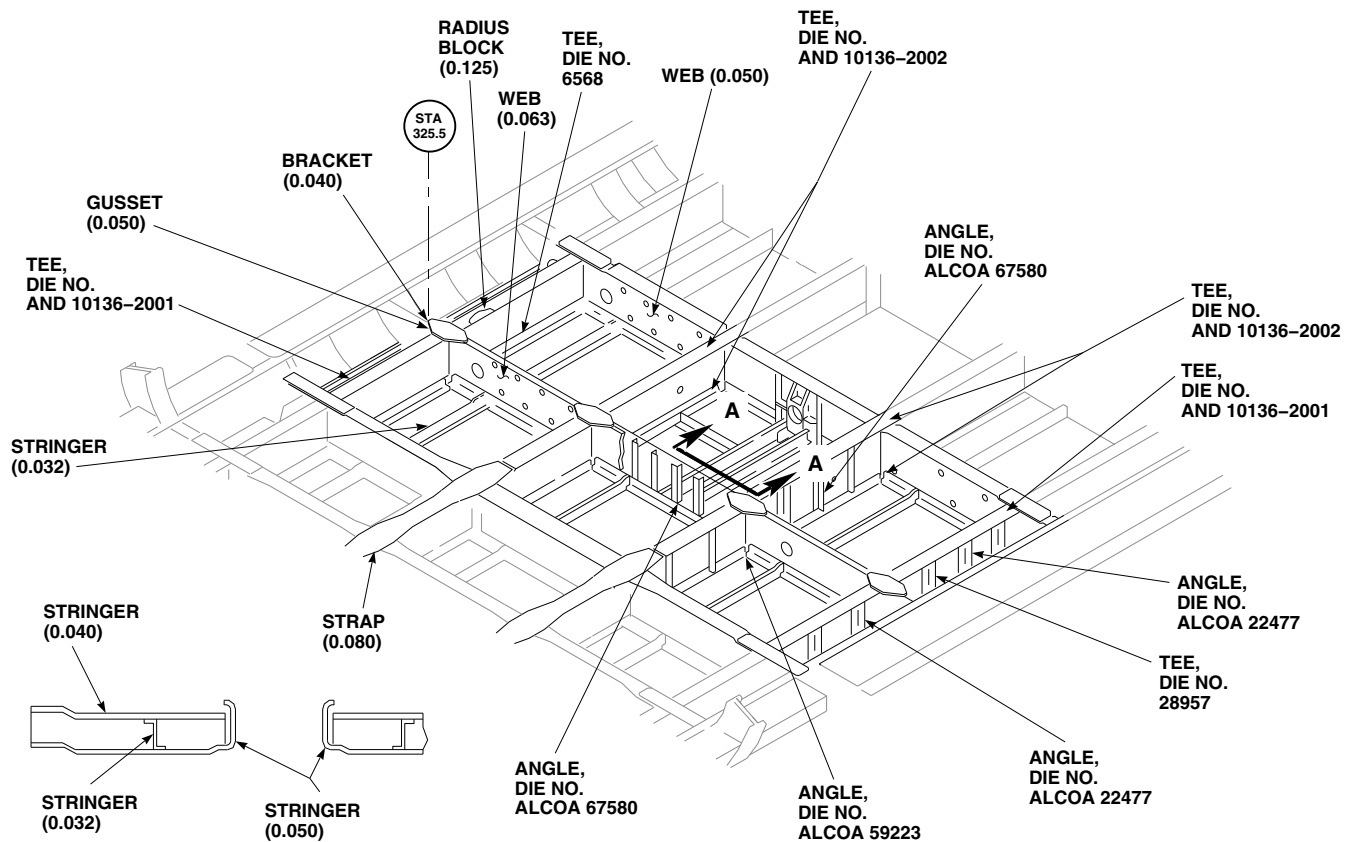
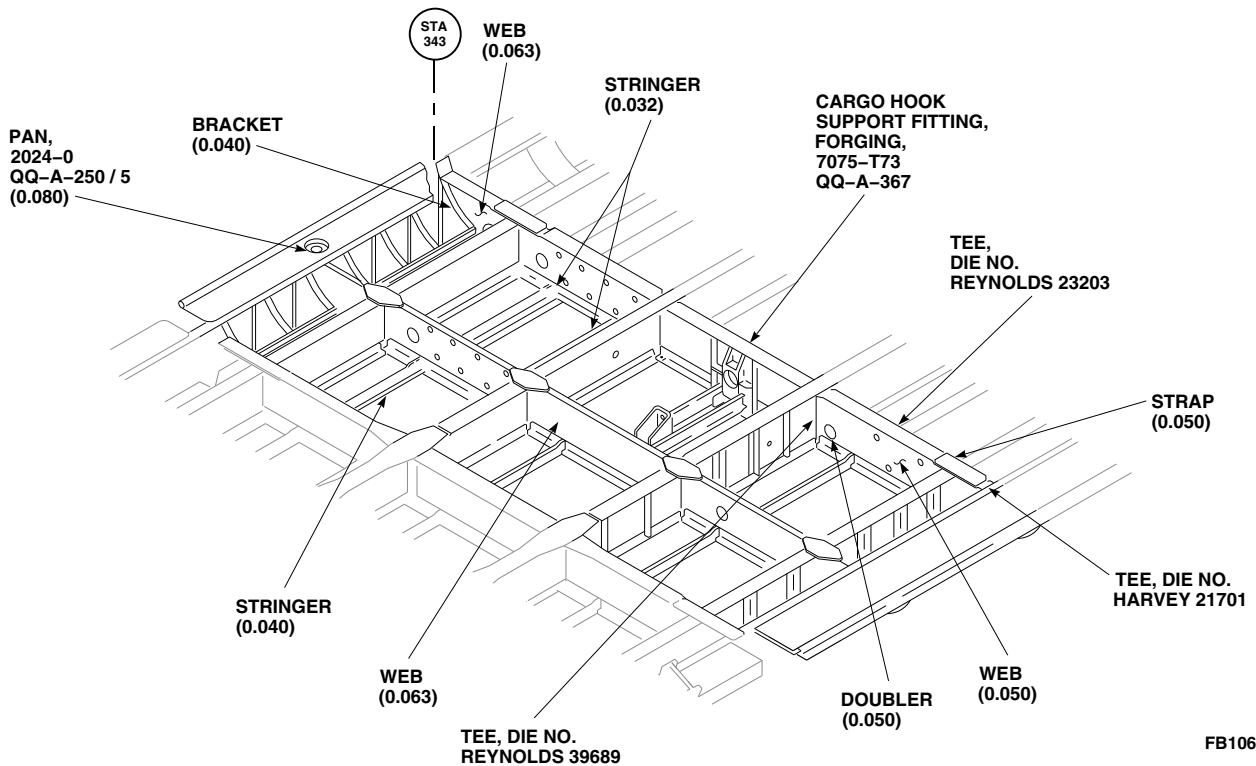


FIGURE 2-6-12-1. REPAIR CABIN SECTION LOWER FRAMING (SHEET 4 OF 10)



SECTION A-A



FB1060_5B
SA

FIGURE 2-6-12-1. REPAIR CABIN SECTION LOWER FRAMING (SHEET 5 OF 10)

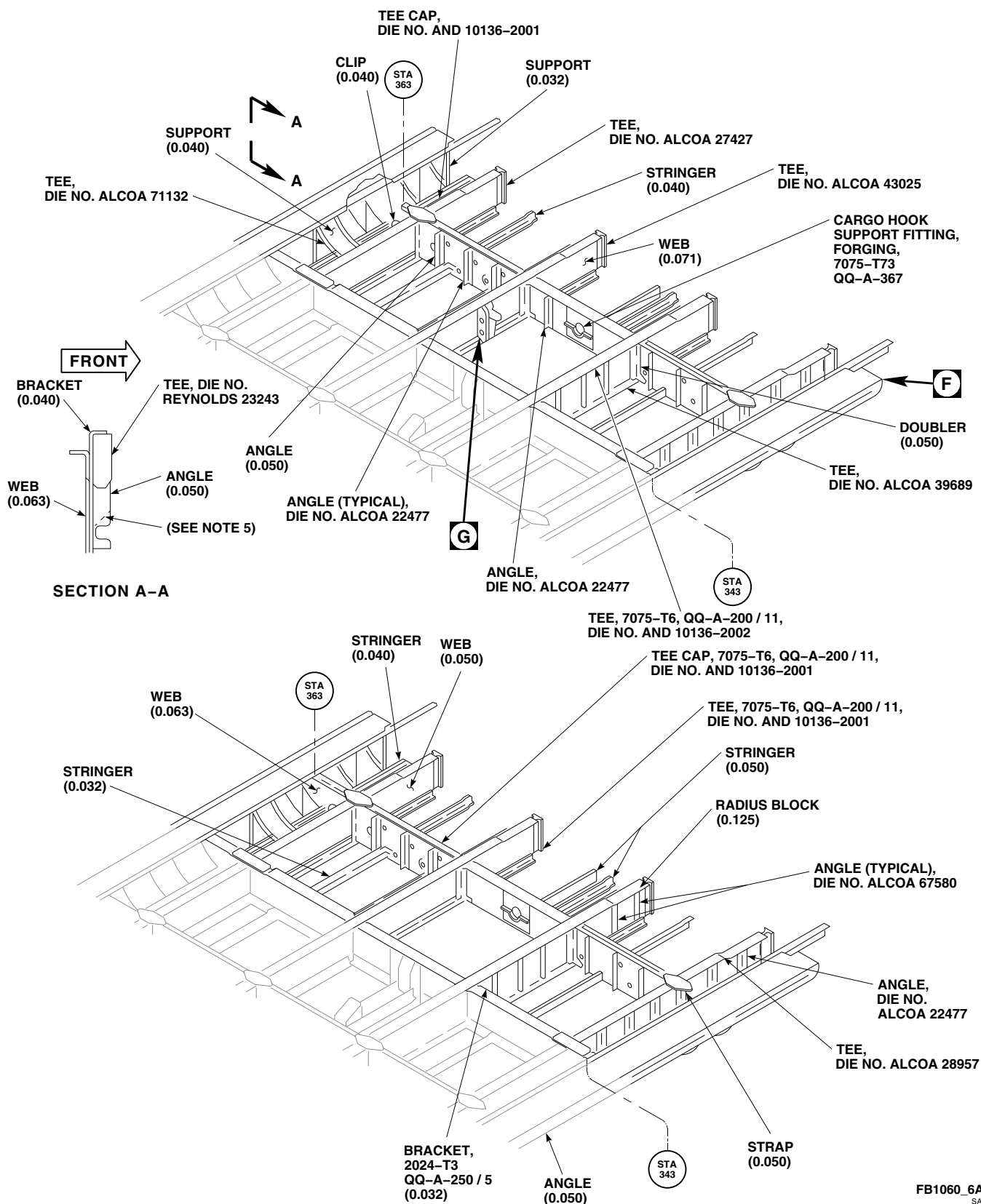


FIGURE 2-6-12-1. REPAIR CABIN SECTION LOWER FRAMING (SHEET 6 OF 10)

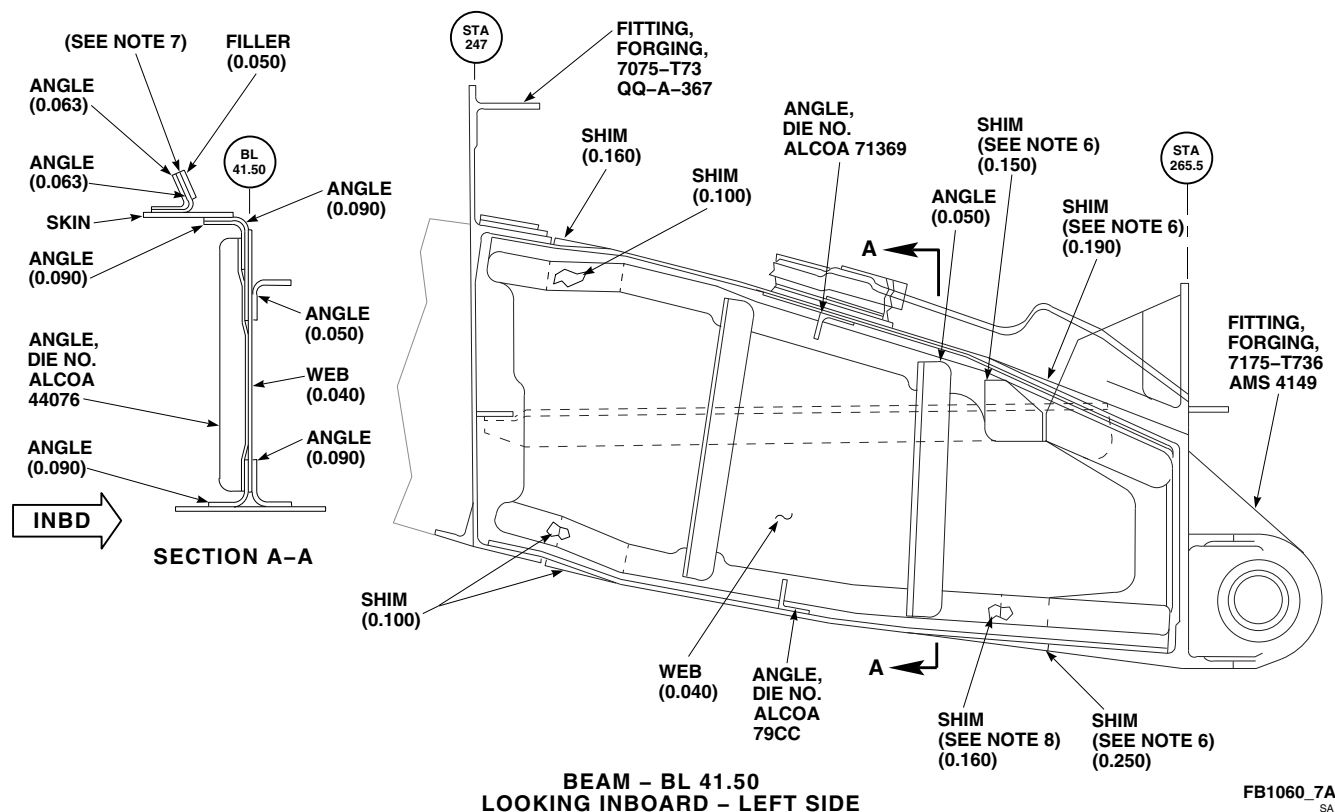
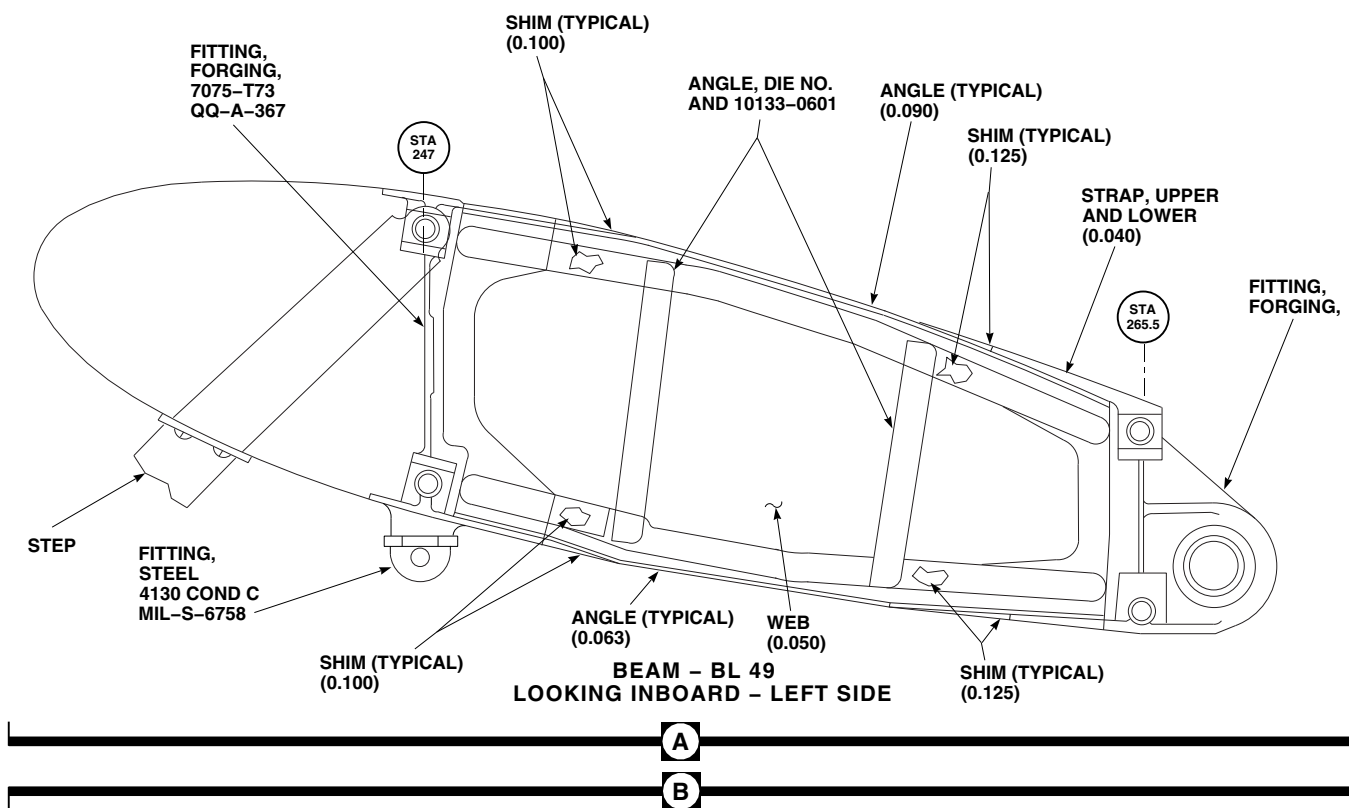
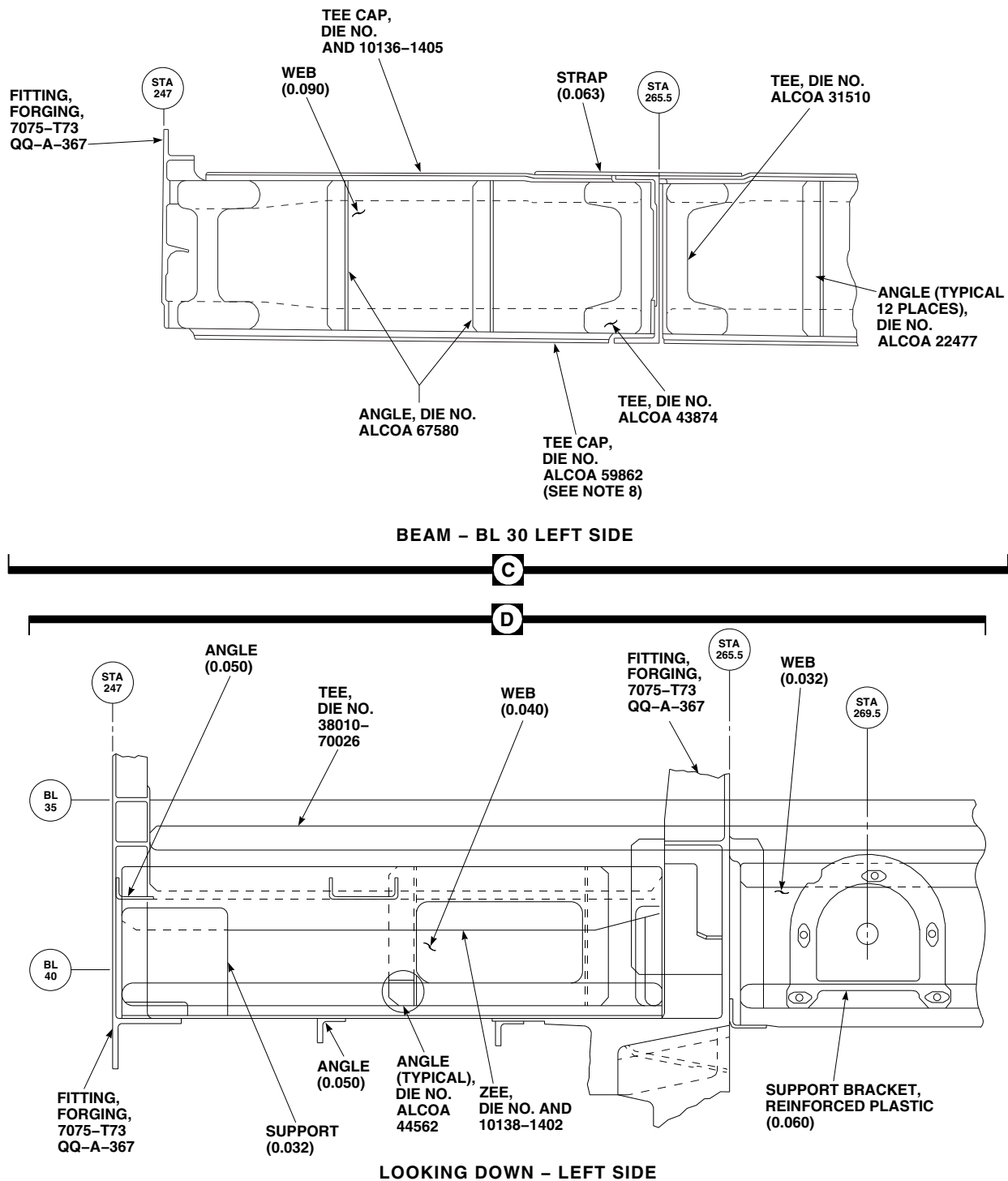


FIGURE 2-6-12-1. REPAIR CABIN SECTION LOWER FRAMING (SHEET 7 OF 10)

FB1060_7A
SA



FB1060_8A
SA

FIGURE 2-6-12-1. REPAIR CABIN SECTION LOWER FRAMING (SHEET 8 OF 10)

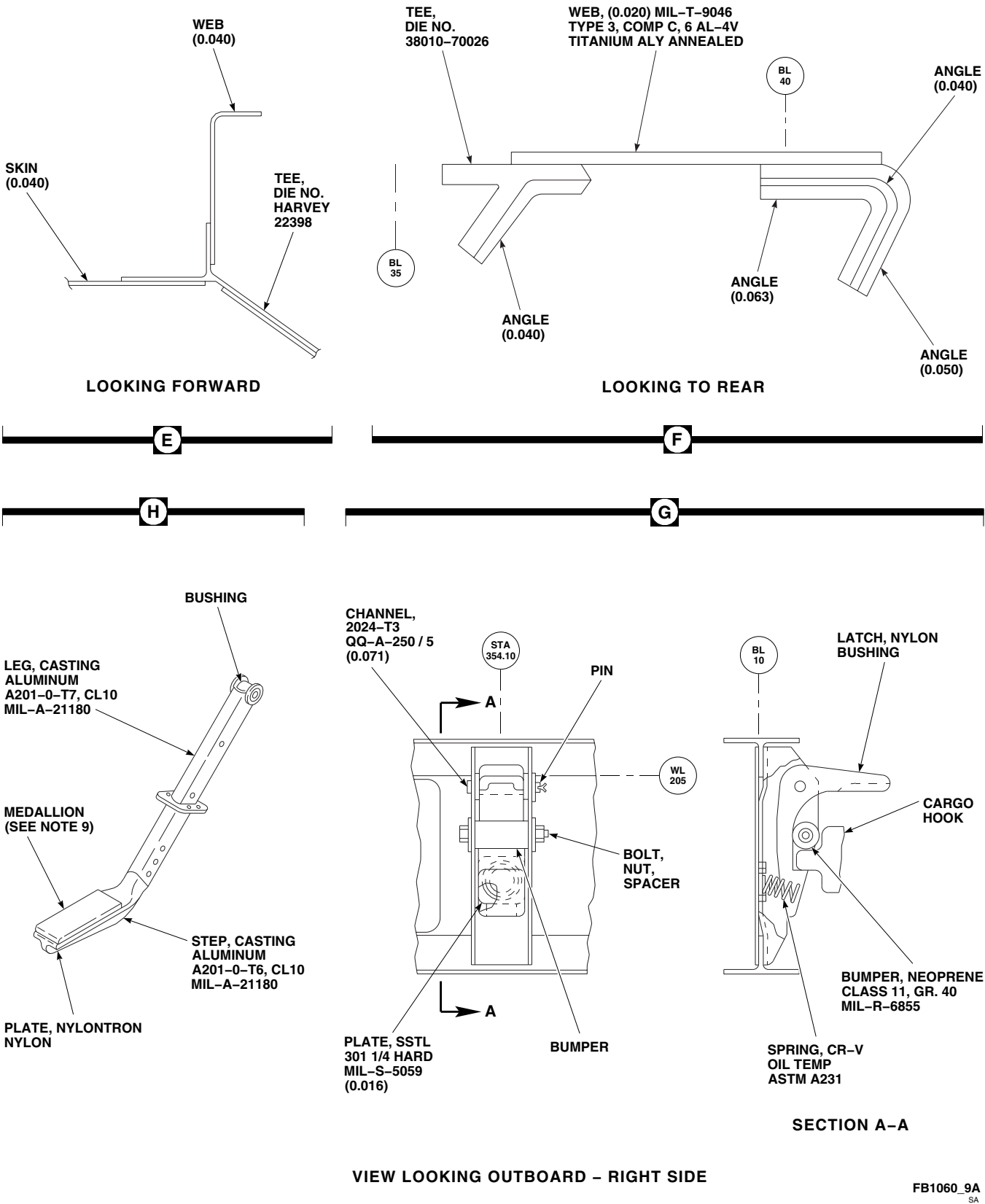


FIGURE 2-6-12-1. REPAIR CABIN SECTION LOWER FRAMING (SHEET 9 OF 10)

FB1060_9A
SA

NOTES

1. UNLESS OTHERWISE NOTED ON SHEETS OF THIS FIGURE, ALL SHEET METAL PARTS SHOWN ARE ALCLAD 7075-T6, QQ-A-250 / 13, AND ALL EXTRUSIONS ARE ALUMINUM ALLOY 7075-T6511, QQ-A-200 / 11.
2. LEFT AND RIGHT SIDES ARE THE SAME, EXCEPT WHEN NOTED.
3. BOND WITH SEALING COMPOUND, ITEM 273, APPENDIX D.
4. **S70-II** DIE NO. AND10136-2004. ◀
5. **W/O ESSS**
6. **701323-SUBQ** DIE NO. ALCOA NO. 33481. ◀
7. EDGE OF ANGLE.
8. BOND SHIM TO STRUCTURE WITH ADHESIVE, ITEM 38, APPENDIX D.
9. SEAL AREA WITH SEALING COMPOUND, ITEM 292, APPENDIX D.
10. **S70-II** DIE NO. ALCOA 33481 ◀
11. BOND TO PLATE WITH ADHESIVE, ITEM 37, APPENDIX D. COAT WITH POLYURETHANE COATING (CLEAR), ITEM 240A, APPENDIX D (0.001 THICK).
12. ALL DIMENSIONS ARE IN INCHES, UNLESS NOTED.

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SA

FIGURE 2-6-12-1. REPAIR CABIN SECTION LOWER FRAMING (SHEET 10 OF 10)

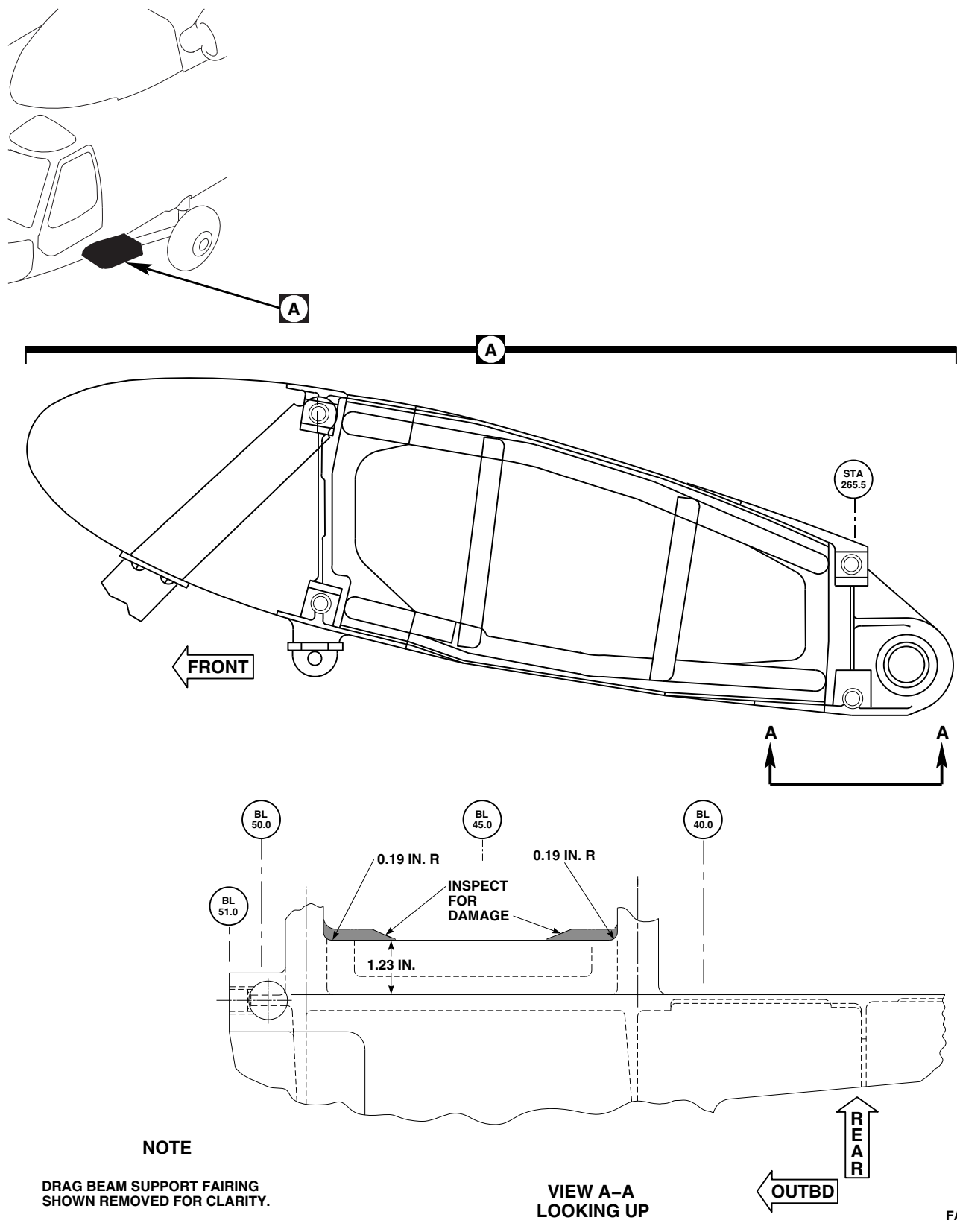


FIGURE 2-6-12-2. REPAIR DRAG BEAM SUPPORT

2-6-13 CABIN SECTION PLATING REPAIR.

PARAGRAPH BREAKDOWN

	PAGE
2-6-13.1 Repair Cabin Section Plating	2-6-13-1

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit

Materials/Parts

- Abrasive Cloth, Item 5, Appendix D
- Adhesive, Item 31, Appendix D
- Adhesive, Item 37, Appendix D
- Sealing Compound, Item 292, Appendix D
- Sealing Compound, Item 295A, Appendix D

References

- AMS-S-5000
- AMS-T-9046
- Appendix D
- MIL-C-7438

- MIL-R-6855
- MIL-S-1030
- MIL-S-18729
- MIL-S-5059
- MIL-T-6855
- PARA 2-6-1
- PARA 2-6-2
- PARA 2-6-3
- PARA 2-6-5
- PARA 2-6-26
- PARA 2-6-27
- QQ-A-200
- QQ-A-250
- QQ-W-470

2-6-13.1 Repair Cabin Section Plating.

- Cabin section plating is Alclad aluminum 7075-T6, QQ-A-250/13, except in high-temperature areas where titanium alloy, AMS-T-9046, is used. Certain fairings and panels are made from reinforced plastic (Figure 2-6-13-1, Sheet 1 through Sheet 10).

- All plating, except for reinforced plastic members, is primary structure. Evaluate damage (PARA 2-6-1).
- Use PARA 2-6-2 for typical plating repairs. When repairing titanium plating, use same gage titanium stock, or the equivalent gage 301/302 stainless steel, MIL-S-5059, for patch-

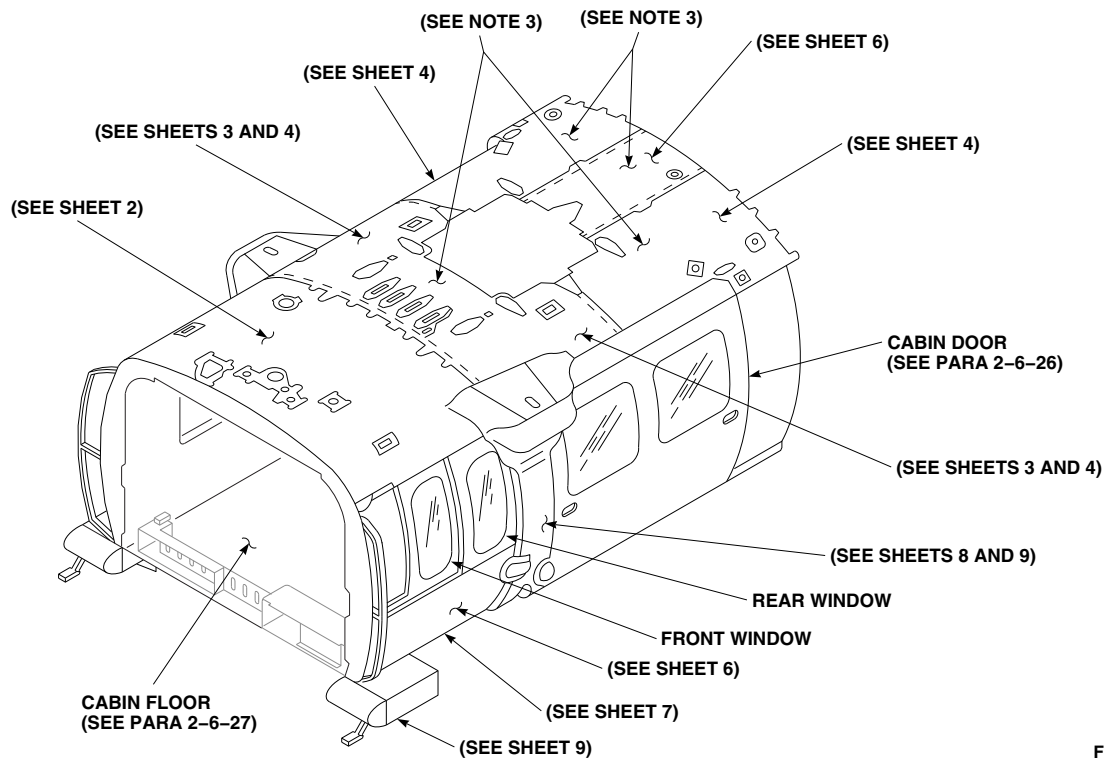
FB1061_1B
SA

FIGURE 2-6-13-1. REPAIR CABIN SECTION PLATING (SHEET 1 OF 10)

- ing. Use Monel or stainless steel fasteners for attaching titanium or stainless steel patches.
- d. Use PARA 2-6-3 for evaluating damage and repairing reinforced plastic.
- e. Make sure drain holes in lower cabin plating are clear of debris.
- f. Inspect lower cabin plating for surface corrosion. Remove surface corrosion using abrasive cloth, Item 5, Appendix D. **CORROSION REMOVAL SHALL NOT EXCEED 5% OF MATERIAL THICKNESS.**
- g. Inspect for fasteners for corrosion. If corrosion is found, proceed as follows:
 - (1) Remove fasteners.
 - (2) Using abrasive cloth, Item 5, Appendix D, remove corrosion beneath fasteners.
 - (3) **CORROSION REMOVAL SHALL NOT EXCEED 5% OF MATERIAL THICKNESS.** If limit is exceeded, replace fasteners.
- h. For corrosion protection and watertightness, seal all repairs on cabin section plating (PARA 2-6-1).
- i. Using sealing compound, Item 292, Appendix D, seal all areas where cabin ceiling is penetrated.
- j. Touch up paint as required (PARA 2-6-5).

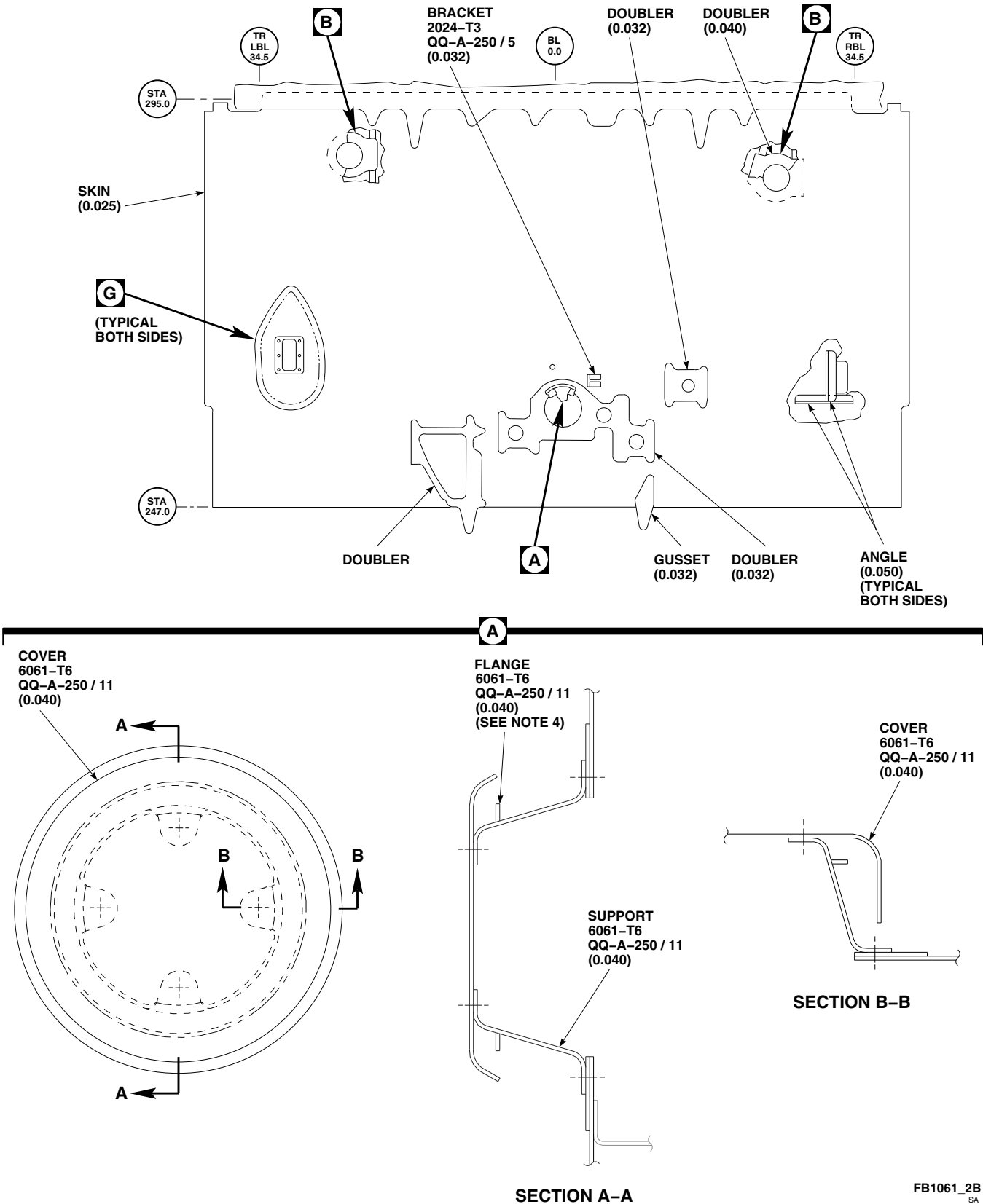


FIGURE 2-6-13-1. REPAIR CABIN SECTION PLATING (SHEET 2 OF 10)

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SA

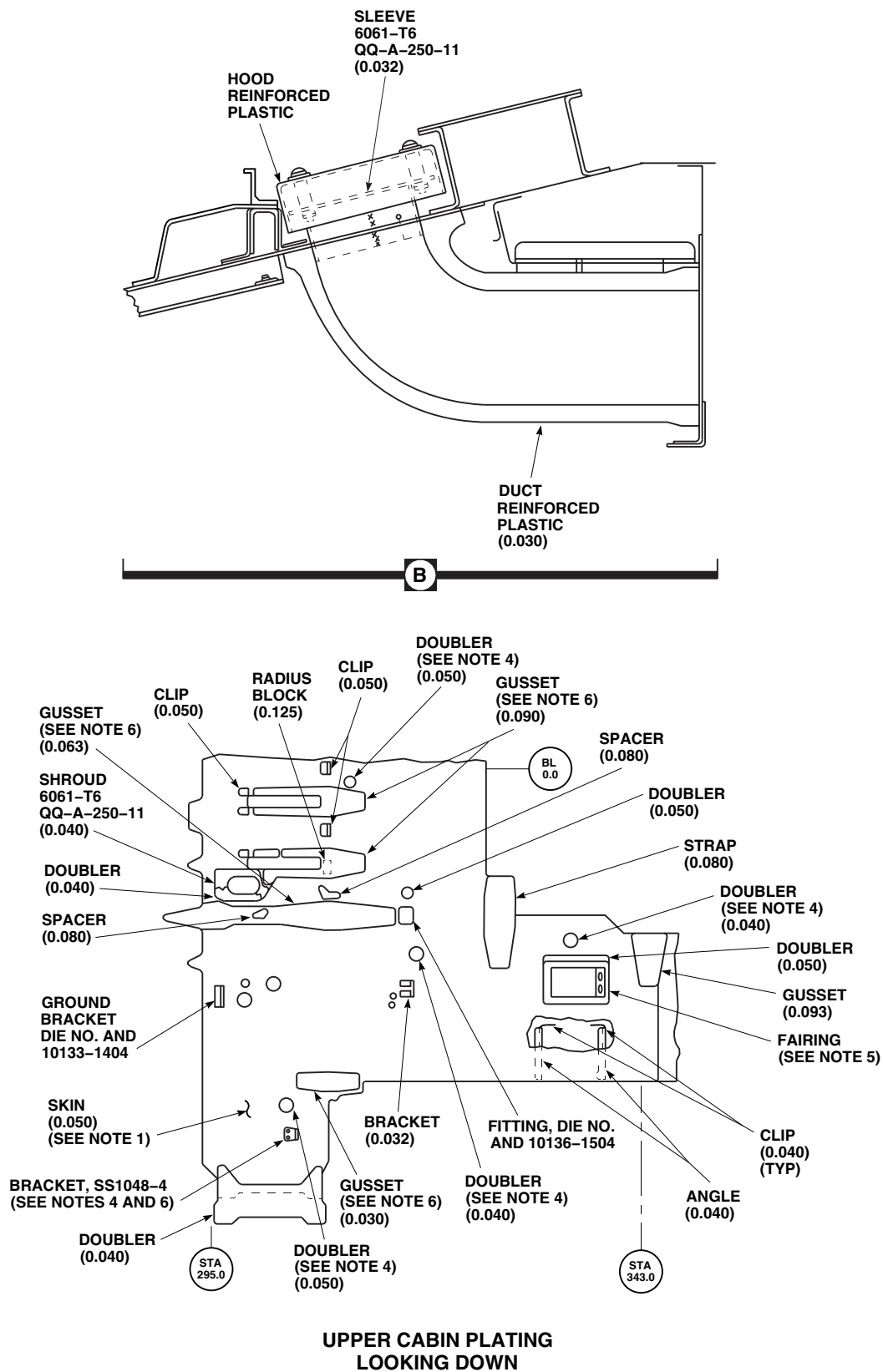
FB1061_3A
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FIGURE 2-6-13-1. REPAIR CABIN SECTION PLATING (SHEET 3 OF 10)

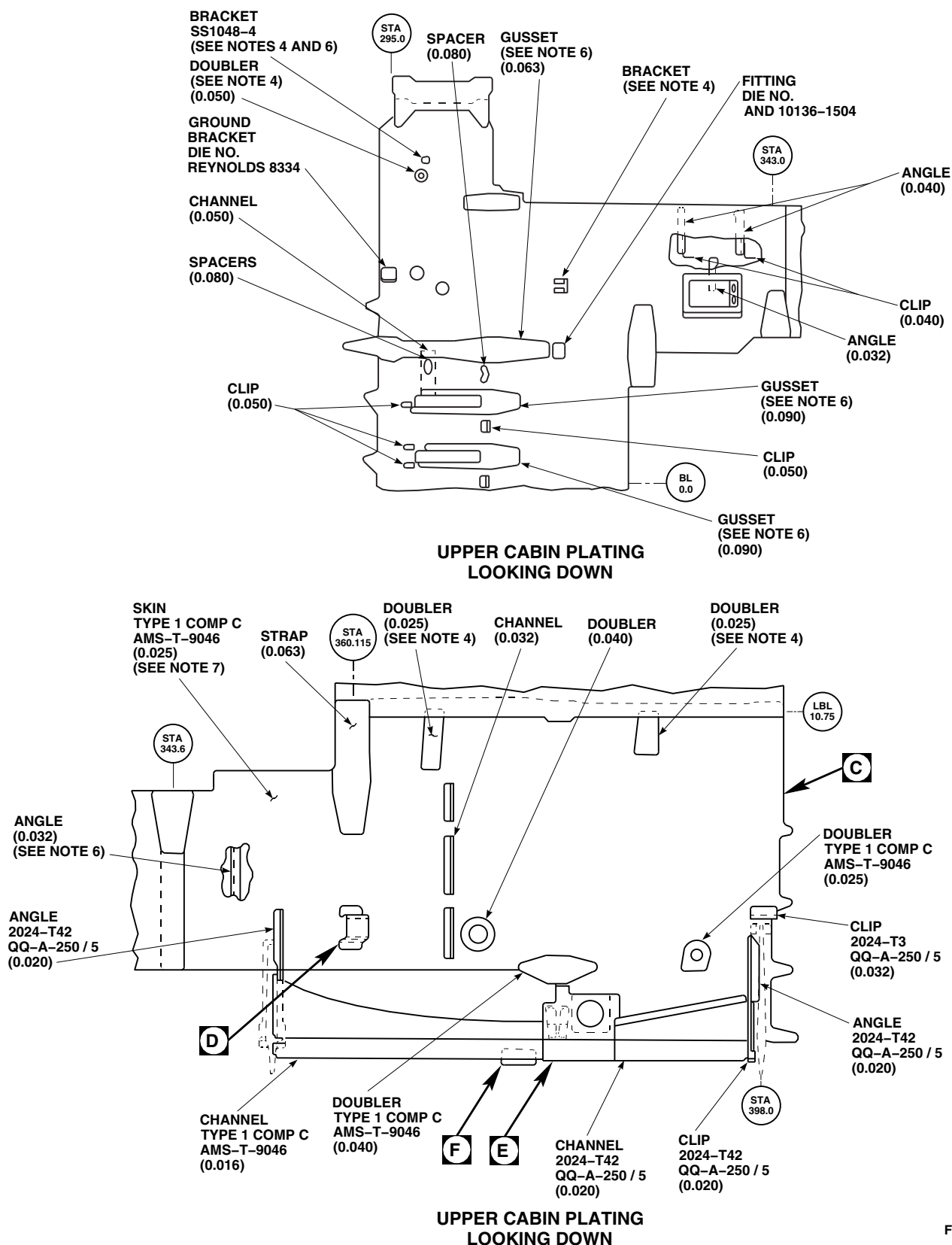


FIGURE 2-6-13-1. REPAIR CABIN SECTION PLATING (SHEET 4 OF 10)

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SA

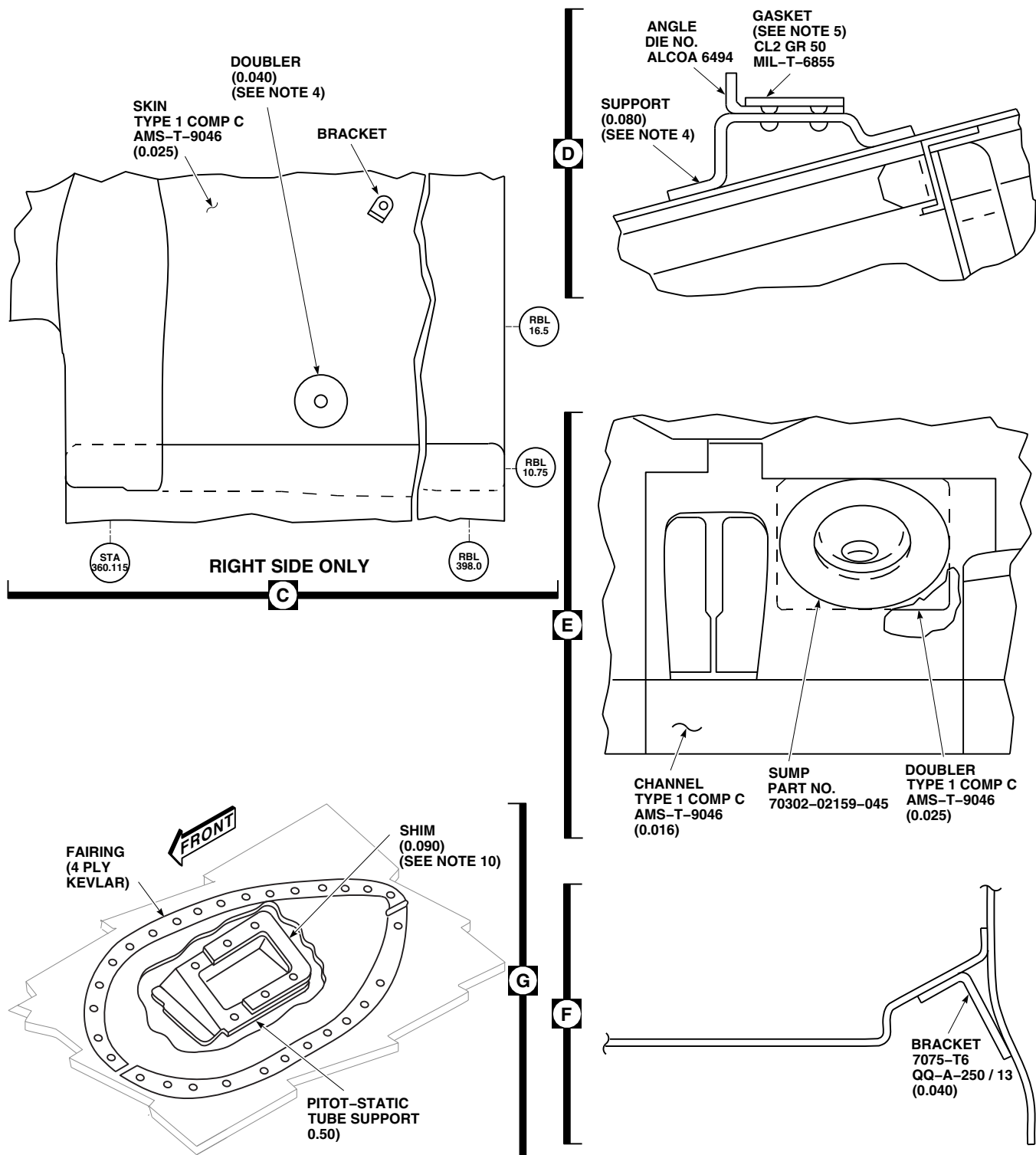
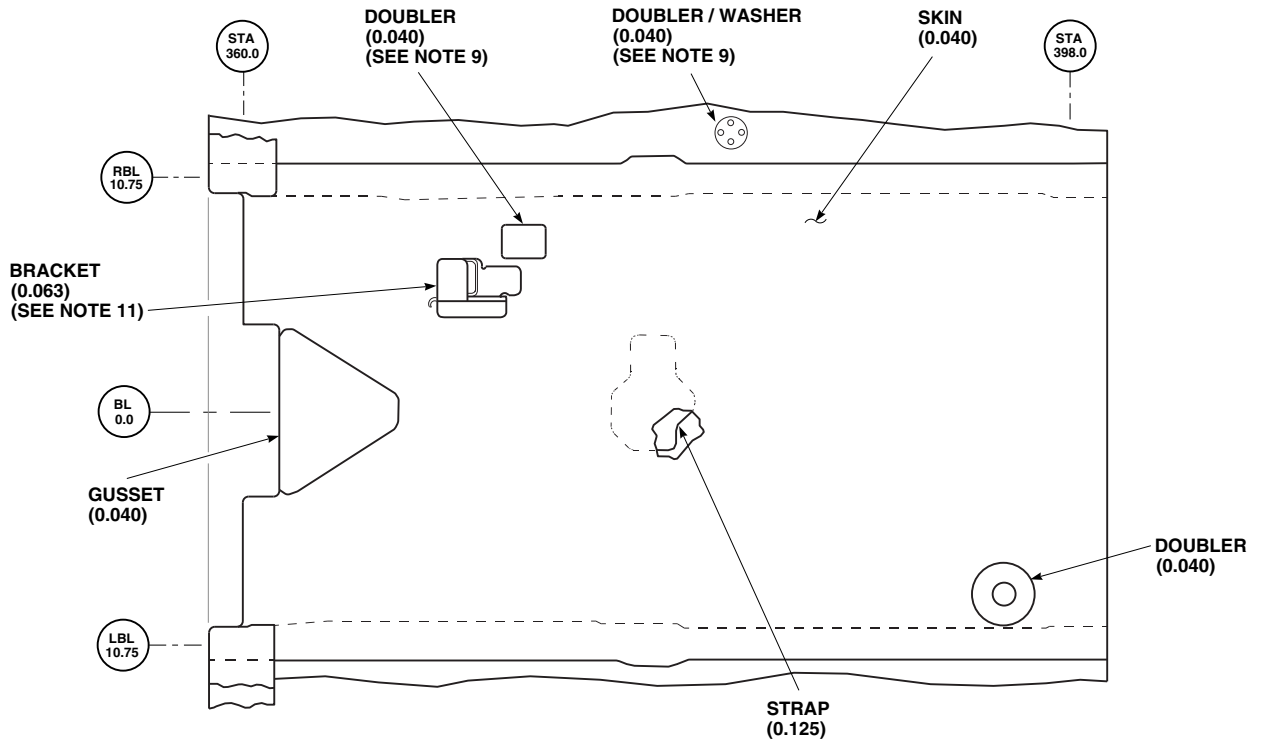
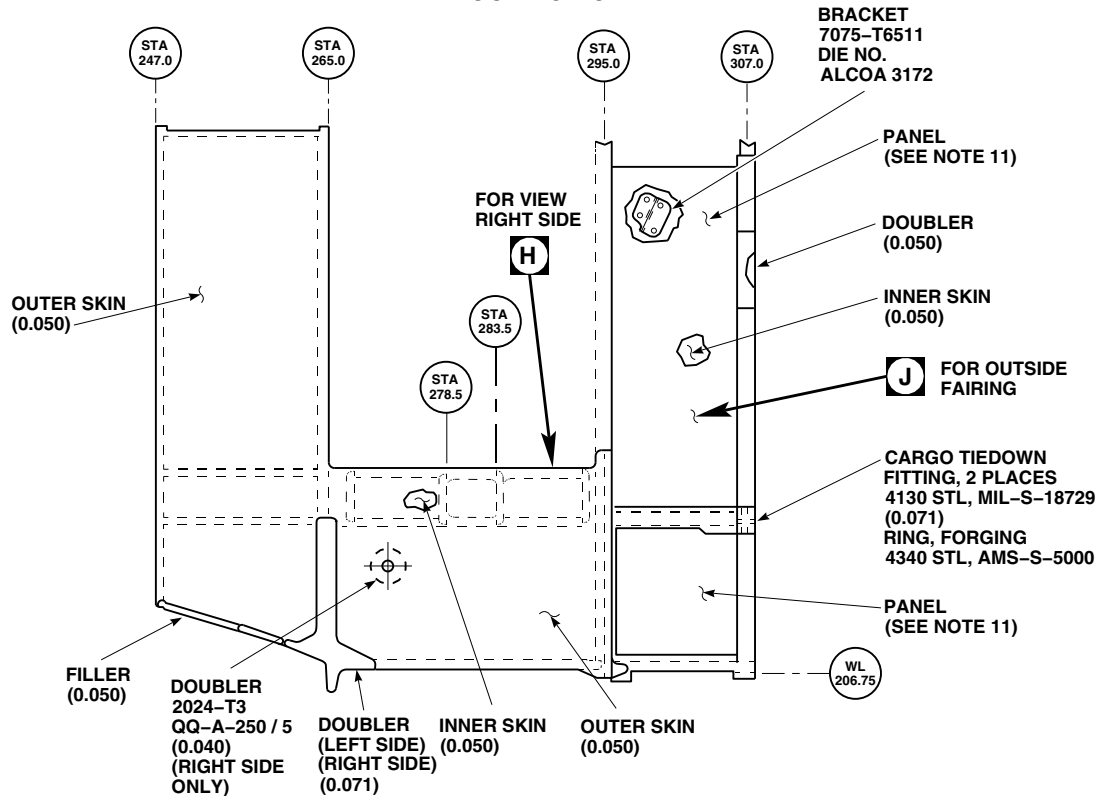
FB1061_5B
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FIGURE 2-6-13-1. REPAIR CABIN SECTION PLATING (SHEET 5 OF 10)



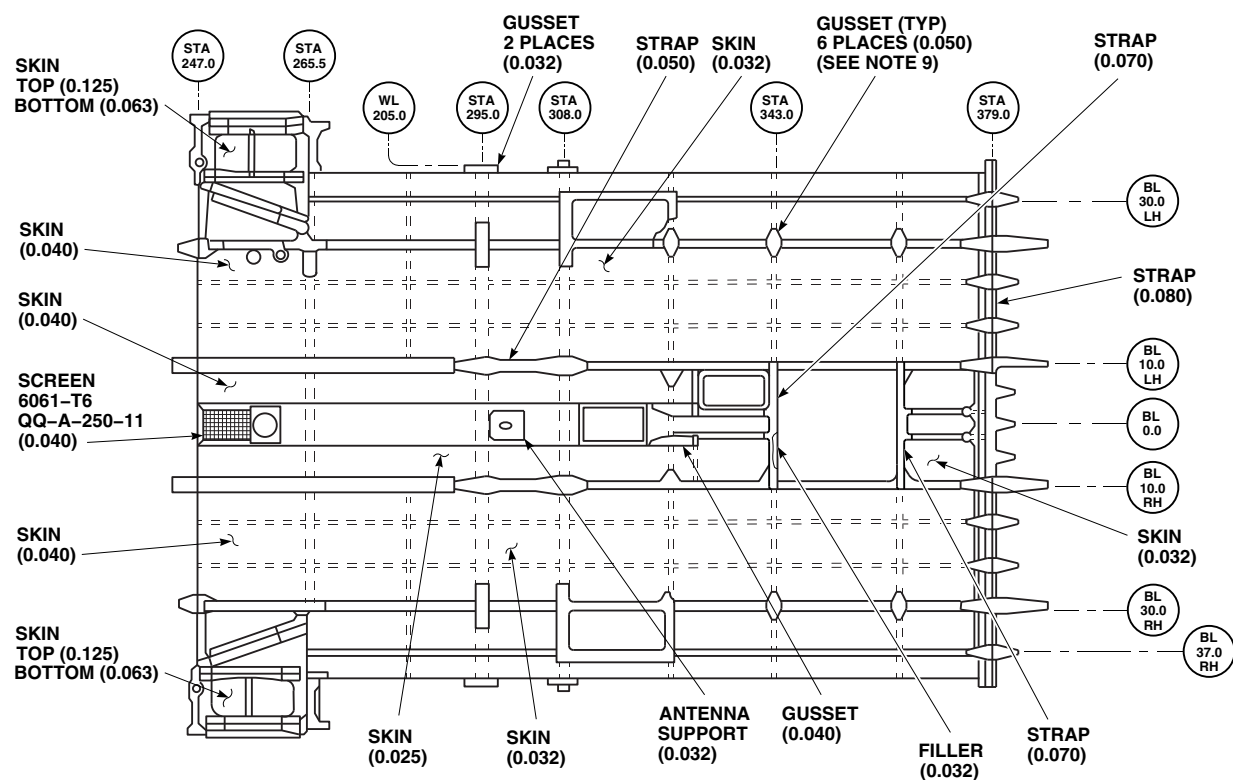
UPPER CABIN PLATING
LOOKING DOWN



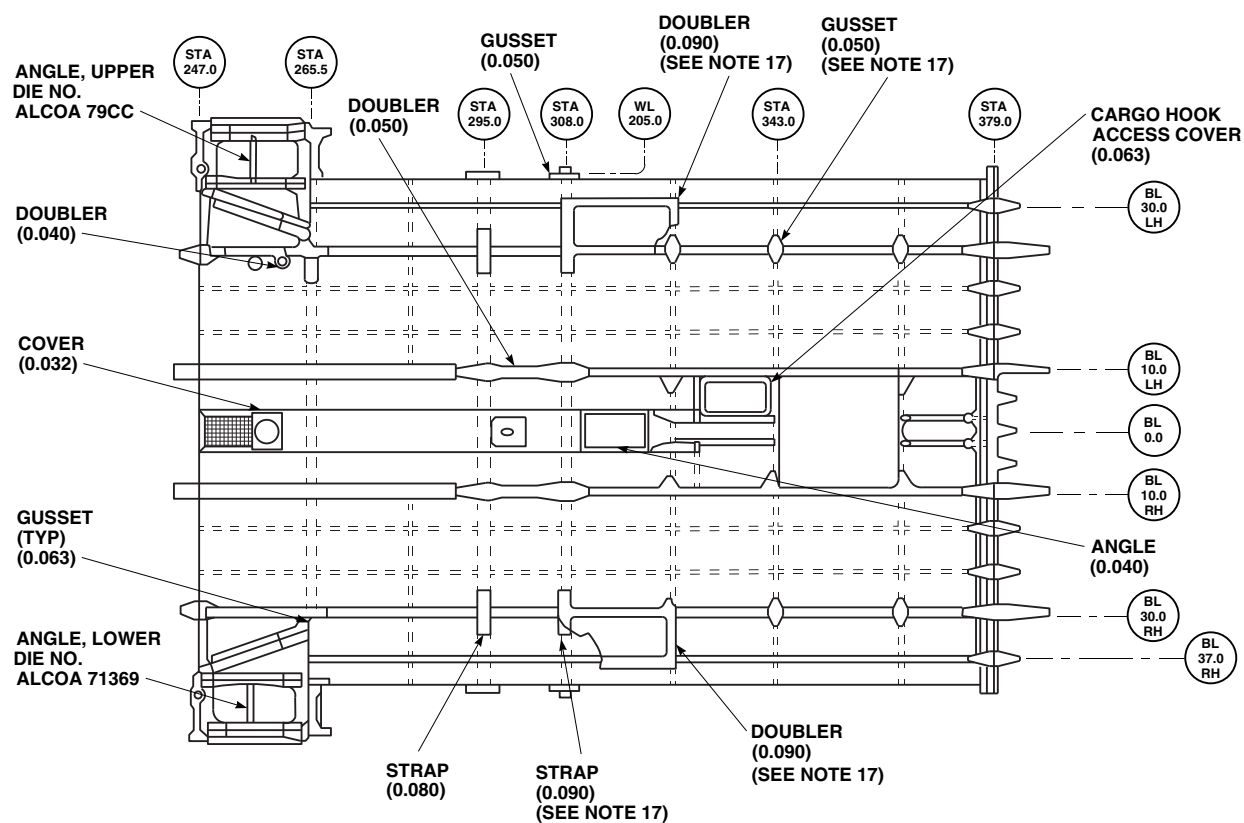
LOOKING INBOARD - LEFT SIDE
RIGHT SIDE SAME EXCEPT AS NOTED

FIGURE 2-6-13-1. REPAIR CABIN SECTION PLATING (SHEET 6 OF 10)

FB1061_6B
SA



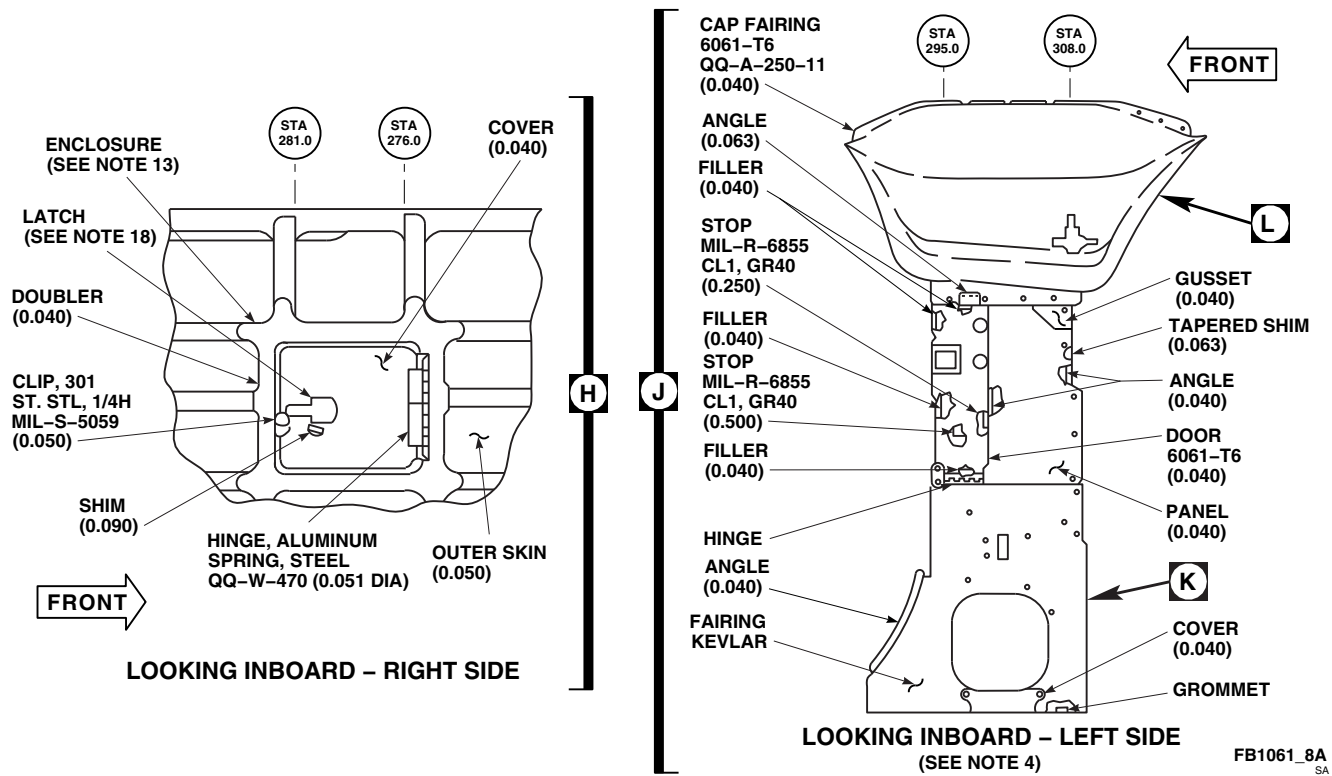
LOWER CABIN PLATING—LOOKING UP

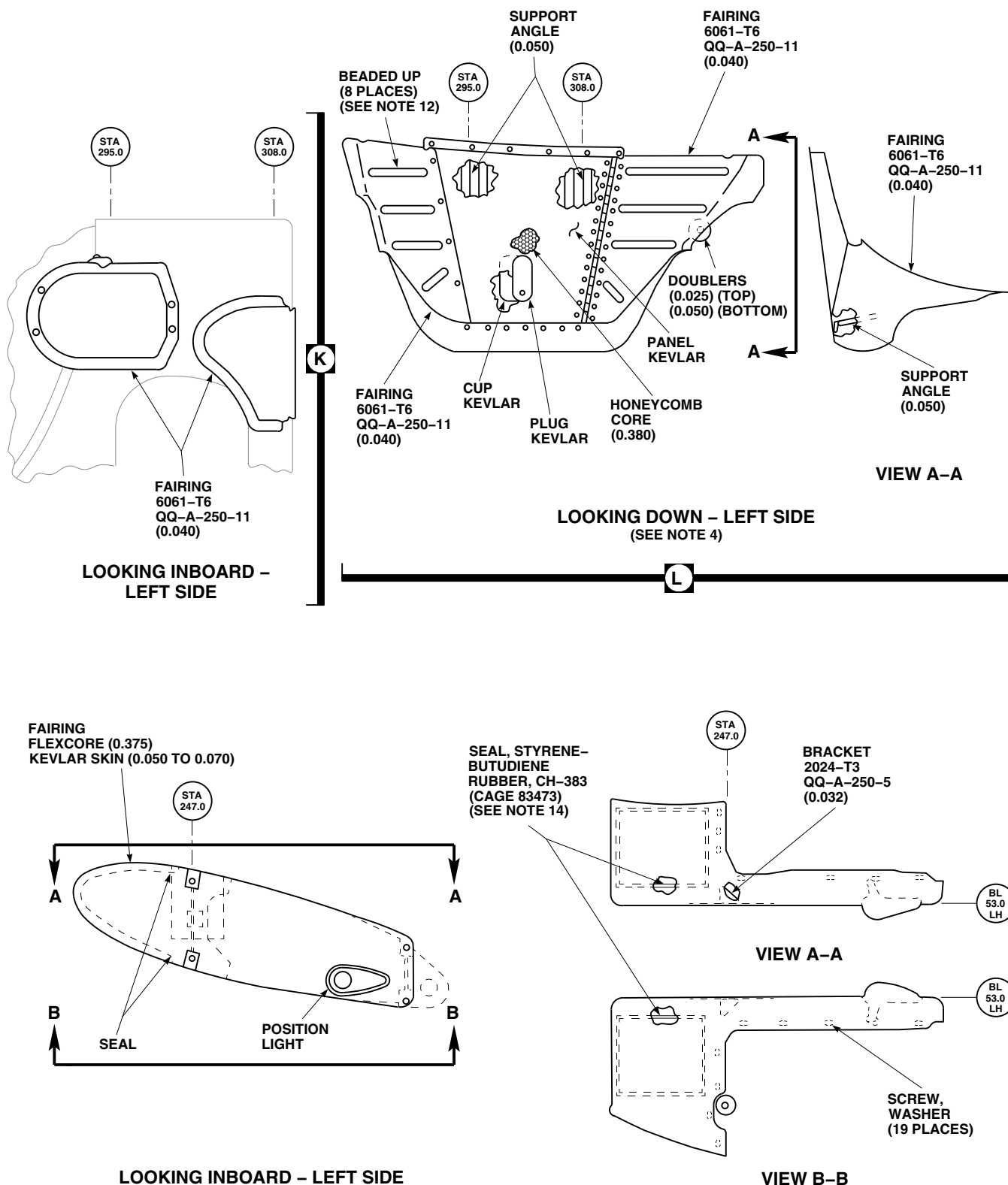


LOWER CABIN PLATING—LOOKING UP

FB1061_7B
SA

FIGURE 2-6-13-1. REPAIR CABIN SECTION PLATING (SHEET 7 OF 10)





FB1061_9A
SA

FIGURE 2-6-13-1. REPAIR CABIN SECTION PLATING (SHEET 9 OF 10)

NOTES

1. UNLESS OTHERWISE NOTED ON SHEETS OF THIS FIGURE, ALL SHEET METAL PARTS SHOWN ARE ALCLAD 7075-T6, QQ-A-250 / 13, AND ALL EXTRUSIONS ARE ALUMINUM ALLOY 7075-T6511, QQ-A-200 / 11.
2. LEFT AND RIGHT SIDES ARE THE SAME, EXCEPT WHERE NOTED.
3. SEAL ALL FAYING SURFACES WITH SEALING COMPOUND, ITEM 295A, APPENDIX D (PARA 2-6-1).
4. **S70-II** 0.058. ◀
5. **70583 - 70692**
6. 2024-T3, QQ-A-250 / 15.
7. REINFORCED PLASTIC 0.030; ALTERNATE MATERIAL: 6061-T6, QQ-A-250 / 11 0.032.
8. STEEL, 4130, COND N, MIL-S-18729.
9. **701539 - SUBQ** 0.020, AMS-T-9046, TYPE III, COMP C. ◀
10. BOND WITH ADHESIVE, ITEM 37, APPENDIX D.
11. CLEAN AND SEAL WITH SEALING COMPOUND, ITEM 292, APPENDIX D (PARA 2-6-1).
12. **S70-II**
13. **701013 - SUBQ**
14. **701264 - SUBQ**
15. DAMPING FOIL SJ2017X (CAGE 76381).
16. PLUG OPENINGS WITH SEALING TAPE, MIL-S-1030. SEAL WITH SEALING COMPOUND, ITEM 292, APPENDIX D. FILLET SEALING COMPOUND ALONG EDGE OF WINDOW FRAME AND SKIN (PARA 2-6-1).
17. **W/O ESSS**
18. H-5010K1941-040-125 (CAGE 83014).
19. FIBERGLASS AND 6061-T6 ENCLOSURES ARE INTERCHANGEABLE.
20. **70583 - 701085**
21. BOND TO FAIRING WITH ADHESIVE, ITEM 31, APPENDIX D.
22. APPLY SEALING COMPOUND, ITEM 295A, APPENDIX D, TO MATING SURFACES AND FASTENERS. INSTALL WHILE WET.

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SA

FIGURE 2-6-13-1. REPAIR CABIN SECTION PLATING (SHEET 10 OF 10)

2-6-14 CABIN SECTION BEADED INNER FUSELAGE PLATING - STA 295.0 THROUGH STA 308.0 REPAIR.

PARAGRAPH BREAKDOWN

		PAGE
2-6-14.1	Repair Cabin Section Beaded Inner Fuselage Plating - STA 295.0 through STA 308.0	2-6-14-1

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit

References

- PARA 2-6-1

2-6-14.1 Repair Cabin Section Beaded Inner Fuselage Plating - STA 295.0 through STA 308.0.

NOTE

Only a total of five beads are permitted to be repaired per panel, after which panel must be replaced.

- Evaluate repairs using PARA 2-6-1.
- Cut away damaged bead 1/16-inch beyond bead bend (Figure 2-6-14-1, Detail B).
- Round corners to 1/4-inch radius.
- Secure narrow doubler (Figure 2-6-14-1, Detail A) to large doubler with single row of rivets (Figure 2-6-14-1).

- With narrow doubler fitted into bead cutout, butt large doubler against existing strap fastened to STA 308.0 frame.
- Secure large doubler to existing skin using double row of rivets on STA 295.0 side of cutout.
- Space rivets 0.38-inch from edge of large doubler and 1-inch apart.
- Position shim over strap where front doubler will attach.
- Secure front doubler over shim and large doubler, picking up existing rivet holes at STA 308.0.

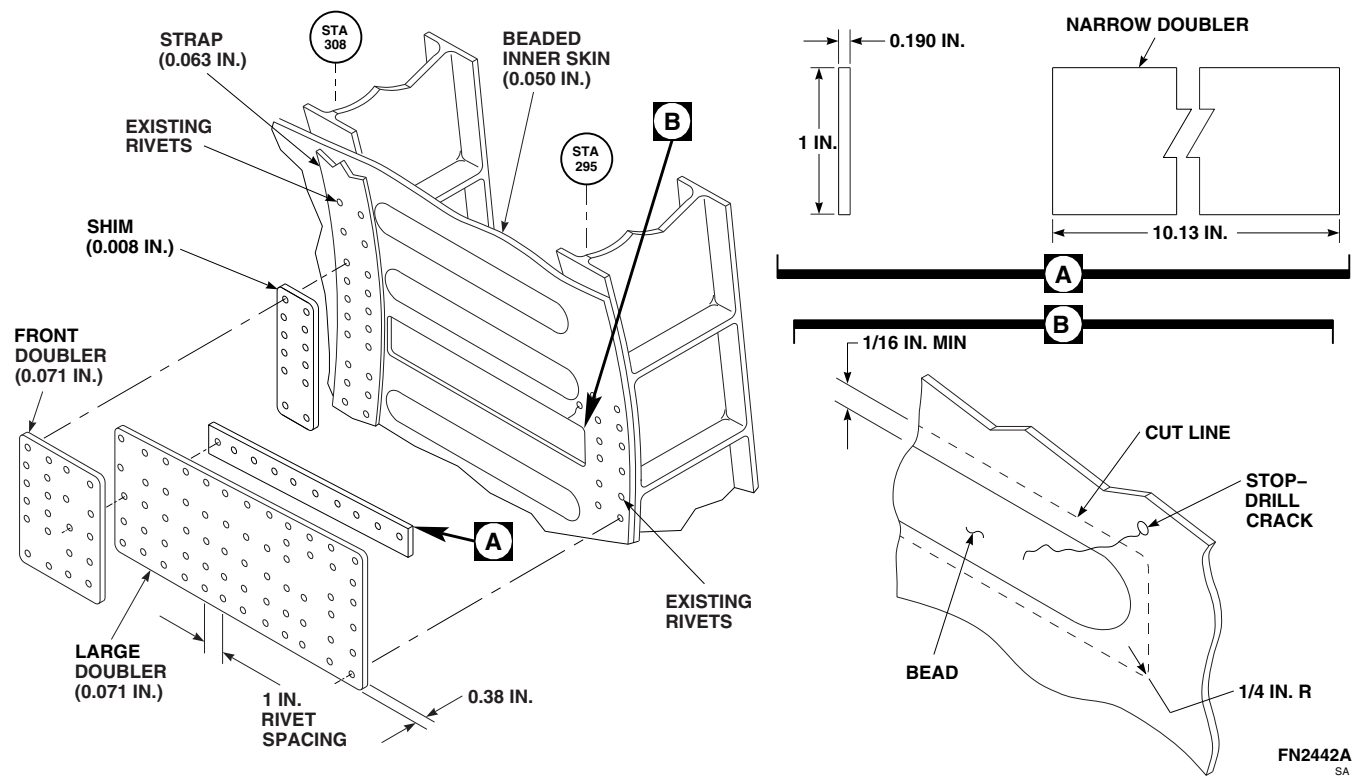


FIGURE 2-6-14-1. REPAIR CABIN SECTION BEADED INNER FUSELAGE PLATING - STA 295.0 - STA 308.0

2-6-15 CABIN DOOR UPPER TRACK REPAIR.

PARAGRAPH BREAKDOWN

	PAGE
2-6-15.1 Repair Cabin Door Upper Track	2-6-15-1

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Clamp
- Drill Set
- Torque Wrench, 30 - 150 in. lbs

Materials/Parts

- Adhesive, Item 32, Appendix D
- Corrosion Preventive Compound, Item 115, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Plastic, Polyvinyl Sheet, Item 237B, Appendix D
- Sealing Compound, Item 292, Appendix D

- Sealing Compound, Item 295A, Appendix D
- Wear Strip - Cabin Door Upper Track, Locally-Made, Figure H-142, Appendix H

References

- AMS 4013
- AMS-DTL-22499
- Appendix D
- Appendix H
- MIL-S-18729
- PARA 2-4-55
- PARA 2-4-65
- PARA 2-4-118
- PARA 2-6-5
- PARA 4-4-7

2-6-15.1 Repair Cabin Door Upper Track.

NOTE

Replacement of left and right cabin door upper tracks is the same.

2-6-15.1.1 Remove.

- a. Remove cabin door (PARA 2-4-55).
- b. Remove cabin door upper fairings (PARA 2-4-65).
- c. Remove engine cowling (PARA 2-4-118).
- d. Remove engine exhaust module (PARA 4-4-7).
- e. Remove screws, washers, and front cabin fairing (Figure 2-6-15-1). Remove sealing compound from fairing and fuselage.

- f. Remove fasteners securing soundproofing to straps riveted on track.
- g. Drill out and remove rivets from bracket and track at STA 314.0 (Figure 2-6-15-1, Sheet 1, Detail B).
- h. Working from front to rear, remove hardware securing track to helicopter structure (Figure 2-6-15-1, Sheet 1 and Sheet 2). Have assistant hold track at front.

NOTE

Note number and location of shims and grip length of bolts and pins for track installation.

- i. Remove fasteners securing upper track to helicopter structure.
- j. Remove upper track from helicopter.

2-6-15.1.2 Inspect.

- a. Visually check track for cracks. **NONE ALLOWED.** If cracked, replace track.
- b. Check for worn or missing wear strips. **WEAR STRIPS SHALL NOT BE WORN TO LESS THAN 0.010-INCH THICK.** If limit is exceeded, replace strips.
- c. Check track for wear. **TRACK SHALL NOT BE WORN TO LESS THAN 0.040-INCH THICK.** If limit is exceeded, replace track.
- d. Inspect areas around fasteners for surface corrosion. Touch up area (PARA 2-6-5).

2-6-15.1.3 Repair.**NOTE**

- Locally make replacement wear strip only when replacing wear strips that vertical rollers rest on. For other wear strips, bond normal replacement wear strip, obtained through supply channels, as follows, but do not rivet to track.
- Repair of left and right cabin door upper track wear strip is the same.

- a. Remove worn wear strip.
- b. Using adhesive, Item 32, Appendix D, fill in, flush with original surface, any worn spots in track not exceeding repair limits.
- c. If required, locally make replacement wear strip (Figure H-142, Appendix H).
- d. Place and center wear strip on area where old wear strip was located.

NOTE

When door is on wear strips (either fully open, or fully closed), rollers will rest on center of strip.

- e. If required, mark and drill rivet holes in track.
- f. Using machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D, clean bonding area on track.
- g. Mix adhesive, Item 32, Appendix D, and apply to bonding area on track, and on wear strip.
- h. Apply wear strip to track.
- i. Cut plastic polyvinyl sheet, Item 237B, Appendix D, to size and place on area where wear strip is to be bonded.
- j. Immediately clamp wear strip in place. Allow adhesive to cure for 1 hour at 160°F (71°C) or 24 hours at 75°F (24°C).

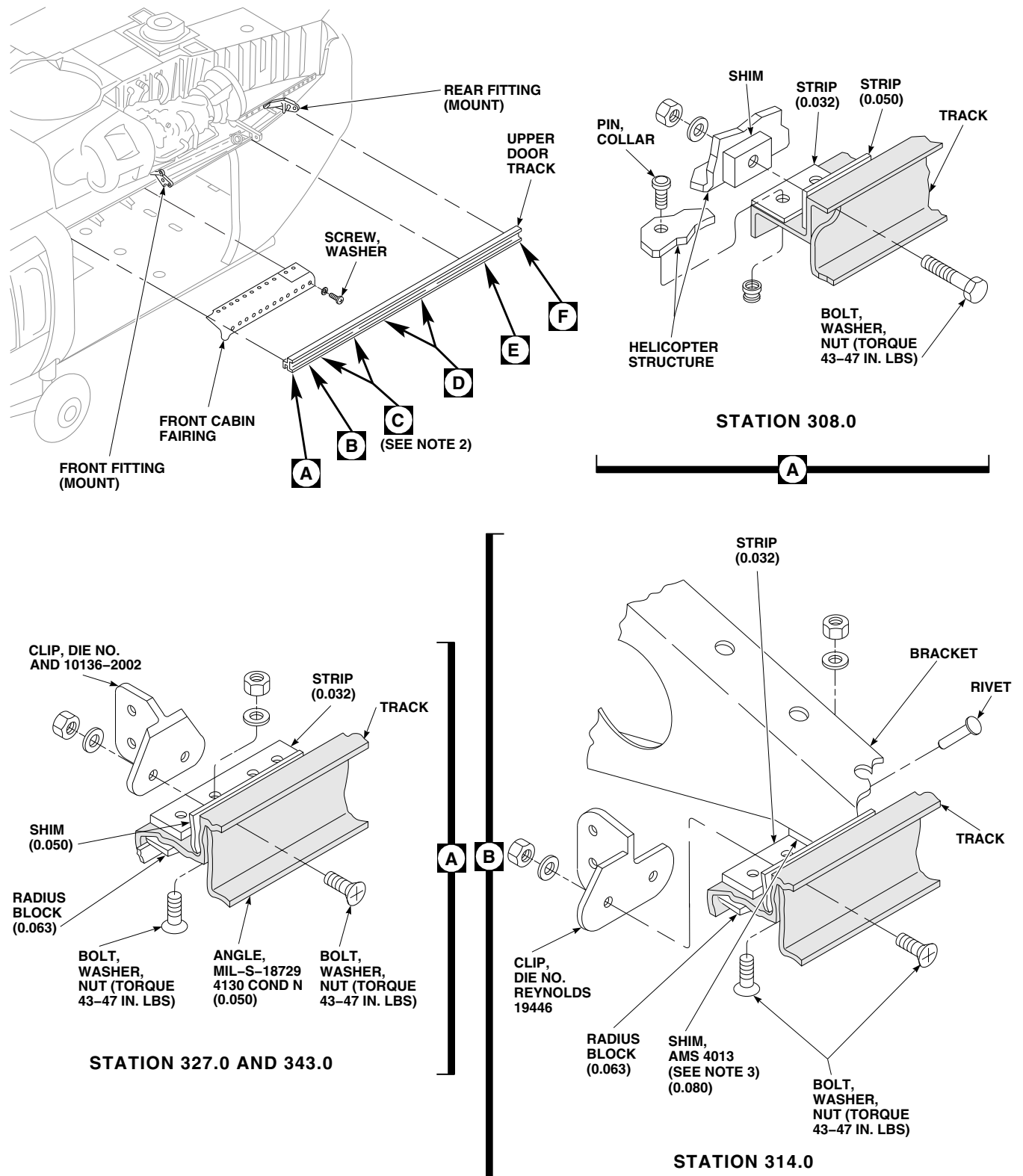
NOTE

Make sure rivets do not protrude above surface of wear strip.

- k. Install rivets in strip.

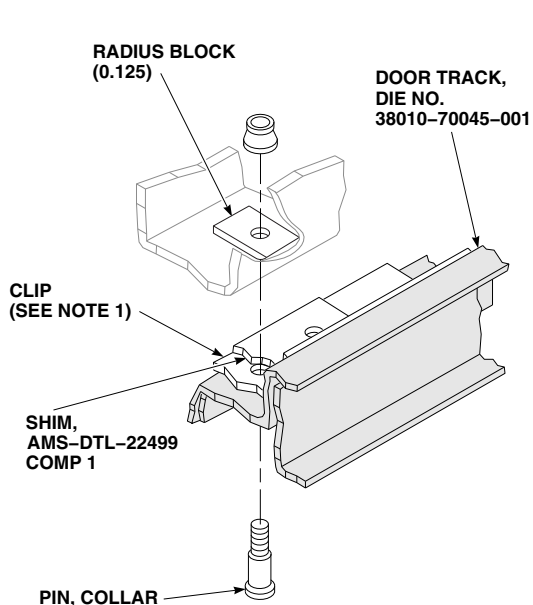
2-6-15.1.4 Install.

- a. Apply sealing compound, Item 295A, Appendix D, to mating surfaces of upper track and fuselage.
- b. Have assistant hold upper track in place on helicopter while sealing compound is still wet.

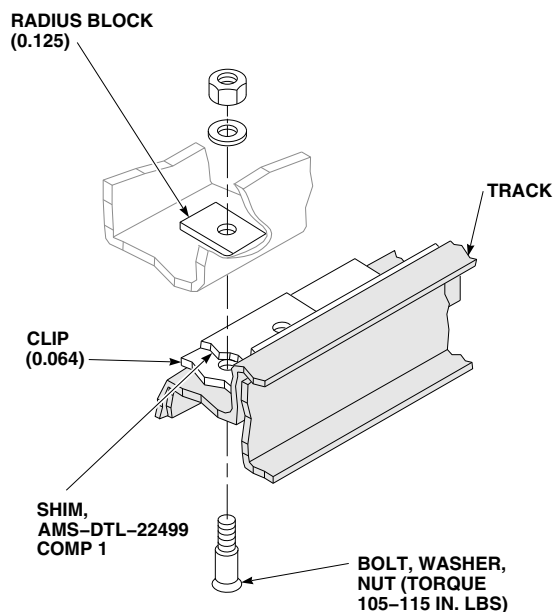


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FIGURE 2-6-15-1. REPAIR CABIN DOOR UPPER TRACK (SHEET 1 OF 3)



STATION 352.0 AND 379.0

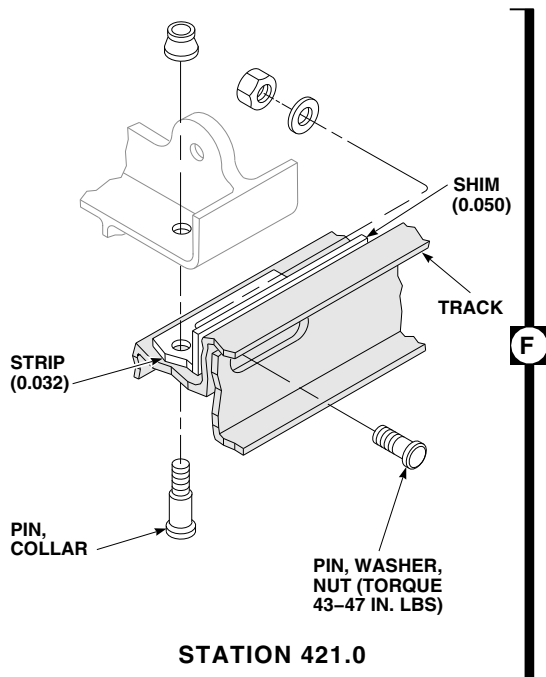


EFFECTIVITY

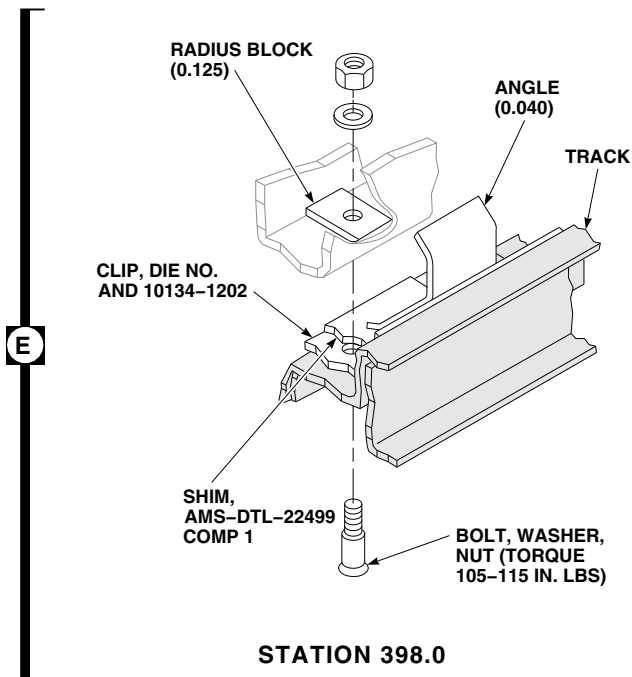
70745-SUBQ

STATION 379.0

D



STATION 421.0



STATION 398.0

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FIGURE 2-6-15-1. REPAIR CABIN DOOR UPPER TRACK (SHEET 2 OF 3)

NOTES

1. DIE NO. AND10134-1202 AT STA 352.0, MIL-S-18729, 4130 COND A / N AT STA 379.0.
2. **ESSS**
3. MAY OR MAY NOT BE PRESENT.
4. **70583-70745**
5. ALL DIMENSIONS ARE IN INCHES, UNLESS NOTED.

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FIGURE 2-6-15-1. REPAIR CABIN DOOR UPPER TRACK (SHEET 3 OF 3)

- c. At STA 308.0, apply sealing compound, Item 295A, Appendix D, to pins, collars, bolt, washer, nut, and shim (Figure 2-6-15-1, Sheet 1, Detail A). Install wet. **TORQUE NUTS TO 43 - 47 INCH-POUNDS.**
- d. At STA 314.0, 327.0, and 343.0, apply sealing compound, Item 295A, Appendix D, to bolts, washers, nuts, shims, and radius blocks (Figure 2-6-15-1, Sheet 1, Details B and C). Install wet. **TORQUE NUTS TO 43 - 47 INCH-POUNDS.**
- e. **70583-70784** At STA 352.0, apply sealing compound, Item 295A, Appendix D, to pins, collars, spacer, shims, and radius blocks (Figure 2-6-15-1, Sheet 2, Detail D). Install wet.
- f. **70785-SUBQ** At STA 379.0, apply sealing compound, Item 295A, Appendix D, to bolts, washers, nuts, shims, and radius blocks. Install wet. **TORQUE NUTS TO 105 - 115 INCH-POUNDS.**
- g. At STA 398.0, apply sealing compound, Item 295A, Appendix D, to bolts, washers, nuts, shims, and radius blocks. Install wet. **TORQUE NUTS TO 105 - 115 INCH-POUNDS** (Figure 2-6-15-1, Sheet 2, Detail E).
- h. At STA 421.0, apply sealing compound, Item 295A, Appendix D, to pins, collars, washers, nuts, and shim (Figure 2-6-15-1, Sheet 2, Detail F). Install wet. **TORQUE NUTS TO 43 - 47 INCH-POUNDS.**
- i. If new track was installed, perform the following:
 - (1) Drill out rivets and remove straps for soundproofing from old track.
 - (2) Install straps on soundproofing with fasteners.
 - (3) Drill and rivet straps to track.
- j. Apply corrosion preventive compound, Item 115, Appendix D to fairing attachment hardware.
- k. Secure front cabin fairing with screws and washers. Seal edges of fairing with sealing compound, Item 292, Appendix D.

- l. Install engine exhaust module (PARA 4-4-7).
- m. Install engine cowling (PARA 2-4-118).
- n. Install cabin door upper fairings (PARA 2-4-65).
- o. Install cabin door (PARA 2-4-55).

2-6-16 CABIN DOOR LOWER TRACK FAIRING REPAIR.

PARAGRAPH BREAKDOWN

	PAGE
2-6-16.1 Repair Cabin Door Lower Track Fairing	2-6-16-1

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Drill Set

Materials/Parts

- Corrosion Preventive Compound, Item 115, Appendix D

References

- Appendix D
- PARA 2-4-55
- PARA 2-4-71
- PARA 2-6-1
- PARA 2-6-5
- QQ-A-250

2-6-16.1 Repair Cabin Door Lower Track Fairing.

NOTE

The following instructions apply to either right or left door lower track fairings.

2-6-16.1.1. Remove.

- Remove cabin door (PARA 2-4-55).
- Remove floor panels (PARA 2-4-71).
- Remove screws and washers attaching fairing to door track (Figure 2-6-16-1).
- Carefully drill out and remove rivets in fairing securing it to fuselage and supports.

2-6-16.1.2. Install.

- Position fairing on fuselage and locate holes.

- Clean and seal under fairing (PARA 2-6-1).
- Install rivets attaching fairing to fuselage.
- Apply corrosion preventive compound, Item 115, Appendix D to fairing attachment hardware.
- Install screws and washers attaching fairing to lower track.
- Make sure area is clean and free of foreign material.
- Touch up paint as required (PARA 2-6-5).
- Install floor panels (PARA 2-4-71).
- Install cabin door (PARA 2-4-55).

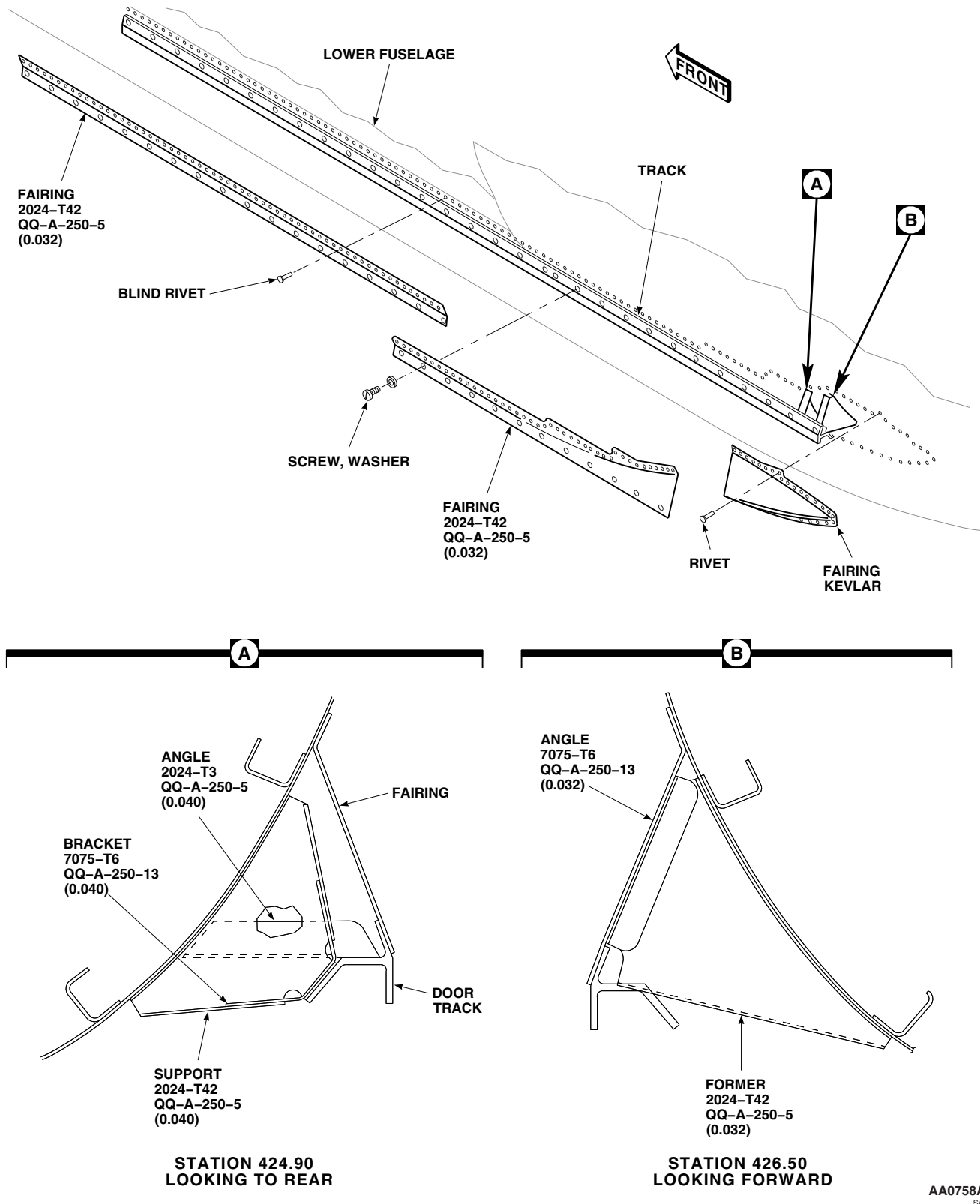


FIGURE 2-6-16-1. REPAIR CABIN DOOR LOWER TRACK FAIRING

2-6-16-2

2-6-17 CABIN DOOR LOWER TRACK WEAR STRIPS REPAIR.

PARAGRAPH BREAKDOWN

	PAGE
2-6-17.1 Repair Cabin Door Lower Track Wear Strips	2-6-17-1

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Clamps
- Drill Set

Materials/Parts

- Adhesive, Item 60, Appendix D
- Machinery Towel, Item 211, Appendix D

- Methyl Ethyl Ketone, Item 217, Appendix D
- Wear Strip - Cabin Door Lower Track, Locally-Made, Figure H-143, Appendix H

References

- Appendix D
- Appendix H

2-6-17.1 Repair Cabin Door Lower Track Wear Strips.

NOTE

Repair of left or right cabin door lower track wear strips is the same.

a. Replace nonriveted wear strips as follows:

- Mark location of wear strip(s). Peel damaged wear strip from track (Figure 2-6-17-1).
- Using machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D, clean remaining adhesive from track. Wipe dry using clean machinery towel, Item 211, Appendix D.
- Locally make replacement wear strips (Figure H-143, Appendix H).

- Position new wear strip on outboard side of track 0.07 inch from bottom edge (Figure 2-6-17-1, Detail A) within location marks made before removing old strip. Using wear strip as a template, drill two 0.125-inch diameter holes through track. Remove wear strip and remove burrs from drilled holes.
- Put a thin coat of adhesive, Item 60, Appendix D, on mating surfaces of track and wear strips.
- Put two wear strips on track, centered on 0.125-inch diameter holes, one on inside and one on outside of track. Rivet wear strips to track with flush rivets. Heads must be flush with wear strip surface (Figure 2-6-17-1, Detail B).
- Using machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D, remove

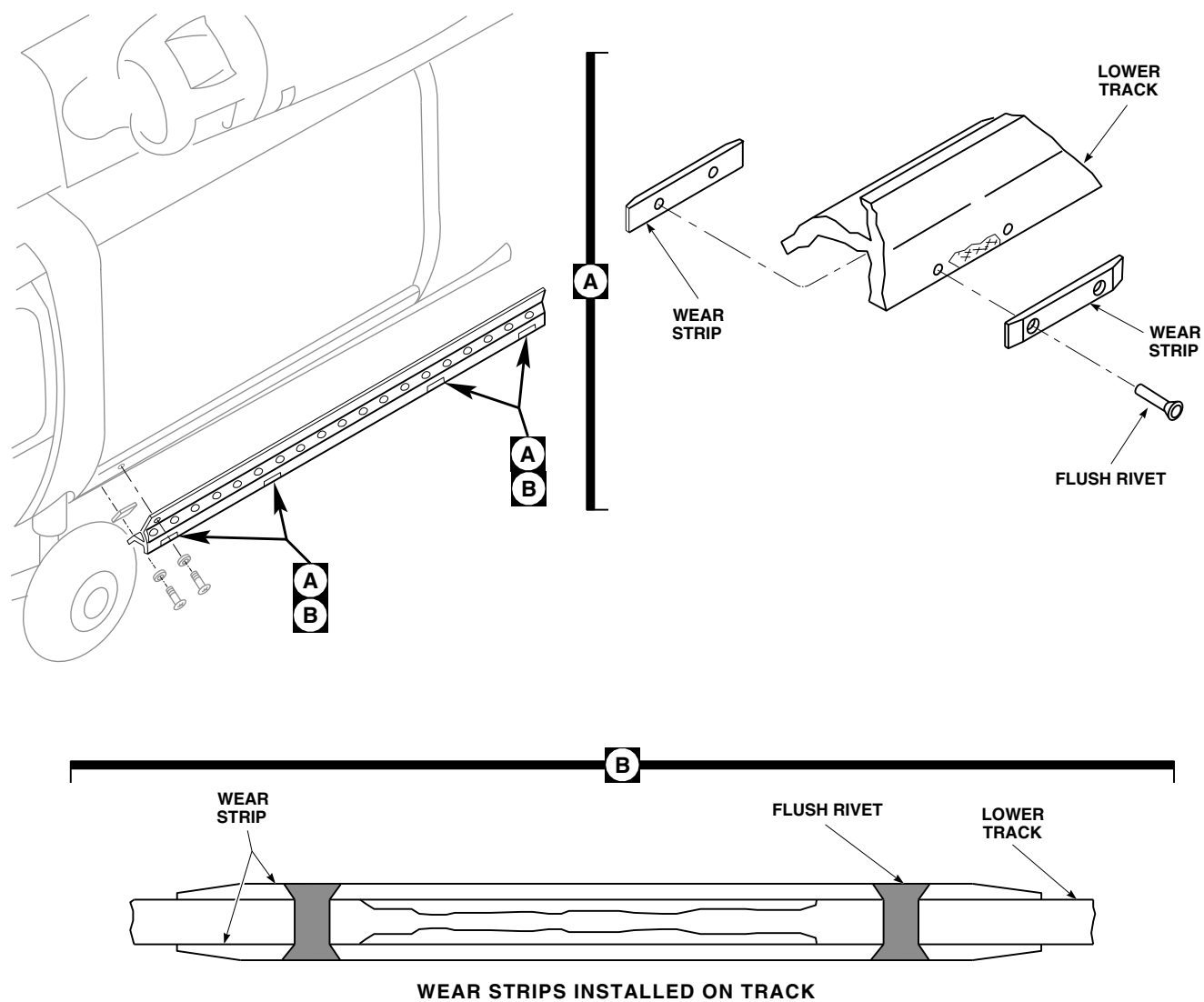
excess adhesive from track and wear strips. Wipe dry using clean machinery towel, Item 211, Appendix D.

b. Replace riveted wear strips as follows:

- (1) Drill out rivets and peel damaged wear strips from track.
- (2) Locally make replacement wear strips (Figure H-143, Appendix H). Do not drill countersunk holes now.
- (3) Position two undrilled wear strips on inside of track, 0.07-inch from bottom edge (Figure 2-6-17-1, Detail A) and

equally spaced each side of rivet holes in track. Clamp wear strips to track.

- (4) Using track rivet holes as pilot holes, drill two holes through wear strips using 0.125-inch diameter drill.
- (5) Remove wear strips from track and countersink drilled holes.
- (6) Position two wear strips on track centered on 0.125-inch diameter holes, one on inside and one on outside of track. Rivet wear strips to track with flush rivets. Heads must be flush with wear strip surface (Figure 2-6-17-1, Detail B).



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FIGURE 2-6-17-1. REPAIR CABIN DOOR LOWER TRACK WEAR STRIPS

2-6-18 EXTERNAL POWER RECEPTACLE LATCH, HINGE, AND COVER.

PARAGRAPH BREAKDOWN

		PAGE
2-6-18.1	Replace External Power Receptacle Latch, Hinge, and Cover	2-6-18-1

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Clecos, 5/32-inch diameter
- Drill

- Drill Bit, No. 24 (0.1520-inch diameter)
- Pin Punch, 1/8-inch diameter
- Rivet Gun

2-6-18.1 Replace External Power Receptacle Latch, Hinge, and Cover.

No. 24 drill bit and 1/8-inch diameter pin punch.

NOTE

Perform steps in procedure to extent necessary to remove and install part(s).

a. Replace latch as follows:

- Using drill with No. 24 drill bit and 1/8-inch diameter pin punch, carefully drill out and remove four rivets securing latch to cover (Figure 2-6-18-1, Detail A).
- Position latch on cover and secure in place with 5/32-inch diameter clecos in two diagonal holes.
- Using rivet gun, install rivets in between clecos. Remove clecos and install remaining rivets.

b. Replace cover as follows:

- Carefully drill out and remove rivets securing cover to hinge using drill with

- Position cover on hinge and install two washers in two end holes between hinge and cover and secure with rivets.

c. Replace hinge as follows:

- If hinge spring is broken, but hinges are serviceable, straighten one crimped end of hinge pin and drive out hinge pin and separate hinges.
- Replace spring, mate hinges and install hinge pin. Crimp ends of hinge pin.
- Drill out rivets securing hinge to inside of receptacle enclosure.
- Drill out rivets securing hinge to cover.
- Install replacement hinge to inside of receptacle enclosure and secure using rivets with washers under the heads of the rivets.

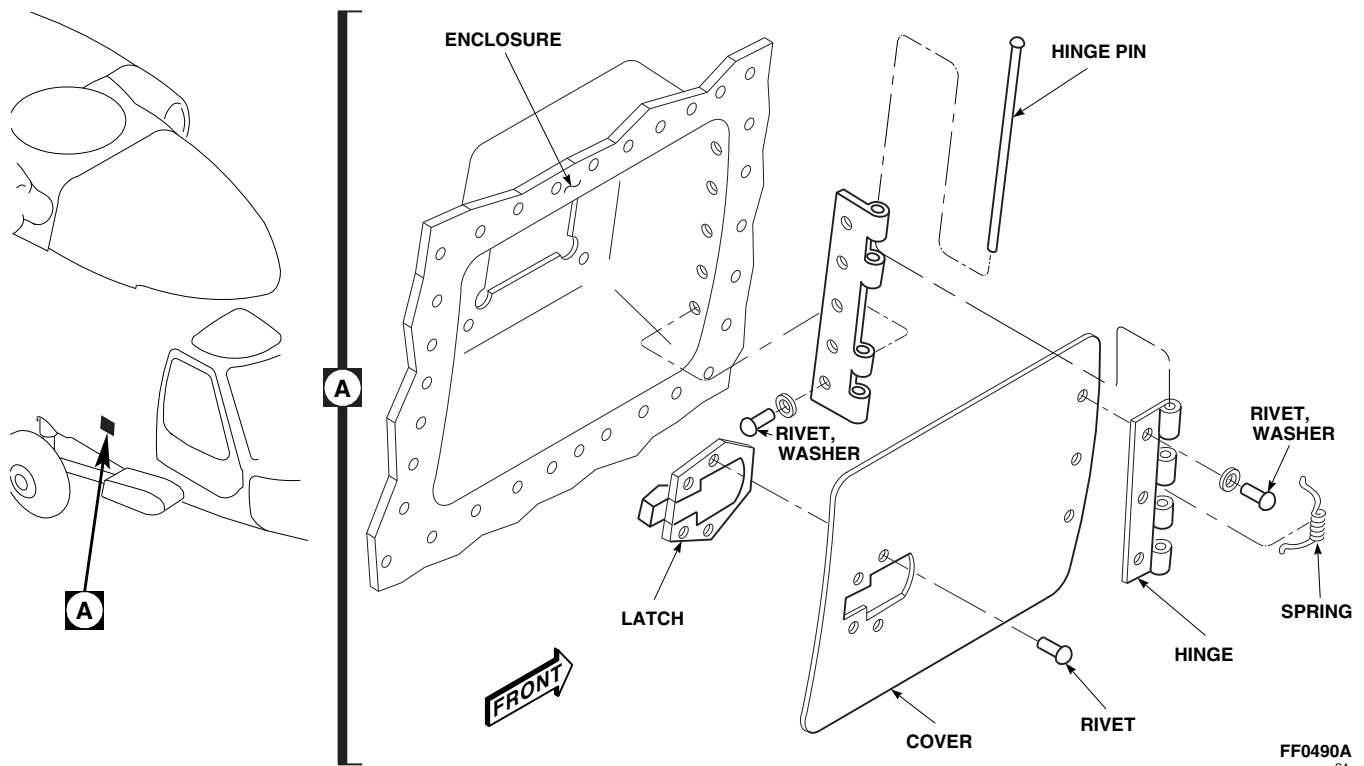


FIGURE 2-6-18-1. REPLACE EXTERNAL POWER RECEPTACLE LATCH, HINGE, AND COVER

- (6) Install cover on hinge.

2-6-19 RESCUE HOIST AIRFRAME FITTING REPAIR.

PARAGRAPH BREAKDOWN

	PAGE
2-6-19.1 Repair Rescue Hoist Airframe Fitting	2-6-19-1

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Fluorescent Penetrant Inspection Kit

Materials/Parts

- Abrasive Cloth, Item 6, Appendix D
- Brush, Acid, Item 84, Appendix D
- Corrosion Resistant Coating, Item 118, Appendix D

- Epoxy Primer Coating Kit, Item 135, Appendix D
- Paint Remover, Item 230, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 2-6-5
- PARA 14-4-1

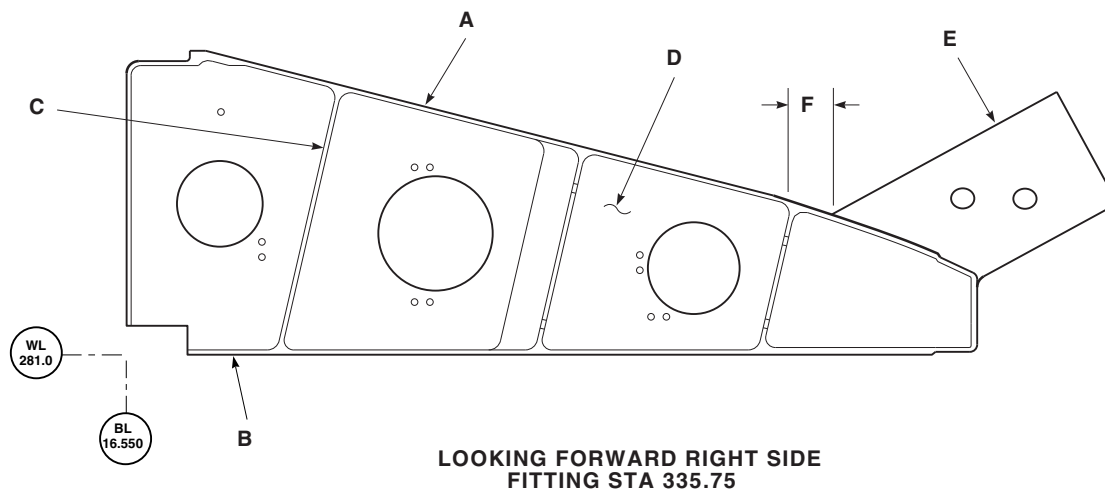
2-6-19.1 Repair Rescue Hoist Airframe Fitting.

- Remove rescue hoist (PARA 14-4-1).
- Remove paint, sealing compound, and primer from inner bore and outside surface area of socket using paint remover, Item 230, Appendix D (Figure 2-6-19-1, Sheet 1).
- Inspect airframe fitting socket Area E for cracks, using eddy current or fluorescent penetrant inspect using fluorescent penetrant inspection kit (ASTM E1417) paying particular attention to bolthole areas. If no cracks are found, proceed to step e. If cracks are found, repair as follows:

NOTE

Damage greater than limits must be reviewed by engineering authority for damage evaluation.

- Damage to Area E may be blended out up to 0.060-inch deep over an area not to exceed 9% of circumferential cross section (total arc length of 0.85-inch). Material may be removed from remaining surface area provided maximum bore diameter does not exceed 2.992-inch (0.002-inch into wall thickness).
- Blend repairable damage using abrasive cloth, Item 6, Appendix D.
- After blending repeat eddy current inspection to verify crack removal.
- Inspect Area A (upper flange) as follows:
 - Check Area A (upper flange) for nicks, dents, scratches, and gouges. Damage may be blended out up to 0.050-inch deep. Blend damage using abrasive cloth, Item 6, Appendix D.



AREA	INSPECTION	MAXIMUM REPAIRABLE LIMITS
A – UPPER FLANGE	NICKS, DENTS, SCRATCHES, OR GOUGES	NOT DEEPER THAN 0.050 INCH (SEE NOTES 1, 2, AND 3)
	CORROSION	NOT DEEPER THAN 0.020 INCH OVER 100% OF CROSS SECTIONAL WIDTH (SEE NOTES 1 AND 5)
B – LOWER FLANGE	NICKS, DENTS, SCRATCHES, OR GOUGES	NOT DEEPER THAN 0.050 INCH (SEE NOTES 1, 2, AND 3)
	CORROSION	NOT DEEPER THAN 0.020 INCH OVER 100% OF CROSS SECTIONAL WIDTH (SEE NOTES 1 AND 5)
C – STIFFENERS	NICKS, DENTS, SCRATCHES, OR GOUGES	NOT DEEPER THAN 0.050 INCH (SEE NOTES 1, 2, AND 3)
	CORROSION	NOT DEEPER THAN 0.020 INCH OVER 100% OF CROSS SECTIONAL WIDTH (SEE NOTES 1 AND 5)
D – WEBS	NICKS, DENTS, SCRATCHES, OR GOUGES	NOT DEEPER THAN 0.040 INCH (SEE NOTES 1, 2, AND 4)
	CORROSION	NOT DEEPER THAN 0.020 INCH OVER A 1.00-INCH DIAMETER CIRCLE (SEE NOTES 1 AND 5)
E – SOCKET ID	NICKS, DENTS, SCRATCHES, OR GOUGES	NOT DEEPER THAN 0.060 INCH OVER 9% OF CIRCUMFERENTIAL CROSS SECTION (TOTAL ARC LENGTH OF 0.85 INCH) (SEE NOTES 1 AND 2)
	CORROSION	NOT DEEPER THAN 0.002 INCH INTO WALL OVER 100% OF ID. MAXIMUM BORE DIAMETER NOT TO EXCEED 2.990 INCH (SEE NOTES 1 AND 5)
F – SOCKET / WEB INTERFACE	NICKS, DENTS, SCRATCHES, OR GOUGES	NOT DEEPER THAN 0.060 INCH OVER A 1.00 INCH LENGTH AND 0.1875 INCH WIDTH (SEE NOTES 1 AND 2)
	CORROSION	NOT DEEPER THAN 0.020 INCH OVER 100% OF CROSS SECTIONAL WIDTH (SEE NOTES 1 AND 5)

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SA**FIGURE 2-6-19-1. REPAIR RESCUE HOIST AIRFRAME FITTING (SHEET 1 OF 2)****2-6-19-2 Change 10**

This Document contains technical data subject to ITAR. See WARNING and classifications on first page.

NOTES

1. THE MAXIMUM REPAIRABLE LIMITS FOR NICKS, DENTS, SCRATCHES, GOUGES, AND CORROSION ARE NOT CUMULATIVE AT ANY ONE CROSS SECTION.
2. SCRATCHES WHICH FALL IN THE CORROSION REPAIRABLE LIMITS ARE ACCEPTABLE.
3. AFTER REWORK FOR NICKS, DENTS, AND GOUGES, BLENDED AREAS FOR FLANGES AND STIFFENERS SHALL NOT EXCEED 0.375-INCH DIAMETER AND BE NO CLOSER THAN ONE HOLE DIAMETER FROM ANY EXISTING FASTENER HOLE.
4. AFTER REWORK FOR NICKS, DENTS, AND GOUGES, BLENDED AREAS FOR WEBS SHALL NOT EXCEED 0.500-INCH DIAMETER AND BE NO CLOSER THAN ONE HOLE DIAMETER FROM ANY EXISTING FASTENER HOLES.
5. ISOLATED CORROSION PITS MAY BE REPAIRED USING THE LIMITS SPECIFIED IN NICKS, DENTS, AND GOUGES FOR THE PARTICULAR ITEM.

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FIGURE 2-6-19-1. REPAIR RESCUE HOIST AIRFRAME FITTING (SHEET 2 OF 2)

- (2) Check Area A (upper flange) for corrosion. Damage may be blended out up to 0.020-inch deep over 100% of cross sectional width. Blend damage using abrasive cloth, Item 6, Appendix D.
- e. Inspect Area B (lower flange) as follows:
 - (1) Check Area B (lower flange) for nicks, dents, scratches, and gouges. Damage may be blended out up to 0.050-inch deep. Blend damage using abrasive cloth, Item 6, Appendix D.
 - (2) Check Area B (lower flange) for corrosion. Damage may be blended out up to 0.020-inch deep over 100% of cross sectional width. Blend damage using abrasive cloth, Item 6, Appendix D.
- f. Inspect Area C (stiffeners) as follows:
 - (1) Check Area C (stiffeners) for nicks, dents, scratches, and gouges. Damage may be blended out up to 0.050-inch deep. Blend damage using abrasive cloth, Item 6, Appendix D.
 - (2) Check Area C (stiffeners) for corrosion. Damage may be blended out up to 0.020-inch deep over 100% of cross sectional width. Blend damage using abrasive cloth, Item 6, Appendix D.
- g. Inspect Area D (webs) as follows:
 - (1) Check Area D (webs) for nicks, dents, scratches, and gouges. Damage may be blended out up to 0.040-inch deep. Blend damage using abrasive cloth, Item 6, Appendix D.
 - (2) Check Area D (webs) for corrosion. Damage may be blended out up to 0.020-inch deep over 1.00-inch diameter circle. Blend damage using abrasive cloth, Item 6, Appendix D.
- h. Inspect Area F (socket/web interface) as follows:
 - (1) Check Area F (socket/web interface) for nicks, dents, scratches, and gouges. Damage may be blended out up to 0.060-inch over a 1.00-inch length and 0.1875-inch width. Blend damage using abrasive cloth, Item 6, Appendix D.
 - (2) Check Area F (socket/web interface) for corrosion. Damage may be blended out up to 0.020-inch deep over 100% of cross sectional width. Blend damage using abrasive cloth, Item 6, Appendix D.
- i. Treat reworked areas on fitting as follows:
 - (1) Apply corrosion resistant coating, Item 118, Appendix D, to reworked areas with acid brush, Item 84, Appendix D.
 - (2) Maintain surface wetness for 3 to 5 minutes.

- (3) Thoroughly rinse treated areas with cold water.
- (4) Allow reworked areas to dry completely.
- (5) Using epoxy primer coating kit, apply one coat of epoxy primer, Item 135, Appendix

D, to all reworked areas. **DO NOT PAINT BOLTHOLES.**

- (6) Touch up finish on all reworked areas (PARA 2-6-5).
- j. Install rescue hoist (PARA 14-4-1).

2-6-20 EXTERNAL STORES SUPPORT SYSTEM FUSELAGE FITTING BEARING.

PARAGRAPH BREAKDOWN

		PAGE
2-6-20.1	Replace External Stores Support System Fuselage Fitting Bearing	2-6-20-2

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Bearing Installation/Removal Tool, 70700-77533-041
- Inside Micrometer
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Cheesecloth, Item 90, Appendix D

- Corrosion Preventive Compound, Item 113, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Grease, Item 144, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Primer, Item 257, Appendix D
- Scouring Pad, Item 270, Appendix D

References

- Appendix D
- PARA 2-6-5

2-6-20.1 Replace External Stores Support System Fuselage Fitting Bearing.

NOTE

Strut fittings at both STA 295.0 and STA 308.0 have similar construction and identical bearings (Figure 2-6-20-1, Sheet 1, Detail A and Detail B).

2-6-20.1.1 Remove.

CAUTION

Damage to fitting chamfer will occur if staked lip of bearing is improperly removed. Be careful not to damage fitting chamfer when machining off staked lip of bearing.

- Apply grease, Item 144, Appendix D, to mandrel of bearing installation/removal tool.
- Machine roll-staked edge of bearing using bearing installation/removal tool (Figure 2-6-20-1, Sheet 2, Step 1).
- Press out bearing using bearing installation/removal tool (Figure 2-6-20-1, Sheet 2, Step 2).
- Using nonmetallic scraper and cheesecloth, Item 90, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D, clean primer from bore and chamfer.

2-6-20.1.2 Inspect.

- Using inside micrometer, measure inside diameter of fitting. **MAXIMUM ALLOWABLE INSIDE DIAMETER IS 1.7495-INCHES.** If limit is exceeded, replace fitting.
- Inspect fitting bore for corrosion. **PITS MUST NOT EXCEED 0.005-INCH DEEP OR COVER MORE THAN 25% OF FITTING INSIDE DIAMETER.** If limits are exceeded, replace fitting.
- If corrosion is within limits, remove using scouring pad, Item 270, Appendix D.

- Inspect fitting bore for damage. **FITTING BORE DAMAGE MUST NOT EXCEED 0.005-INCH DEEP, 0.040-INCH WIDE, AND 0.75-INCH LONG.** If limits are exceeded, replace fitting.
- If damage is within limits, blend out using scouring pad, Item 270, Appendix D. **MAXIMUM BLENDED DEPTH IS 0.005-INCH.** If limit is exceeded, replace fitting.

2-6-20.1.3 Install.

- Make sure fitting bore, chamfers, and adjacent surfaces are clean and dry.
- Apply light coat of primer, Item 257, Appendix D, to inside of fitting bore and to outside surface of outer race of bearing.
- Apply grease, Item 144, Appendix D, to mandrel of bearing installation/removal tool.
- Press bearing into fitting using bearing installation/removal tool (Figure 2-6-20-1, Sheet 2, Step 3).
- Clean primer, if any, from bearing V-grooves, using cheesecloth, Item 90, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D. Wipe dry.
- Swage V-groove lip over chamfer in fitting bore using bearing installation/removal tool (Figure 2-6-20-1, Sheet 2, Step 4).
- Repeat swaging process on other side of bearing using installation/removal tool (Figure 2-6-20-1, Sheet 2, Step 5).
- Inspect staking impressions as follows (Figure 2-6-20-1, Sheet 1, Detail C):
 - Check for gap between staked lip of bearing and bearing bore chamfer (Figure 2-6-20-1, Sheet 1, Detail D). **GAP SHALL NOT EXCEED 0.005-INCH MAXIMUM.** If limit is exceeded, replace bearing.
 - Check for cracks in staked lip (Figure 2-6-20-1, Sheet 1, Detail E). **NONE AL-**

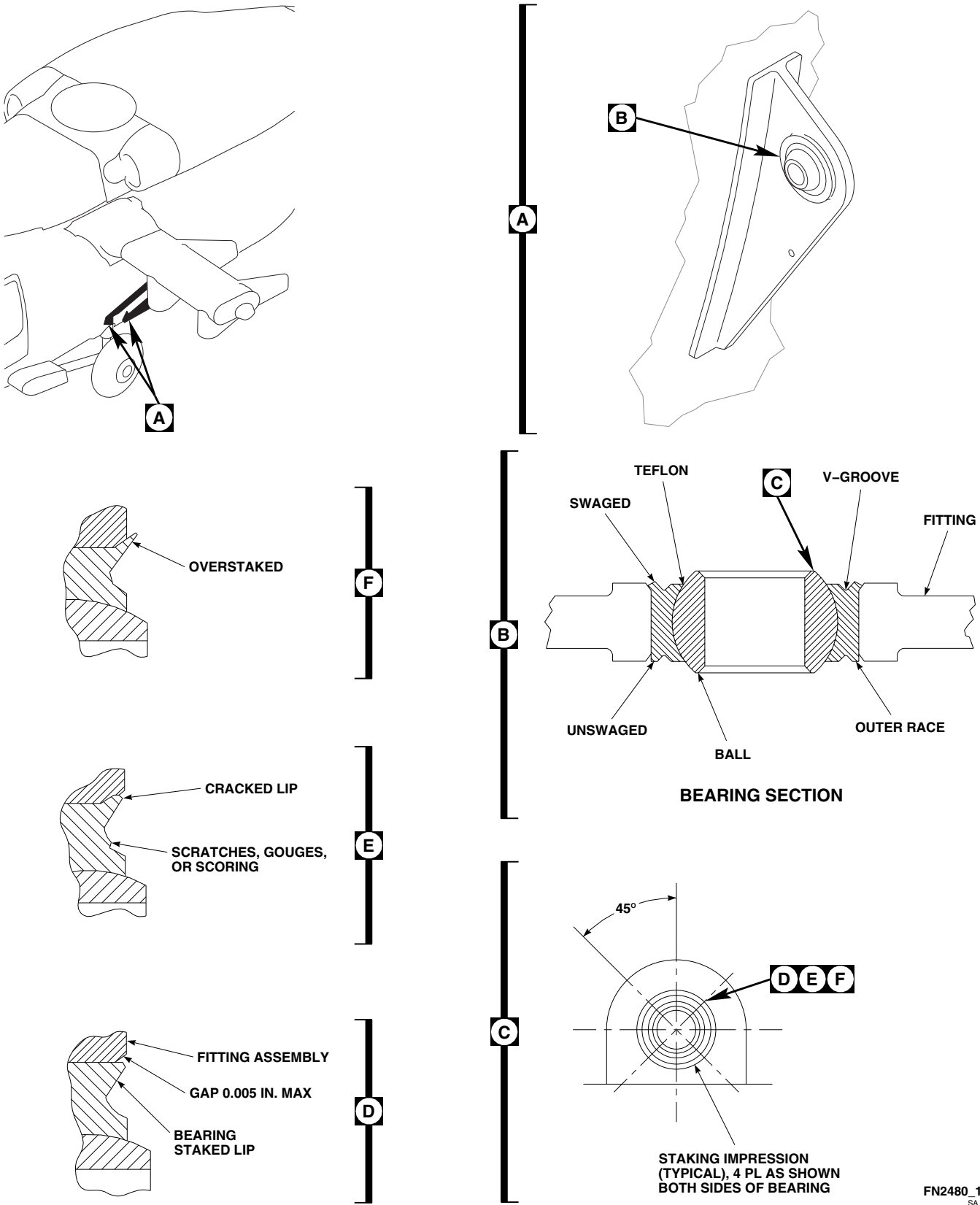
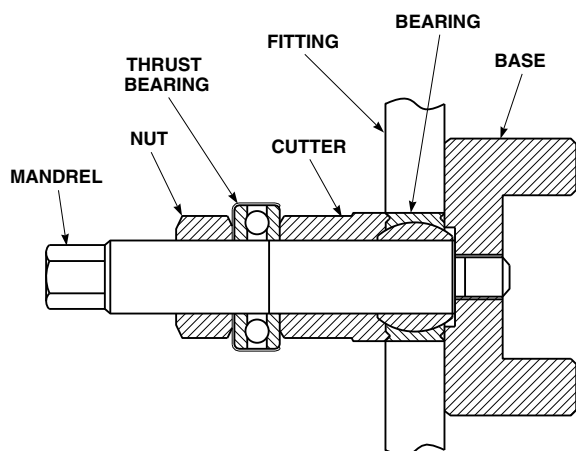
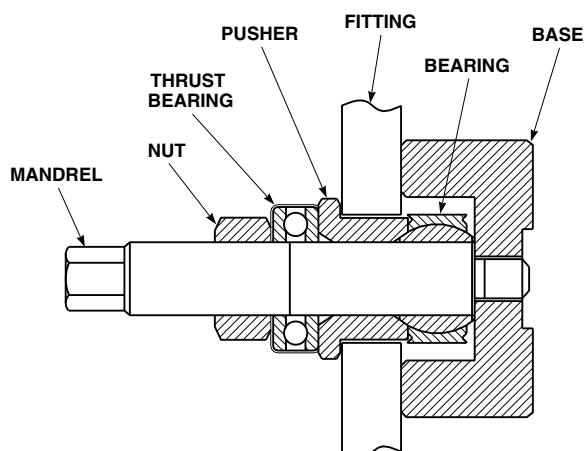
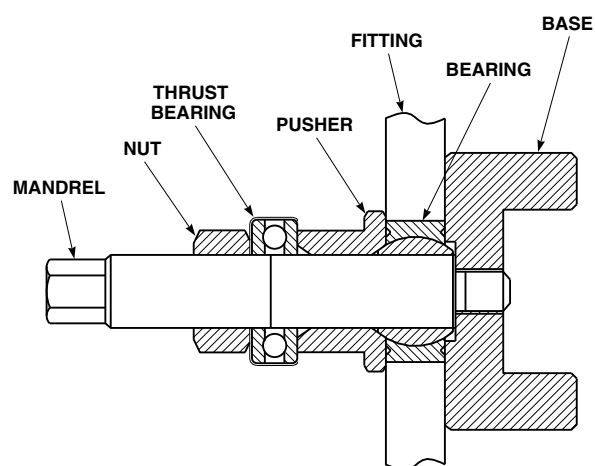
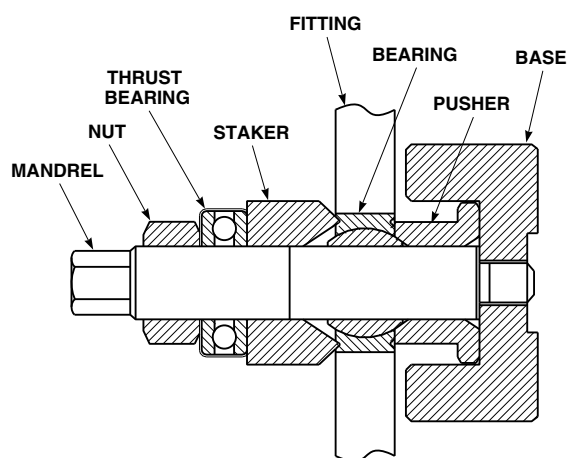
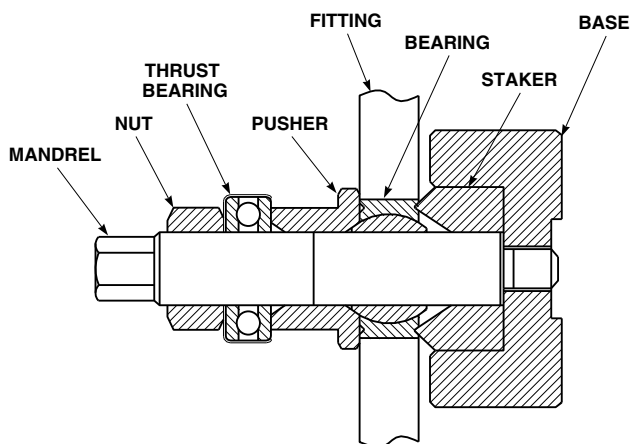


FIGURE 2-6-20-1. REPLACE EXTERNAL STORES SUPPORT SYSTEM FUSELAGE FITTING BEARING (SHEET 1 OF 2)

**STEP 1: REMOVE STAKING****STEP 2: PRESS OUT BEARING****STEP 3: PRESS IN NEW BEARING****STEP 4: STAKE FORWARD SIDE OF FITTING****STEP 5: STAKE REAR SIDE OF FITTING****NOTE**

APPLY GREASE, ITEM 144, APPENDIX D
TO MANDREL PRIOR TO EACH USE.

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FIGURE 2-6-20-1. REPLACE EXTERNAL STORES SUPPORT SYSTEM FUSELAGE FITTING BEARING (SHEET 2 OF 2)

LOWED. If cracks are found, replace bearing.

- (3) Check for scratches, gouges, or scouring caused by staking groove. **NONE ALLOWED.** If damage is found, replace bearing.

- (4) Check staked lip for over staking, forming a feathered or knife edge beyond face of

fitting (Figure 2-6-20-1, Sheet 1, Detail F). **NO OVER STAKING ALLOWED.** If damage is found, replace bearing.

- i. Touch up bearing surfaces with corrosion preventive compound, Item 113, Appendix D.
- j. Touch up fitting surfaces using epoxy primer coating kit, Item 135, Appendix D (PARA 2-6-5).

2-6-21 CREW CHIEF’S WINDOW LOWER TRACK.

PARAGRAPH BREAKDOWN

		PAGE
2-6-21.1	Replace Crew Chief’s Window Lower Track	2-6-21-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Drill Set

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D

- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 2-4-51
- PARA 2-6-1

2-6-21.1 Replace Crew Chief’s Window Lower Track.

2-6-21.1.1 Remove.

- Remove seals from track (PARA 2-4-51).

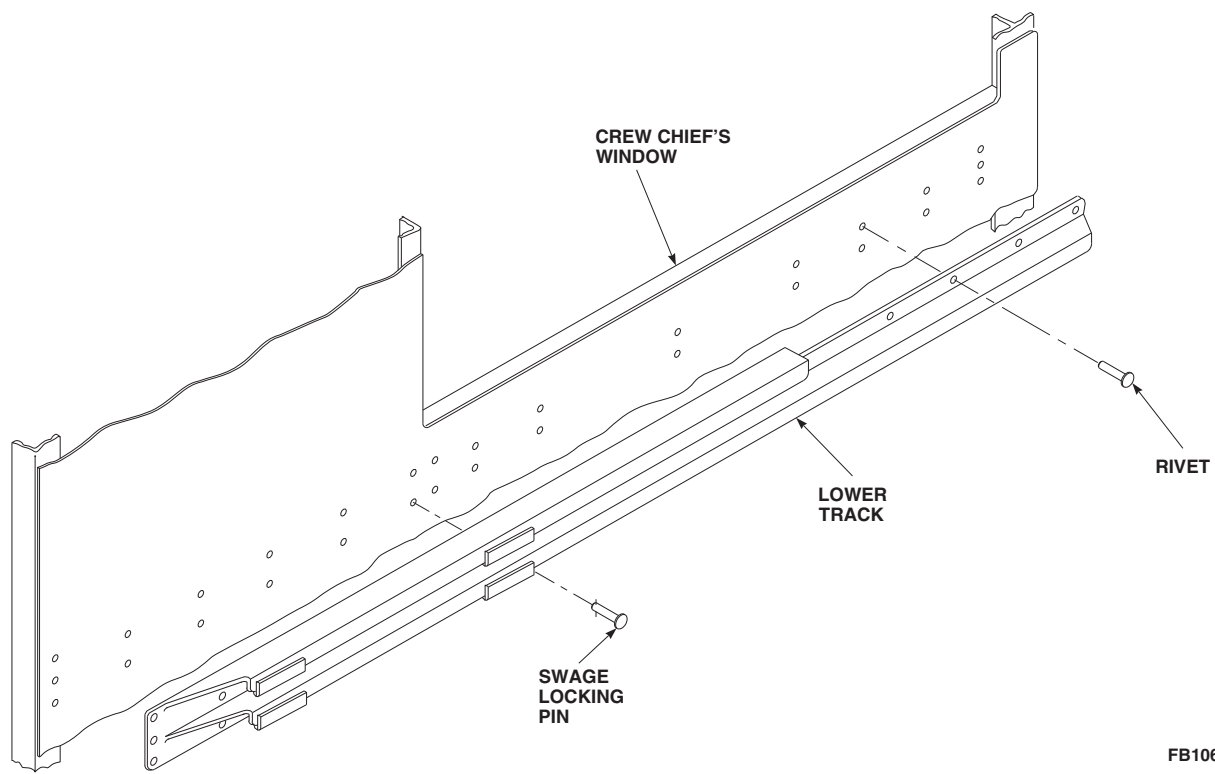
NOTE

Note location, diameter, and length of pins for proper replacement.

- Remove swage locking pins securing lower track to fuselage (Figure 2-6-21-1).
- Using drill, drill out rivets securing track to fuselage.
- Remove lower track from fuselage.

2-6-21.1.2 Install.

- Clean mating surfaces of replacement track and fuselage, using cheesecloth, Item 90, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D.
- Apply even coat of sealing compound, Item 292, Appendix D, to mating surface of lower track.
- With sealing compound still wet, position lower track against fuselage and align fastener holes (Figure 2-6-21-1).
- Apply sealing compound, Item 292, Appendix D, to fastening hardware and install wet.
- Install swage locking pins and collars, and rivets (PARA 2-6-1).
- Install new seals (PARA 2-4-51).

FB1065A
SA**FIGURE 2-6-21-1. REPLACE CREW CHIEF'S WINDOW LOWER TRACK**

2-6-22 CREW CHIEF’S WINDOW LOWER TRACK REPAIR.

PARAGRAPH BREAKDOWN

	PAGE
2-6-22.1 Repair Crew Chief’s Window Lower Track	2-6-22-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Drill Set
- Heat Gun

Materials/Parts

- Abrasive Cloth, Item 7, Appendix D
- Adhesive, Item 32, Appendix D
- Adhesive Primer, Item 68, Appendix D
- Coating Compound, Item 105A, Appendix D

- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Wear Strip, 70207-02727-125, Locally-Made, Figure H-141, Appendix H
- Stirring Stick, Beverage, Item 315C, Appendix D

References

- Appendix D
- Appendix H
- PARA 2-3-6
- PARA 2-4-48

2-6-22.1 Repair Crew Chief’s Window Lower Track.

2-6-22.1.1 Repair.

- Remove crew chief’s window (PARA 2-4-48).
- On helicopters without wear strips, if track wear in area where rear rollers rest is less than limit, refer to PARA 2-3-6 and perform the following:
 - Locally make wear strips, 70207-02727-125 (Figure H-141, Appendix H). Locate wear strip so that rear rollers bear on

center of wear strips when windows are fully open. Mark location on track and remove wear strip.

- Clean mating surfaces of track and wear strips using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D. Wipe area with clean machinery towel.
- Fill damaged track surface with adhesive, Item 32, Appendix D, if necessary. Smooth adhesive even with undamaged track surface, using beverage stirring stick, Item 315C, Appendix D.

- (4) Using one wear strip as a template against outer track surface, drill two $\frac{1}{8}$ -inch diameter holes through track. Remove burrs from drilled holes.
 - (5) Roughen flat surface of wear strips using abrasive cloth, Item 7, Appendix D. Wipe clean with machinery towel, Item 211, Appendix D.
 - (6) Apply light coat of coating compound, Item 105A, Appendix D, to mating surfaces of track and wear strips.
 - (7) Place inside and outside wear strips against track and align rivet holes.
 - (8) Install wear strips to lower track with rivets. Make sure heads are flush with wear strip (Figure 2-6-22-1, Detail C and Detail D).
 - (9) Clean adhesive from track and wear strips using machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D.
- c. If wear strips are loose, worn beyond limits, or missing, perform the following:
- (1) Clean adhesive from lower track using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D. Wipe area with clean machinery towel.
 - (2) Roughen inside edges of wear strip using abrasive cloth, Item 7, Appendix D. Wipe inside edges with clean machinery towel, Item 211, Appendix D.
 - (3) Apply light coat of adhesive primer, Item 68, Appendix D, on lower door track and inside edges of wear strip. Allow primer to cure at 70° - 80°F (21° - 27°C) for 1 hour, followed by 1 hour at 240° - 275°F (116° - 135°C) using heat gun.
 - (4) Apply light coat of adhesive, Item 32, Appendix D, on lower door track and inside edges of wear strip.
 - (5) Install front wear strip on lower track with front end of wear strip $\frac{1}{4}$ -inch from front end of lower track (Figure 2-6-22-1, Detail A and Detail B).
 - (6) Install rear wear strip on lower track with front end of wear strip 11.28-inches from front end of lower track (Figure 2-6-22-1, Detail C).
 - (7) Allow adhesive, Item 32, Appendix D, to cure for at least 1 hour at 160°F (71°C), or 24 hours at 75°F (24°C).
- d. Install crew chief's window (PARA 2-4-48).

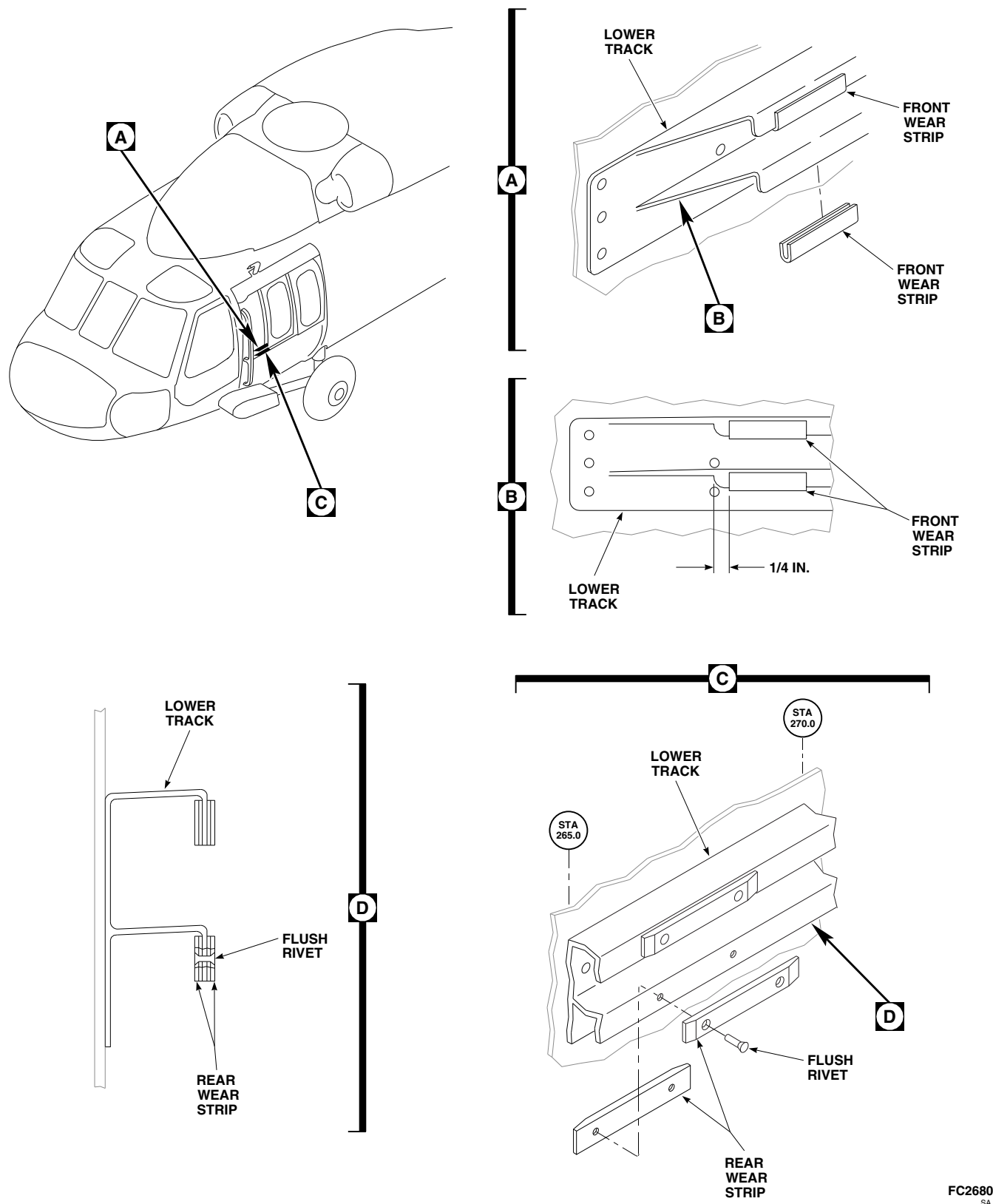


FIGURE 2-6-22-1. REPAIR CREW CHIEF’S WINDOW LOWER TRACK WEAR STRIPS

2-6-23 CREW CHIEF’S WINDOW UPPER TRACK.

PARAGRAPH BREAKDOWN

	PAGE
2-6-23.1 Replace Crew Chief’s Window Upper Track	2-6-23-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Drill Set

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D

- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 2-4-48
- PARA 2-4-52
- PARA 2-6-24

2-6-23.1 Replace Crew Chief’s Window Upper Track.

2-6-23.1.1 Remove.

- Remove crew chief’s windows (PARA 2-4-48).
- Remove upper track fairing (PARA 2-6-24).
- Remove seals from track (PARA 2-4-52).
- Drill out rivets fastening upper track to intercostal support (Figure 2-6-23-1).
- Remove upper track.

2-6-23.1.2 Install.

- Clean mating surfaces of replacement track and intercostal support, using cheesecloth, Item 90,

Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D.

- Apply even coat of sealing compound, Item 292, Appendix D, to mating surface of upper track.
- Place track, with sealing compound still wet, against intercostal support and align fastener holes (Figure 2-6-23-1).
- Coat rivets with sealing compound, Item 292, Appendix D, and install rivets wet.
- Remove excess sealing compound from working surfaces of track.
- Install new seals (PARA 2-4-52).
- Install upper track fairing (PARA 2-6-24).
- Install crew chief’s windows (PARA 2-4-48).

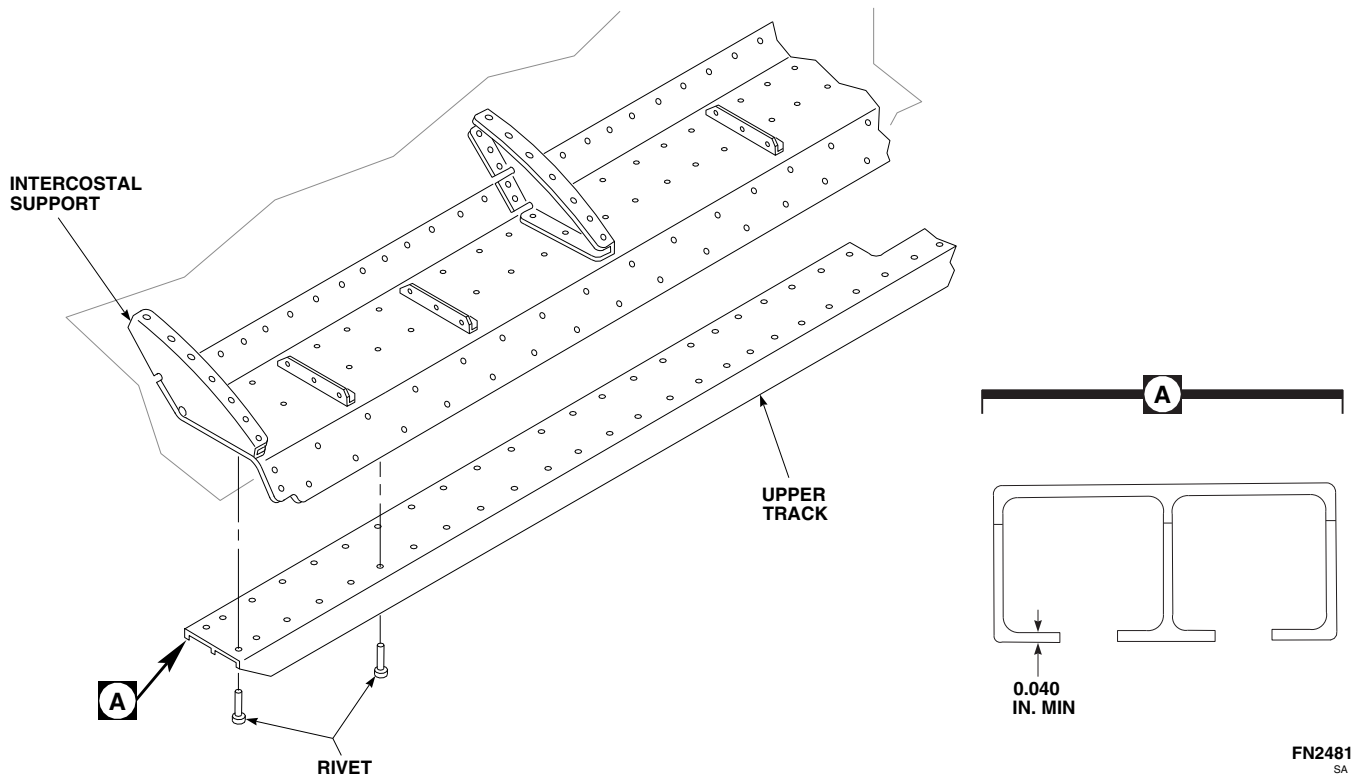


FIGURE 2-6-23-1. REPLACE CREW CHIEF'S WINDOW UPPER TRACK

2-6-24 WINDOW UPPER TRACK FAIRING.

PARAGRAPH BREAKDOWN

	PAGE
2-6-24.1 Replace Window Upper Track Fairing	2-6-24-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Drill Set
- Nonmetallic Scraper, Locally-Made
- Pin Punch

Materials/Parts

- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D

- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 2-6-1
- PARA 2-6-5
- PARA 2-6-13
- PARA 8-4-1

2-6-24.1 Replace Window Upper Track Fairing.

2-6-24.1.1 Remove.

- Remove pitot-static head (PARA 8-4-1).
- Remove pitot-static head support fairing and head support (PARA 2-6-13).
- Using nonmetallic scraper and machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D, remove sealing compound from fairing cap sealed edges, screwheads, and rivets.
- Remove bolt, screws, and washers securing fairing cap to upper track fairing, angle, and fuselage (Figure 2-6-24-1, Detail A).

- Drill out rivet heads holding fairing cap to upper track fairing, strap, and fuselage. Using pin punch, punch out rivets.
- Remove fairing cap.
- Using nonmetallic scraper and machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D, remove sealing compound from upper fairing sealed edges and rivets.
- Drill out rivet heads holding fairing to formers, intercostal support, and fuselage. Using pin punch, punch out rivets.
- Remove upper fairing.

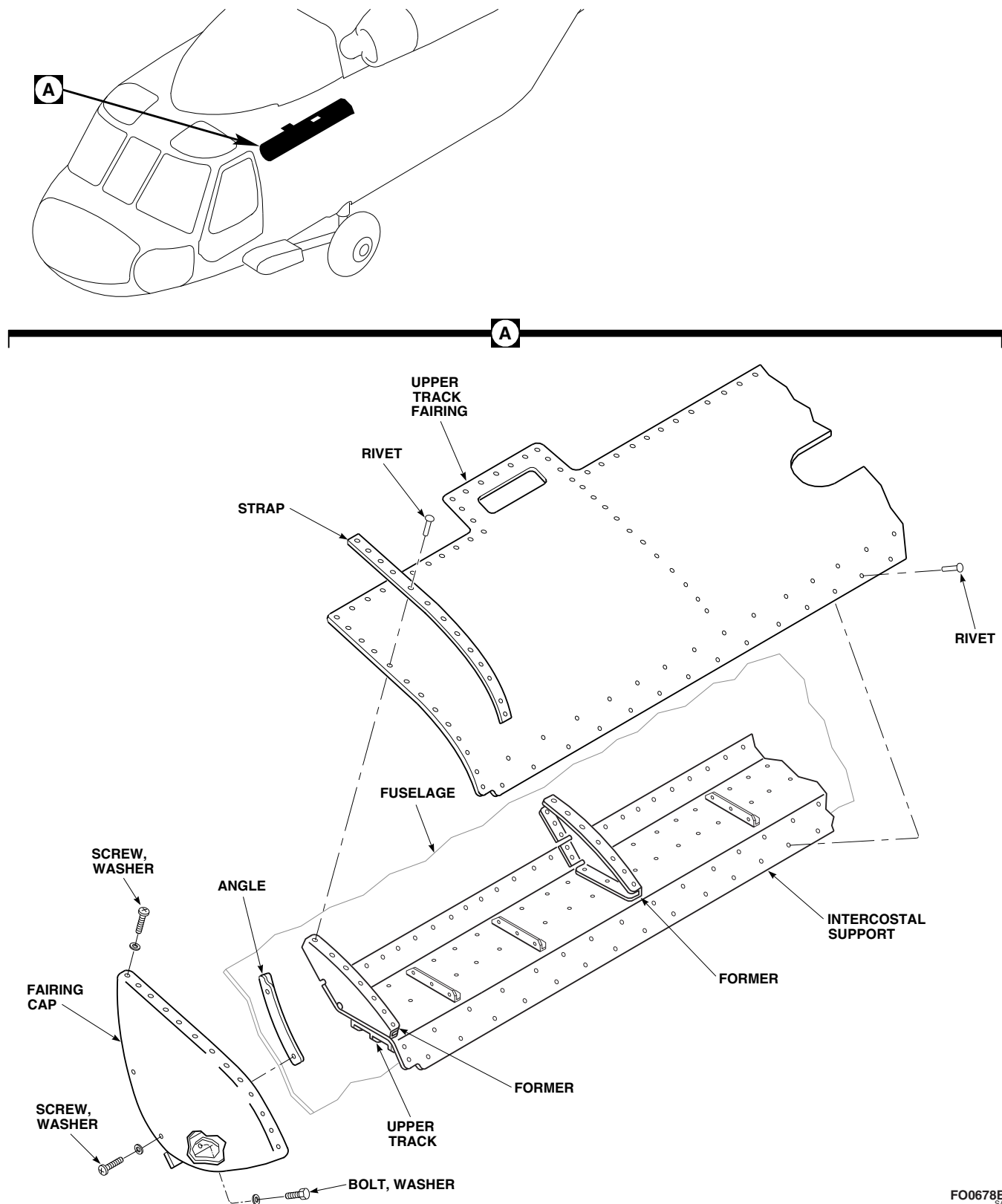
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FIGURE 2-6-24-1. REPLACE WINDOW UPPER TRACK FAIRING

2-6-24.1.2 Install.

- a. Thoroughly clean all mating surfaces of upper track fairing, fairing cap, fuselage, formers, angles, and intercostal support, using nonmetallic scraper and machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D.
- b. Coat all mating surfaces of upper track fairing, fuselage, formers, angles, and intercostal support with liberal amount of sealing compound, Item 292, Appendix D (PARA 2-6-1).
- c. Position upper track fairing in place while sealing compound is wet and align fastener holes (Figure 2-6-24-1, Detail A).
- d. Install rivets wet with sealing compound, Item 292, Appendix D (PARA 2-6-1).
- e. Coat mating surfaces of fairing cap, fuselage, strap, and upper track fairing liberally with sealing compound, Item 292, Appendix D.
- f. While sealing compound is wet, install strap and fairing cap with bolt, screws, washers, and rivets wet with sealing compound, Item 292, Appendix D (PARA 2-6-1).
- g. Seal all around exposed edges of fairing cap and upper track fairing with sealing compound, Item 292, Appendix D (PARA 2-6-1).
- h. Install pitot-static head support fairing and head support (PARA 2-6-13).
- i. Install pitot-static head (PARA 8-4-1).
- j. Touch up paint as required (PARA 2-6-5).

2-6-25 CREW CHIEF’S WINDOW REPAIR.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
2-6-25.1	Repair Crew Chief’s Window	2-6-25-1	2-6-25.2	Replace Plastic Window Panels	2-6-25-6

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Clecos, 1⁄8-inch diameter
- Drill
- Drill Bit, 1⁄8-inch diameter
- Nonmetallic Scraper, Locally-Made
- Pin Punch, 3⁄32-inch diameter drive

Materials/Parts

- Adhesive, Item 61, Appendix D
- Fastener Tape, Pile, Item 137C, Appendix D
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Mild Soap
- Sealing Compound, Item 292, Appendix D

References

- AMS 3195
- AMS 3240

- AMS 3420
- Appendix D
- MIL-P-25690
- MIL-R-6855
- MIL-S-18729
- MIL-S-5059
- PARA 2-4-48
- PARA 2-4-51
- PARA 2-6-1
- PARA 2-6-2
- PARA 2-6-3
- PARA 2-6-4
- PARA 2-6-5
- QQ-A-250

2-6-25.1 Repair Crew Chief’s Window.

- All window structure, except for plastic panels and seals, is aluminum alloy, 2024-T42, QQ-

A-250/5 or 7075-T6, QQ-A-250/13. Windows are stretched Plexiglass and seals are either silicone sponge rubber or neoprene sheet or

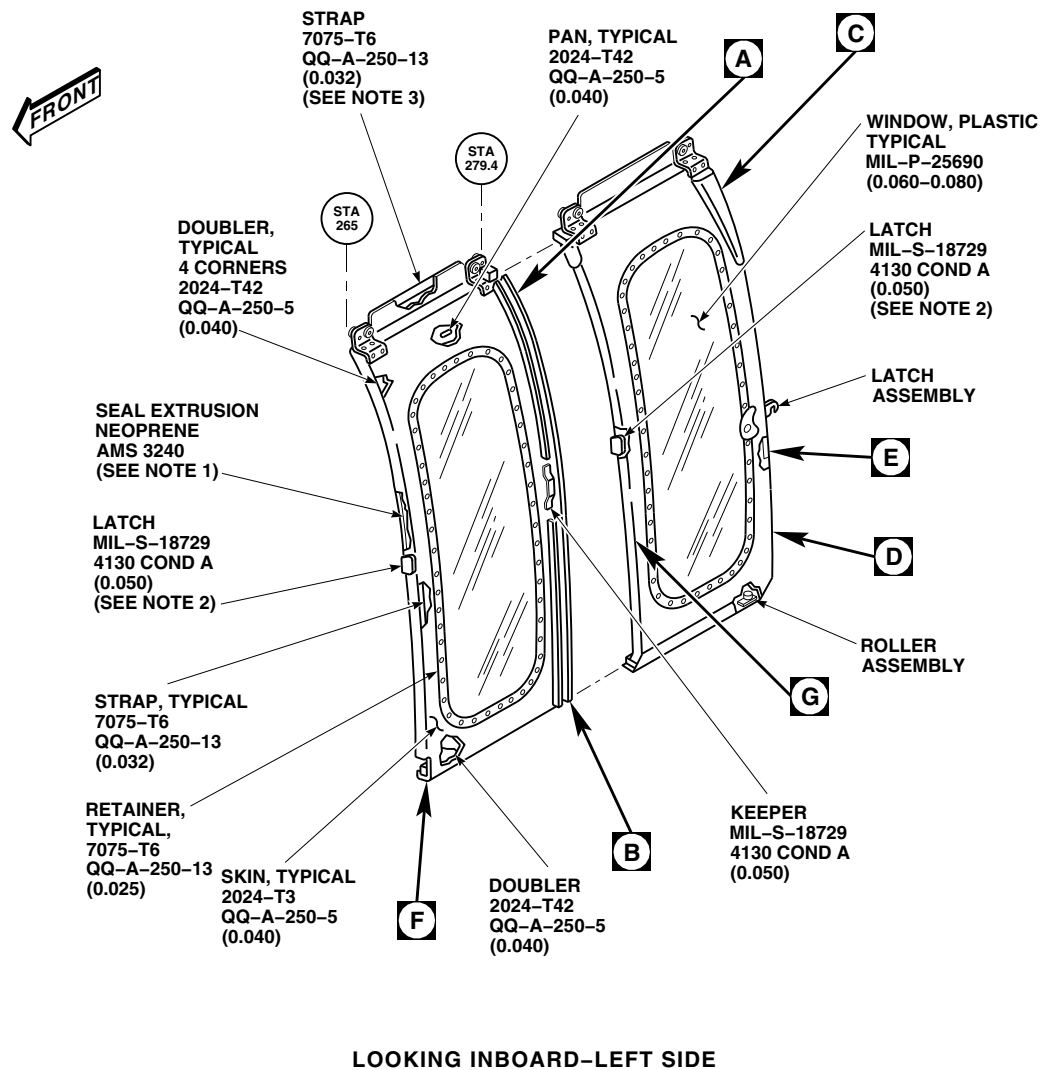
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FIGURE 2-6-25-1. REPAIR CREW CHIEF'S WINDOW (SHEET 1 OF 5)

extrusion charts. Roller supports are made from 4130 steel and rear window fairing is reinforced plastic. Window structure is secondary structure (Figure 2-6-25-1, Sheet 1 through Sheet 5).

- b. If required, remove crew chief's window (PARA 2-4-48).
- c. Evaluate damage to skin and framing (PARA 2-6-1).
- d. Use PARA 2-6-2 and PARA 2-6-4 for typical structural repairs.
- e. Use PARA 2-6-3 for evaluating and repairing reinforced plastic.
- f. If required, rebond seals (PARA 2-4-51).
- g. Seal roller assembly attachment surfaces for dissimilar metals insulation and watertightness using sealing compound, Item 292, Appendix D (PARA 2-6-1).
- h. Seal upper window track mating surfaces for watertightness using sealing compound, Item 292, Appendix D (PARA 2-6-1).

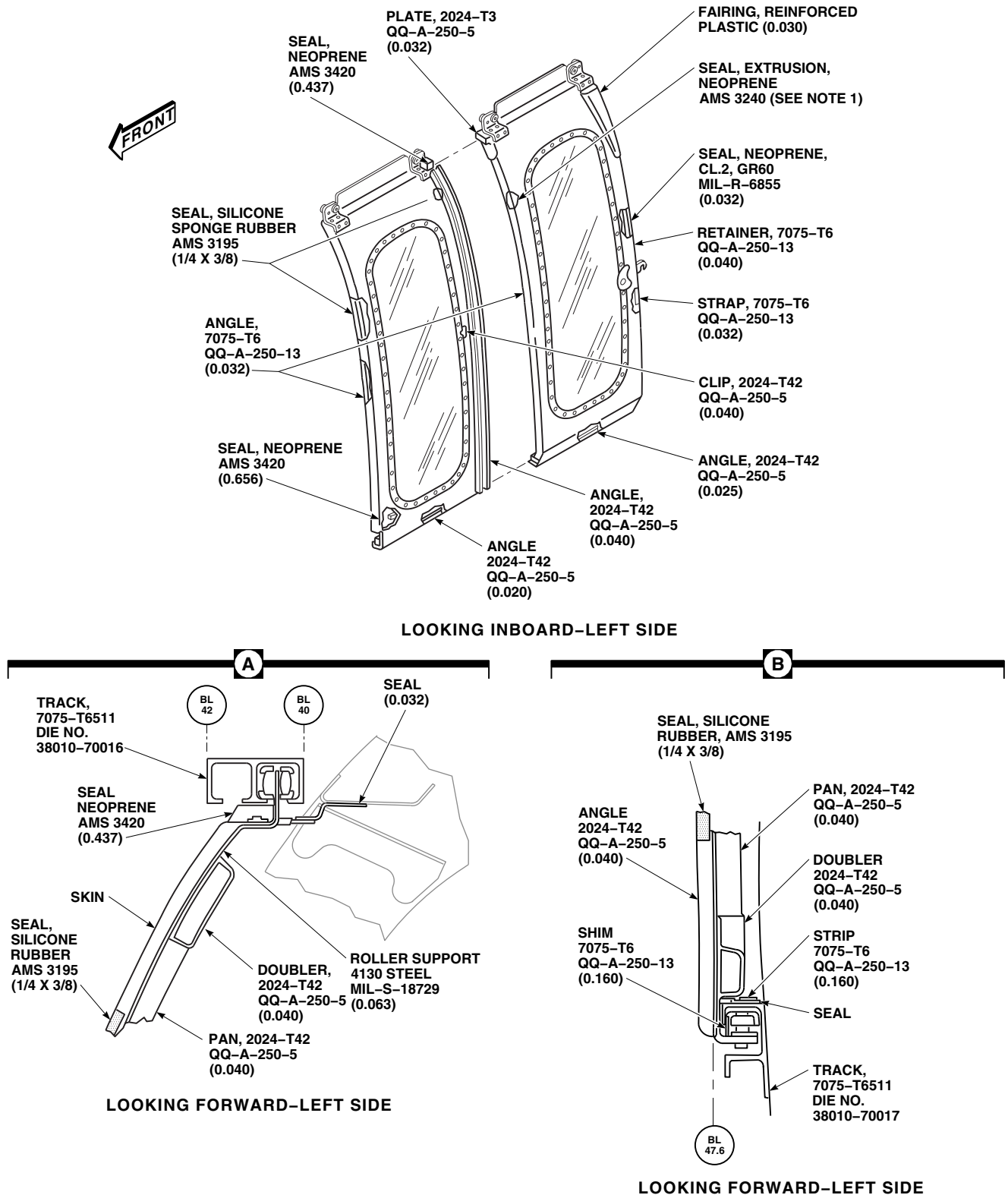


FIGURE 2-6-25-1. REPAIR CREW CHIEF'S WINDOW (SHEET 2 OF 5)

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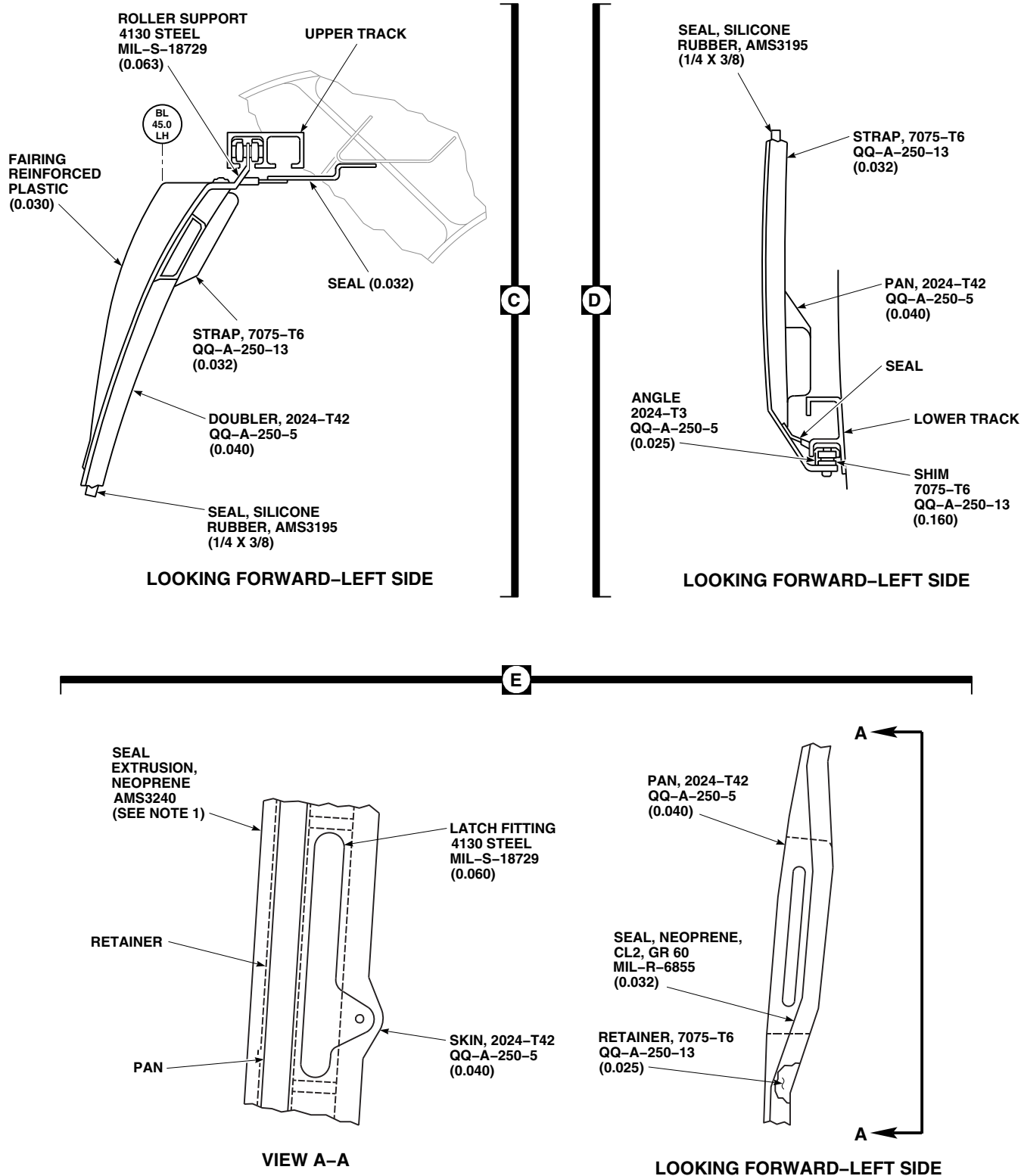
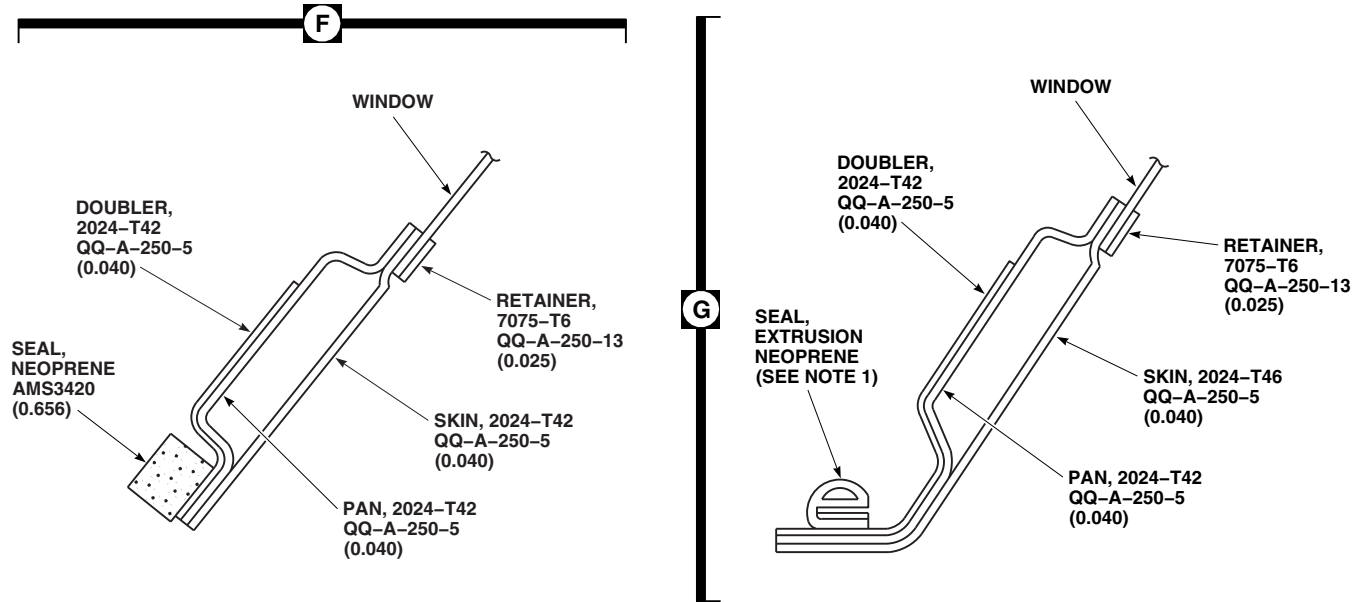
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FIGURE 2-6-25-1. REPAIR CREW CHIEF'S WINDOW (SHEET 3 OF 5)



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FIGURE 2-6-25-1. REPAIR CREW CHIEF'S WINDOW (SHEET 4 OF 5)

NOTES

1. SIKORSKY P / N S6122-82279-2.
2. **70583-70856**
3. **70910-SUBQ**

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FIGURE 2-6-25-1. REPAIR CREW CHIEF'S WINDOW (SHEET 5 OF 5)

- i. Touch up paint in repaired area, as required (PARA 2-6-5).
- j. If required, install crew chief's window (PARA 2-4-48).

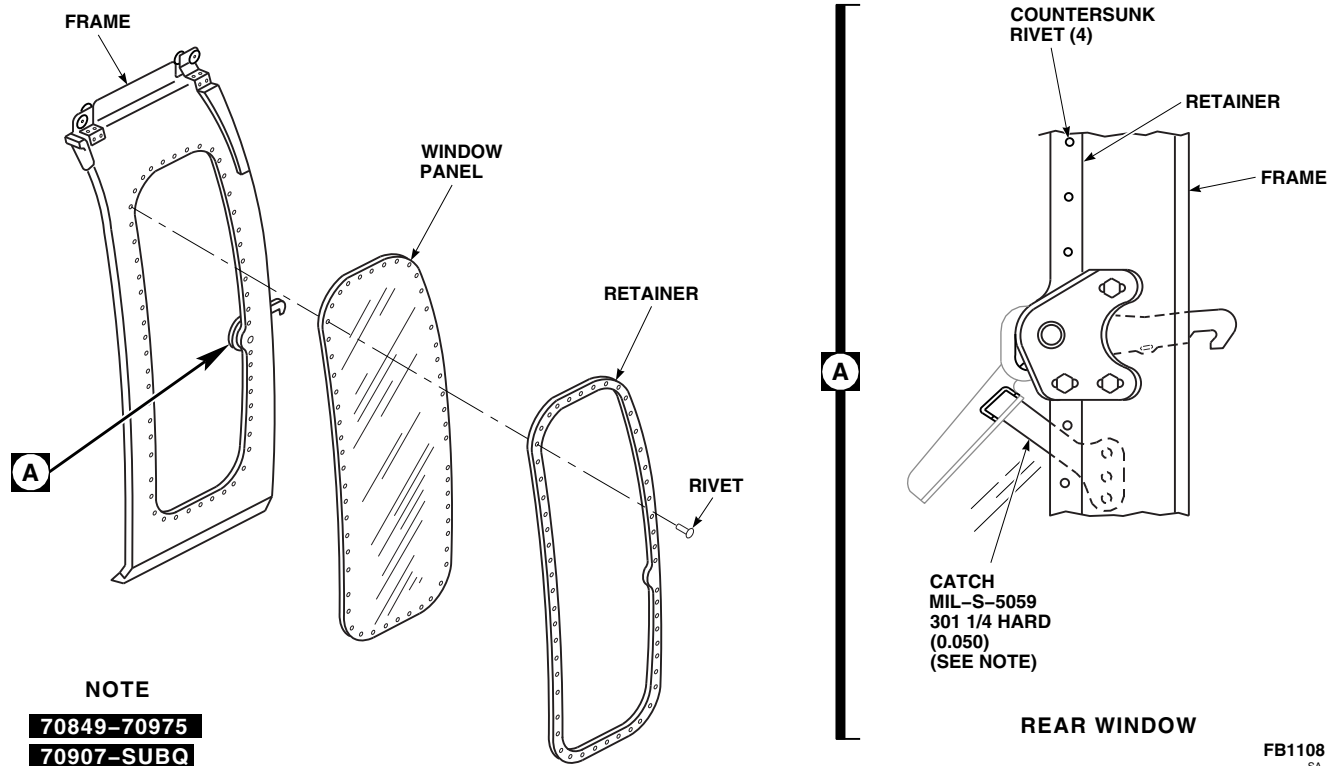
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FIGURE 2-6-25-2. REPLACE PLASTIC WINDOW PANELS

2-6-25.2 Replace Plastic Window Panels.**2-6-25.2.1 Remove.**

- Using nonmetallic scraper, remove pile fastener tape around inside of window opening.
- Carefully drill out and remove rivets securing window panel to skin, using drill with 1/8-inch diameter drill bit and 3/32-inch diameter drive pin punch.
- Using nonmetallic scraper, remove retainer and carefully work window panel loose from skin (Figure 2-6-25-2).
- Clean adhesive from skin and inner surface using machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D, and nonmetallic scraper. Wipe dry with clean machinery towel, Item 211, Appendix D.

2-6-25.2.2 Inspect.

Inspect skin and retainer for damage using PARA 2-6-1 negligible damage limits. If damage is great-

er than negligible damage, replace retainer and repair skin (PARA 2-4-48).

2-6-25.2.3 Install.

- Clean sealing surface of skin using machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D. Wipe dry with clean machinery towel, Item 211, Appendix D.
- Temporarily place window panel over opening. Mark and drill 0.1405-inch holes where rivets will go (Figure 2-6-25-2).
- Wash sealing side of window with mild soap and water. Wipe dry with clean, soft cloth.
- Apply smooth coating of sealing compound, Item 292, Appendix D, on flange of window opening. Apply enough sealing compound for good squeezeout.
- Position window panel in opening and press firmly in place.

- f. Place retainer on outside of window panel, line up holes, and temporarily fasten entire unit with 1/8-inch diameter clecos in every third hole.
- g. Check sealing compound squeezeout on outside of window for continuous bead with no voids. If there are voids, remove window panel and add more sealing compound, Item 292, Appendix D.

NOTE

Four countersunk rivets are installed above latch installation on rear window with head against retainer.

- h. Install universal head rivets in between clecos, advancing around window with each rivet.
- i. Remove clecos and install remaining rivets.

- j. Wipe squeezed-out sealing compound to form a smooth unbroken fillet 1/8 to 1/4-inch wide around entire outside edge of window panel.
- k. Allow sealing compound to cure for 72 hours at room temperature 72°F (22°C) or for 2 hours at 160° - 180°F (71° - 82°C).
- l. Apply even coat of adhesive, Item 61, Appendix D, to bonding surfaces of window frame and pile fastener tape, Item 137C, Appendix D.
- m. Allow adhesive to air dry for 10 - 15 minutes. █
- n. Install pile fastener tape, Item 137C, Appendix D, flush with edge of window opening, pressing it firmly into place.
- o. Allow adhesive to cure at room temperature 72°F (22°C) for 12 - 16 hours. █
- p. Touch up paint in repaired area, as required (PARA 2-6-5).

2-6-26 CABIN DOOR REPAIR.

PARAGRAPH BREAKDOWN

	PAGE
2-6-26.1 Repair Cabin Door	2-6-26-1
INITIAL SETUP	MIL-R-6855
Tools	MIL-S-18729
- Airframe Repairer’s Toolkit - Rivet Gun	- MIL-S-22499 - MIL-STD-670
References	- PARA 2-4-58 - PARA 2-6-3 - PARA 2-6-5 - QQ-A-200 - QQ-A-225 - QQ-A-250
- AMS 3240 - AMS 4229 - AMS-DTL-22499 - L-P-410 - MIL-I-24768 - MIL-PRF-25690	

2-6-26.1 Repair Cabin Door.

- a. Cabin door structure is a bonded sandwich construction of Nomex honeycomb and aluminum skins, reinforced with aluminum angles, straps, and channels. Transparent window panels are riveted to window frames which are secured to door by jettison mechanism. Window and door seals are formed rubber. Rear seal on fuselage is riveted to plat-ing. Jettison/latching mechanism is aluminum alloy and 4130 steel. Pans under windows, housing jettison components, are reinforced plastic. Track extrusion and roller assembly support are aluminum alloy. Forward and rear fairings are Kevlar reinforced plastic. Cabin
- door is secondary structure (Figure 2-6-26-1, Sheet 1 through Sheet 4).
- b. Evaluate and repair damage to honeycomb structure (PARA 2-6-3).
- c. Evaluate and repair damage to reinforced plastic pans and Kevlar fairings (PARA 2-6-3).
- d. Refer to PARA 2-4-58 for door seal installa-tion.
- e. Clean and make minor repairs to transparent plastic windows.
- f. Touch up paint in repaired area as required (PARA 2-6-5).

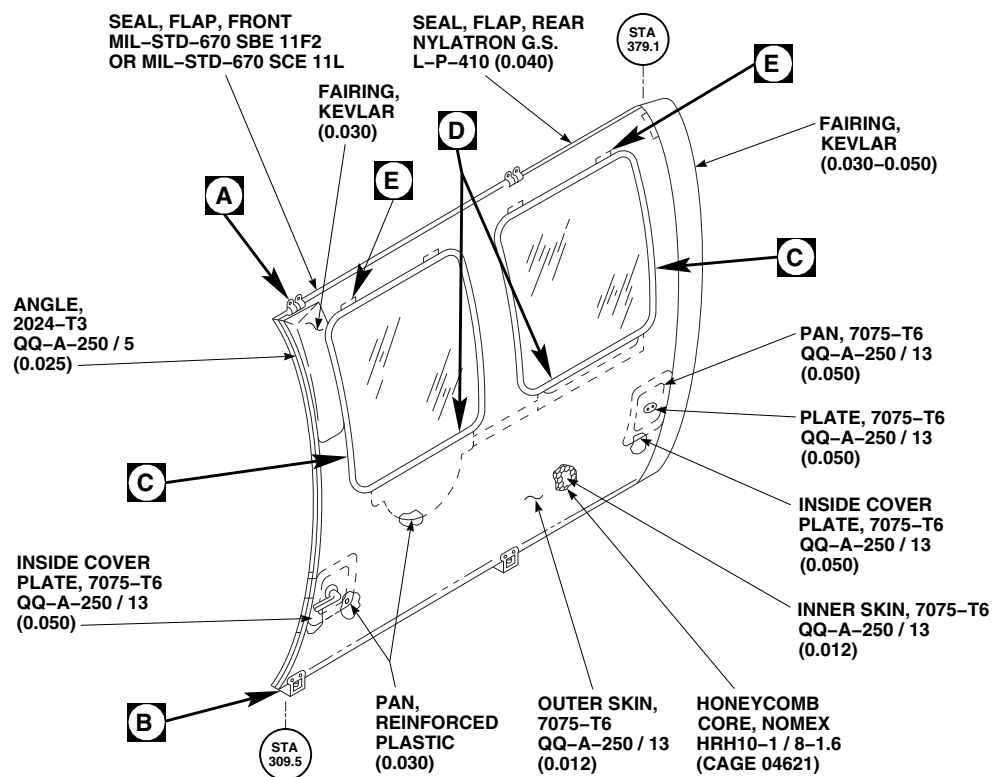
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FIGURE 2-6-26-1. REPAIR CABIN DOOR (SHEET 1 OF 4)

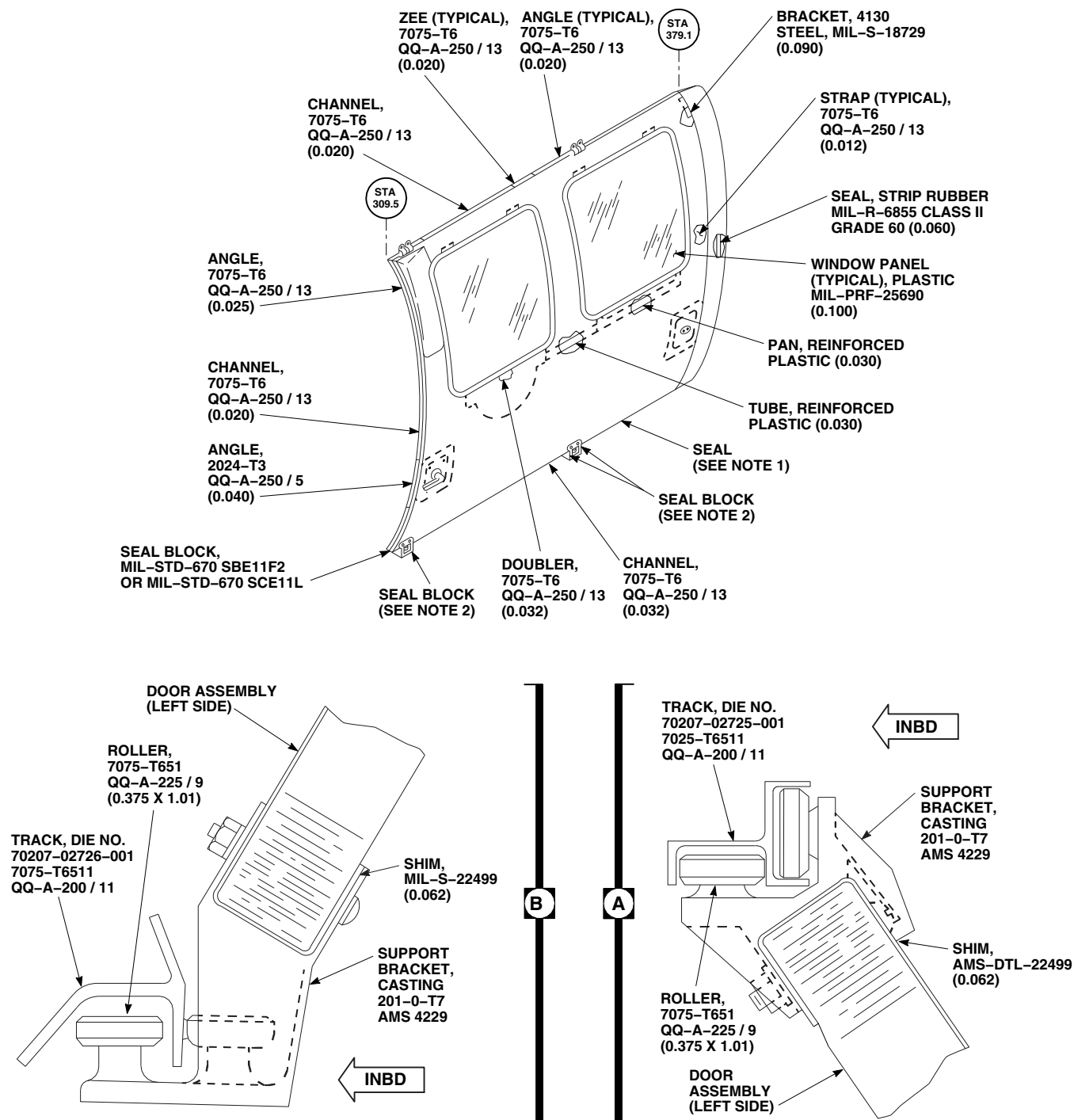


FIGURE 2-6-26-1 REPAIR CABIN DOOR (SHEET 2 OF 4)

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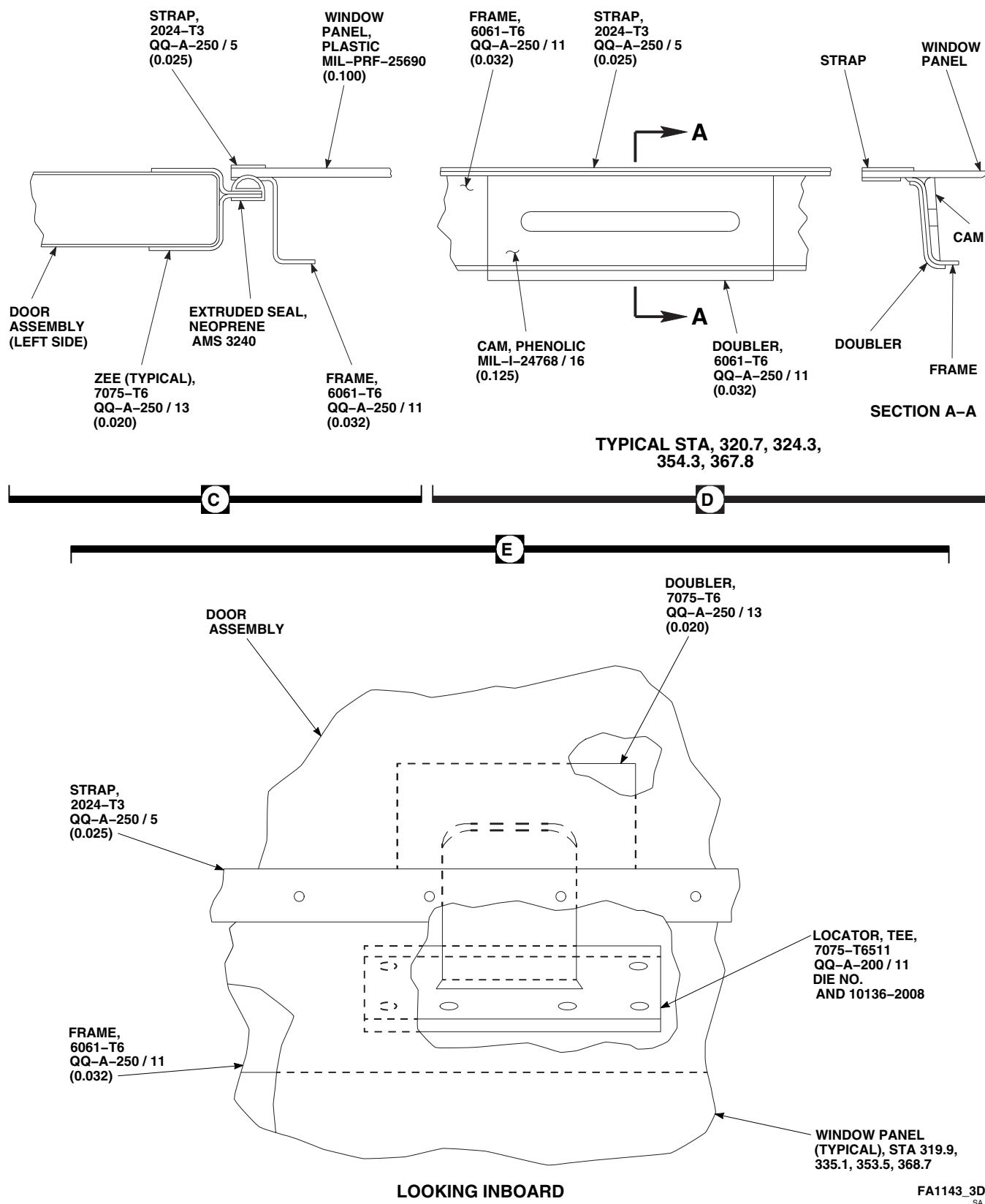


FIGURE 2-6-26-1. REPAIR CABIN DOOR (SHEET 3 OF 4)

NOTES

1. ON HELICOPTERS WITH DOORS, 70217-02701-041 THROUGH -044 SHEATH 7075-T6 QQ-A-250 / 13 (0.020).
ON HELICOPTERS WITH DOORS, 70217-02071-045, -046 SEAL STRIP NYLATRON (0.040).
ON UH-60L HELICOPTERS SERIAL NOS 91-26401 AND SUBSEQUENT, SEAL STRIP URETHANE (0.060).
2. HELICOPTERS WITH DOOR PART NOS 70217-02701-041, -042, -043 AND -044 SEAL BLOCKS MIL-STD-670 SBE 7L.
3. ALL DIMENSIONS ARE IN INCHES, UNLESS NOTED.

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FIGURE 2-6-26-1. REPAIR CABIN DOOR (SHEET 4 OF 4)

2-6-27 CABIN FLOOR REPAIR.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
2-6-27.1	Repair Cabin Floor	2-6-27-1	2-6-27.4	Repair Floor Shims	2-6-27-2
2-6-27.2	Repair Cargo Tiedown Rings and Floor Pans	2-6-27-2	2-6-27.5	Repair Cargo Hook Door	2-6-27-2
2-6-27.3	Repair Floor Attachment Hardware	2-6-27-2			

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Airframe Repairer’s Toolkit
- Hacksaw
- Torque Wrench, Range to Support Seating Torque of Attaching Hardware

Materials/Parts

- Adhesive, Item 48, Appendix D
- Corrosion Compound, Item 108, Appendix D
- Corrosion Preventive Compound, Item 113, Appendix D
- Epoxy Primer Coating Kit, Item 135, Appendix D
- Primer, Item 255, Appendix D
- Primer, Item 257, Appendix D

- Scouring Pad, Item 270, Appendix D

References

- Appendix D
- MIL-R-6855
- MIL-S-18799
- MIL-S-22499
- MIL-STD-670
- MIL-T-9046
- PARA 2-6-1
- PARA 2-6-2
- PARA 2-6-3
- PARA 2-6-4
- PARA 2-6-5
- QQ-A-250

2-6-27.1 Repair Cabin Floor.

- Cabin floor panels and cargo hook access door are of bonded sandwich construction made

from Nomex/Hexcel honeycomb core and fiberglass (reinforced plastic) skins. Provisions for cargo tiedown, passenger’s seat, and litter

fitting installations are incorporated in each panel (Figure 2-6-27-1, Sheet 1 through Sheet 5).

- b. Use PARA 2-6-2 and PARA 2-6-3 for evaluating damage and repairing honeycomb floor panels.
- c. Evaluate damage to metal components of floor panels (PARA 2-6-1). Replace extrusions (PARA 2-6-4).
- d. Pan replacement requires bonding pan to floor panel using adhesive, Item 48, Appendix D. Apply primer, Item 255, Appendix D, to metal pans (aluminum or steel) before applying adhesive.
- e. Floor attachment screws countersunk in dense honeycomb areas are to be seated. **TORQUE IS NOT TO BE MORE THAN 10 INCH-POUNDS ABOVE SEATING TORQUE.**
- f. For corrosion protection and watertightness, seal all repairs on floor panels (PARA 2-6-1).
- g. Touch up paint as required (PARA 2-6-5).
- h. After priming, apply two spray coats of corrosion preventive compound, Item 113, Appendix D, to seal repairs on bottom side of floor panels (PARA 2-6-1).

2-6-27.2 Repair Cargo Tiedown Rings and Floor Pans.

- a. Remove corrosion manually using scouring pad, Item 270, Appendix D. **PITS IN WELD AREA SHALL NOT EXCEED 0.010-INCH DIAMETER AFTER CLEANUP.**
- b. Mix solution of 1 part corrosion compound, Item 108, Appendix D, to 1 part water.
- c. Apply compound solution to corroded areas of tiedown rings and floor pans.
- d. Rinse thoroughly with clean water.

- e. Touch up paint tiedown rings and floor pans using epoxy primer coating kit, Item 135, Appendix D (PARA 2-6-5).

2-6-27.3 Repair Floor Attachment Hardware.

NOTE

If possible, replace corroded screws or bolts if installed. Cadmium-plating will be removed during corrosion removal.

- a. Remove corrosion manually from floor attachment screws, or bolts if installed, using scouring pad, Item 270, Appendix D. **PITS SHALL NOT EXCEED 0.050-INCH DEEP AFTER CLEANUP.**
- b. Mix solution of 1 part corrosion compound, Item 108, Appendix D, to 1 part water.
- c. Apply compound solution to corroded areas of floor attachment screws or bolts.
- d. Rinse thoroughly with water.

2-6-27.4 Repair Floor Shims.

- a. Remove floor pans, if required (Figure 2-6-27-1, Sheet 4, Detail A).
- b. Turn floor pan over. Cut slot in laminated shim $\frac{1}{8}$ -inch wide and $\frac{1}{8}$ -inch deep using hacksaw. Extend slot across entire surface of shim (Figure 2-6-27-1, Sheet 4, Detail B).
- c. Peel laminations from shim to bottom of cut.
- d. Alternate method is to cut slot in laminated shim $\frac{1}{8}$ -inch wide through shim. Extend slot from edge of shim to center hole in shim using hacksaw. Do not cut into floor pan.
- e. Touch up paint floor shims with primer, Item 257, Appendix D (PARA 2-6-5).

2-6-27.5 Repair Cargo Hook Door.

Refer to PARA 2-6-3 to repair any skin or structural damage to door.

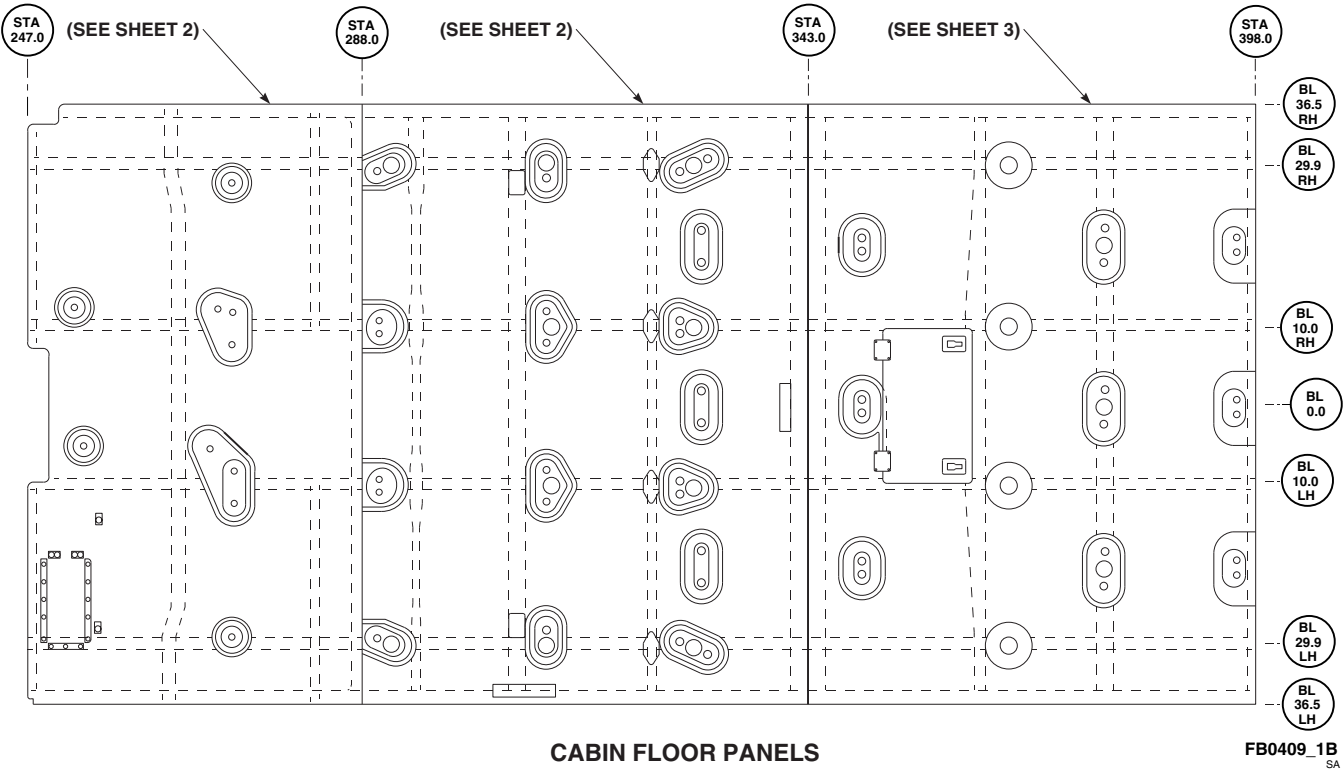
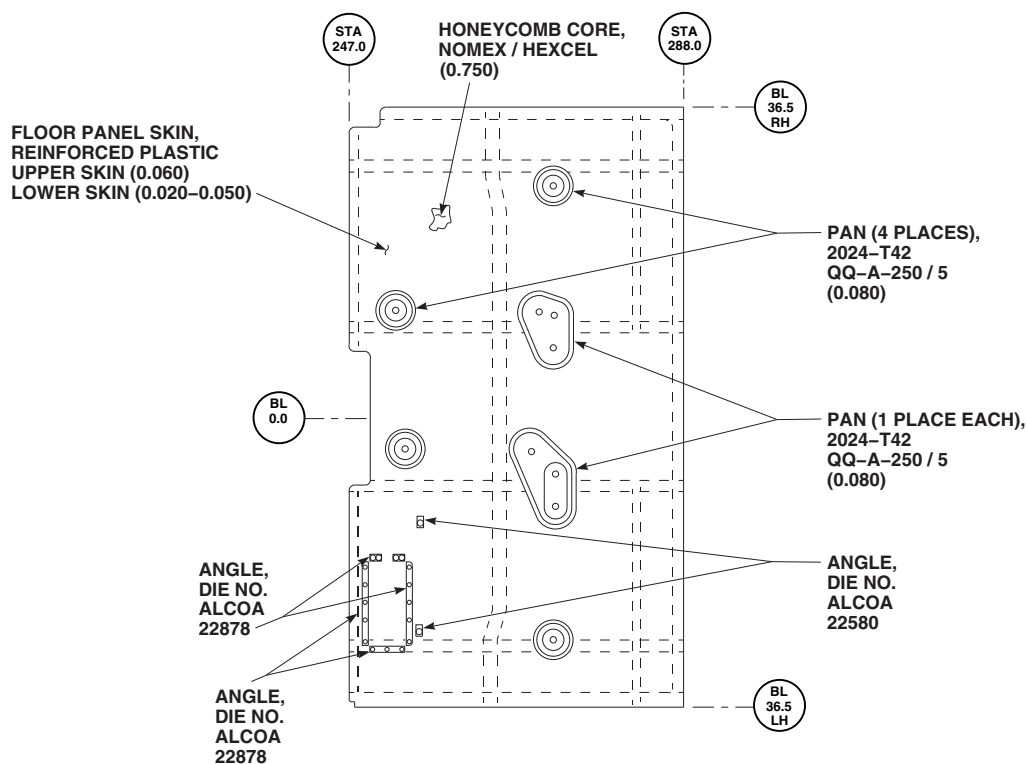
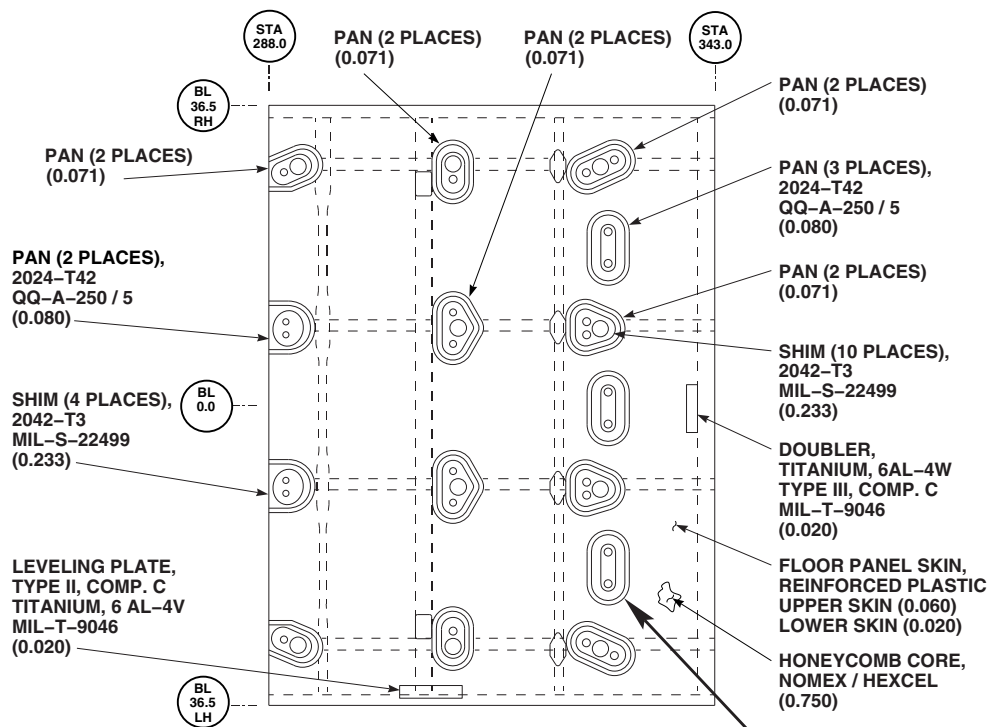


FIGURE 2-6-27-1. REPAIR CABIN FLOOR (SHEET 1 OF 5)



FLOOR PANEL STA 247.0 TO STA 288.0

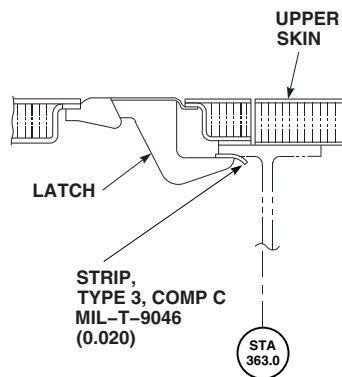
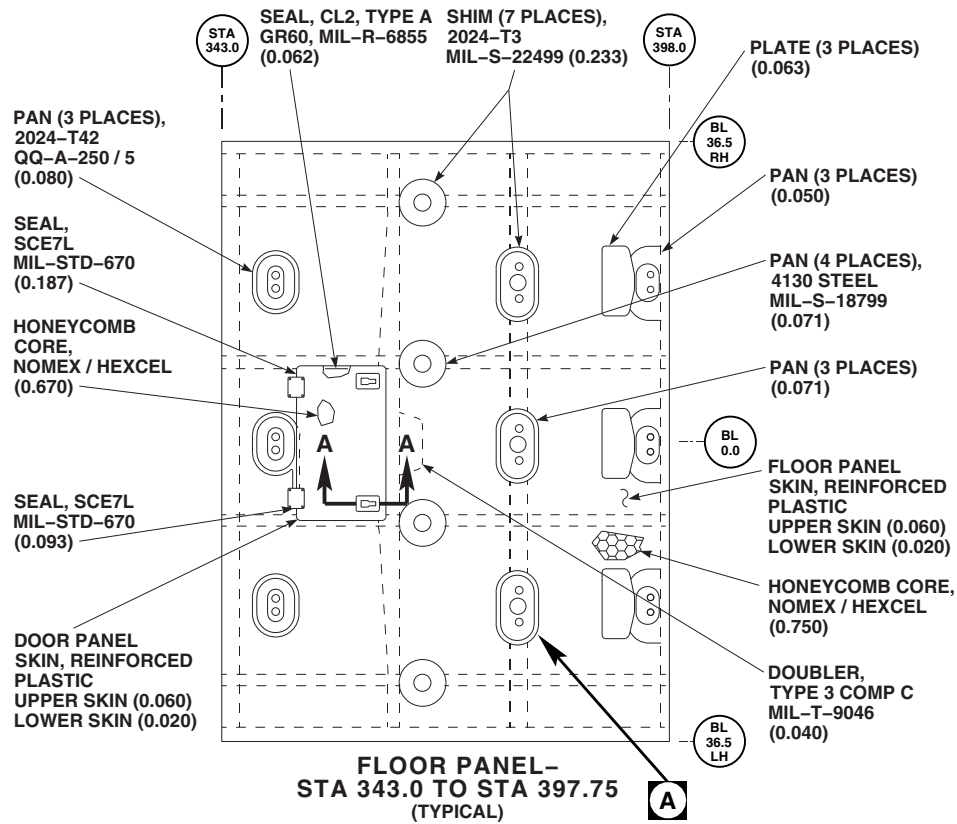


FLOOR PANEL STA 288.0 TO STA 343.0

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FIGURE 2-6-27-1. REPAIR CABIN FLOOR (SHEET 2 OF 5)



SECTION A-A

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FIGURE 2-6-27-1. REPAIR CABIN FLOOR (SHEET 3 OF 5)

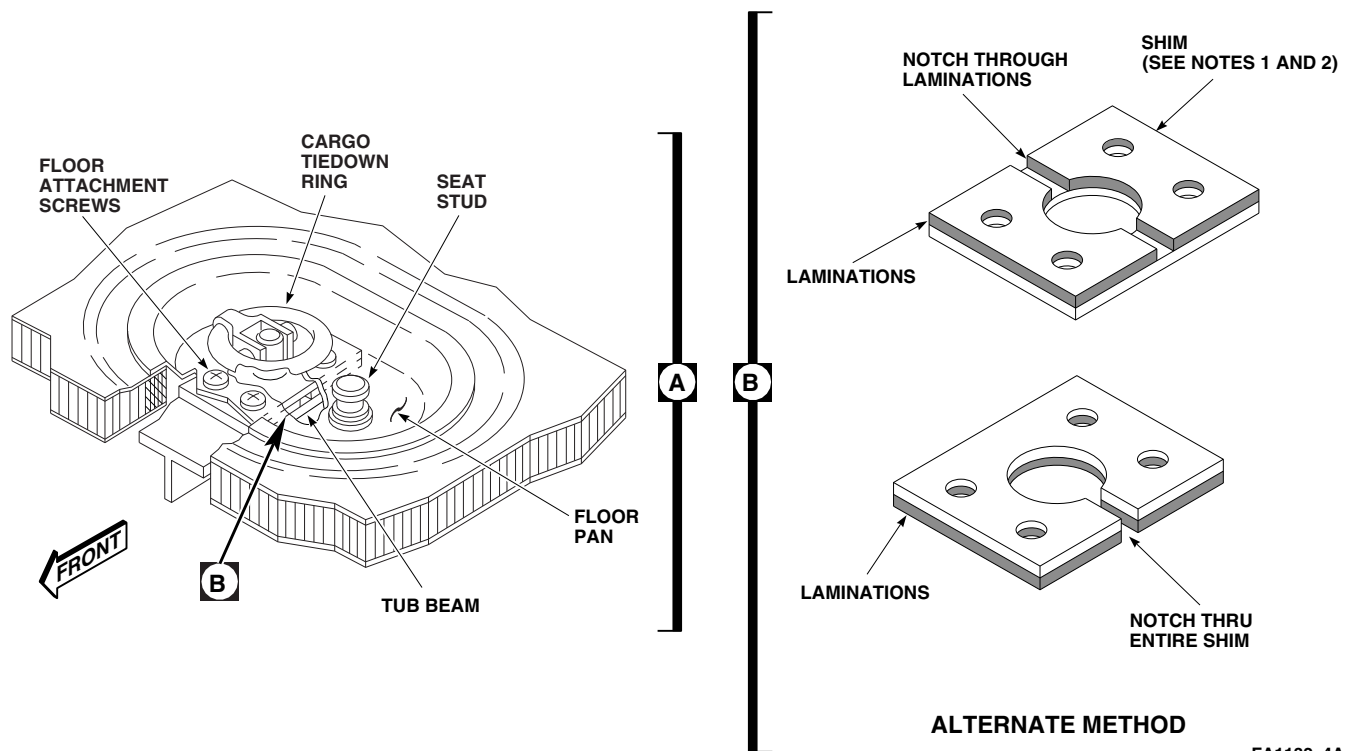
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FIGURE 2-6-27-1. REPAIR CABIN FLOOR (SHEET 4 OF 5)

NOTES

1. SHIM REMOVED FROM FLOOR PAN FOR CLARITY.
2. SHIM SHOWN UPSIDE DOWN. NONLAMINATED PORTION IS BONDED TO FLOOR PAN.
3. ALL DIMENSIONS ARE IN INCHES, UNLESS NOTED.

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FIGURE 2-6-27-1. REPAIR CABIN FLOOR (SHEET 5 OF 5)

2-6-28 MAIN ROTOR PYLON FRAMING REPAIR.

PARAGRAPH BREAKDOWN

	PAGE
2-6-28.1 Repair Main Rotor Pylon Framing	2-6-28-2

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Clecocos, Edge-Grip
- Drill Set
- File
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Abrasive Cloth, Item 4, Appendix D
- Adhesive, Item 30, Appendix D
- Adhesive, Item 32, Appendix D
- Adhesive, Item 42, Appendix D
- Adhesive, Item 56, Appendix D
- Adhesive, Item 57, Appendix D
- Adhesive, Item 66, Appendix D
- Cheesecloth, Item 90, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Rivet, NAS1097AD-4
- Sealing Compound, Item 275, Appendix D
- Sealing Compound, Item 292, Appendix D

- Track Repair Angle, Locally-Made, Figure H-186, Appendix H
- Track Repair Plate, Locally-Made, Figure H-185, Appendix H

References

- AMS 5355
- Appendix D
- Appendix H
- MIL-P-46060
- MIL-R-6855
- MIL-S-18729
- MIL-S-5000
- MIL-S-5059
- MIL-S-6721
- MIL-S-6758
- MIL-S-7947
- MIL-STD-670
- MIL-T-9046
- PARA 2-4-101
- PARA 2-6-1
- PARA 2-6-2

- PARA 2-6-4
- PARA 2-6-5
- QQ-A-200
- QQ-A-225
- QQ-A-250
- QQ-A-367
- WW-T-700

2-6-28.1 Repair Main Rotor Pylon Framing.

2-6-28.1.1 Repair.

- a. Main rotor pylon framing includes formed and extruded members made from 7075-T6, QQ-A-250/13, Alclad sheet stock and 7075-T6511, QQ-A-200/11, aluminum alloy extrusion. Firewalls and heat-resistant components are made from 6AL-4V titanium alloy, MIL-T-9046; 321 corrosion-resistant stainless steel, MIL-S-6721; and 301 corrosion-resistant steel MIL-S-5059 (Figure 2-6-28-1, Sheet 1 through Sheet 12).
- b. Hinge and support fittings for engine cowling/work platform are made from 6AL-4V titanium alloy and forged aluminum alloy, 7075-T73, QQ-A-367. Negligible damage limits to fittings are limited to nicks, scratches, corrosion, and wear, no deeper than 10% of material thickness at point of damage. Blend damage smoothly to surrounding area using long, even strokes. Polish out all surface imperfections and restore protective finish.
- c. Main rotor pylon framing members are considered secondary structure.

NOTE

All structure and hardware passing through cabin skin must be sealed to deck using sealing compound, Item 292, Appendix D.

- d. Evaluate damage to framing members and firewalls (PARA 2-6-1).
- e. Refer to PARA 2-6-2 and PARA 2-6-4 for typical framing repairs. When repairing titanium firewalls use same gage titanium stock

or equivalent gage 301/302 stainless steel, MIL-S-5059, for patching. Use Monel or stainless steel rivets, or spot or fusion welding, for securing titanium and stainless steel patches.

- f. Repair sliding cover track as follows:
 - (1) Locally make track repair plate and track repair angle (Figure H-185 and Figure H-186, Appendix H).
 - (2) Inspect track web for wear from horizontal roller contact. If damage is deeper than 0.020-inch, prepare for installation of repair plate, as follows:
 - (a) Remove nicks and gouges by blending with abrasive cloth, Item 4, Appendix D.
 - (b) Measure remaining thickness of web in damaged area. **WEB THICKNESS SHALL NOT BE LESS THAN 0.060-INCH.** If web thickness is not within limit, replace track.
 - (3) Inspect track flange for wear from vertical roller contact. If damage is deeper than 0.020-inch, prepare for installation of repair angle, as follows:
 - (a) Remove track flange material in wear area, using file, so repair angle seats flush against flange.
 - (b) Measure remaining thickness of flange in repair area. **WEB THICKNESS SHALL NOT BE LESS THAN 0.060-INCH.** If web thickness is not within limit, replace track.
 - (4) Install repair plate and repair angle, as follows:

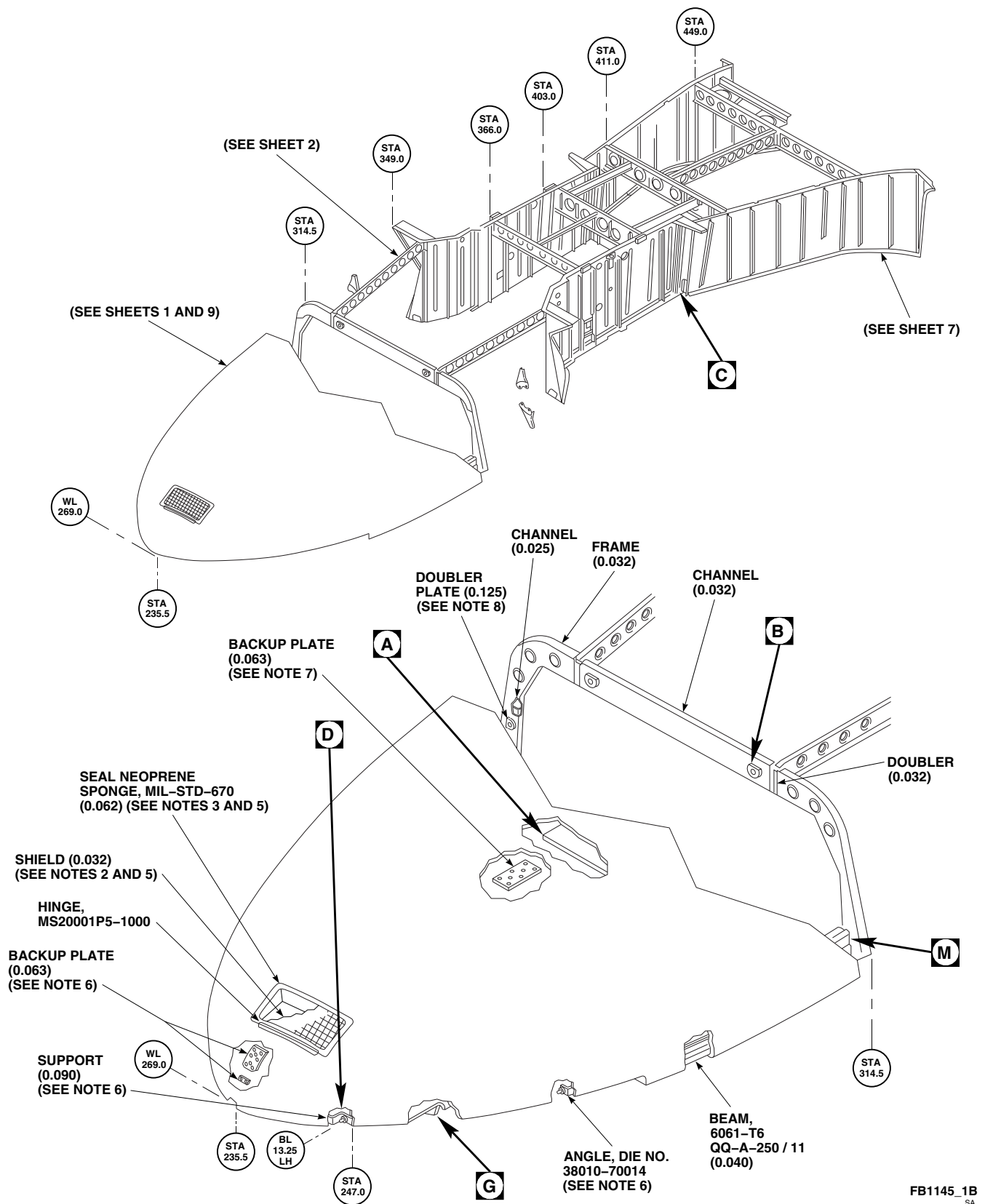
- (a) Chamfer track upper corner the minimum amount needed to let repair angle fit snugly.
- (b) Place repair plate against track web, with wear area between rear fastener holes (Figure 2-6-28-1, Sheet 10, Detail Q). Hold in place with edge-grip clecos.



To avoid weakening track, new fastener holes must be at least 0.375-inch (3 times rivet diameter) from any existing holes in track web.

- (c) Place repair angle against track, aligned with repair plate. Hold in place with edge-grip clecos.
 - (d) Drill $\frac{1}{8}$ -inch holes, centered on repair plate holes, through track web and repair angle.
 - (e) Remove clecos and repair plate and angle.
 - (f) Clean all bonding surfaces using cheesecloth, Item 90, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D.
 - (g) Apply thin coat of adhesive, Item 66, Appendix D, to all bonding surfaces. Let dry for 1 hour.
 - (h) Apply adhesive, Item 32, Appendix D, as liquid shim to web repaired area to bring it level with web surface.
 - (i) Apply even coat of adhesive, Item 32, Appendix D, to all bonding surfaces.
 - (j) Immediately assemble repair plate and angle to track and install three rivets, NAS1097AD-4.
 - (k) Clean up excess adhesive, using cheesecloth, Item 90, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D.
- g. **701901-SUBQ** Remove and replace wear strips as follows (Figure 2-6-28-1, Sheet 10, Detail R):
- (1) Drill out rivets, where required.
 - (2) Remove wear strip and old adhesive, using nonmetallic scraper and cheesecloth, Item 90, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D, or equivalent.
 - (3) Thoroughly clean bonding surfaces of track and wear strip using cheesecloth, Item 90, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D. Wipe dry with clean cheesecloth.
 - (4) Apply light coat of adhesive, Item 66, Appendix D, and let cure for 1 hour at room temperature.
 - (5) Coat bonding surfaces of track and wear strip with adhesive, Item 32, Appendix D. Install in position immediately.
 - (6) Cure at room temperature for 24 hours.
 - (7) Install rivets where required.
- h. Replace sliding cover track as follows:
- (1) Remove main rotor pylon sliding cover (PARA 2-4-101).
 - (2) Drill out rivets securing track, filler, and stop to hat.
 - (3) Rivet replacement track, filler, and stop to hat.
 - (4) Install main rotor pylon sliding cover (PARA 2-4-101).
- i. Bond rubber bumpers to firewalls with adhesive, Item 30 or Item 42, Appendix D.

- j. Seal for corrosion protection and watertightness (PARA 2-6-1).
- k. Touch up paint as required (PARA 2-6-5).



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FIGURE 2-6-28-1. REPAIR MAIN ROTOR PYLON FRAMING (SHEET 1 OF 12)

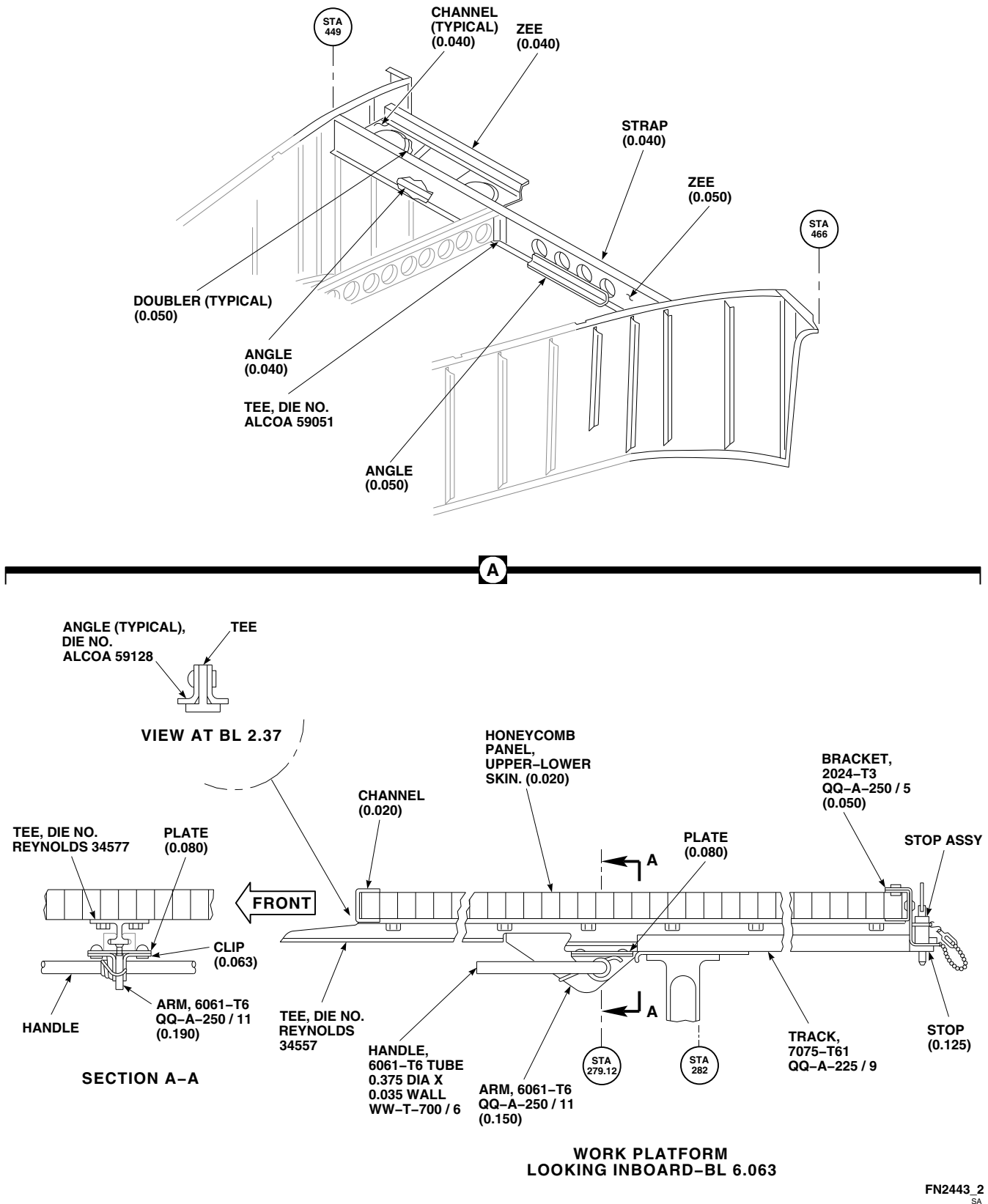
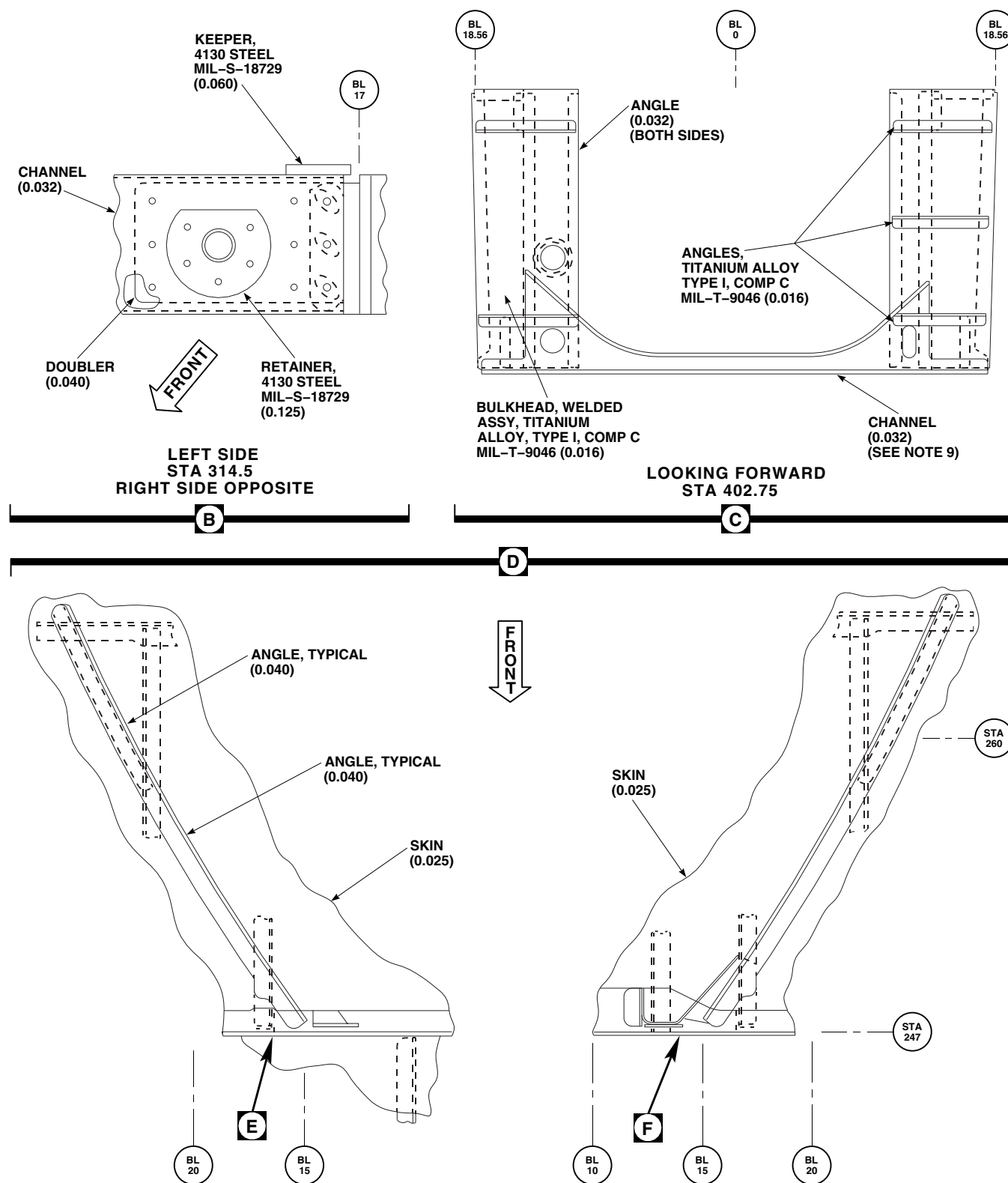


FIGURE 2-6-28-1. REPAIR MAIN ROTOR PYLON FRAMING (SHEET 3 OF 12)



VIEW LOOKING DOWN

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FIGURE 2-6-28-1. REPAIR MAIN ROTOR PYLON FRAMING (SHEET 4 OF 12)

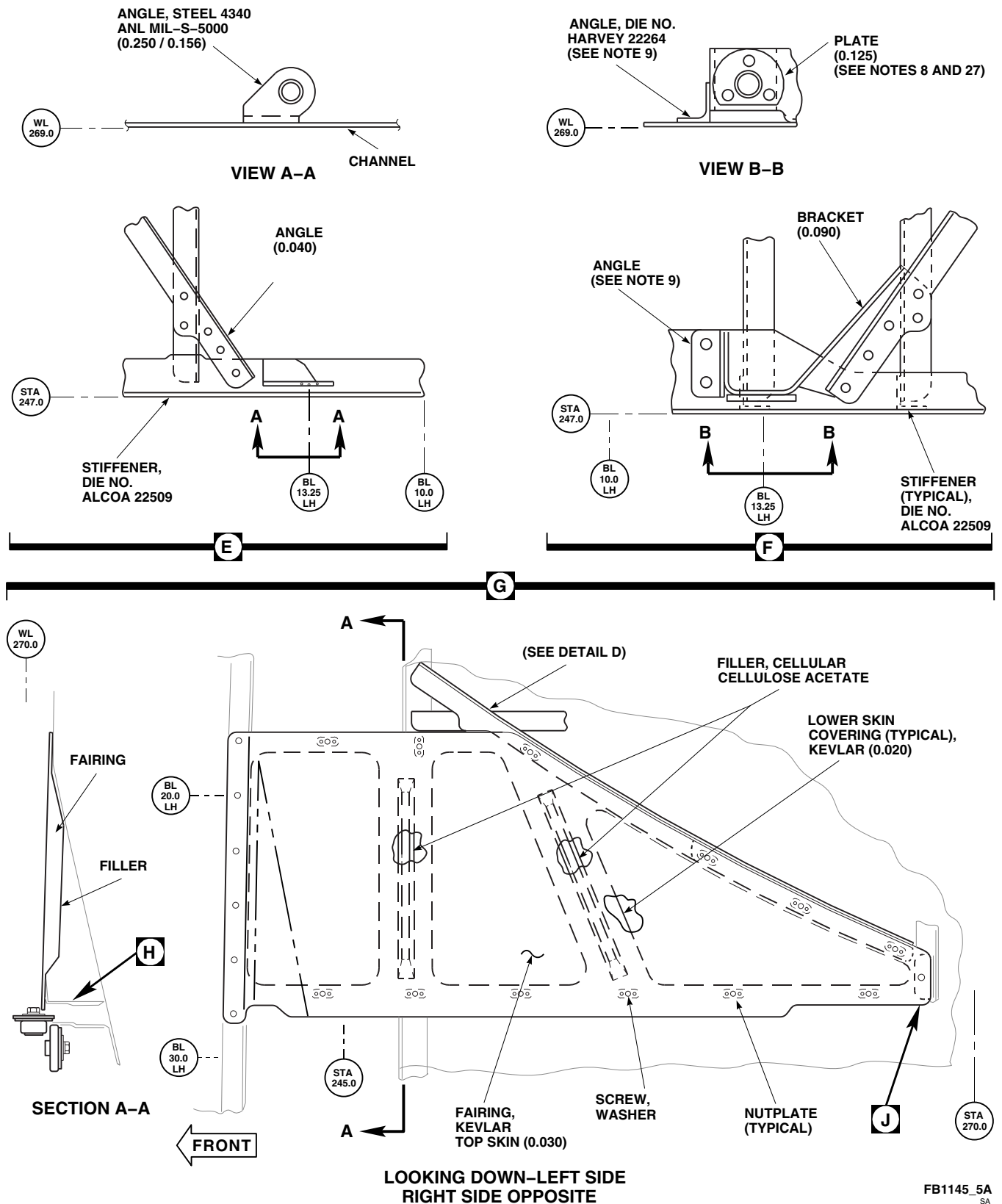


FIGURE 2-6-28-1. REPAIR MAIN ROTOR PYLON FRAMING (SHEET 5 OF 12)

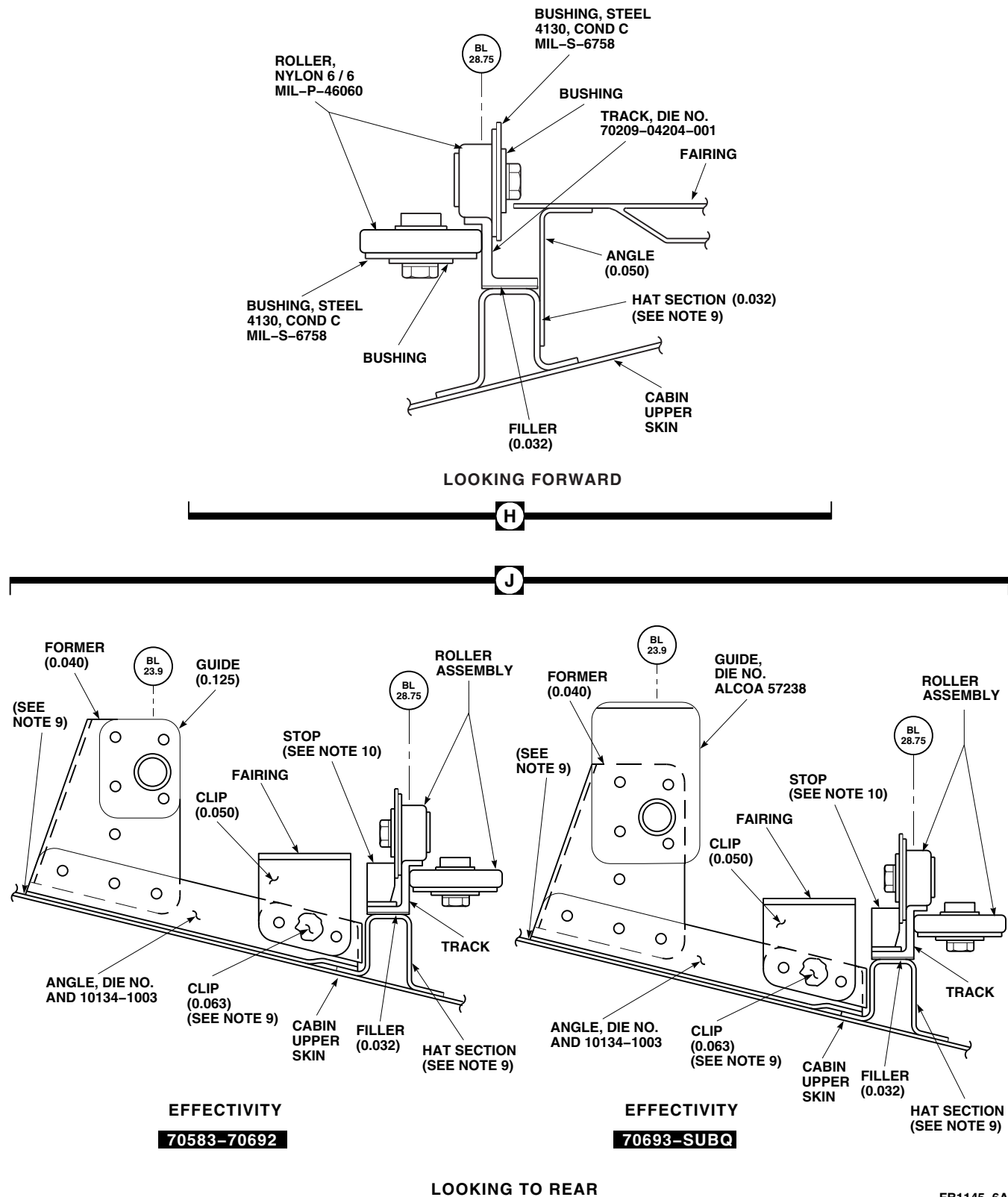
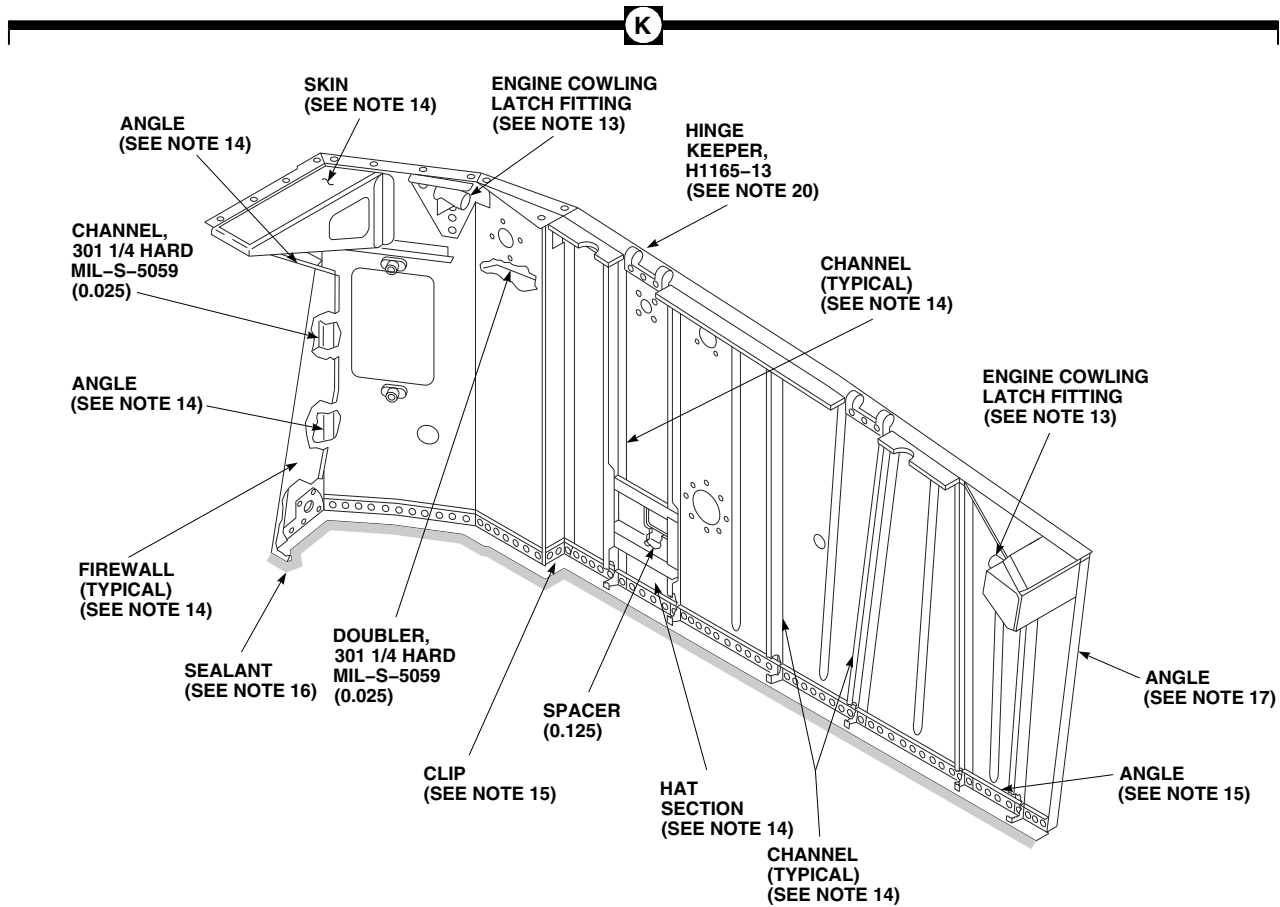


FIGURE 2-6-28-1. REPAIR MAIN ROTOR PYLON FRAMING (SHEET 6 OF 12)



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FIGURE 2-6-28-1. REPAIR MAIN ROTOR PYLON FRAMING (SHEET 7 OF 12)

L

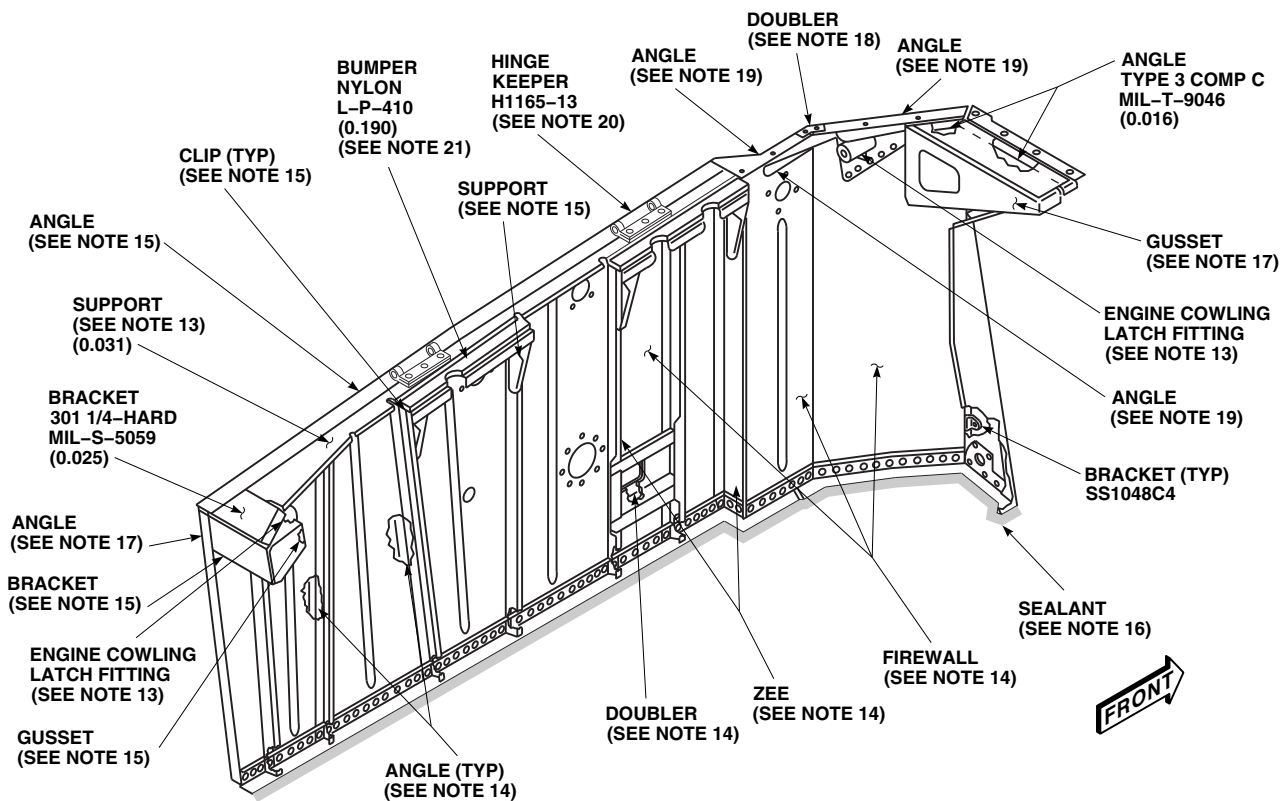
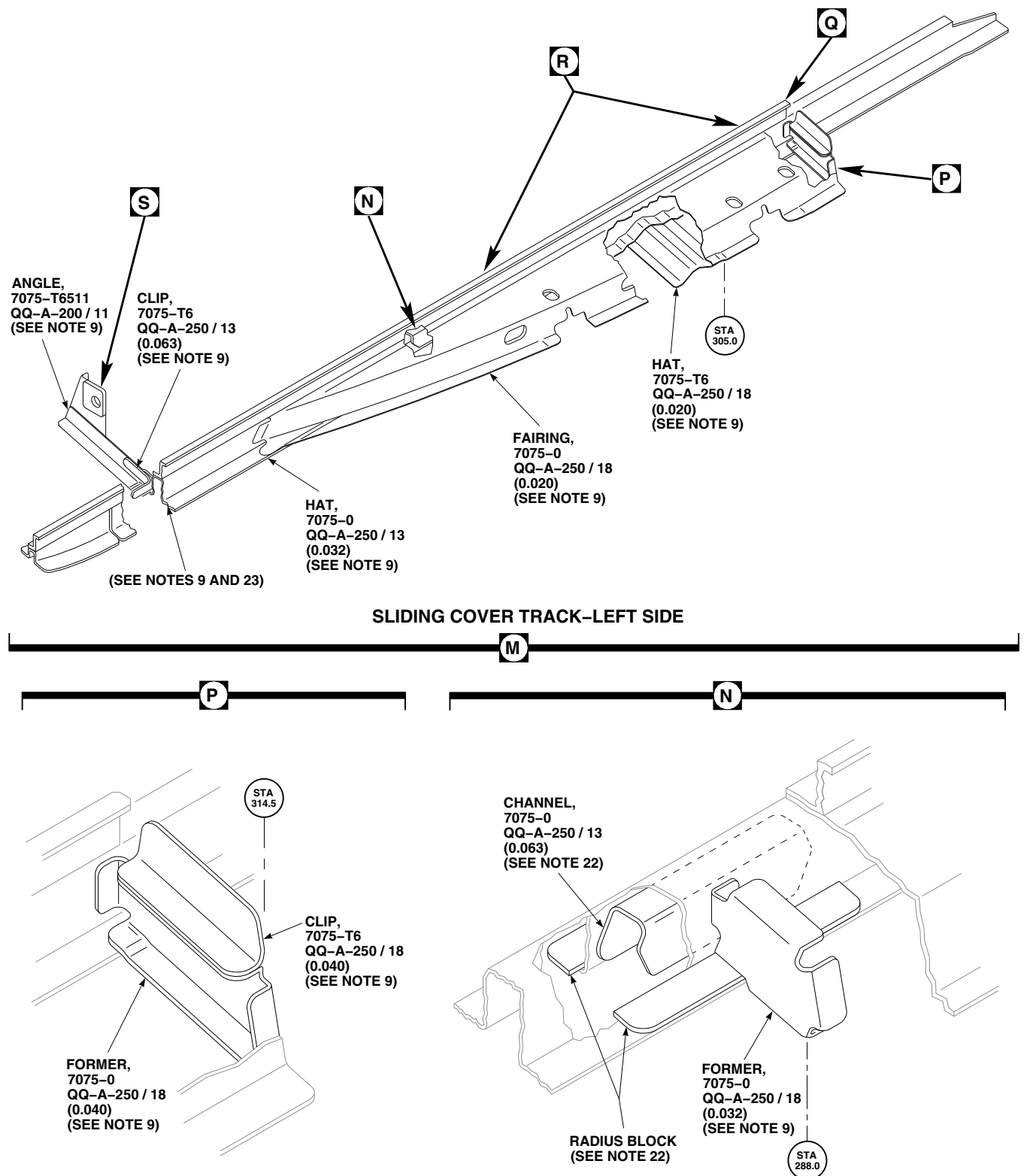
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FIGURE 2-6-28-1. REPAIR MAIN ROTOR PYLON FRAMING (SHEET 8 OF 12)



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FIGURE 2-6-28-1. REPAIR MAIN ROTOR PYLON FRAMING (SHEET 9 OF 12)

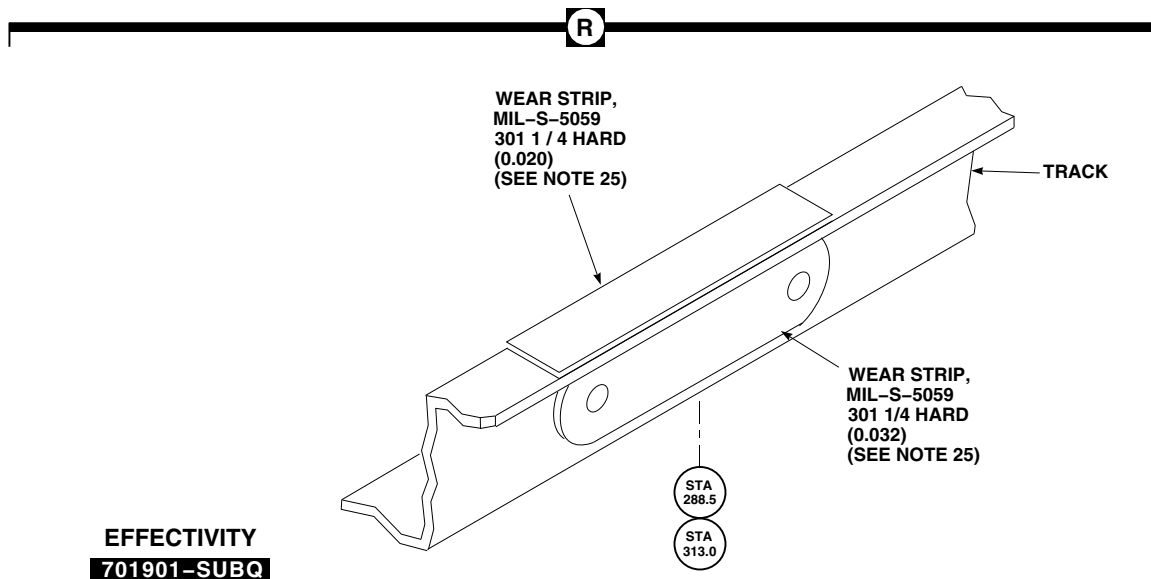
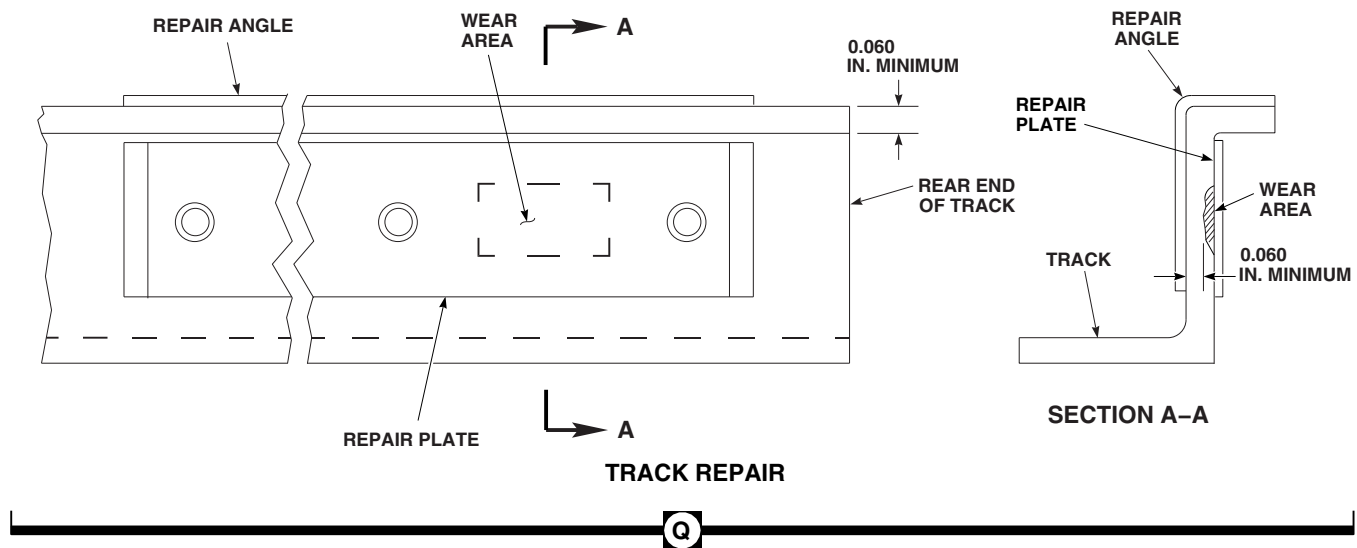
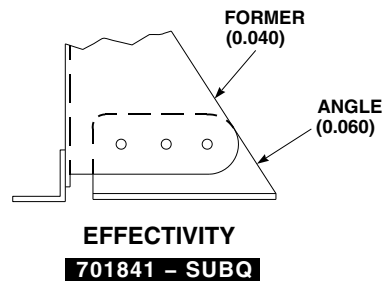
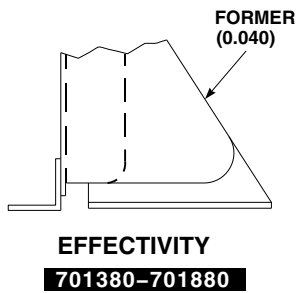
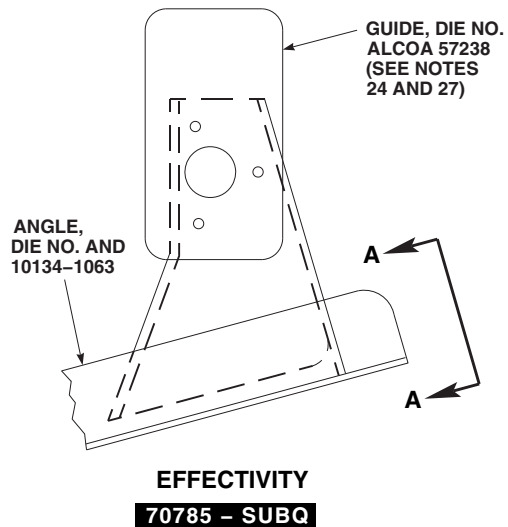
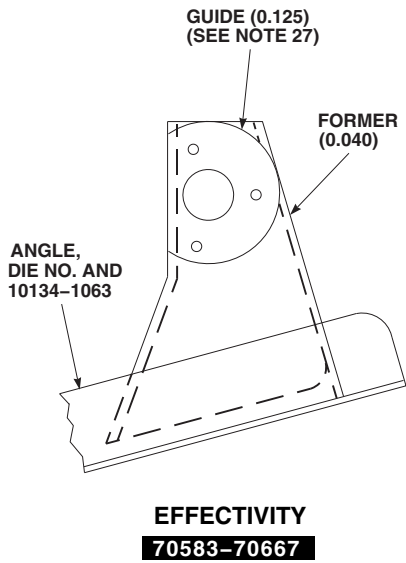
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FIGURE 2-6-28-1. REPAIR MAIN ROTOR PYLON FRAMING (SHEET 10 OF 12)

S



VIEW A-A

VIEW A-A

LOOKING AFT - RIGHT SIDE

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SA

FIGURE 2-6-28-1. REPAIR MAIN ROTOR PYLON FRAMING (SHEET 11 OF 12)

NOTES

1. UNLESS OTHERWISE NOTED ON SHEETS OF THIS FIGURE, SHEET METAL PARTS SHOWN ARE ALCLAD 7075-T6, QQ-A-250 / 13, AND ALL EXTRUSIONS ARE ALUMINUM ALLOY 7075-T6511, QQ-A-200 / 11.
2. LEFT AND RIGHT SIDES ARE THE SAME, EXCEPT WHEN NOTED.
3. BOND TO LIP OF CUTOUT WITH ADHESIVE, ITEM 57, APPENDIX D.
4. BOND SHIELD TO EDGE OF SCREEN WITH ADHESIVE, ITEM 57, APPENDIX D.
5. **70785 - SUBQ**
6. FILL GAP ON MOUNTING SURFACE WITH ADHESIVE, ITEM 32, APPENDIX D.
7. **701029-SUBQ**
8. **701364-SUBQ** 4130 STEEL, MIL-S-18729. ◀
9. CLEAN AND SEAL WITH SEALING COMPOUND, ITEM 292, APPENDIX D.
10. MADE EITHER FROM STEEL, MIL-S-7947, OR FROM CASTING, 356-T6.
11. 6AL-4V TITANIUM ALLOY ANNEALED, TYPE III, COMP C.
12. **701029-SUBQ** 0.016. ◀
13. CASTING AMS 5355 17-4 PH H-1000.
14. TYPE III, COMP C, MIL-T-9046 (0.016).
15. TYPE III, COMP C, MIL-T-9046 (0.025).
16. CLEAN AND BOND WITH ADHESIVE, ITEM 56, APPENDIX D.
17. (0.025).
18. **70583-701387** TYPE III, COMP C, MIL-T-9046 (0.016). ◀
19. **701391-SUBQ** TYPE III, COMP C, MIL-T-9046 (0.032). ◀
20. **70583-701410** TYPE III, COMP C, MIL-T-9046 (0.025). ◀
21. **701411-SUBQ** TYPE III, COMP C, MIL-T-9046 (0.040). ◀
22. PART NO. 70304-02117-101.
23. ALTERNATE MATERIAL: RUBBER, MIL-R-6855, CLASS I, GRADE 40.
24. CLEAN AND SEAL WITH SEALING COMPOUND, ITEM 292, APPENDIX D (PARA 2-6-1).
25. FILL GAPS WITH SEALING COMPOUND, ITEM 275, APPENDIX D, AND SEAL WITH SEALING COMPOUND, ITEM 292, APPENDIX D (PARA 2-6-1).
26. **701901-SUBQ** GUIDE PLATE IS STEEL, MIL-S-5000, 4340 BAR ANL. ◀
27. WEAR STRIPS INSTALLED WITH ADHESIVE, ITEM 32, APPENDIX D.
28. **701901-SUBQ** PLATE OF STEEL, MIL-S-8844. ◀
29. ALUMINUM PART MAY BE REPLACED BY STEEL PLATE FOR REPAIR. WEAR OF LOCATOR PIN CLEARANCE HOLE IN STRUCTURE, UP TO 0.38 INCH IN RADIUS FROM ORIGINAL HOLE CENTER, IS ALLOWED.
30. ALL DIMENSIONS ARE IN INCHES, UNLESS NOTED.

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FIGURE 2-6-28-1. REPAIR MAIN ROTOR PYLON FRAMING (SHEET 12 OF 12)

2-6-29 MAIN ROTOR PYLON FAIRING REPAIR.

PARAGRAPH BREAKDOWN

	PAGE
2-6-29.1 Repair Main Rotor Pylon Fairing	2-6-29-1
INITIAL SETUP	MIL-S-6721
Tools	MIL-S-7047
Airframe Repairer’s Toolkit	MIL-STD-670
Materials/Parts	MIL-T-9043
Adhesive, Item 37, Appendix D	MIL-T-9046
Adhesive, Item 47, Appendix D	MIL-T-9047
Sealing Compound, Item 292, Appendix D	PARA 2-3-11
References	PARA 2-4-113
Appendix D	PARA 2-4-121
AMS 4260	PARA 2-6-1
AMS 5643	PARA 2-6-2
MIL-C-7438	PARA 2-6-3
MIL-R-6855	PARA 2-6-5
MIL-S-18729	QQ-A-250
MIL-S-22499	QQ-A-367
	WW-T-700

2-6-29.1 Repair Main Rotor Pylon Fairing.

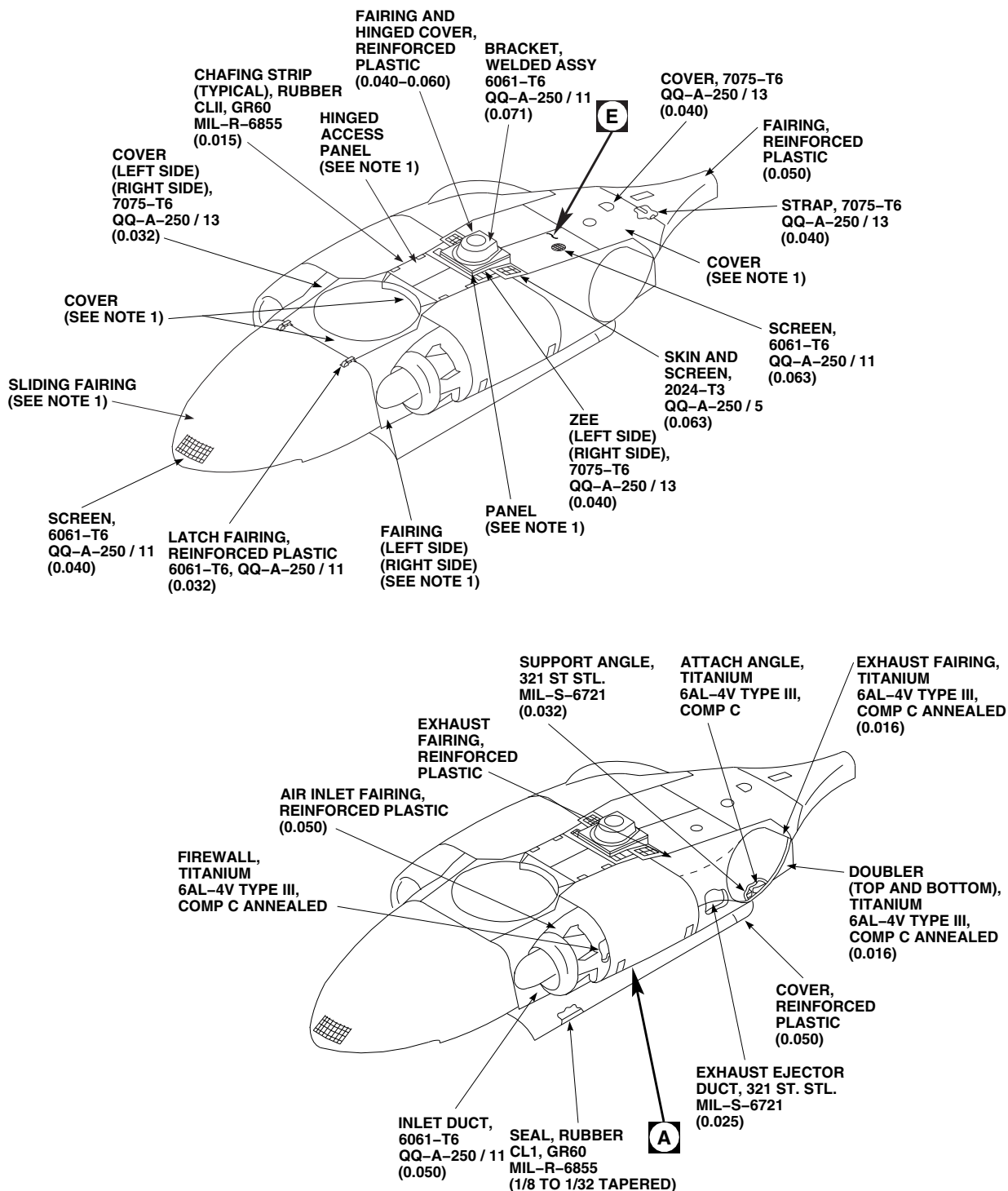
2-6-29.1.1 Repair.

- a. Main rotor pylon fairing and covers are of bonded sandwich construction made from

aluminum honeycomb core and Kevlar laminate skins, except for the latch fairings, IR jammer fairing, and rear fairing, which are Kevlar laminates. Screens and welded bracket assembly for IR jammer fairing are aluminum

alloy, 6061-T6, QQ-A-250/11. Skin cover and screens for oil coolers are aluminum alloy, 2024-T3, QQ-A-250/5. Doublers for hinges and latches on doors are titanium, MIL-T-9046, Type III, Comp C, or graphite laminates (Figure 2-6-29-1, Sheet 1 through Sheet 5).

- b. Repair of graphite is beyond the scope of this manual.
- c. Evaluate damage and make repairs to titanium parts (PARA 2-6-2).
- d. Evaluate damage and make repairs to Kevlar laminates and honeycomb panels (PARA 2-6-3).
- e. Evaluate damage to aluminum alloy parts (PARA 2-6-1) and make repairs using typical repair examples (PARA 2-6-2).
- f. Evaluate damage to engine cowling/work platform hinge and support fitting using limits (PARA 2-4-121 and PARA 2-3-11).
- g. Rebond rubber chafing strips on hinged panels over oil coolers, using adhesive, Item 37, Appendix D, or adhesive, Item 47, Appendix D (PARA 2-4-113).
- h. Seal for corrosion protection and watertightness (PARA 2-6-1).
- i. Touch up paint as required (PARA 2-6-5).



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FIGURE 2-6-29-1. REPAIR MAIN ROTOR PYLON FAIRING (SHEET 1 OF 5)

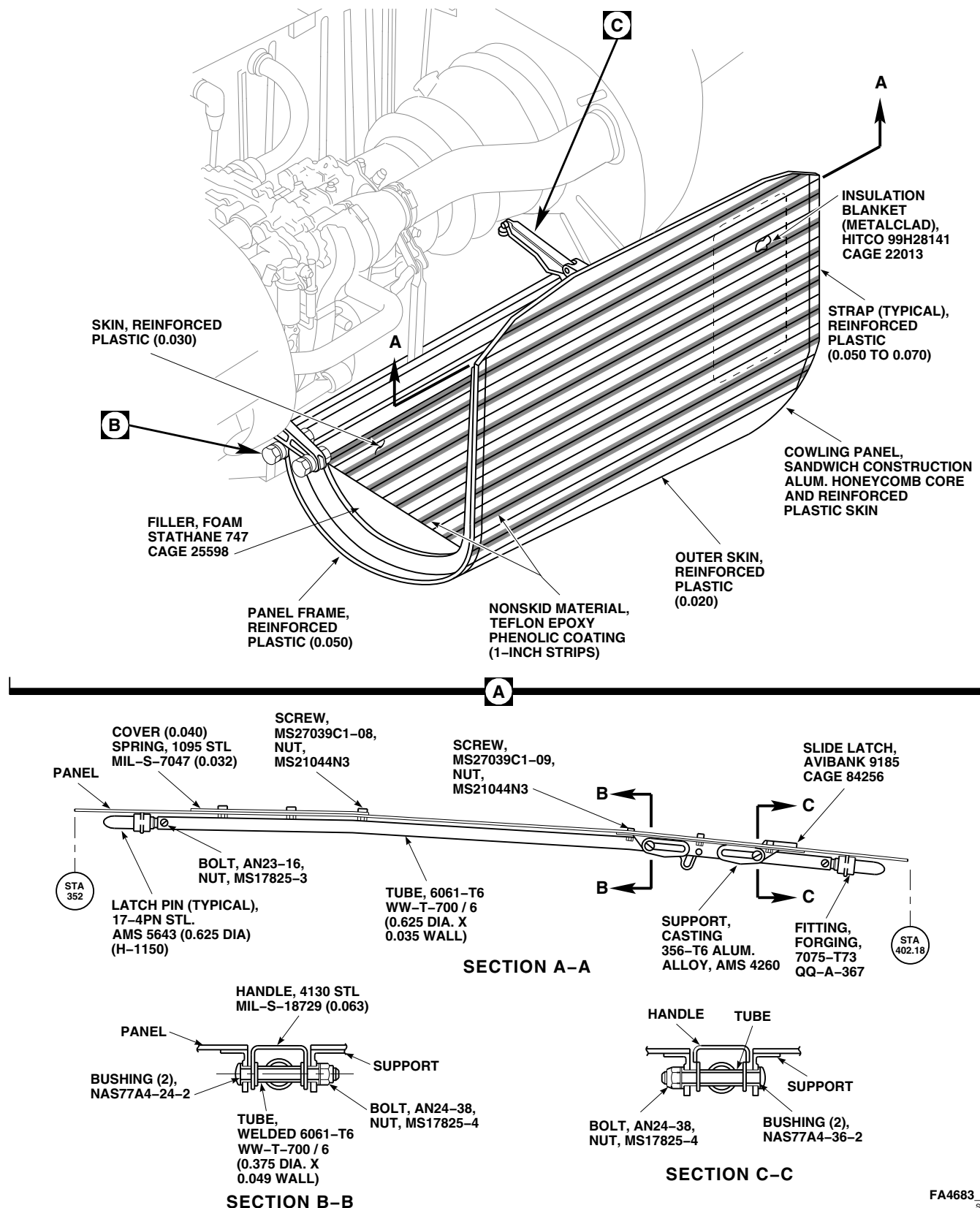


FIGURE 2-6-29-1. REPAIR MAIN ROTOR PYLON FAIRING (SHEET 2 OF 5)

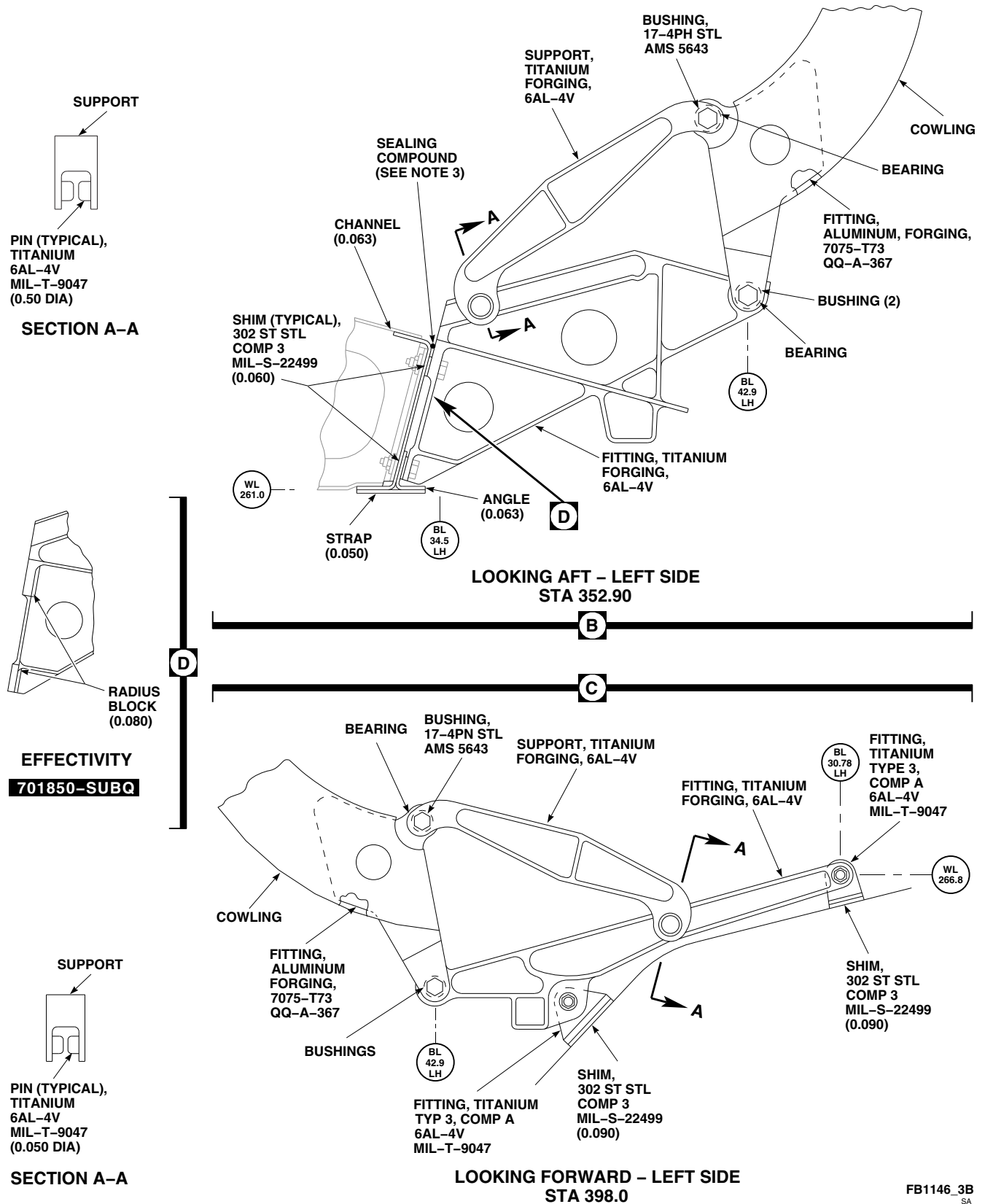


FIGURE 2-6-29-1. REPAIR MAIN ROTOR PYLON FAIRING (SHEET 3 OF 5)

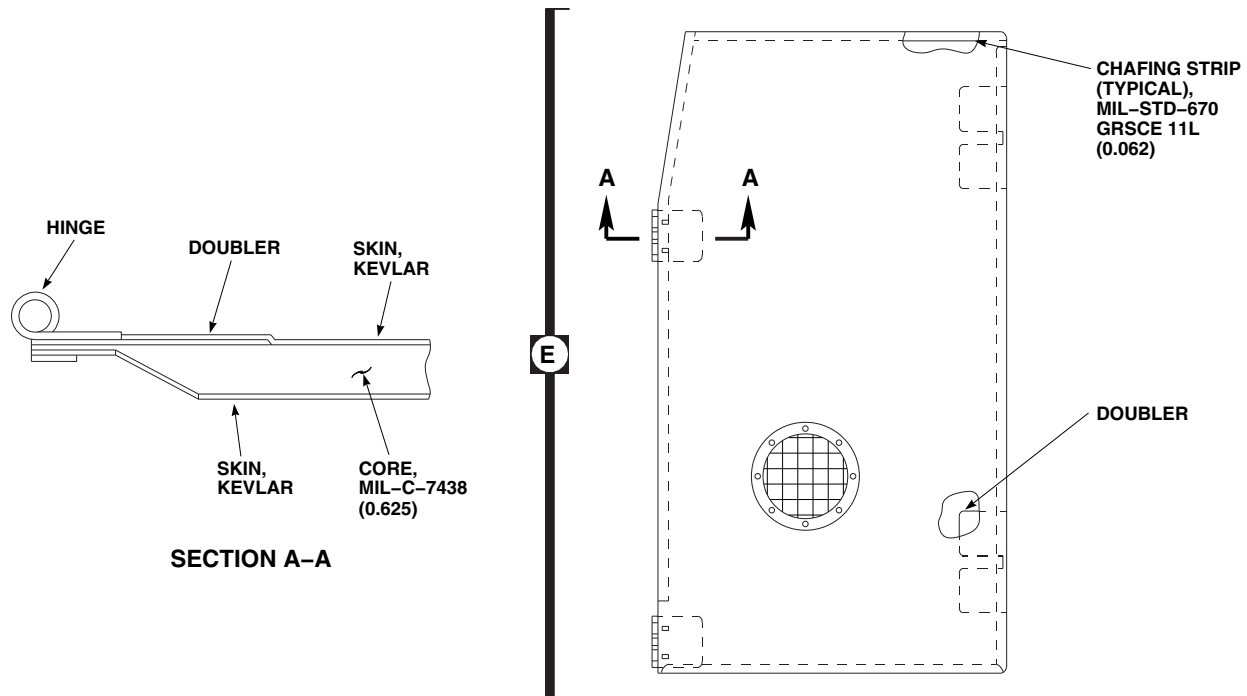
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FIGURE 2-6-29-1. REPAIR MAIN ROTOR PYLON FAIRING (SHEET 4 OF 5)

NOTES

1. SANDWICH CONSTRUCTION ALUMINUM HONEYCOMB (MIL-C-7438) AND KEVLAR SKINS.
2. EITHER TITANIUM, MIL-T-9043 TYPE 111 COMP C, OR GRAPHITE 5 PLIES COVERED BY 2 PLIES KEVLAR.
3. APPLY BEAD OF SEALING COMPOUND, ITEM 292, APPENDIX D.
4. ALL DIMENSIONS ARE IN INCHES, UNLESS NOTED.

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FIGURE 2-6-29-1. REPAIR MAIN ROTOR PYLON FAIRING (SHEET 5 OF 5)

2-6-30 TRANSITION SECTION FRAMING REPAIR.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
2-6-30.1	Transition Section Framing Repair	2-6-30-2	2-6-30.2	Inspect/Replace Fuel Cell Liner	2-6-30-2

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Clecos, No. 40
- Drill
- Drill Bit, 3⁄32-inch diameter
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Adhesion Promoter, Item 25A, Appendix D
- Foam Liner Kit, 70219-03011-041
- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Pressure Sensitive Adhesive Tape, Item 250E, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- AMS 5355
- Appendix D
- MIL-C-7438
- MIL-R-6855
- MIL-S-18729
- PARA 1-7-17
- PARA 2-6-1
- PARA 2-6-2
- PARA 2-6-3
- PARA 2-6-4
- PARA 2-6-5
- PARA 10-4-2
- QQ-A-200
- QQ-A-250
- QQ-A-367

2-6-30.1 Transition Section Framing Repair.**NOTE**

- Transition section framing includes formed and extruded members made from 7075-T6, QQ-A-250/13 Alclad sheet stock and 7075-T6511, QQ-A-200/11 aluminum alloy extrusion. Frame fitting at STA 379.0 and attachment fittings at STA 485.0 are made from forged aluminum alloy, 7075-T73, QQ-A-367.
- All transition section framing, except reinforced plastic/honeycomb core panel, is primary structure. Forged fittings, however, are considered critical primary structure.

- Negligible damage to fittings is limited to nicks, scratches, and corrosion no deeper than 5% of material thickness at point of damage. Blend damage smoothly into surrounding area using long, even strokes. Polish out all surface imperfections and restore protective finish. Damage to fittings greater than negligible limits must be reviewed by engineering authority for damage analysis and, either repair or replacement. All repairs to transition section framing must maintain design strength and alignment (Figure 2-6-30-1, Sheet 1 through Sheet 25).
- Use PARA 2-6-2 and PARA 2-6-4 for typical framing repairs.
- Use PARA 2-6-3 for evaluating damage and repairing reinforced plastic/honeycomb core panels.
- For corrosion protection and watertightness, seal all repairs between frame members and plating (PARA 2-6-1).
- Touch up paint as required (PARA 2-6-5).

2-6-30.2 Inspect/Replace Fuel Cell Liner.**2-6-30.2.1 Inspect.**

Visually inspect fuel cell liner for cracks, corrosion, or degradation. **NONE ALLOWED.** If cracks,

corrosion, or degradation is found, replace fuel cell liner.

2-6-30.2.2 Remove.

- Remove No. 1 or No. 2 main fuel cell (PARA 10-4-2).
- Using nonmetallic scraper, remove fuel cell liners from transition section framing (Figure 2-6-30-1, Sheet 20, Detail AH).
- Using nonmetallic scraper, remove foam blocks from framing structure.

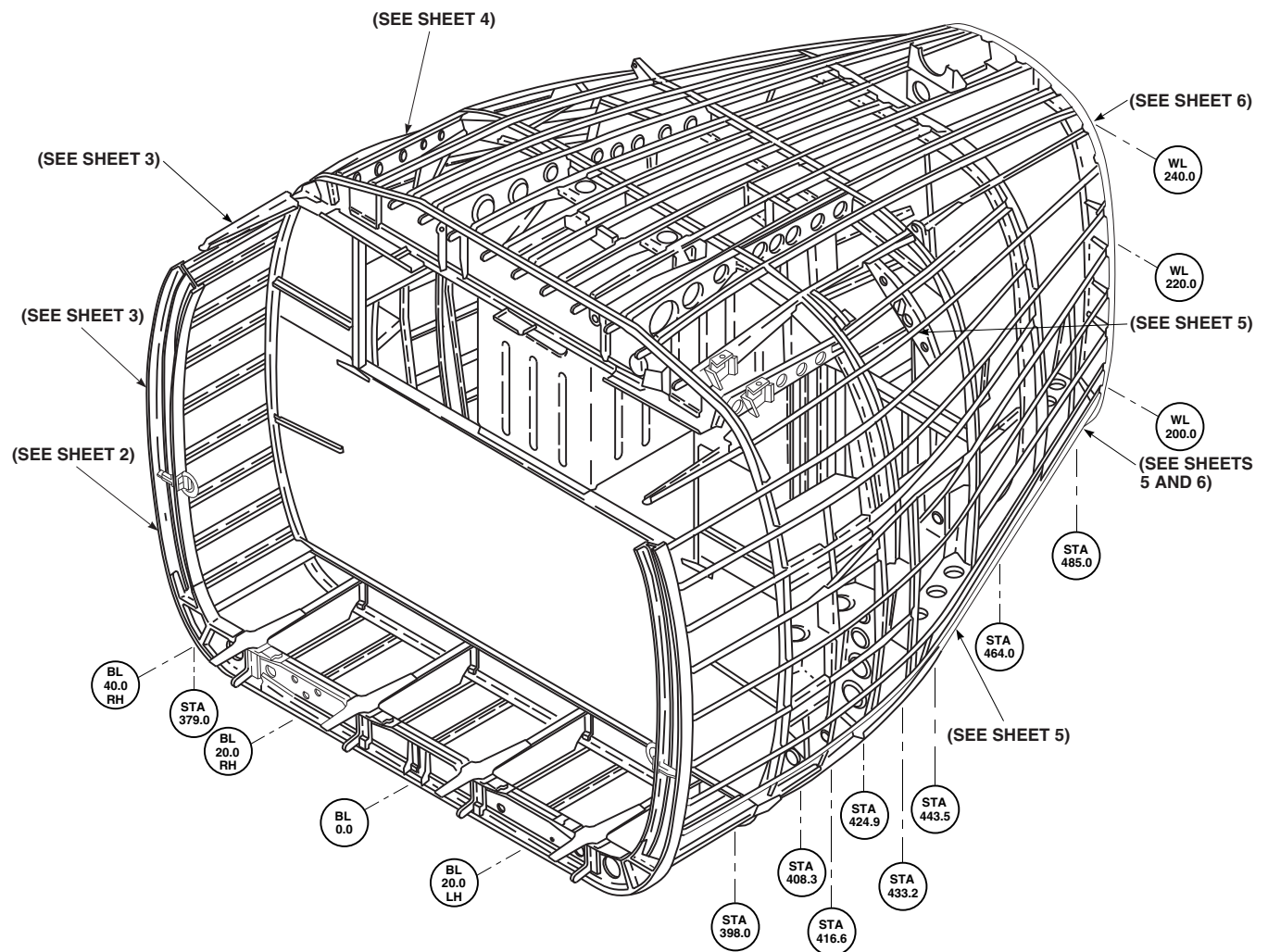
NOTE

Replace all foam blocks at same time.

- Inspect foam blocks for corrosion or deterioration. **NONE ALLOWED.** If corrosion or deterioration is found on any block, replace all foam blocks using foam liner kit, 70219-03011-041.
- Inspect transition section tub (PARA 1-7-17).

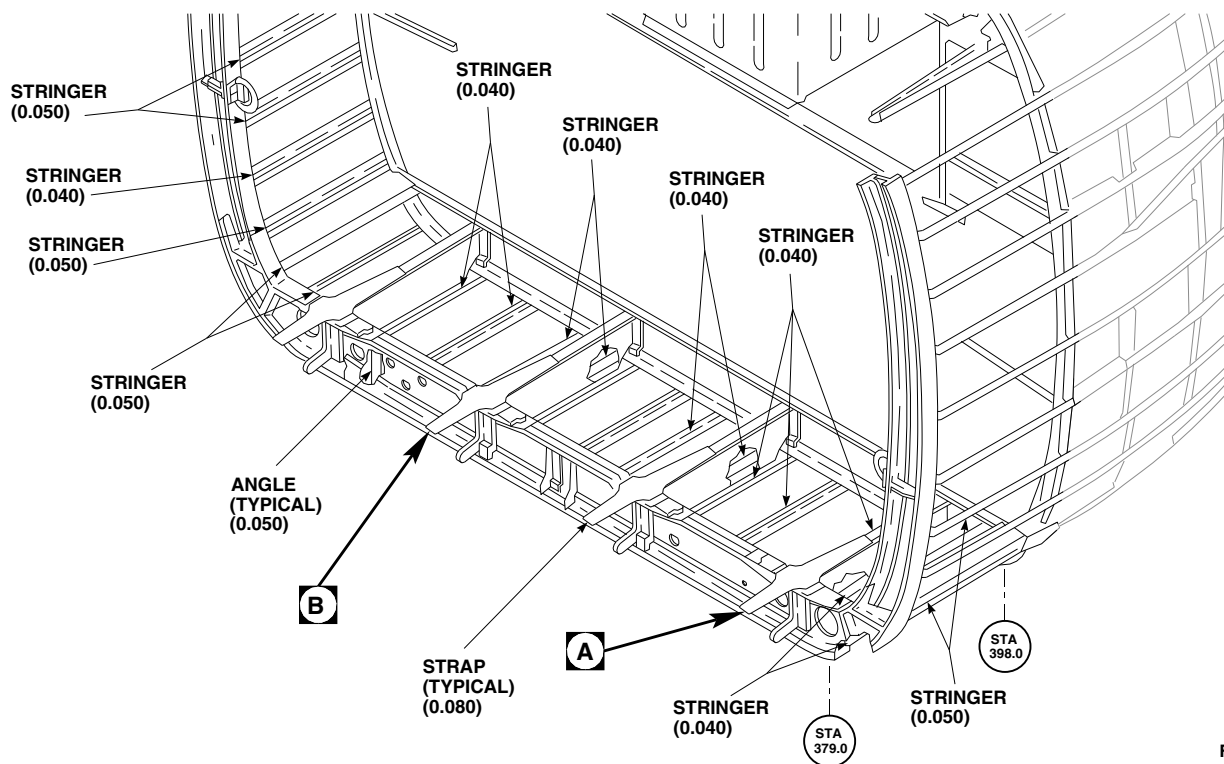
2-6-30.2.3 Install.

- Insert foam blocks into framing structure. Apply sealing compound, Item 292, Appendix D, bonding foam blocks to transition section framing.
- Apply sealing compound, Item 292, Appendix D, to bonding areas of framing. Position liner against framing (Figure 2-6-30-1, Sheet 20, Detail AH).
- Inspect fuel cell liner installation. Sealing compound must be in direct contact with liner and adjoining framing at all points. **NO GAPS ALLOWED.** In locations required to secure liner firmly against framing, perform the following:
 - Using drill with $\frac{3}{32}$ -inch diameter drill bit, drill holes through liner and into framing.
 - Insert No. 40 clecos into holes to secure liner.



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FIGURE 2-6-30-1. REPAIR TRANSITION SECTION FRAMING (SHEET 1 OF 25)

FB1150_2B
SA**FIGURE 2-6-30-1. REPAIR TRANSITION SECTION FRAMING (SHEET 2 OF 25)**

- d. Apply adhesion promoter, Item 25A, Appendix D, to edges of fuel cell liner.
- e. Tape all inside edges of fuel cell liner with pressure sensitive adhesive tape, Item 250E, Appendix D.

NOTE

Holes used to insert clecos may be left unfilled.

- f. Remove clecos.
- g. Using nonmetallic scraper and machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D, remove excess sealing compound.

- h. Inspect bond joints as follows:

- (1) **VOIDS OF SEALING COMPOUND IN JOINT SHALL NOT EXCEED TOTAL OF 3 INCHES IN ANY 6-INCH LENGTH.**
- (2) **VOIDS OF SEALING COMPOUND IN JOINTS SHALL NOT EXCEED, IN LENGTH, 10% OF UPPER EDGE BONDLINE, 25% OF SIDE AND BOTTOM BONDLINE, OR 75% OF CORNER BONDLINE.**

- i. Touch up paint as required (PARA 2-6-5).

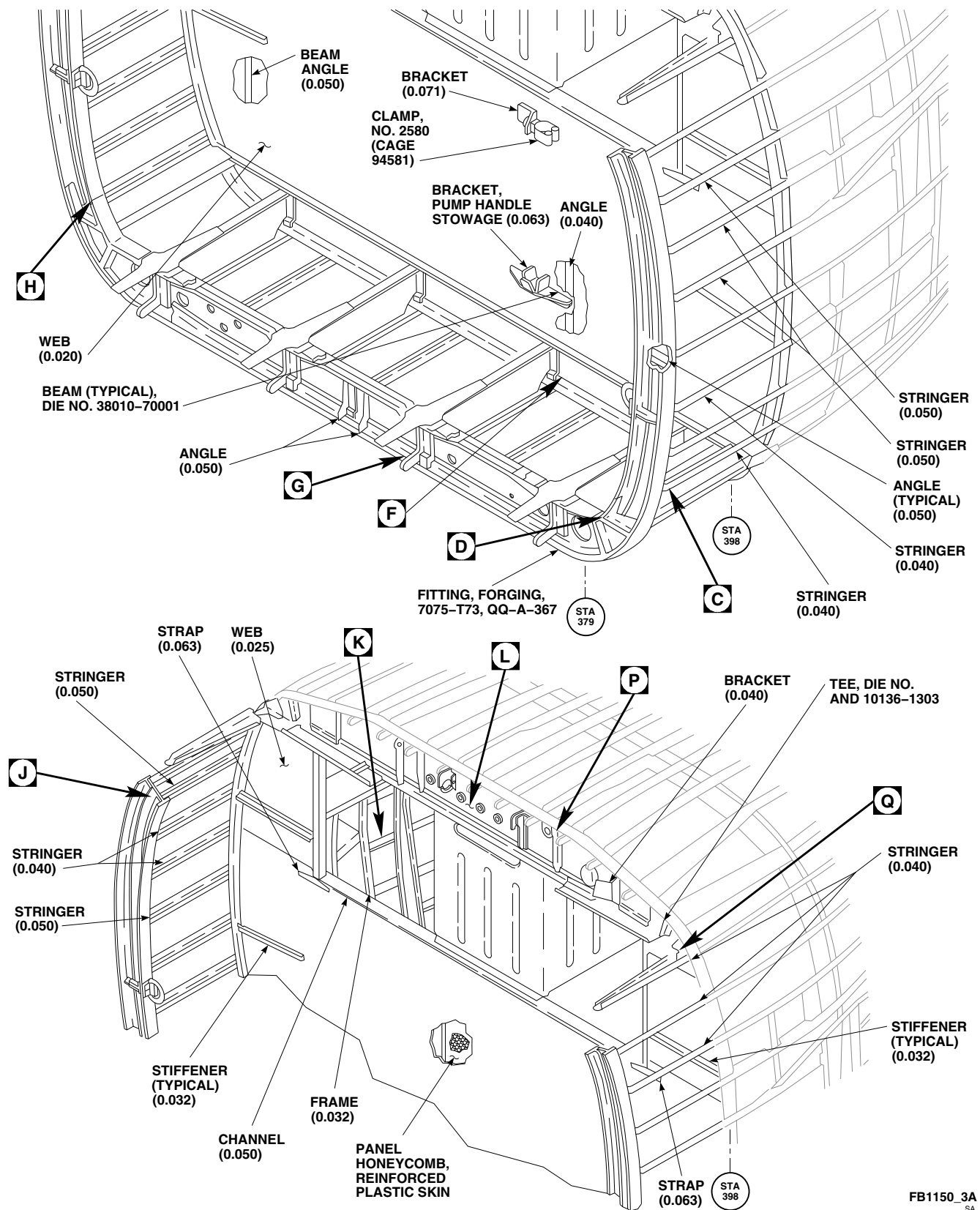


FIGURE 2-6-30-1. REPAIR TRANSITION SECTION FRAMING (SHEET 3 OF 25)

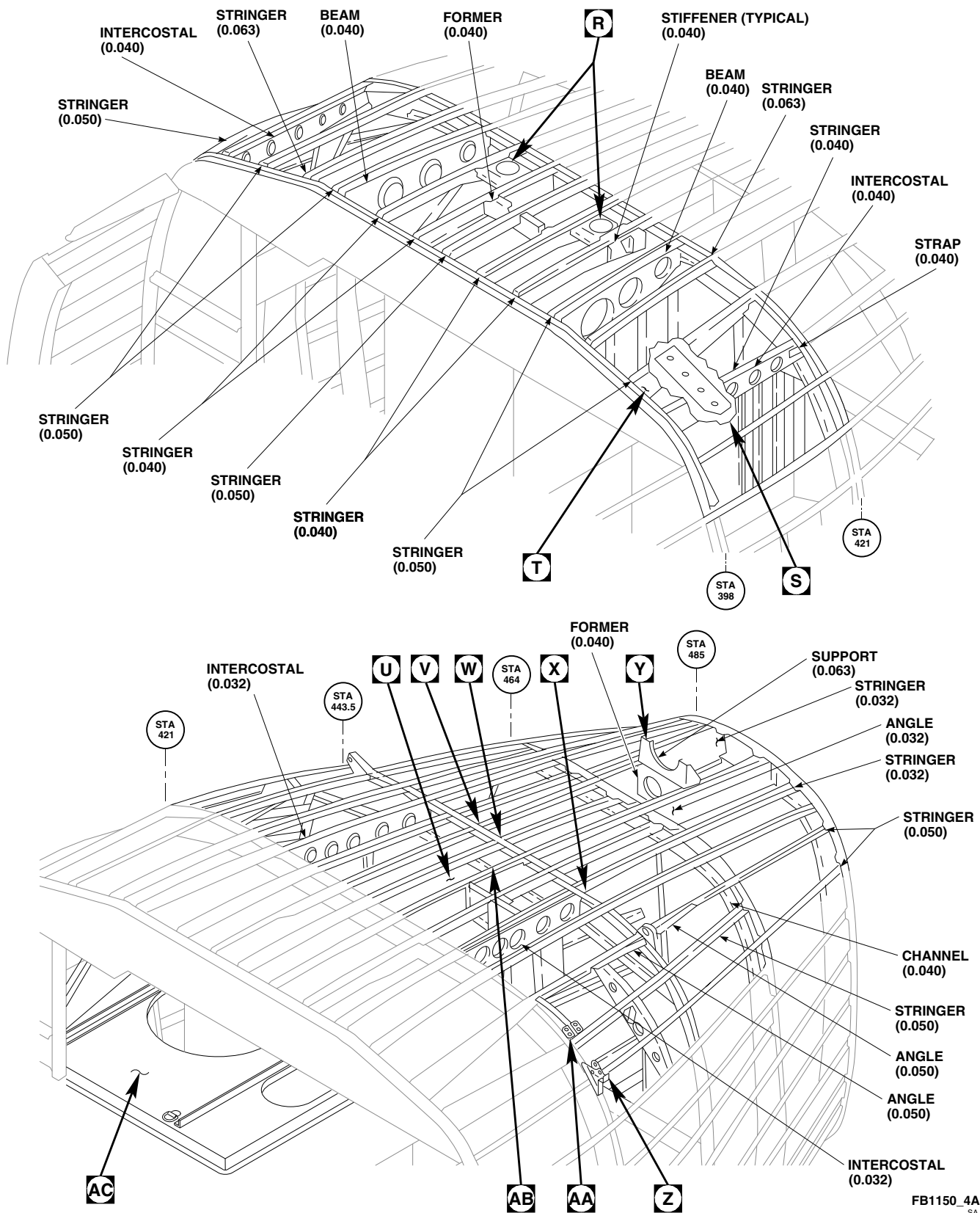
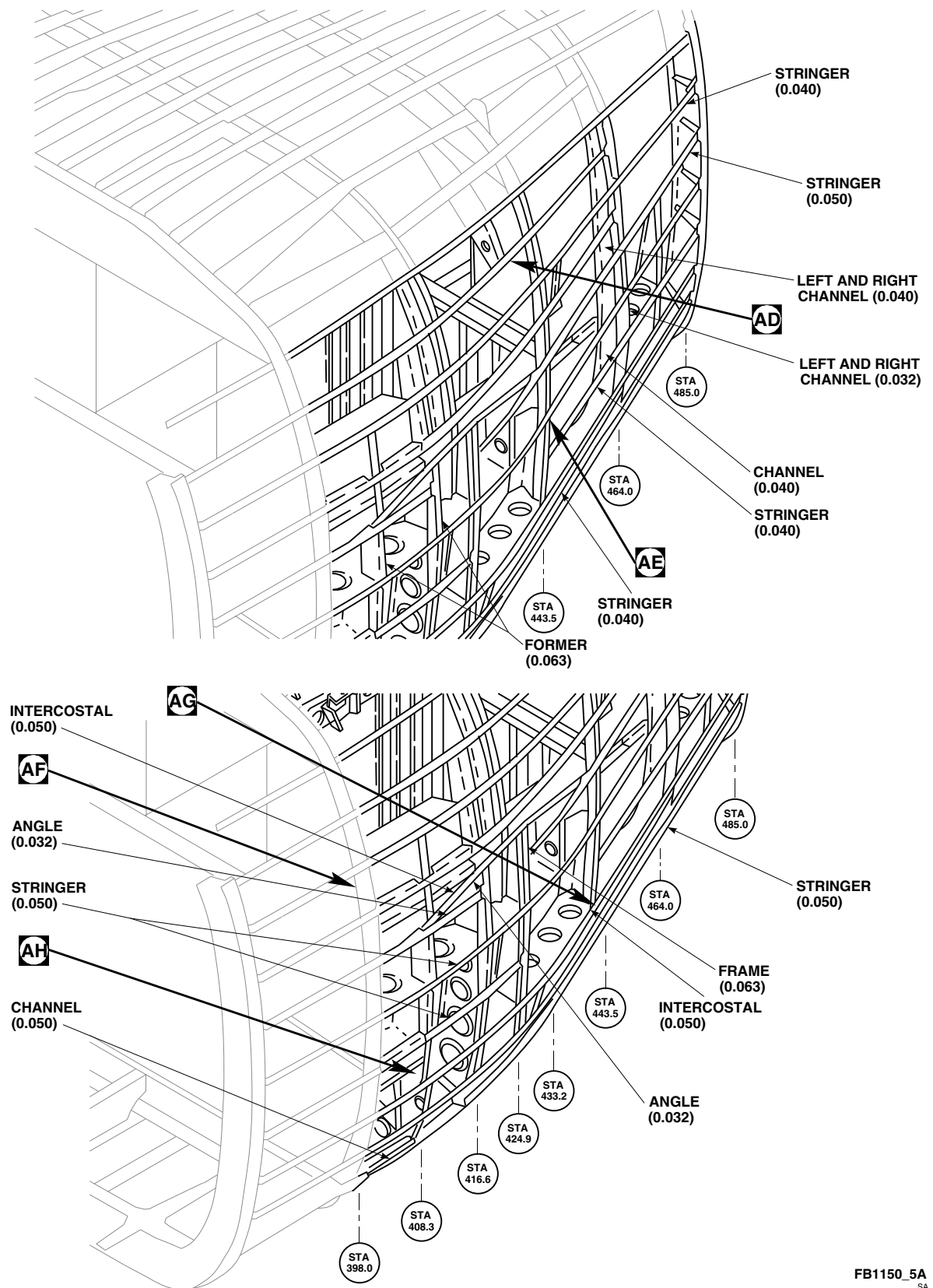


FIGURE 2-6-30-1. REPAIR TRANSITION SECTION FRAMING (SHEET 4 OF 25)



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FIGURE 2-6-30-1. REPAIR TRANSITION SECTION FRAMING (SHEET 5 OF 25)

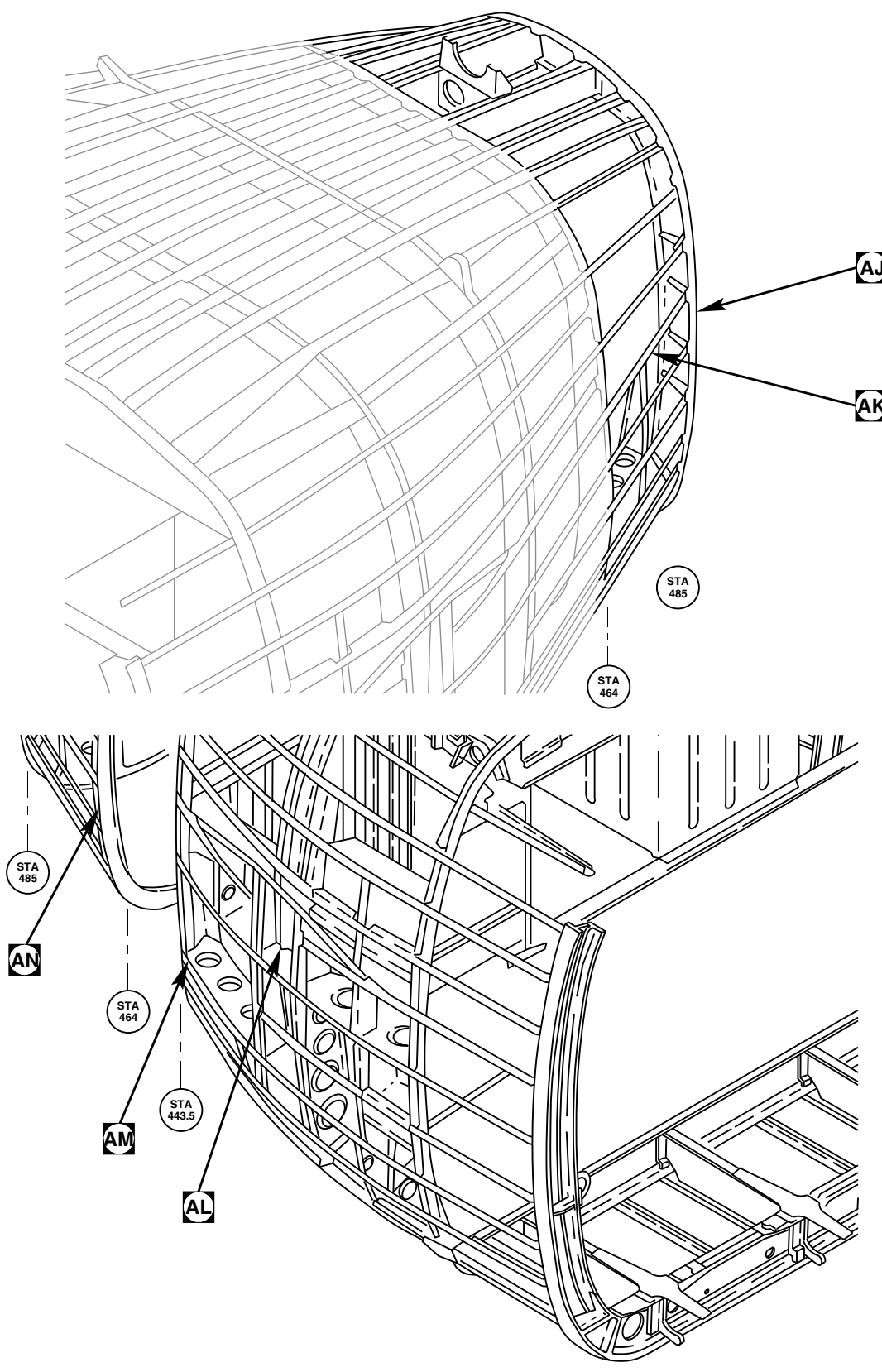
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FIGURE 2-6-30-1. REPAIR TRANSITION SECTION FRAMING (SHEET 6 OF 25)

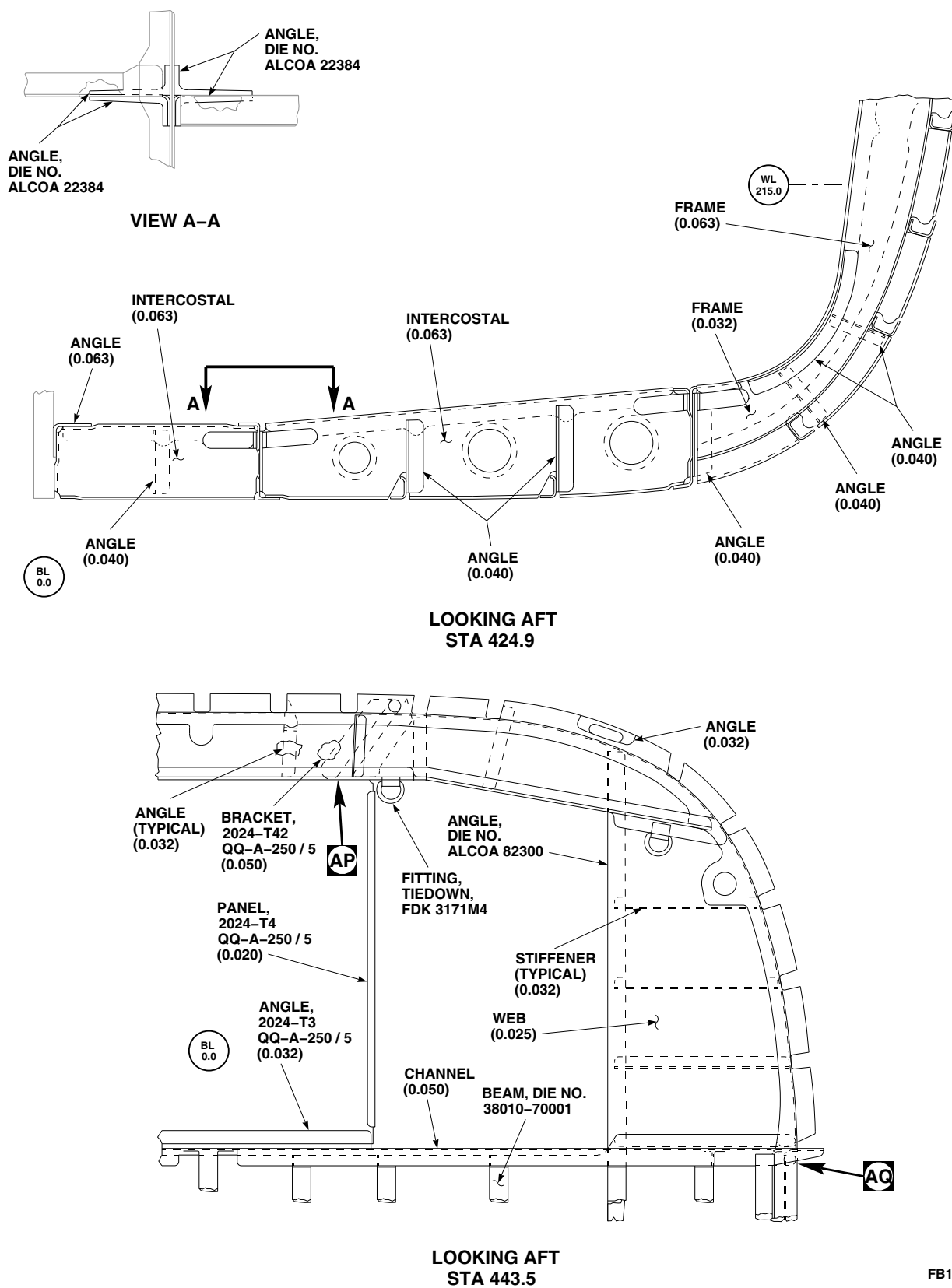


FIGURE 2-6-30-1. REPAIR TRANSITION SECTION FRAMING (SHEET 7 OF 25)

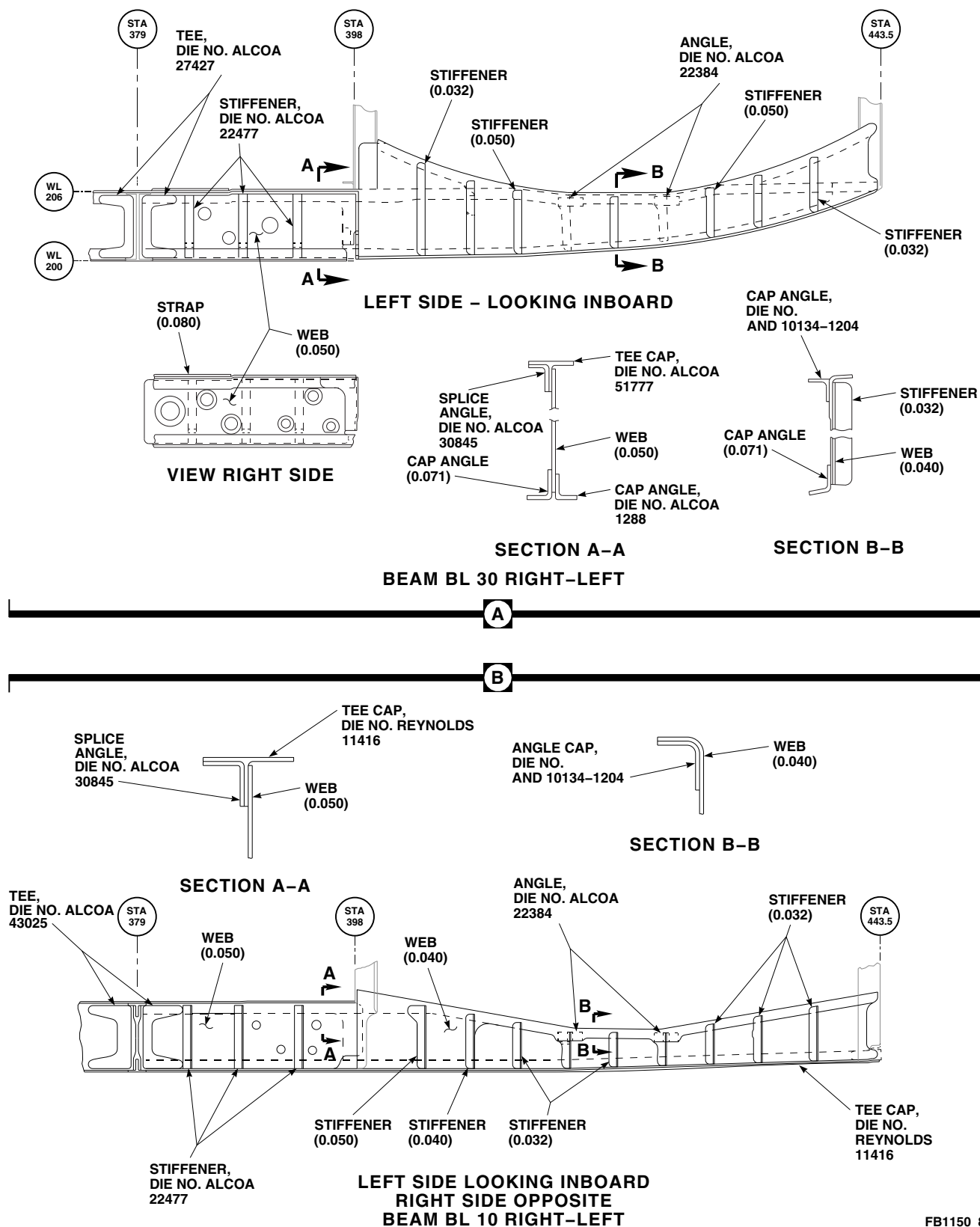
FB1150_8B
SA

FIGURE 2-6-30-1. REPAIR TRANSITION SECTION FRAMING (SHEET 8 OF 25)

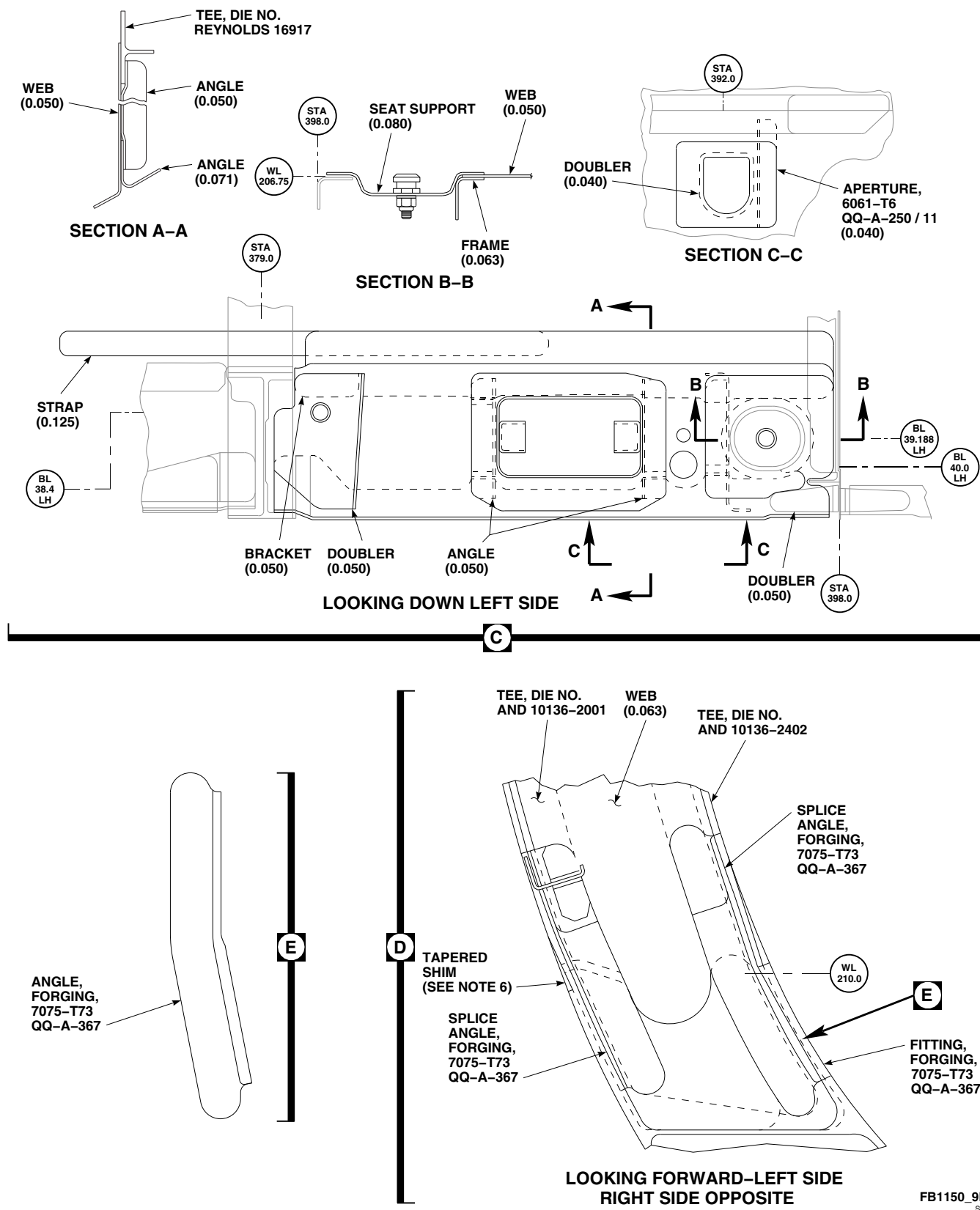


FIGURE 2-6-30-1. REPAIR TRANSITION SECTION FRAMING (SHEET 9 OF 25)

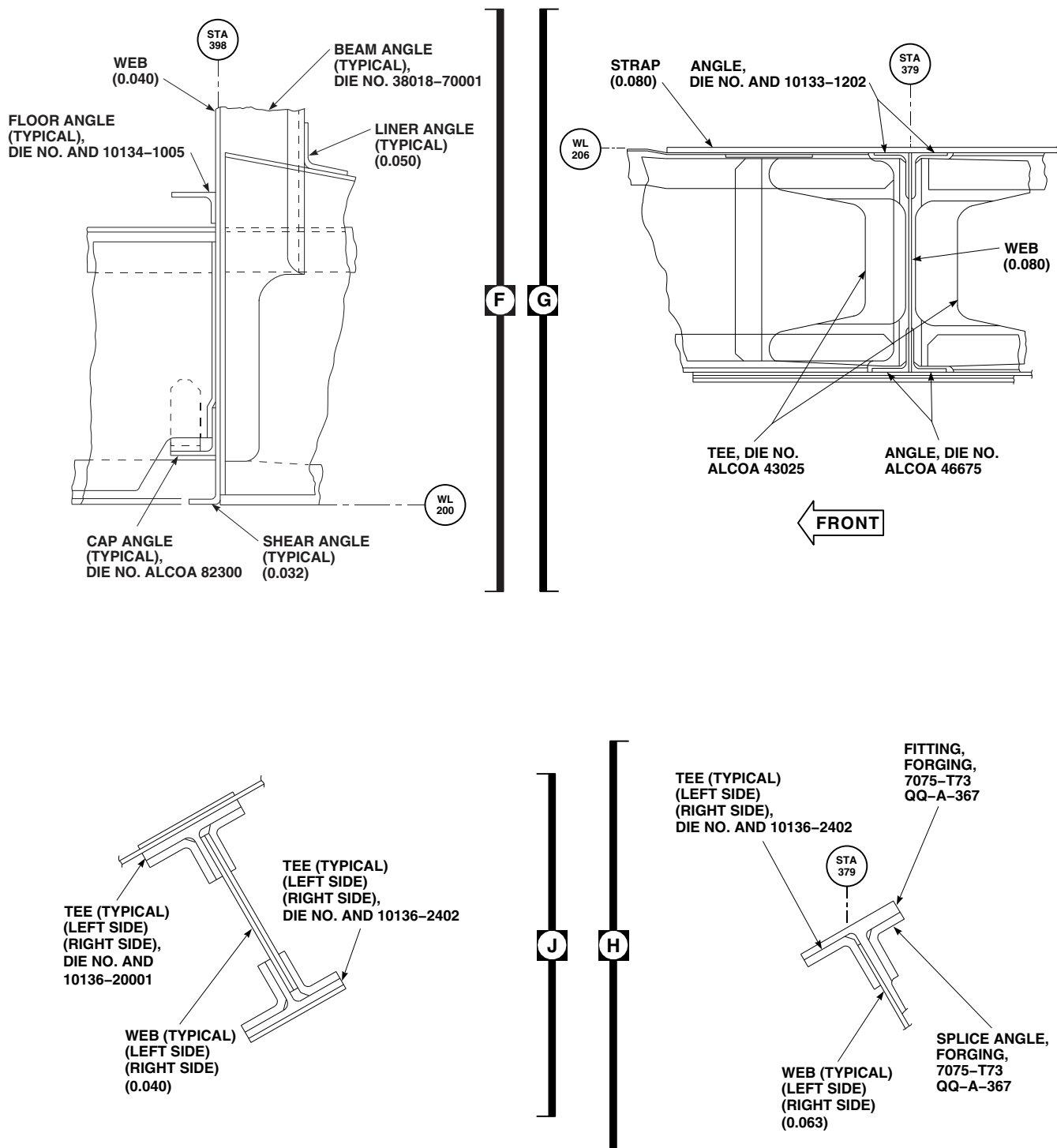
FB1150_10A
SA

FIGURE 2-6-30-1. REPAIR TRANSITION SECTION FRAMING (SHEET 10 OF 25)

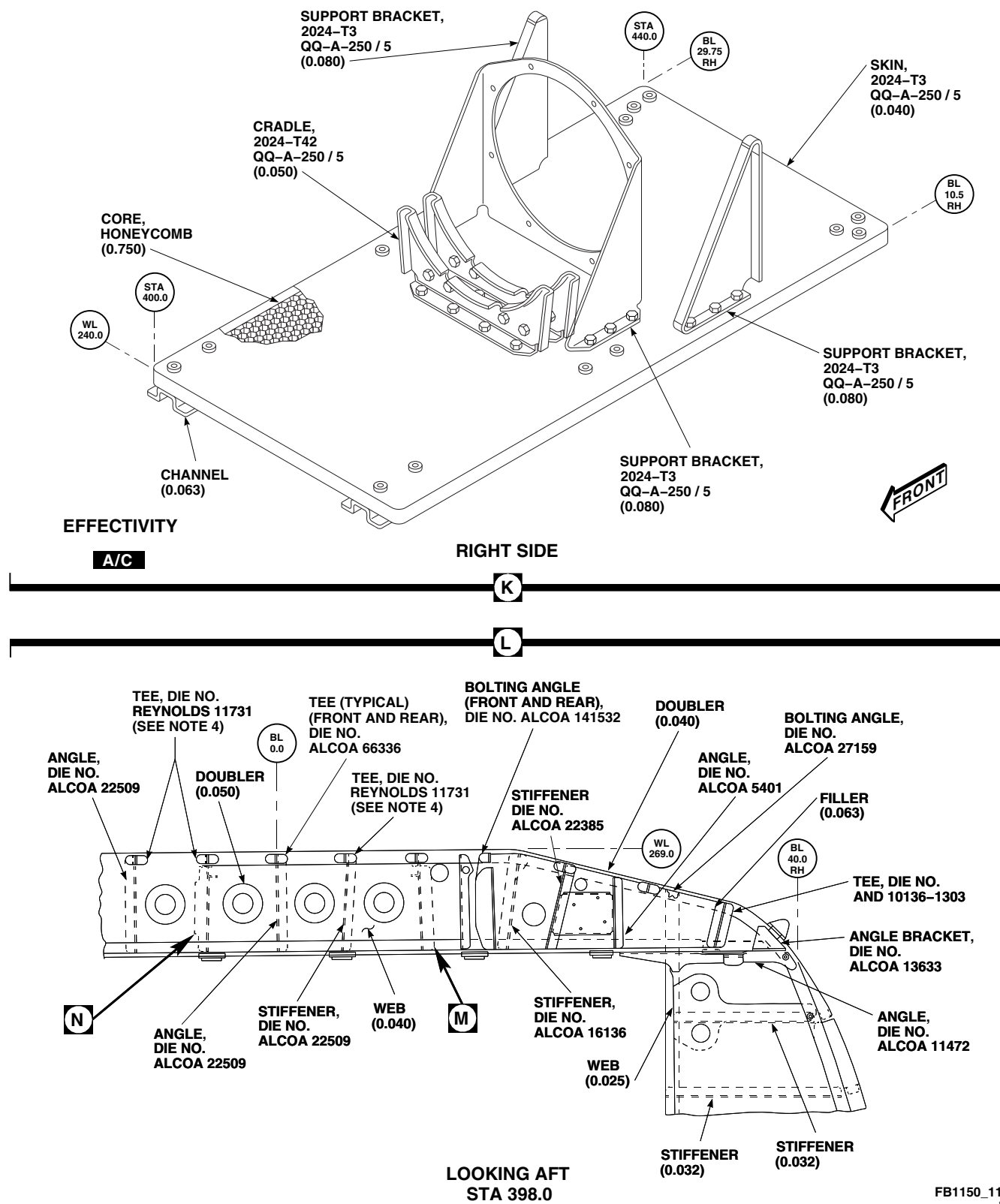


FIGURE 2-6-30-1. REPAIR TRANSITION SECTION FRAMING (SHEET 11 OF 25)

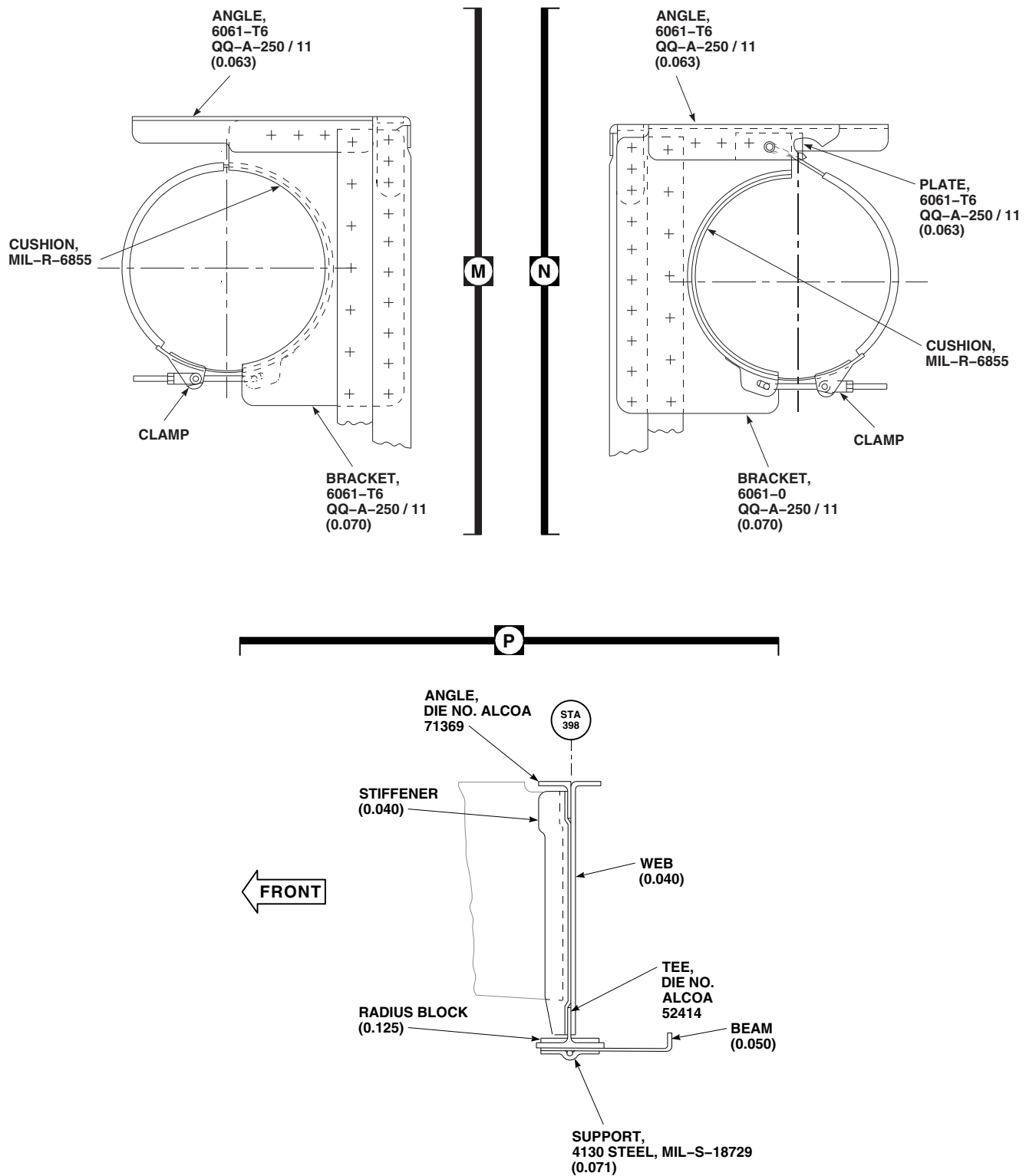
FB1150_12A
SA

FIGURE 2-6-30-1. REPAIR TRANSITION SECTION FRAMING (SHEET 12 OF 25)

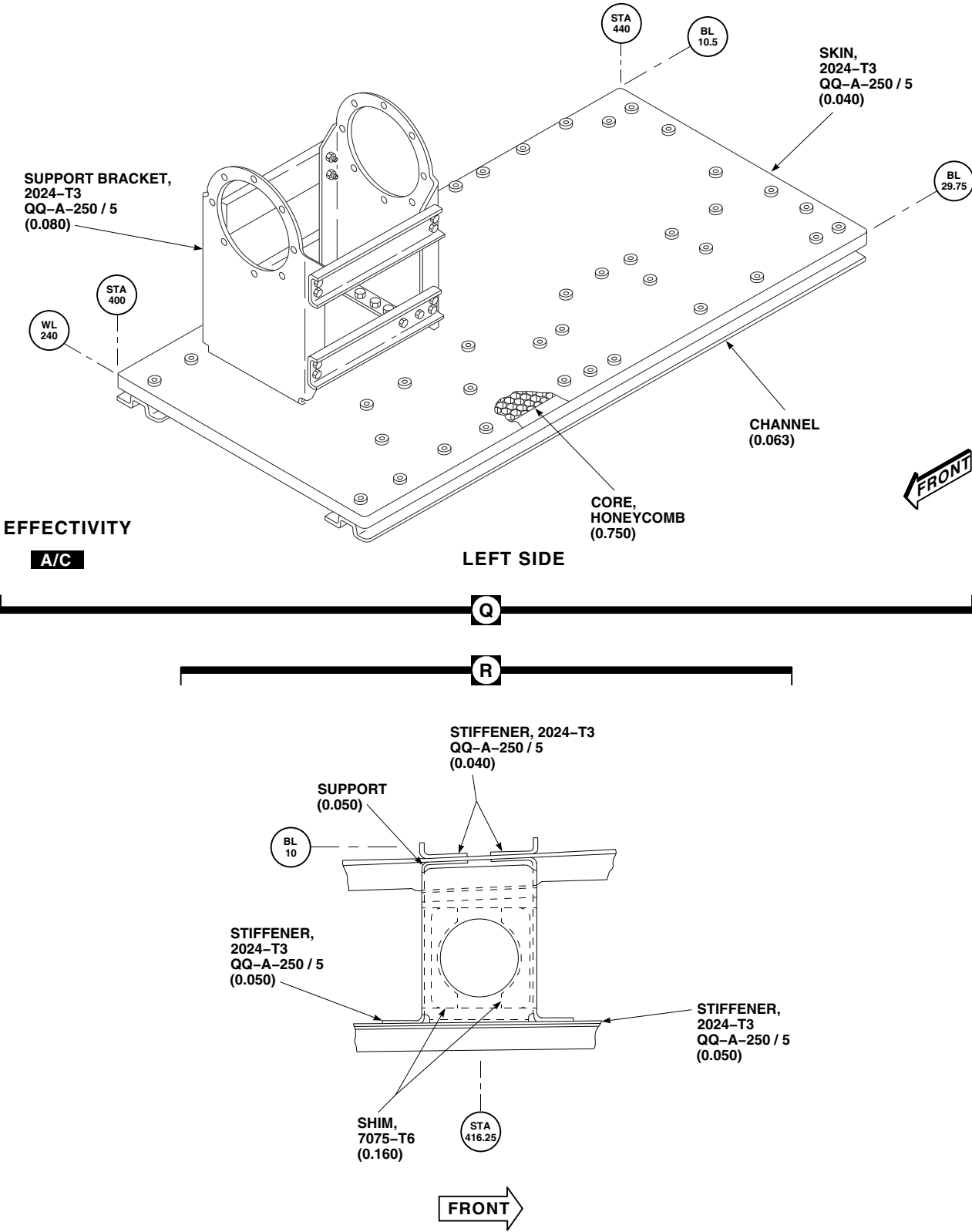
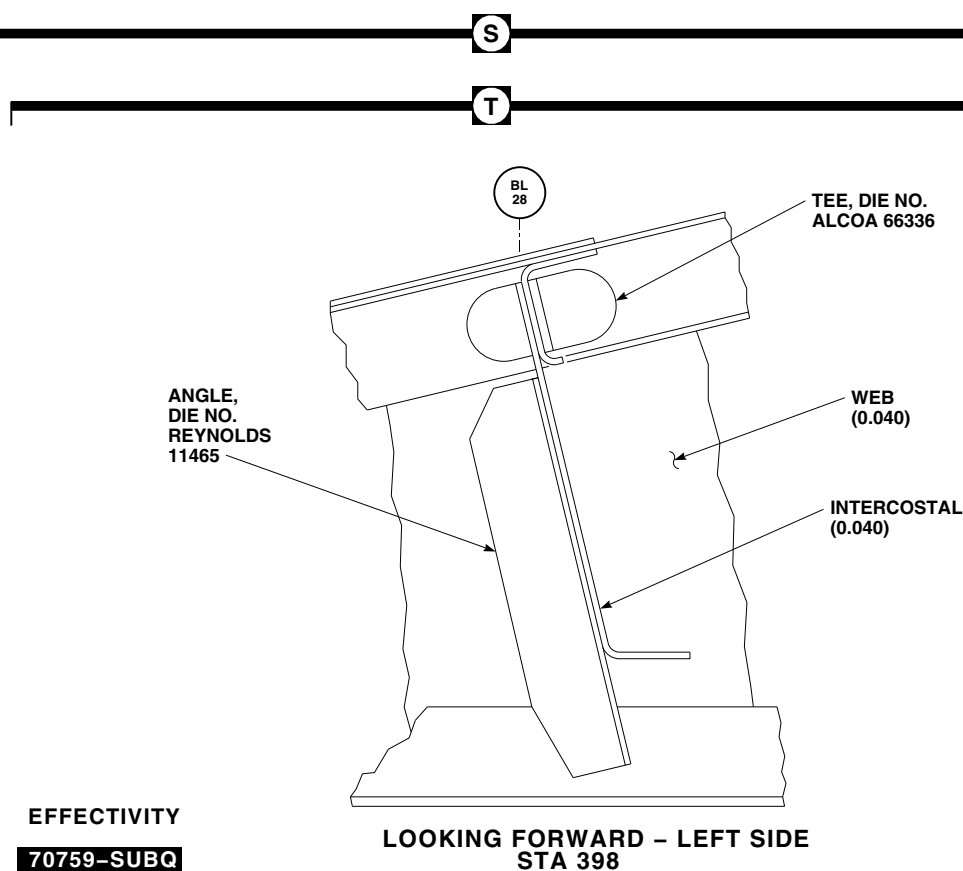
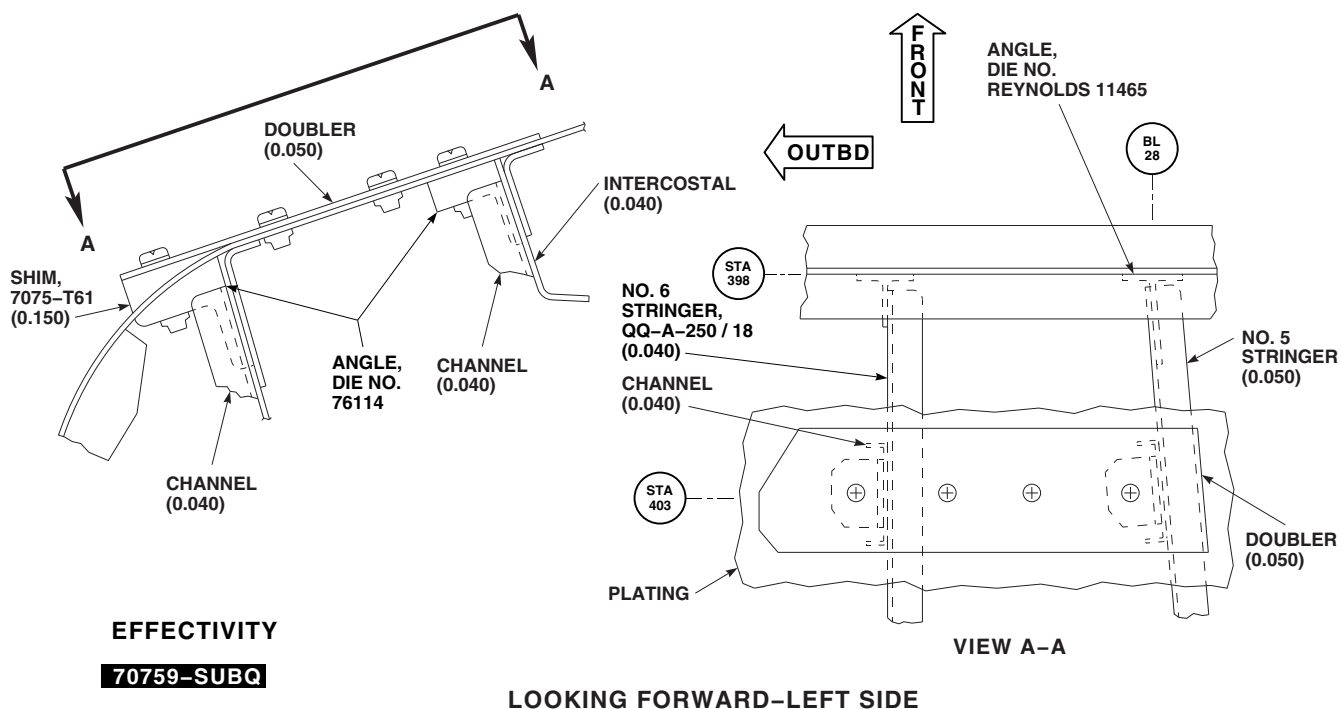


FIGURE 2-6-30-1. REPAIR TRANSITION SECTION FRAMING (SHEET 13 OF 25)



FB1150_14A
SA

FIGURE 2-6-30-1. REPAIR TRANSITION SECTION FRAMING (SHEET 14 OF 25)

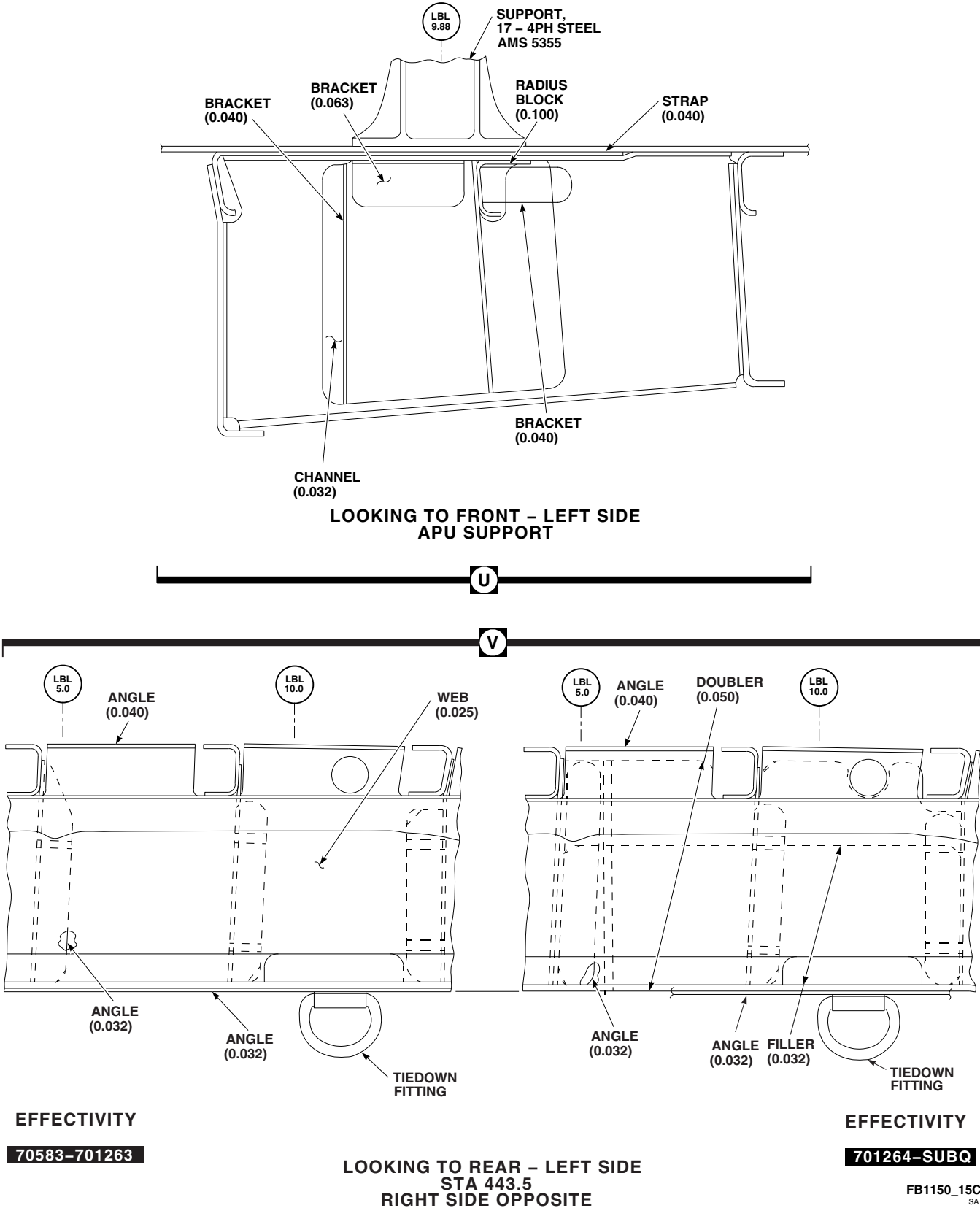


FIGURE 2-6-30-1. REPAIR TRANSITION SECTION FRAMING (SHEET 15 OF 25)

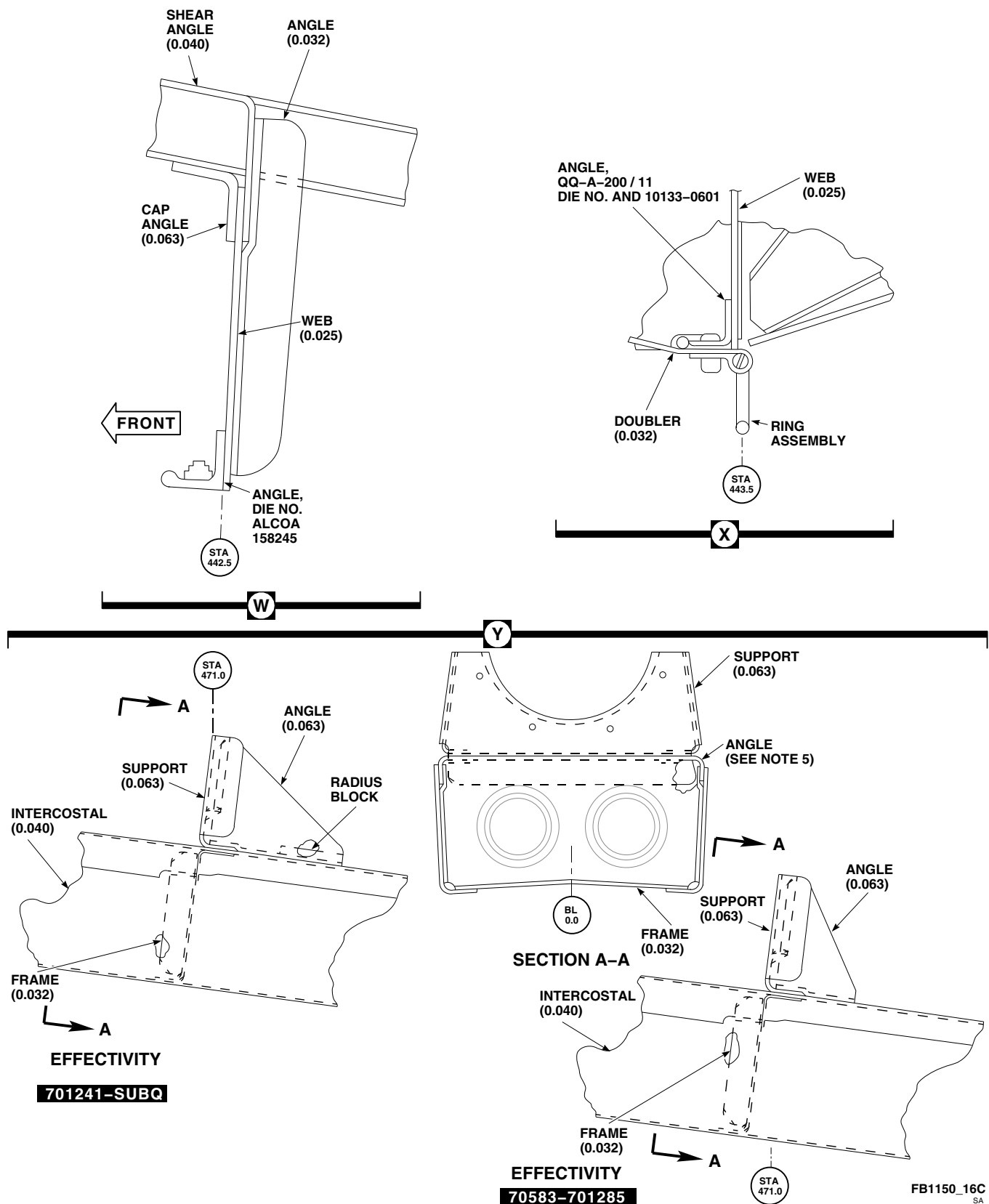


FIGURE 2-6-30-1. REPAIR TRANSITION SECTION FRAMING (SHEET 16 OF 25)

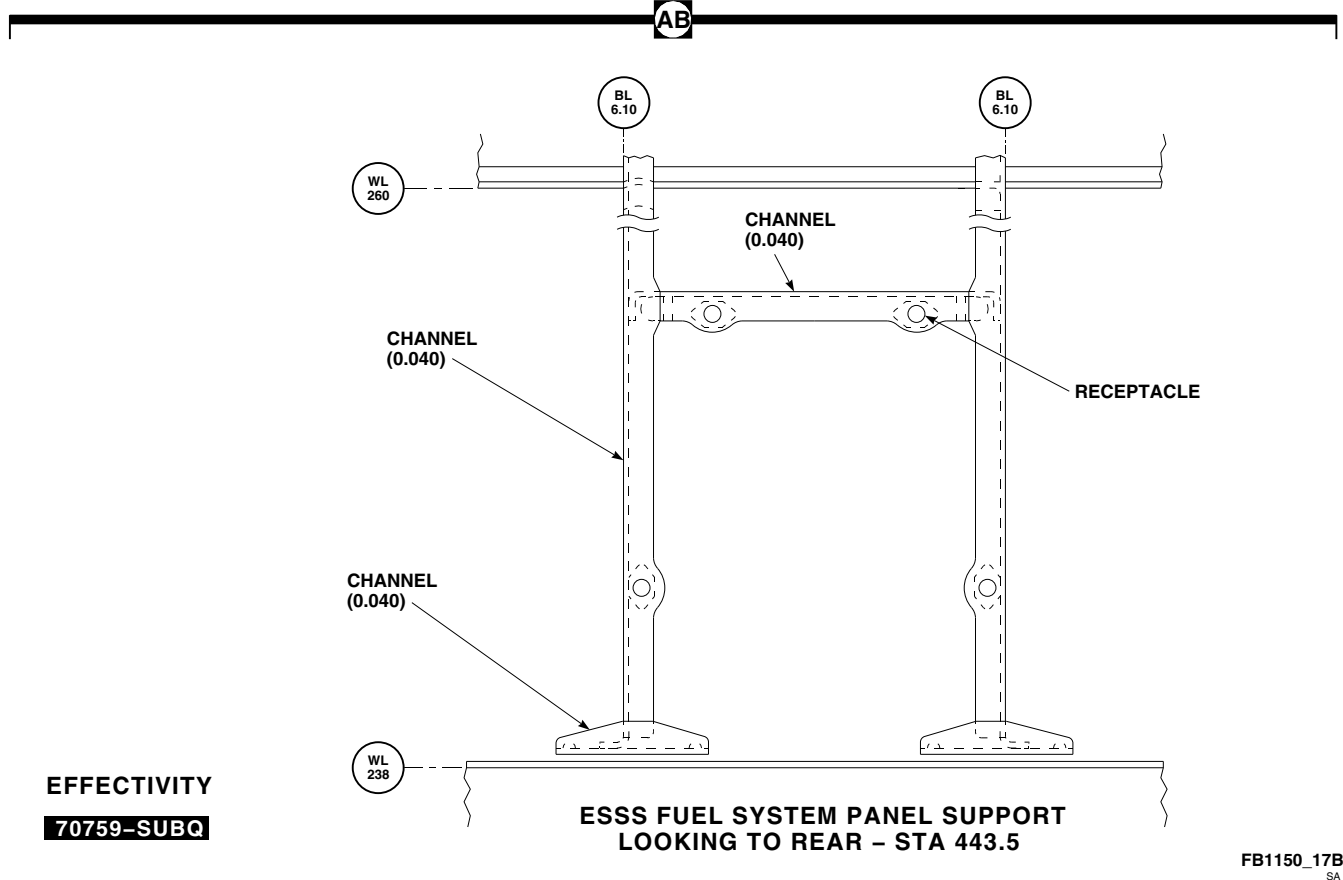
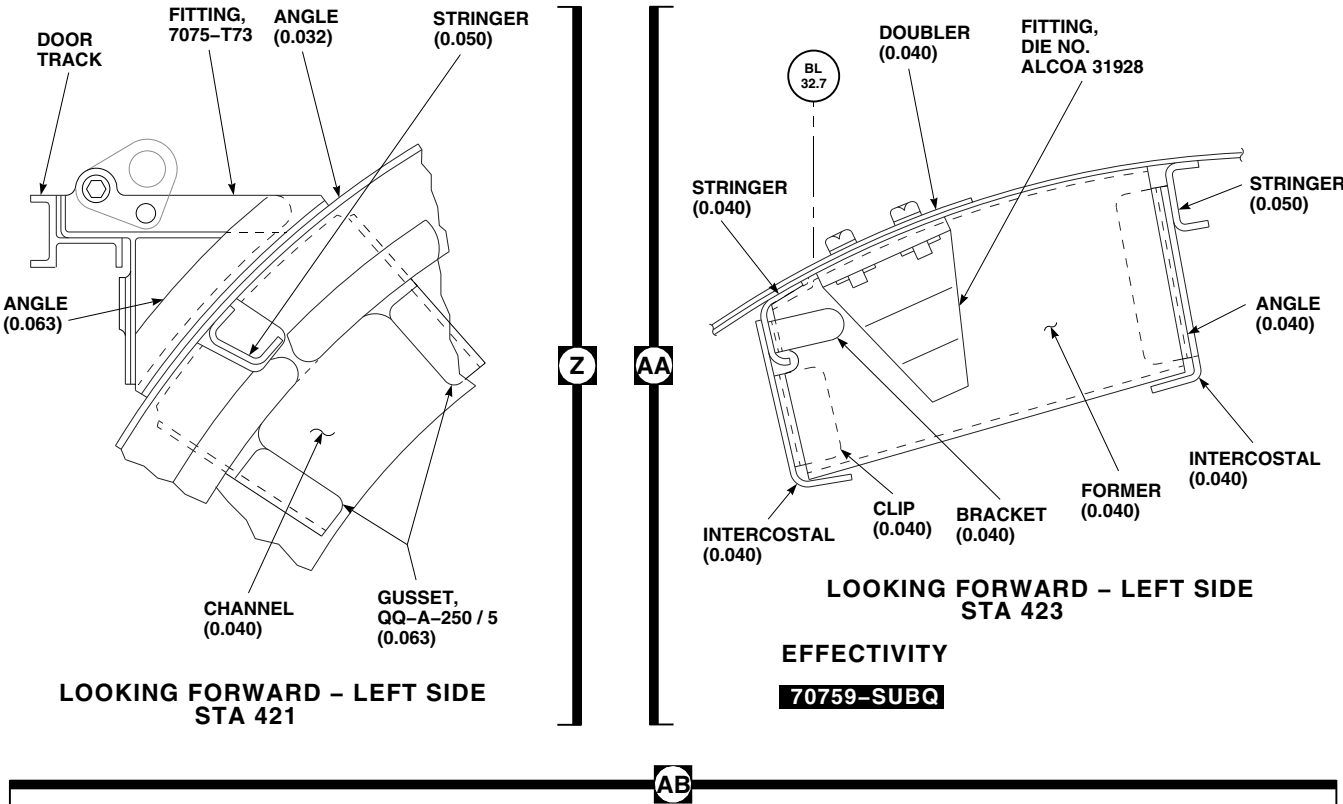


FIGURE 2-6-30-1. REPAIR TRANSITION SECTION FRAMING (SHEET 17 OF 25)

FB1150_17B
SA

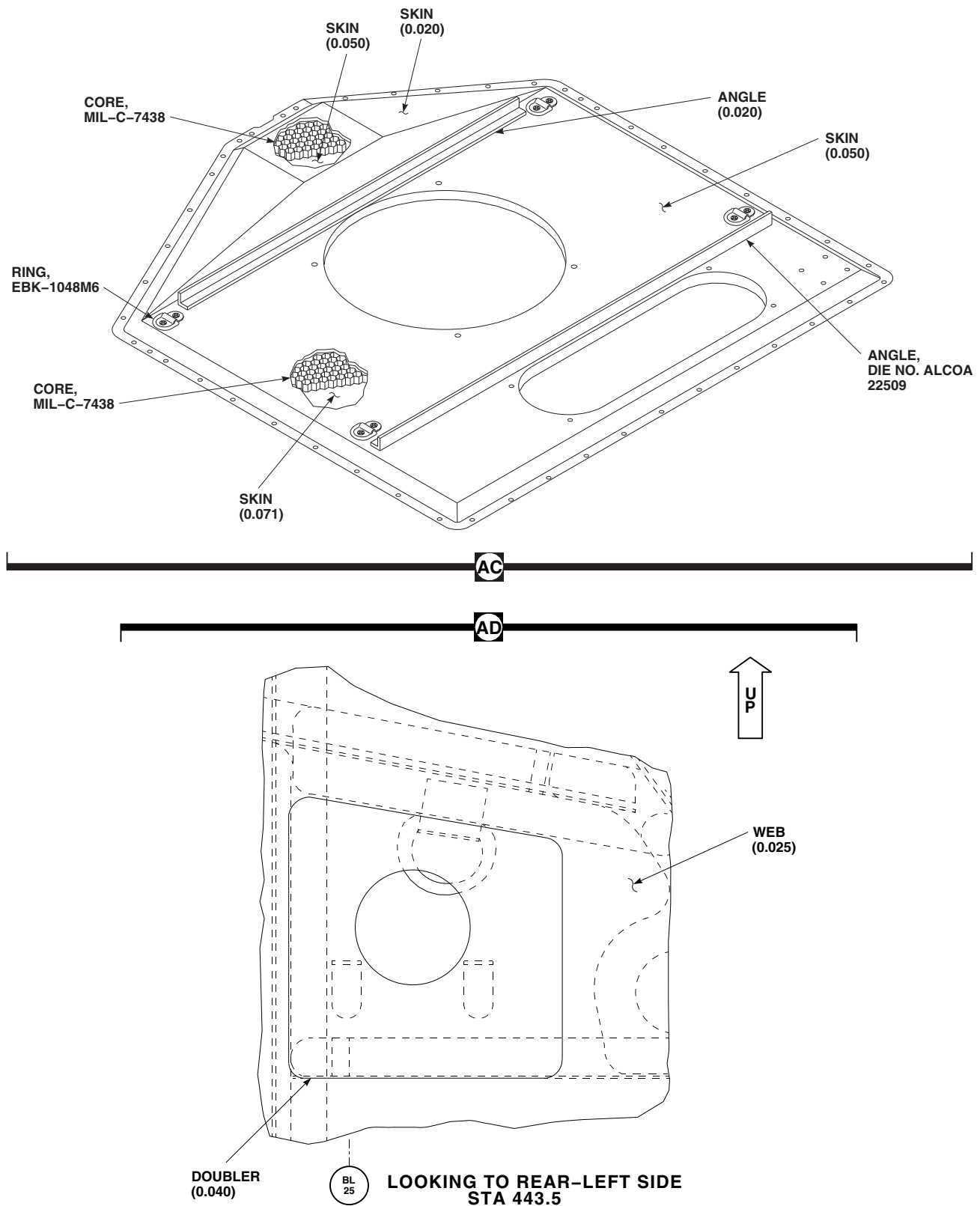
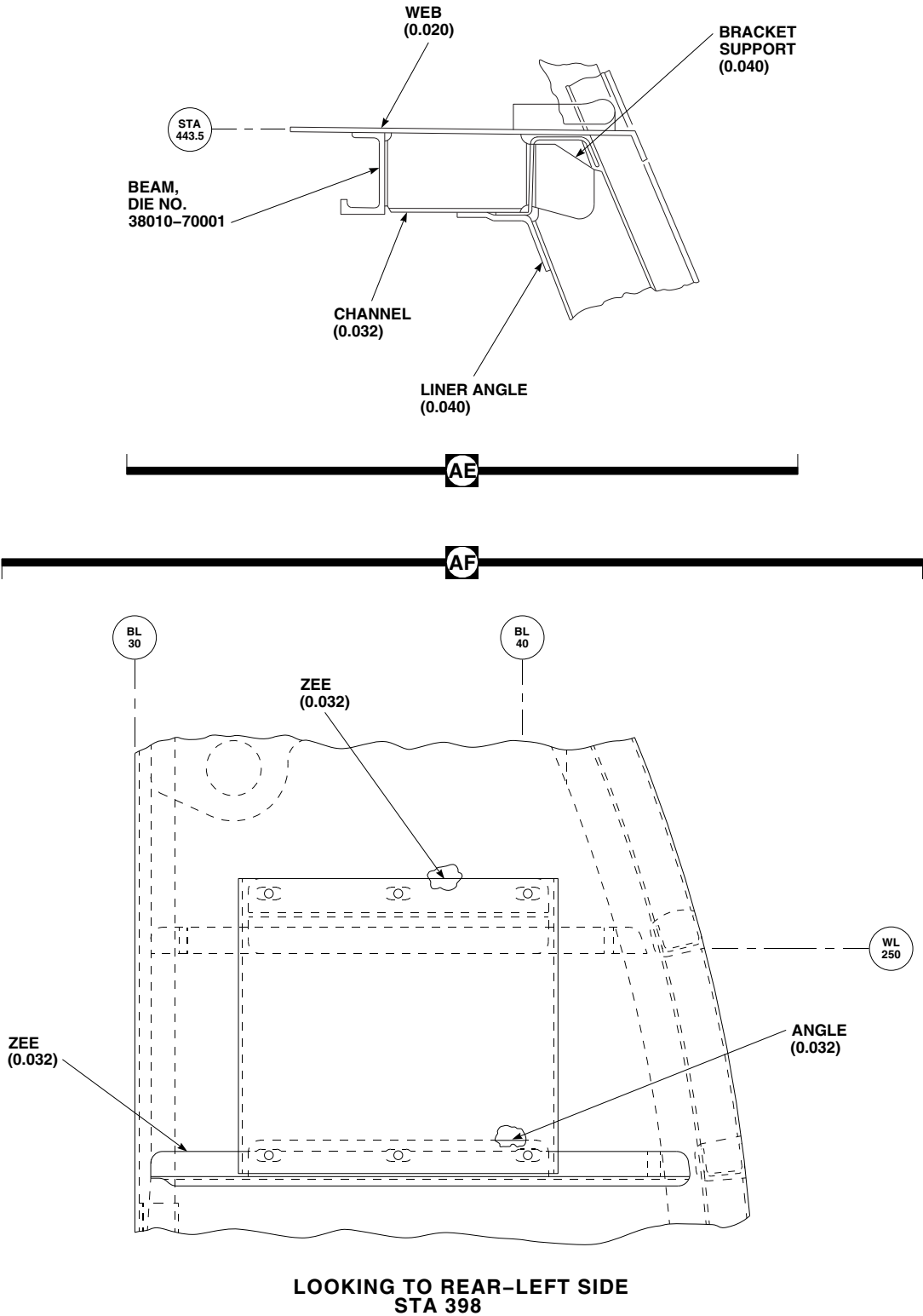


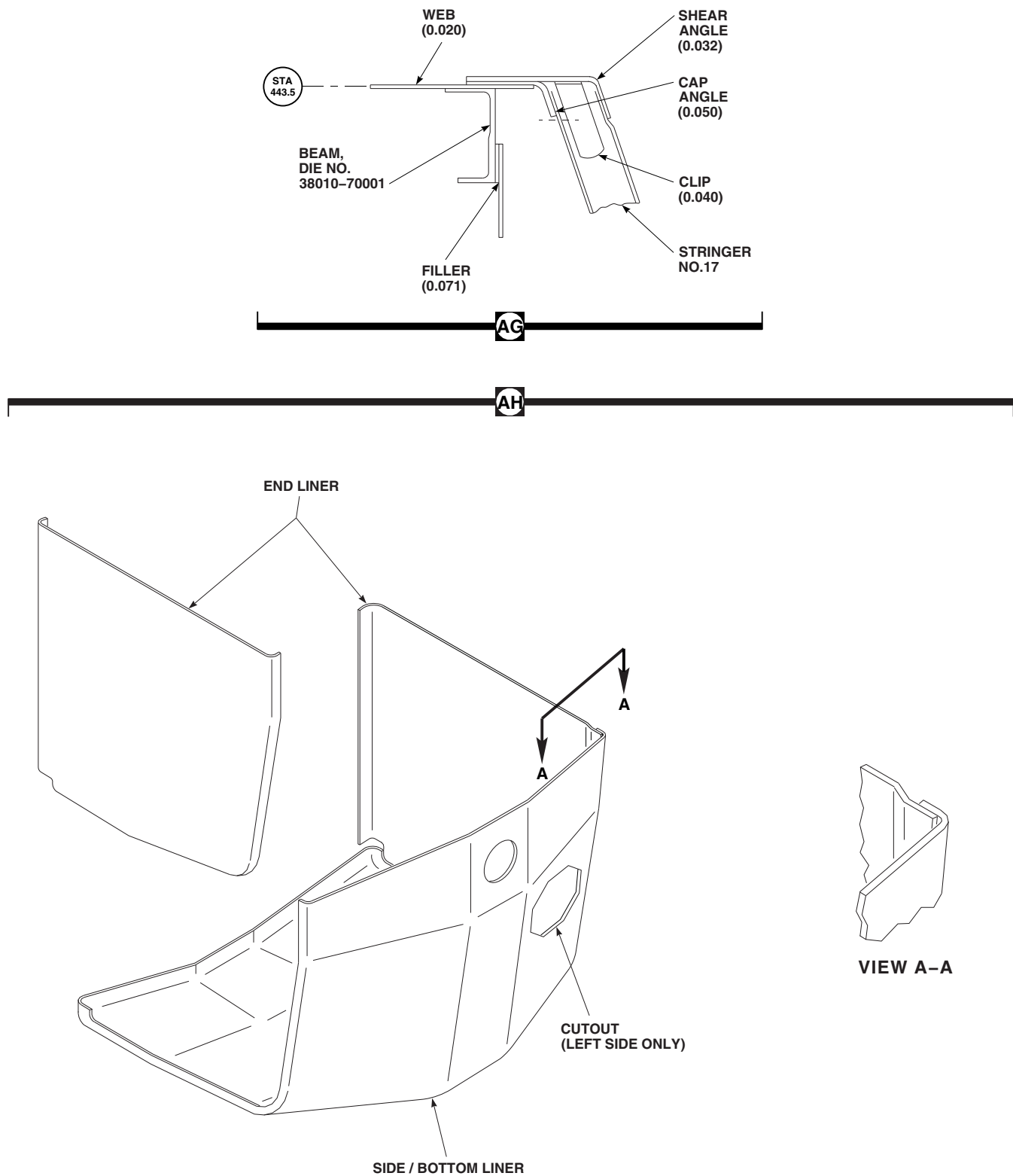
FIGURE 2-6-30-1. REPAIR TRANSITION SECTION FRAMING (SHEET 18 OF 25)

FB1150_18A
SA



FB1150_19A
SA

FIGURE 2-6-30-1. REPAIR TRANSITION SECTION FRAMING (SHEET 19 OF 25)



RIGHT SIDE OPPOSITE, EXCEPT AS SHOWN

FB1150_20A
SA

FIGURE 2-6-30-1. REPAIR TRANSITION SECTION FRAMING (SHEET 20 OF 25)

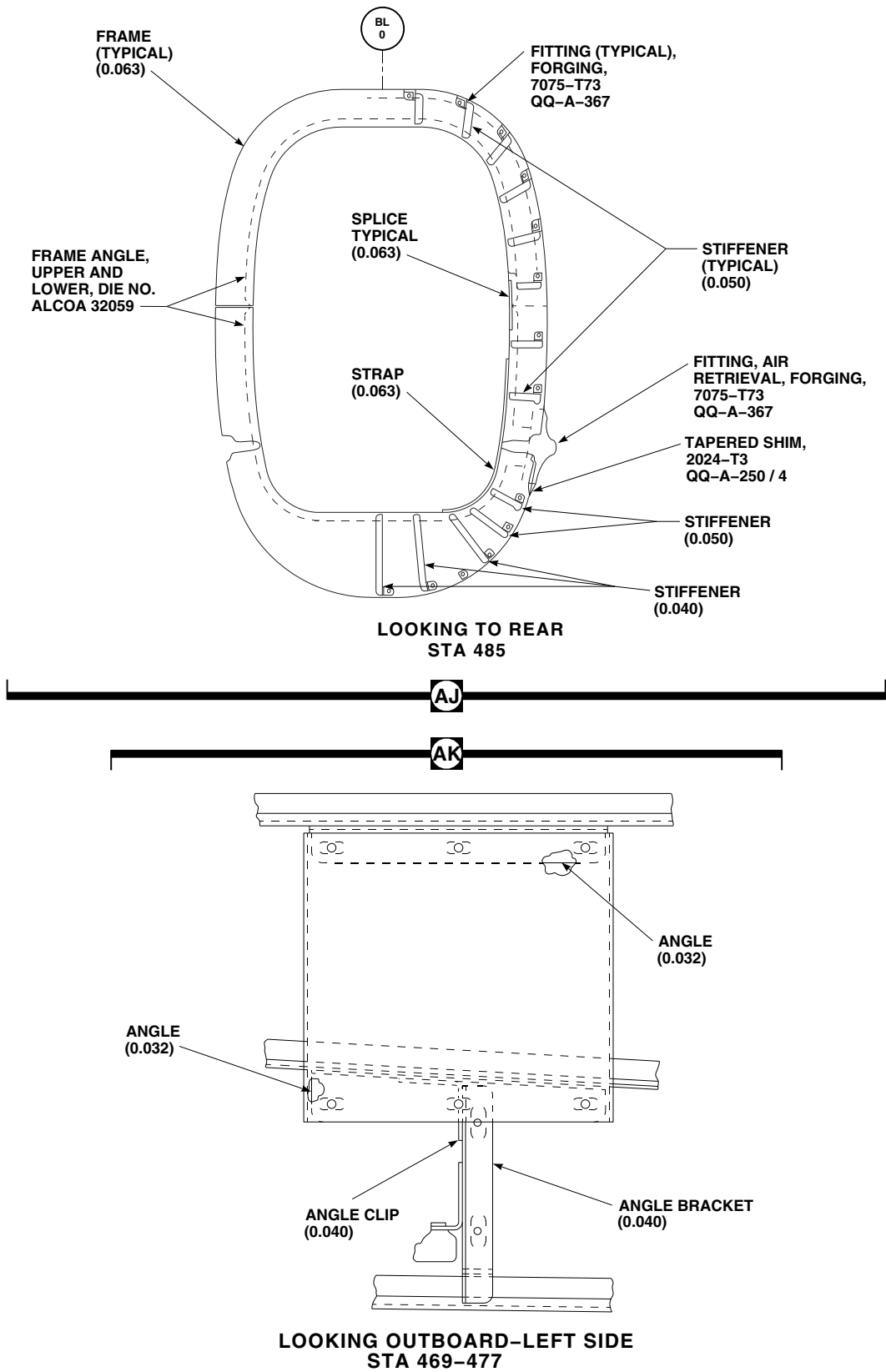
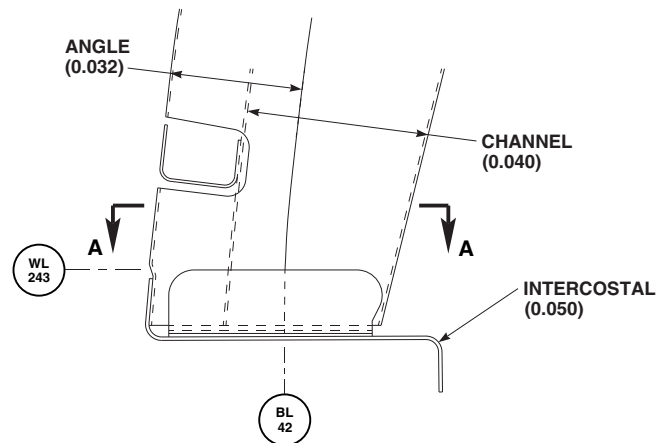


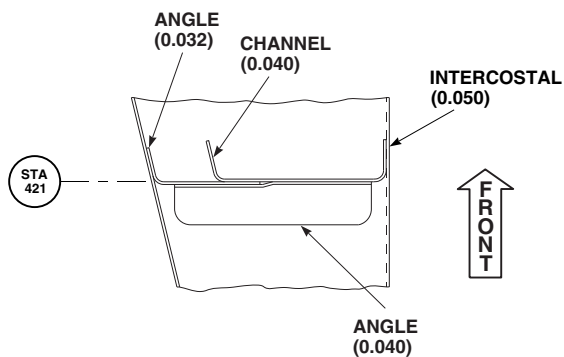
FIGURE 2-6-30-1. REPAIR TRANSITION SECTION FRAMING (SHEET 21 OF 25)

FB1150_21A
SA

AL



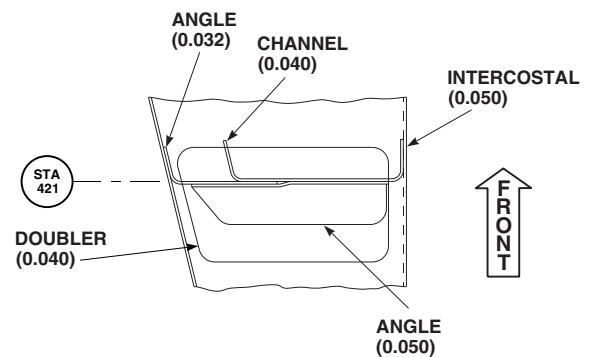
LOOKING FORWARD-LEFT SIDE
STA 421



SECTION A-A

EFFECTIVITY

70583-701285



SECTION A-A

EFFECTIVITY

701438-SUBQ

FB1150_22
SA

FIGURE 2-6-30-1. REPAIR TRANSITION SECTION FRAMING (SHEET 22 OF 25)

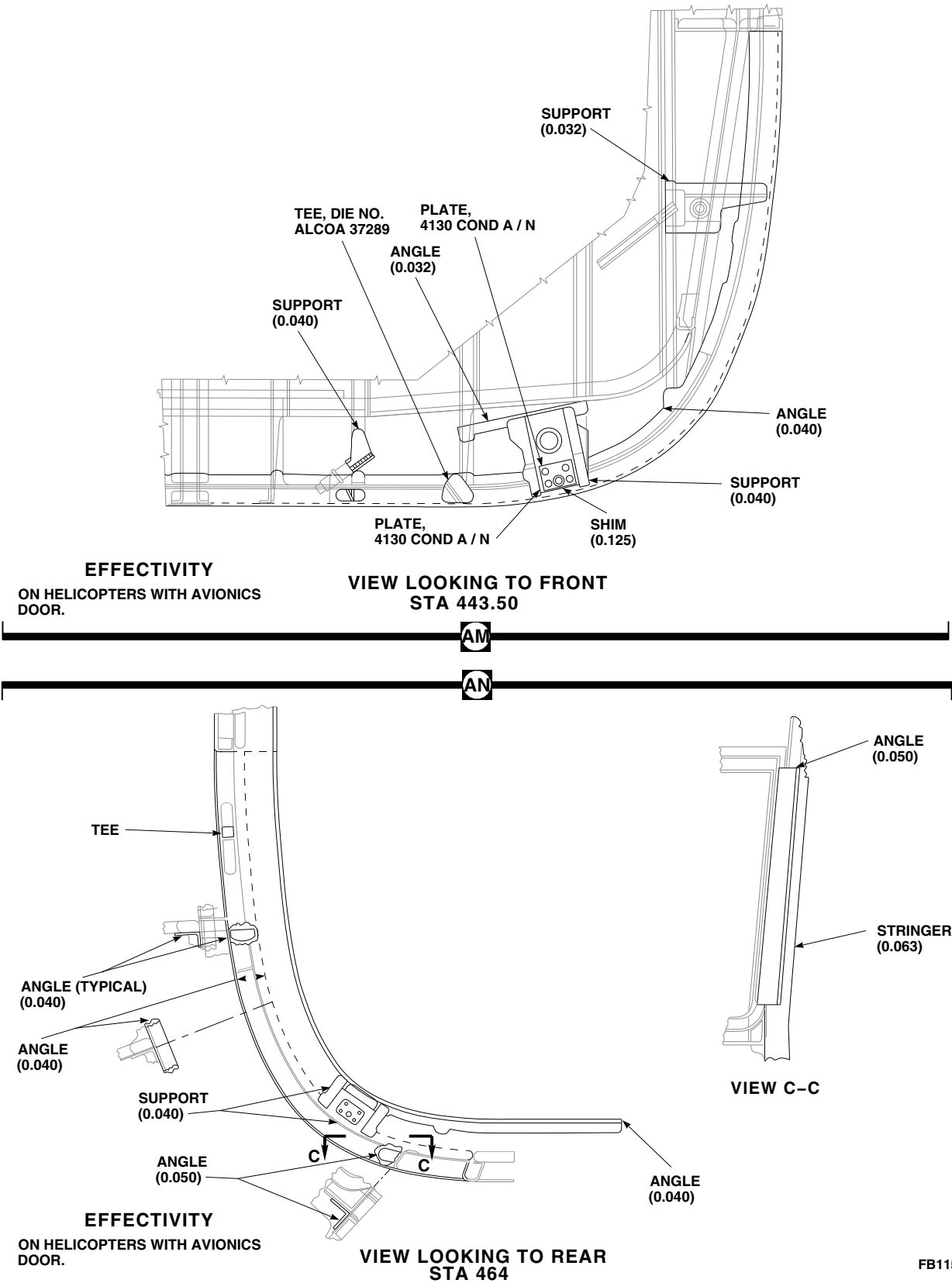
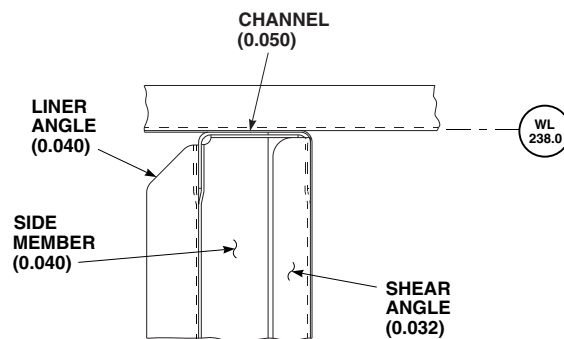
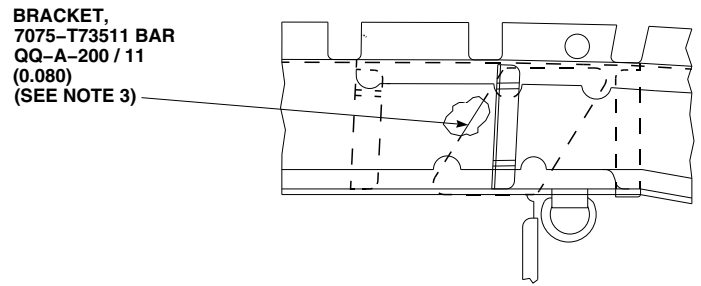


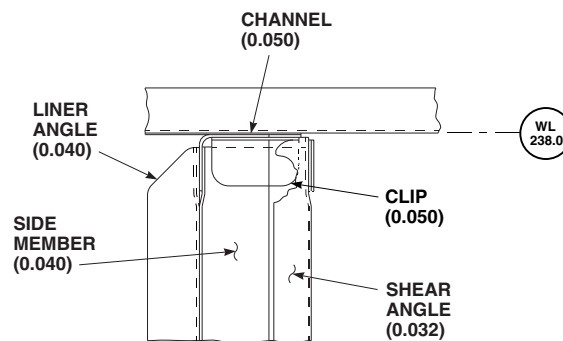
FIGURE 2-6-30-1. REPAIR TRANSITION SECTION FRAMING (SHEET 23 OF 25)



LOOKING INBOARD

EFFECTIVITY

70583-701473



LOOKING INBOARD

EFFECTIVITY

701454-SUBQ

FB1150_24B
SA

FIGURE 2-6-30-1. REPAIR TRANSITION SECTION FRAMING (SHEET 24 OF 25)

NOTES

1. UNLESS OTHERWISE NOTED, ALL SHEET METAL PARTS SHOWN ARE ALCLAD 7075-T6, QQ-A-250 / 13, AND ALL EXTRUSIONS ARE ALUMINUM ALLOY 7075-T6511, QQ-A-200 / 11.
2. LEFT AND RIGHT SIDES ARE THE SAME, EXCEPT WHEN NOTED.
3. 70583 - 701251
4. 701334-SUBQ
5. 701401-SUBQ
6. 0.050
7. ALL DIMENSIONS ARE IN INCHES, UNLESS NOTED.

FB1150_25B
SA

FIGURE 2-6-30-1. REPAIR TRANSITION SECTION FRAMING (SHEET 25 OF 25)

2-6-31 TRANSITION SECTION PLATING REPAIR.

PARAGRAPH BREAKDOWN

	PAGE
2-6-31.1 Repair Transition Section Plating	2-6-31-1
INITIAL SETUP	MIL-C-7637
Tools	MIL-P-15035
Airframe Repairer’s Toolkit	MIL-S-18729
Materials/Parts	MIL-S-5059
Abrasive Cloth, Item 1A, Appendix D	MIL-STD-670
Adhesive, Item 32, Appendix D	MIL-T-9046
Adhesive Primer, Item 68, Appendix D	MIL-W-6858
Sealing Compound, Item 275, Appendix D	PARA 2-6-1
Sealing Compound, Item 292, Appendix D	PARA 2-6-2
References	PARA 2-6-3
AMS 4036	PARA 2-6-5
AMS 4260	QQ-A-200
AMS 5334	QQ-A-230
AMS-T-9046	QQ-A-250
Appendix D	QQ-A-362

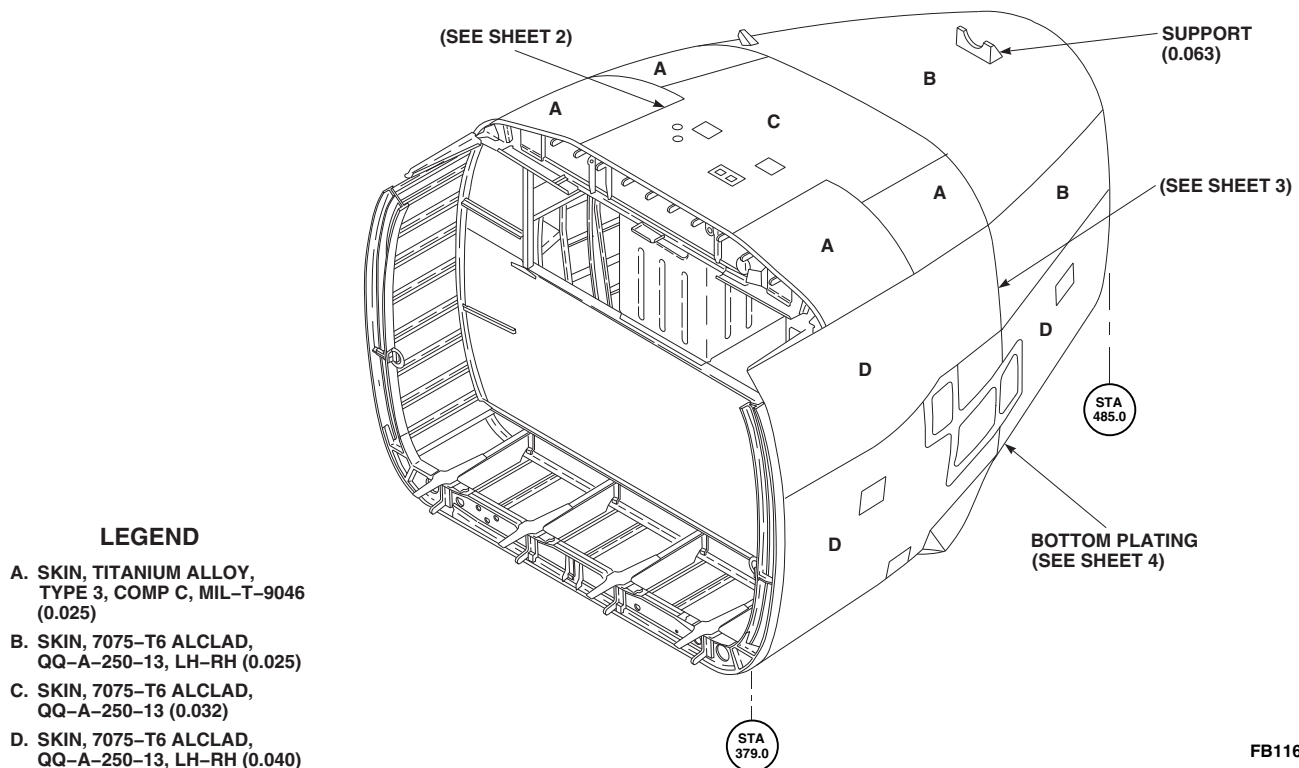
- 2-6-31.1 Repair Transition Section Plating.

a.

Transition section plating is Alclad aluminum 7075-T6, QQ-A-250/13, except in high temperature areas where titanium alloy, MIL-T-9046, is used (Figure 2-6-31-1, Sheet 1 through Sheet 9).
- b.

All plating, with the exception of reinforced plastic members, is primary structure. Evaluate damage (PARA 2-6-1).
- c.

Refer to PARA 2-6-2 for typical plating repairs. When repairing titanium plating, use same gage titanium stock, or the equivalent

FB1162_1
SA**FIGURE 2-6-31-1. REPAIR TRANSITION SECTION PLATING (SHEET 1 OF 9)**

gage 301/302 stainless steel, MIL-S-5059. Use Monel or stainless steel fasteners for attaching titanium or stainless steel patches.

- d. The avionics compartment door is a bonded sandwich-construction assembly made from aluminum alloy honeycomb core and Kevlar (reinforced plastic) skins. Metal components include hinges, straps, latches, and strut fitting.
- e. Evaluate damage in noncritical areas (PARA 2-6-1).
- f. Evaluate damage to latch assemblies and strut fitting (PARA 2-6-1).
- g. Negligible damage to hinge straps is limited to nicks, scratches and corrosion not over 0.005-

inch if point of damage is within 1-inch of hinge line and 0.010-inch if point of damage is beyond 1-inch of hinge line (Figure 2-6-31-1, Sheet 6, Detail F). Blend and polish damage with fine grit abrasive cloth, Item 1A, Appendix D, to same finish as surrounding area.

- h. Use PARA 2-6-3 for typical reinforced plastic repairs.
- i. Refer to Figure 2-6-31-1, Sheet 3 through Sheet 8 for access door panel repair.
- j. For corrosion protection and watertightness, seal all repairs of the transition section plating (PARA 2-6-1).
- k. Touch up paint as required (PARA 2-6-5).

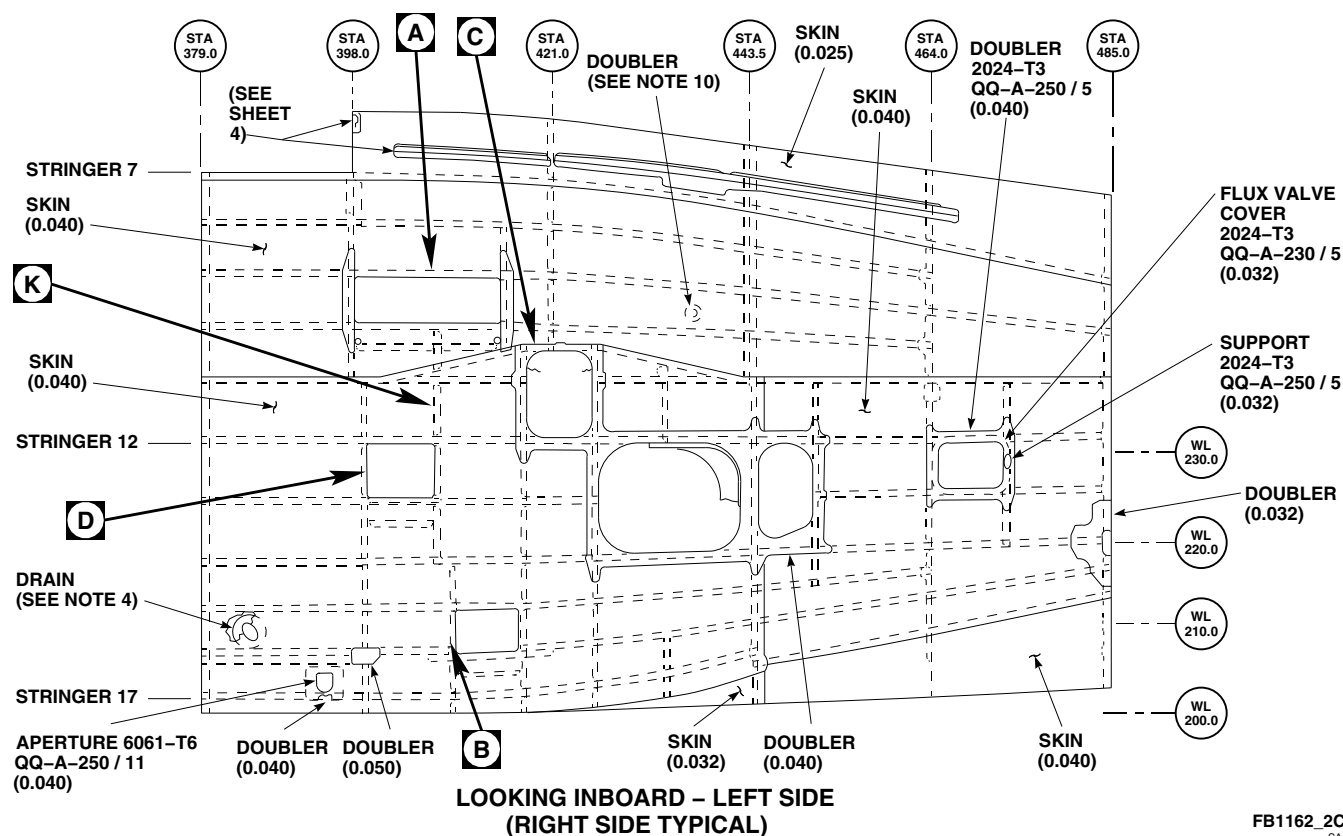
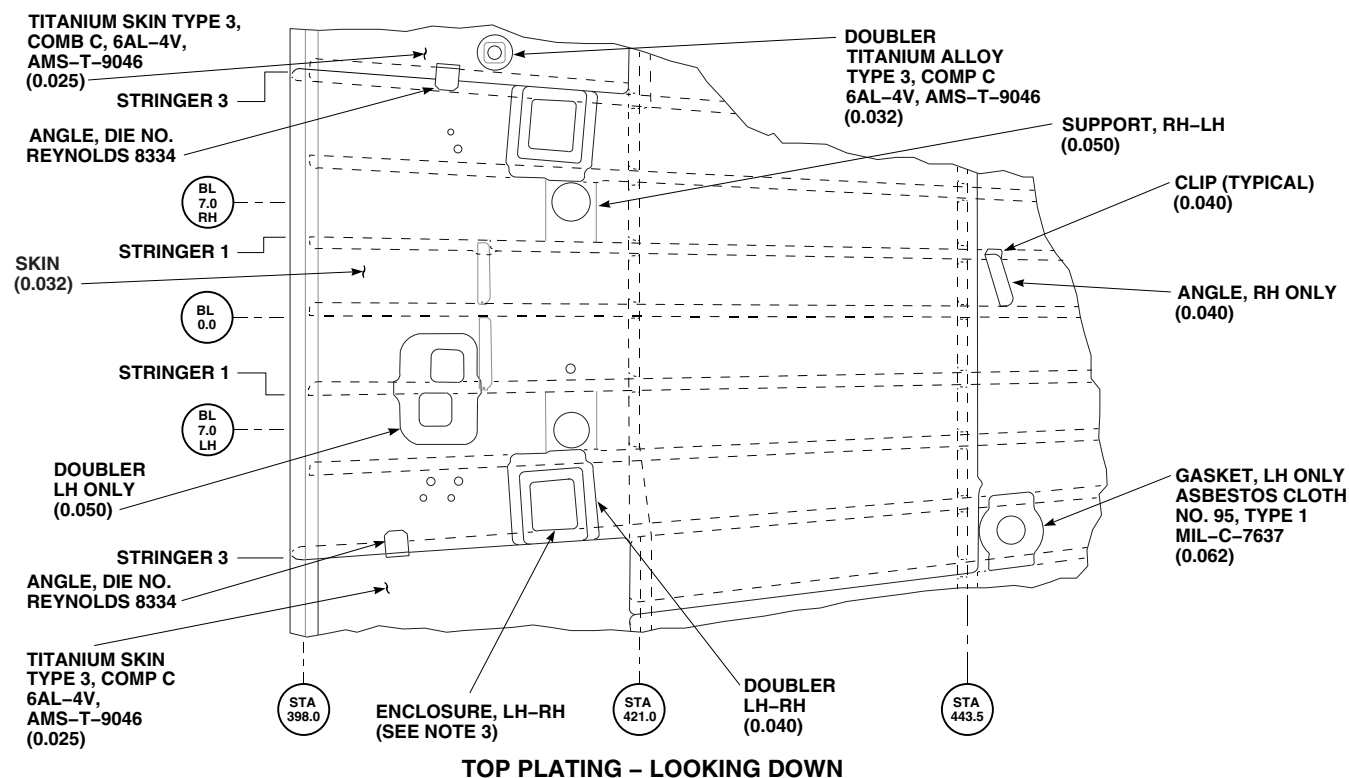
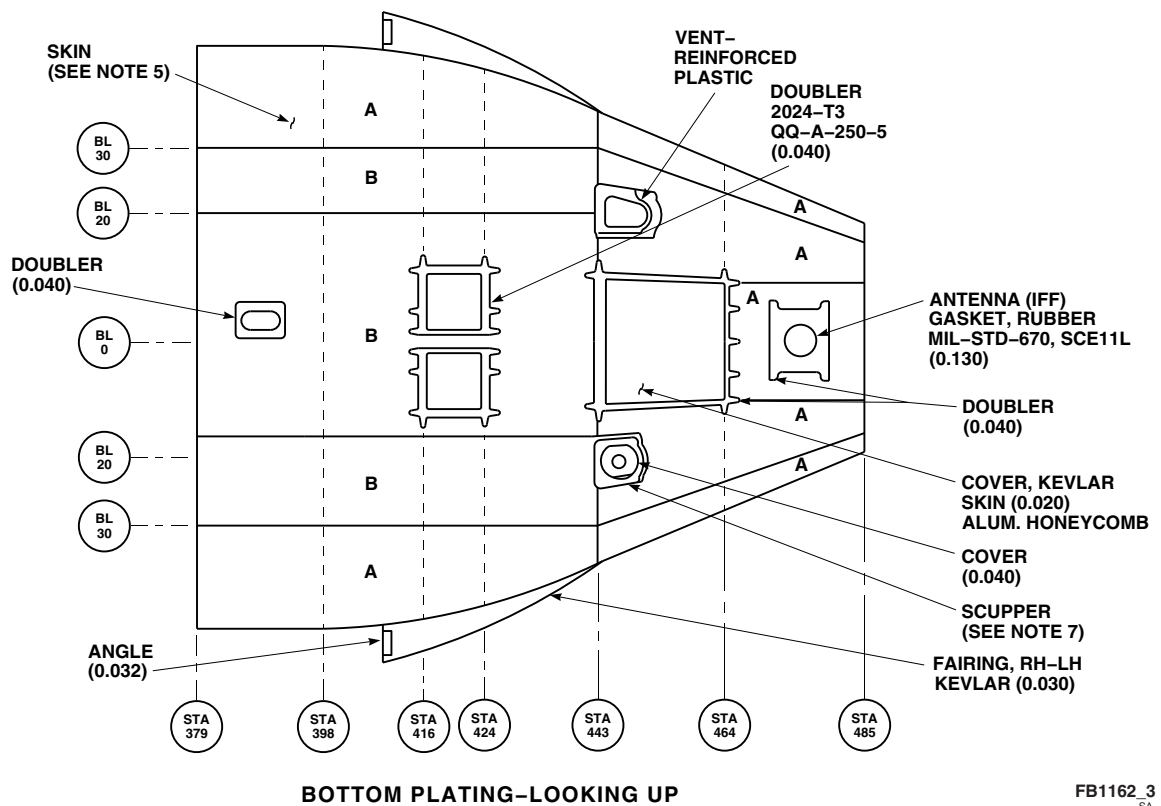
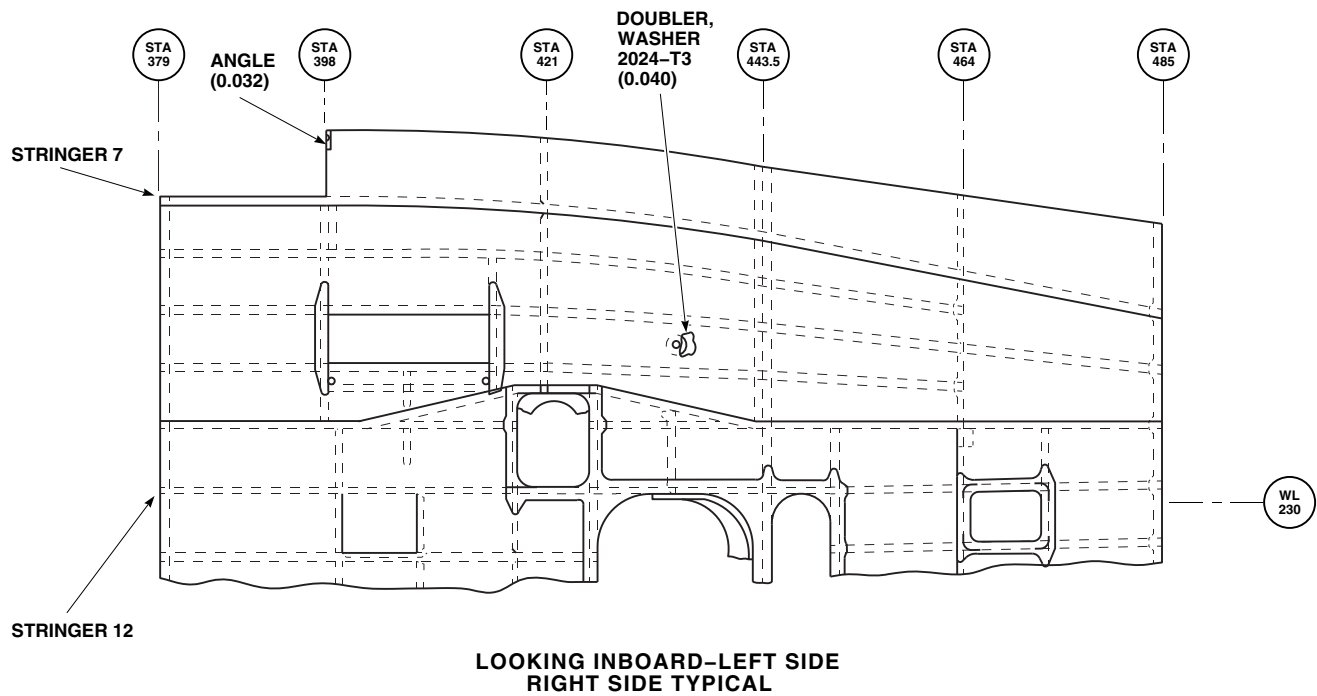


FIGURE 2-6-31-1. REPAIR TRANSITION SECTION PLATING (SHEET 2 OF 9)



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FIGURE 2-6-31-1. REPAIR TRANSITION SECTION PLATING (SHEET 3 OF 9)

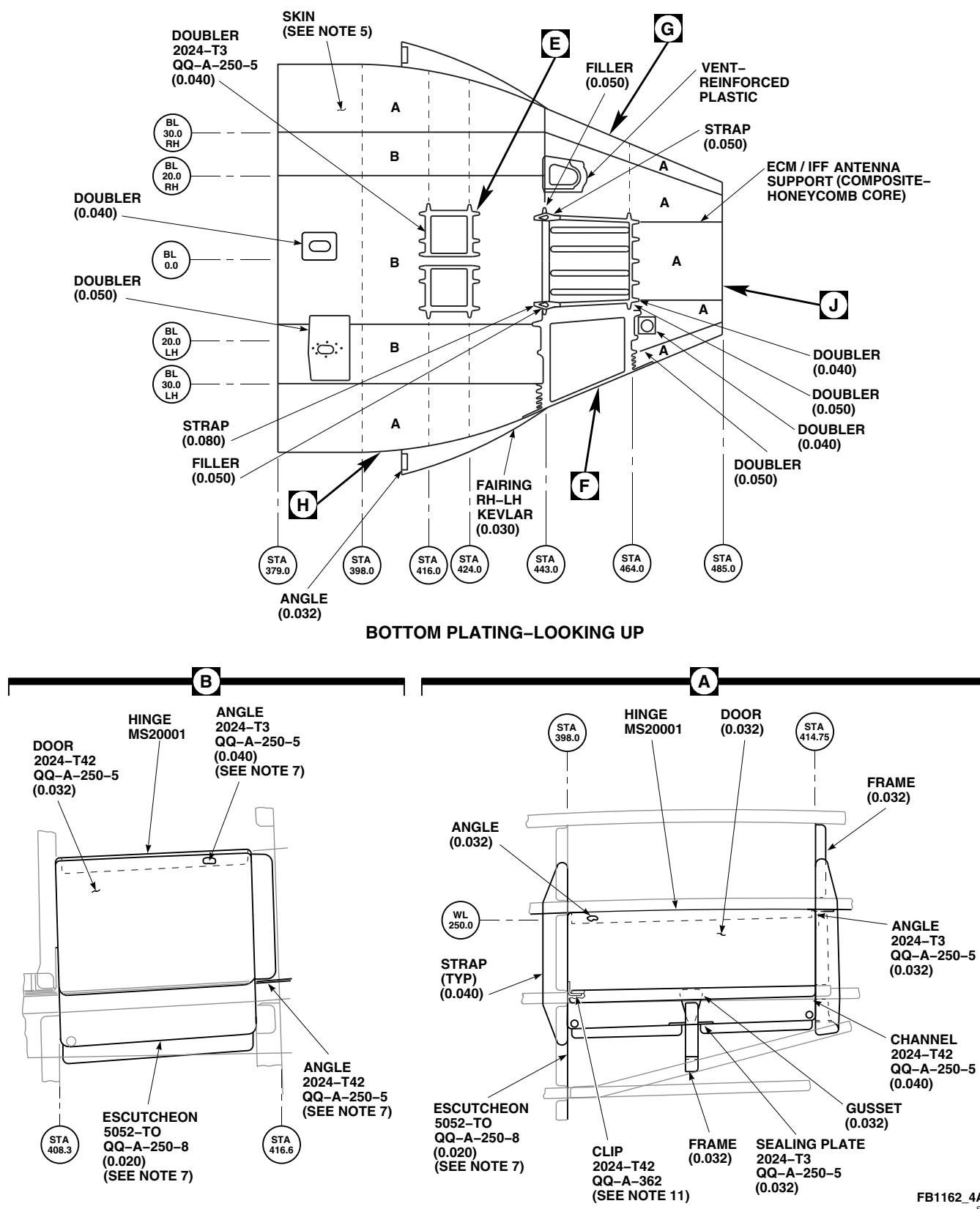


FIGURE 2-6-31-1. REPAIR TRANSITION SECTION PLATING (SHEET 4 OF 9)

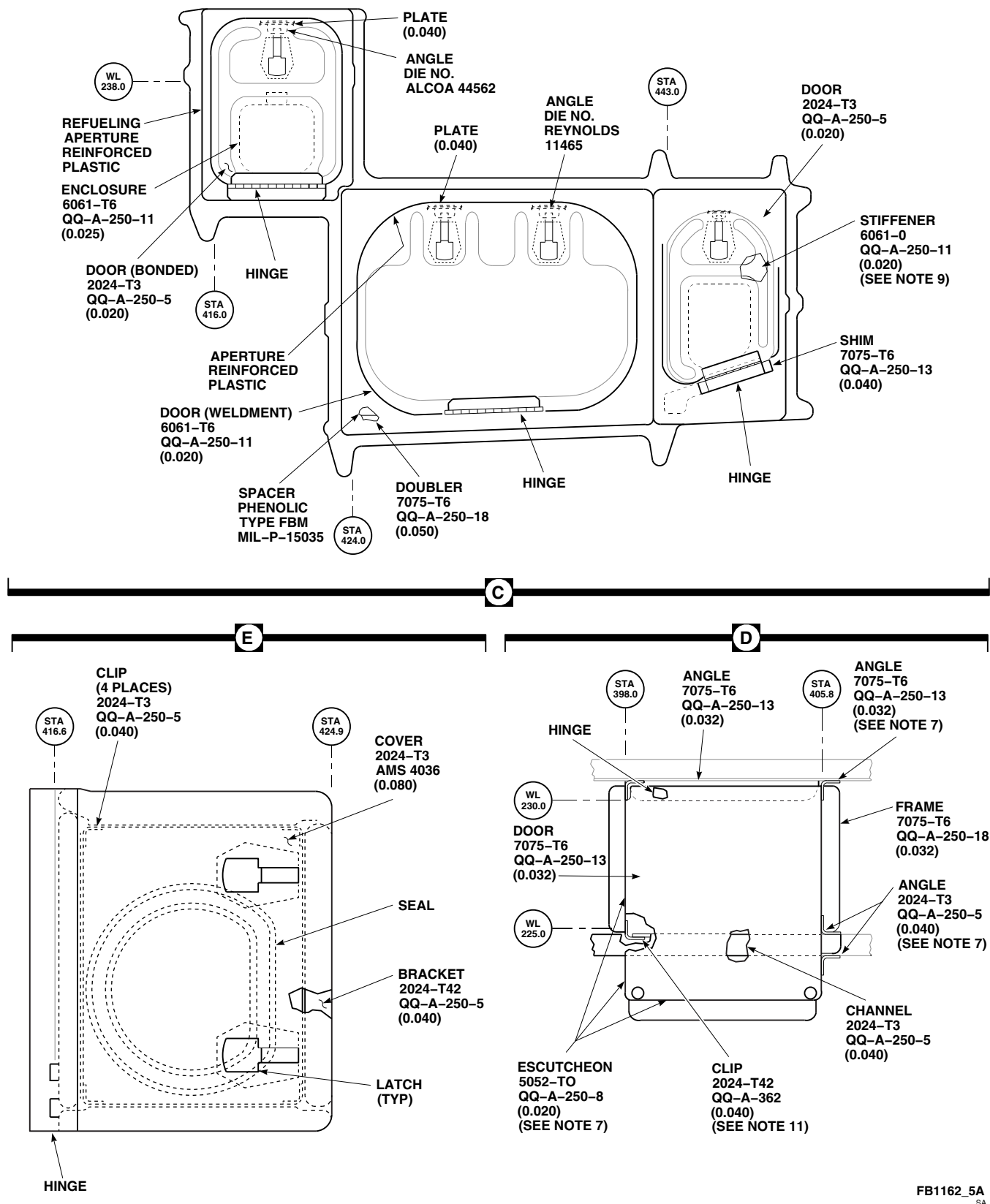


FIGURE 2-6-31-1. REPAIR TRANSITION SECTION PLATING (SHEET 5 OF 9)

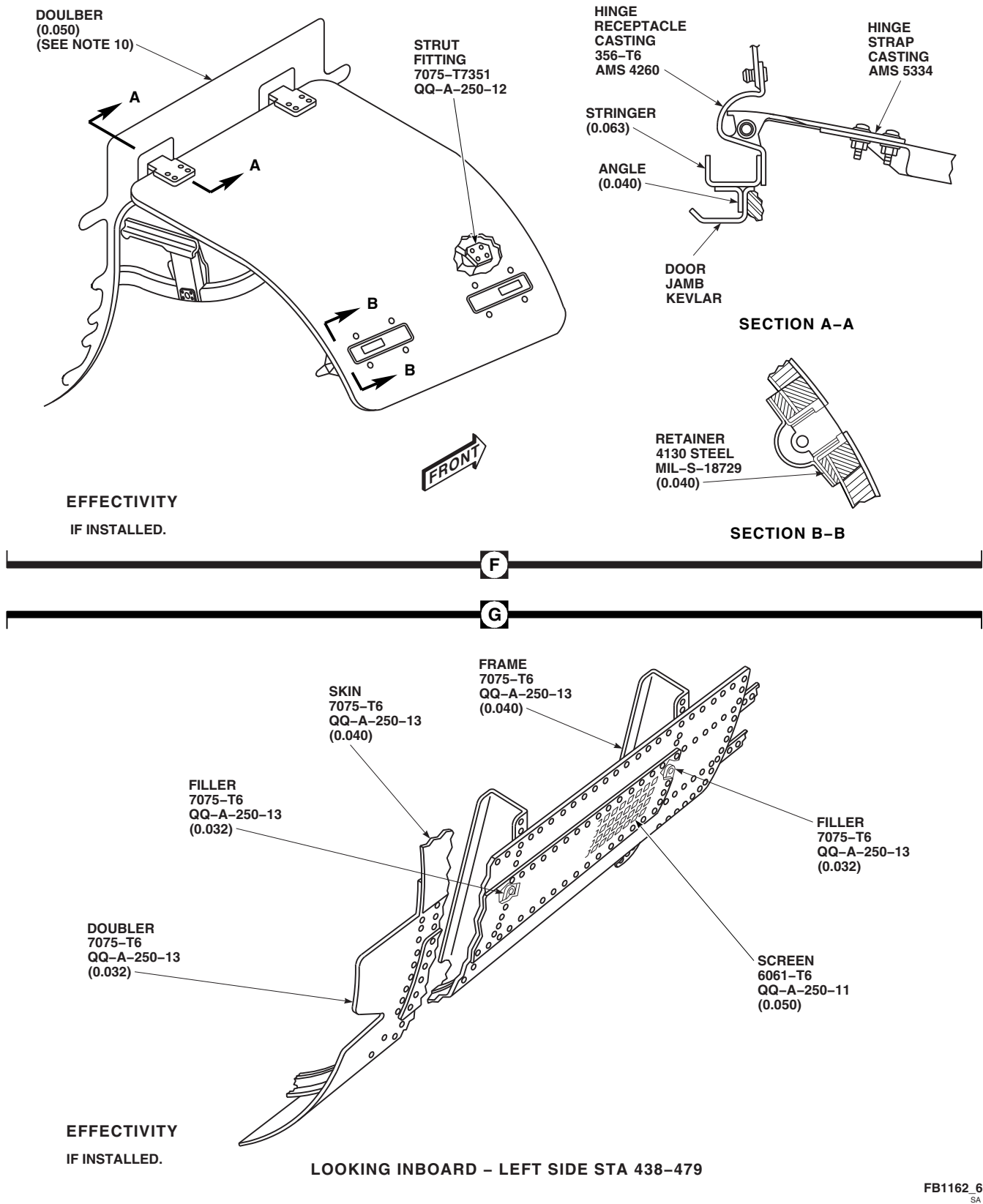


FIGURE 2-6-31-1. REPAIR TRANSITION SECTION PLATING (SHEET 6 OF 9)

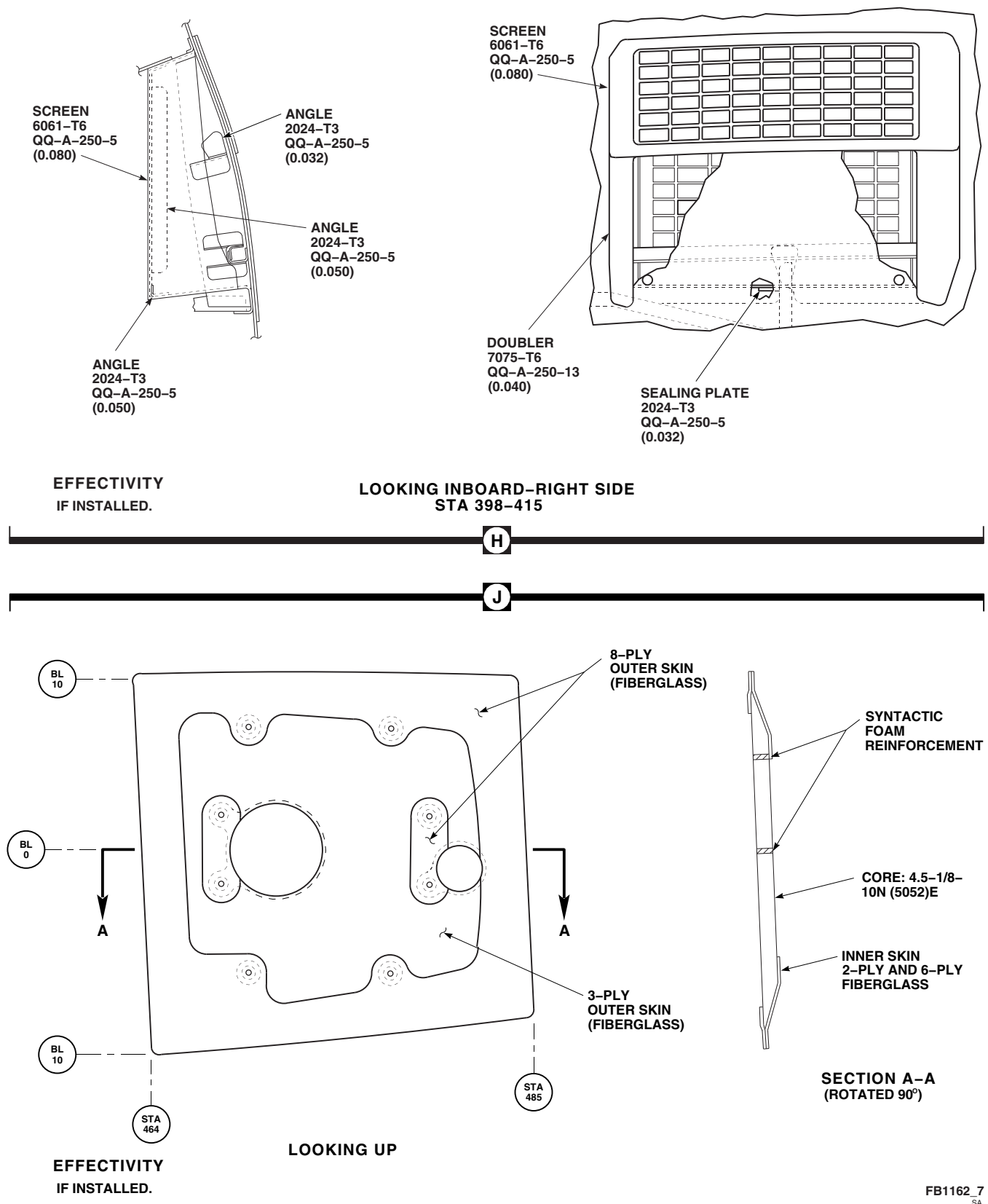


FIGURE 2-6-31-1. REPAIR TRANSITION SECTION PLATING (SHEET 7 OF 9)

FB1162.7
SA

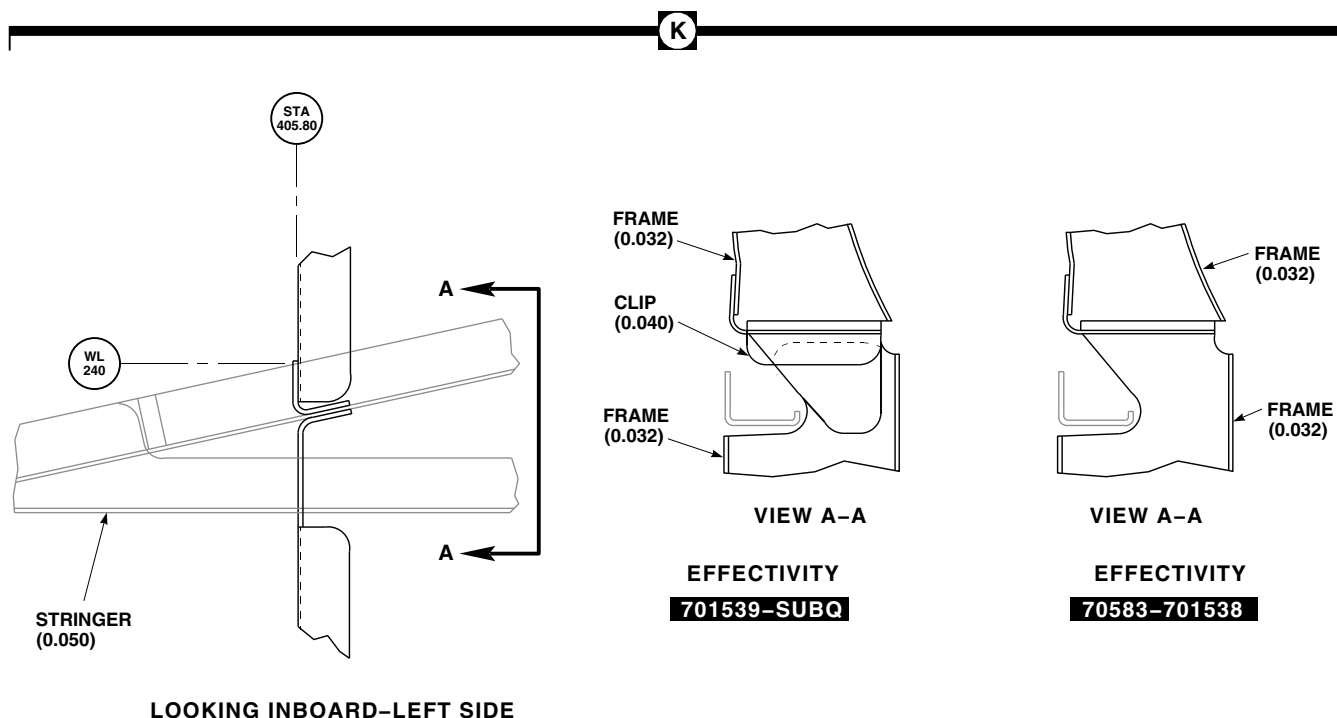


FIGURE 2-6-31-1. REPAIR TRANSITION SECTION PLATING (SHEET 8 OF 9)

FB1162_8
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NOTES

1. UNLESS OTHERWISE NOTED ON SHEETS OF THIS FIGURE, ALL SHEET METAL PARTS SHOWN ARE ALCLAD 7075-T6, QQ-A-250-13, AND ALL EXTRUSIONS ARE ALUMINUM ALLOY, 7075-T6511, QQ-A-200-11.
2. LEFT AND RIGHT SIDES ARE THE SAME, EXCEPT WHEN NOTED.
3. KEVLAR, 0.040 TO 0.060.
4. **70583-70929** WELD ASSEMBLY, 6061-T6, QQ-A-250-11. **70949-SUBQ** CASTING, AMS4260, 356-T6. EITHER WAY MAY BE USED.
5. TITANIUM, 6AL-4YT1, MIL-T-9046.
6. A SKIN, 0.040; B SKIN, 0.032.
7. 6061-T6, QQ-A-250-11, 0.032.
8. CLEAN AND SEAL WITH SEALING COMPOUND, ITEM 292, APPENDIX D. FILL GAPS WITH SEALING COMPOUND, ITEM 275, APPENDIX D AND SEAL WITH SEALING COMPOUND, ITEM 292, APPENDIX D (PARA 2-6-1).
9. PRIMARY CONSTRUCTION: SPOT WELDING PER MIL-W-6858, CLADD C; ALTERNATE CONSTRUCTION: BONDED.
10. GAP OF 0.040 - 0.120 INCH ALLOWED ALL AROUND BETWEEN DOOR AND DOUBLER.
11. **S70-I** SS430X20A010825. **S70-II** QQ-A-250-13, 7075-T6, 0.040.
12. CLEAN BONDING SURFACES, APPLY ADHESIVE PRIMER, ITEM 68, APPENDIX D, AND BOND WITH ADHESIVE, ITEM 32, APPENDIX D, BEFORE FASTENING WITH RIVETS.

FB1162_9D
SA

FIGURE 2-6-31-1. REPAIR TRANSITION SECTION PLATING (SHEET 9 OF 9)

2-6-32 TRANSITION SECTION STEP DOOR HINGE - WL 210.0 AND WL 230.0 REPAIR.

PARAGRAPH BREAKDOWN

	PAGE
2-6-32.1 Repair Transition Section Step Door Hinge - WL 210.0 and WL 230.0	2-6-32-1

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Blind Rivet Kit
- Clecos, 1⁄8-inch diameter
- Drill
- Drill Bit, No. 30 (0.1285-inch diameter)
- Drill Bit, 1⁄8-inch diameter
- Pin Punch, 3⁄32-inch diameter

Materials/Parts

- Nutplate, MS21075-08
- Rivet, NAS1097AD3
- Rivet, NAS1739B4

References

- PARA 2-6-1
- PARA 2-6-5
- QQ-W-470

2-6-32.1 Repair Transition Section Step Door Hinge - WL 210.0 and WL 230.0.

NOTE

Repair of left and right transition section step door hinges at WL 210.0 and WL 230.0 is the same.

2-6-32.1.1 Remove.

- a. Remove four screws securing door to angle and remove door (Figure 2-6-32-1, Detail A).
- b. Open and secure hinge to keep hinges open.
- c. Using drill with 1⁄8-inch diameter drill bit, and 3⁄32-inch diameter pin punch, carefully drill out and remove five rivets securing upper hinge half to step housing and stringer. Remove upper hinge half and angle.
- d. If spring is broken but hinges are serviceable, do not remove angle from lower hinge half. If hinges are to be replaced, drill out rivets in lower hinge half same as upper hinge half.
- e. Straighten hinge pin and remove. Remove spring.
- f. Clean up rivet debris.

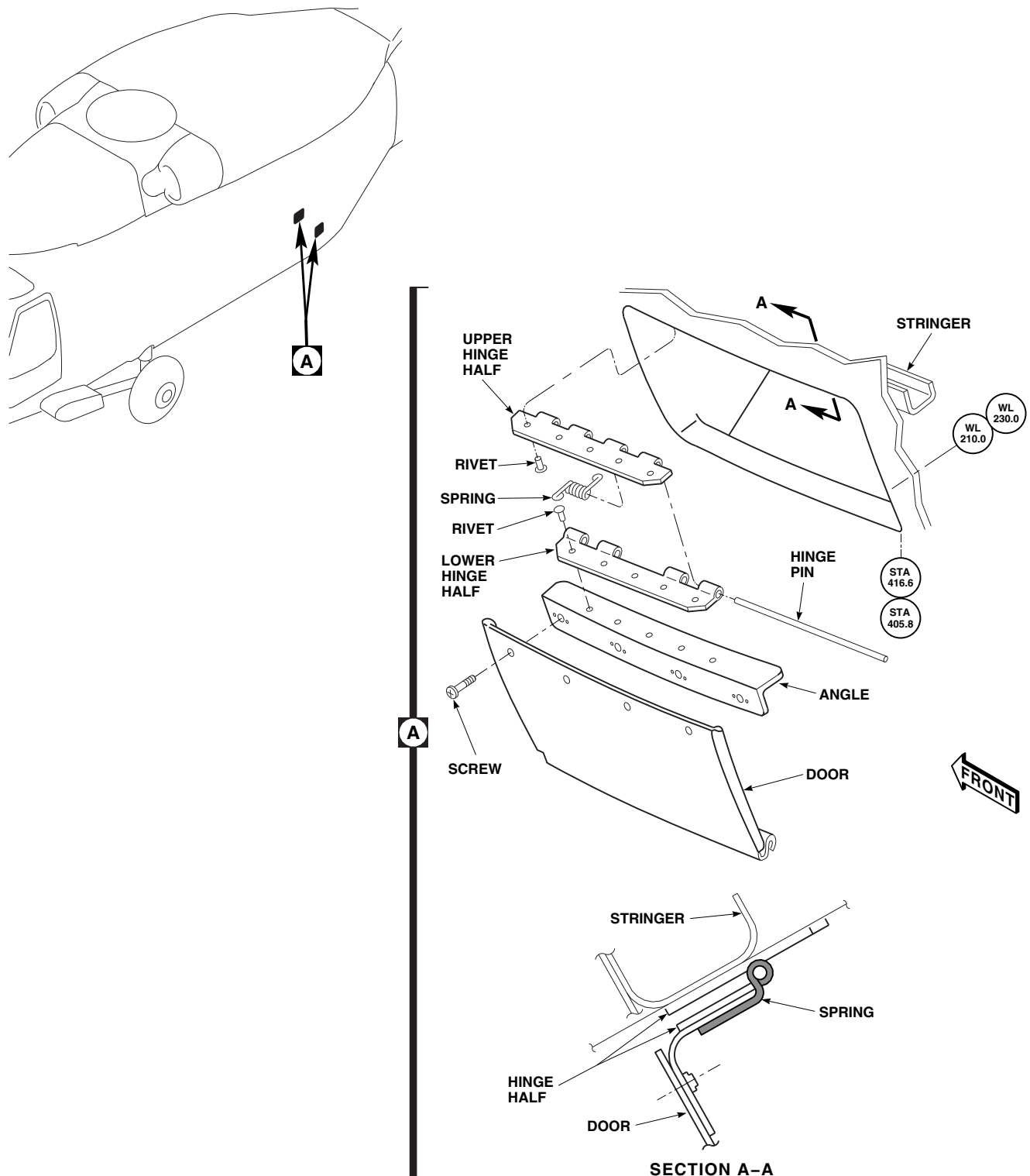
FN2631
SA

FIGURE 2-6-32-1. REPAIR TRANSITION SECTION STEP DOOR HINGE - WL 210.0 AND WL 230.0

2-6-32.1.2 Inspect.

Inspect nutplates on angle for damage. **NONE ALLOWED.** If damaged, replace nutplates using rivets, NAS1097AD3.

2-6-32.1.3 Install.

NOTE

Hinge halves are made from MS20001PY4-0450 and MS20001PX4-0450. Remove center tang on lower hinge half for spring.

- a. Using drill with No. 30 drill bit, lay out holes in replacement hinge halves and drill five holes.
- b. If lower hinge half was removed from angle, position replacement hinge half on angle and secure in place using 1/8-inch diameter clecos in two holes (Figure 2-6-32-1, Detail A).
- c. Install blind rivets, NAS1739B4, in between clecos. Remove clecos and install remaining rivets.

NOTE

Spring, 70209-03023-101, is made from 0.049-inch diameter 1095 steel, QQ-W-470.

- d. Position replacement spring in hinge halves (Figure 2-6-32-1, Section A-A) and insert new hinge pin into hinge halves. Cut pin 1-inch longer than hinge.
- e. Allow hinge pin to extend 1/2-inch beyond hinge on each end for bending.
- f. Bend hinge pin ends enough to secure pin in place.
- g. Clamp hinge to remain open. Position upper hinge against step housing and stringer, and secure in place using 1/8-inch diameter clecos.
- h. Install blind rivets, NAS1739B4, in between clecos. Remove clecos and install remaining rivets.
- i. Release hinge.
- j. For corrosion protection and watertightness, seal edges of hinge installation (PARA 2-6-1).
- k. Position door on angle and secure to angle with four screws, MS27039-0806.
- l. Make sure area is clean and free of foreign material.
- m. Check open/close function of door.
- n. Touch up paint as required (PARA 2-6-5).

2-6-33 TRANSITION SECTION STEP DOOR HINGE - WL 250.0.

PARAGRAPH BREAKDOWN

		PAGE
2-6-33.1	Replace Transition Section Step Door Hinge - WL 250.0	2-6-33-1

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Clecos, 1⁄8-inch diameter
- Drill
- Drill Bit, No. 30 (0.1285-inch diameter)
- Drill Bit, 1⁄8-inch diameter
- Pin Punch, 3⁄32-inch diameter

Materials/Parts

- Nutplate, MS21075-08
- Rivet, NAS1097AD3

- Rivet, NAS1097AD4
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 2-4-45
- PARA 2-4-69
- PARA 2-6-1
- PARA 2-6-5
- QQ-W-470

2-6-33.1 Replace Transition Section Step Door Hinge - WL 250.0.

NOTE

Replacement of both left and right transition section step door hinges at WL 250.0 is the same.

2-6-33.1.1 Remove.

- Remove six screws securing door to angle and remove door (Figure 2-6-33-1, Detail A).
- Open and secure hinge to keep hinges open.
- Using drill with 1⁄8-inch diameter drill bit, and 3⁄32-inch diameter pin punch, carefully drill out and remove ten rivets (five each assembly) securing upper hinge halves to step housing and stringer. Remove upper hinge halves and angle.
- If spring is broken but hinges are serviceable, do not remove angle from lower hinge halves. If hinges are to be replaced, drill out rivets in lower hinge halves, same as upper hinge halves.
- Straighten hinge pin and remove. Remove spring.

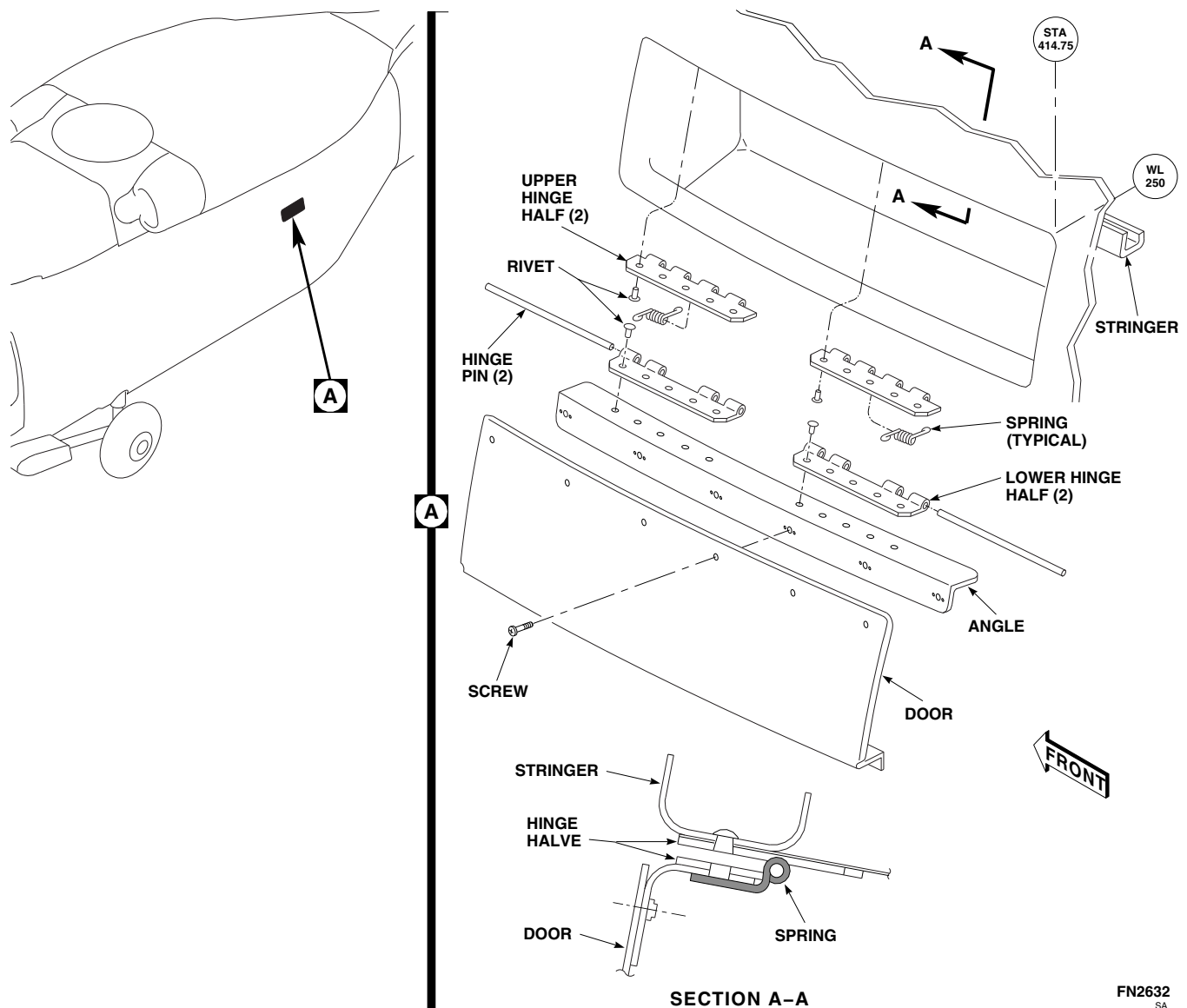


FIGURE 2-6-33-1. REPLACE TRANSITION SECTION STEP DOOR HINGE - WL 250.0

f. Clean up rivet debris.

2-6-33.1.2 Inspect.

Inspect nutplates on angle for damage. **NONE ALLOWED.** If damaged, replace nutplates using rivets, NAS1097AD3.

2-6-33.1.3 Install.

NOTE

Hinge halves are made from MS200001P4-450 and MS20001PX4-

450. Remove center tang on lower hinge halves for springs.

- a. Using drill with No. 30 drill bit, lay out holes in replacement hinge halves and drill five holes in each hinge half.
- b. If lower hinge halves were removed from angle, position replacement hinge halves on angle and secure in place using $\frac{1}{8}$ -inch diameter clecos in end holes of each hinge (Figure 2-6-33-1, Detail A).

- c. Install rivets, NAS1097AD4, in between clecos. Remove clecos and install remaining rivets.

NOTE

Spring, 70209-03023-101, is made from 0.049-inch diameter 1095 steel, QQ-W-470.

- d. Position replacement springs in hinge halves (Figure 2-6-33-1, Section A-A) and insert new hinge pins into hinge halves.
- e. Allow hinge pin to extend at least 1/2-inch beyond hinge on each end for flattening or bending.
- f. Bend hinge pin ends enough to secure pins in place.
- g. Gain access to main fuel cell compartment as follows:
 - (1) Remove rear passenger's seats as required (PARA 2-4-45).
 - (2) Remove soundproofing panels from rear bulkhead covering left and right stowage compartments (PARA 2-4-69).
- h. Have assistant inside main fuel cell compartment buck rivets and clean up rivet debris.

- i. Clamp hinges in open position. Position upper hinges against step housing and stringer, and secure in place using 1/8-inch diameter clecos.
- j. Install rivets, NAS1097AD4 (five each hinge), in between clecos. Remove clecos and install remaining rivets.
- k. Release hinges.
- l. Make sure main fuel cell compartment is clean and free of foreign material.
- m. Close fuel cell compartment as follows:
 - (1) Install soundproofing panels (PARA 2-4-69).
 - (2) Install rear passenger's seats (PARA 2-4-45).
- n. For corrosion protection and watertightness, using sealing compound, Item 292, Appendix D, seal edges of hinge installation (PARA 2-6-1).
- o. Position door on angle and secure to angle with six screws, MS27039-0806.
- p. Make sure area is clean and free of foreign material.
- q. Check open/close function of door.
- r. Touch up paint as required (PARA 2-6-5).

2-6-34 TRANSITION SECTION FUEL SUMP DRAIN DOOR.

PARAGRAPH BREAKDOWN

	PAGE
2-6-34.1 Replace Transition Section Fuel Sump Drain Latch, Hinge, and Door	2-6-34-1

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Clecos, 5/32-inch diameter
- Drill
- Drill Bit, No. 24 (0.1520-inch diameter)
- Nonmetallic Scraper, Locally-Made
- Pin Punch, 1/8-inch diameter
- Rivet Gun

Materials/Parts

- Adhesive, Item 32, Appendix D
- Adhesive, Item 64, Appendix D

- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Primer, Item 254, Appendix D
- Rivet, MS20470AD5
- Rivet, NAS1738B5
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 2-6-1
- PARA 2-6-5

2-6-34.1 Replace Transition Section Fuel Sump Drain Latch, Hinge, and Door.

NOTE

Perform steps in procedure to extent necessary to remove and install part(s).

2-6-34.1.1 Remove.

- Using drill with No. 24 drill bit, and 1/8-inch diameter pin punch, carefully drill out and remove 12 rivets securing hinge to helicopter structure (Figure 2-6-34-1, Detail A).

- Release latches and quarter-turn fasteners and open door.
- Using drill with No. 24 drill bit, and 1/8-inch diameter pin punch, carefully drill out and remove 12 rivets securing hinge to door.
- Remove hinge from door.
- Using drill with No. 24 drill bit, and 1/8-inch diameter pin punch, carefully drill out and remove four rivets securing each latch to door.
- Remove latch or latches, as applicable.

- g. Clean up rivet debris.

2-6-34.1.2 Repair.

- a. Using nonmetallic scraper, remove seal from sump door (Figure 2-6-34-1, Detail B).
- b. Using machinery towel, Item 211, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D, clean bonding surface of door.

NOTE

Seal may also be bonded with adhesive, Item 64, Appendix D. To bond seal with adhesive, Item 64, Appendix D, proceed to step f.

- c. Mix adhesive, Item 32, Appendix D, per manufacturer's instructions.
- d. Apply adhesive to door and new seal. Place seal on door.
- e. Let adhesive cure for 2 hours at 110° - 130°F (43° - 54°C), or for 24 hours at 68°F (20°C).

NOTE

Seal may also be bonded with adhesive, Item 64, Appendix D.

- f. To bond seal to door with adhesive, Item 64, Appendix D, perform the following:
 - (1) Clean area to be bonded using machinery towel, Item 211, Appendix D, dampened with methyl ethyl ketone, Item 217, Appendix D.
 - (2) Apply primer, Item 254, Appendix D, to seal and door. Allow primer to dry at least 3 minutes at 68°F (20°C).
 - (3) Apply ribbon of adhesive, Item 64, Appendix D, to seal bonding area, and firmly press seal in place on door. Allow to cure for 15 minutes at 68°F (20°C).
 - (4) Seal bond joint with sealing compound, Item 292, Appendix D (PARA 2-6-1).

- g. Touch up paint as required (PARA 2-6-5).

2-6-34.1.3 Install.

NOTE

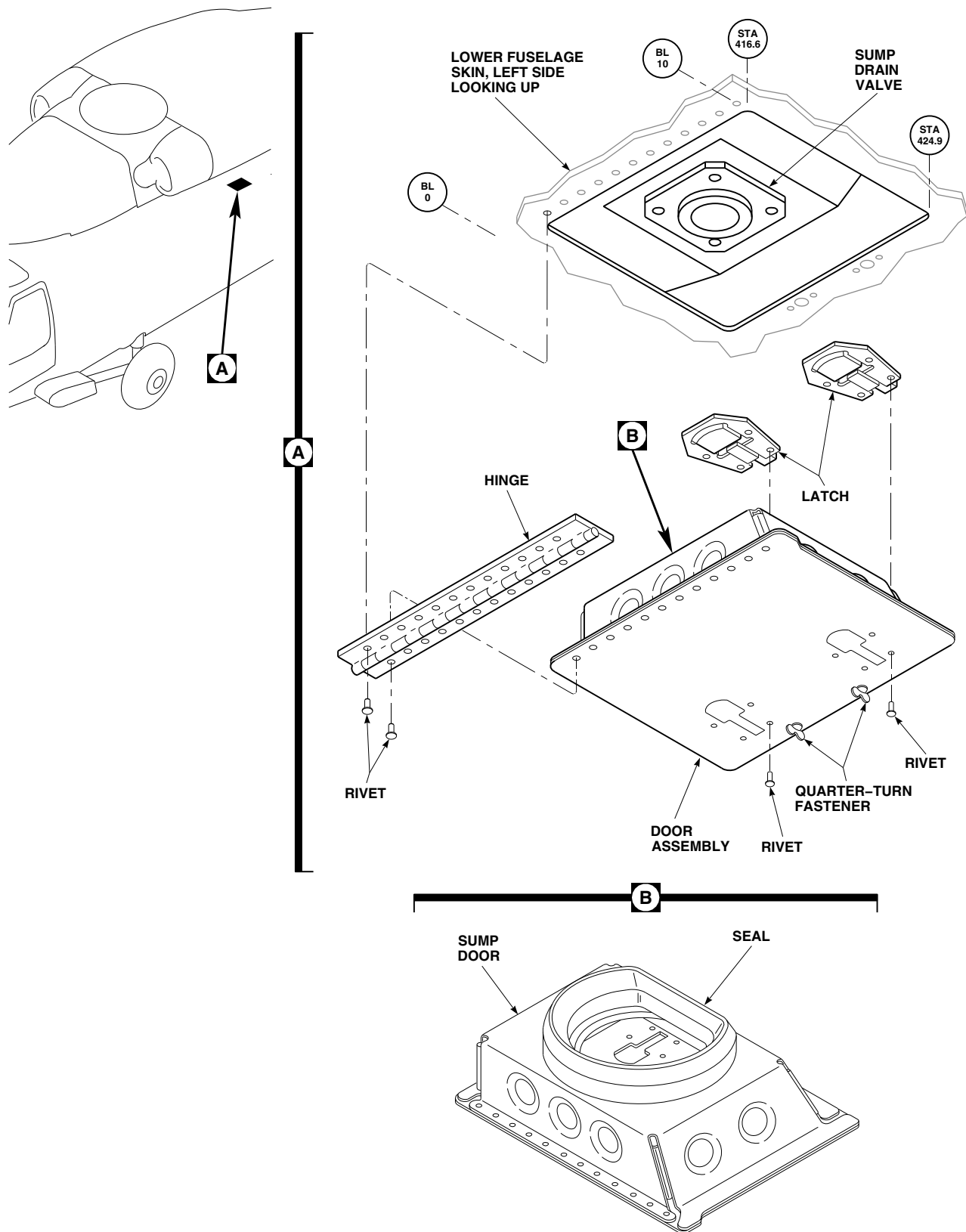
Replace latch, H-5010-081-196, as required.

- a. Position latch on door, and secure in place with $\frac{5}{32}$ -inch diameter clecos in two diagonal corners (Figure 2-6-34-1, Detail A).
- b. Install rivets, MS20470AD5, in between clecos. Remove clecos and install remaining rivets.

NOTE

Replace hinge, MS20001P4-1012, as required.

- c. Position hinge on door, and secure in place with $\frac{5}{32}$ -inch diameter clecos in every third hole.
- d. Using rivet gun, install rivets, MS20470AD5, in between clecos. Remove clecos and install remaining rivets.
- e. Position door on bottom skin structure, secure latches, and secure door in place with $\frac{5}{32}$ -inch diameter clecos in every third hole.
- f. Check open/close function of door. Adjust door, as required.
- g. Install blind rivets, NAS1738B5, in between clecos. Remove clecos and install remaining rivets.
- h. Make sure area is clean and free of foreign material.
- i. Recheck open/close function of door and latches.
- j. Secure quarter-turn fasteners.
- k. Seal edges of assembled parts for corrosion protection and watertightness (PARA 2-6-1).



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FIGURE 2-6-34-1. REPLACE TRANSITION SECTION FUEL SUMP DRAIN LATCH, HINGE, AND DOOR

1. Close and latch door.

2-6-35 TRANSITION SECTION GRAVITY REFUEL DOOR.

PARAGRAPH BREAKDOWN

	PAGE
2-6-35.1 Replace Transition Section Gravity Refueling Latch, Hinge, and Door	2-6-35-1

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Blind Rivet Kit
- Clecos, 5/32-inch diameter
- Drill
- Drill Bit, No. 24 (0.1520-inch diameter)
- Pin Punch, 1/8-inch diameter

Materials/Parts

- Adhesive, Item 37, Appendix D
- Rivet, MS20470AD5

- Rivet, NAS1097AD5
- Rivet, NAS1739B5
- Washer, NAS1149DN816K
- Washer, NAS1149DN832K

References

- Appendix D
- PARA 2-6-1
- PARA 2-6-5

2-6-35.1 Replace Transition Section Gravity Refueling Latch, Hinge, and Door.

NOTE

Perform steps in procedure to extent necessary to remove and install part(s).

2-6-35.1.1 Remove.

- Using drill with No. 24 drill bit and 1/8-inch diameter pin punch, carefully drill out and remove seven rivets securing upper hinge half to door (Figure 2-6-35-1, Detail A). Remove washers.

- Release latch and open door.
- Using drill with No. 24 drill bit and 1/8-inch diameter pin punch, carefully drill out and remove eight rivets securing lower hinge half to gravity refueling aperture.
- Remove hinge, plate, and eight washers. Save washers and plate for reinstallation.
- Using drill with No. 24 drill bit and 1/8-inch diameter pin punch, carefully drill out and remove four rivets securing latch to door.
- Clean up rivet debris.

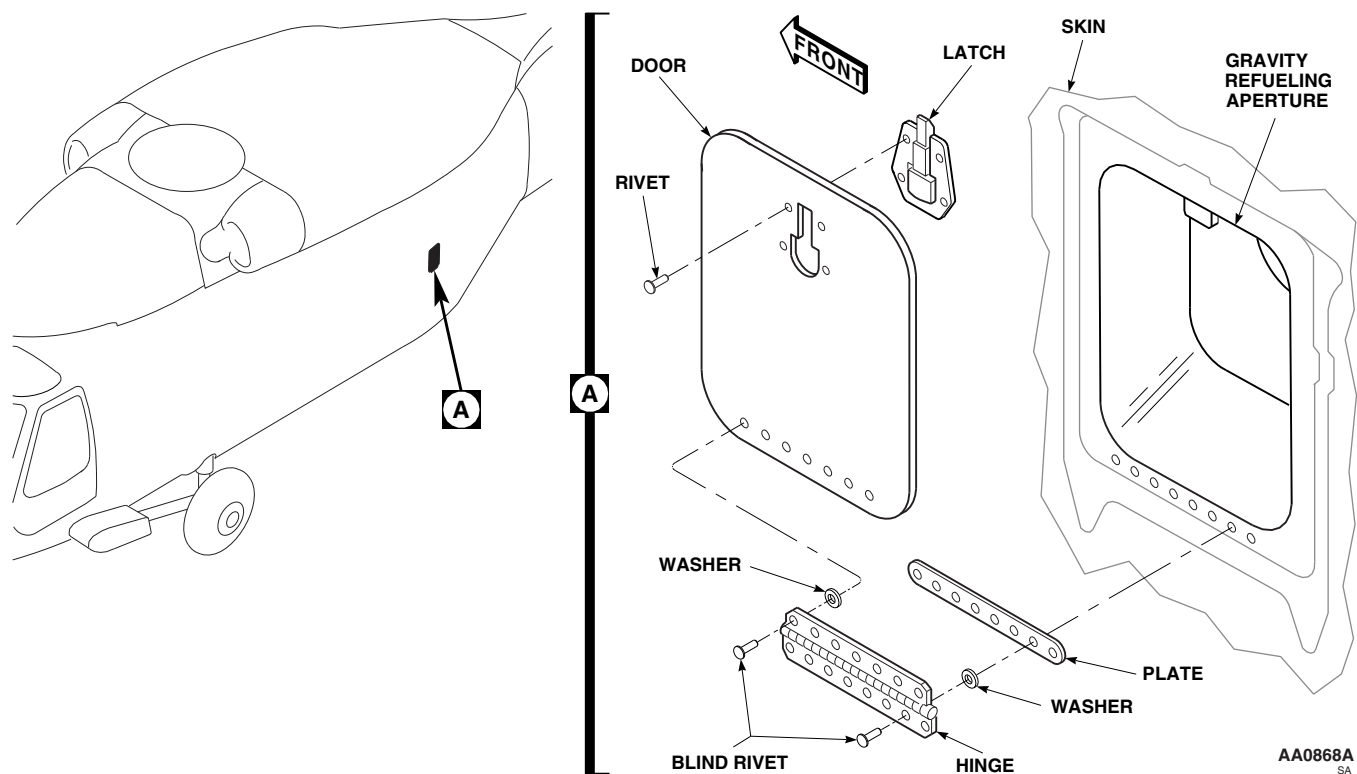


FIGURE 2-6-35-1. REPLACE TRANSITION SECTION GRAVITY REFUELING LATCH/HINGE AND DOOR

2-6-35.1.2 Inspect.

- Inspect part(s) for damage (PARA 2-6-1). If latch or hinge is damaged beyond negligible limits, replace latch or hinge.
- Inspect parts for missing protective finishes. **NONE ALLOWED.** If required, touch up protective finishes (PARA 2-6-5).

2-6-35.1.3 Install.

- For corrosion protection and watertightness, seal parts during assembly (PARA 2-6-1).
- Position latch on door and secure in place with $\frac{5}{32}$ -inch diameter clecos in two diagonal corners (Figure 2-6-35-1, Detail A).
- Install rivets, MS20470AD5, between clecos. Remove clecos and install remaining rivets.

NOTE

Replace hinge, MS20001P4-600, as required.

- Position upper hinge half on door. Position two washers, NAS1149DN832K, under hinge on end holes and two washers, NAS1149DN816K, on holes second from end. Secure in place with $\frac{5}{32}$ -inch diameter clecos.
- Install countersunk rivets, NAS1097AD5, in between clecos. Remove clecos and install remaining rivets.
- Mix adhesive, Item 37, Appendix D, per manufacturer's instructions.
- Apply adhesive to plate and mating area on gravity refueling aperture. Position plate to aperture.
- Position door and four washers over plate on gravity refueling aperture. Latch door to aperture. Position two washers, NAS1149DN832K, between hinge and plate on end holes and two washers, NAS1149DN816K, on holes second from end. Secure in place with $\frac{5}{32}$ -inch diameter clecos.

- i. Check open/close function of door. Adjust, as required.
- j. Install countersunk blind rivets, NAS1739B5, in between clecos. Remove clecos and install remaining rivets.
- k. Recheck open/close function of door.
- l. For corrosion protection and watertightness, seal edges of assembled parts (PARA 2-6-1).
- m. Touch up paint as required (PARA 2-6-5).
- n. Close and latch door.

2-6-36 TRANSITION SECTION PRESSURE REFUEL DOOR.

PARAGRAPH BREAKDOWN

	PAGE
2-6-36.1 Replace Transition Section Pressure Refueling Latch, Hinge, and Door	2-6-36-1

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Blind Rivet Kit
- Clecos, 5⁄32-inch diameter
- Drill
- Drill Bit, No. 24 (0.1520-inch diameter)

- Pin Punch, 1⁄8-inch diameter

Materials/Parts

- Rivet, MS20470AD5
- Rivet, NAS1738B5

References

- PARA 2-6-1
- PARA 2-6-5

2-6-36.1 Replace Transition Section Pressure Refueling Latch, Hinge, and Door.

NOTE

Perform steps in procedure to extent necessary to remove and install part(s).

- e. Using drill with No. 24 drill bit, and 1⁄8-inch diameter pin punch, carefully drill out and remove six rivets securing lower hinge half to pressure refueling aperture. Remove hinge.
- f. Clean up rivet debris.

2-6-36.1.1 Remove.

- a. Using drill with No. 24 drill bit, and 1⁄8-inch diameter pin punch, carefully drill out and remove six rivets securing door to upper hinge half (Figure 2-6-36-1, Detail A).
- b. Release latches and open door.
- c. Using drill with No. 24 drill bit, and 1⁄8-inch diameter pin punch, carefully drill out and remove four rivets securing each latch to door.
- d. Remove latches from door.

2-6-36.1.2 Inspect/Repair.

- a. Inspect parts for damage (PARA 2-6-1). If latch(es) or hinge is damaged beyond negligible limits, replace latch(es) or hinge.
- b. Inspect parts for missing protective finishes. **NONE ALLOWED.** If required, touch up protective finishes (PARA 2-6-5).

2-6-36.1.3 Install.

- a. For corrosion protection and watertightness, seal parts during assembly (PARA 2-6-1).

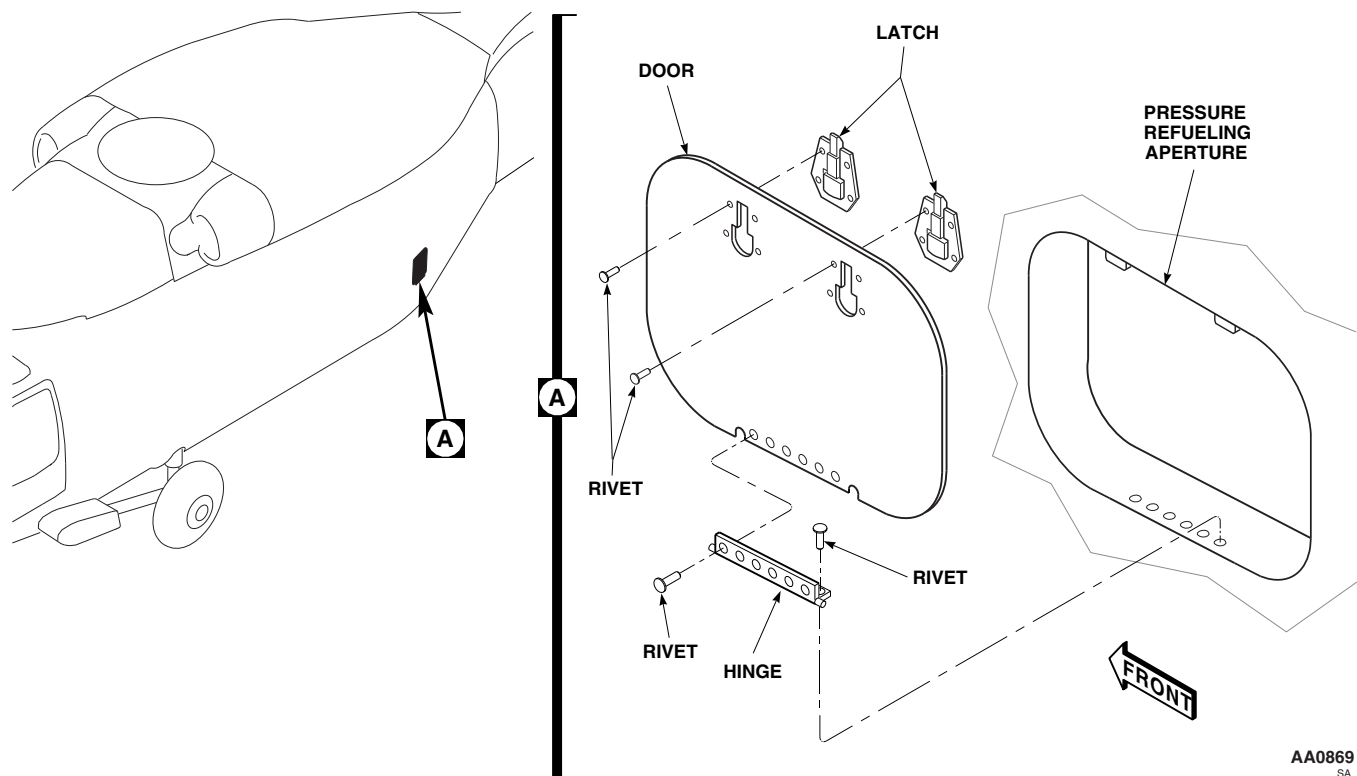


FIGURE 2-6-36-1. REPLACE TRANSITION SECTION PRESSURE REFUELING LATCH, HINGE, AND DOOR

- b. Position latches on door and secure in place with $\frac{5}{32}$ -inch diameter clecos in two diagonal holes (Figure 2-6-36-1, Detail A).
- c. Install rivets, MS20470AD5, in between clecos. Remove clecos and install remaining rivets.
- d. Position lower half of hinge on pressure refueling aperture and secure in place with $\frac{5}{32}$ -inch diameter clecos in end holes.
- e. Install countersunk blind rivets, NAS1738B5, in between clecos. Remove clecos and install remaining rivets.
- f. Position door on pressure refueling aperture and engage latch. Line up upper hinge half with fastener holes in door and secure in place using $\frac{5}{32}$ -inch diameter clecos in end holes.
- g. Check open/close function of door. Adjust hinge, as required.
- h. Install rivets, MS20470AD5, between clecos. Remove clecos and install remaining rivets.
- i. Recheck open/close function of door and latches.
- j. For corrosion protection and watertightness, seal edges of assembled parts (PARA 2-6-1).
- k. Touch up paint as required (PARA 2-6-5).
- l. Close and latch door.

NOTE

Replace hinge, MS20001P4-600, as required.

2-6-37 TRANSITION SECTION PNEUMATIC GROUND START DOOR.

PARAGRAPH BREAKDOWN

	PAGE
2-6-37.1 Replace Pneumatic Ground Start Latch, Hinge, and Door	2-6-37-1

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Clecos, 1⁄8-inch diameter
- Clecos, 5⁄32-inch diameter
- Drill
- Drill Bit, No. 32 (0.1160-inch diameter)
- Pin Punch, 3⁄32-inch diameter

Materials/Parts

- Rivet, MS20470AD4
- Rivet, MS20470AD5
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 2-6-1
- PARA 2-6-5

2-6-37.1 Replace Pneumatic Ground Start Latch, Hinge, and Door.

NOTE

Perform steps in procedure to extent necessary to remove and install part(s).

2-6-37.1.1 Remove.

- Carefully drill out and remove six rivets securing door to upper hinge half (Figure 2-6-37-1, Detail A).
- Release latch and open door.
- Using drill with No. 32 drill bit, and 3⁄32-inch diameter pin punch, carefully drill out and remove four rivets securing lower hinge half to aperture. Remove hinge.

- Carefully drill out and remove four rivets securing latch to door. Remove latch.
- Clean up rivet debris.

2-6-37.1.2 Inspect/Repair.

- Inspect parts for damage (PARA 2-6-1). If latch or hinge is damaged beyond negligible limits, replace latch or hinge.
- Inspect parts for missing protective finishes. **NONE ALLOWED.** If required, touch up protective finishes (PARA 2-6-5).

2-6-37.1.3 Install.

- For corrosion protection and watertightness, seal hinge parts and rivets during assembly (PARA 2-6-1).

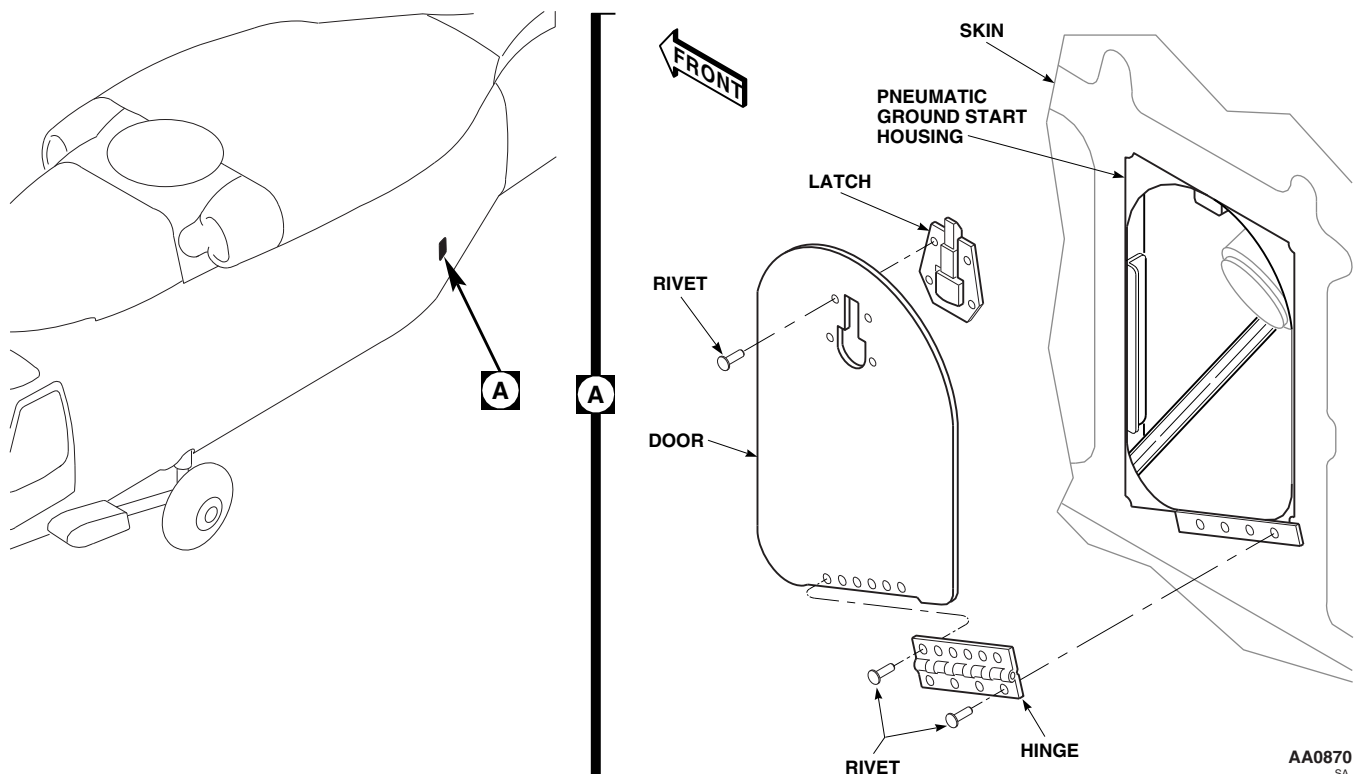


FIGURE 2-6-37-1. REPLACE PNEUMATIC GROUND START LATCH, HINGE, AND DOOR

- b. Position latch on door and secure in place with $\frac{5}{32}$ -inch diameter clecos in two diagonal holes (Figure 2-6-37-1, Detail A).
- c. Install rivets, MS20470AD5, in between clecos. Remove clecos and install remaining rivets.
- f. Position door on housing and engage latch. Line up upper hinge half with fastener holes in door and secure in place with $\frac{1}{8}$ -inch diameter clecos in end holes.
- g. Check open/close function of door. Adjust hinge, as required.
- h. Install rivets, MS20470AD4, in between clecos. Remove clecos and install remaining rivets, check open/close function of door and latch.
- i. For corrosion protection and watertightness using sealing compound, Item 292, Appendix D, seal edges of assembled parts (PARA 2-6-1).
- j. Touch up paint as required (PARA 2-6-5).
- k. Close and latch door.
- d. Position lower hinge half on housing and secure in place with $\frac{5}{32}$ -inch diameter clecos in end holes of hinge.
- e. Install rivets, MS20470AD5, in between clecos. Remove clecos and install remaining rivets.

NOTE

Replace hinge, MS20001P4-375, as required.

2-6-38 TRANSITION SECTION APU ACCESS DOOR SCREWS REPAIR.

PARAGRAPH BREAKDOWN

	PAGE
2-6-38.1 Repair Transition Section APU Access Door Screws	2-6-38-1

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit

Materials/Parts

- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D

- Sealing Compound, Item 283, Appendix D
- Sealing Primer, Item 306, Appendix D

References

- Appendix D
- PARA 2-4-108

2-6-38.1 Repair Transition Section APU Access Door Screws.

- Remove APU access doors (PARA 2-4-108).
- Remove screws and washers securing APU access door hinge fitting to firewall.
- Clean screw and nutplate threads using machinery towel, Item 211, Appendix D, wet with methyl ethyl ketone, Item 217, Appendix D.

- Apply sealing primer, Item 306, Appendix D, to hinge fitting attaching screws and nutplates. Allow primer to dry for 5 minutes.
- Apply sealing compound, Item 283, Appendix D, to screws and nutplates.
- Secure hinge fitting to support firewall with screws and washers.
- Install APU access doors (PARA 2-4-108).

2-6-39 GROUNDING RECEPTACLE.

PARAGRAPH BREAKDOWN

		PAGE
2-6-39.1	Replace Grounding Receptacle	2-6-39-1

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Digital Low-Resistance Ohmmeter

References

- PARA 2-6-5

2-6-39.1 Replace Grounding Receptacle.

NOTE

Replacement procedures are typical for all grounding receptacles.

2-6-39.1.1 Inspect.

- Measure resistance between grounding receptacle and an adjacent point on helicopter skin using digital low-resistance ohmmeter. **RESISTANCE SHALL NOT BE MORE THAN 1.0 OHM.** If resistance is more than 1.0 ohm, perform the following:
 - Remove grounding receptacle (PARA 2-6-39.1.2).
 - Replace chemical conversion coated surfaces (PARA 2-6-5).
 - Install grounding receptacle (PARA 2-6-39.1.3).
 - Measure resistance between grounding receptacle and an adjacent point on helicopter skin using digital low-resistance ohmmeter. **RESISTANCE**

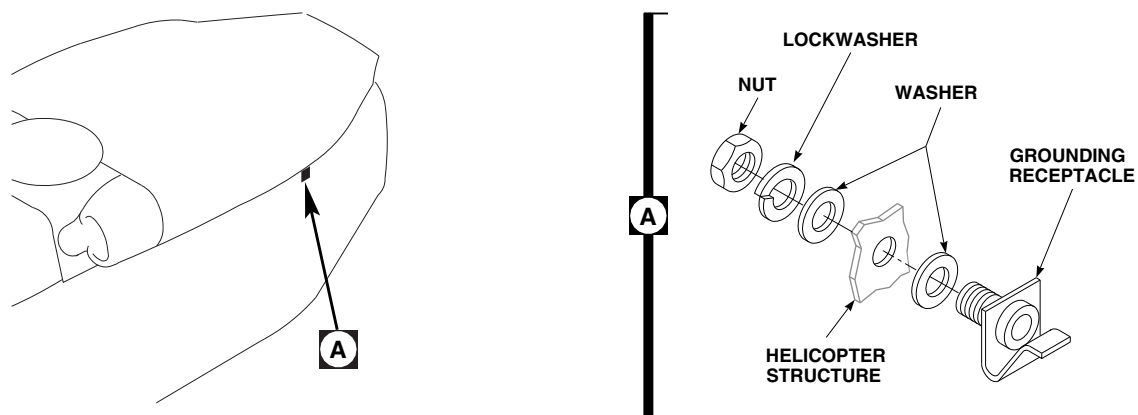
SHALL NOT BE MORE THAN 1.0 OHM. If resistance is more than 1.0 ohm, replace grounding receptacle.

2-6-39.1.2 Remove.

- Remove nut, lockwasher, and washer securing grounding receptacle to helicopter structure (Figure 2-6-39-1, Detail A).
- Remove grounding receptacle from helicopter structure.
- Remove washer from grounding receptacle.

2-6-39.1.3 Install.

- Install washer on grounding receptacle (Figure 2-6-39-1, Detail A).
- Position grounding receptacle through helicopter structure.
- Secure grounding receptacle to helicopter structure with nut, lockwasher, and washer on grounding receptacle. Tighten nut on grounding receptacle.
- Inspect grounding receptacle (PARA 2-6-39.1.1).

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- e. Touch up paint finish, as required (PARA 2-6-5).

2-6-40 TAIL CONE FRAMING REPAIR.

PARAGRAPH BREAKDOWN

	PAGE
2-6-40.1	Repair Tail Cone Framing 2-6-40-2

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Nonmetallic Scraper, Locally-Made
- Torque Wrench, 150 - 1000 in. lbs

Materials/Parts

- Abrasive Cloth, Item 6, Appendix D
- Adhesion Promoter, Item 25A, Appendix D
- Cheesecloth, Item 90, Appendix D
- Corrosion Compound, Item 108, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Pressure Sensitive Adhesive Tape, Item 251, Appendix D
- Scouring Pad, Item 270, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- MIL-S-18729
- MIL-S-5059
- PARA 1-6-7
- PARA 2-6-1
- PARA 2-6-2
- PARA 2-6-4
- PARA 2-6-5
- PARA 2-6-43
- QQ-A-200
- QQ-A-225
- QQ-A-250
- QQ-A-367

2-6-40.1 Repair Tail Cone Framing.**NOTE**

- Tail cone framing includes formed and extruded members made from 7075-T6, QQ-A-250/13, Alclad sheet stock and 7075-T6511, QQ-A-200/11 aluminum alloy extrusion. Fittings at STA 605.0 (landing gear support) and STA 648.0 (canted hinge bulkhead fitting) are made from forged aluminum alloy, 7075-T73, QQ-A-367. Blade stowage fittings (STA 585.0 and STA 624.0) are made from 7075-T6, QQ-A-200/11 (Figure 2-6-40-1, Sheet 1 through Sheet 8).
 - All tail cone framing is primary structure, except tail drive shaft cover supports and formers which are secondary structure. Forged fittings (STA 605.0 and STA 648.0) are considered critical primary structure. Negligible damage to these fittings is limited to 5% of material thickness at point of damage. Blend damage smoothly to surrounding area using long even strokes. Polish with abrasive cloth, Item 6, Appendix D, and restore protective finish. Damage greater than negligible limits to fittings must be reviewed by engineering authority for damage analysis, and either repair or replacement. All repairs to tail cone framing must maintain designed strength and alignment.
- a. Evaluate damage in noncritical areas (PARA 2-6-1).
 - b. Use PARA 2-6-2 and PARA 2-6-4 for typical framing repairs.
 - c. Refer to PARA 2-6-43 for tail rotor drive shaft cover repairs.
 - d. For corrosion protection and watertightness, seal all repairs of mating tail cone framing with plating (PARA 2-6-1).

- e. Touch up paint as required (PARA 2-6-5).
- f. To repair polyethylene abrasion strip on anticollision light support, perform the following:
 - (1) Stow anticollision light.
 - (2) Using nonmetallic scraper, remove old abrasion strip.
 - (3) Clean area using methyl ethyl ketone, Item 217, Appendix D, and wipe area dry with clean cheesecloth, Item 90, Appendix D.
 - (4) Mask area where pressure sensitive tape is to be applied.
 - (5) Spray area with one coat of adhesion promoter, Item 25A, Appendix D.
 - (6) Apply pressure sensitive adhesive tape, Item 251, Appendix D, to primed area.

NOTE

Corrosion is indicated by red film.

- g. Visually inspect exposed area of blade stowage fittings for corrosion. If necessary, remove as follows:
 - (1) Using scouring pad, Item 270, Appendix D, remove corrosion manually.
 - (2) Mix solution of one part corrosion compound, Item 108, Appendix D, to one part water.
 - (3) Apply compound solution to corroded areas.
 - (4) Rinse thoroughly with clean water.
- h. Check that blade support retainer fits securely in fitting. If blade support retainer no longer fits securely in fitting, replace blade stowage fitting (PARA 1-6-7).

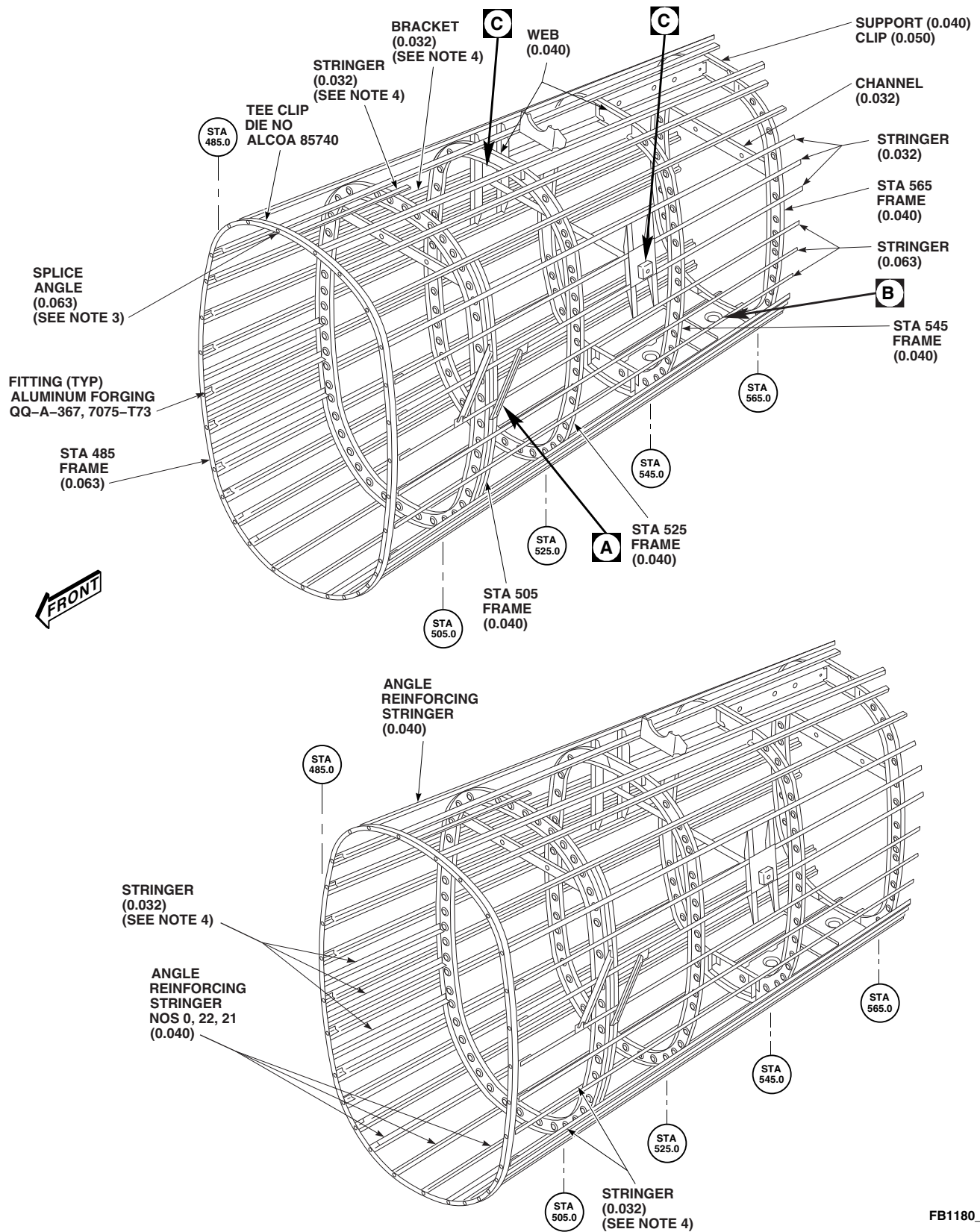


FIGURE 2-6-40-1. REPAIR TAIL CONE FRAMING (SHEET 1 OF 8)

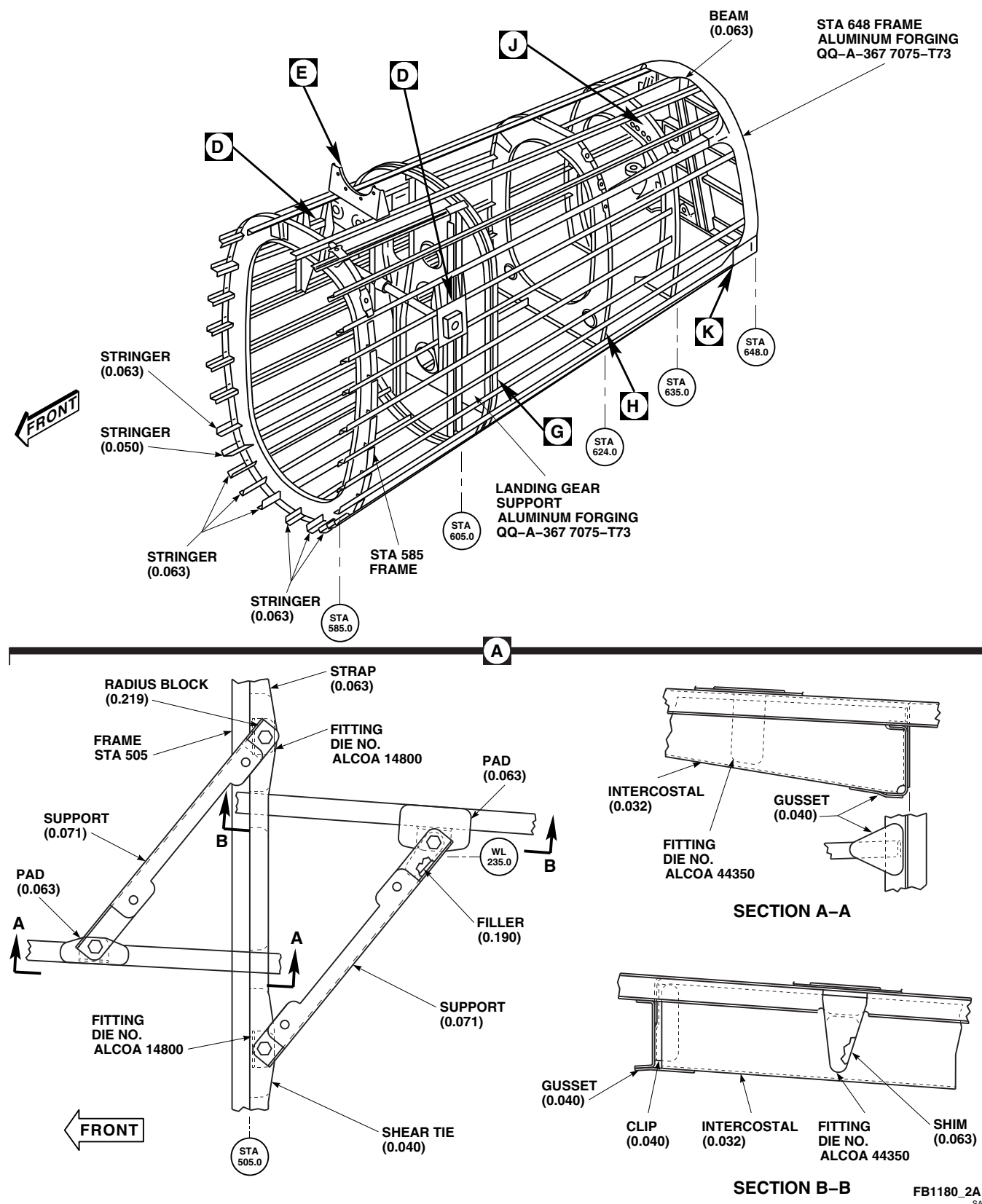


FIGURE 2-6-40-1. REPAIR TAIL CONE FRAMING (SHEET 2 OF 8)

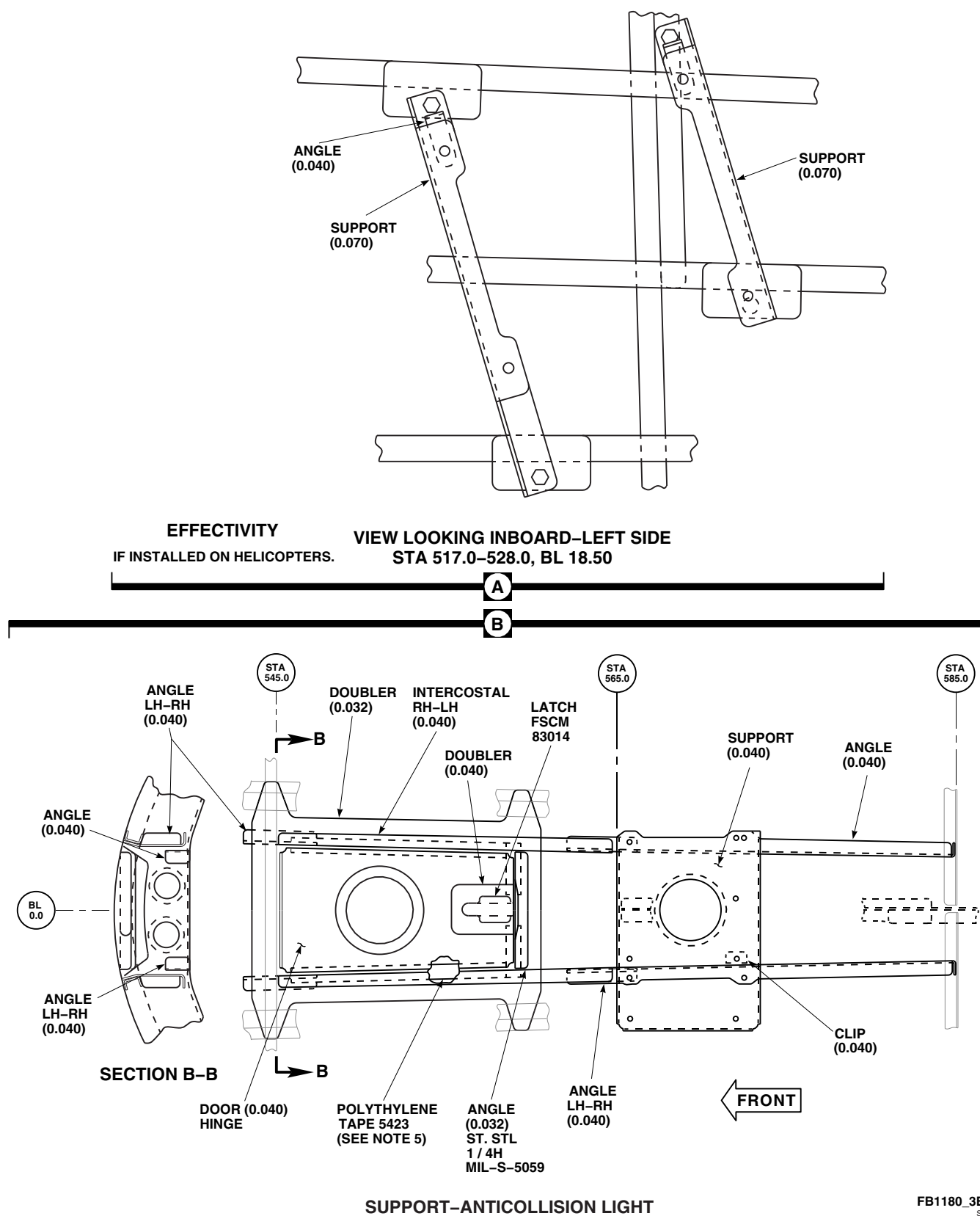
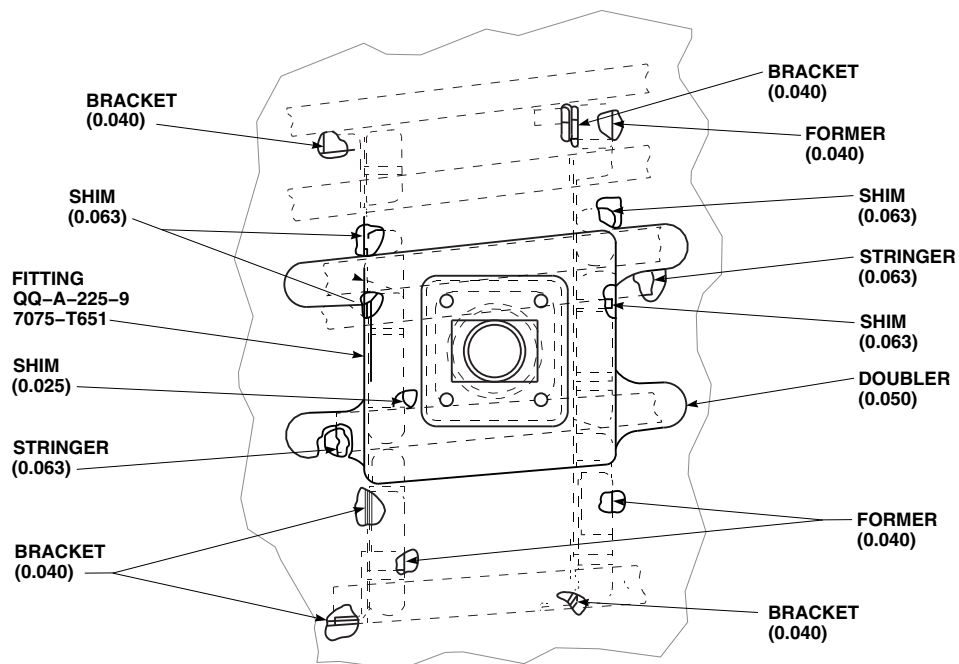


FIGURE 2-6-40-1. REPAIR TAIL CONE FRAMING (SHEET 3 OF 8)

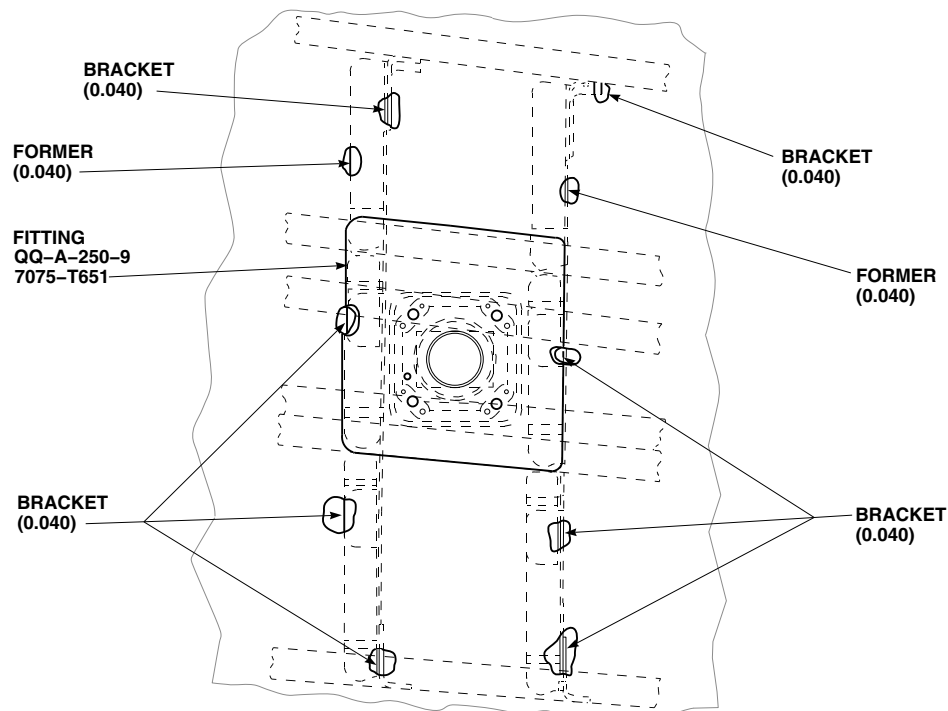


EFFECTIVITY
IF INSTALLED ON HELICOPTERS.

VIEW LOOKING INBOARD-RIGHT SIDE
STA 596.60, WL 236.0
LEFT SIDE OPPOSITE

C

D

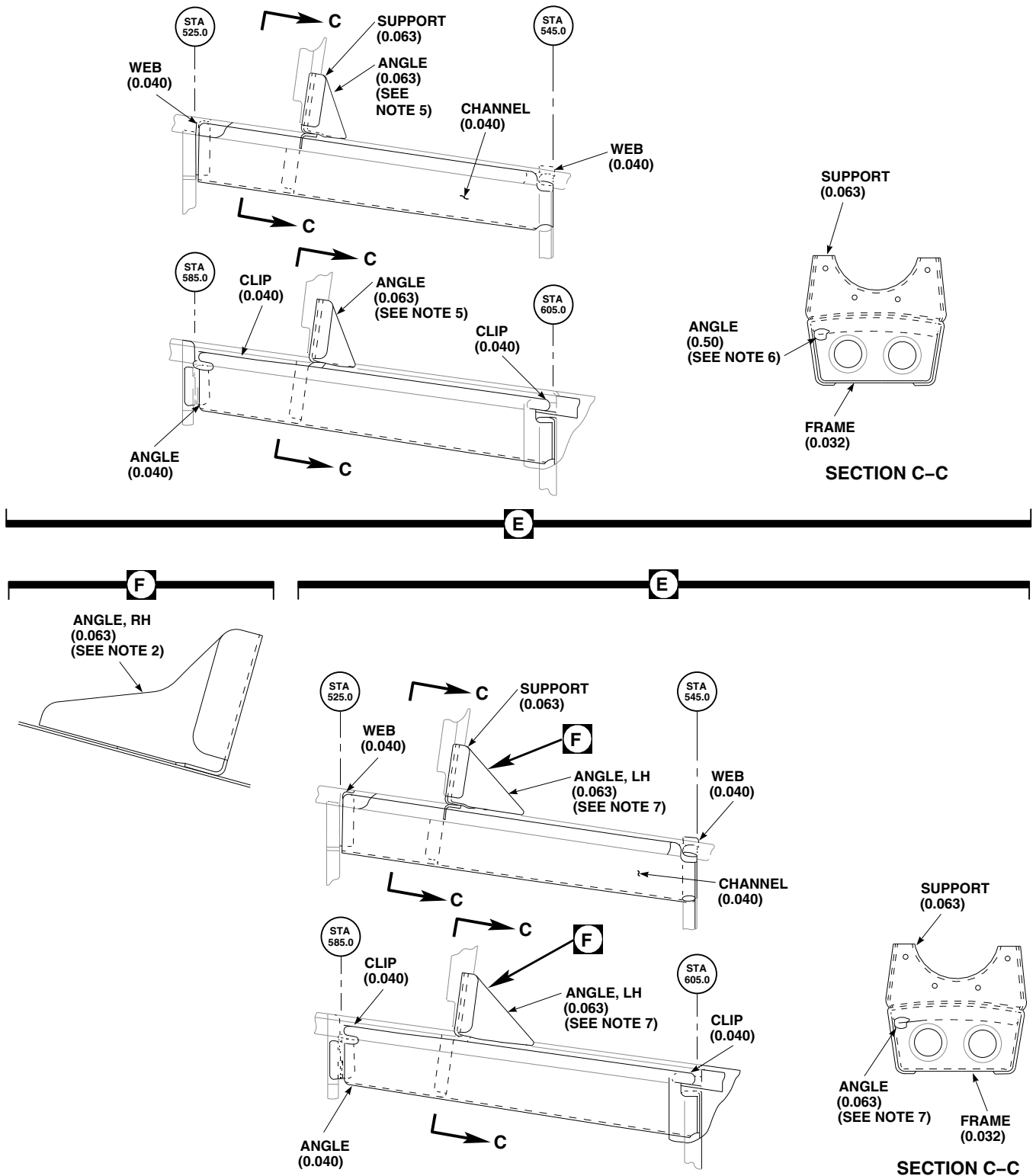


EFFECTIVITY
IF INSTALLED ON HELICOPTERS.

VIEW LOOKING INBOARD-LEFT SIDE
STA 536.0, WL 237.0
RIGHT SIDE OPPOSITE

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FIGURE 2-6-40-1. REPAIR TAIL CONE FRAMING (SHEET 4 OF 8)



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FIGURE 2-6-40-1. REPAIR TAIL CONE FRAMING (SHEET 5 OF 8)

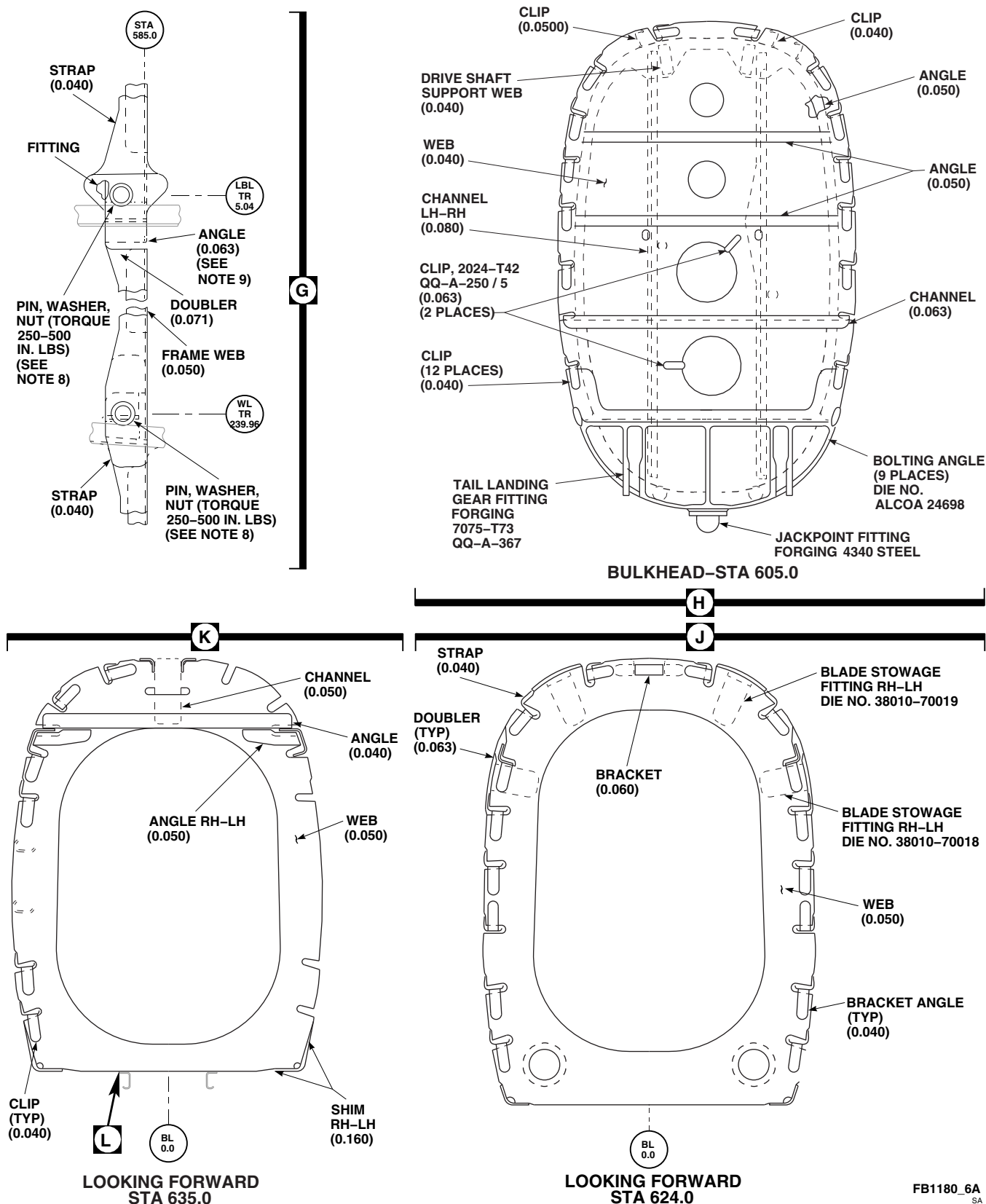


FIGURE 2-6-40-1. REPAIR TAIL CONE FRAMING (SHEET 6 OF 8)

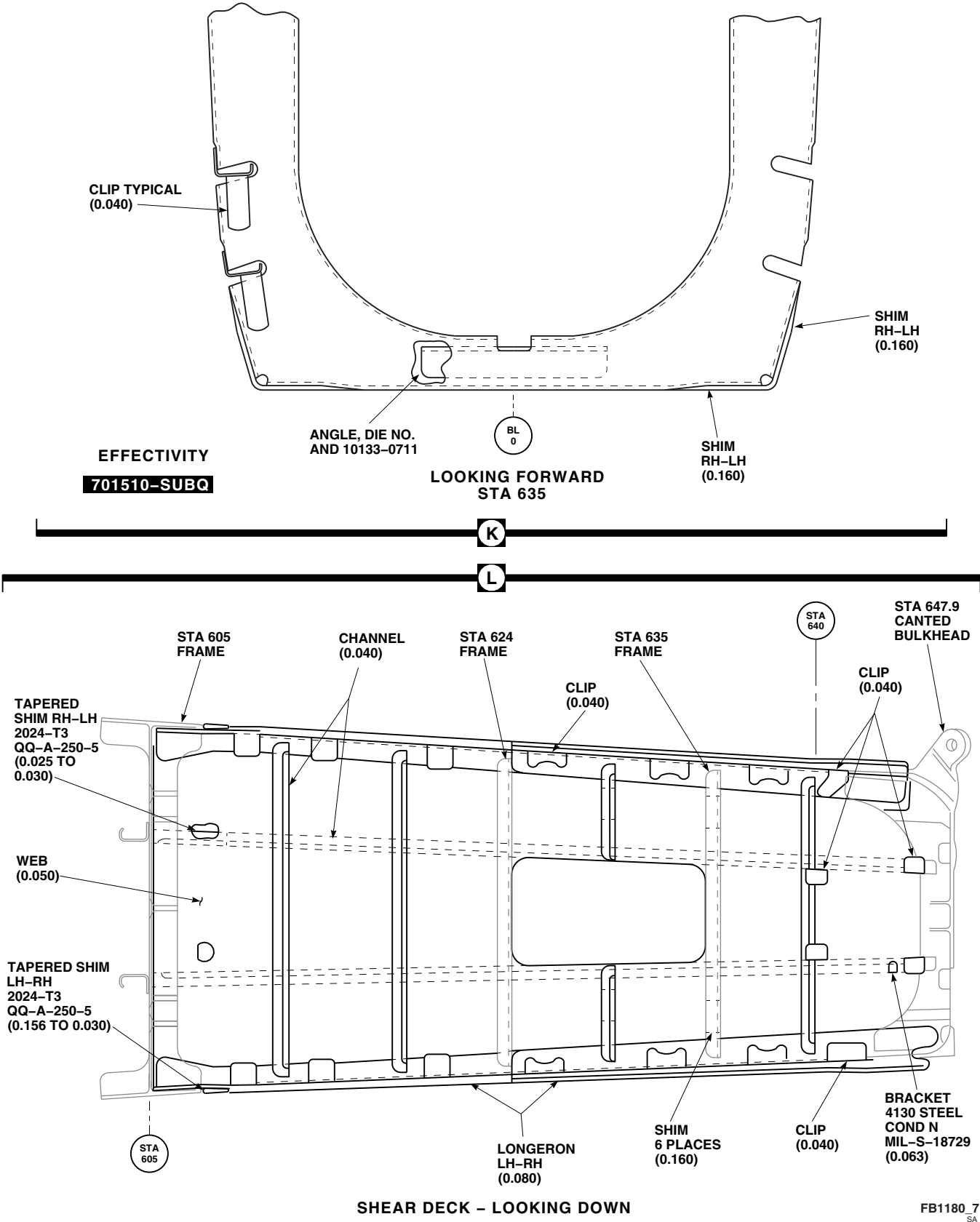


FIGURE 2-6-40-1. REPAIR TAIL CONE FRAMING (SHEET 7 OF 8)

NOTES

1. UNLESS OTHERWISE NOTED ON SHEETS OF THIS FIGURE, ALL SHEET METAL PARTS SHOWN ARE ALCLAD 7075-T6, QQ-A-250-13, AND ALL EXTRUSIONS ARE ALUMINUM ALLOY 7075-T6511, QQ-A-200-11.
2. LEFT AND RIGHT SIDES ARE THE SAME EXCEPT AS NOTED.
3. TOP AND BOTTOM.
4. 701067 – SUBQ
5. 70583 – 701281
6. 701029 – SUBQ
7. 701292 – SUBQ
8. IF HARDWARE IS STAINLESS STEEL, ASSEMBLE WET WITH SEALING COMPOUND, ITEM 292, APPENDIX D.
9. 701391 – SUBQ

FB1180_8B
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FIGURE 2-6-40-1. REPAIR TAIL CONE FRAMING (SHEET 8 OF 8)

2-6-41 TAIL CONE PLATING REPAIR.

PARAGRAPH BREAKDOWN

	PAGE
2-6-41.1 Repair Tail Cone Plating	2-6-41-1
INITIAL SETUP	• PARA 2-6-3
Tools	• PARA 2-6-5
• Airframe Repairer’s Toolkit	• PARA 2-6-43
References	• QQ-A-250
• PARA 2-6-1 • PARA 2-6-2	

2-6-41.1 Repair Tail Cone Plating.

- a. Tail cone plating is Alclad aluminum, 7075-T6, QQ-A-250/13, and except for access panels and reinforced plastic components, is primary structure. Evaluate damage to plating (PARA 2-6-1) (Figure 2-6-41-1, Sheet 1 through Sheet 4).
- b. Refer to PARA 2-6-2 for typical primary skin repairs.

- c. Refer to PARA 2-6-3 for typical reinforced plastic repairs.
- d. Refer to PARA 2-6-43 for tail drive shaft fairing repair information.
- e. For corrosion protection and watertightness, seal all repairs of tail cone plating (PARA 2-6-1).
- f. Touch up paint as required (PARA 2-6-5).

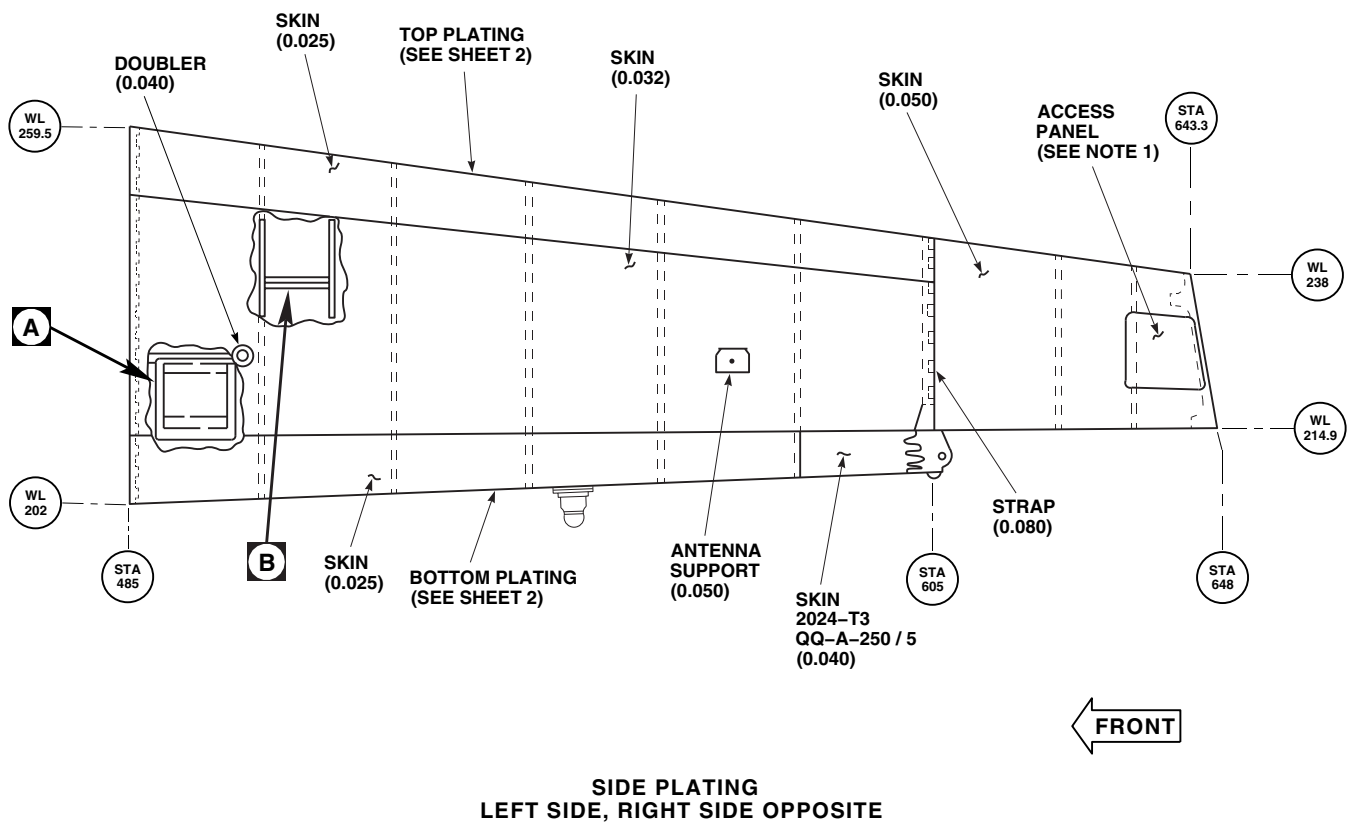
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FIGURE 2-6-41-1. REPAIR TAIL CONE PLATING (SHEET 1 OF 4)

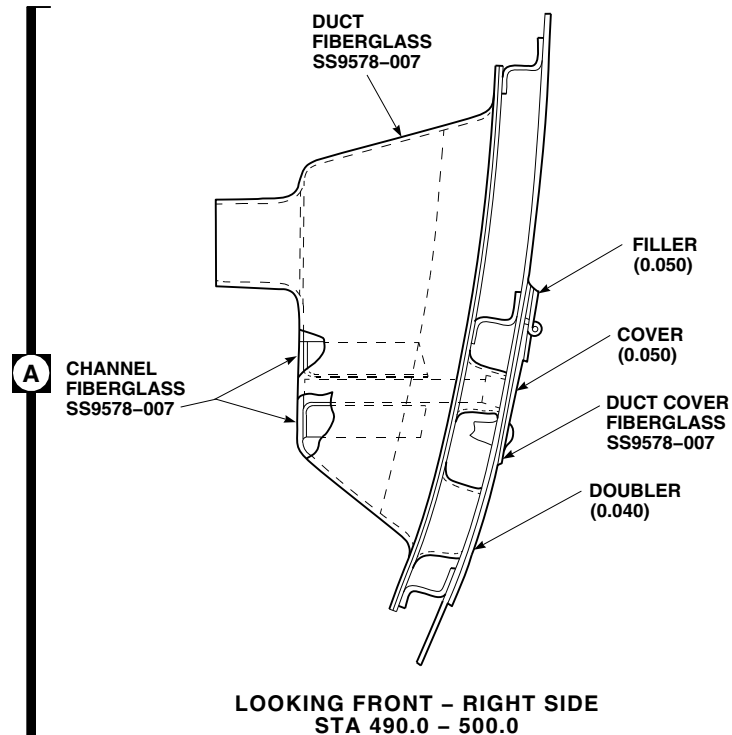
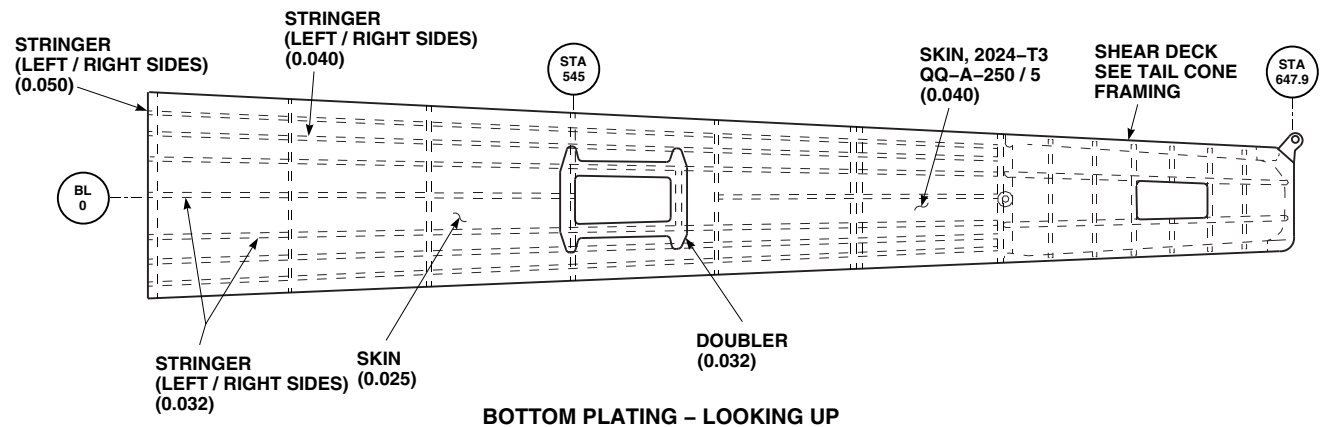
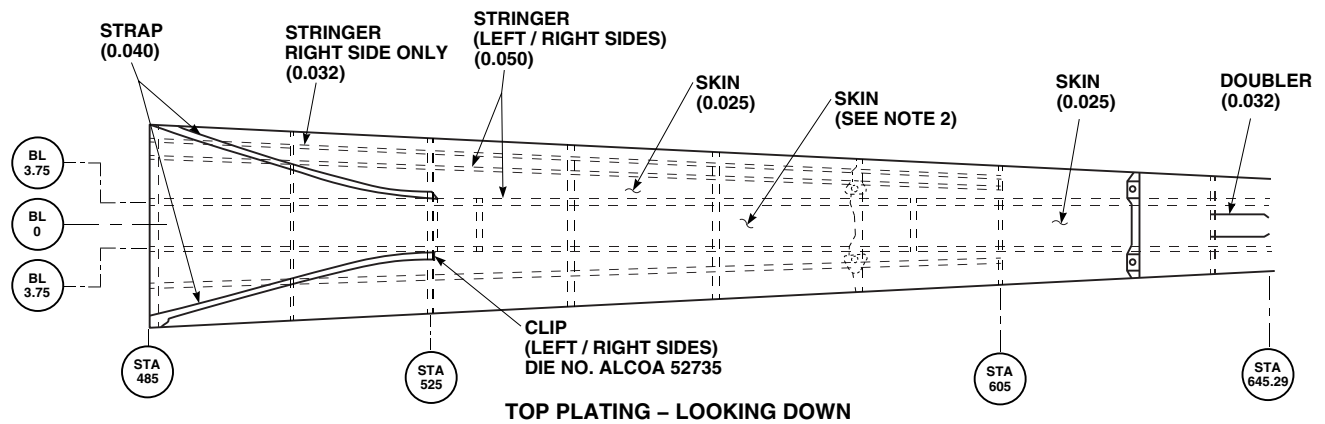


FIGURE 2-6-41-1. REPAIR TAIL CONE PLATING (SHEET 2 OF 4)

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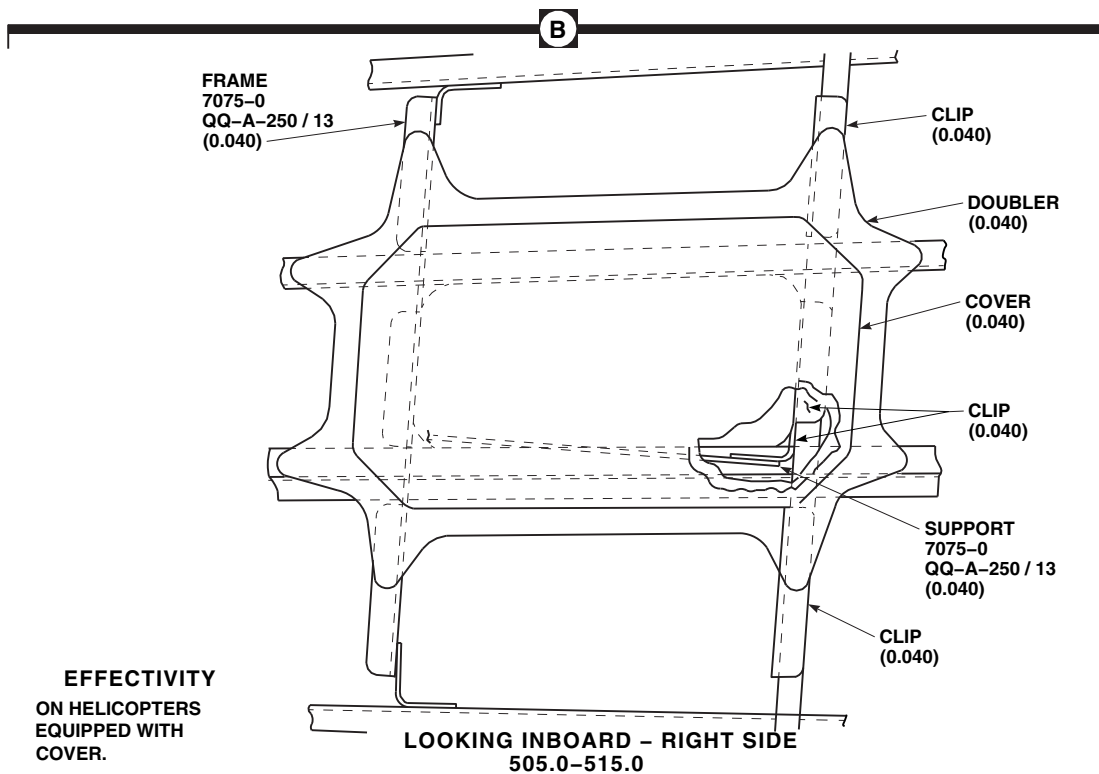


FIGURE 2-6-41-1. REPAIR TAIL CONE PLATING (SHEET 3 OF 4)

NOTES

1. ALUMINUM 7075-T6 0.063
2. 70583-701160 0.025 701172-SUBQ 0.040
3. ALL DIMENSIONS ARE IN INCHES, UNLESS NOTED.

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FIGURE 2-6-41-1. REPAIR TAIL CONE PLATING (SHEET 4 OF 4)

2-6-42 TAIL CONE CANTED HINGE BULKHEAD REPAIR.

PARAGRAPH BREAKDOWN

	PAGE
2-6-42.1 Repair Tail Cone Canted Hinge Bulkhead	2-6-42-2

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Fluorescent Penetrant Inspection Kit
- Inside Micrometer
- Nonmetallic Scraper, Locally-Made

Materials/Parts

- Abrasive Cloth, Item 2, Appendix D
- Abrasive Cloth, Item 6, Appendix D
- Adhesive, Item 33A, Appendix D
- Adhesive, Item 66F, Appendix D
- Alcohol, Denatured, Item 70, Appendix D
- Bolt, AN8-13
- Cheesecloth, Item 90, Appendix D
- Corrosion Compound, Item 108, Appendix D

- Machinery Towel, Item 211, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Nut, AN310-9
- Sealing Compound, Item 292, Appendix D
- Tape, Antiseize, Item 316A, Appendix D
- Washer, NAS1149F0963P

References

- Appendix D
- ASTM E1417
- PARA 1-6-9
- PARA 2-6-1
- PARA 2-6-5
- PARA 5-3-11
- QQ-A-250
- QQ-A-367

2-6-42.1 Repair Tail Cone Canted Hinge Bulkhead.

WARNING

FSCAP

Canted hinge bulkhead fitting has critical areas with critical dimensions. Limits specified must not be exceeded.

- a. Canted hinge bulkhead fitting for tail cone is machined from forged aluminum alloy, 7075-T73, QQ-A-367. Fitting is considered critical primary structure. Bushings, NAS75-6-008, are press-fit into rear side of fitting and inserts, NAS1395-5L, are threaded into front side of fitting. Bushings for upper and lower hinge lugs are cadmium-plated steel and are press-fit into lugs (Figure 2-6-42-1, Sheet 1 through Sheet 3).
- b. Fold tail rotor pylon (PARA 1-6-9).
- c. Inspect/repair canted hinge bulkhead as follows:
 - (1) Negligible damage is limited to nicks, scratches, and corrosion, no deeper than 5% of material thickness at point of damage. Damage greater than negligible limits must be reviewed by engineering authority for damage analysis and recommended repair or replacement.
 - (2) Blend damage smoothly into surrounding area with abrasive cloth, Item 2, Appendix D, using long, even strokes.
 - (3) Polish out all surface imperfections with abrasive cloth, Item 6, Appendix D, and restore protective finish.
 - (4) Using fluorescent penetrant inspection kit, inspect canted hinge bulkhead (ASTM E1417). Cracks and gouges may be repaired per step c. (2) and step c. (3) provided requirements for step c. (1) are met after repair.

- (5) Touch up paint and primer as required (PARA 2-6-5).

WARNING

FSCAP

Installation of radius pads is a critical characteristic. Make sure pads are present and securely bonded to tail cone.

NOTE

- One pad is required at each tail rotor pylon attachment bolthole location.
 - Make sure radius pads are not painted.
- d. Inspect front side of tail cone canted hinge bulkhead for radius pad installation. **LOOSE OR MISSING RADIUS PADS NOT ALLOWED.** If radius pads are loose, remove using nonmetallic scraper.
 - e. Prepare canted hinge bulkhead surfaces for pad installation as follows (Figure 2-6-42-1, Sheet 1):
 - (1) Using nonmetallic scraper, remove adhesive residue.
 - (2) Using machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D, thoroughly remove all remaining adhesive residue. Wipe dry with clean machinery towel, Item 211, Appendix D.
 - f. Prepare radius pad for installation as follows:
 - (1) Using inside micrometer, measure diameter of corresponding bulkhead mounting hole where radius pad will be installed.
 - (2) Open hole in radius pad to match measurement taken in step f. (1).
 - (3) Break all sharp edges.
 - (4) Touch up paint and primer as required (PARA 2-6-5).

- g. Bond radius pad on canted hinge bulkhead as follows (Figure 2-6-42-1, Sheet 2, Section B-B):

- (1) Treat bonded surfaces on bulkhead and radius side of pad with corrosion compound, Item 108, Appendix D, followed by a thorough rinsing with clean water.
- (2) While still wet, check that surfaces are water-break free.
- (3) Repeat step g. (1) if necessary.
- (4) Rinse surfaces with solution of one part denatured alcohol, Item 70, Appendix D, and one part water. Allow to dry.

NOTE

- Application of antiseize tape to bolt is required in order to maintain concentricity during bond cure.
 - Apply tape to bolt grip. Do not apply to threads.
 - When applying tape, make sure it tapers off at end of bolt grip. This will reduce the possibility of tape catching on mounting hole during installation.
- (5) Wrap bolt, AN8-13, with antiseize tape, Item 316A, Appendix D.

NOTE

Make sure radius pad is snug on bolt. If radius pad is loose, remove pad and apply additional antiseize tape.

- (6) Carefully place radius pad on bolt, making sure treated bond surface is not contaminated by antiseizing tape or by touching surface. Make sure bond surface is orientated correctly (bonding surface to face bulkhead).

NOTE

Pot life of adhesives is 30 minutes at 75°F (24°C).

- (7) Prepare adhesive, Item 33A, Appendix D, or adhesive, Item 66F, Appendix D, per manufacturer's instructions.
- (8) Apply coat of adhesive on radius pad.

NOTE

When installing bolt, make sure anti-seize tape does not catch on edge of bulkhead mounting hole. Tape caught on edge of hole may become trapped between radius pad and bulkhead.

- (9) Install bolt, with radius pad, through mounting hole in bulkhead.
- (10) Turn radius pad so radius edges make maximum contact with radius in bulkhead.
- (11) Install washer, NAS1149F0963P, and nut, AN310-9, on bolt and tighten nut just enough to pull radius pad solidly against bulkhead. Do not overtighten.
- (12) Clean up adhesive squeezeout using cheesecloth, Item 90, Appendix D, moistened in methyl ethyl ketone, Item 217, Appendix D.
- (13) If adhesive, Item 33A, Appendix D, was used, allow to cure for 5 days at 70° - 85°F (21° - 30°C), or for 1 hour at 140° - 160°F (60° - 71°C).
- (14) If adhesive, Item 66F, Appendix D, was used, allow to cure for 5 days at 60° - 80°F (16° - 27°C), or for 1 hour at 140° - 250°F (60° - 121°C).

NOTE

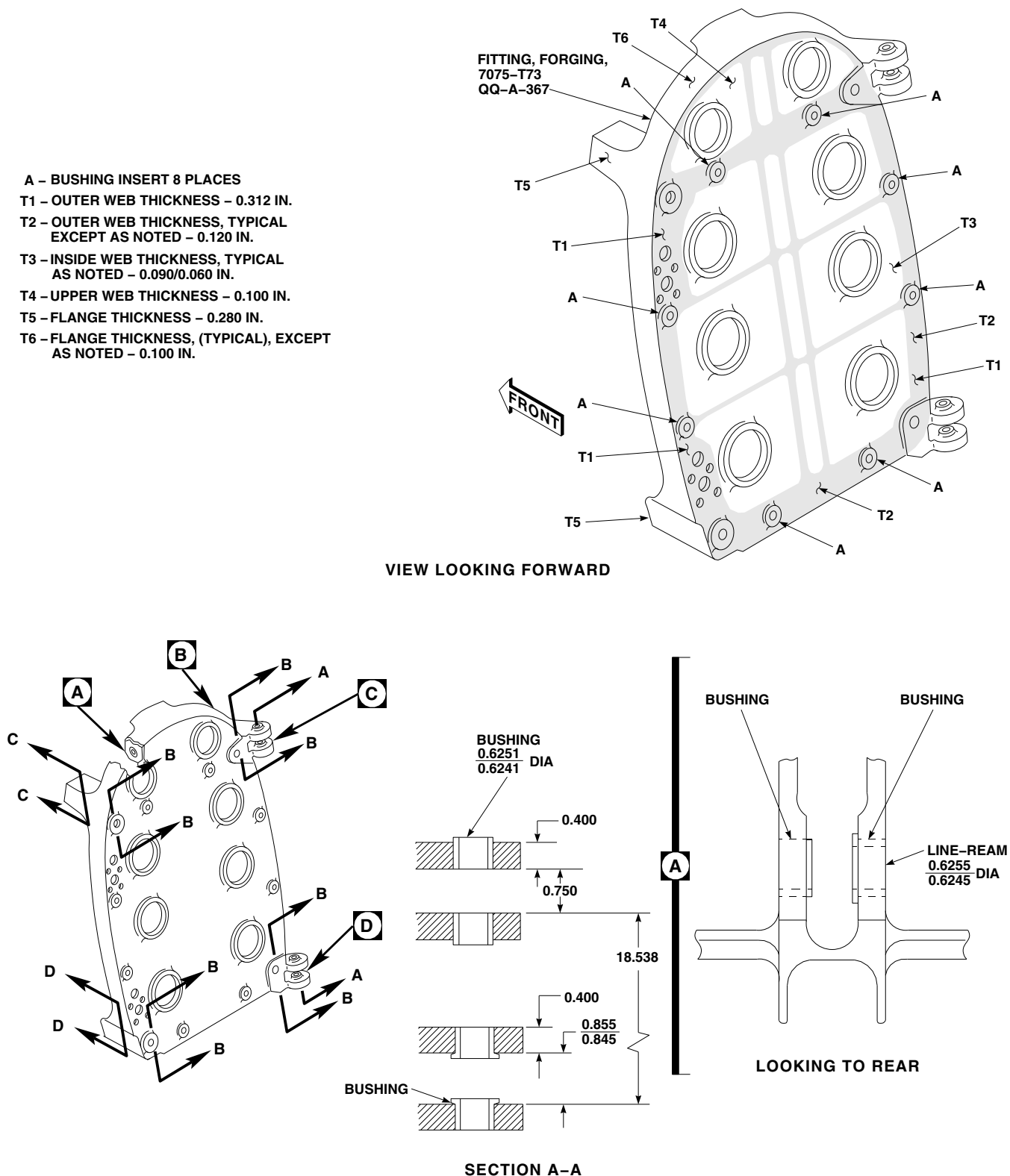
If curing at 70° - 85°F (21° - 30°C) for adhesive, Item 33A, Appendix D, or 60° - 80°F (16° - 27°C) for adhesive, Item 66F, Appendix D, bolt, washer, and nut may be removed after 24 hours.

- (15) After curing, remove bolt, nut, and washer. Make sure any residue from antiseizing tape is removed from bonded assembly.

WARNING**FSCAP**

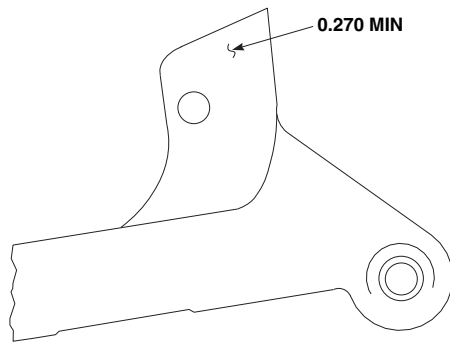
Installation of radius pads is a critical characteristic. Make sure pads are present and securely bonded to tail cone.

- (16) Check bond for voids using coin-tapping method (PARA 5-3-11).
- h. Seal fitting for corrosion protection and watertightness using sealing compound, Item 292, Appendix D (PARA 2-6-1).
- i. Unfold tail rotor pylon (PARA 1-6-9).

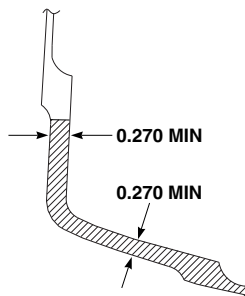
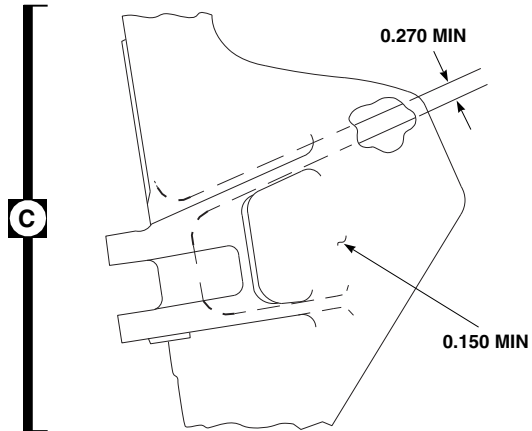


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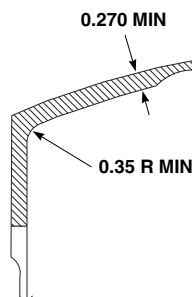
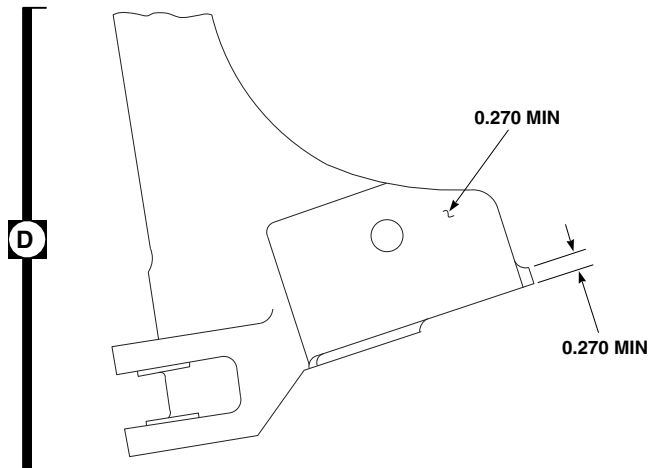
FIGURE 2-6-42-1. REPAIR TAIL CONE CANTED HINGE BULKHEAD (SHEET 1 OF 3)



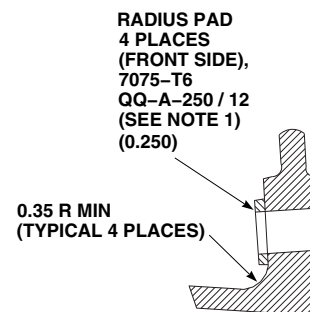
LOOKING DOWN



SECTION D-D



SECTION C-C



SECTION B-B

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FIGURE 2-6-42-1. REPAIR TAIL CONE CANTED HINGE BULKHEAD (SHEET 2 OF 3)

NOTE

1. EACH RADIUS PAD HAS UNIQUE SHAPE TO FIT ITS SEAT IN BULKHEAD.
2. ALL DIMENSIONS ARE IN INCHES, UNLESS NOTED.

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FIGURE 2-6-42-1. REPAIR TAIL CONE CANTED HINGE BULKHEAD (SHEET 3 OF 3)

2-6-43 TAIL ROTOR DRIVE SHAFT COVERS REPAIR.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
2-6-43.1	Repair Tail Rotor Drive Shaft Fairing	2-6-43-1	2-6-43.3	Replace Tail Rotor Drive Shaft Fairing Lower Hinge	2-6-43-3
2-6-43.2	Replace Tail Rotor Drive Shaft Fairing Upper Hinge	2-6-43-1			

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Clecos, 1⁄8-inch diameter
- Cover Indexing Tool
- Drill
- Drill Bit, No. 30 (0.1285-inch diameter)
- Drill Bit, 1⁄8-inch diameter

Materials/Parts

- Rivets, MS20470AD4
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 2-6-1
- PARA 2-6-2
- PARA 2-6-3
- PARA 2-6-5

2-6-43.1 Repair Tail Rotor Drive Shaft Fairing.

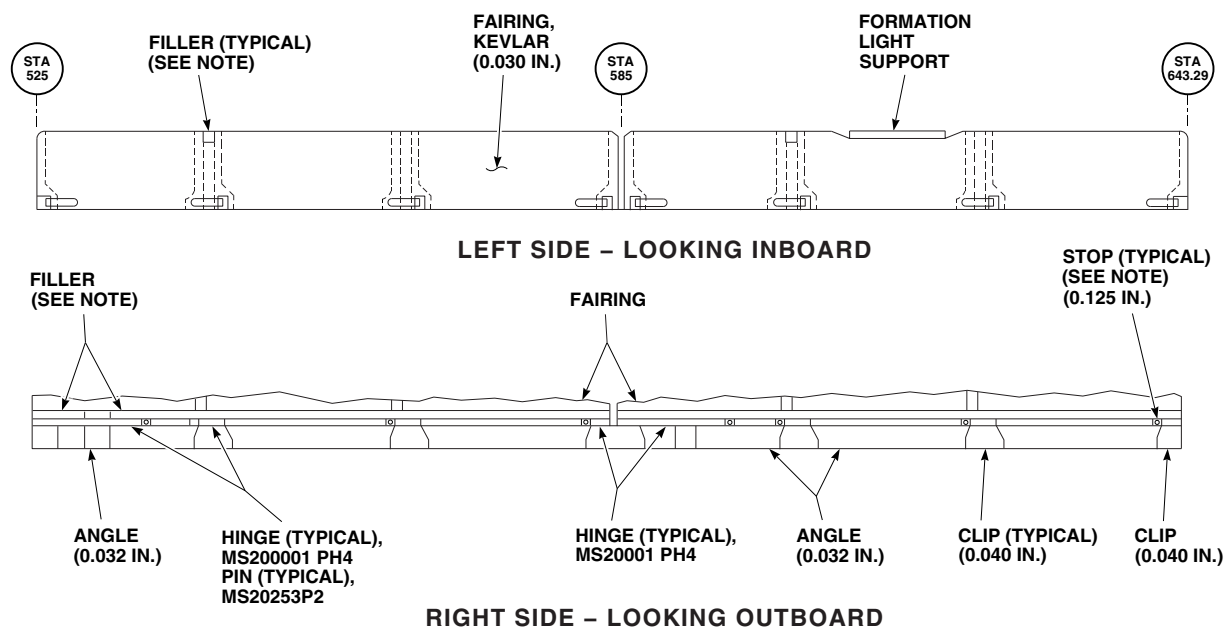
- Hinged fairing covers for tail rotor drive shaft are made of Kevlar reinforced plastic with aluminum alloy clips, angles, hinges, and turn-lock fasteners (Figure 2-6-43-1).
- Evaluate damage and make repairs to Kevlar covers (PARA 2-6-3).
- Evaluate damage and make repairs to metal components (PARA 2-6-2).
- Replace fasteners.

- Make sure area is clean and free of foreign material.
- Touch up paint in repaired area as required (PARA 2-6-5).

2-6-43.2 Replace Tail Rotor Drive Shaft Fairing Upper Hinge.

2-6-43.2.1 Remove.

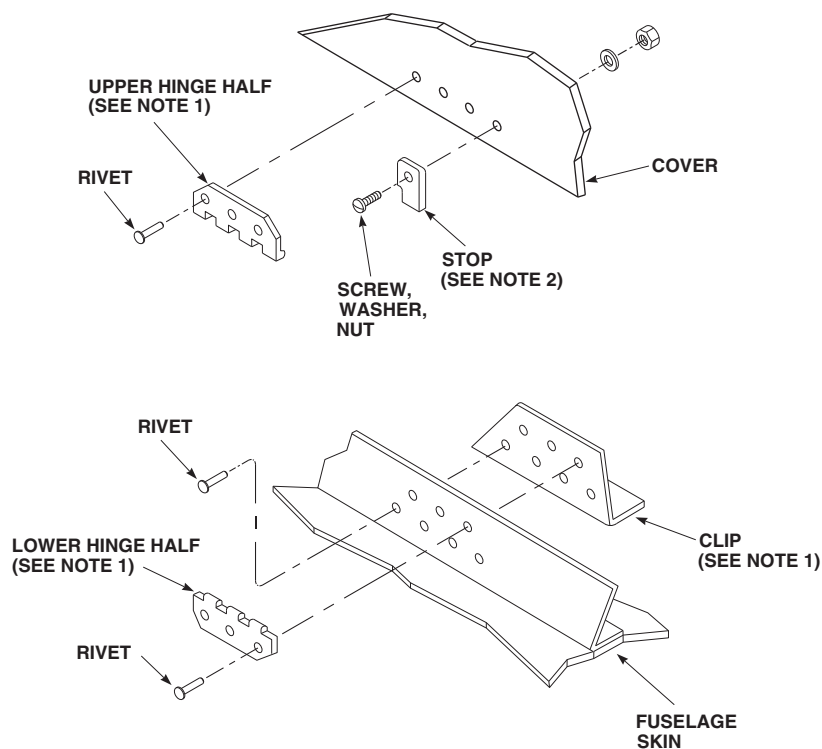
- Using drill with 1⁄8-inch diameter drill bit (Figure 2-6-43-2), carefully drill out and remove rivets securing upper hinge half to cover.

**NOTE**

FILLERS MADE FROM CELLULAR
CELLULOSE ACETATE BONDED INTO
FAIRING.

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FIGURE 2-6-43-1. REPAIR TAIL ROTOR DRIVE SHAFT COVER

**NOTES**

1. TYPICAL, EXCEPT FOR LENGTH.
2. TYPICAL, EXCEPT FOR LOCATION.

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FIGURE 2-6-43-2. REPLACE TAIL ROTOR DRIVE SHAFT COVER UPPER/LOWER HINGE

- b. Open tail rotor drive shaft cover.
- c. Remove nut, washer, and screw securing stop to cover. Remove stop.
- d. Remove hinge pin and upper hinge half.

2-6-43.2.2 Install.

NOTE

Use old hinge half for hole template, if suitable, or cover indexing tool.

- a. Using drill with No. 30 drill bit, lay out and drill holes in replacement hinge half.
- b. For corrosion protection, apply sealing compound, Item 292, Appendix D, under hinge and clip before assembling (PARA 2-6-1).
- c. Using 1/8-inch diameter clecos, position hinge half on fairing and temporarily secure in place.
- d. Line up hinge with lower hinge half and install hinge pin with swaged end on pin stop side (Figure 2-6-43-2).
- e. Coat shank of all fasteners with sealing compound, Item 292, Appendix D, before installing (PARA 2-6-1).
- f. Secure stop to cover with screw, washer, and nut.
- g. Install required rivets between clecos. Remove clecos and install remainder of rivets.
- h. Check open/close function of cover.
- i. Make sure area is clean and free of foreign material.
- j. Touch up paint in repaired area (PARA 2-6-5).

2-6-43.3 Replace Tail Rotor Drive Shaft Fairing Lower Hinge.

2-6-43.3.1 Remove.

NOTE

Rivets securing clip to angle should not be removed unless clip is to be replaced.

- a. Using drill with 1/8-inch diameter drill bit, carefully drill out and remove rivets securing lower hinge half to angle and clip (Figure 2-6-43-2).
- b. Open tail rotor drive shaft cover and remove nut, washer, and screw securing stop to cover. Remove stop.
- c. Remove hinge pin and lower hinge half.
- d. Make sure area is clean and free of foreign material.

2-6-43.3.2 Install.

NOTE

Use old hinge half for hole template, if suitable, or cover indexing tool.

- a. Using drill with No. 30 drill bit, lay out and drill holes in replacement hinge half (Figure 2-6-43-2).
- b. For corrosion protection, apply sealing compound, Item 292, Appendix D, under hinge and clip before assembling (PARA 2-6-1).
- c. Position hinge half on angle and temporarily hold in place, using 1/8-inch diameter clecos.
- d. Line up hinge with upper hinge half and install hinge pin with swaged end on pin stop side.
- e. Coat shank of all fasteners with sealing compound, Item 292, Appendix D, before installing (PARA 2-6-1).
- f. Secure stop to cover with screw, washer, and nut.
- g. Install required rivets, MS20470AD4, between clecos. Remove clecos and install remainder of rivets.
- h. Check open/close function of cover.
- i. Make sure area is clean and free of foreign material.
- j. Touch up paint in repaired area (PARA 2-6-5).

2-6-44 TAIL ROTOR PYLON FRAMING REPAIR.

PARAGRAPH BREAKDOWN

	PAGE
2-6-44.1 Repair Tail Rotor Pylon Framing	2-6-44-1

INITIAL SETUP

- PARA 2-6-1

Tools

- PARA 2-6-2

- Airframe Repairer’s Toolkit

- PARA 2-6-4

Materials/Parts

- PARA 2-6-5

- Abrasive Cloth, Item 6, Appendix D

- PARA 2-6-51

References

- AMS 4260
- AMS 5334
- Appendix D
- MIL-S-18729
- MIL-S-5000

- QQ-A-200
- QQ-A-225
- QQ-A-250
- QQ-A-367
- WW-T-700

2-6-44.1 Repair Tail Rotor Pylon Framing.

- Tail rotor pylon framing includes formed and extruded members made from 7075-T6, QQ-A-250/13 Alclad aluminum sheet stock and 7075-T6511, QQ-A-200/11 aluminum alloy extrusion (Figure 2-6-44-1, Sheet 1 through Sheet 11). Support and attachment fittings made from 7075-T73, QQ-A-367 aluminum alloy forgings, are used for the canted hinge frame, intermediate and tail gear box supports, spar and stabilator attachment, pylon lower step, and stabilator actuator support.
- All tail rotor pylon framing is primary structure. Fittings, however, are considered

critical primary structure. Negligible damage to the fittings is limited to nicks, scratches, and corrosion no deeper than 5% material thickness at point of damage. Blend damage smoothly to surrounding area using long, even strokes. Polish with abrasive cloth, Item 6, Appendix D and restore protective finish. Damage greater than negligible limits must be reviewed by engineering authority for damage evaluation and either repair or replacement. All tail rotor pylon framing repairs must maintain design strength and alignment.

- Evaluate damage in noncritical areas (PARA 2-6-1).

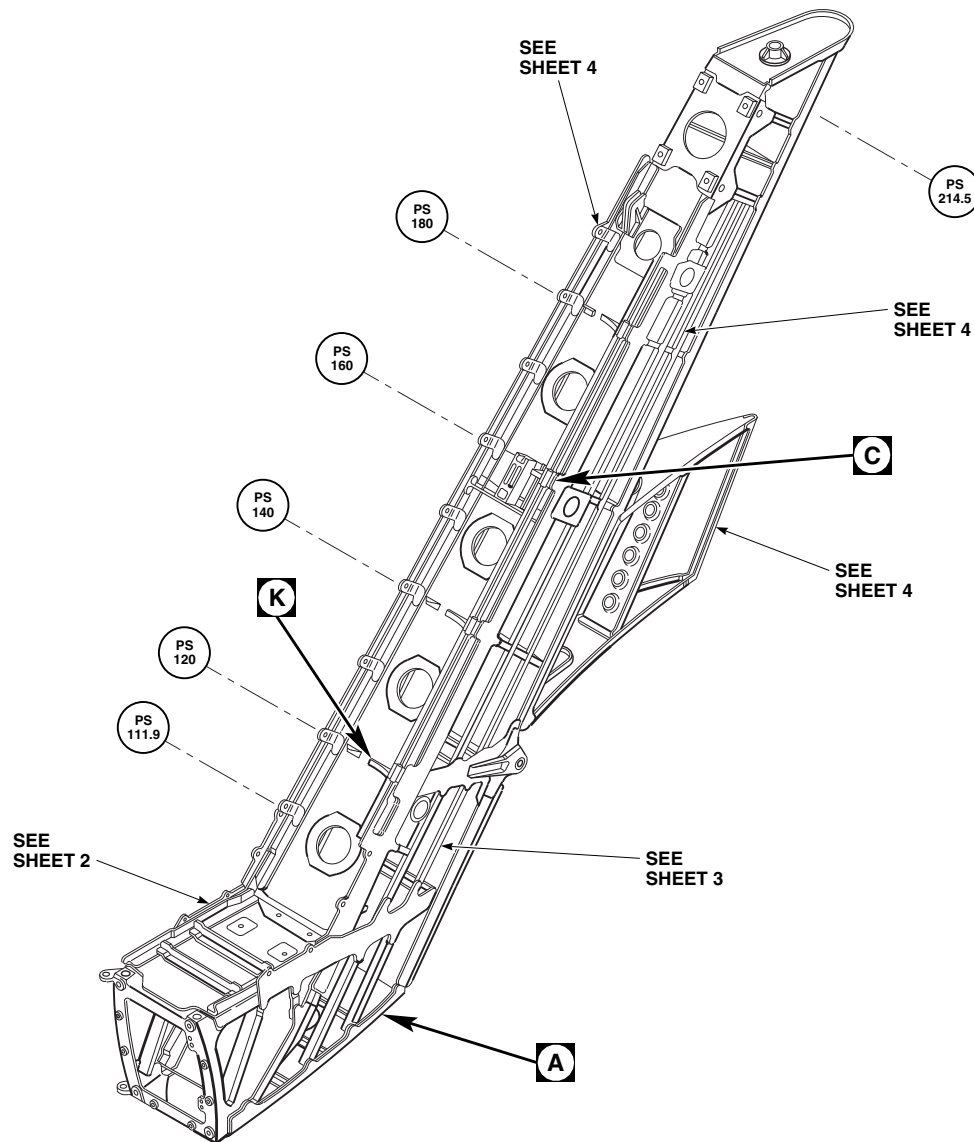
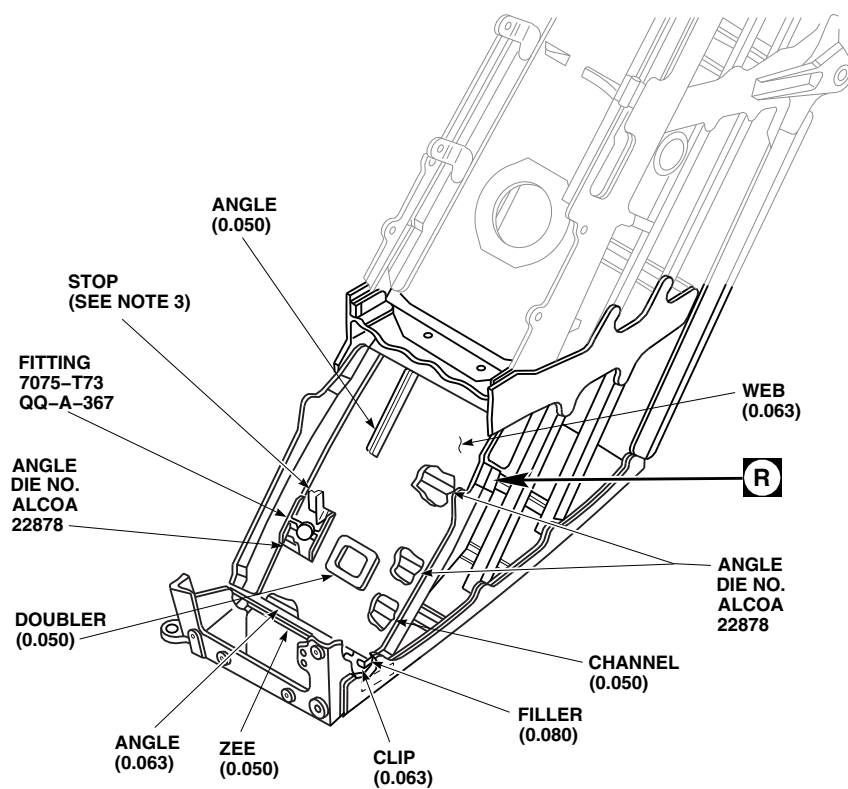
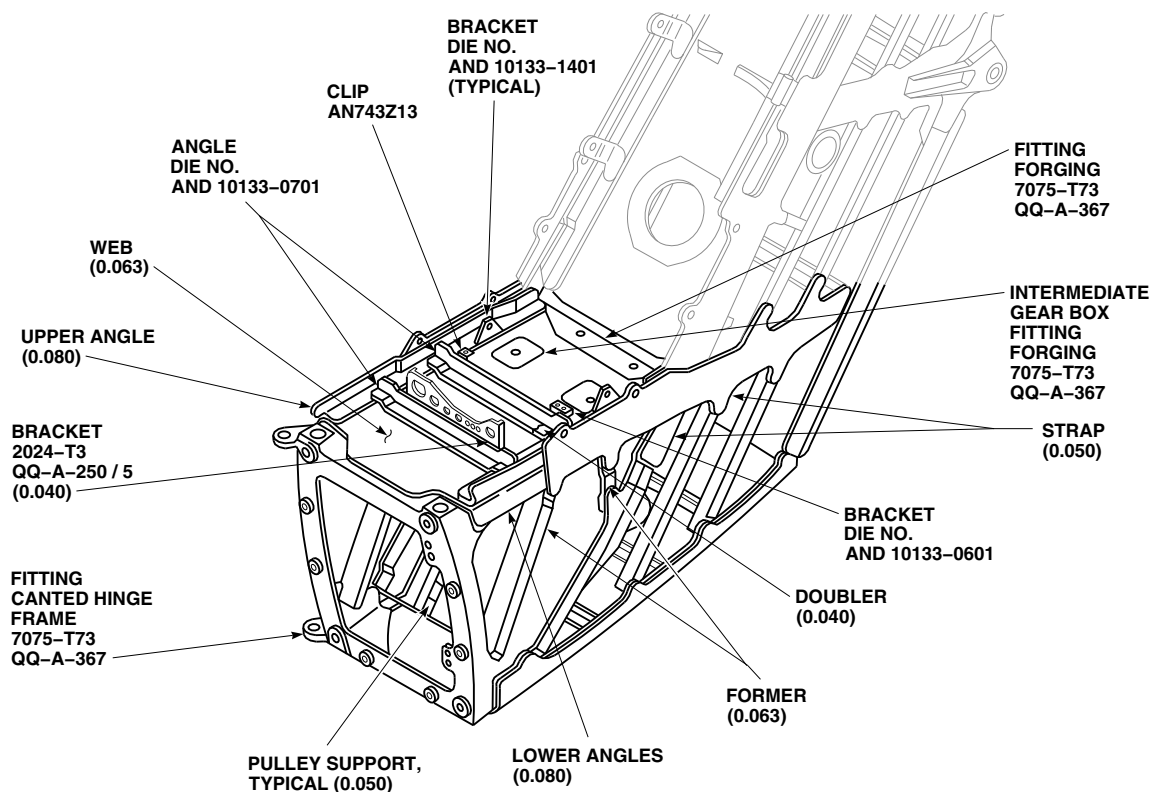
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FIGURE 2-6-44-1. REPAIR TAIL ROTOR PYLON FRAMING (SHEET 1 OF 11)

- d. Use PARA 2-6-2 and PARA 2-6-4 for typical framing repairs.
- e. Refer to PARA 2-6-51 for canted hinge frame fitting repair information.
- f. For corrosion protection and watertightness, seal all repairs between frame members and pylon plating (PARA 2-6-1).
- g. Touch up paint as required (PARA 2-6-5).



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FIGURE 2-6-44-1. REPAIR TAIL ROTOR PYLON FRAMING (SHEET 2 OF 11)

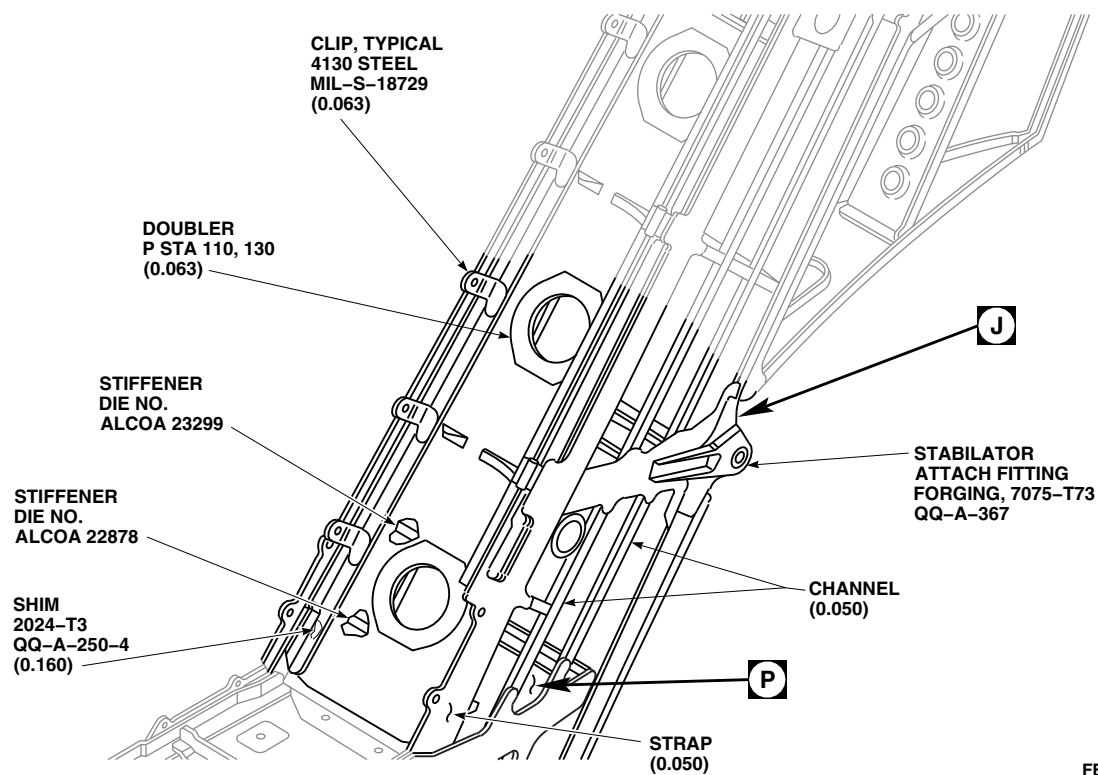
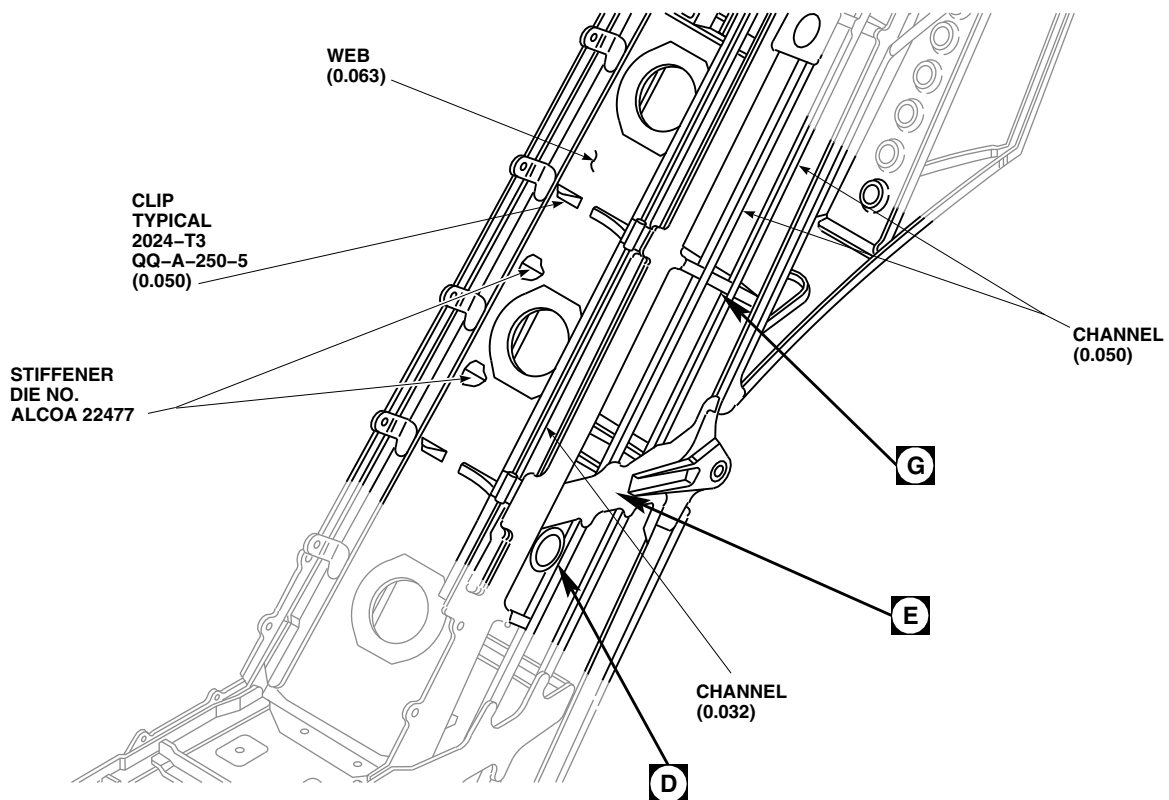
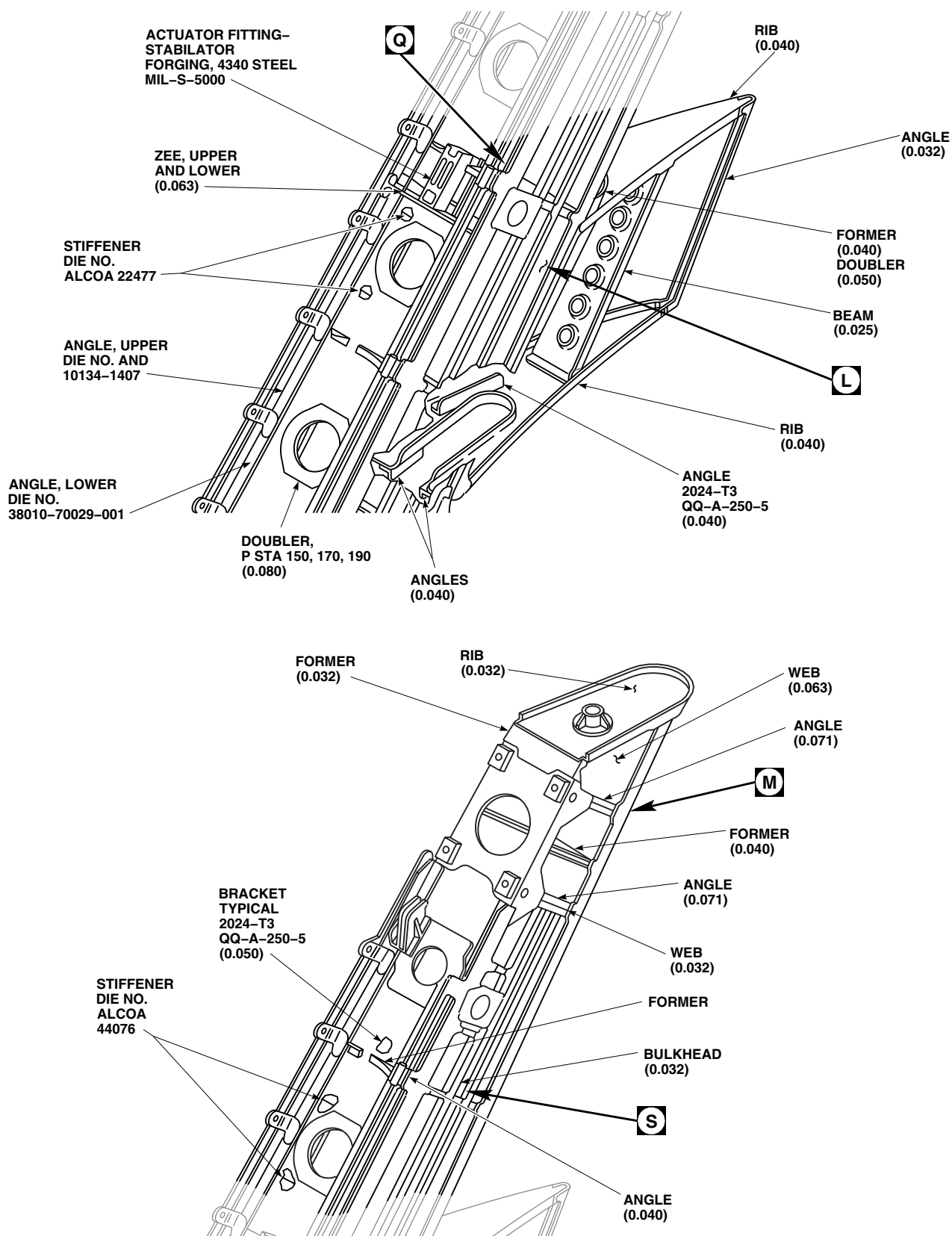
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FIGURE 2-6-44-1. REPAIR TAIL ROTOR PYLON FRAMING (SHEET 3 OF 11)



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FIGURE 2-6-44-1. REPAIR TAIL ROTOR PYLON FRAMING (SHEET 4 OF 11)

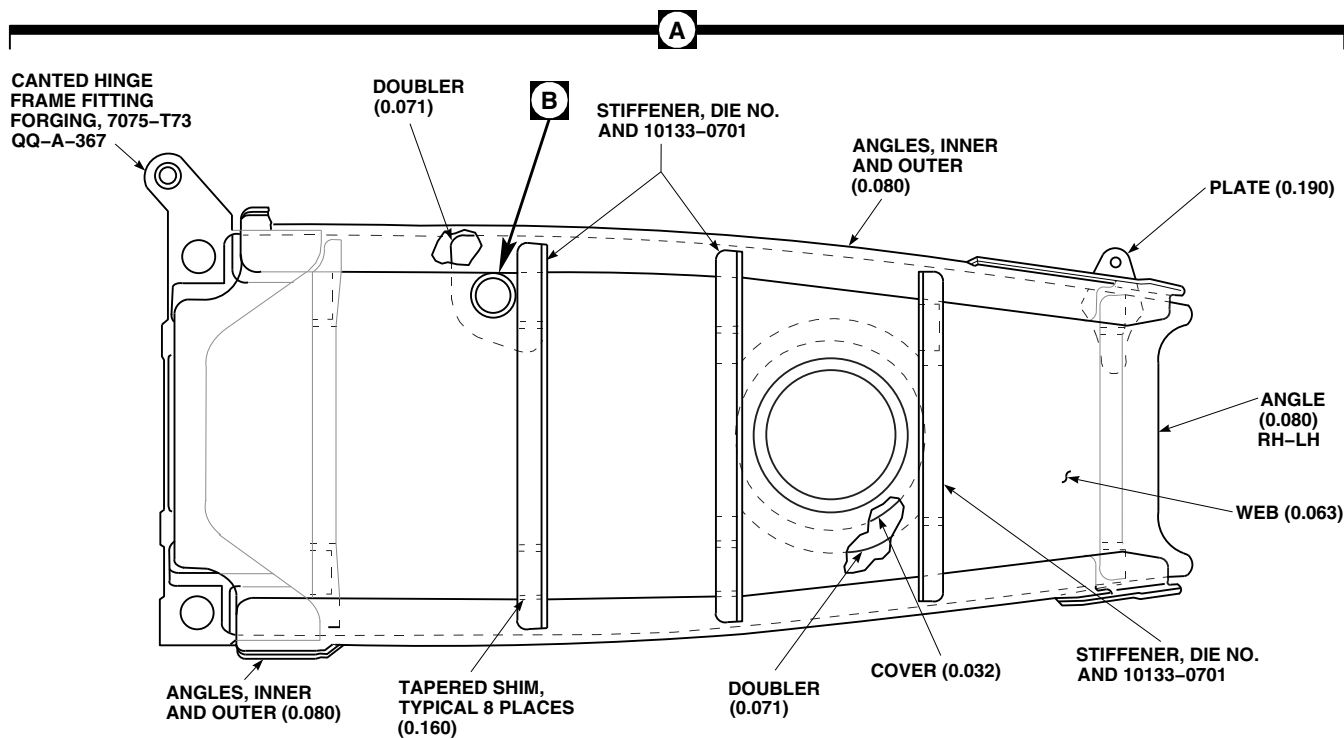
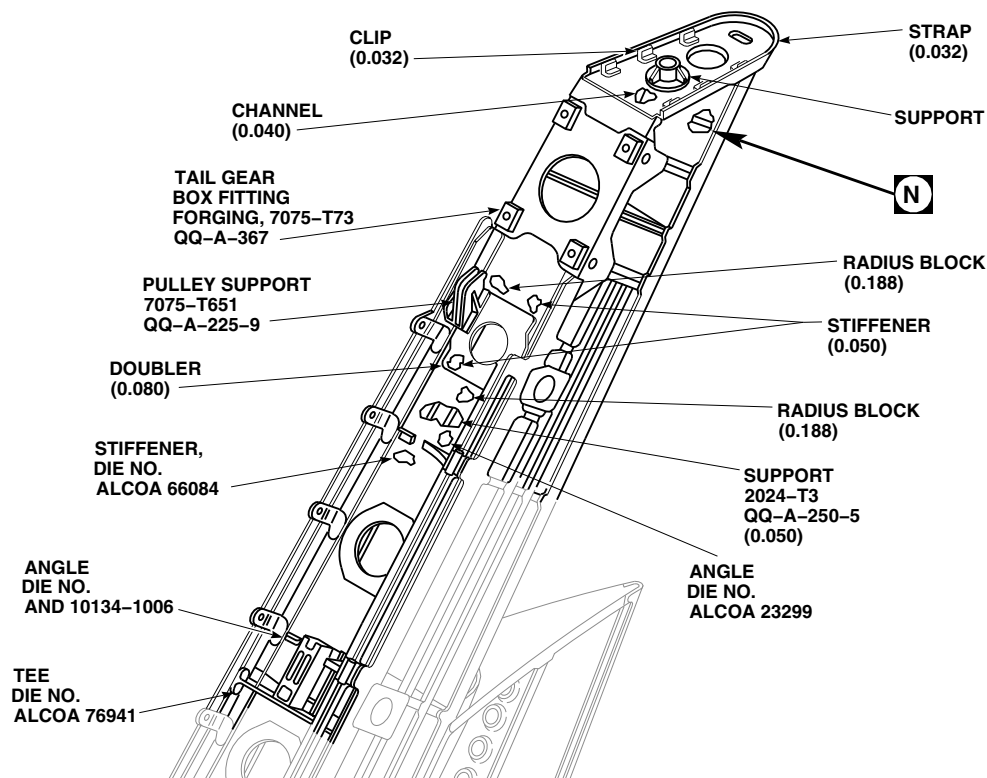
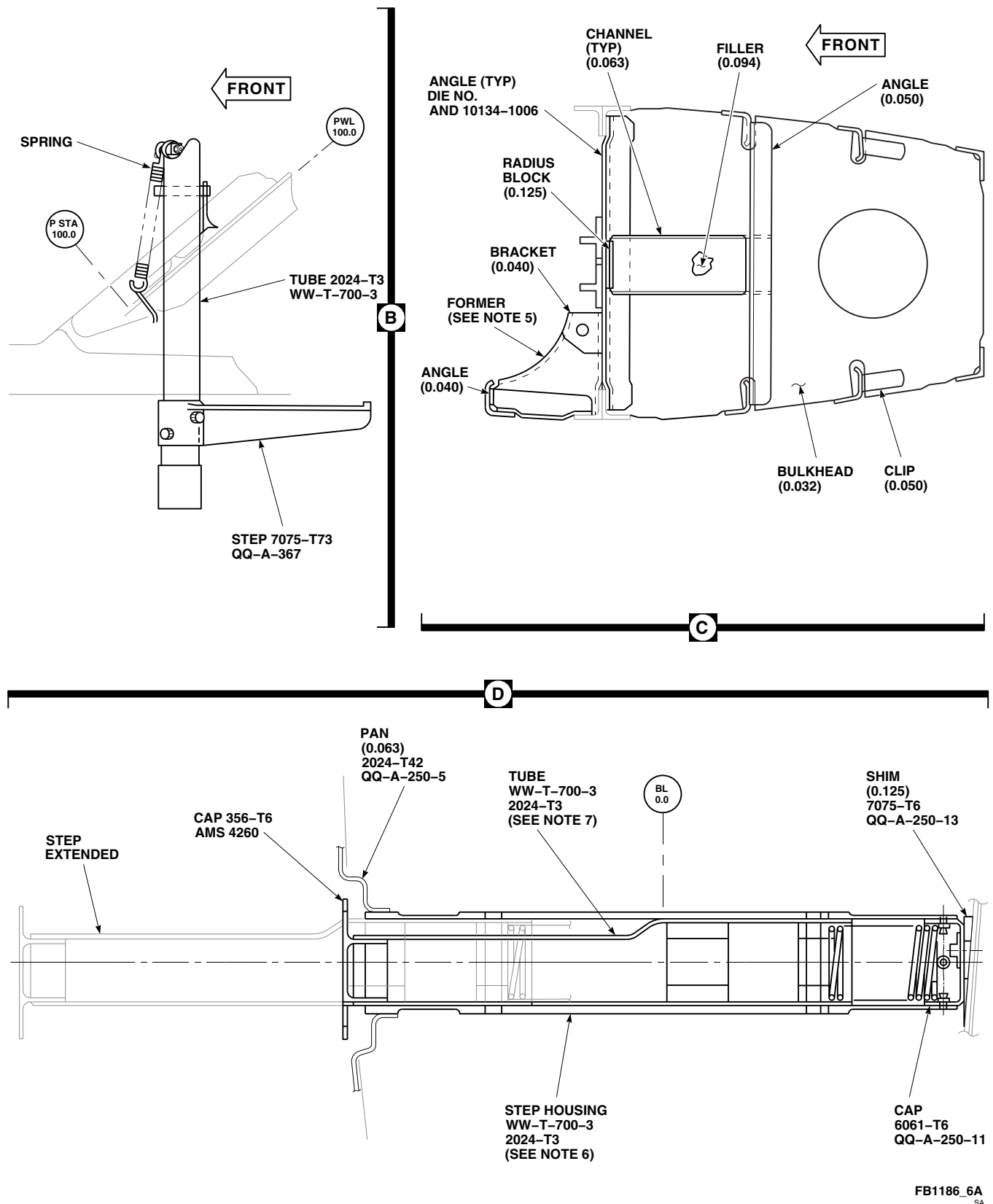
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FIGURE 2-6-44-1. REPAIR TAIL ROTOR PYLON FRAMING (SHEET 5 OF 11)



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FIGURE 2-6-44-1. REPAIR TAIL ROTOR PYLON FRAMING (SHEET 6 OF 11)

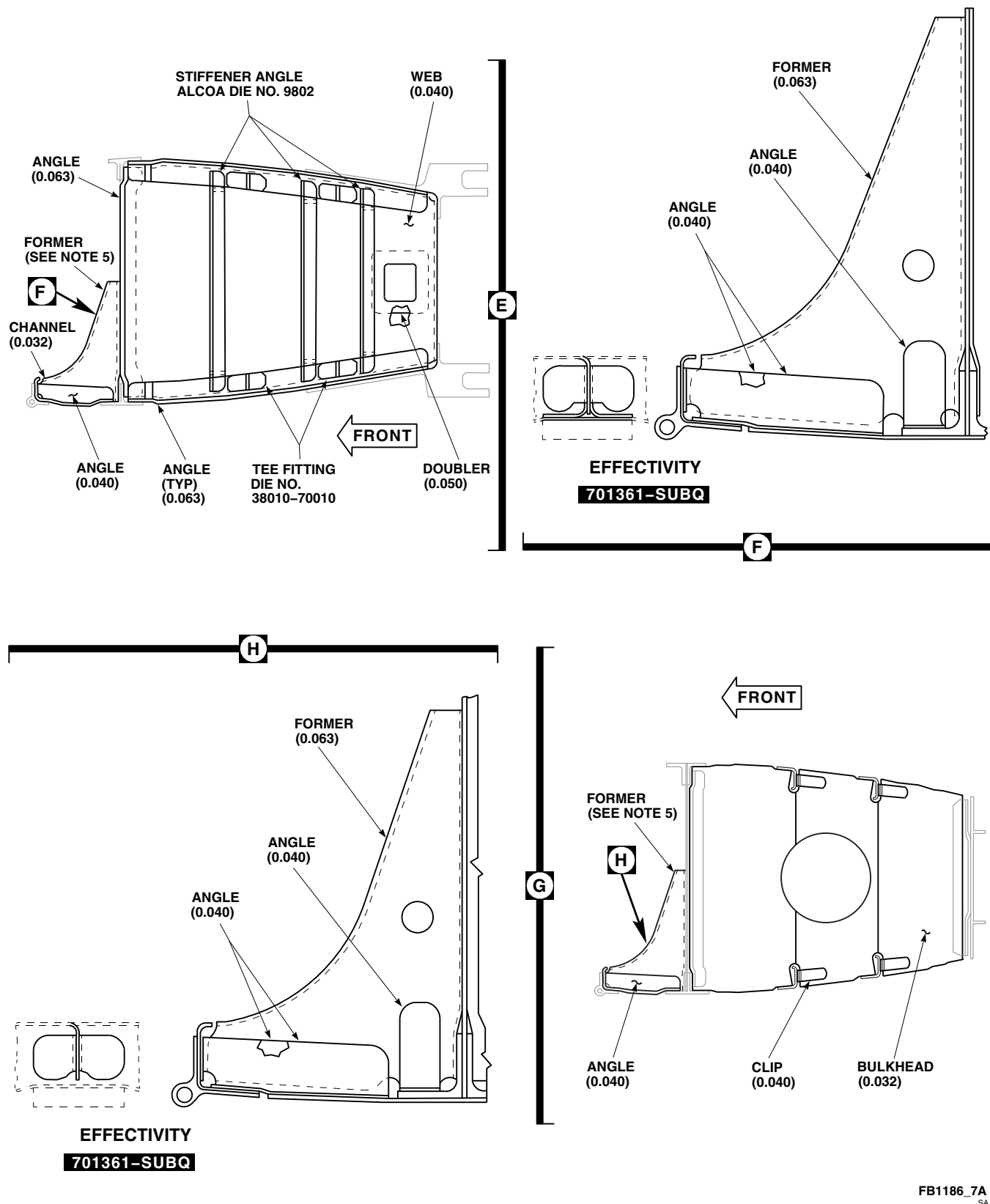
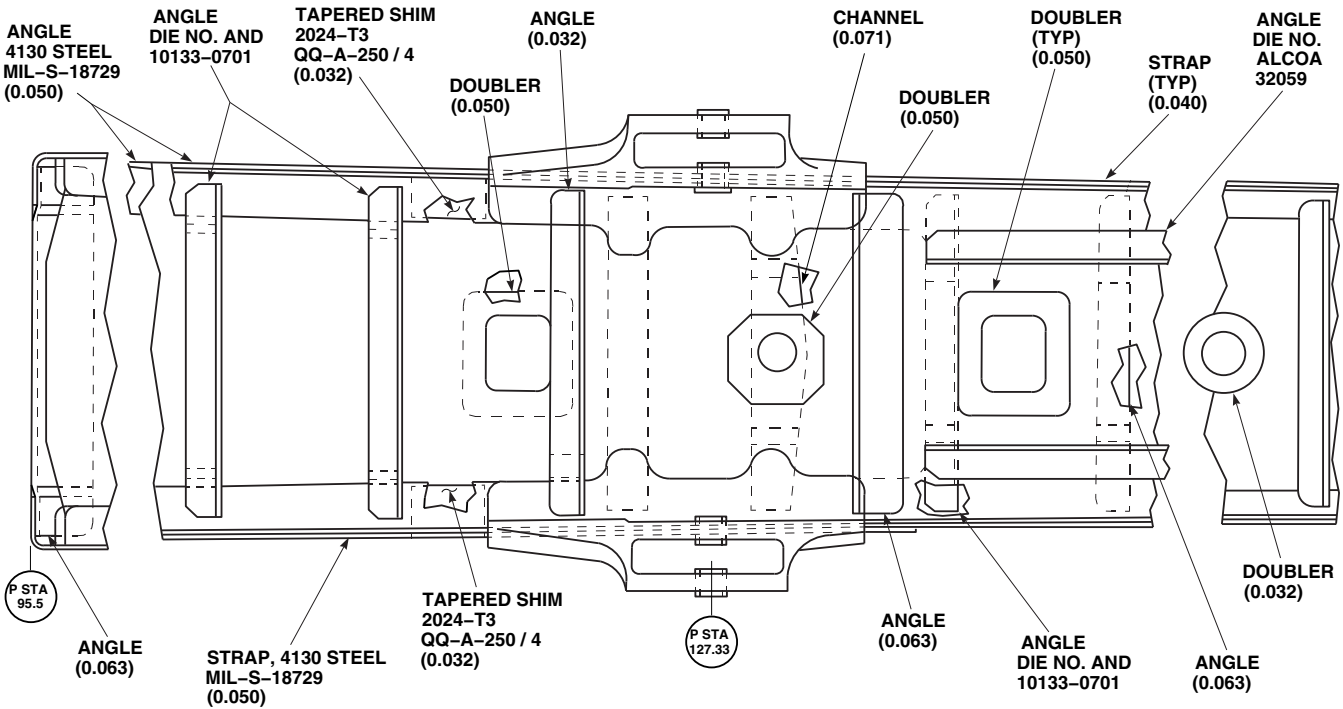


FIGURE 2-6-44-1. REPAIR TAIL ROTOR PYLON FRAMING (SHEET 7 OF 11)

2-6-44-8



REAR SPAR - LOOKING AFT

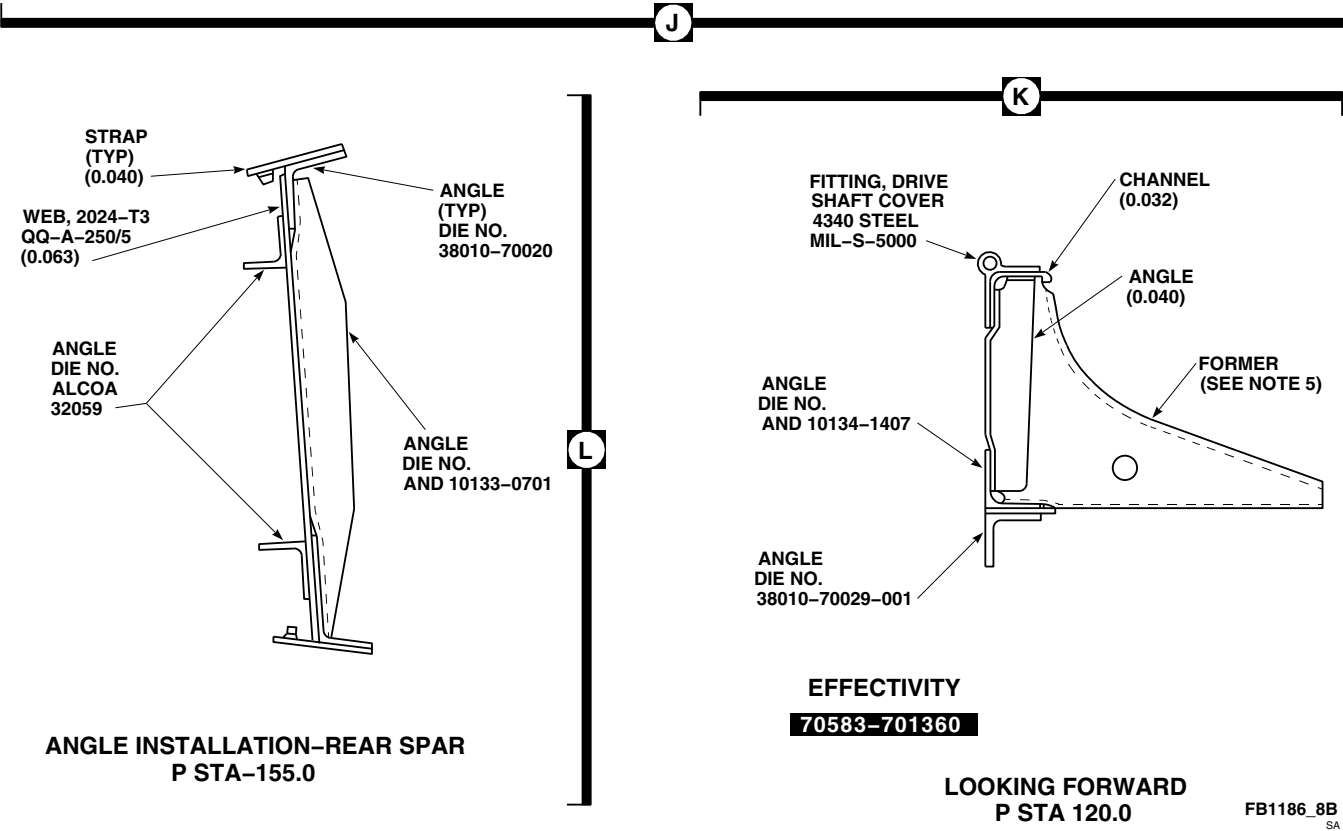


FIGURE 2-6-44-1. REPAIR TAIL ROTOR PYLON FRAMING (SHEET 8 OF 11)

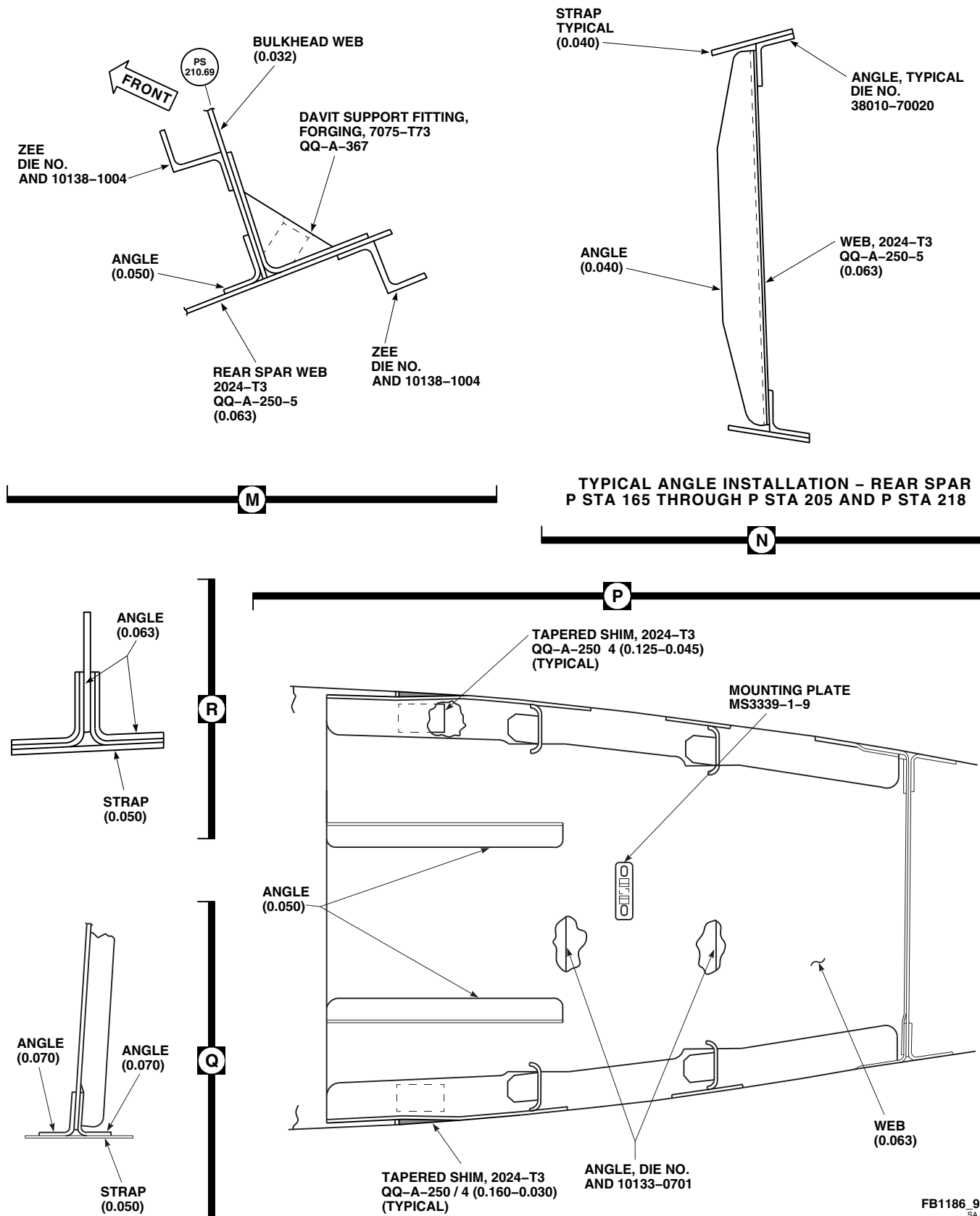


FIGURE 2-6-44-1. REPAIR TAIL ROTOR PYLON FRAMING (SHEET 9 OF 11)

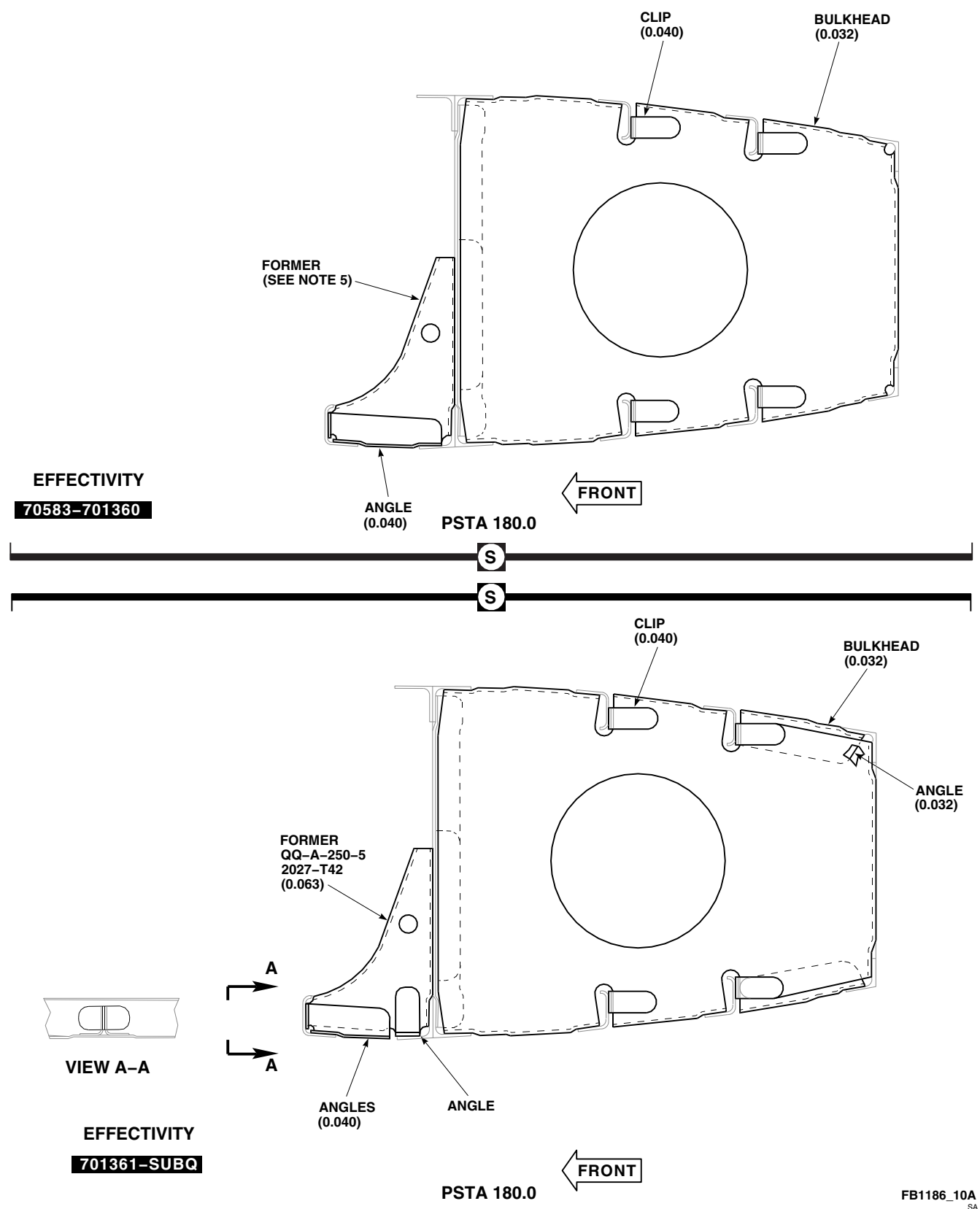


FIGURE 2-6-44-1. REPAIR TAIL ROTOR PYLON FRAMING (SHEET 10 OF 11)

NOTES

1. UNLESS OTHERWISE NOTED, ALL SHEET METAL PARTS SHOWN ARE ALCLAD 7075-T6, QQ-A-250-13, AND ALL EXTRUSION PARTS ARE ALUMINUM ALLOY 7075-T6511, QQ-A-200-11.
2. LEFT AND RIGHT SIDES ARE THE SAME EXCEPT WHEN NOTED.
3. EFFECTIVITY: PYLONS PRIOR TO SERIAL NO. A0330-0816 USE 2024-T3, QQ-A-250-5 (0.050). PYLONS SERIAL NOS A0330-0816 AND SUBSEQUENT USE MIL-S-18729 4130, COND A (0.050).
4. CASTING, AMS 5334.
5. 2024-T42, QQ-A-250-5 (0.063).
6. MADE FROM 38-10-70012-001 SEAMLESS TUBING.
7. MADE FROM TUBING 1.375 IN. OD 0.065 IN WALL.

FB1186_11A
SA**FIGURE 2-6-44-1. REPAIR TAIL ROTOR PYLON FRAMING (SHEET 11 OF 11)**

2-6-45 TAIL ROTOR PYLON PLATING REPAIR.

PARAGRAPH BREAKDOWN

	PAGE
2-6-45.1 Repair Tail Rotor Pylon Plating	2-6-45-1
INITIAL SETUP	MIL-S-5000
Tools	PARA 2-6-1
Airframe Repairer's Toolkit	PARA 2-6-2
Materials/Parts	PARA 2-6-3
Sealing Compound, Item 292, Appendix D	PARA 2-6-5
Steel, Thin Gage	
References	QQ-A-250
Appendix D	
MIL-S-18729	

2-6-45.1 Repair Tail Rotor Pylon Plating.

- a. Tail rotor pylon upper plating, between the forward and rear spars, is Alclad aluminum, 2024-T3, QQ-A-250/5. Lower plating, between spars, is Alclad aluminum 7075-T6, QQ-A-250/13. Leading and trailing edge fairings, and tail rotor and intermediate gear box fairings, are reinforced plastic (Figure 2-6-45-1). Reinforcement straps in the lower pylon are 4130 steel, MIL-S-18729. Upper and lower plating between spars is primary structure. All reinforced plastic or aluminum skin fairings and the cambered fairing are secondary structure. Evaluate damage to the plating, aluminum fairing, and steel straps (PARA 2-6-1).
- b. Use PARA 2-6-2 for typical plating and aluminum fairing repairs. Replace steel straps damaged beyond negligible damage limits.

c. On cambered fairing, wet install any replaced skin fasteners, using sealing compound, Item 292, Appendix D.

d. Use PARA 2-6-3 for evaluating damage and repairing reinforced plastic.

e. For corrosion protection and watertightness, seal all repairs of pylon plating (PARA 2-6-1).

f. Touch up paint as required (PARA 2-6-5).

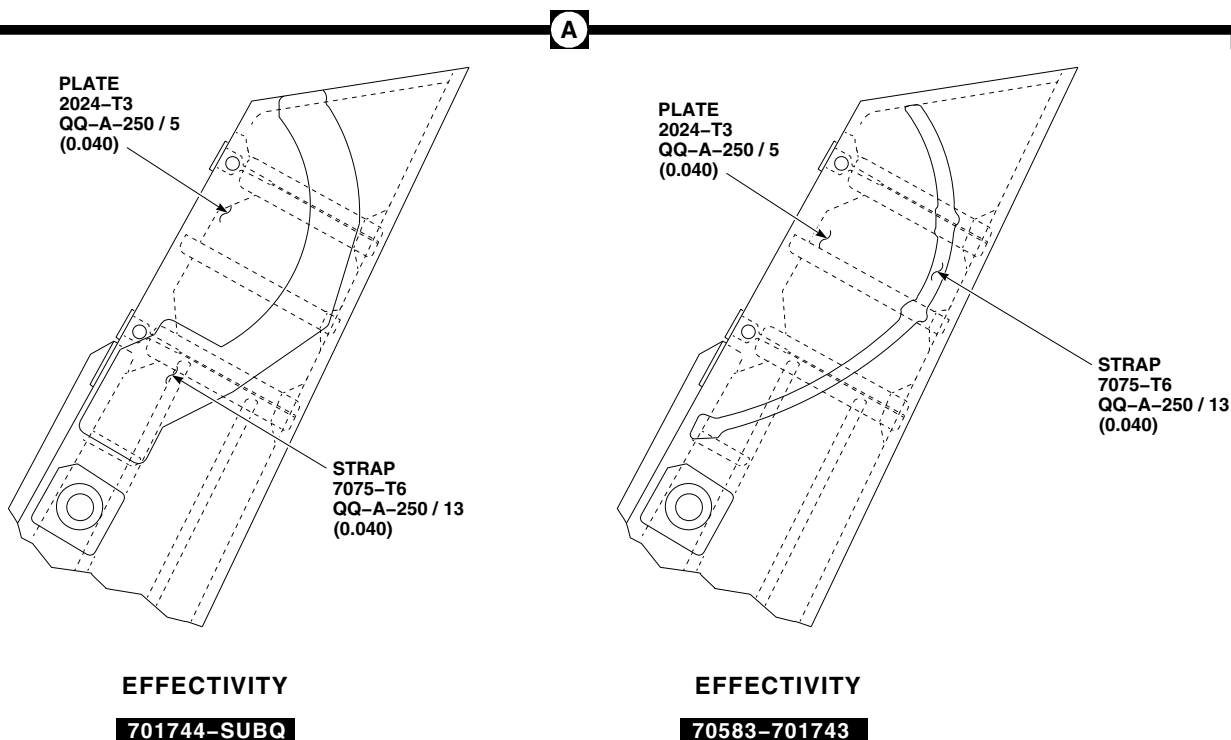
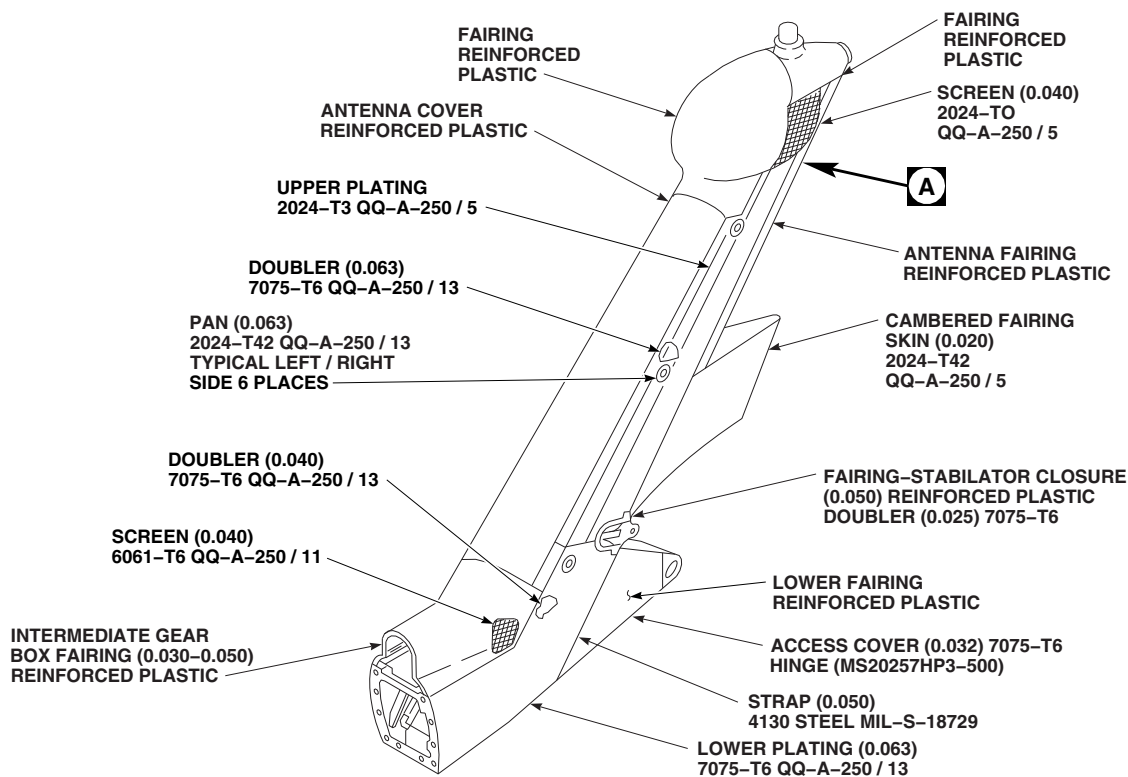
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FIGURE 2-6-45-1. REPAIR TAIL ROTOR PYLON PLATING

2-6-46 TAIL ROTOR PYLON LEFT STABILATOR ATTACHMENT FITTING.

PARAGRAPH BREAKDOWN

		PAGE
2-6-46.1	Replace Left Stabilator Attachment Fitting Bushings and Bearing	2-6-46-1

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Bushing/Bearing Removal/Installation Set, 70700-20129-041
- Feeler Gage, 0.0015-inch

Materials/Parts

- Abrasive Cloth, Item 10, Appendix D
- Cheesecloth, Item 90, Appendix D
- Corrosion Compound, Item 108, Appendix D

- Lubricant, Solid Film, Item 201, Appendix D
- Scouring Pad, Item 270, Appendix D
- Sealing Compound, Item 274, Appendix D
- Trichloroethane 1,1,1, Item 341, Appendix D

References

- Appendix D
- PARA 2-4-132
- PARA 2-4-142

2-6-46.1 Replace Left Stabilator Attachment Fitting Bushings and Bearing.

2-6-46.1.1 Remove.

- Remove stabilator (PARA 2-4-142).
- Remove damaged bushings using bushing/bearing removal/installation set, 70700-20129-041 (Figure 2-6-46-1, Detail A).

- Clean lug bore surfaces using cheesecloth, Item 90, Appendix D moistened with trichloroethane 1,1,1, Item 341, Appendix D.

2-6-46.1.2 Inspect.**WARNING****FSCAP**

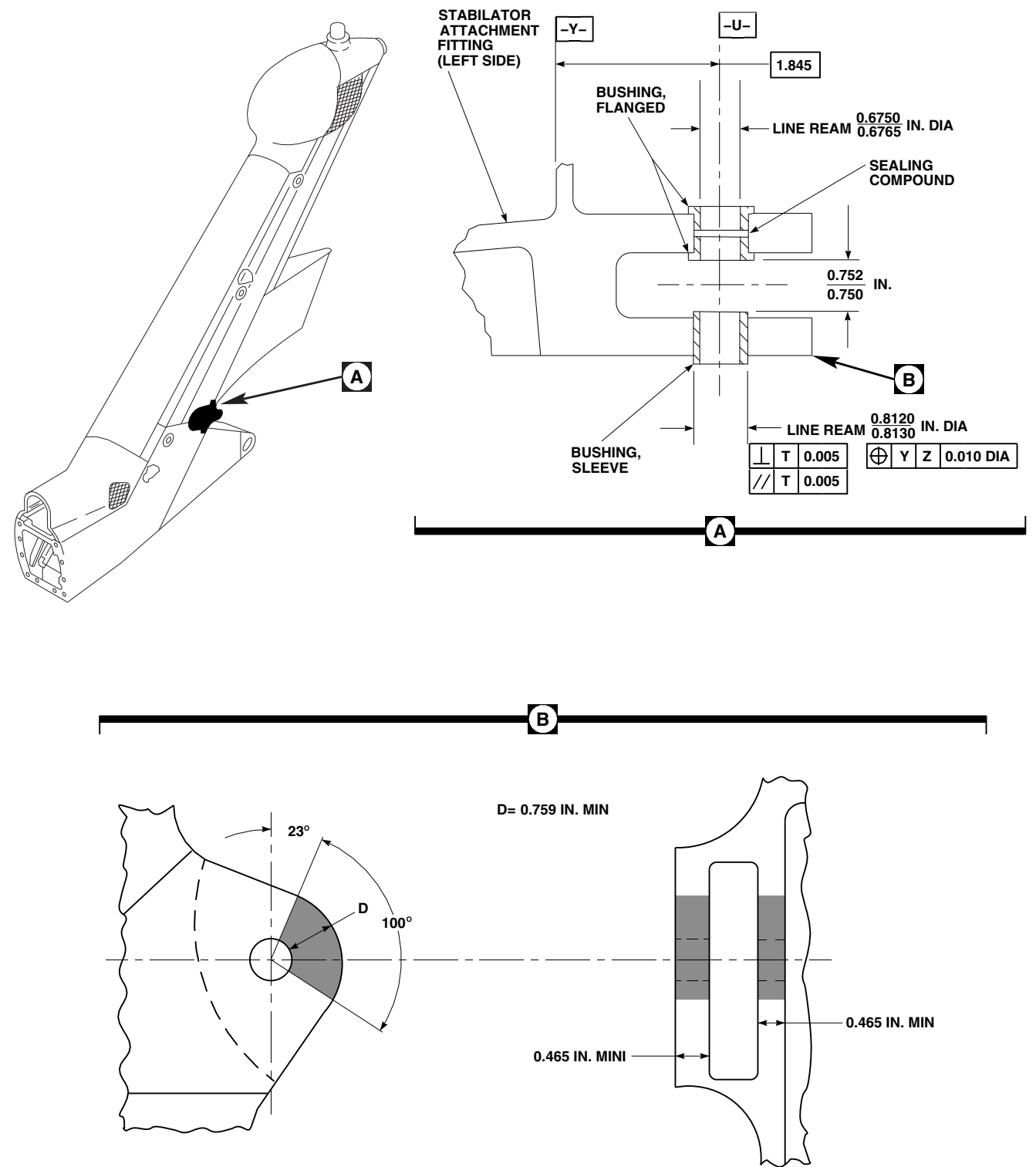
- Left stabilator attachment fittings have critical areas that must not be damaged during inspection or repair. Refer to Figure 2-6-46-1, Detail B for critical characteristic areas.
 - Bonding of liners in fitting is a critical characteristic. Proper bonding must be verified.
 - Damage to fitting will occur if feeler gage is improperly used. Liner may become disbonded. Use care when using feeler gage to check for liner edge separation.
- a. Inspect lug bore surfaces for damage and corrosion as follows:
- (1) Inspect bonding of liners in fitting. Probe for edge separation using 0.0015-inch feeler gage. **VOIDS MUST NOT EXCEED 5% OF JOINT LINE LENGTH.** If limits are exceeded, replace pylon (PARA 2-4-142).
 - (2) Inspect liner bore for corrosion. **PITS MUST NOT EXCEED 0.003-INCH DEEP OR COVER MORE THAN 25% OF LINER BORE.** If limits are exceeded, replace pylon (PARA 2-4-142).
 - (3) If corrosion is within limits remove using scouring pad, Item 270, Appendix D.
 - (4) Inspect liner bore for damage. **LINER BORE DAMAGE MUST NOT EXCEED 0.005-INCH DEEP, 0.40-INCH WIDE, AND 0.75-INCH LONG.** If limits are exceeded, replace pylon (PARA 2-4-142).

- (5) If damage is within limits, blend using scouring pad, Item 270, Appendix D. **MAXIMUM BLENDED DEPTH IS 0.005-INCH.** If limits are exceeded, replace pylon (PARA 2-4-142).

- (6) Blend nicks, dents, scratches, or other damage, using abrasive cloth, Item 10, Appendix D. **MINIMUM THICKNESS BETWEEN BLENDED SURFACE AND OPPOSITE LUG FACE MUST BE 0.465-INCH** (Figure 2-6-46-1, Detail B). If limits are exceeded, replace pylon (PARA 2-4-132). If damage is within limits, blend out using scouring pad, Item 270, Appendix D. **MAXIMUM BLENDED DEPTH IS 0.005-INCH.** If limits are exceeded, replace pylon (PARA 2-4-142).

2-6-46.1.3 Install.

- a. Using bushing/bearing removal/installation set, 70700-20129-041, install bushings wet with sealing compound, Item 274, Appendix D (Figure 2-6-46-1, Detail A).
- b. Apply sealing compound, Item 274, Appendix D, around flange and end of bushing. Pay particular attention to make sure gap between bushings of inboard lug is filled.
- c. Allow sealing compound to cure.
- d. Line-ream bushings 0.6250 - 0.6265-inch diameter.
- e. Deburr bushings.
- f. Apply coating of corrosion compound, Item 108, Appendix D.
- g. Apply coating of solid film lubricant, Item 201, Appendix D.
- h. Install stabilator (PARA 2-4-142).



2-6-47 TAIL ROTOR PYLON RIGHT STABILATOR ATTACHMENT FITTING.

PARAGRAPH BREAKDOWN

	PAGE
2-6-47.1 Replace Right Stabilator Attachment Fitting Bearings	2-6-47-1

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Bushing/Bearing Removal/Installation Set, 70700-20129-041
- Feeler Gage, 0.0015-inch
- Feeler Gage, 0.005-inch
- Fluorescent Penetrant Inspection Kit

Materials/Parts

- Abrasive Cloth, Item 10, Appendix D

- Corrosion Preventive Compound, Item 115A, Appendix D
- Scouring Pad, Item 270, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- ASTM E1417
- PARA 2-4-132
- PARA 2-4-142
- PARA 2-6-5

2-6-47.1 Replace Right Stabilator Attachment Fitting Bearings.

2-6-47.1.1 Remove.



FSCAP

Left and right stabilator attachment fittings on pylon and attachment bolts, and actuator attachment fitting (on stabilator) and attachment bolt, have critical characteristics. Extreme care must be

used to prevent damage during stabilator removal.

- Remove stabilator (PARA 2-4-142).
- Remove spherical bearings from right fitting, using bushing/bearing removal/installation set, 70700-20129-041 (Figure 2-6-47-1, Detail A), as follows:

NOTE

Nuts installed on either end of tools should only be snug.

- If bearing is roller staked, assemble tool (toolkit instructions step 1), and remove

staking flanges on outside of outer race, by rotating arbor and cutter while applying light pressure. Continue until cutter has penetrated staking 0.020-inch.

- (2) Using bushing/bearing installation/removal tool, remove staking flanges from outside of other bearing using procedure in step b. (1).
- (3) Using bushing/bearing installation/removal tool, remove bearings from fitting (toolkit instructions steps 2 and 3).

2-6-47.1.2 Inspect.

WARNING

FSCAP

Left and right stabilator attachment fittings have critical areas that must not be damaged during inspection or repair. Refer to Figure 2-6-47-1 for critical characteristic areas.

- a. Visually inspect critical characteristic areas of left and right fittings (Figure 2-6-47-1, Detail B).
- b. Using abrasive cloth, Item 10, Appendix D, blend out nicks, dents, scratches, or other damage. **MINIMUM THICKNESS BETWEEN BLENDED SURFACE AND OPPOSITE LUG FACE MUST BE 0.485-INCH.** If limits are exceeded, replace pylon (PARA 2-4-132).
- c. Touch up paint as required (PARA 2-6-5).
- d. Check bushing bores on left fitting for galling and corrosion. Blend out any galling or corrosion with abrasive cloth, Item 10, Appendix D. **DAMAGE AFTER BLENDED SHALL NOT EXCEED 0.010-INCH OR 25% OF BORE AREA.**
- e. Using fluorescent penetrant inspection kit, inspect lug area of left and right fitting for cracks (ASTM E1417). **NONE ALLOWED.** If

cracked, replace tail rotor pylon (PARA 2-4-132).

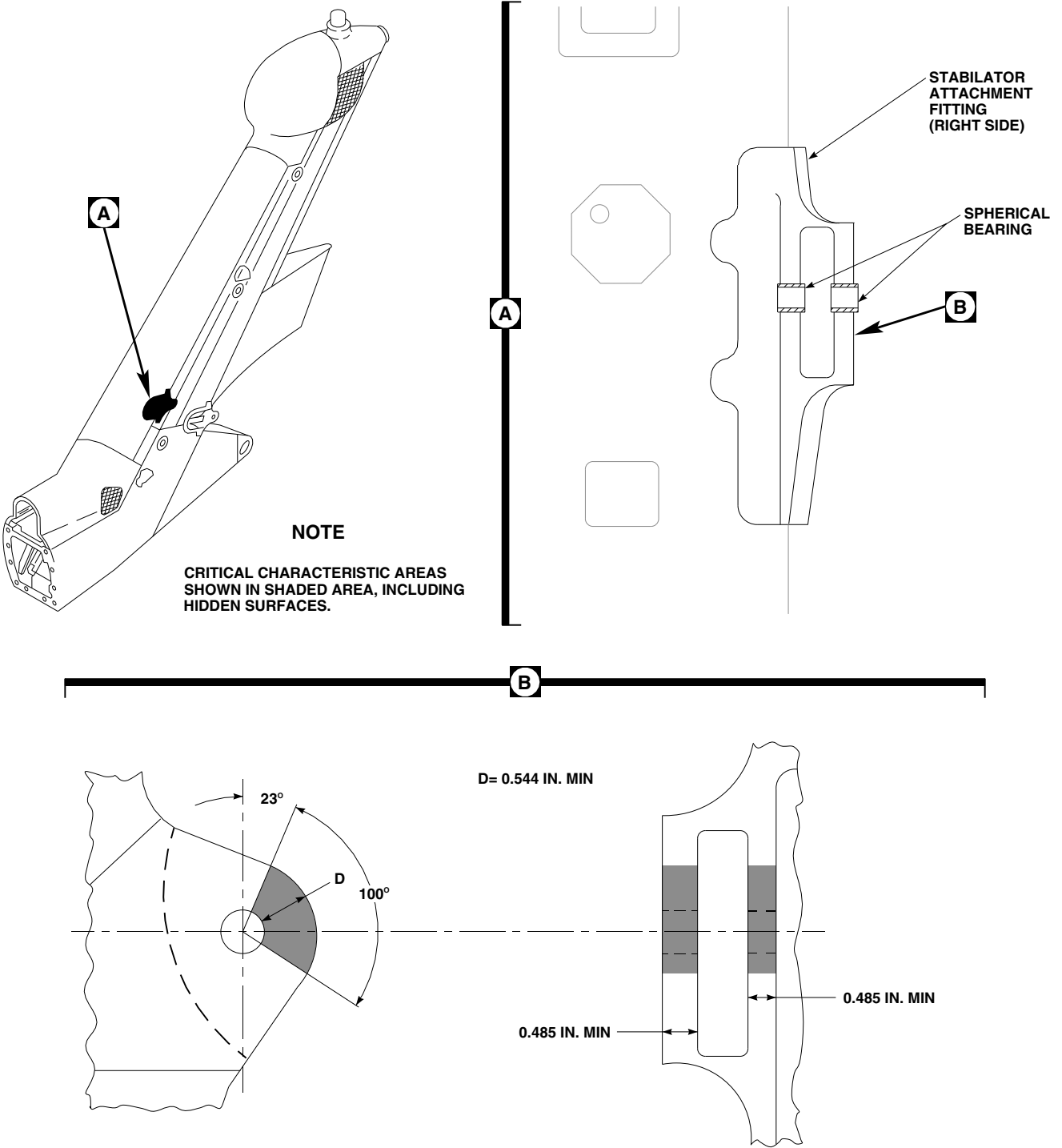
WARNING

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- Bonding of liners in fitting is a critical characteristic. Proper bonding must be verified.
 - Damage to fitting will occur if feeler gage is improperly used. Liner may become disbonded. Use care when using feeler gage to check for liner edge separation.
- f. Inspect bonding of liners in fitting. Using 0.0015-inch feeler gage, probe for edge separation. **VOIDS MUST NOT EXCEED 5% OF JOINT LINE LENGTH.** If limit is exceeded, replace tail rotor pylon (PARA 2-4-132).
 - g. Inspect liner bore for corrosion. **PITS MUST NOT EXCEED 0.003-INCH DEEP OR COVER MORE THAN 25% OF LINER BORE.** If limits are exceeded, replace tail rotor pylon (PARA 2-4-132).
 - h. If corrosion is within limits, remove using scouring pad, Item 270, Appendix D.
 - i. Inspect liner bore for damage. **LINER BORE DAMAGE MUST NOT EXCEED 0.005-INCH DEEP, 0.040-INCH WIDE, AND 0.75-INCH LONG.** If limits are exceeded, replace tail rotor pylon (PARA 2-4-132).
 - j. If damage is within limits, blend out using scouring pad, Item 270, Appendix D. **MAXIMUM BLENDED DEPTH IS 0.005-INCH.** If limit is exceeded, replace tail rotor pylon (PARA 2-4-132).

2-6-47.1.3 Install.

- a. Install spherical bearings (Figure 2-6-47-1, Detail A), as follows:



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FIGURE 2-6-47-1. REPLACE RIGHT STABILATOR ATTACHMENT FITTINGS

NOTE

- For each installation operation, assemble tool from bushing/bearing removal/installation set, 70700-20129-041, according to information plates in tool case.
 - Stabilator attachment fitting liner may need to be chamfered if ball bearings were previously installed. If not previously chamfered, chamfer per steps 4 and 5 of toolkit instructions.
 - Prior to installing bearings, apply corrosion preventive compound, Item 115A, Appendix D, to mating surface of bearing.
- (1) Assemble tool and install bearing following instructions in kit.
 - (2) Using bushing/bearing installation/removal tool, stake both bearings on inner and outer faces (toolkit instructions steps 8 and 9). Stagger stake points 45° from those on first side. Staking flare is complete when 0.005-inch feeler gage cannot be inserted between flare lip and fitting.
- b. On left attachment fitting, using sealing compound, Item 292, Appendix D, seal flanged bushings around flanges.
 - c. Install stabilator (PARA 2-4-142).

2-6-48 TAIL ROTOR PYLON STABILATOR ACTUATOR ATTACHMENT FITTING.

PARAGRAPH BREAKDOWN

		PAGE
2-6-48.1	Replace Tail Rotor Pylon Stabilator Actuator Attachment Fitting	2-6-48-1

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Torque Wrench, 30 - 150 in. lbs

References

- PARA 2-4-140
- PARA 2-4-146
- PARA 6-4-41

2-6-48.1 Replace Tail Rotor Pylon Stabilator Actuator Attachment Fitting.

2-6-48.1.1. Remove.

- Remove tail rotor drive shaft section IV (PARA 6-4-41).
- Remove stabilator actuator (PARA 2-4-146).
- Remove middle tail rotor pylon step (PARA 2-4-140).
- Remove swage lockpins from fitting and zee (Figure 2-6-48-1, Detail A and Detail B).
- Remove small and large pins, bolts, washers, and nuts securing fitting to tail rotor pylon spar.
- Remove fitting and radius blocks from tail rotor pylon spar.

2-6-48.1.2. Install.

- Position fitting on tail rotor pylon spar, with bushings toward top of pylon and flange against zee. Align fastener holes.
- Install radius blocks under heads of four large pins (Figure 2-6-48-1, Detail A).
- Install large and small pins, washers, and nuts (Figure 2-6-48-1, Detail A and Detail B). **TORQUE NUTS TO 43 - 47 INCH-POUNDS.**
- Install bolts, washers, and nuts. **TORQUE NUTS TO 43 - 47 INCH-POUNDS.**
- Install swage lockpins.
- Install middle tail rotor pylon step (PARA 2-4-140).
- Install stabilator actuator (PARA 2-4-146).
- Install tail rotor drive shaft section IV (PARA 6-4-41).

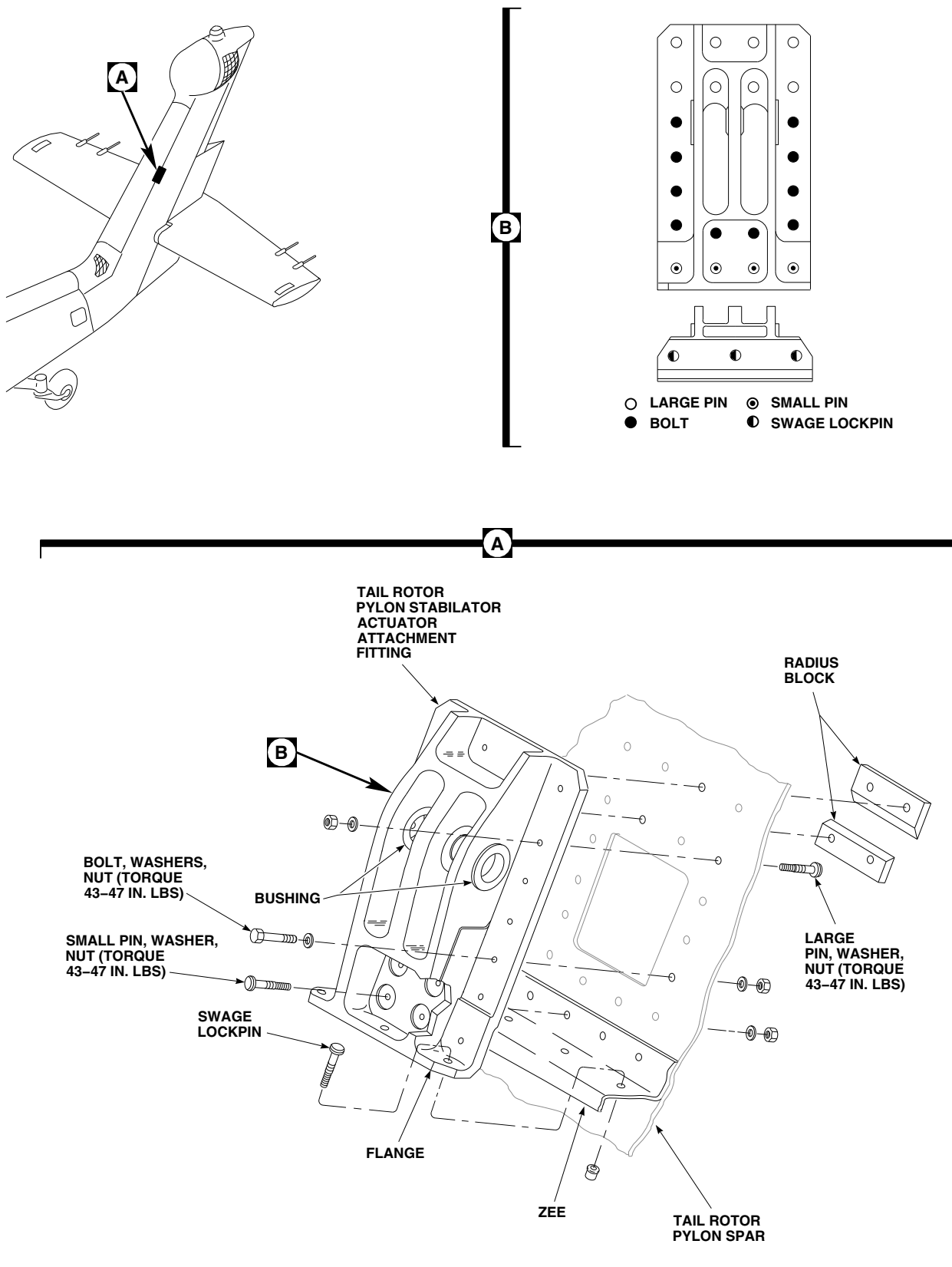
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FIGURE 2-6-48-1. REPLACE TAIL ROTOR PYLON STABILATOR ACTUATOR ATTACHMENT FITTING

2-6-49 TAIL ROTOR PYLON STABILATOR ACTUATOR ATTACHMENT FITTING BUSHINGS AND BEARING.

PARAGRAPH BREAKDOWN

	PAGE
2-6-49.1 Replace Tail Rotor Pylon Stabilator Actuator Attachment Fitting Bushings and Bearing	2-6-49-1

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Bushing/Bearing Removal/Installation Set, 70700-20129-041
- Feeler Gage, 0.0015-inch
- Feeler Gage, 0.005-inch
- Open End Wrench

Materials/Parts

- Abrasive Cloth, Item 8, Appendix D
- Brush, Acid, Item 84, Appendix D

- Cheesecloth, Item 90, Appendix D
- Scouring Pad, Item 270, Appendix D
- Sealing Compound, Item 273, Appendix D
- Sealing Primer, Item 305, Appendix D
- Trichloroethane 1,1,1, Item 341, Appendix D

References

- Appendix D
- PARA 2-4-146
- PARA 2-6-48
- PARA 6-4-41

2-6-49.1 Replace Tail Rotor Pylon Stabilator Actuator Attachment Fitting Bushings and Bearing.

2-6-49.1.1 Remove.

- Remove tail rotor drive shaft section IV (PARA 6-4-41).
- Remove stabilator actuator (PARA 2-4-146).

NOTE

- When replacement of bearing or either bushing is required, replace bearing and both bushings.
- For the following steps refer to instructions in bushing/bearing removal/installation set for removal of bushings and bearing from fitting.

- c. Assemble tool and remove bushings from fitting (toolkit instructions steps 1 and 2) (Figure 2-6-49-1, Detail C).
- d. Assemble tool and remove staking (toolkit instructions steps 3 and 4) from one side of bearing outer race by rotating arbor and cutter, applying light pressure. Continue until cutter has penetrated staking 0.020-inch.
- e. Assemble tool and remove bearing from fitting (toolkit instructions steps 5 and 6) (Figure 2-6-49-1, Detail B).

2-6-49.1.2 Inspect.

WARNING

FSCAP

- Bonding of liners in center lobe of stabilator fitting is a critical characteristic. Proper bonding must be verified.
 - Damage to fitting will occur if feeler gage is improperly used. Liner may become disbonded. Use care when using feeler gage to check for liner edge separation.
- a. Visually inspect bonding of liners in center lobe of fitting. Probe for edge separation using 0.0015-inch feeler gage. **SEPARATIONS MUST NOT EXCEED 5% OF BONDLINE LENGTH.** If limit is exceeded, replace fitting (PARA 2-6-48).
 - b. Inspect liner bore for corrosion. **PITS MUST NOT EXCEED 0.005-INCH DEEP OR COVER MORE THAN 25% OF LINER BORE.** If limits are exceeded, replace fitting (PARA 2-6-48).
 - c. If corrosion is within limits, remove using scouring pad, Item 270, Appendix D.
 - d. Inspect liner bore for damage. **LINER BORE DAMAGE MUST NOT EXCEED 0.005-INCH DEEP, 0.040-INCH WIDE, AND 0.75-**

INCH LONG. If limits are exceeded, replace fitting (PARA 2-6-48).

- e. If damage is within limits blend out using scouring pad, Item 270, Appendix D. **MAXIMUM BLENDED DEPTH IS 0.005-INCH.** If limit is exceeded, replace fitting (PARA 2-6-48).
- f. Measure liner/bearing bore in fitting. **LINER/BEARING BORE IN FITTING MUST BE 0.8128 - 0.8123-INCH DIAMETER** (Figure 2-6-49-1, Detail B). If limit is exceeded, replace fitting (PARA 2-6-48).
- g. Measure bushing bore in fitting. **BUSHING BORE IN FITTING MUST BE 0.8770 - 0.8765-INCH DIAMETER** (Figure 2-6-49-1, Detail C). If limit is exceeded, replace fitting (PARA 2-6-48).

2-6-49.1.3 Install.

CAUTION

To avoid damage to adhesive retaining liners in fitting bore and in bearing, do not allow trichloroethane 1,1,1, to contact these parts.

NOTE

For the following steps refer to bushing/bearing removal/installation set for bushings and bearing installation instructions to fitting.

- a. Clean mating surfaces of bearing outer race and fitting bore, using cheesecloth, Item 90, Appendix D, moistened with trichloroethane 1,1,1, Item 341, Appendix D.
- b. Remove residual sealing compound from fitting surfaces and bores, using abrasive cloth, Item 8, Appendix D. Sand very lightly.
- c. Clean sanded surfaces with cheesecloth, Item 90, Appendix D, moistened with trichloroethane 1,1,1, Item 341, Appendix D. Let air dry.

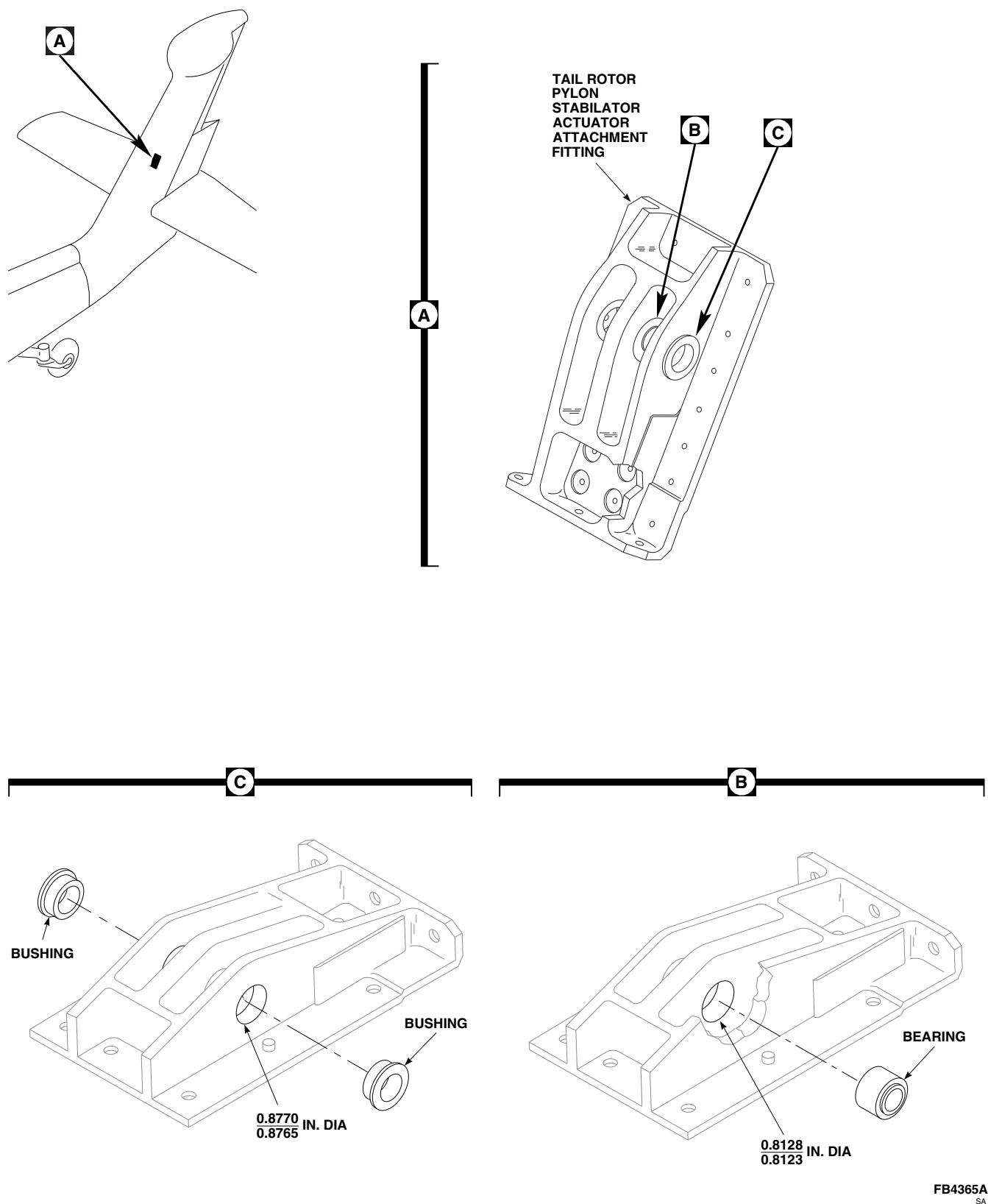


FIGURE 2-6-49-1. REPLACE TAIL ROTOR PYLON STABILATOR ACTUATOR ATTACHMENT FITTING BUSHINGS AND BEARING

- d. Using clean acid brush, Item 84, Appendix D, apply thin coat of sealing primer, Item 305, Appendix D, to bore in fitting center lobe.
- e. Let primer dry for 15 - 30 minutes until visibly dry. Protect drying surface from any contamination or hand contact.

CAUTION

Installation of bearing must be completed promptly to avoid premature curing of sealing compound by the frictional heat of insertion. If bearing is not fully installed within 4 minutes after applying sealing compound, repeat entire cleaning, priming, and compound application procedure.

- f. Using clean acid brush, Item 84, Appendix D, apply even coat of sealing compound, Item 273, Appendix D, to mating surfaces of bearing and fitting bore.

NOTE

For each installation operation, assemble tool from bushing/bearing removal/installation set according to information plates in tool case.

- g. Assemble tool and install bearing (toolkit instructions step 8 and step 9), holding arbor stationary while turning nut with open end wrench until sharp increase in torque is felt (Figure 2-6-49-1, Detail B).
- h. Inspect to verify that bearing is centered in fitting.
- i. Allow 15 minutes for curing of adhesive holding bearing in fitting.
- j. Clean up excess sealing compound, Item 273, Appendix D, using cheesecloth, Item 90, Appendix D, moistened with trichloroethane 1,1,1, Item 341, Appendix D.
- k. Assemble tool and stake bearing (toolkit instructions step 9 and step 10). Staking flare is complete when 0.005-inch feeler gage cannot be inserted between flare lip and fitting.
- l. Assemble tool and install bushings in fittings (toolkit instructions step 11 and step 12).
- m. Install stabilator actuator (PARA 2-4-146).
- n. Install tail rotor drive shaft section IV (PARA 6-4-41).

2-6-50 TAIL ROTOR PYLON CABLE PULLEY SUPPORTS REPAIR.

PARAGRAPH BREAKDOWN

	PAGE
2-6-50.1 Repair Tail Rotor Pylon Cable Pulley Supports	2-6-50-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Bushing Installation Kit
- Hand Reamer Set
- Puller Kit
- Sizing Block, Locally-Made
- Slide Hammer

Materials/Parts

- Cheesecloth, Item 90, Appendix D
- Dry Ice, Item 125, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Primer, Item 257, Appendix D

References

- Appendix D
- PARA 11-4-107

2-6-50.1 Repair Tail Rotor Pylon Cable Pulley Supports.

2-6-50.1.1. Remove.

- Remove pulley wheel from support (PARA 11-4-107).
- Remove bushings from support, using puller set and slide hammer (Figure 2-6-50-1, Detail A or Detail B).
- Thoroughly clean bores in support, using cheesecloth, Item 90, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D.

2-6-50.1.2. Install.

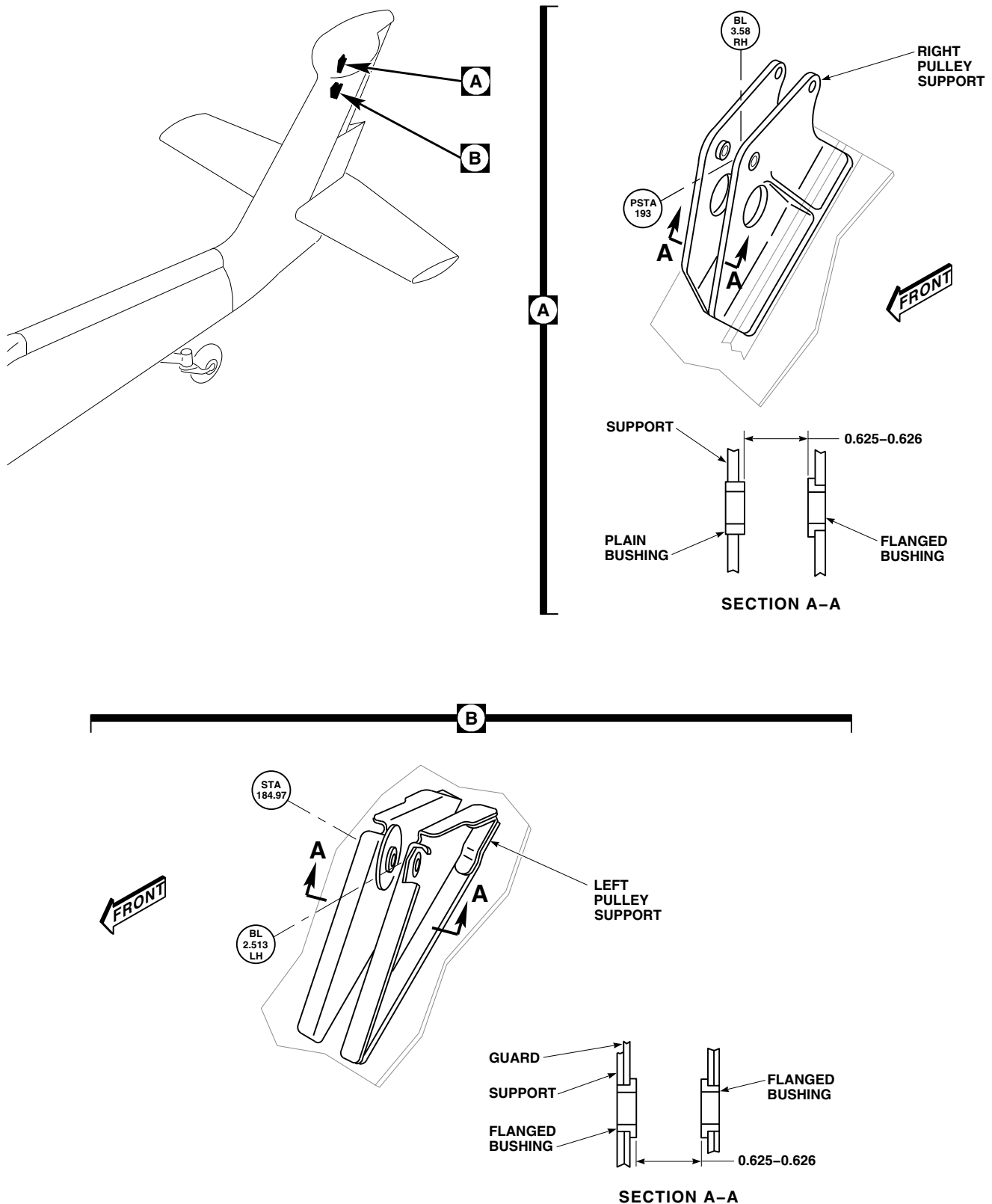
- Put replacement bushings in container of dry ice, Item 125, Appendix D, for 30 minutes.

- Immediately before installing, coat outside of each bushing with primer, Item 257, Appendix D.
- Install bushings in support while primer is still wet, using bushing installation kit and slide hammer (Figure 2-6-50-1, Detail A or Detail B).

NOTE

Obtain correct inside spacing between bushings by using properly sized sizing block, locally-made.

- Line-ream bushings to 0.3120 - 0.3125-inch inside diameter, using hand reamer set.
- Install pulley wheel (PARA 11-4-107).



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FIGURE 2-6-50-1. REPAIR TAIL ROTOR PYLON CABLE PULLEY SUPPORTS

2-6-50-2

2-6-51 TAIL ROTOR PYLON CANTED HINGE FRAME REPAIR.

PARAGRAPH BREAKDOWN

	PAGE
2-6-51.1 Repair Tail Rotor Pylon Canted Hinge Frame	2-6-51-1

INITIAL SETUP

Materials/Parts

- Abrasive Cloth, Item 2, Appendix D
- Abrasive Cloth, Item 6, Appendix D
- Epoxy Primer Coating Kit, Item 135A, Appendix D
- Sealing Compound, Item 292, Appendix D

References

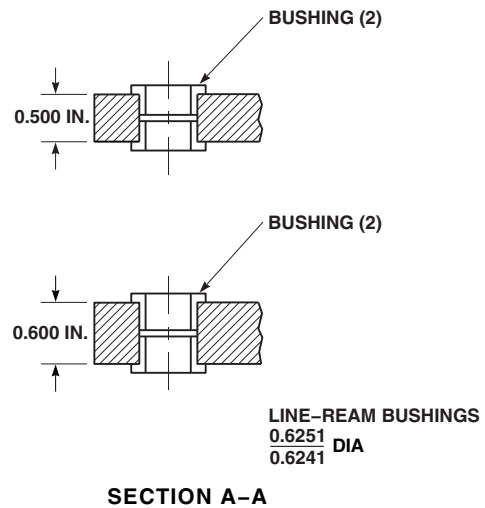
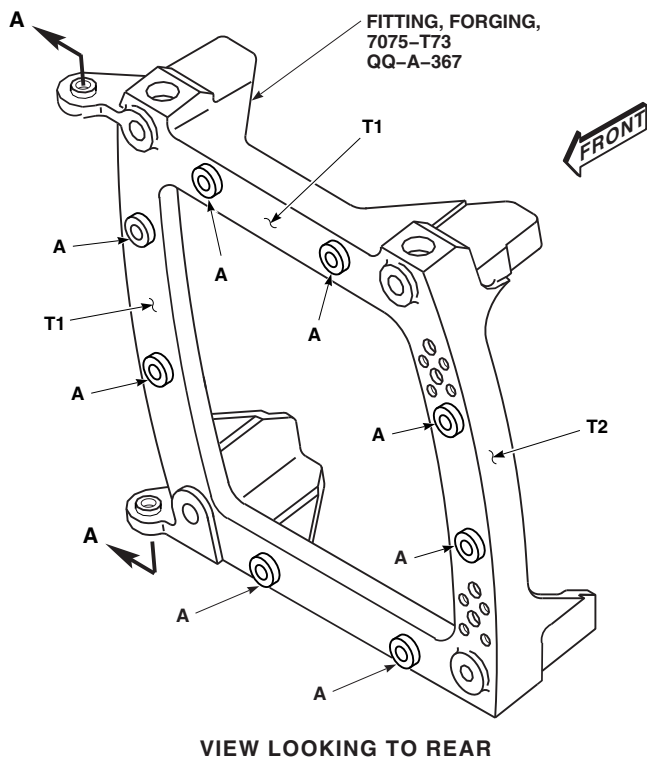
- Appendix D
- PARA 1-6-9
- PARA 2-6-1
- PARA 2-6-5
- QQ-A-367

2-6-51.1 Repair Tail Rotor Pylon Canted Hinge Frame.

- The canted hinge frame for the tail rotor pylon is machined from forged aluminum alloy, 7075-T73, QQ-A-367 (Figure 2-6-51-1). This fitting is considered critical primary structure. Negligible damage to the fitting is limited to nicks, scratches, and corrosion, no deeper than 5% of material thickness at point of damage. Blend damage smoothly to surrounding area with abrasive cloth, Item 2, Appendix D using long, even strokes. Polish with abrasive cloth, Item 6, Appendix D, and restore protective finish. Fitting is prime-coated with primer using, epoxy primer coating kit, Item 135A, Appendix D. Damage to the fitting greater than negligible limits must be reviewed by

engineering authority for damage analysis and recommended repair or replacement. Bushings, eight places, are made from 17-4PH corrosion-resistant steel, heat-treated to H-925 condition and pressed into fitting. Hinge bushings in the upper and lower lugs are cadmium-plated steel and are press-fit into upper and lower side of each lug. Seal hinge bushings around flanges, using sealing compound, Item 292, Appendix D.

- Open and close tail rotor pylon (PARA 1-6-9).
- Seal fitting for corrosion protection and watertightness using sealing compound, Item 292, Appendix D (PARA 2-6-1).
- Touch up paint as required (PARA 2-6-5).



A - BUSHING

T1 - WEB THICKNESS: BETWEEN
BOSSSES, TYPICAL - 0.100 INCH.

T2 - FLANGE THICKNESS,
TYPICAL - 0.160 INCH.

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FIGURE 2-6-51-1. REPAIR TAIL ROTOR PYLON CANTED HINGE FRAME

2-6-52 STABILATOR FRAMING REPAIR.

PARAGRAPH BREAKDOWN

	PAGE
2-6-52.1 Repair Stabilator Framing	2-6-52-1
INITIAL SETUP	MIL-S-18729
Tools	MIL-S-22499
- Airframe Repairer’s Toolkit - Drill Set - Powertrain Repairer’s Toolkit	- MIL-S-5000 - PARA 2-6-1 - PARA 2-6-2
Materials/Parts	- PARA 2-6-4 - PARA 2-6-5
References	- Appendix D - QQ-A-200 - MIL-P-25515 - QQ-A-250

2-6-52.1 Repair Stabilator Framing.

WARNING

Boron strap material presents a health hazard to personnel when cut or drilled. Do not drill, repair, or rework Boron strap.

- a. All stabilator framing between front and rear spars is primary structure. Spars, between Boron reinforcement straps bonded to upper and lower cap angles, are limited to negligible damage repairs (PARA 2-6-1). No repair of Boron straps is authorized. Front and rear spars

are considered critical primary structure. All ribs and beams, between front and rear spars and BL 82.72 ribs, are considered primary structure. Negligible damage in these areas is limited to nicks, scratches, and corrosion, no deeper than 5% of material thickness at point of damage. Blend damage smoothly to surrounding area using long, even strokes. Polish with abrasive cloth, Item 6, Appendix D, and restore protective finish. Damage greater than negligible limits in critical areas must be reviewed by engineering authority for determining repair or replacement. All repairs to stabilator framing must maintain designed strength and alignment criteria (Figure 2-6-52-1, Sheet 1 through Sheet 7).

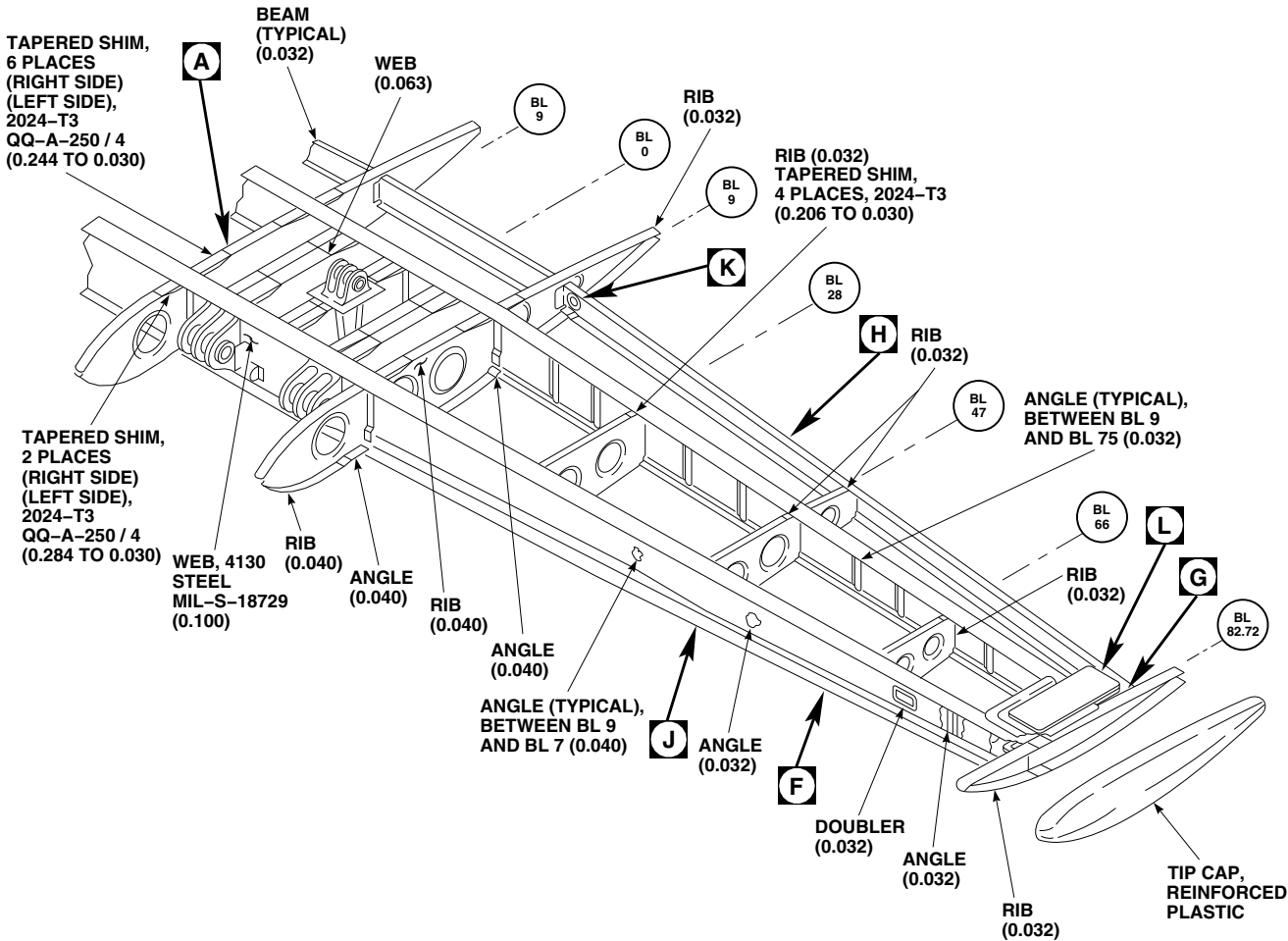
WARNING**FSCAP**

Stabilator actuator attachment fittings have critical characteristic areas. No damage is allowed. Dimension limits must not be exceeded.

- b. Critical surfaces of actuator attachment fitting are shown in Figure 2-6-52-1, Sheet 3, Detail D.
- c. Evaluate damage in noncritical areas (PARA 2-6-1).
- d. Repair nicks, scratches, and gouges in stabilator hinge fitting and actuator attachment fittings

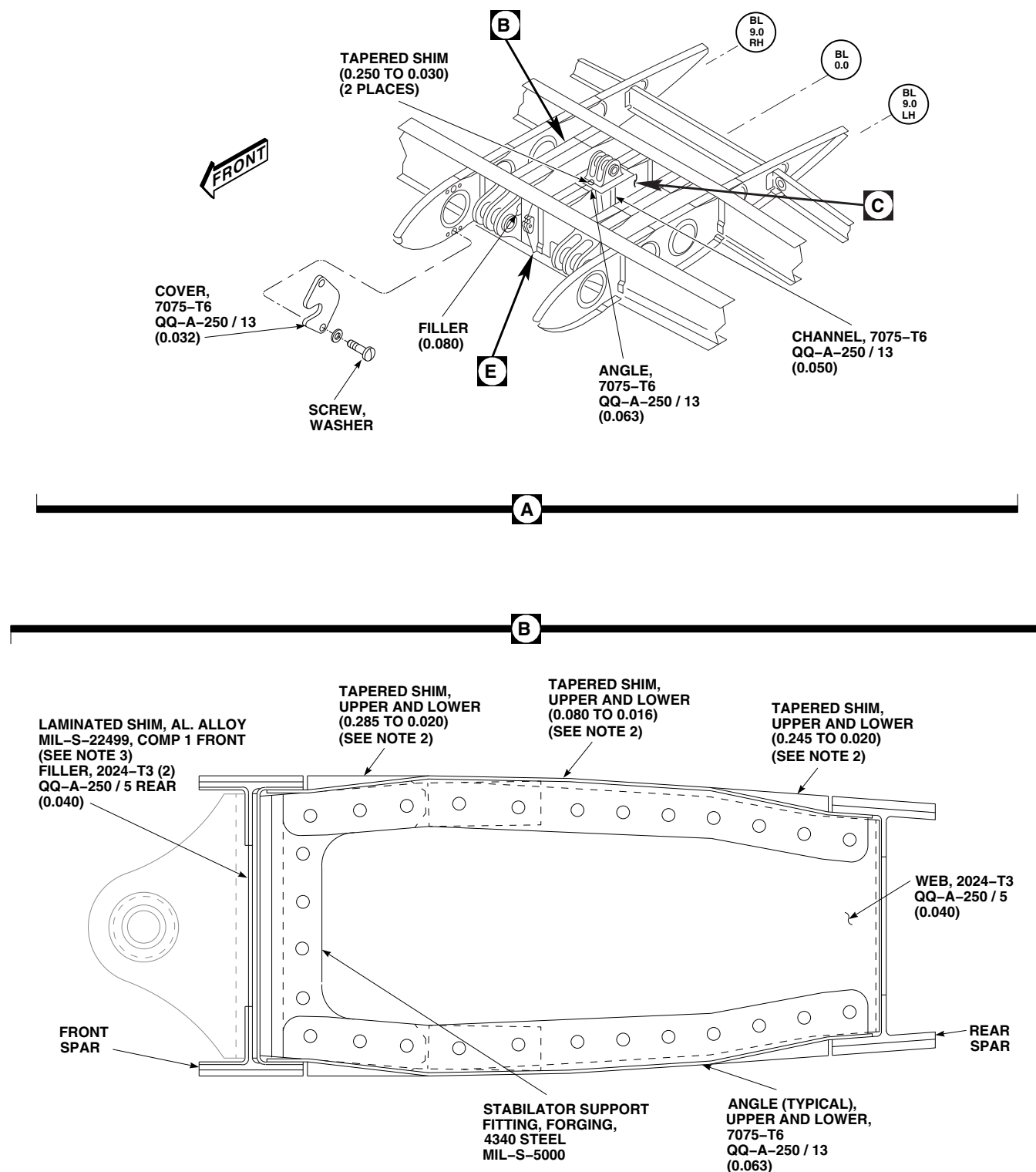
up to 0.005-inch deep, 0.040-inch wide, and 0.500-inch long. Blend damage with minimum blend radius of $\frac{3}{8}$ -inch. Blended area shall be less than 0.008-inch below unrepaired surface after repair.

- e. Use PARA 2-6-2 and PARA 2-6-4 for typical framing repairs.
- f. For corrosion protection and watertightness, seal all repairs between frame and plating (PARA 2-6-1).
- g. Touch up paint as required (PARA 2-6-5).
- h. Stop-drill repairs are a temporary repair, and a follow up repair is required at next 500-hour inspection.



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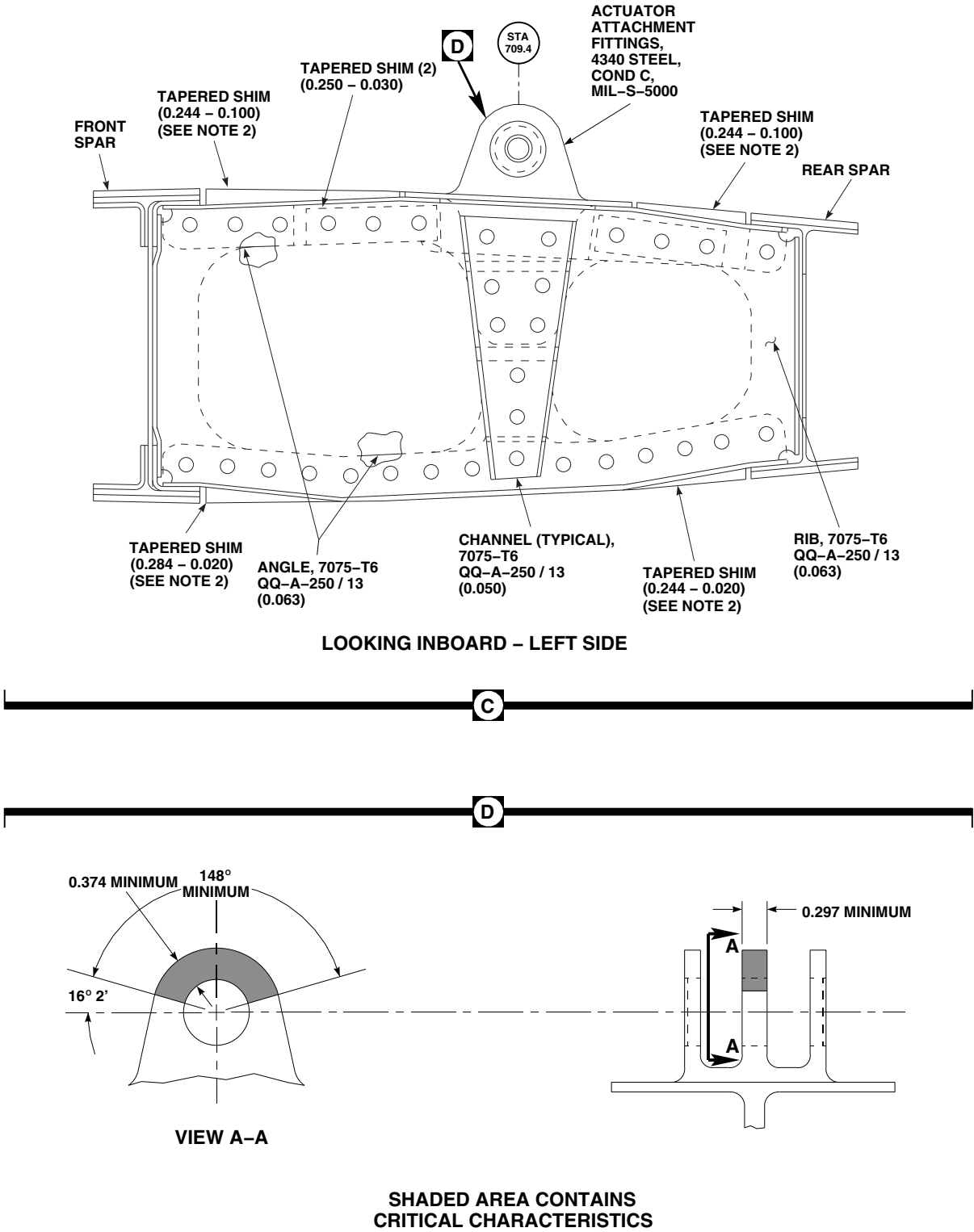
FIGURE 2-6-52-1. REPAIR STABILATOR FRAMING (SHEET 1 OF 7)



LOOKING OUTBOARD - RIGHT/LEFT SIDE

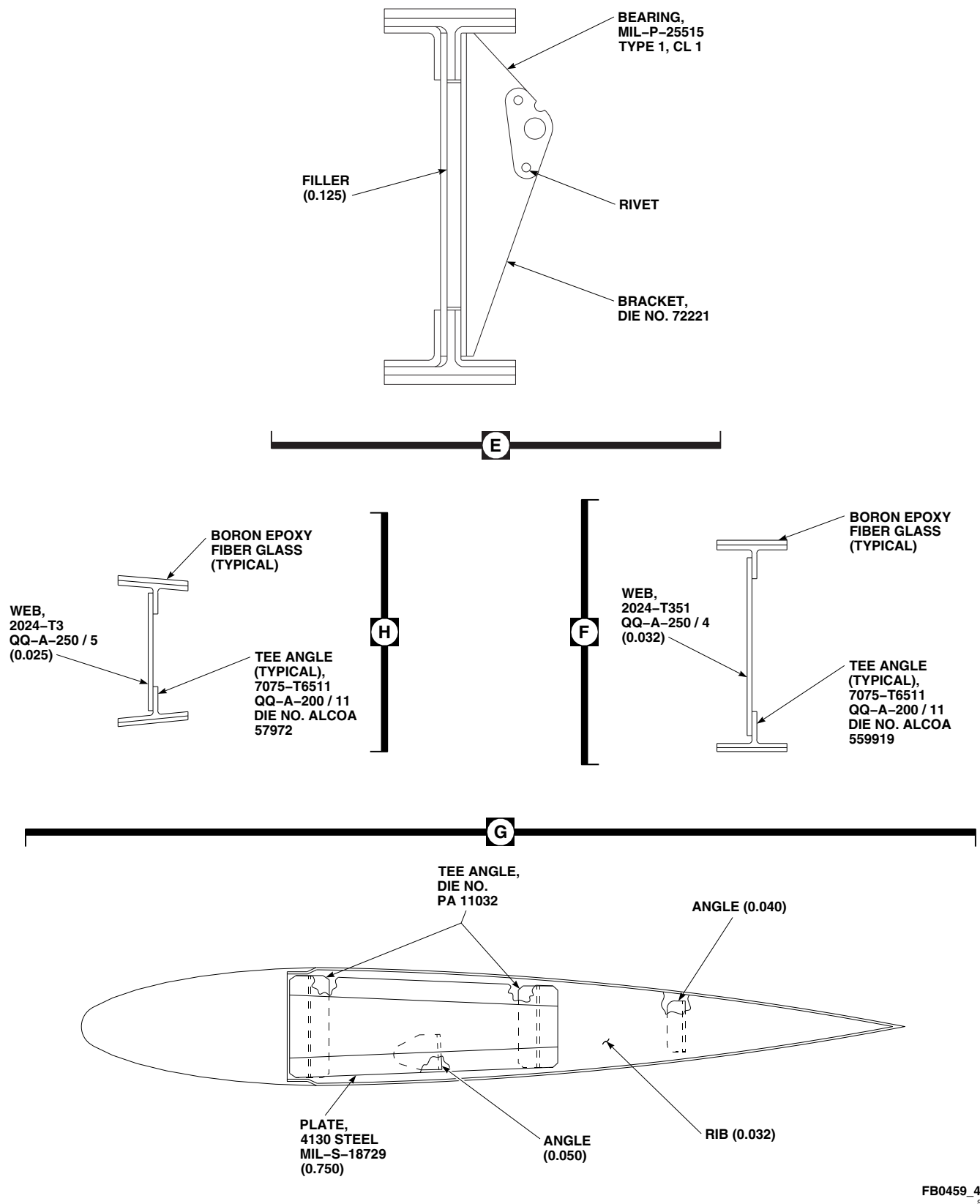
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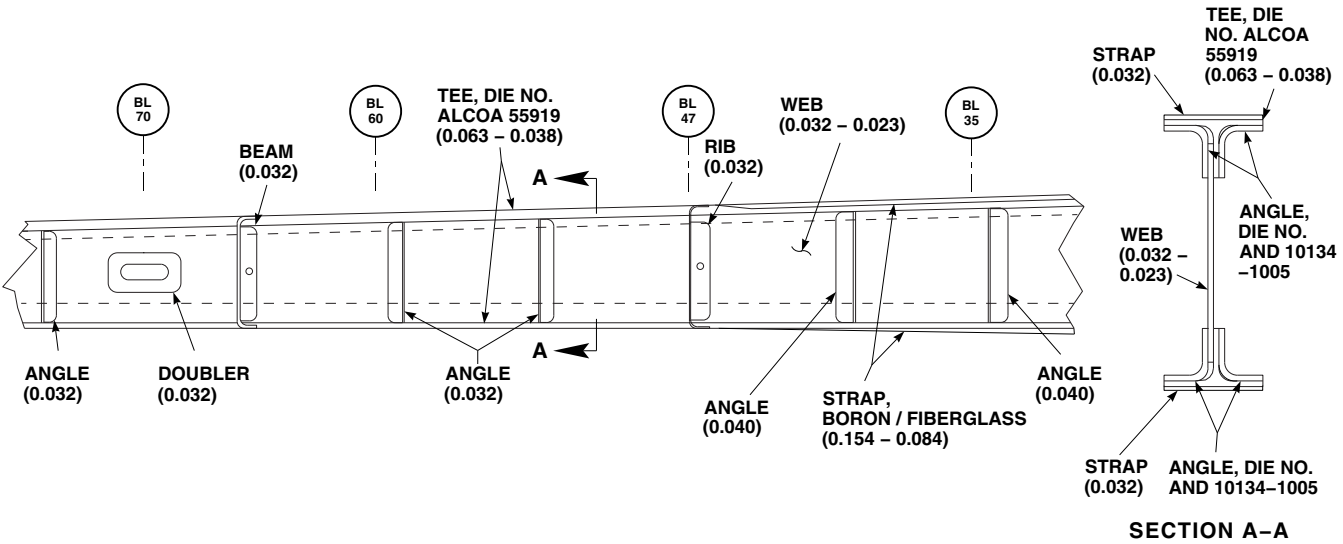
FIGURE 2-6-52-1. REPAIR STABILATOR FRAMING (SHEET 2 OF 7)



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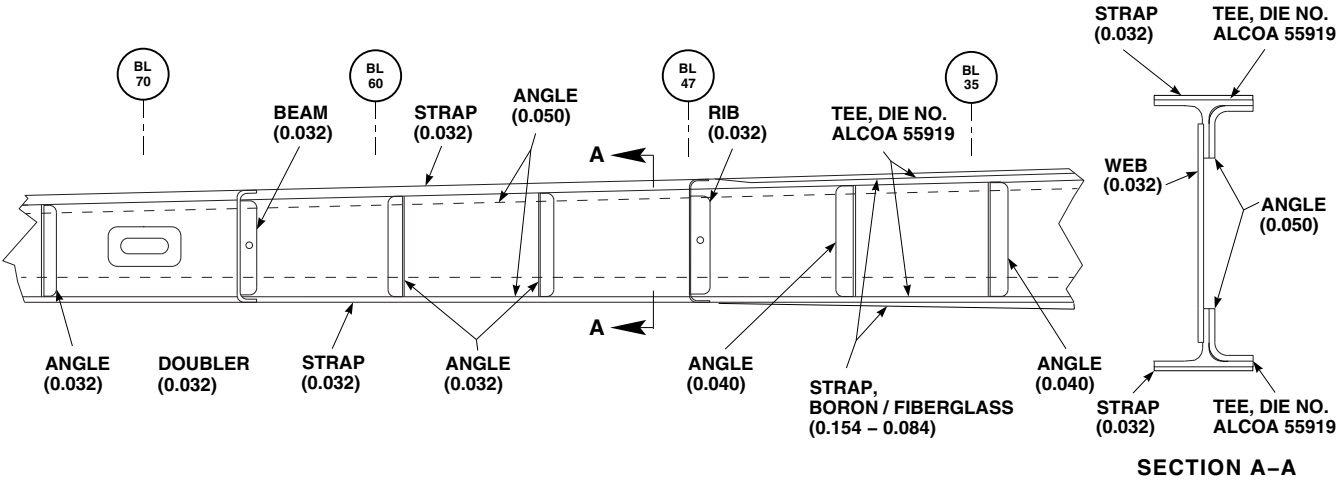
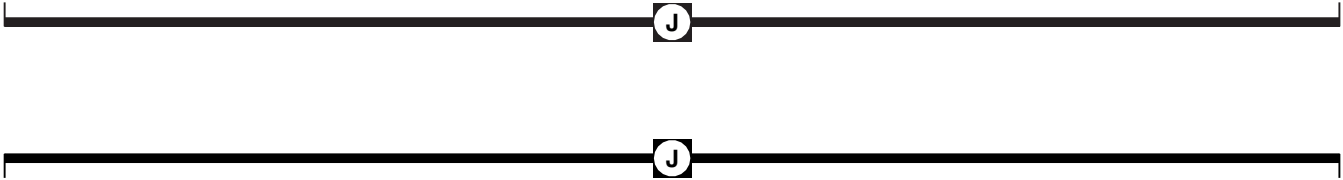
FIGURE 2-6-52-1. REPAIR STABILATOR FRAMING (SHEET 3 OF 7)





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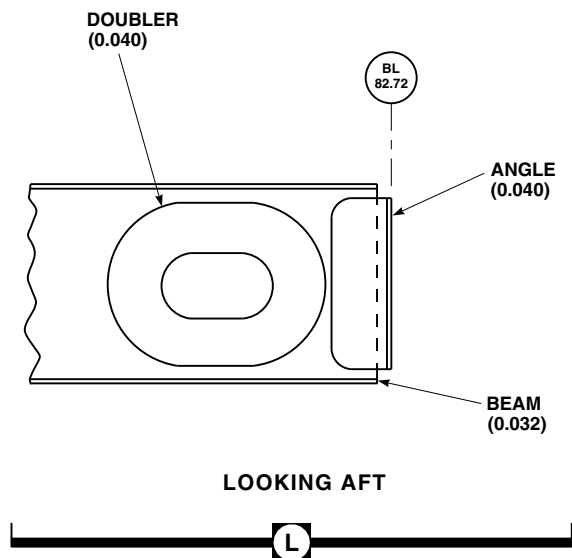
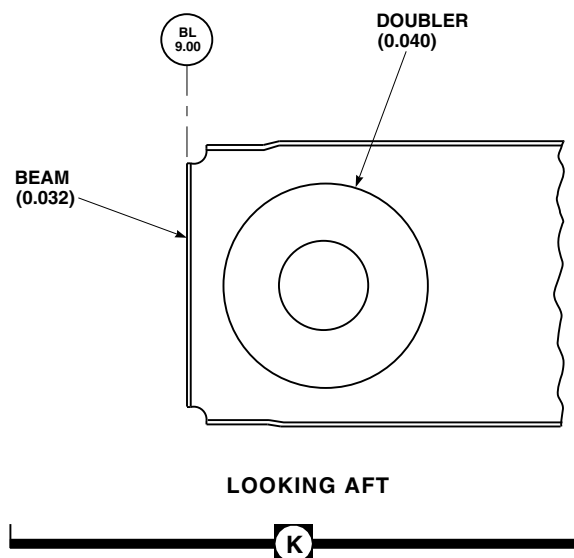


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FIGURE 2-6-52-1. REPAIR STABILATOR FRAMING (SHEET 5 OF 7)



EFFECTIVITY

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701442-SUBQ

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FIGURE 2-6-52-1. REPAIR STABILATOR FRAMING (SHEET 6 OF 7)

NOTES:

1. ALL SHEET METAL PARTS SHOWN ARE 7075-T6 ALCAD. QQ-A-250 / 13, EXCEPT AS NOTED.
2. TAPERED SHIMS ARE 2024-T3, QQ-A-250 / 4.
3. PEEL SHIM TO ELIMINATE ANY GAP BETWEEN FITTING AND SPAR.
4. ALL DIMENSIONS ARE IN INCHES, UNLESS NOTED.

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FIGURE 2-6-52-1. REPAIR STABILATOR FRAMING (SHEET 7 OF 7)

2-6-53 STABILATOR PLATING REPAIR.

PARAGRAPH BREAKDOWN

		PAGE
2-6-53.1	Repair Stabilator Plating	2-6-53-1
2-6-53.2	Repair Trailing Edge Beaded Panel	2-6-53-10

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Drill Bit, No. 40
- Drill Bit, 0.125-inch diameter
- Drill Bit, 0.1875-inch diameter
- Drill Set
- Powertrain Repairer’s Toolkit

Materials/Parts

- Adhesive, Item 32, Appendix D
- Corrosion Resistant Coating, Item 117, Appendix D
- Rivet, M7885
- Rivet, MS20470AD
- Rivet, NAS1738B

- Rivet, NAS1738B-4-1
- Sealing Compound, Item 292, Appendix D
- 0.025-inch 7075-T6 Stock
- 0.040-inch 7075-T6 Stock
- 0.187-inch Stock

References

- Appendix D
- MIL-S-18729
- PARA 2-6-1
- PARA 2-6-2
- PARA 2-6-3
- PARA 2-6-5
- PARA 2-6-53
- PARA 2-6-54
- QQ-A-250

2-6-53.1 Repair Stabilator Plating.

- Stabilator plating is aluminum alloy 7075-T6, QQ-A-250/18, and 6061-T6, QQ-A-250/11, except Kevlar reinforced plastic leading edge and tip cap. Doublers attached to skins are

7075-T6, QQ-A-250/13 (Figure 2-6-53-1, Sheet 1 through Sheet 4).

- All plating, except reinforced plastic members, trailing edge skin, and stabilator mount bolt access covers is primary structure. All repairs

must maintain designed strength and alignment criteria. Evaluate damage (PARA 2-6-1).

- c. To repair damaged stabilator plating inboard of BL 9.0, do this:
 - (1) Remove damaged panel (Figure 2-6-53-1, Sheet 2, Detail D).
 - (2) Make plate support angles of 0.040-inch 7075-T6 stock.
 - (3) Make four clips from 0.040-inch stock.
 - (4) Break all edges and radius corners.
 - (5) Coat parts with primer (PARA 2-6-5).
 - (6) Attach angles to ribs using clips and four rivets, NAS1738B, per clip.
 - (7) Make replacement plating from 0.025-inch stock.
 - (8) Attach plating to angles using rivets, NAS1738B, spaced 1 inch apart.
 - (9) Attach plating to structure with rivets, M7885.
- d. Use PARA 2-6-2 for typical plating repairs.
- e. Use PARA 2-6-3 for typical reinforced plastic repairs.
- f. Refer to PARA 2-6-54 for stabilator tip cap repair.
- g. For corrosion protection and watertightness, seal all repairs on stabilator plating (PARA 2-6-1).
- h. To repair damaged stabilator beaded plating, do this:
 - (1) Cut away damaged plating to at least 1/16-inch beyond any bead bend radius (Figure 2-6-53-1, Sheet 2, Detail C).
 - (2) Make a doubler of 0.187-inch-thick stock 1-1/4-inch wide, and the same length as bead being replaced.
 - (3) Size and cut a patch from 0.025-inch-thick 7075-T6 stock.
 - (4) Break all edges and radius corners on all parts.
 - (5) Coat parts with primer (PARA 2-6-5).
 - (6) Rivet doubler to patch using single row of rivets, MS20470AD, spaced 1 inch apart.
 - (7) Rivet patch to undamaged plating with rivets, NAS1738B. Space rivets 1 inch apart with a 1/4-inch edge distance.
- i. Touch up paint in repaired area as required (PARA 2-6-5).
- j. See Table 2-6-53-1 for damage limits and repair actions.

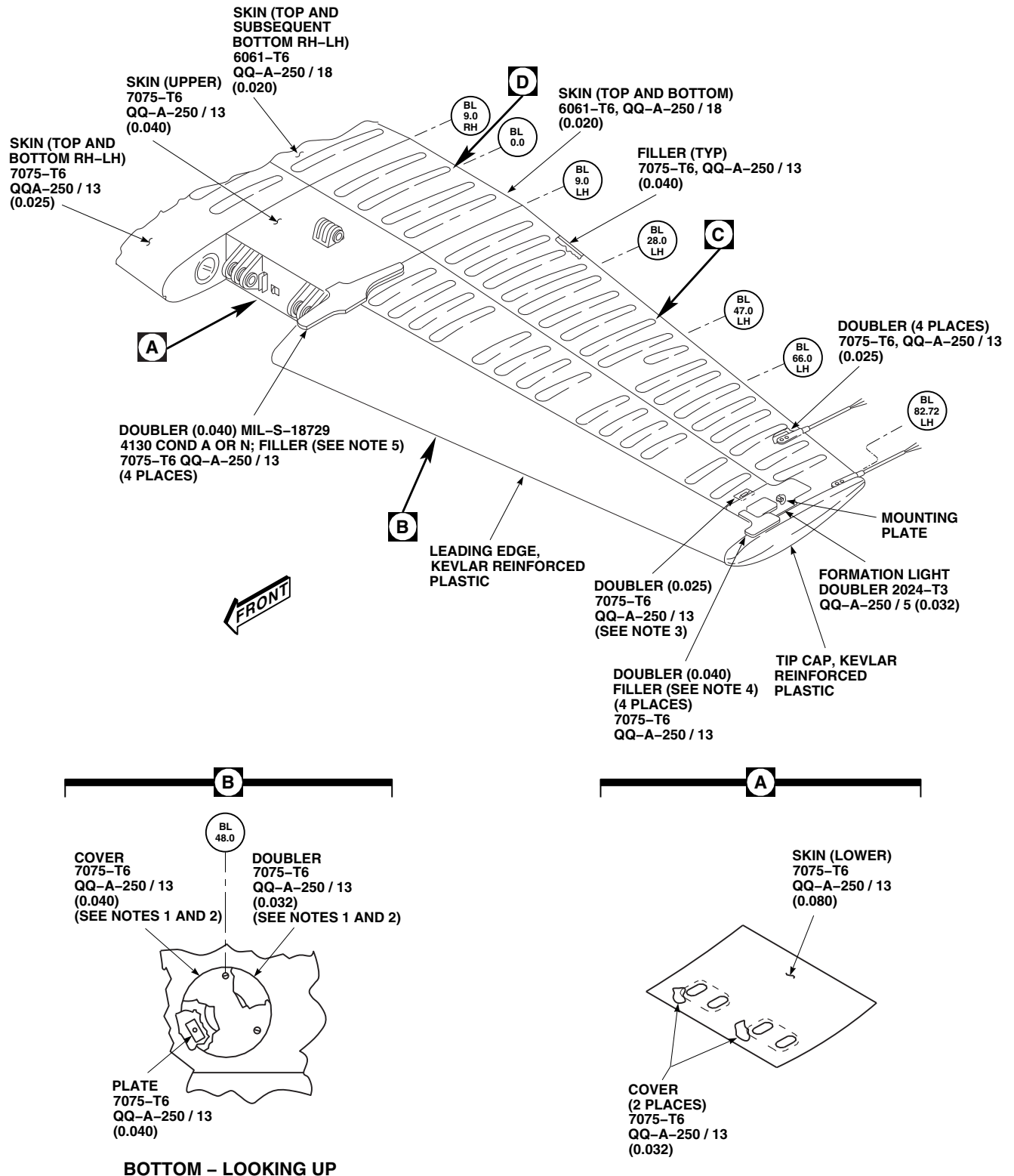


FIGURE 2-6-53-1. REPAIR STABILATOR PLATING (SHEET 1 OF 4)

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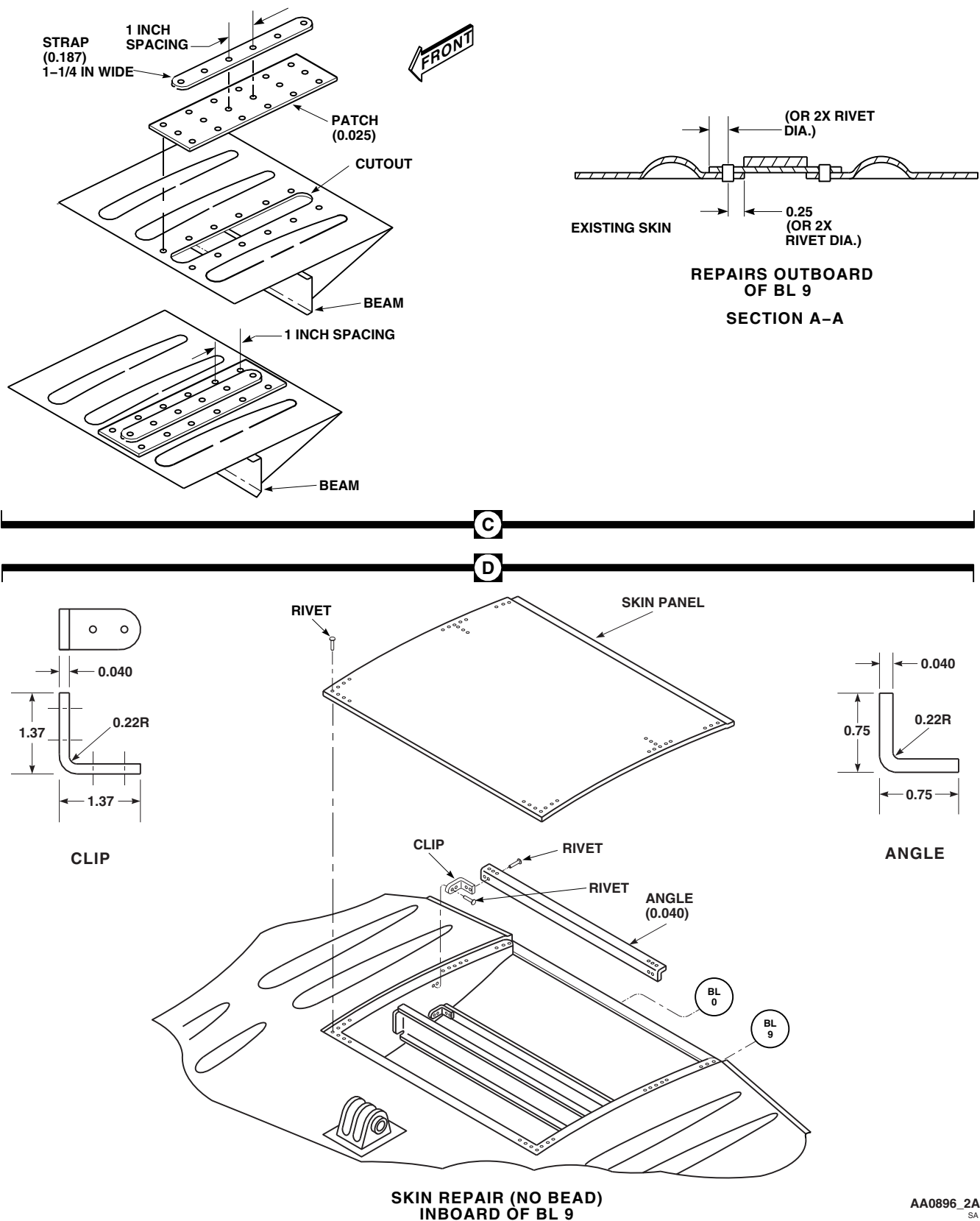
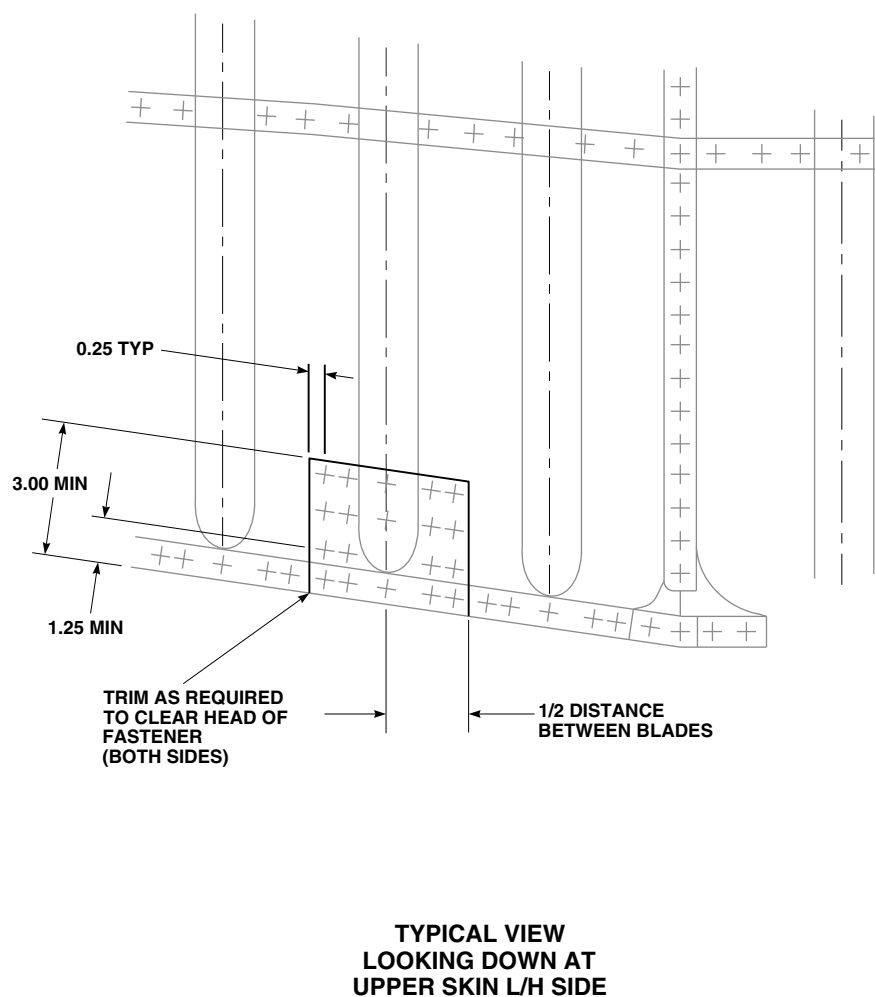


FIGURE 2-6-53-1. REPAIR STABILATOR PLATING (SHEET 2 OF 4)



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FIGURE 2-6-53-1. REPAIR STABILATOR PLATING (SHEET 3 OF 4)

NOTES

1. **70785-SUBQ**
2. BOND WITH ADHESIVE, ITEM 32, APPENDIX D AT ASSEMBLY.
3. **701029 - SUBQ**
4. **70583-701285** (0.025) ◀ **701448-SUBQ** (0.050) ▶
5. **701449-701743** (0.032) ◀ **701059-SUBQ** (0.050) ▶

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FIGURE 2-6-53-1. REPAIR STABILATOR PLATING (SHEET 4 OF 4)

Table 2-6-53-1. Negligible Damage Limits

Component	Type and Size	Repair Action	Repair Location	
			ON ACFT	OFF ACFT
Fiberglass/Epoxy Leading Edge	Scratches - - Surface scratches not penetrating into fabric	No repair required	x	
	- - Scratches penetrat- ing not more than 1 ply, less than 2 inches long	No repair required	x	
	- - Scratches penetrat- ing not more than 2 plies, less than 0.50 inch long, with at least 1 inch between scratches	No repair required	x	
	- - Scratches penetrat- ing all 3 plies but not into honeycomb core	Repair (PARA 2-6-3)	x	
	Damage penetrating into honeycomb core, maximum diameter 6 inches	Repair using fiberglass instead of Kevlar (PARA 2-6-3)	x	

Table 2-6-53-1. Negligible Damage Limits (Cont)

			Repair Location
	Dents - - Up to 0.50 inch diameter and 0.060 inch deep, with at least 2 inches of undamaged surface between dents. No dents allowed within 1 inch of leading edge nose (Figure 2-6-53-2, Detail A)	No repair required. If damage exceeds limits, treat as damage penetrating into core	x
Primary Skin Between Beads	Cracks not ending at fastener holes, maximum length (L) 0.50 inch	Stop-drill using 0.125-inch drill bit	x
	Edge damage, cleaned-up area (C) not more than 0.50 inch deep	Blend damage to smooth, rounded contour not more than twice the depth long	x
	Nicks, scratches, and corrosion cleaned-up area (Figure 2-6-53-2, Section A-A) not to exceed: Depth (T1) 0.004 inch (0.15 x T) Width (b) 0.062 inch (2.5 x T) Length (h) 40% of H	Remove no more material than necessary during cleanup. Treat nicks and scratches as cracks if exceeding T1, b, or h limits (Figure 2-6-53-2)	x
	Dents. Smooth dents, without sharp projections or abrasions, not more than 0.250 inch deep	No repair required. Treat all dents deeper than 0.250 inch, or those with nicks and gouges, as punctures	x
	Punctures, maximum dimension 0.50 inch after cleanup	Trim damage to smooth, rounded contour. Repair using PARA 2-6-2	x

Table 2-6-53-1. Negligible Damage Limits (Cont)

			Repair Location
	Damage exceeding limits (Figure 2-6-53-2)	Repair using PARA 2-6-2 and PARA 2-6-53	x
Primary Skin Beads	Cracks, cleaned-up damage not to exceed: Width (W) 0.1875 inch Length (L) 0.50 inch	Stop-drill using 0.125-inch drill bit	x
	Nicks, scratches, and corrosion cleaned-up area (Figure 2-6-53-2), not to exceed: Depth (T1) 0.004 inch (0.15 x T) Width (b) 0.062 inch (2.5 x T) Width (W) 0.50 inch Length (h) 0.50 inch	Remove no more material than necessary during cleanup. Treat nicks and scratches as cracks if exceeding T1, b, or h limits (Figure 2-6-53-2)	x
	Dents - - Smooth dents without sharp projections or abrasions not deeper than 0.050 inch	Treat all dents deeper than 0.050 inch, or those with nicks and gouges, as punctures	x
	Punctures - - Maximum diameter 0.1875 inch after cleanup	Drill out damage using maximum drill diameter of 0.1875 inch	x
	- - Damage exceeding limits (Figure 2-6-53-2)	Repair using PARA 2-6-2 and PARA 2-6-53	x
Secondary Skin Between Beads	Cracks not ending at fastener holes, maximum length (L) 1 inch	Stop-drill using 0.125-inch drill bit	x
	Edge damage, cleaned-up area (C) not to exceed 1 inch	Blend damage to smooth, rounded contour not longer than twice the depth	x

Table 2-6-53-1. Negligible Damage Limits (Cont)

		Repair Location	
Secondary Skin Beads	Nicks, scratches, and corrosion cleaned-up area (Figure 2-6-53-2, Section A-A) not to exceed: Depth (T1) 0.007 inch (0.35 x T) Width (b) 0.050 inch (2.5 x T) Length (h) 40% of H	Remove no more material than necessary during cleanup. Treat nicks and scratches as cracks if exceeding T1, b, or h limits (Figure 2-6-53-2)	x
	Dents - - Smooth dents without sharp projections or abrasions not deeper than 0.250 inch	Treat all dents deeper than 0.250 inch, or those with nicks and gouges, as punctures	x
	Punctures - - Maximum dimension 1 inch after cleanup	Trim damage to smooth, rounded contour and repair using PARA 2-6-2 and PARA 2-6-53	x
	- - Damage exceeding limits (Figure 2-6-53-2)	Repair using PARA 2-6-2 and PARA 2-6-53	x
	Cracks - - Cleaned-up damage maximum dimensions: Width (W) 0.50 inch Length (L) 1 inch	Stop-drill using 0.125-inch drill bit	x
	Nicks, scratches, and corrosion cleaned-up area (Figure 2-6-53-2, Section A-A), not to exceed: Depth (T1) 0.007 inch (0.35 x T) Width (b) 0.050 inch (2.5 x T)	Remove no more material than necessary during cleanup. Treat nicks and scratches as cracks if exceeding T1, or b limits (Figure 2-6-53-2)	x

Table 2-6-53-1. Negligible Damage Limits (Cont)

		Repair Location
Dents - - Smooth dents without sharp projections or abrasions not deeper than 0.125 inch deep	Treat all dents deeper than 0.125 inch, or those with nicks and gouges, as punctures	x
Punctures - - Maximum dimension 0.50 inch after cleanup	Trim damage to smooth, rounded contour and repair using PARA 2-6-2 and PARA 2-6-53	x
- - Damage exceeding limits (Figure 2-6-53-2)	Repair using PARA 2-6-2 and PARA 2-6-53	x

2-6-53.2 Repair Trailing Edge Beaded Panel.**NOTE**

Use skin panel, 70203-07051-142 (RH) or a -143 (LH), to fabricate a patch. To improve fit, cut the patch from approximately the same area of the skin that the patch will be used on the stabilator.

a. Prepare patch:

- (1) Determine the area and length of the bead crack. Stop drill the end of the crack(s) use a No. 40 drill bit.
- (2) Trim the -142 or -143 to cover the entire crack plus half way between the inboard and outboard bead. The patch should be a minimum of 3 inches long as measured from the trailing edge and extended a minimum of 1 inch beyond crack.
- (3) Use the existing fastener pattern of the trailing edge.
- (4) Cut and size patch for installation maintaining a 0.25-inch edge distance of

either side of patch not to exceed 0.25 - inch distance from the fillet of the radius of bead. No fasteners should be located closer than 1.25-inch of the trailing edge (Figure 2-6-53-1, Sheet 3).

- (5) Break all edges and radius corners of patch.
- (6) Touch up patch with corrosion resistant coating, Item 117, Appendix D.

- b. Apply sealing compound, Item 292, Appendix D, between patch and existing skin.



Care shall be taken not to damage skin on opposite side with tail of fastener.

- c. Fasteners are to be wet installed.
 - (1) Use rivets, MS20470AD, as required at trailing edge.
 - (2) Use rivets, NAS1738B-4-1, as required to fasten patch to existing skin.

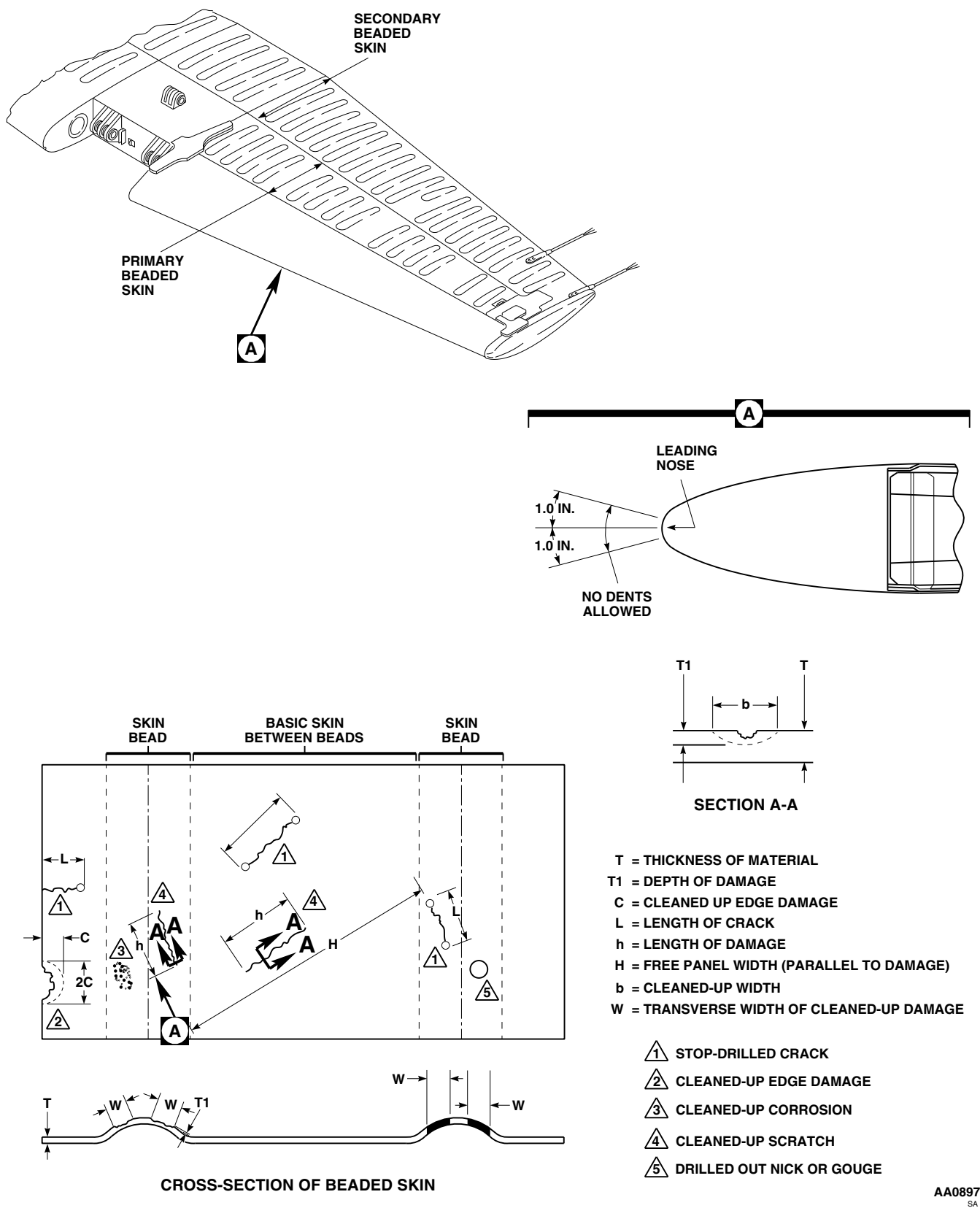


FIGURE 2-6-53-2. REPAIR LIMITS

- d. Touch up paint in repair area as required (PARA 2-6-5).

2-6-54 STABILATOR TIP CAP REPAIR.

PARAGRAPH BREAKDOWN

	PAGE
2-6-54.1 Repair Stabilator Tip Cap	2-6-54-1
INITIAL SETUP	• PARA 2-6-3
Tools	• PARA 2-6-5
• Airframe Repairer’s Toolkit • Powertrain Repairer’s Toolkit	
References	
• PARA 2-4-145 • PARA 2-6-1	
2-6-54.1 Repair Stabilator Tip Cap.	(2) Damage beyond negligible limits will require removal of tip cap for repair or replacement (PARA 2-6-3 and PARA 2-4-145).
a. The stabilator tip cap is made from Kevlar reinforced plastic and is riveted to the outboard ends of the stabilator. Material thickness is 0.030-inch (Figure 2-6-54-1, Detail A).	
b. Evaluate tip cap (PARA 2-6-3), and repair as follows: (1) Negligible damage can be repaired with tip cap installed (PARA 2-6-3).	c. If repair was made to attachment area with tip cap installed, seal tip cap to stabilator for corrosion protection and watertightness (PARA 2-6-1).
	d. Touch up paint finish as required (PARA 2-6-5).

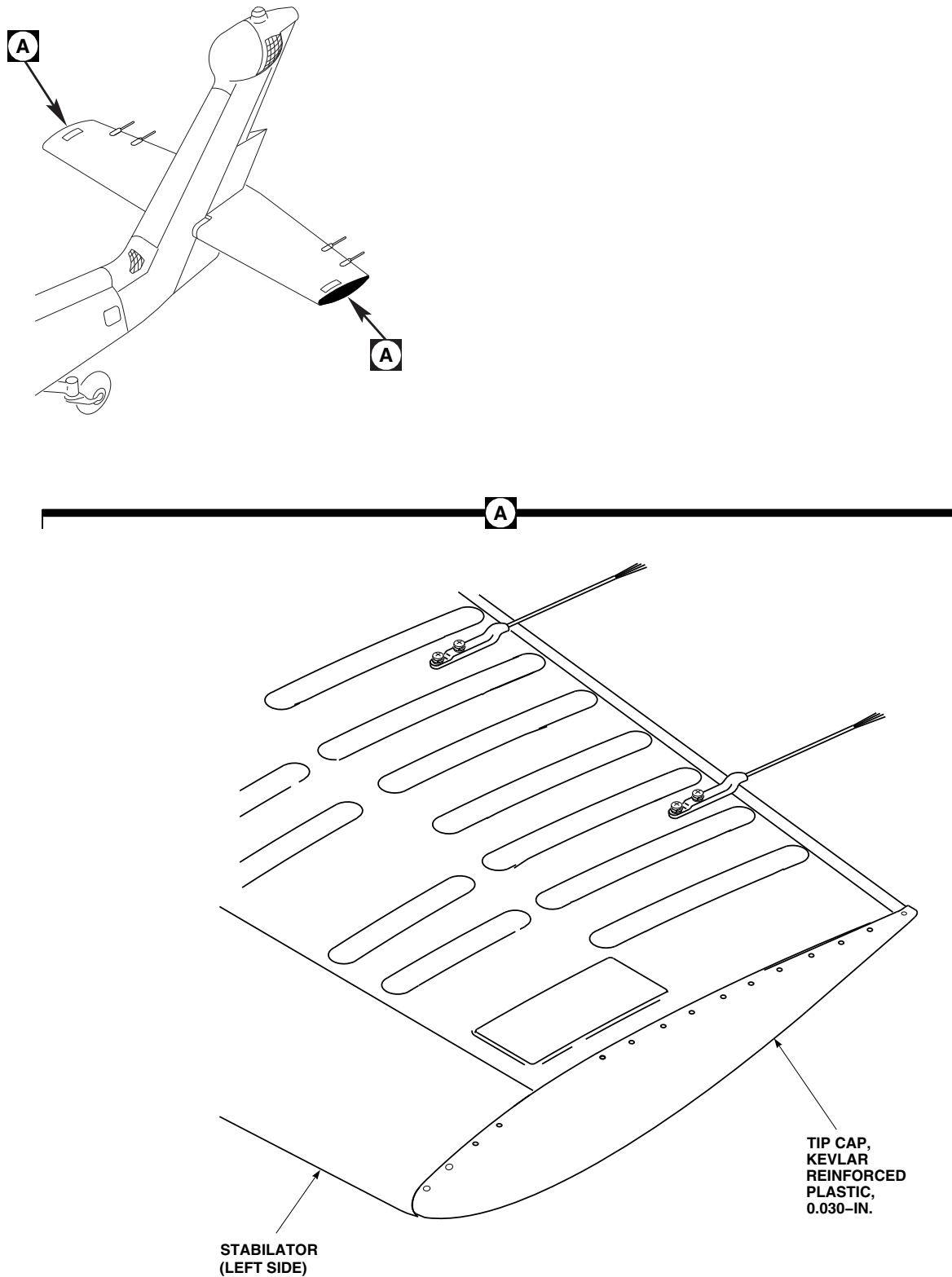


FIGURE 2-6-54-1. REPAIR STABILATOR TIP CAP

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2-6-55 STABILATOR ACTUATOR ATTACHMENT FITTING REPAIR.

PARAGRAPH BREAKDOWN

	PAGE
2-6-55.1 Repair Stabilator Actuator Attachment Fitting	2-6-55-1

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Bushing/Bearing Removal/Installation Set, 70700-20129-041
- Feeler Gage, 0.0015-inch
- Feeler Gage, 0.005-inch
- Padded Stabilator Support
- Powertrain Repairer’s Toolkit
- Wrench

Materials/Parts

- Abrasive Cloth, Item 8, Appendix D
- Acetone, Item 25, Appendix D

- Brush, Acid, Item 84, Appendix D
- Cheesecloth, Item 90, Appendix D
- Scouring Pad, Item 270, Appendix D
- Sealing Compound, Item 273, Appendix D
- Sealing Compound, Item 282, Appendix D
- Sealing Compound, Item 292, Appendix D

References

- Appendix D
- PARA 1-6-16
- PARA 2-4-142
- PARA 2-4-146
- PARA 2-6-48

2-6-55.1 Repair Stabilator Actuator Attachment Fitting.

2-6-55.1.1 Remove.

- Turn on electrical power (PARA 1-6-16).
- On STABILATOR control panel, slew down stabilator.
- Support stabilator with padded support.

- Turn off all electrical power (PARA 1-6-16).
- Disconnect stabilator actuator from stabilator (PARA 2-4-146).
- Lower stabilator slightly and disengage stabilator actuator from stabilator fitting.
- Secure stabilator actuator away from fitting.

NOTE

- When replacement of bearing or either bushing is required, replace bearing and both bushings.
 - For the following steps refer to instructions in bushing/bearing removal/installation set for removal of bushings and bearing from fitting, 70209-06054/70209-07053.
- h. Assemble tool and remove bushings from fitting (toolkit instructions step 1 and step 2) (Figure 2-6-55-1, Detail A).
 - i. If bearing is roller staked, assemble tool and remove staking (toolkit instructions step 3 and step 4) from one side of bearing outer race by rotating arbor and cutter, applying light pressure. Continue until cutter has penetrated staking 0.020-inch.
 - j. Assemble tool and remove bearing from fitting (toolkit instructions step 5 and step 6) (Figure 2-6-55-1, Detail B).

2-6-55.1.2 Inspect.**WARNING****FSCAP**

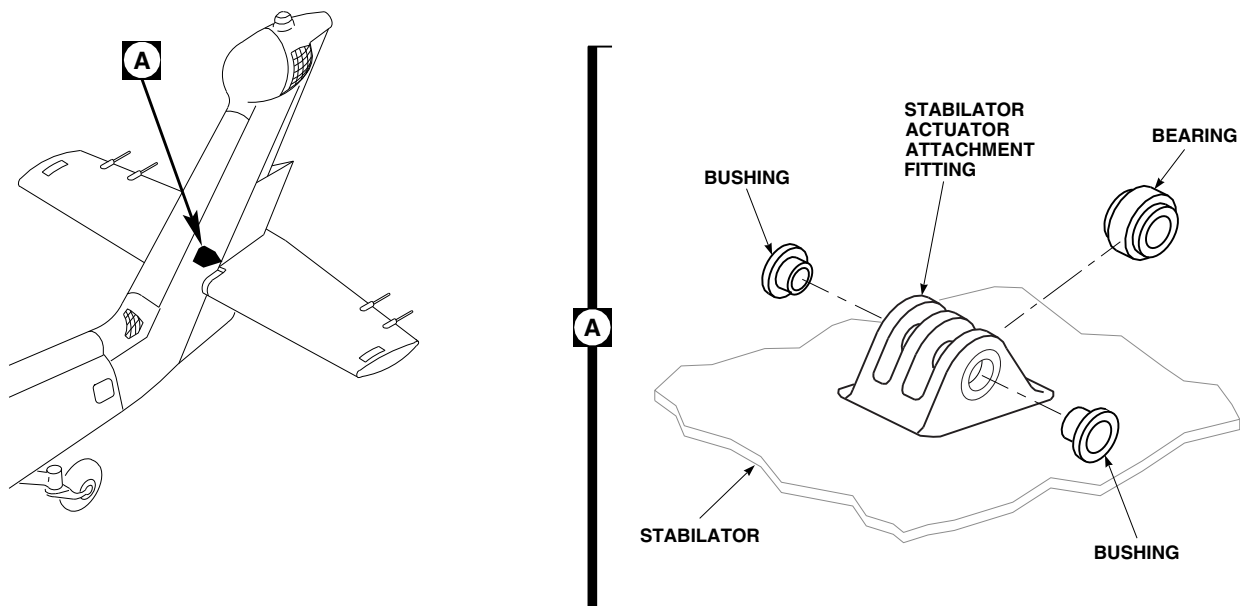
- Bonding of liners in center lobe of stabilator fitting is a critical characteristic. Proper bonding must be verified.
 - Damage to fitting will occur if feeler gage is improperly used. Liner may become disbonded. Use care when using feeler gage to check for liner edge separation.
- a. Visually inspect bonding of liners in center lobe of fitting. Probe for edge separation using 0.0015-inch feeler gage. **SEPARATIONS MUST NOT EXCEED 5% OF BONDLINE LENGTH.** If limit is exceeded, replace fitting (PARA 2-6-48).

- b. Inspect liner bore for corrosion. **PITS MUST NOT EXCEED 0.005-INCH DEEP OR COVER MORE THAN 25% OF LINER BORE.** If limits are exceeded, replace stabilator (PARA 2-4-142).
- c. If corrosion is within limits, remove using scouring pad, Item 270, Appendix D. **MAXIMUM BLENDED DEPTH IS 0.005-INCH.** If limit is exceeded, replace stabilator (PARA 2-4-142).
- d. Inspect liner bore for damage. **LINER BORE DAMAGE MUST NOT EXCEED 0.005-INCH DEEP, 0.040-INCH WIDE, AND 0.62-INCH LONG.** If limits are exceeded, replace stabilator (PARA 2-4-142).
- e. If damage is within limits, blend out using scouring pad, Item 270, Appendix D. **MAXIMUM BLENDED DEPTH IS 0.005-INCH.** If limit is exceeded, replace stabilator (PARA 2-4-142).

2-6-55.1.3 Install.**NOTE**

For the following steps refer to bushing/bearing removal/installation set for bushings and bearing installation instructions to fitting, 70209-06054/70209-07053.

- a. Clean mating surfaces of bearing outer race and fitting bore, using cheesecloth, Item 90, Appendix D, moistened with acetone, Item 25, Appendix D.
- b. Remove residual sealing compound from fitting surfaces and bores, using abrasive cloth, Item 8, Appendix D. Sand very lightly.
- c. Clean sanded surfaces using cheesecloth, Item 90, Appendix D, moistened with acetone, Item 25, Appendix D. Let air dry.
- d. Using clean acid brush, Item 84, Appendix D, apply thin coat of sealing compound, Item 282, Appendix D, to bore in fitting center lobe.



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FIGURE 2-6-55-1. REPAIR STABILATOR ACTUATOR ATTACHMENT FITTING

- e. Let primer dry for 15 - 30 minutes until visibly dry. Protect drying surface from any contamination or hand contact.

CAUTION

Installation of bearing must be completed promptly to avoid premature curing of sealing compound by the frictional heat of insertion. If bearing is not fully installed within four minutes after applying sealing compound, repeat entire cleaning, priming, and compound application procedure.

- f. Using clean acid brush, Item 84, Appendix D, apply even coat of sealing compound, Item 273, Appendix D, to mating surfaces of bearing and fitting bore.

NOTE

For each installation operation, assemble tool from bushing/bearing removal/

installation set according to information plates in tool case.

- g. Assemble tool and install bearing (toolkit instructions step 7 and step 8), holding arbor stationary while turning nut with wrench until sharp increase in torque is felt (Figure 2-6-55-1, Detail A).
- h. Inspect to verify bearing is centered in fitting.
- i. Allow 15 minutes for curing of adhesive holding bearing in fitting.
- j. Clean up excess compound, using cheesecloth, Item 90, Appendix D, moistened with acetone, Item 25, Appendix D.

NOTE

Stagger stake points 45° from those on first side.

- k. Assemble tool and stake bearing (toolkit instructions step 9 and step 10). Staking flare is complete when 0.005-inch feeler gage cannot be inserted between flare lip and fitting.

- l. Assemble tool and install bushings in fitting (toolkit instruction step 11 and step 12).
- m. Apply 1/4-inch minimum fillet of sealing compound, Item 292, Appendix D, around flanged edges of both bushings.
- n. Connect stabilator actuator to stabilator (PARA 2-4-146).
- o. Remove padded stabilator support.

2-6-56 HIRSS FAIRING REPAIR.

PARAGRAPH BREAKDOWN

	PAGE
2-6-56.1 Repair HIRSS Fairing De-	2-6-56-2
laminations	

INITIAL SETUP

Tools

- Airframe Repairer’s Toolkit
- Router Bit
- Squeegee
- Vacuum Bags
- Vacuum Cleaner

Materials/Parts

- Abrasive Cloth, Item 2, Appendix D
- Abrasive Cloth, Item 6, Appendix D
- Acetone, Item 25, Appendix D
- Adhesive, Item 35, Appendix D
- Brush, Acid, Item 84, Appendix D

- Cloth, Satin, Item 105, Appendix D
- Machinery Towel, Item 211, Appendix D
- Masking Tape, Item 212, Appendix D
- Methyl Ethyl Ketone, Item 217, Appendix D
- Plastic, Polyvinyl Sheet, Item 237B, Appendix D
- Sealing Compound, Item 302A, Appendix D
- Stirring Stick, Beverage, Item 315C, Appendix D
- Tack Rag

References

- Appendix D
- PARA 2-6-5
- PARA 5-3-11

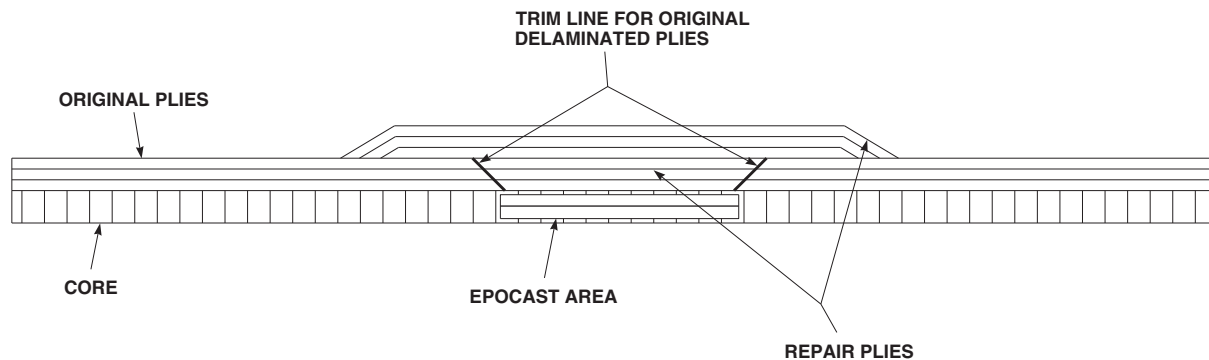
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FIGURE 2-6-56-1. REPAIR LAY-UP

2-6-56.1 Repair HIRSS Fairing Delaminations.**NOTE**

- Repair parts must be a minimum of 1-inch away from cutouts and edges of part.
- Contact Sikorsky Aircraft Corporation if honeycomb core is damaged in excess of 1-inch in diameter.
- This repair is valid for a maximum of 9 x 9-inch skin-to-core delamination.

- Using coin-tapping method, determine extent of delamination (PARA 5-3-11). **DELAMINATION NOT TO EXCEED 9 X 9-INCHES (81 SQUARE-INCHES).**

CAUTION

Honeycomb core can be easily damaged while cutting through skin. Use extreme care to limit damage.

NOTE

Upon removal of cutout skin, note the orientation of the last delaminated ply (ply closest to core). This will be required when performing repair.

- Cut through skin along marked outline of disbonded area, using router bit. Remove and discard cutout skin.

NOTE

A 10:1 scarf angle is required when blending skin plies.

- Blend skin plies using abrasive cloth, Item 2, Appendix D (Figure 2-6-56-1).
- Mark outline around delaminated area 1-inch larger than blended area on all sides. Mark two more consecutive outlines each 1-inch larger than the previous one.

CAUTION

Delamination of honeycomb panel will result if solvent is allowed to seep into panel. Use solvent sparingly to prevent delamination.

- Remove paint from remaining plies inside marked areas using machinery towel, Item 211, Appendix D, moistened with acetone, Item 25, Appendix D.
- Sand overlap area lightly with abrasive cloth, Item 6, Appendix D.
- Using vacuum cleaner, remove loose particles from cavity. Wipe up sanding dust with clean tack rag.

- h. Mask surface just outside marked area with masking tape, Item 212, Appendix D.



Delamination of honeycomb panel will result if solvent is allowed to seep into panel. Use solvent sparingly to prevent delamination.

- i. Clean skin inside masked area using machinery towel, Item 211, Appendix D, moistened with methyl ethyl ketone, Item 217, Appendix D.
- j. Mix sealing compound, Item 302A, Appendix D (Table 2-6-56-1).
- k. Fill exposed core with sealing compound using beverage stirring stick, Item 315C, Appendix D. Force sealing compound into honeycomb to eliminate air pockets, filling core slightly higher than original surface to allow for shrinkage.



Damage to part will result from expansion of wet Epocast when heated. Let Epocast harden to touch before heat-curing.

- l. Cure sealing compound (Table 2-6-56-1).

NOTE

Make sure repair plies are cut perpendicular and parallel (within plus or minus 10°) to fabric weave.

- m. Cut six repair plies from satin cloth, Item 105, Appendix D.

- n. Mix adhesive, Item 35, Appendix D (Table 2-6-56-1).

- o. Spread repair plies on flat surface covered with polyvinyl plastic strip, Item 237B, Appendix D.

- p. Saturate repair plies with mixed adhesive using acid brush, Item 84, Appendix D, or beverage stirring stick, Item 315C, Appendix D.

- q. Apply coat of mixed adhesive to unmasked skin and foam areas of part, using acid brush, Item 84, Appendix D.

NOTE

First repair ply must be placed in direction noted in step b.

- r. Put repair plies in place with each ply overlapping undamaged skin or previous repair ply at least 1-inch in all directions. Fabric orientation of bottom layer must match direction noted in step b. Alternate each repair ply 45°.

- s. Cover repair with polyvinyl plastic strip, Item 237B, Appendix D.

- t. Squeegee trapped air and excess resin out of repair patch.

- u. Apply pressure to repair with vacuum bag.

- v. Cure repair (Table 2-6-56-1).

- w. Remove vacuum bag and plastic strip.

- x. Sand to remove excess resin and blend repair into adjoining surface, using abrasive cloth, Item 6, Appendix D.

- y. Touch up paint, as required (PARA 2-6-5).

Table 2-6-56-1. Resins and Curing Agents

		Mix Ratio, Resin to Curing Agent by Weight			
Resin Base and Curing Agent Type	Resin Base	Curing Agent	Pot Life at 70°F (21°C) Minutes	Notes	Cure Cycle Minimum Time
Resin Base and Curing Agent: Adhesive, Item 35, Ap- pendix D (Type II)	100 ± 2	57 - 59	25 - 30	1, 3	24 hrs at 70°F (21°C) or 1 hr at 240° - 260°F (116° - 126°C)
Resin Base: Sealing Com- pound (Ep- ocast 1652 A), Item 302A, Appendix D (Type II) Curing Agent: Sealing Com- pound (Ep- ocast 1652 B), Item 302A, Appendix D	100 ± 2	11 - 13	40	2, 3	2 hrs at 70°F (21°C) plus 1 - 3 hrs at 145° - 155°F (63° - 68°C)

NOTES

1. High temperature cure follows 1 hour at room temperature.
2. Syntactic foam.
3. Type II (High Temperature). Kevlar/epoxy: 425°F (218°C) maximum service; fiberglass/epoxy: 450°F (232°C) maximum service.